James K. Freericks

Quantum Mechanics Done Right

The Shortest Path from Novice to Researcher

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For Susan, Carl, Samuel and William

Preface

In this book, I will show you how quantum mechanics is "done right." Quantum mechanics has been taught with the same paradigm that was developed in the original textbooks of the 1930s, "modernized" after World War II and then adopted by nearly all textbooks since then. Sure, quantum texts have their differences, but in practice, they are all based on the same methodology—working in the coordinate-space basis using differential equations; this holds for spins-first and Schrödinger-equation-first approaches (because all these two schools do is change the order in which the same materials are presented). Such a practice is similar to having a classical mechanics textbook that uses only the Cartesian coordinate system!

We all know quantum mechanics is a difficult subject. Fortunately, the didactic principles developed in physics education research tell us how to make learning more accessible and more effective. This book is structured with such principles in mind. We begin by developing the conceptual ideas, move onto developing the requisite math and formal ideas, and then apply them to experimental situations. *This book is intended to be accessible but it is not easy.* We only require a high-school-level math background (including algebra, geometry, and trigonometry) and a high-school-level understanding of energy and electromagnetism. All the rest is developed here. If you have ever been told you need to study many years of calculus (and linear algebra) and that you also need multiple physics courses to be ready for quantum mechanics, you now know this is wrong. If you have a high school education, and a commitment to hard work, you are ready.

The foundation for developing such an approach started with Pauli's solution of hydrogen in 1925 and related papers by Dirac in 1926, but it came to the fore with the development of the Schrödinger factorization method in the early 1940s. Herbert Green and Hans Ohanian both developed textbooks that used these principles. Recently, we have expanded this approach to use algebraic methods to determine wavefunctions as well. The spirit of this book is similar to that of Arno Bohm's text, but with a very different focus. It will be hard work, but it is so worth it!

The first quantum revolution (1900–1980) was about understanding quantum mechanics and using its principles to develop technologies (such as the laser and

the transistor) that use these quantum principles. But now we are in the midst of the second quantum revolution (1980–). Here, we can observe, manipulate, and control the individual quanta of atoms, electrons, and light. This is ushering in new technologies, such as the global positioning system (based on atomic clocks) and quantum computers. The second quantum revolution is summarized in the new field of quantum information science and its three pillars of quantum computing, quantum communication, and quantum sensing. This book emphasizes the quantum sensing side of quantum information science, but it will help you prepare for all three areas. It is a new way of teaching quantum mechanics, built for this second quantum revolution. I call it quantum operator mechanics, to differentiate it from the first quantum revolution's matrix mechanics or wave mechanics.

My strategy is to design the materials with accessibility in mind, but not to dilute or modify any of the science; on the contrary, we often cover *more* than what you will find in conventional texts, and certainly we cover more material that is related directly to experiments. *Accessible* does not mean *easy*. This book will take concentrated study and effort to get through all of it.

Who is this book for? First and foremost, it is for those who want to study quantum mechanics, but have not taken the conventional prerequisites. But, anyone who has studied quantum mechanics before will also find fresh new ideas and perspectives in this book. Especially so, because we emphasize a representation-independent way of presenting quantum mechanics. I believe all readers will see something new in the material presented here—even a lifelong quantum mechanic!

Rest assured that quantum mechanics is difficult. But you should never give up on trying to understand something simply because it is hard. I often tell undergraduates in my classes at Georgetown that there is a common method of proving a result that professors, teaching assistants, and even peers will use. It is called *proof by intimidation*. This is when someone of higher authority than you fails to explain something, but asserts it is true. They often end by intimating that you must be dense not to get it. You walk away accepting both that it is true and that you are incapable of understanding it.

This is unacceptable. You should never allow yourself to be academically bullied. My belief is that nothing is beyond being able to be explained and understood by someone with a high school education. So, if you have ever wondered what the quantum world is all about, here is your chance to explore it. There will be no proof by intimidation, only proof by understanding. Yes, you *can* do it.

If you find that scientific and mathematical material was often confusing to you, my pledge is to get you through this material if you can invest the time and effort to ask why and to persist until you will eventually be able to say "I understand."

The first four chapters are dedicated to developing the conceptual ideas of superposition, complementarity, measurement, tagging, and entanglement by examining spin in atoms and the quantum properties of light. Once the conceptual ideas are in hand, we then move into developing some mathematical skills (based on the high school algebra and trigonometry sequences) that will be needed for the more formal work (complex numbers, exponentials, 2×2 matrices, $\cdots$). Chapters 6–12 then cover much of the time-independent material found in conventional undergraduate

and graduate texts, but we also discuss spin models, which are rarely covered in either. Then we have another mathematical interlude in Chap. 13, which discusses some of the more advanced mathematical ideas needed (primarily in linear algebra). The last third of the book focuses on time-dependent phenomena and second quantization of electrons and photons. We even describe how the laser interferometry gravitational-wave observatory works—one of the quantum engineering marvels of the twenty-first century. If you want even more details of what is in the book, please browse the Table of Contents. Every chapter starts with a quote from great thinkers who are not scientists (except for one). Enjoy their wisdom.

If you have been paying attention, you heard me right. We will be taking you through most of the undergraduate and graduate quantum mechanics curriculum in a book that uses no calculus. We will do this without sacrificing rigor, and we will cover more material than is usually found in those texts. There is one small price we do pay for this. We will not cover perturbation theory, scattering theory, or semiclassics here, as they all require some calculus to do it right.

Will you really become a quantum mechanic after finishing this book? You bet you will! By this, I mean that you actually will have the skills to contribute to mentored research on quantum mechanics. How do I know this? Because I have already done this numerous times with students and citizens who have had an interest in quantum mechanics and who have worked through early versions of these materials. So it really does work and it really prepares you to tackle quantum ideas at a high level.

Let me end by discussing my motivation for creating this work. I am a through-and-through quantum mechanic. I find this field fascinating and nearly all of my research is in the area of quantum mechanics. Currently, this field is restricted to experts only, with stark warnings that one must prepare for years to be allowed to sit at the table (just search for the math preparation needed for quantum mechanics on Quora and you will see what I mean). Rubbish! Quantum mechanics is one of the pinnacles of human intellectual development. Everyone should know something about it and those who wish to know more should be able to do so without being forced to spend many years to do so. This is what motivated me to write this book and it also motivated me to create online courses that bring this material to everyone.

In order to bring this work to the widest possible audience, I have made the electronic version open access, available to *everyone*. This comes at a rather significant cost, so I do ask if you find this work useful to you and you have the means to do so, to also purchase the hard cover version of the book. This helps support additional work (and I do have a few more book ideas I am working on). Because the work is open access, all original materials are available for noncommercial re-use. I do include some copyrighted materials in this work and they are clearly described as such where they appear. That material is expressly not available for re-use without requesting permission from the copyright holder. Many of the figures in this book are pictures of the original researchers in the field. I have modified them to look like sketches using the software *graphite*. I did this to make them all have a similar feel as you see them in the work.

The electronic version of the book does have a number of live links in it (tiny urls are available in the print book). We have striven to make links to the most permanent locations we could find, but I apologize that some links may become stale over time. If so, a quick google search will hopefully find what you are looking for. I have a YouTube channel quantum4everyone https://tinyurl.com/bdhssec5, a Quora feed quora-freericks https://tinyurl.com/284mbu6x, and a local site quantum.georgetown.domains https://tinyurl.com/4u9dbmxz that provides a repository for my work on quantum mechanics pedagogy. In addition, the computer tutorials are available for downloading at github https://tinyurl.com/43e487wu. Finally, I created a customized YouTube playlist for the book https://tinyurl.com/yt2895u3, which has a video describing the contents of each chapter, as well as some of its special features. It starts with Quantum Mechanics Done Right https://tinyurl.com/3y2ff55e, The Origins of the Book https://tinyurl.com/hbsjw4zf, and Accessibility versus Easy https://tinyurl.com/2vmmh3zn. I hope you will enjoy these additional resources.

Come and join the fun! Learn how to become a quantum mechanic. You already have all the background you will need.

Acknowledgments This book was a massive undertaking. The early origins of it began in 1996, when I worked with my Georgetown colleague Amy Liu to develop a course for nonscientists called *The Quantum World Around Us*. This course was based heavily on Feynman's *QED* text. We were also influenced by Dan Styer's marvelous text (*The Strange World of Quantum Mechanics*). During this time, we met Ed Taylor, who showed us how the use of animations and tutorials can improve how students learn the conceptual ideas of quantum mechanics. This course ran for about ten years at Georgetown before I moved onto other ventures. In 2014, with the MOOC craze, I started developing a shortened version of the original Georgetown course into a six-week-long MOOC. It launched in 2017, with over 50 high-level computer animations, simulations, and tutorials. Students who completed that course motivated me to carefully re-assess how we teach quantum mechanics and to look for ways to develop the material algebraically instead of via calculus-based methods. Though skeptical at first, finding the resources by Green (*Matrix Mechanics*) and Ohanian (*Principles of Quantum Mechanics*) motivated me to see how far these ideas could really go. It is amazing just how far! This book has taken about seven years to be developed since then. No, I am not a slow writer. Many of the methods that needed to be used had not been developed yet, so we had to invent them. This took long and hard work. But we made it. This book is the fruit of that labor. *It is quantum mechanics done right!*

I had the pleasure to collaborate with many students, faculty, and researchers in developing this work. I wish to thank them all. Amy Liu co-developed the original Georgetown course. Randy Bass and Eddie Maloney supported the early MOOC development, with help from Anna Kruse, Susan Pennestri, Zhuqing Ding, Anne Dumanat, and Yianna Vovides. Two undergraduates, Dylan Cutler and Lucas Vieira-Barbosa, masterfully developed computer simulations, animations, and tutorials in javascript for the MOOC which are linked to throughout the book; images from these codes were extracted by Ben Lobb and Keira Schoolcraft for use in this book. Two other undergraduates, Pia Bhatia and Daphne Maniatis, helped provide a student perspective for the undergraduate course and also helped with many technical details of launching the class on edX. Leanne Doughty, Cece Ochoa, Justin Tran, and Justyna Zwolak have performed education research on materials from the on-line class. The quantum pedagogy research work involved many different people ranging from citizen scientists to students and beyond. This includes Sanchali Banerjee, Sanya Bharti, William Courtney, Maurice Curran, Christina Daniel, Kyla Fraser, Carly Healy, Xinliang (Bruce) Lyu, Eduardo Mungía-González, Livia Guttieres, J. Alex Jacoby, Sunny Kang, Paresh Kenkare, Hansen Lian, Joseph Noonan, Sandro Orjuela, Marko Petrović, Sheldon Rego, Maaz ur Rehman Shah, Michael Rushka, Tomasz Szymanski, Zu-Yao Teoh, Mark Weitzman, David Wolf, and Luogen Xu. In addition, I have had scores of "alpha testers" who have aided the effort by reading and commenting on materials. There are far too many to thank them individually. But, I will mention a few whom I have met through MOOCs or elsewhere who provided particularly useful comments. This includes Apostolos Christou, William Courtney, Sakshi Jha, Michael Johnson,

Image credit: Ruth Monnier. [Converted to look like a sketch by James K. Freericks.]

Jukka Kinkamo, Alan Levin, Sandy Morgenstein, Marco Strasser, Andrew Sutter, Walter Urbanek, Mark Weitzman, and Iman Zabett.

I received numerous grants which helped support this work over the years. They came primarily from the National Science Foundation and Georgetown University. You can find those acknowledgments in the Epilogue (in Part IV of the book).

Professionals have also helped and encouraged me in this project. I would like to acknowledge Rodney Bartlett, Mark Esrick, Jim Franson, Kiko Galvez, Branislav Nikolic, Hans Ohanian, Wolfgang Schleich, Dominika Zgid, and Barton Zwiebach for assistance or helpful ideas during the work on this book.

Carole Sargent from Georgetown University was a continuous source of encouragement and support and her weekly faculty scholarly publishing groups were a source of inspiration to help me complete the project.

Finally, I would like to especially acknowledge Wesley Mathews Jr, a long-time faculty member at Georgetown University, who first introduced me to "operator mechanics." We worked together on developing a number of the techniques discussed in this book. He passed away before it was completed, but his spirit is represented throughout these pages.

Washington, DC, USA
November 30, 2024

James K. Freericks

The original version of the book has been revised. A correction to this book can be found at https://doi.org/10.1007/978-3-031-87845-9

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Part I
Conceptual Ideas

Chapter 1
Introduction to the Quantum World

Let's get small!

Steve Martin, American comedian and actor, 1977

In 1977, Steve Martin released his Grammy-award-winning comedy album entitled *Let's Get Small* (see Fig. 1.1). This is what we do in the quantum world—get small! While everything in the universe is described by the laws of quantum mechanics, the need for quantum mechanics becomes patently clear when we try to determine properties of the ultrasmall, starting with atoms, electrons, and photons. We will not discuss the historical progression that led to this discovery, but instead we will focus on how one interrogates the ultrasmall world. We have a video providing an overview of Part I: Conceptual Ideas https://tinyurl.com/5cf9ry2m and of Chap. 1 https://tinyurl.com/36avn93x.

Quantum mechanics terms are prevalent in popular culture. We often use the "uncertainty principle," or a "quantum leap," in everyday speech, even though we may not have a clear idea of what they really mean. Worse still, most everything you have learned about the real world, and especially about quantum mechanics, has elements that are likely wrong. For example, the following ten statements all have things that are wrong in them:

- Light is a wave, not a particle.
- The electron orbits the nucleus like the Earth orbits the Sun.
- To learn quantum mechanics requires one to learn about differential equations.
- Quantum phenomena only occur inside special laboratories and with expensive equipment.
- The Heisenberg uncertainty principle means we never can know where a particle is.
- Quantum physics is so weird anything can happen and everything is uncertain.
- A quantum leap is akin to a paradigm shift.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/978-3-031-87845-9_1.

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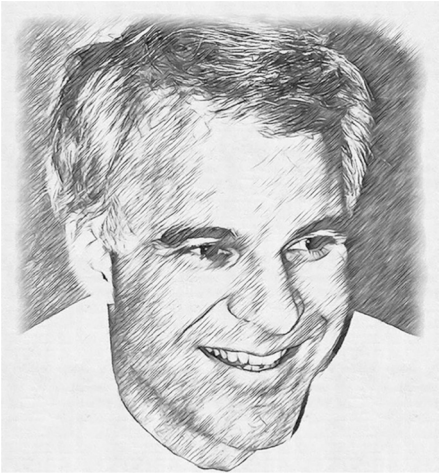

Fig. 1.1 Steve Martin in 1982, shortly after he got small (Courtesy of Towpilot. CC Attribution-Share-Alike 3.0. Converted to look like a sketch by James Freericks)

- An electron decides when to act like a wave and when to act like a particle.
- Quantum mechanics only applies to very small things, not to things we encounter in everyday life.
- Quantum mechanics is a new theory.

By the end of the book, it will be clear to you what is wrong with each of these statements. We have a short video discussing Let's Get Small https://tinyurl.com/2vvt4abc.

The concept of small things, especially objects that are so small we cannot see them, is a difficult one for us to work with. The Vienna Circle of Philosophy, led by Ernst Mach, even went so far as to believe that if they could not see it, it did not exist. We will soon examine how one learns about objects in the ultrasmall domain, how we can eventually see them, interact with them, and even manipulate them. But before we embark on that journey, I want to start with a discussion that dispels the picture most of you have in your head about atoms.

It all started back in elementary school when you were asked to draw a picture of an atom. You were told that protons and neutrons lived in a central region called a nucleus, and electrons lived in shells outside the nucleus. Often, you were given a specific element, like nitrogen, and told how many protons (7) and neutrons (7) to include. Your drawing ended up looking something like the one in Fig. 1.2a. This image is even common on the Internet, when you search for the image of an atom, as in panel (b). Here is what is wrong with this picture. In 1911, Hans Geiger and Ernest Marsden conducted an experiment, suggested by Ernest Rutherford, that shot high-energy helium nuclei at a thin gold foil (Rutherford, 1911). The helium nuclei came from a radioactive source that emits alpha particles (which are helium nuclei), with a high energy. The energy is so high that they were expected to travel nearly straight through the foil, like bullets shot through a piece of paper. But, Marsden and Geiger found that a small fraction bounced back! Rutherford said this was like shooting an artillery shell at a tissue paper and having it bounce back. The conclusion of this experiment was that the nucleus had a size that was 100,000 times smaller than the

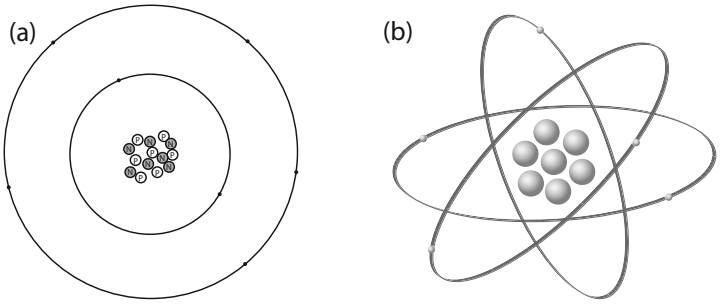

Fig. 1.2 Poor, but common, conceptual images of atoms. Panel (**a**): the picture many of you drew in elementary school, here for nitrogen. Panel (**b**) the common symbol for an atom, arising from the Bohr-Sommerfeld model, inspired by the Rutherford experiment

size of the atom (which was already known to be about 0.1 nanometers (or 1 Å, one Angstrom) large). A nanometer is one-billionth of a meter—a size too small for most of us to comprehend; the smallest object we can see with optical light (with the best microscopes) is about 1000 times larger than one nm. If the atom was blown up to the size of a soccer ball, the nucleus would be 100,000 times smaller—represented by a dot less than the thickness of your hair, or a speck of dust, and needing a magnifying glass just to see it. The electrons that fill the rest of the atom, they are even smaller. All known experiments see them as having no intrinsic size at all. These observations motivate a description of an atom as mostly "empty space." Such a description also has issues with it, but it is a far better image to have in your head than the elementary-school picture. We have a video summarizing the quantum mechanics you experience in everyday life, called The Quantum World Around Us https://tinyurl.com/yxz5z7p2.

If you are going to study ultrasmall objects, you better understand how we can perform experiments on them. The first one we will explore in detail is that of the Stern-Gerlach experiment from 1922. But, to understand how it works requires you to understand a lot about how magnets and current loops interact with magnetic fields. This is a quite complex topic, which we focus on next.

1.1 Classical Stern-Gerlach Experiment

Wait a second. I signed up to learn about quantum mechanics. What is this magnets stuff? It turns out that the simplest quantum problem, which also teaches us one way to observe the ultrasmall world, involves separating atoms by using the fact that they act like small effective magnets. So, we need to understand how this can be done, which requires us to learn about how magnets are affected by magnetic fields in the classical world. It turns out this "simple" problem is still pretty complicated! But don't worry. We will break it down into bite-sized chunks and then assemble it all back together. Ready? Let's go! Many of the ideas developed here are due to Daniel Styer and his remarkable book (Styer, 2000).

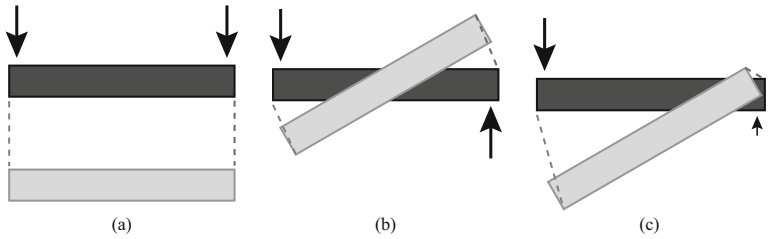

Fig. 1.3 Pushes and pulls, twists and turns. The dark rectangle is at the initial time and the arrows show the pushes or pulls acting on it. The light rectangle is where the dark rectangle evolves to after a short period of time under the given push or pull. (**a**) The push is the same at each edge, so it moves *without* twisting. (**b**) If the push is exactly opposite on each edge, it will *just* twist. (**c**) If one push is stronger, it will twist *and* move

Everyone has played with magnets at some point in their life (if for some reason you never have, please go out and get some to play with before reading further and watch the video Opposites Attract https://tinyurl.com/2anv4fzr). After playing with them a bit, you can observe some facts about magnets: (i) there are two poles to the magnet, called the North pole and the South pole; (ii) the opposite poles attract, while the similar poles repel; and (iii) the strength of the push or pull of the magnets on each other grows as they get closer to each other. You might have also observed that some magnets are stronger than others.

The technical term for the push or pull is a force, but among friends here, it is okay to use push or pull. Let's understand a bit more about how this works. Consider a rectangular object that we push downward *with exactly the same strength* at each of its edges (as shown in Fig. 1.3, left panel). Clearly it will be pushed downward, but you should also intuitively know that it won't twist while being pushed (take out a pencil and push it with two fingers if you are not sure what will happen). Similarly, if we push *oppositely* on each edge with the same strength, it will twist *without* having the center move (see Fig. 1.3, center panel). This twisting action of the two pushes is called a torque, but again, among friends, it is okay to call it a twist or turn. Finally, we have the general case, where we do both at the same time. Here, we push on the two ends, but with different strengths. We will see both motions: the rectangle will twist and its center will move in the direction where the push is the strongest (depicted in Fig. 1.3, right panel). A video depicting this is called Pushes and Pulls, Twists and Turns https://tinyurl.com/8p4azrx8.

We do want to find a way to work with the fact that different magnets have different strengths, but the way the push or pull decreases with distance is always the same. This motivates the use of a concept called a *magnetic field*; the magnetic field is given by the push or pull on the north pole of a standardized magnet—then to find the push or pull on our magnet, we multiply the field by the strength of our magnet. We illustrate it with what we call a *fat arrow*. The arrow points in the direction that the North pole of a small "test magnet" would get pushed, while the width represents the magnitude of the push or pull. Think of this as a graphic mnemonic

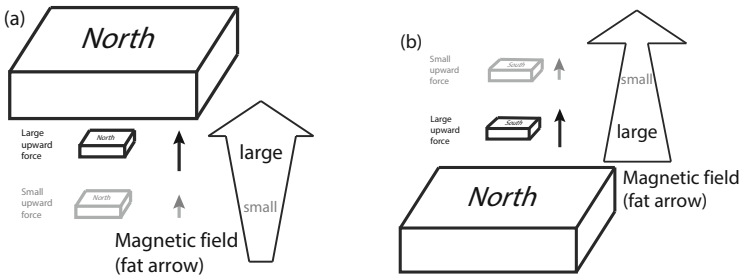

Fig. 1.4 Abstraction of the magnetic field and the fat arrow. On the left, we show two "floating" test magnets and look at the push or pull on the North pole of the test magnet. It is upward and stronger the closer we are to the large magnet. So it is pulled up and the pull decreases as we go further away. This is represented by the fat arrow pointing up and getting narrower as we move down. On the right, we repeat for test magnets above the large magnet (the test magnets have the North pole pointing toward the large magnet). Here the push is upward and decreases the farther we are from the magnet

that helps you visualize the magnetic field from a magnet (which determines the push or pull on a test magnet), since we cannot see the magnetic field with our eyes. More concretely, look at the two illustrations in Fig. 1.4. In the left panel, we look at two locations for a small test magnet, with its North pole directed up toward the large magnet. Since opposite poles attract, it feels a strong upward force that decreases the farther away from the center we go. In the right panel, we approach the magnet from above, again with the North pole pointing toward the large magnet. Here we feel a strong upward push that decreases the farther up we go. We represent this by a fat arrow. The direction of the arrow is the direction of the push on the North pole of the test magnet, while the width of the fat arrow tells us about the strength of the push (wide means strong push, narrow means weak push). It takes some time to get used to this abstract fat arrow device, but it will become quite useful for us. Indeed, once we are used to working with it, we won't need to know what created the magnetic field, we will just need to know what the magnetic field (or the fat arrow) actually is. We have a video discussing this called Fat Arrows are Fields https://tinyurl.com/29fetnaz.

Remember our goal is to perform experiments on objects that are so small we cannot see them with our eyes. Our first approach to model these objects is to assume that they act like small magnetic needles. This is similar to the needle of a compass or can also be thought of as simply being a thin test magnet with a North and South pole. When we model the behavior of these magnetic needles in a magnetic field, we can simplify the behavior to that of forces applied to the North and South poles of the needle only. If we are told what the magnetic field is, via a fat arrow, we can determine the push or pull at each pole of the magnetic needle and from that determine how the center is pushed or pulled and how the needle twists.

In Fig. 1.5, we show what happens to a magnetic needle placed in the magnetic field at the bottom of a large magnet that has its North pole facing up. Recall this

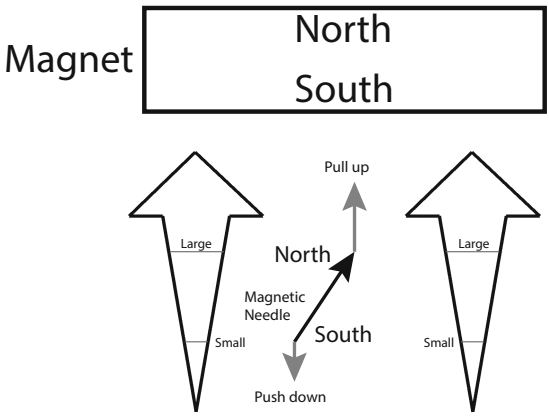

Fig. 1.5 "Floating" magnetic needle in the field below a magnet (illustrated by a fat arrow) with the North pole pointing up. The force on the needle is found by considering the larger force up on the North pole minus the smaller force down on the South pole; the net force will be up. We can see that the needle will also twist in the counter-clockwise direction

is the way the fat arrow looks below the South pole of a magnet. Looking at the fat arrow, the magnitude of the force is determined by the width of the arrow (indicated by the "large" label for the North pole and the "small" label for the South pole). The direction of the force is in the direction of the fat arrow for the North pole and opposite for the South pole (gray arrows). One can see the net force is up and that the magnetic needle will twist toward vertical in the counter-clockwise direction. While you might think the needle will then just align with the magnetic field direction, it always overshoots this twisting motion, because the pushes also cause it to twist in the counter-clockwise direction all the way until the needle is oriented along the vertical direction where the pushes now cause the twisting to be reduced (be sure you understand that the tendency to twist changes as we go past vertical by drawing the arrows for the pushes and pulls on the poles using the fat arrows). This means it will oscillate back and forth forever, unless there is friction to slow it down. In the atomic world, there is no friction, so it really does go forever. If you are having trouble understanding that it overshoots, recall that nothing can stop instantly—everything must be slowed down to stop—here we only start trying to slow down after we pass the vertical orientation. After it overshoots, the push on the S pole increases and the pull on the N pole decreases leading it to eventually stop and then oscillate in the opposite direction until it overshoots again and the process continues. Watch the video Twisting Magnetic Needles https://tinyurl.com/yn86y83c for a visualization of this motion.

You may find having to draw the two arrows and combining them to get the net force to be a bit tedious. We actually have a short-hand way to do this. Okay, it is more of an alternative way to do it rather than a short-hand way, as this method is also a bit tedious. But, it gives us a geometric picture, which is useful to have in your head, so we do need to understand it. It is called the method of projections and is described graphically in Fig. 1.6.

The method of projections provides an alternative way to determine the net push on a magnetic needle. We first take the magnetic field we are working with and extract from the fat arrow what we call the axis of increasing magnetic field. This

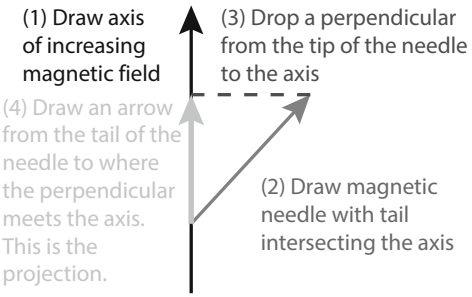

Fig. 1.6 The method of projections. (1) Start by drawing the axis of increasing magnetic field (black). (2) Draw the magnetic needle with its tail on the axis (dark gray). (3) Draw a perpendicular from the tip of the needle to the axis (dashed line). (4) Draw an arrow from the tail of the needle to the perpendicular (light gray). This is the projection of the needle on the axis of increasing magnetic field. It is positive if in the same direction as the axis and negative if opposite. If positive, the force on the needle is in the direction of the *magnetic field* and if negative, opposite

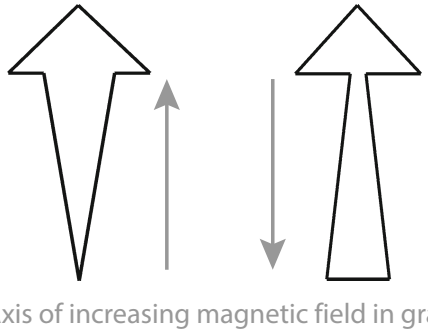

Fig. 1.7 How to determine the axis of increasing magnetic field. Starting from the fat arrow for the magnetic field, we see whether the field increases (arrow gets wider) in the direction of the field or decreases (arrow gets narrower) in the direction of the field. This determines the axis of the increasing magnetic field as being in the same or opposite direction to the direction of the field

axis is either in the same direction as the field (if the fat arrow goes from narrower to wider along the direction of the field) or in the opposite direction (if the fat arrow goes from wider to narrower along the direction of the field). This is illustrated in Fig. 1.7.

The way the method of projections works has five steps. First, we must determine the axis of increasing magnetic field from the fat arrow that describes the field. Second, we draw the magnetic needle in its current orientation, with its tail touching the axis. Third, we draw a perpendicular to the axis of increasing magnetic field. Fourth, we construct the projection by drawing an arrow from the tail of the needle to where the projection intersects the axis. The length of the arrow is the magnitude of the projection. The sign is positive if the arrow points in the direction of the axis of increasing magnetic field and negative if it points opposite. Fifth, the force is along the direction of the magnetic field if the projection is positive and is opposite

if it is negative. Note that the direction of the magnetic field may lie along or opposite to the direction of the axis of increasing magnetic field. Do not conflate the two—see Fig. 1.7 for the relationship between the direction of the field and the axis of increasing field. These rules are complex and not easy to recall, so if you use them, you want to refer to this set of steps and carefully follow each step. We have a video that discusses them in more detail titled Projections to Calculate Forces https://tinyurl.com/nhz6x2sx.

Using either rule to determine the push or pull and the twisting motion on the magnetic needle, we describe the complete motion next. We will look at two cases, but the methods allow you to analyze any case. Assume we place the needle in a magnetic field pointing up and increasing up. Then, if we have a needle that starts with a positive projection (i.e., tending upward), then it will oscillate (similar to a metronome) and be pushed up. The magnitude of the push changes with the orientation, so it is a constantly changing magnitude of the push. Things are even more complicated if it starts pointing somewhat downward. Then the initial projection is negative, but the twisting motion is the same as before (orienting to try to align it upward), so we will transition to a positive projection at some point. Here, we see the same kind of oscillation of the magnetic needle (just over a larger range of angles), but now, the push is initially down, switches to up and then back to down in a periodic way. Most likely the net push, averaged over time, will be up, but one could imagine having an initial setup where there is oscillatory motion *both* for the twisting and for the motion of the center of the needle (think of the initial needle pointing opposite to the magnetic field, where the needle oscillates up and down and rotates back and forth on the circle). This just is very complicated!

Our plan is to describe the classical Stern-Gerlach experiment, but there is one more wrinkle we need to include. Effective magnets can also be created by current loops and we need to analyze this case too. Parts of this motion are even more complex, but the final result is actually quite a bit simpler. So beware, we have a rough road ahead, but in the end it will be smooth sailing.

In 1820, the famous Danish chemist, Hans Christian Ørsted discovered that electrical currents create magnetic fields. Both Ørsted and his law are illustrated in Fig. 1.8. The law he discovered is illustrated schematically on the right. We take our right hand and point the thumb in the direction of the current flow (indicated by the white arrow). Then our fingers curl in the direction of the magnetic field, which changes as we move around the wire. The strength of the magnetic field decreases as we move away from the wire, as you might expect.

We are not interested in current flowing in a straight wire for the Stern-Gerlach experiment. Instead, we want to consider what happens when the wire is wrapped around itself to make a circle. We call such a ring with current flowing through it a *current loop*. The current loop must have a magnetic field associated with it, and it turns out it looks similar to the field of a small bar magnet, so we usually reason about current loops by thinking about the behavior of the *effective magnetic needle associated with the current loop*. This effective needle is found by curling the fingers of your right hand in the direction of the current flow. Then your thumb points in the direction of the North pole of the effective magnetic needle, as shown

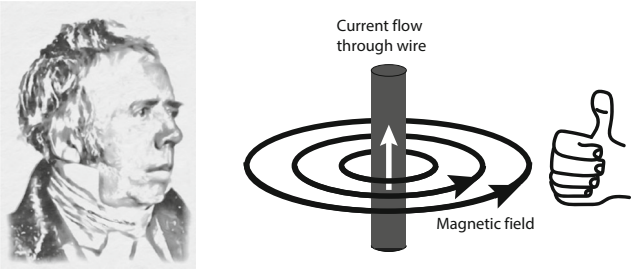

Fig. 1.8 Hans Christian Ørsted and Ørsted's law. Left, image of Ørsted. Right, schematic of Ørsted's law. When current flows through a wire, we point the thumb of our right hand in the direction of the current flow and the fingers curl in the direction of the magnetic field. This magnetic field decreases the farther from the wire we go (Ørsted image from digatalmuseum.se, is in the public domain and was modified to look like a sketch by James Freericks)

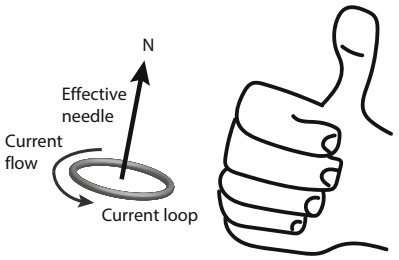

Fig. 1.9 Effective magnetic needle of a current loop. Using the fingers of the right hand, we curl the fingers in the direction of the current flow in the current loop (here, the current flow is counter-clockwise when looking at the current loop from the top). Then the thumb points in the direction of the North pole of the effective magnetic needle associated with the current loop

in Fig. 1.9. As a useful exercise, you should use the right-hand rule for Ørsted's law to verify that the magnetic field of the current loop does indeed look like that field of a magnetic needle (we have the videos Right is Correct https://tinyurl.com/rv4dhzhe and Current Loop Moves a Bar Magnet https://tinyurl.com/bd5euxdh to help). Why a current loop? There was a popular model of atoms at the time, called the Bohr model, built off the results of the Rutherford experiment, that imagined the atoms as a small solar system with electrons orbiting around an ultrasmall nucleus that contained all of the positive charge. While this model ends up not being a particularly good model of the atom, it does motivate thinking of atoms as possibly containing elements that behave like current loops. Indeed, we do find this current-loop-like behavior in atoms (and even in isolated electrons). So, we need to understand how ultrasmall current loops behave in magnetic fields.

When a current loop is placed into a magnetic field, it will feel a push or pull using the effective magnetic needle (and the method of projections) to determine the direction and strength of the force (see Current Loop is an Effective Magnet https://tinyurl.com/udwjrw3f). But, instead of oscillating as a magnetic needle does,

it will undergo motion called precession. There is no other way about it. The motion of precession is difficult to grasp and hard to describe. But let us try to do so in as simple a way as possible because this is the way atoms behave in a magnetic field and we need to be able to understand this to make progress with quantum mechanics. One thing you can do to help you understand this motion is to look at videos of a precessing bicycle wheel or of a gyroscope (you can also purchase toy gyroscopes and play with them). This will make it easier for you to understand the complicated motion we will describe next. We have some videos we have made as well: To and Fro https://tinyurl.com/mtdhdu6j; The Complicated Motion We Call Precession https://tinyurl.com/2pxazb8d; and Motion of Current Loops https://tinyurl.com/3x46tnbw.

If we take a piece of string, put a weight at one end and let it oscillate, we have created a pendulum. It oscillates back and forth, just as our magnetic needles do. But, if I do the same thing to a spinning bicycle wheel, we see different behavior. It does not oscillate anymore, instead it *precesses*. In precession, an axle sticking out, perpendicular to the wheel, traces out a circle in the horizontal plane. There is no simple way to describe why precession acts this way. In a classical mechanics class, you will study this motion in detail, even working out the specific equations of motion of this behavior. But, even after completing that, you often still have no good physical feel for just why it behaves this way; at least I did not. The best I can come up with is that the rotating motion brings in additional constraints due to the conservation of angular momentum, which modifies (and deflects) the motion in this nonintuitive way.

Magnets coming from current loops behave in the same way. The effective needle that represents them acts like the axle of the bicycle wheel and traces out two circles (one for the head and one for the tail). This is illustrated for you in Fig. 1.10. We show half of the period of motion here. The first needle is oriented at around 2 o'clock and ends at around 10 o'clock. The key to this motion is that the projection on the direction of the magnetic field is a constant during this motion. This means the projection on the axis of increasing magnetic field will also be a constant—in other words, the push or pull that the current loop feels in an inhomogeneous field is *constant* in time.

This allows us to describe the motion of the current loop in an inhomogeneous magnetic field. It will have its effective magnetic needle precess, but it also will be pushed up or down, depending on the size of the projection of the effective needle on the axis of increasing magnetic field over time. This force will be constant as long as it is influenced by the magnet. Note how this is different from the magnetic needle, where the strength of the push or pull usually changed with time, because the needle's projection on the field changed with time (watch the video Current Loop in a Magnetic Field https://tinyurl.com/49kc5wjf for a visualization of this).

The Stern-Gerlach experiment involves a little more to its setup. We need a source of current loops. In a real experiment, this is accomplished by placing some source of material (in this case a pellet of nearly pure metallic silver) in a box with a hole in it and heating it up. As we get past the melting point, silver atoms will

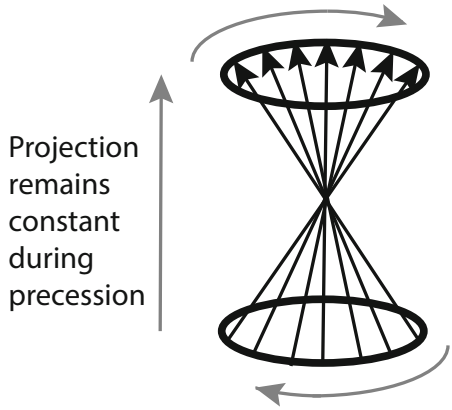

Fig. 1.10 Motion of precession. We focus on the effective magnetic needle that describes the current loop (black arrows above). Then, for precession, both the tip and the tail trace out circles, rotating at constant rates, and with the center of the needle not moving (so they trace out the surface of two cones oriented opposite each other and meeting at the tips). In this case, we also have that the projection of the effective needle on the direction of the magnetic field is a constant. Here we show half of the period of motion of the precession of the effective magnetic needle around the circle

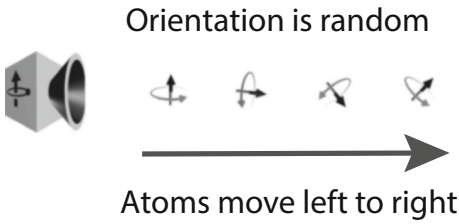

Fig. 1.11 Current-loop beam used in a classical Stern-Gerlach experiment. Initially, you can think of the source as producing a collimated beam of current loops with random initial orientations, as shown. We graphically illustrate the current loop with the tapered, curved arrow. The effective needle is the straight arrow in the center, with the head (darker) at the North pole. Since this is a classical source, we actually will be able to control the direction of the current loop, so in subsequent figures, the current loop direction is determined by the orientation of the source (which will be a polarized source, rather than a random source)

evaporate and come out of the hole in a steady stream, with a high velocity. We use another hole downstream to collimate the beam (i.e., make it into a thin beam) and then we pass it through an inhomogeneous magnet. A classical (macroscopic current loop) setup is shown in Figs. 1.11 (for the source package) and 1.12 (for the rest of the experiment). In the first figure, we show the source and the current-loop beam with random current loops emerging (in subsequent figures the classical current-loop source will be polarized). In the second, we show the inhomogeneous

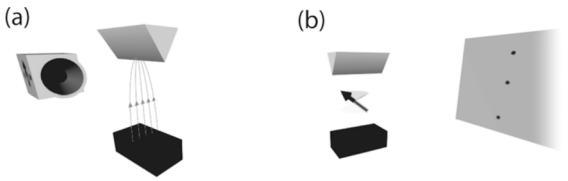

Fig. 1.12 (**a**) Close-up of the polarized current-loop source and the inhomogeneous magnetic field (black is North pole and gray is South pole). The lines with arrows illustrate the direction of the magnetic field and their spacing illustrates the strength (many lines mean a stronger field); the sharp point magnifies the field and makes it more inhomogeneous. (**b**) Close-up of the current loop in the inhomogenenous field. The effective needle precesses and it will feel a push depending on the size and direction of the projection of the effective needle on the vertical direction. Current loops hit the screen at the end of the beam with a vertical location depending on the value of the projection

field and how it affects the current loops. What do you expect to happen? Please think about this using what you know before reading the next paragraph.

Did you make a prediction? Good. We expect the current loops to keep moving in a straight line, except in the region where the field is. There, they will precess and they will feel a push or pull (because they act like little current loops). The strength of the push or pull is given by the magnitude of the projection. If the current loops in the beam are completely random, meaning they have all projections equally likely, we expect the pushes and pulls to separate them out with the largest positive projections at the top and the largest magnitude negative projections at the bottom (see Fig. 1.13 for snapshots of these motions and for the final image on the screen). The inhomogeneous magnet separates the current loops according to their projections by giving them a push that starts them moving vertically and continues to make them grow farther and farther apart, until they reach the screen, where they hit and stick. The screen must be far enough away that the separation is "magnified" enough by this vertical motion so that we can see it. If you have ever seen light pass through a prism, it is the same final effect. The light from a narrow beam is separated according to its color (looking like a rainbow) with the beam spread out enough that we can easily see the different colors. This is what we expect to see when we run the Stern-Gerlach experiment with silver atoms and we have a video entitled The Classical Stern-Gerlach Experiment https://tinyurl.com/3ec4bj2r and a tutorial animation https://bit.ly/4pAdQsB on this. But, that is not what happened. To learn what did, run, do not walk, to the next section of this chapter.

But before you do that, note that we end every section with a set of problems that will help you with mastering the material. If you are working through the book via self-study, it is important that you work through as many of these problems as you can.

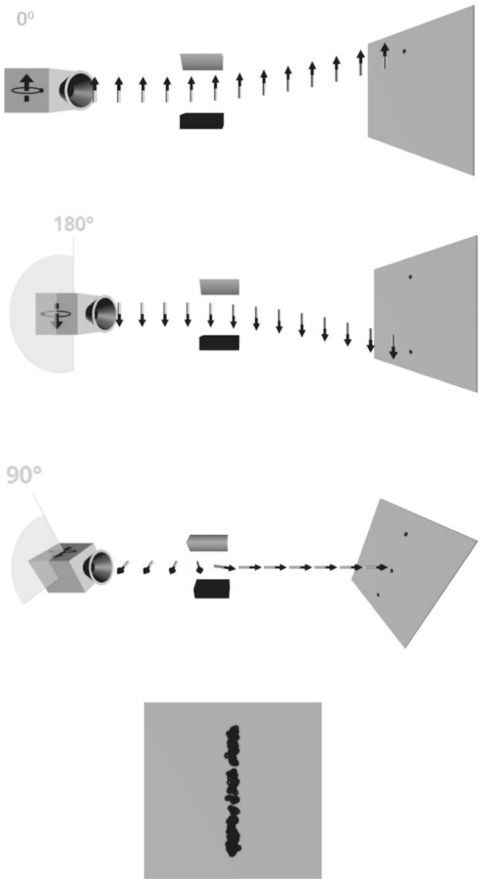

Fig. 1.13 Prediction for the classical Stern-Gerlach experiment with a source of randomly oriented current loops. We show three different initial orientations of the effective needle of the current loop (vertical up, vertical down, and horizontal). The paths show how each case precesses, is deflected, and how it strikes the screen. The bottom panel shows our expected image after many current loops have struck the screen; the position on the screen indicates the value of the projection of the effective needle on the direction of the magnetic field

Chapter 1.1 Problems:

1.1.1 Fat Arrows

Draw the fat arrows for the magnetic field for the following cases: (i) the region above a magnet sitting on a table with the South pole pointing upward and (ii) the region below a magnet sitting on a table with the South pole pointing upward.

1.1.2 Make It Turn

Consider the four depicted cases of pushes on a bar depicted in Fig. 1.14. The arrows all indicate equal magnitude pushes or pulls on the two sides of the bar. Identify

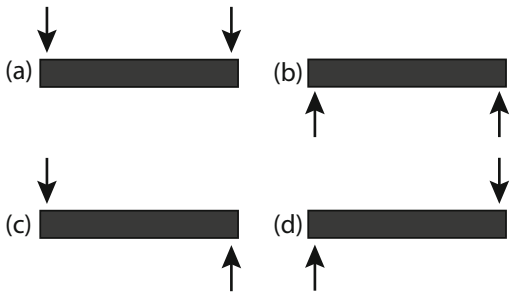

Fig. 1.14 Forces acting on a bar for *Make it turn*

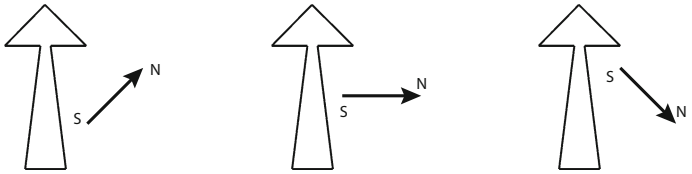

Fig. 1.15 Three initial cases of the orientation of a needle in a magnetic field for *push and twist*. In the problem, you determine the force and how it twists

which cases move the center but do not twist and which twist but do not move the center. Do any of these cases do both? In cases that twist, is it clockwise or counter-clockwise? In cases that it moves, is it up or down?

1.1.3 Push and Twist

Determine the initial direction of the force (push or pull) on the needle given its orientation, as shown in Fig. 1.15; use both the method of drawing the push on each pole and the method of projections to compute your answer. Also determine how it twists (clockwise or counter-clockwise). Repeat this exercise in the case where the magnetic field *increases* going up, rather than the depicted case of decreasing going up. Be sure the two ways of calculating the force agree with each other for both cases. Repeat the analysis again for both cases, but now for current loops, whose effective magnetic needles are the same as drawn in the figure.

1.1.4 Understanding Precession

A free current loop is placed in a magnetic field and there is no friction. Select the best description of what will occur. The arrow discussed below is the effective needle associated with the current loop. Be sure to explain why you chose the answer you chose:

1. The arrow does not move when placed in a magnetic field, so the projection stays the same.
2. The arrow that represents the current loop moves in a magnetic field, but it moves in such a fashion that the projection onto the axis corresponding to the direction of the magnetic field does not change with time.
3. The arrow that represents the current loop oscillates, so the projection changes with time.
4. The arrow that represents the current loop moves in a magnetic field, so its projection on every axis must change with time.

1.1.5 A Stern-Gerlach Experiment That Doesn't Work

Suppose someone suggested to you that they could create a completely uniform magnetic field and that they wanted to use it in the classical Stern-Gerlach experiment to separate current loops according to the projection of their effective magnetic needle onto the direction of the magnetic field.

Politely explain to them in a paragraph or two precisely why their proposed experiment will not work. In doing so, be sure to explain exactly what happens to the current loop as the beam passes through the magnetic field and what kind of pattern will emerge on the screen at the end of the experiment. Describe any differences or similarities to an ordinary classical Stern-Gerlach experiment that is performed on a source of current loops that already have just one definite projection of their effective magnetic needle onto some axis.

Is there a way to differentiate these between a uniform field experiment and a conventional experiment with a source that has just one definite projection?

1.2 Quantum Stern-Gerlach Experiment

When we run the Stern-Gerlach experiment with atoms, we find that the image on the screen is composed of just two dots! One deflected up, and one down, with the deflection magnitudes the same. We see this for *all orientations* of the Stern-Gerlach analyzer (except in very special cases, where we see only one dot). Before jumping into these details, we will start with some history of this important experiment (See an image of Stern and Gerlach in Fig. 1.16).

In 2003, *Physics Today* published an interesting history for the Stern-Gerlach experiment, and I highly recommend it (Friedrich and Hershbach, 2003). The authors argue that the experiment was successful, in part, because the two experimentalists liked to smoke cheap cigars, which had a high sulfur content, and blowing the smoke onto the glass plates made the beam of silver atoms more visible; this is similar in spirit to the photographic development process, but is much faster.

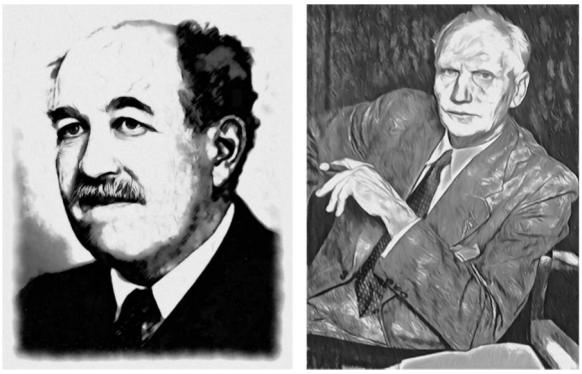

Fig. 1.16 Otto Stern circa 1940 (left) and Walter Gerlach circa 1950 (right). Stern and Gerlach created the first experiment to separate individual atoms in 1922. Note the cigar in Gerlach's hand (Otto Stern image from wikipedia (public domain) and Walter Gerlach image © Springer-Verlag from *Z. Phys.* **155** Iss. 4, 1 (1959) courtesy of AIP Emilio Segrè Visual Archives, E. Scott Barr Collection. Both modified to look like sketches by James Freericks)

The quantum-mechanical theory at that time was based on a concept called *space quantization*, developed by Bohr and refined by Sommerfeld. In this theory, the electrons orbited the nucleus of the atoms like planets around a solar system, but they had their possible orbits restricted—indeed, for a silver atom, one expected two possibilities only, both perpendicular to the magnetic field, but one clockwise and one counter-clockwise. Since these act like little current loops, we anticipate they will separate into two different groupings. Bohr is said to have discouraged Stern and Gerlach from attempting the experiment, to which Gerlach replied "No experiment is so dumb, that it should not be tried," an aphorism, that holds true today and is oft repeated by those who don't even know that Gerlach said it first.

The source for the silver atoms was a small oven that evaporated the atoms and led them through two collimated holes to create a narrow beam, with a velocity of about 660 m/s (measured earlier by Stern). They created the strongest inhomogeneous magnet they could. It was about 5 cm long and had a strength of 0.1 T (magnetic field is measured in units of Tesla); this field is about one thousand times larger than the Earth's magnetic field. The expectation was for a splitting of about 0.1 mm, requiring the beam widths to be significantly smaller than this in order to see the splitting (and requiring a microscope to see it well). This experiment was hard, and it failed many, many times. Finally, once the beams were aligned well enough, Gerlach observed the splitting.

The experiment then became a hallmark for demonstrating that quantum mechanics is real. Unfortunately, the original interpretation was completely wrong, because space quantization was wrong. Within 3 years, a new (and correct) form of quantum mechanics began to be developed. But, it was only in 1927 that the correct interpretation—that the splitting arose from the *spin* of the electron in the silver atom—was proposed.

What is spin? We still do not have a great understanding of what it is. Initially, it was thought that the electron acted like a ball of charge, with a small radius, that spun around an axis and thereby acted like a small current loop. But a simple calculation showed that the surface of the electron would have to move at a velocity ten times the speed of light, if such an interpretation was right. Furthermore, no experiment has ever found a spatial extent for the electron. So, somehow, this current-loop-like behavior occurs, but we do not have a clear image in our head for just how the electron does this. You may find this frustrating, as I do, but we have to live with these frustrating situations in the quantum world. We will find a number of quantum mysteries that we can incorporate into theories, but we have no good way to explain how they occur using the words we have developed for the classical world (which we observe on a day-to-day basis). When you study quantum mechanics, you simply have to get used to this! I do note that when we discuss relativistic quantum mechanics in Chap. 16, we do find that spin arises naturally in the formalism, but it still does not give us a physical picture for precisely *how* it arises.

Let us pause for a moment and visualize what this experiment looks like (see Fig. 1.17 and the tutorial https://bit.ly/3Mps6G1). Similar to the classical

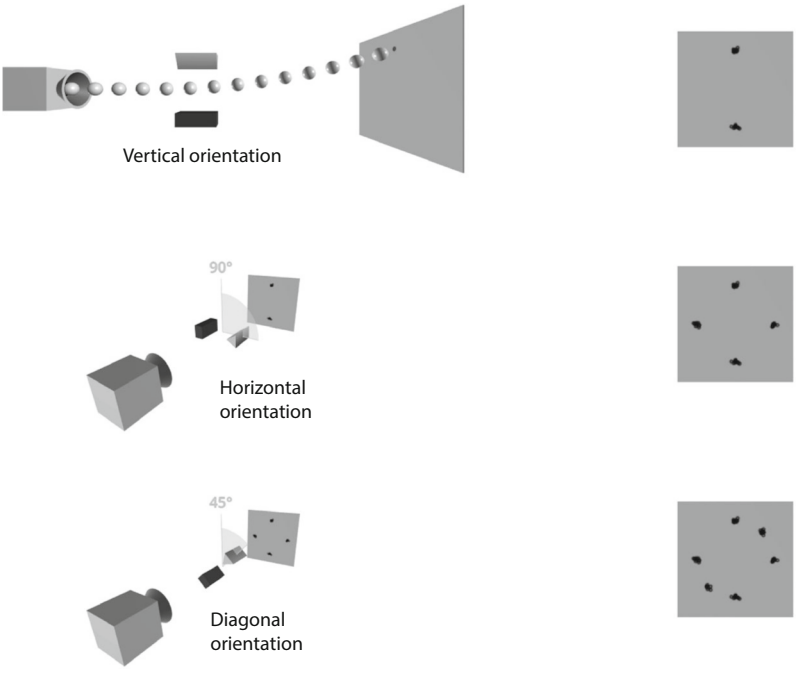

Fig. 1.17 Three different orientations of the Stern-Gerlach experiment. Note how the markings on the screen show that the splitting is always the same, regardless of the orientation of the magnet. Note further that the top image is really schematic. We cannot image the atom as it moves through the device, we can only image the markings on the screen after many atoms have gone through

experiment, we have a source of atoms, which emits them in a narrow beam. But unlike the classical case, we do not know *a priori* just how the effective needles are oriented for these atoms. So, we represent the atoms as spheres, instead of current loops (note that this is a graphic representation we are using—we are not implying that atoms actually *are* hard spheres). The atoms emerge from the source, go through the inhomogeneous magnet, which separates them, and then they hit the screen. The odd thing we find is that no matter how we orient the magnet, the atoms always split *by exactly the same amount*!

At first, you may say, "So what? The results are different than expected. Big deal." But what we will see, as we analyze this further, is that there is something profoundly odd about this result, because it cannot be reconciled with any classical picture for how the current loops could be prepared to have this behavior. For example, one possible explanation is that our source is not creating current loops in random directions, but in well-defined directions. But this hypothesis cannot be reconciled with the fact that no matter how I rotate the apparatus, I always get the same splitting. For if they were in one direction only, there is a direction perpendicular to this, and an experiment with the analyzer in *that* direction we would expect no deflection (because the projection is zero)—but we find the same two equal magnitude projections always. This phenomena is called "the conundrum of projections" and the experimental result is clearly odd; something unexpected has happened. We can try to force it into something understandable with our old (classical) way of thinking, but eventually we run into trouble. Instead, our approach is that we need to embrace that we are *seeing the world in a different way* and we must incorporate this different behavior into a model that we can follow, make predictions for, and ultimately understand.

The other issue that arises is that we have seen our first glimpse into the fact that quantum mechanics is *random*. We will not be able to say what will happen to any particular particle, but instead we will only be able to calculate the probability for what will happen after many runs of the experiment. In this case, we see this because the atoms end up at one of two different spots. If I send one atom through, which one will it go to? We do not know. We only know on average how many go to one and how many go to the other. Such systems are said to be stochastic. Note that some people equate random with unpredictable. But that is not the sense of random used here. Random means I cannot predict the result of any given experiment, like where the next atom will hit, but I can predict how often one thing or another will occur when the experiment is repeated many times. Indeed, quantum mechanics is all about very precise predictions of these probabilities.

To try to understand these odd behaviors, we need some more information, so we will try five new experiments to shed light onto this behavior. I should note that while the simple Stern-Gerlach experiment has been performed many times and by many different groups, these new experiments, where we repeat the experiment one after the other, are only *thought experiments*. I am not aware of any actual experiment that has been done in this fashion by directly stacking Stern-Gerlach analyzers. But experiments similar in spirit to these experiments were performed by Frisch and Segrè in the 1930's (Frisch and Segrè, 1933) and they support everything we will

Fig. 1.18 Packaging of a Stern-Gerlach analyzer. Starting on the left panel with the two paths an atom takes, we package the experiment with "guiding tubes" that carry the atoms after separation and redirect their velocities to be parallel for their exit. Then we put it in a box in the center panel and add a descriptive cover plate in the rightmost panel. The analyzer is depicted with a fat arrow showing how the magnetic field is oriented and how the field increases. We also include + and − labels on the exit paths indicating whether the atom has a positive projection or a negative projection of its effective magnetic needle in the direction of the field. The coordinate system we use has the atomic beam always moving in the y-direction, with vertical in the z direction and horizontal in the x direction

say that follows. So, we present you with these thought experiment results according to the quantum theory, and in the process we will learn something important. These new experiments will help us verify the following questions: How reproducible are the results? How random are they? What can we predict with any certainty about these sets of experiments?

Before jumping in to these experiments, we need some preliminaries. We discovered that atoms are always split into two different groupings when we run an experiment with a particular orientation of the magnet. Since we have only two outcomes, we will turn our Stern-Gerlach experiment into a measurement tool with one input path and two output paths, as shown in Fig. 1.18. We call this device a *Stern-Gerlach analyzer* (which we often shorten to just analyzer). The coordinate system we use has the z-direction as vertical and the x-direction as horizontal (the atomic beam moves in the y-direction). We have a tutorial https://bit.ly/3YlY6x9 that illustrates how the analyzer is constructed.

In our first experiment, we measure the projection of the atom and then we measure it again. This will tell us how *reproducible* the results are. To set this up, we take the output from the first analyzer and send it through the input of the second analyzer. What do you expect will happen? Will they all come out the same exit of the second analyzer, or will the results be random, coming out of both? The setup and results are shown in Fig. 1.19. In the top two, we examine the case of what happens when we put the second analyzer on the + exit of the first—a + atom in the first will exit the + exit of the second, but a − exit of the first never enters the second. Hence, we see the results are reproducible for measuring a + projection. The bottom row shows similar behavior, but examines what happens when a − projection is measured in the first. Again, the results are reproducible—if we enter an analyzer in a definite state, then we emerge from the second analyzer in exactly the same state. We can think of this as the Stern-Gerlach analyzer is preparing the atoms in a definite state of the projection of the effective magnetic needle.

In our second experiment, we examine what happens when we rotate the second analyzer to be oppositely oriented than the first one. Will we still have definite results coming out of the second analyzer? Let us look at Fig. 1.20 to find out. We see that

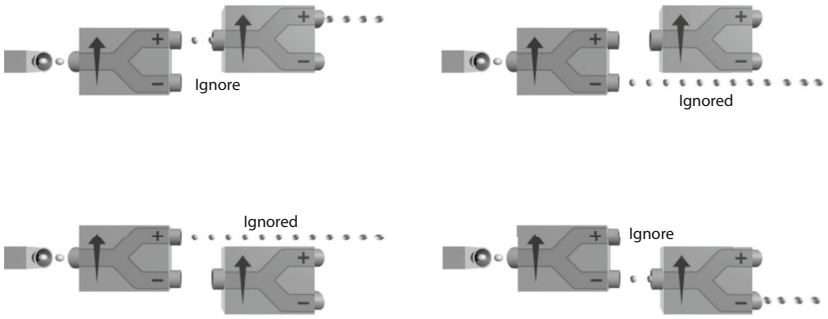

Fig. 1.19 Experiment one: Reproducibility of the Stern-Gerlach experiment—measure the projection, then measure it again. In the top left panel, we measure the projection to be positive on the z-axis, and then measure it again, by taking the output from the $+$ exit as the input to the second analyzer (and ignoring the atoms that exit the $-$ exit of the first analyzer). All atoms that enter the second analyzer emerge from the $+$ exit. Note that we ignore the atoms emerging from the $-$ exit of the first analyzer (shown on the upper right panel). In the bottom two panels, we see the same behavior when we attach the second analyzer to the $-$ exit of the first analyzer

Fig. 1.20 Experiment two: Repeated Stern-Gerlach experiment with a flipped second analyzer. Here, we attach the second analyzer, with its magnetic field in the $-z$ direction to the $-$ exit of the first analyzer. Since we know the direction of the projection (of the effective magnetic needle), we illustrate the projection with "precession cones" that depict the precession of the needles after the atom has exited the first analyzer (we do not know the projection before it enters). Note how atoms with a negative projection on the z axis have a positive projection on the $-z$ axis

the answer is yes. If the atom enters with a definite projection on the z axis, it still has a definite projection on the $-z$ axis. It is just the sign of the projection that changes. Note how we now depict the atoms with "precession cones" on them when we know what the projection is that they have, because they emerged from a definite exit of the Stern-Gerlach analyzer. To be sure you follow how this works, note that the axis of increasing magnetic field changes sign when the axis changes from being $+z$ to $-z$. We have a tutorial https://bit.ly/4iRsZDl to illustrate these concepts.

So, we have clearly established that the results of the Stern-Gerlach analyzer experiment are fully reproducible. Once an atom is measured to be in a particular state, it will remain in that state. And we also saw that if we flip the magnetic field of the analyzer by 180^{o}, the $+$ and $-$ outputs switch roles. These results are what we would hope to see happen for any system that produces predictable (and you might even say logically consistent) results. We will next test what happens if we change the angle of the second analyzer to point in a direction different from vertical, namely, in the horizontal direction.

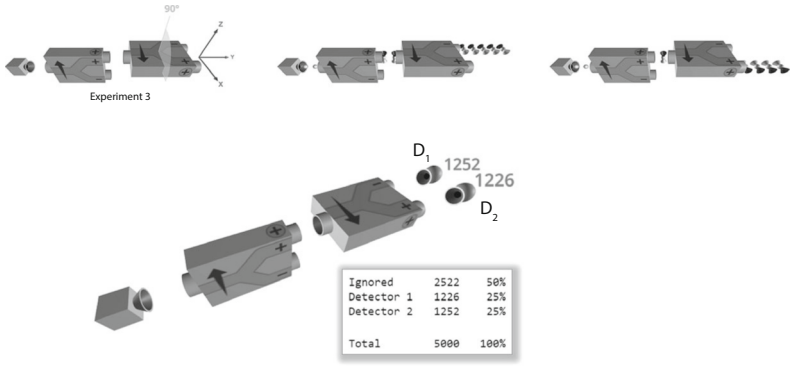

Fig. 1.21 Experiment three: rotate the second analyzer by 90° relative to the first analyzer. This is depicted in the top left panel, here with the second analyzer oriented in the $+x$ direction. The center and right panels illustrate the behavior of the experiment. Atoms enter the vertical analyzer in an unknown state. Consider only the atoms that exit with a positive projection on the z-axis and then enter the second analyzer. They subsequently emerge with a projection that is either positive or negative along the x-axis, as illustrated. In the bottom panel, we show the result expected of a typical run with 5000 atoms sent in. About half of the atoms are ignored, because they exit the $-$ exit of the vertical analyzer. For the remaining atoms, they split, essentially $50-50$ between the two exits of the horizontal analyzer

The experiment is depicted graphically in Fig. 1.21. The top row of panels illustrates the geometry of the experiment (with the first analyzer vertical and the second horizontal). The next two panels illustrate time-lapse "images" of an atom going through the device and emerging at one or the other exit, using the cone analogy to show the direction of the effective magnetic needle. We see that when an atom enters with a positive projection on the vertical direction, it emerges with either a positive or negative projection on the horizontal direction. Which path does the atom choose for any given experiment? We do not know. This choice is *random*, in the sense of flipping a coin: if I land on heads it will be positive, and if I land on tails it will be negative. But is it equally likely? The bottom panel shows what we would register in a typical experimental run that sends 5000 atoms in—half are ignored, because they exit the $-$ exit of the first analyzer. Of the remaining half, half of those emerge from each exit of the second analyzer. This fundamental randomness of quantum mechanics is a better randomness than we can create with a classical computer. It is completely random! Indeed, random number generators have been created that work off of quantum randomness. You can visualize the results in a tutorial https://bit.ly/4tDHxdS that illustrates these concepts.

Since this third experiment is essentially asking the question "How right or left is up or down?", the answer "halfway" makes sense, but probably was not our first guess. What we find we can predict, and predict quite accurately, is the probability for the atom to emerge from either exit after the experiment has been repeated many times. This will soon head us off in the direction of exploring probability in the next section. But before that, we have one more experiment to examine, to see what

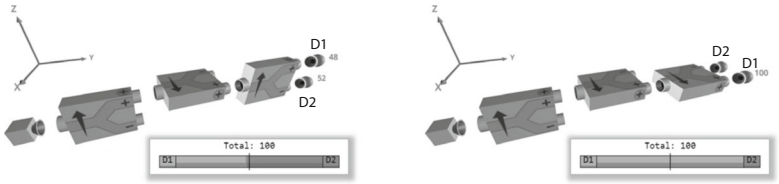

Fig. 1.22 Experiment four: Three-analyzer experiments. (Left) With the third analyzer vertical, we see that the atom no longer has a definite projection on the z-axis. Instead, we have a $50-50$ split. This shows the atom has no memory beyond the last measurement. (Right) With the third analyzer horizontal, we confirm that all atoms that emerge from the final analyzer have a positive projection on the x-axis. Note that many of the atoms in this setup exit through paths that we ignore when we count the exiting atoms

happens when we go through *three* analyzers. As we will see, the results just keep getting stranger . . .

Summarizing our results so far, we have seen that atoms emerging from a random source are in an unknown state before entering the analyzer. Then, after emerging, they split evenly into positive and negative projections, regardless of the orientation of the magnetic field in the analyzer. Next, we saw that if we do prepare them in a definite state, this result is reproducible (experiment one) and it even predicts the projection on an oppositely oriented axis (experiment two). Then, in experiment three, we found that measuring on an axis that is perpendicular to the first analyzer again has random results—the system acts again as if it was a random source input to the second analyzer, even though the state it entered the second analyzer in was completely known. These results suggest that the very act of measuring is what determines the projection of the atom on a specific axis—the state of the atom on other axes becomes indeterminate after it has been measured on one axis. In other words, the atom can have a definite projection on only one axis at any given time. *It cannot have a definite projection on more than one axis at any given time.* We characterize this as "atoms are stupid." They only remember the last axis they are measured on. We now establish that this suggested behavior really occurs. As shown in Fig. 1.22, our next experiment explores this idea more fully.

The first setup has the first analyzer vertical, the second horizontal, and the third vertical. If the second is attached to the positive exit of the first and the third is attached to the positive exit of the second, what do you expect will happen? Make a prediction before reading any further! There are at least two ways we can reason this out. The first way is to say that the atom exits the first analyzer in the $+z$ state, then it exits the second in the $+x$ state (with no memory of what projection it might have had before), so we should get half up and half down, since "up" and "down" are halfway to "right." The second way is to say that atoms emerging from the first analyzer are in the $+z$ state, those emerging from the second analyzer are *also* in the $+x$ state, so the third analyzer inputs atom in the $+z$ and $+x$ state, so all will come out the + exit of the third analyzer. Of course there is a third way you could reason, that quantum mechanics gives an unpredictable randomness, so we get half out of

each exit all the time; the first experiment on reproducibility has already shown this statement is not correct. A quick look at Fig. 1.22 shows that the first way to look at this problem is correct. The atom lost all memory of its projection on the vertical axis.

To ensure this is correct, we rotate the third analyzer to horizontal. Here, we see that yes, all atoms emerge from the + exit. So they are indeed all oriented with a positive projection on the x-axis. A tutorial https://bit.ly/454JA0x illustrates these different results.

We summarize this with a simple statement *Atoms are stupid, but you are not!* Atoms suffer from permanent memory loss. They are stupid! They only remember the last axis that their effective magnetic needle was measured on. I like to tell students that if atoms are this dumb and they can still manage quantum mechanics, then for intelligent folks like you, quantum mechanics should be a snap! Remember this rule. Embrace it. It is important for you to be able to make sense of different experiments. Let us summarize this again. When an atom is measured on a particular axis, it has a projection on that axis. Change the axis and measure again, and it now has a projection on the new axis. It no longer has any definite projection on the previous axis (except for the two special cases of measuring on the same, or on an opposite axis, which we will come to later).

OK. We have learned about perpendicular analyzer configurations and parallel analyzer configurations. What about other angles? This is our last experiment for this section. One idea you might have had is that if the effective needle is somewhat pointing up, we will measure it to be up and similarly when it is pointing somewhat down. But the reality has more of a shade of gray to this than such a simple black-and-white prediction.

Figure 1.23 shows the results for the final experiment in this section. Here, we start from atoms that enter a vertical Stern-Gerlach analyzer. We then measure the number of atoms emerging from a second analyzer oriented at an angle θ relative to the vertical. At 0°, we know all atoms exit the + exit. We also know we find half at 90° and none at 180°. In between, we see that the results lie in between these numbers. For example, at 45°, we have 85 emerging from the + exit and 15 from the − exit for a run with 100 shots. This lies between the numbers we had before, but it clearly is not a linear relation. By adding more angles, we empirically find that the probability to emerge from the + exit when we are at the angle θ is $\cos^2\frac{\theta}{2}$. This means the probability to emerge from the − exit is $\sin^2\frac{\theta}{2}$ (since the total probability must be 1). One can see the smooth curve in the bottom right panel of Fig. 1.23, showing how this behavior looks, and you can explore this yourself in a tutorial https://bit.ly/4iYcsxw.

You might be concerned that the dots did not lay exactly on the $\cos^2\frac{\theta}{2}$ curve. This occurs because random events have fluctuations. I can only measure a good approximation for the probability if I measure the results of many shots. In some cases, 100 measurements are too little to get an accurate result. If we did 1000 times for each, the data would lie closer to the cosine curve, but we would still see deviations. For some physics experiments, measuring the fluctuations provides interesting information about the system. We will examine in detail the properties

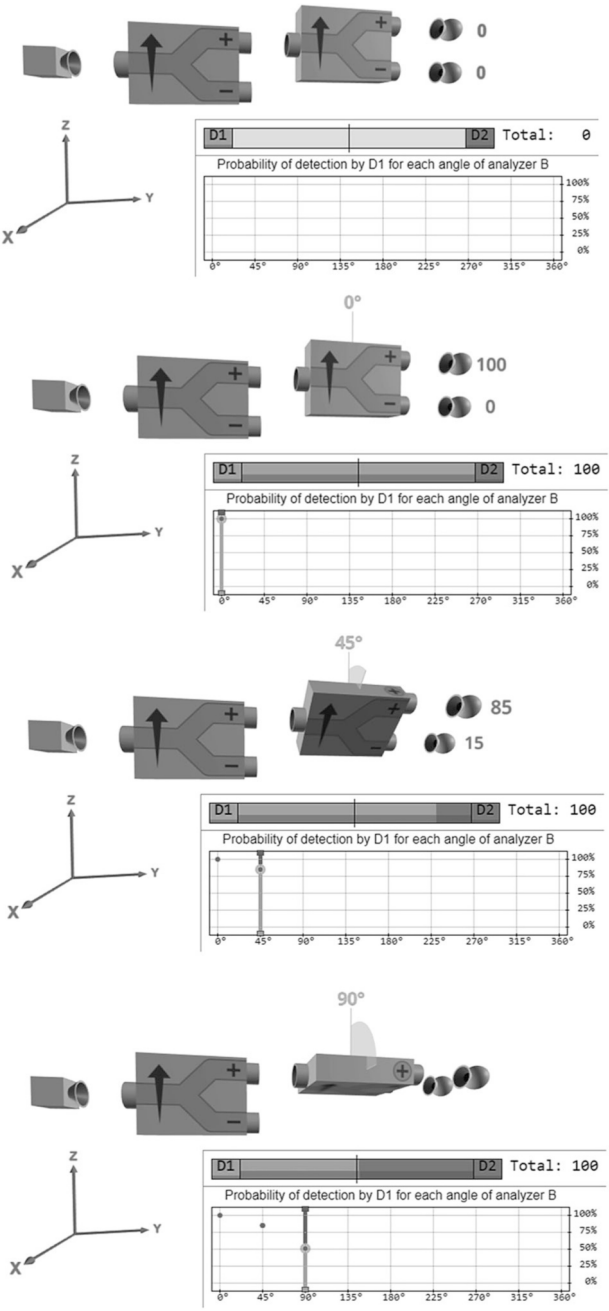

Fig. 1.23 Experiment five: Rotated two-analyzer experiments. The top left panel shows the setup of the experiment. We have a first analyzer that is always vertical and a second one attached to the + exit of the first. We rotate the second analyzer from the vertical and record the probability to measure atoms exiting the + exit of the second analyzer (with 100 shots) in the graph below. In the

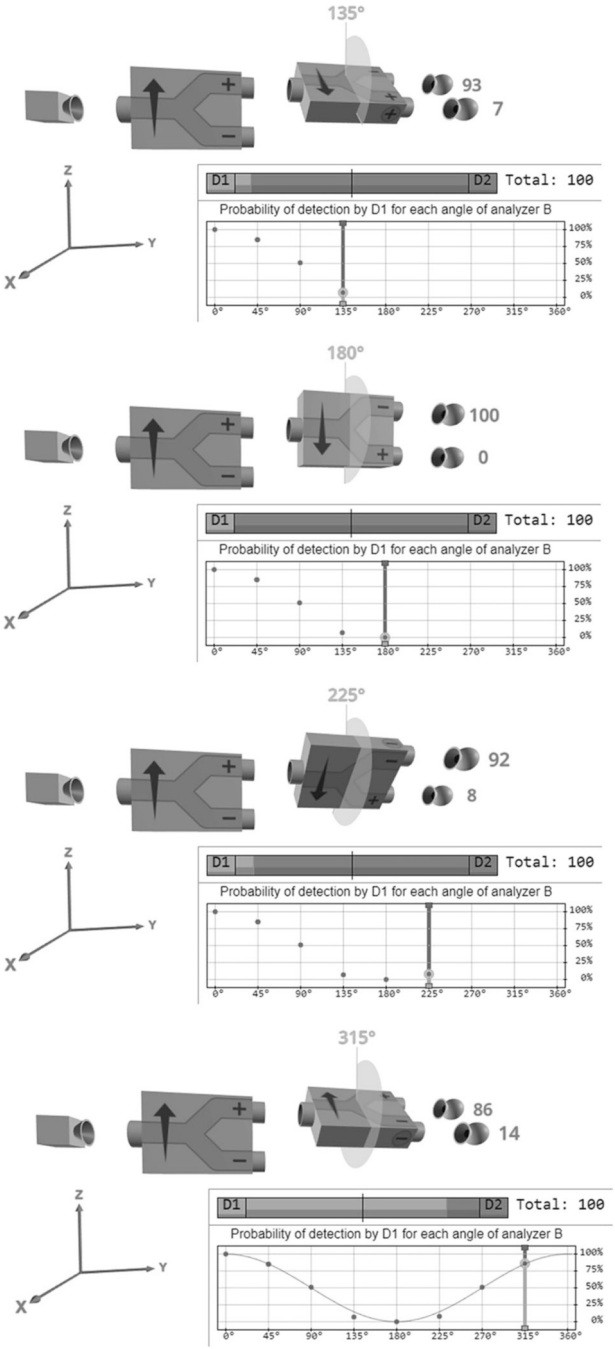

Fig. 1.23 (continued) remaining panels, we show the results for 0°, 45°, 90°, 135°, 180°, and 225°. The lower right panel adds in the points at 270° and 315°. It also shows a smooth curve running through the measured data. This curve is $\cos^2 \frac{\theta}{2}$

of fluctuations when we encounter the Hanbury-Brown and Twiss experiment later in Chap. 4 and for general quantum systems many times throughout the book.

1.3 Probability in the Quantum World

As you can see, we are unfolding this picture for quantum behavior slowly, because it is difficult to fully appreciate at first. But before going too far down that road, we will first take an interlude to learn some probability, because quantum mechanics has an intrinsic random nature to it and the language of random phenomena is the language of probability. After finishing this, we will head right back into the quantum world.

The word *random* is used many different ways in everyday conversation. One way it is used is to convey that results are not predictable in any way. We do not use the word random in this way. There is an aspect of quantum mechanics that is not predictable, but there are other aspects that are very predictable. It is important you understand the distinction. What we cannot predict is what will happen on any given shot of an experiment that has random results (such as a specific run of experiment 3 from the previous section). But, what we can predict very well is the probability for these results to occur. And in this sense, we *do* have real predictive capabilities for random events!

Let us explore this concept of randomness more thoroughly. While the classical version of the Stern-Gerlach experiment might have appeared to be random, in principle, we are able to follow the path of the current loop from start to finish and predict—with complete certainty—precisely where the current loop will hit the screen. Then, the different current loops only appear to hit the screen randomly, due to their random initial state (for the projection of the effective needle that represents the current loop on the axis of increasing magnetic field). The randomness in the classical Stern-Gerlach experiment is not because we were unable to predict how they went from the source to the screen, but it is from the random source of classical current loops.

The quantum Stern-Gerlach experiment is simply different. We cannot detect the atoms until they hit the screen, so we don't know precisely where they are before they are detected. This means that we cannot tell the precise path of the quantum particle; instead, the results for which dot the atom hits the screen at are truly random for each shot of the experiment. What we can do is predict precisely what the probability is for landing in each dot.

This is a real change in the way we interpret the world. We are giving up the microscopic predictability for a "law of large numbers" prediction in aggregate. It is not because we are unwilling to look hard enough or perform the right experiment to see the behavior we are seeking. It is because *the fundamental nature of the quantum world requires this change of viewpoint*. For philosophers, this meant the end of determinism and pre-ordination. But for most of us, this just means small objects behave a bit differently than large objects.

Oftentimes, people mistake the notion that the world is fundamentally random with the conclusion that we cannot predict anything with certainty. Far on the contrary, quantum mechanics is one of the best tested and most accurate theories in all of science. While we might not be able to know a quantum particle's trajectory, we can measure its energy levels, or its effective magnetic needle strength, extremely accurately (more than ten digits of accuracy). And the quantum theory always ends up agreeing with experiment. Quantum mechanics is probably the best tested theory we have for anything. Yes, it has a fundamental randomness to it, but it also has many quantitative properties that can be measured and compared to theory to high accuracy. Do not confuse randomness with irreproducibility, inaccuracy, or impossibility of quantitative testing. As with many subjects, if one asks the right questions, one may answer them with an extraordinary high precision.

We briefly recap the results of the quantum Stern-Gerlach experiment. The rules to describe the atom are simple: (i) Atoms are stupid and remember only the last axis they were measured on; (ii) if they have been measured on one axis and are then measured on another axis (we assume, e.g. that the first measurement resulted in a positive projection on the axis of the first analyzer), then the probability to have a positive projection as measured by the second analyzer is $\cos^2\frac{\theta}{2}$ (if θ is the angle between the Stern-Gerlach analyzers); and (iii) the probability to come out the minus exit is $\sin^2\frac{\theta}{2}$. In addition, the result for any particular shot of the experiment is random, and we see the results of the probabilities only after the experiments are repeated many times. The probabilities are exchanged if we send the atoms from the $-$ exit of the first analyzer into the second analyzer.

So then what is probability? The simplest way to define it, for cases where the total number of options is finite, is that

$$\text{Probability} = \frac{\text{Number of outcomes for which the event successfully occurs}}{\text{Total number of equally likely possible outcomes}}. \tag{1.1}$$

Note that this has reduced the problem of calculating the probability for an event to occur to just a problem in counting. It is important that each possible event that occurs is equally likely. This point is often confused by students. It does not mean that all events must be successful, it means the chance of having each event occur (successful or not) is equal. This rules out using this formula to calculate the probability for rolling weighted (biased) dice. In that case, one number on the die will appear more often than the others. We will focus on calculating the probabilities only for unbiased problems here, hence the simplified definition of probability for an event to occur.

Whenever we encounter something that is new, it is useful to have concrete examples to make sure we understand it. So consider a coin with two sides that we call heads (H) and tails (T). What is the probability that the coin will flip and land on heads? We simply use our rule: there are two options, that it lands on heads or that it lands on tails. One option (heads) is successful, while one option (tails) is not, and each option is equally likely for a fair coin. So the probability is $P(\text{heads}) = \frac{1}{2}$.

This is a fact that many of you already know, but it is useful to carefully show where it comes from using the definition above.

Let's try a somewhat harder problem. What is the probability to get two heads when we flip the coin twice? Now, we need to itemize the different options. We see there are four possible outcomes: (i) HH, (ii) HT, (iii) TH, and (iv) TT. Only one is successful, so the probability is $P(\mathrm{HH}) = \frac{1}{4}$. There are a few subtle points to make here. First, we must note that the two options HT and TH are in fact different. The first says we flip heads first and tails second, while the second says we flip tails first and heads second. You should be able to immediately see that these are two different ways to flip one head and one tail. The second is that it can be tedious to enumerate all possible choices as things become more complicated, yet the result should follow simply from the fact that the probability to flip one coin is $\frac{1}{2}$. Indeed, this is true, and the shortcut to make the calculations of probabilities simpler uses a technique that we call working with *compound probabilities*.

Compound probabilities are constructed from simple probabilities by transcribing the word "and" into multiplication and the word "or" into addition. Let's reword the previous example to What is the probability to flip heads and then flip heads again? We know the probability for heads is $P(\mathrm{H}) = \frac{1}{2}$ so $P(\mathrm{HH}) = P(\mathrm{H}) \times P(\mathrm{H}) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$, which is the same as what we obtained before.

Let us try another example. What is the probability to flip HH or TT? Using compound probabilities, we have $P(\mathrm{HH\ or\ TT}) = \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$. We use "or" as a signal to us to add the probabilities. Now compare the compound probability calculation to the old way

$$\frac{\mathrm{HH} + \mathrm{TT}}{\mathrm{HH} + \mathrm{HT} + \mathrm{TH} + \mathrm{TT}} = \frac{2}{4} = \frac{1}{2}. \tag{1.2}$$

Note that we are assuming that the simple probabilities that we use for constructing the compound probabilities are independent of each other. This is a concept that is subtle and hard to understand. We explain it concretely by showing what it means for two probabilities to be dependent. An example of such a case is the question "What is the probability to roll an even number and a number bigger than three on one die?" By enumeration there are two even numbers bigger than three (4 and 6), so the probability is $\frac{1}{3}$. But the probability for an even number on one die is $\frac{1}{2}$ as is the probability to be a number bigger than 3; one can see these probabilities do not follow the compound probability rule, since $\frac{1}{3} \neq \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$. Why? Because they are not independent of each other! Compare this to the question "What is the probability to roll an even number on one die and a number bigger than three on a second die?" Here they are independent because they each take place on a separate die, and the compound rules can be used. For all the cases we discuss here, the probabilities will be independent, so we can use the compound rules.

We now move on to consider an interesting general question. What is the probability for something to occur *or* the probability for it not to occur? Since the number of possible events is equal to the number of times an event occurs plus the number of times it does not occur (because there are no other options), we have

Fig. 1.24 The 1934 musical, *The Gay Divorcee*, starred Fred Astaire and Ginger Rogers. In this film, Guy Holden (right) explains to Egbert Fitzgerald (left) that "Chance is the fool's name for fate" (Image extracted from the movie and appearing via fair use for commentary about the dialogue. Modified to look like a sketch by James Freericks)

$$\begin{aligned} P(\text{event}) + P(\text{not event}) &= \frac{\text{Number of times the event occurs}}{\text{Total number of events}} \\ &+ \frac{\text{Number of times the event does not occur}}{\text{Total number of events}} \\ &= \frac{\text{Total number of events}}{\text{Total number of events}} \\ &= 1 \end{aligned} \tag{1.3}$$

In other words, the sum of the probability for something to occur plus the probability for it not to occur is equal to one. Sometimes it is easier to calculate the probability for something to occur by computing the probability for it not to occur and subtracting that from one! We will see examples of this soon.

The 1934 musical, *The Gay Divorcee* has a powerful quote in it said by Guy Holden (Fred Astaire), which was "Chance is the fool's name for fate (see Fig. 1.24 for the precise moment in the film)." While this is a fairly catchy phrase, scientists will say that it is actually the opposite statement that is correct—fate is the fool's name for chance. As scientists, we focus on facts, and while one may want to debate and argue about the existence or not of fate, chance is something that is very concrete and real—it is something we can compute—and we do so within the science of probability. The video Fate or Chance https://tinyurl.com/syczwm69 explores this idea further. But to really learn how this works, we need practice, so we now spend time going through some increasingly more complicated problems. All of these examples won't have any quantum mechanics in them, but I do think you will find them fun, and maybe you will even find them mind-blowing! Let's start.

Casino games are a good example to analyze for probability, because the casinos always set them up so that, no matter what you do, their odds of winning are higher than yours. It is a delicate balance, of course, because if people lose all the time they won't play, so the odds cannot be too heavily in favor of the house. One of the casino games is called craps, which has its origins in the early nineteenth century in New Orleans; it rose to high popularity in the early twentieth century and can be

found in most casinos. In craps, a shooter throws two six-sided dice in each turn. Bets are made on the result of the different throws, and people may or may not win after each toss. We will not explain all of the rules, but we will analyze carefully a few of them. In the process of doing this, you will see why the house always has the advantage $\cdots$.

The way the betting works in a casino is that when you place a bet that costs you one chip, then the house takes your chip (if you lose) and the house pays you back some number of chips (if you make the correct prediction) and also returns to you the one chip that you bet. One bet you can make in craps is that the next roll will be a seven. Using our enumeration rules, we know that each die has six sides, so there are $6 \times 6 = 36$ possible rolls of two dice. The pairs of dice that add up to seven are the following: ⚀⚅ ⚁⚄ ⚂⚃ ⚃⚂ ⚄⚁ ⚅⚀. Note how we have to look at all possible pairs (e.g., the roll ⚀⚄ is different from ⚄⚀ since the number on the first die is different from that on the second die). We have to count all these different possibilities. There is a total of six successful rolls that give a seven. The probability is then $\frac{6}{36} = \frac{1}{6}$. The house gives your four chips back, plus your original chip, if you win. Is this a fair bet? Think of what would happen if you played 36 times and bet one chip each time. We expect you to win, on average, a total of six times. If you did win exactly six times, you would get five chips each time (four plus the one you bet), or a total of 30 chips after your six wins. But, you bet 36 chips. This means you lost 6 chips! So, the bet is not fair. How much should the house pay you to make it fair? If they gave you five chips plus the one you bet, then it would have been fair. We will see this behavior of "shorting you one chip" is a recurring theme with the house.

Our next example involves rolling craps on your first roll, in which case you lose (craps is a mispronunciation of the English slang "crabs," which meant two or three). In the modern game, craps is defined to be rolling a total number of 2, 3, or 12 on your first roll. As the shooter (person rolling the dice), you lose if you do this. But others can bet for this to happen. What are the odds that you roll craps on the first roll? Try to work it out before reading on. Again, we have a total of 36 possibilities. Craps corresponds to rolling ⚀⚀, ⚀⚁, ⚁⚀, and ⚂⚂. This is 4 successful cases out of 36 possible, or a probability of $\frac{1}{9}$. If you bet on someone to do this, the house gives you seven chips plus the chip you bet back. Again, it is not a fair bet. If you played this 36 times, you will win 4 times, on average, and end up with 32 chips, or 4 fewer than you started with. They would need to return eight chips (plus the one you bet) to make it fair.

Our last example is called rolling it "the hard way." This bet is to roll a particular even number, say a six, by the pair ⚂⚂ instead of the other ways of ⚀⚅, ⚁⚄, ⚃⚃, and ⚄⚁. You win if the shooter rolls a six the hard way; you lose if the shooter rolls six any other way, or rolls a seven. If anything other than a six or seven is rolled, the bet carries over into the next roll. How many rolls end up losing, how many are successful, and how many are indecisive (note you do not care about the rolls that do nothing)? There is a total of $5 + 6 = 11$ possibilities, and only one is successful, so the probability is $\frac{1}{11}$. You probably guessed that the house pays you back 9 chips plus the chip you bet if you win. Indeed, this is what they do. You may

Fig. 1.25 Happy Birthday! (Original picture from www.pexels.com and modified to look like a sketch by James Freericks)

want to go and try some of the easier probability problems at the end of the section before we jump to more complex probability discussions.

Okay, now we are going to get a bit more advanced. There is a famous question with a surprising answer. How many people do you need to have in a room before there is a better than 50% chance that at least two of the people in the room have the same birthday? We celebrate this question with a birthday cake in Fig. 1.25. While not answering this question directly, let me say that if we have 30 people in the room, the chance of at least two people in the room having the same birthday is over 70%! This is surprising to people because they think the probability should be much smaller. Before tackling the birthday problem in full, let's ask a simpler question. In a room full of six people, what is the probability for at least two people to share the same birth month? Make an educated guess to the answer before proceeding. To get started, what is the probability that two people have the same birth month? Here is a strategy we can take to try to solve this problem. The first person can have a birthday in any month, while the second must match the first month, so it is $\frac{1}{12}$, because only one month matches whatever the first month is. Lets try one more. What is the probability for two people to have birthdays in different months. The first can be any month, while the second can be any of the 11 remaining months that are not the same as the first month, so the probability is $\frac{11}{12}$. Note that since the probability for the two people to have their birthday in the same month or to have their birthday in a different month must be equal to one (remember why?), we have that the sum of those two probabilities is equal to one.

When we get to three people, enumerating the successful cases is much more difficult. We can have the first and second person with the same birth month, but not the third, or the second and third the same, but not the first, or the first and third the same, but not the second, or all three in the same month. Enumerating these cases is quite difficult and it is easy to make a mistake. It is better to instead determine the probability that no two have the same birth month, or stated differently, what is the probability that all three have different birth months? Then one minus this probability yields the answer for the probability that at least two have the same birth month in a group of three people. A further analysis of the birth month problem (and also of the birthday problem) is given in the end of section problems. You might want to look at those before we get to our most advanced problems.

You should have recognized, either by looking at problems, or by flipping coins or rolling dice, that when we run a series of experiments on random events, we do not find that we immediately reproduce the exact probabilities that we calculate. Indeed, the fluctuations from the exact number of times that a result should be successful actually *grows* as the number of tries is increased. But, surprisingly, when we divide those numbers by the total number of tries, the experimental probabilities approach closer and closer to their exact results. This is because the relative fluctuations grow more slowly, proportionately, than the number of tries.

The theorem that proves this is called the *central limit theorem* and it is one of the most important results in statistics and in probability theory. The mathematical details of the proof are a bit beyond us here. But we can be a bit more quantitative. Suppose we want to estimate the probability to flip heads on a coin by flipping it many times and measuring the probability. Suppose we flip it N times. Then the central limit theorem says that most of the time we should expect the number of heads to be between $\frac{N}{2} - \sqrt{N}$ and $\frac{N}{2} + \sqrt{N}$ if it is a fair coin. This says that the width of the interval that we expect for the number of heads grows as N increases. But to get the probability, I have to divide the number of successful events by the total number of tries. This means we divide by N and find that the estimated probability becomes $P_{\text{est}}(H) = \frac{1}{2} \pm \frac{1}{\sqrt{N}}$; the width of the estimated probability interval goes to zero as N gets large! So we expect our probability calculations to look like the data we take once we have amassed enough data that the fluctuations in the probabilities are smaller than the desired accuracy we want to have.

A famous problem in probability is the so-called German tank problem. In World War II, the allies wanted to get precise numbers on how many German tanks were being manufactured each month. The only data they had was from tanks they captured or destroyed. What they discovered was that some parts of the tanks had serial numbers that were numbered sequentially. Hence, if they could infer what was the maximum number of these serial numbers based on the numbers they already knew, they could estimate the number of tanks.

Let's take the simplest example. Suppose we found one tank, and its serial number was 103. Since we don't know anything else, we would infer that the average number we would see for the serial number if we captured many tanks would be one half the total number of tanks. Applying this logic, we would guess the total number of tanks is 206 (this number can change a little depending on where we say the serial number starts, but given that it is a rough estimate, we won't worry about those details here). Now, the allies had more data, and as we get a bigger and bigger list of numbers, then we know two facts. First, the maximum number of tanks is at least as high as the largest number observed. And second, we can use the rest of our data to tell us what we think the spacing between the numbers of tanks is. By adding the maximum observed number plus the average spacing between numbers, we arrive at the maximum predicted number. This leads to a guess (called the minimum-variance unbiased estimator) that is given by the maximum serial number observed added to the maximum number observed divided

by the total number observed and then minus one. Using an analysis like this turned out to produce a quite accurate guess for the number of tanks being produced by the Germans. This has also led to the practice of having randomized serial numbers on weapons parts.

Warning. The remaining material in this section is really hard. It is quite difficult and is not needed for understanding the rest of the book. You can skip it if you want to without losing anything. Really, you can think of it as being optional. But I do encourage you to give it a try, as the result is quite shocking. To give you a sense of the difficulty of this material, I made an error when I first analyzed it and only after some heated discussions with a student (thank you `MIT_SANDY`) did I realize the error I made. We should never be afraid to try something hard. We may not be able to get it on the first try. Nor even the second. But if we persevere, we should eventually be able to get it. However, for Penney's game, you need not worry if you never get it. Just remember to say no if anyone asks you to play!

Walter Penney invented a game in which coins are flipped and the sequence of heads or tails is observed by two players. The first player guesses a sequence of three coin flips in a row, then the second player guesses a different sequence. The guesses are made prior to flipping any coins. Coins are then flipped one after another, until a chosen sequence occurs; the player who picked the correct sequence wins. I have a simple question for you: Is this game fair?

It certainly seems like it is. Consider a simplification with just one coin. You pick heads, I pick tails, and we flip. Each has a chance of 50% to win. This is a fair game. Generalize to two coins, and since the probability to flip HH, HT, TH, and TT is all $\frac{1}{4}$, so it seems like it should be fair as well. Similarly for a sequence of three, which is the official version of Penney's game. But is it? Is it really fair?

This game has a new wrinkle beyond what we have done before, because the enumeration method, of finding the total number of tries and enumerating all successful tries, to find the probability is a challenge, because we do not know how many flips we need to do before the game ends. Let's analyze an example of the simplified two-coin game. I need to pick a combination of two coin flips and so do you. Suppose I picked HH and you picked HT. We know there are four possibilities for two coin flips: HH, HT, TH, and TT. All are equally probable, and so, it seems like any two choices by the players will be fair. But, we need to think about the full game, since it may not end after two flips. So, let us analyze this first case very carefully.

There are four choices for the flip of the first two coins. One is HH. If that occurs, I win. One is HT. If that occurs, you win. One is TH. Here, the next flip determines the winner. If it is H, I win, and if it is T, you win. Finally, if the toss after two coins is TT, we need to keep flipping until an H eventually comes up. Then, we each have an equal chance of winning with the next flip—if it is an H, I win, if it is a T, you win. So, this looks completely fair. And it is.

But, what if you chose TH as your choice instead. Let's go through the analysis again. If the first two flips are HH, I win, if they are TH, you win. If they are HT or TT, you can see there is a trailing T. In order for me to win, I need to get an HH

in a row. But I cannot get that, without you getting your TH first. In other words, in three of the four cases you win! It no longer looks so fair. Now, for the two-coin case, if I picked HT or TH as my first choice, then there is no choice you can pick that will give you an advantage. The advantage only occurs when I pick HH or TT. If we change this to a three-coin flip prediction, it turns out that no matter what I pick, you can always make a choice that gives you an advantage. In fact, if played correctly, the person who picks first will lose, on average, at least two out of three times.

To see the reason why this is so, let's examine a simple case carefully. Suppose I picked TTT as my choice. The optimal winning strategy choice is to take the second coin choice of the first player, flip that coin, and make it your first choice, and then follow it by the first player's first and second choices. So TTT becomes HTT (we take the second coin T, flip it to H, make that our first coin, and follow it by the first and second coins). The analysis of this is simple. We have eight possibilities for the first three coin tosses: HHH, HHT, HTH, HTT, THH, THT, TTH, and TTT. If TTT occurs, I win. For the remaining seven cases, you should see that they will always have an HTT occur before a TTT occurs, because there will always be a leading H. So I lose seven out of eight times. No matter what I choose first, you can always pick the optimal choice, according to the above algorithm, and you will win at least two out of three times. It is amazing how unfair this game is, yet, most people, who play it without thinking carefully, will insist it is still fair, even after they have lost many times! You have an opportunity to analyze some more complex cases in the last problem of this section. Be forewarned, these problems are quite challenging. Do not panic if you cannot do them. If you want to learn more about how to analyze this game, there are numerous videos about it available on the web. Just search them up. I have my own video https://tinyurl.com/yyxkt84t too.

Chapter 1.3 Problems:

1.3.1 Stern-Gerlach Analyzers at an Angle

We set up a pair of Stern-Gerlach analyzers, the first oriented at 45° to the vertical and the second at −45° as shown in Fig. 1.26. The atoms go from the source into

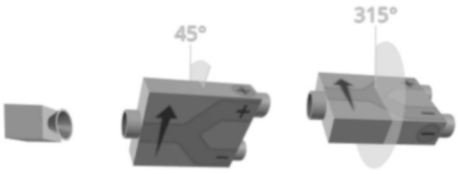

Fig. 1.26 Stern-Gerlach analyzers at an angle. In this problem, the source goes into an analyzer rotated by 45° to the vertical; we take all atoms exiting the + exit and put them into an analyzer oriented at 315° = −45° to the vertical

Fig. 1.27 Example of a license plate in the District of Columbia of the United States. It has three numbers, followed by three numbers. All digits from 0 to 9 are possible (Converted to a sketch by James Freericks)

the first analyzer and then all those exiting the + exit of the first analyzer enter the second analyzer. Your job is to calculate the probabilities to exit from the + and − exits of the second analyzer. Be sure to determine the relative angle between the analyzers to use in the formula for the probabilities.

1.3.2 License Plates

The license plates on automobiles in Washington, DC, USA, have three digits on the left and three on the right, for example, 140–144 shown in Fig. 1.27. What is the probability that the first three digits are identical to the second three digits?

1.3.3 Dice Play

Suppose we have two dice. What is the probability to roll a total of 4 or a total of 11? We answer in stages: How many possible results are there if I roll two dice? Each die has six sides with the numbers 1, 2, 3, 4, 5, and 6 on them. How many different ways can I get a total of 4? How many different ways can I get a total of 11? What is the probability to roll a total of a 4 or a 11?

1.3.4 Coin and Dice Play

Suppose I roll a red die and a green die at the same time. What is the probability to get a two or a four on the red die and a five on the green die? This problem can be solved with either method (direct enumeration or compound probabilities); why not try both ways and see that they agree?

Suppose I have two coins and two dice. I roll and flip them at the same time. What is the probability that the coins will land on different sides, and that the two

dice show different numbers? *Hint*: This is a problem that might be better to solve via compound probabilities using the probability that each will not occur, employing our identity (that the probability for something to occur and to not occur is one) to then find the probability that they will occur. But you are free to use any method you like. *Hint*: the coin tosses are independent of the dice rolls.

You roll three dice. What is the probability that the sum of the three numbers is not a prime? (The prime numbers that you can get here are 3, 5, 7, 11, 13, and 17.) *Hint*: The best way to do this is to enumerate all successful cases. This is not easy! I got it wrong the first time I tried it, so be patient and think of all possibilities, and you will eventually get it.

1.3.5 Three Coins

We are going to work in detail on the problem of flipping three coins in a row. If I flip three coins, how many different possibilities are there? How many have no tails, one tails, two tails, three tails? Compute the probabilities for the different events.

Now it is your turn to get empirical. You will flip three coins eight times and record the number of tails in each set of three coins. After eight times, you can get an approximation for the different probabilities. Determine what this is. Do you expect it to always be the same as the exact probabilities you calculated above? Try flipping 16 times or 24 times. Do your results change? We have a simulator https://bit.ly/3MuDdNS.

Why do we need to go to large numbers of trials to get good approximations for probabilities?

1.3.6 How Many Does It Take?

Two students are having a discussion about the coin flips:

Leonard: I tried the previous problem and my numbers came out close, but I did not get the exact probabilities. I found three tails occurred twice, not once. Something must be wrong with the problem.

Sheldon: My dear Leonard, this could not be farther from the truth. Measured probabilities only start to look like the calculated values when we repeat the experiment many times. What would have happened if I flipped the coins ten times. I could never get any of the probabilities right. This doesn't mean there is anything wrong with the analysis.

Which student is correct? Discuss why and support your argument clearly.

1.3.7 They Say It's Your Birth Month (Day)

In this problem, we will move from solving the birth month problem to the birthday problem.

(a) What is the probability for a group of three people to not have anyone with the same birth month? It may be easier to answer this by stating this problem as what is the probability that two people do not share the same birth month and the third person has yet a different birth month? What is the probability for at least two people to have the same birth month in a group of three? *Hint:* Think about the total probability of this event and the previous event.
(b) Now determine the probability that at least two people have the same birth month in a group of six people.
(c) Now we generalize this to birthdays: how many people do we need for the probability to first be larger than 50% that two people have the same birthday in the group of N people? Note this calculation is a bit lengthy and you will want to use a calculator or computer to help with the arithmetic. (*Ans.* $N = 23$)

We have a simulator https://bit.ly/47QHAKU that you may want to explore on this as well.

1.3.8 Bus Waiting Problem

A problem that fascinated me when I first studied probability was a simply stated problem that had a surprising solution. Suppose a bus arrives at a bus stop at random times. There are either 5 minutes between buses or 10 minutes between buses, and on average, the number of intervals with the period of 5 minutes is equal to the number of intervals with a period of 10 minutes. If the bus stop has a clock that records the time since the last bus, and I arrive at the bus stop at some given time, such as noon, then how long would the period between the last bus and the bus I catch be, on average?

Hint: This problem is not as simple as it might seem at first. Think about what the probabilities are that you arrive during a long interval or arrive during a short interval. To really make this clear, think about a case where the two intervals are 1 minute and 59 minutes. Then you will see that you are 59 times more likely to arrive during a long interval.

Ans. 8.333 minutes.

1.3.9 Penney's Game

Note that this problem is quite challenging. It can be skipped without affecting your ability to complete future problems.

(a) Suppose player 1 picks HHH. Following the optimal strategy, you pick THH. What is the probability for player 1 to win?
(b) Suppose player 1 picks HHT. Following the optimal strategy, you pick THH. What is the probability for player 1 to win?
(c) Now we try a hard one. Player 1 picks HTH, so you pick HHT. What is the probability that player 1 wins?

Hint: For problem (a), you should enumerate all possibilities for the first three coin flips. You will see which cases immediately win. Of the other six, analyze carefully what happens on subsequent flips to see who wins. Then count up the successful cases and divide by the initial number of choices, which is eight. For problem (b), it is best to think of what flips need to be done for the first three flips that guarantee a win for player 1. If player 1 does not get one of those winning rolls, can player 1 ever win? This should allow you to compute the probability exactly. For problem (c), not only do you have to think about the first three flips, but also about finding an equation to relate the probability for you to win in general versus the probability for you to win for one of the eight "first three" tosses. It requires a self-consistent solution, so use the definitions of P_1 as the probability that player 1 wins and $P_2 = 1 - P_1$ as the probability that player 2 wins. Determine a self-consistent equation for P_2 and solve it.

Ans. (a) $P_1 = \frac{1}{8}$, (b) $P_1 = \frac{1}{4}$, and (c) $P_1 = \frac{1}{3}$.

If you wish to explore more, we have a video Penney's Game https://tinyurl.com/yckdjbd6 and a simulator https://bit.ly/3Mv320c.

1.4 The Conundrum of Projections

Recall the results of the Stern-Gerlach experiment when performed with atoms: we found that the pattern separated into two regions, one with a positive projection on the axis and one with a negative projection. Our belief was that the source created atoms with random projections, but then the result that came out did not agree with the classical prediction for a random set of initial projections. These results are summarized in Fig. 1.17.

We told you that this happened because the system was actually a quantum system, but could there be another explanation? What if, by chance, the source only produced atoms with positive and negative projections on some specific axis (and I happened to measure on that axis)? Then I would get the same result as shown in the top panel of Fig. 1.17—but it would be coming from a classical system! How

can we check this? The answer is, we can rotate the analyzer. Then, if the projection is equally positive or negative on one particular direction, rotating the field by 90° should create a projection of zero on this new axis. Be sure you understand this geometry—if the source creates only two effective needles pointing in two specific directions (and having the positive and negative projection on my axis), then there must be a third direction where the projection of both effective needles is always equal to zero. (For those of you who are mathematically inclined, if you pick two needles and they are not parallel, but they have a positive and a negative projection on some axis, then their projection on the axis that points in the direction of the cross product of the two needles will be zero.) However, when the experiment is actually run in different directions, the projections always come out to be the same magnitude. We called this the "conundrum of projections" (following Daniel Styer, who introduced the concept (Styer, 2000)) because there is no arrangement of effective magnetic needles that could create this new pattern. *It is impossible to find two directions for the needles to point in, which always gives the same splitting of the two beams, regardless of the orientation of the magnetic field in the analyzer.* This is why the result must be quantum! We simply have no consistent way to describe it classically for all possible cases.

Atoms are stupid! It turns out that the only way to resolve this problem is to have the projection be determined by the measurement that we make. In other words, until we measure the projection on a particular axis, we do not know what will come out (although we do know precisely what the two options are). It is indeterminate until measured. If we then measure on a second axis, we still do not know whether the projection will be positive or negative, but we can calculate the probability for each result exactly. We describe this by saying atoms are "stupid." They only remember the last axis they were measured on.

Philosophers have another point to make about this, which we will discuss more deeply in Chap. 2. Namely, the projection of the atom is (nearly always) indeterminate before it is measured. The only exception is if we already measured it on one axis and we measure it again on a parallel axis (same direction or opposite direction). Otherwise, we sometimes obtain a projection along and sometimes a projection opposite to the axis measured on. And we cannot say that the atom has any definite projection on that axis until it is measured! This is what indeterminate means.

Furthermore, we have a simple mathematical rule that relates the probability to measure one of the projections for an atom that has had its projection measured on some axis (see Fig. 1.23). We found that the probability for the atom to emerge from the positive exit of the second analyzer, if it came out the positive exit of the first analyzer, is given by

$$\text{Prob}(+, \theta) = \cos^2\left(\frac{\theta}{2}\right), \tag{1.4}$$

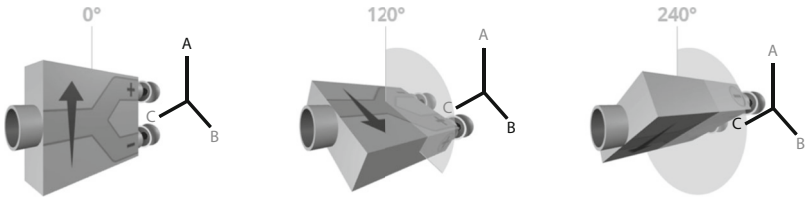

Fig. 1.28 Rotating Stern-Gerlach experiment with three "stop" positions for the analyzers: (**a**) at 0^o or vertical; (**b**) at 120^o; and (**c**) at 240^o. The input to each analyzer is a beam of atoms taken from the + exit of a vertical Stern-Gerlach analyzer to the left (not shown in any panel). The labels **a**, **b**, and **c** show the three different stops

while the probability to come out of the negative exit of the second analyzer was given by

$$\text{Prob}(-,\theta) = \sin^2\left(\frac{\theta}{2}\right), \tag{1.5}$$

where θ is the angle between the two analyzers. This curve is shown in Fig. 1.23. One should note that while it is useful for you to become skilled enough to use this formula for any angle (and using a calculator to do so is highly encouraged), the most important cases are when the angle is $\theta = 0^o$, 90^o, and 180^o. In those three cases, we find $\text{Prob}(+, 0^o) = 1$, $\text{Prob}(+, 90^o) = \frac{1}{2}$, and $\text{Prob}(+, 180^o) = 0$. These three specific cases are well worth remembering. One can work out the similar results, if the atom came out of the negative exit of the first analyzer. I strongly encourage you to do so now.

The last experiments we consider in this chapter are rotating Stern-Gerlach experiments, shown in Fig. 1.28. In this experiment, we have a source of atoms that goes through a Stern-Gerlach analyzer oriented in the vertical direction (0^o) and we take the atoms that exit the + branch of the analyzer. This prepares a beam of atoms with a positive projection on the z-axis. The rotating analyzer rotates between the three stops randomly and instantly, so that when an atom enters the analyzer, it will always find the analyzer at one of the three stops: A, B, or C. We assume it never is seen somewhere in between. The question we want to ask is, under these circumstances, what is the probability for an atom to exit the + exit and what is the probability for it to exit the − exit of the rotating analyzer? We will do this using compound probabilities, and whenever working with compound probabilities, it is useful to express the probability first in words and then convert to an equation. So, for exiting the positive exit of the second analyzer, we want to compute the probability to be at position A and exit the positive exit, or to be at position B and exit the positive exit, or to be at position C and exit the positive exit. Now remember, every time we have an "and," we have to use multiplication for our probability. And every time we have an "or," we have to use plus for our probability.

Our first step is to calculate what the probability is going to be for exiting the positive exit when it's in each of these different positions. So if it's at A, the probability for it to exit, since that's oriented vertically, is going to be 100% or 1. Just recall our formula, where $\cos^2\left(\frac{0°}{2}\right) = 1$. At B, the probability to exit is going to be $\cos^2\left(\frac{120°}{2}\right) = \frac{1}{4}$. While if it's C, the probability to exit from the positive exit is $\cos^2\left(\frac{240°}{2}\right) = \frac{1}{4}$. The total probability (found by converting that formula into an equation) is the probability that it will be at A multiplied by the probability that it will exit the positive exit (if it's at A), plus the probability to be at B multiplied by the probability it will exit the positive exit (if it is at B), plus the probability to be at C multiplied by the probability that it will exit the positive exit (if it is at C).

We still need to determine what the probabilities are that it's going to be at A, B, and C. Since it is just randomly going between them, those probabilities are all equal. So, we have a probability of $\frac{1}{3}$ for each (remember, the total probability must equal one). We just plug into our formula: $\text{Prob}(+) = \frac{1}{3} \times 1 + \frac{1}{3} \times \frac{1}{4} + \frac{1}{3} \times \frac{1}{4} = \frac{1}{2}$. Hence, it emerges from the positive exit half of the time. This is what you might have reasoned without doing any calculations, since the angles are evenly and symmetrically spaced.

What about the probability to exit from the negative exit? We can go through a similar calculation as what we just did (and for practice, it would be good for you to do so). But, because it comes out of one exit or the other, and the total probability is always one, we find the probability to emerge from the $-$ exit is one minus the probability to emerge from the + exit, or $\text{Prob}(-) = 1 - \text{Prob}(+) = \frac{1}{2}$.

What if the rotating analyzer is broken and it never stops at position C? Then, the probability to be at A is $\frac{1}{2}$ as is the probability to be at B (since it is at each stop half of the time now). We find the probability to exit the + exit is now $\text{Prob}(+) = \frac{1}{2} \times 1 + \frac{1}{2} \times \frac{1}{4} + 0 \times \frac{1}{4} = \frac{5}{8}$. It is bigger than before, which you might have also inferred.

Congratulations, these are complex probability calculations. We will find they play an important role when we discuss the Bell experiment in Chap. 2. You will also have the opportunity to examine these types of calculations a bit more in the problems. We also have a video The Rotating Stern-Gerlach Analyzer https://tinyurl.com/3vjeupwh on the rotating Stern-Gerlach analyzer.

So, by now, I hope you will agree with two important facts. The first, is that there are quantum phenomena that we have not found any way to describe with classical physics (this has not yet been unequivocally proven to you, but will be in Chap. 2). Second, quantum events have an unavoidable randomness to them.

These two facts are enough to undermine the philosophy of determinism. This is because the quantum world is manifestly not deterministic. It has randomness inherent in it whenever we perform a measurement. If the universe were unrelentingly deterministic, it could be argued that there is no room in physics for free will; these properties of quantum mechanics re-open the door to a universe that contains

free will. Once again, I don't plan to debate anyone's beliefs here. I just want to point out that we have found phenomena that occur in this universe that cannot be described by classical means and that are required to be random.

So what do we get from a calculation? I am a physicist, and I spend most of my research time working in quantum mechanics and calculating probabilities for things to occur. This is the language of quantum mechanics. We cannot say precisely what will occur, only what the probability is that something can occur. So, how useful is this? We have also discovered when we studied probability that the results of an experiment with a finite number of trials often do not agree with the exact probability. So are these calculations useful?

If all we did was calculate probabilities, perhaps the answer would be only "somewhat so." But it turns out that there are things in quantum mechanics that we can both measure and calculate to very high precision. And these are the things that are the most useful things to measure. In most cases, it turns out that the thing we can measure most accurately is either the time between two events or the frequency/color of a photon of light. We will talk in great detail about the properties of light in Chaps. 3 and 4. Just be aware that the most accurate measurements usually involve time or frequency, and the accuracy is astounding. Our current atomic clocks are so accurate, their precision is similar to measuring the distance across the United States between Washington, DC, and San Francisco to an accuracy of the width of one atom!

The key to attaining high accuracy in quantum mechanics comes from our ability to measure things that are not related to probability. While the probability for different events to occur will always be difficult to attain high accuracy for, because they rely on our having large numbers of tries to achieve high accuracy. The other quantities we can measure, in particular, the difference in energy between two states of an atom, turn out to be things that can both be calculated accurately and be measured accurately, without requiring a similar level of high accuracy for statistical measurements. These types of measurements are called *high-precision measurements*, and they are used to confirm theories, or they show us that our current theory is wrong and needs to be extended.

Hydrogen is a wonderful example for how this process works. If you are interested in this, there is a fabulous book by John Rigden (Rigden, 2003) that goes deep into the history of hydrogen and why it is so important (*Hydrogen: The essential element*). You see, it turns out that the quantum mechanics of an atom with one electron (which is what hydrogen is) can be solved exactly—and we will do it before we have finished this book. As experiments got more and more accurate, this exact theory started to show some problems. In the 1940s, just after World War II, Willis Lamb began performing high-precision measurements of the energy levels of hydrogen. He found that two energy levels—which were predicted to have the same energy—actually differed by a small amount. The reason turned out to require the development of quantum field theory (or the so-called quantum electrodynamics) to fully explain it. Hence, this small discrepancy in the measurement of the colors of light emitted by hydrogen led to the development of

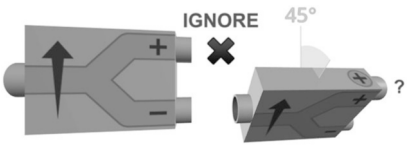

Fig. 1.29 Experiment with two Stern-Gerlach analyzers. Atoms that exit the − exit of the first analyzer are fed into the second analyzer. The angle between them is 45°

a whole new field of quantum mechanics, namely, quantum field theory. And there are numerous other examples that come from different measurements of hydrogen (or other high-precision measurements). Currently, the most interesting one is the measurement of $g - 2$ for the muon, which can be thought of as a heavier and unstable cousin to the electron. The g-factor for a muon (which determines the precise splitting of the two dots in a Stern-Gerlach experiment) is predicted to be exactly 2 in the theory of Dirac—it is modified from two by interaction with the vacuum fluctuations from quantum electrodynamics, weak and strong forces. These effects have been calculated to high accuracy, but the most recent experiment shows deviations, indicating that there may be new physics we are not aware of (and hence did not include in our calculations).

Chapter 1.4 Problems:

1.4.1 At an Angle

We have a Stern-Gerlach analyzer oriented in the vertical direction 0^{o}. We take the beam of atoms that exits the − exit of the first analyzer and put them through a second analyzer, oriented at 45° to the vertical as shown in Fig. 1.29. What is the probability that the atom exits the + exit of the second analyzer?

1.4.2 Rotating Stern-Gerlach Analyzer

(a) Consider the rotating Stern-Gerlach experiment, as shown in Fig. 1.28. Assuming it stops at positions A (left), B (center), and C (right) with equal probability (1/3), what is the probability for an atom to exit the + exit, if the incident atomic beam has a − projection on the vertical axis?

(b) Consider the broken rotating Stern-Gerlach experiment, which only stops at positions B and C (each one equally likely, or a probability of $\frac{1}{2}$). What is the probability for an atom to exit the + exit, if the incident atomic beam has a + projection on the vertical axis?

(c) Consider the broken rotating Stern-Gerlach experiment, which stops at all positions, but spends $\frac{2}{3}$ of its time at A, $\frac{1}{6}$ of its time at B, and $\frac{1}{6}$ of its time at C. What is the probability for an atom to exit the + exit, if the incident atom has a + projection on the vertical axis?

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Chapter 2
Advanced Quantum Mechanics of Spin

I'm still learning things. I'm interested in learning things. I hope I can finally understand physics before I leave the earth.

William Jefferson Clinton, 42nd United States President, 2011

We have learned a lot about quantum mechanics already. We learned it is probabilistic and that it can have indeterminate behavior until measured. The Nobel Prize winner, Richard Feynman, summed this up the best with the following quote from his famous *The Feynman Lectures in Physics*:

> We have implied that in our experimental arrangement (or even in the best possible one) it would be impossible to predict exactly what would happen. We can only predict the odds! This would mean, if it were true, that physics has given up on the problem of trying to predict exactly what will happen in a definite circumstance. Yes! Physics has given up. We do not know how to predict what would happen in a given circumstance, and we believe now that it is impossible—that the only thing that can be predicted is the probability of different events. It must be recognized that this is a retrenchment in our earlier ideal of understanding nature. It may be a backward step, but no one has seen a way to avoid it · · · . So at the present time we must limit ourselves to computing probabilities. We say 'at the present time,' but we suspect very strongly that it is something that will be with us forever—that it is impossible to beat that puzzle—that this is the way nature really is Feynman (1965).

Physicists are troubled that this is the way the real world is, but we have come to accept it so we can learn more about the mysteries of the real world. We provide a video of the overview of Chap. 2 as well https://tinyurl.com/ym7sk5pc.

We start this chapter with a quick recap. We learned a lot in the first chapter. We had our first exposure to the quantum world, where we saw that quantum mechanics is necessarily stochastic and that we cannot determine precisely what happens, only the probability for a set of different events to occur. These results remain with all quantum systems. We also learned that, in some cases, quantum mechanics allows us to measure things with extreme precision. Usually this is the color of light or the ticking of a clock, but the precision with which quantum mechanics has been

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verified is better than any other theory in any science. Amidst all the uncertainty associated with probability is a remarkable certainty of other things, which allow for exceedingly accurate testing of the theory. Not everything measured in quantum mechanics is probabilistic!

One of the themes we came up with is that the "atom is stupid." It only remembers the last axis it had its projection measured on. While this is quite a simple notion, students can often be confused by it, because they want to believe that atoms have a longer-term memory. But they don't! And the sooner this is accepted, the simpler it will be to analyze the different experiments we will be exploring. We have already seen the Stern-Gerlach experiment and we have analyzed it in many different ways.

We will be extending that experiment in a number of different ways, which will lead to quite complex behavior. We will explore an analog to the two-slit experiment, which introduces us to the quantum mystery. We will explore an experiment devised by John Wheeler, which allows us to decide whether we see interference effects or not *after* the particle has passed by the region where it will interfere. Then we explore an experiment devised by Einstein, Podolsky, and Rosen (EPR) in the 1930s, which was one of Einstein's last major works in quantum mechanics. It shows a quantum phenomenon called "spooky action at a distance." We then show how Einstein's proposal of an underlying classical theory that describes the quantum world—local hidden-variable theory—can be proved incorrect using John Bell's ingenious experiment which employs the famous Bell inequalities. We end with a discussion of the application of spins to the real world, with a discussion of nuclear magnetic resonance and magnetic resonance imaging. This is one of the most important examples of quantum mechanics that you encounter in your everyday life.

2.1 The Analyzer Loop

We introduce a new apparatus called the *Stern-Gerlach analyzer loop*. This device will erase the separation of the Stern-Gerlach analyzer. It consists of two pieces. The front end is the conventional Stern-Gerlach analyzer, while the back end is a new device that undoes what the analyzer did. It brings the two paths back to one in such a way that *we cannot tell which path the atom traveled on as it went through the device*. Due to this fact, the measurement of the projection of the effective needle is completely erased, and the atom proceeds as if the analyzer loop was not there. In other words, the analyzer loop does nothing!

We will see that we can create some deep new experiments from an apparatus that does nothing. It will lead us to grapple with some difficult analyses for the behavior of this system. This is because we can select between it doing nothing and it doing something, as we will soon see.

Note, however, that no one has made such an analyzer loop in the way we will describe it to you. Never. We will be constructing so-called thought experiments and we "know" the results of these experiments by applying the rules of quantum mechanics, which describe the real world very accurately. Sometimes textbooks

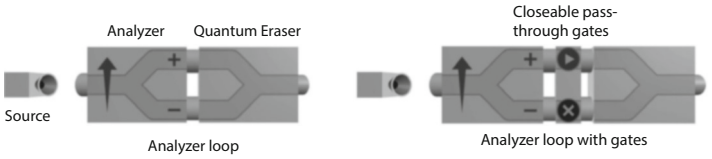

Fig. 2.1 (Left) Analyzer loop constructed from a Stern-Gerlach analyzer on the left half and a quantum eraser on the right half. (Right) Analyzer loop with closeable gates; open is indicated by an arrow with a direction of flow and closed is indicated by an X. Here, the upper path is open and the lower path is closed

present these experiments as if they can actually be done in laboratories. At the moment, they cannot. There are serious technical problems that make it extremely difficult to perform them, but they all can be done in principle. We will proceed as if they can be done in our thought experiments that follow. Note that the EPR and Bell experiments, as well as John Wheeler's quantum eraser (which we will all soon meet), have been carried out, but using light, not atoms. So we are not hoodwinking you in any way. The analysis is ironclad. Furthermore, analyzer loops have been made with atomic chips operating in a somewhat different fashion from the standard Stern-Gerlach apparatus (Machluf et al., 2013), so we do have partial experimental confirmation of these ideas, just not in the specific way we describe below. We can also simulate them using quantum computers, and within these devices, they do behave exactly as we describe. So do not worry, this quantum behavior is completely real.

Now we give the details of the Stern-Gerlach analyzer loop (see Fig. 2.1). The basic premise of the device is rather simple. It starts with a Stern-Gerlach analyzer that will sort an atom according to the projection of its effective magnetic needle on the axis of increasing magnetic field. We then place another device (sometimes called a quantum eraser), which takes the two separated paths of the analyzer and slowly joins them back together until the atom emerges from a single path. The second half undoes what the first half did (hence the name quantum eraser), and does so in such a fashion that the projection that the atom has when it emerges is identical to the projection it had when it entered; in other words, we cannot tell which path the atoms took through the device. At this stage, it appears as if the analyzer loop does nothing. An atom enters it, can move on one of two separate paths (or, more accurately, can move in an indeterminate superposition), and then rejoins, emerging with the same projection it had when it entered.

And if this is all it ever did, it would not be a very interesting device. But, we create two gates in the device, when the atomic paths are at their farthest from each other, where we can introduce an obstruction that the atom will hit if it goes on that branch of the path. Placing the obstruction on one of the paths is the same as measuring the projection. Now the atom will emerge with a projection given by the path it took—the one with no obstruction. Since there are four different choices for how the gates can be opened or closed, this allows us to consider many different possible experiments; of course one case, the situation with both gates closed, is not interesting since nothing comes out when both are blocked.

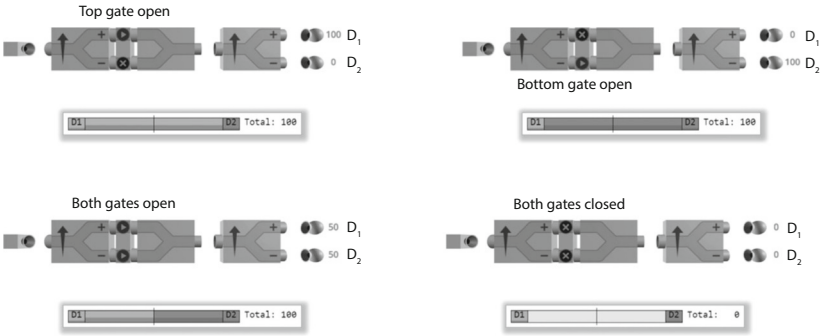

Fig. 2.2 Four different possibilities for the gate configurations in a Stern-Gerlach analyzer loop. (Upper left) Top gate open—atoms that pass through this analyzer loop will take the upper path and emerge with a positive projection on the z-axis. (Upper right) Bottom gate open—atoms that pass through this analyzer loop will take the lower path and emerge with a negative projection on the z-axis. (Lower left) Both gates open—atoms that pass through this analyzer loop emerge with the same state they entered so we find half positive and half negative (because the source is random). (Lower right) Both gates closed—no atoms emerge; this case is a trivial case and will not be considered

We will now investigate some different experiments with these systems. We will be using 100 detected atoms in the illustrations (corresponding to about 200 total atoms, since about half are blocked) and the results do have randomness associated with them. We illustrate the four possible cases in Fig. 2.2. To determine the state of the atoms after exiting the analyzer loop, we take the atoms and pass them through a vertically oriented Stern-Gerlach analyzer, which will count the atoms exiting each path and, thereby, measure the projection of the atoms in the vertical direction. The source is a random source of atoms, with all directions of the projections equally likely. You should predict what happens before looking at the results summarized at the bottom of each panel in the figure. A tutorial simulation https://bit.ly/44T3xYa illustrates these four cases.

Now that we have the basics down, we will be combining a number of our different elements to create ever more complex experiments. It may not seem like it now, but we already have the tools to do some very sophisticated things. To start, we focus on concretely making a new "known-state" source from our devices. This device is constructed from our original random source, which we connect to an analyzer loop with closeable gates. The orientation of the analyzer loop tells us the axis of the states, and the gate that is open tells us whether the projection is positive or negative. Note, we will always have exactly one gate open in this polarized source. As shown in Fig. 2.3, the atoms all emerge in the polarized state (or known state), so we denote such a source as a polarized source with an arrow on the source graphic to indicate the known state the atoms emerge in.

In Fig. 2.4, we show three different experiments with a known-state source of atoms in the state with a positive projection on the z-axis. Note how in the cases where the gates require the atom to take one and only one path (top and middle),

Fig. 2.3 Spin-polarized source of atoms. (Left) Packaging the unpolarized (random) source with an analyzer loop that has the top gate open. (Center) Graphic we use for the polarized source. (Right) Verification that the source produces polarized atoms, where all atoms exit the positive exit

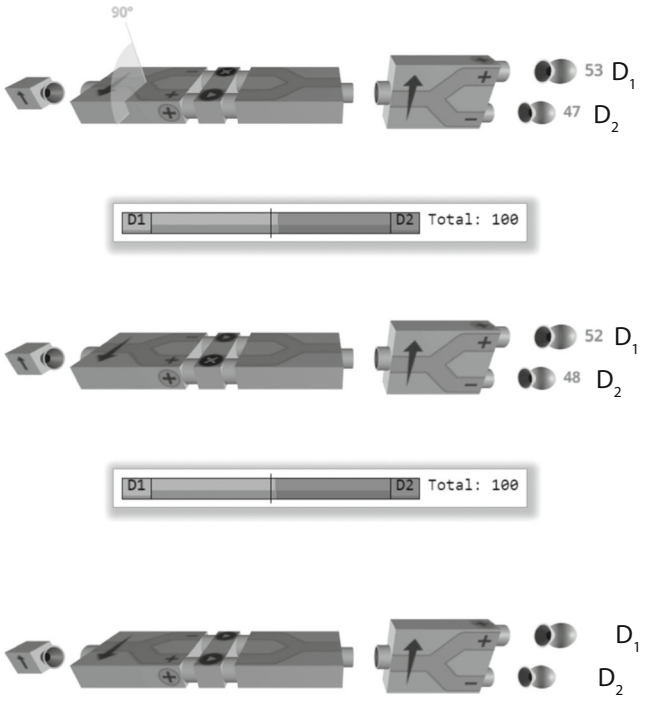

Fig. 2.4 Three experiments with analyzer loops. (Top) Back path is closed on the x-oriented analyzer loop—half come out of each exit of the final analyzer. (Middle) Front path is closed on the x-oriented analyzer loop—half come out of each exit of the final analyzer. (Bottom) Both paths open. The analyzer loop now "does nothing." What output do you expect in the detectors?

we find the same result as we had before. After passing through the half-blocked analyzer loop, the atom lost its memory that it had a definite projection on the z-axis and only knows that it has a definite projection on the x-axis (positive or negative). We have a tutorial simulation https://bit.ly/4oOnYfW illustrating these experimental results. What if both paths are open, as shown at the bottom? Make a prediction, then read on.

In this case, the analyzer loop does nothing, because we have no information about which path the atom goes on—a condition we describe either as the atom takes both paths or, better still, that the path is indeterminate. You might think that

the atom takes one path or the other in the analyzer loop and thereby picks up a definite projection along the x-axis. If that were the case, then the final analyzer would give half the atoms a positive projection and half a negative projection along the z-axis. But that is not what happens. Instead, the atom emerges in the same state it entered, and all of them will come out of the top exit. I always find it a bit hard to swallow that this is what happens—but this is what happens.

In the 1980s and 1990s, there was a popular TV show starring Jerry Seinfeld, which was literally a show about nothing. They did a show on forgetting where a car was parked at the mall, about waiting to be called to dinner at a restaurant, about whether you can back in or pull front in to a parking spot (when parallel parking), and so on. In tribute to this show, and this piece of American culture, we have the analyzer loop, which is literally an experimental apparatus that *does nothing*. At least it does nothing when both gates are open. We will see though that what it does when one gate is closed and what this means for the case when both gates are open will lead to some remarkable behavior. So thanks Jerry. It is true this experiment does nothing, but we will learn a lot about nothing—just like your show taught us to find humor in everything from a show that purported to be about nothing! Now I have to go put up my "Festivus pole" and prepare for the "feats of strength" and the "airing of grievances."

Richard Feynman is again famous for stating that everything you need to know about quantum mechanics is encompassed in the two-slit experiment. I have seen reports that explain how he backtracked a bit from this statement later in his life to include entanglement as second quantum mystery. Nevertheless, the two-slit experiment is an incredibly important experiment to understand, and we will have our first encounter with it now.

As we have already seen, the state of the projection of the spin of an atom is unknown until measured. Once measured, it can only have a definite projection in one specific direction. While a Stern-Gerlach analyzer requires an atom emerging from its exits that has a projection along the orientation of the analyzer (positive or negative), an analyzer loop does not force the atom into a definite state according to its orientation. Instead, an atom going through an analyzer loop with both paths open does not change its original state, regardless of the orientation of the analyzer loop. But, how can this be? How does an atom manage to carry out such a dance? Can we try to observe how this is done? Let us examine each step of this experiment to see if we can figure it out.

We will talk you through two different analyses of the experiment shown in the bottom panel of Fig. 2.4. An atom enters the analyzer loop (which has both paths unblocked) in a state with definite projection along the z-axis. It then goes through the analyzer loop and exits. Finally, it has its spin projection measured on the z-axis. In the classical analysis, the atom definitely travels on one path or the other. When it travels on one and only one path, the atom adopts the spin projection corresponding to the path. This means, upon exit from the analyzer loop, the atom will be in a state with a positive projection on the x-axis (50% of the time) or a negative projection on the x-axis (50% of the time). This is because an atom with a $+z$ projection will be found half of the time with a $+x$ projection and half of the time with a $-x$

projection when we measure along the x-axis. At this stage, regardless of whether it exits in the $+x$ state or the $-x$ state, it will be seen to emerge from the final vertical analyzer half of the time with a positive projection and half of the time with a negative projection. So, the classical analysis concludes that the analyzer loop with both paths open will have half of the atoms have a positive projection on the z-axis and half with a negative projection. We arrive at this conclusion because the eraser half of the analyzer loop simply redirects the atom, which has a definite projection along the x-axis, to the one exit of the analyzer loop, without changing its spin projection.

Our quantum analysis is a bit different. Since we are not observing the atom as it passes through the analyzer loop, we do not know precisely what it is doing. In particular, we do not know how it traverses the split paths of the first half of the analyzer loop. But, because the second half of the analyzer loop erases what the first half did, the atom emerges in the same state as it entered—that is, the analyzer loop did nothing! So it exits the analyzer loop with a $+z$ projection. This means all atoms will emerge from the + exit of the final vertical analyzer.

So which one is right? As you probably suspect, the correct one is the quantum one. But if this is true, what was wrong with the classical analysis? It seemed to be completely logical with solid reasoning. As we go through that analysis, one of the key points is that the atom goes through the front path or the rear path (in the first half of the analyzer loop). Once we know that, then the rest of the analysis completely follows according to our rules, which we have tested and verified. Hence, we must conclude that the atom does not take the positive x path or the negative x path. The only conclusion is that it somehow takes both paths! Now, this may sound completely foreign to you—indeed, it is highly nonintuitive. Nonetheless, as Sherlock Holmes said in *The Sign of Four* (Doyle, 1890) "How often have I said to you that when you have eliminated the impossible, whatever remains, however improbable, must be the truth?" Thus, since the analyzer loop is actually covered and we cannot see what happens inside, we must conclude that the atom goes on both paths! Some philosophers do not like us to describe the situation this way by saying the atom takes two paths. Indeed, we really do not have the right words to describe this situation. So, if you do not like thinking of the atoms as taking two paths, you can also simply say that the path the atom takes is *indeterminate*. What we mean by this is at the heart of the quantum analysis: namely, we simply have no way of knowing precisely how the atom travels through the analyzer loop, yet it somehow "knows" about the fact that there are two paths, because only in this case does it emerge with the same state it entered. We have a tutorial simulation https://bit.ly/4qalt97 that shows these two cases in detail. We have found a completely surprising and confusing result. It is one that requires further study. So, we need to find a way to watch what is going on. This is what we examine next. Don't get upset if you picked the classical analysis as correct; it will come back again, as we modify and clarify the experiments we do.

Our analyzer loop has two gates, one on each arm. What if we swapped out the gates with pass-through detectors? These detectors would watch the atoms as they move through the analyzer loop. We can then see what it means for the atom to "take

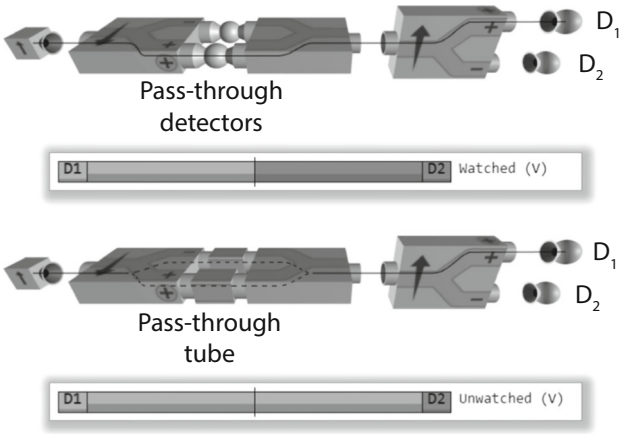

Fig. 2.5 (Top) Analyzer loop experiment with pass-through detectors. Here, we can determine the path the atom takes, so we find half emerge from the $+z$ exit and half from the $-z$ exit of the final analyzer. (Bottom) Analyzer loop experiment with pass-through tubes. Here, we cannot determine how the atom goes through the apparatus, and we find they all emerge from the $+z$ exit of the final analyzer

both paths," or what it means to say "the path is indeterminate." After all, an atom cannot split into two halves and go partially on each path. Or can it? In addition, can the second half of the analyzer loop really erase the splitting onto two paths induced by the first half? Let us see if we can sort out what is happening.

In Fig. 2.5, we show the setup for the two modified experiments we will be considering. In the top panel, we have added pass-through detectors to the analyzer loop. These devices watch and register if an atom passes along a specific arm in the analyzer loop. For now, assume they are 100% efficient—that is, they never fail to work. What do you think will happen? Will they register on both arms when we send an atom through, because it "travels on both paths?" It turns out, when we run this experiment, we always see an atom on one path or the other—we never see the detectors go off on both arms. So, the watched atoms go through definite paths! But now, the result of the experiment has changed. We find that half of the atoms emerge from the $+z$ exit and half from the $-z$ exit of the final vertical analyzer. In other words, *watching at the gates changed the result of the experiment back to our classical analysis*.

Okay, this is a bit disturbing, but we do know measurements can and do make irreversible changes to our system. Let's replace the pass-through detectors with pass-through tubes (bottom panel of Fig. 2.5). Now, we cannot tell which path they take. What is the result? We go back to the quantum analysis—all atoms emerge from the $+z$ exit of the final analyzer!

This is getting pretty devious. But we have one more trick up our sleeve. We will replace one pass-through detector with one pass-through tube. In other words, we only detect the atoms that move along the $+x$ path in the analyzer loop. What

happens now? We go back to the classical analysis, with a 50–50 split in the output from the final analyzer. Why? Well, if the detector goes off, then we know the atom went on the $+x$ path. What if it did not register? Then, we know the atom went on the $-x$ path, since there is no other choice. This means we know the path the atom takes even if we have just one detector, so the classical analysis holds. We have a tutorial simulation https://bit.ly/4962EOp to show all of this behavior.

This is getting frustrating. When an atom has two paths to take and we do not know which path it takes, it can somehow take both! Or, if you don't like that, it emerges *as if it took both*, even if we do not know exactly what it did (i.e., the precise path was *indeterminate*). What we do see is all atoms go to the top detector. When we set up an experiment to look and see how does it go through two paths at once, we don't see that (indeed, that concept of going on two paths is really odd). Instead, we clearly see it goes through one path or another. But then the results at the end of the experiment change. It now registers 50% in each detector! Nature is clever and does not reveal its secrets. When we think we are clever enough to fool her, she changes the results. Watched atoms behave differently than unwatched atoms. Is this frustrating? Yes. We even have a name for it. We call it *the quantum mystery*. Now you know what it is as well.

Overall, this phenomenon goes by the name "quantum interference." This phenomenon is incredibly strange, but completely true. We can describe how nature acts, but we cannot describe why it acts this way. This led Richard Feynman to his famous statement, "No one understands quantum mechanics," whereby he meant that no one understands how nature manages to carry out the behavior of the quantum mystery. No one understands what it means for an atom to take two paths at once, or for the atom's path to be indeterminate. Feynman knew very well that we can understand how to make predictions about quantum phenomena and how to calculate results of experiments with high accuracy. We can even create a mathematical representation of what it means for an atom to take two paths (which we will do later). But we cannot explain *why or how* nature behaves this way. The video Double Analyzer Loops https://tinyurl.com/369wvaje summarizes these issues.

Nevertheless, there is another change to the experiment that we can try. We can set up a pipe on the $-x$ path and a pass-through detector on the $+x$ path. We can then adjust the sensitivity for the pass-through detector so that it only works some fraction of the time (so as to not disturb the atoms too much as they move through the analyzer loop). What do you expect to happen in this case? We will examine this in the problems.

We are going to end this section with a discussion of a class of experiments where we choose different combinations of closed gates in two perpendicular analyzer loops. The analysis is repetitive and tedious, but we are going to discover something really amazing, so stick with me. It is so worth it.

The setup for the experiment is given in Fig. 2.6. We have a polarized source ($+z$) on the left and the atomic beam then passes through a horizontal (A)- and then a vertical (B)-oriented analyzer loop. We can choose which of the four gates to close in the analyzer loops. Our analysis examines all of the different possibilities. Let's

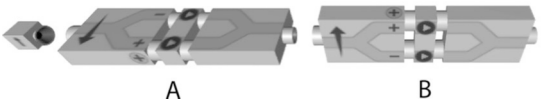

Fig. 2.6 Crossed analyzer loop experiment. We have a polarized source of atoms on the left (preparing the atoms in the $+z$ state). Then the atoms pass through a horizontal-oriented analyzer loop and then through a vertical-oriented analyzer loop. There are four different gates that can be switched open or closed. In the figure, all four gates are open. The gates are labeled by the path in the analyzer loop ($+x$ and $-x$ for A and $+z$ and $-z$ for B)

start with all gates open. In this case the analyzer loops do nothing, so when we go through the first analyzer loop, the path is indeterminate, and it exits in the $+z$ state. Then for the second analyzer loop, we travel on the $+z$ branch and exit in the $+z$ state. One hundred percent of the atoms entering the experiment emerge from the output.

What if we close the $-x$ branch of the first analyzer? Then half of the atoms hit the blockage and are lost, the other half take the $+x$ branch, and we emerge from the first analyzer loop in the $+x$ state. Then the atom path is indeterminate in the second analyzer loop and the atom emerges in the $+x$ state, with only 50% of those entering at the beginning ultimately emerging at the end. The analysis is similar if we block the $+x$ branch, except here atoms take the $-x$ branch and emerge in the $-x$ state, still losing half of the atoms. If instead, we close the $+z$ branch of the second analyzer loop, nothing gets through. Finally, if we close the $-z$ branch of the second analyzer loop, all atoms emerge in the $+z$ state.

Let us now examine what happens if we close two gates. In any case where we close two gates in the same analyzer loop, nothing gets through, so we will close them in different analyzer loops. This gives us four possibilities. We will carefully analyze the case where we close the $-x$ and $-z$ branches. The other cases are similar. Since $-x$ is closed, we travel through the $+x$ branch of the first analyzer loop and emerge in the $+x$ state. Then, because the $-z$ gate is closed, we travel on the $+z$ branch of the second analyzer loop and emerge in the $+z$ state. Half of the atoms are lost in each analyzer loop, so we only have 25% of the atoms that originally entered that exit the final analyzer loop. Similar results occur with all other combinations—25% emerge from the output in the state corresponding to which path is open in the final analyzer loop.

Closing three gates or four gates is not interesting, because in these cases all atoms are always blocked and nothing emerges from the output.

Now, if you look back at these results, something really amazing has happened. Go look at these again and see if you can identify it. If not, we will discuss it next.

Look at the case where only the $+z$ gate is blocked. In that case, we had no atoms make it to the exit—all were blocked. Next, look at the case where we add in additional blockage—the case where both $-x$ and $+z$ are blocked. In this case, we find 25% make it through, all exiting in the $-z$ state. *Adding blockage to the*

experiment let more atoms through! More than that, if we have the experiment set with only $+z$ blocked, we can tell whether someone also added a blockage in the first analyzer loop, with an atom *that does not directly interact with the blockage.* After all, it is the atom that gets through that we detect. So, we can tell that a gate in the first analyzer loop was closed, by detecting an atom that did not hit the blockage. This type of experiment is called an *interaction-free experiment*, and we will see it again multiple times, eventually showing you how to see something *without* shining light on it, which we call "quantum seeing in the dark," in Chap. 4.

One important aspect of all of the material in this section relates to the *information* in the system or, more precisely, the *quantum information*. When we do not know what path the atom took, or the path is indeterminate, we have no information about what the atom did. Normally, in a classical world, when we are given two choices, we would say it did one or it did the other. But quantum mechanics allows a third option, where it somehow is "aware" that it can do both; we can call this the *quantum way*. This phenomenon is called superposition and understanding it is the key to developing an intuition about quantum phenomena.

In the problems below, we examine in more detail many of these different situations one can examine with Stern-Gerlach analyzer loops. These problems are an important step toward you becoming more adept in working with these concepts of superposition and complementarity.

Section 2.1 Problems:

2.1.1 Partially Working Pass-Through Detector

In Fig. 2.7, we show an analyzer loop with one pass-through tube and one pass-through detector. The pass-through detector can be adjusted so that it detects only a fraction of the atoms that go through the device. We know that if the pass-through detector detects all atoms that pass through the device, then we will observe half of the atoms that exit in the $+z$ state and half in the $-z$ state. Similarly, if the pass-through detector is turned off, so it detects no atoms, then we would see all atoms emerge in the state $+z$. Explain what happens when the detector works some of the time, such as half of the time it will detect an atom and half of the time it will not. What will we see in the detectors after we send many atoms through the device? You may find it easiest to think of this scenario as half the time the detector is active and able to measure atoms and half the time it is inactive and is not able to measure atoms.

Fig. 2.7 Analyzer loop experiment with one pass-through detector and one pass-through tube

Hint: there are two different events here—one where the detector is active and one where it is inactive. Your analysis should average over what happens for each event—that is, what happens when it is active and what happens when it is inactive.

2.1.2 Experiments with Three-Analyzer Loops

In the following experiments with three perpendicular analyzer loops, we have closed a range of different gates (see Figs. 2.8, 2.9, 2.10, 2.11, 2.12 and 2.13). For each case, in each figure, determine the state of the atom as it exits the final analyzer loop and determine the percentage of atoms that exit the final analyzer loop relative to the number that enter the first analyzer loop.

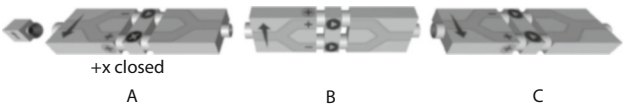

Fig. 2.8 Three-analyzer loop experiment with the $+x$ gate on analyzer loop A closed

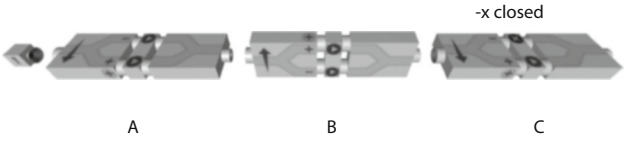

Fig. 2.9 Three-analyzer loop experiment with the $-x$ gate on analyzer loop C closed

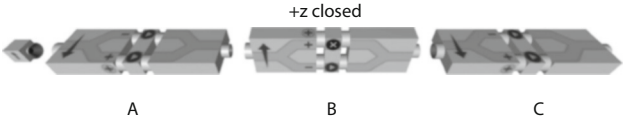

Fig. 2.10 Three-analyzer loop experiment with the $+z$ gate on analyzer loop B closed

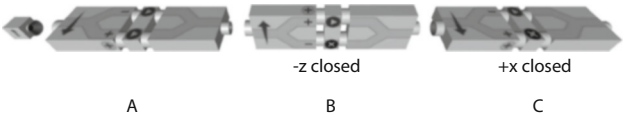

Fig. 2.11 Three-analyzer loop experiment with the $-z$ gate on analyzer loop B closed and the $+x$ gate on analyzer loop C closed

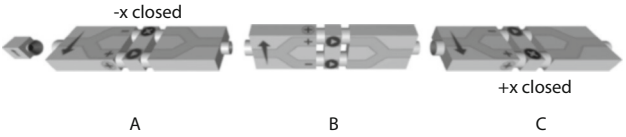

Fig. 2.12 Three-analyzer loop experiment with the $-x$ gate on analyzer loop A closed and the $+x$ gate on analyzer loop C closed

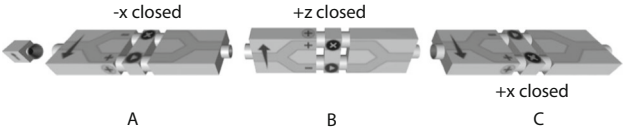

Fig. 2.13 Three-analyzer loop experiment with the $-x$ gate on analyzer loop A closed, the $+z$ gate on analyzer loop B closed, and the $+x$ gate on analyzer loop C closed

2.1.3 Quantum Seeing in the Dark with Atoms

You should have noticed something strange occurred when comparing the setup in Fig. 2.12 with the setup in Fig. 2.13. In particular, adding a blockage in the middle analyzer loop in Fig. 2.13 allows more atoms to exit! Write a paragraph explaining how such a result can occur in the quantum world. You may want to begin by explaining what happens in Fig. 2.12. In fact, it is the atoms that *do not hit* the new obstruction that tell us that the obstruction is there. This behavior is called "quantum seeing in the dark" or an "interaction-free measurement." Be clear, concise, and persuasive in your paragraph.

2.2 Complementarity and Delayed Choice

Niels Bohr is one of the fathers of quantum mechanics (see Fig. 2.14). He developed a theory called *complementarity*, which says that if one develops an experiment that tests for either the "particle" or the "wave" properties of a quantum object, one will not be able to measure the other (complementary) property. We have not yet gotten far enough in the book to discuss Wavefunction@Wave-particle duality in full, but you do know enough quantum mechanics to discuss the simpler notion of complementarity.

We discuss it in a somewhat different light, one that is used much more commonly in describing modern experiments. It relates to whether or not we know what is called "which-way" information. Suppose we have a quantum device with two paths. Then if we know which way the particle goes on those paths, that will determine the quantum properties of the particle to be those associated with the path it took. If we do not have "which-way" information, the particle will retain

Fig. 2.14 Niels Bohr, one of the pioneers of quantum mechanics in the twentieth century (Nobel Prize photo, B. Lagrelius and Westphal, in the public domain and modified to look like a sketch by James Freericks)

whatever properties it had prior to going into the device. In essence, we have already exploited this behavior in describing the way a Stern-Gerlach analyzer loop works. For example, because the analyzer loop has only one exit, if I do not observe at the arms, I have no way of knowing which arm it went through (if that even makes sense, since "which arm" is indeterminate), so I *do not* have which-way information in this case.

It used to be thought that the only way to find out "which-way" information was by disturbing the quantum particle so much that it clearly changed the experiment, and hence explained why the properties of the quantum particle changed when the path it took was observed. But more recent experiments have found ways to determine which path is much more gentle—so gentle that the particle cannot be disturbed in the way previously believed to have happened—and yet the quantum particle still adopts the properties associated with the path it took. Hence, the notion of complementarity, in the sense of "knowing which-way information," is more fundamental and governs the behavior of all objects in the quantum world. We may not fully understand how or why this occurs, but the fact that it does so has been verified numerous times.

We have come up with a word to describe the behavior that we have had forced on us by nature, namely, that a particle can traverse two paths at the same time or, if you prefer, that the path it takes is indeterminate. It is called a *superposition of quantum states*. Instead of thinking of the particle actually separating and rejoining as some "half particle," we think of the particle as simultaneously doing both things, as a whole particle, but with a 50% chance to be found on each path at any given time (although we cannot see how it does it and this does not mean it is half the time moving on just one path or the other). Since there is no classical analog to such behavior, we don't have a simple picture to give you for how this is done, and indeed, the lack of a simple way to visualize or experience this result is one of the things

Fig. 2.15 John Wheeler, the first person to propose delayed-choice experiments (Image credit Ulli Steltzer, courtesy of AIP Emilio Segrè Visual Archives. Modified to look like a sketch by James Freericks)

that can frustrate students who are studying the quantum world. This is not the same as the particle taking one path or the other. The quantum particle experiences *both paths at once*, with equal likelihood, and both as a whole particle. (I get confused every time I read that last sentence, but it is the best I can do to convey the idea.) The motion through both paths at the same time, which cannot be observed, is equivalent to saying that we cannot tell which path or even how a particle moves through the system; the path taken is indeterminate.

In Chap. 3, we will develop a mathematical theory that fully describes these superpositions and allows us to calculate probabilities with them. But for now, we will press on, knowing that this one phenomenon is very strange, but can be described by a coherent theory, and that we will do so shortly.

Enter John Wheeler, who was a distinguished physicist and, among other things, the PhD adviser to Richard Feynman (see Fig. 2.15 to see an image of Wheeler). Late in his career, he began to spend a lot of time thinking about the issue of complementarity. In particular, he wanted to know just when a quantum particle "realizes" it is being watched. In other words, can we set up an experiment, and decide after the atom passes through the two paths, whether or not we want to watch? While this may sound a bit like science fiction, it turns out the answer is yes, and we will discuss more about how to do this with light in Chap. 4.

But for now, let's explore this idea a bit further, because it can lead to some philosophical issues that are difficult to address. For example, Wheeler asked whether we could create a delayed-choice apparatus where the decision of which experiment we perform is made after the particles leave the area of interaction. If so, he further asked: does the choice get communicated to the quantum particle by moving backward in time? While this may seem preposterous, I am not clear on whether there is any simple way to test such a theory. For sure, we know that a delayed-choice experiment can be made. But whether that implies

communication backward in time is another issue. The verdict is still out on that. But the mathematical formulation of quantum mechanics strongly suggests that one can erase without going backward in time. And the reason why is tied to this eerie behavior called superposition.

We will now use Wheeler's delayed-choice experiment to examine the possibility that the analyzer loop results are governed by a classical theory called a *conspiracy theory*. In such a case, the atom knows the arrangement of the analyzer loops and analyzers in an experiment *prior to entering any part of the apparatus*, and the behavior it undergoes is determined by this preknowledge.

For example, if we use our standard arrangement of a source of atoms that are oriented positive in the vertical direction and input them into a horizontal analyzer loop and then into a horizontal or vertical analyzer, the conspiracy theory says that the behavior of the atom as it goes through the analyzer loop is determined by the orientation of the final analyzer.

In other words, if the final analyzer is vertical, then the atom goes through the initial analyzer loop by "taking both paths," but if the final analyzer is horizontal, then the atom chooses one of the two horizontal paths through the initial analyzer loop even if it isn't being watched. We now describe how, by clever choice of experiments, one can perform a delayed-choice experiment to distinguish between these two scenarios. The idea is that switching the orientation of the analyzer after the atom has left the analyzer loop, but before it has entered the analyzer, allows us to see whether the conspiracy theory is true or not.

You may not have noticed this, but this question is directly related to the idea of how an atom can take two paths at the same time, but as a whole particle, or what it means for the path the particle is on to be indeterminate. If the particle conspires with the apparatus to make a choice as to how it will behave, perhaps there is an alternative way to describe the behavior in an analyzer loop without having to invoke the crazy-sounding statements like "the particle goes on both paths." Such conspiracy theories are sometimes called loopholes. A tutorial https://bit.ly/3XQhF0r illustrates four cases to analyze.

Let us try to construct just such a conspiracy theory and see if we can come up with a classical way to describe the situation that the analyzer loop does nothing or that the atom takes both paths. We will consider the situation shown in Fig. 2.16. Here, a vertical polarized source shoots atoms into a horizontal analyzer loop with

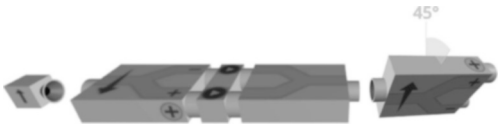

Fig. 2.16 Experimental setup for our attempt at a conspiracy theory. We have a source of $+z$ polarized atoms, a horizontal analyzer loop (oriented in the $+x$ direction), and a final analyzer that can be oriented at any angle relative to the vertical; here, we are showing the orientation at 45°. In a conspiracy theory, we can devise rules that depend on the configuration of the *entire* experimental apparatus

both gates open and we measure the projection of the spin in a final analyzer that is oriented at an angle θ with respect to the vertical; here, we show the case of 45°.

We have a lot of freedom in the rules we can employ to try to describe this experiment classically. The one caveat for the classical description is that the atom must travel on one path or the other in the horizontal analyzer loop. For example, we can choose the probabilities for the atom to pick a path to depend on the final angle θ of the analyzer, even though the atom will not have reached that part of the apparatus when it is traveling through the analyzer loop. Let us give it a try!

We will compute the probability to exit the + exit of the final analyzer loop within such a conspiracy theory. This is given by the probability to take the $+x$ path in the analyzer loop and exit the + exit of the final analyzer or the probability to take the $-x$ path in the analyzer loop and exit the + exit of the final analyzer. In equations, we can write this as

$$P(+) = P(+x;\theta)\cos^2\left(45^\circ - \frac{\theta}{2}\right) + P(-x;\theta)\sin^2\left(45^\circ - \frac{\theta}{2}\right) \tag{2.1}$$

by using our rules for compound probability. The only tricky part of this calculation is that the angle between the horizontal analyzer loop and the final analyzer is given by $90^\circ - \theta$.

It is clear we have to use the conspiracy theory to have the probabilities of traveling on each path in the analyzer loop depend on the angle of the trailing analyzer. This is because if we choose the probability that we know when we observe at the gates, which is $P(+x) = P(-x) = \frac{1}{2}$, we find that $P(+) = \frac{1}{2}$ because $\cos^2\alpha + \sin^2\alpha = 1$ for any angle α. But, the quantum theory tells us the probability is $\cos^2\frac{\theta}{2}$ and this only works when $\theta = 90^\circ$. Using the fact that $P(-x;\theta) = 1 - P(+x;\theta)$ gives us

$$\begin{aligned} P(+) &= P(+x;\theta)\left[\cos^2\left(45^\circ - \frac{\theta}{2}\right) - \sin^2\left(45^\circ - \frac{\theta}{2}\right)\right] + \sin^2\left(45^\circ - \frac{\theta}{2}\right) \\ &= P(+x;\theta)\cos(90^\circ - \theta) + \sin^2\left(45^\circ - \frac{\theta}{2}\right) \\ &= P(+x;\theta)\sin\theta + \frac{1}{2}\left(\cos\frac{\theta}{2} - \sin\frac{\theta}{2}\right)^2 \\ &= \left(P(+x;\theta) - \frac{1}{2}\right)\sin\theta + \frac{1}{2} = \cos^2\left(\frac{\theta}{2}\right). \end{aligned} \tag{2.2}$$

We used the cosine double angle formula to simplify the expression in the first line and the trigonometric addition formulas in the second line. The sine double angle formula is used to convert the third line to the fourth. We also require the final probability to equal the quantum result, given by the second equality on the fourth line. The only parameter that we can vary is $0 \leq P(+x;\theta) \leq 1$. This equation cannot be solved when $\theta \leq 45^\circ$. You should see if you can solve for this yourself

(after using some more trigonometric identities, you should find $P(+x;\theta) = \frac{1}{2}(1 + \text{cotan}\,\theta)$, which can only be solved for $\theta \geq 45°$ because $0 \leq P(+x) \leq 1$). One can easily see that if $\theta = 0$, the left-hand side of the final line equals $\frac{1}{2}$, while the right-hand side equals 1, so no value of $P(+x;\theta)$ will work. But when $\theta = 90°$, we find $P(+x; 90°) = \frac{1}{2}$ does give the quantum result. The video Can We Trick Nature https://tinyurl.com/bde9jss3 covers these results too.

So it seems like we were unsuccessful, but a skeptic might tell us that we can find "magic values" for the probabilities when $\theta \geq 45°$ and this allows us to use a classical description some of the time. Can we rule out using a classical description for those cases too? This is where the genius of Wheeler comes in. He tells us we can run a delayed-choice experiment, where we choose the angle of the final analyzer *after* the atom has passed through the first analyzer loop. The value for the quantum result remains $P(+) = \cos^2\frac{\theta}{2}$, where the angle θ is the final angle of the trailing analyzer. But the conspiracy theory will give the result corresponding to the original value of θ when the atom entered the analyzer loop. Hence, the classical conspiracy theory breaks down for delayed-choice experiments when we change the angle during the running of the experiment. The end result is we cannot make any theory work that has the unwatched atom travel only on one path!

Things have certainly gotten strange, but "you ain't seen nothing yet." The quantum world has much stranger things in store for us. Delayed choice gives us a new window into the quantum world. We employed it to rule out one classical description for quantum results, the so-called conspiracy theory, which has the atom conspire with the experimental apparatus configuration and the combination of the two determines the final measured result. Sometimes these conspiracy theories are called loopholes. We see in the next section that we can rule out even more complex classical theories, like the one Einstein espoused, called local hidden-variable theory.

We are going to analyze the subtleties of "which-way" information with yet another wrinkle. We are going to examine experiments where the atoms are "tagged" in some fashion when they pass through one of the branches of the Stern-Gerlach analyzer and examine what happens when the tag is removed. Unlike a measurement, which irreversibly changes the system, a tag is reversible. But they act in funny ways. If the tag has not been removed, the system will act just as if it was irreversibly measured. But if the tag is removed, then it acts as if it was never tagged in the first place. These kinds of experiments can actually be done, and a modified version with light, where the tag is polarization, will be shown in Chap. 4.

How does tagging work? Our first example of it was the analyzer loop. The atom is tagged on the first half of the analyzer loop. This first half is a Stern-Gerlach analyzer, and if we let the atoms emerge, then just as in an analyzer, we see them emerge from one path or the other, where the spatial location is correlated with the projection of the current loop. If we let the atoms hit a screen at this point, the measurement is irreversible, and this is how we view the analyzer as performing a measurement. But, if we let the atom move through the second half of the analyzer loop, it removes the tag (since both paths return to the same spatial location). This

is how the analyzer loop can do nothing; we can verify the tagging by performing experiments that observe the atoms when they are separated in the middle. Now we discuss a more complex tagging operation that is not a measurement as directly observing them was.

You probably learned previously that an atom is composed of a nucleus made up of protons and neutrons, containing all the positive charge, and the electrons remain primarily outside the nucleus with a negative charge. The electrons are stable in the atom, but by shining light of the right energy onto the atom, we can change the energy that the atom has and change its "state." The atom does this by absorbing the light and increasing its energy. We call this an excited atom, or an atom with electrons in a higher-energy state. Usually atoms with excited electrons that have been placed in a higher-energy state are unstable and they relax by emitting light and having the electrons go back down to a lower-energy state (usually to the lowest-energy state, which we call the ground state). We can assume all atoms emerging from the source are in the ground state, because atomic excitation energies are huge compared to thermal energies. The tagging of atoms involves two operations. The first one, called an exciter, involves shining light onto the atoms and changing the atoms from being in the ground state (lowest-energy state) to being in an excited state. We can tell it is in the excited state because if we watch it for a long enough period of time, the atom will eventually emit light as it relaxes back to the lower-energy state (ground state); this process is called *spontaneous emission*. Similarly, if an atom is in an excited state already, we can shine light onto it to coax it to emit a photon and go to the lower-energy state before it would do so on its own; this process is called *stimulated emission*. Coaxing the atom to emit a photon is called a de-exciter. In order for these experiments to work, we need to be able to count individual photons—this can be done in optical cavities, but it is a difficult experiment. You should keep in mind that we can distinguish an excited atom from a ground-state atom; hence, we use the word tagged to say we have tagged the atoms (meaning we can identify them because they are no longer alike). This is an atomic scale way of putting a label on an atom, so we can watch it and identify it. There is a subtlety to this, however. In order for such an experiment to work, we need to ensure the released photon is not absorbed. This can also be done in an optical cavity which has walls made of mirrors that have very low absorbance. We assume our experiments are all set up in this way, so the absorbing or emitting of a photon is never an irreversible measurement on our system.

In Fig. 2.17, we show the three new attachments for Stern-Gerlach devices. Left is the exciter, indicated by a lightning bolt, which will excite the internal state of an atom from the ground state to one specific excited state. Middle is the de-

Fig. 2.17 Three new attachments to Stern-Gerlach devices: (left) the exciter, (middle) the de-exciter, and (right) the superpositioner

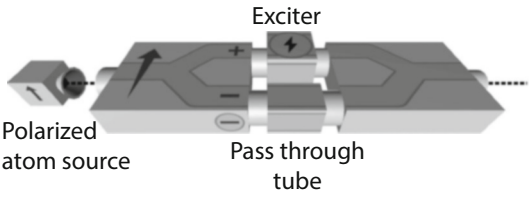

Fig. 2.18 Illustration of tagging. Here, the $+z$ polarized atom goes through an analyzer loop oriented in the $-x$ direction, since tagging is reversible (imagine attaching a second analyzer loop to the output of this one and replacing the exciter with a pass-through tube and the pass-through tube with an exciter; this would remove the tag completely, leaving all atoms in the excited state)

exciter, indicated by the electrical ground symbol, which is a compound device that transforms an atom from an excited state to the ground state, but then blocks it from exiting; however, if the atom is initially in the ground state, it can exit. Hence, it acts like a ground-state pass-through filter. And right is the superpositioner, indicated by the "S," which creates a superposition of the ground and excited states and can be thought of as a "half exciter." In other words, think of this as being the identical device as an exciter, which excites from the ground state to the excited state, but is only run for half of the time. This then creates a superposition of the ground and excited states with a 50% probability to be in one or the other state.

The phenomenon of tagging is illustrated in Fig. 2.18. Here, we have an analyzer loop that has an exciter added to one arm of the analyzer loop. When an atom passes through the loop, the path is indeterminate, which we think of as passing through both arms. But now, when it passes through the arm with the exciter, it will excite the atom on that path. But, because this is not a measurement, our system will be in a state where it is excited and traveled on the positive path (in the $-x$ direction) and it is in the ground state and traveled on the negative path. The tagging operation has created a state that we call an entangled state, because it has created a correlation between *two* degrees of freedom (projection of the spin and ground or excited state of the atom); indeed tagging can be taken to be synonymous with entangling. But, it is not a measurement, because we can remove this correlation by simply passing the atom through a second analyzer loop, with the pass-through tube and exciter exchanged with each other. Then the atom emerges from the second analyzer loop in the state $+z$ projection and excited state. The fact that all the atoms are now in the excited state means that the tag has been erased. The key is that the atomic state is no longer correlated with the projection of the spin of the atom. In other words, we no longer have *entanglement*. Entanglement requires there to be a correlation between spin and the state of the atom—once all atoms are in the excited state, this means there is no more correlation—hence, no more entanglement, and hence no more tag. You might want to think of other ways that the tag can be erased, such as ways for it to end in the $+z$ state and the ground state. Further details of these experiments and how to implement them can be found in Courtney et al. (2020).

The notion of this correlation, and how to keep it straight in your head, is a bit tough. We will be introducing a concrete notation to describe this later in the book, but for now, we will use words instead of formal constructions and notations. It is similar to what happened in the Stern-Gerlach analyzer—there, we correlated the projection of spin with the spatial location of the atom (emerging from the + exit or the − exit). So, you can think of the analyzer as an entangler as well, with the measurement being carried out when we *detect* which exit the atom emerged from on the screen. It is only a tagging before the measurement, because we can "erase" the entanglement by putting the emerging atom through the second half of an analyzer loop. Here, the behavior is similar, but the tag is no longer associated with a spatial position, but with an internal state of the atom.

If I sent the atom that emerged from the analyzer loop in Fig. 2.18 into a vertical analyzer and measured which exit it came out of, I would find half the time the atom is up and half the time it is down. If I instead put it through a device that measures the state of the atom, I would find it is half the time in the ground state and half of the time in the excited state. Even more interesting, if I look at the atomic state of the atoms that emerge from the exits of the analyzer, each will be half the time in the excited state and half the time in the ground state. This result is quite similar to what happens with watched atoms in an analyzer loop, and we want to understand how to analyze such an experiment.

The key to the analysis is that the correlation between the state of the atom and the projection of the spin means that we have which-way information. Hence, we can say that the atom really did go on one path or the other after the measurement of the state of the atom. What if we did not measure the state of the atom (ground or excited)? It does not matter, because we *could* have measured it, so the system acts as if we actually did the measurement, and the system acts classically, with the atom going on one path or the other through the device. In this case, what we have is the atom emerges from the analyzer loop with a definite positive (excited state) or negative (ground state) projection of the spin on the $-x$ axis. Then when it goes through the vertical analyzer, in each case, half go to the + exit and half to the − exit.

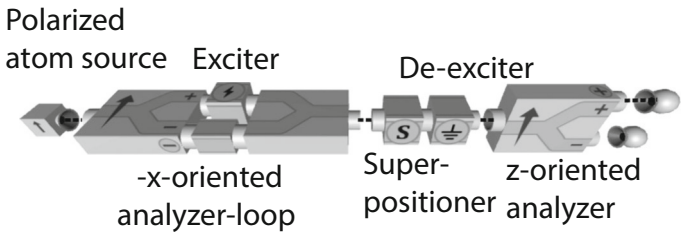

Fig. 2.19 Setup for a quantum eraser experiment with Stern-Gerlach apparatus. We have a polarized source that sends a beam of $+z$ oriented atoms into an analyzer loop in the $-x$ direction, with an exciter on the positive path. Then the beam is sent through a superpositioner and a de-exciter, before being measured in a vertical analyzer

Now, where things really get crazy is if we remove the tag. We cannot completely remove the tag from all atoms if we do not separate them in an analyzer loop again and this results in some seriously complex behavior. Let us analyze this carefully; the experiment is shown in Fig. 2.19. What we will do is first send the atoms through a superpositioner. For now, trust me in what I will tell you happens; we will work out the precise formalism for this later. The ground state goes to a superposition of the ground state plus the excited state, while the excited state goes to a superposition of the ground state minus the excited state. Remember, superposition is the same as "the path is indeterminate." Hence, if I now measure the ground-state atoms only, they will act as if the atom went on both arms of the analyzer loop, and all such atoms will have a $+z$ projection of the spin; similarly, all excited state atoms now have a projection of $-z$. We erased the tag without sending it through another analyzer loop! But we did so at the cost of losing half of the atoms that were in the excited state. Note that the erasure mechanism is to simply remove the correlation between the atomic state and whether it has a positive or negative projection on the $-x$ axis.

How do we measure ground-state atoms only? This is what the de-exciter does, since it acts as a ground-state pass-through filter, because it will de-excite if the atom is in the excited state, but then it will detect the emitted photon and close the exit, blocking an atom that was in the excited state; if no photon is detected, it lets the atom pass through. We will encounter the quantum eraser a few more times, so do not panic if you are having some trouble following all of the subtlety of these arguments. When this was originally introduced by Wheeler, he proposed that the atom might be going backward in time to erase the tag, but, as we will see later, such an assumption is not needed. Tags can be removed without violating the arrow of time.

The difference between a tag and a measurement is subtle. As we have discussed here, a tag is reversible, while a measurement is not. Usually, it is obvious what the difference is. Send an atom through an analyzer and we think we made a measurement, but we really just did a tag (correlating the spin of the atom with the exit it emerges from). This is because we know we can attach the second half of an analyzer loop and undo the tag. The atom is only truly measured when it is detected on the screen. That cannot be reversed, as the atom is stuck to the screen. So, think carefully about when something irreversible happens. Until then, we are usually creating tags (synonymous with making an entangled state), but with the right experiment, we could undo the entanglement and erase the tag.

Section 2.2 Problems:

2.2.1 Is a Human Observer Needed?

We set up a Stern-Gerlach analyzer loop with a pass-through detector on each branch, as shown in the top panel of Fig. 2.5, but we cover up the detectors so we cannot see what the readout is. Will an atom going through such a device keep the projection it had prior to entering, or will it adopt the projection corresponding

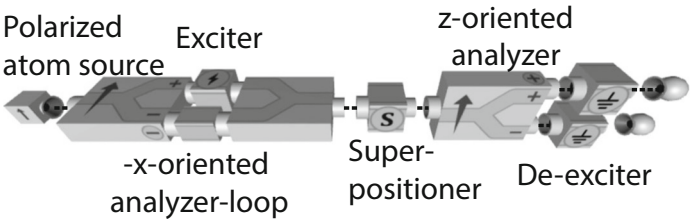

Fig. 2.20 Delayed-choice quantum eraser with the erasure occurring after passing through the final vertical analyzer

to one of the paths of the analyzer loop? Write a short paragraph explaining your position on this.

2.2.2 Erasing the Tag After Measurement

In Fig. 2.20, we place the de-exciters after the atom passes through the vertical analyzer, but before detection. Write a paragraph that discusses the following points: (i) does this arrangement also erase the tag; (ii) does this setup require information to travel backward in time; and (iii) does it restore all atoms to the original state, or just some atoms? Try to be as concrete as possible in discussing all of the different elements of the experiment and what state the atom is in after passing through each piece of the apparatus. Be sure to also carefully describe how different states of the atom are correlated with each other in the different stages of the experiment. Discuss as well what would happen if the de-exciter worked only half the time?

2.3 Einstein, Podolsky, Rosen, and Bell

So far, we have been focusing primarily on what happens to a single atom in an experiment. We repeat the experiment many times, to build up enough statistics to determine the probabilities accurately, but in all cases, the experiments involved just a single atom in the apparatus at any given time. Now we will be exploring experiments that involve two atoms at a time. These experiments bring in a new phenomenon—one that is difficult to describe and involves a lot of current research—*multi-particle entanglement*.

Loosely speaking, entanglement of two particles occurs when they are in a joint quantum-mechanical state that is correlated. This means that if you know what happens to one of the particles, you know what will happen to the other due to the correlation. This has been stated in a somewhat vague manner, but you will see, as we work through this section, that we continually clarify this notion

Fig. 2.21 Albert Einstein (left), Boris Podolsky (middle), and Nathan Rosen (right), the authors of the famous EPR paper in 1935. (Einstein photo by Harris and Ewing, Library of Congress (public domain), Podolsky photo (fair use) from Nordisk Pressefoto, courtesy of AIP Emilio Segrè Visual Archives, Margrethe Bohr Collection, and Rosen photo from Wikipedia (care of Asher Peres). Photos modified to look like sketches by James Freericks)

of entanglement and make it more precise. Because the correlated quantum state requires that the second atom will have a well-defined state governed by what we observe for the first atom, this can lead to bizarre behavior. It might appear as if the atoms somehow communicate with each other instantaneously, which would violate the special theory of relativity, since special relativity says that nothing can be communicated faster than the speed of light. It is weird. Einstein called it "spooky." But no communication is believed to actually occur. Because these arguments are quite subtle, we will take our time to describe them and we will go over the ideas more than once to make sure we have them straight. It makes for a long chapter section, but this material requires it.

Our story begins in 1935, when Albert Einstein, Boris Podolsky, and Nathan Rosen wrote a famous paper that was published in the *Physical Review* of the American Physical Society (Einstein et al., 1935); see Fig. 2.21 for an image of the three authors. The paper has become one of the most important papers in the development of quantum mechanics because of the ideas and issues it brings up regarding how to understand the quantum world. The paper deals with two particles that are entangled and proceeds to discuss what the consequences would be of measurements made on those atoms. The conclusion of the authors was that one either had to adopt an ability for the atoms to know what was happening to each other instantaneously over an arbitrarily long distance (a notion Einstein described as "spooky action at a distance") or that quantum theory was incomplete and the atoms had to have some additional hidden degrees of freedom, instructions, or variables that allowed them to produce the results of the thought experiments that they proposed. The issues raised in the paper have deep implications for the principle of complementarity as well. Bohr responded to the paper with an article that argued that complementarity was the critical principle (Bohr, 1935). It took another 25–30 years before a brilliant theorist, John Stewart Bell, devised a way to test such a hidden-variable theory, and it took quite a bit more time before experimentalists were able to actually run such a test. The results of this line of inquiry will be

discussed later in this section. The video Einstein's Spooky Action at a Distance https://tinyurl.com/2t6z67fd covers many of these ideas.

There is an interesting historical note about the EPR paper. Podolsky leaked the results of the paper to *The New York Times* who ran a front page story that Einstein proves quantum mechanics is wrong. Einstein was furious that Podolsky did this. So much so, that he never spoke to Podolsky again. Einstein felt these issues should be decided among scientists, not among the press.

We start with a modification of the experiment by David Bohm in the 1950s. Bohm's quantum mechanics textbook (Bohm, 1951) was based on lectures given by Robert Oppenheimer at Berkeley after World War II. But, in Chap. 22, on the quantum theory of the measurement process, he introduced a variant of the EPR experiment that is the most common form discussed today. In this experiment, we have a special source of atoms, which creates two atoms that always are created with opposite projections of the spin *on any axis*. So, if the left atom is measured with a positive projection on some axis, the atom on the right will have a negative projection on that same axis, and vice versa. That is, the spin projections are always correlated to be opposite, and on any axis for our Stern-Gerlach analyzers. While it may be hard to comprehend how such a source could be made, if I have H_2 molecules and the spins are in what is called a singlet state, if I can separate the molecule without changing the spins, we would have such a source. The key is that this spin state is indeterminate, because we do not know whether the atom with the positive projection is to the right or the left—in fact, it is not even correct to think it is one or the other, as it is *inseparably both at the same time*! We can only say there is a definite spin to the right and to the left *after* we measure one of the atoms. Hence, it is improper to say it has a definite projection before it is measured. This is called an *entangled state* of atoms, and in particular, it is a spin-singlet state of the pair of atoms. This experimental situation is illustrated in Fig. 2.22 and in a tutorial simulation https://bit.ly/3Mtn7nH.

I want to also review one important, but subtle idea from the theory of special relativity. I do not have the space to develop for you why this is true, but I will state something that you probably know and also point out some important aspects of it. The theory of special relativity tells us that no communication between two people can take place faster than light. For, if it did, then we would be able to change

Fig. 2.22 Two possible results for the Bohm variant of the EPR experiment, where the spin projections are always measured to be opposite to each other. The double-ended source in the middle (labeled with a double-headed arrow) creates EPR pairs, which are then detected in two analyzers, here shown in the same orientation, namely, in the $+z$ direction. Half of the time, the analyzer on the left measures + and half of the time − and the same on the right, but the two results are always correlated—if + on the left, then − on the right, and vice versa. These results are the same for any tilt of the analyzers, as long as both analyzers are at the same tilt angle

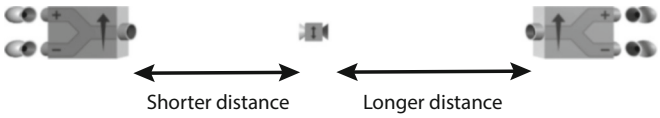

Fig. 2.23 Setup of an EPR experiment where the left analyzer is closer than the right analyzer. But the difference in the distance is too small for light to travel from the left analyzer to the right analyzer in the time between when an atom enters the left analyzer and the time that it enters the right analyzer

"what caused what," and that cannot be done in a world that is causal (since this is equivalent to being able to go backward in time). Note that it does not say "nothing can travel faster than light," just that "nothing can *communicate back and forth* faster than light."

Often when we try to understand the following Bohm experiments, we reason as follows: the basic idea of the Bohm variant of the EPR experiment is to examine how fast atoms communicate the information of their quantum state to each other. To make things "spooky," we set up the experiment so that the analyzer on the left is a little bit closer to the source than the analyzer on the right (see Fig. 2.23). Now, if we run this experiment, we will find the same results as before—half the time a positive projection and half the time a negative projection on both the left and the right, but the two measurements are always correlated to be opposite. But, here is the twist we have by having the left closer than the right. The left is measured first in the lab frame. Let us say that we measure a positive projection on the z-axis. Then this means the atom enters on the right with a negative projection on the z-axis. *How does the atom on the right know that it has a negative projection when no information can be conveyed from the left measurement before the right one is measured?* We can make it even more complicated. If we change the tilt angle of the analyzer on the right after the measurement on the left, what will happen now? We know the atom on the right will act as if it is in the $-z$ state, regardless of the tilt angle of the analyzer on the right. But, how does it know how to do this? Somehow, quantum mechanics acts in a *nonlocal* fashion, where what we mean by this is that correlations can survive the atoms being separated, even by large distances. One could also interpret this as an action at a distance, in the sense that the atom instantly knows what to do, violating the normal rules of causality. Einstein, of course, did not want to have anything that acted like an action at a distance, since he developed his general theory of relativity to get rid of Newton's action-at-a-distance theory for gravity. Einstein, Podolsky, and Rosen instead told us that quantum mechanics was incomplete. The atoms needed to be supplemented by hidden variables, or hidden instructions, that allow it to carry out this spooky behavior.

First, let us discuss whether an instantaneous transfer of information is possible. It may surprise you, but a one-way signal faster than light is possible in relativity, but it cannot be causal; in other words, it cannot make something happen. This is because when we have an arrangement as in Fig. 2.23, with the analyzer to the left closer to the source than the analyzer to the right in the lab frame, if we observe the

experiment in some other frame moving relative to the lab frame, we can set it up in such a way that the right analyzer will be closer to the source than the left. Hence, even if a signal is transferred from one atom to another, we cannot uniquely say which atom sends to which. For example, in the original frame the left analyzer is closer, so we would say the left signals to the right, but in an appropriately moving frame, the right is closer than the left, so we would say the right signals the left. But we cannot have both situations be true. *This means the signal (if it even exists) could not cause anything to occur.* It is more correct to acknowledge that the atoms manage to preserve these correlations without having any communication between themselves. In fact, even if I know on the left, instantly, what the measurement on the right will be once it is completed, I cannot use this to communicate to the right (or have the right communicate to me), because communication requires me to control what the measurement results will be. Quantum mechanics is random, so we do not know what the measurement will be until we observe it. Thus, we cannot send a message to the right by choosing the left measurements to be in a specific pattern; we can only measure what they are and then find a random string of $+$ and $-$ values. Such a random string contains no information and cannot be used to communicate.

The easiest way to understand how the instantaneous correlation works is via a classical analogy. Note that this analogy, like most analogies, is not perfect. In fact, the quantum behavior is richer than this analogy. The classical analogy is to have a coin shop in England send a gold coin and a silver coin every week to two clients, one in the United States and one in Japan. Every week the coins are sent, but which customer gets the gold coin is chosen completely randomly with a 50–50 chance for either. Both customers agree to open their packages at exactly the same time. Now, the moment the customer in Japan opens their package and sees a gold coin, they know instantly that the customer in the United States has a silver coin. While you can never determine what your coin will be until you open it, you know both what your coin is and what your friend's coin is the instant you each open the envelope. You learn this without ever having to ask the friend, because of the correlation of the initial state of the game. Nevertheless, this cannot be used to communicate, because the initial correlations were chosen completely at random by the broker in London. Furthermore, the coins are definitely silver and definitely gold the entire time they travel from London in their individual envelopes to their final destination. Their states do not change instantaneously when the envelope is opened. Can we say that the coins communicated with each other instantaneously the moment they were opened? Of course not. The correlation was set up when the envelopes were packed and addressed; the effect of the correlation was only observed later.

The quantum case is a bit more subtle, because the coin shop does not choose which coin is gold or silver if it is a set of "quantum coins" that are sent. Instead, the state is indeterminate until measured. For every time we run such an experiment, the state of the coin (gold or silver) is only determined at the time the package is opened. It is not determined by the coin shop when they send the coins out. This is one difference between the classical case and the quantum case. This distinction is important, because in the quantum system, we say that the states of the atoms, though entangled, remain indeterminate until measured. It is this complication,

which cannot be represented in any coin-exchange analogy (where the coins always have definite states of silver or gold), that is a critical new behavior that the quantum system has. Furthermore, the whole concept of a delayed-choice variant cannot be fulfilled with a classical analogy. We cannot change the game from gold and silver coins to copper and nickel coins after they have been placed in the envelopes and sent to each customer. But the quantum delayed-choice variants of the EPR experiment can do just that.

So, how does the entangled pair of atoms know how to arrange the measurements so that they are random, but always correlated? We do not know how it does this! This is again the quantum mystery appearing. There are alternatives though. EPR (and Bohm) propose that there are hidden variables that the atoms carry with them. This is similar to the conspiracy theory, but they do not necessarily need to know the arrangement of the experiment before the instruction set is created. The atom moving to the left must know which path it will take when it hits an analyzer at a specific tilt angle on the left, and the same on the right. If we use such hidden variables, then there is no faster than light signal, no quantum mystery, just a set of instructions each atom follows. This really seems like the most reasonable way to resolve and understand how such experiments can work. But, we will find out such theories are untenable. Just like in the conspiracy theory, we will see that there are arrangements of tilt angles that cannot have hidden variables explain the experimental results, no matter how hard we try to. Hence, we will end up with the nonlocal quantum theory—and an unexplainable quantum mystery again.

Let's pause for a moment to recap some of the main ideas of the EPR paper with the Bohm variant. First, we can create quantum states of two particles that are entangled, such that the measurements of the projection one atom has, as measured in a Stern-Gerlach analyzer, is always the opposite to the result that its partner atom has when measured in a Stern-Gerlach analyzer that is oriented along the same axis. But, by combining this idea with a "delayed-choice style" of experiment, we can change the axis that we measure the second atom on, and the results will no longer be predetermined. They will instead give the results as if we entered the second analyzer with the atom having the opposite projection of the first atom on the axis the first atom was measured. How can the second atom know that the first was measured, when no causal signal can travel from the first atom to the second before the second is measured? Some physicists equate what is happening as corresponding to an instantaneous, nonlocal effect, which does not violate special relativity because one cannot communicate back and forth this way, but the quantum state is able to provide such an instantaneous connection. Others think this effect involves no communication between the two particles, the results are what they are, and this is just another quantum mystery that we do not know how nature does what it does. No one has determined an experiment to explain how this works, or even whether such an "instantaneous connection" between the atoms is needed, but the theory of quantum mechanics always makes the correct predictions and does not require any communication. In particular, we know that the quantum theory is upheld because the experiment has been done and it has shown that quantum mechanics is correct (more on this below). Einstein, however, felt that quantum mechanics

was incomplete. It needed to be supplemented by some form of hidden variables or hidden instructions to explain these types of experiments. We will find there are two different ways to summarize this behavior. One is that quantum mechanics is nonlocal, because no local theory can explain the behavior of entanglement. But there are numerous issues from special relativity that cause problems with just espousing the nonlocality. We will have more to say about this later too. The other point of view is called quantum holism—that entangled particles must be thought of as one unit that cannot be broken. We will have more to say about this perspective as well.

In many respects, if the Bohm variant of the EPR experiment was only ever performed on atoms where the axes of the analyzers were always chosen to be the same, we would be able to model the behavior in a simple way. We probably would want to introduce some form of hidden variables to supplement the quantum description and enforce the opposite correlation in the measurement outcomes. But, once we realize that the second atom acts as if it knows what has happened to the first instantly after the first is measured, we are forced to describe this behavior via a new phenomenon called entanglement. This description is not something we are comfortable with. It appears to violate the special theory of relativity, although one can argue that is not an issue due to the fact that communication is not being conveyed instantaneously. But, it definitely does not provide us with a "warm and fuzzy" feeling about the quantum world. Indeed, in special relativity, we have a phenomenon called the relativity of simultaneity. This phenomenon occurs when two events happen, but they are spatially separated by so large a distance that light cannot travel between the two events (an event is a point in spacetime—something that happens at a specified location at a specified time). Then which event occurred first depends on what frame I am observing the two events in. So we cannot even say that one event absolutely occurred before another one in this case. This means if we believe that a signal travels between the two atoms, then we would be forced to also believe that the signal traveled faster than the speed of light. But then, because in different frames the order of the events can be reversed (or even made simultaneous), we would not even be able to say which way the signal traveled—from the atom on the left to the right or vice versa. One should view this as an indication that we should not think of signals traveling between the entangled pairs at all.

Now, think carefully about this. Because the individual measurements are random, but correlated, if we observed in a frame where the right occurred before the left, the results can still be exactly the same as what we observed in a frame with left before right! We can analyze it as the right being measured first, with the left being required to have an opposite projection when it is subsequently measured. In both cases, we obtain the same experimental result. This is really mind-blowing because we cannot say which one occurred first in any absolute sense. And it is completely consistent with special relativity.

Some physicists have adopted a philosophy of "just shut up and calculate" when it comes to these types of philosophical issues. Since we find that the theory we have developed to describe this quantum behavior works well at producing the correct results, and does so every time, let's not argue about the philosophical

Fig. 2.24 John Stewart Bell (Used with permission © Joachim Reinhardt and University of Frankfurt. Modified to look like a sketch by James Freericks)

underpinnings. Instead, let's just find out what happens by properly applying the theory. One feels such a viewpoint is similar to the realization that nature won't let us know how a particle can traverse two paths at once (really to be on indeterminate paths) or how an entangled atom conspires with its partner so that when one is measured and its state is known, then its partner's state is known as well. This perspective avoids worrying about any issues associated with special relativity.

One question you might ask is "how do we know the atoms become instantly anticorrelated when one is measured?" If the second atom is measured in an analyzer oriented perpendicular to the one the first was measured in (let us say the first is along the z-axis), we find that the second atom picks one of the two paths with equal probability. This is not enough information for us to know that there was an anticorrelation of the two atoms. To do that, we rotate the second analyzer to an angle different than $\frac{\pi}{2}$. Then the probabilities that it enters the second analyzer with a positive or a negative projection on the z-axis can be differentiated. This would be a way to carry out such an experiment (by examining the probabilities when the left atom has a $+z$ projection versus when it has a $-z$ projection). Of course for each individual pair, we cannot determine probabilities. Only after we measure many of them.

Yes. There is a lot to sort through in quantum mechanics! We end this section with an extraordinarily clever way to test whether local hidden variables are possible or not.

In 1964, John Stewart Bell wrote a paper about the EPR paradox (see Fig. 2.24) (Bell, 1964). His objective was to see whether or not one could test the incompleteness ideas of EPR and show whether quantum mechanics had nonlocal entanglement or whether it had some hidden variables governing the behavior. This was no small task, for his ideas implied that one could show that a whole class of theories was wrong—namely, the local hidden-variable theory that Einstein believed.

Achieving this goal required going through some complex thought experiments. We will be describing similar experiments here, in order to show you how such a comparison works and how it can rule out hidden variables. Beginning in the 1980s and continuing until at least a recent experiment in 2015, experimentalists devised tests of quantum mechanics that use Bell's ideas. In one of the famous

experiments from the 1980s, they also employed Wheeler's delayed-choice idea, by changing the detectors while the particles were moving through the apparatus. In all cases, it has been found that quantum mechanics, with its spooky action at a distance and its entanglement, is the correct description of the physical world. The 2015 experiment, performed with a number of different apparatuses and different experimental groups, closed all loopholes to any local "conspiracy theory"-based argument (classical or quantum). The conclusion is that quantum mechanics is manifestly nonlocal. The technical term is that local realism is inconsistent with experiment. In particular, this means that the source that creates entangled pairs creates the entangled pairs in indeterminate states. Only when we measure them are the states determined and only when we measure both do we see their correlation due to entanglement. Even after we know this bizarre behavior of quantum mechanics is true, we still struggle with finding a way to come to grips with the fact that we do not have a clear picture for how nature does it. As with most strange quantum behavior, we also struggle to find appropriate words to properly describe the phenomena as well. The video Bell's Theorem https://tinyurl.com/yc3etuck discusses these issues.

Let us first give some examples of how hidden-variable theories would work. Start with the setup in Fig. 2.23. Here, a quantum analysis will give that at each analyzer we will have the measurements be randomly positive or negative, but whenever the left is negative, the right is positive and vice versa. The description via hidden variables is simple. Give the left atom a positive path instruction or a negative path instruction, each with equal probability. Then for the right atom, simply give it the opposite path as the left atom. This will give the exact same results. What happens if we flip the right analyzer to be oriented in the $-z$ direction. Here, the quantum analysis gives half positive and half negative at each analyzer, but the left projection and the right projection now always agree. The instructions here would be simple too. Same as before, but select the right one to now be the same path as the left one. So, we have now seen two examples where hidden variables work. Here is the third: have the right analyzer at 90° to the vertical, or if you like have the right analyzer oriented horizontally. Then the quantum analysis says the atom enters the right analyzer with a positive or negative projection on the z-axis. In either case, we have half the time the atom goes through the $+x$ or $-x$ path. But here there is no correlation between the results from right to left. So, we simply choose the hidden-variable instructions to be left choose $+$ half the time and choose $-$ half the time and the same for the right. This will give identical results as well. What Bell found is if we look at any other orientation of the right analyzer than these three, the hidden variables cannot make the right prediction any more. We will describe this in detail below.

Before doing that though, we want to point out that there already is some "trouble in paradise" for the hidden-variable theories. In particular, the rules are different for different orientations. While it is fine to have the rules depend on the orientation of the analyzers in the experimental arrangement, we can see that if we perform a delayed-choice experiment by rotating the right analyzer loop after the left has measured an atom, then the hidden-variable instructions may have trouble, because

they cannot be changed because we rotated the right analyzer. They have to be set at the time the pair of atoms was created. They are allowed to be as complex as we would like though, so perhaps we can handle even delayed-choice variants.

Let us pause again for a recap of the strange phenomena associated with "spooky action at a distance." Students often are uncomfortable with this notion that the atoms communicate with each other faster than the speed of light. We illustrate that you do not need to think of this as the left influencing what happens on the right, if the left apparatus is slightly closer to the center than the right. For concreteness, let's consider the case where the left analyzer is oriented in the $+z$ direction and the right is oriented in the $+x$ direction (you should be able to generalize this to other cases as well). In a frame where the left measurement occurs first, if we observe the left atom to have a positive projection, then the right atom enters the analyzer with a negative projection along the z-direction, so it will emerge from a horizontal analyzer half the time with a positive projection and half the time with a negative projection on the x-axis. This means there is no correlation between the two. Now, suppose we observe in a frame where the right side is measured first (this may seem odd to you if you never studied special relativity, but trust us, we can find such a frame). Then, if we measure positive along the x-axis, we enter the vertical analyzer on the left in the $-x$ orientation. It will be observed to be half the time positive and half the time negative, along the z-axis, with no correlation between the two measurements. Hence, performing the analysis both ways produces the same result—namely, the measurements are not correlated with each other. This means we do not need to know which one is "actually first" to do this analysis. Actually, we cannot determine which one is first. Which one appears to be first is relative to what frame we use to observe the measurements. Now, knowing this, it is worthwhile thinking about whether we really need to have a faster than light communication between the atoms. Indeed, just like the situation where the idea of "the atom goes on both paths" for the analyzer loop is not something we are comfortable with, the faster than light signaling between the atoms is also something we are not comfortable with. It is perfectly okay to simply say this is what happens in quantum mechanics without trying to determine exactly how quantum mechanics does this. Indeed, just like in the analyzer loop, no one has devised an experiment that can tell us precisely how this occurs. It ends up being more of a philosophical discussion than a physics discussion. This occupies the time of many quantum mechanics and the way they describe how this can occur often sounds crazy. If you want to follow this path further, go look into books that discuss the philosophy of quantum mechanics or that discuss the interpretations of quantum mechanics. They will lay out many of these issues for you, but they will not answer any of the questions, regardless of how persuasive their arguments might be. These issues remain active areas of both research and debate.

There is another perspective that sometimes arises in the philosophical books—this perspective is that we should not bring relativity into a nonrelativistic quantum theory. If we espouse this point of view, then we can always determine which experiment is measured first, and everything is resolved by an instantaneous connection or communication between the two atoms. If you wish to follow this

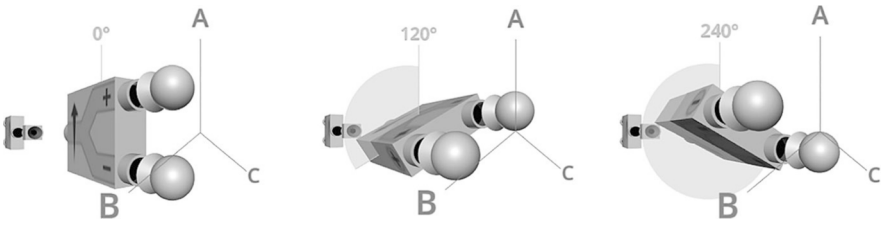

Fig. 2.25 Three positions for the Stern-Gerlach analyzers in the Bell experiment. We have an analyzer on the left and an analyzer on the right. These analyzers can be set in three different positions, denoted A, B, or C, and located at angles of 0° (A), 120° (B), and 240° (C). Each analyzer can be rotated separately

view, you then feel empowered by the fact that when one introduces relativity properly, via what is called a quantum field theory, that the quantum field theory is a relativistically invariant theory and so it is *de facto* correct. In other words, we always have a relativistically invariant theory that can describe this behavior. I am not a big fan of this viewpoint. I am more of the perspective that this is similar to the "going on two paths at once" issue: we do not know how the atom does it there and we do not know how it does it here. This is very close to the "shut up and calculate" viewpoint. In the end you need to decide for yourself just what perspective you feel more comfortable with. No one has found an experiment to differentiate them, and argument never settles physics debates, only experiment does.

On to the main event of the Bell experiment. The experiment is like the Bohm variant of the EPR experiment, with the left analyzer slightly closer to the source than the right analyzer, but the analyzers on each end will now be set in one of three different, equally spaced angles, as shown in Fig. 2.25. The source will release atoms in each direction and the three different positions will be set completely randomly—each position is equally likely, with a probability of $1/3$. Note that some of the time, the two analyzers (left and right) will be in the same position—in this case, the analysis we did before for the Bohm variant of the EPR experiment will hold. When the analyzers are in the same orientation (AA, BB, or CC), they are always measured as opposite "$+-$" or "$-+$," where the symbols denote the projection on the left and right analyzer after the experiment is completed. But when they are not in the same positions, such as a CA or a AB orientation, we will sometimes see the two atoms measured with the same projection "$++$" or "$--$." The question we ask in the Bell experiment is how often are the two projections measured to be the same and how often are they measured to be different? We have a tutorial https://bit.ly/455Smvg covering this experiment.

In order to focus on these same or different measurements, we package the device in a single object called a *Bell analyzer*, which is shown in Fig. 2.26. It has two indicator lights on top that indicate whether the atom exited the + exit or the − exit from the Stern-Gerlach analyzer inside the cylinder. The actual orientation of the analyzer in the given instantiation of the experiment is hidden from view and not recorded.

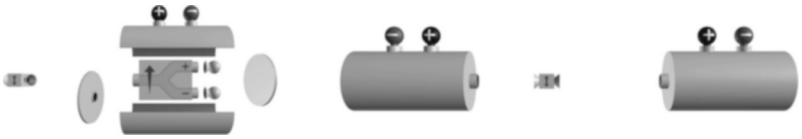

Fig. 2.26 Structure of the Bell analyzer. We package the rotating Stern-Gerlach analyzer into a device with two indicator lights indicating whether the atom was measured with a positive projection or with a negative projection in the analyzer, regardless of which position—A, B, or C—that the analyzer is in. We then have two such analyzers, one to the left and one to the right of the source

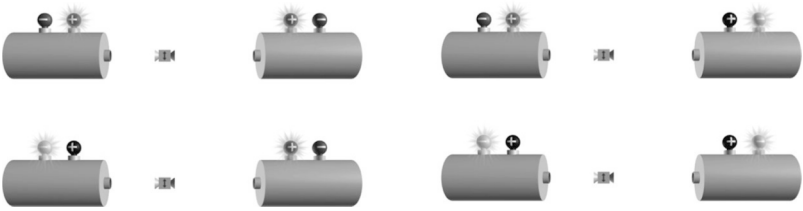

Fig. 2.27 Four different possible results that are measured in the two Bell analyzers for a Bell experiment

In each shot of an experiment, one light turns on in each Bell analyzer, as shown in Fig. 2.27. The experiment simply boils down to how many times do the indicators show " + +" and " − −" versus how often do they show " + −" and " − +." These probabilities can be calculated both for quantum mechanics and for hidden variables.

We start by determining what the probabilities are in the quantum case. Since the only thing that matters in computing the quantum probabilities is the relative angle between the analyzers, and all relative angles we have are $\pm 120°$ for the three position case, we can pick the analyzer on the left to have a definite orientation. We pick it to be in the A position or vertical and we assume, for concreteness, that the atom is measured to have a negative projection in this case (see Fig. 2.28). When in position A, the relative angle is $0°$, so the probability is $\cos^2(0°) = 1$ to exit the + exit and $\sin^2(0°) = 0$ to exit the − exit. When in the B or C position, the relative angle is $\pm 120°$, respectively. In both cases, we find the probability is $\cos^2(60°) = 0.25$ to exit the + exit and $\sin^2(60°) = 0.75$ to exit the − exit. The total probability to exit the + exit on the right is given by the probability that the right analyzer is at position A and it exits the + exit or is at position B and exits the + exit or is in position C and exits the + exit. Converting to equations gives us

$$\text{Prob}(+) = \frac{1}{3} \times 1 + \frac{1}{3} \times \frac{1}{4} + \frac{1}{3} \times \frac{1}{4} = \frac{1}{2}. \tag{2.3}$$

So, half the time it exits the + exit and half the time it exits the − exit in the quantum case. Since the left Bell analyzer flashes − and the right Bell analyzer flashes + half

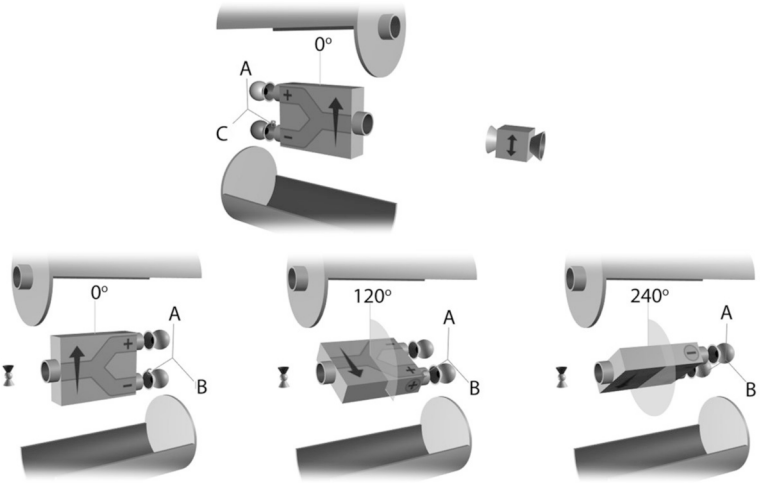

Fig. 2.28 (Top) The left analyzer is in orientation A and measures the atom to have a negative projection. (Bottom left) If the right analyzer is in the A position (vertical), then the atom enters it in the $+z$ state and will exit the + exit. (Bottom middle) If the right analyzer is in the B position (120°), then the atom enters in the $+z$ state and has a 0.25 probability to exit the + exit and a 0.75 probability to exit the $-$ exit. (Bottom right) If the analyzer is in the C position (240°), then the atom enters in the $+z$ state and has a probability of 0.25 to exit the $+$ exit and 0.75 to exit the $-$ exit

the time and $-$ half the time, it is clear that the Bell analyzers flash opposite half the time. One can repeat this analysis for the case where the left analyzer flashes +, and we also find half of the time the two Bell analyzers flash the same and half the time opposite. A second tutorial https://bit.ly/3XQkQW6 performs this quantum analysis.

We now analyze the experiment using local realism, or local hidden variables. Local means that the instruction sets are distributed when the atoms are created and cannot be changed afterward. Realism means that we have instructions for all possible cases. These instructions are hidden in the sense that they are unknown to us. They allow the atoms to exhibit unpredictable yet coordinated behavior, even over large distances. We also assume there is no faster than light communication allowed. These instructions dictate what the outcome will be for atoms emerging from Stern-Gerlach analyzers in any orientation. Note that we do not need to know how the atoms choose these instructions. Just knowing their form is enough for us to analyze the situation. A third tutorial https://bit.ly/4pBJSEC analyzes this hidden-variable case.

In our current experiment, the analyzer can be oriented in one of three positions—A, B, or C—so we need to simply state what exit the atom emerges from. For example, the atom to the left might be given an instruction set, that is, $A = +$, $B = -$, and $C = +$. The instruction set for the right atom must be the opposite, because when both analyzers are in the same position, the results must be opposite,

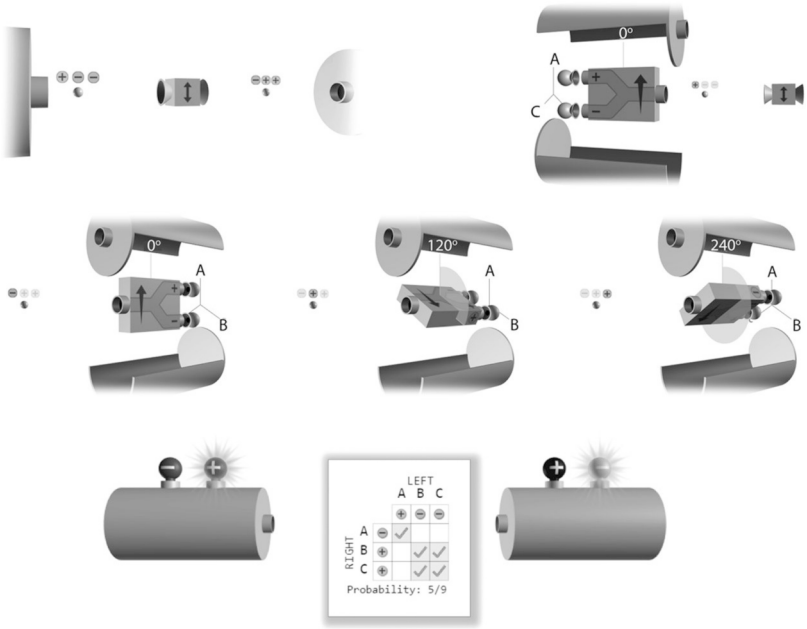

Fig. 2.29 (Top left) The two atoms moving with their instruction sets. (Top right) The atom entering the left analyzer is assumed to be at position A, so the instruction dictates that the atom emerges from the + exit. (Middle row) The three cases for the right analyzer, with the given instructions for position A (left), B (center), and C (right). (Bottom) The instruction set table showing five cases (out of nine) where the two Bell analyzers have opposite lights flash

due to the known anticorrelation of the results. This means we always pick the right atom instructions to be opposite to the left—in this case, we would have $A = -$, $B = +$, and $C = -$. As a shorthand, we just give the ABC instructions in order, such as $+-+$ for the left and $-+-$ for the right. Since the left and right are opposites, we need only report the left instructions. The total number of instruction sets for each atom is eight, because each instruction for a given stop has two choices and we have three stops for the rotating analyzer ($2^3 = 8$). These eight choices are the following: $+++$, $++-$, $+-+$, $+--$, $-++$, $-+-$, $--+$, and $---$. Let us examine carefully what happens if the left instruction is $+--$ as shown in Fig. 2.29. We assume, without loss of generality, that the left analyzer is in the A configuration; the instruction set then tells us that the atom emerges from the + exit. Then we have three choices for the right analyzer. Since everything is predetermined, we can form a table showing the nine choices (since we have three for each atom) of the settings of the analyzers on each side. Of these nine, five of them have atoms exiting opposite exits. So, for this instruction set, we have opposite lights flash with a probability of 5/9. We have a third tutorial https://bit.ly/4jcbaPD that illustrates this case.

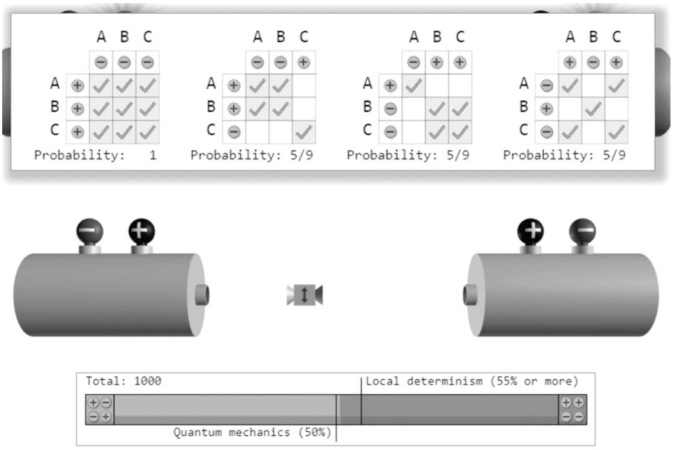

Fig. 2.30 (Top) Table for the different instruction sets showing the cases where the opposite projection lights flash (indicated by the checkmarks). (Bottom) Result of the experiment run for 1000 shots, indicating that the quantum case is correct and hidden variable is incorrect

This is one of the eight instruction sets. In Fig. 2.30, we show all of the tables for the different instruction sets. We only need to show four of them, because the tables are transposed for the opposite instruction set going to the right. Looking at this table, we see all eight instruction sets have a probability of at least 5/9 to flash opposite—the cases +++ and $---$ always are opposite; the cases $++-$; $--+$, $+-+$; $-+-$; $-++$; and $+--$ all have opposite flashes occurring with a probability of 5/9. Now, no matter what probabilities we assign to the frequency of the eight different instruction sets, because every instruction set has a probability that is larger than or equal to 5/9, the overall probability to flash opposite must be at least 5/9.

This means that we have a clear set of predictions. The quantum case says we should have the lights flash opposite 50% of the time, while the local realism prediction is larger than 55.5%. So, what happens? If we test this, we find the quantum case is what is observed and we have a fourth tutorial https://bit.ly/4cihiTl that shows this test. This rules out the local hidden variables. Note that the actual experiments were performed with light rather than Bell analyzers, but the basic analysis is quite similar.

We have found that quantum mechanics is always shown to be correct and local realism or hidden variables are always shown to be incorrect. So how exactly were we able to get different probabilities for each case? In the quantum case, we directly calculate the probabilities, and because the $\cos^2(\theta/2)$ terms vary, we ended up with a situation where the probability was 50% for the lights to be different. For the hidden-variable theory, for each instruction set, there are nine possibilities, and we get opposite lights either nine out of nine times or five out of nine times. The key is that it is always greater than 50%. Hence, no matter what the probability is for any

of the different instruction sets, we must get a probability larger than 50%. And this allows us to make the decision between hidden variables and quantum mechanics. I find this to be a very clever argument that relates these different ideas. And it does so in a very clear way.

Before we leave this section, I want to make clear precisely what Einstein was trying to do. Einstein was not enthusiastic about the random nature of quantum mechanics (in the form of indeterminacy) and its implications on reality. He believed that the real world should always have real, well-defined properties. The indeterminacy of quantum mechanics forced him to back away from everything being determined and predictable, but he did not want to give up on the notion of reality itself, nor did he want to allow something to behave like it had an action at a distance. After all, one of the crowning achievements of his career was replacing Newton's action at a distance in the classical theory of gravity with an alternative general relativity theory, where everything is determined from local properties and there is no action at a distance.

So, he wanted to find a way to keep this notion of reality—that a quantum particle is never in an indeterminate state, it is always determined, but the randomness arises from the hidden variables that we cannot independently measure. The quantum approach allows indeterminate states through the principle of superposition. By allowing for the hidden variables and the complex instruction sets, which can even depend on the setup of the experimental apparatus, Einstein had a neat way to avoid many of the confusing aspects of quantum mechanics. And, indeed, if it turned out to be correct, it would be a marvelous and quite clever theory. But it ended up being wrong. John Bell masterfully found a way to test these ideas and show that only the craziness associated with the nonlocal quantum theory can actually describe the real world. So we have to live with indeterminate states and we have to live with entanglement acting as if there is an action at a distance. (Note that this does not require an action at a distance, just this nonlocal character of quantum mechanics; action at a distance is one way to interpret this behavior, but there are others and most physicists who do not believe in action at a distance.)

Section 2.3 Problems:

2.3.1 Delayed Choice?

We have an EPR experiment setup where the analyzer that measures the state on the left lies closer to the source than the analyzer on the right. In order to see the "spooky" behavior given by the instantaneous transfer of the knowledge of the experimental measurement on the left atom being conveyed to the right atom, must we perform a delayed-choice experiment, where the second analyzer is rotated after the first atom is measured?

Write a short paragraph discussing this point. Be clear and precise.

2.3.2 Cannot Communicate

Suppose we have set up an EPR experiment, where the two analyzers are both oriented vertically. Every time an atom enters the apparatus on the left and is measured, we immediately know exactly what the result will be on the right apparatus. The left apparatus is closer to the source than the right apparatus. Pick which two of the following statements are correct:

1. The experimenter on the left cannot control what the results will be in her apparatus; she can just record what the results are.
2. The experimenter on the right always gets her result after the one on the left, so only the left experimenter can communicate with the right.
3. The experimenter on the right can move her apparatus closer to the source than the experimenter on the left and, by varying the distance back and forth, provide a way to communicate with the experimenter on the left.
4. Because the results of the measurement on the left are random, the results on the right are random as well, even if they are correlated with those on the left, and hence cannot be used to communicate.

2.3.3 Problem with the Coin Analogy

When we discussed the coin analogy, we mentioned that such an analogy would work fine if we always had the detectors oriented in the same direction. This analogy corresponds to a hidden-variable theory, where each atom receives a positive or negative instruction set no matter the orientations of the analyzers. The instructions are always the opposite for the two atoms.

How must this coin analogy be modified if the analyzer on the left is vertical and the one on the right is horizontal? Think about the random nature of the experiments and the correlations between experiments.

2.3.4 Quantum Versus Hidden Variables

In a local hidden-variable theory, we have to decide on the instruction set we give the particles at the time they are created, when they are physically located in the same vicinity. In order for the theory to be able to handle delayed-choice experiments, or experiments where one particle is measured first and then the second is measured, the instruction set must include instructions for all possible experimental setups from the start. Discuss whether this statement is true or false and explain the requirements of the hidden-variable instruction sets.

2.3.5 Requirements of a Bell Experiment

We list some options for a Bell experiment. Determine all of the criteria which must be present for a Bell experiment to test a local hidden-variable theory against quantum mechanics. Ignore loopholes. There are four correct choices.

1. The experiment must create entangled pairs of particles:
2. The experiments can only be done with atoms.
3. There must be a difference between the probabilities for the experimental outcomes of the quantum and of the hidden-variable theories.
4. For experiments with projections of atoms in Stern-Gerlach analyzers, the final detectors can be attached to Stern-Gerlach analyzers at 0°, 90°, and 180°.
5. For experiments with projections of atoms in Stern-Gerlach analyzers, the final detectors must be at angles different from the above angles, at least some of the time.
6. The experiment must change the orientation of one of the analyzers after the particles have been created.
7. The experiment must change the orientation for the second particle after the first has been measured.
8. The experiment must have the two analyzers at different directions, at least some of the time.

2.3.6 Instruction Set

(a) Suppose the Bell theorem experiment is set up with the two analyzers aligned along the same angle, but that angle varies from run to run.
The first symbol of the instruction set (either $+$ or $-$) denotes the path that the left particle takes when it enters the left analyzer (and the exit it emerges from) and the second symbol (again either $+$ or $-$) denotes the same for the right. Determine the rules that will give the same result as a quantum experiment with entangled particles.

(b) Suppose the Bell theorem experiment is set up with the two analyzers always perpendicular (either vertical up in the $+z$ direction or horizontal front in the $+x$ direction), but that direction varies from run to run.

Each answer choice provides the state of the atom measured by the left and right analyzer, depending upon whether the analyzer is oriented along the $+z$ or $+x$ axis. For example, when we say left "$+-$," that means that when the left analyzer is oriented along the z-axis, the atom is measured to be in the $+$ state. When the left analyzer is oriented along the x-axis, then the atom is measured to be in the $-$ state. The first $+$ or $-$ symbol tells us what happens for the z-axis orientation, and the second symbol tells us what happens for the x-axis orientation of the

analyzer. Which choice of hidden-variable instruction sets yields the same result as the quantum analysis?

1. Left "$+\,+$" and right "$-\,-$," left "$+\,-$" and right "$-\,+$," left "$-\,+$" and right "$+\,-$," and left "$-\,-$" and right "$+\,+$," each possibility occurring one-fourth of the time.
2. Left "$+\,+$" and right "$-\,-$" and left "$-\,-$" and right "$+\,+$," each possibility occurring one-half of the time.
3. Left "$+\,-$" and right "$-\,+$" and left "$-\,+$" and right "$+\,-$," each possibility occurring one-half of the time.
4. Left "$+\,+$" and right "$-\,-$" all of the time.

Explain why the choices you did not select are incorrect.

2.3.7 Four Stops: The Quantum Case

Consider a Bell experiment where we have four positions for each analyzer to stop at, at 90° intervals. So there is a stop at $+z$, $+x$, $-z$, and $-x$. Suppose each stop is equally likely and independently chosen. What is the probability that the lights of the two Bell analyzers will flash opposite in the quantum case?

2.3.8 Four Stops: The Local Realism Case

We now go through a series of problems to calculate the expected results of different hidden-variable instruction sets and to compare the predictions of hidden variables to quantum mechanics in a different context from what was covered in the text. Here we consider experiments with four stops of the Stern-Gerlach analyzers, each separated by 90°.

In Fig. 2.31, we explain what the instruction set notation means. Each of the symbols + and − tells us which path the atom takes in the analyzer and which exit it will emerge from. The symbols are ordered according to the possible orientations of the analyzers. In this case, the first symbol is for a $+z$-oriented analyzer, the second for the $+x$ orientation, the third for the $-z$ orientation, and the fourth for the $-x$ orientation. Note how the instruction sets can be decided however we wish to make the rules. The instruction set in the figure, for example, will have the atom emerge from the + exit if the analyzer is oriented in the $+z$, $+x$, or $-x$ directions. Note that we have separate instruction sets for the left-moving atom and for the right-moving atom; those for the right atom are the opposite of that for the left atom.

1. Now, let us compute the probabilities for the hidden variables. We have to assign four projections for the four stops for each atom. The values of these projections are always the opposite for each atom (if the left instruction set is +, then the

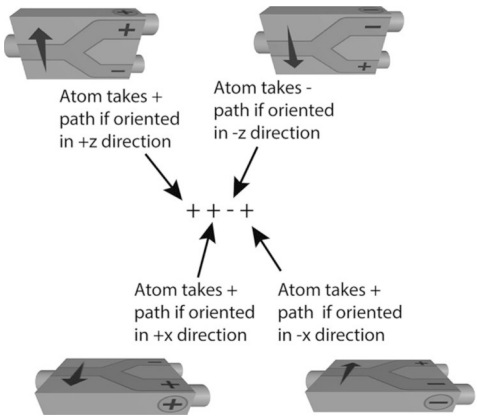

Fig. 2.31 Illustration of how the instruction sets work for a four-stop Bell experiment. The choice for each setting is to exit the + exit or the − exit. The settings are organized in order as $+z$, $+x$, $-z$, and $-x$

right instruction set is $-$). Thus, we only need to report the four values for the left instruction set. There are a total of 16 possibilities for the 4 stops on the left apparatus and the 4 stops on the right apparatus ($4 \times 4 = 16$); these choices are $+z\,+z$, $+z\,+x$, $+z\,-z$, $+z\,-x$, $+x\,+z$, $\cdots$. How many of those 16 possibilities for the stops on each side have opposite projections for the 2 atoms if we have the left instruction set being $+\,+\,+\,+$? This answer is the same as we get if the left instruction set was $-\,-\,-\,-$. Your answer is an integer.
Hint: if the stops are at $+z$ and $-x$, we can see the left instruction is $+$ and the right instruction is $-$. These are opposite, so this is one possibility that is opposite. You need to examine the other 15 choices for the stops.

2. How many of those 16 possibilities have opposite projections for the 2 atoms if we have the left instruction set being $+\,+\,+\,-$? This answer is the same as we get if the left instruction set was $+\,+\,-\,+$, $+\,-\,+\,+$, $-\,+\,+\,+$, $-\,-\,-\,+$, $-\,-\,+\,-$, $-\,+\,-\,-$, and $+\,-\,-\,-$.
3. How many of those 16 possibilities have opposite projections for the 2 atoms if we have the left instruction set being $+\,+\,-\,-$? This answer is the same as we get if the left instruction set was $+\,-\,+\,-$, $+\,-\,-\,+$, $-\,+\,-\,+$, $-\,-\,+\,+$, and $-\,+\,+\,-$.

Based on the analysis in the problems above, it seems like we should again get a probability larger than 50% for hidden variables. However, we cannot always conclude that this is the case. If the source produced instruction sets with two + and two − instructions only (and no other options), then the hidden-variable probability for these experiments would be equal to 50%, exactly the same as in the quantum case. Thus, in the problems above, you showed that in this specific Bell experiment with four positions, we cannot rigorously distinguish the hidden-variable result from the quantum result. Note that this does not mean hidden variables are correct: in our simulated Bell experiment, we showed that the hidden-variable theory is incorrect by examining the case with three positions—in this case, there is a difference between the hidden-variable result and the quantum result, so

they can be differentiated—indeed, this is how we know the quantum case is correct and hidden variables are not. This is how Einstein's idea is proven to be wrong.

2.4 Nuclear Magnetic Resonance and Imaging

Many of you have either had or know someone who has had an MRI image taken of them. But you probably don't know that the origin of this technology goes back to the early days of quantum mechanics, where experiments were being performed on the current loops of atoms. MRI is an acronym which stands for magnetic resonance imaging and it is a shortened form of the original acronym NMRI, which stands for nuclear magnetic resonance imaging. This is due to its relationship with NMR, which is an experimental technique (called nuclear magnetic resonance), which was first developed in the 1950s.

The early researchers in this field had no idea about the potential applications to medicine nor did they predict the availability of an MRI machine in nearly every hospital, as we have today. This is quite common in science. Often the developments in one field of science only become technologically relevant in some other field after years of research and development. One recent example is the Web browser and the protocols that run the Internet, which were first developed by high-energy physicists who needed to transfer data from the experimental facilities to the local universities where the data analysis was being done. It is quite possible that all of the Internet age would either not exist or have been quite delayed had this fundamental high energy physics research not taken place. The development of the MRI is another example that falls into this category. It required three very different basic research areas to be highly developed before the idea of such a diagnostic machine could be developed. These three areas are nuclear magnetic resonance, superconducting magnets, and three-dimensional image reconstruction. Our focus in this section will be on the NMR story, but the other two are equally important in setting the stage for the MRI.

Nuclear magnetic resonance, or NMR, is an acronym which describes the procedure that will be used to change (or as we like to say flip) the orientation of the effective needle of the current loop that represents the nucleus from pointing positive in the vertical direction to negative in the vertical direction. Since we cannot reach in and turn the direction of a nucleus by hand, we have to be clever in developing a methodology that allows us to do this. But before going into the details, just note that nearly all of these experiments involve "flipping the spin" of the nucleus of hydrogen, the simplest element.

So what happened to the nuclear in NMRI? It was dropped because it was felt that patients might not want to be placed in a machine that has nuclear in its name, even if the nuclear refers to the nucleus of hydrogen and has nothing to do with radioactivity.

Let's first start with a brief history of NMR, focusing on the work that resulted in Nobel Prizes on the road toward developing MRI machines. It all started in

1922, when O. Stern and W. Gerlach (University of Frankfurt) performed the first experiment on the current loop behavior of atoms. We are very familiar with this experiment. Fast forward to 1938, when I. I. Rabi (Columbia University) used rotating perpendicular magnetic fields to flip the spin of atoms in Stern-Gerlach experiments from positive to negative projections. In 1943, Stern wins the Nobel Prize in Physics. In 1944, Rabi wins the Nobel Prize in Physics. The next major development was in 1946, when E. Purcell (Harvard University) and F. Bloch (Stanford University) illustrated how to flip the spins of the hydrogen nucleus in a solid (paraffin wax). In 1950, W. Proctor (Stanford), F. Yu (Stanford), and W. Dickinson (MIT) discovered that the size of the magnetic field needed to flip the spins depends on the local chemical environment of the hydrogen atoms, an effect now known as the *chemical shift* and one of the most important aspects of NMR for chemistry. In 1952, Bloch and Purcell win the Nobel Prize in Physics. R. Varian commercialized NMR for chemistry and geology applications in 1955—a relatively quick time for developing commercial applications from basic research. We need to wait until 1966, when R. Ernst and W. Anderson (Varian Laboratories) devised a new experimental technique that increases the sensitivity a hundredfold. Then in 1973, P. Lauterbur (University of Illinois) used NMR to image within a solid the hydrogen nuclei that are having their spins flipped. Also in 1973, P. Mansfield (University of Nottingham) used changing magnetic fields to identify precisely where the different hydrogen nuclei are in slices through the sample. Finally, in 1981, NMR is first used in MRI machines to image patients in a hospital setting. In 1986, J. Haase and A. Frahm (MPI, Göttingen) reduced the data acquisition time by a factor of 100. In 1991, Ernst wins the Nobel Prize in Chemistry. In 2002, Wüthrich wins the Nobel Prize in Chemistry for using NMR for determining molecular structure via the chemical shift; and in 2003, Lauterbur and Mansfield win the Nobel Prize in Medicine.

What this timeline shows you is that science development is often not in a straight-line path and often the final rewards are completely unexpected to the original workers in the field. While we won't do this here, we could come up with a similar list for superconductivity, which is employed for the magnets used in most MRI machines, with a large number of Nobel Prizes for that research area, and one could do the same for tomography and the computer algorithms employed to allow for the data analysis of MRI machines.

A famous physicist, George Pake, who started the Xerox PARC research facility, said (in the late 1950s, well *before* MRI): "That the basic discoveries in science invariably bear fruits which cannot be foretold is an old story to scientists. Even so, it has been a great thrill to see what has grown out of the work of Purcell and Bloch, who were seeking only to get a better understanding of the magnetism of the particles in an atom." It is important to remember this when you wonder what the benefits might be for supporting basic science research. Oftentimes the benefits are remarkable and go far beyond what might have been developed from a targeted research program, with a specific technical goal. This remarkable fact has proved to be true time and time again!

NMR is one of the most important experimental methods in chemistry to determine the structure of molecules. One needs to understand what each of the three words mean to understand the process. Let's begin with a short discussion before developing the ideas more fully. It can be tiring to keep saying things like "the effective needle that represents the current loop" over and over again. So physicists have come up with a shorthand language. We call the effective needle the *spin* of the particle. Many quantum objects, like a silver atom, have a net spin; this means a beam of them splits into more than one beam when placed in a Stern-Gerlach analyzer. For NMR, we focus on the spin of the nucleus, which is a proton for the hydrogen atom.

Our goal is to reach in and control the orientation of the spin of the nucleus. Just pause to think about this for a moment. We are trying to control the direction that the projection of the spin has for the nucleus of the atom, but we cannot touch or handle the nucleus because it's way too small. How can we possibly achieve this? We have seen that when we place a current loop in a magnetic field, it behaves like an effective needle and precesses about the field. So, if we can place the effective needle in a magnetic field such that the field is *perpendicular* to the needle, then the needle will precess about that field and flip its spin!

It turns out that carrying this out is challenging because the current loop precesses about all magnetic fields and is therefore already precessing about the large field we initially placed it in. This means we need to make the perpendicular magnetic field rotate at a rate that matches the rate of the precession of the spin in the original magnetic field. This matching condition is called *resonance*.

With this brief introduction, you are now ready to explore the process in detail. Recall that effective needle that represents the current loop (the spin) precesses in a magnetic field. We apply a large uniform magnetic field in the vertical direction, causing the spins to all precess. Since we want the spin to flip, and the only

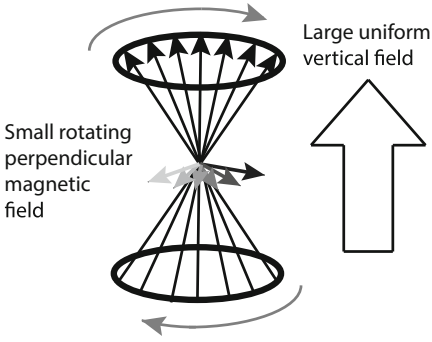

Fig. 2.32 Rotating perpendicular magnetic field (central plane of the figure). The arrows representing the effective magnetic needles of the current loop (spin) precess in the large magnetic field (in the vertical $+z$ direction). If we apply an additional perpendicular magnetic field that rotates at exactly the same rate as the precession, then the spin sees a perpendicular field that always points in the same direction in a frame that rotates at the rate of precession. The spin will then precess about that perpendicular field and eventually flip

way we get the spin direction to change is by precession, we need to have the spin precess about a direction in the x-y plane (see Fig. 2.32). If we wait long enough, the spin will precess all the way around so that it has a negative projection on the z-axis. It will keep going, so if we wait twice the time interval, it will have a positive projection again. This periodic motion, where the spin changes its projection continuously from a positive to a negative and back, is called Rabi flopping (after Isidor Rabi). Note that if we stop precisely when the spin has gone from a positive projection to a negative one, we have achieved our goal of flipping the spin.

Our requirement is that the rate of rotation of the perpendicular field *exactly* matches the precession rate. This condition is called resonance. If it does not exactly match, the direction of the perpendicular magnetic field is changing in the rest frame of the precessing spin (this is the frame that rotates at the precession rate). Once it has rotated by 180° in that rotating frame, it then tries to make the spin precess in the opposite direction, and we will never achieve the flipping of the spin. In fact, because it is constantly changing direction, the spin never comes too close to precessing about anything and just dips down and up a little bit, moving with a low amplitude wobble.

When we achieve the flip of the spin, this really is a remarkable achievement. Just by applying some different magnetic fields, we are able to change the projection of the current loop. We are controlling the spin degree of freedom of the nucleus of hydrogen in the sample. I always find this fact to be completely mind-blowing! So, how do we apply a rotating magnetic field? This is done with radio waves, which have a magnetic field that varies in time. A circularly polarized radio wave has a magnetic field that rotates. We simply need to find the radio wave with the right frequency to match the resonance. We describe how this is done below.

One of the most useful applications of nuclear magnetic resonance is the chemical shift (a change in the resonance condition due to the local chemical environment in a molecule). To see the chemical shift, we need to have a very uniform field and we need to be able to vary the frequency of the radio wave to a very high precision (fractions of one part in a million). But, if we can do this, we discover that the molecule has additional magnetic fields, which we call local fields, surrounding the different hydrogen atoms in the molecule. The nuclear spin responds to the total magnetic field, given by the uniform field plus the local field. As the local fields vary, the frequency for resonance shifts ever so slightly (typically in the part per million range). This shift yields a chemical fingerprint that is unique to different molecules. We illustrate this with one of the first molecules examined in this way—the molecule ethanol, or grain alcohol (a molecule near and dear to many of us).

Ethanol has the chemical formula C_2OH_6 and is represented in a ball and stick model and in an abstract schematic in Fig. 2.33. One should not take the representation too literally. The schematic just shows what atoms are neighbors of each other. The ball and stick model does a little better, representing the geometry of the molecule, with accurate interatomic spacings. The molecule has three different types of hydrogens. On the left, three hydrogens (in group 1) form a triangular

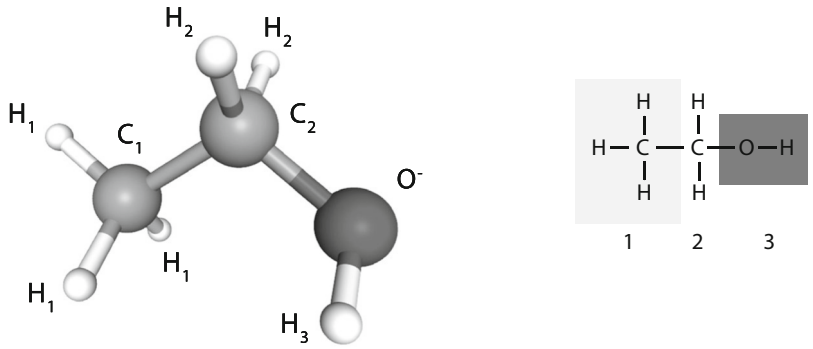

Fig. 2.33 Abstract representations of the ethanol molecule C_2OH_6. (Left) A ball and stick model for ethanol (from the tools at pubchem.ncbi.nlm.nih.gov). The white spheres are hydrogen (grouped into three different groups), the gray spheres are carbon, and the darkest sphere is the oxygen ion (with one extra electron). (Right) A schematic bond illustration emphasizing the three different hydrogen groups in the molecule

pyramidal base for one of the carbons. In the middle of the molecule, two hydrogens bind directly to the other carbon (group 2). Finally, on the right end, the oxygen (which is charged to O^-) bonds to the last hydrogen (group 3) in what is called a *hydrogen bond*.

In the top panel of Fig. 2.34, we see a low-resolution NMR spectroscopy of ethanol, which shows the three different types of hydrogen atoms in the molecule (Blinder, 2011); it is the same resolution of the spectra as was originally measured (Arnold & Packard, 1951). The first and second group peaks always appear at the locations shown. The third group, associated with the hydrogen off the oxygen as a tail on the right side of the molecule, has a peak that depends very strongly on impurities and the solvent that ethanol is in. It tends to move around and can be located at the left edge or in between the other two peaks, depending on its specific environment. In the lower panel, we see a high-resolution image. The group 1 hydrogens split into three subpeaks, called a triplet, because their spins interact with the spins of the hydrogens in the second group, and there are two of them (we will later learn that two hydrogen spins combine to form a spin-one system that has three states, hence the name triplet). The hydrogens in group 2 split into four subpeaks, because they interact with the three hydrogens in group 1 (we will later learn that three hydrogen spins form a spin-3/2, which has four states). Both the large structure of the peaks and the subpeak structure are called the chemical shift. The subpeak structure is particularly important, because it tells us about the chemical bonding in the molecule. Experts in NMR use these types of rules to determine the structure of complex molecules from the NMR signals. It is amazing how accurate this spectroscopy can be.

There is another important method used within NMR, which is primarily a measurement technique used to determine the two decay times of the NMR signal.

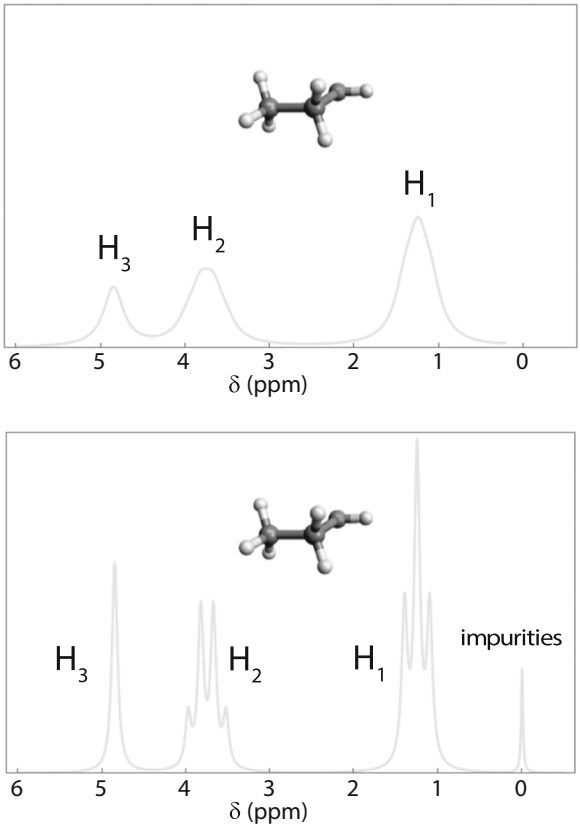

Fig. 2.34 (Top) Original NMR spectra of ethanol showing the chemical shift with three different types of hydrogen groups. (Bottom) Modern NMR spectra of ethanol showing the fine structure of the chemical shift, as discussed in the main text (Blinder, 2011). The horizontal axis measures the shift in the resonance frequency in parts per million of what the average resonance frequency is

The technique is called spin echo, and it was discovered by Erwin Hahn in 1950. We will first explain the procedure and then describe in more detail why it works.

When we discussed the chemical shift, we made it clear that the resonance frequency for different hydrogen nuclei was different, depending on its local chemical environment. This observation brings to light that there are many local magnetic fields near the nucleus, and the resonance frequency, or the rate of precession, depends on what the sum of these local fields are. Normally, we cannot control the level of impurities in our sample, so the local fields become difficult to work with—generically, they make running NMR experiments more difficult. But then in 1950, Erwin Hahn discovered a way to get rid of all of these effects using a clever protocol called a spin echo (see Fig. 2.35) (Hahn, 1950). The experiment starts with the spins aligned to all have a positive projection along the z-axis. In (a) this is illustrated by the white cap, indicating the region where the spins precess around the vertical axis. We then rotate by 90° about the y-axis, to bring the spins down into the $x-y$ plane. We let them evolve for some time (b–c–d) until they have splayed out a bit, because the local magnetic fields are not the same for all spins. The faster spins have moved ahead of the slower spins. In (d), we rotate by

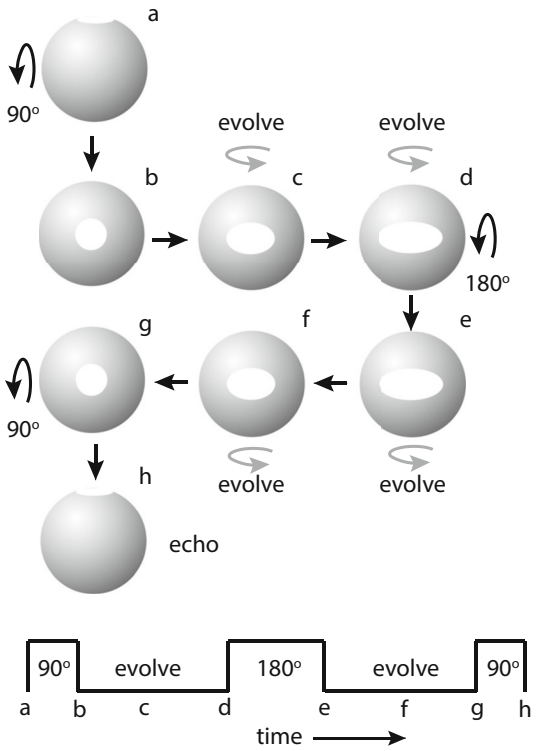

Fig. 2.35 Spin-echo sequence: (**a**) the spins start oriented along the uniform field (represented by the white "cap"); (**b**) we rotate the spins by 90° to put them in the $x - y$ plane; (**c**) we evolve for some time (spins in a larger local field evolve faster, those in a slower field evolve slower, so that they splay out in time; the image shows the spread after multiple complete precessions about the vertical axis); (**d**) we rotate the spins by 180° to invert the order of fast versus slow—now the spins are on the backside of the sphere; (**e**) and (**f**) we evolve the same amount of time as in (**b–c–d**); (**g**) the spins realign by reversing the splaying, we then rotate the spins back by 90° to put them along the magnetic field; and (**h**) we now are ready to measure the absorption of radio waves to determine the density of spins polarized in the up direction. The pulse sequence is indicated below

180° about the y-axis to flip the order, so that the fastest spins are now behind the slowest spins. In (e–f–g), we evolve the spins (by precession) at the same time as in (b–c–d). Then in (g), we rotate by $-90°$ to rotate back on the positive vertical axis. The system is now ready, in (h), to have the number of spins that are up be measured by the NMR absorption spectroscopy. You can see how clever this idea is—by making the 180° rotation, we have reversed the order from slowest to fastest back to fastest to slowest, so that the spins all come together at some later time (see Fig. 2.36). The cleverness of this is it works even if we do not know what the local fields are, because the splaying motion is essentially reversed in the second half of the sequence. The characteristic time that it takes to splay out is called the T_2 time.

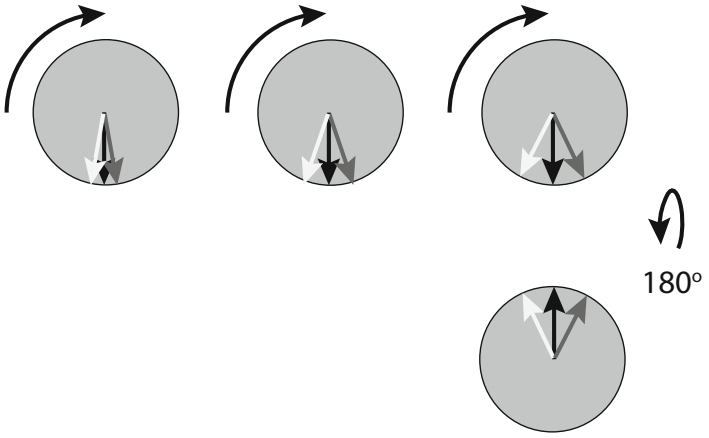

Fig. 2.36 An instantaneous snapshot of the splayed spins at three different times (looking down from above onto the $x-y$ plane; each snapshot is after an integer number of complete precessions for the black arrow). The fastest precessing spins in gray lie ahead of the slowest precessing spins white). The 180° pulse flips the ordering so that the fastest spins (dark gray) are now behind the slowest spins (white). Evolving for the same time now brings all the spins back together again, because the faster spins still precess faster than the slowest spins (not shown)

Obviously, if there is a T_2 time, there must also be a T_1 time. What is that? Well, we gave a little too simplified an explanation of the spreading of the spins in the spin-echo sequence. We only showed the angular spreading in the $x-y$ plane. But they also spread vertically. This occurs because local fields can point in any direction and because the spins are lowest energy when pointing upward (in the direction of the large uniform field), so any energy dissipation will tend to align the spins more vertically. The T_1 time is the characteristic time it takes for the average of the vertical spin to decrease toward zero. It is measured in the spin-echo sequence by extending the time for evolution over (b–c–d) and (e–f–g) to longer and longer times. One might have thought that these two characteristic times should be the same, but they are not always—in solids, the material can have different behaviors in different directions, but in liquids, where the system is spatially homogeneous, we expect the two relaxation times to be equal—indeed, they tend to be equal in liquids. The video Nuclear Magnetic Resonance https://tinyurl.com/56nnukrt describes all of these ideas.

We end the chapter with a discussion about how magnetic resonance imaging works. The basic idea is simple, once you see it. Previously, we explored the reasoning for why we want to have a uniform large vertical magnetic field—namely, since the field governs the rate of precession of the spins of the hydrogen nuclei, we want them all to precess at the same rate to have as sharp (and strong) an NMR absorption line as we can have (this is the jargon for the absorption of radio waves when a spin flips). When we have enough resolution, we can see the chemical shift, which differentiates slightly different precession rates depending on different

chemical environments. Now, for imaging, we take this very uniform field and we add to it a linearly increasing field. This will give a magnetic field profile like our old friend the fat arrows that are tapered at the bottom and increase to the top.

At first, this looks like a bad thing to do because we will have the hydrogen spins precessing at different rates through the sample. But what is important to note is that unlike a poorly controlled nonuniform magnetic field, where the different precession rates will appear almost randomly through the sample, *here we have a correlation between the rate of precession and the vertical height in the sample* because the field varies in a simple way through the 3d space encompassed by the sample. Hence, by observing the strength (amplitude) of the absorption as a function of the frequency of the radio wave, we can measure the density of the hydrogen as a function of position! This is the heart of how we use NMR for imaging.

In an actual machine, the process takes place in a few different steps. We reorient the large uniform field to take slices through the sample in different directions, and then the data is processed by computer-based-tomography algorithms to construct three-dimensional plots of the density of water in the sample. This is in essence how an NMR machine works. Experts will note that the machines usually use the spin-echo sequence discussed above to measure either the T_1 or the T_2 times which correspond to the relaxation from perpendicular to the uniform field to parallel to the uniform field, or to the rate of splaying of the spins at different locations. These images are called T_1 contrast or T_2 contrast images. The magnet used in the MRI machine is usually a superconducting magnet, which requires a closed, cylinder shape, but recently permanent magnet versions, which are more open, have become available. Functional MRI corresponds to taking images when the patient is doing something. For example, you could have an image taken of your brain while you are counting or thinking about shooting a basketball or becoming angry. The MRI machine can measure where the blood flow goes, or if it measures other nuclei, such as phosphorus; it can measure metabolism in the brain. It is remarkable how far the science has already gone; the future is likely to lead to even more remarkable advances. I cannot wait to see what they will be. The video Hey Doc, the MRI Uses Quantum Mechanics https://tinyurl.com/2f6mdv6r further how the MRI machines work.

I hope you now appreciate the complexity of an MRI machine, why quantum mechanics is critical for its operation, and how its development required a series of discoveries in a wide range of different fields before the technology could be envisioned and finally developed. These discoveries were from basic science research, which often has no immediate application to a commercial device, although eventually they all did. The MRI machine is not unique, but is more paradigmatic of the technology developments that we enjoy and often take for granted every day. It is important to think about this from time to time. I find I am always awed and amazed by how sophisticated technology is developed and what the underlying science was that had to be developed first. There is a tale that Michael Faraday (one of the most skilled experts in experimental electromagnetism in the early nineteenth century) was asked by the Chancellor of the Exchequer of England about what electricity would ever be good for. Faraday is said to have replied, “I do

not know, sir, but I suspect that one day you will be able to tax it." It turns out there is no solid evidence that he actually said this, but the spirit of the quote does ring true. We often find science reaps rewards far beyond anyone's initial expectations.

If you have a further interest in NMR, MRI, or the amazing images they produce, there are many resources on the Web that you can search to find further information. Just be careful that many of those sites are written by people who do not understand quantum mechanics as well as you do, and hence they often say incorrect things. You may want to investigate interesting newer developments like functional MRI (which shows what parts of your brain are involved in particular thought processes) that can image things like emotions! Or you may want to investigate some issues like why they recommend you not wear makeup or deodorant when having an MRI taken.

Section 2.4 Problems:

2.4.1 Resonance

Nuclear magnetic resonance involves two magnetic fields: one is a large uniform field and the other is a perpendicular rotating field. Explain what the resonance condition is and why it is needed for this experiment.

2.4.2 The Rotating Field Is On

We set up a nuclear magnetic resonance experiment, where we keep the perpendicular rotating field on for all times. You can assume the perpendicular field is chosen to satisfy the resonance condition.

Explain what happens to the spin as a function of time?

2.4.3 Uniform Field

We must have a large uniform magnetic field for an NMR experiment. This is usually supplied from a superconducting magnet in most devices.

Why is it that we need a uniform field? Think about the fact that a sample is three dimensional and think about the requirements to see the chemical shift.

2.4.4 When to Stop?

We run an NMR experiment and we want to apply the perpendicular field just long enough that a spin that was originally up will now be down (this is called a π-pulse).

How do we determine the duration of the π-pulse? Is it related to the precession rate in the large uniform field or to something else?

2.4.5 Chemical Shift

What happens to the chemical shift if the vertical (large) field becomes more and more nonuniform?

2.4.6 Perturbing Field

When we described the way to image with NMR, we introduced a linearly increasing magnetic field. Must it increase linearly to be able to produce the image, or would some other dependence work just as well?

2.4.7 Blood Flow

The hydrogen in blood will flow in blood vessels during the time of the experiment. Does the fact that it moves have an effect on our ability to image blood flow in an MRI?

2.4.8 Explaining the MRI Machine to Others

An MRI machine has two magnetic fields, one that does not change in time and one that does. Write an essay that discusses the following topics to build up to an explanation of how an MRI machine works:

(1) What happens to the nuclear spin when it is placed in the large static field?
(2) How fast does the rotating magnetic field have to rotate for something interesting to happen?
(3) Describe the interesting thing that happens.
(4) Discuss what would happen if the MRI machine used a uniform static field and explain why it instead uses a linearly varying magnetic field to do imaging.
(5) Summarize what an MRI machine actually measures when it produces an image.

Be sure to write your essay in the fashion that a peer can understand it, so refrain from using jargon, if possible, and use clear and concise language. If it helps, you can make some sketches of figures that would go with the essay and discuss them in the essay.

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Chapter 3
The Quantum Mechanics of Light

For there is always light. If only we're brave enough to see it. If only we're brave enough to be it.

Amanda Gorman, Poet, 2020

Physicists have spent many centuries developing theories for light. It is a problem that was tackled by the best of the best (Newton and Einstein) and many, many others. Understanding the quantum nature of light is complex, but it is something every one of us should understand to be a well-educated individual. It encompasses some of the most bizarre features of quantum mechanics. We begin our encounter with light in this chapter and we provide a video overview of it in https://tinyurl.com/484rsh8c.

3.1 Wave or Particle?

Nothing brings out the question of Wavefunction@Wave-particle duality more clearly than does light. When Newton constructed his original theory, he described light as particles, primarily because it casts sharp lines in shadows and does not curve around objects as waves would. However, he recognized that issues existed with this model, namely, partial reflection of light, which we will get to later (and is a difficult phenomenon to describe with the particle model). Since Newton's views were unquestioned at the time, "light as a particle" ruled the day.

Thomas Young showed that light did have wave properties, just with very small wavelengths, in his famous two-slit experiment which showed light interfering with itself (more about this later). Initially, Newton's particle theory remained strong, but within a decade Fresnel introduced a wave theory of light that Poisson insisted required a bright spot in the center of a shadow cast by a circular object. When

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Arago observed this spot, the wave theory became firmly established. This became the unequivocal proof that light was a wave, and that Newton was wrong. As a result, "light as a wave" ruled the day.

Toward the end of the nineteenth century, it became clear that the simple picture of light as a wave would not work when one studied phenomena like black-body radiation. Black-body radiation is the universal behavior of all objects to glow in color from red to orange to yellow to white as they get hotter. Think about a piece of metal put into a hot fire—it does not matter what it is made of, all metal will start glowing a red-orange color, becoming yellow as it is further heated, and eventually becoming white hot. This phenomenon is the underlying principle for the incandescent light bulb and the photoelectric effect (a truly bizarre quantum effect that we won't be able to discuss in detail right now). Planck used a mathematical trick that was similar to thinking of light as a particle (even though he insisted it was not a particle) to determine a universal law for the colors as a function of the temperature. Einstein took the idea a step further to say light is composed of photons (loosely speaking, a "particle of light," but we will define them much better by the end of the book). A new era was ushered in and "light as a quantum particle" became the law. Experiments could show the particle nature of light or the wave-nature of light depending on the experiment. Somehow, light was both a particle and a wave at the same time!

Perhaps you have heard all of the above before. But have you ever asked yourself, how do I really know what the nature of light is based on my own observations? And if you haven't, can you think of what you might do to determine an answer to this?

Well, it isn't easy. The simplest way I can illustrate this fact to you is a demonstration. Everyone has seen a digital camera before (yes, the one in your smartphone counts as a digital camera). They work with photon detectors in them called charged coupled devices, or CCDs. Expensive CCDs can be triggered when a single photon hits them (your eyes are almost as sensitive, but require a few photons to hit each cell before they fire). We demonstrate the particle nature of light in Fig. 3.1. We use these detectors of light that are so sensitive that they can detect a single quantum particle of light. If we shine very dim light on such a detector, the result looks a lot like how raindrops hit the pavement—all drops are about the same size, but they are discrete and hit in random locations until they cover the street and make it uniformly wet. Same thing with photons. Initially, we have them hit in random locations. Only after many have hit, do we start to see some kind of pattern forming (see Fig. 3.1).

Some of the photons in Fig. 3.1 appear very far from the center of the screen. Because this pattern has no significant wavelike properties, we call it a *particle pattern* or a *classical pattern*. It is similar to the pattern you would see if you sprayed sand onto a screen covered with glue. But remember, these photons are quantum particles. Note that while we can predict the final pattern after many photons hit, we can never tell exactly where a given photon will hit next. This notion of randomness, and the idea that we can only determine the probabilities of events, is one of the most fundamental tenets of quantum mechanics. It underpins everything in the quantum-mechanical explanation of the world. You have already seen this concept in action numerous times already, and this is yet another example. We can predict

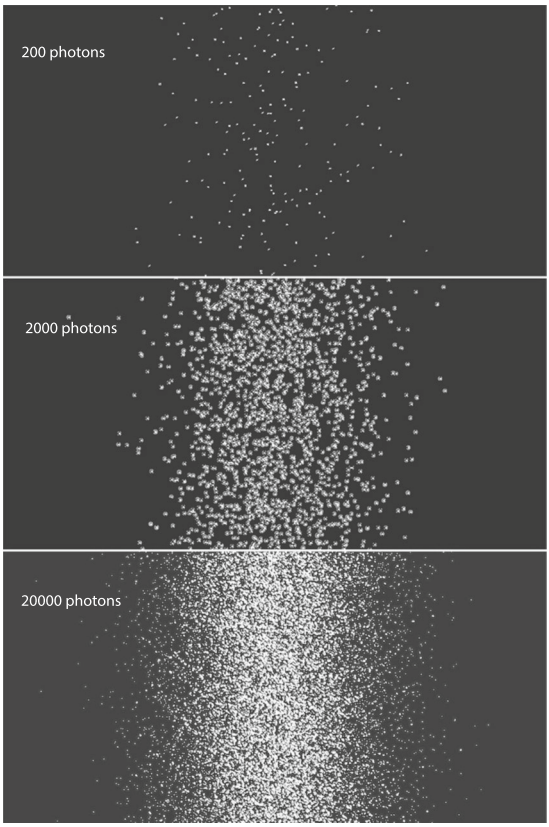

Fig. 3.1 Simulation of photons shone through a single thin slit (the dot is where each individual photon hit the detector screen). (Top) 200 photons: the pattern appears completely random. (Middle) 2000 photons: the pattern is starting to emerge but is not clear. (Bottom) 20,000 photons: one can see the single-slit pattern forming

the probability for a given event to occur, but we cannot predict precisely what the result of a given experiment will be. The University of Vienna has prepared a video https://tinyurl.com/223x3fxm showing how a single photon arrives at a screen one by one and then develops a pattern when all hits are shown together. We also have a simulation https://tinyurl.com/yvvdc22x that allows you to create these patterns yourself.

If you are an expert in optics already, you might know that light traveling through a single slit does display a wavelike interference pattern; it is called single-slit diffraction. These tend to occur near the edges of the pattern, far to the right or far to the left on the screen, and we are ignoring them relative to the overall broad behavior of this particle pattern. Be sure to keep in your mind that there are interference effects even for a single slit as you continue through the chapter. It makes much of what we discuss in this chapter even more bizarre and exciting!

Quantum mechanics requires us to use probability. And these photon experiments directly lead us to ask the question, how many shots of an experiment must be run before observers can determine the general properties of the results? This is an old question that was first answered back in the early 1900s. The Guinness Brewing Company hired William Gosset, who performed research and developed

the statistical rules that relate experiments with small numbers of trials to what would be observed had the experiment been repeated many times. He worked with a famous Irish statistician, Karl Pearson, while on a sabbatical from Guinness, and published the so-called Student's t-test, which worked out the theory for such problems. He had to publish under the pseudonym "Student," so other breweries would not recognize the advantage that his analysis gave to Guinness (he viewed himself as Pearson's student, hence this appropriate pseudonym). Guinness was one of the first companies to engage in industrial-based research and allow their scientists to go on sabbaticals.

But, I digress. Back to quantum mechanics. Richard Feynman devised a simple set of rules for how photons act in different experiments. These rules essentially describe the path-integral method for describing the quantum world. In this section, we begin the development of this theory along the lines described in Feynman's book *QED: The strange theory of light and matter* (Feynman, 1985). Our theory of light follows Feynman's approach quite closely—Feynman goes on to discuss the high-energy physics description of quantum electrodynamics, which we do not cover in this chapter—consult his book if you want details on that. Nothing I will tell you is incorrect, watered down, or simplified. You will be told everything about quantum mechanics the way practicing physicists view quantum mechanics. What you won't be taught in this section is how to solve quantum mechanics problems *efficiently*. That ordinarily takes a number of years of training and sophisticated mathematics. But, this book is not an ordinary one. Within it, I show you how you can understand quantum mechanics as well as many practicing physicists understand it! You will even learn how to do the more complicated calculations with a minimal of math background.

We begin by explaining the different quantum rules and how to apply them. Don't worry if you cannot follow everything at once. We will return to these rules many times until you know them well. But pay particular attention to the notion of an event (defined by the specific measurable initial and final conditions of sources and detectors) and of the alternative ways in which an event can occur (defined by the different ways one can "move" through the apparatus for a specific event). Developing the ability to distinguish these two concepts is critical for success. Note that we use arrows and rules for how the arrows change (rotate, shrink, get added together) to develop theoretical predictions for quantum behavior. This approach is abstract and needs to be thought of as a cookbook that tells you what to do in any given circumstance. At this stage, focus on where the rules lead us as opposed to where the rules come from. We will have more to say about their origin later in the book.

The quantum rules are as follows for a general problem to be solved in this fashion:

(i) Identify all different *events* for a given experiment.
(ii) Find all *alternative ways* that the event can occur.
(iii) For each alternative way, use the quantum rules to assign an *amplitude arrow*.
(iv) Hook arrows head to tail to construct the *final arrow* for the event.

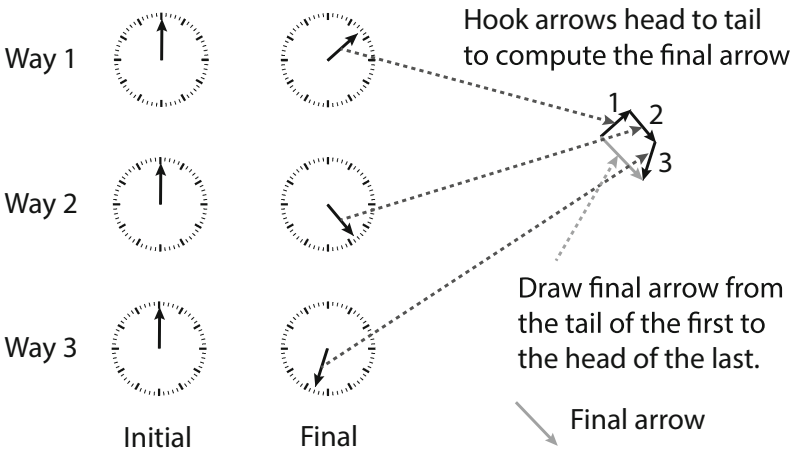

Fig. 3.2 Construction used to combine arrows together via addition and determine the final arrow. We first determine the different final arrows for each alternative way. Starting from the first arrow, we hook them "head to tail" until we have added arrows for each alternative way; note that we cannot change the magnitude or the direction of the arrows when we hook them together. Next, we draw a final arrow from tail of the first arrow to the head of the last arrow (depicted in gray). The square of the length of this arrow is the probability for the event to occur

(v) Square the length of the final arrow to determine the *probability* that the given event will occur.
(vi) Add up the probabilities for each event if you are interested in the *total probability* for a series of events to occur.

Let's parse these different rules out to make sure we understand them. They may appear quite abstract at first, but they will become quite concrete shortly.

An event is defined by the initial state of the system and the final state of the system. This is governed by the sources of light and the detectors of light that we use in the experiment. It also depends on which detectors register the detection of a photon. The alternative ways an event can occur are determined by the different paths a photon can take starting from the source and ending with the specified detectors measuring photons. Our methodology involves determining arrows for each alternative way. This requires us to follow specific quantum rules that we will describe next. Those rules determine an amplitude arrow (which has a definite length and a definite direction) on a two-dimensional plane. Once the amplitude arrows for each alternative way are determined, we hook them head to tail to determine the final amplitude arrow, as depicted in Fig. 3.2.

Now we need to concretely state the rules for light, which we will be using throughout this and the next chapter.

(i) For each alternative way, the arrow starts with unit magnitude and pointing at 12 o'clock at the moment the photon leaves the single-color (monochromatic) light source.

(ii) As the photon travels through air or glass, the arrow rotates at a rate given by the color of the light and the medium it is moving through. For example, in air, red light rotates at 36,000 times per inch that the photon moves; in glass, the photon moves more slowly, so the same color light rotates more times per inch.
(iii) If the photon is transmitted from air to glass (or from glass to air), *shrink* the length of the arrow by a factor of 0.98.
(iv) If the photon is reflected off a glass-air interface, coming from air, then shrink the length of the arrow by a factor of 0.2 and add 6 hours to the clock.
(v) If the photon is reflected from a glass-air interface coming from glass, then shrink the length of the arrow by a factor of 0.2, but do not add any hours to the clock.

Yes, it appears that these rules come out of thin air, but we will see that we can use them to describe quite complex experiments. Please take them as postulates for now. We will describe the origin of these rules in due time. The videos Rule, Rules, Rules, Those Quantum Rules Part I https://tinyurl.com/4rpezvwe and Part II https://tinyurl.com/2zaxwk6m provide some more context for you.

We are going to apply these rules to describe a simple experiment—shine a laser directly onto a photon detector and see if it detects the light. This experiment and the quantum analysis using our rules is depicted in Fig. 3.3. There is one event, the

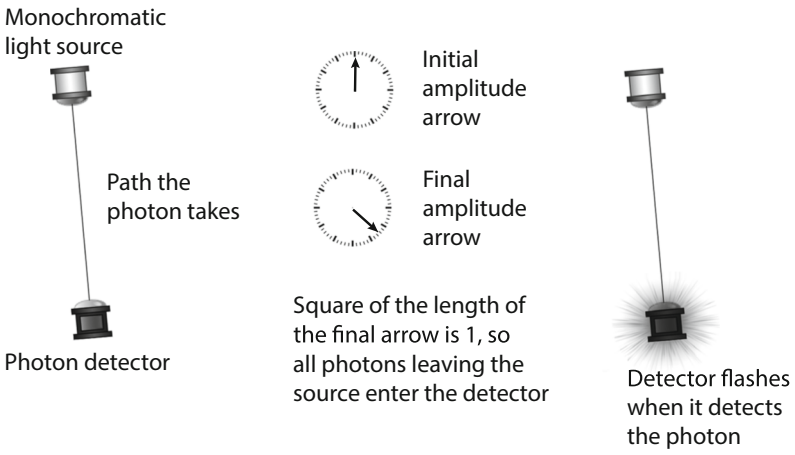

Fig. 3.3 Light leaves the source and enters the detector. The experimental setup is illustrated on the left, where we have a monochromatic light source at the top and a photon detector on the bottom. The line shows the path the photon takes from the source to the detector (there is only one path and hence one alternative way for the event to occur in this example). In the middle top, we show the initial amplitude arrow, which has unit length and points at 12 o'clock. As the photon leaves the source and moves to the detector, the stopwatch hand for the clock revolves at the rate given by the color of the light. It will wind around many, many times and then end at some direction, which is the final arrow, shown on the bottom of the middle of the figure. The length remains 1, so when we square it we find the probability is one to go from the source to the detector. Hence, all photons that leave the source are detected at the detector. In the rightmost panel, we show what this detection looks like (the detector flashes)

photon leaves the source and is detected in the detector. There is only one alternative way for this event to occur: we leave the source and enter the detector by going along a straight-line path from source to detector. The initial arrow has length 1 and is located at 12 o'clock. The photon moves through air from the source to the detector. Our rules tell us that the arrow rotates at a rate determined by the color of light leaving the source as it moves along the path. When it arrives at the source it is still length 1 and points in the direction given by how far the stopwatch hand has rotated. We square the length of the arrow and find a probability of one to go from source to detector. This should come as no surprise. The experiment was a simple one. Note that we use the clock to track how the arrow winds; this becomes particularly useful when we have more than one alternative way to go from source to detector. A video https://tinyurl.com/yrh5aycc introduces you to the light tutorials, including the first tutorial https://bit.ly/4iWdDxl.

We now discuss a much more complex experiment—the phenomena of partial reflection of light. One of the problems with Newton's theory of "light as particles" is that light partially reflects off transparent objects. Everyone has seen this phenomenon for themselves. During the day, we can look through windows, showing us that light is transmitted through glass. But at night, if we look at a window, we see our reflection in it instead. Hence, light also reflects. Does light know that it is day or night? Of course not! The partial reflection is always there. It is just hard to see during the day when it is overwhelmed by the transmitted light from outside. But if you look closely, you can see it. I am often reminded of this is when I let the car wash apply *Armour All* to my dashboard. The shiny dashboard reflects in the windshield and can be distracting while driving, even in daylight. The problem with the Newton's simple particle theory of light is, how does the particle know to transmit or reflect when it hits the surface?

Next up, we consider a slightly more complicated experiment, one where the detector sits inside the glass (such an experiment is contrived, but it is a useful middle step toward learning about partial reflection). There are two events: (i) the photon leaves the source and is detected in the detector or (ii) the photon leaves the source and is not detected in the detector. We focus on the first event, which can occur only one way—yes, we could have reflection instead of transmission, but in that case the detector does not fire, so it is a different event, not an alternative way an event can occur. In Fig. 3.4, we show the experiment in three different snapshots to help us determine the arrow associated with the one alternative way and we have a tutorial https://bit.ly/3KXED2T that shows all of the details. The photon (black dot) starts at the source and our clock hand is at 12 o'clock with an arrow length of 1. As the photon moves from the source to the air-glass interface, the arrow winds according to the clock hand (with the rate determined by the color of the light). At the interface, the arrow shrinks in length from 1 to 0.98. Then as it travels through glass to the detector, it winds with the clock. Note that the photon travels slower in the glass than in air, so the clock hand winds more in glass than in air. Finally, when it reaches the detector, the detector flashes as it detects the light. Note that the probability for this to occur is now 96%, meaning some of the light must have been reflected (the final arrow has length 0.98 and $0.98 \times 0.98 = 0.96$). We did not yet

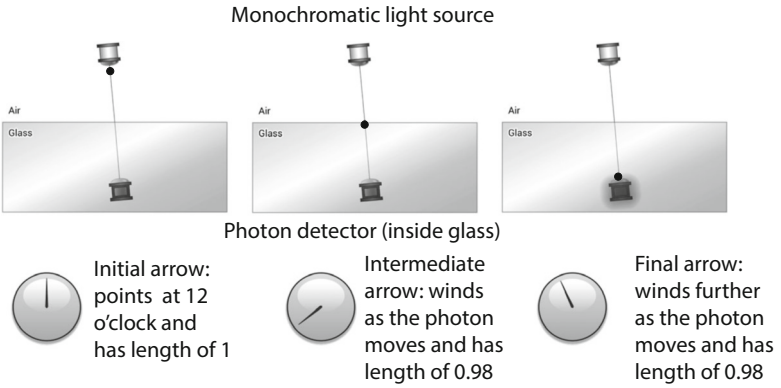

Fig. 3.4 Light leaves the monochromatic light source (top) and is detected with a detector inside glass (middle). Using our rules, we start with the arrow of length 1 and at 12 o'clock (left). As the photon (black circle) reaches the air-glass interface, after turning as recorded on the clock (bottom) during the time it moved through air, it shrinks to a length of 0.98 (middle). Finally, the arrow winds as the photon moves to the detector, as recorded on the clock hand, and ends with a length of 0.98 (right)

examine the probability to be reflected in the only other possible event (but it must arise from all photons *not* transmitted, so it is 4%). Feel free to use our quantum rules to find the arrow for the one alternative way that this other event can occur, and confirm the 4% probability for reflection.

We are now ready for the full experiment. In this case, we have a detector situated near the source on the same side of the glass to detect the reflected light. Our theory is designed for light that comes in at normal incidence, meaning it moves along a line perpendicular to the surface of the glass. But, we cannot have a source and a detector in the same place, so for our diagrams, we will draw the light coming in at a small angle. Please keep in mind this is done as a visual aid, and all the experiments we describe correspond to *normal incidence*.

Let us carefully describe the behavior in Fig. 3.5. We analyze both panels together. There is one event given by the photon leaving the source and being detected in the detector. This event can occur in one of two alternative ways—reflect off the top surface of the glass or reflect off the bottom surface of the glass. For the first alternative, reflecting off the top surface, we start with an arrow of length 1 and pointing at 12 o'clock at the source. As it moves through air, the clock hand winds. When it reaches the air-glass interface, it reflects from air off glass. This requires us to shrink the arrow by a factor of 0.2 (so its length is now 0.2) and to add 6 hours to the clock. Then as it moves from the surface to the detector, it winds again (by the same amount it wound as it moved from source to the surface, since the distance traveled is the same). It ends with a clock hand pointing at about 3:30, indicated by the leftmost clock on both panels. The arrow for this alternative way does not depend on the thickness of the glass, so it is the same for both panels (be sure you understand why). This arrow is copied to the computation panel and is labeled as "top."

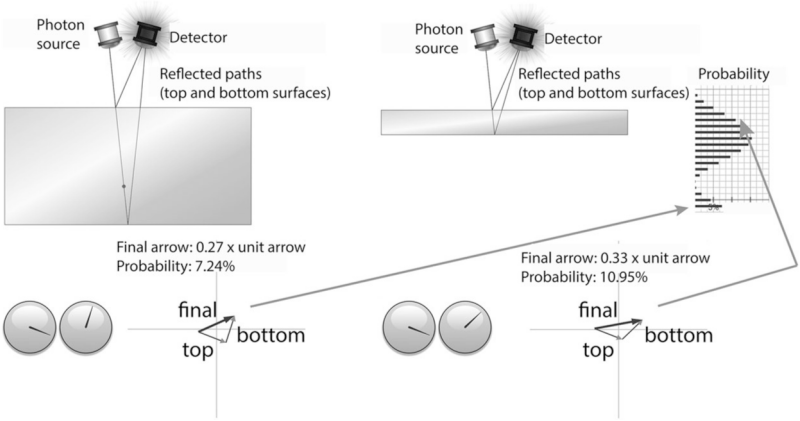

Fig. 3.5 Detecting partial reflection from above the surface of the glass. (Left) a thick slab of glass and (right) a thin slab of glass. The event is a photon leaves the source and is detected in the detector. There are two alternative ways to get there—reflect off the top surface or off the bottom surface. The panels show the results of the arrows for both cases and the combined final arrow. Its square is also shown, giving the probability for reflection, which is plotted as a function of thickness (thinnest at the top) in the rightmost panel

Now, we calculate the second alternative, reflecting off the bottom surface. We start again at the source with a length of 1 and pointing at 12 o'clock. When we get to the air-glass interface, the arrow has wound by the same amount as it wound for the case of reflection off the top at this stage of the process. Now, we transmit, so we multiply the arrow by the factor 0.98, creating an arrow of length 0.98. Next, we move through the glass until we reach the glass-air interface. Here, the arrow winds according to the distance traveled. The photon travels farther through the thicker glass than the thinner glass—here the arrow on the left wound fully around the clock and then some more relative to the thin glass case. At the bottom surface, we shrink by a factor of 0.2, so the arrow length is now $0.2 \times 0.98 = 0.196$. We travel back up to the top air-glass interface, where the arrow winds again the same amount that it did on the downward path through the glass. Now we transmit through the surface, which multiplies the arrow length by 0.98 to become $0.98 \times 0.196 = 0.192$. Finally, the arrow winds as it moves to the detector, winding the same amount as it did on the downward path in the air. The arrow for the bottom is then copied into the computation panel, with its tail placed at the head of the top arrow. The final arrow is also depicted in those panels and squaring its length gives the probability for reflection from this slab of glass. In this case, we see that the thicker glass has a smaller probability to reflect than the thinner glass. The graph on the right shows how the probability of reflection depends on the thickness of the glass. The results for the two cases we have been discussing are explicitly pointed out. We have a tutorial https://bit.ly/3Kvt3vQ that shows more details of how this works.

One can clearly see that the probability for reflection is oscillating as a function of thickness. Why? Let us see if our model explains this phenomena. Recall that the arrow for reflection off the top is fixed to always be the same length (0.2) and pointing in the same direction, regardless of the thickness of the slab of glass (we prepare our experiment so that the top surface of the glass is always the same distance from the source and detector). So, the thickness of the glass simply determines the direction of the second arrow (its length is always 0.192). Now, using these facts, we can see the minimal size of the arrow occurs when the two arrows anti-align (and is of length 0.008), while the maximal size is when they align (and is 0.392). The probabilities must be bounded to run between these two values, which is what we see in the figure. Furthermore, if we slowly increase the thickness, the arrow for the second alternative way slowly winds around more and more. The one remaining question is where do the two arrows point, relative to each other when the thickness approaches zero thickness? (This is a question of *limits*, so it requires careful analysis.) It is one of your problems at the end of the next section.

We have already come across some weirdness. When we looked at light reflecting off a single surface of glass, we found exactly 4% reflect. But with two surfaces, we seem to get a range that runs approximately from 0 to 16%. This must be arising from an interference effect, but it is just plain weird. We will be continuing the discussion of the partial reflection of light in the next section to understand more about this complex behavior.

The work we have completed so far has a nice application to the real world. If you have glasses, or if you use professional quality camera lenses, you may notice that it is difficult to see any partial reflection coming from the surface. This is because many of these lenses are coated with an antireflective coating, which sharply reduces the reflection. These antireflective coatings are constructed from thin layers of transparent material that slows down the speed of photons similar to the way glass does. The thickness can be tuned to work for one color of light, and then to be close to nonreflective for other colors. This is why you can still see a little reflection off the surface of glass that is coated with an antireflective coating. You may even see rainbow colors since some colors reflect more than others!

But, before moving on to the problems, there are a number of points that might need clarification in the material we presented in this section. Let us go through and clarify as many of these as we can. Point 1: what is a source of red photons? That is easy; you have all seen them before. Laser pointers are an excellent source of single color light (also called monochromatic light). If you can bring the intensity down by attenuating the light coming from the laser, you can dim it down to approximately one photon at a time in the experiments. This would then look like the experiment we showed you earlier to illustrate that photons are particles! Note that this dim light is only *approximately* a single-photon source of light, because light tends to bunch together, a phenomenon we will discuss in Chap. 4 (and called the Hong-Ou-Mandel effect). We will not worry about this detail for now since we will treat it in great detail in Chap. 20 when we quantize light. Point 2: what is a photon detector? Also easy. It is the CCD device used in digital cameras or cell phones to detect single photons. A number of years ago, before CCDs, one had to use photomultiplier tubes

to detect photons. We won't discuss them in detail here (but we will later). They have the property of amplifying the signal when one photon strikes them into an avalanche of billions upon billions of electrons traveling through a circuit, which can then be easily detected. Nowadays, CCDs can achieve similar detection efficiencies and are much more convenient. Even more advanced single-photon detectors use superconducting devices tuned close to their superconducting phase transition. Point 3: why does the arrow turn like a stopwatch hand? This is one of our quantum rules. A photon's color is determined by the amount that it rotates for every inch that it moves through air. Blue rotates faster than green which rotates faster than red. The rotation rate varies continuously for all of the colors in the rainbow (and even for ultraviolet or infrared photons beyond the visible spectrum). Of course, we slow the rotation down in our discussions; otherwise, they would be much too fast to comprehend. Point 4: why did the arrow change direction in the experiment for going from a source and being detected in a detector in glass (versus going from the source and being detected by a detector in air) if the distance was the same? In glass, the photon moves more slowly, so it takes more time for the photon to travel the same distance as through air. Hence, the arrow will rotate more as the clock ticks off more time. Point 5: I spent all this time learning probability and now I square arrows. Huh? This is called the *Born rule*, and there is some debate among physicists whether this rule can be derived from something more fundamental, or whether we must take it as a postulate of quantum mechanics. For now, we will take it as a postulate, but we will discuss motivating (or even deriving) it via a simple counting argument in Chap. 13. Remarkably, most of the crazy behavior in quantum mechanics can ultimately be traced back to this rule.

In this section, we learned that light can be thought of as a particle by seeing how it behaved when it was dimmed. We always detect the same "size" photon in each detector, just fewer of them. This tells us that light travels in discrete bundles (photons), all of the same size. But these photons are not like classical particles. They are quantum particles, which possess bizarre properties that we will unveil further in the book. Some questions to think about are as follows: (1) How do photons (quantum particles) interfere with each other to act like waves? Can't I make the interference disappear if I dim the light so much that only one photon at a time is in the experiment? (2) Can a single photon "interfere with itself?" What does that even mean? (3) If light acts like raindrops when they hit a surface, how come I don't feel them hitting my skin? (4) If light is a particle, what does it mean to say it has a wavelength?

I believe the best way to think of this is that the wave characteristics and the particle characteristics emerge as a holistic quantum particle. The two properties are inseparable properties of the whole. The Transnational College of Lex has a beautiful analogy for this in their quantum book (Transnational College of Lex Staff, 2009). Think of the spoken word "photon." The word itself has a particle aspect to it, as the words come in discrete units, just like particles do. But the spoken word is also a sound wave, so it has wave characteristics to it as well. *They cannot be separated!* I cannot have the wave without the particle or vice versa. I always speak the word by displaying both at once, all together. This is how you should think of the

duality of a quantum particle, as *two inseparable parts of the whole*. Do not think of it as "sometimes a particle" and "sometimes a wave," and especially do not think of it as "travels like a wave but is detected like a particle," which is often used to describe photons, but is an incorrect way of thinking about Wavefunction@Wave-particle duality.

Section 3.1 Problems:

3.1.1 How Many Photons?

Take a look at Fig. 3.1, or better, explore the simulation https://bit.ly/4iORtwK. Typically, how many photons does it take to see a pattern appear in a single-photon detection experiment? Do you think this number holds for other types of experiments as well? What determines this number (physicists sometimes discuss this in terms of a visibility)?

3.1.2 How Far Does a Photon Need to Travel?

The clock hand (or arrow) for red light rotates at a rate of 36,000 revolutions per inch that the photon travels in air.

(a) How far does the photon have to move for one full revolution (12 hours) of the clock hand? Report at least four nonzero digits and express your answer in *inches*. (1 inch $= 2.54$ cm)
(b) How far does the photon need to travel for the clock to advance 1 hour? Report your answer in inches.

Think about how these results determine the length scale over which photon interference effects occur for visible light. Does this change for infrared light or ultraviolet light? How so?

3.1.3 How Much More in Glass?

Red light has the clock rotate 36,000 times for each inch the photon travels in air. Note that 1 inch is 0.0254 m, the speed of light in air is approximately 300,000,000 m/s, and light travels at a speed of approximately 200,000,000 m/s in glass.

We have two clocks. Clock A represents a red photon traveling through a thin sheet of glass that is 0.1 inches thick. Clock B represents a red photon traveling the

same distance, 0.1 inches, in air. How many more rotations does Clock A make than Clock B?

Note that the clocks always rotate at the same rate in time; additional rotations occur because it takes more time for a photon to move through glass than air. Your answer should be reported as an integer.

3.1.4 Calculate Those Arrow Lengths!

We considered two alternative ways that the photon could travel from the source to the detector by reflecting off the glass. The first involved reflection off the top surface, while the second corresponded to reflection off the bottom surface. Assuming the initial arrow for each alternative way the event can occur is equal to one, use our quantum rules to calculate the following:

What is the length of the arrow that corresponds to the path for reflection off the top surface? Report your answer to four decimal places.

Assume that the rule for reflection is exact. That is, precisely 4% of the photons reflect off the top surface. Then 96% are transmitted through the surface of the glass. Previously, we used an arrow of length 0.98, with an accuracy of 2 digits in our calculations. Now, and in all remaining calculations in this section, we increase the accuracy to four digits. Compute the length of the arrow for transmission correctly to four decimal places.

Using the number calculated above for transmission, determine the length of the arrow for reflection off the bottom surface, recalling that there are two transmissions and one reflection for that path. Report your answer to four decimal places.

3.1.5 How Much Reflects?

Using the value for the length of the arrow for reflection off the top surface and the length for reflection off the bottom surface, determine the maximum and minimum percentages for partial reflection off the glass as a function of the glass thickness. Consider just these two alternative ways for reflection. Answer to four significant digits.

3.1.6 Explaining the Oscillating Probability

Describe in words why the curve for the probability of reflection as a function of the thickness of glass oscillates. In this experiment, the distance from the source and detector to the top surface of the glass is always the same. The thickness increases by making the glass thicker from below.

To organize your response think about the following elements:

(i) How does the arrow for the alternative way corresponding to reflection off the top surface change as the thickness is increased?
(ii) How does the arrow for the alternative way corresponding to reflection off the bottom surface change as the thickness is increased?
(iii) Using all possible thicknesses, and hence possible results for the arrows corresponding to reflection off the top or bottom surface, determine how the probability varies as a function of the thickness of the glass.

Be sure to write an answer that is clear and in complete sentences. Try to answer both qualitatively and quantitatively.

3.2 Exploring the Quantum Nature of Light

So far, we have only discussed reflection. Now we need to discuss transmission, in order to have a complete story for how partial reflection of light works. But, before jumping into those details, let us remind ourselves of the challenges with the conventional Newtonian particle picture. Could Newtonian particles of light display partial reflection? One way that you might envision this occurring is if the glass has a series of holes in it, like a screen, and the particles of light pass through the holes when they transmit and bounce off the "mesh" when they reflect. Without being able to actually image the structure of glass (at high magnification), it is difficult to explicitly rule such a picture out. Granted, as the glass gets thicker, it becomes difficult to understand how a thicker glass can transmit more than a thinner glass, or how the probability for reflection can oscillate with the thickness (which is what is observed, and which is what we will study in detail below). Furthermore, since glass can be made more reflective by polishing, it is difficult to reconcile that phenomenon with the idea of glass having holes that photons can travel through. So if glass does not have such holes, how does a quantum particle manage to display the phenomenon of partial reflection? The answer is a quantum particle behaves differently from a "billiard ball" particle. In this section, we will determine the explicit theory explaining how the phenomenon of partial reflection of quantum particles takes place.

Now, we focus on how light transmits through the glass. Similar to how we took into account two different alternative ways that light reflected off of the glass in the previous section, we will take into account two alternative ways that light can transmit through glass (see Fig. 3.6). The first way is for light to go straight through. The second way is for light to transmit through the top surface, reflect off the bottom, reflect off the top, and then transmit through the bottom. We will analyze the behavior for transmission by breaking the problem up into a series of steps to determine the probability. You should try to work this out yourself using our quantum rules before reading further. Don't worry if you get stuck. We cover the derivation fully next.

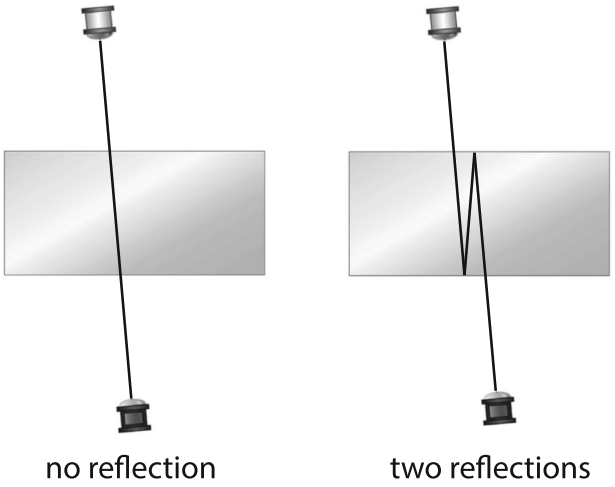

Fig. 3.6 Two alternative ways to transmit through a thin slab of glass. (Left) pure transmission and (right) transmission after two reflections. Recall that the true experiment is at normal incidence, the angles have been distorted to be able to draw the images

In the first alternative way, we start at the source at 12 o'clock and length 1. The clock hand winds as we move from the source to the air-glass interface at the top. Here, the arrow shrinks to 0.9798 (we use our more accurate 4 digit results). It then winds as we move through the glass. Finally, we transmit from glass to air (shrinking by another factor of 0.9798 to a length of 0.96), and then the clock hand winds until we enter the detector. For the second alternative way, we have a winding clock, two transmissions and two reflections, so the arrow length is $0.9798 \times 0.2 \times 0.2 \times 0.9798 = 0.0384$. Both reflections are from glass. The photon travels a distance of twice the thickness of the glass more than the first alternative way. So the arrow winds an additional amount given by how far the clock hand rotates as the photon moves through twice the thickness of the glass. We end up with two arrows. One has length 0.96 and the other has length 0.0384. Their relative orientation is determined by the thickness of glass. You should think about how the two arrows are oriented in the limit as the glass becomes very thin. How thick does the glass need to be for the two arrows to be anti-aligned?

What is clear is that as we continuously increase the thickness of the glass, the arrows both wind around, due to the longer clock times, but the smaller arrow winds faster than the larger arrow because the photon travels a longer distance through the glass. This means the probability will vary continuously between a maximal value (when both arrows point in the same direction) and a minimal value (when both arrows point opposite each other). We can compute the maximal and minimal probabilities. The maximal one is $(0.96 + 0.0384)^2 = 0.9968$ and the minimal one is $(0.96 - 0.0384)^2 = 0.8493$. If you recall, for reflection, it was minimal (with a probability equal to $(0.2 - 0.192)^2 = 0.000064$) when the transmission was

maximal and it was maximal (with a probability equal to $(0.2 + 0.192)^2 = 0.1537$) when the transmission was minimal. The reason why the transmission and reflection are correlated in this fashion is because for both reflection and transmission, the arrow for the second alternative way to reflect or transmit rotates by an additional amount corresponding to the time it takes for the photon to travel *twice* the thickness of the glass (within glass). Be sure you understand this point. *It is important.*

We can immediately see we have a problem. When we add the probability for reflection plus the probability for transmission, we find it is equal to $0.9968 + 0.000064 = 0.996864$ when transmission is maximal and it is equal to $0.8493 + 0.1537 = 1.003$ when reflection is maximal. *The total probability is not equal to one in either case!* This is not an error with accuracy of our calculations. There is something else going on that we will discuss soon. We do want to point out that performing these "sanity checks" of the data and, if they fail, asking why is an important exercise for a quantum mechanic. Take a moment to see if you can figure out what might have gone wrong with our analysis before reading on. This will help you develop independent skills as a problem solver.

Have you discovered what we were missing? If not, maybe think some more about it. Ready for the reveal? It is one of those things that once you see it, it becomes clear, but it can be hard to see it. It is that there are more alternative ways than the two we considered. Actually there is an infinite number of additional ways. They come from additional reflections inside the glass. Recall that our quantum rules are to identify events and then identify all possible alternative ways that an event can occur. For reflection off the glass, we can reflect off the top surface, or off the bottom surface, or we can have a photon which transmits through the top; reflects off the bottom, then off the top, and then off the bottom; and then transmits through to the detector. This involves a more complicated multiple reflection curve whose arrow has a different length, and will point in a different direction (although the direction will be correlated with that of the arrows for other alternative ways). More complex multiple reflection paths are possible as well. The same thing also occurs for transmission. We now guide you through how to calculate these effects that are illustrated in Fig. 3.7.

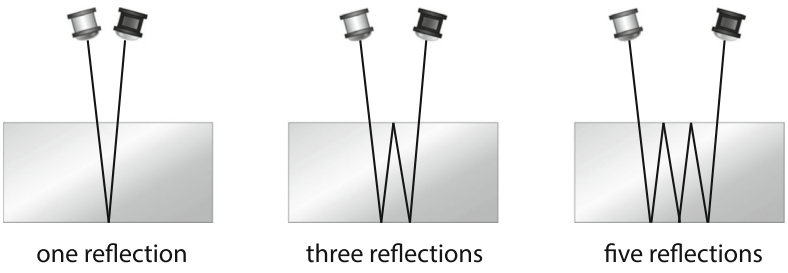

Fig. 3.7 Schematics showing one reflection (left), three reflections (middle), and five reflections (right). Generalizations to 7, 9, 11, and so on reflections are clear

Let us first focus on reflection. Note that there must be an odd number of reflections or the photon will not make it to the detector above the glass slab. The one-reflection curve has a transmission through glass, one reflection, and then a second transmission from glass. The three-reflection curve has the two transmissions and three reflections. The five-reflection curve has two transmissions and five reflections, and so on. We already calculated the length of the one-reflection arrow to be $0.192 = 0.9789 \times 0.2 \times 0.9789$. Each time we add two more reflections, the arrow gets multiplied by $(0.2)^2$. So, the lengths of the arrows become 0.192 (one reflection), 0.00768 (three reflections), 0.0003072 (five reflections), and so on.

So, it should be pretty clear, then, that the formula for the alternative way with $2n+1$ reflections is that the length of the arrow is $0.96 \times (0.2)^{2n+1}$, which obviously gets smaller and smaller as n increases. This means that if we want the calculation to a certain level of accuracy, then it will be safe to neglect arrows that are much smaller than the accuracy we are requiring, and these arrows can be neglected. Now, when we try to compute the probability, we need to know the directions of the arrows as well. Since each additional two reflections will increase the rotation by the same amount, one can see that if the rotation for the one-reflection arrow corresponds to an integer number of rotations, then it is an integer number for all other rotations, and they all will line up in the same direction. The arrow for the reflection off the top will be in the opposite direction, because it is advanced by 6 hours (due to our rule for reflection from air to glass). The total length of all of the arrows that reflect one or multiple times off the bottom will be $0.192 + 0.00768 + 0.0003072 + \cdots \approx 0.19999872$, which seems to be converging to 0.2. Indeed, the full summation is equal to 0.2, which follows from the geometric series (see below). So, when a single transmission through the glass causes the clock hand to advance by 6 hours, adding these two arrows together will get a full cancellation and zero reflection. This is all described in the video Reflections, Reflections, Everywhere https://tinyurl.com/38efnfcp.

We will have a short interlude to discuss the geometric series. This is an incredibly important series to know how to work with. It takes the form

$$S(x) = 1 + x + x^2 + x^3 + x^4 + x^5 + \cdots = \sum_{n=0}^{\infty} x^n. \tag{3.1}$$

It is the sum of increasing powers of x. The second equality shows the abstract representation of a summation, with the Σ symbol being the summation sign. This infinite summation is well-defined if $|x| < 1$.

There is a trick to determining this summation. It is not obvious, so we describe it carefully. We start by examining the sum that includes only N terms, given by

$$S_N(x) = 1 + x + x^2 + \cdots + x^{N-1} = \sum_{n=0}^{N-1} x^n, \tag{3.2}$$

and we multiply it by $1 - x$. We find

$$(1-x)(1+x+x^2+\cdots x^{N-1}) = 1+x+x^2+\cdots+x^{N-1} - (x+x^2+\cdots+x^{N-1}+x^N). \tag{3.3}$$

When we subtract the second line from the first, all terms cancel except the first and the last, so it is equal to $1 - x^N$. This means that we have shown that

$$1+x+x^2+\cdots+x^{N-1} = \frac{1-x^N}{1-x}. \tag{3.4}$$

Now, consider how x^N behaves for large N. If $|x| < 1$, then $\lim_{N\to\infty} x^N = 0$. This is because we keep multiplying by a number whose magnitude is less than 1. Each time we multiply the magnitude gets smaller. Keep repeating and it eventually gets so small, we can neglect it. So we can drop the x^N term when we take the limit of $N \to \infty$ yielding

$$S(x) = 1+x+x^2+x^3+\cdots = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}. \tag{3.5}$$

This result is called the geometric series. Please note that we must have $|x| < 1$; otherwise, the summation does not converge. The video From Here To Infinity https://tinyurl.com/4td2wzkv reviews these results.

Armed with this result, we can perform the summation for the arrow size with multiple reflections as follows:

$$\begin{aligned} &0.96 \times 0.2 \times (1 + (0.2)^2 + (0.2)^4 + (0.2)^6 + \cdots) \\ &\quad = 0.192 \times (1 + 0.04 + 0.04^2 + 0.04^3 + \cdots) \\ &\quad = 0.192 \times \frac{1}{1-0.04} = 0.192 \times \frac{1}{0.96} = 0.2. \end{aligned} \tag{3.6}$$

Once we can see and recognize the geometric series (here with $x = 0.04$), the summation is simple to carry out. Because the arrow from the top is length 0.2 and in the opposite direction, the total is equal to 0, and this is the minimum for the reflection, giving a probability of 0% to reflect.

The case for maximal reflection is more difficult to work out, and we will only sketch the idea here. The issue is that the arrow that reflects off the top has its direction rotated by an additional 6 hours. For maximal reflection, we would want all of the multiple reflected arrows to also be rotated by an additional 6 hours. But,

if the rotation when going from top to bottom back to top in glass corresponds to adding 6 hours to the arrow, then when we reflect off the top and bottom a total of 3 times (top-bottom-top-bottom-top), we will advance by 12 hours because it travels twice as far in the glass (draw a picture), and for 5 reflections we advance by 18 hours again and so on, so the sum of the arrows for multiple reflections lie in the same direction as the top arrow for 1, 5, 9, etc. reflections and in the opposite direction for 3, 7, 11, etc. reflections. The sum from all reflections off the bottom becomes

$$-0.192 \times (1 - 0.04 + 0.04^2 - 0.04^3 + \cdots) = -0.192 \times \frac{1}{1 + 0.04} = -0.1846, \tag{3.7}$$

which gives a total final arrow of length 0.3846 (the minus sign indicates it is pointing in the same direction as the arrow for reflection off the top) or a probability of 0.1479 or 14.79%. This appears to be the highest possible percentage, but the above argument is not a rigorous proof. See if you can achieve a higher percentage (I have had a student show me that one can actually achieve a probability of 0.14935; see if you can determine how). Completing the analogous calculation for the transmission case is a problem given at the end of the section.

The ideas we have been covering have applications to a phenomenon called iridescence and the natural origins of colors. We all have seen soap bubbles and "oil slicks" in puddles. They show a rainbow of different colors. Why? The colors come from partial reflection of light. We have developed all of the quantum theory to understand this phenomena.

Here are the pertinent facts we need to understand this phenomenon:

1. A particular thickness of glass may completely reflect the red light shone on it, but then blue or green light might be transmitted. The color of light that is primarily transmitted or reflected will change according to the thickness of the glass. (Be sure you understand why this is the case.)
2. Soap bubbles or oil slicks behave just like glass due to the fact that the soap film or the thin layer of oil acts similarly to a thin glass layer with the interfaces with air and/or water.
3. The thickness of a soap bubble or of the thin layer of oil may change over time due to movement of the fluids.

The technical name for these rainbows is "iridescence" and it was first studied by Robert Hooke. Newton understood that it required a thin layer of material and used these to explain the colors in peacock feathers. A complete description in terms of the wave theory of light came from Thomas Young. The phenomenon of iridescence is responsible for the colors of bird feathers and butterfly wings and many more naturally occurring colors. It is amazing that the color comes from the thickness of a transparent film (arising from the cells).

There are also some practical uses of this behavior. Suppose you shine light of a single color at a very thin film of an unknown thickness. By analyzing the angle that the light reflects the most, you can infer the thickness of the film (after using some

geometry). Similarly, if you know the thickness of the film, you can determine the wavelength of light from a similar experiment. This technique is often one of the first characterization methods that is used to determine the thickness of a thin film of transparent material. When you think about it, it is a very clever and very precise way to make such a measurement. Multiple layers of thin films can also be used to filter certain colors of light and are called dielectric filters.

Partial reflection has also been used in art. French designer and artist Claire Lavabre uses partial reflection of light in an exhibit (called *Reflet*) to show the difference between reflection and transmission and how we perceive light that shines on glass. Since we know light is a photon, this phenomenon is an interesting exercise in quantum physics.

The art by Lavabre is a rectangle of thin glass about 2 m tall and 3/4 of a meter wide. The glass is both highly reflective and highly transparent. It is leaning against a white wall so that its top is against the wall and its base is about 15 cm from the wall. The wall is white with a dark oval painted on it. When a man stands in front of the glass, he sees his reflection quite strongly where his line of sight through the glass falls on the dark oval. But he sees only a faint reflection where his line of sight through the glass falls on the white wall. This is because the light falling on the oval is mostly absorbed by the dark paint, and so the light returning to the man's eyes comes mostly from his reflection off the glass. But the light passing through the glass and hitting the white wall is mostly reflected back (as a diffuse, non-imaging reflection) and its brightness overwhelms his reflection in the glass making it seem to fade. By moving in front of the glass, you can change the perspective of what is reflected and what is transmitted. This makes the art kinetic.

I end the section with some questions to ponder: (1) One of our rules for reflection has us rotate the arrow by 6 hours. This result seems to come from thin air. It does guarantee that there is no reflection from an infinitesimally thin layer of glass, which is what we might expect, because if there is no glass there, then there cannot be a reflection. So, if we get no reflection for an infinitesimally thin layer, why do we apply antireflective coatings that have some thickness? (2) In real materials, there sometimes is an absorption of light in the material itself. Think about how this absorption effect would modify the different results studied in this section.

Section 3.2 Problems:

3.2.1 Finishing the Transmission Calculation

For transmission, we always have two transmission factors (from the top air-glass interface and from the bottom air-glass interface) and an even number of reflections. So the case with no reflections has a length of $0.9798 \times 0.9798 = 0.96$. The case for two reflections is $0.96 \times 0.2 \times 0.2 = 0.0384$. We work out the lengths for four and six reflections next. You can see the photon paths for each of these alternative ways in Fig. 3.8.

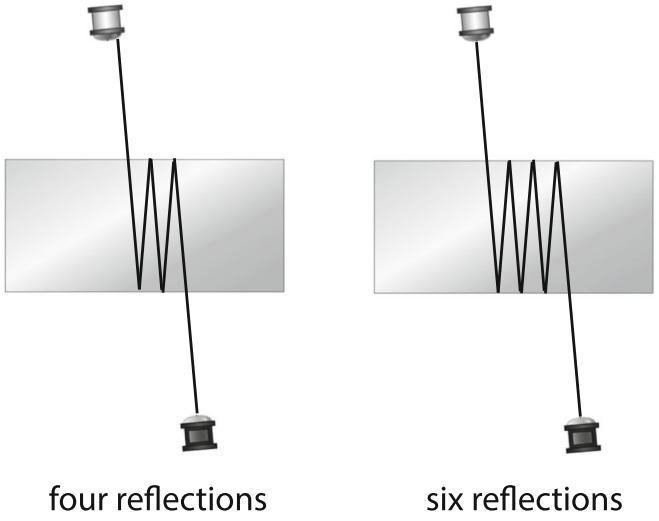

Fig. 3.8 Additional examples of multiple reflections for the alternative ways with four reflections (left) and six reflections (right)

Compute the result for four reflections to six decimal places and the result for six reflections to eight decimal places.

3.2.2 Exploring the Geometric Series with a Simple Example

We will work out some examples of the geometric series for $x = 0.1$. Recall that we showed that

$$S_N(x) = \sum_{n=0}^{N-1} x^n = \frac{1 - x^N}{1 - x}. \tag{3.8}$$

If we pick $x = 0.1$, then $(1 - x) = 0.9$. So, if we have $N = 2$, then $\sum_{n=0}^{1} x^n = 1 + x = 1.1$ and $1 - x^2 = 0.99$, so we see that $1.1 = \frac{0.99}{0.9}$. Repeat this exercise for $N = 3, 4, 5$, and 6. Verify that Eq. (3.8) holds.

Note how it approaches the $N \to \infty$ result by adding an extra digit for each increase in N by one unit. Recall that the exact result is $1/0.9 = 10/9 = 1.111111\cdots$.

3.2.3 Summing the Infinite Number of Terms for Transmission

In this problem we complete the analysis for all of the multiple reflections. Note that this problem is a bit mathematical. I recommend you try it even if you think it might be challenging for you.

Suppose the arrow winds by 6 hours when we travel through glass from top to bottom, then it will wind by an integer number of revolutions plus 6 hours for every reflection path (because we have an odd number of reflection paths). All arrows (no reflection and multiple reflections) point in the same direction at 6 o'clock (plus however much the arrow hand winds in coming from source to the top air-glass interface and the amount to go from the bottom air-glass interface and the detector, but this is the same for all alternative ways). The length of the arrow is then equal to

$$0.96 \times \sum_{n=0}^{\infty} (0.2)^{2n}. \tag{3.9}$$

Use the geometric series to sum this and find that the arrow length is exactly equal to 1. Note that this is precisely the case where we found the reflection probability would be equal to zero, so the total probability is conserved, as it must be. In other words, the total probability of reflection plus transmission is equal to 1.

The other case we analyzed above was if the arrow advanced by 6 hours when traveling from the top of the glass to the bottom and then back to the top. In this case, it advances by 3 hours when traveling from the top to the bottom, so the direct transmission arrow points at three o'clock, the two-reflection arrow points at 9 o'clock, the four-reflection arrow at three o'clock, and so on, so the arrows alternate being parallel and antiparallel, and the sum becomes again

$$0.96 \times \left(1 - (0.2)^2 + (0.2)^4 - (0.2)^6 + \cdots\right) = 0.96 \times \sum_{n=0}^{\infty} (-0.04)^n. \tag{3.10}$$

Use the geometric series to sum this arrow length and square it to get the probability. You should see that when we add up the probability for reflection plus the probability for transmission, we obtain 1 again.

Hence, we have shown with two examples that it looks like the probability problem is solved by including all multiple reflections. Note that the lowest possible value for transmission is a little lower than this. See if you can find it.

Fig. 3.9 Oil slick illuminated by monochromatic light (and converted to a grayscale image) (Image credit: James Freericks)

3.2.4 Illuminating Oil Slicks with Monochromatic Light

We all have seen the rainbow of colors that "dance around" in a soap bubble or in an "oil slick." But how would they look if we illuminated them with monochromatic light. Below we collect a number of facts related to what happens if a soap bubble or an oil slick is illuminated with single color light. The image in Fig. 3.9 approximates this type of illumination.

Only three of the following statements are correct. Choose which three.

1. The thickness of the oil film at a given moment in time is the same everywhere.
2. The thickness of the oil film at a given moment in time varies with position.
3. The thickness of the oil film at a particular location remains the same for all time.
4. The thickness of the oil film at a particular location varies as a function of time.
5. When monochromatic light is shone on the oil slick, it will reflect or transmit, so I will see bands of color and bands of dark regions.
6. When monochromatic light is shone on an oil slick, I will see a single color uniformly over the entire oil slick.

3.2.5 Describing How Iridescence Works

In this exercise, we will explain how the rainbows on soap bubbles and oil slicks occur using our quantum theory of light. Using all of the ideas we have developed so far, write a short essay (1–2 paragraphs) explaining the rainbow colors of a soap bubble or an oil slick and why they vary in time (note that these colors arise when white light is shone on the objects—the white light contains all of the colors of the rainbow). Be sure to explain your ideas clearly. Feel free to use figures if they help in describing this complex behavior.

3.3 Developing the Quantum Model for Light

Now that we have mastered partial reflection and understood what it means to have a photon undergo quantum interference, we move on to a different series of experiments with light, ones which will ultimately unveil the quantum mystery and then allow us to model and predict this bizarre quantum behavior. It is a simple experiment, shining light at a dark screen that has one (or more) thin slit(s) cut into it and looking at the pattern of light that hits a second screen situated far away. This might sound like an odd experiment to examine, but it holds the key to understanding both quantum mechanics and the behavior of light. We start by working through the theory and then show you what the experiment looks like. Light traveling through a single slit can be thought of as an experiment to test the question does light travel in a straight line and to understand why it behaves as it does.

We start our discussion with a simple experiment that has one slit in the middle of the first screen and is depicted in Fig. 3.10. The single-color light source (here blue light) appears at the left. The slit is in the middle and the screen on the right (note that in an actual experiment, the screen will be much farther to the right than what is depicted here). We can draw seven photon paths inside the slit; we consider only the photon paths that change their slope at most once in going from source to detector (as a simplifying approximation). Of course, when a photon leaves the source, not all of the photons leaving the source hit the detectors on the screen on the right. We take the length of the rotating arrow (for each alternative path) to be 0.02 for this set of experiments. Our rule says that a stopwatch winds (at a rate of approximately

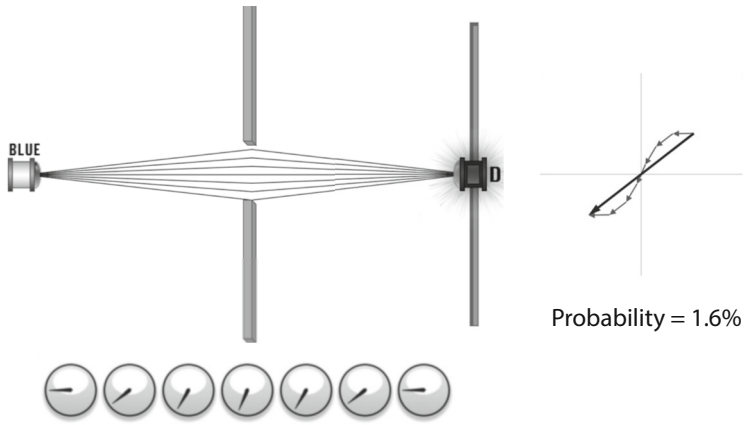

Fig. 3.10 Blue light traveling through a single slit and hitting a detector on a screen behind it (note the figure is not drawn to scale; the detector is much farther to the right in a real setup). The seven lines with a (possible) break in slope inside the slit show the different photon paths through this wide slit. The clock hands represent the time on the clock for each path (clocks running left to right correspond to paths running top to bottom). The arrows, each of length 0.02, are drawn in the right panel as thin arrows aligned head to tail, and then a thick arrow shows the sum of these arrows. Squaring the length of this arrow gives the probability to be detected in the detector, which is equal to 1.6%

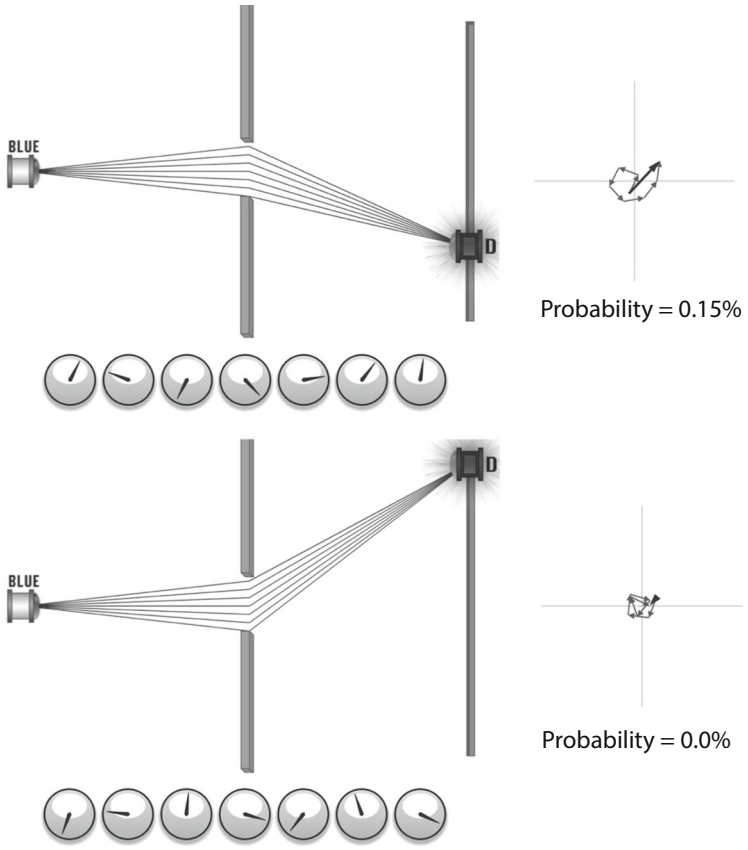

Fig. 3.11 Single-slit experiment with a blue light source and the detectors placed in different locations

72,000 rotations per inch for blue light) as the photon moves through the air. These clocks are shown at the bottom of the figure. The leftmost clock corresponds to the topmost path, the middle clock to the middle path, and the rightmost clock to the bottom path. Since the shortest distance between two points is a straight line, the shortest time is on the middle clock (corresponding to the middle path). The other paths have slightly longer times. We then copy arrows oriented in the direction of the clock hands and hook them head to tail to construct the final arrow. This is shown in the arrow computation window. The arrows nearly line up all together and the sum gives an arrow whose square yields a probability of 1.6% for this case. So, we predict we will see some bright light in the center of the screen just opposite the slit. This is to be expected and agrees with your experience. But note that all photon paths contribute to this light, not just the photon path in the center. So we would say that light nearly travels in a straight line, but not exactly in a straight line.

Now, what happens if we move the detector? This is shown in Fig. 3.11. The setup is the same—source of blue light to the left, screen with one slit to the right of it, and

seven paths drawn in the slit. But now the straight-line path is toward the bottom of the slit, rather than in the middle, for the top panel. Again, we send light along the paths, time how long a photon takes along each path, and then get the sequence of stopped clocks shown at the bottom of the top panel. Note now that the shortest time is the bottom path (right clock) and that the directions vary more here. The clock hands then correspond to the arrows drawn in the computational window, which then gives the final arrow after being summed head to tail. We square that final arrow to find the probability. We can see the clock hands wind more now, so the arrow does not grow as quickly and the final probability is less. The variance of arrow directions is even more in the bottom panel. Hence, in the top panel, we find some light is detected at this angle, but not as much as straight on. In the bottom panel, we instead find, essentially, no light hits the screen there. This then corresponds to a detector located close to the minimum of the diffraction pattern. Note that there is no straight-line path in this final example—we can detect light even if none of the alternative ways is a straight-line path. A tutorial https://bit.ly/4j1rpPm shows all of this behavior.

Let us now summarize this behavior of the single slit. You can immediately see that in the quantum theory, many paths contribute to the light that goes through a narrow slit, not just the path with the shortest length (the straight-line path). This is because the length of the path changes slowly for paths near the shortest distance path, so the arrows do not turn significantly more for those other cases and when we combine them, they are all pointing in about the same direction. Hopefully, you also noticed when there was no straight-line path in the chosen paths, the times for the different paths increased more rapidly, so the arrows (for the different alternative ways the event can occur) tended to spin around a circle and their sum effectively goes nowhere, resulting in a small final arrow (bottom panel of Fig. 3.11). Indeed, in our theory all paths contribute, it is just that some contributions tend to cancel, while others tend to add, and that is how we determine what "tends to contribute" to the final result. If neither of these facts are clear to you, please revisit the figures and reread the text. Pay close attention to these two ideas as you go through these materials, especially when you explore the different detector positions. We have some problems to test your understanding of this at the end of the section.

Our next focus is to determine what happens if we change the slit width and if we change the color of the source of light (say from blue to green to red). We start by looking at what happens as we change the color. Recall that the color of the light determines how fast the stopwatch hand rotates. For visible light, red light rotates the slowest at about 36,000 rotations per inch. Green light rotates about 50% faster, and blue light is about twice as fast. Think about what a slower rate of turning of the clock hands means for the times on the clocks at the end of the experiment. Use that to make a prediction.

It turns out that the energy the photon carries is proportional to the rate at which the clock rotates, so a blue photon has more energy than a green photon, which has more energy than a red photon. This is why ultraviolet light is often used to initiate chemical processes (it has even higher energy than blue light), and why it is more dangerous to cells than infrared light. (It is the ultraviolet light that causes sunburns;

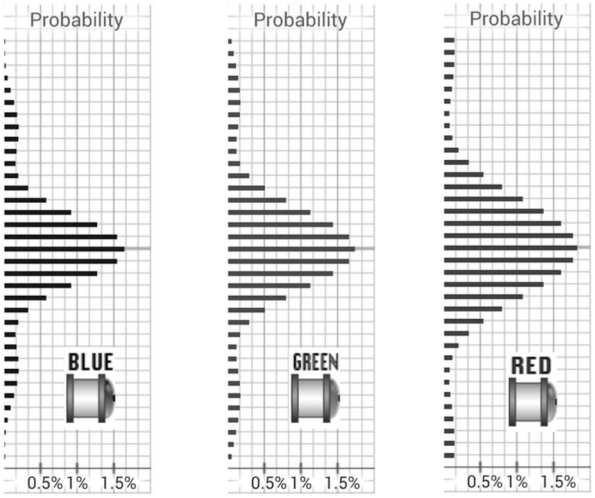

Fig. 3.12 Probability of light hitting the screen in a single-slit diffraction as a function of color of the light source (with the slit width the same for each case). Note how the pattern broadens as we go from blue to green to red. There is also a small increase in the intensity of the light too, but not a huge effect

infrared light, which is lower energy than red light, causes the sunlight to feel hot on your skin.)

While it seems simple to think about the change of color as just a rotation rate, when we combine it with looking at the probability patterns for a specific experiment, the behavior does require careful thought to determine how it evolves. Please make a prediction now.

I hope you predicted that the clock hands would not turn as much. In particular, the arrows we would draw in the computation window would not rotate as far away from the arrow of the straight-line path. This means that one has to move the detector farther away from the center on the screen before we get to the point where the diffraction pattern goes to zero. Hence, the patterns spread out, as we go from blue to green to red. (We also see a small increase in the intensity for red light, which you should also be able to understand for this model. Focus on the detector at the center of the screen to reason this out.) See Fig. 3.12 to see how the patterns change as we change the color of the source. This notion of determining how far up the screen one needs to go in order for the diffraction pattern to reach a minimum is a good way to determine the width of the central peak. The analysis can be complicated, because we need to know how much one clock hand rotates relative to other clock hands. When they rotate enough relative to each other, they act like "spinning wheels" that go nowhere. In other words, they do not contribute much to the final arrow of the system. The video Squeezing Photons Through a Narrow Slit https://tinyurl.com/ypzvwpph and a tutorial https://bit.ly/3XPJ6rq show how changing color affects the pattern.

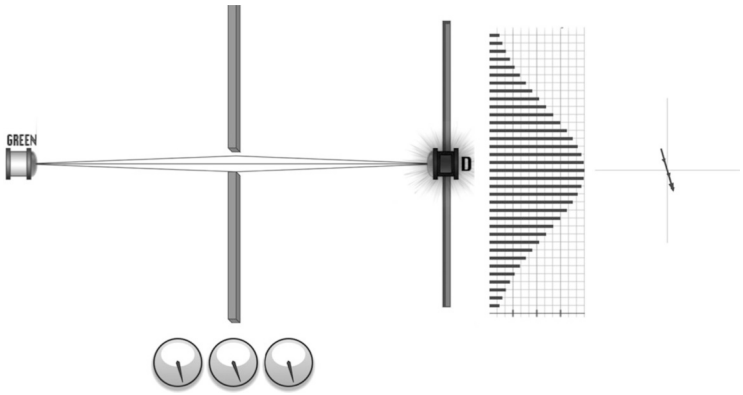

Fig. 3.13 Single-slit diffraction with a green-colored light source and going through a thin slit (three alternative ways through the slit). Note how the clock hands do not change too much for the different ways, meaning they point similarly, leading to a wide probability pattern on the screen

We now examine what happens when we use a single color of light (here, we choose green light) and we change the slit width. The first case we look at is the case where only three alternate ways (three photon paths) fit inside the slit. This result is shown in Fig. 3.13. You can see, for this location of the detector, the clock hands point nearly in the same direction, so the arrows add constructively and the final arrow is large. Because the distance for the top and bottom paths is only slightly longer than the distance for the middle path, as we move the detector up the screen, we need to move far away, before the three arrows form a triangle, when linked head to tail, and the final arrow becomes small. Hence, the width of the diffraction pattern is wide. Think about how wide it becomes if the slit was so narrow we only had one alternative way for the light to go through the slit.

As we make the slit width wider (see Fig. 3.14), we can see the diffraction pattern becomes narrower, as can be predicted by reasoning through our model for light. Again, one does not have to go too far from the center before the different clock hands "spin their wheels" and we have a low probability for detection. This ends up being a general principle, which perhaps is counterintuitive to you—when the slit is made narrow, the pattern becomes wide and when the slit becomes wide, the pattern becomes narrow. The tutorial https://bit.ly/496cpfv shows this behavior.

We have seen two different phenomena—changing the width of the slit with a fixed color and changing the source color with a fixed width of the slit. Narrowing the slit makes the pattern wider, while broadening the slit makes the pattern narrower. Similarly, changing the color from blue to green to red makes the pattern become wider. This gives us a hint that these two behaviors are somehow related. The way to relate them is to think about the wavelength of light. We can think of this as the distance the light travels during one revolution of the clock. For red light, this is about 0.7 microns (or micrometers). Then one can think of the shape of the diffraction pattern as being governed by the ratio of the wavelength of light to the

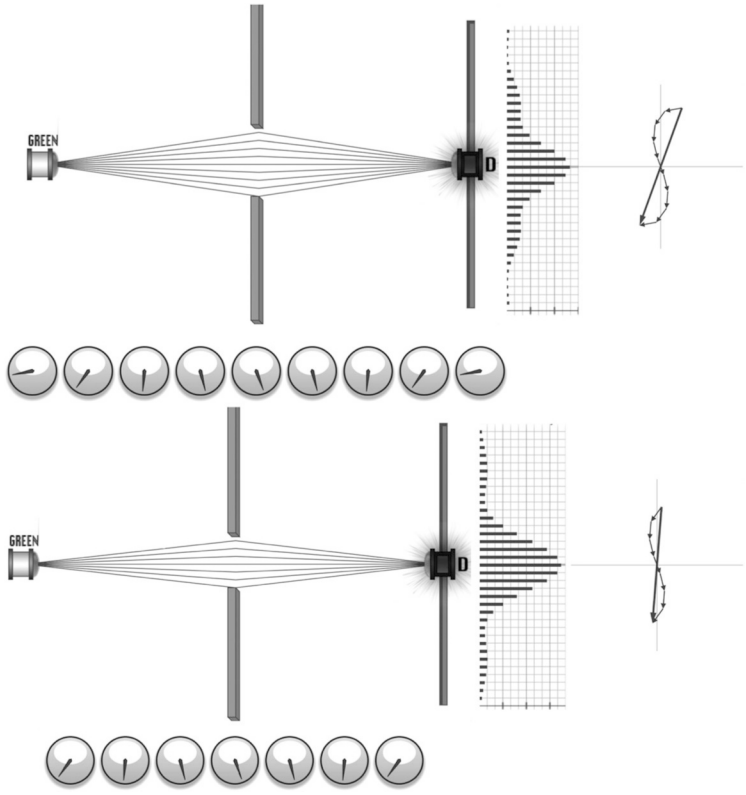

Fig. 3.14 Single-slit diffraction with a green-colored light source and going through a thick slit (nine alternative ways through the slit on top, seven ways on the bottom). Note how a wider slit leads to a narrower probability pattern

slit width. If the wavelength is increased, the ratio increases. If the slit width is decreased, the ratio increases. In both cases, the diffraction pattern gets wider. It really has its origin in the same phenomena, determined by how fast the clock hands rotate relative to neighboring photon paths.

In this section, we examined the properties of light that is shone through a single slit by using the quantum model we have developed. You should have a clear understanding of why light tends to move in straight lines, but if we try to squeeze light by making it move through a narrow slit, it will expand. Changing the color of light acts similarly to changing the width of the slit opening. We end this section with some questions to ponder.

1. How can we say that light tends to travel in straight lines, but when we squeeze it to travel through a very narrow slit, it spreads out? Doesn't this mean the light is not traveling in a straight line anymore? Does the squeezing of the hole and the expansion of the diffraction pattern on the screen relate to Heisenberg's

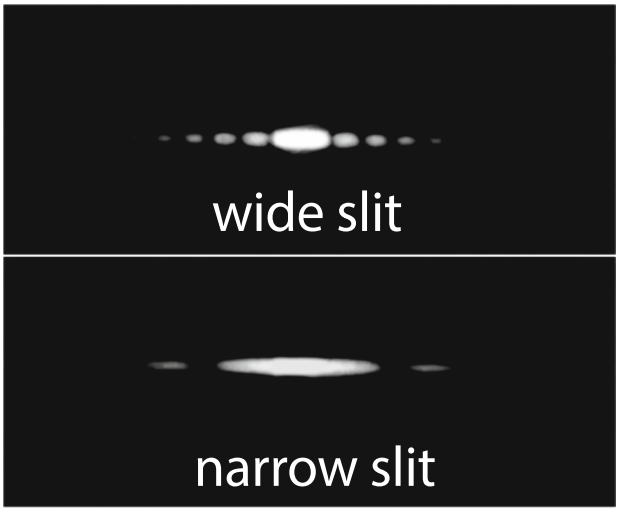

Fig. 3.15 Experimental images of single-slit diffraction from a wide slit (top) and a narrow slit (bottom). One can see the characteristics we discussed clearly present in these images, with the central region expanding in size as the slit is made narrower and with the wavelike diffraction fringes visible for both slits. The central region will become our main focus when we add a second slit—the secondary regions, which are much dimmer, will still be visible, but we will not focus on describing their behavior. These images were taken from a video on single-slit diffraction made by the technical services group of the Massachusetts Institute of Technology (used with permission © Massachusetts Institute of Technology) (Technical Services Group, MIT, 2012)

uncertainty principle (this assumes you have heard of the uncertainty principle—we will discuss it in detail later)?

2. If the color of the light is changed, it acts similar to changing the width of the slit. Explore this relation to understand how increasing the frequency of the light (changing from red to blue) compares with narrowing the slit.
3. In Fig. 3.15, you can see that there are interesting diffraction patterns appearing at the outer edges of the pattern on the screen. Look at the patterns we have with blue light to see if you can understand these wavelike patterns in the diffraction.
4. Since narrowing the slit causes the pattern on the screen to expand, could you devise a way to measure the width of the slit by measuring the size of the pattern on the screen and knowing the wavelength of light? When interpreted this way, the single-slit experiment is, in some sense, like a microscope.
5. What will happen when we add a second slit and consider a two-slit experiment (or a three-slit experiment)?

We have a video Your Turn to Explore the Quantum Rules https://tinyurl.com/mu8saaym and a tutorial https://bit.ly/3MUoPhS that allows you free exploration of these ideas.

Section 3.3 Problems:

3.3.1 Which Paths Contribute the Most?

Please look at the three panels in Figs. 3.10 and 3.11. Can you identify the path that take the least time. Are paths near that path also of least time? Can you determine which paths contribute the most to the final arrow?

3.3.2 Why Does Light Travel in a Straight Line?

You may have learned that light travels in straight lines, especially if you worked with geometrical optics. But, our quantum model requires us to include all possible paths and to weight them the same when we add up the arrows. Given this fact, we now correct the straight-line statement to say that *light often travels in nearly straight lines*. Please clarify why we changed the statement and explain why we state it in this new way.

3.3.3 Arrows that Go Nowhere

Why do the sums of certain collections of nearby paths not contribute significantly to the final probability amplitude arrow? What is needed for a collection of arrows to "spin their wheels" and get nowhere?

3.3.4 Settle the Big Bang Theory Debate

Sheldon and Leonard are having an argument. They are arguing about an arrangement of photon paths that corresponds to a wider slit (about twice as wide) than what we illustrated in our discussion above. This wider slit case has 13 alternative ways (photon paths), three more above and below the seven that we originally considered in Figs. 3.10 and 3.11. Sheldon says that adding three more paths above and below the original seven will significantly change the final probability arrow because you are adding more paths. Leonard claims that because those paths will not make a significant contribution to the final probability arrow, adding those paths hardly changes the probability for the detector when it is located straight behind the slit.

Who is correct?

Write a short response that explains why Sheldon or Leonard is right.

3.3.5 Why Does the Probability Pattern Change?

Recall our discussion for how the probability distribution graph changes when you change the color of the photon source from red to blue. Answer the following questions:

1. Why do red and blue photons produce different probability distribution graphs?
2. We shine light of three different colors (blue, green, and red) through a slit of the same size opening and project them onto a screen that is far away from the slit. What is the order of the patterns of light from widest to narrowest.

3.3.6 Three Diffraction Patterns

Consider a set of three diffraction patterns, one being narrow, one intermediate, and one wide, when they are displayed on a screen. Which probability distribution corresponds to blue light traveling through the thinnest slit? The thickest slit?

3.3.7 Which Is Wider?

Consider two experiments that have the same width for the diffraction pattern on the screen. One uses a blue laser and one uses a red laser that is shone through a single slit. The distance from the slit mask to the screen is the same for both experiments. Which slit is wider? Explain.

3.3.8 Big Bang Theory Argument II

Sheldon and Leonard are having another discussion. They do not agree about what happens when the slit is narrowed almost all the way to zero. Leonard says that there is a problem with the model we are using for light. If we narrow the slit down to almost nothing, the model predicts the pattern will become infinitely wide, when we know no light can travel through a completely blocked slit. Sheldon tells Leonard that he just does not understand the properties of light. Indeed, as the slit is made narrower and narrower, the pattern does get wider and wider. But it also gets dimmer and dimmer, because it is so spread out. Therefore, in the limit as the slit narrows almost all the way to zero, the pattern gets very wide, but it also gets very dim. So it eventually goes dark, as you expect.

Is Sheldon's explanation correct? (Think about whether the model we have developed can explain this behavior.) Explain.

3.3.9 Changing the Color or the Width

Write an essay that describes how changing the slit width or the color of light changes the single-slit pattern. Note that the probability patterns change in similar ways for either effect. Write a one- to two-paragraph essay that explains how these two changes can be thought to relate to one another.

3.4 The Quantum Mystery

> We choose to examine a phenomena which is impossible, absolutely impossible to explain in any classical way, and which has in it the heart of quantum mechanics. In reality, it contains the only mystery ⋯ Any other situation in quantum mechanics, it turns out, can always be explained by saying, 'You remember the case of the experiment with the two holes? It's the same thing.' (Feynman, 1965)

This quote, made by Richard Feynman during a series of six public lectures at Cornell University, has at its heart what we will call the quantum mystery. This section of Chap. 3 describes that quantum mystery. We start our discussion by examining the so-called two-slit experiment—this means that we add one more slit to the mask in the experiments we have been studying. In Fig. 3.16, we show the preliminaries for this experiment. In the center is the mask with two slits in it. The figure shows what happens when one slit is blocked and one slit is open. On average, about 1% of the photons leaving the source hit the detector. Since we have two alternative ways for each slit, this means the length of the arrow for each way is 0.05 (when they point in the same direction, the maximum length is $0.05 + 0.05 = 0.1$, yielding a probability of 1%). The calculation of the precise probability shows that the probability distributions are centered about the straight-line paths from the source to the detector through the upper or lower slit, respectively. In the "scatter plot," we simulate what the pattern would look like for a single-photon source. Here, the photons hit the screen one by one in a pattern that initially looks random, but eventually starts to have its density follow that of the calculated probability distribution. For the full two-slit experiment, we make one change—we open up both slits at the same time by removing the blockage. A video https://tinyurl.com/44dccffw from Leiden University shows how the pattern develops over time. Try to use our rules to predict what you will see when we do this. Think about what the pattern generated by a single-photon source will look like.

We show the results of the two-slit experiment in Fig. 3.17. First, let us focus on how the calculation works. Our event is still the same—a photon leaves the source and arrives in the detector—but now we have four alternative ways that it can happen. Two go through the upper slit and two through the lower one. The times for the alternative ways that go through the same slit are quite close, but those through the different slits can be quite different. So, we will have two sets of arrows, one from each slit. As we move the detector out along the screen, the time becomes more

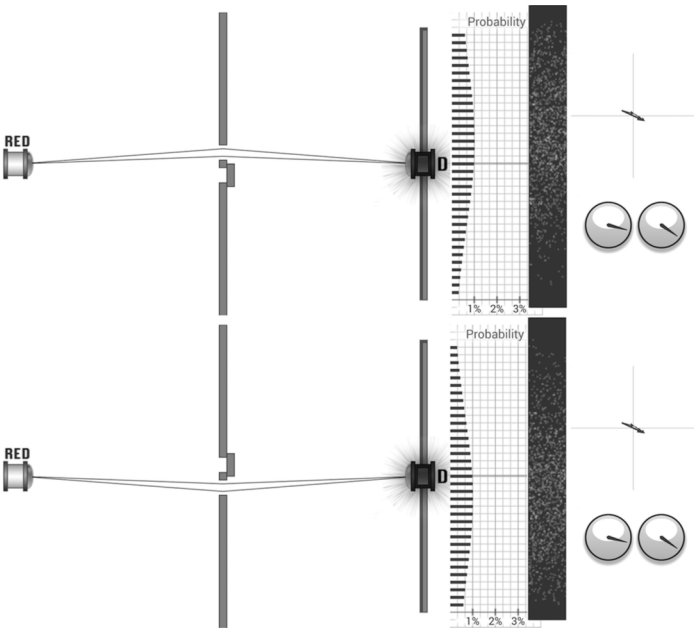

Fig. 3.16 Setup for the two-slit experiment. We show what happens when the upper or lower slit is blocked. In the top panel, we have two alternative ways for the event to occur, as indicated by the two photon paths; each arrow has a length of 0.05 as explained in the text and the clocks time the motion along each path. The two arrows and the final arrow are in the computational panel, which then is squared to show the predicted probability. In a real experiment, the photons appear one by one if we use a single-photon light source and are indicated in the "scatter plot." The photons hit the screen in a seemingly random pattern, but after many have hit, we see the calculated probability distribution emerge. The bottom panel shows the case of the bottom slit opened, where the probability distribution shifts slightly downward

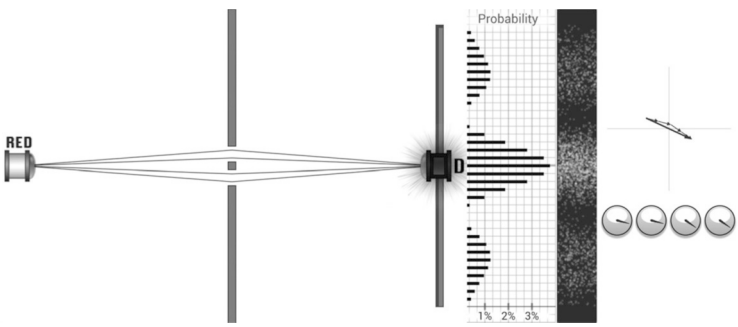

Fig. 3.17 Two-slit experiment with both slits open. Note how the probability pattern has changed significantly, indicating an interference between the paths going through different slits

different, and at some point the results from one slit and the results from the other will end up being opposite to each other and the total probability will go to zero. This can be clearly seen in Fig. 3.17 when we look at the probability distribution (and also with the simulated pattern from a single-photon source). A tutorial https://bit.ly/4rMVB4x shows how the quantum rules are applied and a second one https://bit.ly/48MT811 shows how an experiment works with a single-photon light source. The experiment with a single-photon source brings us to the first part of the quantum mystery. When we use a single-photon source of light, we see individual photons hit the screen, which eventually build up enough to show the interference fringes of the so-called wave pattern. The mystery with this is that the wave patterns usually occur when two sources of waves constructively and destructively interfere to yield an interference pattern. If there is only one photon in the apparatus at a time, how does the photon interfere with itself?

Before moving on, let us pause for a moment to ponder this two-slit behavior. Think about the following four questions:

1. Why is the two-slit image four times brighter than one of the single-slit images? Shouldn't it have been twice as bright since there are two slits? (Compare Figs. 3.16 and 3.17 at the center to see the probability is 1% in the former and 4% in the latter.)
2. Think about how the two-slit image will change if the slits are brought closer together or farther apart. Think about the comparison with the single slit, when the two slits are brought so close together that they will touch.
3. How does the image change as the color of the light changes?
4. How can opening an additional slit make the number of photons hitting a detector go down for some locations of the screen? This happens, for example, at the point where the two-slit probability distribution goes to zero.

So, at this stage you might be thinking that everything is good, we have a set of rules, which we have followed, and they give us the observed behavior. But, we cannot avoid asking the question, does the photon go through both slits at the same time? Now, this is a question that perhaps we should not ask. After all, we have already seen when we discussed the Stern-Gerlach experiments that when a particle moves under superposition, its position is indeterminate. So maybe we cannot ask this question. But, we are curious, so we ask anyway. Is there a way to try to measure this here? The answer is yes, but it requires us to modify the experiment. We need to add detectors at the slits to be able to see how a particle goes through two slits at once (if such a statement even makes any sense at all). In the end, the combination of our first unwatched experiment and our second watched experiment will unveil for us the quantum mystery.

We add two detectors, similar to the pass-through detectors in the Stern-Gerlach analyzer loop, that can observe photons, but do not destroy them. These detectors are oriented toward each slit to allow us to observe the photons at the slits. The setup is shown in Fig. 3.18. Now, when we run such an experiment, we will sometimes see a photon go through one slit or the other, but we never see both detectors fire, as we might have expected if the photon "went through both slits at the same time." But now, instead of seeing the interference pattern of Fig. 3.17, we see the sum of

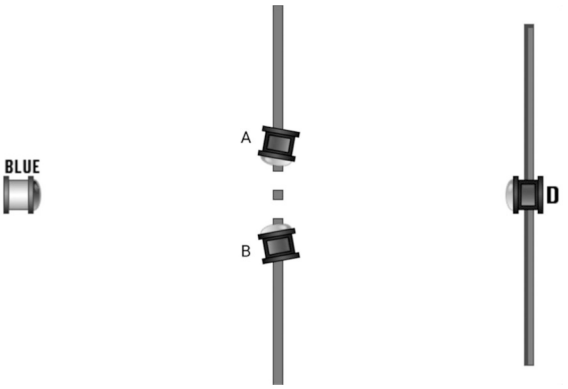

Fig. 3.18 Two-slit experiment with a blue light source and with two extra detectors watching at the slits

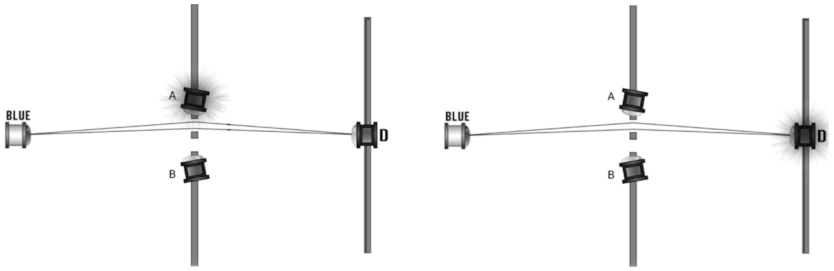

Fig. 3.19 First event for the watched two-slit experiment, where the detector on the upper slit and on the screen both fire; the panels depict two different times in the experiment. There are two alternative ways that this event can occur, as shown with the two paths that go through the upper slit. There is a similar event where the detectors at the lower slit and at the screen both fire, which also has two alternative ways that it can occur. It is not shown here

the particle-like patterns in the two panels of Fig. 3.16. *The result of the experiment has changed due to the fact that we observed at the slits.* How can this occur? Let us analyze the situation carefully using our model. Whenever I think of this experiment, I am reminded of the 1979 movie *Being There*, where Chauncy the gardener (played by Peter Sellers) always says "I like to watch."

Our method requires us to first determine the different events. Recall that an event is determined by the initial conditions (where the sources are) and the final results in the detectors. In Fig. 3.19, we show one of the two possible events—this one has the upper detector and the detector at the screen both fire. This event has two alternative ways it can occur, corresponding to the two paths that go through the upper slit. A similar event also occurs where the detectors at the lower slit and the screen both fire. This event also has two alternative ways it can occur, corresponding to the two photon paths through the lower slit. Note that these two alternative ways *are the same alternative ways when we blocked one of the slits*, which are shown in Fig. 3.16. Hence, the probability distributions are the same as those shown there. The total probability distribution for these two slits open (and being watched) is the sum of these two, which is a relatively flat distribution; it is not an image of two slits as is often shown—light spreads out when it goes through a narrow slit

so the difference in the pattern from each slit is very difficult to see because they mostly overlap. There is a tutorial https://bit.ly/4iVwd8X and video https://tinyurl.com/464fytju that show what happens when we watch. *The interference effects have disappeared because we observed the experiment at the slits!*

Huh!!

Nature does not reveal its secrets to us. We do not see a particle interfering with itself, or breaking into small pieces and reforming; we just mundanely see particles traveling through one slit or the other. But the interference pattern has also disappeared. The act of trying to observe the two-slit experiment changed its results! Note that even though the peak brightness now is only 2% rather than the 4% for the unwatched case, the total number of photons hitting the screen in each time interval is about the same when we watch versus when we do not watch. It is just the pattern on the screen that is different.

This now completes the second part of the quantum mystery. We have a particle that interferes like a wave, but when we try to find out how it does it, it shows only its particle nature and stops showing its wavelike behavior. It is a pernicious behavior that can be seen to be frustratingly unfair. How come we cannot find out how the photon does this?

Our quantum rules clearly show how this occurs. When we have the two-slit experiment and we do not watch the photons, then we have one event and two alternative ways the event can occur (the photon going through the top slit or the bottom slit). To simplify the analysis, consider very narrow slits, where only one path fits through each slit. Then, because I have to add the two arrows, and they can rotate relative to each other, I will have cases where they add together (due to their pointing in the same direction), or they cancel each other out (by lining up in the opposite direction), and I can also have anything in between. Since we square the final probability amplitude arrow to get the probability, the maximum will be four times as large as that of a single slit, while the minimum will be zero. This produces an oscillating amplitude ranging from 0 to 4%.

When I watch with additional detectors at the slits, I have changed the experiment to correspond to two events now, rather than one event. Each event has only one arrow to describe it, so its probability is a constant, and when I add them together, I get 2%, or the average of the above result. The wave pattern has changed into a particle pattern.

While the proper application of the quantum rules clearly show this behavior happening, it still seems like nature is somehow cheating us and not allowing us to properly determine the mystery behind a quantum particle. One can become frustrated or upset by this, but if you want to continue to study quantum mechanics, you ultimately have to accept that this is the way the world really is.

This is not fair!

It may seem like nature is not being fair with us. We have this bizarre behavior, where a particle can act like a wave, but when we try to see how the quantum particle miraculously does this, we see it acts like an ordinary (classical) particle. This has

inspired a lot of thoughts about what happens if we pretend to look, or set up the experiment where we can look, but don't. Does this affect the results? It is easy to get confused, and just remember that, if you follow our quantum rules, you will never have a problem. Just be sure to carefully determine each event and the alternative ways the events can occur, and there will never be an issue with being able to predict the behavior of an experiment. It is how we interpret those results philosophically that can provide us with debates (and there are many philosophers who love these types of debates).

The most common explanation for why the presence of the second detector at the top slit ruins the quantum interference is due to Niels Bohr (and Werner Heisenberg) and is related to the uncertainty principle. The way the argument goes is that if we want to localize the position of the photon well enough that we can tell whether it went through the top slit or the bottom slit, then we have to hit it with some object that has a quantum wavelength small enough that we can differentiate those small distances. When a quantum particle has a small wavelength (or, in our language, when the clock hand spins really fast), then it has a large energy and a large momentum. So, when it scatters off the photon, the photon receives a huge kick (i.e., a huge push) and has been greatly disturbed, ruining its chance for any quantum interference to occur. A simple picture of what this looks like is if we want to see the trajectory of a ping pong ball by shooting BBs at it (BBs are small metal pellets like ball bearings). Each BB has enough force when it strikes the ping pong ball to send it off in some completely different direction, so the process of observing turns out to automatically destroy the quantum interference of the system, because of its huge effect on the photon.

But can we be clever enough to properly set up an experiment that allows us to look at the photon and see what it is doing without losing the interesting quantum interference effects? One option would be to use only one detector and turn its sensitivity way down, so it doesn't disturb the photons as much. But then, some photons might sneak by the detector and go undetected. We can model this by saying the detector has less than perfect efficiency (let's say it will detect on average 36 out of 100 photons that pass by it, or that it has 36% efficiency). Now, we will see some quantum interference, but we will still only see photons that travel through the top slit as traveling through the top slit as a whole photon. The quantum interference is present, but its magnitude is reduced from how large it was in the case with no detector. It occurs, nevertheless, because of the behavior of the photons that are not detected by the detector located at the top slit. In other words, when the photons are detected by the detector at the top slit, they act like a particle and show a constant probability curve, but when they are not, they act like a wave and show the oscillating probability curve. The efficiency determines how we weight each piece of the total probability curve, so it continuously changes from a high-amplitude oscillatory curve (varying from 0 to 4%) down to a zero-amplitude flat curve (at 2%). We call such a situation *partial interference*. We will explore this more thoroughly in the problems at the end of the section. The video Imperfect Detectors https://tinyurl.com/2e2ne7cb shows how to analyze the case with partially working detectors.

It turns out that every experiment that has tried to determine how the quantum particle does what it does to "interfere with itself" has failed to determine how this works. Whenever we can tell which slit the particle goes through, the quantum interference disappears and we see the particle pattern. This result is called *the principle of complementarity* and was introduced by Niels Bohr in 1927. Currently, there is no known way to prove that this principle is correct. To date, no experiment has ever shown that it is violated (some of these interesting new ideas go under the name of the quantum eraser or delayed-choice experiments). You have had a chance to explore some of these in Chap. 2 already, and you will see more in Chap. 4.

So, can we truly understand the quantum mystery?

This question forces us to become a bit philosophical. On the one hand, we have a set of quantum rules which explain this complicated behavior perfectly: identify the event or events, find the different ways they each occur, assign an arrow for each way, sum to get the final arrow, square the final arrow to get the probability for each event, and add the probabilities from all events to get the total probability. Every experiment we have studied so far yields to these rules, and we can calculate accurately the probability or probability ranges for different events to occur. Though we cannot tell what will happen to one particular photon, we can determine what happens, on average, to many photons. This distinction seems reasonable when we realize that quantum mechanics inherently has an element of randomness to it, which requires this probabilistic description. Albert Einstein did not like this and is paraphrased in saying "God does not play dice with the universe." Eventually, however, most people are happy to accept this element of randomness and forget what they were even bothered by in the first place.

Understanding quantum mechanics means we should go one step further. We want to understand how a single particle can produce results expected from a wave interfering with another wave. We have failed to determine exactly how quantum particles exhibit this phenomena. When we try to set up clever experiments probing this question, we find that nature does not reveal her secrets. This led Richard Feynman to say "I think I can safely say that nobody understands quantum mechanics" (from the BBC video "'Probability and Uncertainty—the Quantum Mechanical View of Nature" Feynman, 1964). Ultimately, Feynman is referring to the fact that no one truly understands how the quantum particle manages this feat, which physicists label as "a particle interfering with itself."

Where does this leave us? For me, I liken this to the probability issue described above. I get used to the fact that we have to think of the quantum description as a procedure, or cookbook, of simple rules that the quantum particles follow. Armed with these rules, we can predict a huge range of results of different experiments. This predictive power is an important element of quantum mechanics. Understanding the origin of these quantum rules is another deeper problem, and physicists have not made anywhere near as much progress there. Maybe we will in the future, maybe we will not. But for the moment, we must live with what we can understand and leave for the future what we cannot!

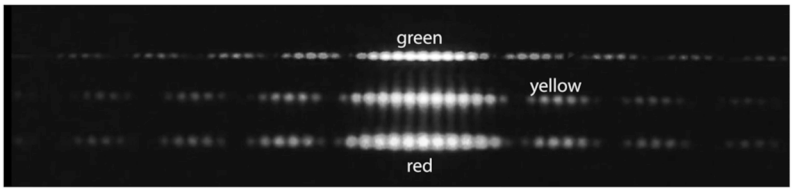

Fig. 3.20 Two-slit experiment using three different lasers—green (top), yellow (middle), and red (bottom). One can clearly see that the central interference pattern (under the green label and above the red label) has secondary (and dimmer) patterns to the right and the left. Those secondary patterns come from the regions where the single-slit diffraction had secondary patterns, as we saw already in Fig. 3.15. When we talk about a particle pattern, we refer to the central pattern of the single-slit diffraction. Note how the widths of the central patterns change for the different colors of light (the slits are the same width for each case) and how each central region is now covered with additional structure of bright and dark features within it. These images were taken from a video on single-slit diffraction made by the technical services group of the Massachusetts Institute of Technology (used with permission © Massachusetts Institute of Technology) (Technical Services Group, MIT, 2012)

The two-slit experiment is the preeminent example of quantum mechanics and contains within it deep mysteries that we may never be able to fully unravel. But we are able to create abstract theories that can predict, with great accuracy, the behavior of quantum particles. You now possess the knowledge about what quantum mechanics is, and how we can describe its behavior. You are ready to understand even more complex behavior like quantum teleportation, quantum computing, quantum encryption, and quantum entanglement. We will discuss some of these advanced ideas later in the book. Needless to say, they all can be related back to the two-slit experiment, just like Richard Feynman averred. What we will cover in the next chapter are bizarre topics like the quantum Zeno effect, quantum seeing in the dark (or an interaction-free measurement), and the Hong-Ou-Mandel experiment (which uses interference to show when two quantum particles are identical).

In Fig. 3.20, we show images of the two-slit experiment for three different color light sources (red, yellow, and green). The slits are the same width for each laser. This is the case for a continuous source (such as a laser). The central region of these images is the region that was the brightest in the single-slit diffraction image, but was continuously bright with no fringes. In the two-slit experiment, the extra interference creates new fringes *inside* this central region. A video https://tinyurl.com/55jt8cx9 summarizes these results. If we used a single-photon light source, the image would build up photon by photon (each arriving randomly), as shown in Fig. 3.17.

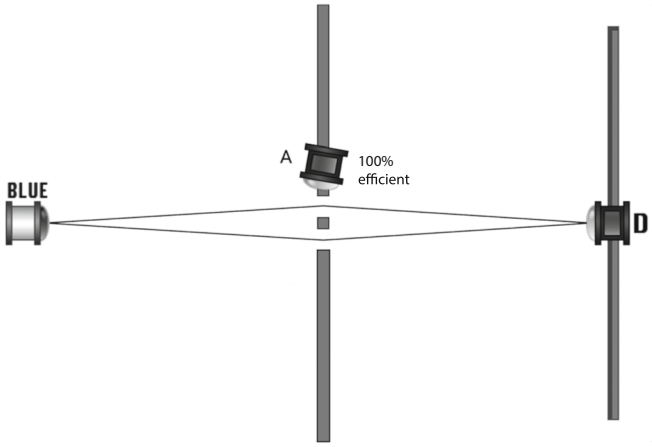

Fig. 3.21 Two-slit detector with one detector at the upper slit. The detector is 100% efficient

Section 3.4 Problems:

3.4.1 One Detector

Suppose we change how we watch the two-slit experiment and put only one detector at the top slit (which still is perfect in the sense that it will detect every photon traveling through the top slit). See Fig. 3.21 for a schematic of the experiment. Note that we use just one alternative way for the photon to travel through each slit, so there is one arrow associated with each slit.

What pattern will we see on the screen with only one detector watching? Be sure to think about what the event or events are for the experiment now, before answering the question.

3.4.2 Imperfect Detector I

Suppose we use one detector (labeled A) only watching at the top slit and that it is 64% efficient (meaning it detects 64 photons for every 100 that pass by it); see Fig. 3.22. We will determine the probability ranges for detecting photons on the screen by using our quantum rules. Note that there are now two events. Event 1: The photon is detected at detector A and D. Event 2: The photon is detected at detector D only. There is one way for event 1 to occur and two ways for event 2. We assume when one of the slits is blocked, 1% of the photons leaving the source arrive at the detector D.

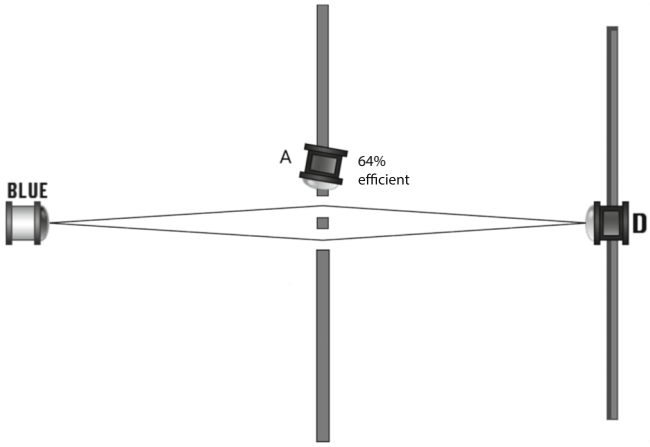

Fig. 3.22 Two-slit detector with one detector at the upper slit. The detector is 64% efficient

What is the probability that a photon leaves the source, goes through the top slit, and is detected in detectors A and D (assuming the bottom slit is blocked)? Note that this requires you to think in terms of compound probabilities, since the detector is 64% efficient.

What is the length of the arrow associated with event 1?

What is the length of the arrow for the alternative way of event 2 that goes through the top slit?

What is the length of the arrow for the alternative way of event 2 that goes through the bottom slit?

3.4.3 Imperfect Detector II

We continue the previous problem.

What is the maximum probability for event 2 to occur?

What is the minimum probability for event 2 to occur?

What is the minimum percentage for the total probability curve summing over both events?

What is the maximum percentage for the total probability curve summing over both events?

3.4.4 Three Slits

Consider a three-slit experiment where the central hole is larger than the two outer holes as shown in Fig. 3.23. So when the other two holes are blocked, 1% of

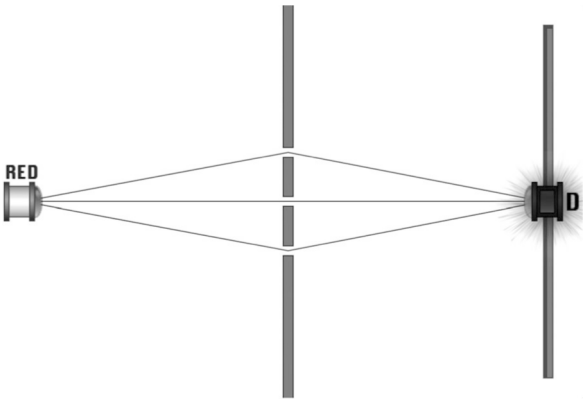

Fig. 3.23 Three-slit detector. The upper and lower slits are described by a single arrow of length 0.1 each (probability of 1% going through each slit). The central slit is described by a single arrow of length 0.2 (probability of 4% going through the central slit)

the photons leaving the source will get through the top (bottom) hole and hit the detector. Similarly, 4% of the photons leaving the source will get through the central hole and hit the detector. Use one path through each hole to represent the different ways the photon can get from the source to the detector (but note that the length of the arrows may be different for the different ways the event or events can occur).

Determine the length of the arrow that represents the path for the photon that goes through the top or bottom hole. You should assume the other holes are blocked to carry out this analysis.

Determine the length of the arrow that represents the path for the photon that goes through the center hole, when the other two holes are blocked.

Assuming the experiment is unwatched, and there are positions for the detector where all possible arrangements between the arrows are possible, determine the largest probability for the photon to leave the source and hit the detector, now with all three holes open. Under the same assumptions, what is the minimal probability?

3.4.5 Three Slits and One Detector I

For this problem, consider the slits arranged as in the previous problem (Problem 3.4.4). Now, a detector is also placed at the top slit that works 100% of the time, as shown in Fig. 3.24. What is the maximum probability for a photon to be detected at the screen now? Under the same assumptions, what is the minimal probability?

3.4.6 Three Slits and One Detector II

For this problem, consider the slits arranged as in Problem 3.4.4. The detector is now moved from the top slit to the middle slit (see Fig. 3.25). What is the maximal probability now? And for the same conditions, what is the minimal probability?

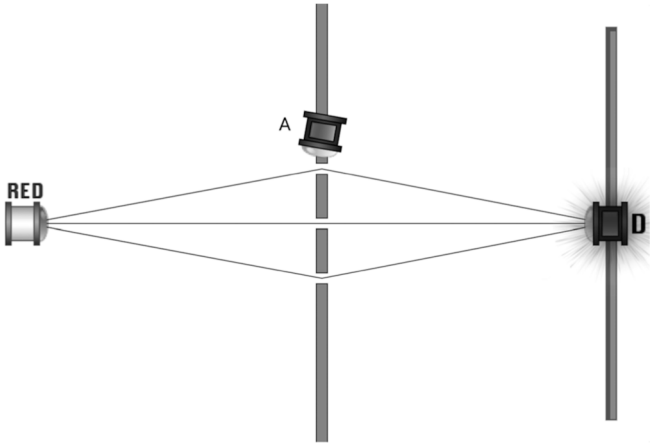

Fig. 3.24 Three-slit detector with a detector (100% efficient) watching at the top slit

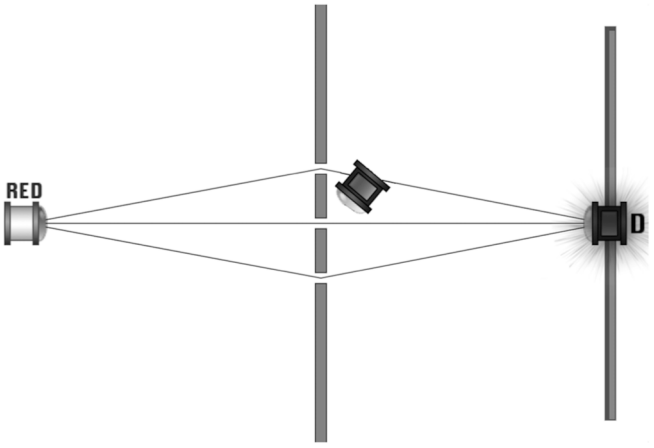

Fig. 3.25 Three-slit detector with a detector (100% efficient) watching at the central slit

We have a video https://tinyurl.com/34p4e6rd and a tutorial https://bit.ly/41pRoYE that allow you to explore the three-slit case further.

3.5 Applications of the Quantum Theory of Light

We will now describe some well-known and maybe not so well-known properties of photons. The first discusses the law of reflection, which states that the angle of incidence of the light ray is equal to the angle of reflection of the light ray when reflecting off a mirror.

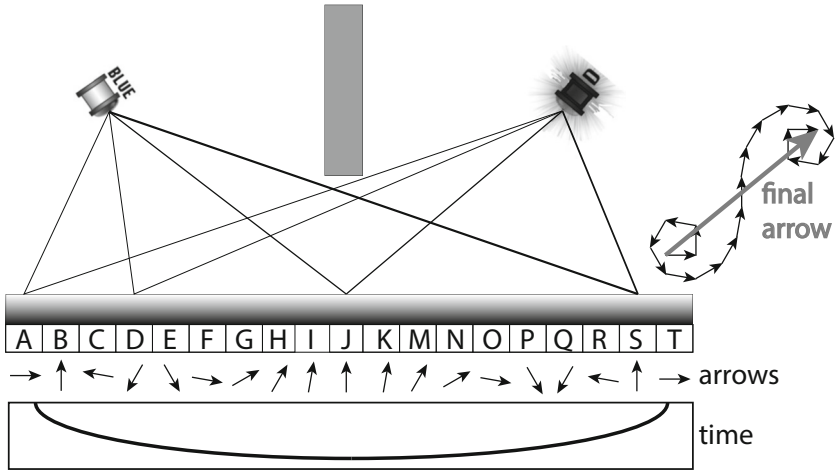

Fig. 3.26 Alternative ways for light to reflect off a mirror. We illustrate the monochromatic source of blue light on the left, a barrier to block light from going straight to the detector in the center, and a single-photon detector to the right. The mirror is divided into different sections. Four paths from source to detector are illustrated, with lines of increasing thickness; these paths start at the source "touch the mirror" on the marked section and then go to the detector. Beneath the mirror, we show the time it takes for the different paths and the arrows we assign for the alternative ways to go from the source to the detector (such as Source-J-Detector, with an arrow pointing at noon). The arrows are added to the right. One can see that far from the center of the mirror, the arrows tend to spin around in place, but near the middle, the arrows accumulate most of the result of the final arrow. But it is not correct to say that the light reflects *only* off the center of the mirror (angle of incidence is equal to angle of reflection)

Our goal is to develop the quantum model that describes reflection. We illustrate the basic principles in Fig. 3.26 using a somewhat simplified description. Our quantum rules tell us that we must first identify the events. There is only one event here, light leaves the source and is detected in the detector, but there are many (really an infinite number) of different ways to do this. We simplify the problem by dividing the mirror up into 19 different sections marked with letters from A to T. Each alternative way corresponds to the light leaving the source, hitting the mirror in a given marked section, and then going to the detector (we use straight-line paths from source to mirror and from mirror to detector). If you learned about reflection in high school, you might have learned the law of reflection that *the angle of incidence is equal to the angle of reflection*. This law would say that light reflects only off section J of the mirror in Fig. 3.26. Our quantum theory is more egalitarian. It says the light can go on all alternative paths, meaning the light can hit off any of the 19 sections of the mirror. Yet, the two descriptions end up being nearly identical. Here is why.

In our rules, we assign an arrow for each alternative way. The arrow simply rotates when the photon travels through air. When it reflects off the mirror surface, we add 6 hours to the clock; we also need to determine the length of the arrow, which should be 1, since no light is absorbed or transmitted by the mirror. The fact

that the total length of the sum of the arrows will then end up being larger than 1 will be clarified when we discuss normalization at the end of the chapter. The panel at the bottom of the figure shows the time it takes as a function of the sector of the mirror that the photon hits. Clearly J is the shortest path. I and K are nearly as short, followed by H and M and so on. So the times for I and K are slightly longer than J, more so for H and M, and similarly for the other sectors. As we move farther from the center sector, the time grows more rapidly and so the arrows for neighboring alternative ways deviate from each other more and more. As we hook the arrows head to tail, starting from sector A, we see the image to the right-hand side of the figure. In the regions far from the center, the nearby arrows spin and go nowhere, while near the center the nearby arrows are about the same direction so they add up to give us most of the final arrow. This is why we can say the angle of incidence is equal to the angle of reflection—the paths contributing the most are centered primarily near the central sector. But the paths nearby contribute a bit as well, similar to what we saw with light traveling through slits. I must point out that we did a simplification for the figure. Because blue light rotates about 72,000 times per inch of travel, to achieve the rotation of arrows as we depict in the figure, we must have sectors whose width is on the order of microns to be consistent. This is why the angle of incidence is equal to the angle of reflection. The J sector in the center must be very narrow. The video Photon Self Reflection https://tinyurl.com/4pkhe7t9 shows how this works.

We can actually show that light from a sector far from the center, say from sector C, also contributes to the reflection. To see how this works, we need to look at sector C close up, where we examine eight subregions with thicknesses on the order of microns in size (see Fig. 3.27). One can see that the time increases rapidly for neighboring subregions. This is why the arrows spin and go nowhere. Here is the clever idea—what if we scrape the mirror so that the arrows that point down or to the left are removed? This leaves just arrows that point up or to the right. Adding together the arrows that point only up or to the right means that our selection process will have the sum of the remaining arrows contribute and no longer spin in place. They add together to create a large final arrow. This means the blue light reflects strongly off of the specially scraped mirror! What happens if we change the color of the light from blue to green, or red? These scrapings are no longer set up to exclude arrows that point down and to the left, so the arrows spin and go nowhere, and the light no longer reflects. But we can make the light reflect again by simply changing the angle to the detector. Hence, the diffraction grating will change white light into a rainbow because different colors reflect off the diffraction grating at different angles. You can see this with a compact disc if you can find one. They are nice diffraction gratings, which easily show the rainbow effect (DVDs also show this, but the effect is not as easy to see, because their line spacing is too narrow).

The technology that can scrape the mirror to form a diffraction grating is nontrivial. While the idea for diffraction gratings dates back to the time of Newton and was related to the discussions of the prism (which also separated light into a rainbow of colors), it was not until engineering had progressed to the point where tens of thousands to a hundred thousand lines per inch could be scraped off a

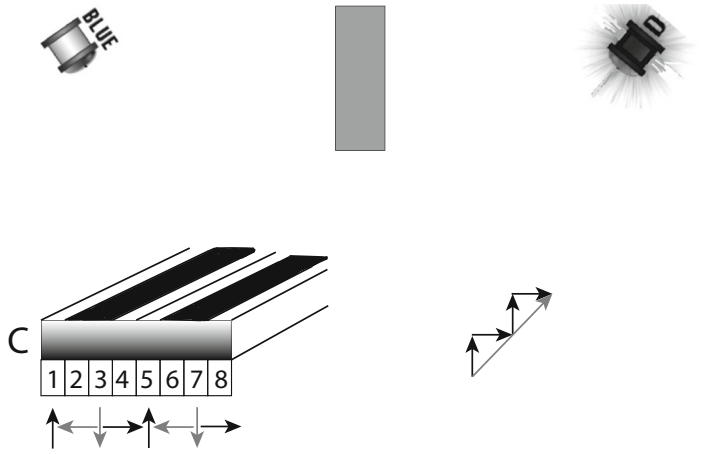

Fig. 3.27 Illustration of how a diffraction grating works. If we blow up sector C on the mirror, we see that the contributions from all arrows will just "spin and go nowhere." But scraping the mirror surface, where the arrows point down or to the left (in subregions 2, 3, 6, and 7), results in the remaining arrows adding together in such a way that we obtain a sizable final arrow. This means the piece of the mirror C will reflect light when it is made into a diffraction grating

mirror surface (in the mid to late 1800s) that they began to be used in science. A high-quality diffraction grating then allowed one to create an instrument called a spectrometer, which essentially shines light onto a diffraction grating and measures the different angles at which the light is strongly reflected. After analyzing some simple geometry, one can then determine the wavelength of the photon (distance it travels for the clock to turn one revolution) by measuring the angle that the light comes off the grating. It was soon found that the elements could be classified according to the discrete lines of color they gave off when they were excited (by directly burning them or by accelerating them to high speed in high-voltage vacuum tubes, and so on). So, the spectral lines of all elements were categorized and tabulated.

When the spectrometer was focused on light from the sun, it was found that there were spectral lines that had not been seen before. This led to the joint discovery of a new element called helium (Helios being the Greek sun god) by British astronomer Norman Lockyer and French astronomer Pierre Janssen (see Fig. 3.28) in 1868. The quantum theory of these spectral lines is an interesting story and we will discuss it more completely when we work on explaining properties of hydrogen. I am blown away by the fact that we discovered an element first on the sun and only later found it on earth! This is an amazing achievement of human intellect.

In the United States, Henry Rowland from Johns Hopkins University was renowned for being the best at making precision diffraction gratings and using them for many applications in astronomy (see Fig. 3.28). He was one of the first leaders of physics in the United States and became the first President of the American Physical Society.

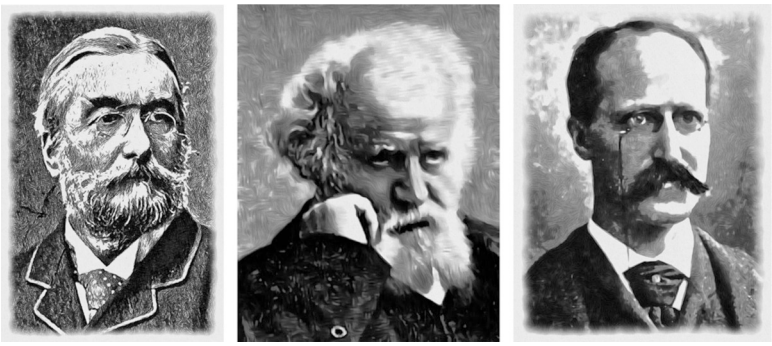

Fig. 3.28 (Left) British astronomer Norman Lockyer, (center) French astronomer Pierre Janssen, and (right) American physicist Henry Rowland. Lockyer and Janssen discovered the element helium in the spectra from the sun. Rowland constructed accurate diffraction gratings for use in spectroscopes (All images from Wikipedia (public domain) and modified to look like sketches by James Freericks)

The colors on a butterfly's wings and bird feathers come from a combination of diffraction and iridescence from thin films made from the cells in the wings themselves. You can even see the colors of butterfly wings change as you shift your head to change the angle from the wing to your eye. Even the animal kingdom uses quantum mechanics!

We now work on describing one final quantum object, the lens. Recall when we looked at light traveling through a single slit, it was the light that went on paths closest to a straight line from the source to the detector that contributed the most to the pattern on the screen. These paths were also the paths that had the shortest distance, or the least time. Hence, we could say light travels in straight lines from a source to a detector. But what would happen if we could slow down the photons traveling close to those straight-line paths, so they took the same time as light traveling on paths that took a longer distance? We could then create a situation where many paths all took the same time, so all the arrows point in the same direction. This is the idea behind a lens.

So how do we slow down photons? You may have learned about Einstein's theory of special relativity, which says the speed of light is a constant, so photons cannot be slowed down! But, indeed, we already know that when photons travel through different materials, they move more slowly. For example, photons move about 50% slower in glass than air. So, if we carefully engineer a sheet of glass to be thicker in the center and thinner at the edges, we can slow the photons down so they take the same time for all paths, and we get a very bright light at the detector. This is how a lens focuses light. (So what was wrong with the relativity argument? Relativity says the speed of light *in a vacuum* is a constant value; it says nothing about the speed of light in glass, or some other medium.)

Our theory for lenses is depicted graphically in Fig. 3.29. When we look at all of the different alternative ways to go from the source to the detector (when the detector

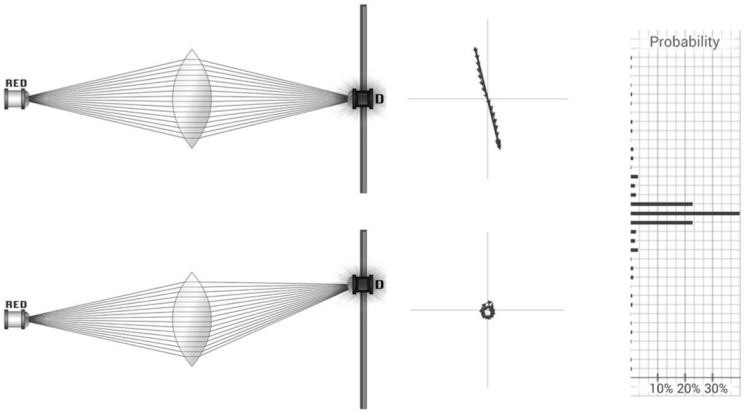

Fig. 3.29 Schematic of how we describe a lens with a detector at the focus (top) and away from the focus (bottom). Note how the lens has the thickness of the glass in it adjusted so that *all* arrows line up in essentially the same direction yielding a large probability. Away from the focus, we find the arrows spin and go nowhere. These results are summarized in the probability plot to the right

is at the focus, in the top panel), we see that the times for all of the alternative ways are essentially the same. This means all arrows point the same way and hence we obtain a huge response. The lens is constructed to provide precisely the delay needed to make the arrows for the different alternative ways all align. This is why it has the shape shown in the figure.

Now, if we move off the focus, we find the times for the different alternative ways are quite different. This means we have arrows that spin and go nowhere. The probability to detect the light in these regions is quite small. The video Through the Looking Glass https://tinyurl.com/4heremhy and a companion tutorial https://bit.ly/3YtZJJb provide additional material on how this works.

The first commercial use of glass lenses occurred in the late thirteenth century in Italy with the creation of eyeglasses. But early lenses often distorted the different colors, because, as we know from our quantum theory of light, all colors can participate and we have to carefully engineer the lens (and properly manufacture it) to ensure that we do not have this so-called chromatic distortion. Indeed, the field really took off when lenses that did not distort colors were invented (in the Netherlands during the late sixteenth century). It is there that the first true optics instruments—the microscope and the telescope—were invented. Galileo popularized the telescope's use and revolutionized our understanding of the solar system. You can even see a movie of how the sunspots that Galileo observed move from day to day. I was fortunate to see a movie of this at a course taught by Edward Tufte. It is well worth looking at.

We end the chapter with one important technical discussion about our calculations—it is called *normalization*. Normalization is an issue we swept under the rug earlier in the book. The basic idea behind it is simple: the total probability

for all possible events must be equal to 1 (or, equivalently, 100%) because nothing else can occur, so this includes all possibilities. This fundamental fact must be incorporated into the description of quantum mechanics. Earlier in this chapter, we would draw a set of lines representing alternative ways that a single event could occur. In doing so, we did not pay attention to the fact that by doubling the number of lines, it would appear that our probability increases by almost a factor of four! *When we employ normalization, this increase in probability does not occur.* Instead, the probability for the event is changed, but only slightly. In reality, we need to continue this process, by adding more and more paths until we can take the limit of an infinite number of paths and arrive at the correct final answer. While this may sound intimidating, we have already done something like this in our discussion of multiple reflections and analysis of geometric series. The procedure for other quantum problems is very similar to what we did when we discussed multiple reflections.

This is not something to be frightened by. Here is how normalization works. If we have a situation where we have a large number of arrows, then normalization is taken into account by computing the probability from the relative length of the final arrow. We first compute the maximum length, by lining all arrows up in the same direction. The relative length is then given by the length computed by adding the arrows for the alternative ways head to tail and then dividing the length of the final arrow by the length of the arrow we computed when we lined all arrows in the same direction. This relative length is always between zero and one and when squared it will give a probability that also lies between zero and one. It is a simple fix to this issue that we ignored previously. We will discuss normalization more thoroughly when we discuss wavefunctions later in the book. This may look to you as something we just added in at the end to fix things up. We did. But, it is not a trick. It is just fixing something we ignored earlier so as to not overload you at the beginning of our treatment of this material.

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Chapter 4
Advanced Quantum Mechanics of Light

It's not dark yet, but it's getting there . . .

Bob Dylan, Singer, Songwriter, Nobel Laureate, 1997

We provide a video overview of Chap. 4 at https://tinyurl.com/3d8vcd3r.

4.1 Interaction-Free Measurement with a Two-Slit Experiment

Psychologists Christopher Chabris and Daniel Simons created a video that they call the selective attention test. If you have never seen it, go watch the selective attention test video https://tinyurl.com/37cw6ndr before reading any further; when I teach this material to students, I always show the video first. Chabris and Simons have even written a book about this Chabris and Simons (2010).

In this video, the viewer is asked to count the number of passes in a basketball game. The video ends by telling people the correct answer and then ask, "did you see the gorilla?" Indeed, a person wearing a gorilla suit strolled through the video, yet most people do not see him. This phenomena can be called *looking without seeing*. In quantum mechanics, we study the converse, *seeing without looking*, which is a very powerful idea that invokes the mysterious behavior of quantum superposition in a remarkable way.

Let's first describe how we actually see anything. In conventional terms, we illuminate something with light. The light, which is composed of many quantum particles called photons, bounces or reflects off of the object and is scattered. We collect some of these photons in detectors (like a camera or our eyes) and construct an image from the number of photons that strike the detector and their different colors.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/978-3-031-87845-9_4.

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https://doi.org/10.1007/978-3-031-87845-9_4

When described in this light, it seems completely obvious that the statement made by Dennis Gabor back in 1961 (1971 Nobel prize winner for inventing the hologram) must be correct: "No observation can be made with less than one photon passing through the object" (Gabor, 1961). We loosely translate this as, "You need at least one photon to interact with an object in order to see it."

Shockingly, quantum interference allows for this idea to be overturned. Indeed, we can see things with quantum particles that do not interact with the object we are looking at. Strange as this may sound, we spend most of the rest of this chapter showing how this can be done and how to make it very efficient.

The idea behind this began with the thought experiment of a superbomb that could be triggered by just one photon hitting it. Could one ever tell whether such a bomb was in a suitcase or not? This was the question posed by Avshalom Elitzur and Lev Vaidman (1993). It began the idea of "interaction-free experiments," or "quantum seeing in the dark," or "seeing without looking." Enter Paul Kwiat, Harald Weinfurter, and Anton Zeilinger, who carried out the first interaction-free experiment and saw an object without having even one photon interact with it. We will discuss their experiment soon, but first we illustrate the idea using the two-slit experiment.

The superbomb and trying to observe it is closely related to the concept of Schrödinger's cat, where a cat is put in a box that has radioactive material and a quantum trigger that will open a vial of cyanide that will kill the cat if just one atom decays. Both of these thought experiments have a quantum trigger that will do something bad. For the superbomb, it will explode; for Schrödinger's cat, it will kill the cat. But they illustrate different concepts. For the superbomb we are interested in an interaction-free measurement, while for Schrödinger's cat we examine superposition states of macroscopic objects. We won't be discussing Schrödinger's cat further here. For me, the cat problem is more a philosophical one than a physics one. The exciting physics lies in the interaction-free measurement as we will soon see.

Before moving further, let's pause to recall the quote from Richard Feynman: *Any other situation in quantum mechanics, it turns out, can be explained by saying, "You remember the case of the experiment with the two holes? It's the same thing"* (Feynman, 1965). We now illustrate the interaction-free measurement using this two-slit experiment. If you don't feel like you have mastered the two-slit experiment and the quantum mystery, please return to Chap. 3 and review it again. It is critical that you understand this in order to be able to generalize that behavior to the interaction-free measurement. Interaction-free measurements are frankly some of the most bizarre behaviors in quantum mechanics. They allow us to detect something by not interacting with it, which is completely counterintuitive. But don't worry, we will parse it out carefully for you so you will become an expert on this.

The two-slit experimental apparatus should be familiar to you—we have a source of single color light to the left, a screen with two holes in the middle, and a screen to the right, where we detect the light (see Fig. 3.17). In Fig. 4.1, we show the probabilities that we found for the case where we have both slits open (left) and either only the top slit open (middle) or the bottom slit open (right). One can see

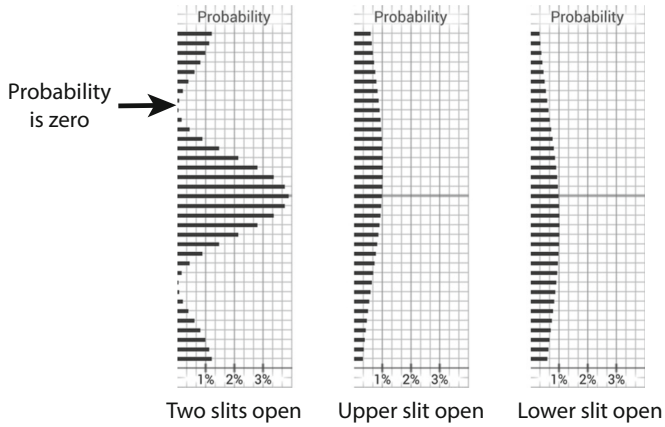

Fig. 4.1 Probability distribution curves needed to analyze an interaction-free measurement with a two-slit experiment. The probability for a successful detection is low, as described in the text, but it does work

the case with two slits open depicts quantum interference effects absent in the cases with just a single slit open. We also have highlighted a position on the detection screen where the probability to detect a photon going through two slits is exactly zero. Now, if we compare the probability for detection at that point for the three different cases, we see the probability is zero to see a photon if both slits are open, but is nonzero if one of the slits is closed. This means, if we put a detector there and send just one photon through the device, we have the opportunity to detect the presence of a blockage in front of one of the slits. This is made possible because if there is a blockage, and we are lucky enough, we will detect the photon in our detector. The chance for this to occur is small—on the order of one out of 100—but it is nonzero. When we do detect the blockage, we do so with a photon that goes through the slit that was not blocked! This means we see the blockage with a photon that did not interact with the blockage. This is what is meant by an interaction-free measurement. While the explanation of how to employ quantum interference to see without looking is relatively simple, we do not know how the particle does this without interacting with the blockage. This is a secret of nature that has not yet been revealed! We have a video https://tinyurl.com/yzk3bwmy and tutorial https://bit.ly/4pGIK2E that explore these ideas further.

The key for interaction-free measurements is that we go from a situation in which photons never enter the detector to a situation in which—when one slit is blocked—there is a chance that a photon will enter the detector and be observed. Hence, when we see a photon, we know there is a blockage with complete certainty! The fundamental requirement to create a "seeing-in-the-dark" experiment is that we have a situation where the probability is zero for detection when the event does not occur and is nonzero when it does occur. Then sending in a single quantum (such as one photon) can be sufficient to detect the blockage. The key question remains—how high a probability for a successful detection can we achieve?

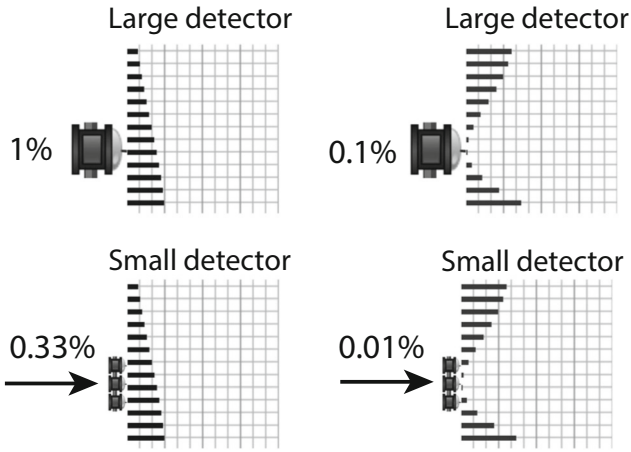

Fig. 4.2 Scale-up of the region near probability zero. In the upper panel, we show how a large detector would work. In the lower panel, how a detector one-third as large would work; the arrow points to the detector that would be used. On the left we have the patterns with one slit open, on the right, with two slits open

It is worthwhile to pause for a moment and examine the efficiency of this experiment a bit more realistically than what we described above, which corresponded to an ideal case. In Fig. 4.2, we show a blowup of the probability curves for the single-slit (left) and two-slit (right) cases with a large detector (top) and small detector (bottom). This figure emphasizes that the detector has a physical detection window of a finite size. First consider the case where there is nothing blocking either slit. Then the probability for the two-slit case goes to zero at one point only, not over a region of finite width. So how does this affect the experiment? Let us first consider the case of a large detector for the single slit case (top left). As we know, in this case, the detector registers 1%. If I replace it by three detectors, each with a detection region one-third as large (bottom left), then the probability for detection in each of those will be one-third as large or 0.33%, because the probability curve is nearly constant, and we are dividing the signal of the large detector among the three smaller detectors—every photon detected by the large detector must be detected by one of the three smaller detectors. Now, the case for the two-slit experiment with a large detector (top right) is expected to be zero, but it is not. It will have a small percentage, found by summing the probability results over the width of the detector. Let us assume this value is one-tenth as large or 0.1%. If we instead use one smaller detector (bottom right), we see that detector will have a percentage smaller than one-third of 0.1%, because the probability curve is not uniform here—it is larger for the two outer detectors. Let us say it is instead going to be 0.01%.

We can now compare the two scenarios. If we use the large detector, we will be successful in detecting a blockage 1% of the time, but we will have a false negative 0.1% of the time (when the two slits are open, but we detect a photon

anyway). We are ten times more likely to detect the blockage than to falsely predict blockage when it is not there. But, we have a huge number of false negatives—this occurs when there is a blockage, but the photon does not hit our detector—it instead hits somewhere else on the screen. If we go to the smaller detectors, our false-negative probability increases, but then we are now 33 times more likely to detect the blockage than have a false positive. It seems like there is no clear way to make these experiments highly efficient at detecting the blockage (and not having false negatives or positives).

Worse, a realistic experiment is even more complicated. Real detectors do not trigger all the time, they have some efficiency to them. Real detectors also will detect photons when they are not there (called a dark current or dark counts). This implies the experiments can never be perfect. This is simply a fact of life. Our goal is to make them work as well as we can, and analyze the results in this light. The video Anything You Can Do I Can Do Better https://tinyurl.com/57jmtpzr discusses these ideas further.

Let us unpack the methodology for an interaction-free measurement a bit further. In the experiment, we are contrasting a system with quantum interference to one without. The probability for the event to occur when we have interference must be exactly zero, and the probability without interference must be nonzero. The hard problem we now face is to determine how to improve the efficiency, or in other words, how to make the probability without interference to be as close to 1 as possible. If the efficiency of an interaction-free experiment is small, it is only an intellectual curiosity and something no one gets excited about. People start to notice if you can get it to work half of the time. Improve it to close to 100% of the time and many people take notice. So, we want to discuss issues related to efficiency and to limitations due to the finite size of a detector for the two-slit experiment version of the interaction-free measurement. We include some warm-up exercises on this topic next.

But, before getting to the exercises, we mention that there is a much better experimental apparatus to use. In Sect. 4.2, we describe the Mach-Zehnder interferometer, which can be used to greatly improve the efficiency. We will find that without doing anything fancy, we can make it work 25% of the time. Adding in advanced quantum-mechanical ideas, we will be able to increase this probability to be as large as we would like it to be.

Section 4.1 Problems:

4.1.1 Details of the Interaction-Free Measurement

The interaction-free measurement considers experiments where a single photon is shot through the device. Sometimes it will never hit the detector, but sometimes it will. The efficiency is how many times the blockage is successfully detected divided by the total number of tries to detect it with a single photon.

Write an essay that discusses issues related to efficiency of the experiment, considering the following points: (i) in Fig. 4.1, we pointed to one point where the detector can be placed where the probability is zero. But there is a second point as well. Is one location better than the other, or are the two equivalent? (ii) How does the experiment change if we include two detectors, one at each point where the probability is zero? Does the efficiency change or stay the same? (iii) Can you tell which slit is blocked, when only one slit is open? (iv) Can the maxima of the two-slit patterns be used instead of the minima for placement of a detector, if we want it to work by measuring a single photon?

Feel free to discuss other aspects about this experiment, especially any ideas that could improve the chance for success in detecting whether one slit is blocked or not.

4.2 The Mach-Zehnder Interferometer

It is clear we need to find another way. While we could understand the principle behind "quantum seeing in the dark," or an interaction-free measurement, by exploring it with the two-slit experiment, we found that the chance of the experiment to actually work was rather low. This problem cannot be fixed within the two-slit experiment. We must look elsewhere.

It turns out that two physicists come to the rescue. In the 1890s, both Ludwig Mach and Ludwig Zehnder created a device called an interferometer, which can be used to greatly improve the probability of success of a quantum-seeing-in-the-dark experiment. Ludwig Mach was the son of Ernst Mach (a very famous physicist and philosopher from the Vienna school) and is said to have lived a troubled life. Ludwig Zehnder was a Swiss physicist who studied under Roentgen (the discoverer of x-rays).

The Mach-Zehnder interferometer uses a new piece of equipment called a beam splitter. Before we describe it in great detail, know that you are already familiar with such a device. It is essentially a pair of partially reflecting sunglasses. When a photon is incident on the sunglasses lens, part of the time it will transmit through the glass, and part of the time it will reflect. We studied similar systems in great detail in Chap. 3. Suppose it was set up in such a way that the paths for the reflected light and the transmitted light differ by exactly 90° and that the probability for reflection (and transmission) was exactly 50%. Then you would have what we call a *balanced beam splitter*, which is depicted for you in the example of mirrored sunglasses in Fig. 4.3.

Beam splitters are different from what we worked with before and the quantum rules for them are modified; the shrink factor is $1/\sqrt{2}$ for both reflection and transmission rather than the values we used before. A graphic of the beam splitter is shown in Fig. 4.4. Note that after the beam splitter, the photon travels in superposition along both output port paths (or equivalently, its output path is indeterminate), until it is detected. The beam splitter creates a superposition—it does not perform a measurement.

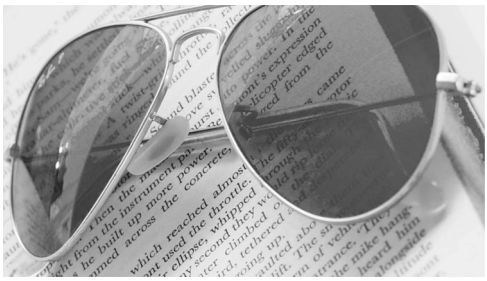

Fig. 4.3 Mirrored sunglasses, showing how a beam splitter transmits and reflects light approximately evenly (source: Jess Bailey Designs from pexels.com, converted to black and white and cropped)

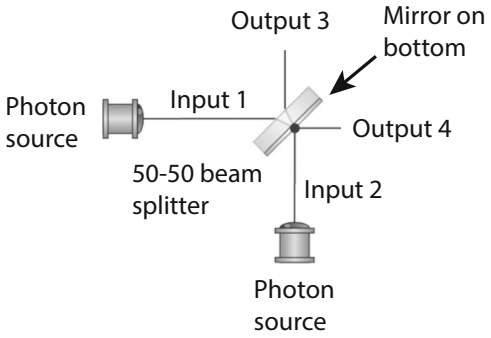

Fig. 4.4 Graphic of a beam splitter showing the configuration where the light beam reflects off the bottom surface (which does not add 6 hours to the clock); if we flip the beam splitter, so it reflects off the top mirrored surface, then we add 6 hours to the clock upon reflection (determined by the input and output ports). The beam splitter has four ports: two input ports and two output ports. In some cases only one input port to the beam splitter is used

It is important to note that the beam splitter has two possible orientations of its mirrored surface. The first is as depicted above, where the light reflects off the bottom surface (from glass) when incident in input port 1. In this case, we do not add 6 hours to the clock when we construct the arrow for the alternative way a photon can move through the device; note that the top surface of the glass has an antireflective coating on it. If, on the other hand, the beam splitter was flipped, so the reflection came off the mirror from air, then we would add 6 hours to the clock, just as we did before. This difference is critical for understanding how the Mach-Zehnder interferometer works.

Consider a single-photon light source shining light into input port 1, and two single-photon detectors on output ports 3 and 4, respectively. Since the beam splitter separates the light equally, we have a 50–50 chance to be detected in either detector. If we repeat the experiment many times, we will find half the time the photon hits the detector on output port 3 and half the time on output port 4. Exactly which port it hits at any given moment is a random event, and we cannot predict it. *This is one of the simplest ways to see the intrinsic randomness in quantum mechanics.*

The diagram in Fig. 4.5 shows a schematic of the Mach-Zehnder interferometer. Similar to the two-slit experiment, the photon path is indeterminate, because it

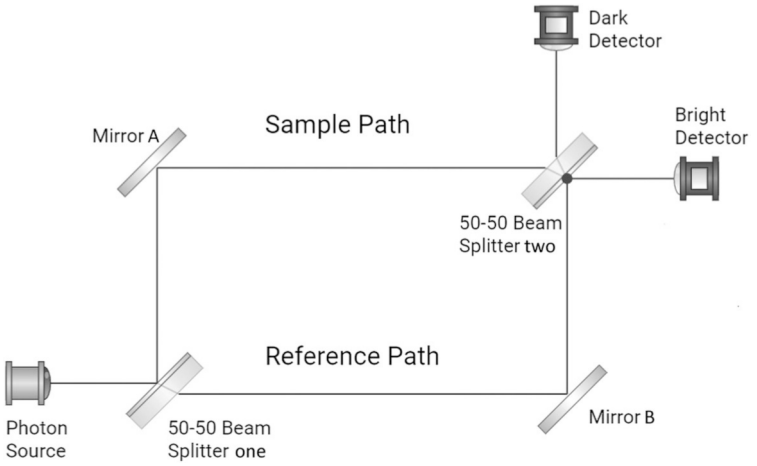

Fig. 4.5 Schematic of a Mach-Zehnder interferometer. The photon source is on the lower left. It shines the light into one input port of the lower left beam splitter, which equally splits the light into two paths—the sample path to the top and the reference path to the bottom. On each path, the photon reflects once and then enters the second beam splitter from both input ports. From there, we attach photon detectors to both output port paths

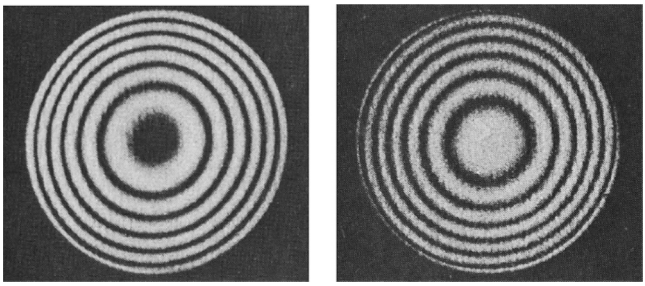

Fig. 4.6 Interference fringes for the bright and dark detectors (from Feinberg, 1983 with permission, © the Optical Society of America, now Optica). Note how the patterns are complementary to each other—they anticorrelate, so when one is bright the other is dark and *vice versa*. We name the detectors via the interference pattern in the center. So the detector that sits at the center of the pattern on the left is the dark detector and the detector that sits at the center of the pattern on the right is the bright detector

moves in superposition. The probabilities at each detector are anticorrelated; when the photon has high probability on one detector, it is low probability on the other detector. For example, the picture of the interference fringes at each detector is shown in Fig. 4.6.

To run an experiment, we put one small detector in the center of each interference pattern, so one detector is "bright" and the other is "dark." Note that, because there are only two possible paths, we expect the quantum-seeing-in-the-dark experiment to be more efficient than with the two-slit experiment, where we had many different

paths. The video Entangling at a Distance, Part I https://tinyurl.com/2jyrp48a and a tutorial https://bit.ly/4mlSBK1 provide additional resources.

Before we get to a seeing-in-the-dark experiment, let us first understand how the Mach-Zehnder interferometer works within our quantum model. We will analyze two events: (i) the photon leaves the source and is detected in the dark detector and (ii) the photon leaves the source and is detected in the bright detector. For detection in the dark detector, we have two alternative ways—one that reflects off the first beam splitter (BS1), travels down the sample path (reflecting off mirror A), and then reflects off the final beam splitter (BS2) and the other, which transmits through the first beam splitter (BS1), travels along the reference path (reflecting off mirror B), and then transmits through the final beam splitter (BS2). For now, assume that the path lengths of the sample and reference paths are exactly the same. As the photon travels along the sample path, we accumulate time along the path, plus we add 6 hours for every reflection coming from air. There are two such reflections—one from the first beam splitter (BS1) and one from mirror A. The second beam splitter (BS2) has a reflection from the glass side, so we add no extra time. We also have to determine the shrink factors. Each beam splitter shrinks by $1/\sqrt{2}$. The mirrors are perfect, so they have no shrinkage. Putting this all together, we have an arrow of length 0.5, pointing in a direction given by the time to run along the sample path plus 12 hours. Next, we analyze the reference path. Here, we have the time to run along the reference path, plus an additional 6 hours for reflection off mirror B. The shrink factors are again $1/\sqrt{2}$ for each beam splitter, so the length of the arrow is 0.5. The final arrow points in the direction given by the time to run along the reference path plus 6 hours. Since we assume the time along the sample path and the reference path is identical, the arrows will be pointed opposite to each other (because of the difference by 6 hours) and will cancel. So, we obtain 0% detection in the dark detector. This is why we call it the dark detector.

We can perform a similar analysis for the bright detector. There are two alternative ways here as well—one for traveling along the sample path and one for traveling along the reference path. For the sample path, we have one reflection from air on the first beam splitter (BS1, 6 hours) and one from mirror A (6 hours). For the reference path, we have one reflection off mirror B (6 hours) and one reflection off the second beam splitter (BS2) from air (6 hours). Hence, for both paths, we have the total time along each path plus 12 hours—the two arrows align in the same direction. Since each path involves going through two beam splitters, the arrow length is 0.5 for each arrow as well. Hence, they add together to yield a final arrow of length 1, or we have 100% of the photons arriving at the bright detector. Hence we understand its name as well.

We can modify the time it takes to go along the sample path by adding something that delays the photon along that path. The simplest thing to use is glass, because light travels slower through glass than through air. By varying the thickness of the glass, we can adjust the path length. When we go back to our calculations with the arrows, we see the arrow from the reference path is unchanged, but from the sample path, we have an extra winding of the arrow given by the extra time to go through the extra glass. This then changes the final arrow—the probability for detection in

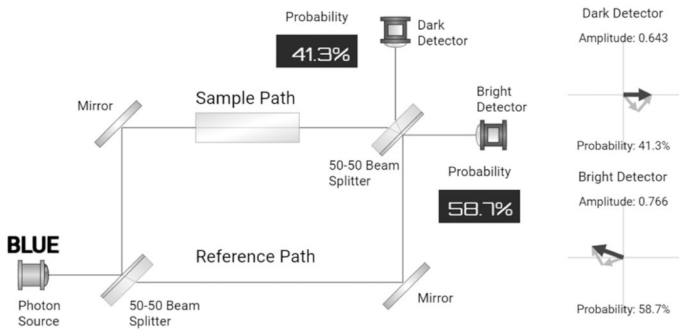

Fig. 4.7 Mach-Zehnder interferometer with an extra piece of glass along the sample path to add time for the photon traveling along that path. To the right, we show the arrows for each alternative way and the final arrow, which is squared to give the final probability. Recall that the reference path arrow is unchanged, but the sample path arrow rotates an additional amount given by the thickness of the glass

the dark detector increases and in the bright detector decreases, until they switch roles, and we have all detection in the dark detector and none in the bright. Make the glass even thicker, and they evolve back toward their original state, with the detection in the dark detector becoming less and in the bright detector becoming more. An example for how this works is shown in Fig. 4.7. Note that for the events of detection in each detector, we have the photon move in superposition, so the path taken is indeterminate.

Note that something interesting happens when the sample path adds exactly 12 hours to the clock for the photon traveling along that path. In this case, we revert back to the dark detector being dark and the bright detector being bright. Hence, we cannot tell the difference between this situation, and the situation where the path lengths along the interferometer are exactly the same for the two paths. Indeed, interferometers are most useful to compare lengths and find small differences (on the order of 10s–100s of nanometers). But they only measure length differences in fractions of a wavelength—if the path difference is a multiple of a wavelength, one cannot tell that there is any length difference. The video Entangling at a Distance, Part II https://tinyurl.com/mrxms4fd and tutorial https://bit.ly/4pzQDqz provide additional information about the interferometer.

Interesting things happen if we change the color of the source. Suppose that the interferometer has a length difference that is exactly a multiple of the wavelength of a blue light source. Change the light to green, and it will no longer exactly match, so we will now register a difference in length. We would need to adjust the sample path length to make the path length difference a multiple of the green-light wavelength. This change of the effective length with a change in color of the source provides some interesting new experiments we can try; in particular, it provides a possibility to determine when the sample and reference paths are identical. You have an essay that explores these ideas further as one of the problems at the end of the section.

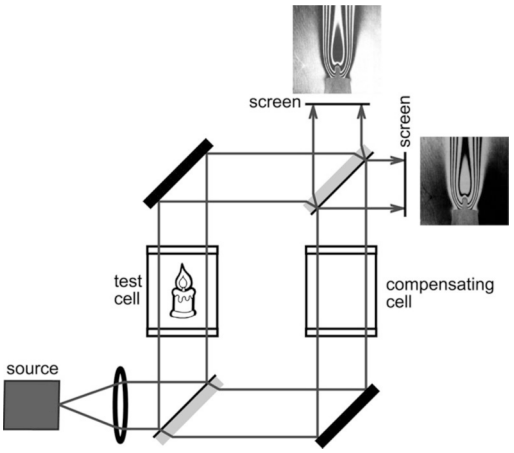

Fig. 4.8 Example of how a Mach-Zehnder interferometer can be used to image objects that have differing densities, here a candle flame (From wikipedia, courtesy of Stigmatella aurantiaca with modification by Kid222r (Creative Commons Attribution-Share Alike 3.0 Unported license))

We have seen how a Mach-Zehnder interferometer can be used to measure distances precisely down to lengths that are a fraction of a micron (a millionth of a meter). Unfortunately, the accuracy is up to an integer multiplied by the wavelength of the light source. While we often talk about the two arms of the interferometer as having equal path lengths, they usually differ by some integer multiple of the wavelength. Interestingly, this causes no problems to the functioning of the interferometer! As an example of an application of the Mach-Zehnder interferometer, see Fig. 4.8. We will explore the use of the interferometer in the laser interferometer gravitational wave observatory later.

One of the most famous experiments ever performed with an interferometer was done by two American physicists Albert Michelson and Edward Morley in the 1880s. Their experiment showed that one cannot measure the presence of the luminiferous aether. You probably have never heard of such a thing because we hardly teach it any more. However, in the 1800s, it was reasoned that, if light was a wave, then it must be the vibrations of some medium, just like water is the medium for water waves when we drop a pebble in a pond. If such an aether exists, then the Earth must move through it at some time during the year, because we orbit the Sun in a circle and move in all different directions as we go through the annual orbit. So, using an interferometer, they decided to measure the motion of the Earth relative to the aether. The Michelson-Morley interferometer has a different geometry than the Mach-Zehnder interferometer, which will be shown when we illustrate the quantum seeing-in-the-dark experiment later. Needless to say, it compares the distance along two different paths. By repeating the measurement over many different days of the year, Michelson and Morley were able to establish that there was no detectable motion through the aether wind. This result eventually led Einstein to postulate that the speed of light was the same in every frame of reference, to develop the special theory of relativity and ultimately to create the general theory of relativity (although it is not clear whether Einstein knew of this experiment or not).

We have learned about this new instrument called an interferometer, which can be employed to precisely compare the lengths of the paths a photon can travel along.

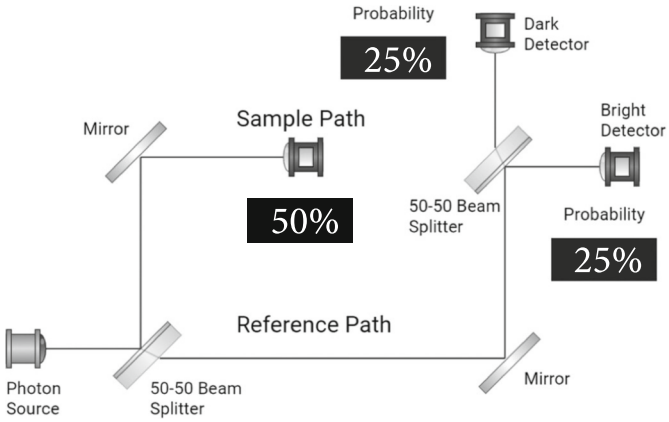

Fig. 4.9 Setup for a quantum-seeing-in-the-dark experiment using a Mach-Zehnder interferometer. We have a blockage (detector) on the sample path. Now there are three events—(i) detection on the sample path, (ii) detection in the dark detector, and (iii) detection in the bright detector. Each event has one way in which it occurs, but the probabilities are different—50% for detection on the sample path and 25% in the bright or dark detectors, respectively

Since it acts, in many respects, like a two-slit experiment, we now want to examine how a quantum-seeing-in-the-dark experiment would work with the interferometer.

The basic idea is that when the interferometer is properly set up, with the path lengths of the two paths equal up to a multiple of the wavelength, there are a dark detector and a bright detector. Focusing on the dark detector, if one of the paths is blocked, then a photon may now arrive there. When it does, we have achieved quantum seeing in the dark. The probability for this to occur is much higher than in the two-slit experiment example. We now discuss all of the details behind this.

The setup for a quantum-seeing-in-the-dark experiment with a Mach-Zehnder interferometer is shown in Fig. 4.9. It is the same as that of the regular Mach-Zehnder interferometer, except we may block the sample path, here showing the blockage via a detector. When there is no blockage, then there is a 0% chance to have the photon go to the dark detector, when the path lengths are adjusted so that the two paths differ by a multiple of the wavelength of the light from the light source. This case has been analyzed already within our quantum model. If there is a blockage on the sample path, then we have an experiment with three events: (i) detection in the detector on the sample path, (ii) detection in the dark detector, and (iii) detection in the bright detector. Each event can occur in only one way. For the detection in the sample path, the arrow has length $1/\sqrt{2}$ (probability of 50%) because the photon went through one beam splitter only. For the bright or dark detectors, the arrow length is $1/2$ (probability of 25%) because the light goes through two beam splitters along the paths. Hence, we observe at the dark detector. If there is no blockage, then we see nothing. If there is blockage, then we see a photon at the dark detector 25% of the time. This is a huge increase over the two-slit experiment example.

This setup for the Mach-Zehnder interferometer will be successful 25% of the time in detecting an obstruction without interacting with it. The video Half Full, or Half Empty https://tinyurl.com/nepmtzr4 and the tutorial https://bit.ly/48TAgxL give more practice in these ideas. Since we went from a huge number of path options in the two-slit experiment, to a minimum of two in the Mach-Zehnder interferometer, it seems unlikely we could improve on this further. But we can. The idea is that we "partially" perform the experiment a number of times in succession, with the failure rate quite small for each attempt. Then the total failure rate will end up being less than in the current case (this is because there is a nonlinear relation entering into the analysis, as we will soon see).

However, this requires us to manipulate a new property of light called polarization. Some of you may be familiar with polarization already. Some may never have heard of the idea. It makes no difference. After the next section, you will all be experts on the idea.

Section 4.2 Problems:

4.2.1 Changing Color in the Mach-Zehnder Interferometer

Suppose we have a Mach-Zehnder interferometer. Discuss the following scenarios in an essay on changing the color of the light. Suppose the interferometer has set a red color source and the path lengths have been adjusted so that the bright and dark detectors receive exactly 100% and 0%, respectively, of the photons that move through the interferometer. What is the difference in the path length between the two arms? If we change the color of the source from red to a different color, without changing the lengths of either of the paths, what will we observe in the bright and dark detectors? Can we determine which arm has the longer path length by changing the color of the source?

Suppose the Mach-Zehnder interferometer has its light source changed to blue, and now the bright detector detects exactly 0% and the dark detector detects exactly 100% of the photons. What is the difference in the path length between the two arms now?

Suppose the path lengths are exactly the same distance for the two arms of the interferometer. When the source is red light, then the bright detector registers 100% and the dark detector registers 0%. Change the color of the light source, keeping the path lengths the same. Do the percentages of photons entering the bright and dark detectors change?

Suppose the path lengths differ by a nonzero integer multiple of the wavelength of red light for the two arms of the interferometer. Then the bright detector registers 100% and the dark detector registers 0%. Change the color of the light source through all of the colors of the rainbow from red to orange to yellow to green to blue to violet, keeping the path lengths unchanged. Do the percentages of photons

entering the bright and dark detectors change? For this question, you can assume the difference in wavelength of red light versus blue light is *exactly* a factor of two.

4.2.2 Quantum Seeing in the Dark with a Mach-Zehnder Interferometer

Discuss the following two points about the quantum-seeing-in-the-dark experiment run with the Mach-Zehnder interferometer.

Suppose the obstruction can go on either path of the interferometer, namely, either the sample path on the top or the reference path on the bottom, but not both. Can we tell which path is blocked if we run the above experiment (which will be 25% efficient)?

Does the probability for success for the interaction-free experiment depend on the color of the light if the interferometer is tuned (for each color) to the correct dark and bright detector probabilities prior to the experiment?

4.2.3 Quantum Silhouette

You are asked to write an essay describing how one can measure a silhouette of a shape with photons that do not hit the shape being imaged. We will use the Mach-Zehnder interferometer, and assume that each time we try, we work with an idealized, perfect interaction-free experiment that will be 100% successful for each photon shot. Assume as well we have an apparatus, which can move the object up, down, left, and right, by exactly specified steps (say steps of 1 mm in size) and it is attached to a computer which remembers precisely what position it is in. The object can be moved into and out of the sample beam path of the interferometer.

In organizing your thinking, consider a systematic way to determine when the object is obstructing the path and when it is not.

Experiments similar to this have actually been performed.

4.3 Polarization of Light and the Quantum Zeno Effect

We start this section with a discussion of the concept of infinity, as it is often poorly taught and the ideas behind it are important for the quantum-seeing-in-the-dark experiment. Our story starts back with the Greek philosophers, in particular Zeno of Elea, who was one of the first philosophers to work with the concept of infinity. He got it wrong. But, looking at his ideas is quite instructive for understanding the concept.

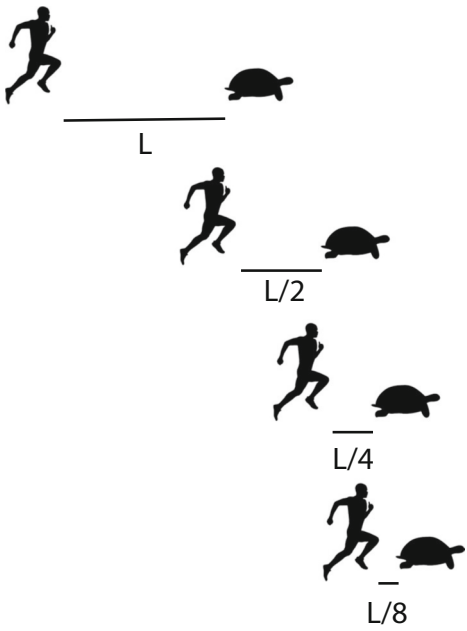

Fig. 4.10 Cartoon of Achilles and the tortoise in a race. Achilles starts a distance L away from the tortoise and then we look at snapshots where the distance is reduced in half during each interval. The total distance Achilles must travel is $2L$ to catch the tortoise (Clip art used with permission from clipartbest.com)

He reasoned that in an unfair race where one racer is given a head start, the person behind can never catch the person who is ahead. Of course, we know that this conclusion is completely false. In a race, the faster one will eventually catch the slower and pass him, if the race is long enough. But it is difficult to find a flaw in Zeno's logic, so we carefully go over it here. See Fig. 4.10 for an illustration of the race between Achilles and the tortoise.

Let's say we have Achilles racing a tortoise and the tortoise has a head start of $L = 100\,\text{m}$; Achilles will need to run 200 m to catch the tortoise. The way Zeno's argument goes is that, before Achilles can cover the full distance of 200 m, he must go 100 m or half the distance, then half the distance again (50 m), and so on and so on. But there are an infinite number of times we can cut the distance in half, so Zeno then argued that Achilles can never meet the position of the tortoise because it takes an infinite number of these steps. He went further to say the fact that we see Achilles catching up to the tortoise and passing it is because our senses cannot be trusted. Can you spot the error in the reasoning? The fallacy in this argument is the assumption that an infinite number of steps also requires an infinite amount of time. It doesn't! And that is why Achilles actually catches the tortoise, in agreement with our common sense. This is because each subsequent step is smaller, so they can be completed more rapidly.

This turns out to be an example of our geometric series again. Recall that we start with the distance of L and we reduce the distance in half with each step. Suppose the first step, which is a distance of L, takes an amount of time Δt. Then the total time is just the sum

$$t = \Delta t \left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots\right), \tag{4.1}$$

because each racer is traveling at constant velocity, so the time to travel half the distance is half the time. This is our old friend the geometric series with $x = 1/2$. We know how to sum it, so we find

$$t = \Delta t \sum_{n=0}^{\infty} \frac{1}{2^n} = \Delta t \frac{1}{1 - \frac{1}{2}} = 2\Delta t. \tag{4.2}$$

Hence, the total time is $t = 2\Delta t$, which should not be surprising, because we are traveling the total distance $2L$ at a constant speed, so if the time to travel half the distance took Δt, then the total time must be twice as much. Indeed, our ability to sum the geometric series shows that the infinite number of steps does take only a finite amount of time.

Zeno's paradox brings up the fundamental question: how do we understand infinity? To a mathematician, understanding different forms of infinity is a joy in the beauty of numbers. To most of the rest of us, it is a confusing conundrum. We will try to keep things simple enough in this book. We care most about situations where we can add together a finite number of terms and eventually reach very close to the infinite limit, like the geometric series example above. We often approximate the infinite limit by a sum over a finite number of terms, and this is good enough for most of our purposes. It makes the infinite limit easier to grasp and understand. The video Catch Me If You Can https://tinyurl.com/9tdecm2j provides some more context on this.

If you are interested in exploring the notion of infinity and other philosophical paradoxes, I highly recommend the book *On the Brink of Paradox* by Agustin Rayo (2019) (there is a MOOC on this as well called "Paradox and Infinity"). Some of that material can be at a high level, but I think everyone can understand Hilbert's hotel and be enriched.

Now that I have convinced you that Zeno was wrong, I will discuss a situation where he is right! There is a way in which the Zeno paradox holds and Achilles never catches the tortoise. It requires Achilles to slow down as they race each other. Then if he slows down at just the right rate, the race can take forever—Achilles never catches the tortoise.

Consider a situation where Achilles slows down as he chases the tortoise (and recall now that the tortoise is moving) such that at the start the distance between them is 100 m, after one time interval it is 50 m, after two intervals it is 33.33 m, after three it is 25 m, and so on. This means that after n intervals it is $100/(n+1)$ m away. The total distance Achilles must travel to catch the tortoise is the total of all of these intervals. This then results in what we call the harmonic series, given by

$$L_{\text{total}} = L\left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots\right) = L\sum_{n=0}^{\infty}\frac{1}{n}. \tag{4.3}$$

Even though this series looks very much like the geometric series, and each term decreases to zero as n is made large, the sum of the series diverges and goes to infinity. The proof, due to Nicole Oresme in 1360, is as follows:

$$L_{\text{total}} = L\left(1 + \underbrace{\frac{1}{2}} + \underbrace{\frac{1}{3} + \frac{1}{4}} + \underbrace{\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}} + \cdots\right) \tag{4.4}$$

$$> L\left(1 + \underbrace{\frac{1}{2}} + \underbrace{\frac{1}{4} + \frac{1}{4}} + \underbrace{\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}} + \cdots\right) \tag{4.5}$$

$$> L\left(1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \cdots\right) = \infty. \tag{4.6}$$

Oresme grouped all the terms from $1/(2^n + 1)$ to $1/(2^{n+1})$ together. Then noting that each of those groups was larger than $2^n \times 1/2^{n+1} = 1/2$, we can underestimate the contribution of those terms by replacing each of them by $1/2$. But we have an infinite number of these groups as $n \to \infty$, and the sum of an infinite number of $1/2$'s is infinity! Hence, Achilles never reaches the tortoise, because he has to run forever. In this case Zeno is right.

We now change the subject to discuss a property of light that we have not discussed yet—the property is called polarization. Zeno's ideas will come back to our discussions before too long.

If you study the classical theory for light, you come across the notion that light is composed of perpendicular electric and magnetic fields that oscillate. The direction that the electric field points in is called the polarization of the light. We have gone to great lengths in this class to teach you that a photon is a particle, not a wave; however, it is a quantum particle which behaves differently than a classical particle. Nevertheless, the quantum particle still has the property of polarization. You can think of this as similar to the spin degree of freedom we discussed with atoms in the first two chapters. But it does have a number of differences, which we will describe in detail.

The phenomenon of polarized light is quite common. It can be illustrated with our old friend the sunglasses again (see Fig. 4.3). Many sunglasses are polarized, because the glare that reflects off surfaces is often polarized, so polarized lenses tend to reduce the glare.

How do we think of this? Start with Fig. 4.11. A polarizer is somewhat like a picket fence (shown on the left), and the photon has a polarization direction it carries with it, like a little rod. If the rod points in the same direction as the picket fence, it goes through. If it is perpendicular to the pickets, it hits them and is stopped. (For other directions it partially goes through, as we describe later, so do not think of the rod as being rigid and getting stuck in the picket fence—this analogy is not perfect.)

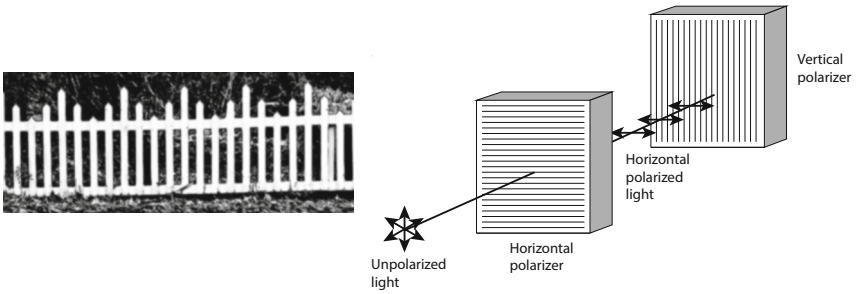

Fig. 4.11 (Left) Image of a picket fence. (Right) Schematic depicting how unpolarized light becomes polarized when passing through a polarizer and how a perpendicular polarizer lets no light through (Picket fence image from Scott Webb on pexels.com, modified to look like a sketch by James Freericks)

Hence, a polarizer is akin to a filter that filters out all light that is not polarized in the same direction, as depicted in the right panel.

Note that light that is polarized up goes through the same polarizer as light that is polarized down, so the direction of the polarization is neither up nor down but instead is something we call "up-down." Similarly, one cannot use a simple vector to describe it, which is an arrow (that would point up or down), but instead we use a line segment with no arrows, which we call a director. While it is true that this kind of looks like an arrow, it is different, which means we won't confuse it with the probability amplitude arrows we are so used to dealing with. So, what direction does a director point in? We would say in two directions. Both up and down, or both right and left, or both in and out. Even though we will be using a line segment with no arrows to represent the director, we could just as well have used a line segment with two arrows to represent it (as we did in the right panel of Fig. 4.11). In any case, we often think of the director as pointing in two directions.

I do want to point out the irony that here we are finding something classical that has two characteristics wrapped up in one, which is something we have become accustomed to in the quantum world. Indeed, asking what direction is the director pointing in is similar to asking what path does an atom take in a Stern-Gerlach analyzer or what path does a photon take in a Mach-Zehnder interferometer. It points in two directions at once!

Obviously, this is not enough for you to understand polarization, yet. We will now go through the concept carefully, step by step. The one thing to keep in mind if you do not know what an electric or magnetic field is is that there is some characteristic of light which points in two directions at once and we will be discussing how this affects the quantum properties of the photons. Keeping this in mind is all you need to move forward. Note further that we have discussed the magnetic field when we discussed the Stern-Gerlach experiment in Chap. 1. Feel free to review that now if you would like to.

Our first setup for experiments on polarizers is given in Fig. 4.12. In panel (a), we see the experiment, with a monochromatic light source to the left, a vertically

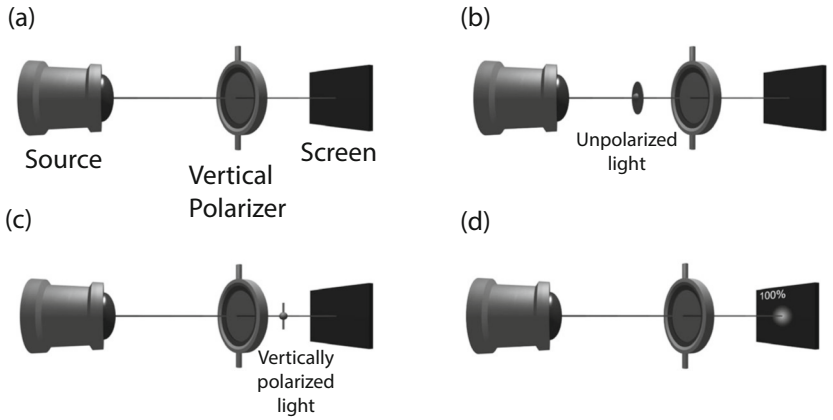

Fig. 4.12 Schematic of a source and polarizer and screen. (**a**) The setup of the apparatus. The polarizer is oriented in the vertical direction as shown from the two bars on the rim of the polarizer, which point vertically. (**b**) Depiction of unpolarized light traveling with a disc to indicate all polarizations are possible. (**c**) Depiction of vertically polarized light, with the director pointing up-down. (**d**) All light that makes it through the polarizer will hit the screen; here 100% gets through

oriented polarizer in the center, and a screen to the right. In panel (b), we illustrate how we depict an unpolarized photon, using a disc, since we do not know what direction the photon is polarized in. In panel (c), after passing through the polarizer, we show the photon now has a director pointing up-down. Finally, in panel (d), all photons that make it through the polarizer also hit the screen, since they go directly from the polarizer to the screen. We have a tutorial https://bit.ly/4abvjTp that helps visualize these experiments.

This experiment with a polarizer might remind you of something we did all the way back in Chap. 1 with the Stern-Gerlach experiment. There we shot a beam of atoms at an analyzer and each atom was emitted from either the $+$ or the $-$ exit, with the effective magnetic arrow pointed in one of two directions. Here, when a photon passes through the polarizer, it emerges with its polarization director along a particular orientation. Both the Stern-Gerlach analyzer and the polarizer can be rotated to change the direction of the axis that the measurement is being made on. The main difference is that the Stern-Gerlach experiment has a vector for the measurement, in that the arrow can point up or down, while the polarization has a director, which points up-down. This means that many of the experiments discussed with atoms earlier can now be performed with photons if we account for their polarization. We will see some examples of this below. But the details, with respect to angles, will be different due to the fact that spin is represented with arrows and the polarization by directors. But first, we describe a new experiment on how the polarization can be rotated. Keep the Stern-Gerlach experiments in mind as we do this.

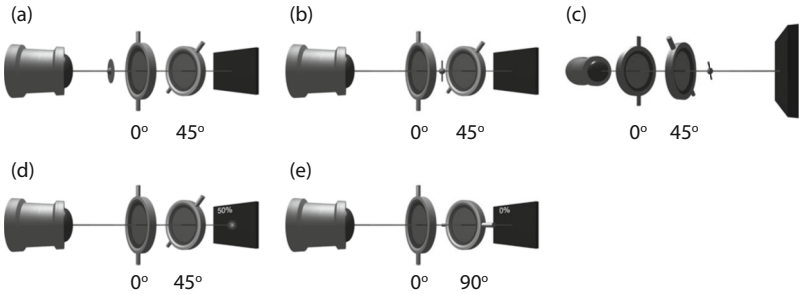

Fig. 4.13 Polarization experiment with two polarizers oriented at 0° and 45° (panels **a**–**d**) and at 0° and 90° (panel **e**); the polarization direction is determined with reference to the vertical. Panel (**a**) shows the unpolarized photon approaching the first polarizer. In panel (**b**), the photon emerges from the first polarizer oriented in the vertical direction. In panel (**c**), it is oriented in the diagonal direction after emerging from the second polarizer. The rules explained in the text describe why only 50% of the photons passing through the first polarizer make it to the screen in panel (**d**). Similarly, if the second polarizer is at 90°, no photons get through both polarizers as in panel (**e**)

We have to generalize our quantum rules for the case with polarization. Now, we have to keep track of an additional degree of freedom for the polarization of a photon, namely, a director, which points in the two directions of the polarization (such as up-down or left-right). The conventional rules of the photon continue to be the same, with the arrow rotating at a rate determined by its color and we have rules for how the arrow changes when we reflect or transmit. Now, in addition, we have to keep track of a director for polarized light. When a photon goes into a polarizer, we need to determine the change in angle $\Delta\theta$ between the director of the photon and the direction of the polarizer. The arrow for the photon is shrunk by a factor of $\cos\Delta\theta$ when we pass through a polarizer, and the director changes its direction according to the direction of the polarizer; be careful not to conflate the arrow with the director for the photon—we need to keep track of both. So, let us use these new rules to determine the arrow for a photon going through the setup with one vertical polarizer and one polarizer at 45° (see Fig. 4.13). If it is not absorbed, the unpolarized photon becomes polarized vertical after going through the vertical polarizer. We take this as the starting point of our analysis with an arrow pointing at 12 o'clock and its length equal to one—we only consider photons that made it through this first polarizer. The polarization director is oriented vertical as well. As the photon moves through air, the arrow simply rotates as the clock hand. When we reach the polarizer at 45°, we focus on the director; it will change from vertical to the diagonal at 45°, and we shrink the arrow by a factor of $\cos 45° = 1/\sqrt{2}$. Hence, when we hit the screen, the arrow length is $1/\sqrt{2}$. Squaring yields a probability of $1/2$ or 50%. Now, consider the new experiment, where the second polarizer is at 90°. Then the arrows shrink by a factor of $\cos 90° = 0$. So no photons reach the screen. Note that this is a general result. Whenever we try to pass vertically polarized photons through a horizontal polarizer, none will get through. So, to summarize, we have an additional rule for treating the polarization of light. We have a shrink factor of $\cos\Delta\theta$ when we pass

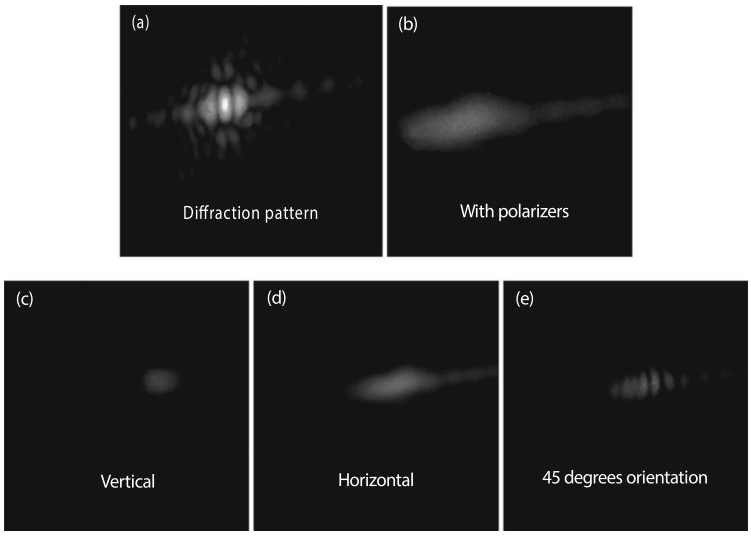

Fig. 4.14 Quantum eraser experiment with light. Panel (**a**) shows the diffraction pattern off of a single thin metal wire stretched vertically through the beam. Panel (**b**) shows the pattern when we add a vertical polarizer on the left and a horizontal polarizer on the right side of the wire. This gives "which-way" information and the interference pattern is gone. Adding a second vertical polarizer (**c**) or a second horizontal polarizer (**d**) after the first set (and before the screen) shows the light from the left and right side of the wire only. If, instead, we use a second polarizer oriented diagonally, then we see the interference pattern again—the classic result for a quantum eraser (Images from Paul Nord video on the quantum eraser Nord, 2011 © Paul Nord and used with permission)

through a polarizer, where $\Delta\theta$ is the angle between the director and the orientation of the polarizer. A tutorial https://bit.ly/4rTNzXB provides some extra visualization of this experiment.

Now for a short interlude. Paul Nord, from Valparaiso University, has created a video https://tinyurl.com/3marjuby that shows a quantum eraser experiment with polarized light (Nord, 2011). The experiment is quite similar to a two-slit experiment, but instead of having two slits, it has a thin metal wire in the center of the laser beam—light can go to the right or to the left around the wire, and we have interference when it travels on both sides. This is shown in Fig. 4.14a, where we can see a clear interference pattern. The light from the laser comes in polarized along the diagonal at 45°. Adjacent to the wire, two polarizers are added: one vertical (left) and one horizontal (right) on both sides. This setup of the polarizers ensures that we have the same amplitude of light going through each side of the wire. This then gives us which-way information by tagging which side the photon has traveled by which polarization the photon has, and on average half are tagged horizontal and half are tagged vertical. Note that it is incorrect at this time to think of the photon as traveling through one side or the other and having a definite polarization. With tagging, the state of the photon remains indeterminate until measured, so it is still traveling in superposition—technically speaking it is an entangled state. In this case, when we

just do a measurement, we see no interference effects (see panel (b)), consistent with the principle of complementarity (because we had which-way information). We can orient polarizers before measuring on the screen to even let through only the light that comes from the left side (panel (c)) or the right side (panel (d)). The real interesting thing is when we use a polarizer oriented along the diagonal before measuring (panel (e)). This removes the tag and the interference pattern comes back when we measure it on the screen. Note that it is not as good an interference as we had when we used no polarizers, but it is clear interference. This is typical of quantum eraser experiments. You should watch the full video the see this excellent demonstration.

This experiment is a remarkable one, for it shows how we can erase quantum tagging (given by the polarization) well after the photon has passed the wire. To me, this is a completely mind-boggling result. We will be discussing it a bit more in the next section. But you should think about how a photon knows to change from a particle pattern to a wave pattern and how it makes the decision when it is already so far away from the place where it interacted with itself, or if there is a better way to describe how this happens. This is related to Wheeler's delayed choice experiment and to what Einstein called "spooky action at a distance." In fact, it is probably the best example of a delayed choice experiment, as the decision to erase the "which-way" information is clearly done well after the light interferes (or does not) at the two slits.

We return from our interlude to re-examine the experiment in Fig. 4.13e. As we saw before, the difference in the angle between the two polarizers is 90°, so the shrink factor is $\cos 90° = 0$, and no photons get through. But now, consider the situation where we add an additional polarizer at 45° in between the vertical and horizontal polarizers (see Fig. 4.15). We can think of this as extra blockage being added to the experiment. But, just as we saw with the Stern-Gerlach analyzers, we now find some photons get through the device! This shows that the photon loses memory of previous values of polarization after it has passed through a polarizer.

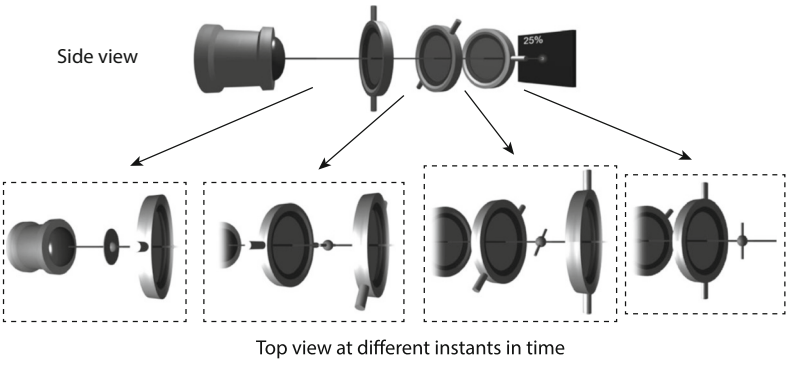

Fig. 4.15 Three-polarizer experiment with polarizers set at 0°, 45°, and 90°. The blowups below show the instants when the photon is in between polarizers using a top view

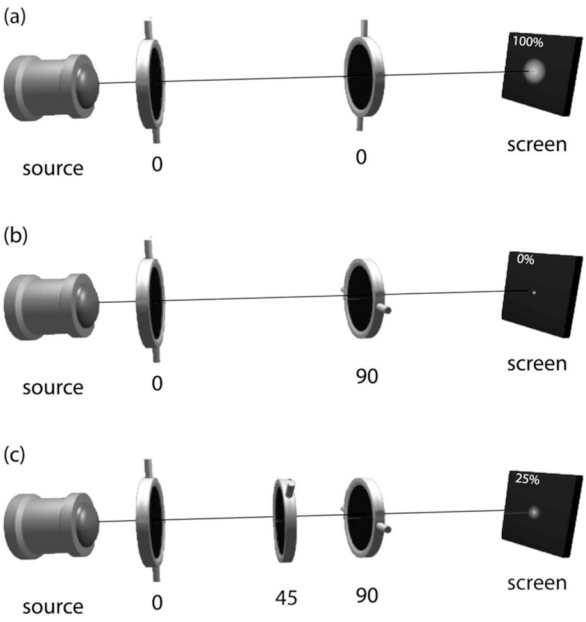

Fig. 4.16 Three experiments that are analogs of the three experiments that we examined in for the Stern-Gerlach experiment in Chap. 1

We analyze this three-polarizer experiment next. At the start, light leaves the source and passes through the vertical polarizer. This is where we describe the light with an arrow of length 1 and pointing at 12 o'clock. The director is up-down. As the photon moves, the arrow rotates, at a rate according to the color of the light. When we reach the diagonal polarizer, the arrow shrinks by a shrink factor given by $\cos \Delta\theta$; here, we have $\theta_i = 0°$ and $\theta_f = 45°$, so the shrink factor is $\cos 45° = 1/\sqrt{2}$. After emerging from the diagonal polarizer, the director is now at 45°. Now, we move to the final polarizer. The arrow continues to rotate at the same rate, and its length is now $1/\sqrt{2}$. Moving through the last polarizer, we find a similar shrink factor, since $90° - 45° = 45°$. The arrow is now shrunk down to $(1/\sqrt{2}) \times (1/\sqrt{2}) = 1/2$. We square to obtain the probability to get through and find it is $1/4$ or 25%.

Back in Chap. 1, just after we determined the Stern-Gerlach experiment displayed quantum, not classical, behavior, we examined three experiments that told us about the repeatability of these experiments. Now, we examine the analogs of these experiments with polarized light in Fig. 4.16. In the first experiment in panel (a), we pass through a vertical polarizer and pass through a vertical one again. Since the light is vertically polarized after the first polarizer, all the light pass through the second polarizer. This establishes that the polarizer creates a well-defined photon state. In the second experiment, we pass through a vertical polarizer and examine how much light goes through a horizontal polarizer. This is similar to the z and $-z$ case for the Stern-Gerlach experiment. But note how the angles are now one-half as large as they were in the Stern-Gerlach case; for Stern-Gerlach, the difference between z and $-z$ is 180°, while here, the difference between V and H is 90°,

but the formulas for the probabilities also have a factor of two difference in the argument, which leads to them having the same numerical value. This is because we are working with directors instead of arrows. Since the polarizers only have one "exit," the experiments are modified as shown here. Finally, in panel (c), we examine the three-polarizer case, which we have seen before. In this case, we get 25% arriving at the screen. This shows that when the light is vertically polarized, it does not remember this polarization when it goes through another polarizer. It changes its polarization instead. Just like the atom, a photon is stupid—it only remembers the last polarizer it went through. We have a video https://tinyurl.com/3r9bumm8 and tutorial https://bit.ly/3XQ7KIh illustrating this experiment. We also have a Free Exploration Tutorial on Polarization https://bit.ly/4pFmaYa to assist you on these ideas. *Since atoms and photons are both stupid, we humans should have no trouble understanding their behavior!*

We end this section with a discussion of the quantum Zeno effect. The quantum Zeno effect underlies the principle behind an *interaction-free measurement.* The basic idea is that we let a quantum system evolve (for us this will mean we let the polarization of a photon rotate). Then we have two choices—we either interrupt the 90° rotation by making measurements after the smaller rotations of each stage to realign the polarization to the vertical—in which case the net effect is that it does not rotate—or we don't make the realignment measurements—and it does rotate. If we take the limit where the number of interruptions approaches infinity, we have a situation where we can have the polarization rotate or not rotate, based on our choice of whether or not to include the interrupting measurements. This allows us to control whether the polarization rotates or not, with essentially 100% fidelity. The quantum Zeno effect is a critically important element in the quantum-seeing-in-the-dark measurement.

In Fig. 4.17, we illustrate a new series of experiments with polarization rotators and polarizers. The polarization rotator rotates the polarization by a specific amount as the light moves through it. In panel (a) the rotation is by 90°, while it is 45° in panel (b) and 22.5° for panel (c). In panel (a), no photons get through, but as we add more stages, the percentage that gets through increases. By the time we have $N \to \infty$, 100% will get through! You may ask, how do we rotate the polarization? It turns out this can be done many ways, but a common one is to let light pass through a vial containing sugar dissolved in water. The length of the vial determines how much the polarization rotates for a given concentration of sugar.

Now, let's look at some examples: The setup involves N stages of polarization rotators and $N + 1$ stages of polarizers, so we have one vertical polarizer for every gap between two adjacent rotators. Figure 4.17 shows $N = 2$ and $N = 4$.

- $N = 1$: Here the polarization rotates to 90°, and 0% get through
- $N = 2$: Here the polarization of each stage rotates by 45°, so 25% get through
- $N = 3$: Here each stage rotates by 30°, so = 42.1875% get through
- $N = 4$: Here each stage rotates by 22.5°, so 53% get through

and so on. The general formula for an N stage case is a probability of $\cos^{2N}(90°/N)$. See if you can derive this yourself.

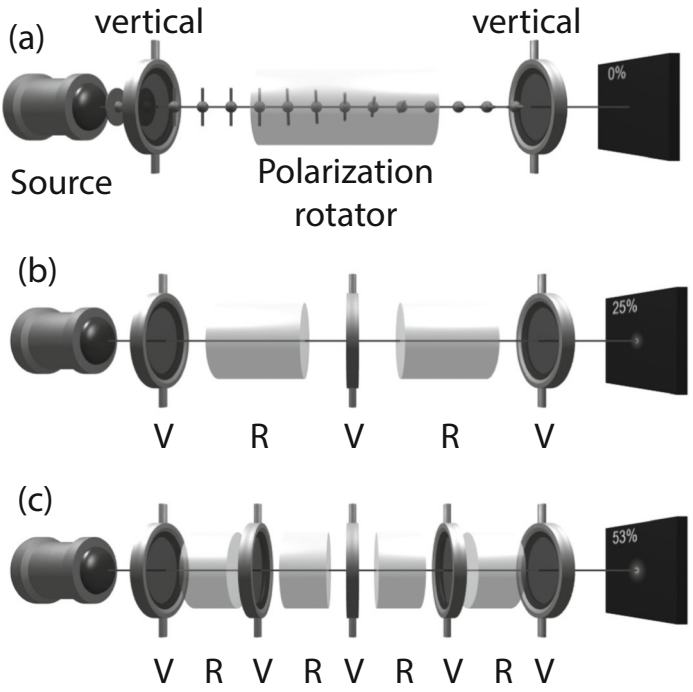

Fig. 4.17 Illustration of the quantum Zeno effect. Panel (**a**): Schematic of a polarization rotator, with snapshots showing how the polarization rotates as it moves through the rotator. This one rotates by 90°, so nothing gets through. In panel (**b**), we break the rotator into two 45° rotators and insert a vertical polarizer between the two; we no longer show the photon snapshots. In this case 25% gets through. In panel (**c**) we break into four 22.5° pieces, with polarizers interspersed. It now allows 53% of the photons to get through. The Zeno effect corresponds to $N \to \infty$ stages

In Fig. 4.18, we plot the probability for a photon to make it through the rotator-polarizer stacks. As $N \to \infty$, the probability goes to 100%. This is what is called the quantum Zeno effect. It occurs in two flavors. If we use the rotator-polarizer sandwiches, then 100% get through, while if we omit the polarizer, 0% gets through. In the former case, we say the measurement (by going through the polarizer) stops the polarization from evolving (or rotating), while in the latter case, we say when we do not measure, the polarization evolves (or rotates). This dichotomy of measurement stopping evolution is called the quantum Zeno effect. The video Show Me the Light https://tinyurl.com/mr2tcrda and companion tutorial https://bit.ly/4pymtDZ provide additional context on this quantum Zeno effect.

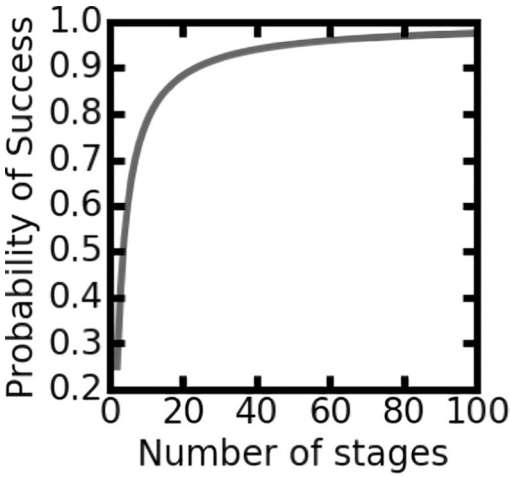

Fig. 4.18 Probability that light gets through the rotator-polarizer system with N stages. As N increases, the probability of transmission approaches 100%. This is because the cosine function is a nonlinear function

Section 4.3 Problems:

4.3.1 Finite Versus Infinite

We are going to compare the difference between finite and infinite expressions. Consider the partial geometric series with $x = 0.1$ as described below.

The first three terms, multiplied by 9, are $9 \times (1 + 0.1 + 0.01) = 9.99$. Work out what it is for the first 6 terms multiplied by 9. Now consider the limit as the number of terms goes to infinity. Is this number equal to 10 or different from it?

4.3.2 To Diverge or Not Diverge

It almost looks like black magic to determine whether or not a series converges to a limit when we include an infinite number of terms. Answer the question below to get guidance on how you can tell!

Under what circumstances do we know a series must *diverge*? Select two correct answers.

- (a) Each term in the series grows without stopping (like $1 + 2 + 3 + 4 + \cdots$).
- (b) Each term in the series can be shown to be larger than the terms in similar series that is known to diverge (like $1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \cdots$ is greater than $1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) + \cdots$).
- (c) Each term is less than the terms in a similar series that converges (like $1 + \frac{1}{10} + \frac{1}{100} + \frac{1}{1000} + \cdots$ is less than $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$).
- (d) The terms in the series go to zero for large n.

4.3.3 When the Tortoise Meets the Hare

Suppose Achilles starts 100 m behind the tortoise and he travels at 11 m per second, while the tortoise travels at 1 m per second. How long does it take before Achilles catches the tortoise?

This is an old brainteaser. Let's compute Zeno's way. The time for half the distance is the time to go 55 m (or, if you like the time for the relative distance to decrease to 50 m; given the speeds of the tortoise and Achilles. Note that Achilles must travel 10% further than the distance between them to catch the tortoise). Then we add in the rest of the half intervals to find total time = 2×(time for Achilles to go 55 m).

Compute the time for Achilles to go 55 m, sum the series, and explain the total time needed.

4.3.4 Don't Point at Me

In physics, we use the technical term *vector* to denote an arrow that points somewhere. We have worked with these arrows, or vectors, quite a bit already. We say the vector points in *one* direction, the direction of the arrow.

Now, we have added the concept of a director, which is how we describe a polarization. How many directions does a director point in?

4.3.5 Do Photons Remember?

Suppose a photon goes through a vertical (up-down) polarizer, then through a diagonal (45°) oriented polarizer, and then through a second diagonal polarizer, oriented at the same angle (45°) as the first diagonal polarizer. Both angles are the same and are with respect to the vertical. What percentage of the photons that got through the first vertical polarizer also get through both diagonal ones? Explain your answer carefully.

4.3.6 Am I Watching?

Let's go back to our friend the two-slit experiment with light. Knowing now that light has a polarization to it, we place a polarizer oriented at 20° from the vertical across both of the slits of the two-slit experiment. You may assume the light incident on the slits is polarized at 0°.

What happens? Two of the following statements are correct.

- (a) The pattern on the screen gets dimmer.
- (b) The pattern on the screen remains the same brightness.
- (c) The pattern on the screen gets brighter.
- (d) The interference pattern (wave pattern) disappears and is replaced by the particle pattern.
- (e) The interference pattern remains a wave pattern with the same positions for the bright and dark bands as before.
- (f) The interference pattern remains a wave pattern, but the distance between dark bands shrinks by the amount of $\cos^2 20°$.

4.3.7 I Am Watching, Right?

Let's go back to our friend the two-slit experiment with light. Knowing now that light has a polarization to it, we place a polarizer oriented at 0° from the vertical across the left slit and one oriented at 90° across the right slit of the two-slit experiment. The light incident on the two slits is oriented at 45°.

What happens to this experiment? Two of the choices below are correct.

- (a) The pattern on the screen gets dimmer.
- (b) The pattern on the screen remains the same brightness.
- (c) The pattern on the screen gets brighter.
- (d) The interference pattern (wave pattern) disappears and is replaced by the particle pattern.
- (e) The interference pattern remains a wave pattern with the same positions for the bright and dark bands as before.
- (f) The interference pattern remains a wave pattern, but the distance between dark bands shrinks by the amount of $\cos^2 45°$.

4.3.8 Now I See

Let's go back to our friend the two-slit experiment with light. Knowing now that light has a polarization to it, we place a polarizer oriented at 0° from the vertical across the left slit and one oriented at 90° across the right slit of the two-slit experiment. Just before the light reaches the screen, I place a polarizer at 45° degrees orientation (which equally lets light of up-down and of left-right polarization through). The light incident on the two slits is oriented at 45°.

What happens to this experiment? Two of the choices below are correct.

- (a) The pattern on the screen gets dimmer.
- (b) The pattern on the screen remains the same brightness.
- (c) The pattern on the screen gets brighter.

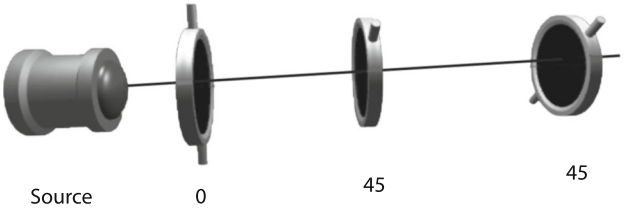

Fig. 4.19 Light leaves the source and goes through polarizers at 0°, 45°, and 45°

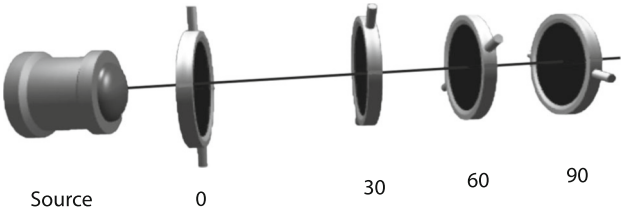

Fig. 4.20 Light leaves the source and goes through polarizers at 0°, 30°, 60°, and 90°

- (d) The interference pattern (wave pattern) disappears and is replaced by the particle pattern.
- (e) The interference pattern remains a wave pattern with the same positions for the bright and dark bands as before.
- (f) The interference pattern remains a wave pattern, but the distance between dark bands shrinks by the amount of $\cos^2 45°$.

4.3.9 Polarizer Madness I

Determine the percentage that makes it from just after the vertical polarizer to after the last polarizer. The experimental setup is in Fig. 4.19.

4.3.10 Polarizer Madness II

Determine the percentage that makes it from just after the vertical polarizer to after the last polarizer. The experimental setup is in Fig. 4.20.

4.3.11 Polarizer Madness III

Determine the percentage that makes it from just after the vertical polarizer to after the last polarizer. The experimental setup is in Fig. 4.21.

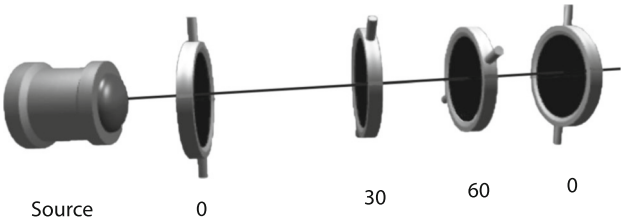

Fig. 4.21 Light leaves the source and goes through polarizers at 0°, 30°, 60°, and 0°

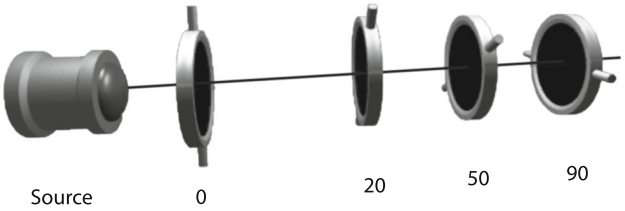

Fig. 4.22 Light leaves the source and goes through polarizers at 0°, 20°, 50°, and 90°

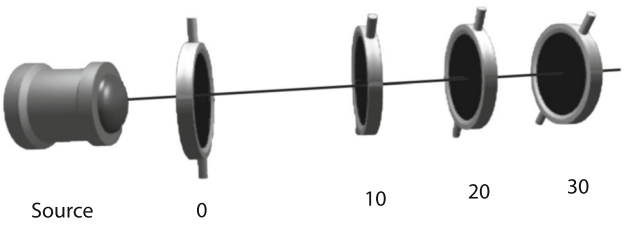

Fig. 4.23 Light leaves the source and goes through polarizers at 0°, 10°, 20°, and 30°

4.3.12 Polarizer Madness IV

Determine the percentage that makes it from just after the vertical polarizer to after the last polarizer. The experimental setup is in Fig. 4.22.

4.3.13 Polarizer Madness V

Determine the percentage that makes it from just after the vertical polarizer to after the last polarizer. The experimental setup is in Fig. 4.23.

4.3.14 Polarized Mach-Zehnder Interferometer

Consider a Mach-Zehnder interferometer that has polarizers placed in both the sample path and the reference path. These two polarizers are placed just after the first beam splitter, one along each path of the interferometer. This placement means the light traveling on each path is polarized according to the direction of the polarizer that is situated on each path. The polarizers can be oriented exactly the same or can be crossed (at 90° to each other). Write an essay that describes what happens to the probability for detection at the bright and dark detectors (assuming the path length for each path is exactly the same). Explain what would happen if one polarizer can be rotated, and starts from being aligned with the other polarizer and then rotates until it is at 90° with respect to the other polarizer. Be sure to write your answer clearly and coherently.

4.3.15 Quantum Zeno Effect

Calculate the percentage to get through the rotator-polarization sandwiches with $N = 6$, $N = 9$, and $N = 15$ stages.

4.4 Quantum Seeing in the Dark

A polarizing beam splitter has the property that it transmits light of one polarization and reflects light of another polarization. So it acts like a beam splitter and a set of crossed polarizers. This unique piece of equipment allows for some interesting experiments to be performed. Before getting into the details of that, we want to describe how one analyzes the behavior of a polarizing beam splitter.

Consider a polarizing beam splitter as in Fig. 4.24, where vertically polarized light (oriented at 0°) is transmitted and horizontally polarized light (oriented at 90°) is reflected. Suppose a photon is incident on the beam splitter and it is polarized at an angle θ with respect to the vertical. Then the probability for the light to be transmitted with vertical polarization is $\cos^2\theta$ and the probability for the light to be reflected with horizontal polarization is $\sin^2\theta$. Of course, because this is a quantum event, we say we have two probability amplitudes: one of length $\cos\theta$ for transmission and one of length $\sin\theta$ for reflection; the photon typically travels in superposition after the beam splitter. But note that if vertically polarized light is incident on the beam splitter, it is always transmitted, and if horizontally polarized light is incident, it is always reflected. If it is polarized in any other direction, we have a superposition of partially transmitted and partially reflected photons, with the arrows for each alternative way determined by the corresponding shrink factor of $\cos\theta$ for transmitted (vertical) light and $\sin\theta$ for reflected (horizontal) light. In this

sense, the polarizing beam splitter is a superpositioner and passing through it does not constitute an irreversible measurement. The tutorial https://bit.ly/3KWXD1p provides a great visualization for how the polarizing beam splitter works.

We want to reiterate that the photon travels in superposition after going through the beam splitter with the horizontal component being reflected and the vertical component being transmitted. If we were to bring back the photons to the beam splitter (say, after reflection on each path), they will recombine after going through the beam splitter a second time to create a photon with the same polarization as it had when it entered. We can think of this as the superposition being re-expressed in a new basis, corresponding to the specific direction the polarization is in (and its perpendicular direction). Hence, the beam splitter acts just like the analyzer loop did with the Stern-Gerlach experiments—it does nothing. However, when the photon is split in superposition, we can manipulate it, just as we did with the analyzer loop. Vijayan Thanasekaran has a great video https://tinyurl.com/4nam6hu9 showing how one works.

The polarizing beam splitter can be employed to create a polarized Mach-Zehnder interferometer. Now, since we can tell which path the photon travels in the interferometer; due to the polarization being different on the sample and reference paths, we will lose the interference effects. If the which-way information is "erased," by passing through a polarizer at 45°, then the interference pattern is restored. So, what this says is when we have "which-way" information, both the bright and dark detectors will receive half of the light uniformly, with no ring-like interference patterns. But, if we place a polarizer at 45° before either detector, the interference pattern returns, because we lost the ability to distinguish between the paths. This is true for both detectors. The price we pay though is that half of the photons are absorbed by each of those 45° polarizers. So, it does not restore the system to the case of an ordinary Mach-Zehnder interferometer without polarizing beam splitters. Instead, the overall intensity is diminished.

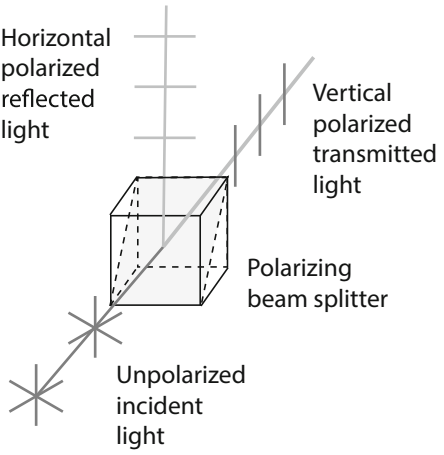

Fig. 4.24 Schematic of a polarizing beam splitter. An unpolarized incident light goes into the input. The transmitted light is vertically polarized and the reflected light is horizontally polarized, for this example

There sometimes is a controversy about these types of experiments, with people having strong opinions about whether these experiments are just classical or truly quantum. The issue arises because these experiments are typically done with lasers that have a high enough intensity that there isn't one photon in the experiment at any given time, but rather there are many, many photons present, and a classical analysis in terms of electric and magnetic fields can be pursued—the light is then described simply in terms of these classical fields and their quantum nature ignored.

So what is right? I am not a philosopher, but I will share my opinion on this subject. First, there are many different experiments where the classical description and the quantum description yield exactly the same results when the intensities are high. One famous example is the Rutherford experiment where alpha particles are shot at a metal foil. But the Mach-Zehnder interferometer and the two-slit experiment are other examples. The probability patterns are the same for a classical and a quantum calculation. So, what does it mean? For me, it means that the system is quantum, and we have a curious coincidence that the similar classical system produces the same results. For others, this implies it is classical. But, in the interferometer and the two-slit experiment, we know that if we use a single-photon source, so at most one photon is in the apparatus at any given moment, then the result is random, and the "classical" probability pattern emerges only after many runs of the experiment. This is a purely quantum result. And for me, most of the complaints that an experiment is not quantum fall under the category where the quantum and classical calculations yield the same result. I cannot see how, in that case, one can ever call such systems classical. *They are quantum!* Everything is quantum—classical mechanics is just a limit of quantum behavior. So, I firmly believe that these illustrations of the quantum eraser are indeed illustrations of quantum experiments. And the fact that they can be illustrated with rather simple equipment is all the better. It makes the quantum phenomena all the more wonderful, accessible, and bewildering!

We are now ready to produce an efficient quantum-seeing-in-the-dark experiment. In principle, it can be made as efficient as desired, but the sample we show will have an $N = 6$ stage setup which is approximately 68% efficient. It uses a Michelson-Morley setup for the interferometer, which has a "tee" shape to it. The basic idea is as follows: We allow photons to run through an experiment, where they are split by a polarizing beam splitter into two paths, and during each cycle through the experiment, we allow the polarization to rotate, starting from the horizontal polarization 90°. If there is a blockage on the path that is vertically polarized, then the polarizing beam splitter will project the polarization back to the polarization of the path that was not blocked (in this case, horizontal), because only horizontally polarized photons avoid the blockage and survive—vertically polarized photons are absorbed by the blockage. So after a set of N stages, either the polarization will have rotated by 90° (to 0° since the polarization started horizontal) or it will not rotate and stay at 90° (horizontal polarization). We pass the final photon through a horizontally oriented polarizer to a detector. If the polarization does not rotate, it goes through the polarizer and hits the screen. If the polarization does rotate, it is absorbed by the final polarizer and does not make it to the screen. Then by detecting

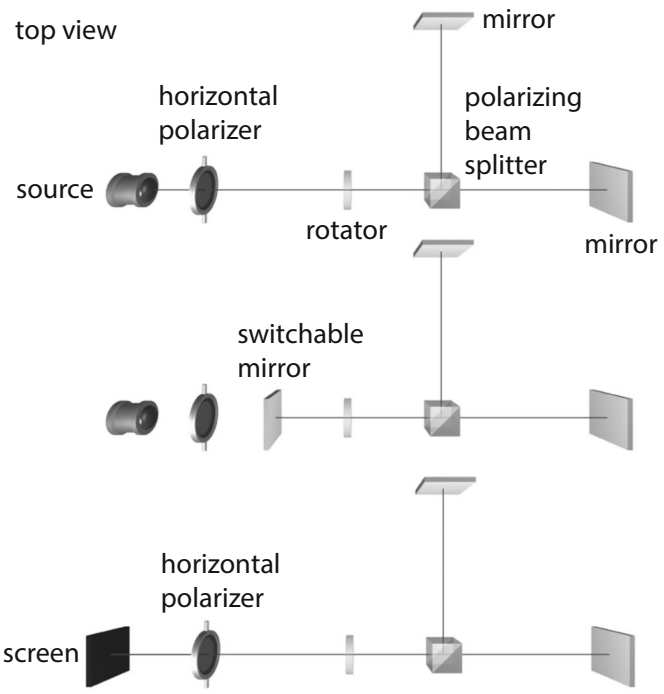

Fig. 4.25 Schematic of the quantum-seeing-in-the-dark experiment from the top, illustrating the three different stages of the experiment, as described in the text

a photon, we know the polarization did not rotate due to the presence of the blockage on one path. But we do this with the photon that went on the other path, so it did not interact with the object it is observing. The photon will go through the system with a success rate that can be made as large as desired by increasing the number of stages.

Now, we will carefully describe how this experiment works. It is lengthy, because we need to put together many of our quantum ideas to make it work. We start with the top schematic in Fig. 4.25. The apparatus has a monochromatic single-photon source to the left, a horizontal polarizer to ensure the initial photon is horizontally polarized, and a polarization rotator that rotates by 7.5° every time the photon passes through it (the advance occurs when traveling through the rotator in both directions because of how the polarization changes upon reflection in the mirror at the end of each path). Then we have the polarizing beam splitter, followed by the two arms of the Michelson-Morley interferometer, with mirrors at the ends of each path (which are equal in length). The photon will traverse through the equipment six times, before it leaves the apparatus. To do this, a switchable mirror (shown in the middle schematic) is turned on while the photon travels six times through the apparatus. Then it is switched off, and the source is replaced by a detector (here a screen), after passing through a horizontal polarizer.

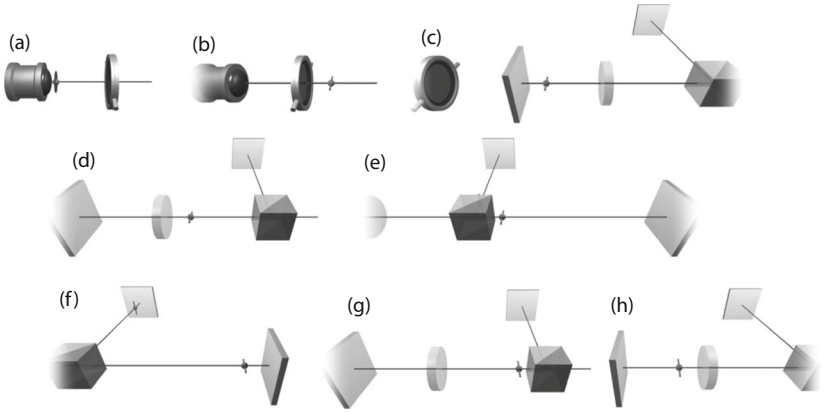

Fig. 4.26 Snapshots of the quantum-seeing-in-the-dark experiment when there is no blockage. The different panels are explained in the text

We now go step by step to describe how this works when there is no blockage on the vertical arm of the interferometer. In this case, the photon travels in superposition along both arms of the interferometer, and when it recombines at the beam splitter, the photon acts as if nothing changed. We look at the different instants shown in Fig. 4.26 of this experiment, for the first pass through the apparatus. In panel (a), the photon leaves the source, with an unknown polarization, but if it is not a vertical polarization, then it has a probability to get through, which is shown in panel (b) with the horizontal polarization. Note that a real experiment works in a somewhat different fashion, using what is called a heralded single photon, created from a process called parametric down conversion, which we will discuss in more detail later in the book. This process guarantees the photon is created with horizontal polarization, so it goes through the first polarizer. In panel (c), we show the switchable mirror turned on after the photon has passed its position. The photon next travels through the polarization rotator and is rotated by 7.5° by the polarization rotator. Next, it impinges on the polarizing beam splitter, which is rotated by 90° from its orientation in Fig. 4.24. This then has the horizontal polarization transmitted (moving to the right) and the vertical polarization reflected (moving upward on the figure). The arrows have their lengths (during this motion in superposition along the arms of the interferometer) determined by the incident polarization— $\sin 7.5°$ on the vertical arm and $\cos 7.5°$ on the horizontal arm. In panel (f), we see the two alternative ways the photons are traveling after reflection off the two mirrors and heading back to the beam splitter. They rejoin in panel (g), constructing the polarization they had before, namely, 7.5°. Finally, in panel (h), the photon travels through the rotator again, rotating another 7.5° to reach 15°. The photon reflects off the switchable mirror and the process repeats again, adding 15° with each double pass through the rotator; after six circuits, we have rotated by 90°. The final polarization of the photon is now vertical. The switchable mirror is removed, and the

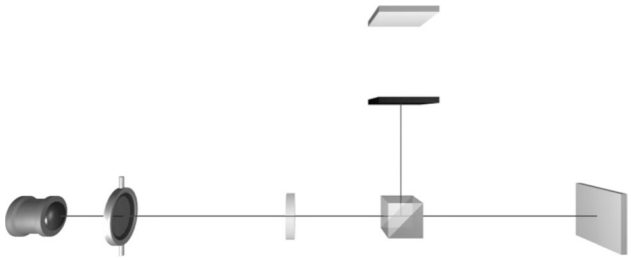

Fig. 4.27 Top view schematic of the quantum-seeing-in-the-dark experiment with blockage. The blockage lies on the vertical path, which has vertical polarization of light

photon goes through a horizontal polarizer before being detected. One can see that because the polarization is vertical, the photon does not get through the polarizer and we detect nothing. Hence, in this first version of the experiment, when there is no blockage on the vertical arm of the interferometer, no photons make it to the detector. We have a tutorial https://bit.ly/3KWjfuQ that goes step by step through this experiment.

Now we look at the second version, where there is a blockage along the vertical path (see Fig. 4.27). In this case, the presence of the blockage provides which-way information, so the photon no longer travels in superposition, but it can only travel on one arm without being absorbed. The experiment starts the same, as in panels (a–e). But when we get to panel (f), the photon is absorbed by the blockage if it moves on the vertical path. It is easiest to calculate the total probability that the photon is absorbed as being one minus the probability it is not absorbed. For that, we focus on the shrink factors when we pass through the polarizing beam splitter, which are $\cos 7.5°$ on the first pass and $\cos 15°$ on the next five passes (it passes through the rotator once for the first pass and twice for the subsequent five passes). As it heads to the final polarizer, it is oriented at 7.5°, giving one final factor of $\cos 7.5°$ (because it passed through the rotator one final time). We then have the total shrinkage of the arrow is $\cos^2 7.5° \times \cos^5 15° = 0.8265$; squaring the length of the arrow gives a total percentage of success of 68.31%. The way we compute these factors arises from the fact that if there is a blockage, the photon no longer travels in superposition, because the events have changed. For the alternative way where the photon is not absorbed, the photon enters the first beam splitter at an angle of 7.5° or 15° depending on the stage it is in. This is the origin of the different cosine factors in the success calculation. Please carefully work through the steps (including first step, second through sixth step, and moving through the final horizontal polarizer) to be sure you understand how to perform this calculation; the rotation of the conventional arrow does not enter into the calculation, because there is only one alternative way for success. The probability for failure is then 31.69%. Note the difference in how the two versions of the experiment work. When there is no blockage, the photon always survives and often is in a superposition state, but when there is blockage, the photon is sometimes absorbed by the blockage or by the polarizers, so its success ratio

falls. Note that by increasing the number of stages, we can increase the successful percentage as close to 100% as we would like. The physical realities of absorption by the optical elements and the efficiency of the detectors will never allow us to fully reach this ideal, but it is clear we can do much better than a 50–50 chance. A tutorial https://bit.ly/4ada9Ej shows step by step how this case works too.

We have succeeded in describing an extraordinary experiment, which is also extremely difficult to understand. But, by applying our quantum model step by step, we can analyze all of the pieces. It is really amazing to be able to see something with photons that do not interact with the object, and all made possible via quantum mechanics! The video This is Just So Weird https://tinyurl.com/45huhzxj explains some more details about this.

There actually is a way to use this in communication that exchanges no photons! It is called *counterfactual communication*. The idea is as follows: the message is encoded in terms of a binary encoding of 0s and 1s in a string. Bob, to whom the message will be sent, holds all of the pieces of the apparatus except the vertical-polarization arm where Alice will put the blockage. Then Alice sends the message by putting in a blockage or not on that path. They need to coordinate their timings to synchronize when photons will be in the device, but, if they have done that then, when a photon goes through the device, it either leaves with one polarization, or another, or it hits the blockage. With a large number of stages, we can make the "hits the blockage" probability as small as we want. Then Alice communicates simply by including or removing the blockage at the right times. Bob measures the polarization of his photon. Depending on what he measures (say choosing 0 for vertical and 1 for horizontal), he gets the binary ones and zeros of Alice's key. And in all cases, Bob just measures his own photons and they never travel on the path that Alice blocks! This seems like complete science fiction. But it follows directly from the quantum-seeing-in-the-dark experiment. Carrying such a communication program out would be quite challenging though. It is more the principle that is so cool.

Suppose you walk up to an electronics store and ask to see the digital cameras and the salesman says "classical or quantum?" This would be a new age for photography! How would the quantum camera work? Why don't we have one available today?

Earlier, in a problem, we asked you to write an essay about how you could capture an image of a silhouette with a Mach-Zehnder interferometer, by focusing on the photons detected at the dark detector and moving the object through the sample beam path of the interferometer. In that case, we asked you to assume the measurement could be made 100% efficient. Now, we have found a way to realize this situation using the quantum-seeing-in-the-dark device. The original quantum-seeing-in-the-dark experiment is more than 20 years old, so why haven't we seen commercial quantum cameras yet?

First off, they would be very slow. We need to move the object through one of the beams of the interferometer and measure whether there is a blockage. We move the object on a grid, so each step on the grid will correspond to a pixel. If we have a binary result (blocked or not blocked), we could create a black-and-white image of the object, which would look like a silhouette. If we change the color of the

light we use, and our object is transparent to some colors and not others, then we could take a color photo, by including more measurements, where we also change the color of the source. Second, they would be very expensive. So far, quantum-seeing-in-the-dark experiments are performed in sophisticated optics labs. Unlike the quantum eraser, which can be performed at home, there doesn't seem to be any simple way to do a seeing-in-the-dark experiment without using sophisticated equipment in a laboratory setting. This is especially true for the single-photon quantum sources. Third, it is unlikely it could be shrunk into a handheld device. Currently, the experiment fills an optical bench, and there is no clear way to shrink it to the size of an integrated circuit, which is needed for a robust handheld device. Fourth, it is not clear that there is a practical need for seeing-in-the-dark photos, so it isn't obvious one could sell such cameras. So, don't expect a quantum camera anytime soon. It just isn't practical at this point in time.

Before leaving the subject, however, I encourage you to watch a remarkable award-winning video by Dag (also known as Dagomir Kaszlikowski) on the quantum-seeing-in-the-dark experiment as part of an action thriller. It is called "Seeing without looking" https://tinyurl.com/5t2suyns (Kazlikowski, 2014). And it effectively shows what it would be like to use a quantum camera to foil the "quantum triggered bomb." I found it very entertaining—you now know enough to fill in the blanks for all of the science.

This material is some of the most bizarre material you can find about quantum mechanics. Not only is it bizarre that nature works this way, but it is truly amazing that experiments can be made to show this behavior and apply it to interesting applications. I hope you have enjoyed learning about it.

Section 4.4 Problems:

4.4.1 Polarizing Beam Splitter

For this problem, look at Fig. 4.24, for the geometry. The angle for the polarization is measured from the vertical.

(a) Light is incident on a polarizing beam splitter. Assume it operates as in the figure, with the vertical polarized light transmitted and horizontal reflected. If the incident light is at 0° (vertical), what percentage of light is transmitted?
(b) If the incident light is at 45°, what percentage of light is transmitted?
(c) If the incident light is at 60°, what percentage of light is transmitted?
(d) If the incident light is at 90° (horizontal), what percentage of light is transmitted?

4.4.2 *Quantum Eraser and Patterns*

There is an important point about the quantum eraser that we have not discussed yet. Namely, the pattern that is seen on the screen without the polarizer appears like the sum of the two interference patterns, similar to the ones shown in Fig. 4.6 for the Mach-Zehnder interferometer, which then has the interference filled in when both patterns are superimposed—that is, it appears uniform. Recall the light is incident on the beam splitter with a 45° polarization; we place a vertical polarizer on the sample path and a horizontal polarizer on the reference path. Then, when we use the polarizer just before the screen, it is as if we pick one of the two uniform particle patterns (when the polarizer is at 0° or 90°). One might think that one can show that the original interference pattern is restored when we orient the polarizer before the screen at 45°. But, the interference pattern has the brightest spots four times brighter than the particle pattern for a single slit (instead of the two times brighter interference pattern that we will see).

Write an essay to describe the behavior in this experiment and explain why it is only twice as bright, as opposed to four times as bright.

4.4.3 *Quantum Seeing in the Dark*

(a) Consider the case with $N = 9$ stages. Following the general formula discussed in the text, determine the probability for success to four-digit accuracy for the case $N = 9$. Work carefully with your calculations. It is easy to make a mistake.
(b) Repeat for the case where $N = 15$.

4.5 Hong-Ou-Mandel Experiment and Photon Bunching

So far in this book we have focused primarily on what happens to a single quantum particle. Sure, we have explored complex phenomena involving quantum interference when there is more than one alternative way an event can occur. But it was always with one particle. This invites the question: What changes when we have more than one particle? This is called *the quantum many-body problem*. It is even richer than the quantum mechanics of single particles. So much so that nearly every quantum physicist works on trying to understand the solutions of the quantum many-body problem. It is the field of research that I work in as well; there are enough open questions that they can occupy thousands of physicists for decades, if not centuries. We don't have the time to discuss the many-body problem in much detail here. But in this section we examine a weird experiment that involves two particles. I have a video entitled The Wide World of Quantum Correlations https://tinyurl.com/3hxrrubn that describes some of my research work.

In the quantum world, there are two classes of particles, fermions and bosons. The next experiment we discuss shows that when two bosons get together, they stay together rather than break up and go their own way. It is called the Hong-Ou-Mandel experiment, and we describe it in detail next. This encounter with entanglement builds on our introduction to multi-particle entanglement in the Einstein-Podolsky-Rosen experiment and the Bell inequality, as well as with the entanglement generated by tagging. We start from a quote from Tom Stoppard's *Hapgood* (Stoppard, 1994), which has at the heart of it the idea of indistinguishability. The play illustrates many ideas of quantum mechanics with an allegory that uses spies. It is an outstanding play that every budding quantum mechanic should read. Here, we focus on one scene, where Blair, a senior spy, talks with Kerner, a physicist-spy after discovering a bag that should have a roll of film in it, but instead had none:

Blair: But why remove our rolls of film? He'd only have to take out what he put in and none would be the wiser.

Kerner: Obviously, because he put in a roll of film and they all look the same; he had to take them all.

Act 1, Scene 2 of Tom Stoppard's *Hapgood*.

The quote describes the situation of indistinguishability. When things are the same, we cannot tell them apart. But when quantum particles are the same, we can create strange quantum phenomena. In the quantum world, there are two types of particles: bosons (like photons and some atoms) are the friendly quantum particles—they like to stay together and do things in parallel; fermions (like electrons and some atoms) are the antisocial particles—they never allow two to share the same quantum numbers or be in the same quantum state. We can say fermions hate each other and bosons love each other. These behaviors arise because we cannot tell the same particles apart, and if we follow the Feynman rules, this means that there are now alternative ways for events to occur that we might not have thought of occurring. They arise from the indistinguishability.

We set the stage for the Hong-Ou-Mandel experiment by first considering the simpler example of the Hanbury Brown and Twiss experiment. This is the experiment that gave birth to the field of quantum optics. Suppose I have two photon sources and two detectors and I shoot two photons (one from each source) at the same time toward the detectors as shown in Fig. 4.28. I can get one of three results: (i) both photons go to detector 1; (ii) both photons go to detector 2; or (iii) one photon goes to each detector. If the last case occurs, can I tell which source the photon that went to detector 1 came from? The answer is no. I cannot, so both options ($A \rightarrow 1$ and $B \rightarrow 2$) and ($A \rightarrow 2$ and $B \rightarrow 1$) are possible. Thus, according to Feynman's rules we must take both alternative ways into account. Now, here is where it gets strange. What if the arrow for the alternative way $A \rightarrow 1$ and $B \rightarrow 2$ is rotated 6 hours from the arrow for the alternative way $A \rightarrow 2$ and $B \rightarrow 1$, but has the same length? Then the two alternative ways have arrows that combine to give 0, and every time I run the experiment, I get two photons in detector 1 or two in detector 2, but they never split to one in each detector. This effect is called "bunching," and

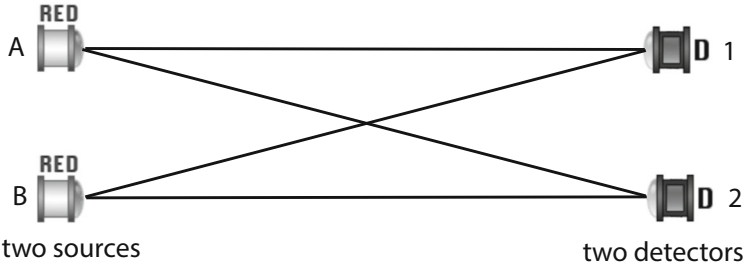

Fig. 4.28 Schematic of the Hanbury Brown and Twiss experiment. Light from two sources (to the left) impinges on two detectors (on the right). The lines indicate the alternative ways light can go from a source to a detector

when the experiment is done with photons, it is called photon bunching. The bosons "love" each other because they "hold hands" and both go to the same detector.

Such an experiment is similar to what Hong, Ou, and Mandel did in the 1980s, where they explicitly showed the quantum interference due to indistinguishability. We will show how this works in detail later in this section. The video Two of the Same are Different https://tinyurl.com/wsjpz932 provides some more introduction into these ideas.

Robert Hanbury Brown and Richard Twiss discovered the concept of photon bunching in the classical physics of radio astronomy. They used the intensity correlations between two radio telescopes to measure the size of the object that was producing the radio waves. They used this method to measure the sizes of stars and other astronomical objects. Although their initial idea was controversial, it soon proved to be correct and, among other applications, was later used in high-energy physics to analyze particle collisions. Now the method has fallen out of usage, as astronomers tend to employ full interferometry to determine the sizes of stars. This is often much harder to do than a Hanbury Brown-Twiss experiment.

The way the Hanbury Brown and Twiss experiment works is as follows. We imagine having light coming from an extended source in the sky, say a star. Think of the two sources in the figure as being at opposite ends of the surface of the star. We have three events—both photons going to one detector, or both going to the other detector, or one photon going to each detector. This third event can occur two ways and so it has interference. By measuring the fluctuations in the intensity of the light in the two detectors, Hanbury Brown and Twiss determined that the fluctuations behave like $\cos 2\pi d\theta/\lambda$, where d is the distance between the detectors, λ is the wavelength of light being observed, and θ is the angular size of the star. Both d and λ are known, allowing us to determine θ. It was first used to determine the angular size of Sirius (the dog star), one of the brightest stars in the galaxy. It turned out to be more accurate than the alternative method that employed Michelson-Morley interferometry at that time. It soon became the standard methodology for measuring the angular size of objects in the sky.

When people tried to describe similar experiments in the quantum world, the initial results also were quite controversial, but Edward Purcell and Leonard Mandel showed that the quantum version also worked, and thereby helped initiate the start of the field of quantum optics.

Previously, we have worked with single-particle quantum mechanics. For any experiment, we analyzed what happened to only one particle as it moved through the experiment. In this case, we found alternative ways that an experiment could occur, which we then translated into words; for example, as the photon goes through the sample path or the reference path, we added the arrows for the different alternative ways together, then squared the sum to get the probability.

Let's go back to refresh our memory about probability in Chap. 1. There, we saw that when we used the word *or*, we added probabilities, and, when we used the word *and*, we multiplied probabilities. In the quantum world, things are similar. When we use the word *or*, we add probability amplitudes, while when we use the word *and*, we multiply probability amplitudes. To date, we have only used the word "or" in quantum-mechanical problems because we had alternative ways for one particle to do something. But, once we start working with multiple particles (the quantum many-body problem), we can have situations where one particle does this and another particle does that. This means we need to carefully describe how to multiply probability amplitudes because we will be using the word "and."

But our amplitudes are arrows. We figured out how to add them by hooking them head to tail. But multiply them? How in the world will we do that? It turns out there is a simple procedure for doing so, and it takes two steps. First, we add the number of hours on the two clocks together to get the total time. The product arrow points in the direction of the sum of the times. Second, we multiply the amplitude of each arrow together. The product arrow has a length given by the product of the amplitudes of each arrow. Those of you familiar with complex numbers may recognize this as the same procedure used to multiply complex numbers together. Indeed it is, as quantum mechanics is described by complex numbers. But even if you don't recognize this, don't worry. Just apply the two rules above and everything will work fine (we will discuss complex numbers in more detail in Chap. 5).

Instead of getting formal with the mathematics at this point, we instead revisit the phenomenon of partial reflection from a thin slab of glass, but now analyze it from a compound probability amplitude perspective. To recap how we work with the probability amplitudes, recall the following: when we have two arrows that we add, we hook them head to tail and draw a final arrow from the tail of the first to the head of the last; when we have two arrows that we multiply, we add the times on the clocks to get the clock time (direction) of the final arrow, and we multiply the amplitudes together to get the final amplitude.

We redo the calculation of partial reflection of a slab of glass, now using probability amplitude multiplication. Recall that the event that the light leaves the source and arrives at the detector occurs in two alternative ways—reflection off the top surface and reflection off the bottom surface. We start in Fig. 4.29a, where the initial arrow for each alternative way has length 1 and is pointed at 12 o'clock. In panel (b), the first alternative has reflection off the top surface and refraction into

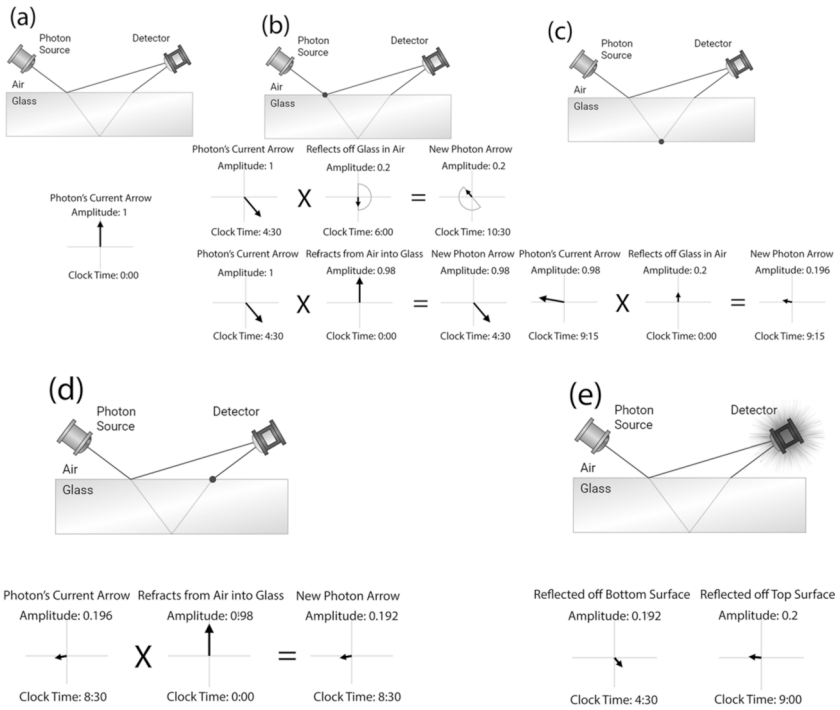

Fig. 4.29 Schematic of how to multiply arrows together using partial reflection of light; note the light reflects vertically, but we change the angles for clarity. (**a**) Initial state, with arrow of length 1 at 12 o'clock. (**b**) Reflection off the top and transmission through the top. Arrows are computed by multiplication. (**c**) Reflection off the bottom. (**d**) Transmission from glass to air. (**e**) Final arrows for each alternative way. The solid dot indicates where we are for a specific arrow computation

the glass. For reflection, we multiply the arrow (of length one and pointing in the direction given by the distance traveled by the photon from the source to the surface and the color of the light) by an arrow of length 0.2 pointing at 6 o'clock (alternative 1, top), and for the refraction, we multiply this arrow by an arrow of length 0.98 pointing at 12 o'clock (alternative 2, bottom). In each case, we multiply following our rules—add the times and multiply the lengths of the arrows. Panel (c) depicts the reflection off the bottom. The refraction arrow (length 0.98) has advanced to 9:15. We multiply it by the reflection arrow (length 0.2 pointing at 12 o'clock). Panel (d) depicts transmission from glass to air (multiply by an arrow of length 0.98 and pointing at 12 o'clock)—after taking into account the rotation as the photon traveled up from the bottom of the glass to the top glass-air interface, we end up with an arrow of length 0.192 pointing at 8:45. Finally, we get to the detector—here we need to add the compound probability arrows we have constructed after multiplying them as we described above and taking into account their winding as they travel from the glass-air interface to the detector. A tutorial https://bit.ly/4oPVId6 will

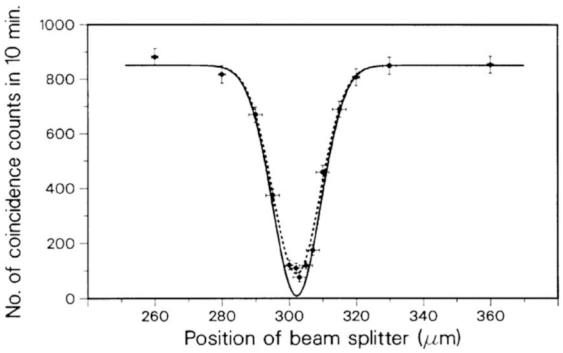

Fig. 4.30 Hong-Ou-Mandel experiment showing photon bunching. The vertical axis is the coincidence detection in both detectors. The experiment is carried out by adjusting the length of one of the photon paths. As the lengths become identical, the coincidence goes to zero, as does the curve. In the experiment, it does not go all the way to zero, but is quite close (Figure taken from Hong et al., 1987 (© American Physical Society, used with permission))

assist you with these calculations. This methodology for multiplying arrows will be used again as we describe the Hong-Ou-Mandel experiment.

We are now ready to describe in detail how the Hong-Ou-Mandel experiment works with one beam splitter and two detectors. The key point is that when the photons are identical (same color, same polarization, and arrive at the same time), then they undergo photon bunching. This means that the photons "hold hands" so that they both arrive in either the top detector or the bottom detector, but never one in each. The experiment is tuned by adjusting the arrival time of each photon at the beam splitter. As it is adjusted so that they are equal, the probability for one photon in each detector goes to zero. This is illustrated in Fig. 4.30 taken from the original paper (Hong et al., 1987). As one can see, there is a dip just as the photons arrive at the same time. This effect shows the indistinguishability of photons!

We analyze the Hong-Ou-Mandel experiment in two stages. First, we look at the case where one photon goes to each detector in Fig. 4.31. There are two alternative ways for this to occur—both photons reflect off the beam splitter, shown in panel (b)—both photons transmit through the beam splitter, shown in panel (c). First consider the reflection case in panel (b). As always, the arrow for each photon starts with unit length and pointing at 12 o'clock. The arrow from source A revolves as it moves through air and is shrunk by a factor of $1/\sqrt{2}$ when reflecting off the mirror from glass (because it is a 50–50 beam splitter). The arrow from source B revolves in air as well and adds 6 hours when it reflects off the beam splitter, since it reflects from air. It does not shrink, because we know it must go to detector D, because the event we are looking at corresponds to photons going to two different detectors, and we already know photon A goes to detector C. Now, to get the final arrow, we multiply the arrow from A by the arrow from B. Recall this involves adding the times and multiplying the lengths. We find the final arrow length is $1/\sqrt{2}$, and in this case, it is pointing at 2:30. Now, we consider the transmission case in panel (c). The

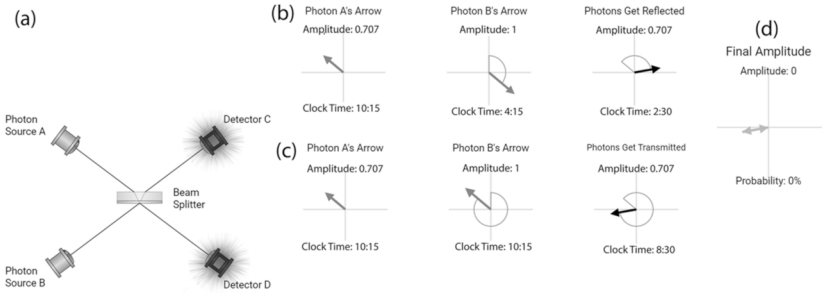

Fig. 4.31 Analysis of the Hong-Ou-Mandel experiment for the case where photons go to different detectors. The schematic of the experiment is shown in panel (**a**); the adjustment of the path lengths is made by moving the beam splitter up or down on a micron length scale. The analysis in terms of the arrows for each photon is given in panel (**b**) for reflection of both photons and in panel (**c**) for the transmission of both photons; the two arrows that are multiplied together are shown in the first two panels and the product arrow is in the third panel. Adding the arrows for each alternative way is shown in panel (**d**) and gives a net probability of 0. The photons never arrive at different detectors

arrows are similar to the previous case. The A arrow shrinks to a length of $1/\sqrt{2}$ and points in the same direction as before. The B arrow remains of length 1 and points in the same direction as arrow A, because both are transmitted. Now, when we multiply the arrows, we find they yield an arrow of length $1/\sqrt{2}$ and pointing at 8:30. Adding the arrows for these two alternative ways is shown in panel (d). One can see they are equal in magnitude, but opposite in direction, so they add to zero. Hence, the probability for the photon to go to detector and have both fire is 0.

The case where both photons go to one detector (either C or D) is shown in Fig. 4.32. In panel (a), we show the case where both go to detector C. The arrows start with a length of one and pointing at 12 o'clock. Arrow A reflects from glass and shrinks to a length of $1/\sqrt{2}$. Arrow B transmits and keeps a length of one because it must go to detector C. Multiplying these arrows together yields the final arrow shown in the third panel. Its length is $1/\sqrt{2}$, so the probability is 50%, because there is only one alternative way. In panel (b), we show the case where both go to detector D. The arrows start with a length of one and pointing at 12 o'clock. Arrow A transmits from glass and shrinks to a length of $1/\sqrt{2}$. Arrow B reflects and keeps a length of one because it must go to detector D—but it does add 6 hours to the clock due to the reflection from air. Multiplying these arrows together yields the final arrow shown in the third panel. Its length is $1/\sqrt{2}$, so the probability is 50%, because there is only one alternative way. Hence half the time both photons go to detector C and half of the time to detector D. This result is called *photon bunching*. The tutorial https://bit.ly/3XMl806 can assist you in the analysis of this experiment.

This experiment shows that by employing a clever design, one can test the remarkable properties of quantum particles that arise from their indistinguishability. In particular, we see from this experiment that photons bunch together or act as

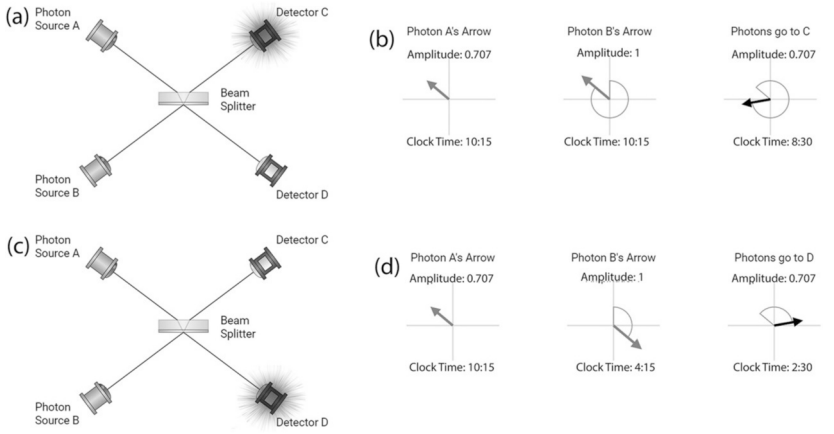

Fig. 4.32 Analysis of the Hong-Ou-Mandel experiment for the case where photons go to the same detector. The schematic of the experiment where both go to detector C is shown in panel (**a**). The analysis in terms of the arrows for each photon is given in panel (**b**); the two arrows are multiplied together for the third panel. The schematic of the experiment where both go to detector D is shown in panel (**c**). The analysis in terms of the arrows for each photon is given in panel (**d**)

bosons. A little thought reveals that this is a truly remarkable result, both in what it shows and in how it does it.

The key for any experiment on indistinguishability is that both of the particles are completely identical. For photons, this means same color, same polarization, and arriving at the same time. Note that this notion of indistinguishability is analogous to the notion of complementarity that we used for single-photon experiments. There, we realized that whenever which-way path information was known, the interference effects disappear. Similarly here, whenever the two photons can be distinguished (since they are not identical), the photon bunching effect goes away.

You might ask why does this effect go away when the photons are distinguishable, since our analysis does not seem to rely on the photons being the same color or same polarization. The reason is because we must have two alternative ways for one photon to go to each detector in order to have interference. If photons are distinguishable, then the analysis of this experiment requires us to have two events for the case where one photon goes to each detector, with each event occurring with just one alternative way, because each detector could be equipped with an ability to measure polarization or the color of light in addition to just counting the photons that enter the detector. Then, even if we have not equipped the detectors in this fashion, because we could have, it is enough to change the analysis (recall when we used tags, they led to distinguishable events even if we did not measure the tag). So, if they are distinguishable, then half the time they bunch and go to one detector and half the time they anti-bunch and one photon goes to each detector. The Hong-Ou-Mandel experiment is very sensitive and really requires the photons to be indistinguishable in order to work.

I have one more technical note for you. You might not have liked that when we analyzed the case where both photons went to detector C or D, we applied the shrink factor only once, because we said the fate of the second photon was known for that event. It is possible to apply the shrink factor to both photons in the analysis, but, before we determine the final arrows, we then have to multiply by the so-called *Bose-enhancement factor*, which is equal to $\sqrt{N!}$ for an N-photon state (so it is $\sqrt{2}$ for the two-photon state here). You can see, in this case, it gives the same answer, and personally, I prefer the way we did it before. But, if you work with N-photon states in this way, you will need to use this Bose enhancement factor for those cases, because it is the general way to work with them, and trying to generalize the way we did it can get problematic (even in the case of an unbalanced beam splitter with two photons). I am telling you this just so you know that this effect is there, in case you run into it in the future. And, you might not be surprised to know that Feynman has a physical way of motivating this Bose enhancement factor as well, and he does. But, I am not going to go into those details here.

We have journeyed far in this book. Starting with classical physics, we slowly showed the different behavior of the quantum world, including the "quantum mystery" of interference. Now, we have moved from single-particle problems to many-body problems. The latter contain unsolved problems that occupy the time of many, many physicists, including myself. Why do we spend so much time working on these problems? Well, the collective quantum effects on many quantum particles bring in a wealth of new behavior, some of which we hope will have practical applications for the technology of the future.

Within the field of quantum information, many of these problems relate to quantum sensing, quantum communications, and quantum computing. Within fields like solid-state physics, they can involve newer, stronger magnets, novel ways to switch materials from metals to insulators, and unique quantum effects like superconductivity. We try hard to understand these systems not only because they are likely to usher in novel bizarre quantum behavior, but also because they may lead to the technology of tomorrow.

Many of the problems that people focus on are how these quantum systems change with time. Understanding the evolution of a quantum system that is driven by a strong electric field is one of the longest standing goals within the field. Recent advances in experiments with ultra-strong and ultrafast lasers are shedding new light onto the behavior of these systems.

Why are they so complicated? It turns out that because quantum systems can interact strongly with each other, we can get into a situation similar to how one solves puzzles like the Rubik's cube—namely, a quantum particle cannot move from one place to another without *taking into account what every other particle is doing*. When we get to the huge number of quantum particles in a piece of metal, it becomes clear that this is a hard problem to solve. We will be discussing many-body-physics problems in the later chapters of this book.

Section 4.5 Problems:

4.5.1 Multiplying Arrows I

We have two arrows. The first has an amplitude of 0.4 and is pointing at 8:00. The second has an amplitude of 0.6 and is pointing at 2:00.

What is the amplitude of the product of the two arrows? What is the time for the product?

4.5.2 Multiplying Arrows II

We have two arrows. The first has an amplitude of 0.8 and is pointing at 5:00. The second has an amplitude of 0.2 and is pointing at 9:00.

What is the amplitude of the product of the two arrows? What is the time for the product?

4.5.3 Polarized Hong-Ou-Mandel Experiment

Consider a Hong-Ou-Mandel experiment like the one we described in the text. If identical polarizers (both oriented along the vertical direction) are placed on each path from the sources to the beam splitter (before the beam splitter), will the results of the experiment change or remain the same? Write an essay to describe your answer.

4.5.4 Erased Hong-Ou-Mandel Experiment

Suppose the beam splitter of a Hong-Ou-Mandel experiment is replaced by a polarizing beam splitter. This changes the polarizations of each photon to always be perpendicular to each other when one photon is transmitted and the other reflects and for the polarizations to be the same when both transmit or both reflect.

Is there a way to add polarizers, so that the polarization information can be erased, and the Hong-Ou-Mandel experiment will still work? Write an essay that describes how one can erase the entanglement created by the polarizing beam splitter.

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Part II
Quantum Operator Mechanics

Chapter 5
Mathematical Interlude

No truth seems true.

Maya Agelou, Poet, 1976

We are now in the second part of the book, covering the material of *Quantum Operator Mechanics*, which spans Chaps. 5–12. We have a video that provides an overview of this second part https://tinyurl.com/bdeu6de9 and of this chapter https://tinyurl.com/mv4za5by.

5.1 Devious Math Tricks

Remember that kid from algebra class who sat across from you and seemed to know the answer to every question faster and more accurately than you? I am going to let you in on a secret—most likely that curve buster was not a genius, nor born to do math—instead, they simply learned a number of devious tricks to get through the material faster and with fewer errors. After finishing this section, you will be able to do so too!

The biggest myth to dispel is that geniuses in algebra have photographic memories and can recall the relevant formulas and apply them at will. Such people may exist, but the braniac across the row from you was not one of these. The reality is you only need to know about ten important equations, and by using them as I am about to describe to you, you will find that you will just know them when you need them. You too can become an algebra master!

Devious trick number 1: $x + 0 = x$.

This is called the "add-zero" trick. It comes up so often that you cannot get through most problems without encountering it. The problem is no one has told you about it, so you often stumble onto it and fail to see its importance. We will see and use

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J. K. Freericks, *Quantum Mechanics Done Right*,
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the add-zero trick again and again. Right now, let me illustrate it to you. Consider the following problem: Factor $x^2 - 25$.

Go ahead and give it a try, but do not just write down the answer if you have memorized it, and see if you can derive it with the add-zero trick.

Here is how it works. An educated guess suggests we should add a number times x as one of the terms, and the 25 suggests that we add $5x - 5x$ to obtain

$$\begin{aligned} x^2 + \underbrace{5x - 5x}_{=0} - 25 &= (x^2 + 5x) - (5x + 25) \\ &= x(x+5) - 5(x+5) \\ &= (x-5)(x+5). \end{aligned} \tag{5.1}$$

All we did was add zero! The creativity was to recognize what zero to add. Indeed, this is how you apply the technique. You will quickly learn what ideas might work for what to add by using the method multiple times. One strategy to follow is to look for a term that is missing in a familiar identity. When this occurs, just add and subtract the missing term and see if using the add-zero identity helps!

Devious trick number 2: $x \cdot 1 = x$.

Just like the "add-zero" trick, the "multiply-by-one" trick can be used to simplify many different expressions. It is often seen in cases where we want to rationalize a denominator, such as the following. Rationalize the denominator of $(1 - \sqrt{3})/(1 + 2\sqrt{3})$.

The strategy is to multiply the numerator and the denominator by the same factor such that the denominator no longer has a square root. Recalling our work above can be written more generally as $a^2 - b^2 = (a+b)(a-b)$ yields the strategy to use $1 = (1 - 2\sqrt{3})/(1 - 2\sqrt{3})$ as the term we multiply by. Now, compute

$$\frac{1-\sqrt{3}}{1+2\sqrt{3}} \times \frac{1-2\sqrt{3}}{1-2\sqrt{3}} = \frac{1-\sqrt{3}-2\sqrt{3}+6}{1-12} = -\frac{7-3\sqrt{3}}{11}. \tag{5.2}$$

This multiply-by-one trick can also be used to simplify products of integers via factorials. Recall the factorial symbol $n! = 1 \times 2 \times 3 \times 4 \cdots \times (n-1) \times n$ for the product of the first n nonzero integers. Then, the following product can be rearranged in terms of factorials as follows:

$$\begin{aligned} (n+1)(n+2)\cdots(2n-1)(2n) &= \frac{1 \cdot 2 \cdot 3 \cdots n}{1 \cdot 2 \cdot 3 \cdots n} \times (n+1)(n+2)\cdots(2n-1)(2n) \\ &= \frac{(2n)!}{n!}. \end{aligned} \tag{5.3}$$

Devious trick number 3: Remember simple identities and use them to construct more complicated identities.

As I mentioned before, there are only a handful of identities worth remembering. Nearly everything else can be derived from these few. As an example, think about

trigonometric identities. There are loads and loads of them. But they all follow from two main results—remembering the definitions of trigonometric functions in terms of sine and cosine and using the fact that $\cos^2\theta + \sin^2\theta = 1$. Let us illustrate how.

Suppose I want to express $\sec^2\theta$ in terms of $\tan\theta$. I simply recall that $\sec\theta = 1/\cos\theta$. Then, I use the multiply-by-one trick with $1 = \cos^2\theta + \sin^2\theta$ to find that

$$\begin{aligned}\sec^2\theta &= \frac{1}{\cos^2\theta} = \frac{1}{\cos^2\theta} \times (\cos^2\theta + \sin^2\theta) \\ &= \frac{\cos^2\theta}{\cos^2\theta} + \frac{\sin^2\theta}{\cos^2\theta} = 1 + \tan^2\theta. \end{aligned} \tag{5.4}$$

Devious trick number 4: Recognize simple identities, when expressed in a complex form.

This devious trick takes a bit more practice to master, as it requires you to use abstraction to recognize the underlying simple identity. Here is an example. We already saw that $x^2 - 25 = (x+5)(x-5)$. We should be able to recognize this in different contexts, where x is more complicated. For example, consider

$$\frac{\sin^2\theta - 25}{\cos^2\theta} = \tan^2\theta - 25\sec^2\theta = (\tan\theta + 5\sec\theta)(\tan\theta - 5\sec\theta). \tag{5.5}$$

You can see the relationship between these two identities rather clearly.

A more complicated example is to factor $x^4 + 2x^2 - 24$. To do this, we first use the add-zero trick and then recognize our previous identity. This is done as follows:

$$\begin{aligned} x^4 + 2x^2 + \underbrace{1-1}_{=0} - 24 &= (x^2+1)^2 - 25 \\ &= (x^2 + 1 + 5)(x^2 + 1 - 5) = (x^2+6)(x^2-4). \end{aligned} \tag{5.6}$$

Note how we had to recognize that $x^4 + 2x^2 + 1 = (x^2+1)^2$ was the square in the expression. To build up practice in doing this, you might find it easier to first let $x^2 = y$ and then factorize $y^2 + 2y - 24$, and then, let $y \to x^2$ at the end.

Devious trick number 5: Use only the information that you know.

This trick also has an added point—learn how to make small changes to what you know to lead to what you want to derive. This trick also takes time and practice to recognize it, but it is worthwhile doing so, as it is a very important trick. We will illustrate it by example. Consider evaluating the following sum:

$$x^2 + x^4 + x^6 + \cdots = \sum_{n=1}^{\infty} x^{2n}. \tag{5.7}$$

Note that the right-hand side is our summation notation. You think of it as a rule for adding numbers. Start with $n = 1$, which gives x^2, and then, add to that the

result with $n = 2$, which gives x^4, and then, add $n = 3$, which gives x^6, and so on, without ever stopping. This sum looks like our friend the geometric series, but it is with respect to x^2 and it is missing the first term. The idea here is to use the geometric series that you know to solve this. One way is by using add zero:

$$\sum_{n=1}^{\infty} x^{2n} = \sum_{n=0}^{\infty} x^{2n} - 1 = \frac{1}{1-x^2} - 1 = \frac{x^2}{1-x^2}. \tag{5.8}$$

We could also proceed another way, by recognizing that

$$\begin{aligned} x^2 + x^4 + x^6 + \cdots &= x^2 \underbrace{(1 + x^2 + x^4 + \cdots)}_{\text{geometric series}} \\ &= x^2 \sum_{n=0}^{\infty} x^{2n} = x^2 \frac{1}{1-x^2} = \frac{x^2}{1-x^2}, \end{aligned} \tag{5.9}$$

as before. Note how we took what we knew, modified it slightly, and discovered something new. I always think about this trick when I think about a scene from Douglas Adams, *The long dark tea-time of the soul*, where the holistic detective Dirk Gently has a curious way of investigating. An example is when he is looking for a woman who lives at a particular address in a town he does not know. Rather than looking up the address, he just gets in the car and starts driving. The car gets into an accident, and lo and behold, the woman he was trying to find was the driver of the other car. He says, "I may not have gone where I intended to go, but I think I have ended up where I needed to be" Adams (1989). Using what you know is similar. Just start from what you know and try to get to where you want to be. *Learning to use what you know, and only what you know will help you a lot in learning quantum mechanics.*

We end by discussing one other technical skill, which involves shifting indices in summations. Students often struggle with this skill. Yet it is straightforward to master it. Doing so takes you to the next level of algebraic skill. Let us look at the last example in Eq. (5.7). We know the geometric series starts with $n = 0$ and not $n = 1$. So, let us shift $n \to \bar{n} + 1$. Then, $\bar{n}$ runs from 0 to infinity (because n ran from 1 to infinity). We obtain

$$\sum_{n=1}^{\infty} x^{2n} = \sum_{\bar{n}=0}^{\infty} x^{2\bar{n}+2} = \sum_{\bar{n}=0}^{\infty} x^{2\bar{n}} \cdot x^2 = \frac{1}{1-x^2} x^2 = \frac{x^2}{1-x^2}, \tag{5.10}$$

just like before. We will see this index-shifting skill becomes critically important in mastering many identities we develop later. I will help you recognize and develop this skill as we move forward.

Table 5.1 Ten mathematical identities you need to know

Identity	Formula
The quadratic equation	$ax^2 + bx + c = 0$ has two roots $x_\pm = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$
Fundamental trigonometric identity	$\cos^2\theta + \sin^2\theta = 1$
Trigonometric definitions	$\sec\theta = \frac{1}{\cos\theta}$, $\tan\theta = \frac{\sin\theta}{\cos\theta}$, $\text{cosec}\theta = \frac{1}{\sin\theta}$, $\text{cotan}\theta = \frac{\cos\theta}{\sin\theta}$
Double-angle formulas	$\cos 2\theta = \cos^2\theta - \sin^2\theta$, $\sin 2\theta = 2\sin\theta\cos\theta$
Factorization of the difference of squares	$a^2 - b^2 = (a+b)(a-b)$
Geometric series	$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$ for $\lvert x\rvert < 1$, $\sum_{n=0}^{N} x^n = \frac{1-x^{N+1}}{1-x}$
Binomial theorem	$(x+y)^N = \sum_{n=0}^{N} \binom{N}{n} x^{N-n} y^n = \sum_{n=0}^{N} \frac{N!}{(N-n)!n!} x^{N-n} y^n$
Power series for the exponential	$e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n$
Relation of trig functions to exponentials	$\cos\theta = \frac{1}{2}(e^{ix} + e^{-ix})$, $\sin\theta = \frac{1}{2i}(e^{ix} - e^{-ix})$
Relation of hyp functions to exponentials	$\cosh x = \frac{1}{2}(e^x + e^{-x})$, $\sinh x = \frac{1}{2}(e^x - e^{-x})$

We end this section with ten important identities that you need to know to work through this book. We will use them freely. We will develop some of these in this chapter if you have never seen them before. They are summarized in Table 5.1. The video Devious Math Tricks https://tinyurl.com/4k4wjxv7 summarizes this discussion.

Section 5.1 Problems:

There are nine problems to practice your devious tricks. Beware, some of them are challenging.

5.1.1 Factoring

Factor $x^2 + 2xy + y^2 - 9$.

5.1.2 Double-Angle Formula

Use the double-angle formula in the form $\cos\theta = \cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}$ to show that $\cos\frac{\theta}{2} = \pm\sqrt{\frac{1+\cos\theta}{2}}$. How do we choose which sign is correct?

5.1.3 Half-Angle Formula

Use the same double-angle formula to find $\sin\frac{\theta}{2}$.

5.1.4 Power Series for Exponentials

Show that $\sum_{n=0}^{\infty}\frac{x^n}{(n+1)!} = \frac{e^x-1}{x}$.

5.1.5 Add Zero Instead of Factorize

Show that $x^2 = 3+2\sqrt{2}$ implies $x = \pm\sqrt{1+\sqrt{2}}$ by using the "add-zero" trick (do not use the quadratic formula).

5.1.6 Double Trouble

Use the double-angle formulas for sine and cosine to show that $\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$.

5.1.7 Geometric Series

Use the finite geometric series formula (in our list of ten formulas) to show that $1+\sqrt[4]{2}+\sqrt{2}+\sqrt[4]{8} = \frac{1}{\sqrt{2}-1}$.

5.1.8 Fun with Factorials

Show that $\frac{(2n)!}{2^n n!} = 1\cdot 3\cdot 5\cdots(2n-3)\cdot(2n-1) = (2n-1)!!$, where we define the double factorial symbol to mean skip every other term in the product of integers. An

even double factorial is a product of all even integers, while an odd double factorial is a product of all odd integers.
Hint: think of how many even integers are in $(2n)!$. How can they be removed?

5.1.9 Trigonometric Summation

Show $\sum_{n=1}^{\infty}(\cos\theta)^{2n} = \frac{1}{\tan^2\theta}$, for $|\theta| < \frac{\pi}{2}$.
Hint: use the geometric series.

5.2 Complex Numbers

Up until now, we have used arrows to represent quantum probability amplitudes. Turns out, those arrows are closely related to complex numbers, and we need to understand complex numbers well to make progress in more sophisticated quantum mechanics calculations. So, we now delve into the world of complex numbers.

You probably already learned that $\sqrt{-1}$ is not a real number but is an imaginary number called i, and we can form complex numbers $z = x + iy$ from these real and imaginary numbers. Complex numbers often appear imposing and foreign, but as you work with them more and more, you will find they become familiar and comfortable. We call x the real part of z ($x = \mathrm{Re}\, z$) and y the imaginary part of z ($y = \mathrm{Im}\, z$); note that *both* x and y are real numbers.

You should already be familiar with working with real numbers in a plane, denoted by an ordered list of two real numbers (x, y). The real plane has both magnitudes and directions to it. We often use polar coordinates of r and θ to denote the radial distance from the origin (at (0,0)) and the angle measured counterclockwise from the horizontal (x) axis. In this context, points on the plane can be thought of as vectors. Complex numbers are also two-dimensional objects that can be represented with an ordered list $x + iy \rightarrow (x, y)$, and we refer to the points as lying in what we now call *the complex plane*. Here, we can still think of representing the points on the complex plane in terms of vectors, but the big difference from the two-dimensional real plane, and the complex plane, is for complex numbers, we have a rule that allows us to multiply two complex numbers to produce a third complex number, the product of the two. We just use the fact that $i^2 = -1$, so that $(x+iy)(x'+iy') = xx'-yy'+i(xy'+yx')$. This difference means we can do much more with complex numbers than we can with a two-dimensional real plane, as you will see in the discussion below.

As an example, we plot the complex number $z = 2 + 3i$ in Fig. 5.1, to illustrate both the Cartesian representation and the polar representation. The real part $x = 2$ and the imaginary part $y = 3$ are the Cartesian representation of the complex number. The polar representation used the radius $r = \sqrt{x^2 + y^2}$ (here given by $r = \sqrt{2^2 + 3^2} = \sqrt{13}$) and the polar angle (measured from the horizontal axis) $\theta = \tan^{-1}(y/x)$ (here given by $\theta = \tan^{-1}(3/2) \approx 56.3^{\circ}$); one has to shift the

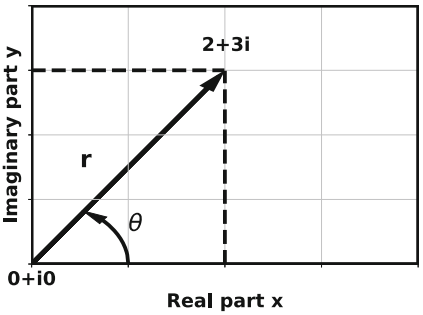

Fig. 5.1 Comparison between Cartesian and polar representations of complex numbers. The complex number $z = 2 + 3i$ can be represented as a Cartesian element (2, 3) or as a polar element $r = \sqrt{2^2 + 3^2}$ and $\theta = \tan^{-1}\left(\frac{3}{2}\right)$. The origin $0 + i0$ is denoted by (0, 0) or $r = 0$

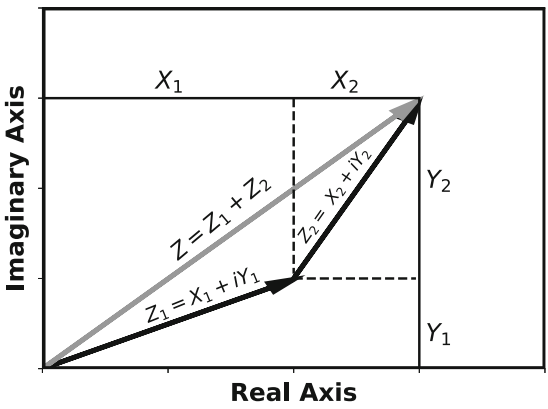

Fig. 5.2 Schematic of how we add two complex numbers by hooking head to tail and drawing a final arrow. Note how this is the same as adding the real parts for the real part of the sum and adding imaginary parts for the imaginary part of the sum

arctangent by $\pm\pi/2$ for complex numbers with $y < 0$. These coordinates represent the distance and angle that a given complex number has relative to the origin (and the x-axis). Of course, which coordinates you choose to use, Cartesian or polar is a matter of taste. Some calculations are easier in one form (addition in the Cartesian form), and others are easier in the other form (multiplication in the polar form). So, we need to gain practice in using *both* forms and be able to convert from one form to another.

If we want to add two complex numbers, the operation is identical to how we add vectors in a two-dimensional plane—we just add the horizontal components and the vertical components. Multiplying complex numbers is something new. The only multiplication with vectors was the dot product or the cross product, neither of which is what we want for multiplication of complex numbers, because we want the product to be a complex number as well. The rule for multiplying complex numbers is most easily stated in polar form—we multiply the radii together and we add the polar angles. This is one of the major differences between complex numbers and the real two-dimensional plane.

Now, let us think about the arrows we have been using. For the arrows, we added them by hooking the head to tail and then finding the final arrow. As Fig. 5.2 shows, this is identical to adding numbers in the plane. It turns out we do exactly the same

thing to add complex numbers. If $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, then the sum is $z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$. For example, add $z_1 = 4 + 7i$ to $z_2 = 3 - 2i$ and see that you obtain $7 + 5i$ for the sum. Recall as well that we also had a procedure for multiplying arrows. We multiplied their lengths and added the hours on the clocks. This is the same as saying the radius of $z_1 \times z_2$ is $r_1 \times r_2$ and the polar angle for $z_1 \times z_2$ is $\theta_1 + \theta_2$. *Hence, the arrows we have been using all along are just complex numbers!* We find the Cartesian form is the best to use when adding them together and the polar form is the best to use when multiplying them. It turns out that the rules we need to follow for working with complex numbers are simple and natural—in particular, we now can add two complex numbers and obtain a complex number as the sum, and we can multiply two complex numbers and obtain a complex number as the product. There is one issue in relating arrows to complex numbers that we do need to address. In polar form, the angle is measured in a counterclockwise direction and from the horizontal axis, while for the arrows, we measured time in a clockwise direction with respect to the vertical axis. The issue of clockwise versus counterclockwise will become clear when we discuss time evolution (they are indeed opposite for good reason). The measurement with respect to the horizontal or vertical axes is a convention and they are just different.

Complex numbers have additional important operations on them, which we will now discuss. We start from the notion of a complex conjugate, denoted with an asterisk superscript: $z^* = (x + iy)^* = x - iy$. It is defined as a reflection of the complex number about the horizontal axis, which changes the sign of the imaginary part. This is illustrated in Fig. 5.3. This rule is simple to implement. Anytime you have an i in a complex number, you change $i \to -i$ for the complex conjugate—said differently, you change the sign of the coefficient of i. Hence, the complex conjugate of a complex number is the original complex number or $(z^*)^* = z$. The proof is direct—$z = x + iy$ implies that $z^* = x - iy$, which implies $(z^*)^* = x + iy$. You can also show this by noting that two reflections in the same mirror returns the original complex number.

The conjugate comes in handy in computing the radius in polar form. We denote the radius by the norm (or absolute value) of z, $|z| = r$. Since $r = \sqrt{x^2 + y^2}$, we have $|z|^2 = r^2 = x^2 + y^2 = zz^*$. Proof: $zz^* = (x + iy)(x - iy) = x^2 + ixy - ixy + i(-i)y^2 = x^2 + y^2$, since $i(-i) = -i^2 = -(-1) = 1$. Note that this

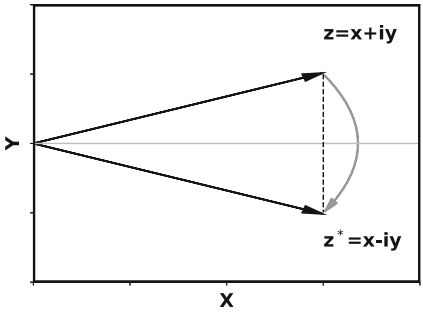

Fig. 5.3 Graphic to illustrate the operation of complex conjugation. Here, we start with a complex number above the horizontal axis and reflect it below when we take the complex conjugate. If we take the complex conjugate again, we return to our original number because a second reflection just reflects back

procedure for the norm produces the result you already know from the conversion between Cartesian and polar coordinates. The complex conjugate in zz^* ensures we get $r^2 = x^2 + y^2$ for the squared radius. For example, $|3 + 4i| = \sqrt{3^2 + 4^2} = 5$.

There will be more to discuss about complex numbers, but we need to know more about exponentials before we can finish this discussion. We will be covering the exponential in the next section. But, before leaving, we have a few more operations to show you. The first is called *rationalizing the denominator* of a complex number. Suppose I have $\frac{1}{z} = \frac{1}{x+iy}$. I can "multiply by one," using the complex conjugate to find

$$\frac{1}{z} \cdot \frac{z^*}{z^*} = \frac{z^*}{|z|^2} = \frac{x - iy}{x^2 + y^2}. \tag{5.11}$$

Note how we got rid of all i's in the denominator. This procedure of "bringing the i up from the denominator" is called rationalizing the denominator. As an example, see if you can verify that $\frac{1}{1-i} = \frac{1}{2}(1+i)$. Remember well the multiply-by-one trick. We will see it return many, many times.

Determining the real and imaginary parts of complex numbers is made easier by using the complex conjugate. For example, you should be able to immediately verify that

$$\mathrm{Re}\, z = \frac{1}{2}(z + z^*), \quad \text{and} \quad \mathrm{Im}\, z = \frac{1}{2i}(z - z^*). \tag{5.12}$$

I hope you are becoming comfortable with complex numbers, especially since you have been working with them already as arrows. They may appear a bit forbidding with the i's around, but if you ever feel you are getting lost, remember that they are just those arrows you worked with in Chaps. 1–4.

There is one final form to write down, which is the polar form of a complex number. A little trigonometry shows that

$$z = x + iy = r(\cos\theta + i \sin\theta) \tag{5.13}$$

is the result. In the next section, we will use exponentials of complex numbers to write this another way; if you are familiar with the Euler identity, you will already recognize it above. But first we need to refresh our memory about exponentials, and we need to derive some useful identities. The next section gets a little technical, and you may find it useful to review proof by induction and the binomial theorem before reading it. The exponential function is important, so let us plunge in once you have finished the practice problems below. The video An Introduction To Complex Numbers https://tinyurl.com/r23yn7nw provides additional context for you.

Section 5.2 Problems:

5.2.1 Complex Addition

Compute $(1 - 3i) + (2 + 4i)$.

5.2.2 Complex Multiplication

Compute $(1-3i)(2+4i)$. Do this first in the Cartesian way using $i^2 = -1$ and then do it the polar way. Verify that the two ways give the same answer.

5.2.3 Norm I

Determine $|6 - 8i|$.

5.2.4 Rationalizing the Denominator

Rationalize the denominator of $\frac{1}{2-7i}$.

5.2.5 Norm II

Compute $\left|\frac{1}{1-i}\right|$ by computing $\sqrt{zz^*}$.

5.2.6 Powers of i

Compute i^2, i^3, and i^4. How does $\frac{1}{i}$ compare to i?

5.2.7 Polar to Cartesian

A complex number has $r = 2$ and $\theta = \frac{\pi}{3}$ (the angle is expressed in radians not degrees). Determine the Cartesian form.

5.2.8 Cartesian to Polar

Convert $4 - 3i$ to polar form.

5.2.9 Complex Conjugate

Which complex numbers are equal to their complex conjugate $z = z^*$? Which complex numbers are the opposite of their complex conjugate $z = -z^*$?

5.3 The Exponential Function

The *exponential* function e^x is one of the most important functions in math, and it is used widely in quantum mechanics, so we want to ensure we understand its properties well. When x is real, it describes exponential growth for $x > 0$ and exponential decay for $x < 0$. When it is imaginary, the function oscillates—all of this behavior is important for quantum mechanics. Let us begin by considering raising a number to a positive integer power. For example,

$$\begin{aligned} 2^{x+y} &= \underbrace{2 \times 2 \times 2 \times 2 \cdots \times 2}_{x+y \text{ terms}} \\ &= \underbrace{2 \times 2 \times \cdots \times 2}_{x \text{ terms}} \times \underbrace{2 \times 2 \times \cdots \times 2}_{y \text{ terms}} \\ &= 2^x \times 2^y = 2^x 2^y. \end{aligned} \tag{5.14}$$

While we showed this explicitly for the number 2 raised to any positive integer, the exponent can be zero ($2^0 = 1$) or negative $2^{-x} = \frac{1}{2^x}$). We can also relax the condition that x and y are integers and use the property $2^{x+y} = 2^x 2^y$ for arbitrary real exponents. We will use this fundamental property of exponents to define what e^x is. Namely, we require $e^{x+y} = e^x e^y$ and we require the slope of e^x to be equal to 1 at $x = 0$. This latter condition will determine the value of e. A plot showing how this works is given in Fig. 5.4. Note how the exponential of e^x lies in between 2^x and 4^x. We adjust the value of e until the slope is exactly 1 (corresponding to the dashed line) at $x = 0$. This ends up yielding a transcendental number approximately equal to $2.718281828\cdots$. We will show how to determine this value for e soon.

Our next goal will be to find what is called a power series (or Taylor series) expansion for e^x. If you have taken calculus, you probably learned a general methodology for how to find power series expansions for arbitrary functions. Here, we will use the binomial theorem to determine the power series for the exponential function without calculus.

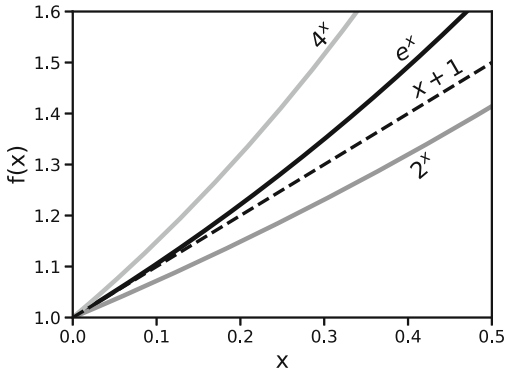

Fig. 5.4 Illustration of how the exponential function is defined. We show 2^x (dark gray), 4^x (light gray), e^x (solid black), and $x + 1$ (dashed black). The function e^x is defined so that its slope at $x = 0$ is 1 (the same slope as the function $x + 1$)

To start, we need to establish the binomial theorem, which is given by

$$(r+s)^n = \sum_{m=0}^{n} \frac{n!}{m!(n-m)!} r^{n-m} s^m, \tag{5.15}$$

where $n! = 1 \times 2 \times 3 \times \cdots \times (n-1) \times n$ is the product of the first n integers and we are using the summation symbol $\sum$, which is a compact way of expressing a summation $\sum_{m=0}^{n} f(m) = f(0) + f(1) + \cdots + f(n)$. Note further that we define $0! = 1$ by convention.

We prove this formula by induction. The way a proof by induction works is to first establish a base case and then assume it holds for all integers from the base case to an integer n and use only those results to prove it also holds for $n + 1$. For example, if the base case corresponds to $n = 0$, then the proof for "$n+1$" establishes it for $n = 1$. Then, the proof for "$n + 1$" establishes it for $n = 2$, and so on, without ever ending. Think about this carefully to see why it works and establishes the proof for all values of n. Bill Courtney gave me a great analogy to think of when proving by induction—consider a line of dominoes all in a row; the base case is like tipping over the first domino and, the proof for $n + 1$ is like we assume the first n dominoes have also fallen; can we show the $n + 1$st will fall as well? if yes, then we know all of the dominoes will fall and the proof is complete. The summation in Eq. (5.15) may seem foreign to you. Whenever we encounter a complex formula, we should examine it for simple cases. For example, when $n = 0$, there is one term in the sum, when $n = 1$, there are two terms, and when $n = 2$, there are three:

$$n = 0 : (r+s)^0 = \sum_{m=0}^{0} \frac{0!}{m!(0-m)!} r^{n-m} s^m = \frac{0!}{0!0!} r^0 s^0 = 1 \tag{5.16}$$

$$n = 1 : (r+s)^1 = \sum_{m=0}^{1} \frac{1!}{m!(1-m)!} r^{n-m} s^m = \frac{1!}{0!1!} r^1 s^0 + \frac{1!}{1!0!} r^0 s^1 = r + s \tag{5.17}$$

$$n = 2 : (r+s)^2 \quad = \sum_{m=0}^{2} \frac{2!}{m!(2-m)!} r^{n-m} s^m = \frac{2!}{0!2!} r^2 s^0 + \frac{2!}{1!1!} r^1 s^1 + \frac{2!}{2!0!} r^0 s^2$$
$$= r^2 + 2rs + s^2. \tag{5.18}$$

Please look carefully at how we go from the summation rules to the explicit terms. Also recall these final results are what you should already know about $(r+s)^n$. You may have even heard of the relationship of these results to Pascal's triangle, but we will not elaborate further on that here.

Now, we go back to our proof by induction. The base case for the binomial theorem corresponds to $n = 0$, which properly expresses $(r+s)^0 = 1$, as we directly evaluated above. Now, we assume the binomial theorem holds for all integers from 0 to n and use that to establish the result for $n+1$. This will then complete the proof.

The proof then proceeds as follows:

$$\begin{aligned}(r+s)^{n+1} &= (r+s)(r+s)^n \\ &= (r+s) \sum_{m=0}^{n} \frac{n!}{m!(n-m)!} r^{n-m} s^m \\ &= \sum_{m=0}^{n} \frac{n!}{m!(n-m)!} r^{n-m+1} s^m + \sum_{m=0}^{n} \frac{n!}{m!(n-m)!} r^{n-m} s^{m+1}, \end{aligned} \tag{5.19}$$

where we used the binomial theorem result for $(r+s)^n$ in the second line and distributed the multiplication in the third line. Next, we use our index shifting to replace $m \to m' - 1$ in the second term to yield

$$(r+s)^{n+1} = \sum_{m=0}^{n} \frac{n!}{m!(n-m)!} r^{n+1-m} s^m + \sum_{m'=1}^{n+1} \frac{n!}{(m'-1)!(n+1-m')!} r^{n+1-m'} s^{m'}. \tag{5.20}$$

Be sure to verify the sum over m' is identical to our previous sum over m for the second term. We now use the "multiply-by-one" trick to rearrange the fractions to look more like the $n+1$ binomial coefficient via

$$\begin{aligned}\frac{n!}{(m'-1)!(n+1-m')!} &= \frac{n!}{(m'-1)!(n+1-m')!} \cdot \frac{n+1}{n+1} \cdot \frac{m'}{m'} \\ &= \frac{(n+1)!}{m'!(n+1-m')!} \cdot \frac{m'}{n+1} \text{ for } 1 \le m' \le n+1, \end{aligned} \tag{5.21}$$

where we used the facts that $n! \cdot (n+1) = (n+1)!$ and $(m'-1)! \cdot m' = m'!$. Note that the summation over m' runs from 1 to $n+1$. Because of the fraction on the right,

we can use the add-zero trick to extend the sum to also include $m' = 0$, because that term in the summation is equal to zero; this results in a summation from 0 to $n + 1$. The second term then becomes

$$\sum_{m'=0}^{n+1} \frac{(n+1)!}{m'!(n+1-m')!} \frac{m'}{n+1} r^{n+1-m'} s^{m'} = \sum_{m=0}^{n+1} \frac{(n+1)!}{m!(n+1-m)!} \frac{m}{n+1} r^{n+1-m} s^{m}, \tag{5.22}$$

because the summation index is a "dummy index" that can be replaced by any symbol we like (here, we choose m for the index). For the first term in Eq. (5.20), we "multiply by one" again (with a different expression for one) to find

$$\frac{n!}{m!(n-m)!} = \frac{n!}{m!(n-m)!} \cdot \frac{n+1}{n+1} \cdot \frac{n+1-m}{n+1-m} = \frac{(n+1)!}{m!(n+1-m)!} \cdot \frac{n+1-m}{n+1}. \tag{5.23}$$

Note that this last step is subtle here. When we multiplied by $(n+1-m)/(n+1-m)$, we restricted to $m \leq n$, so that the denominator is never zero. But, when we look at the term on the right-hand side, because of our convention that $0! = 1$, we can evaluate the expression for $m = n + 1$. We find the result is zero there, so we can add it into the summation, letting the first sum now run to $n + 1$ as well. The final step of the proof is to simply add these two terms in Eqs. (5.22) and (5.23) together with the summation running from $m = 0$ to $m = n + 1$:

$$\begin{aligned}(r+s)^{n+1} &= \sum_{m=0}^{n+1} \frac{(n+1)!}{m!(n+1-m)!} \left(\frac{n+1-m}{n+1} + \frac{m}{n+1} \right) r^{n+1-m} s^{m} \\ &= \sum_{m=0}^{n+1} \frac{(n+1)!}{m!(n+1-m)!} r^{n+1-m} s^{m},\end{aligned} \tag{5.24}$$

which is the result we wanted to show (i.e., the binomial theorem for $n + 1$). This then completes the proof of the binomial theorem. There are a number of steps here, with some of them being a bit subtle, so review this proof carefully to be sure you understand how all these steps fit together. That proof was long! It is one of the pieces we need to establish the power series of the exponential.

We are now ready to verify the power series for the exponential function. We will show that the correct form is given by

$$e^{x} = \sum_{n=0}^{\infty} \frac{1}{n!} x^{n} \tag{5.25}$$

because this power series satisfies the fundamental exponential relation $e^{x+y} = e^{x} \cdot e^{y}$. A sum that has infinity as the upper limit means we continue to sum over all possible values of n without ever ending. The summation must converge to a

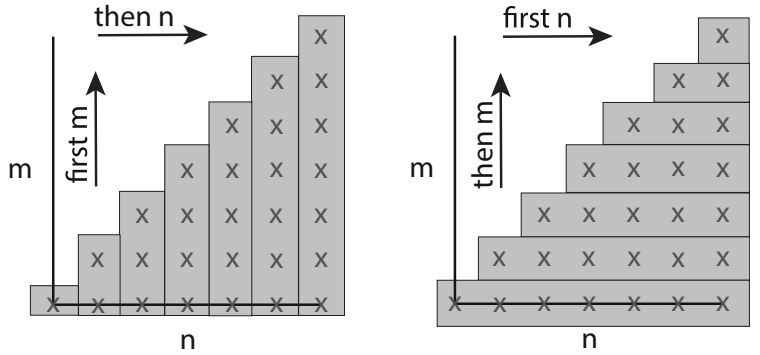

Fig. 5.5 Schematic for how the summations work if m is summed before n (left) or n is summed before m (right). The $\times$ symbols represent the m and n values, while the gray bars indicate how the summation is organized. On the right, we have $0 \leq m \leq n$ and $0 \leq n \leq \infty$, while on the left, we have $m \leq n \leq \infty$ and $0 \leq m \leq \infty$

finite result as we do this in order for such a summation to be meaningful. This will happen with this power series. We start by evaluating the exponential on the left using our power series *ansatz*:

$$e^{x+y} = \sum_{n=0}^{\infty} \frac{1}{n!}(x+y)^n = \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{m=0}^{n} \frac{n!}{m!(n-m)!} x^{n-m} y^m; \tag{5.26}$$

ansatz is a great German word for "educated guess." Note that the two $n!$ terms cancel in the expression on the far right. A graphic depiction of the summation is given in the left panel of Fig. 5.5. The sum over n ranging from 0 to ∞ and over m ranging from 0 to n, with the m sum being performed first, is indicated schematically with the vertical rectangles. The $\times$ marks are the integer m and n values in the summation. For example, pick $n = 0$, then the only value for m is $m = 0$. Next, pick $n = 1$. Then, we have both $m = 0$ and $m = 1$. For $n = 2$, we have $m = 0$, $m = 1$, and $m = 2$. The procedure continues in a similar fashion and is illustrated with the vertical rectangles. We have an infinite number of terms, because we run n from 0 to ∞.

Our next step is straightforward to carry out, but it may seem like it comes from thin air—we switch the order of the summations—summing over n first and then m. When making the switch, we need to ensure that we capture the contribution from every $\times$ mark in the left panel and no more. This is illustrated graphically in the right panel with the horizontal rectangles. For example, if $m = 0$, then $n = 0, 1, 2, \cdots, \infty$, while $m = 1$ gives $n = 1, 2, \cdots, \infty$, and $m = 2$ gives $n = 2, 3, \cdots, \infty$, and so on. It may seem odd that when we sum over m first, we have a finite sum (m runs from 0 to n) followed by an infinite sum (n runs from 0 to ∞), but when we sum over n first, we have two infinite sums—n runs from m to ∞ and m runs from 0 to ∞. One can directly see, by looking at the $\times$'s in the figure

that either way of doing this (m first or n first) *includes exactly the same set of* $\times$*'s*. Hence, the two ways of doing the summation are identical—we can perform the summation either way. Let us convert from an m-first inner summation to an n-first inner summation. We find that we can express e^{x+y} as

$$e^{x+y} = \sum_{m=0}^{\infty} \frac{1}{m!} \sum_{n=m}^{\infty} \frac{1}{(n-m)!} x^{n-m} y^m. \tag{5.27}$$

Note that the way this double summation works is that we start with a particular value of m (which will range from 0 to ∞); then for the given m, we sum n from m to ∞.

We need to massage this formula a bit to get it into the form we need. We will do some index shifting. Let $n' + m = n$, or $n' = n - m$. Then, since $n \geq m$, we have $n' \geq 0$ and

$$e^{x+y} = \underbrace{\sum_{m=0}^{\infty} \frac{1}{m!} y^m}_{e^y} \cdot \underbrace{\sum_{n'=0}^{\infty} \frac{1}{n'!} x^{n'}}_{e^x} = e^x \cdot e^y. \tag{5.28}$$

Note how the first summation is equal to e^y and the second summation is equal to e^x. So, we have shown that this power series satisfies the exponential property $e^{x+y} = e^x \cdot e^y$, which is the first property that the exponential function must satisfy.

To verify the second condition, we need to show the slope of e^x is equal to one at $x = 0$. Recall that the slope of a line is given by the rise divided by the run. When a curve is not a perfect straight line, we find that it often looks closer and closer to a straight line the closer we zoom into a particular point (see Fig. 5.6). Consider the slope for the e^x curve at the points x and $x + \Delta x$, when Δx is small. We find

$$\text{slope} = \frac{\text{rise}}{\text{run}} = \frac{e^{x+\Delta x} - e^x}{(x + \Delta x) - x} = \frac{e^x \left(e^{\Delta x} - 1\right)}{\Delta x}. \tag{5.29}$$

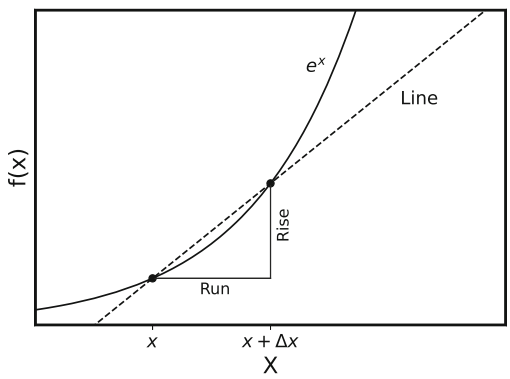

Fig. 5.6 Schematic for how we determine the tangent to a curve at a particular point. The curve e^x is the solid line, while the dashed line is a line that approximates the curve. The line goes through the two function values at x and $x + \Delta x$. By making Δx smaller and smaller, we approach the tangent line. The slope is given by the rise divided by the run

Now, we use the power series we just found to evaluate this result. The second factor in the numerator can be evaluated by noting that

$$e^{\Delta x} - 1 = \sum_{n=0}^{\infty} \frac{1}{n!}(\Delta x)^n - 1 = \cancel{1} + \Delta x + \frac{1}{2}(\Delta x)^2 + \frac{1}{6}(\Delta x)^3 + \cdots - \cancel{1}$$

$$= \Delta x + \frac{1}{2}(\Delta x)^2 + \cdots, \tag{5.30}$$

so the slope at x is equal to $e^x(1 + \frac{1}{2}\Delta x + \frac{1}{6}(\Delta x)^2 + \cdots)$. Now, the next step is similar to what we did when we evaluated the geometric series. We take the limit as $\Delta x \to 0$ to find the tangent of e^x at x (if you like, this is the definition of what the tangent is at x or what the slope is at x). We find

$$\text{slope at } x = e^x. \tag{5.31}$$

I hope you see that the fact that the slope of e^x is equal to e^x is a really cool property of the exponential function. Finally, we set $x = 0$ to find that $e^0 = \sum_{n=0}^{\infty} \frac{1}{n!}0^n = 1$; recall that $0^0 = 1$, $0^n = 0$ for $n \geq 1$, and $0! = 1$. Hence, the slope of e^x at $x = 0$ is 1, as we claimed. This completes the proof that

$$e^x = \sum_{n=0}^{\infty} \frac{1}{n!}x^n. \tag{5.32}$$

Setting $x = 1$ then determines the value for e. We find that

$$e = \sum_{n=0}^{\infty} \frac{1}{n!} = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \cdots \approx 2.71828\cdots. \tag{5.33}$$

The number e is often defined via

$$e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n, \tag{5.34}$$

which originally was given by Bernoulli to help solve the compound interest problem. We show this gives the same result for e as the power series does. As $n \to \infty$, $\frac{1}{n}$ is small, and we have already learned that the exponential of a small number can be approximated from the first few terms of the power series. So

$$e^{\frac{1}{n}} \approx 1 + \frac{1}{n} + \frac{1}{2n^2} + \cdots \tag{5.35}$$

gets closer and closer to $1 + \frac{1}{n}$ as n is made larger (because $\frac{1}{2n^2}$ is then very small). Hence

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = \lim_{n\to\infty}\left(e^{\frac{1}{n}}\right)^n = \lim_{n\to\infty} e^{\frac{n}{n}} = e^1 = e. \tag{5.36}$$

This example for how to take a limit is quite important. The strategy is that we focus on the *leading-order* behavior and drop all higher-order corrections when the leading correction becomes small. This is the methodology you should use when evaluating limits. Our result can be immediately generalized to an alternative form for the exponential function, namely,

$$e^x = \lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n. \tag{5.37}$$

Keeping in mind the idea of leading-order behavior and higher-order corrections, we can compute the slope of the exponential at $x = 0$ from the power series expression as well. Note that

$$e^x = 1 + x + \frac{x^2}{2} + \cdots, \tag{5.38}$$

and so, for small x, we find that $e^x \approx 1 + x$. This is a linear function with an intercept of one and a slope (given by the coefficient of the x term) of one. So, this is yet another way to see that the slope of e^x at $x = 0$ is one. Seeing things in more than one way is often useful and can even lead to new results. For example, we can use this leading-term method to show that the slope of $e^{ax} = 1 + ax + \cdots$ is equal to a at $x = 0$; using $e^{ax+a\Delta x} = e^{ax}(1 + a\Delta x + \cdots)$ shows the slope is equal to ae^{ax} at arbitrary x.

The slope formula is a very important result. It says that e^x has a slope equal to its value at every point. It is the only function that satisfies this property.

We are now able to identify an important identity called the Euler identity, which describes how e^x behaves for imaginary values of x. It is given by

$$e^{i\theta} = \cos\theta + i\sin\theta, \tag{5.39}$$

for real θ. This is a really amazing identity, involving imaginary numbers, exponentials, and trigonometric functions. We prove it next.

First note that $e^{i\theta}$ is a complex number with a norm (or equivalently a length) of 1. This is because

$$|e^{i\theta}| = \sqrt{z^* z|_{z=e^{i\theta}}} = \sqrt{e^{-i\theta}e^{i\theta}} = \sqrt{e^{-i\theta+i\theta}} = \sqrt{e^0} = 1. \tag{5.40}$$

So, for any real number θ, we have $e^{i\theta}$ lies on the unit circle on the complex plane. When $\theta = 0$, $e^{i0} = 1$, which lies on the horizontal axis. As θ increases, we find $e^{i\Delta\theta} \approx 1 + i\Delta\theta - \frac{1}{2}\Delta^2$ (from the power series formula), which has a positive vertical component, so as θ increases, we move around the unit circle in a counterclockwise direction.

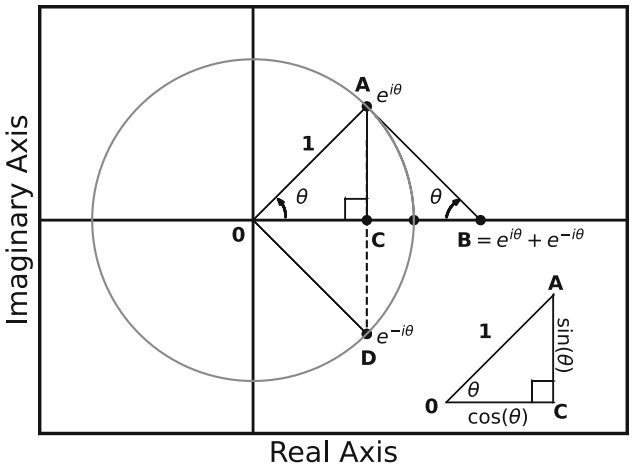

Fig. 5.7 Geometry needed to prove the Euler identity

Now, look at Fig. 5.7. We have drawn $e^{i\theta}$ for a small positive value of θ, located at point A. Then, $e^{-i\theta}$ is the complex conjugate, so it is reflected about the horizontal axis, and then, the sum $e^{i\theta} + e^{-i\theta}$ lies on the horizontal axis (because the imaginary parts cancel in the sum). Note that the triangle OAB is isosceles, since OC=CB (be sure you understand why), and then, the angles θ are the same on the right and on the left. We drop a perpendicular to the horizontal axis from point A, and then, OAC is a right triangle and the line segment OC is half the length of OB. But, OC is defined to be $\cos\theta$, because the hypotenuse of OAC has length 1. Hence, we learn that

$$\cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta}). \tag{5.41}$$

We can go through a similar analysis examining $e^{i\theta} - e^{-i\theta}$ along the imaginary axis and learn that

$$\sin\theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta}). \tag{5.42}$$

Using these two relations, we solve for $e^{i\theta}$ and find the Euler identity, given in Eq. (5.39). Finally, we find

$$e^{i\theta} = \sum_{n=0}^{\infty} \frac{1}{n!}(i\theta)^n = \underbrace{\sum_{m=0}^{\infty} \frac{1}{(2m)!}(i\theta)^{2m}}_{\text{even } m} + \underbrace{\sum_{m=0}^{\infty} \frac{1}{(2m+1)!}(i\theta)^{2m+1}}_{\text{odd } n}, \tag{5.43}$$

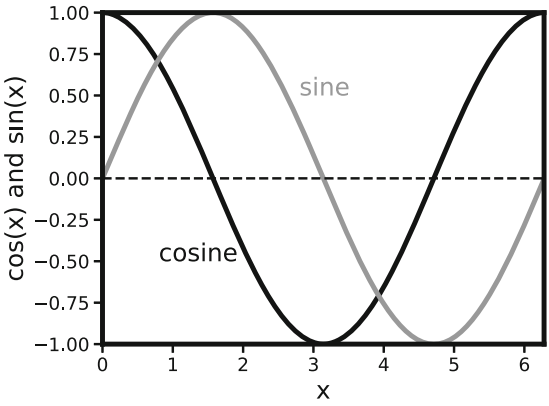

Fig. 5.8 Plots of cosine (black) and sine(gray)

where we separated out the summations over even and odd integers, respectively. Using the fact that $i^2 = -1$, we find the summations have separated into a purely real number and a purely imaginary number, which satisfy

$$\begin{aligned} e^{i\theta} &= \sum_{m=0}^{\infty} \frac{1}{(2m)!}(-1)^m \theta^{2m} + i \sum_{m=0}^{\infty} \frac{1}{(2m+1)!}(-1)^m \theta^{2m+1} \\ &= \cos\theta + i \sin\theta, \end{aligned} \tag{5.44}$$

because the real and imaginary parts are separately equal. Hence, we learn that

$$\cos\theta = \sum_{m=0}^{\infty} \frac{1}{(2m)!}(-1)^m \theta^{2m} = 1 - \frac{1}{2}\theta^2 + \frac{1}{24}\theta^4 + \cdots \tag{5.45}$$

and

$$\sin\theta = \sum_{m=0}^{\infty} \frac{1}{(2m+1)!}(-1)^m \theta^{2m+1} = \theta - \frac{1}{6}\theta^3 + \frac{1}{120}\theta^5 + \cdots . \tag{5.46}$$

Note how $\cos\theta \to 1$ as $\theta \to 0$ and $\sin\theta \to \theta$ as $\theta \to 0$, as shown in Fig. 5.8. The angle θ in this formula is measured in radians, not degrees.

The Euler formula plays a major role in the polar form of complex numbers. We write the Cartesian and polar forms as

$$z = x + iy = re^{i\theta}, \tag{5.47}$$

where $r = \sqrt{zz^*} = \sqrt{|z|^2} = \sqrt{x^2 + y^2}$, $x = r\cos\theta$, $y = r\sin\theta$, and $\tan\theta = \frac{y}{x}$, or $\theta = \tan^{-1}\frac{y}{x}$. In the polar form, we can multiply complex numbers most simply—multiply the magnitude of the arrows and add the times on the clocks. Note that the

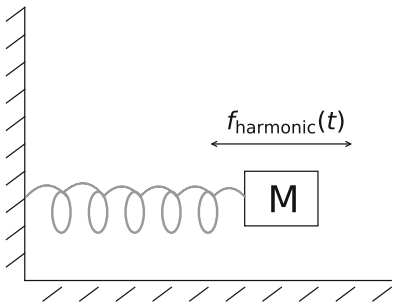

Fig. 5.9 Schematic of harmonic variation with a mass on the end of a spring

polar form for $r = 1$, namely, $e^{i\theta}$ is often called the *complex phase factor* or just *complex phase*, for short.

The final property we discuss in this section is a property called *harmonic variation*. As you have probably seen before, we can represent a harmonically varying function (say, for a mass oscillating on a spring) as a simple cosine function:

$$f_{\text{harmonic}}(t) = A\cos\omega t, \tag{5.48}$$

where A is the amplitude and ω is the angular frequency. We also can write this as

$$f_{\text{harmonic}}(t) = \text{Re}\left(Ae^{-i\omega t}\right); \tag{5.49}$$

here, Re denotes the real part of the complex number, and we require A to be real. The value of f is the position of the center of the mass as it oscillates on the end of the spring with respect to its rest length (see Fig. 5.9).

The mass on the end of the spring oscillates to and fro about the rest length l_0 of the spring. The angular frequency of the oscillation is ω, which is related to the period of the oscillation via $\omega t_{\text{per}} = 2\pi$.

In addition to the position versus time, we also would like to know the velocity versus time. This is given by the rate of change of the position versus time. If you know calculus, this is a simple calculation via taking a derivative with respect to time. But we can also determine it without calculus, because we know the slope of e^x at x. The velocity is given by the slope of the curve of position versus time. Since the power series expansion tells us that $e^{-i\omega t} \approx 1 - i\omega t + \cdots$, we find the slope of a harmonically varying function is given by $-i\omega$ times the function, because the slope of $e^{-i\omega t}$ is $-i\omega e^{-i\omega t}$. Hence, using $x(t) = \text{Re}\left(Ae^{-i\omega t}\right)$, we find that $v(t) = \text{Re}\left(-i\omega e^{-i\omega t}\right) = -A\omega\sin\omega t$; recall that we already showed that the slope of e^{ax} is ae^{ax} at x. This implies that a harmonically varying function (when expressed as an exponential) has a slope given by $-i\omega$ times the function. This property will play a critical role for us in deriving the position-momentum commutation relations in Chap. 7.

One question you might ask is why did we use $e^{-i\omega t}$ instead of $e^{i\omega t}$? It turns out, for this problem, either approach works, but because time advances forward in

one direction, the choice $e^{-i\omega t}$ is preferred because it describes the motion of waves correctly. You can think of this as just a convention we have chosen, but the reasons why we do this are deeper than just a convention.

We have completed our journey into studying properties of the exponential function for now. While there are a number of unique properties that the exponential satisfies, all these properties can be derived from its two defining properties, as we did here. In the next unit, we examine how the corresponding power series can be used to approximate the exponential function, the cosine function, and the sine function.

Section 5.3 Problems:

5.3.1 Binomial Theorem

This problem asks you to verify the binomial theorem concretely by calculating all of the terms in the summation for $n = 5$. Compute

$$(2+3)^5 = \sum_{m=0}^{5} \frac{5!}{m!(5-m)!} 2^{5-m} 3^m = 3125 \tag{5.50}$$

There are six terms you need to compute in the summation.

5.3.2 Binomial Theorem Symmetry

Prove that

$$(r+s)^n = \sum_{m=0}^{n} \frac{n!}{m!(n-m)!} r^{n-m} s^m = \sum_{m'=0}^{n} \frac{n!}{m'!(n-m')!} s^{n-m'} r^{m'} = (s+r)^n \tag{5.51}$$

This shows that the binomial theorem is symmetric in the order of the two powers.

5.3.3 Slope of the Exponential

Use the definition slope = rise over run to compute the slope of $e^{-i\omega t}$ using the points t_0 and $t_0 + \Delta t$. Use the properties of the exponential, not the power series to compute this. Determine what happens to your result as you take the limit $\Delta t \to 0$.

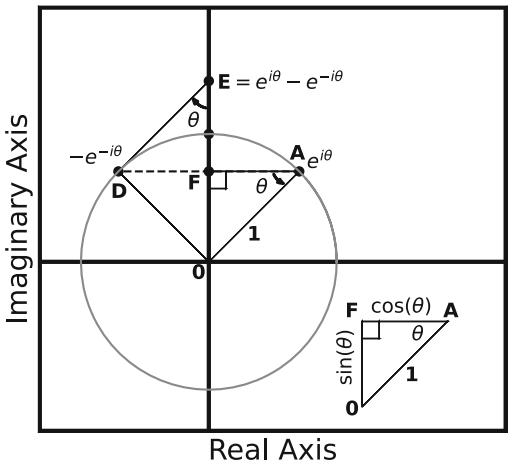

Fig. 5.10 Schematic used for determining the relationship between sine and exponentials

5.3.4 Relationship Between Sine and Exponentials

Using methods similar to those in the main text used to determine the cosine, show that the length of line segment OF is one-half the length of line segment OE and use that to prove that $\sin\theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$. See Fig. 5.10 for the details of the geometry.

5.3.5 Value for e

Using the power series for e^x, compute e to six digits of accuracy, using $x = 1$. Think carefully about how you know when you can stop the calculation because it is accurate enough.

5.3.6 Limits of Powers

Determine $\lim_{n\to\infty}\left(1+\frac{2}{n}\right)^n$.

5.3.7 Trigonometric Identities

Use $\cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$ and $\sin\theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$ to verify the following:

- $\cos^2\theta + \sin^2\theta = 1$
- $\cos^2\theta - \sin^2\theta = \cos 2\theta$

- $2\cos\theta\sin\theta = \sin 2\theta$

5.4 Approximations for Exponentials, Sines, and Cosines

We worked hard to establish the power series for the exponential, the sine and the cosine. Now, we want to use them to approximate those functions and learn more about their behavior. We start with the exponential function. In Fig. 5.11, we show plots of e^x, $1+x$, $1+x+\frac{1}{2}x^2$, and $1+x+\frac{1}{2}x^2+\frac{1}{6}x^3$, for real x. These polynomial approximations to the exponential function are found by truncating the power series found in Ch. 5.3 at a particular order of the polynomial. Note that the more terms we keep in the polynomial, the further in x these curves approximate the exponential. For the smallest values of x near zero, the linear approximation suffices, but for larger values, we need more terms.

In many situations, we find we can derive all we need by approximating the exponential by its linear or quadratic approximation only. Recall how we proved the slope of e^x was equal to e^x by computing the rise over the run and taking the limit as $\Delta x \to 0$. We will return again and again to these approximations throughout the book. It is important for you to master them.

The next case we want to look at is $\sin(x)$ and its approximations of x, $x-\frac{1}{6}x^3$, and $x-\frac{1}{6}x^3+\frac{1}{120}x^5$. The comparison is done in Fig. 5.12. Because $\sin(x)$ only has odd powers of x (making it an odd function), the approximations work better than what we saw for the exponential function (Fig. 5.11).

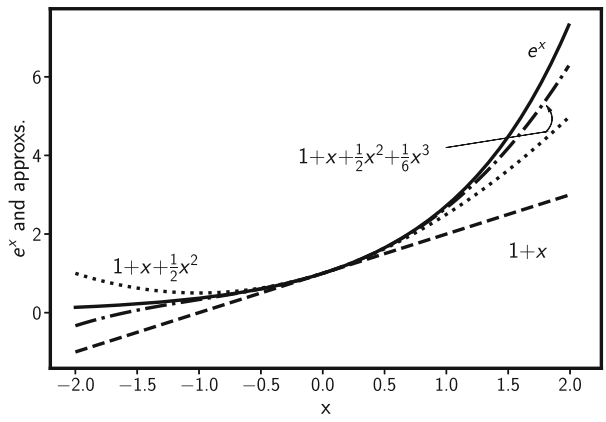

Fig. 5.11 Approximation of the exponential function for real x by truncated polynomials extracted from the power series. The solid line is e^x, the dashed line is $1+x$, the dotted line is $1+x+\frac{1}{2}x^2$, and the chain-dotted line is $1+x+\frac{1}{2}x^2+\frac{1}{6}x^3$

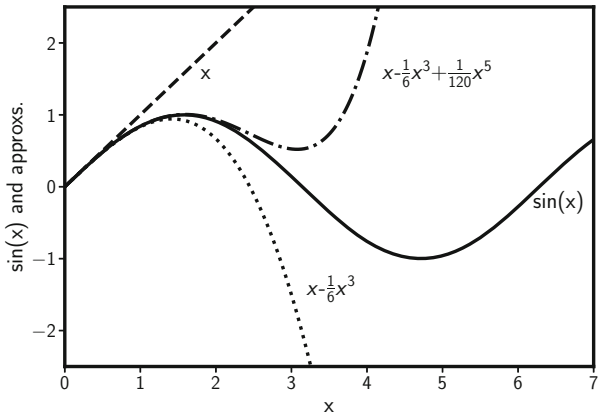

Fig. 5.12 Approximation of $\sin(x)$ for real x by truncated polynomials extracted from the power series. The solid line is $\sin(x)$, the dashed line is x, the dotted line is $x - \frac{1}{6}x^3$, and the chain-dotted line is $x - \frac{1}{6}x^3 + \frac{1}{120}x^5$

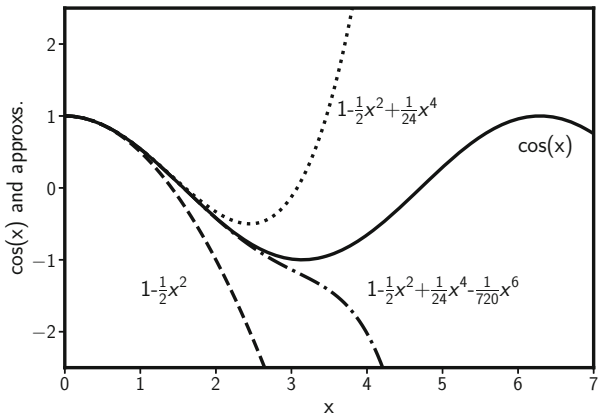

Fig. 5.13 Approximation of $\cos(x)$ for real x by truncated polynomials extracted from the power series. The solid line is $\cos(x)$, the dashed line is $1 - \frac{1}{2}x^2$, the dotted line is $1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$, and the chain-dotted line is $1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 - \frac{1}{720}x^6$

We now move on to $\cos(x)$ and its polynomials of $1 - \frac{1}{2}x^2$ and $1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$ shown in Fig. 5.13. They work well for small x and then fail spectacularly. It looks to me like we are "laying" the approximate polynomial on top of the trigonometric function, and then all of a sudden, it shoots away from it. This spectacular failure, where the polynomial starts to disagree with the trigonometric function, is called a *jet*. Note how the higher-order polynomials for both sine and cosine can start to

produce some of the oscillations of these functions. They produce more and more of these oscillations as we make the degree of the approximating polynomial higher.

This is all we will cover in this short section, but you will want to return to it from time to time to be sure you understand how these functions behave. As you can see, the concepts are straightforward, but it can take some time until you feel comfortable with them and can use them easily. This occurs only with practice. So, let us move on to the problems.

Section 5.4 Problems:

5.4.1 How Accurate?

What polynomial truncation to the exponential function approximates it to an error of at most 1% over the interval $-1 \leq x \leq 1$? Repeat for the interval $-2 \leq x \leq 2$. Note that you should find the error is largest at the end points of the intervals. The percentage error is given by

$$\text{percentage error} = \left| \frac{f_{\text{exact}}(x) - f_{\text{approx}}(x)}{f_{\text{exact}}(x)} \right|. \tag{5.52}$$

You need to verify where the error is the largest and how you know what power to stop the approximation at.

5.4.2 Accuracy for $\sin(x)$

What is the percentage error for the linear, cubic, and quintic approximations to $\sin(x)$ at $x = \frac{\pi}{2}$?

5.4.3 Approximate Trigonometric Identities

One of the important trigonometric identities is $\cos^2 x + \sin^2 x = 1$. If we approximate $\cos x = 1 - \frac{1}{2}x^2$ and $\sin x \approx x$, how large is the error for the trigonometric identity for $0 \leq x \leq \pi$? Repeat this exercise for the higher-order approximations given by $\cos x \approx 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$ and $\sin x \approx x - \frac{1}{6}x^3$.

5.5 Introduction to 2 × 2 matrices

Matrices are arrays of numbers (they can be restricted to only real numbers or allow complex numbers). For now, we will focus on the simplest of all matrices, the 2 × 2 matrices. In their most general form, the matrix is arranged in an array of four numbers in two rows and two columns:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \tag{5.53}$$

where a, b, c, and d, are complex numbers. We use subscripts to denote the elements of the matrix. So, A_{ij} is the matrix element in the ith row and jth column. Or, more concretely, we write

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}, \tag{5.54}$$

with $A_{11} = a$, $A_{12} = b$, $A_{21} = c$, and $A_{22} = d$. We call the matrix elements where $i = j$ the *diagonal elements* (A_{11} and A_{22} or a and d), while the ones with $i \neq j$ are the *off-diagonal elements* (A_{12} and A_{21} or b and c). If $A_{ij} = A_{ji}$ for all i and j, we say we have a *symmetric matrix*; this requires $b = c$ for a 2 × 2 matrix. We can transform a matrix by interchanging the off-diagonal elements (which can be though of as reflecting the elements along the diagonal). This matrix is called the transpose matrix, denote A^T. It takes the form

$$A^T = \begin{pmatrix} a & c \\ b & d \end{pmatrix}. \tag{5.55}$$

You can also think of this as exchanging the rows of the initial matrix with the columns of the transposed matrix. Note that a symmetric matrix satisfies $A = A^T$.

There is a more complex object called the *Hermitian conjugate* of a matrix, which is closely related to the transpose. Indeed, it is formed by transposing the matrix and then taking the complex conjugate of all matrix elements ($A^\dagger_{ij} = A^*_{ji}$). The Hermitian conjugate is denoted with a dagger superscript $A^\dagger$ and it becomes

$$A^\dagger = \begin{pmatrix} A^*_{11} & A^*_{21} \\ A^*_{12} & A^*_{22} \end{pmatrix} = \begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix}. \tag{5.56}$$

A matrix that is equal to its Hermitian conjugate is said to be a Hermitian matrix, and it satisfies $A = A^\dagger$. While it is true that all real symmetric matrices are Hermitian matrices, Hermitian matrices are not always symmetric matrices (they must be real valued to also be symmetric).

Just like we had rules to add and multiply scalar numbers, and vectors, we have the same with matrices. For matrix addition, we add each element "pointwise"

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} + \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} = \begin{pmatrix} a + a' & b + b' \\ c + c' & d + d' \end{pmatrix} \tag{5.57}$$

for the addition of two matrices. Obviously, the sum of two matrices commutes, in that $A + A' = A' + A$. We also can multiply matrices by scalars (similar to how we multiply vectors by scalars)—we simply multiply every matrix element by the scalar.

Before ending this brief introduction, we want to define two invariants of matrices—the determinant and the trace. The determinant of a 2×2 matrix is defined via

$$\det A = A_{11}A_{22} - A_{12}A_{21} = ad - bc. \tag{5.58}$$

The trace is just the sum of the diagonal elements of the matrix:

$$\mathrm{Tr}\, A = A_{11} + A_{22} = a + d. \tag{5.59}$$

We will see later why they are called invariants.

Section 5.5 Problems:

5.5.1 Matrix Addition

Compute the matrix addition:

$$\begin{pmatrix} 1 & 4 \\ -2 & 3 \end{pmatrix} + \begin{pmatrix} 0 & i \\ -i & 2 \end{pmatrix} \tag{5.60}$$

Is your matrix sum a symmetric matrix? Is it a Hermitian matrix?

5.5.2 Determinant and Trace I

Compute the determinant and trace of the following four matrices:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \text{and} \quad \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \tag{5.61}$$

These matrices are called the Pauli spin matrices, and we learn a lot more about them in Chap. 6.

5.5.3 *Determinant and Trace II*

Compute the determinant and trace of the following matrices:

$$\begin{pmatrix} 2 & 1 \\ 1 & \frac{1}{2} \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \text{and} \quad \begin{pmatrix} 1 & -2i \\ 3i & 4 \end{pmatrix}. \tag{5.62}$$

5.6 Matrix Multiplication of 2 × 2 matrices

Now, we will show how we multiply 2×2 matrices. The technique involves a "zipper-like" process with rows and columns of the two matrices, and it is easiest to just illustrate it directly. Consider the product of matrices A and B:

$$AB = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}. \tag{5.63}$$

The A matrix is the unprimed matrix and the B matrix is the primed one. The product is defined to be

$$AB = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} = \begin{pmatrix} aa' + bc' & ab' + bd' \\ ca' + dc' & cb' + dd' \end{pmatrix} = C. \tag{5.64}$$

This looks like an awful mess until you see the secret for how it works. Let's start with the C_{11} matrix element (top left) of the product. It is formed by multiplying the first row of A times the first column of B as follows:

$$(a \; b) \cdot \begin{pmatrix} a' \\ c' \end{pmatrix} = aa' + bc'. \tag{5.65}$$

This acts like a zipper—I pair the first elements of the row (a) and the column (a'), multiply them together and add them to the same pair multiplied for the second element of the row (b) and column (c'). You also can think of this as a dot product (if you are familiar with dot products) between a row vector and a column vector (at least when the matrix has all real matrix elements—when they are complex, we continue to multiply rows and columns as described above without taking an additional complex conjugate).

To get the C_{12} element (top right), I multiply the first row times the second column, and in general, C_{ij} is found by multiplying the ith row of A with the jth column of B.

If you have never seen this before, it may seem like a complicated and confusing thing to do, but it will turn out to be quite an interesting thing to do, because using this technique as a way to multiply two matrices allows us to solve many interesting

problems and it turns out to be the most useful way to define matrix multiplication as a generalization to how we multiply scalar numbers. It turns out that Heisenberg rediscovered this result in 1925 when he invented matrix mechanics—he did not even know what a matrix was and called his objects tableau!

So, is there a rational origin for where this comes from? Well, we can think of the matrix as a collection of vectors. For the 2×2 matrix, it is either two row vectors or two column vectors:

$$\begin{pmatrix} \boxed{a \;\; b} \\ \boxed{c \;\; d} \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} \boxed{a} & \boxed{b} \\ \boxed{c} & \boxed{d} \end{pmatrix}. \tag{5.66}$$

Now, we know how to multiply vectors together. The procedure, called an inner (or dot) product, is precisely this "zipper dance," where we take a row on the left and a column on the right, pair them, multiply them, and add them. So, matrix multiplication actually has its origin in the way we multiply vectors together. This is illustrated schematically as follows:

$$\begin{pmatrix} \boxed{a \;\; b} \\ \boxed{c \;\; d} \end{pmatrix} \begin{pmatrix} \boxed{a'} & \boxed{b'} \\ \boxed{c'} & \boxed{d'} \end{pmatrix} = \begin{pmatrix} (a\ b) \cdot \begin{pmatrix} a' \\ c' \end{pmatrix} & (a\ b) \cdot \begin{pmatrix} b' \\ d' \end{pmatrix} \\ (c\ d) \cdot \begin{pmatrix} a' \\ c' \end{pmatrix} & (c\ d) \cdot \begin{pmatrix} b' \\ d' \end{pmatrix} \end{pmatrix} = \begin{pmatrix} aa' + bc' & ab' + bd' \\ ca' + dc' & cb' + dd' \end{pmatrix}. \tag{5.67}$$

We will illustrate a concrete example of this later. Note that when the numbers in the matrix are complex, this product of vectors still can be thought of as an inner product, because in an inner product we complex conjugate a left column vector before multiplying it by a right column vector (technically speaking for two vectors, we take the Hermitian conjugate, converting a column vector to a row vector). The conversion from a column vector to a row vector includes a complex conjugation, so when we have a row vector and take its inner product with a column vector, the complex conjugation has already been done for us, and we need not repeat it.

For scalar multiplication of numbers, there is an identity element—the number 1—which does not change any number when it multiplies it: $a1 = 1a = a$. Since I can multiply matrices together, there must be an analog of the identity. We call this the identity matrix I and since

$$AI = IA = A, \tag{5.68}$$

we immediately find that the identity matrix is

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \tag{5.69}$$

by looking at the matrix multiplication formula. Note that I is unique and is the only matrix that satisfies both $AI = IA = A$ for all matrices A.

Armed with an identity, we define an inverse matrix (when an inverse exists) to satisfy

$$AA^{-1} = I \tag{5.70}$$

(technically speaking, this is a right inverse). We now show that it is also a left inverse by employing the associative property of matrix multiplication. Multiply the right-inverse equation by A on the right to find

$$(AA^{-1})A = IA. \tag{5.71}$$

But, the left-hand side can be rewritten, using the associative property as

$$(AA^{-1})A = A(A^{-1}A). \tag{5.72}$$

Now, the right-hand side of Eq. (5.71) is equal to A, which is also equal to the right-hand side of Eq. (5.72). Since $A = AI$, we also find that

$$(A^{-1}A) = I, \tag{5.73}$$

establishing that we have a left inverse as well.

Let us now construct the inverse for our 2×2 matrix directly. With a little trial and error, we can find that

$$A^{-1} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \frac{1}{ad - bc}, \tag{5.74}$$

when $\det A = ad - bc \neq 0$. It is easy to check this

$$\begin{aligned} AA^{-1} &= \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \frac{1}{ad - bc} = \begin{pmatrix} ad - bc & 0 \\ 0 & ad - bc \end{pmatrix} \frac{1}{ad - bc} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I, \end{aligned} \tag{5.75}$$

as long as our new friend the determinant is nonzero. This turns out to be a general result for all square matrices—a square matrix is invertible if and only if its determinant is nonzero. If $\det A = 0$, there is no inverse. Note that in the verification that we indeed found the matrix inverse, we used the fact that a scalar λ multiplied by a matrix multiplies all elements of the matrix via

$$\lambda A = \begin{pmatrix} \lambda a & \lambda b \\ \lambda c & \lambda d \end{pmatrix}. \tag{5.76}$$

OK. We have found the matrix identity and matrix inverse (when it exists). We also have used the fact that matrix multiplication is associative $(AB)C = A(BC)$. Is this all we need to know about matrix arithmetic? The answer is no. We have not explored one bizarre, but important, property of matrix multiplication—in general, it is not commutative, meaning we often have $AB \neq BA$; that is, A and B do not commute. Let's check to see if this is true. Recall we already computed

$$AB = \begin{pmatrix} aa' + bc' & ab' + bd' \\ ca' + dc' & cb' + dd' \end{pmatrix}. \tag{5.77}$$

Similarly, one can immediately find that

$$BA = \begin{pmatrix} a'a + b'c & a'b + b'd \\ c'a + d'c & c'b + d'd \end{pmatrix}. \tag{5.78}$$

Now, closely compare the matrix elements. You can see they are not the same. For example, we would need $bc' = b'c$, $ab' + bd' = a'b + b'd$, and $ca' + dc' = c'a + d'c$ for $AB = BA$ to hold, in general. One can see that this will not usually be true. *The order of the factors in matrix multiplication matters.* This is a critical rule you need to know to work in quantum mechanics. This is because noncommutativity also occurs with quantum operators, which we will be exploring soon. *You must get used to working on algebra problems where the order of the factors in terms must be kept track of.* This is one of the challenges you must master on the road to becoming a quantum mechanic. Note, however, that matrix multiplication is distributive. Namely, $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$. One just must ensure that we do not change the matrix product orderings in the distributive law.

Let us go through an example. Let

$$A = \begin{pmatrix} 1 & -1 \\ 2 & 0 \end{pmatrix} \text{ and } B = \begin{pmatrix} 3 & 2 \\ -1 & 1 \end{pmatrix}. \tag{5.79}$$

To multiply out $AB = C$, we need four vector-vector inner products:

$$C_{11}: \quad (1 \ -1) \cdot \begin{pmatrix} 3 \\ -1 \end{pmatrix} = 1 \cdot 3 + (-1) \cdot (-1) = 4$$

$$C_{12}: \quad (1 \ -1) \cdot \begin{pmatrix} 2 \\ 1 \end{pmatrix} = 1 \cdot 2 + (-1) \cdot 1 = 1$$

$$C_{21}: \quad (2 \ 0) \cdot \begin{pmatrix} 3 \\ -1 \end{pmatrix} = 2 \cdot 3 + 0 \cdot (-1) = 6$$

$$C_{22}: \quad (2\ \ 0)\cdot\begin{pmatrix}2\\1\end{pmatrix} = 2\cdot 2 + 0\cdot 1 = 4, \tag{5.80}$$

so

$$C = \begin{pmatrix}4 & 1\\ 6 & 4\end{pmatrix}. \tag{5.81}$$

Now, we repeat for $BA = D$:

$$\begin{aligned}
D_{11}: &\quad (3\ \ 2)\cdot\begin{pmatrix}1\\2\end{pmatrix} = 3\cdot 1 + 2\cdot 2 = 7\\
D_{12}: &\quad (3\ \ 2)\cdot\begin{pmatrix}-1\\0\end{pmatrix} = 3\cdot(-1) + 2\cdot 0 = -3\\
D_{21}: &\quad (-1\ \ 1)\cdot\begin{pmatrix}1\\2\end{pmatrix} = (-1)\cdot 1 + 1\cdot 2 = 1\\
D_{22}: &\quad (-1\ \ 1)\cdot\begin{pmatrix}-1\\0\end{pmatrix} = (-1)\cdot(-1) + 1\cdot 0 = 1,
\end{aligned} \tag{5.82}$$

so

$$D = \begin{pmatrix}7 & -3\\ 1 & 1\end{pmatrix}. \tag{5.83}$$

We clearly have $AB \neq BA$.

If this is the first time you have ever performed matrix multiplication, I recommend that you practice by making some examples yourself or finding them from other sources. You need to be facile with matrix multiplication for the work to come.

There are two special types of matrices we need to discuss. They both require us to apply an operation to the matrices—the Hermitian conjugate, constructed via a complex conjugation and a transpose of the matrix (it is also called the Hermitian adjoint); recall the complex conjugate of $z = x + iy$ is $z^* = x - iy$. For example, we have

$$A = \begin{pmatrix}a & b\\ c & d\end{pmatrix} \rightarrow A^\dagger = \begin{pmatrix}a^* & c^*\\ b^* & d^*\end{pmatrix}, \tag{5.84}$$

which corresponds to interchanging the off-diagonal elements ($b \leftrightarrow c$) and taking the complex conjugate of all matrix elements.

There are two special matrices we can now construct from this "dagger" operation. The first special matrix is called a Hermitian matrix, which satisfies $A^\dagger = A$—in Hermitian matrices, the diagonals are real (be sure you understand

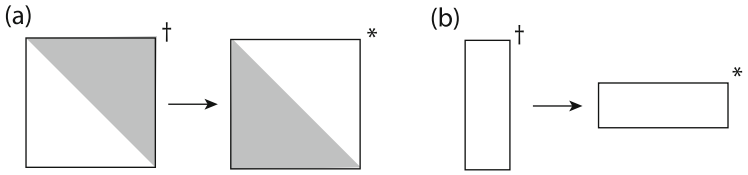

Fig. 5.14 Schematic of Hermitian conjugation. (**a**) Hermitian conjugation of a matrix with transpose and complex conjugation. (**b**) Hermitian conjugation of a vector, which changes from a column to a row, with complex conjugation

why), and the off-diagonals are complex conjugates of each other $A_{ij}^* = A_{ji}$ for all $i \neq j$. The second special matrix is called a unitary matrix, and it satisfies $A^\dagger = A^{-1}$. These matrices play important roles within quantum mechanics (see Fig. 5.14 to see a schematic for how the Hermitian conjugation works for a matrix). An example of a Hermitian matrix is

$$H = \begin{pmatrix} 1 & i \\ -i & -1 \end{pmatrix} = H^\dagger. \tag{5.85}$$

An example of a unitary matrix is

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}. \tag{5.86}$$

Unitary and Hermitian properties are not mutually exclusive—there even are matrices we will meet that are both unitary and Hermitian!

Whenever we work with matrices, we need to also be aware of how the matrices relate to vectors and how vectors relate to each other. We end this section with a discussion of the many different products we will encounter between these different objects.

A matrix multiplied times a vector yields a new vector:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} a\alpha + b\beta \\ c\alpha + d\beta \end{pmatrix}. \tag{5.87}$$

A Hermitian conjugate vector results from taking a column vector, converting it to a row vector via a matrix transpose, and then applying a complex conjugate operation to each element of the row vector (see Fig. 5.14 for a schematic of how this works). So, we have

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix}^\dagger = (\alpha^* \quad \beta^*). \tag{5.88}$$

The operation also works in the opposite direction. The Hermitian conjugate of a row vector is a column vector, with a complex conjugation, so

$$(\gamma \;\; \delta)^\dagger = \begin{pmatrix} \gamma^* \\ \delta^* \end{pmatrix}. \tag{5.89}$$

The inner product of two vectors is formed similarly to matrix multiplication—we simply think of a column vector as a $1 \times n$ vector, while a row vector is a $n \times 1$ vector. Hence, we first convert the left column vector to the Hermitian conjugate row vector (including complex conjugation), and then, we multiply the row times the column. Be sure to take the complex conjugate before multiplying! We find the inner product between a vector $v = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ and $v' = \begin{pmatrix} \alpha' \\ \beta' \end{pmatrix}$ satisfies

$$(\alpha^* \;\; \beta^*) \cdot \begin{pmatrix} \alpha' \\ \beta' \end{pmatrix} = \alpha^*\alpha' + \beta^*\beta' = v^\dagger \cdot v'. \tag{5.90}$$

If the inner product of two vectors is equal to 0, then the two vectors are said to be *orthogonal*. The inner product of a vector with itself is always nonnegative—$v^\dagger \cdot v = |\alpha|^2 + |\beta|^2 \geq 0$. This is called the norm squared of the vector (the norm is its square root). The vector-matrix-vector product using a row vector on the left and a column vector on the right becomes a scalar number:

$$\begin{aligned}(\alpha^* \;\; \beta^*) \cdot \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \begin{pmatrix} \alpha' \\ \beta' \end{pmatrix} = (\alpha^* \;\; \beta^*) \cdot \begin{pmatrix} a\alpha' + b\beta' \\ c\alpha' + d\beta' \end{pmatrix} = \alpha^*(a\alpha' + b\beta') \\ + \beta^*(c\alpha' + d\beta') = v^\dagger \cdot A \cdot v'. \end{aligned} \tag{5.91}$$

Finally, an outer product of two vectors (denoted by the $\otimes$ symbol) creates a matrix out of products of the elements of each vector as follows:

$$v \otimes v'^\dagger = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \otimes (\alpha'^* \;\; \beta'^*) = \begin{pmatrix} \alpha\alpha'^* & \alpha\beta'^* \\ \beta\alpha'^* & \beta\beta'^* \end{pmatrix}. \tag{5.92}$$

We will use these relations throughout the remainder of the book. Note that it takes practice and discipline to get the complex conjugations right.

Section 5.6 Problems:

5.6.1 Is It an Inverse?

Verify that the formula for the inverse works in the form $A^{-1}A = I$ for a general 2×2 matrix that has $ad - bc \neq 0$.

5.6.2 Do They Commute?

Determine whether the following matrices commute (meaning $AB = BA$), by calculating both AB and BA and comparing them:

$$A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \text{ and } B = \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}. \tag{5.93}$$

5.6.3 Find the Inverse

Calculate the inverse of

$$\begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix} \tag{5.94}$$

and verify that it is indeed the inverse by showing both $AA^{-1} = A^{-1}A = I$.

5.6.4 Is It Unitary?

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}. \tag{5.95}$$

What is the condition on the matrix elements such that A is unitary ($A^\dagger = A^{-1}$)? You will find four conditions. Use the unitary matrix properties to reduce this to two relations between the matrix elements and one relation for the determinant.

5.6.5 Is It Unitary II?

Verify that

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \text{ and } \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \tag{5.96}$$

are unitary matrices (recall how a scalar multiplies a matrix and that $1/i = -i$).

5.6.6 Determinant of a Product Is the Product of the Determinants

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}. \tag{5.97}$$

Show that $\det(AB) = \det(A)\det(B) = \det(BA)$.

This is a long calculation. It is best to compute AB first, take its determinant, then compute $\det(A)$ and $\det(B)$, and compute their product. Then, compare both expressions to verify they are the same. Next, compute the product BA, take its determinant, and compare it as well.

This result can be directly generalized to show that the determinant of a product of matrices is equal to the product of the determinants of those matrices. If you are ambitious, you can use the results above in a proof by induction to show that

$$\det(ABC\cdots N) = \det(A)\det(B)\det(C)\cdots\det(N). \tag{5.98}$$

5.6.7 Vector Gymnastics

Let

$$v = \begin{pmatrix} 1 \\ 4 \end{pmatrix}, \quad v' = \begin{pmatrix} 2 \\ -1 \end{pmatrix} \quad \text{and} \quad A = \begin{pmatrix} 0 & -1 \\ -1 & 2 \end{pmatrix}. \tag{5.99}$$

Compute $v^\dagger \cdot v'$, $v \otimes v'^\dagger$, and $v^\dagger \cdot A \cdot v'$.

Hint: $v'^\dagger = (2 \ -1)$.

5.6.8 Orthogonality

Which pairs of vectors are orthogonal

$$v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad v_w = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad \text{and} \quad v_4 = \begin{pmatrix} 1 \\ -i \end{pmatrix}. \tag{5.100}$$

5.6.9 Order of Vectors in the Inner Product

If

$$v = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \quad \text{and} \quad v' = \begin{pmatrix} \alpha' \\ \beta' \end{pmatrix}. \tag{5.101}$$

Determine how to convert $v^\dagger \cdot v'$ into $v'^\dagger \cdot v$? Consider the elements of v and v' as being complex numbers (which includes the possibility that they are real).

Hint: the relationship involves one mathematical operation.

5.6.10 Factorizing a Unitary

Consider the unitary matrix given by

$$U = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta e^{i\phi} & \cos\theta e^{i\phi} \end{pmatrix}, \tag{5.102}$$

with both θ and ϕ real. Show that U can be factorized in the form

$$U = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} \begin{pmatrix} b & 0 \\ 0 & \frac{1}{b} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix} \tag{5.103}$$

and in the form

$$U = \begin{pmatrix} 1 & 0 \\ \alpha & 1 \end{pmatrix} \begin{pmatrix} \beta & 0 \\ 0 & \frac{1}{\beta} \end{pmatrix} \begin{pmatrix} 1 & \gamma \\ 0 & 1 \end{pmatrix}. \tag{5.104}$$

Determine a, b, and c, as well as α, β, and γ in terms of θ and ϕ.
Hint: multiply out the right-hand side of the matrices and force it to work.

5.7 Eigenvectors and Eigenvalues

In this section, we discuss an interesting property of matrices that is an integral part of quantum mechanics—namely, the determination of eigenvalues and eigenvectors. Some people might say that all of quantum mechanics boils down to analyzing eigenvalues and eigenvectors. Why? Because anything we can measure about a quantum system can be represented by a matrix, and the result of the measurement must be one of the eigenvalues associated with that matrix! In general, the matrix is a Hermitian matrix. But, before we describe the details for how this works, we

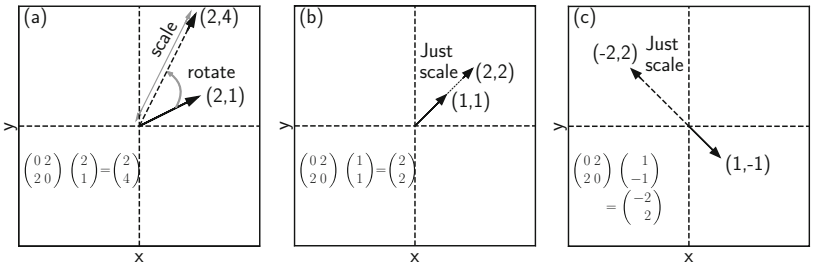

Fig. 5.15 Schematic of how a matrix can rotate and scale a vector after matrix multiplication. In panel (**a**), we show the general case of both scaling and rotation. Panel (**b**) shows a case of just scaling and panel (**c**) also shows just scaling, but here with a negative-valued scaling. The vectors that just scale are called eigenvectors

have to first get through some heavy math to be able to understand this concept. Our approach only uses algebra and the quadratic equation, but it takes time and is long. You have been warned. I do promise that in the end all this heavy lifting will be worth it, but we must get through it first.

The multiplication of matrices times vectors is a physical process—there are two things that can happen to a vector—it can rotate or it can be scaled. In Fig. 5.15, we show works in the general case in panel (a), for a positive scaling in panel (b), and for a negative scaling in panel (c). The case where the matrix just scales a vector with no rotation is a special case we will be discussing in detail in this section. The vectors that just scale are called *eigenvectors*.

In this part of the book, we work primarily with 2×2 (and sometimes 3×3 matrices), but we will focus entirely on 2×2 matrices in this section, which requires us to understand the quadratic equation. We start by deriving the quadratic equation to make sure everyone is familiar with it.

Let $P(x) = ax^2 + bx + c$ be a quadratic polynomial in x, with a, b, and c complex numbers. We want to find the two (possibly complex) solutions to the equation $P(x) = 0$; we denote the solutions as x_+ and x_-. Since $a \neq 0$ (otherwise we have a linear, not a quadratic equation), we divide both sides by a to obtain

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0. \tag{5.105}$$

The next step is to *complete the square* via our add-zero trick

$$x^2 + \frac{b}{a}x + \frac{c}{a} = \left(x + \frac{b}{2a}\right)^2 + \frac{c}{a} - \frac{b^2}{4a^2} = 0. \tag{5.106}$$

Now, we move the remaining constants to the right-hand side

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2} \tag{5.107}$$

and then take the square root of both sides, recalling that there are two square roots—one the negative of the other—yields

$$x + \frac{b}{2a} = \pm\frac{1}{2a}\sqrt{b^2 - 4ac}. \tag{5.108}$$

The two roots are then found by moving the last constant to the right-hand side, which gives us

$$x_\pm = -\frac{b}{2a} \pm \frac{1}{2a}\sqrt{b^2 - 4ac} \tag{5.109}$$

as the two solutions to $P(x) = 0$. Since we can write this equation now as $(x - x_+)(x - x_-) = 0$, we immediately find that $x_+ + x_- = -\frac{b}{a}$ and $x_+ x_- = \frac{c}{a}$.

Now that we reviewed the quadratic equation, we come back to eigenvectors. An *eigenvector* is defined as a vector v associated with a matrix A such that the matrix multiplied by the vector (Av) produces a number (called the *eigenvalue*) times the original vector. While this definition is straightforward, I find this concept is often *very* difficult for students to understand when they first see it. We will go through a number of examples carefully, but it is useful to repeat the definition—*an eigenvector satisfies a matrix times an eigenvector is equal to a number times the original vector*—as many times as you need to to remember it. The strategy we will adopt is to first find the eigenvalue, denoted λ, which satisfies the eigenvalue-eigenvector relationship for the given matrix A, and then, afterward, we will find the eigenvector v. Then, we will have found a solution that solves the relation $Av = \lambda v$.

Recall that multiplying a matrix times a vector is exactly the same as what we did for the first column only of matrix multiplication. Hence

$$\underbrace{\begin{pmatrix} a & b \\ c & d \end{pmatrix}}_{\text{matrix}} \underbrace{\begin{pmatrix} \alpha \\ \beta \end{pmatrix}}_{\text{vector}} = \begin{pmatrix} a\alpha + b\beta \\ c\alpha + d\beta \end{pmatrix} = \underbrace{\lambda}_{\text{number}} \underbrace{\begin{pmatrix} \alpha \\ \beta \end{pmatrix}}_{\text{original vector}} = \begin{pmatrix} \lambda\alpha \\ \lambda\beta \end{pmatrix} \tag{5.110}$$

for an eigenvector $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$, with λ called the eigenvalue. This appears to be quite hard to solve, since we need both $a\alpha + b\beta = \lambda\alpha$ and $c\alpha + d\beta = \lambda\beta$ to hold simultaneously. So, we proceed carefully.

First notice that if $\alpha = \beta = 0$, the equations we are trying to solve are automatically satisfied for any λ. This is a trivial solution that we are not interested in. To find the nontrivial solution, we use the multiply-by-one trick, in the form $Iv = v$ for the 2×2 identity matrix I, to rewrite the eigenvalue-eigenvector relation as

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}. \tag{5.111}$$

We next, multiply the identity matrix by λ and subtract the right-hand side from both sides of the equation to yield

$$\begin{pmatrix} a-\lambda & b \\ c & d-\lambda \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \tag{5.112}$$

We still have the trivial solution for all λ. We want to find the nontrivial solution, in order to determine the eigenvector. If the matrix on the left-hand side was invertible (meaning it has an inverse), then we could multiply both sides of the equation by the inverse matrix, which would yield the trivial solution $\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. The nontrivial solution for the eigenvector occurs *only when the determinant is zero*, implying that the matrix is not invertible. Hence, we require

$$\det \begin{pmatrix} a-\lambda & b \\ c & d-\lambda \end{pmatrix} = (a-\lambda)(d-\lambda) - bc = 0, \tag{5.113}$$

or

$$\lambda^2 - (a+d)\lambda + ad - bc = 0 \tag{5.114}$$

(note that it is not a coincidence that $a+d = \mathrm{Tr}A$ and $ad-bc = \det A$ both appear in this equation). This is solved by the quadratic equation giving

$$\lambda_\pm = \frac{a+d}{2} \pm \frac{1}{2}\sqrt{(a+d)^2 - 4ad + 4bc}. \tag{5.115}$$

Using $(a+d)^2 - 4ad = a^2 + 2ad + d^2 - 4ad = a^2 - 2ad + d^2 = (a-d)^2$ yields

$$\lambda_\pm = \frac{a+d}{2} \pm \frac{1}{2}\sqrt{(a-d)^2 + 4bc}. \tag{5.116}$$

Note that if A is Hermitian (meaning a and d are real and $c = b^*$), then the eigenvalue is always real, because $(a-d)^2 + 4bc = (a-d)^2 + 4|b|^2$ is a real nonegative number. This turns out to be a general property of Hermitian matrices of any dimension.

Let us examine four concrete examples. The first is

$$A = \begin{pmatrix} 0 & 1 \\ -3 & 1 \end{pmatrix}. \tag{5.117}$$

It is not a Hermitian matrix. The eigenvalues are

$$\lambda_{\pm} = \frac{1}{2} \pm \frac{1}{2}\sqrt{1-12} = \frac{1}{2} \pm i\frac{\sqrt{11}}{2}, \tag{5.118}$$

which are complex. The second example is

$$A = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}, \tag{5.119}$$

which is Hermitian and satisfies

$$\lambda_{\pm} = \pm\frac{1}{2}\sqrt{4+16} = \pm\sqrt{5}. \tag{5.120}$$

This has real eigenvalues. Note that there are two roots, so we expect there to be two eigenvectors associated with the two eigenvalues for this matrix. The third example is a complex-valued matrix, which is Hermitian

$$A \begin{pmatrix} 1 & i \\ -i & -1 \end{pmatrix}; \tag{5.121}$$

its eigenvalues are $\pm\sqrt{2}$. Our final example is a non-Hermitian matrix:

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \tag{5.122}$$

it has one eigenvalue equal to 0. This occurs for the vector $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$—all other vectors of the form $\begin{pmatrix} a \\ b \end{pmatrix}$, with $b \neq 0$, will rotate and scale (pure rotation only occurs if $a = 0$).

Henceforth, we assume A is Hermitian, meaning both a and d are real and $b = c^*$. Then, we find the eigenvalues are real and given by

$$\lambda_{\pm} = \frac{a+d}{2} \pm \frac{1}{2}\sqrt{(a-d)^2 + 4|b|^2}. \tag{5.123}$$

A particularly simple case occurs when $b = 0$. Then, the eigenvalues are equal to a and d and the corresponding eigenvectors can be read off as $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$, because the matrix is diagonal. Please check these by matrix-vector multiplication with $A = \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix}$.

How do we find the eigenvectors for the general case, when $b \neq 0$? We go back to the starting point in Eq. (5.112) and solve for α and β after plugging in what the

eigenvalue λ is

$$\begin{pmatrix} a-\lambda_\pm & b \\ b^* & d-\lambda_\pm \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} (a-\lambda_\pm)\alpha + b\beta \\ b^*\alpha + (d-\lambda_\pm)\beta \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \tag{5.124}$$

Looking at the top row, we find a relation between the coefficients as

$$\beta = -\frac{a-\lambda_\pm}{b}\alpha. \tag{5.125}$$

Plugging this into the bottom row yields

$$b^*\alpha - \frac{(a-\lambda_\pm)(d-\lambda_\pm)}{b}\alpha = 0. \tag{5.126}$$

This only has a solution for $\alpha \neq 0$ if

$$\frac{|b|^2 - (a-\lambda_\pm)(d-\lambda_\pm)}{b}\alpha = 0. \tag{5.127}$$

But the term in the numerator is always equal to zero, because it was the original quadratic equation condition for the eigenvalues. So, the equation holds.

Something is odd though. We have not found both α and β, just a relationship between the two. This is a general result and always occurs. It is because an eigenvector is not uniquely determined—if I multiply the eigenvector by any nonzero number, it is still an eigenvector. This occurs because the eigenvalue-eigenvector relationship $Av = \lambda v$ is a homogenenous equation. If I multiply both sides by a scalar c, I find $A(cv) = \lambda(cv)$, so that cv is also an eigenevector with the same eigenvalue. This extra freedom can easily lead to confusion, so we have developed a convention—we normalize the eigenvectors. This means we multiply them by a constant so that the norm of the vector is one, or $|\alpha|^2 + |\beta|^2 = 1$. The method used to solve this is simple—just divide by the length. So, for example, if the vector was equal to $(4 \quad -3)$, then the norm is $\sqrt{4^2 + (-3)^2} = \sqrt{25} = 5$. Hence, our normalized vector is $(\frac{4}{5} \quad -\frac{3}{5})$. So, we need to do the same thing for our general eigenvector. We find

$$|\alpha|^2 + |\beta|^2 = |\alpha|^2 + \underbrace{\left|\frac{a-\lambda_\pm}{b}\right|^2 |\alpha|^2}_{|\beta|^2} = 1. \tag{5.128}$$

Rearranging this equation gives

$$|\alpha|^2 \left(1 + \left|\frac{a-\lambda_\pm}{b}\right|^2\right) = |\alpha|^2 \frac{|b|^2 + |a-\lambda_\pm|^2}{|b|^2} = 1. \tag{5.129}$$

Now, we solve by taking a square root and then using the solution for α to find β. This gives us

$$\alpha = \frac{|b|e^{i\theta}}{\sqrt{|b|^2 + |a - \lambda_\pm|^2}} \text{ and } \beta = -\frac{(a - \lambda_\pm)|b|e^{i\theta}}{b\sqrt{|b|^2 + |a - \lambda_\pm|^2}}. \tag{5.130}$$

You might be surprised by the appearance of the $e^{i\theta}$ factor in these expressions. This is called a complex phase, and the normalized eigenvectors are still determined only up to a complex phase (i.e., a complex number $e^{i\theta}$) that multiplies both α and β. It is conventional to pick $\theta = 0$ so that $e^{i\theta} = 1$, in order to make our life easier; this means α is real and β is possibly complex. Once we use this convention, the eigenvectors are completely determined.

We discuss this a bit more, because it is a subtle point, which is often difficult to understand when you first see it. Suppose $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ is an eigenvector of our general Hermitian matrix:

$$\begin{pmatrix} a & b \\ b^* & d \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \lambda \begin{pmatrix} \alpha \\ \beta \end{pmatrix}. \tag{5.131}$$

Then, if we consider a new vector $\begin{pmatrix} \gamma e^{i\theta}\alpha \\ \gamma e^{i\theta}\beta \end{pmatrix}$, with γ and θ real, we find that

$$\begin{pmatrix} a & b \\ b^* & d \end{pmatrix} \underbrace{\gamma e^{i\theta} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}}_{\text{eigenvector}} = \lambda \underbrace{\gamma e^{i\theta} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}}_{\text{eigenvector}} \tag{5.132}$$

is also an eigenvector with the same eigenvalue. This implies that the eigenvector-eigenvalue relationship only determines the ratio of α/β. To get control over these extra degrees of freedom, we require the eigenvector to be unit norm. Hence, if $|\alpha|^2 + |\beta|^2 = 1$, then we must have $\gamma = 1$ to ensure that $\gamma^2(|\alpha|^2 + |\beta|^2) = 1$. Hence, we still have the uncertainty of the complex phase $e^{i\theta}$. The official math language for this is that the eigenvector is a "ray in the Hilbert space." If you are familiar with the concept of an equivalence relation, you can think of the ray as the equivalence class of unit norm vectors—we only need one representative from the equivalence class to represent it, so the vectors related by a complex phase are all "essentially the same."

What this implies is that eigenvectors such as $\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $-\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\frac{i}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\frac{1+i}{2}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and all other choices of the form $\frac{e^{i\theta}}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ all refer to the same ray or as we will say in quantum mechanics to the same "state." In other words, two quantum states are the same if they are related by a complex phase $e^{i\theta}$. Similarly, we

say two eigenvectors are the same if $v_1 = \begin{pmatrix}\alpha\\ \beta\end{pmatrix}$ and $v_2 = e^{i\theta}\begin{pmatrix}\alpha\\ \beta\end{pmatrix}$, and then, both v_1 and v_2 *represent the same eigenvector*! The general formulas for the eigenvectors are complicated. The idea is not to memorize them, as it is often easier to rederive them for any specific example that needs to be worked out.

This was a lot of work, but we now need to verify that it is correct. This will require even more work! So, take a deep breath as we check our calculation. We start by multiplying the matrix A times our eigenvector, expressed in the "polite way" (norm equal to one, first nonzero element positive). Using Eq. (5.130), we find

$$\begin{aligned}\begin{pmatrix} a & b\\ b^* & d\end{pmatrix}\begin{pmatrix}\alpha\\ \beta\end{pmatrix} &= \begin{pmatrix} a & b\\ b^* & d\end{pmatrix}\frac{|b|}{\sqrt{|b|^2+|a-\lambda_\pm|^2}}\begin{pmatrix}1\\ -\frac{a-\lambda_\pm}{b}\end{pmatrix}\\ &= \frac{|b|}{\sqrt{|b|^2+|a-\lambda_\pm|^2}}\begin{pmatrix} a-(a-\lambda_\pm)\\ b^*-\frac{d}{b}(a-\lambda_\pm)\end{pmatrix}. \end{aligned} \tag{5.133}$$

Yes, this looks like a mess. What is worse, we need to verify the second line is equal to $\lambda\begin{pmatrix}\alpha\\ \beta\end{pmatrix}$. Fortunately, we can ignore the fraction $|b|/\sqrt{|b|^2+|a-\lambda_\pm|^2}$, when we verify the eigenvalue-eigenvector relationship, because eigenvectors can be multiplied by any nonzero number, and they still remain eigenvectors. Hence, the two equations that must hold are

$$a-(a-\lambda_\pm) = \lambda_\pm \tag{5.134}$$

$$b^*-\frac{d}{b}(a-\lambda_\pm) = -\frac{\lambda_\pm}{b}(a-\lambda_\pm). \tag{5.135}$$

The first equation obviously holds, because $a - a = 0$. As for the second, we multiply both sides by $b \neq 0$, so that the left-hand side becomes

$$\begin{aligned}|b|^2-d(a-\lambda_\pm) &= |b|^2-ad+\frac{ad+d^2}{2}\pm\frac{d}{2}\sqrt{(a-d)^2+4|b|^2}\\ &= |b|^2-\frac{ad}{2}+\frac{d^2}{2}\pm\frac{d}{2}\sqrt{(a-d)^2+4|b|^2}. \end{aligned} \tag{5.136}$$

Now, we simplify the right-hand side of Eq. (5.135). After multiplying by b, it is

$$\begin{aligned}&-\left(\frac{a+d}{2}\pm\frac{1}{2}\sqrt{(a-d)^2+4|b|^2}\right)\left(\frac{a-d}{2}\mp\frac{1}{2}\sqrt{(a-d)^2+4|b|^2}\right)\\ &\quad= -\frac{a^2-d^2}{4}\pm\frac{d}{2}\sqrt{(a-d)^2+4|b|^2}+\frac{1}{4}\left((a-d)^2+4|b|^2\right)\\ &\quad= -\frac{ad}{2}+\frac{d^2}{2}+|b|^2\pm\frac{d}{2}\sqrt{(a-d)^2+4|b|^2}, \end{aligned} \tag{5.137}$$

which is the same. This means we have verified the eigenvalue-eigenvector relationship.

So, we have shown that these two eigenvectors are

$$e_+ = \frac{|b|}{\sqrt{|b|^2 + (a-\lambda_+)^2}} \begin{pmatrix} 1 \\ -\frac{a-\lambda_+}{b} \end{pmatrix}$$
$$e_- = \frac{|b|}{\sqrt{|b|^2 + (a-\lambda_-)^2}} \begin{pmatrix} 1 \\ -\frac{a-\lambda_-}{b} \end{pmatrix}. \tag{5.138}$$

It turns out that the eigenvectors of Hermitian matrices, corresponding to different eigenvalues, are always orthogonal. This means that $e_+^\dagger \cdot e_- = 0$—recall the $\dagger$ denotes the Hermitian conjugate, which takes the vector $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ to its transpose conjugate $(\alpha^* \ \beta^*)$. We check the orthogonality next, but we suppress the common factor $|b|^2/\sqrt{(|b|^2 + (a-\lambda_+)^2)(|b|^2 + (a-\lambda_-)^2)}$ in our manipulations. This gives

$$e_+^\dagger \cdot e_- \propto 1 + \frac{(a-\lambda_+)(a-\lambda_-)}{|b|^2} \propto |b|^2 + a^2 - a(\lambda_+ + \lambda_-) + \lambda_+\lambda_-, \tag{5.139}$$

where $\propto$ mean "proportional to" and we removed the common factor of $1/|b|^2$ from the second line. But the eigenvalues of 2×2 matrices satisfy special properties—their sum is $\mathrm{Tr}\, A$ and their product is $\det A$. This is most easily seen by looking at Eq. (5.114), which shows the linear coefficient is $-\mathrm{Tr}\, A$ and the constant term is $\det A$. We can also write that quadratic as $(\lambda - \lambda_+)(\lambda - \lambda_-)$. Expanding the quadratic out and equating the coefficients then tells us that $\lambda_+ + \lambda_- = a + d$ and $\lambda_+\lambda_- = ad - |b|^2$. Using these results in the inner product of the eigenvectors then tells us it is equal to $|b|^2 + a^2 - a(a+d) + ad - |b|^2 = 0$. Hence, the two eigenvectors are indeed orthogonal.

Note that we need to take the Hermitian conjugate of the vector e_+ before performing the inner product, because that is the requirement for the inner product of vectors that have components that are complex valued. The Hermitian conjugate ensures that the norm squared of a vector is positive, as it must be. The fact that eigenvectors form an orthonormal set (they have norm equal to one and are mutually orthogonal) is an important result from linear algebra, which we won't prove in general here, but you can see that it comes out naturally when we examine 2×2 matrices.

Again, that was a lot of work! But we got through it. We will work on building up stamina in algebraic manipulations as we work through this book. Eventually, you will be comfortable working out very lengthy calculations. We have one final topic we need to show about eigenvectors and that is that the set of eigenvectors is also *complete*. This means that the set of eigenvectors spans the vector space. Or, without the jargon, that any vector can be written as a sum of numbers multiplied times the different eigenvectors. When such a construction exists for all vectors in

the vector space, we say we have a basis for the vector space. Here, we will show completeness using a different approach that involves the outer product. One way to state what completeness is, which is the way we will use it in quantum mechanics, is that

$$\mathbb{I}_2 = \begin{pmatrix}\alpha_+\\ \beta_+\end{pmatrix} \otimes (\alpha_+^* \;\; \beta_+^*) + \begin{pmatrix}\alpha_-\\ \beta_-\end{pmatrix} \otimes (\alpha_-^* \;\; \beta_-^*). \tag{5.140}$$

Don't freak out! I will explain what these funny symbols mean and how we verify this identity. The identity matrix $\mathbb{I}_2$ is just the 2×2 identity matrix, given by $\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$, while the outer product satisfies

$$\begin{pmatrix}\alpha\\ \beta\end{pmatrix} \otimes (\alpha^* \;\; \beta^*) = \begin{pmatrix}\alpha\alpha^* & \alpha\beta^*\\ \beta\alpha^* & \beta\beta^*\end{pmatrix}. \tag{5.141}$$

You can think of the outer product as being the first element of the first column vector (α) multiplied by the second row vector, forming the first row of the outer product, and similarly, the second element of the column vector (β) is multiplied by the second row vector, forming the lower row of the outer product. Schematically, this looks like

$$\begin{pmatrix}\alpha\times\boxed{\alpha^* \;\; \beta^*}\\ \beta\times\boxed{\alpha^* \;\; \beta^*}\end{pmatrix}. \tag{5.142}$$

All outer products (also called tensor products) are formed this way.

After recalling that we are adding two matrices together, the 1 1 element of the sum on the right-hand side can be immediately seen to be

$$|\alpha_+|^2 + |\alpha_-|^2 = \frac{|b|^2}{|b|^2 + (a-\lambda_+)^2} + \frac{|b|^2}{|b|^2 + (a-\lambda_-)^2} = 1 \tag{5.143}$$

$$\frac{|b|^2\left(2|b|^2 + 2a^2 - 2a\,(\lambda_+ + \lambda_-) + \lambda_+^2 + \lambda_-^2\right)}{|b|^4 + |b|^2\left(2a^2 - 2a(\lambda_+ + \lambda_-) + \lambda_+^2 + \lambda_-^2\right) + \left(a^2 - a(\lambda_+ + \lambda_-) + \lambda_+\lambda_-\right)^2} = 1$$

$$\frac{|b|^2\left(2|b|^2 + 2a^2 - 2a(a+d) + a^2 + d^2 + 2|b|\right)^2}{|b|^4 + |b|^2\left(2a^2 - 2a(a+d) + a^2 + d^2 + 2|b|^2\right) + \left(a^2 - a(a+d) + ad - |b|^2\right)^2}$$

$$= \frac{4|b|^4 + |b|^2(a-d)^2}{4|b|^4 + |b|^2(a-d)^2} = 1. \tag{5.144}$$

In these algebraic manipulations, we used the facts that $\lambda_+ + \lambda_- = a + d$, $\lambda_+\lambda_- = ad - |b|^2$, and $\lambda_+^2 + \lambda_-^2 = (a+d)(\lambda_+ + \lambda_-) - 2(ad - |b|^2) = a^2 + d^2 + 2|b|^2$.

Similarly, the 1 2 element of the sum of the two matrices is

$$\alpha_+\beta_+^* + \alpha_-\beta_-^* = |b|^2\left(-\frac{\frac{a-\lambda_+}{b^*}}{|b|^2+(a-\lambda_+)^2} - \frac{\frac{a-\lambda_-}{b^*}}{|b|^2+(a-\lambda_-)^2}\right) \tag{5.145}$$

$$= -\frac{|b|^2}{b^*\left(|b|^2+(a-\lambda_+)^2\right)\left(|b|^2+(a-\lambda_-)^2\right)} \times \Big((a-\lambda_+)\big[|b|^2+(a-\lambda_-)^2\big] + (a-\lambda_-)\big[|b|^2+(a-\lambda_+)^2\big]\Big). \tag{5.146}$$

Next, we remove the common factors and substitute in for $\lambda_\pm$ in the squared terms, which gives

$$\begin{aligned}
&\propto (a-\lambda_+)\left[|b|^2+\left(\frac{a-d}{2}+\frac{1}{2}\sqrt{(a-d)^2+4|b|^2}\right)^2\right] \\
&\quad + (a-\lambda_-)\left[|b|^2+\left(\frac{a-d}{2}-\frac{1}{2}\sqrt{(a-d)^2+4|b|^2}\right)^2\right] \\
&\propto (a-\lambda_+)\left[|b|^2+\frac{(a-d)^2}{4}+\frac{a-d}{2}\sqrt{(a-d)^2+4|b|^2}+\frac{(a-d)^2}{4}+|b|^2\right] \\
&\quad + (a-\lambda_-)\left[|b|^2+\frac{(a-d)^2}{4}-\frac{a-d}{2}\sqrt{(a-d)^2+4|b|^2}+\frac{(a-d)^2}{4}+|b|^2\right]
\end{aligned} \tag{5.147}$$

Now, we substitute in for the remaining $\lambda_\pm$ terms

$$\begin{aligned}
&\propto \left[\frac{a-d}{2}-\frac{1}{2}\sqrt{(a-d)^2+4|b|^2}\right]\left[2|b|^2+\frac{(a-d)^2}{2}+\frac{a-d}{2}\sqrt{(a-d)^2+4|b|^2}\right] \\
&+ \left[\frac{a-d}{2}+\frac{1}{2}\sqrt{(a-d)^2+4|b|^2}\right]\left[2|b|^2+\frac{(a-d)^2}{2}-\frac{a-d}{2}\sqrt{(a-d)^2+4|b|^2}\right].
\end{aligned} \tag{5.148}$$

When we FOIL these terms, we see that the terms proportional to the square root cancel, leaving us with

$$\begin{aligned}
&\propto (a-d)|b|^2+\frac{(a-d)^3}{4}-\frac{(a-d)^3}{4}-(a-d)|b|^2 \\
&\quad + (a-d)|b|^2+\frac{(a-d)^3}{4}-\frac{(a-d)^3}{4}-(a-d)|b|^2 \\
&\propto 0.
\end{aligned} \tag{5.149}$$

Similarly, one can show that the 2 1 element is zero and the 2 2 element is 1. The former follows since it is the complex conjugate of the 1 2 element, while the latter is a problem for you. While the fact that the sum of the outer product of eigenvectors equals the unit matrix may seem like some novelty at this point, it is quite important, and we will use it heavily in the remainder of the book.

OK. Time to relax. We are done with the long algebra for a while. We now examine how these ideas work with an explicit example. Note that once we set specific values for a, b, and d, we could substitute into our general equations and get all the results, but it often is easier to rederive the results for a particular case. So, as an example, let us choose

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{5.150}$$

The eigenvalues $\lambda_\pm$ satisfy $\det \begin{pmatrix} -\lambda & 1 \\ 1 & -\lambda \end{pmatrix} = \lambda^2 - 1 = 0$, so that $\lambda_\pm = \pm 1$. The eigenvectors are found by solving

$$(A - \lambda_\pm \mathbb{I}_2) \begin{pmatrix} \alpha_\pm \\ \beta_\pm \end{pmatrix} = 0 \;\rightarrow\; \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \alpha_+ \\ \beta_+ \end{pmatrix} = 0 \text{ and } \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \alpha_- \\ \beta_- \end{pmatrix} = 0. \tag{5.151}$$

So, we find the eigenvectors to be

$$e_+ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ and } e_- = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \tag{5.152}$$

You can check these results, because $-\alpha_+ + \beta_+ = 0$ and $|\alpha_+|^2 + |\beta_+|^2 = 1$ and $\alpha_- + \beta_- = 0$ and $|\alpha_-|^2 + |\beta_-|^2 = 1$.

We can check that the eigenvectors form an orthonormal set, since $e_+ \cdot e_+ = 1$, $e_- \cdot e_- = 1$ and $e_+ \cdot e_- = 0$ hold. For completeness, we see that

$$e_+ \otimes e_+^\dagger + e_- \otimes e_-^\dagger = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{5.153}$$

So, the completeness relation also holds. Since the eigenvectors form a complete set, any two-dimensional vector can be expanded as $v = re_+ + se_-$, with r and s complex numbers. In addition, the two eigenvectors are linearly independent, meaning one cannot express the first in terms of the second or *vice versa*.

Note that there is a special shortcut to finding eigenvectors of Hermitian 2×2 matrices. Since the eigenvectors form an orthonormal set, if $e_+ = \begin{pmatrix} \alpha_+ \\ \beta_+ \end{pmatrix}$, then we immediately find that $e_- = \begin{pmatrix} \beta_+^* \\ -\alpha_+^* \end{pmatrix}$. Or, in other words, $\alpha_- = \beta_+^*$ and $\beta_- = -\alpha_+^*$.

We use eigenvalues and eigenvectors all the time in quantum mechanics. For 2×2 matrices, we need to recall that all Hermitian matrices have two real eigenvalues, the eigenvectors can be chosen to form an orthonormal set, and the eigenvectors are complete. Eigenvectors remain eigenvectors after multiplying the vector by an arbitrary nonzero number—if they are normalized, the arbitrariness is only a complex phase $e^{i\theta}$.

Finally, let us reiterate why we focus so much on Hermitian matrices. It turns out that most measurable quantities in quantum mechanics are associated with a Hermitian matrix (in a particular basis). It is often said this is *required* for all measurable quantities, because measurements always produce real numbers, but the situation is a bit more subtle than that, as we will see as we develop the quantum-mechanics formalism further.

Now, it is your turn for some practice with problems.

Section 5.7 Problems:

5.7.1 Eigenvalues

Find the eigenvalues of

$$A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix} \quad \text{and} \quad C = \begin{pmatrix} 1 & i \\ -i & -1 \end{pmatrix} \tag{5.154}$$

5.7.2 Eigenvalues and Eigenvectors

Find the eigenvalues and eigenvectors of

$$\begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix} \tag{5.155}$$

and verify that the eigenvectors you find are orthogonal.

5.7.3 Eigenvector Trick

Because eigenvectors of Hermitian matrices are orthonormal, we should have that if $e_+ = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$, then e_- must be $\begin{pmatrix} \beta^* \\ -\alpha^* \end{pmatrix}$, since that is the unit vector orthogonal to e_+ (recall that $*$ denotes the complex conjugate). Use this trick to find the eigenvalues and eigenvectors of

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}. \tag{5.156}$$

Verify that $Ae_+ = \lambda_+ e_+$ and $Ae_- = \lambda_- e_-$. Note that when using this trick, the second eigenvector may not have the first component being a real nonnegative number. This can be fixed, if desired, by multiplying the eigenvector by an overall complex phase to ensure the first component of the vector is a positive real number.

5.7.4 Completeness

Finish the derivation started in the text by calculating 2 2 matrix element of the outer product of the general eigenvectors to show that it is equal to 1.

5.7.5 Completeness Trick

Using the "trick" in Problem 5.7.3, let $e_+ = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ and $e_- = \begin{pmatrix} \beta^* \\ -\alpha^* \end{pmatrix}$, with $|\alpha_+|^2 + |\beta_+|^2 = 1$. Use these results and these results only, to show that

$$\mathbb{I}_2 = e_+ \otimes e_+^\dagger + e_- \otimes e_-^\dagger \tag{5.157}$$

holds.

5.7.6 Hermitian Conjugate

Suppose that e_+ and e_- are eigenvectors of A with eigenvalues λ_+ and λ_-. Then

$$\sum_{j=1}^{2} A_{ij}(e_+)_j = \lambda_+ (e_+)_i \tag{5.158}$$

and

$$\sum_{j=1}^{2} A_{ij}(e_-)_j = \lambda_- (e_-)_i. \tag{5.159}$$

Show that the Hermitian conjugate equation also holds:

$$(e_+)^\dagger A = \lambda_+ (e_+)^\dagger \text{ and } (e_-)^\dagger A = \lambda_- (e_-)^\dagger, \tag{5.160}$$

by taking the Hermitian conjugate of the eigenvalue equations. Use the facts that $\lambda_\pm$ are real and the A matrix is Hermitian, so $A_{ji} = A^*_{ij}$. Note that if the eigenvectors are complex valued, then the left eigenvectors are *different* from the right eigenvectors, because of the complex conjugate.

Hint: if you find the abstract problem to be challenging, try some concrete examples like $A = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & i \\ -i & -2 \end{pmatrix}$ and then try the general case.

5.7.7 Eigenvectors and Complex Phases

Identify the eigenvectors that correspond to the same state (or equivalently the same ray). This means the vectors are identical up to multiplication by a complex phase $e^{i\theta}$ with real θ.

(a) $\begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$

(b) $\begin{pmatrix} \frac{1+i}{2} \\ \frac{1-i}{2} \end{pmatrix}$

(c) $\begin{pmatrix} \frac{i}{\sqrt{2}} \\ -\frac{i}{\sqrt{2}} \end{pmatrix}$

(d) $\begin{pmatrix} \frac{1}{2\sqrt{2}} + \frac{\sqrt{3}i}{2\sqrt{2}} \\ -\frac{1}{2\sqrt{2}} - \frac{\sqrt{3}i}{2\sqrt{2}} \end{pmatrix}$

(e) $\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1-i}{2} \end{pmatrix}$

(f) $\begin{pmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$

Hint: find the factor $e^{i\theta}$ to multiply by that makes the top component of the vector real and positive.

5.7.8 Decomposition to Find Eigenvalues

Consider a matrix of the form

$$A = \begin{pmatrix} E_1 & V \\ V^* & E_2 \end{pmatrix} \tag{5.161}$$

with E_1 and E_2 both real. Define the average $\bar{E} = \frac{E_1+E_2}{2}$ and the difference $\Delta = \frac{E_1-E_2}{2}$. Write the matrix as

$$A = \bar{E}\mathbb{I}_2 + \begin{pmatrix} \Delta & V \\ V^* & -\Delta \end{pmatrix} \tag{5.162}$$

From this decomposition, you should be able to immediately show that the eigenvalues are $\bar{E} \pm \sqrt{\Delta^2 + |V|^2}$. This decomposition can be employed on any Hermitian 2×2 matrix to quickly obtain the eigenvalues.

5.7.9 Scaling and Rotating

Consider the matrix:

$$A = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}. \tag{5.163}$$

Show by drawing some pictures that the vector $\begin{pmatrix} a \\ b \end{pmatrix}$ is mapped to $\begin{pmatrix} 2b \\ 2a \end{pmatrix}$ when it is multiplied by A. Use picture to show for many different choices of a and b the vector that is multiplied by A is both rotated and scaled. However, if you pick $a = b = 1$, or $a = -b = 1$, those vectors are only scaled, not rotated. Show as well that these eigenvectors are orthogonal and that they span the two-dimensional vector space.

5.7.10 Viète's Formulas

A student of mine, Tomasz Szymanski, pointed out to me that the two results $\lambda_+ + \lambda_- = a + d$ and $\lambda_+\lambda_- = ad - |b|^2$ for Hermitian matrices can be employed to simplify some of the torturous algebra we employed in computing eigenvectors and the completeness relations. Viète's idea is to never have to work with a square root in the algebraic manipulations. For example, the expression $\lambda_+^2 + \lambda_-^2$ can be rewritten using the add-zero trick as

$$\lambda_+^2 + \lambda_-^2 + 2\lambda_+\lambda_- - 2\lambda_+\lambda_- = (\lambda_+ + \lambda_-)^2 - 2\lambda_+\lambda_-. \tag{5.164}$$

Using Viète's formulas, we find that

$$\begin{aligned}(\lambda_+ + \lambda_-)^2 - 2\lambda_+\lambda_- &= (a+d)^2 - 2(ad - |b|^2) \\ &= a^2 + 2ad + d^2 - 2ad + 2|b|^2 \\ &= a^2 + d^2 + 2|b|^2. \end{aligned} \tag{5.165}$$

In this way, we avoid having to ever work with the square roots. Revisit one of the problems, such as computing α_{12} for completeness and rearrange the algebra using Viète's rules.

Do you find this approach to be simpler than the standard way?

As a historical note, Francois Viète was a French mathematician from the sixteenth century.

5.8 Dirac bra and ket Notation

We now introduce a commonly used notation for quantum mechanics, which was invented by Paul Dirac in 1939 Dirac (1939). It is called the "bra" and "ket" notation to associate a bra-ket (sometimes also called a bracket) with a quantum expectation value or measurement. Until you have seen and understood this notation, it looks quite foreign. But you need to always remember, it is just a notation, nothing more. There is no scientific content to it; it is just a convenience, which converts otherwise long expressions into shorter more manageable ones, without losing any content! Sounds great. Let's get started.

To begin, we focus on the problems we have just been working on—the eigenvectors of 2×2 Hermitian matrices. Recall we had two eigenvectors, which we wrote as

$$e_+ = \begin{pmatrix} \alpha_+ \\ \beta_+ \end{pmatrix} \text{ and } e_- = \begin{pmatrix} \alpha_- \\ \beta_- \end{pmatrix} = \begin{pmatrix} \beta_+^* \\ -\alpha_+^* \end{pmatrix}, \tag{5.166}$$

where the last result follows from the orthogonality of the eigenvectors. The Dirac "bra" and "ket" are just new notation for representing row and column vectors. So, we use a ket for column vectors, where, in this example, $e_+ = |+\rangle = \begin{pmatrix} \alpha_+ \\ \beta_+ \end{pmatrix}$ and $e_- = |-\rangle = \begin{pmatrix} \alpha_- \\ \beta_- \end{pmatrix}$. The ket notation is always of the form, vertical line, label, angle bracket. Next, the row vectors are given by "bras" (Hermitian conjugates of kets) as follows:

$$e_+^\dagger = (\alpha_+^* \;\; \beta_+^*) = \langle +|, \;\; e_-^\dagger = (\alpha_-^* \;\; \beta_-^*) = (\beta_+ \;\; -\alpha_+) = \langle -|. \tag{5.167}$$

Note how the bra has an angle bracket to the left, a label that describes the state in the center and a vertical bar on the right that "closes" the bra—this is like the mirror image of a ket.

We combine "bras" and "kets" as a bra-ket (dropping the double line to a single line) to describe inner (or dot) products as

$$\langle +|+\rangle = e_+^\dagger \cdot e_+ = (\alpha_+^* \;\; \beta_+^*) \cdot \begin{pmatrix} \alpha_+ \\ \beta_+ \end{pmatrix} = |\alpha_+|^2 + |\beta_+|^2 = 1$$

$$\langle +|-\rangle = e_+^\dagger \cdot e_- = (\alpha_+^* \;\; \beta_+^*) \cdot \begin{pmatrix} \beta_+^* \\ -\alpha_+^* \end{pmatrix} = \alpha_+^* \beta_+^* - \beta_+^* \alpha_+^* = 0. \tag{5.168}$$

Please work out for yourself what $\langle -|+\rangle$ and $\langle -|-\rangle$ are equal to. The idea of this notation is that the bra-ket looks like a "closed" object $\langle a|b\rangle$, which indicates that the final result is a scalar (complex) number. Note that a bra $\langle a|$ is the Hermitian conjugate of a ket $|a\rangle$, when we represent them as row and column vectors. The bra and ket notation works for any vector, not just for eigenvectors.

We can use this notation to indicate matrix-vector multiplication. For example, the eigenvalue-eigenvector relationship, given by $Ae_\pm = \lambda_\pm e_\pm$, translates into

$$\underbrace{\hat{A}}_{\text{operator}} \underbrace{|\pm\rangle}_{\text{state}} = \underbrace{\lambda_\pm}_{\text{number}} \underbrace{|\pm\rangle}_{\text{state}}. \tag{5.169}$$

Here, we used $\hat{A}$ ("A-hat") to denote the matrix A, but thought of now as an abstract operator. We describe what an abstract operator is in the next paragraph. Here, the symbol $\lambda_\pm$ is still the eigenvalue (real number) for the eigenvalue equation. The vector is now represented by the ket and we often call bras and kets states instead of vectors. The Hermitian conjugate equation is $\langle \pm|\hat{A} = \lambda_\pm \langle \pm|$, where we used the fact that $\hat{A} = \hat{A}^\dagger$, for a Hermitian operator (and the fact that eigenvalues of Hermitian operators are real). Note here that the operator acts to the left on the bra. In general, in Dirac notation, an operator can act to the left or to the right. This freedom can be viewed as a nice aspect of the notation, although mathematicians are bothered by the ambiguity that it brings, and for this reason, they do not like using Dirac notation. But don't worry, it won't get you into trouble here; we will point out all the subtleties for you as they occur.

We use the term abstract operator (often shortened to just operator) to indicate something that acts on a state (or ket) and maps it to a new state (or ket), multiplied by a possibly complex number. We can always think of representing abstract operators via matrices and the states as vectors, so that an operator acting on a state can be synonymous with a matrix multiplying a vector. In the bra-ket notation, operators can act to the right or to the left—this is like a matrix being multiplied from the left by a row vector or from the right by a column vector. If it is multiplied from the right and from the left, then we create a number again—in Dirac notation, we write $v^\dagger A v'$ as $\langle v|\hat{A}|v'\rangle$. You can think of this as a sandwich—the bra and ket are the "bread" and the operator is the "filling."

It is useful to take a little break here and recap what we have done. We use the Dirac bra and ket notation to indicate different vectors (or states). We use a hat to indicate an abstract operator (which can also be schematically thought of as a matrix), and we sandwich bras and kets together to indicate inner (dot) products between vectors (which is a complex number). It is just a more compact way of representing everything we have been working with so far.

So, for example, we can use this notation to describe the Stern-Gerlach states we worked with earlier. Let $|\uparrow\rangle$ denote a positive projection on the $+z$ axis, $|\downarrow\rangle$ is a negative projection on the $+z$ axis, $|\rightarrow\rangle$ is a positive projection on the $+x$ axis, and $|\leftarrow\rangle$ is a negative projection on the $+x$ axis. Then, $|\nearrow\rangle$ would indicate a positive projection on an axis at an angle θ to the vertical. One can see it is a little difficult to tell precisely what the angle θ is, so it is more common to use a notation where $|\uparrow\rangle$ denotes a positive projection and $|\downarrow\rangle$ denotes a negative projection. Then, we use a subscript to indicate the axis. So, we would have notations like $|\uparrow\rangle_z$ or $|\uparrow\rangle_x$ or $|\downarrow\rangle_\theta$ to indicate different states oriented on different axes. Using this notation, we have identities such as

$$|\uparrow\rangle_z = |\downarrow\rangle_{-z} \text{ and } |\downarrow\rangle_z = |\uparrow\rangle_{-z}, \tag{5.170}$$

which means, in words, the positive projection on the z-axis is the same as a negative projection on the minus z-axis, and so on.

Similarly, we can describe the probability amplitude on the screen for single and two-slit experiments via a generic symbol $|\psi\rangle$. You can think of $|\psi\rangle$ as the abstract set of rules we use to describe all alternative ways to get from the source to the detector, expressed generically as the rules, without evaluating them at a specific position of the detector. Because the amplitude arrow changes with position, we need to introduce a bra to indicate the position where we want the probability arrow to be evaluated for a specific detector. Then, $\langle x|\psi\rangle$ is defined to be $\psi(x)$, which is the probability amplitude (arrow) at position x and is given by applying the abstract rules to determine the final arrow after hooking all the specific arrows for the alternative ways to leave the source and arrive at the specific position x. This object $\psi(x)$ is called the *wavefunction*; it and its modulus squared, which is the *probability density*, are both plotted in Fig. 5.16. Note the difference between the quantum state, which is the set of rules, and the wavefunction, which involves evaluating those rules at a particular spatial location, resulting in a number (arrow). It is important to understand properly how these different objects are determined and what they mean.

Of course, we square the length of the arrows to determine the probability at x: $P(x) = |\psi(x)|^2 = |\langle x|\psi\rangle|^2$. The object $\psi(x) = \langle x|\psi\rangle$, which we call the wavefunction, is an inner product. It is the projection of the quantum state $|\psi\rangle$ onto the eigenvectors of position $\langle x|$. While it is the mainstay of essentially all quantum textbooks, we will not focus on it as much as they do, because we will not need it as much in the way we develop the quantum theory.

So, to summarize, the Dirac bra and ket notation is just a compact and elegant way of describing objects we have been working with already. It turns out to be a

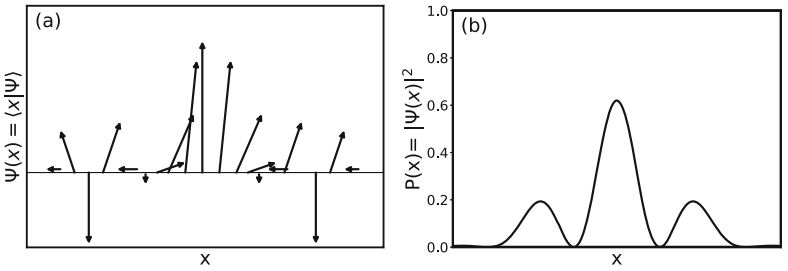

Fig. 5.16 Schematic for how the final arrows become the wavefunction (left) and how the square of the wavefunction becomes the probability density (right)

common language often used to describe quantum mechanics and is a notation we will use freely for the remainder of the book. We will develop its use slowly, to ensure you are comfortable using it.

Here is an executive summary of Dirac notation: a state $|\psi\rangle$ is a normalized state, if it satisfies $\langle\psi|\psi\rangle = 1$. The closed bra-ket $\langle\phi|\psi\rangle$ is always a complex number. An operator $\hat{A}$ maps a state to a state, which we denote as $\hat{A}|\psi\rangle$. Here, we include a hat to indicate that we can think of it as an abstract operator (representation independent). The product $\hat{A}|\psi\rangle$ is a ket. Similarly, the product $\langle\psi|\hat{A}$ is a bra. Then, $\langle\phi|\hat{A}|\psi\rangle$ is a complex number (also called a matrix element), because we "collapsed" a bra with a ket.

We have been discussing abstract operators and matrices and acting as if they can be simply interchanged. But they are different, so we want to carefully unpack them here. The difference is similar to the difference between a vector and the expansion of a vector in a particular basis. The abstract operator will map a vector into another vector. To define it, we simply need to say what the map is. At this stage, it is not a matrix. To represent it as a matrix, we need to convert from operators to numbers. The way we do this is to use our completeness relation (from the end of the previous section), written out in Dirac notation as $|e_1\rangle\langle e_1| + |e_2\rangle\langle e_2| = \mathbb{I}_2$, where $\{|e_1\rangle, |e_2\rangle\}$ is the orthonormal basis set, with $\langle e_i|e_j\rangle = \delta_{ij}$; δ_{ij} is the Kronecker delta symbol—it is 1 if $i = j$ and is 0 if $i \neq j$. In Dirac representation, we suppress the tensor product, so the ket-bra product is really analogous to the tensor product of the column vector with the row vector, which we saw led to a matrix, previously. In Dirac notation, we recognize that a ket-bra product is an operator, because it maps a ket vector to another ket vector. Continuing in the derivation, we take our original multiplication $\hat{A}|\psi\rangle$ and multiply by one to the right and the left of the operator:

$$\begin{aligned}\hat{A}|\psi\rangle &= \mathbb{I}_2\hat{A}\mathbb{I}_2|\psi\rangle \\ &= \sum_i |e_i\rangle\langle e_i|\hat{A}|\sum_j |e_j\rangle\langle e_j|\psi\rangle \\ &= |e_1\rangle\langle e_1|\hat{A}|e_1\rangle\langle e_1|\psi\rangle + |e_1\rangle\langle e_1|\hat{A}|e_2\rangle\langle e_2|\psi\rangle \\ &+ |e_2\rangle\langle e_2|\hat{A}|e_1\rangle\langle e_1|\psi\rangle + |e_2\rangle\langle e_2|\hat{A}|e_2\rangle\langle e_2|\psi\rangle. \end{aligned} \tag{5.171}$$

All of the bra-kets are complex numbers. If we denote $\langle e_1|\psi\rangle = \alpha$ and $\langle e_2|\psi\rangle = \beta$, then we have the correspondence:

$$|\psi\rangle = |e_1\rangle\alpha + |e_2\rangle\beta \;\leftrightarrow\; \begin{pmatrix}\alpha\\ \beta\end{pmatrix}. \tag{5.172}$$

Note how the vector $|\psi\rangle$ is abstractly expanded in the basis $|e_i\rangle$ and how this is isomorphic to the standard column vector. It is often useful to think of the inner product formed by a bra $\langle\phi|$ and a ket $|\psi\rangle$ as determining how much of the ket $|\phi\rangle$ is "in" the ket $|\psi\rangle$—in other words, it is the *projection* of the ket $|\psi\rangle$ onto the ket $|\phi\rangle$. The matrix works in the same way. We denote $\langle e_1|\hat{A}|e_1\rangle = a$, $\langle e_1|\hat{A}|e_2\rangle = b$, $\langle e_2|\hat{A}|e_1\rangle = c$, and $\langle e_2|\hat{A}|e_2\rangle = d$. If the operator $\hat{A}$ is Hermitian, then we have $c = b^*$. This can be understood as follows. Since a bra-ket is an inner product, according to our rules, we have $\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*$. When we have operators in the bra-ket, complex conjugation of the bra-ket "sandwich" changes the operator in the "filling" to its Hermitian conjugate according to

$$\langle\phi|\hat{A}|\psi\rangle^* = \langle\psi|\hat{A}^\dagger|\phi\rangle. \tag{5.173}$$

Hence, we have the equivalence:

$$\hat{A} = |e_1\rangle\langle e_1|a + |e_1\rangle\langle e_2|b + |e_2\rangle\langle e_1|c + |e_2\rangle\langle e_2|d \;\leftrightarrow\; \begin{pmatrix}a & b\\ c & d\end{pmatrix}. \tag{5.174}$$

Recall how we said before that the outer product given by a ket-bra product is an operator, and here, you can see how using these outer products with a set of basis vectors allows us to completely specify the operator. In addition, the matrix elements of the Hermitian conjugate operator correspond to the complex conjugate transpose of the matrix representation of the original operator. Finally, one should see that Eq. (5.171) has the following equivalence:

$$\begin{aligned}\hat{A}|\psi\rangle &= \big(|e_1\rangle\langle e_1|a + |e_1\rangle\langle e_2|b + |e_2\rangle\langle e_1|c + |e_2\rangle\langle e_2|d\big)\big(|e_1\rangle\alpha + |e_2\rangle\beta\big)\\ &= |e_1\rangle\big(a\alpha + b\beta\big) + |e_2\rangle\big(c\alpha + d\beta\big)\\ &\leftrightarrow \begin{pmatrix}a & b\\ c & d\end{pmatrix}\begin{pmatrix}\alpha\\ \beta\end{pmatrix} = \begin{pmatrix}a\alpha + b\beta\\ c\alpha + d\beta\end{pmatrix}.\end{aligned} \tag{5.175}$$

I hope these correspondences are clear to you.

We end the section with a concrete description for how we go from an abstract operator to a matrix representation. Suppose the operator $\hat{A}$ is defined via taking the unit norm ket $|\phi\rangle$ and multiplying it by a and taking the orthogonal unit norm ket $|\psi\rangle$ and multiplying it by d. It then is represented by

$$\hat{A} = |\phi\rangle\langle\phi|a + |\psi\rangle\langle\psi|d. \tag{5.176}$$

Now, we can directly compute how $\hat{A}$ acts on a general vector given by $|\chi\rangle = \alpha|\phi\rangle + \beta|\psi\rangle$. We find

$$\begin{aligned}\hat{A}|\chi\rangle &= \hat{A}\big(\alpha|\phi\rangle + \beta|\psi\rangle\big)\\ &= \alpha\big(\underbrace{\hat{A}|\phi\rangle}_{a|\phi\rangle}\big) + \beta\big(\underbrace{\hat{A}|\psi\rangle}_{d|\psi\rangle}\big)\\ &= a\alpha|\phi\rangle + d\beta|\psi\rangle. \end{aligned} \tag{5.177}$$

Our second example considers the two vectors:

$$|\phi\rangle \leftrightarrow \frac{1}{\sqrt{5}}\begin{pmatrix} 2\\ -1\end{pmatrix} \text{ and } |\psi\rangle \leftrightarrow \frac{1}{\sqrt{5}}\begin{pmatrix} 1\\ 2\end{pmatrix}. \tag{5.178}$$

You can immediately verify that $\langle\phi|\phi\rangle = \langle\psi|\psi\rangle = 1$ and $\langle\phi|\psi\rangle = 0$. Suppose the operator $\hat{B}$ is represented by the matrix $\begin{pmatrix} 4 & -i\\ i & 2\end{pmatrix}$. We want to change the basis from the current basis (given by the vectors $\left\{\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix}\right\}$ into the $\phi - \psi$ basis. To do this, we need to compute the four matrix elements $\langle\phi|\hat{B}|\phi\rangle$, $\langle\phi|\hat{B}|\psi\rangle$, $\langle\psi|\hat{B}|\phi\rangle$, and $\langle\psi|\hat{B}|\psi\rangle$. Direct computation gives

$$\begin{aligned}\langle\phi|\hat{B}|\phi\rangle &= \frac{1}{5}(2\ \ -1)\cdot\begin{pmatrix}4 & -i\\ i & 2\end{pmatrix}\cdot\begin{pmatrix}2\\ -1\end{pmatrix} = \frac{18}{5}\\ \langle\phi|\hat{B}|\psi\rangle &= \frac{1}{5}(2\ \ -1)\cdot\begin{pmatrix}4 & -i\\ i & 2\end{pmatrix}\cdot\begin{pmatrix}1\\ 2\end{pmatrix} = \frac{4}{5} - i\\ \langle\psi|\hat{B}|\phi\rangle &= \frac{1}{5}(1\ \ 2)\cdot\begin{pmatrix}4 & -i\\ i & 2\end{pmatrix}\cdot\begin{pmatrix}2\\ -1\end{pmatrix} = \frac{4}{5} + i\\ \langle\psi|\hat{B}|\psi\rangle &= \frac{1}{5}(1\ \ 2)\cdot\begin{pmatrix}4 & -i\\ i & 2\end{pmatrix}\cdot\begin{pmatrix}1\\ 2\end{pmatrix} = \frac{12}{5}. \end{aligned} \tag{5.179}$$

We say that the operator $\hat{B}$ is given by the following in the $\{|\phi\rangle, |\psi\rangle\}$ basis:

$$\begin{pmatrix}\langle\phi|\hat{B}|\phi\rangle & \langle\phi|\hat{B}|\psi\rangle\\ \langle\psi|\hat{B}|\phi\rangle & \langle\psi|\hat{B}|\psi\rangle\end{pmatrix} = \begin{pmatrix}\frac{18}{5} & \frac{4}{5} - i\\ \frac{4}{5} + i & \frac{12}{5}\end{pmatrix}. \tag{5.180}$$

Note that in this new basis, the matrix remains Hermitian, just like the original matrix. In addition, the trace is 6 for the original and transformed matrix and the determinant is 7 for both as well. These are general results. The change of basis of a matrix preserves both the determinant and the trace (as well as special properties such as Hermiticity or unitarity). The operator is always referred to as $\hat{B}$ and is

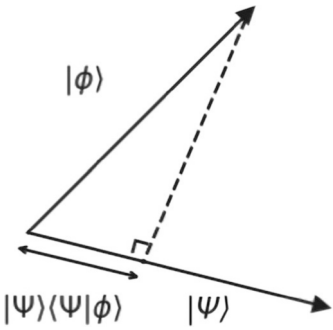

Fig. 5.17 Schematic of the projection of the vector $|\phi\rangle$ on $|\psi\rangle$. As can be seen from the double arrow highlighting $|\psi\rangle\langle\psi|\phi\rangle$, the projection (or, if you like, the inner product of $\langle\psi|\phi\rangle$) shows how much of $|\psi\rangle$ is in $|\phi\rangle$ (we assume both vectors are unit norm)

independent of the basis. Once a basis is chosen, we can compute the representation of the operator as a matrix in that basis.

I have found students often struggle with making sense of how to convert from an abstract operator to the matrix that represents it. It is an important task to master, so please take the time to understand it.

Section 5.8 Problems:

5.8.1 Projection Operator

The operator $\hat{P}_\psi = |\psi\rangle\langle\psi|$ is called the projection operator onto state $|\psi\rangle$ (see Fig. 5.17). Show that if $|\psi\rangle$ is normalized, then $\hat{P}_\psi^2 = \hat{P}_\psi$. Show that $\hat{P}_\psi|\phi\rangle$ projects the vector $|\phi\rangle$ onto the vector $|\psi\rangle$, meaning the result is the inner product of $|\psi\rangle$ with $|\phi\rangle$ multiplied by the vector $|\psi\rangle$, or $\left(\psi^\dagger \cdot \phi\right)\psi$ in vector notation.

5.8.2 Complex Conjugation in Dirac Notation

Since $\langle\phi|\hat{A}|\psi\rangle$ is a complex number, we can take its complex conjugate. The rule is that

$$\langle\phi|\hat{A}|\psi\rangle^* = \langle\psi|\hat{A}^\dagger|\phi\rangle, \tag{5.181}$$

where we interchange bra to ket and ket to bra and we take the Hermitian conjugate of the operator. Let

$$|\psi\rangle = \begin{pmatrix}\alpha\\ \beta\end{pmatrix}, \quad |\phi\rangle = \begin{pmatrix}\gamma\\ \delta\end{pmatrix}, \quad \text{and} \quad A = \begin{pmatrix}a & b\\ c & d\end{pmatrix}, \tag{5.182}$$

where all symbols are complex numbers. Compute $\langle\phi|\hat{A}|\psi\rangle$, take its complex conjugate, and show it is equal to $\langle\psi|\hat{A}^\dagger|\phi\rangle$.

Hint: recall if $|\phi\rangle = \begin{pmatrix}\gamma\\ \delta\end{pmatrix}$, then $\langle\phi| = (\gamma^* \;\; \delta^*)$.

5.8.3 Completeness

The completeness relation is written as

$$\mathbb{I} = |+\rangle\langle+| + |-\rangle\langle-|, \tag{5.183}$$

which is the sum of the projection operators corresponding to the $|+\rangle$ and $|-\rangle$ states. Use $|+\rangle = \begin{pmatrix}\alpha\\ \beta\end{pmatrix}$ and $|-\rangle = \begin{pmatrix}\beta^*\\ -\alpha^*\end{pmatrix}$, and show that

$$\big(|+\rangle\langle+| + |-\rangle\langle-|\big)|\phi\rangle = |\phi\rangle, \tag{5.184}$$

for $|\phi\rangle = \begin{pmatrix}\gamma\\ \delta\end{pmatrix}$, by converting the bra and ket notation to 2×2 matrices and performing the matrix multiplications and sums. Recall that $|\alpha|^2 + |\beta|^2 = 1$.

Hint: $|+\rangle\langle+| = \begin{pmatrix}\alpha\\ \beta\end{pmatrix} \otimes (\alpha^* \;\; \beta^*)$, the outer product.

5.8.4 Representation of an Operator as a Matrix in a Specific Basis

Let $|+\rangle$ and $|-\rangle$ be the eigenvectors of $\hat{A}$, which is represented by a 2×2 Hermitian matrix with eigenvalues λ_+ and λ_-. Compute

$$\langle+|\hat{A}|+\rangle, \;\; \langle+|\hat{A}|-\rangle, \;\; \langle-|\hat{A}|+\rangle, \;\; \langle-|\hat{A}|-\rangle. \tag{5.185}$$

Assume $\langle+|+\rangle = \langle-|-\rangle = 1$ and $\langle+|-\rangle = 0$. Your answers should depend only on λ_+ and λ_-.

5.8.5 Eigenvectors of the Projection Operator

Suppose $\hat{P}_+ = |+\rangle\langle+|$ is the projection operator onto the $|+\rangle$ eigenstate. Show that $|+\rangle$ and $|-\rangle$ are eigenvectors of $\hat{P}_+$ with eigenvalues 1 and 0, respectively.

5.8.6 Ray Versus Vector

We have mentioned already that multiplying a vector by a nonzero complex number does not change the quantum state. In essence, just the direction of the vector is important, not its length or its complex phase. To show how this enters into calculations, one often writes a quantum state in an operator form (also called a density matrix) that is denoted by ρ_ψ and is given by

$$\rho_\psi = \frac{|\psi\rangle\langle\psi|}{\langle\psi|\psi\rangle}. \tag{5.186}$$

Show that if you define a new state from a rescaled vector $|\phi\rangle = c|\psi\rangle$, that the density matrix is unchanged. That is, show that we have $\rho_\psi = \rho_\phi$ for $c \neq 0$.

5.8.7 Nondegenerate Eigenvectors of a Hermitian Operator Must Be Orthogonal

In this problem, we will consider a general case, rather than just 2×2 matrices. Suppose we have a set of eigenvectors $\{|\psi_n\rangle\}$ that satisfy

$$\hat{O}|\psi_n\rangle = \lambda_n|\psi_n\rangle \tag{5.187}$$

for a Hermitian operator $\hat{O}$, with λ_n the (real-valued) eigenvalues. We assume we are in the *nondegnerate* case, where $\lambda_n \neq \lambda_m$ for all $m \neq n$.

Show that $\langle\psi_n|\psi_m\rangle = 0$ for $m \neq n$.

Hint: consider evaluating $\langle\psi_m|\hat{O}|\psi_n\rangle$ two different ways for $n \neq m$ by operating the operator to the left or to the right. Use the fact that these states are eigenvectors and that the eigenvalues are different to conclude the orthogonality of the vectors. Work abstractly using only what you know.

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Chapter 6
Spin One-Half and Polarized Light

There is really nothing more to say—except why. But since why is difficult to handle, one must take refuge in how.

Toni Morrison, Writer, Nobel Laureate, 1969

6.1 Quantum Formalism for Spin

We now have all the tools we need to describe quantum systems using a formal mathematical language. Our strategy will be to apply this formalism to quantum systems you are already familiar with—the Stern-Gerlach experiment and light. We start by returning to our old friend the Stern-Gerlach experiment, which turns out to be the simplest quantum system to describe. We will also use the proper language to describe these systems—we say we are working with particles that have spin (or sometimes we are working with just spins), rather than saying we are working with quantum objects that "act like current loops." A summary of Chapter 6 is given in the following video https://tinyurl.com/58p4p7yb.

As we described before, when we introduced the bra and ket notation in Sect. 5.8, we will use an up arrow ($\uparrow$) to denote a positive projection, which we will call "spin-up" and a down arrow ($\downarrow$) to denote a negative projection, which we will call "spin-down." We use a subscript on the bra or ket to denote the axis of increasing magnetic field, which is always in the same direction as the magnetic field for Stern-Gerlach analyzers, so $|\uparrow\rangle_z$ describes the quantum state of an atom with a positive projection on the z-axis and $|\downarrow\rangle_x$ describes the quantum state of an atom with a negative projection on the x-axis.

Quantum mechanics is all about employing operators to describe measurable quantities and determining how they modify quantum states (represented by bras and kets). We will describe some highlights for just how next. This is really the crux of how the quantum formalism works and is critically important for you to learn. At this stage, we think of it as a postulate for measurement first and then

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work through the consequences. The operators that represent a measurable quantity will be Hermitian (the technically correct phrase is self-adjoint, which is even more restrictive), and the expectation value for a system in a state $|\psi\rangle$ with respect to an operator $\hat{A}$ will be $\langle\psi|\hat{A}|\psi\rangle$; $\hat{A}$ can be the operator for energy, momentum, position, and so on. The quantity $\langle\psi|\hat{A}|\psi\rangle$ represents the average result we will find if we repeat an experiment many times with each shot starting from the same initial state $|\psi\rangle$—the actual result measured in each shot is an eigenvalue of the operator $\hat{A}$—weighting the values of the eigenvalues by the probabilities that we will measure them results in the *expectation value*. Because eigenvalues of Hermitian operators are real, most measurements will measure real quantities, but this is not a requirement, and there are circumstances where a measured object is complex valued—in that case, we usually determine it by separately measuring the real and imaginary parts. So, this result can be thought of as a sum over the probability that $|\psi\rangle$ contains contributions from a specific eigenvector of $\hat{A}$ multiplied by the eigenvalue of $\hat{A}$ for that eigenvector. Note that the quantum state $|\psi\rangle$ is always a unit norm state $\langle\psi|\psi\rangle = 1$, which is the way we guarantee that the sum of the probabilities of all the possible outcomes of the measurement is equal to one—a rule we are quite familiar with.

In particular, we said the operator corresponding to an observable (which is just jargon for something that can be measured) is usually Hermitian. We use a dagger (†) to denote the Hermitian conjugate of an operator, so that $\hat{A}^\dagger$ is the Hermitian conjugate of $\hat{A}$. A Hermitian operator satisfies $\hat{A}^\dagger = \hat{A}$; its matrix representation will be Hermitian as well (technically speaking, when we work with infinite-dimensional quantum systems, the details for this equivalence gives the difference between a Hermitian and a self-adjoint operator—more on that later). Determining the Hermitian conjugate of an operator is not a trivial thing to do. If you get into the details of Hilbert space theory, you will find it can be one of the more complex calculations you will engage in. Fortunately for us, we have a simpler way to treat these issues here, because when we work with spin, there are no subtle complexities. The way in which we proceed is that if we know what happens when $\hat{A}$ acts on a ket $\hat{A}|\psi\rangle =$ something, then we take the Hermitian conjugate via $\langle\psi|\hat{A}^\dagger =$ Hermitian conjugate of that something. When an operator is Hermitian, it can act to the right on a bra or to the left on a ket, because we immediately know its action both ways (since $\hat{A} = \hat{A}^\dagger$). If it is not Hermitian, we need to know what the Hermitian conjugate does to allow the operator to act opposite to its "natural" direction.

If we have more than one quantum operator, then we will want to determine the *commutator* between the two operators, denoted $[\hat{A}, \hat{B}]$. The commutator is defined to be

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}. \tag{6.1}$$

This object is important, because two operators need not commute with each other, and when they do not, the commutator often tells us important information about both operators—we use the commutator to evaluate many matrix elements, as you will learn shortly. One important property of the commutator is it will tell us

whether two operators can have simultaneous eigenvectors, meaning the same set of eigenvectors for one operator is also the set of eigenvectors for the other operator. If $[\hat{A}, \hat{B}] = 0$, then we can find simultaneous eigenvectors for both. In this case, if we measure the property associated with the operator $\hat{A}$, it does not disturb the property associated with the operator $\hat{B}$. One other point to note is that operators can be represented by matrices (as we discussed in the mathematical interlude) and quantum states can be represented by vectors. So, matrix and vector algebra is also used to describe quantum systems. We will see many examples for how this works, starting in the next section.

In fact, the fact that the commutator between some operators is not zero is essentially the property that makes quantum mechanics different from the classical world. This may not be obvious just yet, but you will eventually see that everything we do stems from the fundamental commutation relations. Our first experience will be with the operators for spin. Later, we will look at the commutator between position and momentum. Both are key concepts for revealing the properties of the quantum world.

Please do not be frightened about the notion of an abstract operator. Just because it is a new concept does not mean it has to be intimidating. You will find your comfort level with them grows as you become more familiar with them.

We are now ready to create our first set of quantum operators, which are the spin operators. Note that an abstract operator transforms (or maps) one quantum state vector to another, that is, it maps a ket to a ket. When we represent an operator by a matrix, and the ket by a vector, then we will use matrix-vector multiplication as the map, transforming the initial vector to the final vector (recall the transformation of the vector is then a rotation and a scaling).

The first operator we will define is called the $\hat{S}_z$ "ess-sub-zee" operator. This operator simply measures the projection of the spin on the z-axis. This means the states $|\uparrow\rangle_z$ and $|\downarrow\rangle_z$ must be the eigenstates of this operator. Measurements on silver atoms show that the magnitude of the eigenvalue is $\frac{\hbar}{2}$, where $\hbar = \frac{h}{2\pi}$ is called "h-bar" and h is Planck's constant. The magnitude of $\hbar$ is determined by how large the splitting of the two beams is for a given rate of change of the strength of the magnetic field. Since the atoms deflect up or down by the same amount, the eigenvalues must be precisely $\pm \frac{\hbar}{2}$.

Whenever you see Planck's constant (or $\hbar$), you know we are dealing with a quantum phenomenon. As we discussed earlier, the Stern-Gerlach experiment is a quantum experiment, so the appearance of $\hbar$ should not be surprising. Nevertheless, we need to discuss this object a bit more. It has the dimensions of something that we call an *action*, which is something most people have never heard of. The dimensions of an action are the same dimensions as position times momentum or of angular momentum, which is mass $\times$ (length)^2/time. We will discuss the numerical value of $\hbar$ as we need it. For now, just think of it as a small number with dimensions of an action (or an angular momentum), and now you know what those dimensions mean. Note that many people are more comfortable thinking of the dimensions of $\hbar$ as being the dimensions of angular momentum. If this is you, this is fine. In time, we will learn better just what the size of $\hbar$ really means.

OK. Now, we come back to our operator $\hat{S}_z$. We have argued that $\hat{S}_z$ transforms the up and down states along the z-axis via a scaling with no rotation (since they are eigenvectors). Hence, we have

$$\hat{S}_z|\uparrow\rangle_z = \frac{\hbar}{2}|\uparrow\rangle_z \text{ and } \hat{S}_z|\downarrow\rangle_z = -\frac{\hbar}{2}|\downarrow\rangle_z. \tag{6.2}$$

Note that we only know what this operator does to states oriented along the z-axis. At this point, we do not know what $\hat{S}_z|\uparrow\rangle_x$ is (we will soon determine it though). Because the two eigenvalues are real and we know how the operator transforms the two basis vectors (up and down spin along the z-axis), we can conclude that the operator is Hermitian, and it is an observable. Of course, we set it up to be an observable from the start, so we had to obtain this result.

We have determined one operator, by determining how it transformed two states. Can we think of others? It would be nice to find an operator that can convert a $|\uparrow\rangle_z$ state to a $|\downarrow\rangle_z$ and *vice versa*. So, let us define one! We define $\hat{S}_+$, the spin-raising operator, and $\hat{S}_-$, the spin-lowering operator, to satisfy

$$\hat{S}_+|\downarrow\rangle_z = \hbar|\uparrow\rangle_z \text{ and } \hat{S}_+|\uparrow\rangle_z = 0 \tag{6.3}$$

$$\hat{S}_-|\uparrow\rangle_z = \hbar|\downarrow\rangle_z \text{ and } \hat{S}_-|\downarrow\rangle_z = 0. \tag{6.4}$$

You might ask why did we choose $\hbar$ and not $2\hbar$ or $-\hbar$? At this stage, we could choose whatever we want, but it will turn out that the choice made here leads to special properties of these operators that are quite important, so we pick this special value and press on; it is not a coincidence that the difference of the two eigenvalues of $\hat{S}_z$ is also equal to $\hbar$. Note further that with the two operators that we introduced, we put in a condition that we cannot raise the spin of an up projection or lower the spin of a down projection. We set it up so that trying to raise an up spin destroys the state, similar for the lowering of a down spin. You can view this as $|\uparrow\rangle_z$ is an eigenstate of $\hat{S}_+$ with eigenvalue 0 and the same for $|\downarrow\rangle_z$ and $\hat{S}_-$. But note that $|\uparrow\rangle_z$ is not an eigenstate of $\hat{S}_-$, nor is $|\downarrow\rangle_z$ of $\hat{S}_+$. This tells us these operators are not Hermitian, because a Hermitian operator would have two eigenvectors and two real eigenvalues.

The facts that $\hat{S}_+|\uparrow\rangle_z = 0$ and $\hat{S}_-|\downarrow\rangle = 0$ are not just conveniences. They are required for us to have a consistent description of the quantum mechanics of spin. If you find this description too abstract for you, you can always ground these results by recognizing that these operators are going to be represented in terms of the 2×2 matrices we have been working with, and this is the main subject of the next section. We will find that neither $\hat{S}_+$ nor $\hat{S}_-$ are Hermitian, but $\hat{S}_+^\dagger = \hat{S}_-$ and *vice versa*. Because the state space is two dimensional (because the Stern-Gerlach analyzer always has two spots), the operator is completely determined by how it transforms the two basis vectors of the two-dimensional space. In addition, we will find in our future work that finding states annihilated by operators is one of the most

important results that we use to perform quantum calculations and here we have our first example of this.

One other object that naturally arises in quantum mechanics is the commutator $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$ of two operators. At first, working with commutators may feel quite foreign, but as we develop the materials for describing quantum mechanics, you will find that they are a natural object to work with and they are related to what happens when one performs one measurement after another. In any case, we have to start sometime, so let us dive in! We have found three operators so far: $\hat{S}_z$, which is the projection of spin on the z-axis; $\hat{S}_+$, which is the spin-raising operator; and $\hat{S}_-$, which is the spin-lowering operator. So, we want to determine the three commutators:

$$[\hat{S}_z, \hat{S}_+] = \hat{S}_z\hat{S}_+ - \hat{S}_+\hat{S}_z \tag{6.5}$$

$$[\hat{S}_z, \hat{S}_-] = \hat{S}_z\hat{S}_- - \hat{S}_-\hat{S}_z \tag{6.6}$$

$$[\hat{S}_+, \hat{S}_-] = \hat{S}_+\hat{S}_- - \hat{S}_-\hat{S}_+. \tag{6.7}$$

How can we do this? Remember, we can only use what we know. All we know is how these operators act on the spin states in the z-direction. So, we need to have the commutator, as an abstract operator, act on the states $|\uparrow\rangle_z$ and $|\downarrow\rangle_z$, which form the basis for our two-dimensional quantum system. The commutator requires us to apply the product of two operators onto a state. How do we do this? We first apply the operator on the right and then apply the operator on the left. Since an operator maps a ket to a ket, we can apply chains of operators onto a ket, and the final result will always be a ket. Hence, $\hat{S}_z\hat{S}_+$ means apply $\hat{S}_+$ first and then apply $\hat{S}_z$ second. If we represent the operators as matrices, then we can express the product of two operators as the product of the two corresponding matrices. Recall that the quantum spin space is two dimensional (even though the spin can point anywhere in three-dimensional space) because the output of a Stern-Gerlach experiment always gave two results. This means all quantum states can be described in terms of the $|\uparrow\rangle_z$ and $|\downarrow\rangle_z$ states—we will see precisely how very soon.

Don't worry if calculating a commutator seems unfamiliar to you. It will pay great dividends. We start with the commutator $[\hat{S}_z, \hat{S}_+]$. We have to act it on the two basis kets. First on $|\uparrow\rangle_z$:

$$\begin{aligned}[\hat{S}_z, \hat{S}_+]|\uparrow\rangle_z &= \hat{S}_z\hat{S}_+|\uparrow\rangle_z - \hat{S}_+\hat{S}_z|\uparrow\rangle_z \\ &= 0 - \hat{S}_+\frac{\hbar}{2}|\uparrow\rangle_z \\ &= 0. \end{aligned} \tag{6.8}$$

We applied our rules where $\hat{S}_+|\uparrow\rangle_z = 0$ and that numbers can always be interchanged with operators, because numbers commute with everything. Similarly

$$[\hat{S}_z, \hat{S}_+]|\downarrow\rangle_z = \hat{S}_z\hat{S}_+|\downarrow\rangle_z - \hat{S}_+\hat{S}_z|\downarrow\rangle_z$$
$$= \hat{S}_z\hbar|\uparrow\rangle_z - \hat{S}_+\left(-\frac{\hbar}{2}\right)|\downarrow\rangle_z$$
$$= \frac{\hbar^2}{2}|\uparrow\rangle_z + \frac{\hbar^2}{2}|\uparrow\rangle_z = \hbar^2|\uparrow\rangle_z. \tag{6.9}$$

These two results say that $[\hat{S}_z, \hat{S}_+]$ annihilates $|\uparrow\rangle_z$ and transforms $|\downarrow\rangle_z$ to a number times $|\uparrow\rangle_z$. We already know an operator that does this—it is the raising operator. So, we can summarize these results via

$$[\hat{S}_z, \hat{S}_+] = \hbar\hat{S}_+. \tag{6.10}$$

Note that this result follows because $\hat{S}_+|\uparrow\rangle_z = 0$ and $\hat{S}_+|\downarrow\rangle_z = \hbar|\uparrow\rangle_z$ (which was our definition of what the raising operator did to those two states). Because our spin states take on only two values, and because we have determined the action of the commutator on both of them, we have determined the operator relation $[\hat{S}_z, \hat{S}_+] = \hbar\hat{S}_+$. A similar calculation, which is a problem for you, shows that $[\hat{S}_z, \hat{S}_-] = -\hbar\hat{S}_-$ (note the minus sign). Finally, we have our third commutator $[\hat{S}_+, \hat{S}_-] = \hat{S}_+\hat{S}_- - \hat{S}_-\hat{S}_+$. We find that

$$[\hat{S}_+, \hat{S}_-]|\uparrow\rangle_z = (\hat{S}_+\hat{S}_- - \hat{S}_-\hat{S}_+)|\uparrow\rangle_z = \hat{S}_+\hbar|\downarrow\rangle_z - 0 = \hbar^2|\uparrow\rangle_z \tag{6.11}$$

$$[\hat{S}_+, \hat{S}_-]|\downarrow\rangle_z = (\hat{S}_+\hat{S}_- - \hat{S}_-\hat{S}_+)|\downarrow\rangle_z = 0 - \hat{S}_-\hbar|\uparrow\rangle_z = -\hbar^2|\downarrow\rangle_z, \tag{6.12}$$

because $\hat{S}_-|\downarrow\rangle_z = \hat{S}_+|\uparrow\rangle_z = 0$. Note that the minus sign comes from the minus sign in the commutator. The $\hat{S}_z$ operator acts the same way, so we find $[\hat{S}_+, \hat{S}_-] = 2\hbar\hat{S}_z$. It turns out that these three commutators, summarized by

$$[\hat{S}_+, \hat{S}_-] = 2\hbar\hat{S}_z \text{ and } [\hat{S}_z, \hat{S}_\pm] = \pm\hbar\hat{S}_\pm, \tag{6.13}$$

are all we need to determine all of the properties of spin, but it is convenient to introduce some new operators and evaluate some additional commutators to make it easier to work with spin problems. This is what we now work on.

To start, we have an asymmetry in that we have a spin operator for the z-axis projection, but not for the x- or y-axes. This is inconvenient. So, we define

$$\hat{S}_x = \frac{1}{2}\left(\hat{S}_+ + \hat{S}_-\right) \text{ and } \hat{S}_y = \frac{1}{2i}\left(\hat{S}_+ - \hat{S}_-\right). \tag{6.14}$$

Inverting these relations gives us

$$\hat{S}_+ = \hat{S}_x + i\hat{S}_y \text{ and } \hat{S}_- = \hat{S}_x - i\hat{S}_y. \tag{6.15}$$

It seems like these relations came out of thin air. Let us see why they make sense by going back to the Stern-Gerlach experiment again. Recall that if we measured on the z-axis and took atoms from the positive exit and then fed them into an analyzer along the x-axis, we found half the time they were positive and half the time they were negative. The operator $\frac{1}{2}(\hat{S}_+ + \hat{S}_-)$ certainly seems like it should be consistent with such a picture, and we will see it precisely is later in the section when we write the states $|\uparrow\rangle_x$ and $|\downarrow\rangle_x$ in terms of $|\uparrow\rangle_z$ and $|\downarrow\rangle_z$ as our Stern-Gerlach intuition tells us we must be able to do. Note that the choice we made earlier—that $\hat{S}_+|\downarrow\rangle_z = \hbar|\uparrow\rangle_z$—was made in order to have the formulas we use here be simple. The other motivation for this is that we want the spin operator in the x-, y-, and z-directions to all appear similar, with no direction preferred in free space. We chose to label them how we wanted to, with no specific direction determined by anything; hence, there is no preferred direction. So, let us look at the commutators:

$$[\hat{S}_x, \hat{S}_y] = \Big[\underbrace{\frac{1}{2}\left(\hat{S}_+ + \hat{S}_-\right)}_{\hat{S}_x}, \underbrace{\frac{1}{2i}\left(\hat{S}_+ - \hat{S}_-\right)}_{\hat{S}_y}\Big]$$

$$= \frac{1}{4i}\left([\hat{S}_+, \hat{S}_+] - [\hat{S}_+, \hat{S}_-] + [\hat{S}_-, \hat{S}_+] - [\hat{S}_-, \hat{S}_-]\right). \tag{6.16}$$

Note that $[\hat{S}_+, \hat{S}_+] = \hat{S}_+\hat{S}_+ - \hat{S}_+\hat{S}_+ = 0 = [\hat{S}_-, \hat{S}_-]$ and $[\hat{S}_+, \hat{S}_-] = 2\hbar\hat{S}_z = -[\hat{S}_-, \hat{S}_+]$. So, we immediately find that

$$[\hat{S}_x, \hat{S}_y] = -\frac{1}{2i}[\hat{S}_+, \hat{S}_-] = i\hbar\hat{S}_z. \tag{6.17}$$

This then tells us that $[\hat{S}_y, \hat{S}_x] = -i\hbar\hat{S}_z$, since $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A} = -[\hat{B}, \hat{A}]$. Next, we examine

$$[\hat{S}_x, \hat{S}_z] = \frac{1}{2}[\hat{S}_+ + \hat{S}_-, \hat{S}_z] = \frac{1}{2}\left(-\hbar\hat{S}_+ + \hbar\hat{S}_-\right) = -i\hbar\frac{1}{2i}\left(\hat{S}_x - \hat{S}_-\right) = -i\hbar\hat{S}_y, \tag{6.18}$$

so that $[\hat{S}_z, \hat{S}_x] = i\hbar\hat{S}_y$. You may see that a pattern is emerging here. Let us confirm this by computing our last commutator:

$$[\hat{S}_y, \hat{S}_z] = \frac{1}{2i}[\hat{S}_+ - \hat{S}_-, \hat{S}_z] = \frac{\hbar}{2i}(-\hat{S}_+ - \hat{S}_-) = i\hbar\hat{S}_x, \tag{6.19}$$

and $[\hat{S}_z, \hat{S}_y] = -i\hbar\hat{S}_x$. These results are summarized in Table 6.1.

If we stare at the Table 6.1 for some time, we see there is a definite pattern, each expression in the top row is ordered in a ring as in the left panel of Fig. 6.1 and the bottom row is ordered in a ring according to the right panel. The way we use the table is, if the commutator is in the top row, then the result has a plus sign, and if the commutator is in the bottom row, then the result has a minus sign. The

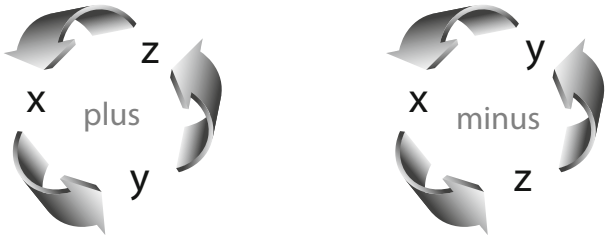

Fig. 6.1 Mnemonic for remembering the signs for the commutation relations. The figure on the left shows the commutators that have a plus sign and the one on the right shows those that have a minus sign. For example, $[\hat{S}_x, \hat{S}_y] = +i\hbar\hat{S}_z$ follows from using the left panel which has a plus sign for the commutator

figure illustrates the same thing. In it, we look for the arrow that flows from the first operator to the second. If we find the arrow in the "plus" part of the illustration, then the result has a plus sign, and if we find it in the "minus" part, then the result has a minus sign. Also note that the subscripts x, y, and z each appear just once in each formula. So, for example, if the subscripts of the commutator terms are x and z, then the subscript of the result will be y. And the table or the figure tells you the sign of the result. We can summarize these results with one compact formula:

$$[\hat{S}_i, \hat{S}_j] = i\hbar \sum_k \epsilon_{ijk}\hat{S}_k. \tag{6.20}$$

Note that the i that multiplies $\hbar$ is the imaginary i, not an index of x, y, or z. The symbol ϵ_{ijk} is called the *antisymmetric tensor* (or the Levi-Cevita symbol), which satisfies

$$\epsilon_{xyz} = \epsilon_{yzx} = \epsilon_{zxy} = 1 \tag{6.21}$$

$$\epsilon_{xzy} = \epsilon_{zyx} = \epsilon_{yxz} = -1, \tag{6.22}$$

and $\epsilon_{ijk} = 0$ if any two indices are the same. This means that if i and j are fixed, then there is only one nonzero index k for which $\epsilon_{ijk} \neq 0$ in the summation. Note how this formula treats all Cartesian spin indices equally. This is what we must have, since no direction is preferred. Note further, that because we can measure spin in any direction, $\hat{S}_x$, $\hat{S}_y$, and $\hat{S}_z$ must all be Hermitian. Using $\hat{S}_\pm = \hat{S}_x \pm i\hat{S}_y$ then shows immediately that $\hat{S}_\pm^\dagger = \hat{S}_\mp$. If you are familiar with the so-called right-hand rule, you can also use it to summarize these rules. Try it!

Table 6.1 Commutation relations of the spin operators in Cartesian coordinates

$[\hat{S}_x, \hat{S}_y] = i\hbar\hat{S}_z$	$[\hat{S}_y, \hat{S}_z] = i\hbar\hat{S}_x$	$[\hat{S}_z, \hat{S}_x] = i\hbar\hat{S}_y$
$[\hat{S}_y, \hat{S}_x] = -i\hbar\hat{S}_z$	$[\hat{S}_z, \hat{S}_y] = -i\hbar\hat{S}_x$	$[\hat{S}_x, \hat{S}_z] = -i\hbar\hat{S}_y$

This derivation and summary was a lot of work. Let us take a moment to digest what it means before we jump into our last commutator calculation. What these relations say is that the different Cartesian coordinates of spin do not commute (i.e., $[\hat{S}_i, \hat{S}_j] \neq 0$). When two operators do not commute, they cannot be simultaneously diagonalized, which means that we cannot find a set of eigenvectors that simultaneously works for any two pairs of Cartesian spin operators. We have already seen this in the Stern-Gerlach experiment, as there is no state with a positive projection on *both* the x- and z-axes. This fact is not immediately obvious, but the derivation is simple, so let us derive it next.

Suppose we have two operators $\hat{A}$ and $\hat{B}$ and there exists a state $|\psi\rangle$ that is a simultaneous eigenvector of $\hat{A}$ *and* $\hat{B}$. This means both

$$\hat{A}|\psi\rangle = a|\psi\rangle \text{ and } \hat{B}|\psi\rangle = b|\psi\rangle, \tag{6.23}$$

with a and b numbers that are the eigenvalues (if $\hat{A}$ and $\hat{B}$ are Hermitian operators, then both a and b are real). To show that $\hat{A}$ and $\hat{B}$ must commute (meaning $[\hat{A}, \hat{B}] = 0$), we simply compute

$$\hat{A}\hat{B}|\psi\rangle = \hat{A}b|\psi\rangle = b\hat{A}|\psi\rangle = ba|\psi\rangle \tag{6.24}$$

$$\hat{B}\hat{A}|\psi\rangle = \hat{B}a|\psi\rangle = a\hat{B}|\psi\rangle = ab|\psi\rangle. \tag{6.25}$$

But $ab = ba$, since they are numbers. Hence

$$[\hat{A}, \hat{B}]|\psi\rangle = (\hat{A}\hat{B} - \hat{B}\hat{A})|\psi\rangle = 0. \tag{6.26}$$

We have shown this for only one state $|\psi\rangle$, but we saw when we discussed eigenvalues and eigenvectors that the set of all eigenvectors of any Hermitian operator forms a complete set, meaning any vector in the space can be written as a linear combination of the eigenvectors. So, the argument we just gave for $|\psi\rangle$ will hold for all eigenvectors and so will hold for every linear combination of them. This means it holds for all vectors. This is the same as saying that $[\hat{A}, \hat{B}] = 0$ must hold as an operator equation for the simultaneous eigenstates to exist. This is a general result in quantum mechanics—two operators can be simultaneously diagonalized (meaning we can find one set of eigenvectors for both operators) only if the two operators commute with each other (note the terms eigenvector and eigenstate are synonymous and will be freely used).

Back to our spin problem, we found that we cannot simultaneously diagonalize any two components of spin (since $[\hat{S}_i, \hat{S}_j] \neq 0$ for $i \neq j$). However, there are two operators that mutually commute, and we now show how to find a second commuting operator. It turns out that $\hat{S}^2 = \hat{S}_x^2 + \hat{S}_y^2 + \hat{S}_z^2$ commutes with $\hat{S}_x$, with $\hat{S}_y$, and with $\hat{S}_z$. In other words, we want to show that

$$[\hat{S}^2, \hat{S}_j] = 0, \tag{6.27}$$

for $j = x,\ y,$ and z. We need one more identity about commutators before we can proceed. We need to determine how to evaluate the commutator with a product of operators:

$$[\hat{A}\hat{B}, \hat{C}] = \hat{A}\hat{B}\hat{C} - \hat{C}\hat{A}\hat{B}. \tag{6.28}$$

We use our "add-zero" trick and rearrange

$$\begin{aligned}[\hat{A}\hat{B}, \hat{C}] &= \hat{A}\hat{B}\hat{C} \underbrace{-\hat{A}\hat{C}\hat{B} + \hat{A}\hat{C}\hat{B}}_{\text{add zero}} - \hat{C}\hat{A}\hat{B} \\ &= \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B}. \end{aligned} \tag{6.29}$$

The easy way to remember this is you "pull" the operator out from the product to the right or to the left depending on which side of the product it is on. For example, we can think of the above identity as

$$[\underset{\leftarrow}{\hat{A}}\ \underset{\rightarrow}{\hat{B}}, \hat{C}] = \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B}. \tag{6.30}$$

The identity holds for the product on the other side as well

$$[\hat{A}, \underset{\leftarrow}{\hat{B}}\ \underset{\rightarrow}{\hat{C}}] = \hat{B}[\hat{A}, \hat{C}] + [\hat{A}, \hat{B}]\hat{C}, \tag{6.31}$$

which you should verify by adding zero as we did above (the arrows show you a mnemonic for how to remember the rule). These two identifies are called the *Leibniz rule* or the *product rule*. It is the first of four fundamental operator identities we will be using throughout the book. These four identities empower you do to many complex calculations.

We are now ready to compute the commutator:

$$\begin{aligned}[\hat{S}^2, \hat{S}_j] &= \sum_i [\hat{S}_i^2, \hat{S}_j] = \sum_i \left(\hat{S}_i [\hat{S}_i, \hat{S}_j] + [\hat{S}_i, \hat{S}_j]\hat{S}_i \right) \\ &= i\hbar \sum_i \hat{S}_i \sum_k \epsilon_{ijk} \hat{S}_k + i\hbar \sum_i \sum_k \epsilon_{ijk} \hat{S}_k \hat{S}_i \\ &= i\hbar \sum_i \sum_k \epsilon_{ijk} \left(\hat{S}_i \hat{S}_k + \hat{S}_k \hat{S}_i \right). \end{aligned} \tag{6.32}$$

Let's get concrete and look at the terms. Suppose $j = x$. Then, we can have $i = y$ and $k = z$ ($\epsilon_{yxz} = -1$) or $i = z$ and $k = y$ ($\epsilon_{zxy} = 1$). This means

$$[\hat{S}^2, \hat{S}_x] = i\hbar \underbrace{(-\hat{S}_y \hat{S}_z - \hat{S}_x \hat{S}_y)}_{i=y \text{ and } k=z} + \underbrace{(\hat{S}_z \hat{S}_y + \hat{S}_y \hat{S}_z)}_{i=z \text{ and } k=y} = 0. \tag{6.33}$$

Looking at $j = y$ and $j = z$ gives the same result as well. So, we learn that $[\hat{S}^2, \hat{S}_j] = 0$. This implies that we can simultaneously find eigenstates of $\hat{S}^2$ and either $\hat{S}_x$, or $\hat{S}_y$, or $\hat{S}_z$. It is customary to take the last case and find simultaneous eigenstates of $\hat{S}^2$ and $\hat{S}_z$. We will find the basis $|\uparrow\rangle_z$ and $|\downarrow\rangle_z$ is the simultaneous eigenbasis for both.

We already know that $|\uparrow\rangle_z$ has an eigenvalue of $\frac{\hbar}{2}$ with respect to $\hat{S}_z$ and $|\downarrow\rangle_z$ has an eigenvalue of $-\frac{\hbar}{2}$ with respect to $\hat{S}_z$. Since $\hat{S}^2$ and $\hat{S}_z$ commute, these states should also be eigenstates with respect to $\hat{S}^2$. We need to compute the corresponding eigenvalue. To do this, we first write $\hat{S}^2$ in terms of the raising and lowering operators, because we know what they do when they act on the spin states in the z-direction. Using the definitions of the Cartesian components, we find that

$$\begin{aligned}\hat{S}^2 &= \hat{S}_x^2 + \hat{S}_y^2 + \hat{S}_z^2 = \frac{1}{4}(\hat{S}_+ + \hat{S}_-)^2 - \frac{1}{4}(\hat{S}_+ - \hat{S}_-)^2 + \hat{S}_z^2 \\ &= \frac{1}{4}(\hat{S}_+^2 + \hat{S}_+\hat{S}_- + \hat{S}_-\hat{S}_+ + \hat{S}_-^2 - \hat{S}_+^2 + \hat{S}_+\hat{S}_- + \hat{S}_-\hat{S}_+ - \hat{S}_-^2) + \hat{S}_z^2 \\ &= \frac{1}{2}(\hat{S}_+\hat{S}_- + \hat{S}_-\hat{S}_+) + \hat{S}_z^2, \end{aligned} \tag{6.34}$$

where we have to be careful to preserve the order of products of operators that do not commute. Now, using the action of these operators on the states, we find that

$$\hat{S}^2|\uparrow\rangle_z = \left[\frac{1}{2}(\hat{S}_+\hat{S}_- + \hat{S}_-\hat{S}_+) + \hat{S}_z^2\right]|\uparrow\rangle_z = \left(\frac{1}{2}\hbar^2 + 0 + \frac{1}{4}\hbar^2\right)|\uparrow\rangle_z = \frac{3}{4}\hbar^2|\uparrow\rangle_z, \tag{6.35}$$

and the same eigenvalue for $|\downarrow\rangle_z$. Note that this result could have been anticipated. We already know the z-component of spin contributed $\frac{\hbar^2}{4}$ to the eigenvalue—we should expect the x and y contributions to be the same, if there is no preferred direction (and there is not).

We have one more set of problems to examine before we are finished with this section. We have already shown all of the spin commutation relations. The one thing remaining for us to do is to find out how to express the states $|\uparrow\rangle_x$, $|\downarrow\rangle_x$, $|\uparrow\rangle_y$, and $|\downarrow\rangle_y$ in terms of the basis states $|\uparrow\rangle_z$ and $|\downarrow\rangle_z$. To do this, we need to solve the eigenvalue-eigenvector relations. Since the z-axis states are a basis, we can write $|\uparrow\rangle_x = \alpha|\uparrow\rangle_z + \beta|\downarrow\rangle_z$, with $|\alpha|^2 + |\beta|^2 = 1$. Then, we need to solve

$$\begin{aligned}\hat{S}_x|\uparrow\rangle_x &= \frac{1}{2}(\hat{S}_+ + \hat{S}_-)|\uparrow\rangle_x = \frac{1}{2}(\hat{S}_+ + \hat{S}_-)\big(\alpha|\uparrow\rangle_z + \beta|\downarrow\rangle_z\big) \\ &= \frac{\hbar}{2}(\alpha|\downarrow\rangle_z + \beta|\uparrow\rangle_z) = \frac{\hbar}{2}|\uparrow\rangle_x = \frac{\hbar}{2}\big(\alpha|\uparrow\rangle_z + \beta|\downarrow\rangle_z\big). \end{aligned} \tag{6.36}$$

The last equality is the enforcement of the eigenvalue-eigenvector relationship. One immediately sees that we must have $\alpha = \beta = \frac{1}{\sqrt{2}}$ (using our "polite rules" for the overall complex phase of the eigenvector). Recall our trick for the eigenvectors of

a two-dimensional system, if one is $(\alpha \;\; \beta)$, then the other is $(-\beta^* \;\; \alpha^*)$. Using this trick, we find the two eigenstates are

$$|\uparrow\rangle_x = \frac{1}{\sqrt{2}}|\uparrow\rangle_z + \frac{1}{\sqrt{2}}|\downarrow\rangle_z \text{ and } |\downarrow\rangle_x = \frac{1}{\sqrt{2}}|\uparrow\rangle_z - \frac{1}{\sqrt{2}}|\downarrow\rangle_z \tag{6.37}$$

after changing the sign on the second eigenvector to be "polite." One can almost see how we got half positive projection on the x-axis and half negative projection on the x-axis in the Stern-Gerlach experiment (when we sent the atoms from one exit of a z-oriented analyzer into a x-oriented analyzer), but we will work out those details a bit later. The eigenvectors on the y-axis will be a problem at the end of the section.

In summary, we have shown that the two forms of the commutation relations for spin (either $[\hat{S}_z, \hat{S}_\pm] = \pm\hbar\hat{S}_\pm$ and $[\hat{S}_+, \hat{S}_-] = 2\hbar\hat{S}_z$, or, equivalently, $[\hat{S}_i, \hat{S}_j] = i\hbar\sum_k \epsilon_{ijk}\hat{S}_k$, with $\hat{S}_\pm = \hat{S}_x \pm i\hat{S}_y$) can be employed to show that $[\hat{S}^2, \hat{S}_j] = 0$. This means we can find simultaneous eigenstates of $\hat{S}^2$ and $\hat{S}_z$. We will soon see that these commutation relations can be used to learn much more about quantum systems with spin. As we end the section, we leave with a question—how should you think about the states $|\uparrow\rangle$ and $|\downarrow\rangle$? These are the states of definite projection of spin along the axis of a Stern-Gerlach analyzer. In quantum mechanics, we will often seek out quantum states that have a definite property. We will find when we want to compute the results of experiments, we combine these states with others in an inner product (or bra-ket) or in a matrix element. Sometimes, we will prefer to represent these abstract states as concrete vectors. But, it is always best to think of these states as simply the quantum states corresponding to exiting from the $+$ or $-$ exit of a Stern-Gerlach analyzer. In other words, the abstract bra or ket represents a *real physical result* via the notion of a quantum state associated with that result. Hopefully, this will help keep the abstraction at a level that you will be comfortable with.

Section 6.1 Problems:

6.1.1 Lowering Operator Commutator

Show that

$$[\hat{S}_z, \hat{S}_-] = -\hbar\hat{S}_-. \tag{6.38}$$

6.1.2 Commutator of the Total Square of the Spin Operator

Show that

$$[\hat{S}^2, \hat{S}_y] = 0 \text{ and } [\hat{S}^2, \hat{S}_z] = 0, \tag{6.39}$$

where $\hat{S}^2 = \hat{S}_x^2 + \hat{S}_y^2 + \hat{S}_z^2$.

6.1.3 Eigenstates in the y-Direction

Verify that

$$|\uparrow\rangle_y = \frac{1}{\sqrt{2}}(|\uparrow\rangle_z + i|\downarrow\rangle_z) \text{ and } |\downarrow\rangle_y = \frac{1}{\sqrt{2}}(|\uparrow\rangle_z - i|\downarrow\rangle_z), \tag{6.40}$$

when expressed in "polite" form.

6.1.4 A Useful Relation

Show that

$$(\hat{S}_x^2 + \hat{S}_y^2)|\uparrow\rangle_z = \frac{\hbar^2}{2}|\uparrow\rangle_z \text{ and } (\hat{S}_x^2 + \hat{S}_y^2)|\downarrow\rangle_z = \frac{\hbar^2}{2}|\downarrow\rangle_z. \tag{6.41}$$

Hint: write $\hat{S}_x^2 + \hat{S}_y^2 = \hat{S}^2 - \hat{S}_z^2$ and use the eigenvalue-eigenvector relations that we already know.

6.1.5 Eigenvalue of $\hat{S}^2$

Show that $\hat{S}^2|\uparrow\rangle_x = \frac{3}{4}\hbar^2|\uparrow\rangle_x$, $\hat{S}^2|\downarrow\rangle_x = \frac{3}{4}\hbar^2|\downarrow\rangle_x$, $\hat{S}^2|\uparrow\rangle_y = \frac{3}{4}\hbar^2|\uparrow\rangle_y$, and $\hat{S}^2|\downarrow\rangle_y = \frac{3}{4}\hbar^2|\downarrow\rangle_y$.
Hint: note that you do not need to use any operator manipulations if you use the facts that
$\hat{S}^2|\uparrow\rangle_z = \frac{3}{4}\hbar^2|\uparrow\rangle_z$ and $\hat{S}^2|\downarrow\rangle_z = \frac{3}{4}\hbar^2|\downarrow\rangle_z$ and the expansions of the x- and y-eigenstates in terms of the z-eigenstates.

6.2 Pauli Spin Matrices and Spinors

We mentioned earlier that one can interchangeably work with operators and brakets, or one can work with matrices and vectors. We illustrate this now, developing

the theory for matrix representations of spin and of spinors. We start by organizing a two-dimensional vector that we call a spinor via

$$|\psi\rangle \leftrightarrow \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \tag{6.42}$$

which uses a two-dimensional vector to represent the quantum state $|\psi\rangle = \alpha|\uparrow\rangle_z + \beta|\downarrow\rangle_z$, with $|\alpha|^2 + |\beta|^2 = 1$. The spinor is a vector that we will be working with. Its upper component is the amount of the $|\uparrow\rangle_z$ in the state $|\psi\rangle$, and its lower component is the amount of state $|\downarrow\rangle_z$ in $|\psi\rangle$. You can also think of this as a formal way of working just with the coefficients of the different basis vectors for the states of the spin system.

Our next step is to work out what the matrix representations of the spin operators are. As we discussed in Sect. 5.8, when we showed how to determine the matrix that represents the operator in a particular basis. The way we did this was to pick all bra-ket pairs of the basis states and organize them in a "table" corresponding to the matrix elements of the matrix that represents the operator (a sandwich of bra-operator-ket). For example, for the operator $\hat{S}_z$ and the basis $\{|\uparrow\rangle_z, |\downarrow\rangle_z\}$, we have

$$\hat{S}_z \leftrightarrow \begin{pmatrix} {}_z\langle\uparrow|\hat{S}_z|\uparrow\rangle_z & {}_z\langle\uparrow|\hat{S}_z|\downarrow\rangle_z \\ {}_z\langle\downarrow|\hat{S}_z|\uparrow\rangle_z & {}_z\langle\downarrow|\hat{S}_z|\downarrow\rangle_z \end{pmatrix}. \tag{6.43}$$

This means we need to compute those four bra-operator-ket "sandwiches." Recalling that $\hat{S}_z|\uparrow\rangle_z = \frac{\hbar}{2}|\uparrow\rangle_z$ and $\hat{S}_z|\downarrow\rangle_z = -\frac{\hbar}{2}|\downarrow\rangle_z$ and the orthonormal properties of the basis: ${}_z\langle\uparrow|\uparrow\rangle_z = {}_z\langle\downarrow|\downarrow\rangle_z = 1$ and ${}_z\langle\downarrow|\uparrow\rangle_z = {}_z\langle\uparrow|\downarrow\rangle_z = 0$ allows us to complete the calculation quickly, yielding

$$\hat{S}_z \leftrightarrow \begin{pmatrix} \frac{\hbar}{2} & 0 \\ 0 & -\frac{\hbar}{2} \end{pmatrix} \tag{6.44}$$

as the representation for $\hat{S}_z$.

We work similarly for $\hat{S}_\pm$:

$$\hat{S}_z \leftrightarrow \begin{pmatrix} {}_z\langle\uparrow|\hat{S}_+|\uparrow\rangle_z & {}_z\langle\uparrow|\hat{S}_+|\downarrow\rangle_z \\ {}_z\langle\downarrow|\hat{S}_+|\uparrow\rangle_z & {}_z\langle\downarrow|\hat{S}_+|\downarrow\rangle_z \end{pmatrix} = \begin{pmatrix} 0 & \hbar \\ 0 & 0 \end{pmatrix}, \tag{6.45}$$

and for $\hat{S}_-$, we have

$$\hat{S}_z \leftrightarrow \begin{pmatrix} {}_z\langle\uparrow|\hat{S}_-|\uparrow\rangle_z & {}_z\langle\uparrow|\hat{S}_-|\downarrow\rangle_z \\ {}_z\langle\downarrow|\hat{S}_-|\uparrow\rangle_z & {}_z\langle\downarrow|\hat{S}_-|\downarrow\rangle_z \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ \hbar & 0 \end{pmatrix}. \tag{6.46}$$

The other spin operators can be constructed from these matrices, because $\hat{S}_x = \frac{1}{2}(\hat{S}_x + \hat{S}_-)$ and $\hat{S}_y = \frac{1}{2i}(\hat{S}_+ - \hat{S}_-)$, so we find that

$$\hat{S}_x \leftrightarrow \begin{pmatrix} 0 & \frac{\hbar}{2} \\ \frac{\hbar}{2} & 0 \end{pmatrix} \quad \text{and} \quad \hat{S}_y \leftrightarrow \begin{pmatrix} 0 & -\frac{i\hbar}{2} \\ \frac{i\hbar}{2} & 0 \end{pmatrix}. \tag{6.47}$$

Note that all $\hat{S}_j$, with $j = x$, y, and z, are represented by Hermitian matrices, but $\hat{S}_\pm$ are not Hermitian. However, we do have the matrices that represent the raising and lowering operators satisfy $(\hat{S}_+)^\dagger = \hat{S}_-$ and *vice versa*. You should explicitly verify all of these claims directly from the matrix forms.

Now, simple matrix multiplication shows that

$$\hat{S}_x^2 \leftrightarrow \begin{pmatrix} \frac{\hbar^2}{4} & 0 \\ 0 & \frac{\hbar^2}{4} \end{pmatrix}, \quad \hat{S}_y^2 \leftrightarrow \begin{pmatrix} \frac{\hbar^2}{4} & 0 \\ 0 & \frac{\hbar^2}{4} \end{pmatrix}, \quad \text{and} \quad \hat{S}_z^2 \leftrightarrow \begin{pmatrix} \frac{\hbar^2}{4} & 0 \\ 0 & \frac{\hbar^2}{4} \end{pmatrix}. \tag{6.48}$$

From this, we immediately find that

$$\hat{S}^2 \leftrightarrow \begin{pmatrix} \frac{3\hbar^2}{4} & 0 \\ 0 & \frac{3\hbar^2}{4} \end{pmatrix}; \tag{6.49}$$

that is, it is $\frac{3}{4}\hbar^2$ times the identity matrix $\mathbb{I}_2$.

Armed with these matrices, we want to see how the eigenvalue-eigenvector relations work (recall that this is a matrix times vector equals a number times the vector). The z case was already worked out, because the $\hat{S}_z$ operator is represented by a diagonal matrix. The x and y relations are a little more complicated. Taking the relations we worked out in Sect. 6.1 in terms of operators and states, and rewriting them in terms of matrices and spinors, we have

$$\hat{S}_x|\uparrow\rangle_x \leftrightarrow \begin{pmatrix} 0 & \frac{\hbar}{2} \\ \frac{\hbar}{2} & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \tag{6.50}$$

$$\hat{S}_x|\downarrow\rangle_x \leftrightarrow \begin{pmatrix} 0 & \frac{\hbar}{2} \\ \frac{\hbar}{2} & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix} = -\frac{\hbar}{2} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix} \tag{6.51}$$

and

$$\hat{S}_y|\uparrow\rangle_y \leftrightarrow \begin{pmatrix} 0 & -\frac{i\hbar}{2} \\ \frac{i\hbar}{2} & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} \end{pmatrix} \tag{6.52}$$

$$\hat{S}_y|\downarrow\rangle_y \leftrightarrow \begin{pmatrix} 0 & -\frac{i\hbar}{2} \\ \frac{i\hbar}{2} & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{i}{\sqrt{2}} \end{pmatrix} = -\frac{\hbar}{2} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{i}{\sqrt{2}} \end{pmatrix}. \tag{6.53}$$

Note that the set of eigenvalues for the x, y, and z directions are identical (they are all equal to $\frac{\hbar}{2}$). This is to be expected from the Stern-Gerlach experiment discussion,

where we see the same separation of the two lines, regardless of the direction of the analyzer.

There is one more spin operator identity we need to work out. We start with the commutator $[\hat{S}_x, \hat{S}_y] = i\hbar\hat{S}_z$ and multiply on the left by $\hat{S}_x$ for one equation and on the right for a second equation:

$$\hat{S}_x^2\hat{S}_y - \hat{S}_x\hat{S}_y\hat{S}_x = i\hbar\hat{S}_x\hat{S}_z \tag{6.54}$$

$$\hat{S}_x\hat{S}_y\hat{S}_x - \hat{S}_y\hat{S}_x^2 = i\hbar\hat{S}_z\hat{S}_x. \tag{6.55}$$

Now, add the two equations to find that

$$\hat{S}_x^2\hat{S}_y - \hat{S}_y\hat{S}_x^2 = i\hbar(\hat{S}_x\hat{S}_z + \hat{S}_z\hat{S}_x). \tag{6.56}$$

But recall that the representation of the square of a spin operator is proportional to the identity matrix—$\hat{S}_x^2 \leftrightarrow \frac{\hbar^2}{4}\mathbb{I}_2$ and the identity matrix commutes with every matrix—this implies that the left-hand side of the equation is equal to zero. This then gives us that

$$\hat{S}_x\hat{S}_z + \hat{S}_z\hat{S}_x = \{\hat{S}_x, \hat{S}_z\} = 0, \tag{6.57}$$

where the curly braces signifies a new object called the *anticommutator*

$$\{\hat{A}, \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A}. \tag{6.58}$$

One can immediately generalize this calculation to the spin operators in the other directions, which finally shows that

$$\{\hat{S}_i, \hat{S}_j\} = 0, \tag{6.59}$$

for all $i \neq j$, while $\{\hat{S}_i, \hat{S}_i\} = 2\hat{S}_i^2 = \frac{\hbar^2}{2}\mathbb{I}_2$, so we finally have

$$\{\hat{S}_i, \hat{S}_j\} = \frac{\hbar^2}{2}\delta_{ij}\mathbb{I}_2, \tag{6.60}$$

where δ_{ij} is the Kronnecker delta function, which satisfies $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$. Be sure to verify all cases for yourself, to be sure you understand these relations. You can work directly with the matrices, multiplying the matrices that represent each operator, if you want to.

Wolfgang Pauli (see Fig. 6.2) introduced three matrices, which are called the Pauli spin matrices. They are given by the matrices that represent the spin operators along each Cartesian direction, divided by $\frac{\hbar}{2}$. They are defined by

Fig. 6.2 Image of Wolfgang Pauli, who first created the matrices that represent spin and are called the Pauli spin matrices. He won the Nobel Prize in Physics in 1945. [Image from the Swedish Royal Academy (public domain) and modified to look like a sketch by James Freericks]

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \ \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \ \text{and} \ \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{6.61}$$

They satisfy $\hat{S}_i \leftrightarrow \frac{\hbar}{2}\sigma_i$. We immediately find that

$$[\sigma_i, \sigma_j] = 2i \sum_k \epsilon_{ijk}\sigma_k \tag{6.62}$$

$$\{\sigma_i, \sigma_j\} = 2\delta_{ij}\mathbb{I}_2. \tag{6.63}$$

In the next section, we will discuss a number of important relations for the Pauli spin matrices. The video The Matrix Representation of Spin https://tinyurl.com/2yk2bkn9 helps explain these ideas further.

We have been working with the matrix and spinor relations for spin. The question we address now is how do we represent a spin state that is up in some arbitrary direction given by the two spherical angles θ and ϕ, as shown in Fig. 6.3. Recall that we represent a point on the unit sphere by the angle θ, called the polar angle, that it makes relative to the $+z$-axis ($0 \leq \theta \leq \pi$), and the angle ϕ, called the azimuthal angle, that the projection of the vector makes in the x-y-plane, relative to the x-axis ($0 \leq \phi \leq 2\pi$). A simple calculation shows that $\vec{e}_n$, the unit vector in the θ-ϕ direction, is given by

$$\vec{e}_n = \sin\theta\cos\phi\,\vec{e}_x + \sin\theta\sin\phi\,\vec{e}_y + \cos\theta\,\vec{e}_z, \tag{6.64}$$

where $\vec{e}_i$ is the unit vector in the $i = x$, y, and z direction. We say a spin is in an up state along $\vec{e}_n$ if

$$\vec{e}_n \cdot \hat{\vec{S}}|\uparrow\rangle_{\theta\text{-}\phi} = (\sin\theta\cos\phi\,\hat{S}_x + \sin\theta\sin\phi\,\hat{S}_y + \cos\theta\,\hat{S}_z)|\uparrow\rangle_{\theta\text{-}\phi} = \frac{\hbar}{2}|\uparrow\rangle_{\theta\text{-}\phi}, \tag{6.65}$$

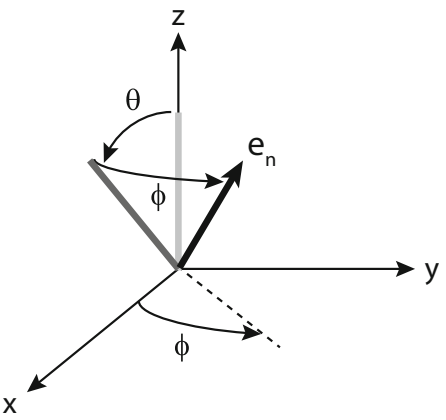

Fig. 6.3 Direction $\vec{e}_n$ of the magnetic field of the Stern-Gerlach analyzer. It is oriented in the θ-ϕ direction. Starting from the z-axis, the angle θ is rotated about the y-axis, followed by a rotation by ϕ about the z-axis, to end in the θ-ϕ direction, denoted here by the unit vector $\vec{e}_n$

which says that when the projection of the spin along the $\vec{e}_n$ axis acts on the eigenstate, it produces $\frac{\hbar}{2}$ times that state. Note that if $\vec{e}_n = (0, 0, 1)$, corresponding to $\theta = 0$, then we have the usual definition of $|\uparrow\rangle_z$, and so on. We find the solution by working with the matrix representation. We have

$$\vec{e}_n \cdot \hat{\vec{S}} = \sin\theta\cos\phi\,\hat{S}_x + \sin\theta\sin\phi\,\hat{S}_y + \cos\theta\,\hat{S}_z \leftrightarrow \frac{\hbar}{2}\begin{pmatrix} \cos\theta & \sin\theta\, e^{-i\phi} \\ \sin\theta\, e^{i\phi} & -\cos\theta \end{pmatrix}, \tag{6.66}$$

which you will verify in the problems below.

Note that this matrix is Hermitian, so it has real eigenvalues, as it must. We represent the state with a spinor via $|\uparrow\rangle_{\theta\text{-}\phi} \leftrightarrow \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$. Then, the eigenvalue satisfies the two relations:

$$\frac{\hbar}{2}\left(\cos\theta\,\alpha + \sin\theta\, e^{-i\phi}\,\beta\right) = \frac{\hbar}{2}\alpha, \tag{6.67}$$

$$\frac{\hbar}{2}\left(\sin\theta\, e^{-i\phi}\,\alpha - \cos\theta\,\beta\right) = \frac{\hbar}{2}\beta. \tag{6.68}$$

They are both solved by

$$\alpha = \frac{\sin\theta}{1-\cos\theta}e^{-i\phi}, \qquad \beta = \sqrt{\frac{1+\cos\theta}{1-\cos\theta}}\,e^{-i\phi}\,\beta, \tag{6.69}$$

and $|\alpha|^2 + |\beta|^2 = 1$. The solution then becomes

$$|\uparrow\rangle_{\theta\text{-}\phi} \leftrightarrow \frac{1}{\sqrt{2}}\begin{pmatrix} \sqrt{1+\cos\theta} \\ \sqrt{1-\cos\theta}\, e^{i\phi} \end{pmatrix} = \begin{pmatrix} \cos\frac{\theta}{2} \\ \sin\frac{\theta}{2}\, e^{i\phi} \end{pmatrix}, \tag{6.70}$$

which you will verify in the problems below (you may want to review Sect. 5.2 again and note that $|\beta| \neq \beta$ because of the $e^{i\phi}$ term). Note that $\sqrt{1-\cos\theta}$ is always real. As a check, let us verify that the general result agrees with the results we have already found along the different Cartesian indices. Setting $\theta = 0$ gives us $|\uparrow\rangle_z \leftrightarrow \begin{pmatrix}1\\0\end{pmatrix}$; setting $\theta = \frac{\pi}{2}$ and $\phi = 0$ gives $|\uparrow\rangle_x \leftrightarrow \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}$; and $\theta = \frac{\pi}{2}$ and $\phi = \frac{\pi}{2}$ gives $|\uparrow\rangle_y \leftrightarrow \frac{1}{\sqrt{2}}\begin{pmatrix}1\\i\end{pmatrix}$. This is a nice check that everything is working correctly. The solution for the down-spin state is worked out in Problem 6.2.5.

One thing you might be bothered by is how can $|\uparrow\rangle_z$ and $|\downarrow\rangle_z$ be orthogonal to each other, meaning ${}_z\langle\uparrow|\downarrow\rangle_z = 0$? After all, one state has a positive projection on the z-axis and the other has a negative projection on the z-axis—how can they be orthogonal? The answer is, of course, that the up-spin and down-spin states are eigenfunctions of $\hat{S}_z$ with different eigenvalues and represented as spinors via $|\uparrow\rangle_z \leftrightarrow \begin{pmatrix}1\\0\end{pmatrix}$ and $|\downarrow\rangle_z \leftrightarrow \begin{pmatrix}0\\1\end{pmatrix}$, so they are clearly orthogonal. Indeed, eigenvectors with different eigenvalues *must always be orthogonal*, and this is easy to see from the spinor representation. The confusion arises from the fact that one should not interpret the quantum state $|\uparrow\rangle_z$ as being proportional to a unit vector in the z-direction. That is in a three-dimensional space—the quantum states are represented by spinors that live in an abstract two-dimensional space and *are not represented by vectors in the three-dimensional world*. Trying to represent them in this way will always lead to confusion. Do not do so. Only compare like objects to like objects—spinors to spinors, quantum states to quantum states—do not compare the labels of the bras and kets with the state vectors. We never perform calculations with the labels.

We have accomplished a lot. In addition to determining how to work with spin in terms of spinors and matrices, we also now know how to find a spin-up (or spin-down) state in any direction in three dimensions. This is pretty advanced stuff already. I often find graduate students in physics struggle with calculating the spin eigenstates in an arbitrary direction. But you won't!

Section 6.2 Problems:

6.2.1 Anticommutators

Show that

$$\{\hat{S}_x, \hat{S}_y\} = 0 \text{ and } \{\hat{S}_y, \hat{S}_z\} = 0. \tag{6.71}$$

6.2.2 Product of Pauli Matrices

Using matrix multiplication for 2×2 matrices, show the Pauli spin matrix product formula given by

$$\sigma_i \sigma_j = \delta_{ij} \mathbb{I}_2 + i \sum_k \epsilon_{ijk} \sigma_k \tag{6.72}$$

for all i and j.

6.2.3 Spin Operator in an Arbitrary Direction

Verify the 2×2 matrix representation in Eq. (6.66) for $\vec{e}_n \cdot \hat{\vec{S}}$, with $\vec{e}_n = \sin\theta \cos\phi \, \vec{e}_x + \sin\theta \sin\phi \, \vec{e}_y + \cos\theta \, \vec{e}_z$ and $\hat{\vec{S}} = \frac{\hbar}{2} \left(\sigma_x \, \vec{e}_x + \sigma_y \, \vec{e}_y + \sigma_z \, \vec{e}_z \right)$.

6.2.4 Double-Angle Formula

Show that

$$\sqrt{1 - \cos\theta} = \sqrt{2} \sin\frac{\theta}{2} \quad \text{and} \quad \sqrt{1 + \cos\theta} = \sqrt{2} \cos\frac{\theta}{2}, \tag{6.73}$$

both follow from the double-angle formula $\cos\theta = \cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}$, where we restrict to $0 \leq \theta \leq \pi$, so all square roots are of nonnegative numbers.

6.2.5 Down-Spin State in an Arbitrary Direction

Modify the calculation completed in the text that found the up-spin state pointing in an arbitrary direction, to find the down-spin state pointing in an arbitrary direction. In other words, find the state $| \downarrow \rangle_{\theta\text{-}\phi}$, such that

$$\vec{e}_n \cdot \hat{\vec{S}} | \downarrow \rangle_{\theta\text{-}\phi} = -\frac{\hbar}{2} | \downarrow \rangle_{\theta\text{-}\phi}, \tag{6.74}$$

by finding the spinor representation for $| \downarrow \rangle_{\theta\text{-}\phi}$; use our "polite rule" to ensure the first nonzero component of the spinor is positive and real. Once you have found your vector, show that
${}_{\theta\text{-}\phi}\langle \uparrow | \downarrow \rangle_{\theta\text{-}\phi} = 0$ by taking the inner product of the two spinors (don't forget to complex conjugate the representation for the bra when taking the inner product).

6.2.6 Working with Spin States in Arbitrary Directions

This problem uses the results of the previous problem. Using the formulas for $|\uparrow\rangle_{\theta\text{-}\phi}$ and $|\downarrow\rangle_{\theta\text{-}\phi}$, pick appropriate θ and ϕ values to find the following quantum states:

(a) $|\downarrow\rangle_z$, $|\downarrow\rangle_x$, and $|\downarrow\rangle_y$
(b) $|\uparrow\rangle_{-z}$, $|\uparrow\rangle_{-x}$, and $|\uparrow\rangle_{-y}$
(c) $|\downarrow\rangle_{-z}$, $|\downarrow\rangle_{-x}$, and $|\downarrow\rangle_{-y}$

Hint: $-x$ is $\theta = \frac{\pi}{2}$ and $\phi = \pi$; $-y$ is $\theta = \frac{\pi}{2}$ and $\phi = \frac{3\pi}{2}$, and so on.

6.2.7 Inversion of Spin States

This problem uses the results of the last two problems. Note that $\vec{e}_n = \sin\theta\cos\phi\,\vec{e}_x + \sin\theta\sin\phi\,\vec{e}_y + \cos\theta\,\vec{e}_z$, so that $\vec{e}_n \rightarrow -\vec{e}_n$ requires us to take $\theta \rightarrow \pi - \theta$ and $\phi \rightarrow \pi + \phi$. Since we know that

$$|\uparrow\rangle_{\vec{e}_n} = |\uparrow\rangle_{\theta\text{-}\phi}, \tag{6.75}$$

then we must have that

$$|\downarrow\rangle_{-\vec{e}_n} = |\uparrow\rangle_{\pi-\theta\text{-}\pi+\phi}. \tag{6.76}$$

Using the results from Problem 6.2.5, verify this identity making the transformation on the angles for the result from Problem 6.2.5 and comparing it to the expression for the up spin at θ and ϕ. Be sure to be polite in your answer!

6.2.8 Operator Derivation of the Spin Eigenstate

We do not need to form the matrix to find the spin eigenstates. Start from the operators, and how the operators act on spin states, to find the spin eigenstates at an arbitrary angle. Act the operator $\vec{e}_n \cdot \hat{\vec{S}}$ onto the arbitrary state $|\psi\rangle = \alpha|\uparrow\rangle_z + \beta|\downarrow\rangle_z$, and use the relations for the raising, lowering, and z-component of spin operators acting on each term in this state and the fact that they must equal $\frac{\hbar}{2}|\psi\rangle$ to derive Eqs. (6.67) and (6.68) without matrices. The same can be done for the down-spin projection eigenstate on an arbitrary axis.

6.3 Pauli Spin-Matrix Identities

Some may find this section a bit intimidating with many matrix formulas and symbols. If this is you, just relax and move slowly through the material. It is all self-contained and you can make it through with complete understanding! Note that nothing presented here is well motivated, it is more of the form, here is an identity, and let's calculate it. We will find the motivation for and the usefulness of these identities soon. For now, think of it more as just exploring properties of Pauli matrices and gaining practice in manipulating abstract expressions. So, we continue our discussion of the mathematics of 2×2 matrices (or equivalently spin operators) and derive a few high-level identities before we jump into discussing quantum mechanics again in the next section.

Recall that the identity and the three Pauli spin matrices are

$$\mathbb{I}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \text{and} \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{6.77}$$

They satisfy the relations:

$$\{\sigma_i, \sigma_j\} = 2\delta_{ij}\mathbb{I}_2, \quad [\sigma_i, \sigma_j] = 2i \sum_k \epsilon_{ijk}\sigma_k, \tag{6.78}$$

and $\sigma_i\sigma_j = \delta_{ij}\mathbb{I}_2 + i\sum_k \epsilon_{ijk}\sigma_k$. We want to examine some of these identities concretely via matrix multiplication. For example, we have

$$\sigma_x\sigma_y = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \tag{6.79}$$

$$\sigma_y\sigma_x = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}. \tag{6.80}$$

So, $\sigma_x\sigma_y + \sigma_y\sigma_x = 0$ and $\sigma_x\sigma_y - \sigma_y\sigma_x = \begin{pmatrix} 2i & 0 \\ 0 & -2i \end{pmatrix} = 2i\sigma_z$. You can work these out for other explicit cases for practice, and I strongly recommend that you do so, to verify all of these identities.

Another interesting identity is the Pauli triple product:

$$\sigma_x\sigma_y\sigma_z = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix} = i\mathbb{I}_2, \tag{6.81}$$

which you should also verify by multiplying out the matrices. It turns out this is a generic identity, since

$$\sigma_i\sigma_j\sigma_k = i\epsilon_{ijk}\mathbb{I}_2, \tag{6.82}$$

holds when i, j, and k are all different. The proof is simple. Using the identity for the product of two Pauli matrices, we multiply on the right by σ_k. If all are different, we find that $(\sigma_i\sigma_j)\sigma_k = (i\epsilon_{ijk}\sigma_k)\sigma_k = i\epsilon_{ijk}\mathbb{I}_2$, because $\sigma_k^2 = \mathbb{I}_2$ for all k. Note, when at least two of i, j, and k are the same, the result is different and can be easily worked out.

It turns out that we can write any 2×2 matrix in terms of the identity matrix and the three Pauli spin matrices. After all, a 2×2 matrix has four degrees of freedom (a, b, c, d), and we have four linearly independent matrices $\{\mathbb{I}_2, \sigma_x, \sigma_y.\sigma_z\}$, so this should work. So, we set

$$\alpha\mathbb{I}_2 + \beta\sigma_x + \gamma\sigma_y + \delta\sigma_z = \begin{pmatrix} \alpha+\delta & \beta - i\gamma \\ \beta + i\gamma & \alpha - \delta \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = A. \tag{6.83}$$

Solving gives

$$\alpha + \delta = a, \quad \beta - i\gamma = n, \quad \beta + i\gamma = c, \quad \text{and} \quad \alpha - \delta = d. \tag{6.84}$$

So, we have

$$\alpha = \frac{a+d}{2}, \quad \beta = \frac{b+c}{2}, \quad \gamma = i\frac{b-c}{2}, \quad \text{and} \quad \delta = \frac{a-d}{2}. \tag{6.85}$$

We often write this in a compact form as

$$A = \alpha\mathbb{I}_2 + \vec{v}\cdot\vec{\sigma}, \tag{6.86}$$

where $\alpha = \frac{a+d}{2}$ and $\vec{v} = \left(\frac{b+c}{2}, i\frac{b-c}{2}, \frac{a-d}{2}\right)$. Here, we form a "vector of matrices," given by

$$\vec{\sigma} = \sigma_x\vec{e}_x + \sigma_y\vec{e}_y + \sigma_z\vec{e}_z = (\sigma_x, \sigma_y, \sigma_z). \tag{6.87}$$

Don't be intimidated or confused by the vector of matrices, given by $\vec{\sigma}$. You will see how to work with it and your comfort level will rise. Note finally that α and $\vec{v}$ are real for Hermitian 2×2 matrices (but the identity works for general matrices).

We now use this machinery to show how one can evaluate different expressions with Pauli spin matrices. Our first goal is to evaluate $(\vec{a}\cdot\vec{\sigma})(\vec{b}\cdot\vec{\sigma})$ with $\vec{a} = (a_x, a_y, a_z)$ and $\vec{b} = (b_x, b_y, b_z)$ both real vectors. We will use the identity for the product of two Pauli spin matrices to do this. First, we write each inner product as $\vec{a}\cdot\vec{\sigma} = a_x\sigma_x + a_y\sigma_y + a_z\sigma_z = \sum_i a_i\sigma_i$ and $\vec{b}\cdot\vec{\sigma} = \sum_j b_j\sigma_j$. Then, the product of the two becomes

$$(\vec{a}\cdot\vec{\sigma})(\vec{b}\cdot\vec{\sigma}) = \sum_i\sum_j a_i b_j \sigma_i\sigma_j$$

$$= \sum_i \sum_j a_i b_j \left(\delta_{ij} \mathbb{I}_2 + i \sum_k \epsilon_{ijk} \sigma_k \right)$$
$$= \sum_i a_i b_i \mathbb{I}_2 + i \sum_{ijk} \epsilon_{ijk} a_i b_j \sigma_k$$
$$= \vec{a} \cdot \vec{b}\, \mathbb{I}_2 + i \vec{a} \cdot (\vec{b} \times \vec{\sigma}) = \vec{a} \cdot \vec{b} \mathbb{I}_2 + i \vec{\sigma} \cdot (\vec{a} \times \vec{b}). \tag{6.88}$$

Here, we used the representation of the scalar triple product in terms of ϵ_{ijk} and the fact that it is unchanged under an even permutation of the "vectors." Note that the δ_{ij} disappeared in the third line, along with the sum over j, because it forces $i = j$, so we need not include a separate sum over j anymore. If $\vec{a} = \vec{b}$, then we have $(\vec{a} \cdot \vec{\sigma})^2 = |\vec{a}|^2 \mathbb{I}_2$, since the scalar triple product $\vec{a} \cdot (\vec{b} \times \vec{\sigma}) = \vec{\sigma} \cdot (\vec{a} \times \vec{b})$ vanishes when $\vec{b}$ is parallel to $\vec{a}$ (because of the antisymmetric tensor). This identity looks complex, but it just involves 2×2 matrices on each side.

Our next identity may look quite foreboding—it is the "matrix exponential" defined by $e^{i\vec{v} \cdot \vec{\sigma}}$—where we work with a real vector $\vec{v}$. What does this even mean? The argument of an exponential is supposed to be a number, not a matrix. How can we calculate it? The answer is simple—we use the power series expansion, because we know how to multiply matrices together and how to add them. So, the matrix exponential means

$$e^{i\vec{v} \cdot \vec{\sigma}} = \sum_{m=0}^{\infty} \frac{(i)^m}{m!} (\vec{v} \cdot \vec{\sigma})^m \,. \tag{6.89}$$

When we looked at the product of two inner products of Pauli matrices, we found that $(\vec{v} \cdot \vec{\sigma})^2 = |\vec{v}|^2 \mathbb{I}_2$, for a real vector $\vec{v}$. Hence, $(\vec{v} \cdot \vec{\sigma})^{2m} = |\vec{v}|^{2m} \mathbb{I}_2$ and $(\vec{v} \cdot \vec{\sigma})^{2m+1} = |\vec{v}|^{2m} \vec{v} \cdot \vec{\sigma}$. In other words, even powers are proportional to $\mathbb{I}_2$ and odd powers are proportional to $\vec{v} \cdot \vec{\sigma}$. This yields

$$e^{i\vec{v} \cdot \vec{\sigma}} = \underbrace{\sum_{n=0}^{\infty} \frac{(i)^{2n}}{(2n)!} |\vec{v}|^{2n} \mathbb{I}_2}_{\text{even}} + \underbrace{\sum_{n=0}^{\infty} \frac{(i)^{2n+1}}{(2n+1)!} |\vec{v}|^{2n} \vec{v} \cdot \vec{\sigma}}_{\text{odd}}$$
$$= \underbrace{\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} |\vec{v}|^{2n}}_{\cos |\vec{v}|} \mathbb{I}_2 + i \frac{\vec{v} \cdot \vec{\sigma}}{|\vec{v}|} \underbrace{\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} |\vec{v}|^{2n+1}}_{\sin |\vec{v}|} \,. \tag{6.90}$$

Hence, we have

$$e^{i\vec{v} \cdot \vec{\sigma}} = \cos |\vec{v}|\, \mathbb{I}_2 + i \frac{\vec{v} \cdot \vec{\sigma}}{|\vec{v}|} \sin |\vec{v}|, \tag{6.91}$$

which is a generalization of the Euler formula for Pauli spin matrices (compare it to $e^{i\theta} = \cos\theta + i\sin\theta$). You will find this identity is invaluable for solving many problems we work on. Note that we can use this identity to concretely show that $e^{i\vec{v}\cdot\vec{\sigma}}e^{-i\vec{v}\cdot\vec{\sigma}} = \mathbb{I}_2$. Just multiply them together:

$$\begin{aligned} e^{i\vec{v}\cdot\vec{\sigma}}e^{-i\vec{v}\cdot\vec{\sigma}} &= \left(\cos|\vec{v}|\mathbb{I}_2 + i\frac{\vec{v}\cdot\vec{\sigma}}{|\vec{v}|}\sin|\vec{v}|\right)\left(\cos|\vec{v}|\mathbb{I}_2 - i\frac{\vec{v}\cdot\vec{\sigma}}{|\vec{v}|}\sin|\vec{v}|\right) \\ &= \cos^2|\vec{v}|\mathbb{I}_2 + i\frac{\vec{v}\cdot\vec{\sigma}}{|\vec{v}|}\cos|\vec{v}|\sin|\vec{v}| - i\frac{\vec{v}\cdot\vec{\sigma}}{|\vec{v}|}\cos|\vec{v}|\sin|\vec{v}| \\ &\quad + \frac{(\vec{v}\cdot\vec{\sigma})^2}{|\vec{v}|^2}\sin^2|\vec{v}| \\ &= \cos^2|\vec{v}|\mathbb{I}_2 + \frac{|\vec{v}|^2\mathbb{I}_2}{|\vec{v}|^2}\sin^2|\vec{v}| = \mathbb{I}_2. \end{aligned} \tag{6.92}$$

So, we explicitly have shown that $e^{-i\vec{v}\cdot\vec{\sigma}}$ is the inverse of $e^{i\vec{v}\cdot\vec{\sigma}}$!

Our next identity involves what is called a similarity transformation. This is when we sandwich a matrix (or operator) between another matrix and its inverse. We will later learn that the quantity we will calculate can be worked out via a formula called the Hadamard lemma, which is another one of our fundamental identities. But, for now, we will use the exponential of a Pauli matrix formula to work out this result. We calculate (for $\vec{v}$ a real vector)

$$e^{i\vec{v}\cdot\vec{\sigma}}\sigma_j e^{-i\vec{v}\cdot\vec{\sigma}}, \tag{6.93}$$

which has the Pauli spin matrix multiplied by a matrix from the left and the inverse of that matrix from the right, so it is a similarity transformation. Now, we substitute again the matrix exponential formula, to find

$$\begin{aligned} e^{i\vec{v}\cdot\vec{\sigma}}\sigma_j e^{-i\vec{v}\cdot\vec{\sigma}} &= \left(\cos|\vec{v}|\mathbb{I}_2 + i\frac{\vec{v}\cdot\vec{\sigma}}{|\vec{v}|}\sin|\vec{v}|\right)\sigma_j\left(\cos|\vec{v}|\mathbb{I}_2 - i\frac{\vec{v}\cdot\vec{\sigma}}{|\vec{v}|}\sin|\vec{v}|\right) \\ &= \cos^2|\vec{v}|\,\sigma_j + i\sum_i\frac{v_i}{|\vec{v}|}\cos|\vec{v}|\sin|\vec{v}|\,[\sigma_i,\sigma_j] \\ &\quad + \sum_i\sum_k\frac{v_i v_k}{|\vec{v}|^2}\sin^2|\vec{v}|\,\sigma_i\sigma_j\sigma_k \\ &= \cos^2|\vec{v}|\,\sigma_j - \sum_i\sum_k\frac{v_i}{|\vec{v}|}\sin 2|\vec{v}|\,\epsilon_{ijk}\,\sigma_k \\ &\quad + \sum_i\sum_k\frac{v_i v_k}{|\vec{v}|^2}\sin^2|\vec{v}|\sigma_i\sigma_j\sigma_k. \end{aligned} \tag{6.94}$$

The triple product of Pauli spin matrices can be further simplified, but it is cumbersome to take into account all possible cases. Nevertheless, we will find this result useful when we pick some special cases for $\vec{v}$.

We end this section with a short discussion of a Lie group. Lie group elements are of the form of an exponential of sums of numbers multiplied by matrices, which are called the Lie group generators. The group properties are determined by the commutators of the generators, which is called the Lie algebra. Here, the Pauli matrices are generators of the Lie group called SU(2) for special unitary group of 2×2 matrices. It involves unitary matrices and special means their determinant is one. A group has a few properties that it must satisfy. It has a multiplication operation, and the product of two group elements is always another group element. The multiplication operation must have an inverse as well. Groups satisfy the associative law of multiplication, but they need not satisfy the commutative law, and SU(2) does not satisfy the commutative law for multiplication. How do we check we have a group? We already showed that elements of the form $e^{i\vec{v}\cdot\vec{\sigma}}$, with real $\vec{v}$ have an inverse. Now, if we look carefully at Eq. (6.91), we find

$$e^{i\vec{v}\cdot\vec{\sigma}} = \begin{pmatrix} \cos|\vec{v}| + i\frac{v_z}{|\vec{v}|}\sin|\vec{v}| & i\frac{v_x - iv_y}{|\vec{v}|}\sin|\vec{v}| \\ i\frac{v_x + iv_y}{|\vec{v}|}\sin|\vec{v}| & \cos|\vec{v}| - i\frac{v_z}{|\vec{v}|}\sin|\vec{v}| \end{pmatrix}. \tag{6.95}$$

The determinant of a matrix is equal to one, as required by a SU(2) group elements, and the matrix is unitary as well (which can be seen because the Hermitian conjugate is the matrix corresponds to $\vec{v} \to -\vec{v}$, which we know to be the inverse). Since the determinant of the product of two matrices is the product of the determinants, the determinant of the product of two group elements also has a determinant of one. The final result we need to check is if I multiply two group elements together, can I write the product in the exponential form with a real vector $\vec{v}$? This is also true but is not quite so obvious. We will not prove it here, but you will in the problems. What is nice though is that our explicit formula for the matrix exponential allows us to work concretely with this Lie group, and we will find we do so many times in the rest of the book. Groups, especially Lie groups, are quite important in physics and quantum mechanics. We won't be able to delve too deeply into group theory here, but with what we have already shown you, it is a concept that can be mastered. The video Pauli Spin Matrix Identities https://tinyurl.com/muxfhxuk summarizes the material discussed here.

Section 6.3 Problems:

6.3.1 Linear Independence

Determine the α and $\vec{v}$ that represent $\begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$ as $\alpha\mathbb{I}_2 + \vec{v}\cdot\vec{\sigma}$. Repeat for $\begin{pmatrix} 1 & 0 \\ 1 & 3 \end{pmatrix}$.

6.3.2 Product of Inner Products with Pauli Spin Matrix Vectors

Let $\vec{a} = (1, 0, 1)$ and $\vec{b} = (0, 1, 1)$. Compute $(\vec{a} \cdot \vec{\sigma})(\vec{b} \cdot \vec{\sigma})$ two different ways. First, find the matrix $\vec{a} \cdot \vec{\sigma}$ and then the matrix $\vec{b} \cdot \vec{\sigma}$, and then, multiply the two matrices together. Second, use our formula $(\vec{a} \cdot \vec{\sigma})(\vec{b} \cdot \vec{\sigma}) = \vec{a} \cdot \vec{b}\, I_2 + i\vec{a} \cdot (\vec{b} \times \vec{\sigma})$. Compare the two results and make sure they agree! Recall that the scalar triple product is $\vec{a} \cdot (\vec{b} \times \vec{c}) = \sum_i \sum_j \sum_k \epsilon_{ijk} a_i b_j c_k$.

6.3.3 Matrix Exponential

Let $\vec{v} = \frac{\pi}{2}(1, 0, 1)$. Compute $e^{i\vec{v}\cdot\vec{\sigma}}$ from the formula we developed in the text.

6.3.4 Products of Matrix Exponentials

Let $\vec{v} = \frac{\pi}{2}(1, 0, 1)$ and $\vec{v}' = \frac{\pi}{2}(0, 1, 1)$ and compute

$$e^{i\vec{v}\cdot\vec{\sigma}} e^{i\vec{v}'\cdot\vec{\sigma}}. \tag{6.96}$$

Also, compute

$$e^{i(\vec{v}+\vec{v}')\cdot\vec{\sigma}}. \tag{6.97}$$

You should find the two calculations disagree.

Under what circumstances for $\vec{v}$ and $\vec{v}'$ do we have that

$$e^{i\vec{v}\cdot\vec{\sigma}} e^{i\vec{v}'\cdot\vec{\sigma}} = e^{i(\vec{v}+\vec{v}')\cdot\vec{\sigma}}. \tag{6.98}$$

Note that, in general, matrix exponentials do not satisfy this rule because matrices, in general, do not commute.

6.3.5 Similarity Transformations

Let $\vec{v} = \frac{\pi}{2}(1, 0, 0)$ and let $j = x$, y, and z. Compute

$$e^{i\vec{v}\cdot\vec{\sigma}} \sigma_j e^{-i\vec{v}\cdot\vec{\sigma}}. \tag{6.99}$$

Can you explain why the answers are what they are?

6.3.6 Properties of Pauli Spin Matrices

Answer the following questions about the properties of Pauli matrices.

(a) A Hermitian matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ satisfies a and d are real and $c = b^*$. Are the Pauli matrices Hermitian?
(b) A unitary matrix satisfies $U^\dagger = U^{-1}$. Multiply $\sigma_i^\dagger \sigma_i$ and see if the result is $\mathbb{I}_2$. Are the Pauli matrices unitary?
(c) The determinant of a 2×2 matrix is $ad - bc$. Compute the determinant of each of the Pauli matrices.
(d) The trace of a 2×2 matrix is $a + d$. Compute the trace of the Pauli matrices.
(e) We learned earlier that the determinant is the product of the two eigenvalues $\det A = \lambda_+ \lambda_-$ and the trace is the sum of the two eigenvalues $\mathrm{Tr}\, A = \lambda_+ + \lambda_-$. Use these two facts and the values for the determinant and trace to find the eigenvalues of the Pauli matrices.

These properties are rather unique properties of the Pauli matrices.

6.3.7 SU(2) Rotations

Let $\vec{v}_z = \left(0, 0, -\frac{\theta}{2}\right)$. The transformation

$$\sigma_i' = e^{-i\frac{\theta}{2}\sigma_z} \sigma_i e^{i\frac{\theta}{2}\sigma_z}, \tag{6.100}$$

corresponds to a rotation by an angle θ about the z-axis.

(a) Verify that

$$\sigma_x' = \cos\theta \sigma_x + \sin\theta \sigma_y \tag{6.101}$$

$$\sigma_y' = -\sin\theta \sigma_x + \cos\theta \sigma_y \tag{6.102}$$

$$\sigma_z' = \sigma_z, \tag{6.103}$$

(b) Choose $\vec{v}_x = \left(-\frac{\theta}{2}, 0, 0\right)$ and $\vec{v}_y = \left(0, -\frac{\theta}{2}, 0\right)$ and verify that the transformations they generate correspond to the rotations about the x- and y-axis, respectively, by the angle θ.

Note that we use the angle $\frac{\theta}{2}$ in the exponential. This is a special case for the 2×2 matrices. These are not rotations of three-dimensional vectors.

6.3.8 Physical Significance of $(\vec{e}_n \cdot \vec{\sigma})^2 = \mathbb{I}_2$

Consider the eigenvalues $\{\lambda_+, \lambda_-\}$ and eigenvectors $\{\vec{v}_+, \vec{v}_-\}$ of the 2×2 Hermitian matrix A. Verify that the eigenvectors of the squared matrix A^2 are the same vectors and the eigenvalues are now λ_+^2 and λ_-^2.

Let $A = \vec{e}_n \cdot \vec{\sigma}$, with $\vec{e}_n$ the unit vector pointing in the θ-ϕ direction. It is a Hermitian matrix, so it satisfies the property shown in the first part of the problem. Compute $A^2 = (\vec{e}_n \cdot \vec{\sigma})^2$ and show it is equal to $\mathbb{I}_2$, so both of its eigenvalues are equal to 1.

Use this result to explain why the eigenvalues in the Stern-Gerlach experiment are $\pm \frac{\hbar}{2}$ for any direction θ-ϕ of the Stern-Gerlach analyzer.

6.3.9 The Special Case of Eigenvalues Equal to Zero or One

The matrix for $\sigma_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ is not Hermitian. Determine the vector that satisfies $\sigma_+ \vec{v} = 0$. When an operator annihilates a vector (the same as having a zero eigenvalue), we can use this fact to do powerful calculations. For example, we can multiply by one via $e^{\alpha\sigma_+}\vec{v} = \vec{v}$, which you should also verify.

The matrix $H = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$ has an eigenvector with eigenvalue 1 and an eigenvector with eigenvalue 0. Find both. For this matrix, we can do a multiply by one in two ways: when H acts on the eigenvector that has a unit eigenvalue, it is multiplied by one, and when e^H acts on the eigenvector with zero eigenvalue, it is multiplied by one.

In general, zero and one eigenvalues can lead to interesting things, especially the zero eigenvalues. You will discover this in great detail soon.

6.3.10 Factorization of the Matrix Exponential

In Problem 5.6.10, we showed how to factorize a unitary matrix written in a specific form. Here, we will show how to factorize a unitary matrix exponential. Consider the matrix $e^{-i\frac{\theta}{2}\sigma_y}$. Compute the matrix using our formula. Then, factorize it into the form

$$e^{-i\frac{\theta}{2}\sigma_y} = e^{\alpha\sigma_+}e^{\beta\sigma_z}e^{\gamma\sigma_-}. \tag{6.104}$$

Repeat for the form

$$e^{-i\frac{\theta}{2}\sigma_y} = e^{\alpha'\sigma_-}e^{\beta'\sigma_z}e^{\gamma'\sigma_+}. \tag{6.105}$$

It turns out you can work out these identities for any ordering of the three factors.

Hint: $e^{a\sigma_+} = \begin{pmatrix} 1 & 2a \\ 0 & 1 \end{pmatrix}$ and a similar identity for the lowering spin matrix.

6.3.11 Matrix Exponentials Do Form a Lie Group

In this problem, we show that we can write the product of two matrix exponentials as a third matrix exponential. We need to show that

$$e^{i\vec{v}\cdot\vec{\sigma}}e^{i\vec{v}'\cdot\vec{\sigma}} = e^{i\vec{v}''\cdot\vec{\sigma}} \tag{6.106}$$

for some vector $\vec{v}''$ that depends on the original vectors $\vec{v}$ and $\vec{v}'$.

Verify that the product on the left is given by

$$\left(\cos|\vec{v}|\cos|\vec{v}'| - \sin|\vec{v}|\sin|\vec{v}'|\frac{\vec{v}\cdot\vec{v}'}{|\vec{v}|\,|\vec{v}'|}\right)\mathbb{I}_2 + i\left(\cos|\vec{v}|\sin|\vec{v}'|\frac{\vec{v}'}{|\vec{v}'|} + \sin|\vec{v}|\cos|\vec{v}'|\frac{\vec{v}}{|\vec{v}|} - \sin|\vec{v}|\sin|\vec{v}'|\frac{\vec{v}\times\vec{v}'}{|\vec{v}|\,|\vec{v}'|}\right)\cdot\vec{\sigma}. \tag{6.107}$$

The term multiplying the identity matrix is a scalar, and it must be a cosine of an angle α. The term dotted into $\vec{\sigma}$ is a unit vector multiplied by sine of α. To verify such an angle exists, we need to verify the coefficients take this form. The simplest way to do this is to show that $\cos^2\alpha + \sin^2\alpha = 1$. This is done by squaring the coefficient of the first term and adding it to the norm squared of the vector on the second line and verifying the sum is equal to one.

Hint: recall that $\vec{v}\cdot\vec{v}' = |\vec{v}|\,|\vec{v}'|\cos\theta$, where θ is the angle between $\vec{v}$ and $\vec{v}'$. Recall as well that $|\vec{v}\times\vec{v}'| = |\vec{v}\,|\vec{v}'|\sin\theta$, with the θ same angle as in the dot product identity.

In principle, you can work out the algebra to determine $\vec{v}''$, but it is a cumbersome formula that will not be particularly enlightening. The formula for this vector is not unique, although the matrix exponential is. This is because $\cos(|\vec{v}''| + 2\pi) = \cos(|\vec{v}''|)$ and similarly for sine.

6.4 Stern-Gerlach Experiment Revisited

We go back to revisit our old friend the Stern-Gerlach experiment, but we are now armed with Dirac notation, spin operators, and their matrix and spinor representations. So, we are able to reexamine these systems with a more sophisticated approach. You may want to review Chaps. 1 and 2 to be sure you recall the Stern-Gerlach experiments before proceeding.

To begin, we must decide on our coordinate system—we choose the z-direction to be up, x to be toward us (out of the page), and y to the right. Once the axes are determined, we now imagine creating an atom with a random direction for its spin corresponding to an angle θ relative to the z-axis and an angle ϕ in the x-y plane. The quantum state of the atom is denoted $|\uparrow\rangle_{\theta\text{-}\phi}$ for the atom whose spin is pointing in the direction of $\vec{e}_n = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$, such that

$$\vec{e}_n \cdot \hat{\vec{S}}|\uparrow\rangle_{\theta\text{-}\phi} = \frac{\hbar}{2}\cos\frac{\phi}{2}|\uparrow\rangle_z + \frac{\hbar}{2}\sin\frac{\theta}{2}e^{i\phi}|\downarrow\rangle_z \leftrightarrow \frac{\hbar}{2}\begin{pmatrix}\cos\frac{\theta}{2}\\ \sin\frac{\theta}{2}e^{i\phi}\end{pmatrix}, \tag{6.108}$$

when expressed first as a superposition of up and down states in the z-direction and second as a spinor.

Let us pause for a moment to be sure we understand the spinor and superposition representations. To begin, how is it that a spin pointing in the θ-ϕ direction in three dimensions is represented by a two-dimensional spinor? Recall we have only two states in the Stern-Gerlach experiment—the one giving the $+\frac{\hbar}{2}$ projection and the one having the $-\frac{\hbar}{2}$ projection. *All spin states can be expressed as a superposition of these two states.* This is why the spin pointing in the θ-ϕ direction in three dimensions is represented as a two-dimensional spinor (for the up and down components)—or as a superposition of up and down states. Both representations yield the exact same quantum state. In addition, please do not confuse the eigenvalue of the spin operator, which has a magnitude proportional to $\hbar$, with probabilities we will soon calculate and are always dimensionless numbers between 0 and 1. The eigenvalue equation is often used to determine the state, while an inner product with a bra (followed by a modulus squared) is employed in the probability calculation, which we will detail soon. Having gone through this quick review, we are now ready to go back and analyze the Stern-Gerlach experiments with this new fancy quantum notation.

If we measure an atom in the spin state $|\uparrow\rangle_{\theta\text{-}\phi}$ with a Stern-Gerlach analyzer oriented in the $+z$ direction, then the probability to exit the $+$ exit is

$$P(+,z) = |{}_z\langle\uparrow|\uparrow\rangle_{\theta\text{-}\phi}|^2 = |{}_z\langle\uparrow|\cos\frac{\theta}{2}|\uparrow\rangle_z + {}_z\langle\uparrow|\sin\frac{\theta}{2}e^{i\phi}|\downarrow\rangle_z|^2 = \cos^2\frac{\theta}{2}, \tag{6.109}$$

while the probability to emerge from the $-$ exit is

$$P(-,z) = |{}_z\langle\downarrow|\uparrow\rangle_{\theta\text{-}\phi}|^2 = \sin^2\frac{\theta}{2}. \tag{6.110}$$

In other words, we take the inner product of the eigenstate corresponding to the exit being measured (bra; ${}_z\langle\uparrow|$) with the state the system is prepared in (ket; $|\uparrow\rangle_{\theta\text{-}\phi}$) and then take the modulus squared of the inner product to find the probability. We use the same rule to analyze all of our Stern-Gerlach experiments—this is the same rule we used in Chap. 2 (although it might not have looked like it at the time).

Now, we return to the three basic measurements shown in Figs. 1.19, 1.20, and 1.21: (i) measure on the $+z$ axis and measure on $+z$ again; (ii) measure on the $+z$ axis and then measure on the $-z$ axis; and measure on the $+z$ axis and then on the $+x$ axis. Let us analyze each in turn. For the first, we have the atom in the state $|\uparrow\rangle_z$ when emerging from the + exit of the first analyzer. The probability to subsequently emerge from the + exit of the second analyzer is given by the modulus squared of the inner product of the ket state that the atom is currently in $|\uparrow\rangle_z$ and the bra state corresponding to the exit of the analyzer, or $P(+,z) = |_z\langle\uparrow|\uparrow\rangle_z|^2 = 1$ and $P(-,z) = |_z\langle\downarrow|\uparrow\rangle_z|^2 = 0$. Hence, all atoms emerge from the + exit. For the second experiment, we must note that $|\uparrow\rangle_z = |\downarrow\rangle_{-z}$ and $|\downarrow\rangle_z = |\uparrow\rangle_{-z}$, as we saw when we first encountered the Stern-Gerlach experiment. Note that in this experiment, we take the atoms emerging from the − exit of the first analyzer. Then, the probability to emerge from the + exit of the second analyzer is given by $|_{-z}\langle\uparrow|\downarrow\rangle_z|^2$, which is the same as $|_z\langle\downarrow|\downarrow\rangle_z|^2 = 1$, and from the − exit is $|_{-z}\langle\downarrow|\downarrow\rangle_z|^2$, which is the same as $|_z\langle\uparrow|\downarrow\rangle_z|^2 = 0$. Finally, for our last experiment, we need to evaluate the probability to emerge from the + exit of the x analyzer is $|_x\langle\uparrow|\uparrow\rangle_z|^2$. Since we already determined that $|\uparrow\rangle_x = \frac{1}{\sqrt{2}}(|\uparrow\rangle_z + |\downarrow\rangle_z)$, we immediately find the probability to emerge from the + exit is

$$|_x\langle\uparrow|\uparrow\rangle_z|^2 = \left|\left(\frac{1}{\sqrt{2}}\,{}_z\langle\uparrow| + \frac{1}{\sqrt{2}}\,{}_z\langle\downarrow|\right)|\uparrow\rangle_z\right|^2 = \left|\frac{1}{\sqrt{2}}\cdot 1 + \frac{1}{\sqrt{2}}\cdot 0\right|^2 = \frac{1}{2}. \tag{6.111}$$

As you can see, by using the different spin states and calculating the relevant inner products and squaring them, we determine all of the probabilities for any given experiment. Let's use this approach to analyze the three-analyzer experiment with $+z$, $+x$, and $+z$ analyzers in a row (see Fig. 1.22). In this example, we have a more complex analysis, because we have multiple experiments to account for. This requires us to carefully account for how the states transform, using the ket-bra products of the projection operators, as described next. The way we analyze these experiments is to treat the passage through each analyzer as a measurement at the time the atom exits each analyzer. This is a shortcut analysis, as opposed to the more correct way to describe these experiments by having the passage through each device create an entangled (or tagged) state, after exiting each analyzer, and evaluating the probability only at the end. This shortcut can always be used in cases where nothing is untagged. Otherwise, you may analyze the experiment improperly. This shortcut is quite helpful and simplifies the analysis, so it is worthwhile to use it when we can (that is when there is no untagging in the procedure). Furthermore, if we block one exit of a Stern-Gerlach analyzer (that is not part of the rest of the apparatus), it really is a measurement at that given analyzer, because we have now performed an irreversible measurement. So, whenever we block the second exit, the following projective approach is the correct way to proceed, and not just a shortcut.

Recall the experiment has a $+z$ oriented analyzer first, a $+x$ oriented analyzer second, and a $+z$ oriented analyzer third. Each subsequent analyzer is attached to

the + exit of the previous one. Here is how we analyze these experiments, which requires us to use the projectors for the projective measurement, because we have intermediate states before the final measurement. The atom emerging from the + exit of the first $+z$ oriented analyzer is obviously in the $|\uparrow\rangle_z$ state. Passing through the second $+x$ oriented analyzer and looking at atoms emerging from the + exit requires us to apply the ket-bra product $|\uparrow\rangle_{x\,x}\langle\uparrow|$ to the state $|\uparrow\rangle_z$. This transforms the state as follows:

$$\left(|\uparrow\rangle_{x\,x}\langle\uparrow|\right)|\uparrow\rangle_z = |\uparrow\rangle_x\left({}_x\langle\uparrow||\uparrow\rangle_z\right) = |\uparrow\rangle_x\left(\frac{1}{\sqrt{2}}\,{}_z\langle\uparrow|\uparrow\rangle_z + \frac{1}{\sqrt{2}}\,{}_z\langle\downarrow|\uparrow\rangle_z\right)$$
$$= \frac{1}{\sqrt{2}}|\uparrow\rangle_x. \tag{6.112}$$

So, the state $\frac{1}{\sqrt{2}}|\uparrow\rangle_x$ enters the third analyzer. We use our rule to determine the probability to exit each exit of the final $+z$ oriented analyzer: the probability to exit the + exit is given by

$$\left|{}_z\langle\uparrow|\frac{1}{\sqrt{2}}|\uparrow\rangle_x\right|^2 = \frac{1}{2}\left|{}_z\langle\uparrow|\left(\frac{1}{\sqrt{2}}|\uparrow\rangle_z + \frac{1}{\sqrt{2}}|\downarrow\rangle_z\right)\right|^2 = \frac{1}{2}\times\frac{1}{2} = \frac{1}{4}. \tag{6.113}$$

This is the same result as we obtained in Chap. 2 using our old rules.

It may seem confusing that the quantum state that originally had a norm of one has shrunk its norm to $\frac{1}{2}$ in the intermediate state $\left(\frac{1}{\sqrt{2}}|\uparrow\rangle_x\right)$. This often happens when we take our shortcut in calculations by using the projections for each measurement in a sequence. This is because this methodology only calculates the probability amplitude for *part* of the original quantum state, the part that has not been blocked or ignored, which is why it has a coefficient less than one, yielding a norm less than one. We did this routinely with our probability arrows in our earlier work, and these quantum states can be directly related to those arrows. We will see when we analyze tagging later that the norm of the complete state (not just the part that is ultimately measured) remains one at all times, and we obtain the same final answer as well.

Now, we reexamine another class of problems, shown for one specific case in Fig. 1.29—the situation where an atom exits one exit of a vertically oriented ($+z$) Stern-Gerlach anaylzer and is then fed into a second analyzer at an angle θ relative to the vertical. Suppose we capture the atoms exiting the + exit of the z-oriented analyzer (note this is different from the figure). Then, the state of the atom entering the second analyzer is $|\uparrow\rangle_z$. The probability to emerge from the + exit of the second analyzer is $P(+,\theta) = |_{\theta\text{-}\phi=0}\langle\uparrow|\uparrow\rangle_z|^2$. Recall that we already worked out this eigenstate in Eq. (6.70), so we have $|\uparrow\rangle_{\theta\text{-}\phi=0} = \cos\frac{\theta}{2}|\uparrow\rangle_z + \sin\frac{\theta}{2}|\downarrow\rangle_z$. Substituting into the probability formula gives us $P(+,\theta) = \cos^2\frac{\theta}{2}$; a similar analysis, which you should do, gives us that $P(-,\theta) = \sin^2\frac{\theta}{2}$. Note that we have now calculated one of the quantum rules we simply told you previously. In experiments, we often

are interested in the average value of the experiment. Here, the average measured value of the projection along the θ axis is given by the probability to be measured in one or the other exit, multiplied by the projection corresponding to that exit, or

$$P(+,\theta)\cdot\left(\frac{\hbar}{2}\right)+P(-,\theta)\cdot\left(-\frac{\hbar}{2}\right)=\frac{\hbar}{2}\left(\cos^2\frac{\theta}{2}-\sin^2\frac{\theta}{2}\right)=\frac{\hbar}{2}\cos\theta. \tag{6.114}$$

Think of what this last formula tells us about the special cases of $\theta = 0$, $\frac{\pi}{2}$, and π. Does the formula make sense? In quantum mechanics, we often want to know what the average value of an observable is (such as the projection of the spin on the z-axis) for a specific experiment. This is because this is what is actually measured after the experiment is repeated many times. For the example we just discussed, the simple result is $\frac{\hbar}{2}\cos\theta$.

Now, we will move from discussing Stern-Gerlach analyzers with our new formalism to discussing analyzer loops in this new formalism. Recall for the analyzer, we would project onto the state corresponding to the exit. For an analyzer loop, we project onto the sum of the two states corresponding to the paths of the analyzer loop. So, if I had a generic state $|\psi\rangle$, the procedure of going through an analyzer loop oriented in the $+z$-direction maps it to

$$|\psi\rangle \rightarrow |\uparrow\rangle_z\,{}_z\langle\uparrow|\psi\rangle + |\downarrow\rangle_z\,{}_z\langle\downarrow|\psi\rangle. \tag{6.115}$$

Note that this map produces a superposition of states. Now, because our completeness relation is $\hat{\mathbb{I}} = |\uparrow\rangle_z\,{}_z\langle\uparrow| + |\downarrow\rangle_z\,{}_z\langle\downarrow|$, we see that when an atom in state $|\psi\rangle$ goes through an unwatched analyzer loop, it emerges in the same state that it entered, because $|\psi\rangle \rightarrow \hat{\mathbb{I}}|\psi\rangle = |\psi\rangle$. Hence, the unwatched analyzer loop does nothing, just as we already knew! We also now have an explicit formula, in Eq. (6.115), for what is meant by "the atom goes on both paths"—it forms a superposition with respect to the states corresponding to each path, or we can say "the atom moves in superposition." If a path is blocked, we have performed an irreversible measurement, which is modeled by removing the projection onto the blocked path state. So, if the spin-up path is blocked, we have that moving through the analyzer loop corresponds to $|\psi\rangle \rightarrow |\downarrow\rangle_z\,{}_z\langle\downarrow|\psi\rangle$, while if the down-spin path is blocked, we have $|\psi\rangle \rightarrow |\uparrow\rangle_z\,{}_z\langle\uparrow|\psi\rangle$. The probability that an atom emerges from an analyzer loop with a blocked path is just the modulus squared of the state vector that it emerges with: $|{}_z\langle\downarrow|\psi\rangle|^2$, if the $\uparrow$ path is blocked, and $|{}_z\langle\uparrow|\psi\rangle|^2$ if the $\downarrow$ path is blocked. This analysis now allows us to solve analyzer-loop problems, by applying these rules. It also gives us insight about what the "experiment that does nothing" really is doing. One can think of it as a *spin-basis changer*. A state comes in in some state, and the analyzer loop makes us expand that state in the spin basis corresponding to the orientation of the analyzer loop. Then, if something happens at the arms, we can immediately apply the effect to our state, because it is already in the proper basis for the analysis.

This is a good point to pause and recap our rules. To start, if an atom enters an analyzer in an arbitrary state $|\psi\rangle_{\theta\text{-}\phi}$ expressed in a basis in the θ-ϕ direction, then

the probability to emerge from the + exit of an (arbitrarily oriented) analyzer is $P(+) = |\langle\uparrow\ |\psi\rangle_{\theta\text{-}\phi}|^2$ and the probability to emerge from the − exit is $P(-) = |\langle\downarrow\ |\psi\rangle_{\theta\text{-}\phi}|^2$. Here, the up and down bras are in the orientation of the analyzer that is making the measurement. This can be thought of as the analyzer projecting the initial state onto the analyzer state corresponding to the + exit, which can be written as

$$|\psi\rangle_{\theta\text{-}\phi} \rightarrow |\uparrow\rangle\langle\uparrow\ |\psi\rangle_{\theta\text{-}\phi}, \tag{6.116}$$

where we suppressed the angle that the analyzer lies in, to cover all cases. The probability is then $|\langle\uparrow\ |\uparrow\rangle\langle\uparrow\ |\psi\rangle_{\theta\text{-}\phi}|^2 = |\langle\uparrow\ |\psi\rangle_{\theta\text{-}\phi}|^2$. For the − exit, we just change the $\uparrow$ to $\downarrow$. Similarly, for the analyzer loop, we have the sum over both projections, so $|\psi\rangle_{\theta\text{-}\phi} \rightarrow |\uparrow\rangle\langle\uparrow\ |\psi\rangle_{\theta\text{-}\phi} + |\downarrow\rangle\langle\downarrow\ |\psi\rangle_{\theta\text{-}\phi}$, which is identical to $|\psi\rangle_{\theta\text{-}\phi}$. However, if we observe the analyzer loop, by having which-way information due to a blocked path, for example, the $\uparrow$ path, then we remove the projector onto the blocked path so that we have $|\psi\rangle_{\theta\text{-}\phi} \rightarrow \cancel{|\uparrow\rangle\langle\uparrow|\psi\rangle_{\theta\text{-}\phi}} + |\downarrow\rangle\langle\downarrow\ |\psi\rangle_{\theta\text{-}\phi}$.

In our analysis so far, we have been using the so-called *Copenhagen interpretation* to analyze our experiments as projective measurements. In this case, when a measurement occurs, we model this by applying a projection operator corresponding to the set of states determined by the measurement. In the Copenhagen interpretation, the result of a measurement is one of the eigenvalues corresponding to the operator being measured (here being $\vec{e}_n \cdot \hat{\vec{S}}$), and after the measurement, we project the quantum state onto the state that corresponds to the eigenvalue that was measured. This is called the "collapse of the wavefunction", and it is an irreversible process, in the sense that if we know the result of the measurement, we cannot reconstruct the original quantum state, except in special cases. If the quantum state is not destroyed in the measurement, as it often is, then we need to renormalize the new state, so its squared modulus is one; as we have seen in examples discussed already, when we apply the projection operator to a state vector, the norm of the state vector is usually reduced (in special cases, it can be preserved); this is why we need to renormalize the projected state. Note that only certain types of experiments correspond to projective measurements—if the given experiment does not correspond to such a measurement, then it must be analyzed in a different fashion. How can we tell? It sometimes is subtle, but the key point is when one makes the irreversible measurement, was the eigenvalue of an operator directly measured, or was its result inferred? If the latter, then the analysis of the experiment is more complex. It is clear that the Stern-Gerlach analyzer and the analyzer loop are both devices that are typically described by projective measurements, but since the analyzer loop can also be employed for tagging of atoms, which is not a measurement, the situation is case dependent. We will carefully explain how one analyzes different experiments as we encounter them again with our new formalism.

Our next example is the analog of the two-slit experiment, carried out with a horizontal analyzer loop followed by a vertical analyzer, and shown in the top panel of Fig. 2.5. The system has an $|\uparrow\rangle_z$ state enter the analyzer loop from the left. If we

do not watch, then the quantum state is transformed as follows as it moves through the different components of the experiment:

$$\begin{aligned}
|\uparrow\rangle_z &\rightarrow |\uparrow\rangle_x\, {}_x\langle\uparrow|\uparrow\rangle_z + |\downarrow\rangle_x\, {}_x\langle\downarrow|\uparrow\rangle_z && \text{after the analyzer loop} \\
&= |\uparrow\rangle_x \cdot \frac{1}{\sqrt{2}} + |\downarrow\rangle_x \cdot \frac{1}{\sqrt{2}} && \text{evaluate the bra-kets} \\
&= \frac{1}{2}\Big(|\uparrow\rangle_z + |\downarrow\rangle_z\Big) + \frac{1}{2}\Big(|\uparrow\rangle_z - |\downarrow\rangle_z\Big) && \text{change to } z\text{-basis} \\
&= |\uparrow\rangle_z && \text{exiting the analyzer loop.}
\end{aligned} \tag{6.117}$$

So, the state emerges as it entered, in the state $|\uparrow\rangle_z$. It then takes the upper path in the final analyzer, and emerges from the $+$ exit.

But what happens if we watch at the arms of the analyzer loop? The simplest way to do this is to block one of the paths, say the $+$ path on the x-axis. Then, the up state entering the blocked analyzer loop transforms as

$$|\uparrow\rangle_z \rightarrow |\downarrow\rangle_x\, {}_x\langle\downarrow|\uparrow\rangle_z = \frac{1}{\sqrt{2}}|\downarrow\rangle_x, \tag{6.118}$$

and as we enter the vertical analyzer, we have the following probabilities: $P(+,z) = \left|\frac{1}{\sqrt{2}}\, {}_z\langle\uparrow|\downarrow\rangle_x\right|^2 = \frac{1}{4}$ to emerge from the $+$ exit and $P(-,z) = \left|\frac{1}{\sqrt{2}}\, {}_z\langle\downarrow|\downarrow\rangle_x\right|^2 = \frac{1}{4}$ to emerge from the $-$ exit. Recall that one-half of the incident atoms hit the blocked $+x$ path). The analysis is similar if we instead used a 100% efficient pass-through detector on the $+x$ branch. There is a problem on this at the end of the section.

The next example we analyze is the quantum seeing in the dark scenario, as shown in Figs. 2.12 and 2.13. Recall that we need to consider two cases, corresponding to the two figures. The input from the source on the left is atoms in the $|\uparrow\rangle_z$ state. The analyzer loops are oriented in the $+x$, $+z$, and $+x$ directions, respectively. For the first experiment, we block the $-x$ path on the first analyzer loop and the $+x$ path on the third analyzer loop. The transformation of the state from the entrance into the first analyzer loop to its exit is given by

$$|\uparrow]\rangle_z \rightarrow |\uparrow\rangle_x\, {}_x\langle\uparrow|\uparrow\rangle_z = \frac{1}{\sqrt{2}}|\uparrow\rangle_x. \tag{6.119}$$

Going through the second analyzer loop transforms as

$$\frac{1}{\sqrt{2}}|\uparrow\rangle_x \rightarrow \frac{1}{\sqrt{2}}|\uparrow\rangle_x, \tag{6.120}$$

because $|\uparrow\rangle_z\, {}_z\langle\uparrow| + |\downarrow\rangle_z\, {}_z\langle\downarrow| = \hat{\mathbb{I}}$. Finally, going through the third analyzer loop, we have

$$\frac{1}{\sqrt{2}}|\uparrow\rangle_x \rightarrow \frac{1}{\sqrt{2}}|\downarrow\rangle_x\, {}_x\langle\downarrow|\uparrow\rangle_x = 0. \tag{6.121}$$

So, nothing exits this string of three analyzer loops.

However, if we now add blockage by also blocking the $-z$ path on the middle analyzer loop, we find the transmission through the first analyzer loop is the same $|\uparrow\rangle_z \rightarrow \frac{1}{\sqrt{2}}|\uparrow\rangle_x$. Through the middle analyzer loop, we now have

$$\frac{1}{\sqrt{2}}|\uparrow\rangle_x \rightarrow \frac{1}{\sqrt{2}}|\uparrow\rangle_z\, {}_z\langle\uparrow|\uparrow\rangle_x = \frac{1}{2}|\uparrow\rangle_z, \tag{6.122}$$

and through the final analyzer loop, we have

$$\frac{1}{2}|\uparrow\rangle_z \rightarrow \frac{1}{2}|\downarrow\rangle_x\, {}_x\langle\downarrow|\uparrow\rangle_z = \frac{1}{2\sqrt{2}}|\downarrow\rangle_x. \tag{6.123}$$

Hence, the probability to exit the final analyzer loop is $\left(\frac{1}{2\sqrt{2}}\right)^2 = \frac{1}{8}$, and they exit in the $|\downarrow\rangle_x$ state. This means that adding blockage resulted in *more* atoms emerging from the exit. So, by setting up an experiment with the $-x$ path of the first analyzer loop and $+x$ path of the third analyzer loop, we simply search for atoms to exit the apparatus. Any atoms that do signify that the middle analyzer loop has a blockage (in either of the paths). We detect the blockage by an atom that does not interact with the blockage! Note that this is an example of an inferred measurement—we determine the blockage by measuring the presence of an atom that does not interact with the blockage.

Our last example is a problem. We have a beam of atoms created in the $|\uparrow\rangle_z$ state. They are input into a Stern-Gerlach analyzer oriented at 15° from the vertical. We then take the atoms exiting the $-$ exit and place them into a second Stern-Gerlach analyzer oriented at 60° to the vertical. What is the probability to make it out of the $+$ exit of the second Stern-Gerlach analyzer.

Solution Our initial state is $|\uparrow\rangle_z$. Going through the first analyzer projects onto $|\downarrow\rangle_{\theta=15^\circ\text{-}\phi=0}$ state. Recall that

$$|\downarrow\rangle_{\theta=15^\circ\text{-}\phi=0} = -\sin 7.5^\circ|\uparrow\rangle_z + \cos 7.5^\circ|\downarrow\rangle_z, \tag{6.124}$$

so the probability to exit the $-$ exit is

$$P(-, 15^\circ) = |{}_{\theta=15^\circ\text{-}\phi=0}\langle\downarrow|\uparrow\rangle_z|^2 = \sin^2 7.5^\circ \tag{6.125}$$

and the atom is in the state $\sin 7.5^\circ|\downarrow\rangle_{\theta=15^\circ\text{-}\phi=0}$. For the remainder of the problem, we will suppress the $\phi = 0$ notation on the states—all states we work with have $\phi = 0$. Next, we project onto the state $|\uparrow\rangle_{\theta=60^\circ}$ and take the modulus squared to find the probability:

$$
\begin{aligned}
P(+, 60^\circ) &= |_{\theta=60^\circ}\langle\uparrow|\sin 7.5^\circ|\downarrow\rangle_{\theta=15^\circ}|^2 \\
&= \sin^2 7.5^\circ \Big| \Big({}_z\langle\uparrow|\cos 30^\circ + {}_z\langle\downarrow|\sin 30^\circ \Big) \\
&\quad \times \Big(-\sin 7.5^\circ|\uparrow\rangle_z + \cos 7.5^\circ|\downarrow\rangle_z \Big) \Big|^2 \\
&= \sin^2 7.5^\circ \Big| -\cos 30^\circ \sin 7.5^\circ + \sin 30^\circ \cos 7.5^\circ \Big|^2 \\
&= \sin^2 7.5^\circ \sin^2 22.5^\circ = 0.00249. \qquad (6.126)
\end{aligned}
$$

In the last line, we use the trigonometric identity for the sum of two angles. This calculation shows that $P(+, 60^\circ) \approx 0.25\%$.

Before ending this section, we now have all the formal machinery to discuss two important quantum information science topics—quantum key distribution and the no-cloning theorem. Before you can understand quantum key distribution, you need to know what a one-time pad is. While the pad will be sent with binary encoding (i.e., a string of ones and zeros, selected randomly), it is easiest to understand it with a conventional alphabet code. The idea is that we send 26 letters repeatedly in random order. The 26 letters are the mapping of abcd...xyz to a different ordering. So if"j" was the first letter in the 26 letters of the first instantiation of the pad, then we would replace "a" by "j" in our cypher, if the first letter we were encoding was "a". If we just used 26 letters and one mapping instantiation for the entire message, it could eventually get decoded, because letters do not appear with the same frequency in language (think about the most popular guessed letters in *Wheel of Fortune*). With a one-time pad, we remap the letters *for each letter of our message*. This seems like it is overkill, but it guarantees no one else can read the message unless they have the same one-time pad. So, it is very secure. The challenge is distributing the pad securely. This is where quantum mechanics comes in.

One of the hallmarks of quantum information is the ability to send encrypted messages securely using quantum mechanics. Not only can the transfer be completely secure, one can tell whether an eavesdropper is trying to listen in to the key distribution while the process of distributing the key is occurring! Charlie Bennett and Gilles Brassard developed a protocol to securely do this back in 1984 Bennett and Brassard (1984)! Yes, 1984!! It is now called the BB84 protocol.

Here is how it works. The key that will be distributed is going to be in binary form, a string of zeroes and ones, like $100101101\cdots$. We will send the key using atoms and Stern-Gerlach analyzers. We have the two parties, which are traditionally called *Alice* and *Bob*; if you need a mental image, I like to think of Alice Cooper and Bob Dylan, but feel free to choose whomever you want. Here is how they do it:

1. Alice first generates a random key of zeroes and ones. This is the key that is desired to be sent (but as we will see only part of the key will ultimately be used). This list of zeroes and ones is called the key string, and it is the basis for the one-time pad that we want to send to Bob.

2. Alice generates a second random key that we will call the control key. It is also a series of zeroes and ones.
3. Alice has a source that can emit atoms in the $+z$, $-z$, $+x$, or $-x$ states. We have a question discussing how this can be done below.
4. Alice now maps her two bit streams, the key stream and the control stream. The key stream determines whether Alice sends a positive projection (corresponding to bit 0) or a negative projection (corresponding to bit 1). The control stream determines whether it is in the z (0) or x (1) direction.
5. Bob has his own control bit stream that he also created from random numbers. *It is completely unrelated to Alice's control bit stream.*
6. Bob sets his analyzer in the z position or the x position depending on whether his control bit stream has a 0 or 1, respectively. He measures each atom sent by Alice.
7. After the entire bit stream has been processed, Alice and Bob send each other what their measurement settings were for their control bit streams. This can be done publicly.
8. Alice and Bob create the key for their shared one-time pad only from the subset of measurements they made where they agreed on the experimental settings. To determine this shared key, Alice and Bob compare their control strings for matches. The shared key is generated by writing down the corresponding term in Alice's key string only for the measurements where Alice and Bob agreed. This is identical to the results Bob measured for those cases.

OK. Eight steps to do this, two bit streams each. Clearly a complicated procedure. Is it all that is needed? Yes, it is! We explore it more in the problems below.

Is this protocol secure with respect to an eavesdropper? It is common to call the eavesdropper *Eve*. She intercepts the atoms before they get to Bob and sends a copy of what she measures to Bob. While standard key distribution is susceptible to such an attack, with quantum key distribution, this approach is guaranteed to fail. Let's see if we can sort out why.

If Eve measures on the same axis as Bob and Alice, then she will get the same result and will be able to steal a copy of the key whenever this happens. The way Eve is detected is due to what happens when she measures *on the wrong axis*. In particular, in the situation where Alice and Bob agree on their measurement axis, but Eve chooses the wrong one, we will have Eve send an x projected state to Bob's z-oriented analyzer or *vice versa*. In either case, Bob will measure the correct result only half of the time. How do they use this consequence to detect Eve? There is an essay question below that will lead you through how this is done.

The key to the foiling of eavesdropping relies on the *no-cloning theorem*. Eve cannot copy the quantum state, so she cannot both detect it and pass it on to Bob. This theorem is often hard to understand, because it has some obvious counterexamples. If I know a state is in a particular orientation, say the z-direction, I simply measure it with an analyzer in the z-direction, and I can make as many copies as I would like. The key of the no-cloning theorem is that this is the only

case where this happens. If we do not know the axis that the state is prepared in, then we cannot make just one measurement and determine the state.

We now look more deeply into the theorem. Let us assume that we can clone a state. To do so, there is a unitary "copy" operator $\hat{U}$, which satisfies $\hat{U}^\dagger \hat{U} = \hat{\mathbb{I}}$ and can copy quantum states in the following way:

$$\hat{U}\Big(|\psi\rangle \otimes |\uparrow\rangle_z\Big) = |\psi\rangle \otimes |\psi\rangle. \tag{6.127}$$

The copy operator $\hat{U}$ is defined to be the operator that copies some arbitrary unit norm state $|\psi\rangle$ from the left term in the tensor product to the right term, without changing the left term, as illustrated in Eq. (6.127). In other words, we are copying the first state into the second state of the tensor product without changing the first state. For definiteness, we assume the right state is initially in the up state along the z-axis, but we find our analysis will be independent of the details of this initial state on the right.

Because the $\hat{U}$ operator works to copy arbitrary states, it also satisfies $\hat{U}\big(|\phi\rangle \otimes |\uparrow\rangle_z\big) = |\phi\rangle \otimes |\phi\rangle$ too, copying a second arbitrary state from the left term to the right without disturbing the left term. Our proof by contradiction simply looks at the inner product of these two states. We have

$$\Big(\langle\phi| \otimes {}_z\langle\uparrow|\Big)\Big(|\psi\rangle \otimes |\uparrow\rangle_z\Big) = \langle\phi|\psi\rangle\, {}_z\langle\uparrow|\uparrow\rangle_z = \langle\phi|\psi\rangle. \tag{6.128}$$

But now, we use the "multiply-by-one" trick and include the operator $\hat{U}^\dagger \hat{U} = \hat{\mathbb{I}}$ in the center of the inner product; note that this operator acts on both terms in the tensor product. This gives

$$\Big(\langle\phi| \otimes_z \langle\uparrow|\Big)\hat{U}^\dagger \hat{U}\Big(|\psi\rangle \otimes |\uparrow\rangle_z\Big) = \Big(\langle\phi| \otimes {}_z\langle\uparrow|\Big)\Big(|\psi\rangle \otimes |\uparrow\rangle_z\Big). \tag{6.129}$$

Now, we let the $\hat{U}$ operator act to the right, where it makes a copy creating $|\psi\rangle \otimes |\psi\rangle$, and $\hat{U}^\dagger$ act to the left, where it also makes a copy creating $\langle\phi| \otimes \langle\phi|$. Evaluating the inner product gives us

$$\Big(\langle\phi| \otimes \langle\phi|\Big)\Big(|\psi\rangle \otimes |\psi\rangle\Big) = \big(\langle\phi|\psi\rangle\big)^2. \tag{6.130}$$

But these two inner products must be the same. Hence, we have $\langle\phi|\psi\rangle = (\langle\phi|\psi\rangle)^2$. The only numbers which satisfy this are 0 and 1. But recall that we picked the states $|\psi\rangle$ and $|\phi\rangle$ to be arbitrary states. We cannot have two arbitrary states always be either parallel to each other or orthogonal to each other! Hence, no such unitary operator can be found and our proof is complete. Take some time to carefully go through the argument, as it is subtle. We will derive it a different way in a problem below.

Section 6.4 Problems:

In all of the problems below, use the formalism in terms of the quantum state vectors and the projection hypothesis, as developed in this section, to solve the following three problems.

6.4.1 Probability at an Angle

Analyze and determine the probability to enter and then emerge from each of the final exits of the second analyzer of an apparatus that has a vertical analyzer followed by an analyzer that is rotated at an angle θ from the vertical. The second analyzer is attached to the $-$ exit of the first analyzer. An image of the apparatus for a specific angle is given in Fig. 1.29.

6.4.2 Rotating Stern-Gerlach Experiment

We have a rotating Stern-Gerlach experiment as shown in Fig. 1.28, which can stop on three positions—vertical (position A), 120° from the vertical (position B), and 240° from the vertical (position C). Determine the probability to exit the $+$ exit of the rotating Stern-Gerlach analyzer if the atoms enter the rotating Stern-Gerlach analyzer in the $|\downarrow\rangle_z$ state, and the rotating analyzer spends one-third of the time, on average, in each of the three positions. Repeat for the input state of $|\uparrow\rangle_x$.

6.4.3 Two-Slit Experiment with Spin

Reanalyze the two-slit experiment with spin using our formalism. Consider the case with a single pass-through detector. An image of the apparatus is shown in Fig. 2.7. Analyze the case where the pass-through detector is 100% efficient. Analyze it again in the case where the pass-through detector is 64% efficient. If you are having trouble with the partially working detector, it will be covered in more detail in the next two subsections.

6.4.4 A Single-Atom Source for Quantum Key Distribution

Write an essay describing the possibilities and challenges with creating a single-atom source of silver atoms polarized in the $\pm z$ or $\pm x$ directions. Consider the following issues. If we use an analyzer loop, we can block a path to select a positive

or negative projection on a specific axis. How often will we be successful for the projection we want? What should be input into the analyzer loop? How do we pick atoms with spin projections along the z-axis? Along the x-axis? Can we guarantee a single atom is emitted in a specific spin projection at a given time? What if we had a single-atom source that could produce randomly oriented spins for the atoms? How can you modify the protocol if we cannot guarantee atoms being emitted in any specific time interval? How can you know an atom was emitted?

6.4.5 Constructing the One-Time Pad

Alice's key string is	01101	01100	10100	11011
Alice's control string is	11010	10100	01011	10001
Bob's control string is	00100	01001	10111	01011

Assume the measurements are perfect. What is the shared key that they generated from this protocol?

6.4.6 Detecting Eve

Write an essay discussing a strategy that Alice and Bob can use to determine the presence of Eve. Think about how they can sacrifice some fraction of the one-time pad to compare. How should they compare to determine if there is an eavesdropper? What is different if there is no eavesdropper? What happens if one can only produce a single atom for the quantum key distribution of the one-time pad only some of the time, as opposed to always? Does that affect the ability to detect Eve?

6.4.7 No-Cloning Theorem

Now, we prove the theorem using a different approach. Suppose $|\psi\rangle = \alpha|\uparrow\rangle_z + \beta|\downarrow\rangle_z$. The copying operation, given by

$$|\psi\rangle \otimes |\uparrow\rangle_z \rightarrow \hat{U}\Big(|\psi\rangle \otimes |\psi\rangle\Big). \tag{6.131}$$

Using the expression for $|\psi\rangle$, employ the following steps to complete the proof:

1. Expand the state $|\psi\rangle \otimes |\psi\rangle$ in terms of the four basis vectors $|\uparrow\rangle_z \otimes |\downarrow\rangle_z$, $|\uparrow\rangle_z \otimes |\uparrow\rangle_z$, $|\downarrow\rangle_z \otimes |\uparrow\rangle_z$, and $|\downarrow\rangle_z \otimes |\downarrow\rangle_z$. This is the result of $\hat{U}(|\psi\rangle \otimes |\uparrow\rangle_z)$
2. Repeat the expansion of the state in terms of the four basis functions, but now with the state $|\psi\rangle \otimes |\uparrow\rangle_z$.

3. Apply the copy operation $\hat{U}$ to the expansion you just calculated for $|\psi\rangle \otimes |\uparrow\rangle_z$, by operating $\hat{U}$ onto each term in the expansion separately.
4. Compare the two calculations of the copy operation. You should find they only work when either $\alpha = 0$ or $\beta = 0$.

6.4.8 *Superposition Versus Entanglement*

In this section, we have described the behavior of the Stern-Gerlach experiment solely in terms of superposition. We had one degree of freedom, the spin projection of the atom on a specific axis, and we used that to describe the different states. But we could have instead described these states as *entangled* states, by introducing a set of states for the position of the atom, given by |upper⟩, |central⟩, and |lower⟩. Upper is the spatial position associated with the + exit, central is the spatial position associated with the input to the analyzer, and lower is associated with the − exit. We can use the same three names without confusion even if the analyzer is rotated by 90° by associating upper with the spatial location of + and lower with the spatial location of −. Then, when we send a $+x$-oriented spin, starting in the state $\frac{1}{\sqrt{2}}(|\uparrow\rangle_z + |\downarrow\rangle_z) \otimes |\text{center}\rangle$ (an unentangled state) through a z-analyzer, it emerges in a state $\frac{1}{\sqrt{2}}|\uparrow\rangle_z \otimes |\text{upper}\rangle + \frac{1}{\sqrt{2}}|\downarrow\rangle_z \otimes |\text{lower}\rangle$, which is entangled. When we measure it, we can think now of projecting with respect to the position state on the screen rather than the spin state of the exit. Write a short discussion about how interpreting the experiments this way gives all the same results as before.

Now, for the analyzer loop, we find that if it is unwatched, then the second half of the tensor product is always in the state |center⟩ as the system emerges from the exit. But if we have the possibility of watching at one gate in the middle of the loop, then we add an extra set of states to our tensor product of |seen⟩, |unseen⟩, or |not watched⟩, for example. Use this discussion to describe what happens for analyzer loops, which now need triple tensor product to describe the states.

Finally, discuss the more philosophical question of whether it is preferable to describe these experiments as simple superpositions or as entangled states. What are the pros and cons of each method. Is there ever a circumstance where the two approaches will give different results?

6.5 Polarization of Light Revisited

We have worked quite hard to develop the quantum mechanics of systems that have two states, such as spin, which has $|\uparrow\rangle$ and $|\downarrow\rangle$ states. But, it turns out that these so-called two-state models are ubiquitous in quantum mechanics and many interesting systems can be described using the same formalism! This means our hard work up

to now goes much further, because we can describe many more systems with this same formalism.

Some of the additional two-state systems are systems that employ the polarization of light in determining their properties. Let us recall the facts we worked out in Chap. 4. We know light is composed of quantum particles called photons. Light has another "internal property" called polarization, which we described with a director; the two independent polarizations were up-down (vertical) and left-right (horizontal). Light acts like every other quantum particle—it is "stupid" and remembers only the last axis it was measured on. The polarization property is described precisely in this fashion.

Light can be polarized in a linear fashion or in a circular fashion. Circularly polarized light is somewhat more complicated to describe than linearly polarized light, and we will not need it in this book, so we will focus solely on linear polarization. Just like the spin states, polarization is found in two orthogonal states called vertical polarization $|V\rangle$ (which we also call up-down) and horizontal polarization $|H\rangle$ (which we also call left-right). Light polarized in the θ-direction is expressed as

$$|\theta\rangle = \cos\theta|V\rangle + \sin\theta|H\rangle, \tag{6.132}$$

in analogy to what we had for spin in the θ-ϕ direction. But note that here the angular dependence is θ not $\frac{\theta}{2}$. This is because up polarization is identical to down polarization and is just called $|V\rangle$. When we talk about linear polarization, it corresponds to the direction of the electric field associated with the photon (an electric field is similar to a magnetic field—it measures the push or pull on a charged object). Since the electric field oscillates sinusoidally, this is why the up and down directions are the same for polarization. Clearly, $|\theta{=}0\rangle = |V\rangle$ and $|\theta{=}\frac{\pi}{2}\rangle = |H\rangle$. But, now, we have $|\theta{=}\pi\rangle = -|V\rangle$ is viewed as the same as $|V\rangle$, because changing the sign does not change the state. This leads to the rule that whenever we work with polarization, the state $-|P\rangle$ is equivalent to $|P\rangle$, or $|\theta + \pi\rangle = -|\theta\rangle$ is equivalent to $|\theta\rangle$. At first this may seem bizarre, but if you look back at Sect. 5.7, you will recall that a normalized eigenvector has a degree of freedom, which allowed it to be multiplied by any complex phase $e^{i\phi}$, including $\phi = \pi$, which has $e^{i\pi} = -1$. So, generically, any eigenvector $|\psi\rangle$ is identified with the eigenvector $-|\psi\rangle$. Changing the sign (or even the complex phase) does not change any probabilities. But the relative signs of phases in superpositions are important. Hence, the state in Eq. (6.132) is a different polarization state from

$$|-\theta\rangle = \cos\theta|V\rangle - \sin\theta|H\rangle, \tag{6.133}$$

but is the same as

$$|\theta + \pi\rangle = -\cos\theta|V\rangle - \sin\theta|H\rangle. \tag{6.134}$$

We also can use our spinor representation. The state $|\theta\rangle$ is represented as follows:

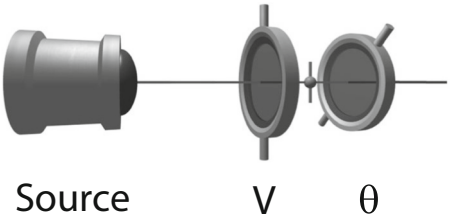

Fig. 6.4 Experiment that takes a vertically polarized photon and shoots it through a polarizer oriented at an angle of θ with respect to the vertical. The photon leaves the source and goes through a vertical polarizer V to prepare it in the $|V\rangle$ state before going through the next polarizer. The vertically polarized photon is depicted schematically with the vertical director and is ready to go into the polarizer oriented at θ

$$|\theta\rangle \leftrightarrow \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix} \leftarrow \begin{pmatrix} \text{coefficient of } |V\rangle \\ \text{coefficient of } |H\rangle \end{pmatrix}, \tag{6.135}$$

when written as a spinor.

Okay, we have determined how to represent our polarization states. Now, we need to determine how to represent the action of the polarizer on a quantum state. As before with spins, we want to find how the polarizer transforms the photon state—the rule is analogous. In particular, the polarizer acts like a projection operator onto its polarization axis (just as a Stern-Gerlach analyzer does), and the probability to transmit through the polarizer is the modulus squared of this polarization quantum state. It is an irreversible transition—one cannot construct the polarization before passing through a polarizer from the value of the polarization afterward. In this case, the experiment is nondestructive for the photons that make it through the polarizer.

For example, take a photon that is polarized in the up-down direction $|V\rangle$ and pass it through a polarizer oriented with its axis at an angle θ to the vertical, as shown in Fig. 6.4. We find $|V\rangle$ is transformed as

$$|V\rangle \to |\theta\rangle\,\langle\theta|V\rangle = |\theta\rangle\Big(\cos\theta\,\underbrace{\langle V|V\rangle}_{=1} + \sin\theta\,\underbrace{\langle H|V\rangle}_{=0}\Big) = \cos\theta|\theta\rangle, \tag{6.136}$$

because $\langle V|V\rangle = 1$ and $\langle H|V\rangle = 0$. So, if $\theta = 0$, we find that $|V\rangle \to |V\rangle$, if $\theta = \frac{\pi}{2}$, $|V\rangle \to 0|H\rangle = 0$, and as $\theta \to \frac{\pi}{2}$, the amplitude of the transmitted photon gets smaller and smaller. The probability to be transmitted is $\big|\cos\theta|\theta\rangle\big|^2 = \cos^2\theta\langle\theta|\theta\rangle = \cos^2\theta$. This follows because

$$\begin{aligned}\langle\theta|\theta\rangle &= \Big(\cos\theta\langle V| + \sin\theta\langle H|\Big)\Big(\cos\theta|V\rangle + \sin\theta|H\rangle\Big) \\ &= \cos^2\theta\langle V|V\rangle + \cos\theta\sin\theta\langle V|H\rangle + \sin\theta\cos\theta\langle H|V\rangle + \sin^2\theta\langle H|H\rangle \\ &= \cos^2\theta + \sin^2\theta = 1.\end{aligned} \tag{6.137}$$

Of course this is as it should be, because all quantum states are normalized to one. When light goes through a polarizer, the norm of the state is reduced using our shortcut method for performing calculations—this is because the part of the photon that does not make it through is absorbed by the polarizer, which is an irreversible process, and hence, it constitutes a measurement.

Now that we have developed the formalism to describe the quantum states of polarization and how to describe the action of a polarizer on such states, we next reexamine the experiments we first analyzed in Chap. 4, but now using our new quantum formalism. The first experiment we will discuss is shown in panel (d) of Fig. 4.13, where a vertical polarized photon is passed through a horizontal oriented polarizer. Our analysis starts from the photon after it has emerged from the first vertically oriented polarizer. The photon is in the state $|V\rangle$. Now, it arrives at the H polarizer. We use our rule $|V\rangle \rightarrow |H\rangle\langle H|V\rangle = |H\rangle \cdot 0 = 0$ and find that the photon is blocked—none get through to the screen. Now, we add more "blockage," corresponding to an additional polarizer oriented along the diagonal D, at 45° from the vertical, situated in between the V and H polarizers. This is shown in panel (b) of Fig. 4.13. We find the state of the photon transforms from $|V\rangle$ to $|45°\rangle\langle 45°|V\rangle = \frac{1}{\sqrt{2}}|V\rangle$ after passing through the 45° oriented polarizer, because

$$|45°\rangle = \cos 45°|V\rangle + \sin 45°|H\rangle = \frac{1}{\sqrt{2}}\big(|V\rangle + |H\rangle\big). \tag{6.138}$$

Hence, $|45°\rangle\langle 45°|V\rangle = \frac{1}{\sqrt{2}}|45°\rangle(\langle V|V\rangle + \langle H|V\rangle) = \frac{1}{\sqrt{2}}|45°\rangle$. Then, passing through the following horizontal polarizer transforms the system as

$$|45°\rangle \rightarrow \frac{1}{\sqrt{2}}|H\rangle\langle H|45°\rangle = \frac{1}{2}|H\rangle, \tag{6.139}$$

and the probability to hit the screen is $P = \left|\frac{1}{2}|H\rangle\right|^2 = \frac{1}{4}\langle H|H\rangle = \frac{1}{4}$, just as we saw before. This is yet another example showing how adding more blockage can actually increase the probability for a quantum particle to move through a device.

Our next step is to examine the quantum Zeno effect, but to do so requires us to incorporate polarization rotators into our quantum formalism. These objects act simply. A polarization rotator will rotate the polarization by and angle α. We model this by transforming a state $|\theta\rangle$ to $|\theta + \alpha\rangle$. It is that easy!

We now analyze a four-stage quantum Zeno effect, which has four 22.5° polarization rotators sandwiched between five vertical polarizers, as shown in Fig. 4.17, panel (c). We emerge from the first polarizer in the $|V\rangle$ state, which has $\theta = 0$. We will label it as $|0\rangle$ for this exercise. Then, the rotators and polarizers transform as

$$\underbrace{|0\rangle}_{\text{polarizer 1}} \rightarrow \underbrace{|22.5°\rangle}_{\text{rotator 1}} \rightarrow \underbrace{|0\rangle\langle 0|22.5°\rangle}_{\text{polarizer 2}} \rightarrow \underbrace{|22.5°\rangle\langle 0|22.5°\rangle}_{\text{rotator 2}} \rightarrow \underbrace{|0\rangle\big(\langle 0|22.5°\rangle\big)^2}_{\text{polarizer 3}} \rightarrow$$

$$\rightarrow \underbrace{|22.5°\rangle\left(\langle 0|22.5°\rangle\right)^2}_{\text{rotator 3}} \rightarrow \underbrace{|0\rangle\left(\langle 0|22.5°\rangle\right)^3}_{\text{polarizer 4}} \rightarrow \underbrace{|22.5°\rangle\left(\langle 0|22.5°\rangle\right)^3}_{\text{rotator 4}}$$

$$\rightarrow \underbrace{|0\rangle\left(\langle 0|22.5°\rangle\right)^4}_{\text{polarizer 5}}. \tag{6.140}$$

But $\langle 0|22.5°\rangle = \langle V|V\rangle \cos 22.5° + \langle V|H\rangle \sin 22.5° = \cos 22.5°$. So, the probability for a photon to emerge from this four-state quantum Zeno effect is

$$P = \left||V\rangle\left(\langle 0|22.5°\rangle\right)^4\right|^2 = \left(\cos 22.5°\right)^8 \langle V|V\rangle \approx 0.5308, \tag{6.141}$$

or about 53.5% of photons make it through the device. You will evaluate some more complex cases in the end of section problems.

We have two more items we need to cover about polarization of light. The first is how to describe a polarizing beam splitter, and the second is to show how a quantum seeing in the dark experiment works. A polarizing beam splitter is shown in Fig. 4.24. The figure shows that the reflected light is horizontally polarized, and the transmitted light is vertically polarized (but this can change depending on the orientation of the polarizing beam splitter). We describe this quantum mechanically, by taking the input light and separating its $|V\rangle$ and $|H\rangle$ components to travel on the transmitted and reflected paths, respectively. So, if the initial photon state $|\psi\rangle$ is given by the superposition $|\psi\rangle = \alpha|V\rangle + \beta|H\rangle$, with $|\alpha|^2 + |\beta|^2 = 1$, then the polarizing beam splitter transforms $|\psi\rangle$ into a transmitted photon amplitude given by $\alpha|V\rangle$ and a reflected photon amplitude given by $\beta|H\rangle$, so

$$|\psi\rangle \rightarrow \begin{cases} \alpha|V\rangle & \text{(transmitted)} \\ \beta|H\rangle & \text{(reflected)} \end{cases}. \tag{6.142}$$

Similarly, the beam splitter also does the opposite. It takes an $\alpha|V\rangle$ photon amplitude, which is transmitted, and a $\beta|H\rangle$ photon amplitude, which is reflected, and combines both amplitudes into the superposition $|\psi\rangle = \alpha|V\rangle + \beta|H\rangle$ into the output channel of the beam splitter. Schematically, we have

$$\left.\begin{matrix} \alpha|V\rangle & \text{(transmitted)} \\ \beta|H\rangle & \text{(reflected)} \end{matrix}\right\} \rightarrow |\psi\rangle. \tag{6.143}$$

Note that the horizontal or vertical polarization orientations depend on the orientation of the polarizing beam splitter—for the quantum seeing in the dark, it is rotated by 90° which makes the transmitted light horizontally polarized and the reflected light vertically polarized.

In Fig. 4.25, we show a schematic for the quantum seeing in the dark experiment (the reflected path of the Michelson-Morley interferometer is the path where a blockage would be placed). Our polarizing beam splitter is rotated by 90° relative

to how it is oriented in Fig. 4.24, so the transmitted light is horizontally polarized and the reflected light is vertically polarized in this experiment. The light enters in the $|H\rangle$ state, with a horizontally oriented polarization. The rotator transforms to the $|7.5°\rangle = \sin 7.5°|V\rangle + \cos 7.5°|H\rangle$ state. The polarizing beam splitter reflects the $\sin 7.5°|V\rangle$ up and transmits the $\cos 7.5°|H\rangle$ photon to the right. If there is no blockage on the up-down path, the photon reflects off the mirrors in both arms and recombines in the polarizing beam splitter into a $|-7.5°\rangle$ state, now moving to the left—the photon emerges with the same magnitude of polarization it had when it entered, but the sign changes due to the way polarization reflects off of mirrors ($\theta \to -\theta$). If, instead, there is a blockage on the up-down path, then the $|H\rangle$ component of the photon state is removed, and the photon emerges from the beam splitter, heading left, in the state $\cos 7.5°|H\rangle$, with a reduced amplitude.

In the meantime, the source and polarizer on the left are replaced by a mirror, so that subsequent passes through the device rotate the polarization by 7.5° twice, leading to a net 15° rotation of the polarization. As we mentioned, while going left, it has a negative polarization, the rotator rotates it more negative, and then, upon reflection, it changes sign and then is rotated in the positive direction as it passes through to the right. We pass through five more times, then remove the mirror, and pass the device through a horizontal polarizer. If there is no object blocking the path, then the polarization rotated by a total of $7.5° + 5 \times 15° + 7.5° = 90°$ and is now in the $|V\rangle$ state, so it is blocked by the polarizer. If there is an object blocking the path, then the photon is in the $|-7.5°\rangle$ state and has an amplitude for the $|H\rangle$ component of $\cos 7.5° \times \cos^5 15° \times \cos 7.5°$, so that the probability to pass through the polarizer is $P = \cos^4 7.5° \cos^{10} 15° = 0.6831$ or 68.3% of the time the experiment can detect the blockage without interacting with it. This is the same as what we analyzed before; the calculation is examined further in the problems.

We end this section with a discussion of measurement on a beam splitter and the so-called *separation fallacy*. Many quantum resources describe these experiments incorrectly, conflating a measurement with tagging. Let us recall, that tagging corresponds to the creation of a superposition (which can be to an entangled state and can be reversed), while measurement involves an irreversible change to the system. The beam splitter provides a nice way to discuss this misconception. The discussion is subtle, so pay close attention.

When the photon comes from the source, it is moving in the input channel toward the beam splitter. We use a ket $|\psi\rangle$ to describe the state of the photon, given by its color and polarization (it will turn out that the photon does not have a definite color, but only a *nearly definite* color, but we save that story for later). So the photon traveling from the source to the detector is described by the state $|\psi\rangle \otimes |\text{input}\rangle$. After going through the beam splitter, the state transforms as follows:

$$|\psi\rangle \otimes |\text{input}\rangle \to \frac{1}{\sqrt{2}}\Big(|\psi\rangle \otimes |A\rangle + |\psi\rangle \otimes |B\rangle\Big). \tag{6.144}$$

Here, the kets $|A\rangle$ and $|B\rangle$ denote the paths the photon is on. The state the beam splitter transforms to is a superposition state that is not entangled here (a polarizing

beam splitter would make it entangled, but an ordinary beam splitter does not). This is what a tagging operation does. (The outer product notation was introduced when we discussed matrices back in Sect. 5.7. The tensor product is similar and will be discussed more thoroughly in the next section.) Note that it is incorrect to say the beam splitter measures the photon in some fashion and it travels on path A or on path B. It is correct to say that the path the photon travels on is indeterminate, because it is evolving in a superposition state. This is the process that we call tagging. If it is reversible (as we know if we add two mirrors, have two equal-length paths, and direct the photon back onto a second beam splitter as in a Mach-Zehnder interferometer), then the photon combines back into one state, $|\psi\rangle \otimes |\text{bright}\rangle$, which moves along the single path to the bright detector. So, the tagging operation can be reversed. This is the signal to us that going through a beam splitter is not a measurement, it is not the same as going through a polarizer, which is a measurement.

So, what happens when I do measure in a detector just after the beam splitter? Since a single photon can only be measured once, only one detector will "fire" when the photon goes through the apparatus. After we have measured the photon, the collapse argument of the Copenhagen interpretation tells us that the photon was on the definite path that the detector was on. But it is incorrect to say that simply because we have a detector in place on the path behind a beam splitter (and before we have measured anything) the photon travels on one path or the other. It travels in superposition, and when we measure, we project either with the operator $\hat{\mathbb{I}} \otimes |A\rangle\langle A|$ or $\hat{\mathbb{I}} \otimes |B\rangle\langle B|$. Of course, the photon is destroyed after the measurement.

Now, we get philosophical. If we measured the photon in detector A, then can we say the photon traveled through the beam splitter on path A and was detected in the detector, or do we say it traveled in superposition and was collapsed to detector A after measurement? This question is a bit ill-posed, because both interpretations are valid (in other words, the measurement does not tell us one or the other occurred and both lead to the same measurement prediction). What we cannot say is the following. If we have the setup as in Fig. 6.5, so the photon travels on path A or path B when it is detected, but, now after the photon travels through the beam splitter but before it gets to the detectors, we remove the detectors (as in a delayed-choice experiment). Then, because the detectors were removed, information travels backward in time to tell the photon to change from moving on definite paths to moving in a superposition. This is a *retrocausal argument*, and it is incorrect. The proper interpretation is that the photon travels in superposition until measured, but, once we have measured it on a particular path, we can infer that it did definitely travel along that path to the detector in this "shot" of the experiment. Such a statement is validated by the fact that the measurement in the detector changes the facts we know, and it is impossible to travel along B and be detected in A. Hence, the measured photon did travel on a definite path to the detector. Quantum mechanics is complicated!

To summarize, we have shown that the same methods used to describe the Stern-Gerlach experiment can also be used to describe experiments that describe polarized

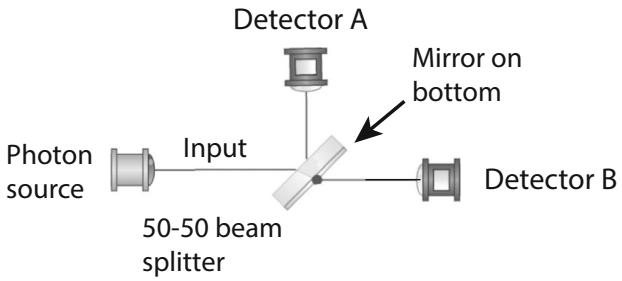

Fig. 6.5 Beam splitter used to illustrate the separation fallacy. A photon is incident from the left. After passing through the beam splitter, it is transformed into the superposition (but not entangled) state $|\psi\rangle \otimes |\text{input}\rangle \rightarrow \frac{1}{\sqrt{2}}(|\psi\rangle \otimes |A\rangle + |\psi\rangle \otimes |B\rangle)$. It can be measured in only one detector. After measurement, we can infer which path it took, but it is incorrect to say the path becomes definite after going through the beam splitter and before measurement. Only after measurement can we make such an inference. Colloquially, we say the measurement collapsed the wavefunction, but what we mean by that is our information about the photon has changed due to the measurement

light. But this is not all. The two-level quantum systems can be used to describe many more quantum examples. We will use it again many times in the remainder of the book.

Section 6.5 Problems:

6.5.1 Change in Basis for Polarization

The state $|\theta\rangle = \cos\theta|V\rangle + \sin\theta|H\rangle$, for $0 \leq \theta \leq 90°$, is orthogonal to the state $|\theta + 90°\rangle = -\sin\theta|V\rangle + \cos\theta|H\rangle$. Determine how to expand both states $|V\rangle$ and $|H\rangle$ in terms of the orthonormal basis
$\{|\theta\rangle,\ |\theta + 90°\rangle\}$. That is, find α, β, γ, and δ such that

$$|V\rangle = \alpha|\theta\rangle + \beta|\theta + 90°\rangle \ \text{ and } \ |H\rangle = \gamma|\theta\rangle + \delta|\theta + 90°\rangle. \tag{6.145}$$

6.5.2 Multiple Polarizers

Use our quantum formalism to analyze the the multiple polarizer experiment shown in Fig. 4.22 and determine the probability to make it through the three polarizers. If you want some more practice, analyze the other "polarizer madness" questions in a similar fashion.

6.5.3 Quantum Zeno Effect

Determine the probability that a photon gets through an N-stage quantum Zeno effect with sandwiches of polarizers and rotators, with $N = 6, 9, 15$, and 45 stages. *Hint*: The $N = 4$ case was analyzed in the text and gave a 53.08% chance for success.

6.5.4 Quantum Seeing in the Dark

Assume we have a six-stage (polarization rotator of 7.5°) quantum seeing in the dark experiment, as described in the text above. Show that the probability to emerge from the experiment at the end is 0% if there is no blockage on the up-down arm.

For the same quantum seeing in the dark experiment, go through all the steps to show that the probability to detect an obstruction is 68.3%.

How many stages are there if the polarization rotator rotates by 3°? Determine the probability to detect the obstruction for this experiment.

6.5.5 Inferring Versus Measuring

Write an essay about the beam splitter discussion on the separation fallacy and discuss the differences between what we actually measure and what we can infer from what we measured. In particular, compare and contrast the idea of moving in superposition versus inferring motion along a definite path after a measurement has been completed. Do your conclusions have anything to say about whether we should think of the quantum states we use to describe these systems as being real or being computational tools that allow us to predict the results of the measurements?

6.5.6 Quantum Key Distribution II

The original BB84 proposal for quantum key distribution involved using polarization states of light measured on two different directions, either horizontal and vertical or on two perpendicular diagonals D and D' (45° and $-45°$). Write an essay that describes how the quantum key distribution works in this case. One of the benefits of using light for quantum key distribution is we can use a process called parametric down conversion, which creates two photons at the same time, each which can be set up to have opposite polarization. This can then be used as a source for a quantum key distribution setup that is likely to be much more efficient than a similar atom experiment. Discuss why in your essay as well.

6.6 Delayed Choice and Quantum Erasers Revisited

We end the chapter by examining our last set of spin and light experiments—those that involve Wheeler's delayed-choice option. Recall this was a situation where the properties of an apparatus are changed while the quantum particle is moving through the device. In Chap. 2, we saw our first quantum version of the experiment, which involved a "conspiracy theory" version based on instruction sets and on the particle knowing the experimental setup ahead of time. Since we found the delayed-choice experiment was actually described by the quantum theory, we will revisit these classes of theories using our new quantum formalism. Our first task will be to examine the spin-based systems and then we will examine a delayed-choice variant for the two-slit experiment of light. Let's get started.

The top panel of Fig. 6.6 shows the initial setup for a delayed-choice experiment. If an atom enters in a state $|\uparrow\rangle_z$, it goes through the horizontal analyzer loop and emerges with the same state it had as it entered, namely, $|\uparrow\rangle_z$; we can think of it as being expressed in the x-basis when in the arms of the analyzer loop as $\frac{1}{\sqrt{2}}(|\uparrow\rangle_x + |\downarrow\rangle_x)$, where it travels in superposition. It then enters the trailing analyzer and takes the + path, exiting with probability $P(+;z) = |_z\langle\uparrow|\uparrow\rangle_z|^2 = 1$.

If, instead, we watch at the arms of the analyzer loop with a pass-through detector, the state transforms from $|\uparrow\rangle_z$ to either $|\uparrow\rangle_x$ or $|\downarrow\rangle_x$, depending on which arm the atom was observed to be on as it moved through the first half of the analyzer loop. It then enters the + path of the final vertical analyzer with probability $P(+;z) = |_z\langle\uparrow|\uparrow\rangle_x|^2 = \frac{1}{2}$, or $P(+;z) = |_z\langle\uparrow|\downarrow\rangle_x|^2 = \frac{1}{2}$, because both the up state on the x-axis and the down state on the x-axis can go to the + exit of a vertical analyzer (indeed, half of the time, they go to the + exit and half of the time they go to the − exit).

If, instead, the analyzer is flipped to lie in the horizontal orientation, as shown in the bottom panel of Fig. 6.6, we analyze the quantum case as follows. The atom still enters in the $|\uparrow\rangle_z$, and if we do not observe at the arms, it emerges in the same

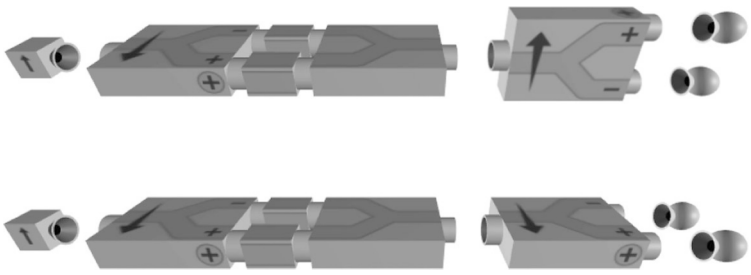

Fig. 6.6 The setup of a delayed-choice experiment with spin. The analyzer loop has pass-through tubes, which let the atoms move through without detecting anything. The analyzer loop is horizontal, in the $+x$ direction. The initial setup is in the top panel where the trailing analyzer is vertical. The alternative setup is in the bottom panel, where the trailing analyzer is horizontal

$|\uparrow\rangle_z$ state. Entering the horizontal analyzer, it will exit the + exit half of the time ($P(+;x) = |_x\langle\uparrow|\uparrow\rangle_z|^2 = \frac{1}{2}$) and the − exit half of the time ($P(-;x) = |_x\langle\downarrow|\uparrow\rangle_z|^2 = \frac{1}{2}$). On the other hand, if we do watch at the analyzer loop, we will see the atom move on the $+x$ arm half of the time and along the $-x$ arm half of the time. If it is on the $+x$ arm, it emerges from the analyzer loop in the $|\uparrow\rangle_x$ state, and hence, all atoms will exit the + exit of the analyzer (this occurs half of the time). If it took the − arm, the situation is opposite—it exits the analyzer loop in the $|\downarrow\rangle_x$ state, so it always exits the − exit of the final analyzer (this occurs half of the time). This result is identical to what is seen in the unwatched case, in terms of the probabilities at the detectors.

The above analysis was rather straightforward, with no surprises. A more interesting case, which we looked at in Chap. 3, is shown in Fig. 2.18 and corresponds to the case with an exciter on one arm. Note that the analyzer loop is oriented in the $-x$ direction here, and the exciter lies on the positive projection arm of the analyzer loop. We continue to have a perpendicular trailing analyzer and we now tag the atoms to determine which-way information. The tagging is done by shining light of a specific color onto the positive projection arm of the analyzer loop. This light will be absorbed by the atom and excite it from the ground state (denoted $|GS\rangle$) to the excited state ($|ES\rangle$); we assume only one excited state can be excited by the light. When the atoms are tagged, we have which-way information and the analysis proceeds exactly as above for the case where the atoms are watched. The new wrinkle we can add is to erase the which-way information after passing through the loop. If we do this, then the experiment should revert back to the unwatched case. We want to show, in detail, how the untagging works as a quantum eraser, using our new formalism.

Before we can jump into doing this, we need to develop a notation to describe the atom since it can be in a ground state or an excited state, and it has a spin projection on some axis. These two properties are independent of each other. We form the tensor product of the two states, which is like an outer product (and uses the same symbol), but it is for vectors, not matrices, so it is a bit different. Let us show what the notation is and then explain how it is used.

We express the atomic state first ($|GS\rangle$ or $|ES\rangle$) and the spin state second ($|\uparrow\rangle_z$ or $|\downarrow\rangle_{-x}$, *etc.*), so it looks like $|GS\rangle \otimes |\uparrow\rangle_z$. The circle-times symbol is used to keep the two states differentiated from each other, because they describe completely different and independent properties of the atom. The tensor product allows us to describe quantum states with different properties in this fashion, where we keep track of the separate properties in each factor of the tensor product. Quantum operators will typically act on only one of the two kets, but it is possible to have operators that act on both. The tensor-product bras are written in the same order, so we have the Hermitian conjugate of the state above is $\langle GS| \otimes_x \langle\downarrow|$, with the atomic state first and the spin state second. When we take the dot product of tensor-product bras and kets, it is equal to the ordinary product of the inner product bra-kets for like states with like states. For example,

$$\Big(\langle ES| \otimes {}_{-x}\langle\uparrow|\Big) \cdot \Big(|ES\rangle \otimes |\uparrow\rangle_z\Big) = \langle ES|ES\rangle \times {}_{-x}\langle\uparrow|\uparrow\rangle_z \tag{6.146}$$

is the inner product of two tensor product states, resulting in the product of two bra-kets. Note that if one pair of states in the tensor product is orthogonal, then the tensor product states are orthogonal—though this might be surprising, it is a fundamental property of tensor products. We used this same tensor-product methodology when we analyzed the watched two-slit experiments in Sect. 3.4. This product of bra-kets acts like all bra-kets—the norm is always positive—when the inner product vanishes, the vectors are orthogonal, and so on. You might ask why did we pick the atomic state to be first in the tensor product—there is no specific reason; we need to pick one of the vectors to be first and then to stick with that notation. In many cases, it is customary to put the spin state second.

We need to describe the different operations involved with tagging and untagging to perform our full analysis. As before, we will find it convenient to use the operators and bras and kets some of the time and matrices that represent the operators and spinors for the states other times. For example, we will use $\begin{pmatrix}1\\0\end{pmatrix}$ to describe the $|GS\rangle$ state and $\begin{pmatrix}0\\1\end{pmatrix}$ to describe the $|ES\rangle$ state—these states must be orthogonal, because they are energy eigenstates of the atom with different energies. The exciter maps the $|GS\rangle$ to excited state and *vice versa*, so we describe in two equivalent ways. The state and operator language is

$$|GS\rangle \rightarrow |ES\rangle \text{ and } |ES\rangle \rightarrow |GS\rangle, \tag{6.147}$$

while the matrix-spinor language uses the unitary matrix $\begin{pmatrix}0&1\\1&0\end{pmatrix}$ to describe the exciting operation via

$$\begin{pmatrix}1\\0\end{pmatrix} \rightarrow \begin{pmatrix}0\\1\end{pmatrix} \text{ and } \begin{pmatrix}0\\1\end{pmatrix} \rightarrow \begin{pmatrix}1\\0\end{pmatrix}. \tag{6.148}$$

We are now ready to describe the experiment shown in Fig. 2.18. We start in the $|GS\rangle$ and the $|\uparrow\rangle_z$ state of the atom or, using our tensor-product notation, in the state $|GS\rangle \otimes |\uparrow\rangle_z$. Going through the analyzer loop with the exciter on the positive projection branch, the atom will be excited from the ground state ($|GS\rangle \rightarrow |ES\rangle$) when we go on the positive projection branch, while the atom will stay in the ground state ($|GS\rangle$) when we go on the negative projection branch. The spin-state $|\uparrow\rangle_z$ must be expressed in the $-x$-basis, as $\frac{1}{\sqrt{2}}(|\uparrow\rangle_{-x} + |\downarrow\rangle_{-x})$ for us to determine the action of the tagging. So, we have that the analyzer loop transforms the states via

$$|GS\rangle \otimes |\uparrow\rangle_z \rightarrow \frac{1}{\sqrt{2}}\Big(|ES\rangle \otimes |\uparrow\rangle_{-x} + |GS\rangle \otimes |\downarrow\rangle_{-x}\Big). \tag{6.149}$$

It is important to recognize that this transformation is not a measurement. It is a tagging—we went from a product state to a superposition that is an entangled state.

This superposition state then goes into the next stage of the experimental apparatus. Suppose we had a vertical analyzer attached to the output of the analyzer loop to measure the z-component of the spin. Using our projective measurement rule, we apply the projection operator $\hat{I} \otimes |\uparrow\rangle_z\, {}_z\langle\uparrow|$ to our superposition state and then take the modulus squared of the projected state to determine the probability. The calculation is straightforward but has many steps. It is given by

$$
\begin{aligned}
P(+;z) &= \Big| |ES\rangle \otimes |\uparrow\rangle_z\, {}_z\langle\uparrow| \uparrow\rangle_{-x}\, {}_{-x}\langle\uparrow| \uparrow\rangle_z \\
&+ |GS\rangle \otimes |\uparrow\rangle_z\, {}_z\langle\uparrow| \downarrow\rangle_{-x}\, {}_{-x}\langle\downarrow| \uparrow\rangle_z \Big|^2 \\
&= \langle ES|ES\rangle\, {}_z\langle\uparrow| \uparrow\rangle_z \Big|\, {}_z\langle\uparrow| \uparrow\rangle_{-x}\, {}_{-x}\langle\uparrow| \uparrow\rangle_z \Big|^2 \\
&+ \langle GS|GS\rangle\, {}_z\langle\uparrow| \uparrow\rangle_z \Big|\, {}_z\langle\uparrow| \downarrow\rangle_{-x}\, {}_{-x}\langle\downarrow| \uparrow\rangle_z \Big|^2 \\
&+ \Big(\langle ES|GS\rangle + \langle GS|ES\rangle \Big)\, {}_z\langle\uparrow| \uparrow\rangle_z\, {}_z\langle\uparrow| \uparrow\rangle_{-x}\, {}_{-x}\langle\uparrow| \uparrow\rangle_z\, {}_z\langle\uparrow| \downarrow\rangle_{-x}\, {}_{-x}\langle\downarrow| \uparrow\rangle_z \\
&= \frac{1}{4} + \frac{1}{4} + 0 = \frac{1}{2}.
\end{aligned}
\tag{6.150}
$$

This results because $\langle ES|GS\rangle = \langle GS|ES\rangle = 0$ and $\langle GS|GS\rangle = \langle ES|ES\rangle = 1$. A similar calculation, which you should try, shows that $P(-;z) = \frac{1}{2}$. This result is the same result that we had when the atoms were watched, because we had which-way information. Here, however, we only tagged and then measured, and we did not even measure the tag. This is what happens with tagged systems—if we measure them without removing the tag, the end result is identical to what we would have gotten had we measured the tag. Technically, this follows from the orthogonality of the states involved in the tagged superposition state. This is an important property of quantum systems and being able to analyze experiments like this is an important skill to develop.

We need to describe a superpositioner operation to be able to remove the tag; it is shown in two different forms of the quantum eraser experiment in Figs. 2.19 and 2.20. The superpositioner operation is "half" of an exciter and is described by the unitary matrix $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, so that

$$
\begin{pmatrix} 1 \\ 0 \end{pmatrix} \to \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{or} \quad |GS\rangle \to \frac{1}{\sqrt{2}} \big(|GS\rangle + |ES\rangle \big) \tag{6.151}
$$

$$
\begin{pmatrix} 0 \\ 1 \end{pmatrix} \to \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad \text{or} \quad |ES\rangle \to \frac{1}{\sqrt{2}} \big(|GS\rangle - |ES\rangle \big). \tag{6.152}
$$

This sum or difference of states means that the atom no longer has definite energy. We call this the superposition of two energy eigenstates. But the superpositions are

different, because one has a + in it and one has a −, and we can set up an experiment to distinguish them. Note that applying the unitary matrix twice returns us back to the original state—so this matrix is its own inverse as well—a property called *involution*. In other words, we have

$$\begin{pmatrix}1\\0\end{pmatrix} \to \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix} \to \begin{pmatrix}1\\0\end{pmatrix} \text{ and } \begin{pmatrix}0\\1\end{pmatrix} \to \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix} \to \begin{pmatrix}0\\1\end{pmatrix}, \tag{6.153}$$

or

$$|GS\rangle \to \frac{1}{\sqrt{2}}\big(|GS\rangle + |ES\rangle\big) \to |GS\rangle \text{ and } |ES\rangle \to \frac{1}{\sqrt{2}}\big(|GS\rangle - |ES\rangle\big) \to |ES\rangle. \tag{6.154}$$

This means we can undo the superpositioner operation by applying it twice in a row.

We use the entangler operation to untag the system via the following two-step process: The atom starts in the state

$$\frac{1}{\sqrt{2}}\Big(|ES\rangle \otimes |\uparrow\rangle_{-x} + |GS\rangle \otimes |\downarrow\rangle_{-x}\Big), \tag{6.155}$$

after emerging from the analyzer loop (we substituted in ${}_{-x}\langle\uparrow|\uparrow\rangle_z = {}_{-x}\langle\downarrow|\uparrow\rangle_z = \frac{1}{\sqrt{2}}$). Now, we apply the superpositioner to get

$$\frac{1}{2}\Big((|GS\rangle - |ES\rangle) \otimes |\uparrow\rangle_{-x} + (|GS\rangle + |ES\rangle) \otimes |\downarrow\rangle_{-x}\Big), \tag{6.156}$$

and then we filter the system to measure only the atoms in the ground state (how we do this is described next). This means we project onto the state $|GS\rangle$ using the projection operator $|GS\rangle\langle GS| \otimes \hat{\mathbb{I}}$ to obtain

$$\frac{1}{2}|GS\rangle \otimes \big(|\uparrow\rangle_{-x} + |\downarrow\rangle_{-x}\big) = \frac{1}{\sqrt{2}}|GS\rangle \otimes |\uparrow\rangle_z. \tag{6.157}$$

The probability to measure the atom in a definite z-state of spin after applying the ground-state filter is given by $P(+;z) = \frac{1}{2}\langle GS|GS\rangle\,{}_z\langle\uparrow|\uparrow\rangle_z = \frac{1}{2}$ and $P(-;z) = 0$. So, where did the rest of the probability go? Because we used a ground-state filter, we only measured the atoms in the ground state, which is half of the total atoms (since the other half are in the $|ES\rangle$ state). This is why the total probability is only $\frac{1}{2}$! Since all of the atoms we measure after the ground-state filter are measured to be in the $+z$ state, we have removed the tag and restored all of the measured atoms to the $+z$ state. This shows clearly that the tagging operation is reversible.

Perhaps, you noticed that if we instead applied an excited-state filter, and projected onto $|ES\rangle$ before measuring, our final projected state becomes $\frac{1}{\sqrt{2}}|ES\rangle \otimes |\downarrow\rangle_z$—in this case, the probability to exit the + exit is zero and from the − exit is

$\frac{1}{2}$. This seems very odd, because we started in the $|\uparrow\rangle_z$ state and all our experiment did was to tag and untag. But because we did this by expanding the state in terms of the $|\uparrow\rangle_x$ and $|\downarrow\rangle_x$ states, we can transform the tagged state into *either* the $|\uparrow\rangle_z$ state (by filtering to the ground state) or onto the $|\downarrow\rangle_z$ state (by filtering to the excited state). The ground-state filter is our quantum eraser. The excited-state filter is something else, allowed and different from the quantum eraser, but consistent with the rules of quantum mechanics. Indeed, the presence of these states underlies why we cannot completely remove the tag, and instead, we can only remove the tag half of the time. This appears to be a common property of delayed-choice quantum eraser experiments. If we think of the analyzer loop as a quantum eraser, that is an exception to this rule, as it really does nothing, allowing 100% of the atoms to go through the device—the tagging and untagging really is complete. But the analyzer loop is not a delayed-choice apparatus because it tags and then immediately untags.

There are many ways to project onto one of the atomic states $|GS\rangle$ or $|ES\rangle$. We will describe two methods here. One way is to put the atoms into a highly polished metal cavity, which can be tuned to coax the atom to lose a photon and make a transition $|ES\rangle \rightarrow |GS\rangle$. The transition can be detected by measuring the number of photons in the cavity nondestructively. So, we measure the number of photons before the atom enters, let the atom enter, and then measure the number of photons afterward. If the number is the same, we had a $|GS\rangle$ atom. If it is one more, we had a $|ES\rangle$ atom. Then, we use post-selection, based on the final number of photons in the cavity, to determine which results we keep, depending on whether we want a ground-state or an excited-state filter. Note that the cavities are not perfect, so photons can be absorbed (meaning we transform from the ground state to the excited state), the $|ES\rangle$ atom might not be coaxed to emit the photon, or a $|GS\rangle$ atom may absorb a photon. Some, but not all, of these issues can be corrected by the photon count measurements. A second way is to shine laser light onto the atom that allows the excited state to be excited to an even higher energy excited state, but which does not excite the $|GS\rangle$. Then, by continuing to shine the light on the atom, we will create what is called a cycling transition; the atom will repeatedly absorb and emit light of the frequency of the laser as the atom transitions between the first and second excited state and back. The emitted light is emitted in all directions, and so the light emitted from the cycling transition can be easily measured. This makes the excited-state atom bright and the ground-state atom dark. Such detection of atomic states is used in ion-trap-based quantum computers to detect the different states of the qubits. I am always awed when I think about how this works. It always reminds me of the Dr. Seuss book *Horton Hears a Who* and how we can detect objects too small to be seen directly by using clever methods to amplify our ability to "see" them.

Now wait a minute, you might say. We did not really erase anything. When atoms go through the analyzer loop and are watched (or tagged), we already know that half of the atoms will exit through the + exit and half through the − exit of a perpendicular analyzer. All we are saying with the eraser is "just look at the ones going through the + exit." So what is the big deal? But it is deeper than this, because

atoms going through the + exit are now all $|GS\rangle$ atoms and those through the − exit are now all $|ES\rangle$ atoms. Yet, when we tagged them, all $|GS\rangle$ atoms were $|\downarrow\rangle_{-x}$ and all $|ES\rangle$ atoms were $|\uparrow\rangle_{-x}$. So, if we did not erase the tag, the atoms exiting the + exit would be $|GS\rangle$ half of the time and $|ES\rangle$ half the time. So, the erasure operation clearly is doing something. What appears to be a required result for these types of delayed-choice experiments is that the erasure only works half of the time—when the $|GS\rangle$ atom exits the + exit. We will see the same result happen again when we look at a delayed-choice two-slit experiment with light. To a quantum eraser purist, we would say the quantum erasure experiment is successful only half the time (when it produces $|GS\rangle \otimes |\uparrow\rangle_z$) and the other half of the time it produces a state that is different from any of the states we started with. So, we actually erase only half of the time, and we must discard the wrong states that we created, via post-selection at the end of the experiment.

It is important to understand the difference between observing the atoms at the positive projection and negative projection arms with a pass-through detector, versus tagging the atoms. When the atoms are observed, they go into either the $|\uparrow\rangle_{-x}$ or $|\downarrow\rangle_{-x}$ state immediately, and half of the time a $|\uparrow\rangle_{-x}$ atom and half of the time a $|\downarrow\rangle_{-x}$ atom emerges from the analyzer loop. These atoms are not in superposition; they are in what is called a *mixed state*. We have encountered mixed states before when we had multiple events we were analyzing, each distinct event was analyzed with a different component of the mixed state. On the other hand, when we tag the atoms, our state is a superposition of tensor-product states, $\frac{1}{\sqrt{2}}\big(|ES\rangle \otimes |\uparrow\rangle_{-x} + |GS\rangle \otimes |\downarrow\rangle_{-x}\big)$, when it emerges from the analyzer loop. When we measure the spin with a vertical analyzer after the analyzer loop, we cannot tell the difference between these two states. In both cases, we see half of the atoms emerge from the + exit and half of the time they emerge from the − exit. The difference arises because the measurement is irreversible—it cannot be erased after going through the pass-through detector. In the Copenhagen interpretation, we say this happens because we collapsed the wavefunction at the moment of the measurement. The collapse is modeled by the projective measurement. You will see next that the two-slit experiment with light behaves in the same way—when tagged, the tag can be erased, but when measured, it cannot be undone.

We next treat the two-slit experiment with polarized light at each slit. We will assume the slits are so small that we can use just one arrow or one quantum state, to represent the photon traveling through each slit. We will use the notation $|\psi_A\rangle$ to denote the amplitude arrow associated with slit A and $|\psi_B\rangle$ to denote the amplitude arrow associated with slit B. We use the same notation developed in Sect. 6.5, namely, $|\theta\rangle$ denotes light polarized at an angle θ relative to the vertical. Now, our photon has two quantum properties: (i) the amplitude arrow and (ii) its polarization. We use the tensor-product symbol $\otimes$ again to create the notation for the full quantum state. A generic amplitude-polarization pair is then denoted by $|\psi\rangle \otimes |\theta\rangle$, where we place the probability amplitude first and the polarization state second (again, this order is up to us to decide). Recall that we call the position on the screen x and then denote the amplitude arrow at site x on the screen via $\langle x|\psi\rangle = \psi(x)$,

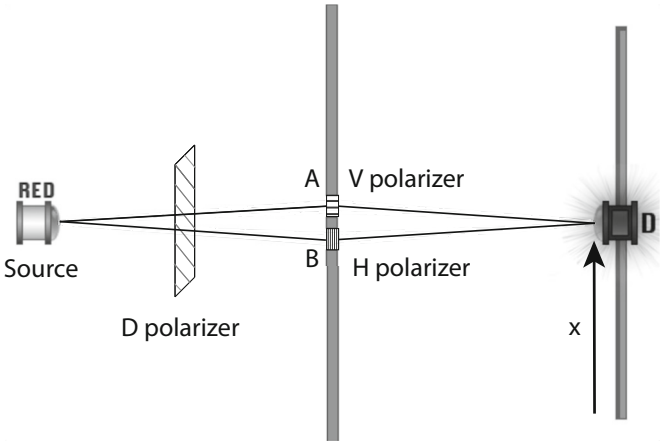

Fig. 6.7 In a two-slit experiment with polarizers, we send the source through a 45° oriented diagonal polarizer and then allowed to go through two slits, the top (A) with a vertical polarizer and the bottom (B) with a horizontal polarizer. We then measure the light on a screen behind the slits

which is an arrow or, equivalently, a complex number (modulus times a phase). So, $\psi(x) = re^{i\phi}$, with both r and ϕ typically changing as a function of x (position on the screen). We also can denote $\psi(x)$ via an arrow (clock hand) for each alternative way, as done in Chap. 3. The experiment is illustrated in Fig. 6.7. We now have all the machinery needed to analyze the details of the two-slit experiment with polarizers.

Before going any further, let us describe in words what these symbols mean. The state $|\psi\rangle = |\psi_A\rangle + |\psi_B\rangle$ is the set of rules we use to determine the arrows for each alternative way the event can occur and then combined to make the final arrow. For example, $|\psi_A\rangle$ would be the following rules: start at the source with an arrow of length 1 and pointing at 12 o'clock. As we move through the air to the top slit, wind the arrow according to the color of the photon, and then when we reach the top slit (labeled A), reduce the magnitude to 0.1. Finally, as we travel through air to the screen, continue to rotate the arrow. Note that we have not yet specified where on the screen we are going. This is because the state $|\psi_A\rangle$ is the abstract state constructed from the set of rules we just stated. Now, we combine the two arrows, by forming $|\psi_A\rangle + |\psi_B\rangle$, which is the set of rules for constructing the two arrows for the two alternative ways that travel through each slit. Finally, the bra $\langle x|$ tells us where on the screen we should compute these arrows and $\psi(x) = \langle x|\psi\rangle$ is the complex number identified with the final arrow. It is important to keep in mind the difference between the abstract quantum state (here, described by the set of rules) from the complex number (or arrow) that we find only when we specify where to determine the final arrow (wavefunction).

Let's repeat how our old calculation is done before showing how to do it with our new formalism. First, we work with the case with no polarizers. We assume the slits

are so narrow that there is only one arrow needed to represent the alternative way through each slit. We first note that 1% of the photons leaving the source and going through one slit (with the other slit closed) reach the detector. Hence, our arrow has a length of $\sqrt{0.01} = 0.1$. We have two arrows for the two alternative ways to go from the source to the detector, so the final arrow is of length ranging between 0 and 0.2, and the probability to be detected ranges from 0% to 4%, depending on how the two arrows are oriented.

Now, let us try this again with our new formalism. We will neglect the polarization at first. So, to convert a quantum state to a probability amplitude, which is a complex number, we need to form a bra-ket. As we mentioned before, the bra-ket is a wavefunction, denoted $\langle x|\psi\rangle = \psi(x) = re^{i\phi}$ for the detector of finite width, centered at position x. In Chap. 3, we called $\psi(x)$ the final probability amplitude arrow for the event of leaving the source and registering in the detector. We form $|\psi\rangle$ by adding the alternative way through A and through B. Hence, $|\psi\rangle = |\psi_A\rangle + |\psi_B\rangle$ and $\langle x|\psi\rangle = \langle x|\psi_A\rangle + \langle x|\psi_B\rangle$, which is equal to the arrow for the alternative way through slit A + the arrow for the alternative way through slit B to create the final arrow for detection in D. Now, these arrows will be represented by complex numbers with a modulus that satisfies $r = 0.1$ and with a phase ϕ determined by the color of the light and how long it took the photon to go from the source to the detector. We denote these respective phases by ϕ_A and ϕ_B. Then, we find

$$\langle x|\psi\rangle = \langle x|\psi_A\rangle + \langle x|\psi_B\rangle = 0.1e^{i\phi_A} + 0.1e^{i\phi_B} = 0.1e^{i\phi_A}\left(1 + e^{i(\phi_B - \phi_A)}\right). \tag{6.158}$$

The probability is

$$P(x) = |\langle x|\psi\rangle|^2 = 0.01\left|1 + e^{i(\phi_B - \phi_A)}\right|^2. \tag{6.159}$$

But $|1 + e^{i\alpha}|^2 = (1 + e^{i\alpha})(1 + e^{-i\alpha}) = 2 + 2\cos\alpha = 4\cos^2\frac{\theta}{2}$. So, we have that $P(x) = 0.04\cos^2\left(\frac{\phi_B - \phi_A}{2}\right)$, which runs from 0% to 4%, just as before. The pattern changes with x because ϕ_A and ϕ_B both vary with x.

When we add in polarization, we have to work a little bit harder. If we send a photon through the device and there are no polarizers, then the photon will have some definite polarization, but we do not know what it is. But, for definiteness, we will assume that we send the light through a diagonal polarizer D that is oriented at a 45° angle before moving to the screen with the two slits. Then, we write the quantum state as $|\psi\rangle \otimes |45°\rangle$. We project onto $|x\rangle$ via the operator $|x\rangle\langle x| \otimes \hat{\mathbb{I}}$, which applies $|x\rangle\langle x|$ to the first ket and $\hat{\mathbb{I}}$ to the second ket and becomes $|x\rangle\langle x|\psi\rangle \otimes |45°\rangle$. Since the detector measures all polarizations, it acts like a projection $|V\rangle\langle V| + |H\rangle\langle H| = \hat{\mathbb{I}}$ acting on the polarization part of the tensor-product state, so we find the measurement involves taking the square of the norm of the final state, which gives

$$P(x) = \langle x|x\rangle |\langle x|\psi\rangle|^2 \langle 45^\circ|45^\circ\rangle = |\psi(x)|^2 = 0.04 \cos^2 \frac{\phi_B - \phi_A}{2}, \tag{6.160}$$

the same as before. Note that ϕ_A and ϕ_B depend on x, so this creates an oscillating pattern on the screen. Note that we are using the fact that $\langle x|x\rangle = 1$ in this derivation. This assumption requires the detectors on the screen to have a small width centered on x, and we only calculate the probability for each detector at the x value corresponding to the center of the detector. Of course, this is what we always have because detectors must have a finite width—we are just reiterating the assumption for you here that we have been using all along. There is a similar assumption about the polarizers that they filter out the polarization in a narrow window of angles about the specified angle. This is the way real experiments work.

Now, if we add polarizers at the slits, say a vertical polarizer on the top slit A and a horizontal one on the bottom slit B, then if the photon starts polarized at 45°, we have the quantum state for measurement in a detector at position x is

$$|x\rangle \langle x|\psi_A\rangle \otimes |V\rangle \langle V|45^\circ\rangle + |x\rangle \langle x|\psi_B\rangle \otimes |H\rangle \langle H|45^\circ\rangle, \tag{6.161}$$

and the probability for detection is the norm squared of this state, which is

$$\begin{aligned} P(x) &= \Big(\langle x|\psi_A\rangle^* \langle x| \otimes \langle V|45^\circ\rangle^* \langle V| + \langle x|\psi_B\rangle^* \langle x| \otimes \langle H|45^\circ\rangle^* \langle H|\Big) \\ &\quad \cdot \Big(|x\rangle \langle x|\psi_A\rangle \otimes |V\rangle \langle V|45^\circ\rangle + |x\rangle \langle x|\psi_B\rangle \otimes |H\rangle \langle H|45^\circ\rangle\Big) \\ &= |\langle x|\psi_A\rangle|^2 |\langle V|45^\circ\rangle|^2 \langle x|x\rangle \langle V|V\rangle + |\langle x|\psi_B\rangle|^2 |\langle H|45^\circ\rangle|^2 \langle x|x\rangle \langle H|H\rangle \\ &\quad + \langle x|\psi_A\rangle^* \langle x|\psi_B\rangle \langle V|45^\circ\rangle^* \langle H|45^\circ\rangle \langle x|x\rangle \langle V|H\rangle \\ &\quad + \langle x|\psi_B\rangle^* \langle x|\psi_A\rangle \langle H|45^\circ\rangle^* \langle V|45^\circ\rangle \langle x|x\rangle \langle H|V\rangle \\ &= |\langle x|\psi_A\rangle|^2 |\langle V|45^\circ\rangle|^2 + |\langle x|\psi_B\rangle|^2 |\langle H|45^\circ\rangle|^2 \\ &= 0.01 \cos^2 45^\circ + 0.01 \sin^2 45^\circ = 0.01, \end{aligned} \tag{6.162}$$

or 1% (note that the result is independent of the orientation of the initial polarizer, even though we explicitly calculated it for a polarizer along the diagonal). The interference effects are gone! This happens because the cross terms vanish due to $\langle V|H\rangle = \langle H|V\rangle = 0$. You might be confused as to why this is not 2%. This is because half of the time, the photons are absorbed by the polarizers at the slits.

That was the first phase of our delayed-choice experiment. The next phase is to introduce a polarizer at the detector. If we put a V polarizer in front of the detector, we only see light from slit A. If we put an H polarizer, we only see light from slit B. But, if we put the polarizer at 45°, we see light equally from both slits, and we should see the interference come back.

Let's work this out. Our quantum state at the detector centered at x is

$$\langle x|\psi_A\rangle\,\langle V|45^\circ\rangle|x\rangle\otimes|V\rangle+\langle x|\psi_B\rangle\,\langle H|45^\circ\rangle|x\rangle\otimes|H\rangle, \tag{6.163}$$

just before the detector. If the polarizer is vertical in front of the detector, then we project the state onto $|V\rangle\,\langle V|$ and find

$$P(x)=|\langle x|\psi_A\rangle|^2|\langle V|45^\circ\rangle|^2=0.01\cos^2 45^\circ=0.005. \tag{6.164}$$

This is exactly one-half the result for photons going through slit A only, which we expect, because half of the photons are blocked by the vertical polarizer at slit A. Note that the wavefunction contribution $\langle x|\psi_A\rangle$ can vary with x, but the polarization contribution $\langle V|45^\circ\rangle$ is always independent of x. You are asked to analyze the case of a horizontal polarizer placed just before the detector in the problems—it yields the same numerical result.

Now, we move on to the 45° orientation for the polarizer. The quantum state is the same as before, but now we project onto the $|45^\circ\rangle$ state, which means we act the operator

$$|45^\circ\rangle\,\langle 45^\circ|=\frac{1}{2}\Big(|V\rangle\,\langle V|+|H\rangle\,\langle V|+|V\rangle\,\langle H|+|H\rangle\,\langle H|\Big), \tag{6.165}$$

onto the state as a projector before we take the norm of the state for the probability. Projecting onto this diagonal polarization state then yields

$$\begin{aligned}&\frac{1}{2}\langle x|\psi_A\rangle\,\langle V|45^\circ\rangle\,|x\rangle\otimes\Big(|V\rangle+|H\rangle\Big)+\frac{1}{2}\langle x|\psi_B\rangle\,\langle H|45^\circ\rangle\,|x\rangle\otimes\Big(|V\rangle+|H\rangle\Big)\\ =&\frac{1}{2}\Big(\langle x|\psi_A\rangle\,\langle V|45^\circ\rangle+\langle x|\psi_B\rangle\,\langle H|45^\circ\rangle\Big)\,|x\rangle\otimes\Big(|V\rangle+|H\rangle\Big).\end{aligned} \tag{6.166}$$

Taking the norm squared gives

$$\begin{aligned}P(x)&=\frac{1}{2}\Big|\langle x|\psi_A\rangle\,\langle V|45^\circ\rangle\\&\quad+\langle x|\psi_B\rangle\,\langle H|45^\circ\rangle\Big|^2\langle x|x\rangle\Big(\underbrace{\langle V|V\rangle+\langle V|H\rangle+\langle H|V\rangle+\langle H|H\rangle}_{=2}\Big)\\&=\frac{1}{2}\times 0.01\times\left|\frac{1}{\sqrt{2}}+e^{i(\phi_B-\phi_A)}\frac{1}{\sqrt{2}}\right|^2\\&=0.005\Big(1+\cos(\phi_B-\phi_A)\Big)=0.01\cos^2\left(\frac{\phi_B-\phi_A}{2}\right).\end{aligned} \tag{6.167}$$

This shows the expected interference effects, but the amplitude of the oscillations is one-fourth the amplitude without any polarizers. This is because half of the photons are lost at the polarizers at the slits and half at the polarizer just before the detector.

The quantum eraser does restore the interference pattern, but at the expense of losing half of the photons hitting the screen. This experiment can even be done at home. There are YouTube videos that show it Vijayan (2012); see the video https://tinyurl.com/48vrzz5e. In some videos, you won't see a polarizer between the source and the two slits. This is possible because many laser light sources emit polarized light, so by properly rotating the laser source, one can achieve a diagonal polarization and then not require the diagonal polarizer. You have an essay below to discuss what happens if the polarizer is rotated away from 45°. You should find that as the angle of the initial polarizer approaches $\theta \to 0$ or $\theta \to 90°$, the interference effects disappear as well. Think about why this is so.

Be sure to note that this is a delayed-choice experiment—the choice of the orientation of the final polarizer (just before the detector) can be made after the photon travels through the two-slit experiment. As you can see from our analysis above, there is no need for any retrocausal (backward in time) argumentation to make this work. The formalism we developed naturally leads to the delayed-choice results. Unlike the beam splitter, we see this is true here even though the polarizers at the slits really do constitute an irreversible measurement. One can construct delayed-choice experiments either way.

It is comforting that our new quantum formalism shows how all of the weird quantum behavior we covered before is explained by a straightforward mathematical model. In the delayed-choice experiments, we see a reduction of the final probability by 50%—a result that appears to be generic for all delayed-choice experiments. Let us now explain why one needs this to happen if we want to think of the final polarizer as a simple filter. As we rotate the polarizer, we can see the photons coming from slit A only, from slit B only, or from a combination of both. We can think of the total pattern as being the sum of these two results. At 0°, we filter out the results from slit B (leaving the results from slit A), and at 90°, we filter out the results from slit A (leaving the results from slit B). When we are at a 45° angle for the polarizer, we filter out a different combination—we get contributions from both slits and the filtered pattern now has oscillations in it. We can think of the part filtered out as being proportional to $|\psi_A(x) - \psi_B(x)|^2$, leaving behind $|\psi_A(x) + \psi_B(x)|^2$. If the polarizer is oriented at $-45°$, the filtered and transmitted results switch. This situation is reminiscent of the quantum-seeing-in-the-dark experiment with the Michelson-Morley interferometer, which at first worked only half of the time at best. We were able to improve the efficiency by adding in an interaction-free measurement component, based on the quantum Zeno effect. It would be interesting to see if such ideas could be incorporated into these delayed-choice two-slit experiments and raise the amplitude of the interference pattern.

We want to reiterate that at no moment did our analysis require us to go backward in time. Yet, the quantum eraser experiment still allows us to choose, *after the quantum particles emerged from the device*, whether we have which-way information or not. One interesting point to ponder is developed in the problems for the Stern-Gerlach setup. Namely, what happens if we do the projection measurement to determine the state that the atoms is in *after* we measure its projection on the

vertical axis? You will find that all the atoms exiting the + exit are in the $|GS\rangle$ and all exiting the − exit are in the $|ES\rangle$. Hence, we can choose whether we want to erase the which-way information long after the spin measurements are done! There are subtle implications for the separation fallacy as well—in the Stern-Gerlach experiments, we tag but do not measure, when going through an analyzer loop, but in the two-slit experiment, we measure when we go through a polarizer because this is an irreversible transformation, and we cannot reconstruct the original polarization after moving though the polarizer. But we still create an entangled superposition, because we have two alternative ways to go from source to detector—through slit A or slit B—so we emerge from the polarization measurement in an entangled state, which is identical to what we normally refer to as a tagged state. The measurement at the polarizers acts primarily to reduce the overall probability by 50%. We should think of this as a partial measurement that creates an entangled state. It may sound like this is the separation fallacy again, but the key is that we end up in an entangled state. In the separation fallacy, we end up in a mixed state. The latter occurs when one performs a partial measurement on an entangled state. But here, because we started from a product state as we move from source to the screen with the slits (as a photon of definite polarization), we end up in an entangled state. This is somewhat like an "inverse separation fallacy."

We end this section with a discussion of how we can use the same methodology to analyze the watched two-slit experiments we explored in Chap. 3 and shown in Figs. 3.21 and 3.22. This setup has a detector at slit A only. The detector can be described in terms of two states $|Y\rangle$ to indicate it detected a particle and $|N\rangle$ to indicate it has not yet detected a particle. The detector works by being "primed" to trigger and waiting for a trigger. It stays in the primed state until triggered. The first state $|Y\rangle$ can be thought of as "yes" the detector did fire and $|N\rangle$ denotes "no" the detector did not fire. Of course, these states are orthogonal, and we take them to be unit norm, so $\langle Y|Y\rangle = \langle N|N\rangle = 1$ and $\langle Y|N\rangle = \langle N|Y\rangle = 0$. Then, a two-slit experiment with a detector at slit A, which works 100% of the time, is described by the entangled state:

$$|x\rangle \langle x|\psi_A\rangle \otimes |Y\rangle + |x\rangle \langle x|\psi_B\rangle \otimes |N\rangle, \tag{6.168}$$

for the detector at position x. If, on the other hand, the detector works only 64% of the time, the state is instead given by

$$|x\rangle \langle x|\psi_A\rangle \otimes \Big(0.8|Y\rangle + 0.6|N\rangle\Big) + |x\rangle \langle x|\psi_B\rangle \otimes |N\rangle. \tag{6.169}$$

We need to carefully explain where this comes from. First, the second piece of the superposition involving the arrow for the alternative way through slit B is unchanged, since there is no detector there. As for the part of the quantum state from slit A, since it works 64% of the time, this corresponds to an arrow of length 0.8, because $0.8^2 = 0.64$. Then, the other component, where the detector does not fire, has a length of 0.6, because $0.6^2 = 0.36$. You will have a problem below to

calculate the probabilities for different experiments. You will find we get the same results as we determined previously.

In this section, we have discussed superposition, mixed states, and entangled states. Thinking back to our earlier model, with drawing arrows on a piece of paper, the superposition state corresponded to a situation where we had more than one alternative way for an event to occur. A mixed state corresponded to the case where we had multiple independent events, but each occurred with their own probabilities. Entangled states are superpositions of quantum states with more than one degree of freedom that cannot be separated into different factors. They correspond to superpositions that have which-way information. Superpositions lead to interference effects. Mixed states tend to reduce interference effects. Entanglement creates a system that has properties where the whole is different from the sum of the parts. They often suppress interference effects. All are intrinsic quantum properties, which make the world a marvelous and bizarre place.

Section 6.6 Problems:

You should assume for all problems involving two-slit experiments that the slits are so narrow that we can use a single path, corresponding to a single arrow, corresponding to a single $\langle x|\psi\rangle$, for the alternative way that goes through that slit.

6.6.1 Horizontal Polarizer at the Detector

Analyze the two-slit experiment with V and H polarizers at the slits, in the case where a H polarizer is placed just before the detector.

6.6.2 Diagonal Polarizers at the Slits

Describe how to carry out a two-slit experiment with polarizers that has the polarizer on slit A in the diagonal (45°) and the polarizer at slit B in the other diagonal ($-45°$). What direction should the polarizer between the source and the slit mask be? How do the results depend on the orientation of the polarizer placed just before the detector.

6.6.3 Unpolarized Delayed-Choice Two-Slit Experiment

Consider a delayed-choice two-slit experiment that has no polarizer between the source and detector and which creates unpolarized light (light that has a random polarization for each photon). Run the experiment many times and average over the polarization θ for each "shot" of the experiment. Use the fact that the average of

$\cos^2\theta = \frac{1}{2}$ when averaged over randomly chosen θ. This latter result can be found from recalling that $\cos^2\theta + \sin^2\theta = 1$ and that the average of $\cos^2\theta$ and of $\sin^2\theta$ are the same, when averaged over random θ.

6.6.4 Rotating the Initial Polarizer

Write an essay about the delayed-choice two-slit experiment, with the V polarizer at slit A and the H polarizer at slit B. Describe what happens to the experiment when we rotate the orientation of the polarizer between the source and the slit mask.

6.6.5 Partially Working Detector

Use the formalism we developed to calculate the pattern on the screen for a two-slit experiment with a detector at slit A that works 64% of the time. Repeat for a detector that works 36% of the time.

6.6.6 How to Create Entangled States

Creating entangled quantum states is not simple. In this section, we learned about two ways to create entangled states. One was with the analyzer loop that had an exciter attached on one of its arms. The other was with a two-slit experiment that has polarizers in it. In one case, the entanglement creation was done reversibly, in the other, irreversibly, by including a measurement. Explain how these two ways of creating entanglement are similar and how they are different. Can you think of other ways to create entangled states?

6.6.7 Two Detectors

Modify our formalism to describe the two-slit experiment with detectors that work 100% of the time and are placed at slits A and B. Note that we have to modify our quantum state to include an extra tensor product for the detector at slit B, so a general quantum state involves sums over terms like $|x\rangle\langle x|\psi_{A \text{ or } B}\rangle \otimes |Y \text{ or } N\rangle_A \otimes |Y \text{ or } N\rangle_B$.

6.6.8 Three Slits

Examine a three-slit experiment with detectors that work 100% of the time and all slits are the same size. Consider the case with one detector at slit A only and the case where there are two detectors—one at slit A and one at slit B. If you are brave, generalize to detectors that work only some of the time.

6.6.9 Delayed-Choice Stern-Gerlach

Complete the analysis of the Stern-Gerlach experiment shown in Fig. 2.20, with the de-exciters (which project onto the $|GS\rangle$ or $|ES\rangle$ states) positioned well after the vertical analyzer. You should find that the atom is in the state $\frac{1}{\sqrt{2}}|GS\rangle \otimes |\uparrow\rangle_z$ or the state $\frac{1}{\sqrt{2}}|ES\rangle \otimes |\downarrow\rangle_z$ depending on whether it exits the $+$ exit or the $-$ exit. Use the fact that a state of the form $|\psi_1\rangle \otimes |\uparrow\rangle_z + |\psi_2\rangle \otimes |\uparrow\rangle_z$ can be rewritten as $\big(|\psi_1\rangle + |\psi_2\rangle\big) \otimes |\uparrow\rangle_z$.

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Chapter 7
The Simple Harmonic Oscillator

A while back I bought an idiot's guide to [quantum mechanics], from which I learned only that I hadn't yet reached the rank of an idiot.

Joe Bennett, Journalist, 2007

7.1 Position, Momentum, and Energy Operators

Until this point, we have been solving problems that involve just two states or are reducible to two-state problems. While we could examine a plethora of complex quantum phenomena, it is not sufficient to describe the behavior of electrons in atoms or how quantum particles interact with each other. That all changes now as we work to set up the quantum mechanics of the atom. There is much for us to do before we get to the hydrogen atom. We start, in this chapter, with the simple harmonic oscillator in one dimension. An overview of the chapter is also available at https://tinyurl.com/mryj2m6y.

To begin, we need to determine the important quantum operators. It turns out quantum operators that represent observables are self-adjoint and have representations in terms of Hermitian matrices (even infinite-dimensional ones). Self-adjoint means the Hermitian conjugate of the operator (or its matrix representation) is equal to the operator (or its matrix) and that the domain of the operator and its Hermitian adjoint are the same; Hermitian, or symmetric, means just that the operator is equal to its Hermitian conjugate when it acts on states that both operators can act on. The domain is simply the set of vectors that we are allowed to apply the operator onto where the result is a vector that is still in our Hilbert space (recall operators map vectors to vectors). For a finite-dimensional system, we need not worry about the domain issue, so self-adjoint and Hermitian are synonymous. For an infinite-dimensional system, resolving the domain issue is a complex topic that requires math far beyond the math used in this book. So, unless told otherwise, you can think

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of self-adjoint and Hermitian as being essentially synonymous, but just keep in the back of your head that they are different, and sometimes the difference is important. We will also encounter operators that are not self-adjoint—we can always work with such operators, but they are not observables which are guaranteed to have real eigenvalues.

So, what are the important operators for a quantum particle? We have position $\hat{x}$, momentum $\hat{p}$, and energy $\hat{H}$, where the hats all denote that the objects are operators. Just like we did before, we start with the eigenstates (equivalent to eigenfunctions and eigenvectors). Remember, an eigenstate of an operator is a state that has a definite value for that operator. For position, the eigenstate satisfies

$$\hat{x}|x\rangle = x|x\rangle, \tag{7.1}$$

which is a state $|x\rangle$ that has a definite position x. At first, you might think this equation looks like a silly equation, but what it says is a state $|x\rangle$ will return a number x times that state, when the operator $\hat{x}$ acts on the state. This is the classical eigenvalue-eigenvector relationship, since an operator acting on a state produces a number times that state. Similarly, for momentum, we have

$$\hat{p}|p\rangle = p|p\rangle. \tag{7.2}$$

Finally, for the energy, we have

$$\hat{H}|n\rangle = E_n|n\rangle, \tag{7.3}$$

where we use the notation $|n\rangle$ for an eigenstate of the Hamiltonian operator $\hat{H}$, which has an energy eigenvalue given by E_n.

We need to construct the Hamiltonian and discuss how to make predictions for what will be measured in experiment. Since the Hamiltonian is the operator for the total energy, it is equal to the kinetic energy operator plus the potential energy operator. Many of you know that kinetic energy is given by $\frac{1}{2}mv^2$. The formula for kinetic energy is often derived in a mechanics class from the work-energy theorem. It also can be derived from some simple assumptions within special relativity. But both of these are beyond the scope of this book. We assume the formula is known and familiar to you. The specific form is motivated by the facts that because larger objects are more difficult to move, the kinetic energy should be proportional to the mass of the object m. In addition, energy is a scalar, not a vector, and the simplest scalar is v^2, so it should be proportional to v^2. Note that this is just a motivation for what the kinetic energy is. It is not a formal derivation. In the end, we have arrived at the kinetic energy is equal to $\frac{1}{2}mv^2 = \frac{p^2}{2m}$, when we use the fact that momentum satisfies $p = mv$. This formula is written in one dimension, but a simple extension holds in three dimensions. If we denote the Cartesian components of the momentum operators as $\hat{p}_x$, $\hat{p}_y$, and $\hat{p}_z$, for the x-, y-, and z-directions, then the kinetic energy becomes (KE stands for kinetic energy)

$$\widehat{KE} = \frac{1}{2m}\left(\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2\right) = \frac{1}{2m}\hat{\vec{p}}^{\,2}. \tag{7.4}$$

The potential energy, on the other hand, is just a function of the Cartesian coordinates $\hat{r}_x$, $\hat{r}_y$, and $\hat{r}_z$, given by $V(\hat{r}_x, \hat{r}_y, \hat{r}_z)$; we use PE to abbreviate potential energy. The total energy operator, and hence the Hamiltonian, is

$$\hat{H} = \widehat{KE} + \widehat{PE} = \frac{1}{2m}\left(\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2\right) + V(\hat{r}_x, \hat{r}_y, \hat{r}_z). \tag{7.5}$$

Note that the hats on the arguments of V show that it is an operator and we will be using the notation $\hat{r}_\alpha$ instead of the simpler $\hat{x}$, $\hat{y}$, and $\hat{z}$ because it makes some of the formulas we work with simpler (especially summations over the Cartesian coordinates, which are indexed by α). But, for a one-dimensional system, it is much easier to work with $\hat{x}$ and $\hat{p}$, which is what we will use.

Let's review how we use quantum mechanics to predict the results of a measurement. If the system has been prepared in a quantum state $|\psi\rangle$, then the expectation value of an operator $\hat{O}$ is given by $\langle\psi|\hat{O}|\psi\rangle$, which is what one would find after averaging over many measurements on identically prepared systems. It is also interesting to examine the fluctuations about the mean for an experiment, denoted $\Delta\hat{O}$, which satisfies

$$\begin{aligned}\left(\Delta\hat{O}\right)^2 &= \langle\psi|\left(\hat{O} - \langle\psi|\hat{O}|\psi\rangle\right)^2|\psi\rangle \\ &= \langle\psi|\hat{O}^2|\psi\rangle - 2\langle\psi|\hat{O}\langle\psi|\hat{O}|\psi\rangle|\psi\rangle + \left(\langle\psi|\hat{O}|\psi\rangle\right)^2 \\ &= \langle\psi|\hat{O}^2|\psi\rangle - 2\left(\langle\psi|\hat{O}|\psi\rangle\right)^2 + \left(\langle\psi|\hat{O}|\psi\rangle\right)^2 \\ &= \langle\psi|\hat{O}^2|\psi\rangle - \left(\langle\psi|\hat{O}|\psi\rangle\right)^2. \end{aligned} \tag{7.6}$$

$\left(\Delta\hat{O}\right)^2$ is also called the variance and $\Delta\hat{O}$ is the standard deviation. Note that the fluctuations vanish if $|\psi\rangle$ is an eigenvector of $\hat{O}$, as you can easily verify—this is a nice way to verify that you have prepared the system in an eigenstate. As an example, consider measuring the energy in a quantum state $|\psi\rangle$. The expectation value of the energy is

$$E_{\text{ave}} = \langle\psi|\hat{H}|\psi\rangle = \langle\psi|\hat{H}\hat{\mathbb{I}}|\psi\rangle = \sum_n \langle\psi|\hat{H}|n\rangle\langle n|\psi\rangle, \tag{7.7}$$

by using the completeness of energy eigenstates given by $\sum_n |n\rangle\langle n| = \hat{\mathbb{I}}$, which is a generalization of our result for 2×2 matrices to any square matrix or operator. The completeness relation was inserted between the $\hat{H}$ and the state as our multiply-by-one trick. But, because of the energy eigenvalue relation in Eq. (7.3) and because $\langle n|\psi\rangle = \langle\psi|n\rangle^*$, we have

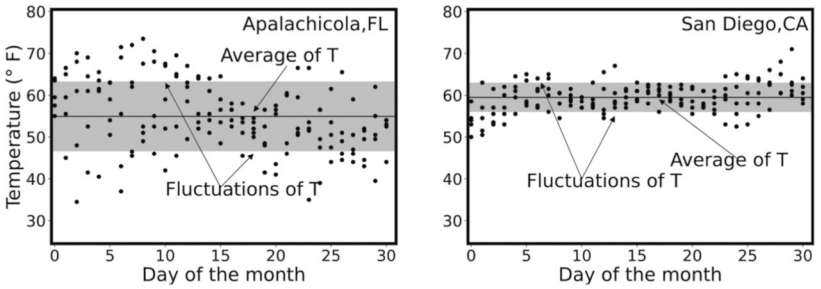

Fig. 7.1 Example of data with the mean and the variance displayed graphically. The data is for temperatures in the same month. In San Diego, the average temperature is a bit higher than in Apalachicola, but the variance in Apalachicola is significantly higher

$$E_{\text{ave}} = \sum_n \langle\psi|\hat{H}|n\rangle\,\langle n|\psi\rangle = \sum_n E_n |\langle n|\psi\rangle|^2, \tag{7.8}$$

which can be thought of as a summation $E_{\text{ave}} = \sum_n E_n P_n$, with $P_n = |\langle n|\psi\rangle|^2$ the probability that we find E_n as the result when we measure the energy for a system in the state $|\psi\rangle$. Next, we look at the formula for the quantum fluctuations (which is a jargon used for the variance about the mean) and is given by

$$\left(\Delta\hat{H}\right)^2 = \langle\psi|\hat{H}|\psi\rangle - E_{\text{ave}}^2 = \sum_n E_n^2 P_n - \left(\sum_n E_n P_n\right)^2. \tag{7.9}$$

Because we need to perform many measurements to determine the expectation value of an operator for a quantum system in the state $|\psi\rangle$, we can also measure the quantum fluctuations. These two quantities often give us interesting information about the system.

In Fig. 7.1, we plot the average temperature and the fluctuations about the mean of the temperature for one month in two different cities in the United States—Apalachicola on the left and San Diego on the right. The average temperatures are different, as can be seen by comparing the solid lines. The fluctuations are different too, as indicated by the gray regions. So, knowing the average and the fluctuations about the mean tells us more information about the data. We find this is especially true for a quantum system. For example, the fluctuations of different measurements are related by the Heisenberg uncertainty principle, which we will discuss in Sect. 7.7. The fluctuations can also tell us how close the state we prepared the system is to an eigenstate, because eigenstates have no fluctuations. We will be talking a lot about expectation values and quantum fluctuations throughout the book, and these figures provide us with some insight into why we will find this to be a useful thing to do.

In our discussion, we have been talking about expectation values and fluctuations after many experiments. The Copenhagen interpretation of quantum mechanics says we have a collapse of the state for each individual measurement, which means if we prepare the system in a state $|\psi\rangle$ and measure its energy, we find one specific value, given by one of the energy eigenvalues, say E_n, for which $\langle n|\psi\rangle \neq 0$. Furthermore, the quantum state is said to collapse from being in one state $|\psi\rangle$ to being in the state $|n\rangle$ for all times after the measurement (unless something else happens to the system). This state collapse is a formal way of describing a phenomenon we already know: "the atom is stupid" and only remembers the last experiment performed on it. This notion of collapse, odd as it may sound, helps us understand many of the bizarre results of quantum mechanics. The probability to measure E_n is given by $|\langle n|\psi\rangle|^2$. Note that the collapse of the state need not be a *physical event*. It can simply be thought of as an update of information about the system. We will have more to say about this throughout the book. A video https://tinyurl.com/4e72su9j discusses these ideas from a different perspective.

One more issue we need to point out. There is no ambiguity in what is meant by the Hamiltonian in Eq. (7.5), because $\hat{p}_i$ commutes with itself ($[\hat{p}_i, \hat{p}_i] = \hat{p}_i^2 - \hat{p}_i^2 = 0$ and also $[\hat{p}_i, \hat{p}_j] = 0$), so there are no issues with the ordering of the factors in the kinetic energy. We will also find that the position operators commute with themselves as well ($[\hat{r}_i, \hat{r}_j] = 0$), so the potential energy is also well defined, with no ordering issues, no matter how complicated the potential is.

Finally, you might be confused by questions like "what is the relationship between the actual position of an atom and the position operators?" or "what does $\hat{H}$ have to do with the energy my atom has?" In other words, what is the real meaning of these operators versus what the quantum particle is actually doing? As much as we would like to be able to answer these types of questions, we cannot. For there is no—this is what the particle is actually doing—it is all about the operators, the expectation values in quantum states, and how they interrelate. Thus, if we want to know the position of an atom in state $|\psi\rangle$, we don't find one answer, but we find many different answers, as the position is indeterminate. In the instance where we measure the position to be x_0, then the atom collapses into the state $|x_0\rangle$. If $|\psi\rangle$ was a state of definite energy, $|x_0\rangle$ is not, so measuring energy after measuring position will give a range of different energy values, rather than the one value we would have gotten had we measured energy in $|\psi\rangle$ before we measured position. And so on. It is time to leave the classical thinking along the lines of "but what is the particle really doing" behind us. We are moving on to thinking quantum mechanically from now on. If we want to measure something, we need the Hermitian operator corresponding to the measurement, and we follow the Copenhagen rules to tell us what to do. This methodology was coined by David Mermin as "shut up and calculate." Now, you are ready to shut up and calculate too, but you are also ready to start building up your quantum reasoning and quantum intuition as we develop the subject further. I do want to emphasize, however, that this discussion is idealized—we can never measure a position with perfect precision—so the real world must be analyzed in a more complex way, and we will carefully describe many of these complexities as we develop the quantum theory further. In addition, we have to carefully design

experiments that will yield measurable results—just because I have a Hermitian operator does not mean that I know how to construct an experiment that measures it. We will see in the world of the ultrasmall that it is quite difficult to create an experiment that can measure properties of individual quantum objects. We need to always remember this.

In this subsection, we have introduced three new operators, position, momentum, and energy, and discussed the expectation value and the fluctuations about the mean. Our next step will be to look at their commutators.

Section 7.1 Problems:

7.1.1 Fluctuations of an Eigenstate

If $|\psi\rangle = |n\rangle$, an energy eigenstate with energy E_n, which satisfies $\hat{H}|n\rangle = E_n|n\rangle$. Compute $\langle n|\hat{H}|n\rangle$ and $\langle n|\hat{H}^2|n\rangle - \Big(\langle n|\hat{H}|n\rangle\Big)^2$. Recall that $\langle n|n\rangle = 1$.

7.1.2 Probability and Norm Squared of Inner Products of Quantum States

We showed that in the expectation value $P_n = |\langle n|\psi\rangle|^2$. Show that this is a valid probability distribution by showing that $\sum_n P_n = 1$.

7.1.3 Repeated Measurements

Suppose we write $|\psi\rangle$ in an energy eigenbasis expansion, and we use completeness and multiply by one in the form

$$|\psi\rangle = \sum_n |n\rangle\,\langle n|\psi\rangle = \sum_n c_n|n\rangle, \tag{7.10}$$

with $c_n = \langle n|\psi\rangle$. An experiment is performed to measure the energy at time $t = 0$. The result is E_m. If we wait until time $t = 1$ and measure the energy again, do we know exactly what the result will be, or can it be one of many different results? Explain. *Hint*: use the state collapse in your argument and note that energy eigenstates are stationary states that do not change with time.

7.1.4 Expectation Values and Fluctuations in Superposition States

A state $|\psi\rangle = \frac{1}{2}|2\rangle + \frac{\sqrt{3}}{2}|4\rangle$ is prepared, where $|n\rangle$ is an eigenstate of $\hat{H}$ with energy E_n. Compute $\langle\psi|\hat{H}|\psi\rangle$ and $\langle\psi|\hat{H}^2|\psi\rangle - \left(\langle\psi|\hat{H}|\psi\rangle\right)^2$. Your answer will depend on E_2 and E_4.

7.1.5 Ordering of Operators

If $V(\hat{r}_x, \hat{r}_y, \hat{r}_z) = \frac{1}{2}k\hat{r}_x^2 + k\hat{r}_x\hat{r}_y + \frac{1}{2}k\hat{r}_y^2 + \frac{1}{2}k\hat{r}_z^2$, does it matter whether the second term is written as $k\hat{r}_x\hat{r}_y$ or $k\hat{r}_y\hat{r}_x$? Explain.

7.1.6 Vanishing Fluctuations Implies an Eigenstate

If $\left(\Delta\hat{H}\right)^2 = \langle\psi|\hat{H}^2|\psi\rangle - \langle\psi|\hat{H}|\psi\rangle^2 = 0$, can we conclude that $|\psi\rangle$ is an eigenstate of $\hat{H}$? *Hint*: recall that if the norm of a vector is zero, then the vector is the zero vector.

7.2 Deriving the Canonical Commutation Relations

We have now introduced three critically important operators: position $\hat{\vec{r}} = (\hat{r}_x, \hat{r}_y, \hat{r}_z)$; momentum $\hat{\vec{p}} = (\hat{p}_x, \hat{p}_y, \hat{p}_z)$; and energy $\hat{H} = \frac{1}{2m}\hat{\vec{p}}\cdot\hat{\vec{p}} + V(\hat{r}_x, \hat{r}_y, \hat{r}_z)$. These objects are operators because in quantum mechanics we must associate an operator like $\hat{r}_x$ with the x-position of the quantum particle. If the quantum particle is in the state $|\psi\rangle$, then, on average, the position will be found to be equal to $\langle\psi|\hat{r}_x|\psi\rangle$. Hence, it is useful to think of $\hat{r}_x$ *as the position* in some abstract sense, but determining its quantitative value requires us to compute an expectation value (and it usually fluctuates about the mean, so it does not have a definite value—it is instead indeterminate). This is a bit different from the classical situation where we don't have to introduce an operator, and we have no uncertainty. In quantum mechanics, everything involves working with these operators. Another source of confusion is a question along the lines of "OK, I have these operators, but I don't know what to do with them, or how to evaluate expectation values." This is fine. We are going to develop the rules you need to work with them in turn, just as we did for spin. But you already know the main rule—when an operator acts on its eigenvector, it returns the eigenvalue multiplied by the eigenvector. This property,

plus evaluating and manipulating commutators, will be the "bread and butter" for most of our work. So, we need to now go and determine some commutators!

Most textbooks will relate the commutators that we use in quantum mechanics to something called a Poisson bracket, which is developed in advanced classical mechanics courses (this is the way Dirac first motivated using commutators in quantum mechanics). But this requires you to have an advanced knowledge of classical mechanics to appreciate it and it sends the wrong message, in my opinion, indicating that we should think of quantum mechanics as a modification of classical mechanics. The real world is actually the other way around. Everything is quantum! Classical mechanics emerges as a limiting behavior of the quantum world. So, it is okay if you have no idea what a Poisson bracket is. We don't need it. Other textbooks will postulate a representation of the momentum operator in terms of a derivative with respect to x and then compute the commutator. This is also not needed, and since we are not using calculus, we will not proceed this way either. Finally, some textbooks motivate it from de Broglie's ideas relating to matter waves, but those discussions are often confusing and hard to follow. In the end, students find the introduction of the commutator to be confusing and poorly motivated.

What is most unfortunate is that it does not need to be this way. Herbert Green discussed how to physically motivate what the commutator is from experiment in his book *Matrix Mechanics* (Green, 1965). We follow the spirit of Green's approach here and we motivate what the commutator between position and momentum must be by some simple physics reasoning and the results of some simple experiments.

Our starting point are the operators for position, given by $\hat{r}_x$, $\hat{r}_y$, and $\hat{r}_z$. Our first result is that the position operators all commute with each other ($[\hat{r}_i, \hat{r}_j] = 0$). This follows, because we should be able to determine the position of a quantum particle in all three spatial dimensions, and only commuting operators can have simultaneous eigenfunctions (discussed in Sect. 6.1). The reason why we insist upon this is simple—if we measure a position in the x-direction, it should have no bearing on the measurement in the y-direction, or the z-direction, because we should be able to have states that have definite position in our theory. If you do not like this argument, then just take as a fact that the position operators all mutually commute as a postulate of quantum mechanics. In either case, the conclusion is the same, $[\hat{r}_x, \hat{r}_y] = 0$, $[\hat{r}_y, \hat{r}_z] = 0$, and $[\hat{r}_z, \hat{r}_x] = 0$. Note that this further implies that the position operators commute with any function of position. While this may seem obvious to you (or may seem completely unclear), it follows from a simple argument that we will develop next.

To begin, if $[\hat{r}_x, \hat{r}_y] = 0$, then $[\hat{r}_x, \hat{r}_y^2] = 0$ by using the Leibniz rule for commutators of products of operators

$$[\hat{r}_x, \hat{r}_y^2] = \hat{r}_y[\hat{r}_x, \hat{r}_y] + [\hat{r}_x, \hat{r}_y]\hat{r}_y = 0, \tag{7.11}$$

since each commutator on the right-hand side vanishes. By repeating this and applying to all combinations of directions, we find $[\hat{r}_i, \hat{r}_j^n] = 0$ for all n. Hence, the position operators, and their arbitrary powers and products with each other, all commute. This then implies that $[\hat{r}_i, V(\hat{r}_x, \hat{r}_y, \hat{r}_z)] = 0$. The reasoning here borrows

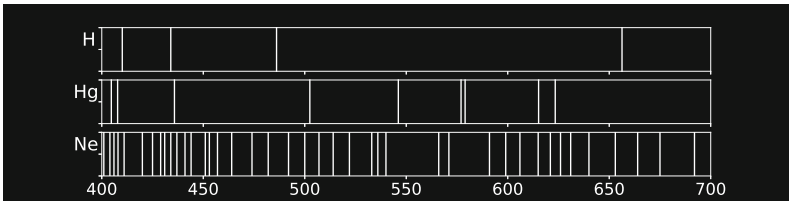

Fig. 7.2 Atomic spectra of three different atoms in the visible: hydrogen (H), mercury (Hg), and neon (Ne). Atoms release light in very sharply defined lines that are a "fingerprint" for each element

a result from calculus, that we will state but not prove, that V can be expressed in a power series in $\hat{r}_x$, $\hat{r}_y$, and $\hat{r}_z$ for all potentials we consider in this book. Hence, we learn that the position operators commute with the potential energy piece of the Hamiltonian.

In the late 1800s, physicists used spectrometers to separate light by bouncing it off of a diffraction grating and using the precise measurement of the angle it reflected to determine the wavelength of light or, equivalently, the rate at which the stopwatch hand rotates in the quantum model for light that we developed in Chap. 3. The sharp lines observed in this spectra (see Fig. 7.2) were ubiquitous, appearing in the spectra for all atoms. Understanding those lines was a challenge, because classical physics predicted a continuous rainbow of colors to be emitted as the electrons spiralled into the nucleus of the atom (with a lifetime on the order of a nanosecond). Niels Bohr developed a model for the atoms, which simply postulated that such a spiralling behavior could not occur. He interpreted the sharp lines as arising from the light that was emitted from stable atomic states, with the energy of the light, given by the Planck-Einstein relation as the product of $\hbar$ with the angular frequency ω, equal to the difference in energy of the atomic states, or

$$E_i - E_f = \hbar\omega, \tag{7.12}$$

with E_i the initial energy the atom has and E_f the final energy the atom has. Let $|i\rangle$ be the initial energy eigenstate of the atom and $|f\rangle$ be the final energy eigenstate, so that $\hat{H}|i\rangle = E_i|i\rangle$ and $\hat{H}|f\rangle = E_f|f\rangle$, with E_i and E_f the aforementioned initial and final energy eigenvalues, respectively, of the atom. The atom has many energy levels, each which satisfies $\hat{H}|n\rangle = E_n|n\rangle$; here, we focus only on the two levels corresponding to one observed spectral line. We are going to analyze the commutator between $\hat{H}$ and $\hat{r}_x$, but we are going to do it sandwiched between the initial and final states in a matrix element; hence, we examine $\langle f|[\hat{r}_x, \hat{H}]|i\rangle$. For concreteness, we pick the hydrogen atom and $\hat{H}$ describes the energy of the electron in the atom, so m will be the mass of the electron. Using the fact that the Hamiltonian operator is a Hermitian operator and that the states we are working with are energy eigenstates, we find that

$$\langle f|[\hat{r}_x, \hat{H}]|i\rangle = \langle f|\hat{r}_x\hat{H} - \hat{H}\hat{r}_x|i\rangle = \langle f|\hat{r}_x\hat{H}|i\rangle - \langle f|\hat{H}\hat{r}_x|i\rangle$$
$$= \left(E_i - E_f\right)\langle f|\hat{r}_x|i\rangle = \hbar\omega\langle f|\hat{r}_x|i\rangle, \tag{7.13}$$

after using the Planck-Einstein relation. Note that we also used the fact that $\hat{H}|f\rangle = E_f|f\rangle$ implies that $\langle f|\hat{H} = E_f\langle f|$, because $\hat{H}$ is a Hermitian operator.

In the following discussion, we will mix classical and quantum reasoning to arrive at a final formula for the commutator of $\hat{r}_x$ and $\hat{p}_x$. *It is not a formal derivation.* The argument can be made more rigorous and even extended to derive the Heisenberg equation of motion and the time-dependent Schrödinger equation (details are in Green's book), but we do not do so here, because those arguments require calculus. The main message you should gain from the following analysis is that one can motivate what $[\hat{r}_i, \hat{p}_j]$ is by (i) noting that there are sharp lines in the atomic spectra, postulating there are stable energy states in the atom, and requiring that the light emitted by the atoms is a photon, whose energy is equal to the energy difference between the two atomic states; (ii) using the Planck-Einstein relation to relate the frequency of light to the photon's energy; (iii) assuming that the atomic matrix element oscillates at the same frequency as the emitted light (this was Heisenberg's breakthrough in 1925); and (iv) requiring a correspondence between the time rate of change of a position matrix element with the corresponding momentum matrix element.

Now, let us recall what we learned about harmonically varying functions in Sect. 5.3. There we found that if the position varies as $Ae^{-i\omega t}$, then the velocity varies as $-i\omega Ae^{-i\omega t}$. Here, we assume that the matrix element between the initial and final states has a time dependence that oscillates at the same frequency as that of the emitted light. This is motivated by the fact that oscillating charges create light and the frequency of light is given by the frequency of the oscillating charges. Hence, we have that

$$\langle f|[\hat{r}_x, \hat{H}]|i\rangle = \hbar\omega\langle f|\hat{r}_x|i\rangle = \hbar\omega Ae^{-i\omega t}, \tag{7.14}$$

from the calculation completed above in Eq. (7.13). Here, the amplitude of the oscillation A is still unknown to us. But since $\langle f|\hat{r}_x|i\rangle = Ae^{-i\omega t}$, then $-i\omega Ae^{-i\omega t}$ is equal to the rate of change of $\hat{r}_x$ with respect to time, which is just the velocity $\hat{v}_x$ (note that this argument is avoiding dealing with the question of how operators or quantum states change with time—we will cover that later in the book). So, we need to require our quantum system to satisfy

$$\hbar Ae^{-i\omega t} = i\hbar\left(-i\omega Ae^{-i\omega t}\right) = i\hbar\langle f|\hat{v}_x|i\rangle. \tag{7.15}$$

Since $\hat{v}_x = \frac{\hat{p}_x}{m}$, we require

$$\langle f|[\hat{r}_x, \hat{H}]|i\rangle = \frac{i\hbar}{m}\langle f|\hat{p}_x|i\rangle. \tag{7.16}$$

Since this relation will hold for all initial and final states, we elevate the relation to one that holds over the operators directly, or

$$[\hat{r}_x, \hat{H}] = \frac{i\hbar}{m}\hat{p}_x. \tag{7.17}$$

Let's pause for a moment to recap once more what went into this "derivation." It used the fact that atomic spectra have sharp lines, that the matrix element between initial and final states varies harmonically with time, and that harmonically varying position relates to velocity with an additional factor of $-i$ times the frequency. That is all there is to it.

Now, we have one more step before we get to the canonical commutation relation. Let us evaluate $[\hat{r}_x, \hat{H}]$ directly for a one-dimensional problem first:

$$[\hat{r}_x, \hat{H}_{1d}] = \left[\hat{r}_x, \frac{\hat{p}_x^2}{2m} + V(\hat{r}_x)\right] = \left[\hat{r}_x, \frac{\hat{p}_x^2}{2m}\right], \tag{7.18}$$

because $[\hat{r}_x, V(\hat{r}_x)] = 0$. Next, we determine that

$$\left[\hat{r}_x, \frac{\hat{p}_x^2}{2m}\right] = \frac{1}{2m}[\hat{r}_x, \hat{p}_x^2] = \frac{1}{2m}\Big(\hat{p}_x[\hat{r}_x, \hat{p}_x] + [\hat{r}_x, \hat{p}_x]\hat{p}_x\Big), \tag{7.19}$$

from the Leibniz identity. But from Eq. (7.16), we find that the commutator above must also equal $\frac{i\hbar}{m}\hat{p}_x$. The simplest way to make this work is to set

$$[\hat{r}_x, \hat{p}_x] = i\hbar, \tag{7.20}$$

because a number commutes with an operator. We assume that this is true, which gives us our final result, called the canonical commutation relation between position and momentum. It says these two operators do not commute and hence are what we call incompatible operators, implying we cannot simultaneously find eigenvalues for position and momentum with a single eigenvector. That is, no state $|\psi\rangle$ exists, such that $\hat{r}_x|\psi\rangle = x|\psi\rangle$ and $\hat{p}_x|\psi\rangle = p|\psi\rangle$. I want to make clear that what we have done in this section is a *motivation* for the canonical commutation relation, not a rigorous derivation. In quantum mechanics, the canonical commutation relation $[\hat{r}_x, \hat{p}_x] = i\hbar$ is a *postulate* that cannot be derived. But it can be strongly motivated, as we did here. It really does come from a connection to atomic spectra. In fact, one can even reorganize Heisenberg's matrix mechanics formalism to derive the canonical commutation relation under two assumptions—the properties of atomic spectra require us to use matrices to describe the behavior and the matrices satisfy the classical equations of motion (often called Eherenfest's theorem) (Tran et al., 2025). So, the motivation sketched above can be made more rigorous, but it requires knowledge we are not assuming here, and so, we stick with the motivation argument for our purposes.

Now, let's move on to three dimensions. We need to assume that $[\hat{r}_x, \hat{p}_y] = [\hat{r}_x, \hat{p}_z] = 0$ as well, because measuring position or momentum in perpendicular directions should be independent of each other. Then, the simple generalization of the one-dimensional derivation shows that

$$[\hat{r}_i, \hat{p}_j] = i\hbar\delta_{ij}, \tag{7.21}$$

where $\delta_{ij} = 1$ if $i = j$ and is 0 if $i \neq j$ (this symbol is called the Kronnecker delta function). The i in front of the $\hbar$ is the imaginary number i. You will complete this derivation in the problems.

We have found two commutators relating our three new operators: (i) $[\hat{r}_j, \hat{H}] = \frac{i\hbar}{m}\hat{p}_j$ and (ii) $[\hat{r}_i, \hat{p}_j] = i\hbar\delta_{ij}$. We have one left, $[\hat{p}_i, \hat{H}]$. Since we assume the components of the momentum operator commute with themselves, just as we assumed for position, we have that $[\hat{p}_i, \hat{p}_j] = 0$, which then yields

$$[\hat{p}_i, \hat{H}] = \left[\hat{p}_i, \frac{\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2}{2m} + V(\hat{r}_x, \hat{r}_y, \hat{r}_z)\right] = [\hat{p}_i, V(\hat{r}_x, \hat{r}_y, \hat{r}_z)]. \tag{7.22}$$

It turns out that one can find an exact expression for this commutator, if one uses calculus. But we will not do that here. Fortunately, we do not need the general result, and instead, we will carefully calculate this result in a case-by-case basis as we need it. So, for now, we leave this last commutator as unresolved. The video The Canonical Commutation Relation https://tinyurl.com/25pa2wd9 explains these ideas further.

In summary, we have showed that the canonical commutation relation, namely, that $[\hat{r}_i, \hat{p}_j] = i\hbar\delta_{ij}$, is a consequence of some simple assumptions that we used to reconcile the experiments on atomic spectra, which show sharp lines. Curiously, we can derive this without having a complete theory for how atomic spectra works, just the notion of the harmonic variation of matrix elements (introduced by Heisenberg in 1925). Now, armed with this canonical commutation relation (which Heisenberg called the "quantum condition"), we are ready to solve our first problem in the quantum mechanics of particles moving in space with forces acting on them. But first, you have some problems to work on.

Section 7.2 Problems:

7.2.1 Canonical Commutation Relation in Three Dimensions

Complete the derivation started in the text for the three-dimensional case with

$$\hat{H} = \frac{1}{2m}\left(\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2\right) + V(\hat{r}_x, \hat{r}_y, \hat{r}_z). \tag{7.23}$$

Use it to show that $[\hat{r}_j, \hat{H}] = \frac{i\hbar}{m}\hat{p}_j$ and $[\hat{r}_i, \hat{p}_j] = i\hbar\delta_{ij}$.

We use the one-dimensional notation with $\hat{x} = \hat{r}_x$ and $\hat{p} = \hat{p}_x$ for the remaining problems.

7.2.2 Commutation Relation of Powers of Momentum

We have already shown that $[\hat{x}, \hat{p}] = i\hbar$ and $[\hat{x}, \hat{p}^2] = 2i\hbar\hat{p}$. Use induction to prove that

$$[\hat{x}, \hat{p}^n] = i\hbar n\hat{p}^{n-1}. \tag{7.24}$$

7.2.3 Commutation Relation of Powers of Position

Show $[\hat{x}^2, \hat{p}] = 2i\hbar\hat{x}$. Then, use induction to prove that

$$[\hat{x}^n, \hat{p}] = i\hbar n\hat{x}^{n-1}. \tag{7.25}$$

7.2.4 Commutation Relation with an Exponential

Use the power-series expansion for the exponential that we derived in Sect. 5.3 to show that

$$\left[\hat{x}, e^{-\frac{i}{\hbar}x\hat{p}}\right] = xe^{-\frac{i}{\hbar}x\hat{p}}, \tag{7.26}$$

for x a real number. Do not confuse x the number, with $\hat{x}$ the operator.

Hint: $e^{-\frac{i}{\hbar}x\hat{p}} = \sum_{n=0}^{\infty}\left(\frac{-ix}{\hbar}\right)^n \frac{1}{n!}\hat{p}^n$. Work carefully—it is easy to make a mistake.

7.2.5 Similarity Transformation from a Commutator with an Exponential

Use the identity derived in Problem 7.2.4 to show that

$$e^{\frac{i}{\hbar}x\hat{p}}\hat{x}e^{-\frac{i}{\hbar}x\hat{p}} = \hat{x} + x. \tag{7.27}$$

Since $e^{-\frac{i}{\hbar}x\hat{p}}$ is a unitary transformation, the unitary transformation that takes $\hat{x} \rightarrow \hat{U}^{\dagger}\hat{x}\hat{U} = \hat{x} + x$ translates every $\hat{x}$ operator by x. We say $e^{-\frac{i}{\hbar}x\hat{p}}$ is the unitary operator for translations.

Hint: no new commutators need to be computed.

Note that since $e^{-\hat{A}}$ is the inverse of $e^{\hat{A}}$ because $[\hat{A}, \hat{A}] = 0$ implies that $e^{\hat{A}}e^{-\hat{A}} = e^{\hat{A}-\hat{A}} = \hat{\mathbb{I}}$. This means the unitary transformation is also a similarity transformation, which takes the form $\hat{x} \rightarrow \hat{O}^{-1}\hat{x}\hat{O}$.

7.2.6 *Another Commutator with an Exponential and Its Similarity Transformation*

Rework problems 7.2.4 and 7.2.5 for the unitary operator $e^{\frac{i}{\hbar}p\hat{x}}$ (with p a real number, not to be confused with $\hat{p}$) and the unitary transformation of $\hat{p}$. Show that

$$e^{-\frac{i}{\hbar}p\hat{x}}\hat{p}e^{\frac{i}{\hbar}p\hat{x}} = \hat{p} + p, \tag{7.28}$$

and that $e^{\frac{i}{\hbar}p\hat{x}}$ is the unitary operator for the translation of momentum.

Note that one can create the general translation operator of momentum in three dimensions by taking the products of the corresponding translation operators for the x-, y-, and z-directions. The order of the factors does not matter, because the three translation operators all mutually commute. A similar construction works for translation of position.

7.3 Free Particle

It is difficult to describe quantum-mechanical behavior without discussing the case of free particles. This corresponds to the situation where $V(\hat{x}) = 0$, and hence, in one dimension, we have $\hat{H} = \frac{\hat{p}^2}{2m}$. Obviously, momentum eigenstates $|p\rangle$, which satisfy $\hat{p}|p\rangle = p|p\rangle$, are also the energy eigenstates of $\hat{H}$. This is because

$$\hat{H}|p\rangle = \frac{1}{2m}\hat{p}^2|p\rangle = \frac{1}{2m}\hat{p}(\hat{p}|p\rangle) = \frac{p}{2m}\hat{p}|p\rangle = \frac{p^2}{2m}|p\rangle; \tag{7.29}$$

recall that numbers always commute with operators Hence, we learn that $E_p = \frac{p^2}{2m}$.

Recall that we said that $\langle x|\psi\rangle = \psi(x)$ is the wavefunction in position space. You may have heard a lot about wavefunctions. While they have their place in the quantum world, when you work with a representation-independent form of

quantum mechanics, as we do in this book, they are less important. Nevertheless, they do provide opportunities to visualize the wave properties of quantum particles, so they are useful to obtain a more complete understanding of the systems we will be studying. Unlike the methodology used in most quantum textbooks, we will compute wavefunctions here using operator manipulations. The position-space wavefunction of the free-particle energy eigenstate (or, equivalently, the momentum eigenstate) is given by $\psi_p(x) = \langle x|p\rangle$. How do we find this overlap?

We start by working with a special state, the momentum eigenstate at the origin, denoted $|0_p\rangle$, which satisfies $\hat{p}|0_p\rangle = 0|0_p\rangle = 0$. In other words, the momentum eigenstate at the origin has momentum equal to 0, so its eigenvalue is 0, and the momentum operator *annihilates the state*. We already told you that if we can find states annihilated by an operator, it allows us to do marvelous things. We will see how this works here. We claim that $|p\rangle = e^{\frac{i}{\hbar}p\hat{x}}|0_p\rangle$. How can we prove this? We use what we know, of course! The only thing we know is the eigenvalue-eigenvector relationship for the momentum eigenstate. Hence, we need to show that $\hat{p}|p\rangle = p|p\rangle$. Let us check this directly. We have

$$\hat{p}|p\rangle = \hat{p}e^{\frac{i}{\hbar}p\hat{x}}|0_p\rangle. \tag{7.30}$$

Our next step is to multiply by one—we want to do this in a way so that we have created a similarity transformation of the $\hat{p}$ operator, as we worked out in Problem 7.2.6. So, we have

$$\hat{p}|p\rangle = e^{\frac{i}{\hbar}p\hat{x}}\underbrace{e^{-\frac{i}{\hbar}p\hat{x}}\hat{p}e^{\frac{i}{\hbar}p\hat{x}}}_{=\hat{p}+p}|0_p\rangle. \tag{7.31}$$

Be sure you see how using a multiply-by-one trick ($e^{\hat{A}}e^{-\hat{A}} = e^{\hat{A}-\hat{A}} = e^0 = 1$) allows us to create a similarity transformation and how we can immediately evaluate the similarity transformation using the results of Problem 7.2.6. Now, using the similarity transformation and the fact that $\hat{p}|0_p\rangle = 0$ leads us to

$$\hat{p}|p\rangle = e^{\frac{i}{\hbar}p\hat{x}}(\hat{p}+p)|0_p\rangle = e^{\frac{i}{\hbar}p\hat{x}}(0+p)|0_p\rangle = pe^{\frac{i}{\hbar}p\hat{x}}|0_p\rangle = p|p\rangle. \tag{7.32}$$

So, the eigenvalue-eigenvector relationship is verified. For the next step of the derivation, we need to use an important operator identity that a function of an operator acting on an eigenstate of that operator is equal to the function evaluated at the eigenvalue multiplied by the eigenvector. In equations, this is

$$f(\hat{A})|a\rangle = f(a)|a\rangle. \tag{7.33}$$

Here, we have $|a\rangle$ is an eigenvector of $\hat{A}$, which satisfies $\hat{A}|a\rangle = a|a\rangle$. One reason why this works is that every well-behaved function $f(\hat{A})$ can be expanded in a power series, and when $\hat{A}^n|a\rangle$, we obtain $a^n|a\rangle$. But what about poorly behaved

functions? It holds for them too. One way to justify this is a poorly behaved function is always arbitrarily close to a well-behaved function, so the identity must hold for them as well. This is a critically important identity that you must remember. Functions of operators acting on eigenstates of that operator are always simple—we just replace the operator in the argument of the function by the corresponding eigenvalue!

Now, we apply the rule here to compute the wavefunction. We have

$$\langle x|p\rangle = \langle x|e^{\frac{i}{\hbar}p\hat{x}}|0_p\rangle = e^{\frac{i}{\hbar}px}\langle x|0_p\rangle. \tag{7.34}$$

Our last step requires us to express the position eigenstate in terms of an operator acting on the position eigenstate at the origin. In the problems, you will show that

$$|x\rangle = e^{-\frac{i}{\hbar}x\hat{p}}|0_x\rangle \text{ and } \langle x| = \langle 0_x|e^{\frac{i}{\hbar}x\hat{p}}. \tag{7.35}$$

Here, the state $|0_x\rangle$, the position eigenstate at the origin, satisfies $\hat{x}|0_x\rangle = 0|0_x\rangle = 0$. So, we now use this result. We find

$$\psi_p(x) = \langle x|p\rangle = e^{\frac{i}{\hbar}px}\langle 0_x|e^{\frac{i}{\hbar}x\hat{p}}|0_p\rangle = e^{\frac{i}{\hbar}px}\langle 0_x|0_p\rangle. \tag{7.36}$$

Here, we used the fact that $f(\hat{p})|0_p\rangle = f(0)|0_p\rangle$. Note that the term $\langle 0_x|0_p\rangle$ is just a number. We call it the *normalization constant*. Determining it usually requires calculus, and this is one of the few aspects of quantum mechanics that we cannot derive for you without calculus. So, when we need these normalization constants, we will just tell you what they are, having done the calculus calculation for you. But, in this case, we don't really need the number, so, we will simply call $\langle 0_x|0_p\rangle = c$ and analyze the form of the momentum (and free-particle Hamiltonian) eigenstate. We will discuss more about the normalization constant shortly, as in this case, there are some subtleties.

Our operator calculation required us to work with two similarity transformations, which were problems worked out in the previous section of the book. The similarity transformations worked out here are special cases of a more general similarity transformation that we will use later in the book to compute wavefunctions. Indeed, this general similarity transformation is one of our four fundamental operator identities, and we will be working with it soon. It is called the *Hadamard lemma*. The approach we used to determine the wavefunction here is an approach we will repeat many times. We first find states annihilated by a specific operator and then we use that fact, along with the operator forms of the position or momentum eigenstates, to compute the wavefunction using operator manipulations alone. This is the strategy we will always use for a representation-independent methodology for computing wavefunctions. After you gain enough practice, you will be able to easily rearrange operators as we did here—we primarily use the similarity transformation, the fact that special states are annihilated by particular operators, and "multiply by

one" to compute them. You may even feel like you are physically moving operators around as you carry out calculations. I think you are going to like how this works.

We worked hard to get the wavefunction; let's now investigate its properties. To start, note that the wavefunction we found holds for any p value and it is also an energy eigenstate with energy $E_p = \frac{p^2}{2m}$. This result holds as we *continuously* change p, and such energy eigenstates are called *continuum solutions*. The energy also has a degeneracy, as the energy eigenvalue for p is the same as the energy eigenvalue for $-p$ in one dimension; in higher dimensions, all states with the same $|\vec{p}|$ have the same energy. We found that the momentum eigenstate wavefunction is

$$\psi_p(x) = ce^{\frac{i}{\hbar}xp} = c\Big(\cos\left(\frac{ixp}{\hbar}\right) + i\sin\left(\frac{ixp}{\hbar}\right)\Big), \tag{7.37}$$

which has wavelike behavior with a constant wavelength given by

$$\lambda = \frac{2\pi\hbar}{p} = \frac{h}{p}, \tag{7.38}$$

due to the periodicity of the cosine and sine functions. This wavelength is called the *de Broglie wavelength*. We say the free particle is a *matter wave*. We have just showed that the solution of the free particle has wavelike behavior that we expect for quantum particles.

We now discuss the constant c more thoroughly. The modulus squared of the wavefunction $|\psi_p(x)|^2 = |c|^2$ is a constant. If we add this up over all space (technically this would be an integral, but you can think of lining up an infinite number of detectors and summing their constant percentage of detection), we would find the total probability is always infinity. The only way out of this is if $c \to 0$, but then the wavefunction vanishes. If we replace infinite space by a box with a finite size of volume V, then $c \propto \frac{1}{\sqrt{V}}$, which would be a small, but nonzero number. I did not discuss this in order to confuse you. Instead, I wanted to show you the issues that arise in trying to create a rigorous theory that can handle functions of this form (which was supposed to be the "simplest" quantum problem to solve). In "wavefunction first" approaches, many of these details are swept under the rug, because treating them properly requires advanced methods from analysis. As you will see in this book, we will not need to worry about many of these issues, and as a result, we will not run into any problems. But, to ease your mind that this can be done, we will go over the example of using a finite box in one dimension in the next section.

So far, we have been working with the momentum eigenstate wavefunction in position space, but we want to also be able to compute the position space eigenstate wavefunction in momentum space. The former is given by $\langle x|p\rangle$, while the latter is $\langle p|x\rangle$. The two are related by a complex conjugate, because $\langle p|x\rangle = \langle x|p\rangle^*$. We will denote momentum-space wavefunctions with a tilde, so we have both

$$\psi_p(x) = ce^{\frac{i}{\hbar}xp} = \langle x|p\rangle \text{ and } \tilde{\psi}_x(p) = c^* e^{-\frac{i}{\hbar}xp} = \langle p|x\rangle. \tag{7.39}$$

Usually, we choose the constant factor to be real, so $c = c^*$. This result shows there is a simple connection between the free-particle wavefunction in position space and the position eigenstate wavefunction in momentum space.

We do end the section with one word of warning. We will not be working with the position eigenstate wavefunction in position space $\langle x|x'\rangle$, nor will we work with the momentum eigenstate wavefunction in momentum space $\langle p|p'\rangle$. Both of these objects are complicated and given by what is called a Dirac delta function. It is difficult to work with such objects without using calculus. Good thing, we won't ever need them for anything we do in this book!

Section 7.3 Problems:

7.3.1 Position Eigenfunction in Operator Form

Show all the steps to verify that

$$|x\rangle = e^{-\frac{i}{\hbar}x\hat{p}}|0_x\rangle. \tag{7.40}$$

7.3.2 Relation Between Position Eigenfunction in Momentum Space and Momentum Eigenfunction in Position Space

Use the commutators

$$\left[\hat{p}, e^{\frac{i}{\hbar}p\hat{x}}\right] = pe^{\frac{i}{\hbar}p\hat{x}}, \quad \text{and} \quad \left[\hat{x}, e^{-\frac{i}{\hbar}x\hat{p}}\right] = xe^{-\frac{i}{\hbar}x\hat{p}} \tag{7.41}$$

to derive $|p\rangle$ and $|x\rangle$ respectively. Use the ansatz given for the states of definite momentum and position and use the commutators to verify that they are states of definite momentum and position, respectively. Then, use these definitions to compute $\tilde{\psi}(p) = \langle p|x\rangle$. Verify that proceeding this way does indeed yield the same answer as we found by complex conjugation.

7.3.3 De Broglie Wavelength

Using the definition that the wavelength λ is the shortest distance that satisfies

$$\langle x + \lambda|p\rangle = \langle x|p\rangle, \tag{7.42}$$

prove that $\lambda = \frac{2\pi\hbar}{p} = \frac{h}{p}$, which is called the *de Broglie wavelength* of a particle.

7.3.4 *Free Particle in Three Dimensions*

Generalize the result we found for one dimension to find the free-particle wavefunction $\psi_{\vec{p}}(\vec{r})$ via

$$\psi_{\vec{p}}(\vec{r}) = \langle \vec{r} | \vec{p} \rangle = \langle r_x | p_x \rangle \, \langle r_y | p_y \rangle \, \langle r_z | p_z \rangle, \tag{7.43}$$

which follows since we can write

$$|\vec{p}\rangle = |p_x\rangle \otimes |p_y\rangle \otimes |p_z\rangle \;\text{ and }\; |\vec{r}\rangle = |r_x\rangle \otimes |r_y\rangle \otimes |r_z\rangle. \tag{7.44}$$

Also compute $\tilde{\psi}_{\vec{r}}(\vec{p}) = \langle \vec{p} | \vec{r} \rangle$.

7.4 Free Particle on a Circle

We will now explore a similar problem to the free particle, but rather than directly construct the eigenstates, we will use an advanced methodology first developed by Schrödinger in 1940 (Schrödinger, 1940), which is called the *factorization method*. The full technique will be developed in Chap. 11. For now, just think of what we are doing as a series of "inspired guesses" that lead us to the solution.

We will consider a one-dimensional particle that moves on a circle with perimeter L. Figure 7.3 shows the system. While the circle wraps around itself and has no ends, we can also think of it as a straight line segment, where we identify the endpoints as the same point, that is, we identify the point 0 with the point L. The Hamiltonian is

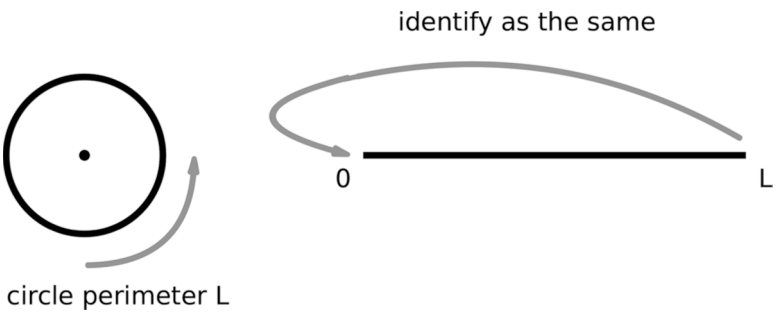

Fig. 7.3 Schematic of two different ways to think of a particle on a circle with perimeter L. Left, the circular domain with no edges, and right, the linear domain, with the endpoints 0 and L identified as being the same

just the free-particle Hamiltonian, which we write as $\hat{H} = \frac{\hat{p}^2}{2m}$. There is just a one-dimensional degree of freedom (seen most easily from the straight-line depiction with identified edges), even if we think of the circle as a curve embedded in a two-dimensional plane. The inspired guess we will use comes from two alternative forms, which we will now show. The first is written as follows:

$$\hat{H} = \hat{A}_k^\dagger \hat{A}_k + E_k, \tag{7.45}$$

with

$$\hat{A}_k = \frac{1}{\sqrt{2m}}\big(\hat{p} - i\hbar k \tan k\hat{x}\big) \quad \text{and} \quad E_k = \frac{\hbar^2 k^2}{2m}. \tag{7.46}$$

To check that this is the free-particle Hamiltonian, as it looks anything but the free-particle Hamiltonian, we must calculate

$$\begin{aligned} \hat{A}_k^\dagger \hat{A}_k &= \frac{1}{2m}\big(\hat{p} + i\hbar k \tan k\hat{x}\big)\big(\hat{p} - i\hbar k \tan k\hat{x}\big) \\ &= \frac{\hat{p}^2}{2m} - \frac{i\hbar k}{2m}[\hat{p}, \tan k\hat{x}] + \frac{\hbar^2 k^2}{2m}\tan^2 k\hat{x} \end{aligned} \tag{7.47}$$

(recall that both $\hat{x}$ and $\hat{p}$ are Hermitian operators). Computing the commutator $[\hat{p}, \tan k\hat{x}]$ is a bit long, so we tackle it next.

We begin with the similarity identity we worked out in the problems of Sect. 7.2

$$e^{-ik\hat{x}}\hat{p}e^{ik\hat{x}} = \hat{p} + \hbar k, \tag{7.48}$$

where $\hbar k$ is a number, not an operator. Multiplying from the left by $e^{ik\hat{x}}$ yields

$$\underbrace{e^{ik\hat{x}}e^{-ik\hat{x}}}_{1}\hat{p}e^{ik\hat{x}} = e^{ik\hat{x}}\big(\hat{p} + \hbar k\big), \tag{7.49}$$

or

$$\Big[\hat{p}, e^{ik\hat{x}}\Big] = e^{ik\hat{x}}\hbar k. \tag{7.50}$$

Recall Euler's formula to find

$$[\hat{p}, \cos k\hat{x} + i\sin k\hat{x}] = \hbar k\big(\cos k\hat{x} + i\sin k\hat{x}\big). \tag{7.51}$$

Equating the real and imaginary parts of both sides (after recalling that $[\hat{p}, \hat{x}] = -i\hbar$ is imaginary) yields

$$[\hat{p}, \cos k\hat{x}] = i\hbar k \sin k\hat{x}, \tag{7.52}$$

and

$$[\hat{p}, \sin k\hat{x}] = -i\hbar k \cos k\hat{x}. \tag{7.53}$$

If you are feeling uneasy about the i's on each side of the equation, just look at the first few terms of the power-series expansions on each side, and you should be able to verify that both sides are equal. This isn't the commutator with the tangent, yet—we need to work up to that. Next, we find the commutators with the secant and cosecant by using the Leibniz rule. Note that

$$\begin{aligned}[\hat{p}, \hat{\mathbb{I}}] = 0 &= [\hat{p}, \cos k\hat{x} \sec k\hat{x}] \\ &= [\hat{p}, \cos k\hat{x}] \sec k\hat{x} + \cos k\hat{x}[\hat{p}, \sec k\hat{x}].\end{aligned} \tag{7.54}$$

Solving for the commutator with the secant gives us

$$[\hat{p}, \sec k\hat{x}] = -i\hbar k \tan k\hat{x} \sec k\hat{x}. \tag{7.55}$$

Proceeding in exactly the same way with $\operatorname{cosec} k\hat{x}$ and $\sin k\hat{x}$ yields

$$[\hat{p}, \operatorname{cosec} k\hat{x}] = i\hbar k \operatorname{cotan} k\hat{x} \operatorname{cosec} k\hat{x}. \tag{7.56}$$

Finally, we work out

$$\begin{aligned}[\hat{p}, \tan k\hat{x}] &= [\hat{p}, \sin k\hat{x} \sec k\hat{x}] \\ &= -i\hbar k \cos k\hat{x} \sec k\hat{x} - i\hbar k \tan^2 k\hat{x} \\ &= -i\hbar k\left(1 + \tan^2 k\hat{x}\right) \\ &= -i\hbar k \sec^2 k\hat{x},\end{aligned} \tag{7.57}$$

and, similarly

$$[\hat{p}, \operatorname{cotan} k\hat{x}] = i\hbar k \operatorname{cosec}^2 k\hat{x}. \tag{7.58}$$

Now, back to Eq. (7.47), using the commutator with the tangent, we find

$$\begin{aligned}\hat{H} = \hat{A}_k^\dagger \hat{A}_k + E_k &= \frac{\hat{p}^2}{2m} + \frac{\hbar^2 k^2}{2m}\left(\tan^2 k\hat{x} - \sec^2 k\hat{x}\right) + E_k \\ &= \frac{\hat{p}^2}{2m} - \frac{\hbar^2 k^2}{2m} + E_k = \frac{\hat{p}^2}{2m},\end{aligned} \tag{7.59}$$

since $E_k = \frac{\hbar^2 k^2}{2m}$. We used the fact that $\tan^2 k\hat{x} - \sec^2 k\hat{x} = \hat{\mathbb{I}}$.

The second form of $\hat{H}$ that we will use is

$$\hat{H} = \hat{B}_k^\dagger \hat{B}_k + E_k \tag{7.60}$$

with

$$\hat{B}_k = \frac{1}{\sqrt{2m}}\big(\hat{p} + i\hbar k \operatorname{cotan} k\hat{x}\big) \text{ and } E_k = \frac{\hbar^2 k^2}{2m}. \tag{7.61}$$

We find by direct calculation that

$$\begin{aligned}\hat{H} = \hat{B}_k^\dagger \hat{B}_k + E_k &= \frac{\hat{p}^2}{2m} + \frac{i\hbar k}{2m}[\hat{p}, \operatorname{cotan} k\hat{x}] + \frac{\hbar^2 k^2}{2m}\operatorname{cotan}^2 k\hat{x} + E_k \\ &= \frac{\hat{p}^2}{2m} + \frac{\hbar^2 k^2}{2m}\big(\operatorname{cotan}^2 k\hat{x} - \operatorname{cosec}^2 k\hat{x}\big) + E_k \\ &= \frac{\hat{p}^2}{2m},\end{aligned} \tag{7.62}$$

because $\operatorname{cosec}^2 k\hat{x} - \operatorname{cotan}^2 k\hat{x} = \hat{\mathbb{I}}$.

We have found two ways to "factorize" the Hamiltonian. Factorizing a Hamiltonian often is not unique. As a general rule, for one-dimensional potentials, if the potential is even, it usually can be factorized two ways for a given energy. If it is not even, there is usually only one factorization. You may now say, so what. We worked extremely hard to factorize the Hamiltonian. Why work so hard? We are going to show you next how we can use this factorization to find the eigenvalues and eigenstates of $\hat{H}$. The method uses the rule we stated before about special states—we first find states that satisfy

$$\hat{A}_k|\psi_k\rangle = 0 \text{ and } \hat{B}_k|\phi_k\rangle = 0. \tag{7.63}$$

For either of these states, we have

$$\hat{H}|\psi_k\rangle = E_k|\psi_k\rangle \text{ or } \hat{H}|\phi_k\rangle = E_k|\phi_k\rangle, \tag{7.64}$$

implying that *both* of these special states are energy eigenstates. Let us look more carefully at the defining conditions for these states (which are called the *subsidiary conditions*)

$$\hat{A}_k|\psi_k\rangle = 0 \text{ implies } \hat{p}|\psi_k\rangle = i\hbar k \tan k\hat{x}|\psi_k\rangle \tag{7.65}$$

and

$$\hat{B}_k|\phi_k\rangle = 0 \text{ implies } \hat{p}|\phi_k\rangle = -i\hbar k \operatorname{cotan} k\hat{x}|\phi_k\rangle. \tag{7.66}$$

It looks like any energy is possible, but we have not yet enforced the condition that we are on a circle. For this, we must have

$$\hat{A}_k(\hat{x} + L) = \hat{A}_k(\hat{x}), \tag{7.67}$$

and similarly for $\hat{B}_k$, because the points x and $x + L$ are the same. This condition tells us that $\tan(k\hat{x} + kL) = \tan k\hat{x}$ (or $kL = \pi n$) and $\operatorname{cotan}(k\hat{x} + kL) = \operatorname{cotan} k\hat{x}$ (or $kL = \pi n$), because both tangent and cotangent have period π. So, we learn that energy is quantized:

$$E_n = \frac{\pi^2 \hbar^2 n^2}{2mL^2} \quad n = 0, 1, \cdots . \tag{7.68}$$

We will find below that only even integer solutions are allowed if we require the wavefunction to be periodic on the circle. Let us find those wavefunctions and evaluate what happens as $x \to x + L$.

We first note that from the form of $\hat{H}$, we have

$$\hat{p}^2|\psi_k\rangle = \hbar^2 k^2 |\psi_k\rangle \text{ and } \hat{p}^2|\phi_k\rangle = \hbar^2 k^2 |\phi_k\rangle. \tag{7.69}$$

So, we immediately have

$$\hat{p}^{2m}|\psi_k\rangle = \left(\hbar^2 k^2\right)^m |\psi_k\rangle \text{ and } \hat{p}^{2m}|\phi_k\rangle = \left(\hbar^2 k^2\right)^m |\phi_k\rangle, \tag{7.70}$$

and the relation in Eqs. (7.65) and (7.66) shows that

$$\begin{aligned} \hat{p}^{2m+1}|\psi_k\rangle &= \left(\hbar^2 k^2\right)^m i\hbar k \tan k\hat{x}|\psi_k\rangle \text{ and} \\ \hat{p}^{2m+1}|\phi_k\rangle &= \left(\hbar^2 k^2\right)^m i\hbar k \operatorname{cotan} k\hat{x}|\phi_k\rangle. \end{aligned} \tag{7.71}$$

The wavefunction is the inner product between the position eigenstate and the energy eigenstate, so

$$\langle x|\psi_k\rangle = \psi_k(x) \text{ and } \langle x|\phi_k\rangle = \phi_k(x). \tag{7.72}$$

Using

$$\langle x| = \langle 0_x|e^{\frac{i}{\hbar}x\hat{p}} = \sum_{m=0}^{\infty} \left(\frac{ix}{\hbar}\right)^m \frac{1}{m!} \langle 0_x|\hat{p}^m, \tag{7.73}$$

then yields

$$\psi_k(x) = \sum_{m=0}^{\infty} \left(\frac{ix}{\hbar}\right)^{2m} \frac{1}{(2m)!} \langle 0_x| \left(\hbar^2 k^2\right)^m |\psi_k\rangle$$
$$+ \sum_{m=0}^{\infty} \left(\frac{ix}{\hbar}\right)^{2m+1} \frac{1}{(2m+1)!} \langle 0_x| \left(\hbar^2 k^2\right)^m i\hbar k \tan k\hat{x} |\psi_k\rangle. \tag{7.74}$$

But $\langle 0_x| \tan k\hat{x} = \langle 0_x| \tan 0 = 0$. So, we find

$$\psi_k(x) = \sum_{m=0}^{\infty} (-1)^m \frac{(kx)^{2m}}{(2m)!} \langle 0_x|\psi_k\rangle = \cos kx \langle 0_x|\psi_k\rangle. \tag{7.75}$$

The number $\langle 0_x|\psi_k\rangle$ is the normalization constant, which is equal to $\sqrt{\frac{2}{L}}$, unless $k = 0$, where it is $\frac{1}{\sqrt{L}}$.

We are now in an odd situation, because we find that

$$\psi_k(x + L) = \sqrt{\frac{2}{L}} \cos(kx + kL) = \sqrt{\frac{2}{L}} \cos\left(\frac{\pi n x}{L} + \pi n\right) = (-1)^n \psi_k(x). \tag{7.76}$$

The wavefunction is periodic if the integer n is even but is antiperiodic if the integer n is odd. Many textbooks will say that the wavefunction must be continuous, so we must choose n to be even, but the actual condition that we have is $|\psi_k(x)|^2$ must be continuous (the probability density must be continuous), and this is true for either choice. A similar issue will arise in Chap. 9 when we discuss orbital angular momentum. There, one can show that n must be even to have consistent orbital angular momentum properties, yielding periodic boundary conditions and a continuous wavefunction on the circle. We will invoke the same requirement here, having $n = 2n'$ and periodic boundary conditions for the wavefunction. This requires us to throw out half of our solutions. You will see the argument for why, when we discuss angular momentum later. Could we have chosen the antiperiodic wavefunctions? We could have, in fact, considered other boundary conditions as well, but that is a more advanced topic that we will not discuss further here. In summary, the energy is

$$E_{n'} = \frac{2\pi^2 \hbar^2 n'^2}{mL^2}, \tag{7.77}$$

and

$$\psi_{n'}(x) = \sqrt{\frac{2}{L}} \cos\left(\frac{2\pi n' x}{L}\right) \quad \text{for } n' = 1, 2, 3 \cdots, \tag{7.78}$$

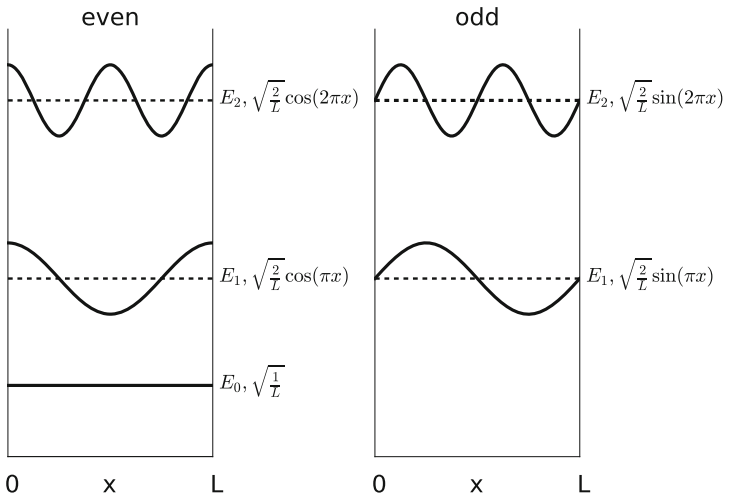

Fig. 7.4 Even (left) and odd (right) wavefunctions for a free particle on a circle with periodic boundary conditions. Note how the number of times the wavefunction vanishes increases as the energy gets higher. Shorter wavelengths correspond to higher energies

and $\psi_0(x) = \frac{1}{\sqrt{L}}$. These functions are *even* functions, defined to satisfy $\psi_{n'}(x) = \psi_{n'}(-x)$. We will discuss why this happens in more general terms in Sect. 7.8.

Because $\operatorname{cotan}(0) = \infty$, we need a different strategy to solve for these wavefunctions. This is done in the problems. The result of that calculation yields a set of odd wavefunctions

$$\phi_{n'}(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{2\pi n' x}{L}\right) \quad \text{for } n' = 1, 2, 3 \cdots , \tag{7.79}$$

with the same energy $E_{n'}$. Note that for $n' \geq 1$, the energy eigenstates are degenerate. The first few low-energy wavefunctions are plotted in Fig. 7.4, separated into even and odd wavefunctions. The wavefunctions oscillate more as the energy increases.

The subsidiary condition shows that neither of the energy eigenstates we have found are also eigenstates of the momentum operator (except for the state with $E = 0$ or $k = 0$). But, because we have two degenerate eigenstates for each energy above $E = 0$, we can take linear combinations of these eigenstates to create joint eigenstates of energy and momentum. This is because $[\hat{p}, \hat{H}] = 0$. To start, we claim that

$$\tan k\hat{x}|\psi_k\rangle = |\phi_k\rangle \;\text{ and }\; \operatorname{cotan} k\hat{x}|\phi_k\rangle = |\psi_k\rangle. \tag{7.80}$$

We prove the first relation next; the second relation is a problem for you. Note that if $\tan k\hat{x}|\psi_k\rangle = |\phi_k\rangle$, then the subsidiary condition, given by $\hat{p}|\phi_k\rangle = -i\hbar k \operatorname{cotan} k\hat{x}|\phi_k\rangle$, becomes

$$\hat{p}\underbrace{\tan k\hat{x}|\psi_k\rangle}_{|\phi_k\rangle} = -i\hbar k \operatorname{cotan} k\hat{x}(\tan k\hat{x}|\psi_k\rangle) = -i\hbar k|\psi_k\rangle. \tag{7.81}$$

We can also establish this directly using the subsidiary condition as follows:

$$\begin{aligned}\hat{p}\tan k\hat{x}|\psi_k\rangle &= [\hat{p}, \tan k\hat{x}]|\psi_k\rangle + \tan k\hat{x}\ \hat{p}|\psi_k\rangle \\ &= -i\hbar k \sec^2 k\hat{x}|\psi_k\rangle + i\hbar k \tan^2 k\hat{x}|\psi_k\rangle \\ &= -i\hbar k(\sec^2 k\hat{x} - \tan^2 k\hat{x})|\psi_k\rangle \\ &= -i\hbar k|\psi_k\rangle. \end{aligned} \tag{7.82}$$

So, the identity holds. We can also check the original identity by multiplying on the left with the position bra. We find $\tan k\hat{x}|\psi_k\rangle = |\phi_k\rangle$ becomes $\langle x|\tan k\hat{x}|\psi_k\rangle = \langle x|\phi_k\rangle$, which becomes $\tan(kx)\sqrt{\frac{2}{L}}\cos(kx) = \sqrt{\frac{2}{L}}\sin(kx)$, which obviously holds. We now use these relations to find the momentum eigenstates. We claim that $|p = \pm\hbar k\rangle = \frac{1}{\sqrt{2}}(|\psi_k\rangle \pm i|\phi_k\rangle)$. Check:

$$\begin{aligned}\hat{p}|\pm\hbar k\rangle &= \frac{1}{\sqrt{2}}\Big(i\hbar k \tan k\hat{x}|\psi_k\rangle \pm i(-i\hbar k \operatorname{cotan} k\hat{x}|\phi_k\rangle)\Big) \\ &= \pm\hbar k\frac{1}{\sqrt{2}}(\pm i|\phi_k\rangle + |\psi_k\rangle) = \pm\hbar k|\pm\hbar k\rangle. \end{aligned} \tag{7.83}$$

So, $|\pm\hbar k\rangle = \frac{1}{\sqrt{2}}(|\psi_k\rangle \pm i|\phi_k\rangle)$ are eigenstates of $\hat{p}$ with eigenvalues $\pm\hbar k$. They also are eigenstates of $\hat{H}$, with eigenvalues $\frac{\hbar^2k^2}{2m}$. For the circle with periodic boundary conditions, we have $k = \frac{2\pi n}{L}$.

We end the section with an important technical discussion about the operator $\hat{x}$ on the circle. It turns out this operator is not a valid operator and cannot be used on the circle. The simpler heuristic discussion is that if $\hat{x}$ was a self-adjoint operator on the circle, it would have a set of eigenvectors. Then, we would have a state $|0_x\rangle$, which satisfies $\hat{x}|0_x\rangle = 0$ and $\hat{x}|L\rangle = L|L\rangle$. But we must identify $|0_x\rangle \leftrightarrow |L\rangle$. This cannot be done, because $0 \neq L$. One might be tempted to say that this is okay, because we can have the operator $\hat{x}$ be Hermitian, but not self-adjoint, on the circle. But that does not work either. For, if we did, then we would be able to examine states of the form $e^{\frac{i}{\hbar}p_0\hat{x}}|\hbar k\rangle$. One can use our operator identity, after a multiply by one, to show that

$$\hat{p}e^{\frac{i}{\hbar}p_0\hat{x}}|\hbar k\rangle = e^{\frac{i}{\hbar}p_0\hat{x}}(\hat{p} + p_0)|\hbar k\rangle = (\hbar k + p_0)e^{\frac{i}{\hbar}p_0\hat{x}}|\hbar k\rangle. \tag{7.84}$$

So, this state is a momentum eigenstate with momentum $p_0 + \hbar k$. But the only allowed momentum values are $2\pi\hbar n/L$, so only if $p_0 = 2\pi\hbar m/L$ is this a valid translation of the momentum eigenstate. Note that for those values of p_0, the

operator $e^{\frac{i}{\hbar}p_0\hat{x}} = e^{\frac{2\pi mi}{L}\hat{x}}$. In this case, when $x \to x + L$, the exponential operator remains the same. Hence, it is well defined. What this means is that while $\hat{x}$ is not well defined on a circle, the operator $e^{\frac{2\pi mi}{L}\hat{x}}$ is well defined. Now, if we go back and look at all of the commutators we worked with, they all involved trigonometric functions that are also periodic on the circle. Hence, all we need for the analysis we did to hold is the postulate that on the circle

$$\left[\hat{p}, e^{\frac{2\pi mi}{L}\hat{x}}\right] = \frac{2\pi\hbar m}{L} e^{\frac{2\pi mi}{L}\hat{x}}, \tag{7.85}$$

instead of $[\hat{x}, \hat{p}] = i\hbar$. Thus, all of the calculations we did above are fine. It is just that on a circle, $\hat{x}$ is not a valid operator. Only the proper exponential of $\hat{x}$ is a valid operator. Quantum mechanics can be subtle! The video Particle On a Circle https://tinyurl.com/ysa5y8w7 discusses these ideas as well.

Congratulations. We have solved our first nontrivial quantum problem. We explored many of its properties. In particular, we found that energy eigenvalues are usually discrete. Here, the discreteness comes from applying a boundary condition to the problem. You might even say, only the allowed wavelengths "fit" on the circle. We found the wavefunctions oscillate, with more oscillations the higher their energy. We determined all of this from clever use of commutators. While this result might have been difficult for you to work out on your own, it is part of a more general method that we will develop further in the book. As we develop it, you will find it will become more and more easy for you to follow and even to develop new results on your own. Eventually, you will become an expert with it.

Section 7.4 Problems:

7.4.1 Alternative Proof of the Energy Eigenstate

In the text, we showed that $\hat{p}^2|\psi_k\rangle = \hbar^2k^2|\psi_k\rangle$ and $\hat{p}^2|\phi_k\rangle = \hbar^2k^2|\phi_k\rangle$, because they are energy eigenstates of $\hat{H}$. Here, we show that we can obtain the same result from the subsidiary conditions in Eqs. (7.65) and (7.66). For $|\psi_k\rangle$, we have

$$\hat{p}^2|\psi_k\rangle = \hat{p}(\hat{p}|\psi_k\rangle) = \hat{p}(i\hbar k \tan k\hat{x}|\psi_k\rangle = i\hbar k[\hat{p}, \tan k\hat{x}]|\psi_k\rangle + i\hbar k \tan k\hat{x}\ \hat{p}|\psi_k\rangle. \tag{7.86}$$

Complete the calculation to show that $\hat{p}^2|\psi_k\rangle = \hbar^2k^2|\psi_k\rangle$. Repeat for $|\phi_k\rangle$.

Finally, show that

$$\hat{p}^{2m+1}|\psi_k\rangle = i\hbar k(\hbar^2k^2)^m \tan k\hat{x}|\psi_k\rangle, \tag{7.87}$$

via

$$\hat{p}^{2m+1}|\psi_k\rangle = \hat{p}(\hat{p}^2)^m|\psi_k\rangle = \hat{p}(\hbar^2k^2)^m|\psi_k\rangle = \cdots. \tag{7.88}$$

7.4.2 Solving for $\phi_k(x)$

We established that $|\phi_k\rangle = \tan k\hat{x}|\psi_k\rangle$. Use the result for $\psi_k(x)$ to determine $\phi_k(x)$ by multiplying from the left by $\langle x|$.

7.4.3 Finding Energy Eigenstates from Momentum Eigenstates

An alternative way to solve this problem is to take the momentum eigenfunctions found in Sect. 7.3 and select only those eigenfunctions that also satisfy $\psi(x+L) = \psi(x)$ (periodic boundary conditions). Verify that you get the same energies and degeneracies as we found above.

7.4.4 Energy Wavefunctions from Momentum Wavefunctions

In this continuation of the previous problem, we will compute the wavefunctions of the energy eigenstates. How do they compare to the even and odd solutions found in the text. How can we have two different forms for eigenfunctions with the same energy?
Hint: consider a linear superposition $|\chi\rangle = a|\psi_1\rangle + b|\psi_2\rangle$ with $\hat{H}|\psi_1\rangle = E|\psi_1\rangle$ and $\hat{H}|\psi_2\rangle = E|\psi_2\rangle$. Is $|\chi\rangle$ and eigenstate of $\hat{H}$?

7.4.5 Relationship Between $|\psi_k\rangle$ and $|\phi_k\rangle$

Verify that $\text{cotan}\, k\hat{x}|\phi_k\rangle = |\psi_k\rangle$ by following the strategy outlined in the text.

7.4.6 Position Eigenstates on the Circle

Verify that the operator $\hat{U} = e^{ik\hat{x}}$ is unitary, by showing that $\hat{U}^\dagger\hat{U} = \hat{\mathbb{I}}$. Show that the states $|x\rangle$, which satisfy $\hat{U}|x\rangle = e^{ikx}|x\rangle$, correspond to a well-defined eigenvalue-eigenvector relation if $k = 2\pi n'/L$ for $n' = 0, 1, 2, 3, \cdots$. These states play the role of position eigenstates on the circle and we can find the equivalent position expectation values by computing $\langle\psi|\text{Re}(\hat{U})|\psi\rangle$ and taking $x = \frac{1}{k}\cos^{-1}[\langle\psi|\text{Re}(\hat{U})|\psi\rangle]$ as the expectation value of x on the circle.

These states are the states we use to compute the wavefunctions. You can see that they are, in fact, the states we did use as the bras that we multiplied by from the left.

7.4.7 Ladder Operators for Momentum and Energy

We work with the simultaneous momentum and energy eigenstates $|\pm\hbar k\rangle = \frac{1}{\sqrt{2}}(|\psi_k\rangle \pm i|\phi_k\rangle)$. Since $k = 2\pi n/L$, we will denote the wavenumbers k by k_n using the integer in the wavenumber. Then, the operator $\hat{U} = e^{\frac{i2\pi\hat{x}}{L}}$ is a raising operator for momentum (and energy) and $\hat{U}^\dagger = e^{-\frac{i2\pi\hat{x}}{L}}$ is the lowering operator for momentum.

Show that

$$\hat{p}\,\hat{U}|\hbar k_n\rangle = \left(\hbar k_n + \frac{2\pi\hbar}{L}\right)\hat{U}|\hbar k_n\rangle, \tag{7.89}$$

and

$$\hat{H}\hat{U}|\hbar k_n\rangle = \frac{\hbar^2\left(k_n + \frac{2\pi\hbar}{L}\right)^2}{2m}\hat{U}|\hbar k_n\rangle, \tag{7.90}$$

so that $\hat{U}$ raises n by 1: $\hat{U}|\hbar k_n\rangle = |\hbar k_{n+1}\rangle$. Verify that $\hat{U}^\dagger$ is a lowering operator, whereby $\hat{U}^\dagger|\hbar k_n\rangle = |\hbar k_{n-1}\rangle$.

Using these results, we can create all states as follows: we let $|0\rangle = |\hbar k_0\rangle$ denote the state with zero momentum. Then

$$|\hbar k_n\rangle = (\hat{U})^n|0\rangle \text{ and } |-\hbar k_n\rangle = (\hat{U}^\dagger)^n|0\rangle. \tag{7.91}$$

Verify as well that

$$\hat{H}|\hbar k_n\rangle = \frac{\hbar^2 k_n^2}{2m}|\hbar k_n\rangle. \tag{7.92}$$

7.5 Introduction to the Simple Harmonic Oscillator

The simple harmonic oscillator is one of the best examples of an exactly solvable problem in quantum mechanics. We can think of it as a quantum particle attached to a spring, but that does not seem so physical (see Fig. 7.5), because we cannot make springs that are a similar size as elementary quanta.

Even though there is no physical spring that we can attach to an electron, the simple harmonic oscillator remains one of the most important problems in physics. Why? Because it also describes the behavior of quantum particles moving in any potential that has a minimum to it.

This isn't just a result that the minimum looks like a simple harmonic oscillator, it actually can be proven to be true for any potential that curves up from a minimum. Unfortunately, the proof involves calculus, so I won't give it here (but if you know

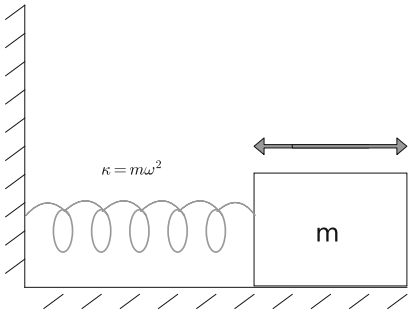

Fig. 7.5 Schematic of a mass m attached to a spring with spring constant $\kappa = m\omega^2$ and oscillating back and forth. Such a depiction seems odd in the quantum world, but the model is nevertheless quite important

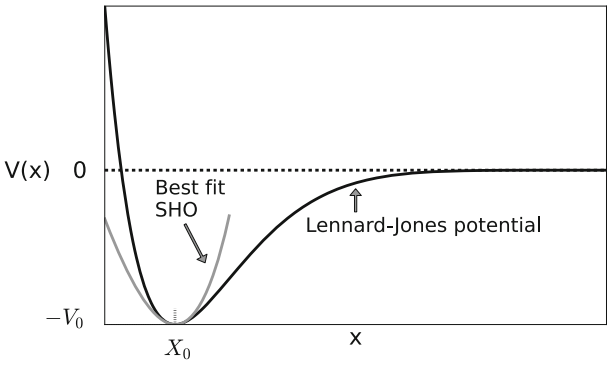

Fig. 7.6 Example of how a simple harmonic oscillator potential (gray) approximates a general potential (black) with a minimum (here given by the Lennard-Jones 6–12 potential)

calculus, it involves a Taylor series expansion about the minimum of the potential energy function). An example illustrating this result for the Lennard-Jones potential is shown in Fig. 7.6.

So, it turns out that the simple harmonic oscillator is ubiquitous for nearly all problems of a particle moving in a potential. But it goes further than that. We use harmonic oscillators to also describe light (as an oscillation of the electromagnetic field). The entire field of quantum optics is based on this.

Let's just list some of the types of problems that can be described by the simple harmonic oscillator close to its minimum in energy: (i) the hydrogen atom states with nonzero angular momentum; (ii) the oscillations of a molecular dimer; (iii) the Lennard-Jones 6–12 potential for interatomic interactions; (iv) the minima of the optical lattice potential, which traps atoms in an egg-carton-like potential; and much, much more.

To get started, we need to write down the Hamiltonian for the simple harmonic oscillator as

$$\hat{H}_{SHO} = \frac{\hat{p}^2}{2m} + \frac{1}{2}\kappa\hat{x}^2 = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2, \tag{7.93}$$

where κ = spring constant and $\omega = \sqrt{\frac{\kappa}{m}}$. The first thing we notice is this is a quadratic function of momentum and of position. This means the Hamiltonian is unchanged if $\hat{p} \rightarrow -\hat{p}$ or if $\hat{x} \rightarrow -\hat{x}$. Furthermore, aside from some different constants in front, the Hamiltonian has the same dependence on momentum as it has on position—namely, a quadratic dependence. This will have implications for the behavior of many of our results, as we expect similar behavior with respect to momentum and with respect to position.

As before, we will let $|n\rangle$ denote the energy eigenstates for $\hat{H}_{SHO}$, so that

$$\hat{H}_{SHO}|n\rangle = E_n|n\rangle. \tag{7.94}$$

The wavefunction of the energy eigenstate in the position basis is then $\psi_n(x) = \langle x|n\rangle$ and in the momentum basis is then $\tilde{\psi}_n(p) = \langle p|n\rangle$.

To expand on the relationship between x and p for the simple harmonic oscillator, let $\hat{P} = \frac{\hat{p}}{\sqrt{m}}$ and $\hat{X} = \sqrt{m}\omega\hat{x}$, then

$$\hat{H}_{SHO} = \frac{1}{2}\hat{P}^2 + \frac{1}{2}\hat{X}^2, \tag{7.95}$$

where the symmetric relationship between position and momentum now becomes clearer. For example, $\psi_n(X) = \langle X|n\rangle$ has the same functional dependence on X as $\tilde{\psi}_n(P) = \langle P|n\rangle$ has on P. Converting back to $\hat{x}$ and $\hat{p}$ gives $\hat{p} = \sqrt{m}\hat{P}$ and $\hat{x} = \hat{X}/\sqrt{m}\omega$, which implies that $\hat{x} \leftrightarrow \frac{\hat{p}}{m\omega}$; hence, $\psi_n(x)$ and $\tilde{\psi}\left(\frac{p}{m\omega}\right)$ have the same functional form or, equivalently, the same functional dependence on their arguments. Of course, x and $\frac{p}{m\omega}$ have the same dimensions, which they must in order for the functional form to be the same. As an example of this relationship, if

$$\psi_n(x) = ce^{-\frac{m\omega}{2\hbar}x^2}, \tag{7.96}$$

then

$$\tilde{\phi}_n(p) = c'e^{-\frac{1}{2\hbar m\omega}p^2}, \tag{7.97}$$

with c and c' constants.

Section 7.5 Problems:

7.5.1 Shifted Simple Harmonic Oscillator

If we have a shifted harmonic oscillator, with a Hamiltonian given by

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 + m\omega^2 x_0\hat{x}. \tag{7.98}$$

Define $\hat{\tilde{x}} = \hat{x} + x_0$. Then, $[\hat{x}, \hat{p}] = [\hat{\tilde{x}}, \hat{p}]$, since $[x_0, \hat{p}] = 0$. Rewrite $\hat{H}$ in terms of $\hat{p}$ and $\hat{\tilde{x}}$. What are the differences between $\hat{H}_{SHO}$ and the shifted harmonic oscillator? What are the similarities?

7.5.2 *Approximation of the Lennard-Jones 6–12 Potential*

Consider the Lennard-Jones 6–12 potential given by

$$V_{L-J}(x) = V_0\left[\left(\frac{x_0}{x}\right)^{12} - 2\left(\frac{x_0}{x}\right)^{6}\right]. \tag{7.99}$$

The minimum occurs at $x = x_0$ and the minimum value of the potential is $V(x_0) = -V_0$. The curvature (or "spring constant") is $V_0\frac{72}{x_0^2} = \kappa$. So, we have

$$\begin{aligned} V_{L-J}(x) &\approx -V_0 + \frac{1}{2}V_0\frac{72}{x_0^2}(x - x_0)^2 + \cdots \quad \text{for } x \text{ near } x_0 \\ &\approx -V_0 + 36V_0\left(1 - \frac{x}{x_0}\right)^2 + \cdots . \end{aligned} \tag{7.100}$$

Determine the percentage error for V_{L-J} when $x = 0.95x_0$ and when $x = 1.05x_0$ by computing

$$\left|\frac{V_{L-J}^{harmonic}(x) - V_{L-J}(x)}{V_{L-J}(x)}\right| \times 100. \tag{7.101}$$

How accurate do you expect the harmonic approximation to be for this potential?

7.6 Operator Method to Determine Energy Eigenstates of the Simple Harmonic Oscillator

We will work with an abstract operator method to determine the energy eigenstates of the simple harmonic oscillator. This method has its origin in the matrix mechanics of Heisenberg, Born, and Jordan from 1925; see Fig. 7.7 for these three inventors of matrix mechanics. The approach has been modified somewhat since then and is a standard in all quantum textbooks. It is often attributed to Dirac, since it

Fig. 7.7 Werner Heisenberg (left), Max Born (middle), and Pascual Jordan (right) all circa 1925 (Heisenberg image photographed by Friedrich Hund, AIP Emilio Segrè Visual Archives, Physics Today Collection; Born image courtesy of AIP Emilio Segrè Visual Archives, Lande Collection; and Jordan image courtesy of AIP Emilio Segrè Visual Archives, Segrè Collection. Images converted to look like sketches by James Freericks)

was written in his 1930 textbook (Dirac, 1930), but it also appears in Born and Jordan's earlier 1930 textbook (Born and Jordan, 1930), in essentially identical form. Dirac's contribution is primarily in thinking of these objects as operators as opposed to matrices, which is an important intellectual abstraction. We will modify the standard argument here to be more in line with Schrödinger's factorization method (Schrödinger, 1940), which we will develop further throughout the book. We start from the simple harmonic oscillator Hamiltonian:

$$\hat{H}_{SHO} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2, \tag{7.102}$$

and notice that, aside from the numerical factors out front, it looks reminiscent of $\hat{S}_x^2 + \hat{S}_y^2$. Recall when we worked with spin, that we rewrote the sum of the squares as $\frac{1}{2}\left(\hat{S}_+\hat{S}_- + \hat{S}_-\hat{S}_+\right)$, with $\hat{S}_\pm = \hat{S}_x \pm i\hat{S}_y$. This is like factoring the sum of squares as $\alpha^2 + \beta^2 = (\alpha - i\beta)(\alpha + i\beta)$—except here, we need to worry about operator ordering. It has also become standard to work with operators that are dimensionless for the simple harmonic oscillator. So, after a little trial and error, we find that the correct operators to use are

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + i\frac{\hat{p}}{m\omega}\right) \text{ and } \hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - i\frac{\hat{p}}{m\omega}\right). \tag{7.103}$$

These operators are *not* Hermitian, but $(\hat{a})^\dagger = \hat{a}^\dagger$ and $(\hat{a}^\dagger)^\dagger = \hat{a}$, which immediately follows because the position and momentum operators are Hermitian. Note how $\hat{x}$ and $\frac{\hat{p}}{m\omega}$ both appear in a similar way, as we predicted in Sect. 7.5. As with all operators we have worked with in the past, one of the first things we do is

look at the commutator. We have

$$\begin{aligned}[\hat{a}, \hat{a}^\dagger] &= \frac{m\omega}{2\hbar}\left[\hat{x} + i\frac{\hat{p}}{m\omega}, \hat{x} - i\frac{\hat{p}}{m\omega}\right] \\ &= \frac{m\omega}{2\hbar}[\hat{x}, \hat{x}] + \frac{i}{2\hbar}\Big(-[\hat{x}, \hat{p}] + [\hat{p}, \hat{x}]\Big) + \frac{1}{2\hbar m\omega}[\hat{p}, \hat{p}] \\ &= \frac{i}{2\hbar}(-2i\hbar) = 1. \end{aligned} \tag{7.104}$$

So, $[\hat{a}, \hat{a}^\dagger] = 1$.

Next, we invert our operator relations to

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}\big(\hat{a} + \hat{a}^\dagger\big) \text{ and } \hat{p} = -i\sqrt{\frac{\hbar m\omega}{2}}\big(\hat{a} - \hat{a}^\dagger\big). \tag{7.105}$$

and substitute into $\hat{H}_{SHO}$, being mindful of the operator ordering, to find

$$\begin{aligned}\hat{H}_{SHO} &= \frac{1}{2m}(-i)^2\frac{\hbar m\omega}{2}(\hat{a} - \hat{a}^\dagger)(\hat{a} - \hat{a}^\dagger) + \frac{1}{2}m\omega^2\frac{\hbar}{2m\omega}(\hat{a} + \hat{a}^\dagger)(\hat{a} + \hat{a}^\dagger) \\ &= -\frac{\hbar\omega}{4}\Big(\hat{a}^2 - \hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a} + (\hat{a}^\dagger)^2\Big) + \frac{\hbar\omega}{4}\Big(\hat{a}^2 + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} + (\hat{a}^\dagger)^2\Big) \\ &= \frac{\hbar\omega}{2}\Big(\hat{a}^\dagger\hat{a} + \hat{a}\hat{a}^\dagger\Big) = \frac{\hbar\omega}{2}\Big(2\hat{a}^\dagger\hat{a} + [\hat{a}, \hat{a}^\dagger]\Big) \\ &= \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right). \end{aligned} \tag{7.106}$$

We can use this to immediately find the ground state. First, we recall that the norm squared of a vector $\langle\psi|\psi\rangle \geq 0$, and if the norm equals zero, then it must be the zero vector. If $|\psi\rangle$ is the ground state of the simple harmonic oscillator Hamiltonian, then $\hat{H}_{SHO}|\psi\rangle = E_{gs}|\psi\rangle$, or $E_{gs} = \langle\psi|\hat{H}_{SHO}|\psi\rangle$. Using our form for the Hamiltonian in Eq. (7.106) then gives us $E_{gs} = \hbar\omega(\langle\psi|\hat{a}^\dagger\hat{a}|\psi\rangle + \frac{1}{2})$, because $\langle\psi|\psi\rangle = 1$. The first term involves the norm squared of $\hat{a}|\psi\rangle$, so that the energy is minimized for $\hat{a}|\psi\rangle = 0$, and the ground-state energy is then $E_{gs} = \hbar\omega/2$. We denote the ground state as $|0\rangle$, so that $\hat{a}|0\rangle = 0$, is the subsidiary condition for this system; note that the right-hand side is the 0 vector.

Now, we want to find excited states. We know that $\hat{a}|0\rangle = 0$, so we cannot use powers of $\hat{a}$ acting on the ground state to create excited states. What about $\hat{a}^\dagger|0\rangle$. Well, this state cannot vanish, because $|0\rangle = [\hat{a}, \hat{a}^\dagger]|0\rangle = \hat{a}\hat{a}^\dagger|0\rangle$, which implies $\hat{a}^\dagger|0\rangle \neq 0$. Similarly, one can show $\big(\hat{a}^\dagger\big)^n|0\rangle \neq 0$. Our claim is $\big(\hat{a}^\dagger\big)^n|0\rangle$ are the (unnormalized) excited energy eigenstates with $n = 1, 2, 3, \cdots$. To check this, we need to compute $\hat{H}_{SHO}(\hat{a}^\dagger)^n|0\rangle$, which we will do by working out some commutators. We want to compute $[\hat{a}, (\hat{a}^\dagger)^n]$. We work out the first few with the

Leibniz rule. For $n = 1$, we already have $[\hat{a}, \hat{a}^\dagger] = 1$. For $n = 2$, we have

$$[\hat{a}, (\hat{a}^\dagger)^2] = \hat{a}^\dagger[\hat{a}, \hat{a}^\dagger] + [\hat{a}, \hat{a}^\dagger]\hat{a}^\dagger = \hat{a}^\dagger \cdot 1 + 1 \cdot \hat{a}^\dagger = 2\hat{a}^\dagger. \tag{7.107}$$

It is easy to work out the $n = 3$ case as $[\hat{a}, (\hat{a}^\dagger)^3] = 3(\hat{a}^\dagger)^2$. This motivates an induction hypothesis that $[\hat{a}, (\hat{a}^\dagger)^n] = n(\hat{a}^\dagger)^{n-1}$. We already have shown the base case, with $n = 1$. Assume it holds for $n - 1$ and then we use Leibniz for the n case, with a breakup as $\hat{a}^\dagger(\hat{a}^\dagger)^{n-1}$:

$$\begin{aligned}
[\hat{a}, (\hat{a}^\dagger)^n] &= \hat{a}^\dagger[\hat{a}, (\hat{a}^\dagger)^{n-1}] + [\hat{a}, \hat{a}^\dagger](\hat{a}^\dagger)^{n-1} \\
&= \hat{a}^\dagger(n-1)(\hat{a}^\dagger)^{n-2} + (\hat{a}^\dagger)^{n-1} \\
&= n(\hat{a}^\dagger)^{n-1},
\end{aligned} \tag{7.108}$$

which completes the proof by induction. We use this to show that $(\hat{a}^\dagger)^n|0\rangle$ is an eigenstate as follows:

$$\begin{aligned}
\hat{H}_{SHO}(\hat{a}^\dagger)^n|0\rangle &= \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)(\hat{a}^\dagger)^n|0\rangle \\
&= \hbar\omega\left(\hat{a}^\dagger[\hat{a}, (\hat{a}^\dagger)^n] + \frac{1}{2}(\hat{a}^\dagger)^n\right)|0\rangle \\
&= \hbar\omega\left(n + \frac{1}{2}\right)(\hat{a}^\dagger)^n|0\rangle.
\end{aligned} \tag{7.109}$$

We used the add-zero trick, because $\hat{a}|0\rangle = 0$, to create the commutator in the second line. These are all the energy eigenstates.

Normalization follows simply as well. We denote the nth normalized eigenstate as $|n\rangle$. Then, because we have

$$\begin{aligned}
\langle 0|\hat{a}^n(\hat{a}^\dagger)^n|0\rangle &= \langle 0|\hat{a}^{n-1}\hat{a}(\hat{a}^\dagger)^n|0\rangle \\
&= \langle 0|\hat{a}^{n-1}[\hat{a}, (\hat{a}^\dagger)^n]|0\rangle \\
&= n\langle 0|\hat{a}^{n-1}(\hat{a}^\dagger)^{n-1}|0\rangle \\
&= n(n-1)\langle 0|\hat{a}^{n-2}(\hat{a}^\dagger)^{n-2}|0\rangle \\
&= n!\langle 0|0\rangle = n!.
\end{aligned} \tag{7.110}$$

We again use the add-zero trick to create the commutator and then iterate the procedure to get the $n!$. Then, the normalized eigenstate is $\frac{1}{\sqrt{n!}}(\hat{a}^\dagger)^n|0\rangle$.

One of the biggest questions we need to address in quantum mechanics is why do bound states only exist at a discrete set of allowed energies? This is one of the phenomena that is highly nonclassical—classical systems can have any energy

along a continuum, but bound states in the quantum realm cannot. They only are allowed to have discrete values, as we see with the energy eigenvalues of the simple harmonic oscillator. Note that this does not hold for all quantum states. As we saw when we discussed momentum eigenstates, they have a continuum of possible momentum values and hence, for a free particle Hamiltonian, a continuum of energy eigenvalues. It is the bound states that always have discrete energy levels. Why? There must be something special that requires this. Indeed, there is. Let's try to trace where this happened in our current example. Let's go back and look at how we calculated the excited energy eigenvalues. It is the fact that $[\hat{a}, (\hat{a}^\dagger)^n] = n(\hat{a}^\dagger)^{n-1}$ increases by one as the exponent n increases by 1. It is this discrete, nonzero increase that leads to the discreteness of the allowed energy eigenvalues. Again, we can trace that all back to the commutator of position with momentum being nonzero! When we discuss the factorization method in its full generality, we will see explicitly why we must have this discreteness. You have gotten your first taste of this here.

We have one more property to work out before we are done. We want to compute $\hat{a}|n\rangle$ and $\hat{a}^\dagger|n\rangle$. We have

$$\hat{a}|n\rangle = \frac{1}{\sqrt{n!}}\hat{a}(\hat{a}^\dagger)^n|0\rangle = \frac{1}{\sqrt{n!}}[\hat{a}, (\hat{a}^\dagger)^n]|0\rangle = \frac{n}{\sqrt{n!}}(\hat{a}^\dagger)^{n-1}|0\rangle = \sqrt{n}|n-1\rangle. \tag{7.111}$$

Similarly, we have

$$\hat{a}^\dagger|n\rangle = \frac{1}{\sqrt{n!}}\hat{a}^\dagger(\hat{a}^\dagger)^n|0\rangle = \frac{1}{\sqrt{n!}}(\hat{a}^\dagger)^{n+1}|0\rangle = \sqrt{n+1}|n+1\rangle. \tag{7.112}$$

We say that the $\hat{a}$ operator lowers the energy eigenstate and is the Dirac *lowering operator* and that $\hat{a}^\dagger$ operator raises the energy eigenstate and $\hat{a}^\dagger$ is the Dirac *raising operator*. Together, the pair of operators is called the Dirac *ladder operators*. We can step up the energy eigenstates with the raising operator and step down them with the lowering operator. The relations are well worth remembering, and as a mnemonic, the coefficient is always the largest number in the equation. Note as well that the subsidiary condition is the lowering condition with $n = 0$—the ground state cannot have its energy lowered. The video Factorization Method for the Simple Harmonic Oscillator https://tinyurl.com/3wnp6v79 summarizes these discussions.

When we develop the full Schrödinger factorization method, we will revisit the simple harmonic oscillator calculation and solve it in a somewhat different fashion by using the so-called *intertwining relation*.

Section 7.6 Problems:

7.6.1 Commutator of Ladder Operators

Prove that $[\hat{a}^\dagger, \hat{a}^n] = -n\hat{a}^{n-1}$.

7.6.2 Inner Products of Energy Eigenstates

Using $|n\rangle = \frac{1}{\sqrt{n!}}(\hat{a}^\dagger)^n|0\rangle$, show that $\langle m|n\rangle = 0$ unless $m = n$.
Hint: $\langle 0|\hat{a}|0\rangle = 0$, because $\hat{a}|0\rangle = 0$. Similarly, $\langle 0|\hat{a}^\dagger|0\rangle = 0$, because $\langle 0|\hat{a}^\dagger = 0$, by taking the Hermitian conjugate of the equation $\hat{a}|0\rangle = 0$.

7.6.3 Matrix Elements of the Ladder Operators

Determine $\langle m|\hat{a}^\dagger|n\rangle$ and $\langle m|\hat{a}|n\rangle$. You should find only one m value gives a nonzero result for fixed n for both cases. You can also use the complex conjugation operation and the fact that $\langle m|\hat{a}|n\rangle^* = \langle n|\hat{a}^\dagger|m\rangle$ to check your results.

7.6.4 Number Operator

We define the number operator by $\hat{n} = \hat{a}^\dagger\hat{a}$. Show that the state $|n\rangle = \frac{1}{\sqrt{n!}}(\hat{a}^\dagger)^n|0\rangle$ satisfies $\hat{n}|n\rangle = n|n\rangle$. That is, it is a normalized eigenstate of the number operator, with eigenvalue n.

7.6.5 Powers

Show the following three relations (α is a complex number):

$$(\alpha\hat{a}^\dagger + \alpha^*\hat{a})^2 = \alpha^2(\hat{a}^\dagger)^2 + 2|\alpha|^2\hat{a}^\dagger\hat{a} + (\alpha^*)^2\hat{a}^2 + |\alpha|^2; \tag{7.113}$$

$$\begin{aligned}(\alpha\hat{a}^\dagger + \alpha^*\hat{a})^3 = {} & \alpha^3(\hat{a}^\dagger)^3 + 3\alpha|\alpha|^2(\hat{a}^\dagger)^2\hat{a} + 3\alpha^*|\alpha|^2\hat{a}^\dagger\hat{a}^2 + (\alpha^*)^3\hat{a}^3 \\ & + 3\alpha|\alpha|^2\hat{a}^\dagger + 3\alpha^*|\alpha|^2\hat{a};\end{aligned} \tag{7.114}$$

and

$$\begin{aligned}(\alpha\hat{a}^\dagger + \alpha^*\hat{a})^4 = {} & \alpha^4(\hat{a}^\dagger)^4 + 4\alpha^2|\alpha|^2(\hat{a}^\dagger)^3\hat{a} + 6|\alpha|^4(\hat{a}^\dagger)^2\hat{a}^2 + 4(\alpha^*)^2|\alpha|^2\hat{a}^\dagger\hat{a}^3 \\ & + (\alpha^*)^4\hat{a}^4 + 6\alpha^2|\alpha|^2(\hat{a}^\dagger)^2 + 12|\alpha|^4\hat{a}^\dagger\hat{a} + 6(\alpha^*)^2|\alpha|^2\hat{a}^2 + 3|\alpha|^4.\end{aligned} \tag{7.115}$$

You will need to use commutators to put the operator expressions into the desired expressions above. The form of the operators in these results, with the raising operators all to the left of the lowering operators, is called *normal-ordered form*.

7.7 Heisenberg Uncertainty Relations

Suppose we perform a series of measurements of the values of some Hermitian operator $\hat{A}$ in some quantum state $|\psi\rangle$. We know, on average, we will measure $\langle\psi|\hat{A}|\psi\rangle$. But do all operators and states behave the same? The answer is no! One important property to examine is the "fluctuations about the mean," or how much do the measured values vary about the mean, yet produce the same result, on average.

Consider the plots in Fig. 7.8. The vertical axis is the value of the measurement, while the horizontal axis is the experimental measurement trials (or shots). One can see the first curve is a horizontal line with no fluctuations. This is what we would see if $|\psi\rangle$ was an eigenstate of $\hat{A}$. The middle shows moderate fluctuations and the right large fluctuations. In these cases, the state $|\psi\rangle$ is not an eigenstate of $\hat{A}$, but the fluctuations are largest in the right figure. Clearly, these three cases are different, so there is information about the state we measure that is hidden in these fluctuations. As we already saw, the fluctuations vanish if the state is an eigenstate, but what if the fluctuations are moderate, versus large. What does this tell us?

Seeing fluctuations is not unexpected. We know some things fluctuate more than others (think of daily temperatures in New York City, versus those at an island close to the equator). But in quantum mechanics, we have incompatible operators that cannot be simultaneously diagonalized. Now, we turn the question around—what is the best quantum state I can find to minimize the *product* of the fluctuations of both incompatible operators. The answer to this leads to Heisenberg's uncertainty relations. Note that because these operators are incompatible, we cannot find a simultaneous eigenstate for both operators. So, there must be some fluctuations. We are asking how small can we make them? As an example, we will calculate the fluctuations of position and momentum in the different simple harmonic oscillator eigenstates and relate this to Heisenberg's uncertainty relation. To begin, the fluctuation (or variance) of an operator $\hat{A}$ in a state $|\psi\rangle$ is defined to be

$$(\Delta A_\psi)^2 = \langle\psi|(\hat{A} - \langle\psi|\hat{A}|\psi\rangle)^2|\psi\rangle = \|\big(\hat{A} - \langle\psi|\hat{A}|\psi\rangle\big)|\psi\rangle\|^2. \tag{7.116}$$

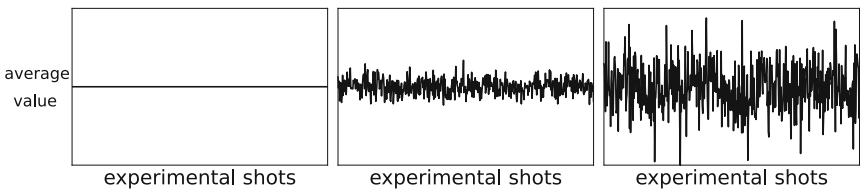

Fig. 7.8 Example of experimental data with the same average value, but different fluctuations. On the left, we have a system with no fluctuations, in the middle with moderate fluctuations, and on the right, with large fluctuations. We take the square root of the sum of the squares of the deviations from the mean to quantitatively measure the size of the fluctuations

This is the square of the deviation of the operator from its average value (also called the *standard deviation*); the second form shows it in terms of the norm of a state, after a specific operator acts on it. Why do we define it this way? If $|\psi\rangle$ is an eigenstate of $\hat{A}$, then $\hat{A}|\psi\rangle = A|\psi\rangle$ with A a number and $\langle\psi|\hat{A}|\psi\rangle = A$, because $\langle\psi|\psi\rangle = 1$. Hence, $(\Delta A_\psi)^2 = \langle\psi|(\hat{A} - A)^2|\psi\rangle$, which becomes $\langle\psi|(\hat{A} - A)(\hat{A} - A)|\psi\rangle = 0$, because $\hat{A}|\psi\rangle = A|\psi\rangle$ implies $(\hat{A} - A)|\psi\rangle = 0$. So, the fluctuation of an operator in an eigenstate is zero. This is good. It says there are no fluctuations when measuring an operator in an eigenstate, from the eigenvalue-eigenvector relation. It is also consistent with saying the state is a state of definite A.

What about when $|\psi\rangle$ is not an eigenvector of $\hat{A}$? Then, we have to work to perform the calculation. But there is a simplification—we can rewrite $(\Delta A_\psi)^2$ as follows:

$$\begin{aligned}(\Delta A_\psi)^2 &= \langle\psi|(\hat{A} - \langle\psi|\hat{A}|\psi\rangle)^2|\psi\rangle \\ &= \langle\psi|\hat{A}^2 - 2\hat{A}\langle\psi|\hat{A}|\psi\rangle + \big(\langle\psi|\hat{A}|\psi\rangle\big)^2|\psi\rangle \\ &= \langle\psi|\hat{A}^2|\psi\rangle - 2\big(\langle\psi|\hat{A}|\psi\rangle\big)^2 + \big(\langle\psi|\hat{A}|\psi\rangle\big)^2 \\ &= \langle\psi|\hat{A}^2|\psi\rangle - \big(\langle\psi|\hat{A}|\psi\rangle\big)^2. \end{aligned} \tag{7.117}$$

So, we need to compute only two terms to get the fluctuation of an operator in a given state.

But why sum the squares of the deviations? This is because, if we sum the deviations only, they vary positively and negatively about the average value and sum to zero, regardless of how big the magnitude of the fluctuations are. Hence, when we sum the squares, all terms are positive, and we can clearly tell the difference between a small and a large fluctuation. We usually take a square root at the end to make sure the fluctuations have the same units as the average or mean value has.

The uncertainty has a geometric relationship that is straightforward to visualize. If we start with the state $|\psi\rangle$, then $\langle\psi|\hat{A}|\psi\rangle|\psi\rangle$ is a real number multiplying $|\psi\rangle$ and represents the projection of $\hat{A}|\psi\rangle$ onto $|\psi\rangle$. The vector $\hat{A}|\psi\rangle$ scales and rotates the vector $|\psi\rangle$. Then, the difference $\hat{A}|\psi\rangle - \langle\psi|\hat{A}|\psi\rangle|\psi\rangle$ is perpendicular to $|\psi\rangle$, which is easily seen by multiplying by the bra $\langle\psi|$, where we have

$$\langle\psi|\hat{A}|\psi\rangle - \langle\psi|\hat{A}|\psi\rangle\langle\psi|\psi\rangle = 0, \tag{7.118}$$

because $\langle\psi|\psi\rangle = 1$. This means the vector $\hat{A}|\psi\rangle - \langle\psi|\hat{A}|\psi\rangle|\psi\rangle$ is perpendicular to $|\psi\rangle$, as shown in Fig. 7.9. Now, the norm of that perpendicular component is equal to the fluctuations $\Delta A_\psi = \|(\hat{A} - \langle\psi|\hat{A}|\psi\rangle)|\psi\rangle\|$. This geometrical picture is often a useful way to picture what the variance, or uncertainty, or fluctuations are of an operator with respect to a particular state. In particular, if $|\psi\rangle$ is an eigenstate, then when $\hat{A}$ acts on it, it only scales the state—then there is no rotation, and the variance vanishes.

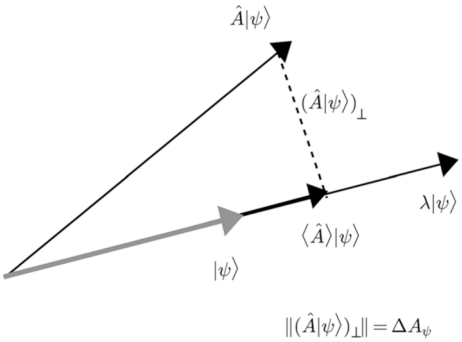

Fig. 7.9 Geometrical relationship to understand uncertainty. The state vector $|\psi\rangle$ is highlighted in gray and the expectation value $\langle\hat{A}\rangle = \langle\psi|\hat{A}|\psi\rangle$ multiplied by $|\psi\rangle$ is a vector that has just scaled $|\psi\rangle$ by a real number since $\hat{A}$ is Hermitian. Of course, operating $\hat{A}$ on $|\psi\rangle$ will scale and rotate the vector. The length of the perpendicular component $\left(\hat{A}|\psi\rangle\right)_\perp$ is the uncertainty ΔA_ψ

In Sect. 5.3, we discussed the classical harmonic motion of a mass on a spring, which is given by

$$x(t) = A\cos(\omega t + \phi) \text{ and } p(t) = -m\omega A\sin(\omega t + \phi). \tag{7.119}$$

These two functions are plotted in Fig. 5.8. You can clearly see from them that both the cosine and the sine are as much negative as positive, meaning that the average of the position and momentum over one period in time is zero. Similarly, if we square both quantities and use the facts that the averages of $\cos^2$ and $\sin^2$ are each $\frac{1}{2}$ (which follows from the fact that $\cos^2\theta + \sin^2\theta = 1$ and each average is the same over one period—just shift the two curves and see they track each other perfectly), this allows us to compute the fluctuations of position and momentum as a function of time—the average position and momentum are both zero—so we find

$$\Delta x = \frac{A}{\sqrt{2}} \text{ and } \Delta p = \frac{Am\omega}{\sqrt{2}}. \tag{7.120}$$

These two quantities can be used to find the average energy over the period as well, which is

$$E_{\text{ave}} = \frac{(\Delta p)^2}{2m} + \frac{m\omega^2(\Delta x)^2}{2} = \frac{A^2m\omega^2}{4} + \frac{A^2m\omega^2}{4} = \frac{A^2m\omega^2}{2}. \tag{7.121}$$

Now, we express the fluctuations in position and momentum in terms of the average energy:

$$\Delta x = \frac{\sqrt{E_{\text{ave}}}}{\sqrt{m}\omega} \text{ and } \Delta p = \sqrt{E_{\text{ave}}m}. \tag{7.122}$$

This means the product of the uncertainties in position and momentum are

$$\Delta x \Delta p = \frac{E_{\text{ave}}}{\omega}. \tag{7.123}$$

So, we learn that for a classical oscillator, if we look at the fluctuations over time (there is no equivalent of fluctuations when we measure for a classical system, as we have with a quantum system—the only classical fluctuations are over time), we find the product of the fluctuations in position and momentum is equal to the average energy divided by the frequency. We will find this same formula holds for the quantum system (but now for the fluctuations in quantum measurements made in energy eigenstates, as opposed to classical fluctuations over time). It is amazing that such a correspondence will occur.

Now, we are ready to calculate the quantum fluctuations in the energy eigenstates (given by $|n\rangle = \frac{1}{\sqrt{n!}}(\hat{a}^\dagger)^n|0\rangle$). We start with computing $(\Delta x_n)^2$. First, recall that $\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a} + \hat{a}^\dagger)$. Then

$$\langle n|\hat{x}|n\rangle = \langle n|\sqrt{\frac{\hbar}{2m\omega}}(\hat{a} + \hat{a}^\dagger)|n\rangle = \frac{1}{n!}\sqrt{\frac{\hbar}{2m\omega}}\langle 0|\hat{a}^{n+1}(\hat{a}^\dagger)^n + \hat{a}^n(\hat{a}^\dagger)^{n+1}|0\rangle. \tag{7.124}$$

We need to compute these expectation values. We can only use what we know. How do we do it? Think about this before reading any further.

Recall that $[\hat{a}, (\hat{a}^\dagger)^n] = n(\hat{a}^\dagger)^{n-1}$. We can then use our add-zero trick to replace $\hat{a}(\hat{a}^\dagger)^n|0\rangle = [\hat{a}, (\hat{a}^\dagger)^n]|0\rangle$, since $\hat{a}|0\rangle = 0$. So, we have

$$\begin{aligned}\langle 0|\hat{a}^{n+1}(\hat{a}^\dagger)^n|0\rangle &= \langle 0|\hat{a}^n n(\hat{a}^\dagger)^{n-1}|0\rangle = \langle 0|\hat{a}^{n-1}n(n-1)(\hat{a}^\dagger)^{n-2}|0\rangle = \cdots \\ &= n!\langle 0|\hat{a}|0\rangle = 0,\end{aligned} \tag{7.125}$$

since $\hat{a}|0\rangle = 0$ by the subsidiary condition. We used the add-zero trick to convert the expectation values to commutators, by peeling an $\hat{a}$ operator off, one at a time, and forming a commutator with the remaining powers of $\hat{a}^\dagger$. Similarly, we have

$$\langle 0|\hat{a}^n(\hat{a}^\dagger)^{n+1}|0\rangle = n!\langle 0|\hat{a}^\dagger|0\rangle = 0, \tag{7.126}$$

since $\langle 0|\hat{a}^\dagger = 0$. This is because it is the Hermitian conjugate of the equation $\hat{a}|0\rangle = 0$. In quantum mechanics, the operators can act to the right on a ket or to the left on a bra. To determine how they act to the left on a bra, we first determine how the Hermitian conjugate operator acts to the right on a ket and then take the Hermitian conjugate of the equation. So, we have $\hat{a}|0\rangle = 0 \implies \langle 0|\hat{a}^\dagger = 0$. We have seen in this example that when the number of raising operators and the number of lowering operators in a ground-state expectation value differ by 1, then the expectation value vanishes. This turns out to be generalized to the expectation value vanishing if the number of raising operators is not equal to the number of lowering operators, which

is a very useful fact to remember. After all of this long work, we arrive at the fact that $\langle n|\hat{x}|n\rangle = 0$.

Next up, we need to examine $\langle n|\hat{x}^2|n\rangle$. This becomes

$$\begin{aligned}\langle n|\hat{x}^2|n\rangle &= \frac{\hbar}{2m\omega}\frac{1}{n!}\langle 0|\hat{a}^n(\hat{a}+\hat{a}^\dagger)^2(\hat{a}^\dagger)^n|0\rangle \\ &= \langle 0|\Big(\hat{a}^{n+2}(\hat{a}^\dagger)^n + \hat{a}^{n+1}(\hat{a}^\dagger)^{n+1} + \hat{a}^n\hat{a}^\dagger\hat{a}(\hat{a}^\dagger)^n + \hat{a}^n(\hat{a}^\dagger)^{n+2}\Big)|0\rangle.\end{aligned} \tag{7.127}$$

The first and last terms vanish, since the number of raising and lowering operators do not balance. The middle terms become

$$\langle n|\hat{x}^2|n\rangle = \frac{\hbar}{2m\omega}\frac{1}{n!}\Big((n+1)! + n\times n!\Big) = \frac{\hbar}{m\omega}\left(n+\frac{1}{2}\right). \tag{7.128}$$

Hence, $(\Delta x_n)^2 = \frac{\hbar}{m\omega}\left(n+\frac{1}{2}\right)$. So, we have that $\Delta x_n = \sqrt{\frac{\hbar}{m\omega}}\sqrt{n+\frac{1}{2}}$.

Now, let's do the momentum case. Recall $\hat{p} = \sqrt{\frac{\hbar m\omega}{2}}(-i)(\hat{a}-\hat{a}^\dagger)$. So

$$\langle n|\hat{p}|n\rangle = -i\sqrt{\frac{\hbar m\omega}{2}}\frac{1}{n!}\langle 0|\hat{a}^n(\hat{a}-\hat{a}^\dagger)(\hat{a}^\dagger)^n|0\rangle = 0, \tag{7.129}$$

because the $\hat{a}$ and $\hat{a}^\dagger$ factors are unbalanced. Next, we have

$$\begin{aligned}\langle n|\hat{p}^2|n\rangle &= -\frac{\hbar m\omega}{2}\frac{1}{n!}\langle 0|\hat{a}^n\Big(\hat{a}^2 - \hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a} + (\hat{a}^\dagger)^2\Big)(\hat{a}^\dagger)^n|0\rangle \\ &= -\frac{\hbar m\omega}{2}\frac{1}{n!}\Big(0 - (n+1)! - n\times n! + 0\Big) = \hbar m\omega\left(n+\frac{1}{2}\right).\end{aligned} \tag{7.130}$$

So, $(\Delta p_n)^2 = \hbar m\omega\left(n+\frac{1}{2}\right)$ and $\Delta p_n = \sqrt{\hbar m\omega}\sqrt{n+\frac{1}{2}}$.

We define the uncertainty for a state $|\psi\rangle$ to be

$$\Delta\psi = \sqrt{(\Delta x_\psi)^2}\sqrt{(\Delta p_\psi)^2} = \Delta x_\psi \Delta p_\psi. \tag{7.131}$$

For the harmonic oscillator energy eigenstates, we take $|\psi\rangle = |n\rangle$, and then

$$\Delta\psi_n = \sqrt{\frac{\hbar}{m\omega}\left(n+\frac{1}{2}\right)\hbar m\omega\left(n+\frac{1}{2}\right)} = \hbar\left(n+\frac{1}{2}\right). \tag{7.132}$$

This result is exactly the same as the classical result, because the right-hand side is equal to E_n/ω. It is amazing that we can relate a classical time average to a quantum fluctuation in the state in this fashion. It does not always work out this way—the

harmonic oscillator is special. We will find below that the Heisenberg uncertainty principle says that

$$\Delta\psi_n \geq \frac{\hbar}{2}. \tag{7.133}$$

We see that this holds for the simple harmonic oscillator energy eigenstates, with equality holding only for the ground state. We say the ground state has *minimal uncertainty*.

The fact that even the ground state has some uncertainty follows from the fact that $E_0 = \frac{1}{2}\hbar\omega$. This means the lowest-energy state has nonzero energy. We call this energy the *zero-point energy*, and it implies that the ground state has some "motion" in it (in the sense that it has nonzero kinetic and potential energy, on average). But the zero point motion is not the regular oscillations of a spring. We will describe the regular oscillations when we discuss coherent states later.

We can go further with this idea by examining what is called the *virial theorem*. To derive it, we simply take the commutator of $\hat{H}$ with the product of operators $\hat{x}\,\hat{p}$ (called the virial) and evaluate the expectation value in an energy eigenstate $|n\rangle$. Recall that we can evaluate the commutator with $\hat{H}$ easily, because $\hat{H}|n\rangle = E_n|n\rangle$ and similarly for its Hermitian conjugate. We find

$$\langle n|[\hat{H}, \hat{x}\,\hat{p}]|n\rangle = \langle n|\hat{H}\hat{x}\,\hat{p} - \hat{x}\,\hat{p}\hat{H}|n\rangle = E_n\langle n|\hat{x}\,\hat{p} - \hat{x}\,\hat{p}|n\rangle = 0. \tag{7.134}$$

Now, we evaluate the commutator by directly computing it via the Leibniz rule

$$\begin{aligned}\langle n|[\hat{H}, \hat{x}\,\hat{p}]|n\rangle &= \langle n|\hat{x}[\hat{H}, \hat{p}] + [\hat{H}, \hat{x}]\hat{p}|n\rangle \\ &= \langle n|\hat{x}\left[\frac{1}{2}m\omega^2\hat{x}^2, \hat{p}\right] + \left[\frac{1}{2m}\hat{p}^2, \hat{x}\right]\hat{p}|n\rangle \\ &= i\hbar\langle n|m\omega^2\hat{x}^2 - \frac{\hat{p}^2}{m}|n\rangle = 0;\end{aligned} \tag{7.135}$$

we established that this vanishes with the previous calculation. We define the average kinetic energy in the energy eigenstate labeled by n to be $K_n = \langle n|\frac{\hat{p}^2}{2m}|n\rangle$ and the average potential energy in the energy eigenstate labeled by n to be $V_n = \langle n|\frac{1}{2}m\omega^2\hat{x}^2|n\rangle$. Then, the virial theorem tells us $K_n = V_n$ for the simple harmonic oscillator. Of course, because position and momentum are treated equally in the simple harmonic oscillator, this is what must happen. It is nice to see that we can derive it using just our operator manipulations. Classically, we know that the energy is continuously transferred from potential energy to kinetic energy and back during an oscillation so that the energy is half of the time kinetic and half of the time potential, on average—quantum mechanically, we learn that the average kinetic and potential energies in energy eigenstates are the same. Furthermore, since $E_n = K_n + V_n$, we have

$$K_n = \frac{1}{2}E_n = \frac{1}{2}\hbar\omega\left(n + \frac{1}{2}\right) = V_n, \tag{7.136}$$

for the simple harmonic oscillator.

The final task we have for this section is to derive the Heisenberg uncertainty principle for all states, not just the simple harmonic oscillator eigenstates. This requires us to first derive the Schwarz inequality, which says for any two quantum states $|\psi\rangle$ and $|\phi\rangle$, we have

$$|\langle\phi|\psi\rangle| \leq \|\phi\|\,\|\psi\| = \sqrt{\langle\phi|\phi\rangle\,\langle\psi|\psi\rangle}. \tag{7.137}$$

The proof is both subtle and ingenious, so pay attention! The "divine inspiration" is to look at

$$\|\,|\phi\rangle - \lambda|\psi\rangle\|^2 = \Big(\langle\phi| - \lambda^*\langle\psi|\Big)\cdot\Big(|\phi\rangle - \lambda|\psi\rangle\Big) \geq 0, \tag{7.138}$$

for complex λ. Expanding out the norm squared gives us

$$\langle\phi|\phi\rangle - \lambda\langle\phi|\psi\rangle - \lambda^*\langle\psi|\phi\rangle + |\lambda|^2\langle\psi|\psi\rangle \geq 0. \tag{7.139}$$

Rearranging, we obtain

$$\lambda\langle\phi|\psi\rangle + \lambda^*\langle\psi|\phi\rangle \leq \langle\phi|\phi\rangle + |\lambda|^2\langle\psi|\psi\rangle. \tag{7.140}$$

This holds for all λ. But one value of λ leads to a remarkable cancellation. Do you see what it is? We choose $\lambda = \frac{\langle\psi|\phi\rangle}{\langle\psi|\psi\rangle}$. Then, we have

$$\frac{|\langle\phi|\psi\rangle|^2 + |\langle\phi|\psi\rangle|^2}{\langle\psi|\psi\rangle} \leq \langle\phi|\phi\rangle + \frac{|\langle\psi|\phi\rangle|^2}{\langle\psi|\psi\rangle}, \tag{7.141}$$

or

$$|\langle\phi|\psi\rangle|^2 \leq \langle\phi|\phi\rangle\,\langle\psi|\psi\rangle. \tag{7.142}$$

Taking the square root of both sides gives

$$|\langle\phi|\psi\rangle| \leq \sqrt{\langle\phi|\phi\rangle\,\langle\psi|\psi\rangle}, \tag{7.143}$$

which is the Schwarz inequality. You may know it in the form of the absolute value of the dot product between two vectors is less than or equal to the product of the lengths of the two vectors. You might wonder where the second "divine inspiration" to choose the special value of λ came from—it is the value of λ that minimizes the norm we were calculating—this might not be so obvious but can be shown simply with calculus. However, that fact is not needed to appreciate the beauty of

the Schwarz inequality. You might even be able to guess this special value just by staring at the equations.

Okay. We are ready for uncertainty! We begin by defining

$$p_0 = \langle\psi|\hat{p}|\psi\rangle \text{ and } x_0 = \langle\psi|\hat{x}|\psi\rangle, \tag{7.144}$$

for a unit norm state ($\langle\psi|\psi\rangle = 1$). Now, we compute the commutator:

$$\langle\psi|[\hat{x} - x_0, \hat{p} - p_0]|\psi\rangle = i\hbar, \tag{7.145}$$

because subtracting constants inside a commutator does not change the commutator. We also can expand the commutator directly via

$$\begin{aligned}\langle\psi|[\hat{x} - x_0, \hat{p} - p_0]|\psi\rangle &= \langle\psi(\hat{x} - x_0)(\hat{p} - p_0)|\psi\rangle - \langle\psi|(\hat{p} - p_0)(\hat{x} - x_0)|\psi\rangle \\ &= \langle\psi|(\hat{x} - x_0)(\hat{p} - p_0)|\psi\rangle - \langle\psi|(\hat{x} - x_0)(\hat{p} - p_0)|\psi\rangle^* \\ &= 2i\,\mathrm{Im}\Big(\langle\psi|(\hat{x} - x_0)(\hat{p} - p_0)|\psi\rangle\Big).\end{aligned} \tag{7.146}$$

Now, we define two new vectors from our old ones as

$$|\phi\rangle = (\hat{p} - p_0)|\psi\rangle \text{ and } |\chi\rangle = (\hat{x} - x_0)|\psi\rangle. \tag{7.147}$$

The Schwarz inequality then says that

$$|\langle\chi|\phi\rangle| \leq \sqrt{\langle\chi|\chi\rangle\,\langle\phi|\phi\rangle}, \tag{7.148}$$

or

$$|\langle\psi|(\hat{x} - x_0)(\hat{p} - p_0)|\psi\rangle| \leq \sqrt{\langle\psi|(\hat{x} - x_0)^2|\psi\rangle\,\langle\psi|(\hat{p} - p_0)^2|\psi\rangle}. \tag{7.149}$$

But we already defined

$$\langle\psi|(\hat{x} - x_0)^2|\psi\rangle = (\Delta x_\psi)^2 \text{ and } \langle\psi|(\hat{p} - p_0)^2|\psi\rangle = (\Delta p_\psi)^2. \tag{7.150}$$

So, we find

$$\frac{\hbar}{2} = \mathrm{Im}\Big(\langle\psi|(\hat{x} - x_0)(\hat{p} - p_0)|\psi\rangle\Big) \leq |\langle\psi|(\hat{x} - x_0)(\hat{p} - p_0)|\psi\rangle| \leq \Delta x_\psi\,\Delta p_\psi, \tag{7.151}$$

or

$$\Delta x_\psi\,\Delta p_\psi \geq \frac{\hbar}{2} \tag{7.152}$$

as the general result. This is called the Heisenberg uncertainty principle, and it follows solely from the commutator of position with momentum and the Schwarz inequality! It says measurements of the position and the momentum of a particle that is in a quantum state $|\psi\rangle$ must fluctuate. It even can be used to tell how accurately we can know one or the other, on average, according to the explicit value of $\hbar$. It turns out this restriction only becomes important for small objects. Note that it is often misstated as "we cannot determine the position and momentum of a particle in a quantum state at the same time." We can, and we discuss this when we cover the time-of-flight measurement later. Uncertainty is about the fluctuations after many experiments, or it is a geometrical statement about a single quantum state, but it does not tell us what happens if we try to measure both quantities at once. That very interesting experiment will be discussed in detail once we have enough background to fully understand it, so stay tuned. The video Uncertainty Principle https://tinyurl.com/389yeem5 discusses these issues.

Section 7.7 Problems:

7.7.1 Uncertainty and Atoms

Atoms are on the order of one Angstrom (Å) in size, which is 0.1 nm $= 10^{-10}$ m. The mass of an electron is $m_e c^2 = 511{,}000$ eV, with $c \approx 3 \times 10^8$ m/s, the speed of light. (Here, m is meter and eV is electron volt.) Using the identity $hc = 12{,}400$ eV-Å, find the uncertainty in the velocity of an electron confined to a one-dimensional box the size of an Angstrom; that is, assume that $\Delta x_\psi = 1$ Å. What fraction of the speed of light is this uncertainty in the velocity? Note that the h in the hc identity is $h = 2\pi\hbar$. Use only the constants given in the problem here to formulate your answer. Recall the multiply-by-one trick!

7.7.2 Expectation Value of Cubes

Prove that $\langle n|\hat{x}^3|n\rangle = 0$. Prove also that $\langle n|\hat{p}^3|n\rangle = 0$ for $|n\rangle$ an energy eigenstate of a simple harmonic oscillator.

7.7.3 Expectation Values of Quartics

Compute $\langle n|\hat{x}^4|n\rangle$. Be careful with your operator orderings. Recall that we calculated $(\hat{a} + \hat{a}^\dagger)^4$ in Problem 7.6.5.

7.7.4 Matrix Elements of Position and Momentum

Prove that $\langle m|\hat{x}|n\rangle$ and $\langle m|\hat{p}|n\rangle$ are nonzero for $m = n \pm 1$. Compute $\langle n+1|\hat{x}|n\rangle$, $\langle n-1|\hat{x}|n\rangle$, $\langle n+1|\hat{p}|n\rangle$, and $\langle n-1|\hat{p}|n\rangle$.

7.7.5 Matrix Elements of Powers of Position and Momentum

Prove, in general, that

$$\langle m|\hat{x}^{\alpha}|n\rangle \text{ and } \langle m|\hat{p}^{\alpha}|n\rangle \text{ for } \alpha = \text{ positive integer} \tag{7.153}$$

are nonzero for

$$m = n-\alpha,\ n-\alpha+2,\ n-\alpha+4, \cdots, n+\alpha. \tag{7.154}$$

You do not need to compute the nonzero values!

7.7.6 Generalized Heisenberg Uncertainty Principle

Consider two Hermitian operators $\hat{A}$ and $\hat{B}$ and their commutator $\hat{C} = -i[\hat{A}, \hat{B}]$. Use techniques similar to what we did for the Heisenberg uncertainty principle to show that

$$\Delta A_{\psi}\,\Delta B_{\psi} \geq \frac{|\langle\psi|\hat{C}|\psi\rangle|}{2}, \tag{7.155}$$

where

$$(\Delta A_{\psi})^2 = \langle\psi|(\hat{A}-A_0)^2|\psi\rangle \text{ and } (\Delta B_{\psi})^2 = \langle\psi|(\hat{B}-B_0)^2|\psi\rangle \tag{7.156}$$

with $A_0 = \langle\psi|\hat{A}|\psi\rangle$ and $B_0 = \langle\psi|\hat{B}|\psi\rangle$.

This is the generalized Heisenberg uncertainty principle (first worked out by Robertson) for arbitrary operators that do not commute with each other.

7.7.7 Schrödinger's Version of Uncertainty

In the derivation of the generalized Heisenberg uncertainty principle above, you needed to use a relationship where you considered only the imaginary part of an

expectation value in the inequality. Schrödinger showed how to use both the real and imaginary parts. Use the add-zero trick, to show that

$$\langle\psi|\hat{A}\hat{B}|\psi\rangle = \frac{1}{2}\langle\psi|[\hat{A},\hat{B}]|\psi\rangle + \frac{1}{2}\langle\psi|\{\hat{A},\hat{B}\}|\psi\rangle = \frac{1}{2}\Big(i\langle\psi|\hat{C}|\psi\rangle + \langle\psi|\hat{D}|\psi\rangle\Big), \tag{7.157}$$

where $\hat{C} = -i[\hat{A},\hat{B}]$ and $\hat{D} = \{\hat{A},\hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A}$. Take this result and show the Schrödinger form of the uncertainty relation, which is given by

$$\Delta A_\psi \Delta B_\psi \geq \frac{\sqrt{|\langle\psi|\hat{C}|\psi\rangle|^2 + |\langle\psi|\hat{D}|\psi\rangle|^2}}{2}. \tag{7.158}$$

Comment: It is much more common to use the generalized Heisenberg uncertainty principle than Schrödinger's form. This is, in part, because it is easier to compute the commutator of two operators than the anticommutator. Other issues that arise when discussing the uncertainty principle are issues associated with the domains of the operators. This issue requires advanced analysis to properly treat. We will not need to worry about this issue unless you are told it is a problem. But it is something to keep in mind for the future as you continue your studies of quantum mechanics. The domain issues and their consequences can be subtle.

7.8 Parity

When we studied cosines and sines, we noted that $\sin(-x) = -\sin(x)$ and $\cos(-x) = \cos(x)$. The former is called an *odd* function and the latter is an *even* function. You have probably heard of even and odd functions before. Physically, you can think of putting a mirror along the vertical axis—if the function maps to itself in the mirror it is even, if it maps to its negative, it is odd.

In quantum mechanics, the generalization of even and odd functions is made with the *parity operator* which is denoted $\hat{\Pi}$. It is defined by how it acts on both position and momentum eigenstates, via

$$\hat{\Pi}\,|x\rangle = |-x\rangle \quad \text{and} \quad \hat{\Pi}\,|p\rangle = |-p\rangle. \tag{7.159}$$

It reflects the eigenstates about the origin—$\hat{\Pi}$ transforms the position (momentum) eigenstate at x (p) to the position (momentum) eigenstate at $-x$ ($-p$). What happens when we apply the parity operator twice. Clearly, we have

$$\hat{\Pi}^2|x\rangle = \hat{\Pi}(\hat{\Pi}\,|x\rangle) = \hat{\Pi}\,|-x\rangle = |x\rangle, \tag{7.160}$$

and similarly for $|p\rangle$. So, since $\{|x\rangle\}$ and $\{|p\rangle\}$ are both complete sets, we must have $\hat{\Pi}^2 = \hat{\mathbb{I}}$. This implies that the parity operator has eigenvalues equal to ± 1. Now, we take the Hermitian conjugate of Eq. (7.159):

$$\langle x|\,\hat{\Pi}^\dagger = \langle -x| \;\text{ and }\; \langle p|\,\hat{\Pi}^\dagger = \langle -p|. \tag{7.161}$$

Then

$$\langle x|\,\hat{\Pi}^\dagger\hat{\Pi}\,|p\rangle = \langle -x| - p\rangle. \tag{7.162}$$

But recall that $\langle x|p\rangle = \langle 0_x|0_p\rangle e^{\frac{i}{\hbar}xp}$, so we have $\langle x|p\rangle = \langle -x| - p\rangle$ for all x and p. This means

$$\hat{\Pi}^\dagger\hat{\Pi} = \hat{\mathbb{I}}, \tag{7.163}$$

or $\hat{\Pi}$ is unitary. Now, take $\hat{\Pi}^\dagger\hat{\Pi} = \hat{\mathbb{I}}$ and multiply from the right on both sides by $\hat{\Pi}$ and use the fact that $\hat{\Pi}^2 = \hat{\mathbb{I}}$, to learn that $\hat{\Pi}^\dagger = \hat{\Pi}$, or the parity operator is Hermitian. We have learned that the parity operator is an *involution* (squares to 1) that is unitary and Hermitian. That is a lot of special properties.

Finally, we compute $\hat{\Pi}\,\hat{x} + \hat{x}\,\hat{\Pi}$ by acting on the state $|x\rangle$. Then

$$\hat{\Pi}\,(\hat{x}|x\rangle) = x\,\hat{\Pi}\,|x\rangle = x|-x\rangle \;\text{ and }\; \hat{x}\,(\hat{\Pi}\,|x\rangle) = \hat{x}|-x\rangle = -x|x\rangle, \tag{7.164}$$

so $(\hat{\Pi}\,\hat{x}+\hat{x}\,\hat{\Pi})|x\rangle = 0$ for all x, which implies $\hat{\Pi}\,\hat{x}+\hat{x}\,\hat{\Pi} = 0$ as an operator relation. Using similar logic, we can also find that $\hat{\Pi}\,\hat{p} + \hat{p}\,\hat{\Pi} = 0$ as well. Multiplying both of these relations from the right by $\hat{\Pi}$ then gives us

$$\hat{\Pi}\,\hat{x}\,\hat{\Pi} = -\hat{x} \;\text{ and }\; \hat{\Pi}\,\hat{p}\,\hat{\Pi} = -\hat{p}. \tag{7.165}$$

We have worked out many properties of the parity operator. Let's now represent it in terms of our raising and lowering operators for the simple harmonic oscillator.

We define the number operator from the raising and lowering operators via $\hat{N} = \hat{a}^\dagger\hat{a}$. Then

$$[\hat{N}, \hat{a}] = [\hat{a}^\dagger\hat{a}, \hat{a}] = [\hat{a}^\dagger, \hat{a}]\hat{a} = -\hat{a}, \tag{7.166}$$

and

$$[\hat{N}, \hat{a}^\dagger] = [\hat{a}^\dagger\hat{a}, \hat{a}^\dagger] = \hat{a}^\dagger[\hat{a}, \hat{a}^\dagger] = \hat{a}^\dagger. \tag{7.167}$$

From our work on the simple harmonic oscillator in Sect. 7.6, we know that the harmonic oscillator energy eigenstates are also the eigenstates of the number operator, because

$$\hat{N}|n\rangle = \hat{a}^\dagger\hat{a}|n\rangle = \sqrt{n}\,\hat{a}^\dagger|n-1\rangle = n|n\rangle. \tag{7.168}$$

Our claim is that

$$\hat{\Pi} = (-1)^{\hat{N}} = e^{i\pi\hat{N}}. \tag{7.169}$$

The operator squares to $\hat{\mathbb{I}}$ because $e^{2i\pi\hat{N}} = \hat{\mathbb{I}}$—recall that all the eigenvalues of $\hat{N}$ are integers and $e^{2i\pi n} = 1$ for all integers n. Similarly $\left(e^{i\pi\hat{N}}\right)^{\dagger} = e^{-i\pi\hat{N}}$, because $\hat{N}^{\dagger} = (\hat{a}^{\dagger}\hat{a})^{\dagger} = \hat{a}^{\dagger}(\hat{a}^{\dagger})^{\dagger} = \hat{a}^{\dagger}\hat{a} = \hat{N}$. This is the inverse of $e^{i\pi\hat{N}}$. Hence, this operator is unitary. Finally, because $e^{i\pi\hat{N}} = (-1)^{\hat{N}}$, one can see it is Hermitian as well by taking the Hermitian conjugate. So, it satisfies all of the special properties of the parity operator. Finally, we need to verify that it inverts the position and momentum eigenstates. This can be done by verifying Eq. (7.165), which is an equivalent relation (that you should be able to verify). So, we have

$$\hat{\Pi}\,\hat{x}\,\hat{\Pi} = e^{i\pi\hat{N}}\,\hat{x}\,e^{-i\pi\hat{N}} = \sqrt{\frac{\hbar}{2m\omega}}e^{i\pi\hat{N}}(\hat{a}+\hat{a}^{\dagger})e^{-i\pi\hat{N}}. \tag{7.170}$$

We have to evaluate these Hadamard lemmas. They do not truncate but are an infinite series given in Eq. (7.284). Even though we have not derived the general formula yet, you can use it. It involves evaluating a number of nested commutators; they are evaluated by starting from the innermost commutator and moving outward. However, because $[\hat{N},\hat{a}] = -\hat{a}$, the n-fold nested commutator "repeats" and is equal to $(-1)^n\hat{a}$, and because $[\hat{N},\hat{a}^{\dagger}] = \hat{a}^{\dagger}$, the n-fold nested commutator also "repeats" and is equal to $\hat{a}^{\dagger}$. Hence, the Hadamard lemma acting on each these operators simplifies to $e^{i\pi\hat{N}}\hat{a}e^{-i\pi\hat{N}} = e^{-i\pi}\hat{a} = -\hat{a}$ and $e^{i\pi\hat{N}}\hat{a}^{\dagger}e^{-i\pi\hat{N}} = e^{i\pi}\hat{a}^{\dagger} = -\hat{a}^{\dagger}$ (you will verify in the problems). Plugging into our above result then gives

$$\hat{\Pi}\,\hat{x}\,\hat{\Pi} = -\hat{x}, \tag{7.171}$$

and similarly $\hat{\Pi}\,\hat{p}\,\hat{\Pi} = -\hat{p}$. This then finishes the proof that $\hat{\Pi} = (-1)^{\hat{N}}$.

If the eigenvalue of a state under $\hat{\Pi}$ is $+1$, we say the state is *even*, while if it is -1, we say the state is *odd*. This means $|n\rangle$ is even if n is even and it is odd if n is odd (see the problems for a proof). One can immediately see that $[\hat{H},\hat{N}] = 0$, because $\hat{H} = \hbar\omega\left(\hat{N}+\frac{1}{2}\right)$, so these states are simultaneous eigenstates of both operators.

Any operator that commutes with $\hat{\Pi}$ is said to be an *even operator*, so $\hat{H}$ is an even operator, while $\hat{x}$ and $\hat{p}$ are odd operators (they both anticommute with $\hat{\Pi}$). One important result of the parity operator is that states of opposite parity must be orthogonal. This is just our old result that eigenvectors with different eigenvalues are orthogonal. The proof is straight forward. If $\hat{\Pi}\,|a\rangle = |a\rangle$, so that $\langle a|\,\hat{\Pi}^{\dagger} = \langle a|$, and $\hat{\Pi}\,|b\rangle = -|b\rangle$. Then, we have that $\hat{\mathbb{I}} = \hat{\Pi}^{\dagger}\hat{\Pi}$ allows us to use a multiply by one inside the bra-ket to yield

$$\langle a|b\rangle = \langle a|\hat{\mathbb{I}}|b\rangle = \langle a|\,\hat{\Pi}^{\dagger}\hat{\Pi}\,|b\rangle = \langle a|(+1)(-1)|b\rangle = -\langle a|b\rangle. \tag{7.172}$$

But the only number that is equal to its negative is zero, so we learn that $\langle a|b\rangle = 0$. Applying this to the harmonic oscillator energy eigenstates tells us $\langle m|n\rangle = 0$ for m even and n odd and *vice versa*. Of course, this does not tell us anything new, since we already knew that the energy eigenstates with different energy eigenvalues are orthogonal. But if $\hat{O}_E$ is an even operator, such that $\hat{\Pi}\,\hat{O}_E\,\hat{\Pi} = \hat{O}_E$, then a "multiply by one" tells us that

$$\langle m|\hat{O}_E|n\rangle = \langle m|\,\hat{\Pi}^\dagger\,\hat{\Pi}\,\hat{O}_E\,\hat{\Pi}\,\hat{\Pi}\,|n\rangle = (-1)^{m+n}\langle m|\hat{O}_E|n\rangle. \tag{7.173}$$

This is because $\langle m|\,\hat{\Pi}^\dagger = \langle m|(-1)^m$, $\hat{\Pi}\,|n\rangle = (-1)^n|n\rangle$, and $\hat{\Pi}\,\hat{O}_E\,\hat{\Pi} = \hat{O}_E$. This matrix element vanishes if $m + n$ is odd. This is a very powerful result (and there is a similar one for an odd operator, whose expectation value vanishes if $m + n$ is even). You will prove that in the problems.

We showed that if an operator, like the Hamiltonian $\hat{H}$, is an even operator, then the eigenfunctions $|n\rangle$ are either even or odd. This is a quite common result. If the Hamiltonian has some symmetry, then there will be eigenfunctions that share that symmetry, but there usually will be eigenfunctions of a different symmetry as well. So, one *cannot* assume that if the Hamiltonian has some symmetry, then all of the eigenstates share the same symmetry. Many of them will have a different symmetry. In the more general cases than just parity, the same thing occurs. If some operator is unchanged after a similarity transformation (which is equivalent to the operator commuting with the operator of the similarity transformation), we say the similarity transformation is a symmetry of the system. In this case, we can often infer that many matrix elements of the original operator vanish due to this symmetry. This is another very powerful result that is used often in quantum mechanics. We definitely will see these ideas emerge again.

Section 7.8 Problems:

7.8.1 Odd Parity Operators

Prove that for an odd operator $\hat{O}_O$, which satisfies $\hat{\Pi}\,\hat{O}_O\,\hat{\Pi} = -\hat{O}_O$, we have $\langle m|\hat{O}_O|n\rangle = 0$ if $m + n$ is even. Here, the bra and ket are energy eigenstates of the harmonic oscillator.

7.8.2 Even and Odd Parity Position Eigenstates

Construct position-based eigenfunctions of $\hat{\Pi}$ by finding the coefficients α and β such that $\alpha|x\rangle + \beta|-x\rangle$ are eigenstates of $\hat{\Pi}$:

$$\hat{\Pi}\left(\alpha|x\rangle + \beta|-x\rangle\right) = \pm\left(\alpha|x\rangle + \beta|-x\rangle\right). \tag{7.174}$$

Note that you must solve for the coefficients for the two cases + and −. Be sure to identify which state is even and which is odd. You do not need to normalize the states.

Repeat for momentum states using $\gamma|p\rangle + \delta|-p\rangle$.

Hint: $\hat{\Pi}|x\rangle = |-x\rangle$ and $\hat{\Pi}|p\rangle = |-p\rangle$.

7.8.3 Commutator of the Number Operator with Powers

Prove that $[\hat{N}, (\hat{a}^\dagger)^m] = m(\hat{a}^\dagger)^m$ and use this result to show that $\hat{N}|m\rangle = m|m\rangle$.

7.8.4 Parity of Trigonometric Operator Functions

Show that $\sin(\alpha\hat{x})$ is an odd operator and $\cos(\beta\hat{x})$ is an even operator.

Hint: use the appropriate power series and the multiply-by-one trick to show that

$$\hat{\Pi}\,\hat{x}^n\,\hat{\Pi} = \underbrace{\hat{\Pi}\,\hat{x}\,\hat{\Pi}\,\hat{\Pi}\,\hat{x}\,\hat{\Pi}\cdots\hat{\Pi}\,\hat{x}\,\hat{\Pi}}_{n \text{ factors}}, \tag{7.175}$$

along with the power-series expressions for sine and cosine.

7.8.5 Eigenvectors with Different Eigenvalues Are Orthogonal

The fact that opposite parity states are orthogonal follows from a more general result that the eigenstates of any Hermitian operator $\hat{O}$ are orthogonal if they have different eigenvalues. The proof is found by evaluating $\langle a|\hat{O}|b\rangle$ (with $\hat{O}|a\rangle = O_a|a\rangle$ and $\hat{O}|b\rangle = O_b|b\rangle$) in two different ways. First, let $\hat{O}$ act to the right on $|b\rangle$ and then let it act to the left on $\langle a|$. You should find that we must have either $O_a = O_b$ or $\langle a|b\rangle = 0$. So, when eigenvalues are different, we must have $\langle a|b\rangle = 0$.

7.8.6 Parity Operator Gymnastics

Write the parity operator $\hat{\Pi} = (-1)^{\hat{N}} = e^{i\pi\hat{N}}$ and express the exponential function as a power series in the operators. Show that if it acts on a state $|m\rangle$ that we have $\hat{\Pi}|m\rangle = (-1)^m|m\rangle$.

Compute $e^{i\pi\hat{N}}\hat{a}e^{-i\pi\hat{N}}$ by using the Hadamard lemma. This requires using the formula in Eq. (7.284) and noting that all nested commutators contribute, but their

evaluation is simple because the "repeat." Note that you can return to this problem after seeing the full derivation of the Hadamard lemma, if desired.

7.8.7 *Action of the Parity Operator on Position and Momentum States*

Show that the operator identities $\hat{\Pi}\,\hat{x}\,\hat{\Pi} = -\hat{x}$ and $\hat{\Pi}\,\hat{p}\,\hat{\Pi} = -\hat{p}$ each imply that $\hat{\Pi}\,|x\rangle = |-x\rangle$ and $\hat{\Pi}\,|p\rangle = |-p\rangle$.

Hint: apply $\hat{x}$ onto $\hat{\Pi}\,|x\rangle$ and $\hat{p}$ onto $\hat{\Pi}\,|p\rangle$ and carefully interpret your results.

7.9 The Baker-Campbell-Hausdorff Formula

This section is a long and technical one. It discusses two of our four fundamental operator identities—the Hadamard lemma and the Baker-Campbell-Hausdorff formula. We won't prove either of these in their full generality. Instead, we will provide you with proofs of simpler cases and illustrate for you how you can use the general results, if you need to (the Hadamard lemma, will be proved in its full generality later in Sect. 7.11). These simpler cases are actually the most used cases, so they take us quite far.

I mentioned earlier that you could go from knowing just high school algebra to the research frontier by mastering the material in this book. This advanced training comes at a cost—we will be going through some sophisticated algebra. It is all straightforward material, meaning it does not use any advanced techniques. But it does require perseverance and stamina. If you cannot make it through on your first try, *do not panic*! You can always return to it later when you feel your algebra skills are stronger. We have relegated many of the more technical aspects to the problems. You can feel free to skip them at first if they seem to be too formidable. But please try to return to them later when you have built up more stamina and skill in your algebra.

Often, when dealing with complex technical formulas, it is useful to first encounter them in an explicit form, work with them in that form, and then revisit the result in a more general and abstract fashion. We begin by generalizing the problems in Sect. 7.2, where we analyzed the commutators $\left[e^{\frac{i}{\hbar}p_0\hat{x}},\,\hat{p}\right]$ and $\left[e^{\frac{i}{\hbar}x_0\hat{p}},\,\hat{x}\right]$. Now, we revisit this and examine the general result $\left[e^{\hat{A}},\,\hat{B}\right]$ for the case where $[\hat{A},\,\hat{B}]$ commutes with $\hat{A}$. Stated mathematically, this means $[\hat{A},\,[\hat{A},\,\hat{B}]] = 0$. Let's look at some examples to be sure we understand this nested commutator condition. If $\hat{A} = \hat{x}$ and $\hat{B} = \hat{p}$, then $[\hat{A},\,\hat{B}] = [\hat{x},\,\hat{p}] = i\hbar$, which is a number. We know numbers commute with everything (numbers or operators), so $[\hat{x},\,[\hat{x},\,\hat{p}]] = 0$ and so this pair of operators can be used in the identity we will derive below. What about

$[\hat{p}, \hat{x}^2] = \hat{x}[\hat{p}, \hat{x}] + [\hat{p}, \hat{x}]\hat{x} = -2i\hbar\hat{x}$. But $[\hat{p}, [\hat{p}, \hat{x}^2]] = [\hat{p}, -2i\hbar\hat{x}] = -2\hbar^2 \neq 0$. So, this would be an operator pair that does not satisfy $[\hat{A}, [\hat{A}, \hat{B}]] = 0$. However, $[\hat{p}, [\hat{p}, \hat{x}^2]]$ is a number, so it commutes with everything, as does $[\hat{x}^2, [\hat{x}^2, \hat{p}]]$, so this pair does meet a criterion we will encounter later in this section.

The formula we will now prove is

$$\left[e^{\hat{A}}, \hat{B}\right] = [\hat{A}, \hat{B}]e^{\hat{A}} \text{ if } [\hat{A}, [\hat{A}, \hat{B}]] = 0. \tag{7.176}$$

This is the second of our four fundamental identities (but in a special case, the so-called Weyl form). The proof requires a few steps. First, we need to show that $[\hat{A}^n, \hat{B}] = n[\hat{A}, \hat{B}]\hat{A}^{n-1}$, which is similar to the formulas we already derived for the commutators of powers of position with momentum or powers of momentum with position. We use induction: the case $n = 1$, which is the base case, is obvious, since $[\hat{A}, \hat{B}] = [\hat{A}, \hat{B}]$. Now, to complete the proof by induction, we assume it holds for $n - 1$ and prove it holds for n. So, we assume $[\hat{A}^{n-1}, \hat{B}] = (n-1)[\hat{A}, \hat{B}]\hat{A}^{n-2}$ and use the Leibniz rule to calculate

$$\begin{aligned}
[\hat{A}^n, \hat{B}] &= [\hat{A} \cdot \hat{A}^{n-1}, \hat{B}] = \hat{A}[\hat{A}^{n-1}, \hat{B}] + [\hat{A}, \hat{B}]\hat{A}^{n-1} \\
&= \hat{A}(n-1)[\hat{A}, \hat{B}]\hat{A}^{n-2} + [\hat{A}, \hat{B}]\hat{A}^{n-1} \\
&= (n-1)[\hat{A}, \hat{B}]\hat{A}^{n-1} + [\hat{A}, \hat{B}]\hat{A}^{n-1} = n[\hat{A}, \hat{B}]\hat{A}^{n-1},
\end{aligned} \tag{7.177}$$

because $[\hat{A}, [\hat{A}, \hat{B}]] = 0$, allowing us to move $\hat{A}$ through $[\hat{A}, \hat{B}]$ from the second to the third line of the proof. It is very important for you to note that this result is not a general result—it only holds when $[\hat{A}, [\hat{A}, \hat{B}]] = 0$.

Now, on to the main act, we have to compute the commutator of the exponential by expanding it in a power series and evaluating the commutators term by term using the commutator with powers that we just worked out:

$$\begin{aligned}
\left[e^{\hat{A}}, \hat{B}\right] &= \sum_{n=0}^{\infty} \frac{1}{n!}\left[\hat{A}^n, \hat{B}\right] = \sum_{n=0}^{\infty} \frac{1}{n!} n[\hat{A}, \hat{B}]\hat{A}^{n-1} \\
&= \sum_{n=1}^{\infty} \frac{1}{(n-1)!}[\hat{A}, \hat{B}]\hat{A}^{n-1} = \sum_{n=0}^{\infty} \frac{1}{n!}[\hat{A}, \hat{B}]\hat{A}^n \\
&= [\hat{A}, \hat{B}]e^{\hat{A}} = e^{\hat{A}}[\hat{A}, \hat{B}].
\end{aligned} \tag{7.178}$$

In the second line, we made the substitution $n \to n+1$, after noticing that the $n = 0$ term does not contribute to the summation. So, we have shown that $\left[e^{\hat{A}}, \hat{B}\right] = [\hat{A}, \hat{B}]e^{\hat{A}}$ if $[\hat{A}, [\hat{A}, \hat{B}]] = 0$. If you are careful, you might ask whether we were justified to also use the facts that $\left[\hat{A}^n, [\hat{A}, \hat{B}]\right] = 0$ and $\left[e^{\hat{A}}, [\hat{A}, \hat{B}]\right] = 0$, but you

should be able to show that both follow from $[\hat{A}, [\hat{A}, \hat{B}]] = 0$. This result can be rearranged as follows by multiplying from the right (on both sides) by $e^{-\hat{A}}$:

$$\begin{aligned} e^{A}\hat{B} - \hat{B}e^{\hat{A}} &= [\hat{A}, \hat{B}]e^{\hat{A}} \\ e^{\hat{A}}\hat{B}e^{-\hat{A}} - Be^{\hat{A}}e^{-\hat{A}} &= [\hat{A}, \hat{B}]e^{\hat{A}}e^{-\hat{A}}, \end{aligned} \tag{7.179}$$

or

$$e^{\hat{A}}\hat{B}e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}], \tag{7.180}$$

when $[\hat{A}, [\hat{A}, \hat{B}]] = 0$. This result is a very important one. It is the *Weyl form of the Hadamard lemma*, which is the second of our four fundamental operator identities. We will derive the general formula later.

Now, we move to a more complex identity built off of this special form of the Hadamard lemma. We will show that

$$e^{\hat{A}}e^{\hat{B}} = e^{\hat{B}}e^{\hat{A}}e^{[\hat{A}, \hat{B}]} \text{ if } [\hat{A}, [\hat{A}, \hat{B}]] = 0. \tag{7.181}$$

The proof relies on using $n - 1$ multiply-by-one tricks in between each $\hat{B}$ factor in the power-series expansion for the exponential. We have

$$\begin{aligned} e^{\hat{A}}e^{\hat{B}}e^{-\hat{A}} &= e^{\hat{A}}\left(\sum_{n=0}^{\infty}\frac{1}{n!}(\hat{B})^n\right)e^{-\hat{A}} = \sum_{n=0}^{\infty}\frac{1}{n!}e^{\hat{A}}\hat{B}^n e^{-\hat{A}} \\ &= \sum_{n=0}^{\infty}\frac{1}{n!}\underbrace{e^{\hat{A}}\hat{B}e^{-\hat{A}}e^{\hat{A}}\hat{B}e^{-\hat{A}}\cdots e^{\hat{A}}\hat{B}e^{-\hat{A}}}_{n \text{ factors}} \\ &= \sum_{n=0}^{\infty}\frac{1}{n!}\left(e^{\hat{A}}\hat{B}e^{-\hat{A}}\right)^n = e^{e^{\hat{A}}\hat{B}e^{-\hat{A}}} \\ &= e^{\hat{B}+[\hat{A}, \hat{B}]} = e^{\hat{B}}e^{[\hat{A}, \hat{B}]}. \end{aligned} \tag{7.182}$$

Hence, we have shown that $e^{\hat{A}}e^{\hat{B}} = e^{\hat{B}}e^{\hat{A}}e^{[\hat{A}, \hat{B}]}$ if $[\hat{A}, [\hat{A}, \hat{B}]] = 0$ and $[\hat{B}, [\hat{B}, \hat{A}]] = 0$. This identity is another powerful one, which we use when we want to interchange the order of exponentials of operators. It is called the *Weyl form of the exponential reordering identity*. It is a corollary of our Hadamard lemma and a special case of the *braiding identity*. We will derive the general braiding identity later as well. In the generalization of both identities, we will find that they become infinite series of nested commutators. The braiding identity is extremely powerful, because it "reaches inside" the arguments of arbitrary functions (not just exponentials).

We now move on to calculate the Baker-Campbell-Hausdorff formula, which is a "half-way braiding relation" and is the third of our four fundamental operator

identities. In the exponential reordering identity, we interchange the orders of the exponentials of the two different operators, and we have a correction term given by the commutator of the two operators. In the Baker-Campbell-Hausdorff formula, instead of reordering the two exponential factors, we put all terms into one exponent. Here, the correction term is one half of the commutator of the two operators. This is why we say it is a "half-way braiding relation." The proof of this relation is more complicated. We will be manipulating expressions of exponentials written in terms of power series of operators. It is a good idea to review Sect. 5.3 before diving into the remainder of this section (especially the discussion about interchanging the order of the summations in infinite power series).

Our goal is to generalize the expression we worked out in Sect. 5.3 (for numbers x and y) and given by $e^x e^y = e^{x+y}$ to an operator identity. Because operators do not commute with each other, we expect the result to change. Indeed, it does. The way it changes is summarized in the Baker-Campbell-Hausdorff formula. There is a way to write down the general formula (first found by Dynkin in 1947), which we will show at the end of the section, but even in the general form, it is a difficult formula to work with. This is quite advanced material.

The first few terms look like

$$e^{\hat{A}} e^{\hat{B}} = e^{\hat{A}+\hat{B}+\frac{1}{2}[\hat{A},\hat{B}]+\frac{1}{12}\left([\hat{A},[\hat{A},\hat{B}]]+[\hat{B},[\hat{B},\hat{A}]]\right)+\cdots}. \tag{7.183}$$

Note that if $[\hat{A}, [\hat{A}, \hat{B}]] = [\hat{B}, [\hat{B}, \hat{A}]] = 0$, then all of the higher-order terms vanish, and we have the following forms of the formula:

$$e^{\hat{A}+\hat{B}} = e^{\hat{A}} e^{\hat{B}} e^{-\frac{1}{2}[\hat{A},\hat{B}]} = e^{\hat{B}} e^{\hat{A}} e^{\frac{1}{2}[\hat{A},\hat{B}]}. \tag{7.184}$$

This is called the *Weyl form of the Baker-Campbell-Hausdorff formula* and it is the form that is most commonly used. We will use it many times in this book.

We will show how one can prove this simplified version exactly. The Baker-Campbell-Hausdorff formula is really a calculus result, so it is challenging to work out its details without calculus. The simplified Weyl form can be fully derived, but the path is long and complicated. We will mainly motivate and coach you on how to perform the full calculation within the main text, leaving some of the more detailed calculations for the problems. This work is long and requires a lot of stamina. If you do not feel up to it now, just do as much as you can and then return later when you are more adept at working through algebra. *But be sure to remember the result.* We can use the result even if we have not verified every step of the derivation.

For our first derivation, we will assume that $[\hat{A}, [\hat{A}, \hat{B}]] = 0$ and $[\hat{B}, [\hat{B}, \hat{A}]] = 0$. The starting point is to generalize the binomial theorem (which we covered in Sect. 5.3). Recall, for numbers, we had

$$(x+y)^n = \sum_{m=0}^{n} \frac{n!}{m!(n-m)!} x^{n-m} y^m. \tag{7.185}$$

Does this hold for operators? We expect it won't because operators need not commute with each other. Let's look at the powers $n = 0$, 1, and 2 to see what it looks like. The case with $n = 0$ works with operators, because

$$(A + B)^0 = \frac{0!}{0!0!}\hat{A}^0\hat{B}^0 = 1. \tag{7.186}$$

Similarly, the case with $n = 1$ also works:

$$(\hat{A} + \hat{B})^1 = \frac{1!}{0!1!}\hat{A}^1\hat{B}^0 + \frac{1!}{1!0!}\hat{A}^0\hat{B}^1 = \hat{A} + \hat{B}, \tag{7.187}$$

which also checks. But, the case $n = 2$ shows some differences, because

$$(\hat{A} + \hat{B})^2 = (\hat{A} + \hat{B})(\hat{A} + \hat{B}) = \hat{A}^2 + \hat{A}\hat{B} + \hat{B}\hat{A} + \hat{B}^2, \tag{7.188}$$

but

$$\sum_{m=0}^{2} \frac{2!}{m!(2-m)!}\hat{A}^{2-m}\hat{B}^m = \hat{A}^2 + 2\hat{A}\hat{B} + \hat{B}^2. \tag{7.189}$$

Note that the binomial expansion always has the $\hat{A}$ terms to the left of the $\hat{B}$ terms, while the powers of $(\hat{A} + \hat{B})$ have all different possible orderings of the monomial terms (so it has $\hat{A}\hat{B} + \hat{B}\hat{A}$ instead of $2\hat{A}\hat{B}$). The difference is, of course, the commutator (because it must vanish if we use numbers instead of operators), so we find that

$$(\hat{A} + \hat{B})^2 = \sum_{m=0}^{2} \frac{2!}{m!(2-m)!}\hat{A}^{2-m}\hat{B}^m + [\hat{B}, \hat{A}]. \tag{7.190}$$

The last term is the correction term due to operators that do not commute.

Let us continue. For $n = 3$, we have

$$(\hat{A} + \hat{B})^3 = \hat{A}^3 + \hat{A}^2\hat{B} + \hat{A}\hat{B}\hat{A} + \hat{B}\hat{A}^2 + \hat{A}\hat{B}^2 + \hat{B}\hat{A}\hat{B} + \hat{B}^2\hat{A} + \hat{B}^3, \tag{7.191}$$

while the binomial theorem result is

$$\sum_{m=0}^{3} \frac{3!}{m!(3-m)!}\hat{A}^{3-m}\hat{B}^m = \hat{A}^3 + 3\hat{A}^2\hat{B} + 3\hat{A}\hat{B}^2 + \hat{B}^3. \tag{7.192}$$

Some simple algebra then shows that

$$(\hat{A} + \hat{B})^3 = \sum_{m=0}^{3} \frac{3!}{m!(3-m)!}\hat{A}^{3-m}\hat{B}^m + 3[\hat{B}, \hat{A}](\hat{A} + \hat{B}). \tag{7.193}$$

It looks like a pattern may be forming, but it is complicated and not easy to identify yet. Let's look at $n = 4$ to see. Take a deep breath. For $n = 4$, we have

$$(\hat{A} + \hat{B})^4 = \hat{A}^4 + \hat{A}^3\hat{B} + \hat{A}^2\hat{B}\hat{A} + \hat{A}\hat{B}\hat{A}^2 + \hat{B}\hat{A}^3 + \hat{A}^2\hat{B}^2 + \hat{A}\hat{B}\hat{A}\hat{B} + \hat{B}\hat{A}^2\hat{B} + \hat{A}\hat{B}^2\hat{A} + \hat{B}\hat{A}\hat{B}\hat{A} + \hat{B}^2\hat{A}^2 + \hat{A}\hat{B}^3 + \hat{B}\hat{A}\hat{B}^2 + \hat{B}^2\hat{A}\hat{B} + \hat{B}^3\hat{A} + \hat{B}^4, \tag{7.194}$$

while the binomial theorem gives

$$\sum_{m=0}^{4} \frac{4!}{m!(4-m)!} \hat{A}^{4-m}\hat{B}^m = \hat{A}^4 + 4\hat{A}^3\hat{B} + 6\hat{A}^2\hat{B}^2 + 4\hat{A}\hat{B}^3 + \hat{B}^4. \tag{7.195}$$

To determine the relation between them, we subtract one from the other and find we can factor out a $[\hat{B}, \hat{A}]$ term to find

$$(\hat{A} + \hat{B})^4 = \sum_{m=0}^{4} \frac{4!}{m!(4-m)!} \hat{A}^{4-m}\hat{B}^m + [\hat{B}, \hat{A}](6\hat{A}^2 + 9\hat{A}\hat{B} + 3\hat{B}\hat{A} + 6\hat{B}^2). \tag{7.196}$$

To rearrange the second term so that all of the operators have $\hat{A}$ to the left of $\hat{B}$, we have to bring in another commutator term, which gives

$$(\hat{A} + \hat{B})^4 = \sum_{m=0}^{4} \frac{4!}{m!(4-m)!} \hat{A}^{4-m}\hat{B}^m + [\hat{B}, \hat{A}](6\hat{A}^2 + 12\hat{A}\hat{B} + 6\hat{B}^2) + 3\big([\hat{B}, \hat{A}]\big)^2. \tag{7.197}$$

It looks like things are getting complicated. But we will show below that this does fit a well-defined pattern.

But, first, let's expand the Baker-Campbell-Hausdorff formula, in its Weyl form, and compare the two sides to see if the result makes sense. Recall we are trying to verify that $e^{\hat{A}+\hat{B}} = e^{\hat{A}}e^{\hat{B}}e^{\frac{1}{2}[\hat{B},\hat{A}]}$. The left term becomes

$$\begin{aligned} e^{\hat{A}+\hat{B}} &= 1 + (\hat{A} + \hat{B}) + \frac{1}{2}\big(\hat{A}^2 + \hat{A}\hat{B} + \hat{B}\hat{A} + \hat{B}^2\big) \\ &+ \frac{1}{6}\big(\hat{A}^3 + \hat{A}^2\hat{B} + \hat{A}\hat{B}\hat{A} + \hat{B}\hat{A}^2 + \hat{A}\hat{B}^2 + \hat{B}\hat{A}\hat{B} + \hat{B}^2\hat{A} + \hat{B}^3\big) \\ &+ \frac{1}{24}\big(\hat{A}^4 + \hat{A}^3\hat{B} + \hat{A}^2\hat{B}\hat{A} + \hat{A}\hat{B}\hat{A}^2 + \hat{B}\hat{A}^3 + \hat{A}^2\hat{B}^2 + \hat{A}\hat{B}\hat{A}\hat{B} + \hat{B}\hat{A}^2\hat{B} \\ &+ \hat{A}\hat{B}^2\hat{A} + \hat{B}\hat{A}\hat{B}\hat{A} + \hat{B}^2\hat{A}^2 + \hat{A}\hat{B}^3 + \hat{B}\hat{A}\hat{B}^2 + \hat{B}^2\hat{A}\hat{B} + \hat{B}^3\hat{A} + \hat{B}^4\big) + \cdots, \end{aligned} \tag{7.198}$$

while the right-hand side becomes

$$\begin{aligned}
&\Big(1+\hat{A}+\frac{1}{2}\hat{A}^2+\frac{1}{6}\hat{A}^3+\frac{1}{24}\hat{A}^4+\cdots\Big)\Big(1+\hat{B}+\frac{1}{2}\hat{B}^2+\frac{1}{6}\hat{B}^3+\frac{1}{24}\hat{B}^4+\cdots\Big)\\
&\qquad\times\Big(1+\frac{1}{2}[\hat{B},\hat{A}]+\frac{1}{2}\Big(\frac{1}{2}[\hat{B},\hat{A}]\Big)^2+\frac{1}{6}\Big(\frac{1}{2}[\hat{B},\hat{A}]\Big)^3+\cdots\Big)\\
&=1+\hat{A}+\hat{B}+\frac{1}{2}(\hat{A}^2+2\hat{A}\hat{B}+\hat{B}^2)+\frac{1}{6}(\hat{A}^3+3\hat{A}^2\hat{B}+3\hat{A}\hat{B}^2+\hat{B}^3)\\
&\qquad+\frac{1}{24}(\hat{A}^4+4\hat{A}\hat{B}^3+6\hat{A}^2\hat{B}^2+4\hat{A}\hat{B}^3+\hat{B}^4)+\cdots\\
&\qquad+\frac{1}{2}[\hat{B},\hat{A}]\Big(1+\hat{A}+\hat{B}+\frac{1}{2}(\hat{A}^2+2\hat{A}\hat{B}+\hat{B}^2)+\cdots\Big)+\cdots\\
&\qquad+\frac{1}{2}\Big(\frac{1}{2}[\hat{B},\hat{A}]\Big)^2\Big(1+\cdots\Big)+\cdots,
\end{aligned}\tag{7.199}$$

where we have grouped together the terms according to powers of $[\hat{B},\hat{A}]$. This can be rewritten as

$$\begin{aligned}
&1+\hat{A}+\hat{B}+\frac{1}{2}\big(\hat{A}^2+2\hat{A}\hat{B}+\hat{B}^2+[\hat{B},\hat{A}]\big)\\
&\quad+\frac{1}{6}\big(\hat{A}^3+3\hat{A}^2\hat{B}+3\hat{A}\hat{B}^2+\hat{B}^3+3[\hat{B},\hat{A}](\hat{A}+\hat{B})\big)\\
&\quad+\frac{1}{24}\Big(\hat{A}^4+4\hat{A}^3\hat{B}+6\hat{A}^2\hat{B}^2+4\hat{A}\hat{B}^3+\hat{B}^4+6[\hat{B},\hat{A}]\big(\hat{A}^2+2\hat{A}\hat{B}+\hat{B}^2\big)\\
&\quad+3\big([\hat{B},\hat{A}]\big)^2\Big)+\cdots,
\end{aligned}\tag{7.200}$$

which agrees with our expansions through $n=4$ given above. If this is the first time you have worked on a complex problem like this, you might think it is enough, we have found the pattern, checked it through the first few terms, it must work for the rest. While in this case, that is true, we really have to *establish* that it works to all orders. We do that next.

This is another place where we pause for a moment, as you can move to the next section if the above analysis is enough to convince you that the Weyl form holds. Then, you can come back when you are ready to see the remainder of the proof.

Sometimes when you are faced with a long and complex calculation, it may seem overwhelming and difficult to complete. For me, it reminds me of French cooking. I love to watch Jacques Pepin's cooking show. In one episode, he made a Paris-Brest cake, which celebrates a bicycle race from Paris to Brest. I decided to make it. It first requires you to prepare a large puff pastry in the shape of a bicycle wheel. Then, you have to make an almond brittle. Next, you crush the brittle in a mortar and pestle to make it into a ground-up praline. Then, you need to make a vanilla custard (with vanilla beans of course). Finally comes the whipped cream (whipped

by hand, of course) and you are ready for assembly—cut the pastry in half, layer the custard, praline, and whipped cream, and put the top back on. A little powdered sugar and voilá you are done. Each step is really simple. Nothing is complicated, and you need only modest training to complete them. But there are just *so many steps*—it took me nearly two days to complete it! Well, solving hard problems is the same way. In most cases, each step is simple, but they have to be combined in the proper order, and you have to do them all. This can take time. So, as you work through a long problem, just remember, *it is just French cooking!* And, by the way, the cake was fantastic. Remember the concept of French cooking and your algebra will be fantastic too.

We now show how this gives the Weyl form of the Baker-Campbell-Hausdorff formula when $[\hat{A}, \hat{B}]$ commutes with $\hat{A}$ and $\hat{B}$. First, we introduce the notion of *normal ordering*, denoted with two colons. The symbol $: \hat{O} :$ means the *normal-ordered form* of $\hat{O}$, which is defined, for this calculation, as all $\hat{A}$ factors to the left of all $\hat{B}$ factors. In other words

$$: (\hat{A} + \hat{B})^n := \sum_{m=0}^{n} \frac{n!}{m!(n-m)!} \hat{A}^{n-m} \hat{B}^m, \tag{7.201}$$

and

$$\begin{aligned} : e^{(\hat{A}+\hat{B})} : &= \sum_{n=0}^{\infty} \frac{1}{n!} : (\hat{A} + \hat{B})^n := \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{m=0}^{n} \frac{n!}{m!(n-m)!} \hat{A}^{n-m} \hat{B}^m \\ &= \sum_{n=0}^{\infty} \sum_{m=0}^{n} \frac{1}{(n-m)!} \hat{A}^{n-m} \frac{1}{m!} \hat{B}^m = \sum_{m=0}^{\infty} \frac{1}{m!} \sum_{n=m}^{\infty} \frac{1}{(n-m)!} \hat{A}^{n-m} \hat{B}^m \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \hat{A}^n \sum_{m=0}^{\infty} \frac{1}{m!} \hat{B}^m = e^{\hat{A}} e^{\hat{B}}. \end{aligned} \tag{7.202}$$

In the third line, we switched the order of the summations, just as we did in Fig. 5.5. Then, we shift $n \to n + m$ in the second sum on the same line. The summations then become independent of each other, and we obtain the two exponentials. Note that we use the normal-ordering notation just as a shorthand. We can only bring it in if the operators have already been commuted through into the normal-ordered form. This means you cannot simply introduce normal ordering as you like, but if you have operators in a normal-ordered form, they often act just as numbers do, which is why it is a nice notation to us, when appropriate.

We use the normal-ordering notation to rewrite the binomial theorem identity for operators where $[\hat{A}, \hat{B}]$ commutes with $\hat{A}$ and $\hat{B}$. We find

$$\frac{1}{n!}(\hat{A}+\hat{B})^n = \frac{1}{0!}\frac{:(\hat{A}+\hat{B})^n:}{n!} + \frac{1}{1!}\left(\frac{1}{2}[\hat{B},\hat{A}]\right)\frac{:(\hat{A}+\hat{B})^{n-2}:}{(n-2)!}$$
$$+ \frac{1}{2!}\left(\frac{1}{2}[\hat{B},\hat{A}]\right)^2\frac{:(\hat{A}+\hat{B})^{n-4}:}{(n-4)!} + \frac{1}{3!}\left(\frac{1}{2}[\hat{B},\hat{A}]\right)^3\frac{:(\hat{A}+\hat{B})^{n-6}:}{(n-6)!}$$
$$+ \cdots + \frac{1}{\left[\frac{n}{2}\right]!}\left(\frac{1}{2}[\hat{B},\hat{A}]\right)^{\left[\frac{n}{2}\right]}\frac{:(\hat{A}+\hat{B})^{n-2\left[\frac{n}{2}\right]}:}{\left(n-2\left[\frac{n}{2}\right]\right)!}, \tag{7.203}$$

which you verify by induction in Problem 7.9.2. Okay, this is it; we are at the last set of steps. We have

$$e^{\hat{A}+\hat{B}} = \sum_{n=0}^{\infty}\frac{1}{n!}(\hat{A}+\hat{B})^n = \sum_{n=0}^{\infty}\sum_{m=0}^{\left[\frac{n}{2}\right]}\frac{1}{m!(n-m)!}\left(\frac{1}{2}[\hat{B},\hat{A}]\right)^m\frac{:(\hat{A}+\hat{B})^{n-2m}:}{(n-2m)!}$$
$$= \sum_{m=0}^{\infty}\frac{1}{m!}\left(\frac{1}{2}[\hat{B},\hat{A}]\right)^m\left[\sum_{\substack{n=2m\\ n=\text{even}}}^{\infty}\frac{1}{(n-2m)!}:(\hat{A}+\hat{B})^{n-2m}:\right.$$
$$\left. + \sum_{\substack{n=2m+1\\ n=\text{odd}}}^{\infty}\frac{1}{(n-2m)!}:(\hat{A}+\hat{B})^{n-2m}:\right]$$
$$= e^{\frac{1}{2}[\hat{B},\hat{A}]}e^{\hat{A}}e^{\hat{B}}. \tag{7.204}$$

Note that changing the order of the summation, from the first line to the second and third lines, is a highly nontrivial operation. We direct you how to do it in Problem 7.9.3. The two summations in the big square bracket spanning lines two and three look like they depend on m, but they do not. They can be combined to yield an exponential, and because of the normal ordering, it is the product of the two exponentials $e^{\hat{A}}e^{\hat{B}}$. Of course, once this is all derived, the factor $e^{\frac{1}{2}[\hat{B},\hat{A}]}$ commutes with $e^{\hat{A}}$ and $e^{\hat{B}}$, so it can be placed anywhere in the product of the three operators.

So, this was a lot. But it is not the end of the story. The derivation was only the Weyl form of the Baker-Campbell-Hausdorff formula. There are more corrections if $[\hat{A},\hat{B}]$ does not commute with $\hat{A}$ or $\hat{B}$. We will not go through the full derivation, but we will show how you can motivate the result, similar to our expansion through $n=4$ above for the Weyl form. Just be aware this is not a complete derivation. So, the case we will look at is when $[\hat{A},[\hat{A},\hat{B}]]$ and $[\hat{B},[\hat{B},\hat{A}]]$ both commute with $\hat{A}$ and $\hat{B}$.

To determine the next term, we will motivate it, but not derive it in full. We will just perform an expansion of both sides of the equation to determine the correction term. This is similar to what we did first for deriving the Weyl form of the Baker-Campbell-Hausdorff formula. Recalling that $[\hat{A},\hat{B}]$ no longer commutes with $\hat{A}$ or

$\hat{B}$, we need to be careful about the orderings when we expand the exponential in a power series via

$$e^{\hat{A}+\hat{B}+\frac{1}{2}[\hat{A},\hat{B}]} = \sum_{n=0}^{\infty} \frac{1}{n!} \left(\hat{A} + \hat{B} + \frac{1}{2}[\hat{A}, \hat{B}] \right)^n . \tag{7.205}$$

We now group together terms that have the same total number of $\hat{A}$ and $\hat{B}$ factors in each monomial. The first few terms are simple, but then it gets complex:

$$\begin{aligned}
n = 0 \;&:\; 1 \\
n = 1 \;&:\; \hat{A} + \hat{B} \\
n = 2 \;&:\; \frac{1}{2}\big(\hat{A}^2 + \hat{A}\hat{B} + \hat{B}\hat{A} + \hat{B}^2\big) + \frac{1}{2}[\hat{A}, \hat{B}] \\
n = 3 \;&:\; \frac{1}{6}\big(\hat{A}^3 + \hat{A}^2\hat{B} + \hat{A}\hat{B}\hat{A} + \hat{B}\hat{A}^2 + \hat{A}\hat{B}^2 + \hat{B}\hat{A}\hat{B} + \hat{B}^2\hat{A} + \hat{B}^3\big) \\
&\quad + \frac{1}{4}(\hat{A} + \hat{B})[\hat{A}, \hat{B}] + \frac{1}{4}[\hat{A}, \hat{B}](\hat{A} + \hat{B}) \\
n = 4 \;&:\; \frac{1}{24}\big(\hat{A}^4 + \hat{A}^3\hat{B} + \hat{A}^2\hat{B}\hat{A} + \hat{A}\hat{B}\hat{A}^2 + \hat{B}\hat{A}^3 + \hat{A}\hat{B}\hat{A}\hat{B} + \hat{A}\hat{B}^2\hat{A} \\
&\quad + \hat{B}\hat{A}\hat{B}\hat{A} + \hat{B}\hat{A}^2\hat{B} + \hat{B}^2\hat{A}^2 + \hat{A}\hat{B}^3 + \hat{B}\hat{A}\hat{B}^2 + \hat{B}^2\hat{A}\hat{B} + \hat{B}^3\hat{A}\big) \\
&\quad + \frac{1}{12}\big(\hat{A}^2 + \hat{A}\hat{B} + \hat{B}\hat{A} + \hat{B}^2\big)[\hat{A}, \hat{B}] + \frac{1}{12}(\hat{A} + \hat{B})[\hat{A}, \hat{B}](\hat{A} + \hat{B}) \\
&\quad + \frac{1}{12}[\hat{A}, \hat{B}]\big(\hat{A}^2 + \hat{A}\hat{B} + \hat{B}\hat{A} + \hat{B}^2\big) + \frac{1}{8}[\hat{A}, \hat{B}]\,[\hat{A}, \hat{B}].
\end{aligned} \tag{7.206}$$

We want to compare this to the similar terms in $e^{\hat{A}}e^{\hat{B}}$ given by

$$e^{\hat{A}}e^{\hat{B}} = \left(1 + \hat{A} + \frac{1}{2}\hat{A}^2 + \frac{1}{6}\hat{A}^3 + \cdots\right)\left(1 + \hat{B} + \frac{1}{2}\hat{B}^2 + \frac{1}{6}\hat{B}^3 + \cdots\right). \tag{7.207}$$

We expand it out again and then group according to the total exponent, just as before. This gives

$$\begin{aligned}
n = 0 \;&:\; 1 \\
n = 1 \;&:\; \hat{A} + \hat{B} \\
n = 2 \;&:\; \frac{1}{2}\big(\hat{A}^2 + 2\hat{A}\hat{B} + \hat{B}^2\big) \\
n = 3 \;&:\; \frac{1}{6}\big(\hat{A}^3 + 3\hat{A}^2\hat{B} + 3\hat{A}\hat{B}^2 + \hat{B}^3\big)
\end{aligned}$$

$$n = 4 \quad : \quad \frac{1}{24}\big(\hat{A}^4 + 4\hat{A}^3\hat{B} + 6\hat{A}^2\hat{B}^2 + 4\hat{A}\hat{B}^3 + \hat{B}^4\big), \tag{7.208}$$

for the first few terms.

Comparing the two expressions, we immediately see that the $n = 0$, 1, and 2 expressions are identical, as expected. For $n = 3$, we find when we subtract the corresponding $n = 3$ terms in Eq. (7.208) from those in Eq. (7.206) that the result becomes

$$\begin{aligned}
\text{difference} &= \frac{1}{6}(\hat{A}\hat{B}\hat{A} + \hat{B}\hat{A}^2) + \frac{1}{6}(\hat{B}\hat{A} + \hat{B}^2\hat{A}) + \frac{1}{4}(\hat{A} + \hat{B})(\hat{A}\hat{B} - \hat{B}\hat{A}) \\
&\quad + \frac{1}{4}(\hat{A}\hat{B} - \hat{B}\hat{A})(\hat{A} + \hat{B}) - \frac{1}{3}\hat{A}^2\hat{B} - \frac{1}{3}\hat{A}\hat{B}^2 \\
&= \frac{1}{12}\big(2\hat{A}\hat{B}\hat{A} + 2\hat{B}\hat{A}^2 + 2\hat{B}\hat{A} + 2\hat{B}^2\hat{A}) + 3\hat{A}^2\hat{B} - 3\hat{A}\hat{B}\hat{A} + 3\hat{B}\hat{A}\hat{B} \\
&\quad - 3\hat{B}^2\hat{A} + 3\hat{A}\hat{B}\hat{A} - \hat{B}\hat{A}^2 + 3\hat{B}\hat{A}^2 - 3\hat{B}\hat{A}\hat{A} - 4\hat{A}^2\hat{B} - 4\hat{A}\hat{B}^2\big) \\
&= \frac{1}{12}\big(2\hat{A}\hat{B}\hat{A} - \hat{B}\hat{A}^2 + 2\hat{B}\hat{A}\hat{B} - \hat{B}^2\hat{A} - \hat{A}^2\hat{B} - \hat{A}\hat{B}^2\big) \\
&= -\frac{1}{12}[\hat{A}, [\hat{A}, \hat{B}]] - \frac{1}{12}[\hat{B}, [\hat{B}, \hat{A}]],
\end{aligned} \tag{7.209}$$

where the last equality is easiest to see by fully expanding out the last line and comparing it to the line above it. You will have the opportunity to work out the result for the $n = 4$ case in Problem 7.9.8. The $n = 4$ result becomes

$$-\frac{1}{12}\big([\hat{A}, [\hat{A}, \hat{B}]] + [\hat{B}, [\hat{B}.\hat{A}]]\big)(\hat{A} + \hat{B}). \tag{7.210}$$

Collecting these terms shows that through fourth order, we have demonstrated that

$$\begin{aligned}
e^{\hat{A}+\hat{B}+\frac{1}{2}[\hat{A},\hat{B}]} &= \big(1 + \hat{A} + \hat{B} + \frac{1}{2}(\hat{A}^2 + 2\hat{A}\hat{B} + \hat{B}^2) + \cdots\big) \\
&\quad \times \left(1 - \frac{1}{12}([\hat{A}, [\hat{A}, \hat{B}]] + [\hat{B}, [\hat{B}, \hat{A}]]) \right. \\
&\quad \left. + \frac{1}{2}\left\{\frac{1}{12}([\hat{A}, [\hat{A}, \hat{B}]] + [\hat{B}, [\hat{B}, \hat{A}]])\right\}^2 + \cdots\right) \\
&= e^{\hat{A}} e^{\hat{B}} e^{-\frac{1}{12}\left([\hat{A},[\hat{A},\hat{B}]]+[\hat{B},[\hat{B},\hat{A}]]\right)},
\end{aligned} \tag{7.211}$$

at least through the first two terms in the expansion of the correction term. While, in principle, we can continue as we did before and prove this for all the terms, it rapidly becomes too much to handle, even for a seasoned algebra master (this is because a calculus-based approach can arrive at the answer quite a bit more easily,

even if it remains a hard problem). My advice if you really want to see how to derive this to higher orders is to wait until you have mastered enough calculus to follow the standard derivation. In any case, it should be clear how we have derived the next set of terms in the Baker-Campbell-Hausdorff formula, even if this argument is not completely rigorous.

While Baker, Campbell, and Hausdorff made significant progress in developing the formula that bears their name, it was not until 1947 that Dynkin found the general formula, which we will write down next. Even writing it down is a bit complicated. It is

$$e^{\hat{A}} e^{\hat{B}} = e^{\eta(\hat{A},\hat{B})}, \tag{7.212}$$

$$\eta(\hat{A},\hat{B}) = \sum_{m=1}^{\infty} \frac{(-1)^{m-1}}{m} \times \sum_{\substack{p_i \geq 0,\ q_i \geq 0 \\ p_i + q_i \geq 1}} \frac{[\overbrace{\hat{A}\cdots\hat{A}}^{p_1}\overbrace{\hat{B}\cdots\hat{B}}^{q_1}\cdots\overbrace{\hat{A}\cdots\hat{A}}^{p_m}\overbrace{\hat{B}\cdots\hat{B}}^{q_m}]}{\left(\sum_j (p_j + q_j)\right) p_1! q_1! \cdots p_m! q_m!} \tag{7.213}$$

$$[\hat{C}\hat{D}\hat{E}\cdots\hat{J}] = [\cdots[[\hat{C},\hat{D}],\hat{E}],\cdots,\hat{J}]; \tag{7.214}$$

note that the nested commutators here start with the innermost operator in the nesting to the left, as opposed to how we typically write it with the innermost one on the right. The first few terms correspond to $m = 1$, $\sum_i (p_i + q_i) = 1$ ($p_1 = 1$, $q_1 = 0$ and $p_1 = 0$, $q_1 = 1$), $m = 1$, $\sum_i (p_i + q_i) = 2$ ($p_1 = 2$, $q_1 = 0$, and $p_1 = 1$, $q_1 = 1$, and $p_1 = 0$, $q_1 = 2$), and $m = 2$, $\sum_i (p_i + q_i) = 2$ ($p_1 = 1$, $q_1 = 0$, $p_2 = 1$, $q_2 = 0$, and $p_1 = 1$, $q_1 = 0$, $p_2 = 0$, $q_2 = 1$, and $p_1 = 0$, $q_1 = 1$, $p_2 = 1$, $q_2 = 0$, and $p_1 = 0$, $q_1 = 1$, $p_2 = 0$, $q_2 = 1$), and so on. We must take into account all possibilities for the formula, but many of the commutators vanish (e.g., $p_1 = 2$, $q_1 = 0$ vanishes and all $m = 2$, $\sum(p + q) = 2$ terms vanish as well). The net contribution becomes

$$\eta(\hat{A},\hat{B}) = \underbrace{\hat{A} + \hat{B}}_{m=1,\ \sum(p+q)=1} + \underbrace{\frac{1}{2}[\hat{A},\hat{B}]}_{m=1,\ \sum(p+q)=2} + \underbrace{\frac{1}{12}[[\hat{A},\hat{B}],\hat{B}] + \frac{1}{12}[[\hat{B},\hat{A}],\hat{A}]}_{m=2,\ \sum(p+q)=3} + \cdots . \tag{7.215}$$

From this, we can immediately see the issues with it—it does not directly give us the final coefficients of each nested commutator term—in general, many different terms contribute to give the final coefficient. Hence, determining each contribution still requires a lot of work! Nevertheless, it is amazing that an exact expression

for the Baker-Campbell-Hausdorff formula exists. In fact, I had spent many years of my career not even aware that one did. It is only when I started writing this book that I discovered the general result even exists! What does not exist is a simple and direct way to determine the coefficient of a given nested commutator term in the expansion. So, while this is the general result for our third fundamental operator identity (the other two being the Leibniz identity and the Hadamard lemma), it is not easy to use—we instead encounter the Baker-Campbell-Hausdorff formula most often in the Weyl form, or in slightly more complicated forms, for everything we will work on in this book. If you know some calculus, the video Baker-Campbell-Hausdorff Theorem https://tinyurl.com/us836z7b provides an alternative derivation that does show all of the details from a calculus perspective. It is still hard. You may find it interesting to look through even if you do not know calculus.

Below, we include problems detailing the technical details of the derivation of the Weyl form that we left out, showing alternate ways to derive the expansion, and applying it to a physical problem, to see how it works in practice.

Section 7.9 Problems:

7.9.1 Nested Commutators

Determine for the following pairs of operators whether $[\hat{A}, \hat{B}]$ commutes with $\hat{A}$ and $\hat{B}$ and whether $[\hat{A}, [\hat{A}, \hat{B}]]$ and $[\hat{B}, [\hat{B}, \hat{A}]]$ both commute with $\hat{A}$ and $\hat{B}$.

(a) $\hat{A} = \hat{a}$ and $\hat{B} = \hat{a}^\dagger$
(b) $\hat{A} = \hat{a}^2$ and $\hat{B} = \left(\hat{a}^\dagger\right)^2$
(c) $\hat{A} = \hat{n} = \hat{a}^\dagger\hat{a}$ and $\hat{B} = \hat{a}^\dagger + \hat{a}$
(d) $\hat{A} = \hat{n} = \hat{a}^\dagger\hat{a}$ and $\hat{B} = \left(\hat{a}^\dagger\right)^2 + \hat{a}^2$

7.9.2 Special Case of the Operator-Based Binomial Theorem

This problem is particularly challenging. Try it only if you are ready. For pairs of operators where $[\hat{A}, \hat{B}]$ commutes with both $\hat{A}$ and $\hat{B}$, we will prove the generalized operator-based binomial theorem by induction. The base case with $n = 1$ is trivial.

(a) Assume that

$$\frac{1}{n!}(\hat{A} + \hat{B})^n = \sum_{m=0}^{\left[\frac{n}{2}\right]} \frac{1}{m!(n-2m)!} \left(\frac{1}{2}[\hat{B}, \hat{A}]\right)^m : (\hat{A} + \hat{B})^{n-2m} :, \qquad (7.216)$$

holds for all $n \leq 2k$. Prove that it holds for $n = 2k + 1$.

(b) Assume this result holds for all $n \leq 2k+1$ and prove that it holds for $n = 2k+2$.

This then establishes the generalized binomial theorem for all n. Recall that $[x]$ is equal to the largest integer less than or equal to x.

7.9.3 Switching the Order of the Summations with Operators

Show that

$$\sum_{n=0}^{\infty}\sum_{m=0}^{[\frac{n}{2}]}\frac{x^m}{m!}\frac{y^{n-2m}}{(n-2m)!} = \sum_{m=0}^{\infty}\left\{\sum_{\substack{n=2m\\ n\text{ even}}}^{\infty}\frac{x^m}{m!}\frac{y^{n-2m}}{(n-2m)!} + \sum_{\substack{n=2m+1\\ n\text{ odd}}}^{\infty}\frac{x^m}{m!}\frac{y^{n-2m}}{(n-2m)!}\right\}$$

$$= \sum_{m=0}^{\infty}\frac{x^m}{m!}\left\{\sum_{\substack{n=0\\ n\text{ even}}}^{\infty}\frac{y^n}{n!} + \sum_{\substack{n=1\\ n\text{ odd}}}^{\infty}\frac{y^n}{n!}\right\} = e^x e^y, \tag{7.217}$$

where x and y can be operators, because we never change the ordering of any product of x with y.

Hint: Carefully examine the case with even versus odd n.

7.9.4 Double Nested Commutators

(a) Show that

$$[\hat{A},[\hat{A},\hat{B}]] = \hat{A}^2\hat{B} - 2\hat{A}\hat{B}\hat{A} + \hat{B}\hat{A}^2. \tag{7.218}$$

(b) Show that

$$[\hat{B},[\hat{B},\hat{A}]] = \hat{B}^2\hat{A} - 2\hat{B}\hat{A}\hat{B} + \hat{A}\hat{B}^2. \tag{7.219}$$

7.9.5 Baby Zassenhaus Formula I

The Baker-Campbell-Hausdorff formula says that

$$e^{\hat{A}+\hat{B}+\frac{1}{2}[\hat{A},\hat{B}]} = e^{\hat{A}}e^{\hat{B}}e^{-\frac{1}{12}\{[\hat{A},[\hat{A},\hat{B}]]+[\hat{B},[\hat{B},\hat{A}]]\}}, \tag{7.220}$$

when $\hat{A}$ and $\hat{B}$ commute with both nested double commutators. Let $\hat{\alpha} = \hat{A} + \hat{B}$ and $\hat{\beta} = \frac{1}{2}[\hat{A}, \hat{B}]$. Then, $[\hat{\alpha}, \hat{\beta}]$ commutes with $\hat{\alpha}$ and $\hat{\beta}$, so

$$e^{\hat{\alpha}+\hat{\beta}} = e^{\hat{\alpha}} e^{\hat{\beta}} e^{-\frac{1}{2}[\hat{\alpha},\hat{\beta}]}. \tag{7.221}$$

Use this result to show that

$$e^{\hat{A}+\hat{B}} = e^{\hat{A}} e^{\hat{B}} e^{\frac{1}{2}[\hat{B},\hat{A}]} e^{-\frac{1}{6}[\hat{A},[\hat{B},\hat{A}]]-\frac{1}{3}[\hat{B},[\hat{B},\hat{A}]]}. \tag{7.222}$$

7.9.6 Baby Zassenhaus Formula II

Interchange $\hat{A}$ and $\hat{B}$ in the above result and rearrange it to show that

$$e^{\hat{A}} e^{\hat{B}} = e^{\hat{B}} e^{\hat{A}} e^{-[\hat{B},\hat{A}]} e^{\frac{1}{2}[(\hat{A}+\hat{B}),[\hat{B},\hat{A}]]} \tag{7.223}$$

when $\hat{A}$ and $\hat{B}$ both commute with $[\hat{A}, [\hat{A}, \hat{B}]]$ and $[\hat{B}, [\hat{B}, \hat{A}]]$. Be sure not to reorder operators that do not commute in your derivation, without correcting for their commutator.

7.9.7 Generating Function

We will use the Baker-Campbell-Hausdorff formula to evaluate the expectation value of $\hat{x}^{2n}$ in the ground state of a harmonic oscillator. Recall that this is equal to

$$\langle 0|\hat{x}^{2n}|0\rangle = \left(\frac{\hbar}{2m\omega}\right)^n \langle 0| \left(\hat{a} + \hat{a}^\dagger\right)^{2n} |0\rangle. \tag{7.224}$$

This is difficult to evaluate in general, so we will use a trick, which is called a *generating function method*. We adapt the method a bit to use it without calculus.

We evaluate

$$\langle 0|e^{\alpha(\hat{a}^\dagger+\hat{a})}|0\rangle, \tag{7.225}$$

using the Baker-Campbell-Hausdorff formula in its Weyl form. Choose $\hat{A} = \alpha\hat{a}^\dagger$ and $\hat{B} = \alpha\hat{a}$, with α = real. Then, $[\hat{A}, \hat{B}] = \alpha^2[\hat{a}^\dagger, \hat{a}] = -\alpha^2$ is a number, so it commutes with all operators. Note that we chose the operators as we did because

we know that $\hat{a}|0\rangle = 0$ and $\langle 0|\hat{a}^\dagger = 0$. Then, the exponentials of the corresponding operators evaluate to 1, since $e^{\alpha\hat{a}}|0\rangle = |0\rangle$, and we remove them from the formula. This uses the fact that we found an operator that annihilates the state—such a theme will always allow us to simplify calculations with operators.

This means we can evaluate the expectation value of the exponential function using the Weyl form of the Baker-Campbell-Hausdorff formula. But we can also evaluate it by expressing the exponential function as its power series. So

$$\langle 0|e^{\alpha(\hat{a}^\dagger+\hat{a})}|0\rangle = \sum_{n=0}^{\infty} \frac{\alpha^n}{n!} \langle 0| \left(\hat{a}^\dagger + \hat{a}\right)^n |0\rangle. \tag{7.226}$$

But odd powers of $\hat{a}^\dagger + \hat{a}$ vanish when we take their expectation values, because we can never balance the number of raising and lowering operators, so we actually find that

$$\langle 0|e^{\alpha(\hat{a}^\dagger+\hat{a})}|0\rangle = \sum_{n=0}^{\infty} \frac{\alpha^{2n}}{(2n)!} \langle 0| \left(\hat{a}^\dagger + \hat{a}\right)^{2n} |0\rangle. \tag{7.227}$$

Now, take the answer you got from using the Baker-Campbell-Hausdorff formula and expand it in a power series in α. Then, by equating powers of α on both sides, you should be able to show that

$$\langle 0|\hat{a}^{2n}|0\rangle = \left(\frac{\hbar}{2m\omega}\right)^n \frac{(2n)!}{2^n n!}. \tag{7.228}$$

Think about why it is okay to do this. In particular, does the operator expectation value of powers of $\hat{x}$ depend on α?

Note that a similar formula can be worked out for the expectation value of $\hat{p}^{2n}$ in the ground state, but we do not need that for the applications we will show you, so we do not compute it here. We will use this formula in approximately discussing properties of the deuteron.

This method can be extended to higher energy eigenstates, but that requires more work (think about how you could do it). The difficulty arises in evaluating the expectation value in terms of the Baker-Campbell-Hausdorff formula.

7.9.8 The Case with n = 4

Work out the algebra for the $n = 4$ coefficients to show that the Baker-Campbell-Hausdorff formula holds in the case where the double nested commutators commute with $\hat{A}$ and $\hat{B}$. Note that this problem requires a significant amount of algebra and significant care.

7.9.9 Alternative Derivation of the Baker-Campbell-Hausdorff Formula

The original work of Baker, Campbell, and Hausdorff did not rigorously derive the identity. That had to wait fifty years for Dynkin to do so. In this problem, we proceed as the original three did and provide a nonrigorous derivation, which quickly gets many terms in the expansion.

We write $e^{\hat{A}}e^{\hat{B}} = e^{\hat{A}+\hat{B}+\hat{C}}$, where $\hat{C}$ is the operator we are trying to find. We expand both sides in a power series as

$$\sum_{m=0}^{\infty}\sum_{n=0}^{\infty}\frac{\hat{A}^m}{m!}\frac{\hat{B}^n}{n!} = \sum_{r=0}^{\infty}\frac{(\hat{A}+\hat{B}+\hat{C})^r}{r!}, \tag{7.229}$$

and then equate like powers.

Let's examine the lowest four terms on each side. The left-hand side is

$$\begin{aligned}1 + \hat{A} + \hat{B} + \frac{1}{2}\hat{A}^2 + \hat{A}\hat{B} + \frac{1}{2}\hat{B}^2 + \frac{1}{6}\hat{A}^3 + \frac{1}{2}\hat{A}^2\hat{B} + \frac{1}{2}\hat{A}\hat{B}^2 + \frac{1}{6}\hat{B}^3 \\ + \frac{1}{24}\hat{A}^4 + \frac{1}{6}\hat{A}^3\hat{B} + \frac{1}{4}\hat{A}^2\hat{B}^2 + \frac{1}{6}\hat{A}\hat{B}^3 + \frac{1}{24}\hat{B}^4 + \cdots,\end{aligned} \tag{7.230}$$

and the right-hand side is

$$\begin{aligned}1 + \hat{A} + \hat{B} + \hat{C} + \frac{1}{2}(\hat{A}^2 + \hat{B}^2 + \hat{C}^2 + \hat{A}\hat{B} + \hat{A}\hat{C} + \hat{B}\hat{A} + \hat{B}\hat{C} + \hat{C}\hat{A} + \hat{C}\hat{B}) \\ + \frac{1}{6}(\hat{A}^3 + \hat{B}^3 + \hat{C}^3 + \hat{A}\hat{B}^2 + \hat{A}\hat{C}^2 + \hat{B}\hat{A}^2 + \hat{B}\hat{C}^2 + \hat{C}\hat{A}^2 + \hat{C}\hat{B}^2 \\ + \hat{A}^2\hat{B} + \hat{A}^2\hat{C} + \hat{B}^2\hat{A} + \hat{B}^2\hat{C} + \hat{C}^2\hat{A} + \hat{C}^2\hat{B} + \hat{A}\hat{B}\hat{A} + \hat{A}\hat{B}\hat{C} + \hat{A}\hat{C}\hat{A} \\ + \hat{A}\hat{C}\hat{B} + \hat{B}\hat{A}\hat{B} + \hat{B}\hat{A}\hat{C} + \hat{B}\hat{C}\hat{A} + \hat{B}\hat{C}\hat{B} + \hat{C}\hat{A}\hat{B} + \hat{C}\hat{A}\hat{C} + \hat{C}\hat{B}\hat{A} \\ + \hat{C}\hat{B}\hat{C}) + \cdots.\end{aligned} \tag{7.231}$$

The term $\hat{C}$ is the "correction" term we are trying to solve for. We organize it in terms of total powers of the operators $\hat{A}$ and $\hat{B}$. So, we have $\hat{C} = \hat{C}_2 + \hat{C}_3 + \cdots$. We look at each order on the left and right sides. Zeroth order gives $1 = 1$. First order gives $\hat{A} + \hat{B} = \hat{A} + \hat{B}$. Second order gives $\frac{1}{2}\hat{A}^2 + \hat{A}\hat{B} + \frac{1}{2}\hat{B}^2 = \hat{C}_2 + \frac{1}{2}(\hat{A}^2 + \hat{A}\hat{B} + \hat{B}\hat{A} + \hat{B}^2)$. Solving for $\hat{C}_2$ yields $\hat{C}_2 = \frac{1}{2}[\hat{A}, \hat{B}]$. Continue to third order, and verify that $\hat{C}_3 = \frac{1}{12}([\hat{A}, [\hat{A}, \hat{B}]] + [\hat{B}, [\hat{B}, \hat{A}]])$. If you are brave, solve for $\hat{C}_4$ as well.

You can see that this methodology also gets quite complicated for higher orders. Can you explain why this approach is not rigorous, but is making an assumption?

7.9.10 Working with Dynkin's Exact Formula

In the text, we showed how to use Dynkin's formula to find the first and second correction terms in the Baker-Campbell-Hausdorff formula. Use it to find the fourth-order correction. Note that multiple terms contribute to the fourth-order contribution. You need to find all of them. Show that the fourth correction term is $\frac{1}{24}[\hat{B},[\hat{A},[\hat{B},\hat{A}]]]$.

7.10 Simple Harmonic Oscillator Wavefunctions

Most treatments of quantum mechanics focus on wavefunctions in coordinate space. Because the wavefunction is a tool to be used to predict probabilities for experiments (as we did with the spin and light experiments in Chaps. 1–4, its overuse can lead to misconceptions as it may suggest relating the wavefunction to other physical notions, which might either be untrue or might involve an interpretation that can neither be verified nor shown to be incorrect. What is important to realize is that a wavefunction is just the projection of a quantum state $|\psi\rangle$, represented by a ket (and often an eigenstate of some operator), onto some complete set of bra vectors that are convenient to use. We typically use one of two complete sets: (i) the coordinate (or position) space representation, with $\langle x|$, and (ii) the momentum-space representation, with $\langle p|$. While the coordinate-space basis is often the most popular one to use, both are equally valid and are useful. One way to think about the wavefunction is that it asks how much of $\langle x|$ (or $\langle p|$) is found in the specific state $|\psi\rangle$. Mod-squaring the wavefunction then gives the probability distribution. It should be clear this is particularly useful if the system has been prepared in a state $|\psi\rangle$ and I want to find the probability that it is located in a region near x or p.

We start by considering the ground-state wavefunction in position space, formed by taking the overlap of $|0\rangle$ with $\langle x|$ or $\psi_0(x) = \langle x|0\rangle$. To determine it, we have to use what we know. The first thing is to represent the position bra in terms of its translation operator acting on the position state at the origin $\langle 0_x|$ via

$$\psi_0(x) = \langle x|0\rangle = \langle 0_x|e^{\frac{i}{\hbar}x\hat{p}}|0\rangle. \tag{7.232}$$

So, what do we do next? We only know two things: (i) the harmonic oscillator ground state is annihilated by the lowering operator, or $\hat{a}|0\rangle = 0$, and (ii) the position eigenstate at the origin is annihilated by the position operator, or $\langle 0_x|\hat{x} = 0$. How do we use this here? The operator in the exponent is the momentum operator, not the position or lowering operator. Our first step is to rewrite it in terms of the raising and lowering operators via $\hat{p} = -i\sqrt{\frac{\hbar m\omega}{2}}(\hat{a} - \hat{a}^\dagger)$, which gives us

$$\psi_0(x) = \langle 0_x|e^{\sqrt{\frac{m\omega}{2\hbar}}x(\hat{a}-\hat{a}^\dagger)}|0\rangle. \tag{7.233}$$

We cannot do anything with these operators when both are in the same exponent, so we can use the Baker-Campbell-Hausdorff formula (which we will call the BCH formula from now on), to split the one exponential into two exponentials. We want the $\hat{a}$ operator to the right, because it annihilates against the ground state. So, we choose $\hat{A} = -\sqrt{\frac{m\omega}{2\hbar}}x\hat{a}^\dagger$ and $\hat{B} = \sqrt{\frac{m\omega}{2\hbar}}x\hat{a}$. We then have $-\frac{1}{2}[\hat{A}, \hat{B}] = -\frac{m\omega x^2}{4\hbar}$, which is a number, so we find that

$$\psi_0(x) = e^{-\frac{m\omega x^2}{4\hbar}} \left\langle 0_x | e^{-\sqrt{\frac{m\omega}{2\hbar}}x\hat{a}^\dagger} e^{\sqrt{\frac{m\omega}{2\hbar}}x\hat{a}} | 0 \right\rangle. \tag{7.234}$$

Now, we expand the exponential on the right in a power series and use the fact that $\hat{a}|0\rangle = 0$, to learn we can replace the exponential of the operator by 1. In other words, $e^{\alpha\hat{a}}|0\rangle = \sum_{n=0}^{\infty} \frac{\alpha^n}{n!}\hat{a}^n|0\rangle = 1|0\rangle = |0\rangle$. This yields

$$\psi_0(x) = e^{-\frac{m\omega x^2}{4\hbar}} \left\langle 0_x | e^{-\sqrt{\frac{m\omega}{2\hbar}}x\hat{a}^\dagger} | 0 \right\rangle. \tag{7.235}$$

Unfortunately, we obtain no simplification when $\hat{a}^\dagger$ acts on the right and we do not know how it acts to the left on a position eigenstate. So, what do we do? The only thing we know is how $\hat{x}$ acts on $\langle 0_x|$. How can we get an exponential of $\hat{x}$ in the exponent? This requires two steps. First, we can create exponentials of any number multiplied by $\hat{a}$ in front of $|0\rangle$, and then, we can use BCH to move the operator up to the exponent. So what constant? A little thought shows we use the following:

$$\psi_0(x) = e^{-\frac{m\omega x^2}{4\hbar}} \left\langle 0_x | e^{-\sqrt{\frac{m\omega}{2\hbar}}x\hat{a}^\dagger} e^{-\sqrt{\frac{m\omega}{2\hbar}}x\hat{a}} | 0 \right\rangle. \tag{7.236}$$

Then, we use BCH in reverse to put both terms in the same exponent, or

$$\psi_0(x) = e^{-\frac{m\omega x^2}{2\hbar}} \left\langle 0_x | e^{-\sqrt{\frac{m\omega}{2\hbar}}x(\hat{a}^\dagger + \hat{a})} | 0 \right\rangle, \tag{7.237}$$

note how the exponent of the Gaussian has increased by a factor of two. Now, replace the sum of the raising and lowering operator by the position operator and evaluate it on the $\langle 0_x|$ bra, to give

$$\psi_0(x) = e^{-\frac{m\omega x^2}{2\hbar}} \left\langle 0_x | e^{-\frac{m\omega}{\hbar}x\hat{x}} | 0 \right\rangle = e^{-\frac{m\omega x^2}{2\hbar}} \langle 0_x | 0 \rangle. \tag{7.238}$$

This is the wavefunction up to its normalization constant $\langle 0_x|0\rangle = \psi_0(x{=}0)$, which is a number. As I mentioned before, the normalization constant is the one thing we cannot independently calculate, so I have to tell you what it is—it is $\langle 0_x|0\rangle = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}}$. So, we have that the ground-state wavefunction is given by

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{m\omega x^2}{2\hbar}}. \tag{7.239}$$

This is a Gaussian function, which is maximal at $x = 0$. The probability to have a particle detected by a detector located at position x and with width Δx is given by $P(x) \approx |\psi_0(x)|^2 \Delta x = \sqrt{\frac{m\omega}{\pi\hbar}} e^{-\frac{m\omega x^2}{\hbar}} \Delta x$. The detector must have a narrow width Δx, for this formula to be accurate. It is similar to how we calculated the probability to be detected on the screen for our two-slit experiment.

The Gaussian function is an interesting function, as can be seen by looking at the ground-state wavefunction in Fig. 7.10. A Gaussian has a maximum and a width associated with it. The square of a Gaussian function remains a Gaussian function, but with a narrower width. When we look at the probability to find the particle anywhere, we see that the wavefunction is nonzero for all x, even though it gets very small for large $|x|$. This is quite different from a classical oscillator, which only oscillates between minus and plus some fixed amplitude.

The derivation of the momentum-space wavefunction is similar and is Problem 7.10.1 It becomes

$$\tilde{\phi}_0(p) = \left(\frac{1}{\pi\hbar m\omega}\right)^{\frac{1}{4}} e^{-\frac{p^2}{2\hbar m\omega}}. \tag{7.240}$$

Note that the functional form is identical (both are Gaussians) and they are related by $x \leftrightarrow \frac{p}{m\omega}$. The only thing that changes is the normalization constant.

We just computed the ground state. What about the higher-energy states? We proceed in the same way, just replace $|0\rangle \rightarrow \frac{1}{\sqrt{n!}}(\hat{a}^\dagger)^n|0\rangle$. The first step, using the BCH to separate out the two operators, is identical, and we find

$$\psi_n(x) = \langle x|n\rangle = e^{-\frac{m\omega x^2}{4\hbar}} \frac{1}{\sqrt{n!}} \left\langle 0_x | e^{-\sqrt{\frac{m\omega}{2\hbar}} x\hat{a}^\dagger} e^{\sqrt{\frac{m\omega}{2\hbar}} x\hat{a}} (\hat{a}^\dagger)^n |0\right\rangle. \tag{7.241}$$

We need to move the exponential of $\hat{a}$ to the right past $(\hat{a}^\dagger)^n$. This can be done in two different ways. The first is to use the general commutator we worked out in Eq. (7.178) to create an intertwining relation. Recall that $[\hat{a}, \hat{a}^\dagger] = 1$, so the pair of operators satisfy the criteria for using $\left[e^{\hat{A}}, \hat{B}\right] = [\hat{A}, \hat{B}]e^{\hat{A}}$. We choose $\hat{A} = \sqrt{\frac{m\omega}{2\hbar}} x\hat{a}$ and $\hat{B} = \hat{a}^\dagger$. Then, we have

$$\left[e^{\sqrt{\frac{m\omega}{2\hbar}} x\hat{a}}, \hat{a}^\dagger\right] = \sqrt{\frac{m\omega}{2\hbar}} x e^{\sqrt{\frac{m\omega}{2\hbar}} x\hat{a}}$$

$$e^{\sqrt{\frac{m\omega}{2\hbar}} x\hat{a}} \hat{a}^\dagger - \hat{a}^\dagger e^{\sqrt{\frac{m\omega}{2\hbar}} x\hat{a}} = \sqrt{\frac{m\omega}{2\hbar}} x e^{\sqrt{\frac{m\omega}{2\hbar}} x\hat{a}}$$

$$e^{\sqrt{\frac{m\omega}{2\hbar}}x\hat{a}}\hat{a}^\dagger = \left(\hat{a}^\dagger + \sqrt{\frac{m\omega}{2\hbar}}x\right)e^{\sqrt{\frac{m\omega}{2\hbar}}x\hat{a}}. \tag{7.242}$$

This is an *intertwining relation*. It says every time we move the exponential factor past an $\hat{a}^\dagger$ operator, by moving it to the right, we shift it by the constant in the above equation. So, when we shift through $(\hat{a}^\dagger)^n$, we have to do it n times, yielding n shifts by the constant, one for each $\hat{a}^\dagger$ operator. The second way to do this is to use the special case of the Hadamard lemma in Eq. (7.180). We invoke it by using a multiply by one on the right to create a Hadamard. This yields

$$\underbrace{e^{\sqrt{\frac{m\omega}{2\hbar}}x\hat{a}}(\hat{a}^\dagger)^n e^{-\sqrt{\frac{m\omega}{2\hbar}}x\hat{a}}}_{\text{Hadamard lemma}} e^{\sqrt{\frac{m\omega}{2\hbar}}x\hat{a}} = \left(\hat{a}^\dagger + \sqrt{\frac{m\omega}{2\hbar}}x\right)^n e^{\sqrt{\frac{m\omega}{2\hbar}}x\hat{a}}, \tag{7.243}$$

which is the same result we would get after using the intertwining relation n times.

Now that we know how to move the exponential of the lowering operator past the power of the raising operator, it then acts on the harmonic oscillator ground state, where we can replace it by 1. Next, we introduce a similar exponential factor to the right of the ground-state ket but with the opposite sign of the exponent (a multiply-by-one trick), so we can ultimately create an $\hat{x}$ operator in the exponential operator to the left. We need to move this through the power of the raising operator, and we can use the intertwining relation or the Hadamard lemma to do so. It produces another shift by the same amount giving us

$$\psi_n(x) = e^{-\frac{m\omega x^2}{4\hbar}}\frac{1}{\sqrt{n!}}\left\langle 0_x|e^{-\sqrt{\frac{m\omega}{2\hbar}}x\hat{a}^\dagger}e^{-\sqrt{\frac{m\omega}{2\hbar}}x\hat{a}}\left(\hat{a}^\dagger + \sqrt{\frac{2m\omega}{\hbar}}x\right)^n|0\right\rangle. \tag{7.244}$$

Now, we use the BCH formula in reverse, to put both raising and lowering operator terms into the same exponent, and then operate the exponential of a constant multiplied by $\hat{x}$ onto $\langle 0_x|$, where it can be replaced by 1. Then, we have

$$\psi_n(x) = e^{-\frac{m\omega x^2}{2\hbar}}\frac{1}{\sqrt{n!}}\left\langle 0_x|\left(\hat{a}^\dagger + \sqrt{\frac{2m\omega}{\hbar}}x\right)^n|0\right\rangle. \tag{7.245}$$

The claim we make is that the matrix element is a polynomial multiplied by the Gaussian function. But we have to do some more work to establish this.

We start by defining the function H_n by

$$H_n\left(x\sqrt{\frac{m\omega}{\hbar}}\right) = \frac{\sqrt{2^n}}{\langle 0_x|0\rangle}\left\langle 0_x|\left(\hat{a}^\dagger + x\sqrt{\frac{2m\omega}{\hbar}}\right)^n|0\right\rangle, \tag{7.246}$$

so that

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} \frac{1}{\sqrt{2^n n!}} H_n\left(x\sqrt{\frac{m\omega}{\hbar}}\right) e^{-\frac{m\omega x^2}{2\hbar}}. \tag{7.247}$$

If we set $n = 0$, we find from Eq. (7.246) that $H_0\left(x\sqrt{\frac{m\omega}{\hbar}}\right) = 1$. When we set $n = 1$, we find that

$$\begin{aligned} H_1\left(x\sqrt{\frac{m\omega}{\hbar}}\right) &= \frac{\sqrt{2}}{\langle 0_x|0\rangle}\left\langle 0_x|\hat{a}^\dagger + x\sqrt{\frac{2m\omega}{\hbar}}|0\right\rangle = \frac{\sqrt{2}}{\langle 0_x|0\rangle}\langle 0_x|(-\hat{a})|0\rangle + 2x\sqrt{\frac{m\omega}{\hbar}} \\ &= 2x\sqrt{\frac{m\omega}{\hbar}}. \end{aligned} \tag{7.248}$$

From the first to second equality, we replace $\langle 0_x|\hat{a}^\dagger$ by $\langle 0_x|(-\hat{a})$, because $\langle 0_x|\hat{x} = 0$, implies that $\langle 0_x|(\hat{a} + \hat{a}^\dagger) = 0$. Now, we determine a two-term recurrence relation for the general H_n. We peel off one of the operator factors in the definition of H_n on the left in Eq. (7.246). Then, the constant term is proportional to H_{n-1}, while the raising operator term can be replaced by the negative of the lowering operator, which becomes a commutator by the add-zero trick and then becomes proportional to H_{n-2} after evaluating the commutator. In equations, this is given by

$$\begin{aligned} H_n\left(x\sqrt{\tfrac{m\omega}{\hbar}}\right) &= \frac{\sqrt{2^n}}{\langle 0_x|0\rangle}\langle 0_x|\left(\hat{a}^\dagger + x\sqrt{\tfrac{2m\omega}{\hbar}}\right)\left(\hat{a}^\dagger + x\sqrt{\tfrac{2m\omega}{\hbar}}\right)^{n-1}|0\rangle \\ &= \frac{\sqrt{2^n}}{\langle 0_x|0\rangle}\langle 0_x|(-\hat{a})\left(\hat{a}^\dagger + x\sqrt{\tfrac{2m\omega}{\hbar}}\right)^{n-1}|0\rangle \\ &\quad + x\sqrt{\tfrac{2m\omega}{\hbar}}\frac{\sqrt{2^n}}{\langle 0_x|0\rangle}\langle 0_x|\left(\hat{a}^\dagger + x\sqrt{\tfrac{2m\omega}{\hbar}}\right)^{n-1}|0\rangle \\ &= -\frac{\sqrt{2^n}}{\langle 0_x|0\rangle}\langle 0_x|\left[\hat{a}, \left(\hat{a}^\dagger + x\sqrt{\tfrac{2m\omega}{\hbar}}\right)^{n-1}\right]|0\rangle + 2x\sqrt{\tfrac{m\omega}{\hbar}}H_{n-1}\left(x\sqrt{\tfrac{m\omega}{\hbar}}\right) \\ &= -(n-1)\frac{\sqrt{2^n}}{\langle 0_x|0\rangle}\langle 0_x|\left(\hat{a}^\dagger + x\sqrt{\tfrac{2m\omega}{\hbar}}\right)^{n-2}|0\rangle + 2x\sqrt{\tfrac{m\omega}{\hbar}}H_{n-1}\left(x\sqrt{\tfrac{m\omega}{\hbar}}\right) \\ &= 2x\sqrt{\tfrac{m\omega}{\hbar}}H_{n-1}\left(x\sqrt{\tfrac{m\omega}{\hbar}}\right) - 2(n-1)H_{n-2}\left(x\sqrt{\tfrac{m\omega}{\hbar}}\right). \end{aligned} \tag{7.249}$$

This is the recurrence relation for the *"physicist's" Hermite polynomials*. The recurrence relation is

$$H_n(z) = 2zH_{n-1}(z) - 2(n-1)H_{n-2}(z) \;\text{ with }\; z = x\sqrt{\frac{m\omega}{\hbar}}. \tag{7.250}$$

Table 7.1 First six Hermite polynomials

n	$H_n(z)$
0	1
1	$2z$
2	$4z^2 - 2$
3	$8z^3 - 12z$
4	$16z^4 - 48z^2 + 12$
5	$32z^5 - 160z^3 + 120z$

Since we already found H_0 and H_1 directly, we substitute into the recurrence relation to find H_2. Then, we know H_1 and H_2, so we can use the recurrence relation to find H_3 and so on. The first six Hermite polynomials are shown in Table 7.1. You will have the opportunity to use these same techniques to determine $\tilde{\phi}_n(p) = \langle p|n\rangle$. This calculation was originally worked out by Rushka and Freericks (2020).

There is another technique we can use to compute the wavefunctions and it is called the Rodrigues' formula. In calculus, this involves a complex formula of multiple derivatives. The operator-based generalization instead will involve nested commutators. It works with the position and momentum operators rather than the raising and lowering operators and was developed by Noonan et al. (2024).

The key identity comes from the special case of the Hadamard lemma with $\hat{A} = \frac{m\omega}{2\hbar}\hat{x}^2$ and $\hat{B} = -\frac{i}{\sqrt{2\hbar m\omega}}\hat{p}$. The commutator is $[\hat{A}, \hat{B}] = \sqrt{\frac{m\omega}{2\hbar}}\hat{x}$, which commutes with $\hat{A}$, so we have that

$$\hat{a}^\dagger = -\frac{i}{\sqrt{2\hbar m\omega}} e^{\frac{m\omega}{2\hbar}\hat{x}^2}\,\hat{p}\,e^{-\frac{m\omega}{2\hbar}\hat{x}^2}. \tag{7.251}$$

Recall that in order to remove operators that act on states, we need to find a state that is annihilated by the operator. Here, we know that $\hat{p}|0\rangle = im\omega\hat{x}|0\rangle$, from the subsidiary condition $\hat{a}|0\rangle = 0$. Then, we have

$$\begin{aligned}\hat{p}e^{\frac{m\omega}{2\hbar}\hat{x}^2}|0\rangle &= \underbrace{e^{\frac{m\omega}{2\hbar}\hat{x}^2}e^{-\frac{m\omega}{2\hbar}\hat{x}^2}}_{=1}\hat{p}e^{\frac{m\omega}{2\hbar}\hat{x}^2}|0\rangle = e^{\frac{m\omega}{2\hbar}\hat{x}^2}(\hat{p} - im\omega\hat{x})|0\rangle \\ &= e^{\frac{m\omega}{2\hbar}\hat{x}^2}\sqrt{\frac{2\hbar}{m\omega}}\hat{a}|0\rangle = 0.\end{aligned} \tag{7.252}$$

We use a multiply by one, followed by n add zeros, to allow us to convert powers of $\hat{p}$ into nested commutators of $\hat{p}$—the nested commutators are the operator equivalent of the conventional Rodrigues' formula. The multiply by one is is used to create the state annihilated by $\hat{p}$, and then, the add zero just creates the commutators, one by one, starting from the rightmost $\hat{p}$ operator. So, we have

$$(\hat{a}^\dagger)^n|0\rangle = \left(-\frac{i}{\sqrt{2\hbar m\omega}}\right)^n e^{\frac{m\omega}{2\hbar}\hat{x}^2}\hat{p}^n e^{-\frac{m\omega}{2\hbar}\hat{x}^2}\underbrace{e^{-\frac{m\omega}{2\hbar}\hat{x}^2}e^{\frac{m\omega}{2\hbar}\hat{x}^2}}_{=1}|0\rangle \tag{7.253}$$

$$= \left(-\frac{i}{\sqrt{2\hbar m\omega}}\right)^n e^{\frac{m\omega}{2\hbar}\hat{x}^2}\left[\hat{p},\left[\hat{p},\cdots,\left[\hat{p},e^{-\frac{m\omega}{\hbar}\hat{x}^2}\right]\cdots\right]\right]_n \underbrace{e^{\frac{m\omega}{2\hbar}\hat{x}^2}|0\rangle}_{\text{annihilated by } \hat{p}}.$$

The commutator is nested n-fold times and we can move the Gaussian factor next to the ground-state ket to the left through the nested commutators, because the nested commutators produce just a function of $\hat{x}$ which commutes with it. We use this form to define the Hermite-polynomial operator via the analog of the Rodrigues' formula as

$$(-1)^n e^{\frac{m\omega}{\hbar}\hat{x}^2}\left[\frac{i\hat{p}}{\sqrt{\hbar m\omega}},\left[\frac{i\hat{p}}{\sqrt{\hbar m\omega}},\cdots,\left[\frac{i\hat{p}}{\sqrt{\hbar m\omega}},e^{-\frac{m\omega}{\hbar}\hat{x}^2}\right]\cdots\right]\right]_n|0\rangle$$
$$= H_n\left(\hat{x}\sqrt{\frac{m\omega}{\hbar}}\right)|0\rangle = \sqrt{2^n}(\hat{a}^\dagger)^n|0\rangle. \tag{7.254}$$

Note that this identity requires the operators to act on the simple harmonic oscillator ground state, because we have used the subsidiary condition in our derivation. To check that this agrees with our old derivation, one can immediately see that $n = 0$ has $H_0\left(\hat{x}\sqrt{\frac{m\omega}{\hbar}}\right) = 1$ and $n = 1$ has

$$H_1\left(\hat{x}\sqrt{\frac{m\omega}{\hbar}}\right)|0\rangle = -e^{\frac{m\omega}{\hbar}\hat{x}^2}\left[\frac{i\hat{p}}{\sqrt{\hbar m\omega}},e^{-\frac{m\omega}{\hbar}\hat{x}^2}\right]|0\rangle = 2\hat{x}\sqrt{\frac{m\omega}{\hbar}}|0\rangle. \tag{7.255}$$

These both agree with our previous work. Now, we must determine the recurrence relation.

Using the definition of the operator-valued Hermite polynomial in Eq. (7.254), we have

$$H_n\left(\hat{x}\sqrt{\frac{m\omega}{\hbar}}\right)|0\rangle = -e^{\frac{m\omega}{\hbar}\hat{x}^2}\left[\frac{i\hat{p}}{\sqrt{\hbar m\omega}},H_{n-1}\left(\hat{x}\sqrt{\frac{m\omega}{\hbar}}\right)e^{-\frac{m\omega}{\hbar}\hat{x}^2}\right]|0\rangle. \tag{7.256}$$

Using the Leibniz rule, we find

$$H_n\left(\hat{x}\sqrt{\frac{m\omega}{\hbar}}\right)|0\rangle$$
$$= \left(2\hat{x}\sqrt{\frac{m\omega}{\hbar}}H_{n-1}\left(\hat{x}\sqrt{\frac{m\omega}{\hbar}}\right) - \left[\frac{i\hat{p}}{\sqrt{\hbar m\omega}},H_{n-1}\left(\hat{x}\sqrt{\frac{m\omega}{\hbar}}\right)\right]\right)|0\rangle. \tag{7.257}$$

We need to determine the commutator. For that, we use a trick, motivated by how we determined the recurrence relation using our first method to calculate the wavefunction, and examine what happens when we operate a lowering operator onto $\sqrt{2^n}(\hat{a}^\dagger)^n|0\rangle$. We have

$$\sqrt{2^n}\hat{a}(\hat{a}^\dagger)^n|0\rangle = \sqrt{2^n}\left[\hat{a}, (\hat{a}^\dagger)^n\right]|0\rangle = n\sqrt{2^n}(\hat{a}^\dagger)^{n-1}|0\rangle. \tag{7.258}$$

Translating this into the Hermite-polynomial operators by comparing to Eq. (7.254) gives us

$$\hat{a}\,H_n\left(\hat{x}\sqrt{\frac{m\omega}{\hbar}}\right)|0\rangle = \sqrt{2}n\,H_{n-1}\left(\hat{x}\sqrt{\frac{m\omega}{\hbar}}\right)|0\rangle. \tag{7.259}$$

But the $\hat{a}$ operator multiplying the Hermite-polynomial operator can be replaced by a commutator (by the add-zero trick), then replacing the lowering operator by its definition in terms of the position and momentum operators, and noting that the position operator commutes with H_n gives us

$$\frac{i}{\sqrt{2\hbar m\omega}}\left[\hat{p}, H_n\left(\hat{x}\sqrt{\frac{m\omega}{\hbar}}\right)\right]|0\rangle = \sqrt{2}n\,H_{n-1}\left(\hat{x}\sqrt{\frac{m\omega}{\hbar}}\right)|0\rangle. \tag{7.260}$$

Rearranging and taking $n \to n-1$, we find

$$\left[\frac{i\,\hat{p}}{\sqrt{\hbar m\omega}}, H_{n-1}\left(\hat{x}\sqrt{\frac{m\omega}{\hbar}}\right)\right]|0\rangle = 2(n-1)H_{n-2}\left(\hat{x}\sqrt{\frac{m\omega}{\hbar}}\right)|0\rangle. \tag{7.261}$$

This then yields a similar Hermite-polynomial recurrence relation, given by

$$H_n\left(\hat{x}\sqrt{\frac{m\omega}{\hbar}}\right)|0\rangle = 2\hat{x}\sqrt{\frac{m\omega}{\hbar}}\,H_{n-1}\left(\hat{x}\sqrt{\frac{m\omega}{\hbar}}\right)|0\rangle - 2(n-1)H_{n-2}\left(\hat{x}\sqrt{\frac{m\omega}{\hbar}}\right)|0\rangle. \tag{7.262}$$

Note the similarity with the earlier recurrence relation in Eq. (7.249)—except here it is for operator-valued polynomials acting on the harmonic oscillator ground state. If we multiply the last line of Eq. (7.254) by the $\langle x|$ bra from the left, we immediately see that we obtain the same final wavefunction in position space. Note that this Rodrigues' formula in this particular form is not useful in calculating the wavefunction in momentum space. Problem 7.10.5 discusses this in more detail at the end of the section and which describes how you can construct a different form of the Rodrigues' formula for the momentum wavefunction.

Now that we have the wavefunctions, let's look at them. Note that the wavefunctions for $\psi_n(x)$ and $\tilde{\phi}_n(p)$ are of the same form, except for the different normalization constants ($\langle 0_x|0\rangle$ versus $\langle 0_p|0\rangle$) and the substitution $x \to \frac{p}{m\omega}$. In Fig. 7.10, we plot the first 12 position-space wavefunctions centered on their respective energies, overlaid by the harmonic oscillator potential $V(x) = \frac{1}{2}m\omega^2x^2$; this can also be viewed as a momentum-space wavefunction overlaid with the kinetic energy. The probability density is the modulus squared of the wavefunction.

When we look at the wavefunctions, we can identify some general behaviors that are generic for wavefunctions in one dimension: (i) the number of nodes (points

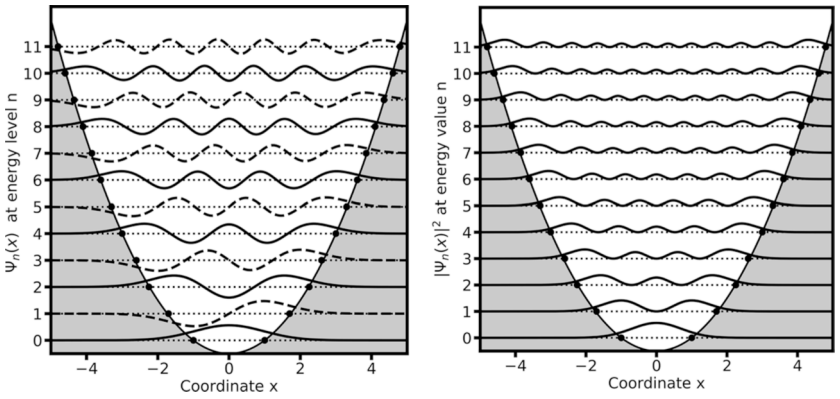

Fig. 7.10 (Left) lowest 12 simple harmonic oscillator wavefunctions centered on their respective energy. (Right) modulus squared of the 12 lowest simple harmonic oscillator wavefunctions. The potential is also plotted and the allowed region is white (kinetic energy greater than zero, while the forbidden region is gray (kinetic energy less than zero). The solid lines (in the left panel) are even wavefunctions, while the dashed lines are odd wavefunctions. These plots can also be interpreted as the momentum-space wavefunctions, as described in the text

where $\psi(x) = 0$) for the energy wavefunctions starts at zero for the ground state and increases by one for each higher energy state; (ii) the coordinate-space wavefunctions oscillate when $E_n > V(x)$ and rapidly decay when $E_n < V(x)$; (iii) the ground state is an even function of x ($\psi(x) = \psi(-x)$), and higher-energy states alternate between odd ($\psi(x) = -\psi(-x)$) and even as the energy increases; and (iv) the amplitude and probability are usually largest near the classical turning point, where $V(x) = E_n = \hbar\omega\left(n + \frac{1}{2}\right)$, especially for large n. This is expected, because the particle spends more time near the turning points (the places where $V(x) = E_n$), since it slows down and turns around there. One can say similar statements about the momentum wavefunctions, which we discuss in Problem 7.10.7.

One should also note that the period of oscillations gets longer as the turning point is approached and this becomes clearer as n gets large. In Fig. 7.11, we plot the wavefunction and probability distribution function for $\psi_{100}(x)$. This function is even and has 100 nodes.

Let's see if we can understand some of these rules better. The number of nodes increases because this is needed to ensure energy eigenfunctions for different energies are all orthogonal. But it is difficult to motivate, or even explain, without calculus. So, just think of this as an interesting observation here. We will explain how one can prove the node theorem using our operator methods later. The oscillations can be understood by recalling that the free particle oscillated with a wavelength $\lambda = \frac{h}{p}$. We can think of the wavefunction for the harmonic oscillator as roughly oscillating with a wavelength that continuously changes as p changes with x. This will also help us explain why the amplitude is largest near the turning point. The wavefunctions are even or odd, because the potential is an even function

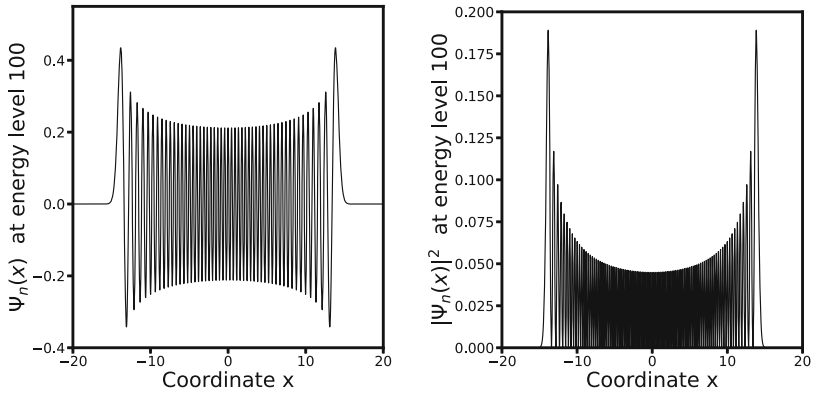

Fig. 7.11 (Left) wavefunction and (right) modulus squared of the wavefunction for $n = 100$. Note how the maximal probability near the origin is fairly flat and how it is peaked near the turning point (at about ± 14.18). You can also see how the period of the oscillations increases as one approaches the turning point

of x, implying parity is a conserved quantity (commutes with $\hat{H}$), and hence, we can simultaneously diagonalize energy and parity.

The way to understand some more of this qualitative behavior is to go back to the motion of a classical simple harmonic oscillator, namely, a mass on a spring. Give it a push, and it will oscillate back and forth forever, if there is no friction. This is illustrated schematically in Fig. 7.5. If we look at the position as a function of time, it is given by $x(t) = \bar{x}\cos\omega t$ and the momentum is $p(t) = -m\omega\bar{x}\sin\omega t$. Recall that the cosine and sine curves are plotted in Fig. 5.8. When $|x|$ is a maximum, the cosine is equal to ± 1 and the sine is equal to 0, so $p = 0$. When $|p|$ is a maximum, the sine is equal to ± 1 and the cosine is 0, so $x = 0$. In quantum mechanics, $|\psi_n(x)|^2\Delta x$ determines the probability to find the particle that is in the nth energy eigenstate to be located in the small region around point x (Δx is the width of the detector). So, where should this probability distribution be the largest? The classical answer is where it is moving the slowest. This is because it spends the most time in the region where it is moving most slowly. For oscillatory motion, it moves the slowest at the classical turning point, because this is where the particle stops instantaneously and turns around. Indeed, the quantum probability distribution is peaked here as well. Now, here is the surprise. Where is the momentum distribution peaked? You might think near $p = 0$ again, because it is moving the slowest there and so should spend the most time there, but that is actually incorrect! Indeed, I was stymied by this reasoning for a long time too. The momentum distribution is maximal where $|p|$ is the largest, which occurs near $x = 0$. Why? Because momentum changes the slowest when it is "turning around" and that happens near its maximum and minimum (just like position changed the slowest near its maximum and minimum). This is true for all smooth functions. And this is what you see if you recall that the quantum result for the position wavefunction (and probability distribution) look the

same as the momentum one. Clearly, the momentum is peaked around the points $\pm P_{max}$. Even though it may seem counterintuitive, it is correct, as you can see by looking at Fig. 7.11.

Let us look at some problems.

Section 7.10 Problems:

7.10.1 Simple Harmonic Oscillator Ground-State Wavefunction in Momentum Space

Use the momentum translation operator to write the momentum-space ground-state wavefunction as

$$\tilde{\phi}_0(p) = \langle p|0\rangle = \langle 0_p|e^{-\frac{i}{\hbar}p\hat{x}}|0\rangle, \tag{7.263}$$

with $\langle 0_p|\hat{p} = 0$. Use a similar operator-based methodology to what we did in the text, with the BCH formula (not the operator Rodrigue's formula), to determine the momentum-space ground-state wavefunction. Use $\langle 0_p|0\rangle = \left(\frac{1}{\pi\hbar m\omega}\right)^{\frac{1}{4}}$.

7.10.2 Simple Harmonic Oscillator Wavefunction in Momentum Space

Working off of the results from the previous problem, compute the momentum-space wavefunctions for arbitrary energy eigenstates. You will find the argument of the Hermite polynomial is $z = \frac{p}{\sqrt{\hbar m\omega}}$. You may find that you end up with an additional factor of i^n in your wavefunction. This can be removed, because the global phase of any wavefunction is not determined.

7.10.3 Hermite Polynomials

Work out the next three entries in Table 7.1 to find $H_6(z)$, $H_7(z)$, and $H_8(z)$.

7.10.4 Shortcut Derivation in Position Space

In our first derivation of the harmonic oscillator wavefunctions, we split the translation operator into exponentials of raising and lowering operators, moved

the exponential of the lowering operator to the right until it could be removed by operating against the ground state. Then, we multiplied by 1 to introduce another exponential of the lowering operator, which is moved to the right until it can join with the exponential of the raising operator to make an exponential of the position operator. This then is removed when it acts on the position eigenstate at the origin.

This approach can be shortcut in two ways. First, we could leave the momentum operator in the exponent and just introduce an exponential of the lowering operator at the ground-state ket and move it to the left so it can be combined into the exponent with the raising operator to create the position operator. What is the operator that needs to be introduced of the form $e^{\alpha\hat{a}}$? Verify you get the same wavefunction using this shortcut.

There is another shortcut we can use. Instead of using a BCH formula to break into exponentials of raising and lowering operators, we can instead break into exponentials of $\hat{x}$ and $\hat{a}$. Determine how to write $\hat{p} = \alpha\hat{x} + \beta\hat{a}$, so that you can carry out this shortcut. Then, carry it out and verify you get the correct position-space wavefunction.

These types of shortcuts could also be used in calculating the momentum-space wavefunction. But you are not asked to do that in this problem.

7.10.5 *Rodrigues' Formula for Momentum-Space Wavefunctions*

The form of the Rodrigues' formula that was derived in the text is not so useful to derive the wavefunctions in momentum space. This is because the momentum-space wavefunction would take the form

$$\langle p|\sqrt{2^n}H_n\left(\hat{x}\sqrt{\frac{m\omega}{\hbar}}\right)|0\rangle. \tag{7.264}$$

The Hermite-polynomial operator is a polynomial in $\hat{x}$. But the calculation of $\langle p|\hat{x}^m|0\rangle$ is complicated—one can use the subsidiary condition to replace the rightmost $\hat{x}$ factor by a constant multiplied by $\hat{p}$, but moving it to the left, where we can evaluate it against the momentum eigenstate, has a correction term in the form of a commutator. We find many correction terms for large powers m, as we have to repeat this procedure many times. One can try to work out how to calculate the results this way, but it is quite complicated to do so.

Instead, the best way to proceed is to use a different Rodrigues' formula, by writing the raising operator as

$$\hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}}e^{\frac{1}{2\hbar m\omega}\hat{p}^2}\hat{x}e^{-\frac{1}{2\hbar m\omega}\hat{p}^2}. \tag{7.265}$$

This then allows you to construct the Rodrigues' formula as a sequence of nested commutators with respect to the position operator. Show that this procedure can work and you obtain the correct momentum-space wavefunction. Note that the Hermite polynomials you will work with will have an argument given by $z = \frac{p}{\sqrt{\hbar m\omega}}$ and that you now have to determine the state that vanishes when $\hat{x}$ acts on it.

7.10.6 Even or Odd?

A harmonic oscillator energy eigenstate wavefunction has 17 nodes. Is it an even function or an odd function or neither?

7.10.7 Rules That Momentum-Space Wavefunctions Satisfy

Restate the rules that the wavefunctions satisfy for momentum-space wavefunctions, keeping in mind that the turning points in momentum space are the points where $E_n = \frac{p^2}{2m}$.

7.10.8 Probability of Position and of Momentum

We already saw that the probability density is given by $|\psi_n(x)|^2$ or $|\tilde{\phi}_n(p)|^2$ in position and momentum space, respectively. For this problem, we focus on position space, but the approach is similar in momentum space.

We ask the question, what is the probability density for the position and for the momentum? We know the expectation value of position or momentum in an energy eigenstate is given by $\langle n|\hat{x}|n\rangle$ or $\langle n|\hat{p}|n\rangle$. To get to the probability distributions, we insert the complete set of states $\sum_x |x\rangle\langle x|$ and focus on the results for each individual x, as x is varied. In other words, the probability density for position is

$$P_x(x) = \langle n|x\rangle\langle x|\hat{x}|n\rangle = x|\psi_n(x)|^2, \tag{7.266}$$

and the probability distribution for momentum is

$$P_p(x) = \langle n|x\rangle\langle x|\hat{p}|n\rangle. \tag{7.267}$$

The probability distribution for momentum cannot be immediately computed. We must simplify $\frac{1}{\sqrt{n!}}\hat{p}(\hat{a}^\dagger)^n|0\rangle$ in order to determine it. Note that the momentum operator can be expanded in terms of raising and lowering operators—the raising operator will change the energy eigenstate, and the lowering operator can be

replaced by a commutator, which removes a raising operator, also changing the energy eigenstate. Show that

$$P_p(x) = -i\sqrt{\frac{\hbar m\omega}{2}}\psi_n(x)^*\big(\sqrt{n}\psi_{n-1}(x) - \sqrt{n+1}\psi_{n+1}(x)\big). \tag{7.268}$$

This probability distribution for momentum is also called the *particle current density*.

7.11 Coherent and Squeezed States

So far, we have been working with states of definite energy. Even though definite energy seems like a natural concept, these states appear quite foreign to our everyday experience—the energy values are discrete, they have energy spacings between them of a fixed size, and the lowest-energy state still has nonzero energy in it. In short, they don't really look anything like a mass on a spring that we are used to. They even can be quite challenging to create within a laboratory. Is there an analog in quantum mechanics to pulling the mass, letting it go, and watching the mass oscillate to and fro? The answer is yes, and it was Erwin Schrödinger who first examined them back in 1926 (Schrödinger, 1926b).

We have operators that correspond to pulling the spring and letting it go (translation of position, given by $e^{-\frac{i}{\hbar}x\hat{p}}$) and giving it a push (translation of momentum, given by $e^{\frac{i}{\hbar}p\hat{x}}$) or an arbitrary combination of the two, which we write as

$$e^{-\frac{i}{\hbar}x\hat{p}}e^{\frac{i}{\hbar}p\hat{x}} = e^{\frac{i}{\hbar}(-x\hat{p}+p\hat{x})}e^{-\frac{i}{2\hbar}xp}, \tag{7.269}$$

after using the BCH formula in it's Weyl form. Note that we plan to apply this operator to the ground state, but the global phase of a state does not change any of its properties, so we drop the extra complex phase when we define the *displacement operator*. But to properly write down the displacement operator requires us to convert from position and momentum into raising and lowering operators. Recalling that $\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a}+\hat{a}^\dagger)$ and $\hat{p} = \sqrt{\frac{\hbar m\omega}{2}}\frac{1}{i}(\hat{a}-\hat{a}^\dagger)$, we have

$$\hat{D}(\alpha) = e^{\frac{i}{\hbar}(-x\hat{p}+p\hat{x})} = e^{\sqrt{\frac{m\omega}{2\hbar}}x(-\hat{a}+\hat{a}^\dagger)+\frac{i}{\sqrt{2\hbar m\omega}}p(\hat{a}+\hat{a}^\dagger)} = e^{-\alpha^*\hat{a}+\alpha\hat{a}^\dagger} \tag{7.270}$$

with the displacement parameter α equal to

$$\alpha = \sqrt{\frac{m\omega}{2\hbar}}x + \frac{i}{\sqrt{2\hbar m\omega}}p. \tag{7.271}$$

You will show in Problem 7.11.1 that the displacement operator is a unitary operator. We call the *coherent state* $|\alpha\rangle$ the state that results after applying the displacement operator onto the harmonic oscillator ground state, or

$$\hat{D}(\alpha)|0\rangle = e^{-\alpha^*\hat{a}+\alpha\hat{a}^\dagger}|0\rangle = |\alpha\rangle. \tag{7.272}$$

Since the displaced state has shifted the position or momentum of the ground state, this is exactly how we pull the ground state or give it a push.

The coherent state has some interesting properties. Perhaps, one of the most interesting ones is that is it an eigenstate of the lowering operator. We can verify this from our simplified form of the Hadamard lemma. We simply use a multiply by one on the left, recognize the Hadamard lemma, evaluate it, and then use the fact that the ground state is annihilated by the lowering operator. In equations, we have

$$\begin{aligned}
\hat{a}|\alpha\rangle &= \hat{a}\hat{D}(\alpha)|0\rangle = \hat{a}e^{-\alpha^*\hat{a}+\alpha\hat{a}^\dagger}|0\rangle \\
&= \underbrace{e^{-\alpha^*\hat{a}+\alpha\hat{a}^\dagger}e^{\alpha^*\hat{a}-\alpha\hat{a}^\dagger}}_{=1}\hat{a}e^{-\alpha^*\hat{a}+\alpha\hat{a}^\dagger}|0\rangle \\
&= e^{-\alpha^*\hat{a}+\alpha\hat{a}^\dagger}\underbrace{e^{\alpha^*\hat{a}-\alpha\hat{a}^\dagger}\hat{a}e^{-\alpha^*\hat{a}+\alpha\hat{a}^\dagger}}_{\text{Hadamard lemma}}|0\rangle \\
&= e^{-\alpha^*\hat{a}+\alpha\hat{a}^\dagger}(\hat{a}+\alpha)|0\rangle = \alpha e^{-\alpha^*\hat{a}+\alpha\hat{a}^\dagger}|0\rangle = \alpha\hat{D}(\alpha)|0\rangle = \alpha|\alpha\rangle.
\end{aligned} \tag{7.273}$$

This means that when the lowering operator acts on a coherent state, it can replace the operator by α via $\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$. By taking the Hermitian conjugate, we find that $\langle\beta|\hat{a}^\dagger = \langle\beta|\beta^*$. So, the bra is an eigenstate of the raising operator. You will show in Problem 7.11.2 that the coherent state ket is not an eigenstate of the raising operator and the coherent state bra is not an eigenstate of the lowering operator.

The fact that the coherent state is an eigenvector of the lowering operator is something that can be understood by defining "shifted" raising and lowering operators for a Hamiltonian shifted in position and momentum. Consider the shifted Hamiltonian

$$\hat{H}' = \frac{(\hat{p}-p)^2}{2m} + \frac{1}{2}m\omega^2(\hat{x}-x)^2 = \frac{\hat{p}'^2}{2m} + \frac{1}{2}m\omega^2\hat{x}'^2, \tag{7.274}$$

where we have shifted the position by x and the momentum by p. Then, we define a shifted lowering operator via

$$\hat{a}' = \sqrt{\frac{m\omega}{2\hbar}}(\hat{x}-x) + \frac{i}{\sqrt{2\hbar m\omega}}(\hat{p}-p). \tag{7.275}$$

Then, one can show, as before, that $\hat{H}' = \hbar\omega\left(\hat{a}'^\dagger\hat{a}' + \frac{1}{2}\right)$ and the ground state satisfies $\hat{a}'|0'\rangle = 0$. If we convert back to our original lowering operator, we find that

$$\hat{a}'|0'\rangle = (\hat{a} - \alpha)|0'\rangle = 0 \implies \hat{a}|0'\rangle = \alpha|0'\rangle. \tag{7.276}$$

This means that $|0'\rangle = |\alpha\rangle$. This shows again that the coherent state is just a displaced ground state, but it also shows that the coherent state is an eigenstate of the lowering operator because that is what is needed to have the subsidiary condition for the shifted Hamiltonian. So, the eigenvalue-eigenvector condition is required simply to find the ground state of the shifted Hamiltonian.

We have found a new harmonic oscillator state. It is not an energy eigenstate. It instead corresponds to stretching and pushing the mass on the "spring." Can we relate these new states to the energy eigenstates? We should be able to, because the set of eigenstates of a self-adjoint operator forms a complete set, so any state we create should be able to be expanded in terms of the energy eigenstates. How do we do this? It follows from the the application of the displacement operator onto the ground state to determine the coherent state. If we use the BCH formula, in its Weyl form, to separate the single exponential factor into two factors (plus a correction), with the lowering operator to the right, we have

$$|\alpha\rangle = \hat{D}(\alpha)|0\rangle = e^{-\alpha^*\hat{a}+\alpha\hat{a}^\dagger}|0\rangle = e^{\alpha\hat{a}^\dagger}e^{-\alpha^*\hat{a}}e^{\frac{1}{2}|\alpha|^2[\hat{a}^\dagger,\hat{a}]}|0\rangle, \tag{7.277}$$

where the third term is the correction term equal to $e^{-\frac{1}{2}|\alpha|^2}$ after calculating the commutator. Now, because $\hat{a}|0\rangle = 0$, the exponential term with $\hat{a}$ can be replaced by 1, when it operates on the ground state. We then expand the remaining exponential in its power series, to give

$$|\alpha\rangle = e^{-\frac{1}{2}|\alpha|^2}\sum_{n=0}^{\infty}\frac{\alpha^n}{n!}\big(\hat{a}^\dagger\big)^n|0\rangle = e^{-\frac{1}{2}|\alpha|^2}\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}|n\rangle. \tag{7.278}$$

This is the expansion of the coherent state in terms of the energy eigenstates. In Problems 7.11.3 and 7.11.4, we verify that the coherent state is normalized, and that it is an eigenstate of the lowering operator, from this expansion. You also will show that two coherent states are never orthogonal to each other.

Because we have only translated the ground-state position and momentum, we anticipate that the shape of the wavefunction is unchanged, so its uncertainty relations should be the same as well. We verify this next. Start with the average of position, which we expect to be equal to x, because that is what we translated to. Let's check. The average is given by

$$\langle\alpha|\hat{x}|\alpha\rangle = \sqrt{\frac{\hbar}{2m\omega}}\langle\alpha|\big(\hat{a}+\hat{a}^\dagger\big)|\alpha\rangle = \sqrt{\frac{2\hbar}{m\omega}}\mathrm{Re}\,\alpha = x. \tag{7.279}$$

In computing the expectation value, we used the facts that $\langle\alpha|\hat{a}^\dagger = \langle\alpha|\alpha^*$ and $\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$. We find it is equal to x, by using the explicit value of α in Eq. (7.271). Now, for the expectation value of $\hat{x}^2$, we find

$$\begin{aligned}\langle\alpha|\hat{x}^2|\alpha\rangle &= \frac{\hbar}{2m\omega}\langle\alpha|\left(\hat{a}+\hat{a}^\dagger\right)\left(\hat{a}+\hat{a}^\dagger\right)|\alpha\rangle \\ &= \frac{\hbar}{2m\omega}\langle\langle\alpha|\left(\hat{a}^2+\hat{a}^\dagger\hat{a}+\hat{a}\hat{a}^\dagger+(\hat{a}^\dagger)^2\right)|\alpha\rangle \\ &= \frac{\hbar}{2m\omega}\langle\langle\alpha|\left(\hat{a}^2+2\hat{a}^\dagger\hat{a}+(\hat{a}^\dagger)^2+1\right)|\alpha\rangle \\ &= \frac{\hbar}{2m\omega}\left((\alpha+\alpha^*)^2+1\right) = x^2+\frac{\hbar}{2m\omega}. \end{aligned} \tag{7.280}$$

In computing this result, we needed to use the commutation relation to convert $\hat{a}\hat{a}^\dagger = \hat{a}^\dagger\hat{a}+1$ before we can compute the average. So, we find that

$$(\Delta x)_\alpha^2 = \frac{\hbar}{2m\omega} \quad \text{or} \quad (\Delta x)_\alpha = \sqrt{\frac{\hbar}{2m\omega}}. \tag{7.281}$$

In Problem 7.11.7, you will show that

$$(\Delta p)_\alpha^2 = \frac{\hbar m\omega}{2} \quad \text{or} \quad (\Delta p)_\alpha = \sqrt{\frac{\hbar m\omega}{2}}. \tag{7.282}$$

Hence, we have

$$(\Delta x)_\alpha(\Delta p)_\alpha = \frac{\hbar}{2}, \tag{7.283}$$

just as we claimed it should be. So, coherent states are always minimum uncertainty states. The video Coherent States https://tinyurl.com/eexecxr8 discusses some additional ideas.

Before we can move onto our next topic, which is squeezed states, we need to derive the general form for the Hadamard lemma. It takes the form

$$e^{\hat{A}}\hat{B}e^{-\hat{A}} = \sum_{n=0}^{\infty}\frac{1}{n!}\underbrace{[\hat{A},[\hat{A},[\cdots[\hat{A},\hat{B}]\cdots]]]}_{n \text{ nested commutators with } \hat{A}}. \tag{7.284}$$

The n-fold nested commutators have n $\hat{A}$ operators and one $\hat{B}$ operator in the nested commutator, with the $\hat{B}$ operator in the center. Note that the case with $n = 0$ is given by just the operator $\hat{B}$. This is our second fundamental operator identity in its full generality (the first fundamental identity is the Leibniz formula). The proof is straightforward and proceeds by induction. We start by expanding the exponentials in a power series:

$$\left(\sum_{m'=0}^{\infty}\frac{\hat{A}^{m'}}{m'!}\right)\hat{B}\left(\sum_{m=0}^{\infty}\frac{(-1)^m\hat{A}^m}{m!}\right). \tag{7.285}$$

Next, we group the terms into definite powers of $\hat{A}$. The terms with exactly n $\hat{A}$ operators in the expansion are given by

$$\sum_{m=0}^{n} \frac{\hat{A}^{n-m}}{(n-m)!} \hat{B} \frac{(-1)^m \hat{A}^m}{m!}, \tag{7.286}$$

as can be worked out directly from the previous expression by including all terms where $m' = n - m$. In order to complete the proof, we need to show that this is equal to $\frac{1}{n!}[\hat{A}, [\hat{A}, [\cdots [\hat{A}, \hat{B}] \cdots]\,]\,]_n$, where the subscript n reminds us that we have n nested commutators. One can immediately see that the $n = 0$ term is $\hat{B}$ and the $n = 1$ term is $\hat{A}\hat{B} - \hat{B}\hat{A} = [\hat{A}, \hat{B}]$, which we can take as our base case. Just as a check, we consider the $n = 2$ case, which is

$$\frac{1}{2}\hat{A}^2\hat{B} - \hat{A}\hat{B}\hat{A} + \frac{1}{2}\hat{B}\hat{A}^2 = \frac{1}{2}\big(\hat{A}[\hat{A}, \hat{B}] - [\hat{A}, \hat{B}]\hat{A}\big) = \frac{1}{2}[\hat{A}, [\hat{A}, \hat{B}]\,]. \tag{7.287}$$

This also checks, so we are ready for the general case. We assume it works for all cases up to $n - 1$ and then prove the result for n. So, our induction hypothesis is that

$$\sum_{m=0}^{n-1} \frac{\hat{A}^{n-1-m}}{(n-1-m)!} \hat{B} \frac{(-1)^m \hat{A}^m}{m!} = \frac{1}{(n-1)!}[\hat{A}, [\hat{A}, [\cdots [\hat{A}, \hat{B}] \cdots]\,]\,]_{n-1}. \tag{7.288}$$

Our strategy for proof is to work out a formula for the n-fold nested commutator, which we find by commuting both sides of Eq. (7.288) with $\hat{A}$ and multiplying both sides by $(n-1)!$. This gives us

$$[\hat{A}, [\hat{A}, [\cdots [\hat{A}, \hat{B}],]]]_n = \sum_{m=0}^{n-1} \frac{(n-1)!(-1)^m}{(n-1-m)!m!} [\hat{A}, \hat{A}^{n-1-m}\hat{B}\hat{A}^m]. \tag{7.289}$$

Expanding out the commutator on the right-hand side gives us

$$\sum_{m=0}^{n-1} \frac{(n-1)!}{(n-1-m)!m!} \left((-1)^m \hat{A}^{n-m}\hat{B}\hat{A}^m + (-1)^{m+1}\hat{A}^{n-1-m}\hat{B}\hat{A}^{m+1}\right). \tag{7.290}$$

Now, we let $m \to m - 1$ in the second summation, which gives

$$\begin{aligned} &\sum_{m=0}^{n-1} \frac{(n-1)!}{(n-1-m)!m!} (-1)^m \hat{A}^{n-m}\hat{B}\hat{A}^m \\ &+ \sum_{m=1}^{n} \frac{(n-1)!}{(n-m)!(m-1)!} (-1)^m \hat{A}^{n-m}\hat{B}\hat{A}^m. \end{aligned} \tag{7.291}$$

Now, we perform an operation similar to what we did when working on the power series of the exponential—we convert the pre-factors in both summations into $\frac{(n-1)!}{(n-m)!m!}$ as follows:

$$\sum_{m=0}^{n-1} \frac{(n-1)!(n-m)}{(n-m)!m!}(-1)^m \hat{A}^{n-m}\hat{B}\hat{A}^m + \sum_{m=1}^{n} \frac{(n-1)!m}{(n-m)!m!}(-1)^m \hat{A}^{n-m}\hat{B}\hat{A}^m. \tag{7.292}$$

In the first term, we can extend the upper limit of the summation to $m = n$, because the numerator is proportional to $(n - m) = 0$. Similarly, for the second term, we can extend the summation to $m = 0$, because the numerator has an m in it and the denominator is nonzero, due to $0! = 1$. Hence, we have

$$\begin{aligned} &\sum_{m=0}^{n} \frac{(n-1)!(n-m+m)}{(n-m)!m!}(-1)^m \hat{A}^{n-m}\hat{B}\hat{A}^m \\ &= \sum_{m=0}^{n} \frac{n!}{(n-m)!m!}(-1)^m \hat{A}^{n-m}\hat{B}\hat{A}^m. \end{aligned} \tag{7.293}$$

Putting this back into the right-hand side of Eq. (7.289) and dividing both sides by $n!$ completes the proof by induction. So, we have established the general form of the Hadamard lemma. We can see, when $[\hat{A}, \hat{B}]$ commutes with $\hat{A}$, we obtain the special case we had been using previously. The video Hadamard Lemma https://tinyurl.com/4vvnpe3w provides some alternative ways to derive these formulas that employ calculus.

Squeezed states are constructed by applying the squeezing operator onto the harmonic oscillator ground state. They were created by Kennard in 1927 (Kennard, 1927) but were not used extensively until Glauber emphasized working with them in the 1960s (Glauber, 1965); see Fig. 7.12. If we recall that the displacement operator was the exponential of a linear combination of raising and lowering operators, the squeezing operator is the exponential of a quadratic in the raising and lowering operators. It is defined to be

$$\hat{S}(\xi, \eta) = e^{-\frac{\xi}{2}\left(\hat{a}^\dagger\right)^2 + i\frac{\eta}{2}\left(\hat{a}^\dagger\hat{a} + \hat{a}\hat{a}^\dagger\right) + \frac{\xi^*}{2}\hat{a}^2}, \tag{7.294}$$

where ξ can be a complex number, but η must be real. One can immediately see that $\hat{S}(\xi, \eta)^\dagger = \hat{S}(-\xi, -\eta) = \hat{S}^{-1}(\xi, \eta)$, so $\hat{S}$ is unitary. The last equality comes from the fact that $e^x e^{-x} = 1$ even with operators, allowing us to identify the inverse operator because it is written in the exponential form. Since the squeezing operator is the exponential of a quadratic in the raising and lowering operators, it acts like a change in the oscillator frequency or mass. For both parameters, changing them can make the harmonic oscillator more squeezed toward the origin (e.g., by making the spring constant much larger) or expanded away from the origin (e.g., by making the spring constant much smaller).

Fig. 7.12 Earle Kennard and Roy Glauber in the late 1950s (Kennard image courtesy of AIP Emilio Segré Visual Archives and Glauber image courtesy of AIP Emilio Segrè Visual Archives, Physics Today Collection. Converted to look like a sketch by James Freericks)

The first thing we will do is examine the *unitary transformation*, mediated by the squeezing operator, on the raising and lowering operators. A unitary transformation is a conjugation operation—we multiply the transformed operator by $\hat{U}^\dagger$ from the left and $\hat{U}$ from the right—it is useful, because when we transform a state $|\psi\rangle \to \hat{U}|\psi\rangle$, then the expectation value $\langle\psi|\hat{O}|\psi\rangle \to \langle\psi|\hat{U}^\dagger\hat{O}\hat{U}|\psi\rangle$. This is why the unitary transformation $\hat{O} \to \hat{U}^\dagger\hat{O}\hat{U}$ is important. Because a unitary operator has its Hermitian conjugate equal to its inverse, we can also write the unitary transformation as a similarity transformation, given by $\hat{O} \to \hat{U}^{-1}\hat{O}\hat{U}$. Indeed, we now compute

$$\hat{S}(\xi,\eta)^\dagger\hat{a}\hat{S}(\xi,\eta) = \hat{S}^{-1}(\xi,\eta)\hat{a}\hat{S}(\xi,\eta) = \hat{S}(-\xi,-\eta)\hat{a}\hat{S}(\xi,\eta). \tag{7.295}$$

This last form is the one we want to work with, because it is in the form of the Hadamard lemma. Hence, we compute

$$\hat{S}(-\xi,-\eta)\hat{a}\hat{S}(\xi,\eta) = \sum_{n=0}^{\infty}\frac{1}{n!}[\hat{A},[\hat{A},[\cdots,[\hat{A},\hat{B}],\cdots]]]_n, \tag{7.296}$$

with $\hat{A} = \frac{\xi}{2}\left(\hat{a}^\dagger\right)^2 - i\frac{\eta}{2}\left(\hat{a}^\dagger\hat{a} + \hat{a}\hat{a}^\dagger\right) - \frac{\xi^*}{2}\hat{a}^2$ and $\hat{B} = \hat{a}$. We find

$$\begin{aligned}
[\hat{A},\hat{B}] &= -\xi\hat{a}^\dagger + i\eta\hat{a},\\
[\hat{A},[\hat{A},\hat{B}]] &= \left(|\xi|^2 - \eta^2\right)\hat{a},\\
[\hat{A},[\hat{A},[\hat{A},\hat{B}]]] &= \left(|\xi|^2 - \eta^2\right)(-\xi\hat{a}^\dagger + i\eta\hat{a}),
\end{aligned}$$

$$[\hat{A},[\hat{A},[\hat{A},[\hat{A},\hat{B}]]]] = \left(|\xi|^2-\eta^2\right)^2\hat{a}, \tag{7.297}$$

and so on. The Hadamard lemma requires an infinite number of terms, but they repeat in a pattern. We can see $\hat{a}^\dagger$ has coefficients in the power series equal to $\xi\left(|\xi|^2-\eta^2\right)^m$ for odd $n = 2m+1$, while $\hat{a}$ has a power series with coefficients $\left(|\xi|^2-\eta^2\right)^m$ for even $n = 2m$ and $i\eta\left(|\xi|^2-\eta^2\right)^m$ for odd $n = 2m+1$.

Before we can write down the answer, we need to review the definition of the hyperbolic trigonometric functions. In particular, $\cosh x = \frac{e^x+e^{-x}}{2}$ and $\sinh x = \frac{e^x-e^{-x}}{2}$. Using the power series for e^x, we find that $\cosh x = \sum_{m=0}^{\infty}\frac{x^{2m}}{(2m)!}$ and $\sinh x = \sum_{m=0}^{\infty}\frac{x^{2m+1}}{(2m+1)!}$, which looks just like the cosine and sine series, but without the $(-1)^m$ terms. So, we find that

$$\begin{aligned}\hat{S}(-\xi,-\eta)\hat{a}\hat{S}(\xi,\eta) = &\left(\cosh\sqrt{|\xi|^2-\eta^2}+\frac{i\eta}{\sqrt{|\xi|^2-\eta^2}}\sinh\sqrt{|\xi|^2-\eta^2}\right)\hat{a}\\ &-\frac{\xi}{\sqrt{|\xi|^2-\eta^2}}\sinh\sqrt{|\xi|^2-\eta^2}\,\hat{a}^\dagger.\end{aligned} \tag{7.298}$$

Taking the Hermitian conjugate gives us the transformation of the raising operator:

$$\begin{aligned}\hat{S}(-\xi,-\eta)\hat{a}^\dagger\hat{S}(\xi,\eta) = &-\frac{\xi^*}{\sqrt{|\xi|^2-\eta^2}}\sinh\sqrt{|\xi|^2-\eta^2}\,\hat{a}\\ &\left(\cosh\sqrt{|\xi|^2-\eta^2}-\frac{i\eta}{\sqrt{|\xi|^2-\eta^2}}\sinh\sqrt{|\xi|^2-\eta^2}\right)\hat{a}^\dagger.\end{aligned} \tag{7.299}$$

In this calculation, we recall that the Hermitian conjugate of the product of three operators requires us to reverse the order, so $(\hat{A}\hat{B}\hat{C})^\dagger = \hat{C}^\dagger\hat{B}^\dagger\hat{A}^\dagger$; the derivation of this is done in Problem 7.11.6. This transformation resembles closely similar similarity transformations of Pauli spin matrices, except the trigonometric functions are replaced by hyperbolic functions ($\sin \to \sinh$ and $\cos \to \cosh$) and so on. This is not just a coincidence. We will see that we can use similar Lie algebraic methods for identities of the squeezing operator.

Before doing anything else, it is now time to calculate the uncertainty in the squeezed vacuum state, defined by

$$|\xi,\eta\rangle = \hat{S}(\xi,\eta)|0\rangle. \tag{7.300}$$

We calculate the uncertainty by recognizing that the expectation values correspond to the expectation value of the unitarily transformed operators, which can be expressed in terms of the raising and lowering operators. In particular, we have that

$$\langle \xi, \eta | \hat{x} | \xi, \eta \rangle = \sqrt{\frac{\hbar}{2m\omega}} \langle 0 | \hat{S}(\xi, \eta)^\dagger \hat{x} \hat{S}(\xi, \eta) | 0 \rangle. \tag{7.301}$$

Note that the squeezed vacuum, like the displaced vacuum, is no longer an eigenstate of the harmonic oscillator Hamiltonian—it is not a state with definite energy. We let

$$\kappa = \cosh \sqrt{|\xi|^2 - \eta^2} + i\eta \frac{\sinh \sqrt{|\xi|^2 - \eta^2}}{\sqrt{|\xi|^2 - \eta^2}} \text{ and } \lambda = \xi \frac{\sinh \sqrt{|\xi|^2 - \eta^2}}{\sqrt{|\xi|^2 - \eta^2}}. \tag{7.302}$$

Then, we have $\hat{S}(\xi, \eta)^\dagger \hat{a} \hat{S}(\xi, \eta) = \kappa \hat{a} - \lambda \hat{a}^\dagger$ and $\hat{S}(\xi, \eta)^\dagger \hat{a}^\dagger \hat{S}(\xi, \eta) = -\lambda^* \hat{a} + \kappa^* \hat{a}^\dagger$. Plugging into the above equation yields

$$\langle \xi, \eta | \hat{x} | \xi, \eta \rangle = \sqrt{\frac{\hbar}{2m\omega}} \langle 0 | \kappa \hat{a} - \lambda \hat{a}^\dagger - \lambda^* \hat{a} + \kappa^* \hat{a}^\dagger | 0 \rangle = 0, \tag{7.303}$$

because $\hat{a}|0\rangle = 0$ and $\langle 0 | \hat{a}^\dagger = 0$. Similarly, we have

$$\begin{aligned}
\langle \xi, \eta | \hat{x}^2 | \xi, \eta \rangle &= \frac{\hbar}{2m\omega} \langle 0 | \left\{ (\kappa - \lambda^*) \hat{a} + (\kappa^* - \lambda) \hat{a}^\dagger \right\}^2 | 0 \rangle \\
&= \frac{\hbar}{2m\omega} \langle 0 | (\kappa - \lambda^*)^2 \hat{a}^2 + |\kappa - \lambda^*|^2 (\hat{a}\hat{a}^\dagger + \hat{a}^\dagger \hat{a}) + (\kappa^* - \lambda)^2 (\hat{a}^\dagger)^2 | 0 \rangle \\
&= \frac{\hbar}{2m\omega} |\kappa - \lambda^*|^2 \langle 0 | \big(\hat{a}\hat{a}^\dagger + \hat{a}^\dagger \hat{a} \big) | 0 \rangle \\
&= \frac{\hbar}{2m\omega} \big(|\kappa|^2 - 2\mathrm{Re}(\kappa\lambda) + |\lambda|^2 \big).
\end{aligned} \tag{7.304}$$

In Problem 7.11.11, you will show that $\langle \xi, \eta | \hat{p} | \xi, \eta \rangle = 0$ and

$$\langle \xi, \eta | \hat{p}^2 | \xi, \eta \rangle = \frac{\hbar}{2m\omega} \big(|\kappa|^2 + 2\mathrm{Re}(\kappa\lambda) + |\lambda|^2 \big). \tag{7.305}$$

Let's analyze in detail for $\eta = 0$ and $\xi = re^{i\phi}$. Then, $\kappa = \cosh r$ and $\lambda = e^{i\phi} \sinh r$. The uncertainties become

$$\begin{aligned}
(\Delta x)^2_{re^{i\phi}, \eta = 0} &= \frac{\hbar}{2m\omega} (\cosh^2 r + \sinh^2 r - 2 \cos\phi \cosh r \sinh r) \\
&= \frac{\hbar}{2m\omega} (\cosh 2r - \cos\phi \sinh 2r),
\end{aligned} \tag{7.306}$$

after using the hyperbolic "double angle" identities that are worked out in Problem 7.11.10. We also find that

$$(\Delta p)^2_{re^{i\phi}, \eta = 0} = \frac{\hbar m\omega}{2} (\cosh 2r + \cos\phi \sinh 2r). \tag{7.307}$$

Now, choose $\phi = 0$ and use the fact that $\cosh 2r \pm \sinh 2r = e^{\pm 2r}$ to find that

$$(\Delta x)^2_{re^{i\phi},\eta=0} = \frac{\hbar}{2m\omega} e^{-2r} \quad \text{and} \quad (\Delta p)^2_{re^{i\phi},\eta=0} = \frac{\hbar m\omega}{2} e^{2r}. \tag{7.308}$$

So, the position uncertainty is squeezed and the momentum uncertainty is enhanced for $r > 0$, but the product of the two remains minimal uncertainty. Let $\phi = \pi$, and the situation reverses—the position uncertainty is enhanced and the momentum uncertainty is squeezed, but the product remains minimal uncertainty. This ends up being one of the most important properties of squeezed states—they can reduce the uncertainty in the measurement of one quantity, at the expense of a higher uncertainty in the conjugate quantity—if you desire to measure just one of the two conjugate quantities, this can lead to a huge advantage arising solely from quantum-mechanical behavior.

It is an interesting exercise to plot these results in phase space. This corresponds to a plot with scaled position horizontal and scaled momentum vertical as shown in Fig. 7.13—we take $x \to \sqrt{m\omega}x$ and $p \to p/\sqrt{m\omega}$. We define two new operators:

$$\hat{x}_\theta = \cos\theta\sqrt{m\omega}\hat{x} + \sin\theta\frac{\hat{p}}{\sqrt{m\omega}} \quad \text{and} \quad \hat{p}_\theta = -\sin\theta\sqrt{m\omega}\hat{x} + \cos\theta\frac{\hat{p}}{\sqrt{m\omega}}. \tag{7.309}$$

These operators exchange the roles of position and momentum as θ changes from 0 to 2π. Because these operators now have the same dimension, we can compute uncertainty for these operators at any angle θ. We find, for a coherent state, the uncertainty does not depend on the angle θ, so a plot of the uncertainty, centered about the average value of the scaled position and momentum, is just a circle, shown in the left panel of Fig. 7.13. On the other hand, if we look at a general squeezed

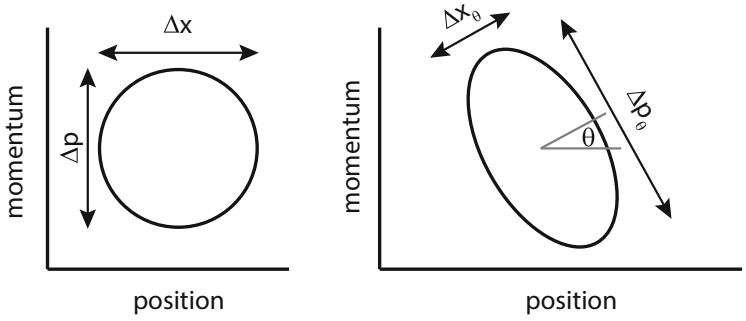

Fig. 7.13 Uncertainty relations for a coherent state (left) and for a squeezed state (right). In this plot, we work with normalized position and momentum, where $x \to \sqrt{m\omega}x$ and $p \to p/\sqrt{m\omega}$. In these units, the coherent state has $\Delta x = \sqrt{\frac{\hbar}{2}}$ and $\Delta p = \sqrt{\frac{\hbar}{2}}$, while the squeezed state has $\Delta x_\theta = \sqrt{\frac{\hbar}{2}}e^{-r}$ and $\Delta p_\theta = \sqrt{\frac{\hbar}{2}}e^{r}$

state, because the uncertainty scales by e^{-r} for one operator and e^{r} for the conjugate operator, we distort the circle into an ellipse (of the same area) that can have its major axis at any angle (depending on the properties of the squeezed state), which is illustrated in the right panel of Fig. 7.13. What this says is for a squeezed state, calculating the uncertainty at different θ does not always give the same amount, and in general, both uncertainties are larger than that of a coherent state—only when the angle is close to the major or minor axes do we find a suppression of one of the uncertainties.

When we discussed the Pauli spin matrices, we had a problem (6.3.10) that showed how one can factorize the exponential of one of the Pauli matrices into the product of an exponential of the raising matrix, the z-matrix, and the lowering matrix. This identity is a special case of the *exponential disentangling identity* (Arechi et al., 1972). The exponential disentangling identity is our fourth and final fundamental operator identity. We will work it out for the symplectic group here. We will also be working with it for the SU(2) group as well—that discussion will be in Chap. 8. The way that we proceed involves a few steps. First, we will compute the operator algebra for three operators defined as quadratic operators of the raising and lowering operators. Second, we will find a set of 2×2 matrices that satisfy the same algebra. Third, we will compute the exponential of these 2×2 matrices exactly and then factorize the exact result in terms of the product of three exponentials. Finally, we elevate this identity, which was derived for 2×2 matrices only, into an identity for the operators. This last step relies on there being a Lie algebra and Lie group structure behind our identities. It only works because the 2×2 matrices form what is called a *faithful representation*. We will explain what this means and why it works once we have completed the derivation.

To start, we need to define our operators; we have

$$\hat{K}_+ = \frac{1}{2}\left(\hat{a}^\dagger\right)^2, \quad \hat{K}_- = \frac{1}{2}\hat{a}^2, \quad \text{and} \quad \hat{K}_0 = \frac{1}{4}\left(\hat{a}^\dagger\hat{a} + \hat{a}\hat{a}^\dagger\right). \tag{7.310}$$

Note that $\hat{K}_+^\dagger = \hat{K}_-$ and $\hat{K}_0^\dagger = \hat{K}_0$. Let's compute some commutators! This requires only the Leibniz rule and the fact that $[\hat{a}, \hat{a}^\dagger] = 1$. We find that

$$[\hat{K}_0, \hat{K}_+] = \frac{1}{8}\hat{a}^\dagger\left[\hat{a}, \left(\hat{a}^\dagger\right)^2\right] + \frac{1}{8}\left[\hat{a}, \left(\hat{a}^\dagger\right)^2\right]\hat{a}^\dagger = \frac{1}{2}\left(\hat{a}^\dagger\right)^2 = \hat{K}_+. \tag{7.311}$$

Note that the Hermitian conjugate of a commutator satisfies

$$[\hat{A}, \hat{B}]^\dagger = (\hat{A}\hat{B} - \hat{B}\hat{A})^\dagger = \hat{B}^\dagger\hat{A}^\dagger - \hat{A}^\dagger\hat{B}^\dagger = [\hat{B}^\dagger, \hat{A}^\dagger] = -[\hat{A}^\dagger, \hat{B}^\dagger]. \tag{7.312}$$

That is, when we take the Hermitian conjugate of a commutator, we reverse the order of the operators and take their Hermitian conjugate. We use this to immediately see that

$$[\hat{K}_0, \hat{K}_+]^\dagger = \hat{K}_+^\dagger \quad \Longrightarrow \quad [\hat{K}_0, \hat{K}_-] = -\hat{K}_-. \tag{7.313}$$

Finally, we have that

$$[\hat{K}_+, \hat{K}_-] = \frac{1}{4}[(\hat{a}^\dagger)^2, \hat{a}^2] = \frac{1}{4}\hat{a}[(\hat{a}^\dagger)^2, \hat{a}] + \frac{1}{4}[(\hat{a}^\dagger)^2, \hat{a}]\hat{a} = -2\hat{K}_0. \tag{7.314}$$

In summary, we have

$$[\hat{K}_0, \hat{K}_\pm] = \pm\hat{K}_\pm \text{ and } [\hat{K}_+, \hat{K}_-] = -2\hat{K}_0. \tag{7.315}$$

These commutation relations look almost identical to the spin commutation relations, given by $[\hat{S}_z, \hat{S}_\pm] = \pm\hbar\hat{S}_\pm$ and $[\hat{S}_+, \hat{S}_-] = 2\hbar\hat{S}_z$. Aside from the $\hbar$, the difference arises from the changed sign in front of the commutator between the raising and lowering operator. This change is what makes these commutators the commutators of the symplectic group rather than the special unitary group. We will not define the symplectic group in general, but its operators satisfy the above algebra.

Now that we have identified the algebra between the operators, we need to represent them in terms of 2×2 matrices. We find that

$$K_0 = \frac{1}{2}\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad K_+ = \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix}, \quad \text{and} \quad K_- = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \tag{7.316}$$

are three matrices that satisfy the symplectic group commutator algebra. You will check this via matrix multiplication in Problem 7.11.9, but as a quick check, note that

$$K_0K_+ - K_+K_0 = \frac{1}{2}\begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix} - \frac{1}{2}\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix} = K_+ \checkmark. \tag{7.317}$$

Using the relationship between the $\hat{K}$ operators and the K matrices, we see we can compute the representation of the squeezing operator for the 2×2 matrices via

$$S(\xi, \eta) = e^{-\xi K_+ + 2i\eta K_0 + \xi^* K_-} = \exp\begin{pmatrix} -i\eta & \xi^* \\ \xi & i\eta \end{pmatrix}. \tag{7.318}$$

This exponential can be computed by using the power series for the exponential. We compute the first few powers of the matrix:

$$\begin{aligned}
\begin{pmatrix} -i\eta & \xi^* \\ \xi & i\eta \end{pmatrix}^2 &= \begin{pmatrix} |\xi|^2 - \eta^2 & 0 \\ 0 & |\xi|^2 - \eta^2 \end{pmatrix} = (|\xi|^2 - \eta^2)\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\
\begin{pmatrix} -i\eta & \xi^* \\ \xi & i\eta \end{pmatrix}^3 &= (|\xi|^2 - \eta^2)\begin{pmatrix} -i\eta & \xi^* \\ \xi & i\eta \end{pmatrix} \\
\begin{pmatrix} -i\eta & \xi^* \\ \xi & i\eta \end{pmatrix}^4 &= (|\xi|^2 - \eta^2)^2\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.
\end{aligned} \tag{7.319}$$

This behavior is the same as what we saw when we computed the unitary transformation of the raising and lowering operators. The pattern is easy to identify, and then summing the series gives us

$$S(\xi,\eta) = \cosh\sqrt{|\xi|^2-\eta^2}\,\mathbb{I} + \frac{2i\eta}{\sqrt{|\xi|^2-\eta^2}}\sinh\sqrt{|\xi|^2-\eta^2}\,K_0 - \frac{\xi}{\sqrt{|\xi|^2-\eta^2}}\sinh\sqrt{|\xi|^2-\eta^2}\,K_+ + \frac{\xi^*}{\sqrt{|\xi|^2-\eta^2}}\sinh\sqrt{|\xi|^2-\eta^2}\,K_-, \tag{7.320}$$

which is an exact result for the 2×2 matrices. The exponential disentangling identity uses exponentials of each of the symplectic matrices. One can immediately see that

$$e^{aK_+} = \begin{pmatrix} 1 & 0 \\ -a & 1 \end{pmatrix}, \quad e^{bK_0} = \begin{pmatrix} e^{-\frac{b}{2}} & 0 \\ 0 & e^{\frac{b}{2}} \end{pmatrix}, \quad \text{and} \quad e^{cK_-} = \begin{pmatrix} 1 & c \\ 0 & 1 \end{pmatrix}. \tag{7.321}$$

We first compute a disentangling identity with the exponentials ordered as K_+, K_0, and K_-. Multiplying the three matrices

$$\begin{pmatrix} 1 & 0 \\ -a & 1 \end{pmatrix}\begin{pmatrix} e^{-\frac{b}{2}} & 0 \\ 0 & e^{\frac{b}{2}} \end{pmatrix}\begin{pmatrix} 1 & c \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} e^{-\frac{b}{2}} & ce^{-\frac{b}{2}} \\ -ae^{-\frac{b}{2}} & -ace^{-\frac{b}{2}} + e^{\frac{b}{2}} \end{pmatrix}. \tag{7.322}$$

Now, we equate matrix elements to find that $e^{-\frac{b}{2}} = \kappa^*$, $a = -\frac{\lambda}{\kappa^*}$, and $c = \frac{\lambda^*}{\kappa^*}$, using the definitions in Eq. (7.302). One can verify that the lower right element satisfies

$$-ace^{-\frac{b}{2}} + e^{\frac{b}{2}} = \frac{|\lambda|^2+1}{\kappa^*} = \frac{|\kappa|^2}{\kappa^*} = \kappa \ \checkmark. \tag{7.323}$$

So, the factorization works. This means we have verified that

$$e^{-\xi K_+ + 2i\eta K_0 + \xi^* K_-} = e^{-\frac{\lambda}{\kappa^*}K_+} e^{-2\ln(\kappa^*)K_0} e^{\frac{\lambda^*}{\kappa^*}K_-}, \tag{7.324}$$

which is our first exponential disentangling identity. In Problem 7.11.14, you will verify the identity in the opposite order, namely,

$$e^{-\xi K_+ + 2i\eta K_0 + \xi^* K_-} = e^{\frac{\lambda^*}{\kappa}K_-} e^{2\ln(\kappa)K_0} e^{-\frac{\lambda}{\kappa}K_+}. \tag{7.325}$$

Now, we use the group theory. Because the 2×2 matrices are a faithful representation of the symplectic group, which means each matrix representation maps uniquely onto one and only one group element. Then, a powerful theorem from Lie algebra tells us that the identity holds when we replace the 2×2 matrices by the operators. So, we have

$$e^{-\frac{\xi}{2}\left(\hat{a}^\dagger\right)^2+\frac{i}{2}\eta\left(\hat{a}^\dagger\hat{a}+\hat{a}\hat{a}^\dagger\right)+\frac{\xi^*}{2}\hat{a}^2} = e^{-\frac{\lambda}{2\kappa^*}\left(\hat{a}^\dagger\right)^2}e^{-\frac{1}{2}\ln(\kappa^*)\left(\hat{a}^\dagger\hat{a}+\hat{a}\hat{a}^\dagger\right)}e^{\frac{\lambda^*}{2\kappa^*}\hat{a}^2}, \tag{7.326}$$

$$= e^{\frac{\lambda^*}{2\kappa}\hat{a}^2}e^{\frac{1}{2}\ln(\kappa)\left(\hat{a}^\dagger\hat{a}+\hat{a}\hat{a}^\dagger\right)}e^{-\frac{\lambda}{2\kappa}\left(\hat{a}^\dagger\right)^2}, \tag{7.327}$$

which is the operator form of the two symplectic disentangling identities.

The disentangling identity lets us find alternative expressions for the squeezed state, which allow us to determine the overlap of two squeezed states, among other things. We work this out for a simplified case of the squeezing operator—the case $\xi = re^{i\phi}$ and $\eta = 0$, which corresponds to $\kappa = \cosh r$ and $\lambda = e^{i\phi}\sinh r$. We start by examining a simpler operator form of the squeezed state. We have

$$\begin{aligned}\hat{S}(\xi{=}re^{i\phi},\,\eta{=}0)|0\rangle &= e^{-\frac{re^{i\phi}}{2}\left(\hat{a}^\dagger\right)^2+\frac{re^{-i\phi}}{2}\hat{a}^2}|0\rangle \\ &= e^{-\frac{e^{i\phi}\tanh r}{2}\left(\hat{a}^\dagger\right)^2}e^{-\frac{1}{2}\ln(\cosh r)\left(\hat{a}^\dagger\hat{a}+\hat{a}\hat{a}^\dagger\right)}e^{\frac{e^{-i\phi}\tanh r}{2}\hat{a}^2}|0\rangle \\ &= \frac{1}{\sqrt{\cosh r}}e^{-\frac{e^{i\phi}\tanh r}{2}\left(\hat{a}^\dagger\right)^2}|0\rangle,\end{aligned} \tag{7.328}$$

since $\hat{a}|0\rangle = 0$. We can use the power series for the exponential to rewrite this expression in terms of the energy eigenstates:

$$\hat{S}(\xi{=}re^{i\phi},\,\eta{=}0)|0\rangle = \sum_{n=0}^{\infty}\left(-\frac{1}{2}e^{i\phi}\tanh r\right)^n\frac{\sqrt{(2n)!}}{n!}|2n\rangle. \tag{7.329}$$

Note that only even energy eigenstates appear in the sum. This should be a hint that creating such a state might be challenging, as we have to ensure the amplitude for every odd energy eigenstate is exactly zero. This form can now be used to find the overlap between two different squeezed states. We have

$$\begin{aligned}\langle\xi = r'e^{i\phi'},\,\eta{=}0|\xi = re^{i\phi},\,\eta{=}0\rangle &= \frac{1}{\sqrt{\cosh r'\cosh r}} \\ &\times\sum_{m=0}^{\infty}\sum_{n=0}^{\infty}\left(-\frac{1}{2}e^{-i\phi'}\tanh r'\right)^m\left(-\frac{1}{2}e^{i\phi}\tanh r\right)^n\frac{\sqrt{(2m)!}\sqrt{(2n)!}}{m!n!}\langle 2m|2n\rangle \\ &= \frac{1}{\sqrt{\cosh r\cosh r'}}\sum_{n=0}^{\infty}\left(\frac{1}{4}e^{i\phi-i\phi'}\tanh r\tanh r'\right)^n\frac{(2n)!}{(n!)^2}.\end{aligned} \tag{7.330}$$

The exponential disentangling identity is a very powerful identity. Most people studying quantum optics have not see it, so they would have significant difficulty in computing the overlap between two squeezed states; in fact, only a handful of quantum optics textbooks discuss the identity. You will see we can do wonderful things with it as we use it again and again. The lecture Squeezed States https://tinyurl.com/c98vhr2v provides some additional discussions on this topic.

We have studied squeezed states and coherent states; can we combine them? Yes, we can. The combination is called a squeezed-coherent state, and it applies the squeezing operator first and then the displacement operator second (the two different orderings are easily related by a Hadamard lemma, as you will show in Problem 7.11.17). So, we have the most general squeezed-coherent state is

$$\hat{D}(\alpha)\hat{S}(\xi,\eta)|0\rangle. \tag{7.331}$$

One can compute the uncertainty relations for such states, but we will not do that now.

Section 7.11 Problems:

7.11.1 The Displacement Operator Is Unitary

By taking the Hermitian conjugate, show that $\hat{D}(\alpha)$ is a unitary operator or that $\hat{D}(\alpha)^\dagger = \hat{D}(\alpha)^{-1}$.

7.11.2 Coherent State Is Not an Eigenvector of the Raising Operator

Modify the argument given in Eq. (7.273) to examine whether the coherent state is an eigenstate of the raising operator. After performing similar operations, you should find that

$$\hat{a}^\dagger|\alpha\rangle = \alpha^*|\alpha\rangle + \text{ another state.} \tag{7.332}$$

Determine what that other state is. Because it is nonzero, the coherent state is not an eigenstate of the raising operator for any α.

7.11.3 No State Is the Eigenvector of the Raising Operator

In this problem, we will work out why the raising operator has no eigenvector. To start, let's try to form one. We want to find $|\beta\rangle$ such that $\hat{a}^\dagger|\beta\rangle = \beta|\beta\rangle$. Let $|\beta\rangle = \sum_{n=0}^{\infty} c_n|n\rangle$. Show that an eigenvector requires

$$c_1 = \frac{\sqrt{1}}{\beta}c_0, \quad c_2 = \frac{\sqrt{1\cdot 2}}{\beta^2}c_0, \quad \cdots \quad c_n = \frac{\sqrt{n!}}{\beta^n}c_0. \tag{7.333}$$

Normalization then would require $1 = \sum_{n=0}^{\infty} |c_n|^2$, which implies that

$$c_0 = \frac{1}{\sqrt{\sum_{n=0}^{\infty} \frac{n!}{\beta^{2n}}}}. \tag{7.334}$$

Show that this is equal to 0, because $n!$ always grows faster than β^{2n}, as n becomes large. If c_0 is required to vanish, then we do not have a normalized state. So, there is no eigenfunction.

7.11.4 *Normalization, Overlap, and Eigenstate of the Lowering Operator via the Expansion in Terms of Energy Eigenstates*

(a) Show that $|\alpha\rangle$ is normalized by computing

$$\langle\alpha|\alpha\rangle = e^{-|\alpha|^2} \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \frac{\alpha^{*m}\alpha^n}{\sqrt{m!n!}} \langle m|n\rangle. \tag{7.335}$$

Use the fact that the energy eigenstates form an orthonormal set to verify that the coherent state is normalized.

(b) Verify that the overlap of two coherent states satisfies

$$\langle\beta|\alpha\rangle = e^{-\frac{1}{2}|\alpha|^2 + \beta^*\alpha - \frac{1}{2}|\beta|^2}. \tag{7.336}$$

This result shows that coherent states with different displacement parameters are never orthogonal, but the larger the difference between α and β, the closer to orthogonal they become.

(c) Use the expansion in terms of energy eigenstates to verify that $\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$ and that $|\alpha\rangle$ is not an eigenstate of $\hat{a}^\dagger$. For the lowering operator, start by examining

$$\hat{a}|\alpha\rangle = \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} \hat{a}|n\rangle. \tag{7.337}$$

Recall that $\hat{a}|n\rangle = \sqrt{n}|n-1\rangle$ to complete your argument. Be sure to explain clearly why one operator yields an eigenstate, while the other cannot.

7.11.5 Bra Eigenstate via the BCH Formula

Use the Baker-Campbell-Hausdorff formula to show that $\langle\alpha|\hat{a}^\dagger = \langle\alpha|\alpha^*$. Pick the ordering of the two exponentials such that the exponential of $\hat{a}^\dagger$ is on the left and recall that $\langle 0|\hat{a}^\dagger = 0$. Then, use the Hadamard lemma to derive the final result. Do not just take the Hermitian conjugate of $\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$.

7.11.6 Hermitian Conjugate of a Product of Operators

We will work this problem out first for matrices. So far, we have focused on just working with 2×2 matrices, but many results take essentially the same form for $n \times n$ matrices. For example, the ik matrix element of the product of two $n \times n$ matrices A and B is given by

$$(AB)_{ik} = \sum_{j=1}^{n} A_{ij} B_{jk}. \tag{7.338}$$

Make sure this equation is precisely what we used for 2×2 matrices, and be sure you are clear that it uses the same idea of the dot product of a row vector with a column vector to form each matrix element of the product.

Using the fact that the Hermitian conjugate of a matrix is the complex conjugate transpose, or $(M^\dagger)_{ij} = M^*_{ji}$, prove that

$$(AB)^\dagger_{ij} = \sum_{j=1}^{n} B^*_{ji} A^*_{kj} = \sum_{j=1}^{n} (B^\dagger)_{ij} (A^\dagger)_{jk} = (B^\dagger A^\dagger)_{ik}. \tag{7.339}$$

This relation, which holds for $n \times n$ matrices, also holds for operators, so we learn that $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger \hat{A}^\dagger$. This leads to the rule for taking the Hermitian conjugate of a product of operators, which is to take the Hermitian conjugate of each term, and then invert the order of the terms in the final product.

7.11.7 Uncertainty of Momentum in a Coherent State

Complete the calculation started in the text, and show that the uncertainty in momentum is $(\Delta p)^2_\alpha = \frac{\hbar m \omega}{2}$ in the coherent state $|\alpha\rangle$.

7.11.8 Wavefunction of a Coherent State

The wavefunction of a coherent state in coordinate space is defined to be

$$\psi_\alpha(x) = \langle x|\alpha\rangle = \langle x|\hat{D}(\alpha)|0\rangle. \tag{7.340}$$

We find the wavefunction by using the BCH formula. The simplest way to do this is to write

$$\alpha\hat{a}^\dagger - \alpha^*\hat{a} = \underbrace{\alpha(\hat{a}+\hat{a}^\dagger)}_{\hat{A}} - \underbrace{(\alpha+\alpha^*)\hat{a}}_{\hat{B}}, \tag{7.341}$$

and use the BCH formula in terms of the $\hat{A}$ and $\hat{B}$ operators. You should find that $\hat{B}$ annihilates against the harmonic oscillator ground state, while $\hat{a}+\hat{a}^\dagger = \sqrt{\frac{2m\omega}{\hbar}}\hat{x}$ yields $\sqrt{\frac{2m\omega}{\hbar}}x$ when it acts on $\langle x|$ from the right.

This should allow you to compute the wavefunction of the coherent state. Note that the symbol x is overloaded, being used as the coordinate for the wavefunction and being used as the displacement in the coherent state, so write $\alpha = \sqrt{\frac{m\omega}{2\hbar}}x_0 + \frac{i}{\sqrt{2\hbar m\omega}}p_0$, to resolve this issue. Replace α in terms of x_0 and p_0 and simplify your final answer in terms of the position and momentum shifts in the coherent state. You should find the final result is easy to understand why it has the form it has.

7.11.9 Symplectic Group Algebra

Complete the derivation of the commutation relations for the symplectic algebra in terms of the 2×2 matrices.

There is a 3×3 matrix representation of the symplectic algebra as well. It is

$$K_+ = \begin{pmatrix} 0 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix}, \quad K_- = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad \text{and} \quad K_0 = \frac{1}{2}\begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{7.342}$$

Verify that these matrices do satisfy the symplectic algebra.

You should find a pattern develop as well when you look at powers of these matrices, in such a way that the powers take repeating forms, multiplied by numbers. This allows you to compute both sides of the exponential disentangling identity. Do this, and verify, via matrix multiplication that the identity holds. Note that it holds because of the group theory theorem, because in this case, one does not have $(K_+)^2 = 0$ as we had for the 2×2 matrix representation.

7.11.10 Hyperbolic Cosine and Sine Identities

Using $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ and $e^{-x} = \sum_{n=0}^{\infty} (-1)^n \frac{x^n}{n!}$, show that

$$\cosh x = \frac{e^x + e^{-x}}{2} = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}, \tag{7.343}$$

and

$$\sinh x = \frac{e^x - e^{-x}}{2} = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}. \tag{7.344}$$

Use the definition of the hyperbolic functions in terms of the exponential functions to show that $\cosh^2 x - \sinh^2 x = 1$, $\cosh^2 x + \sinh^2 x = \cosh 2x$, and $2 \cosh x \sinh x = \sinh 2x$.

7.11.11 Uncertainty of Momentum in the Ground State

Using the methods developed in the main text, and setting $\xi = re^{i\phi}$ and $\eta = 0$, compute the uncertainty in momentum, given by

$$(\Delta p)^2_{re^{i\phi},\eta=0}\Big|_{\text{gs}} = \langle 0|\hat{S}(re^{i\phi}, 0)^\dagger \hat{p}^2 \hat{S}(re^{i\phi}, 0)|0\rangle - \Big(\langle 0|\hat{S}(re^{i\phi}, 0)^\dagger \hat{p} \hat{S}(re^{i\phi}, 0)|0\rangle\Big)^2. \tag{7.345}$$

7.11.12 Uncertainties in an Excited Squeezed State

Generalize the previous problem to an arbitrary energy state. Compute both uncertainties, determined by

$$(\Delta x)^2_{re^{i\phi},\eta=0}\Big|_n = \langle n|\hat{S}(re^{i\phi}, 0)^\dagger \hat{x}^2 \hat{S}(re^{i\phi}, 0)|n\rangle - \Big(\langle n|\hat{S}(re^{i\phi}, 0)^\dagger \hat{x} \hat{S}(re^{i\phi}, 0)|n\rangle\Big)^2. \tag{7.346}$$

and

$$(\Delta p)^2_{re^{i\phi},\eta=0}\Big|_n = \langle n|\hat{S}(re^{i\phi}, 0)^\dagger \hat{p}^2 \hat{S}(re^{i\phi}, 0)|n\rangle - \Big(\langle n|\hat{S}(re^{i\phi}, 0)^\dagger \hat{p} \hat{S}(re^{i\phi}, 0)|n\rangle\Big)^2. \tag{7.347}$$

Comment on why the results come out the way they do, based on our earlier work on uncertainty in the unsqueezed states.

Hint: calculate the unitary transformation of the position or momentum operator (and their powers) by the squeezing operator first, and then compute the expectation value.

7.11.13 Operator That Annihilates a Squeezed State

Show that

$$\left(\kappa\hat{a} + \lambda\hat{a}^{\dagger}\right)\hat{S}(\xi, \eta{=}0)|0\rangle = 0. \tag{7.348}$$

Hint: use the multiply-by-one trick, in the form $\hat{S}\hat{S}^{\dagger} = \hat{\mathbb{I}}$ and the value we already computed for $\hat{S}^{\dagger}\hat{a}\hat{S}$, appropriately.

7.11.14 Second Exponential Disentangling Identity

Verify the formula in Eq. (7.325) by working with our 2×2 matrix methods.

7.11.15 Exponential Disentangling Identity for a Special Limit of the Squeezing Operator

Exponentials of $\hat{x}^2$ and $\hat{p}^2$ have particularly simple exponential disentangling identities. Verify that

$$e^{ik\left(\hat{a}^{\dagger}\pm\hat{a}\right)^2} = e^{\frac{ik}{1\mp 2ik}\left(\hat{a}^{\dagger}\right)^2} e^{-\frac{1}{2}\ln(1\mp 2ik)\left(\hat{a}^{\dagger}\hat{a}+\hat{a}\hat{a}^{\dagger}\right)} e^{\frac{ik}{1\mp 2ik}\hat{a}^2}. \tag{7.349}$$

7.11.16 Wavefunction of a Simplified Squeezed State

The wavefunction of an $\eta = 0$ squeezed state takes the form

$$\psi_{\xi,0}(x) = \langle x|e^{-\frac{\xi}{2}\left(\hat{a}^{\dagger}\right)^2 + \frac{\xi^*}{2}\hat{a}^2}|0\rangle. \tag{7.350}$$

First, use our exponential disentangling identity with the exponential of $\hat{a}^2$ on the right and operate on the harmonic ground state. You should now have an exponential

of the square of the raising operator sandwiched between the $\langle x|$ bra and the $|0\rangle$ ket. We can determine this expectation value if we can convert the exponential of the square of a raising operator into the exponential of a position operator squared. This is done in the previous problem. Use that identity to complete this problem and determine the squeezed-state wavefunction. Note that you will need the add-zero identity $\hat{a}\hat{a}^\dagger = \hat{a}^\dagger\hat{a} + 1$ in your derivation.

7.11.17 Order of the Operators in a Squeezed-Coherent State

Use a multiply by one and the Hadamard lemma to compare the squeezed-coherent state with the state defined by a displacement first, followed by a squeezing. This is easiest to carry out by noting that the braiding relation allows the Hadamard to enter all the way inside the arguments of the functions, that you just need to work out the Hadamard lemma on the raising and lowering operators only; it even goes inside the powers of the arguments.

7.11.18 Squeezing with the Virial

Consider the unitary operator:

$$\hat{U}_r = e^{-i\frac{r}{\hbar}(\hat{x}\hat{p}+\hat{p}\hat{x})}, \tag{7.351}$$

which is an exponential of the virial operator. It satisfies

$$\hat{U}_r^\dagger \hat{x}\hat{U}_r = \hat{x}e^r \quad \text{and} \quad \hat{U}_r^\dagger \hat{p}\hat{U}_r = \hat{p}e^{-r}. \tag{7.352}$$

First, express the position and momentum operators in terms of the raising and lowering operators. Does the $\hat{U}_r$ have a representation as a squeezing operator? If so, determine what ξ and η are in terms of r. Second, verify Eq. (7.352). Third, compute the uncertainties in position and momentum for the ground state.

7.11.19 Revisiting the Translation Operators of Position and Momentum

Earlier, we argued that the translation operator for position was $e^{-\frac{i}{\hbar}x\hat{p}}$ and for momentum it was $e^{\frac{i}{\hbar}p\hat{x}}$. But there is nothing in our previous work that forbade us from shifting $\hat{p} \to \hat{p} + f(\hat{x})$ in the exponent for the position translation and shifting $\hat{x} \to \hat{x} + g(\hat{p})$ in the exponent for the momentum translation operator.

This is because we stated what the translation operator for position did to $\hat{x}$ and what the translation operator for momentum did to $\hat{p}$ only by considering the corresponding Hadamard lemmas. Now that we have the exact Hadamard lemma formula, we can complete the analysis. First, verify that if the momentum or position operators are shifted as we propose with f or g in the exponents of the corresponding translation operators, they can still be employed to define the position and momentum eigenstates, just as we did before.

But our earlier discussion was incomplete. Using the full Hadamard lemma formula, we can now complete it. We want the translation operator for position (momentum) to translate the position (momentum) operator, but we also require it to *not* transform the remaining momentum (position) operator at the same time. Show that this second condition constrains both functions f and g to be constant functions. Since a constant function can always be reabsorbed into the position and momentum state vectors (because they are only defined up to a complex phase), show that the position and momentum translation operators must be uniquely given by the operators we have been working with all along.

7.12 Copenhagen Interpretation Applied to the Simple Harmonic Oscillator

In the previous section, we illustrated two new quantum states that can be formed from the simple harmonic oscillator energy eigenstates: (i) the coherent state, which is a ground state, translated in position and momentum (and has minimal uncertainty) and (ii) a squeezed vacuum state, where the uncertainty is traded off between position and momentum as the squeezing parameter is changed (we often call the harmonic oscillator ground state $|0\rangle$ the vacuum state). These two states are fundamentally different—coherent states do not change their shape and maintain uniform uncertainty, while squeezed states change their shape and trade off uncertainty of x with uncertainty of p and *vice versa*.

We have now discussed a large number of different types of quantum states, and it is easy to lose a sense of what they mean and why we constructed them, so let us pause to quickly review them before moving on. Classically, a harmonic oscillator state always has definite position, momentum, and energy. In quantum mechanics, we cannot have all three be definite. So, an energy eigenstate has no uncertainty in the energy (and is called a stationary state). These states are not so simple to create, but experimental work on trapped ions can manipulate and control such states, and we will discuss these types of experiments later in the book, once we understand the dynamics of quantum systems. The energy eigenstates form a complete orthonormal set, making them useful as a basis set for calculations. But there is no clear classical analog of these states. They also have odd concepts associated with them—they are said to be stationary, but the average kinetic energy is nonzero. They have minimal uncertainty in the ground state, but the uncertainty increases for higher

energy states. There is no clear notion of position or momentum with these states, but if we take the overlap of these states with position or momentum bras, then we construct the position-space or momentum-space wavefunctions, which are Gaussians multiplied by Hermite polynomials. They have probability distributions for position and momentum—the position and momentum is indefinite in these states.

The coherent states are the quantum analogs of pulling or pushing a classical mass on a spring. They correspond to position and momentum translations of the ground state. They are not energy eigenstates and their completeness relation is complicated (they are said to be overcomplete). They have minimal uncertainty, and they are the most classical of the different quantum states.

The squeezed states can be thought of in the following odd fashion: they are like using the ground state for a problem with a different mass or a different frequency of the harmonic oscillator. Surprisingly, this allows us to trade off uncertainty in one operator for uncertainty in its canonically conjugate operator. These states are also difficult to make, as they require some kind of nonlinear interaction to make them. But they are quite useful for quantum metrology and can greatly improve the accuracy of measurements, going beyond what is called the standard quantum limit. They cannot form a complete set, because they only form a superposition of even energy eigenstates.

We now examine the squeezed states a bit more. Recall that for the squeezed vacuum state, which we found by setting $\xi = r$ ($\phi = 0$ and $\eta = 0$), we have

$$(\Delta x)_{r,0} = \sqrt{\frac{\hbar}{2m\omega}}e^{-r} \quad \text{and} \quad (\Delta p)_{r,0} = \sqrt{\frac{\hbar m\omega}{2}}e^{r}. \tag{7.353}$$

In Problem 7.11.13, we found an operator that annihilated the squeezed vacuum state. As you know, states that are annihilated by an operator are "golden" for us—they always lead to something useful. If we take the results of that problem, with $\phi = 0$, and we divide both sides by $\cosh r$, which does not change the relation, since $\cosh r \neq 0$, we find that

$$\left(\hat{a} + \tanh r\, \hat{a}^{\dagger}\right) \hat{S}(r, 0)|0\rangle = 0. \tag{7.354}$$

But, as $r \to \infty$, we find that this relation says that $\hat{a} + \hat{a}^{\dagger}$ annihilates the corresponding squeezed vacuum state or, after multiplying by $\sqrt{\frac{\hbar}{2m\omega}}$, we find that

$$\hat{x}\left(\hat{S}(r\to\infty, 0)|0\rangle\right) = 0. \tag{7.355}$$

Remember, we have seen this state before, it is the position eigenstate at the origin! So, $\hat{S}(r \to \infty, 0)|0\rangle \propto |0_x\rangle$ (we use the "proportional to" symbol, because there could be constant factors in the relation and we will sort those out later). Similarly, if we take $r \to -\infty$, then $\hat{a} - \hat{a}^{\dagger}$ annihilates the corresponding squeezed vacuum;

so, $\hat{S}(r \to -\infty, 0)|0\rangle \propto |0_p\rangle$. We can then determine the position eigenstates at any x ($|x\rangle$) and the momentum eigenstates at any p ($|p\rangle$) by applying the appropriate position (or momentum) translation operator to the respective state at the origin.

We will work this out explicitly for the momentum case, leaving the position case as Problem 7.12.1 at the end of the section. The momentum translation operator (on a ket) is $e^{\frac{i}{\hbar}p\hat{x}}$. Here, p is the momentum eigenvalue (i.e., a real number), while $\hat{x}$ is an operator. So, we learn that

$$|p\rangle \propto e^{\frac{i}{\hbar}p\hat{x}}\hat{S}(r \to -\infty, 0)|0\rangle = e^{\frac{ip(\hat{a}+\hat{a}^\dagger)}{\sqrt{2\hbar m\omega}}}\left(\lim_{r\to-\infty} e^{-\frac{r}{2}\left(\hat{a}^\dagger\right)^2+\frac{r}{2}\hat{a}^2}\right)|0\rangle, \tag{7.356}$$

where we use the proportionality symbol, because we will normalize this state at the end. We now use the exponential disentangling identity in Eq. (7.326) to simplify the expression to

$$\begin{aligned}|p\rangle &\propto e^{\frac{ip(\hat{a}+\hat{a}^\dagger)}{\sqrt{2\hbar m\omega}}}\lim_{r\to-\infty} e^{-\frac{\tanh r}{2}\left(\hat{a}^\dagger\right)^2}\frac{1}{(\cosh r)^{\hat{a}^\dagger\hat{a}+\frac{1}{2}}}e^{\frac{\tanh r}{2}\hat{a}^2}|0\rangle \\ &\propto \lim_{r\to-\infty}\left(\frac{1}{\sqrt{\cosh r}}e^{\frac{ip(\hat{a}+\hat{a}^\dagger)}{\sqrt{2\hbar m\omega}}}e^{-\frac{\tanh r}{2}\left(\hat{a}^\dagger\right)^2}\right)|0\rangle.\end{aligned} \tag{7.357}$$

To simplify further, we use the BCH formula, in the Weyl form, ($e^{\hat{A}+\hat{B}} = e^{\hat{A}}e^{\hat{B}}e^{\frac{1}{2}[\hat{B},\hat{A}]}$, with $\hat{A}$ being the raising operator term and $\hat{B}$ being the lowering operator term) on the leftmost term, and move the commutator term outside the limit, since it is independent of r, to give

$$|p\rangle \propto e^{-\frac{p^2}{4\hbar m\omega}}\lim_{r\to-\infty}\left(\frac{1}{\sqrt{\cosh r}}e^{i\frac{p}{\sqrt{2\hbar m\omega}}\hat{a}^\dagger}e^{i\frac{p}{\sqrt{2\hbar m\omega}}\hat{a}}e^{-\frac{\tanh r}{2}\left(\hat{a}^\dagger\right)^2}\right)|0\rangle. \tag{7.358}$$

Now, we use a multiply by one at the ground state, to create a Hadamard with the lowering operator (and the exponential of the square of the raising operator), that is, the two rightmost operators. This goes inside the exponent and then inside the square of the raising operator and gives us

$$|p\rangle \propto e^{-\frac{p^2}{4\hbar m\omega}}\lim_{r\to-\infty}\left(\frac{1}{\sqrt{\cosh r}}e^{i\frac{p}{\sqrt{2\hbar m\omega}}\hat{a}^\dagger}e^{-\frac{\tanh r}{2}\left(\hat{a}^\dagger+i\frac{p}{\sqrt{2\hbar m\omega}}\right)^2}\right)|0\rangle. \tag{7.359}$$

Expanding out the square and combining all remaining operator terms into one exponent yields

$$|p\rangle \propto \lim_{r\to-\infty}\left(e^{-\frac{p^2}{4\hbar m\omega}(1-\tanh r)}\frac{1}{\sqrt{\cosh r}}e^{i\frac{p}{\sqrt{2\hbar m\omega}}(1-\tanh r)\hat{a}^\dagger-\frac{\tanh r}{2}\left(\hat{a}^\dagger\right)^2}\right)|0\rangle. \tag{7.360}$$

Now, we have to take the limit $r \to -\infty$. It looks like it gives 0 due to the $\cosh r$ term in the denominator! But we first *regularize* before we take the limit. This is often done in physics and you will become comfortable doing it yourself after you have seen it a sufficient number of times. At first, though, it looks like we are cheating. But we really are not. Here is how it goes. We multiply by $\sqrt{\cosh r}$ inside the limit before we take the limit. At first, you will protest—this seems like it changes the answer. But we are multiplying by a function of r, not an operator, and we still need to normalize at the end, so this will only change our result by a constant factor at the end of the limit. This is why we can do this. Recalling that $\lim_{r\to-\infty} \tanh r = -1$, we find that

$$|p\rangle \propto e^{-\frac{p^2}{2\hbar m\omega}} e^{i\frac{2p}{\sqrt{2\hbar m\omega}}\hat{a}^\dagger + \frac{1}{2}(\hat{a}^\dagger)^2}|0\rangle. \tag{7.361}$$

Our final step is normalization. You might be tempted to multiply by a $\langle p|$ bra. But we cannot do that, because we cautioned you not to work with wavefunctions of momentum eigenstates in momentum space. So, instead, we multiply by the harmonic oscillator ground state. Then, using the fact that we already solved for the momentum-space wavefunction, we have that

$$\tilde{\phi}_0^*(p) = \langle 0|p\rangle = c\, e^{-\frac{p^2}{2\hbar m\omega}} \left\langle 0|e^{i\frac{2p}{\sqrt{2\hbar m\omega}}\hat{a}^\dagger + \frac{1}{2}(\hat{a}^\dagger)^2}|0\right\rangle, \tag{7.362}$$

where c is the normalization constant we are seeking. Now, the momentum-space wavefunction was found in Eq. (7.240), and we evaluate the matrix element by letting the raising operators act to the left on $\langle 0|$, where they annihilate, implying the matrix element is equal to 1. Then, we find that

$$c = \tilde{\phi}_0(p) e^{\frac{p^2}{2\hbar m\omega}} = \left(\frac{1}{\pi\hbar m\omega}\right)^{\frac{1}{4}}. \tag{7.363}$$

The final result is then

$$|p\rangle = \left(\frac{1}{\pi\hbar m\omega}\right)^{\frac{1}{4}} e^{-\frac{p^2}{2\hbar m\omega}} e^{i\frac{2p}{\sqrt{2\hbar m\omega}}\hat{a}^\dagger + \frac{1}{2}(\hat{a}^\dagger)^2}|0\rangle. \tag{7.364}$$

In Problem 7.12.1, you will show that

$$|x\rangle = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{m\omega x^2}{2\hbar}} e^{i\sqrt{\frac{2m\omega}{\hbar}}x\hat{a}^\dagger - \frac{1}{2}(\hat{a}^\dagger)^2}|0\rangle. \tag{7.365}$$

This result allows us to derive a serious identity that the Hermite polynomials satisfy. By introducing a complete set of harmonic oscillator energy eigenstates, we have

$$|x\rangle = \sum_{n=0}^{\infty} |n\rangle \langle n|x\rangle = \sum_{n=0}^{\infty} \psi_n(x)^* |n\rangle. \tag{7.366}$$

Using

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} \frac{1}{\sqrt{2^n n!}} e^{-\frac{m\omega x^2}{2\hbar}} H_n\left(\sqrt{\frac{m\omega}{\hbar}}x\right) = \psi_n^*(x), \tag{7.367}$$

we find that

$$|x\rangle = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{m\omega x^2}{2\hbar}} \sum_{n=0}^{\infty} \frac{1}{n!} H_n\left(\sqrt{\frac{m\omega}{\hbar}}x\right) \frac{\left(\hat{a}^\dagger\right)^n}{\sqrt{2^n}} |0\rangle, \tag{7.368}$$

after using the fact that $|n\rangle = \frac{1}{\sqrt{n!}}\left(\hat{a}^\dagger\right)^n |0\rangle$. Setting $y = \frac{\hat{a}^\dagger}{\sqrt{2}}$ then suggests that the following identity must hold (after comparing to the form we derived for the position eigenstate above)

$$\sum_{n=0}^{\infty} \frac{1}{n!} H_n\left(\sqrt{\frac{m\omega}{\hbar}}x\right) y^n = e^{-y^2 + 2\sqrt{\frac{m\omega}{\hbar}}xy}. \tag{7.369}$$

Indeed, this is an identity that the Hermite polynomials satisfy (called the generating function identity), but we will not derive it independently here. You can find it in any serious work on Hermite polynomials.

This preliminary work shows us how to find the position and momentum eigenstates in terms of the simple harmonic oscillator eigenstates. This resolves the issue we had before when we had to assume that $|0_x\rangle$ and $|0_p\rangle$ existed—here, we now express them in terms of harmonic oscillator quantities, so they definitely exist! We will use these results to analyze how one determines predictions for a series of experiments that follow the Copenhagen paradigm. We will next state the Copenhagen interpretation rules for analyzing measurements. These rules cannot be derived (although some will be motivated later in the book). You can think of them as postulates of quantum mechanics, within this interpretation. There are many other interpretations of quantum mechanics that have their own rules. Some examples are the many world's interpretation or the Bohm pilot-wave theory. For now, we press on as a "Copenhagist." If you are not happy with this, we will invoke the "shut up and calculate" dogma—quantum mechanics results must agree for predictions of measurements for all different interpretations—otherwise, we could rule some out by performing measurements that differentiate them. The philosophical meaning of a specific interpretation is something physicists and philosophers still work on (and debate) today.

The Copenhagen measurement paradigm is postulated to abide by the following four steps:

1. Prepare an initial quantum state $|\psi\rangle$ and choose a self-adjoint operator $\hat{O}$ that represents the quantity being measured.
2. Measure $\hat{O}$. Note that one has to design an experiment that will measure $\hat{O}$, which may not be easy to do; but for now, we assume it can be done. The observed result must be one of the eigenvalues of $\hat{O}$. Suppose the corresponding eigenstate is denoted $|O_n\rangle$ and the eigenvalue is O_n. Then, the measurement process is modeled by the projection onto the state $|O_n\rangle$: $|\psi\rangle \rightarrow |O_n\rangle \langle O_n|\psi\rangle$. This is called *state collapse*.
3. The probability to measure the value O_n, if we repeated the experiment many times (with the same initial state $|\psi\rangle$), is given by $P(O_n) = \left\| |O_n\rangle \langle O_n|\psi\rangle \right\|^2 = |\langle O_n|\psi\rangle|^2$, since $\langle O_n|O_n\rangle = 1$.
4. Immediately after the experiment is over, the state changes from $|\psi\rangle$ to $|O_n\rangle$, until another experiment is performed or until the system has dynamically evolved to another state (if the measured state is not an eigenstate of the Hamiltonian). Note that this is a renormalization of the post-measurement state, so the system conserves total probability again.

There are many comments to make about this procedure. First, it is often said that we must have a self-adjoint operator (or many conflate this and say a Hermitian operator), because experiments can only measure real quantities. But this is actually not the case; experiments can measure complex quantities, so the chosen operator can be a unitary operator (it even can be what is called a normal operator). Some quantities that are measured that are studied in classical electromagnetism are indeed complex (like the ac impedance of a circuit or a dielectric constant). But we will stick with the self-adjoint operator for the statement of the rules. Second, just having an appropriate operator does not mean we know how to measure it. We will even find that most measurements do not fit into this paradigm. In counting measurements, we count quanta and do not measure an operator—instead, we use the counting results to infer properties about the measurement. We will have more to say about this later and how counting measurements are analyzed. Third, probability predictions mean we must repeat the measurement, starting from the same state $|\psi\rangle$ many times to determine the probability distributions of the results. One measurement is not enough, unless the system was already in an eigenstate. Fourth, the projection hypothesis tells us what the state goes into immediately after the measurement. It may evolve into a different state over time. Some measurements destroy the state or modify the state in the process of the measurement—in these cases, the rule (4) does not apply. Fifth, the procedure needs to be modified if our measuring device is not so precise to differentiate between all possible eigenvalues but instead simultaneously measures a range of eigenvalues (such as a detector of width Δx that approximately measures the position). And sixth, the wavefunction collapse need not be thought of as a physical event—it can be thought of as an update to the information we have about the system—in this sense, it is similar to how Bayesian statistics works. We will spend quite some time sorting through these subtleties throughout the rest of the book.

There is often a lot of confusion surrounding the relationship between points (2), (3), and (4). One reason is that for rules (2) and (3), we use a projection operator $|O_n\rangle\langle O_n|\psi\rangle$ and $|\langle O_n|\psi\rangle|^2$, whereas, in point (4), we say $|\psi\rangle \rightarrow |O_n\rangle$, with no factor of $\langle O_n|\psi\rangle$. How can this be? It is because they refer to two different things even though it appears like they are contradictory. Let's illustrate this with a simple example.

Start the system in the ground state of a simple harmonic oscillator, so $|\psi\rangle = |0\rangle$. Now measure position, so $\hat{O} = \hat{x}$ (and assume the measurement is infinitesimally accurate). Suppose in one shot of the experiment, we observe the particle to be at x_0. This means $|\psi\rangle = |0\rangle \rightarrow |x_0\rangle$ from rule (4) for this shot of the experiment. This is because we always observe an entire particle at a single location when we do this measurement. It is never split among multiple detectors. To find the probability distribution, we need to repeat the measurement many times. In this case, we use rules (2) and (3):

$$P(x_0) = |\langle x_0|0\rangle|^2 = |\psi_0(x_0)|^2 = \sqrt{\frac{m\omega}{\pi\hbar}}e^{-\frac{m\omega x_0^2}{\hbar}}, \tag{7.370}$$

since $\hat{x}|x_0\rangle = x_0|x_0\rangle$. See how the rules are applied in a different fashion—one applies to each shot (4), while the others (2 and 3) require many shots to build up a probability distribution.

Let us consider another example. This one involves measuring the number of quanta in the harmonic oscillator. The quanta are associated with the number operator $\hat{N} = \hat{a}^\dagger\hat{a}$, which counts the number of excitations above the ground state. This number is called the quanta above the ground state. Suppose we start in a coherent state $|\psi\rangle = |\alpha\rangle$. As we worked out earlier, this state is a linear superposition over all energy eigenstates $|n\rangle$. I measure the number operator to determine the number of quanta excited in the coherent state. For each measurement, I will observe a nonnegative integer as the number of quanta excited. Let's say for a particular shot of the experiment, I observe n_0 quanta. This means $|\alpha\rangle \rightarrow |n_0\rangle$ after the measurement for that shot. Now, I measure the number of quanta again (without starting over). Do I obtain the same number as before, or are there many different possibilities? Since the number states are eigenstates of the Hamiltonian, they do not change after the experiment, so there is only one possible measurement result when repeated, namely, n_0 again. What is the probability to measure a specific number of quanta in the first measurement? We use our rule to find that

$$P(n_0) = |\langle n_0|\alpha\rangle|^2 = |\langle n_0|\sum_{m=0}^{\infty}\frac{\alpha^m}{\sqrt{m!}}e^{-\frac{1}{2}|\alpha|^2}|m\rangle|^2 = \frac{|\alpha|^{2n_0}}{n_0!}e^{-|\alpha|^2}. \tag{7.371}$$

A plot of this result for $\alpha = 3$ is shown in Fig. 7.14.

What happens if I measure $\hat{N}$ and then measure $\hat{H}$? Does the state change with the second measurement? The answer is no. This is because $|n_0\rangle$ is already an eigenstate of both $\hat{N}$ and $\hat{H}$ (recall that $[\hat{N}, \hat{H}] = 0$). Suppose we start in

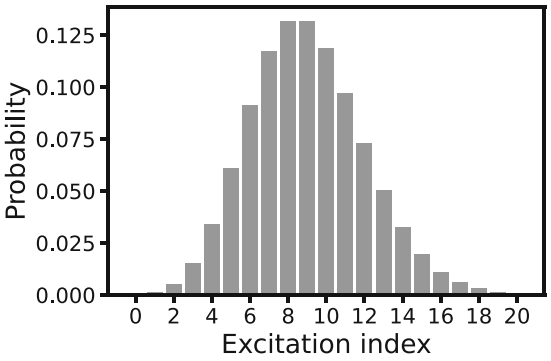

Fig. 7.14 Histogram showing the probability to measure a particular quanta in a coherent state with $\alpha = 3$

the state $|n_0\rangle$, measure position $\hat{x}$ and then measure energy $\hat{H}$. Is the energy still $\hbar\omega\left(n_0 + \frac{1}{2}\right)$? No, because the state $|x_0\rangle$ generically requires a superposition over all energy eigenstates, so we have the possibility to observe any energy afterward (but not equally likely for each energy value). The probability to measure a specific energy is a problem you will solve in Problem 7.12.3 at the end of the section.

In addition to finding the probability for a particular result to be measured, we can also ask what is the average value that will be measured after many measurements and what are the fluctuations about the mean? For any operator $\hat{O}$, with eigenvalues O_n and eigenstates $|O_n\rangle$, we have already determined that the probability to measure O_n is $P(O_n) = |\langle O_n|\psi\rangle|^2$ from rule (3). We use this probability to find the average value and the fluctuations. Namely, we have

$$\langle\hat{O}\rangle = \sum_{n=0}^{\infty} P(O_n)O_n = \sum_{n=0}^{\infty} |\langle O_n|\psi\rangle|^2 O_n = \langle\psi|\hat{O}|\psi\rangle, \tag{7.372}$$

since $\sum_{n=0}^{\infty} |O_n\rangle\langle O_n| = \hat{\mathbb{I}}$, from our completeness relation for self-adjoint operators. Note that this is the same as what we had been using previously. The fluctuations are

$$(\Delta O)_\psi^2 = \sum_{n=0}^{\infty} P(O_n)O_n^2 - \left(\sum_{n=0}^{\infty} P(O_n)O_n\right)^2 = \langle\psi|\hat{O}^2|\psi\rangle - |\langle\psi|\hat{O}|\psi\rangle|^2. \tag{7.373}$$

As an example, let's look at the coherent state case again. We found the probability in Eq. (7.371). Using that result, we have

$$\begin{aligned}\langle\hat{N}\rangle &= \sum_{n=0}^{\infty} P(n)n = \sum_{n=0}^{\infty} \frac{|\alpha|^{2n}}{n!} e^{-|\alpha|^2} n = \sum_{n=1}^{\infty} \frac{|\alpha|^{2n}}{(n-1)!} e^{-|\alpha|^2} \\ &= |\alpha|^2 \sum_{n=0}^{\infty} \frac{|\alpha|^{2n}}{n!} e^{-|\alpha|^2} = |\alpha|^2,\end{aligned} \tag{7.374}$$

which is a manipulation we have done before (we shifted the summation index in going to the second line). The fluctuations are a bit harder to calculate. We have

$$
\begin{aligned}
(\Delta N)_\alpha^2 &= \sum_{n=0}^{\infty} P(n)n^2 - \left(\sum_{n=0}^{\infty} P(n)n\right)^2 = \sum_{n=0}^{\infty} \frac{|\alpha|^{2n}}{n!} n^2 e^{-|\alpha|^2} - |\alpha|^4 \\
&= \sum_{n=0}^{\infty} \frac{|\alpha|^{2n}}{n!} \big(n(n-1)+n\big) e^{-|\alpha|^2} - |\alpha|^4 \\
&= \sum_{n=2}^{\infty} \frac{|\alpha|^{2n}}{(n-2)!} e^{-|\alpha|^2} + |\alpha|^2 - |\alpha|^4 \\
&= |\alpha|^4 \sum_{n=0}^{\infty} \frac{|\alpha|^{2n}}{n!} e^{-|\alpha|^2} + |\alpha|^2 - |\alpha|^4 = |\alpha|^2. \qquad (7.375)
\end{aligned}
$$

So, $\langle \hat{N} \rangle = |\alpha|^2$ and $(\Delta N) = |\alpha|$. In the calculation of the fluctuations, we first used "add zero" to convert $n^2 \to n^2 - n + n = n(n-1) + n$ and then dropped the terms in the summation that were zero. Then, we shifted the index and summed the remaining summation, which was just an exponential. Note that the expectation values and fluctuations for position and momentum were already computed in Sect. 7.11.

We plot the probability to observe the state $|n\rangle$ when we measure the quanta for a system in a squeezed state with $\xi = r = 2$ and $\eta = 0$ in Fig. 7.15. You have an opportunity to determine the average value and the fluctuations about the mean in Problems 7.12.5 and 7.12.6.

We have developed a lot of complex math and we have seen we can use just a few techniques (such as the four fundamental operator identities and their corollaries) to perform many different quantum calculations. This methodology is the way many quantum calculations are performed, especially for more advanced topics. We hope you appreciate the great strides in sophistication that you have already made in

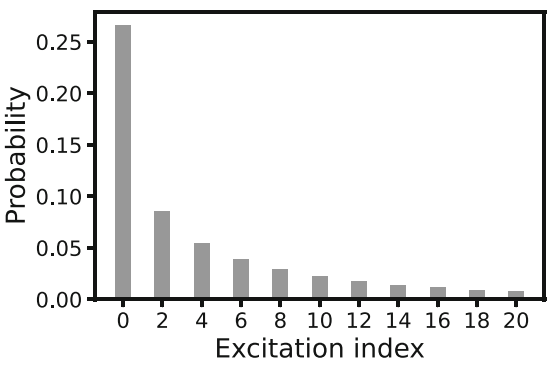

Fig. 7.15 Histogram of the probability to measure a particular quanta in a squeezed state with $\xi = r = 2$, $\phi = 0$, and $\eta = 0$. Note how only even quanta are present

mastering this material. In most of the rest of the book, we simply apply these same identities and use these same methodologies to learn even more new things about the quantum world.

A question you might have with all of this work on coherent and squeezed states is can anyone actually make these states? The answer is yes! It is a large portion of the field called quantum optics, since many of these states are formed with light (they also have been examined with ions in a trap). For example, a laser creates coherent states of light. Squeezed states are more difficult to make and require a nonlinear medium to make them—we will not discuss the details of how this is done here. But they have been made, and one of the ways you can tell you have made them is to measure the probability distribution for the number of quanta excited in the light or examine the mean and the fluctuations about the mean. It is amazing that these complex quantum states can now be made routinely in the lab. Coherent states are already in commercial products, and squeezed states are likely to be there soon. But note that in both of these states there are limitations to how large the different parameters α and ξ and η can be. We cannot make *arbitrary* coherent and squeezed states yet.

Another point we want to reemphasize is just because we have found a self-adjoint operator does not mean that we know how to measure it in an experiment. Finding experiments that can measure properties of quantum systems is a hard problem that requires a lot of creative thinking. There are not so many different options that we have for measuring quantum objects. So, do not expect that every self-adjoint operator can easily be measured. We need to very cleverly design an experiment to measure each operator that we propose to measure.

Section 7.12 Problems:

7.12.1 Constructing the Position Eigenstate from the Harmonic Oscillator States

Construct $|x\rangle$ as an operator acting on the simple harmonic oscillator ground state $|0\rangle$ by noting that the state $|0_x\rangle \propto \hat{S}(r \to \infty, \eta{=}0)|0\rangle$. Express your final result as a single exponential of an operator that is a quadratic function of the raising operator and properly normalize your final result. Note that you need to use the translation operator for position and you will need to regularize the result.

7.12.2 Constructing the Momentum Eigenstates from the Harmonic Oscillator Wavefunctions

Construct $|p\rangle$ via $\sum_{n=0}^{\infty} |n\rangle \langle n|p\rangle = |p\rangle$ by noting that $\langle n|p\rangle = \psi_n^*(p)$ and using the identity $\sum_{n=0}^{\infty} \frac{1}{n!} H_n(z) y^n = e^{-y^2+2zy}$ for an appropriate z and y.

7.12.3 Measure Position Followed by Energy

Start with the state $|\psi\rangle = |n_0\rangle$ for some nonnegative integer n_0. Measure $\hat{x}$ and you find the particle located at x_0. Then, immediately measure the energy (operator is $\hat{H}$) and determine the probability that you will measure E_n.

Hint: recall that the wavefunction satisfies $\langle x_0|n\rangle = \psi_n(x_0) = \langle n|x_0\rangle^*$.

7.12.4 Measure Energy, Momentum, and Then Energy Again

Prepare a harmonic oscillator in the equal superposition:

$$|\psi\rangle = \frac{1}{2}\sum_{n=0}^{3}|n\rangle. \tag{7.376}$$

What are the possible energy eigenvalues we can measure with nonzero probability if we measure $\hat{H}$. Determine $\langle\psi|\hat{H}|\psi\rangle$ and $(\Delta H)_\psi$.

If, instead, momentum is measured in the state $|\psi\rangle$ and the result is found to be p_0, and then $\hat{H}$ is measured afterward, what is the probability to measure E_n? Recall the definition of the wavefunction.

7.12.5 Average Number of Quanta in a Squeezed State

Prepare the state $\hat{S}(r,0)|0\rangle$ and determine the average value of $\hat{N}$ in this state: $\langle\hat{N}\rangle = \langle 0|\hat{S}(r,0)^\dagger\hat{N}\hat{S}(r,0)|0\rangle$. Note that the multiply-by-one trick, with $\hat{S}\hat{S}^\dagger = 1$, may be useful here.

7.12.6 Fluctuations About the Mean for the Quanta in a Squeezed State

Determine the fluctuations about the mean for the state described in the previous problem.

7.12.7 Challenge Problem: Wavefunction of a Coherent-Squeezed State

While you have all of the techniques to compute the wavefunction of the coherent-squeezed state, given by $\psi_{\xi,\alpha}(x) = \langle x|\hat{D}(\alpha)\hat{S}(\xi,0)|0\rangle$, with $\xi = re^{i\phi}$, the

calculation is a serious challenge to complete. See whether you can establish that the wavefunction is

$$\psi_{\xi,\alpha}(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} \frac{1}{\sqrt{|\cosh r - e^{i\phi}\sinh r|}} \frac{\sqrt{\cosh r - e^{-i\phi}\sinh r}}{\sqrt{|\cosh r - e^{-i\phi}\sinh r|}} \times e^{-\frac{i}{2\hbar}x_0 p_0} e^{\frac{i}{\hbar}p_0 x} \exp\left(-\frac{\cosh r + e^{i\phi}\sinh r}{\cosh r - e^{i\phi}\sinh r}\frac{m\omega}{2\hbar}(x - x_0)^2\right). \tag{7.377}$$

In this problem, we use $\alpha = \sqrt{\frac{m\omega}{2\hbar}}x_0 + \frac{i}{\sqrt{2\hbar m\omega}}p_0$. Replace the $\langle x|$ bra by the Hermitian conjugate of the result in Eq. (7.365), so that the calculation has simple harmonic oscillator ground states as the bra and the ket. Then, find a strategy to remove all of the operators using the operator identities we have available. You need all of the different disentangling identities we have derived to complete the calculation. Good luck. If you need help, consult (Munguia-González et al., 2021).

7.13 Simple Harmonic Oscillator in Higher Dimensions

We have been working primarily in a one-dimensional world. But the real world has three spatial dimensions. What changes as we move from one-d to three-d? The answer is, curiously, not very much. There are some new phenomena, such as degeneracy of energy eigenstates, and new symmetries, but those are relatively minor changes. Nearly everything we have already done carries over to higher dimensions. This is good. We have worked very hard in one dimension.

The three-dimensional simple harmonic oscillator can be written as

$$\hat{H} = \frac{\hat{p}_x^2}{2m} + \frac{\hat{p}_y^2}{2m} + \frac{\hat{p}_z^2}{2m} + \frac{1}{2}m\omega_x^2\hat{r}_x^2 + \frac{1}{2}m\omega_y^2\hat{r}_y^2 + \frac{1}{2}m\omega_z^2\hat{r}_z^2. \tag{7.378}$$

It is the sum of three independent one-dimensional harmonic oscillators (note we use a subscript $\alpha \in \{x, y, z\}$ for the frequencies, which can be different in different directions); we also switched to the $\hat{r}_\alpha$ notation for the spatial position operators. How do we find the energy eigenvalues and the energy eigenstates for this Hamiltonian? We do the same thing as we did with the factorization method in Sect. 7.6, only now, we have raising and lowering operators in each of the three coordinate directions $\alpha = x, y, z$. They are constructed in the same way as before because each oscillator is independent of the others:

$$\hat{a}_x = \sqrt{\frac{m\omega_x}{2\hbar}}\left(\hat{r}_x + i\frac{\hat{p}_x}{m\omega_x}\right), \quad \hat{a}_x^\dagger = \sqrt{\frac{m\omega_x}{2\hbar}}\left(\hat{r}_x - i\frac{\hat{p}_x}{m\omega_x}\right),$$

$$\hat{a}_y = \sqrt{\frac{m\omega_y}{2\hbar}}\left(\hat{r}_y + i\frac{\hat{p}_y}{m\omega_y}\right), \quad \hat{a}_y^\dagger = \sqrt{\frac{m\omega_y}{2\hbar}}\left(\hat{r}_y - i\frac{\hat{p}_y}{m\omega_y}\right),$$
$$\hat{a}_z = \sqrt{\frac{m\omega_z}{2\hbar}}\left(\hat{r}_z + i\frac{\hat{p}_z}{m\omega_z}\right), \quad \hat{a}_z^\dagger = \sqrt{\frac{m\omega_z}{2\hbar}}\left(\hat{r}_z - i\frac{\hat{p}_z}{m\omega_z}\right). \tag{7.379}$$

They satisfy the same operator algebra, noting that the raising and lowering operators for different directions commute:

$$[\hat{a}_i, \hat{a}_j] = 0, \quad [\hat{a}_i, \hat{a}_j^\dagger] = \delta_{ij}, \quad \text{and} \quad [\hat{a}_i^\dagger, \hat{a}_j^\dagger] = 0, \tag{7.380}$$

for $i, j \in \{x, y, z\}$. We form the states as tensor products (recall we covered those in Sect. 6.6 to express entangled states). We write

$$|\psi\rangle = |n_x\rangle \otimes |n_y\rangle \otimes |n_z\rangle, \tag{7.381}$$

where the $\hat{a}_i$ ($\hat{a}_i^\dagger$) operators act only on the corresponding ket of the tensor product. If you like, we can write $\hat{a}_x \otimes \hat{\mathbb{I}} \otimes \hat{\mathbb{I}}$, $\hat{\mathbb{I}} \otimes \hat{a}_y \otimes \hat{\mathbb{I}}$, and $\hat{\mathbb{I}} \otimes \hat{\mathbb{I}} \otimes \hat{a}_z$ for the operators, to make clear which tensor-product term they act on. But this approach rapidly becomes too cumbersome to use. Instead, we just let the subscript tell us what the operator acts on. As for the states, we often use the shortcut:

$$|\psi\rangle = |n_x\rangle \otimes |n_y\rangle \otimes |n_z\rangle = |n_x, n_y, n_z\rangle. \tag{7.382}$$

Note that this is just a simplification in the notation. In terms of the raising and lowering operators, the Hamiltonian is reexpressed as

$$\hat{H} = \hbar\omega_x\left(\hat{a}_x^\dagger\hat{a}_x + \frac{1}{2}\right) + \hbar\omega_y\left(\hat{a}_y^\dagger\hat{a}_y + \frac{1}{2}\right) + \hbar\omega_z\left(\hat{a}_z^\dagger\hat{a}_z + \frac{1}{2}\right), \tag{7.383}$$

and from this, we can immediately read off that the energy eigenvalue associated with the state $|n_x, n_y, n_z\rangle$ is

$$E_{n_x n_y n_z} = \hbar\omega_x\left(n_x + \frac{1}{2}\right) + \hbar\omega_y\left(n_y + \frac{1}{2}\right) + \hbar\omega_z\left(n_z + \frac{1}{2}\right). \tag{7.384}$$

Similarly, a wavefunction is found by multiplying the state ket with the bra $\langle r_x, r_y, r_z| = \langle r_x| \otimes \langle r_y| \otimes \langle r_z|$ to give

$$\begin{aligned}\langle r_x, r_y, r_z | n_x, n_y, n_z\rangle &= \Psi_{n_x n_y n_z}(\vec{r}) \\ &= \Big(\langle r_x| \otimes \langle r_y| \otimes \langle r_z|\Big) \cdot \Big(|n_x\rangle \otimes |n_y\rangle \otimes |n_z\rangle\Big) \\ &= \langle r_x|n_x\rangle \times \langle r_y|n_y\rangle \times \langle r_z|n_z\rangle \\ &= \psi_{n_x}(r_x) \times \psi_{n_y}(r_y) \times \psi_{n_z}(r_z),\end{aligned} \tag{7.385}$$

and similarly for the momentum wavefunctions. The wavefunctions are just products of the wavefunctions in each dimension. As you can see, it is relatively simple to make this generalization to higher dimensions. Let us work out the ground-state wavefunction, which is simply a product of the ground-state wavefunctions in each dimension. This gives

$$\psi_{gs}(r_x, r_y, r_z) = \left(\frac{m\omega_x}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{m\omega_x}{2\hbar}r_x^2} \left(\frac{m\omega_y}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{m\omega_y}{2\hbar}r_y^2} \left(\frac{m\omega_z}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{m\omega_z}{2\hbar}r_z^2} \tag{7.386}$$

with $E_{gs} = \frac{\hbar}{2}(\omega_x + \omega_y + \omega_z)$. The video The Three Dimensional Simple Harmonic Oscillator https://tinyurl.com/2xbzc9kc provides some further discussion.

It is interesting to reconsider parity, because we can contrast the parity in one spatial dimension, with that in another spatial dimension. The parity operator for three dimensions is

$$\hat{\Pi} = \hat{\Pi}_x \otimes \hat{\Pi}_y \otimes \hat{\Pi}_z = (-1)^{\hat{N}_x} \otimes (-1)^{\hat{N}_y} \otimes (-1)^{\hat{N}_z} = (-1)^{\hat{N}_x + \hat{N}_y + \hat{N}_z}. \tag{7.387}$$

In the problems, you will show that $\hat{\Pi}\hat{H}\hat{\Pi} = \hat{H}$, so $\hat{H}$ has definite parity. The parity of an energy eigenstate is still even or odd, but the total parity is now the product of the individual parities in each dimension. In other words, the eigenvalue of the parity operator is $(-1)^{n_x+n_y+n_z}$ for the energy eigenstate $|n_x, n_y, n_z\rangle$. So, we can have an overall odd parity state that is even parity in x, even parity in y, and odd parity in z. This relationship between total parity, and the parity in each dimension, is simply a consequence of the fact that

$$(-1)^{n_x+n_y+n_z} = (-1)^{n_x} \times (-1)^{n_y} \times (-1)^{n_z}. \tag{7.388}$$

Things become more interesting when we make some of the frequencies the same. If $\omega_x = \omega_y = \omega_z = \omega$, then we have the *isotropic simple harmonic oscillator*, whose Hamiltonian is

$$\hat{H} = \frac{1}{2m}(\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2) + \frac{1}{2}m\omega^2(\hat{r}_x^2 + \hat{r}_y^2 + \hat{r}_z^2). \tag{7.389}$$

One can see that the Hamiltonian depends only on the radial distance between the origin and the particle for both position and momentum, since

$$\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2 = |\hat{\vec{p}}^{\,2}| = \hat{p}^2 \quad \text{and} \quad \hat{r}_x^2 + \hat{r}_y^2 + \hat{r}_z^2 = |\hat{\vec{r}}^{\,2}| = \hat{r}^2. \tag{7.390}$$

This means both the kinetic energy and the potential energy look like "spheres" and hence are the same in any direction. This then means that the system has full rotational symmetry about any axis that goes through the origin. This added symmetry forces some of the energy eigenvalues to coincide. This phenomenon is called *degeneracy*. In fact, one of the main consequences of having extra symmetry in a problem is that the degeneracy of the eigenvalues increases.

Table 7.2 Exact enumeration of the lowest energy eigenstates of the isotropic simple harmonic oscillator to illustrate the degeneracy of the different energy eigenvalues

State $n_x\, n_y\, n_z$	Energy	Degeneracy
0 0 0	$\frac{3}{2}\hbar\omega$	None
1 0 0	$\frac{5}{2}\hbar\omega$	Threefold
0 1 0	$\frac{5}{2}\hbar\omega$	
0 0 1	$\frac{5}{2}\hbar\omega$	
2 0 0	$\frac{7}{2}\hbar\omega$	Sixfold
0 2 0	$\frac{7}{2}\hbar\omega$	
0 0 2	$\frac{7}{2}\hbar\omega$	
1 1 0	$\frac{7}{2}\hbar\omega$	
1 0 1	$\frac{7}{2}\hbar\omega$	
0 1 1	$\frac{7}{2}\hbar\omega$	

Let us examine the isotropic harmonic oscillator carefully in three dimensions. Recall that $E_{n_x n_y n_z} = \hbar\omega\left(n_x + n_y + n_z + \frac{3}{2}\right)$. Any time we can find three nonnegative integers that sum to the same value, we have degenerate energy eigenstates. An example of this for the lowest few energy eigenstates is summarized in Table 7.2. As the energy becomes larger, the degeneracy becomes more difficult to determine. The most straightforward way to determine them is by enumeration of the different cases. Geometrically, one can think of them as being the set of states with the same distance from the origin for a lattice, where the lattice points are labelled by nonnegative integers.

The ground state now has added symmetry. We find that

$$\psi_{gs}(r_x, r_y, r_z) = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{3}{4}} e^{-\frac{m\omega}{2\hbar}r^2} \quad \text{and} \quad \psi_{gs}(p_x, p_y, p_z) = \left(\frac{1}{\pi\hbar m\omega}\right)^{\frac{3}{4}} e^{-\frac{1}{2\hbar m\omega}p^2} \tag{7.391}$$

are both functions only of $r^2 = r_x^2 + r_y^2 + r_z^2$ and $p^2 = p_x^2 + p_y^2 + p_z^2$. Now, in general, not all wavefunctions are just functions of r and p. We already saw this phenomenon when we discussed parity—there, the Hamiltonian was even, but the wavefunctions were either even or odd. For the isotropic simple harmonic oscillator, the symmetry is more complex, and we will develop it thoroughly in Chap. 9. But there are cases where we can find the r^2 only dependence for some wavefunctions. To do so requires us to form the proper linear superposition of the degenerate eigenstates. Of course, all of the components in these superpositions have the same energy, so the energy of the superposition is the same as well.

We show how to do this with a nontrivial example. We will work this out for the states that have $n_x + n_y + n_z = 4$ and construct the spherically symmetric

solution, which is a function only of r. The energy is $\frac{11}{2}\hbar\omega$, and the specific wavefunctions we will use are those labeled by the triplet of indices in the set {(400), (040), (004), (022), (202), (220)}. Note that this is not all eigenfunctions that have this energy—it is just the subset of degenerate wavefunctions that have even powers of the Cartesian coordinates. These are the only ones we need for this demonstration. To construct them, we need to use the wavefunction components for each dimension, given by Eq. (7.247), with $n = 0$, 2, and 4. Namely, we use

$$\begin{aligned}
(n=4):\ & \frac{1}{8\sqrt{6}} H_4\left(\sqrt{\frac{m\omega}{\hbar}} r_x\right) e^{-\frac{m\omega}{2\hbar} r_x^2} \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} \\
(n=2):\ & \frac{1}{2\sqrt{2}} H_2\left(\sqrt{\frac{m\omega}{\hbar}} r_x\right) e^{-\frac{m\omega}{2\hbar} r_x^2} \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} \\
(n=0):\ & e^{-\frac{m\omega}{2\hbar} r_x^2} \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}}.
\end{aligned} \tag{7.392}$$

When we take the product of three wavefunctions, with one for each dimension, we will have a common factor of $e^{-\frac{m\omega}{2\hbar} r^2} \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{3}{4}}$, which we will suppress in the following formulas. Also, recall that $H_2(z) = 4z^2 - 2$ and $H_4(z) = 16z^4 - 48z^2 + 12$. We construct the spherically symmetric wavefunction in steps. First, add together the 400, 040, and 004 terms, which is proportional to

$$\left(16\left(\frac{m\omega}{\hbar}\right)^2 (r_x^4 + r_y^4 + r_z^4) - 48\frac{m\omega}{\hbar}(r_x^2 + r_y^2 + r_z^2) + 12\right) \frac{1}{8\sqrt{6}}. \tag{7.393}$$

The constant term and the quadratic terms are already written in terms of r. The quartic terms can be written in terms of $r^4 = r_x^4 + r_y^4 + r_z^4 + 2r_x^2 r_y^2 + 2r_y^2 r_z^2 + 2r_z^2 r_x^2$, if we can remove the cross terms. So, we find that $\psi_{400} + \psi_{040} + \psi_{004}$ is proportional to

$$\left(16\left(\frac{m\omega}{\hbar}\right)^2 r^4 - 48\frac{m\omega}{\hbar} r^2 + 12\right) \frac{1}{8\sqrt{6}} - \left(\frac{m\omega}{\hbar}\right)^2 (r_x^2 r_y^2 + r_y^2 r_z^2 + r_z^2 r_x^2) \frac{32}{8\sqrt{6}}. \tag{7.394}$$

We can form the cross terms from the 220, 202, and 022 functions. We find that $\psi_{220} + \psi_{202} + \psi_{022}$ is proportional to

$$\begin{aligned}
&\left(4\frac{m\omega}{\hbar} r_x^2 - 2\right)\left(4\frac{m\omega}{\hbar} r_y^2 - 2\right)\frac{1}{8} + \left(4\frac{m\omega}{\hbar} r_y^2 - 2\right)\left(4\frac{m\omega}{\hbar} r_z^2 - 2\right)\frac{1}{8} \\
&+ \left(4\frac{m\omega}{\hbar} r_z^2 - 2\right)\left(4\frac{m\omega}{\hbar} r_x^2 - 2\right)\frac{1}{8}.
\end{aligned} \tag{7.395}$$

We can see that using the proper coefficient will allow us to cancel the cross terms, while the remaining terms are functions of r, so canceling the cross terms will achieve our goal. Hence, we learn that

$$\psi_{400} + \psi_{040} + \psi_{004} + \frac{1}{8\sqrt{6}} \left(\psi_{220} + \psi_{202} + \psi_{022}\right), \tag{7.396}$$

will be a function of r only, and hence, it will be rotationally symmetric.

We have sketched how one can find spherically symmetric eigenfunctions by forming appropriate linear combinations of the Cartesian number eigenstates. Let's now investigate some more fundamental relations, such as uncertainty, for the isotropic oscillator. We define

$$(\Delta\vec{r})_{\vec{n}} = \sqrt{(\Delta r_x)^2_{n_x} + (\Delta r_y)^2_{n_y} + (\Delta r_z)^2_{n_z}} = \sqrt{\frac{\hbar}{2m\omega}}\sqrt{2n_x + 2n_y + 2n_z + 3}, \tag{7.397}$$

which is the position uncertainty for the states $|n_x, n_y, n_z\rangle$. Similarly

$$(\Delta\vec{p})_{\vec{n}} = \sqrt{(\Delta p_x)^2_{n_x} + (\Delta p_y)^2_{n_y} + (\Delta p_z)^2_{n_z}} = \sqrt{\frac{\hbar m\omega}{2}}\sqrt{2n_x + 2n_y + 2n_z + 3}. \tag{7.398}$$

So, we find that

$$(\Delta\vec{r})_{\vec{n}}(\Delta\vec{p})_{\vec{n}} = \frac{\hbar}{2}\Big(2(n_x + n_y + n_z) + 3\Big), \tag{7.399}$$

and the minimal uncertainty is now

$$(\Delta\vec{r})_{gs}(\Delta\vec{p})_{gs} = \frac{3}{2}\hbar. \tag{7.400}$$

This is larger than the one-dimensional result due to the three different spatial dimensions.

Essentially, the remainder of all of the rest of the work that we did for the simple harmonic oscillator in one dimension extends to higher dimensions in a similar fashion. We just apply the relevant operators to the different components of the states written in the Cartesian number basis using the tensor products. Hence, we can find the position-space and momentum-space wavefunctions, the coherent and squeezed states, and so on. There is a wealth of new information that we can work with by simply building off all of the previous work we have already completed. We will illustrate some of this in the problems, and we will return to the isotropic oscillator in later chapters.

Section 7.13 Problems:

7.13.1 Parity in Three Dimensions

Use the position and momentum operator basis for the three-dimensional simple harmonic oscillator Hamiltonian $\hat{H}$, and verify that $\hat{\Pi}\hat{H}\hat{\Pi} = \hat{H}$, signifying that $\hat{H}$ is an even operator. Repeat the calculation in the raising and lowering operator form of the Hamiltonian.

7.13.2 Degenerate States

Consider the isotropic three-dimensional simple harmonic oscillator. Determine all 15 degenerate states, by listing the different triples (n_x, n_y, n_z) that have energy $\frac{11}{2}\hbar\omega$.

7.13.3 Ground-State Wavefunction in Three Dimensions

Use the three-dimensional translation operator $e^{\frac{i}{\hbar}(r_x\hat{p}_x + r_y\hat{p}_y + r_z\hat{p}_z)}$ to calculate $\psi_{gs}(r_x, r_y, r_z) = \langle r_x, r_y, r_z|0, 0, 0\rangle$ for the Hamiltonian in Eq. (7.383).

Hint: first review Sect. 7.10 and develop a similar strategy to use but now for the three-dimensional case.

7.13.4 Linear Combination Is a Function of r

Construct the linear combination of ψ_{220}, ψ_{202}, and ψ_{022}, for the isotropic oscillator that is a function only of r.

7.13.5 Two-Dimensional Simple Harmonic Oscillator

Consider the general two-dimensional simple harmonic oscillator, whose Hamiltonian is

$$\hat{H} = \frac{\hat{p}_x^2}{2m} + \frac{\hat{p}_y^2}{2m} + \frac{1}{2}m\omega_x^2\hat{r}_x^2 + \frac{1}{2}m\omega_y^2\hat{r}_y^2. \tag{7.401}$$

(a) Determine all allowed energy levels.
(b) For the lowest five energy eigenvalues, determine the degeneracies for all states (and enumerate the states) for the isotropic case, with $\omega_x = \omega_y = \omega$.

7.13.6 *Uncertainty for the Two-Dimensional Isotropic Oscillator*

For the two-dimensional isotropic oscillator, determine the uncertainties:

$$(\Delta\vec{r})_{\vec{n}} = \sqrt{\langle\vec{n}|\hat{r}_x^2|\vec{n}\rangle + \langle\vec{n}|\hat{r}_y^2|\vec{n}\rangle - \langle\vec{n}|\hat{r}_x|\vec{n}\rangle^2 - \langle\vec{n}|\hat{r}_y|\vec{n}\rangle^2}$$
$$(\Delta\vec{p})_{\vec{n}} = \sqrt{\langle\vec{n}|\hat{p}_x^2|\vec{n}\rangle + \langle\vec{n}|\hat{p}_y^2|\vec{n}\rangle - \langle\vec{n}|\hat{p}_x|\vec{n}\rangle^2 - \langle\vec{n}|\hat{p}_y|\vec{n}\rangle^2}. \tag{7.402}$$

Determine the uncertainty product $(\Delta\vec{r})_{\vec{n}}\,(\Delta\vec{p})_{\vec{n}}$. What is the minimal uncertainty for the ground state?

7.14 Applications of the Simple Harmonic Oscillator

Our journey into the quantum-mechanical behavior of quantum particles has taken us a long way. We have developed many complex identities, and we have created a number of interesting quantum states. Nevertheless, much of this work was quite formal with complex algebraic manipulations that sometimes seemed to have no end. In such a situation, it is easy to "lose the forest for the trees." This section is intended to ground us back into physics again. We will examine three applications of the simple harmonic oscillator to quantum systems that describe important physical phenomena. This will help us relate the abstract math back to the real world.

All three of the examples we will develop involve the interaction of two quantum particles with each other. Hence, we must examine the so-called *two-body problem*, which involves the quantum behavior of two particles that interact with each other. The Hamiltonian is

$$\hat{H} = \frac{\hat{\vec{p}}_1^{\,2}}{2m_1} + \frac{\hat{\vec{p}}_2^{\,2}}{2m_2} + V(\hat{\vec{r}}_1, \hat{\vec{r}}_2), \tag{7.403}$$

where we work in three dimensions and the particles have mass m_1 and m_2, respectively. It is very common for the potential to depend only on the difference in the positions of the two particles—$V(\hat{\vec{r}}_1 - \hat{\vec{r}}_2)$—which is an assumption we make about the potential from now on.

If you have studied classical mechanics in physics before, you have probably already encountered the center of mass (COM) and the relative (REL) coordinates.

You can think of the COM coordinate as describing the motion of the entire system on average, while the relative coordinates determine the motion of each particle relative to the COM. If the potential depends just on the relative coordinates, then the COM motion is that of a free particle, which greatly simplifies the analysis.

Okay, enough talk. How do we make this separation? The COM coordinate is the "mass-weighted" position coordinate, which becomes

$$\hat{X}_{COM} = \frac{m_1\hat{x}_1 + m_2\hat{x}_2}{m_1 + m_2}, \tag{7.404}$$

for one dimension. Note how the mass-weighted position is divided by the total mass to ensure it has the right dimensions. Now, the momentum operator is already weighted by the mass, so we might conjecture the sum of the two momenta gives the COM momentum, but we proceed a bit more cautiously, because we need to require that the COM momentum has a canonical commutation relation with the COM position and we require it commutes with the relative position. This latter condition means our new degrees of freedom (COM and REL) will be independent of each other, just as the 1 and 2 degrees of freedom were independent of each other. So, we make an ansatz, that $\hat{P}_{COM}$ is a linear combination of the two momenta, or $\hat{P}_{COM} = \alpha\hat{p}_1 + \beta\hat{p}_2$, with α and β numbers. We require $[\hat{x}_{COM}, \hat{p}_{COM}] = i\hbar$. Using the facts that $[\hat{x}_1, \hat{p}_1] = i\hbar$ and $[\hat{x}_2, \hat{p}_2] = i\hbar$ and that the "1" operators commute with the "2" operators then tells us that

$$[\hat{X}_{COM}, \hat{P}_{COM}] = \frac{m_1}{m_1 + m_2}\alpha[\hat{x}_1, \hat{p}_1] + \frac{m_2}{m_1 + m_2}\beta[\hat{x}_2, \hat{p}_2], \tag{7.405}$$

so, we must have

$$\frac{m_1\alpha + m_2\beta}{m_1 + m_2} = 1, \tag{7.406}$$

in order to satisfy the commutation relation. While there are an infinite number of solutions to this, we need to also require that this COM momentum also commutes with the relative coordinate, given by $\hat{x}_{REL} = \hat{x}_1 - \hat{x}_2$. Then

$$[\hat{x}_{REL}, \hat{P}_{COM}] = (\alpha - \beta)i\hbar = 0. \tag{7.407}$$

This implies that $\alpha = \beta = 1$, or $\hat{P}_{COM} = \hat{p}_1 + \hat{p}_2$, as we anticipated earlier. Similarly, we have the relative momentum satisfies $\hat{p}_{REL} = \alpha'\hat{p}_1 + \beta'\hat{p}_2$. We require both

$$[\hat{X}_{COM}, \hat{p}_{REL}] = 0 \text{ and } [\hat{x}_{REL}, \hat{p}_{REL}] = i\hbar. \tag{7.408}$$

This means we have $m_1\alpha' + m_2\beta' = 0$ and $\alpha' - \beta' = 1$. This is solved by $\alpha' = \frac{m_2}{m_1+m_2}$ and $\beta' = -\frac{m_1}{m_1+m_2}$. Hence, we have

$$\hat{p}_{REL} = \frac{m_2\hat{p}_1 - m_1\hat{p}_2}{m_1 + m_2}. \tag{7.409}$$

Next, we need to reexpress $\hat{H}$ in terms of our new coordinates. We express $\hat{p}_i$ in terms of the COM and REL momentum as

$$\hat{p}_1 = \frac{m_1}{m_1 + m_2}\hat{P}_{COM} + \hat{p}_{REL} \text{ and } \hat{p}_2 = \frac{m_2}{m_1 + m_2}\hat{P}_{COM} - \hat{p}_{REL}, \tag{7.410}$$

for each spatial coordinate (but we focus on one dimension here). The kinetic energy becomes

$$\frac{\hat{p}_1^2}{2m_1} + \frac{\hat{p}_2^2}{2m_2} = \frac{\hat{P}_{COM}^2}{2(m_1 + m_2)} + \frac{\hat{P}_{REL}^2}{2\frac{m_1 m_2}{m_1+m_2}} = \frac{\hat{P}_{COM}^2}{2M_{TOT}} + \frac{p_{REL}^2}{2\mu_{RED}}, \tag{7.411}$$

where $M_{TOT} = m_1 + m_2$ is the total mass and $\mu_{RED} = \frac{m_1 m_2}{m_1+m_2}$ is the reduced mass. The two-particle Hamiltonian in three dimensions becomes

$$\hat{H} = \frac{\hat{\vec{P}}_{COM}^{\,2}}{2M_{TOT}} + \frac{\hat{\vec{P}}_{REL}^{\,2}}{2\mu_{RED}} + V(\hat{\vec{r}}_{REL}), \tag{7.412}$$

where we now sum over all three spatial directions. Note how the COM motion is decoupled from the relative motion; this is called separation of variables. The COM motion is that of a free particle (since $V = 0$), while the REL motion is that of an effective particle of mass μ_{RED} moving in a potential that depends only on the relative coordinate. If we write the wavefunction as a tensor product of the COM and REL motions as

$$|\psi\rangle = |\psi_{COM}\rangle \otimes |\psi_{REL}\rangle, \tag{7.413}$$

the COM motion is decoupled from the REL motion; meaning COM operators act only on the first component of the tensor product and REL operators act only on the second component. Because the COM motion is that of the free particle, we often ignore this motion. For completeness, notice that the COM state is a momentum eigenstate, so we have that

$$|\psi_{COM}\rangle = |p_{COM}\rangle \text{ and } \frac{\hat{P}_{COM}^{\,2}}{2M_{TOT}}|\psi_{COM}\rangle = \frac{P_{COM}^{\,2}}{2M_{TOT}}|\psi_{COM}\rangle. \tag{7.414}$$

Now that we have finished these preliminaries, we work on the *harmonium* atom Hamiltonian, which is an approximation to a conventional atom, but the interactions of the electrons are treated as harmonic oscillators, and we focus on one dimension for simplicity. Note that this is a completely made up problem that can be solved

exactly, but it has interesting properties shared by the solution of the hydrogen atom. The harmonium Hamiltonian then is

$$\hat{H}_{harmonium} = \frac{\hat{p}_1^2}{2m} + \frac{\hat{p}_2^2}{2m} + \frac{1}{2}k\hat{x}_1^2 + \frac{1}{2}k\hat{x}_2^2 + \frac{1}{2}k(\hat{x}_1 - \hat{x}_2)^2. \tag{7.415}$$

We are assuming equal masses of the particles and the potential does not depend solely on the relative coordinates. But a transformation to COM and REL coordinates still allows us to solve the problem, because it separates the variables. This is because the potential energy is a quadratic in the position variables. To see how, we set $\hat{p}_1 = \frac{1}{2}\hat{P}_{COM} + \hat{p}_{REL}$, $\hat{p}_2 = \frac{1}{2}\hat{P}_{COM} - \hat{p}_{REL}$, $\hat{x}_1 = \hat{X}_{COM} + \frac{1}{2}\hat{x}_{REL}$, and $\hat{x}_2 = \hat{X}_{COM} - \frac{1}{2}\hat{x}_{REL}$. Then

$$\hat{p}_1^2 + \hat{p}_2^2 = \frac{1}{2}\hat{P}_{COM}^2 + 2\hat{p}_{REL}^2, \tag{7.416}$$

$$\hat{x}_1^2 + \hat{x}_2^2 = 2\hat{X}_{COM}^2 + \frac{1}{2}\hat{x}_{REL}^2, \tag{7.417}$$

and we find that

$$\hat{H}_{harmonium} = \frac{1}{4m}\hat{P}_{COM}^2 + k\hat{X}_{COM}^2 + \frac{1}{m}\hat{P}_{REL}^2 + \frac{3}{4}k\hat{x}_{REL}^2 = \hat{H}_{COM} + \hat{H}_{REL}. \tag{7.418}$$

Both of the Hamiltonians (COM and REL) are simple harmonic oscillators, with

$$\omega_{COM} = \sqrt{\frac{k_{COM}}{m_{COM}}} = \sqrt{\frac{2k}{2m}} = \sqrt{\frac{k}{m}} = \omega, \tag{7.419}$$

and

$$\omega_{REL} = \sqrt{\frac{k_{REL}}{m_{REL}}} = \sqrt{\frac{\frac{3}{2}k}{\frac{1}{2}m}} = \sqrt{\frac{3k}{m}} = \sqrt{3}\omega. \tag{7.420}$$

The energy $|n_{COM}\rangle \otimes |n_{REL}\rangle = |n_{COM}, n_{REL}\rangle$ is then

$$E_{n_{COM},n_{REL}} = \hbar\omega\left(n_{COM} + \frac{1}{2} + \sqrt{3}n_{REL} + \frac{\sqrt{3}}{2}\right). \tag{7.421}$$

The transformation to COM and REL coordinates changed the problem from a coupled two-particle problem to two decoupled effective single particle problems! Solving the latter was easy. If we want to find the solution in terms of the original variables, we have to convert from COM and REL back to 1 and 2.

Our next problem is a real physical problem, the problem of the deuteron, which is the nucleus of the isotope of hydrogen that has one proton and one neutron. It

is a critically important particle in the evolution of the universe after the big bang, because it ultimately allowed heavier atoms to form and become stable. This is because, initially, the universe contained free protons, neutrons, and electrons in a superhot soup. As the system cooled, the protons bound to the electrons to form hydrogen, and neutrons bound to protons to form the deuteron. If the latter did not happen, the neutrons, which are unstable, would shortly have decayed (free neutrons are unstable with a half-life of about 15 minutes) and none of the heavier atoms could have formed, as there would have been no neutrons left. The deuteron is also an intermediate step toward forming the He^4 nucleus, and in the big bang, the main elements that formed were hydrogen and helium. Heavier elements formed primarily in stars (the heaviest in neutron stars). So, the fact that a proton and a neutron bind together to form a deuteron is a critical step toward the formation of all elements.

Nuclear physics is a complicated field and the interaction between nucleons is not as well known as it is between charged particles. So, it is common to try different forms of potentials to see if they provide appropriate descriptions of the nucleons. We neglect the mass difference between a proton and a neutron, because they are so similar. We try a potential that decays like a Gaussian, so that

$$\hat{H}_{deuteron} = \frac{\hat{\vec{p}}_1^{\,2}}{2m} + \frac{\hat{\vec{p}}_2^{\,2}}{2m} - Ve^{-\beta^2(\hat{\vec{r}}_1 - \hat{\vec{r}}_2)^2}, \tag{7.422}$$

with β a constant with dimensions of inverse length. If β is small, then we can approximate the exponential by the first two terms of the power series for the exponential, which is a harmonic oscillator potential in three dimensions:

$$-Ve^{-\beta^2(\hat{\vec{r}}_1 - \hat{\vec{r}}_2)^2} \approx -V + V\beta^2\left(\hat{r}_{x,REL}^2 + \hat{r}_{y,REL}^2 + \hat{r}_{z,REL}^2\right). \tag{7.423}$$

The transformed Hamiltonian is then

$$\hat{H}_{deuterium} = \frac{\hat{\vec{P}}_{COM}^{\,2}}{4m} + \frac{\hat{\vec{p}}_{REL}^{\,2}}{m} - V + V\beta^2\left(\hat{r}_{x,REL}^2 + \hat{r}_{y,REL}^2 + \hat{r}_{z,REL}^2\right). \tag{7.424}$$

The COM motion is that of a free particle, while the relative motion is that of a simple harmonic oscillator, with a ground-state energy of $\frac{3}{2}\hbar\omega - V$. The frequency satisfies $\omega_{REL} = \sqrt{\frac{2V\beta^2}{\frac{1}{2}m}}$. Hence, we find the binding energy (which is the negative of the ground-state energy) is

$$E_{bind} = V - 3\hbar\beta\sqrt{\frac{V}{m}}. \tag{7.425}$$

This is a rough estimate of the binding energy. It turns out that evaluating

$$\langle 000| - \frac{\hat{\vec{p}}_{REL}^{\,2}}{m} + Ve^{-\beta^2(\hat{\vec{r}}_1 - \hat{\vec{r}}_2)^2}|000\rangle \approx E_{bind} \tag{7.426}$$

gives a more accurate estimate to the binding energy. Let us now see why.

There is a principle in quantum mechanics called the *variational princple*, which says that if we make a guess to the ground-state wavefunction $|\psi_{guess}\rangle$, then

$$E_{guess} = \langle \psi_{guess}|\hat{H}|\psi_{guess}\rangle \tag{7.427}$$

produces an estimate for the ground-state energy that is guaranteed to be equal to or higher than the ground-state energy. In the variational principle's most common usage, the guess wavefunction depends on some parameters, and by minimizing the energy with respect to those parameters produces the best guess for the ground-state energy (within the class of wavefunctions considered in the guess). The minimization principle requires calculus to carry out, so we will not be discussing that aspect here. But we can prove that

$$E_{gs} \leq \langle \psi_{guess}|\hat{H}|\psi_{guess}\rangle \tag{7.428}$$

rather easily and the proof is an important result in quantum mechanics.

Suppose we let $|n\rangle$ denote the energy eigenstates of $\hat{H}$, such that $\hat{H}|n\rangle = E_n|n\rangle$ is the energy eigenvalue (Schrödinger) equation. Then, because $\{|n\rangle\}$ is a complete set, we can expand

$$|\psi_{guess}\rangle = \sum_n c_n|n\rangle, \tag{7.429}$$

with $\langle \psi_{guess}|\psi_{guess}\rangle = 1$, implying that $\sum_n |c_n|^2 = 1$. We evaluate

$$\begin{aligned}\langle \psi_{guess}|\hat{H}|\psi_{guess}\rangle &= \sum_m \sum_n c_m^* \langle m|\hat{H}|n\rangle c_n = \sum_m \sum_n c_m^* c_n E_n \langle m|n\rangle \\ &= \sum_m \sum_n c_m^* c_n E_n \delta_{mn} = \sum_n |c_n|^2 E_n. \end{aligned} \tag{7.430}$$

But, because $\sum_n |c_n|^2 = 1$, we have $E_{gs} = E_{gs}\sum_n |c_n|^2$. Then, we have

$$\langle \psi_{guess}|\hat{H}|\psi_{guess}\rangle - E_{gs} = \sum_n |c_n|^2 (E_n - E_{gs}). \tag{7.431}$$

But $E_n \geq E_{gs}$, $\forall n$, so $E_n - E_{gs} \geq 0$. Since $|c_n|^2 \geq 0$ too, we have that the summation is over all nonnegative terms, so

$$\langle \psi_{guess}|\hat{H}|\psi_{guess}\rangle - E_{gs} \geq 0, \tag{7.432}$$

or

$$E_{gs} \leq \langle \psi_{guess} | \hat{H} | \psi_{guess} \rangle, \tag{7.433}$$

which is what we wanted to show.

We can use our variational principle to improve the estimate of the energy of our deuteron using the exact Hamiltonian, via

$$E_{bind} \approx -\left\langle 000 | \frac{\hat{\vec{p}}_{REL}^{\,2}}{m} | 000 \right\rangle + \langle 000 | V e^{\beta^2 (\hat{\vec{r}}_1 - \hat{\vec{r}}_2)^2} | 000 \rangle. \tag{7.434}$$

The first expectation value of the kinetic energy can be determined to be one half of the previous total energy estimate (when we used the approximate simple harmonic oscillator Hamiltonian rather than the exact one) because the virial theorem tells us a harmonic oscillator splits its average energy exactly between the potential and kinetic energies. You will evaluate the second expectation value in Problem 7.14.3 and compare your final result to the simple harmonic oscillator Hamiltonian approximation.

Our final problem is to examine a simple model for the vibrational spectra of a molecule that uses the Morse potential. The Morse potential can be solved exactly using operator methods, and we will return to it later, after we have fully developed the operator methods. Here, we will apply the harmonic oscillator approximation to the Morse potential and compare it to the exact solution. The Morse potential is used to describe the potential that a diatomic molecule is in and describe its vibrational motion. Most molecules have an anharmonic potential, because it is more difficult to compress them, than it is to stretch them. So, the potential has a longer tail for large extension of the molecule (before the molecule dissociates). This is precisely the behavior of the Morse potential, shown in Fig. 7.16.

The electrons in molecules form a "glue" that holds the molecule together and acts like an effective spring that connects the two ions of a diatomic molecule. As the ions get pushed close together, the electrons can no longer stay between the ions, so they are pushed away—the interaction then becomes strongly repulsive because the two ions have the same sign of their electric charge. Similarly, as we pull them far apart, the chemical bond disappears, because the electrons separate and go to the neutral atomic constituents—neutral atoms do not bond strongly together. So, we expect the potential to be large and positive for small r and approach 0 for large r, with some minimum in between. The molecule is stable near the minimum and is held together by the chemical bond. We consider an isoatomic diatomic molecule, such as H_2 or O_2, so $\mu_{RED} = \frac{1}{2}m$.

We use the Morse potential to describe the one-dimensional degree of freedom given by the distance between the two ions:

$$V_{Morse}(\hat{r}) = D_e \left(e^{-2a(\hat{r}-r_e)} - 2e^{-a(\hat{r}-r_e)} \right) = D_e \left(1 - e^{-a(\hat{r}-r_e)} \right)^2 - D_e. \tag{7.435}$$

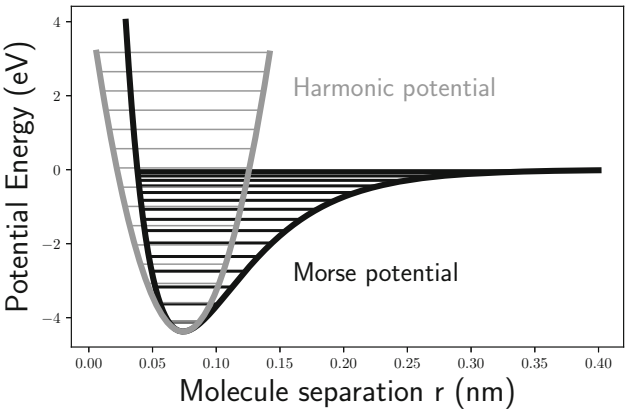

Fig. 7.16 The Morse potential (black) and its harmonic approximation (gray). We use $D_e = 4.75$ eV, $r_e = 0.074$ nm, and $a = 19.3$ nm^{-1} for the Morse potential, and the harmonic potential approximates the Morse potential near its minimum; we have $\mu_{RED}c^2 = 469$ MeV. See how the harmonic oscillator potential is symmetric about its origin, but the Morse potential is asymmetric—the latter is closer to the true potential of diatomic atoms in a molecule. The corresponding energy levels are also plotted. Note how the harmonic oscillator levels are evenly spaced and approximate the Morse potential levels well near the ground state. The Morse potential levels get closer to each other for higher excited states

In the second form, one can immediately see that the minimum occurs when $r = r_e$ and the minimum value of the potential is $-D_e$. In fitting the Morse potential to a given molecule, we have three parameters we need to determine a, r_e, and D_e. The spring constant can be found by expanding about $r = r_e$, which gives

$$V_{Morse}(\hat{r}) \approx -D_e + D_e a^2(\hat{r} - r_e)^2, \tag{7.436}$$

for a spring constant equal to $k = 2D_e a^2$. Hence, the frequency for the effective harmonic oscillator is $\omega = \sqrt{\frac{k}{\frac{1}{2}m}} = \sqrt{\frac{4D_e a^2}{m}} = 2a\sqrt{\frac{D_e}{m}}$. The energy levels, in the harmonic approximation, are then

$$E_n^{SHO}(Morse) \approx 2\hbar a\sqrt{\frac{D_e}{m}}\left(n + \frac{1}{2}\right) - D_e. \tag{7.437}$$

The exact energy levels are

$$E_n^{exact}(Morse) = 2\hbar a\sqrt{\frac{D_e}{m}}\left(n + \frac{1}{2}\right) - \frac{\hbar^2 a^2}{m}\left(n + \frac{1}{2}\right)^2 - D_e. \tag{7.438}$$

For small n, the approximate and exact energies are quite close to each other. But they start to deviate more and more as n is increased. We work out some

comparisons quantitatively in Problems 7.14.4 and 7.14.5. Figure 7.16 also shows how this works for a specific example. Note that the Morse energies only hold for n values where $E_n < 0$ or $E_{n+1} > E_n$. The potential only has a finite number of bound states.

These examples provide some circumstances where the simple harmonic oscillator is employed to describe real quantum systems. The other system that is described by the simple harmonic oscillator has already been mentioned before—it is the way we describe photons. Indeed, this is the most important application of the simple harmonic oscillator, and we will discuss this more fully later.

Section 7.14 Problems:

7.14.1 Reduced Mass in Atomic Units

In atomic units, we measure masses in units of energy, so $m \rightarrow mc^2$, with c the speed of light (you can think of this as a multiply by one if we multiply and divide by c^2). We use electron volts as the energy scale. In these units, the mass of an electron is $m_e c^2 \approx 511{,}000$ eV and the mass of a proton is $m_p c^2 \approx 938{,}000{,}000$ ev, and so on. Compute the reduced mass:

$$\mu_{RED} c^2 = \frac{m_p c^2 m_e c^2}{m_p c^2 + m_e c^2}, \tag{7.439}$$

for the hydrogen atom with one proton and one electron. How much less than the electron mass is the reduced mass? Repeat for positronium, which is composed of an electron and an antielectron, each having the same mass.

In general, we have $\frac{1}{2} m_{min} \leq \mu_{RED} \leq m_{min}$, where m_{min} is the smaller of the two masses. Can you prove this general result? The two above examples show the limiting behaviors.

7.14.2 Bogoliubov Transformation

Consider the Hamiltonian:

$$\hat{H} = \frac{D_1}{2}\left(\hat{a}^\dagger \hat{a}^\dagger + \hat{a}\hat{a}\right) + \frac{D_2}{2}\left(\hat{a}^\dagger \hat{a} + \hat{a}\hat{a}^\dagger\right) + D_3, \tag{7.440}$$

where all D_i are real. Use the Bogoliubov transformation, which we used when we described squeezing, to define

$$\hat{A}^\dagger = \alpha \hat{a}^\dagger + \beta \hat{a} \quad \text{and} \quad \hat{A} = \beta^* \hat{a}^\dagger + \alpha^* \hat{a}. \tag{7.441}$$

Find the condition on α and β such that

$$[\hat{A}^\dagger, \hat{A}] = 1, \tag{7.442}$$

and write α and β in terms of hyperbolic functions for the general form that preserves the commutator. Recall that $\cosh^2 x - \sinh^2 x = 1$. Pick appropriate α and β so that you can convert the Hamiltonian into

$$\hat{H} = c\hat{A}^\dagger \hat{A} + c', \tag{7.443}$$

and determine both c and c'. Finally, find the energy eigenvalues of $\hat{H}$, by converting to the form $E_n = cn + c'$ to finally express how it depends on D_1, D_2, and D_3.

Hint: after finding α and β, solve for the original raising and lowering operators in terms of $\hat{A}^\dagger$ and $\hat{A}$ and substitute into the original form of $\hat{H}$ to find c and c'.

7.14.3 Deuteron Variational Energy

In this problem, we compute the deuteron variational energy:

$$E_{gs}^{var} = \left\langle 000 \middle| \frac{\hat{\vec{p}}^{\,2}}{2\mu_{RED}} \middle| 000 \right\rangle + \langle 000| - V e^{-\beta^2(\hat{\vec{r}}_1 - \hat{\vec{r}}_2)^2} |000\rangle. \tag{7.444}$$

Using our tensor product notation, show that

$$\begin{aligned} E_{gs}^{var} &= \frac{3}{2}\hbar\beta\sqrt{\frac{V}{m}} - V\,\langle 0|e^{-\beta^2\hat{r}_x^2}|0\rangle\,\langle 0|e^{-\beta^2\hat{r}_y^2}|0\rangle\,\langle 0|e^{-\beta^2\hat{r}_z^2}|0\rangle \\ &= \frac{3}{2}\hbar\beta\sqrt{\frac{V}{m}} - V\left(\langle 0|e^{-\beta^2\hat{r}_x^2}|0\rangle\right)^3. \end{aligned} \tag{7.445}$$

Note that the kinetic energy is computed via the viral theorem for the simple harmonic oscillator, as described in the text.

The expectation value is computed by recalling Problem 7.9.7, which used a "generating function" approach to computing the expectation value. We found that

$$\langle 0|e^{-\beta^2\hat{r}_x^2}|0\rangle = \sum_{n=0}^{\infty} \frac{(-\beta^2)^n}{n!}\langle 0|\hat{r}_x^{2n}|0\rangle \tag{7.446}$$

and recalling that we found $\langle 0|\hat{r}_x^{2n}|0\rangle = \left(\frac{\hbar}{4m\omega}\right)^n \frac{(2n)!}{n!}$.

Use the identity:

$$\sum_{n=0}^{\infty}\left(\frac{z}{2}\right)^n \frac{(2n)!}{(n!)^2} = \frac{1}{\sqrt{1-2z}}, \tag{7.447}$$

in your computation. Determine how much lower the proper calculation of the variational energy is than what we found for the harmonic oscillator approximation of the Hamiltonian (which gave $-V+3\hbar\beta\sqrt{\frac{V}{m}}$). The deuteron size is approximately 2×10^{-6} nm. Use $\beta = 2 \times 10^5$ nm^{-1}, $V = 19.1$ MeV, and $mc^2 = 938$ MeV. Note that the deuteron binding energy is about 2.2 MeV. You may find using $\hbar c = \frac{1240}{2\pi}$ eV-nm will be helpful to you.

Note that this is an illustrative problem. The potential is not a particularly good potential for the deuteron. The real deuteron has only one weakly bound state, as does this one. But the shape of the potential is not so accurate. We chose this potential to allow you to see the difference between the harmonic oscillator energy for the ground state and a better approximation to the ground-state energy for a nontrivial problem that we found by using the variational principle.

7.14.4 Morse Potential for the Hydrogen Molecule H_2

Consider the Morse potential for an H_2 molecule with $D_e = 4.75$ eV, $r_e = 0.074$ nm, and $a = 19.3$ nm^{-1}. Using the approximation of a harmonic oscillator, determine the lowest-energy state. How many harmonic oscillator levels lie below $E = 0$? For each level with $E < 0$, compare the harmonic oscillator energy to the exact energy and compute the relative error:

$$\left| \frac{E_n^{SHO} - E_n^{Morse}}{E_n^{Morse}} \right|, \tag{7.448}$$

for all states with $E < 0$.

7.14.5 Comparing the Morse Potential Energies to the Simple Harmonic Oscillator Energies

Using the same parameters for the Morse potential as used in the previous problem, take the ground state of the simple harmonic oscillator and compute the exact variational energy given by

$$E_{var}^{Morse} = \left\langle 0| \frac{\hat{p}_x^2}{2\mu_{RED}} |0 \right\rangle + \langle 0|V_{Morse}(\hat{r}_x)|0\rangle, \tag{7.449}$$

by noting the simple harmonic oscillator results for the kinetic energy and by expanding the Morse potential using the power series for the exponential. Compute the relative error:

$$\left| \frac{E_{var}^{SHO} - E_0^{Morse}}{E_0^{Morse}} \right|, \tag{7.450}$$

for the ground state. Comment on the accuracy of the variational calculation versus the simple harmonic oscillator prediction. If you are adventurous, repeat for all eigenstates with $E_{SHO} < 0$.

7.14.6 Trapped Quantum Particle

In this problem, we consider a charged quantum particle trapped in a simple harmonic oscillator potential and placed in an uniform electric field. The Hamiltonian (in one-dimension) is

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 - q\varepsilon\hat{x}, \tag{7.451}$$

since the potential energy of a particle with charge q in an electric field of strength ε is $-q\varepsilon\hat{x}$. Complete the square of the potential energy and noting that adding a constant to the definition of $\hat{a}$ and $\hat{a}^\dagger$ does not change the commutators, deduce that

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right) - \frac{q^2\varepsilon^2}{2m\omega^2}. \tag{7.452}$$

Hence, a charged particle in an electric field just has its energy shifted downward due to the field and has the equilibrium position shifted by $x_0 = \frac{q\varepsilon}{m\omega^2}$. Note that the simple harmonic oscillator is also used to describe the motion of quantum particles in a magnetic field, but to do so requires a fair amount of calculus, so we will not discuss that further here.

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Chapter 8
Summary of the Fundamental Operator Identities

The abstraction is often the most definite form for the intangible thing in myself.

Georgia O'Keefe, Artist

This chapter is a short one. We summarize here all of the operator identities we have already derived and that we will use throughout the book. We describe them in general and in simplified forms. This chapter is to be used primarily as a reference, so there are no problems at the end of each section. It is the place to go if you are looking to use an identity but forgot the precise form. A summary of the chapter is also available at https://tinyurl.com/48y7uarc.

8.1 First Fundamental Identity: Leibniz Product Rule for Commutators

The commutator rule has two forms:

$$[\hat{A}, \hat{B}\hat{C}] = \hat{B}[\hat{A}, \hat{C}] + [\hat{A}, \hat{B}]\hat{C} \quad \text{and} \quad [\hat{A}\hat{B}, \hat{C}] = \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B}. \tag{8.1}$$

The mnemonic to remember these is that we pull out the factors in the product to the left or the right depending on the order of the terms in the product.

There is a special case, when $[\hat{A}, \hat{B}]$ commutes with $\hat{A}$. Then we have

$$[\hat{A}^n, \hat{B}] = n[\hat{A}, \hat{B}]\hat{A}^{n-1}. \tag{8.2}$$

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A second commutator identity is the so-called Jacobi identity. It takes the form

$$[\hat{A}, [\hat{B}, \hat{C}]] + [\hat{B}, [\hat{C}, \hat{A}]] + [\hat{C}, [\hat{A}, \hat{B}]] = 0. \tag{8.3}$$

The proof for this is done by simply expanding out each nested commutator and then showing all terms cancel.

8.2 Second Fundamental Identity: Hadamard Lemma

The Hadamard lemma is an exponential-like series but with increasingly higher-rank nested commutators that represent the similarity transformation of an operator $\hat{B}$:

$$e^{\hat{A}}\hat{B}e^{-\hat{A}} = \hat{B} + \sum_{n=1}^{\infty} \frac{1}{n!}[\hat{A}, [\hat{A}, \cdots, [\hat{A}, \hat{B}]\cdots]]_n. \tag{8.4}$$

The n subscript is to remind us that the nested commutator is n-fold nested. This identity has a few different corollaries. The first is the exponential-re-ordering identity

$$e^{\hat{A}}e^{\hat{B}} = e^{e^{\hat{A}}\hat{B}e^{-\hat{A}}}e^{\hat{A}} = e^{\hat{B}+\sum_{n=1}^{\infty}\frac{1}{n!}[\hat{A},[\hat{A},\cdots,[\hat{A},\hat{B}]\cdots]]_n}e^{\hat{A}}. \tag{8.5}$$

In the Weyl form, when $[\hat{A}, \hat{B}]$ commutes with $\hat{A}$ and $\hat{B}$, we have the special case

$$e^{\hat{A}}e^{\hat{B}} = e^{\hat{B}}e^{\hat{A}}e^{[\hat{A},\hat{B}]}. \tag{8.6}$$

The second is the braiding identity, which holds for any function f that can be expanded in a power series:

$$e^{\hat{A}}f(\hat{B})e^{-\hat{A}} = f\left(e^{\hat{A}}\hat{B}e^{-\hat{A}}\right) = f\left(\hat{B} + \sum_{n=1}^{\infty}\frac{1}{n!}[\hat{A}, [\hat{A}, \cdots, [\hat{A}, \hat{B}]\cdots]]_n\right). \tag{8.7}$$

The special cases correspond to when the nested commutators truncate, or when they repeat in a pattern. The former is already covered by the general result; just the summation is finite. For the repeating pattern, there are three possibilities we will cover: (i) the exponential case, where $[\hat{A}, \hat{B}] = \hat{B}$; (ii) the "trigonometric" case $[\hat{A}, \hat{B}] = i\hat{C}$ and $[\hat{A}, \hat{C}] = -i\hat{B}$; and (iii) the "hyperbolic" case, where $[\hat{A}, \hat{B}] = \hat{C}$ and $[\hat{A}, \hat{C}] = \hat{B}$. In the exponential case, we have

$$e^{\alpha\hat{A}}\hat{B}e^{-\alpha\hat{A}} = e^{\alpha}\hat{B}; \tag{8.8}$$

in the trigonometric case, we have

$$e^{i\alpha\hat{A}}\hat{B}e^{-i\alpha\hat{A}} = \cos\alpha\ \hat{B} - \sin\alpha\ \hat{C}; \tag{8.9}$$

and for the hyperbolic case, we have

$$e^{\alpha\hat{A}}\hat{B}e^{-\hat{A}} = \cosh\alpha\ \hat{B} + \sinh\alpha\ \hat{C}. \tag{8.10}$$

8.3 Third Fundamental Identity: Baker-Campbell-Hausdorff Formula

While the Baker-Campbell-Hausdorff formula does have a general expression, it is quite complicated and will not be repeated here. That exact formula does not appear to be particularly useful in calculations. Instead, we report the first few terms of the BCH formula as follows:

$$e^{\hat{A}}e^{\hat{B}} = e^{\hat{A}+\hat{B}+\frac{1}{2}[\hat{A},\hat{B}]+\frac{1}{12}\left([\hat{A},[\hat{A},\hat{B}]]+[\hat{B},[\hat{B},\hat{A}]]\right)-\frac{1}{24}[\hat{B},[\hat{A},[\hat{A},\hat{B}]]]+\cdots}. \tag{8.11}$$

The most important special case is the Weyl-form of the BCH formula, which applies when $[\hat{A},[\hat{A},\hat{B}]] = [\hat{B},[\hat{A},\hat{B}]] = 0$. In this case, we have

$$e^{\hat{A}}e^{\hat{B}} = e^{\hat{A}+\hat{B}}e^{\frac{1}{2}[\hat{A},\hat{B}]}, \tag{8.12}$$

because the commutator can be removed from the exponent, because it commutes with $\hat{A}$ and $\hat{B}$.

There is another form of the BCH formula, where the terms are organized according to their factors rather than putting everything into one exponent. This is called the Zassenhaus formula. The special case is precisely the Weyl-form of the BCH formula, given in the equation above. In the general case, it takes the form

$$e^{\hat{A}+\hat{B}} = e^{\hat{A}}e^{\hat{B}}e^{-\frac{1}{2}[\hat{A},\hat{B}]}e^{\frac{1}{3}[\hat{B},[\hat{A},\hat{B}]]+\frac{1}{6}[\hat{A},[\hat{A},\hat{B}]]}\cdots, \tag{8.13}$$

for the lowest-order terms.

Note that there are some other special cases where the BCH formula can be explicitly determined, but we will not discuss them further here, as they are not required for the work in this book. But you can refer to Wilcox (1967) and Foulis (2017) for more details.

8.4 Fourth Fundamental Identity: Exponential Disentangling Identities

The exponential disentangling identities arise from properties of Lie algebras and Lie groups and were originally worked out, in the physics context, in Arechi et al.

(1972). In this book, we focus on identities for the SU(2) and Sp(2) groups. There are two main identities that we use, and the Sp(2) group has two special cases that are important.

In the SU(2) case, we summarize the identities in terms of the spin operators. The form also holds for angular momentum vectors, which we will meet in Chap. 9. The two main identities we need are the following:

$$e^{\frac{i}{\hbar}\theta\hat{S}_y} = e^{-\tan\frac{\theta}{2}\frac{\hat{S}_-}{\hbar}} e^{\ln\left(\cos^2\frac{\theta}{2}\right)\frac{\hat{S}_z}{\hbar}} e^{\tan\frac{\theta}{2}\frac{\hat{S}_+}{\hbar}} \tag{8.14}$$

$$= e^{\tan\frac{\theta}{2}\frac{\hat{S}_+}{\hbar}} e^{-\ln\left(\cos^2\frac{\theta}{2}\right)\frac{\hat{S}_z}{\hbar}} e^{-\tan\frac{\theta}{2}\frac{\hat{S}_-}{\hbar}}. \tag{8.15}$$

Note the sign change in the middle factor for the two different identities. These identities require $-\pi < \theta < \pi$

In the symplectic group, we have that the raising and lowering operators map to the symplectic matrices in the following way:

$$\frac{1}{2}\left(\hat{a}^\dagger\right)^2 \leftrightarrow \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix} = K_+ \tag{8.16}$$

$$\frac{1}{4}\left(\hat{a}^\dagger\hat{a} + \hat{a}\hat{a}^\dagger\right) \leftrightarrow \frac{1}{2}\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = K_0 \tag{8.17}$$

$$\frac{1}{2}\hat{a}^2 \leftrightarrow \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = K_-. \tag{8.18}$$

The exponential operator depends on two parameters ξ, which can be complex and η, which must be real. We also must have $|\xi|^2 \geq \eta^2$. We define two new parameters from these:

$$\lambda = \xi\frac{\sinh\sqrt{|\xi|^2 - \eta^2}}{\sqrt{|\xi|^2 - \eta^2}}, \tag{8.19}$$

and

$$\kappa = \cosh\sqrt{|\xi|^2 - \eta^2} + i\eta\frac{\sinh\sqrt{|\xi|^2 - \eta^2}}{\sqrt{|\xi|^2 - \eta^2}}. \tag{8.20}$$

Then, we have the two main exponential disentangling identities are:

$$e^{-\xi K_+ + 2i\eta K_0 + \xi^* K_-} = e^{-\frac{\lambda}{\kappa^*}K_+} e^{-2\ln(\kappa^*)K_0} e^{\frac{\lambda^*}{\kappa^*}K_-} \tag{8.21}$$

$$= e^{\frac{\lambda}{\kappa}K_-} e^{2\ln(\kappa)K_0} e^{-\frac{\lambda}{\kappa}K_+}, \tag{8.22}$$

where we have the sign change on the middle term again. Converting to the operator form then gives us

$$e^{-\frac{\xi}{2}(\hat{a}^\dagger)^2+\frac{i}{2}\eta(\hat{a}^\dagger\hat{a}+\hat{a}\hat{a}^\dagger)+\frac{\xi^*}{2}\hat{a}^2} = e^{-\frac{\lambda}{2\kappa^*}(\hat{a}^\dagger)^2}e^{-\frac{1}{2}\ln(\kappa^*)(\hat{a}^\dagger\hat{a}+\hat{a}\hat{a}^\dagger)}e^{\frac{\lambda^*}{2\kappa^*}\hat{a}^2} \tag{8.23}$$

$$= e^{\frac{\lambda}{2\kappa}\hat{a}^2}e^{\frac{1}{2}\ln(\kappa)(\hat{a}^\dagger\hat{a}+\hat{a}\hat{a}^\dagger)}e^{-\frac{\lambda}{2\kappa}(\hat{a}^\dagger)^2}. \tag{8.24}$$

The special case is the situation where the raising and lowering operators correspond to operators proportional to $\hat{x}^2$ or $\hat{p}^2$. This means $\xi = -ik$ and $\eta = \pm k$; it is complicated, because $|\xi|^2 = \eta^2$, which requires a limiting procedure to compute. Instead, one can just compute directly for that case. We find that

$$e^{ik(\hat{a}^\dagger\pm\hat{a})^2} = e^{\frac{ik}{1\mp 2ik}(\hat{a}^\dagger)^2}e^{-\frac{1}{2}\ln(1\mp 2ik)(\hat{a}^\dagger\hat{a}+\hat{a}\hat{a}^\dagger)}e^{\frac{ik}{1\mp 2ik}\hat{a}^2}. \tag{8.25}$$

This is the last identity we have for this chapter. We have a video https://tinyurl.com/yyak645w summarizing them too.

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Chapter 9
Angular Momentum and Spin

Keep exploring. Keep dreaming. Keep asking why. Don't settle for what you already know. Never stop believing in the power of your ideas, your imagination, your hard work to change the world.

Barack Obama, 44th President of the United States, Nobel Laureate, 2015

In this chapter, we develop the background needed to move our quantum mechanical formalism fully into three dimensions. We discuss angular momentum, which is the motional complement to spin angular momentum, that we developed earlier. There are some critical differences between orbital and spin angular momentum, and we need to carefully discuss how they relate and how they differ. Why is angular momentum so important? Because in problems where we have central forces (forces that depend only on the distance between two particles and act along the line connecting the two particles), the angular momentum is conserved. So, it becomes an integral part of how we solve the problem. The most important problem we have to solve, the hydrogen atom, is a central force problem; hence, we need this work before we can move forward to some really interesting physics that will come in Chap. 11. We have a video tour of the chapter as well. See https://tinyurl.com/ydv5yvnx.

9.1 Commutation Relations for Position, Momentum, and Angular Momentum

Angular momentum is a difficult concept to master. On the one hand, we have all seen and experienced rotational motion—from a merry-go-round to a car tire. Clearly, it takes some impetus to get objects to start to rotate and angular momentum

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relates to the rotational motion just as linear momentum relates to the motion along a straight line. But, on the other hand, quantitative descriptions of angular motion require mastering complex vector dynamics, with forces replaced by torques, and cross products appearing all over the place. You are not expected to be a master of that material, which usually is taught as an advanced undergraduate class. Nevertheless, the bottom line is we just don't tend to have a good intuitive feel for angular momentum, and so it is just much more challenging for us to work with.

Fortunately, in quantum mechanics, many of these challenges do not create significant roadblocks. This is because we work with operators, commutators, and states. While these are abstract, they tend not to require as sophisticated visualizations of vectors in three dimensions, as the classical mechanics analysis does.

But, before we jump into calculating operators and their commutators, let us first focus on why angular momentum is important in the first place. Its importance stems from the fact that Coulomb's law is a central force law for the attraction of opposite charges. This means the force of attraction between a negatively charged electron and a positively charged nucleus depends only on the distance between the electron and the nucleus and lies along the line connecting the relative positions of the electron and the nucleus. Such forces are called *central forces* (gravity is also a central force), and angular momentum is conserved for particles moving in a central force field. Hence, understanding angular momentum is critical for understanding atoms. This is why we must delve into this topic in quantum mechanics.

In this section, we will explore the theory of angular momentum, focusing on the commutation relations of the angular momentum algebra. We employ a similar operator method as we used for spin in Chap. 6 and the simple harmonic oscillator in Chap. 7. This general case lies in between the two extremes we studied in those chapters—for spin-one-half, we could raise a spin at most once, while for the simple harmonic oscillator, we can raise the excitations of a state an infinite number of times. For angular momentum, we can raise a finite number of times, but often more than once. The details for how this works will be derived in Sect. 9.3.

Recall from Chap. 1 that in order to cause something to rotate, we must apply a push on one side and a pull on the other, as depicted in Fig. 1.3. These pushes and pulls must be located a distance away from each other in order to cause a rotation (called a *lever arm*). The angular momentum operator $\hat{\vec{L}}$ is then defined to be the cross product between the position and momentum operators, or

$$\hat{\vec{L}} = \hat{\vec{r}} \times \hat{\vec{p}} \text{ and } \hat{L}_i = \sum_{jk} \epsilon_{ijk} \hat{r}_j \hat{p}_k, \tag{9.1}$$

which ends up "measuring" how much an object rotates (recall that the Levi-Civita totally antisymmetric tensor ϵ_{ijk} vanishes if any two indices are the same and is one or minus one, depending on whether the permutation ijk is an even or an odd permutation of $x\ y\ z$). To start, note that $\hat{\vec{L}}$ is well defined, because it involves

position and momentum operators in perpendicular directions (which commute with each other). So we have

$$\hat{L}_x = \hat{r}_y \hat{p}_z - \hat{r}_z \hat{p}_y, \quad \hat{L}_y = \hat{r}_z \hat{p}_x - \hat{r}_x \hat{p}_z, \quad \text{and} \quad \hat{L}_z = \hat{r}_x \hat{p}_y - \hat{r}_y \hat{p}_x, \tag{9.2}$$

and the order of the operators in the factors is unimportant, because they all commute with each other.

As with all of our operators, we need to determine the commutators of angular momentum with position and momentum. Once we know all commutators, then we can work out different operator identities and determine the eigenvalues and eigenstates algebraically, just as we have done before. So, let us start with evaluating

$$[\hat{r}_i, \hat{L}_j] = \left[\hat{r}_i, \sum_{kl} \epsilon_{jkl} \hat{r}_k \hat{p}_l \right]. \tag{9.3}$$

First, we distribute the commutator over the rightmost product of the operators using the Leibniz rule

$$[\hat{r}_i, \hat{L}_j] = \sum_{kl} \epsilon_{jkl} \left(\hat{r}_k [\hat{r}_i, \hat{p}_l] + [\hat{r}_i, \hat{r}_k] \hat{p}_l \right). \tag{9.4}$$

The first commutator is $i\hbar\delta_{il}$, while the second vanishes, to yield

$$[\hat{r}_i, \hat{L}_j] = \sum_{kl} \epsilon_{jkl} i\hbar \hat{r}_k \delta_{il} = i\hbar \sum_{k} \epsilon_{ijk} \hat{r}_k. \tag{9.5}$$

This result follows, because the sum over l forces $l = i$, and then we cyclically permuted the indices of the Levi-Cevita tensor via $jki \to ijk$ in ϵ_{ijk}. In Problem 9.1.1, you will show the analogous commutator with momentum is given by

$$[\hat{p}_i, \hat{L}_j] = i\hbar \sum_{k} \epsilon_{ijk} \hat{p}_k. \tag{9.6}$$

To be concrete, these commutators look like $[\hat{r}_x, \hat{L}_y] = i\hbar \hat{r}_z$ and $[\hat{p}_x, \hat{L}_z] = -i\hbar \hat{p}_y$, and so on.

We denote the operators $\hat{r}^2 = \hat{\vec{r}} \cdot \hat{\vec{r}}$ and $\hat{p}^2 = \hat{\vec{p}} \cdot \hat{\vec{p}}$. Then the commutators of angular momentum with position and momentum derived above imply that

$$[\hat{r}^2, \hat{\vec{L}}] = 0 \quad \text{and} \quad [\hat{p}^2, \hat{\vec{L}}] = 0. \tag{9.7}$$

We show the derivation of the first one here; the second one is left as Problem 9.1.2 for you. Using the Leibniz rule, we have

$$[\hat{r}^2, \hat{L}_j] = \left[\sum_i \hat{r}_i^2, \hat{L}_j\right] = \sum_i \Big(\hat{r}_i[\hat{r}_i, \hat{L}_j] + [\hat{r}_i, \hat{L}_j]\hat{r}_i\Big)$$
$$= i\hbar \sum_{ik} \epsilon_{ijk}\hat{r}_i\hat{r}_k + i\hbar \sum_{ik} \epsilon_{ijk}\hat{r}_k\hat{r}_i$$
$$= 2i\hbar \sum_{ik} \epsilon_{ijk}\hat{r}_i\hat{r}_k, \tag{9.8}$$

because $[\hat{r}_i, \hat{r}_k] = 0$. But $\epsilon_{ijk} = -\epsilon_{kji}$, while $\hat{r}_i\hat{r}_k = \hat{r}_k\hat{r}_i$. This means one of the factors in the product is odd when we exchange $i \leftrightarrow k$, while the other factor is even, implying the summand is odd, and thereby vanishes. Often this argument is hard to follow, so let us look concretely at these results—for $j = x$, we have

$$[\hat{r}^2, \hat{L}_x] = 2i\hbar \sum_{il} \epsilon_{ixk}\hat{r}_i\hat{r}_k = 2i\hbar(\hat{r}_z\hat{r}_y - \hat{r}_y\hat{r}_z) = 0; \tag{9.9}$$

you can easily verify this also holds true for $j = y$ and for $j = z$.

We have one more commutator to evaluate. It is angular momentum with itself, given by

$$[\hat{L}_i, \hat{L}_j] = \sum_{kl}\sum_{k'l'} \epsilon_{ikl}\epsilon_{jk'l'}[\hat{r}_k\hat{p}_l, \hat{r}_{k'}\hat{p}_{l'}]$$
$$= \sum_{kl}\sum_{k'l'} \epsilon_{ikl}\epsilon_{jk'l'}\Big(\cancel{\hat{r}_k\hat{r}_{k'}[\hat{p}_l, \hat{p}_{l'}]} + \hat{r}_k[\hat{p}_l, \hat{r}_{k'}]\hat{p}_{l'} + \hat{r}_{k'}[\hat{r}_k, \hat{p}_{l'}]\hat{p}_l$$
$$+ \cancel{[\hat{r}_k, \hat{r}_{k'}]\hat{p}_l\hat{p}_{l'}}\Big)$$
$$= \sum_{kl}\sum_{k'l'} \epsilon_{ikl}\epsilon_{jk'l'}\Big(-i\hbar\hat{r}_k\hat{p}_{l'}\delta_{lk'} + i\hbar\hat{r}_{k'}\hat{p}_k\delta_{kl'}\Big)$$
$$= -i\hbar \sum_{kll'} \epsilon_{ikl}\epsilon_{jll'}\hat{r}_k\hat{p}_{l'} + i\hbar \sum_{kk'l} \epsilon_{ikl}\epsilon_{jk'k}\hat{r}_{k'}\hat{p}_l. \tag{9.10}$$

Now, it looks like we are stuck, but the totally antisymmetric tensor satisfies a useful identity that we will invoke next. The identity is

$$\sum_k \epsilon_{ijk}\epsilon_{i'j'k} = \delta_{ii'}\delta_{jj'} - \delta_{ij'}\delta_{ji'}. \tag{9.11}$$

Here is how we establish this identity; we have seen it before, but we review it again here. The argumentation is a bit subtle, so follow closely. To begin, the tensor vanishes unless i, j, and k are different and similarly for i', j', k. Hence, either $i = i'$ and $j = j'$ or $i = j'$ and $j = i'$. Pause to be sure you follow that argument. For example, if $k = x$, then either $i = y$ and $j = z$ or $i = z$ and $j = y$. We have a similar result for i' and j'. Now, suppose $i = i'$ and $j = j'$. Then $\epsilon_{ijk}\epsilon_{i'j'k} = (\epsilon_{ijk})^2 = 1$. If

$i = j'$ and $j = i'$, then $\epsilon_{ijk}\epsilon_{i'j'k} = \epsilon_{ijk}\epsilon_{jik} = -(\epsilon_{ijk})^2 = -1$. This then establishes the identity.

Now, we apply this identity to our calculation and find that

$$\begin{aligned}
[\hat{L}_i, \hat{L}_j] &= i\hbar \sum_{kk'l'} \big(\delta_{ij}\delta_{kl'} - \delta_{il'}\delta_{kj}\big)\hat{r}_k \hat{p}_{l'} - i\hbar \sum_{k'l} \big(\delta_{ij}\delta_{lk'} - \delta_{ik'}\delta_{jl}\big)\hat{r}_{k'} \hat{p}_l \\
&= i\hbar\delta_{ij}\hat{\vec{r}} \cdot \hat{\vec{p}} - i\hbar\hat{r}_j \hat{p}_i - i\hbar\delta_{ij}\hat{\vec{r}} \cdot \hat{\vec{p}} + i\hbar\hat{r}_i \hat{p}_j \\
&= i\hbar\big(\hat{r}_i \hat{p}_j - \hat{r}_j \hat{p}_i\big) \\
&= i\hbar \sum_{i'j'} \big(\delta_{ii'}\delta_{jj'} - \delta_{ij'}\delta_{ji'}\big)\hat{r}_{i'} \hat{p}_{j'}. \qquad (9.12)
\end{aligned}$$

We wrote the last line, so we can use the identity with the antisymmetric tensors in reverse. It yields

$$\begin{aligned}
[\hat{L}_i, \hat{L}_j] &= i\hbar \sum_{i'j'k} \epsilon_{ijk}\epsilon_{i'j'k}\hat{r}_{i'} \hat{p}_{j'} \\
&= i\hbar \sum_k \epsilon_{ijk}\hat{L}_k. \qquad (9.13)
\end{aligned}$$

So, we find that $[\hat{L}_i, \hat{L}_j] = i\hbar \sum_k \epsilon_{ijk}\hat{L}_k$. Note that this is exactly the same form as the commutator we found for spin $[\hat{S}_i, \hat{S}_j] = i\hbar \sum_k \epsilon_{ijk}\hat{S}_k$. Just like we did in Sect. 6.1, we can show that $[\hat{L}_i, \hat{\vec{L}} \cdot \hat{\vec{L}}] = 0$. We have

$$\begin{aligned}
[\hat{L}_i, \hat{\vec{L}} \cdot \hat{\vec{L}}] &= \left[\hat{L}_i, \sum_j \hat{L}_j \hat{L}_j\right] = \sum_j \Big(\hat{L}_j[\hat{L}_i, \hat{L}_j] + [\hat{L}_i, \hat{L}_j]\hat{L}_j\Big) \\
&= i\hbar \sum_{jk} \epsilon_{ijk}\big(\hat{L}_j \hat{L}_k + \hat{L}_k \hat{L}_j\big) = 0, \qquad (9.14)
\end{aligned}$$

because $\big(\hat{L}_j \hat{L}_k + \hat{L}_k \hat{L}_j\big)$ is symmetric under the interchange of j with k, while ϵ_{ijk} is antisymmetric. So, we have established that

$$[\hat{L}_i, \hat{L}_j] = i\hbar \sum_k \epsilon_{ijk}\hat{L}_k \text{ and } [\hat{L}_i, \hat{\vec{L}} \cdot \hat{\vec{L}}] = 0. \qquad (9.15)$$

These are also the same relations we found for spin. Hence, we can find states that are simultaneously eigenstates of any component of $\hat{\vec{L}}$ and of $\hat{\vec{L}}^2$. By convention, we always choose $\hat{L}_z$ and $\hat{\vec{L}}^2$. Working out the eigenvalues and eigenstates and their properties will be done in Sect. 9.3.

Similar to the case with spins, we can also define

$$\hat{L}_{\pm} = \hat{L}_x \pm i\hat{L}_y \tag{9.16}$$

and immediately show the alternative form of the SU(2) algebra given by

$$[\hat{L}_z, \hat{L}_{\pm}] = \pm\hbar\hat{L}_{\pm} \text{ and } [\hat{L}_+, \hat{L}_-] = 2\hbar\hat{L}_z. \tag{9.17}$$

You will derive the relations in Problem 9.1.3. There are two more commutators you need:

$$[\hat{r}_i, \hat{\vec{L}} \cdot \hat{\vec{L}}] = i\hbar \sum_k \epsilon_{ijk}\left(\hat{L}_j\hat{r}_k + \hat{r}_k\hat{L}_j\right) \text{ and}$$
$$[\hat{p}_i, \hat{\vec{L}} \cdot \hat{\vec{L}}] = i\hbar \sum_k \epsilon_{ijk}\left(\hat{L}_j\hat{p}_k + \hat{p}_k\hat{L}_j\right). \tag{9.18}$$

Both are in Problem 9.1.4 for you to finish. We have made a video on The Commutation Relations of Angular Momentum https://tinyurl.com/282jn4wr to provide some additional review and context.

Section 9.1 Problems:

9.1.1 Commutator of Momentum with Angular Momentum

Show that $[\hat{p}_i, \hat{L}_j] = i\hbar \sum_k \epsilon_{ijk}\hat{p}_k$.

9.1.2 Commutator of $\hat{p}^2$ with Angular Momentum

Using the results from the previous problem, show that $[\hat{p}^2, \hat{L}_j] = 0$.

9.1.3 SU (2) Algebra with Raising and Lowering Operators

Verify the $SU(2)$ algebra given in Eq. (9.17).

9.1.4 Commutation Relation of Position and Momentum with the Square of the Angular Momentum Operator

Show that the commutation relations in Eq. (9.18) hold.

9.2 Relationship of Angular Momentum Operators to Rotations

In Sect. 9.1, we derived the commutation relations between position and angular momentum, between momentum and angular momentum, and also between angular momentum and angular momentum. These commutation relations are the relations of a vector operator. A *vector operator* $\hat{v}_i$ satisfies, by definition, the following commutation relation with angular momentum

$$[\hat{v}_i, \hat{L}_j] = i\hbar \sum_k \epsilon_{ijk} \hat{v}_k, \tag{9.19}$$

and, in particular, we established that $\hat{\vec{r}}$, $\hat{\vec{p}}$, and $\hat{\vec{L}}$ are all vector operators in the previous section. Vector operators satisfy a number of important properties—they are the analog of a three-dimensional vector in quantum mechanics, and they transform the same way under rotations. In this section, we will show how angular momentum relates to the rotations of vector operators.

In mathematics, the set of rotations forms what is called an *abstract group*, which means the rotations are closed under multiplication (two rotations applied one after another can be reexpressed as a single rotation), there is an identity element (no rotation), and every rotation has an inverse rotation (applying a rotation and then its inverse results in no rotation). Rotations in three dimensions form what is called a *non-Abelian group*, because there exist rotations where changing the order of the product of two rotations results in a different final rotation. For example, take a book and rotate it 90° about an axis through its center (z-axis) and then 90° about an axis through its spine (y-axis). Then reverse the order of the rotations (to rotate first about the lower edge via the y-axis and then about the spine via the z-axis). The final orientation is different, as shown in Fig. 9.1.

Continuous Lie groups, such as the rotation group, have all rotations that are continuously connected to the identity group element being able to be written in the form

$$\hat{U}_{\vec{e}_n}(\alpha) = e^{-\frac{i}{\hbar}\alpha \vec{e}_n \cdot \hat{\vec{L}}}, \tag{9.20}$$

where $\vec{e}_n$ is a vector pointing in the direction of the axis of the rotation and α is the magnitude of the rotation, measured in radians. If you compare this to an earlier problem with spin (see Problem 6.3.7), where we had a $\theta/2$ instead of α, we see this arises simply due to the fact that the rotation is expressed in terms of the Pauli matrices rather than the spin operators, where $\hat{\vec{S}} = \frac{\hbar}{2}\vec{\sigma}$. If the rotation is expressed in terms of the spin operators, then it takes the same form as what is written above in Eq. (9.20). Note that the angular-momentum operators $\{\hat{L}_i\}$ are called the *generators of the Lie algebra*, and here they satisfy the operator commutation relations given by $[\hat{L}_i, \hat{L}_j] = i\hbar \sum_k \epsilon_{ijk} \hat{L}_k$.

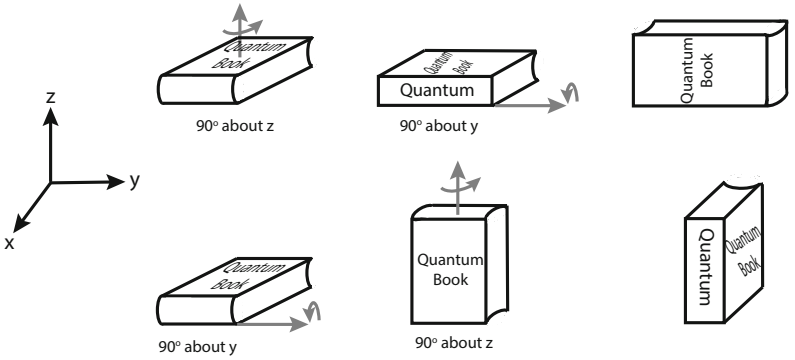

Fig. 9.1 Orientations of a book after the different rotations illustrate how the rotations do not always commute. Here we rotate by 90° about the z and then y axes, or *vice versa*. Note how the final orientations on the right are not the same

To verify that this operator really corresponds to a rotation, we need to evaluate the transformation of the vector operators under the rotation. Note that states and operators transform differently. The state transforms as $|\psi\rangle \rightarrow \hat{U}_{\vec{e}_n}(\alpha)|\psi\rangle = |\psi\rangle_{rot}$. Now consider an operator $\hat{O}$ acting on the original state $|\psi\rangle$ and given by $\hat{O}|\psi\rangle$. Since an operator acting on a state is a state, we find the operator acting on the state transforms as

$$\hat{U}_{\vec{e}_n}(\alpha)\hat{O}|\psi\rangle = \underbrace{\hat{U}_{\vec{e}_n}(\alpha)\hat{O}\hat{U}^{\dagger}_{\vec{e}_n}(\alpha)}_{\hat{O}_{rot}}\underbrace{\hat{U}_{\vec{e}_n}(\alpha)|\psi\rangle}_{|\psi\rangle_{rot}}, \tag{9.21}$$

after using a multiply-by-one in between the operator and the state. This means the operators transform via

$$\hat{O} \rightarrow e^{-\frac{i}{\hbar}\alpha\vec{e}_n\cdot\hat{\vec{L}}}\hat{O}e^{\frac{i}{\hbar}\alpha\vec{e}_n\cdot\hat{\vec{L}}}, \tag{9.22}$$

under a rotation of α about an axis $\vec{e}_n$.

For spin one-half, we could work out the unitary operators $\hat{U}_{\vec{e}_n}(\alpha)$ exactly. You will later see that for angular momentum $l = 1$, we can work them out exactly as well. The general case is a bit messy and will be worked out after we examine some special cases. We will also see how to do this for the general case for one specific rotation using the exponential disentangling identity later in the book, when we discuss spherical harmonics. First, we can compute an exact formula for how vector operators transform to the rotated frame. While the general result is of the form

$$\hat{v}_{i,rot} = e^{-\frac{i}{\hbar}\alpha\vec{e}_n\cdot\hat{\vec{L}}}\hat{v}_i e^{\frac{i}{\hbar}\alpha\vec{e}_n\cdot\hat{\vec{L}}}, \tag{9.23}$$

we will content ourselves to first work this out concretely for simpler cases where the rotation axis is along one of the coordinate axes. For example, with $\vec{e}_n = \vec{e}_z$, we want to compute

$$\hat{v}_{i,rot} = e^{-\frac{i}{\hbar}\alpha\vec{e}_z\cdot\hat{\vec{L}}}\hat{v}_i e^{\frac{i}{\hbar}\alpha\vec{e}_z\cdot\hat{\vec{L}}} = e^{-\frac{i\alpha}{\hbar}\hat{L}_z}\hat{v}_i e^{\frac{i\alpha}{\hbar}\hat{L}_z}. \tag{9.24}$$

There are three cases to work out. The simple one is the transformation of $\hat{v}_z$, because $[\hat{v}_z, \hat{L}_z] = 0$. This then means that

$$\hat{v}_{z,rot} = e^{-\frac{i\alpha}{\hbar}\hat{L}_z}\hat{v}_z e^{\frac{i\alpha}{\hbar}\hat{L}_z} = \hat{v}_z. \tag{9.25}$$

This result is exactly what we expect, because rotating an ordinary three-dimensional vector about an axis that goes through the vector does nothing to the vector—so we expect the rotation of a vector operator about an axis parallel to that operator should also leave it unchanged. Indeed, this is what we found. For $i = x$ and $i = y$, we have to work harder. Consider $i = x$ first. The transformation is given by the Hadamard lemma, so we need to compute some commutators. We find that $[\hat{L}_z, \hat{v}_x] = i\hbar\hat{v}_y$. Then

$$[\hat{L}_z, [\hat{L}_z, \hat{v}_x]] = i\hbar[\hat{L}_z, \hat{v}_y] = \hbar^2\hat{v}_x, \tag{9.26}$$

and the higher-order nested commutators will repeat. In particular, $[\hat{L}_z, [\hat{L}_z, [\hat{L}_z, \hat{v}_x]]] = i\hbar^3\hat{v}_y$, and so on. This means the Hadamard lemma separates its infinite power series into one series for even numbers of nested commutators and a different one for odd numbers of nested commutators. The final result becomes

$$\begin{aligned}\hat{v}_{x,rot} &= \sum_{n=0}^{\infty}\frac{(-1)^n\alpha^{2n}}{(2n)!}\hat{v}_x + \sum_{n=0}^{\infty}\frac{(-1)^n\alpha^{2n+1}}{(2n+1)!}\hat{v}_y \\ &= \cos\alpha\,\hat{v}_x + \sin\alpha\,\hat{v}_y.\end{aligned} \tag{9.27}$$

Please work carefully to verify the Hadamard lemma result in this case on your own, as these kinds of calculations are an important skill you need to develop. Our final result shows that the vector operator $\hat{v}_x$ has rotated by an angle of α in the counterclockwise direction in the x-y plane. This is precisely what a rotation by α about the z-axis should do! In Problem 9.2.2, you will establish that

$$\begin{aligned}\hat{v}_{y,rot} &= \sum_{n=0}^{\infty}\frac{(-1)^n\alpha^{2n}}{(2n)!}\hat{v}_y - \sum_{n=0}^{\infty}\frac{(-1)^n\alpha^{2n+1}}{(2n+1)!}\hat{v}_x \\ &= -\sin\alpha\,\hat{v}_x + \cos\alpha\,\hat{v}_y.\end{aligned} \tag{9.28}$$

This is again exactly what a rotation by α should do for a vector that originally pointed in the y-direction. To see a visualization of this, look at Fig. 9.2.

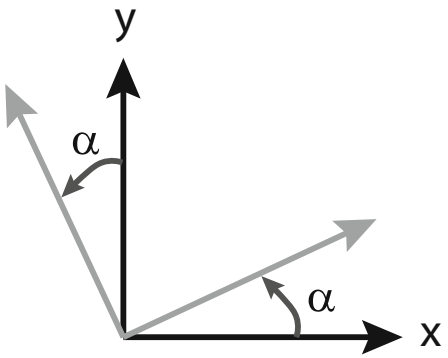

Fig. 9.2 Image showing vectors originally oriented along the x- and y-axes that have been rotated by $\alpha = 25^\circ$ degrees. One can use trigonometry to see that the two gray arrows correspond to $\vec{e}_x{}^{\text{rot}} = \cos\alpha\,\vec{e}_x + \sin\alpha\,\vec{e}_y$ and $\vec{e}_y{}^{\text{rot}} = -\sin\alpha\,\vec{e}_x + \cos\alpha\,\vec{e}_y$. This is the same transformation that the vector operators $\hat{v}_x$ and $\hat{v}_y$ undergo when rotated by an angle α about the z-axis, as we derived in the text

What about rotations about the x- or y-axes? There are two ways you can work those results out. Go through the same brute force calculation, as we did above for the z-axis rotation, or using the cyclic permutations of $x\ y\ z$ and the behavior of ϵ_{ijk}, we can infer the results for those two cases as follows:

$$\begin{aligned} x\text{-axis}: &\quad \hat{v}_x \to \hat{v}_x, \quad \hat{v}_y \to \cos\alpha\,\hat{v}_y + \sin\alpha\,\hat{v}_z, \quad \hat{v}_z \to -\sin\alpha\,\hat{v}_y + \cos\alpha\,\hat{v}_z \\ y\text{-axis}: &\quad \hat{v}_x \to \cos\alpha\,\hat{v}_x - \sin\alpha\,\hat{v}_z, \quad \hat{v}_y \to \hat{v}_y, \quad \hat{v}_z \to \sin\alpha\,\hat{v}_x + \cos\alpha\,\hat{v}_z. \end{aligned} \tag{9.29}$$

Sometimes students will find such an argument hard to follow. If this is you, try to do it concretely first, then look again at the permutation argument, and see if you can make sense of it. We ask you to spend some time investigating this argument in more detail in Problem 9.2.3.

Now, we wish to work with the general case of $\hat{U}_{\vec{e}_n}(\alpha) = e^{-\frac{i}{\hbar}\alpha\vec{e}_n\cdot\hat{\vec{L}}}$. We have the $\vec{e}_n$ vector lie in the θ-ϕ direction. This means the operator takes the form $\exp\big(-\frac{i}{\hbar}\alpha(\sin\theta\,\cos\phi\,\hat{L}_x + \sin\theta\,\sin\phi\,\hat{L}_y + \cos\theta\,\hat{L}_z)\big)$. Using the general result for a rotation about the z-axis then gives us

$$\hat{U}_{\vec{e}_n}(\alpha) = e^{-\frac{i}{\hbar}\phi\hat{L}_z} e^{-\frac{i}{\hbar}\alpha(\sin\theta\hat{L}_x + \cos\theta\hat{L}_z)} e^{\frac{i}{\hbar}\phi\hat{L}_z}, \tag{9.30}$$

because the Hadamard lemma goes into the exponent via the braiding relation, and then we use the rotation of the vector operator $\hat{L}_x$ and $\hat{L}_z$ about the z-axis to establish the identity. Now, we repeat again, with a rotation about the y-axis, to give us

$$\hat{U}_{\vec{e}_n}(\alpha) = e^{-\frac{i}{\hbar}\phi\hat{L}_z} e^{-\frac{i}{\hbar}\theta\hat{L}_y} e^{-\frac{i}{\hbar}\alpha\hat{L}_z} e^{\frac{i}{\hbar}\theta\hat{L}_y} e^{\frac{i}{\hbar}\phi\hat{L}_z}. \tag{9.31}$$

So, we have factorized the general rotation operator into five terms via the application of two Hadamards. Note that this is the form expected by the Euler angle decomposition of a general rotation. The way to envision this is we start by rotating the vector $\vec{e}_n$ to the z-axis (via two rotations), then rotate by α about the z-axis (via one rotation), and then rotate the axis back to the $\vec{e}_n$ direction (via two rotations).

At this stage, calculating how a vector operator transforms is a straightforward calculation, but it is really serious "French cooking." We have five different similarity transformations we have to carry out to accomplish the transformation. They are of the following form (remember the Hermitian conjugate of a product or operators reverses the order:

$$\hat{O}_{rot} = e^{-\frac{i}{\hbar}\phi\hat{L}_z} e^{-\frac{i}{\hbar}\theta\hat{L}_y} e^{-\frac{i}{\hbar}\alpha\hat{L}_z} e^{\frac{i}{\hbar}\theta\hat{L}_y} e^{\frac{i}{\hbar}\phi\hat{L}_z} \hat{O} e^{-\frac{i}{\hbar}\phi\hat{L}_z} e^{-\frac{i}{\hbar}\theta\hat{L}_y} e^{\frac{i}{\hbar}\alpha\hat{L}_z} e^{\frac{i}{\hbar}\theta\hat{L}_y} e^{\frac{i}{\hbar}\phi\hat{L}_z}. \tag{9.32}$$

Note how we have five nested Hadamard lemmas to evaluate to perform the general rotation! But before we discuss the general case, let us first look at our special cases. If $\theta = \phi = 0$, then the operator is just a rotation by α about the z-axis. This is the same as what we did before. What about if $\theta = \pi/2$ and $\phi = 0$? This corresponds to the x-rotation. It decomposes to a $-\pi/2$ rotation about y first, then α about z and then $\pi/2$ about y. Let us look at the three vector operators and see how they transform. We have

$$\hat{v}_x \underset{-\frac{\pi}{2} \text{ about } y}{\rightarrow} \hat{v}_z \underset{\alpha \text{ about } z}{\rightarrow} \hat{v}_z \underset{\frac{\pi}{2} \text{ about } y}{\rightarrow} \hat{v}_x \quad \checkmark \tag{9.33}$$

for the x direction;

$$\hat{v}_y \underset{-\frac{\pi}{2} \text{ about } y}{\rightarrow} \hat{v}_y \underset{\alpha \text{ about } z}{\rightarrow} -\sin\alpha\, \hat{v}_x + \cos\alpha\, \hat{v}_y \underset{\frac{\pi}{2} \text{ about } y}{\rightarrow} \cos\alpha\, \hat{v}_y + \sin\alpha\, \hat{v}_z \quad \checkmark \tag{9.34}$$

for the y direction; and

$$\hat{v}_z \underset{-\frac{\pi}{2} \text{ about } y}{\rightarrow} -\hat{v}_x \underset{\alpha \text{ about } z}{\rightarrow} -\cos\alpha\, \hat{v}_x - \sin\alpha\, \hat{v}_y \underset{\frac{\pi}{2} \text{ about } y}{\rightarrow} -\sin\alpha\, \hat{v}_y + \cos\alpha\, \hat{v}_z \quad \checkmark \tag{9.35}$$

for the z direction. You will look at the y-axis rotation, with $\theta = \pi/2$ and $\phi = \pi/2$ in Problem 9.2.4.

The general case is really too painful to show all of the results for it (especially since we won't need them for any of our future work). But, to show you that it can be done, we will show how $\hat{v}_x$ transforms, by carrying out all five similarity transformations:

$$\begin{aligned} \hat{v}_x &\underset{-\phi \text{ about } z}{\rightarrow} \cos\phi\, \hat{v}_x - \sin\phi\, \hat{v}_y \\ &\underset{-\theta \text{ about } y}{\rightarrow} \cos\theta\, \cos\phi\, \hat{v}_x - \sin\phi\, \hat{v}_y + \sin\theta\, \cos\phi\, \hat{v}_z \end{aligned}$$

$$
\begin{aligned}
&\underset{\alpha \text{ about } z}{\rightarrow} (\cos\alpha\ \cos\theta\ \cos\phi + \sin\alpha\ \sin\phi)\hat{v}_x \\
&\quad + (-\cos\alpha\ \sin\phi + \sin\alpha\ \cos\theta\ \cos\phi)\hat{v}_y + \sin\theta\ \cos\phi\ \hat{v}_z \\
&\underset{\theta \text{ about } y}{\rightarrow} (\cos\alpha\ \cos^2\theta\ \cos\phi + \sin\alpha\ \cos\theta\ \sin\phi + \sin^2\theta\ \cos\phi)\hat{v}_x \\
&\quad + (-\cos\alpha\ \sin\phi + \sin\alpha\ \cos\theta\ \cos\phi)\hat{v}_y \\
&\quad + ((1-\cos\alpha)\cos\theta\ \sin\theta\ \cos\phi - \sin\alpha\ \sin\theta\ \cos\phi)\hat{v}_z \\
&\underset{\phi \text{ about } z}{\rightarrow} \left(\cos\alpha + (1-\cos\alpha)\sin^2\theta\ \cos^2\phi\right)\hat{v}_x \\
&\quad + \left(\sin\alpha\ \cos\theta + (1-\cos\alpha)\sin^2\theta\ \cos\phi\ \sin\phi\right)\hat{v}_y \\
&\quad + \left((1-\cos\alpha)\cos\theta\ \sin\theta\ \cos\phi - \sin\alpha\ \sin\theta\ \sin\phi\right)\hat{v}_z. \qquad (9.36)
\end{aligned}
$$

Yes. It is a real mess (and a great example of "French cooking"). But, you can do it if you need to. Don't worry, you are not asked to do $\hat{v}_y$ or $\hat{v}_z$ in the problems. One check of the answer above that you can perform is to set θ and ϕ equal to the values needed for the x, y, or z rotations and verify that this formula gives the right answer.

Section 9.2 Problems:

9.2.1 Vector Operators

It is a curious fact that the commutator relation for the vector operator, given in Eq. (9.19), holds with the angular momentum operator on either position in the commutator, without changing the sign, so it can never be forgotten! Use Eq. (9.19) to show that we also have

$$[\hat{L}_i, \hat{v}_j] = i\hbar \sum_k \epsilon_{ijk}\hat{v}_k, \qquad (9.37)$$

holds. Be sure that you get your indices right, as they are different in the two expressions!

9.2.2 Rotation of $\hat{v}_y$ Under a z Rotation

Compute the single Hadamard transformation by summing the infinite number of terms after finding the repeating pattern of the nested commutators to show that

$$e^{-\frac{i\alpha}{\hbar}\hat{L}_z}\hat{v}_y e^{\frac{i\alpha}{\hbar}\hat{L}_z} = -\sin\alpha\ \hat{v}_x + \cos\alpha\ \hat{v}_z. \qquad (9.38)$$

9.2.3 Rotations About the x and y Axes

Using the cyclic symmetry of $\epsilon_{ijk} = \epsilon_{jki} = \epsilon_{kij}$, provide a cogent argument and derivation that $e^{-\frac{i\alpha}{\hbar}\hat{L}_x}$ and $e^{-\frac{i\alpha}{\hbar}\hat{L}_y}$ both correspond to rotations by α radians about the x- and y-axes, respectively. That is, show that

$$e^{-\frac{i\alpha}{\hbar}\hat{L}_x}\hat{v}_z e^{\frac{i\alpha}{\hbar}\hat{L}_x} = -\sin\alpha\,\hat{v}_y + \cos\alpha\,\hat{v}_z, \tag{9.39}$$

and

$$e^{-\frac{i\alpha}{\hbar}\hat{L}_y}\hat{v}_x e^{\frac{i\alpha}{\hbar}\hat{L}_y} = \cos\alpha\,\hat{v}_x - \sin\alpha\,\hat{v}_z, \tag{9.40}$$

and so on.

9.2.4 Special Case of the General Rotation in Eq. (9.31)

Using the factorized form of the general rotation, given in Eq. (9.31), set $\theta = \pi/2$ and $\phi = \pi/2$ to describe a rotation by α about the y-axis. Compute the relevant Hadamard lemma transformations to show how $\hat{v}_x$, $\hat{v}_y$, and $\hat{v}_z$ transform under this rotation. Compare your answers to the known results we worked out directly in the text using the single Hadamard lemma for these special cases.

9.2.5 Rotating the North-Pole State to an Arbitrary Direction

Consider the composite rotation by an angle θ about the y-axis followed by a rotation by an angle ϕ about the z-axis. It is given by

$$\hat{U}(\theta,\phi) = \hat{U}_{\vec{e}_z}(\phi)\hat{U}_{\vec{e}_y}(\theta). \tag{9.41}$$

Define a general "angular" position-basis state to satisfy

$$\cos\hat{\theta}\,|\theta,\phi\rangle = \cos\theta\,|\theta,\phi\rangle \;\text{ and }\; \sin\hat{\theta}\,|\theta,\phi\rangle = \sin\theta\,|\theta,\phi\rangle \tag{9.42}$$

and

$$\cos\hat{\phi}\,|\theta,\phi\rangle = \cos\phi\,|\theta,\phi\rangle \;\text{ and }\; \sin\hat{\phi}\,|\theta,\phi\rangle = \sin\phi\,|\theta,\phi\rangle. \tag{9.43}$$

Note that we use the cosine and sine of the angle to define these eigenvalue-eigenvector relations rather than the angles themselves, because angles are ill-

defined operators in quantum mechanics, while trigonometric functions of angle operators are well defined (recall the particle on the circle work from Sect. 7.4).

Suppose we have a "north-pole" state that satisfies $\cos\hat{\theta}\,|0_\theta, 0_\phi\rangle = |0_\theta, 0_\phi\rangle$ and $\sin\hat{\theta}\,|0_\theta, 0_\phi\rangle = 0$. Use the eigenvalue-eigenvector relation along with the Hadamard lemma to establish that the state at an arbitrary direction can be constructed from the north-pole state via

$$|\theta, \phi\rangle = e^{-\frac{i}{\hbar}\phi\hat{L}_z} e^{-\frac{i}{\hbar}\theta\hat{L}_y} |0_\theta, 0_\phi\rangle. \tag{9.44}$$

Note that for the north-pole state, the angle ϕ is actually arbitrary, but we find it convenient to set it equal to zero anyway.

9.3 Angular Momentum Eigenstates (Orbital and Spin)

We now work on one of the first solved quantum-mechanics problems—the eigenvalues and eigenstates of angular momentum. This problem appears already in Heisenberg's original paper on matrix mechanics (Heisenberg, 1925) and also in the follow-up "Dreimannarbeit" paper by Born, Heisenberg, and Jordan. (Born et al., 1926). It is a classic matrix-mechanics calculation. It is also very elegant in how it is carried out. I hope you will agree.

You know the drill. We have already done it twice (in Secs. 6.1 and 7.6). We use the commutation relations and the properties of the eigenstates to determine all eigenvalues and eigenstates (and their properties) solely by clever applications of logic, the operator algebra, and the properties of the eigenstates.

We begin by summarizing the commutation relations for angular momentum, which take either the Cartesian form

$$[\hat{L}_i, \hat{L}_j] = i\hbar \sum_k \epsilon_{ijk}\hat{L}_k \;\text{ and }\; [\hat{L}_i, \hat{\vec{L}}^{\,2}] = 0, \tag{9.45}$$

or the raising-lowering form

$$[\hat{L}_z, \hat{L}_\pm] = \pm\hbar\hat{L}_\pm, \;\; [\hat{L}_+, \hat{L}_-] = 2\hbar\hat{L}_z, \;\; [\hat{L}_z, \hat{\vec{L}}^{\,2}] = 0, \;\text{ and }\; [\hat{L}_\pm, \hat{\vec{L}}^{\,2}] = 0. \tag{9.46}$$

All of these relations were derived in Sect. 9.1.

The convention we use is to find simultaneous eigenstates of $\hat{L}_z$ and $\hat{\vec{L}}^2$, which is the maximal commuting set of operators for angular momentum. We label the eigenvalue for $\hat{L}_z$ (which we will see equals $\hbar m$) by m and for $\hat{\vec{L}}^{\,2}$ (which we will see equals $\hbar^2 l(l+1)$) by l. Note that in both cases, the label will not be the eigenvalue. So the state $|l, m\rangle$ satisfies the following eigenvalue-eigenvector relations:

$$\hat{\vec{L}}^2|l,m\rangle = \hbar^2 l(l+1)|l,m\rangle \tag{9.47}$$

$$\hat{L}_z|l,m\rangle = \hbar m|l,m\rangle. \tag{9.48}$$

Even though it looks like the relationship between the choice of the label and the eigenvalue is weird and possibly restricted, we will see it is actually natural. Note also that because m and l are arbitrary, the z-component of angular momentum can be any number multiplied by $\hbar$, while the total-squared angular momentum eigenvalue is an arbitrary positive number multiplied by $\hbar^2$ for $l \geq 0$; the eigenvalue of the sum of the squares of operators is a positive-semidefinite operator, so there is no loss of generality.

The classic way that matrix mechanics works is that we find operators that raise or lower eigenvalues and then we show that they cannot be raised or lowered too much, providing a condition that leads us to determine the allowed eigenvalues. It is a very ingenious approach, and it should be clear after you learn how to do this as to why Heisenberg won the Nobel prize for this work. The first thing we need to show is that $\hat{L}_\pm$ raises or lowers the $\hat{L}_z$ eigenvalue by $\hbar$ and leaves the l eigenvalue unchanged; this is a reasonable thing to try given our earlier work on spin, where this was the case. The latter result follows from the fact that $[\hat{L}_\pm, \hat{\vec{L}}^2] = 0$ as can be seen from direct calculation

$$\hat{\vec{L}}^2\left(\hat{L}_\pm|l,m\rangle\right) = \hat{L}_\pm\left(\hat{\vec{L}}^2|l,m\rangle\right) = \hbar^2 l(l+1)\hat{L}_\pm|l,m\rangle. \tag{9.49}$$

So, $\hat{L}_\pm|l,m\rangle$ has the same eigenvalue with respect to $\hat{\vec{L}}^2$ as $|l,m\rangle$ does. But, we also have that

$$\hat{L}_z\left(\hat{L}_\pm|l,m\rangle\right) = \hat{L}_\pm\left(\hat{L}_z|l,m\rangle\right) + [\hat{L}_z, \hat{L}_\pm]|l,m\rangle = \hbar(m\pm 1)\hat{L}_\pm|l,m\rangle, \tag{9.50}$$

after using the commutation relation; so the state $\hat{L}_\pm|l,m\rangle$ is proportional to $|l,m\pm 1\rangle$. For this reason, we call $\hat{L}_+$ the *angular momentum raising operator* and $\hat{L}_-$ the *angular momentum lowering operator*.

Now comes our first very clever step. We recall that we can write the sums of the squares of the x and y components of angular momentum two different ways, since $\hat{L}_x^2 + \hat{L}_y^2 = \frac{1}{2}(\hat{L}_+\hat{L}_- + \hat{L}_-\hat{L}_+)$ (if you do not recall seeing that result, please verify it again). This sum of operators is positive semidefinite—this is because its expectation value is the sum of the norm-squared of $\hat{L}_x|l,m\rangle$ and $\hat{L}_y|l,m\rangle$, or one-half the sum of the norm-squared of $\hat{L}_+|l,m\rangle$ and $\hat{L}_-|l,m\rangle$—in either case the sum of norm-squareds is positive semidefinite. But $\hat{L}_x^2 + \hat{L}_y^2 = \hat{\vec{L}}^2 - \hat{L}_z^2$, so we have that

$$0 \leq \langle l,m|\left(\hat{L}_x^2 + \hat{L}_y^2\right)|l,m\rangle = \langle l,m|\left(\hat{\vec{L}}^2 - \hat{L}_z^2\right)|l,m\rangle = \hbar^2\Big(l(l+1) - m^2\Big), \tag{9.51}$$

because $\langle l, m|l, m\rangle = 1$ (we work with a normalized eigenbasis). This tells us that $|m| \leq \sqrt{l(l+1)}$. *This means there is a maximal and a minimal value of* m! Call the maximal value $m_{\max}$ and the minimal value $m_{\min}$. Then, we must have

$$\hat{L}_+|l, m_{\max}\rangle = 0 \text{ and } \hat{L}_-|l, m_{\min}\rangle = 0, \tag{9.52}$$

because the maximal m value cannot be raised, and the minimal value cannot be lowered. We will use these results to determine the maximal and minimal values next.

So, let us look at the first result $\hat{L}_+|l, m_{\max}\rangle = 0$. Then, if we multiply from the left by $\hat{L}_-$, this cannot change anything, so we have that $\hat{L}_-\hat{L}_+|l, m_{\max}\rangle = 0$. But

$$\hat{L}_-\hat{L}_+ = \frac{1}{2}\left(\hat{L}_-\hat{L}_+ + \hat{L}_+\hat{L}_-\right) + \frac{1}{2}[\hat{L}_-, \hat{L}_+] = \hat{\vec{L}}^{\,2} - \hat{L}_z^2 - \frac{1}{2} \times 2\hbar\hat{L}_z, \tag{9.53}$$

so, we find that $\hat{L}_-\hat{L}_+|l, m_{\max}\rangle = 0$ becomes

$$\left(\hat{\vec{L}}^{\,2} - \hat{L}_z^2 - \hbar\hat{L}_z\right)|l, m_{\max}\rangle = \hbar^2\Big(l(l+1) - m_{\max}\big(m_{\max}+1\big)\Big)|l, m_{\max}\rangle = 0. \tag{9.54}$$

This is a quadratic equation for $m_{\max}$, whose two roots are $m_{\max} = l$ and $m_{\max} = -l-1$. We now do a similar analysis for $\hat{L}_-|l, m_{\min}\rangle = 0$. Multiply by $\hat{L}_+$ from the left and use $\hat{L}_+\hat{L}_- = \frac{1}{2}\big(\hat{L}_+\hat{L}_- + \hat{L}_-\hat{L}_+\big) + \frac{1}{2}[\hat{L}_+, \hat{L}_-]$, which equals $\hat{\vec{L}}^{\,2} - \hat{L}_z^2 + \hbar\hat{L}_z$, then leads us to $l(l+1) - m_{\min}^2 + m_{\min} = 0$, or $m_{\min} = l+1$, or $-l$. But, we must have $m_{\max} \geq m_{\min}$. This requires us to have $m_{\max} = l$ and $m_{\min} = -l$. This means that $m_{\max} - m_{\min} = 2l$. It also shows why we chose to write the eigenvalue for $\hat{\vec{L}}^{\,2}$ as $\hbar^2 l(l+1)$.

Let us now start from the state $|l, m_{\min}\rangle$. If we apply $\hat{L}_+$ $2l$ times to this state, it must be proportional to $|l, m_{\max}\rangle$. It must end on $m_{\max}$, where another application annihilates the state; otherwise, I would be able to apply the raising operator one more time and find a state with $m > m_{\max}$, which is impossible. But I can only apply the raising operator an *integer* number of times. So, we learn that $2l$ must be an integer! This means either l is an integer or l is a half-odd integer. For orbital angular momentum, we derive in Problem 9.3.9 below that l must be an integer. In addition, because this analysis for the commutators is identical to what we did for spin, we find that we can also describe abstract spin states with l equal to a half-odd integer as well (l = integer is also possible for a spin state). The formulas we will work out here apply to all cases of angular momentum: those originating from orbital motion about a central point and those arising from the intrinsic spin properties of a quantum particle.

We now will work out the numerical factors that arise when we apply raising or lowering operators. We assume all $|l, m\rangle$ states are normalized, so $\langle l, m|l', m'\rangle = \delta_{ll'}\delta_{mm'}$. Then we set

$$\hat{L}_+|l, m\rangle = c_{lm}|l, m+1\rangle. \tag{9.55}$$

Computing the norm gives

$$\langle l, m|\hat{L}_+^\dagger \hat{L}_+|l, m\rangle = |c_{lm}|^2\langle l, m+1|l, m+1\rangle = |c_{lm}|^2. \tag{9.56}$$

But,

$$\begin{aligned}\langle l, m|\hat{L}_+^\dagger \hat{L}_+|l, m\rangle &= \langle l, m|\hat{L}_-\hat{L}_+|l, m\rangle \\ &= \langle l, m|\left(\hat{\vec{L}}^2 - \hat{L}_z^2 - \hbar\hat{L}_z\right)|l, m\rangle \\ &= \hbar^2\big(l(l+1) - m^2 - m\big) = \hbar^2(l-m)(l+m+1).\end{aligned} \tag{9.57}$$

So, we find that

$$c_{lm} = \hbar\sqrt{l(l+1) - m(m+1)} = \hbar\sqrt{(l-m)(l+m+1)}, \tag{9.58}$$

where the most commonly used form is the last form. Note that we have chosen the global phase of the wavefunctions by always choosing the positive root. The $\hat{L}_-$ case is worked out in a similar way. We define

$$\hat{L}_-|l, m\rangle = c'_{lm}|l, m-1\rangle. \tag{9.59}$$

But, $\hat{L}_-^\dagger \hat{L}_- = \hat{L}_+\hat{L}_- = \hat{\vec{L}}^2 - \hat{L}_z^2 + \hbar\hat{L}_z$. So, we find that

$$\langle l, m|\hat{L}_-^\dagger \hat{L}_-|l, m\rangle = \langle l, m|\left(\hat{\vec{L}}^2 - \hat{L}_z^2 + \hbar\hat{L}_z\right)|l, m\rangle = \hbar^2\big(l(l+1) - m(m-1)\big). \tag{9.60}$$

Hence, we conclude that

$$c'_{lm} = \hbar\sqrt{l(l+1) - m(m-1)} = \hbar\sqrt{(l+m)(l-m+1)}, \tag{9.61}$$

where the last equation is the one most commonly used. Note that these formulas do satisfy $c_{ll} = c'_{l\,-l} = 0$, as they must (because $|l, l\rangle$ cannot be raised and $|l, -l\rangle$ cannot be lowered.

Our final task in this section is to construct the eigenstates as we did for the simple harmonic oscillator. We start from the state $|l, -l\rangle$, which is called the *lowest weight state* and is the analog of the harmonic oscillator vacuum state $|0\rangle$, because $\hat{L}_-|l, -l\rangle = 0$. Then, we find that

$$|l, -l+1\rangle = \frac{1}{\sqrt{2l\cdot 1}}\frac{\hat{L}_+}{\hbar}|l, -l\rangle$$

$$|l, -l+2\rangle = \frac{1}{\sqrt{(2l-1)\cdot 2\cdot(2l)\cdot 1}}\left(\frac{\hat{L}_+}{\hbar}\right)^2|l, -l\rangle$$

$$|l,-l+3\rangle = \frac{1}{\sqrt{(2l-2)\cdot 3\cdot(2l-1)\cdot 2\cdot(2l)\cdot 1}}\left(\frac{\hat{L}_+}{\hbar}\right)^3|l,-l\rangle$$

$$\vdots$$

$$|l,-l+k\rangle = \frac{1}{\sqrt{(2l+1-k)\cdots(2l-1)\cdot(2l)\cdot k!}}\left(\frac{\hat{L}_+}{\hbar}\right)^k|l,-l\rangle. \tag{9.62}$$

Or, if we set $k = l + m$, we find the general result

$$|l,m\rangle = \sqrt{\frac{(l-m)!}{(2l)!(l+m)!}}\left(\frac{\hat{L}_+}{\hbar}\right)^{l+m}|l,-l\rangle. \tag{9.63}$$

We have solved another quantum problem! This time, it was the angular momentum problem. We explicitly constructed angular-momentum eigenstates and their eigenvalues. In this case, we have simultaneous eigenstates with respect to the operators $\hat{\vec{L}}^2$ and $\hat{L}_z$. Note that we move up or down the angular-momentum multiplet with the raising and lowering operators, respectively. For each l, we have $2l + 1$ "m" states $|l,m\rangle$ with $-l \leq m \leq l$. The states satisfy $\hat{\vec{L}}^2|l,m\rangle = \hbar^2 l(l+1)|l,m\rangle$ and $\hat{L}_z|l,m\rangle = \hbar m|l,m\rangle$. We also have that l is an integer or a half-odd integer. The eigenstates are constructed from the lowest-weight state $|l,-l\rangle$ via Eq. (9.63).

One other useful exercise is to construct the matrix representation for angular momentum. We already did this for spin one-half in Chap. 7. Here, we sketch how to do it in general. Note that we will be working with matrices that are larger than the 2×2 matrices we have been working with. As you will see, this does not cause any problems. But, it might be new to some of you.

Physicists are quite creative when it comes to naming different things. Let's see how this is done for angular-momentum multiplets. If $l = 0$, we have one state only, so we call it a *singlet*. For $l = \frac{1}{2}$, we have $2l+1 = 2$ states, which we call a *doublet*. For $l = 1$, there are three states, and it is a *triplet*. For $l = \frac{3}{2}$, you guessed it—four states and a *quartet*. Very clever, don't you think!

I have found, over many years of teaching quantum mechanics, that students often really struggle in determining the matrix representation of an operator. So, we will try to do it carefully. Whenever we want to make a matrix representation of an operator, we need the operator, and we need the basis states that we represent the matrix in. The matrices for $l = 0$ are trivial, because $\hat{\vec{L}}^2|0,0\rangle = \hat{L}_z|0,0\rangle = \hat{L}_\pm|0,0\rangle = 0$. Hence, all matrices will just be the scalar 0. For $l = \frac{1}{2}$, we get 2×2 matrices. Each matrix is formed via varying the m values for the expectation values

$$O_{mm'} = \langle l,m|\hat{O}|l,m'\rangle, \tag{9.64}$$

where $-l \leq m, m' \leq l$ and $\hat{O} = \hat{\vec{L}}^2, \hat{L}_z, \hat{L}_\pm, \hat{L}_x$, and $\hat{L}_y$. Note that the l is fixed, and only the m values vary. So, for $l = \frac{1}{2}$, we have $m = \frac{1}{2}$ and $m = -\frac{1}{2}$. Then, because $\hat{\vec{L}}^2|\frac{1}{2}, m\rangle = \hbar^2 \frac{3}{4}|\frac{1}{2}, m\rangle$ and $\hat{L}_z|\frac{1}{2}, m\rangle = \hbar m|\frac{1}{2}, m\rangle$, those two matrices are diagonal:

$$\hat{\vec{L}}^2\left(l=\frac{1}{2}\right) = \hbar^2 \begin{pmatrix} \frac{3}{4} & 0 \\ 0 & \frac{3}{4} \end{pmatrix} \quad \text{and} \quad \hat{L}_z\left(l=\frac{1}{2}\right) = \hbar \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}. \tag{9.65}$$

The raising and the lowering operators are not diagonal. Recall that $\hat{L}_+|\frac{1}{2}, -\frac{1}{2}\rangle = \hbar|\frac{1}{2}, \frac{1}{2}\rangle$ and $\hat{L}_+|\frac{1}{2}, \frac{1}{2}\rangle = 0$. So, we have

$$\hat{L}_+\left(l=\frac{1}{2}\right) = \begin{pmatrix} \langle\frac{1}{2}, \frac{1}{2}|\hat{L}_+|\frac{1}{2}, \frac{1}{2}\rangle & \langle\frac{1}{2}, \frac{1}{2}|\hat{L}_+|\frac{1}{2}, -\frac{1}{2}\rangle \\ \langle\frac{1}{2}, -\frac{1}{2}|\hat{L}_+|\frac{1}{2}, \frac{1}{2}\rangle & \langle\frac{1}{2}, -\frac{1}{2}|\hat{L}_+|\frac{1}{2}, -\frac{1}{2}\rangle \end{pmatrix} = \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \tag{9.66}$$

and similarly $\hat{L}_-(\frac{1}{2}) = \hbar \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. Finally, we construct

$$\hat{L}_x\left(\frac{1}{2}\right) = \frac{1}{2}\left(\hat{L}_+\left(\frac{1}{2}\right) + \hat{L}_-\left(\frac{1}{2}\right)\right) = \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{9.67}$$

and similarly

$$\hat{L}_y\left(\frac{1}{2}\right) = \frac{1}{2i}\left(\hat{L}_+\left(\frac{1}{2}\right) - \hat{L}_-\left(\frac{1}{2}\right)\right) = \frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \tag{9.68}$$

just as we found before. Before moving on, let me give you a pep talk about how to think when constructing matrix representations of operators. We work in a basis that is complete, so any state can be expanded in terms of this basis. We start with one of the basis vectors and operate the operator on it. This creates a new vector, because operators map vectors to vectors. But the new vector can then be expanded in our basis (because it is complete). The matrix representation of the operator is just a coherent book-keeping device to store the coefficients of the new vector as it is expanded in the given basis and then repeating for the operator acting on each basis vector. I hope this mindset will help you compute the matrix representation of operators more easily.

In the problems below, we show many different physical applications of angular momentum, and we investigate many of the ideas related to formal manipulations of angular momentum and properties of angular-momentum representations. The video Angular Momentum Eigenstates https://tinyurl.com/yza66a2n describes this algebra in detail.

Section 9.3 Problems:

9.3.1 Angular Momentum Matrices for $l = 1$

There are three $l = 1$ states: $|1, 1\rangle$, $|1, 0\rangle$, and $|1, -1\rangle$. They are all eigenstates of $\hat{\vec{L}}^2$ and $\hat{L}_z$: We have $\hat{\vec{L}}^2|1, m\rangle = 2\hbar^2|1, m\rangle$ and $\hat{L}_z|1, m\rangle = \hbar m|1, m\rangle$. The raising operator satisfies $\hat{L}_+|1, -1\rangle = \sqrt{2}\hbar|1, 0\rangle$, $\hat{L}_+|1, 0\rangle = \sqrt{2}\hbar|1, 1\rangle$, and $\hat{L}_+|1, 1\rangle = 0$. You can work out the relations for $\hat{L}_-$ by using the general results for the c'_{1m}. Use all of these results to determine the $l = 1$ angular momentum matrices for $\hat{\vec{L}}^2$, $\hat{L}_z$, $\hat{L}_+$, $\hat{L}_-$, $\hat{L}_x$, and $\hat{L}_y$. You should find, for example, that

$$\hat{L}_x(l = 1) = \hbar \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & 0 \end{pmatrix} = \langle 1, m|\hat{L}_x|1, m'\rangle, \tag{9.69}$$

for $m, m' \in \{-1, 0, 1\}$.

Take your three Cartesian results, and show for each one that $\left(\hat{L}_i(l{=}1)\right)^3 = \hbar^2 \hat{L}_i(l{=}1)$ for $i = x$, y, and z. This allows you to compute the matrix exponential of powers of these operators as follows:

$$e^{-\frac{i\alpha}{\hbar}\hat{L}_i(l=1)} = \mathbb{I}_3 - i \sin\alpha \frac{\hat{L}_i(l{=}1)}{\hbar} + (\cos\alpha - 1)\left(\frac{\hat{L}_i(l{=}1)}{\hbar}\right)^2. \tag{9.70}$$

Please verify this formula.

9.3.2 Fluctuations of Angular Momentum Measurements

Let $\vec{e}_r = \sin\theta\ \cos\phi\, \vec{e}_x + \sin\theta\ \sin\phi\, \vec{e}_y + \cos\theta\, \hat{e}_z$. Determine $\vec{e}_r \cdot \hat{\vec{L}}|l, m\rangle$ in terms of $|l, m + 1\rangle$, $|l, m\rangle$, and $|l, m - 1\rangle$. Use $\hat{L}_x = \frac{1}{2}(\hat{L}_+ + \hat{L}_-)$ and $\hat{L}_y = \frac{1}{2i}(\hat{L}_+ - \hat{L}_-)$. You should find that $\langle l, m|\vec{e}_r \cdot \hat{\vec{L}}|l, m\rangle = \hbar m \cos\theta$. Continue to find $\left(\vec{e}_r \cdot \hat{\vec{L}}\right)^2 |l, m\rangle$, and show that

$$\langle l, m|\left(\vec{e}_r \cdot \hat{\vec{L}}\right)^2 |l, m\rangle = \hbar^2 \left(\frac{\sin^2\theta}{2}(l^2 + l - m^2) + \cos^2\theta\, m^2\right). \tag{9.71}$$

Use these two results to compute the fluctuation

$$\left(\Delta\vec{e}_r \cdot \hat{\vec{L}}\right)^2_{|l,m\rangle} = \langle l,m| \left(\vec{e}_r \cdot \hat{\vec{L}}\right)^2 |l,m\rangle - \left(\langle l,m|\vec{e}_r \cdot \hat{\vec{L}}|l,m\rangle\right)^2. \tag{9.72}$$

You should find that when $\theta = 0$ or $\theta = \pi$, that $\left(\vec{e}_r \cdot \hat{\vec{L}}\right)^2_{|l,m\rangle} = 0$. Why does this occur? What does it say about uncertainty? Note that the fluctuation never vanishes for $\theta \neq 0,\ \pi$ because $l(l+1) > m^2$, except for the case where $l = 0$, where the result always vanishes for all angles.

Hint: Recall what an eigenvector is.

9.3.3 Angular Momentum and Uncertainty

Recall our generalized uncertainty relation that we derived in Problem 7.7.6:

$$\left(\Delta\hat{A}_\psi\right)\left(\Delta\hat{B}_\psi\right) \geq \frac{1}{2}\left|\langle\psi|(-i)[\hat{A},\hat{B}]|\psi\rangle\right|. \tag{9.73}$$

Use the results from the previous problem for appropriately chosen θ and ϕ to show that

$$(\Delta L_x)_{|l,m\rangle}\left(\Delta L_y\right)_{|l,m\rangle} = \frac{\hbar^2}{2}\left(l(l+1) - m^2\right), \tag{9.74}$$

and hence that the cases $m = \pm l$ have the lowest uncertainty product of the different angular momentum eigenstates.

9.3.4 Uncertainty and the Stern-Gerlach Experiment

Examine the results of the previous problem for the case where $l = \frac{1}{2}$, as we have in the Stern-Gerlach experiment. If a state has a definite projection on the z-axis, then $(\Delta L_x)(\Delta L_y) = \frac{\hbar^2}{4}$, which is minimal uncertainty. Explain why this result implies that we know nothing about the projection on the x and y axes, when the atom has a definite projection on the z-axis.

9.3.5 Spherically Symmetric Rotor

A spherically symmetric rotor has a moment of inertia I associated with it. For example, the spherical surface (like a hollow ball) of radius a has a moment of inertia given by $\frac{2}{3}ma^2 = I$. Just like $\frac{\hat{P}^2}{2m}$ is the contribution to the kinetic energy

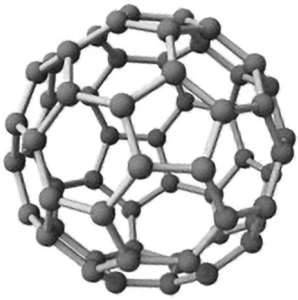

Fig. 9.3 Schematic of a "Buckyball," which has 60 carbon atoms located approximately at the locations of the vertices of a soccer ball. Note how it closely resembles a hollow sphere. The Buckyball is named in honor of the architect Buckminster Fuller

by the linear momentum of the object, $\frac{\hat{\vec{L}}^2}{2I}$ is the contribution from the angular momentum (rotational kinetic energy). In nature, there is a molecule called a "Buckeyball," a fullerene, with formula C_{60}, formed by 60 carbon atoms that take the shape of a soccer ball. The atoms are not exactly on the surface of a sphere, but they are quite close. So, we take $I = \frac{2}{3} \times 60 m_c \times a^2$, where m_c is the mass of a carbon atom (see Fig. 9.3).

Compute the energy of the fullerene molecule in an angular momentum state l. Your answer will look like $E_{\text{rot}} l(l+1)$. Determine E_{rot} in eV. Note that $a = 0.355\,\text{nm}$ for a fullerene molecule.

You should find that $E_{\text{rot}} = 3.45 \times 10^{-7}$ eV. The rotational spectral lines are given by the *difference* in energy between two adjacent energy levels, or $E = E_{\text{rot}}\big((l+1)(l+2) - l(l+1)\big) = E_{\text{rot}} 2(l+1)$. This means the *difference* between adjacent spectral lines (for different values of l) is given by $\Delta E = 2E_{\text{rot}}$. Show that $\Delta E \approx 0.0055\,\text{cm}^{-1}$ after using $8071\,\text{cm}^{-1} = 1$ eV. The experimental measurements shows a splitting between the spectral lines of $0.007\,\text{cm}^{-1}$. Not bad for such a simple model!

9.3.6 Rotational Properties of Ammonia

Most molecules are not rotationally symmetric. Consider an ammonia molecule NH_3. It has the nitrogen in the center and three hydrogen atoms below in a trigonal pyramidal shape. The moment of inertia I_z about the vertical axis (through the pyramid) is different from the moment of inertia about the x or y axis ($I_x = I_y = I$). This situation is called a symmetric top, and the Hamiltonian becomes

$$\hat{H} = \frac{\hat{L}_x^2 + \hat{L}_y^2}{2I} + \frac{\hat{L}_z^2}{2I_z}, \tag{9.75}$$

in what is called the *body frame* of the molecule. Determine the energy levels of a state $|l, m\rangle$. *Hint*: Use $\hat{L}_x^2 + \hat{L}_y^2 = \hat{\vec{L}}^2 - \hat{L}_z^2$ (see Fig. 9.4).

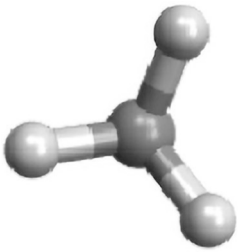

Fig. 9.4 Ball and stick model for the ammonia molecule NH_3. The central ball is nitrogen and the outer balls are hydrogens

For the remainder of the problem, assume that $I_z > I$, which is the case of an oblate top in the shape of a pancake; ammonia is an oblate top. The selection rules for the absorption of light are $\Delta l = \pm 1$ and $\Delta m = 0$. This means the spacing of the absorption lines is due to I only; I_z plays no role. If the distance in frequency between adjacent rotational transitions is $\Delta f = 19.89\ \text{cm}^{-1}$, determine the inverse moment of inertia in units of eV in the form $\frac{\hbar^2}{2I}$. Use the fact that 8071 cm^{-1} is equivalent to 1 eV when we multiply by $2\pi\hbar c$.

9.3.7 *Rotational Spectra of Diatomic Molecules*

Heteronuclear diatomic molecules (such as HCl shown in Fig. 9.5) are shaped like dumbbells. Since they are composed of two different atoms, as they rotate, the dipole along the molecule oscillates back and forth. An oscillating dipole couples to light, so the molecule can absorb or emit light. In this problem, we work in what is called the "body-fixed frame."

(a) Show that if I_z is very small, then we must have $\Delta m = 0$ for absorption by low-energy light.

(b) If $\Delta m = 0$ and $\Delta l = \pm 1$ only, then an absorption line appears at energies given by $E(l+1) - E(l)$. Find the shift in energy between two adjacent absorption lines as a function of the angular momentum l.

(c) Compute the moment of inertia by following the diagram in Fig. 9.6. The origin is at the center of the hydrogen nucleus. (i) Find the center-of-mass coordinate $r_{COM} = \frac{m_{cl} r}{m_h + m_{cl}} = \alpha r$ along the molecular axis. (ii) Compute $I = r_{COM}^2 m_h + (r - r_{COM})^2 m_{cl}$ in terms of α, t, and the masses. (iii) Show that $I = \mu r^2 = \frac{m_h m_{cl}}{m_h + m_{cl}} r^2$. (iv) Express $\frac{\hbar^2}{2I}$ as an energy in eV and as a frequency f in cm^{-1} via $2\pi\hbar c f = \frac{\hbar^2}{2I}$. (v) Use the rotational line spacing of the HCl molecular rotational spectra, given by 21.18 cm^{-1} to determine the molecular bond length r in Angstroms.

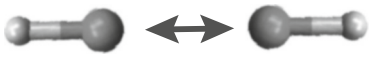

Fig. 9.5 Diatomic rotational spectra of a simple molecule hydrogen chloride. The hydrogen is the small ball, and the chlorine is the large ball. We show two extreme snapshots of the rotation. Since the charge on the hydrogen and on the chlorine is different, the spinning molecule looks like an oscillating dipole, which can absorb and emit light

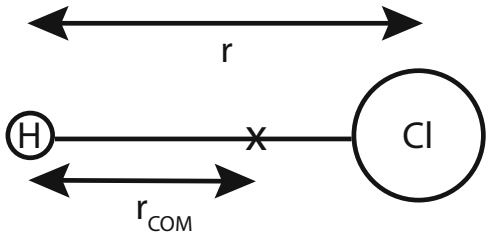

Fig. 9.6 Schematic diagram used to compute the center of mass and the moment of inertia of the diatomic molecule HCl. The origin lies at the center of the hydrogen nucleus. The symbol r is the internuclear distance

9.3.8 Higher Angular Momentum Stern-Gerlach Experiment

Suppose we shoot a beam of neutral atoms with angular momentum l through a Stern-Gerlach analyzer. Work through the problem below to determine what happens as we will now have more than just two spots.

(a) Choose the coordinate axes such that z is oriented along the magnetic field direction in the analyzer. How many different projections of the angular momentum $\hat{L}_z$ onto the axis of increasing magnetic field do we have for the set of states $|l, m\rangle$ with $-l \leq m \leq l$.
(b) How does the push or pull on the atom behave for states $|l, m\rangle$ which have different projections $\hbar m$ of $\hat{L}_z$ onto the magnetic field?
(c) How many spots will we have on the screen?
(d) What values must l have to have one of the spots correspond to atoms that go straight through with no deflection? Your answer will be integers, half-odd integers, or both.
(e) As l gets very large, what happens? Is it reasonable to assume that as $l \to \infty$, the system becomes classical? Why or why not?

Hint: You may assume that as $l \to \infty$, the spot corresponding to $m = l$ will hit the screen at a *finite* distance above the centerline.

9.3.9 Orbital Angular Momentum Must Have l Be an Integer

Most textbooks use an incorrect argument about the single-valuedness of the spherical harmonics to explain why the orbital angular momentum must have l be an integer. We will show that the angular momentum must be an integer three

different ways with none of the ways relying on this incorrect assumption. The first way is in this problem, and it follows an argument from Ballentine's wonderful textbook (Ballentine, 2015).

Recall that $\hat{L}_z = \hat{r}_x \hat{p}_y - \hat{r}_y \hat{p}_x$. We will show the eigenvalues of $\hat{L}_z$ must be integers multiplied by $\hbar$. First, define $\hat{q}_1 = \frac{\hat{r}_x + \hat{p}_y}{\sqrt{2}}$, $\hat{q}_2 = \frac{\hat{r}_x - \hat{p}_y}{\sqrt{2}}$, $\hat{p}_1 = \frac{\hat{p}_x - \hat{r}_y}{\sqrt{2}}$, and $\hat{p}_2 = \frac{\hat{p}_x + \hat{r}_y}{\sqrt{2}}$. Verify that $[\hat{q}_1, \hat{q}_2] = 0$, $[\hat{p}_1, \hat{p}_2] = 0$, and $[\hat{q}_\alpha, \hat{p}_\beta] = i\hbar\delta_{\alpha\beta}$, with α, $\beta \in \{1, 2\}$.

Show that

$$\hat{L}_z = \frac{1}{2}\left(\hat{p}_1^2 + \hat{q}_1^2\right) - \frac{1}{2}\left(\hat{p}_2^2 + \hat{q}_2^2\right), \tag{9.76}$$

which is the difference of two simple harmonic oscillators, each with $\omega = 1$. Both harmonic oscillators commute with each other, so the two harmonic oscillators can be diagonalized independently. The difference of these energy eigenvalues is given by $\hbar(n_1 + \frac{1}{2}) - \hbar(n_2 + \frac{1}{2})$, which is $\hbar$ multiplied by an integer. This establishes that the orbital angular momentum must have integral l. We will show this result in two additional ways later in this chapter.

A purist might be concerned that this difference of two harmonic oscillators has unbounded energies in both directions (positive and negative "energy"). Most quantum theories require the energy eigenvalues to be bounded from below. Indeed, this should be a concern and is the reason why we present two other derivations of this result. For cases where $l < \infty$, the $\hat{L}_z$ eigenvalues are always bounded, so there is no problem with this argument in those cases.

9.3.10 *Polynomial Equation Satisfied by Angular Momentum Matrices*

We define angular momentum matrices (not to be conflated with rotation matrices, which we will discuss later in the chapter) as

$$\hat{L}_i(l)_{m'm} = \langle l, m' | \hat{L}_i | l, m \rangle. \tag{9.77}$$

For spin $\frac{1}{2}$, these are $\frac{\hbar}{2}\sigma_i$. For $l = 1$, we found that

$$\hat{L}_x(l{=}1) = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \hat{L}_y(l{=}1) = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad \text{and}$$

$$\hat{L}_z(l{=}1) = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{9.78}$$

The general case is a $(2l+1)\times(2l+1)$ matrix as $\hat{L}_i(l)_{m'm} = \langle l, m'|\hat{L}_i|l, m\rangle$ for the matrices with a total angular momentum of l, with $-l \le m, m' \le l$.

We have already seen for spin-one-half that

$$\left(\hat{S}_i\left(l{=}\frac{1}{2}\right)\right)^2 = \frac{\hbar^2}{4}\mathbb{I}_2, \tag{9.79}$$

and for $l = 1$, we have

$$\left(\hat{L}_i(l{=}1)\right)^3 = \hbar^2\hat{L}_i(l{=}1). \tag{9.80}$$

The question is, what happens for arbitrary l? Our claim is that for angular momentum l (integer or half-odd integer), we have

$$\left(\frac{1}{\hbar}\hat{L}_i(l) - l\,\mathbb{I}_{2l+1}\right)\left(\frac{1}{\hbar}\hat{L}_i(l) - (l-1)\mathbb{I}_{2l+1}\right)\cdots\left(\frac{1}{\hbar}\hat{L}_i(l) - (-l+1)\mathbb{I}_{2l+1}\right)$$
$$\times\left(\frac{1}{\hbar}\hat{L}_i(l) - (-l)\mathbb{I}_{2l+1}\right) = 0, \tag{9.81}$$

holds. Verify that the spin-$\frac{1}{2}$ case and the spin-1 case obey these formulas, by moving all terms to the left-hand side, and factorizing for $l = 1$.

We prove this general result in two steps. We operate the product of angular momentum operators (not the matrices) on arbitrary angular momentum states, with fixed l. Since the states are arbitrary, the proof will be general at the operator level. Then we will convert the operators into the corresponding matrices.

(a) Show that if we take an arbitrary state in a total angular momentum l multiplet: $|\psi\rangle = \sum_{m=-l}^{l} C_m|l, m\rangle$, with C_m arbitrary complex numbers, then we have

$$\left(\frac{\hat{L}_z}{\hbar} - l\right)\left(\frac{\hat{L}_z}{\hbar} - (l-1)\right)\cdots\left(\frac{\hat{L}_z}{\hbar} - (-l)\right)|\psi\rangle = 0. \tag{9.82}$$

Hint: show that the terms in the product commute with each other and evaluate $\left(\frac{\hat{L}_z}{\hbar}\right)|l, m\rangle$. After establishing this result, then we can insert complete sets of states within the given multiplet between each of the operator factors, which converts them into matrices, and then establishes that

$$\prod_{m=-l}^{l}\left(\frac{1}{\hbar}\hat{L}_z(l) - m\,\mathbb{I}_{2l+1})\right) = 0. \tag{9.83}$$

(b) To prove the same result for the x and y matrices, we go back to the product of operators again and note that

$$\hat{L}_x = \hat{U}_y\left(\frac{\pi}{2}\right)\hat{L}_z\hat{U}_y^\dagger\left(\frac{\pi}{2}\right) \quad \text{and} \quad \hat{L}_y = \hat{U}_x\left(\frac{\pi}{2}\right)\hat{L}_z\hat{U}_x^\dagger\left(\frac{\pi}{2}\right). \tag{9.84}$$

The operator $\hat{U}_i(\alpha) = e^{-\frac{i}{\hbar}\alpha\vec{e}_i\cdot\hat{\vec{L}}}$ is a unitary operator that satisfies $\hat{U}_i(\alpha)\hat{U}_i^\dagger(\alpha) = \hat{\mathbb{I}}$. The similarity transformation with the same unitaries moves out through all products of operators (essentially this is the multiply-by-one trick in reverse), so

$$\begin{aligned}\prod_{m=-l}^{l}\left(\frac{\hat{L}_x}{\hbar} - m\right) &= \prod_{m=-l}^{l}\left(\hat{U}_y\left(\frac{\pi}{2}\right)\hat{L}_z\hat{U}_y^\dagger\left(\frac{\pi}{2}\right) - \hat{U}_y\left(\frac{\pi}{2}\right)m\hat{U}_y^\dagger\left(\frac{\pi}{2}\right)\right)\\ &= \hat{U}_y\left(\frac{\pi}{2}\right)\prod_{m=-l}^{l}\left(\frac{\hat{L}_z}{\hbar} - m\right)\hat{U}_y^\dagger\left(\frac{\pi}{2}\right).\end{aligned} \tag{9.85}$$

Now, we simply recognize that $\hat{U}_y^\dagger\left(\frac{\pi}{2}\right)|\psi\rangle = \sum_{m=-l}^{l} C'_m|l,m\rangle$ for some constants C'_m (which are related to C_m by multiplication by a unitary matrix). Then the argument proceeds similar to how it was done in (a). Finally repeat with $\hat{U}_x\left(\frac{\pi}{2}\right)$ for the product of $\hat{L}_y$ factors.

If you like working with symmetry arguments, you should try to argue how if the result holds for $\hat{L}_z$ must also hold for $\hat{L}_x$ and $\hat{L}_y$.

(c) Work out the representation for $l = \frac{3}{2}$ for x, y, and z, and show that they verify

$$\left(\frac{1}{\hbar^2}\hat{L}_i^2\left(l = \frac{3}{2}\right) - \frac{9}{4}\mathbb{I}_4\right)\left(\frac{1}{\hbar^2}\hat{L}_i^2\left(l = \frac{3}{2}\right) - \frac{1}{4}\mathbb{I}_4\right) = 0. \tag{9.86}$$

Unfortunately, one cannot easily use this to sum the exponential of angular momentum operators for $l = \frac{3}{2}$ as we did for $l = \frac{1}{2}$ and $l = 1$. But, it does turn out that other methods can be used to sum the exponentials for arbitrary l; however, the procedure is quite complicated. For details, see Curtright et al. (2014).

9.4 Addition of Angular Momentum

Addition of angular momentum is a rather advanced topic, and it requires both abstraction and a fair amount of algebra. It is extremely important, however, because we often have independent angular momenta that need to be combined. This can be the spin of an electron with its orbital angular momentum, or the angular momentum of the electrons with the angular momentum of the nucleus, and so on. The key is that the two angular momenta must be *independent* of each other, which we take to mean they commute with each other. Let $\hat{\vec{J}}^{(1)}$ and $\hat{\vec{J}}^{(2)}$ denote these two angular momenta, which satisfy

$$\left[\hat{J}_i^{(1)}, \hat{J}_j^{(1)}\right] = i\hbar \sum_k \epsilon_{ijk} \hat{J}_k^{(1)}, \left[\hat{J}_i^{(2)}, \hat{J}_j^{(2)}\right] = i\hbar \sum_k \epsilon_{ijk} \hat{J}_k^{(2)}, \text{ and } \left[\hat{J}_i^{(1)}, \hat{J}_j^{(2)}\right] = 0. \tag{9.87}$$

The last commutator is the one that tells us the two angular momenta are independent of each other.

Combining the two angular-momentum operators is straightforward—simply add them together $\hat{J}_i = \hat{J}_i^{(1)} + \hat{J}_i^{(2)}$; Problem 9.4.1 shows this is the only nontrivial choice that can be made. The total angular-momentum operator $\hat{\vec{J}}$ satisfies the commutation relations of angular momentum:

$$\begin{aligned}
[\hat{J}_i, \hat{J}_j] &= \left[\hat{J}_i^{(1)} + \hat{J}_i^{(2)}, \hat{J}_j^{(1)} + \hat{J}_j^{(2)}\right] \\
&= \left[\hat{J}_i^{(1)}, \hat{J}_j^{(1)}\right] + \left[\hat{J}_i^{(2)}, \hat{J}_j^{(2)}\right] \\
&= i\hbar \sum_k \epsilon_{ijk} \hat{J}_k^{(1)} + i\hbar \sum_k \epsilon_{ijk} \hat{J}_k^{(2)} = i\hbar \sum_k \epsilon_{ijk} \hat{J}_k.
\end{aligned} \tag{9.88}$$

This derivation simply uses the facts that the (1) operators commute with the (2) operators and the (1) and (2) operators are each angular-momentum operators. Our next step is to derive a maximal set of commuting operators to use to describe the eigenstates of the combined angular momentum. There are two ways to do this, and we start with the first way. To begin, note that $\left(\hat{\vec{J}}^{(1)}\right)^2$, $\hat{J}_z^{(1)}$, $\left(\hat{\vec{J}}^{(2)}\right)^2$, and $\hat{J}_z^{(2)}$ all mutually commute. The proof is direct. Since $\left[\hat{J}_i^{(1)}, \hat{J}_j^{(2)}\right] = 0$ and $\left[\left(\hat{\vec{J}}^{(i)}\right)^2, \hat{J}_z^{(i)}\right] = 0$ for $i = 1, 2$, we can see that all operators commute, as claimed. Hence, we can label the eigenstates via $|j_1, m_1, j_2, m_2\rangle$, and they satisfy

$$\left(\hat{\vec{J}}^{(1)}\right)^2 |j_1, m_1, j_2, m_2\rangle = \hbar^2 j_1(j_1+1)|j_1, m_1, j_2, m_2\rangle \tag{9.89}$$

$$\hat{J}_z^{(1)}|j_1, m_1, j_2, m_2\rangle = \hbar m_1 |j_1, m_1, j_2, m_2\rangle \tag{9.90}$$

$$\left(\hat{\vec{J}}^{(2)}\right)^2 |j_1, m_1, j_2, m_2\rangle = \hbar^2 j_2(j_2+1)|j_1, m_1, j_2, m_2\rangle \tag{9.91}$$

$$\hat{J}_z^{(2)}|j_1, m_1, j_2, m_2\rangle = \hbar m_2 |j_1, m_1, j_2, m_2\rangle. \tag{9.92}$$

This one was quite straightforward. Now, on to the second way. Note that $\hat{\vec{J}}^2$, $\hat{J}_z$, $\left(\hat{\vec{J}}^{(1)}\right)^2$, and $\left(\hat{\vec{J}}^{(2)}\right)^2$, all mutually commute. Let's check. Obviously $\left[\hat{\vec{J}}^2, \hat{J}_z\right]$ commutes, because $\hat{\vec{J}}$ is an angular momentum. Similarly, $\left[\left(\hat{\vec{J}}^{(i)}\right)^2, \hat{J}_z\right] = 0$ for $i = 1, 2$, because 1's commute with 2's and we can write $\hat{J}_z = \hat{J}_z^{(1)} + \hat{J}_z^{(2)}$. The

commutators that require more work are $\left[\hat{\vec{J}}^2, \left(\hat{\vec{J}}^{(i)}\right)^2\right] = 0$, for $i = 1, 2$. We verify this is true by writing

$$\hat{\vec{J}}^2 = \left(\hat{\vec{J}}^{(1)}\right)^2 + 2\hat{\vec{J}}^{(1)} \cdot \hat{\vec{J}}^{(2)} + \left(\hat{\vec{J}}^{(2)}\right)^2 = \sum_i \left(\hat{J}_i^{(1)}\hat{J}_i^{(1)} + 2\hat{J}_i^{(1)}\hat{J}_i^{(2)} + \hat{J}_i^{(2)}\hat{J}_i^{(2)}\right). \tag{9.93}$$

Since $\left[\left(\hat{\vec{J}}^{(i)}\right)^2, \hat{J}_z^{(i)}\right] = 0$ for $i = 1, 2$, we find both commutators vanish after using the Leibniz rule. This means that we can also write the states as $|j, m, j_1, j_2\rangle$. These states satisfy

$$\hat{\vec{J}}^2|j, m, j_1, j_2\rangle = \hbar^2 j(j+1)|j, m, j_1, j_2\rangle \tag{9.94}$$

$$\hat{J}_z|j, m, j_1, j_2\rangle = \hbar m|j, m, j_1, j_2\rangle \tag{9.95}$$

$$\left(\hat{\vec{J}}^{(1)}\right)^2|j, m, j_1, j_2\rangle = \hbar^2 j_1(j_1+1)|j, m, j_1, j_2\rangle \tag{9.96}$$

$$\left(\hat{\vec{J}}^{(2)}\right)^2|j, m, j_1, j_2\rangle = \hbar^2 j_2(j_2+1)|j, m, j_1, j_2\rangle. \tag{9.97}$$

Both representations are equally valid. How do they relate to each other? In particular, we know what j_1 and j_2 are, but what are the allowed values for j? Next, how do we relate results in one basis to results in another basis? The transformation coefficients between the two bases are called *Clebsch-Gordon coefficients*. We will constructively determine them next. But note that generically, one cannot have that a state $|j_1, m_1, j_2, m_2\rangle$ always has definite j, because $\hat{J}_z^{(i)}$ does not commute with $\hat{\vec{J}}^2$, although there are always two instances where the state does have definite j, which we will describe below. It is easy to "lose yourself in the forest" when we do this. Try to focus or reread this material to be sure you master it.

We begin by working with the uncoupled basis $|j_1, m_1, j_2, m_2\rangle$. This state is also an eigenstate of $\hat{J}_z$, with eigenvalue $\hbar(m_1 + m_2) = \hbar m$. Hence, we learn that $m = m_1 + m_2$. Unfortunately, this state is often not an eigenstate of $\hat{\vec{J}}^2$. There is one state where we do know the value of j. It is the state $|j_1, m_1{=}j_1, j_2, m_2{=}j_2\rangle$. This is because

$$\hat{J}_+|j_1, m_1{=}j_1, j_2, m_2{=}j_2\rangle = \left(\hat{J}_+^{(1)} + \hat{J}_+^{(2)}\right)|j_1, m_1{=}j_1, j_2, m_2{=}j_2\rangle = 0, \tag{9.98}$$

because $|j_1, m_1{=}j_1\rangle$ and $|j_2, m_2{=}j_2\rangle$ both cannot be raised. This means the state is a maximal m state as well, so the state with $m = j_1 + j_2$ cannot be raised. Hence, it has $j = j_1 + j_2$. Congratulations, we have identified one of the states! How do we find the other states? The procedure is straightforward, but tedious. Let us sketch

it next. The strategy uses only the properties of angular momentum raising and lowering operators and the orthogonality of eigenstates with different eigenvalues.

We have already established that, up to a phase,

$$|j{=}j_1 + j_2, m{=}j_1 + j_2, j_1, j_2\rangle = |j_1, m_1{=}j_1, j_2, m_2{=}j_2\rangle, \tag{9.99}$$

and we equate the two, by convention, setting the phase equal to 1. How do we find another state? Well we know that we can lower the m value by applying $\hat{J}_-$. This will keep j fixed and reduce m by one to $j_1 + j_2 - 1$. So, we have that

$$\hat{J}_-|j{=}j_1{+}j_2, m{=}j_1{+}j_2, j_1, j_2\rangle \propto |j{=}j_1{+}j_2, m{=}j_1{+}j_2{-}1, j_1, j_2\rangle, \tag{9.100}$$

and the state with $j = j_1 + j_2 - 1$ and $m = j_1 + j_2 - 1$ is orthogonal to the above state within the basis of states that have $m = j_1 + j_2 - 1$ (given by $|j_1, m_1{=}j_1, j_2, m_2{=}j_2 - 1\rangle$ and $|j_1, m_1{=}j_1 - 1, j_2, m_2{=}j_2\rangle$). Then we repeat the procedure—we hit each new state we found in the previous step with $\hat{J}_-$, which reduces m by 1 while keeping j fixed. If we have enough basis states that a state with j reduced by one from the lowest j value in the previous step, then it is found to be the vector orthogonal to all of the other vectors with the same m value that we constructed from the lowering operator. Eventually, one state in the next step cannot be lowered, and it is annihilated by $\hat{J}_-$. From then on, one state is annihilated by $\hat{J}_-$ until we finally reach the last state $|j{=}j_1 + j_2, m = -j_1 - j_2, j_1, j_2\rangle = |j_1, m_1{=}-j_1, j_2, m_2{=}-j_2\rangle$. It appears as if we will generate all possible j values down to 0 (or $\frac{1}{2}$), depending on j_1 and j_2. But, the minimal j value turns out to be $|j_1 - j_2|$. This is because at the kth step, we are working with states that have $m = j_1 + j_2 - k$. We construct them from $m_1 = j_1, m_2 = j_2 - k$; $m_1 = j_1 - 1, m_2 = j_2 - k + 1$; $\cdots$; $m_1 = j_1 - k, m_2 = j_2$. But, if $j_1 - k < -j_1$ or $j_2 - k < -j_2$, then there is no new state entering at this step, because $\hat{J}_-^{(i)}|j_i, m_i = -j_i\rangle = 0$ for $i = 1$ or 2, and so we simply apply the lowering operator onto all the states from the previous step, and there is no new j value created at this step; that is, the minimal value of j cannot be reduced.

It is easiest to see how this works with a concrete example, which is shown in Fig. 9.7. We pick $j_1 = \frac{5}{2}$ and $j_2 = \frac{7}{2}$. The maximal j is $\frac{5}{2} + \frac{7}{2} = 6$. There is one state with $m = 6$, corresponding to $|\frac{5}{2}, \frac{5}{2}, \frac{7}{2}, \frac{7}{2}\rangle$ in the $|j_1, m_1, j_2, m_2\rangle$ notation. For $m = 5$, there are two states: $|\frac{5}{2}, \frac{5}{2}, \frac{7}{2}, \frac{5}{2}\rangle$ and $|\frac{5}{2}, \frac{3}{2}, \frac{7}{2}, \frac{7}{2}\rangle$. It has a $j = 6$ and $j = 5$ state. The next step has three states with $m = 4$, given by $|\frac{5}{2}, \frac{5}{2}, \frac{7}{2}, \frac{3}{2}\rangle$, $|\frac{5}{2}, \frac{3}{2}, \frac{7}{2}, \frac{5}{2}\rangle$, and $|\frac{5}{2}, \frac{1}{2}, \frac{7}{2}, \frac{7}{2}\rangle$. It contains $j = 4, 5, 6$. This continues until we reach the step with $m = 0$. Here, there are only six states, so it has $j = 1, 2, 3, 4, 5, 6$. There is no state with $j = 0$. For lower m, we start removing states, because we will have the lowering operator annihilate states with $j = m$. So, the allowed j values range from $j_1 + j_2 = 6$ to $|j_1 - j_2| = 1$. Let's count the total states. The j_1 multiplet has $2\times\frac{5}{2}+1 = 6$ states, while the j_2 multiplet has $2\times\frac{7}{2}+1 = 8$ states. The total number of states is then $6 \times 8 = 48$. This is the number of black circles in the figure. Now, count the number of states with $j = 1$ to $j = 6$. It is $3 + 5 + 7 + 9 + 11 + 13 = 48$.

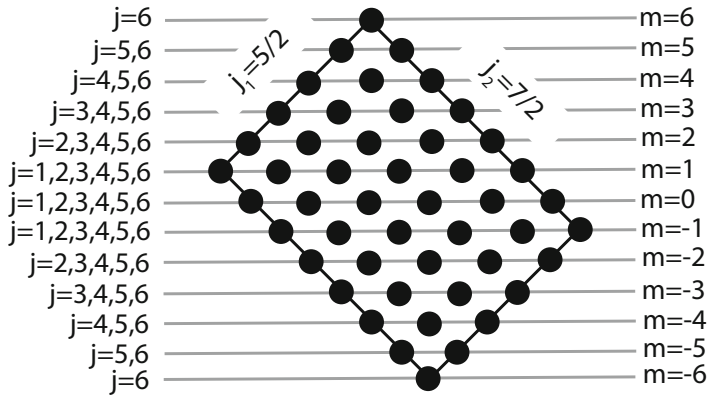

Fig. 9.7 Schematic for how angular momentum $j_1 = \frac{5}{2}$ is added to $j_2 = \frac{7}{2}$. The maximal j value is $j = 6$. The labels on the right show the $n = m$ value, while those on the left show the j values at that value of m. As we move down each step, new j values enter until we have $j = 1 - 6$. At this stage, j_1 cannot be lowered anymore, so there is no state with $j = 0$

So, the counting works! Now we check the general case. The total number of states is $N_{\text{tot}} = (2j_1 + 1)(2j_2 + 1)$. We need to sum

$$\sum_{j=|j_1-j_2|}^{j_1+j_2} (2j+1) = \sum_{j=0}^{j_1+j_2} (2j+1) - \sum_{j=0}^{|j_1-j_2|-1} (2j+1) \tag{9.101}$$

to find the total number of states. We used the add-zero trick to rewrite it in terms of two summations. To determine each sum, we need to compute $\sum_{j=0}^{n} 2j$, because $\sum_{j=0}^{n} 1 = n + 1$. Twice the sum of the first n integers is found by writing the summation twice, but changing the order of the second sum, so we have

$$\begin{aligned} &1 + \quad 2 \quad + \quad 3 \quad + \cdots + (n-1) + n \\ &+ n + (n-1) + (n-2) + \cdots + \quad 2 \quad + 1 \\ &= \underbrace{(n+1) + (n+1) + (n+1) + \cdots + (n+1) + (n+1)}_{n \text{ terms}} = n(n+1). \end{aligned} \tag{9.102}$$

It is often said that Euler invented this approach when he was a child to avoid a mathematical assignment given as a punishment to sum the first 100 integers. In any case, we now use this result to find that

$$\begin{aligned} \sum_{j=|j_1-j_2|}^{j_1+j_2} (2j+1) &= (j_1+j_2)(j_1+j_2+1) + j_1 + j_2 + 1 \\ &\quad - (|j_1-j_2|-1)|j_1-j_2| - |j_1-j_2| \end{aligned}$$

$$= 4j_1 j_2 + 2(j_1 + j_2) + 1 = (2j_1 + 1)(2j_2 + 1) = N_{\text{tot}}. \tag{9.103}$$

So it checks in the general case. We used the fact that $(a + b)^2 - (a - b)^2 = a^2 + 2ab + b^2 - a^2 + 2ab - b^2 = 4ab$ in deriving this result.

Hence, to summarize at this stage, we have shown that we can write

$$|j, m, j_1, j_2\rangle = \sum_{m_1=-j_1}^{j_1} |j_1, m_1, j_2, m - m_1\rangle \underbrace{\langle j_1, m_1, j_2, m - m_1 | j, m, j_1, j_2\rangle}_{\text{number}}, \tag{9.104}$$

which relates the coupled basis as a linear superposition of the uncoupled basis, with the Clebsch-Gordon coefficients, given by $\langle j_1, m_1, j_2, m - m_1 | j, m, j_1, j_2\rangle$, the numbers that multiply the different kets in the superposition. The coefficients are zero, of course, if $|m - m_1| > j_2$. Similarly, we have

$$|j_1, m_1, j_2, m_2\rangle = \sum_{j=|j_1-j_2|}^{j_1+j_2} |j, m_1+m_2, j_1, j_2\rangle \langle j, m_1+m_2, j_1, j_2 | j_1, m_1, j_2, m_2\rangle, \tag{9.105}$$

with $\langle j, m_1 + m_2, j_1, j_2 | j_1, m_1, j_2, m_2\rangle$ the Clebsch-Gordon coefficients for this superposition. The phases are always chosen to be real, so the only ambiguity is in the signs of the coefficients, which are fixed by a convention.

This has been a lot of formalism. Let us practice with some concrete examples. The simplest nontrivial example has $j_1 = \frac{1}{2}$ and $j_2 = \frac{1}{2}$. Then $j = 0$ or $j = 1$. The $j = 1$ state is called a triplet, and the $j = 0$ state is called a singlet. Obviously, we always have $j_1 = j_2 = \frac{1}{2}$, so we can use a shorthand notation, where $|j, m, j_1, j_2\rangle \to |j, m\rangle$ and $|j_1, m_1, j_2, m_2\rangle \to |m_1, m_2\rangle$. We find it is easiest to write $|m_1, m_2\rangle$ in terms of up and down arrows, so $|\frac{1}{2}, \frac{1}{2}\rangle = |\uparrow\rangle_1 \otimes |\uparrow\rangle_2 = |\uparrow\uparrow\rangle$, $|\frac{1}{2}, -\frac{1}{2}\rangle = |\uparrow\rangle_1 \otimes |\downarrow\rangle_2 = |\uparrow\downarrow\rangle$, and so on. The state with $m = 1$ is unique and is given by

$$|j{=}1, m{=}1\rangle = |\uparrow\uparrow\rangle. \tag{9.106}$$

We have two states with $m = 0$: $|\uparrow\downarrow\rangle$ and $|\downarrow\uparrow\rangle$. The state with $|j{=}1, m{=}0\rangle$ is a linear combination of these two, so it is given by $|j{=}1, m{=}0\rangle = \alpha|\uparrow\downarrow\rangle + \beta|\downarrow\uparrow\rangle$. How do we find the coefficients α and β? We apply the lowering operators $\hat{J}_- = \hat{J}_-^{(1)} + \hat{J}_-^{(2)}$. Recall that $\hat{J}_-^{(i)}|\uparrow\rangle_i = \hbar|\downarrow\rangle_i$ and $\hat{J}_-^{(i)}|\downarrow\rangle_i = 0$. So we have that

$$|j{=}1, m{=}0\rangle \propto \hat{J}_-|\uparrow\uparrow\rangle = \left(\hat{J}_-^{(1)} + \hat{J}_-^{(1)}\right)|\uparrow\uparrow\rangle = \hbar|\downarrow\uparrow\rangle + \hbar|\uparrow\downarrow\rangle. \tag{9.107}$$

Note how the state is not normalized. To normalize it, we have to multiply by $\frac{1}{\sqrt{2}\hbar}$, so we find that

$$|j{=}1, m{=}0\rangle = \frac{1}{\sqrt{2}}|\uparrow\uparrow\rangle + \frac{1}{\sqrt{2}}|\downarrow\uparrow\rangle. \tag{9.108}$$

Next, we find the state $|j{=}0, m{=}0\rangle$. How do we find it? It must be perpendicular to $|j{=}1, m{=}0\rangle$. But, we know how to do this—switch the coefficients, and make one of them negative, so we have that

$$|j{=}0, m{=}0\rangle = \frac{1}{\sqrt{2}}|\uparrow\downarrow\rangle - \frac{1}{\sqrt{2}}|\downarrow\uparrow\rangle. \tag{9.109}$$

Finally, to determine $|j{=}1, m{=}-1\rangle$, we use the lowering operator one more time. This produces the state $|\downarrow\downarrow\rangle$, as follows

$$\hat{J}_-|j{=}1, m{=}0\rangle = \frac{1}{\sqrt{2}}(\hat{J}_-^{(1)}|\uparrow\downarrow\rangle + \cancel{\hat{J}_-^{(2)}|\uparrow\downarrow\rangle} + \cancel{\hat{J}_-^{(1)}|\downarrow\uparrow\rangle} + \hat{J}_-^{(2)}|\downarrow\uparrow\rangle) = \hbar\sqrt{2}|\downarrow\downarrow\rangle. \tag{9.110}$$

So we have

$$|j{=}1, m{=}-1\rangle = |\downarrow\downarrow\rangle. \tag{9.111}$$

In summary, we have found that

$$\begin{aligned} |j{=}1, m{=}1\rangle &= |\uparrow\uparrow\rangle, \\ |j{=}1, m{=}0\rangle &= \frac{1}{\sqrt{2}}\big(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle\big), \\ |j{=}1, m{=}-1\rangle &= |\downarrow\downarrow\rangle, \end{aligned} \tag{9.112}$$

and

$$|j{=}0, m{=}0\rangle = \frac{1}{\sqrt{2}}\big(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle\big). \tag{9.113}$$

Now, we work out the general case with j_1 and $j_2 = \frac{1}{2}$. Then $j = j_1 + \frac{1}{2}$ and $j = j_1 - \frac{1}{2}$ are the two possibilities. Clearly

$$|j{=}j_1 + \tfrac{1}{2}, m{=}j_1 + \tfrac{1}{2}\rangle = |m_1{=}j_1, m_2{=}\tfrac{1}{2}\rangle, \tag{9.114}$$

where we suppress the j_1, and j_2 labels from the coupled and decoupled kets. We found this state, because it is the unique state with the maximal m value. To get the next state in the $j = j_1 + \frac{1}{2}$ multiplet, we hit the first state with a $\hat{J}_-$ operator. Using the lowering operator, we have

$$\begin{aligned}
&\hat{J}_-|j{=}j_1+\tfrac{1}{2}, m{=}j_1+\tfrac{1}{2}\rangle \\
&\quad=\hbar\sqrt{\left(j_1+\tfrac{1}{2}+j_1+\tfrac{1}{2}\right)\left(j_1+\tfrac{1}{2}-j_1-\tfrac{1}{2}+1\right)}|j{=}j_1+\tfrac{1}{2}, m{=}j_1-\tfrac{1}{2}\rangle \\
&\quad=\hbar\sqrt{(j_1+j_1)(j_1-j_1+1)}|m_1{=}j_1-1, m_2{=}\tfrac{1}{2}\rangle+\hbar|m_1{=}j_1, m_2{=}-\tfrac{1}{2}\rangle,
\end{aligned} \tag{9.115}$$

where on the first line, we use the standard $|l, m\rangle$ formula, and in the second line, we rewrite the lowering operator as the sum of the lowering operators for the first and second angular momentum. Then we have

$$\begin{aligned}
|j{=}j_1+\tfrac{1}{2}, m{=}j_1-\tfrac{1}{2}\rangle=&\sqrt{\frac{2j_1}{2j_1+1}}|m_1{=}j_1-1, m_2=\tfrac{1}{2}\rangle \\
&+\frac{1}{\sqrt{2j_1+1}}|m_1{=}j_1, m_2{=}-\tfrac{1}{2}\rangle.
\end{aligned} \tag{9.116}$$

The state $|j{=}j_1-\frac{1}{2}, m{=}j_1-\frac{1}{2}\rangle$ is orthogonal to this state, so we have

$$\begin{aligned}
|j{=}j_1-\tfrac{1}{2}, m{=}j_1-\tfrac{1}{2}\rangle=&-\frac{1}{\sqrt{2j_1+1}}|m_1{=}j_1-1, m_2=\tfrac{1}{2}\rangle \\
&+\sqrt{\frac{2j_1}{2j_1+1}}|m_1{=}j_1, m_2{=}-\tfrac{1}{2}\rangle.
\end{aligned} \tag{9.117}$$

To find the next lower m value, we hit with $\hat{J}_-$ again. We find

$$\begin{aligned}
&\hat{J}_-|j{=}j_1+\tfrac{1}{2}, m{=}j_1-\tfrac{1}{2}\rangle \\
&\quad=\hbar\sqrt{\left(j_1+\tfrac{1}{2}+j_1-\tfrac{1}{2}\right)\left(j_1+\tfrac{1}{2}-j_1+\tfrac{1}{2}+1\right)}|j{=}j_1+\tfrac{1}{2}, m{=}j_1-\tfrac{3}{2}\rangle \\
&\quad=\hbar\sqrt{(j_1+j_1)(j_1-j_1+1)}\frac{1}{\sqrt{2j_1+1}}|m_1{=}j_1-1, m_2=-\tfrac{1}{2}\rangle \\
&\quad+\hbar\sqrt{(j_1+j_1+1)(j_1-j_1+1+1)}\sqrt{\frac{2j_1}{2j_1+1}}|m_1{=}j_1-2, m_2=\tfrac{1}{2}\rangle \\
&\quad+\hbar\sqrt{\frac{2j_1}{2j_1+1}}|m_1{=}j_1-1, m_2=-\tfrac{1}{2}\rangle,
\end{aligned} \tag{9.118}$$

or

$$|j{=}j_1+\tfrac{1}{2}, m=j_1-\tfrac{3}{2}\rangle = \sqrt{\frac{2j_1-1}{2j_1+1}}|m_1{=}j_1-2, m_2{=}\tfrac{1}{2}\rangle + \sqrt{\frac{2}{2j_1+1}}|m_1{=}j_1-1, m_2{=}-\tfrac{1}{2}\rangle. \tag{9.119}$$

You might be able to see the overall pattern. In general, we find that

$$|j{=}j_1+\tfrac{1}{2}, m\rangle = \sqrt{\frac{j_1+m+\frac{1}{2}}{2j_1+1}}|m_1{=}m-\tfrac{1}{2}, m_2{=}\tfrac{1}{2}\rangle + \sqrt{\frac{j_1-m+\frac{1}{2}}{2j_1+1}}|m_1{=}m+\tfrac{1}{2}, m_2{=}-\tfrac{1}{2}\rangle, \tag{9.120}$$

and

$$|j{=}j_1-\tfrac{1}{2}, m\rangle = -\sqrt{\frac{j_1-m+\frac{1}{2}}{2j_1+1}}|m_1{=}m-\tfrac{1}{2}, m_2{=}\tfrac{1}{2}\rangle + \sqrt{\frac{j_1+m+\frac{1}{2}}{2j_1+1}}|m_1{=}m+\tfrac{1}{2}, m_2{=}-\tfrac{1}{2}\rangle. \tag{9.121}$$

As a check, we can derive this another way. We form a linear combination of the two states with fixed total m, and we solve the eigenvalue problem for the operator of total angular momentum $\hat{\vec{J}}^2$, which we expand as

$$\hat{\vec{J}}^2 = \left(\hat{\vec{J}}^{(1)}\right)^2 + \hat{J}_+^{(1)}\hat{J}_-^{(2)} + \hat{J}_-^{(1)}\hat{J}_+^{(2)} + 2\hat{J}_z^{(1)}\hat{J}_z^{(2)} + \left(\hat{\vec{J}}^{(2)}\right)^2. \tag{9.122}$$

Be sure to verify this expansion yourself. We use coefficients α and β in our expansion of the state with definite j

$$|j, m\rangle = \alpha|m_1{=}m-\tfrac{1}{2}, m_2{=}\tfrac{1}{2}\rangle + \beta|m_1{=}m+\tfrac{1}{2}, m_2{=}-\tfrac{1}{2}\rangle. \tag{9.123}$$

Then, we have

$$\frac{\hat{\vec{J}}^2}{\hbar^2}|j, m\rangle = \alpha\left(j_1(j_1+1)+\tfrac{3}{4}+m-\tfrac{1}{2}\right)|m_1{=}m-\tfrac{1}{2}, m_2{=}\tfrac{1}{2}\rangle + \alpha\sqrt{\left(j_1-m+\tfrac{1}{2}\right)\left(j_1+m-\tfrac{1}{2}+1\right)}|m_1{=}m+\tfrac{1}{2}, m_2{=}-\tfrac{1}{2}\rangle$$

$$+\beta\left(j_1(j_1+1)+\tfrac{3}{4}-m-\tfrac{1}{2}\right)|m_1{=}m+\tfrac{1}{2},m_2{=}-\tfrac{1}{2}\rangle$$
$$+\beta\sqrt{\left(j_1+m+\tfrac{1}{2}\right)\left(j_1-m-\tfrac{1}{2}+1\right)}|m_1{=}m-\tfrac{1}{2},m_2{=}\tfrac{1}{2}\rangle. \tag{9.124}$$

We can rewrite this as an eigenvalue-eigenvector equation

$$j(j+1)\begin{pmatrix}\alpha\\ \beta\end{pmatrix}=\begin{pmatrix} j_1(j_1+1)+m+\frac{1}{4} & \sqrt{\left(j_1+\frac{1}{2}\right)^2-m^2}\\ \sqrt{\left(j_1+\frac{1}{2}\right)^2-m^2} & j_1(j_1+1)-m+\frac{1}{4}\end{pmatrix}\begin{pmatrix}\alpha\\ \beta\end{pmatrix}. \tag{9.125}$$

The eigenvalues λ satisfy

$$\lambda^2-2\left(j_1+\tfrac{1}{2}\right)^2\lambda+\left(j_1+\tfrac{1}{2}\right)^4-m^2-\left(j_1+\tfrac{1}{2}\right)^2+m^2=0. \tag{9.126}$$

Solving for the roots gives us

$$\begin{aligned}\lambda_\pm&=\left(j_1+\tfrac{1}{2}\right)^2\pm\sqrt{\left(j_1+\tfrac{1}{2}\right)^4-\left(j_1+\tfrac{1}{2}\right)^4+\left(j_1+\tfrac{1}{2}\right)^2}\\ &=\left(j_1+\tfrac{1}{2}\right)^2\pm\left(j_1+\tfrac{1}{2}\right).\end{aligned} \tag{9.127}$$

This is summarized as

$$\lambda_\pm=\begin{cases}\left(j_1+\frac{1}{2}\right)\left(j_1+\frac{3}{2}\right)=j(j+1)\text{ for }j=j_1+\frac{1}{2}\\ \left(j_1+\frac{1}{2}\right)\left(j_1-\frac{1}{2}\right)=j(j+1)\text{ for }j=j_1-\frac{1}{2}\end{cases}. \tag{9.128}$$

So, we do get the correct eigenvalues for $j_1+\frac{1}{2}$ and $j_1-\frac{1}{2}$.

To find the coefficients α and β, we must find the eigenvectors. This will be completed by you in Problem 9.4.4. These results are summarized in Table 9.1 for the case where we add spin one-half to j_1. One can work out other cases, like $j_2=1$, or $j_2=\frac{3}{2}$, etc. They all become more complicated to work out. So tables have been created to summarize these results. But, in many cases, proper repeated use of the spin-one-half table is all that is needed to work out the general case. This algebra is long, so the video Addition of Angular Momentum I https://tinyurl.com/5dc45nvt should help. The specific case of adding spin-one-half is covered in Addition of Angular Momentum II https://tinyurl.com/2xx2entc.

Table 9.1 Table of the Clebsch-Gordon coefficients for adding angular momentum j_1 to angular momentum $j_2 = \frac{1}{2}$. The expansion is $|j, m, j_1, j_2 = \frac{1}{2}\rangle = c_+|j_1, m_1 = m - \frac{1}{2}, j_2 = \frac{1}{2}, m_2 = \frac{1}{2}\rangle + c_-|j_1, m_1 = m + \frac{1}{2}, j_2 = \frac{1}{2}, m_2 = -\frac{1}{2}\rangle$. The coefficients $c_\pm$ are given in the table for each of the two allowed j values

	$m_2 = \frac{1}{2}$	$m_2 = -\frac{1}{2}$
$j = j_1 + \frac{1}{2}$	$\sqrt{\frac{j_1+m+\frac{1}{2}}{2j_1+1}}$	$\sqrt{\frac{j_1-m+\frac{1}{2}}{2j_1+1}}$
$j = j_1 - \frac{1}{2}$	$-\sqrt{\frac{j_1-m+\frac{1}{2}}{2j_1+1}}$	$\sqrt{\frac{j_1+m+\frac{1}{2}}{2j_1+1}}$

Section 9.4 Problems:

9.4.1 Possible Forms for the Total Angular Momentum Operators

In the text, we simply formed $\hat{\vec{J}}$ as the equal sum of $\hat{\vec{J}}^{(1)}$ and $\hat{\vec{J}}^{(2)}$. In this problem, we explore why we made that choice. Let

$$\hat{\vec{J}}(\alpha, \beta) = \alpha \hat{\vec{J}}^{(1)} + \beta \hat{\vec{J}}^{(2)}. \tag{9.129}$$

Show that

$$[\hat{J}_i(\alpha, \beta), \hat{J}_j(\alpha, \beta)] = i\hbar \sum_k \epsilon_{ijk} \left(\alpha^2 \hat{J}_k^{(1)} + \beta^2 \hat{J}_k^{(2)}\right), \tag{9.130}$$

which are the commutation relations for an angular momentum for four cases of α and β. One of them ($\alpha = 0,\ \beta = 0$) is trivial and uninteresting. See if you can find the other three cases and identify what they correspond to.

Hint: the solutions to $x^2 = x$ are $x = 0$ and $x = 1$.

9.4.2 Tensor-Product Notation for Combining Angular Momentum

In our discussion of combining angular momentum, we suppressed the proper use of tensor products, introduced in Chap. 6. This is done, because it saves space in the formulas, but it can be confusing. In this problem, we illustrate the proper use of the tensor products. In particular, we write

$$\hat{\vec{J}} = \hat{\vec{J}}^{(1)} \otimes \mathbb{I} + \mathbb{I} \otimes \hat{\vec{J}}^{(2)}. \tag{9.131}$$

The commutator of the components of $\hat{\vec{J}}$ with itself is then

$$[\hat{J}_i, \hat{J}_j] = \left[\hat{J}_i^{(1)} \otimes \mathbb{I} + \mathbb{I} \otimes \hat{J}_i^{(2)}, \hat{J}_j^{(1)} \otimes \mathbb{I} + \mathbb{I} \otimes \hat{J}_j^{(2)}\right] = \left[\hat{J}_i^{(1)}, \hat{J}_j^{(2)}\right] \otimes \mathbb{I} + \mathbb{I} \otimes \left[\hat{J}_i^{(2)}, \hat{J}_j^{(2)}\right], \tag{9.132}$$

because $\hat{I}$ commutes with everything. Use this result to show that $\hat{\vec{J}}$ is an angular momentum operator. Also show that the four operators in the set

$$\left\{\left(\hat{\vec{J}}^{(1)}\right)^2 \otimes \mathbb{I},\ \hat{J}_z^{(1)} \otimes \mathbb{I},\ \mathbb{I} \otimes \left(\hat{\vec{J}}^{(2)}\right)^2,\ \mathbb{I} \otimes \hat{J}_z^{(2)}\right\} \tag{9.133}$$

all commute. Repeat and show the following operators all commute:

$$\left\{\left(\hat{\vec{J}}^{(1)}\right)^2 \otimes \mathbb{I} + 2\sum_i \hat{J}_i^{(1)} \otimes \hat{J}_i^{(2)} + \mathbb{I} \otimes \left(\hat{\vec{J}}^{(2)}\right)^2,\ \hat{J}_z^{(1)} \otimes \mathbb{I} + \mathbb{I} \otimes \hat{J}_z^{(2)},\ \left(\hat{\vec{J}}^{(1)}\right)^2 \otimes \mathbb{I},\ \mathbb{I} \otimes \left(\hat{\vec{J}}^{(2)}\right)^2\right\}. \tag{9.134}$$

9.4.3 Angular Momentum for Many Spin-One-Half Particles

We are going to examine the results for combining one to eight independent spin-one-half particles (such as electrons). One particle is spin $\frac{1}{2}$. For two particles, we must add spin $\frac{1}{2}$ to spin $\frac{1}{2}$, which gives a spin 0 and a spin 1. For three, we can add spin 0 and spin $\frac{1}{2}$, which gives spin $\frac{1}{2}$, and a spin 1 and spin $\frac{1}{2}$, which gives spin $\frac{1}{2}$ and spin $\frac{3}{2}$. Hence, there are two spin $\frac{1}{2}$ states and one spin $\frac{3}{2}$. Graphically, this looks like the schematic in Fig. 9.8. In this problem, you continue this diagram out to eight particles.

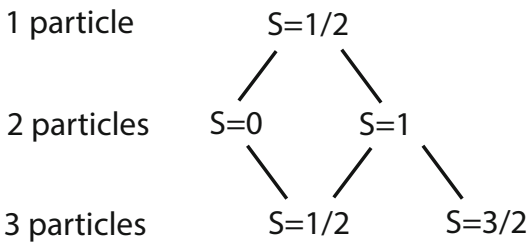

Fig. 9.8 Schematic of how spin-$\frac{1}{2}$ is combined multiple times

To check your results, note that n particles have 2^n states, so the sum of all angular momentum states must add up to 2^n for each n. For example,

$$n = 1: \; S = \tfrac{1}{2} \implies 2 \cdot \tfrac{1}{2} + 1 = 2 \text{ states} = 2^1 \checkmark \tag{9.135}$$

$$n = 2: \; \left.\begin{array}{ll} S = 0 & \implies 2 \cdot 0 + 1 = 1 \text{ state} \\ S = 1 & \implies 2 \cdot 1 + 1 = 3 \text{ states} \end{array}\right\} \; 4 \text{ states} = 2^2 \checkmark \tag{9.136}$$

$$n = 3: \; \left.\begin{array}{ll} S = \frac{1}{2} \text{ (twice)} & \implies 2\left(2 \cdot \frac{1}{2} + 1\right) = 4 \text{ states} \\ S = \frac{3}{2} & \implies 2 \cdot \frac{3}{2} + 1 \;\; = 4 \text{ states} \end{array}\right\} \; 8 \text{ states} = 2^3 \checkmark \tag{9.137}$$

and so on.

Ans.: You should find that $n = 8$ has 14 $S = 0$, 28 $S = 1$, 20 $S = 2$, 7 $S = 3$, and 1 $S = 4$ states.

9.4.4 Eigenvectors for the "Add Spin-One-Half" Table

Complete the following algebra for the "add spin-one-half" table. Find the eigenvectors (α, β) for $j = j_1 \pm \frac{1}{2}$ by solving the eigenvalue-eigenvector equation in Eq. (9.125).

Hint: Once you have found one solution, the other one follows by interchanging $\alpha \to -\beta$ and $\beta \to \alpha$ for the one solution you already found. This is because the eigenvectors must be orthogonal, and there are only two of them.

9.4.5 Concrete Example of "Add Spin-One-Half"

We consider adding $j_1 = \frac{5}{2}$ and $j_2 = \frac{1}{2}$ by using the "add spin-one-half" table that we derived above. Work out all of the different states. For example, we have

$$\begin{aligned} |j{=}3, m{=}3\rangle &= |m_1{=}\tfrac{5}{2}, m_2{=}\tfrac{1}{2}\rangle, \\ |j{=}3, m{=}2\rangle &= \sqrt{\tfrac{5}{6}}|m_1{=}\tfrac{3}{2}, m_2{=}\tfrac{1}{2}\rangle + \sqrt{\tfrac{1}{6}}|m_1{=}\tfrac{5}{2}, m_2{=}-\tfrac{1}{2}\rangle, \end{aligned} \tag{9.138}$$

and so on. Use this problem to practice using the table.

9.4.6 "Add Spin-One" Table I

Consider the "add spin-one" Clebsch-Gordon coefficient table. We want to solve the problem using the matrix method. We express

$$|j, m, j_1, j_2\rangle = \alpha|j_1, m_1{=}m-1, j_2{=}1, m_2{=}1\rangle + \beta|j_1, m_1{=}m, j_2{=}1, m_2=0\rangle + \gamma|j_1, m_1{=}m+1, j_2{=}1, m_2{=}-1\rangle. \tag{9.139}$$

Then we strive to find the coefficients α, β, and γ such that

$$\begin{pmatrix} j_1(j_1{+}1){+}2m & \sqrt{2(j_1{+}m)(j_1{-}m{+}1)} & 0 \\ \sqrt{2(j_1{+}m)(j_1{-}m{+}1)} & j_1(j_1{+}1){+}2 & \sqrt{2(j_1{-}m)(j_1{+}m{+}1)} \\ 0 & \sqrt{2(j_1{-}m)(j_1{+}m{+}1)} & j_1(j_1{+}1){-}2m \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = j(j{+}1) \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}. \tag{9.140}$$

This matrix uses the identity in Eq. (9.122). Verify that the matrix is correct. Note that one must have $j = j_1 + 1$, $j = j_1$, or $j = j_1 - 1$. Next, find the eigenvalues by setting $\det(A - j(j+1)\mathbb{I}_3) = 0$, which also determines the allowed values for j; the matrix A is given in the equation above. Note that the determinant of a 3×3 matrix of the form we have is given by

$$\det \begin{pmatrix} a & b & 0 \\ b & c & d \\ 0 & d & e \end{pmatrix} = ace - ad^2 - eb^2. \tag{9.141}$$

Hint: You already know what the allowed values for j must be. Use that information to help you verify the calculations. This problem requires a lot of stamina to get through it.

9.4.7 "Add Spin-One" Table II

As a continuation of the previous problem, we now determine the normalized eigenvectors, which are summarized in Table 9.2.

9.4.8 "Add Spin-One" Table III

In this problem, we also develop the add $j_2 = 1$ table, but now using the first method with the lowering operators. Start by creating $|j{=}j_1 + 1, m{=}j_1 + 1\rangle = |m_1{=}j_1, m_2{=}1\rangle$ and hit it with $\hat{J}_-$ to find $|j{=}j_1 + 1, m{=}j_1\rangle$. Then you can use orthogonality to find $|j{=}j_1, m{=}j_1\rangle$.

Table 9.2 Clebsch-Gordon coefficients for the "add-one" case. The coefficients α, β, and γ are the three elements in each row for the corresponding total j value, respectively

	$m_2 = 1$	$m_2 = 0$	$m_2 = -1$
$j = j_1 + 1$	$\sqrt{\frac{(j_1+m)(j_1+m+1)}{(2j_1+1)(2j_1+2)}}$	$\sqrt{\frac{(j_1+m+1)(j_1-m+1)}{(j_1+1)(2j_1+1)}}$	$\sqrt{\frac{(j_1-m)(j_1-m+1)}{(2j_1+1)(2j_1+2)}}$
$j = j_1$	$-\sqrt{\frac{(j_1+m)(j_1-m+1)}{2j_1(j_1+1)}}$	$\frac{m}{\sqrt{j_1(j_1+1)}}$	$\sqrt{\frac{(j_1-m)(j_1+m+1)}{2j_1(j_1+1)}}$
$j = j_1 - 1$	$\sqrt{\frac{(j_1-m)(j_1-m+1)}{2j_1(2j_1+1)}}$	$-\sqrt{\frac{(j_1-m)(j_1+m)}{j_1(2j_1+1)}}$	$\sqrt{\frac{(j_1+m)(j_1+m+1)}{2j_1(2j_1+1)}}$

Continue one more step to find $|j{=}j_1 + 1, m{=}j_1 - 1\rangle$ and $|j{=}j_1, m{=}j_1 - 1\rangle$ by using $\hat{J}_-$ again. Then, we find $|j{=}j_1 - 1, m{=}j_1 - 1\rangle$ by orthogonality. Recall that a vector orthogonal to two three-dimensional vectors $\vec{v}_1$ and $\vec{v}_2$ is $\vec{v}_1 \times \vec{v}_2$. Verify that the results you have found agree with the table (you may need to change overall signs, since they are only set by convention). If you are bold, see if you can intuit the results of the table by continuing to apply $\hat{J}_-$.

9.4.9 Heisenberg Exchange Interaction

Suppose I have a particle of spin j_1 interacting with a second particle of spin j_2 via the so-called Heisenberg exchange interaction:

$$\hat{H} = -c\hat{\vec{S}}^{(1)} \cdot \hat{\vec{S}}^{(2)}. \tag{9.142}$$

Determine the possible energy eigenvalues using results from combining angular momentum. In addition, if we add a magnetic field, so that

$$\hat{H} \to -c\hat{\vec{S}}^{(1)} \cdot \hat{\vec{S}}^{(2)} - gB\left(\hat{S}_z^{(1)} + \hat{S}_z^{(2)}\right). \tag{9.143}$$

Hint: If you are stuck, use the identity that is found by defining $\hat{\vec{S}} = \hat{\vec{S}}^{(1)} + \hat{\vec{S}}^{(2)}$ and using the fact that

$$\hat{\vec{S}}^{(1)} \cdot \hat{\vec{S}}^{(2)} = \frac{1}{2}\left(\hat{\vec{S}}^2 - \left(\hat{\vec{S}}^{(1)}\right)^2 - \left(\hat{\vec{S}}^{(2)}\right)^2\right). \tag{9.144}$$

Ans. $E = -\frac{c}{2}\big(S(S+1) - S_1(S_1(+1) - S_2(S_2+1)\big) - gBm$, with $S = |S_1 - S_2|$, $|S_1 - S_2| + 1, \cdots, S_1 + S_2$ and $-S \le m \le S$.

9.5 Entangled States, EPR, and Bell

The simplest case for combining angular momentum is the case of two spin-$\frac{1}{2}$ states. In this case, we obtain a spin singlet and a spin triplet state. The analysis of Sect. 9.4 gave us

$$\begin{aligned} |j=0, m=0\rangle &= \tfrac{1}{\sqrt{2}}\big(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle\big) \\ |j=1, m=1\rangle &= |\uparrow\uparrow\rangle, \\ |j=1, m=0\rangle &= \tfrac{1}{\sqrt{2}}|\big(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle\big), \\ |j=1, m=-1\rangle &= |\downarrow\downarrow\rangle. \end{aligned} \tag{9.145}$$

Two of these states are product states—$|\uparrow\uparrow\rangle$ and $|\downarrow\downarrow\rangle$—while two are entangled states, $\frac{1}{\sqrt{2}}\big(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle\big)$ and $\frac{1}{\sqrt{2}}\big(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle\big)$. In fact, these two states are *maximally entangled states*. They play a central role in our demonstration of Bohm's version of the EPR experiment. A spin singlet state ($j=0$) is a rotationally invariant state. This means it looks the same in any direction. We wrote it as $|j{=}0, m{=}0\rangle = \frac{1}{\sqrt{2}}\big(|\uparrow\downarrow\rangle_z - |\downarrow\uparrow\rangle_z\big)$, where the subscript z is to denote that we are measuring in the z-basis ($|\uparrow\rangle_z$ is the state with a positive projection on the z-axis and $|\downarrow\rangle_z$ is the state with the negative projection in the z-axis). Because the $j=0$ state is rotationally invariant, we expect it to be written the same way in any basis: $|j{=}0, m{=}0\rangle = \frac{1}{\sqrt{2}}\big(|\uparrow\downarrow\rangle_{\theta\phi} - |\downarrow\uparrow\rangle_{\theta\phi}\big)$, where θ, ϕ denotes that we are measuring with respect to an axis oriented in the θ, ϕ directions. We show this next.

To start, recall the solution for the up and down spin states oriented in the direction θ, ϕ, which we worked out in Sect. 6.2, and found that

$$|\uparrow\rangle_{\theta\phi} = \cos\tfrac{\theta}{2}|\uparrow\rangle_z + \sin\tfrac{\theta}{2}e^{i\phi}|\downarrow\rangle_z \quad \text{and} \quad |\downarrow\rangle_{\theta\phi} = \sin\tfrac{\theta}{2}e^{-i\phi}|\uparrow\rangle_z - \cos\tfrac{\theta}{2}|\downarrow\rangle_z. \tag{9.146}$$

Inverting these relations, we also find

$$\begin{aligned} |\uparrow\rangle_z &= \cos\tfrac{\theta}{2}|\uparrow\rangle_{\theta\phi} + \sin\tfrac{\theta}{2}e^{i\phi}|\downarrow\rangle_{\theta\phi} \quad \text{and} \\ |\downarrow\rangle_z &= -\sin\tfrac{\theta}{2}e^{-i\phi}|\uparrow\rangle_{\theta\phi} + \cos\tfrac{\theta}{2}|\downarrow\rangle_{\theta\phi}. \end{aligned} \tag{9.147}$$

Now, we substitute the above results into the expression for the singlet state, to reexpress it in the basis at the angles θ, ϕ, which gives

$$\begin{aligned} &|j=0, m=0\rangle \\ &= \frac{1}{\sqrt{2}}\Big[\Big(\cos\tfrac{\theta}{2}|\uparrow\rangle_{\theta\phi} + \sin\tfrac{\theta}{2}e^{i\phi}|\downarrow\rangle_{\theta\phi}\Big) \otimes \Big(-\sin\tfrac{\theta}{2}e^{-i\phi}|\uparrow\rangle_{\theta\phi} + \cos\tfrac{\theta}{2}|\downarrow\rangle_{\theta\phi}\Big) \end{aligned}$$

$$-\left(-\sin\tfrac{\theta}{2}e^{-i\phi}|\uparrow\rangle_{\theta\phi}+\cos\tfrac{\theta}{2}|\downarrow\rangle_{\theta\phi}\right)\otimes\left(\cos\tfrac{\theta}{2}|\uparrow\rangle_{\theta\phi}+\sin\tfrac{\theta}{2}e^{i\phi}|\downarrow\rangle_{\theta\phi}\right)\Big]. \tag{9.148}$$

Multiplying out the four terms in each tensor product and using our shorthand notation which suppresses the $\otimes$ symbol yield

$$\begin{aligned}
&|j=0,m=0\rangle\\
&=\frac{1}{\sqrt{2}}\Big[\cancel{\left(-\cos\tfrac{\theta}{2}\sin\tfrac{\theta}{2}e^{-i\phi}+\cos\tfrac{\theta}{2}\sin\tfrac{\theta}{2}e^{-i\phi}\right)}|\uparrow\uparrow\rangle_{\theta\phi}+\left(\cos^2\tfrac{\theta}{2}+\sin^2\tfrac{\theta}{2}\right)|\uparrow\downarrow\rangle_{\theta\phi}\\
&\quad+\left(-\sin^2\tfrac{\theta}{2}-\cos^2\tfrac{\theta}{2}\right)|\downarrow\uparrow\rangle_{\theta\phi}+\cancel{\left(\cos\tfrac{\theta}{2}\sin\tfrac{\theta}{2}e^{i\phi}-\cos\tfrac{\theta}{2}\sin\tfrac{\theta}{2}e^{i\phi}\right)}|\downarrow\downarrow\rangle_{\theta\phi}\Big]\\
&=\frac{1}{\sqrt{2}}\Big(|\uparrow\downarrow\rangle_{\theta\phi}-|\downarrow\uparrow\rangle_{\theta\phi}\Big).
\end{aligned} \tag{9.149}$$

So, the form of the singlet state is unchanged for any direction of the basis vectors! This result requires the minus sign in the expression for the singlet state—with the plus sign in the $|j{=}1,m{=}0\rangle$ state, we find that it transforms to a more complex superposition of states, so it is *not* invariant in different bases. This is explored more in the problems.

We have derived a really powerful result. Let us see how we can use it to analyze EPR experiments with our Dirac notation. Recall how the experiment worked, as shown in Fig. 2.23. The source prepares a pair of atoms in the $|j{=}0,m{=}0\rangle$ state. Recall that our measurement scheme for this experiment involves projecting onto the state corresponding to the exit the atom goes through (in the basis corresponding to the direction of the Stern-Gerlach analyzer) and finally squaring the probability amplitude to determine the probability for the event to occur. Because we have two atoms and two analyzers, there are four possible experimental outcomes. So, we project onto these four states using the following projection operators: $|\uparrow\uparrow\rangle\langle\uparrow\uparrow|$, $|\uparrow\downarrow\rangle\langle\uparrow\downarrow|$, $|\downarrow\uparrow\rangle\langle\downarrow\uparrow|$, and $|\downarrow\uparrow\rangle\langle\downarrow\uparrow|$ to find the probability for a specific measurement to occur; for example, $|\downarrow\uparrow\rangle\langle\downarrow\uparrow|$ would be used for exiting the negative exit on the left analyzer and the positive exit from the right analyzer. Using the specific form for the singlet state, we find that $P(++)=|\langle\uparrow\uparrow|j{=}0,m{=}0\rangle|^2=0$, $P(+-)=|\langle\uparrow\downarrow|j{=}0.m{=}0\rangle|^2=\frac{1}{2}$, $P(-+)=|\langle\downarrow\uparrow|j{=}0,m{=}0\rangle|^2=\frac{1}{2}$, and $P(--)=|\langle\downarrow\downarrow|j{=}0,m{=}0\rangle|^2=0$. So, we find the atoms always emerge from opposite exits of the two analyzers: half of the time it emerges + from the left and − from the right and half of the time − from the left and + from the right. It is important to recognize that we did not specify the axis of the analyzer in the projection operator. This is because the singlet state has exactly the same form, regardless of the orientation of the analyzer—so the result is the same for any orientation! Note that we also said nothing about which measurement occurred first. This is because the time of the measurement does not matter—after all, the theory of relativity tells us the order of the events depends on the frame we work in. But,

even if we were to say we measure on the left first, we still find the right result will be the opposite of the left result and both possibilities occur half of the time. To be concrete, if the measurement is a + on the left, then we project via

$$|j{=}0, m{=}0\rangle \rightarrow |\uparrow\rangle_L\, {}_L\langle\uparrow|j{=}0, m{=}0\rangle = \tfrac{1}{\sqrt{2}}|\uparrow\downarrow\rangle. \tag{9.150}$$

The probability for a + measurement on the left is 50%. Once this occurs, the measurement on the right is guaranteed to be −. A similar analysis holds when we measure a − on the left. Well, what about if the measurement is on the right first? Nothing changes! The final results are exactly the same. *There is no need to invoke relativity to send information about the result of the measurement from the right to the left for this to occur.* We always get the same result, but we do not learn "how the system knows" what the result will be. This is again one of our quantum mysteries. We should be familiar with such things by now—some questions in quantum mechanics can be answered, others cannot.

Let us now get more complicated. How do the probabilities change if we rotate the left analyzer by the angle θ relative to the vertical (with $\phi = 0$) as shown in Fig. 9.9 (where $\theta = 240^{\text{o}}$). We need to work out the projectors, which now require us to specify the axes of the states, which are the following: $|\uparrow\rangle_\theta \otimes |\uparrow\rangle_z$, $|\uparrow\rangle_\theta \otimes |\downarrow\rangle_z$, $|\downarrow\rangle_\theta \otimes |\uparrow\rangle_z$, and $|\downarrow\rangle_\theta \otimes |\downarrow\rangle_z$. Now we find that $P(++) = \frac{1}{2}|_\theta\langle\uparrow|\downarrow\rangle_z|^2 = \frac{1}{2}\sin^2\frac{\theta}{2}$, $P(+-) = \frac{1}{2}|_\theta\langle\uparrow|\uparrow\rangle_z = \frac{1}{2}\cos^2\frac{\theta}{2}$, $P(-+) = \frac{1}{2}|_\theta\langle\downarrow|\downarrow\rangle_z|^2 = \frac{1}{2}\cos^2\frac{\theta}{2}$, and $P(--) = \frac{1}{2}|_\theta\langle\downarrow|\uparrow\rangle_z|^2 = \frac{1}{2}\sin^2\frac{\theta}{2}$. As an example, we illustrate how one computes $P(++)$.

$$\begin{aligned} P(++) &= |\big(|\uparrow\rangle_\theta \otimes |\uparrow\rangle_z\big)\big({}_\theta\langle\uparrow| \otimes_z {}\langle\uparrow|\big)|j{=}0.m{=}0\rangle|^2 \\ &= |{}_\theta\langle\uparrow| \otimes_z {}\langle\uparrow|\tfrac{1}{\sqrt{2}}\big(|\uparrow\rangle_z \otimes |\downarrow\rangle_z - |\downarrow\rangle_z \otimes |\uparrow\rangle_z\big)|^2 \\ &= \tfrac{1}{2}|_\theta\langle\uparrow|\downarrow\rangle_z|^2 = \tfrac{1}{2}|\sin\tfrac{\theta}{2}e^{i\phi}|^2 = \tfrac{1}{2}\sin^2\tfrac{\theta}{2}. \end{aligned} \tag{9.151}$$

You should verify the other three cases yourself. These probabilities are needed to analyze the Bell experiment. But, before we do that, it is important to note that if $\theta = 90^{\text{o}}$, corresponding to the left analyzer oriented along the $+x$ direction, then the probability is equally $\frac{1}{2}$ to emerge from the + or − exit of the right analyzer and *there is no correlation with what is measured on the left!* We already saw this in Chap. 2, but now we have the full quantum derivation of this result. For any other angle of the analyzer, there is some correlation between the measurements on the left and the right, which become complete at $\theta = 0^{\text{o}}$ and 180^{o}.

Let us pause for a moment and recap what our full quantum derivation has given us for the EPR experiment. If we look at the measurements of the projections in any pair of Stern-Gerlach analyzers, with any relative orientation θ between them, then we find that half of the time each analyzer measures a positive projection and half of the time a negative projection. But, if we look closer at the *correlations* between the two measurements, we find that

Fig. 9.9 EPR experiment with the left analyzer rotated by an angle θ relative to the vertical (here by 240°). In this case, one can register a positive projection on both analyzers, as shown

$$\frac{P(\text{opposite})}{P(\text{opposite}) + P(\text{same})} = \frac{\frac{1}{2}\cos^2\frac{\theta}{2} + \frac{1}{2}\cos^2\frac{\theta}{2}}{\frac{1}{2}\cos^2\frac{\theta}{2} + \frac{1}{2}\cos^2\frac{\theta}{2} + \frac{1}{2}\sin^2\frac{\theta}{2} + \frac{1}{2}\sin^2\frac{\theta}{2}} = \cos^2\frac{\theta}{2}. \tag{9.152}$$

So, when the relative angle is 0°, they are anticorrelated; when the angle is 180°, they are completely correlated; and when the angle is 90°, we have a completely random result—one-half anticorrelated and one-half correlated. By choosing the relative angle, we can choose the level of correlation. What this means is regardless of which analyzer is measured first, we always see the correlations. This does not require a faster than light communication from one analyzer to the other. Instead, we see that it naturally occurs due to the quantum state and has no dependence on the distance between the analyzers. This is simply one of the quantum mysteries that we have to learn to live with.

We are now at the stage where we can discuss the Bell experiment more thoroughly. First, we recap how the experiment worked. Similar to the EPR experiment, we have a source which makes pairs of atoms in the singlet state. Now, the analyzers are packaged in a cylinder, and they rotate between one of N different allowed positions, equally spaced on a circle. We analyzed the case with $N = 3$ in detail. In this case, the analyzers are at 0°, 120°, and 240°. See Fig. 2.27. We now work out the quantum analysis for the N-stop Bell experiment. The angles θ are 0°, $\frac{360^\circ}{N}$, $2 \cdot \frac{360^\circ}{N}$, $\cdots$, $(N-1) \cdot \frac{360^\circ}{N}$. If we measure a + on the left, then the probability to measure a − on the right is

$$P(+-) = \frac{1}{N}\sum_{n=0}^{N-1}\frac{1}{2}|_z\langle\downarrow \mid \downarrow\rangle_{\theta_n}|^2 = \frac{1}{N}\sum_{n=0}^{N-1}\frac{1}{2}\cos^2\left(\frac{360^\circ n}{2N}\right). \tag{9.153}$$

Similarly, we have

$$P(-+) = \frac{1}{N}\sum_{n=0}^{N-1}\frac{1}{2}|_z\langle\uparrow \mid \uparrow\rangle_{\theta_n}|^2 = \frac{1}{N}\sum_{n=0}^{N-1}\frac{1}{2}\cos^2\left(\frac{360^\circ n}{2N}\right). \tag{9.154}$$

Hence, we have

$$P(+-)+P(-+) = \frac{1}{N}\sum_{n=0}^{N-1}\cos^2\left(\frac{\pi n}{N}\right) = \frac{1}{2N}\sum_{n=0}^{N-1}\left(1+\cos\frac{2\pi n}{N}\right)$$
$$= \frac{1}{2}+\frac{1}{2N}\mathrm{Re}\sum_{n=0}^{N-1}e^{\frac{2\pi i n}{N}} = \frac{1}{2}+\frac{1}{2N}\mathrm{Re}\left(\frac{1-e^{2\pi i}}{1-e^{\frac{2\pi i}{N}}}\right) = \frac{1}{2}. \tag{9.155}$$

The last line follows from $\sum_{n=0}^{N-1} x^n = \frac{1-x^N}{1-x}$, as we derived when we worked out the geometric series. The quantum calculation predicts a probability of $\frac{1}{2}$ for the two lights to be opposite and $\frac{1}{2}$ for the two lights to be the same. In other words, half the time we have the atoms exit the same exit and half the time they exit the opposite exits.

A hidden variable theory must pass an instruction set, and the set is chosen to always be the opposite for the left and right analyzers, when the analyzer is on the same stop point. We already worked out the probability for $N = 3$ in Sect. 2.3, and we found that $P(+-)+P(-+) \geq \frac{5}{9}$. For $N = 4$, we found that $P(+-)+P(-+) \geq \frac{1}{2}$, so a Bell experiment does not work, in the sense that a quantum probability always is $\frac{1}{2}$, and we cannot differentiate from the hidden variables if the lower bound for the instruction sets is also $\frac{1}{2}$. In the problems, you will prove that one cannot conclude anything from a Bell experiment when N is even, while for odd N, we will find that $P(+-)+P(-+) \geq \frac{N^2+1}{2N^2}$, which implies the experiment works best for $N = 3$. As an example, we create the hidden-variable instruction sets for the $N = 5$ case. Recall for an instruction set, the right choices are always opposite to the left choices. For $N = 5$, we have $2^5 = 32$ different possibilities for the instruction sets. We have five stops A, B, C, D, and E. There are three inequivalent tables, displayed below with checkmarks showing the cases where the two measured spins are opposite:

(I)	A	B	C	D	E
	$-$	$-$	$-$	$-$	$-$
A $+$	✓	✓	✓	✓	✓
B $+$	✓	✓	✓	✓	✓
C $+$	✓	✓	✓	✓	✓
D $+$	✓	✓	✓	✓	✓
E $+$	✓	✓	✓	✓	✓

(II)	A	B	C	D	E
	$-$	$-$	$-$	$-$	$+$
A $+$	✓	✓	✓	✓	×
B $+$	✓	✓	✓	✓	×
C $+$	✓	✓	✓	✓	×
D $+$	✓	✓	✓	✓	×
E $-$	×	×	×	×	✓

(III)	A	B	C	D	E
	$-$	$-$	$-$	$+$	$+$
A $+$	✓	✓	✓	×	×
B $+$	✓	✓	✓	×	×
C $+$	✓	✓	✓	×	×
D $-$	×	×	×	✓	✓
E $-$	×	×	×	✓	✓

(9.156)

In total, there are 32 tables, but they can all be constructed from the 3 tables above. The (I) table has two equivalent ones, found by interchanging $+ \leftrightarrow -$. Table (II) has ten equivalent ones, found by placing the single opposite sign ($-$ in the column, $+$ in the row) in any of the five slots (but it must be in the corresponding column for a given row) and then a factor of two for $+ \leftrightarrow -$. Table III has 20 equivalent ones; since we can place the 2 opposite signs in any of 10 different arrangements

because 5 choose 2 is 10. Then the $+ \leftrightarrow -$ correspondence gives an extra factor of two. Now, for each of the different tables, we need to compute the probability that the spins will be measured oppositely. These are given by the checkmarks. Hence for Table I, $P(\text{opposite}) = 1$, for Table II, $P(\text{opposite}) = \frac{17}{25}$, and for Table III, $P(\text{opposite}) = \frac{13}{25}$. This means we must have $P(\text{opposite}) \geq \frac{13}{25}$. Hence, one could use a five-stop experiment for a Bell test. You will analyze the general case in the problems.

Before ending this section, we discuss a truly bizarre phenomenon called teleportation. To begin, we must describe the four Bell states. They are all entangled states. The spin-singlet state is one of them, while the other $m = 0$ state is another one, and the final two are positive and negative linear superpositions of the all up and all down states. The four states are

$$|\Phi_0\rangle = \frac{1}{\sqrt{2}}\big(|\uparrow\rangle \otimes |\uparrow\rangle + |\downarrow\rangle \otimes |\downarrow\rangle\big) = \mathbb{I} \otimes \mathbb{I}|\Phi_0\rangle, \tag{9.157}$$

$$|\Phi_1\rangle = \frac{1}{\sqrt{2}}\big(|\uparrow\rangle \otimes |\downarrow\rangle + |\downarrow\rangle \otimes |\uparrow\rangle\big) = \mathbb{I} \otimes \sigma_x|\Phi_0\rangle, \tag{9.158}$$

$$|\Phi_2\rangle = \frac{i}{\sqrt{2}}\big(|\uparrow\rangle \otimes |\downarrow\rangle - |\downarrow\rangle \otimes |\uparrow\rangle\big) = \mathbb{I} \otimes \sigma_y|\Phi_0\rangle, \tag{9.159}$$

$$|\Phi_3\rangle = \frac{1}{\sqrt{2}}\big(|\uparrow\rangle \otimes |\uparrow\rangle - |\downarrow\rangle \otimes |\downarrow\rangle\big) = \mathbb{I} \otimes \sigma_z|\Phi_0\rangle, \tag{9.160}$$

which we write out explicitly using the tensor-product notation; we have suppressed a subscript on the two kets to indicate a label for the spaces the spins in the tensor product live in. One should be able to immediately see that every state is entangled. But for this type of two-state entanglement that involves a single tensor product, there is a simple rule one can use to determine whether a state is entangled. We write the entangled state in terms of four coefficients

$$\begin{aligned}|\psi\rangle = \sum_{ij} c_{ij}|i\rangle \otimes |j\rangle = c_{\uparrow\uparrow}|\uparrow\rangle \otimes |\uparrow\rangle + c_{\uparrow\downarrow}|\uparrow\rangle \otimes |\downarrow\rangle + c_{\downarrow\uparrow}|\downarrow\rangle \otimes |\uparrow\rangle \\ + c_{\downarrow\downarrow}|\downarrow\rangle \otimes |\downarrow\rangle,\end{aligned} \tag{9.161}$$

and then rearrange the coefficients into a 2×2 matrix:

$$C = \begin{pmatrix} c_{\uparrow\uparrow} & c_{\downarrow\uparrow} \\ c_{\downarrow\uparrow} & c_{\downarrow\downarrow} \end{pmatrix}. \tag{9.162}$$

Then, the state is entangled whenever $\det C \neq 0$. This brings in a geometric nature to entanglement. If the vectors that make up the matrix, thought of as the two rows of C, span a nonzero area, then the state is entangled. You should be able to see that any case of a product state will be guaranteed to have zero determinant. One

can now check directly that the determinants corresponding to each Bell state are indeed nonzero, verifying that they are entangled states.

The Bell states actually form an orthonormal basis for the four-dimensional tensor-product spin space. This has to be verified directly, although using some Pauli matrix identities makes the calculation a bit simpler. First note that the state $|\Phi_0\rangle$ is normalized and it is orthogonal to the states $|\Phi_i\rangle$ for $i = 1$, 2, and 3. This can be verified by just computing the inner products of the different vectors. Next, using the fact that $\sigma_i\sigma_j = \delta_{ij}\mathbb{I}_2 + i\sum_k \epsilon_{ijk}\sigma_k$, we find that

$$\langle\Phi_i|\Phi_j\rangle = \langle\Phi_0|\big(\mathbb{I}\otimes\sigma_i\big)\big(\mathbb{I}\otimes\sigma_j\big)|\Phi_0\rangle = \langle\Phi_0|\mathbb{I}\otimes\Big(\delta_{ij}\mathbb{I} + i\sum_k \epsilon_{ijk}\sigma_k\Big)|\Phi_0\rangle$$

$$= \delta_{ij}\underbrace{\langle\Phi_0|\Phi_0\rangle}_{=1} + i\sum_k \epsilon_{ijk}\underbrace{\langle\Phi_0|\Phi_k\rangle}_{=0} = \delta_{ij}, \tag{9.163}$$

for $1 \le i, j \le 3$. This shows that they do indeed form an orthonormal set, and hence they are an orthonormal basis for the space. One can express the total spin states in terms of the Bell basis—this shown in Problem 9.5.7.

We are almost ready to discuss teleportation. The last item we need is how to describe three-particle entanglement, as teleportation requires us to work with such a state. The key for teleportation is that we can write a three-particle entangled state in two different ways. Then a measurement on two of the spins "collapses" the state and allows a state to be teleported. The details are a bit tricky, so we describe them carefully next. Note that the original state is *destroyed* during the teleportation, so we are not copying the state (which is not possible due to the no-cloning theorem). In addition, we never measure or know what the state is. Nevertheless, we exactly transfer the state from one place to another, in a way where it seems like the transfer is instantaneous. But, we need to transfer two bits of information, using conventional means (which implies it is not instantaneous), in order to complete the teleportation.

Here is how it goes. Suppose I have a quantum superposition of a single spin, which can be written as $|\psi\rangle_a = \alpha|\uparrow\rangle_a + \beta|\downarrow\rangle_a$; I am using the a subscript to denote the space that my spin lies in. This is the state I wish to teleport to you and I do not even know myself what α and β are. In order to make this transfer, you and I must share a Bell state—we choose to share the $|\Phi_0\rangle$ state over the spin spaces b and c. We denote this as $|\Phi_0\rangle_{bc} = \frac{1}{\sqrt{2}}(|\uparrow\rangle_b \otimes |\uparrow\rangle_c + |\downarrow\rangle_b \otimes |\downarrow\rangle_c)$. Finally, we form the tensor product state of the state to be transferred and the Bell state as $|\psi\rangle_a \otimes |\Phi_0\rangle_{bc}$. I will work with the a and b spins, while you will work with the c spin. The way that teleportation proceeds is I have the state $|\psi\rangle_a$ prepared on one of my qubits, and I am given the b spin of the entangled Bell state, while you are given the c spin to hold. Now, if we expand our tensor-product state, we find that

$$|\psi\rangle_a \otimes |\Phi_0\rangle_{bc} = \frac{1}{\sqrt{2}}\Big(\alpha|\uparrow\rangle_a \otimes |\uparrow\rangle_b \otimes |\uparrow\rangle_c + \alpha|\uparrow\rangle_a \otimes |\downarrow\rangle_b \otimes |\downarrow\rangle_c$$

$$+ \beta|\downarrow\rangle_a \otimes |\uparrow\rangle_b \otimes |\uparrow\rangle_c + \beta|\downarrow\rangle_a \otimes |\downarrow\rangle_b \otimes |\downarrow\rangle_c\Big). \tag{9.164}$$

The goal is to reexpress this in terms of Bell states with respect to the ab spaces, which are both in my lab. This is done by using the results of Problem 9.5.7 that expresses the different product states $|\sigma\rangle_a \otimes |\sigma'\rangle_b$, with $\sigma,\ \sigma' \in \{\uparrow, \downarrow\}$ in terms of the Bell states. Substituting in these results gives us the following:

$$
\begin{aligned}
&|\psi\rangle_a \otimes |\Phi_0\rangle_{bc} \\
&= \frac{1}{2}\Big(\alpha|\Phi_0\rangle_{ab} \otimes |\uparrow\rangle_c + \alpha|\Phi_3\rangle_{ab} \otimes |\uparrow\rangle_c + \alpha|\Phi_1\rangle_{ab} \otimes |\downarrow\rangle_c - i\alpha|\Phi_2\rangle_{ab} \otimes |\downarrow\rangle_c \\
&\quad + \beta|\Phi_1\rangle_{ab} \otimes |\uparrow\rangle_c + i\beta|\Phi_2\rangle_{ab} \otimes |\uparrow\rangle_c + \beta|\Phi_0\rangle_{ab} \otimes |\downarrow\rangle_c - \beta|\Phi_3\rangle_{ab} \otimes |\downarrow\rangle_c\Big) \\
&= \frac{1}{2}\Big(|\Phi_0\rangle_{ab} \otimes \big(\alpha|\uparrow\rangle_c + \beta|\downarrow\rangle_c\big) + |\Phi_1\rangle_{ab} \otimes \big(\alpha|\downarrow\rangle_c + \beta|\uparrow\rangle_c\big) \\
&\quad + |\Phi_2\rangle_{ab} \otimes \big(-i\alpha|\downarrow\rangle_c + i\beta|\uparrow\rangle_c\big) + |\Phi_3\rangle_{ab} \otimes \big(\alpha|\uparrow\rangle_c - \beta|\downarrow\rangle_c\big)\Big). \qquad (9.165)
\end{aligned}
$$

This identity can be written in the form

$$
|\psi\rangle_a \otimes |\Phi_0\rangle_{bc} = \frac{1}{2}\sum_i |\Phi_i\rangle_{ab} \otimes \sigma_i|\Psi\rangle_c, \qquad (9.166)
$$

where I am using the notation $\sigma_0 = \mathbb{I}_2$. Now, to complete the teleportation, I just measure my two spins a and b in the Bell basis. Note that I have to be able to carefully set up an experiment that can measure in this Bell basis, which is by no means simple. For now, we assume that I can do it. I will find, whenever I do it, that I will observe one of the Bell states, and they are each equally likely, appearing one-fourth of the time. But, once I have measured a Bell state, then your spin has collapsed to the corresponding state in parenthesis, as follows: (i) if I measure $|\Phi_0\rangle_{ab}$, then you have $\alpha|\uparrow\rangle_c + \beta|\downarrow\rangle_c$; if I measure $|\Phi_1\rangle_{ab}$, then you have $\alpha|\downarrow\rangle_c + \beta|\uparrow\rangle_c$; if I measure $|\Phi_2\rangle_{ab}$, then you have $-i\alpha|\downarrow\rangle_c + i\beta|\uparrow\rangle_c$; and if I measure $|\Phi_3\rangle_{ab}$, then you will have $\alpha|\uparrow\rangle_c - \beta|\downarrow\rangle_c$. It looks like we teleported the correct state only one in four times—when you measured $|\Phi_0\rangle_{ab}$. But, it is not as bad as that. All I need to do is tell you precisely which state I measured. If I measured $|\Phi_0\rangle_{ab}$, then you have the teleported state. If I measured $|\Phi_1\rangle_{ab}$, you apply σ_x to your state, and it becomes the correct state. If I measured $|\Phi_2\rangle_{ab}$, then you apply σ_y to your state. And, finally, if I measured $|\Phi_3\rangle_{ab}$, then you apply σ_z to your state. So, after I send you which state I measured, which must be sent slower than the speed of light, you then know what to do to prepare the correct teleported state. Note again that at no time do we ever need to know what α or β are. Also note that my measurement destroys the entangled state, as is usually the case. I don't know about you, but I find this concept of teleportation to be mind blowing! It is not Star Trek transporter stuff, but it is still remarkable. All I need to have to make it work is an ability to share the original Bell state between you and I and an ability to measure my state in the Bell basis. That is all.

Section 9.5 Problems:

9.5.1 Spin Triplet State in an Arbitrary Direction

(a) Compute the wavefunction for the $|j{=}1, m{=}1\rangle = |\uparrow\uparrow\rangle_z$ state in the θ, ϕ basis. Note that this state is not invariant, because it is not, in general, of the form $|\uparrow\uparrow\rangle_{\theta\phi}$.
(b) Verify that the $|j{=}1, m{=}1\rangle_{\theta\phi}$ state is orthogonal to the $|j{=}0, m{=}0\rangle_{\theta\phi}$ state by directly computing the inner product between the two states.

9.5.2 Spin Singlet State Is an Eigenstate of All Cartesian Spin Operators

Take the spin-singlet state, and show that $\hat{S}_i|j{=}0, m{=}0\rangle = 0$ for $i = x, y,$ and z. This means that this state is an eigenstate of all three operators, with eigenvalue 0. Does this cause a problem with our general discussion of there being no simultaneous eigenvectors of noncommuting operators?

9.5.3 N-Stop Bell Experiment

Compute $P(++) + P(--)$ for the N-stop Bell experiment following similar steps to how we computed $P(+-) + P(-+)$. Do not use the rule $P(++) + P(+-) + P(-+) + P(--) = 1$. Instead, work out all of the algebra directly. Verify that the total probability is equal to 1 afterward.

9.5.4 Even N Bell Experiment Cannot Differentiate Between Hidden Variables and Quantum Mechanics

For an even-N Bell experiment, show that there is at least one configuration for which the hidden variable instruction set gives $p = \frac{1}{2}$; hence, we can only conclude that

$$P(\text{opposite}) \geq \frac{1}{2}, \tag{9.167}$$

for the hidden variables case. This is insufficient to conclude that hidden variables give a different answer from quantum mechanics.

Hint: consider the instruction set for half + and half − projections.

9.5.5 Odd N Bell Experiment Can Differentiate

Prove that for odd N, that

$$P(\text{opposite}) \geq \frac{N^2+1}{2N^2} \tag{9.168}$$

for the hidden variables prediction.

Hint: show that the minimal probability occurs for $\frac{N+1}{2}$ positive and $\frac{N-1}{2}$ negative projections (or *vice versa*). Be careful to get your counting right and review the $N = 3$ and $N = 5$ cases first.

9.5.6 Exploring Properties of Bell States

For the Bell states in the z basis, show that the probability for both spins to be measured the same is 1 for $|\Phi_0\rangle$ and $|\Phi_3\rangle$. Similarly, show the probability to measure the two spins to be opposite is 1 for $|\Phi_1\rangle$ and $|\Phi_2\rangle$.

Calculate the probability to measure $\hat{S}_x$ to be the same for both spins in each of the four Bell states.

Calculate the probability to measure $\hat{S}_y$ to be the same for the four Bell states.

9.5.7 Bell States to Spin States

Expand the four states $|\uparrow\uparrow\rangle$, $|\uparrow\downarrow\rangle$, $|\downarrow\uparrow\rangle$, and $|\downarrow\downarrow\rangle$ in the Bell basis.

Using those results, expand the four states $|j{=}1, m\rangle$ and $|j{=}0, m{=}0\rangle$ in terms of the four Bell states.

9.5.8 Essay on Teleportation

The teleportation procedure is an interesting one to use. Write an essay that explains why it does not violate the no-cloning theorem. Is there a way to use teleportation to copy a state?

9.6 Further Discussions on Angular Momentum States

In this section, we discuss further the properties of angular momentum states. We start from the ferromagnetic state, where j is maximal. For example, if we look at N spin-one-half particles, the maximal possible total spin is $\frac{N}{2}$. The state $|j{=}\frac{N}{2}, m{=}\frac{N}{2}\rangle$ is the ground state of the system in a magnetic field

$$\hat{H} = -g\mu_B B \sum_{i=1}^{N} \hat{S}_z^i, \tag{9.169}$$

if we pick the z-axis to lie along the direction of the magnetic field. In Problem 9.6.1, you will find all of the energy eigenvalues and the eigenstates for the ferromagnetic case.

The state $|j{=}\frac{N}{2}, m{=}\frac{N}{2}\rangle$ is obviously a product state, because it is formed as a tensor product of the different spin states for each particle via

$$|j{=}\frac{N}{2}, m{=}\frac{N}{2}\rangle = \underbrace{|\uparrow\rangle \otimes |\uparrow\rangle \otimes \cdots \otimes |\uparrow\rangle}_{N \text{ factors}} = |\uparrow\uparrow\cdots\uparrow\rangle. \tag{9.170}$$

So, this state should remain a product state if it is rotated into any other direction. For example, if I look at this state in the x-direction, then, because $|\uparrow\rangle_x = \frac{1}{\sqrt{2}}(|\uparrow\rangle_z + |\downarrow\rangle_z)$, we have

$$\begin{aligned} |\uparrow\uparrow\cdots\uparrow\rangle_x &= \frac{1}{2^{\frac{N}{2}}}(|\uparrow\rangle + |\downarrow\rangle) \otimes (|\uparrow\rangle + |\downarrow\rangle) \otimes \cdots \otimes (|\uparrow\rangle + |\downarrow\rangle) \\ &= \frac{1}{2^{\frac{N}{2}}} \sum_{\sigma_i \in \{\uparrow,\downarrow\}} |\sigma_i, \sigma_2, \cdots, \sigma_N\rangle, \end{aligned} \tag{9.171}$$

which is an equally weighted sum over all of the eigenstates of $\hat{S}_z^{\text{tot}} = \sum_{i=1}^{N} \hat{S}_z^i$. Note that in this last form, it is a sum over 2^N terms, and it is by no means obvious that this is a product state, until we look at the top line of Eq. (9.171), which clearly shows the product-state form. In general, one cannot just look at a quantum state expressed as a linear combination over some set of states and tell whether it is a product state or an entangled state. There are many applications where the state polarized along the x-axis is useful, because every $\hat{S}_z^{\text{tot}}$ eigenstate has a representative in the sum over the states. We will encounter this later in the book.

Keeping with the ferromagnetic states, we can also easily generate the states with $|j{=}\frac{N}{2}, m\rangle$ for all $-\frac{N}{2} \leq m \leq \frac{N}{2}$. We do this by starting with the $|j{=}\frac{N}{2}, m{=}\frac{N}{2}\rangle$ state $|\uparrow\uparrow\cdots\uparrow\rangle$ and hit it with the $\hat{S}_-^{\text{tot}}$ operator $\frac{N}{2} - m$ times. Let us see what happens when we do this:

$$\hat{S}_-^{\text{tot}}|j{=}\tfrac{N}{2}, m{=}\tfrac{N}{2}\rangle = \hat{S}_-^1 \otimes \mathbb{I} \otimes \cdots \otimes \mathbb{I}|\uparrow\uparrow\cdots\uparrow\rangle + \mathbb{I} \otimes \hat{S}_-^2 \otimes \cdots \otimes \mathbb{I}|\uparrow\uparrow\cdots\uparrow\rangle$$
$$+ \cdots + \mathbb{I} \otimes \mathbb{I} \cdots \otimes \hat{S}_-^N|\uparrow\uparrow\cdots\uparrow\rangle$$
$$= |\downarrow\uparrow\cdots\uparrow\rangle + |\uparrow\downarrow\cdots\uparrow\rangle + \cdots + |\uparrow\uparrow\cdots\downarrow\rangle, \tag{9.172}$$

where we have exactly N terms, each with coefficient 1, and with one down spin located at all of the N-possible positions. To normalize, we need to multiply by $\frac{1}{\sqrt{N}}$. Hitting the state with two $\hat{S}_-^{\text{tot}}$ operators will create all possible states with two $\downarrow$ spins and $\frac{N}{2} - 2$ $\uparrow$ spins; they also all have coefficient of 1. There are $\binom{N}{2} = \frac{N(N-1)}{2}$ terms, so the normalization constant is $\sqrt{\frac{2}{N(N-1)}}$. Continuing, if we apply $\hat{S}_-^{\text{tot}}$ n times, then we have $\binom{N}{n} = \frac{N!}{n!(N-n)!}$ terms, each with coefficient 1, corresponding to n $\downarrow$ spins and $N - n$ $\uparrow$ spins. The normalization constant is $\sqrt{\frac{n!(N-n)!}{N!}}$. Hence, we have the general formula given by

$$|j{=}\frac{N}{2}, m\rangle = \sqrt{\frac{\left(\frac{N}{2}-m\right)!\left(\frac{N}{2}+m\right)!}{N!}} \sum_{\sigma_i:\, \frac{N}{2}-m \downarrow \text{ and } \frac{N}{2}+m \uparrow} |\sigma_1, \sigma_2, \cdots, \sigma_N\rangle. \tag{9.173}$$

We already saw that the states at the extremes, $|j{=}\frac{N}{2}, m{=}\frac{N}{2}\rangle$ and $|j{=}\frac{N}{2}, m{=}-\frac{N}{2}\rangle$, are product states and not entangled. It turns out that all other states $|j{=}\frac{N}{2}, m{\neq}\frac{N}{2}, -\frac{N}{2}\rangle$ are entangled, and the level of entanglement grows as m heads toward 0 (or $\pm\frac{1}{2}$). These states are also called *Dicke states*. Note that we are not introducing a quantitative measure of the level of entanglement at this stage, so the discussion is really qualitative—as we discussed before, it is not so simple to determine when states are entangled or are product states. But, for here, we will think of the level of entanglement as correlated with the size of the fluctuation

$$\left(\Delta\hat{\vec{S}}^{\text{tot}}\right)^2_{|j=\frac{N}{2},m\rangle} = \langle j{=}\tfrac{N}{2}, m|\left(\hat{\vec{S}}^{\text{tot}}\right)^2|j{=}\tfrac{N}{2}, m\rangle$$
$$- \langle j{=}\tfrac{N}{2}, m|\hat{\vec{S}}^{\text{tot}}|j{=}\tfrac{N}{2}, m\rangle \cdot \langle j{=}\tfrac{N}{2}, m|\hat{\vec{S}}^{\text{tot}}|j{=}\tfrac{N}{2}, m\rangle. \tag{9.174}$$

It turns out that this fluctuation is fairly simple to calculate. Recall that

$$\left(\hat{\vec{S}}^{\text{tot}}\right)^2|j{=}\tfrac{N}{2}, m\rangle = \hbar^2\frac{N}{2}\left(\frac{N}{2}+1\right)|j{=}\tfrac{N}{2}, m\rangle, \tag{9.175}$$

and

$$\hat{\vec{S}}^{\text{tot}}|J{=}\tfrac{N}{2},m\rangle = \frac{1}{2}c_+(\vec{e}_x - i\vec{e}_y)|J{=}\tfrac{N}{2},m+1\rangle + \hbar m\vec{e}_z|j{=}\tfrac{N}{2},m\rangle + \frac{1}{2}c_-(\vec{e}_x + i\vec{e}_y)|j{=}\tfrac{N}{2},m-1\rangle, \tag{9.176}$$

where the coefficients $c_\pm$ were worked out earlier when we discussed the algebraic determination of the angular momentum eigenstates in Sect. 9.3. Note that they do not contribute to the final answer, so we do not need them here. In this derivation, we used the fact that

$$\hat{S}_x^{\text{tot}}\vec{e}_x + \hat{S}_y^{\text{tot}}\vec{e}_y = \frac{1}{2}\hat{S}_+^{\text{tot}}(\vec{e}_x - i\vec{e}_y) + \frac{1}{2}\hat{S}_-^{\text{tot}}(\vec{e}_x + i\vec{e}_y), \tag{9.177}$$

which follows from the add-zero trick. Hence, we obtain

$$\left(\Delta\hat{\vec{S}}^{\text{tot}}\right)^2_{|j=\frac{N}{2},m\rangle} = \hbar\sqrt{\frac{N}{2}\left(\frac{N}{2}+1\right) - m^2}, \tag{9.178}$$

which is maximized for minimal $|m|$.

We end this section by illustrating how we can build up the different total spin multiplets just by using the "add-spin-one-half" table. We begin with a figure that illustrates how we build up states, shown in Fig. 9.10. Recall for $N = 2$ is given in Eq. (9.145). For $N = 3$, we build off the $N = 2$ results. The figure shows we have two spin one-half and one spin three-half states. The first spin one-half state is found by adding a spin one-half to the singlet state from $N = 2$ given by $\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$, which we get by just adding the new spin one-half degree of freedom to the end of the spin-singlet state:

$$s = \tfrac{1}{2},\ m = \tfrac{1}{2}:\ \frac{1}{\sqrt{2}}(|\uparrow\downarrow\uparrow\rangle - |\downarrow\uparrow\uparrow\rangle) \text{ and } m = -\tfrac{1}{2}:\ \frac{1}{\sqrt{2}}(|\uparrow\downarrow\downarrow\rangle - |\downarrow\uparrow\downarrow\rangle). \tag{9.179}$$

The other one comes from the spin-triplet state, which has $s = 1$. For this, we have to use our add-spin-one-half table in Table 9.1. We find

$$-\sqrt{\frac{j_1 - m + \frac{1}{2}}{2j_1+1}} = \begin{cases} -\sqrt{\frac{1}{3}} & j_1 = 1,\ m = \frac{1}{2} \\ -\sqrt{\frac{2}{3}} & j_1 = 1,\ m = -\frac{1}{2} \end{cases}$$

and

$$\sqrt{\frac{j_1 + m + \frac{1}{2}}{2j_1+1}} = \begin{cases} \sqrt{\frac{2}{3}} & j_1 = 1,\ m = \frac{1}{2} \\ \sqrt{\frac{1}{3}} & j_1 = 1,\ m = -\frac{1}{2}. \end{cases} \tag{9.180}$$

This then gives us the other spin one-half states:

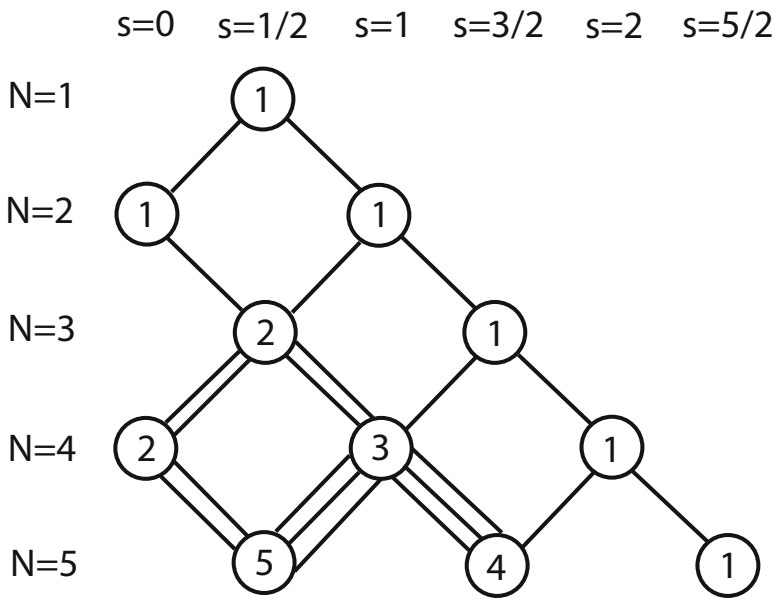

Fig. 9.10 Schematic of how spin-one-half particles are combined into total spin states. The total spin columns are labeled at the top, and the particle number is labeled on the left. Each circle represents a state with definite N and definite s. The numbers in the circles denote the total number of spin states for that given N and s values. The lines show how the add-spin-one-half table works—it can add or subtract $s = \frac{1}{2}$, but it cannot go below 0

$$
\begin{aligned}
m = \tfrac{1}{2}: \quad & -\sqrt{\tfrac{2}{3}}|m_1{=}0, m_2{=}\tfrac{1}{2}\rangle + \sqrt{\tfrac{1}{3}}|m_1{=}1, m_2{=}-\tfrac{1}{2}\rangle \\
&= \frac{1}{\sqrt{6}}(-|\uparrow\downarrow\uparrow\rangle - |\downarrow\uparrow\uparrow\rangle + 2|\uparrow\uparrow\downarrow\rangle) \qquad (9.181) \\
m = -\tfrac{1}{2}: \quad & -\sqrt{\tfrac{2}{3}}|m_1{=}-1, m_2{=}\tfrac{1}{2}\rangle + \sqrt{\tfrac{1}{3}}|m_1{=}0, m_2{=}-\tfrac{1}{2}\rangle \\
&= \frac{1}{\sqrt{6}}(-2|\downarrow\downarrow\uparrow\rangle + |\uparrow\downarrow\downarrow\rangle + |\uparrow\downarrow\uparrow\rangle). \qquad (9.182)
\end{aligned}
$$

You should try to work out the case with $s = \frac{3}{2}$ using the table as well. The solution is

$$
\begin{aligned}
|j{=}\tfrac{3}{2}, m{=}\tfrac{3}{2}\rangle &= |\uparrow\uparrow\uparrow\rangle \\
|j{=}\tfrac{3}{2}, m{=}\tfrac{1}{2}\rangle &= \frac{1}{\sqrt{3}}(|\uparrow\uparrow\downarrow\rangle + |\uparrow\downarrow\uparrow\rangle + |\downarrow\uparrow\uparrow\rangle) \\
|j{=}\tfrac{3}{2}, m{=}-\tfrac{1}{2}\rangle &= \frac{1}{\sqrt{3}}(|\uparrow\downarrow\downarrow\rangle + |\downarrow\uparrow\downarrow\rangle + |\downarrow\downarrow\uparrow\rangle) \\
|j{=}\tfrac{3}{2}, m{=}-\tfrac{3}{2}\rangle &= |\downarrow\downarrow\downarrow\rangle. \qquad (9.183)
\end{aligned}
$$

One can verify that all of these states are orthogonal to each other and are normalized. Note that the $s = \frac{3}{2}$ state satisfies the general results we described at the beginning of the section—the state for a given m value involves a uniform superposition of all states that have the given m value and then multiplied by a normalization constant. We can now continue in this fashion with the add-spin-one-half tables to create the results for $N = 4$, which include two $s = 0$, three $s = 1$ and one $s = 2$ state. This is Problem 9.6.2 at the end of the section. You can see that continuing in this way gets quite tedious as N increases, but it could be easily programmed into a computer to generate the states for many more N values (at least until you run out of computer memory, as the scope of the problem doubles in size each time N is increased by one).

Section 9.6 Problems:

9.6.1 Energy Eigenstates for the Zeeman Hamiltonian

For N spin-one-half particles, find all eigenstates and eigenvalues of

$$\hat{H} = -g\mu_B B \sum_{i=1}^{N} \hat{S}_z^i. \tag{9.184}$$

Note that one can describe all the eigenstates easily using descriptive words and that it is better to work with the simplest states that are eigenstates of $\hat{S}_z^{\text{tot}}$ and not of $(\hat{\vec{S}}^{\text{tot}})^2$.

9.6.2 Spin States for $N = 4$

Using the add-spin-one-half table, construct the two total spin zero, three total spin one, and one total spin two states from the $N = 4$ states constructed in the text. Verify that these states are an orthonormal set.

9.7 Spherical Harmonics

Spherical harmonics are the position-space wavefunctions of the states with definite orbital angular momentum and definite z-component of orbital angular momentum. When I first learned spherical harmonics, I found the procedure to be torturous. I had to work through lengthy calculus manipulations to extract the angular part of the Laplacian. Then I had to solve the differential equation for the Legendre

polynomials (and in the end the final solution was just told to me with a "trust us" smile from the instructor). It was a lot of complex manipulations with what seemed like a limited return for the effort. So, when I set out to write this book, I did not plan to include a section on spherical harmonics. But then I realized that knowing what spherical harmonics are does indeed provide many benefits—it allows you to visualize the spatial patterns of atomic orbitals, and it allows you to talk to people about angular momentum who has taken more conventional quantum courses. So I knew this section had to be in the book. As I delved into the subject further, I found the operator-based method to determine spherical harmonics had both a physical nature to it that helped me understand what these wavefunctions really are and that make the calculation much clearer, especially since I could directly eye the prize at the end throughout the effort. Now, don't get me wrong, it still is a bit torturous—some quantum problems are just hard. But, I hope you enjoy this new way to look at spherical harmonics and that you will find it insightful, just as I do. And if this is the first time you encounter them, I hope you will feel that the effort to learn them was indeed worth it. It is one of the derivations I am most proud of. Grab your chef's hat, because we have some French cooking to do!

Wavefunctions involve the overlap of one quantum state with another. Here, the spherical harmonic is defined to be the overlap of the state with definite squared angular momentum and definite z-component of angular momentum, with a spherical coordinate position state that lies in the θ, ϕ direction on the surface of a sphere. We described the angular momentum state $|l, m\rangle$ in Sect. 9.3 and the angular position state $|\theta, \phi\rangle$ in Problem 9.2.5. The spherical harmonic $Y_{lm}(\theta, \phi)$ is then defined via

$$Y_{lm}(\theta, \phi) = \langle \theta, \phi | lm \rangle. \tag{9.185}$$

As a wavefunction, we keep l and m fixed and vary θ and ϕ. Note that we have $-l \leq m \leq l, 0 \leq \theta \leq \pi$, and $0 \leq \phi \leq 2\pi$ for the spherical harmonic. The angle θ is the polar angle from the vertical z-axis to the point on the circle and ϕ is the axial angle for how far the projection of the point in the x-y-plane has rotated away from the x-axis (see Fig. 6.3); note that physicists and mathematicians invert which angle is θ and which is ϕ—we use the physicists convention here. We take the Hermitian conjugate of the ket $|\theta, \phi\rangle = e^{-\frac{i}{\hbar}\phi \hat{L}_z} e^{-\frac{i}{\hbar}\theta \hat{L}_y} |\theta{=}0, \phi{=}0\rangle$, which is the rotation of the north-pole-state $|\theta{=}0, \phi{=}0\rangle$ to the θ, ϕ direction. Note that the north-pole state can have an arbitrary value of the axial angle ϕ and it describes the same state; we nevertheless write is as $\phi = 0$, because the canonical way we construct the $|\theta, \phi\rangle$ state gives it $\phi = 0$ after the rotation about the y axis. We will use this fact of no ϕ dependence of the north-pole state shortly. Now, the spherical harmonic becomes (after converting the $|\theta, \phi\rangle$ ket into a bra)

$$Y_{lm}(\theta, \phi) = \langle \theta{=}0, \phi{=}0 | e^{\frac{i}{\hbar}\theta \hat{L}_y} e^{\frac{i}{\hbar}\phi \hat{L}_z} | l, m \rangle. \tag{9.186}$$

Note the physical nature of this state—the north-pole state is rotated to an arbitrary direction θ, ϕ, and then we compute the overlap with the state that has definite l and m. Because the angular momentum state $|l, m\rangle$ is an eigenstate of $\hat{L}_z$, we can immediately evaluate the action of the first operator onto it, and it is replaced by the factor $e^{im\phi}$, because the eigenvalue of $\hat{L}_z$ is $\hbar m$. This means the spherical harmonic boils down to calculating the expectation value of the rotation about the y-axis

$$Y_{lm}(\theta, \phi) = e^{im\phi} \langle \theta{=}0, \phi{=}0 | e^{\frac{i}{\hbar}\theta \hat{L}_y} | l, m \rangle. \tag{9.187}$$

This calculation is challenging. The state $|l, m\rangle$ does not have a definite value with respect to $\hat{L}_y$, and this operator cannot act to the left on the north-pole state and be immediately calculated. So, we must go back to our basic principles—what do we know? Well, we know what the raising and lowering operators do when they act on $|l, m\rangle$, so we need to convert the exponential of $\hat{L}_y$ into exponentials of $\hat{L}_\pm$. How do we evaluate the exponential involving the raising and lowering operators? We have to use one of our four fundamental identities. After some thought, we find the right identity to use is the exponential disentangling identity, because it will isolate the raising and lowering operators into different exponentials, which can then be evaluated.

But before we do that, we want to point out one important identity that arises from the fact that the north-pole state is unchanged by a rotation about the z-axis. In other words, we have that $e^{-\frac{i}{\hbar}\phi \hat{L}_z} |\theta{=}0, \phi{=}0\rangle = |\theta{=}0, \phi{=}0\rangle$, because the north-pole state is independent of ϕ. This means that the north-pole state has a nonzero overlap only with a definite angular-momentum state that has $m = 0$. To see this, we use multiply by one and use the property of the north-pole state

$$\begin{aligned} \langle \theta{=}0, \phi{=}0 | l, m \rangle &= \langle \theta{=}0, \phi{=}0 | \underbrace{e^{\frac{i}{\hbar}\phi \hat{L}_z} e^{-\frac{i}{\hbar}\phi \hat{L}_z}}_{=1} l, m \rangle \\ &= e^{-im\phi} \langle \theta{=}0, \phi{=}0 | l, m \rangle, \end{aligned} \tag{9.188}$$

because we let the operator on the left act to the left on the north-pole state, where it does nothing, and we let the operator operator on the right act on the right, which is an eigenstate and can be immediately evaluated. So, we learn that $\langle \theta{=}0, \phi{=}0 | l, m \rangle = 0$ unless $m = 0$. There is a very important conclusion that we obtain from this—it tells us that m must be an integer (and thereby l must be an integer) for orbital angular momentum eigenstates. This is another way that we prove that orbital angular momentum must have integer l.

We do one more manipulation before calculating the final result. We introduce a sum over the complete set of angular momentum states (for fixed l) as follows

$$Y_{lm}(\theta, \phi) = \langle \theta{=}0, \phi{=}0 | \sum_{m'=-l}^{l} |l, m'\rangle \langle l, m' | e^{\frac{i}{\hbar}\theta \hat{L}_y} | l, m \rangle$$

$$= \underbrace{\langle \theta{=}0, \phi{=}0 | l, m'{=}0 \rangle}_{\text{normalization constant}} \langle l, m'{=}0 | e^{\frac{i}{\hbar}\theta \hat{L}_y} | l, m \rangle. \tag{9.189}$$

We are left with only one term from the sum, because all terms with $m' \neq 0$ have a vanishing coefficient.

Okay. We are now ready for the main event. We use the exponential disentangling identity. The identity we use is the SU(2) identity given in Eq. (8.14), with the raising operator on the right and the lowering operator on the left. So, we have

$$\langle l, m'{=}0 | e^{\frac{i}{\hbar}\theta \hat{L}_y} | l, m \rangle = \langle l, m{=}0 | e^{-\tan\frac{\theta}{2}\frac{\hat{L}_-}{\hbar}} e^{\ln\left(\cos^2\frac{\theta}{2}\right)\frac{\hat{L}_z}{\hbar}} e^{\tan\frac{\theta}{2}\frac{\hat{L}_+}{\hbar}} | l, m \rangle, \tag{9.190}$$

which is proved in Problem 9.7.6. There is a second way to use the exponential disentangling identity, using the opposite order of the factors; you will work with that identity in the problems to compute the spherical harmonics for $m < 0$.

Now, we just need to evaluate the different terms, which are illustrated schematically in Fig. 9.11. Each exponential term can now be directly evaluated by using a power series expansion for the exponential and determining what each operator term does when it acts on the state to the right. We already know the result for the raising and lowering operators from Sect. 9.3:

$$\hat{L}_+ | l, m \rangle = \hbar \sqrt{(l-m)(l+m+1)} | l, m+1 \rangle$$

and

$$\hat{L}_- | l, m \rangle = \hbar \sqrt{(l+m)(l-m+1)} | l, m-1 \rangle. \tag{9.191}$$

One can see that when $\hat{L}_+$ acts on $|l, m{=}l\rangle$, it annihilates the state. Hence, the exponential $e^{\tan\frac{\theta}{2}\frac{\hat{L}_+}{\hbar}}$ acting on $|l, m\rangle$ has only $l - m + 1$ nonzero terms in the power series when they act on the state to the right. The net result is given by

$$e^{\tan\frac{\theta}{2}\frac{\hat{L}_+}{\hbar}} | l, m \rangle = \sum_{n=0}^{l-m} \frac{\left(\tan\frac{\theta}{2}\right)^n}{n!} \left[\prod_{r=1}^{n} \sqrt{(l+m+r)(l-m-r+1)} \right] | l, m+n \rangle. \tag{9.192}$$

The next term in the operator product is an exponential of $\hat{L}_z$, but the states $|l, m\rangle$ are eigenstates of this operator, so it can be applied immediately on each term. We find

$$e^{\ln\left(\cos^2\frac{\theta}{2}\right)\frac{\hat{L}_z}{\hbar}} e^{\tan\frac{\theta}{2}\frac{\hat{L}_+}{\hbar}} | l, m \rangle = \sum_{n=0}^{l-m} \frac{\left(\tan\frac{\theta}{2}\right)^n \left(\cos\frac{\theta}{2}\right)^{2m+2n}}{n!}$$

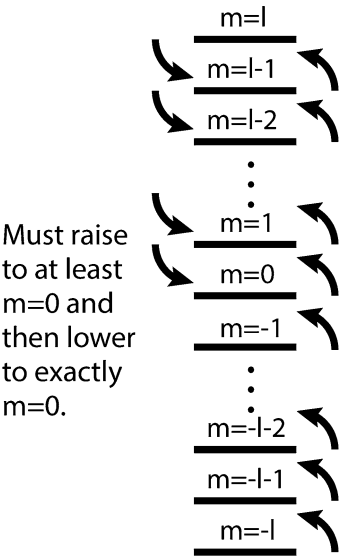

Fig. 9.11 Schematic of the ladder operator terms that arise when we use the exponential disentangling identity. The exponential of the raising operator creates a polynomial in powers of the raising operator, which includes all possible excitations up to the highest state. The exponential of the lowering operator is similar, but, here, we must end the lowering precisely at $m = 0$ when computing the spherical harmonic

$$\times \left[\prod_{r=1}^{n} \sqrt{(l+m+r)(l-m-r+1)} \right] |l, m+n\rangle. \tag{9.193}$$

Now, we lower until the ket gets to $m = 0$. Hence, we must have $m + n \geq 0$, and we only apply the $\hat{L}_-$ operator $m + n$ times on each corresponding term in the summation. This then yields

$$\begin{aligned} \langle l, m'{=}0 | e^{\frac{i}{\hbar}\theta \hat{L}_y} | l, m \rangle = & \sum_{n=\max(0,-m)}^{l-m} \frac{\left(\tan\frac{\theta}{2}\right)^n \left(\cos\frac{\theta}{2}\right)^{2m+2n} \left(-\tan\frac{\theta}{2}\right)^{m+n}}{n!(m+n)!} \\ & \times \left[\prod_{s=1}^{m+n} \sqrt{(l-m-n+s)(l+m+n-s+1)} \right] \\ & \times \left[\prod_{r=1}^{n} \sqrt{(l+m+r)(l-m-r+1)} \right], \end{aligned} \tag{9.194}$$

where we used the fact that $\langle l, m'{=}0 | l, m{=}0 \rangle = 1$. Essentially, we are done, but we always go through another exercise to relate our results to those of dead French or German mathematicians. In this case, our dead French mathematician is Legendre. So, we now massage and simplify the formula.

The first step to pay homage to Legendre is to collect the trigonometric factors and use the half-angle formulas: $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$ and $1 - \cos\theta = 2\sin^2\frac{\theta}{2}$. Then we find that

$$\left(\tan\tfrac{\theta}{2}\right)^n \left(\cos\tfrac{\theta}{2}\right)^{2m+2n} \left(-\tan\tfrac{\theta}{2}\right)^{m+n} = \left(-\frac{1}{2}\right)^{m+n} (\sin\theta)^m (1-\cos\theta)^n. \tag{9.195}$$

The products of the integers become

$$\sqrt{\frac{(l+m+n)!(l-m)!l!(l+m+n)!}{(l+m)!(l-m-n)!(l-m-n)!l!}} = \frac{(l+m+n)!}{(l-m-n)!}\sqrt{\frac{(l-m)!}{(l+m)!}} \tag{9.196}$$

The results with $m \geq 0$ can now be put into the standard form. The associated Legendre function is defined to be

$$P_l^m(\cos\theta) = \frac{1}{2^m}(\sin\theta)^m \sum_{n=0}^{l-m}(-1)^n \frac{(l+m+n)!}{(l-m-n)!(m+n)!n!}\left(\frac{1-\cos\theta}{2}\right)^n, \tag{9.197}$$

so that we have

$$Y_{lm}(\theta,\phi) = \sqrt{\frac{2l+1}{4\pi}}(-1)^m \sqrt{\frac{(l-m)!}{(l+m)!}} P_l^m(\cos\theta) e^{im\phi} \tag{9.198}$$

for $m \geq 0$. The normalization factor requires calculus to determine, and we put in its specific value given by $\langle\theta{=}0, \phi{=}0|l, m{=}0\rangle = \sqrt{\frac{2l+1}{4\pi}}$ in the formula above. The case with $m < 0$ is discussed in Problem 9.7.6. There is a general formula that relates positive to negative m, given by $Y_{l-|m|}(\theta,\phi) = (-1)^{|m|}[Y_{l|m|}(\theta,\phi)]^*$. This operator-based derivation presented here was only discovered in 2018 (Weitzman and Freericks, 2018). Note that if we wanted to, we could have computed the overlap with the bra $\langle l, m'|$ instead of $\langle l, m'{=}0|$. Doing so would have determined the so-called *rotation matrices*, which we will discuss next as an alternative way to compute the spherical harmonics using operators. If you really like algebra, you can consult the Weitzman and Freericks article to see a discussion about some of the subtleties involved with the different ways one can compute the spherical harmonics for $m < 0$.

There is a completely different way to calculate spherical harmonics that also uses operators, and this approach is due to Wigner (originally worked out in German in 1931 and translated into English in 1959) (Wigner, 1959). Wigner's strategy is to use $2l$ spin one-half particles to determine the spherical harmonics. The defining formula remains the same, but the exponential of $\hat{L}_y$ is now rewritten as a sum of Pauli matrices that act on the spins in each of the tensor-product spaces. Because we know what the exponential of any Pauli matrix is, we can immediately calculate the result. Then, because $2l$ spin one-half particles, with maximal angular momentum, is isomorphic to orbital angular momentum with quantum number l, it allows us to determine the spherical harmonics in an alternative fashion. We describe the details for how this works next.

Our starting point is Eq. (9.173), where we showed that we could construct a state with total angular momentum l out of a collection of $2l$ spin-one-half degrees of freedom. The superposition in Eq. (9.173) has $l+m$ $\uparrow$ spins and $l-m$ $\downarrow$ spins, with each term in the superposition having exactly the same coefficient. But, we actually do not need to worry about the specific locations of the up or down spins, because the spin operators are always summed over all of the different spins in the tensor product. So, thinking in terms of equivalence classes, we need only keep track of one of the elements in the sum, because each of the terms in the superposition is completely equivalent to every other term. However, we do need to get the counting right, so we need to multiply the one representative state by the number of states in the summation, which is given by $\frac{(2l)!}{(l+m)!(l-m)!}$. Then, we describe our "equivalence class" of the angular-momentum state as

$$|l,m\rangle = \sqrt{\frac{(2l)!}{(l+m)!(l-m)!}} \underbrace{|\uparrow\rangle \otimes |\uparrow\rangle \otimes \cdots \otimes |\uparrow\rangle}_{l+m \text{ terms}} \underbrace{|\downarrow\rangle \otimes |\downarrow\rangle \otimes \cdots \otimes |\downarrow\rangle}_{l-m \text{ terms}}. \tag{9.199}$$

Before doing anything else, we must verify that this representation satisfies the correct properties of the $|l,m\rangle$ state.

There are three properties we need to verify: the action of the operators $\hat{L}_z$ and $\hat{L}_\pm$ onto $|l,m\rangle$. Recall, we know what the exact results of the action of these three operators are on the state $|l,m\rangle$ from the work in Sect. 9.3. Since we are working with the equivalence classes, the calculations of the operators acting on the one equivalent state are straightforward to work out, but we have to be careful to get the counting right. Since we have

$$\hat{L}_z = \hat{S}_z \otimes \mathbb{I} \otimes \cdots \otimes \mathbb{I} + \mathbb{I} \otimes \hat{S}_z \otimes \cdots \otimes \mathbb{I} + \cdots + \mathbb{I} \otimes \mathbb{I} \otimes \cdots \hat{S}_z, \tag{9.200}$$

we see that the z-component of the spin operator acts on each spin separately and is then totalled. When it acts on an $\uparrow$ spin, it gives back $\frac{\hbar}{2}$ (we have $l+m$ up spins) and when it acts of $\downarrow$, it gives a $-\frac{\hbar}{2}$ (we have $l-m$ down spins), so, we find that

$$\hat{L}_z|l,m\rangle = \frac{\hbar}{2}(l+m-l+m)|l,m\rangle = \hbar m|l,m\rangle, \tag{9.201}$$

which is what it needs to be. For the $\hat{L}_+$ operator, we can have the individual spin raising operators act on the up spins or the down spins—they annihilate the up spins and convert the down spins to up spins. So the number of up spins increases by 1 and the number of down spins decreases by 1. Since we have $l-m$ down spins, we find that

$$
\begin{aligned}
&\hat{L}_+|l,m\rangle \\
&= \sqrt{\frac{(2l)!}{(l+m)!(l-m)!}}(l-m)\underbrace{|\uparrow\rangle\otimes|\uparrow\rangle\otimes\cdots\otimes|\uparrow\rangle}_{l+m+1\text{ terms}}\underbrace{|\downarrow\rangle\otimes|\downarrow\rangle\otimes\cdots\otimes|\downarrow\rangle}_{l-m-1\text{ terms}} \\
&= \sqrt{\frac{(2l)!}{(l+m+1)!(l-m-1)!}}\sqrt{(l-m)(l+m+1)} \\
&\qquad\times\underbrace{|\uparrow\rangle\otimes|\uparrow\rangle\otimes\cdots\otimes|\uparrow\rangle}_{l+m+1\text{ terms}}\underbrace{|\downarrow\rangle\otimes|\downarrow\rangle\otimes\cdots\otimes|\downarrow\rangle}_{l-m-1\text{ terms}} \\
&= \sqrt{(l-m)(l+m+1)}|l,m+1\rangle,
\end{aligned}
\tag{9.202}
$$

which also is what it needs to be. Similarly, one can show that $\hat{L}_-|l,m\rangle = \sqrt{(l+m)(l-m+1)}|l,m-1\rangle$. So the angular momentum operators, written in this spin-one-half tensor-product representation, do indeed behave as they are required to, in order to determine the angular momentum states $|l,m\rangle$.

So, now we can apply the rotation by θ about the y-axis onto the state $|l,m\rangle$. First, we write how the transformation works in terms of the Wigner rotation matrices. Wigner defined these matrices for rotations about axes pointing in arbitrary directions, but here we will work just with rotation matrices for a single rotation about the y-axis. Wigner defines the *rotation matrix* $d^{(l)}_{m'm}(\theta)$ via

$$
e^{\frac{i}{\hbar}\theta\hat{L}_y}|l,m\rangle = \sum_{m'=-l}^{l}|l,m'\rangle\, d^{(l)}_{m'm}(\theta) = \sum_{m'=-l}^{l}|l,m'\rangle\,\langle|l,m'|e^{\frac{i}{\hbar}\theta\hat{S}_y}|l,m\rangle,
\tag{9.203}
$$

which follows again from completeness in the fixed-l sector and a definition of the coefficients as the rotation matrices. Recall that the rotation operator now takes a similar tensor-product form, given by

$$
e^{\frac{i}{\hbar}\theta\hat{L}_y} = e^{\frac{i}{\hbar}\theta\hat{S}_y}\otimes e^{\frac{i}{\hbar}\theta\hat{S}_y}\otimes\cdots\otimes e^{\frac{i}{\hbar}\theta\hat{S}_y},
\tag{9.204}
$$

and the action of $e^{\frac{i}{\hbar}\theta\hat{S}_y}$ on the up (or down) spins states is given by the transformation $\cos\frac{\theta}{2}|\uparrow\rangle - \sin\frac{\theta}{2}|\downarrow\rangle$ ($\cos\frac{\theta}{2}|\downarrow\rangle + \sin\frac{\theta}{2}|\uparrow\rangle$). Hence, we have

$$
\begin{aligned}
e^{\frac{i}{\hbar}\theta\hat{L}_y}|l,m\rangle = &\sqrt{\frac{(2l)!}{(l+m)!(l-m)!}}\left(\cos\tfrac{\theta}{2}|\uparrow\rangle - \sin\tfrac{\theta}{2}|\downarrow\rangle\right)^{l+m} \\
&\times\left(\sin\tfrac{\theta}{2}|\uparrow\rangle + \cos\tfrac{\theta}{2}|\downarrow\rangle\right)^{l-m}.
\end{aligned}
\tag{9.205}
$$

Note that we have temporarily suppressed the tensor-product symbols in the powers of the states. Now, we expand the powers using the binomial theorem to find that

$$
\begin{aligned}
e^{\frac{i}{\hbar}\theta\hat{L}_y}|l,m\rangle &= \sqrt{\frac{(2l)!}{(l+m)!(l-m)!}}\sum_{i=0}^{l+m}\frac{(l+m)!}{i!(l+m-i)!}\left(\cos\tfrac{\theta}{2}\right)^{i}\left(-\sin\tfrac{\theta}{2}\right)^{l+m-i} \\
&\quad\times\underbrace{|\uparrow\rangle\otimes|\uparrow\rangle\otimes\cdots\otimes|\uparrow\rangle}_{i\text{ terms}}\underbrace{|\downarrow\rangle\otimes|\downarrow\rangle\otimes\cdots\otimes|\downarrow\rangle}_{l+m-i\text{ terms}} \\
&\times\sum_{j=0}^{l-m}\frac{(l-m)!}{j!(l-m-j)!}\left(\sin\tfrac{\theta}{2}\right)^{l-m-j}\left(\cos\tfrac{\theta}{2}\right)^{j} \\
&\quad\times\underbrace{|\uparrow\rangle\otimes|\uparrow\rangle\otimes\cdots\otimes|\uparrow\rangle}_{l-m-j\text{ terms}}\underbrace{|\downarrow\rangle\otimes|\downarrow\otimes\cdots\otimes|\downarrow\rangle}_{j\text{ terms}} \\
&= \sum_{m'=-l}^{l}\sqrt{\frac{(2l)!}{(l+m')!(l-m')!}}\,d^{(l)}_{m'm}(\theta) \\
&\quad\times\underbrace{|\uparrow\rangle\otimes|\uparrow\rangle\otimes\cdots\otimes|\uparrow\rangle}_{l+m'\text{ terms}}\underbrace{|\downarrow\rangle\otimes|\downarrow\rangle\otimes\cdots\otimes|\downarrow\rangle}_{l-m'\text{ terms}}. \qquad (9.206)
\end{aligned}
$$

We extract the result for the rotation matrix by fixing $i = m + m' + j$, then we find that

$$
\begin{aligned}
d^{(l)}_{m'm}(\theta) &= \sqrt{\frac{(l+m')!(l-m')!}{(l+m)!(l-m)!}}\sum_{j=\max(0,-m-m')}^{\min(l-m,l-m')}\frac{(l+m)!}{(m+m'+j)!(l-m'-j)!} \\
&\quad\times(-1)^{l-m'-j}\frac{(l-m)!}{j!(l-m-j)!}\left(\cos\tfrac{\theta}{2}\right)^{m+m'+2j}\left(\sin\tfrac{\theta}{2}\right)^{2l-m-m'-2j}. \qquad (9.207)
\end{aligned}
$$

For the spherical harmonics, we require $m' = 0$, so that

$$
\begin{aligned}
Y_{lm}(\theta,\phi) &= \langle\theta{=}0,\phi{=}0|l,m{=}0\rangle e^{im\phi}\frac{l!}{\sqrt{(l+m)!(l-m)!}} \\
&\quad\times\sum_{j=\max(0,-m)}^{\min(l-m,l)}\frac{(l+m)!}{(l-j)!(m+j)!}\frac{(l-m)!}{j!(l-m-j)!} \\
&\quad\times(-1)^{l-j}\left(\cos\tfrac{\theta}{2}\right)^{m+2j}\left(\sin\tfrac{\theta}{2}\right)^{2l-m-2j}. \qquad (9.208)
\end{aligned}
$$

This result can be converted into the standard result in terms of associated Legendre functions. You will examine this conversion in the problems. Before we move into the problems, we summarize the spherical harmonics from these formulas

Table 9.3 Spherical harmonics for $l = 0$, 1, and 2. The spherical harmonics require a convention for the signs of the different terms. This table uses the Condon-Shortley sign convention. You can also see that they satisfy $Y_{l-|m|}(\theta, \phi) = (-1)^{|m|} \left[Y_{l|m|}(\theta, \phi)\right]^*$

$l = 0$	$Y_{0m}(\theta, \phi)$
$m = 0$	$\frac{1}{\sqrt{4\pi}}$
$l = 1$	$Y_{1m}(\theta, \phi)$
$m = 1$	$-\sqrt{\frac{3}{8\pi}} \sin\theta\, e^{i\phi}$
$m = 0$	$\sqrt{\frac{3}{4\pi}} \cos\theta$
$m = -1$	$\sqrt{\frac{3}{8\pi}} \sin\theta\, e^{-i\phi}$
$l = 2$	$Y_{2m}(\theta, \phi)$
$m = 2$	$\sqrt{\frac{15}{32\pi}} \sin^2\theta\, e^{2i\phi}$
$m = 1$	$-\sqrt{\frac{15}{8\pi}} \cos\theta \sin\theta\, e^{i\phi}$
$m = 0$	$\sqrt{\frac{5}{16\pi}} \left(2\cos^2\theta - \sin^2\theta\right)$
$m = -1$	$\sqrt{\frac{15}{8\pi}} \cos\theta \sin\theta\, e^{-i\phi}$
$m = -2$	$\sqrt{\frac{15}{32\pi}} \sin^2\theta\, e^{-2i\phi}$

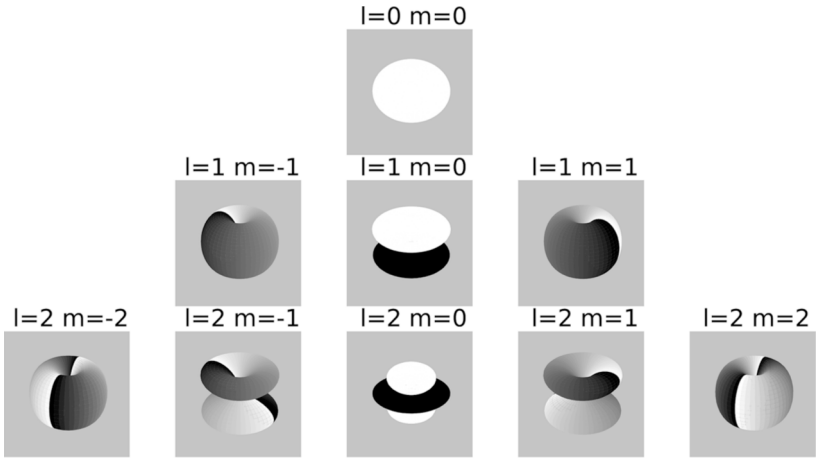

Fig. 9.12 False grayscale image of the spherical harmonics with $l = 0$, 1, and 2. The surface plots the amplitude of Y_{lm}, and the grayscale color shows the complex phase. So, for example, the $l = 0, m = 0$ case is just a sphere with uniform phase. One can see the phase for positive m is opposite that of negative m and that there are more nodes when l increases for fixed m (Thanks to Alan Levin for assistance on making this image)

for $l = 0$, 1, and 2 in Table 9.3. We end with showing some figures of what spherical harmonics look like in Fig. 9.12. The video Spherical Harmonics Without Derivatives https://tinyurl.com/e3jrueer provides a physical picture of what these functions mean.

Section 9.7 Problems:

9.7.1 Exponential Disentangling I

We derive the exponential disentangling identity in a general form for spin one-half and then for the angular-momentum operators.

(a) Show that for $\hat{J}_i = \frac{\hbar}{2}\sigma_i$, we can rewrite

$$-\xi\frac{\hat{J}_+}{\hbar} + \xi^*\frac{\hat{J}_-}{\hbar} = -\frac{\xi}{2}(\sigma_x + i\sigma_y) + \frac{\xi^*}{2}(\sigma_x - i\sigma_y) = -i\mathrm{Im}\xi\ \sigma_x - i\mathrm{Re}\xi\ \sigma_y. \tag{9.209}$$

Use this result to compute the generalized Euler identity for the Pauli matrices:

$$\begin{aligned} e^{-\frac{\xi}{\hbar}\hat{J}_+ + \frac{\xi^*}{\hbar}\hat{J}_-} &= \cos|\xi|\mathbb{I} - i\left(\mathrm{Im}\xi\ \sigma_x + \mathrm{Re}\,\sigma_y\right)\frac{\sin|\xi|}{|\xi|} \\ &= \begin{pmatrix} \cos|\xi| & -\frac{\xi}{|\xi|}\sin|\xi| \\ \frac{\xi^*}{|\xi|}\sin|\xi| & \cos|\xi| \end{pmatrix}. \end{aligned} \tag{9.210}$$

(b) Using $\hat{J}_+ = \frac{\hbar}{2}\sigma_+ = \frac{\hbar}{2}\begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$ and $\hat{J}_- = \frac{\hbar}{2}\begin{pmatrix} 0 & 0 \\ 2 & 0 \end{pmatrix}$, show that

$$e^{\frac{\alpha}{\hbar}\hat{J}_+} = \mathbb{I} + \tfrac{\alpha}{2}\sigma_+ = \begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix}, \tag{9.211}$$

and

$$e^{\frac{\beta}{\hbar}\hat{J}_-} = \mathbb{I} + \tfrac{\beta}{2}\hat{J}_- = \begin{pmatrix} 1 & 0 \\ \beta & 1 \end{pmatrix}. \tag{9.212}$$

Hint: compute $(\sigma_+)^2$ and $(\sigma_-)^2$ and use that in the power-series expansion for the exponential.

(c) Factorize the result in part (a) as follows:

$$\begin{pmatrix} \cos|\xi| & -\frac{\xi}{|\xi|}\sin|\xi| \\ \frac{\xi^*}{|\xi|}\sin|\xi| & \cos|\xi| \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \beta & 1 \end{pmatrix}\begin{pmatrix} b & 0 \\ 0 & c \end{pmatrix}\begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix}. \tag{9.213}$$

You should find $b = \frac{1}{c}$ and $\alpha = -\frac{\xi}{|\xi|}\tan|\xi|$.

(d) Using the results from (b) show that

$$e^{-\frac{\xi}{\hbar}\hat{J}_+ + \frac{\xi^*}{\hbar}\hat{J}_-} = e^{\frac{\xi^*}{|\xi|}\tan|\xi|\frac{\hat{J}_-}{\hbar}}\, e^{\ln\left(\cos^2|\xi|\right)\frac{\hat{J}_z}{\hbar}}\, e^{-\frac{\xi}{|\xi|}\tan|\xi|\frac{\hat{J}_+}{\hbar}}. \tag{9.214}$$

You will need the fact that $1 + \tan^2 x = \sec^2 x$.

9.7.2 *Exponential Disentangling II*

We repeat the previous problem to compute the exponential disentangling identity in the opposite order:

$$\begin{pmatrix} \cos|\xi| & -\frac{\xi}{|\xi|}\sin|\xi| \\ \frac{\xi^*}{|\xi|}\sin|\xi| & \cos|\xi| \end{pmatrix} = \begin{pmatrix} 1 & \alpha' \\ 0 & 1 \end{pmatrix} \begin{pmatrix} b' & 0 \\ 0 & c' \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \beta' & 1 \end{pmatrix}. \tag{9.215}$$

This should then yield

$$e^{-\frac{\xi}{\hbar}\hat{J}_+ + \frac{\xi^*}{\hbar}\hat{J}_-} = e^{-\frac{\xi}{|\xi|}\tan|\xi|\frac{\hat{J}_+}{\hbar}} e^{-\ln\left(\cos^2|\xi|\right)\frac{\hat{J}_z}{\hbar}} e^{\frac{\xi^*}{|\xi|}\tan|\xi|\frac{\hat{J}_-}{\hbar}}. \tag{9.216}$$

9.7.3 3×3 *Representation for* $l = 1$

The $l = 1$ angular momentum can be represented by 3×3 matrices of the form

$$\hat{L}_i \leftrightarrow \begin{pmatrix} \langle 1,1|\hat{L}_i|1,1\rangle & \langle 1,1|\hat{L}_i|1,0\rangle & \langle 1,1|\hat{L}_i|1,-1\rangle \\ \langle 1,0|\hat{L}_i|1,1\rangle & \langle 1,0|\hat{L}_i|1,0\rangle & \langle 1,0|\hat{L}_i|1,-1\rangle \\ \langle 1,-1|\hat{L}_i|1,1\rangle & \langle 1,-1|\hat{L}_i|1,0\rangle & \langle 1,-1|\hat{L}_i|1,-1\rangle \end{pmatrix}. \tag{9.217}$$

Verify that

$$\hat{L}_x = \hbar \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & 0 \end{pmatrix}, \quad \hat{L}_y = \hbar \begin{pmatrix} 0 & -\frac{i}{\sqrt{2}} & 0 \\ \frac{i}{\sqrt{2}} & 0 & -\frac{i}{\sqrt{2}} \\ 0 & \frac{i}{\sqrt{2}} & 0 \end{pmatrix}, \quad \text{and} \quad \hat{L}_z = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{9.218}$$

Using these matrices, verify that $[\hat{L}_i, \hat{L}_j] = i\hbar \sum_k \epsilon_{ijk}\hat{L}_k$. Note that the matrix multiplication for 3×3 matrices is similar to the 2×2 matrices. We multiply the row times the corresponding column. It is just that there are nine multiplications now instead of the four needed for the 2×2 matrices, because the matrices have nine matrix elements. You should be able to also show that $\hat{\hat{L}} \cdot \hat{\hat{L}} = 2\hbar^2 \mathbb{I}$.

Verify that

$$\left(\frac{\hat{L}_i}{\hbar}\right)^3 = \left(\frac{\hat{L}_i}{\hbar}\right), \tag{9.219}$$

for $i = x$, y, and z.

Using the last result, show that

$$e^{\frac{i\theta}{\hbar}\hat{L}_i} = \mathbb{I}_3 + i\sin\theta\left(\frac{\hat{L}_i}{\hbar}\right) + (\cos\theta - 1)\left(\frac{\hat{L}_i}{\hbar}\right)^2. \tag{9.220}$$

Use that result to also show that

$$(\hat{L}_i - \hbar\mathbb{I}_3)\hat{L}_i(\hat{L}_i + \hbar\mathbb{I}_3) = 0. \tag{9.221}$$

Explain why this equation must be true based on the fact that the eigenvalues of $\hat{L}_i$ are $\{-\hbar, 0, \hbar\}$ only.

9.7.4 *Euler Identity for $l = 1$ Angular Momentum Matrices*

This problem is a more difficult version of the previous problem, but it is a straightforward French-cooking problem. Try it if you are ready.

Compute

$$e^{\frac{i\theta}{\hbar}\vec{n}\cdot\hat{\vec{L}}} \text{ with } \vec{n} = (\sin\alpha\cos\beta, \sin\alpha\sin\beta, \cos\alpha), \tag{9.222}$$

which is a unit vector.

Hint: First show that

$$\left(\frac{\vec{n}\cdot\hat{\vec{L}}}{\hbar}\right)^3 = \left(\frac{\vec{n}\cdot\hat{\vec{L}}}{\hbar}\right). \tag{9.223}$$

9.7.5 *Verification of the Exponential Disentangling Identity for $l = 1$*

Because the exponential disentangling identity holds for all representations, it must work for the $l = 1$ representation worked out above. This problem requires you to have completed the previous problem. It also involves a significant amount of algebra.

(a) Compute $e^{-\frac{\xi}{\hbar}\hat{L}_+ + \frac{\xi^*}{\hbar}\hat{L}_-}$ using the power series expansion and the relation $\left(\frac{\hat{L}_i}{\hbar}\right)^3 = \left(\frac{\hat{L}_i}{\hbar}\right)$ for the Cartesian operators.

Hint: convert to $\hat{L}_x$ and $\hat{L}_y$, and use the results for exponentials computed above. You should find

$$
\begin{aligned}
e^{-\frac{\xi}{\hbar}\hat{L}_+ + \frac{\xi^*}{\hbar}\hat{L}_-} &= \mathbb{I} - i\sin 2|\xi| \left(\frac{\mathrm{Im}\xi\,\hat{L}_x + \mathrm{Re}\xi\,\hat{L}_y}{|\xi|}\right) \\
&\quad + (\cos 2|\xi| - 1)\left(\frac{\mathrm{Im}\xi\,\hat{L}_x + \mathrm{Re}\xi\,\hat{L}_y}{|\xi|}\right)^2 \\
&\quad \begin{pmatrix} \frac{1+\cos 2|\xi|}{2} & -\frac{1}{\sqrt{2}}\frac{\xi}{|\xi|}\sin 2|\xi| & \frac{\xi^2}{|\xi|^2}\frac{1-\cos 2|\xi|}{2} \\ \frac{1}{\sqrt{2}}\frac{\xi^*}{|\xi|}\sin 2|\xi| & \cos 2|\xi| & -\frac{1}{\sqrt{2}}\frac{\xi}{|\xi|}\sin 2|\xi| \\ \frac{\xi^{*2}}{|\xi|^2}\frac{1-\cos 2|\xi|}{2} & \frac{1}{\sqrt{2}}\frac{\xi^*}{|\xi|}\sin 2|\xi| & \frac{1+\cos 2|\xi|}{2} \end{pmatrix}.
\end{aligned}
\tag{9.224}
$$

(b) Compute the three factors

$$
e^{\frac{\xi^*}{|\xi|}\tan|\xi|\frac{\hat{L}_-}{\hbar}}, \quad e^{2\ln\cos|\xi|\frac{\hat{L}_z}{\hbar}}, \quad \text{and} \quad e^{-\frac{\xi}{|\xi|}\tan|\xi|\frac{\hat{L}_+}{\hbar}}. \tag{9.225}
$$

as 3×3 matrices.

(c) Verify that

$$
e^{-\frac{\xi}{\hbar}\hat{L}_+ + \frac{\xi^*}{\hbar}\hat{L}_-} = e^{\frac{\xi^*}{|\xi|}\tan|\xi|\frac{\hat{L}_-}{\hbar}}\, e^{2\ln\cos|\xi|\frac{\hat{L}_z}{\hbar}}\, e^{-\frac{\xi}{|\xi|}\tan|\xi|\frac{\hat{L}_+}{\hbar}}, \tag{9.226}
$$

by multiplying the three matrices together. You should review the trigonometric half-angle formulas when working these results out.

This shows that the identity we derived for spin one-half works for spin one as well. Indeed, it works for arbitrary spin, but we will not independently verify that.

9.7.6 Spherical Harmonics for $m < 0$

The calculation of spherical harmonics for $m < 0$ using the exponential disentangling identity is best done by using the form of the exponential disentangling identity with the raising operator on the left and the lowering operator on the right. One can lower the state down to at most the state $|l, -l\rangle$. It then needs to be raised to $m = 0$ for the spherical harmonic. Follow closely the way we worked out the

results for $m > 0$ in the text, but now perform the calculation for $m < 0$ using the second form of the exponential disentangling identity. Verify that the identity $Y_{l-|m|}(\theta, \phi) = (-1)^{|m|} \left[Y_{l|m|}(\theta, \phi)\right]^*$ holds.

9.7.7 $l = 0$, 1, and 2 from Each Formula

Using the concrete formula for the spherical harmonics that we found in the text, verify explicitly the results summarized in Table 9.3.

9.7.8 Converting the Rotation Matrix into Associated Legendre Functions

In the text, we found the formula for the spherical harmonics from the rotation matrix by setting $m' = 0$ in Eq. (9.208). Convert this result into the result that uses the associated Legendre polynomials. Your result should be exactly the same as the final result found from the exponential disentangling identity. This problem requires significant algebra as well.

9.7.9 Parity

Parity involves $\vec{r} \rightarrow -\vec{r}$. For spherical coordinates, this means $\theta \rightarrow \pi - \theta$ and $\phi \rightarrow \phi + \pi$. Use the concrete result for the spherical harmonic to show that

$$Y_{lm}(\pi - \theta, \phi + \pi) = (-1)^l Y_{lm}(\theta, \phi). \tag{9.227}$$

This then implies that $\hat{\Pi}|l, m\rangle = (-1)^l |l, m\rangle$.

9.7.10 Angle Operators Are Ill-Defined

Many quantum textbooks say that an angle operator such as $\hat{\phi}$, which satisfies $[\hat{\phi}, \hat{L}_z] = i\hbar$, is a valid quantum operator. But such an operator cannot be well defined for the following reason. If $\hat{L}_z|l, m\rangle = \hbar m|l, m\rangle$ then, you should show that

$$\hat{L}_z e^{i\alpha\hat{\phi}}|l, m\rangle = \hbar(m + \alpha)e^{i\alpha\hat{\phi}}|l, m\rangle. \tag{9.228}$$

This means that if we have a state that has eigenvalues of $\hbar m$ under $\hat{L}_z$, then we can construct states with eigenvalues $\hbar(m+\alpha)$ as well. But our study of the algebraic properties of angular momentum showed that m can only be an integer or a half-odd integer. So, this is not allowed for arbitrary α. Indeed, if we allow only $\cos\hat{\phi}$ and $\sin\hat{\phi}$ as valid operators, then the shift of the eigenvalue that we can produce from those operators is to shift m by an integer. This is similar to the discussion we had about the position operator when we worked on the particle on a circle.

So we don't make the mistake of working directly with angle operators in this book. Note that there is a way to make the angle operator well-defined, but it is quite complex and technical, so we do not discuss it further here. In this book, we will be content to work only with the sines and cosines of the angle operators.

9.8 Schwinger's Representation of Angular Momentum

Julian Schwinger is one of the most accomplished physicists of the twentieth century (see Fig. 9.13 for an image of Schwinger). He shared a Nobel prize with Richard Feynman and Shinichiro Tomonaga for the development of field theory methods of quantum electrodynamics. He was also the Ph. D. advisor to four more Nobel prize winners. In 1952, Schwinger devised an ingenious way to represent angular momentum in terms of simple harmonic oscillators. The original publication was as a government report for the Atomic Energy Commission that was very difficult to obtain; it has now been reprinted as a Dover paperback (Schwinger, 2015). Using this formalism unifies the integer and half-odd-integer representations and provides an alternative methodology to derive all of the angular momentum identities that we have been working with.

We begin by reviewing the properties of the simple-harmonic oscillator in two dimensions, which are required for this formalism. Working with a common frequency $\omega = 1$ for each spatial dimension, we find that the Hamiltonian becomes

$$\hat{H} = \hbar\left(\hat{a}_1^\dagger\hat{a}_1 + \hat{a}_2^\dagger\hat{a}_2 + 1\right), \tag{9.229}$$

with $[\hat{a}_i\,,\hat{a}_j^\dagger] = \delta_{ij}$, $[\hat{a}_i\,,\hat{a}_j] = 0$, and $[\hat{a}_i^\dagger\,,\hat{a}_j^\dagger] = 0$. The normalized eigenstates are

$$|n_1, n_2\rangle = \frac{1}{\sqrt{n_1!n_2!}}\left(\hat{a}_1^\dagger\right)^{n_1}\left(\hat{a}_2^\dagger\right)^{n_2}|0\rangle. \tag{9.230}$$

We know that $\hat{a}_1^\dagger|n_1, n_2\rangle = \sqrt{n_1+1}|n_1+1, n_2\rangle$, $\hat{a}_2^\dagger|n_1, n_2\rangle = \sqrt{n_2+1}|n_1, n_2+1\rangle$, $\hat{a}_1|n_1, n_2\rangle = \sqrt{n_1}|n_1-1, n_2\rangle$, and $\hat{a}_2|n_1, n_2\rangle = \sqrt{n_2}|n_1, n_2-1\rangle$.

We use $\hat{J}$ to denote the angular momentum, because this representation works for both integer and half-odd-integer representations, and so is generic. We define

Fig. 9.13 Julian Schwinger in the early 1960s (Photo credit: AIP Emilio Segrè Visual Archives, Physics Today Collection. Modified to look like a sketch by James Freericks)

$$\hat{J}_z = \frac{\hbar}{2}\left(\hat{a}_1^\dagger \hat{a}_1 - \hat{a}_2^\dagger \hat{a}_2\right), \quad \hat{J}_+ = \hbar \hat{a}_1^\dagger \hat{a}_2, \quad \text{and} \quad \hat{J}_- = \hbar \hat{a}_2^\dagger \hat{a}_1. \tag{9.231}$$

Let's check. The calculation primarily requires using the Leibniz rule many times. The results for the SU(2) commutation relations are as follows:

$$\begin{aligned}[\hat{J}_z, \hat{J}_+] &= \frac{\hbar^2}{2}\left[\hat{a}_1^\dagger \hat{a}_1 - \hat{a}_2^\dagger \hat{a}_2, \hat{a}_1^\dagger \hat{a}_2\right] = \frac{\hbar^2}{2}\left(\left[\hat{a}_1^\dagger \hat{a}_1, \hat{a}_1^\dagger\right]\hat{a}_2 - \hat{a}_1^\dagger\left[\hat{a}_2^\dagger \hat{a}_2, \hat{a}_2\right]\right) \\ &= \frac{\hbar^2}{2}\left(\hat{a}_1^\dagger[\hat{a}_1, \hat{a}_1^\dagger]\hat{a}_2 - \hat{a}_1^\dagger[\hat{a}_2^\dagger, \hat{a}_2]\hat{a}_2\right) = \hbar^2 \hat{a}_1^\dagger \hat{a}_2 = \hbar \hat{J}_+, \end{aligned} \tag{9.232}$$

$$\begin{aligned}[\hat{J}_z, \hat{J}_-] &= \frac{\hbar^2}{2}\left[\hat{a}_1^\dagger \hat{a}_1 - \hat{a}_2^\dagger \hat{a}_2, \hat{a}_2^\dagger \hat{a}_1\right] = \frac{\hbar^2}{2}\left(\hat{a}_2^\dagger\left[\hat{a}_1^\dagger \hat{a}_1, \hat{a}_1\right] - \left[\hat{a}_2^\dagger \hat{a}_2, \hat{a}_2^\dagger\right]\hat{a}_1\right) \\ &= \frac{\hbar^2}{2}\left(\hat{a}_2^\dagger[\hat{a}_1^\dagger, \hat{a}_1]\hat{a}_1 - \hat{a}_2^\dagger[\hat{a}_2, \hat{a}_2^\dagger]\hat{a}_1\right) = -\hbar^2 \hat{a}_2^\dagger \hat{a}_1 = -\hbar \hat{J}_-, \end{aligned} \tag{9.233}$$

and

$$\begin{aligned}[\hat{J}_+, \hat{J}_-] &= \hbar^2\left[\hat{a}_1^\dagger \hat{a}_2, \hat{a}_2^\dagger \hat{a}_1\right] = \hbar^2\left(\hat{a}_1^\dagger\left[\hat{a}_2, \hat{a}_2^\dagger\right]\hat{a}_1 + \hat{a}_2^\dagger\left[\hat{a}_1^\dagger, \hat{a}_1\right]\hat{a}_2\right) \\ &= \hbar^2\left(\hat{a}_1^\dagger \hat{a}_1 - \hat{a}_2^\dagger \hat{a}_2\right) = 2\hbar \hat{J}_z. \end{aligned} \tag{9.234}$$

So, these operators do satisfy the angular momentum commutation relations. We also can show that $\hat{N} = \hat{a}_1^\dagger \hat{a}_1 + \hat{a}_2^\dagger \hat{a}_2$ commutes with all of the angular momentum

operators:

$$[\hat{N}, \hat{J}_{\pm}] = [\hat{N}, \hat{J}_z] = 0. \tag{9.235}$$

Let us interpret these results. Since $\hat{N}$ commutes with the angular-momentum operators, a representation of the angular momentum has a fixed total quantum number with respect to $\hat{N}$, which is equal to $n_1 + n_2$. The angular-momentum raising operators keep the total $n_1 + n_2$ fixed but transfer from the 2 oscillator to the 1 oscillator, while the angular-momentum lowering operator does the opposite. Furthermore, $\hat{J}_z$ measures the difference in excitations between the two oscillators. Let us investigate this in more detail. Suppose that $n_1 + n_2 = 2j$. Then one can see that

$$\hat{J}_z|n_1, n_2\rangle = \hbar\frac{n_1 - n_2}{2}|n_1, n_2\rangle = \hbar(j - n_2)|n_1, n_2\rangle. \tag{9.236}$$

Since we have $0 \leq n_2 \leq 2j$, because $n_1 + n_2 = j$, we see that the $\hat{J}_z$ eigenvalue ranges from $-j\hbar$ to $j\hbar$, indicating that the angular momentum squared eigenvalue is equal to $\hbar^2 j(j+1)$; we denote the $\hat{J}_z$ eigenvalue as $\hbar\frac{\hat{n}_1 - \hat{n}_2}{2} = \hbar m$. The action of the angular-momentum raising operators can be computed directly from harmonic oscillator raising or lowering operators, which give

$$\begin{aligned}\hat{J}_+|n_1, n_2\rangle &= \hbar\hat{a}_1^\dagger\hat{a}_2|n_1, n_2\rangle = \hbar\sqrt{n_1 + 1}\sqrt{n_2}|n_1 + 1, n_2 - 1\rangle \\ &= \hbar\sqrt{(j + m + 1)(j - m)}|n_1, n_2\rangle,\end{aligned} \tag{9.237}$$

and

$$\begin{aligned}\hat{J}_-|n_1, n_2\rangle &= \hbar\hat{a}_2^\dagger\hat{a}_1|n_1, n_2\rangle = \hbar\sqrt{n_1}\sqrt{n_2 + 1}|n_1, n_2\rangle \\ &= \hbar\sqrt{(j + m)(j - m + 1)}|n_1, n_2\rangle.\end{aligned} \tag{9.238}$$

When we note that $|n_1, n_2\rangle = |j, m\rangle$, where the state on the left is written in the simple-harmonic-oscillator notation and the state on the right is written in the squared angular momentum and z-component of angular momentum notation. We see that these relations are the expected correct relations for the action of the angular-momentum raising and lowering operators on the given eigenstates. Using these relations, following the steps worked out in Sect. 9.3, we find that $\hat{\vec{J}}^2|j, m\rangle = \hbar^2 j(j+1)|j, m\rangle$ for all states in the multiplet. Using $\hat{\vec{J}}^2 = \frac{1}{2}(\hat{J}_+\hat{J}_- + \hat{J}_-\hat{J}_+) + \hat{J}_z^2$ yields

$$\begin{aligned}\hat{\vec{J}}^2 &= \hbar^2\left(\tfrac{1}{2}\hat{a}_1^\dagger\hat{a}_2\hat{a}_2^\dagger\hat{a}_1 + \tfrac{1}{2}\hat{a}_2^\dagger\hat{a}_1\hat{a}_1^\dagger\hat{a}_2 + \tfrac{1}{4}\left(\hat{a}_1^\dagger\hat{a}_1 - \hat{a}_2^\dagger\hat{a}_2\right)^2\right) \\ &= \hbar^2\left(\hat{a}_1^\dagger\hat{a}_1\hat{a}_2^\dagger\hat{a}_2 + \tfrac{1}{2}\hat{a}_1^\dagger\hat{a}_1 + \tfrac{1}{2}\hat{a}_2^\dagger\hat{a}_2 + \tfrac{1}{4}(\hat{a}_1^\dagger)^2(\hat{a}_1)^2 + \tfrac{1}{4}(\hat{a}_2^\dagger)^2(\hat{a}_2)^2\right.\end{aligned}$$

$$+\tfrac{1}{4}\hat{a}_1^\dagger\hat{a}_1+\tfrac{1}{4}\hat{a}_2^\dagger\hat{a}_2-\tfrac{1}{2}\hat{a}_1^\dagger\hat{a}_1\hat{a}_2^\dagger\hat{a}_2\Big)$$
$$=\tfrac{\hbar^2}{2}\left(\hat{a}_1^\dagger\hat{a}_1\hat{a}_2^\dagger\hat{a}_2+\tfrac{1}{2}(\hat{a}_1^\dagger)^2(\hat{a}_1)^2+\tfrac{1}{2}(\hat{a}_2^\dagger)^2(\hat{a}_2)^2+\tfrac{3}{2}\hat{a}_1^\dagger\hat{a}_1+\tfrac{3}{2}\hat{a}_2^\dagger\hat{a}_2\right). \tag{9.239}$$

As a check, operate it onto $|n_1, n_2\rangle$ to find

$$\begin{aligned}\hat{\vec{J}}^2|n_1,n_2\rangle &= \tfrac{\hbar^2}{2}\left(n_1n_2+\tfrac{1}{2}n_1(n_1-1)+\tfrac{1}{2}n_2(n_2-1)+\tfrac{3}{2}n_1+\tfrac{3}{2}n_2\right)|n_1,n_2\rangle\\ &= \frac{\hbar^2}{4}(n_1^2+2n_1n_2+n_2^2+2n_1+2n_2)|n_1,n_2\rangle\\ &= \frac{\hbar^2}{4}(n_1+n_2+2)(n_1+n_2)|n_1,n_2\rangle=\hbar^2(j+1)j|n_1,n_2\rangle,\end{aligned} \tag{9.240}$$

as it must.

So, what does this all mean? We have a new representation we can use for angular momentum. But unlike the old way, where states of different j were independent of each other, this representation yields a unified picture for us of how the representations are interrelated. We illustrate this in Fig. 9.14, showing the harmonic oscillator excitations, the j and m eigenvalues, and how $\hat{J}_\pm$ steps through them. But now, we also have the option of stepping from one j value to another. This requires us to examine some new operators, which satisfy a symplectic algebra (just like the operators we used to examine squeezing). They are sometimes also called the *hyperbolic algebra*. We will see a similar algebra also arise for fermions when we discuss them in Chap. 10. Note how the half-odd integer representations have no $m=0$ states. The $\hat{J}_\pm$ operators move along the diagonals with slope -1 and carry us through all the states in an angular momentum multiplet with fixed j. The hyperbolic operators change the j value by ± 1.

One of the hyperbolic operators involves the number operator, the other two create or remove two simple harmonic oscillator excitations. They are

$$\hat{K}_0=\frac{\hbar}{2}\left(\hat{a}_1^\dagger\hat{a}_1+\hat{a}_2^\dagger\hat{a}_2-1\right),\quad \hat{K}_+=\hbar\hat{a}_1^\dagger\hat{a}_2^\dagger,\quad\text{and}\quad \hat{K}_-=\hbar\hat{a}_1\hat{a}_2. \tag{9.241}$$

We see that

$$\begin{aligned}[\hat{K}_0,\hat{K}_+] &= \frac{\hbar^2}{2}\left[\hat{a}_1^\dagger\hat{a}_1+\hat{a}_2^\dagger\hat{a}_2,\hat{a}_1^\dagger\hat{a}_2^\dagger\right]\\ &= \frac{\hbar^2}{2}\left(\hat{a}_1^\dagger\left[\hat{a}_1,\hat{a}_1^\dagger\right]\hat{a}_2^\dagger+\hat{a}_1^\dagger\hat{a}_2^\dagger\left[\hat{a}_2,\hat{a}_2^\dagger\right]\right)=\hbar\hat{K}_+,\end{aligned} \tag{9.242}$$

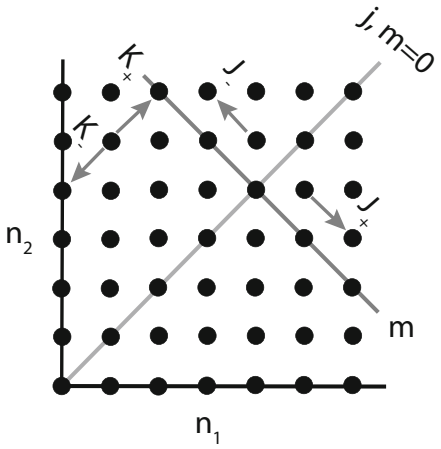

Fig. 9.14 Schematic of Schwinger's representation of angular momentum. The dots represent a given $|n_1, n_2\rangle$ state, which has $j = \frac{n_1+n_2}{2}$ and $m = \frac{n_1-n_2}{2}$. The m values change along the diagonal with slope -1 and this is where the $\hat{J}_\pm$ operators shift the z-component of angular momentum. The $\hat{K}_\pm$ operators shift up or down along the diagonal with slope 1. They connect the different integer angular-momentum representations or the half-odd-integer angular-momentum representations

$$
\begin{aligned}
[\hat{K}_0, \hat{K}_-] &= \frac{\hbar^2}{2}\left[\hat{a}_1^\dagger \hat{a}_1 + \hat{a}_2^\dagger \hat{a}_2, \hat{a}_1 \hat{a}_2\right] \\
&= \frac{\hbar^2}{2}\left(\left[\hat{a}_1^\dagger, \hat{a}_1\right]\hat{a}_1 \hat{a}_2 + \left[\hat{a}_2^\dagger, \hat{a}_2\right]\hat{a}_1 \hat{a}\ \right) = -\hbar \hat{K}_-,
\end{aligned}
\tag{9.243}
$$

and

$$
\begin{aligned}
[\hat{K}_+, \hat{K}_-] &= \hbar^2\left[\hat{a}_1^\dagger \hat{a}_2^\dagger, \hat{a}_1 \hat{a}_2\right] = \hbar^2\left(\hat{a}_1^\dagger\left[\hat{a}_2^\dagger, \hat{a}_1 \hat{a}_2\right] + \left[\hat{a}_1^\dagger, \hat{a}_1 \hat{a}_2\right]\hat{a}_2^\dagger\right) \\
&= \hbar^2\left(\hat{a}_1^\dagger \hat{a}_1\left[\hat{a}_2^\dagger, \hat{a}_2\right] + \left[\hat{a}_1^\dagger, \hat{a}_1\right]\hat{a}_2 \hat{a}_2^\dagger\right) = \hbar^2\left(-\hat{a}_1^\dagger \hat{a}_1 - \hat{a}_2^\dagger \hat{a}_2 + 1\right) \\
&= -2\hbar \hat{K}_0.
\end{aligned}
\tag{9.244}
$$

These commutators look just like the angular-momentum commutation relations, except for the minus sign in the last commutator. Note that $\hat{K}_0$ is the Hamiltonian of the two simple harmonic oscillators minus a constant.

Now, we can see that the $\hat{K}_\pm$ operators step us along the diagonal (with slope 1) by creating two simple harmonic oscillator excitations. So, we find we can step through all integer angular-momentum representations or all half-odd-integer angular-momentum representations. Problem 7.8.1 shows that the hyperbolic operators commute with the angular-momentum operators and Problem 9.8.5 computes the factors $\hat{K}_\pm |j, m\rangle = C_\pm^{j,m} |j \pm 1, m\rangle$.

We can also use this representation to compute Wigner's rotation matrices and the spherical harmonics. For integer j, we begin with the definition of the rotation matrices found in Sect. 9.7, and written here as

$$d^{(j)}_{m'm}(\theta) = \langle j, m' | e^{\frac{i}{\hbar}\theta \hat{J}_y} | j, m \rangle. \tag{9.245}$$

We will work out the rotation matrices in two different ways—the first is by using the Hadamard lemma and the second is via the exponential disentangling identity—both can be used to complete the calculation. We start with the representation of $|j, m\rangle$ in terms of the harmonic oscillator raising operators

$$|j, m\rangle = \frac{1}{\sqrt{(j+m)!(j-m)!}} \left(\hat{a}_1^\dagger\right)^{j+m} \left(\hat{a}_2^\dagger\right)^{j-m} |0\rangle. \tag{9.246}$$

We also use the fact that $\left(\hat{J}_y = \frac{1}{2i}(\hat{J}_+ - \hat{J}_-) = \frac{\hbar}{2i}\left(\hat{a}_1^\dagger \hat{a}_2 - \hat{a}_2^\dagger \hat{a}_1\right)\right)$.

Using the fact that $\hat{J}_y|0\rangle = 0$, we find that

$$\begin{aligned} d^{(j)}_{m'm}(\theta) = \frac{1}{\sqrt{(j+m')!(j-m')!(j+m)!(j-m)!}} \langle 0| \left(\hat{a}_1\right)^{j+m'} \left(\hat{a}_2\right)^{j-m'} \\ \times e^{\frac{i}{\hbar}\theta \hat{J}_y} \left(\hat{a}_1^\dagger\right)^{j+m} \left(\hat{a}_2^\dagger\right)^{j-m} \underbrace{e^{-\frac{i}{\hbar}\theta \hat{J}_y}|0\rangle}_{|0\rangle}. \end{aligned} \tag{9.247}$$

One can recognize the Hadamard lemma on the second line. It can be moved inside each power of each operator. For example, we have

$$e^{\frac{i}{\hbar}\theta \hat{J}_y} \left(\hat{a}_1^\dagger\right)^{j+m} e^{-\frac{i}{\hbar}\theta \hat{J}_y} = \left(e^{\frac{i}{\hbar}\theta \hat{J}_y} \hat{a}_1^\dagger e^{-\frac{i}{\hbar}\theta \hat{J}_y}\right)^{j+m}, \tag{9.248}$$

and then the Hadamard lemma has a repeating infinite sum given by

$$\begin{aligned} e^{\frac{i}{\hbar}\theta \hat{J}_y} \hat{a}_1^\dagger e^{-\frac{i}{\hbar}\theta \hat{J}_y} &= \hat{a}_1^\dagger + \tfrac{i}{\hbar}\theta \left[\hat{J}_y, \hat{a}_1^\dagger\right] + \cdots = \hat{a}_1^\dagger - \tfrac{\theta}{2}\hat{a}_2^\dagger - \tfrac{1}{2}\left(\tfrac{\theta}{2}\right)^2 \hat{a}_1^\dagger + \cdots \\ &= \cos\tfrac{\theta}{2}\hat{a}_1^\dagger - \sin\tfrac{\theta}{2}\hat{a}_2^\dagger. \end{aligned} \tag{9.249}$$

In Problem 9.8.3, you will show that

$$e^{\frac{i}{\hbar}\theta \hat{J}_y} \hat{a}_2^\dagger e^{-\frac{i}{\hbar}\theta \hat{J}_y} = \sin\tfrac{\theta}{2}\hat{a}_1^\dagger + \cos\tfrac{\theta}{2}\hat{a}_2^\dagger. \tag{9.250}$$

This then yields

$$d^{(j)}_{m'm}(\theta) = \frac{1}{\sqrt{(j+m')!(j-m')!(j+m)!(j-m)!}} \langle 0| \left(\hat{a}_1\right)^{j+m'} \left(\hat{a}_2\right)^{j-m'} \times \left(\cos\tfrac{\theta}{2}\hat{a}_1^\dagger - \sin\tfrac{\theta}{2}\hat{a}_2^\dagger\right)^{j+m} \left(\sin\tfrac{\theta}{2}\hat{a}_1^\dagger + \cos\tfrac{\theta}{2}\hat{a}_2^\dagger\right)^{j-m} |0\rangle. \tag{9.251}$$

Now, we expand the factors that involve trigonometric functions with the binomial theorem and then evaluate the matrix element in terms of the raising and lowering operators. This gives

$$\begin{aligned} d^{(j)}_{m'm}(\theta) &= \frac{1}{\sqrt{(j+m')!(j-m')!(j+m)!(j-m)!}} \\ &\times \sum_{n'=0}^{j+m} \frac{(j+m)!}{n'!(j+m-n')!} \left(\cos\tfrac{\theta}{2}\right)^{n'} \left(-\sin\tfrac{\theta}{2}\right)^{j+m-n'} \\ &\times \sum_{n=0}^{j-m} \frac{(j-m)!}{n!(j-m-n)!} \left(\sin\tfrac{\theta}{2}\right)^{j-m-n} \left(\cos\tfrac{\theta}{2}\right)^{n} \\ &\times \langle 0| \left(\hat{a}_1\right)^{j+m'} \left(\hat{a}_2\right)^{j-m'} \left(\hat{a}_1^\dagger\right)^{j-m-n+n'} \left(\hat{a}_2^\dagger\right)^{j+m+n-n'} |0\rangle \\ &= \frac{1}{\sqrt{(j+m')!(j-m')!(j+m)!(j-m)!}} \\ &\times \sum_{n'=0}^{j+m} \frac{(j+m)!}{n'!(j+m-n')!} \left(\cos\tfrac{\theta}{2}\right)^{n'} \left(-\sin\tfrac{\theta}{2}\right)^{j+m-n'} \\ &\times \sum_{n=0}^{j-m} \frac{(j-m)!}{n!(j-m-n)!} \left(\sin\tfrac{\theta}{2}\right)^{j-m-n} \left(\cos\tfrac{\theta}{2}\right)^{n} \\ &\times (j+m')!(j-m')!\delta_{m',n'-m-n}\delta_{n'-m-n,m'}. \end{aligned} \tag{9.252}$$

Simplifying the final result yields

$$d^{(j)}_{m'm}(\theta) = \sqrt{\frac{(j+m')!(j-m')!}{(j+m)!(j-m)!}} \sum_{n=\max(0,-m-m')}^{\min(j-m,j-m')} \frac{(j+m)!}{(n+m'+n)!(j-m'-n)!} \times \frac{(j-m)!}{n!(j-m-n)!}(-1)^{j-m'-n} \left(\cos\tfrac{\theta}{2}\right)^{m+m'+2n} \left(\sin\tfrac{\theta}{2}\right)^{2j-m-m'-2n}, \tag{9.253}$$

which is exactly what we found with Wigner's method in the previous section. The derivation for the spherical harmonics then proceeds in exactly the same way by simply setting $m' = 0$. So, we do not repeat it here.

We also illustrate how to complete this calculation with the exponential disentangling identity. Since the Schwinger method is a representation of angular momentum, we already have the exponential disentangling identity given by

$$e^{\frac{\theta}{2\hbar}(\hat{J}_+ - \hat{J}_-)} = e^{\frac{\theta}{2}\left(\hat{a}_1^\dagger \hat{a}_2 - \hat{a}_2^\dagger \hat{a}_1\right)} = e^{-\tan\frac{\theta}{2}\hat{a}_2^\dagger \hat{a}_1} e^{\ln\left(\cos\frac{\theta}{2}\right)\left(\hat{a}_1^\dagger \hat{a}_1 - \hat{a}_2^\dagger \hat{a}_2\right)} e^{\tan\frac{\theta}{2}\hat{a}_1^\dagger \hat{a}_2}. \tag{9.254}$$

The derivation is a bit long, so we will separate into pieces. First, we substitute in the exponential disentangling identity and find

$$\begin{aligned} d^{(j)}_{m'm}(\theta) = {} & \frac{1}{\sqrt{(j+m')!(j-m')!(j+m)!(j-m)!}} \langle 0| \left(\hat{a}_1\right)^{j+m'} \left(\hat{a}_2\right)^{j-m'} \\ & \times e^{-\tan\frac{\theta}{2}\hat{a}_2^\dagger \hat{a}_1} e^{\ln\left(\cos\frac{\theta}{2}\right)\left(\hat{a}_1^\dagger \hat{a}_1 - \hat{a}_2^\dagger \hat{a}_2\right)} e^{\tan\frac{\theta}{2}\hat{a}_1^\dagger \hat{a}_2} \left(\hat{a}_1^\dagger\right)^{j+m} \left(\hat{a}_2^\dagger\right)^{j-m} |0\rangle. \end{aligned} \tag{9.255}$$

Now, we expand the exponentials of the raising and lowering operators in their power series, yielding

$$\begin{aligned} d^{(j)}_{m'm}(\theta) = {} & \frac{1}{\sqrt{(j+m')!(j-m')!(j+m)!(j-m)!}} \langle 0| \left(\hat{a}_1\right)^{j+m'} \left(\hat{a}_2\right)^{j-m'} \\ & \times \sum_{n=0}^{\infty} \frac{(-\tan\frac{\theta}{2})^n}{n!} \left(\hat{a}_2^\dagger \hat{a}_1\right)^n e^{\ln\left(\cos\frac{\theta}{2}\right)\left(\hat{a}_1^\dagger \hat{a}_1 - \hat{a}_2^\dagger \hat{a}_2\right)} \\ & \times \sum_{n'=0}^{\infty} \frac{(\tan\frac{\theta}{2})^{n'}}{n'!} \left(\hat{a}_1^\dagger \hat{a}_2\right)^{n'} \left(\hat{a}_1^\dagger\right)^{j+m} \left(\hat{a}_2^\dagger\right)^{j-m} |0\rangle. \end{aligned} \tag{9.256}$$

In the next step, we take each $\hat{a}_2^\dagger$ operator from the summation that is on the left and use the add-zero trick to make it into a commutator with the $\hat{a}_2$ operators, and then we evaluate the n commutators. We do the same with the $\hat{a}_2$ operators on the right. In either case, if the power n or n' is too high, then those operators annihilate against the vacuum state, so the summations are limited to $n \leq j - m'$ and $n' \leq j - m$. Including the numerical factors from the commutators, we find that

$$\begin{aligned} d^{(j)}_{m'm}(\theta) = {} & \frac{1}{\sqrt{(j+m')!(j-m')!(j+m)!(j-m)!}} \sum_{n=0}^{j-m'} \sum_{n'=0}^{j-m} \frac{(-\tan\frac{\theta}{2})^n (\tan\frac{\theta}{2})^{n'}}{n!n'!} \\ & \times \langle 0| \left(\hat{a}_1\right)^{j+m'+n} \left(\hat{a}_2\right)^{j-m'-n} \frac{(j-m')!}{(j-m'-n)!} e^{\ln\left(\cos\frac{\theta}{2}\right)\left(\hat{a}_1^\dagger \hat{a}_1 - \hat{a}_2^\dagger \hat{a}_2\right)} \\ & \times \frac{(j-m)!}{(j-m-n')!} \left(\hat{a}_1^\dagger\right)^{j+m+n'} \left(\hat{a}_2^\dagger\right)^{j-m-n'} |0\rangle. \end{aligned} \tag{9.257}$$

The exponential of the logarithm of the cosine term now can be evaluated, because the state to the right has a definite number of excitations for both oscillators. It becomes $(\cos\frac{\theta}{2})^{2m+2n'}$. Then, we can evaluate the matrix element. Hence, we find

$$d^{(j)}_{m'm}(\theta) = \sqrt{\frac{(j-m')!(j-m)!}{(j+m')!(j+m)!}} \sum_{n=\max(0,m-m')}^{j-m'} \frac{\left(\sin\frac{\theta}{2}\right)^{m'-m+2n}\left(\cos\frac{\theta}{2}\right)^{m+m'}}{n!(m'-m+n)!} \times \frac{(j+m'+n)!}{(j-m'-n)!}(-1)^n. \tag{9.258}$$

This does not look anything like the result we had before for the rotation matrix. We need to massage the formula. Rearranging the trigonometric factors gives

$$d^{(j)}_{m'm}(\theta) = \sqrt{\frac{(j-m')!(j-m)!}{(j+m')!(j+m)!}} \sum_{n=\max(0,m-m')}^{j-m'} \frac{\left(\sin\frac{\theta}{2}\cos\frac{\theta}{2}\right)^{m'-m+2n}\left(\cos^2\frac{\theta}{2}\right)^{m-n}}{n!(m'-m+n)!} \times \frac{(j+m'+n)!}{(j-m'-n)!}(-1)^n. \tag{9.259}$$

Now, we use $2\sin\frac{\theta}{2}\cos\frac{\theta}{2} = \sin\theta$ and $2\cos^2\frac{\theta}{2} = 1+\cos\theta$. Hence,

$$d^{(j)}_{m'm}(\theta) = \sqrt{\frac{(j-m')!(j-m)!}{(j+m')!(j+m)!}} \sum_{n=\max(0,m-m')}^{j-m'} \frac{(\sin\theta)^{m'-m+2n}\,(1+\cos\theta)^{m-n}}{2^{m'+n}n!(m'-m+n)!} \times \frac{(j+m'+n)!}{(j-m'-n)!}(-1)^n. \tag{9.260}$$

Next, we use $\sin^2\theta/(1+\cos\theta) = (1-\cos^2\theta)/(1+\cos\theta) = 1-\cos\theta$ to give us

$$d^{(j)}_{m'm}(\theta) = \sqrt{\frac{(j-m')!(j-m)!}{(j+m')!(j+m)!}} \sum_{n=\max(0,m-m')}^{j-m'} \frac{(\sin\theta)^{m'-m}\,(1+\cos\theta)^{m}}{2^{m'+n}n!(m'-m+n)!} \times \frac{(j+m'+n)!}{(j-m'-n)!}(-1)^n(1-\cos\theta)^n. \tag{9.261}$$

Rearranging, we have

$$d^{(j)}_{m'm}(\theta) = \sqrt{\frac{(j+m)!(j-m)!}{(j+m')!(j-m')!}} \frac{1}{2^{m'}}\,(\sin\theta)^{m'-m}\,(1+\cos\theta)^{m}$$

$$\times \sum_{n=\max(0,m-m')}^{j-m'} \left(-\frac{1}{2}\right)^n (1-\cos\theta)^n \frac{(j-m')!}{n!(j-m'-n)!}$$
$$\times \frac{(j+m'+n)!}{(j+m)!(m'-m+n)!}. \tag{9.262}$$

This is an alternative form for the rotation matrix; it does not look like our previous formulas. One way to check this result is to compute the spherical harmonics from it. Setting $m' = 0$, and restricting $m \geq 0$, we find

$$Y_{lm}(\theta,\phi) = \sqrt{\frac{2l+1}{4\pi}} e^{im\phi} \sqrt{\frac{(j-m)!}{(j+m)!}} (\sin\theta)^m \sum_{n=m}^{j} \left(-\frac{1}{2}\right)^n (1-\cos\theta)^{n-m}$$
$$\times \frac{(j+n)!}{n!(n-m)!(j-n)!}. \tag{9.263}$$

Finally, we shift $n \to n+m$ to obtain the formula we had before for the spherical harmonics with $m \geq 0$ (expressed in terms of the associated Legendre function):

$$Y_{lm}(\theta,\phi) = \sqrt{\frac{2l+1}{4\pi}} (-1)^m \sqrt{\frac{(j-m)!}{(j+m)!}} \frac{1}{2^m} (\sin\theta)^m$$
$$\times \sum_{n=0}^{j-m} (-1)^n \frac{(j+m+n)!}{(j-m-n)!(m+n)!n!} \left(\frac{1-\cos\theta}{2}\right)^n e^{im\phi}. \tag{9.264}$$

This illustrates that the formula we have derived from the exponential disentangling identity is indeed correct even though the intermediate form looked different.

We end with an explicit construction of the parity operator, which we could not do before. We find $\hat{\Pi} = \exp\left[i\frac{\pi}{2}\left(\hat{a}_1^\dagger \hat{a}_1 + \hat{a}_2^\dagger \hat{a}_2\right)\right] = e^{i\frac{\pi}{2}\hat{N}}$ for integer j representations, which have an even number $(2j)$ of oscillator excitations. Note that we cannot define parity for half-odd-integer spins, because there is no scalar coordinate space wavefunction. Recall that $Y_{lm}(\theta,\phi)$ required us to compute the $m' = 0$ component of the rotation matrix $d^{(j)}_{m'm}(\theta)$. But half-odd-integer representations have no $m' = 0$ component. Hence, there are no coordinates θ and ϕ for us to take to $\pi - \theta$ and $\pi + \phi$. This means we cannot calculate the parity inverted wavefunctions for half-odd-integral angular momentum. This makes the parity operator defined above a well-defined operator for the integer spin representations only.

I hope you have found some new insight from looking at angular momentum with this new perspective. If you found the math exhausting, take a break and then come back and look at it again to try to see the "big picture." Schwinger discovered these results in 1952 and published them in a hard-to-find internal government report. They were republished 10 years later in a reprint volume and are now available in

book form Schwinger (2015). You should take a look. It is an interesting read, even today.

Section 9.8 Problems:

9.8.1 Hyperbolic Operators

Show the following commutation relations between the hyperbolic operators and the angular momentum operators:

$$[\hat{J}_{\pm}, \hat{K}_0] = 0, \quad [\hat{J}_z, \hat{K}_{\pm}] = 0, \quad \text{and} \quad [\hat{J}_z, \hat{K}_0] = 0. \tag{9.265}$$

Note that the remaining commutation relations, namely, $[\hat{K}_{\pm}, \hat{J}_{\pm}]$ involve operators that are not in the $\hat{K}$ or $\hat{J}$ sets. So, in general, the commutator algebra is not closed within these sets of operators.

9.8.2 Schwinger's Form of the Cartesian Angular-Momentum Operators

Using Schwinger's representation, construct the Cartesian operators $\hat{J}_x = \frac{1}{2}(\hat{J}_+ + \hat{J}_-)$ and $\hat{J}_y = \frac{1}{2i}(\hat{J}_+ - \hat{J}_-)$, and verify using the harmonic oscillator commutation relations, that we have

$$[\hat{J}_i, \hat{J}_j] = i\hbar \sum_k \epsilon_{ijk} \hat{J}_k. \tag{9.266}$$

9.8.3 Hadamard Lemma

Verify the identity given in Eq. (9.250).

9.8.4 Position and Momentum in Schwinger's Approach

Schwinger's representation includes all angular momentum operators, what about the position and momentum operators? Can we think of

$$\sqrt{\frac{\hbar}{2}}\left(\hat{a}_j+\hat{a}_j^\dagger\right) \text{ and } \sqrt{\frac{\hbar}{2}}\frac{1}{i}\left(\hat{a}_j-\hat{a}_j^\dagger\right), \tag{9.267}$$

as position and momentum operators, as we did for the conventional simple harmonic oscillator?

If we use the following mapping:

$$\begin{aligned}\hat{r}_z&=\sqrt{\frac{\hbar}{2}}\left(\hat{a}_1+\hat{a}_1^\dagger\right),\quad \hat{p}_z=\sqrt{\frac{\hbar}{2}}\frac{1}{i}\left(\hat{a}_1-\hat{a}_1^\dagger\right),\\ \hat{r}_x&=\sqrt{\frac{\hbar}{2}}\left(\hat{a}_2+\hat{a}_2^\dagger\right),\quad \hat{p}_x=\sqrt{\frac{\hbar}{2}}\frac{1}{i}\left(\hat{a}_2-\hat{a}_2^\dagger\right),\end{aligned} \tag{9.268}$$

then, compute $\hat{J}_y=\hat{r}_z\hat{p}_x-\hat{r}_x\hat{p}_z$ and compare it to the result of the Schwinger mapping, given by $\hat{J}_y=\frac{\hbar}{i}\left(\hat{a}_1^\dagger\hat{a}_2-\hat{a}_2^\dagger\hat{a}_1\right)$.

Unfortunately, there is no obvious combination to make $\hat{r}_x$ and $\hat{p}_y$. So, we cannot interpret the simple harmonic oscillators in terms of the three-dimensional positions and momenta. This is perhaps obvious, since we only have two oscillators, and we would expect we need three. So, this representation can describe angular momentum, the hyperbolic operators, parity, but not all the coordinates and momenta. However, if one works with three oscillators, then we can determine the three different position and momentum operators and all the angular momentum operators as well. This system has even more symmetry, namely, it has $SU(3)$ symmetry. We will discuss this system a bit more in Chap. 11.

9.8.5 Ladder Operators That Change j

Using the harmonic oscillator algebra, compute the constants in

$$\hat{K}_+|j,m\rangle=C_+^{j,m}|j+1,m\rangle \text{ and } \hat{K}_-|j,m\rangle=C_-^{j,m}|j-1,m\rangle. \tag{9.269}$$

These operators step between different j values. We don't often have a need to use these operators, because we do not often calculate matrix elements between states with different total squared angular momentum.

9.8.6 Parity of an Orbital Angular-Momentum State

Verify that

$$\hat{\Pi}|j,m\rangle=(-1)^j|j,m\rangle \tag{9.270}$$

for integer angular-momentum representations.

9.8.7 Three-Dimensional Matrix Representation

Using the three basis vectors in the harmonic oscillator $|n_1, n_2\rangle$ notation and given by $|2,0\rangle = \frac{1}{\sqrt{2}}\left(\hat{a}_1^\dagger\right)^2|0,0\rangle$, $|1,1\rangle = \hat{a}_1\hat{a}_2|0,0\rangle$, and $|0,2\rangle = \frac{1}{\sqrt{2}}\left(\hat{a}_2^\dagger\right)^2|0,0\rangle$, construct the matrices for $\hat{J}_\pm$, $\hat{J}_z$, $\hat{K}_\pm$, and $\hat{K}_0$.

Note that the matrices each have nine matrix elements in them of the form

$$\left(\hat{J}_+\right)_{m'm} = \langle m'|\hat{J}_+|m\rangle, \tag{9.271}$$

where $|m{=}-1\rangle = |2,0\rangle$, $|m{=}0\rangle = |1,1\rangle$, and $|m{=}1\rangle = |0,2\rangle$.

Do the calculation entirely in the harmonic oscillator basis—do not use the algebra of the J's or the K's. Think carefully about why your results are what they are, especially for the $\hat{K}_\pm$ matrices.

9.8.8 Hyperbolic Casimir Operator

Compute

$$\hat{\vec{K}}^2 = \frac{1}{2}\left(\hat{K}_+\hat{K}_- + \hat{K}_-\hat{K}_+\right) + \left(\hat{K}_0\right)^2, \tag{9.272}$$

in terms of $\hat{a}_1^\dagger$, $\hat{a}_1$, $\hat{a}_2^\dagger$, and $\hat{a}_2$. Compare your result to the results for $\hat{\vec{J}}^2$, and show that

$$\hat{\vec{K}}^2 - \hat{\vec{J}}^2 = \left(\hat{a}_1^\dagger\hat{a}_1 - \tfrac{1}{2}\right)\left(\hat{a}_2^\dagger\hat{a}_2 - \tfrac{1}{2}\right). \tag{9.273}$$

9.8.9 Matrix Element that Shifts j

Compute

$$\langle j, m|\hat{K}_-\hat{\vec{J}}^2\hat{K}_+|j, m\rangle, \tag{9.274}$$

from the simple harmonic oscillator representation. Explain why the answer you find is what it is.

9.8.10 The Other Form of the Exponential Disentangling Identity

Use the alternative form of the exponential disentangling identity to compute the spherical harmonics $Y_{lm}(\theta, \phi)$. That is, start from the identity with $\hat{J}_-$ to the right.

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Chapter 10
Ising and XY Models of Interacting Spins

I want to perceive and understand the hidden powers and laws of things, in order to have them in my power.

Salvador Dali, Artist

So far, we have primarily focused on the spin or angular momentum of an individual quantum particle. In this chapter, we examine the simplest models of spins interacting with each other. The Ising model was created by Wilhelm Lenz—he gave the problem to his Ph. D. student Ernst Ising, who solved the one-dimensional model. Hence, a statistical-mechanical model for a simplified model of quantum spins was solved *even before the quantum theory for spin was fully developed!* In this chapter, we will investigate a number of different spin models. They are all exactly solvable, but some are more complicated than others. One of the key techniques we will use is *fermionization*, which relates the spins to fermions. This will allow us to introduce the ideas of second quantization and of fermions. I am sure you will find it fun and interesting. A quick preview is also available at https://tinyurl.com/mur6hzsn.

10.1 Introduction to Lattices

Before we jump into the problem of the quantum mechanics of interacting spins, we start by providing an introduction to periodic lattices. We do this, because all of the spin models that we will work with lie on periodic lattices, and we need some basic information about them before we get started. This introduction will be rather brief and will focus on the simplest periodic lattice—the one-dimensional lattice.

In Sect. 7.4, we solved the problem of a particle moving on a circle. We discovered that if we used periodic boundary conditions, we could reexpress our solutions as states that were also momentum eigenstates. This is not surprising,

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Fig. 10.1 Schematic of a periodic lattice. The lattice is infinite in extent in both directions, but the lattice repeats after counting N lattice sites. In other words, we identify N with 0 and $N+1$ with 1 and -1 with $N-1$, and so on. The gray rectangle illustrates the set of N inequivalent lattice sites and is labeled by the N different equivalence classes from 0 to $N-1$

because the free-particle Hamiltonian is proportional to the square of the momentum operator, so every eigenstate of momentum is also an eigenstate of the free-particle Hamiltonian on a circle. In this section, we focus on lattices. They share a lot of properties with the particle on a circle, but there are differences too. On the circle, the particle could be found anywhere. On a lattice, the space is divided up into discrete locations, called lattice sites, and the particles spend their time in the vicinity of these lattice sites, unless they are able to move to other lattice sites. But, we describe the problem only with respect to these discrete, countable lattice sites (see Fig. 10.1). Doing so provides a number of simplifications that we will discuss in detail below.

Your first question might be, why in the world would anyone want to do this? The reason lies in the quantum mechanics of solids. Many materials condense into solids that form a regular lattice structure of ion cores (formerly atoms, but some electrons have been stripped off), and a sea of electrons that move through these ion-core potentials. You should think of this as similar to an egg carton (without the eggs) or a muffin tin (without the muffins). It is a periodic arrangement of the ion cores, through which the electrons move. In some cases, the electrons remain bound to the ion cores, and the solid is an insulator, with no free electrons moving around. If the atom has a spin degree of freedom though, we can describe the low-energy behavior via a model of spins at the lattice sites that can interact with their neighboring spins. This is the physical system we will study in this chapter. We need to understand some more properties of lattices first, and we will describe them in terms of a single quantum particle moving through the potential of the ion cores. In subsequent sections, we will work on the spin models in much more detail.

Since we work with the discretized lattice sites as our degree of freedom, let us first discuss the notation we will use. Suppose the spacing between the centers of each site is a. Then, a one-dimensional lattice can be described by lattice sites located at $R_j = aj$ for $j \in \mathbb{Z}$. If the lattice is infinite in size, all possible integers are allowed. But, we instead often describe finite-sized lattices, with N lattice sites. To make them "appear" to be infinite in size, we use periodic boundary conditions, just like we did on the circle, and we will describe shortly how this is modified for the discrete lattices. First, we discuss the notation, though.

Since the lattice sites are labeled by an integer, we label the quantum states localized at those sites by $|j\rangle$, with $0 \leq j \leq N-1$. What periodic boundary conditions means is the state $|j+N\rangle = |j\rangle$; that is, we identify the states according to the reduced index (by adding or subtracting multiples of N) until the index lies in the range $0 \leq j \leq N-1$. This is similar to what we did with the particle on

the circle. If you have heard of an equivalence class, this would be how the integers are divided into equivalence classes modulo N. One can see this is a discretized description of the states for a particle on a circle. What are the consequences of the periodic boundary conditions? Recall for the particle on a circle that the periodic boundary conditions produced a set of allowed momentum eigenvalues, given by $p_n = \frac{2\pi\hbar n}{L}$ for integers n. We define the wavenumbers k_n in an analogous fashion as $k_n = \frac{2\pi n}{aN}$ for the lattice with N sites. These wavenumbers satisfy the property that

$$e^{ik_n aN} = e^{2\pi i n} = 1. \tag{10.1}$$

This is the property we want for all allowed wavenumbers in the discretized lattice. It is solved by $k_n = \frac{2\pi n}{aN}$.

We now describe a periodic Hamiltonian that describes particles moving on the lattice. It is described by

$$\hat{H} = -t \sum_{j=0}^{N-1} (|j\rangle\langle j+1| + |j+1\rangle\langle j|), \tag{10.2}$$

where, again, the state $|N\rangle$ is identified with $|0\rangle$. The parameter t is called the *hopping matrix element*, and the Hamiltonian is called a *tight-binding Hamiltonian*. We have not discussed Hamiltonians of this form yet. Recall the basis states for the lattice are orthonormal, so $\langle i|j\rangle = \delta_{ij}$. This means that if we start in a state $|i\rangle$, then we find

$$\hat{H}|i\rangle = -t \sum_{j=0}^{N-1} (|j\rangle\langle j+1| + |j+1\rangle\langle j|)|i\rangle = -t|i-1\rangle - t|i+1\rangle. \tag{10.3}$$

That is, the Hamiltonian takes a particle that is originally at lattice site i and moves it to a state that is in a superposition of being on the site to the immediate left $|i-1\rangle$ and the site to the immediate right $|i+1\rangle$. If we let the Hamiltonian act again, we will obtain $t^2(|i-2\rangle + 2|i\rangle + |i+2\rangle)$ and so on. The state becomes an increasingly more complicated superposition as we apply the Hamiltonian again and again. Note further that the Hamiltonian is translationally invariant when we translate by a lattice spacing. In other words, we have the Hamiltonian transforms to

$$\begin{aligned}\hat{H} &\to -t \sum_{j=0}^{N-1} \big(|j+1\rangle\langle j+2| + |j+2\rangle\langle j+1|\big) \\ &= -t \sum_{j=1}^{N-1} \big(|j\rangle\langle j+1| + |j+1\rangle\langle j|\big) - t\big(|N\rangle\langle N+1| + |N+1\rangle\langle N|\big) = \hat{H},\end{aligned} \tag{10.4}$$

because the state $|N\rangle = |0\rangle$ and the state $|N+1\rangle = |1\rangle$.

Let us find the energy eigenstates. They must be expressible as a superposition of the lattice states, or $\sum_{j=0}^{N-1} c_j |j\rangle$. There is a theorem, called Bloch's theorem, which tells us that the states that have the coefficients given by $c_j = e^{ik_n aj}/\sqrt{N}$ are the eigenstates, because the Hamiltonian is translationally invariant. We label them by the wavenumber k_n as

$$|k_n\rangle = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} e^{ik_n aj} |j\rangle. \tag{10.5}$$

Before verifying that these are the energy eigenstates, we first verify another important property of these states—if we shift the index of each ket by one unit, we can express this as the original state multiplied by a complex phase determined by k_n. We check this first

$$\begin{aligned} |k_n\rangle_{|j\rangle \to |j+1\rangle} &= \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} e^{ik_n aj} |j+1\rangle \\ &= \frac{1}{\sqrt{N}} \left(e^{-ik_n a} \sum_{j=1}^{N-1} e^{ik_n aj} |j\rangle + e^{ik_n a(N-1)} |0\rangle \right) \\ &= e^{-ik_n a} |k_n\rangle. \end{aligned} \tag{10.6}$$

In this calculation, we first shift the index of each ket by one unit (but not the coefficient). This can be thought of as a translation operator by one unit to the right being applied to the state. Then, we shift the index in the summation to $j \to j-1$. Finally, we note that $e^{ik_n a(N-1)} = e^{-ik_n a}$. So, when the state is translated to the right by one unit, the state picks up a phase factor of $e^{-ik_n a}$. Similarly, if we shift to the left, then we have it picks up a phase factor of $e^{ik_n a}$. Okay. We can now use this information to show it is an energy eigenstate of the Hamiltonian. Recall, when $\hat{H}$ acts on this state, it creates a superposition that shifted the state to the right and to the left (and was multiplied by $-t$). Hence, we have

$$\hat{H}|k_n\rangle = -t \left(e^{ik_n a} |k_n\rangle + e^{-ik_n a} |k_n\rangle \right) = -2t \cos(k_n a) |k_n\rangle = E_n |k_n\rangle. \tag{10.7}$$

This means the energy eigenvalue for the state $|k_n\rangle$ is given by $E_n = -2t \cos(k_n a)$. This relation between the wavenumber and the energy is called the *bandstructure*. Note that if we take the limit where $N \to \infty$, then we find that $\epsilon(k) = -2t \cos(ka)$, where the symbol ϵ is used to denote the energy associated with the bandstructure.

Similar to how we can look at wavefunctions in position space and momentum space, when we discuss lattices, we can work in *real space*, which we have been

doing, or we can work in the so-called *reciprocal space*, where our orthonormal basis vectors are $\{|k_n\rangle | n = 0, \ldots, N-1\}$. First we need to verify that the states $|k_n\rangle$ are normalized:

$$\begin{aligned}\langle k_n|k_n\rangle &= \frac{1}{N}\sum_{j=0}^{N-1}\sum_{j'=0}^{N-1} e^{-ik_naj}e^{ik_naj'}\langle j|j'\rangle \\ &= \frac{1}{N}\sum_{j=0}^{N-1}\sum_{j'=0}^{N-1} e^{-ik_naj}e^{ik_naj'}\delta_{jj'} = \frac{1}{N}\sum_{j=0}^{N-1} 1 = 1. \end{aligned} \tag{10.8}$$

We also can verify orthogonality for $n \neq n'$:

$$\begin{aligned}\langle k_n|k_{n'}\rangle &= \frac{1}{N}\sum_{j=0}^{N-1}\sum_{j'=0}^{N-1} e^{-ik_naj}e^{ik_{n'}aj'}\langle j|j'\rangle = \frac{1}{N}\sum_{j=0}^{N-1}\sum_{j'=0}^{N-1} e^{-ik_naj}e^{ik_naj'}\delta_{jj'} \\ &= \frac{1}{N}\sum_{j=0}^{N-1} e^{-i(k_n-k_{n'})aj} = \frac{1}{N}\frac{1-e^{-i(k_n-k_{n'})aN}}{1-e^{-i(k_n-k_{n'})a}} = 0, \end{aligned} \tag{10.9}$$

where we used the finite summation formula in Eq. (3.4) from our unit on the geometric series. The result vanishes, because $e^{-i(k_n-k_{n'})aN} = e^{-2\pi i(n-n')N} = 1$ and $e^{-i(k_n-k_{n'})a} \neq 1$. Hence, we have verified that $\langle k_n|k_{n'}\rangle = \delta_{nn'}$. We also have proven the identity

$$\frac{1}{N}\sum_{j=0}^{N-1} e^{-i(k_n-k_{n'})aj} = \delta_{nn'}. \tag{10.10}$$

One can see that if $n = n'$, that the exponentials are all equal to 1, and the sum becomes N, which then becomes 1 when divided by N. If $n \neq n'$, then we have a sum over exponentials evenly spaced around the unit circle, which cancels and gives zero. This is how you can think of this identity.

Before we show how to transform the Hamiltonian into reciprocal space, we examine the analog of the wavefunction. We have

$$\langle j|k_n\rangle = \frac{1}{\sqrt{N}}\sum_{j'=0}^{N-1} e^{ik_naj'}\underbrace{\langle j|j'\rangle}_{\delta_{jj'}} = \frac{1}{\sqrt{N}}e^{ik_naj}. \tag{10.11}$$

If we recall that the wavefunction of a momentum eigenstate satisfied $\langle x|p\rangle = ce^{\frac{i}{\hbar}xp}$, we see that the discretized version is essentially the same, when we recall that the equivalent momentum is equal to $\hbar k_n$ and the equivalent position is $x = aj$. Now, to transform the Hamiltonian to reciprocal space, we use the resolution of the

identity operator on both sides and then the results from Eq. (10.11) to find that

$$\begin{aligned}
\hat{H} &= \sum_{n=0}^{N-1}\sum_{n'=0}^{N-1} |k_n\rangle \langle k_n|\hat{H}|k_{n'}\rangle \langle k_{n'}| \\
&= -t\sum_{n=0}^{N-1}\sum_{n'=0}^{N-1}\sum_{j=0}^{N-1} |k_n\rangle \langle k_n|\big(|j\rangle \langle j+1| + |j+1\rangle \langle j|\big)|k_{n'}\rangle \langle k_{n'}| \\
&= -\frac{t}{N}\sum_{n=0}^{N-1}\sum_{n'=0}^{N-1}\sum_{j=0}^{N-1} |k_n\rangle \left(e^{-ik_n aj}e^{ik_{n'}(j+1)} + e^{-ik_n a(j+1)}e^{ik_{n'}aj}\right) \langle k_{n'}| \\
&= -t\sum_{n=0}^{N-1}\sum_{n'=0}^{N-1} |k_n\rangle \langle k_{n'}| \left(e^{ik_{n'}a} + e^{-ik_n a}\right) \underbrace{\frac{1}{N}\sum_{j=0}^{N-1} e^{i(k_{n'}-k_n)aj}}_{\delta_{nn'}} \\
&= \sum_{n=0}^{N-1}(-2t\cos k_n a)\,|k_n\rangle \langle k_n|, \qquad (10.12)
\end{aligned}$$

where the sum over n' simply sets $n' = n$ due to the $\delta_{nn'}$ factor. So, you can see that the Hamiltonian becomes diagonal in reciprocal space. This means the reciprocal space basis is the energy eigenfunction basis for a particle moving on a lattice.

Note that in this section we focused on the absolutely simplest lattice—the one-dimensional periodic lattice. All of this work can be generalized to higher dimensions, but some of the generalizations require additional technical sophistication, so we do not go into those details here. Instead, we launch right into the problems.

Section 10.1 Problems:

10.1.1 Energy Eigenvalues for Hopping on a Lattice

Compute and then plot energy levels for the hopping Hamiltonian for the allowed k values for the cases with $N = 2, 4, 6, 8$. You should see that if two of the wavevectors satisfy $k_n a = -k'_n a \mod 2\pi$, then those energy levels are degenerate. You should note that for general wavevectors, the energies come in pairs, but the extreme energy levels are nondegenerate. Determine these extreme wavevectors. Does the number of extreme wavevectors change for even versus odd N?

10.1.2 Completeness of the Real and Reciprocal Space Bases

The Hilbert space we work with is finite dimensional, because we have N states. The real space basis $|j\rangle$ (with $0 \leq j \leq N-1$) forms a complete set, such that

$$\mathbb{I}_N = \sum_{j=0}^{N-1} |j\rangle \langle j|. \tag{10.13}$$

We now check whether $\sum_{n=0}^{N-1} |k_n\rangle \langle k_n|$ is the identity operator. Use the relationship between the reciprocal-space basis vectors and the real-space basis vectors to show that the sum over reciprocal ket-bra products is equal to the sum over real ket-bra products.

10.1.3 How to Determine if a Hamiltonian Is Periodic

In the text, we showed that the nearest-neighbor hopping Hamiltonian was unchanged, if we shifted the indices by one step to the right. In general, a Hamiltonian is *periodic* if it can be decomposed into a sum $\hat{H} = \sum_j \hat{H}_j$, with each $\hat{H}_j$ isomorphic to each other. For example, in the nearest-neighbor hopping model, we have $\hat{H}_j = -\frac{t}{2}\big(|j\rangle \langle j+1| + |j+1\rangle \langle j| + |j-1\rangle \langle j| + |j\rangle \langle j-1|\big)$. Write a short essay that describes a procedure that can be used to identify how we determine the decomposed pieces $\hat{H}_i$ for a Hamiltonian to see if the Hamiltonian is periodic.

10.1.4 Open Boundary Conditions and Allowed Wavevectors

We can also solve the problem with open boundary conditions, but it is a bit complicated, so we talk you through how to do it. We now have states $|j\rangle$ with $0 \leq j \leq N+1$, but the coefficient of the $|0\rangle$ and $|N+1\rangle$ states must both vanish in any wavefunction. So, we have

$$\hat{H} = -t \sum_{j=0}^{N} \big(|j\rangle \langle j+1| + |j+1\rangle \langle j|\big). \tag{10.14}$$

Note that there are no terms that connect $N+1$ to 0, so the Hamiltonian is on a line, not a ring. Then the claim is

$$|\psi_k\rangle = \sum_{j=1}^{N} \sin(kaj)|j\rangle \text{ with } ka(N+1) = n\pi. \tag{10.15}$$

Your first challenge is to show that $|\psi_k\rangle$ is an energy eigenstate with $E_k = -2t\cos(ka)$ for $k = \frac{\pi n}{a(N+1)}$ for $n = 1, 2, \ldots, N$.

Note that establishing this result is much more complicated than for the periodic case. You will need to use the fact that $\sin\big(ka(N+1)\big) = 0$ to handle the boundary terms properly.

10.1.5 Open Boundary Conditions and Energy Eigenvalues

Taking the solutions from the previous problem, plot the energy eigenvalues for $N = 4$, 5, 6, 7, and 8. Discuss the degeneracy of the energy levels for the open boundary conditions case.

10.2 Introduction to the Ising Model

In 1920, a German physicist, Wilhelm Lenz, introduced a model of spins interacting with neighboring spins on a lattice (Lenz, 1920). No one paid much attention to it. He then assigned it to one of his Ph.D. students, Ernst Ising, with the problem being to show how these spins become ferromagnetically oriented (all aligned in the same direction) at low temperature for a one-dimensional chain; see Fig. 10.2 for images of these two. This problem was important, because independently both Niels Bohr (1911) and Hendrika Johanna Van Leeuwen (1921) proved that no permanent magnets exist in a classical world. Ising, however, failed in showing the phase transition. He instead published results showing there was no permanent ferromagnetism (Ising, 1925). The story might have ended there, except that Rudolph Peierls proved that in two dimensions the Ising model does become a permanent magnet (more precisely a permanent ferromagnet) in 1936 (Peierls, 1936). This was followed by an exact solution for the transition temperature in 1944 by Lars Onsager (1944) and for the spontaneous magnetization (how strong the magnet is as a function of temperature below the transition temperature) in 1949 (Onsager, 1949) (with all details published by C. N. Yang in 1952). All of this work is focused on the statistical mechanics of the Ising model. In this section, we are interested in and focus on the quantum-mechanical properties.

In many respects, the Ising model is the simplest quantum model, and it forms a bridge between the quantum and classical worlds. This is because the spins of the Ising model all commute with each other, so they appear as classical degrees of freedom do (although when we consider the statistical mechanics, we use the discrete quantum energy eigenvalues in computing the thermodynamic properties). But, we will see in Chap. 14 that the dynamics of the model can be fully quantum in nature, and hence, much interest remains in the model and its quantum behavior, even today.

Fig. 10.2 Images of Ernst Ising in the early 1930s and Wilhelm Lenz in the early 1920s. (The Ising image is a gift of Jane Ising and courtesy of the Emilo Segrè Visual Archive and the Lenz image is from Prof. Wilhelm Westfall and courtesy of the Emilo Segrè Visual Archive. Both have been converted to look like sketches by Jim Freericks)

We will be tackling the Ising model and its variants many times in the remainder of this book. Here, in our first encounter, we will focus on the symmetries and on the eigenvalues and eigenstates of the model. Later, when we develop how quantum systems evolve in time, we will examine the dynamics and some of the contemporary studies of this model, which remain exciting research problems today. Unfortunately, we will not be able to spend much time on the thermodynamics of this model, so we will not be able to discuss the phase-transition temperature or the spontaneous magnetization, because those topics require a full development of quantum statistical mechanics, which we will not pursue in this book.

Enough with all the preliminaries. Let us dive in and define the Ising model Hamiltonian. We begin with a description in terms of the Pauli spin matrices from Chap. 6, and we write

$$\hat{H}_{\text{Ising}} = -\sum_{i=1}^{N}\sum_{j=1}^{N} J_{ij}\sigma_i^z\sigma_j^z, \tag{10.16}$$

where we now use a superscript to denote which Pauli spin matrix we use (here it is the z-matrix) and we use a subscript to denote the lattice site where the spin is located (there are N lattice sites in total). The symbol J_{ij}, with dimensions of energy, is the spin-exchange coefficient, which couples the spins at sites i and j together. For a Hermitian Hamiltonian, we require the J_{ij} to be real and to satisfy

$J_{ij} = J_{ji}$. Hence, it is a real and symmetric matrix. This Hamiltonian can be thought of as describing a series of 1's and -1's distributed on lattice sites, with connections or "bonds" between any two lattice sites where $J_{ij} \neq 0$. In such a picture, we are working in the customary basis where the angular momentum eigenstates are expressed along the z-axis. Sometimes, it is convenient to transform from the Pauli spin matrix picture to a real spin picture. Then, we use the identity $\hat{S}_i^z = \frac{\hbar}{2}\sigma_i^z$ and $\tilde{J}_{ij} = \frac{4}{\hbar^2} J_{ij}$ to obtain

$$\hat{\tilde{H}}_{\text{Ising}} = -\sum_{i=1}^{N}\sum_{j=1}^{N} \tilde{J}_{ij}\hat{S}_i^z\hat{S}_j^z. \tag{10.17}$$

See Fig. 10.3 for a graphical depiction of the Ising model, illustrating different bonds.

The first question you might ask is what does this Ising model have to do with real magnets that I have seen before? There are a number of magnets made from rare earths elements, such as samarium or yttrium. These elements have localized electrons (from their f-electrons) that can act like little bar magnets and are described by a local spin variable at each lattice site of the crystal that is occupied by a rare-earth element. In many cases, the bar magnets have spin components possible in more than one direction and hence are described by more complicated models

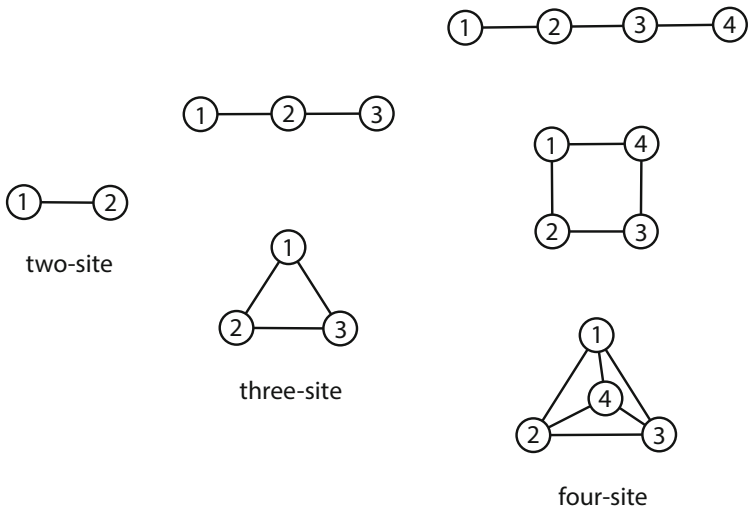

Fig. 10.3 Examples of graphs representing lattice sites (numbered circles) for two-site, three-site, and four-site lattices, with bonds indicated by solid lines that connect two lattice sites together when $J_{ij} \neq 0$. Note the word "bond" is a jargon for these models and simply means that site i and site j have a bond between them if and only if $J_{ij} \neq 0$. It does not mean they form a molecule in the chemistry sense

than the Ising model (we discuss some of these models later). But if the spins are highly anisotropic, then the Ising model is the best model for their behavior. Some examples of materials that behave like the Ising model are complicated materials like $Dy(C_2H_5SO_4)_3(H_2O)$ and $LiHOF_4$. We show some simple graphs with nearest-neighbor spin interactions only in Fig. 10.3.

Our next step is to examine the symmetry of this model. Recall that symmetries arise from operators that commute with the Hamiltonian. There are many symmetries. As an example, the operator σ_i^z (or $\hat{S}_i^z$) commutes with $\hat{H}$ for all i. This means that the total z-component of spin

$$\sigma^z = \sum_{i=1}^{N} \sigma_i^z \quad \text{or} \quad \hat{S}^z = \sum_{i=1}^{N} \hat{S}_i^z, \tag{10.18}$$

commutes with $\hat{H}_{\text{Ising}}$, so we can simultaneously diagonalize both. It turns out that states of definite $\hat{S}^z$ are eigenstates of $\hat{H}_{\text{Ising}}$ with energy given by

$$E_\psi = -\sum_{i=1}^{N}\sum_{j=1}^{N} \tilde{J}_{ij} S_i^z S_j^z \quad \text{for} \quad |\psi\rangle = |S_1^z, S_2^z, \ldots, S_N^z\rangle, \tag{10.19}$$

and $S_i^z = \pm\frac{\hbar}{2}$. We prefer to use a shorthand and write $E_\psi = -\sum_{i=1}^{N}\sum_{j=1}^{N} J_{ij}\sigma_i\sigma_j$, with $|\psi\rangle = |\sigma_1, \sigma_2 \cdots, \sigma_N\rangle$ and $\sigma_i = \uparrow \ (1)$ or $\downarrow \ (-1)$, when we calculate the energy. It can be pretty disconcerting reconciling these different notations at first, but you get the hang of how to do it pretty quickly. Note that $|\psi\rangle$ is automatically an eigenstate of $\hat{S}^z$ (or σ^z) since $\hat{S}^z|\psi\rangle = \left(\sum_{i=1}^{N} S_i^z\right)|\psi\rangle$ and $\sigma^z|\psi\rangle = \left(\sum_{i=1}^{N} \sigma_i^z\right)|\psi\rangle$. As an example, consider the $N = 3$ nearest-neighbor Ising model, given by

$$\hat{H} = -2J_{12}\sigma_1\sigma_2 - 2J_{23}\sigma_2\sigma_3 - 2J_{13}\sigma_1\sigma_3. \tag{10.20}$$

Here, we have combined the terms with $i < j$ with those that have $i > j$. Then the state $|\uparrow\uparrow\uparrow\rangle$ has definite z-component of spin (with $m = \frac{3}{2}\hbar$) and has $E = -2J_{12} - 2J_{23} - 2J_{13}$. Similarly, the state $|\uparrow\downarrow\uparrow\rangle$ has energy $2J_{12} + 2J_{23} - 2J_{13}$. And so on, for other examples.

The Ising Hamiltonian has an even deeper symmetry to it, which is related to a unitary transformation $\hat{U} = e^{-i\theta\sigma^z}$ and

$$\hat{H}_{\text{Ising}} \to \hat{U}\hat{H}_{\text{Ising}}\hat{U}^\dagger = \hat{H}_{\text{Ising}}. \tag{10.21}$$

This follows immediately, since each σ_i^z commutes with $\hat{H}_{\text{Ising}}$ and the fact that $\hat{U}\hat{U}^\dagger = 1$. But if we perform the unitary transformation (rotation) on the other Pauli matrices, we must perform the full calculation. Recall from Sect. 6.3 that

$$\hat{U} = e^{-i\theta\sigma^z} = \prod_{j=1}^{N} e^{-i\theta\sigma_j^z} = \prod_{j=1}^{N} \left(\cos\theta\, \mathbb{I}_j - i\sin\theta\, \sigma_j^z\right). \tag{10.22}$$

Then

$$\begin{aligned}\hat{U}\sigma_j^x\hat{U}^\dagger &= \left(\cos\theta\, \mathbb{I}_j - i\sin\theta\, \sigma_j^z\right)\sigma_j^x\left(\cos\theta\, \mathbb{I}_j + i\sin\theta\, \sigma_j^z\right)\\ &= \cos^2\theta\, \sigma_j^x - i\sin\theta\, \cos\theta\left[\sigma_j^z, \sigma_j^x\right] + \sin^2\theta\, \sigma_j^z\sigma_j^x\sigma_j^z\\ &= \cos(2\theta)\, \sigma_j^x + \sin(2\theta)\, \sigma_j^y. \end{aligned}\tag{10.23}$$

Similarly,

$$\hat{U}\sigma_j^y\hat{U}^\dagger = -\sin(2\theta)\, \sigma_j^x + \cos(2\theta)\, \sigma_j^y, \tag{10.24}$$

which corresponds to a rotation about the z-axis in spin space by an angle of 2θ, as shown in Fig. 10.4.

We proved the spin-rotation identity explicitly for spin one-half. It turns out that the exponential of angular momentum operators generate rotations about the axis determined by the direction of those operators, as we already illustrated in Sect. 9.2. It is important to note as well that unitary transformations do not change the form of the commutators. For example, consider the rotated spin operators designated by the $'$ symbol:

$$\hat{S}^\alpha \to \hat{U}\hat{S}^\alpha\hat{U}^\dagger = \hat{S}'^\alpha. \tag{10.25}$$

Then,

$$\left[\hat{S}'^\alpha, \hat{S}'^\beta\right] = \left[\hat{U}\hat{S}^\alpha\hat{U}^\dagger, \hat{U}\hat{S}^\beta\hat{U}^\dagger\right]$$

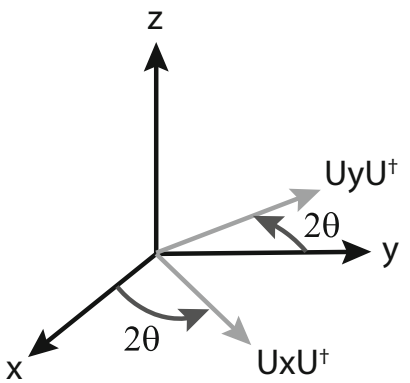

Fig. 10.4 Schematic that shows how the operator $\hat{U} = e^{-i\theta\sigma^z}$ produces a rotation by 2θ in the x-y plane with a unitary transformation $\sigma_i^\alpha \to \hat{U}\sigma_i^\alpha\hat{U}^\dagger = \sigma_i'^\alpha$

$$= \hat{U}\hat{S}^{\alpha}\underbrace{\hat{U}^{\dagger}\hat{U}}_{=\mathbb{I}}\hat{S}^{\beta}\hat{U}^{\dagger} - \hat{U}\hat{S}^{\beta}\underbrace{\hat{U}^{\dagger}\hat{U}}_{=\mathbb{I}}\hat{S}^{\alpha}\hat{U}^{\dagger} = \hat{U}\left[\hat{S}^{\alpha}, \hat{S}^{\beta}\right]\hat{U}^{\dagger}$$

$$= i\hbar\sum_{\gamma}\epsilon_{\alpha\beta\gamma}\hat{U}\hat{S}^{\gamma}\hat{U}^{\dagger} = i\hbar\sum_{\gamma}\epsilon_{\alpha\beta\gamma}\hat{S}'^{\gamma}. \tag{10.26}$$

So, the angular momentum commutation relations are preserved. This is a universal property of unitary transformations (it actually holds for the more general similarity transformations). You will explore this in another way in the problems.

The Ising model Hamiltonian has another symmetry, namely, a *spin-reflection parity* symmetry, where we transform the spin as follows:

$$\hat{S}^{x} \to \hat{S}^{x}, \quad \hat{S}^{y} \to -\hat{S}^{y}, \quad \text{and} \quad \hat{S}^{z} \to -\hat{S}^{z}. \tag{10.27}$$

If you look carefully at this transformation, it is a spin rotation by $\theta = \frac{\pi}{2}$ about the x-axis and given by

$$\hat{U}_{sr} = e^{-i\frac{\pi}{2}\sigma^{x}}. \tag{10.28}$$

We can immediately see that the Hamiltonian is unchanged by this transformation, because changing the sign of all z-component spin operators does not change the Hamiltonian, since it is quadratic in these spin operators. As we examine generalizations of the Ising model, we will see some of these symmetries remain, while some are lost.

The spin-reflection parity symmetry maps a state with z-component of spin $\frac{\hbar}{2}m$ into one with z-component of spin $-\frac{\hbar}{2}m$. Suppose a state $|\psi\rangle$ with z-component eigenvalue m also has an energy eigenvalue $E(m)$ under $\hat{H}_{\text{Ising}}$, so that

$$\hat{H}_{\text{Ising}}|\psi(m)\rangle = E(m)|\psi(m)\rangle; \tag{10.29}$$

note that the energy can depend on more than just the eigenvalue m, but we have not explicitly shown such a dependence here. Multiply both sides of the equation by $\hat{U}_{sr}$, and introduce $\mathbb{I} = \hat{U}_{sr}^{\dagger}\hat{U}_{sr}$ between $\hat{H}_{\text{Ising}}$ and $|\psi(m)\rangle$. This yields

$$\hat{U}_{sr}\hat{H}_{\text{Ising}}\hat{U}_{sr}^{\dagger}\hat{U}_{sr}|\psi(m)\rangle = E(m)\hat{U}_{sr}|\psi(m)\rangle$$

$$\hat{H}_{\text{Ising}}\hat{U}_{sr}|\psi(m)\rangle = E(m)\hat{U}_{sr}|\psi(m)\rangle. \tag{10.30}$$

Look carefully at this. What it says is that the state $\hat{U}_{sr}|\psi(m)\rangle$ has the same energy eigenvalue as the state $|\psi(m)\rangle$. But $\hat{U}_{sr}$ will flip all spins $\uparrow \to \downarrow$ and $\downarrow \to \uparrow$. Hence, $\hat{U}_{sr}|\psi(m)\rangle = |\psi(-m)\rangle$. This means that the Ising model always has degenerate energy eigenvalues. This even occurs when $m = 0$, because the state $\hat{U}_{sr}|\psi(m{=}0)\rangle \neq |\psi(m{=}0)\rangle$ after we perform the transformation for the product-state basis we have been using.

So, why do we call it parity? Because we can form a linear combination of definite parity states given by

$$|\psi_+\rangle = \frac{1}{\sqrt{2}}\left(|\psi(m)\rangle + \hat{U}_{sr}|\psi(m)\rangle\right) \text{ and}$$
$$|\psi_-\rangle = \frac{1}{\sqrt{2}}\left(|\psi(m)\rangle - \hat{U}_{sr}|\psi(m)\rangle\right), \tag{10.31}$$

because $\left(\hat{U}_{sr}\right)^2 = \mathbb{I}$, as shown in the problems. Hence, the energy eigenstates can be combined to form states that are either even or odd under the spin-reflection parity. But note that we cannot simultaneously diagonalize $\hat{U}_{sr}$ and $\hat{S}_z$, because they do not commute). Let us look at an example. If I have the state $|\uparrow\uparrow\uparrow\rangle$, then $\hat{U}_{sr}|\uparrow\uparrow\uparrow\rangle = |\downarrow\downarrow\downarrow\rangle$, so the even and odd parity states are then $\frac{1}{\sqrt{2}}(|\uparrow\uparrow\uparrow\rangle \pm |\downarrow\downarrow\downarrow\rangle)$. One can immediately see that these states are eigenstates under the $\hat{U}_{sr}$ operator. The energy eigenstates are always at least twofold degenerate, so we can always have even parity and odd parity energy eigenstates by forming these linear combinations.

We have now fully characterized the general symmetries of $\hat{H}_{\text{Ising}}$ and the implications for degenerate energy eigenstates and their spin-reflection parity. In some cases, there is additional spatial symmetry, if the lattice can be rotated or reflected back to itself (this requires the J_{ij} values to be the same for bonds that are rotated or reflected to each other; we will discuss these symmetries later). It is now time to discuss the properties of the ground state in general. We will see some surprises lay ahead of us, even for this simple model.

We begin our discussion of eigenstates by considering the ferromagnetic Ising model, where $J_{ij} \geq 0$ for all i and j. Then, the energy for the state $|\psi\rangle = |\sigma_1, \sigma_2, \ldots, \sigma_N\rangle$ is given by $E(\psi) = -\sum_{ij} J_{ij}\sigma_i\sigma_j$. Note that the energy is minimized if all σ_i's are chosen to be the same sign, so $|\uparrow\uparrow\cdots\uparrow\rangle$ and $|\downarrow\downarrow\cdots\downarrow\rangle$ are both energy eigenstates with the lowest possible energy. This optimization occurs, because any spin flip will change the sign of the contribution of some of the J_{ij} terms and increase the energy. You might be surprised that the energy is negative. This is possible, and when it occurs, then the ground state is the most negative energy state. These degenerate ground states have all of the spins aligned up, or all aligned down, which is called the *ferromagnetic state*. The other case that we can immediately find the ground state is the case where the lattice is *bipartite*, which means it separates into an A sublattice and a B sublattice, such that $J_{ij} \neq 0$ only when i and j belong to opposite sublattices. See Fig. 10.5 for an example of a bipartite lattice. In addition, $J_{ij} < 0$ is required for us to find the ground state. We find a state called the *Néel state*, which has all the up spins on the A sublattice and all the down spins on the B sublattice (or the opposite state). This state minimizes the energy. We denote these two states as $|\uparrow\downarrow\uparrow\cdots\downarrow\rangle$ and $|\downarrow\uparrow\downarrow\cdots\uparrow\rangle$. It is critical that the exchange constants J_{ij} be nonpositive for the antiferromagnetic state to be the ground state.

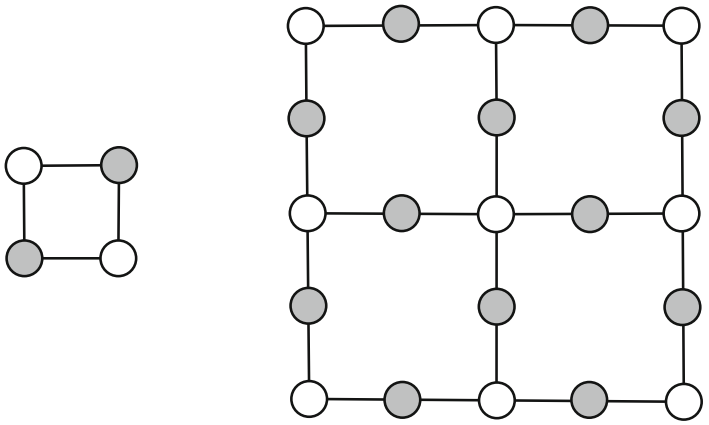

Fig. 10.5 Example of bipartite lattices. On the left, we have a bipartite lattice with the A sublattice in white and the B sublattice in gray. The number of sites is the same on each sublattice. On the right, we have another bipartite lattice where the number of lattice sites in each sublattice is not the same (such a lattice is called a Lieb lattice). The lines signify nonzero J_{ij}'s, and they are always connected between white and gray sites only. Both lattices could be continued into infinite-sized lattices by extending them in the plane

It should come as no surprise to you that the reason why we can find an exact ground state for both the ferromagnetic and antiferromagentic cases is arising from an underlying symmetry. This symmetry is called a *partial spin-parity transition*—we apply the spin-parity transformation only for $i \in A$ sublattice only (and we assume we have a case where the lattice is bipartite). This then transforms the ferromagnetic Hamiltonian to the antiferromagnetic Hamiltonian:

$$-\sum_{i=1}^{N}\sum_{j=1}^{N} J_{ij}\sigma_i^z\sigma_j^z \rightarrow \sum_{i=1}^{N}\sum_{j=1}^{N} J_{ij}\sigma_i^z\sigma_j^z, \tag{10.32}$$

via $H_{\text{Ising}}^{AF} = \hat{U}_{sr}(A)\hat{H}_{\text{Ising}}^{F}\hat{U}_{sr}^{\dagger}(A)$, where the A label indicates we apply the spin-reflection parity operation only to the A sublattice. This changes the sign of the spin exchange because $J_{ij} \neq 0$ only when i and j are on opposite sublattices. The transformation is a unitary transformation, which means the eigenvalues for the ferromagnetic and for the antiferromagnetic cases *are exactly the same* (but the eigenstates are not). This partial spin-reflection transformation allows us to relate both eigenstates and eigenvalues between the ferromagnetic and the antiferromagnetic cases.

It turns out that these are the only cases where the ground state can be found easily. An open problem is the so-called *spin-glass* problem, where J_{ij} are chosen at random, with a distribution that allows both positive and negative values. It then becomes a very hard problem to find the minimal energy state. This may be surprising to you, because we do know all the eigenstates (which are just the product

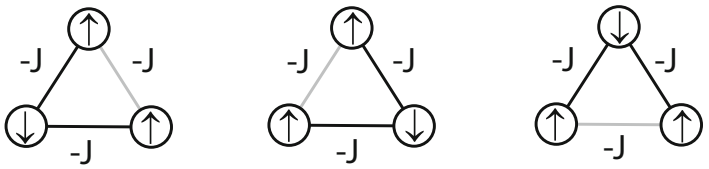

Fig. 10.6 Three different illustrations of frustration. A lattice with triangles that has antiferromagnetic exchange integrals ($J < 0$) will exhibit frustration. There is no way to place spins on the lattice that have antiferromagnetic interactions between all three pairs of spins (the spins on the black bonds are antiferromagnetic, but the spins on the gray one are ferromagnetic). This leads to the phenomenon of *frustration*, where one often also finds a degeneracy between different spin configurations. Here, the two spin configurations will have the same energy and hence be degenerate

states $|\sigma_1, \sigma_2, \ldots, \sigma_N\rangle$). But, there are 2^N of them, which becomes a huge number of states even for moderately sized N. The reason the minimization is difficult is due to a phenomenon called *frustration*, illustrated in Fig. 10.6, where for any odd numbered ring, with uniform negative spin exchange, we cannot choose between the three configurations, which are degenerate in energy. Having a system with large amounts of frustration makes it quite difficult to find the minimum energy state.

One other interesting phenomena in one-dimensional antiferromagnets is the phenomenon of *fractionalization* of excitations. What we mean by this is the excitation given by a defect becomes two "half defects" as it is made larger and larger. This phenomena typically occurs only in one-dimensional lattices. Consider an antiferromagnetic Ising model, with all spin exchange terms equal to $J < 0$ for nearest neighbors, i and $i \pm 1$, and $J_{ij} = 0$ otherwise. Next, take one of the Néel states and flip the spin in the center. This forms a defect, which has two bonds changed, as emphasized in the figure. The energy cost is given by $4|J|$ to do this, due to the two bonds that are "broken" due to the spin flip and the fact that the bond interaction appears twice in the summation. Now, flip additional spins, in turn, moving outward from the center. You can clearly see that the excitation is now split in half—one part is to the left and one is to the right. This separation, which can be made as large as desired and has only one bond broken, is called a fractional excitation. It is illustrated graphically in Fig. 10.7. As you will see in the problems, the fractionalization is primarily a phenomenon that occurs in one-dimensional chains. In higher dimensions, one instead creates a string of excitations as the spin excitations are separated, and this leads to a phenomenon called *confinement*, which occurs with quarks. The quarks are always bound in pairs to form a meson or in triples to form baryons. The idea of a potential energy increasing linearly as a string is a common feature for quark confinement, and this behavior is most easily seen (in analogy) in these higher-dimensional spin models.

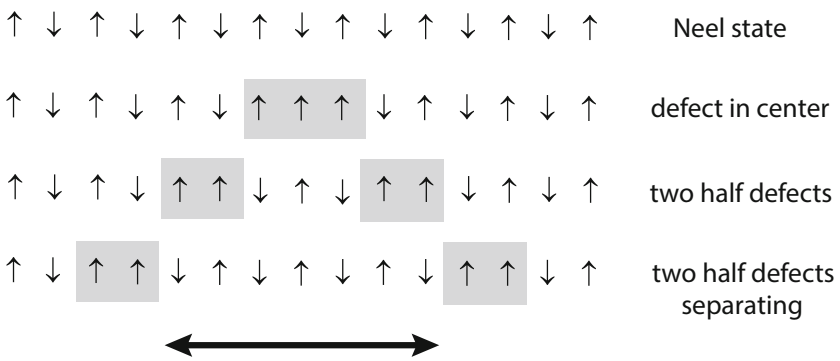

Fig. 10.7 Fractional spin excitations. Note how the fractional spin excitation (one broken bond) can separate with no extra bonds being broken (in one dimension). This does not occur in two dimensions. The top line shows the Nèel state. The next line flips two spins at each boundary, "breaking" two bonds. As we flip spins to move the defects apart, we create two half defects, which separate further as we move them outward

Section 10.2 Problems:

10.2.1 Commutators of Spin Operators After a Rotation

Suppose we have spin operators that satisfy $\left[\hat{S}^{\alpha}, \hat{S}^{\beta}\right] = i\hbar \sum_{\gamma} \epsilon_{\alpha\beta\gamma} \hat{S}^{\gamma}$. We transform the operators via a rotation by $\frac{\theta}{2}$ about the z-axis, so that

$$\begin{aligned} \hat{S}^{x} &\to \cos\theta\, \hat{S}^{x} + \sin\theta\, \hat{S}^{y} = \hat{S}'^{x}, \\ \hat{S}^{y} &\to -\sin\theta\, \hat{S}^{x} + \cos\theta\, \hat{S}^{y} = \hat{S}'^{y}, \quad \text{and} \\ \hat{S}^{z} &\to \hat{S}^{z} = \hat{S}'^{z}. \end{aligned} \tag{10.33}$$

Show, via direct computation, that $\left[\hat{S}'^{\alpha}, \hat{S}'^{\beta}\right] = i\hbar \sum_{\gamma} \epsilon_{\alpha\beta\gamma} \hat{S}'^{\gamma}$.

10.2.2 Spin-Reflection Parity Operator Defines an Involution

The spin-reflection parity operator corresponds to $\hat{S}^{x} \to \hat{S}^{x}$, $\hat{S}^{y} \to -\hat{S}^{y}$, and $\hat{S}^{z} \to -\hat{S}^{z}$. Show that this transformation is an involution, in the sense that if the transformation is applied twice, the operators map to themselves, and the states map to themselves.

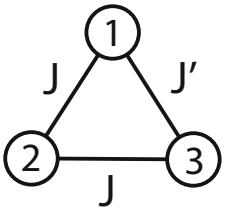

Fig. 10.8 Three-site lattice. Two bonds are J and one is J'

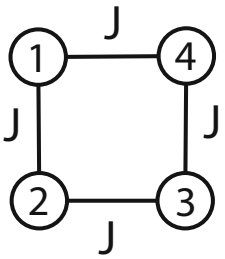

Fig. 10.9 Four-site lattice. All bonds are equal strength and equal to J

10.2.3 Exact Diagonalization of a Three-Site Ising Model

Consider the three-site Ising model depicted in the Fig. 10.8

$$\hat{H}_{\text{Ising}} = -J\sigma_1^z\sigma_2^z - J\sigma_2^z\sigma_3^z - J'\sigma_1^z\sigma_3^z. \tag{10.34}$$

There are eight energy eigenstates of the form $|\sigma_1, \sigma_2, \sigma_3\rangle$ with $\sigma_i = \pm 1$. Find the lowest energy states when $J > 0$ and $J' > 0$. Repeat for $J = J' < 0$. Note that the set of all energy eigenvalues for these two cases are *not* the same, because the lattice is not bipartite.

10.2.4 Exact Diagonalization of a Four-Site Ising Model

Consider the four-site cluster shown in the Fig. 10.9 with

$$\hat{H}_{\text{Ising}} = -J\sigma_1^z\sigma_2^z - J\sigma_2^z\sigma_3^z - J\sigma_3^z\sigma_4^z - J\sigma_4^z\sigma_1^z. \tag{10.35}$$

This has $2^4 = 16$ states. Find all energy eigenvalues, verifying their degeneracy, both for $J > 0$ and for $J < 0$. Note that the set of eigenvalues is the same for both cases, because this is a bipartite lattice.

10.2.5 Power-Law Interactions

Consider a one-dimensional chain that satisfies

$$J_{ij} = -\frac{J}{2|i-j|^{\alpha}}, \tag{10.36}$$

with N even and $i \neq j$. Show that the ground state remains the Nèel state if $\alpha > 1$ and $J > 0$.

10.2.6 Fluctuations of Spin in Definite Parity States

Consider an Ising model with $J_{ij} \geq 0$ for all i and j, which has $|\uparrow\uparrow\cdots\uparrow\rangle$ and $|\downarrow\downarrow\cdots\downarrow\rangle$ as the two ground states. Compute

$$\left(\Delta S^z\right)^2_{\psi} = \langle\psi|(\hat{S}^z)^2|\psi\rangle - \left(\langle\psi|\hat{S}^z|\psi\rangle\right)^2, \tag{10.37}$$

for $\hat{S}^z = \sum_{i=1}^{N} \hat{S}_i^z$ and $|\psi\rangle = \frac{1}{\sqrt{2}}(|\text{all up}\rangle \pm |\text{all down}\rangle)$. Compare those fluctuations to the fluctuations of $|\psi\rangle = |\text{all up}\rangle$ and $|\psi\rangle = |\text{all down}\rangle$.

10.2.7 Fractionalization in Two Dimensions

Consider a two-dimensional Ising model on a square lattice with nearest-neighbor antiferromagnetic interactions $J < 0$) and uniform for all bonds. Use pictures to show that a spin flip at the center no longer separates into a collection of two independent fractional excitations (as the defect is "pulled apart)." But, instead, it creates a string of broken bonds that remain and increase the energy as the spin excitations are separated along a line as we did in one dimension. This shows that fractional spin excitations are bound by a "string" of broken bonds in two dimensions. This simple picture can be used as a "bare-bones" model to describe the origin of quark confinement.

10.3 The Ising Model in a Longitudinal Field

We now add a longitudinal magnetic field (in the z-direction), so that

$$\hat{H}_{\text{Ising}} = -\sum_{i=1}^{N}\sum_{j=1}^{N} J_{ij}\sigma_i^z\sigma_j^z - \sum_{i=1}^{N} B_i\sigma_i^z. \tag{10.38}$$

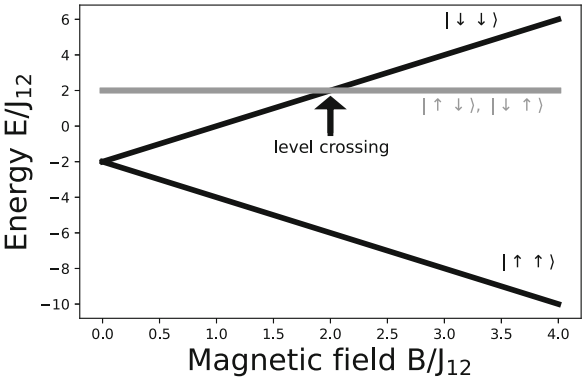

Fig. 10.10 Energy levels of the ferromagnetic Ising model in a longitudinal field with $N = 2$. The black lines are the two ferromagnetic states that are degenerate at $B = 0$ and split in opposite directions when the magnetic field is increased. The gray lines are the degenerate energy level for the other two states. The level crossing is marked where the black and gray lines cross

Now, we no longer have any spin-reflection parity, but the z-component of spin still commutes with $\hat{H}_{\text{Ising}}$. So the eigenstates remain the product states in the z-basis, but now there is no longer degeneracy between spin-reflection–parity partners. We say the degeneracy from the parity symmetry is *lifted* by the longitudinal field.

Let's see how this works for the two-site model (with $J_{12} = J_{21}$). We have

$$\hat{H}_{\text{Ising}} = -2J_{12}\sigma_1^z\sigma_2^z - B_1\sigma_1^z - B_2\sigma_2^z. \tag{10.39}$$

The eigenstates and energies are

$$\begin{aligned} &|\uparrow\uparrow\rangle : E_{\uparrow\uparrow} = -2J_{12} - B_1 - B_2; \quad |\uparrow\downarrow\rangle : \; E_{\uparrow\downarrow} = 2J_{12} - B_1 + B_2; \\ &|\downarrow\uparrow\rangle : E_{\downarrow\uparrow} = 2J_{12} + B_1 - B_2; \quad |\downarrow\downarrow\rangle : \; E_{\downarrow\downarrow} = -2J_{12} + B_1 + B_2. \end{aligned} \tag{10.40}$$

Assume $J_{12} > 0$ and $B_1 = B_2 > 0$. The energy levels are plotted in Fig. 10.10. Note that we often choose $J_{12} = \frac{J}{2}$ in order to get rid of extra factors of 2.

There are two interesting behaviors in the plot of these energy levels as a function of the magnetic field. The first is the level crossing, where the $m = -1$ state becomes degenerate with the $m = 0$ states at a particular value of B. Level crossings are allowed when eigenstates have different quantum numbers (here, the specific quantum number is for the operator $\hat{S}^z$), because eigenstates with different quantum numbers are not coupled together by $\hat{H}$. The proof of this is simple. Suppose $\hat{O}$ is the self-adjoint symmetry operator and $[\hat{O}, \hat{H}] = 0$. Then, if $|a\rangle$ is a joint eigenvector of both $\hat{O}$ and $\hat{H}$, with eigenvalues O_a and E_a, respectively (and similarly for $|b\rangle$ and O_b and E_b). Then, we find whenever $O_a - O_b \neq 0$, we have $(O_a - O_b)\langle a|\hat{H}|b\rangle = \langle a|\hat{O}\hat{H}|b\rangle - \langle a|\hat{H}\hat{O}|b\rangle$, which is equal to $\langle a|[\hat{O}, \hat{H}]|b\rangle$,

which vanishes because $[\hat{O}, \hat{H}] = 0$. But, this then implies that $\langle a|\hat{H}|b\rangle = 0$. Or, in other words, the states $|a\rangle$ and $|b\rangle$ are not coupled by $\hat{H}$. In fact, only the set of states that have the same eigenvalues under $\hat{O}$ can be coupled together via $\hat{H}$. This means the energy levels for different quantum numbers can cross (we already saw in Sect. 5.7 for 2×2 matrices that if two states are coupled, their energy eigenvalues are never degenerate; one of the problems below discusses this further. When these two uncoupled states do cross, we have what is called an *accidental degeneracy*. It occurs without there being an extra symmetry that requires it to occur.

Note, however, that $|\uparrow\downarrow\rangle$ and $|\downarrow\uparrow\rangle$ remain degenerate! This is the second interesting observation in the figure. The degeneracy of these two states is not lifted, because the eigenvalues are *independent of the magnetic field*. Hence, the degeneracy required at $B = 0$ continues for $B \neq 0$, because the eigenvalues are independent of the magnetic field. This actually occurs because $B_1 = B_2$, which allows an extra spatial symmetry in the problem (interchange the lattice sites $1 \leftrightarrow 2$ and the Hamiltonian is unchanged). We see this simply by letting $B_1 \neq B_2$ and noting that none of the energies are degenerate any more.

That is about it for the longitudinal-field Ising model—the main new result is that the symmetry is changed, so we can have level crossings with accidental degeneracies and that most spin-reflection-parity-based degeneracies are lifted unless we are in a special case, such as the extra spatial symmetry discussed above. This does not make this model too interesting, but in the cases where either the fields, or the spin-exchange constants, or both, are random, then, the problem of finding the lowest energy state becomes a complicated search problem. This is even an active field of research into the physics of glassy materials.

Section 10.3 Problems:

10.3.1 Three Sites in a Longitudinal Field

Consider the $N = 3$ Ising model with a longitudinal field that is on a triangular graph. We have $J_{12} = J_{21} = J_{23} = J_{32} = J_{13} = J_{31} = \frac{J}{2}$ and $B_1 = B_2 = B_3 = B$. Solve for the energy levels for arbitrary B when $J > 0$. Determine the accidental degeneracies and the states that do not have their degeneracy lifted. Repeat for $J < 0$. In both cases, plot the energy levels for the different states versus the magnetic field.

10.3.2 Four Sites in a Longitudinal Field

Compute all energy levels for the $N = 4$ Ising model with a longitudinal field on a square graph, meaning there are bonds only between the nearest-neighbors on the square. All nonzero J_{ij} are the same and equal to $\frac{J}{2}$. The magnetic field is also

uniform $B_i = B$. Consider both the case with $J > 0$ and $J < 0$. Identify all accidental degeneracies, level crossings, and degeneracies that are not lifted. Plot the results for the energy versus the magnetic field.

10.3.3 Fluctuations of the z-Component of Spin

Consider the spin fluctuations again:

$$\left(\Delta \hat{S}^z\right)^2 = \langle\psi|\left(\hat{S}^z\right)^2|\psi\rangle - \left(\langle\psi|\hat{S}^z|\psi\rangle\right)^2. \tag{10.41}$$

Look at the solutions for the Ising model in a longitudinal field for $N = 3$ and $N = 4$ above. Identify all states that can have a nonzero fluctuation for the z-component of spin. What is required of the state in order to maximize the fluctuation?

10.3.4 Avoided Crossing

Consider the two-state quantum system with a Hamiltonian given by

$$\hat{H} = \begin{pmatrix} \epsilon_0 & \Delta \\ \Delta & -\epsilon_0 \end{pmatrix}. \tag{10.42}$$

Show that as long as the matrix element coupling the two states, given by Δ, is nonzero, then the two energy levels never cross for $-\infty < \epsilon_0 < \infty$. This is called an *avoided crossing*. Plot the two energy levels as a function of ϵ_0. Determine the separation at the value for ϵ_0 where the separation is minimal.

10.3.5 A Simple Randomly Coupled Ising Model

Consider the $N = 3$ model on a triangle. Choose $J_{ij} = J_{ji}$ and $J_{12} = -0.7$, $J_{23} = -0.8$, and $J_{31} = -0.1$. Let $B_1 = 0.5B_2 = 0.7B_3$. As B_1 varies, identify the ground state of this model for all B_1. This is a simple random coupling and random field Ising model.

10.4 Spins, Fermions, and the Jordan-Wigner Transformation

So far, we have focused on the Ising model, where every term in the Hamiltonian commutes with any other term in the Hamiltonian (this holds also for the Ising model in a longitudinal field). The next most complex set of spin models is called XY models, which are of the form

$$\hat{H}_{\mathrm{XY}} = -\frac{4J}{\hbar^2}\sum_{i=1}^{N}\left(\hat{S}_i^x\hat{S}_{i+1}^x + \hat{S}_i^y\hat{S}_{i+1}^y\right) = -J\sum_{i=1}^{N}\left(\sigma_i^x\sigma_{i+1}^x + \sigma_i^y\sigma_{i+1}^y\right), \tag{10.43}$$

where we are working with the *isotropic* version of this model. One has to specify the boundary conditions, because the summation has terms with $i = N+1$. For open boundary conditions, we set any operator with an index of $N+1$ equal to 0. For periodic boundary conditions, we set an operator with index $N+1$ equal to the same operator with index 1. If we reexpress the XY model in terms of the raising and lowering operators, given by $\sigma_i^x = \frac{1}{2}(\sigma_i^+ + \sigma_i^-)$ and $\sigma_i^y = \frac{1}{2i}(\sigma_i^+ - \sigma_i^-)$, then we find that

$$\hat{H}_{\mathrm{XY}} = -\frac{J}{2}\sum_{i=1^N}\left(\sigma_i^+\sigma_{i+1}^- + \sigma_i^-\sigma_{i+1}^+\right), \tag{10.44}$$

where we have chosen the spins to be interacting on a one-dimensional lattice. We will be working with periodic boundary conditions.

In order to be concrete and ensure you understand what this model is, we will focus on the $N = 2$ model and construct its matrix representation concretely. When we use periodic boundary conditions, each term appears twice in the Hamiltonian, so

$$\hat{H}_{\mathrm{XY}}(N=2) = -J\left(\sigma_1^+\sigma_2^- + \sigma_1^-\sigma_2^+\right). \tag{10.45}$$

We have four states:

$$|1\rangle = |\uparrow\uparrow\rangle;\quad |2\rangle = |\uparrow\downarrow\rangle;\quad |3\rangle = |\downarrow\uparrow\rangle;\quad \text{and}\quad |4\rangle = |\downarrow\downarrow\rangle. \tag{10.46}$$

To construct the matrix representation of the Hamiltonian, we act the Hamiltonian on each basis vector $|i\rangle$ and then express it in terms of those basis vectors. So $\hat{H}_{\mathrm{XY}}|1\rangle = 0$, because neither spin can be raised. Similarly $\hat{H}_{\mathrm{XY}}|4\rangle = 0$ as well. We also have $\hat{H}_{\mathrm{XY}}|2\rangle = -J|3\rangle$ and $\hat{H}_{\mathrm{XY}}|3\rangle = -J|2\rangle$. This means the matrix representation of the Hamiltonian is

$$\langle i|\hat{H}_{\mathrm{XY}}|j\rangle = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -J & 0 \\ 0 & -J & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{10.47}$$

The energy eigenvalues are the eigenvalues of this matrix. Whenever a matrix has a row of zeros, this corresponds to a zero eigenvalue. We know both the state $|\uparrow\uparrow\rangle$ and $|\downarrow\downarrow\rangle$ are eigenstates with eigenvalues equal to 0. To compute the other two eigenvalues, we can see that it corresponds to the 2×2 matrix in the center of the Hamiltonian matrix, and its eigenvalues and eigenvectors are $-J$: $\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$ and J: $\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$, which you can directly check. Note that this is the kind of calculation you should be able to do yourself. If you are having any trouble with this, you might want to review the matrix computations discussed in Chap. 5.

In the Ising model case, the set of basis states with spins along the z axis for each tensor-product component was an eigenbasis, so the Hamiltonian was already diagonalized. For random spin exchange coefficients, it is difficult to find the ground states without looking at all states, but on the whole, the quantum effects were minimal. Here, a z-basis will not always be an eigenbasis, because the operators flip an up spin to a down spin and *vice versa* on neighboring sites. An example for how one finds the eigenstates was given above for $N = 2$. But, how do we do this in general? We have already seen one way that we can diagonalize Hamiltonians when we worked with the simple harmonic oscillator. There, we found that if we could factorize the Hamiltonian into the form $\hat{H}_{\mathrm{SHO}} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)$, then we could show that the energy eigenstates became $|n\rangle = \frac{1}{\sqrt{n!}}\left(\hat{a}^\dagger\right)^n|0\rangle$, with energy $\hbar\omega\left(n + \frac{1}{2}\right)$. The commutator of the Hamiltonian with the raising operator is

$$\left[\hat{H}_{\mathrm{SHO}}, \hat{a}^\dagger\right] = \hbar\omega\hat{a}^\dagger. \tag{10.48}$$

This is the generic form for the commutator of a raising operator with a Hamiltonian. So, we will see if we can find such an operator for the XY model.

The raising operator for each individual spin is σ_j^+, so, for the XY model, we might anticipate that an operator formed by a linear combination of σ_j^+ could also be a raising operator for the entire Hamiltonian. Let us test out this idea by computing the commutator

$$\begin{aligned}\left[\hat{H}_{\mathrm{XY}}, \sum_{j=1}^{N}\alpha_j\sigma_j^+\right] &= -\frac{J}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}\alpha_j\left\{\sigma_i^+\left[\sigma_{i+1}^-, \sigma_j^+\right] + \left[\sigma_i^-, \sigma_j^+\right]\sigma_{i+1}^+\right\} \\ &= J\sum_{j=1}^{N}\alpha_j\left(\sigma_{j-1}^+\sigma_j^z + \sigma_j^z\sigma_{j+1}^+\right).\end{aligned} \tag{10.49}$$

Fig. 10.11 Elliott Lieb (left), Theodore Schultz (middle), and Daniel Mattis (right) who wrote a 1961 paper on the fermionization of the XY model. (Lieb image courtesy of the Emilio Segrè Visual Archives, Physics Today Collection. Schultz image also from the Physics Today Collection. Mattis image courtesy of Ray Howze (originally appearing in the *Salt Lake City Weekly*, January 17, 2018). All images have been modified to look like sketches by James Freericks)

This is almost of the form we need, but it has those extra σ^z terms that we cannot get rid of. It does suggest that adding in some σ_j^z terms into the raising operator might help, because $(\sigma_i^z)^2 = \mathbb{I}$ allows for extra simplifications. But, rather than just guessing, we will try to proceed a bit more systematically.

The strategy we will use is based on an original idea of Pascual Jordan and Eugene Wigner, first worked out in the late 1920s (Jordan & Wigner, 1928), but clarified and extended by Elliott Lieb et al. in 1961 (Lieb et al., 1961). These three appear in Fig. 10.11. The method is now called *fermionization*. To begin the systematic way we will be proceeding, we recall that for a Pauli matrix, we have that

$$e^{i\frac{\pi}{2}(\sigma^z+1)} = e^{i\frac{\pi}{2}}\left(\cos\left(\tfrac{\pi}{2}\right)\mathbb{I}_2 + i\sin\left(\tfrac{\pi}{2}\right)\sigma^z\right) = -\sigma^z. \tag{10.50}$$

We define a set of new operators, which we will call $\hat{c}_i$ and $\hat{c}_i^\dagger$. They have a string of $-\sigma^z = e^{i\frac{\pi}{2}(\sigma^z+1)}$ factors running from the first site to site $i-1$. This string of factors is called the *Jordan-Wigner string*. The new operators are

$$\hat{c}_i = e^{i\frac{\pi}{2}\sum_{j=1}^{i-1}(\sigma_j^z+1)}\frac{\sigma_i^-}{2} \quad \text{and} \quad \hat{c}_i^\dagger = e^{-i\frac{\pi}{2}\sum_{j=1}^{i-1}(\sigma_i^z+1)}\frac{\sigma_i^+}{2}. \tag{10.51}$$

Note that these operators *anticommute* with each other, with the following relations:

$$\{\hat{c}_i\,,\hat{c}_j\}_+ = 0, \quad \{\hat{c}_i^\dagger\,,\hat{c}_j^\dagger\}_+ = 0, \quad \text{and} \quad \{\hat{c}_i\,,\hat{c}_j^\dagger\}_+ = \delta_{ij}. \tag{10.52}$$

Now we prove these three results. For the first, we have

$$\{\hat{c}_i\,,\hat{c}_j\}_+ = \frac{1}{4}e^{i\frac{\pi}{2}\sum_{k=1}^{i-1}(\sigma_k^z+1)}\sigma_i^- e^{i\frac{\pi}{2}\sum_{k'=1}^{j-1}(\sigma_{k'}^z+1)}\sigma_j^- + \frac{1}{4}e^{i\frac{\pi}{2}\sum_{k'=1}^{j-1}(\sigma_{k'}^z+1)}\sigma_j^- e^{i\frac{\pi}{2}\sum_{k=1}^{i-1}(\sigma_k^z+1)}\sigma_i^-. \tag{10.53}$$

For concreteness, we choose $i < j$ (the case $i = j$ is clear, because $(\sigma_i^-)^2 = 0$). Then, because the σ^z's commute with all Pauli matrices on different lattice sites, we can move them past the raising or lowering Pauli and combine them together to cancel out. Then we are left with

$$\{\hat{c}_i\,,\hat{c}_j\}_+ = \frac{1}{4}(-1)^{i-1}\left(\sigma_i^- e^{i\frac{\pi}{2}\sum_{k=i}^{j-1}(\sigma_k^z+1)}\sigma_j^- + e^{i\frac{\pi}{2}\sum_{k=i}^{j-1}(\sigma_k^z+1)}\sigma_j^-\sigma_i^-\right). \tag{10.54}$$

In the first term, we move all exponential factors to the left (using a multiply by one)

$$\begin{aligned}\sigma_i^- e^{i\frac{\pi}{2}\sigma_i^z} &= e^{i\frac{\pi}{2}\sigma_i^z}e^{-i\frac{\pi}{2}\sigma_i^z}\sigma_i^- e^{i\frac{\pi}{2}\sigma_i^z}\\ &= e^{i\frac{\pi}{2}\sigma_i^z}(-i\sigma_i^z)\sigma_i^-(i\sigma_i^z) = e^{i\frac{\pi}{2}\sigma_i^z}(-\sigma_i^-),\end{aligned} \tag{10.55}$$

so that

$$\{\hat{c}_i\,,\hat{c}_j\} = \frac{1}{4}(-1)^{i-1}e^{i\frac{\pi}{2}\sum_{k=i}^{j-1}\sigma_k^z}[\sigma_j^-,\sigma_i^-] = 0. \tag{10.56}$$

The anticommutator has been converted into a commutator for the spin operators, and because spin operators on different sites commute, we find the $\hat{c}$ operators anticommute. Using this result, we immediately find that $\{\hat{c}_i^\dagger,\hat{c}_j^\dagger\} = 0$ as well, by simply taking the Hermitian conjugate of the relation we just derived. This then leaves the last identity we need to prove

$$\{\hat{c}_i\,,\hat{c}_j^\dagger\}_+ = \frac{1}{4}e^{i\frac{\pi}{2}\sum_{k=1}^{i-1}(\sigma_k^z+1)}\sigma_i^- e^{-i\frac{\pi}{2}\sum_{k'=1}^{j-1}(\sigma_{k'}^z+1)}\sigma_j^+ + \frac{1}{4}e^{-i\frac{\pi}{2}\sum_{k'=1}^{j-1}(\sigma_{k'}^z+1)}\sigma_j^+ e^{i\frac{\pi}{2}\sum_{k=1}^{i-1}(\sigma_k^z+1)}\sigma_i^-. \tag{10.57}$$

If $i < j$, the manipulations are just like before for $\{\hat{c}_i,\hat{c}_j\}_+$, and we find this anticommutator is also zero because $[\sigma_j^+,\sigma_i^-] = 0$ when $i \neq j$. The case with $i > j$ immediately follows by taking the Hermitian conjugate of the identity for $i < j$. So, we are left with working out the case with $i = j$. In this case, the exponential factors commute through the $\sigma^\pm$ (because none of the exponential factors has a Pauli matrix from site i) and then cancel each other out. So, in this case, we still have an anticommutator left. We just evaluate it

$$\begin{aligned}\{\sigma_i^-,\sigma_i^+\}_+ &= \{\sigma_i^x - i\sigma_i^y,\sigma_i^x + i\sigma_i^y\}_+\\ &= \{\sigma_i^x,\sigma_i^x\}_+ - i\{\sigma_i^y,\sigma_i^x\}_+ + i\{\sigma_i^x,\sigma_i^y\}_+ + \{\sigma_i^y,\sigma_i^y\}_+\end{aligned}$$

$$= 4\mathbb{I}_2, \tag{10.58}$$

so, we finally have $\{\hat{c}_i\,, \hat{c}_j^\dagger\}_+ = \delta_{ij}$.

It turns out that operators that have the anticommutation relations in Eq. (10.52) create ($\hat{c}^\dagger$) and annihilate ($\hat{c}$) fermions. This is because fermionic states are required to be antisymmetric under the interchange of any two fermions (meaning the state changes sign when two fermions are interchanged). Fermions also must satisfy the Pauli exclusion principle, which says that no two fermions can be in the same quantum state. Indeed, this is in essence the definition of what a fermion is. One often encounters this definition in a high-school chemistry class, where you fill in atomic or molecular states according to rules for fermions. Here, we have been a bit more specific about the properties including the one for antisymmetry. This property is not so obvious, but if you exchange the two fermions twice, then the state must return back to its original, so one can have antisymmetry, which is what we have for fermions, or the state can be symmetric, which is what we have for bosons. We will discuss these properties in more detail later, when we describe the concept of *indistinguishability* of quantum objects.

To see how these operators and states work, we start from the vacuum state, which is a tensor product state of the state with no fermions for each lattice site and is given by $|0\rangle = |0\rangle_1 \otimes |0\rangle_2 \cdots \otimes |0\rangle_N$. For fermions, we can have either zero fermions or one fermion on each lattice site. We denote these states as $|0\rangle$ and $|1\rangle$. Then, a state with three fermions, one on each of the first three lattice sites, is given by

$$\begin{aligned} |\psi_{123}\rangle &= \hat{c}_1^\dagger \hat{c}_2^\dagger \hat{c}_3^\dagger |0\rangle \\ &= \hat{c}_1^\dagger |0\rangle_1 \otimes \hat{c}_2^\dagger |0\rangle_2 \otimes \hat{c}_3^\dagger |0\rangle_3 \otimes |0\rangle_4 \cdots |0\rangle_N \\ &= |123\rangle, \end{aligned} \tag{10.59}$$

where the last line is a shorthand notation listing the lattice sites occupied by the fermions, with an ordering given by the ordering of the creation operators from left to right. The state is indicated schematically in Fig. 10.12. Now, to see that it satisfies the Pauli exclusion principle, we see that

$$\hat{c}_1^\dagger |\psi\rangle = \underbrace{\hat{c}_1^\dagger \hat{c}_1^\dagger}_{=0} \hat{c}_2^\dagger \hat{c}_3^\dagger |0\rangle = 0, \tag{10.60}$$

because $(\hat{c}_i^\dagger)^2 = 0$, which follows from the anticommutation relation when $i = j$. One can also think of this as

$$\hat{c}_1^\dagger \hat{c}_1^\dagger |0\rangle_1 = \hat{c}_1^\dagger |1\rangle_1 = 0, \tag{10.61}$$

because we cannot put two fermions in the same quantum state. Similarly, the state $|213\rangle = \hat{c}_2^\dagger \hat{c}_1^\dagger \hat{c}_3^\dagger |0\rangle$ is equal to $-|123\rangle$, which follows because $\{\hat{c}_1^\dagger, \hat{c}_2^\dagger\}_+ = 0$. We

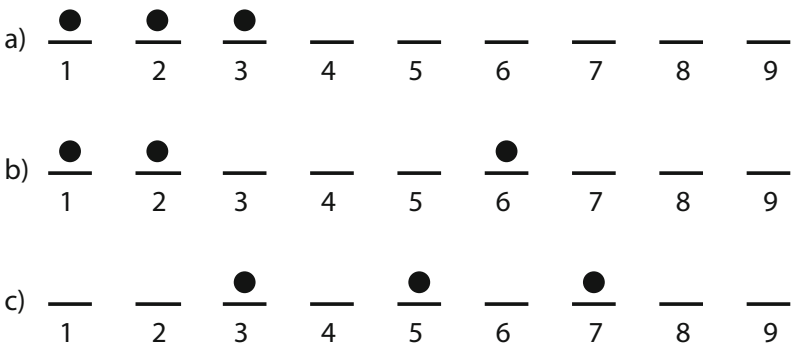

Fig. 10.12 Schematics of different fermionic states on an $N = 9$ lattice. Each state has three electrons in it. In panel (**a**), they occupy sites 1, 2, and 3. In (**b**), sites 1, 2, and 6. In (**c**), sites 3, 5, and 7. The black circles indicate the electrons, and the lines indicate the lattice sites

will have a lot more to say, and we will explore these ideas much more in a future chapter. Note that most of the challenge in working with fermions is keeping track of the signs. It can be very frustrating at times trying to get them correct.

We still have more work to do with this transformation, though. To begin, we can invert the transformation. To do this, we need to first introduce the fermion number operator $\hat{n}_i = \hat{c}_i^\dagger \hat{c}_i$. This operator counts the number of fermions. Using the facts that $\hat{c}|0\rangle = 0$, $\hat{c}|1\rangle = |0\rangle$, $\hat{c}^\dagger|0\rangle = |1\rangle$, and $\hat{c}^\dagger|1\rangle = 0$, we can immediately see that $\hat{n}|0\rangle = 0$ and $\hat{n}|1\rangle = |1\rangle$. (For example, $\hat{n}|1\rangle = \hat{n}\hat{c}^\dagger|0\rangle = \hat{c}^\dagger\{\hat{c}, \hat{c}^\dagger\}_+|0\rangle = \hat{c}^\dagger|0\rangle = |1\rangle$.) So the eigenvalues of the number operator are the number of fermions in a given state. This is equivalent to the spin operator via

$$\hat{n} \leftrightarrow \tfrac{1}{2}\left(\sigma^z + 1\right), \tag{10.62}$$

as can be seen by making the correspondence between $|\downarrow\rangle \leftrightarrow |0\rangle$ and $|\uparrow\rangle \leftrightarrow |1\rangle$. This is the standard physics correspondence for fermionization, which maps fermions to spins. We will discuss the quantum information science mapping, used in quantum computing later in this section—unfortunately, that mapping is the *opposite* of this one. Using the equivalence of the fermionic and spin operators immediately leads us to the inverse transformation as

$$\sigma_i^+ = e^{i\pi \sum_{j=1}^{i-1} \hat{n}_j} \hat{c}_i^\dagger \quad \text{and} \quad \sigma_i^- = \hat{c}_i \, e^{-i\pi \sum_{j=1}^{i-1} \hat{n}_j}. \tag{10.63}$$

The result in Eq. (10.62) allows us to map the magnetic field onto the fermions. We need to map the spin-exchange terms as well. This requires us to map $\sigma_i^+\sigma_{i+1}^- + \sigma_i^-\sigma_{i+1}^+$ onto the fermionic representation. Using the result from Eq. (10.63) immediately gives us

$$\begin{aligned}\sigma_i^+\sigma_{i+1}^- + \sigma_i^-\sigma_{i+1}^+ &= e^{i\pi\sum_{j=1}^{i-1}\hat{n}_j}\hat{c}_i^\dagger\hat{c}_{i+1}e^{-i\pi\sum_{k=1}^{i}\hat{n}_k} \\ &\quad + \hat{c}_i\, e^{-i\pi\sum_{j=1}^{i-1}\hat{n}_j}e^{i\pi\sum_{k=1}^{i}\hat{n}_k}\hat{c}_{i+1}^\dagger \\ &= \hat{c}_i^\dagger e^{-i\pi\hat{n}_i}\hat{c}_{i+1} + \hat{c}_i\, e^{i\pi\hat{n}_i}\hat{c}_{i+1}^\dagger. \end{aligned} \tag{10.64}$$

Next, we use the fact that $\hat{n}^2 = \hat{n}$, which follows from $\hat{n}^2 = \hat{c}^\dagger\hat{c}\hat{c}^\dagger\hat{c} = \hat{c}^\dagger\{\hat{c}, \hat{c}^\dagger\}_+\hat{c} = \hat{c}^\dagger\hat{c} = \hat{n}$. We used the add-zero trick in the center to create the anticommutator, because $\hat{c}^2 = 0$. Using this result, one can expand the exponential in a power series and find that

$$e^{i\theta\hat{n}} = \sum_{j=0}^{\infty}\frac{(i\theta)^j}{j!}\hat{n}^j = 1 + \left(e^{i\theta} - 1\right)\hat{n}, \tag{10.65}$$

because $\hat{n}^j = \hat{n}$ for $j \geq 1$ (another application of the add-zero trick). Setting $\theta = \pi$ or $\theta = -\pi$ gives the same result $1 - 2\hat{n}$. We use this next. We find that

$$\hat{c}_i^\dagger e^{-i\pi\hat{n}_i} = \hat{c}_i^\dagger\left(1 - 2\hat{n}_i\right) = \hat{c}_i^\dagger, \tag{10.66}$$

since $\hat{c}^\dagger\hat{n} = \hat{c}^\dagger\hat{c}^\dagger\hat{c} = 0$. Similarly, we have

$$\hat{c}_i\, e^{i\pi\hat{n}_i} = \hat{c}_i\,(1 - 2\hat{n}_i) = \hat{c}_i - 2\hat{c}_i\,\hat{c}_i^\dagger\hat{c}_i = \hat{c}_i - 2\{\hat{c}_i\,, \hat{c}_i^\dagger\}_+\hat{c}_i = -\hat{c}_i\,. \tag{10.67}$$

This then gives us

$$\sigma_i^+\sigma_{i+1}^- + \sigma_i^-\sigma_{i+1}^+ = \hat{c}_i^\dagger\hat{c}_{i+1} - \hat{c}_i\,\hat{c}_{i+1}^\dagger = \hat{c}_i^\dagger\hat{c}_{i+1} + \hat{c}_{i+1}^\dagger\hat{c}_i\,. \tag{10.68}$$

We will use this identity to solve the XY model in the next section.

You might have noticed that the Jordan-Wigner string disappeared from the XY model after fermionization. This occurred, because it is a one-dimensional model with nearest-neighbor interactions. In this case, when we do not have periodic boundary conditions, there are no Jordan-Wigner strings in the final Hamiltonian (but see Problem 10.4.3). In general, however, the strings survive and form a chain of exponentials of number operators between the fermionic operators. This makes the fermion terms in the Hamiltonian into nonlocal terms, which can complicate calculations with them, as we will see in the next section. This issue often arises in quantum computation. Quantum computers work with qubits, which are usually described in terms of spin-one-half degrees of freedom. So, if we want to solve a fermionic model on a quantum computer, we need to convert from the fermionic representation to the spin representation. This process is called *fermionic encoding*. The Jordan-Wigner transformation is one way to do this, but there are others. In this case, the fermions become spin states, but the fermionic operators become spin raising and lowering operators along with the Jordan-Wigner strings. Another source of confusion is quantum information uses a $|0\rangle$ to denote the up state and a

$|1\rangle$ to denote the down state, so the mapping to spin is opposite to the one we just worked out. This is just an unfortunate reality that we have to live with. It is now time to jump into the problems.

Section 10.4 Problems:

10.4.1 Fermionic Operator Gymnastics

Using the representation of $\hat{c}_i^\dagger$ and $\hat{c}_i$ in terms of Pauli spin matrices with Jordan-Wigner strings, verify that

$$[\hat{c}_i\,,\hat{n}_j] = \hat{c}_i\,\delta_{ij} \quad \text{and} \quad [\hat{c}_i^\dagger\,,\hat{n}_j] = -\hat{c}_i^\dagger\delta_{ij}. \tag{10.69}$$

Verify that $(-1)^{\hat{n}_i} = \left(1-2\hat{n}_i\right)$.
Verify that

$$\hat{c}_i\,(-1)^{\hat{n}_i} = -\hat{c}_i \quad \text{and} \quad \hat{c}_i^\dagger(-1)^{\hat{n}_i} = \hat{c}_i^\dagger. \tag{10.70}$$

10.4.2 Fermionization Examples

Construct σ_i^z in terms of Fermi operators.
Determine $\sigma_i^z\sigma_{i+1}^z$ in terms of Fermi operators.

10.4.3 Jordan-Wigner String for Periodic Boundary Conditions

Consider a lattice with periodic boundary conditions. Show that $\sigma_N^+\sigma_{N+1}^- + \sigma_N^-\sigma_{N+1}^+ = \sigma_N^+\sigma_1^- + \sigma_N^-\sigma_1^+$ can be written as

$$\left(\hat{c}_N^\dagger\hat{c}_1 + \hat{c}_1^\dagger\hat{c}_N\right)(-1)^{1+\sum_{i=1}^N\hat{n}_i}. \tag{10.71}$$

This result implies that the term corresponding to the link that completes the "ring" of the periodic lattice looks like the other bonds whenever $\sum_{i=1}^N\hat{n}_i$ is odd.
Hint: The identity $1 = (-1)^{\hat{n}_i}(-1)^{\hat{n}_i}$ (because the eigenvalues of the number operator are always the integers 0 or 1) and the identities in Problem 10.4.1 should help you.

10.4.4 Quadratic Operators Acting on Fermionic States

Recall that we use an integer label in a ket to say which states are occupied in our fermionic notation. So, for example,

$$\hat{c}_i^\dagger \hat{c}_j^\dagger \hat{c}_k^\dagger |0\rangle = |ijk\rangle. \tag{10.72}$$

The states are always put into a canonical form, where the indices in the raising operator products are ordered, so we have $i < j < k$.

Now, suppose we act a quadratic operator, say $\hat{c}_m^\dagger \hat{c}_j$ onto the state $|ijk\rangle$, and we have $m < i$. Then, the state can be put into canonical form as follows:

$$\hat{c}_m^\dagger \hat{c}_j |ijk\rangle = \hat{c}_m^\dagger \hat{c}_j \hat{c}_i^\dagger \hat{c}_j^\dagger \hat{c}_k^\dagger |0\rangle = -\hat{c}_m^\dagger \hat{c}_i^\dagger \{\hat{c}_j, \hat{c}_j^\dagger\}_+ \hat{c}_k^\dagger |0\rangle = |mik\rangle. \tag{10.73}$$

Here, in the second step, we can replace $\hat{c}_j \hat{c}_j^\dagger |0\rangle \to \{\hat{c}_j, \hat{c}_j^\dagger\}_+ |0\rangle$ because $\hat{c}_j |0\rangle = 0$.

Your job is to evaluate the following quadratic operators acting on the given states. Note that the hardest part of this work is getting the signs right when putting the state into canonical form.

$$\hat{c}_1^\dagger \hat{c}_1 |135\rangle, \quad \hat{c}_8^\dagger \hat{c}_6 |246\rangle, \quad \text{and} \quad \hat{c}_2^\dagger \hat{c}_3 |134\rangle. \tag{10.74}$$

Hint: a quadratic of fermion operators can move past any fermionic operator that it shares no indices with, without changing any signs, because there are two anticommutations. This rule can help make your calculation a bit easier.

10.4.5 Fermionic Parity Projection Operators

Consider the two operators

$$\hat{P}_i = \tfrac{1}{2}\left(1 + (-1)^{\hat{n}_i}\right) \quad \text{and} \quad \hat{Q}_i = \tfrac{1}{2}\left(1 - (-1)^{\hat{n}_i}\right). \tag{10.75}$$

Show that these two operators are projection operators, which satisfy $\hat{P}_i^2 = \hat{P}_i$, $\hat{Q}_i^2 = \hat{Q}_i$, $\hat{P}_i + \hat{Q}_i = \mathbb{I}$, and $\hat{P}_i \hat{Q}_i = \hat{Q}_i \hat{P}_i = 0$.
Verify the following operators are also projection operators

$$\hat{P} = \tfrac{1}{2}\left(1 + (-1)^{\sum_{i=1}^N \hat{n}_i}\right) \quad \text{and} \quad \hat{Q} = \tfrac{1}{2}\left(1 - (-1)^{\sum_{i=1}^N \hat{n}_i}\right). \tag{10.76}$$

We will use these operators in the next section to solve the XY model exactly.

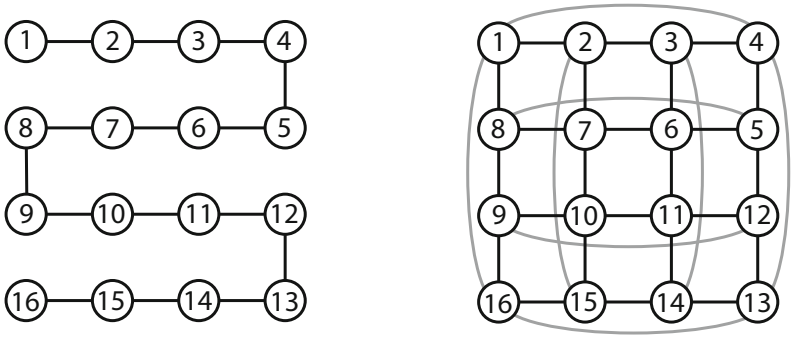

Fig. 10.13 (Left) Snake-like path used to determine the Jordan-Wigner strings for the Jordan-Wigner transformation. (Right) The square lattice with periodic boundary conditions. The black bonds are direct nearest neighbor bonds. The gray bonds are the bonds we have due to the periodic boundary conditions

10.4.6 Jordan-Wigner Transformation on a Square Lattice

Consider a 4×4 square lattice with periodic boundary conditions. The lattice sites are numbered in an ordered way along a one-dimensional "snake" path as shown in the Fig. 10.13.

Suppose the XY model is written as

$$\hat{H}_{\mathrm{XY}} = -\tfrac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} J_{ij} \left(\sigma_i^+ \sigma_j^- + \sigma_i^- \sigma_j^+ \right), \tag{10.77}$$

with all nonzero J_{ij} values equal to J. The ij pairs that have bonds are 1,2; 1,4; 1,8; 1,16; 2,3; 2,7; 2,15; 3,4; 3,6; 3,14; 4,5; 4,13; 5,6; 5,8; 5,12; 6,7; 6,11; 7,8; 7,10; 8,9; 9,10; 9,12; 9,16; 10,11; 10,15; 11,12; 11,14; 12,13; 13,14; 13,16; 14,15; and 15,16. We list each pair of indices for a bond just once.

Show that, unlike in the one-dimensional chain, here we have that all of the terms $\sigma_i^+ \sigma_j^- + \sigma_i^- \sigma_j^+$ create Jordan-Wigner strings running between sites i and j when $j \neq i \pm 1$. Draw some pictures to show what these strings look like (recall the "snake" path). Identify the longest string.

10.5 The XY Model

We already wrote down the XY model, but we repeat it again here for a periodic lattice in one dimension:

$$\hat{H}_{\mathrm{XY}} = -\frac{J}{2}\sum_{i=1}^{N}\left(\sigma_i^+\sigma_{i+1}^- + \sigma_i^-\sigma_{i+1}^+\right) - B\sum_{i=1}^{N}\sigma_i^z$$
$$= -J\sum_{i=1}^{N}\left(\sigma_i^x\sigma_{i+1}^x + \sigma_i^y\sigma_{i+1}^y\right) - B\sum_{i=1}^{N}\sigma_i^z, \tag{10.78}$$

where the second form makes clear why it is called the XY model. Note that this is the *isotropic* form of the XY model, which is the most common case considered (the J value in the x- and y-directions is the same).

Let us start our discussion by examining the symmetry of the XY model Hamiltonian. To begin, the spin-reflection parity symmetry now requires $\sigma^x \to -\sigma^x$, $\sigma^y \to -\sigma^y$, and $\sigma^z \to \sigma^z$, which tells us nothing about the parity of the states when written in the z-basis, because the z states do not change under this operation. So, this spin-reflection-parity operation does not provide us with any additional information. If we instead use the spin-reflection parity operator, call it $\hat{\Pi}_y$, which transforms as $\sigma^x \to -\sigma^x$, $\sigma^y \to \sigma^y$, and $\sigma^z \to -\sigma^z$, then we find $\hat{\Pi}_y\hat{H}_{\mathrm{XY}}\hat{\Pi}_y^{-1}$ is the XY Hamiltonian with $B \to -B$. Note that the similarity transformation does not change eigenvalues, which can be easily proven as follows by using the multiply-by-one trick:

$$\begin{aligned}\hat{H}_{\mathrm{XY}}(B)|\psi\rangle = E|\psi\rangle \;&\Longrightarrow\; \hat{\Pi}_y\hat{H}_{\mathrm{XY}}(B)|\psi\rangle = E\hat{\Pi}_y|\psi\rangle \\ &\Longrightarrow\; \underbrace{\hat{\Pi}_y\hat{H}_{\mathrm{XY}}(B)\hat{\Pi}_y^{-1}}_{=\hat{H}_{\mathrm{XY}}(-B)}\hat{\Pi}_y|\psi\rangle = E\hat{\Pi}_y|\psi\rangle \\ &\Longrightarrow\; \hat{H}_{\mathrm{XY}}(-B)\hat{\Pi}_y|\psi\rangle = E\hat{\Pi}_y|\psi\rangle. \end{aligned}\tag{10.79}$$

This says that if E is an eigenvalue of $\hat{H}_{\mathrm{XY}}(B)$ with eigenvector $|\psi\rangle$, then E is also an eigenvalue of $\hat{H}_{\mathrm{XY}}(-B)$ with eigenvector $\hat{\Pi}_y|\psi\rangle$. Hence, the eigenvalue spectrum is symmetric about $B = 0$ when plotted as a function of B. The total z-component of spin does commute with $\hat{H}_{\mathrm{XY}}$, as can be found by direct computation. It obviously commutes with the magnetic field term. For the spin-exchange term, we have

$$\begin{aligned}&\left[\sum_{i=1}^{N}\sigma_i^z,\; -\frac{J}{2}\sum_{j=1}^{N}\left(\sigma_j^+\sigma_{j+1}^- + \sigma_j^-\sigma_{j+1}^+\right)\right] \\ &= -\frac{J}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}\left[\sigma_i^z, \sigma_j^+\sigma_{j+1}^- + \sigma_j^-\sigma_{j+1}^+\right] \\ &= -J\sum_{i=1}^{N}\left(\sigma_i^+\sigma_{i+1}^- - \sigma_{i-1}^+\sigma_i^- - \sigma_i^-\sigma_{i+1}^+ + \sigma_{i-1}^-\sigma_i^+\right) = 0, \end{aligned}\tag{10.80}$$

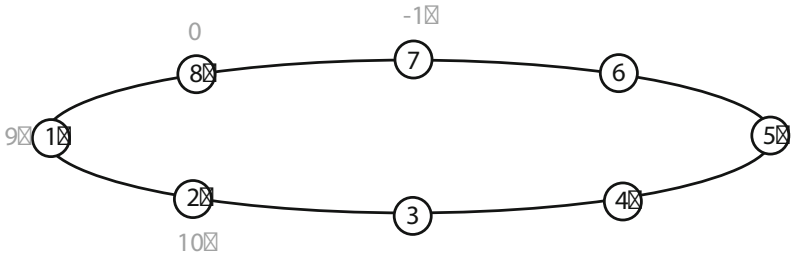

Fig. 10.14 Schematic of how a one-dimensional chain with periodic boundary conditions is constructed on a circle. Note that this eight-site ($N = 8$) example maps site $9 \to 1$, $10 \to 2$, etc. and $0 \to 8$, $-1 \to 7$, etc. In other words, the site labels are always mapped into the range $1 \leq i \leq 8$ by adding or subtracting multiples of 8; labels outside the fundamental range are in gray. This is generalized to any size chain by adding or subtracting multiples of N

which follows after shifting the index of summation for the second and fourth terms by $i \to i + 1$. So, the quantum eigenstates can also be classified by their z-component of the total spin.

We are considering only the case of the XY model on a one-dimensional chain with periodic boundary conditions (which can be thought of as all sites sitting on a circle as shown in the Fig. 10.14). This model has an additional translational symmetry, which tells us that the system is unchanged by rotating around the circle by any multiple of lattice sites. Effectively, this means that if we shift $i \to i + m$ for any integer m, the Hamiltonian is unchanged. Whenever we have such a translational symmetry, we will find that working in momentum space is a better choice than working in position space. What is meant by this will be made clear below.

The XY model can be diagonalized by going to the fermionic representation. We find, in terms of the fermions, that we have

$$\hat{H}_{\mathrm{XY}} = -\frac{J}{2}\sum_{i=1}^{N-1}\left(\hat{c}_i^\dagger \hat{c}_{i+1} + \hat{c}_{i+1}^\dagger \hat{c}_i\right) + \frac{J}{2}\left(\hat{c}_1^\dagger \hat{c}_N + \hat{c}_N^\dagger \hat{c}_1\right)(-1)^{\hat{N}_e} - B\sum_{i=1}^{N}\left(2\hat{n}_i - 1\right), \tag{10.81}$$

where $\hat{n}_i = \hat{c}_i^\dagger \hat{c}_i$ is the electron number operator for site i and $\hat{N}_e = \sum_{i=1}^{N} \hat{n}_i$ is the total electron number operator. In Problem 10.6.1, you will show that $[\hat{N}_e, \hat{H}_{\mathrm{XY}}] = 0$. This occurs, because the magnetic field term only counts the total number of electrons, while the spin-exchange term only moves electrons from one lattice site to its neighboring site. So the total number of electrons is conserved. This is equivalent to the result we already showed that $\hat{S}_z$ commutes with the original Hamiltonian and m was shown to be a good quantum number. In Problem 10.4.5, we define an even $\hat{P} = \frac{1}{2}\left(1 + (-1)^{\hat{N}_e}\right)$ and odd $\hat{Q} = \frac{1}{2}\left(1 - (-1)^{\hat{N}_e}\right)$ parity projection operator.

Because $\hat{P} + \hat{Q} = \hat{\mathbb{I}}$, we have

$$\hat{H}_{\rm XY} = \hat{P}\hat{H}_{\rm XY}\hat{P} + \hat{Q}\hat{H}_{\rm XY}\hat{P} + \hat{P}\hat{H}_{\rm XY}\hat{Q} + \hat{Q}\hat{H}_{\rm XY}\hat{Q} = \hat{P}\hat{H}_{\rm XY}\hat{P} + \hat{Q}\hat{H}_{\rm XY}\hat{Q}. \tag{10.82}$$

The second identity follows, because both projection operators commute with $\hat{H}_{\rm XY}$ (because $[\hat{N}_e, \hat{H}_{\rm XY}] = 0$) and $\hat{P}\hat{Q} = 0$ (and $\hat{Q}\hat{P} = 0$). Now, because the $\hat{P}$ operator projects onto states where $(-1)^{\hat{N}_e}|\psi\rangle = |\psi\rangle$ and the $\hat{Q}$ operator projects onto states where $(-1)^{\hat{N}_e}|\psi\rangle = -|\psi\rangle$, we have that

$$\hat{H}_{\rm XY} = \hat{P}\hat{H}_+\hat{P} + \hat{Q}\hat{H}_-\hat{Q}, \tag{10.83}$$

with

$$\hat{H}_\pm = -\frac{J}{2}\sum_{i=1}^{N-1}\left(\hat{c}_i^\dagger\hat{c}_{i+1} + \hat{c}_{i+1}^\dagger\hat{c}_i\right) \pm \frac{J}{2}\left(\hat{c}_1^\dagger\hat{c}_N + \hat{c}_N^\dagger\hat{c}_1\right) - B\sum_{i=1}^{N}\left(2\hat{n}_i - 1\right). \tag{10.84}$$

This means that for an even number of fermions, we diagonalize $\hat{H}_+$, while for an odd number of fermions, we diagonalize $\hat{H}_-$. How we diagonalize them is discussed next.

We now go to momentum space. We alluded to the fact that we can diagonalize the XY model Hamiltonian in momentum space. And we can do this. But the procedure is a bit technical, so we need to describe it carefully. The procedure uses a transformation of the fermionic operators from position space to momentum space. We define the momentum space creation and annihilation operators via (the momentum-space operators will always have a k subscript)

$$\hat{c}_k = \frac{1}{\sqrt{N}}\sum_{j=1}^{N} e^{-ikj}\hat{c}_j \quad \text{and} \quad \hat{c}_k^\dagger = \frac{1}{\sqrt{N}}\sum_{j=1}^{N} e^{ikj}\hat{c}_j^\dagger. \tag{10.85}$$

It is only defined for N equally spaced momentum numbers k, which will be chosen below to satisfy either periodic or antiperiodic boundary conditions. The choices for the momenta are

$$\{k\} = \begin{cases} \frac{2\pi}{N}, \frac{4\pi}{N}, \frac{6\pi}{N}, \cdots, \frac{2N\pi}{N} & \text{periodic} \\ \frac{\pi}{N}, \frac{3\pi}{N}, \frac{5\pi}{N}, \cdots, \frac{(2N-1)\pi}{N} & \text{antiperiodic.} \end{cases} \tag{10.86}$$

We will verify why these are the correct choices below. These relations can be inverted via

$$\hat{c}_j = \frac{1}{\sqrt{N}}\sum_{k} e^{ikj}\hat{c}_k \quad \text{and} \quad \hat{c}_j^\dagger = \frac{1}{\sqrt{N}}\sum_{k} e^{-ikj}\hat{c}_k^\dagger. \tag{10.87}$$

We prove the first result for both cases. The strategy is to substitute the definitions in Eq. (10.85) into the expression in Eq. (10.87) and then use the geometric series. Noting that $k_{\text{per}} = \frac{2\pi m}{N}$ for $1 \leq m \leq N$ and $k_{\text{antiper}} = k_{\text{per}} - \frac{\pi}{N}$, we have for the periodic case that

$$\hat{c}_j = \frac{1}{\sqrt{N}} \sum_{m=1}^{N} e^{\frac{i2\pi mj}{N}} \underbrace{\frac{1}{\sqrt{N}} \sum_{j'=1}^{N} e^{-\frac{i2\pi mj'}{N}} \hat{c}_{j'}}_{\hat{c}_k}. \tag{10.88}$$

The summation over the integer m can be done immediately by the geometric series with $z = e^{i\frac{2\pi}{N}(j-j')}$. First, if $j = j'$, then each term in the summation over m is equal to 1, and we have N of them. Second, if $j \neq j'$, then we have

$$\sum_{m=1}^{N} z^m = \frac{1 - z^{N+1}}{1 - z} - 1 = -\frac{1 - z^N}{1 - \frac{1}{z}}. \tag{10.89}$$

The numerator then becomes $1 - e^{i2\pi(j-j')} = 1 - 1 = 0$. Hence, the sum over m becomes $N\delta_{jj'}$ and we find that

$$\hat{c}_j = \sum_{j'=1}^{N} \frac{N}{N} \delta_{jj'} \hat{c}_{j'} = \hat{c}_j \quad \checkmark. \tag{10.90}$$

Note how this result required $\left(e^{\frac{i2\pi(j-j')}{N}}\right)^N = 1$ in order for the identity to hold. This is the condition required for periodic boundary conditions. What about the antiperiodic case? If you follow the steps carefully, we will have exactly the same result when $j = j'$, and when $j \neq j'$, we have an extra factor of $e^{-\frac{i\pi(j-j')}{N}}$ that multiplies the geometric sum. But because the sum is 0, we still get the same result as for the periodic case.

Now, we replace the fermionic operators in position space, with those in momentum space to convert the XY model to its momentum-space form. We find

$$\begin{aligned}
\hat{H}_{\text{XY}} = &-\frac{J}{2} \sum_{j=1}^{N} \sum_{k} \sum_{k'} \frac{1}{N} \hat{c}_k^\dagger \hat{c}_{k'} \left(e^{ikj} e^{-ik'(j+1)} + e^{ik(j+1)} e^{-ik'j} \right) \\
&+ \frac{J}{2} \sum_{k} \sum_{k'} \frac{1}{N} \hat{c}_k^\dagger \hat{c}_{k'} \left(e^{ikN} e^{-ik'} \left((-1)^{\hat{N}_e} + e^{-ik'N} \right) \right. \\
&\left. + e^{ik} e^{-ik'N} \left((-1)^{\hat{N}_e} + e^{ikN} \right) \right)
\end{aligned}$$

$$- B \sum_{j=1}^{N} \sum_{k} \sum_{k'} \frac{2}{N} \hat{c}_k^\dagger \hat{c}_{k'} e^{ikj} e^{-ik'j} + BN, \tag{10.91}$$

where we added and subtracted a boundary term to make the summations on the top line run from $j = 1$ to $j = N$. The summations over j on the top and bottom lines can be done, and they are given by $N\delta_{kk'}$. Then the sum over k' can also be done, which sets $k = k'$. Now, we have for an odd number of fermions $(-1)^{\hat{N}_e} = -1$; here $\hat{N}_e = \sum_{j==1}^{N} \hat{c}_j^\dagger \hat{c}_j$ is the total fermion number operator, which counts the number of fermions. In this case, we use periodic boundary conditions, where $e^{\pm ikN} = 1$, so the boundary term vanishes. Similarly, for an even number of fermions, we have $(-1)^{\hat{N}_e} = 1$, but we use antiperiodic boundary conditions, so that $e^{\pm ikN} = -1$, so the boundary term vanishes again. This means we end up with

$$\hat{H}_{\mathrm{XY}} = -J \sum_{k} \cos(k) \hat{c}_k^\dagger \hat{c}_k - B \sum_{k} \left(2\hat{c}_k^\dagger \hat{c}_k - 1 \right). \tag{10.92}$$

Note that we need to know whether we are in the even or odd sector for the fermions, because the k values in the summations are different for those two cases (but there are exactly N terms in each summation). Note further that the Hamiltonian is now diagonal, as we claimed it would be.

Since the total z-component of spin commutes with the Hamiltonian $\hbar m$ is a good quantum number, so we can use it to organize our different energy eigenstates. This quantum number is equivalent to the number of fermions, because, if we let N_e denote the number of fermions, then $m = N_e - \frac{1}{2}N$. If the number of lattice sites is even, then m is an integer, while if N is odd, then m is a half-odd integer. When we add a fermion to the system, the angular momentum increases by one. This means in this case, the fermions carry $s = 1$. This is allowed, because they are abstract fermions.

To find the lowest energy state, we use the so-called bathtub principle. As you fill the water in a bathtub, it goes to the lowest part of the tub first and fills in from there. Here, we do the same thing. The ground state has the lowest total energy. Since the energy of the fermionic system is given by the sum of the occupied energy levels (because the expectation value of $\hat{c}^\dagger \hat{c}$ is 0 or 1), which is the coefficient of the $\hat{c}_k^\dagger \hat{c}_k$ term in the Hamiltonian, the bathtub principle is illustrated schematically in Fig. 10.15. We choose all of the k values where $-J\cos(k) - 2B < 0$. If there are any terms where $-J\cos(k) - 2B = 0$, then the ground state will have degeneracy. Otherwise, it is unique, unless the highest occupied level has two different k values that give it the same energy. Now, as B changes its magnitude, the number of k points where the above coefficient is less than zero will change. This will provide what are called *level crossings*, where the number of fermions in the ground state changes with B. But, note that when the number of fermions changes by 1, the allowed k values changes as we go from even to odd or *vice versa*. It is best to illustrate this principle next with an example. We examine the case with $N = 4$

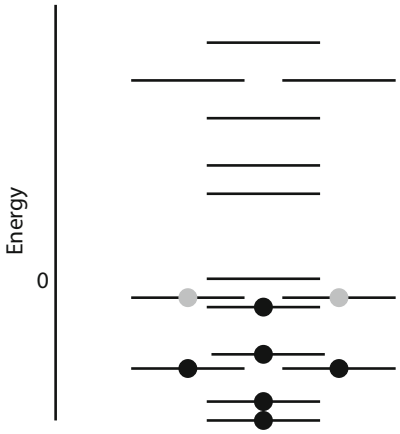

Fig. 10.15 Schematic of the bathtub principle. The (spinless) fermions can occupy only one energy level (levels indicated by the solid lines—degenerate levels have two lines). We fill up all available levels with energy less than zero to find the ground state. If there is a degeneracy of energy levels at the highest occupied level, then the ground state is degenerate. Here, we have seven electrons indicated by the circles, and the last electron can occupy either of the two highest energy occupied states (note that the ground state has eight electrons with both grayed states occupied)

explicitly. When $N_e = 0$, 2, or 4, the allowed k values are $k = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}$, and $\frac{7\pi}{4}$. When $N_e = 1$ or 3, then the allowed k values are $\frac{2\pi}{4}, \frac{4\pi}{4}, \frac{6\pi}{4}$, and $\frac{8\pi}{4}$. The even energies are then $-\frac{J}{\sqrt{2}} - 2B$ ($\frac{\pi}{4}$ and $\frac{5\pi}{4}$), $\frac{J}{\sqrt{2}} - 2B$ ($\frac{3\pi}{4}$ and $\frac{7\pi}{4}$). This means, for $N_e = 0$, the total energy is 0. For $N_e = 2$, the total energy is $-\sqrt{2}J - 4B$. While for $N_e = 4$, the total energy is $-8B$. Similarly, for $N_e = 1$ or 3, the allowed energies are $-J - 2B$ (2π), $-2B$ ($\frac{\pi}{2}$ and $\frac{3\pi}{2}$), and $J - 2B$ (π). For $N_e = 1$, the total energy is $-J - 2B$ and for $N_e = 3$, the total energy is $-J - 6B$. The energy eigenvalues for the ground state are plotted in Fig. 10.16. For $B > \frac{J}{2}$, the ground state has $N_e = 4$ ($m = 2$) for $\frac{\sqrt{2}-1}{2}J < B < \frac{J}{2}$, then the $N_e = 3$ ($m = 1$) state is the ground state. For $-\frac{\sqrt{2}-1}{2}J < B < \frac{\sqrt{2}-1}{2}J$, the state with $N_e = 2$ ($m = 0$) is the ground state. For $-\frac{J}{2} < B < -\frac{\sqrt{2}-1}{2}J$, the state with $N_e = 1$ ($m = -1$) is the ground state. And finally, for $B < -\frac{J}{2}$, the state with $N_e = 0$ ($m = -2$) is the ground state. One can see that these boundaries are symmetrically displaced about $B = 0$. One can construct the higher-energy states by occupying other energy levels, but we will not enumerate all possibilities here. However, they are included in gray in Fig. 10.16. You will have a chance to work out the case for $N = 6$ in the problems. The energy levels are plotted in Fig. 10.17.

In general, one needs calculus to evaluate the energy per lattice site in the limit as $N \to \infty$. But, we can play a trick to solve this by using the geometric series, which will be elaborated on in Chap. 13. Experts in calculus will readily see that this is a valid way to evaluate the resulting integral. For $B = 0$, the energies are

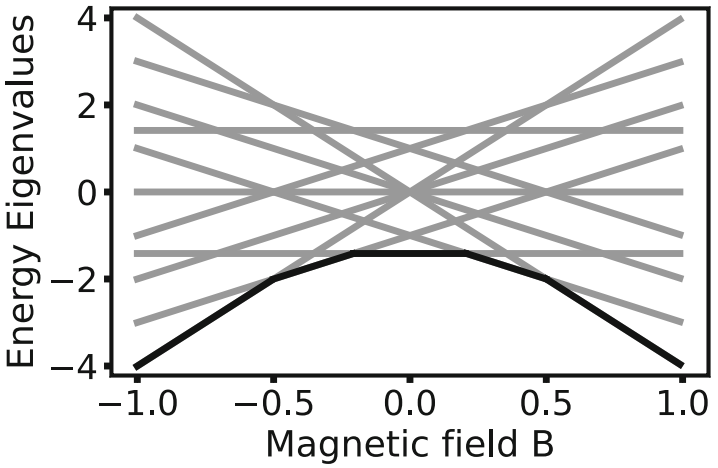

Fig. 10.16 Energy levels for the XY model with $N = 4$. The ground state is indicated in black and the excited states in gray. Note how the energy levels are symmetric about a mirror plane that runs along the line $B = 0$, and how the degeneracy of the energy levels is higher at $B = 0$. The kinks in the ground-state energy mark the level crossings where the fermion number (z-component of spin quantum number) changes

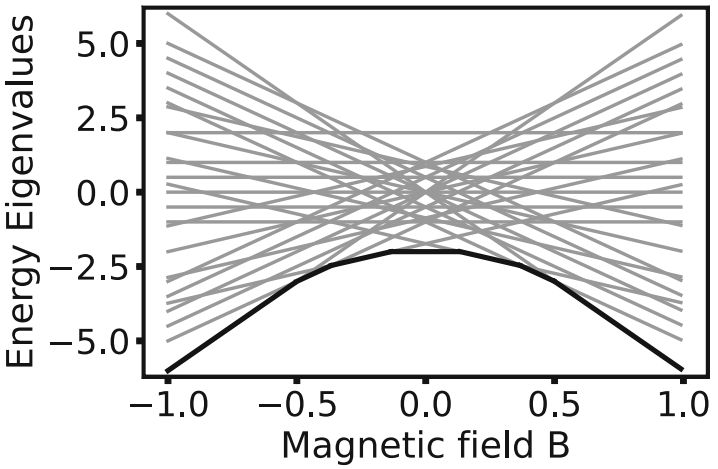

Fig. 10.17 Energy levels for the XY model with $N = 6$. The ground state is indicated in black and the excited states in gray. Note the added degeneracy when $B = 0$ and the reflection symmetry about the line $B = 0$. The ground-state energy is $6B$ for $B < -0.5J$, $-J + 4B$ for $-0.5J < B < -\frac{1}{2}(\sqrt{3} - 1)J$, $-\sqrt{3}J + 2B$ for $-\frac{1}{2}(\sqrt{3} - 1)J < B < -\frac{1}{2}(2 - \sqrt{3})J$, $-2J$ for $-\frac{1}{2}(2 - \sqrt{3})J < B < \frac{1}{2}(2 - \sqrt{3})J$, $-\sqrt{3}J - 2B$ for $\frac{1}{2}(2 - \sqrt{3})J < B < \frac{1}{2}(\sqrt{3} - 1)J$, $-J - 4B$ for $\frac{1}{2}(\sqrt{3} - 1)J < B < 0.5J$, and $-6B$ for $0.5J < B$. Many of the energy levels are degenerate. This is not shown in the figure

$-J\cos\left(\frac{2\pi m}{N}-\frac{\pi}{N}\right)$ for $N_e=$ even and $-\frac{N}{4}<m<\frac{N}{4}$, which is half of the allowed k values. Note that we have shifted the range of allowed momentum values due to the periodicity of the cosine function. Since the number of electrons must be even, we work with N being an even multiple of 2, so that $N_e=\frac{N}{2}=$ even. Then we have

$$\begin{aligned}\frac{E_{gs}}{N}&=-\frac{J}{N}\sum_{m=-\frac{N}{4}+1}^{\frac{N}{4}}\cos\left(\frac{2\pi m}{N}-\frac{\pi}{N}\right)=-\frac{2J}{N}\sum_{m=0}^{\frac{N}{4}-1}\cos\left(\frac{2\pi m}{N}+\frac{\pi}{N}\right)\\&=-\frac{2J}{N}\mathrm{Re}\,e^{\frac{i\pi}{N}}\sum_{m=0}^{\frac{N}{4}-1}e^{\frac{i2\pi m}{N}}=-\frac{2J}{N}\mathrm{Re}\left(e^{\frac{i\pi}{N}}\frac{1-e^{\frac{i2\pi N}{4N}}}{1-e^{\frac{i2\pi}{N}}}\right)\\&\approx-\frac{2J}{N}\mathrm{Re}\left(\left(1+i\tfrac{\pi}{N}\right)\frac{1-i}{1-1-\frac{i2\pi}{N}}\right)=-\tfrac{J}{\pi}+\mathcal{O}\left(\tfrac{1}{N}\right),\end{aligned}\tag{10.93}$$

which approaches $E_{gs}/N=-J/\pi$ as $N\to\infty$. In the problems, you will extend this analysis for $B\neq 0$.

We have provided an exact solution for the XY model by carefully determining the ground-state energy as a function of B and finding the level crossings with respect to changing B. So, what is interesting about the XY model? There is a lot! One can study it in higher dimensions. As we will see in the problems, it cannot be solved by fermionization in higher dimensions, because the Jordan-Wigner strings do not disappear, and they complicate the analysis. Much work has been spent to understand these solutions. Similarly, if we allow $J_{ij}\neq 0$ beyond just nearest neighbors, then the fermionization method no longer works either. One interesting case of this form is power law decaying exchange constants given by $J_{ij}=\frac{J}{|i-j|^{\alpha}}$ for some power $\alpha\geq 0$, because these systems have been studied extensively with quantum simulators (which are the precursors of quantum computers). Early work with quantum simulators focused on the range $0\leq\alpha\leq 3$, and we will discuss this problem later in the book.

Section 10.5 Problems:

10.5.1 Fermionic Operator Gymnastics

We will work out a number of commutators and anticommutators in this problem.

(a) Show that

$$\left[\sum_{i=1}^{N} \hat{c}_i^\dagger \hat{c}_i \, , \, \hat{H}_{\rm XY}\right] = 0 \tag{10.94}$$

using the position-space representation for the Hamiltonian with fermions.

(b) Show that

$$\{\hat{c}_k, \hat{c}_{k'}\}_+ = 0, \;\; \{\hat{c}_k^\dagger, \hat{c}_{k'}^\dagger\}_+ = 0, \;\; \text{and} \;\; \{\hat{c}_k, \hat{c}_{k'}^\dagger\}_+ = \delta_{kk'}, \tag{10.95}$$

using the definition of the momentum-space fermionic operators in terms of an expansion with respect to the position-space fermionic operators.

(c) If the Hamiltonian is diagonalized in terms of $\hat{H} = \sum_k \epsilon(k)\hat{c}_k^\dagger \hat{c}_k$, then $[\hat{c}_k^\dagger, \hat{H}] = -\epsilon(k)\hat{c}_k^\dagger$. In this problem, we will compute

$$[\hat{c}_i^\dagger, \hat{H}_{\rm XY}] \tag{10.96}$$

in the position-space representation and then sum the results, following the definition of the momentum-space operators, to determine that

$$[\hat{c}_k^\dagger, \hat{H}_{\rm XY}] = (J\cos(k) + 2B)\hat{c}_k^\dagger, \tag{10.97}$$

which is another way to show that the Hamiltonian is diagonalized by going to momentum space.

10.5.2 The Bathtub Principle

Suppose I have a Hamiltonian given by $\hat{H} = \sum_k \epsilon(k)\hat{c}_k^\dagger \hat{c}_k$. Prove the bathtub principle: the ground-state energy for N_e fermions is the sum of the N_e lowest values of $\epsilon(k)$ for the k values in the summation that defines the Hamiltonian.

10.5.3 $N = 4$ XY Model Solved in the Spin Language

Consider the four-site XY model in the spin representation and given by

$$\hat{H}_{\rm XY} = -\frac{J}{2}\sum_{i=1}^{4}\left(\sigma_i^+\sigma_{i+1}^- + \sigma_i^-\sigma_{i+1}^+\right) - B\sum_{i=1}^{4}\sigma_i^z. \tag{10.98}$$

Note that there are five different quantum numbers m for the z-component of spin. We organize them in linear combinations that have definite symmetry as follows:

$$
\begin{aligned}
m = 2: \quad & |\uparrow\uparrow\uparrow\uparrow\rangle \\
m = 1: \quad & \tfrac{1}{2}(|\uparrow\uparrow\uparrow\downarrow\rangle + |\uparrow\uparrow\downarrow\uparrow\rangle + |\uparrow\downarrow\uparrow\uparrow\rangle + |\downarrow\uparrow\uparrow\uparrow\rangle) \\
& \tfrac{1}{2}(|\uparrow\uparrow\uparrow\downarrow\rangle - |\uparrow\uparrow\downarrow\uparrow\rangle + |\uparrow\downarrow\uparrow\uparrow\rangle - |\downarrow\uparrow\uparrow\uparrow\rangle) \\
& \tfrac{1}{2}(|\uparrow\uparrow\uparrow\downarrow\rangle + |\uparrow\uparrow\downarrow\uparrow\rangle - |\uparrow\downarrow\uparrow\uparrow\rangle - |\downarrow\uparrow\uparrow\uparrow\rangle) \\
& \tfrac{1}{2}(|\uparrow\uparrow\uparrow\downarrow\rangle - |\uparrow\uparrow\downarrow\uparrow\rangle - |\uparrow\downarrow\uparrow\uparrow\rangle + |\downarrow\uparrow\uparrow\uparrow\rangle) \\
m = 0: \quad & \tfrac{1}{2}(|\uparrow\uparrow\downarrow\downarrow\rangle + |\uparrow\downarrow\downarrow\uparrow\rangle + |\downarrow\downarrow\uparrow\uparrow\rangle + |\downarrow\uparrow\uparrow\downarrow\rangle) \\
& \tfrac{1}{\sqrt{2}}(|\uparrow\downarrow\uparrow\downarrow\rangle + |\downarrow\uparrow\downarrow\uparrow\rangle) \\
& \tfrac{1}{2}(|\uparrow\uparrow\downarrow\downarrow\rangle - |\uparrow\downarrow\downarrow\uparrow\rangle + |\downarrow\downarrow\uparrow\uparrow\rangle - |\downarrow\uparrow\uparrow\downarrow\rangle) \\
& \tfrac{1}{2}(|\uparrow\uparrow\downarrow\downarrow\rangle + |\uparrow\downarrow\downarrow\uparrow\rangle - |\downarrow\downarrow\uparrow\uparrow\rangle - |\downarrow\uparrow\uparrow\downarrow\rangle) \\
& \tfrac{1}{2}(|\uparrow\uparrow\downarrow\downarrow\rangle - |\uparrow\downarrow\downarrow\uparrow\rangle - |\downarrow\downarrow\uparrow\uparrow\rangle + |\downarrow\uparrow\uparrow\downarrow\rangle) \\
& \tfrac{1}{\sqrt{2}}(|\uparrow\downarrow\uparrow\downarrow\rangle - |\downarrow\uparrow\downarrow\uparrow\rangle) \\
m = -1: \quad & \tfrac{1}{2}(|\downarrow\downarrow\downarrow\uparrow\rangle + |\downarrow\downarrow\uparrow\downarrow\rangle + |\downarrow\uparrow\downarrow\downarrow\rangle + |\uparrow\downarrow\downarrow\downarrow\rangle) \\
& \tfrac{1}{2}(|\downarrow\downarrow\downarrow\uparrow\rangle - |\downarrow\downarrow\uparrow\downarrow\rangle + |\downarrow\uparrow\downarrow\downarrow\rangle - |\uparrow\downarrow\downarrow\downarrow\rangle) \\
& \tfrac{1}{2}(|\downarrow\downarrow\downarrow\uparrow\rangle - |\downarrow\downarrow\uparrow\downarrow\rangle - |\downarrow\uparrow\downarrow\downarrow\rangle + |\uparrow\downarrow\downarrow\downarrow\rangle) \\
& \tfrac{1}{2}(|\downarrow\downarrow\downarrow\uparrow\rangle - |\downarrow\downarrow\uparrow\downarrow\rangle - |\downarrow\uparrow\downarrow\downarrow\rangle + |\uparrow\downarrow\downarrow\downarrow\rangle) \\
m = -2: \quad & |\downarrow\downarrow\downarrow\downarrow\rangle. \qquad (10.99)
\end{aligned}
$$

Using the fact that the spin-exchange terms interchange neighboring up and down spins ($\uparrow\downarrow \leftrightarrow \downarrow\uparrow$), while annihilating up-up or down-down combinations, and the magnetic field term simply counts the number of up spins minus the number of down spins, compute the energies of these states by finding the matrix representation for $\hat{H}_{XY}$ using the basis above. This basis incorporates the spatial symmetry to make computing the energies easier. Many of the states are already eigenstates, but if a state is not an eigenstate, then it will be coupled to other states, and one will need to diagonalize the sub-block matrix of the Hamiltonian. The worst matrix you will encounter is a 2×2 matrix.

You should find the following 16 energies: $-4B$, $-J-2B$, $J-2B$, $-2B$, $-2B$, $\sqrt{2}J$, $-\sqrt{2}J$, 0, 0, 0, 0, $-J+2B$, $J+2B$, $2B$, $2B$, and $4B$. You should find the ground state energies we computed in the main text agree with the ones found here.

Hint: Consider the state $|\downarrow\downarrow\downarrow\downarrow\rangle$. It has no up-down nearest-neighbor pairs, so the spin-exchange term annihilates against it. The magnetic field term counts the number up minus number down, so, we find the state is an eigenstate with an energy given by $4B$. Consider the state $\frac{1}{2}(|\downarrow\downarrow\downarrow\uparrow\rangle + |\downarrow\downarrow\uparrow\downarrow\rangle + |\downarrow\uparrow\downarrow\downarrow\rangle + |\uparrow\downarrow\downarrow\downarrow\rangle)$. It is also an eigenstate. The spin-exchange term maps the state to itself, so that

contributes a $-J$ to the eigenvalue, and the magnetic field term gives a $2B$ contribution, so this state is also an eigenstate, with $E = -J + 2B$.

10.5.4 $N = 6$ XY Model

Work out the energies for the $N = 6$ one-dimensional ring using fermionization. Compare your results to those shown in Fig. 10.17. In particular, make sure all of the level crossings that you predict are as seen in the figure.

10.5.5 Thermodynamic Limit with $B \neq 0$

Using the geometric summation trick we used in the main text to find the energy in the thermodynamic limit when $B = 0$, compute $E_{gs}(B)/N$ as a function of the magnetic field B. Take $-\pi \leq k < \pi$, and define $k_{\max} = \cos^{-1}\left(-\frac{2B}{J}\right)$. Take the limit $N \to \infty$, and show that

$$\frac{E_{gs}}{N} = -\frac{1}{\pi}\sqrt{J^2 - 4B^2} - \frac{2B}{\pi}\cos^{-1}\left(-\frac{2B}{J}\right) + B, \tag{10.100}$$

where $N_e/N = \frac{1}{\pi}\cos^{-1}\left(-\frac{2B}{J}\right)$.

10.5.6 Jordan-Wigner Strings

Write an essay that discusses why, in the cases of higher spatial dimensions, or longer-range spin-exchange interactions in one dimension, that the Jordan-Wigner strings do not cancel out anymore. Discuss further why this makes a solution via fermionization problematic, because we do not know how to properly handle the Jordan-Wigner strings.

10.5.7 Antiferromagnetic Case for the XY Model

Consider the antiferromagnetic case (nearest-neighbor spin-exchange on a one-dimensional ring with N sites), where $J < 0$. In the case where $B = 0$, you should be able to show that the two Neél states, given by $|\uparrow\downarrow\uparrow\downarrow\cdots\uparrow\downarrow\rangle$ and $|\downarrow\uparrow\downarrow\uparrow\cdots\downarrow\uparrow\rangle$ are the ground states for N even. When N is odd, there will be one site that has either two up spins next to each other or two down spins next to each

other. Since these defects can occur on any lattice site, show that the degeneracy of the ground state increases significantly for odd N.

If you are bold, work out the ground-state energy for $N = 4$ and for $N = 5$ as a function of the magnetic field B.

10.5.8 Why Fermionization Works

In this problem, we explore why a spin excitation does not create energy eigenstates, but a fermionic excitation does. The spin excitation operator in momentum space is given by

$$\sigma_k^+ = \frac{1}{\sqrt{N}} \sum_{j=1}^{N} \sigma_j^+ e^{ikj}, \tag{10.101}$$

and resembles the fermionic excitation

$$\hat{c}_k^\dagger = \frac{1}{\sqrt{N}} \sum_{j=1}^{N} \hat{c}_j^\dagger e^{ikj}. \tag{10.102}$$

The difference between these two operators can be seen more clearly if we compare

$$\left(\sigma_k^+\right)^2 | \downarrow\downarrow\downarrow \cdots \downarrow\rangle \text{ to } \left(\hat{c}_k^\dagger\right)^2 |0\rangle. \tag{10.103}$$

Compute these operators acting on the respective states. Using the facts that $(\sigma_j^+)^2 = 0$ and $(\hat{c}_j^\dagger)^2 = 0$ determine why the fermionic state cannot be excited twice but the spin state can. This difference summarizes why the fermionization allows the XY model to be solved, while applying many spin excitation operators does not create energy eigenstates. See if you can explain why this is so.

10.5.9 Frustration

Consider the $N = 3$ periodic chain with $J < 0$. This is a frustrated antiferromagnet. Determine all eight energy levels. Determine the degeneracy of the ground state. Often a frustrated antiferromagnet will have degenerate levels either for the ground state or for the low-lying excited states. Use either the fermionization method, or directly work with the spin representation to solve this problem.

10.6 The Transverse-Field Ising Model

We examine the so-called transverse-field Ising model, where the magnetic field does not point in the direction of the spins that interact via spin exchange. Similar to the XY model, we choose the spin exchange between spin components lying in the x direction and the field in the z direction. It can also be thought of as the XY model without the Y components:

$$H_{\text{TF-Ising}} = -J \sum_{i=1}^{N} \sigma_i^x \sigma_{i+1}^x - B \sum_{i=1}^{N} \sigma_i^z. \tag{10.104}$$

We now replace $\sigma_i^x \to \frac{1}{2}(\sigma_i^+ + \sigma_i^-)$ to find

$$H_{\text{TF-Ising}} = -\frac{J}{4} \sum_{i=1}^{N} \left(\sigma_i^+ \sigma_{i+1}^+ + \sigma_i^+ \sigma_{i+1}^- + \sigma_i^- \sigma_{i+1}^+ + \sigma_i^- \sigma_{i+1}^-\right) - B \sum_{i=1}^{N} \sigma_i^z. \tag{10.105}$$

Note how now we have $\sigma_i^+ \sigma_{i+1}^+$ and $\sigma_i^- \sigma_{i+1}^-$ terms, which cancelled for the XY model.

We will fermionize this model soon, but before we do, we want to discuss the symmetry that this model has. Similar to the XY model on a periodic chain, we have translations (rotational) symmetry, where shifting all lattice sites $i \to i + n$ does not change the Hamiltonian. Experts anticipate that this means we will use a Fourier transformation again to solve the problem (corresponding to the change from position space to momentum space). We also have the spin-reflection-parity symmetry, where $\sigma^x \to -\sigma^x$, $\sigma^y \to -\sigma^y$, and $\sigma^z \to \sigma^z$. The parity of eigenstates with respect to this symmetry will be important in discussing certain types of entangled states, but only when expressed in the x or y basis. We also can look at the spin parity symmetry $\hat{\Pi}_y$ again. It shows again that the set of energy eigenvalues for positive B is the same as the set of energy eigenvalues for negative B.

We are now ready for fermionization, which will take three technical steps. The first step uses the fermionization in real space, given by

$$\sigma_i^+ \to 2e^{i\pi \sum_{j=1}^{i-1} \hat{n}_j} \hat{c}_i^\dagger \quad \text{and} \quad \sigma_i^- \to 2e^{-i\pi \sum_{j=1}^{i-1} \hat{n}_j} \hat{c}_i \,. \tag{10.106}$$

Substituting into $\hat{H}_{\text{TF-Ising}}$ yields

$$\begin{aligned} \hat{H}_{\text{TF-Ising}} = -J \sum_{i=1}^{N-1} \left(\hat{c}_i^\dagger e^{i\pi \hat{n}_i} \hat{c}_{i+1}^\dagger + \hat{c}_i^\dagger e^{-i\pi \hat{n}_i} \hat{c}_{i+1} + \hat{c}_i \, e^{i\pi \hat{n}_i} \hat{c}_{i+1}^\dagger + \hat{c}_i \, e^{-i\pi \hat{n}_i} \hat{c}_{i+1}\right) \\ - J \left(\hat{c}_N^\dagger e^{i\pi(\hat{N}_e - \hat{n}_N)} \hat{c}_1^\dagger + \hat{c}_N^\dagger e^{i\pi(\hat{N}_e - \hat{n}_N)} \hat{c}_1 + \hat{c}_N e^{-i\pi(\hat{N}_e - \hat{n}_N)} \hat{c}_1^\dagger \right. \end{aligned}$$

$$+\,\hat{c}_N e^{-i\pi(\hat{N}_e-\hat{n}_N)}\hat{c}_1\Big) - B\sum_{i=1}^{N}(2\hat{n}_i-1), \tag{10.107}$$

where we use the fact that $e^{\pm i2\pi\hat{n}_j} = \hat{\mathbb{I}}$, because the number operator has integer eigenvalues. Now we have to do some technical manipulations. First, we note that $\hat{n}^2 = \hat{c}^\dagger\hat{c}\hat{c}^\dagger\hat{c} = \hat{c}^\dagger\{\hat{c},\hat{c}^\dagger\}_+\hat{c} = \hat{c}^\dagger\hat{c} = \hat{n}$. This then means that

$$e^{i\pi\hat{n}} = \sum_{m=1}^{\infty}\frac{(i\pi)^m}{m!}\hat{n}^m = 1 + \left(e^{i\pi}-1\right)\hat{n} = e^{-i\pi\hat{n}}. \tag{10.108}$$

This means that $e^{\pm i\pi\hat{n}}\hat{c}^\dagger = (1-2\hat{n})\hat{c}^\dagger = -\hat{c}^\dagger$ and $e^{\pm i\pi\hat{n}}\hat{c} = (1-2\hat{n})\hat{c} = \hat{c}$. We use these two identities in simplifying the Hamiltonian to

$$\begin{aligned}
\hat{H}_{\text{TF-Ising}} &= -J\sum_{i=1}^{N-1}\left(\hat{c}_i^\dagger\hat{c}_{i+1}^\dagger + \hat{c}_i^\dagger\hat{c}_{i+1} - \hat{c}_i\,\hat{c}_{i+1}^\dagger - \hat{c}_i\,\hat{c}_{i+1}\right)\\
&\quad - J\left(-\hat{c}_N^\dagger\hat{c}_1^\dagger - \hat{c}_N^\dagger\hat{c}_1 + \hat{c}_N\hat{c}_1^\dagger + \hat{c}_N\hat{c}_1\right)e^{i\pi\hat{N}_e} - B\sum_{i=1}^{N}(2\hat{n}_i-1)\\
&= -J\sum_{i=1}^{N-1}\left(\hat{c}_i^\dagger\hat{c}_{i+1}^\dagger + \hat{c}_i^\dagger\hat{c}_{i+1} - \hat{c}_i\,\hat{c}_{i+1}^\dagger - \hat{c}_i\,\hat{c}_{i+1}\right)\\
&\quad + J\left(\hat{c}_N^\dagger\hat{c}_1^\dagger + \hat{c}_N^\dagger\hat{c}_1 - \hat{c}_N\hat{c}_1^\dagger - \hat{c}_N\hat{c}_1\right)e^{i\pi\hat{N}_e} - B\sum_{i=1}^{N}(2\hat{n}_i-1).
\end{aligned} \tag{10.109}$$

Because $e^{i\pi\hat{N}_e}$ commutes with the product of any two fermionic operators (creation, or annihilation, or a mixed product), the operators $\hat{P}$ and $\hat{Q}$, which project onto the even and odd parity fermion sectors, also commute with the Hamiltonian. So, similar to the XY model, we have the decomposition $\hat{H}_{\text{TF-Ising}} = \hat{P}\hat{H}_+\hat{P} + \hat{Q}\hat{H}_-\hat{Q}$, with

$$\begin{aligned}
\hat{H}_\pm &= -J\sum_{i=1}^{N-1}\left(\hat{c}_i^\dagger\hat{c}_{i+1}^\dagger + \hat{c}_i^\dagger\hat{c}_{i+1} - \hat{c}_i\,\hat{c}_{i+1}^\dagger - \hat{c}_i\,\hat{c}_{i+1}\right)\\
&\quad \pm J\left(\hat{c}_N^\dagger\hat{c}_1^\dagger + \hat{c}_N^\dagger\hat{c}_1 - \hat{c}_N\hat{c}_1^\dagger - \hat{c}_N\hat{c}_1\right) - B\sum_{i=1}^{N}(2\hat{n}_i-1).
\end{aligned} \tag{10.110}$$

This means we will use periodic boundary conditions when the number of fermions is odd and antiperiodic boundary conditions when the number of fermions is even, just as we had before. The way this is implemented is to simply solve for the

Hamiltonian using the appropriate boundary conditions depending on the number of fermions.

Now, the second step is to go to momentum space. We use the same definitions of the momentum-space fermions and the same k choices for periodic or antiperiodic boundary conditions as before. Then we find that

$$\begin{aligned}\hat{H}_{\pm} = &-\frac{J}{N}\sum_{j=1}^{N}\sum_{kk'}\Big(e^{-i(k+k')j}\hat{c}_k^{\dagger}\hat{c}_{k'}^{\dagger}e^{-ik'} + e^{-i(k-k')j}\hat{c}_k^{\dagger}\hat{c}_{k'}e^{ik'} \\ &+ e^{-i(k-k')j}\hat{c}_k^{\dagger}\hat{c}_{k'}e^{-ik} + e^{i(k+k')j}\hat{c}_k\hat{c}_{k'}e^{ik}\Big) \\ &- \frac{B}{N}\sum_{j=1}^{N}\sum_{kk'}\left(e^{-i(k-k')j}2\hat{c}_k^{\dagger}\hat{c}_{k'} - \frac{1}{N}\right). \end{aligned} \tag{10.111}$$

The sum over momentum corresponds to the allowed momenta for periodic or antiperiodic boundary conditions, corresponding to which projected Hamiltonian we are working with. The sum over j gives $N\delta_{k,-k'}$ for the first and fourth terms, while it gives $N\delta_{kk'}$ for the second, third, and fifth terms. In both cases, we can then sum over k' as well because of the Kronnecker delta function. This gives

$$\begin{aligned}\hat{H}_{\pm} = &-J\sum_{k}{}'\left(\hat{c}_k^{\dagger}\hat{c}_{-k}^{\dagger}e^{ik} + \hat{c}_k^{\dagger}\hat{c}_k e^{ik} + \hat{c}_k^{\dagger}\hat{c}_k e^{-ik} + \hat{c}_{-k}\hat{c}_k e^{-ik}\right) \\ &- 2(J+B)\hat{c}_0^{\dagger}\hat{c}_0 + 2(J-B)\hat{c}_{\pi}^{\dagger}\hat{c}_{\pi} - B\sum_{k}{}'(2\hat{c}_k^{\dagger}\hat{c}_k - 1). \end{aligned} \tag{10.112}$$

The $-k$ denotes the shifted momentum term, where $k + k' = 2\pi$, for the original set of momenta that are all positive, but which can be also thought of as k and $-k$ pairs; for simplicity, we will use k and $-k$ now instead of k and $2\pi - k$—the two situations are completely equivalent, but the notations is much simpler with the former method. We have separated out the special points $k = 0$ and $k = \pi$, which appear in the odd parity sector only—the primes on the summations means that we do not include those points. Note that these special terms do not enter in the even parity sector (and the sums are over all momentum points in that case). The two middle terms look familiar, but the double creation and double annihilation operators mean that we have not yet diagonalized the Hamiltonian.

We now play some index gymnastics to make the expression for the Hamiltonian a bit simpler. For the middle two terms in the first sum, we see that we can combine the exponentials as $e^{ik} + e^{-ik} = 2\cos k$ and then, because we can switch $k \to -k$ in the sum, and we can replace the $2\hat{c}_k^{\dagger}\hat{c}_k \to \hat{c}_k^{\dagger}\hat{c}_k + \hat{c}_{-k}^{\dagger}\hat{c}_{-k}$. For the double creation and annihilation terms, we write $\hat{c}_k^{\dagger}\hat{c}_{-k}^{\dagger} = \frac{1}{2}(\hat{c}_k^{\dagger}\hat{c}_{-k}^{\dagger} - \hat{c}_{-k}^{\dagger}\hat{c}_k^{\dagger})$ and then let $k \to -k$ in the second term, which is equivalent to $e^{ik} \to i\sin k$ and similarly for the annihilation operators. Hence, we have

Fig. 10.18 Nikolay Bogoliubov in the late 1950s. (Image courtesy of the Emilo Segrè Visual Archive, Physics Today Collection. Image converted to look like a sketch by Jim Freericks)

$$\hat{H}_{\pm} = -J \sum_{k}{}' \left(\hat{c}_k^{\dagger} \hat{c}_{-k}^{\dagger} i \sin k + (\hat{c}_k^{\dagger} \hat{c}_k + \hat{c}_{-k}^{\dagger} \hat{c}_{-k}) \cos k - \hat{c}_{-k} \hat{c}_k i \sin k \right)$$
$$- 2(J+B) \hat{c}_0^{\dagger} \hat{c}_0 + 2(J-B) \hat{c}_{\pi}^{\dagger} \hat{c}_{\pi} - B \sum_{k}{}' (\hat{c}_k^{\dagger} \hat{c}_k + \hat{c}_{-k}^{\dagger} \hat{c}_{-k}) + BN. \tag{10.113}$$

This leaves us in a quandary. We have not yet diagonalized the Hamiltonian, except for the points where $k = 0$ and $k = \pi$ (which enter only when we have odd parity for the fermions). Now, by making the summations symmetric with respect to k and $-k$, we are ready for the third step of the fermionization—performing the so-called Bogoliubov transformation. This was first discovered by the Ukrainian physicist Nikolay Bogoliubov (see Fig. 10.18), and the transformation is often used in the theory of superconductivity (which we will not be able to discuss in this book). We encountered a variant of it already in a problem on the simple harmonic oscillator.

We now describe this third step. What we will do is redefine the fermion creation and annihilation operators in terms of a linear combination of the current creation and annihilation operators as a means to remove the pure quadratics in the creation and annihilation operators, respectively. It is a bit easier to show how this works first and then explain it. We form linear combinations of the old fermions to make new ones. The key is that we allow the new fermions to be mixtures of the old creation and annihilation operators. Namely, we take

$$\hat{d}_k^{\dagger} = \alpha_k^* \hat{c}_k^{\dagger} + \beta_{-k} \hat{c}_{-k} \quad \text{and} \quad \hat{d}_k = \alpha_k \hat{c}_k + \beta_{-k}^* \hat{c}_{-k}^{\dagger} \tag{10.114}$$

and for $k = 0, \pi$ we set $\alpha_k = 1$ and $\beta_{-k} = 0$. In order for this to be a fermion-to-fermion transformation, we insist that the transformation preserves the anticommutation relations of fermions, now in terms of the $\hat{d}$ operators. So, we

require

$$\{\hat{d}_k^\dagger, \hat{d}_{k'}^\dagger\}_+ = 0, \quad \{\hat{d}_k, \hat{d}_{k'}\}_+ = 0, \quad \text{and} \quad \{\hat{d}_k, \hat{d}_{k'}^\dagger\}_+ = \delta_{kk'}. \tag{10.115}$$

In the problems, you will verify that these anticommutation relations do hold as long as we have that

$$\alpha_k^* \beta_k + \alpha_{-k}^* \beta_{-k} = 0 \quad \text{and} \quad |\alpha_k|^2 + |\beta_{-k}|^2 = 1. \tag{10.116}$$

We parameterize the second equation using

$$\alpha_k = e^{i\phi_k} \cos\theta_k \quad \text{and} \quad \beta_{-k} = e^{i\phi_k'} \sin\theta_k, \tag{10.117}$$

which automatically satisfies the second equation (since $\cos^2\theta_k + \sin^2\theta_k = 1$). Then, the first relation becomes

$$e^{-i\phi_k} \cos\theta_k e^{i\phi'_{-k}} \sin\theta_{-k} + e^{-i\phi_{-k}} \cos\theta_{-k} e^{i\phi_k'} \sin\theta_k = 0, \tag{10.118}$$

which holds by choosing $\phi_{-k} = \phi_k$, $\phi'_{-k} = \phi_k'$, and $\theta_{-k} = -\theta_k$. Hence, the anticommutation relations are unchanged if we have $\alpha_k^* = e^{-i\phi_k} \cos\theta_k = \alpha_{-k}^*$ and $\beta_k^* = -e^{-i\phi_k'} \sin\theta_k = -\beta_{-k}^*$.

Using the freedom we have to choose the three angles, we substitute the transformation into the Hamiltonian, and we force the coefficients of the double creation and double annihilation operators to vanish. To do this, we first need the relationship between the $\hat{c}$ and $\hat{d}$ operators. We have

$$\begin{aligned}
\alpha_k \hat{d}_k^\dagger + \beta_k \hat{d}_{-k} &= \left(\alpha_k \alpha_k^* + \beta_k \beta_k^*\right) \hat{c}_k^\dagger + \left(\alpha_k \beta_{-k} + \beta_k \alpha_{-k}\right) \hat{c}_{-k} \\
&= \left(\cos^2\theta_k + \sin^2\theta_k\right) \hat{c}_k^\dagger \\
&\quad + \left(e^{i\phi_k + i\phi_k'} \cos\theta_k \sin\theta_k - e^{i\phi_k + i\phi_k'} \cos\theta_k \sin\theta_k\right) \hat{c}_{-k} \\
&= \hat{c}_k^\dagger.
\end{aligned} \tag{10.119}$$

Similarly, we find that $\alpha_k^* \hat{d}_k + \beta_k^* \hat{d}_{-k}^\dagger = \hat{c}_k$ by taking the Hermitian conjugate. Now, we can substitute into the transverse-field Ising model Hamiltonian, where we find

$$\begin{aligned}
\hat{H}_\pm = -J \sum_k{}' \Big[&\left(\alpha_k \hat{d}_k^\dagger + \beta_k \hat{d}_{-k}\right) \left(\alpha_{-k} \hat{d}_{-k}^\dagger + \beta_{-k} \hat{d}_k\right) i \sin k \\
&+ \Big\{ \left(\alpha_k \hat{d}_k^\dagger + \beta_k \hat{d}_k\right) \left(\alpha_k^* \hat{d}_k + \beta_k^* \hat{d}_{-k}^\dagger\right) \\
&+ \left(\alpha_{-k} \hat{d}_{-k}^\dagger + \beta_k \hat{d}_k\right) \left(\alpha_{-k}^* \hat{d}_{-k} + \beta_{-k}^* \hat{d}_k^\dagger\right) \Big\} \cos k
\end{aligned}$$

$$
\begin{aligned}
&- \left(\alpha^*_{-k}\hat{d}_{-k} + \beta^*_{-k}\hat{d}^\dagger_k\right)\left(\alpha^*_k\hat{d}_k + \beta^*_k\hat{d}^\dagger_{-k}\right) i\sin k\Big] \\
&- B\sum_k{}' \Big[\left(\alpha_k\hat{d}^\dagger_k + \beta_k\hat{d}_{-k}\right)\left(\alpha^*_k\hat{d}_k + \beta^*_k\hat{d}^\dagger_{-k}\right) \\
&+ \left(\alpha_{-k}\hat{d}^\dagger_{-k} + \beta_{-k}\hat{d}_k\right)\left(\alpha^*_{-k}\hat{d}_{-k} + \beta^*_{-k}\hat{d}^\dagger_k\right)\Big] + BN \\
&- 2(J+B)\hat{d}^\dagger_0\hat{d}_0 + 2(J-B)\hat{d}^\dagger_\pi\hat{d}_\pi, \qquad (10.120)
\end{aligned}
$$

where the last line only enters for the $\hat{H}_-$ operator. Now we need to combine the coefficients of all of the terms that involve the same operators. We note that we are allowed to change $k \to -k$ in the summations, and in some cases, the operators form anticommutators, which we can replace by their values. The algebra is a bit long, definitely a French-cooking exercise. Using the expression for the transformation coefficients in terms of the angles, we find the following results. The coefficient of the $\hat{d}^\dagger_k\hat{d}^\dagger_{-k}$ term is given by

$$
\begin{aligned}
&- J\left(\alpha_k\alpha_{-k} - \beta^*_{-k}\beta^*_k\right) i\sin k - J\left(\alpha_k\beta^*_k - \alpha_{-k}\beta^*_{-k}\right)\cos k \\
&\quad - B\left(\alpha_k\beta^*_k - \alpha_{-k}\beta^*_{-k}\right) \\
&= -J\left(e^{i2\phi_k}\cos^2\theta_k + e^{-i2\phi'_k}\sin^2\theta_k\right) i\sin k - 2Je^{-i\phi_k+i\phi'_k}\cos\theta_k\sin\theta_k\cos k \\
&\quad - 2Be^{i\phi_k-i\phi'_k}\cos\theta_k\sin\theta_k, \qquad (10.121)
\end{aligned}
$$

while the coefficient of the $\hat{d}_{-k}\hat{d}_k$ is

$$
\begin{aligned}
&- J\left(\beta_k\beta_{-k} - \alpha^*_{-k}\alpha^*_k\right) i\sin k - J\left(\beta_k\alpha^*_k - \beta_{-k}\alpha^*_{-k}\right)\cos k \\
&\quad - B\left(\beta_k\alpha^*_k - \beta_{-k}\alpha^*_{-k}\right) \\
&= -J\left(-e^{i2\phi_k}\sin^2\theta_k - e^{-i2\phi'_k}\cos^2\theta_k\right) i\sin k - 2Je^{-i\phi_k+i\phi'_k}\cos\theta_k\sin\theta_k\cos k \\
&\quad - 2Be^{-i\phi_k+i\phi'_k}\cos\theta_k\sin\theta_k. \qquad (10.122)
\end{aligned}
$$

We need both of these coefficients to simultaneously vanish. We first choose $\phi_k = \frac{\pi}{4} = \phi'_k$ in order to make all of the expressions real. Then, we require

$$
J\left(\cos^2\theta_k - \sin^2\theta_k\right)\sin k - (J\cos k + B)(2\cos\theta_k\sin\theta_k) = 0. \qquad (10.123)
$$

This is solved by

$$
\tan 2\theta_k = \frac{J\sin k}{J\cos k + B}. \qquad (10.124)
$$

Because $\tan 2\theta_k$ ranges from $-\infty$ to ∞, we can always find a θ_k that solves this for each allowed k value. But, we need to be careful with the ranges of the angles. If $k > 0$, then $\sin k > 0$ and for B large, θ_k is near 0. As B decreases toward 0, the angle increases. If $\cos k < 0$, then we reach a point where the denominator vanishes and $\theta_k = \frac{\pi}{2}$. By continuity, we want to ensure that θ_k becomes larger than $\frac{\pi}{2}$ in this case, which means the sine function is positive and the cosine function is negative. One analyzes the case $\sin k < 0$ in a similar way. In addition, the special points $k = 0,\ \pi$ have $\theta_k = 0$. The remaining terms can now be found with some tedious algebra. We have that the coefficient of the $\hat{d}_k^\dagger \hat{d}_k$ is

$$\begin{aligned}
&-2J\left(\alpha_k\beta_{-k} - \alpha_k^*\beta_{-k}^*\right) i\sin k - 2J\left(\alpha_k\alpha_k^* - \beta_{-k}\beta_{-k}^*\right)\cos k \\
&\quad - 2B\left(\alpha_k\alpha_k^* - \beta_{-k}\beta_{-k}^*\right) \\
&= J(-2\cos\theta_k\sin\theta_k)\sin k - J(\cos^2\theta_k - \sin^2\theta_k)\cos k - B(\cos^2\theta_k - \sin^2\theta_k) \\
&= -J\sin 2\theta_k\sin k - (J\cos k + B)\cos 2\theta_k. \qquad (10.125)
\end{aligned}$$

To simplify this, we use the expression for $\tan 2\theta_k$ in Eq. (10.124) to find that

$$\cos 2\theta_k = \pm\frac{1}{\sqrt{1+\tan^2\theta_k}} = \frac{J\cos k + B}{\sqrt{J^2 + 2JB\cos k + B^2}}, \qquad (10.126)$$

because of the continuity argument we made above, and

$$\sin 2\theta_k = \frac{J\sin k}{\sqrt{J^2 + 2JB\cos k + B^2}}. \qquad (10.127)$$

Note that the sign of $\cos\theta_k$ is determined by the sign of $J\cos k + B$, while the sign of the $\sin\theta_k$ term is determined by the sign of $\sin k$. We then find the coefficient for general k becomes

$$\frac{-J^2\sin^2 k - (J\cos k + B)^2}{\sqrt{J^2 + 2JB\cos k + B^2}} = -\sqrt{J^2 + 2JB\cos k + B^2}. \qquad (10.128)$$

The coefficient of the $\hat{d}_{-k}^\dagger \hat{d}_{-k}$ term is the same. For the special terms, we have the coefficient for $k = 0$ is $-2(J + B)$, while for $k = \pi$, it is $2(J - B)$. The last piece we need to compute is the constant term (arising from the anticommutators). For general k, it is given by

$$\begin{aligned}
&-J\left(\beta_k\alpha_{-k} - \alpha_{-k}^*\beta_k^*\right) i\sin k - J\left(\beta_k\beta_k^* + \beta_{-k}\beta_{-k}^*\right)\cos k \\
&\quad - B\left(\beta_k\beta_k^* + \beta_{-k}\beta_{-k}^*\right) + B \\
&= J(2\cos\theta_k\sin\theta_k)\sin k - J(2\sin^2\theta_k)\cos k - B(2\sin^2\theta_k) + B \\
&= \sqrt{J^2 + 2JB\cos k + B^2} - J\cos k, \qquad (10.129)
\end{aligned}$$

after using the fact that $\cos 2\theta_k - 1 = -2\sin^2\theta_k$. The last constant term vanishes when summed over k, which you will show in the problems. Note that for the odd parity case, there is a left over $2B$ in the constant term not accounted for, so we add it into the extra terms with $k = 0,\ \pi$. So, we end up with

$$\begin{aligned}\hat{H}_{\pm} &= -{\sum_k}' \sqrt{J^2 + 2JB\cos k + B^2}\left(\hat{d}_k^\dagger \hat{d}_k + \hat{d}_{-k}^\dagger \hat{d}_{-k} - 1\right) \\ &\quad - (J+B)\left(\hat{d}_0^\dagger \hat{d}_0 - \tfrac{1}{2}\right) + (J-B)\left(\hat{d}_\pi^\dagger \hat{d}_\pi - \tfrac{1}{2}\right) \\ &= -2{\sum_k}' \sqrt{J^2 + 2JB\cos k + B^2}\left(\hat{d}_k^\dagger \hat{d}_k - \tfrac{1}{2}\right) \\ &\quad - 2(J+B)\left(\hat{d}_0^\dagger \hat{d}_0 - \tfrac{1}{2}\right) + 2(J-B)\left(\hat{d}_\pi^\dagger \hat{d}_\pi - \tfrac{1}{2}\right). \end{aligned} \tag{10.130}$$

Well, that was a lot of work! Let us see how we use it.

Let us determine the eigenstates. There are a number of new issues we must deal with to work with this fermionized Hamiltonian. First, we construct a vacuum state, denoted by $|0\rangle$, which satisfies

$$\hat{d}_k|0\rangle = 0 \ \ \forall k, \tag{10.131}$$

which is a complicated state when expressed in terms of the original fermion states or the original spin states. Fortunately, for us, we will always assume such a vacuum state exists and construct our states from it. We do not need to know the relationship between the vacuum state and the original fermion states (or the original spin states) for anything we will be doing.

When we look at the Hamiltonian, it is important to note that the summation over the k values is determined by the number of fermions in the *original* fermions, not the ones after the Bogoliubov transformation. But, because the Hamiltonian mixes together states with even numbers of c fermions, the parity of the even c states is the same parity as the even d states and similarly for the odd parity. Hence, we can choose the parity based on an even or odd number of d fermions. Next, since the square root term is always positive, and the coefficient in front is negative, we want to occupy all of the d fermion states to find the ground state. This is the bathtub principle again—we occupy all negative energy states and no positive energy states, and then the energy is as negative as we can make it. Hence, the vacuum state ends up being the highest energy state of the energy spectrum. This means the ground state becomes

$$|\psi_{gs}\rangle = \prod_k \hat{d}_k^\dagger|0\rangle \ \text{ with } \ E_{gs}(B) = -\sum_k \sqrt{J^2 + 2JB\cos k + B^2}. \tag{10.132}$$

This formula assumes the ground state is in the even parity sector; a simple modification of this formula holds in the odd parity sector and in the large N limit

one need not worry about the differences. Unfortunately, we cannot perform the summation any more using our geometric series trick. We need calculus to do this, and it becomes an elliptic integral in the $N \to \infty$ limit.

While the energy appears to be a smooth function of B, one can show, via something called a *duality transformation*, that this model has a phase transition at $B = J$. We can describe what the phase transition looks like without discussing the duality transformation. Note that when the field is large, $B \gg J$, the ground state in the spin language is oriented opposite to the field, so $|\psi_{gs}\rangle \to |\uparrow\uparrow\uparrow\cdots\uparrow\rangle_z$ as $B/J \to \infty$. The subscript z indicates the state is written in the z basis (which is the direction the field is in). Note that as $B \to \infty$, we have $\theta_k \to 0$, so that $\hat{d}_k^\dagger = \frac{1+i}{\sqrt{2}}\hat{c}_k^\dagger$. This means that the $\hat{d}_k$ operator is actually proportional to the $\hat{S}_k^-$ operator (plus a Jordan-Wigner string), and this means that $|0\rangle = |\downarrow\downarrow\downarrow\cdots\downarrow\rangle_z$ and the ground state (up to an overall phase) becomes $|\psi_{gs}\rangle = |\uparrow\uparrow\uparrow\cdots\uparrow\rangle_z$. Note that this argument uses the fact that filling all up states in momentum space is the same as filling all up states in position space, as discussed in the problems. Now, let us investigate the limit $B \to 0$. Here, we have $\theta_k = k$, so that $\hat{d}_k^\dagger = \frac{1+i}{\sqrt{2}}(\cos k\, \hat{c}_k^\dagger - \sin k\, \hat{c}_{-k})$ and $\hat{d}_k = \frac{1+i}{\sqrt{2}}(\cos k\, \hat{c}_k - \sin k\, \hat{c}_{-k}^\dagger)$. This leads to a complex set of equations. If we go back to the original Hamiltonian, we see it is just the ferromagnetic Ising model in the x-direction, which has $|\uparrow\uparrow\uparrow\cdots\uparrow\rangle_x$ and $|\downarrow\downarrow\downarrow\cdots\downarrow\rangle_x$ as the two degenerate ground states. But, in the x-basis, parity is a good quantum number, so the two degenerate states that are also parity eigenstates are

$$\frac{1}{\sqrt{2}}\left(|\uparrow\uparrow\uparrow\cdots\uparrow\rangle_x \pm |\downarrow\downarrow\downarrow\cdots\downarrow\rangle_x\right). \tag{10.133}$$

In the problems, we investigate this relationship further. The phase transition involves the change of the states from the large B to the small B limits.

The exact solution by fermionization is an interesting technique for solving the nearest-neighbor transverse-field Ising model. Unfortunately, it cannot be extended to longer-range Ising model, which we discuss next (the problem is that the Jordan-Wigner strings cannot be gotten rid of, and we do not know how to work with them). The Hamiltonian becomes

$$\hat{H} = -\sum_{i>j}^{N} J_{ij}\sigma_i^x\sigma_j^x - B\sum_{i=1}^{N}\sigma_i^z, \tag{10.134}$$

with J_{ij} nonzero for all i and j. For long-range interactions, we usually have

$$J_{ij} = \frac{J}{|i-j|^\alpha}, \tag{10.135}$$

where J is an energy and $0 \le \alpha \le 3$ are the most commonly studied cases, because they are the cases most easily studied in ion-trap quantum simulators. The limit $\alpha \to$

0 has an equal interaction between all spins and is called the all-to-all interaction. This is discussed in Sect. 10.7. The application to ion traps is discussed in Chap. 21.

We end the section with a discussion of a small cluster calculation, similar to what we did for the XY model. We will work with a 4-site cluster and find the 16 energy levels. Recall, for $N_e = 0$, 2, and 4, we use $k = \pm\frac{\pi}{4}$ and $\pm\frac{3\pi}{4}$. While for $N_e = 1$ and 3, we use $k = 0$, $\pm\frac{\pi}{2}$, and π. Note that since the number of fermions is given by the d fermions, they no longer correspond to a definite value for the z component of spin, as they did with the XY model. The energies are given by summing the individual energies of the fermions multiplied by $\pm\frac{1}{2}$ depending on whether we have a fermion or we have no fermion. Let us work out the energies $-\sqrt{J^2 + 2B\cos k + B^2}$ for these k values. We find the following results for $B > 0$ (after recalling the special results for $k = 0$ and $k = \pi$ which are not always given by the negative square root)

$$-\sqrt{J^2 + 2B\cos k + B^2} = \begin{cases} -\sqrt{J^2 + \sqrt{2}JB + B^2} & \text{for } k = \pm\frac{\pi}{4} \\ -\sqrt{J^2 - \sqrt{2}JB + B^2} & \text{for } k = \pm\frac{3\pi}{4} \\ -(J + B) & \text{for } k = 0 \\ -\sqrt{J^2 + B^2} & \text{for } k = \pm\frac{\pi}{2} \\ (J - B) & \text{for } k = \pi. \end{cases} \tag{10.136}$$

To determine the different energy levels, we pick the number of occupied energy values, and then sum up all of the terms using the allowed k values—do not forget that there are constant terms in the Hamiltonian that need to be added in as well. We find the following results when $B > 0$. For $N_e = 0$, we have $-2\sqrt{J^2 + \sqrt{2}JB + B^2} - 2\sqrt{J^2 - \sqrt{2}JB + B^2}$. For $N_e = 1$, there are four energies given by $-2J - 2\sqrt{J^2 + B^2}$, $-2B$, $-2B$, and $2J - 2\sqrt{J^2 + B^2}$. For $N_e = 2$, there are six levels: $-2\sqrt{J^2 + \sqrt{2}JB + B^2} + 2\sqrt{J^2 - \sqrt{2}JB + B^2}$, 0, 0, 0, 0, and $2\sqrt{J^2 + \sqrt{2}JB + B^2} - 2\sqrt{J^2 - \sqrt{2}JB + B^2}$. For $N_e = 3$, we have four levels again, given by $-2J + 2\sqrt{J^2 + B^2}$, $2B$, $2B$, $2J + 2\sqrt{J^2 + B^2}$. Finally, for $N_e = 4$, we have $2\sqrt{J^2 + \sqrt{2}JB + B^2} + 2\sqrt{J^2 - \sqrt{2}JB + B^2}$. As expected, the energy levels come in $\pm$ pairs. You should check that you can reproduce the calculation of these energy levels yourself, by occupying the relevant energy levels. Note that there is no clear signal of the phase transition in small-cluster calculations, but if you plot the energy levels, you will see that there is a nondegenerate ground state for $B > J$, but two different parity states become nearly degenerate for $B < J$. This is the precursor of the quantum phase transition in the thermodynamic limit.

As a check on our calculations, we can determine the energy levels for two states fairly quickly in the spin representation. Using

$$|\psi\rangle = \frac{1}{2}(|\downarrow\uparrow\uparrow\uparrow\rangle_z + |\uparrow\downarrow\uparrow\uparrow\rangle_z + |\uparrow\uparrow\downarrow\uparrow\rangle_z + |\uparrow\uparrow\uparrow\downarrow\rangle_z)$$

$$|\phi\rangle = \frac{1}{2}(|\uparrow\downarrow\downarrow\downarrow\rangle_z + |\downarrow\uparrow\downarrow\downarrow\rangle_z + |\downarrow\downarrow\uparrow\downarrow\rangle_z + |\downarrow\downarrow\downarrow\uparrow\rangle_z), \tag{10.137}$$

we can compute the Hamiltonian matrix as

$$\hat{H}_{\text{TF-Ising}} = \begin{pmatrix} -2J - 2B & -2J \\ -2J & -2J + 2B \end{pmatrix}, \tag{10.138}$$

which follows from the direct application of $\hat{H}$ onto each state and then computing the overlap with these two basis states. For example, the magnetic field term just counts the number of up spins minus the number of down, multiplied by $-B$, so it is diagonal. The spin-exchange term, involves an interchanging of neighboring spins (which map each state to itself, resulting in the $-2J$ diagonal terms, since each interchangeable spin has two neighbors). It also has the terms that flip two neighboring spins and these map the two states to each other, producing the off-diagonal terms (also with a factor of 2 due to the two neighbors). The energy is found in the standard way, and it satisfies $E^2 + 4JE + 4J^2 - 4B^2 - 4J^2 = 0$. This leads to $E = -2J \pm 2\sqrt{J^2 + B^2}$, which are precisely two of the solutions with $N_e = 1$ and $N_e = 3$ from our fermionic solution. While not an extensive test, this is a good check that the algebra is working.

Let us now solidify your knowledge of this material by trying some additional problems.

Section 10.6 Problems:

10.6.1 Details of the Bogoliubov Transformation

Using the definitions in Eq. (10.114), prove that the fundamental anticommutation relations $\{\hat{d}_k^\dagger, \hat{d}_{k'}^\dagger\}_+ = 0$, $\{\hat{d}_k, \hat{d}_{k'}\}_+ = 0$, and $\{\hat{d}_k, \hat{d}_{k'}^\dagger\}_+ = \delta_{kk'}$ require $\alpha_k^* \beta_k + \alpha_{-k}^* \beta_{-k} = 0$ and $|\alpha_k|^2 + |\beta_{-k}|^2 = 1$. Recall that the definitions for $k = 0$ and $k = \pi$ are fixed at the beginning.

10.6.2 Show that the $\cos k$ Constant Term Vanishes

Prove that $\sum_{j=1}^{N} \cos \frac{2\pi j}{N} = 0$. It is easiest to proceed by rewriting the expression in terms of Re e^{ix} and then evaluating the sum of the exponentials via the geometric series.

10.6.3 Relationship Between Momentum-Space and Position-Space Fermion States

In this problem, you are to prove that

$$\left(\prod_k \hat{c}_k^\dagger\right)|0\rangle \propto \left(\prod_{j=1}^{N} \hat{c}_j^\dagger\right)|0\rangle, \tag{10.139}$$

where $\hat{c}_j^\dagger = \frac{1}{\sqrt{N}} \sum_k e^{-ikj} \hat{c}_k^\dagger$, with the k values corresponding to either periodic, or antiperiodic boundary conditions.

Hint: Use the result that the square of a fermionic creation operator is equal to 0 liberally in your arguments. You do not need to get the constant factor in front correct, just argue that it must be proportional to this form.

10.6.4 $N = 2$ Small-Cluster Calculation (Spin Language)

Consider the two-site transverse-field Ising model in the spin language, given by

$$\hat{H}_2 = -2J\sigma_1^x\sigma_2^x - B(\sigma_1^z + \sigma_2^z) \tag{10.140}$$

The factor of 2 is introduced to count the fact that the ring connects site 1 to 2 and then also from 2 to 1.

For this problem, we find it is best to work in the x basis in position space. We have four states, which we arrange into parity eigenstates via

$$\begin{aligned}
|1\rangle &= \tfrac{1}{\sqrt{2}}(|\uparrow\uparrow\rangle_x + |\downarrow\downarrow\rangle_x) \\
|2\rangle &= \tfrac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle_x + |\downarrow\uparrow\rangle_x) \\
|3\rangle &= \tfrac{1}{\sqrt{2}}(|\uparrow\uparrow\rangle_x - |\downarrow\downarrow\rangle_x) \\
|4\rangle &= \tfrac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle_x - |\downarrow\uparrow\rangle_x).
\end{aligned} \tag{10.141}$$

The interaction due to the spin exchange is simple here, because each state is an eigenstate of the spin-exchange term (this is why we work in the x-basis). The eigenvalues are as follows: for $|1\rangle$ and $|3\rangle$, it is $-2J$ and for $|2\rangle$ and $|4\rangle$, it is $2J$. The magnetic field is more complicated, because the σ^z matrix in the x-basis is given by

$${}_x\langle m'|\sigma^z|m\rangle_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \tag{10.142}$$

the B-field term couples the states $|1\rangle$ and $|2\rangle$ together and also the states $|3\rangle$ and $|4\rangle$ are coupled together. Construct the 2×2 matrices for the small cluster in the even and odd sectors. You should find the odd sector is diagonal, while the even sector is not. Determine all four eigenvalues.
Ans. $\pm 2J$ and $\pm 2\sqrt{J^2 + B^2}$.

10.6.5 $N = 2$ Small-Cluster Calculation (Fermionic Language)

Let us solve the same problem in the fermionic language. You should be able to verify that the fermionic Hamiltonian is given by

$$\hat{H}_2 = -2J(\hat{c}_1^\dagger \hat{c}_2^\dagger + \hat{c}_1^\dagger \hat{c}_2 + \hat{c}_2^\dagger \hat{c}_1 + \hat{c}_2 \hat{c}_1) - 2B(\hat{c}_1^\dagger \hat{c}_1 + \hat{c}_2^\dagger \hat{c}_2 - 1). \tag{10.143}$$

Convert the Hamiltonian into momentum space using $\hat{c}_0 = \frac{1}{\sqrt{2}}(\hat{c}_1 + \hat{c}_2)$ and $\hat{c}_\pi = \frac{1}{\sqrt{2}}(\hat{c}_1 - \hat{c}_2)$. We have four energy eigenstates to find. Rather than doing the Bogoliubov transformation, you should be able to verify that $\hat{c}_0^\dagger|0\rangle$ and $\hat{c}_\pi^\dagger|0\rangle$ are both eigenstates and that the state $|0\rangle$ is coupled to the state $\hat{c}_0^\dagger \hat{c}_\pi^\dagger|0\rangle$. Determine the 2×2 matrix, and diagonalize it to find the remaining two eigenvalues. You should find the same answers as in the previous problem, because the language we use does not change the eigenvalues.

10.6.6 Exploring $N = 2$ Small-Cluster Parity States

We saw from the spin representation that the ground state when $B = 0$ is degenerate and consisted of definite parity states given by

$$\frac{1}{\sqrt{2}}(|\uparrow\uparrow\uparrow\cdots\uparrow\rangle_x \pm |\downarrow\downarrow\downarrow\cdots\downarrow\rangle_x). \tag{10.144}$$

In this problem, we will investigate this from the fermionic perspective. In the two-site case, both

$$\hat{c}_0^\dagger|0\rangle \text{ and } \frac{1}{\sqrt{2}}(|0\rangle - \hat{c}_0^\dagger \hat{c}_\pi^\dagger|0\rangle) \tag{10.145}$$

are the two ground states. Using the fact that $|0\rangle = |\downarrow\downarrow\rangle_z$ in position space (down spin on both sites in the z-direction), and that

$$\hat{c}_0^\dagger|0\rangle = \tfrac{1}{2}(\sigma_1^+ + \sigma_2^+)|\downarrow\downarrow\rangle_z. \tag{10.146}$$

Show that

$$\frac{1}{\sqrt{2}}(|0\rangle - \hat{c}_0^\dagger \hat{c}_\pi^\dagger |0\rangle) \propto \frac{1}{\sqrt{2}}(|\uparrow\uparrow\rangle_x + |\downarrow\downarrow\rangle_x). \tag{10.147}$$

These are the two parity-related all up plus or minus all down states (in the x-direction) that we expect. But we can see the relationship between the fermionic, and the spin representations is complicated.
Hint: recall that $\hat{c}_1^\dagger = \frac{1}{2}\sigma_1^+$ and $\hat{c}_2^\dagger = \frac{1}{2}(-1)^{(\sigma_1^z+1)}\sigma_2^+$, since we must include the Jordan-Wigner string. Also recall that $|\uparrow\rangle_z = \frac{1}{\sqrt{2}}(|\uparrow\rangle_x + |\downarrow\rangle_x)$ and $|\downarrow\rangle_z = \frac{1}{\sqrt{2}}(|\uparrow\rangle_x - |\downarrow\rangle_x)$.

10.7 Model

In 1965, the Lipkin-Meshkov-Glick model was introduced as a simplified model for nuclear physics (Lipkin et al., 1965). In this section, we discuss a special case of that model, one described by a transverse-field Ising model in another limit, where it can be exactly solved, namely, the limit where every spin interacts equally with all other spins. The Hamiltonian becomes

$$\hat{H}_{\text{LMG}} = -\sum_{i=1}^{N}\sum_{j=1}^{N} \tilde{J}_{ij}\hat{S}_i^z\hat{S}_j^z - B\sum_{i=1}^{N}\hat{S}_i^x. \tag{10.148}$$

There are a few important points to make about this model.

1. The Hamiltonian is written in the language where the spin exchange is along the z direction and the field along the x direction. This is different from what we used before, but the requirement for this model is only that they are perpendicular to each other.
2. We added in terms where $i = j$, because $\left(\hat{S}_i^z\right)^2 = \frac{\hbar^2}{4}\mathbb{I}$ is proportional to the identity operator and can be added without loss of generality.
3. We pick $\tilde{J}_{ij} = \frac{J}{N}$, so that the energy scales with the number of lattice sites N.

The last point is a subtle one, which requires more explanation. In physical systems, when the system is made larger, the energy is expected to grow linearly with the system size (we say the energy is an *extensive* quantity). For example, in the Ising model, we studied earlier, with periodic boundary conditions on a one-dimensional chain of length N, we found that the energy of the ferromagnetic ground state was equal to $-2JN\hbar^2/4$, because we have $2N$ pairs of spins with bonds between them, and each bond contributes $-J$ to the total energy. This energy grows linearly with N as we claimed it must. But for the all-to-all model, we have $N(N-1)$ bonds (not counting $i = j$), so the energy becomes $-\tilde{J}N(N-1)\hbar^2/4$, which

grows quadratically in N. If we choose $\tilde{J} = \frac{J}{N}$, then the energy will depend linearly on N as we require for a physical system. Let us pause for a moment to think this through. As N increases, the interaction $\frac{J}{N}$ goes to zero, while the number of spins increases like N. This scenario of many spins interacting and the interaction becoming infinitesimal and between all spins is the situation that we call a *mean-field theory*. This is often the first approximation used to describe a complex quantum system. In the Lipkin-Meshkov-Glick model case, the complex model is ultimately described exactly by a mean field theory!

Using an equal interaction between all spins, $\tilde{J}_{ij} = \frac{J}{N}$, we rewrite the Hamiltonian as

$$\hat{H}_{\text{LMG}} = -\frac{J}{N}\left(\hat{S}^z\right)^2 - B\hat{S}^x, \tag{10.149}$$

where $\hat{S}^\alpha = \sum_{i=1}^N \hat{S}_i^\alpha$ is the total spin operator in the α direction. Written in this form, we immediately see that the total spin $\hat{S}^2 = \left(\hat{S}^x\right)^2 + \left(\hat{S}^y\right)^2 + \left(\hat{S}^z\right)^2$ is a conserved quantity, because $[\hat{H}_{\text{LMG}}, \hat{S}^2] = 0$. This follows, simply because $[\hat{S}^\alpha, \hat{S}^2] = 0$ for all α. As a result, we can examine the solutions for each possible value of the total spin, which ranges from 0 (or $\frac{1}{2}$) up to $\frac{N}{2}$; the minimum value is 0 for even N and is $\frac{1}{2}$ for odd N.

The original Lipkin model is

$$\hat{H}_{\text{Lipkin}} = -\frac{J}{2N}\left[\left(\hat{S}^+\right)^2 + \left(\hat{S}^-\right)^2\right] - B\hat{S}^z. \tag{10.150}$$

This is closely related to the all-to-all transverse-field Ising model. Note that

$$\begin{aligned}\left(\hat{S}^+\right)^2 &= \sum_{i=1}^N \sum_{j=1}^N \left(\hat{S}_i^x + i\hat{S}_i^y\right)\left(\hat{S}_j^x + i\hat{S}_j^y\right) \\ &= \sum_{i=1}^N \sum_{j=1}^N \left(\hat{S}_i^x\hat{S}_j^x + i\hat{S}_i^x\hat{S}_j^y + i\hat{S}_i^y\hat{S}_j^x - \hat{S}_i^y\hat{S}_j^y\right),\end{aligned} \tag{10.151}$$

and

$$\begin{aligned}\left(\hat{S}^-\right)^2 &= \sum_{i=1}^N \sum_{j=1}^N \left(\hat{S}_i^x - i\hat{S}_i^y\right)\left(\hat{S}_j^x - i\hat{S}_j^y\right) \\ &= \sum_{i=1}^N \sum_{j=1}^N \left(\hat{S}_i^x\hat{S}_j^x - i\hat{S}_i^x\hat{S}_j^y - i\hat{S}_i^y\hat{S}_j^x - \hat{S}_i^y\hat{S}_j^y\right).\end{aligned} \tag{10.152}$$

This yields

$$\hat{H}_{\text{Lipkin}} = -\frac{J}{N}\left[\left(\hat{S}^x\right)^2 - \left(\hat{S}^y\right)^2\right] - B\hat{S}^z, \tag{10.153}$$

which is a specific limit of another more general Hamiltonian, given by

$$\hat{H} = -\frac{J}{N}\left[\left(\hat{S}^x\right)^2 + \lambda\left(\hat{S}^y\right)^2\right] - B\hat{S}^z, \tag{10.154}$$

with $-1 \leq \lambda \leq 1$. The case $\lambda = -1$ is the original Lipkin model, and the case $\lambda = 0$ is the transverse-field Ising model (we would relabel $x \leftrightarrow z$ to put into our standard form). The case of arbitrary λ is often called the Lipkin-Meshkov-Glick model.

Our focus here is on the all-to-all transverse-field Ising model (which is also called the single-axis twisting model). We work in the form with the exchange in the z-direction and the field in the x-direction. There is one more symmetry this model has, which is the spin-reflection parity symmetry, which we have worked with before. In this symmetry, we take $\hat{S}^x \to \hat{S}^x$, $\hat{S}^y \to -\hat{S}^y$, and $\hat{S}^z \to -\hat{S}^z$. It corresponds to a rotation by π about the x-axis. The details for how this works are given in Problem 10.7.1. Because this is a parity symmetry, the eigenstates can be organized to have definite spin-reflection parity (even or odd) when represented in the z-basis.

We now examine some limits where the eigenfunctions can be immediately found. If $J = 0$ ($\frac{B}{J\hbar} \to \infty$), the Hamiltonian approaches $\hat{H} = -B\hat{S}^x$, which has the state with all spins aligned along the $+x$-axis as the ground state:

$$|\psi_{gs}\rangle = |1, \uparrow\rangle_x \otimes |2, \uparrow\rangle_x \otimes \cdots \otimes |N, \uparrow\rangle_x, \tag{10.155}$$

or, because $|\uparrow\rangle_x = \frac{1}{\sqrt{2}}(|\uparrow\rangle_z + |\downarrow\rangle_z)$, we also have that

$$\begin{aligned} |\psi_{gs}\rangle &= \frac{1}{2^{\frac{N}{2}}}(|1, \uparrow\rangle_z + |1, \downarrow\rangle_z) \otimes (|2, \uparrow\rangle_z + |2, \downarrow\rangle_z) \otimes \cdots \otimes (|N, \uparrow\rangle_z + |N, \downarrow\rangle_z) \\ &= \frac{1}{2^{\frac{N}{2}}} \sum_{m_1,\ldots,m_N} |m_1, m_2, \ldots m_N\rangle_z, \end{aligned} \tag{10.156}$$

which is an even homogeneous superposition of every possible product state in the z-basis. The ground-state energy is $E_{gs}\left(\frac{B}{J\hbar} \to \infty\right) = -B\hbar\frac{N}{2}$. This ground state is even parity and nondegenerate.

The other simple limit is $B \to 0$, where the ground state is either $|\uparrow\uparrow\uparrow\cdots\uparrow\rangle_z$ or $|\downarrow\downarrow\downarrow\cdots\downarrow\rangle_z$, or any linear combination of these two states. The even and odd parity solutions are

$$\frac{1}{\sqrt{2}}\left(|\uparrow\uparrow\uparrow\cdots\uparrow\rangle_z \pm |\downarrow\downarrow\downarrow\cdots\downarrow\rangle_z\right), \tag{10.157}$$

respectively. The ground-state energy is

$$E_{gs}\left(\frac{B}{J\hbar}=0\right)=-\frac{J}{N}\frac{\hbar^2}{4}N^2=-J\frac{\hbar^2}{4}N. \tag{10.158}$$

This ground state is degenerate and has both even and odd parity solutions.

As J is increased from 0 to $J\hbar\approx B$, to $J\hbar\gg B$, the ground state changes from being oriented along the x-axis to being an entangled superposition of all up and all down along the z-axis. Since the ground state is degenerate at $B=0$, either the degeneracy occurs only at $B=0$, or there is a critical $\frac{\hbar}{B}$ where the transition to the ground state occurs. We will discuss this more below.

In order to understand this model better, let's start with the simplest case of $N=1$. Here, we have

$$\hat{H}_1=-J\tfrac{\hbar^2}{4}\left(\sigma^z\right)^2-B\tfrac{\hbar}{2}\sigma^x, \tag{10.159}$$

which can be written in matrix form (by using the Pauli matrices) as

$$H_1=\begin{pmatrix}-J\frac{\hbar^2}{4} & -B\frac{\hbar}{2}\\ -B\frac{\hbar}{2} & -J\frac{\hbar^2}{4}\end{pmatrix}. \tag{10.160}$$

The eigenvectors are

$$\psi_\pm=\frac{1}{\sqrt{2}}\begin{pmatrix}1\\ \pm 1\end{pmatrix}, \tag{10.161}$$

with eigenvalues

$$E_\pm=-J\tfrac{\hbar^2}{4}\mp B\tfrac{\hbar}{2}, \tag{10.162}$$

which is plotted in Fig. 10.19. One can see in this case that there are only two energy levels—the even-parity state is the ground state for all B and becomes degenerate with the odd-parity state only at $B=0$.

We look next at $N=2$. We have

$$\hat{H}_2=-\frac{J}{2}\frac{\hbar^2}{4}\left(\left(\sigma_1^z\right)^2\otimes\mathbb{I}_2+2\sigma_1\otimes\sigma_2+\mathbb{I}_1\otimes\left(\sigma_2^z\right)^2\right)-\frac{\hbar}{2}\left(\sigma_1^z\otimes\mathbb{I}_2+\mathbb{I}_1\otimes\sigma_2^z\right), \tag{10.163}$$

as the general Hamiltonian. But, recall that the states can be classified in terms of their total spin. Here, that would be $S=1$ and $S=0$. The ground state is always in the maximal spin sector, so we can work just in the $S=1$ sector. Then we can represent $\hat{H}_2$ via a matrix in the z-basis as

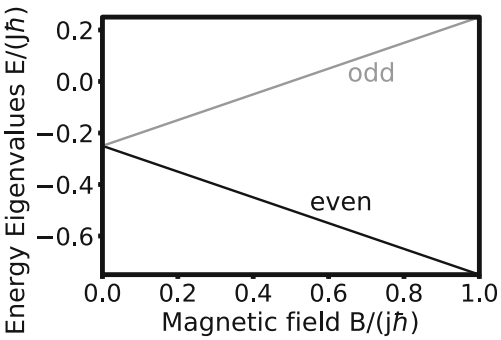

Fig. 10.19 Eigenvalues of the $N = 1$ all-to-all transverse-field Ising model as a function of the magnetic field. Even parity states are in black and odd parity in gray

$$ {}_z\langle m|\hat{H}_2|m'\rangle_z = \begin{pmatrix} -\frac{J}{2}\hbar^2 & -B\frac{\hbar}{2}\sqrt{2} & 0 \\ -B\frac{\hbar}{2}\sqrt{2} & 0 & -B\frac{\hbar}{2}\sqrt{2} \\ 0 & -B\frac{\hbar}{2}\sqrt{2} & -\frac{J}{2}\hbar^2 \end{pmatrix}, \tag{10.164} $$

because $\hat{S}^x = \frac{1}{2}(\hat{S}^+ + \hat{S}^-)$ and the matrix elements of $\hat{S}^z$ and $\hat{S}^\pm$ were worked out in an arbitrary $|lm\rangle$ basis in Sect. 9.3. In particular, $\hat{S}^z|m\rangle = \hbar m|m\rangle$, $\hat{S}^+|m\rangle = \hbar\sqrt{2}|m+1\rangle$ for $m = 0, -1$, and $\hat{S}^-|m\rangle = \hbar\sqrt{2}|m-1\rangle$ for $m = 1, 0$. We will discuss in the second mathematical interlude (Chap. 13) how one finds eigenvalues and eigenvectors for matrices bigger than 2×2 matrices, but in this case, we can reduce the problem to a 1×1 matrix and a 2×2 matrix by reorganizing the states in terms of their spin-reflection-parity symmetry. There is one odd state given by

$$ \psi_{\text{odd}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \tag{10.165} $$

which must be an eigenstate (do you remember why?) and two even eigenstates expressed in terms of the even basis, given by

$$ \psi^1_{\text{even}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \quad \text{and} \quad \psi^2_{\text{even}} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \tag{10.166} $$

which will create a 2×2 matrix. By acting the Hamiltonian on the odd state, we immediately can verify that $E_{\text{odd}} = -\frac{J}{2}\hbar^2$, which is independent of B. The 2×2 matrix for the even case is

$$ \begin{pmatrix} \langle\psi^1_{\text{even}}|\hat{H}_2|\psi^1_{\text{even}}\rangle & \langle\psi^1_{\text{even}}|\hat{H}_2|\psi^2_{\text{even}}\rangle \\ \langle\psi^2_{\text{even}}|\hat{H}_2|\psi^1_{\text{even}}\rangle & \langle\psi^2_{\text{even}}|\hat{H}_2|\psi^2_{\text{even}}\rangle \end{pmatrix} = \begin{pmatrix} -\frac{J}{2}\hbar^2 & -B\hbar \\ -B\hbar & 0 \end{pmatrix}. \tag{10.167} $$

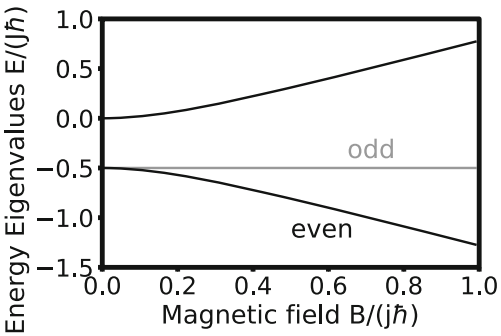

Fig. 10.20 Eigenvalues of H_2 for $S = 1$. Even parity states are in black and odd parity in gray

The eigenvalues are $E^{\pm}_{\text{even}} = -\frac{J}{4}\hbar^2 \pm \sqrt{\left(\frac{J}{4}\hbar^2\right)^2 + B^2\hbar^2}$, which are plotted in Fig. 10.20. One can see that the two lowest energy levels are closer for nonzero B than what we had for $N = 1$. But the qualitative behavior is similar with degeneracy only at $B = 0$.

To proceed further for larger N requires us to get numerical. The Hamiltonian matrix, before separating it into even and odd sectors, is easy to compute from the results of Sect. 9.3. The matrix is tridiagonal. Along the main diagonal, we have

$$H_{N,mm} = -\frac{J}{N}\hbar^2 m^2 \quad \text{for} \quad -\tfrac{N}{2} \leq m \leq \tfrac{N}{2}, \tag{10.168}$$

and the off-diagonals are

$$H_{N,mm+1} = -B\tfrac{\hbar}{2}\sqrt{\left(\tfrac{N}{2} - m\right)\left(\tfrac{N}{2} + m + 1\right)} = H_{N,m+1m}. \tag{10.169}$$

We then plug this into a computer and ask it to compute the energy levels. A typical plot is given in Fig. 10.21. There, we show the energy levels for $N = 500$ and the excitation energy $E - E_{gs}$. The solid lines are even-parity eigenvalues, and the dashed lines are odd-parity eigenvalues. Note how there is a minimal gap in the even sector near $J = B$ and that there is a near degeneracy of the lowest even and odd states for $B < J\hbar$. As N increases, the gap decreases, and as $N \to \infty$, it closes. The infinite-N case has a quantum phase transition at $B = J\hbar$. For $B > J\hbar$, the ground state is even and nondegenerate, while for $B < J\hbar$, it is degenerate with both even and odd representatives. One can show that the gap closes slowly, $E_{\text{gap}} \approx \frac{C}{N^{1/3}}$ as $N \to \infty$.

We end with a direct calculation of the mean-field theory result, which is the same as solving the classical spin problem. As $N \to \infty$, the ratio of the energy level spacing to the total spin $S = \frac{N}{2}$ gets smaller and smaller, so the energies become a continuous function of the angle of the spin. We let $S^z = \hbar S \cos\theta$ and $\hat{S}^x = \hbar S \sin\theta$. Then we have

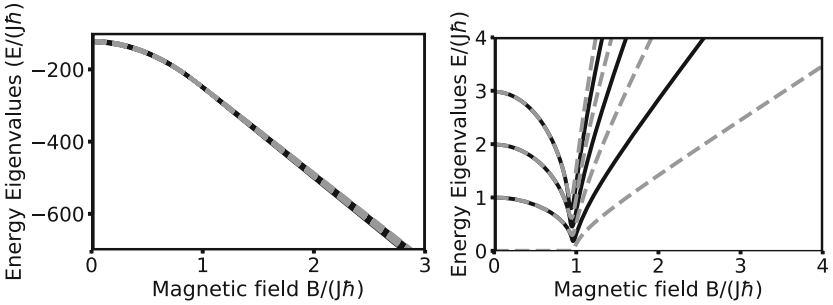

Fig. 10.21 (Left) Lowest five energy levels of even (solid black) and odd (dashed gray) parity. (Right) Energy difference from the ground state. Note how the transition is at $B \approx J\hbar$ and how the system becomes nearly twofold degenerate for all energy eigenvalues when $B < J\hbar$. Here, we are examining the case with $N = 500$

$$\begin{aligned} E(\theta) &= -\frac{J}{N}\hbar^2 S^2 \cos^2\theta - B\hbar S \sin\theta = -\hbar S\left(\frac{J\hbar}{2}\cos^2\theta + B\sin\theta\right) \\ &= -\hbar S\left(J\hbar - \frac{J}{2}\hbar(1+\sin^2\theta) + B\sin\theta\right) \\ &= -\tfrac{1}{2}\hbar^2 SJ\left(2-(1+x^2)+\frac{2B}{J\hbar}x\right), \end{aligned} \tag{10.170}$$

with $0 \leq x = \sin\theta \leq 1$. Using calculus, we immediately find that the energy is minimized at $\theta = \frac{\pi}{2}$ for $B > J\hbar$. At $B < J\hbar$, the minimum moves to $\theta = \sin^{-1}\left(\frac{B}{J\hbar}\right)$ and runs to 0 as B is decreased. The point $B = J\hbar$ is the critical point, where the ground state changes from a nondegenerate state oriented along the x-axis to a degenerate ground state, which is a superposition that rotates in the $x - z$ plane to both all up along z and all down along z. For finite N, the character of the system only approximately changes at $B = J\hbar$, because the gap to the excited state does not close and the states are only *nearly* degenerate.

One can analyze the minimization problem without calculus by looking at different plots of the curves we are minimizing. This is shown in Fig. 10.22. Let $\frac{B}{J\hbar} = \alpha$. Then we have the scenarios shown in Fig. 10.22. One can easily see why the transition occurs, as the character of the curves change as α changes, but it is difficult to verify that the minimum is at $x = \alpha$ without using calculus (if you know calculus, we just differentiate our expression with respect to x and set the derivative equal to zero, as is done for minimization problems).

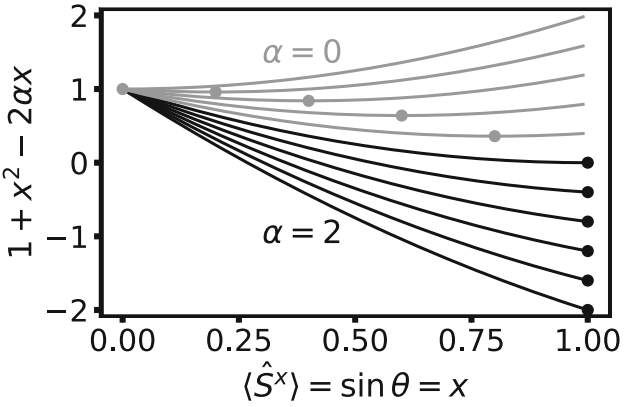

Fig. 10.22 Scenarios for the minimization of the energy as $N \to \infty$. The lines plot $1 + x^2 - 2\alpha x$ for $0 \leq \alpha \leq 2$ in steps of 0.2. The black lines have the minimum at $x = 1$, while the gray curves have the minimum at $x = \alpha$ (for $0 \leq \alpha \leq 1$)

Section 10.7 Problems:

10.7.1 The Spin-Reflection-Parity Transformation

Show that the spin-reflection-parity operation is given by $\hat{H} \to \hat{U}^\dagger \hat{H} \hat{U}$ with $\hat{U} = e^{i\pi \hat{S}_x/\hbar}$. Show that $\hat{U}^\dagger \hat{H} \hat{U} = \hat{H}$ (i.e., the Hamitonian is unchanged by the transformation) and that $\hat{U}^\dagger \hat{S}^x \hat{U} = \hat{S}^x$, $\hat{U}^\dagger \hat{S}^y \hat{U} = -\hat{S}^y$, and $\hat{U}^\dagger \hat{S}^z \hat{U} = -\hat{S}^z$. Verify that

$$[\hat{S}^\alpha, \hat{S}^\beta] = i\hbar \sum_\gamma \epsilon_{\alpha\beta\gamma} \hat{S}^\gamma, \tag{10.171}$$

for the *transformed* spin operators.

10.7.2 The $N = 3$ All-to-All Transverse-Field Ising Model

Consider the Hamiltonian matrix for $N = 3$. Show that it is equal to

$$H_3 = \begin{pmatrix} -J\hbar^2 \frac{3}{4} & -B\frac{\sqrt{3}}{2} & 0 & 0 \\ -B\frac{\sqrt{3}}{2} & -J\hbar^2 \frac{1}{12} & -B & \\ 0 & -B & -J\hbar^2 \frac{1}{12} & -B\frac{\sqrt{3}}{2} \\ 0 & 0 & -B\frac{\sqrt{3}}{2} & -J\hbar^2 \frac{3}{4} \end{pmatrix}. \tag{10.172}$$

10.7.3 The $N = 3$ All-to-All Transverse-Field Ising Model II

The transformation to even and odd parity states is made by using the following unitary matrix, which creates the linear combinations for even and odd parity states

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & 0 & 0 & -1 \end{pmatrix} \tag{10.173}$$

The transformed Hamiltonian matrix is found via $H_3 \rightarrow U^\dagger H_3 U$. Compute the transformed matrix, and show it reduces to two 2×2 blocks

$$U^\dagger H_3 U = \left(\begin{array}{c|c} H_3(even) & 0 \\ \hline 0 & H_3(odd) \end{array} \right). \tag{10.174}$$

Find the eigenvalues of H_3 by finding the eigenvalues of each 2×2 matrix. You might want to plot the results as a function of B as well and compare with the $N = 1$ and $N = 2$ results.

10.7.4 The $N = 4$ Transverse-Field Ising Model

Verify for $N = 4$ that we have

$$H_4 = \begin{pmatrix} -J\hbar^2 & -B & 0 & 0 & 0 \\ -B & -J\hbar^2 \frac{1}{4} & -B\frac{\sqrt{6}}{2} & 0 & 0 \\ 0 & -B\frac{\sqrt{6}}{2} & 0 & -B\frac{\sqrt{6}}{2} & 0 \\ 0 & 0 & -B\frac{\sqrt{6}}{2} & -J\hbar^2 \frac{1}{4} & -B \\ 0 & 0 & 0 & -B & -J\hbar^2 \end{pmatrix}. \tag{10.175}$$

10.7.5 Numerical Solution for Small N

If you are numerically inclined, you can write a program to compute H_N for various N values, find the B field (near $B = J\hbar$) where the energy gap in the even symmetry sector is minimal, and verify that the gap decreases as $N^{\frac{1}{3}}$. This is a challenging problem.

10.7.6 The Quantum Number for $\hat{S}^z$

Write a short essay that explains why the eigenvalues of $\hat{S}^z$ are not good quantum numbers for the energy eigenstates of the Hamiltonian for a finite-N system.

10.7.7 Entanglement for $B \to 0$

Explain why the ground state in the limit $B \to 0$ is given by the entangled state

$$|\psi_{gs}\rangle = \frac{1}{\sqrt{2}}\left(|\uparrow\uparrow\uparrow\cdots\uparrow\rangle_z + |\downarrow\downarrow\downarrow\cdots\downarrow\rangle_z\right) \tag{10.176}$$

instead of the odd superposition of all up and all down.
Hint: recall that the ground state for $B \neq 0$ is nondegenerate, so it has a linear combination determined by the parity symmetry that it has at any positive B value. Is there a special B value that makes determining the parity especially simple?

10.8 The Dicke Model

The two main systems we have studied so far are spin systems and the simple harmonic oscillator. In this section, we combine the two in what is called the *Dicke model*, which was created in the 1950s (Dicke, 1954). The original version and related models like the Jaynes-Cummings model (Jaynes & Cummings, 1963) (one spin/atom) and the Tavis-Cummings model (Tavis & Cummings, 1968) (many spins/atoms) were created later in the 1960s. They involve the interaction of a photon with an atom. The atom was approximated as a two-level system with a ground state and an excited state. A photon with an energy close to the energy difference of the atomic states is placed in a cavity (a box with highly reflective walls) and interacts with the atom. The system becomes correlated, and if we have many atoms and many photons, it is a many-body problem in quantum mechanics. Serge Haroche worked hard at realizing these systems with atoms and photons in cavities and ultimately shared the Nobel prize for this work in 2012. See Fig. 10.23 for images of Dicke and Haroche.

In this section, we emphasize the pure spin language for this model, and we will discuss the atomic mapping later. The Hamiltonian resembles that of the Lipkin model, in that it has effective all-to-all interactions and the total spin operator $\hat{S}^2$ commutes with $\hat{H}$, so it is a good quantum number again. Without further ado, here is the Hamiltonian:

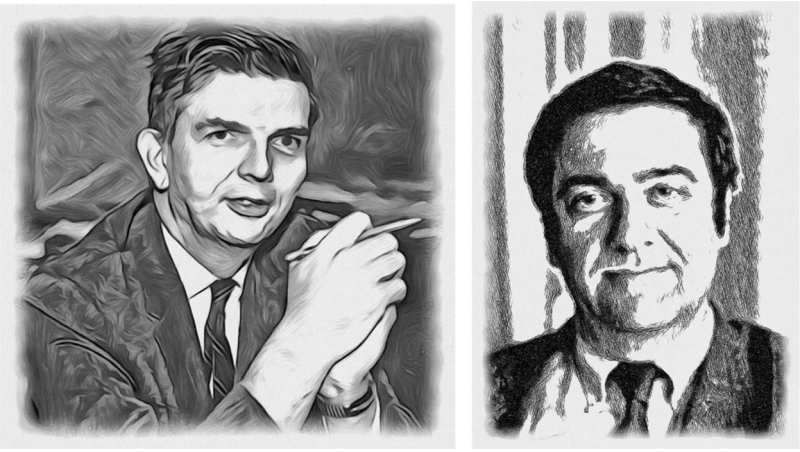

Fig. 10.23 (Left) Robert Dicke in the 1950s. (AIP Emilio Segrè Visual Archive, Physics Today Collection. Converted to look like a sketch by James Freericks.) (Right) Serge Haroche in the early 1990s. (AIP Emilio Segrè Visual Archives, Physics Today Collection, W. F. Meggers Gallery of Nobel Laureates Collection. Converted to look like a sketch by James Freericks)

$$\hat{H}_{\text{Dicke}} = \hbar\omega\hat{a}^\dagger\hat{a} - \frac{2g}{\sqrt{N}}\left(\hat{a} + \hat{a}^\dagger\right)\hat{S}^z\cos(\mu t) - B_x(t)\hat{S}^x, \tag{10.177}$$

where the spin operators are total spin operators for N spin-one-half particles (so $S = \frac{N}{2}$) and $\hat{a}^\dagger$ ($\hat{a}$) are the raising (lowering) operators for a simple harmonic oscillator ($[\hat{a}, \hat{a}^\dagger] = 1$). This model involves the coupling of a single photon (or phonon) to a collection of N spins, with the coupling being identical for all spins. Note that in the cavity QED experiments of Haroche, the harmonic oscillator is given by a photon. But there is another platform where we can examine this behavior and that is with trapped ions. Here, the ions are trapped into a lattice structure, and they have small oscillations, called *normal modes* about the equilibrium positions. These normal modes are also described by simple harmonic oscillators, but now we call them *phonons*. In this section, we will focus on using the phonon language and thinking of the systems we examine in terms of trapped ions.

The Dicke model is a classic model to apply the rotating wave approximation (RWA) to. This approximation requires us to go to the rotating frame of reference (similar to what we did when we examined nuclear magnetic resonance in Sect. 2.4 and which we will revisit in Chap. 21). We then keep the terms that vary slowly in the rotating frame and drop terms that vary rapidly, because their effect averages to zero. In this sense, it is an approximation that people often need to work with for a while before they start to feel comfortable with it, because it seems counterintuitive. One way to try to wrap your head around it is that if you have something that rapidly oscillates, then sometimes it is more convenient to follow the averaged motion than trying to follow each oscillation.

We carry out the transformation to the rotating frame by applying a similarity transformation to the Hamiltonian via a unitary operator $\hat{U}$, which rotates the states of the system and then subtracting the operator used in the exponent of the unitary operator that is used to rotate the system. Hence, we compute

$$\hat{H} \to \hat{U}^\dagger \hat{H} \hat{U} - \hat{H}_0, \quad \text{with } \hat{U} = e^{-\frac{i}{\hbar}\hat{H}_0 t}. \tag{10.178}$$

If you are becoming more comfortable with similarity transformations, then you might be confused by the subtraction of the term $\hat{H}_0$. But, if we look at a rotating system, we have the kinetic energy of the rotation in the lab frame, not in the rotating frame. Subtracting $\hat{H}_0$ is how we remove the kinetic energy of rotation in going from the lab frame to the rotating frame. This procedure will become clear later, when we discuss time evolution in Chap. 14. For now, take Eq. (10.178) as a rule for how we transform into the rotating frame.

For the Dicke model, we pick $\hat{H}_0 = \hbar\mu\hat{a}^\dagger\hat{a}$. Obviously $\hat{U}^\dagger \hat{S}^\alpha \hat{U} = \hat{S}^\alpha$, since the spin degree of freedom is independent of the harmonic oscillator. We need to compute $\hat{U}^\dagger \hat{a} \hat{U}$ and $\hat{U}^\dagger \hat{a}^\dagger \hat{U}$. For both, we use the Hadamard lemma as follows:

$$\begin{aligned}
\hat{a}(t) = \hat{U}^\dagger \hat{a} \hat{U} &= e^{i\mu\hat{a}^\dagger\hat{a}t}\hat{a}e^{-i\mu\hat{a}^\dagger\hat{a}t} \\
&= \hat{a} + i\mu t[\hat{a}^\dagger\hat{a}, \hat{a}] + \frac{(i\mu t)^2}{2}[\hat{a}^\dagger\hat{a}, [\hat{a}^\dagger\hat{a}, \hat{a}]] + \cdots \\
&= \hat{a} - i\mu t\hat{a} + \frac{(i\mu t)^2}{2}\hat{a} + \cdots = e^{-i\mu t}\hat{a},
\end{aligned} \tag{10.179}$$

and

$$\begin{aligned}
\hat{a}^\dagger(t) = \hat{U}^\dagger \hat{a}^\dagger \hat{U} &= e^{i\mu\hat{a}^\dagger\hat{a}t}\hat{a}^\dagger e^{-i\mu\hat{a}^\dagger\hat{a}t} \\
&= \hat{a}^\dagger + i\mu t[\hat{a}^\dagger\hat{a}, \hat{a}^\dagger] + \frac{(i\mu t)^2}{2}[\hat{a}^\dagger\hat{a}, [\hat{a}^\dagger\hat{a}, \hat{a}^\dagger]] + \cdots \\
&= \hat{a}^\dagger + i\mu t\hat{a}^\dagger + \frac{(i\mu t)^2}{2}\hat{a}^\dagger + \cdots = e^{i\mu t}\hat{a}^\dagger.
\end{aligned} \tag{10.180}$$

Substituting these results into Eq. (10.178) yields

$$\begin{aligned}
\hat{H} &= \hbar\omega\hat{a}^\dagger e^{i\mu t}\hat{a}e^{-i\mu t} - \frac{2g}{\sqrt{N}}\left(\hat{a}e^{-i\mu t} + \hat{a}^\dagger e^{i\mu t}\right)\hat{S}^z \cos\mu t - B^x(t)\hat{S}^x - \hbar\mu\hat{a}^\dagger\hat{a} \\
&= \hbar(\omega - \mu)\hat{a}^\dagger\hat{a} - \frac{g}{\sqrt{N}}\left(\hat{a}(1 + e^{-2i\mu t}) + \hat{a}^\dagger(1 + e^{2i\mu t})\right)\hat{S}^z - B^x(t)\hat{S}^x.
\end{aligned} \tag{10.181}$$

Note that all of the time dependence is gone in the Hamiltonian except for the two exponential terms $e^{\pm 2i\mu t}$. The RWA drops those terms, because they oscillate rapidly (if μ is large) and their effect averages to zero. This finally gives us the rotating wave approximation Hamiltonian

$$\hat{H}_{\text{RWA}} = -\hbar\delta\hat{a}^\dagger\hat{a} - \frac{g}{\sqrt{N}}\left(\hat{a} + \hat{a}^\dagger\right)\hat{S}^z - B^x(t)\hat{S}^x, \tag{10.182}$$

with $\delta = \mu - \omega$, called the *detuning*. For the remainder of the section, we will assume that we are working with a red detuned case, where $\mu < \omega$ or $\delta < 0$. In this case, the first term of $\hat{H}_{\text{RWA}}$ is positive.

Before doing anything else, we need to determine the symmetries of $\hat{H}_{\text{RWA}}$. Just like in the Lipkin model, we have $[\hat{S}^2, \hat{H}_{\text{RWA}}] = 0$ (which you should be able to immediately verify), so the total squared spin is a symmetry, and we can work with states that have definite total spin S (in this case, the ground state is again in the sector with $S = \frac{N}{2}$). Unfortunately, the spin-reflection-parity symmetry, given by $\hat{S}^x \to \hat{S}^x$, $\hat{S}^y \to -\hat{S}^y$, and $\hat{S}^z \to -\hat{S}^z$, is not a symmetry anymore, because the spin-phonon interaction term changes sign under this transformation. But we can find a new symmetry, which we call the *spin-phonon reflection symmetry*, where we transform $\hat{S}^x \to \hat{S}^x$, $\hat{S}^y \to -\hat{S}^y$, $\hat{S}^z \to -\hat{S}^z$, $\hat{a} \to -\hat{a}$, and $\hat{a}^\dagger \to -\hat{a}^\dagger$. One can see that in this transformation, all commutators are preserved, as is $\hat{H}_{\text{RWA}}$. The symmetry is given by the unitary operator

$$\hat{U}_{\text{sp-ph}} = e^{-i\pi\hat{a}^\dagger\hat{a}}e^{-i\pi\hat{S}^x}. \tag{10.183}$$

One can compute that $\hat{U}^\dagger_{\text{sp-ph}}\hat{H}_{\text{RWA}}\hat{U}_{\text{sp-ph}} = \hat{H}_{\text{RWA}}$, which implies that $[\hat{U}_{\text{sp-ph}}, \hat{H}_{\text{RWA}}] = 0$, so this parity operator is a symmetry of the Hamiltonian. These two symmetries allow us to work with the maximal spin multiplet $S = \frac{N}{2}$ and with states of even or odd parity with respect to this spin-phonon reflection.

Before we start looking at eigenstates, it is interesting to investigate the structure of $\hat{H}_{\text{RWA}}$ a bit more as it still has some surprises for us. The main surprise is that it is an infinite-dimensional matrix for any N value! The reason why is that our basis states are organized as tensor products

$$|n\rangle \otimes |S = \tfrac{N}{2}, m\rangle, \tag{10.184}$$

where n is the phonon number ($n \geq 0$) and m is the z-component of the total spin ($-\frac{N}{2} \leq m \leq \frac{N}{2}$). Note that n can run all the way to infinity! Hence, the number of basis vectors is infinite and so is the Hamiltonian matrix, given by

$$\langle n| \otimes \langle S = \tfrac{N}{2}, m|\ \hat{H}_{\text{RWA}}\ |n'\rangle \otimes |S = \tfrac{N}{2}, m'\rangle = \hat{H}^{\text{RWA}}_{mn,m'n'}, \tag{10.185}$$

since the indices n and n' are unbounded.

There is one more aspect of the Dicke model that we want to investigate before looking at the eigenstates. This is the relationship between the Dicke model and the Lipkin model. At first, they look completely different, but if we define new operators given by

$$\hat{b} = \hat{a} - \frac{g}{\sqrt{N}\hbar\delta}\hat{S}^z \text{ and } \hat{b}^\dagger = \hat{a}^\dagger - \frac{g}{\sqrt{N}\hbar\delta}\hat{S}^z, \tag{10.186}$$

then we find that $[\hat{b}, \hat{b}^\dagger] = [\hat{a}, \hat{a}^\dagger] = 1$ and

$$\hat{H}_{\text{RWA}} = -\hbar\delta\hat{b}^\dagger\hat{b} + \frac{g^2}{N\hbar\delta}\left(\hat{S}^z\right)^2 - B^X(t)\hat{S}^x, \tag{10.187}$$

which looks like the Lipkin model plus a harmonic oscillator described by $\hat{b}$ and $\hat{b}^\dagger$. But, because $\hat{b}$ and $\hat{b}^\dagger$ do not commute with $\hat{S}^x$, it is difficult to use the raising operator $\hat{b}^\dagger$ to create eigenstates of $\hat{H}_{\text{RWA}}$. This transformation tells us that in some circumstances, we should expect the behavior of the Dicke model to be similar to the Lipkin model. We will elaborate on this theme below.

Now, we are ready to examine the ground states for $B \to 0$ and $B \to \infty$. It is easiest to see what these states are by examining the form of the Hamiltonian, when expressed in terms of the b operators. When $B \to \infty$, the state is polarized along the x-direction, and the phonon is in the ground state. On the other hand, as $B \to 0$, the fully polarized state is along the z-direction (either all up or all down). When the system has a nonzero expectation value for $\hat{S}^z$, the phonon is translated to the right or to the left, with the shift proportional to the projection of $\hat{\vec{S}}$ onto the z-axis. Recall the problem when we shifted the harmonic oscillator in Chap. 7 and how the shifting of the oscillator is accomplished in the coherent state. The parity state, however, requires the $B \to 0$ ground state to be an even or odd superposition of these states. Hence, we find that

$$\lim_{B\to 0}|\psi_{gs}\rangle = \frac{1}{\sqrt{2}}\left(|x_0\rangle \otimes |\uparrow\uparrow\uparrow\cdots\uparrow\rangle \pm |-x_0\rangle \otimes |\downarrow\downarrow\downarrow\cdots\downarrow\rangle\right), \tag{10.188}$$

where $|\pm x_0\rangle$ is the coherent state centered at $\pm x_0 = \pm g\sqrt{\frac{2N}{m\hbar|\delta|}}$. These states are, of course, an entangled superposition and are famously called the *Schrödinger cat states* or more simply the *cat states*. For intermediate B values, the ground state is complicated, involving a superposition over many m states and many phonon states. Nevertheless, we still expect a transition when $B \approx \frac{g^2}{\hbar|\delta|}$, just as in the Lipkin model. As $N \to \infty$, the even and odd parity states will become degenerate in the low B-field regime (the so-called *superradiant phase*) and the ground state remains nondegenerate for the high-field regime, where one is in the normal phase.

In the normal phase, the low-energy excitations are phonon-like excitations, because the field B^x is much larger than the phonon frequency (in the rotating frame; $B^x \gg \hbar|\delta|$). In the superradiant regime, the excitations can be spin-like

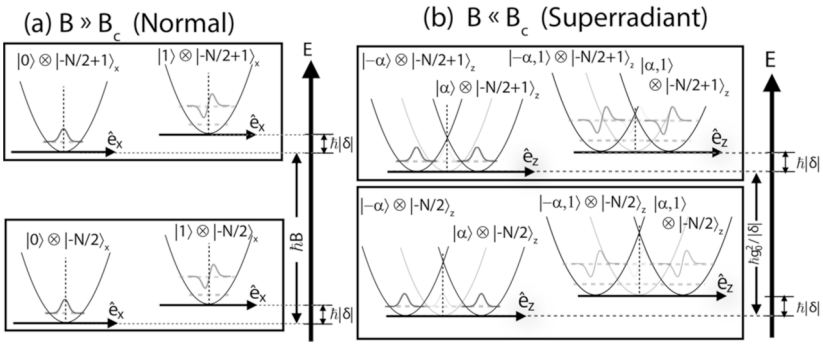

Fig. 10.24 Schematic of the excitation spectrum of the Dicke model in the normal (left) and superradiant (right) phases. Excitations are spin flips or phonon excitations, and the lowest energy excitations depend on the relative sizes of B^x, $\hbar|\delta|$, and $g^2/\hbar|\delta|$. The phonon excitations change the character of the wavefunction (Gaussian like to a state with one node for the lowest excitation). In the normal phase, the phonon site is centered about the coordinate 0, while in the superradiant phase, they are pushed to the right or to the left (Figure adapted from Cohn et al., 2018 CC BY license)

or phonon-like, depending on whether $\hbar|\delta|$ is larger than or smaller than $\frac{g^2}{\hbar|\delta|}$. The phonons in this regime are centered around the shifted minima $\pm x_0$, representative of the phonon shifts in the superradiant phase. A schematic for how this works is given in Fig. 10.24.

As with the Lipkin model, we can examine the $N \to \infty$ limit, where the spin becomes classical. Then the Hamiltonian becomes

$$\lim_{N\to\infty} \hat{H}_{\text{RWA}} = -\hbar\delta\hat{b}^\dagger\hat{b} - \frac{g^2}{|\delta|}\frac{\hbar^2 N}{4}\cos^2\theta - B\frac{\hbar N}{2}\sin\theta. \tag{10.189}$$

Since the spins are proportional to N, they dominate the total energy, and the transition is just like what we saw in the Lipkin model. The phonon will be in the coherent state centered at the shifted origin given by the spin component of the wavefunction. As with the Lipkin model, the spin is oriented along the x-direction until the minimum of the energy (with respect to θ) occurs at $\theta < \frac{\pi}{2}$. Then the ground state is degenerate and the system smoothly changes to $\theta \to 0$. As the phonon state translates, the number of harmonic oscillator excitations created by $\hat{a}^\dagger$ become large.

Before ending this section, we want to discuss the phenomenon called *reservoir engineering*. This corresponds to a situation where one entangles a system (here the spins) with the reservoir (here the phonons) in order to achieve some specific goal. When we create such an entangled ground state, one must be able to measure both the spins and the phonons at the same time in order to see (and use) the entanglement. This is because we have a tagged state, and if we do not measure the tag (the phonons), then we cannot see the entanglement. Let us examine a bit

more closely. If we measure only the spins, which are common for most ion-trap-based realizations of the Dicke model, then we will see the spins as being all up half of the time and all down half of the time, but the entanglement is not seen. This is analogous to the measurement in a Stern-Gerlach analyzer loop that is oriented vertically and has horizontally oriented spins enter it. If we watch at the branches, then the superposition is broken, and we find the spins to be half the time up and half the time down. Measuring the spins without measuring the phonons requires us to project onto the measured spin state and sum over all possible phonon states. This creates what is called the *maximally mixed state*. It requires something called a density matrix to properly describe it (this corresponds to the situation where we analyze multiple events each possibly having more than one alternative). This is typically what happens when one entangles with a reservoir—the system is fully entangled, but if you only measure the system, you lose all of that entanglement. This is discussed further in the problems.

But, if we can engineer the reservoir state prior to the measurement, then we can transfer the entanglement from the system and reservoir to the system only. The idea for how to do this is the following: The state we are in at $B = 0$ is given in Eq. (10.188). We then evolve the reservoir such that the harmonic oscillator coherent states both evolve from $\pm x_0$ to 0. Then the state becomes

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle \otimes (|\uparrow\uparrow\uparrow\cdots\uparrow\rangle \pm |\downarrow\downarrow\downarrow\cdots\downarrow\rangle)\,. \tag{10.190}$$

The state has been changed from an entangled spin-phonon state to a product state between the spin and the phonons. And, the spin state is now an entangled superposition. Hence, this transfers the entanglement from the spin-phonon system to the spin-only system. We will discuss reservoir engineering more in Chap. 21.

Section 10.8 Problems:

10.8.1 Hamiltonian of the Dicke Model in Matrix Form

The Hamiltonian matrix for the Dicke model can be constructed as described in Eq. (10.185). Using the form for the Dicke Hamiltonian in the rotating wave approximation, given in Eq. (10.182), show that

$$\begin{aligned} H^{\mathrm{RWA}}_{mn,m'n'} = &-\hbar\delta\delta_{mm'}\delta_{nn'} - \frac{g}{\sqrt{N}}\hbar m\delta_{mm'}(\sqrt{n'+1}\,\delta_{nn'+1} + \sqrt{n'}\,\delta_{nn'-1}) \\ &- B^x(t)\tfrac{1}{2}\left(\sqrt{S(S+1)-m'(m'-1)}\,\delta_{mm'+1}\delta_{nn'}\right. \\ &\left.+\sqrt{S(S+1)-m'(m'-1)}\,\delta_{mm'-1}\delta_{nn'}\right). \end{aligned} \tag{10.191}$$

Note that the matrix is infinite dimensional, even for $N = 2$. For any practical numerical calculation, the matrix must be truncated. Care must be taken to find the best way to terminate the matrix and to ensure that the finite matrix remains Hermitian.

10.8.2 Phonon Cutoff in the Cat State

Recall that the simple harmonic oscillator coherent state $|\alpha\rangle$ satisfies $\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$ and

$$|\alpha\rangle = \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} e^{-\frac{1}{2}|\alpha|^2} |n\rangle. \tag{10.192}$$

Suppose we are in the $B^x = 0$ ground state for N spins given by

$$|\psi_{gs}\rangle = \frac{1}{\sqrt{2}} \left(|\alpha_{up} = \tfrac{g\sqrt{N}}{2\hbar|\delta|}\rangle \otimes |m = \tfrac{N}{2}\rangle + |\alpha_{down} = -\tfrac{g\sqrt{N}}{2\hbar|\delta|}\rangle \otimes |m = -\tfrac{N}{2}\rangle \right). \tag{10.193}$$

What size cutoff in the maximum phonon number is needed to have the probability for phonon occupation at the cutoff be less than 0.0001? Express your answer as a function of N. You should find that $n_{\text{cutoff}} \approx 143$ when $\frac{g^2 N}{4\hbar^2\delta^2} = 200$.
Hint: Stirling's formula, given by $\ln n! \approx n \ln n - n$, is helpful in solving this problem. As a general rule, the cutoff is close to $\frac{g^2 N}{4\hbar^2\delta^2}$.

10.8.3 Transferring Entanglement from Spin-Phonon Entanglement to Spin-Only Entanglement

In this problem, we will discover how measurements of spin states can provide information about the ground state. We consider the $B^x = 0$ ground state $|\psi_{gs}\rangle = \frac{1}{\sqrt{2}}(|\alpha_0\rangle \otimes |m{=}\frac{N}{2}\rangle + |-\alpha_0\rangle \otimes |m{=}-\frac{N}{2}\rangle)$ and the entangled spin state $|\psi_{\text{spin}}\rangle = \frac{1}{\sqrt{2}}|\alpha{=}0\rangle \otimes (|m{=}\frac{N}{2}\rangle + |m{=}-\frac{N}{2}\rangle)$, which we get by evolving the harmonic oscillator states of the reservoir back to the origin.

(a) What is the probability to measure $|\frac{N}{2}\rangle$ if we measure just the spins for both states? Use the fact that if we do not measure the simple harmonic oscillator states, then the probability to measure a particular spin state $|m\rangle$ is given by

$$P(m) = \sum_{n=0}^{\infty} |\langle n| \otimes \langle m| \, |\psi\rangle|^2. \tag{10.194}$$

Use the fact that $\sum_{n=0}^{\infty} |n\rangle \langle n| = \hat{\mathbb{I}}$ in your derivation.

(b) The spin entanglement can only be measured by determining what is called an off-diagonal matrix element of the density matrix. Without going into details, the matrix element we need to compute is found from $\langle \psi | \hat{S}_1^x \otimes \hat{S}_2^x \otimes \cdots \otimes \hat{S}_N^x | \psi \rangle$. You will need the overlap of two coherent states $\langle \alpha | \beta \rangle$ derived in Sect. 7.11. You may assume that $\alpha_0 \gg 1$.

This problem illustrates the difference between a maximally mixed state (corresponding to the $B^x = 0$ ground state when we do not measure the phonons) versus a spin-entangled state. You should find that the $B^x = 0$ ground state and the spin entangled state are found to be half the time all up and half the time all down. But when we measure the product of $\hat{S}^x$ operators, one case vanishes, while the other is of order one.

10.8.4 Mean-Field Theory ($N \to \infty$) for the Dicke Model

In this problem, we investigate the mean-field theory limit of $N \to \infty$. While the spins become classical, the harmonic oscillator remains quantum, so we cannot solve it exactly. We will calculate an approximation to the ground-state energy. Note that if $|\psi\rangle$ is an eigenstate of $\hat{H}$, then $\frac{\langle \psi \hat{H} | \psi \rangle}{\langle \psi | \psi \rangle} = E$ is the ground-state energy. If $|\psi\rangle$ is not a ground state, then we say $\langle \psi | \hat{H} | \psi \rangle / \langle \psi | \psi \rangle$ is the variational energy for the state $|\psi\rangle$. We discuss the variational principle in more detail in Sect. 12.7. Here, we simply need to know that it is a good approximation to the true ground-state energy. Using the fact that the coherent state is an eigenstate of $\hat{a}$, we can compute the variational energy for

$$|\psi\rangle = \frac{1}{\sqrt{2}} \left(|\alpha\rangle \otimes |\theta\rangle \pm |-\alpha\rangle \otimes |\pi - \theta\rangle \right), \tag{10.195}$$

where θ is the angle the spin makes relative to the z-axis in the $x - z$ plane and α is real. Note that we can assume that $\langle \theta | \pi - \theta \rangle = 0$ and $\langle \alpha | -\alpha \rangle = e^{-\alpha^2}$. Also, when $\theta = \frac{\pi}{2}$ and $\alpha = 0$, only the $+$ solution exists. Keeping all of these details in mind, show that

$$\frac{E}{N} = -\hbar\delta\alpha_0^2 + g\alpha_0(\cos\theta - \cos(\pi - \theta)) - \frac{B^x}{4}(\sin\theta + \sin(\pi - \theta)) \tag{10.196}$$

after assuming that $\alpha = \alpha_0\sqrt{N}$. This is a classic minimization problem, which is best done with calculus. If we first focus on the α_0 dependence, it has the form $a\alpha_0^2 + b\alpha_0$. By completing the square, this is minimized (for $a > 0$) by minimizing

$$a\left(\alpha_0 + \tfrac{b}{2a}\right)^2 - \tfrac{b^2}{4a^2}, \tag{10.197}$$

which is clearly minimized at $\alpha_0 = -\frac{b}{2a}$. Do this and then show that the minimization over θ requires $\alpha_0 = -\frac{g\cos\theta}{\hbar|\delta|}$ and the function to minimize is

$$-\frac{g^2}{\hbar|\delta|}\cos^2\theta - \frac{B^x}{2}\sin\theta. \tag{10.198}$$

Establish that the function is minimized by $\theta = \frac{\pi}{2}$ for large B^x until $\frac{\hbar|\delta|B^x}{4g^2} < 1$. This is where the phase transition is approximately located. The phase transition is from a nondegenerate ground state to a doubly degenerate ground state. Explain as well why this analysis is approximate and $|\psi\rangle$ is not the actual ground state.

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Chapter 11
Nonrelativistic Hydrogen Atom

I am here to give voice to the invisible.

Carlos Santana, Singer, Songwriter

In this chapter, we focus on three-dimensional problems that have rotational symmetry, with a concentrated effort on describing the properties of the hydrogen atom. This is an exactly solvable problem that is extremely important. To get there, we first have to describe how we work in spherical coordinate systems, and we need to develop the general rules for the factorization method. There is a lot of exciting material here, so hold onto your hat, as we go for a wild ride! Catch a quick overview as you chase down your hat at https://tinyurl.com/5x2kz5hp.

11.1 Scalar Operators, Vector Operators, and Quantum Corrections to Vector Identities

In many quantum textbooks, the author will say something about how the Planck constant $\hbar$ is so small, that quantum effects only occur for small particles. While it is true that $\hbar$ is represented by a small number when expressed in SI units ($\hbar = 1.05457 \times 10^{-34}$ J-s)—such a statement has no physical content—any physical constant is as big or as small as you want it to be depending on how you define your units. After all, 1 m is both 10^{10} Å and is 3×10^{-9} light-s; is it a large number or a small one?

Many texts will also discuss how quantum systems become classical (the the so-called *classical limit*) when we let $\hbar \to 0$. This is also a difficult result to work with, because the semiclassical approximation, which we will not discuss in this book, depends inversely on $\hbar$ (as $\frac{1}{\hbar}$), so the limit $\hbar \to 0$ must be performed very carefully.

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We won't provide any stories here about how "small" $\hbar$ is, nor will we focus on examining limits where $\hbar \to 0$. But, we will find it very useful to employ $\hbar$ as a way to determine when we have quantum corrections to classical objects. In this section, for example, we will work with operators that are vectors, in the sense that they live in a three-dimensional vector space as

$$\hat{\vec{A}} = \hat{A}_x \vec{e}_x + \hat{A}_y \vec{e}_y + \hat{A}_z \vec{e}_z = (\hat{A}_x, \hat{A}_y, \hat{A}_z). \tag{11.1}$$

We will investigate under what situations classical vector identities continue to hold and in what situations they have quantum corrections.

We begin our journey by defining *scalar* and *vector* operators. We already started the discussion in Chap. 9 when we examined the properties of $\hat{\vec{r}}$, $\hat{\vec{p}}$, and $\hat{\vec{L}} = \hat{\vec{r}} \times \hat{\vec{p}}$. A scalar operator $\hat{O}$ satisfies

$$[\hat{O}, \hat{\vec{L}}] = 0; \tag{11.2}$$

that is, it commutes with $\hat{\vec{L}}$. We already saw that $\hat{\vec{r}}^{\,2}$, $\hat{\vec{p}}^{\,2}$, and $\hat{\vec{L}}^2$ all commute with $\hat{\vec{L}}$, meaning they are all scalar operators. We investigate this more thoroughly next. But, before doing that, let us recall the definition of a vector operator. A vector operator satisfies

$$[\hat{A}_i, \hat{L}_j] = i\hbar \sum_k \epsilon_{ijk} \hat{A}_k. \tag{11.3}$$

It is curious that one does not need to worry about the order of the terms in this identity, as we also have that

$$[\hat{L}_i, \hat{A}_j] = i\hbar \sum_k \epsilon_{ijk} \hat{A}_k, \tag{11.4}$$

which you will prove in Problem 11.1.1.

The first nontrivial result we want to show is that the dot product of two vector operators is a scalar operator. To show this, we just compute

$$\begin{aligned} [\hat{\vec{A}} \cdot \hat{\vec{B}}, \hat{L}_j] &= \sum_i [\hat{A}_i \hat{B}_i, \hat{L}_j] = \sum_i \left(\hat{A}_i [\hat{B}_i, \hat{L}_j] + [\hat{A}_i, \hat{L}_j] \hat{B}_i \right) \\ &= \sum_i \sum_k i\hbar \epsilon_{ijk} \left(\hat{A}_i \hat{B}_k + \hat{A}_k \hat{B}_i \right), \end{aligned} \tag{11.5}$$

by using the Leibniz rule. If we interchange the i and k dummy indices in the second term, we obtain

$$\sum_i \sum_k i\hbar \epsilon_{ijk} \hat{A}_k \hat{B}_i = i\hbar \sum_i \sum_k \epsilon_{kji} \hat{A}_i \hat{B}_k = -i\hbar \sum_i \sum_k \epsilon_{ijk} \hat{A}_i \hat{B}_k, \tag{11.6}$$

which cancels with the first term and shows that $[\hat{\vec{A}} \cdot \hat{\vec{B}}, \hat{\vec{L}}] = 0$, or that $\hat{\vec{A}} \cdot \hat{\vec{B}}$ is a scalar operator if $\hat{\vec{A}}$ and $\hat{\vec{B}}$ are both vector operators. This is a nice result, which agrees with the classical results of vectors (a dot product of two vectors is a scalar).

But, there are some surprises! As an operator, the order of two vector operators in the dot product matters, because

$$\hat{\vec{A}} \cdot \hat{\vec{B}} = \hat{\vec{B}} \cdot \hat{\vec{A}} + \sum_i [\hat{A}_i, \hat{B}_i], \tag{11.7}$$

and the second term is nonzero if $\hat{A}_i$ and $\hat{B}_i$ do not commute. A common example of this is position and momentum. We immediately see that

$$\hat{\vec{r}} \cdot \hat{\vec{p}} = \hat{\vec{p}} \cdot \hat{\vec{r}} + \sum_i [\hat{r}_i, \hat{p}_i] = \hat{\vec{p}} \cdot \hat{\vec{r}} + 3i\hbar. \tag{11.8}$$

Dot products of two Hermitian vector operators need not be Hermitian either—indeed, we have

$$\left(\hat{\vec{r}} \cdot \hat{\vec{p}}\right)^\dagger = \hat{\vec{p}}^\dagger \cdot \hat{\vec{r}}^\dagger = \hat{\vec{p}} \cdot \hat{\vec{r}} = \hat{\vec{r}} \cdot \hat{\vec{p}} - 3i\hbar \neq \hat{\vec{r}} \cdot \hat{\vec{p}}. \tag{11.9}$$

This leads us to a new operation, called *Hermitizing a product of operators*. We take the average of the two orderings to do this. In other words, we Hermitize $\hat{\vec{r}} \cdot \hat{\vec{p}}$ by forming the operator

$$\tfrac{1}{2}\left(\hat{\vec{r}} \cdot \hat{\vec{p}} + \hat{\vec{p}} \cdot \hat{\vec{r}}\right) = \hat{\vec{r}} \cdot \hat{\vec{p}} - \tfrac{3}{2}i\hbar. \tag{11.10}$$

The left-hand side shows us that the operator is manifestly Hermitian (just check by taking its Hermitian conjugate).

So, we have learned that the dot product of two vector operators is a scalar operator. This fact can save us a lot of algebra. For example, after we showed that $\hat{\vec{r}}$, $\hat{\vec{p}}$, and $\hat{\vec{L}}$ were all vector operators, then we immediately can conclude that $\hat{\vec{r}} \cdot \hat{\vec{r}} = \hat{r}^2$, $\hat{\vec{p}} \cdot \hat{\vec{p}} = \hat{p}^2$, $\hat{\vec{L}} \cdot \hat{\vec{L}} = \hat{L}^2$, $\hat{\vec{r}} \cdot \hat{\vec{p}}$, $\hat{\vec{r}} \cdot \hat{\vec{L}}$, and $\hat{\vec{p}} \cdot \hat{\vec{L}}$ are all scalar operators that commute with $\hat{\vec{L}}$. Of course, we do have that $\hat{\vec{r}} \cdot \hat{\vec{L}} = \sum_i \sum_j \sum_k \epsilon_{ijk} \hat{r}_i \hat{r}_j \hat{p}_k = 0$, because the product $\hat{r}_i \hat{r}_j = \hat{r}_j \hat{r}_i$ is symmetric when we interchange $i \leftrightarrow j$, but $\epsilon_{ijk} = -\epsilon_{jik}$ changes sign under such an interchange. We see that $\hat{\vec{p}} \cdot \hat{\vec{L}}$ vanishes for the same reason.

$\hat{\vec{L}}^2$ is a particularly interesting operator, because we can write

$$\hat{\vec{L}}^2 = \sum_i \sum_{i'} \sum_j \sum_{j'} \sum_k \epsilon_{ijk} \hat{r}_i \hat{p}_j \epsilon_{i'j'k} \hat{r}_{i'} \hat{p}_{j'}$$

$$
\begin{aligned}
&= \sum_i \sum_{i'} \sum_j \sum_{j'} \left(\sum_k \epsilon_{ijk}\epsilon_{i'j'k} \right) \hat{r}_i \hat{p}_j \hat{r}_{i'} \hat{p}_{j'} \\
&= \sum_i \sum_{i'} \sum_j \sum_{j'} \left(\delta_{ii'}\delta_{jj'} - \delta_{ij'}\delta_{ji'} \right) \hat{r}_i \hat{p}_j \hat{r}_{i'} \hat{p}_{j'} \\
&= \hat{r}^2 \hat{p}^2 + \sum_i \sum_j \hat{r}_i [\hat{p}_j, \hat{r}_i] \hat{p}_j - \left(\hat{\vec{r}} \cdot \hat{\vec{p}} \right) \left(\hat{\vec{p}} \cdot \hat{\vec{r}} \right) - \sum_i \sum_j \hat{r}_i \hat{p}_j [\hat{r}_j, \hat{p}_i] \\
&= \hat{r}^2 \hat{p}^2 - \left(\hat{\vec{r}} \cdot \hat{\vec{p}} \right) \left(\hat{\vec{p}} \cdot \hat{\vec{r}} \right) - 2i\hbar \hat{\vec{r}} \cdot \hat{\vec{p}} \\
&= \hat{r}^2 \hat{p}^2 - \left(\hat{\vec{r}} \cdot \hat{\vec{p}} \right)^2 + i\hbar \left(\hat{\vec{r}} \cdot \hat{\vec{p}} \right).
\end{aligned} \tag{11.11}
$$

Note that we have to be very careful to keep track of the order of the different operators in this derivation. We used the antisymmetric tensor identity from Eq. (9.11) in the third line, we used the canonical commutation relation $[\hat{r}_j, \hat{p}_i] = i\hbar\delta_{ij}$, and we used Eq. (11.9) in the last line. This identity can be rewritten as

$$
\hat{p}^2 = \frac{1}{\hat{r}^2} \left(\hat{\vec{r}} \cdot \hat{\vec{p}} \right)^2 - \frac{i\hbar}{\hat{r}^2} \left(\hat{\vec{r}} \cdot \hat{\vec{p}} \right) + \frac{1}{\hat{r}^2} \hat{L}^2, \tag{11.12}
$$

by rearranging the equality and multiplying both sides by $\frac{1}{\hat{r}^2}$ from the left. This is an important identity that we will revisit in Sect. 11.3. The term proportional to $\hbar$ is the *quantum correction*. It appears because the quantum operators do not commute—it would not be present in a classical calculation with vectors. Note that we are using a shortcut notation here: $\hat{r}^2 = \hat{r}_x^2 + \hat{r}_y^2 + \hat{r}_z^2 = \left(\hat{\vec{r}} \right)^2$. This notation will be used throughout the remainder of the book for the position operator, the momentum operator, and the angular momentum operator.

We will also work with cross products of operators. Here, we find quantum corrections too. In a classical system, we always have $\vec{A} \times \vec{A} = 0$. Not so for all vector operators. Instead, we have

$$
\left(\hat{\vec{A}} \times \hat{\vec{A}} \right)_k = \sum_i \sum_j \epsilon_{ijk} \hat{A}_i \hat{A}_j. \tag{11.13}
$$

Don't be fooled by this equation. While it certainly looks like it should vanish because when we interchange i and j, we get a sign change from the Levi–Civita tensor. But, it only vanishes is $[\hat{A}_i, \hat{A}_j] = 0$. In other words, we can write the cross product of the vector operator with itself as

$$
\begin{aligned}
\left(\hat{\vec{A}} \times \hat{\vec{A}} \right)_k &= \tfrac{1}{2} \sum_i \sum_j \left(\epsilon_{ijk} \hat{A}_i \hat{A}_j + \epsilon_{jik} \hat{A}_j \hat{A}_i \right) \\
&= \tfrac{1}{2} \sum_i \sum_j \epsilon_{ijk} [\hat{A}_i, \hat{A}_j].
\end{aligned} \tag{11.14}
$$

This immediately tells us that $\hat{\vec{r}} \times \hat{\vec{r}} = 0$ and $\hat{\vec{p}} \times \hat{\vec{p}} = 0$, but for angular momentum, we have that

$$\left(\hat{\vec{L}} \times \hat{\vec{L}}\right)_k = \tfrac{1}{2} \sum_i \sum_j \epsilon_{ijk} [\hat{L}_i, \hat{L}_j] = \tfrac{i\hbar}{2} \sum_i \sum_j \sum_{k'} \epsilon_{ijk} \epsilon_{ijk'} \hat{L}_{k'}$$
$$= \tfrac{i\hbar}{2} \sum_j \sum_{k'} \left(\delta_{jj}\delta_{kk'} - \delta_{jk'}\delta_{jk}\right) \hat{L}_{k'} = i\hbar \hat{L}_k. \tag{11.15}$$

So, we have derived the curious identity

$$\hat{\vec{L}} \times \hat{\vec{L}} = i\hbar \hat{\vec{L}}, \tag{11.16}$$

where the right-hand side is again a quantum correction.

We end this section with a discussion about how rotations affect vector operators. This is a first glance at the powerful *Wigner-Ekhardt theorem*. In Sect. 9.2, we showed that a vector operator $\hat{\vec{v}}$ transformed under rotations as vectors. For example, under a rotation about the z-axis, we have

$$e^{-\frac{i}{\hbar}\alpha \hat{L}_z} \hat{v}_x e^{\frac{i}{\hbar}\alpha \hat{L}_z} = \cos\alpha \hat{v}_x + \sin\alpha \hat{v}_y$$
$$e^{-\frac{i}{\hbar}\alpha \hat{L}_z} \hat{v}_y e^{\frac{i}{\hbar}\alpha \hat{L}_z} = -\sin\alpha \hat{v}_x + \cos\alpha \hat{v}_y$$
$$e^{-\frac{i}{\hbar}\alpha \hat{L}_z} \hat{v}_z e^{\frac{i}{\hbar}\alpha \hat{L}_z} = \hat{v}_z. \tag{11.17}$$

This can be summarized in matrix form as

$$e^{-\frac{i}{\hbar}\alpha \hat{L}_z} \hat{\vec{v}} e^{\frac{i}{\hbar}\alpha \hat{L}_z} = \begin{pmatrix} \cos\alpha & \sin\alpha & 0 \\ -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \hat{v}_x \\ \hat{v}_y \\ \hat{v}_z \end{pmatrix}. \tag{11.18}$$

This relation says that when we apply the similarity transformation for the rotation of the vector operator $\hat{\vec{v}}$, it is equal to applying the rotation matrix to the vector operator. This result also holds for the general rotation, about an axis in the $\vec{e}_n$ direction, which is proven in Problem 11.1.6.

We will work later with a different basis for the vector operators that allows us to re-express the relation in Eq. (11.18) in terms of vector components related to $l = 1$ angular momentum and the $m = 1, 0, -1$ values. This then brings in the rotation matrices that we developed in Sects. 9.7 and 9.8.

Section 11.1 Problems:

11.1.1 Ordering of the Operators in the Vector Operator Definition Does Not Matter

We defined a vector operator to satisfy

$$[\hat{A}_i, \hat{L}_j] = i\hbar \sum_k \epsilon_{ijk} \hat{A}_k. \tag{11.19}$$

Show that this formula is identical to

$$[\hat{L}_i, \hat{A}_j] = i\hbar \sum_k \epsilon_{ijk} \hat{A}_k. \tag{11.20}$$

This is curious, because we interchanged terms in the commutator, which you might think should change the sign, but if you look carefully, you will see that is not the only thing we interchanged.

Hint: If having any trouble, write out a concrete case for specific i and j values.

Thanks to Mark Weitzman for pointing out this result to me.

11.1.2 Cross Products of Operators

Show the following cross products of vector operators

$$\begin{aligned}
\hat{\vec{r}} \times \hat{\vec{L}} &= \left(\hat{\vec{r}} \cdot \hat{\vec{p}}\right) \hat{\vec{r}} - \hat{r}^2 \hat{\vec{p}} + i\hbar \hat{\vec{r}}, \\
\hat{\vec{p}} \times \hat{\vec{L}} &= \hat{p}^2 \hat{\vec{r}} - \left(\hat{\vec{p}} \cdot \hat{\vec{r}}\right) \hat{\vec{p}} + i\hbar \hat{\vec{p}}, \\
\hat{\vec{r}} \times \hat{\vec{L}} &= -\hat{\vec{L}} \times \hat{\vec{r}} + 2i\hbar \hat{\vec{r}}, \\
\hat{\vec{p}} \times \hat{\vec{L}} &= -\hat{\vec{L}} \times \hat{\vec{p}} + 2i\hbar \hat{\vec{p}}, \\
\hat{\vec{r}} \times \hat{\vec{p}} &= -\hat{\vec{p}} \times \hat{\vec{r}}.
\end{aligned} \tag{11.21}$$

Show further that the Hermitization of $\hat{\vec{r}} \times \hat{\vec{L}}$ and $\hat{\vec{p}} \times \hat{\vec{L}}$ are

$$\begin{aligned}
\tfrac{1}{2}\left(\hat{\vec{r}} \times \hat{\vec{L}} - \hat{\vec{L}} \times \hat{\vec{r}}\right) &= \hat{\vec{r}} \times \hat{\vec{L}} - i\hbar \hat{\vec{r}}, \\
\tfrac{1}{2}\left(\hat{\vec{p}} \times \hat{\vec{L}} - \hat{\vec{L}} \times \hat{\vec{p}}\right) &= \hat{\vec{p}} \times \hat{\vec{L}} - i\hbar \hat{\vec{p}}.
\end{aligned} \tag{11.22}$$

11.1.3 The Cross Product of Two Vector Operators Is a Vector Operator

If $\hat{\vec{A}}$ and $\hat{\vec{B}}$ are both vector operators, show that $\hat{\vec{A}} \times \hat{\vec{B}}$ is also a vector operator.

Hint: if you get stuck trying to work out the derivation all the way through to the end, try evaluating the goal you are seeking to establish ($i\hbar \sum_k \epsilon_{ijk} \left(\hat{\vec{A}} \times \hat{\vec{B}}\right)_k$) and see if it can be put into the form of the result you calculated when you computed the required commutators for the vector operator.

11.1.4 Hermitization of the Cross Product of Two Vector Operators

Show that the Hermitization of $\hat{\vec{A}} \times \hat{\vec{B}}$ for $\hat{\vec{A}}$ and $\hat{\vec{B}}$ both Hermitian vector operators is

$$\frac{1}{2}\left(\hat{\vec{A}} \times \hat{\vec{B}} - \hat{\vec{B}} \times \hat{\vec{A}}\right) = \hat{\vec{A}} \times \hat{\vec{B}} - \frac{1}{2}\sum_{ijk} \epsilon_{ijk}[\hat{A}_i, \hat{B}_j]\vec{e}_k, \tag{11.23}$$

where we recall that $\vec{e}_k$ is a unit vector pointing in the kth Cartesian direction (and it is *not* an operator).

11.1.5 Rotation of the Dot Product of Two Vector Operators

For vector operators $\hat{\vec{A}}$ and $\hat{\vec{B}}$, show that

$$e^{-\frac{i}{\hbar}\alpha \hat{L}_z} \hat{\vec{A}} \cdot \hat{\vec{B}}\, e^{\frac{i}{\hbar}\alpha \hat{L}_z} = \hat{\vec{A}} \cdot \hat{\vec{B}}, \tag{11.24}$$

or, in other words, verify explicitly that $\hat{\vec{A}} \cdot \hat{\vec{B}}$ is a scalar operator.

11.1.6 General Rotation of a Vector Operator

In the text, we worked out the rotation identity for a vector operator rotated by an angle α about the z-axis. Work out the rotation identity for how a vector operator changes for a rotation by an angle α about the axis $\vec{e}_n$, which is oriented in the θ-ϕ direction (that is, the radial unit vector direction in spherical coordinates for an axis pointing along $\vec{r}$).

11.2 Central Forces and Spherical Coordinates

It is common to study quantum mechanics after studying intermediate classical mechanics. This book, however, assumes only high-school-level knowledge of classical mechanics. We have now reached the point where we will start to develop the quantum theory for atoms (with a primary focus on the simplest atom, hydrogen). So, we need to be sure you understand the background for the forces that are at play in atoms. These forces are of a special type, called *central forces*. Such a force is directed in the radial direction, and it depends on the distance from the force center (which we will take to be the origin of our coordinate system). We show how a central force works in Fig. 11.1.

Let's start by recalling the discussion of forces that we had in Chap. 1. We pointed out that a force is a push or a pull on something. We are familiar with contact forces when we push our hand on a table, or pat someone on the back. But in quantum mechanics, the two most important forces are noncontact forces—Hooke's law (force exerted by a spring), which grows linearly as you move further from the center and Coulomb's law (electrostatic force on charged particles), which decreases as $\frac{1}{r^2}$ as you move further and further from the force center. In both cases, the force is attractive when directed toward the origin, and it is always directed in the radial direction. Its strength only depends on the distance from the center of force.

While the words describing central forces are simple and clear, I have found that working with these concepts can be challenging for students. The main reason is due to a lack of familiarity with spherical coordinates. Understanding these coordinates

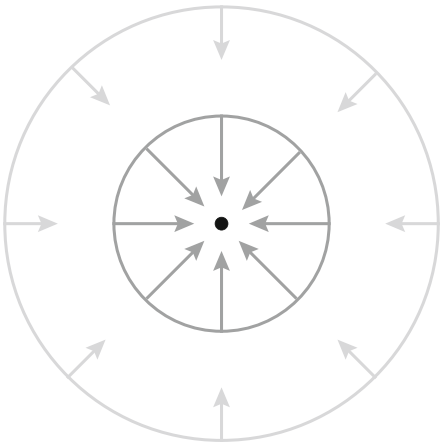

Fig. 11.1 Example of how a central force works. The solid black circle is the force center. The larger circles are drawn with a fixed radius about the force center. The arrows show the central forces (here attractive) acting on a particle located at each of their respective tails. The forces for one radii all have the same length (because the force depends only on the distance to the center), and they all point toward the center (for an attractive force). The strength of the force can change for different radii—here, it gets smaller as the radius gets larger

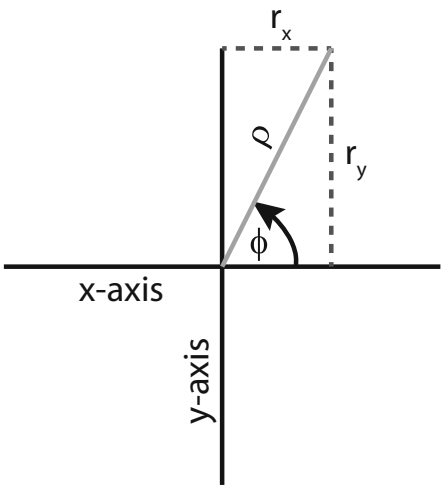

Fig. 11.2 Cartesian and polar coordinates for two dimensions. The Cartesian coordinates are r_x and r_y along the x- and y-axes, respectively (dashed lines). The polar coordinates are $\rho = \sqrt{r_x^2 + r_y^2}$ and $\phi = \tan^{-1}\left(\frac{r_y}{r_x}\right)$. The polar angle ϕ is measured with respect to the x-axis

and working with them are critical to many aspects of quantum mechanics, so we will slowly develop the full theory for them next.

We begin by considering the simpler case of two dimensions, where everything can be more easily visualized and worked out, moving to three dimensions afterward. Let's begin by reminding ourselves about polar coordinates. We have discussed them already in the context of complex numbers in Sect. 5.2 when we converted from the Cartesian to the polar form and back. But now we focus on two-dimensional systems described by real coordinates. The Cartesian form is the vector (r_x, r_y) or $\vec{\rho} = r_x\vec{e}_x + r_y\vec{e}_y$, where r_α are real numbers corresponding to the Cartesian coordinates of the vector. The polar coordinates are the radius $\rho = \sqrt{r_x^2 + r_y^2}$ and the polar angle $\phi = \tan^{-1}\frac{r_y}{r_x}$ (measured relative to the x-axis), as shown in Fig. 11.2. The angular coordinate requires some additional comments. It is not well defined just from the arctangent, because that function only produces angles that satisfy $-\frac{\pi}{2} \le \phi \le \frac{\pi}{2}$ and the angular coordinate runs over a domain of length 2π. We discuss the details for how to do this in Problem 11.2.1. Here, we simply note that there is an alternative way one can specify the coordinate, which we already encountered when we discussed the particle on a circle in Sect. 7.4, namely to use the cosine and sine of the angle. In other words, by setting $\cos\phi = \frac{r_x}{\rho}$ *and* $\sin\phi = \frac{r_y}{\rho}$, all trigonometric functions of ϕ can be immediately determined, and we will find that usually that is what we need—trigonometric functions of ϕ. In addition, this will be critical when we consider angle operators defined in terms of Cartesian operators.

Most students are familiar with this aspect of polar coordinates. Where many students struggle is when we express the position vectors in polar form as $\vec{\rho} = \rho\vec{e}_\rho$, that is in terms of the unit vector $\vec{e}_\rho$ in the radial direction. The first question is why is there only one unit vector (in the Cartesian form, we had two unit vectors)? This is because of the way the radial unit vector is defined. It is a unit vector in

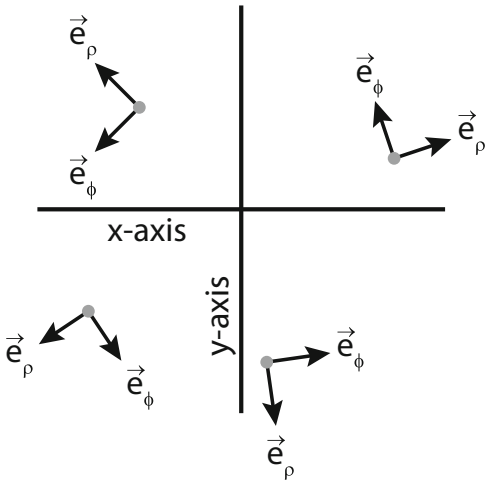

Fig. 11.3 Illustration of the radial and polar unit vectors for four different points in the two-dimensional plane. Note how the radial unit vector is always in the radial direction, the polar unit vector is always perpendicular to it, and how they change directions for different points in the plane. Such behavior is generic for unit vectors in non Cartesian coordinate systems. Think about what this means when we construct vector operators using these unit vectors

the direction from the origin to (r_x, r_y); hence, it is in the same direction as $\vec{\rho}$. In other words, $\vec{e}_\rho = \frac{\vec{\rho}}{\rho}$; one can easily check that this is a unit vector. The key difference from the Cartesian case is that the Cartesian unit vectors always point along the same direction (along the positive Cartesian axes), while the radial unit vector *changes* direction depending on what direction the vector (r_x, r_y) is pointing in the plane. Using the facts that $r_x = \rho\cos\phi$ and $r_y = \rho\sin\phi$, we immediately find that $\vec{e}_\rho = (\cos\phi, \sin\phi)$. The polar unit vector is perpendicular to the radial unit vector and points in the direction where ϕ increases with ρ held constant. Hence, we have that $\vec{e}_\phi = (-\sin\phi, \cos\phi)$. In Fig. 11.3, we plot the radial and polar unit vectors for some different points in the plane. This is to solidify for you that these unit vectors change directions as we move to different points in the plane. Be sure you understand this point. It is important.

Armed with the results for two dimensions, we are now ready to jump into the realm of three dimensions. Things are quite similar, but it is harder to draw figures to visualize the results. So be prepared to focus.

To start, a point in three dimensions is given by $(r_x, r_y, r_z) = r_x\vec{e}_x + r_y\vec{e}_y + r_z\vec{e}_z$, with the Cartesian unit vectors $\vec{e}_\alpha$ and the Cartesian coordinates r_α. The spherical coordinates are the radius r, the polar angle θ, and the azimuthal angle ϕ. The polar angle is the tilt of the radial vector relative to the z-axis, while the azimuthal angle is the angle of the radial vector measured from the x-axis and projected onto the x-y plane. The radial coordinate is $r = \sqrt{r_x^2 + r_y^2 + r_z^2}$. In spherical coordinates, the z-axis is vertical (see Fig. 11.4). Then the polar angle θ is the angle between (r_x, r_y, r_z) relative to the z-axis. We compute its cosine via the dot product

$$\cos\theta = \frac{\vec{r}\cdot\vec{e}_z}{r}. \tag{11.25}$$

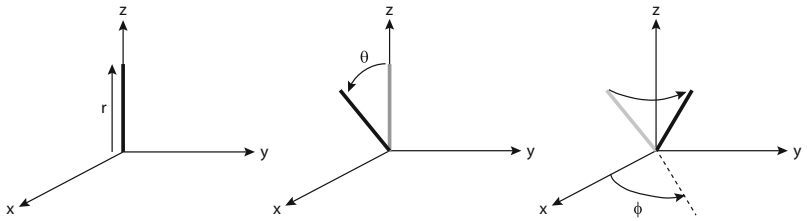

Fig. 11.4 Spherical coordinate system. The radius is $r = \sqrt{r_x^2 + r_y^2 + r_z^2}$. The polar angle θ is the angle from the z-axis to the vector $\vec{r}$, while the azimuthal angle ϕ is analogous to the angle we used in two dimensions. In the three panels, we show how you get to the desired point by moving a distance r along the z-axis, rotating by θ about the y-axis and then rotating by ϕ about the z-axis. Then we have $r_x = r\sin\theta\cos\phi$, $r_y = r\sin\theta\sin\phi$, and $r_z = r\cos\theta$

Here, $\theta = \cos^{-1}\frac{r_z}{r}$. The angle runs from $0 \le \theta \le \pi$. I find the following analogy useful. Stand up and hold your right arm out at an angle with your elbow tucked against your side. Extend your index finger. The tip of your finger is the (r_x, r_y, r_z) point and the angle θ is the angle between your body (vertical and in the z-direction) and the point at your index finger. The azimuthal angle ϕ is the same angle as the one we used in two dimensions. It is unfortunate that mathematicians customarily interchange these two angles, calling the polar angle ϕ and the azimuthal angle θ. Since this is a physics book, we use the physics convention throughout.

The azimuthal angle ϕ is then defined in the same way as we did in two dimensions—$\phi = \tan^{-1}\frac{r_x}{r_y}$. Now, going back to our analogy, point your arms with the index finger outstretched and the elbow tucked into your side and pointing straight in front of you. This is $\phi = 0$. Now, use your feet to turn counter clockwise, keeping your body straight up and your arm pointing to your front. As you turn all the way around, you trace out the angle $0 \le \phi \le 2\pi$. We consider only the two-dimensional plane to describe this angle. The two-dimensional vector $\vec{\rho} = r_x\vec{e}_x + r_y\vec{e}_y$ has a length that satisfies $\sqrt{r_x^2 + r_y^2} = r\sin\theta$. How do we know this? Recall that $\cos\theta = \frac{r_z}{r}$. This implies that

$$\sin\theta = \sqrt{1-\cos^2\theta} = \sqrt{1-\frac{r_z^2}{r^2}} = \sqrt{\frac{r_x^2+r_y^2}{r^2}} = \frac{\sqrt{r_x^2+r_y^2}}{r}. \tag{11.26}$$

Using these results yields $r_x = r\sin\theta\cos\phi$, $r_y = r\sin\theta\sin\phi$, and $r_z = r\cos\theta$.

We define the radial unit vector in the same way as we did in two dimensions:

$$\vec{e}_r = \frac{\vec{r}}{r} = \sin\theta\cos\phi\,\vec{e}_x + \sin\theta\sin\phi\,\vec{e}_y + \cos\theta\,\vec{e}_z. \tag{11.27}$$

It points in the direction from the origin to (r_x, r_y, r_z). The $\vec{e}_\theta$ direction lies in a plane that goes through the z-axis and the points (r_x, r_y, r_z). It satisfies

$$\vec{e}_\theta = \cos\theta\cos\phi\,\vec{e}_x + \cos\theta\sin\phi\,\vec{e}_y - \sin\theta\,\vec{e}_z. \tag{11.28}$$

Finally, the unit vector in the ϕ-direction is

$$\vec{e}_\phi = \vec{e}_r \times \vec{e}_\theta = -\sin\phi\,\vec{e}_x + \cos\phi\,\vec{e}_y. \tag{11.29}$$

Just like in the two-dimensional case, these unit vectors do change direction as one changes the coordinates and moves to different positions on the sphere.

It does take some practice and some comfort level getting used to working with these two coordinate systems. We will continue to practice as we develop further results in spherical coordinates within this chapter.

Okay. Now that we have finished with the geometry, we are ready to investigate central forces in quantum mechanics. Because the central forces depend only on $\hat{r} = \sqrt{\hat{r}_x^2 + \hat{r}_y^2 + \hat{r}_z^2}$ and are oriented along (or opposite to) the radial direction, their potentials are just functions of $\hat{r}$—they have no angular dependence. Hence, the potential is constant on the surface of spheres and thereby is invariant under rotations about any axis that goes through the origin. There are three primary potentials we will consider. The first is the simple-harmonic oscillator $V(\hat{r}) = \frac{1}{2}\kappa\hat{r}^2$. The second is the Coulomb potential $V(\hat{r}) = -\frac{e^2}{4\pi\epsilon_0 r}$. The third is the particle in spherical box, which we will work with in Chap. 12 and defer until then to describe.

The Hamiltonian for a central potential becomes

$$\hat{H} = \frac{1}{2m}\left(\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2\right) + V(\hat{r}) = \frac{\hat{p}^2}{2m} + V(\hat{r}), \tag{11.30}$$

where we define $\hat{p}^2 = \hat{\vec{p}}^{\,2} = \hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2$. Recall from Sect. 11.1 that we showed that scalar operators commute with angular momentum $\hat{L}_i$ and that dot products of vector operators are scalar operators. Hence, we immediately learn that $[\hat{H}, \hat{L}_i] = 0$ for $i = x$, y, and z. This means that both $\hat{\vec{L}}^2$ and $\hat{L}_z$ commute with $\hat{H}$ for problems with central forces. We will see later that we can employ this to write quantum states in central potentials in a factorized form with a tensor product as

$$|\psi\rangle = |\psi_r\rangle \otimes |l, m\rangle. \tag{11.31}$$

Here, $|\psi_r\rangle$ is a quantum state that is only acted on by the radial component of the momentum operator and the radial coordinate operator. All of the angle dependence of the quantum states is encompassed in the angular momentum eigenstates $|l, m\rangle$ (of course, to see the angular dependence, we need to form the position-space wavefunctions by taking the overlap with $\langle\theta, \phi|$ as we did when we formed the spherical harmonics $Y_{lm}(\theta, \phi) = \langle\theta, \phi\,|\,l, m\rangle$. We expect there to be a consequence to the fact that the central potential is rotationally invariant. Indeed there is, but as we saw with even potentials that have even and odd wavefunctions, here rotationally invariant potentials do have rotational invariant eigenfunction (s-waves), but they also have many other wavefunctions that have broken rotational symmetry (p-waves, d-waves, f-waves, and so on).

Section 11.2 Problems:

11.2.1 Complete Definition of the Polar Angle in Two Dimensions

In the two-dimensional example of polar coordinates, we loosely defined the angle ϕ via $\phi = \tan^{-1}\frac{r_y}{r_x}$. This gives angles that range over two quadrants, with $r_x > 0$. The angle ϕ satisfies $-\frac{\pi}{2} \leq \phi \leq \frac{\pi}{2}$. In this problem, we explore some different ways to extend the definition to include points with $r_x < 0$.

(a) By adding or subtracting angles like $\frac{\pi}{2}$ or π, determine a definition for ϕ that holds for $-\pi \leq \phi \leq \pi$.
Hint: Use the unit step function to incorporate different results for ϕ when $r_x > 0$ and when $r_x < 0$. Be sure your definitions provide a continuous angle ϕ over the range.

(b) Alternatives like $\cos^{-1}\frac{r_x}{r}$ or $\sin^{-1}\frac{r_y}{r}$ can also be employed to define ϕ. Do this using the arc cosine over the range $0 \leq \phi \leq 2\pi$.
Hint: You will need unit step functions here as well for the two different cases you will encounter.

(c) If you define ϕ implicitly through both $\cos\phi$ and $\sin\phi$, then you can define it for all four quadrants, without explicitly determining ϕ, but working exclusively with the sine and cosine functions. Explain how this can be done. It is this third option that is used when we construct position-space operators in spherical polar coordinates.

11.2.2 Relationship Between Cartesian and Spherical Polar Coordinates

In the text, we showed that θ was defined by $\cos\theta = \frac{\vec{r}\cdot\vec{e}_z}{r} = \frac{r_z}{r}$. This implies that $r_z = r\cos\theta$. Use the "loose" definition of $\phi = \tan^{-1}\frac{r_y}{r_x}$ to verify that $r_x = r\sin\theta\cos\phi$ and $r_y = r\sin\theta\sin\phi$.

11.2.3 Unit Vectors in Spherical Polar Coordinates

Using the results for r_x, r_y, and r_z in terms of r, θ, and ϕ described in the previous problem, show that

$$\vec{e}_r = \frac{(r_x, r_y, r_z)}{r} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta) \text{ and}$$
$$\vec{e}_\phi = \frac{1}{\rho}(-r_y, r_x, 0) = (-\sin\phi, \cos\phi, 0). \tag{11.32}$$

Then compute $\vec{e}_\theta = \vec{e}_\phi \times \vec{e}_r$ and show that it satisfies $\vec{e}_\theta = (\cos\theta\cos\phi, \cos\theta\sin\phi, -\sin\theta)$. Verify that these three vectors form an orthonormal set.

11.2.4 Commutators of Central Potential and Kinetic Energy with Angular Momentum

Explain why the fact that $[\hat{r}, \hat{L}_i] = 0$ implies that $[V(\hat{r}), \hat{L}_i] = 0$. Similarly, explain why the kinetic energy commutes with all $\hat{L}_i$. Finally, explain why $\hat{L}^2$ and $\hat{L}_z$ commute with a central-force Hamiltonian. This result means we can work with states that have definite l and definite m when solving central potential problems.

11.2.5 Isotropic Harmonic Oscillator

Show that the simple harmonic oscillator potential in three dimensions, given by

$$V(\hat{r}_x, \hat{r}_y, \hat{r}_z) = \tfrac{1}{2}\kappa_x\hat{r}_x^2 + \tfrac{1}{2}\kappa_y\hat{r}_y^2 + \tfrac{1}{2}\kappa_z\hat{r}_z^2 \tag{11.33}$$

becomes a central potential only if $\kappa_x = \kappa_y = \kappa_z = \kappa$ (recall that $\kappa = m\omega^2$ is the spring constant). This is called the *isotropic harmonic oscillator*.

11.3 Radial Momentum and Angular Momentum

Central-force problems in quantum mechanics are best solved in polar or spherical polar coordinates. In a calculus-based course, one uses a complicated procedure to convert the coordinates, derivatives, and unit vectors from the Cartesian coordinate system to the polar or spherical polar coordinate system. We take a different approach here following closely the geometry and using no calculus. The technical procedure we will work out is called *separation of variables*. In essence, we simply want to separate the motion into radial-directed motion and angular motion on the surface of the sphere. It should be intuitive that such a separation is possible, but we need to do it in a mathematically precise manner.

We begin in two dimensions. In Cartesian coordinates, we have the operators $\hat{r}_x$, $\hat{r}_y$, $\hat{p}_x$, and $\hat{p}_y$. We want to work in polar coordinates, where naively, we would have

$\hat{\rho}, \hat{\phi}, \hat{p}_\rho$ and $\hat{p}_\phi$, but there are some subtleties involved with defining these operators, similar to how the position operator was not well defined when we discussed the particle on the circle in Sect. 7.4. Our first step is to figure out the precise definitions of these operators and determine their commutation relations.

We begin with $\hat{\rho}$, which is defined via $\hat{\rho}^2 = \hat{r}_x^2 + \hat{r}_y^2$. One might be concerned about how we take an operator square root. The way to do this is to work with eigenstates of $\hat{\rho}^2$ and then take the positive square root of the eigenvalues—in other words, the square root operator has the same eigenvectors as the original operator, but has different eigenvalues. We also can formally work with the square root, as we show you next in computing commutators. We compute the commutators of the radial position operator with Cartesian momentum, by using the Leibniz rule:

$$[\hat{p}_x, \hat{\rho}^2] = [\hat{p}_x, \hat{r}_x^2 + \hat{r}_y^2] = [\hat{p}_x, \hat{r}_x^2] = \hat{r}_x[\hat{p}_x, \hat{r}_x] + [\hat{p}_x, \hat{r}_x]\hat{r}_x = -2i\hbar\hat{r}_x. \tag{11.34}$$

Similarly, we have $[\hat{p}_y, \hat{\rho}^2] = -2i\hbar\hat{r}_y$. Now comes the trick. We use Leibniz again to compute $[\hat{p}_x, \hat{\rho}^2]$ in the form

$$[\hat{p}_x, \hat{\rho}^2] = \hat{\rho}[\hat{p}_x, \hat{\rho}] + [\hat{p}_x, \hat{\rho}]\hat{\rho} = -2i\hbar\hat{r}_x. \tag{11.35}$$

We would like to divide by $\hat{\rho}$, but to do this, we need to know that $[\hat{\rho}, [\hat{\rho}, \hat{p}_x]] = 0$ so we can shift the $\hat{\rho}$ on the right to the left of the product in the second term (and then multiply both sides of the equation by $\frac{1}{\hat{\rho}}$). The idea for how to show this comes from Arno Bohm's quantum textbook (Bohm, 1994). Because the eigenvectors for $\hat{\rho}$ and $\hat{\rho}^2$ are identical, if the squared operator commutes with another operator, so does the linear power of the operator, because commutation relations are determined by the eigenvectors. So, we compute the nested commutator with $\hat{\rho}^2$ and show it vanishes. We have

$$\begin{aligned}
[\hat{\rho}^2, [\hat{\rho}, \hat{p}_x]] &= \hat{\rho}^2[\hat{\rho}, \hat{p}_x] - [\hat{\rho}, \hat{p}_x]\hat{\rho}^2 \\
&= \hat{\rho}\left(\hat{\rho}[\hat{\rho}, \hat{p}_x]\right) \underbrace{- \left(\hat{\rho}[\hat{\rho}, \hat{p}_x]\right)\hat{\rho} + \hat{\rho}\left([\hat{\rho}, \hat{p}_x]\hat{\rho}\right)}_{=0} - \left([\hat{\rho}, \hat{p}_x]\hat{\rho}\right)\hat{\rho} \\
&= [\hat{\rho}, \hat{\rho}[\hat{\rho}, \hat{p}_x]] + [\hat{\rho}, [\hat{\rho}, \hat{p}_x]\hat{\rho}] \\
&= [\hat{\rho}, [\hat{\rho}^2, \hat{p}_x]] = 2i\hbar[\hat{\rho}, \hat{r}_x] = 0.
\end{aligned} \tag{11.36}$$

In the first line, we expand out the commutator. Then, we add zero and regroup the terms, recognizing that we actually have a commutator of $\hat{\rho}$ with either $\hat{\rho}[\hat{\rho}, \hat{p}_x]$ or $[\hat{\rho}, \hat{p}_x]\hat{\rho}$. Then, we recognize we can use the Leibniz rule in reverse to combine the sum of the two commutators into a single commutator with $\hat{\rho}^2$. Finally, we substitute in the known commutator $[\hat{\rho}^2, \hat{p}_x]$. This then establishes that $[\hat{\rho}^2, [\hat{\rho}, \hat{p}_x]] = 0$ and thereby, we also have that $[\hat{\rho}, [\hat{\rho}, \hat{p}_x]] = 0$ too. This means we can move the $\hat{\rho}$ operator through the commutator in Eq. (11.35) to place both $\hat{\rho}$ operators on the left. Then we multiply both sides by $\frac{1}{\hat{\rho}}$ from the left, and we finally find that

$[\hat{p}_x, \hat{\rho}] = -i\hbar\frac{\hat{r}_x}{\hat{\rho}}$ and using similar steps $[\hat{p}_y, \hat{\rho}] = -i\hbar\frac{\hat{r}_y}{\hat{\rho}}$. We also can compute $[\hat{p}_x, \frac{1}{\hat{\rho}}]$ using Leibniz again. We have

$$0 = [\hat{p}_x, 1] = [\hat{p}_x, \hat{\rho}\,\tfrac{1}{\hat{\rho}}] = \hat{\rho}[\hat{p}_x, \tfrac{1}{\hat{\rho}}] + [\hat{p}_x, \hat{\rho}]\tfrac{1}{\hat{\rho}}. \tag{11.37}$$

Hence, we have

$$[\hat{p}_x, \tfrac{1}{\hat{\rho}}] = -\tfrac{1}{\hat{\rho}}[\hat{p}_x, \hat{\rho}]\tfrac{1}{\hat{\rho}} = i\hbar\frac{\hat{r}_x}{\hat{\rho}^3}. \tag{11.38}$$

Similarly, we have $[\hat{p}_y, \frac{1}{\hat{\rho}}] = i\hbar\frac{\hat{r}_y}{\hat{\rho}^3}$. Let us pause for a moment to see what we have done. We have figured out how to find the commutators of momentum with functions of $\hat{\rho} = \sqrt{\hat{r}_x^2 + \hat{r}_y^2}$ by judiciously using our commutation rules. If you were given this as a problem to do without any hints, it would seem to be impossible. But after employing a few simple ideas in the right order, we find that these commutators are manageable to compute.

We move onto one of our next goals—determining the radial momentum $\hat{p}_\rho$, that is, the component of the momentum that lies in the radial direction, corresponding to the radial motion of the particle. In a classical world, we would simply project $\vec{p}$ onto the radial unit vector $\vec{e}_\rho = \frac{\vec{\rho}}{\rho}$. But in quantum mechanics, we have to worry about operators that do not commute with each other. Hence, we try to follow a similar definition, but we have to worry about preserving operator ordering. In other words, the unit vector in the radial direction becomes a *unit vector operator* in the radial direction. Our first requirement is that $\hat{p}_\rho$ be Hermitian (note this is not the same as self-adjoint; in general, the radial momentum is *not* self-adjoint). We first try the simple choice of multiplying by the radial unit vector from the left

$$\hat{p}_\rho \stackrel{?}{=} \frac{\hat{\vec{\rho}}}{\hat{\rho}} \cdot \hat{\vec{p}} = \frac{1}{\hat{\rho}}\left(\hat{r}_x\hat{p}_x + \hat{r}_y\hat{p}_y\right). \tag{11.39}$$

Is this Hermitian? The conjugate is (remember that $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger\hat{A}^\dagger$ and position and momentum operators are both Hermitian)

$$\left(\frac{\hat{\vec{\rho}}}{\hat{\rho}} \cdot \hat{\vec{p}}\right)^\dagger = \left(\hat{p}_x\hat{r}_x + \hat{p}_y\hat{r}_y\right)\frac{1}{\hat{\rho}}, \tag{11.40}$$

since all position and momentum operators are Hermitian. To convert back to the original form, we need to commute the $\hat{r}_x$ and $\hat{r}_y$ to the left as well as the $\frac{1}{\hat{\rho}}$. While there could be a miracle and these become equal, this seems to be quite unlikely (and indeed doing the calculation shows that this operator is *not* Hermitian). Does this mean we are stuck and cannot define the radial momentum? No! We simply start with a manifestly Hermitian version of the radial momentum, obtained by symmetrizing the expression. Namely, we choose

$$\hat{p}_\rho = \frac{1}{2\hat{\rho}}\hat{\vec{\rho}}\cdot\hat{\vec{p}} + \hat{\vec{p}}\cdot\hat{\vec{\rho}}\frac{1}{2\hat{\rho}} = \frac{1}{2\hat{\rho}}\left(\hat{r}_x\hat{p}_x + \hat{r}_y\hat{p}_y\right) + \left(\hat{p}_x\hat{r}_x + \hat{p}_y\hat{r}_y\right)\frac{1}{2\hat{\rho}}. \tag{11.41}$$

Now, when we take the Hermitian conjugate, we find that

$$\hat{p}_\rho^\dagger = \left(\hat{p}_x\hat{r}_x + \hat{p}_y\hat{r}_y\right)\frac{1}{2\hat{\rho}} + \frac{1}{2\hat{\rho}}\left(\hat{r}_x\hat{p}_x + \hat{r}_y\hat{p}_y\right) = \hat{p}_\rho, \tag{11.42}$$

so we will use this as our definition of the radial momentum. Note that the radial momentum is Hermitian, but it turns out not to be self-adjoint. The latter point is not so simple to show without calculus, but what it implies is the radial momentum does not have a complete set of eigenstates—in fact, *we never work with eigenstates of the radial momentum operator*. The difference between Hermitian and self-adjoint operators is determined by the domains that the operators have when they act on the bras versus the kets. We do not have the mathematical tools at our disposal to determine these properties ourselves at this stage. But this issue only arises when the operator in question is unbounded. Then we can either have that a Hermitian operator is essentially self-adjoint, which is, for all intents and purposes, a self-adjoint operator, or it is not self adjoint. The Cartesian position and momentum operators are essentially self-adjoint, but the radial momentum is not. This will not create any issues with how we work with the operator though. We just will never introduce eigenstates of the operator.

We find it more convenient for many calculations to move all coordinate operators to the left. Doing so gives us

$$\begin{aligned}\hat{p}_\rho &= \frac{1}{\hat{\rho}}\left(\hat{r}_x\hat{p}_x + \hat{r}_y\hat{p}_y\right) + \tfrac{1}{2}\left([\hat{p}_x,\hat{r}_x] + [\hat{p}_y,\hat{r}_y]\right)\frac{1}{\hat{\rho}} + \tfrac{1}{2}\left(\hat{r}_x\left[\hat{p}_x,\tfrac{1}{\hat{\rho}}\right] + \hat{r}_y\left[\hat{p}_y,\tfrac{1}{\hat{\rho}}\right]\right)\\ &= \frac{1}{\hat{\rho}}\hat{\vec{\rho}}\cdot\hat{\vec{p}} - \frac{i\hbar}{\hat{\rho}} + \tfrac{1}{2}\left(\hat{r}_x\frac{\hat{r}_x}{\hat{\rho}^3} + \hat{r}_y\frac{\hat{r}_y}{\hat{\rho}^3}\right) = \frac{1}{\hat{\rho}}\hat{\vec{\rho}}\cdot\hat{\vec{p}} - \frac{i\hbar}{2\hat{\rho}},\end{aligned} \tag{11.43}$$

after using the fact that $\hat{r}_x^2+\hat{r}_y^2 = \hat{\rho}^2$. This is our final form for the radial momentum operator. Now, let's find the commutator with $\hat{\rho}$. We have

$$[\hat{\rho},\hat{p}_\rho] = \left[\hat{\rho},\tfrac{1}{\hat{\rho}}\hat{\vec{\rho}}\cdot\hat{\vec{p}}\right] = \frac{1}{\hat{\rho}}[\hat{\rho},\hat{r}_x\hat{p}_x + \hat{r}_y\hat{p}_y] = \frac{1}{\hat{\rho}}\left(\hat{r}_x\frac{i\hbar\hat{r}_x}{\hat{\rho}} + \hat{r}_y\frac{i\hbar\hat{r}_y}{\hat{\rho}}\right) = i\hbar, \tag{11.44}$$

after using $\hat{r}_x^2 + \hat{r}_y^2 = \hat{\rho}^2$ again. Hence, the radial coordinate and the radial momentum have the same canonical commutation relation as the Cartesian coordinates and Cartesian momenta! Note, however, that $\hat{p}_\rho$ is a complicated function of Cartesian position operators and Cartesian momentum operators. In particular, a Cartesian momentum eigenstate is not an eigenstate of $\hat{p}_\rho$. Indeed, the radial momentum does not have a complete set of eigenstates.

Now we need to define the momentum in the $\vec{e}_\phi$ direction. In classical mechanics, we have that $\vec{e}_\phi = -\frac{r_y}{\rho}\vec{e}_x + \frac{r_y}{\rho}\vec{e}_y$. So, we try to define the momentum in the angular direction as follows:

$$\hat{p}_\phi \stackrel{?}{=} \hat{\vec{e}}_\phi \cdot \hat{\vec{p}} = \frac{1}{\hat{\rho}} \left(-\hat{r}_y \hat{p}_x + \hat{r}_x \hat{p}_y \right). \tag{11.45}$$

Note that there is no ordering ambiguity here, because $[\hat{r}_x, \hat{p}_y] = [\hat{r}_y, \hat{p}_x] = 0$. Note further that we require here that the momentum operator dimensions are those of momentum. Furthermore, we have that

$$[\hat{\rho}, -\hat{r}_y \hat{p}_x + \hat{r}_x \hat{p}_y] = -\hat{r}_y [\hat{\rho}, \hat{p}_x] + \hat{r}_x [\hat{\rho}, \hat{p}_y] = -i\hbar \frac{\hat{r}_y \hat{r}_x}{\hat{\rho}} + i\hbar \frac{\hat{r}_x \hat{r}_y}{\hat{\rho}} = 0. \tag{11.46}$$

Hence, $\hat{p}_\phi$ commutes with $\hat{\rho}$ and with any function of $\hat{\rho}$. This implies that

$$\left(\hat{p}_\phi\right)^\dagger = \left(-\hat{p}_x \hat{r}_y + \hat{p}_y \hat{r}_x\right) \frac{1}{\hat{\rho}} = \frac{1}{\hat{\rho}} (-\hat{r}_y \hat{p}_x + \hat{r}_x \hat{p}_y) = \hat{p}_\phi, \tag{11.47}$$

and the operator is Hermitian. There is one final point to note. We have already seen this operator before, but you might not recognize it. Recall that $\hat{L}_k = \sum_{ij} \epsilon_{ijk} \hat{r}_i \hat{p}_j$, so that $\hat{L}_z = \hat{r}_x \hat{p}_y - \hat{r}_y \hat{p}_x = \hat{\rho} \hat{p}_\phi = \hat{p}_\phi \hat{\rho}$. So, we learn that

$$\hat{p}_\phi = \frac{\hat{L}_z}{\hat{\rho}}. \tag{11.48}$$

Note that because $[\hat{L}_z, \hat{\rho}] = 0$, we have no operator ordering issues with representing $\hat{p}_\phi$ in this manner. If you have a strong classical mechanics background, you might notice that this momentum in the angular direction is not the canonical momentum conjugate to the angular coordinate. This is fine. We are working here with the components of momentum in two orthogonal directions. That is all.

The final object we need to work with is the angular coordinate (or phase) operator $\hat{\phi}$, but just like the position operator on a circle, it is not a well-defined operator. Only the cosine and sine of that operator are well-defined, because they are consistent with periodic boundary conditions as $\phi \to \phi + 2\pi$. We actually can determine the sine and cosine in terms of the Cartesian operators and the radial coordinate operator, which then allows us to determine the relevant commutators we will need. For example, $\cos\hat{\phi} = \frac{\hat{r}_x}{\hat{\rho}}$, $\sin\hat{\phi} = \frac{\hat{r}_y}{\hat{\rho}}$, and $e^{i\hat{\phi}} = \frac{\hat{r}_x}{\hat{\rho}} + i\frac{\hat{r}_y}{\hat{\rho}}$. We can compute the commutator of the momentum with this operator by using this mapping. We have

$$\begin{aligned} \left[\hat{p}_\rho, e^{i\hat{\phi}}\right] &= \left[\hat{p}_\rho, \frac{\hat{r}_x}{\hat{\rho}} + i\frac{\hat{r}_y}{\hat{\rho}}\right] \\ &= \frac{1}{\hat{\rho}} \left[\hat{r}_x \hat{p}_x + \hat{r}_y \hat{p}_y - \frac{i\hbar}{2}, \frac{\hat{r}_x}{\hat{\rho}} + i\frac{\hat{r}_y}{\hat{\rho}}\right] \end{aligned}$$

$$= \frac{\hat{r}_x}{\hat{\rho}}\left(-\frac{i\hbar}{\hat{\rho}} + i\hbar\frac{\hat{r}_x^2}{\hat{\rho}^3} + i\hbar\frac{\hat{r}_y^2}{\hat{\rho}^3}\right) + i\frac{\hat{r}_y}{\hat{\rho}}\left(-\frac{i\hbar}{\hat{\rho}} + i\hbar\frac{\hat{r}_y^2}{\hat{\rho}^3} + i\hbar\frac{\hat{r}_x^2}{\hat{\rho}^3}\right)$$
$$= 0. \tag{11.49}$$

Even though we know the operator $\hat{\phi}$ is not well defined, this result allows us to think in terms of the commutator of the radial momentum operator with the angular coordinate. Here is how. We can rearrange the commutator from the form $\hat{p}_\rho e^{i\hat{\phi}} = e^{i\hat{\phi}}\hat{p}_\rho$ by multiplying on the left by $e^{-i\hat{\phi}}$ to the form $e^{-i\hat{\phi}}\hat{p}_\rho e^{i\hat{\phi}} = \hat{p}_\rho$. Now, expanding the Hadamard lemma gives us

$$\hat{p}_\rho - i[\hat{\phi}, \hat{p}_\rho] + \tfrac{1}{2}(-i)^2[\hat{\phi}, [\hat{\phi}, \hat{p}_\rho]] + \cdots = \hat{p}_\rho. \tag{11.50}$$

Clearly, this equality will only hold if we have that $[\hat{\phi}, \hat{p}_\rho] = 0$. This is the reason why we can work with the commutator of the radial momentum operator with the angular coordinate operator and not get into any trouble. Hence, because the commutator of the radial momentum operator with the exponential of the angular coordinate operator is zero, this then implies that we can think of it loosely as $[\hat{\phi}, \hat{p}_\rho] = 0$.

We have one more commutator to work out: $[\hat{\phi}, \hat{p}_\phi]$. Given that this momentum operator in an angular direction has a relationship with the z-component of the angular momentum operator, we first work out the Hadamard lemma with $\hat{L}_z$ as follows:

$$e^{-i\hat{\phi}}\hat{L}_z e^{i\hat{\phi}} = \left(\frac{\hat{r}_x}{\hat{\rho}} - i\frac{\hat{r}_y}{\hat{\rho}}\right)\hat{L}_z\left(\frac{\hat{r}_x}{\hat{\rho}} + i\frac{\hat{r}_y}{\hat{\rho}}\right)$$
$$= \frac{1}{\hat{\rho}^2}\left(\hat{r}_x\hat{L}_z\hat{r}_x + \hat{r}_y\hat{L}_z\hat{r}_y + i(\hat{r}_x\hat{L}_z\hat{r}_y - \hat{r}_y\hat{L}_z\hat{r}_x\right)$$
$$= \frac{1}{\hat{\rho}^2}\left\{(\hat{r}_x^2 + \hat{r}_y^2)\hat{L}_z + \hat{r}_x[\hat{L}_z, \hat{r}_x] + \hat{r}_y[\hat{L}_z, \hat{r}_y] + i\hat{r}_x[\hat{L}_z, \hat{r}_y] - i\hat{r}_y[\hat{L}_z, \hat{r}_x]\right\}$$
$$= \frac{1}{\hat{\rho}^2}\left(\hat{\rho}^2\hat{L}_z + i\hbar\hat{r}_x\hat{r}_y - i\hbar\hat{r}_x\hat{r}_y + \hbar\hat{r}_x^2 + \hbar\hat{r}_y^2\right) = \hat{L}_z + \hbar. \tag{11.51}$$

In this derivation, we used add-zero on the third line and then evaluated the commutators on the fourth line using the fact that these are vector operators. Comparing this result to the Hadamard lemma, we find it can only hold if we have

$$[\hat{\phi}, \hat{L}_z] = i\hbar, \tag{11.52}$$

in the same "loose" sense we discussed earlier. This is often shown as the canonical commutation relation for an angular coordinate operator, but one needs to recognize

the restrictions on this, given the inability to define an angle operator by itself. Continuing with this caution, we can also write $[\hat{\phi}, \hat{p}_\phi] = \frac{i\hbar}{\hat{\rho}}$.

Armed with these results, we can now calculate the separation of variables for the kinetic-energy operator

$$\hat{K} = \frac{1}{2m}\left(\hat{p}_x^2 + \hat{p}_y^2\right), \tag{11.53}$$

in polar coordinates. The algebra becomes a bit tedious, but is something every physicist should do at least once. So, get out your French cooking mindset. Here we go. To begin, we recall that $\hat{p}_\rho$ and $\hat{p}_\phi$ are the radial and tangential components of the momentum, so we anticipate that the kinetic energy should involve the sum of the squares of both. But, we have to be careful because of the operator ordering issues, which can lead to quantum corrections. We start with $\hat{p}_\rho^2$:

$$\left(\hat{p}_\rho\right)^2 = \frac{1}{\hat{\rho}}\left(\hat{r}_x\hat{p}_x + \hat{r}_y\hat{p}_y - \tfrac{i\hbar}{2}\right)\frac{1}{\hat{\rho}}\left(\hat{r}_x\hat{p}_x + \hat{r}_y\hat{p}_y - \tfrac{i\hbar}{2}\right). \tag{11.54}$$

The first step is to bring the $\frac{1}{\hat{\rho}}$ term from the middle all the way to the left. To do so requires us to move past $\hat{p}_\rho$, and because $\left[\hat{p}_\rho, \frac{1}{\hat{\rho}}\right] = \frac{i\hbar}{\hat{\rho}^2}$, we find that

$$\left(\hat{p}_\rho\right)^2 = \frac{1}{\hat{\rho}^2}\left(\hat{r}_x\hat{p}_x + \hat{r}_y\hat{p}_y + \tfrac{i\hbar}{2}\right)\left(\hat{r}_x\hat{p}_x + \hat{r}_y\hat{p}_y - \tfrac{i\hbar}{2}\right). \tag{11.55}$$

Expanding further gives

$$\begin{aligned}\left(\hat{p}_\rho\right)^2 &= \frac{1}{\hat{\rho}^2}\left(\hat{r}_x\hat{p}_x\hat{r}_x\hat{p}_x + 2\hat{r}_x\hat{r}_y\hat{p}_x\hat{p}_y + \hat{r}_y\hat{p}_y\hat{r}_y\hat{p}_y + \tfrac{\hbar^2}{4}\right)\\ &= \frac{1}{\hat{\rho}^2}\left(\hat{r}_x^2\hat{p}_x^2 + 2\hat{r}_x\hat{r}_y\hat{p}_x\hat{p}_y + \hat{r}_y^2\hat{p}_y^2 - i\hbar(\hat{r}_x\hat{p}_x + \hat{r}_y\hat{p}_y) + \tfrac{\hbar^2}{4}\right).\end{aligned} \tag{11.56}$$

Now, we examine $\hat{p}_\phi^2$ (and we recall that $\left[\hat{p}_\phi, \frac{1}{\hat{\rho}}\right] = 0$):

$$\begin{aligned}\left(\hat{p}_\phi\right)^2 &= \frac{1}{\hat{\rho}^2}\left(-\hat{r}_y\hat{p}_x + \hat{r}_x\hat{p}_y\right)\left(-\hat{r}_y\hat{p}_x + \hat{r}_x\hat{p}_y\right)\\ &= \frac{1}{\hat{\rho}^2}\left(\hat{r}_y^2\hat{p}_x^2 - \hat{r}_y\hat{p}_x\hat{r}_x\hat{p}_y - \hat{r}_x\hat{p}_y\hat{r}_y\hat{p}_x + \hat{r}_x^2\hat{p}_y^2\right)\\ &= \frac{1}{\hat{\rho}^2}\left(\hat{r}_y^2\hat{p}_x^2 - 2\hat{r}_x\hat{r}_y\hat{p}_x\hat{p}_y + \hat{r}_x^2\hat{p}_y^2 + i\hbar(\hat{r}_x\hat{p}_x + \hat{r}_y\hat{p}_y)\right).\end{aligned} \tag{11.57}$$

Hence, we find that

$$\left(\hat{p}_\rho\right)^2 + \left(\hat{p}_\phi\right)^2 = \hat{p}_x^2 + \hat{p}_y^2 + \tfrac{\hbar^2}{4\hat{\rho}^2}, \tag{11.58}$$

or

$$\hat{K} = \frac{\hat{p}_x^2 + \hat{p}_y^2}{2m} = \frac{\hat{p}_\rho^2 + \hat{p}_\phi^2}{2m} - \frac{\hbar^2}{8m\hat{\rho}^2} = \frac{\hat{p}_\rho^2}{2m} + \frac{\hat{L}_z^2 - \frac{\hbar^2}{4}}{2m\hat{\rho}^2}. \tag{11.59}$$

Note that the correction term is purely quantum mechanical, because it is proportional to $\hbar^2$. There is one final issue we need to address. In separation of variables, we want the operators we separate to commute; it is customary to use the right-hand equation which immediately satisfies this requirement because $[\hat{L}_z, \hat{\rho}] = 0$. We will see how this separation simplifies the energy eigenvalue problem and allows us to write the eigenstate in a tensor-product format later.

This completes our work on polar coordinates. We have determined the position and momentum operators and their commutation relations. We have also determined how to rewrite the kinetic energy in terms of these new operators. Now, we simply repeat this exercise for three-dimensions using spherical polar coordinates.

We work with spherical coordinates, which take the following form as operators:

$$\begin{aligned} \hat{r} &= \sqrt{\hat{r}_x^2 + \hat{r}_y^2 + \hat{r}_z^2} \text{ and } \hat{\rho} = \sqrt{\hat{r}_x^2 + \hat{r}_y^2}, \\ \cos\hat{\theta} &= \frac{\hat{r}_z}{\hat{r}} \text{ and } \sin\hat{\theta} = \frac{\hat{\rho}}{\hat{r}}, \\ \cos\hat{\phi} &= \frac{\hat{r}_x}{\hat{\rho}} \text{ and } \sin\hat{\phi} = \frac{\hat{r}_y}{\hat{\rho}}. \end{aligned} \tag{11.60}$$

Note that we do not write a formula for the angular-coordinate operators, because they are not well defined—they are only properly defined via the cosine and sine of the angular operators, which relates to the Cartesian operators. We begin with the radial momentum, which is defined in a manifestly Hermitian fashion via

$$\hat{p}_r = \frac{1}{2\hat{r}}\hat{\vec{r}} \cdot \hat{\vec{p}} + \hat{\vec{p}} \cdot \hat{\vec{r}}\frac{1}{2\hat{r}}, \tag{11.61}$$

just as before. This operator is Hermitian because it is an average of the two possible orderings and the momentum and position operators are each Hermitian (please verify the Hermiticity if you are unsure about this). For convenience, we would like to move all position operators to the left, so it looks like $\hat{p}_r = \frac{1}{\hat{r}}\hat{\vec{r}} \cdot \hat{\vec{p}} + \text{something}$, where "something" is the quantum correction due to noncommutativity of operators. We find that

$$\begin{aligned}\hat{p}_r &= \frac{1}{\hat{r}}\hat{\vec{r}}\cdot\hat{\vec{p}} + \tfrac{1}{2}\left([\hat{p}_x,\hat{r}_x]+[\hat{p}_y,\hat{r}_y]+[\hat{p}_z,\hat{r}_z]\right)\frac{1}{\hat{r}} \\ &+ \frac{1}{2}\left(\hat{r}_x\left[\hat{p}_x,\frac{1}{\hat{r}}\right]+\hat{r}_y\left[\hat{p}_y,\frac{1}{\hat{r}}\right]+\hat{r}_z\left[\hat{p}_z,\frac{1}{\hat{r}}\right]\right) \\ &= \frac{1}{\hat{r}}\hat{\vec{r}}\cdot\hat{\vec{p}} - \frac{3i\hbar}{2\hat{r}} + \frac{i\hbar}{2}\left(\hat{r}_x^2+\hat{r}_y^2+\hat{r}_z^2\right)\frac{1}{\hat{r}^3} = \frac{1}{\hat{r}}\hat{\vec{r}}\cdot\hat{\vec{p}} - \frac{i\hbar}{\hat{r}}\end{aligned} \tag{11.62}$$

and see that there is a quantum correction. Next, we work out the commutator of the radial coordinate operator with the radial momentum:

$$[\hat{r},\,\hat{p}_r] = \left[\hat{r},\,\frac{1}{\hat{r}}(\hat{r}_x\hat{p}_x+\hat{r}_y\hat{p}_y+\hat{r}_z\hat{p}_z)\right] = \frac{i\hbar}{\hat{r}}\,\frac{\hat{r}_x^2+\hat{r}_y^2+\hat{r}_z^2}{\hat{r}} = i\hbar. \tag{11.63}$$

So, we have the canonical commutation relation $[\hat{r},\,\hat{p}_r] = i\hbar$.

Before moving onto $\hat{p}_\theta$, we work with $\hat{p}_\phi$, which turns out to be identical to what we just did for two dimensions. Hence, we have that

$$\hat{p}_\phi = \frac{\hat{L}_z}{\hat{\rho}},\;\; [\hat{\phi},\,\hat{p}_\phi] = \frac{i\hbar}{\hat{\rho}},\;\; \text{and}\;\; [\hat{\phi},\,\hat{L}_z] = i\hbar, \tag{11.64}$$

where we recall that the commutation relations involving angular operators are "schematic," as we discussed earlier. Note that it is the planar radius $\hat{\rho}$ and not the spherical radius $\hat{r}$ in these formulas. Furthermore, a direct computation shows that $[\hat{r},\,\hat{p}_\phi] = 0$. Now, we work with $\hat{p}_\theta$. Recall that the operator-valued unit vector in the θ direction is

$$\begin{aligned}\hat{\vec{e}}_\theta &= \cos\hat{\theta}\cos\hat{\phi}\,\vec{e}_x + \cos\hat{\theta}\sin\hat{\phi}\,\vec{e}_y - \sin\hat{\theta}\,\vec{e}_z \\ &= \frac{\hat{r}_x\hat{r}_z}{\hat{\rho}\hat{r}}\vec{e}_x + \frac{\hat{r}_y\hat{r}_z}{\hat{\rho}\hat{r}}\vec{e}_y - \frac{\hat{\rho}}{\hat{r}}\vec{e}_z.\end{aligned} \tag{11.65}$$

We pick the symmetrized form of the dot product $\hat{p}_\theta = \frac{1}{2}\hat{\vec{e}}_\theta\cdot\hat{\vec{p}} + \frac{1}{2}\hat{\vec{p}}\cdot\hat{\vec{e}}_\theta$, which becomes

$$\hat{p}_\theta = \frac{\hat{r}_x\hat{r}_z}{2\hat{\rho}\hat{r}}\hat{p}_x + \frac{\hat{r}_y\hat{r}_z}{2\hat{\rho}\hat{r}}\hat{p}_y - \frac{\hat{\rho}}{2\hat{r}}\hat{p}_z + \hat{p}_x\frac{\hat{r}_x\hat{r}_z}{2\hat{\rho}\hat{r}} + \hat{p}_y\frac{\hat{r}_y\hat{r}_z}{2\hat{\rho}\hat{r}} - \hat{p}_z\frac{\hat{\rho}}{2\hat{r}}. \tag{11.66}$$

This can be rewritten, with all of the coordinate operators to the left, as

$$\hat{p}_\theta = \frac{\hat{r}_x\hat{r}_z}{\hat{\rho}\hat{r}}\hat{p}_x + \frac{\hat{r}_y\hat{r}_z}{\hat{\rho}\hat{r}}\hat{p}_y - \frac{\hat{\rho}}{\hat{r}}\hat{p}_z - \frac{i\hbar}{2}\frac{\hat{r}_z}{\hat{\rho}\hat{r}}, \tag{11.67}$$

which you will verify in one of the problems. Using $\hat{L}_x = \hat{r}_y\hat{p}_z - \hat{r}_z\hat{p}_y$ and $\hat{L}_y = \hat{r}_z\hat{p}_x - \hat{r}_x\hat{p}_z$, allows us to rewrite $\hat{p}_\theta$ as

$$\hat{p}_\theta = \frac{\hat{r}_x \hat{L}_y}{\hat{\rho}\hat{r}} - \frac{\hat{r}_y \hat{L}_x}{\hat{\rho}\hat{r}} - \frac{i\hbar}{2}\frac{\hat{r}_z}{\hat{\rho}\hat{r}}. \tag{11.68}$$

This form immediately tells us that $[\hat{r}, \hat{p}_\theta] = 0$, because $\hat{r}$ is a scalar operator. We next check the commutator with $\cos\hat{\phi}$ vanishes:

$$\begin{aligned}[\cos\hat{\phi}, \hat{p}_\theta] &= \frac{1}{\hat{\rho}\hat{r}}\left[\cos\hat{\phi}, \hat{r}_x\hat{L}_y - \hat{r}_y\hat{L}_x\right] \\ &= \frac{\hat{r}_z}{\hat{\rho}\hat{r}}[\cos\hat{\phi}, \hat{r}_x\hat{p}_x + \hat{r}_y\hat{p}_y],\end{aligned} \tag{11.69}$$

because $\hat{r}_z$ commutes with $\cos\hat{\phi} = \frac{\hat{r}_x}{\hat{\rho}}$. Now, we recognize that $\hat{r}_x\hat{p}_x + \hat{r}_y\hat{p}_y = \hat{\rho}\hat{p}_\rho$ after ignoring the quantum correction term, which commutes with $\cos\hat{\phi}$. This planar radial momentum operator commutes with $\cos\hat{\phi}$, so we immediately obtain $[\cos\hat{\phi}, \hat{p}_\theta] = 0$. A similar argument shows that $[\sin\hat{\phi}, \hat{p}_\theta] = 0$. The next commutator we need to work out is the schematic $[\hat{\theta}, \hat{p}_\theta]$. We do this by forming the Hadamard with the exponential, just as before:

$$\begin{aligned}e^{-i\hat{\theta}}(\hat{r}_x\hat{L}_y - \hat{r}_y\hat{L}_x)e^{i\hat{\theta}} &= \left(\frac{\hat{r}_z}{\hat{r}} - i\frac{\hat{\rho}}{\hat{r}}\right)(\hat{r}_x\hat{L}_y - \hat{r}_y\hat{L}_x)\left(\frac{\hat{r}_z}{\hat{r}} + i\frac{\hat{\rho}}{\hat{r}}\right) \\ &= \left(\frac{\hat{r}_z}{\hat{r}} - i\frac{\hat{\rho}}{\hat{r}}\right)\left(\frac{\hat{r}_z}{\hat{r}} + i\frac{\hat{\rho}}{\hat{r}}\right)(\hat{r}_x\hat{L}_y - \hat{r}_y\hat{L}_x) \\ &+ \left(\frac{\hat{r}_z}{\hat{r}} - i\frac{\hat{\rho}}{\hat{r}}\right)\left(\frac{\hat{r}_x}{\hat{r}}[\hat{L}_y, \hat{r}_z] - \frac{\hat{r}_y}{\hat{r}}[\hat{L}_x, \hat{r}_z] + i\frac{\hat{r}_x}{\hat{r}}[\hat{L}_y, \hat{\rho}] - i\frac{\hat{r}_y}{\hat{r}}[\hat{L}_x, \hat{\rho}]\right).\end{aligned} \tag{11.70}$$

But, we know that $[\hat{L}_y, \hat{r}_z] = i\hbar\hat{r}_x$ and $[\hat{L}_x, \hat{r}_z] = -i\hbar\hat{r}_y$. Also, $[\hat{L}_x, \hat{\rho}^2] = 2i\hbar\hat{r}_y\hat{r}_z$, which implies that $[\hat{L}_x, \hat{\rho}] = i\hbar\frac{\hat{r}_y\hat{r}_z}{\hat{\rho}}$ and $[\hat{L}_y, \hat{\rho}] = -i\hbar\frac{\hat{r}_x\hat{r}_z}{\hat{\rho}}$. Substituting these results gives us

$$\begin{aligned}e^{-i\hat{\theta}}(\hat{r}_x\hat{L}_y - \hat{r}_y\hat{L}_x)e^{i\hat{\theta}} &= \hat{r}_x\hat{L}_y - \hat{r}_y\hat{L}_x + \hbar\hat{\rho}\left(\frac{\hat{r}_z}{\hat{r}} - i\frac{\hat{\rho}}{\hat{r}}\right)\left(\frac{\hat{r}_z}{\hat{r}} + i\frac{\hat{\rho}}{\hat{r}}\right) \\ &= \hat{r}_x\hat{L}_y - \hat{r}_y\hat{L}_x + \hbar\hat{\rho}.\end{aligned} \tag{11.71}$$

This tells us that

$$[\hat{\theta}, \hat{p}_\theta] = \frac{i\hbar}{\hat{r}}, \tag{11.72}$$

as we might have suspected. The only commutator left to calculate is $[\cos\hat{\theta}, \hat{p}_\phi] = 0$ (and similarly for the sine), which we assign as Problem 11.3.3.

At this stage, we have determined the spherical coordinates and momenta and their commutation relations. Now, we need to determine the kinetic energy in spherical coordinates. One might think we should just compute the square of each of the components of momentum and then look for quantum corrections, but it turns out to be better to consider the sum of $\hat{p}_r^2 + \hat{p}_{r,\perp}^2$, summing the radial momentum and the component of the momentum vector perpendicular to the radial momentum, squared. How do we find the perpendicular part? Well, classically, we have that $|\vec{p} \cdot \vec{e}_r| = |p| \cos\theta$ and $|\vec{p} \times \vec{e}_r| = |p| \sin\theta$. In quantum mechanics, we note that $\hat{\vec{e}}_r \times \hat{\vec{p}} = \frac{1}{\hat{r}} \hat{\vec{L}}$. So, we compute these two terms. First we have

$$\left(\hat{p}_r\right)^2 = \frac{1}{\hat{r}}(\hat{r}_x \hat{p}_x + \hat{r}_y \hat{p}_y - i\hbar)\frac{1}{\hat{r}}(\hat{r}_x \hat{p}_x + \hat{r}_y \hat{p}_y - i\hbar). \tag{11.73}$$

As before, we move the $\frac{1}{\hat{r}}$ and all $\hat{r}_\alpha$ terms to the left and recall that $\left[\hat{p}_r, \frac{1}{\hat{r}}\right] = \frac{i\hbar}{\hat{r}^2}$, so that

$$\begin{aligned}
\left(\hat{p}_r\right)^2 &= \frac{1}{\hat{r}^2}(\hat{r}_x \hat{p}_x + \hat{r}_y \hat{p}_y)(\hat{r}_x \hat{p}_x + \hat{r}_y \hat{p}_y - i\hbar) \\
&= \frac{1}{\hat{r}^2}\big(\hat{r}_x^2 \hat{p}_x^2 + \hat{r}_y^2 \hat{p}_y^2 + \hat{r}_z^2 \hat{p}_z^2 + 2\hat{r}_x \hat{r}_y \hat{p}_x \hat{p}_y + 2\hat{r}_x \hat{r}_z \hat{p}_x \hat{p}_z + 2\hat{r}_z \hat{r}_y \hat{p}_z \hat{p}_y \\
&\quad + \hat{r}_x[\hat{p}_x, \hat{r}_x]\hat{p}_x + \hat{r}_y[\hat{p}_y, \hat{r}_y]\hat{p}_y + \hat{r}_z[\hat{p}_z, \hat{r}_z]\hat{p}_z - i\hbar\hat{r}_x\hat{p}_x - i\hbar\hat{r}_y\hat{p}_y - i\hbar\hat{r}_z\hat{p}_z\big) \\
&= \frac{1}{\hat{r}^2}\big(\hat{r}_x^2 \hat{p}_x^2 + \hat{r}_y^2 \hat{p}_y^2 + \hat{r}_z^2 \hat{p}_z^2 + 2\hat{r}_x \hat{r}_y \hat{p}_x \hat{p}_y + 2\hat{r}_x \hat{r}_z \hat{p}_x \hat{p}_z + 2\hat{r}_z \hat{r}_y \hat{p}_z \hat{p}_y \\
&\quad - 2i\hbar(\hat{r}_x \hat{p}_x + \hat{r}_y \hat{p}_y + \hat{r}_z \hat{p}_z)\big).
\end{aligned} \tag{11.74}$$

Similarly, we find that

$$\begin{aligned}
&\frac{1}{\hat{r}}\left(\hat{\vec{r}} \times \hat{\vec{p}}\right) \cdot \frac{1}{\hat{r}}\left(\hat{\vec{r}} \times \hat{\vec{p}}\right) = \frac{1}{\hat{r}^2}\big[(\hat{r}_y \hat{p}_z - \hat{r}_z \hat{p}_y)(\hat{r}_y \hat{p}_z - \hat{r}_z \hat{p}_y) \\
&\quad + (\hat{r}_z \hat{p}_x - \hat{r}_x \hat{p}_z)(\hat{r}_z \hat{p}_x - \hat{r}_x \hat{p}_z) + (\hat{r}_x \hat{p}_y - \hat{r}_y \hat{p}_x)(\hat{r}_x \hat{p}_y - \hat{r}_y \hat{p}_x)\big] \\
&\quad = \frac{1}{\hat{r}^2}\big[(\hat{r}_y^2 + \hat{r}_z^2)\hat{p}_x^2 + (\hat{r}_x^2 + \hat{r}_z^2)\hat{p}_y^2 + (\hat{r}_x^2 + \hat{r}_y^2)\hat{p}_z^2 \\
&\qquad - 2\hat{r}_x \hat{r}_y \hat{p}_x \hat{p}_y - 2\hat{r}_y \hat{r}_z \hat{p}_y \hat{p}_z - 2\hat{r}_x \hat{r}_z \hat{p}_x \hat{p}_z \\
&\qquad - \hat{r}_y[\hat{p}_z, \hat{r}_z]\hat{p}_y - \hat{r}_z[\hat{p}_y, \hat{r}_y]\hat{p}_z - \hat{r}_x[\hat{p}_z, \hat{r}_z]\hat{p}_x \\
&\qquad - \hat{r}_z[\hat{p}_x, \hat{r}_x]\hat{p}_z - \hat{r}_x[\hat{p}_y, \hat{r}_y]\hat{p}_x - \hat{r}_y[\hat{p}_x, \hat{r}_x]\hat{p}_y\big] \\
&\quad = \frac{1}{\hat{r}^2}\big[(\hat{r}_y^2 + \hat{r}_z^2)\hat{p}_x^2 + (\hat{r}_x^2 + \hat{r}_z^2)\hat{p}_y^2 + (\hat{r}_x^2 + \hat{r}_y^2)\hat{p}_z^2 \\
&\qquad - 2\hat{r}_x \hat{r}_y \hat{p}_x \hat{p}_y - 2\hat{r}_y \hat{r}_z \hat{p}_y \hat{p}_z - 2\hat{r}_x \hat{r}_z \hat{p}_x \hat{p}_z + 2i\hbar(\hat{r}_x \hat{p}_x + \hat{r}_y \hat{p}_y + \hat{r}_z \hat{p}_z)\big].
\end{aligned} \tag{11.75}$$

One can see that when we combine these two results together, the cross terms cancel, while the "diagonal" terms combine, and we have a $\hat{r}^2$ canceling in the numerator and denominator. So we have just shown that

$$\hat{\vec{p}}^{\,2} = \hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2 = \left(\hat{p}_r\right)^2 + \frac{\hat{\vec{L}}^{\,2}}{\hat{r}^2}, \tag{11.76}$$

and there is no quantum correction for the separation of variables result in three dimensions! Note that we already could have gone directly to this separation of variables using a result from our vector manipulations in Sect. 11.1. Namely, Eq. (11.12) can be used to show this result as well. We have Problem 11.3.6 verifying this directly too.

There is one final exercise we must do in this section. We need to convert the translation operator from Cartesian to spherical coordinates. We show in detail how to do this for the position translation operator in polar coordinates, followed by the derivation in spherical polar coordinates. The extension to the momentum translation operator should be clear, but we will not go through those details here.

The position translation operator in two dimensions is

$$\hat{T}(\vec{\rho}) = e^{-\frac{i}{\hbar}\vec{\rho}\cdot\hat{\vec{p}}} = e^{-\frac{i}{\hbar}(r_x\hat{p}_x + r_y\hat{p}_y)}, \tag{11.77}$$

where $r_x = \rho\cos\phi$ and $r_y = \rho\sin\phi$, and they are all numbers, not operators. Now, we can also think of the vector $\vec{\rho} = (r_x, r_y)$ as the vector $(\rho, 0)$ that is rotated by an angle ϕ in the counter-clockwise direction. So, we should also be able to write the translation operator as

$$\hat{T}(\vec{\rho}) = e^{-\frac{i}{\hbar}\phi\hat{L}_z} e^{-\frac{i}{\hbar}\rho\hat{p}_x} e^{\frac{i}{\hbar}\phi\hat{L}_z} = e^{-\frac{i}{\hbar}\rho e^{-\frac{i}{\hbar}\phi\hat{L}_z}\hat{p}_x e^{\frac{i}{\hbar}\phi\hat{L}_z}}, \tag{11.78}$$

after using the braiding relation. Now, because the momentum operator is a vector operator, it is rotated by an angle ϕ by the similarity transformation, so

$$e^{-\frac{i}{\hbar}\phi\hat{L}_z}\hat{p}_x e^{\frac{i}{\hbar}\phi\hat{L}_z} = \cos\phi\ \hat{p}_x + \sin\phi\ \hat{p}_y. \tag{11.79}$$

This then proves that the proposed alternative form of the translation operator is indeed correct. To finish putting the translation operator into polar coordinates, we need to rewrite $\hat{p}_x$ in terms of $\hat{p}_\rho$ and $\hat{p}_\theta$. This is a simple algebra exercise, which gives

$$\hat{p}_x = \left(\hat{p}_\rho - \frac{i\hbar}{2\hat{\rho}}\right)\cos\hat{\phi} - \hat{p}_\phi\sin\hat{\phi}. \tag{11.80}$$

In this case, the ordering of the operators in the second term must be maintained, because those operators do not commute. To get our final formula for the position translation operator in polar coordinates, we substitute in for $\hat{p}_x$ to yield

$$\hat{T}(\vec{\rho}) = e^{-\frac{i}{\hbar}\phi\hat{L}_z} e^{-\frac{i}{\hbar}\rho\left(\hat{p}_\rho - \frac{i\hbar}{2\hat{\rho}}\right)\cos\hat{\phi} - \hat{p}_\phi \sin\hat{\phi}} e^{\frac{i}{\hbar}\phi\hat{L}_z}. \tag{11.81}$$

Note that this is an *exact operator identity*. In Problem 11.3.7, you will prove this in an alternative manner.

The procedure for the translation operator in spherical polar coordinates is similar. We have that the translation operator is

$$\hat{T}(\vec{r}) = e^{-\frac{i}{\hbar}\vec{r}\cdot\hat{\vec{p}}} = e^{-\frac{i}{\hbar}(r_x\hat{p}_x + r_y\hat{p}_y + r_z\hat{p}_z)}, \tag{11.82}$$

in Cartesian form (the numbers r_x, r_y, and r_z denote the translation vector $\vec{r} = r_x\vec{e}_x + r_y\vec{e}_y + r_z\vec{e}_z$). In spherical polar coordinates, we know that we can start with a vector of length $r = \sqrt{r_x^2 + r_y^2 + r_z^2}$ pointing along the z-direction. Then rotate by θ about the y-axis, followed by a rotation by ϕ about the z-axis. Hence, we should have that

$$\vec{r}\cdot\hat{\vec{p}} = e^{-\frac{i}{\hbar}\phi\hat{L}_z} e^{-\frac{i}{\hbar}\theta\hat{L}_y} r\,\hat{p}_z e^{\frac{i}{\hbar}\theta\hat{L}_y} e^{\frac{i}{\hbar}\phi\hat{L}_z}. \tag{11.83}$$

We verify it by computing two Hadamards. First, we have

$$e^{-\frac{i}{\hbar}\theta\hat{L}_y}\hat{p}_z e^{\frac{i}{\hbar}\theta\hat{L}_y} = \sin\theta\,\hat{p}_x + \cos\theta\,\hat{p}_z, \tag{11.84}$$

and then we have

$$e^{-\frac{i}{\hbar}\phi\hat{L}_z}\hat{p}_x e^{\frac{i}{\hbar}\phi\hat{L}_z} = \cos\phi\,\hat{p}_x + \sin\phi\,\hat{p}_y, \tag{11.85}$$

and $\hat{p}_z$ is unchanged by the ϕ rotation, because it points along the rotation axis. Putting these two results together, we indeed see that Eq. (11.83) is satisfied, because $r_x = r\sin\theta\cos\phi$, $r_y = r\sin\theta\sin\phi$, and $r_z = r\cos\theta$. To completely put it into the spherical polar coordinate form, we need to replace $\hat{p}_z$ in terms of the momenta in spherical coordinates. The algebra is straightforward, and we have

$$\hat{p}_z = \left(\hat{p}_r - \frac{i\hbar}{2\hat{r}}\right)\cos\hat{\theta} - \hat{p}_\theta \sin\hat{\theta}. \tag{11.86}$$

This yields the final form for the translation operator:

$$e^{-\frac{i}{\hbar}\phi\hat{L}_z} e^{-\frac{i}{\hbar}\theta\hat{L}_y} e^{-\frac{i}{\hbar}r\left\{\left(\hat{p}_r - \frac{i\hbar}{2\hat{r}}\right)\cos\hat{\theta} - \hat{p}_\theta\sin\hat{\theta}\right\}} e^{\frac{i}{\hbar}\theta\hat{L}_y} e^{\frac{i}{\hbar}\phi\hat{L}_z}. \tag{11.87}$$

We will use these translation operators in two and three dimensions to aid in determining wavefunctions in polar and spherical polar coordinates. Further details

about converting translation operators into these coordinate systems can be found in Rushka et al. (2021).

We are now ready for the problems.

Section 11.3 Problems:

11.3.1 Quantum Correction for the Radial Momentum in Two Dimensions

In two dimensions, we tried to construct the radial momentum via $\frac{1}{\hat{\rho}}\hat{\vec{\rho}} \cdot \hat{\vec{p}}$, but we found that

$$\left(\frac{1}{\hat{\rho}}\hat{\vec{\rho}} \cdot \hat{\vec{p}}\right)^{\dagger} = \hat{\vec{\rho}} \cdot \hat{\vec{p}}\frac{1}{\hat{\rho}}. \tag{11.88}$$

Move the coordinate terms on the right-hand side of the aforementioned equation to the left, to show that

$$\left(\frac{1}{\hat{\rho}}\hat{\vec{\rho}} \cdot \hat{\vec{p}}\right)^{\dagger} = \frac{1}{\hat{\rho}}\hat{\vec{\rho}} \cdot \hat{\vec{p}} + \frac{i\hbar c}{\hat{\rho}} \tag{11.89}$$

Determine the constant c.

11.3.2 Commutator $[\sin\hat{\phi}, \hat{L}_z]$ in Polar Coordinates

Using the strategy we have for converting trigonometric functions of $\hat{\phi}$ into operator relations involving Cartesian coordinate operators, compute $[\sin\hat{\phi}, \hat{L}_z]$ for the case of two-dimensional polar coordinates.

11.3.3 Commutators $[\cos\hat{\theta}, \hat{p}_\phi]$ and $[\sin\hat{\theta}, \hat{p}_\phi]$

Using the strategy, for converting trigonometric functions of angle operators into operator relations involving Cartesian coordinate operators, compute $[\cos\hat{\theta}, \hat{p}_\phi]$ and $[\sin\hat{\theta}, \hat{p}_\phi]$ in spherical polar coordinates.

11.3.4 Commutator of Radial Momentum and Powers of the Radial Coordinate

We showed that the radial momentum in two dimensions satisfies $[\hat{\rho}, \hat{p}_\rho] = i\hbar$ and in three dimensions, satisfies $[\hat{r}, \hat{p}_r] = i\hbar$. This is the standard canonical commutation relation between position and momentum. Use that relation to further show that both

$$[\hat{p}_\rho, \hat{\rho}^n] = -i\hbar n\hat{\rho}^{n-1} \text{ and } [\hat{p}_r, \hat{r}^n] = -i\hbar n\hat{r}^{n-1}. \tag{11.90}$$

11.3.5 A Conundrum About Separation of Variable in Three Dimensions

In the text, we showed concretely how

$$\hat{\vec{p}}^{\,2} = \hat{p}_r^2 + \frac{1}{\hat{r}^2}\left(\hat{\vec{r}} \times \hat{\vec{p}}\right)^2. \tag{11.91}$$

But $\hat{\vec{r}} \times \hat{\vec{p}}$ is perpendicular to both $\hat{\vec{r}}$ and $\hat{\vec{p}}$ (at least the classical vectors are). Explain how such a decomposition can occur given this orthogonality?

11.3.6 Separation of Variables Revisited with Vector Operator Identities

In this problem, we derive the separation of variables in three dimensions a bit differently than how we proceeded in the text. Instead, we use the operator vector identities that we derived earlier. Assume that $\hat{\vec{A}}$ and $\hat{\vec{B}}$ are both vector operators and that $[\hat{A}_i, \hat{A}_j] = [\hat{B}_i, \hat{B}_j] = 0$. Then, we have that

$$\begin{aligned}
\left(\hat{\vec{A}} \times \hat{\vec{B}}\right)^2 &= \sum_{ijlm}\left(\sum_k \epsilon_{ijk}\epsilon_{klm}\right)\hat{A}_i\hat{B}_j\hat{A}_l\hat{B}_m \\
&= \sum_{ijlm}\left(\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}\right)\hat{A}_i\hat{B}_j\hat{A}_l\hat{B}_m \\
&= \hat{\vec{A}}^{\,2}\hat{\vec{B}}^{\,2} + \sum_{ij}\hat{A}_i[\hat{B}_j, \hat{A}_i]\hat{B}_j - \left(\hat{\vec{A}} \cdot \hat{\vec{B}}\right)^2 \\
&\quad - \sum_{ij}\left(\hat{A}_i\hat{B}_j[\hat{A}_j, \hat{B}_i] + \hat{A}_i\hat{B}_i[\hat{A}_j, \hat{B}_j]\right).
\end{aligned} \tag{11.92}$$

If you are unsure about the aforementioned identity, just rederive it for yourself.

(a) Using $\hat{\vec{A}} = \hat{\vec{r}}$ and $\hat{\vec{B}} = \hat{\vec{p}}$, show that

$$\left(\hat{\vec{r}} \times \hat{\vec{p}}\right)^2 = \hat{r}^2 \hat{p}^2 - \left(\hat{\vec{r}} \cdot \hat{\vec{p}}\right)^2 + i\hbar \hat{\vec{r}} \cdot \hat{\vec{p}}, \tag{11.93}$$

or $\hat{\vec{p}}^{\,2} = \hat{p}_r^2 + \frac{1}{\hat{r}^2}\hat{\vec{L}}^{\,2}$.

Hint: you need to compute $\frac{1}{\hat{r}^2}(\hat{\vec{r}} \cdot \hat{\vec{p}})\frac{1}{\hat{r}}(\hat{\vec{r}} \cdot \hat{\vec{p}})$+ a correction term.

(b) Show that

$$\hat{r}^2 = \frac{1}{\hat{p}}\left(\hat{\vec{p}} \cdot \hat{\vec{r}}\right)\frac{1}{\hat{p}}\left(\hat{\vec{p}} \cdot \hat{\vec{r}}\right) + \frac{\hat{\vec{L}}^{\,2}}{\hat{p}^{\,2}} = \hat{r}_p^2 + \frac{\hat{\vec{L}}^{\,2}}{\hat{p}^2}, \tag{11.94}$$

where $\hat{r}_p = \frac{1}{\hat{p}}(\hat{\vec{p}} \cdot \hat{\vec{r}} + i\hbar)$ is the *momential coordinate*, analogous to the radial momentum. The result of this calculation can be used to determine the properties of a harmonic oscillator in momentum space using the aforementioned separation of variables of the potential energy contribution from $\hat{r}^2$; other problems in momentum space do not separate as easily.

11.3.7 Alternative Derivation of the Translation Operator in Polar Coordinates

In this problem, we will show an alternative way to determine the operator expression for the translation operator in polar coordinates.

(a) The first step is to express the Cartesian momenta in terms of the polar coordinate momenta. Using the results for the momenta in polar coordinates to show that we have both

$$\hat{p}_x = \left(\hat{p}_\rho - \frac{i\hbar}{2\hat{\rho}}\right)\cos\hat{\phi} - \hat{p}_\phi \sin\hat{\phi} \quad \text{and} \quad \hat{p}_y = \left(\hat{p}_\rho - \frac{i\hbar}{2\hat{\rho}}\right)\sin\hat{\phi} + \hat{p}_\phi \cos\hat{\phi}. \tag{11.95}$$

Note that the operator ordering must be exactly as written for the second terms in each equation.

(b) Use the facts that $r_x = \rho\cos\phi$ and $r_y = \rho\sin\phi$ (all numbers, not operators) along with the trigonometric sum and difference formulas to show that

$$r_x\hat{p}_x + r_y\hat{p}_y = \rho\Bigg[\left(\hat{p}_\rho - \frac{i\hbar}{2\hat{\rho}}\right)(\cos\phi\cos\hat{\phi} + \sin\phi\sin\hat{\phi})$$

$$- \hat{p}_\phi(\cos\phi \sin\hat{\phi} - \sin\phi\cos\hat{\phi})\Big]$$

$$= \rho\Big[\Big(\hat{p}_\rho - \frac{i\hbar}{2\hat{\rho}}\Big)\cos(\hat{\phi}-\phi) - \hat{p}_\phi \sin(\hat{\phi}-\phi)\Big]. \qquad (11.96)$$

(c) Finally, show that

$$r_x\hat{p}_x + r_y\hat{p}_y = \rho e^{-\frac{i}{\hbar}\phi\hat{L}_z}\Big[\Big(\hat{p}_\rho - \frac{i\hbar}{2\hat{\rho}}\Big)\cos\hat{\phi} - \hat{p}_\phi \sin\hat{\phi})\Big]e^{\frac{i}{\hbar}\phi\hat{L}_z}. \qquad (11.97)$$

(d) Substituting this last result into the formula $T(\vec{\rho}) = e^{-\frac{i}{\hbar}(r_x\hat{p}_x + r_y\hat{p}_y)}$ and using the inverse of the braiding relation will then reproduce the same exact operator identity for the translation operator in polar coordinates. Verify this.

11.4 Energy Scales for Hydrogen and Separation of Variables for Quantum States

Before 1900, we already knew a lot about atoms. The most important quantities were the size of the atom, coming in at about 1 Å or 10^{-8} cm $= 10^{-10}$ m and the energy of light emitted from the atoms, which was primarily visible light, with energies on the order of a few electron volts (eV) (technically speaking we knew the wavelengths, which only later were related to energy after Planck and Einstein discussed photons and the relationship $E = \hbar\omega$ between energy and frequency of a photon). An electron volt (eV) is how much energy an electron gains when accelerated by a 1 V potential difference (slightly less than a common 1.5 V battery). It is an important energy unit for quantum mechanics. By the way, an electron moving with one eV of energy may not seem like it is moving very fast, but its speed is already more than one-thousandth the speed of light!

Physicists have a lazy nature to themselves. We love to estimate things, without doing detailed calculations. We call this "back of the envelope" calculations. So, let us focus our "back of the envelope" onto atoms and see what we can come up with. We start with what we know makes up an atom—we have electrons, whose mass we call m (sometimes m_e if there is any ambiguity) and its electric charge $-e$ (the symbol e is always positive). We also have the speed of light $c = 3 \times 10^8$ m/s $= 3 \times 10^{10}$ m/s $= 2 \times 10^{18}$ Å/s. Using just these three constants, what energy, length, and timescales can we find? We need to recall that kinetic energy is proportional to the mass times the square of the velocity, so $mc^2 \approx 511{,}000$ eV is an energy scale. We also note that $e^2/4\pi\epsilon_0 r$ is an energy scale, with r some length scale. Equating this to the rest-mass energy gives a length scale $r_e = e^2/4\pi\epsilon_0 mc^2 = 2.8179 \times 10^{-5}$ Å (called the classical radius of the electron). A timescale is found

by dividing by c: $t = r_e/c \approx 10^{-23}$ s. The problem is, none of these scales are close to what is seen in an atom. The length scale is 100,000 times too small, and the energy scale is 100,000 times too large. We clearly need another physical constant to work with.

In 1900, we were greeted by a new physical constant $\hbar$. In terms of the size of this constant, I find it most useful to recall that

$$2\pi\hbar c = 12{,}400 \text{ ev-Å}, \tag{11.98}$$

as the way to work with Planck's constant. It has the dimensions of momentum times length or angular momentum. Using it, along with e, m, and c, we can find some new energy and length scales—we can even find a *dimensionless* coupling constant!

To begin, let's first find a length scale. Recall that $\frac{\hbar^2}{ma^2}$, with a a length, has dimensions of energy (check this if you do not recall it). So, $\hbar^2/m$ has dimensions of energy times length^2. But $e^2/4\pi\epsilon_0$ has dimensions of energy times length, so we find a new length scale from their ratios, namely

$$a_0 = \frac{4\pi\epsilon_0\hbar^2}{me^2} = 0.529 \text{ Å}. \tag{11.99}$$

This length scale is on the order of the size of the atom. It is called the *Bohr radius*. The energy scale follows as

$$\frac{e^2}{4\pi\epsilon_0 a_0} = \frac{me^4}{16\pi^2\epsilon_0^2\hbar^2} \text{ or } \frac{\hbar^2}{ma_0^2} = \frac{me^4}{16\pi^2\epsilon_0^2\hbar^2}. \tag{11.100}$$

Both give the same energy scale, called the *Hartree*: $me^4/16\pi^2\epsilon_0^2\hbar^2 = 27.2$ eV. We often prefer to work with the *Rydberg*, which is one half of a Hartree or Ry $= 13.6\,\text{eV} = e^2/8\pi\epsilon_0 a_0$. These energy scales are also similar to atomic energy scales. Finally, we can form a dimensionless constant via

$$\frac{e^2}{4\pi\epsilon_0 a_0 mc^2} = \frac{e^4}{16\pi^2\epsilon_0^2\hbar^2c^2}. \tag{11.101}$$

We call this constant α^2, because we prefer to work with its square root

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \approx \frac{1}{137}. \tag{11.102}$$

It is called the *fine-structure constant*. One can see that the fine-structure constant relates many of these scales. For example, we have

Fig. 11.5 Mary-Kay Gaillard teaching in a high-energy phenomenology class at the University of California, Berkeley, in the early 1990s. (Photo courtesy of the AIP Emilio Segrè Visual Archives, Physics Today Collection. Converted to look like a sketch by James Freericks.)

$$\frac{e^2}{4\pi\epsilon_0 a_0} = \alpha^2 mc^2 \text{ and } a_0 = \frac{r_e}{\alpha^2}. \tag{11.103}$$

What does this mean? It means the binding energy of the atom is α^2 times smaller than the electron's rest-mass energy. Hence, atom formation involves a *small energy scale* relative to the rest-mass energy. This means most atoms can be treated nonrelativistically. It also means that atom formation is a low-energy effect.

I know when I first saw discussions about finding energy, length, and timescales from constants, I thought it was a waste of time. I mused to myself, enough of this "jibber-jabber," let's just calculate with the real theory. But then, in my second year of graduate school, I took a high-energy phenomenology class with Mary-Kay Gaillard, one of the world's experts on the charm quark and shown in Fig. 11.5; she has written an autobiography (Gaillard, 2015), which I highly recommend. We were all very excited to be in this class and to be able to finally calculate something "real" when our first homework assignment came. The problem set was full of lots of complex and high-level material. I believe we were calculating the scattering of light off of an electron (the so-called *Compton scattering*). We all struggled to complete the work. We spent hours upon hours. Days even. But this was fine. It was our passion. And we were calculating something "real." Finally, we came up with our answers.

I recall that most of us had obtained different answers. But we each thought we were right, and everyone else was wrong. Well, we turned in the assignment, and Mary K, as she liked to be called, got up to the blackboard and showed us how you could get the answer up to a factor of 2 (or maybe even $\sqrt{2}$) by simple dimensional analysis. All those hours upon hours of hard work was just to determine the coefficient in front! This was a really important lesson for me and it is why I am sharing it with you. *Dimensional analysis should always be done first*!! It provides you with a useful angle to understand the problem you are trying to solve. But, it

does not solve everything. In the end, we must do the hard work of the calculation to find the number in front, even if it is a small correction. This is the life of a physicist.

Before ending this section, we have one more important task. It is separation of variables. We do it twice for central force problems—first to reduce the two-particle problem to a single effective particle problem and second to reduce to independent radial and angular dependences. Then, we will use this in solving the hydrogen atom problem.

We already showed how to reduce from a two-particle problem to a single-particle problem in Sect. 7.14. We briefly summarize this result and emphasize how it leads to a factorization of the quantum state.

To be concrete, we work with the electron and proton operators for hydrogen, indicated by an e or a p superscript, respectively. Our four operators are then $\hat{\vec{r}}^{\,e}$, $\hat{\vec{p}}^{\,e}$, $\hat{\vec{r}}^{\,p}$, and $\hat{\vec{p}}^{\,p}$. The operators all commute with each other, except for the canonical commutation relation pairs, given by $[\hat{r}^e_\alpha, \hat{p}^e_\beta] = i\hbar\delta_{\alpha\beta}$ and $[\hat{r}^p_\alpha, \hat{p}^p_\beta] = i\hbar\delta_{\alpha\beta}$. To separate variables, we work in center of mass and relative coordinates. These are defined by

$$\hat{\vec{R}}_{cm} = \frac{m_e\hat{\vec{r}}^{\,e} + m_p\hat{\vec{r}}^{\,p}}{m_e + m_p} = \frac{m_e\hat{\vec{r}}^{\,e} + m_p\hat{\vec{r}}^{\,p}}{M}, \tag{11.104}$$

where $M = m_e + m_p$ is the total mass of the system. In the center-of-mass coordinates, the position operators are weighted by the relative masses of the particles. The center-of-mass momentum, given by $\hat{\vec{P}}_{cm} = \hat{\vec{p}}^{\,e} + \hat{\vec{p}}^{\,p}$, is also a weighted sum of the velocities (recall that $\vec{p} = m\vec{v}$). We already showed that these combinations are canonical, meaning that $[\hat{R}_{cm\,\alpha}, \hat{P}_{cm\,\beta}] = i\hbar\delta_{\alpha\beta}$. We also work on the relative quantities. The relative position is simple. It is $\hat{\vec{r}}_{rel} = \hat{\vec{r}}^{\,e} - \hat{\vec{r}}^{\,p}$. The canonically conjugate momentum is given by the relative velocity, or $\hat{\vec{p}}_{rel} = \frac{M}{m_e}\hat{\vec{p}}^{\,e} - \frac{M}{m_p}\hat{\vec{p}}^{\,p}$. Using $\mu = \frac{m_e m_p}{m_e+m_p}$, which is the reduced mass, we find the relative position and momentum are also canonical: $[\hat{r}_{rel\,\alpha}, \hat{p}_{rel\,\beta}] = i\hbar\delta_{\alpha\beta}$. We also verified previously that the center of mass and relative operators all commute.

Using these definitions of the center of mass and relative operators, we can transform the Hamiltonian via

$$\hat{H} = \frac{(\hat{\vec{p}}^{\,p})^2}{2m_p} + \frac{(\hat{\vec{p}}^{\,e})^2}{2m_e} + V(\hat{\vec{r}}^{\,e} - \hat{\vec{r}}^{\,p}) = \frac{\hat{\vec{P}}^{\,2}_{cm}}{2M} + \frac{\hat{\vec{p}}^{\,2}_{rel}}{2\mu} + V(\hat{\vec{r}}_{rel}). \tag{11.105}$$

Note that this is the sum of a free center-of-mass Hamiltonian $\hat{H}_{cm} = \hat{\vec{P}}^{\,2}_{cm}/2M$ and a relative motion Hamiltonian $\hat{H}_{rel} = \hat{\vec{p}}^{\,2}_{rel}/2\mu + V(\hat{\vec{r}}_{rel})$. Whenever a Hamiltonian can be written as a *sum* over terms that commute with each other, then the wavefunction can be constructed as a *tensor product* of independent terms, where

the commuting parts of the Hamiltonian act only on the corresponding tensor product factors of the wavefunction. The total energy is then the sum of the energies of each tensor-product energy eigenstate. Hence, we have

$$|\psi\rangle = |\psi_{cm}\rangle \otimes |\psi_{rel}\rangle. \tag{11.106}$$

The center-of-mass Hamiltonian only acts on the center-of-mass wavefunction and similarly for the relative motion Hamiltonian. Thus, because the center-of-mass is a free particle, we take it to be an eigenstate of the center-of-mass momentum operator, or $\hat{\vec{P}}_{cm}|\psi_{cm}\rangle = \vec{P}_{cm}|\psi_{cm}\rangle$. Then we have

$$\begin{aligned}\hat{H}|\psi\rangle &= \hat{H}_{cm}|\psi_{cm}\rangle \otimes |\psi_{rel}\rangle + |\psi_{cm}\rangle \otimes \hat{H}_{rel}|\psi_{rel}\rangle \\ &= \frac{\hat{\vec{P}}_{cm}^{\,2}}{2M}|\psi_{cm}\rangle \otimes |\psi_{rel}\rangle + |\psi_{cm}\rangle \otimes \hat{H}_{rel}|\psi_{rel}\rangle.\end{aligned} \tag{11.107}$$

The center-of-mass motion is that of a free particle, given by its total kinetic energy, and it is trivial to describe. So, we typically drop keeping track of the center-of-mass motion when we describe the relative motion and focus on solving the relative motion only. Just bear in mind for a two-particle system, we still have the center-of-mass motion to keep track of. Another way of thinking of this is we work in the center-of-mass frame. Then the total energy, given by $E = E_{cm} + E_{rel} = E_{rel}$, because the center-of-mass energy vanishes in the center-of-mass frame. As a result, we rarely discuss or even keep track of the center-of-mass motion, since it is so simple to add back in if needed. The video Two-Body Problem https://tinyurl.com/y6ddzdjt describes how to perform the separation too. For now, we focus all of our efforts on the relative motion problem.

Note that the two-particle problem nicely reduces to an effective single-particle problem when the force depends just on the distance between the particles. If we add a third particle, then no such reduction is possible—we have to take into account all degrees of freedom of the many-body problem. We will discuss some simple many-body problems in later chapters (and we already have discussed some for spin models). Most of current research is focused on trying to solve the quantum many-body problem. It is a challenge.

Now that we have reduced to considering the relative motion only, we still have another separation of variables to do—we need to separate into the radial and angular motions. This requires us to carefully describe how to re-express the Cartesian position state in three dimensions $|r_x\rangle \otimes |r_y\rangle \otimes |r_z\rangle = |r_x, r_y, r_z\rangle$ in terms of spherical coordinates, given by $|r\rangle \otimes |\theta, \phi\rangle$. The procedure is somewhat technical and requires care to work out. The starting point is the identity, given in Eq. (11.87) from the previous section. This is an exact operator identity relating the Cartesian translation operator to the translation operator expressed in spherical coordinates. The proper construction of the states in spherical coordinates requires a limiting procedure. Recall that translations always commute with each other, so, we can

always write

$$|r_x, r_y, r_z\rangle = e^{\frac{i}{\hbar}\vec{\delta}\cdot\hat{\vec{p}}} e^{-\frac{i}{\hbar}\vec{r}\cdot\hat{\vec{p}}} e^{-\frac{i}{\hbar}\vec{\delta}\cdot\hat{\vec{p}}} |0_x, 0_y, 0_z\rangle, \tag{11.108}$$

where we have added and removed a translation by $\vec{\delta}$. Note that here, we have $\vec{r} = r_x\vec{e}_x + r_y\vec{e}_y + r_z\vec{e}_z$ is a number, not an operator, and similarly, we have $\vec{\delta} = \delta_x\vec{e}_x + \delta_y\vec{e}_y + \delta_z\vec{e}_z$ is also a number. We will be picking the vector $\vec{\delta}$ to be an infinitesimal vector, and we will eventually take the limit that it approaches zero. But having a nonzero $\vec{\delta}$ is critical for a number of the steps we need to take to construct the tensor product of position states in spherical coordinates. We take the length $\delta = \sqrt{\delta_x^2 + \delta_y^2 + \delta_z^2}$ to be infinitesimal, and we take it to be the state given by

$$|\vec{\delta}\rangle = |\delta_x, \delta_y, \delta_z\rangle = e^{-\frac{i}{\hbar}\phi\hat{L}_z} e^{-\frac{i}{\hbar}\theta\hat{L}_y} |0, 0, \delta\rangle, \tag{11.109}$$

which is a rotation of the state along the north pole by θ and ϕ, respectively, to be oriented in the θ, ϕ direction in spherical coordinates; that is, $\delta_x = \delta\sin\theta\cos\phi$, $\delta_y = \delta\sin\theta\sin\phi$, and $\delta_z = \delta\cos\theta$. We make this choice, so that when we apply the translation operator in spherical coordinates, from Eq. (11.87), the rotations on the right cancel out to give us the following action of the remaining operators onto the north-pole state:

$$|\vec{r}\rangle = e^{\frac{i}{\hbar}\vec{\delta}\cdot\hat{\vec{p}}} e^{-\frac{i}{\hbar}\phi\hat{L}_z} e^{-\frac{i}{\hbar}\theta\hat{L}_y} e^{-\frac{i}{\hbar}r\left\{\left(\hat{p}_r - \frac{i\hbar}{2\hat{r}}\right)\cos\hat{\theta} - \hat{p}_\theta\sin\hat{\theta}\right\}} |0, 0, \delta\rangle. \tag{11.110}$$

Note that the ket $|0, 0, \delta\rangle$ satisfies $\cos\hat{\theta}\,|0, 0, \delta\rangle = \frac{\hat{r}_z}{\hat{r}}|0, 0, \delta\rangle = |0, 0, \delta\rangle$, because the state is an eigenstate of both $\hat{r}_z$ and of $\hat{r}$ with the eigenvalue being δ in both cases. Similarly, we find $\sin\hat{\theta}\,|0, 0, \delta\rangle = \frac{\hat{\rho}}{\hat{r}}|0, 0, \delta\rangle = 0$, because the state is an eigenstate with respect to $\hat{\rho} = \sqrt{\hat{r}_x^2 + \hat{r}_y^2}$ with eigenvalue 0. The north-pole state has no definite eigenvalues for the operators $\cos\hat{\phi}$ and $\sin\hat{\phi}$, as we discussed previously.

We now simplify the form of the rightmost operator term acting on a north-pole state. The way we do this is to expand the exponential into a power series and operate each term one-by-one onto the north-pole state. In the leftmost term, we can replace the $\cos\hat{\theta}$ and $\sin\hat{\theta}$ terms by their respective eigenvalues of 1 and 0. Then we have the nth term look like

$$\frac{1}{n!}\left\{\left(\hat{p}_r - \frac{i\hbar}{2\hat{r}}\right)\cos\hat{\theta} - \hat{p}_\theta\sin\hat{\theta}\right\}^{n-1}\left(\hat{p}_r - \frac{i\hbar}{2\hat{r}}\right)|0, 0, \delta\rangle. \tag{11.111}$$

Next, we separate one of the $n-1$ factors on the left out, and because the $\cos\hat{\theta}$ and $\sin\hat{\theta}$ terms commute with $\hat{p}_r$ and $\hat{r}$, we can bring them through to the right and act them onto the north-pole state, and replace them by their eigenvalues. This removes the $\hat{p}_\theta$ term and changes the $\cos\hat{\theta}$ term to 1. We repeat this until all factors

are simplified, and then we resum the series. We end up with the simplified form

$$|\vec{r}\rangle = e^{\frac{i}{\hbar}\vec{\delta}\cdot\hat{\vec{p}}} e^{-\frac{i}{\hbar}\phi\hat{L}_z} e^{-\frac{i}{\hbar}\theta\hat{L}_y} e^{-\frac{i}{\hbar}r\left(\hat{p}_r - \frac{i\hbar}{2\hat{r}}\right)} |0, 0, \delta\rangle. \tag{11.112}$$

Now, we can move the angular momentum terms to the right, because they commute with the rightmost exponential factor, and we can write the position-space eigenstate as

$$e^{-\frac{i}{\hbar}\phi\hat{L}_z} e^{-\frac{i}{\hbar}\theta\hat{L}_y} |0, 0, \delta\rangle = |\delta_x, \delta_y, \delta_z\rangle = |r{=}\delta\rangle \otimes |\theta, \phi\rangle. \tag{11.113}$$

Here, the angular operator satisfies the following relations:

$$\begin{aligned} \cos\hat{\theta}\,|\theta, \phi\rangle &= \cos\theta\,|\theta, \phi\rangle, \quad \sin\hat{\theta}\,|\theta, \phi\rangle = \sin\theta\,|\theta, \phi\rangle \\ \cos\hat{\phi}\,|\theta, \phi\rangle &= \cos\phi\,|\theta, \phi\rangle, \quad \sin\hat{\phi}\,|\theta, \phi\rangle = \sin\phi\,|\theta, \phi\rangle, \end{aligned} \tag{11.114}$$

except when $\theta = 0$ or π, where the ϕ eigenvalue–eigenvector equations are indeterminate. Now, the tensor-product state will create the separation of variables that we desire, but we need to be careful in understanding how different operators act on different terms. The Cartesian operators do not separate into tensor-product operators, as they act on both terms in the tensor product. But by converting back and forth to the Cartesian form, we can determine what these operators do. In fact, we can use a simple analysis to show that the operators $\hat{r}$ and $\hat{p}_r$ act only on the radial state, while the operators $\cos\hat{\theta}$, $\sin\hat{\theta}$, $\cos\hat{\phi}$, and $\sin\hat{\phi}$ (as well as the angular momentum operators) act only on the angular state. We illustrate this concretely for $\cos\hat{\theta}$ next. We have

$$\begin{aligned} \cos\hat{\theta}|r{=}\delta\rangle \otimes |\theta, \phi\rangle &= \frac{\hat{r}_z}{\hat{r}}|r{=}\delta\rangle \otimes |\theta, \phi\rangle \\ &= \underbrace{e^{-\frac{i}{\phi}\hbar\hat{L}_z} e^{-\frac{i}{\hbar}\theta\hat{L}_y} e^{\frac{i}{\hbar}\theta\hat{L}_y} e^{\frac{i}{\hbar}\phi\hat{L}_z}}_{=1} \hat{r}_z e^{-\frac{i}{\hbar}\phi\hat{L}_z} e^{-\frac{i}{\hbar}\theta\hat{L}_y} \frac{1}{\hat{r}}|0, 0, \delta\rangle \\ &= e^{-\frac{i}{\hbar}\phi\hat{L}_z} e^{-\frac{i}{\hbar}\theta\hat{L}_y} (-\sin\theta\,\hat{r}_x + \cos\theta\,\hat{r}_z)\frac{1}{\hat{r}}|0, 0, \delta\rangle \\ &= e^{-\frac{i}{\hbar}\phi\hat{L}_z} e^{-\frac{i}{\hbar}\theta\hat{L}_y} \cos\theta \frac{\delta}{\sqrt{0^2 + 0^2 + \delta^2}}|0, 0, \delta\rangle \\ &= \cos\theta|\delta_x, \delta_y, \delta_z\rangle \\ &= \cos\theta|r{=}\delta\rangle \otimes |\theta, \phi\rangle. \end{aligned} \tag{11.115}$$

Note how it is the rotation of the operator $\hat{r}_z$ that produces the $\cos\theta$ term and how we can evaluate the operators acting directly on the states most easily in the Cartesian

basis. This is the strategy used for all operators expressed in the Cartesian basis—a similar argument works for $\sin\hat{\theta}$, $\cos\hat{\phi}$, and $\sin\hat{\phi}$. For the angular momentum operators, we already know that they produce rotations, and hence only affect the angular terms in the tensor products, and similarly, the radial momentum, which produces translation in the radial direction, only acts on the radial component. So, we can think of this precisely as you would like to—trigonometric functions of angle operators (and angular momentum operators) act like the identity on the radial part of the tensor product and act nontrivially on the angular part. The opposite is true for the radial operator and the radial momentum. Keeping the way the operators act on these states clear, we can now take the limit where $\delta \to 0$, and note that for the angular state, this means, we can express it as two rotation operators acting on the north-pole state, with unit magnitude along the z-axis, just as we did in Sect. 9.7. This then brings us to the tensor product decomposition of the position eigenstate in spherical coordinates. It is

$$|\vec{r}\rangle = e^{-\frac{i}{\hbar}r\left(\hat{p}_r - \frac{i\hbar}{\hat{r}}\right)}|0_r\rangle \otimes e^{-\frac{i}{\hbar}\phi\hat{L}_z}e^{-\frac{i}{\hbar}\theta\hat{L}_z}|0_\theta, 0_\phi\rangle, \tag{11.116}$$

or, as a bra, we have

$$\langle\vec{r}| = \langle 0_r|e^{\frac{i}{\hbar}\left(\hat{p}_r + \frac{i\hbar}{\hat{r}}\right)} \otimes \langle\theta, \phi|, \tag{11.117}$$

which is the form we will most commonly use.

This brings us to the final separation of variables step. When working with solving for energy eigenstates in a central-force problem, we know that our three mutually commuting operators are the Hamiltonian $\hat{H}$, the square of the total angular momentum $\hat{\vec{L}}^2$, and the z-component of angular momentum $\hat{L}_z$. So, we write the energy eigenstate as $|\psi\rangle = |\psi_r\rangle \otimes |l, m\rangle$, where $\hat{r}$ and $\hat{p}_r$ act on the $|\psi_r\rangle$ ket only, while the angular momentum operators act only on the $|l, m\rangle$ state. The wavefunction becomes

$$\psi(r, \theta, \phi) = \langle\vec{r}|\psi\rangle = \langle r|\psi_r\rangle\langle\theta, \phi|l, m\rangle, \tag{11.118}$$

which becomes $\psi(r, \theta, \phi) = \psi_r(r)Y_{lm}(\theta, \phi)$. Here, we have

$$\psi_r(r) = \langle 0_r|e^{\frac{i}{\hbar}\left(\hat{p}_r + \frac{i\hbar}{\hat{r}}\right)}|\psi_r\rangle. \tag{11.119}$$

We will use this final relation heavily when we determine wavefunctions for central force problems. A video Radial Momentum and the Radial Translation Vector https://tinyurl.com/sbu4my69 provides more details.

Section 11.4 Problems:

11.4.1 Numerical Estimates for Hydrogen

In this problem, we will show how one can use just the quantities we have already found—$a_0 = \frac{4\pi\epsilon_0\hbar^2}{m_e e^2} = 0.529$ Å, Ry $= \frac{e^2}{8\pi\epsilon_0 a_0} = 13.6$ eV, $\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c}$, and $2\pi\hbar c = 12\,400$ eV-Å—in computing other quantities relevant for atoms.

To begin, we show how you can determine the numerical value of the fine-structure constant. First use the multiply-by-one trick to find

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} = \frac{e^2}{4\pi\epsilon_0 a_0}\frac{a_0}{\hbar c} = \frac{e^2}{4\pi\epsilon_0 a_0}2\pi a_0\frac{1}{2\pi\hbar c} = \frac{27.2\text{ eV}\cdot 2\cdot\pi\cdot 0.529\text{Å}}{12\,400\text{ eV-Å}}$$
$$\approx \frac{1}{137}. \tag{11.120}$$

Calculate this to higher accuracy and show it is equal to $\alpha = 1/137.0359991$, using $2\pi\hbar c = 12398.41984$ ev-Å, Ry $= 13.605693122$ eV, and $a_0 = 0.5291772109$ Å. The current error is in the last digit of the fine structure constant.

(a) Using the constants discussed earlier and $c \approx 3\times 10^{18}$ eV-Å, compute the time $\frac{\hbar^6 c^3 (4\pi\epsilon_0)^5}{me^{10}}$ in seconds, which is related to the lifetime of excited states of hydrogen. *Ans.* 6.2×10^{-11} s.

(b) Use the same techniques to compute the wavelength of a photon that has energy

$$E_{\text{phot}} = \frac{2e^2 g_e g_p \hbar^2}{12\pi\epsilon_0 m_e m_p a_0^3 c^2}, \tag{11.121}$$

with $g_e \approx 2$, $g_p \approx 5.56$, $m_e c^2 \approx 511$ MeV, and $m_p c^2 \approx 938$ MeV. Note that the wavelength is found from $\lambda = \frac{hc}{E_{\text{phot}}}$. This is the hyperfine splitting of the ground-state of hydrogen, critical for radio astronomy. *Ans.* 5.86×10^{-6} eV, $\lambda \approx 21$ cm.

(c) Compute the wavelength of light emitted from hydrogen with an energy given by $\frac{3e^2}{32\pi\epsilon_0 a_0}$. *Ans.* 1215 Å

11.4.2 Center-of-Mass and Relative Position and Momentum Commutators

Verify that $[\hat{R}_{cm\,\alpha}, \hat{p}_{rel\,\beta}] = 0$ and $[\hat{r}_{rel\,\alpha}, \hat{P}_{cm\,\beta}] = 0$.

11.4.3 Separation of Variables for the Kinetic Energy of Hydrogen

Determine the kinetic-energy terms of the Hamiltonian for hydrogen by first expressing $\hat{\vec{p}}_p$ and $\hat{\vec{p}}_e$ in terms of $\hat{\vec{P}}_{cm}$ and $\hat{\vec{p}}_{rel}$ and showing that

$$\hat{H}_{\text{kin}} = \frac{\hat{\vec{p}}_p^{\,2}}{2m_p} + \frac{\hat{\vec{p}}_e^{\,2}}{2m_e} = \frac{\hat{\vec{P}}_{cm}^{\,2}}{2M} + \frac{\hat{\vec{p}}_{rel}^{\,2}}{2\mu}. \tag{11.122}$$

11.4.4 Proton and Electron Motion

We can reconstruct the motion for the two particles in hydrogen by solving for $\hat{\vec{r}}_p$ and $\hat{\vec{r}}_e$ in terms of $\hat{\vec{R}}_{cm}$ and $\hat{\vec{r}}_{rel}$. Do this, and think of the classical regime, where the electron and proton have well-defined trajectories in space. Draw pictures to examine what the trajectories are if

$$\vec{R}_{cm} = \begin{pmatrix} vt \\ 0 \\ 0 \end{pmatrix} \text{ and } \vec{r}_{rel} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}. \tag{11.123}$$

Repeat for

$$\vec{R}_{cm} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \text{ and } \vec{r}_{rel} = \begin{pmatrix} r\cos\omega t \\ r\sin\omega t \\ 0 \end{pmatrix}. \tag{11.124}$$

Finally sketch for

$$\vec{R}_{cm} = \begin{pmatrix} vt \\ 0 \\ 0 \end{pmatrix} \text{ and } \vec{r}_{rel} = \begin{pmatrix} r\cos\omega t \\ r\sin\omega t \\ 0 \end{pmatrix}. \tag{11.125}$$

11.4.5 Separation of Variables for Two Particles

Using the factorized form for the state $|\psi\rangle = |\psi_{cm}\rangle \otimes |\psi_{rel}\rangle$ to show that the coordinate-space wavefunction can be written as the product

$$\psi(\vec{R}_{cm}, \vec{r}_{rel}) = e^{\frac{i}{\hbar}\vec{P}_{cm}\cdot\vec{R}_{cm}} \langle \vec{R}_{cm}{=}0 | \vec{P}_{cm}\rangle \psi_{rel}(\vec{r}_{rel}). \tag{11.126}$$

11.4.6 Reduced Mass

The reduced mass is given by $\mu = \frac{m_1 m_2}{m_1+m_2}$. Compare the reduced mass for hydrogen, deuterium (hydrogen with a proton and a neutron in the nucleus), positronium (hydrogen from a positron and an electron), and muonic hydrogen (hydrogen with a muon orbiting a proton). Use $m_p c^2 \approx 938$ MeV, $m_d = 2.01 m_p$, $m_e c^2 = 511$ KeV, and $m_\mu c^2 = 105.6$ MeV. Use the mass of the positron as being exactly equal to the mass of the electron.

11.4.7 Rydberg Constant

The Rydberg constant is given by

$$R_\infty = \frac{Ry}{2\pi\hbar c} = \frac{e^2}{16\pi^2 \epsilon_0 \hbar c a_0} = 1.097373 \times 10^9 \text{ cm}^{-1}. \tag{11.127}$$

The wavelengths of light emitted by hydrogen satisfy

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right), \tag{11.128}$$

where $R_H = \frac{m_p}{m_e+m_p} R_\infty$ and n_i and n_f are both positive integers.

Using $R_d = \frac{m_d}{m_d+m_e} R_\infty$, compute the difference in cm^{-1} of the inverse wavelength of the transition from $n_i = 3$ to $n_f = 2$ for hydrogen and deuterium. This energy difference can be easily measured and provided the first proof that deuterium existed. Note that doing so uses spectroscopy to measure the mass of an atom! I always find it remarkable that one can "weigh" an atom in this fashion!

11.4.8 Action of $\hat{r}$ in the Tensor-Product Space

In the text, we showed how $\cos\hat{\theta}$ acted on a tensor-product state by converting to the Cartesian vector and evaluating the operators on that state. This then showed that the angle operator really only acted on the angular part of the tensor product state. Repeat that calculation, but now for the radial operator $\hat{r}$ and show that it acts only on the radial part of the tensor-product state for position.

11.4.9 Spherical Coordinates in Momentum Space

Be advised that this is a long problem. Repeat the calculations that converted a position eigenstate from a Cartesian basis to a spherical basis for momentum space, converting the momentum space state from a Cartesian form to a spherical form. Note that we use $\hat{p}^2 = \hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2$ and develop the angles in the same fashion as we did in position space, by taking ratios of Cartesian operators and radial operators, just now for momentum.

11.5 Schrödinger Factorization Method

In 1926, Erwin Schrödinger discovered the famous Schrödinger equation (Schrödinger, 1926a), which is a differential equation for the coordinate-space wavefunction and is studied in pretty much every quantum-mechanical textbook. It is also the main reason why most people will claim that one needs to know significant amounts of calculus to be able to master quantum mechanics.

Much less well known is Schrödinger's work from 15 years later (1940–1941) (Schrödinger, 1940)—Schrödinger was in Ireland during the Second World War and developed these methods while in exile. He showed how one can solve all of these quantum problems *without using any calculus*! This new method is called *the factorization method*, and it has been developed more thoroughly by Infeld and Hull (1951), Green (1965), and Ohanian (1990). It has not made an appearance in most quantum texts except for the solution of the simple-harmonic oscillator, which we worked out in Chap. 7 and which is the simplest example of this technique. It turns out that the general method is a bit more abstract, but we will find it is straightforward to employ it to solve a wide array of problems. Every problem that can be solved analytically using Schrödinger's differential equation can also be solved by the factorization method. But, with the factorization method, we can find the energy eigenvalues without finding the corresponding wavefunctions! We will illustrate many examples of the factorization method in the remainder of the book. Schrödinger is shown in Fig. 11.6 around the time he created the factorization method.

The method starts by factorizing the Hamiltonian to make it look like the operator approach to the harmonic oscillator. But then it does something odd—it constructs a set of auxiliary Hamiltonians, which usually are different from the initial Hamiltonian. Nevertheless, these auxiliary Hamiltonians are exactly what we need to make Schrödinger's factorization method work. You may find it necessary to read this section a few times before the method makes sense, due to its abstract nature. So proceed slowly.

We are ready to get started. Let's go!

The method starts by factorizing our Hamiltonian $\hat{H}$, which we will denote as $\hat{H}_0$, with a subscript 0, because we will be working with many auxiliary Hamiltonians,

Fig. 11.6 Sketch of Erwin Schrödinger in 1945. (Illustrated by Gide Gennaro, courtesy of AIP Emilio Segrè Visual Archives, Weber Collection.)

each with their own subscript. So, we assume we can determine a a pair of operators and a constant ($\hat{A}_0$, $\hat{A}_0^\dagger$ and E_0), such that the initial Hamiltonian can be written as

$$\hat{H} = \hat{H}_0 = \hat{A}_0^\dagger \hat{A}_0 + E_0. \tag{11.129}$$

When we worked with the simple harmonic oscillator, we found there was a fairly simple way to "guess" the factorization and force it to work. In general, it is highly nontrivial to determine how to perform this factorization, but, for all analytically solvable problems in quantum mechanics, the factorization can be done with relatively simple ansatzes for the raising and lowering operators. Note that the factorization has the freedom of including an extra constant term E_0. This degree of freedom is important. In some cases, the sign of E_0 will be critical in allowing us to solve a given problem, and we will discuss that further in some examples later.

The next step, however, deviates from what we did in the simple-harmonic oscillator case (although, if you look carefully, we did actually do this in a hidden way, and in the next section, we will show how that works concretely). We define a new *auxiliary* Hamiltonian, denoted $\hat{H}_1$ from the factorized form by switching the order of the raising and lowering operators in the factorized form of the Hamiltonian:

$$\hat{H}_1 = \hat{A}_0 \hat{A}_0^\dagger + E_0. \tag{11.130}$$

While this may seem like an innocuous thing to do, it can dramatically change the potential, so that the Hamiltonian $\hat{H}_1$ does not appear to be related at all to $\hat{H}_0$. Or, they may appear very closely related. It is what it is, and the construction procedure is clear and simple. Now, because the potential does change (and it always changes, because $[\hat{A}_0, \hat{A}_0^\dagger] \neq 0$), we set out to factorize the auxiliary Hamiltonian and put it into the form $\hat{H}_1 = \hat{A}_1^\dagger \hat{A}_1 + E_1$, allowing for the constant term to change as well (we will find the constant term always increases). You may recall that I mentioned that factorizing a Hamiltonian was hard. Now, we have to factorize two Hamiltonians. This must be doubly hard! Well, yes and no. In the general case, the answer is yes. But, for a class of problems that have what are called *shape-invariant* potentials, the form of the factorization stays the same from one step to the next. The only thing that changes is the parameters in the formulas for the raising and lowering operators. This makes it possible to carry out the factorization in a simple way.

Now, we continue. We form what is called a *factorization chain*, given by a sequence of auxiliary Hamiltonians, in the following form

$$\begin{aligned}
\hat{H}_0 &= \hat{A}_0^\dagger \hat{A}_0 + E_0 \\
\hat{H}_1 &= \hat{A}_0 \hat{A}_0^\dagger + E_0 = \hat{A}_1^\dagger \hat{A}_1 + E_1 \\
\hat{H}_2 &= \hat{A}_1 \hat{A}_1^\dagger + E_1 = \hat{A}_2^\dagger \hat{A}_2 + E_2 \\
\hat{H}_3 &= \hat{A}_2 \hat{A}_2^\dagger + E_2 = \hat{A}_3^\dagger \hat{A}_3 + E_3, \qquad (11.131) \\
&\vdots \qquad (11.132)
\end{aligned}$$

with a chain-like structure that can continue out to an infinite-sized chain (in many cases, the chain ends after a finite number of terms). The general step for the auxiliary Hamiltonian is

$$\hat{H}_i = \hat{A}_{i-1} \hat{A}_{i-1}^\dagger + E_{i-1} = \hat{A}_i^\dagger \hat{A}_i + E_i. \tag{11.133}$$

The key to the method is finding each of the new factorizations (right-hand side equations), since each auxiliary Hamiltonian has a different potential from the previous one. Doing so is a hard problem (one might even say it is currently an unsolved problem), but for now, we will assume that we can do it in the general case, to understand better the structure for how this works—rest assured, that for all analytically solvable problems, it can be done for each step, as we will see when we go through some examples.

Back when we studied the simple harmonic oscillator, we developed an operator identity that we called the intertwining relation. It involved moving a raising operator through its corresponding Hamiltonian to the left. In equations, we want to prove the identity

$$\hat{H}_i \hat{A}_i^\dagger = \hat{A}_i^\dagger \hat{H}_{i+1}. \tag{11.134}$$

You can clearly see that as the raising operator is moved past its corresponding Hamiltonian, it increases the index of the auxiliary Hamiltonian. The proof follows simply by using the definitions of the different Hamiltonians. We start by substituting in the definition of the ith auxiliary Hamiltonian using its second form: $\hat{H}_i \hat{A}_i^\dagger = (\hat{A}_i^\dagger \hat{A}_i + E_i)\hat{A}_i^\dagger$. Next, we distribute the rightmost raising operator into the parenthesis $\hat{H}_i \hat{A}_i^\dagger = (\hat{A}_i^\dagger \hat{A}_i \hat{A}_i^\dagger + E_i \hat{A}_i^\dagger)$. Since operators commute with numbers, we can move it past the number in the second term $\hat{H}_i \hat{A}_i^\dagger = (\hat{A}_i^\dagger \hat{A}_i \hat{A}_i^\dagger + \hat{A}_i^\dagger E_i)$. Now you can see that we can factor a raising operator out to the left, to give us $\hat{H}_i \hat{A}_i^\dagger = \hat{A}_i^\dagger (\hat{A}_i \hat{A}_i^\dagger + E_i)$. But the term in the parenthesis is just $\hat{H}_{i+1}$ using the first form for the definition of the auxiliary Hamiltonian. So we have established Eq. (11.134). This is a very important relationship. We will use it to determine all of the energy eigenstates of the original Hamiltonian. We will discover, along the way, all of the energy eigenstates of the auxiliary Hamiltonians as well. Hence, this approach will solve for the energy eigenvalues and eigenvectors of the original and all auxiliary Hamiltonians. Pretty cool, right?

Our next step is to find the ground state and the ground-state energy. This follows from what we call the *subsidiary condition*. When we have a Hamiltonian expressed in the form $\hat{H} = \hat{A}^\dagger \hat{A} + E$, then, if we take the expectation value of the Hamiltonian in any normalized state $|\psi\rangle$, we find that

$$\langle \psi | \hat{H} | \psi \rangle = \langle \psi | \hat{A}^\dagger \hat{A} | \psi \rangle + E = \| \hat{A} | \psi \rangle \|^2 + E; \tag{11.135}$$

note that this expectation value becomes equal to the energy eigenvalue if $|\psi\rangle$ is an energy eigenstate. Now, the norm of a vector is bigger or equal to zero, and the only state with zero norm is the zero state. Hence, the energy-expectation value is minimized if

$$\hat{A} | \psi \rangle = 0, \tag{11.136}$$

a condition called the *subsidiary condition*. The ground-state energy is then equal to E. Hence, the ground state of our original Hamiltonian is given by $\hat{A}_0 | \psi_0 \rangle = 0$ with energy E_0. We will use ψ to label the energy eigenstates of the original Hamiltonian $\hat{H} = \hat{H}_0$. The subscript determines which state—$i = 0$ is the ground state $i = 1$ is the first excited state, and so on. Similarly, the ith auxiliary Hamiltonian has a ground state given by $\hat{A}_i | \phi_i \rangle = 0$ with energy E_i. We will use ϕ to label the ground state of an auxiliary Hamiltonian, with the index denoting which auxiliary Hamiltonian. Using these subsidiary conditions, we immediately find the ground states of our original problem and all auxiliary problems too.

Now we simply state what the (unnormalized) excited states of the original Hamiltonian are. One could, with trial and error, discover their particular forms. The key is we need to have the set of operators change the Hamiltonian acting from the left from $\hat{H}_0$ to $\hat{H}_i$ as it moves through each operator in the chain (via

the intertwining relation), until we reach the ith ground state on the right. Since this state is an energy eigenstate of $\hat{H}_i$, we can determine the energy eigenvalue. In equations, we claim the ith (unnormalized) excited state of $\hat{H}$, $|\psi_i\rangle$ is given by

$$|\psi_i\rangle = C_i \hat{A}_0^\dagger \hat{A}_1^\dagger \hat{A}_2^\dagger \cdots \hat{A}_{i-1}^\dagger |\phi_i\rangle, \tag{11.137}$$

where C_i is an as yet undetermined normalization constant. We verify this by showing it is an energy eigenstate with respect to $\hat{H}_0$ after using the intertwining relation i times:

$$\begin{aligned}
\hat{H}_0|\psi_i\rangle &= C_i \underbrace{\hat{H}_0 \hat{A}_0^\dagger} \hat{A}_1^\dagger \hat{A}_2^\dagger \cdots \hat{A}_{i-1}^\dagger |\phi_i\rangle \\
&= C_i \hat{A}_0 \underbrace{\hat{H}_1 \hat{A}_1^\dagger} \hat{A}_2 \cdots \hat{A}_{i-1}^\dagger |\phi_i\rangle \\
&= C_i \hat{A}_0^\dagger \hat{A}_1^\dagger \underbrace{\hat{H}_2 \hat{A}_2^\dagger} \cdots \hat{A}_{i-1}^\dagger |\phi_i\rangle \\
&= C_i \hat{A}_0^\dagger \hat{A}_1^\dagger \hat{A}_2^\dagger \cdots \hat{A}_{i-1}^\dagger \hat{H}_i |\phi_i\rangle \\
&= C_i \hat{A}_0^\dagger \hat{A}_1^\dagger \hat{A}_2^\dagger \cdots \hat{A}_{i-1}^\dagger E_i |\phi_i\rangle \\
&= E_i C_i \hat{A}_0^\dagger \hat{A}_1^\dagger \hat{A}_2^\dagger \cdots \hat{A}_{i-1}^\dagger |\phi_i\rangle = E_i |\psi_i\rangle.
\end{aligned} \tag{11.138}$$

The energy of the ith excited state of $\hat{H}$ is given by the constant E_i in the factorization of the ith auxiliary Hamiltonian!

We now show that for a given factorization chain, we must have the energies form a strictly increasing sequence given by $E_0 < E_1 < E_2 < \cdots$. If we look at the construction for the auxiliary Hamiltonians, where we have $\hat{H}_i = \hat{A}_{i-1}\hat{A}_{i-1}^\dagger + E_{i-1} = \hat{A}_i^\dagger \hat{A}_i + E_i$, one might ask, why can't I set $\hat{A}_i = \hat{A}_{i-1}^\dagger$ and $E_i = E_{i-1}$? The reason why, is if we do this, then the factorization chain terminates at the $(i-1)$st step. Why? We look at the subsidiary condition if we make this choice. We have $\hat{A}_i|\phi_i\rangle = \hat{A}_{i-1}^\dagger|\phi_i\rangle = 0$. But then when we form $|\psi_i\rangle = C_i \hat{A}_0^\dagger \hat{A}_1^\dagger \cdots \underbrace{\hat{A}_{i-1}^\dagger |\phi_i\rangle}$, we find that it must vanish due to the subsidiary condition (focus on the underbrace and recall that $\hat{A}_i|\phi_i\rangle = \hat{A}_{i-1}^\dagger|\phi_{i-1}\rangle = 0$ if we made the "trivial" choice for the factorization). Now, it is possible that a factorization chain can terminate, because the Hamiltonian has only a finite number of bound states. This occurs, for example, with the Morse potential. This is a mechanism whereby the factorization chain can terminate. If it has terminated, then we do not have a $|\psi_i\rangle$ state. But, if it does not terminate, then we must have that $\hat{A}_{i-1}^\dagger|\phi_i\rangle \neq 0$, otherwise $|\psi_i\rangle = 0$. Now using that condition, we have that

$$E_i = \langle\phi_i|\hat{H}_i|\phi_i\rangle = \langle\phi_i|(\hat{A}_{i-1}\hat{A}_{i-1}^\dagger|\phi_i\rangle + E_{i-1} = \|\hat{A}_{i-1}^\dagger|\phi_i\rangle\|^2 + E_{i-1}. \tag{11.139}$$

But the norm squared $\|\hat{A}_{i-1}^{\dagger}|\phi_i\rangle\|^2$ is strictly greater than zero, because $\hat{A}_{i-1}^{\dagger}|\phi_i\rangle \neq 0$ implies the norm is nonzero. This means $E_i > E_{i-1}$; hence, the energies for a single factorization chain must form an *increasing* sequence. Note that *this requirement means that the bound-state energy levels are discrete, because there is a nonzero jump from one energy level to the next one*. This is one of the hallmarks of quantum mechanics—that energy is not continuous in bound states. Tracking down the source of the discreteness, we see that it arises from the fact that the position and momentum operators do not commute. This implies a direct link between the canonical commutation relation and the discreteness of the energy eigenvalues in bound states. In this book, we first motivated the canonical commutation relation from the sharp lines in atomic spectra. We see that discrete energy levels in bound states result from this commutation relation. An alternative approach is proposed by Terry Robinson (2021). He starts from the postulate that quantum mechanics is the theory of discrete measurements and from that assumption he derives the canonical commutation relation (along with the Schrödinger factorization method). Both approaches are equally valid, and you might want to investigate Robinson's alternative viewpoint.

We can use intertwining to determine the normalization constant C_i. This is found by calculating the norm

$$\langle\psi_i|\psi_i\rangle = |C_i|^2\langle\phi_i|\hat{A}_{i-1}\cdots\hat{A}_1\underbrace{\hat{A}_0\hat{A}_0^{\dagger}}\hat{A}_1^{\dagger}\cdots\hat{A}_{i-1}^{\dagger}|\phi_i\rangle. \tag{11.140}$$

Focus on the terms highlighted by the underbrace. It can be written as $\hat{H}_1 - E_0$. After doing this, it can be moved to the right by using the intertwining relation, until it becomes $\hat{H}_i - E_0$ acting directly on $|\phi_i\rangle$, which can be replaced by $(E_i - E_0)|\phi_i\rangle$. After doing this, we have $\hat{A}_1\hat{A}_1^{\dagger} = \hat{H}_2 - E_1$ in the center, and we can do the same thing—move it to the right using the intertwining relation until it reaches the right-hand side as $\hat{H}_i - E_1$ acting on $|\phi_i\rangle$, which gives another factor of $(E_i - E_1)$. Continuing in this manner, we can remove all operators, and we find we have

$$C_i = \frac{1}{\sqrt{(E_i - E_0)(E_i - E_1)(E_i - E_2)\cdots(E_i - E_{i-1})}} \tag{11.141}$$

as the normalization constant. Note that in this analysis, we assume that all auxiliary Hamiltonian ground states are normalized, or $\langle\phi_i|\phi_i\rangle = 1$.

We are now ready to discuss the form of the raising and lowering operators more specifically. We describe them in terms of a so-called *superpotential* $W_i(k_i'x)$. The lowering operators take the form

$$\hat{A}_i = \frac{1}{\sqrt{2m}}\big(\hat{p} - i\hbar k_i W_i(k_i'\hat{x})\big). \tag{11.142}$$

The momentum dependence is the same for all of these operators, while the superpotential varies for each of them. The constants k_i and k_i' are wavevectors

with dimensions of inverse length. In a shape-invariant potential, the form of the superpotential will be the same, but the parameters in the superpotential will vary with i. We now show that the superpotential must be real. Assume it is complex, which we write schematically as $\hat{B} + i\hat{C}$. Then, we find that

$$\hat{A}_i^\dagger \hat{A}_i = \frac{\hat{p}^2}{2m} - i\frac{\hbar k_i}{2m}[\hat{p} + \hbar k_i \hat{C}, \hat{B}] + \underbrace{\frac{\hbar k_i}{2m}(\hat{p}\hat{C} + \hat{C}\hat{p})}_{\text{linear in momentum}} + \frac{\hbar^2 k_i^2}{2m}(\hat{B}^2 + \hat{C}^2). \tag{11.143}$$

Hence, if we do not have any linear terms in momentum in the Hamiltonian (which we do not have when there are no external fields), then we must have the imaginary part $\hat{C}$ vanishes. Then, because W_i must be real, we have that

$$\hat{A}_i^\dagger \hat{A}_i + E_i = \frac{\hat{p}^2}{2m} - i\frac{\hbar k_i}{2m}[\hat{p}, W_i(k_i'\hat{x})] + \frac{\hbar^2 k_i^2}{2m} W_i^2(k_i'\hat{x}) + E_i, \tag{11.144}$$

which means we must satisfy the equation

$$V_i(\hat{x}) = -i\frac{\hbar k_i}{2m}[\hat{p}, W_i(k_i'\hat{x})] + \frac{\hbar^2 k_i^2}{2m} W_i^2(k_i'\hat{x}) + E_i. \tag{11.145}$$

This is the complicated nonlinear equation that we need to solve in order to factorize each auxiliary Hamiltonian. There is no general way to solve it, but we do know its solutions in special cases. Note that the solution need not be unique, because it is a nonlinear equation. We will discuss the consequences of this when we discuss an example in the next section. In essence, the whole complication in working with the factorization method boils down to finding methods that allow us to solve this nonlinear equation. We will work on this in more detail throughout the remainder of the book. We do note one important property of the superpotential—$k_i W_i(k_i' x)$ must be positive as $x \to \infty$, and it must be negative as $x \to -\infty$. This is required in order for the bound state to have a wavefunction that vanishes as $|x| \to \infty$. The proof of this requires calculus, but we motivate it in Problem 11.5.6. Note that there are potentials, for which one may find a factorization that does not satisfy this criterion. This case is called a case that has *broken supersymmetry*—we will not discuss such problems in detail in this book. These systems should still be able to be factorized following the general procedures we use, while the supersymmetry-breaking factorization is an additional factorization, that cannot be used to form a factorization chain, but instead shows that the eigenvalues for the problem with $V(\hat{x})$ are the same as the eigenvalues for $V(-\hat{x})$ when the potential is not even. This is a result that should always hold. You should convince yourself of this by thinking about a unitary transformation of the Hamiltonian and of the energy eigenstates with the parity operator.

The subsidiary condition implies that the superpotential diverges if the ground-state wavefunction vanishes. To see this, we have that $\langle x|\hat{A}_i|\phi_i\rangle = 0$ implies that

$$\langle x|\hat{p}|\phi_i\rangle = i\hbar k_i W_i(k_i' x)\langle x|\phi_i\rangle, \tag{11.146}$$

or

$$W_i(k_i' x) = \frac{\langle x|\hat{p}|\phi_i\rangle}{i\hbar k_i \phi_i(x)}. \tag{11.147}$$

This implies that if there is a zero in the ground-state wavefunction, given by a position $\bar{x}$ where $\phi_i(\bar{x}) = 0$, then $W_i(k_i'\bar{x})$ diverges. This statement requires that $\langle\bar{x}|\hat{p}|\phi_i\rangle \neq 0$. This question is answered in Problem 11.5.2. The wavefunction often vanishes on the boundary of the domain. We do not call those zeros "nodes." But any zero that occurs *inside* the domain (typically $-\infty < x < \infty$ for a one-dimensional problem) is called a *node*. In one dimension, there is a node theorem, which says the ground state has no nodes, and the ith excited state has exactly i nodes. We will sketch a proof of this theorem later. There is another implication of this relationship. Two requirements of a wavefunction in quantum mechanics are that the probability density, given by $|\psi_i(x)|^2 = \langle x|\psi_i\rangle\,\langle\psi_i|x\rangle$ and the probability current density, given by $\langle x|\hat{p}|\psi_i\rangle\,\langle\psi_i|x\rangle$, must be continuous for all x. Taking their ratio, we find that the ratio of the probability current density to the probability density is proportional to the superpotential (as shown in the relation in Eq. (11.147)). Hence, if we have a potential that is discontinuous at a point x_0, then the superpotential will still be continuous at that point. This is an important requirement we will use when we solve for the problem of a particle moving in a finite square well, which has a discontinuous potential. We treat such problems in Chap. 12.

We end the section with a brief summary of the main points related to the factorization method:

1. The factorization method seeks to write a Hamiltonian and the associated auxiliary Hamiltonians in the form $\hat{H}_i = \hat{A}_{i-1}\hat{A}_{i-1}^\dagger + E_{i-1} = \hat{A}_i^\dagger\hat{A}_i + E_i$, with $\hat{H} = \hat{H}_0$ at the start of the factorization chain.
2. The energies $\{E_i\}$ are the energy eigenvalues of $\hat{H}$.
3. The subsidiary condition $\hat{A}_i|\phi_i\rangle = 0$ determines the auxiliary ground states for each auxiliary Hamiltonian; when $i = 0$, we use ψ to denote the energy eigenstates of the original Hamiltonian $\hat{H}$, so the ground state of $\hat{H}$ satisfies $\hat{A}_0|\psi_0\rangle = 0$.
4. The normalized eigenstates are written as $|\psi_i\rangle = C_i\hat{A}_0^\dagger\hat{A}_1^\dagger\cdots\hat{A}_{i-1}^\dagger|\phi_i\rangle$, with $C_i = 1/\sqrt{\prod_{j=0}^{i-1}(E_i - E_j)}$.
5. The energies form a strictly increasing sequence $E_0 < E_1 < E_2 < \cdots$, which leads to discrete energy eigenvalues for the bound states. The discreteness arises from the canonical commutation relation $[\hat{x}, \hat{p}] = i\hbar$.
6. The lowering operator is written as $\hat{A}_i = \frac{1}{\sqrt{2m}}\big(\hat{p} - i\hbar k_i W_i(k_i'\hat{x})\big)$, with the superpotential W_i being a real-valued function.
7. The superpotential is related to the potential via a nonlinear relation given by $V_i(\hat{x}) = -i\frac{\hbar k_i}{2m}[\hat{p}, W_i(k_i'\hat{x})] + \frac{\hbar^2 k_i^2}{2m}W_i^2(k_i'\hat{x}) + E_i$.

8. The superpotential $k_i W_i(k_i' x)$ must be positive when $x \to \infty$ and must be negative when $x \to -\infty$ to have a ground-state wavefunction that vanishes as $|x| \to \infty$.
9. The superpotential diverges when the wavefunction vanishes.
10. The node theorem states that the ith excited state has exactly i nodes. Verifying this property implies that we have found all of the energy eigenstates of $\hat{H}$.
11. The factorization method might not be unique. In some cases, it is possible to find more than one factorization chain for a given problem. This is how degeneracy can enter into the energy eigenvalues in special cases. The second factorization may produce only some of the solutions already found from another factorization chain. It is also possible that an additional factorization cannot be used to make a factorization chain.

In this section, we have focused on the factorization method for bound states, which have discrete energy levels. Continuum problems will be discussed later, after we have more experience in working with this formalism, but we will not be able to solve those problems in general. The video The Schrödinger Factorization Method https://tinyurl.com/484kz8zb provides additional discussion on this.

Section 11.5 Problems:

11.5.1 Changing the Phase of the Raising and Lowering Operators

Suppose we have created a factorization chain with operators $\hat{A}_j$ and $\hat{A}_j^\dagger$. Consider the new set of operators defined as

$$\hat{B}_j = e^{i\alpha_j}\hat{A}_j \quad \text{and} \quad \hat{B}_j^\dagger = e^{-i\alpha_j}\hat{A}_j^\dagger. \tag{11.148}$$

Here α_j are a set of arbitrary real numbers. Show that these operators also satisfy being a factorization chain. What happens to the eigenvalues E_j and the ground-state eigenvectors $|\phi_j\rangle$ if we use the B operators instead of the A operators. What about the excited states $|\psi_i\rangle$? How does this relate to the statement that the superpotential is real?

11.5.2 Divergence of the Superpotential at a Zero of the Energy Eigenstate Wavefunction

Our claim is that at a point $\bar{x}$, where $\langle \bar{x}|\psi\rangle = \psi(\bar{x})$ vanishes, we cannot also have $\langle \bar{x}|\hat{p}|\psi\rangle = 0$ as well. This is a well know result from Sturm–Liouville theory of differential equations. It applies when we have a differential equation in Sturm–

Liouville form and the potential does not diverge at $\bar{x}$, and we are not expanding about a boundary point of the domain. We also require in our argument that $V(x)$ is a smooth function, which we will define more carefully as follows. Now, in this book, we work without calculus, so we will give you the flavor for how the argument goes, but we will not complete the argument in its full generality. The basic idea is that if *both* $\langle \bar{x}|\psi\rangle = 0$ and $\langle \bar{x}|\hat{p}|\psi\rangle = 0$, then we must have $\psi(x) = 0$ for all x. We will use the fact that $|\psi\rangle$ is an energy eigenstate of the Hamiltonian $\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x})$ such that $\hat{H}|\psi\rangle = E|\psi\rangle$.

We write $\psi(x) = \langle \bar{x}|e^{\frac{i}{\hbar}(x-\bar{x})\hat{p}}|\psi\rangle$, and we expand the exponential in a power series. Such an expansion is possible for all finite $\bar{x}$, which is why this approach does not work for $|\bar{x}| \to \infty$, or any other boundary point of the domain. Hence, we have that

$$\psi(x) = \sum_{n=0}^{\infty} \frac{i^n (x-\bar{x})^n}{n!\,\hbar^n} \langle \bar{x}|\hat{p}^n|\psi\rangle. \tag{11.149}$$

We assume that both $\langle \bar{x}|\psi\rangle = 0$ and $\langle \bar{x}|\hat{p}|\psi\rangle = 0$. Our claim is that from this, we can show that $\langle \bar{x}|\hat{p}^n|\psi\rangle = 0$ for all n. In this problem, you will investigate how this works for some small n values and learn the general idea for why it will work for all n.

(a) Use the energy eigenvalue–eigenvector equation to show that

$$\langle \bar{x}|\hat{p}^2|\psi\rangle = 2m\langle \bar{x}|\big(E - V(\hat{x})\big)|\psi\rangle. \tag{11.150}$$

Operate the terms on the right-hand-side equation to the left onto the position bra to show that $\langle \bar{x}|\hat{p}^2|\psi\rangle = 0$.

(b) Evaluate the next moment of the momentum via

$$\langle \bar{x}|\hat{p}^3|\psi\rangle = 2m\langle \bar{x}|\hat{p}(E - V(\hat{x}))|\psi\rangle. \tag{11.151}$$

Use the add-zero trick to move the $\hat{p}$ operator to the right so that $E - V(\hat{x})$ can be removed from the expectation value by acting to the left. Next, notice that $[\hat{p}, V(\hat{x})]$ is a well-defined function of $\hat{x}$ due to the smoothness criterion we stated earlier. Call this function $[\hat{p}, V(\hat{x})] = -i\hbar V'(\hat{x})$. Using this function, verify that $\langle \bar{x}|\hat{p}^3|\psi\rangle = 0$.

(c) Evaluate the quartic moment of the momentum. Define a new function given by $[\hat{p}, [\hat{p}, V(\hat{x}]] = -\hbar^2 V''(\hat{x})$, and use it, plus the previous results to show that $\langle \bar{x}|\hat{p}^4|\psi\rangle = 0$.

Note that these calculations are complicated, because we can only replace $\hat{p}^2|\psi\rangle = 2m(E - V(\hat{x}))|\psi\rangle$, which leaves behind a trail of increasingly more complicated commutators. There is no way out of this situation, because we only know what happens when $\hat{p}^2$ acts on $|\psi\rangle$.

I hope this algebra convinces you that it can be repeated for increasingly more complicated cases, and that, when all the dust settles, we will be able to conclude that $\langle \bar{x}|\hat{p}^n|\psi\rangle = 0$ for all n because of the smoothness of the potential and all of its nested commutators with momentum. Then, we have established the result we need.

11.5.3 Relationship Between Adjacent Potentials in the Factorization Chain

The factorization chain says that

$$\frac{\hat{p}^2}{2m} + V_i(\hat{x}) = \hat{A}_i^\dagger \hat{A}_i + E_i \tag{11.152}$$

and

$$\frac{\hat{p}^2}{2m} + V_{i+1}(\hat{x}) = \hat{A}_i \hat{A}_i^\dagger + E_i\,. \tag{11.153}$$

Hence, we have

$$V_{i+1}(\hat{x}) = V_i(\hat{x}) + [\hat{A}_i\,, \hat{A}_i^\dagger]. \tag{11.154}$$

Verify the aforementioned equation, and re-express it in terms of the superpotential, to show how V_{i+1} relates to V_i and W_i.

11.5.4 The Case Study of a Particle on a Circle

Go back to the particle in a circle problem in Sect. 7.4.

(a) Determine the superpotentials for the $\hat{A}_k$ and $\hat{B}_k$ based factorizations.
(b) Compute the potential in each case. Note that we have two different factorizations here.
(c) Using the A factorization, see whether you can construct a factorization chain by computing $\hat{A}_k\hat{A}_k^\dagger + E_k$. Is this also the potential for a particle on a circle? Can you factorize the Hamiltonian with this new potential?
(d) We stated that the energies of the factorization chain are strictly increasing. Explain how the particle on a circle has degenerate energy eigenvalues yet it still can be factorized with a factorization chain.

11.5.5 A Simple Criterion of the Superpotential for Shape-Invariant Potentials

Our formulas for the potentials are given by

$$V_i(\hat{x}) = -i\frac{\hbar k_i}{2m}[\hat{p}, W_i(k_i'\hat{x})] + \frac{\hbar^2 k_i^2}{2m}W_i^2(k_i'\hat{x}) + E_i, \tag{11.155}$$

and

$$V_{i+1}(\hat{x}) = i\frac{\hbar k_i}{2m}[\hat{p}, W_i(k_i'\hat{x})] + \frac{\hbar^2 k_i^2}{2m}W_i^2(k_i'\hat{x}) + E_i. \tag{11.156}$$

(a) Consider a superpotential of the form $W_i(k_i'\hat{x}) = \left(1 + \frac{\alpha_i}{k_i'\hat{x}}\right)$. Determine the general form of the two potentials V_i and V_{i+1}. Are they schematically the same?
(b) Consider a superpotential of the form $W_i(k_i'\hat{x}) = \left(k_i'\hat{x} + \frac{\alpha_i}{k_i'\hat{x}}\right)$. Determine the general form of the two potentials V_i and V_{i+1}. Are they schematically the same?

You should find that they are schematically similar in these two cases, and that means these are cases where one has a shape-invariant potential. We will see examples of problems we can solve using these superpotentials later.

11.5.6 Limiting Behavior of the Superpotential as $|x| \to \infty$

In this problem, we will motivate why we choose the superpotential to be positive as $x \to \infty$ and negative when $x \to -\infty$. For simplicity, we will assume for large $|x|$ we have $W(x) \to \pm\alpha$ with $\alpha > 0$ and the sign chosen to be the same as the sign of x. We will solve the problem for $x \to \infty$ first.

We have $\hat{p}|\psi\rangle = i\hbar\alpha|\psi\rangle$ in the $x \to \infty$ limit, but we assume it holds for all x in our analysis. Show that

$$\hat{p}e^{\alpha\hat{x}}|\psi\rangle = 0. \tag{11.157}$$

Hint: multiply on the left by $e^{-\alpha\hat{x}}$ and use the Hadamard lemma.

The state annihilated by $\hat{p}$ is $|0_p\rangle$ and we know $\langle x|0_p\rangle = \langle 0_x|0_p\rangle$, which is a constant (just calculate it directly if you do not recall this result). Use this to show that

$$\psi(x) = \langle 0_x|0_p\rangle e^{-\alpha x}, \tag{11.158}$$

which goes to zero as $x \to \infty$. We claim that this is also the behavior of the wavefunction as $x \to \infty$ (when the superpotential approaches a positive constant α as $x \to \infty$) even though we only did the analysis for a constant superpotential.

Repeat the analysis for $x \to -\infty$ to show that if the superpotential approaches a negative constant, then the wavefunction will go to zero as $x \to -\infty$.

This result motivates why we have the rule about the sign of the superpotential as $|x| \to \infty$. It is only a "motivation" because we have not rigorously shown that if the superpotential is a function of x, and it approaches this limiting behavior for large $|x|$, then the wavefunction will decay exponentially. This is the best we can do without calculus. It is a pretty convincing argument though.

11.5.7 Terminating the Chain

In the text, we showed you that if we chose $\hat{A}_{i+1} = \hat{A}_i^\dagger$, then the factorization chain would be required to terminate because the $i+1$st subsidiary condition would guarantee that $\hat{A}_i^\dagger|\phi_{i+1}\rangle = 0$. But, if we did not make this choice and found a valid factorization, then we assumed that $\hat{A}_i^\dagger|\phi_{i+1}\rangle \neq 0$. In this problem, you will prove that this is so.

It is best to prove this by contradiction. Start by assuming that $\hat{A}_i^\dagger|\phi_{i+1}\rangle = 0$. Then, the $i+1$st subsidiary condition implies that $\hat{A}_i^\dagger|\phi_{i+1}\rangle = \hat{A}_{i+1}|\phi_{i+1}\rangle = 0$. Use this result to show a relationship between the ith and $i+1$st superpotential operators acting on $|\phi_{i+1}\rangle$. Multiply from the left with $\langle x|$ and use the fact that $\phi_{i+1}(x) \neq 0$ inside the domain because it is a ground state. Finally, show that the assumption that $\hat{A}_i^\dagger|\phi_{i+1}\rangle = 0$ implies that $\hat{A}_{i+1} = \hat{A}_i^\dagger$, which is not allowed. Hence, we must have $\hat{A}_i^\dagger|\phi_{i+1}\rangle \neq 0$.

This problem is thanks to Joseph Noonan, a student who worked with me on research on Rodrigues formula in the summer of 2022.

11.5.8 Expectation Value of Momentum Vanishes in an Energy Eigenstate

We start this problem in the ground state. Our first step is to show that $\langle\psi_0|\hat{p}|\psi_0\rangle = 0$. To do this, first note that the subsidiary condition can be written as $\hat{p}|\psi_0\rangle = i\hbar k_0 W_0(k_0'\hat{x})|\psi_0\rangle$ or as $\langle\psi_0|\hat{p} = -i\hbar k_0\langle\psi_0|W_0(k_0'\hat{x})$. Take $\hat{p} = \frac{1}{2}\hat{p} + \frac{1}{2}\hat{p}$ and act one piece to the left and one to the right. Use this to show that $\langle\psi_0|\hat{p}|\psi_0\rangle = 0$ (recall the superpotential is real-valued).

Next, let's evaluate $\langle\psi_0|\hat{p}^3|\psi_0\rangle$. We do this by again breaking into two pieces and then evaluating $\hat{p}^2$ to the right for one piece and to the left for the other. You should find that

$$\langle\psi_0|\hat{p}^3|\psi_0\rangle = m\big(\langle\psi_0|\big(E - V(\hat{x})\big)\hat{p}|\psi_0\rangle + \langle\psi_0|\hat{p}\big(E - V(\hat{x})\big)|\psi_0\rangle\big). \tag{11.159}$$

Use the subsidiary condition to act the momentum operator to the right or to the left and verify that the expectation value vanishes (recall that functions of the position operator commute).

Discuss how this generalizes to all odd powers. You should be able to come up with a cogent and clear argument that the expectation of all odd powers of momentum must vanish for any ground state. You do not need to make the proof completely rigorous.

Use the previous result to show that the expectation value of momentum vanishes in all energy eigenstates. Again, a clear argument justifying why is all that is needed not a rigorous proof.

Note that there are some technical issues related to whether or not these odd powers of momentum have expectation values that are well-defined, but we will not worry about that here.

11.6 Application of the Schrödinger Factorization Method to the Simple Harmonic Oscillator

We already solved the simple harmonic oscillator via factorizing the Hamiltonian back in Sect. 7.6. We repeat the calculation here, but now using the new Schrödinger factorization methodology. This will use different raising and lowering operators than before, and the procedure is also a bit different from how we did it the first time. We also have a surprise at the end of a second factorization, which is a general result for even potentials. We will discuss that further as well.

The Hamiltonian for the simple harmonic oscillator is

$$\hat{H} = \frac{\hat{p}^2}{2m} + \tfrac{1}{2}m\omega^2\hat{x}^2, \tag{11.160}$$

with $\omega = \sqrt{\frac{\kappa}{m}}$. We need to find the superpotential that satisfies

$$\tfrac{1}{2}m\omega^2\hat{x}^2 = -\frac{i\hbar k_0}{2m}[\hat{p}, W_0(k_0'\hat{x})] + \frac{\hbar^2 k_0^2}{2m}W_0^2(k_0'\hat{x}) + E_0. \tag{11.161}$$

Since $[\hat{p}, \hat{x}] = -i\hbar$ is a constant, it is clear that choosing $W_0(k_0'\hat{x}) = k_0'\hat{x}$ should provide us with enough freedom to solve the problem. We must have $k_0 k_0' > 0$ to satisfy the sign condition on the superpotential as $|x| \to \infty$. Substituting into the nonlinear equation gives us

$$\tfrac{1}{2}m\omega^2\hat{x} = -\frac{\hbar^2 k_0 k_0'}{2m} + \frac{\hbar^2 k_0^2 k_0'^2}{2m}\hat{x}^2 + E_0. \tag{11.162}$$

This implies that $k_0 k_0' = \frac{m\omega}{\hbar}$ and $0 = -\frac{\hbar\omega}{2} + E_0$, or $E_0 = \frac{1}{2}\hbar\omega$. We also have that

$$\hat{A}_0 = \frac{1}{\sqrt{2m}}(\hat{p} - im\omega\hat{x}) \text{ and } \hat{A}_0^\dagger = \frac{1}{\sqrt{2m}}(\hat{p} + im\omega\hat{x}). \tag{11.163}$$

Note that there is no ambiguity in the sign of the superpotential due to the sign rule that we have for the superpotential.

Now, the procedure says we form

$$\hat{H}_1 = \hat{A}_0\hat{A}_0^\dagger + E_0 = \frac{1}{2m}(\hat{p}^2 + im\omega[\hat{p},\hat{x}] + m^2\omega^2\hat{x}^2) + \tfrac{1}{2}\hbar\omega = \hat{H}_0 + \hbar\omega. \tag{11.164}$$

The first auxiliary Hamiltonian is the same as the original Hamiltonian, but shifted upward by the constant $\hbar\omega$. This is easy to factorize via

$$\hat{A}_1 = \frac{1}{\sqrt{2m}}(\hat{p} - im\omega\hat{x}) \text{ and } \hat{A}_1^\dagger = \frac{1}{\sqrt{2m}}(\hat{p} + im\omega\hat{x}) \tag{11.165}$$

and $E_1 = \frac{3}{2}\hbar\omega$. It is also easy to see that indeed all the auxiliary raising and lowering operators are the same, and the auxiliary Hamiltonians satisfy $\hat{H}_j = \hat{H}_0 + j\hbar\omega$. Note that in this case, we could not even think of choosing $\hat{A}_{i+1} = \hat{A}_i^\dagger$ because the superpotential would not satisfy the required sign rule. The states follow simply as well. We have $\hat{A}_0|\psi_0\rangle = 0$ and similarly $\hat{A}_j|\phi_j\rangle = 0$; since the lowering operators are all the same, each one of these states is the same, and given by our old ground state $|0\rangle$. This might be a surprise to you, but there is nothing in our procedure that said the auxiliary ground states need to be different from each other (and from the ground state of $\hat{H}$). In all other cases they are—this is the one exceptional case. Finally, we form the states using our formula. Note that $(E_j - E_i) = (j - i)\hbar\omega$ so that

$$|\psi_j\rangle = \frac{1}{\sqrt{j!}}\frac{1}{(\hbar\omega)^{\frac{j}{2}}}\hat{A}_0^\dagger\hat{A}_1^\dagger\cdots\hat{A}_{j-1}|0\rangle = \frac{1}{\sqrt{j!}}\left(\frac{\hat{A}_0^\dagger}{\sqrt{\hbar\omega}}\right)^j|0\rangle, \tag{11.166}$$

which is precisely our old form (up to a global phase), because $\hat{a}^\dagger = -\frac{i}{\sqrt{\hbar\omega}}\hat{A}_0^\dagger$. Since the global phase of each raising and lowering operator can be modified by complex conjugate phases, without changing any of the states (up to global phases) and energies, it should be clear that the solutions are exactly the same as before!

The big surprise here is that there is a second factorization of the original simple-harmonic-oscillator Hamiltonian. We will find that this alternate factorization produces only half of the energy eigenstates—all of the Fock states with odd n are produced. We will see that the reason why such a factorization chain exists is because the original potential is an even function of x. Whenever this happens,

we can always find an alternative factorization that produces half of the energy eigenstates. We will describe why this is possible too.

Let's look back at the nonlinear equation for the superpotential of the simple harmonic oscillator in Eq. (11.161). If we chose a superpotential of the form $k_0 W_0(k_0'\hat{x}) = k_0\big(k_0'\hat{x} + \frac{\alpha_0}{k_0'\hat{x}}\big)$, then you can see that the square of this function gives as a term that goes like $\hat{x}^2$, a constant term, and a term that goes like $\frac{1}{\hat{x}^2}$. The commutator of momentum with the superpotential gives a constant term plus a term that goes like $\frac{1}{\hat{x}^2}$. If we can cancel the two contributions that go like $\frac{1}{\hat{x}^2}$, then we can construct the simple-harmonic-oscillator potential plus a constant! This will then be sufficient to have a valid factorization. Let's see how this works, by performing the calculation carefully.

We use the symbol $\hat{B}$ for the ladder operators so as to not confuse them with the $\hat{A}$ operators, and we use $\hat{\hat{H}}$ to denote the Hamiltonians for the same reason. Using $k_0 W_0(k_0'\hat{x}) = k_0(k_0'\hat{x} + \frac{\alpha_0}{k_0'\hat{x}})$ in Eq. (11.161), we find we must have

$$\tfrac{1}{2}m\omega^2\hat{x}^2 = -\frac{\hbar^2 k_0 k_0'}{2m} + \frac{\hbar^2 k_0\alpha_0}{2mk_0'\hat{x}^2} + \frac{\hbar^2 k_0^2}{2m}\left(k_0'^2\hat{x}^2 + 2\alpha_0 + \frac{\alpha_0^2}{k_0'^2\hat{x}^2}\right) + E_0. \tag{11.167}$$

Setting $\alpha_0 = -\frac{k_0'}{k_0}$ and $\hbar k_0 k_0' = m\omega$ yields $E_0 = \frac{3}{2}\hbar\omega$ and

$$\hat{B}_0 = \frac{1}{\sqrt{2m}}\left(\hat{p} - im\omega\hat{x} + \frac{i\hbar}{\hat{x}}\right) \quad \text{and} \quad \hat{B}_0^\dagger = \frac{1}{\sqrt{2m}}\left(\hat{p} + im\omega\hat{x} - \frac{i\hbar}{\hat{x}}\right). \tag{11.168}$$

This yields $\hat{\hat{H}}_0 = \hat{B}_0^\dagger\hat{B}_0 + \frac{3}{2}\hbar\omega$. The ground-state energy for this factorization is $\frac{3}{2}\hbar\omega$, which is the first excited state. As we stated earlier, this alternative factorization does not give us all of the energy eigenstates.

Now, to find the first auxiliary Hamiltonian, we compute

$$\hat{\hat{H}}_1 = \hat{\hat{H}}_0 + [\hat{B}_0, \hat{B}_0^\dagger], \tag{11.169}$$

which becomes

$$\hat{\hat{H}}_1 = \frac{\hat{p}^2}{2m} + \frac{\hbar^2}{m\hat{x}^2} + \tfrac{1}{2}m\omega^2\hat{x}^2 + \hbar\omega. \tag{11.170}$$

We can see, in this case, the first auxiliary Hamiltonian indeed has a different potential than the original Hamiltonian. But the presence of the term proportional to $\frac{1}{\hat{x}^2}$ does not change the form of the superpotential we need to use to perform the factorization. Hence, we can immediately work on finding the superpotential for the factorization of the first auxiliary Hamiltonian $k_1 W_1(k_1'\hat{x})$, using the form in Eq. (11.167) with $0 \to 1$ (and adding $\hbar\omega$). Equating coefficients tells us that $\alpha_1 = -2\frac{k_1'}{k_1}$, or $\frac{k_1'}{k_1}$ and $k_1 k_1' = \hbar m\omega$. We see we must choose $\alpha_1 = -2\frac{k_1'}{k_1}$, because

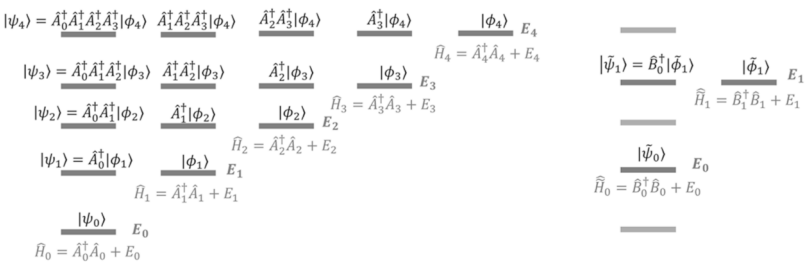

Fig. 11.7 On the left, we show a schematic with the energy levels (gray lines), the Hamiltonians (below the lines), and the states (above the lines) for the traditional factorization method applied to the simple harmonic oscillator. On the right, we show the same for the singular factorization (which includes only one-half of the states, shown in darker gray)

the other choice (of $\alpha_1 = \frac{k_1'}{k_1}$) will yield $\hat{A}_1 = \hat{A}_0^\dagger$, which is not allowed. Hence, we have $E_1 = \frac{7}{2}\hbar\omega$. The ladder operators become

$$\hat{B}_1 = \frac{1}{\sqrt{2m}}\left(\hat{p} - im\omega\hat{x} + \frac{2i\hbar}{\hat{x}}\right) \quad \text{and} \quad \hat{B}_1^\dagger = \frac{1}{\sqrt{2m}}\left(\hat{p} + im\omega\hat{x} - \frac{2i\hbar}{\hat{x}}\right). \tag{11.171}$$

You can see that the superpotential diverges at $x = 0$, which means the wavefunction must have a node at $x = 0$. You will work out the general case in Problem 11.7.2. We also show a schematic of how the energies and states look for the simple harmonic oscillator with the ordinary factorization (left) and the singular factorization (right) in Fig. 11.7.

So, why does this occur? It occurs because when we have an even potential, the wavefunctions alternate between even and odd, with the ground state being even. Hence, the alternate factorization is producing all of the odd solutions. The odd solutions have the interesting property of always have a node at $x = 0$. This divides the real axis into two pieces—one the positive real axis and the other the negative real axis. These domains are identical to each other (after reflection about the origin), and it turns that this means that we can always find an alternative (singular) factorization for these odd wavefunctions. We have not shown you why this always works, but we will find that it does. Furthermore, we will find whenever it does work, there is an equivalent central force problem that is solved by exactly the same factorization, as we will see in the next section. In other cases than the even potential, a singular factorization does not generically work, with one other exception. The energy eigenstates of the free particle in one dimension also have factorization chains because its nodes are evenly spaced. The video Schrödinger Factorization Chain https://tinyurl.com/5h3xnsze discusses both the simple harmonic oscillator and the particle in a box.

Section 11.6 Problems:

11.6.1 Transformation of Singular Ladder Operators

Consider the singular factorization lowering operator $\hat{B}_0$. We can write it as a similarity transformation of the original lowering operator $\hat{A}$ (we suppress the index, since all of those lowering operators are the same). Use the fact that $\hat{x}^\gamma \hat{p} \hat{x}^{-\gamma} = \frac{i\hbar\gamma}{\hat{x}}$. You can prove this from the fact that $[\ln \hat{x}, \hat{p}] = \frac{i\hbar}{\hat{x}}$ and the Hadamard lemma, or simply by moving the momentum operator to the right and correcting with a commutator of momentum with a power of position. The commutator can be found from the fact that $\hat{x} = e^{\ln \hat{x}}$ and $[\hat{x}, \hat{p}] = i\hbar$. (There are other ways to verify it as well.)

Find the exponent γ such that

$$\hat{B}_0 = \hat{x}^\gamma \hat{A}\, \hat{x}^{-\gamma}. \tag{11.172}$$

Rewrite the subsidiary condition for the singular factorization, given by $\hat{B}_0|\tilde{\psi}_0\rangle = 0$ in terms of $\hat{A}$ and verify that $|\tilde{\psi}_0\rangle \propto 2\sqrt{\frac{m\omega}{\hbar}}\hat{x}|0\rangle$, with $|0\rangle$ the ground state of the harmonic oscillator. This problem requires you to do some creative algebra.

Hint: Use the fact that $\hat{A}|0\rangle = 0$ to convert the operator acting on the state into a commutator acting on the state. You may find that writing $\hat{B}_0 = \hat{A} + \frac{1}{\sqrt{2m}}\frac{i\hbar}{\hat{x}}$ to be helpful too.

11.6.2 General Result for Singular Ladder Operators and Auxiliary Hamiltonians

Continue the process of determining the terms in the factorization chain and verify that the general lowering operator can be expressed as

$$\hat{B}_j = \frac{1}{\sqrt{2m}}\left(\hat{p} - im\omega\hat{x} + \frac{(j+1)i\hbar}{\hat{x}}\right) \tag{11.173}$$

and the auxiliary Hamiltonian is given by

$$\hat{\tilde{H}}_j = \frac{\hat{p}^2}{2m} + \frac{\hbar^2 j(j+1)}{2m\hat{r}^2} + \tfrac{1}{2}m\omega^2\hat{x}^2 + j\hbar\omega. \tag{11.174}$$

The auxiliary Hamiltonian ground-state energies are given by $E_j = \hbar\omega\left(2j + \frac{3}{2}\right)$.

You may want to work out one or two more steps before jumping to the general result. You should be able to prove it by induction. If you feel ready, you can prove

the induction step from the start, because we already established this result for $j = 0$ and $j = 1$ in the main text.

11.6.3 Energy Eigenstates of the Singular Factorization

In this problem, we ignore the normalization constants of the energy eigenstates.

Find the general exponent γ_j such that

$$\hat{B}_j = \hat{x}^{\gamma_j} \hat{A}\, \hat{x}^{-\gamma_j}. \tag{11.175}$$

Taking the Hermitian conjugate, we also have that

$$\hat{B}_j^\dagger = \hat{x}^{-\gamma_j} \hat{A}^\dagger\, \hat{x}^{\gamma_j}. \tag{11.176}$$

The state $|3\rangle \propto \left(\hat{A}^\dagger\right)^3 |0\rangle \propto H_3\left(\hat{x}\sqrt{\frac{m\omega}{\hbar}}\right)|0\rangle$ is the first excited state of the singular factorization chain, so it is also equal to $\hat{B}_0^\dagger|\tilde{\phi}_1\rangle$, where the state $|\tilde{\phi}_1\rangle$ satisfies $\hat{B}_1|\tilde{\phi}_1\rangle = 0$. We want to establish that these results are all true.

Show that $|\tilde{\phi}_1\rangle$ satisfies

$$|\tilde{\phi}_1\rangle \propto \hat{x}^2 \frac{m\omega}{\hbar}|0\rangle. \tag{11.177}$$

Then use this to show that

$$\hat{B}_0^\dagger|\tilde{\phi}_1\rangle \propto H_3\left(\hat{x}\sqrt{\frac{m\omega}{\hbar}}\right)|0\rangle. \tag{11.178}$$

You may find the identities $\hat{B}_1 = \hat{A} + \frac{1}{\sqrt{2m}}\frac{2i\hbar}{\hat{x}}$ and $\hat{B}_0^\dagger = \hat{A} + \frac{1}{\sqrt{2m}}(2im\omega\hat{x} - \frac{i\hbar}{\hat{x}})$ to be helpful as well. Recall that $H_3(z) = 8z^3 - 12z$.

This supports our claim that the singular factorization chain does produce all the odd Fock states.

If you are brave investigate the next case, for the state $|5\rangle$, and think about how one would show the result in general. In principle, this should be possible, but the algebra and the identities required may be complicated to work with.

11.7 The Isotropic Simple Harmonic Oscillator

We are now ready to apply the Schrödinger factorization method to our first central force problem—the isotropic simple harmonic oscillator—whose Hamiltonian is given by

$$\hat{H} = \frac{\hat{\vec{p}}^{\,2}}{2M} + \tfrac{1}{2}M\omega^2\hat{\vec{r}}^{\,2}. \tag{11.179}$$

We use the symbol M for the mass so as to not confuse it with an index m that we will use heavily in central force problems. Note that we already solved this problem in Sect. 7.13 in the Cartesian basis. We now revisit it in the case with spherical symmetry (the frequencies in each Cartesian direction are now identical), and we use the spherical symmetry explicitly in the solution. As before, we will be dropping the vector sign on the squared operators to make the notation simpler.

The strategy we use is the same as what we use with all central force problems we will solve. First, we employ separation of variables to rewrite the Hamiltonian in terms of the radial motion and the angular motion. Then, we act the Hamiltonian onto a tensor-product state appropriate for this separation of variables. Then, we seek the appropriate factorization of the radial energy eigenvalue equation. Next we form the factorization chain, with its auxiliary Hamiltonians and their ladder operators. From this, we form the energy eigenstates and determine all of the energies. Finally, we project the energy eigenstate onto the position or momentum eigenstates to construct the wavefunctions in position or momentum space. Note that this last step is not always needed in our approach, although it is critical in the more traditional approaches to quantum mechanics (because wavefunctions and energies are determined simultaneously in conventional approaches). We also use a shortcut method to find the energy eigenstates for some problems with extra symmetry (the isotropic oscillator and the Coulomb problem).

We start by looking at the maximal commuting set of operators for central force problems to determine the initial tensor-product state. We first use separation of variables to rewrite the Hamiltonian as

$$\hat{H} = \frac{\hat{p}_r^2}{2M} + \frac{\hat{L}^2}{2M\hat{r}^2} + \tfrac{1}{2}M\omega^2\hat{r}^2. \tag{11.180}$$

Since $\hat{L}_\alpha$ commutes with $\hat{r}$ and with $\hat{p}_r$, we immediately find that $\hat{H}$, $\hat{L}^2$, and $\hat{L}_z$ form a maximally commuting set of operators. This allows us to use separation of variables to factorize the wavefunction into a tensor product given by

$$|\psi\rangle = |\psi_r\rangle \otimes |l, m\rangle, \tag{11.181}$$

where $\hat{L}^2|l, m\rangle = \hbar^2 l(l+1)|l, m\rangle$ and $\hat{L}_z|l, m\rangle = \hbar m|l, m\rangle$. Substituting into the Schrödinger equation transforms $\hat{H}|\psi\rangle = E|\psi\rangle$ into

$$\hat{H}_l|\psi_r\rangle = \left(\frac{\hat{p}_r^2}{2M} + \frac{\hbar^2 l(l+1)}{2M\hat{r}^2} + \tfrac{1}{2}M\omega^2\hat{r}^2\right)|\psi_r\rangle = E_{nl}|\psi_r\rangle, \tag{11.182}$$

where we dropped the $|l, m\rangle$ states from the tensor product in our formulas to save space and focused solely on the radial part of the equation. In this equation, we used the fact that $|l, m\rangle$ is an eigenstate of $\hat{L}^2$, which produces for us an infinite

number of radial problems we need to solve, indexed by the total angular momentum eigenvalue l. Note that this energy eigenvalue equation is independent of m, so the energy eigenvalue is a function of the principal quantum number n (which is yet to be determined) and l, the total angular momentum.

Looking at the Hamiltonian for each definite l value, we see that the effective potential for the radial problem has an $\hat{r}^2$ term and a $\frac{1}{\hat{r}^2}$ term. This looks just like the form we obtained with the singular factorization of the one-dimensional simple harmonic oscillator from the previous section. Hence, this motivates an ansatz given by the following superpotential

$$k_{\bar{l}}^{(l)} W_{\bar{l}} \left(k_{\bar{l}}^{(l)\prime} \hat{r} \right) = k_{\bar{l}}^{(l)} \left(k_{\bar{l}}^{(l)\prime} \hat{r} + \frac{\alpha_{\bar{l}}}{k_{\bar{l}}^{(l)\prime} \hat{r}} \right), \tag{11.183}$$

where (l) denotes the angular momentum quantum number, and $\bar{l}$ is the index for the corresponding link in the factorization chain. It is important not to confuse the two. Because it is a little painful to work with so many constants, we define new constants to work with such that

$$\hbar k_{\bar{l}}^{(l)} W_{\bar{l}} \left(k_{\bar{l}}^{(l)\prime} \hat{r} \right) = \beta_{\bar{l}}^{(l)} \hat{r} + \frac{\hbar \alpha_{\bar{l}}^{(l)}}{\hat{r}}; \tag{11.184}$$

this will streamline the algebra we have to do next. This means the lowering operator takes the form

$$\hat{A}_{\bar{l}}^{(l)} = \frac{1}{\sqrt{2M}} \left(\hat{p}_r - \frac{i\hbar \alpha_{\bar{l}}^{(l)}}{\hat{r}} - i\beta_{\bar{l}}^{(l)} \hat{r} \right). \tag{11.185}$$

As you may recall, when we square the superpotential, we will obtain a quadratic term, a constant term, and an inverse quadratic term. When we take the commutator with $\hat{p}_r$, we obtain a constant term and an inverse quadratic term. Hence, we should have enough degrees of freedom to solve for the coefficients and perform the factorization. If the form of the auxiliary Hamiltonians remains the same (which we also anticipate from our work in the previous section), then we should be able to use this form to factorize all of the Hamiltonians. Hence, this problem is a shape-invariant potential problem that can be easily solved with the factorization method.

We start with the case of $l = 0$. To avoid excess notation, we remove the superscript (0) for simplicity, but in any case where there may be confusion, we will bring it back. Now, we form

$$\begin{aligned} \hat{H}_0 &= \hat{A}_0^\dagger \hat{A}_0 + E_0 \\ &= \frac{1}{2M} \left(\hat{p}_r^2 - i \left[\hat{p}_r, \frac{\hbar \alpha_0}{\hat{r}} + \beta_0 \hat{r} \right] + \frac{\hbar^2 \alpha_0^2}{\hat{r}^2} + 2\hbar \alpha_0 \beta_0 + \beta_0^2 \hat{r}^2 \right) + E_0 \end{aligned}$$

$$= \frac{\hat{p}_r^2}{2M} + \frac{\hbar^2(\alpha_0^2+\alpha_0)}{2M\hat{r}^2} + \frac{\beta_0^2}{2M}\hat{r}^2 + \frac{\hbar\beta_0(2\alpha_0-1)}{2M} + E_0. \tag{11.186}$$

We see that we must have $\alpha_0^2+\alpha_0=0$ and $\beta_0=\pm M\omega$, which yields four possible cases: $\alpha_0=0$, or -1 and $\beta_0=M\omega$, or $-M\omega$. Since the superpotential must be positive as $r\to\infty$, we choose $\beta_0=M\omega$. There is a new rule we need to determine the choice for the α_0 parameter. We look at the behavior of the wavefunction near $r=0$. The wavefunction is given by

$$\psi_r(r) = \langle r|\psi_r\rangle = \langle 0_r|e^{\frac{i}{\hbar}\left(\hat{p}_r+\frac{i\hbar}{\hat{r}}\right)}|\psi_r\rangle. \tag{11.187}$$

As $r\to 0$, we can consider just the diverging term of the superpotential, so the subsidiary condition says $\left(\hat{p}_r-\frac{i\hbar\alpha_0}{\hat{r}}\right)|\psi_r\rangle=0$. If $\alpha_0=-1$, then the state $|\psi_r\rangle$ is an eigenstate of the operator in the exponential with eigenvalue 0 as $r\to 0$, so the wavefunction approaches a constant. If, instead, we have $\alpha_0=0$, then the state $|\psi_r\rangle$ is an eigenstate of $\hat{p}_r$ with eigenvalue 0, and the wavefunction will diverge as $r\to 0$ (as can be seen by performing the power-series expansion of the exponential). So, when we have a diverging superpotential as $r\to 0$, we will want to choose the divergence to have a negative sign in the superpotential (or $\alpha_0<0$ in order to guarantee that the wavefunction will be finite at the origin). So, this leads us to choose $\alpha_0=-1$. Note that this is *the same choice* as what we used for the singular factorization of the simple harmonic oscillator in one-dimension. This is the relationship we referred to previously between a singular factorization and the central-force problems. We also find that $E_0=\frac{3}{2}\hbar\omega$, which agrees with the Cartesian solution. Note further that had we chosen $\alpha_0=0$, then we would have the factorization used for the one-dimensional simple harmonic oscillator and this would give the wrong ground-state energy.

Now, we compute in the product of the ladder operators in the opposite order to determine the first auxiliary Hamiltonian

$$\hat{H}_1^{\text{aux}} = \hat{H}_0 + [\hat{A}_0, \hat{A}_0^\dagger] = \frac{\hat{p}_r^2}{2M} + \frac{\hbar^2}{M\hat{r}^2} + \tfrac{1}{2}M\omega^2\hat{r}^2 + \hbar\omega. \tag{11.188}$$

We append a superscript "aux" in order to avoid confusion when we are solving these problems with different angular momentum; note that this means $\hat{H}_l=\hat{H}_0^{\text{aux}(l)}$ for the first Hamiltonian in the chain. We factorize this with $\alpha_1^2+\alpha_1=2$, whose solutions are -2 and 1. If we choose 1, then $\hat{A}_1=\hat{A}_0^\dagger$, which is not allowed, so we must choose $\alpha_1=-2$. As before, we have $\beta_1=M\omega$. This yields $E_1=\frac{7}{2}\hbar\omega$. Note that the first auxiliary Hamiltonian for $l=0$ is the same as the physical Hamiltonian we want to solve for $l=1$, except for an extra shift of the potential by the constant $\hbar\omega$. This fact will allow us to determine the solution of all of the angular momentum problems "at the same time" in a shortcut. But, before doing that, we want to discuss in more detail how the conventional solution works.

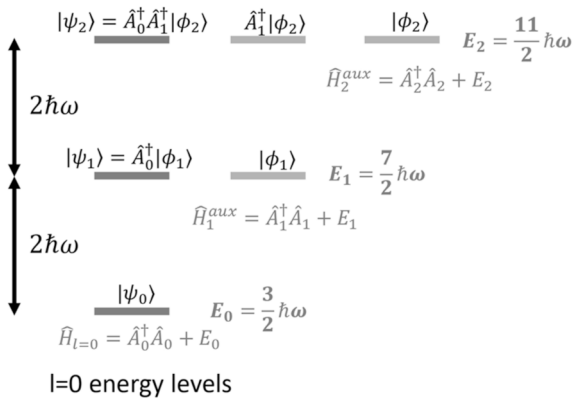

Fig. 11.8 The conventional factorization method as applied to $\hat{H}_0$, producing the energy levels for the $l = 0$ case. We show the physical Hamiltonian energy eigenvalues and states along with the auxiliary Hamiltonian energy levels and states (lighter shade of gray)

After continuing to repeat the conventional factorization chain process, we find, for the $l = 0$ Hamiltonian, we have

$$\begin{aligned}
\hat{A}_{\bar{l}}^{(0)} &= \frac{1}{\sqrt{2M}}\left(\hat{p}_r + \frac{i\hbar(\bar{l}+1)}{\hat{r}} - iM\omega\hat{r}\right), \\
\hat{H}_{\bar{l}}^{\text{aux}} &= \frac{\hat{p}_r^2}{2M} + \frac{\hbar^2\bar{l}(\bar{l}+1)}{2M\hat{r}^2} + \tfrac{1}{2}M\omega^2\hat{r}^2 + \bar{l}\hbar\omega, \quad \text{and} \\
E_{\bar{l}}^{(0)} &= \left(2\bar{l} + \tfrac{3}{2}\right)\hbar\omega.
\end{aligned} \tag{11.189}$$

We also have $\hat{A}_{\bar{l}}^{(0)}|\phi_{\bar{l}}^{(0)}\rangle = 0$ and $|\psi_{\bar{l}}^{(0)}\rangle = \hat{A}_0^{(0)\dagger}\hat{A}_1^{(0)\dagger}\cdots\hat{A}_{\bar{l}-1}^{(0)\dagger}|\phi_{\bar{l}}^{(0)}\rangle$ (unnormalized). This is the conventional way to solve the problem, and you should verify each of the general steps, whose results are shown here. Note that the factorization method constructs the infinite number of energy eigenstates for a fixed l by constructing the factorization chain for each l and all of the associated auxiliary Hamiltonians (for each l). The details for extending this result to arbitrary angular momentum l are discussed in Problem 11.7.3. A figure of the energy levels for the different l values is given in Fig. 11.8.

For the shortcut method, we need to derive a new intertwining relation, which does not relate auxiliary Hamiltonians to each other but, instead, relates the physical Hamiltonians we are trying to solve (the $\hat{H}_l$'s) to each other. It is a small change in the intertwining relation, because the auxiliary and physical Hamiltonians are related by a constant shift in the potential energy. The conventional intertwining relation is

$$\hat{H}_{\bar{l}}^{\text{aux}(l)}\left(\hat{A}_{\bar{l}}^{(l)}\right)^\dagger = \left(\hat{A}_{\bar{l}}^{(l)}\right)^\dagger \hat{H}_{\bar{l}+1}^{\text{aux}(l)}, \tag{11.190}$$

which can be rewritten as

$$\hat{H}_l \hat{A}_l^\dagger = \hat{A}_l^\dagger \left(\hat{H}_{l+1} + \hbar\omega\right) \text{ and } \hat{A}_l = \frac{1}{\sqrt{2M}} \left(\hat{p}_r + \frac{i\hbar(l+1)}{\hat{r}} - M\omega\hat{r}\right). \tag{11.191}$$

Here, we use the notation $\hat{A}_l^\dagger = \left(\hat{A}_{l-l'}^{(l')}\right)^\dagger$, since the raising operator $\left(\hat{A}_{\bar{l}}^{(l)}\right)^\dagger$ depends only on $l + \bar{l}$. In fact, for the shortcut method, we do not need the more complicated notation, as we solve each physical Hamiltonian for the different l values "at once." This is another nice simplification for the shortcut method.

Using the form for the $\hat{A}_l$, we can immediately determine the factorization for $\hat{H}_l$. It is

$$\hat{H}_l = \hat{A}_l^\dagger \hat{A}_l + E_l, \tag{11.192}$$

with $E_l = \left(l + \frac{3}{2}\right)\hbar\omega$. This is the ground-state energy for the state with angular momentum l, which satisfies $\hat{A}_l |\psi_{n=l,l}\rangle = 0$. Here, we introduce the principal quantum number, which determines the energy eigenvalue of the given state via $E_n = \left(n + \frac{3}{2}\right)\hbar\omega$. We can easily see that $n = 0, 1, 2, \cdots$ is a counting number. Our next step is to compute the other states than the ground states for each l. They are formed as follows:

$$|\psi_{n=m+l,l-m}\rangle \propto \hat{A}_{l-m}^\dagger \hat{A}_{l-m+1}^\dagger \cdots \hat{A}_{l-1}^\dagger |\phi_{n=l,l}\rangle, \tag{11.193}$$

where we have not included the normalization factor, yet. Note that the m index used here should not be confused with the angular momentum quantum number for the z-component, which does not enter into any results for the isotropic harmonic oscillator. By acting $\hat{H}_{l-m}$ onto this state, and using the intertwining relation, we find that $E_{n=m+l,l-m} = \left(m + l + \frac{3}{2}\right)\hbar\omega$. This state has principal quantum number $n = m + l$ and angular momentum $l - m$. Because the $\hat{A}_l$ operators are defined down to $l = 0$, the chain of raising operators cannot include more than $m = l$ terms. One can see each raising operator increases n by 1 and decreases l by 1; hence, the states are connected as shown by the diagonal arrows. This leads to an energy spacing for the fixed l columns of $2\hbar\omega$. A figure showing the energy eigenvalues and the states is given in Fig. 11.9. That is it! We have found the energy eigenvalues and the energy eigenstates using the short-cut method. It is much more efficient than the longer method. The reason why we have such a short-cut method arises from the fact that this problem has extra symmetry in it. The d-dimensional isotropic simple harmonic oscillator has the symmetry of SU(d), the special unitary matrix group of $d \times d$ matrices. We will discuss why this symmetry arises in Problem 11.7.9.

We can verify that we have consistent results by doing some counting from the energy levels we found from the Cartesian solution, where $E_{n_x,n_y,n_z} = \left(n_x + n_y + n_z + \frac{3}{2}\right)\hbar\omega$. All of these energies are present in Fig. 11.9, but we do not so easily see the degeneracy across l, nor that states of fixed l have an energy

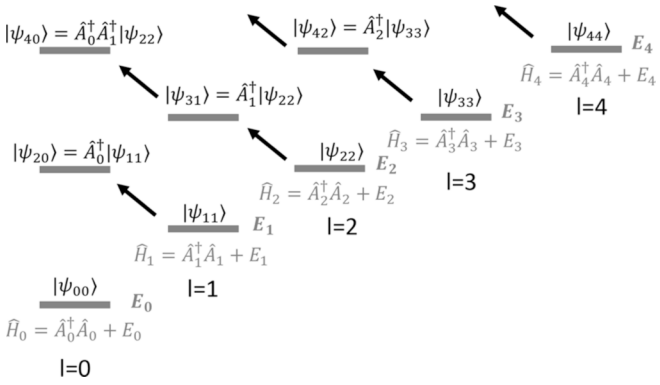

Fig. 11.9 Energy spectra of the isotropic simple harmonic oscillator as determined by the shortcut method. Lines are drawn at the energy eigenvalues for the allowed energies as a function of the angular momentum. Starting from the lowest-energy state for a given angular momentum $|\psi_{n=l,l}\rangle$, the arrows show how the raising operators create each $|\psi_{n=l+m,l-m}\rangle$ state for $0 \leq m \leq l$. Each diagonal chain ends when $m = l$, and we reach the zero-angular-momentum energy eigenstates. In the shortcut method, there are no auxiliary Hamiltonians

increment of $2\hbar\omega$. To see how this occurs, we do some counting. Consider the states with energy $E = \frac{7}{2}\hbar\omega$. Using the Cartesian energy formula, we see that these states are formed by the following six choices for (n_x, n_y, n_z): (2,0,0), (0,2,0), (0,0,2), (1,1,0), (1,0,1), and (0,1,1). So the degeneracy is sixfold. Now, look at the angular momentum. We see that $E = \frac{7}{2}\hbar\omega$ has an $l = 2$ state and a $l = 0$ state both degenerate. Using the fact that an angular momentum multiplet has $2l + 1$ different m states, we find that the degeneracy is $(2 \times 2 + 1) + (2 \times 0 + 1) = 6$, which agrees. This counting argument can be employed as a consistency check for all rows of Fig. 11.9. We illustrate this further in Problem 11.7.2.

We next calculate the wavefunctions. We will be doing this from our generalized Rodrigues formula approach (Noonan et al., 2024). This requires us to do the following: (i) find a similarity transformation for each raising operator and convert the product of the chain of operators into a similarity transformation of something related to powers of momentum; (ii) find the state annihilated by $\hat{p}_r$; (iii) change the similarity transformed expression into nested commutators involving momentum; and (iv) prove that the resulting nested commutators satisfy the recurrence relations of associated Laguerre polynomials. This approach has slight modifications for different Hamiltonians—we now provide the details for the isotropic oscillator in three dimensions.

We start with the similarity transformation of the raising operator. We use a Hadamard lemma to get the term linear in $\hat{r}$. This is similar to what we did earlier with the one-dimensional harmonic oscillator. Using $e^{\alpha\hat{r}^2}\hat{p}_r e^{-\alpha\hat{r}^2} = \hat{p}_r + 2i\hbar\alpha\hat{r}$, we can write

$$e^{\frac{M\omega}{2\hbar}\hat{r}^2}\hat{p}_r e^{-\frac{M\omega}{2\hbar}\hat{r}^2} = \hat{p}_r + iM\omega\hat{r}, \tag{11.194}$$

by using $\alpha = \frac{M\omega}{2\hbar}$. We need to also use a different similarity transformation to obtain the $-\frac{i\hbar(l+1)}{\hat{r}}$ term. Recall that $[\hat{r}^n, \hat{p}_r] = i\hbar n\hat{r}^{n-1}$, so that

$$[\hat{r}^{-(l+1)}, \hat{p}_r] = \hat{r}^{-(l+1)}\hat{p}_r - \hat{p}_r\hat{r}^{-(l+1)} = -i\hbar(l+1)\hat{r}^{-(l+2)}$$
$$\implies \quad \hat{r}^{-(l+1)}\hat{p}_r\hat{r}^{(l+1)} = \hat{p}_r - \frac{i\hbar(l+1)}{\hat{r}}, \tag{11.195}$$

after multiplying both sides of the equation by $\hat{r}^{(l+1)}$ from the right and doing some rearranging. Combining these two transformations, which commute with each other, we have that

$$\hat{A}_l^\dagger = \frac{1}{\sqrt{2M}}\hat{r}^{-(l+1)}e^{\frac{M\omega}{2\hbar}\hat{r}^2}\hat{p}_r e^{-\frac{M\omega}{2\hbar}\hat{r}^2}\hat{r}^{(l+1)}. \tag{11.196}$$

Now, we will use the relation

$$|\psi_{n=l+m,l-m}\rangle = C_{l+m,l-m}\hat{A}_{l-m}^\dagger\hat{A}_{l-m+1}^\dagger\cdots\hat{A}_{l-1}^\dagger|\psi_{n=l,l}\rangle, \tag{11.197}$$

with

$$\frac{1}{C_{l+m,l-m}^2} = \langle\psi_{l,l}|\hat{A}_{l-1}\cdots\hat{A}_{l-m}\hat{A}_{l-m}^\dagger\cdots\hat{A}_{l-1}^\dagger|\psi_{l,l}\rangle. \tag{11.198}$$

We use $\hat{A}_l\,\hat{A}_l^\dagger = \hat{H}_l - E_l + [\hat{A}_l, \hat{A}^\dagger]$. Evaluating the commutator gives us $\hat{A}_l\,\hat{A}_l^\dagger = \hat{H}_{l+1} - \hbar\omega\left(l - 1 - \frac{3}{2}\right)$. Then we use intertwining to move the operator to the right to act on the state. We find that

$$C_{l+m,l-m} = \frac{1}{\sqrt{2m\hbar\omega\cdot 2(m-1)\hbar\omega\cdots 2(1)\hbar\omega}} = \frac{1}{\sqrt{(\hbar\omega)^m 2^m m!}}. \tag{11.199}$$

Plugging in the similarity-transformed form of the raising operators yields

$$|\psi_{l+m,l-m}\rangle = \frac{1}{\sqrt{(4\hbar M\omega)^m m!}}\hat{r}^{-(l-m+1)}e^{\frac{M\omega}{2\hbar}\hat{r}^2}\left(\hat{p}_r\frac{1}{\hat{r}}\right)^m e^{-\frac{M\omega}{2\hbar}\hat{r}^2}\hat{r}^{(l+1)}|\psi_{l,l}\rangle. \tag{11.200}$$

Our next step is to find the state that $\hat{p}_r$ annihilates. We find this by performing the Hermitian conjugate of the similarity transformation onto the subsidiary condition given by $\hat{A}_l|\psi_{l,l}\rangle = 0$, which can be rewritten as $\hat{r}^{(l+1)}e^{-\frac{M\omega}{2\hbar}\hat{r}^2}\hat{p}_r\hat{r}^{-(l+1)}e^{\frac{M\omega}{2\hbar}\hat{r}^2}|\psi_{l,l}\rangle = 0$, or

$$\hat{p}_r\hat{r}^{-(l+1)}e^{\frac{M\omega}{2\hbar}\hat{r}^2}|\psi_{l,l}\rangle = 0, \tag{11.201}$$

We use this result to change the expression for the energy eigenstates into nested commutators, which is the operator analog of the Rodrigues formula for the orthogonal polynomial; the first step is to multiply by 1 to create the state annihilated by $\hat{p}_r$ on the right. Converting the $\hat{p}_r$ operators into nested commutators, then yields

$$|\psi_{l+m,l-m}\rangle = \frac{1}{\sqrt{(4\hbar M\omega)^m m!}}\,\hat{r}^{-(l-m+1)}e^{\frac{M\omega}{2\hbar}\hat{r}^2} \times \left[\hat{p}_r, \tfrac{1}{\hat{r}}\left[\hat{p}_r, \tfrac{1}{\hat{r}}\left[\hat{p}_r, \cdots, \tfrac{1}{\hat{r}}\left[\hat{p}_r, \hat{r}^{2l+1}e^{-\frac{M\omega}{\hbar}\hat{r}^2}\right]\cdots\right]\right]\right]_m \hat{r}^{-(l+1)}e^{\frac{M\omega}{2\hbar}\hat{r}^2}|\psi_{l,l}\rangle. \tag{11.202}$$

Note how the $\frac{1}{\hat{r}}$ from each product of operators gets multiplied by the commutator to the right when we form the nested commutators. The derivation of this result starts from the right-most factor of $\hat{p}_r\frac{1}{\hat{r}}$ and uses add-zero to convert it to a commutator (because $\hat{p}_r$ annihilates the state on the right). This is then repeated for each momentum factor until we obtain the final result aforementioned. We next move all of the $\frac{1}{\hat{r}}$ factors into the commutators (and one factor from the $\hat{r}$ factors on the left) by employing the Leibniz identity and then we also move the operators on the far right to the far left, which is possible because the nested commutator is a polynomial in $\hat{r}$ multiplied by a power of $\hat{r}$ and a Gaussian. This becomes

$$|\psi_{l+m,l-m}\rangle = \frac{1}{\sqrt{(4\hbar M\omega)^m m!}}\,\hat{r}^{-(2l+1-m)}e^{\frac{M\omega}{\hbar}\hat{r}^2} \times \left[\tfrac{1}{\hat{r}}\hat{p}_r, \left[\tfrac{1}{\hat{r}}\hat{p}_r, \cdots, \left[\tfrac{1}{\hat{r}}\hat{p}_r, \hat{r}^{2l+1}e^{-\frac{M\omega}{\hbar}\hat{r}^2}\right]\cdots\right]\right]_m |\psi_{l,l}\rangle. \tag{11.203}$$

To be concrete, we examine the two simplest cases of this formula here. For $m = 0$, the result is simple and given by $|\psi_{l,l}\rangle$. For $m = 1$, we have

$$\begin{aligned}|\psi_{l+1,l-1}\rangle &= \frac{1}{\sqrt{4\hbar M\omega}}\hat{r}^{-(2l+1)}e^{\frac{M\omega}{\hbar}\hat{r}^2}\left[\hat{p}_r, \hat{r}^{2l+1}e^{-\frac{M\omega}{\hbar}\hat{r}^2}\right]|\psi_{l,l}\rangle \\ &= \frac{1}{\sqrt{4\hbar M\omega}}\left(-\frac{2i\hbar}{\hat{r}}\right)\left[\left(l+\tfrac{1}{2}\right) - \tfrac{M\omega}{\hbar}\hat{r}^2\right]|\psi_{l,l}\rangle.\end{aligned} \tag{11.204}$$

In the first line, we used the fact that the commutator is a function of $\hat{r}$, so it commutes with other functions of $\hat{r}$, allowing us to move the rightmost factors to the left. This is clearly a polynomial, as we claimed it had to be. For the general case, and using 20–20 hindsight, we define the result of the nested polynomials in terms of two indices—the first is m, which is the degree of the polynomial (in $\hat{r}^2$), and the second is one half the exponent of $\hat{r}$ in the innermost commutator minus the number of commutators, or $l - m + \frac{1}{2}$. That this is the proper choice will become clear soon. Hence, we have

Table 11.1 First few associated Laguerre polynomials that appear in the wavefunctions for the isotropic simple harmonic oscillator

m	$L_m^{\left(l+\frac{1}{2}\right)}(x)$
0	1
1	$l+\frac{3}{2}-x$
2	$\frac{1}{2}\left(l+\frac{5}{2}\right)\left(l+\frac{3}{2}\right)-\left(l+\frac{3}{2}\right)x+\frac{x^2}{2}$
3	$\frac{1}{6}\left(l+\frac{7}{2}\right)\left(l+\frac{5}{2}\right)\left(l+\frac{3}{2}\right)-\frac{1}{2}\left(l+\frac{7}{2}\right)\left(l+\frac{5}{2}\right)x+\frac{1}{2}\left(l+\frac{7}{2}\right)x^2-\frac{x^3}{6}$

$$\left[\tfrac{1}{\hat{r}}\hat{p}_r,\left[\tfrac{1}{\hat{r}}\hat{p}_r,\left[\tfrac{1}{\hat{r}}\hat{p}_r,\cdots,\left[\tfrac{1}{\hat{r}}\hat{p}_r,\hat{r}^{2l+1}e^{-\frac{M\omega}{\hbar}\hat{r}^2}\right]\cdots\right]\right]\right]_m$$
$$=m!\left(-\frac{2i\hbar}{\hat{r}^2}\right)^m\hat{r}^{2l+1}e^{-\frac{M\omega}{\hbar}\hat{r}^2}L_m^{\left(l-m+\frac{1}{2}\right)}\left(\frac{M\omega}{\hbar}\hat{r}^2\right). \tag{11.205}$$

It turns out that this polynomial is the associated Laguerre polynomial, which is defined to be

$$L_n^{(\alpha)}(x)=\sum_{m=0}^{n}(-1)^m\binom{n+\alpha}{n-m}\frac{x^m}{m!}. \tag{11.206}$$

For example, we calculated aforementioned with $m=1$, we have $n=1$ and $\alpha=l-\frac{1}{2}$, because $L_1^{(\alpha)}=\alpha+1-x$. Note that the $m=0$ result can be written as $|\psi_{l,l}\rangle=L_0^{\left(l+\frac{1}{2}\right)}\left(\frac{M\omega}{\hbar}\hat{r}^2\right)|\psi_{l,l}\rangle$ too, since $L_0^{(\alpha)}=1$. These two results are the first two results of the recurrence relation that we derive next. But first, we summarize the associated Laguerre polynomials of the form $L_m^{\left(l+\frac{1}{2}\right)}(x)$ in Table 11.1—it is precisely polynomials of this form that enter for the solutions of the isotropic simple harmonic oscillator.

This last step in our derivations for $m=0$ and $m=1$ was our "paying homage to dead French or German mathematicians." Here, the lucky mathematician is Edmond Laguerre, a French mathematician from the nineteenth century. In this case, I want to emphasize that this exercise is actually quite worthwhile. Trying to work out the results one wavefunction at a time by taking multiple nested commutators is a quite challenging prospect. If we can instead find a way to determine all of the results in a closed form, it is a real simplification of this procedure. So, we need to verify our claimed relation via a proof by induction. We have already established this result for $m=0$ and $m=1$ (check it if unsure and recall the result is for the nested commutators only, not for the full set of operators). Verifying it for the general case involves a bit of algebra, in our French-cooking style. We use a proof by induction, assuming that the general result in Eq. (11.205) holds up to m.

We take the $m+1$-fold nested commutator

$$\left[\tfrac{1}{\hat{r}}\hat{p}_r, \left[\tfrac{1}{\hat{r}}\hat{p}_r, \left[\tfrac{1}{\hat{r}}\hat{p}_r, \cdots, \left[\tfrac{1}{\hat{r}}\hat{p}_r, \hat{r}^{2l+1}e^{-\frac{M\omega}{\hbar}\hat{r}^2}\right]\cdots\right]\right]\right]_{m+1}, \tag{11.207}$$

and evaluate the innermost commutator, which gives

$$\left[\tfrac{1}{\hat{r}}\hat{p}_r, \hat{r}^{2l+1}e^{-\frac{M\omega}{\hbar}\hat{r}^2}\right] = -i\hbar\left((2l+1)\hat{r}^{2l-1}e^{-\frac{M\omega}{\hbar}\hat{r}^2} - \frac{2M\omega}{\hbar}\hat{r}^{2l+1}e^{-\frac{M\omega}{\hbar}\hat{r}^2}\right). \tag{11.208}$$

Substituting this back into the nested commutator yields

$$\begin{aligned}
&-i\hbar(2l+1)\left[\tfrac{1}{\hat{r}}\hat{p}_r, \left[\tfrac{1}{\hat{r}}\hat{p}_r, \cdots, \left[\tfrac{1}{\hat{r}}\hat{p}_r, \hat{r}^{2l-1}e^{-\frac{M\omega}{\hbar}\hat{r}^2}\right]\cdots\right]\right]_m \\
&+\frac{2i\hbar M\omega}{\hbar}\left[\tfrac{1}{\hat{r}}\hat{p}_r, \left[\tfrac{1}{\hat{r}}\hat{p}_r, \cdots, \left[\tfrac{1}{\hat{r}}\hat{p}_r, \hat{r}^{2l+1}e^{-\frac{M\omega}{\hbar}\hat{r}^2}\right]\cdots\right]\right]_m.
\end{aligned} \tag{11.209}$$

Now we use our definition that relates the nested commutator to our associated Laguerre polynomial in Eq. (11.205). This gives

$$\begin{aligned}
&-i\hbar(2l+1)m!\left(-\frac{2i\hbar}{\hat{r}^2}\right)^m \hat{r}^{2l-1}e^{-\frac{M\omega}{\hbar}\hat{r}^2}L_m^{\left(l-m-\frac{1}{2}\right)}\left(\frac{M\omega}{\hbar}\hat{r}^2\right) \\
&+\frac{2i\hbar M\omega}{\hbar}m!\left(-\frac{2i\hbar}{\hat{r}^2}\right)^m \hat{r}^{2l+1}e^{-\frac{M\omega}{\hbar}\hat{r}^2}L_m^{\left(l-m+\frac{1}{2}\right)}\left(\frac{M\omega}{\hbar}\hat{r}^2\right) \\
&= 2i\hbar m!\left(-\frac{2i\hbar}{\hat{r}^2}\right)^m \hat{r}^{2l-1}e^{-\frac{M\omega}{\hbar}\hat{r}^2} \\
&\times\left(\frac{M\omega}{\hbar}\hat{r}^2L_m^{\left(l-m+\frac{1}{2}\right)}\left(\frac{M\omega}{\hbar}\hat{r}^2\right) - (l+\tfrac{1}{2})L_m^{\left(l-m-\frac{1}{2}\right)}\left(\frac{M\omega}{\hbar}\hat{r}^2\right)\right).
\end{aligned} \tag{11.210}$$

But the associated Laguerre polynomials satisfy the following recurrence relation

$$\begin{aligned}
&\frac{M\omega}{\hbar}\hat{r}^2L_m^{\left(l-m+\frac{1}{2}\right)}\left(\frac{M\omega}{\hbar}\hat{r}^2\right) - (l+\tfrac{1}{2})L_m^{\left(l-m-\frac{1}{2}\right)}\left(\frac{M\omega}{\hbar}\hat{r}^2\right) \\
&= -(m+1)L_{m+1}^{\left(l-m-\frac{1}{2}\right)}\left(\frac{M\omega}{\hbar}\hat{r}^2\right),
\end{aligned} \tag{11.211}$$

which you will prove in Problem 11.7.4. Now, if we substitute this into our expression, we find the $(m+1)$-fold nested commutator in Eq. (11.207) becomes

$$(m+1)!\left(-\frac{2i\hbar}{\hat{r}^2}\right)^{m+1} \hat{r}^{2l+1}e^{-\frac{M\omega}{\hbar}\hat{r}^2}L_{m+1}^{\left(l-m-\frac{1}{2}\right)}\left(\frac{M\omega}{\hbar}\hat{r}^2\right), \tag{11.212}$$

which is exactly what it needs to be to correspond to the $m+1$ result in Eq. (11.205). This completes the proof by induction.

Using this identity for the product of operators acting on the ground state for angular momentum l, then gives us the wavefunction $\langle r|\psi_{l+m,l-m}\rangle$ via

$$\psi_{l+m,l-m}(r) = \sqrt{\left(\frac{\hbar}{M\omega}\right)^m m!}\,(-i)^m r^{-m} L_m^{\left(l-m+\frac{1}{2}\right)}\left(\frac{M\omega}{\hbar}r^2\right)\langle r|\psi_{l,l}\rangle. \tag{11.213}$$

We just need to determine the ground-state wavefunction. This is done using our translation operator. The subsidiary condition says that

$$\hat{p}_r|\psi_{l,l}\rangle = \left(-\frac{i\hbar(l+1)}{\hat{r}} + iM\omega\hat{r}\right)|\psi_{l,l}\rangle. \tag{11.214}$$

Now, bring $-\frac{i\hbar}{\hat{r}}$ to the left-hand side, to find that

$$\left(\hat{p}_r + \frac{i\hbar}{\hat{r}}\right)|\psi_{l,l}\rangle = \left(-\frac{i\hbar l}{\hat{r}} + iM\omega\hat{r}\right)|\psi_{l,l}\rangle. \tag{11.215}$$

We can use this result to find a state that is annihilated by the operator on the left-hand side:

$$\begin{aligned}\left(\hat{p}_r + \frac{i\hbar}{\hat{r}}\right)\hat{r}^{-l}e^{\frac{M\omega}{2\hbar}\hat{r}^2}|\psi_{l,l}\rangle &= \hat{r}^{-l}e^{\frac{M\omega}{\hbar}\hat{r}^2}\left(\hat{p}_r + \frac{i\hbar}{\hat{r}}\right)|\psi_{l,l}\rangle \\ &\quad + \left[\hat{p}_r + \frac{i\hbar}{\hat{r}}, \hat{r}^{-l}e^{\frac{M\omega}{2\hbar}\hat{r}^2}\right]|\psi_{l,l}\rangle \\ &= \hat{r}^{-l}e^{\frac{M\omega}{2\hbar}\hat{r}^2}\left(\underbrace{-\frac{i\hbar l}{\hat{r}} + iM\omega\hat{r}}_{\text{subsidiary cond.}} + \underbrace{\frac{i\hbar}{\hat{r}} - iM\omega\hat{r}}_{\text{commutator}}\right)|\psi_{l,l}\rangle \\ &= 0.\end{aligned} \tag{11.216}$$

This allows us to compute the wavefunction from the spherical translation operator as follows:

$$\begin{aligned}\psi_{l,l}(r) = \langle r|\psi_{l,l}\rangle &= r^l e^{-\frac{M\omega}{2\hbar}r^2}\langle r|\hat{r}^{-l}e^{\frac{M\omega}{2\hbar}\hat{r}^2}|\psi_{l,l}\rangle \\ &= r^l e^{-\frac{M\omega}{2\hbar}r^2}\langle 0_r|e^{\frac{i}{\hbar}r\left(\hat{p}_r+\frac{i\hbar}{\hat{r}}\right)}\hat{r}^{-l}e^{\frac{M\omega}{2\hbar}\hat{r}^2}|\psi_{l,l}\rangle \\ &= r^l e^{-\frac{M\omega}{2\hbar}r^2}\langle 0_r|\tfrac{1}{\hat{r}^l}|\psi_{l,l}\rangle.\end{aligned} \tag{11.217}$$

In the first line, we multiplied by 1, using the fact that $\langle r|\hat{r} = \langle r|r$. The second line re-expresses the radial bra in terms of the radial translation operator. Finally, we used the fact that $\hat{p}_r + \frac{i\hbar}{\hat{r}}$ annihilates the state to the right, so the exponential can be replaced by 1 (we also evaluated the Gaussian operator on the bra to the left replacing it by 1 as well). We still have a normalization constant $\langle 0_r|\frac{1}{\hat{r}^l}|\psi_{l,l}\rangle$ to determine. It may seem like this expectation value diverges, but because the ground state behaves like r^l as $r \to 0$, we need to divide by $\frac{1}{r^l}$ to have a constant term result after evaluating in the limit as $r \to 0$. As before, this requires calculus, so we will just tell you the answer. It is

$$\langle 0_r|\frac{1}{\hat{r}^l}|\psi_{l,l}\rangle = \frac{1}{\pi^{\frac{1}{4}}}\sqrt{\frac{2^{l+2}}{(2l+1)!!}}\left(\frac{M\omega}{\hbar}\right)^{\frac{3}{4}}\left(\frac{M\omega}{\hbar}\right)^{\frac{l}{2}}. \tag{11.218}$$

Putting this all together gives us

$$\begin{aligned}\psi_{l+m,l-m}(r) = &\frac{(-i)^m}{\pi^{\frac{1}{4}}}\sqrt{\frac{2^{l+2}m!}{(2l+1)!!}}\left(\frac{M\omega}{\hbar}\right)^{\frac{3}{4}}\left(\frac{M\omega}{\hbar}r^2\right)^{\frac{l-m}{2}} \\ &\times L_m^{\left(l-m+\frac{1}{2}\right)}\left(\frac{M\omega}{\hbar}r^2\right)e^{-\frac{M\omega}{2\hbar}r^2}.\end{aligned} \tag{11.219}$$

To put this into standard form, we drop the global phase, and we replace $l+m \to n$ and $l-m \to l$, to write the result in terms of the principal quantum number and the angular momentum quantum number; recall that $n-l = 2m$ is always even. This gives the final result for the wavefunction:

$$\begin{aligned}\psi_{n,l,m}(r,\theta,\phi) = &\frac{1}{\pi^{\frac{1}{4}}}\sqrt{\frac{2^{\frac{n+l}{2}+2}\left(\frac{n-l}{2}\right)!}{(n+l+1)!!}}\left(\frac{M\omega}{\hbar}\right)^{\frac{3}{4}}\left(\frac{M\omega}{\hbar}r^2\right)^{\frac{l}{2}} \\ &\times L_{\frac{n-l}{2}}^{\left(l+\frac{1}{2}\right)}\left(\frac{M\omega}{\hbar}r^2\right)e^{-\frac{M\omega}{2\hbar}r^2}Y_{lm}(\theta,\phi),\end{aligned} \tag{11.220}$$

with $E_{n,l} = \left(n+\frac{3}{2}\right)\hbar\omega$. That was a lot of work! The final result looks intimidating, but it is a constant, multiplied by a power of r times a Gaussian, a polynomial in r, and the spherical harmonic. So, it really is not so bad.

The solution of the wavefunction in position space and in momentum space is quite similar, because the Hamiltonian is symmetric under an interchange of position and momentum. But to concretely work out the results, we need to modify how we work on the problem. We will work with the magnitude of momentum $\hat{p} = \sqrt{\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2}$ (not the radial momentum) and the momential radius (analog of the radial momentum) given by

$$\hat{r}_p = \tfrac{1}{2}\left(\frac{\hat{\vec{p}}}{\hat{p}}\cdot\hat{\vec{r}} + \hat{\vec{r}}\cdot\frac{\hat{\vec{p}}}{\hat{p}}\right) = \frac{1}{\hat{p}}\left(\hat{p}_x\hat{r}_x + \hat{p}_y\hat{r}_y + \hat{p}_z\hat{r}_z\right) + \frac{i\hbar}{\hat{p}}. \tag{11.221}$$

Next, we write $\hat{r}^2 = \hat{r}_p^2 + \frac{\hat{L}^2}{\hat{p}^2}$. The Hamiltonian then becomes

$$\hat{H} = \tfrac{1}{2}M\omega^2\hat{r}_p^2 + \frac{1}{2}\frac{M\omega^2\hat{L}^2}{\hat{p}^2} + \frac{\hat{p}^2}{2M}, \tag{11.222}$$

and one can see that aside from the rearrangement of some constants, this has the same form as the problem did in position space. So, to solve it, we write a tensor product again $|\tilde{\phi}\rangle = |\tilde{\phi}_p\rangle \otimes |l, m\rangle$ and the momential Hamiltonian for each angular momentum l takes the form

$$\hat{H}_l' = \tfrac{1}{2}M\omega^2\hat{r}_p^2 + \tfrac{1}{2}\frac{M\omega^2\hbar^2 l(l+1)}{\hat{p}^2} + \frac{\hat{p}^2}{2M}. \tag{11.223}$$

This can then be factorized to

$$\hat{H}_l' = \hat{B}_l^\dagger\hat{B}_l + E_l', \tag{11.224}$$

with

$$\hat{B}_l = \sqrt{\frac{M}{2}}\omega\left(\hat{r}_p + \frac{i\hbar\alpha_l'}{\hat{p}} - i\beta_l'\hat{p}\right). \tag{11.225}$$

As before, one selects the coefficients to produce $\hat{H}_l'$, choosing the constants so that the superpotential is positive as $p \to \infty$ and so on. Next, one forms the intertwining relation and uses a shortcut method to find all of the eigenstates. The steps are sketched in the problems. Nevertheless, you should be able to see that the momentum-space wavefunctions will have the same form, with a Laguerre polynomial in $\frac{p^2}{\hbar M\omega}$ multiplied by a power of p and an exponential $e^{-\frac{p^2}{2\hbar M\omega}}$. More details are discussed in the problems. Note that in order to have a separation of variables, we had to express our Hamiltonian as a function of $\hat{r}_p$ and $\hat{p}$ only. This creates a challenge for solving other problems that cannot be separated into this form when determining the momentum-space wavefunction. We discuss this in more detail when we work on the Coulomb problem in momentum space.

We now return to the discussion about the hidden SU(3) symmetry of the isotropic harmonic oscillator in three dimensions. The degeneracy of the energy eigenvalues is no accident. It is required by the symmetry of the problem (as are most degeneracies). Unfortunately, it is not so simple to describe this symmetry in its full glory. Instead, we will discuss some aspects of it to give you a taste of its full behavior. This will provide you with enough background for further study in the future. Recall that a two-dimensional oscillator can describe the SU(2) symmetry,

Table 11.2 First few radial wavefunctions of the isotropic simple harmonic oscillator in three dimensions

n	l	$\psi_{n,l}(r)$
0	0	$\frac{2}{\pi^{1/4}}\left(\frac{M\omega}{\hbar}\right)^{3/4} e^{-\frac{M\omega}{2\hbar}r^2}$
1	1	$\frac{2\sqrt{2}}{\sqrt{3}\pi^{1/4}}\left(\frac{M\omega}{\hbar}\right)^{3/4}\sqrt{\frac{M\omega}{\hbar}}\, r e^{-\frac{M\omega}{2\hbar}r^2}$
2	0	$\frac{2\sqrt{2}}{\sqrt{3}\pi^{1/4}}\left(\frac{M\omega}{\hbar}\right)^{3/4}\left(\frac{3}{2}-\frac{M\omega}{\hbar}r^2\right) e^{-\frac{M\omega}{2\hbar}r^2}$
2	2	$\frac{4}{\sqrt{15}\pi^{1/4}}\left(\frac{M\omega}{\hbar}\right)^{3/4}\left(\frac{M\omega}{\hbar}r^2\right) e^{-\frac{M\omega}{2\hbar}r^2}$

as we worked out in detail in Sect. 9.8. In the following problem, we extend part of that analysis to an N-dimensional oscillator, which has SU(N) symmetry (special unitary group of $N \times N$ matrices).

We can look at the wavefunction in position space to understand properties of the energy eigenstates. We look at the amplitude $\psi_{n,l}(r)$, the probability density per unit volume $|\psi_{n,l}(r)|^2$, and the probability density per spherical shell $r^2|\psi_{n,l}(r)|^2$. As a reference, we show the first few wavefunctions in Table 11.2. In Fig. 11.10, we plot the radial wavefunction, its modulus squared, and r^2 times its modulus squared for the first four eigenstates. There are a few things to note. First, in the left panel, we see the s-wave states ($l = 0$) go to a constant value as $r \to 0$, while p-wave ($l = 1$), d-wave ($l = 2$), and higher angular momentum states go to zero. The behavior as $r \to 0$ goes like r^l, as we can easily see from the final wavefunction because every Laguerre polynomial has a nonzero constant term. Second, as $r \to \infty$, they all decay very fast. This is characteristic of a Gaussian tail to the wavefunction. Higher-order polynomials multiplying the Gaussian may delay the decay for a little while, but eventually the Gaussian decay always wins.

You may recall the property of energy eigenstates, that states with different energy eigenvalues are always orthogonal and that we can always arrange our degenerate energy eigenstates so that they are also orthogonal. The overlap of two states $|\psi_{n,l}\rangle \otimes |l, m\rangle$ and $|\psi_{n',l'}\rangle \otimes |l', m'\rangle$ is given by the product $\langle\psi_{n',l'}|\psi_{n,l}\rangle\, \langle l', m'|l, m\rangle$. The first thing to see is that if we do not have both $l = l'$ and $m = m'$, then the overlap is zero from the angular part of the wavefunction, regardless of the radial wavefunctions. So when either $l \neq l'$ or $m \neq m'$, then $\langle\psi_{n',l'}|\psi_{n,l}\rangle$ need not be zero, even though the full wavefunctions are orthogonal (in fact, it is almost always *not* zero). The orthogonality is enforced by the angular degrees of freedom in this case. But, when both $l = l'$ and $m = m'$, then we must have $\langle\psi_{n',l'}|\psi_{n,l}\rangle = 0$. We see precisely this behavior in the left panel. Note how the 00, 11, and 22 states all have just one sign. They can never be orthogonal (and they need not be). But the 00 and 20 wavefunctions are clearly different. The 00 is always one sign, while the 20 changes sign. It does so in just the right way to be orthogonal. Since we mapped the three-dimensional problem onto a one-dimensional problem by separating the radial and angular degrees of freedom, the radial part satisfies the node theorem. This means that each higher-energy wavefunction (with fixed l) has one more node than the previous one (recall that we do not count nodes at $r = 0$ or $r \to \infty$). Finally,

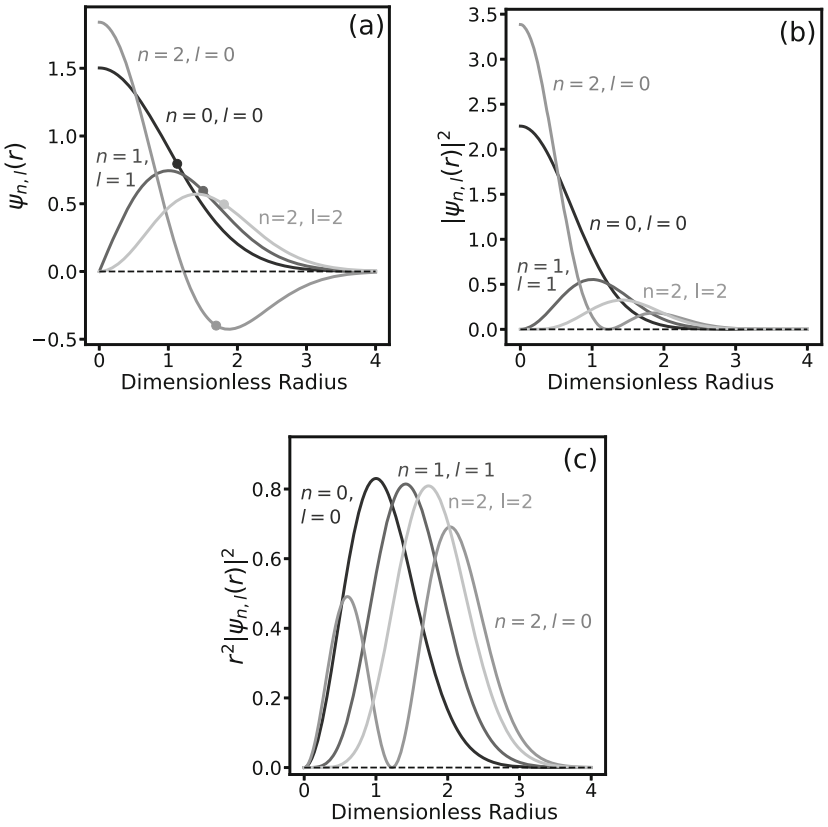

Fig. 11.10 Plot of (**a**) $\psi_{n,l}(r)$, (**b**) $|\psi_{n,l}(r)|^2$, and (**c**) $r^2|\psi_{n,l}(r)|^2$ for the first three energy levels of the isotropic harmonic oscillator in three dimensions. The average radius for each wavefunction is indicated by the solid dot in panel (**a**)

if one studies this problem the conventional way via differential equations, then one usually notes that the wavefunctions satisfy important boundary conditions—as $r \to 0$, $r\psi_{n,l}(r) \to 0$ and as $r \to \infty$, $r\psi_{n,l}(r) \to 0$. Both conditions only hold when the energy is equal to one of the energy eigenvalues. It is said then that the oscillations of the wavefunction are chosen to ensure the boundary condition holds. I find the more concrete operator way of constructing energy eigenstates easier to understand and follow than the "fitting the boundary conditions" argument (because the latter has spatially varying wavelengths), but you need to decide for yourself which argument you prefer.

The last thing we will talk about is the probability. The modulus squared of the wavefunction tells us about the *probability distribution*. This means we have to think about partitioning space into little pieces (like cubes of some small side length) and then the probability to be found in a cube of size L centered at position r_x, r_y, and r_z is approximately $|\psi(r_x, r_y, r_z)|^2 L^3$. The smaller L is, the better the approximation

Table 11.3 Average radius and fluctuations of the radial coordinate for different energy eigenfunctions of the isotropic simple harmonic oscillator in three dimensions. Note that the average radius moves out as the energy increases and wavefunctions with multiple peaks have much higher fluctuations

n	l	$\langle\psi_{n,l}\|\hat{r}\|\psi_{n,l}\rangle\sqrt{\frac{\hbar}{M\omega}}$	$(\Delta r)_{\|\psi_{n,l}\rangle}$
0	0	$\frac{2}{\sqrt{\pi}} = 1.128$	$\sqrt{\frac{3\pi-8}{2\pi}} = 0.476$
1	1	$\frac{8}{2\sqrt{\pi}} = 1.504$	$\sqrt{\frac{45\pi-128}{18\pi}} = 0.486$
2	0	$\frac{3}{\sqrt{\pi}} = 1.693$	$\sqrt{\frac{7\pi-18}{2\pi}} = 0.797$
2	2	$\frac{16}{5\sqrt{\pi}} = 1.805$	$\sqrt{\frac{175\pi-512}{50\pi}} = 0.490$

is. So we now have a question to answer. Where is the probability the highest? I had an argument with Walter Urbaniak, who is an engineer and a quantum enthusiast, who insisted that for s-wave wavefunctions, it is most likely to find the particle at the origin, because $|\psi(r_x, r_y, r_z)|^2$ is peaked there (as can be seen in the panel (b)). But, I knew this was misleading, because we find the probability in spherically symmetric problems not by looking at little cubes, but by looking as little spherical shells of width δR and radius R. Then the probability to be found in one of those shells is given by the volume of the shell times the probability density, which is $4\pi R^2 \delta R|\psi(R)|^2$. As you can see from panel (c), this is zero at the origin *for all wavefunctions*. So who is right? The answer, as you might have guessed with many quantum questions is both of us! You see, if I want to focus on the probability in little cubes, I focus on the panel (b). If in little spherical shells on the panel (c). The problem with the panel (c) is that the shell shrinks to zero as $r \to 0$. So, it isn't very useful in answering the question how likely is the particle to be found at the origin. Indeed, for s-wave states, they are most likely to be at the origin, even though their average radius (indicated in panel (a) with a solid dot and given by $\langle\psi_{n,l}|\hat{r}|\psi_{n,l}\rangle$) is not zero. Higher l angular momentum states, however, always avoid the origin and are never found there. One needs to be careful about precisely what question you are asking when examining probability distributions in three dimensions in order to know which one is the right one to use. We end with a table that reports the expectation values of $\hat{r}$ and its fluctuations in Table 11.3. A video https://tinyurl.com/54nz8me3 covers much of this algebra too.

Section 11.7 Problems:

11.7.1 Direct Computation of the Second Laguerre Polynomial

In the text, we calculated the $m = 0$ and $m = 1$ polynomials. Compute the next one for $m = 2$, given by $\hat{A}^\dagger_{l-2}\hat{A}^\dagger_{l-1}|\psi_{l,l}\rangle$. Express as a polynomial in $\frac{M\omega}{\hbar}\hat{r}^2$. Show that it agrees with the general form of the Laguerre polynomial.

Hint: it may be easier to compute this after converting to the nested commutators with respect to radial momentum.

11.7.2 Counting of States

Using the Cartesian form of the energy eigenstates, count how many states have $E = \frac{17}{2}\hbar\omega$. You should find that this number is the same as the number of states with $l = 7$, $l = 5$, $l = 3$, and $l = 1$, as suggested by the degeneracy pattern for the isotropic harmonic oscillator and depicted in the main text. Do your counting carefully! It is easy to make a mistake.

11.7.3 Conventional Schrödinger Factorization

In this problem, we compute the energy eigenvalues and energy eigenstates the original Schrödinger way, forming a factorization chain for each l value. We start from

$$\hat{H}_l = \hat{H}_0^{(l)} = \hat{A}_0^{(l)\dagger}\hat{A}_0^{(l)} + E_0^{(l)}, \tag{11.226}$$

which is factorized by the operator

$$\hat{A}_0^{(l)} = \frac{1}{\sqrt{2M}}\left(\hat{p}_r + \frac{i\hbar(l+1)}{\hat{r}} - iM\omega\hat{r}\right), \tag{11.227}$$

and has $E_0^{(l)} = \left(l + \frac{3}{2}\right)\hbar\omega$. This starting point is the same as what we had with the shortcut method. But now, things change a little bit as we form the factorization chain for this specific l value. The factorization chain satisfies

$$\hat{H}_j^{(l)} = \hat{A}_{j-1}^{(l)}\hat{A}_{j-1}^{(l)\dagger} + E_{j-1}^{(l)} = \hat{A}_j^{(l)\dagger}\hat{A}_j^{(l)} + E_j^{(l)}. \tag{11.228}$$

Determine the operators $\hat{A}_j^{(l)}$ that are created in this factorization chain for $j \geq 1$. You should find that they are expressed in the same form as what we have seen before and that they are functions of $l + j$ rather than l and j separately. You should find that the energy, however, takes the form

$$E_j^{(l)} = \left(2j + l + \frac{3}{2}\right) \tag{11.229}$$

for the jth excited state. Look carefully at Fig. 11.8 to verify that this step of $2\hbar\omega$ is correct when l is fixed. Determine the formula for the normalized energy eigenstates as well. They are written as operators acting on $|\phi_j^{(l)}\rangle$, which satisfies $\hat{A}_0^{(l)}|\phi_0^{(l)}\rangle = 0$. Verify that we indeed have that

$$\hat{H}_l|\psi_j\rangle = E_j^{(l)}|\psi_j\rangle \tag{11.230}$$

by employing the operator algebra for the factorization method (you should determine the intertwining relation to help with these manipulations).

The computation of the wavefunctions will be the same as what was done before and need not be repeated. One of the benefits of the shortcut method that we used in the text is that it makes the notation much more streamlined and less confusing. But, if there is ever any doubt, one can always use the ordinary Schrödinger factorization method.

11.7.4 Laguerre Polynomial Identity

The Laguerre polynomial satisfies

$$L_n^{(\alpha)}(x) = \sum_{m=0}^{n} (-1)^m \frac{(n+\alpha)!}{(n-m)!(m+\alpha)!} \frac{x^m}{m!}, \tag{11.231}$$

and in cases where α is not an integer, we evaluate the factorials via the corresponding gamma function.

Use this definition to prove the identity we needed:

$$(\alpha + n) L_n^{(\alpha-1)}(x) - x L_n^{(\alpha)}(x) = (n+1) L_{n+1}^{(\alpha-1)}(x). \tag{11.232}$$

This identity can be proven by manipulating the definition of the associated Laguerre polynomials using the power series aforementioned. You may find it easier to work from one side or the other and recall the add-zero trick, as it will be needed to get summations to extend over the proper ranges.

11.7.5 Momentum-Space Wavefunctions

This is a long problem, but if you can complete it, you truly are a master of this material. Your task is to redo the aforementioned derivations to compute the momentum space eigenstates and wavefunctions using the $\hat{B}_l^\dagger$ operators and the form of the Hamiltonian in terms of the momentum $\hat{p}$ and the momential radius $\hat{r}_p$. In your derivation, you will need to find the ground state for angular momentum l, which you should be able to show satisfies $\langle p|\psi_{l,l}\rangle = p^l e^{-\frac{p^2}{2\hbar M\omega}} \langle 0_p|\frac{1}{\hat{p}^l}|\psi_{l,l}\rangle$. The normalization constant is

$$\langle 0_p|\frac{1}{\hat{p}^l}|\psi_{l,l}\rangle = \frac{2^{l+\frac{3}{2}}\sqrt{(l+1)!}}{\pi^{1/4}\sqrt{(2l+2)!}} \left(\frac{1}{\hbar M\omega}\right)^{\frac{l}{2}+\frac{3}{4}}. \tag{11.233}$$

11.7.6 Two-Dimensional Isotropic Simple Harmonic Oscillator

One can apply the same strategies as we did here for the two-dimensional isotropic simple-harmonic oscillator. Find its energy spectrum, energy eigenstates, and wavefunctions in coordinate space. This is also a long problem. Some highlights are as follows: The Hamiltonian is separated into the radial motion and the angular motion as follows:

$$\hat{H} = \frac{\hat{p}_\rho^2}{2M} = \frac{\left(\hat{L}_z^2 - \frac{\hbar^2}{4}\right)}{2M\hat{\rho}^2} + \tfrac{1}{2}M\omega^2\hat{\rho}^2. \tag{11.234}$$

The state is written as a tensor product, given by $|\psi_\rho\rangle \otimes |m\rangle$. You should find that the radial eigenstate can be written in terms of products of raising operators acting on the ground state for a given z-component of angular momentum. Here, there is a subtlety, because we have the eigenvalue for the z-component of spin can go negative. One can work with just the nonnegative values of the angular momentum, because the Hamiltonian is independent of the sign of the z-component of angular momentum. Or, you can let the raising operators range from m to $-m$. Either method has some subtleties associated with it, and if you are having trouble, you can consult Rushka et al. (2021) for more details on how it works. To complete the wavefunction calculation, you will need to compute the wavefunction for the ground state with a given value of angular momentum. Use the same idea as we did in the text—find a state annihilated by the operator that appears in the exponent for the polar coordinate translation operator. You do not need to worry about the normalization constant of the final result.

11.7.7 Expectation Values of $\hat{r}$ and $\hat{p}_r$

In this problem, you will investigate computing expectation values of powers of $\hat{r}$ and $\hat{p}_r$. We work with the labels $|\psi_{n=l+m,l-m}\rangle = |\psi_{l+m,l-m}\rangle$ notation instead of the $|\psi_{n,l}\rangle$ notation.

(a) Recall that the hypervirial relation says that $\langle\psi_{l+m,l-m}|[\hat{r}, \hat{H}_{l-m}]|\psi_{l+m,l-m}\rangle = 0$, because we can act the Hamiltonian to the right or to the left on the energy eigenstate, and in either case, it gives E_{l+m}. Use this result to show that $\langle\psi_{l+m,l-m}|\hat{p}_r|\psi_{l+m,l-m}\rangle = 0$.

(b) Use $\langle\psi_{l+m,l-m}|[\hat{r}\,\hat{p}_r, \hat{H}_{l-m}]|\psi_{l+m,l-m}\rangle = 0$ to show that

$$\langle\psi_{l_m,l-m}|\frac{\hat{p}_r^2}{2M} + \frac{\hbar^2(l-m)(l-m+1)}{2M\hat{r}^2}|\psi_{l+m,l-m}\rangle = \langle\psi_{l+m,l-m}|\tfrac{1}{2}M\omega^2\hat{r}^2|\psi_{l+m,l-m}\rangle. \tag{11.235}$$

This is called the *virial theorem*. Use this to prove that

$$\langle \psi_{l+m,l-m} | \hat{r}^2 | \psi_{l+m,l-m} \rangle = \frac{\hbar}{M\omega} \left(l + m + \tfrac{3}{2} \right). \tag{11.236}$$

(c) One can find a relationship between the expectation values of $\hat{r}$ and $\frac{1}{\hat{r}}$ by noting that

$$\hat{A}^\dagger_{l-m-1} - \hat{A}_{l-m-1} = \sqrt{\tfrac{2}{M}} \left(\frac{i\hbar(l-m)}{\hat{r}} - iM\omega\hat{r} \right), \tag{11.237}$$

using the following strategy. In the following, the states $|\psi_{l+m,l-m}\rangle$ are all normalized. If $\langle \psi_{l+m+1,l-m-1} | \psi_{l+m,l-m} \rangle = 0$, we could develop a nice identity using the expectation value

$$\langle \psi_{l+m,l-m} | \left(\hat{A}^\dagger_{l+m-1} - \hat{A}_{l-m-1} \right) | \psi_{l+m,l-m} \rangle. \tag{11.238}$$

But the overlap of those two states is not zero, because the orthogonality of those states comes from the angular momentum part of the tensor product, not the radial part. Indeed, one can show that they are not zero, in general. But this is not the end of the world. The strategy will work for the first state in the series, $|\psi_{l,l}\rangle$, because $\hat{A}_l |\psi_{l,l}\rangle = 0$. Use these ideas to show that

$$\langle \psi_{l,l} | \hat{r} | \psi_{l,l} \rangle = \frac{\hbar}{M\omega} (l+1) \langle \psi_{l,l} | \tfrac{1}{\hat{r}} | \psi_{l,l} \rangle. \tag{11.239}$$

There does not seem to be a way to extend this to the general $|\psi_{l+m,l-m}\rangle$ state. However, one can find recurrence relations with a lot more work by commuting each lowering operator in the state $\langle \psi_{l+m,l-m}|$ to the right through the difference in the ladder operators, to meet the raising operators on the right. This will create a recurrence relation between different expectation values, but it does not seem like the recurrence relations can be closed to find all of the desired expectation values.

(d) Evaluate $\langle \psi_{l+m,l-m} | [\hat{p}_r, \hat{H}_{l-m}] | \psi_{l+m,l-m} \rangle$ in two different ways to show that

$$\langle \psi_{l+m,l-m} | \hat{r} | \psi_{l+m,l-m} \rangle = \left(\frac{\hbar}{M\omega} \right)^2 (l-m)(l-m+1) \langle \psi_{l+m,l-m} | \frac{1}{\hat{r}^3} | \psi_{l+m,l-m} \rangle, \tag{11.240}$$

which requires $l > m$. The identity does not hold for $l = m$, because this is an s-wave state, with zero angular momentum, and $\langle \psi_{l+m,l-m} | \frac{1}{\hat{r}^3} | \psi_{l+m,l-m} \rangle$ is not well defined for such states.

(e) In this part of the problem, we will use the identity

$$\left(\hat{A}_{l-1} - \hat{A}_l^\dagger\right)\left(\hat{A}_l - \hat{A}_{l-1}^\dagger\right) = -\frac{\hbar^2(2l+1)^2}{2M\hat{r}^2} + 2\hbar\omega(2l+1) - 2M\omega^2\hat{r}^2. \tag{11.241}$$

Evaluate

$$\langle\psi_{l,l}|\left(\hat{A}_{l-1} - \hat{A}_l^\dagger\right)\left(\hat{A}_l - \hat{A}_{l-1}^\dagger\right)|\psi_{l,l}\rangle \tag{11.242}$$

in two different ways—one using the identity we developed earlier, and the second way by converting the raising and lowering operator pairs into Hamiltonians, or by using the subsidiary condition (to the right or to the left). Use this to show that

$$\langle\psi_{l,l}|\frac{1}{\hat{r}^2}|\psi_{l,l}\rangle = \frac{M\omega}{\hbar}\frac{1}{l+\frac{1}{2}}. \tag{11.243}$$

Hint: You need the expectation value of $\hat{r}^2$ from part (b) and $\hat{A}_{l-1}\hat{A}_{l-1}^\dagger = \hat{H}_l - \hbar\omega\left(l - \frac{1}{2}\right)$. This result also allows us to compute

$$\langle\psi_{l,l}|\hat{p}_r^2|\psi_{l,l}\rangle = M\hbar\omega\frac{l+\frac{3}{4}}{l+\frac{1}{2}}, \tag{11.244}$$

when it is combined with the result in part (b).

Unfortunately, there does not seem to be any simple algebraic way to compute the expectation value of $\langle\psi_{l+m,l-m}|\hat{r}|\psi_{l+m,l-m}\rangle$. So, we are not able to examine uncertainty relations in spherical coordinates purely algebraically. It can be done with calculus. Similarly, no simple way exists to extend these results to nonzero m. But, a clever idea might work. If you discover one, let me know!

11.7.8 Fluctuations of the Radial Coordinate

Using calculus, we find that (Robinett, 2006; Commins, 2014)

$$\langle\psi_{l,l}|\hat{r}|\psi_{l,l}\rangle = \sqrt{\frac{\hbar}{M\omega}}\frac{\left(2^{l+1}(l+1)!\right)^2}{(2l+2)!}\frac{1}{\sqrt{\pi}}. \tag{11.245}$$

There are a few things to note about this result. First, the factor of $\frac{1}{\sqrt{\pi}}$ tells us that it is highly unlikely that we can find this result algebraically. There is no simple way to obtain such factors via commutators and the operation of operators onto states.

Second, one factor of $2^{l+1}(l+1)!$ removes all even integer factors from the factorial in the denominator. Then we are left with the other factor of $2^{l+1}(l+1)!$ divided by $(2l+1)!!$. Since each factor in the denominator is less than the corresponding integer in the numerator, this product is always larger than 1. Third, if we use Stirling's approximation for the factorial, given by $n! \approx \exp(n \ln n - n)$, we find, for large l, this factor is approximately

$$\exp[2(l+1)\ln 2 + 2(l+1)\ln(l+1) - 2(l+1) - 2(l+1)\ln(2(l+1) + 2(l+1)] \approx 1. \tag{11.246}$$

A more accurate analysis using $n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$ gives

$$\lim_{l\to\infty} \frac{[2^{l+1}(l+1)!]^2}{(2l+2)!} \approx \sqrt{\frac{l+1}{2}}\sqrt{2\pi}. \tag{11.247}$$

Use the result for $\langle\psi_{l,l}|\hat{r}|\psi_{l,l}\rangle$ and the expectation values for $\hat{r}^2$, $\hat{p}_r$, and $\hat{p}_r^2$ found in the previous problem to

(a) Compute $(\Delta r)_{|\psi_{l,l}\rangle} = \sqrt{\langle\psi_{l,l}|\hat{r}^2|\psi_{l,l}\rangle - (\langle\psi_{l,l}|\hat{r}|\psi_{l,l}\rangle)^2}$.
(b) Compute $(\Delta p_r)_{|\psi_{l,l}\rangle} = \sqrt{\langle\psi_{l,l}|\hat{p}_r^2|\psi_{l,l}\rangle - (\langle\psi_{l,l}|\hat{p}_r|\psi_{l,l}\rangle)^2}$.
(c) Verify that

$$(\Delta r)_{|\psi_{l,l}\rangle}(\Delta p_r)_{|\psi_{l,l}\rangle} = \hbar\sqrt{l+\tfrac{3}{2} - \tfrac{1}{\pi}\left(\frac{2^{l+1}(l+1)!}{(2l+1)!!}\right)^2}\sqrt{\frac{l+\frac{3}{4}}{l+\frac{1}{2}}}. \tag{11.248}$$

(d) This asymptotic analysis is still not accurate enough. It shows that $\lim_{l\to\infty}(\Delta r)_{|\psi_{l,l}\rangle}(\Delta p_r)_{|\psi_{l,l}\rangle} \approx \frac{\hbar}{\sqrt{2}}$, when the true answer approaches $\frac{\hbar}{2}$. One needs an even more accurate approximation to the factorial to get the right answer. Compute the exact results numerically for $l = 0$, $l = 1$, $l = 10$, and $l = 100$ (*Ans.* $0.5832\hbar$, $0.5252\hbar$, $0.5031\hbar$, and $0.5003\hbar$).
Note that for the one-dimensional simple harmonic oscillator, we had that the uncertainty was minimal for a Gaussian, corresponding to the ground state with $n = 0$. The higher n values had increasing uncertainty. Here, we have the opposite. For the radial coordinate and momentum and the lowest-energy states for each angular momentum l, we find the uncertainty is lowest for the largest l values! Yet, uncertainty for the Cartesian coordinates is lowest for the ground state! This simply is the way it is. The spherical coordinate system acts differently than the Cartesian, and this is borne out here. It is due to the operators we compute uncertainty for, because, in some cases, the wavefunctions we compute the uncertainty for *are the same*. This is weird, but this is the quantum world.
(e) *Bonus question* A longer asymptotic expansion for the factorial is

$$\ln n! \approx n\ln n - n + \tfrac{1}{2}\ln(2\pi) + \tfrac{1}{2}\ln\left(n + \tfrac{1}{6} + \tfrac{1}{72n} + \cdots\right). \tag{11.249}$$

Can you verify that as $l \to \infty$, we have that $(\Delta r)_{|\psi_{l,l}\rangle}(\Delta p_r)_{|\psi_{l,l}\rangle} \to \frac{\hbar}{2}\left(1+\frac{5}{144l}+\cdots\right)$?

Hint: The algebra is long and complicated. You can verify the accuracy by comparing with your exact numerics, to ensure that you did not make a mistake. You should find the accuracy is excellent for this approximation.

11.7.9 N-Dimensional Isotropic Simple Harmonic Oscillator and the Hidden SU(N) Symmetry

This problem was inspired by Zhang Rudong.

The n-dimensional isotropic simple harmonic oscillator can be written as

$$\hat{H} = \hbar\omega\left(\sum_{j=1}^{n}\hat{a}_j^\dagger\hat{a}_j + \tfrac{n}{2}\right), \tag{11.250}$$

with $[\hat{a}_i^\dagger, \hat{a}_j^\dagger] = 0$, $[\hat{a}_i\,, \hat{a}_j] = 0$, and $[\hat{a}_i\,, \hat{a}_j^\dagger] = \delta_{ij}$.

Let U_{ij} be an $n \times n$ unitary matrix. This means $UU^\dagger = U^\dagger U = \mathbb{I}_n$, or

$$\sum_{k=1}^{n} U_{ki}^* U_{kj} = \sum_{k=1}^{n} U_{ik}U_{kj}^* = \delta_{ij}. \tag{11.251}$$

Show that the new ladder operators defined by

$$\hat{b}_i = \sum_{k=1}^{n} U_{ik}\hat{a}_k \;\text{ and }\; \hat{b}_i^\dagger = \sum_{k=1}^{n}\hat{a}_k^\dagger U_{ki}^\dagger, \tag{11.252}$$

satisfy

$$[\hat{b}_i^\dagger, \hat{b}_j^\dagger] = 0, \;\; [\hat{b}_i\,, \hat{b}_j] = 0, \;\; \text{and} \;\; [\hat{b}_i\,, \hat{b}_j^\dagger] = \delta_{ij}. \tag{11.253}$$

Show as well that

$$\hat{H} = \hbar\omega\left(\sum_{j=1}^{n}\hat{b}_j^\dagger\hat{b}_j + \tfrac{n}{2}\right). \tag{11.254}$$

The generators of the SU(N) symmetry are

$$\hat{J}_{ij} = \tfrac{1}{\hbar}[\hat{x}_i\hat{p}_j - \hat{x}_j\hat{p}_i] \;\text{ for }\; i \neq j \tag{11.255}$$

$$\hat{S}_{ij} = \frac{2}{\hbar\omega}\left[\frac{1}{2M}\hat{p}_i\hat{p}_j + \frac{1}{2}M\omega^2\hat{x}_i\hat{x}_j - \frac{1}{n}\hat{H}\delta_{ij}\right]. \tag{11.256}$$

Express these operators in terms of the ladder operators $\hat{a}_i$ and $\hat{a}_i^\dagger$. Verify that they all commute with $\hat{H}$. Show also that $\sum_{j=1}^{n}\hat{S}_{jj} = 0$.

For $N = 2$, there are three independent operators $\hat{J}_{12}$, $\hat{S}_{11}$ and $\hat{S}_{12}$. Show that $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$ can be expressed in terms of them. The algebra for higher N is more complicated and goes beyond what we will pursue further here. The original idea for this work comes from Baker (1956).

11.8 Hydrogen

Congratulations! You have now reached what is probably the most important single topic in a quantum mechanics course—hydrogen. The hydrogen atom has an exact solution, and we can perform experiments on hydrogen exceedingly accurately. By closely comparing the solutions with experiments, we can learn many new things. There is a wonderful book on this theme that I highly recommend—John Rigden's *Hydrogen: The essential element* (Rigden, 2003). You see, hydrogen has been the mainstay of quantum mechanics for over a century. It is unfortunate in many respects that the Bohr model worked so well for it, because it led physicists astray for about 15 years. Nevertheless, with hydrogen, you can (i) verify the accuracy of energy level calculations; (ii) measure the mass of the proton; (iii) discover the deuteron; (iv) determine that electrons have spin; (v) understand why the spin g-factor is anomalous and almost equal to 2; (vi) establish measurable effects of quantum electrodynamics; (viii) discover nuclear spin; (viii) measure the charge radius of the proton; and much, much more. It is the entry point for nearly all developments in modern physics and touches on a whole slew of different fields of physics, including radio astronomy, quantum field theory, diagnostic medical imaging, and much, much more.

So, we will have a lot to say about hydrogen. To give you a little road map of what is to come, we use the factorization method to find the energy eigenstates and the operator-generalized Rodrigues formula to determine an operator identity for those eigenstates in this section. In the next section, we use spherical symmetry to find the coordinate wavefunctions. In the section after that, we work without using spherical symmetry and determine the momentum wavefunctions (which can be directly measured). We re-encounter Laguerre polynomials and discover new friends—reverse Bessel polynomials and Gegenbauer polynomials. In the following sections, we show the full power of the algebraic method, which can determine expectation values of many operators, and we end with some of the applications—exploring Rydberg atoms and hyperfine structure. Fine structure will be discussed later when we discuss relativistic quantum mechanics and the Dirac equation.

Nuclear magnetic resonance and photo-production of light will be covered when we develop time evolution of quantum systems.

The Hamiltonian for hydrogen is the sum of the kinetic energy plus the Coulomb interaction between the electron and the proton. We use the reduction from a two-body problem to a one-body problem as described in Sects. 7.14 and 11.4. We also decompose the kinetic energy into its radial and angular components, So, we have that the Hamiltonian for the effective particle is

$$\hat{H} = \frac{\hat{p}_r^2}{2M} + \frac{\hat{L}^2}{2M\hat{r}^2} - \frac{e^2}{4\pi\epsilon_0 \hat{r}}. \tag{11.257}$$

Here, M is the reduced mass of the electron, given by

$$M = \frac{m_N m_e}{m_N + m_e} = \frac{m_e}{1 + \frac{m_e}{m_N}}, \tag{11.258}$$

with m_e the mass of the electron and m_N the mass of the nucleus, which we will take to be m_p the mass of the proton, unless we are considering a more complicated atom than common hydrogen. The interaction is weighted by e^2, since this is the product of the magnitude of the charge of the electron and the proton. In some cases, we will consider *hydrogenic ions*, where a heavier nucleus, with a higher charge (denoted Ze) captures a single electron. We will not clutter our formulas with the extra Z factor. Instead, we set $Z = 1$ for now.

Just like what we did in the previous section, we first recall the maximal set of commuting operators. They are $\hat{H}$, $\hat{L}^2$, and $\hat{L}_z$ (as they are for any central potential). Angular momentum is a good quantum number because the potential depends only on $\hat{r}$ and $[\hat{L}_i, \hat{r}] = 0$. We form the tensor-product eigenstate, just like before as $|\psi\rangle = |\psi_r\rangle \otimes |l, m\rangle$, where $|l, m\rangle$ is the angular momentum eigenstate with definite quantum numbers of total angular momentum l and of z-component of angular momentum m. Acting the full Hamiltonian onto the tensor-product state, then gives us an infinite number of Hamiltonians for each total angular momentum, given by

$$\hat{H}_l = \frac{\hat{p}_r^2}{2M} + \frac{\hbar^2 l(l+1)}{2M\hat{r}^2} - \frac{e^2}{4\pi\epsilon_0 \hat{r}}. \tag{11.259}$$

and the energy eigenvalue equation is $\hat{H}_l|\psi_r\rangle = E|\psi_r\rangle$.

Because this system has extra symmetry (which will be discussed further), we can use the short-cut method to do the factorization and find all energy eigenvalues and eigenstates "at once." We will use $\hat{B}_l$ to denote these radial ladder operators so as to not confuse them with the Cartesian operators that we will discuss later. Hence, we have

$$\hat{H}_l = \hat{B}_l^\dagger \hat{B}_l + E_l. \tag{11.260}$$

In order to perform this factorization, we need to obtain terms that go like $\frac{1}{\hat{r}}$, $\frac{1}{\hat{r}^2}$, and constants when we form the product of the raising and lowering operators. It is easy to determine that an appropriate ansatz for these ladder operators is given by

$$\hat{B}_l = \frac{1}{\sqrt{2M}}\left(\hat{p}_r - \frac{i\hbar\alpha_l}{a_0} - \frac{i\hbar\beta_l}{\hat{r}}\right), \tag{11.261}$$

with $a_0 = \frac{4\pi\epsilon_0\hbar^2}{Me^2}$ the reduced Bohr radius, introduced to keep the real numbers α_l and β_l dimensionless. We call it the reduced Bohr radius, because we use the reduced mass of hydrogen rather than the electron mass. A quick calculation yields

$$\begin{aligned}\hat{B}_l^\dagger \hat{B}_l &= \frac{\hat{p}_r^2}{2M} - \frac{i\hbar}{2M}\left[\hat{p}_r, \frac{\alpha_l}{a_0} + \frac{\beta_l}{\hat{r}}\right] + \frac{\hbar^2}{2M}\left(\frac{\alpha_l^2}{a_0^2} + \frac{2\alpha_l\beta_l}{a_0\hat{r}} + \frac{\beta_l^2}{\hat{r}^2}\right) \\ &= \frac{\hat{p}_r^2}{2M} + \frac{\hbar^2}{2M}\frac{\beta_l(\beta_l+1)}{\hat{r}^2} + \frac{\hbar^2}{Ma_0}\frac{\alpha_l\beta_l}{\hat{r}} + \frac{\hbar^2\alpha_l^2}{2Ma_0^2} = \hat{H}_l - E_l. \end{aligned} \tag{11.262}$$

This is precisely in our desired form. Our choices are either $\beta_l = l$ or $\beta_l = -(l+1)$ and $\alpha_l = -\frac{1}{\beta_l}$ because $\frac{\hbar^2}{Ma_0} = \frac{e^2}{4\pi\epsilon_0}$. But, we must have the superpotential be positive as $r \to \infty$. This means we must have $\alpha_l > 0$, hence we must have $\beta_l < 0$. This means we must choose $\beta_l = -l - 1$ and $\alpha_l = \frac{1}{l+1}$. Note that this choice will also produce a wavefunction that is finite as $r \to 0$, as given by the rule we worked out in the previous section when performing the factorization for the isotropic harmonic oscillator. The energy becomes

$$E_l = -\frac{\hbar^2\alpha_l^2}{2Ma_0^2} = -\frac{\hbar^2}{2Ma_0^2(l+1)^2} = -\frac{e^2}{8\pi\epsilon_0 a_0(l+1)^2}. \tag{11.263}$$

It is customary to let $n = l + 1$ be the principal quantum number and to label the energy with respect to the principal quantum number. To avoid confusion with the energies that appear in the factorization chain, we will use $\varepsilon_n = E_{n-1} = E_l$ to avoid any confusion. The lowering operator then becomes

$$\hat{B}_l = \frac{1}{\sqrt{2M}}\left(\hat{p}_r - \frac{i\hbar}{(l+1)a_0} + \frac{i\hbar(l+1)}{\hat{r}}\right). \tag{11.264}$$

Next, we need to find the intertwining relation. A simple calculation shows that

$$\begin{aligned}\hat{B}_l\,\hat{B}_l^\dagger &= \frac{\hat{p}_r^2}{2M} + \frac{i\hbar}{2M}\left[\hat{p}_r, \frac{1}{(l+1)a_0} - \frac{(l+1)}{\hat{r}}\right] \\ &\quad + \frac{\hbar^2}{2M}\left(\frac{1}{(l+1)^2a_0^2} - \frac{2}{a_0\hat{r}} + \frac{(l+1)^2}{\hat{r}^2}\right)\end{aligned}$$

$$= \frac{\hat{p}_r^2}{2M} + \frac{\hbar^2(l+1)(l+2)}{2M\hat{r}^2} - \frac{e^2}{4\pi\epsilon_0\hat{r}^2} + \frac{e^2}{8\pi\epsilon_0 a_0(l+1)^2} = \hat{H}_{l+1} - E_l. \tag{11.265}$$

The intertwining relation then becomes

$$\hat{H}_l\hat{B}_l^\dagger = (\hat{B}_l^\dagger\hat{B}_l + E_l)\hat{B}_l^\dagger = \hat{B}_l^\dagger(\hat{B}_l\,\hat{B}_l^\dagger + E_l) = \hat{B}_l^\dagger\hat{H}_{l+1}. \tag{11.266}$$

So, this is exactly like our regular intertwining relation, with no constant shifts added in. We can now use this intertwining relation to directly determine all the energy eigenstates for each $\hat{H}_l$, similar to what we did for the isotropic harmonic oscillator, using our short-cut method.

Just as we did for the isotropic harmonic oscillator, we now employ the intertwining relation to find all of the energy eigenstates. We already know the ground states for each $\hat{H}_l$. We can denote them in one of two ways (we now suppress the ψ label we were using before and label with the quantum numbers only): $|n{=}l+1, l\rangle$ or $|n, l{=}n-1\rangle$; we suppress the m quantum number dependence, because the radial equation is independent of m. We will tend to use the latter notation, because it is more common to denote the states in terms of the principal quantum number n than the angular momentum l. These ground states satisfy $\hat{B}_l\,|n{=}l+1, l\rangle = 0$ or equivalently $\hat{B}_{n-1}|n, l{=}n-1\rangle = 0$. Because of this subsidiary condition, we have that the ground-state energy for a state with definite angular momentum is given by $E_l = \varepsilon_n$, with $n = l + 1$. Again, we will tend to use the latter notation. Armed with this information, one can directly construct a total of n energy eigenstates from $|n, l{=}n-1\rangle$ simply by forming the right chain of raising operators acting on the state. We have

$$|n, l\rangle = C_{n,l}\hat{B}_l^\dagger\hat{B}_{l+1}^\dagger\cdots\hat{B}_{n-2}^\dagger|n, l{=}n-1\rangle \tag{11.267}$$

is an energy eigenstate with energy ε_n. The set of states determined by the string of raising operators acting on a ground state with definite angular momentum must terminate with the state that has the string extending to $l = 0$, because the raising operator is not defined for $l = -1$ due to a division by zero. Here, we have introduced the normalization factor $C_{n,l}$ that we will determine shortly. But first, we verify that this state is an energy eigenstate by employing the intertwining relation to move the Hamiltonian operator to the right through the different raising operators. We find

$$\hat{H}_l|n, l\rangle = \hat{H}_l\ \overbrace{\hat{B}_l^\dagger\ \ \hat{B}_{l+1}^\dagger}\ \cdots\hat{B}_{n-2}^\dagger\,|n, l{=}n-1\rangle$$

$$= \hat{B}_l^\dagger\ \hat{H}_{l+1}\ \overbrace{\hat{B}_{l+1}^\dagger\ \ \hat{B}_{l+2}^\dagger}\ \cdots\hat{B}_{n-2}^\dagger\,|n, l{=}n-1\rangle$$

$$= \hat{B}_l^\dagger \hat{B}_{l+1}^\dagger \cdots \hat{H}_{n-2} \overset{\hat{B}_{n-2}^\dagger}{\curvearrowright} |n, l{=}n-1\rangle$$

$$= \hat{B}_l^\dagger \hat{B}_{l+1}^\dagger \cdots \hat{B}_{n-2}^\dagger \hat{H}_{n-1} |n, l{=}n-1\rangle$$

$$= \varepsilon_n |n, l{=}n-1\rangle. \tag{11.268}$$

We used the intertwining relation $n - l - 1$ times, and we used the fact that $|n, l{=}n-1\rangle$ is an energy eigenstate of $\hat{H}_{n-1}$ with energy $\varepsilon_n = E_{n-1}$. Note how the energy for the energy eigenstate $|n, l\rangle$ does not depend on l. This means all of these states are degenerate in energy for $0 \leq l \leq n-1$. As we discussed before, this must arise from a hidden symmetry, which we will describe later. Note as well how much simpler this shortcut method is than the standard Schrödinger factorization method, which would require us to set up a factorization chain for each angular momentum quantum number. Of course, you will have a chance to do it the standard way in the problems. One question you might ask is how do I know the state $|n, l\rangle$ corresponds to the state with angular momentum l? The answer is that it is an eigenstate of $\hat{H}_l$, which was constructed to be the Hamiltonian when the angular momentum quantum number was precisely equal to l.

We still need to determine the normalization constant $C_{n,l}$. This is found via direct calculation using intertwining. Unlike for the isotropic harmonic oscillator, here we have the usual intertwining relationship, so we can immediately read off the constant via the set of energy eigenvalues, which gives us

$$\begin{aligned} C_{n,l} &= \frac{1}{\sqrt{(\varepsilon_n - \varepsilon_{n-1})(\varepsilon_n - \varepsilon_{n-2}) \cdots (\varepsilon_n - \varepsilon_{l+2})(\varepsilon_n - \varepsilon_{l+1})}} \\ &= \left(\frac{8\pi\epsilon_0 a_0 n^2}{e^2}\right)^{\frac{n-l-1}{2}} \frac{(n-1)!}{l!} \sqrt{\frac{(n+l)!}{(2n-1)!(n-l-1)!}}. \end{aligned} \tag{11.269}$$

Note that we used the fact that $n^2 - j^2 = (n + j)(n - j)$ in the derivation.

You might be surprised at how easy it was to find the energy eigenvalues and eigenstates for hydrogen, since it is such a hallmark problem in physics. But, this is how it is. You see, we have developed so many quantum skills that you are now able to perform powerful calculations quite efficiently.

Now that we have solved the problem, let us look carefully at the energy spectrum (given by $\{E : E_{n,l} = -\frac{e^2}{8\pi\epsilon_0 a_0 n^2}$, for $n = 1, 2, 3, \cdots$ and $l = 0, 1, \cdots n-1\}$). We have found that for every l value, the lowest-energy state is E_l, and it is degenerate with states that we constructed that all have angular momentum less than l. We plot the spectrum in Fig. 11.11.

One thing to see in the figure of the spectrum is that the energy levels pile up and become quite dense as they approach $E = 0$ from below ($n \to \infty$). In addition, the degeneracy across l is also striking. Of course, this behavior is atypical. Most spectra for quantum systems are nondegenerate. The degeneracy arises because of a

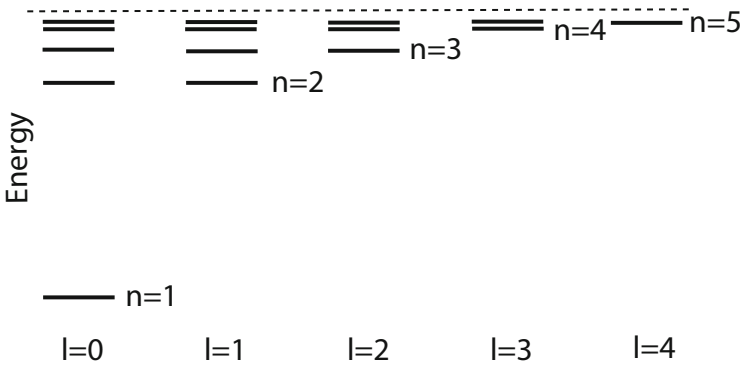

Fig. 11.11 The energy eigenvalue spectrum of (nonrelativistic) hydrogen plotted versus the angular momentum of the different quantum states. You can clearly see the degeneracy for fixed n, and you can also see how the states pile up as we approach $E = 0$ ($n \to \infty$). The short-cut method finds the energy levels for fixed n (moving horizontally along a fixed energy line), while the traditional Schrödinger factorization method finds them for fixed l (moving vertically, with a fixed l value)

hidden symmetry in the system. The symmetry was *de facto* used by Wolfgang Pauli when he solved for the hydrogen spectra using a methodology that is often described as matrix mechanics in 1926 (Pauli, 1926). In fact, Pauli's approach primarily employed commutation relations of operators, so it is more apt to describe it as *operator mechanics*, a third form of quantum mechanics (after matrix mechanics and wave mechanics). Indeed, this book is an exposition of operator mechanics throughout. Pauli's calculation exploited an additional vector that is conserved in Coulomb's law, the so-called Runge–Lenz vector. Then by manipulating operators constructed from angular momentum and the Runge–Lenz vector, one can infer the spectrum. It was not fully appreciated at that time, but the operator algebra Pauli worked out is the so-called SO(4) symmetry, which stands for special orthogonal group of matrices of dimension 4. This symmetry was worked out explicitly by Vladimir Fock in 1935 (Fock, 1935). The SO(4) group has two independent SU(2) symmetries—one is the standard angular momentum symmetry, the other is related to the Runge–Lenz vector. We will not describe how to do this calculation in detail, because the commutator algebra is rather torturous, and it does not lead to a general methodology that can be used in other problems. Of course, Pauli's article is still quite readable today; I encourage you to look there if you want to see how it was done; Zwiebach also solves hydrogen this way (Zwiebach, 2022), as do a small number of other textbook authors. But you have been warned that it is a long journey through the different commutators. In my opinion, the factorization method is much more elegant in how it works.

It is now a good time to recall the energies of atomic systems. The rest mass energy of an electron is given by $m_e c^2$, and it is about 511,000 eV. The ground-state energy of hydrogen, given by $E_{gs} = -\frac{e^2}{8\pi\epsilon_0 a_0}$ is about 13.6 eV, or some

40,000 times smaller. Atomic energies are generically small compared to relativistic energies, except for very heavy atoms. This is why it is a good approximation to ignore relativistic effects in a first look at atomic structure. The atomic energy unit $-\frac{e^2}{8\pi\epsilon_0 a_0}$ is so often encountered, that we have given it its own name—it is called the *Rydberg*. Sometimes people like to use a unit twice as large—$\frac{e^2}{4\pi\epsilon_0 a_0} \approx 27.2$ eV. This unit is called the *Hartree*. By looking at the concrete energy levels, we see that the energy magnitude drops precipitously. We have for the first few energy levels $\varepsilon_1 \approx -13.6$ eV, $\varepsilon_2 \approx 3.4$ eV, $\varepsilon_2 \approx 1.5$ eV, $\varepsilon_3 \approx 0.85$ eV, and so on. Energy differences on the order of eVs correspond to optical transitions, as you can see. This is why we can make bright lights from gases of neon (orange), sodium (yellow), and argon (red). It is also why chemists use flame tests to determine the atoms in a molecule. We will discuss how atoms produce light more as we learn more about them.

Now we will use the Rodrigues formula strategy to convert the product of the raising operators acting on the ground state with angular momentum l into a Laguerre polynomial operator acting on that state (Noonan et al., 2024). We use the same strategy as we used previously. It will not be repeated here. We will just jump into carrying out the different steps. We start by finding a similarity transformation of the radial momentum operator that generates the lth raising operator $\hat{B}_l^\dagger$. The constant term is found by using a Hadamard lemma with an operator in the exponent that is linear in $\hat{r}$, while the inverse power of $\hat{r}$ is constructed in the same way as before—using a power-law similarity transformation. We find that

$$\hat{B}_l^\dagger = \frac{1}{\sqrt{2M}} \hat{r}^{-(l+1)} e^{\frac{\hat{r}}{(l+1)a_0}} \hat{p}_r e^{-\frac{\hat{r}}{(l+1)a_0}} \hat{r}^{l+1}. \tag{11.270}$$

The order of the two operators that determine the similarity transformation can be interchanged because they commute with each other. Substituting into the expression for the energy eigenstate yields

$$|n, l\rangle = C_{n,l} \left(\frac{1}{\sqrt{2M}}\right)^{n-l-1} \hat{r}^{-(l+1)} e^{\frac{\hat{r}}{(l+1)a_0}} \prod_{m=1}^{n-l-1} \left(\hat{p}_r e^{-\frac{\hat{r}}{(l+m)(l+m+1)a_0}} \frac{1}{\hat{r}}\right)$$
$$\times e^{-\frac{\hat{r}}{na_0}} r^n |n, l{=}n-1\rangle, \tag{11.271}$$

after combining neighboring similarity transformation factors together. Next, we find the state that $\hat{p}_r$ annihilates. Using the subsidiary condition, one can quickly find that

$$\hat{p}_r e^{\frac{\hat{r}}{na_0}} \hat{r}^{-n} |n, l{=}n-1\rangle = 0. \tag{11.272}$$

This allows us to convert the string of raising operators into nested commutators (starting from the rightmost and moving to the left) as follows:

$$|n,l\rangle = C_{n,l}\left(\frac{1}{\sqrt{2M}}\right)^{n-l-1}\hat{r}^{-(l+1)}e^{\frac{\hat{r}}{(l+1)a_0}} \times\left[\hat{p}_r, e^{-\frac{\hat{r}}{(l+1)(l+2)a_0}}\frac{1}{\hat{r}}\left[\hat{p}_r, e^{-\frac{\hat{r}}{(l+2)(l+3)a_0}}\frac{1}{\hat{r}}\left[\hat{p}_r,\cdots\right.\right.\right.$$
$$\left.\left.\left.\cdots, e^{-\frac{\hat{r}}{(n-2)(n-1)a_0}}\frac{1}{\hat{r}}\left[\hat{p}_r, e^{-\frac{(2n-1)\hat{r}}{(n-1)na_0}}\hat{r}^{2n-1}\right]\cdots\right]\right]\right]_{n-l-1} e^{\frac{\hat{r}}{na_0}}\hat{r}^{-n}|n,l{=}n-1\rangle. \tag{11.273}$$

Once again, we move the $\frac{1}{\hat{r}}$ term inside the commutators and move the far right operators to the left to find

$$|n,l\rangle = C_{n,l}\left(\frac{1}{\sqrt{2M}}\right)^{n-l-1}\hat{r}^{-(n+l)}e^{\frac{(n+l+1)\hat{r}}{n(l+1)a_0}} \times\left[\frac{1}{\hat{r}}\hat{p}_r, e^{-\frac{\hat{r}}{(l+1)(l+2)a_0}}\left[\frac{1}{\hat{r}}\hat{p}_r, e^{-\frac{\hat{r}}{(l+2)(l+3)a_0}}\left[\frac{1}{\hat{r}}\hat{p}_r,\cdots\right.\right.\right.$$
$$\left.\left.\left.\cdots, e^{-\frac{\hat{r}}{(n-2)(n-1)a_0}}\left[\frac{1}{\hat{r}}\hat{p}_r, e^{-\frac{(2n-1)\hat{r}}{(n-1)n}}\hat{r}^{2n-1}\right]\cdots\right]\right]\right]_{n-l-1}|n,l{=}n-1\rangle. \tag{11.274}$$

The case $l = n-1$ is trivial and gives $|n,l{=}n-1\rangle$ (since $C_{n,n-1} = 1$). The case $l = n-2$ gives

$$|n,l{=}n-2\rangle = C_{n,n-2}\frac{1}{\sqrt{2M}}\hat{r}^{-(2n-1)}e^{\frac{(2n-1)\hat{r}}{(n-1)na_0}}\left[\hat{p}_r, e^{-\frac{(2n-1)\hat{r}}{(n-1)na_0}}\hat{r}^{2n-1}\right]|n,l{=}n-1\rangle$$
$$= C_{n,n-2}\left(-\frac{i\hbar}{\sqrt{2M}}\right)\frac{2n-1}{2n-2}\frac{1}{\hat{r}}\left(2n-2-\frac{2\hat{r}}{na_0}\right)|n,l{=}n-1\rangle. \tag{11.275}$$

In Problem 11.8.8, you will be asked to work out the next two cases.

After one has worked out a few cases, it becomes clear that this procedure is straightforward, but very tedious to carry out. So, as before, we search for a way to find a recurrence relation for the nested commutators, by defining them as a polynomial indexed by two integers. After a lot of staring and thinking, we can determine what the general result is. This becomes the hypothesis for a proof by induction, which reads

$$\left[\frac{1}{\hat{r}}\hat{p}_r, e^{-\frac{\hat{r}}{(l+1)(l+2)a_0}}\left[\frac{1}{\hat{r}}\hat{p}_r, e^{-\frac{\hat{r}}{(l+2)(l+3)a_0}}\left[\frac{1}{\hat{r}}\hat{p}_r,\cdots\right.\right.\right.$$

$$\cdots, e^{-\frac{\hat{r}}{(n-2)(n-1)a_0}}\left[\tfrac{1}{\hat{r}}\hat{p}_r, e^{-\frac{(2n-1)\hat{r}}{(n-1)na_0}}\hat{r}^{2n-1}\right]\cdots\Big]\Big]\Big]_{n-l-1}$$

$$= \left(-\frac{i\hbar}{2}\right)^{n-l-1}\frac{(2n-1)!}{(n+l)!}\,\frac{(n-l-1)!\,l!}{(n-1)!}$$

$$\times \hat{r}^{2l+1}e^{-\frac{(n+l+1)\hat{r}}{n(l+1)a_0}}L_{n-l-1}^{(2l+1)}\left(\frac{2\hat{r}}{na_0}\right), \tag{11.276}$$

and can be viewed as defining the polynomial $L_{n-l-1}^{(2l+1)}\left(\frac{2\hat{r}}{na_0}\right)$—it will turn out that this is an associated Laguerre polynomial, but we do not know that just yet. The hard task in determining this formula is to find the correct constant factors that appear in front. It is a bit of trial and error to land on the right result. The order of the associated Laguerre polynomial is given by the number of raising operators, or the nesting of the commutators, both of which are equal to $n-l-1$; this is the subscript index for the associated Laguerre polynomial. For the superscript, it is given by twice the angular momentum plus one. It is not obvious why it should be that at this time, but this is what it is. We summarize the first few polynomials in Table 11.1. Using the table, we see that the $l = n-1$ case involves the Laguerre polynomial $L_0^{(2n-1)}$, while the $l = n-2$ case involves $L_1^{(2n-3)}$. These are the first two terms of our recurrence relation.

We use a similar strategy as before to establish that the nested commutators are the associated Laguerre polynomials. We take the $n-l$-fold nested commutator

$$\left[\tfrac{1}{\hat{r}}\hat{p}_r, e^{-\frac{\hat{r}}{l(l+1)a_0}}\left[\tfrac{1}{\hat{r}}\hat{p}_r, e^{-\frac{\hat{r}}{(l+1)(l+2)a_0}}\left[\tfrac{1}{\hat{r}}\hat{p}_r, \cdots\right.\right.\right.$$

$$\left.\left.\left.\cdots, e^{-\frac{\hat{r}}{(n-2)(n-1)a_0}}\left[\tfrac{1}{\hat{r}}\hat{p}_r, e^{-\frac{(2n-1)\hat{r}}{(n-1)na_0}}\hat{r}^{2n-1}\right]\cdots\right]\right]\right]_{n-l}, \tag{11.277}$$

and rewrite it using the replacement of the $n-l-1$ nested commutators by their corresponding polynomial (from the induction step), which gives us

$$|n, l-1\rangle = C_{n,l-1}\frac{1}{\sqrt{2M}}\left(-\frac{i\hbar}{\sqrt{2M}}\right)^{n-l-1}\left(\frac{1}{2}\right)^{n-l-1}\frac{(2n-1)!}{(n+l)!}\,\frac{(n-l-1)!\,l!}{(n-1)!}$$

$$\hat{r}^{-(n+l)}e^{\frac{(n+l)\hat{r}}{nla_0}}\left[\hat{p}_r, \hat{r}^{2l+1}\,e^{-\frac{(n+l)\hat{r}}{nla_0}}L_{n-l-1}^{(2l+1)}\left(\frac{2\hat{r}}{na_0}\right)\right]|n, l{=}n-1\rangle, \tag{11.278}$$

after some algebra to simplify the expression. The Leibniz rule tells us that the commutator has three terms in it. We use the fact that

$$\left[\hat{p}_r, L_{n-l-1}^{(2l+1)}\left(\tfrac{2\hat{r}}{na_0}\right)\right] = i\hbar\,\frac{2}{na_0} L_{n-l-2}^{(2l+2)}\left(\tfrac{2\hat{r}}{na_0}\right), \tag{11.279}$$

which is established by writing the Laguerre polynomial out in its power series, and taking the commutator term by term, and finally, rewriting as an associated Laguerre polynomial using its definition; you will verify this in Problem 11.8.9. Then we find that

$$\begin{aligned}|n, l-1\rangle = C_{n,l-1}\tfrac{1}{\sqrt{2M}}\left(-\frac{i\hbar}{\sqrt{2M}}\right)^{n-l-1}\left(\frac{1}{2}\right)^{n-l-1}\frac{(2n-1)!}{(n+l)!}\,\frac{(n-l-1)!\,l!}{(n-1)!}\\ \hat{r}^{-(n-l)}\left((2l+1)L_{n-l-1}^{(2l+1)}\left(\frac{2\hat{r}}{na_0}\right) - \frac{(n+l)\hat{r}}{nla_0}L_{n-l-1}^{(2l+1)}\left(\frac{2\hat{r}}{na_0}\right)\right.\\ \left. - \frac{2\hat{r}}{na_0}L_{n-l-2}^{(2l+2)}\left(\frac{2\hat{r}}{na_0}\right)\right)|n, l{=}n-1\rangle.\end{aligned} \tag{11.280}$$

Factoring out $\frac{1}{2l}$ and simplifying, we find

$$\begin{aligned}|n, l-1\rangle = C_{n,l-1}\left(-\frac{i\hbar}{\sqrt{2M}}\right)^{n-l}\left(\frac{1}{2}\right)^{n-l}\frac{(2n-1)!}{(n+l)!}\,\frac{(n-l-1)!\,(l-1)!}{(n-1)!}\\ \hat{r}^{-(n-l)}\left((2l)(2l+1)L_{n-l-1}^{(2l+1)}\left(\frac{2\hat{r}}{na_0}\right) - (n+l)\,\frac{2\hat{r}}{na_0}L_{n-l-1}^{(2l+1)}\left(\frac{2\hat{r}}{na_0}\right)\right.\\ \left. - (2l)\,\frac{2\hat{r}}{na_0}L_{n-l-2}^{(2l+2)}\left(\tfrac{2\hat{r}}{na_0}\right)\right)|n, l{=}n-1\rangle.\end{aligned} \tag{11.281}$$

Using $x = \frac{2\hat{r}}{na_0}$, the terms inside the large parentheses can be summarized by

$$(2l)(2l+1)L_{n-l-1}^{(2l+1)}(x) - (n+l)xL_{n-l-1}^{(2l+1)}(x) - (2l)xL_{n-l-2}^{(2l+2)}(x). \tag{11.282}$$

We now manipulate them using the associated Laguerre polynomial recurrence relations. First we replace $-xL_{n-l-2}^{(2l+2)}(x)$ by $(n-l)L_{n-l}^{(2l)}(x) - (2l+1-x)L_{n-l-1}^{(2l+1)}(x)$, which transforms the expression to $(n-l)\big(2lL_{n-l}^{(2l)}(x) - xL_{n-l-1}^{(2l+1)}(x)\big)$. Next, we use the same identity again, in the form where we replace $-xL_{n-l-1}^{(2l+1)}(x)$ by $(n-l+1)L_{n-l+1}^{(2l-1)}(x) - (2l-x)L_{n-l}^{(2l)}(x)$. This transforms the expression to

$$(n-l)\left(2lL_{n-l}^{(2l)}(x) + (n-l+1)L_{n-l+1}^{(2l-1)}(x) - (2l-x)L_{n-l}^{(2l)}(x)\right). \tag{11.283}$$

We use another identity, which replaces $(n-l+1)L_{n-l-1}^{(2l-1)}(x)$ by $(n+l)L_{n-l}^{(2l-1)}(x) - xL_{n-l}^{(2l)}(x)$. Substituting this in gives our final result $(n-l)(n+l)L_{n-l}^{(2l-1)}(x)$. Replacing the nested commutator terms in the expression for $|n, l-1\rangle$ then yields

$$|n, l-1\rangle = C_{n,l-1}\left(-\frac{i\hbar}{\sqrt{2M}}\right)^{n-l}\left(\frac{1}{2}\right)^{n-l}\frac{(2n-1)!}{(n+l-1)!}\frac{(n-l)!\,(l-1)!}{(n-1)!}$$
$$\times L_{n-l}^{(2l-1)}\left(\frac{2\hat{r}}{na_0}\right)|n, l{=}n-1\rangle, \tag{11.284}$$

which is exactly what is needed to complete the proof by induction. This completes the identity to replace the product of raising operators by an operator function of $\hat{r}$; but see Problem 11.8.10 for one final rearrangement including all numerical factors. If we compare this to the isotropic oscillator, we see that the subscript of the Laguerre polynomial is the same for both (given by the number of raising operators applied to the state and determining the degree of the polynomial) and the upper parameters are related by a factor of 2—for the isotropic oscillator, it is one half that of the Coulomb problem—but, interestingly, the argument of the polynomial is a quadratic for the isotropic oscillator and is linear for the Coulomb problem, so they do share a number of similarities. This was a lot of work, and you deserve a break before moving on!

The hydrogen energy eigenvalue problem has one more twist. The bound states that we found earlier are *not* the only energy eigenstates. There exist energy eigenstates that are unbound and have a continuum of energies with $E > 0$. Hence, we have *both* discrete energy eigenvalues (corresponding to the bound states) and a continuum of energy eigenvalues (corresponding to the unbound states). This scenario is generic for potentials that do not go to infinity as the radial coordinate goes to infinity. We will not solve for these continuum energy eigenstates here. It is virtually impossible to do that without calculus. The video The Coulomb Problem for Hydrogen https://tinyurl.com/2rpuzj6k illustrates the bound-state calculation.

Section 11.8 Problems:

11.8.1 Standard Schrödinger Factorization Method for Hydrogen

In this problem, we work through the standard Schrödinger approach for the factorization method. This determines the energy eigenstates for constant l, as opposed to our shortcut method, which found them for constant n. We have

$$\hat{B}_0^{(l)} = \frac{1}{\sqrt{2M}}\left(\hat{p}_r - \frac{i\hbar}{(l+1)a_0} + \frac{i\hbar(l+1)}{\hat{r}}\right), \tag{11.285}$$

and

$$\hat{H}_l = \hat{H}_0^{(l)} = \hat{B}_0^{(l)\dagger}\hat{B}_0^{(l)} + E_0^{(l)}. \tag{11.286}$$

Here, the superscript (l) denotes the angular momentum value, while the subscript is the link in the chain (starting from 0, as is customary). Compute $E_0^{(l)}$ and verify that the factorization works, as claimed. Note that the lowest energy state $|\psi_{n=l+1,l}\rangle$ satisfies $\hat{B}_0^{(l)}|\psi_{n=l+1,l}\rangle = 0$.

We next construct our first auxiliary Hamiltonian

$$\hat{H}_1^{(l)} = \hat{B}_0^{(l)}\hat{B}_0^{(l)\dagger} + E_0^{(l)} = \hat{B}_1^{(l)\dagger}\hat{B}_1^{(l)} + E_1^{(l)}. \tag{11.287}$$

Determine $\hat{H}_1^{(l)}$ and how to factorize it in terms of the auxiliary ladder operators. Also compute $E_1^{(l)}$. The auxiliary ground state is $|\phi_1^{(l)}\rangle$, which satisfies $\hat{B}_1^{(l)}|\phi_1^{(l)}\rangle = 0$. The first excited state of $\hat{H}_l$ then becomes (unnormalized) $\hat{B}_0^{(l)\dagger}|\phi_1^{(l)}\rangle$. Show that this construction does produce an eigenstate of $\hat{H}_l$ (you will need to work with the standard intertwining relation to do so) and that $E_{n=l+2,l} = E_1^{(l)}$.

Note that this is a book-keeping exercise because all of the algebra has already been done before. Work out how the chain continues, determining the energy eigenstates and the energies of the rest of the states with fixed l. Describe how one can re-express the eigenstates in terms of a Laguerre polynomial operators acting on auxiliary Hamiltonian ground states, including the appropriate normalization factors. This requires almost no algebra on your part, just a comparison with what we did before.

11.8.2 The Two-Dimensional Coulomb Problem

Consider the two-dimensional Coulomb problem (note that this is not the standard Coulomb interaction in two dimensions, but this is what the problem is called), given by

$$\hat{H} = \frac{\hat{p}_\rho^2}{2M} + \frac{\hat{L}_z^2 - \frac{\hbar^2}{4}}{2M\hat{\rho}^2} - \frac{e^2}{4\pi\epsilon_0\hat{\rho}}. \tag{11.288}$$

Solve this problem for the bound-state energy spectrum using the factorization method. Determine the energy eigenstates only as products of raising operators acting on specific states. If you get stuck, consult Rushka et al. (2021). Note that this is a long problem.

11.8.3 Degeneracy of Energy Eigenvalues

If we look at the energy diagram, we see that the ground state for angular momentum l is degenerate with one of the excited states for all angular momentum values less than l. Recall that each angular momentum state is $2l + 1$-fold degenerate, from

the different m quantum numbers. For example, the $l = 2$ state is $2 \times 2 + 1 = 5$-fold degenerate from m. Similarly, $l = 1$ is threefold degenerate and $l = 0$ is not degenerate. This gives a total number of degenerate states for $n = 3$ as $5 + 3 + 1 = 9 = 3^2$. Show that in general, for a given n value, the degeneracy is n^2.

11.8.4 Hydrogenic Ions

In the text, we computed the the energy levels for hydrogen. Retrace the the steps and perform the calculations for the energies again, now including the Z atomic number factor for hydrogenic ions (atoms that have had all electrons stripped from them except for one)

How large does Z need to be before the energy of the ground state is equal to $m_e c^2 \approx 511$ KeV? It is for atoms of this atomic number and higher where relativistic effects become quite important.

11.8.5 Photoproduction of Hydrogen

In this problem, we investigate some of the transitions giving rise to spectral lines emitted from hydrogen. The most prominent transitions in hydrogen have a selection rule that says the l value changes by ± 1. Hence, a transition from a 2s state to a 1s state is said to be *forbidden* (which is a poor word choice, because it can occur as a two-photon process, which we will discuss later). A transition from 2p to 1s is *allowed*. Transitions that end in 1s are called the *Lyman series* and are in the ultraviolet; those that end in 2s are called the *Balmer series*, and they are in the visible, and those that end in 3s are called the *Paschen series* and are in the infrared. Higher series exist too (4s is the Brackett series, 5s is the Pfund series, and 6s is the Humphreys series). After 6s, they are no longer named. In this problem, we will investigate some of the transitions in the Lyman, Balmer, and Paschen series. This is illustrated in Fig. 11.12, which is called a *Grotrian diagram*.

Compute the energies, frequency in Hz (not the angular frequency), and the wavelength for the following transitions: Lyman (2p-1s, 3p-1s, 4p-1s); Balmer (3p-2s, 4p-2s, 5p-2s); and Paschen (4p-3s, 5p-3s, 6p-3s). In addition, tell us whether the light is ultraviolet, visible, or infrared. As an example, we do the calculation for the first one for you. In the 2p-1s transition, you should find the energy is 10.204 eV, the frequency is 2.468×10^{15} Hz, and the wavelength is 121.5 nm. This is in the ultraviolet regime.

Use constants in the following form with $m_p \to \infty$: $\varepsilon_n = -\frac{e^2}{8\pi\epsilon_0 n^2} = -\frac{13.6057}{n^2}$ eV; $2\pi\hbar f = 2\pi\hbar c \frac{f}{c} = \frac{2\pi\hbar c}{\lambda} = \frac{12400 \text{ ev-Å}}{\lambda}$; and $c = 2.998 \times 10^{18}$ Å/s. Find a resource to help you classify when light is ultraviolet, visible, and infrared.

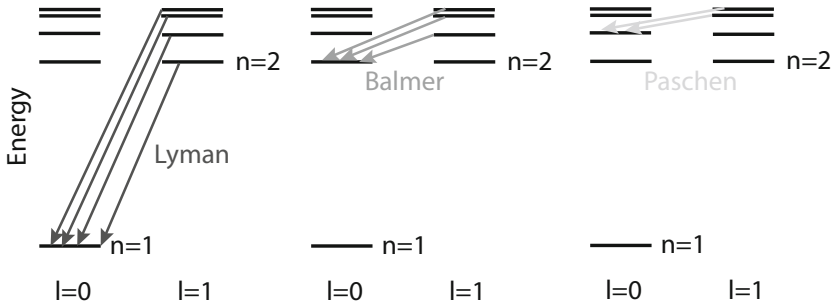

Fig. 11.12 Grotrian diagrams for the photoproduction of hydrogen. The arrows indicate the allowed transitions for the highest intensity spectral lines. We show just the first three series of spectral lines (Lyman, Balmer, and Paschen)

11.8.6 Proton and Deuteron Masses

(a) When the proton mass is taken to be infinitely large, the 1s energy level of hydrogen is given by $E_{1s}(m_p \to \infty) = -13.605693123$ eV. Please verify this result using $E_{1s}(m_p \to \infty) = -\frac{e^2}{8\pi\epsilon_0 a_0} = -\frac{m_e e^4}{32\pi^2\epsilon_0^2\hbar^2}$. Note that it is linear in the mass of the effective single particle (here it is m_e). When $m_p \to \infty$, we have $M \to m_e$, with $m_e c^2 = 510998.9500$ eV. But, for hydrogen, we have $M = \frac{m_e m_p}{m_e+m_p} = \frac{m_e}{1+\frac{m_p}{m_e}}$. The measured value for one of the two possible 3p to 2s transitions is given by light with a frequency of $456.681555 \times 10^{12}$ Hz. This can be converted to electron volts by using the relations

$$2\pi\hbar c = 12398.4198 \text{ ev-Å} \quad \text{and} \quad c = 2.99792458 \times 10^{18} \text{ Å/s}. \tag{11.289}$$

The way to do it is via $E_{3p} - E_{2s} = 2\pi\hbar f = 2\pi\hbar c\frac{f}{c}$, so we find that

$$E_{3p} - E_{2s} = 12398.4198 \text{ ev-Å} \times \frac{1}{2.99792458 \text{ Å/s}} \times 456.68155 \times 10^{12}\text{s}^{-1}, \tag{11.290}$$

or $E_{3p} - E_{2s} = 1.88868313$ eV, where the last digit is uncertain. In our theory, we are ignoring fine-structure effects (although we did take real experimental data for one of the fine-structure transitions). This can be improved using more accurate formulas that include fine structure, as we will see later when we discuss relativistic effects. Determine the mass of the proton from this measurement. How does it compare to the accepted value of 938.2720882×10^6 eV=$m_p c^2$. The error comes primarily from our not using the proper formula for the fine-structure corrections and the Lamb shift. We will discuss these corrections later. I do not know about you, but I am amazed at how accurately one can "weigh" the proton by determining the energy of the light emitted by hydrogen!

(b) How does the analysis change if we want to detect a deuteron? Using $m_D c^2 = 1875.612942$ MeV, determine the change in frequency of light for the 3p-2s transition. Use the values given in part (a) for $E_{1s}(m_p \to \infty)$ $2\pi\hbar c$, and c to compute your answer. Find both the change in frequency and the change in wavelength for a hydrogen atom versus a deuterium atom. You should find the difference in wavelength is on the order of 2 Å. This wavelength difference (on the order of one part in 10,000) could be measured in the 1930s. The problem was seeing the signal because the natural occurrence of deuterium is in an abundance of 0.0156% (it was known the abundance was low, but not how low before this was first measured). So, the challenge was to increase the abundance of deuterium to get the signal high enough that it can be seen. This is easiest to accomplish by electrochemistry, which can increase the percentage of heavy water. But, when Harold Urey first did this experiment in 1931, he enhanced the D concentration by evaporating liquid hydrogen near its melting point. The lighter H evaporates more rapidly, leaving behind the heavier D. An interesting historical account appears in Brickwedde (1982). Urey won the Nobel prize for this work!

If you have ever heard of mass spectrometry, you might ask why not do that to find Deuterium? The problem is whenever you have hydrogen around, you also have the hydrogen molecule, and that will have such a large signal in a mass spectrometer, that it will completely wipe out any signal coming from D, so a different approach was required to make the discovery.

11.8.7 Testing the Validity of Coulomb's Law

One of the possible uses of high-precision measurements of the spectrum of hydrogen is to test the validity of Coulomb's law. Suppose the potential was actually of the form

$$-\frac{e^2}{4\pi\epsilon_0\hat{r}}\left(1+\delta\frac{a_0}{\hat{r}}\right) \tag{11.291}$$

with $\delta \ll 1$. Use this form for the potential and rewrite the effective Hamiltonian (after acting on the angular momentum part of the tensor product) in the form

$$\hat{H}_l = \frac{\hat{p}_r^2}{2M} + \frac{\hbar^2\lambda(\lambda+1)}{2M\hat{r}^2} - \frac{e^2}{4\pi\epsilon_0\hat{r}}, \tag{11.292}$$

where you need to determine λ as a function of l and δ, which approaches l as $\delta \to 0$. You must solve this problem using the conventional Schrödinger factorization method to get the energy eigenvalues, because it does not have the symmetry needed for the short-cut method.

Hint: Use the lowering operator $\hat{B}_\lambda$ in your construction, where we let $l \to \lambda$, which is no longer an integer. Show that the energy eigenstates satisfy

$$\varepsilon_{n,l}(\delta) = -\frac{e^2}{8\pi\epsilon_0 a_0} \frac{1}{\left[n + \sqrt{\left(l+\frac{1}{2}\right)^2 - 2\delta} - \left(l+\frac{1}{2}\right)\right]^2}, \tag{11.293}$$

and that the correction to the energy, for small δ is

$$\varepsilon_{n,l}(\delta) - \varepsilon_n = -\frac{e^2}{4\pi\epsilon_0 a_0} \frac{\delta}{n^3\left(l+\frac{1}{2}\right)} + O(\delta^2). \tag{11.294}$$

You may find the expansion $(1+\epsilon)^\alpha = 1 + \alpha\epsilon + \cdots$ to be helpful for small ϵ.

Note how the energy now depends on both n and l here, because we have broken the extra symmetry that led to the extra degeneracy for $\delta = 0$. Indeed, you should find there no longer is any degeneracy when $\delta \neq 0$.

11.8.8 Calculation of Low-Order Laguerre Polynomials for Energy Eigenstates

In the text, we computed the first two nontrivial Laguerre polynomials by explicitly calculating the nested commutators. Continue this exercise to compute the third and fourth nontrivial Laguerre polynomial by continuing the procedure two more steps. Be careful, it is very easy to make a mistake and the calculations do become long.

11.8.9 Verifying the Associated Laguerre Polynomial Identities

The associated Laguerre polynomial satisfies

$$L_n^{(\alpha)}(x) = \sum_{m=0}^{n} (-1)^m \frac{(n+\alpha)!}{(n-m)!(m+\alpha)!} \frac{x^m}{m!}, \tag{11.295}$$

and in cases where α is not an integer, we evaluate the factorials via the corresponding gamma function; for hydrogen, we always have that α is an integer.

Use this definition to prove the following four identities that we will need. The first is the commutator identity with the radial momentum:

$$\left[\hat{p}_r, L_n^{(\alpha)}\left(\frac{2\hat{r}}{na_0}\right)\right] = i\hbar\frac{2}{na_0} L_{n-1}^{(\alpha+1)}\left(\frac{2\hat{r}}{na_0}\right). \tag{11.296}$$

The second is

$$-xL_{n-l-2}^{(2l+2)}(x) = (n-l)L_{n-l}^{(2l)}(x) - (2l+1-x)L_{n-l-1}^{(2l+1)}(x). \tag{11.297}$$

The third is

$$-xL_{n-l-1}^{(2l+1)}(x) = (n-l+1)L_{n-l+1}^{(2l-1)}(x) - (2l-x)L_{n-l}^{(2l)}(x). \tag{11.298}$$

The fourth and final one is

$$(n-l+1)L_{n-l-1}^{(2l-1)}(x) = (n+l)L_{n-l}^{(2l-1)}(x) - xL_{n-l}^{(2l)}(x). \tag{11.299}$$

All of these identities can be proven by manipulating the definition of the associated Laguerre polynomials using the power series aforementioned. You may find it easier to work from one side or the other and recall the add-zero trick, as it will be needed many times to get summations to extend over the proper ranges.

11.8.10 Normalization of the Final Identity for the Energy Eigenstates

Use the explicit value of $C_{n,l}$ to show that

$$|n,l\rangle = (-i)^{n-l-1}\left(\frac{na_0}{2\hat{r}}\right)^{n-l-1}\sqrt{\frac{(2n-1)!(n-l-1)!}{(n+l)!}} \times L_{n-l-1}^{(2l+1)}\left(\frac{2\hat{r}}{na_0}\right)|n,l{=}n-1\rangle. \tag{11.300}$$

This is our final formula for the product of the raising operators acting on the ground state for $l = n-1$. Because a quantum state can have any global phase applied to it, without changing the state, we usually remove the $(-i)^{n-l-1}$ factor. In the next section, we will use this identity to determine the position-space wavefunctions for hydrogen.

11.9 Coordinate Wavefunctions for Hydrogen

Now, we determine the position-space wavefunctions for hydrogen. They are given by $\psi_{nlm}(r,\theta,\phi) = (i)^{n-l-1}\langle r| \otimes \langle\theta,\phi| \cdot |n,l\rangle \otimes |l,m\rangle$, where $|n,l\rangle$ is the radial energy eigenstate we derived in the previous section, and we added a global phase factor to agree with the standard form of the wavefunction. If we note that $Y_{lm}(\theta,\phi) = \langle\theta,\phi|l,m\rangle$, we can re-express the wavefunction as

Table 11.4 First few associated Laguerre polynomials that appear in the wavefunctions for hydrogen. One can see they become more and more cumbersome as the lower index is increased

m	$L_m^{(2l+1)}(x)$
0	1
1	$2l+2-x$
2	$\frac{1}{2}(2l+3)(2l+2)-(2l+3)x+\frac{x^2}{2}$
3	$\frac{1}{6}(2l+4)(2l+3)(2l+2)-\frac{1}{2}(2l+4)(2l+3)x+\frac{1}{2}(2l+4)x^2-\frac{x^3}{6}$

$$\begin{aligned}\psi_{nlm}(r,\theta,\phi) &= C_{n,l}\langle r|\hat{B}_l^\dagger \hat{B}_{l+1}^\dagger \cdots \hat{B}_{n-2}^\dagger |n, l{=}n-1\rangle Y_{lm}(\theta,\phi)\\ &= \langle r|\sqrt{\frac{(2n-1)!(n-l-1)!}{(n+l)!}}\left(\frac{na_0}{2\hat{r}}\right)^{n-l-1} L_{n-l-1}^{(2l+1)}\left(\frac{2\hat{r}}{na_0}\right)|n, l{=}n-1\rangle\\ &\times Y_{lm}(\theta,\phi)\\ &= \sqrt{\frac{(2n-1)!(n-l-1)!}{(n+l)!}}\left(\frac{na_0}{2r}\right)^{n-l-1} L_{n-l-1}^{(2l+1)}\left(\frac{2r}{na_0}\right)\langle r|n, l{=}n-1\rangle\\ &\times Y_{lm}(\theta,\phi), \end{aligned} \tag{11.301}$$

where the last line follows because $\langle r|\hat{r} = \langle r|r$ since the bra is an eigenstate for the radial position operator. We summarize the first few associated Laguerre polynomials in the factorization chain in Table 11.4. This leaves us with the task of computing the normalized ground-state wavefunction for each angular momentum l in position space; that is, the first state in the chain (to the right) given by $|n, l{=}n-1\rangle$.

To do this, we use the translation operator in spherical coordinates. But first, we need to work on the subsidiary condition to find the state that is annihilated by $\hat{p}_r + \frac{i\hbar}{\hat{r}}$. The subsidiary condition tells us that

$$\hat{B}_{n-1}|n, l{=}n-1\rangle = 0, \text{ or } \hat{p}_r|n, n-1\rangle = \left(-\frac{i\hbar n}{\hat{r}} + \frac{i\hbar}{na_0}\right)|n, n-1\rangle. \tag{11.302}$$

Moving one term of $\frac{i\hbar}{\hat{r}}$ to the left then gives

$$\left(\hat{p}_r + \frac{i\hbar}{\hat{r}}\right)|n, n-1\rangle = i\hbar\left(-\frac{n-1}{\hat{r}} + \frac{1}{na_0}\right)|n, n-1\rangle. \tag{11.303}$$

Next, we find the state annihilated by this operator. It is

$$\left(\hat{p}_r + \frac{i\hbar}{\hat{r}}\right)\hat{r}^{-(n-1)}e^{\frac{\hat{r}}{na_0}}|n, n-1\rangle = 0, \tag{11.304}$$

which you should be able to directly verify. Now the computation of the ground-state wavefunction proceeds by using the multiply-by-one trick (which is employed to create the state that $\hat{p}_r + \frac{i\hbar}{\hat{r}}$ annihilates):

$$\begin{aligned}
\psi_{n,n-1}(r) = \langle r|n, n-1\rangle &= \langle r| \underbrace{\hat{r}^{n-1} e^{-\frac{\hat{r}}{na_0}}}_{\text{operate on } \langle r|} \hat{r}^{-(n-1)} e^{\frac{\hat{r}}{na_0}} |n, n-1\rangle \\
&= r^{n-1} e^{-\frac{r}{na_0}} \langle r|\hat{r}^{-(n-1)} e^{\frac{\hat{r}}{na_0}} |n, n-1\rangle \\
&= r^{n-1} e^{-\frac{r}{na_0}} \langle 0_r| e^{\frac{i}{\hbar} r \left(\hat{p}_r + \frac{i\hbar}{\hat{r}}\right)} \underbrace{\hat{r}^{-(n-1)} e^{\frac{\hat{r}}{na_0}} |n, n-1\rangle}_{\text{annihilated by } \hat{p}_r + \frac{i\hbar}{\hat{r}}} \\
&= r^{n-1} e^{-\frac{r}{na_0}} \langle 0_r|\hat{r}^{-(n-1)} e^{\frac{\hat{r}}{na_0}} |n, n-1\rangle \\
&= r^{n-1} e^{-\frac{r}{na_0}} \langle 0_r|\hat{r}^{-(n-1)} |n, n-1\rangle. \qquad (11.305)
\end{aligned}$$

This is the final form. The term $\langle 0_r| \frac{1}{\hat{r}^{n-1}} |n, n-1\rangle$ is the normalization constant, because the ground-state wavefunction with $l = n-1$, behaves like $r^l = r^{n-1}$ as $r \to 0$ (as can be easily seen by the form above). The normalization constant can be easily determined, but it requires calculus. The result is

$$\langle 0_r| \frac{1}{\hat{r}^{n-1}} |n, n-1\rangle = \left(\frac{2}{na_0}\right)^{n+\frac{1}{2}} \frac{1}{\sqrt{(2n)!}}. \qquad (11.306)$$

Finally, putting everything together, we find that

$$\begin{aligned}
\psi_{nlm}(r, \theta, \phi) = &\left(\frac{2}{na_0}\right)^{\frac{3}{2}} \sqrt{\frac{(n-l-1)!}{2n(n+l)!}} \left(\frac{2r}{na_0}\right)^l L_{n-l-1}^{(2l+1)}\left(\frac{2r}{na_0}\right) e^{-\frac{r}{na_0}} \\
&\times Y_{lm}(\theta, \phi), \qquad (11.307)
\end{aligned}$$

which is the standard form of the energy eigenstate wavefunctions for hydrogen. The wavefunctions for $n \leq 3$ are summarized in Table 11.5. Note that this uses the modern notation for the associated Laguerre polynomials, so it will disagree by a factor involving $(n + l)!$ from older-style textbooks; this is the form that appears in Wikipedia and in symbolic manipulation programs, such as mathematica. This more modern notation is also cleaner. The video Position Space Wavefunctions of Hydrogen https://tinyurl.com/mzk79dzj discusses how to do this calculation as well.

We plot the radial wavefunctions and probabilities given by $\psi_{nl}(r)$, $|\psi_{nl}(r)|^2$, and $r^2|\psi_{nl}(r)|^2$ for the s-wave ($l = 0$) (Fig. 11.13), p-wave ($l = 1$) (Fig. 11.14),

Table 11.5 Hydrogen wavefunctions in position space for $n \leq 3$

n	l	$\psi_{nlm}(r,\theta,\phi)$
1	0	$\frac{2}{a_0^{3/2}} e^{-\frac{r}{a_0}} Y_{00}(\theta,\phi)$
2	1	$\frac{1}{2\sqrt{6}a_0^{3/2}} \left(\frac{r}{a_0}\right) e^{-\frac{r}{2a_0}} Y_{1m}(\theta,\phi)$
2	0	$\frac{1}{\sqrt{2}a_0^{3/2}} \left(1-\frac{r}{2a_0}\right) e^{-\frac{r}{2a_0}} Y_{00}(\theta,\phi)$
3	2	$\frac{4}{81\sqrt{30}a_0^{3/2}} \left(\frac{r}{a_0}\right)^2 e^{-\frac{r}{3a_0}} Y_{2m}(\theta,\phi)$
3	1	$\frac{8}{27\sqrt{6}a_0^{3/2}} \left(1-\frac{r}{6a_0}\right) \left(\frac{r}{a_0}\right) e^{-\frac{r}{3a_0}} Y_{1m}(\theta,\phi)$
3	0	$\frac{2}{3\sqrt{3}a_0^{3/2}} \left[1-\frac{2r}{3a_0}+\frac{2}{27}\left(\frac{r}{a_0}\right)^2\right] e^{-\frac{r}{3a_0}} Y_{00}(\theta,\phi)$

d-wave ($l = 2$), and f-wave ($l = 4$) (Fig. 11.15) and grouped according to their angular momentum for $n \leq 4$ (we group the d-wave and f-wave curves together).

There is a lot of interesting behavior in these plots. To start, note that only the s-wavefunctions have nonzero amplitude at the origin (with the amplitude shrinking as n increases). The other angular momentum states all have wavefunctions that vanish at the origin (due to the r^l behavior). As n increases, we see the number of nodes increases. So, for each angular momentum, the lowest energy state (with $n = l + 1$) has no nodes (we do not count the zeros as $r \to 0$ or $r \to \infty$), and the number increases by 1 as n increases by 1. Looking at the square of the wavefunction, we find the probability density to be found in small cubes (for s-wave functions, the interpretation is more complicated for higher l states because of the angular dependence coming from Y_{lm}). The final panels show the probability density to be found in spherical shells of radius r. The probability densities push close to the origin for small n values and spread further out as n increases. Higher l states have a "centrifugal barrier" pushing them further from the origin as well. We will be able to put quantitative numbers to these qualitative discussions in Sect. 11.11, where we compute expectation values $\langle nl|\hat{r}^m|nl\rangle$. Finally, we can describe the behavior as $r \to 0$ and $r \to \infty$. For large r, the wavefunctions always go to zero. This is because of the exponential factor, suppresses the wavefunction faster than any polynomial in r grows. As $r \to 0$, the wavefunctions behave like r^l. They are "pushed" farther and farther from the origin as l increases. This is what we expect as the "centrifugal force" increases for higher angular momentum.

Many textbooks like to say the the two probability distributions (in position space) represent the likelihood to find an electron in a small cubical box (or on a spherical shell) in some region of space when we measure where the electron is in the atom. The only problem with this is that there is no known experiment that can measure the position of an electron within an atom. We have no such experimental device. Most experiments on single quanta require us to have a magnification step, which magnifies the quanta to the point where they can be observed. No such device exists to determine where an electron is in an atom. But, if the atom is placed in a strong electric field, careful experiments can magnify the probability density

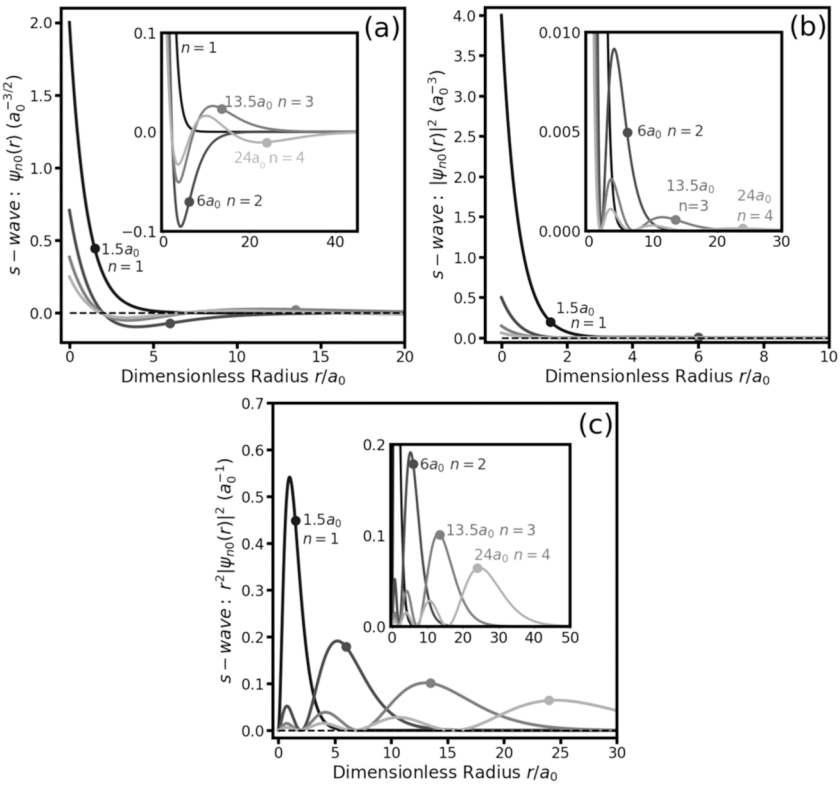

Fig. 11.13 (**a**) The wavefunction $\psi(r)$, (**b**) modulus squared $|\psi(r)|^2$ (probability density for cubes), and (**c**) probability density for spherical shells $r^2|\psi(r)|^2$ for $n = 1$, 2, 3, and 4 for s-wave orbitals ($l = 0$). The dot marks the average radius for the electron in the different energy eigenstates $\langle nl|\hat{r}|nl\rangle$. Note how it increases like n^2 and how it always lies close to the outermost peak in the curve in panel (**c**); the wavefunctions are all pushed to the right as n increases. All s-wave orbitals are nonzero at $r = 0$

profile to the point where it can be measured. The experimental technique is called *photoionization microscopy* (Stodolna et al., 2013). The experiment is challenging, and the theory to describe it goes beyond what we cover in this book. But, the results are marvelous. One can image the transverse nodal structure of hydrogen in an electric field. We don't show these results here, because they do not resemble the results for hydrogen in zero electric field. Experiments that measure the momentum probability distribution (in the absence of an electric field) do show the probability distribution for the electron in momentum space. We compute these profiles in the next section. The other items that can be measured, albeit indirectly, are the average radius and average radius squared. These enter into calculations of the lifetimes of different excited states and, hence, are measured via spectroscopy. We will discuss such averages in detail later in Sect. 11.11.

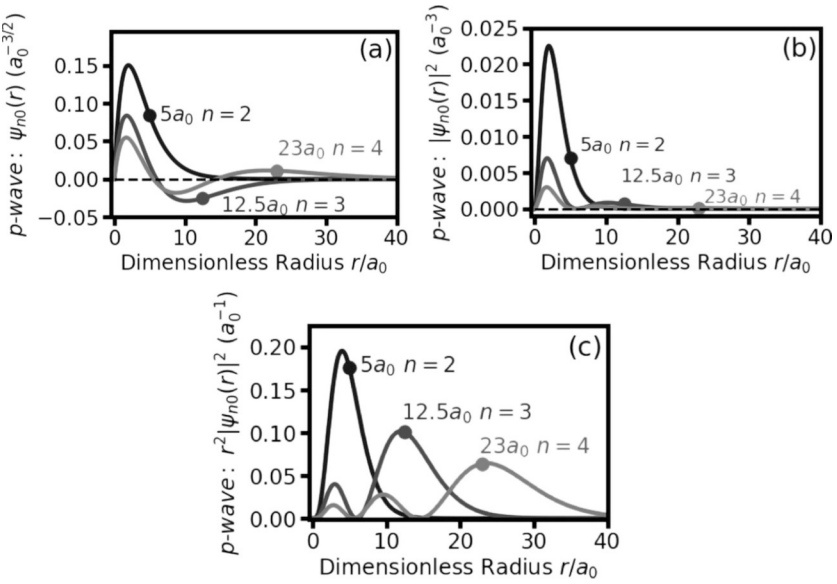

Fig. 11.14 (**a**) The wavefunction $\psi(r)$, (**b**) modulus squared $|\psi(r)|^2$ (probability density for cubes), and (**c**) probability density for spherical shells $r^2|\psi(r)|^2$ for $n = 2$, 3, and 4 for p-wave orbitals ($l = 1$). The dot marks the average radius again. Note how all these wavefunctions vanish linearly as $r \to 0$

We now discuss how the wavefunctions, energies, and everything else change when we examine hydrogenic ions (atoms with all electrons except one removed by ionization). The Hamiltonian only changes by $e^2 \to Ze^2$, with Z the charge of the nucleus in the hydrogenic ion. This can be translated into a modification of the Bohr radius $a_0 = \frac{4\pi\epsilon_0\hbar^2}{Me^2} \to \frac{a_0}{Z}$. The energy $\varepsilon_n = -\frac{e^2}{8\pi\epsilon_0 a_0 n^2}$ becomes $\varepsilon_n = -\frac{Z^2e^2}{8\pi a_0 n^2}$. In all other formulas (like the wavefunctions and ladder operators), one simply replaces $a_0 \to \frac{a_0}{Z}$ to find the hydrogenic results.

In addition to hydrogenic ions, there is a large zoo of other hydrogen-like atoms. This includes antimatter hydrogen (an antiproton and a positron), positronium (an electron and a positron), muonium (a positively charged muon and an electron), muonic hydrogen (a proton and a negatively charged muon), and even donor and acceptor states in semiconductors. We will discuss these exotic systems a bit more in Sect. 11.13, when we also discuss Rydberg atoms—hydrogen atoms in large n and large l states.

One might be disappointed that we have not discussed what an electron is "really doing" in an atom between the moments when we do experiments. This is a common misconception that we often bring with us from the classical world if we do not adopt a more quantum viewpoint. As we have seen many times before, we do not know what an electron is precisely doing when we are not measuring it. While such a statement seems obvious to make, it is particularly important in the quantum

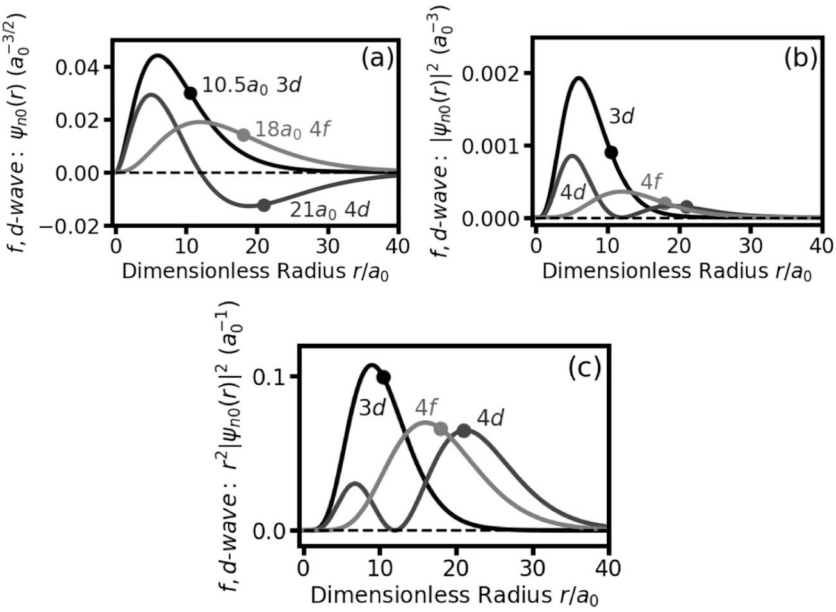

Fig. 11.15 The wavefunction $\psi(r)$, modulus squared $|\psi(r)|^2$ (probability density for cubes), and probability density for spherical shells $r^2|\psi(r)|^2$ for $n = 3$ and 4 for d- and f-wave orbitals ($l = 2$ and 3)

world, because if the electron is in a superposition, then the precise state it is in is indeterminate, and there is no way to know what it is doing in such a state. Similarly, other common questions are do electrons orbit like planets around a "nuclear sun" or do they move in these energy eigenstates? Again, there is no simple way to answer such questions. One the one hand, the probability distributions are static, because eigenstates are stable states with constant (in time) probability distributions. On the other hand, the electron cannot be motionless, as that would violate the uncertainty principle (a motionless particle is in a state with zero momentum and zero uncertainty in momentum, while it is localized where it sits motionless). Furthermore, if we could measure the position of an electron in an atom (which we cannot, as described earlier), we would see random positions for the measurements made on atoms prepared in the same state and at different times (since measurements destroy the state the electron is in, they need to be repeated in this fashion). Because of the probabilistic nature of these results, one would not see electrons moving in orbits, nor can one infer an average momentum from subsequent measurements. But, all is not lost. As we show in Problem 11.9.6, one can determine the average kinetic energy, and by taking the square root, the root-mean-square velocity. Doing this for hydrogen finds that it is on the order of one one-hundredth the speed of light! So the electrons "move" very fast indeed. Just not in the classical way you are imagining they do—and certainly not like little planets orbiting a "nuclear" sun. We

just don't have the best words to describe how this works. This is something you have already encountered—the molecules in the air in a room have zero average velocity, but the root-mean-square velocity is often about 1000 km per hour. We don't know exactly how those molecules are moving either.

We end this section by framing a discussion on how one can estimate the size of the proton from measurements of the light emitted by hydrogen. But first, we must discuss a technical result for how a small perturbation of the Hamiltonian will change the corresponding energy eigenvalues. Suppose the problem we wish to study is hydrogen plus a small change due to the finite size of the proton in the nucleus. We know this must be a small correction, because the nucleus is 100,000 times smaller than the atom. We write the full Hamiltonian as

$$\hat{H}(\lambda) = \hat{H}_0 + \lambda \hat{H}_1. \tag{11.308}$$

Here, λ is a small dimensionless number, and $\hat{H}_1$ is the additional operator that we have added to the Hamiltonian. Before the perturbation, we have $\hat{H}(0)|\psi_n(0)\rangle = \hat{H}_0|\psi(0)\rangle = E_n(0)|\psi_n(0)\rangle$, where we use a subscript n to denote the label for the energy eigenvalue. In our problem of interest, λ will have a specific value, but it is easier to consider the family of Hamiltonians $\hat{H}(\lambda)$ determined by varying the specific value of λ. Under the assumption that the energy eigenvalues are normalized for all λ, we have that

$$E_n(\lambda) = \langle \psi_n(\lambda)|\hat{H}(\lambda)|\psi_n(\lambda)\rangle, \tag{11.309}$$

with $\hat{H}(\lambda)|\psi_n(\lambda)\rangle = E_n(\lambda)|\psi_n(\lambda)\rangle$. We will now show that

$$E_n(\lambda) = E_n(0) + \lambda\langle \psi_n(0)|\hat{H}_1|\psi_n(0)\rangle + O(\lambda^2). \tag{11.310}$$

This says that initially the change in energy is determined by the expectation value of the perturbation with respect to the unperturbed energy eigenstates. Figure 11.16 shows that the eigenvector at λ is composed of a component along $|\psi_n(0)\rangle$ and a component perpendicular to $|\psi_n(0)\rangle$, which we call $|\Delta\psi_n(\lambda)\rangle$. Because $\langle \Delta\psi_n(\lambda)|\psi_n(0)\rangle = 0$, we find that we must have

$$|\psi_n(\lambda)\rangle = \sqrt{1-|\alpha|^2}|\psi_n(0)\rangle + |\Delta\psi_n(\lambda)\rangle, \tag{11.311}$$

(where $|\alpha|^2 = \langle \Delta\psi_n(\lambda)|\Delta\psi_n(\lambda)\rangle$) is at least of order λ (since $\alpha \to 0$ as $\lambda \to 0$, because $\lim_{\lambda\to 0}|\psi(\lambda)\rangle \to |\psi(0)\rangle$), and we expect the behavior to be linear in λ because the perturbation of the Hamiltonian is linear in λ. This means that α is $O(\lambda)$, and then to lowest order in λ, we have that $|\psi(\lambda)\rangle = |\psi(0)\rangle + \lambda|\Delta\psi(0)\rangle + O(\lambda^2)$, where $|\Delta\psi(0)\rangle = \lim_{\lambda\to 0}|\Delta\psi(\lambda)\rangle$. Now, we compute

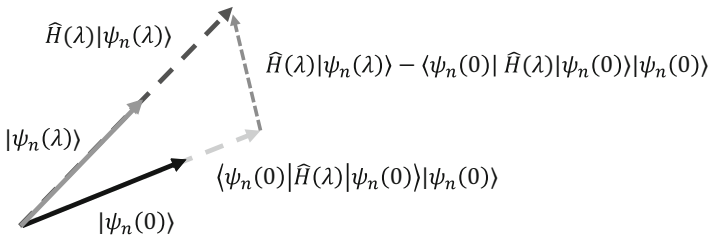

Fig. 11.16 Schematic for how an energy eigenstate $|\psi_n(0)\rangle$ (solid black arrow) is perturbed by a small change $\lambda\hat{H}_1$ added to the Hamiltonian, yielding the new eigenstate of $\hat{H}(\lambda) = \hat{H}_0 + \lambda\hat{H}_1$, which we denote by $|\psi_n(\lambda)\rangle$ (solid gray arrow). The component of the energy eigenvector (multiplied by the Hamiltonian) in the direction of $|\psi_n(0)\rangle$ is given by $\langle\psi_n(0)|\hat{H}(\lambda)|\psi_n(0)\rangle$ to first-order in λ (long-dashed light gray arrow). The remaining component of the energy eigenvector must be perpendicular to $|\psi_n(0)\rangle$ (short-dashed gray arrow) and called $|\Delta\psi(\lambda)\rangle$ in the text. We neglect terms of order λ^2 in the figure

$$\begin{aligned}
E_n(\lambda) &= \langle\psi_n(\lambda)|\big(\hat{H}_0 + \lambda\hat{H}_1\big)|\psi_n(\lambda)\rangle \\
&= \big(\langle\psi_n(0)| + \lambda\langle\Delta\psi(0)|\big)\big(\hat{H}_0 + \lambda\hat{H}_1\big)\big(|\psi(0)\rangle + \lambda|\Delta\psi(0)\rangle\big) \\
&= \langle\psi_n(0)|\hat{H}_0|\psi_n(0)\rangle + \lambda\langle\psi_n(0)|\hat{H}_0|\Delta\psi(0)\rangle + \lambda\langle\Delta\psi_n(0)|\hat{H}_0|\psi_n(0)\rangle \\
&+ \lambda\langle\psi_n(0)|\hat{H}_1|\psi_n(0)\rangle + O(\lambda^2) \\
&= E_n(0) + \lambda\langle\psi_n(0)|\hat{H}_1|\psi_n(0)\rangle + O(\lambda^2). \qquad (11.312)
\end{aligned}$$

Here, we used the facts that $\hat{H}_0|\psi_n(0)\rangle = E_n(0)|\psi_n(0)\rangle$ and $\langle\Delta\psi_n(0)|\psi_n(0)\rangle = 0$. This is then the result we wanted to show. We will see later in Sect. 12.7 that this result can be interpreted in terms of the variational principle. The result we have here is the result from first-order perturbation theory, so named, because it is the linear correction in λ; we have a video https://tinyurl.com/3adebkpm that discusses this a bit more formally and from a calculus point of view. We will not discuss higher-order perturbation theory in this book.

OK, now we get to apply this to the proton radius problem. We do not know anything *a priori* about how the nucleus is structured. We do know that hydrogen has a proton in its nucleus, and we also know an old experiment, called Rutherford scattering, which shows that the nucleus is approximately 100,000 times smaller than the atom. We also know it is charged and that the proton charge and the electron charge are exactly the same in magnitude, but opposite in sign. There is one more fact that we need to know from electromagnetism—the potential from a spherically symmetric charged sphere acts exactly like that of a point particle when one is outside of the sphere. This means that $V(\hat{r}) = -\frac{e^2}{4\pi\epsilon_0\hat{r}}$ for $\hat{r} > r_p$, the charge radius of the proton; do not conflate the proton charge radius with the momential

radius. What about inside the sphere? One choice would be to pick $V(\hat{r}) = -\frac{e^2}{4\pi\epsilon_0 r_p}$ for $\hat{r} < r_p$. That case corresponds to all of the charge lying on the surface of the sphere or radius r_p. We could pick instead

$$V(\hat{r}) = -\frac{e^2}{8\pi\epsilon_0 r_p}\left(3 - \frac{\hat{r}^2}{r_p^2}\right), \quad \text{for } \hat{r} < r_p. \tag{11.313}$$

This is the potential of a uniformly charged sphere of radius r_p. Other choices are possible too, if the charge distribution changes as we move through the sphere. The point is, we do not really know what the correct potential is. So, we can either find arguments independent of the specific form of the potential inside the sphere, or we can try a few different choices and see how they compare to experiment. Carrying out the full calculation will require calculus, but we can get very close to the correct final answer without using any calculus.

We structure the problem as $\hat{H} = \hat{H}_0 + \Delta\hat{H}$, with $\hat{H}_0$ equal to the ordinary hydrogen Hamiltonian and

$$\Delta\hat{H} = \theta(0 \le \hat{r} \le r_p)\left(-\frac{e^2}{4\pi\epsilon_0 r_p}\right)\Delta V(\hat{r}), \tag{11.314}$$

the perturbing potential. We choose $\Delta V(\hat{r}) = 1$ for the case where all the charge is on the surface of the sphere and $\Delta V(\hat{r}) = \frac{3}{2} - \frac{1}{2}\frac{\hat{r}^2}{r_p^2}$ for a uniformly charged sphere. Using the general result for the energy shifts that we derived earlier, we have

$$\varepsilon_{nl} = -\frac{e^2}{8\pi\epsilon_0 a_0 n^2} - \frac{e^2}{4\pi\epsilon_0 r_p}\langle n, l|\theta(0 \le \hat{r} \le r_p)\Delta V(\hat{r})|n, l\rangle. \tag{11.315}$$

Note that $r_p \ll a_0$. This means we can take the wavefunction to be a constant value, equal to $\psi_{nl}(0)$ when we calculate the expectation value. Hence, only s-wave energy levels $l = 0$ will be shifted by the finite size of the nucleus, because higher angular momentum states have a vanishing wavefunction at $r = 0$. The expectation value is then multiplied by the volume of a sphere the size of the nucleus $\left(\frac{4}{3}\pi r_p^3\right)$ (this is the case of thinking of computing the probability from the probability density of a small sphere centered at the origin and of the size of the nucleus). Then, there will be a correction factor c, which is of order 1, that takes into account the distribution of the charge inside the nucleus. For a spherical shell of charge all on the surface, we have $c = 1$, while for a uniform sphere of charge we have $c = \frac{6}{5}$. Hence, one can see the constant does not depend strongly on the charge distribution. In the problems, you will show for s-wave states that $|\psi_{n00}(0)|^2 = \frac{1}{\pi}\frac{1}{n^3 a_0^3}$. This means the shift in energy is

$$\varepsilon_{n00} = -\frac{e^2}{8\pi\epsilon_0 n^2 a_0} - \frac{e^2}{4\pi\epsilon_0 r_p}\frac{4}{3}c\left(\frac{r_p}{n a_0}\right)^3 = -\frac{e^2}{8\pi\epsilon_0 n^2 a_0}\left(1 + \frac{4}{3}c\frac{1}{n}\left(\frac{r_p}{a_0}\right)^2\right). \tag{11.316}$$

It is customary to assume the charge is distributed uniformly through the sphere with $c = \frac{6}{5}$. The shift in energy of the ground state is then approximately 13.6 eV $\frac{8}{5}\left(\frac{r_p}{a_0}\right)^2$, which is of the order of 10^{-9} eV for $\frac{r_p}{a_0} = 0.00001$. Accurate measurements of hydrogen find the radius of the proton to be approximately 8.77×10^{-16} m (Pohl et al., 2010). We will return to this problem again later in the book.

Section 11.9 Problems:

11.9.1 Identifying Quantum Numbers from the Wavefunction

Suppose someone gave you a hydrogen wavefunction, but forgot to label it. How could you determine the n, l, and m values? Write a short description of your procedure and then use it to determine n, l, and m for the following wavefunction:

$$\begin{aligned}\psi_{nlm}(\vec{r}) = &\frac{1}{\sqrt{\pi}a_0^{3/2}}\sqrt{\frac{5\cdot 7\cdot 17}{6\cdot 20!}}\frac{1}{2^{17}}\frac{1}{3^8}\\ &\times\left(41040\left(\frac{r}{a_0}\right)^8 - 1140\left(\frac{r}{a_0}\right)^9 + \left(\frac{r}{a_0}\right)^{10}\right)e^{-\frac{r}{12a_0}}\\ &\times e^{2i\phi}\sin^2\theta(143\cos^6\theta - 143\cos^4\theta + 33\cos^2\theta - 1).\end{aligned} \tag{11.317}$$

11.9.2 Size of the Atom

One of the odd results of atomic physics is that almost all atoms are the same size. Why is it that uranium, with 92 electrons, has a radius of 1.56 Å, which is about the same size as hydrogen, with 1 electron, which has a radius of 0.53 Å? In this problem, we will explore this mystery.

(a) We start with the virial theorem (which is derived in Prob. 11.9.6). It tells us that

$$\langle n, l|\tfrac{1}{\tilde{r}}|n, l\rangle = \frac{1}{a_0 n^2} \tag{11.318}$$

for hydrogen. If we use this as a measure of the average of the inverse radius, then the average radius is approximately $n^2 a_0$ (we show how to make this more precise in Sect. 11.11). Generalize to uranium (with $Z = 92$) and use the fact that uranium has 2(1s), 2(2s), 6(2p), 2(3s), 6(3p), 2(4s), 10(3d), 6(4p), 2(5s), 10(4d), 6(5p), 2(6s), 14(4f), 10(5d), 6(6p), 2(7s), 3(5f), and 1(6d) electrons to estimate the size of its outermost orbital 7s. Recall that the Bohr radius depends on Z for a hydrogenic ion.

(b) The result worked out in (a) neglected the effect of screening. Since the atom is organized in shells of increasing radius, where the electrons remain fairly localized

within the shells, the outermost electrons do not see a $Z = 92$, but rather see a much smaller Z value. Choose $Z \approx 6$ for the outermost shell. How large is the atom now? The true radius of uranium is about $3a_0$.

11.9.3 Pickering–Fowler Lines

Edward Charles Pickering was a US astronomer from the late nineteenth and early twentieth century. His career in astronomy involved a tedious analysis of photographs and spectra from stars. He recruited over 80 women to work with him, primarily doing computations. In 1896, he discovered a spectrum from the star ζ-Puppis, which had lines identified as belonging to hydrogen, plus some additional lines never seen on earth. These additional lines in the spectrum are called Pickering lines or Pickering–Fowler lines; Fowler did experiments on earth to try to reproduce these lines. While a few elements were discovered astronomically (the most important being helium), it is odd to have new hydrogen lines seen only in some, but not all stars. In addition, these lines corresponded to half-odd-integer quantum numbers.

When Niels Bohr formulated his theory for hydrogen in 1913, he had a different proposal. He thought that the Pickering lines came from singly ionized helium He_4^+. It turned out that a careful analysis of these lines showed that the ionized helium explanation must be correct.

Let's first work out the wavelengths of light for transitions from hydrogen. Recall that the energy levels are given by

$$\varepsilon_n = -\frac{e^2}{8\pi\epsilon_0 n^2 a_0} \tag{11.319}$$

so that the light, arising from transitions between two energy levels, will have energy given by

$$\hbar\omega = \varepsilon_m - \varepsilon_n = \frac{e^2}{8\pi\epsilon_0 a_0}\left(\frac{1}{n^2} - \frac{1}{m^2}\right), \tag{11.320}$$

where the initial state has index m, which is larger than the final state index n. The wavelength is $\lambda = \frac{2\pi c}{\omega}$, or

$$\lambda_{mn} = \frac{16\pi^2\epsilon_0 a_0 \hbar c}{e^2}\,\frac{m^2 n^2}{m^2 - n^2} = \frac{1}{R_H}\,\frac{m^2 n^2}{m^2 - n^2}. \tag{11.321}$$

The Rydberg constant R_H for an electron, is defined to be $R_\infty = \frac{e^4 m_e}{64\pi^3\epsilon_0^2\hbar^3 c} = 10973731.568160(21)\ \mathrm{m}^{-1}$. The Rydberg constant for hydrogen (corrected for

the reduced mass) is $R_H = R_\infty \frac{M}{m_e} = R_\infty \frac{1}{1+\frac{m_e}{m_p}} = R_\infty \times \frac{1}{1.00054}$. Note that even though we discussed a strategy for computing physical quantities of atoms in terms of the Bohr radius, the rest mass of the electron, and the ground-state energy of hydrogen (Rydberg of energy), spectroscopists prefer to work with the Rydberg constant, which is not to be conflated with the Rydberg of energy; yes, it is overloaded.

Calculate the wavelength for the Balmer series $n = 2$, with transitions from $m = 3, 4, 5, 6$. Calculate in angstroms (0.1 nm) to an accuracy of 0.1Å.

The Pickering–Fowler hypothesis was that the Balmer series also has wavelengths given by

$$\lambda_{m2} = \frac{1.00054}{10973732\text{m}^{-1}} \frac{4m^2}{m^2 - 4} \tag{11.322}$$

with half-odd integers ($m = 3.5, 4.5, 5.5, 6.5$). Compute the wavelength (in Angstrom) for these cases too.

Two of the observed Pickering lines are at 5412 Å and 4542 Å. Bohr proposed they come from ionized helium. In this case, our formula changes by multiplying the Rydberg constant by 4 to obtain it for ionized helium (the nuclear charge of helium is 2 and the nuclear charge enters as a square in the Rydberg constant) and changing the reduced mass to

$$M = \frac{m_e m_{He}}{m_e + m_{He}} = \frac{m_e}{1 + \frac{m_e}{M_{He}}} = m_e \times \frac{1}{1.000137}. \tag{11.323}$$

We examine transitions to the $n = 4$ level. The Pickering lines correspond to transitions from odd energy levels, particularly 7 and 9.

Show that

$$\lambda_{m4} = \frac{1.0001374}{10973732\text{ m}^{-1}} \frac{16m^2}{m^2 - 16}. \tag{11.324}$$

Calculate the results for $n = 7$ and $n = 9$. You should find 5413.1 Å and 4542.9 Å, respectively.

Even though the agreement is not perfect, the fact that the Bohr theory, with the effective mass for helium instead of hydrogen, approaches quite a bit closer to the measured result, made people believe the Bohr theory (and his explanation of the Pickering lines) must be correct. Einstein is said to have said that this analysis confirms Bohr's theory as being correct (of course we now know it is not correct, and the treatment we worked out in the main text is the proper analysis).

One point I want to leave you with on this is that it is the *small differences* that allow us to accept one theory over another and again points to the importance of precision measurements.

11.9.4 S-Wave Wavefunction at the Origin

Using the fact that $Y_{00}(\theta, \phi) = \frac{1}{\sqrt{4\pi}}$, determine $\psi_{n,l=0,m=0}(r{=}0)$. Note that you will need to determine the constant term in the general associated Laguerre polynomial to solve this problem.

11.9.5 Electron Capture

In our discussion of the proton radius problem, and the effect of the proton charge radius on energy levels, we saw that the electrons do spend some time in the nucleus. Given the fact that an electron can interact with a proton and form a neutron and an electron neutrino via

$$e^- + p^+ \rightarrow n + \nu_e. \tag{11.325}$$

Why aren't all atoms unstable to this process, called *electron capture*? You can safely ignore the mass of the electron neutrino in your analysis.

11.9.6 Virial Theorem

Use the commutator of $[\hat{r}\,\hat{p}_r, \hat{H}]$ to determine the virial theorem

$$\langle n, l|\tfrac{e^2}{4\pi\epsilon_0\hat{r}}|n, l\rangle = \tfrac{1}{M}\langle n, l|\hat{p}^2|n, l\rangle, \tag{11.326}$$

or, the negative of the expectation value of the potential energy is equal to twice the expectation value of the kinetic energy in every energy eigenstate. Note that here we have $\hat{p}^2 = \hat{p}_r^2 + \frac{\hat{L}^2}{\hat{r}^2}$. Use this result to determine the root-mean-square velocity of an electron in the state $|n, l\rangle$. Explain to what level of accuracy one expects relativistic effects to enter for the hydrogen atom. What about the innermost (1s) electrons of uranium? How relativistic are they?

Hint: consider the total energy, which you know, is the sum of the kinetic and potential energies.

11.10 The Hydrogen Atom in Cartesian Coordinates and Momentum-Space Wavefunctions

Nearly every quantum mechanics textbook derives the wavefunctions of hydrogen (graduate texts usually derive all of them, while undergraduate texts usually derive

just a subset of them). Surprisingly, only two advanced monographs derive the momentum-space wavefunctions—the textbook by De Lange and Raab (1991) uses operator methods to derive the results, while the textbook by Judd (1975) uses the SO(4) symmetry found by Fock (1935) and an integral equation discovered by Hylleraas (1932) to solve it. Even advanced treatises such as Bethe and Salpeter (1977) or Bransden and Joachim (2003) do not derive the momentum-space wavefunctions, they instead just quote them. But, we do so here. Naturally, this material is a bit more complex than some of the other topics we work on, but not overly so. We employ a new method that we developed for directly solving this problem *without using* the spherical symmetry of central forces (Lyu et al., 2022). If you can master the material in this section, then you will know something that only a handful of other experts in the world know. Isn't that cool!

We begin by first discussing *harmonic polynomials*. Harmonic polynomials are an alternative way to determine the angular part of the wavefunction in a central potential (that is, they do not use spherical harmonics). They were first used by Kramers in his quantum mechanics textbook from the 1930s (Kramers, 1957) (which was only translated into English in 1957) and have recently been revived by Weinberg (2012). Our approach follows closely the work of Avery (2010), but generalized to an operator language. First, we need to define the polynomials. They are *homogeneous* polynomials of degree l. This means that each monomial term of the polynomial is a number multiplied by a product of any number of $\hat{r}_x$, $\hat{r}_y$, and $\hat{r}_z$ factors, such that the sum of the exponents of all of the position operators in each monomial term is equal to l. We denote such a polynomial by $P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z)$. The l denotes the homogeneous degree of the polynomial, and the h subscript tells us it is a harmonic polynomial. The homogeneous property can be summarized by the relation

$$[\hat{r}\,\hat{p}_r,\, P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z)] = -i\hbar l\, P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z). \tag{11.327}$$

To see why this relation guarantees that we have a homogeneous polynomial of degree l, we first rewrite $\hat{r}\,\hat{p}_r = \hat{r}_x\hat{p}_x + \hat{r}_y\hat{p}_y + \hat{r}_z\hat{p}_z - i\hbar$. The constant term commutes with everything, so we can drop it in the commutator. The other terms will have the $\hat{p}_\alpha$ remove each $\hat{r}_\alpha$ factor in each monomial and replace it by $-i\hbar$, but then be multiplied by the corresponding $\hat{r}_\alpha$. Hence, each individual operator factor in a monomial is replaced by $-i\hbar$ times itself. Since each monomial has a total of l factors in it, we get a sum of l copies of $-i\hbar$ times the monomial. So, for a homogeneous polynomial, these factors are common to each monomial term and can be factored out of the entire polynomial. Hence, the aforementioned identity holds. If even one monomial is not homogeneous, then the identity fails, so it is a true test for homogeneity of the polynomial.

The other requirement of a harmonic polynomial is that it solves Laplace's equation. We will not go through an extended discussion of Laplace's equation. Instead, we will just state what it is in operator form. We require the harmonic polynomials to also satisfy

$$\sum_{\alpha=x,y,z} \left[\hat{p}_\alpha, \left[\hat{p}_\alpha, P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z)\right]\right] = 0. \tag{11.328}$$

Note that you do not need to know what Laplace's equation even is to use this property. We simply require any polynomial that we call a harmonic polynomial to satisfy the two properties in Eqs. (11.327) and (11.328). We will only use these two results in all of our work with these polynomials. If you like, you can simply take these two properties as the definition of what we call a harmonic polynomial.

Let's look at some examples. I find this is always useful when you have a new property that you are working with. The simplest examples of harmonic polynomials are the following: (i) degree 0—1; (ii) degree 1—$\hat{r}_x$, $\hat{r}_y$, and $\hat{r}_z$; and (iii) degree 2—$\hat{r}_x\hat{r}_y$, $\hat{r}_y\hat{r}_z$, $\hat{r}_z\hat{r}_x$, $\hat{r}_x^2 - \hat{r}_y^2$, and $\hat{r}_x^2 + \hat{r}_y^2 - 2\hat{r}_z^2$. You should verify that all of these examples satisfy the two properties that are the defining relations for the harmonic polynomials. More examples are given in the problems. Finally, note that if we use spherical coordinates to write $\hat{r}_x = \hat{r}\sin\hat{\theta}\cos\hat{\phi}$, $\hat{r}_y = \hat{r}\sin\hat{\theta}\sin\hat{\phi}$, and $\hat{r}_z = \hat{r}\cos\hat{\theta}$, and substitute it into the harmonic polynomial, then one can immediately show that $\hat{P}_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z)\frac{1}{\hat{r}^l}$ is a function only of angle operators. We will see later that this is an important observation.

We next investigate a Cartesian factorization of the hydrogen Hamiltonian, which was discovered by Ioffe and coworkers in 1984 (Andrianov et al., 1984). I first learned of it when I took the edX MOOC 8.05x from MIT by Barton Zwiebach and Jolyon Bloomfield. It is remarkable that $\hat{H}$ can be factorized in terms of a *sum* of noncommuting Cartesian ladder operators

$$\hat{H}(\lambda) = \sum_{\alpha=x,y,z} \hat{A}_\alpha^\dagger(\lambda)\hat{A}_\alpha(\lambda) + \tilde{\varepsilon}_\lambda = \sum_{\alpha=x,y,z} \frac{\hat{p}_\alpha^2}{2M} - \frac{e^2}{4\pi\epsilon_0\lambda\hat{r}}, \tag{11.329}$$

with hydrogen corresponding to the case with $\lambda = 1$. The ladder operators are

$$\hat{A}_\alpha(\lambda) = \frac{1}{\sqrt{2M}}\left(\hat{p}_\alpha - \frac{i\hbar}{\lambda a_0}\frac{\hat{r}_\alpha}{\hat{r}}\right) \quad \text{and} \quad \hat{A}_\alpha^\dagger(\lambda) = \left(\hat{A}_\alpha(\lambda)\right)^\dagger. \tag{11.330}$$

The proof comes from a direct calculation:

$$\begin{aligned}\hat{A}_\alpha^\dagger(\lambda)\hat{A}_\alpha(\lambda) &= \frac{1}{2M}\left(\hat{p}_\alpha^2 - \frac{i\hbar}{\lambda a_0}\left[\hat{p}_\alpha, \frac{\hat{r}_\alpha}{\hat{r}}\right] + \frac{\hbar^2}{\lambda^2 a_0^2}\frac{\hat{r}_\alpha^2}{\hat{r}^2}\right) \\ &= \frac{\hat{p}_\alpha^2}{2M} - \frac{\hbar^2}{2M\lambda a_0}\left(\frac{1}{\hat{r}} - \frac{\hat{r}_\alpha^2}{\hat{r}^3}\right) + \frac{\hbar^2}{2M\lambda^2 a_0^2}\frac{\hat{r}_\alpha^2}{\hat{r}^2}.\end{aligned} \tag{11.331}$$

Summing over α and recalling that $\hat{r}^2 = \hat{r}_x^2 + \hat{r}_y^2 + \hat{r}_z^2$, yields

$$\sum_{\alpha=x,y,z} \hat{A}_\alpha^\dagger(\lambda)\hat{A}_\alpha(\lambda) = \frac{1}{2M}\left(\hat{p}_x^2+\hat{p}_y^2+\hat{p}_z^2\right) - \frac{e^2}{4\pi\epsilon_0\lambda\hat{r}} + \frac{e^2}{8\pi\epsilon_0\lambda^2 a_0}, \tag{11.332}$$

which is the Hamiltonian in Eq. (11.329), if we choose $\tilde{\varepsilon}_\lambda = -\frac{e^2}{8\pi\epsilon_0\lambda^2 a_0}$. The restriction to $\lambda = 1$, then gives the factorization of the hydrogen Hamiltonian. Note that $\tilde{\varepsilon}_\lambda = E_{l=\lambda-1}$ in the notation we used earlier for solving hydrogen (when we pick $\lambda = n$).

Since each $\hat{A}_\alpha^\dagger(\lambda)\hat{A}_\alpha(\lambda)$ is a positive semidefinite operator, we immediately learn that the ground state must be annihilated by all of the lowering operators, or

$$\hat{A}_\alpha(\lambda)|\phi_\lambda\rangle = 0, \quad \text{for } \alpha = x,\ y,\ z. \tag{11.333}$$

We use $|\phi_\lambda\rangle$ to denote the ground state of $\hat{H}(\lambda)$. But, $\lambda = 1$, is special, because it is the hydrogen Hamiltonian, so we denote the energy eigenstates (ground and excited) for the $\lambda = 1$ using $|\psi\rangle$. The ground state $|\psi_{gs}\rangle$ is given by $|\phi_{\lambda=1}\rangle$. The subsidiary condition can be rewritten as

$$\hat{p}_\alpha|\phi_1\rangle = \frac{i\hbar}{a_0}\frac{\hat{r}_\alpha}{\hat{r}}|\phi_1\rangle \quad \text{for } \alpha = x,\ y,\ z. \tag{11.334}$$

The ground-state energy is $\tilde{\varepsilon}_{\lambda=1} = -\frac{e^2}{8\pi\epsilon_0 a_0}$, as we already know. We similarly define ground states of all possible auxiliary Hamiltonians via the subsidiary condition for arbitrary λ,

$$\hat{A}_\alpha(\lambda)|\phi_\lambda\rangle = 0, \quad \text{for } \alpha = x,\ y,\ z, \tag{11.335}$$

which then yields

$$\hat{p}_\alpha|\phi_\lambda\rangle = \frac{i\hbar}{\lambda a_0}\frac{\hat{r}_\alpha}{\hat{r}}|\phi_\lambda\rangle. \tag{11.336}$$

The auxiliary ground-state energy is then $\tilde{\varepsilon}_\lambda = -\frac{e^2}{8\pi\epsilon_0\lambda^2 a_0}$. Note that λ can be any real value here.

Our task is to develop a theory for hydrogen in this Cartesian basis. But, we do not do this in a "vacuum." Instead, we use the results we have already developed to help motivate the arguments here. We are also strongly motivated by a calculus-based approach to this problem, worked out by Fowles (1962), who showed that one needs to work with the Cartesian operators *and* the radial operator in position space. We will not rely on using angular momentum in this work. All calculations will only employ Cartesian operators and operators constructed from them, such as $\hat{r}$, $\hat{p}_r$, and radial ladder operators, which we will denote as $\hat{B}_r^\dagger(\lambda)$ and $\hat{B}_r(\lambda)$.

Our first step is similar to the separation of variables into the radial and angular components of the kinetic energy. But, we don't use angular momentum. Instead,

we use the "add-zero" trick to write

$$\hat{T} = \frac{1}{2M}\left(\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2\right) = \frac{\hat{p}_r^2}{2M} + \hat{T}_\perp, \tag{11.337}$$

with the perpendicular kinetic energy defined to be

$$\hat{T}_\perp = \frac{1}{2M}\left(\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2 - \hat{p}_r^2\right). \tag{11.338}$$

The designation of "perpendicular" means perpendicular to the radial direction, and we know it corresponds to the angular contributions to the kinetic energy. Knowing what we do about separation of variables, we expect $\hat{T}_\perp$ to be $\hat{L}^2/2M\hat{r}^2$. But, because we are not working with states of definite angular momentum, we cannot immediately evaluate what the square of the angular momentum operator is when it acts on a state. Nevertheless, we will derive an identity that should seem familiar to you given this intuition. It is

$$\hat{T}_\perp P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z)|\phi_\lambda\rangle = \frac{\hbar^2 l(l+1)}{2M\hat{r}^2} P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z)|\phi_\lambda\rangle. \tag{11.339}$$

Note that this implies that a harmonic polynomial of order l carries total angular momentum given by the quantum number l. Since this result must follow from the two defining relations of the harmonic polynomials, it should be clear that deriving this result is highly nontrivial. Indeed, it is a "French cooking" exercise, that we undertake now.

The proof is long and complex. Since P_h^0 is just a constant, the special case of $l = 0$ is identical to showing that $\hat{T}_\perp$ annihilates all auxiliary ground states, or $\hat{T}_\perp|\phi_\lambda\rangle = 0$. We always find good things, when we find a state that is annihilated by an operator. So, we will now prove this in two steps. The first is to evaluate the action of $\hat{T}_\perp$ by relating it to the fact that $|\phi_\lambda\rangle$ is an energy eigenstate of $\hat{H}(\lambda)$. We find that

$$\begin{aligned}\hat{T}_\perp|\phi_\lambda\rangle &= \left(\hat{T} - \frac{\hat{p}_r^2}{2M}\right)|\phi_\lambda\rangle = \left(\hat{H}(\lambda) + \frac{e^2}{4\pi\epsilon_0\lambda\hat{r}} - \frac{\hat{p}_r^2}{2M}\right)|\phi_\lambda\rangle \\ &= \left(-\frac{e^2}{8\pi\epsilon_0\lambda^2 a_0} + \frac{e^2}{4\pi\epsilon_0\lambda\hat{r}} - \frac{\hat{p}_r^2}{2M}\right)|\phi_\lambda\rangle.\end{aligned} \tag{11.340}$$

But, we have that

$$\begin{aligned}\hat{p}_r|\phi_\lambda\rangle &= \left(\frac{1}{\hat{r}}\left(\hat{r}_x\hat{p}_x + \hat{r}_y\hat{p}_y + \hat{r}_z\hat{p}_z - i\hbar\right)\right)|\phi_\lambda\rangle \\ &= \left(\frac{i\hbar}{\lambda a_0\hat{r}}\left(\frac{\hat{r}_x^2}{\hat{r}} + \frac{\hat{r}_y^2}{\hat{r}} + \frac{\hat{r}_z^2}{\hat{r}}\right) - \frac{i\hbar}{\hat{r}}\right)|\phi_\lambda\rangle\end{aligned}$$

$$= i\hbar \left(\frac{1}{\lambda a_0} - \frac{1}{\hat{r}} \right) |\phi_\lambda\rangle, \tag{11.341}$$

which follows from employing the Cartesian form of the subsidiary condition to replace momentum operators by position operators. Now we use this result twice to determine $\frac{1}{2M}\hat{p}_r^2|\phi_\lambda\rangle$. We have

$$\begin{aligned}
\tfrac{1}{2M}\hat{p}_r^2|\phi_\lambda\rangle &= \frac{i\hbar}{2M}\hat{p}_r \left(\frac{1}{\lambda a_0} - \frac{1}{\hat{r}} \right) |\phi_\lambda\rangle \\
&= -\frac{\hbar^2}{2M} \left\{ \left(\frac{1}{\lambda a_0} - \frac{1}{\hat{r}} \right)^2 - \frac{1}{i\hbar} \left[\hat{p}_r, \frac{1}{\hat{r}} \right] \right\} |\phi_\lambda\rangle \\
&= -\frac{\hbar^2}{2M} \left(\frac{1}{\lambda^2 a_0^2} - \frac{2}{\lambda a_0 \hat{r}} \right) |\phi_\lambda\rangle = \left(-\frac{e^2}{8\pi\epsilon_0 \lambda^2 a_0} + \frac{e^2}{4\pi\epsilon_0 \lambda \hat{r}} \right) |\phi_\lambda\rangle,
\end{aligned} \tag{11.342}$$

after using $a_0 = 4\pi\epsilon_0\hbar^2/Me^2$. Substituting into the previous expression then gives $\hat{T}_\perp|\phi_\lambda\rangle = 0$.

Since the state is annihilated by this operator, it allows us to convert products of operators acting on the $|\phi_\lambda\rangle$ ket into a commutator acting on the state, via

$$\begin{aligned}
\hat{T}_\perp P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z)|\phi_\lambda\rangle &= \left[\hat{T}_\perp, \hat{P}_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z) \right] |\phi_\lambda\rangle \\
&= \left[\hat{T} - \frac{\hat{p}_r^2}{2M}, P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z) \right] |\phi_\lambda\rangle.
\end{aligned} \tag{11.343}$$

This requires us to calculate two commutators. The first one is dealt with as follows:

$$\begin{aligned}
&\left[\hat{T}, P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z) \right] |\phi_\lambda\rangle = \tfrac{1}{2M} \sum_\alpha \left[\hat{p}_\alpha^2, P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z) \right] |\phi_\lambda\rangle \\
&= \tfrac{1}{2M} \sum_\alpha \left\{ \hat{p}_\alpha \left[\hat{p}_\alpha, P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z) \right] + \left[\hat{p}_\alpha, P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z) \right] \hat{p}_\alpha \right\} |\phi_\lambda\rangle \\
&= \tfrac{1}{2M} \left\{ \underbrace{\sum_\alpha \left[\hat{p}_\alpha, \left[\hat{p}_\alpha, P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z) \right] \right]}_{\text{harmonic so } =0} |\phi_\lambda\rangle + 2 \sum_\alpha \left[\hat{p}_\alpha, P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z) \right] \underbrace{\hat{p}_\alpha |\phi_\lambda\rangle}_{\text{subsidiary}} \right\} \\
&= \frac{i\hbar}{M\lambda a_0} \sum_\alpha \underbrace{\left[\hat{p}_\alpha, P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z) \right] \frac{\hat{r}_\alpha}{\hat{r}}}_{\text{Leibniz rule}} |\phi_\lambda\rangle \\
&= \frac{i\hbar}{M\lambda a_0} \left[\hat{p}_r, P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z) \right] |\phi_\lambda\rangle.
\end{aligned} \tag{11.344}$$

In the first step, we use the Leibniz rule to write the commutator as the sum of two terms. Next, we use add zero to write the two terms as a doubly nested commutator plus a correction (think of this as moving the $\hat{p}_\alpha$ factor to the right in the first term). The doubly nested commutator vanishes due to the second property of harmonic polynomials. Then we act the Cartesian momentum operator onto the auxiliary ground state, and we use the subsidiary condition to interchange it with a position operator acting on that state. Finally, we move it inside the commutator using the Leibniz rule and the fact that the commutator of a position operator with a harmonic polynomial vanishes to rewrite it as a radial momentum operator commuted with the harmonic polynomial.

We need to compute the commutator of the radial momentum with the harmonic polynomial. In general, this is complicated to carry out, but we handle it via a trick. First, we show that the radial momentum commutes with $\frac{\hat{r}_\alpha}{\hat{r}}$, which we anticipate to be true simply because the radial momentum commutes with all angle operators—$\frac{\hat{r}_\alpha}{\hat{r}}$ is expressed in terms of angle operators. So, we directly compute

$$\left[\hat{p}_r, \frac{\hat{r}_\alpha}{\hat{r}}\right] = \hat{r}_\alpha \left[\hat{p}_r, \frac{1}{\hat{r}}\right] + [\hat{p}_r, \hat{r}_\alpha]\frac{1}{\hat{r}} = i\hbar\left(\frac{\hat{r}_\alpha}{\hat{r}^2} - \frac{\hat{r}_\alpha}{\hat{r}^2}\right) = 0, \tag{11.345}$$

and verify the claimed result. Now, because P_h^l is a homogeneous polynomial of degree l, $\frac{1}{\hat{r}^l} P(\hat{r}_x, \hat{r}_y, \hat{r}_z)$ is a polynomial function of $\frac{\hat{r}_\alpha}{\hat{r}}$, so it commutes with the radial momentum operator. Hence,

$$\left[\hat{p}_r, \frac{1}{\hat{r}^l} P(\hat{r}_x, \hat{r}_y, \hat{r}_z)\right] = 0 = \frac{1}{\hat{r}^l}\left[\hat{p}_r, P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z)\right] + \left[\hat{p}_r, \frac{1}{\hat{r}^l}\right] P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z), \tag{11.346}$$

or

$$\left[\hat{p}_r, P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z)\right] = -i\hbar l \frac{1}{\hat{r}} P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z). \tag{11.347}$$

Combining with Eq. (11.344), we find that

$$\left[\hat{T}, P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z)\right] = \frac{e^2 l}{4\pi\epsilon_0 \lambda \hat{r}} P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z). \tag{11.348}$$

We also can use this result to determine the commutator of the harmonic polynomial with the square of the radial momentum operator. This uses a simple application of the Leibniz rule, followed by a second commutator to move the radial momentum operator to the far right. In equations, we have

$$\begin{aligned}\left[\hat{p}_r^2, P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z)\right] &= -i\hbar l \left(\hat{p}_r P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z)\tfrac{1}{\hat{r}} + \tfrac{1}{\hat{r}} P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z)\hat{p}_r\right) \\ &= -2i\hbar l P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z)\tfrac{1}{\hat{r}}\hat{p}_r - \frac{\hbar^2 l(l-1)}{\hat{r}^2} P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z).\end{aligned} \tag{11.349}$$

Hence,

$$\begin{aligned}\hat{T}_\perp P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)|\phi_\lambda\rangle &= \left(\frac{e^2 l}{4\pi\epsilon_0\lambda\hat{r}}+\frac{\hbar^2 l(l-1)}{2M\hat{r}^2}-\frac{\hbar^2 l}{M\lambda a_0\hat{r}}+\frac{\hbar^2 l}{M\hat{r}^2}\right)P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)|\phi_\lambda\rangle \\ &= \frac{\hbar^2 l(l+1)}{2M\hat{r}^2}P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)|\phi_\lambda\rangle. \end{aligned} \tag{11.350}$$

This is one of the critically important identities we will use to determine the energy eigenstates for hydrogen.

On the one hand, we just showed that $\hat{T}_\perp - \frac{\hbar^2 l(l+1)}{2M\hat{r}^2}$ annihilates the state $P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)|\phi_\lambda\rangle$. This will be required along with our intertwining relation to verify the energy eigenstate. On the other hand, you may suspect that this result also shows that $P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)|\phi_\lambda\rangle$ is a state with total angular momentum quantum number l. Here is how we verify this. Recall that we showed in Sect. 11.3 that $\hat{T}_\perp = \frac{\hat{L}^2}{2M\hat{r}^2}$. So, the identity in Eq. (11.350) can be rearranged into

$$\hat{L}^2 P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)|\phi_\lambda\rangle = \hbar^2 l(l+1)P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)|\phi_\lambda\rangle. \tag{11.351}$$

This tells us that the harmonic polynomial acting on an auxiliary ground-state ket generates a linear combination of states all with total angular momentum quantum number l. In general, it is *not* a simultaneous eigenstate of $\hat{L}_z$ (or $\hat{L}_x$, or $\hat{L}_y$).

Okay, onto the intertwining relation. We will be moving some of the lengthier derivations into the problems and only sketch the main ideas in the text. We will also be using the same form for the radial ladder operators of hydrogen as we used before (a superpotential being equal to a constant plus the inverse power of $\hat{r}$). So, we use

$$\hat{B}_r(\lambda) = \frac{1}{\sqrt{2M}}\left(\hat{p}_r - \frac{i\hbar}{\lambda a_0}+\frac{i\hbar\lambda}{\hat{r}}\right) \text{ and } \hat{B}_r^\dagger(\lambda) = \frac{1}{\sqrt{2M}}\left(\hat{p}_r + \frac{i\hbar}{\lambda a_0}-\frac{i\hbar\lambda}{\hat{r}}\right). \tag{11.352}$$

A direct calculation then shows that

$$\begin{aligned}\hat{H}(\lambda{=}1) &= \hat{B}_r^\dagger(\lambda)\hat{B}_r(\lambda)+\hat{T}_\perp-\frac{\hbar^2\lambda(\lambda+1)}{2M\hat{r}^2}+\tilde{\varepsilon}_{\lambda+1} \\ &= \hat{B}_r(\lambda)\hat{B}_r^\dagger(\lambda)+\hat{T}_\perp-\frac{\hbar^2(\lambda+1)(\lambda+2)}{2M\hat{r}^2}+\tilde{\varepsilon}_{\lambda+1}. \end{aligned} \tag{11.353}$$

Be sure to go through this algebra yourself! Note that in the identities, the Hamiltonian has $\lambda = 1$, but the raising and lowering operators are for *arbitrary* nonzero λ. Furthermore, evaluating a commutator and moving terms to the opposite side of the equation shows that

$$\hat{T}_\perp \hat{B}_r^\dagger(\lambda) = \left(\hat{B}_r^\dagger(\lambda) - \frac{2i\hbar}{\sqrt{2M}\hat{r}}\right)\hat{T}_\perp \tag{11.354}$$

$$\frac{1}{\hat{r}^2}\hat{B}_r^\dagger(\lambda) = \left(\hat{B}_r^\dagger(\lambda) - \frac{2i\hbar}{\sqrt{2M}\hat{r}}\right)\frac{1}{\hat{r}^2}. \tag{11.355}$$

This indicates that we can move $\hat{T}_\perp$ and $\frac{1}{\hat{r}^2}$ through $\hat{B}_r^\dagger(\lambda)$ operators at the expense of a shift being applied to the radial raising operator. In Problem 11.10.2, you will show that the base case of the intertwining relationship is

$$\begin{aligned}\hat{H}(\lambda{=}1)\hat{B}_r^\dagger(\lambda) = \hat{B}_r^\dagger(\lambda)&\left(\hat{H}(\lambda{=}1) + \frac{\hbar^2(\lambda+1)}{M\hat{r}^2}\right)\\ &- \frac{2i\hbar}{\sqrt{2M}\hat{r}}\left(\hat{T}_\perp - \frac{\hbar^2\lambda(\lambda+1)}{2M\hat{r}^2}\right).\end{aligned} \tag{11.356}$$

The full intertwining relation that we need is then shown to be

$$\begin{aligned}\hat{H}(\lambda{=}1)\hat{B}_r^\dagger(l)\hat{B}_r^\dagger(l+1)\cdots\hat{B}_r^\dagger(n-2) &= \hat{B}_r^\dagger(l)\cdots\hat{B}_r^\dagger(n-2)\\ &\times\left(\hat{H}(\lambda{=}1) + \frac{\hbar^2}{M\hat{r}^2}\sum_{j=1}^{n-l-1}(n-j)\right)\\ +\left\{\left(\hat{B}_r^\dagger(l) - \tfrac{2i\hbar}{\sqrt{2M}\hat{r}}\right)\left(\hat{B}_r^\dagger(l+1) - \tfrac{2i\hbar}{\sqrt{2M}\hat{r}}\right)\cdots\left(\hat{B}_r^\dagger(n-2) - \tfrac{2i\hbar}{\sqrt{2M}\hat{r}}\right)\right.&\\ \left. - \hat{B}_r^\dagger(l)\hat{B}_r^\dagger(l+1)\cdots\hat{B}_r^\dagger(n-2)\right\}&\left(\hat{T}_\perp - \frac{\hbar^2 l(l+1)}{2M\hat{r}^2}\right).\end{aligned} \tag{11.357}$$

The ansatz we use for the wavefunction involves a string of raising operators multiplied by a power of $\hat{r}$ and a harmonic polynomial operator that all act onto an auxiliary ground-state ket. We need to choose the parameters in each raising operator and the index for the auxiliary ground state. The choices of these parameters are determined by the requirement that we have an energy eigenstate. The procedure is straightforward, but a bit technical, so we just jump to the result, which is

$$|\psi_{nl}\rangle = \hat{B}_r^\dagger(l)\hat{B}_r^\dagger(l+1)\cdots\hat{B}_r^\dagger(n-2)\hat{r}^{n-l-1}P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)|\phi_n\rangle, \tag{11.358}$$

for the unnormalized energy eigenstate. This form might be what you expected, from our previous work, except that the power of the operator $\hat{r}$ is not in our previous eigenstate ansatz (the difference arises because the kets here are different from the kets we used before, as we will see later). To verify that this is an eigenstate, we evaluate $\hat{H}(\lambda{=}1)$ onto the state. As we do, we will see why the precise choices for

the parameters must be what they are in order for the state to be an energy eigenstate. Using the intertwining relation gives us

$$\begin{aligned}\hat{H}(\lambda{=}1)|\psi_{nl}\rangle &= \hat{B}_r^\dagger(l)\hat{B}_r^\dagger(l+1)\cdots\hat{B}_r^\dagger(n-2)\\ &\quad\times\left(\hat{H}(\lambda{=}1)+\frac{\hbar^2}{M\hat{r}^2}\sum_{j=1}^{n-l-1}(n-j)\right)\hat{r}^{n-l-1}P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)|\phi_n\rangle\\ &\quad+\left\{\left(\hat{B}_r^\dagger(l)-\tfrac{2i\hbar}{\sqrt{2}M\hat{r}}\right)\cdots\left(\hat{B}_r^\dagger(n-2)-\tfrac{2i\hbar}{\sqrt{2}M\hat{r}}\right)-\hat{B}_r^\dagger(l)\cdots\hat{B}_r^\dagger(n-2)\right\}\\ &\quad\times\left(\hat{T}_\perp-\frac{\hbar^2 l(l+1)}{2M\hat{r}^2}\right)\hat{r}^{n-l-1}P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)|\phi_n\rangle. \end{aligned}\tag{11.359}$$

But $\hat{T}_\perp-\frac{\hbar^2 l(l+1)}{2M\hat{r}^2}$ commutes with $\hat{r}^{n-l-1}$ and annihilates $P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)|\phi_n\rangle$, so we find

$$\begin{aligned}&\hat{H}(\lambda{=}1)|\psi_{nl}\rangle = \hat{B}_r^\dagger(l)\hat{B}_r^\dagger(l+1)\cdots\hat{B}_r^\dagger(n-2)\\ &\times\left(\hat{H}(\lambda{=}1)+\frac{\hbar^2}{M\hat{r}^2}\sum_{j=1}^{n-l-1}(n-j)\right)\hat{r}^{n-l-1}P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)|\phi_n\rangle\\ &=\hat{B}_r^\dagger(l)\hat{B}_r^\dagger(l+1)\cdots\hat{B}_r^\dagger(n-2)\\ &\times\left(\frac{\hat{p}_r^2}{2M}+\hat{T}_\perp-\frac{e^2}{4\pi\epsilon_0\hat{r}}+\frac{\hbar^2}{2M\hat{r}^2}(n+l)(n-l-1)\right)\hat{r}^{n-l-1}P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)|\phi_n\rangle\\ &=\hat{B}_r^\dagger(l)\hat{B}_r^\dagger(l+1)\cdots\hat{B}_r^\dagger(n-2)\\ &\times\left(\frac{\hat{p}_r^2}{2M}+\frac{\hbar^2}{2M\hat{r}^2}n(n-1)-\frac{e^2}{4\pi\epsilon_0\hat{r}}\right)\hat{r}^{n-l-1}P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)|\phi_n\rangle, \end{aligned}\tag{11.360}$$

where we used the fact that the "perpendicular" kinetic energy operator minus the right operator annihilates the state to the right. Note that only the choices of integers that we used produces the right constants so that we get the form aforementioned. Also, this choice works for all $0\le l<n-1$. This is one of the reasons for the choice we made. In our next step, we work out the radial-momentum term

$$\begin{aligned}&\hat{p}_r^2\hat{r}^{n-l-1}P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)|\phi_n\rangle=\frac{1}{\hat{r}^l}P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)\hat{p}_r^2\hat{r}^{n-1}|\phi_n\rangle\\ &=\frac{1}{\hat{r}^l}P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)\left(\hat{r}^{n-1}\hat{p}_r^2-2i\hbar(n-1)\hat{r}^{n-2}\hat{p}_r-\hbar^2(n-1)(n-2)\hat{r}^{n-3}\right)|\phi_n\rangle, \end{aligned}\tag{11.361}$$

because $\hat{r}_\alpha/\hat{r}$ commutes with $\hat{p}_r$. Plugging this result in yields

$$\begin{aligned}
&\hat{H}(\lambda{=}1)|\psi_{nl}\rangle = \hat{B}_r^\dagger(l)\hat{B}_r^\dagger(l+1)\cdots\hat{B}_r^\dagger(n-2)\,P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)\hat{r}^{n-l-1}\\
&\times\left(\frac{\hat{p}_r^2}{2M}-\frac{i\hbar}{M\hat{r}}(n-1)\hat{p}_r-\frac{\hbar^2}{2M\hat{r}^2}(n-1)(n-2)+\frac{\hbar^2}{2M\hat{r}^2}n(n-1)-\frac{e^2}{4\pi\epsilon_0\hat{r}}\right)|\phi_n\rangle\\
&=\hat{B}_r^\dagger(l)\hat{B}_r^\dagger(l+1)\cdots\hat{B}_r^\dagger(n-2)\,P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)\hat{r}^{n-l-1}\\
&\quad\times\left(-\frac{e^2}{8\pi\epsilon_0 n^2 a_0}+\frac{e^2}{4\pi\epsilon_0 n\hat{r}}+\frac{e^2(n-1)}{4\pi\epsilon_0 n\hat{r}}-\frac{\hbar^2 n(n-1)}{2M\hat{r}^2}+\frac{\hbar^2 n(n-1)}{2M\hat{r}^2}-\frac{e^2}{4\pi\epsilon_0\hat{r}}\right)|\phi_n\rangle\\
&=\tilde{\varepsilon}_n|\psi_{nl}\rangle. \qquad (11.362)
\end{aligned}$$

So, this is an energy eigenstate of the hydrogen Hamiltonian! Normalization of this state is left as Problem 11.10.3. The normalized wavefunction is written as

$$|\psi_{nl}\rangle_{\text{norm}} = \gamma_{nl}\hat{B}_r^\dagger(l)\hat{B}_r^\dagger(l+1)\cdots\hat{B}_r^\dagger(n-2)\hat{r}^{n-l-1}P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)|\phi_n\rangle. \qquad (11.363)$$

At this stage, and after normalization, we have produced the normalized energy eigenstates of hydrogen from the Cartesian factorization! We did not use angular momentum in the derivation. We next discuss the position-space wavefunction before moving on to the momentum space wavefunction. The algebra continues to be a bit tedious, but it is just "French cooking," and the final result will be worth it, as we get to discuss some wonderful physics with momentum-space wavefunctions.

Okay, let's dive into the coordinate-space wavefunction derivation. It is quite close to the generalized Rodrigues formula approach that we already did. In fact, all operator identities carry through exactly the same. The only thing different is we are working with a different state on the right, and we have the harmonic polynomial operators. Our first step is to move the factor $\frac{1}{\hat{r}^l}P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)$ to the left, because it commutes with each $\hat{B}_r^\dagger(\lambda)$. Hence, we have

$$|\psi_{nl}\rangle = \gamma_{nl}\frac{1}{\hat{r}^l}P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)\hat{B}_r^\dagger(l)\cdots\hat{B}_r^\dagger(n-2)\hat{r}^{n-1}|\phi_n\rangle. \qquad (11.364)$$

The next step is the same as before—we express the raising operators in terms of a similarity transformation, involving a power and an exponential term. Then, we need to find the state that $\hat{p}_r$ annihilates. Using the subsidiary conditions, we find that the radial momentum operator satisfies Eq. (11.341), which means the radial momentum operator annihilates the following state

$$\hat{p}_r e^{\frac{\hat{r}}{\lambda}}\frac{1}{\hat{r}}|\phi_n\rangle = 0. \qquad (11.365)$$

Using this result to convert products of operators acting on this state into nested commutators, we find that

$$|\psi_{nl}\rangle = \gamma_{nl}\left(\frac{1}{\sqrt{2M}}\right)^{n-l-1}\hat{r}^{-(2l+1)}P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)e^{\frac{(n+l+1)\hat{r}}{n(l+1)a_0}}$$
$$\times\left[\frac{1}{\hat{r}}\hat{p}_r, e^{-\frac{\hat{r}}{(l+1)(l+2)a_0}}\left[\frac{1}{\hat{r}}\hat{p}_r, e^{-\frac{\hat{r}}{(l+2)(l+3)a_0}}\left[\frac{1}{\hat{r}}\hat{p}_r,\cdots\right.\right.\right.$$
$$\left.\left.\left.\cdots, e^{-\frac{\hat{r}}{(n-2)(n-1)a_0}}\left[\frac{1}{\hat{r}}\hat{p}_r, e^{\frac{(2n-1)\hat{r}}{(n-1)na_0}}\hat{r}^{2n-1}\right]\cdots\right]\right]\right]_{n-l-1}|\phi_n\rangle. \tag{11.366}$$

The nested commutators can be replaced by a Laguerre polynomial. Using Eq. (11.276) and performing some significant algebra, finally yields

$$|\psi_{nl}\rangle = (-i)^{n-l-1}\left(\frac{na_0}{2}\right)^{n-1}\sqrt{\frac{(2n-1)!}{(n+l)!(n-l-1)!}}\frac{1}{\hat{r}^l}P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)$$
$$\times\left(\frac{2\hat{r}}{na_0}\right)^l L_{n-l-1}^{(2l+1)}\left(\frac{2\hat{r}}{na_0}\right)|\phi_n\rangle. \tag{11.367}$$

This is the normalized energy eigenstate. From here, the coordinate-space wavefunction is trivial to find. We just multiply by the position bra $\langle r_x, r_y, r_z|$ and find that

$$\psi_{nl}(r_x,r_y,r_z) = (-i)^{n-l-1}\left(\frac{na_0}{2}\right)^{n-1}\sqrt{\frac{(2n-1)!(n-l-1)!}{(n+l)!}}$$
$$\times\frac{1}{r^l}P_h^l(r_x,r_y,r_z)\left(\frac{2r}{na_0}\right)^l L_{n-l-1}^{(2l+1)}\left(\frac{2r}{na_0}\right)\langle r_x,r_y,r_z|\phi_n\rangle. \tag{11.368}$$

Computing $\phi_n(r_x,r_y,r_z)$ is simple. We rearrange Eq. (11.341) into

$$\left(p_r+\frac{i\hbar}{\hat{r}}\right)|\phi_n\rangle = \frac{i\hbar}{na_0}|\phi_n\rangle, \tag{11.369}$$

which is an eigenvalue–eigenvector relationship. Now, the state $|r_x,r_y,r_z\rangle$ can also be written in spherical coordinates as $|r\rangle\otimes|\theta,\phi\rangle$. Then, using the spherical translation operator gives us

$$\begin{aligned}\phi_n(r_x,r_y,r_z) &= \langle 0_r|e^{\frac{ir}{\hbar}\left(\hat{p}_r+\frac{i\hbar}{\hat{r}}\right)}|\phi_n\rangle\\ &= \langle 0_r|\phi_n\rangle e^{-\frac{r}{na_0}}.\end{aligned} \tag{11.370}$$

There is no angular dependence, because $\hat{L}_z$ and $\hat{L}_y$ both annihilate $|\phi_n\rangle$, as can be verified by using the subsidiary condition. Next, the constant satisfies $\langle 0_r|\phi_n\rangle =$

$\left(\frac{2}{na_0}\right)^{n+\frac{1}{2}} \frac{1}{\sqrt{(2n)!}}$ so we have

$$\psi_{nl}(r_x, r_y, r_z) = (-i)^{n-l-1} \frac{1}{\sqrt{2}} \left(\frac{2}{na_0}\right)^{\frac{3}{2}} \sqrt{\frac{(n-l-1)!}{n(n+l)!}} \left(\frac{2r}{na_0}\right)^l \times L_{n-l-1}^{(2l+1)}\left(\frac{2r}{na_0}\right) e^{-\frac{r}{na_0}} \frac{1}{r^l} P_h^l(r_x, r_y, r_z), \tag{11.371}$$

which is the same result as we had before, except for the extra phase factor, which can be dropped. Note that when we calculate the normalization factor, it is done so that the corresponding ψ_{nn-1} is normalized; in other words, $\phi_n(r_x, r_y, r_z)$ is not normalized here.

Our last task is the momentum-space wavefunction. This is a bit more complicated because the energy eigenstate is written in terms of position operators, and we need to evaluate them against momentum eigenstates. The procedure to do this heavily uses the Cartesian subsidiary condition $\hat{A}_\alpha|\phi_n\rangle = 0$ in the derivation. I want to prepare you now—this is tough stuff. It is an extreme French-cooking problem. But, you can make it through with some "divide and conquer" approaches.

The initial strategy is to simply multiply by the momentum bra $\langle p_x, p_y, p_z|$ from the left to find (we drop the phase factor)

$$\begin{aligned}\phi_{nl}(p_x, p_y, p_z) &= \left(\frac{na_0}{2}\right)^{n-1} \sqrt{\frac{(2n-1)!(n-l-1)!}{(n+l)!}} \\ &\times \langle p_x, p_y, p_z| \left(\frac{2\hat{r}}{na_0}\right)^l L_{n-l-1}^{2l+1}\left(\frac{2\hat{r}}{na_0}\right) \frac{1}{\hat{r}^l} P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z)|\phi_n\rangle \\ &= \left(\frac{na_0}{2}\right)^{n-l-1} \sqrt{\frac{(2n-1)!(n-l-1)!}{(n+l)!}} \\ &\times \langle p_x, p_y, p_z| L_{n-l-1}^{2l+1}\left(\frac{2\hat{r}}{na_0}\right) P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z)|\phi_n\rangle. \end{aligned} \tag{11.372}$$

The operators in the matrix element are polynomials of $\hat{r}$ and $\hat{r}_\alpha$. We use the subsidiary condition of the auxiliary Hamiltonian ground state to replace position operators by momentum operators. The momentum operators need to then move to the far left, where they can act on the momentum state bra. But, in the process, there are numerous additional terms that arise from commutation relations, and all of those terms need to be gathered. Since the matrix element still has position operators left over, we repeat until they are all gone. The procedure can become quite messy when we attempt to complete it concretely, so we need a strategy to follow. We describe the details both in the text and in the problems.

We start with a warm-up problem—compute the matrix element of $\hat{r}^m$:

$$\mu_m(\vec{p}, n) = \langle \vec{p}\,|\hat{r}^m|\phi_n\rangle = \langle p_x, p_y, p_z|\hat{r}^m|\phi_n\rangle. \tag{11.373}$$

Here is how we proceed. First, we write $\hat{r} = \sum_\alpha \frac{\hat{r}_\alpha^2}{\hat{r}}$ and recall that $\frac{\hat{r}_\alpha}{\hat{r}}|\phi_n\rangle = \frac{na_0}{i\hbar}\hat{p}_\alpha|\phi_n\rangle$, so

$$\mu_m(\vec{p},n) = \sum_\alpha \langle \vec{p}\,|\hat{r}^{m-1}\hat{r}_\alpha \tfrac{na_0}{i\hbar}\hat{p}_\alpha|\phi_n\rangle. \tag{11.374}$$

Move the $\hat{p}_\alpha$ operator all the way to the left

$$\begin{aligned}\mu_m(\vec{p},n) &= \tfrac{na_0}{i\hbar}\sum_\alpha \left(\langle \vec{p}\,|\hat{p}_\alpha\hat{r}^{m-1}\hat{r}_\alpha|\phi_n\rangle + \langle \vec{p}\,|\left[\hat{r}^{m-1}\hat{r}_\alpha,\,\hat{p}_\alpha\right]|\phi_n\rangle\right)\\ &= \tfrac{na_0}{i\hbar}\sum_\alpha \left(p_\alpha\langle \vec{p}\,|\hat{r}^{m-1}\hat{r}_\alpha|\phi_n\rangle + i\hbar\langle \vec{p}\,|\left((m-1)\hat{r}^{m-3}\hat{r}_\alpha^2 + \hat{r}^{m-1}\right)|\phi_n\rangle\right),\end{aligned} \tag{11.375}$$

since $\langle \vec{p}\,|\hat{p}_\alpha = \langle \vec{p}\,|p_\alpha$ and after evaluating the commutator. So, we now have (the 3 below comes from the sum over α for the last term)

$$\mu_m(\vec{p},n) = 3na_0\mu_{m-1}(\vec{p},n) + \sum_\alpha \tfrac{na_0}{i\hbar}p_\alpha\langle \vec{p}\,|\hat{r}^m\tfrac{\hat{r}_\alpha}{\hat{r}}|\phi_n\rangle + na_0(m-1)\langle \vec{p}\,|\hat{r}^{m-1}|\phi_n\rangle, \tag{11.376}$$

where we used $\hat{r}^{m-3}\sum_\alpha \hat{r}_\alpha^2 = \hat{r}^{m-1}$. Combining the first and last terms and using the subsidiary condition on the middle term give

$$\mu_m(\vec{p},n) = na_0(m+2)\langle \vec{p}\,|\hat{r}^{m-1}|\phi_n\rangle + \sum_\alpha \left(\tfrac{na_0}{i\hbar}\right)^2 p_\alpha\langle \vec{p}\,|\hat{r}^m\hat{p}_\alpha|\phi_n\rangle. \tag{11.377}$$

Again, we move the momentum operator to the left, picking up the extra terms from the commutator, to yield

$$\begin{aligned}\mu_m(\vec{p},n) &= na_0(m+2)\mu_{m-1}(\vec{p},n) - \left(\frac{na_o p}{\hbar}\right)^2\mu_m(\vec{p},n)\\ &\quad + \sum_\alpha \frac{(na_0)^2}{i\hbar}p_\alpha m\langle \vec{p}\,|\hat{r}^{m-1}\tfrac{\hat{r}_\alpha}{\hat{r}}|\phi_n\rangle.\end{aligned} \tag{11.378}$$

Note how the second term on the right-hand side can be moved to the left-hand side of the equation and combined with that term. In Problem 11.10.4, you will iterate this process until it ends, to find

$$\begin{aligned}\left(1+\left(\tfrac{na_o p}{\hbar}\right)^2\right)\mu_m(\vec{p},n) &= na_0(m+2)\mu_{m-1}(\vec{p},n)\\ &\quad - \left(\tfrac{na_o p}{\hbar}\right)^2\sum_{k=2}^{m}\frac{m!}{(m+k)!}(na_0)^l\mu_{m-k}(\vec{p},n).\end{aligned} \tag{11.379}$$

The m-term recurrence relation can be converted into a three-term recurrence relation, which is worked out in Problem 11.10.4 and solved in Problem 11.10.5. To write down the final result, we first express

$$\mu_m(\vec{p}, n) = \frac{Q_n(\xi_n^2)(na_0)^m}{\left(1+\xi_n^2\right)^m}\langle p_x, p_y, p_z|\phi_n\rangle, \tag{11.380}$$

for $\xi_n = \frac{na_o p}{\hbar}$. The polynomial Q_m is

$$Q_m(\xi_n^2) = \frac{(m+2)!}{2}\sum_{j=0}^{\left\lfloor\frac{m+1}{2}\right\rfloor}(-1)^j\frac{(m+1)!}{(m+1-2j)!(2j+1)!}\xi_n^{2j}. \tag{11.381}$$

With the moments $\mu_m(\vec{p}, n)$ completely determined, we now move on to the next phase of the derivation—extracting the harmonic polynomial from the matrix element.

The operator we need to evaluate takes the generic form $\langle\vec{p}\,|\hat{r}^{m-l}P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z)|\phi_n\rangle$; we write the exponent as $m-l$ to make some formulas that follow simpler. We use the same strategy as before. For each monomial in $P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z)$, we can replace a factor of $\hat{r}_\alpha$ by $\frac{na_0}{i\hbar}\hat{r}\,\hat{p}_\alpha$ (after moving the $\hat{r}_\alpha$ factor to the far right), because this is what we obtain after acting $\hat{r}_\alpha$ onto $|\phi_n\rangle$. The harmonic polynomial has had a factor of $\hat{r}_\alpha$ removed, which can be done via $-\frac{1}{i\hbar l}\sum_\alpha[\hat{p}_\alpha, P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z)]$, as we can see from Eq. (11.327). In other words, each term in the sum over α of the commutator has removed one $\hat{r}_\alpha$ term from the harmonic polynomial, because

$$-\frac{1}{i\hbar l}\sum_\alpha\hat{r}_\alpha[\hat{p}_\alpha, P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z)] = P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z). \tag{11.382}$$

Hence, the sum of the commutator without the $\hat{r}_\alpha$ is equal to a sum over terms that each have an $\hat{r}_\alpha$ factor removed from the original monomial (try some examples with harmonic polynomials if this is not clear). After noting that $\hat{r}$ commutes with $[\hat{p}_\alpha, P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z)]$, we find that

$$\begin{aligned}\langle\vec{p}\,|\hat{r}^{m-l}P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z)|\phi_n\rangle &= -\tfrac{1}{i\hbar l}\sum_\alpha\langle\vec{p}\,|\hat{r}^{m-l}[\hat{p}_\alpha, P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z)]\hat{r}_\alpha|\phi_n\rangle \\ &= -\frac{na_0}{(i\hbar)^2 l}\sum_\alpha\langle\vec{p}\,|\hat{r}^{m-l+1}[\hat{p}_\alpha, P_h^l(\hat{e}_x, \hat{r}_y, \hat{r}_z)]\hat{p}_\alpha|\phi_n\rangle.\end{aligned} \tag{11.383}$$

Now, we move $\hat{p}_\alpha$ to the left and act on $\langle\vec{p}\,|$. Remember that we have to keep track of all commutators. We have

$$\langle \vec{p}\,|\hat{r}^{m-l} P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)|\phi_n\rangle = -\frac{na_0}{(i\hbar)^2 l}\sum_\alpha \Big\{ p_\alpha \langle \vec{p}\,|\hat{r}^{m-l+1}[\hat{p}_\alpha, P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_y)]|\phi_n\rangle + \langle \vec{p}[\hat{r}^{m-l+1}, \hat{p}_\alpha][\hat{p}_\alpha, P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)]|\phi_n\rangle - \langle \vec{p}\,|\hat{r}^{m-l+1}[\hat{p}_\alpha,[\hat{p}_\alpha, P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)]]|\phi_n\rangle \Big\}. \tag{11.384}$$

The last term vanishes because P_h^l is a harmonic polynomial, as shown in Eq. (11.328). Evaluating the remaining commutator gives

$$\langle \vec{p}\,|\hat{r}^{m-l} P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)|\phi_n\rangle = -\frac{na_0}{(i\hbar)^2 l}\sum_\alpha \Big\{ p_\alpha \langle \vec{p}\,|\hat{r}^{m-l+1}[\hat{p}_\alpha, P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_y)]|\phi_n\rangle + i\hbar(m-l+1)\langle \vec{p}\,|\hat{r}^{m-l}[\hat{p}_\alpha, P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)]\tfrac{\hat{r}_\alpha}{\hat{r}}|\phi_n\rangle \Big\}. \tag{11.385}$$

Again, we replace $\frac{\hat{r}_\alpha}{\hat{r}}$ by $\frac{na_0\hat{p}_\alpha}{i\hbar}$ and move it to the left, picking up relevant commutators:

$$\begin{aligned}
&\langle \vec{p}\,|\hat{r}^{m-l} P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)|\phi_n\rangle \\
&= -\frac{na_0}{(i\hbar)^2 l}\sum_\alpha \Big\{ p_\alpha \langle \vec{p}\,|\hat{r}^{m-l} \underbrace{\big(\hat{r} + na_0(m-l+1)\big)}_{\text{polynomial}}[\hat{p}_\alpha, P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_y)]|\phi_n\rangle \\
&\quad + i\hbar n a_0 (m-l)(m-l+1)\langle \vec{p}|\hat{r}^{m-l-2}[\hat{p}_\alpha, P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)]\hat{r}_\alpha|\phi_n\rangle \Big\} \\
&= \frac{na_0}{\hbar^2 l}\langle \vec{p}\,|\hat{r}^{m-l}\big(\hat{r} + na_0(m-l+1)\big)[\hat{p}_\alpha, P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)]|\phi_n\rangle \\
&\quad + (na_0)^2(m-l)(m-l-1)\langle \vec{p}\,|\hat{r}^{m-l-2} P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)|\phi_n\rangle,
\end{aligned} \tag{11.386}$$

where we used Eq. (11.327) again. Let's pause for a moment to digest what we have done. We have removed one $\hat{r}_\alpha$ term from $P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)$ and replaced it with a p_α. Doing so adds a polynomial in $\hat{r}$ to the expectation value and creates a recursion relation with the expectation value with two lower powers of $\hat{r}$. You can see how we proceed from here. Each iteration will remove additional $\hat{r}_\alpha$ terms and replace them with p_α, complicating the expectation value with additional polynomials. Clearly, this is cumbersome, and we need to be extra careful. So, let's start with the simplest case $m = n - 1 = l$. We just have to recall that with the removal of each $\hat{r}_\alpha$ term from $P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)$, the power of the monomials in the polynomial decreases by 1. After iterating $j < n$ times, we have

$$\langle \vec{p}\,|P_h^{n-1}(\hat{r}_x,\hat{r}_y,\hat{r}_z)|\phi_n\rangle = \frac{(na_0)^{2j}(n-1-j)!}{\hbar^{2j}(n-1)!}\sum_{\alpha_1\alpha_2\cdots\alpha_j} p_{\alpha_1}p_{\alpha_2}\cdots p_{\alpha_j}$$

$$\times\,\langle \vec{p}\,|\theta_j\left(\frac{\hat{r}}{na_0}\right)[\hat{p}_{\alpha_j},[\hat{p}_{\alpha_{j-1}},[\hat{p}_{\alpha_{j-2}},\cdots,[\hat{p}_{\alpha_1},P_h^{n-1}(\hat{r}_x,\hat{r}_y,\hat{r}_z)]\cdots]]]|\phi_n\rangle, \tag{11.387}$$

with θ_j a yet to be determined polynomial; we already know that $\theta_0 = 1$. In Problem 11.10.6, you will show that θ_{j+1} satisfies a recurrence relation

$$\theta_{j+1}\left(\frac{\hat{r}}{na_0}\right) = \sum_{j=0}^{j+1}\left(\frac{na_0}{i\hbar}\right)^k\left[\cdots\left[\frac{\hat{r}}{na_0}\theta_j\left(\frac{\hat{r}}{na_0}\right),\hat{p}_r\right],\cdots,\hat{p}_r\right]_k, \tag{11.388}$$

where the k subscript indicates that there are k nested commutators. After j reaches $n-1$, we have our final answer

$$\begin{aligned}\langle \vec{p}\,|P_h^{n-1}(\hat{r}_x,\hat{r}_y,\hat{r}_z)|\phi_n\rangle &= \frac{(na_0)^{2n-2}}{\hbar^{2n-2}(n-1)!}\sum_{\alpha_1\cdots\alpha_{n-1}} p_{\alpha_1}\cdots p_{\alpha_{n-1}}\langle \vec{p}\,|\theta_{n-1}\left(\frac{\hat{r}}{na_0}\right)\\ &\quad\times[p_{\alpha_{n-1}},[\hat{p}_{n-2},\cdots,[\hat{p}_{\alpha_1},P_h^{n-1}(\hat{r}_x,\hat{r}_y,\hat{r}_z)]\cdots]]_{n-1}|\phi_n\rangle\\ &= \frac{(na_0)^{2n-2}}{(i\hbar)^{n-1}}P_h^{n-1}(p_x,p_y,p_z)\langle \vec{p}\,|\theta_{n-1}\left(\frac{\hat{r}}{na_0}\right)|\phi_n\rangle.\end{aligned} \tag{11.389}$$

The appearance of $P_h^{n-1}(p_x,p_y,p_z)$ may seem odd, as if it came out of nowhere. Let's look at this formula concretely for a simple example. Take $P_h^2(\hat{r}_x,\hat{r}_y,\hat{r}_z) = \hat{r}_x^2 - \hat{r}_y^2$. Then

$$\sum_{\alpha_1}[\hat{p}_{\alpha_1},\hat{r}_x^2-\hat{r}_y^2]p_{\alpha_1} = \sum_{\alpha_1}p_{\alpha_1}(-2i\hbar)(\delta_{\alpha_1,x}\hat{r}_x - \delta_{\alpha_1,y}\hat{r}_y) = -2i\hbar(p_x\hat{r}_x - p_y\hat{r}_y). \tag{11.390}$$

Computing the second nested commutator, we find

$$\begin{aligned}\sum_{\alpha_1\alpha_2}p_{\alpha_1}p_{\alpha_2}[\hat{p}_{\alpha_2},[\hat{p}_{\alpha_1},P_h^2(\hat{r}_x,\hat{r}_y,\hat{r}_z)]] &= (-2i\hbar)(-i\hbar)(p_x^2-p_y^2)\\ &= 2!(-i\hbar)^2P_h^2(p_x,p_y,p_z).\end{aligned} \tag{11.391}$$

The way to think of this is that multiple commutators eventually end in a number, which is the coefficient of the corresponding monomial of P_h^l times $l!(-i\hbar)^l$. Replacing each of the original $\hat{r}_\alpha$ by the corresponding p_α then reconstructs the harmonic polynomial, but now as a function of $\vec{p}$. See Problem 11.10.7 for more examples.

We have produced a remarkable identity. We have shown that the matrix element of $P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z)$ becomes $P_h^l(p_x, p_y, p_z)$ times the matrix element of the polynomial θ_l. We need to know more about these polynomials, which are called reverse Bessel polynomials. Their explicit formula is worked out in Problem 11.10.8.

So, now we need to do the general case. I suspect you are thinking that it was pretty cool how we "razzle-dazzled" the harmonic polynomial as a function of $\hat{r}_\alpha$ into a function of p_α. And it was! Now I also know what you are thinking—let's go back and add in the power $\hat{r}^{m-l}$ into the matrix element and work that result out. But it will fail. Badly.

Why? Because there is no recurrence relation between powers and harmonic polynomials. So now you go back and think—that doesn't sound right—we just worked out some recursions with harmonic polynomials. You see, each of the commutators of a harmonic polynomial with different $\hat{p}_\alpha$ was also a harmonic polynomial. So that means we already worked out the recurrence—it is with respect to reverse Bessel polynomials! Indeed we have. What we have already verified is that

$$\langle \vec{p}\,|\theta_{m-l}\left(\frac{\hat{r}}{na_0}\right)P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z)|\phi_n\rangle = \frac{(na_0)^{2l}}{(i\hbar)^l}P_h^l(p_x, p_y, p_z)\langle \vec{p}\,|\theta_m\left(\frac{\hat{r}}{na_0}\right)|\phi_n\rangle. \tag{11.392}$$

So, now the plan is clear—we need to rewrite the Laguerre polynomial in an expansion in terms of reverse Bessel polynomials, then we can evaluate each matrix element. Of course, we first need to evaluate the expectation value of a reverse Bessel polynomial to complete the job. It turns out this is remarkably simple. We have

$$\langle p_x, p_y, p_z|\theta_m\left(\frac{\hat{r}}{na_0}\right)|\phi_n\rangle = \frac{(m+1)!2^m}{\left(1+\xi_n^2\right)^m}\langle p_x, p_y, p_z|\phi_n\rangle, \tag{11.393}$$

but the proof is quite involved, so it is relegated to Problem 11.10.9 as well.

For the next part, we bring in a fact about the reverse Bessel polymomials. A polynomial of the form $\sum_{j=0}^n a_j z^j$ is expanded in terms of reverse Bessel polynomials via

$$\sum_{j=0}^n a_j z^j = \sum_{j=0}^n c_j \theta_j(z) \tag{11.394}$$

with

$$c_j = \sum_{k=0}^j \frac{(-1)^k}{2^k}\frac{(j+k)!}{k!(j-k)!}\left(a_{j+k} - (j+k+1)a_{j+k+1}\right), \tag{11.395}$$

as summarized in the treatise for reverse Bessel polynomials (Grosswald, 1978). For Laguerre polynomials, we have an explicit formula for the a_j for $L_{n-l-1}^{(2l+1)}$:

$$a_j = (-1)^j \frac{(n+l)!}{j!} \binom{n+l}{2l+j+1} 2^j \tag{11.396}$$

for $0 \leq j \leq n-l-1$. The final formula for c_j is worked out in Problem 11.10.10.

We are getting close to the big reveal! At this stage, we have all identities in place, and we can compute as follows with the phase factor included to be consistent with the standard form of the wavefunction. Note that the work here is mainly bookkeeping to make sure each step is followed correctly. We find

$$\begin{aligned}
\tilde{\psi}_{nl}(p_x, p_y, p_z) &= \left(\tfrac{na_0}{2}\right)^{n-l-1} \sqrt{\tfrac{(2n-1)!(n-l-1)!}{(n+l)!}} \\
&\quad \times \langle p_x, p_y, p_z | L_{n-l-1}^{(2l+1)} \left(\tfrac{2\hat{r}}{na_0}\right) P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z) | \phi_n \rangle \\
&= \left(\tfrac{na_0}{2}\right)^{n-l-1} \sqrt{\tfrac{(2n-1)!(n-l-1)!}{(n+l)!}} \sum_{j=0}^{n-l-1} (-1)^j 2^j \tfrac{2n(n+l+j)!}{j!(n-l-j-1)!(2l+2j+2)!} \\
&\quad \times \langle p_x, p_y, p_z | \theta_j \left(\tfrac{\hat{r}}{na_0}\right) P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z) | \phi_n \rangle \\
&= \left(\tfrac{na_0}{2}\right)^{n-l-1} \sqrt{\tfrac{(2n-1)!(n-l-1)!}{(n+l)!}} \sum_{j=0}^{n-l-1} (-1)^j 2^j \tfrac{2n(n+l+j)!}{j!(n-l-j-1)!(2l+2j+2)!} \tfrac{(na_0)^{2l}}{(i\hbar)^l} \\
&\quad \times P_h^l(p_x, p_y, p_z) \langle p_x, p_y, p_z | \theta_{j+l} \left(\tfrac{\hat{r}}{na_0}\right) | \phi_n \rangle \\
&= 2^{l+1-n} (na_0)^{n+l-1} \sqrt{\tfrac{(2n-1)!(n-l-1)!}{(n+l)!}} \sum_{j=0}^{n-l-1} (-1)^j 2^j \tfrac{2n(n+l+j)!}{j!(n-l-j-1)!(2l+2j+2)!} \\
&\quad \times \tfrac{(l+j+1)!2^{l+j}}{(1+\xi_n^2)^{l+j}(i\hbar)^l} P_h^l(p_x, p_y, p_z) \langle p_x, p_y, p_z | \phi_n \rangle.
\end{aligned} \tag{11.397}$$

The steps we used were as follows: (i) we first multiplied the energy eigenstate by the momentum eigenstate bra from the left; (ii) then we expanded the Laguerre operator polynomial in terms of the reverse Bessel operator polynomial; (iii) next we evaluated the expectation value of a reverse Bessel polynomial multiplied by a harmonic polynomial; and (iv) finally, we evaluated the expectation value of the reverse Bessel polynomial. Now, aside from the need to determine the auxiliary Hamiltonian ground-state wavefunction in momentum space, we are done. But, as we have found before, it is customary to re-express the results in terms of polynomials studied previously by dead French and German mathematicians. In this case, it is the Gegenbauer polynomials, which satisfy the expression

$$C_{n-l-1}^{l+1}\left(\frac{-1+\xi_n^2}{1+\xi_n^2}\right) = \frac{1}{l!}\sum_{j=0}^{n-l+1}(-1)^j 2^{2j+1}\frac{(n+l+j)!(l+j+1)!}{j!(n-l-j-1)!(2l+2j+2)!} \times \frac{1}{\left(1+\xi_n^2\right)^j}. \tag{11.398}$$

We can then re-express out result in terms of the Gegenbauer polynomials as

$$\tilde{\psi}_{nl}(p_x, p_y, p_z) = 2^{2l+1-n}(na_0)^{n+l-1}nl!\sqrt{\frac{(2n-1)!(n-l-1)!}{(n+l)!}} \times C_{n-l-1}^{l+1}\left(\frac{-1+\xi_n^2}{1+\xi_n^2}\right)\frac{1}{\left(1+\xi_n^2\right)^l}\langle p_x, p_y, p_z|\phi_n\rangle\frac{P_h^l(p_x, p_y, p_z)}{(i\hbar)^l}. \tag{11.399}$$

The last item needed to finish the calculation is determining $\langle p_x, p_y, p_z|\phi_n\rangle$. We do so with the Cartesian translation operator for momentum, We have that

$$\begin{aligned}\langle p_x, p_y, p_z|\phi_n\rangle &= \langle 0_{p_x}, 0_{p_y}, 0_{p_z}|e^{-\frac{i}{\hbar}(p_x\hat{r}_x+p_y\hat{r}_y+p_z\hat{r}_z)}|\phi_n\rangle \\ &= \sum_{k=0}^{\infty}\frac{1}{k!}\left(\frac{i}{\hbar}\right)^k\langle 0_{p_x}, 0_{p_y}, 0_{p_z}|(p_x\hat{r}_x+p_y\hat{r}_y+p_z\hat{r}_z)^k|\phi_n\rangle.\end{aligned} \tag{11.400}$$

Note that here, we have no simple way to find a state that is an eigenstate of the operator in the exponent anymore, so we need to solve the problem by "brute force." Consider first the case with $k = 1$:

$$\begin{aligned}&\langle 0_{p_x}, 0_{p_y}, 0_{p_z}|(p_x\hat{r}_x+p_y\hat{r}_y+p_z\hat{r}_z)|\phi_n\rangle \\ &= \frac{na_0}{i\hbar}\langle 0_{p_x}, 0_{p_y}, 0_{p_z}|\hat{r}(p_x\hat{p}_x+p_y\hat{p}_y+p_z\hat{p}_z)|\phi_n\rangle \\ &= \frac{na_0}{i\hbar}\langle 0_{p_x}, 0_{p_y}, 0_{p_z}|\sum_\alpha p_\alpha[\hat{r}, \hat{p}_\alpha]|\phi_n\rangle \\ &= na_0\langle 0_{p_x}, 0_{p_y}, 0_{p_z}|\sum_\alpha p_\alpha\frac{\hat{r}_\alpha}{\hat{r}}|\phi_n\rangle \\ &= \frac{(na_0)^2}{i\hbar}\langle 0_{p_x}, 0_{p_y}, 0_{p_z}|\sum_\alpha p_\alpha\hat{p}_\alpha|\phi_n\rangle = 0.\end{aligned} \tag{11.401}$$

We used the subsidiary condition twice in this derivation. In general, all odd powers vanish and the even powers give $k!(\frac{k}{2}+1)!(na_0p)^k$, as worked out in Problem

11.10.12. So, we find that

$$\langle p_x, p_y, p_z|\phi_n\rangle = \sum_{k=0}^{\infty}(-1)^k\left(\frac{na_0p}{\hbar}\right)^{2k}(k+1)\langle 0_{p_x}, 0_{p_y}, 0_{p_z}|\phi_n\rangle. \tag{11.402}$$

This series is very similar to the geometric series. To determine what it is, we explore a bit further. The geometric series is

$$\frac{1}{1+z} = \sum_{k=0}^{\infty}(-1)^k z^k \tag{11.403}$$

and its square is

$$\frac{1}{(1+z)^2} = \sum_{k=0}^{\infty}(-1)^k z^k \sum_{j=0}^{\infty}(-1)^j z^j = \sum_{k=0}^{\infty}\sum_{j=0}^{\infty}(-1)^{j+k} z^{j+k}. \tag{11.404}$$

If we go back to Sect. 5.3, where we reorganized the double sum to a sum over the "diagonal" terms with $m = j + k$, running from 0 to infinity, and over $m' = j - k$, with $-m \leq m' \leq m$ (because both j and k are non-negative). Since the sum does not depend on m', we find a contribution given by the number of allowed m' terms. You might immediately say, I can count those, and they are $2m + 1$, but you need to be careful. Suppose $m = 3$, then we have (j, k) values of $(3, 0)$, $(2, 1)$, $(1, 2)$, and $(0, 3)$. There are four terms. The m' values are then -3, -1, 1, and 3. There are four of them, because the change in steps of 2. Hence, the number of m' terms is equal to $m + 1$, not $2m + 1$. Hence,

$$\sum_{m=0}^{\infty}(-1)^m z^m (m+1) = \frac{1}{(1+z)^2}, \tag{11.405}$$

and we find that

$$\langle p_x, p_y, p_z|\phi_n\rangle = \frac{1}{\left(1+\left(\frac{na_0p}{\hbar}\right)^2\right)^2}\langle 0_{p_x}, 0_{p_y}, 0_{p_z}|\phi_n\rangle. \tag{11.406}$$

The constant $\langle 0_{p_x}, 0_{p_y}, 0_{p_z}|\phi_n\rangle$ is found via normalization of the associated $\langle\vec{p}|\psi_{nn-1}\rangle$ wavefunction. It is given by

$$\langle 0_{p_x}, 0_{p_y}, 0_{p_z}|\phi_n\rangle = \left(\frac{na_0}{\hbar}\right)^{\frac{3}{2}}\frac{1}{(na_0)^{n-1}}\frac{2^{n+2}}{\sqrt{\pi(2n)!}}. \tag{11.407}$$

Putting everything together finally gives us the normalized momentum wavefunction for hydrogen:

$$\tilde{\psi}_{nl}(p_x, p_y, p_z) = \sqrt{\frac{2(n-l-1)!}{\pi(n+l)!}} n^2 \left(\frac{na_0 p}{\hbar}\right)^l \frac{P_h^l(p_x, p_y, p_z)}{(ip)^l} \times C_{n-l-1}^{l+1}\left(\frac{-1+\xi_n^2}{1+\xi_n^2}\right) \frac{2^{2l+2} l!}{\left(1+\xi_n^2\right)^{l+2}} \left(\frac{a_0}{\hbar}\right)^{\frac{3}{2}}. \tag{11.408}$$

This is the standard result for the hydrogen wavefunction in momentum space.

We did it! We did it! We did it! Making it through this derivation (especially if you worked through all of the problems) is quite an achievement. I am still in awe that such a complex result arises solely from the fact that $[\hat{r}_\alpha, \hat{p}_\beta] = i\hbar\delta_{\alpha\beta}$ and algebra. And, while a challenge, it really is possible to get all the way through to the end. Very few people have ever worked out this result explicitly. You are now a member of this select group! This material is tough, so the video https://tinyurl.com/ah8wsfut may help too.

So, why did we work so hard? I told you there would be a big payoff and here it is—this wavefunction that we worked so hard to determine is the one that can actually be measured and observed! The technique is called electron momentum spectroscopy, and it has been an experimental tool for over 50 years. It is often called (e,2e) spectroscopy because one shoots an energetic electron onto a target containing hydrogen. The electron ionizes the hydrogen and ejects an electron, so the "reaction" can be written as

$$e^- + H(1s) \rightarrow 2e^- + H^+. \tag{11.409}$$

In other words, we shoot an energetic electron onto a hydrogen atom, and it strips an electron off the atom to ionize it—one electron in and two electrons out. We can detect that such an event occurred, by measuring the coincidence for the detection of both of the outgoing electrons. The analysis is nontrivial (books have been written about it (Weigold, 1999)), and it requires a quantum theory of scattering, which we will not develop in this book. But the formal analysis is rather simple—the scattering is proportional to the probability to find the initial electron with momentum p, namely $|\tilde{\psi}_{nl}(\vec{p})|^2$. The experiments are usually set up to detect electrons emitted with no recoil of the atomic nucleus. Then the momentum is determined by measuring the angles that the electrons are detected at. This then measures directly the momentum distribution for the electron in the initial state inside the atom (which is, invariably, the 1s ground state). From the general result for the hydrogen wavefunction in momentum space, we find that the probability distribution is given by

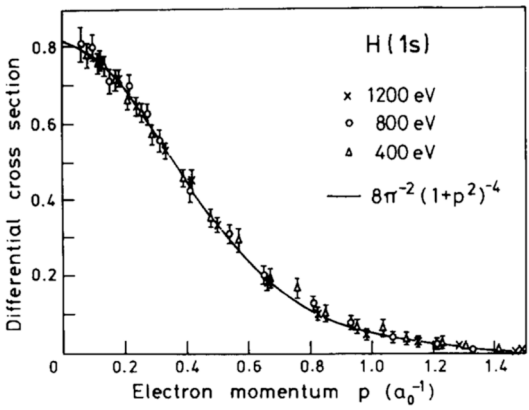

Fig. 11.17 Probability distribution in momentum space for electrons in the 1s orbital of hydrogen as measured in (e,2e) spectroscopy from Lohmann and Weigold (1981). (© Elsevier (1981) and reproduced with permission of *Physics Letters A*)

$$\text{probability} = \frac{8}{\pi^2}\frac{1}{\left(1+\left(\frac{a_0 p}{\hbar}\right)^2\right)^4}, \tag{11.410}$$

after using the fact that the normalized $P_h^0(p_x, p_y, p_z) = \frac{1}{\sqrt{4\pi}}$. A comparison of this result with experiments performed in 1981 by Lohmann and Weigold (1981) is shown in Fig. 11.17. You can see that the agreement is excellent.

Section 11.10 Problems:

11.10.1 Examples of Harmonic Polynomials

Verify that the following are all harmonic polynomials:

(a) $\hat{r}_x\hat{r}_y\hat{r}_z$
(b) $\hat{r}_x^3 - 3\hat{r}_x\hat{r}_y^2$
(c) $\hat{r}_x(\hat{r}_x^2 + \hat{r}_y^2 - 4\hat{r}_z^2) = \hat{r}_x^3 + \hat{r}_x\hat{r}_y^2 - 4\hat{r}_x\hat{r}_z^2$
(d) $(\hat{r}_x^2 - \hat{r}_y^2)(6\hat{r}_z^2 - \hat{r}_x^2 - \hat{r}_y^2) = -\hat{r}_x^4 + \hat{r}_y^4 + 6\hat{r}_x^2\hat{r}_z^2 - 6\hat{r}_y^2\hat{r}_z^2$

11.10.2 Intertwining Relation

(a) Verify that $\hat{T}_\perp \hat{B}_r^\dagger(\lambda) = \left(\hat{B}_r^\dagger(\lambda) - \frac{2i\hbar}{\sqrt{2M}\hat{r}}\right)\hat{T}_\perp$ and $\frac{1}{\hat{r}^2}\hat{B}_r^\dagger(\lambda) = \left(\hat{B}_r^\dagger(\lambda) - \frac{2i\hbar}{\sqrt{2M}\hat{r}}\right)\frac{1}{\hat{r}^2}$
(b) Show that the intertwining relation is

$$\hat{H}(\lambda{=}1)\hat{B}_r^\dagger(\lambda) = \hat{B}_r^\dagger(\lambda)\left(\hat{H}(\lambda{=}1) + \tfrac{\hbar^2(\lambda+1)}{M\hat{r}^2}\right) - \tfrac{2i\hbar}{\sqrt{2M\hat{r}}}\left(\hat{T}_\perp - \tfrac{\hbar^2\lambda(\lambda+1)}{2M\hat{r}^2}\right) \tag{11.411}$$

(c) Use induction to show that

$$\begin{aligned}
&\hat{H}(\lambda{=}1)\hat{B}_r^\dagger(l)\hat{B}_r^\dagger(l+1)\cdots\hat{B}_r^\dagger(n-2) \\
&= \hat{B}_r^\dagger(l)\cdots\hat{B}_r^\dagger(n-2)\left(\hat{H}(\lambda{=}1) + \tfrac{\hbar^2}{M\hat{r}^2}\sum_{j=1}^{n-l-1}(n-j)\right) \\
&\quad + \left\{\left(\hat{B}_r^\dagger(l) - \tfrac{2i\hbar}{\sqrt{2M\hat{r}}}\right)\left(\hat{B}_r^\dagger(l+1) - \tfrac{2i\hbar}{\sqrt{2M\hat{r}}}\right)\cdots\left(\hat{B}_r^\dagger(n-2) - \tfrac{2i\hbar}{\sqrt{2M\hat{r}}}\right)\right. \\
&\quad \left. - \hat{B}_r^\dagger(l)\hat{B}_r^\dagger(l+1)\cdots\hat{B}_r^\dagger(n-2)\right\}\left(\hat{T}_\perp - \tfrac{\hbar^2 l(l+1)}{2M\hat{r}^2}\right)
\end{aligned} \tag{11.412}$$

Hint: it is easiest to assume this induction relation holds for this case and consider the case with an additional $\hat{B}_r^\dagger(l-1)$ added on the left.

11.10.3 Normalization of Energy Eigenstates

Using our standard method of replacing the central $\hat{B}_r(\lambda)\hat{B}_r^\dagger(\lambda)$ by $\hat{H}(\lambda{=}1) + \cdots$, show that the norm satisfies

$$\begin{aligned}
\langle\psi_{nl}|\psi_{nl}\rangle &= \langle\phi_n|P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)\hat{r}^{n-l-1}\hat{B}_r(n-2)\cdots\hat{B}_r(l)\hat{B}_r^\dagger(l) \\
&\qquad \cdots\hat{B}_r^\dagger(n-2)\hat{r}^{n-l-1}P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)|\phi_n\rangle \\
&= \prod_{j=l+1}^{n-1}(\tilde{\varepsilon}_n - \tilde{\varepsilon}_j)\langle\phi_n|\hat{r}^{2n-2}\left(\tfrac{1}{\hat{r}^l}P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)\right)^2|\phi_n\rangle.
\end{aligned} \tag{11.413}$$

We choose the auxiliary ground state to be normalized by $\langle\phi_n|\hat{r}^{2n-2}|\phi_n\rangle = 1$, which yields $\langle 0_{r_x}, 0_{r_y}, 0_{r_z}|\phi_n\rangle = \left(\frac{2}{na_0}\right)^{n+\frac{1}{2}}\frac{1}{\sqrt{(2n)!}}$ (but this normalization factor does not enter into the result further). We also assume that the harmonic polynomial term, given by $\frac{1}{\hat{r}^l}P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)$ involves only angles and is always chosen to have unit norm over the angles. Then, you need to compute the normalization factor γ_{nl} for the energy eigenstate and show it is equal to

$$\gamma_{nl} = \left(\frac{8\pi\epsilon_0 a_o n^2}{e^2}\right)^{\frac{n-l-1}{2}}\frac{(n-1)!}{l!}\sqrt{\frac{(n+l)!}{(2n-1)!(n-l-1)!}}. \tag{11.414}$$

11.10.4 Recurrence Relation for the Moments

Starting from where we left off in the derivation of the moments with

$$\left(1+\left(\frac{na_0p}{\hbar}\right)^2\right)\mu_m(\vec{p},n)=na_0(m+2)\mu_{m-1}(\vec{p},n) + \sum_\alpha \frac{(na_0)^3}{(i\hbar)^2}mp_\alpha\langle\vec{p}\,|\hat{r}^{m-1}\,\hat{p}_\alpha|\phi_n\rangle \tag{11.415}$$

(a) Go through one more iteration to find

$$\left(1+\left(\frac{na_0p}{\hbar}\right)^2\right)\mu_m(\vec{p},n)=na_0(m+2)\mu_{m-1}(\vec{p},n)-\left(\frac{na_0p}{\hbar}\right)^2 na_0m\mu_{m-2}(\vec{p},n) + \sum_\alpha \frac{(na_0)^4}{(i\hbar)^2}m(m-1)p_\alpha\langle\vec{p}\,|\hat{r}^{m-2}\,\hat{p}_\alpha|\phi_n\rangle. \tag{11.416}$$

(b) Complete the proof by induction to show that

$$\mu_m(\vec{p},n)=\frac{na_0(m+2)}{1+\left(\frac{na_0p}{\hbar}\right)^2}\,\mu_{m-1}(\vec{p},n) - \frac{\left(\frac{na_0p}{\hbar}\right)^2}{1+\left(\frac{na_0p}{\hbar}\right)^2}\sum_{k=1}^{m}\frac{m!}{(m-k)!}(na_0)^k\mu_{m-k}(\vec{p},n). \tag{11.417}$$

(c) The result for part (b) is an m-term recurrence relation for $\mu_m(\vec{p},n)$. This is inconvenient to use. Convert it into a three-term recurrence relation by taking the relation for $m-1$, multiply it by mna_0, and replace the terms with $2\le k\le m$ in the summation. Show that

$$\mu_m(\vec{p},n)=\frac{2na_0(m+1)}{1+\left(\frac{na_0p}{\hbar}\right)^2}\,\mu_{m-1}(\vec{p},n)-\frac{(na_0)^2m(m+1)}{1+\left(\frac{na_0p}{\hbar}\right)^2}\,\mu_{m-2}(\vec{p},n). \tag{11.418}$$

11.10.5 Solving the Recurrence Relation for the Moments

We will use a proof by induction to solve for the recurrence relation. The base case is found from

$$\mu_0(\vec{p}, n) = \langle p_x, p_y, p_z | \phi_n \rangle \tag{11.419}$$

and has $Q_{m=0}(\xi_n^2) = 1$.

(a) Solve for $Q_{m=1}(\xi_n^2)$ by finding $\mu_1(\vec{p}, n) = \langle p_x, p_y, p_z | \hat{r} | \phi_n \rangle$. Use the same approach as used in the text: (i) $\hat{r} = \sum_\alpha \frac{\hat{r}_\alpha^2}{\hat{r}}$; (ii) $\frac{\hat{r}_\alpha}{\hat{r}} | \phi_n \rangle = \frac{na_0}{i\hbar} \hat{p}_\alpha | \phi_n \rangle$; (iii) $\langle \vec{p} \, | \hat{p}_\alpha = \langle \vec{p} \, | p_\alpha$; and so on, to find that $Q_1(\xi_n^2) = 3 - \xi_n^2$.

(b) Use the three-term recurrence relation for $\mu_m(\vec{p}, n)$ and the definition of $\mu_m(\vec{p}, n)$ in terms of $Q_m(\xi_n^2)$ to find

$$Q_m(\xi_n^2) = 2(m+1) Q_{m-1}(\xi_n^2) - m(m+1)(1+\xi_n^2) Q_{m-2}(\xi_n^2). \tag{11.420}$$

(c) Verify that Q_0 and Q_1 have the form of Eq. (11.381), given in the text. Note that $\lfloor x \rfloor$ is the *floor* of x, which is the largest integer less than or equal to x.

(d) The proof by induction requires two separate proofs. One for Q_{2k+1} and one for Q_{2k}. In each case, we assume that the form in Eq. (11.381) holds for all m less than $2k+1$ (or less than $2k$) and then show it holds for $m = 2k+1$ (or $m = 2k$). The proof requires you to work with factorials and to simplify complex terms. This is a classic "French-cooking" exercise. Just be sure to remove as many common factors as you can from the terms being added together. You will also have to consider the case with $(\xi_n^2)^0$ and $(\xi_n^2)^{\max \exp}$ separately from the general case (so each proof has some subcases that need to be verified).

11.10.6 Recurrence Relation for Reverse Bessel Polynomials

(a) Starting from the defining relation

$$\langle \vec{p} \, | P_h^l(\hat{r}_x, \hat{r}_y, \hat{r}_z) | \phi_n \rangle = \frac{(na_0)^{2j}(n-1-j)!}{\hbar^{2j}(n-1)!} \sum_{\alpha_1 \alpha_2 \cdots \alpha_j} p_{\alpha_1} p_{\alpha_2} \cdots p_{\alpha_j}$$
$$\times \langle \vec{p} \, | \theta_j \left(\frac{\hat{r}}{na_0} \right) \left[\hat{p}_{\alpha_j}, \left[\cdots, \left[\hat{p}_{\alpha_1}, P_h^{n-1}(\hat{r}_x, \hat{r}_y, \hat{r}_z) \right] \cdots \right] \right]_j | \phi_n \rangle, \tag{11.421}$$

remove one more term from the harmonic polynomial to find

$$\langle \vec{p}\,|P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)|\phi_n\rangle = \frac{(na_0)^{2j+2}(n-2-j)!}{\hbar^{2j+2}(n-1)!}\Bigg\{\sum_{\alpha_1\alpha_2\cdots\alpha_j} p_{\alpha_1}p_{\alpha_2}\cdots p_{\alpha_{j+1}}$$
$$\times\langle\vec{p}\,|\tfrac{\hat{r}}{na_0}\theta_j\left(\tfrac{\hat{r}}{na_0}\right)\Big[\hat{p}_{\alpha_{j+1}},\Big[\cdots,\Big[\hat{p}_{\alpha_1},P_h^{n-1}(\hat{r}_x,\hat{r}_y,\hat{r}_z)\Big]\cdots\Big]\Big]_{j+1}|\phi_n\rangle$$
$$+\sum_{\alpha_1\cdots\alpha_{j+1}} p_{\alpha_1}\cdots p_{\alpha_j}\langle\vec{p}\,|\Big[\tfrac{\hat{r}}{na_0}\theta_j\left(\tfrac{\hat{r}}{na_0}\right),\hat{p}_r\Big]\Big[\hat{p}_{\alpha_{j+1}},\Big[$$
$$\cdots,\Big[\hat{p}_{\alpha_1},P_h^{n-1}(\hat{r}_x,\hat{r}_y,\hat{r}_z)\Big]\cdots\Big]\Big]_{j+1}\tfrac{\hat{r}_{\alpha_{j+1}}}{\hat{r}}|\phi_n\rangle\Bigg\}. \tag{11.422}$$

You will need to use the fact that $[\hat{p}_\alpha, f(\hat{r})] = [\hat{p}_r, f(\hat{r})]\frac{\hat{r}_\alpha}{\hat{r}}$ (which you should be able to derive for yourself). Also recall that the j-fold nested commutator of the $n-1$ degree harmonic polynomial is a harmonic polynomial of degree $n-1-j$.

(b) Note the "trailing" $\frac{\hat{r}_\alpha}{\hat{r}}$ term on the far right of the second term. It can be converted to $\hat{p}_{\alpha_{j+1}}$ (times a constant) and then moved to the left. It commutes with the nested commutator due to the fact that a harmonic polynomial satisfies Laplace's equation. When it passes through the commutator of the reverse Bessel polynomial, we can convert the Cartesian momentum commutator into a radial momentum commutator multiplied by the same $\frac{\hat{r}_\alpha}{\hat{r}}$. It also moves to the far left and acts on the momentum eigenstate, bringing in a factor of $p_{\alpha_{j+1}}$. The process then continues until the nested commutator of the reverse Bessel polynomial is zero. This then allows us to determine a formula for the reverse Bessel polynomial, which is given by

$$\theta_{j+1}\left(\tfrac{\hat{r}}{na_0}\right) = \sum_{k=0}^{j+1}\left(\tfrac{na_0}{i\hbar}\right)^k\Big[\cdots\Big[\Big[\left(\tfrac{\hat{r}}{na_0}\right)\theta_j\left(\tfrac{\hat{r}}{na_0}\right),\hat{p}_r\Big],\hat{p}_r\Big],\cdots,\hat{p}_r\Big]_k. \tag{11.423}$$

Verify this result.

11.10.7 Conversion of the Harmonic Polynomial from Position Operators to Momentum

Show that the final formula for the harmonic polynomial, given by

$$P_h^l(p_x,p_y,p_z)=\frac{1}{(i\hbar)^l l!}\sum_{\alpha_1\cdots\alpha_l} p_{\alpha_1}\cdots p_{\alpha_l}[\hat{p}_{\alpha_l},[\hat{p}_{\alpha_{l-1}},\cdots[\hat{p}_{\alpha_1},P_h^l(\hat{r}_x,\hat{r}_y,\hat{r}_z)]\cdots]], \tag{11.424}$$

holds for the following cases

(a) $\hat{r}_x\hat{r}_y\hat{r}_z$
(b) $\hat{r}_x^2 + \hat{r}_y^2 - 2r_z^2$
(c) $\hat{r}_x^3 - 3\hat{r}_x\hat{r}_y^2$

11.10.8 Proof of the Explicit Formula for Reverse Bessel Polynomials

Starting from the base case of the first two reverse Bessel polynomials, given by $\theta_0\left(\frac{\hat{r}}{na_0}\right) = 1$ and $\theta_1\left(\frac{\hat{r}}{na_0}\right) = \frac{\hat{r}}{na_0} + 1$, and the general recurrence relation, carry out a proof by induction to show that the reverse Bessel polynomials are given by

$$\theta_j\left(\frac{\hat{r}}{na_0}\right) = \sum_{k=0}^{j} \frac{(j+k)!}{(j-k)!k!}\,\frac{1}{2^k}\left(\frac{\hat{r}}{na_0}\right)^{j-k}. \tag{11.425}$$

Note that this is a challenging problem to complete.

11.10.9 Matrix Element of a Reverse Bessel Polynomial

In this problem, we evaluate the matrix element of a reverse Bessel polynomial. It will require a few identities, which I only know how to verify using calculus and complex analysis. I will just tell you those identities as we need them.

(a) Using the exact result for the moments in terms of $Q_m(\xi_n^2)$ and the definition of the reverse Bessel polynomials, show that

$$\langle p_x, p_y, p_z|\theta_m\left(\frac{\hat{r}}{na_0}\right)|\phi_n\rangle = \sum_{j=0}^{m} \frac{(m+j)!}{(m-j)!j!2^j}\,\frac{1}{(1+\xi_n^2)^{m-j}} \times \sum_{k=0}^{\left\lfloor\frac{m-j+1}{2}\right\rfloor} (-1)^k \frac{(m-j+2)!(m-j+1)!}{2(m-j+1-2k)!(2k+1)!}\xi_n^{2k}\langle p_x, p_y, p_z|\phi_n\rangle. \tag{11.426}$$

(b) The innermost sum can be performed because it takes the form

$$\sum_{k=0}^{\left\lfloor \frac{m-j+1}{2} \right\rfloor} (-1)^k \binom{n}{2k+1} \xi_n^{2k}, \tag{11.427}$$

multiplied by a term independent of k. Use the binomial theorem for $(1+i\xi_n)^\alpha$ and $(1-i\xi_n)^\alpha$ in the combination

$$\frac{1}{2\xi_n}\left((1+i\xi_n)^{m-j+2} - (1-i\xi_n)^{m-j+2}\right), \tag{11.428}$$

and add the expression into the summation we need to perform. Use this to determine the innermost summation.

(c) After completing the innermost sum, the outermost sum takes the form

$$\sum_{j=0}^{m} \frac{(m+j)!(m-j+1)!}{j!2^j} \frac{(1+\xi_n^2)^{j-m}}{2\xi_n} \operatorname{Im}\left[(1+\xi_n)^{m-j+2}\right]. \tag{11.429}$$

Show that this result is equal to

$$\operatorname{Im}\left\{ \frac{m!(1+i\xi_n)^{m+2}}{2\xi_n(1+\xi_n^2)^m} \sum_{j=0}^{m} \binom{m+j}{j} (m-j+1)\left(\tfrac{1-i\xi_n}{2}\right)^j \right\}. \tag{11.430}$$

(d) The summation in (c) has the form

$$\sum_{j=0}^{m} \binom{m+j}{j} (m-j+1) z^j. \tag{11.431}$$

See if you can determine what the sum is equal to. I could only do this using complicated calculus and complex analysis identities. In fact, I could only find the result of the sum after taking the imaginary part. You should complete the problem using the identity, as I outline further, if you, like me cannot calculate the summation algebraically. Using $z = \left(\frac{1-i\xi_n}{2}\right)$, we have

$$\sum_{j=0}^{m} \binom{m+j}{j} (m-j+1) z^j = \frac{z^{m+2}}{m!} f(z), \tag{11.432}$$

with $\operatorname{Im} f(z) = (m+1)!2\left(\frac{2}{1+\xi_n^2}\right)^{m+2}$. Use this to show the outermost sum becomes $(m+1)!\left(\frac{2}{1+\xi_n^2}\right)^m$. This then establishes the matrix element used in Eq. (11.393).

11.10.10 Expansion of the Laguerre Polynomial in Terms of Reverse Bessel Polynomials

We compute the coefficients for the reverse Bessel polynomial expansion of the Laguerre polynomial $L_{n-l-1}^{(2l+1)}$. First show that

$$a_{j+k} - (j+k+1)a_{j+k+1} = (-1)^{j+k}2^{j+k}\frac{(n+l)!}{(j+k)!}\frac{(n+l)!(2n-j-k)}{(2l+j+k+2)!(n-l-k-1)!}. \tag{11.433}$$

Use this result to further show that

$$c_j = (-1)^j 2^j[(n+l)!]^2 \times \sum_{k=0}^{\min(j,n-l-j-1)} \frac{(2n-j-k)}{k!(j-k)!(2l+j+k+2)!(n-l-j-k-1)!}. \tag{11.434}$$

Finally, use the van der Monde identity, given by

$$\sum_{j=0}^{n}\binom{n}{j}\binom{m}{k-j} = \binom{n+m}{k}, \tag{11.435}$$

to shown that

$$c_j = (-1)^j 2^j \frac{2n(n+l)!(n+l+j)!}{j!(n-l-j-1)!(2l+2j+2)!}. \tag{11.436}$$

11.10.11 Gegenbauer Polynomials

The standard form for a Gegenbauer polynomial (with m a positive integer) is

$$C_m^{\nu}(x) = \sum_{k=0}^{m}\frac{\Gamma(m+k+2\nu)\Gamma(\nu+\frac{1}{2})}{k!(m-k)!\Gamma(2\nu)\Gamma(\nu+k+\frac{1}{2})}\left(\frac{x-1}{2}\right)^k, \tag{11.437}$$

expressed in terms of the gamma function $\Gamma(\nu)$. Using $\nu = l+1$, $m = n-l-1$, and $x = (-1+\xi_n^2)/(1+\xi_n^2)$, show that this result does give Eq. (11.398). Recall that $\Gamma(z+1) = z!$ for z an integer and

$$\frac{\Gamma(\nu+k+\frac{1}{2})}{\Gamma(\nu+\frac{1}{2})} = (\nu+k-\tfrac{1}{2})(\nu+k-+\tfrac{3}{2})\cdots(\nu+\tfrac{3}{2})(\nu+\tfrac{1}{2}) \tag{11.438}$$

when both ν and k are integers.

11.10.12 Auxiliary Hamiltonian Ground State

In this problem, we will prove in general that

$$\langle p_x, p_y, p_z|(p_x\hat{r}_x + p_y\hat{r}_y + p_z\hat{r}_z)^{2m'+1}|\phi_n\rangle = 0. \tag{11.439}$$

We will also prove

$$\langle 0_{p_x}, 0_{p_y}, 0_{p_z}|(p_x\hat{r}_x + p_y\hat{r}_y + p_z\hat{r}_z)^{2m'}|\phi_n\rangle = (2m')!(m'+1)(na_0p)^{2m'} \times \langle 0_{p_x}, 0_{p_y}, 0_{p_z}|\phi_n\rangle. \tag{11.440}$$

Start by working out the product

$$\begin{aligned}
&\langle p_x, p_y, p_z|(p_x\hat{r}_x + p_y\hat{r}_y + p_z\hat{r}_z)^m|\phi_n\rangle \\
&\quad= \sum_{\alpha_1\cdots\alpha_m} p_{\alpha_1}p_{\alpha_2}\cdots p_{\alpha_m}\langle 0_{p_x}, 0_{p_y}, 0_{p_z}|\hat{r}_{\alpha_1}\cdots\hat{r}_{\alpha_m}|\phi_n\rangle \\
&\quad= \sum_{\alpha_1\cdots\alpha_m} p_{\alpha_1}p_{\alpha_2}\cdots p_{\alpha_m}\frac{na_0}{i\hbar}\langle 0_{p_x}, 0_{p_y}, 0_{p_z}|\hat{r}_{\alpha_1}\cdots\hat{r}_{\alpha_{m-1}}\hat{r}\,\hat{p}_{\alpha_m}|\phi_n\rangle \\
&\quad= \sum_{\alpha_1\cdots\alpha_m} p_{\alpha_1}p_{\alpha_2}\cdots p_{\alpha_m}\frac{na_0}{i\hbar}\langle 0_{p_x}, 0_{p_y}, 0_{p_z}|[\hat{r}_{\alpha_1}\cdots\hat{r}_{\alpha_{m-1}}\hat{r},\, \hat{p}_{\alpha_m}]|\phi_n\rangle.
\end{aligned} \tag{11.441}$$

Continue in this manner to verify the following recurrence relation between the matrix element for m and $m-2$:

$$\begin{aligned}
&\langle 0_{p_x}, 0_{p_y}, 0_{p_z}|(p_x\hat{r}_x + p_y\hat{r}_y + p_z\hat{r}_z)^m|\phi_n\rangle \\
&\quad= (m-1)(m+2)(na_0p)^2\langle 0_{p_x}, 0_{p_y}, 0_{p_z}|(p_x\hat{r}_x + p_y\hat{r}_y + p_z\hat{r}_z)^{m-2}|\phi_n\rangle.
\end{aligned} \tag{11.442}$$

Use this, along with the fact that $\langle 0_{p_x}, 0_{p_y}, 0_{p_z}|p_x\hat{r}_x + p_y\hat{r}_y + p_z\hat{r}_z|\phi_n\rangle = 0$ (which was worked out in the main text) to show that all odd m moments vanish and all even m moments are given by the formula above.

Note that the proof of the recurrence relation does require a number of additional steps not outlined above.

11.10.13 Concrete Results for the First Few Momentum Wavefunctions of Hydrogen

Verify that the hydrogen wavefunctions for the first three n values are

$$\tilde{\psi}_{10}(\vec{p}\,) = \frac{1}{\sqrt{\pi}} \frac{4\sqrt{2}}{\left(1+\left(\frac{a_0 p}{\hbar}\right)^2\right)^2} \left(\frac{a_0}{\hbar}\right)^{3/2} P_h^0(p_x, p_y, p_z)$$

$$\tilde{\psi}_{21}(\vec{p}\,) = \frac{64}{\sqrt{3\pi}} \left(\frac{2a_0 p}{\hbar}\right) \frac{1}{\left(1+\left(\frac{2a_0 p}{\hbar}\right)^2\right)^3} \left(\frac{a_0}{\hbar}\right)^{3/2} \frac{P_h^1(p_x, p_y, p_z)}{(ip)}$$

$$\tilde{\psi}_{20}(\vec{p}\,) = \frac{32}{\sqrt{\pi}} \frac{-1+\left(\frac{2a_0 p}{\hbar}\right)^2}{\left(1+\left(\frac{2a_0 p}{\hbar}\right)^2\right)^3} \left(\frac{a_0}{\hbar}\right)^{3/2} P_h^0(p_x, p_y, p_z)$$

$$\tilde{\psi}_{32}(\vec{p}\,) = \frac{576}{\sqrt{15\pi}} \left(\frac{3a_0 p}{\hbar}\right)^2 \frac{1}{\left(1+\left(\frac{3a_0 p}{\hbar}\right)^2\right)^4} \left(\frac{a_0}{\hbar}\right)^{3/2} \frac{P_h^2(p_x, p_y, p_z)}{(ip)^2}$$

$$\tilde{\psi}_{31}(\vec{p}\,) = \frac{288}{\sqrt{3\pi}} \left(\frac{3a_0 p}{\hbar}\right) \frac{-1+\left(\frac{3a_0 p}{\hbar}\right)^2}{\left(1+\left(\frac{3a_0 p}{\hbar}\right)^2\right)^4} \left(\frac{a_0}{\hbar}\right)^{3/2} \frac{P_h^1(p_x, p_y, p_z)}{(ip)}$$

$$\tilde{\psi}_{30}(\vec{p}\,) = \frac{72}{\sqrt{6\pi}} \frac{3-10\left(\frac{3a_0 p}{\hbar}\right)^2+3\left(\frac{3a_0 p}{\hbar}\right)^4}{\left(1+\left(\frac{3a_0 p}{\hbar}\right)^2\right)^4} \left(\frac{a_0}{\hbar}\right)^{3/2} P_h^0(p_x, p_y, p_z) \tag{11.443}$$

by carefully evaluating the formula in Eq. (11.408).

11.11 Expectation Values of $\hat{r}$ and the Hypervirial Theorem

It is common to calculate expectation values of $\hat{r}$ for different hydrogen energy eigenstates. The expectation values can be employed to examine average values of the radius, its square, and its uncertainty relation. They can also be used to find the potential energy and the expectation value of the centrifugal term in the kinetic energy. In addition, they are used in calculating first-order perturbation theory corrections due to relativistic effects (which we will treat exactly rather

than perturbatively in Chap. 16). Many of these applications (and others) will be discussed later.

These expectation values are computed in many quantum textbooks. In others, they are only stated, with a note that the calculations are too difficult to present. This is definitely true if you attempt to perform these calculations using calculus. While the integrals needed are fairly straightforward, the procedure is cumbersome and highly prone to error. An operator method approach is instead quite simple. This is what we use here.

These identities are computed using an approach called the *hypervirial theorem*. This is relatively simple to understand, but has some subtlety that we have already seen in Prob. 11.7.7, when we computed expectation values of powers of $\hat{r}$ for the isotropic harmonic oscillator in three dimensions. We will now state the theorem in its most common usage, where the expectation value is with respect to energy eigenstates $|n\rangle$ which satisfy $\hat{H}|n\rangle = E_n|n\rangle$. The theorem simply states that

$$\langle n|[\hat{O}, \hat{H}]|n\rangle = \langle n|\hat{O}\hat{H}|n\rangle - \langle n|\hat{H}\hat{O}|n\rangle = (E_n - E_n)\langle n|\hat{O}|n\rangle = 0. \tag{11.444}$$

The energy eigenvalues emerge by acting the Hamiltonian operator on the corresponding bra or ket. The subtlety that arises is that the states $|n\rangle$ must be well-defined when both $\hat{O}$ and $\hat{H}$ act on them in either order. The technical description of this is that both operators are Hermitian with respect to $|n\rangle$ and $\hat{O}|n\rangle$. Verifying this requires calculus, so we will point out situations where one needs to worry about this.

To apply this to hydrogen, we want to determine expectation values of powers of $\hat{r}$ with respect to energy eigenstates $|\psi_{nl}\rangle = |nl\rangle$. The strategy is to use the hypervirial theorem to determine a recurrence relation between the expectation values for different powers of $\hat{r}$. Then, by explicitly determining these expectation values $\langle nl|\hat{r}^m|nl\rangle$ for $m = 0, -1$, and -2, allows us to determine all matrix elements algebraically. This problem was first solved using calculus by Kramers (1957) and Pasternack (1937). It was only recently that the completely algebraic solution was worked out; it was started in Epstein and Epstein (1962) and completed in Szymanski and Freericks (2021).

We start with the hypervirial theorem using the Hermitian symmetric operator $\hat{O} = \hat{r}^m\hat{p}_r + \hat{p}_r\hat{r}^m$. We find that the commutator of $\hat{O}$ with the hydrogen Hamiltonian for definite angular momentum $\hat{H}_l = \frac{\hat{p}^2}{2M} + \frac{\hbar^2 l(l+1)}{2M\hat{r}^2} - \frac{e^2}{4\pi\epsilon_0\hat{r}}$ is given by

$$\begin{aligned}\left[\hat{r}^m\hat{p}_r + \hat{p}_r\hat{r}^m, \hat{H}\right] &= \tfrac{i\hbar}{2M}m\left(\hat{r}^{m-1}\hat{p}_r^2 + 2\hat{p}_r\hat{r}^{m-1}\hat{p}_r + \hat{p}_r^2\hat{r}^{m-1}\right)\\ &\quad + \tfrac{i\hbar}{2M}4\hbar^2 l(l+1)\hat{r}^{m-3} - \tfrac{i\hbar e^2}{4\pi\epsilon_0}\hat{r}^{m-2}.\end{aligned} \tag{11.445}$$

Focusing on the top line on the right-hand side, we take each of the $\hat{p}_r\hat{r}^{m-1}\hat{p}_r$ terms and for one of them we move the $\hat{p}_r$ term on the left to the right, while for the other, we move the $\hat{p}_r$ term on the right to the left, remembering to keep the commutator

corrections. Then we have

$$\hat{r}^{m-1}\hat{p}_r^2 + 2\hat{p}_r\hat{r}^{m-1}\hat{p}_r + \hat{p}_r^2\hat{r}^{m-1} = 2\hat{p}_r^2\hat{r}^{m-1} + 2\hat{r}^{m-1}\hat{p}_r^2 - i\hbar(m-1)\hat{r}^{m-2}\hat{p}_r + i\hbar(m-1)\hat{p}_r\hat{r}^{m-2}. \tag{11.446}$$

The last two terms are a commutator, which can be computed to give us our final result:

$$\hat{r}^{m-1}\hat{p}_r^2 + 2\hat{p}_r\hat{r}^{m-1}\hat{p}_r + \hat{p}_r^2\hat{r}^{m-1} = 2\hat{p}_r^2\hat{r}^{m-1} + 2\hat{r}^{m-1}\hat{p}_r^2 + \hbar^2(m-1)(m-2)\hat{r}^{m-3}. \tag{11.447}$$

We substitute this into our original commutator expression and use an "add-zero" trick to construct the Hamiltonian, which yields

$$\left[\hat{r}^m\hat{p}_r + \hat{p}_r\hat{r}^m, \hat{H}\right] = i\hbar\Big\{2m\hat{H}_l\hat{r}^{m-1} + 2m\hat{r}^{m-1}\hat{H}_l + \tfrac{\hbar^2}{2M}(m-1)[m(m-2) - 4l(l+1)]\hat{r}^{m-3} + 2(2m-1)\tfrac{e^2}{4\pi\epsilon_0}\hat{r}^{m-2}\Big\}. \tag{11.448}$$

Recalling that $\hat{H}_l|nl\rangle = \varepsilon_n|nl\rangle = -\frac{e^2}{8\pi\epsilon_0 a_0 n^2}|nl\rangle$, we obtain

$$\begin{aligned}\langle nl|\left[\hat{r}^m\hat{p}_r + \hat{p}_r\hat{r}^m, \hat{H}_l\right]|nl\rangle &= 0\\ = i\hbar\Big(&-2m\tfrac{e^2}{4\pi\epsilon_0 a_0 n^2}\langle nl|\hat{r}^{m-1}|nl\rangle - 2(1-2m)\tfrac{e^2}{4\pi\epsilon_0}\langle nl|\hat{r}^{m-2}|nl\rangle\\ &- \tfrac{\hbar^2}{2M}(m-1)\Big((2l+1)^2 - (m-1)^2\Big)\langle nl|\hat{r}^{m-3}|nl\rangle\Big),\end{aligned} \tag{11.449}$$

after using the fact that $4l(l+1) - m(m-2) = 4l^2 + 4l + 1 - m^2 + 2m - 1$. Finally, we simplify by replacing e^2 by $\frac{1}{a_0}$ and constants, to get the final result for the Kramers–Pasternack relation:

$$0 = -\frac{2m}{n^2}\frac{1}{a_0^2}\langle nl|\hat{r}^{m-1}|nl\rangle + 2(2m-1)\frac{1}{a_0}\langle nl|\hat{r}^{m-2}|nl\rangle - \frac{1}{2}(m-1)\Big((2l+1)^2 - (m-1)^2\Big)\langle nl|\hat{r}^{m-3}|nl\rangle. \tag{11.450}$$

This leads to a two-term recurrence relation—if we know two adjacent moments, we can then construct all of the rest.

We start by determining the expectation values for $m > 0$. To do so, we first set $m = 1$. In this case, the coefficient of the third term in the recurrence vanishes, and

we find

$$\langle nl|\frac{1}{\hat{r}}|nl\rangle = \frac{1}{a_0 n^2}. \tag{11.451}$$

This is the familiar virial theorem, which we derived earlier. Next, we set $m = 2$. This gives

$$\langle nl|\hat{r}|nl\rangle = \tfrac{1}{2}\big(3n^2 - l(l+1)\big)a_0, \tag{11.452}$$

showing that the average radius (for fixed l) grows like n^2 as we reasoned earlier. Setting $m = 3$, yields

$$\langle nl|\hat{r}^2|nl\rangle = a_0^2\frac{n^2}{2}\left(5n^2 + 1 - 3l(l+1)\right). \tag{11.453}$$

Computing the results for $m = 4$ and $m = 5$ are given in Problem 11.11.2.

Before pushing on to the negative m values, we first look more carefully at the results for the average values of $\hat{r}$ and $\hat{r}^2$ for the low-lying eigenstates in Table 11.6. In particular, we find that

$$\sqrt{(\Delta r)^2_{|nl\rangle}} = \frac{a_0}{2}\sqrt{n^4 + 2n^2 - l^2(l+1)^2}. \tag{11.454}$$

The fluctuations of momentum are (from the virial theorem and the fact that $\langle nl|\hat{p}_r|nl\rangle = 0$, which is worked out in the problems)

$$\sqrt{(\Delta p_r)^2_{|nl\rangle}} = \frac{\hbar}{a_0}\frac{1}{n}\sqrt{1 - \frac{l(l+1)}{n\left(l+\frac{1}{2}\right)}}. \tag{11.455}$$

So we have

$$\sqrt{(\Delta r)^2_{|nl\rangle}}\sqrt{(\Delta p_r)^2_{|nl\rangle}} = \frac{\hbar}{2}\sqrt{\left(n^2 + 2 - \tfrac{l^2(l+1)^2}{n^2}\right)\left(1 - \tfrac{l(l+1)}{n\left(l+\frac{1}{2}\right)}\right)}. \tag{11.456}$$

We see that the uncertainty for higher l states for a fixed value of n is lower. This is because these states have fewer nodes. More nodes means a more spread out wavefunction in position space, which typically means more uncertainty.

We are now ready to move onto the negative moments. We immediately see an issue with $m = 0$. The term in the recurrence relation proportional to $\langle nl|\frac{1}{\hat{r}^2}|nl\rangle$ has a coefficient of zero. So we need to find this matrix element from some other independent calculation. Once we have that, we find that

Table 11.6 Expectation values for $\langle nl|\hat{r}|nl\rangle$ and $\langle nl|\hat{r}^2|nl\rangle$ for $n \leq 3$. One can see that the uncertainty significantly increases for larger n. Examination of the plots in Figs. 11.13, 11.14, and 11.15, show that in general, $\langle nl|\hat{r}|nl\rangle$ is systematically larger than the most likely value of $\hat{r}$ (found from the maximum of $r^2|\langle nl|\hat{r}|nl\rangle|^2$)

n	l	$\langle nl\|\hat{r}\|nl\rangle$	$\langle nl\|\hat{r}^2\|nl\rangle$
1	0	$\frac{3}{2}a_0$	$3a_0^2$
2	1	$3a_0$	$30a_0^2$
2	0	$6a_0$	$42a_0^2$
3	2	$\frac{9}{2}a_0$	$68a_0^2$
3	1	$\frac{21}{2}a_0$	$180a_0^2$
3	0	$\frac{27}{2}a_0$	$207a_0^2$

$$\langle nl|\tfrac{1}{\hat{r}^3}|nl\rangle = \frac{1}{a_0}\frac{1}{l(l+1)}\langle nl|\tfrac{1}{\hat{r}^2}|nl\rangle \tag{11.457}$$

In most quantum textbooks, the expectation value for $\langle nl|\frac{1}{\hat{r}^2}|nl\rangle$ is determined by using the Feynman–Hellmann theorem and calculus. I do not like this derivation, because it involves taking a derivative with respect to a parameter that is always presented as an integer (the total angular momentum quantum number). While it is possible to formulate the problem in such a way that everything is fine in a formal mathematical sense, it can be quite confusing. In any case, we are interested in a purely algebraic derivation, and it turns out this is possible. We go through the details now.

We start with a variant of the "multiply-by-one" trick. Recall that $\hat{B}_r^\dagger(l)\hat{B}_r(l) = \hat{H}_l - \varepsilon_{l+1}$ and $\hat{H}_l|nl\rangle = \varepsilon_n|nl\rangle$. Hence,

$$\langle nl|\tfrac{1}{\hat{r}^2}|nl\rangle = \langle nl|\hat{B}_r^\dagger(l)\hat{B}_r(l)\tfrac{1}{\hat{r}^2}|nl\rangle\frac{1}{(\varepsilon_n - \varepsilon_{l+1})}. \tag{11.458}$$

This does not immediately seem to help, but if we can move the $\hat{B}_r(l)$ factor past the $\frac{1}{\hat{r}^2}$ term, it will meet a $\hat{B}_r^\dagger(l)$ term, and we can use the fact that $\hat{B}_r(l)\hat{B}_r^\dagger(l) = \hat{H}_{l+1} - \varepsilon_{l+1}$ on both sides of the $\frac{1}{\hat{r}^2}$ term. We also need to take into account some normalization factors. We have that $|nl\rangle = C_{nl}\hat{B}_r^\dagger(l)\cdots\hat{B}_r^\dagger(n-2)|n, l{=}n-1\rangle$ and $|nl+1\rangle = C_{nl+1}\hat{B}_r^\dagger(l+1)\cdots\hat{B}_r^\dagger(n-2)|n, l{=}n-1\rangle$ are related by

$$|nl\rangle = \frac{1}{\sqrt{\varepsilon_n - \varepsilon_{l+1}}}\hat{B}_r^\dagger(l)|nl+1\rangle. \tag{11.459}$$

So, we compute

$$\begin{aligned}\langle nl|\tfrac{1}{\hat{r}^2}|nl\rangle &= \frac{1}{\varepsilon_n-\varepsilon_{l+1}}\langle nl|\hat{B}_r^\dagger(l)\tfrac{1}{\hat{r}^2}\hat{B}_r(l)|nl\rangle\\ &+\frac{1}{\varepsilon_n-\varepsilon_{l+1}}\langle nl|\hat{B}_r^\dagger(l)\left[\hat{B}_r(l),\tfrac{1}{\hat{r}^2}\right]|nl\rangle\\ &=\frac{1}{(\varepsilon_n-\varepsilon_{l+1})^2}\langle nl+1|\hat{B}_r(l)\hat{B}_r^\dagger(l)\tfrac{1}{\hat{r}^2}\hat{B}_r(l)\hat{B}_r^\dagger(l)|nl+1\rangle\\ &+\frac{1}{\varepsilon_n-\varepsilon_{l+1}}\frac{2i\hbar}{\sqrt{2M}}\langle nl|\hat{B}_r^\dagger(l)\hat{B}_r(l)\tfrac{1}{\hat{r}^3}|nl\rangle. \end{aligned}\tag{11.460}$$

Now, we use the fact that $\hat{B}_r(l)\hat{B}_r^\dagger(l)=\hat{H}_{l+1}-\varepsilon_{l+1}$ and commute the $\hat{H}_{l+1}$ through, using the intertwining relation, until it hits the maximal angular momentum bra on the left and ket on the right. We also write out the raising operator concretely in the second term. This gives

$$\begin{aligned}\langle nl|\tfrac{1}{\hat{r}^2}|nl\rangle &= \langle nl+1|\tfrac{1}{\hat{r}^2}|nl+1\rangle\\ &+\frac{2i\hbar}{(\varepsilon_n-\varepsilon_{l+1})2M}\langle nl|\hat{p}_r+i\hbar\left(\tfrac{1}{(l+1)a_0}-\tfrac{l+1}{\hat{r}}\right)\tfrac{1}{\hat{r}^3}|nl\rangle. \end{aligned}\tag{11.461}$$

The last two terms are simply expectation values of $\frac{1}{\hat{r}^3}$ and $\frac{1}{\hat{r}^4}$. The term with the radial momentum is dispatched using the hypervirial theorem as follows:

$$\begin{aligned}0=\langle nl|\left[\hat{H}_l,\tfrac{1}{\hat{r}^2}\right]|nl\rangle &= \frac{2i\hbar}{2M}\langle\left(\hat{p}_r\tfrac{1}{\hat{r}^3}+\tfrac{1}{\hat{r}^3}\hat{p}_r\right)|nl\rangle\\ &=\frac{i\hbar}{M}\langle nl|\left(2\hat{p}_r\tfrac{1}{\hat{r}^3}-\tfrac{3i\hbar}{\hat{r}^4}\right)|nl\rangle. \end{aligned}\tag{11.462}$$

This can be rearranged to

$$\langle nl|\hat{p}_r\tfrac{1}{\hat{r}^3}|nl\rangle=\frac{3i\hbar}{2}\langle nl|\tfrac{1}{\hat{r}^4}|nl\rangle.\tag{11.463}$$

OK, we are nearly there. We need the Kramers-Pasternack relation for $m=-1$ as well. It is

$$-\tfrac{2}{n^2a_0^2}\langle nl|\tfrac{1}{\hat{r}^2}|nl\rangle+\tfrac{6}{a_0}\langle nl|\tfrac{1}{\hat{r}^3}|nl\rangle=(4l^2+4l-3)\langle nl|\tfrac{1}{\hat{r}^4}|nl\rangle.\tag{11.464}$$

We use this equation to remove the $\frac{1}{\hat{r}^4}$ expectation value and then Eq. (11.457) to relate the expectation value of $\frac{1}{\hat{r}^3}$ to the expectation value of $\frac{1}{\hat{r}^2}$. When all the dust settles, we find that

$$\langle nl|\tfrac{1}{\hat{r}^2}|nl\rangle = \frac{l+\frac{3}{2}}{l+\frac{1}{2}}\langle nl+1|\tfrac{1}{\hat{r}^2}|nl+1\rangle, \tag{11.465}$$

or, after extending it all the way out to the end, we have

$$\langle nl|\tfrac{1}{\hat{r}^2}|nl\rangle = \frac{n-\frac{1}{2}}{l+\frac{1}{2}}\langle nn-1|\tfrac{1}{\hat{r}^2}|nn-1\rangle. \tag{11.466}$$

We now need to evaluate the expectation value on the right for the maximal angular momentum state. We use the facts that $\hat{B}_r\,(n-1)|n, l{=}n-1\rangle = 0$ and

$$\hat{B}_r^\dagger(n-1) - \hat{B}_r\,(n-1) = \sqrt{\tfrac{2}{M}}i\hbar\left(\tfrac{1}{na_0} - \tfrac{n}{\hat{r}}\right), \tag{11.467}$$

to see that

$$\begin{aligned}\left(\hat{B}_r^\dagger(n-1) - \hat{B}_r\,(n-1)\right)^2 &= -\tfrac{2\hbar^2}{M}\left(\tfrac{1}{n^2a_0^2} - \tfrac{2}{a_0\hat{r}} + \tfrac{n^2}{\hat{r}^2}\right)\\ &= \hat{B}_r^\dagger(n-1)\hat{B}_r^\dagger(n-1) - \hat{B}_r^\dagger(n-1)\hat{B}_r\,(n-1)\\ &\quad - \hat{B}_r\,(n-1)\hat{B}_r^\dagger(n-1) + \hat{B}_r\,(n-1)\hat{B}_r\,(n-1).\end{aligned} \tag{11.468}$$

Using the subsidiary condition on the bra and on the ket leaves just one nonzero expectation value for the product of ladder operators; hence, by multiplying on the left with the energy eigenstate bra and similarly on the right, we obtain

$$\begin{aligned}&-\tfrac{2\hbar^2}{M}\left(\tfrac{1}{n^2a_0^2} - \tfrac{2}{a_0}\langle nn-1|\tfrac{1}{\hat{r}}|nn-1\rangle + n^2\langle nn-1|\tfrac{1}{\hat{r}^2}|nn-1\rangle\right)\\ &\quad = -\langle nn-1|\hat{B}_r\,(n-1)\hat{B}_r^\dagger(n-1)|nn-1\rangle.\end{aligned} \tag{11.469}$$

But,

$$\hat{B}_r\,(n-1)\hat{B}_r^\dagger(n-1) = \hat{H}_n - \varepsilon_n = \hat{H}_{n-1} + \tfrac{\hbar^2 n}{M\hat{r}^2} - \varepsilon_n. \tag{11.470}$$

So,

$$\begin{aligned}\langle nn-1|\hat{B}_r\,(n-1)\hat{B}_r^\dagger(n-1)|nn-1\rangle &= \langle nn-1|\hat{H}_{n-1} + \tfrac{\hbar^2 n}{M\hat{r}^2} - \varepsilon_n|nn-1\rangle\\ &= \tfrac{\hbar^2 n}{2M}\langle nn-1|\tfrac{1}{\hat{r}^2}|nn-1\rangle.\end{aligned} \tag{11.471}$$

Equating the two sides of Eq. (11.469) then gives us

$$\tfrac{2\hbar^2}{M}\left(\tfrac{1}{n^2a_0^2} - \tfrac{2}{n^2a_0^2} + n^2\langle nn-1|\tfrac{1}{\hat{r}^2}|nn-1\rangle\right) = \tfrac{\hbar^2 n}{M}\langle nn{-}1|\tfrac{1}{\hat{r}^2}|nn{-}1\rangle, \tag{11.472}$$

or

$$\langle nn-1|\tfrac{1}{\hat{r}^2}|nn-1\rangle = \frac{1}{a_0^2 n^2\left(n-\frac{1}{2}\right)}. \tag{11.473}$$

Combining with Eq. (11.466), we obtain our final result, namely

$$\langle nl|\tfrac{1}{\hat{r}^2}|nl\rangle = \frac{1}{a_0^2 n^3\left(l+\frac{1}{2}\right)}, \tag{11.474}$$

which is the well-known result for this expectation value. Using this, we can now compute the expectation values of $\frac{1}{\hat{r}^3}$, $\frac{1}{\hat{r}^4}$ and more negative m values. Note that the expectation values of $\hat{r}^m$ for $m = -3$ and $m = -4$ are only well defined for $l \geq 1$. You will get infinity for $l = 1$. Higher negative powers (like $m = -5$ and $m = -6$) are only well defined for $l \geq 2$. The formulas actually show the infinities if the l value is too small. But, with these three results ($m = 0$, $m = -1$, and $m = -2$), we can use the Kramers–Pasternack recurrence relation to calculate the results for all negative m.

Section 11.11 Problems:

11.11.1 Uncertainty

Using the relations for uncertainty

(a) Compute the uncertainty product $\sqrt{(\Delta r)^2_{|nl\rangle}}\sqrt{(\Delta p_r)^2_{|nl\rangle}}$ for the maximal angular momentum states ($l = n-1$) and show they have minimal uncertainty as $n \to \infty$.
(b) Repeat this exercise for $l = 0$ and show that the uncertainty relation is approximately linear in n, increasing as $n \to \infty$.
(c) Comment on the dichotomy between these two results.

11.11.2 Matrix Elements for $m = 3$ and $m = 4$

Derive the following expectation values:

(a) $m = 3$

$\langle nl|\hat{r}^3|nl\rangle$

$$= a_0^3 \frac{n^2}{8} \left(35n^2(n^2-1) - 30n^2(l+2)(l-1) + 3(l+2)(l+1)l(l-1)\right) \tag{11.475}$$

(b) $m = 4$

$$\langle nl|\hat{r}^4|nl\rangle = a_0^4 \frac{n^4}{8} \left(64n^4 - 35n^2(2l^2+2l-3) + 5l(l+1)(3l^2+3l-10) + 12\right). \tag{11.476}$$

One can see that these expectation values become complicated for large m.

11.11.3 Moments for $m = -3$ and $m = -4$

Derive the following expectation values:

(a) $m = -3$

$$\langle nl|\tfrac{1}{\hat{r}^3}|nl\rangle = \frac{1}{a_0^3} \frac{1}{n^3(l+1)\left(l+\frac{1}{2}\right)l} \tag{11.477}$$

(b) $m = -4$

$$\langle nl|\tfrac{1}{\hat{r}^4}|nl\rangle = \frac{1}{a_0^4} \frac{3n^2 - l(l+1)}{2n^5\left(l+\frac{3}{2}\right)(l+1)\left(l+\frac{1}{2}\right)l\left(l-\frac{1}{2}\right)} \tag{11.478}$$

These also get complicated for large m.

Note that when m is too large for a given l, the results diverge. Note the divergence for $m = 3$ and $l = 0$.

11.11.4 Kramers–Pasternack-Like Relations for the Isotropic Simple Harmonic Oscillator in Three Dimensions

One can derive similar recurrence relations for the isotropic simple harmonic oscillator in three dimensions. Find the analog of the Kramers–Pasternack relations for that system. Note that it connects even moments to even moments and odd moments to odd moments only. That is, there are two different recurrence relations.

Ans: The even relation is

$$(m+2)\langle nl|\hat{r}^{m+2}|nl\rangle = (m+1)(2n+3)\langle nl|\hat{r}^{m}|nl\rangle + m\left(\tfrac{m^2-9}{4} - (l-1)(l+m-1)\right)\langle nl|\hat{r}^{m-2}|nl\rangle. \tag{11.479}$$

11.11.5 Pasternack Inversion Relation

Pasternack also derived the so-called *inversion relation* given by

$$\langle nl|\tfrac{1}{\hat{r}^{m+2}}|nl\rangle = \left(\tfrac{2}{na_0}\right)^{2m+1}\frac{(2l-m)!}{(2l+m+1)!}\langle nl|\hat{r}^{m-1}|nl\rangle. \tag{11.480}$$

Hint: use $m = 0$ as a base case, and then induction, assuming that the Kramers–Pasternack recurrence relations hold to move outward in both directions.

11.11.6 Radial Momentum Expectation Values

(a) Show that $\langle nl|\hat{p}_r|nl\rangle = 0$ by using the hypervirial theorem with $\hat{O} = \hat{r}$.
(b) Determine $\langle nl|\hat{p}_r^2|nl\rangle$ by using the energy eigenvalues and the radial moment with $m = -2$.
(c) Discuss whether there is a way to determine $\langle nl|\hat{p}_r^m|nl\rangle$ for nonnegative m using similar ideas as were used for the Kramers–Pasternack problem. Can such a recurrence relation be found? Can anything be done for moments of the radial momentum with $m < 0$?

11.12 Exotic Hydrogen Analogs and the Proton Charge Radius Problem

The hydrogen atom is an exactly solvable problem. This allows us to test the results against small corrections and to even use those small corrections to look for new physical phenomena. Indeed, precision experiments do precisely that. By measuring something that can be described exactly, we discover new physical phenomena when the measurements disagree with the predictions. This is why physicists get excited when a high-precision measurement yields unexpected results.

Let us examine how this works for hydrogen. In the hydrogen problem, we use the effective mass to map the two-particle problem into an effective single-

particle problem with the effective mass. These corrections enter at the level of one part in 10^3. There are a few more corrections one needs to take into account. The first are the effects of relativity. Our quantum mechanical formalism is a nonrelativistic formalism. Introducing relativity, in its full glory, requires quantum field theory, which allows for particle–antiparticle creation. But, one can calculate small relativistic corrections by generalizing quantum mechanics to being a relativistically invariant theory. This was first done by Dirac when he created the so-called Dirac equation. We will discuss this equation in detail in Chap. 16. But, before doing that, we can discuss the effects of relativity. The first one is a kinetic energy effect, arising from the fact that in relativity, the kinetic energy is given by $\sqrt{m^2c^4 + p^2c^2} - mc^2$, whose lowest-order term (for low velocity) is given by $p^2/2m$. But there obviously are higher-order corrections. There is a second relativistic effect, called spin–orbit coupling (and the Darwin term). which arises from the fact that the nucleus is in motion (as well as the electron) and a moving charge creates a magnetic field, which affects the spin of the electron. These corrections enter to the order of one part in 10^6.

Another effect occurs when the nucleus has a spin (such as the proton in hydrogen, which is a spin one half particle). Then there can be a direct interaction between the nuclear spin and the electron spin. For hydrogen, this is a one part in 10^7 effect. We discuss it in detail in Sect. 11.14.

Yet another effect is the so-called Lamb shift. This effect occurs because the charged particles in hydrogen interact with the fluctuating electric fields in the vacuum of free space. This interaction changes the energy of the hydrogen atom on the order of one part in 10^7 as well. It was discovered because two energy levels, the $2s$ and $2p$ levels, are degenerate even after relativistic effects are taken into account. But very precise measurements in the mid-1940s show that they had a small energy splitting. This discovery then led to the development of quantum electrodynamics, as it was one of the important experimental results that could be compared to calculations.

In addition, we have the proton charge radius, which is an effect that enters at the one part in 10^9 level. This requires extraordinarily precise measurements to see it. For a number of years, two different types of experiments that measured it gave different results. As we mentioned before, when this happens, it is usually due to new physics, as we saw with the Lamb shift. But very recently, it has become clear that there is no discrepancy. We will discuss this problem in more detail later in this section.

Finally, in our discussions of hydrogen, we have assumed that the g-factor for the electron is equal to 2 (this is what arises from the Dirac equation). But, again, quantum electrodynamics effects change the g factor from exactly 2 to a number slightly larger than 2. These corrections affect any of the energies that involve the electron spin (spin–orbit coupling and hyperfine splitting) and can be though of as small corrections on already small corrections.

There are even more subtle small effects that might be observable, and these motivate a number of new experiments with exotic hydrogen-like atoms. In this section, we will discuss a number of examples of hydrogen-like atoms and how

performing high-precision measurements on them can lead to interesting tests for new physics. I hope this short introduction makes clear to you why high-precision experiments are so important. They are one way that we can discover new physical behavior!

With this prelude, we now discuss the different hydrogen analogs. There are eight broad categories. They are (i) hydrogenic ions (and isotopic effects); (ii) muonic hydrogen; (iii) leptonic hydrogen; (iv) hadronic hydrogen; (v) anti-hydrogen; (vi) donors in semiconductors; (vi) excitons; and (viii) Rydberg atoms. We now briefly discuss each of these systems.

Previously, when we discussed the Pickering–Fowler lines in astronomy and the discovery of deuterium, we showed how the effective mass changes the spectral lines in hydrogen by about one part in 10^3. The latter example is an isotopic effect, while the former is a hydrogenic ion. *Hydrogenic ions* are defined to be atoms that have had all but one of their electrons stripped off. In this case, the system looks just like hydrogen, except it has a nucleus with a charge Z times larger (Z is the atomic number of the atom), and it has a nuclear mass that differs by the atomic weight of the atomic isotope. It is straightforward, in principle, to create such ions by shining light with the high enough frequencies, we can remove electrons from the atom one by one. But this is really only practical for atoms that need a few electrons removed, such as helium, lithium, and beryllium. The reason why is that the ionization potential for the last few electrons in an atom with a large atomic number becomes very large. It is difficult to find light sources with enough energy to remove those final few electrons. But there are some other methods that can be used that remove electrons from atoms. If we place the atom in a large enough electric field, it bends the potential energy curve that binds the electrons to the atoms, allowing them to tunnel out. If we wait long enough, we should be able to remove all but one of the electrons via a tunneling effect across the barrier. But, the electric fields needed are huge and typically only available in high powered lasers. This means they will oscillate back and forth rather than statically change the binding potential curve. Another process that can work is to remove a deep core electron with X-ray light. Then, either a high energy electron drops into this core state and emits a photon, or it drops nonradiatively to fill in the core electron level (a so-called *Auger process*) where it transfers energy to the higher-energy electrons (instead of creating a photon), which is sufficient enough to remove some of those electrons from the atom. Finally, intense laser light can remove electrons via *multiphoton processes* where the intensity of the light is high enough that the individual photons act together to provide enough energy to ionize the electrons. But, no matter how you look at it, preparing hydrogenic ions for high atomic number atoms is very difficult to do, so most studies of these systems focus on the lower atomic weight systems. The main changes for the hydrogenic (or isotopic) atoms from the results for hydrogen are the changes in the energy levels due to the increased nuclear charge and the change in the reduced mass and a similar modification of the Bohr radius. A quantitative investigation of this is done in Problem 11.12.2.

Muonic hydrogen is an interesting system, because it replaces the electron by a muon. The muon is the heavy cousin of the electron, and its mass is about 207 times

higher than the electron. This makes its Bohr radius about 200 times smaller than for ordinary hydrogen, and it is thereby much more sensitive to the proton charge radius. The creation of muonic hydrogen is not so difficult. One shoots a muon beam into a gas of hydrogen. It loses energy as it moves through the atomic gas and eventually is captured by a hydrogen. Since it is heavier than the electron, it moves to the low energy state and rapidly causes the electron to be ejected (via an Auger process, most likely). Note that the muon is unstable, with a lifetime on the order of a microsecond, so these atoms do not live too long. In addition to their interest in muonic hydrogen, muons have been used to examine hydrogen molecules, where it has been hypothesized that they can catalyze nuclear fusion by bringing the nuclei close enough to each other when the muon is the "glue" for the molecule. But, this has never been firmly established experimentally. They also are interesting for studying capture effects by the nucleus, because the muon again is more likely to be found in the nucleus than the electron.

Leptonic hydrogen replaces the nucleus by a positively charged lepton (positron, anti-muon, etc.). Note that leptons are electrons, muons, and tau particles, which are not composed of quarks. In these systems, the reduced mass is significantly changed from the reduced mass in hydrogen, because the leptons are much lighter than the proton. Examples include positronium (an electron and a positron) and muonium (an electron orbiting a positive muon "nucleus"). The most interesting aspect of leptonic hydrogen though is that it has no particles that directly interact by the strong force. So, they are a perfect test system for testing quantum electrodynamics. But, because particles like the muon decay in about a microsecond, leptonic hydrogen can be unstable and last only a short time. So, experiments on them must be done quickly. Positronium has an additional issue, if the electron and positron meet, they can annihilate each other and create photons.

Hadrons are elementary particles constructed from quarks. They have two types—*baryons*, which are fermions (examples include protons and neutrons) and *mesons*, which are bosons (examples include pions, kaons, and the sigma particle). *Hadronic hydrogen* is a hydrogen analogue composed entirely of hadrons. One can replace the electron by a negatively charged meson; one could also compose the analog from positively and negatively charged mesons, and more. The most common hadronic hydrogen analog replaces the electron with a negatively charged meson. These systems are interesting because the strong interaction effects are accentuated and can be studied more easily (at least before the unstable hadrons decay).

Antihydrogen is the antiparticle analog of hydrogen. It is very difficult to make, because it requires an antiproton and a positron. Experiments on it have been carried out at CERN and more are planned. The way we make them is straightforward, but requires some luck. A high energy antiproton is shot through a beam of high atomic number atoms. If the antiproton passes close to the highly charged nucleus, there can be enough energy to create electron–positron pairs. Then, in some circumstances, the antiproton can capture a positron and make antihydrogen. A much harder task is to try to trap these antihydrogen atoms once created. So far, only a handful of these exotic systems have been created, and they have not been sufficient to perform

extensive spectroscopy on them. The interest here is to see whether antimatter has different spectra than matter, which might lead to insight as to why most of the universe is composed of matter instead of antimatter.

An interesting place for finding hydrogen analogs is in semiconductors. We are not going to explain semiconductors in any detail in this book, but the one fact you need to know is that semiconductors have an energy gap. The electrons fill all energy eigenstates states below the energy gap and none above. Now, if we bring in a donor atom of phosphorus, which has one more electron than silicon does, then this defect has an extra electron to place. Rather than put it above the energy gap, it remains bound to the donor, which has a net positive charge (because the phosphorus nucleus has one more proton than silicon). This creates a hydrogen-like atom, but the environment is not that of free space, instead it is inside a semiconductor (which has a different dielectric constant, for example). One can also find the analog of antihydrogen here, by bringing in an aluminum donor, with one fewer electron, and then the nucleus has a net negative charge relative to silicon, and it binds a hole (absence of an electron in the lower band, which is similar to a bubble in water). These donor and acceptor states are very important in the field of semiconductors, but we will not be able to study them in depth here.

Semiconductors have another interesting hydrogen analog, here it is an analog of positronium. The idea behind this is that we excite an electron from the filled lower band to the excited upper band. This leaves behind a hole in the lower band, which acts like a bubble of air in water. You can think of a bubble of air in water as having negative mass, because it is pushed upward to the surface by gravity (in reality, it is the water moving downward that "fills the hole"). Similarly, in a semiconductor, the holes act like particles with negative mass and with positive electronic charge. This allows them to bind to the electron created in the upper band. Such a system is called an *exciton*, and it is easiest to see when we shine light on a semiconductor as low-weight absorption peaks inside the band gap. This is because when the exciton binds, it reduces its energy relative to a free hole at the top of the lower band and a free electron at the bottom of the upper band.

Our last hydrogen analog is the *Rydberg atom*. This system is both so important and also so amenable to the methods we have developed that we devote the next section to their discussion. Rydberg atoms are formed by taking an outer electron (usually from an alkali atom) and exciting it to a very high-energy eigenstate. As we know from our work on hydrogen, as n increases, the average radius increases to $n^2 a_0$. This means the highly excited electron is far from the atom. With an $n = 100$, it is micron-sized! The electron is very weakly bound here, which means spectroscopy on it is typically at microwave or radio wave frequencies. In addition, when electrons are excited to high principal quantum number states, they behave semiclassically according to Bohr's correspondence principle. What this means is for these electrons, we can even think of them, sometimes, as orbiting the nucleus as in a mini solar system (here think of pluto orbiting the sun for a reasonable analog). In addition, Rydberg atoms have strong influences on neighboring Rydberg atoms, because they are so large in size. This is currently being used to make quantum computers.

So, why are they similar to hydrogen? Because all of the other particles sit down in the range of a few angstroms. Since there are Z protons and $Z-1$ electrons, the net charge that the outermost electron sees is just one positive charge. Hence, for the outermost electron, the system looks remarkably like hydrogen! We will discuss these systems in much greater detail in the next section.

Before we end this section, we want to discuss the proton charge radius puzzle and how it was resolved by comparing high-precision measurements on muonic hydrogen with the two-photon $2s \rightarrow 1s$ transition in ordinary hydrogen. The puzzle arose from two independent measurements of the proton charge radius having significantly different results. But before we describe that in detail, we need to talk a bit about spectroscopy of atoms.

We saw when we solved the hydrogen atom that we found many degenerate energy levels. This degeneracy arose from an extra symmetry of the system, although we did not explore that symmetry in any detail. Symmetric or not, when we shine light on a hydrogen atom in its ground state, if the light has the right energy, it can cause an electron to absorb the photon and move to a higher energy level, where the difference in the energy levels is equal to the energy of the absorbed photon. If the atom is in an excited state, we need only wait, and it will emit a photon and drop down to a lower energy level. These transitions satisfy the so-called *selection rules*. These rules tell us which transitions are allowed. The most common transitions are dipole transitions, where the angular momentum quantum number changes by one unit. If the transition is not dipole allowed, it is said to be forbidden, even though it often occurs due to other types of transitions. As an example, consider the transition from the $n = 2$ shell to the $n = 1$ shell. If we are in a 2p state, it can immediately go to the 1s ground state (the transition occurs in less than a nanosecond). But if we are in the 2s state, it is not a dipole allowed transition to the 1s state. It turns out the most common way to transition to the ground state is a two-photon transition, where two photons are released. This is a much more difficult transition, and the atom stays in the 2s state for the order of one second before making the transition. This is a billion times longer!

We are almost ready to describe how the proton charge radius arises, but we need a little more history first. After Pauli and Schrödinger solved for the energy levels of hydrogen, spectroscopy was done very accurately, and it was found that the energy of the 2p states were not all the same. We will describe in detail how this happens when we discuss the Dirac equation later in the book, but for now, just know that the Dirac equation picks some of the 2p states to remain degenerate amongst themselves and the rest of the 2p's and the 2s states are also degenerate. This splitting of the 2p states is called the atomic fine structure, and the size of the splitting was about 10 GHz. The problem is that as the spectroscopy improved, more splittings were found—indeed, it was believed that the 2s state lay slightly above the 2p state that the Dirac equation said it must be degenerate with. This splitting of about 1 GHz is called the *Lamb shift* after Willis Lamb, who discovered it with Retherford just after World War II ended.

The Lamb shift is surprising, because it does not come from the quantum mechanical description of the hydrogen atom, even after including relativistic

effects. It is instead due to the interaction of the nucleus with the electromagnetic field vacuum fluctuations. So, it is a quantum field theory effect. We will not describe how to calculate the Lamb shift in detail here, as that requires us to perform a perturbation theory calculation, and we are not covering that material in this work. But, we will describe the experiment, which is truly ingenious. Indeed, Lamb won the Nobel prize for this work.

The Lamb-shift experiment uses a very clever approach to focus on the small energy differences. The way it works is to start with a beam of molecular hydrogen and heat it up to over 3000^{o} K where the molecules dissociate to atomic hydrogen. Then a beam of electrons is shot onto the atomic beam to excite the electrons to the $n = 2$ shell. Any electron excited to any of the 2p states will immediately transition to the ground state, so after a few ns, the beam is all 1s and 2s hydrogen. Then, they shine microwaves of nearly constant frequency onto the beam. If the energy is just right, it will stimulate the electrons in the 2s state to transition to the 2p state, and then they transition almost immediately back to the ground state. Finally, the beam hits a thin foil. This was chosen, because if the hydrogen atom is in the 2s state, then it causes an Auger transition releasing electrons from the film. These electrons are then gathered and counted as a current. The experiment compares the current when the microwaves are present versus when they are not. When present, and of the right frequency to cause the transition from 2s to 2p, the electron current goes down, because we deplete the hydrogen atoms in the 2p state. This then allows us to identify where the resonance condition is for the $2\text{s} \rightarrow 2\text{p}$ transition. Note how clever this experiment is! It measures the resonance condition by counting electrons. No wonder Lamb won a Nobel prize. I took the time to explain this experiment in detail to you because it illustrates that precision experiments need to be well thought out. They did not directly measure the $2\text{s} \rightarrow 1\text{s}$ transition and the $2\text{p} \rightarrow 1\text{s}$ transition and compare the energies (which would agree to at least 6 digits), but rather directly measured the energy difference. This strategy is often used in high precision work.

Okay. With this experiment clear, we are now ready to focus on the proton charge radius problem. When we covered it earlier (in Sect. 11.9), we worked with two specific charge distributions. But, the modern approach is to not consider the distribution itself, but to determine the shift in the s-wave energy levels due to the fact that the electric potential changes at the proton charge radius r_p. Note that this means that some of the Lamb shift is due to the finite size of the proton charge radius (it is about one part in 1000).

There are two ways to measure the shifts in the energy levels for the s-wave states due to the proton charge radius. The first is to measure the precise energy of the transition from the 2s state to the 1s state. Because the lifetime is so incredibly long, the transition can be determined very accurately (one part in 10^{15}); one also needs to measure a second transition accurately to determine the proton charge radius contributions. The second is to measure the Lamb shift of muonic hydrogen. Here, because the energy levels are increased significantly in magnitude (due to the change in the effective mass), the transition between the 2s and 2p levels is now in the optical regime, and we can use conventional lasers to do the experiment. Because the muon orbit is so much closer to the nucleus, it is affected much more strongly

by the proton charge radius, which makes the effect larger, and thereby easier to measure. The way the experiment works is a beam of muons is shot into a dilute molecular hydrogen gas. When a muon is captured, it drops to the lowest energy state. Except the transition from 2s $\rightarrow$1s is very long lived, so a fraction of the muonic hydrogen atoms are in the excited 2s state. Then light is shone onto the system (it is pulsed laser light, timed to be right at the time the muon has fallen to the 2s state). Next, the light emitted from the 2p $\rightarrow$1s transition, which is now in the X-ray regime, is detected to see that the 2s $\rightarrow$2p transition took place. One compares the number of photons emitted when the laser is on versus when it is off, to find the resonance condition (more photons when on resonance). The experiment is hard, because it requires the timing coincidences, and it requires seeing an increase in photons of only about 1%.

The error bars for the muonic hydrogen experiment are about ten times smaller than from measuring the transition in conventional hydrogen. The two results disagreed for many years, and this was called the proton charge radius puzzle. How could it be? One conjecture was that the standard model of particle physics was wrong, and muons act differently than electrons for their Coulomb interactions. Recently, the experiments in ordinary hydrogen have been repeated by other groups, and they are finding answers more in agreement with the muonic hydrogen results. So, it appears that the proton charge-radius puzzle may now be resolved, and there is no violation of the standard model. It is amazing how high-precision measurements can examine these deep questions!

I thank Randolf Pohl for interesting discussions about the proton radius problem that helped develop the discussion presented here.

Section 11.12 Problems:

11.12.1 Energies and Sizes of Exotic Hydrogen Systems

Calculate the effective mass, ground-state energy, and effective Bohr radius of the following exotic hydrogen systems: (i) muonic hydrogen; (ii) positronium; (iii) muonium; and (iv) hadronic hydrogen with a proton and a Π^- meson.

11.12.2 High Z Hydrogen Analogs

Work out how the Bohr radius and ground-state energy change with Z in formulas that hold for arbitrary Z. Determine quantitatively what the Bohr radius and ground-state energy are for (a) helium, (b) lithium, (c) beryllium, and (d) uranium.

11.12.3 Proton Charge Radius Problem

Focus on the muonic hydrogen experiment for the proton charge radius and use the facts that in hydrogen, the ground state is at approximately 13.6 eV, and the Lamb shift is about 1 GHz. Assuming all energies scale, determine the 2s—1s transition energy and the Lamb shift for muonic hydrogen by employing simple scaling of the energies.

11.12.4 High-Precision Measurements

Write a brief essay describing what high-precision measurements are and why they are important for discovering new physical behavior. For example, the discovery of the Lamb shift played a large role in the development of quantum electrodynamics, the quantum field theory for the electromagnetic forces.

11.13 Semiclassical Hydrogen and Rydberg Atoms

When we solved the hydrogen atom problem, we found that the energy eigenstates were degenerate for all allowed angular momenta with a fixed value of the principal quantum number n. In our solution for hydrogen, we ignored spin, because the Hamiltonian was independent of spin. But, as we add in relativistic effects, the eigenstate degeneracies are determined by the total j angular momentum quantum number, not just the principal quantum number. This is because of an effect called spin–orbit coupling, where the spin of the electron feels the effect of the magnetic field due to the orbital motion of the nucleus (recall in our reduction to an effective single particle motion, we solve for the relative coordinate—when we go back to the coordinates of the nucleus and the electron, we find *both* move). This has led to a description of the states of hydrogen in terms of the so-called *term symbols*. The term symbols are easy to construct. They are the principal quantum number n followed by a letter, which denotes the orbital angular momentum state (in order: s, p, d, f, $\cdots$). Then we use a subscript, which says what the total angular momentum is (formed by combining the electron spin with the orbital angular momentum).

Let's look at some examples. The ground-state term symbol is $1s_{\frac{1}{2}}$, because the $l = 0$ s state has a total angular momentum $j = \frac{1}{2}$. For $n = 2$, we have $2s_{\frac{1}{2}}$, $2p_{\frac{3}{2}}$, and $2p_{\frac{1}{2}}$. This is because the two options when we add angular momentum for $l = 1$ with $s = \frac{1}{2}$, we can make $j = \frac{1}{2}$ and $j = \frac{3}{2}$. Let us do one more. For $n = 3$, we have $3s_{\frac{1}{2}}$, $3p_{\frac{3}{2}}$, $3p_{\frac{1}{2}}$, $3d_{\frac{5}{2}}$, and $3d_{\frac{3}{2}}$. When we solve the Dirac equation for hydrogen

Table 11.7 Approximate quantum defect values for the alkalies (from Bransden and Joachim (2003))

Atom	$l_{\text{core}}^{\max}$	$l=0$	$l=1$	$l=2$	$l=3$
Li	1	0.40	0.04	0.00	0.00
Na	2	1.35	0.85	0.01	0.00
K	3	2.19	1.71	0.25	0.00
Rb	4	3.13	2.66	1.34	0.01
Cs	5	4.06	3.59	2.46	0.02

later, we will find that the energy eigenvalues depend only on n and j, so states with term symbols that have the same n and j will remain degenerate. We have already seen that the Lamb shift often lifts this degeneracy.

Previously, we discussed the exact solution for hydrogen and for exotic hydrogen analogs. It turns out the hydrogen solution is even more versatile, as it also describes the behavior of the alkali atoms (Li lithium, Na sodium, K potassium, Rb rubidium, Cs cesium, and Fr francium). These are all atoms that have filled shells of electrons, with one remaining electron in the outermost shell. As we saw earlier when we discussed hydrogenic ions, the outermost electron sees a nucleus that has an effective charge of approximately 1, because the nucleus plus the other electrons are bound much closer to the nucleus than the outermost electron is. This means that naively, we expect the outermost electron to act just like the solution of the hydrogen atom. But, it is not that simple. This is because the atoms in a given orbital spend some of their time inside the shells of the innermost electrons (recall all s electrons even spend some time inside the nucleus). This causes a shift in the energy of the electrons, but, remarkably, the shift appears to depend only on the angular momentum and not on the principal quantum number! Both experimental measurements and theoretical calculations show us that we can describe the energy levels with a *quantum defect* δ_l, which is given by

$$E_{nl} = -\frac{e^2}{8\pi\epsilon_0(n-\delta_l)^2}. \tag{11.481}$$

It is remarkable that the quantum defect does not depend on the principal quantum number, but just the angular momentum. As a general rule, if a specific angular momentum is not in the filled states of the innermost core electrons, then that quantum defect is almost zero. Consult Table 11.7 to see what these numbers are for the alkalies. One can see that the quantum defects increase approximately by 1 as the electron core includes higher angular momentum. The quantum defect model was first proposed by the famous British physicist Douglas Hartree.

In addition to the alkalies, there is another system described well by the quantum defect model. This is the so-called Rydberg atom. Rydberg atoms correspond to essentially any atom, where we take one of the outermost electrons and promote it to a very high quantum number state, typically $n \approx 100$–300. In this case, the inner core of electrons completely screens the nucleus (most of the time), leaving an effective charge of 1, but the low angular momentum states still spend some time

Fig. 11.18 Johannes Rydberg was a pioneer of spectroscopy and is the namesake of Rydberg atoms. (Photo by Per Bagge and courtesy of the University of Lund library (public domain). Image converted to look like a sketch by Jim Freericks)

inside the ion core, so we still need to use the quantum defect model. But, if we place them in a high angular momentum state, the energy levels behave essentially like hydrogen with a vanishing quantum defect. In Fig. 11.18, you see an image of Johannes Rydberg for whom the Rydberg atoms are named.

Let's start by discussing some of the properties of such atoms. We will assume, for simplicity, that our principal quantum number is $n \approx 100$ for this discussion. First off, since the average radius scales like n^2, the radius of the outermost electron, and hence, of the Rydberg atom, is about 0.5 μm, which is huge. So, a gas of Rydberg atoms will continually collide with each other, unless they are very dilute. But, when they collide, they can strip the outermost electron off the atom, because the electrons are weakly bound. The energy of the atom scales inversely with n^2, so it is at the meV level (which corresponds to radio wave frequencies). Finally, one might be surprised that these atoms are relatively long lived. While we have not calculated the lifetime of an excited state of an atom yet, the lifetime scales like n^3. One typically finds the atomic lifetime of low-n excited states is about a nanosecond, so Rydberg atoms can live for a ms or longer. While this may seem like a short time, it is an eternity with regard to atomic time scales. In the problems, we will concretely calculate some of these quantities for Rydberg atoms.

We can also calculate the energy-level spacing between different energy levels with nearby principal quantum numbers. To be concrete, lets assume l is fixed, and

we compute the energy difference between a state with $n = n_0$ and $n = n_0 + \Delta n$. We find

$$\Delta E \approx \frac{e^2}{8\pi\epsilon_0} \frac{\Delta n}{(n_0 - \delta_l)^3}, \tag{11.482}$$

for $n_0 \gg 1$. This means the energy spacing is nearly constant, and is small, as it scales inversely with the cube of n_0. If it was exactly evenly spaced, then we could form coherent states just as we did for the simple harmonic oscillator, since the energy spectrum would be identical. It turns out that we can approximately form coherent states, and they last for quite a long time, before the energy differences cause the wavepacket to start to change shape (recall the coherent state of harmonic oscillator maintained the Gaussian shape always). When we form a coherent state, we can localize the electron in a specific region of space about the atom, and it will act similar to a particle orbiting the nucleus, like a microscopic solar system. Even though we told you not to think of the motion in atoms like microscopic solar systems, in Rydberg atoms, this is actually okay. We will discuss how this works more in Problem 11.13.5 as well.

Rydberg atoms are pretty cool. They allow us to see the crossover between quantum and classical mechanics. They also allow us to make atoms much larger in physical size, allowing us to carefully control their interactions. So, how do we make them? We describe three different ways. First, one can simply make a beam of atoms and shoot a beam of electrons at the atoms. The collisions of the electrons with the atoms will give energy to the atoms (we saw this type of experiment with the Lamb shift previously). If the electron beam energy is tuned properly, we can excite the electrons in the atoms to be almost ionized and hence to become Rydberg atoms. But this approach is not very efficient, and it does not consistently produce Rydberg atoms with the same principal quantum numbers. So, we have a second method. This method uses tunable dye lasers that can have their light frequency adjusted within a small range. For an alkali such as lithium, we would use the dye laser to excite from the 2s to the 3s level (this requires a two-photon transition, just as we discussed in the proton charge radius problem). Then the electron is excited from the 3s to a np state with a large n value. This approach is very well controlled and efficient. Finally, there is a third approach, which first applies an electric field to the atom. As you might imagine, the electric field will distort the orbitals in the atom and essentially moves the Rydberg levels lower in energy. After the atom is polarized by the electric field, one uses a laser to photoexcite into a precise Rydberg level (or a coherent state of nearby levels) and then the electric field is turned off. What about detecting the Rydberg atoms? One of the easiest ways to do this is to put them in a small electric field, which is large enough to ionize the atoms. Since the ionized atom is then in an electric field, it feels a force and can be moved to a detector. This turns out to be a very sensitive way of counting the Rydberg atoms that have been prepared in a given Rydberg state.

So, what do we use these curious things for? It turns out there are a lot of different applications. In astronomy, Rydberg atoms contribute to radio wave spectroscopy. This is because in outer space, the atomic clouds are very dilute, so collisions tend not to shorten the Rydberg atom lifetimes. Hence, they form a significant contribution to the radio-wave spectra. Rydberg atoms also are used to carefully study the behavior of quantum systems as we transition from the quantum to the semiclassical regime. In particular, one can observe orbiting wavepackets that look like micro solar systems. But, what looks like the most important application is in quantum information and quantum computing. By using a technique called optical tweezers, individual Rydberg atoms can be trapped and moved from one spatial location to another. Because they are so large, when they are brought close to one another, one can make the two atoms interact. By carefully controlling how this is done, one can create a quantum computer from Rydberg atoms. They have shown a lot of promise in being able to do real quantum computations and only time will tell how important they will become in this arena.

Section 11.13 Problems:

11.13.1 Angular Momentum and the Shape of Orbitals

Recall that the Hamiltonian for hydrogen in a state with definite total angular momentum l is

$$\hat{H}_l = \frac{\hat{p}_r^2}{2M} + \frac{\hbar^2 l(l+1)}{2M\hat{r}^2} - \frac{e^2}{4\pi\epsilon_0 \hat{r}}. \tag{11.483}$$

We call the combination of the last two terms the *effective potential*. If we were to solve this system classically, we would plot the effective potential, then we can draw a horizontal line corresponding to the energy of the system, and for allowed classical energies, it will intersect the effective potential curve at two points called the turning points. These turning points correspond to the minimum and maximum radii of the elliptical orbit. So, the more different the radii are, the more elliptical is the orbit. Note that for s-wave systems, with $l = 0$, the left turning point is always at $r = 0$, because the particle cannot have a smaller radius than 0. See Fig. 11.19 for some examples.

What this says, is that $l = 0$ states have the most elliptical orbits, and when the states with the highest allowed angular momentum value (for the given energy) have the most circular orbits. I must admit, when I first was asked which angular momentum has corresponds to circular orbits, I thought $l = 0$ of course, because it is spherically symmetric, but it is actually the highest allowed angular momentum state.

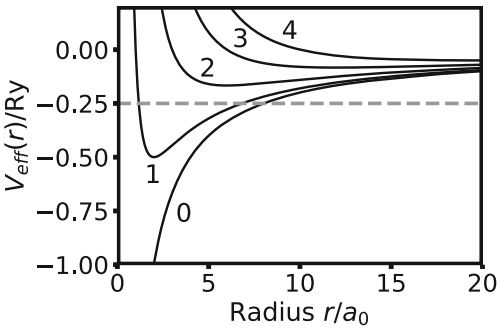

Fig. 11.19 Effective potential for different angular momenta and the corresponding turning points for the given total energy. Here, we show the energy corresponding to the first excited state of hydrogen, and we see classically only the $l = 0$ and $l = 1$ effective potential curves support bound states, and both turning points correspond to quite elliptical orbits

11.13.2 The Quantum Defect Model in Rydberg Atoms

Using these results, write a short essay that describes, in terms of semiclassical reasoning, why the quantum defect for large n Rydberg states will become zero once l is large enough. Note that this semiclassical reasoning is very accurate for Rydberg atoms, because the ionic core looks like an extended size nucleus that has charge $+1$. If you would like to make your discussion more quantitative, compute the turning points for an energy corresponding to the hydrogen energy for $n = 100$, where you should find that the turning points converge toward circular orbits for high angular momentum.

11.13.3 Size of Rydberg Atoms

Assume that the Rydberg atom has a principal quantum number given by n. Ignoring the quantum defect, compute the average radius of the atom $\langle \hat{r} \rangle$ for $n = 100$, 200, and 300.

11.13.4 Energy Scales of Rydberg Atoms

Compute the ionization energy for a Rydberg atom (ignore the quantum defect) for $n = 100$, 200, and 300. Recall the ionization energy is minus the bound state energy of the electron.

11.13.5 Coherent States of Rydberg Atoms

Compute the next term, proportional to $(\Delta n)^2$, in Eq. (11.482) for the energy level spacing. Ignore the quantum defect and pick $n_0 = 100$. What is the difference in the energy level spacing from $n = 101$ to $n = 100$ from the energy level spacing from $n = 102$ to $n = 101$?

Because the energy level spacings are nearly uniformly spaced, we can construct coherent states formed from superpositions of these states. Also, because the effective potential for the Rydberg atom is nearly quadratic at the minimum energy (corresponding to the maximal angular momentum allowed), we can think of forming coherent states from this system, which looks almost like a harmonic oscillator. If one mixes in a range of n values, and a mix of l values, and a mix of m values, then we will find we have a localized wave packet that is localized in space about a region small compared to the size of the atom. In such a scenario, we can measure the electron orbiting in a nearly elliptical orbit. Such experiments have been actually carried out. Note further that the main way to create coherent states is to start the system in a specific pure state (for the harmonic oscillator) and give it a small push, which automatically gives it the linear combination of states corresponding to the coherent state. Assume we have done this. Write a short essay describing what you think happens afterward. Can you estimate how long the wave packet will remain coherent and orbit along an elliptical path. If we try to image the wave packet with optical light (which will eject the electron after the light hits it), can we observe this motion? How big would the Rydberg atom need to be to see it? How could we try to observe the motion?

11.14 Hyperfine Structure and Radio Astronomy

We have already introduced the fine structure, which corresponds to the relativistic effects that lift the degeneracy of the hydrogen energy levels (we will finish our treatment later when we discuss the Dirac equation). We also discussed an additional splitting that arises from quantum electrodynamics effects called the Lamb shift (we will not fully derive that result here). There is one more atomic effect that is important. In addition to the relativistic spin–orbit correction that is part of the fine structure, the nucleus may contain spin. In this case, there is an additional interaction between the spin of the nucleus and the spin of the electron. The shift in the energies due to this effect is called the *hyperfine structure*. We discuss this effect here, along with its application to both radio astronomy and high-precision measurements.

One of the curiosities in quantum mechanics is that elementary particles have the so-called *g factors*, which can be thought of as a "fudge factor" used to make sure that the interaction between the magnetic moment of a quantum particle and an external magnetic field has the right magnitude. For an electron, we found that the g factor is predicted to be 2 (we will derive this when we discuss the Dirac equation).

But, quantum electrodynamic effects renormalize the g factor and make it slightly larger than 2. The magnetic moment of the electron becomes $g\mu_B$, with $\mu_B = \frac{e\hbar}{m_e}$ the Bohr magneton. One can see that the magnetic moment is inversely proportional to the mass, and because the electron charge is negative, the magnetic moment of an electron is negative. The proton has a magnetic moment that is positive, with a magnitude $\frac{m_e}{m_p}$ smaller and a g factor of $g_p = 5.59$, which is very different from the electron. This gives the magnetic moment equal to $g_p \frac{m_e}{m_p}\mu_B$. This result is an indication that the proton is a composite particle (we now know it is composed of three quarks plus gluons) because it is significantly different from 2; recall the Dirac equation predicts a g factor of 2 for elementary particles. The fact that the neutron has a magnetic moment at all indicates that it is also a composite particle (because a neutral particle should not have a magnetic moment), which was known even before the quark theory. But, this is a digression, and we need to go back to the main event.

To calculate the hyperfine structure, we must first determine the magnetic field due to the magnetic dipole of the nucleus at the specific location of the electron. Then we can determine the energy by taking the inner product between the electron magnetic dipole moment and the effective field generated by the proton. To determine this magnetic field requires us to work with Maxwell's equations from the classical theory of electromagnetism. But, because I promised you a book without calculus, we will not be able to carry out the derivation here (for those of you who are familiar with Maxwell's equations, you can find a derivation in most quantum mechanics textbooks; I find Zwiebach's derivation to be quite nice (Zwiebach, 2022)). Instead, we will determine the Hamiltonian for hydrogen in the ground state (1s) and then show how to solve this problem.

In the course of calculating the change to the Hamiltonian due to the hyperfine interaction, we need to average the magnetic field over the locations where the electron is. Because the electron in the 1s orbital is found to be uniformly distributed across the surface of a sphere, we find that the magnetic dipole only affects the electron when it lies inside the nucleus! This then gives us a contribution to Hamiltonian of the form

$$\Delta\hat{H} = \frac{2gg_p\mu_B^2 m_e \mu_0}{3m_p\hbar^2}\hat{\vec{S}}_e \cdot \hat{\vec{S}}_p |\psi_{100}(0)|^2, \tag{11.484}$$

which includes the square of the wavefunction at the origin, indicating the dependence on the probability that the electron is located in the nucleus. Here, μ_0 is the permeability of the vacuum. We calculated the wavefunction earlier, and we have

$$|\psi_{100}(0)|^2 = \frac{4}{a_0^3}\frac{1}{4\pi} = \frac{1}{\pi a_0^3}, \tag{11.485}$$

where the the first factor is the radial contribution, and the second is the spherical harmonic. This then gives us the final result:

$$\Delta\hat{H} = \frac{2g g_p \mu_B^2 m_e \mu_0}{3\pi m_p \hbar^2 a_0^3} \hat{\vec{S}}_e \cdot \hat{\vec{S}}_p. \tag{11.486}$$

We need to recognize that because the nucleus of hydrogen is a proton, which is a spin-one-half particle, and the electron is also a spin-one-half particle, which we have two degrees of freedom for the proton and two for the electron, resulting in a Hamiltonian matrix that is a 4×4 matrix. Our goal is to find the energy levels of this matrix, which will tell us the splitting between the hyperfine energy levels of hydrogen. Because we know two independent spin-one-half degrees of freedom produce a spin singlet ($s_{\text{tot}} = 0$) and three spin triplet ($s_{\text{tot}} = 1$) states. The difference between these energy levels will be the hyperfine splitting.

There are two ways to solve this problem. We will sketch the first method, which can be completed in Problem 11.14.1, and then we will solve it the second way. In the first way, we work in the basis where the spin states are tensor products in the z-basis, or

$$\left\{ |\uparrow\rangle_z^e \otimes |\uparrow\rangle_z^p,\ |\uparrow\rangle_z^e \otimes |\downarrow\rangle_z^p,\ |\downarrow\rangle_z^e \otimes |\uparrow\rangle_z^p,\ |\downarrow\rangle_z^e \otimes |\downarrow\rangle_z^p \right\}. \tag{11.487}$$

To perform the calculation, we have to recall that

$$\hat{\vec{S}}_e \cdot \hat{\vec{S}}_p = \frac{1}{2}\left(\hat{S}_+^e \hat{S}_-^p + \hat{S}_-^e \hat{S}_+^p\right) + \hat{S}_z^e \hat{S}_z^p. \tag{11.488}$$

Then, it becomes a straightforward exercise to compute the 4×4 matrix. For example, you should find

$$\hat{\vec{S}}_e \cdot \hat{\vec{S}}_p |\uparrow\rangle_z^e \otimes |\uparrow\rangle_z^p = \frac{\hbar^2}{4} |\uparrow\rangle_z^e \otimes |\uparrow\rangle_z^p \tag{11.489}$$

and

$$\hat{\vec{S}}_e \cdot \hat{\vec{S}}_p |\uparrow\rangle_z^e \otimes |\downarrow\rangle_z^p = \frac{\hbar^2}{2} |\downarrow\rangle_z^e \otimes |\uparrow\rangle_z^p - \frac{\hbar^2}{4} |\uparrow\rangle_z^e \otimes |\downarrow\rangle_z^p. \tag{11.490}$$

Computing the other two cases allows you to find the 4×4 matrix. Diagonalizing it solves the problem, which we discuss in Problem 11.14.1.

The second way to solve the problem is to work in the basis of definite total angular momentum, given by

$$\begin{aligned} |1,1\rangle &= |\uparrow\rangle_z^e \otimes |\uparrow\rangle_z^p,\ |1,0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle_z^e \otimes |\downarrow\rangle_z^p + |\downarrow\rangle_z^e \otimes |\uparrow\rangle_z^p), \text{ and} \\ |1,-1\rangle &= |\downarrow\rangle_z^e \otimes |\downarrow\rangle_z^p \end{aligned} \tag{11.491}$$

and

$$|0,0\rangle = \frac{1}{\sqrt{2}}\left(|\uparrow\rangle_z^e \otimes |\downarrow\rangle_z^p - |\downarrow\rangle_z^e \otimes |\uparrow\rangle_z^p\right). \tag{11.492}$$

One might want to compute the matrix elements directly and see that the matrix is already diagonal, but there is a simpler way to proceed. We use the identity about the square of the total spin operators, to find that

$$\hat{\vec{S}}_e \cdot \hat{\vec{S}}_p = \frac{1}{2}\left(\hat{\vec{S}}_{tot}^2 - \hat{\vec{S}}_e^2 - \hat{\vec{S}}_p^2\right). \tag{11.493}$$

If we recall that the state $|j, m_j, l_1, l_2\rangle$ is precisely the set of states we are working with, then we find states given by $|j, m_j\rangle$ are eigenstates of $\hat{\vec{S}}_e \cdot \hat{\vec{S}}_p$ with eigenvalues $\frac{1}{4}\hbar^2$ for the triplet states and $-\frac{3}{4}\hbar^2$ for the singlet state. So, we can use this result to find that the singlet state is the lowest-energy state and the energy splitting is given by

$$\Delta E_{\text{triplet-singlet}} = \frac{2g g_p \mu_B^2 m_e \mu_0}{3\pi m_p a_0^3}. \tag{11.494}$$

You evaluate this splitting in Problem 11.14.3 and show that it corresponds to the famous 21 cm radio astronomy line.

In 1944, Hendrik van de Hulst (see Fig. 11.20, left) predicted that hydrogen would emit radio waves at a frequency of 1420 MHz coming from the hyperfine transition in the ground-state of hydrogen. It took until 1951 before Purcell (see Fig. 11.20, right) and Ewen measured it in the universe. By 1952, a radio astronomy map of the milky way showed that it had a spiral structure. Why is it so useful? Most hydrogen is not excited optically, because the energies required are on the order of 10 eV, which is a relatively high energy. The energy spacing for the hyperfine structure is so low that the thermal energy of deep space (the 3K background radiation) is high enough for the system to always exist in the excited state. So, there is always hydrogen available that is ready to make a transition. The hydrogen photons travel through essentially everything without being absorbed, so it can be seen nearly everywhere. It has an exceedingly long lifetime (around 10 million years), so you need a lot of it around to see the emissions. Because the line is so sharp, it can be used to show how fast the objects emitting the light are moving due to the red shift from the Doppler effect. The radio astronomy images often provide much more detail than the visible light, as shown in the image of Fig. 11.21), comparing the visible to the radio wave images. This shows you how important quantum mechanics is to radio astronomy.

The energy splitting in hydrogen is one of the most accurately measured results in all of science. We have measured it to be 1420.405751767(1) MHz, which is about 13 digits of accuracy! It is measured in the hydrogen maser, which is like a laser but uses much longer wavelengths. The maser was invented about 7 years before the laser. The first one was made from ammonia molecules. The hydrogen maser,

Fig. 11.20 Hendrik van de Hulst (left), who first proposed that the hydrogen hyperfine transitions could be seen astronomically, and Edward Purcell (right), who first observed the spectra, which started the field of radio astronomy. (Hulst photo courtesy of the AIP Emilio Segrè Visual Archives, Physics Today Collection and Purcell photo from National Archives and Records Administration, courtesy of AIP Emilio Segrè Visual Archives. Both images converted to look like sketches by James Freericks)

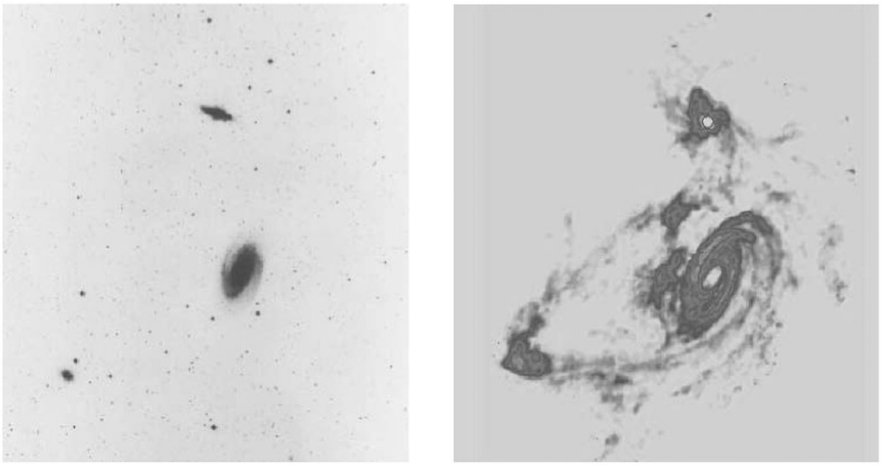

Fig. 11.21 Optical (left) versus radio astronomy (right) images of the M81 galaxy. You can see the central regions in the visible spectra do not extend too far out, but the radio astronomy features extend much farther out and show the large-scale structures of the galaxies. (These images are based on work in Sun et al. (1994) and are used by permission. Left image © Nature Publishing Group, right image is public domain from the National Radio Astronomy Observatory)

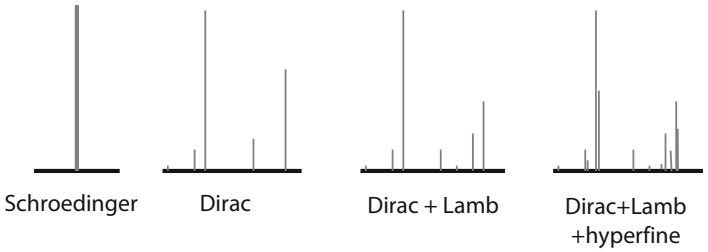

Fig. 11.22 Schematic of how spectral lines show additional structure as higher-precision measurements can be made. On the far left, we have the Balmer line for a 3p to 2s transition. Dirac shows the additional fine structure from relativistic effects (with the splittings requiring about five digits of accuracy to see). Next comes the Lamb shift with splittings about an order of magnitude smaller. Finally, the rightmost panel shows the structure when hyperfine effects are also included. (This image was inspired by Kramida (2010))

which is used for measuring the hyperfine splitting, is an atomic clock standard. The operating principle is simple: a beam of atomic hydrogen is shot through an inhomogeneous magnetic field (just like in the Stern–Gerlach experiment, to create a beam with more excited states than ground states). Then we wait for a transition to occur, which reflects back and forth between two mirrors. The light stimulates more light, and eventually, we obtain a coherent beam of radio frequency light that has a very stable frequency associated with it. By using the measured frequency as an atomic clock, we can measure the frequency to an incredibly high accuracy.

What can we do with such a device? If we measure the frequency accurately enough, we can determine the charge radius of the proton, or we can determine the fine structure constant. Sometimes experiments are done with a positive muon at the nucleus rather than a proton, because the muon acts like a point particle, allowing the fine structure constant to be determined more accurately. It is measured in antihydrogen (antiproton and antielectron atom) to determine violation of the so-called CPT theorem. If you would like to learn more, you should read John Rigden's book about high-precision experiments (Rigden, 2003).

Before ending this section, we show an image of what the spectra from hydrogen looks like as we take into account the fine structure, Lamb shift, and hyperfine structure (Fig. 11.22). You can see that as we are able to measure the frequencies with higher and higher precision, we can differentiate the different peaks and from that determine the level spacings for these small energy splittings.

Section 11.14 Problems:

11.14.1 Solving the Hyperfine Structure in the z-Basis

In the main text, we started you on the road to compute the matrix elements of the 4×4 matrix, using the z-basis, given by the four states

$$|1\rangle = |\uparrow\rangle_z^e \otimes |\uparrow\rangle_z^p, \quad |2\rangle = |\uparrow\rangle_z^e \otimes |\downarrow\rangle_z^p,$$
$$|3\rangle = |\downarrow\rangle_z^e \otimes |\uparrow\rangle_z^p, \quad \text{and} \quad |4\rangle = |\downarrow\rangle_z^e \otimes |\downarrow\rangle_z^p. \tag{11.495}$$

Recall the way that we compute the matrix: (i) we operate the operator $\hat{\vec{S}}_e \cdot \hat{\vec{S}}_p$ onto each basis state in turn, and then (ii) determine the expansion of the action of the operator onto the state in terms of the basis states. Each coefficient is an entry in the row of the matrix. Then the operator $\Delta\hat{H}$ is formed by multiplying by an overall constant. In making this construction, you should find that two of the rows of the matrix are already diagonal and two are coupled together. This means to find the eigenvalues and eigenvectors of the matrix requires you to just read off the results from the diagonal pieces and then find the eigenvalues and eigenvectors of the remaining 2×2 matrix. Of course, the answers must be the same as what we found in the text.

11.14.2 Solving the Hyperfine Structure in the Total Spin Basis

For this problem, we compute the matrix representation for $\Delta\hat{H}$ in the definite total spin basis. Use the states

$$|a\rangle = |\uparrow\rangle_z^e \otimes |\uparrow\rangle_z^p, \quad |b\rangle = \frac{1}{\sqrt{2}}\left(|\uparrow\rangle_z^e \otimes |\downarrow\rangle_z^p + |\downarrow\rangle_z^e \otimes |\uparrow\rangle_z^p\right),$$
$$|c\rangle = |\downarrow\rangle_z^e \otimes |\downarrow\rangle_z^p, \quad \text{and} \quad |d\rangle = \frac{1}{\sqrt{2}}\left(|\uparrow\rangle_z^e \otimes |\downarrow\rangle_z^p - |\downarrow\rangle_z^e \otimes |\uparrow\rangle_z^p\right). \tag{11.496}$$

You should find that the matrix is diagonal, so the eigenvalues and eigenvectors are easy to read off of it.

11.14.3 Energy, Frequency, and Wavelength of the Hyperfine Transition

Compute the energy, frequency, and wavelength of the hyperfine transition for hydrogen in the 1s state. the energy splitting between the singlet and triplet state is

$$\Delta E_{\text{singlet-triplet}} = \frac{2gg_p\mu_B^2 m_e \mu_0}{3\pi m_p a_0^3}. \tag{11.497}$$

Now, substitute in what these constants are: $g = 2$, $g_p = 5.59$, $\mu_B = \frac{e\hbar}{2m_e}$, $\mu_0 = \frac{4\pi\alpha\hbar}{e^2c}$, $\alpha = 0.007297 \approx \frac{1}{137}$ (the fine-structure constant), $a_0 = \frac{\hbar}{m_e c\alpha}$ (the Bohr radius), $m_e c^2 = 511$ keV, and $m_p c^2 = 938$ MeV. Work carefully. It is easy to make a mistake!

11.14.4 Lifetime of the Hyperfine Transition

When we discussed transitions between different atomic states, we indicated that the longer the lifetime is, the narrower the spread of the possible emission frequencies of light would be. The hyperfine transition is an extreme case of this, with a lifetime of about 10 million years. The spread in energy of the light relates to the lifetime by an approximate relationship given by $\Delta E\tau \approx \hbar$. Use this fact in your discussions and write a short essay about why one needs a long lifetime in order to have an accurate measurement of the frequency of an atomic transition.

11.14.5 $g - 2$ Measurement of the Electron

The frequency of the hydrogen maser is measured so accurately that it appears as if we could use it to measure $g - 2$ for the electron. But there is a challenge with doing this. Explain what that challenge is and why such an experiment would not directly work.

Hint: think about the accuracy of the parameters that are used to calculate the energy splitting of the hyperfine transition and how accurately we know them.

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Chapter 12
Particles in Boxes and Their Applications

You can't dream unless you know what the possibilities are.

Sonia Sotomajor, United States Supreme Court Justice

In this chapter, we introduce the particle in a box. This problem has become a standard in most quantum texts, even though it does not accurately represent too many real physical systems. Instead, it is used to illustrate quantum principles because it is viewed as one of the simplest quantum systems to discuss. Within our approach, however, this problem is a bit more challenging to solve. Nevertheless, it does provide a number of interesting insights, and it also allows us to prove an important result about the existence of bound states in one, two, and three dimensions. Finally, no student of quantum mechanics can become a true quantum mechanic without knowing at least a little bit about the particle in a box. So, watch an overview video at https://tinyurl.com/kdkfe8hx, and let us get started!

12.1 Particle in a One-dimensional Box

The one-dimensional particle in a box is often viewed as the simplest problem in quantum mechanics. We have a particle trapped in a region of space of length L. There are two ways to think of this problem: (i) in the first, our universe is just of length L, and since we do not have periodic boundary conditions, we require the wavefunction to vanish on the boundaries of our universe; and (ii) our universe runs from $-\infty < x < \infty$, and the potential is infinitely large outside of the box also requiring the wavefunction to vanish at the boundaries of the box. While these two viewpoints seem like they should be the same, we will see they have fundamental differences as we describe the behaviors found in the particle in a box. For now, we

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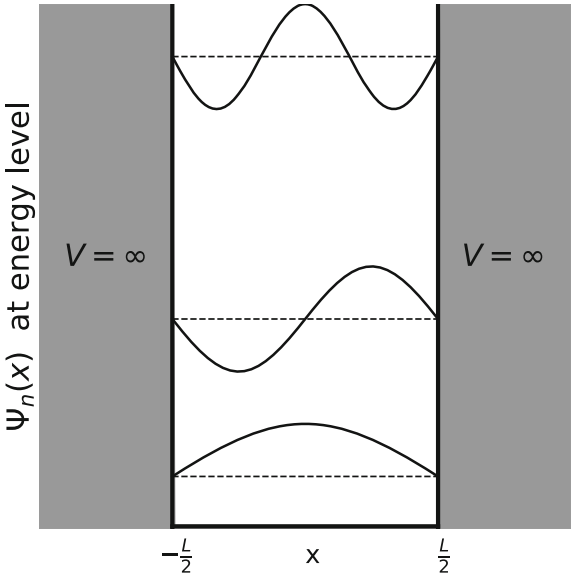

Fig. 12.1 Particle in a box potential, with wavefunctions overlaid at their energies. If one works in a universe that is just the size of the box, the shaded area should be ignored. If one works on the entire real axis, then the shaded area represents the infinite potential outside the box. The first three wavefunctions are also plotted (centered on their energies depicted by the dashed lines)

take the second viewpoint, so the potential is chosen to be an even potential (for convenience only) and is given by

$$V(\hat{x}) = \begin{cases} \infty & \text{if } |x| \geq \frac{L}{2}, \\ 0 & \text{if } |x| < \frac{L}{2}. \end{cases} \tag{12.1}$$

It is sketched in Fig. 12.1. Note that because the minimum of the potential is equal to 0, all energy eigenvalues must be larger than 0. It turns out that this potential is difficult to solve with the factorization method, but as a result, it is an excellent exercise in illustrating the details of the technique.

We begin by thinking about how we might be able to solve the nonlinear operator equation that must be solved to determine the superpotential in the factorization method. We focus on functions W_j that are well-defined for $-\frac{L}{2} \leq x \leq \frac{L}{2}$ and are able to incorporate the hard-wall boundary condition, which requires the wavefunction to vanish for all $|x| \geq \frac{L}{2}$; otherwise, the energy will be infinite for these eigenfunctions (because we are multiplying a nonzero wavefunction value by an infinite potential). Since we already saw that the energy wavefunctions for a free particle are of the form $e^{\pm ikx}$, we are motivated to find a trigonometric function that will diverge at $|x| = \frac{L}{2}$ to restrict the wavefunction to remain inside the box. Before proceeding further, we want to recall how momentum commutes with trigonometric functions of $\hat{x}$. We have $[\hat{p}, \cos(k\hat{x})] = i\hbar k \sin(k\hat{x})$ and $[\hat{p}, \sin(k\hat{x})] = -i\hbar k \cos(k\hat{x})$ for cosine and sine. Then we have $[\hat{p}, \sec(k\hat{x})] = -i\hbar k \tan(k\hat{x}) \sec(k\hat{x})$ and $[\hat{p}, \text{cosec}(k\hat{x})] = i\hbar k \text{cot}(k\hat{x})\text{cosec}(k\hat{x})$ for the secant and cosecant. Finally, we have $[\hat{p}, \tan(k\hat{x})] = -i\hbar k \sec^2(k\hat{x})$ and $[\hat{p}, \text{cotan}(k\hat{k})] = i\hbar k \text{cosec}^2(k\hat{x})$ for the tangent and cotangent. We also need to recall some of our

trigonometric identities, including $\sec^2\theta - \tan^2\theta = 1$ and $\mathrm{cosec}^2\theta - \mathrm{cotan}^2\theta = 1$ (which both follow directly from $\cos^2\theta + \sin^2\theta = 1$) in our work that follows.

To find the superpotential that will allow us to write the Hamiltonian in the form $\hat{H} = \hat{A}^\dagger\hat{A} + E$, with $E > 0$, we must solve the nonlinear equation given by

$$-\frac{i\hbar k}{2M}[\hat{p}, W(k'\hat{x})] + \frac{\hbar^2 k^2}{2M}W^2(k'\hat{x}) = V(\hat{x}) - E, \tag{12.2}$$

which is a constant for $|x| < \frac{L}{2}$. The most obvious choice to take is simply $W(k'\hat{x}) = 1$. Then $[\hat{p}, 1] = 0$ and the operator equation simplifies to

$$\frac{\hbar^2 k^2}{2M} = V(\hat{x}) - E. \tag{12.3}$$

But, when we pick $V = 0$, we find that we are forced to have $E \leq 0$. So there is no real k that we can find to solve this equation. This means this simplest choice will not work. We will find that this factorization does work when $E < 0$, and we will use it in the next section.

Since that did not work, we need to try harder. Another choice we can make, keeping the trigonometric and commutator identities in mind, is $W(k'\hat{x}) = \tan(k'\hat{x})$. Then we find that the nonlinear equation becomes

$$-\frac{i\hbar k}{2M}(-i\hbar k')\sec^2(k'\hat{x}) + \frac{\hbar^2 k^2}{2M}\tan^2(k'\hat{x}) = V(\hat{x}) - E. \tag{12.4}$$

Now, we must make the choice $k = k'$ and then we must satisfy

$$-\frac{\hbar^2 k^2}{2M} = V(\hat{x}) - E, \tag{12.5}$$

which is immediately solved for $V = 0$ with $E > 0$. Furthermore, since $\tan(k'\hat{x})$ diverges whenever $x = \pm\frac{\pi}{2k'}$, this function will give the right choice for the superpotential when we tune k (and k') such that the superpotential diverges when $x = \pm\frac{L}{2}$. It naturally gives us a potential that vanishes, just as we need for the particle in a box.

But, we are not done! There is another choice we can make, namely we can also solve the nonlinear operator equation with $W(k'\hat{x}) = \mathrm{cotan}(k'\hat{x})$. The nonlinear equation now becomes

$$-\frac{i\hbar k}{2M}(i\hbar k')\mathrm{cosec}^2(k'\hat{x}) + \frac{\hbar^2 k^2}{2M}\cot^2(k'\hat{x}) = V(\hat{x}) - E. \tag{12.6}$$

Here, we must choose $k = -k'$, and then we arrive at the same final equation as we had for the tangent case. But now, because $\cot(k'\hat{x})$ diverges at $x = 0$ and $x = \frac{\pi}{k'}$, this superpotential is the right one to use for a particle in a box that has boundaries

at $x = 0$ and $x = L$. You will have an opportunity to explore this case further in the problems.

Okay. We have found that the particle in a box Hamiltonian can be factorized with a lowering operator given by

$$\hat{A}_0 = \frac{1}{\sqrt{2M}}\big(\hat{p} - i\hbar k_0 \tan(k_0\hat{x})\big), \tag{12.7}$$

which yields $\hat{H} = \hat{A}_0^\dagger \hat{A}_0 + E_0$ with $E_0 = \frac{\hbar^2 k_0^2}{2M}$ and the requirement that $k_0 L = \pi$. One might wonder, what about choosing $k_0 L = n\pi$ for other integers n? This will satisfy all of our required properties too. But, because the wavefunction is equal to zero whenever the superpotential diverges, this will create nodes inside the box. The ground state is known, by the node theorem, not to have any nodes. This requires us to stick with our original choice with $n = 1$. Hence, we have $E_0 = \frac{\hbar^2\pi^2}{2ML^2}$. There is a modification of the factorization method, called *single-shot factorization* that does work with a single factorization to obtain all eigenvalues and eigenstates, but we will not discuss that method further here; it does enter in the next section.

Our next step is to form the rest of the factorization chain and find the excited states. The procedure is essentially a cookbook. We interchange the order of the raising and lowering operators, which is easiest to do by computing their commutator, given by

$$\begin{aligned}\left[\hat{A}_0, \hat{A}_0^\dagger\right] &= \tfrac{1}{2M}\left[\hat{p} - i\hbar k_0 \tan k_0\hat{x},\ \hat{p} + i\hbar k_0 \tan k_0\hat{x}\right] \\ &= \frac{\hbar^2 k_0^2}{2M} 2\sec^2 k_0\hat{x} = \frac{\hbar^2 k_0^2}{2M}\left(2 + 2\tan^2 k_0\hat{x}\right).\end{aligned} \tag{12.8}$$

Then we have the first auxiliary Hamiltonian, given by $\hat{H}_1 = \hat{A}_0\hat{A}_0^\dagger + \frac{\hbar^2 k_0^2}{2M}$, becomes

$$\hat{H}_1 = \hat{A}_0\hat{A}_0^\dagger + \frac{\hbar^2 k_0^2}{2M} = \frac{\hat{p}^2}{2M} + \frac{\hbar^2 k_0^2}{2M}\left(2 + 2\tan^2 k_0\hat{x}\right), \tag{12.9}$$

which we must refactorize into the form $\hat{H}_1 = \hat{A}_1^\dagger \hat{A}_1 + E_1$. The nonlinear operator equation for W_1 is then

$$-\frac{i\hbar k_1}{2M}[\hat{p}, W_1(k_1'\hat{x})] + \frac{\hbar^2 k_1^2}{2M} W_1^2(k_1'\hat{x}) = \frac{\hbar^2 k_0^2}{2M}(2 + 2\tan^2 k_0\hat{x}) - E_1. \tag{12.10}$$

After a little thought, we try a similar form as before, with $W_1(k_1'\hat{x}) = \tan k_1'\hat{x}$. Then the nonlinear operator equation becomes

$$-\frac{i\hbar k_1}{2M}(-i\hbar k_1')\sec^2(k_1'\hat{x}) + \frac{\hbar^2 k_1^2}{2M}\tan^2(k_1'\hat{x}) = \frac{\hbar^2 k_0^2}{2M}(2 + 2\tan^2(k_0\hat{x})) - E_1. \tag{12.11}$$

This requires $k_1' = k_0$, because we need to have an auxiliary Hamiltonian ground state with no nodes, so there is only one choice here. Then we find that $-k_1k_0+k_1^2 = 2k_0^2$ and $E_1 = 2\frac{\hbar^2k_0^2}{2M} + \frac{\hbar k_1k_0}{2M}$. This then implies that $k_1 = 2k_0$ or $k_1 = -k_0$. If we choose the latter case, then we have $\hat{A}_1 = \hat{A}_0^\dagger$, which is not allowed, so we must choose the former solution. Then, we find that $E_1 = 4\frac{\hbar^2\pi^2}{2ML^2}$.

After staring at this result a bit, we see that we can solve the general case as well. For the general case, we have

$$\hat{A}_j = \tfrac{1}{\sqrt{2M}}\left(\hat{p} - i\hbar k_j \tan(k_0\hat{x})\right). \tag{12.12}$$

Then, the factorization chain requires that we also satisfy

$$\begin{aligned}
\hat{A}_j^\dagger\hat{A}_j &= \frac{\hat{p}^2}{2M} - \frac{\hbar^2k_jk_0}{2M}\sec^2(k_0\hat{x}) + \frac{\hbar^2k_j^2}{2M}\tan^2(k_0\hat{x}) \\
&= \frac{\hat{p}^2}{2M} - \frac{\hbar^2k_jk_0}{2M} + \frac{\hbar^2k_j(-k_0+k_j)}{2M}\tan^2(k_0\hat{x}) \\
\hat{A}_j\hat{A}_j^\dagger &= \frac{\hat{p}^2}{2M} + \frac{\hbar^2k_jk_0}{2M} + \frac{\hbar^2k_j(k_0+k_j)}{2M}\tan^2(k_0\hat{x})
\end{aligned} \tag{12.13}$$

and

$$\hat{H}_j = \hat{A}_j^\dagger\hat{A}_j + E_j = \hat{A}_{j-1}\hat{A}_{j-1}^\dagger + E_{j-1}. \tag{12.14}$$

These requirements are solved by $k_j(k_j-k_0) = k_{j-1}(k_{j-1}+k_0)$ and $E_j - E_{j-1} = \frac{\hbar^2k_0}{2M}k_0(k_j + k_{j-1})$. Finally, we arrive at the conclusion that $k_j = k_{j-1} + k_0$ or $k_j = -k_{j-1}$. The latter is not allowed by our rules, so we must have the former. Then $E_j = E_{j-1} + \frac{\hbar^2k_0}{2M}(2k_{j-1} + k_0)$. Now, recall that $k_1 = 2k_0$. This then implies that $k_2 = 3k_0$ and $E_2 = E_1 + \frac{\hbar^2k_0}{2M}(2k_1 + k_0) = \frac{\hbar^2\pi^2}{2ML^2}(4+5) = 9\frac{\hbar^2\pi^2}{2ML^2}$. The general solution is $k_j = (j+1)k_0$ and $E_j = (j+1)^2\frac{\hbar^2\pi^2}{2ML^2}$.

The corresponding eigenstates $|\psi_j\rangle$ are constructed from the following procedure. First, we must find the auxiliary ground state $|\phi_j\rangle$, which satisfies $\hat{A}_j|\phi_j\rangle = 0$, or

$$\hat{p}|\phi_j\rangle = i\hbar(j+1)k_0\tan(k_0\hat{x})|\phi_j\rangle, \tag{12.15}$$

with $\langle\phi_j|\phi_j\rangle = 1$ (recall that the ground state is identified with the initial factorization, so $|\psi_0\rangle = |\phi_0\rangle$ are the same states). The jth energy eigenstate $|\phi_j\rangle$ is found from Schrödinger's construction as

$$|\psi_j\rangle = c_j\hat{A}_0^\dagger\hat{A}_1^\dagger\cdots\hat{A}_{j-1}^\dagger|\phi_j\rangle, \tag{12.16}$$

with

$$c_j = \frac{1}{\sqrt{(E_j - E_0)(E_j - E_1)\cdots(E_j - E_{j-1})}}. \tag{12.17}$$

Finally, the wavefunction in position space is found by taking the inner product with the position bra $\langle x|$. Hence, we find that

$$\psi_j(x) = \langle x|\psi_j\rangle = c_j \langle x|\hat{A}_0^\dagger \hat{A}_1^\dagger \cdots \hat{A}_{j-1}^\dagger|\phi_j\rangle. \tag{12.18}$$

It turns out that there are two ways to calculate the wavefunctions, and we will discuss both of them. The first one is a "brute-force" approach, where we write $|x\rangle = e^{-\frac{i}{\hbar}x\hat{p}}|0_x\rangle$, so that

$$\psi_j(x) = c_j \left\langle 0_x|e^{\frac{i}{\hbar}x\hat{p}} \hat{A}_0^\dagger \hat{A}_1^\dagger \cdots \hat{A}_{j-1}^\dagger|\phi_j\right\rangle. \tag{12.19}$$

This looks challenging to work out. One way to do it is to expand the exponential. Then, we first replace all $\hat{p}$ operators in the ladder operators by functions of $\hat{x}$ operating on $|\phi_j\rangle$ via the subsidiary condition. Next, we do the same for the powers of $\hat{p}$ coming from the expansion of the exponential. Finally, all $\hat{x}$ operators are acted on the bra state to the left. It is not easy to work this out for the general case. Instead, we look at the first few eigenstates, starting from the ground state,

The ground state satisfies the subsidiary condition given by $\hat{A}_0|\phi_0\rangle = 0$, or $\hat{p}|\phi_0\rangle = i\hbar k_0 \tan(k_o\hat{x})|\phi_0\rangle$. Using the fact that $c_0 = 1$, we find that

$$\psi_0(x) = \langle x|\phi_0\rangle = \langle 0_x|e^{\frac{i}{\hbar}x\hat{p}}|\phi_0\rangle = \sum_{n=0}^{\infty} \frac{1}{n!}\left(\frac{i}{\hbar}\right)^n \langle 0_x|\hat{p}^n|\phi_0\rangle; \tag{12.20}$$

recall that $|\psi_0\rangle = |\phi_0\rangle$ in our notation. We can get rid of each $\hat{p}$ operator by using the subsidiary condition, but each time we do so, we obtain a function of $\hat{x}$ to the right, and we must commute the next $\hat{p}$ operator to the right through it, before we can use the subsidiary condition. But, there turns out to be a nice simplification to make our lives a bit easier. We find that

$$\begin{aligned}\hat{p}^2|\phi_0\rangle &= \hat{p}i\hbar k_0 \tan(k_0\hat{x})|\phi_0\rangle = i\hbar k_0 \left([\hat{p}, \tan k_0\hat{x}] + \tan(k_0\hat{x})\,\hat{p}\right)|\phi_0\rangle \\ &= i\hbar k_0 \left(-i\hbar k_0 \sec^2(k_0\hat{x}) + i\hbar k_0 \tan^2(k_0\hat{x})\right)|\phi_0\rangle = \hbar^2 k_0^2|\phi_0\rangle. \end{aligned} \tag{12.21}$$

Using this result, we immediately find that

$$\hat{p}^{2m}|\phi_0\rangle = (\hbar k_0)^{2m}|\phi_0\rangle \;\text{ and }\; \hat{p}^{2m+1}|\phi_0\rangle = (\hbar k_0)^{2m+1} i \tan(k_0\hat{x})|\phi_0\rangle. \tag{12.22}$$

Now, since $\langle 0_x|\hat{x} = 0$, we find that $\langle 0_x|\tan(k_0\hat{x}) = 0$, and the wavefunction becomes

$$\psi_0(x) = \langle 0_x|\phi_0\rangle\langle 0_x|\phi_0\rangle \sum_{m=0}^{\infty}(-1)^m \frac{(k_0x)^{2m}}{(2m)!} = \langle 0_x|\phi_0\rangle \cos(k_0x). \tag{12.23}$$

This is the ground state for the particle in a box up to its normalization factor, which satisfies $\langle 0_x|\phi_0\rangle = \sqrt{\frac{2}{L}}$. Hence, we have

$$\psi_0(x) = \sqrt{\frac{2}{L}} \cos\left(\frac{\pi x}{L}\right). \tag{12.24}$$

Before computing the first excited state, we want to illustrate the alternative way to compute the wavefunction. Recall that this followed by finding a state, constructed from $|\phi_0\rangle$, that is annihilated by the momentum operator. Then the translation operator acts like the identity on that state. With a little trial and error, we can quickly find that

$$\begin{aligned}\hat{p}\sec(k_0\hat{x})|\phi_0\rangle &= \big([\hat{p}, \sec(k_0\hat{x})] + \sec(k_0\hat{x})\hat{p}\big)|\phi_0\rangle \\ &= i\hbar k_0\big(-\tan(k_0\hat{x})\sec(k_0\hat{x}) + \sec(k_0\hat{x})\tan(k_0\hat{x})\big)|\phi_0\rangle = 0.\end{aligned} \tag{12.25}$$

So, we use this to find the wavefunction in the standard way, with multiplying by 1:

$$\begin{aligned}\psi_0(x) &= \langle x|\cos(k_0\hat{x})\sec(k_0\hat{x})|\psi_0\rangle = \cos(k_0x)\langle 0_x|e^{\frac{i}{\hbar}x\hat{p}}\sec(k_0\hat{x})|\phi_0\rangle \\ &= \cos(kx)\langle 0_x|\phi_0\rangle,\end{aligned} \tag{12.26}$$

just like before. Here, our first step is multiplying by 1. Then we let the $\cos(k\hat{x})$ operator act to the left against the position bra and move it out of the expectation value. Then, because $\hat{p}$ annihilates $\sec(k_0\hat{x})|\phi_0\rangle$, the exponential is replaced by 1. Finally, we can act the secant term against the position bra at the origin, replacing it by 1. This approach is elegant, because it does not require any expansion of the exponential, but it requires us to find the state annihilated by the momentum operator, which may not be so simple to do.

Now, onto the first excited state. It is found by using both $\hat{A}_1|\phi_1\rangle = 0$ and $|\psi_1\rangle = \hat{A}_0^\dagger|\phi_1\rangle$. Then, we have that $\psi_1(x) = c_1\langle x|\hat{A}_0^\dagger|\phi_1\rangle$. Our first step is to use the subsidiary condition to obtain

$$\hat{A}_1|\phi_1\rangle = 0 \quad\Longrightarrow\quad \hat{p}|\phi_1\rangle = 2i\hbar k_0\tan(k_0\hat{x})|\phi_1\rangle. \tag{12.27}$$

Then, if we work with the first approach, we start with

$$\hat{A}_0^\dagger|\phi_1\rangle = \frac{3i\hbar k_0}{\sqrt{2M}}\tan(k_0\hat{x})|\phi_1\rangle. \tag{12.28}$$

We next have to compute the effect of powers of momentum on this state. As before, we will find that the state is an eigenstate of $\hat{p}^2$, which makes the computation relatively simple to undertake. In particular, we find that

$$\begin{aligned}\hat{p}\hat{A}_0^\dagger|\phi_1\rangle &= \frac{3i\hbar k_0}{\sqrt{2M}}\left([\hat{p},\tan(k_0\hat{x})] + \tan(k_0\hat{x})\hat{p}\right)|\phi_1\rangle \\ &= \frac{3i\hbar k_0}{\sqrt{2M}}\big(-i\hbar k_0\sec^2(k_0\hat{x}) + 2i\hbar k_0\tan^2(k_0\hat{x})\big)|\phi_1\rangle \\ &= \frac{3(\hbar k_0)^2}{\sqrt{2M}}\big(1-\tan^2(k_0\hat{x})\big)|\phi_1\rangle,\end{aligned} \tag{12.29}$$

and

$$\begin{aligned}\hat{p}^2\hat{A}_0^\dagger|\phi_1\rangle &= \frac{3(\hbar k_0)^2}{\sqrt{2M}}\big(\hat{p} - 2\tan(k_0\hat{x})[\hat{p},\tan(k_0\hat{x})] - \tan^2(k_0\hat{x})\hat{p}\big)|\phi_1\rangle \\ &= \frac{3(\hbar k_0)^2}{\sqrt{2M}}\big(2i\hbar k_0\tan(k_0\hat{x}) + 2i\hbar k_0\tan(k_0\hat{x})\sec^2(k_0\hat{x}) \\ &\quad - 2i\hbar k_0\tan^3(k_0\hat{x})\big)|\phi_1\rangle \\ &= \frac{3(\hbar k_0)^2}{\sqrt{2M}}(2i\hbar k_0)2\tan(k_0\hat{x})|\phi_1\rangle.\end{aligned} \tag{12.30}$$

Because this is a number times the original state $\hat{A}_0^\dagger|\phi_1\rangle$, we immediately find that

$$\begin{aligned}(\hat{p})^{2m}\hat{A}_0^\dagger|\phi_1\rangle &= \frac{3i\hbar k_0}{\sqrt{2M}}(2\hbar k_0)^{2m}\tan(k_0\hat{x})|\phi_1\rangle, \\ (\hat{p})^{2m+1}\hat{A}_0^\dagger|\phi_1\rangle &= \frac{3\hbar k_0}{2\sqrt{2M}}(2\hbar k_0)^{2m+1}\big(1-\tan^2(k_0\hat{x})\big)|\phi_1\rangle.\end{aligned} \tag{12.31}$$

This allows us to evaluate the wavefunction by expanding the translation operator as a power series, evaluating each term separately, and then resumming the series. Using the fact that $\langle 0_x|\tan(k_0\hat{x}) = 0$, we find that

$$\psi_1(x) = c_1\langle 0_x|e^{\frac{i}{\hbar}x\hat{p}}\hat{A}_0^\dagger|\phi_1\rangle = c_1\langle 0_x|\sum_{n=0}^{\infty}\frac{1}{n!}\left(\frac{ix}{\hbar}\right)^n(\hat{p})^n\hat{A}_0^\dagger|\phi_1\rangle$$

$$= c_1 \langle 0_x|\phi_1\rangle \sum_{m=0}^{\infty} \frac{3i\hbar k_0}{2\sqrt{2M}} (-1)^m \frac{(2k_0 x)^{2m+1}}{(2m+1)!}$$

$$= c_1 \langle 0_x|\phi_1\rangle \frac{3i\hbar k_0}{2\sqrt{2M}} \sin(2k_0 x). \tag{12.32}$$

Using $c_1 = 1/\sqrt{E_1 - E_0} = \sqrt{\frac{2M}{3\hbar^2 k_0^2}}$ gives

$$\psi_1(x) = \frac{\sqrt{3}}{2} i \langle 0_x|\phi_1\rangle \sin(2k_0 x). \tag{12.33}$$

This still requires normalization, which finally yields

$$\psi_2(x) = \sqrt{\frac{2}{L}} \sin(2k_0 x). \tag{12.34}$$

As one can see, it is becoming more difficult to determine the higher-order wavefunctions with this approach. This is not because it requires complex abstract details. It is just because the algebra gets increasingly long and messy. However, if we bring in some new ideas, we can still rapidly get to the general result. We will show the general approach next, but before going there, we just want to comment that one can also find the first excited state using our other approach. This is Problem 12.1.3.

The new idea that we bring in is that the excited states of the particle-in-a-box Hamiltonian are eigenstates, or

$$\hat{p}^2 |\psi_j\rangle = \big(\hbar(j+1)k_0\big)^2 |\psi_j\rangle. \tag{12.35}$$

Then, we find that

$$\langle 0_x|(\hat{p})^{2m}|\psi_j\rangle = \big(\hbar(j+1)k_0\big)^{2m} \langle 0_x|\psi_j\rangle \text{ and}$$

$$\langle 0_x|(\hat{p})^{2m+1}|\psi_j\rangle = \big(\hbar(j+1)k_0\big)^{2m+1} \langle 0_x|\hat{p}|\psi_j\rangle; \tag{12.36}$$

Note that this is an identity for the excited states, not for the auxiliary Hamiltonian ground states. Finally, we note that either $\langle 0_x|\psi_j\rangle = 0$, or $\langle 0_x|\hat{p}|\psi_j\rangle = 0$, as we saw with the ground and first excited state. In fact, our claim is that for even indices, we have that $\langle 0_x|\hat{p}|\psi_{2j}\rangle = 0$ and for odd indices $\langle 0_x|\psi_{2j+1}\rangle = 0$.

We can reason both of these results out in a straightforward manner. Consider the state $|\phi_j\rangle$, which satisfies the subsidiary condition, given by $\hat{p}|\phi_j\rangle = i\hbar(j+1)|\tan(k_0\hat{x})|\phi_j\rangle$. this tells us that $\langle 0_x|\hat{p}|\phi_j\rangle = i\hbar(j+1)\langle 0_x|\tan(k_0\hat{x})|\phi_j\rangle = 0$ and $\langle 0_x|\phi_j\rangle \neq 0$ for all auxiliary Hamiltonian ground states. Now, the string of raising

operators acting on the auxiliary ground state creates a polynomial in $\tan(k_0\hat{x})$ that is an even polynomial for even j and an odd polynomial for odd j. This is enough to show the desired relation, which you will complete in the problems.

We end this section with one final comment. You might have noticed that we have not discussed uncertainty here. In fact, the concept is somewhat ill-defined for the situation where the box is the entire "universe" and there is no region outside the box. This is because momentum is not well defined as an operator, even though $\hat{p}^2$ is. If, instead, our "universe" is the entire real line, then momentum is well defined, and one can compute the uncertainty in momentum. But, our operator methods do not have any simple way to compute the uncertainty in position. This can be done with calculus, in a fairly straightforward way, and the states do satisfy uncertainty, as expected, but because there is no simple way to perform the calculation with operators here, we choose not to discuss this further.

Section 12.1 Problems:

In a number of the problems, we will work with a particle in a box, but shifted to lie between $x = 0$ and $x = L$. Hence,

$$V(x) = \begin{cases} \infty & \text{for } x < 0 \text{ and } x > L \\ \\ 0 & \text{for } 0 \leq x \leq L. \end{cases} \tag{12.37}$$

As before, the wavefunction vanishes for $x \leq 0$ and $x \geq L$.

12.1.1 Particle in a Box Ranging from $0 \leq x \leq L$

(a) Solve the nonlinear operator equation to determine the superpotential. It is

$$-\frac{i\hbar k}{2M}[\hat{p}, W(k'\hat{x})] + \frac{\hbar^2 k^2}{2M} W(k'\hat{x}) = V(\hat{x}) - E, \tag{12.38}$$

with $E > 0$ and which does not diverge for $0 < x < L$.

(b) Find the ground-state energy by employing the factorization method.

(c) Construct $\hat{A}_1$ and find E_1.

(d) Generalize to find $\hat{A}_j$ and E_j.

(e) Compare your results to the ones for the particle in a box with $|x| \leq \frac{L}{2}$ and explain the similarities and differences.

Hint: cotangent and cosecant will play prominent roles in your work.

12.1.2 Wavefunctions for the General Energy Eigenstate for the Particle in a Box Using the Brute-Force Method

Complete the derivation of $\psi_j(x) = \langle x|\psi_j\rangle$ for the particle in a box with $|x| \leq \frac{L}{2}$. After fixing the normalization constant, you should find that $\psi_{2j}(x) = \sqrt{\frac{2}{L}}\cos\left(\frac{(2j+1)\pi x}{L}\right)$ and $\psi_{2j+1}(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{2j\pi x}{L}\right)$. Note that the normalization constants must be found by calculus, so they are just given to you here.

12.1.3 Wavefunctions for Excited States Using the State Annihilated by $\hat{p}$

Show that, in general, we have $\hat{p}\frac{1}{\left(\cos(k_0\hat{x})\right)^{j+1}}|\phi_j\rangle = 0$ for the auxiliary Hamiltonian ground states.

Next, focus on the first excited state. Multiplying by 1, in the form of $\frac{\cos^2 k_0\hat{x}}{\cos^2 k_0\hat{x}}$, inserted just to the right of the $\langle x|$, show that the first excited state is proportional to $2\cos(k_0\hat{x})\sin(k_0\hat{x}) = \sin(2k_0\hat{x})$.

Repeat for the second excited state, and show that its wavefunction is proportional to $\cos(3k_0\hat{x})$.

See whether you can find a way to generalize this to an arbitrary excited state. Note that this last part is quite challenging.

12.1.4 Ground State Wavefunction for the Particle in a Box Ranging from $0 \leq x \leq L$

Using the brute-force method, modeled on what we did in the text, find the ground-state wavefunction when the box runs from $0 \leq x \leq L$.

12.1.5 First Excited State Wavefunction for the Particle in a Box Ranging from $0 \leq x \leq L$

Using the brute-force method, modeled on what we did in the text, find the first excited state wavefunction when the box runs from $0 \leq x \leq L$.

12.1.6 General Wavefunction for the Particle in a Box with $0 \leq x \leq L$

Using the brute-force method, modeled on what we did in the text, find the general wavefunction when the box runs from $0 \leq x \leq L$.

Answer: $\psi_j(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{j\pi x}{L}\right)$.

12.1.7 Subtle Mathematical Issues with the Particle in a Box Problem

When we worked with the particle on a circle, we found that the position operator was not well defined, because we could use it to translate the momentum and create states with any energy. Here, If we consider a world that lives only inside the box, and no position outside, then position is perfectly well defined, but momentum is not (because we cannot translate the system outside the box). Discuss what the implications are. For example, is $\hat{p}^2$ a well-defined operator? Are there consequences that affect our using the commutation relation $[\hat{p}, \hat{x}] = -i\hbar$? What commutation relations did we need for the calculations? Are they valid? Write a short essay that discusses these issues. You may also want to think about how this changes if we live in the space $-\infty < x < \infty$. In this case, the momentum operator is a well-defined concept, because we can translate anywhere, but there still are subtleties to think about. In particular, one needs to think carefully about the Hamiltonian, because it is not equal to $\hat{p}^2$ everywhere, only inside the box.

12.1.8 Singular Factorization of the Particle in a Box with $|x| < \frac{L}{2}$

Recall that we discussed that even potentials should always have a singular factorization in addition to the nonsingular factorization chain (with the singularity occurring at $x = 0$). The particle in a box, with zero potential for $|x| \leq \frac{L}{2}$, is an even potential. Work out its singular factorization by thinking carefully about the solution from $0 < x < L$ and making the substitution $L \to \frac{L}{2}$. Determine all of the energy eigenvalues in the factorization chain. Note how this singular factorization only produces the odd wavefunction solutions (which correspond to half of the solutions). If you are feeling adventurous, determine the wavefunctions and verify that they are odd functions.

Hint: you will want to work with a superpotential that depends on $\text{cotan}(k'\hat{x})$ with $k' = \frac{2\pi}{L}$.

12.2 Particle in a One-dimensional Square Well

We now want to consider a particle in a finite square-well potential. It is customary to have the potential vanish as $|x| \to \infty$, so it takes the form

$$V(\hat{x}) = \begin{cases} 0 & \text{for } x \leq -\frac{L}{2} \\ -V_0 & \text{for } -\frac{L}{2} \leq x \leq \frac{L}{2} \\ 0 & \text{for } x \geq \frac{L}{2}, \end{cases} \tag{12.39}$$

which is shown in Fig. 12.2. We will find that this potential supports only a finite number of bound states. While most factorization chains have an infinite number of bound states, we have already seen that the Morse potential only has a finite number. But this case is not like that one. Indeed, this is a problem that cannot be solved with a factorization chain in the ordinary way we have done it before. We will explain why shortly. Instead, we will solve this problem using what we call a *"single-shot" factorization*. Such an approach is an alternative to the Schrödinger factorization chain, and it is most valuable in solving problems with continuous energy eigenvalues. Unfortunately, the general methodology for single-shot factorization requires us to use calculus, so we will not develop it in full here.

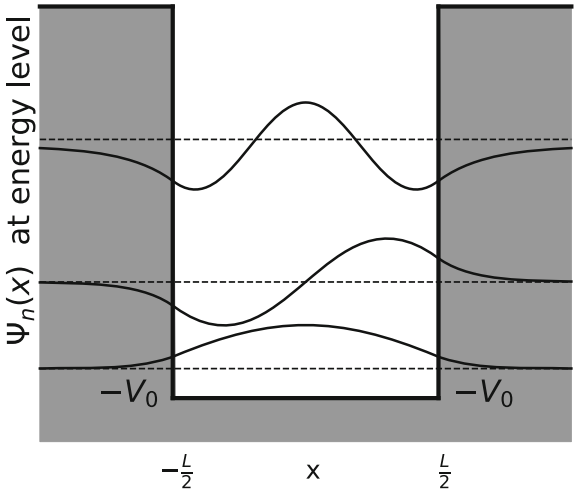

Fig. 12.2 Finite square-well potential in one dimension for a box running from $-\frac{L}{2}$ to $\frac{L}{2}$. The depth of the potential is V_0 (corresponding to $\phi_0 = \frac{3\pi}{2}$) and the potential vanishes as $|x| \to \infty$. This potential supports three bound states and is right at the transition where a fourth state enters. The wavefunctions are plotted with centers on their energy levels. Note how the energies are closer to the bottom of the well than in the particle in a box and how they leak outside the box, with the leakage more extensive the higher the energy level

However, it will work admirably for all of the finite square-well problems we solve in this chapter.

Before we jump into the details of single-shot factorization, we want to pause for just a moment to discuss *supersymmetric quantum mechanics*. It was developed by Ed Witten in the 1980s (Witten, 1982) as an example of how supersymmetry works in quantum mechanics. Without going into great details about how supersymmetry works, the formulation relates two Hamiltonians—the original one, factorized as $\hat{A}^\dagger \hat{A} + E$ and its superpartner $\hat{A}\hat{A}^\dagger + E$, which you should recognize, as our original Hamiltonian and the first auxiliary Hamiltonian. Even though the potentials of the two Hamiltonians can be quite different, the energy levels all agree, except that the original Hamiltonian has one more energy level not found in the partner, namely the ground state. Whenever this occurs, one is said to have a supersymmetric pair of Hamiltonians. A huge amount of work has gone into studying these supersymmetric pairs, and there is a nice book that summarizes much of this work (Cooper et al., 2001). We won't go into much more detail about this here, except to say that many of the ideas developed within supersymmetric quantum mechanics have already been discussed within the different frameworks we have developed here.

We now move on to our main event, showing how single-shot factorization works. We start similar to how we started the particle in a box solution, by finding a factorization of the form

$$\hat{A} = \frac{1}{\sqrt{2M}}\left(\hat{p} - i\hbar k \tan(k\hat{x})\right), \tag{12.40}$$

where we found that

$$\begin{aligned}
\hat{A}^\dagger \hat{A} &= \frac{\hat{p}^2}{2M} - \frac{i\hbar k}{2M}[\hat{p}, \tan(k\hat{x})] + \frac{\hbar^2 k^2}{2M}\tan^2(k\hat{x}) \\
&= \frac{\hat{p}^2}{2M} - \frac{\hbar^2 k^2}{2M}\sec^2(k\hat{x}) + \frac{\hbar^2 k^2}{2M}\tan^2(k\hat{x}) \\
&= \frac{\hat{p}^2}{2M} - \frac{\hbar^2 k^2}{2M} = -V_0 - E,
\end{aligned} \tag{12.41}$$

for $|x| \leq \frac{L}{2}$. We define the parameter ϕ via

$$\phi = \frac{kL}{2} = \sqrt{\frac{2M(V_0 + E)}{\hbar^2}}\frac{L}{2}, \tag{12.42}$$

and the factorized Hamiltonian is summarized as

$$\hat{H} = \hat{A}^\dagger \hat{A} + E = \frac{\hat{p}^2}{2M} - V_0, \tag{12.43}$$

for $|x| \leq \frac{L}{2}$. For $|x| > \frac{L}{2}$, we have $\hat{H} = \frac{\hat{p}^2}{2M}$ and $E < 0$. Hence, we choose

$$\hat{A} = \begin{cases} \frac{1}{\sqrt{2M}}(p + i\hbar\kappa) & \text{for } x < -\frac{L}{2} \\ \frac{1}{\sqrt{2M}}(p - i\hbar\kappa) & \text{for } x > \frac{L}{2}, \end{cases} \tag{12.44}$$

so that

$$\hat{A}^\dagger \hat{A} = \frac{\hat{p}^2}{2M} + \frac{\hbar^2\kappa^2}{2M} = \hat{H} - E, \tag{12.45}$$

where we recall that $E < 0$. Solving for κ, we find

$$\kappa = \sqrt{-\frac{2ME}{\hbar^2}} \tag{12.46}$$

and the Hamiltonian is

$$\hat{H} = \hat{A}^\dagger \hat{A} + E = \frac{\hat{p}^2}{2M} \tag{12.47}$$

for $|x| \geq \frac{L}{2}$.

At this stage, it seems like we can factorize for any value of E, but we still have one more requirement—we must ensure that the superpotential is continuous at $x = \pm\frac{L}{2}$. This tells us that we must have

$$-i\hbar k \tan\left(\tfrac{kL}{2}\right) = -i\hbar\kappa, \tag{12.48}$$

or

$$\tan\phi = \frac{\kappa}{k}. \tag{12.49}$$

You should check that the requirement is the same at $x = -\frac{L}{2}$. Now, we rearrange the equations a bit. We define $\phi_0 = \sqrt{2MV_0}L/2\hbar$, and then we have that

$$\frac{\kappa L}{2} = \sqrt{\phi_0^2 - \phi^2}, \tag{12.50}$$

and the transcendental equation becomes

$$\tan\phi = \sqrt{\frac{\phi_0^2}{\phi^2} - 1}. \tag{12.51}$$

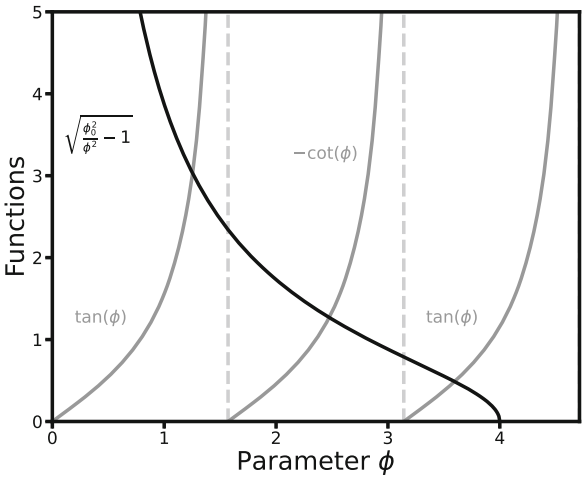

Fig. 12.3 Graphical method of solution for the transcendental equations in Eqs. (12.51) and (12.61). The black curve is the square-root function, the grays are the tangent and the cotangent. The solutions are marked by the points where the lines cross. Here, we have $\phi_0 = 4$, where there are two tangent solutions and one cotangent solution. Note that each tangent solution corresponds to an even wavefunction, while each cotangent solution corresponds to an odd wavefunction

To solve this equation, we plot each side of the equation and look for the points of intersection, which each correspond to a solution. This is done in Fig. 12.3 with the symbols illustrating the solutions. Focus on the tangent curves (the other curves will come into play later).

We need to think physically about what will happen with the solutions of the particle in the finite square well versus the infinite potential box. Since quanta like to spread out if they can (recall that the free particle has an equal probability to be found anywhere), we expect that the finite square well will spread out a bit more than the infinite box. This means the wavefunction will no longer vanish at the boundaries of the wall, but will spread out beyond it (we will see this is the case later when we discuss wavefunctions). But, this does bring up an issue with trying to form the Schrödinger factorization chain. This is because inside the box, the factorization is the same in the two cases, so we will get the same factorization chain as before. But, as we argued earlier, we expect the nodes to move farther away from each other. Why does this fail? It is because our potential is discontinuous, with a jump at the box walls. The jump is not consistent with the factorization method, so we will not be able to form the factorization chain. Nevertheless, we are able to go forward with single-shot factorization, so we do so. This is because any time we find a solution ϕ^* of the transcendental equation, then we have a factorization given by $\hat{H} = \hat{A}^\dagger(\phi^*)\hat{A}(\phi^*) + E(\phi^*)$, with $k^* = \frac{2\phi^*}{L}$ and $E(\phi^*) = \frac{\hbar^2 k^{*2}}{2M} - V_0$. Of course, the raising and lowering operators (and the energy) change for the different solutions ϕ^* of the transcendental equation. But, in each factorization (for each separate solution ϕ^*) the energy eigenstate satisfies $\hat{A}(\phi^*)|\psi(\phi^*)\rangle = 0$, with eigenvalue $E(\phi^*)$; note that ϕ^* is just a *label* for each of these states. Looking at the different spatial regions, we find that the subsidiary condition (for each ϕ^*) becomes

$$\hat{p}|\psi(\phi^*)\rangle = i\hbar k^* \tan(k^*\hat{x})|\psi(\phi^*)\rangle \quad \text{for} \quad |x| \leq \tfrac{L}{2}, \tag{12.52}$$

and

$$\hat{p}|\psi(\phi^*)\rangle = -i\hbar\kappa^*|\psi(\phi^*)\rangle \text{ for } x < -\tfrac{L}{2}$$
$$\hat{p}|\psi(\phi^*)\rangle = i\hbar\kappa^*|\psi(\phi^*)\rangle \text{ for } x > \tfrac{L}{2}, \tag{12.53}$$

with $\kappa^* = 2\sqrt{\phi_0^2 - \phi^{*2}}/L$. The first equation is one we have seen before for the particle in the box, and we know the wavefunction there is a cosine. The second set of equations are eigenvalue–eigenvector relations, but with imaginary eigenvalues. How can this be with a Hermitian operator $\hat{p}$? It arises because it holds only in a limited region of space. But, the solution is the same wavefunction for a momentum eigenstate, but now with a complex wavenumber, so it takes the form $e^{\pm\kappa x}$ in the two different regions. Putting this together gives us the wavefunction as

$$\langle x|\psi(\phi^*)\rangle = \psi(x;\phi^*) = \begin{cases} Be^{\kappa^* x} & \text{for } x < -\frac{L}{2} \\ A\cos(k^* x) & \text{for } |x| \leq \frac{L}{2} \\ Be^{-\kappa^* x} & \text{for } x > \frac{L}{2} \end{cases}. \tag{12.54}$$

The wavefunction is chosen to be even here (hence, B is the same on the left and the right) because the tangent superpotential solutions are even functions of x. The ratio $\frac{B}{A}$ is found from the fact that the wavefunction must be continuous; hence, it is given by

$$\frac{B}{A} = \cos\left(\frac{k^* L}{2}\right) e^{\frac{\kappa^* L}{2}}, \tag{12.55}$$

and the overall constant in front of the wavefunction must be found by normalization. In the problems, you will work out the algebra independently for the wavefunction outside the box.

One might be concerned that this procedure is not well-defined because the ladder operators diverge when $k^* x = (2n+1)\frac{\pi}{2}$. But, this is precisely the locations where $\psi(x;\phi^*) = 0$, so the divergence is required, because these points are nodes of the wavefunction. This procedure finds all of the even wavefunctions, which occur for each value of ϕ^* that solves the transcendental equation (and there always is only a finite number of solutions).

Now, we move on to the odd solutions, which require the cotangent superpotential. We use a $\hat{B}$ to denote the ladder operators here, so as to not confuse with the even wavefunction case. Then, by using the results for the particle in a box, we find that the lowering operator (in the region $|x| \leq \frac{L}{2}$) is given by

$$\hat{B} = \frac{1}{\sqrt{2M}}\left(\hat{p} + i\hbar k \text{cotan}(k\hat{x})\right) \text{ for } |x| \leq \tfrac{L}{2}, \tag{12.56}$$

which yields

$$\begin{aligned}\hat{B}^\dagger \hat{B} &= \frac{\hat{p}^2}{2M} + \frac{i\hbar k}{2M}[\hat{p}, \operatorname{cotan}(k\hat{x})] + \frac{\hbar^2 k^2}{2M}\operatorname{cotan}^2(k\hat{x}) \\ &= \frac{\hat{p}^2}{2M} - \frac{\hbar^2 k^2}{2M}\operatorname{cosec}^2(k\hat{x}) + \frac{\hbar^2 k^2}{2M}\operatorname{cotan}^2(k\hat{x}) \\ &= \frac{\hat{p}^2}{2M} - \frac{\hbar^2 k^2}{2M} = \frac{\hat{p}^2}{2M} - V_0 - E, \end{aligned} \tag{12.57}$$

with $\frac{kL}{2} = \phi = \sqrt{\frac{2M(V_0+E)}{\hbar^2}}\frac{L}{2}$. In the exterior regions, we then have

$$\hat{B} = \begin{cases} \frac{1}{\sqrt{2M}}(p + i\hbar\kappa) & \text{for } x < -\frac{L}{2}, \\ \frac{1}{\sqrt{2M}}(\hat{p} - i\hbar\kappa) & \text{for } x > \frac{L}{2}, \end{cases} \tag{12.58}$$

with $\hat{B}^\dagger \hat{B} = \frac{\hat{p}^2}{2M} + \frac{\hbar^2 k^2}{2M}$ in the exterior region as well. We define

$$\frac{\kappa L}{2} = \sqrt{-\frac{2ME}{\hbar^2}}\frac{L}{2} = \sqrt{\phi_0^2 - \phi^2}. \tag{12.59}$$

Continuity of $\hat{B}$ at the boundaries gives the transcendental equation

$$i\hbar \operatorname{cotan}\left(\frac{kL}{2}\right) = -i\hbar\kappa \tag{12.60}$$

or

$$-\operatorname{cotan}(\phi) = \sqrt{\frac{\phi_0^2}{\phi^2} - 1}. \tag{12.61}$$

Figure 12.3 shows the graphical solution of this transcendental equation as well. We let $\tilde{\phi}$ denote a solution to this transcendental equation. The subsidiary condition is given by $\hat{B}|\psi(\tilde{\phi})\rangle = 0$, and then the wave functions are $\psi(x;\tilde{\phi}) = \langle x|\psi(\tilde{\phi})\rangle$. In Problem 12.2.4, you will determine the wavefunctions, which are given by the following odd functions

$$\psi(x;\tilde{\phi}) = \begin{cases} -\beta e^{\tilde{\kappa}x} & \text{for } x < -\frac{L}{2} \\ \alpha \sin(\tilde{k}x) & \text{for } |x| \le \frac{L}{2} \\ \beta e^{-\tilde{\kappa}x} & \text{for } x > \frac{L}{2}, \end{cases} \tag{12.62}$$

where $\tilde{k} = \frac{2\tilde{\phi}}{L}$ and $\tilde{\kappa} = 2\sqrt{\phi_0^2 - \tilde{\phi}^2}/L$. Every root of the transcendental equation will yield a valid solution for a different bound state of this problem. But, we need

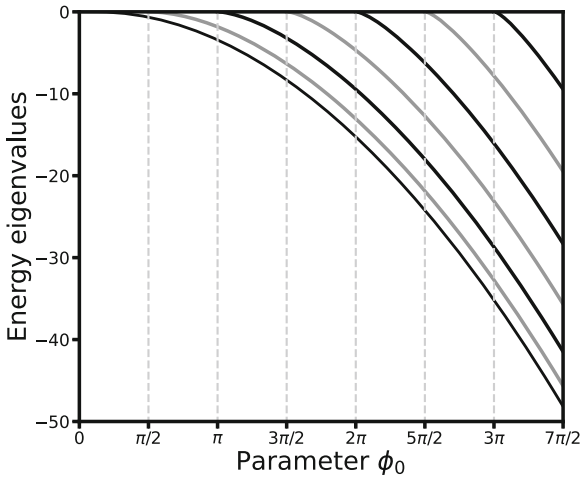

Fig. 12.4 Bound-state energies for the one-dimensional finite square well as a function of ϕ_0. The dashed lines mark where a new bound state enters, with an energy at the top of the box $E \approx 0$. In this calculation, we work in a box of width π, and the energies are given as $E = \frac{4}{\pi^2}(\phi^{*2} - \phi_0^2)$ for each root ϕ^*

$\phi_0 \geq \frac{\pi}{2}$ or $V_0 \geq \frac{\hbar^2\pi^2}{2ML^2}$ for there to be at least one odd solution. Recall that the even and odd solutions alternate due to the node theorem. This example shows us that one can still find the eigenvalues and eigenvectors even when the ordinary factorization method fails.

We now look a bit closer into the solutions of the transcendental equations, which are depicted graphically in Fig. 12.3. There is one solution only for $0 < \phi_0 < \frac{\pi}{2}$. Two for $\frac{\pi}{2} < \phi_0 < \pi$, three for $\pi < \phi_0 < \frac{3\pi}{2}$, and so on. In the limit as $\phi_0 \to \infty$, we find the even solutions occur at $\phi^* = \frac{(2n+1)\pi}{2}$ and the odd ones at $\tilde{\phi} = n\pi$. This is exactly what we found for the particle in a box previously, which shows that these results are connected (in the infinite potential limit) with those of the particle in a box, as we might have suspected. The case $V_0 \to 0$ is more challenging to analyze. Here, we will have just one even solution, so we focus on the transcendental equation $\tan\phi = \sqrt{\phi_0^2 - \phi^2}/\phi$. We want to expand both sides for small arguments. We know ϕ_0 is small, and we write $\phi^* = \phi_0 - \delta$, with δ very small, since we anticipate from the graph that the root is close to ϕ_0. Then $\tan\phi^* \approx \tan\phi_0$ and $\sqrt{(\phi_0 - \phi^*)(\phi_0 + \phi^*)} \approx \sqrt{2\phi_0\delta}$. The transcendental equation becomes $\phi_0 = \frac{\sqrt{2\phi_0\delta}}{\phi_0}$, or $\delta = \phi_0^3/2$, which is very small, if ϕ_0 is small. Solving for the energy, we find that $E = -V_0^2\frac{mL^2}{2\hbar^2}$. Hence, the energy increases quadratically with V_0 when $V_0 \to 0$. This means the finite square well always has at least one bound state solution, no matter how small V_0 is. We plot the bound-state energies versus V_0 in Fig. 12.4. Note how each new energy eigenstate has a very flat dependence on V_0 initially and then increases faster.

We end this section with a semiquantitative discussion about what happens when we try to observe a particle in the forbidden region. Recall, in the forbidden region, the wavefunction is exponentially decaying and if we calculate the average kinetic energy, restricted to that region, it will be negative, because the wavefunction there

acts like a momentum eigenstate with imaginary momentum. So, what does a particle with negative energy look like? It turns out we can never find out. This is because the act of measuring the particle localized to the forbidden region gives it enough energy that the observed kinetic energy must be positive. This argument, as far as we know, originates with Winter (1979). To start, if we are to find the particle in the forbidden region, because the wavefunction decays exponentially, we expect to find it in a region whose uncertainty is on the order of $\Delta x \approx \frac{1}{\kappa}$. This arises, because the exponential function decays very fast when its argument is larger than 1, so we have essentially no chance to find the particle lying far outside of this region. Then the uncertainty in the momentum is $\Delta p \approx \frac{\hbar}{\Delta x} = \hbar\kappa$. The energy of the state, is given by $E = -V_0 + \frac{2\hbar^2\phi^2}{mL^2}$, while the extra energy added to localize in the forbidden region is $\Delta E \approx (\Delta p)^2/2M \approx V_0 - \frac{2\hbar^2\phi^2}{mL^2}$. When we add this energy to observe the particle to its current energy, we see it will be larger than zero. Since there is no potential energy in the forbidden region, this means it has positive kinetic energy. The physical principle involved is that it takes energy to localize a particle in some region of space. To find the particle in a forbidden region requires us to localize it into the part of the forbidden region that is accessible to it. When we do this, we find that the energy we must add to localize the particle is just enough to give it a total energy that is positive, so it has a positive kinetic energy. Nature does not allow us to observe what a negative kinetic energy looks like. This is similar in spirit to the fact that nature does not allow us to directly observe what it means when quanta travel in superposition. This may be frustrating to learn, but, on the other hand, it is pretty neat that this result makes a lot of physical sense, and it arises simply from analyzing the quantum behavior. We have a video https://tinyurl.com/5bn8y8dt that describes some of the more technical issues (using calculus) and also provides a quasi-derivation of the more conventional Schrödinger equation in it differential equation format.

Section 12.2 Problems:

12.2.1 Wavefunctions in the Forbidden Region

Consider the region where $|x| > \frac{L}{2}$. The lowering operator is written as $\hat{A} = \frac{1}{\sqrt{2M}}(\hat{p} \pm i\hbar\kappa)$, with $\kappa = \sqrt{-\frac{2ME}{\hbar^2}}$, a real number. In this problem, we will determine the wavefunctions and also fix which sign is correct for each region external to the box. Recall that in this region, the eigenstate satisfies $\hat{A}|\psi(\phi^*)\rangle = 0$. Determine the wavefunction by expanding the translation operator in a power series. That is, write

$$\psi(x;\phi^*) = \langle x|\psi(\phi^*)\rangle = \langle 0_x|e^{\frac{i}{\hbar}x\hat{p}}|\psi(\phi^*)\rangle. \tag{12.63}$$

Expand the exponential in a power series and use the subsidiary condition to show that

$$\psi(x;\phi^*) = \langle 0_x|\psi(\phi^*)\rangle e^{\pm\kappa x}. \tag{12.64}$$

Noting that the wavefunction must be finite everywhere, determine which sign is correct for $x < -\frac{L}{2}$ and for $x > \frac{L}{2}$.

12.2.2 Solving the Transcendental Equation

Using a calculator or a computer, solve the transcendental equation for the bound-state energies when $\phi_0 = 0.5$, 2, 4, and 6. Identify which eigenvalue has an even wavefunction and which has an odd wavefunction.

Hint: You will need to come up with a strategy for solving the transcendental equation. You can use the graphical picture to bracket your roots and then use a simple method such as bisection to find the roots by computer. If you use a calculator, you may want to solve by iteration. Think about how you can do this.

12.2.3 Halo States

When the potential is weak ($V_0 \ll 1$), we expect the bound state to be weakly bound and to extend over a large region of position space. The best way to examine this is to determine the uncertainty in position Δx, But there is no simple way to do this with our operator methods. Instead, we recall that the energy can be expressed as

$$|E| = \frac{\hbar^2}{2ML_{\text{eff}}^2}, \tag{12.65}$$

where L_{eff} is the effective size of the region where the particle is bound. Use the result derived in the text for the energy in the limit as $V_0 \to 0$, to determine L_{eff} for the weakly bound state. You should find that the effective length grows as $V_0 \to 0$. These large weakly bound states are called halo states. Similar halo states are found in cold neutral atom states near a Feshbach resonance and in Rydberg atoms.

Answer: $L_{\text{eff}} = \frac{\hbar^2}{mLV_0}$

12.2.4 Odd Solutions

In the text, we showed how to determine the wavefunctions for the even solutions. Modeling the approach we used for finding the even solutions, show that the odd solutions are indeed given by the formulas shown in the main text in Eq. (12.62).

12.3 Particle in a Two-dimensional Circular Box

We now move on to two dimensional (circular) boxes and describe a particle moving inside an infinite circular-shaped two-dimensional box

$$V(\rho,\phi)=\begin{cases}0 & \text{for } \rho\le R\\ \infty & \text{for } \rho>R.\end{cases} \tag{12.66}$$

To begin, we need to remind you how we describe the kinetic energy in polar coordinates using separation of variables. We have

$$\hat{H}=\frac{\hat{p}_x^2}{2M}+\frac{\hat{p}_y^2}{2M}=\frac{\hat{p}_\rho^2}{2M}+\frac{\hat{p}_\phi^2}{2M}-\frac{\hbar^2}{8M\hat{\rho}^2}=\frac{\hat{p}_\rho^2}{2M}+\frac{(\hat{L}_z)^2-\frac{1}{4}\hbar^2}{2M\hat{\rho}^2}, \tag{12.67}$$

after using $\hat{p}_\phi=\frac{\hat{L}_z}{\hat{\rho}}$. Note that there is a quantum correction to the kinetic energy in two dimensions. We can factorize the energy eigenstate into a radial component and an angular component ($|\psi\rangle=|\psi_\rho\rangle\otimes|m\rangle$), the latter being the state $|m\rangle$ that has definite z-component of angular momentum ($\hat{L}_z|m\rangle=\hbar m|m\rangle$). Then, we create an infinite number of radial Hamiltonians, which must satisfy $\hat{H}_m|\psi_\rho\rangle=E|\psi_\rho\rangle$, with

$$\hat{H}_m=\begin{cases}\frac{\hat{p}_\rho^2}{2M}+\frac{\hbar^2\left(m^2-\frac{1}{4}\right)}{2M\hat{\rho}^2} & \text{for } \rho\le R\\ \infty & \text{for } \rho>R.\end{cases} \tag{12.68}$$

We focus first on the case with $m=0$, given by

$$\hat{H}_0=\frac{\hat{p}_\rho^2}{2M}-\frac{\hbar^2}{8M\hat{\rho}^2}, \tag{12.69}$$

for $\rho\le R$. You might be saying to yourself that this should be easy to find a way to factorize it, because a $\frac{i\hbar k}{\hat{\rho}}$ superpotential will square to $\frac{1}{\hat{\rho}^2}$ and the commutator with $\hat{p}_\rho$ has the same form, so we can find the factorization immediately. But, then you would have acted too quickly, and we now show you why. Let's try to use

$$\hat{A}_0=\frac{1}{\sqrt{2M}}\left(\hat{p}_\rho-\frac{i\hbar k}{\hat{\rho}}\right). \tag{12.70}$$

Then we find that

$$\hat{A}_0^\dagger\hat{A}_0=\frac{\hat{p}_\rho^2}{2M}+\frac{\hbar^2k(k+1)}{2M\hat{\rho}^2}, \tag{12.71}$$

which requires $k = -\frac{1}{2}$ to produce the Hamiltonian. Note that there is no constant term generated by the product of the operators. This means that this state has energy $E = 0$. This is indeed our problem, even though the statement is correct. We should have many energies, but here we find only one. This does not mean that there is only one energy. No, it means our factorization was not sufficiently general to allow us to find all of the possible solutions. What we need to do is to formulate single-shot factorization in a more general way and then use it to find the most general factorization.

The approach for single-shot factorization is to write the superpotential in the general form

$$i\hbar k W(k'\hat{x}) = i\hbar k g(k'\hat{x}) + \frac{[\hat{p}, f(k'\hat{x})]}{f(k'\hat{x})}, \tag{12.72}$$

where we work generally for the one-dimensional case before translating to the radial problem at hand. Recalling that we can calculate the commutator of the denominator from the Leibniz rule as

$$\left[\hat{p}, \frac{1}{f(k'\hat{x})}\right] = -\frac{[\hat{p}, f(k'\hat{x})]}{f^2(k'\hat{x})}, \tag{12.73}$$

we find that

$$\begin{aligned}\hat{A}^\dagger \hat{A} = \frac{\hat{p}^2}{2M} - \frac{i\hbar k}{2M}[\hat{p}, g(k'\hat{x})] - \frac{1}{2M}\frac{[\hat{p}, [\hat{p}, f(k'\hat{x})]]}{f(k'\hat{x})} + \frac{1}{2M}\left(\frac{[\hat{p}, f(k'\hat{x})]}{f(k'\hat{x})}\right)^2 \\ + \frac{\hbar^2 k^2}{2M} g^2(k'\hat{x}) - \frac{i\hbar k}{M}\frac{g(k'\hat{x})[\hat{p}, f(k'\hat{x})]}{f(k'\hat{x})} - \frac{1}{2M}\left(\frac{[\hat{p}, f(k'\hat{x})]}{f(k'\hat{x})}\right)^2.\end{aligned} \tag{12.74}$$

One can immediately see that the last terms on each line cancel, leaving us with

$$\begin{aligned}\hat{A}^\dagger \hat{A} = \frac{\hat{p}^2}{2M} - \frac{i\hbar k}{2M}[\hat{p}, g(k'\hat{x})] + \frac{\hbar^2 k^2}{2M} g^2(k'\hat{x}) - \frac{1}{2M}\frac{[\hat{p}, [\hat{p}, f(k'\hat{x})]]}{f(k'\hat{x})} \\ - \frac{i\hbar k}{M}\frac{g(k'\hat{x})[\hat{p}, f(k'\hat{x})]}{f(k'\hat{x})}.\end{aligned} \tag{12.75}$$

Your first reaction to this might be that it is making things worse. But, there are special functions, called confluent hypergeometric functions, for which the expression earlier greatly simplifies. We will not work with these functions in complete generality, but for this problem we need to work with a special case of them, called Bessel functions, which take the form

$$J_m(x) = \left(\frac{x}{2}\right)^m \sum_{n=0}^{\infty} (-1)^n \left(\frac{x}{2}\right)^{2n} \frac{1}{n!(n+m)!}. \tag{12.76}$$

This expression is a power series that looks just like a cosine series (times a power), except each term has an extra factor of $\frac{1}{(n+m)!}$ in it. So, one might anticipate that it behaves similarly to the cosine, but perhaps decaying to zero as the argument is made large due to the extra factor. This is the case for these functions. We will see by direct computation that these functions do indeed simplify the complicated operator expression for the product of the two ladder operators and are the needed addition to our superpotential to solve the problem at hand.

Let's work out the commutation relation for this Bessel function. We have

$$[\hat{p},\, J_m(k'\hat{x})] = \sum_{n=0}^{\infty} (-1)^n \left(\frac{k'}{2}\right)^{2n+m} \frac{1}{n!(n+m)!} \left[\hat{p},\, \hat{x}^{2n+m}\right]. \tag{12.77}$$

Using the result for the commutator of momentum with a power of position, we find

$$[\hat{p},\, J_m(k'\hat{x})] = -i\hbar \sum_{n=0}^{\infty} (-1)^n \left(\frac{k'}{2}\right)^{2n+m} \frac{1}{n!(n+m)!} (2n+m)\hat{x}^{2n+m-1}. \tag{12.78}$$

We write $2n + m = n + n + m$ and note that the n term can cancel one factor in $\frac{1}{n!}$, while the $n + m$ term cancels one factor in $\frac{1}{(n+m)!}$. Then we have two contributions (we have dropped terms that vanish in the summations)

$$\begin{aligned}[\hat{p},\, J_m(k'\hat{x})] = &-i\hbar \sum_{n=1}^{\infty} (-1)^n \left(\frac{k'}{2}\right)^{2n+m} \frac{1}{(n-1)!(n+m)!} \hat{x}^{2n+m-1} \\ &- i\hbar \sum_{n=0}^{\infty} (-1)^n \left(\frac{k'}{2}\right)^{2n+m} \frac{1}{n!(n+m-1)!} \hat{x}^{2n+m-1}.\end{aligned} \tag{12.79}$$

Finally, we shift $n \to n + 1$ in the first summation to yield

$$\begin{aligned}[\hat{p},\, J_m(k'\hat{x})] = &\, i\hbar \sum_{n=0}^{\infty} (-1)^n \left(\frac{k'}{2}\right)^{2n+m+2} \frac{1}{n!(n+m+1)!} \hat{x}^{2n+m+1} \\ &- i\hbar \sum_{n=0}^{\infty} (-1)^n \left(\frac{k'}{2}\right)^{2n+m} \frac{1}{n!(n+m-1)!} \hat{x}^{2n+m-1}.\end{aligned} \tag{12.80}$$

Now we use the definition of the Bessel function again, to find that

$$[\hat{p},\, J_m(k'\hat{x})] = \frac{i\hbar k'}{2} \left(J_{m+1}(k'\hat{x}) - J_{m-1}(k'\hat{x})\right). \tag{12.81}$$

This holds for $m > 0$. The case with $m = 0$ is examined in Problem 12.3.1. Solving it shows that $[\hat{p}, J_0(k'\hat{x})] = i\hbar k' J_1(k'\hat{x})$.

We are now ready to do the general factorization of $\hat{H}_0$. After some trial and error, we find the proper choice is

$$\hat{A}_0 = \frac{1}{\sqrt{2M}} \left(\hat{p}_\rho - i\hbar k \frac{J_1(k\hat{\rho})}{J_0(k\hat{\rho})} + \frac{i\hbar}{2\hat{\rho}} \right), \tag{12.82}$$

where we evaluated the commutator directly for the choice $f = J_0$, and we used $g = -\frac{1}{2k\hat{\rho}}$ from our first factorization attempt. Now, we compute

$$\hat{A}_0^\dagger \hat{A}_0 = \frac{\hat{p}_\rho^2}{2M} - \frac{\hbar^2}{8M\hat{\rho}^2} + \frac{\hbar^2 k^2}{4M} \frac{J_2(k\hat{x}) - J_0(k\hat{x})}{J_0(k\hat{x})} - \frac{\hbar^2 k}{2M\hat{\rho}} \frac{J_1(k\hat{x})}{J_0(k\hat{x})}. \tag{12.83}$$

It looks like we are stuck at this point, but there is a recurrence relation between Bessel functions that you will derive in Problem 12.3.3. It is given by

$$\frac{2m}{x} J_m(x) = J_{m+1}(x) + J_{m-1}(x). \tag{12.84}$$

Using it to write $J_2(k\hat{\rho}) = \frac{2}{k\hat{\rho}} J_1(k\hat{\rho}) - J_0(k\hat{\rho})$, we find the remarkable simplification, which yields

$$\hat{A}_0^\dagger \hat{A}_0 = \frac{\hat{p}_\rho^2}{2M} - \frac{\hbar^2}{8M\hat{\rho}^2} - \frac{\hbar^2 k^2}{2M}, \tag{12.85}$$

or

$$\hat{H}_0 = \hat{A}_0^\dagger \hat{A}_0 + E_0, \tag{12.86}$$

with $E_0 = \frac{\hbar^2 k^2}{2M}$. The state with energy E_0 satisfies the subsidiary condition, given by $\hat{A}_0(k)|\psi(k)\rangle = 0$, where we have introduced the index k to make everything clear. This result holds for any k, so in a single shot, we have factorized the Hamiltonians that give us the continuum of energy eigenvalues, indexed by k. For the free particle we are done. This is a continuum of energies, and we get all of them simply by changing k in the raising and lowering operators and in the state.

However, we are interested in solving the particle in a box. So, for the $\hat{A}_0(k)$ operator, we must pick k in a manner such that the superpotential has no divergences for $\rho > 0$ except right at $\rho = R$ (because this guarantees the wavefunction goes to zero at the boundary of the box, which is what we need to have for a valid solution). As we mentioned before, these Bessel functions look like sines or cosines, with amplitudes that decay for large arguments. In particular, they never diverge. So, to satisfy our requirement, the ground state must correspond to $k = \frac{x_1^{(0)}}{R}$, where $x_1^{(0)}$ is

the first zero of $J_0(x)$. This tells us that $k_0 = \frac{x_1^{(0)}}{R}$ and $E_0 = \frac{\hbar^2 (x_1^{(0)})^2}{2MR^2}$. What about excited states? We still require the wavefunction to vanish on the boundary of the box, so the higher energy states for $m = 0$ must correspond to the higher zeros of the Bessel function. Indeed, we have $k_n = \frac{x_{n+1}^{(0)}}{R}$ and $E_n = \frac{\hbar^2 (x_{n+1}^{(0)})^2}{2MR^2}$, will give us all of the energy eigenvalues. Note that the Bessel function has an infinite number of zeros, so there are an infinite number of energy eigenstates for $m = 0$, as we suspected. This is how single-shot factorization works—we find a factorization that works for any parameter, and then we set the parameter equal to the values needed to satisfy the boundary conditions. In this way, we find all of the eigenstates "at once."

Now, we need to compute the wavefunctions for $m = 0$. To do this, we use the Rodrigues formula idea and rewrite the subsidiary condition as a similarity transformation acting on the eigenstate. This follows straightforwardly as

$$\hat{A}_0 |\psi(k)\rangle = \sqrt{\hat{\rho}} J_0(k\hat{\rho}) \hat{p}_\rho \frac{1}{J_0(k\hat{\rho})} \frac{1}{\sqrt{\hat{\rho}}} |\psi(k)\rangle = 0, \tag{12.87}$$

which tells us that the state

$$\frac{1}{J_0(k\hat{\rho})} \frac{1}{\sqrt{\hat{\rho}}} |\psi(k)\rangle \tag{12.88}$$

is annihilated by $\hat{p}_\rho$. Our next step is to recall how we use the translation operator in polar coordinates to compute the wavefunction. Our approach is

$$\psi_0(\rho, \phi; k) = \langle \rho | \otimes \langle \phi | \, |\psi(k)\rangle \otimes |m\rangle = \langle 0_\rho | e^{\frac{i}{\hbar}\rho\left(\hat{p}_\rho + \frac{i\hbar}{2\rho}\right)} |\psi(k)\rangle \frac{1}{\sqrt{2\pi}} e^{im\phi}. \tag{12.89}$$

The radial matrix element is simple to evaluate. We just note that the state $\frac{1}{J_0(k\hat{\rho})}|\psi(k)\rangle$ is annihilated by $\hat{p}_\rho + \frac{i\hbar}{2\hat{\rho}}$. So, we multiply by 1 to immediately find that

$$\psi_0(\rho, \phi; k) = \frac{1}{\sqrt{2\pi}} J_0(k\rho) e^{im\phi} \langle 0_\rho | \psi(k)\rangle, \tag{12.90}$$

with the last term the normalization constant for the radial part of the wavefunction. Here, we used the fact that $\frac{1}{J_0(0)} = 1$. There is no simple way to find the normalization constant. It needs to computed numerically for these problems.

Note, that if we try to form the factorization chain, in this case, we will find it fails. This is something you will analyze in Problem 12.3.5. This result might be surprising to you, because we could form a factorization chain for the one-dimensional particle-in-a-box problem. The reason why it was successful there and will not be successful here is because the roots of the cosine or sine are evenly spaced, while the roots of the Bessel functions are not (see Fig. 12.5). The uniform

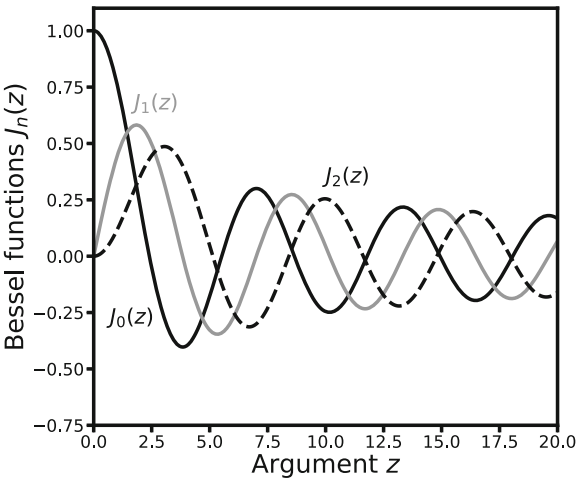

Fig. 12.5 The first three Bessel functions $J_0(z)$ (solid black), $J_1(z)$ (gray), and $J_2(z)$ (dashed black). These functions oscillate like sines or cosines, but they also decay for large argument. Note how the zeros are not uniformly spaced. In addition, they approach the origin as z^n

spacing allowed us to create the factorization chain for one dimension. The lack of even spacing here prevents us from doing so in two dimensions.

But, we are not finished yet. We still need to find the solutions for $m \neq 0$. The way this works is similar to how we do conventional factorization, but it has a few differences. We start by defining indexed ladder operators, which we denote as $\hat{B}$ so as to not confuse with the $\hat{A}$ operators we discussed earlier. Here, we have

$$\hat{B}_m = \frac{1}{\sqrt{2M}} \left(\hat{p}_\rho + i\hbar \frac{\left(m + \frac{1}{2}\right)}{\hat{\rho}} \right), \tag{12.91}$$

then

$$\begin{aligned}
\hat{B}_m^\dagger \hat{B}_m &= \frac{\hat{p}_\rho^2}{2M} + \frac{i\hbar}{2M} \left[\hat{p}_\rho, \frac{\left(m + \frac{1}{2}\right)}{\hat{\rho}} \right] + \frac{\hbar^2 \left(m + \frac{1}{2}\right)^2}{2M\hat{\rho}^2} \\
&= \frac{\hat{p}_\rho^2}{2M} - \frac{\hbar^2 \left(m + \frac{1}{2}\right)}{2M\hat{\rho}^2} + \frac{\hbar^2 \left(m + \frac{1}{2}\right)^2}{2M\hat{\rho}^2} = \frac{\hat{p}_\rho^2}{2M} + \frac{\hbar^2 \left(m^2 - \frac{1}{4}\right)}{2M\hat{\rho}^2} \\
\hat{B}_m \hat{B}_m^\dagger &= \frac{\hat{p}_\rho^2}{2M} + \frac{\hbar^2 \left((m+1)^2 - \frac{1}{4}\right)}{2M\hat{\rho}^2}.
\end{aligned} \tag{12.92}$$

Hence, we have the intertwining relation

$$\hat{H}_{m+1} \hat{B}_m = \hat{B}_m \hat{B}_m^\dagger \hat{B}_m = \hat{B}_m \hat{H}_m. \tag{12.93}$$

This means that if we have found a state that satisfies $\hat{A}_0(k)|\psi(k)\rangle = 0$, then

$$|\psi_m(k)\rangle = \hat{B}_{m-1}\hat{B}_{m-2}\cdots\hat{B}_1\hat{B}_0|\psi(k)\rangle \tag{12.94}$$

is an (unnormalized) eigenstate of $\hat{H}_m$ with eigenvalue $E_m = \frac{\hbar^2k^2}{2M}$. The proof follows from just applying the Hamiltonian to the state as follows:

$$\begin{aligned}
\hat{H}_m|\psi_m(k)\rangle &= \underbrace{\hat{B}_{m-1}\hat{B}_{m-1}^\dagger}_{\hat{H}_m}\underbrace{\hat{B}_{m-1}\hat{B}_{m-2}\cdots\hat{B}_1\hat{B}_0|\psi(k)\rangle}_{|\psi_m(k)\rangle} \\
&= \hat{B}_{m-1}\underbrace{\hat{B}_{m-1}^\dagger\hat{B}_{m-1}}_{\hat{H}_{m-1}}\hat{B}_{m-2}\cdots\hat{B}_1\hat{B}_0|\psi(k)\rangle \\
&= \hat{B}_{m-1}\underbrace{\hat{H}_{m-1}}_{\hat{B}_{m-2}\hat{B}_{m-2}^\dagger}\hat{B}_{m-2}\cdots\hat{B}_1\hat{B}_0|\psi(k)\rangle \\
&= \hat{B}_{m-1}\hat{B}_{m-2}\underbrace{\hat{B}_{m-2}^\dagger\hat{B}_{m-2}}_{H_{m-2}}\cdots\hat{B}_1\hat{B}_0|\psi(k)\rangle \\
&= \hat{B}_{m-1}\hat{B}_{m-2}\cdots\hat{B}_1\hat{B}_0\hat{H}_0|\psi(k)\rangle \\
&= \frac{\hbar^2k^2}{2M}\hat{B}_{m-1}\hat{B}_{m-2}\cdots\hat{B}_1\hat{B}_0|\psi(k)\rangle \\
&= \frac{\hbar^2k^2}{2M}|\psi_m(k)\rangle.
\end{aligned} \tag{12.95}$$

Please be sure to pay very careful attention to the differences here from the conventional way that we found eigenstates. There is no auxiliary Hamiltonian, and we apply lowering, not raising, operators to our state. Also note that we have not yet fixed the value of k.

Let's compute the eigenvectors with $m = 1$. We have

$$\begin{aligned}
|\psi_1(k)\rangle = \hat{B}_0|\psi(k)\rangle &= \frac{1}{\sqrt{2M}}\left(\hat{p}_\rho + \frac{i\hbar}{2\hat{\rho}}\right)|\psi(k)\rangle \\
&= \frac{1}{\sqrt{2M}}\left(i\hbar k\frac{J_1(k\hat{x})}{J_0(k\hat{x})} - \frac{i\hbar}{2\hat{\rho}} + \frac{i\hbar}{2\hat{\rho}}\right)|\psi(k)\rangle \\
&= \frac{i\hbar k}{\sqrt{2M}}\frac{J_1(k\hat{x})}{J_0(k\hat{x})}|\psi(k)\rangle.
\end{aligned} \tag{12.96}$$

Then

$$\psi_1(\rho;k) = \langle\rho|\psi_1(k)\rangle = \frac{i\hbar k}{\sqrt{2M}}\frac{J_1(k\rho)}{J_0(k\rho)}\langle\rho|\psi(k)\rangle = \frac{i\hbar k}{\sqrt{2M}}cJ_1(k\rho), \tag{12.97}$$

with c a normalization constant. Our boundary condition remains $J_1(kR) = 0$. The roots are $x_n^{(1)} = k_n^{(1)}R$. So the eigenfunctions for $m = 1$ become

$$\psi_1(\rho;k_n^{(1)}) = c'J_1(k_n^{(1)}\rho) = c'J_1\left(x_n^{(1)}\frac{\rho}{R}\right), \tag{12.98}$$

with c' the normalization constant (which can depend on n).

We work out one more case explicitly. For $m = 2$, we have

$$\begin{aligned}
\hat{B}_1\hat{B}_0|\psi(k)\rangle &= \frac{1}{\sqrt{2M}}\left(\hat{p}_\rho + \frac{3i\hbar}{2\hat{\rho}}\right)\frac{i\hbar k}{\sqrt{2M}}\frac{J_1(k\hat{\rho})}{J_0(k\hat{\rho})}|\psi(k)\rangle \\
&= \frac{i\hbar k}{2M}\left(\left[\hat{p}_\rho, \frac{J_1(k\hat{\rho})}{J_0(k\hat{\rho})}\right] + \frac{J_1(k\hat{\rho})}{J_0(k\hat{\rho})}\hat{p}_\rho + \frac{3i\hbar}{2\hat{\rho}}\frac{J_1(k\hat{\rho})}{J_0(k\hat{\rho})}\right)|\psi(k)\rangle \\
&= \frac{i\hbar k}{2M}\left(\frac{i\hbar k}{2}\frac{J_2(k\hat{\rho}) - J_0(k\hat{\rho})}{J_0(k\hat{\rho})} - i\hbar k\frac{J_1^2(k\hat{\rho})}{J_0^2(k\hat{\rho})}\right. \\
&\quad \left. + \frac{J_1(k\hat{\rho})}{J_0(k\hat{\rho})}\left(i\hbar k\frac{J_1(k\hat{\rho})}{J_0(k\hat{\rho})} - \frac{i\hbar}{2\hat{\rho}}\right) + \frac{3i\hbar}{2\hat{\rho}}\frac{J_1(k\hat{\rho})}{J_0(k\hat{\rho})}\right)|\psi(k)\rangle \\
&= \frac{(i\hbar k)^2}{2M}\frac{J_2(k\hat{\rho}) + \frac{2}{k\hat{\rho}}J_1(k\hat{\rho}) - J_0(k\hat{\rho})}{2J_0(k\hat{\rho})}|\psi(k)\rangle \\
&= \frac{(i\hbar k)^2}{2M}\frac{J_2(k\hat{\rho})}{J_0(k\hat{\rho})}|\psi(k)\rangle.
\end{aligned} \tag{12.99}$$

In the first line, we used the result for $\hat{B}_0|\psi(k)\rangle$ that we already worked out. In the second line, we shift the momentum to the right at the cost of adding a commutator. In the third line, we evaluate the commutator and also the action of the momentum onto the state. Finally, we use the identity $\frac{2}{k\hat{\rho}}J_1(k\hat{\rho}) = J_2(k\hat{\rho}) + J_0(k\hat{\rho})$. The final result is quite simple, so we see that $\psi_2(\rho;k) = c''J_2(k\rho)$, and we can apply the boundary condition as before to find the allowed k values where $\psi_2(R;k) = 0$. The general result follows by induction and is Problem 12.3.7.

The single-shot factorization works in this general fashion. We first find a factorization for arbitrary k values. Then we find the allowed energies and wavefunctions by applying the boundary conditions, which select discrete k values. For the particle-in-a-box in two dimensions, we need only factorize the first problem, with $m = 0$, and then we can use an approach similar to general factorization to generate all of the results for $m > 0$. The results for $m < 0$ are identical to those for the corresponding $m > 0$, except the angular factor changes from $e^{i|m|\phi}$ to $e^{-i|m|\phi}$,

which we immediately see because the Hamiltonians $\hat{H}_m$ depend only on m^2, so the radial solutions must be identical for positive and negative m. The key to this approach is that we can find a parameterized factorization. You might ask why doesn't the general factorization method work? The answer is that it *does* work. But, the potentials are not shape-invariant potentials, so we cannot solve the problem analytically. Nevertheless, the single-shot factorization method comes to the rescue and works very well for these types of problems.

Section 12.3 Problems:

12.3.1 Compute the Commutator $[\hat{p}, J_0(k\hat{x})]$

Use the power series expression and the definition of the Bessel functions to directly compute the commutator $[\hat{p}, J_0(k\hat{x})]$.

12.3.2 Commutator $\left[\hat{p}, \frac{1}{J_0(k\hat{x})}\right]$

Use the Leibniz rule, and the general result for the commutator of momentum with a Bessel function, to compute $\left[\hat{p}, \frac{1}{J_0(k\hat{x})}\right]$.

Hint: Recall that $[\hat{p}, 1] = 0$ and our trick of multiplying by 1.

12.3.3 Recurrence Relation of Bessel Functions

Prove the recurrence relation given in Eq. (12.84). You should use $m = n + m - n$ in your derivation, and you will need to shift indices in the power-series expansions.

12.3.4 A Subtle Issue in the Single-shot Factorization

Examine carefully the circumstances where $\hat{A}_0 = \frac{1}{\sqrt{2M}}\left(\hat{p}_\rho - i\hbar k \frac{J_1(k\hat{\rho})}{J_0(k\hat{\rho})} + \frac{i\hbar}{2\hat{\rho}}\right)$ diverges, when it acts on $|\psi(k)\rangle$. Explain why the procedure we used by relating the allowed wavevector to the zeros of the Bessel functions is the correct procedure to find the allowed k values. You can assume that $J_1(x)$ is always nonzero when $J_0(x)$ vanishes.

12.3.5 Why the Conventional Factorization Does Not Work for These Problems

The standard factorization procedure should work for this problem, but we have no clear way to carry it out, because the auxiliary potential is not a shape-invariant potential. Take the raising and lowering operators $\hat{A}_0^\dagger$ and $\hat{A}_0$ and show that when we compute the first auxiliary Hamiltonian $\hat{A}_0\hat{A}_0^\dagger + E_0$, we find that the potential is complicated and not of a form that one can easily factorize it. Note that this does not mean a factorization does not exist, it just means we don't have an analytic way to compute it.

12.3.6 An Important Commutator

Compute the commutator

$$\left[\hat{p}_\rho, \frac{J_m(k\hat{\rho})}{J_0(k\hat{\rho})}\right] = \frac{i\hbar k}{2}\frac{J_{m+1}(k\hat{\rho}) - J_{m-1}(k\hat{\rho})}{J_0(k\hat{\rho})} - i\hbar k\frac{J_m(k\hat{\rho})J_1(k\hat{\rho})}{J_0^2(k\hat{\rho})}. \tag{12.100}$$

Note that when $m = 0$, the left-hand side vanishes, so the right side must vanish as well. This yields an identity between J_1 and J_{-1}. Determine what it is.

12.3.7 Determining the Wavefunctions for the General m Sector by Induction

For the proof by induction, we first assume that

$$|\psi_m(k)\rangle = \hat{B}_{m-1}\hat{B}_{m-2}\cdots\hat{B}_1\hat{B}_0|\psi)k)\rangle = \frac{(i\hbar k)^m}{(2M)^{\frac{m}{2}}}\frac{J_m(k\hat{\rho})}{J_0(k\hat{\rho})}|\psi(k)\rangle \tag{12.101}$$

holds (with $m = 0$ as the base case). Then show that

$$|\psi_{m+1}(k)\rangle = \hat{B}_m\hat{B}_{m-1}\cdots\hat{B}_1\hat{B}_0|\psi(k)\rangle = \frac{(i\hbar k)^m}{(2M)^{\frac{m}{2}}}\hat{B}_m\frac{J_m(k\hat{\rho})}{J_0(k\hat{\rho})}|\psi(k)\rangle. \tag{12.102}$$

Use the commutator derived in the previous problem to move the momentum operator to the right, act it against the state using the subsidiary condition, and use the recurrence relation to finally show that

$$|\psi_{m+1}(k)\rangle = \frac{(i\hbar k)^{m+1}}{(2M)^{\frac{m+1}{2}}} \frac{J_{m+1}(k\hat{\rho})}{J_0(k\hat{\rho})} |\psi(k)\rangle, \tag{12.103}$$

which completes the proof by induction.

12.3.8 Brute-Force Calculation of the Wavefunction (Part 1)

These next two problems are quite challenging, but it is sometimes useful to calculate results in more than one way to verify they are correct.

We start by using the result of Problem 12.3.6 to directly compute the wavefunction in a brute-force manner from the subsidiary condition. In two dimensions, we have that

$$\langle \rho | = \langle 0_\rho | e^{\frac{i}{\hbar}\rho\left(\hat{p}_\rho + \frac{i\hbar}{2\hat{\rho}}\right)}. \tag{12.104}$$

Hence, we have the wavefunction $\psi_0(\rho; k)$ is given by

$$\psi_0(\rho; k) = \langle 0_\rho | e^{\frac{i}{\hbar}\rho\left(\hat{p}_\rho + \frac{i\hbar}{2\hat{\rho}}\right)} |\psi(k)\rangle. \tag{12.105}$$

The subsidiary condition is

$$\frac{1}{\sqrt{2M}} \left(\hat{p}_\rho - i\hbar k \frac{J_1(k\hat{\rho})}{J_0(k\hat{\rho})} + \frac{i\hbar}{2\hat{\rho}} \right) |\psi(k)\rangle = 0. \tag{12.106}$$

In the brute-force method, we expand the exponential as a power series and evaluate each term, one by one. We have

$$\langle \rho | \psi(k) \rangle = \sum_{n=0}^{\infty} \left(\frac{i\rho}{\hbar} \right)^n \frac{1}{n!} \langle 0_\rho | \left(\hat{p}_\rho + \frac{i\hbar}{2\hat{\rho}} \right)^n |\psi(k)\rangle. \tag{12.107}$$

We evaluate powers of the operator acting on $|\psi(k)\rangle$ as follows: first, show that

$$\left(\hat{p}_\rho + \frac{i\hbar}{2\hat{\rho}} \right) |\psi(k)\rangle = i\hbar k \frac{J_1(k\hat{\rho})}{J_0(k\hat{\rho})} |\psi(k)\rangle; \tag{12.108}$$

then show that

$$\left(\hat{p}_\rho + \frac{i\hbar}{2\hat{\rho}} \right)^2 |\psi(k)\rangle = i\hbar k \left\{ \left[\hat{p}_\rho + \frac{i\hbar}{2\hat{\rho}}, \frac{J_1(k\hat{\rho})}{J_0(k\hat{\rho})} \right] \right.$$

$$+ i\hbar k \frac{J_1(k\hat{\rho})}{J_0(k\hat{\rho})} \left(\hat{p}_\rho + \frac{i\hbar}{2\rho} \right) \Bigg\} |\psi(k)\rangle$$

$$= (i\hbar k)^2 \left(\frac{J_2(k\hat{\rho})}{2J_0(k\hat{\rho})} - \tfrac{1}{2} \right) |\psi(k)\rangle; \tag{12.109}$$

third, show that

$$\left(\hat{p}_\rho + \frac{i\hbar}{2\hat{\rho}} \right)^3 |\psi(k)\rangle = (i\hbar k)^3 \left(\frac{J_3(k\hat{\rho}) - 3J_1(k\hat{\rho})}{4J_0(k\hat{\rho})} \right) |\psi(k)\rangle; \tag{12.110}$$

and fourth

$$\left(\hat{p}_\rho + \frac{i\hbar}{2\hat{\rho}} \right)^4 |\psi(k)\rangle = (i\hbar k)^4 \left(\frac{J_4(k\hat{\rho}) - 4J_2(k\hat{\rho})}{8J_0(k\hat{\rho})} + \tfrac{3}{8} \right) |\psi(k)\rangle. \tag{12.111}$$

Now, use the fact that $\langle 0_\rho | J_m(k\hat{\rho}) = 0$ unless $m = 0$, where it equals 1, to determine the first few terms in the power series

$$\langle \rho | \psi(k) \rangle = 1 - \left(\tfrac{k\rho}{2} \right)^2 + \tfrac{1}{4} \left(\tfrac{k\rho}{2} \right)^4 - \cdots, \tag{12.112}$$

which are the first three terms in the power-series expansion for $J_0(k\rho)$.

12.3.9 Brute-Force Calculation of the Wavefunction (Part 2)

This part is the really hard part. Use induction to determine

$$\left(\hat{p}_\rho + \frac{i\hbar}{2\hat{\rho}} \right)^m |\psi(k)\rangle \tag{12.113}$$

to complete the derivation. You should find that

$$\langle 0_\rho | \left(\hat{p}_\rho + \frac{i\hbar}{2\hat{\rho}} \right)^{2m+1} |\psi(k)\rangle = 0 \text{ and}$$

$$\langle 0_\rho | \left(\hat{p}_\rho + \frac{i\hbar}{2\hat{\rho}} \right)^{2m} |\psi(k)\rangle = (-1)^m \left(\tfrac{i\hbar k}{2} \right)^{2m} \frac{(2m)!}{m!m!}. \tag{12.114}$$

You need to find an appropriate strategy to compute the general term, and it is not simple. Be creative and see if you can do it.

12.3.10 Particle in a Two-Dimensional Rectangular Box

In this problem, we solve for a particle in a two-dimensional rectangular box of dimensions L_x and L_y, respectively. The potential is

$$V(\hat{r}_x, \hat{r}_y) = \begin{cases} 0 & 0 \le x \le L_x \text{ and } 0 \le y \le L_y \\ \infty & \text{otherwise.} \end{cases} \tag{12.115}$$

Our Hamiltonian is

$$\hat{H} = \frac{\hat{p}_x^2}{2M} + \frac{\hat{p}_y^2}{2M} + V(\hat{r}_x, \hat{r}_y), \tag{12.116}$$

which can be factorized as the sum $\hat{H} = \hat{H}_x + \hat{H}_y$, with

$$\hat{H}_x = \hat{A}_0^\dagger(x)\hat{A}_0(x) + \frac{\hbar^2 k_x^2}{2M} \text{ and } \hat{H}_y = \hat{A}_0^\dagger(y)\hat{A}_0(y) + \frac{\hbar^2 k_y^2}{2M}. \tag{12.117}$$

(a) Show that

$$\hat{A}_0(x) = \frac{1}{\sqrt{2M}}\left(\hat{p}_x + i\hbar k_x \operatorname{cotan}(k_x \hat{r}_x)\right) \tag{12.118}$$

and

$$\hat{A}_0(y) = \frac{1}{\sqrt{2M}}\left(\hat{p}_y + i\hbar k_y \operatorname{cotan}(k_y \hat{r}_y)\right). \tag{12.119}$$

(b) Verify that the ground state has $k_x = \frac{\hbar\pi}{L_x}$ and $k_y = \frac{\hbar\pi}{L_y}$ and that the ground-state energy is the sum $E_x = \frac{\hbar^2 k_x^2}{2M} + \frac{\hbar^2 k_y^2}{2M}$.

(c) Verify that the ground-state wavefunction is given by

$$\psi_{gs}(x, y) = \langle x|\psi_x\rangle\langle y|\psi_y\rangle \propto \sin\left(\frac{\pi x}{L_x}\right)\sin\left(\frac{\pi y}{L_y}\right), \tag{12.120}$$

which is the product of the one-d wavefunctions in each direction.

(d) Show that the excited state energies and wavefunctions are

$$E(n_x, n_y) = \frac{\hbar^2\pi^2}{2M}\left(\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2}\right) \text{ and } \psi_{n_x n_y}(x, y) \propto \sin\left(\frac{n_x \pi x}{L_x}\right)\sin\left(\frac{n_y \pi y}{L_y}\right). \tag{12.121}$$

12.4 Particle in a Two-dimensional Circular Well

We now focus on the particle in a square-well potential for a circular box, given by

$$V(\rho,\phi) = \begin{cases} 0 & \text{for } \rho > R \\ -V_0 & \text{for } 0 \le \rho \le R. \end{cases} \tag{12.122}$$

The analysis inside the box (for the factorization) is identical to what we just did before, with the only difference being a new k value needs to be chosen for each eigenstate. In particular, we have $k = \sqrt{\frac{2M(E+V_0)}{\hbar^2}}$, with $E < 0$.

Outside the box, we need to factorize $\hat{H} = \frac{\hat{p}_\rho^2}{2M} - \frac{\hbar^2}{8M\hat{\rho}^2}$ with an energy eigenvalue $E < 0$. Unfortunately, there is no way to motivate what this solution is. We can say that similar to the one-dimensional case, where the solutions went from being oscillating sines or cosines inside the box to exponentially decaying functions outside the box, that here we must change from oscillating J_m Bessel functions to exponentially decaying K_m Bessel functions. But the K_m Bessel functions have no simple power-series expansion, so we cannot concretely perform the calculation as we did before. They do satisfy similar recurrence relations. It turns out that knowing these recurrence relations and the behavior for large argument is all we need to find the solution.

While we cannot prove the recurrence relations using the power series as we did for the J_m Bessel functions, we can state them. They are

$$\frac{2m}{x} K_m(x) = K_{m+1}(x) - K_{m-1}(x) \tag{12.123}$$

and the commutator is

$$[\hat{p}, K_m(\hat{x})] = \frac{i\hbar}{2}\big(K_{m+1}(\hat{x}) + K_{m-1}(\hat{x})\big), \tag{12.124}$$

which is similar to the ones for J_m, but with some changes of signs. We plot the first few $K_m(x)$ functions in Fig. 12.6. Note that K_0 diverges as $x \to 0$ (which is part of the reason why it does not have a power-series expansion there).

Now, we are ready to construct the $\hat{A}_0$ operator outside the box. We have

$$\hat{A}_0 = \frac{1}{\sqrt{2M}} \left(\hat{p}_\rho - i\hbar\kappa \frac{K_1(\kappa\hat{\rho})}{K_0(\kappa\hat{\rho})} + \frac{i\hbar}{2\hat{\rho}} \right), \tag{12.125}$$

which is the same as the operator for $\rho < R$, except for $J \to K$. We compute

$$\hat{A}_0^\dagger \hat{A}_0 = \frac{\hat{p}_\rho^2}{2M} - \frac{i\hbar\kappa}{2M} \left[\hat{p}_\rho, \frac{K_1(\kappa\hat{\rho})}{K_0(\kappa\hat{\rho})} - \frac{1}{2\kappa\hat{\rho}} - \frac{1}{2\kappa\hat{\rho}} \right] + \frac{\hbar^2\kappa^2}{2M} \left(\frac{K_1(\kappa\hat{\rho})}{K_0(\kappa\hat{\rho})} - \frac{1}{2\kappa\hat{\rho}} \right)^2$$

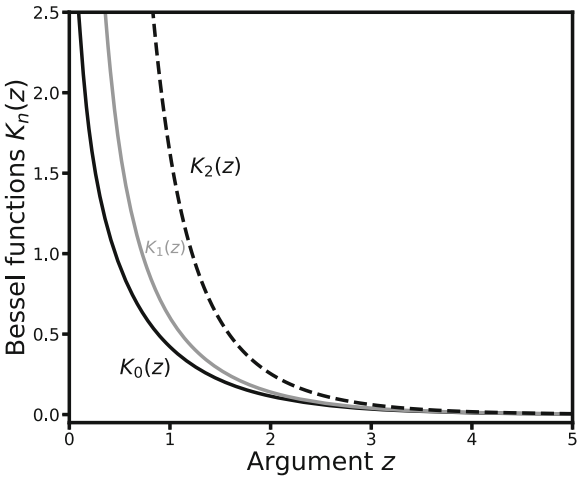

Fig. 12.6 Exponentially decaying Bessel functions $K_0(z)$ (solid black), $K_1(z)$ (gray), and $K_2(z)$ (dashed black). These functions look like exponentials and do not oscillate

$$= \frac{\hat{p}_\rho^2}{2M} - \frac{i\hbar\kappa}{2M}\left(\frac{i\hbar\kappa}{2}\frac{K_2(\kappa\hat{\rho}) + K_0(\kappa\hat{\rho})}{K_0(\kappa\hat{\rho})} - i\hbar\kappa\frac{K_1^2(\kappa\hat{\rho})}{K_0^2(\kappa\hat{\rho})} - \frac{i\hbar}{2\kappa\hat{\rho}^2}\right)$$
$$+ \frac{\hbar^2\kappa^2}{2M}\left(\frac{K_1^2(\kappa\hat{\rho})}{K_0^2(\kappa\hat{\rho})} - \frac{K_1(\kappa\hat{\rho})}{\kappa\hat{\rho}K_0(\kappa\hat{\rho})} + \frac{1}{4\kappa^2\hat{\rho}^2}\right). \tag{12.126}$$

We substitute in $K_2(\kappa\hat{\rho}) = K_0(\kappa\hat{\rho}) + \frac{2}{\kappa\hat{\rho}}K_1(\kappa\hat{\rho})$ and find that

$$\hat{A}_0^\dagger\hat{A}_0 = \frac{\hat{p}_\rho^2}{2M} + \frac{\hbar^2\kappa^2}{2M} - \frac{\hbar^2}{8M\hat{\rho}^2}. \tag{12.127}$$

So, $\hat{A}_0^\dagger\hat{A}_0 + E_0 = \hat{H}$, with $E_0 = -\frac{\hbar^2\kappa^2}{2M}$.

Now,we require $\hat{A}_0$ to be continuous at $\rho = R$. This implies that

$$-i\hbar\kappa\frac{K_1(\kappa R)}{K_0\kappa R)} + \frac{i\hbar}{2R} = -i\hbar k\frac{J_1(kR)}{J_0(kR)} + \frac{i\hbar}{2R}. \tag{12.128}$$

Hence,

$$\frac{J_1(kR)}{J_0(kR)} = \frac{\kappa}{k}\frac{K_1(\kappa R)}{K_0(\kappa R)}, \tag{12.129}$$

is the transcendental equation we need to solve to find the solution. The problem is solved graphically in Fig. 12.7 (we use $kR = \theta$ and $\sqrt{\frac{2MV_0}{\hbar^2}}R = \theta_0$). All κ and k that solve this equation produce energy eigenstates.

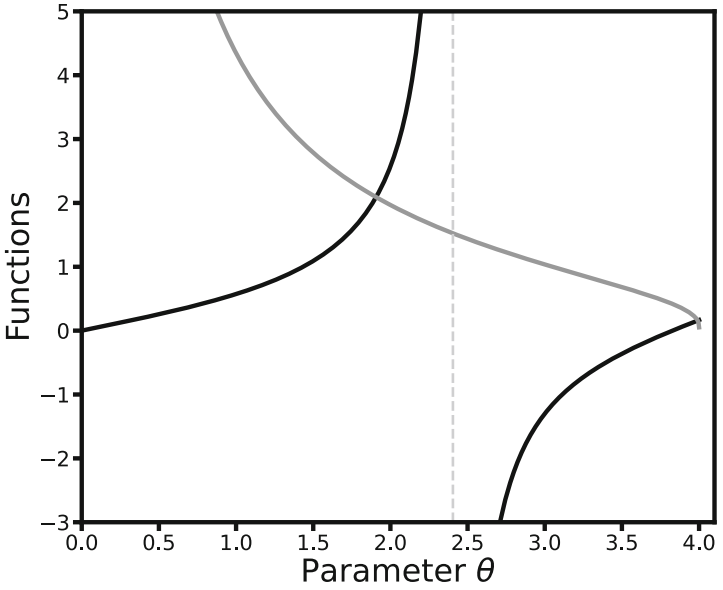

Fig. 12.7 Graphical solution to the transcendental equation needed to solve the two-dimensional square-well problem. Here we have $\theta_0 = 4$. Note that there is always at least one solution, even for small V_0. In this case, there are two solutions. The black curve is $\frac{J_1(kR)}{J_0(kR)}$, while the gray curve is $\frac{\kappa}{k}\frac{K_1(\kappa R)}{K_0(\kappa R)}$. The dashed line marks where the Bessel function $J_0(kR)$ has a zero

We can see that there is always at least one solution. Let's analyze the solution when V_0 is small. In this case, we already know $J_0(\theta) \approx 1$, $J_1(\theta) \approx \frac{\theta}{2}$, $K_0\left(\sqrt{\theta_0^2 - \theta^2}\right) \approx -\ln\sqrt{\theta_0^2 - \theta^2}$, and $K_1\left(\sqrt{\theta_0^2 - \theta^2}\right) \approx \frac{1}{\sqrt{\theta_0^2 - \theta^2}}$. Using these results then gives

$$\frac{\theta}{2} = -\frac{\sqrt{\theta_0^2 - \theta^2}}{\theta}\frac{1}{\sqrt{\theta_0^2 - \theta^2}\ln\sqrt{\theta_0^2 - \theta^2}} \tag{12.130}$$

or

$$\ln\left(\theta_0^2 - \theta^2\right) = -\frac{2}{\theta^2}. \tag{12.131}$$

But $\theta_0^2 - \theta^2 = -\frac{2MR^2E}{\hbar^2}$ and $\theta \approx \theta_0 = \sqrt{\frac{2MV_0}{\hbar^2}}R$. This gives us

$$E = -\frac{\hbar^2}{2MR^2}e^{-\frac{\hbar^2}{MR^2V_0}}, \tag{12.132}$$

which is exponentially small in V_0 (for small V_0), but always exists. We will use this result to show that all attractive potentials in two-dimensions have at least one bound state in Sect. 12.7.

Just like we did for the infinite well, we can now find solutions for $m \neq 0$, similar to how we did this in Sect. 12.4. We develop this formalism in the problems.

Section 12.4 Problems:

12.4.1 Energy Eigenstates for the Circular Square Well with $m \neq 0$

Assume we have a state $|\psi(k)\rangle$ that satisfies

$$\left(\hat{p}_\rho - i\hbar k \frac{J_1(k\hat{\rho})}{J_0(k\hat{\rho})} + \frac{i\hbar}{2\hat{\rho}}\right)|\psi(k)\rangle = 0, \tag{12.133}$$

for $\rho < R$ and

$$\left(\hat{p}_\rho - i\hbar\kappa \frac{K_1(\kappa\hat{\rho})}{K_0(\kappa\hat{\rho})} + \frac{i\hbar}{2\hat{\rho}}\right)|\psi(k)\rangle = 0, \tag{12.134}$$

for $\rho > R$. Show that if we define

$$\hat{B}_m = \frac{1}{\sqrt{2M}}\left(\hat{p}_\rho + \frac{i\hbar\left(m+\frac{1}{2}\right)}{\hat{\rho}}\right), \tag{12.135}$$

then

$$|\psi_m(k)\rangle = \hat{B}_{m-1}\hat{B}_{m-2}\cdots\hat{B}_1\hat{B}_0|\psi(k)\rangle \tag{12.136}$$

is an energy eigenstate of $\hat{H}_m$ with an eigenvalue given by $E = \frac{\hbar^2k^2}{2M} - V_0 = -\frac{\hbar^2\kappa^2}{2M}$.

12.4.2 Energy Eigenstates with $m = 1$

Consider the case with $m = 1$. Find the general wavefunction, which depends on k and κ. Use continuity of $\langle\rho|\psi_1(k)\rangle$ and $\langle\rho|\hat{p}_\rho|\psi_1(k)\rangle$ to find the transcendental equation

$$\frac{J_2(kR)}{J_1(kR)} = \frac{\kappa}{k}\frac{K_2(\kappa R)}{K_1(kR)}. \tag{12.137}$$

This equation can only be solved if V_0 is large enough. See if you can determine how large V_0 must be for there to be at least one solution.

Hint: you may need to do this numerically.

12.4.3 Energy Eigenstates with m = 2

Repeat the previous problem for $m = 2$.

12.5 Particle in a Three-dimensional Spherical Box

We now examine the potential in three dimensions, with

$$V(r) = \begin{cases} 0 & \text{for } r < R \\ \infty & \text{for } r > R. \end{cases} \tag{12.138}$$

Recall the separation of variables discussed in Sect. 11.3, where we found that

$$\hat{H} = \frac{\hat{p}_r^2}{2M} + \frac{\hat{L}^2}{2M\hat{r}^2} + V(\hat{r}), \tag{12.139}$$

in spherical coordinates. We examine the tensor-product solutions of the form $|\psi\rangle = |\psi_r\rangle \otimes |lm\rangle$. Then

$$\hat{H}|\psi\rangle = \left(\frac{\hat{p}_r^2}{2M} + \frac{\hbar^2 l(l+1)}{2M\hat{r}^2} + V(\hat{r})\right)|\psi_r\rangle \otimes |lm\rangle, \tag{12.140}$$

since $\hat{L}^2|lm\rangle = \hbar^2 l(l+1)|lm\rangle$. We focus first on the $l = 0$ solutions, which solve

$$\left(\frac{\hat{p}_r^2}{2M} + V(\hat{r})\right)|\psi_r\rangle = E|\psi_r\rangle. \tag{12.141}$$

This problem behaves identically to the particle in a one-dimensional box that runs from 0 to L, except here, we have $L = R$. The solution, as before, is to set

$$\hat{A}_0(k) = \frac{1}{\sqrt{2M}}\big(\hat{p}_r + i\hbar k \operatorname{cotan}(k\hat{r})\big), \tag{12.142}$$

and $\hat{H}_0 = \hat{A}_0^\dagger(k)\hat{A}_0(k) + E_0(k)$, with $E_0(k) = \frac{\hbar^2 k^2}{2M}$. We pick a k value such that the superpotential does not diverge inside the box, but does at $r = R$, so $\hat{A}_0 = \hat{A}_0(k_0)$, $k_0 = \frac{\pi}{R}$, and the ground-state energy is $E_0(k_0) = \frac{\hbar^2\pi^2}{2MR^2}$.

We continue to form the rest of the factorization chain precisely as before. We have

$$\hat{A}_n = \frac{1}{\sqrt{2M}}\big(\hat{p}_r + i\hbar k_n \text{cotan}(k_0\hat{r})\big). \tag{12.143}$$

Then

$$\hat{H}_n = \hat{A}_{n-1}\hat{A}_{n-1}^\dagger + E_{n-1} = \hat{A}_n^\dagger\hat{A}_n + E_n, \tag{12.144}$$

with

$$\begin{aligned}\hat{A}_{n-1}\hat{A}_{n-1}^\dagger &= \frac{\hat{p}_r^2}{2M} - \frac{i\hbar k_{n-1}}{2M}[\hat{p}_r, \text{cotan}(k_0\hat{r})] + \frac{\hbar^2 k_{n-1}^2}{2M}\text{cotan}^2(k_0\hat{r}) \\ &= \frac{\hat{p}_r^2}{2M} + \frac{\hbar^2 k_{n-1}(k_{n-1}+k_0)}{2M}\text{cosec}^2(k_0\hat{r}) - \frac{\hbar^2 k_{n-1}^2}{2M},\end{aligned} \tag{12.145}$$

and

$$\hat{A}_n^\dagger\hat{A}_n = \frac{\hat{p}_r^2}{2M} + + \frac{\hbar^2 k_n(k_n - k_0)}{2M}\text{cosec}^2(k_0\hat{r}) - \frac{\hbar^2 k_n^2}{2M}. \tag{12.146}$$

This implies that $E_n = \frac{\hbar^2 k_n^2}{2M}$ and $k_{n-1}(k_{n-1} + k_0) = k_n(k_n - k_0)$. There are two possible solutions: either $k_n = -k_{n-1}$ or $k_n = k_{n-1} + k_0$. The former would set $\hat{A}_n = \hat{A}_{n-1}^\dagger$, which is not allowed, so the latter is the solution, and then we have $k_n = (n+1)k_0$. This then shows that $E_n = \frac{\hbar^2\pi^2(n+1)^2}{2M}$ and

$$\hat{A}_n = \frac{1}{\sqrt{2M}}\big(\hat{p}_r + i\hbar(n+1)k_0\text{cotan}(k_0\hat{r})\big). \tag{12.147}$$

Now, we construct the eigenstates. We focus first on the ground state for each auxiliary Hamiltonian. The strategy is identical to the one-dimensional case, except that we need to find the state annihilated by $\hat{p}_r + \frac{i\hbar}{\hat{r}}$. It is much easier to solve for the wavefunction by finding the state annihilated by this operator than to expand the translation operator in a power series. Motivated by our Rodrigues formula work, we transform the subsidiary condition into

$$\frac{1}{\hat{r}}\sin^{n+1}(k_0\hat{r})\left(\hat{p}_r + \frac{i\hbar}{\hat{r}}\right)\frac{1}{\sin^{n+1}(k_0\hat{r})}\hat{r}|\phi_n\rangle = 0. \tag{12.148}$$

Hence,

$$\begin{aligned}\phi_n(r) = \langle r|\phi_n\rangle &= \frac{\sin^{n+1}(k_0 r)}{r}\langle r|\frac{\hat{r}}{\sin^{n+1}(k_0\hat{r})}|\phi_0\rangle \\ &= \frac{\sin^{n+1}(k_0 r)}{r}\langle 0_r|e^{\frac{i}{\hbar}r\left(\hat{p}_r+\frac{i\hbar}{\hat{r}}\right)}\frac{\hat{r}}{\sin^{n+1}(k_0\hat{r})}|\phi_0\rangle \\ &= \frac{\sin^{n+1}(k_0 r)}{r}\langle 0_r|\frac{\hat{r}}{\sin^{n+1}(k_0\hat{r})}|\phi_0\rangle, \end{aligned} \tag{12.149}$$

where the last factor is the normalization constant (by now, you should be able to quickly recognize that it is a finite number). Using the power series for the sine function, you should be able to see that $\lim_{r\to 0}\frac{\sin(kr)}{r} = k$, by expanding just the first term in the series, and seeing all higher terms vanish. This then tells us that the ground state is

$$\psi_0(r) = \frac{\sin(k_0 r)}{k_0 r}\langle 0_r|\psi(k_0)\rangle. \tag{12.150}$$

The standard way to find the wavefunctions for the excited states is to compute

$$\psi_n(r) = c\left\langle r|\hat{A}_{n-1}^{\dagger}\hat{A}_{n-2}^{\dagger}\cdots\hat{A}_1^{\dagger}\hat{A}_0^{\dagger}|\psi_n\right\rangle, \tag{12.151}$$

with c a normalization constant. You will investigate this for some of the low-lying excited states Problem 12.5.1. It is rather tedious to proceed this way. Instead, using the single-shot factorization idea, we see that we have a solution for any k value, given by $c\frac{\sin(kr)}{kr}$, so if we choose k values such that $\psi_n(R) = 0$, and we have exactly n nodes for $0 < r < R$, then we must have the correct excited-state solution to the problem. Indeed, this is the case, and we do have that

$$\psi_n(r) = c\frac{\sin\left((n+1)k_0 r\right)}{(n+1)k_0 r}, \tag{12.152}$$

with c a normalization constant. You will verify this in Problem 12.5.3 another way for the lowest few energy eigenstates.

We still need to find the solutions for $l \neq 0$. This is done in a similar way to what we did in two dimensions. We construct the operators

$$\hat{B}_n = \frac{1}{\sqrt{2M}}\left(\hat{p}_r + \frac{i\hbar n}{\hat{r}}\right). \tag{12.153}$$

One finds that

$$\hat{B}_n^\dagger \hat{B}_n = \frac{\hat{p}_r^2}{2M} + \frac{\hbar n(n-1)}{2M\hat{r}^2} \tag{12.154}$$

and

$$\hat{B}_n \hat{B}_n^\dagger = \frac{\hat{p}_r^2}{2M} + \frac{\hbar^2 n(n+1)}{2M\hat{r}^2}. \tag{12.155}$$

This implies that $\hat{H}_l = \hat{B}_l\, \hat{B}_l^\dagger = \hat{B}_{l+1}^\dagger \hat{B}_{l+1}$ holds inside the box and the intertwining relation becomes $\hat{H}_l \hat{B}_l = B_l\, \hat{H}_{l-1}$. As before, we can use these relations to find the energy eigenstates for $l \neq 0$. We have

$$|\psi^{(l)}(k)\rangle = c\hat{B}_l\, \hat{B}_{l-1} \cdots \hat{B}_2\, \hat{B}_1 |\psi(k)\rangle, \tag{12.156}$$

where c is a normalization constant and $\hat{p}_r|\psi(k)\rangle = -i\hbar k \mathrm{cotan}(k\hat{r})|\psi(k)\rangle$. By operating $\hat{H}_l$ onto this state from the left, and using the intertwining relation, you should be able to see that it is an energy eigenstate, with an energy given by $E(k) = \frac{\hbar^2 k^2}{2M}$. Be sure to do this to verify that they are indeed energy eigenstates.

Let us take a look at what these states look like. The case with $l = 0$ was already worked out and is given by $|\psi(k)\rangle$, which satisfies the subsidiary condition and $\psi^{(0)}(r;k) = c\frac{\sin(kr)}{kr}$. Then for $l = 1$, we have

$$|\psi^{(1)}(k)\rangle = \frac{1}{\sqrt{2M}}\left(\hat{p}_r + \frac{i\hbar}{\hat{r}}\right)|\psi(k)\rangle = \frac{i\hbar k}{\sqrt{2M}}\left(\frac{1}{k\hat{r}} - \mathrm{cotan}(k\hat{r})\right)|\psi(k)\rangle. \tag{12.157}$$

Hence,

$$\psi^{(1)}(r;k) = c\frac{i\hbar k}{\sqrt{2M}}\left(\frac{\sin(kr)}{(kr)^2} - \frac{\cos(kr)}{kr}\right). \tag{12.158}$$

For $l = 2$, we find

$$\begin{aligned}
|\psi^{(2)}(k)\rangle &= \frac{1}{2M}\left(\hat{p}_r + \frac{2i\hbar}{\hat{r}}\right)\left(\hat{p}_r + \frac{i\hbar}{\hat{r}}\right)|\psi(k)\rangle \\
&= \frac{i\hbar k}{2M}\left(\hat{p}_r + \frac{2i\hbar}{\hat{r}}\right)\left(\frac{1}{k\hat{r}} - \mathrm{cotan}(k\hat{r})\right)|\psi(k)\rangle \\
&= \frac{(i\hbar k)^2}{2M}\left\{\left(\frac{2}{k\hat{r}} - \mathrm{cotan}(k\hat{r})\right)\left(\frac{1}{k\hat{r}} - \mathrm{cotan}(k\hat{r})\right)\right. \\
&\quad \left. + \frac{1}{(k\hat{r})^2} - \mathrm{cosec}^2(k\hat{r})\right\}|\psi(k)\rangle \\
&= \frac{(i\hbar k)^2}{2M}\left(\frac{3}{(k\hat{r})^2} - \frac{3}{k\hat{r}}\mathrm{cotan}(k\hat{r}) - 1\right)|\psi(k)\rangle.
\end{aligned} \tag{12.159}$$

The wavefunction becomes

$$\psi^{(2)}(r;k) = c\left(\frac{3\sin(kr)}{(kr)^3} - \frac{3\cos(kr)}{(kr)^2} - \frac{\sin(kr)}{kr}\right). \tag{12.160}$$

Of course, for each l, we must choose k so that the wavefunction vanishes at $r = R$. This can be done for the ground state in each angular-momentum sector, as well as each excited state. The functions that emerge from this procedure are well-known special functions, which are called *spherical Bessel functions*. In Problem 12.5.2, you will derive a recurrence relation for these functions, starting from $l = 0$.

Section 12.5 Problems:

12.5.1 Alternative Derivation for the $l = 0$ Wavefunctions from the Factorization Chain

In the text, we derived the auxiliary ground states and showed that the wavefunction is given by Eq. (12.151). Determine the first excited state by using the subsidiary condition of the factorization chain. Repeat for the second and third excited states. In all cases, use trigonometric identities to verify that the results are all of the generic form $\frac{\sin(kr)}{kr}$, with $k = (n+1)k_0$ for the nth excited state.

12.5.2 Recurrence Relation for the Spherical Bessel Functions

Using the state $|\psi(k)\rangle$, which satisfies $\hat{p}_r|\psi(k)\rangle = -i\hbar k\,\mathrm{cotan}(k\hat{r})|\psi(k)\rangle$, we define a function $g_l(k\hat{r})$ that satisfies

$$|\psi^{(l)}(k)\rangle = \frac{(i\hbar k)^l}{(2M)^{\frac{l}{2}}} g_l(k\hat{r})|\psi(k)\rangle. \tag{12.161}$$

In terms of the operators, we then have that

$$g_l(k\hat{r})|\psi(k)\rangle = \frac{1}{(i\hbar k)^l}\left(\hat{p}_r + \frac{i\hbar l}{\hat{r}}\right)\left(\hat{p}_r + \frac{i\hbar(l-1)}{\hat{r}}\right)\cdots\left(\hat{p}_r + \frac{i\hbar}{\hat{r}}\right)|\psi(k)\rangle. \tag{12.162}$$

The recurrence relation is found by considering the eigenstate for $l+1$ and expressing it in terms of the l result. First, verify a base case, and then show that

$$g_{l+1}(k\hat{r}) = \frac{l+1}{k\hat{r}} g_l(k\hat{r}) - \mathrm{cotan}(k\hat{r})g_l(k\hat{r}) + \frac{1}{i\hbar k}[\hat{p}_r, g_l(k\hat{r})]. \tag{12.163}$$

12.5.3 Spherical Bessel Functions for $l = 3$ and $l = 4$

Using the method shown in the text, continue to find the general results for $l = 3$ and $l = 4$.

12.5.4 Cubical Particle in a Box

We extend the results for the particle in a square box from the last subsection to a particle in a cubic box. The potential is

$$V(x, y, z) = \begin{cases} 0 & \text{for } 0 \le x \le L_x;\ 0 \le y \le L_y;\ 0 \le z \le L_z \\ \infty & \text{for other positions.} \end{cases} \tag{12.164}$$

The Hamiltonian is

$$\hat{H} = \frac{\hat{p}_x^2}{2M} + \frac{\hat{p}_y^2}{2M} + \frac{\hat{p}_z^2}{2M} + V(\hat{r}_x, \hat{r}_y, \hat{r}_z). \tag{12.165}$$

Use the factorization method to show that

$$E_{n_x,n_y,n_z} = \frac{\hbar^2\pi^2}{2M}\left(n_x^2 + n_y^2 + n_z^2\right), \tag{12.166}$$

and

$$\psi(x, y, z) \propto \sin\left(\frac{n_x\pi x}{L_x}\right)\sin\left(\frac{n_y\pi y}{L_y}\right)\sin\left(\frac{n_z\pi z}{L_z}\right). \tag{12.167}$$

12.6 Particle in a Three-dimensional Spherical Well

This is the last of our particle-in-a-box problems. The potential is now

$$V(r) = \begin{cases} -V_0 & \text{for } 0 \le r \le R \\ 0 & \text{for } r > R. \end{cases} \tag{12.168}$$

As before, we will need to construct ladder operators in both regions of r, insisting on continuity of the superpotential at $r = R$. We start by considering the $l = 0$ case.

In the region $0 \leq r \leq R$, we have the same result we used in the infinite box, namely

$$\hat{A}_0 = \frac{1}{\sqrt{2M}}\big(\hat{p}_r + i\hbar k \text{cotan}(k\hat{r})\big). \tag{12.169}$$

Then

$$\hat{A}_0^\dagger \hat{A}_0 = \frac{\hat{p}_r^2}{2M} - \frac{\hbar^2 k^2}{2M}. \tag{12.170}$$

Since $\hat{H} = \frac{\hat{p}_r^2}{2M} - V_0$, we have $\hat{A}_0^\dagger \hat{A}_0 + E_0 = \hat{H}$, which requires

$$E_0 = -V_0 + \frac{\hbar^2 k^2}{2M}. \tag{12.171}$$

In the region where $r > R$, we have $\hat{H} = \frac{\hat{p}_r^2}{2M}$. So, we choose

$$\hat{A}_0 = \frac{1}{\sqrt{2M}}(\hat{p}_r - i\hbar\kappa) \tag{12.172}$$

which gives us that

$$\hat{A}_0^\dagger \hat{A}_0 = \frac{\hat{p}_r^2}{2M} + \frac{\hbar^2 \kappa^2}{2M}. \tag{12.173}$$

This implies that $\hat{H} = \frac{\hat{p}_r^2}{2M} = \hat{A}_0^\dagger \hat{A}_0 + E_0$, with $E_0 = -\frac{\hbar^2\kappa^2}{2M}$. Continuity of $\hat{A}_0$, at $r = R$, requires

$$\frac{\kappa}{k} = -\text{cotan}(kR). \tag{12.174}$$

As before, we let

$$\phi = \sqrt{\frac{2MR^2}{\hbar^2}(E + V_0)} \text{ and } \phi_0 = \sqrt{\frac{2MR^2}{\hbar^2}V_0}. \tag{12.175}$$

The transcendental equation is transformed into

$$\sqrt{\frac{\phi_0^2}{\phi^2} - 1} = -\text{cotan}(\phi). \tag{12.176}$$

If you recall from the work on the one-dimensional case, this equation requires us to have $\phi_0 > \frac{\pi}{2}$ to have at least one solution (it corresponds to the odd solutions

for the one-dimensional problem). This has a great significance—it implies that *an attractive three-dimensional potential does not always have a bound state*. Only when $\phi_0 > \frac{\pi}{2}$, will there be at least one solution to the transcendental equation of the square-well potential in three dimensions.

The energy, expressed in terms of k, is the same as we found for the one-dimensional case. The wavefunction is different now. For $0 \leq r \leq R$, we have $\psi_0(r;k) = \langle r|\psi(k)\rangle$ is given by $\psi_0(r;k^*) = A\sin(k^*r)/(k^*r)$, with k^* a root of the transcendental equation. For $r > R$, we must first find the state annihilated by $\hat{p}_r + \frac{i\hbar}{\hat{r}}$. We find that

$$\frac{1}{\hat{r}}e^{-\kappa\hat{r}}\left(\hat{p}_r + \frac{i\hbar}{\hat{r}}\right)e^{\kappa\hat{r}}\hat{r}|\psi(k)\rangle = 0. \tag{12.177}$$

This then implies that $\psi_0(r;k^*) = Be^{-\kappa^* r}/(\kappa^* r)$ for $r > R$, where we have the transcendental equation satisfied for k^* and κ^*. The ratio of A/B is determined by requiring the wavefunction to be continuous, which says

$$\frac{A}{B} = \frac{k^*}{\kappa^*}\frac{e^{-\kappa^* R}}{\sin(k^* R)}. \tag{12.178}$$

The final constant is fixed by normalization (and requires calculus to do so). We will not determine it here.

The main takeaway from this section is that the particle in a finite square well in a spherical box requires a minimal strength of attraction to have a bound state. This was discovered by Rudolph Peierls in 1929 at the suggestion of Heisenberg. For us, this is a crucial piece of information that we will use to determine when attractive potentials have at least one bound state and when they do not. These questions will be answered in the next section. But first, we must further develop the variational method, which provides us with interesting inequalities for approximate expectation values (usually of the energy). This tool will then be used to discuss the question of whether or not an attractive potential has a bound state as a function of the spatial dimensions.

We still need to show how to find the solutions for $l \neq 0$, when they exist. For $r < R$, the procedure is identical to what we did in the previous section, except the energies are shifted down by V_0. In other words, we have a string of $\hat{B}$ lowering operators, with the appropriate subscripts applied to the $l = 0$ solution. The energy eigenstates have the same form of the energy as the $l = 0$ counterpart (but because we solve a different transcendental equation, the numerical values are different). For $r > R$, we make the same construction, with the same $\hat{B}$ operators, the only change is the subsidiary condition for the $l = 0$ solution, which changes outside the box. The wavefunctions also follow similarly. Inside the box, the algebra is identical to what we did previously. So, we find the wavefunctions are spherical Bessel functions again for $r < R$. For $r > R$, we need to work out the results. But the algebra is much simpler here.

We start with $l = 1$. For $0 \leq r \leq R$, we already know that

$$\psi^{(1)}(r; k^*) = A\left(\frac{\sin(k^*r)}{(k^*r)^2} - \frac{\cos(k^*r)}{k^*r}\right), \tag{12.179}$$

while for $r > R$, we have that the wavefunction satisfies

$$\begin{aligned}\psi^{(1)}(r; k^*) &= B\frac{\sqrt{2M}}{i\hbar\kappa^*}\langle r|\hat{B}_1|\psi(k)\rangle \\ &= B\frac{1}{i\hbar\kappa^*}\langle r|\left(\hat{p}_r + \frac{i\hbar}{\hat{r}}\right)|\psi(k)\rangle \\ &= B\left(1 + \frac{1}{\kappa^*r}\right)\frac{e^{-\kappa^*r}}{\kappa^*r},\end{aligned} \tag{12.180}$$

with a conveniently chosen amplitude B. Note that the constants k^* and κ^* do not necessarily solve the original transcendental equation. Our requirement is that the wavefunction and $\langle r|\hat{p}_r|\psi^{(1)}(k^*)\rangle$ are both continuous at $r = R$. Since we have the exact wavefunctions in terms of the parameters k^* and κ^*, we find that for continuity of the wavefunction, we require

$$\frac{A}{B} = \frac{e^{-\kappa^*R}\left(\frac{1}{\kappa^*R} + \frac{1}{(\kappa^*R)^2}\right)}{\frac{\sin(k^*R)}{(k^*R)^2} - \frac{\cos(k^*R)}{k^*R}}. \tag{12.181}$$

In Problem 12.6.2, we work out continuity of $\langle r|\hat{p}_r|\psi^{(1)}(k^*)\rangle$, which yields

$$\frac{A}{B} = \frac{\kappa^*}{k^*}\frac{e^{-\kappa^*}\left(\frac{1}{\kappa^*R} + \frac{1}{(\kappa^*R)^2} + \frac{1}{(\kappa^*R)^3}\right)}{\sin(k^*R)\left(-\frac{1}{k^*R} + \frac{1}{(k^*R)^3} - \frac{\text{cotan}(k^*R)}{(k^*R)^2}\right)}. \tag{12.182}$$

The transcendental equation is found by equating these two results. After some significant rearranging, we find it becomes

$$\frac{1}{\phi^*} - \text{cotan}\phi^* = -\frac{\phi^*}{\sqrt{\phi_0^2 - \phi^{*2}}}\left(1 + \frac{1}{\sqrt{\phi_0^2 - \phi^{*2}}}\right), \tag{12.183}$$

using the same notation that we used before (you will work this out in Problem 12.6.3). In general, the energies decrease for the finite square well versus the box, while the wavefunctions spread out a bit more.

The procedure for higher l values follows in a similar manner. We derive a recurrence relation for the solutions for higher l in Problem 12.6.4. We also illustrate how to solve them. The solutions inside the box remain spherical Bessel functions,

while the ones outside are modified spherical Bessel functions. The details become increasingly tedious, and since we do not need these results in great detail, we stop here.

Section 12.6 Problems:

12.6.1 Wavefunction for $l = 0$ and $r > R$

Verify that Eq. (12.177) holds, and show how it can be used to derive the wavefunction for $r > R$ with $l = 0$.

12.6.2 Continuity Requirements for the $l = 0$ Wavefunction

In the text, we showed in Eq. (12.178) the requirements for the wavefunction to be continuous. Using the subsidiary condition, compute $\langle r|\hat{p}_r|\psi(k)\rangle$ both for $r < R$ and for $r > R$, and find the condition, expressed as the ratio of A/B for the continuity of this term. Then, combine the two relations and show that when they both hold, we must have the transcendental equation for $l = 0$ holds. Note that you must choose the A and B coefficients as done in the text, which absorbs some constants into their definitions.

12.6.3 Calculation of $\langle r|\hat{p}_r|\psi^{(1)}(k^*)\rangle$

Compute $\langle r|\hat{p}_r\hat{B}_1|\psi(k)\rangle$ for both $r < R$ and $r > R$. Then verify both Eqs. (12.182) and (12.183). Note that some constants are absorbed into the definitions of A and B. You will need to follow the convention from the text here.
Ans. You should find

$$\langle r|\hat{p}_r\hat{B}_1|\psi(k)\rangle = \frac{(\hbar k^*)^2}{\sqrt{2M}}\left(1 - \frac{1}{(k^*r)^2} + \frac{\operatorname{cotan}(k^*r)}{k^*r}\right)\langle r|\psi(k)\rangle, \tag{12.184}$$

for $r < R$ and

$$\langle r|\hat{p}_r\hat{B}_1|\psi(k)\rangle = \frac{(\hbar\kappa^*)^2}{\sqrt{2M}}\left(\frac{1}{(\kappa^*r)^2} + \frac{1}{\kappa^*r} + 1\right)\langle r|\psi(k)\rangle, \tag{12.185}$$

for $r > R$.

12.6.4 Recurrence Relations for $l > 0$

We consider the general case for $l \geq 1$. Recall that

$$\langle r|\psi^{(l)}(k)\rangle = \langle r|\hat{B}_l\,\hat{B}_{l-1}\cdots\hat{B}_1|\psi(k)\rangle. \tag{12.186}$$

Define

$$|\psi^{(l)}(k^*)\rangle = (i\hbar k^*)^l g_l(k^*r)|\psi(k^*)\rangle \tag{12.187}$$

and show that

$$g_l(k^*\hat{r}) = \frac{1}{i\hbar k^*}[\hat{p}_r, g_{l-1}(k^*\hat{r})] + \left(-\text{cotan}(k^*\hat{r}) + \frac{l}{k\hat{r}}\right) g_{l-1}(k^*\hat{r}). \tag{12.188}$$

Work out the first three cases and show that

$$g_1(k^*\hat{r}) = -\text{cotan}(k^*\hat{r}) + \frac{1}{k^*\hat{r}}, \tag{12.189}$$

$$g_2(k^*\hat{r}) = -\frac{3}{k^*\hat{r}}\text{cotan}(k^*\hat{r}) + \left(-1 + \frac{3}{(k^*\hat{r})^2}\right), \tag{12.190}$$

and

$$g_3(k^*\hat{r}) = \left(1 - \frac{15}{(k^*\hat{r})^2}\right)\text{cotan}(k^*\hat{r}) + \left(-\frac{6}{k^*\hat{r}} + \frac{15}{(k^*\hat{r})^3}\right). \tag{12.191}$$

Show that the first three wavefunctions are

$$\psi^{(1)}(k^*r) = \frac{1}{(k^*r)^2}\sin(k^*r) - \frac{\cos(k^*r)}{k^*r} = j_1(k^*r), \tag{12.192}$$

$$\psi^{(2)}(k^*r) = \left(-\frac{1}{k^*r} + \frac{3}{(k^*r)^3}\right)\sin(k^*r) - \frac{3}{(k^*r)^2}\cos(k^*r) = j_2(k^*r), \tag{12.193}$$

and

$$\begin{aligned}\psi^{(3)}(k^*r) &= \left(-\frac{6}{(k^*r)^2} + \frac{15}{(k^*r)^4}\right)\sin(k^*r) + \left(\frac{1}{k^*r} - \frac{15}{(k^*r)^3}\right)\cos(k^*r)\\ &= j_3(k^*r).\end{aligned} \tag{12.194}$$

These wavefunctions are called *spherical Bessel functions*.

Next, we work on $r > R$. We define $|\psi^{(l)}(k)\rangle = (i\hbar\kappa^*)^l h_l(\kappa^*r)|\psi(k)\rangle$. Show that

$$h_l(\kappa^*\hat{r}) = \frac{1}{i\hbar\kappa^*}[\hat{p}_r, h_{l-1}(\kappa^*\hat{r})] + \left(1 + \frac{l}{\kappa^*\hat{r}}\right) h_{l-1}(\kappa^*\hat{r}). \tag{12.195}$$

Compute the first three functions and verify that

$$h_1(\kappa^*\hat{r}) = 1 + \frac{1}{\kappa^*\hat{r}}, \tag{12.196}$$

$$h_2(\kappa^*\hat{r}) = 1 + \frac{3}{\kappa^*\hat{r}} + \frac{3}{(\kappa^*\hat{r})^2}, \tag{12.197}$$

and

$$h_3(\kappa^*\hat{r}) = 1 + \frac{6}{\kappa^*\hat{r}} + \frac{15}{(\kappa^*\hat{r})^2} + \frac{15}{(\kappa^*\hat{r})^3}. \tag{12.198}$$

Finally, compute the first three wavefunctions to verify that

$$\psi^{(1)}(\kappa^* r) = e^{-\kappa^* r}\left(\frac{1}{\kappa^* r} + \frac{1}{(\kappa^* r)^2}\right) = \frac{2}{\pi}k_1(\kappa^* r), \tag{12.199}$$

$$\psi^{(2)}(\kappa^* r) = e^{-\kappa^* r}\left(\frac{1}{\kappa^* r} + \frac{3}{(\kappa^* r)^2} + \frac{3}{(\kappa^* r)^3}\right) = \frac{2}{\pi}k_2(\kappa^* r), \tag{12.200}$$

and

$$\psi^{(3)}(\kappa^* r) = e^{-\kappa^* r}\left(\frac{1}{\kappa^* r} + \frac{6}{(\kappa^* r)^2} + \frac{15}{(\kappa^* r)^3} + \frac{15}{(\kappa^* r)^4}\right) = \frac{2}{\pi}k_3(\kappa^* r). \tag{12.201}$$

To solve the problem for arbitrary l, one needs to continue the recurrence relations to larger l values. This is beyond what we will do here. Note further that the relation between k^* and κ^* is determined by the transcendental equation (found by applying the continuity criterion at the interface) and then the last step is to normalize the wavefunctions.

12.7 Existence of Bound States

We have solved *many* quantum problems. The variety of solutions was large—simple harmonic oscillators have an infinite number of bound states, as does the hydrogen atom (the latter also having a continuum of unbound states as well)—but some only have a finite number of bound states, like the particle in a finite square well and the Morse potential. Some might have no bound states, like the particle in a spherical square-well potential with V_0 small enough. How do we make sense

of this? Is there some underlying principle at play? The reality is, even after years of work, we cannot say *a priori* how many bound states a potential will have just from its shape. But, we can prove a powerful result about attractive potentials in one or two dimensions—they will always have at least one bound state. While in three dimensions, there is no such guarantee; that is, the potential needs to be deep enough to have at least one bound state.

The proof relies on using a powerful method called the *variational principle*. This principle has broad applications, but we will focus on only one here—estimating the ground-state energy. We have discussed this principle earlier, but here we go into more detail. To begin, we need some notation. We employ a notation inspired from the simple harmonic oscillator and denote an energy eigenstate by $|n\rangle$, such that

$$\hat{H}|n\rangle = E_n|n\rangle, \tag{12.202}$$

with E_n the corresponding energy eigenvalue of the Hamiltonian $\hat{H}$. We order the labels such that the energies are nondecreasing; that is, $E_n \geq E_m$ if $n > m$. Note that this notation is somewhat schematic, in the sense that the continuum energy eigenstates cannot be indexed by an integer. But, we do not need to go into that issue here, and it only complicates the notation. Our next step is to invoke the fact that the energy eigenfunctions of $\hat{H}$ form a complete set. This tells us any quantum state $|\psi\rangle$ can be expressed as a superposition of the states $|n\rangle$ with complex coefficients c_n. In equations, we have

$$|\psi\rangle = \sum_n c_n|n\rangle \tag{12.203}$$

as the superposition of $|\psi\rangle$ in terms of the eigenstates of the Hamiltonian $\hat{H}$ (it is here where the complexity of using a different notation for continuum and discrete energy eigenstates would need to be brought in, but as you will see, this complication won't change any of our derivations). Note that the coefficients c_n are easy to find. Since the eigenstates satisfy $\langle n|m\rangle = \delta_{mn}$, we find that

$$\langle m|\psi\rangle = \sum_n c_n\langle m|n\rangle = \sum_n c_n\delta_{mn} = c_m. \tag{12.204}$$

So, we have an explicit formula for c_m for any given $|\psi\rangle$.

We use this representation to calculate the expectation value for the energy when in the state $|\psi\rangle$, given by $E_\psi = \langle\psi|\hat{H}|\psi\rangle$. We find that

$$\begin{aligned} E_\psi &= \langle\psi|\hat{H}|\psi\rangle = \sum_m\sum_n c_m^* c_n\langle m|\hat{H}|n\rangle = \sum_m\sum_n c_m^* c_n E_n\langle m|n\rangle \\ &= \sum_m\sum_n c_m^* c_n E_n\delta_{mn} = \sum_n |c_n|^2 E_n. \end{aligned} \tag{12.205}$$

We assume $|\psi\rangle$ is a normalized quantum state, so we have that

$$\langle\psi|\psi\rangle = \sum_m \sum_n c_m^* c_n \langle m|n\rangle = \sum_m \sum_n c_m^* c_n \delta_{mn} = \sum_m |c_m|^2 = 1. \tag{12.206}$$

We want to compare the energy E_ψ to the ground-state energy E_0. Recall that $E_m \geq E_0$ for all $m > 0$ and $|c_m|^2 \geq 0$. This yields

$$E_\psi = \sum_{m=0}^{\infty} |c_m|^2 E_m \geq \sum_{m=0}^{\infty} |c_m|^2 E_0 = E_0. \tag{12.207}$$

Hence, we have found that $\langle\psi|\hat{H}|\psi\rangle \geq E_0$. This result tells us that no matter what the wavefunction $|\psi\rangle$ is, we must have E_ψ is larger than the ground-state energy. While this may not sound like an important result, it is. One way it is used is to guess approximate ground-state wavefunctions. The variational principle tells us that whatever guess $|\psi\rangle$ that gives the lowest value for E_ψ will be the best prediction of the ground-state energy.

For us, we use it in a different way. We are going to examine problems where the potential $\hat{V}$ is always less than or equal to zero, meaning

$$\langle\vec{r}|V(\hat{\vec{r}})|\vec{r}\rangle \leq 0, \tag{12.208}$$

which holds whenever $V(\vec{r}) \leq 0$. Furthermore, the potential will approach zero for large $|\vec{r}|$. In other words, we require $V(\vec{r}) \to 0$ as $|\vec{r}| \to \infty$, in any direction. A *bound state* is any energy eigenstate that has $E_n < 0$ under these conditions (bound states generally have more broad properties than just this, but this is all we need). The notion of a bound state arises because we anticipate that the state will be localized around the regions where the potential is nonzero (although we will not prove this localization here). It usually decays exponentially for large $|\vec{r}|$, as we saw with our particle-in-a-box examples, although the technical term is that it decays faster than any power. To prove that a given potential has at least one bound state, all we have to do is show there is some $|\psi\rangle$ for which $E_\psi < 0$. Then we know that $E_0 \leq E_\psi < 0$, and hence, it is a bound state. We use this criterion in our proof.

We start with the one-dimensional case. The potential we consider is nonpositive and goes to zero as $|x| \to \infty$. No matter what its shape, as long as it is piecewise continuous, then we can prove it always has at least one bound state. The strategy is to show that a square-well potential, which vanishes everywhere except for $x_0 < x < x_0 + L$ (where it is equal to $-V_0 < 0$), can be found such that $V_{\text{sq}} \geq V(x)$ for all x. An example of this is shown in Fig. 12.8. Note that this simply says we can always inscribe a square well inside any piecewise continuous nonpositive well. Given such a square well, we know that it has at least one bound state. The energy of that bound state is

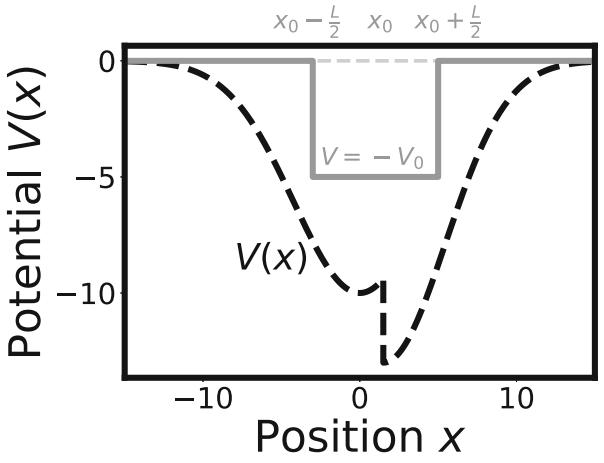

Fig. 12.8 Example that shows how to inscribe a square-well potential into a general potential. We simply require that the square-well potential always lies above the general potential (and lies below 0). Such an inscribed potential can always be found for any piecewise continuous potential

$$E_{\text{gs}}^{\text{sq}} = \langle \psi_{\text{gs}}^{\text{sq}} | \hat{H}_{\text{sq}} | \psi_{\text{gs}}^{\text{sq}} \rangle = \langle \psi_{\text{gs}}^{\text{sq}} | \tfrac{\hat{p}}{2M} + V_{\text{sq}}(\hat{x}) + \underbrace{V(\hat{x}) - V_{\text{sq}}(\hat{x})}_{\delta V(\hat{x})} | \psi_{\text{gs}}^{\text{sq}} \rangle. \tag{12.209}$$

But, since $\delta V(x) = V(x) - V_{\text{sq}}(x) \leq 0$ for all x, we have

$$\left\langle \psi_{\text{gs}}^{\text{sq}} | \delta V(\hat{x}) | \psi_{\text{gs}}^{\text{sq}} \right\rangle \leq 0. \tag{12.210}$$

Hence,

$$E_{\psi_{\text{gs}}^{\text{sq}}} = E_{\text{gs}}^{\text{sq}} + \left\langle \psi_{\text{gs}}^{\text{sq}} | \delta V(\hat{x}) | \psi_{\text{gs}}^{\text{sq}} \right\rangle \leq E_{\text{gs}}^{\text{sq}} < 0. \tag{12.211}$$

This implies that $E_{\text{gs}} \leq E_{\text{gs}}^{\text{sq}} < 0$, which implies that the Hamiltonian with potential $V(\hat{x})$ has at least one bound state! This completes the proof for the one-dimensional case.

The proof used the fact that if $\delta V(x) \leq 0$ for all x, this then implies that $\langle \psi | \delta V(\hat{x}) | \psi \rangle \leq 0$ must hold as well. This result seems quite reasonable. Stated in words, it says the average value of a nonpositive potential must be nonpositive. But, we want to establish it more clearly. If we were using calculus, the result becomes completely clear. Here, we will use an approximation to the calculus result, which should also make the result clear. We divide up position space into discrete blocks of width Δx, which we denote as $x_j = j\Delta x$, with j an integer. Then, we approximately find that

$$\langle\psi|\delta V(\hat{x})|\psi\rangle = \sum_i \sum_j \langle\psi|x_i\rangle \langle x_i|\delta V(\hat{x})|x_j\rangle \langle x_j|\psi\rangle. \tag{12.212}$$

Note that the potential is diagonal in the position basis, so that $\langle x_i|\delta V(\hat{x})|x_j\rangle = \delta_{ij}\langle x_i|\delta V(\hat{x})|x_i\rangle$, and then we have that

$$\langle\psi|\delta V(\hat{x})|\psi\rangle = \sum_i \psi^*(x_i)\langle x_i|\delta V(\hat{x})|x_i\rangle\psi(x_i) = \sum_i |\psi(x_i)|^2\langle x_i|\delta V(\hat{x})|x_i\rangle. \tag{12.213}$$

The first factor is positive and the second factor is nonpositive, so every term in the sum is nonpositive, and so is their sum. Hence, this then proves that $\langle\psi|\delta V(\hat{x})|\psi\rangle \leq 0$.

The proof for the 2d case is similar. We simply find a circular disc potential, centered about some point (x_0, y_0) in the plane and of radius R, with a potential value $-V_0$ that fits inside the attractive potential $V(x, y)$ such that

$$\langle\psi|\delta V(\hat{r}_x, \hat{r}_y)|\psi\rangle = \langle\psi|V(\hat{r}_x, \hat{r}_y) - V_{\text{disk}}(\hat{r}_x, \hat{r}_y)|\psi\rangle \leq 0. \tag{12.214}$$

Then, following the same steps as before, using the ground state of the square well on the disk, we find

$$E_{\text{gs}} \leq \left\langle\psi_{\text{gs}}^{\text{disk}}|\hat{H}|\psi_{\text{gs}}^{\text{disk}}\right\rangle = E_{\text{gs}}^{\text{disk}} + \left\langle\psi_{\text{gs}}^{\text{disk}}|\delta V(\hat{r}_x, \hat{r}_y)|\psi_{\text{gs}}^{\text{disk}}\right\rangle < 0. \tag{12.215}$$

So there is always at least one bound state.

For the three-dimensional case, we have a simple counter example. The spherical square-well potential has no bound state until the potential is sufficiently deep. This is enough to prove that one does not always have at least one bound state in three dimensions.

This then concludes the proofs. We have shown that any attractive potential in 1-d and 2-d has at least one bound state, while for the 3-d case, this is no longer guaranteed—some potentials must be strong enough to have at least one bound state. It is remarkable that such a simple analysis allows us to conclude such a wide reaching result!

Section 12.7 Problems:

12.7.1 Variational Principle for the First Excited State

Suppose we have a state $|\psi\rangle$ that is guaranteed to be perpendicular to the ground state $\langle 0|\psi\rangle = 0$. Such a condition usually arises due to symmetry, such as looking at odd states for an even potential, which has an even ground state. Use arguments similar to what we did on the variational principle to show that $E_\psi = \langle\psi|\hat{H}|\psi\rangle \geq$

E_1, which is the first-excited-state energy. In other words, E_ψ is guaranteed to be at least as large as the first excited state.

12.7.2 Variational Principle for Unnormalized States

Suppose we work with a state $|\phi\rangle$ that is *not normalized*: $\langle\phi|\phi\rangle \neq 1$. Prove that

$$E_\phi = \frac{\langle\phi|\hat{H}|\phi\rangle}{\langle\phi|\phi\rangle}. \tag{12.216}$$

12.7.3 Variational Principle Example for the Isotropic Harmonic Oscillator (Part 1)

In general, one needs to use calculus to evaluate variational energies. But we can use some of the results we have found previously to determine the variational energy of special cases. The first one we examine is using the ground state of the hydrogen atom as an approximation for the isotropic harmonic oscillator. Previously, we used the virial theorem to compute $\langle\psi_{1s}|\frac{\hat{p}^2}{2M}|\psi_{1s}\rangle$, and we also computed $\langle\psi_{1s}|\hat{r}^2|\psi_{1s}\rangle$. This allows us to compute the variational energy of the isotropic oscillator

$$E_{\psi_{1s}} = \left\langle\psi_{1s}|\frac{\hat{p}^2}{2M} + \tfrac{1}{2}k\hat{r}^2|\psi_{1s}\right\rangle. \tag{12.217}$$

Compute this variational energy exactly and verify that it is larger than the ground-state energy, which is $\frac{3}{2}\hbar\omega$.

12.7.4 Variational Principle Example for the Isotropic Harmonic Oscillator (Part 2)

Repeat the previous problem for a 2p state and compare $E_{\psi_{2p}}$ to the first excited state of the 3d simple harmonic oscillator, since $\langle\psi_{2p}|\psi_{\text{gs}}^{\text{SHO}}\rangle = 0$ by symmetry.

12.7.5 Variational Principle for the Morse Potential

Recall that the Morse potential is given by

$$V_{\text{Morse}}(\hat{x}) = D_e\left(e^{-2a\hat{x}} - 2e^{-a\hat{x}}\right). \tag{12.218}$$

We compute the variational energy for a simple harmonic oscillator ground state with Hamiltonian

$$\hat{H}_{SHO} = \frac{\hat{p}^2}{2M} + 2a^2 D_e \hat{x}^2 - D_e. \tag{12.219}$$

In other words, compute

$$E_{\psi_{\text{SHO gs}}} = \langle \psi_{\text{SHO gs}} | \hat{H}_{\text{Morse}} | \psi_{\text{SHO gs}} \rangle \tag{12.220}$$

and compare to the Morse potential ground-state energy given by

$$E_{\text{gs}}^{\text{Morse}} = a\sqrt{\frac{D_e}{M}} - \frac{\hbar^2 a^2}{4M} - D_e. \tag{12.221}$$

12.7.6 Qualitative Behavior of the Particle in a Box

Write an essay that describes what happens in a particle in a box when we make the box wider and wider and also what happens when we make it narrower and narrower. Focus on the behavior of the ground-state energy. Think about what happens as the box is made so narrow that the ground state energy is larger than twice the particle energy. At this point, one has enough energy to create particle–antiparticle pairs, which is a relativistic field theory effect. So, at some point even a particle in a box requires field theory ideas.

12.8 Uncertainty in Boxes and Other Mathematical Subtleties

One of the issues we have been discussing about particles in boxes is that some of the operators we like to work with are not self-adjoint. The biggest issue we face with such operators is that the spectral theorem does not hold—this implies that we cannot use a completeness relation to expand in terms of a sum over eigenstates—in fact, eigenstates don't even exist in some cases.

We discussed this already for the particle on a circle, where we learned that $\hat{x}$ was not self-adjoint (in fact, it was not even well-defined). Similarly, for a particle in an infinite square well with just a finite-sized box, $\hat{p}$ is not self-adjoint. Finally, for the two and three-dimensional boxes, $\hat{p}_\rho$ and $\hat{p}_r$ are not self-adjoint. We can

still work with such operators, we just have to be extra careful, as they do produce surprises. This is what we want to discuss in this section. In fact, some of these issues can be resolved algebraically, but often we need calculus to be able to handle them properly. We won't be treating any of the calculus issues here, but we will point out tips for how to identify when you might be in a situation that can lead to trouble.

Recall that the uncertainty relations involve $\Delta x_\psi = \sqrt{\langle\psi|\hat{x}^2|\psi\rangle - \langle\psi|\hat{x}|\psi\rangle^2}$ and $\Delta p_\psi = \sqrt{\langle\psi|\hat{p}^2|\psi\rangle - \langle\psi|\hat{p}|\psi\rangle^2}$. So, we need to calculate four expectation values to look at uncertainty. It will turn out that three of these are pretty easy for a particle in a box. One is a bear.

Let us start with the case where the operators $\hat{x}$ and $\hat{p}$ are both essentially self-adjoint, which is the case of a free particle on an infinite one-dimensional line. We know the eigenstates (up to an undetermined normalization constant) are $|p\rangle = e^{\frac{i}{\hbar}p\hat{x}}|0_p\rangle$. This is an eigenstate of momentum, so $\hat{p}|p\rangle = p|p\rangle$ and $\hat{p}^2|p\rangle = p^2|p\rangle$, where p is a number, not an operator. You may recall that we cautioned against working with the inner products $\langle p|p'\rangle$, because we said they are ill defined. The proper way to work with them involves working with delta functions, which we will not be doing in this book. For now, we assume that $\langle p|p\rangle = 1$ is a normalized state, which is what we would find if we discretize momentum space, as we have discussed many times before. Using that result, we immediately find $\Delta p_{|p\rangle} = 0$, which is the result we expect for an eigenstate of the operator we are computing the uncertainty for—such a case has no uncertainty. Now, what about $\Delta x_{|p\rangle}$? Here we need to invoke some calculus-like arguments in pictures. Using the translation operator, it is easy to see that we must have $\langle p|\hat{x}|p\rangle = \langle 0_p|\hat{x}|0_p\rangle$. Now, we introduce a completeness relation for the position eigenstates, which is okay because the position operator is essentially self-adjoint. Note that we think of this in terms of the discretization into small "boxes" that we can count. This gives

$$\langle p|\hat{x}|p\rangle = \langle 0_p|\hat{x}|0_p\rangle = \sum_x \langle 0_p|x\rangle x \Delta x \langle x|0_p\rangle = \sum_x |\langle 0_p|x\rangle|^2 x \Delta x, \tag{12.222}$$

where Δx is the discretization of position space. The squared matrix element is the square of the position eigenstate wavefunction evaluated in momentum space at $p = 0$. This is just a number, which is independent of x. Hence, the sum is a positive area minus an equal magnitude negative area, as illustrated in Fig. 12.9.

One can see that the sum vanishes if we evaluate the limit to $|x| \to \infty$ symmetrically to the left and to the right. This means $\langle p|\hat{x}|p\rangle = 0$. Now, in a similar manner, we can evaluate $\langle p|\hat{x}^2|p\rangle = \sum_x |\langle 0_p|x\rangle|^2 x^2 \Delta x$. As one can see from Fig. 12.10, this summation diverges and becomes infinite when $|x| \to \infty$. This implies that $\Delta x_{|p\rangle} = \infty$, which *can* be consistent with the uncertainty relation because, depending on how we evaluate the limit $0 \times \infty$ can be a finite number, which is larger than $\frac{\hbar}{2}$. To check to see whether this does actually work, we need to evaluate the uncertainties in a finite length box and take the limit as the length goes to infinity.

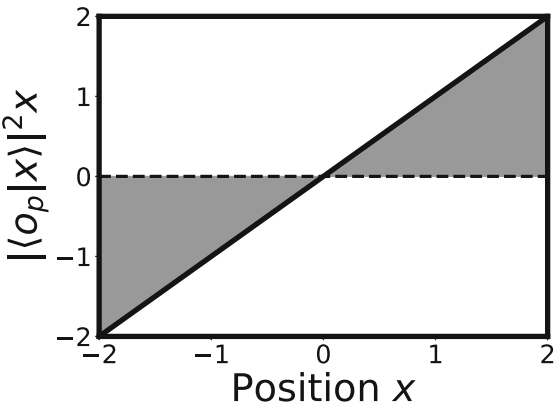

Fig. 12.9 Note how the line x has a slope of 1, so the positive area (right) is the area of the triangle, or $\frac{1}{2}x^2$, while the negative area (left) is the same magnitude. Hence, they cancel when they are added together, which is easiest to see in this visual depiction

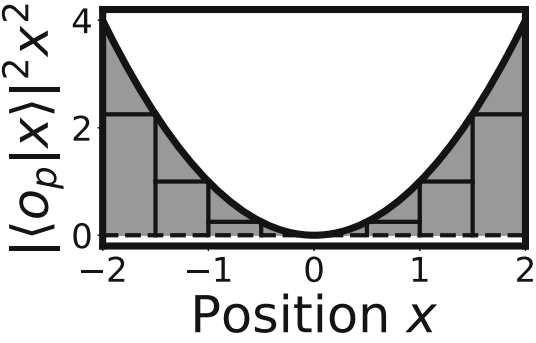

Fig. 12.10 Plot of $|\langle 0_p|x\rangle|^2 x^2$. We add up the area under the parabola (shaded region). This can be approximated by adding up the areas of the rectangles shown. It clearly diverges when we sum over all x

We can contrast the previous calculation with the calculation for a particle on a circle. Here, the eigenfunctions are of the same form, except p must be chosen to be consistent with the periodic boundary conditions. Hence, we will have $\Delta p_{|p\rangle} = 0$ again. We also have $\langle p|\hat{x}|p\rangle = 0$, if we take the circle to run from $-\frac{L}{2}$ to $\frac{L}{2}$, which is an even potential. But now $\langle p|\hat{x}^2|p\rangle$ must be *finite*, because it must be less than $\frac{L^2}{4}$. This means that on the circle, we have that $\Delta x_{|p\rangle}\Delta p_{|p\rangle} = 0$. This result is completely correct and can be traced back to the fact that the general derivation we performed for uncertainty no longer holds if $\hat{x}$ is not self-adjoint. One can correct the the uncertainty relation derivation on the circle and the corrected relation becomes $\Delta x_{|p\rangle}\Delta p_{|p\rangle} \geq 0$, which does hold here. It requires calculus to evaluate the uncertainty relation properly on a circle, so we do not go into the details here.

Our main task is to evaluate the uncertainty relation for the particle in an infinite square well potential in one dimension. We can nearly do this entirely with operators, but it does require a *small* amount of calculus. This result is simple enough that we can describe it even to people who have never seen calculus.

We will compute the uncertainty for the ground state $|\psi\rangle$, which satisfies $\hat{p}|\psi\rangle = i\hbar k \tan(k\hat{x})|\psi\rangle$ with $k = \frac{\pi}{L}$ for a box that runs from $-\frac{L}{2}$ to $\frac{L}{2}$. We will employ a number of tricks to evaluate the uncertainty algebraically. These tricks, although they appear to be "bulletproof," only hold on normalizable states that do not extend

to infinity (the so-called bound states). Since the ground state of a particle in a box (and all excited states) is normalizable, the approach we give is valid. Our first step is to use the fact that $\hat{p}^2|\psi\rangle = \hbar^2 k^2|\psi\rangle$. Hence, we have

$$\begin{aligned}\langle\psi|[\hat{H},\hat{x}]|\psi\rangle &= \frac{1}{2M}\langle\psi|[\hat{p}^2,\hat{x}]|\psi\rangle \\ &= \frac{1}{2M}\left(\langle\psi|\hat{p}^2\hat{x}|\psi\rangle - \langle\psi|\hat{x}\hat{p}^2|\psi\rangle\right) \\ &= \frac{\hbar^2k^2}{2M}\big(\langle\psi|\hat{x}|\psi\rangle - \langle\psi|\hat{x}|\psi\rangle\big) = 0. \end{aligned} \tag{12.223}$$

But, $[\hat{H},\hat{x}] = -\frac{i\hbar}{M}\hat{p}$, so we learn that $\langle\psi|\hat{p}|\psi\rangle = 0$. Note that this is true for any potential, as long as $|\psi\rangle$ is a bound state. The issue is that we need $\hat{x}|\psi\rangle$ to be well-defined and while this works for bound states, it does not work for momentum eigenstates in the infinite one-dimensional system. But it does work for all finite one-dimensional boxes.

The calculation of $\langle\psi|\hat{p}^2|\psi\rangle$ is simple, because $|\psi\rangle$ is an eigenstate of $\hat{p}^2$ (for the case when the "universe" is just the finite box). So, we have that $\langle\psi|\hat{p}^2|\psi\rangle = \hbar^2k^2$. This means that $\Delta p_{|\psi\rangle} = \hbar k = \frac{\hbar\pi}{L}$.

To calculate $\langle\psi|\hat{x}|\psi\rangle$, we use the fact that the potential is even $V(\hat{x}) = V(-\hat{x})$, and hence, the parity operator commutes with $\hat{H}$. This means that the ground state has definite parity $\hat{\Pi}|\psi\rangle = \pm|\psi\rangle$. So $\hat{\Pi}^2|\psi\rangle = |\psi\rangle$, and we compute

$$\begin{aligned}\langle\psi|\hat{x}|\psi\rangle &= \langle\psi|\hat{\Pi}^2\hat{x}\hat{\Pi}^2|\psi\rangle \\ &= \langle\psi|\hat{\Pi}\big(\hat{\Pi}\hat{x}\hat{\Pi}\big)\hat{\Pi}|\psi\rangle \\ &= (\pm1)^2\langle\psi|(-\hat{x})|\psi\rangle = -\langle\psi|\hat{x}|\psi\rangle. \end{aligned} \tag{12.224}$$

So, we conclude that $\langle\psi|\hat{x}|\psi\rangle = 0$. Note that this conclusion holds for any nondegenerate energy eigenstate in a potential that is even.

The final operator we need to evaluate is $\langle\psi|\hat{x}^2|\psi\rangle$. This is the hard expectation value to calculate. It is a challenge, but it is doable. It requires us to calculate one area (integral), and at that point, I will just tell you what it is, because it is difficult to calculate without being fluent in the calculus. But, the procedure for interrogating that matrix element with operator methods will provide us with clues about just how to tell when we have to be extra careful in our operator manipulations, because they might lead to problems. In our approach, we will need an auxiliary state $|\phi\rangle = \tan(k\hat{x})|\psi\rangle$. Note that this state does not vanish at the boundary of the box (where $x = \pm\frac{L}{2}$, because $\langle x|\psi\rangle = \sqrt{\frac{2}{L}}\cos kx$, so $\langle x|\phi\rangle = \sqrt{\frac{2}{L}}\sin kx$). Indeed, this fact is extremely important, as it will be the key to identifying the subtle issue we will encounter.

We need to know some additional properties of $|\psi\rangle$ and $|\phi\rangle$. First, from the definition, and the subsidiary condition, we immediately see that $\hat{p}|\psi\rangle =$

$i\hbar k \tan(k\hat{x})|\psi\rangle = i\hbar k|\phi\rangle$. Next, we find that

$$\begin{aligned}\hat{p}|\phi\rangle &= \hat{p}\tan(k\hat{x})|\psi\rangle = \big(\tan(k\hat{x})\hat{p} + [\hat{p}, \tan(k\hat{x})]\big)|\psi\rangle \\ &= i\hbar k\big(\tan^2(k\hat{x}) - \sec^2(k\hat{x})\big)|\psi\rangle \\ &= -i\hbar k|\psi\rangle. \end{aligned} \tag{12.225}$$

Our "clever" way to calculate $\langle\psi|\hat{x}^2|\psi\rangle$ is to start with $\langle\psi|[\hat{p}, \hat{x}^2]|\phi\rangle$. This can be evaluated explicitly using the action of $\hat{p}$ on each state to get

$$\langle\psi|[\hat{p}, \hat{x}^2]|\phi\rangle = -i\hbar k\langle\phi|\hat{x}^2|\phi\rangle + i\hbar k\langle\psi|\hat{x}^2|\psi\rangle. \tag{12.226}$$

We can also evaluate the commutator directly, to find

$$\langle\psi|[\hat{p}, \hat{x}^2]|\phi\rangle = -2i\hbar\langle\psi|\hat{x}|\phi\rangle. \tag{12.227}$$

This gives one of our identities:

$$\langle\psi|\hat{x}^2|\psi\rangle - \langle\phi|\hat{x}^2|\phi\rangle = -\frac{2}{k}\langle\psi|\hat{x}|\phi\rangle. \tag{12.228}$$

But, we also have

$$\langle\psi|\hat{x}^2|\psi\rangle + \langle\phi|\hat{x}^2|\phi\rangle = \langle\psi|\big(1+\tan^2(k\hat{x})\big)\hat{x}^2|\psi\rangle = \langle\psi|\frac{\hat{x}^2}{\cos^2(k\hat{x})}|\psi\rangle. \tag{12.229}$$

To evaluate this, we introduce a complete set of states $\hat{\mathbb{I}} = \sum_x |x\rangle\langle x|$ to obtain

$$\langle\psi|\hat{x}^2|\psi\rangle + \langle\phi|\hat{x}^2|\phi\rangle = \sum_x |\langle x|\psi\rangle|^2 \frac{x^2}{\cos^2(kx)}\Delta x = \sum_x \frac{2}{L}x^2\Delta x. \tag{12.230}$$

The summation is actually an integral, but it can be evaluated rather easily, as we show in Problem 12.8.1. The result is $\sum_x \frac{2}{L}x^2\Delta x = \frac{L^2}{6}$ (when we take the limit as $\Delta x \to 0$). We also need the result $\langle\psi|\hat{x}|\phi\rangle$ to be able to find $\langle\psi|\hat{x}^2|\psi\rangle$. This last result requires calculus to determine, and it is equal to $\frac{L}{2\pi}$ (one might think it could be calculated algebraically—Problem 12.8.2 illustrates why this does not work and also illustrates how to identify situations where the algebraic methods break down). Putting everything together, we find that

$$\langle\psi|\hat{x}^2|\psi\rangle = L^2\left(\frac{1}{12} - \frac{1}{2\pi^2}\right). \tag{12.231}$$

Hence, $\Delta x_{|\psi\rangle} = L\sqrt{\frac{1}{12} - \frac{1}{2\pi^2}}$. The uncertainty relation becomes

$$\Delta x_{|\psi\rangle}\Delta p_{|\psi\rangle} = \hbar\sqrt{\frac{\pi^2}{12} - \frac{1}{2}} \approx 0.567\hbar, \tag{12.232}$$

which is larger than $\frac{\hbar}{2}$ as it must be.

So, this example shows some of the subtleties that arise for particle-in-a-box problems, but what it may also do is make us feel uneasy. How do we know when our algebraic methods will fail? The answer lies in a careful analysis of the functions we are working with. For example, $\langle x|\psi\rangle = \sqrt{\frac{2}{L}}\cos\frac{\pi x}{L}$ is a function that *vanishes* at the end points, while $\langle x|\phi\rangle = \sqrt{\frac{2}{L}}\sin\frac{\pi x}{L}$ is a function that *does not vanish* at the endpoints. This latter function cannot be expressed in terms of the eigenfunctions for the particle in a box, since all the eigenfunctions vanish at the endpoints. We show in Problem 12.8.2 how this fact can lead to problems when calculating matrix elements of the momentum operator that mix the two functions. In calculus, we call the corrections *boundary terms*, and it is precisely these boundary terms that need to taken into account. When we work with functions that do vanish at the endpoints (like all of our energy eigenstates except those in systems with periodic boundary conditions), there never is any problem. But, if we work on a finite domain and we introduce functions that do not vanish on the endpoints, then we need to be very cautious about how we proceed.

The particle in a box has even more surprises for us. First off, as we mentioned before, $\hat{p}$ is not self-adjoint in the finite domain $-\frac{L}{2} \leq x \leq \frac{L}{2}$. There are no momentum eigenstates $|p\rangle$, and I cannot construct a momentum-space wavefunction. I can compute moments of the square of the momentum $\langle\psi\left(\hat{p}^2\right)^n|\psi\rangle$, though, since $\hat{p}^2|\psi(k)\rangle = \hbar^2k^2|\psi(k)\rangle$. Indeed, we find that $\langle\psi(k)|\left(\hat{p}^2\right)^n|\psi(k)\rangle = (\hbar k)^{2n}$. This is all correct if I restrict to the finite domain. But, what if I want to work with a momentum-space wavefunction? Well, I can think of any function living on the finite domain as also living on $-\infty < x < \infty$. In this infinite domain, $\hat{p}$ is an essentially self-adjoint operator and $|p\rangle$ exists. Hence, one can compute $\langle p|\psi(k)\rangle$. The calculation requires calculus and yields

$$\psi(p) \propto \frac{\cos\frac{pL}{2\hbar}}{1 - \left(\frac{p}{\pi\hbar}\right)^2}. \tag{12.233}$$

This wavefunction has an expectation value of $\hat{p}^2$, which is equal to $(\hbar k)^2$ as we expect, but *all higher moments are infinite*! In addition, If we multiply the wavefunction by p^2 it does not give the eigenvalue $(\hbar k)^2$ multiplied by the wavefunction, so this wavefunction is not an eigenvector under $\hat{p}^2$. This is because the Hamiltonian is not just $\frac{\hat{p}^2}{2M}$, it has an infinite potential outside the box, so the energy eigenstates are eigenstates under $\hat{H}$, not $\hat{p}^2$.

So what is right? You may be shocked that the answer is both! If we restrict to the finite domain, all moments of $\hat{p}^2$ are finite and given by $(\hbar k)^{2n}$. If, we instead

consider the box to live in the infinite domain, then $\langle\psi(k)|(\hat{p}^2)^n|\psi(k)\rangle = \infty$ for $n \geq 2$. So, the choice of the domain, and how we work with the functions plays a critical role in determining the behavior of the system. This is the price we pay when working with non-self-adjoint operators.

So, how do we proceed without running into trouble? First off, if the operators you are working with are self-adjoint, or essentially self-adjoint, there is no problem, as long as the operators acting on the states are well defined. Second, if you try to work with states that have the wrong boundary conditions, like $\tan(k\hat{x})|\psi\rangle = |\phi\rangle$, beware! Often you will find identities won't work that you think should because they have corrections due to boundary terms. Third, never assume you can expand in terms of an eigenbasis of a non-self-adjoint operator. This is guaranteed to go poorly.

Rest assured, these subtleties do not occur for "nice" systems like the simple harmonic oscillator or the hydrogen atom. It is usually the particle-in-a-box problems that have issues. This is why they are not as simple as other texts might lead you to believe. Also note that when you do learn calculus and explore other quantum resources, you will encounter something called a Dirac delta function. This object, when manipulated properly, can treat many of these issues. But, it is easy to use it improperly, so again, caution and care are needed with these types of problems.

We end with one final comment. Throughout the work we have done in this book, we always assumed there was a state that existed, from which we built the rest of our states. For position and momentum states, we assumed there was a state at the origin. For the factorization method, we assumed there was a state that satisfied each subsidiary condition. Showing that such states exist, rather than assuming, as we did, that they exist, is a quite complex mathematics problem. It requires delving deeply into real analysis and Hilbert space theory. Obviously, we will not do that here. But, I mention it, simply to let you know that there are subtleties associated with these assumptions too. So, one needs to be careful, sometimes with them.

Section 12.8 Problems:

12.8.1 A Poor-man's Approach to Finding the Area Under a Curve

In the main text, we gave a brief description of how one can calculate the signed area under a curve. Here we work on three concrete examples. Technically speaking, what you are about to see is calculus, but you will see it in a simplified way, which allows us to establish the few critical results we need for the discussions in the text.

(a) In Fig. 12.11, we show how to determine the area under a constant curve, by breaking it up into N pieces. Verify the results in the figure for the area of the rectangle, that is, show that the two results are the same.

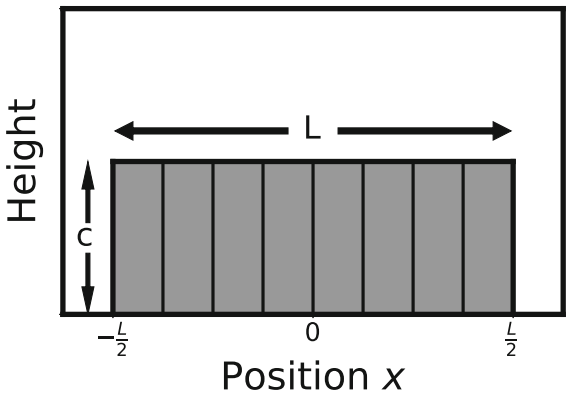

Fig. 12.11 We show a rectangle of length L and height c, corresponding to the function $f(x) = c$ for $-\frac{L}{2} \leq x \leq \frac{L}{2}$. We split it up into N smaller rectangles, each of width $\Delta x = \frac{L}{N}$. Now, the area is $N \times c \times \Delta x = Lc$, giving the same result

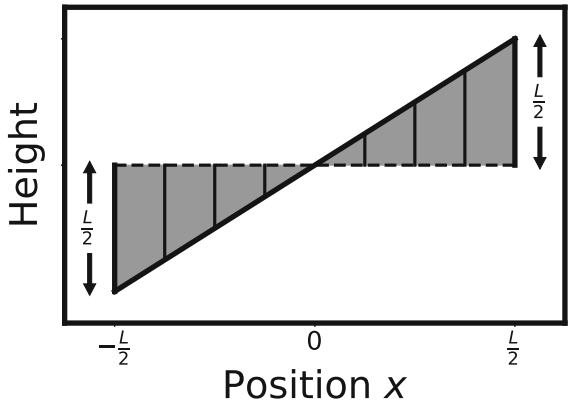

Fig. 12.12 In this figure, we apply the same procedure to a linear function, which looks like two triangles (one with positive area, the other with negative area). We see the two triangles, with each triangle having an area given by $\frac{1}{2}$ base $\times$ height. The triangle on the left side lies below the x-axis, so we say its area is negative (because the function value is negative). One can see the sum of the two areas is zero, because the triangles have the same area, but opposite signs. Dividing the system into N pieces yields the same areas

(b) Using Fig. 12.12 as a guide, compute the area for the left and right triangles in the left panel and show that they give exactly the same result using the smaller N areas on the left. You should find the magnitude of the area of each triangle is $\frac{L^2}{8}$.

(c) We now examine a case that can only be summed by making the little rectangles—the case of $f(x) = x^2$. We focus on one half of the area, ranging from $x = 0$ to $x = \frac{L}{2}$. The area of the piece running from $-\frac{L}{2}$ to 0 is exactly the same. We label each point with an integer. At point j, we have $x_j = j\Delta x$ and $f(x_j) = (j\Delta x)^2$. We approximate the area by two rectangles. One that goes through the leftmost point and one that goes through the rightmost point of the function for each narrow "rectangle." Clearly, the total area is larger than the

sum of the smaller rectangles and smaller than the sum of the larger rectangles. So, using the so-called squeeze theorem, we have

$$\sum_{j=l}^{N}(j-1)^2(\Delta x)^3 \leq \text{ area } \leq \sum_{j=1}^{N} j^2(\Delta x)^3. \tag{12.234}$$

To complete the calculation, we need to compute the sum of the squares of the first N counting numbers. We do this in stages. First sum N copies of the number 1:

$$\sum_{j=1}^{N} 1 = \underbrace{1+1+1+\cdots+1}_{N \text{ terms}} = N. \tag{12.235}$$

Next, we want to sum the first N counting numbers $\sum_{j=1}^{N} j = 1+2+3+\cdots+N$. We do this by a neat trick. First, we note that

$$\sum_{j=1}^{N}\left\{(j+1)^2 - j^2\right\} = 2^2 - 1^2 + 3^2 - 2^2 + \cdots + (N+1)^2 - N^2. \tag{12.236}$$

Show that many terms cancel, and we are left with the sum equalling $(N+1)^2 - 1 = N^2 + 2N = N(N+2)$. But, if we expand out the summand, we find the j^2 terms cancel and we have

$$\sum_{j=1}^{N}\left\{j^2 + 2j + 1 - j^2\right\} = \sum_{j=1}^{N}(2j+1) = 2\sum_{j=1}^{N} j + \sum_{j=1}^{N} 1. \tag{12.237}$$

We already worked out the last sum, while the first sum is twice what we want. Show that

$$\sum_{j=1}^{N} j = \frac{N(N+1)}{2}. \tag{12.238}$$

Use a similar argument with $\sum_{j=1}^{N}\left\{(j+1)^3 - j^3\right\} = (N+1)^3 - 1$ to show that

$$\sum_{j=1}^{N} j^2 = \frac{(2N+1)N(N+1)}{6}. \tag{12.239}$$

(d) We now put this together with the previous work to compute the area. Show that

$$\sum_{j=1}^{N}(j-1)^2(\Delta x)^3 = \frac{(2N-1)(N-1)N}{6}(\Delta x)^3 = \frac{L^3}{24}\left(1-\frac{3}{2N}+\frac{1}{2N^2}\right), \tag{12.240}$$

and

$$\sum_{j=1}^{N} j^2(\Delta x)^3 = \frac{(2N+1)N(N+1)}{6}(\Delta x)^3 = \frac{L^3}{24}\left(1+\frac{3}{2N}+\frac{1}{2N^2}\right). \tag{12.241}$$

(e) Now comes the part that often causes trouble for people the first time they see it. We are going to take the limit as $N \to \infty$. Since $\frac{1}{2N^2} \to 0$ and $\frac{3}{2N} \to 0$ in this limit, we find that the area is given by $\frac{L^3}{24}$ by the squeeze theorem. We multiply by 2 to get the sum of the area of the two curves. This gives the answer in the text. Please carefully verify all of these statements.

Congratulations, you have evaluated your first nontrivial integral!

12.8.2 Subtleties Involving $\langle\psi|\hat{x}|\phi\rangle$

If we introduce the complete set of position states (with some spread Δx), we have

$$\langle\psi|\hat{x}|\phi\rangle = \sum_x \sum_{x'} \langle\psi|x\rangle\,\langle x|\hat{x}|x'\rangle\,\langle x'|\phi\rangle = \frac{2}{L}\sum_x \cos(kx)x\,\sin(kx)\Delta x, \tag{12.242}$$

because the matrix element $\langle x|\hat{x}|x'\rangle = x\Delta x$ if $x = x'$ and vanishes otherwise. We do not have any simple way to compute this sum, but we can note that because cosine is an even function and x and sine are odd functions, the summation is an even function. In addition, one can easily see that for the ground state, every term in the sum is nonnegative. So the matrix element must be positive.

However, recalling the $\hat{p}^2|\psi\rangle = \hbar^2k^2$, we find that

$$\begin{aligned}\langle\psi|\hat{x}|\phi\rangle &= \frac{1}{\hbar^2k^2}\langle\psi|\hat{p}^2\hat{x}|\phi\rangle \\ &= \frac{1}{\hbar^2k^2}\langle\psi|\hat{p}\hat{x}\,\hat{p}|\phi\rangle + \frac{1}{\hbar^2k^2}\langle\psi|\hat{p}[\hat{p},\hat{x}]|\phi\rangle \\ &= -\langle\phi|\hat{x}|\psi\rangle - \frac{\hbar^2k}{\hbar^2k^2}\langle\psi|\psi\rangle.\end{aligned} \tag{12.243}$$

On the first line, we use the fact that $\langle\psi|$ is an eigenstate of $\hat{p}^2$. In the second line, we move one momentum operator factor to the right past the position operator (and

add in the correction for doing so). In the third line, we recall what happens when the momentum operator acts on $\langle\psi|$ and $|\phi\rangle$, and substitute in. Finally, we note that the explicit form as a sum tells us that $\langle\psi|\hat{x}|\phi\rangle$ is *real*, so we find that

$$\langle\psi|\hat{x}|\phi\rangle = -\frac{1}{2k} < 0. \tag{12.244}$$

This must have an error, since we established before that the matrix element must be positive. See if you can find where the error is.

Hint: boundary terms arise when we change from a momentum operator acting to the left to one acting to the right when the functions do not vanish at the boundary. But the algebraic method cannot account for these so-called boundary terms.

12.8.3 Extensions to Radial Momentum

This question is quite exploratory. It asks you to consider whether we can use similar methods to determine uncertainty relations for the radial momentum and the radius in 2d and 3d? Do you need to have some integrals evaluated to do this? Are there subtleties with the evaluation of some of the matrix elements using algebraic methods? Note that this is more of a project than a homework problem.

12.8.4 Uncertainty on a Circle

We already saw on the circle that the uncertainty principle was apparently violated, because energy eigenstates can be written as eigenstates of $\hat{p}$ as well, so that $\Delta p_{\psi_p} = 0$ for such states. We will call these momentum states ψ_p. Since $\Delta x_{\psi_p} \neq \infty$ for those states (because the circle is finite in size), we have $\Delta x_{\psi_p} \Delta p_{\psi_p} = 0$, which is not larger than $\frac{\hbar}{2}$. To trace the error, first evaluate

$$\langle\psi_p|[\hat{x}, \hat{p}]|\psi_p\rangle, \tag{12.245}$$

by evaluating the commutator to show it equals $i\hbar$. This is originally how we did the calculation in Chap. 8. Now, evaluate it directly via

$$\langle\psi_p|\hat{x}\hat{p}|\psi_p\rangle - \langle\psi_p|\hat{p}\hat{x}|\psi_p\rangle, \tag{12.246}$$

and using the fact that ψ_p is an eigenstate of the momentum operator. You should find that evaluated this way, we have that $\langle\psi_p|[\hat{x}, \hat{p}]|\psi_p\rangle = 0$. It turns out this latter calculation is the one that is correct. If we use it to compute the uncertainty relation, then we find the uncertainty relation on the circle is

$$\Delta x_\psi \Delta x_\psi \geq 0, \tag{12.247}$$

which does not provide any lower limit!

Ok, this is all well and good, but how do we know how to proceed? If both operators are self-adjoint, or essentially self-adjoint, there never is any ambiguity, because all ways of calculating the expectation value will agree. If they are not well-behaved operators, then we need to resort back to calculus so we can properly treat the boundary terms. Fortunately, these subtle issues only enter quite rarely in quantum mechanics (and often are glossed over). So, in general, you do not need to worry about them most of the time. At least now you know they do exist and have been given a sense of how to detect them, even if we do not provide you with an ironclad way to know how to do so. It just should be motivation for you to develop more mathematical skills to study quantum mechanics using calculus as well.

12.9 Why Are Many Eigenvalues Quantized (While Some Are Not)?

One of the defining characteristics of quantum mechanics is the phenomenon of *quantization*. This is the fact that certain observables can only be measured with discrete values. We have seen this in a number of different examples through the book already. We have seen enough examples that it makes sense to discuss the origin of this behavior, as seen from an operator mechanics perspective. As with many complex phenomena, there is more than one way in which it can occur.

The first way that we can have quantization arises from symmetry. The best example of this is angular momentum, which was treated in Chap. 9. Angular momentum is related to rotational symmetry. Indeed, we have already seen that the rotation operators are all generated from exponentiating the angular momentum operators (multiplied by the rotation angles). This is the classic way that a Lie group is generated by its corresponding Lie algebra. We do not go into great detail about Lie groups here, because the topic is too advanced for this work (see Corry (2017); Adams (1994) for an interesting perspective on this). But the point we need to keep in mind is that the Lie algebra operators have raising and lowering operators in their algebra—in this case, they are $\hat{L}_\pm = \hat{L}_x \pm i\hat{L}_y$. Because the commutation relation is $[\hat{L}_z, \hat{L}_\pm] = \pm\hbar\hat{L}_z$, the raising and lowering operators change the $\hat{L}_z$ eigenvalue by $\pm\hbar$, respectively. Hence, the $\hat{L}_z$ eigenvalues are discrete and shift by integer multiples of $\hbar$. This is a consequence of the symmetry properties of the Lie algebra. All Lie algebras have raising and lowering operators, which require discrete quantum numbers, just like $\hat{L}_z$. This is how symmetry can lead to discrete eigenvalues and to quantization.

To be more concrete, recall that we were able to both derive that the eigenvalues of $\hat{L}_z$ were equal to integers or half-odd integers (multiplied by $\hbar$) and that the eigenvalue of $\hat{\vec{L}}^2$ was $\hbar^2 l(l+1)$, with l an integer, or half-odd integer. All of

this followed straightforwardly from the angular momentum commutator algebra $[\hat{L}_i, \hat{L}_j] = i\hbar \sum_k \epsilon_{ijk} \hat{L}_k$. This is precisely how it works in all cases where quantization arises from symmetry. The quantization is forced upon us by the consistency relations of the algebra of the operators. The hard part is figuring out the details behind precisely how this works for each case.

The next way that quantization occurs is when raising and lowering operators have constraints. Let's start by recalling how we found the eigenstates for the free particle, which satisfied $|p\rangle = \exp(-\frac{i}{\hbar} p\hat{x})|0_p\rangle$. In this case, we think of $\frac{\hat{p}}{\sqrt{2M}}$ as the raising and lowering operator of the free-particle Hamiltonian $\hat{H} = \hat{p}^2/2M$, and we constructed each eigenstate of $\hat{H}$ by finding the eigenfunctions of the operator $\hat{p}$. Aside from the requirement that the state $|0_p\rangle$ be normalizable in some mathematical sense (which we did not go into details on, but were warned that it involves much mathematical subtlety), there are no other constraints on this system. Hence, we find a continuum of allowed momentum states, and there is no momentum (or energy) quantization.

Contrast this to the case of a particle in a box, where we had $\hat{A} = \frac{1}{\sqrt{2M}}[\hat{p} - i\hbar k \tan(k\hat{x})]$. Here again, we could find eigenvalues and eigenvectors for any possible k value, but we required that $\hat{A}$ diverged as $\hat{x}$ approaches the boundary of the box, which led us to the condition $k = \frac{(2n+1)\pi}{L}$. It was this extra boundary condition constraint that required k values to be chosen from a discrete set of allowed values. This then led to a discrete set of allowed energies and to energy quantization. This is the most common way that quantization is taught. From a differential equation perspective, it is always a boundary condition, often tied to the requirement of normalizability, because the wave function must go to zero at infinity to be normalizable (in a crude heuristic sense; mathematicians make this notion very precise, and it is a little different than just vanishing at infinity). From the operator perspective, it is a constraint on the ladder operators—for the particle on a circle, we required the superpotential to be periodic or antiperiodic; for the particle in a box, we required the superpotential to diverge at the boundary; for the particle in a finite square well, we required the superpotential to be continuous. In all cases, we provided a specific constraint. This then led to only discrete allowed parameters for the valid raising and lowering operators and finally led to discrete energies.

The way to see these boundary conditions in pictures from the wavefunction perspective is that the wavelengths are selected so that the wavefunctions "fit" inside the boundary. We have seen this numerous times before and comment on this now.

The best place to start is for the particle in a box. This is plotted in Fig. 12.1. One can see that the requirement that the wavefunction vanishes at the edges of the box shows that only specific values of k produce a wave with exactly the right wavelength to vanish at both ends. Similarly, in Fig. 7.4, we plot the wavefunctions for the particle on a circle—there the wavefunction is continuous at the edges (because we use periodic boundary conditions, and the left edge is wrapped to meet the right edge). Now moving to the particle in a finite square well in Fig. 12.2, we see the wavelength is slightly longer than that of the box, and we have a smooth

transition to the exponentially decaying part of the wavefunction. This is so it can "look" slightly outside the box, and we have a smooth transition to the exponentially decaying part of the wavefunction. This is how we have a boundary condition of the wavefunction approaching zero at infinity. One can see similar behavior for the simple harmonic oscillator in Fig. 7.10 and the hydrogen atom in Figs. 11.13, 11.14, and 11.15. Before ending this discussion from the position-space viewpoint, we want to make one more comment. It is often stated in many textbooks that we can understand "orbits" in hydrogen by having a wavelength that fits around the orbit (which is usually displayed as a circular orbit). I do not recommend espousing this picture, as it brings us too close to the incorrect "solar system analog" of hydrogen.

Is this the only way it can happen? No! There is one more way, which we have employed many times when solving problems using Schrödinger's factorization method—it is the case where we require a string of operators acting on an auxiliary ground state to create an excited state. Recall the general Schrödinger procedure—we form the nth eigenstate $|\psi_n\rangle$ via a product of $n-1$ (usually different) raising operators acting on the ground state $|\psi_n\rangle \propto \hat{A}_n^\dagger \hat{A}_{n-1}^\dagger \cdots \hat{A}_0^\dagger |\psi_{\text{gs}}^{\text{aux}}\rangle$. Then the intertwining relation shows that the energy of the excited state is the same as that of the auxiliary ground state. But recall we have two different relations that define the auxiliary Hamiltonian, one with the raising operator first and one with it last. Using nonegativity of the norm, we then showed that the energy of each auxiliary Hamiltonian forms an increasing chain of energies. But, such a chain requires a discrete energy increase from one auxiliary Hamiltonian to the next and, hence, from one excited state to the next. So, factorization chains always have discrete increases in energy.

This ends our summary on quantization. It is one of the most important aspects of quantum mechanics. While it is not due to just one cause, the behavior is quite ubiquitous and arises in many different ways. As with everything in this operator mechanics point of view, the behavior can always be traced back to the fact that the position and momentum operators do not commute—essentially all bound states we found have their behavior result from the fact that $[\hat{x}, \hat{p}] = i\hbar$.

Before we leave the topic, it is useful to discuss briefly what happens when eigenvalues are not quantized, but are continuous. As we showed before, we usually handle such cases using the single-shot factorization method (or some other means to find the eigenstates) where we do not impose any additional constraints. Because the eigenvalues are continuous, we do not expect there to be a Schrödinger factorization chain. But, the free particle in one dimension does have one, which we constructed earlier in this chapter. The key reason why this is possible is that the nodes of the wavefunctions are evenly spaced. Hence, when we work with a larger k value, we can exactly match the nodes for a smaller k value. These are the wavevectors that we will see occurring in each factorization chain. So, the chain brings in quantized energy eigenvalues, but because we can form an infinite number of these chains, we will have a continuum of energy eigenvalues. Weird. Why does it not work if the nodes are not evenly spaced? Because the superpotential has singularities in them and this makes the system act like a set of independent problems in each domain (between the singularities). If the superpotential is not the

same in each domain, then the energy levels need not match in different domains. This means we do not have an energy eigenstate over all space. When evenly spaced, the superpotential identically repeats in each domain. Then the energies for solving the problem in each domain must be exactly equal, and we have an energy eigenstate over all space. It is very odd. Udiprod has made a wonderful video https://tinyurl.com/26j2azj4 visualizing many ideas of measurement in one-dimensional problems, and this seems like a good place to show it.

This section was about synthesizing the ideas behind quantization. We have no problems at the end of the section, but instead provide a few essay topics to help you carefully think through these ideas and recognize why quantization occurs. Reviewing some of the examples we already solved should greatly help you in organizing your thoughts. After completing this, you are ready for the third part of the book, which focuses on advanced topics. Just like the second part, it will begin with a math interlude. This time, most of the material will be new to you, but it is critical to familiarize yourself with this to be able to move forward. Many exciting and new quantum ideas follow afterward. Be prepared. The journey is about to go to an even higher level.

Section 12.9 Problems:

12.9.1 Quantization from Symmetry

Write an essay explaining how quantitzation arises due to symmetry using an example from the problems we solved already of this form.

12.9.2 Quantization from the Canonical Commutation Relation

Write an essay that carefully describes how the canonical commutation relation leads to quantization when we have a factorization chain. Be sure to discuss nonsingular and singular factorization chains.

12.9.3 Continuum States

Describe how systems end up with continuum eigenvalues. Be sure to use examples in your discussion. Explain whether the existence of a Schrödinger factorization chain is possible in the continuum.

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Part III
Advanced Quantum Operator Methods

Chapter 13
Advanced Matrix Properties

When the subject is strong, simplicity is the only way to treat it.

Jacob Lawrence, Artist

As we move into more advanced properties of quantum mechanics, we need to extend our understanding of matrices from 2×2 matrices to arbitrary dimension matrices. The properties are quite similar, but there are a number of important ideas that need to be mastered to handle these more general cases. In most circumstances we will be working with these advanced concepts and illustrating them, but we will not be proving them to you in the most general case. Doing so would take us too far from our main task. Instead, we focus on your being able to understand and to work with these concepts. We will also discuss two other advanced mathematics ideas related to summations and counting in the final sections of the chapter. Preview all of this content at https://tinyurl.com/ymv97c34. In addition, you can also see a summary of Part III on Advanced Quantum Operator Methods at https://bit.ly/4rRxn9l.

13.1 Matrix Arithmetic

Arbitrary dimension matrices are similar to 2×2 matrices. They are collections of columns (or rows) organized into an array. We will focus on working with *square matrices* of dimension n, which we call $n \times n$ matrices. These consist of a table of n columns of entries, each column containing n elements. One can similarly think of it as a collection of n rows of entries, each row containing n elements. We denote the entries as A_{ij} with the first index i denoting the row and the second index j denoting the column. Hence, we have an $n \times n$ matrix takes the following form:

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$$A = \begin{pmatrix} A_{11} & A_{12} & A_{13} & \cdots & A_{1n-1} & A_{1n} \\ A_{21} & A_{22} & A_{23} & \cdots & A_{2n-1} & A_{2n} \\ A_{31} & A_{32} & A_{33} & \cdots & A_{3n-1} & A_{3n} \\ \vdots & \vdots & \vdots & \cdots & \vdots & \vdots \\ A_{n1} & A_{n2} & A_{n3} & \cdots & A_{nn-1} & A_{nn} \end{pmatrix}. \tag{13.1}$$

In order to work with these matrices, we need to understand the rules we need to use for arithmetic, corresponding to addition and multiplication. Because we can multiply a matrix by a scalar number or by another matrix, there are a few different options we need to cover. We begin with addition, which is quite simple to understand—we just add the matrix elements entrywise. The only requirement is that the shape of the matrices must be the same. So, if we add two $n \times n$ matrices together, we find that

$$A + B = \begin{pmatrix} A_{11} + B_{11} & A_{12} + B_{12} & A_{13} + B_{13} & \cdots & A_{1n-1} + B_{1n-1} & A_{1n} + B_{1n} \\ A_{21} + B_{21} & A_{22} + B_{22} & A_{23} + B_{23} & \cdots & A_{2n-1} + B_{2n-1} & A_{2n} + B_{2n} \\ A_{31} + B_{31} & A_{32} + B_{32} & A_{33} + B_{33} & \cdots & A_{3n-1} + B_{3n-1} & A_{3n} + B_{3n} \\ \vdots & \vdots & \vdots & \cdots & \vdots & \vdots \\ A_{n1} + B_{n1} & A_{n2} + B_{n2} & A_{n3} + B_{n3} & \cdots & A_{nn-1} + B_{nn-1} & A_{nn} + B_{nn} \end{pmatrix}, \tag{13.2}$$

or $(A + B)_{ij} = A_{ij} + B_{ij}$. One can immediately see that adding two matrices is commutative, in that $A + B = B + A$. Multiplying a matrix by a scalar λ is also simple. We just multiply each matrix element by the same scalar, so λA has matrix elements given by λA_{ij}. Clearly multiplication by a scalar is also commutative: $\lambda A = A\lambda$.

Matrix multiplication is more complicated. It is interesting to note that in 1925, when Heisenberg discovered matrix mechanics, he also rediscovered the matrix multiplication rule! He did not even know that what he was working with were called matrices—instead, he called them tableau. As we noted before, if $C = AB$, with all matrices $n \times n$ matrices, then the product matrix element C_{ij} is found by computing the dot product between the vector in the ith row of A, with the jth column of B. In equations, this becomes

$$C_{ij} = (AB)_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj}. \tag{13.3}$$

Matrix multiplication is associative $(AB)C = A(BC)$ and distributive $(A + B)C = AC + BC$, as can be verified by writing out the matrix elements for the terms on each side of the equation and comparing them. But, in general, it is not commutative. For we have $(AB)_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj}$, which need not be equal to $(BA)_{ij} = \sum_{k=1}^{n} B_{ik} A_{kj}$. Some matrices will commute with each other, but this is a special case, not a general result. So, you should always assume that $AB \neq BA$,

unless you know you are in one of these special cases. For example, a diagonal matrix D is a matrix where $D_{ij} = 0$ if $i \neq j$. Two diagonal matrices D and D' do commute under matrix multiplication, because DD' is also diagonal and $(DD')_{ii} = D_{ii}D'_{ii} = (D'D)_{ii}$.

It is easy to verify that the *identity matrix*, which is a diagonal matrix with 1's on each diagonal matrix element, is the multiplicative identity. We have $\mathbb{I}_{ij} = \delta_{ij}$, so that $\mathbb{I}A = A\mathbb{I} = A$. Be sure to verify these results yourself. Since we have a multiplicative identity, we can ask about the inverse of a matrix. Indeed, the matrix inverse A^{-1} satisfies both $AA^{-1} = \mathbb{I}$ and $A^{-1}A = \mathbb{I}$. You will prove that a left inverse implies a right inverse in Problem 13.1.3. Determining the inverse matrix of a general matrix is a difficult problem to solve. If the matrix is invertible, precise algorithms exist to carry out the matrix inversion, but they are extremely tedious to carry out "by hand." We will not develop the algorithm here, we will just use the fact that it can be done.

There is a deep relationship between operators and their matrix representations, which can be shown most clearly when we use Dirac notation. If we work in a specific basis $\{|\psi_i\rangle\}$ for our vector space, then an operator $\hat{A}$ has a matrix representation given by

$$\hat{A} \leftrightarrow A_{ij} = \langle\psi_i|\hat{A}|\psi_j\rangle. \tag{13.4}$$

Recall the way that we compute this is to pick a basis vector $|\psi_j\rangle$ and operate the operator $\hat{A}$ onto the vector (which produces another vector, since operators map vectors to vectors). We then expand this mapped vector in terms of our basis vectors, with the coefficients becoming the matrix elements. Using Dirac notation, we have an interesting new way to think of matrix multiplication by using the completeness relation. In particular, if we are interested in the matrix of the product of two operators $(AB)_{ij} = \langle\psi_i|\hat{A}\hat{B}|\psi_j\rangle$, then by introducing the completeness relation in between the two operators, we "derive" the matrix multiplication formula. In equations, we have

$$(AB)_{ij} = \langle\psi_i|\hat{A}\hat{B}|\psi_j\rangle = \sum_k \langle\psi_i|\hat{A}|\psi_k\rangle\,\langle\psi_k|\hat{B}|\psi_j\rangle = \sum_k A_{ik}B_{kj}. \tag{13.5}$$

There are a number of special matrices. We summarize three of them here. The first one is a *Hermitian* matrix. It satisfies $A_{ij} = A^*_{ji}$, where the $*$ symbol is complex conjugation. Not too surprisingly, Hermitian operators are represented by Hermitian matrices. Next, we have *unitary* matrices. These are matrices whose Hermitian conjugate is equal to their inverse. In other words, $U^{-1}_{ij} = U^*_{ji}$. The Hermitian conjugate is the matrix transpose (interchanging the order of the indices of the matrix) followed by a complex conjugation. The third special matrix we have is an *orthogonal* matrix, which is a real-valued unitary matrix. It has its inverse equal to its matrix transpose.

The matrices associated with operators are expressed in a specific basis. One of the most important processes to understand is what happens to the matrices if we change the basis. We will describe this procedure first within the Dirac notation and then from that extract the matrix expressions. Assume we start in a basis given by $\{|\psi_i\rangle\}$ and we wish to work instead in the basis $\{|\phi_i\rangle\}$. Then we can use the completeness relation to relate the two and define a unitary matrix that connects the two bases, as follows:

$$|\phi_i\rangle = \sum_j |\psi_j\rangle \underbrace{\langle\psi_j|\phi_i\rangle}_{\text{numbers}} = \sum_j |\psi_j\rangle U^*_{ij} = \sum_j |\psi_j\rangle U^\dagger_{ji}, \tag{13.6}$$

where we defined the unitary matrix $U_{ij} = \langle\phi_i|\psi_j\rangle$. We can verify that the matrix is unitary by multiplying $UU^\dagger$ via

$$(UU^\dagger)_{ij} = \sum_k U_{ik}U^\dagger_{kj} = \sum_k \langle\phi_i|\psi_k\rangle\langle\psi_k|\phi_j\rangle = \langle\phi_i|\phi_j\rangle = \delta_{ij}, \tag{13.7}$$

where we assume that the bases we are working with are orthonormal bases and we used the completeness relation to get rid of the summation. With the transformation matrix in hand, we can compute the transformation of the matrix by using completeness twice

$$A^\phi_{ij} = \langle\phi_i|\hat{A}|\phi_j\rangle = \sum_{kk'} \langle\phi_i|\psi_k\rangle\langle\psi_k|\hat{A}|\psi_{k'}\rangle\langle\psi_{k'}|\phi_j\rangle = \sum_{kk'} U_{ik}A^\psi_{kk'}U^\dagger_{k'i}. \tag{13.8}$$

Hence the relationship between the matrix in the ϕ representation and in the ψ representation is given by a similarity transformation: $A^\phi = UA^\psi U^\dagger$. In quantum mechanics, we often choose to work in a specific basis—the energy eigenbasis, or the position eigenbasis, and so on. But, there are many different bases that we could work in. In general, two different bases are related by the unitary transformation worked out above. For example, the procedure to convert from some arbitrary basis to the basis where the Hamiltonian matrix is diagonal is called diagonalizing the Hamiltonian—it is accomplished by applying a specific unitary transformation to the system.

The unitary transformations (and orthogonal transformations) are special cases of *similarity transformations*, which are of the form $A \to OAO^{-1}$. This is because $U^\dagger = U^{-1}$. Similarity transformations have one important property, given by the transformation of a product of matrices equal to the product of the transformed matrices (because the "interior" $O^{-1}O = \mathbb{I}$ terms cancel in the interior of the products). We have seen this type of behavior previously with the braiding relation where we could move a Hadamard transformation of a function into the argument of the function.

Section 13.1 Problems:

13.1.1 Practice with Matrix Arithmetic

Let

$$A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 2 \\ 1 & 2 & 1 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} -1 & 0 & 1 \\ 0 & 2 & 1 \\ -1 & 0 & 1 \end{pmatrix}. \tag{13.9}$$

Compute $C = A + B$. Compute $D = AB$. Compute $E = BA$. Does A commute with B?

13.1.2 Commutativity and Diagonal Matrices

Consider a diagonal matrix, whose matrix elements are $D_{ij} = d_i \delta_{ij}$, and a general matrix, whose matrix elements are A_{ij}. Show that $DA \neq AD$ except in special cases. Determine those special cases.

13.1.3 A Left Inverse Implies a Right Inverse

Let A_L^{-1} be a left inverse, so that $A_L^{-1} A = \mathbb{I}$. Prove that $A A_L^{-1} = \mathbb{I}$.

Hint: Assume a right inverse exists and consider an appropriate product of matrices to show that $A_L^{-1} = A_R^{-1}$.

13.1.4 Example of a Similarity Transformation

The Hadamard matrix is defined to be

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \tag{13.10}$$

Show that $H^{-1} = H$. Next, recall the Pauli matrices are

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \text{and} \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{13.11}$$

Compute the three similarity transformations $H\sigma_\alpha H^{-1}$ for $\alpha = x,\ y,\ z$. Comment on the results of this calculation.

13.1.5 Relationship Between Products of Operators and Products of Matrices

The matrix representing the product of two operators $\hat{A}\hat{B} = \hat{C}$ satisfies $C_{ij} = \langle\psi_i|\hat{A}\hat{B}|\psi_j\rangle$ in the basis $\{|\psi_i\rangle\}$. Use the completeness relation in terms of the ket-bra "butterfly" to show that $C_{ij} = \sum_k A_{ik}B_{kj}$. Generalize this result to powers of a single operator to verify that powers of operators are represented by powers of matrices.

13.1.6 Calculate a Commutator Using Operators and Using Matrices

The matrix representation of the commutator between $\hat{A}$ and $\hat{B}$ is given by $\langle\psi_i|\hat{A}\hat{B}|\psi_j\rangle - \langle\psi_i|\hat{B}\hat{A}|\psi_j\rangle$. Determine what the matrix representation of the commutator is in terms of the matrices that represent the operators $\hat{A}$ and $\hat{B}$.

13.1.7 Unitary Transformations Preserve Commutators

If $\hat{U}$ is a unitary operator, then the unitary transformation of an operator $\hat{A}$ is $\hat{U}\hat{A}\hat{U}^\dagger$. When expressed in terms of matrices, we find $A_{ij} \to \sum_{kk'} U_{ik}A_{kk'}U^\dagger_{k'j}$.

Show that the unitary transformation does not modify the result for the commutator $[\hat{x},\ \hat{p}]$, both in terms of operators and in terms of matrices.

13.1.8 Compute the Commutator of $\hat{H}$ with $[\hat{x},\ \hat{p}]$

Assume the Hamiltonian is of the form $\frac{\hat{p}^2}{2M} + V(\hat{x})$. Use the Jacobi identity $[\hat{A},[\hat{B},\hat{C}]] = -[\hat{B},[\hat{C},\hat{A}]] - [\hat{C},[\hat{A},\hat{B}]]$ to compute the nested commutator $[\hat{H},[\hat{x},\hat{p}]]$ and verify that it vanishes. Next, working in the basis of energy eigenstates, where the Hamiltonian matrix is diagonal, use the matrix version of the vanishing of the nested commutator to show that the matrix representation of $[\hat{x},\ \hat{p}]$ must be a diagonal matrix. This is a variant of how Born and Jordan established the

matrix representation of the canonical commutation relation when matrix mechanics was invented.

13.1.9 U Versus $U^\dagger$

Depending on how the operator (or matrix) is defined for the unitary transformation, we will see operators transform as $\hat{U}\hat{A}\hat{U}^\dagger$ or $\hat{U}^\dagger\hat{A}\hat{U}$. Show that if we were to define the transformation matrix as $\tilde{U} = \langle\psi_i|\phi_j\rangle$, then the transformed matrix would be $\tilde{U}^\dagger A\tilde{U}$. Hence, one needs to be very careful in determining the transformed operators and matrices based on the definition of the transformation.

13.2 Determinants

If you have never heard of a determinant before, it might seem like a very odd object to construct. But, it turns out the determinant is used to understand a lot of complex properties of matrices, so it is a very important object to work with when discussing them. This becomes clearer as we determine the properties of the determinant.

The *determinant* of a matrix A with matrix elements A_{ij} is defined to be

$$\det A = \sum_{\substack{\text{permutations} \\ P \text{ of } n \text{ objects}}} (-1)^p A_{p_1 1} A_{p_2 2} A_{p_3 3} ... A_{p_n n}, \tag{13.12}$$

where $(p_1, p_2, ..., p_n)$ is a permutation of $(1, 2, ..., n)$ and

$$(-1)^p = \begin{cases} 1 & \text{if } p \text{ is an even permutation} \\ -1 & \text{if } p \text{ is an odd permutation.} \end{cases} \tag{13.13}$$

We illustrate this with some examples. The permutations are all possible reorderings of the set of numbers. For example, the six permutations of 3 numbers are 123, 231, 312, 132, 213, and 321. How do we determine whether a permutation is even or odd? Consider 312. To get back to 123, we need to

$$\begin{aligned} &\text{interchange 31 so then } 312 \rightarrow 132 \\ &\text{interchange 32 so then } 132 \rightarrow 123. \end{aligned} \tag{13.14}$$

In this case, these are two pair permutations. The number of pair permutations determines whether a permutation is classified as even or odd. An even permutation involves an even number of pair permutations. This implies that the even permuta-

tions make up one half of the $n!$ permutations of n numbers and odd permutations are the other half.

While this form for the definition of the determinant is simple to use for small matrices, it rapidly becomes painful to work with. Start with a 2×2 matrix to see that

$$\det \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} = \underbrace{A_{11}A_{22}}_{12\to 12} - \underbrace{A_{21}A_{12}}_{12\to 21}, \tag{13.15}$$

where the 12 permutation is even $(+1)$ and the 21 permutation is odd (-1). The 3×3 matrix determinant satisfies

$$\begin{aligned} \det \begin{pmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{pmatrix} &= \underbrace{A_{11}A_{22}A_{33}}_{123\to 123} + \underbrace{A_{31}A_{12}A_{23}}_{123\to 312} + \underbrace{A_{21}A_{32}A_{13}}_{123\to 231} \\ &\quad - \underbrace{A_{11}A_{32}A_{23}}_{123\to 132} - \underbrace{A_{31}A_{22}A_{13}}_{123\to 321} - \underbrace{A_{21}A_{12}A_{33}}_{123\to 213} \end{aligned} \tag{13.16}$$

where the permutations 123, 312, 231 are even and the permutations 132, 321, 213 are odd.

We now examine the properties of the determinant. It turns out that the definition of the determinant implies a number of different powerful properties. These properties allow us to calculate determinants much more easily than having to sum $n!$ products of n terms. We start with some notation by introducing the completely antisymmetric tensor ϵ (we have worked with the three-dimensional completely antisymmetric tensor before—here, we generalize to arbitrary dimensions). We write

$$\det A = \sum_{p} \epsilon_{p_1 p_2 \ldots p_n} A_{p_1 1} A_{p_2 2} \ldots A_{p_n n} \tag{13.17}$$

with the completely antisymmetric tensor satisfying

$$\epsilon_{p_1 p_2 \ldots p_n} = \begin{cases} 0 & \text{any two } p_i\text{'s are the same} \\ 1 & p_1 \ldots p_n \text{ is an even permutation} \\ -1 & p_1 \ldots p_n \text{ is an odd permutation.} \end{cases} \tag{13.18}$$

The determinant satisfies the following ten properties:

(1) If a row is all zeros, then $\det A = 0$ (because an element from each row appears in the product of each term, so each product has a zero in it). As a result, the determinant ends up as the sum of $n!$ terms that are all zero and the determinant is 0.

(2) If a column has all zeros, then $\det A = 0$ (the argument for this is similar to the one above).
(3) If a row is multiplied by a constant λ, the determinant is multiplied by λ (since each row has an element that appears in one factor of each term in the sum for the determinant). The same thing happens for an entire column multiplied by a constant.
(4) If two rows of a matrix are interchanged, the sign of the determinant changes ($\det A \rightarrow -\det A$). The interchange of a row is like adding a permutation to each term in the determinant. This changes the even permutations to odd permutations, and vice versa—hence, the sign of the determinant changes.
(5) If two rows are proportional to each other, then $\det A = 0$. Take one row and multiply by a constant so it is equal to the second row. This changes the determinant by that factor according to property 4 above. Then, interchange the two rows—the matrix doesn't change, but $\det A = -\det A$, which implies that $\det A = 0$.
(6) If row i can be written as $A_{ij} = B_{ij} + C_{ij}$ for $i \leq j \leq n$, then $\det A = \det B + \det C$ where B is the matrix A with the ith row replaced by B_i and C is the matrix A with the ith row replaced by C_i.
Proof: Every place A_{ix} appears in a term in the determinant (one factor for each term), we replace the A_{ix} factor with $B_{ix} + C_{ix}$. Then we group all the B terms together and all the C terms together in the summation and we immediately see that $\det A = \det B + \det C$.
(7) If we take α times row i and add to row j, it does not change the determinant. This is because we think of $B_{jx} = A_{jx}$ and $C_{jx} = A_{jx}\alpha$ but C has two rows proportional to one another, so $\det C = 0$ and $\det A = \det B + \det C$ imply that the determinant is unchanged.
(8) The transpose of a matrix A^T interchanges the columns of the matrix with its rows—so $(A^T)_{ij} = A_{ji} = (A)_{ji}$. The determinant of $A^T = \det A$. This follows, since we can permute a product $A_{p_1 1}A_{p_2 2}A_{p_3 3}...A_{p_n n}$ into the product $A_{1p_1}A_{2p_2}A_{3p_3}...A_{np_n}$ by reordering the terms in the sum. The permutations remain even or odd, so overall, the determinant stays the same.
(9) The ij cofactor of a matrix is defined to be $(-1)^{i+j}$ times the det of the $(n-1) \times (n-1)$ matrix whose elements are found by removing the ith row and the jth column of A. The determinant satisfies

$$\det A = \sum_{j=1}^{n} A_{ij} C_{ij} \tag{13.19}$$

and similarly, we have

$$\det A = \sum_{i=1}^{n} A_{ij} C_{ij}. \tag{13.20}$$

This is often a useful way to find determinants if a row or column has many zeros.

(10) If $C = AB$, such that $C_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj}$, then $\det C = \det A \det B$.
Proof:

$$\det C = \sum_{p} \epsilon_{p_1 \cdots p_n} C_{p_1 1} C_{p_2 2} \cdots C_{p_n n} \tag{13.21}$$

$$= \sum_{p} \epsilon_{p_1 \ldots p_n} \sum_{k_1=1}^{n} A_{p_1 k_1} B_{k_1 1} \sum_{k_2=1}^{n} A_{p_2 k_2} B_{k_2 2} \cdots \sum_{k_n=1}^{n} A_{p_n k_n} B_{k_n n} \tag{13.22}$$

$$= \sum_{p_1 \cdots p_n} \sum_{k_1 \cdots k_n} \epsilon_{p_1 \cdots p_n} A_{p_1 k_1} A_{p_2 k_2} \cdots A_{p_n k_n} B_{k_1 1} B_{k_2 2} \cdots B_{k_n n}. \tag{13.23}$$

So, we end up with

$$\sum_{k_1 \cdots k_n} \det A \cdot \epsilon_{k_1 \cdots k_n} B_{k_1 1} B_{k_2 2} \cdots B_{k_n n} \tag{13.24}$$

or

$$\det A \cdot \det B \tag{13.25}$$

because

$$\sum_{p_1 \cdots p_n} \epsilon_{p_1 \cdots p_n} A_{p_1 k_1} A_{p_2 k_2} \cdots A_{p_n k_n} = \det A \cdot \epsilon_{k_1 \ldots k_n}. \tag{13.26}$$

This last identity summarizes how the permutation $k_1 k_2 \cdots k_n$ of $12 \cdots n$ is even or odd according to whether the completely antisymmetric tensor is 1 or -1. It is a little subtle, so think carefully about it (maybe try it for some small matrices concretely if having any trouble).

Let us use these properties to determine interesting results for matrices. Probably the most important use of a determinant is to separate matrices into those that have an inverse and those that do not. If we recall that the determinant of a product of matrices is the product of the determinants and that the determinant of an identity matrix is 1, then, we learn that $AA^{-1} = \mathbb{I}$ implies that $\det A \det A^{-1} = \det \mathbb{I} = 1$. Or, $\det A^{-1} = \frac{1}{\det A}$. Hence, a matrix is invertible only if $\det A \neq 0$. This is a remarkable result. It tells us the condition for a matrix to be invertible. Just looking at a general matrix provides us with no clue as to whether an inverse can be found. Computing the determinant immediately tells us whether it can be found. Furthermore, one can quickly show that the determinant of the similarity

transformation of a matrix is equal to the determinant of the original matrix. See if you can reason out why.

Similarly, the fact that $\det(AB) = \det A \det B$ allows us to determine matrix groups. Recall that a group is a collection of objects with a multiplication law that is closed under multiplication, has an identity, and has an inverse for each element of the group. Using the multiplication property of a matrix immediately tells us that the set of matrices of a fixed size (say $n \times n$ matrices) with $\det A = 1$ forms a group. Because the determinant is 1, the matrix is invertible. The product of two determinant 1 matrices also has determinant 1. And we just need to ensure that the collection is closed under multiplication. But, because the product of these matrices is another matrix with determinant equal to 1, it should be clear that this set is closed. Such groups are typically called "special" groups.

Finally, we want to explicitly compute the determinant of a diagonal matrix. One can see from the expansion definition of the determinant that all products vanish except for one, the one corresponding to the identity permutation, so a diagonal matrix $D_{ij} = d_i \delta_{ij}$ will have a determinant given by $\det D = \prod_{i=1}^{n} d_i$. One can use this to immediately verify that the identity matrix has determinant equal to 1, as we stated above. Furthermore, if any diagonal matrix element of a diagonal matrix vanishes, then the determinant does too. We have a video called Determinants https://tinyurl.com/3pjuafmu that discusses some more advanced ideas related to determinants.

Section 13.2 Problems:

13.2.1 Determinants of Matrices with Vanishing Elements Below or Above the Diagonal

Suppose we have a matrix that has vanishing matrix elements below the diagonal. So, $A_{ij} = 0$ for $j > i$. Show that the determinant of this matrix is given by the product of its diagonal matrix elements, even if there are nonzero matrix elements above the diagonal. Does this also hold for matrices with vanishing matrix elements above the diagonal and nonvanishing below?

13.2.2 Efficient Calculation of a Determinant via the Method of Cofactors

You will need the result of the previous problem to work on this problem.

Compute the determinant of the following $n \times n$ matrix using the method of cofactors. You should find two nonzero cofactors contribute if you evaluate along the top row, and each cofactor can have its determinant evaluated immediately.

$$\begin{pmatrix} 1 & 0 & 0\,0\,0\cdots & 0 & 1-\beta E/n \\ -1+\beta E/n & 1 & 0\,0\,0\cdots & 0 & 0 \\ 0 & -1+\beta E/n & 1\,0\,0\cdots & 0 & 0 \\ \vdots & \vdots & \vdots\,\vdots\,\vdots\,\vdots & \vdots & \vdots \\ 0 & 0 & 0\,0\,0\cdots & -1+\beta E/n & 1 \end{pmatrix}. \tag{13.27}$$

The matrix has 1 along the diagonal, $-1+\frac{\beta E}{n}$ on the first lower subdiagonal, and $1+\frac{\beta E}{n}$ on the upper right corner. All other matrix elements are 0. Take your result and evaluate it in the limit as $n \to \infty$.

Ans. $1+e^{-\beta E}$.

13.2.3 Hilbert Matrices

The *Hilbert matrices* are a special class of matrices useful for testing computer algorithms, because they are difficult to work with. They are defined as follows: $H_1 = 1$,

$$H_2 = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{3} \end{pmatrix}, \quad H_3 = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{pmatrix} \quad \text{and} \quad H_4 = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \frac{1}{6} \\ \frac{1}{4} & \frac{1}{5} & \frac{1}{6} & \frac{1}{7} \end{pmatrix}. \tag{13.28}$$

Compute the determinants of the Hilbert matrices using the definition in terms of permutations. Note that this is a pretty long calculation for H_4, since it includes 24 terms. You should find the determinants are 1, $\frac{1}{12}$, $\frac{1}{1240}$, and $\frac{1}{6048000}$. One can see these determinants approach zero very rapidly, but are always nonzero. Matrices with determinants close to zero can be difficult to work with numerically, especially when trying to find their matrix inverse.

13.2.4 A Neat Rule About Factors in the Determinant

Consider the matrix A given by

$$A = \begin{pmatrix} 1 & 4 & 7 \\ 1 & 8 & 9 \\ 2 & 1 & 0 \end{pmatrix} \tag{13.29}$$

and look at the prime factors of the numbers you read off from the rows. For example, the first row is 147, which has prime factors of 3, 7, and 7. Find the prime factors of the other numbers read off the rows and determine the greatest common divisor of the three numbers. The determinant will have that greatest common divisor as a factor.

Check this carefully for the matrix A. As a bonus, see if you can use the 10 rules for determinants to understand why this must be so.

13.2.5 Other Special Matrix Groups

Do matrices where the determinant is equal to $e^{i\theta}$ form a group? What about the set of matrices with determinants given by 1 and -1? Explain.

13.2.6 The Determinant of a Hermitian Matrix Must Be Real

Prove that the determinant of a Hermitian matrix must be real.

Hint: Since the determinant is unchanged by a similarity transformation, use that to compute the determinant in the most appropriate basis.

13.3 Eigenvalues and Eigenvectors

When a matrix multiplies a vector it can do two things: (i) it rotates the vector and (ii) it changes its length. Indeed, what else can it do, as matrix multiplication maps a vector to another vector? What is *absolutely amazing* is that for a class of matrices called Hermitian matrices, one can find exactly n different orthogonal basis vectors for an $n \times n$ matrix such that when the matrix multiplies each of those basis vectors, there is no rotation, only scaling.

Hence, eigenvalues and eigenvectors are found by choosing the basis for the matrix such that when the matrix M multiplies one of the basis vectors e^{λ} it results in a number λ times the basis vector e^{λ}. In equations, we have

$$\sum_{ij} M_{ij} e_j^{\lambda} = \lambda e_i^{\lambda}. \tag{13.30}$$

In words we summarize as follows:

"A matrix times a vector equals a number times that vector."

Keep this mantra in mind to help you always understand what an eigenvalue and an eigenvector are. It is probably the most misunderstood concept by students of physics until they finally get it.

Again.

"A matrix times a vector equals a number times that vector."

How do we find these solutions? We rearrange the equations to have a shifted matrix annihilating a vector as follows:

$$\sum_{j=1}^{n} M_{ij} e_j^\lambda = \lambda e_i^\lambda \implies (M - \lambda \mathbb{I}) e^\lambda = 0. \tag{13.31}$$

This has a nontrivial solution only if $\det(M - \lambda \mathbb{I}) = 0$, which is an n-degree polynomial in λ. It will have n solutions in the complex plane.

If M is real and symmetric, it has exactly n real eigenvalues (this is called a symmetric matrix; the results actually hold a bit more broadly to include Hermitian matrices when matrix elements are complex valued). In addition, there might be multiple roots where $\lambda_i = \lambda_j$. When this occurs we say there is a *degeneracy*.

The eigenvectors are found by solving the equation

$$\sum_{j=1}^{n} M_{ij} e_j^\lambda = \lambda e_i^\lambda \tag{13.32}$$

for the specific λ value chosen (we already found the possible λ values when we found the roots to $\det(M - \lambda \mathbb{I}) = 0$). We have not explained how to solve this, but the condition for a solution to exist is $\det(M - \lambda \mathbb{I}) = 0$. In almost all cases, we solve these problems numerically, because they are too cumbersome to solve by hand.

If M is Hermitian, it also has exactly n real solutions. (A Hermitian matrix satisfies $M^\dagger = M$, which means we take the transpose and the complex conjugate of all elements; a symmetric matrix has all real values and satisfies $M^T = M$—obviously all symmetric matrices are also Hermitian). This result is called the spectral theorem. We will not prove the spectral theorem here.

One important property is that the eigenvectors are orthogonal to each other if they have different eigenvalues. In the derivation below, we use a λ superscript to denote the eigenvalue associated with the eigenvector e^λ. To see that the corresponding eigenvectors are orthogonal, note that the relation $\sum_{j=1}^{n} M_{ij} e_j^\lambda = \lambda e_i^\lambda$ can be rewritten as $\sum_{j=1}^{n} e_j^\lambda M_{ji} = \lambda e_i^\lambda$ because the matrix M is symmetric ($M_{ij} = M_{ji}$). So, let's compute $\sum_{ij} e_i^\lambda M_{ij} e_j^{\lambda'}$. By summing over j first, this becomes $\lambda' e^\lambda \cdot e^{\lambda'}$. But summing over i first, we have $\lambda e^\lambda \cdot e^{\lambda'}$. Of course, these two must be equal, so we learn that

$$(\lambda - \lambda') e^\lambda \cdot e^{\lambda'} = 0. \tag{13.33}$$

Hence, if $\lambda \neq \lambda'$, then we must have $e^{\lambda} \cdot e^{\lambda'} = 0$, implying the two vectors are orthogonal. It turns out that we can always find orthogonal eigenvectors when the eigenvalues are degenerate as well, so you can assume the eigenvectors are always orthogonal. The procedure to do this is called *Gram-Schmidt orthogonalization*; we will describe it in detail shortly.

Here is an example with a 3×3 matrix. Find the eigenvalues of the following matrix:

$$M = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 0 & -1 \\ 2 & -1 & -1 \end{pmatrix} \tag{13.34}$$

We start by shifting and taking the determinant

$$\det(M - \lambda \mathbb{I}) = 0, \tag{13.35}$$

which becomes

$$\det \begin{pmatrix} 1-\lambda & 0 & 2 \\ 0 & -\lambda & -1 \\ 2 & -1 & -1-\lambda \end{pmatrix} = 0. \tag{13.36}$$

There are many ways to evaluate the determinant. However you prefer to do it, you will end up with the result

$$\lambda^3 - 6\lambda + 1 = 0. \tag{13.37}$$

From here, we must go numerical to solve for the three different λ's and for the eigenvectors. This is often the case for these kinds of problems. You might have thought, wait this is a cubic, we can find the roots *analytically*. While true, often the formulas for the roots are so complicated that computing them numerically is better.

How do we make the transformation to "diagonalize" the matrix? It is a similarity transformation (which means multiply by one matrix on the left and by its inverse on the right). We form U via

$$U^{-1} = \begin{pmatrix} | & | & \vdots & | \\ e_1 & e_2 & \dots & e_n \\ | & | & \vdots & | \end{pmatrix} \tag{13.38}$$

$$U = \begin{pmatrix} \cdots & - & e_1^* & - & \cdots \\ \cdots & - & e_2^* & - & \cdots \\ \vdots & & \vdots & & \vdots \\ \cdots & - & e_n^* & - & \cdots \end{pmatrix} = U^\dagger. \tag{13.39}$$

Then multiplying out and recalling that the eigenvectors are an orthonormal set yields

$$U^M U^{-1} = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & \cdots \\ \vdots & & \vdots & \vdots \\ 0 & \ldots & 0 & \lambda_n \end{pmatrix}. \tag{13.40}$$

This is why the matrix becomes a diagonal matrix in the eigenvector basis. It also shows that we can always diagonalize the matrix by a similarity transformation (unless it does not have n eigenvalues, where things become trickier). This is why the spectral theorem is important—it guarantees we can always do this for Hermitian matrices.

We now discuss the *Perron-Frobenius theorem*, which states important properties for the eigenvector corresponding to the maximal eigenvalue for nonnegative matrices. This proof is a bit technical and it might require reading it a few times before it sticks. You do not need to know every detail, but the basic idea is important. It is summarized at the end of the proof. A nonnegative matrix has all matrix elements real and $M_{ij} \geq 0$ for $i \neq j$; note that there are no requirements for the diagonal elements, except that they are real. Furthermore, if M is *connected*, then the maximal eigenvalue is nondegenerate, and the maximal eigenvector has all of its vector components positive.

Before going through the proof of this theorem, we must define what a connected matrix is. We say a $n \times n$ matrix is connected if for every pair of different indices $i \neq j$, there is a chain of indices $k_1, k_2, \cdots, k_m$ such that the product

$$M_{ik_1} M_{k_1 k_2} M_{k_2 k_3} \cdots M_{k_m j} \neq 0. \tag{13.41}$$

In words, this says that we can find a path of nonzero matrix elements in the matrix that takes me from index i to index j. The number of elements in the chain, given by $m + 1$, must be less than or equal to the matrix dimension n; m depends on the choice of i and j and need not be the same for each path.

Now, we move on to the proof. Let D be the minimal value for the diagonal matrix elements: $D = \min M_{ii}$ for $1 \leq i \leq n$. We will consider the matrix $M' = M + D\mathbb{I}$. Following Problem 13.3.2, we have that the eigenvalues of M' are given by each of the eigenvalues of M shifted by D; schematically, we write this as $\lambda' = \lambda + D$. The matrix M' has all of its matrix elements nonnegative: $M'_{ij} \geq 0$. Now, we assume $\Lambda' > 0$ is the maximal eigenvalue of M' and we let ψ be its normalized

(real) eigenvector (so that $\sum_{i=1}^{n} \psi_i^2 = 1$). Then, we have that

$$\sum_{j=1}^{n} M'_{ij} \psi_j = \Lambda' \psi_i, \tag{13.42}$$

for all i. Suppose some of the elements of ψ are negative. Then we consider the vector $\phi = |\psi|$, which satisfies $\phi_i = |\psi_i|$. By taking the absolute value of Eq. (13.42), we find that

$$\Lambda' \phi_i = \Lambda' |\psi_i| = \left| \sum_{j=1}^{n} M'_{ij} \psi_j \right| \leq \sum_{j=1}^{n} |M'_{ij}| \, |\psi_j| = \sum_{j=1}^{n} M'_{ij} |\psi_j| = \sum_{j=1}^{n} M'_{ij} \phi_j, \tag{13.43}$$

because ψ_i could be less than zero and $M'_{ij} \geq 0$. Hence,

$$\sum_{j=1}^{n} M'_{ij} \phi_j \geq \Lambda' \phi_i. \tag{13.44}$$

If we multiply by ϕ_i and sum over i, we find that $\sum_{ij=1}^{n} \phi_i M'_{ij} \phi_j \geq \Lambda' \sum_{i=1}^{n} \phi_i^2 = \Lambda'$. This means that if we expand the vector ϕ in the eigenvector basis of M', it includes an eigenvector with an eigenvalue that is equal to Λ' or larger. But, because Λ' is the largest eigenvalue, we must have that $\phi = |\psi|$ is also an eigenvector with eigenvalue Λ'. Hence, among the eigenvectors with the largest eigenvalue of M', there is at least one eigenvector that has all of its nonzero components the same sign.

To prove that the largest eigenvalue is nondegenerate, we must use the connectivity condition. We will show that if M' is connected, then $\phi_i > 0$ for all i and then that the eigenvalue is nondegenerate. We start with a simple case. Suppose $\phi_k = 0$. Because M' is connected, there is some k' such that $M'_{kk'} \neq 0$. Assume that $\phi_{k'} \neq 0$. Then, the eigenvalue equation for the kth component becomes

$$\sum_{j=1}^{n} M'_{kj} \phi_j = \underbrace{\sum_{j=1, j \neq k'}^{n} M'_{kj} \phi_j}_{\text{nonnegative}} + \underbrace{M_{kk'} \phi_{k'}}_{\text{positive}} = \Lambda' \phi_k = 0. \tag{13.45}$$

This cannot hold, because the left-hand side is positive, while the right-hand side vanishes, so the assumption that $\phi_k = 0$ is impossible. For the general case, we look at the kth component of the equation $(M')^{m+1} \phi = (\Lambda')^{m+1} \phi$. Then, the connectivity condition guarantees at least one term in the summations on the left-hand side is nonzero (because we must have at least one $\phi_i \neq 0$; otherwise the eigenvector is not a unit norm vector, and we assume that k and i are connected by an $m + 1$ long chain). Then, again, we have the left-hand side is positive, while the

right-hand side vanishes, which is a contradiction. Hence, if the matrix is connected, then the eigenvector with the maximal eigenvalue must have *all* of its components positive. We show that the maximal eigenvalue is nondegenerate in Problem 13.3.6.

Sometimes it is easy to "lose the forest for the trees" when working on a proof like this. The basic idea is rather simple. If my matrix has nonnegative elements, and I want to find the largest eigenvector, I expect it should be a vector with nonnegative elements, because any negative element in the eigenvector leads to a subtraction in the eigenvalue-eigenvector relationship, which will reduce the eigenvalue. Furthermore, a connected matrix tells me the eigenvector will have nonzero elements everywhere, because each element of the eigenvector is eventually connected to all other elements. Converting such ideas into a rigorous proof is a bit harder to accomplish, but it is important to focus on the ideas. The Perron-Frobenius theorem will be quite important when we discuss magnetism in the Hubbard model.

We end the discussion in this section with the *configuration interaction* approximation for finding energy eigenstates. We already know how we construct the matrix that represents an operator. If we construct the matrix that represents the Hamiltonian and diagonalize it, then we will determine all of the energy eigenvalues and energy eigenvectors. But, for many problems, the Hamiltonian matrix is too large to be stored, even on a supercomputer. What can we do then? This is where the idea of configuration interaction comes in. If we truncate the matrix, throwing away many parts of it, leaving behind a much smaller matrix (that does fit on our computer), then we have an approximation to the exact matrix representing the Hamiltonian. If we diagonalize this matrix, it becomes a variational approximation to the true energy eigenvalues. This approach is very common to use in chemistry, albeit more accurate and more efficient methods have been developed and the method is not widely used anymore. Why is it a variational approximation? Because the eigenvector of the truncated matrix can be expressed as an expansion in terms of all of the eigenvectors of the exact solution. Hence, it provides an upper bound on the ground-state energy. We will discuss this idea a bit more when we discuss quantum chemistry in Chap. 19. We have a video https://tinyurl.com/5337h6tf that discusses eigenvalues and eigenvectors in a somewhat different way.

Section 13.3 Problems:

13.3.1 Eigenvalues and Eigenvectors of Hermitian Matrices

Prove that the eigenvalues of a Hermitian matrix are real. Prove that the eigenvectors of a Hermitian matrix (with nondegenerate eigenvalues) are orthogonal.

Note that we assumed the eigenvalues were nondegenerate for simplicity. The orthogonality result can be extended to include the case with degenerate eigenvalues as well.

13.3.2 Shifting Eigenvalues by Shifting the Diagonal of a Matrix

Prove that the eigenvalues of $M + \alpha\mathbb{I}$ are each shifted by α to become $\lambda + \alpha$.

13.3.3 Eigenvalues and Eigenvectors

Determine the eigenvalues and their corresponding eigenvectors for the following Hermitian matrix. Verify that the eigenvalues are real and the eigenvectors are orthogonal.

$$\begin{pmatrix} U+1 & 1 & -\sqrt{2} \\ 1 & 1 & \sqrt{2} \\ -\sqrt{2} & \sqrt{2} & U \end{pmatrix}. \tag{13.46}$$

Note that U is a real number. You will find a cubic equation for the eigenvalues, but it will factorize into a quadratic and linear equation.

13.3.4 Determinant and Eigenvalues

Show for a Hermitian matrix that the determinant is the product of its n eigenvalues, or $\det M = \prod_{i=1}^{n} \lambda_i$.

13.3.5 Properties of a Connected Matrix

If a matrix is disconnected, we can separate the n indices into connected subsets. Suppose we have a matrix with two connected subsets of indices. Describe what the matrix looks like if we reorder it so all the connected indices in the first connected set are the first indices of the matrix and all the indices of the second set are the second set of the indices. This type of a matrix is called *block diagonal*.

13.3.6 Uniqueness of the Maximal Eigenvalue of a Positive Semidefinite Matrix

Prove that the maximal eigenvalue of a connected nonnegative matrix is nondegenerate.

Hint: If there is a second eigenvector, it cannot have all positive elements. But, its absolute value must be a maximal eigenvector, so it must be the eigenvector with all positive elements. By subtracting the two eigenvector equations, you should be able to find a contradiction.

13.3.7 Using the Perron-Frb̈enius Theorem to Find the Ground State

Suppose you have a matrix that represents the Hamiltonian of a system that is nonpositive. Write an essay describing how you can use the Perron-Frobenius theorem to show that the ground state is nondegenerate.

Hint: Consider $-M$.

13.4 Functions of Matrices

There are two ways to define functions of matrices. The first way is via a power series—we simply take the power series in terms of numbers and replace it by a power series in terms of matrices. The second way starts with diagonal matrices and applies the function to each diagonal element, in turn. To use it on a general matrix, we first diagonalize the matrix, then apply the function, and then use the inverse transformation back to the original basis for the matrix. Both approaches yield the same function.

As an example, consider the exponential function. We have

$$e^M = U^\dagger \begin{pmatrix} e^{\lambda_1} & 0 & \dots & 0 \\ 0 & e^{\lambda_2} & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & e^{\lambda_{n-1}} & 0 \\ 0 & \cdots & 0 & e^{\lambda_n} \end{pmatrix} U \tag{13.47}$$

because

$$e^M = 1 + M + \frac{1}{2}M^2 + \frac{1}{6}M^3 + \frac{1}{24}M^4 + \cdots \tag{13.48}$$

$$= U^\dagger U \left(1 + M + \frac{1}{2}M^2 + \frac{1}{6}M^3 + \dots\right) U^\dagger U \tag{13.49}$$

$$= U^\dagger \left(1 + UMU^\dagger + \frac{1}{2}(UMU^\dagger)^2 + \cdots\right) U \tag{13.50}$$

$$= U^{\dagger} \begin{pmatrix} e^{\lambda_1} & 0 & \cdots & 0 \\ 0 & e^{\lambda_2} & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & e^{\lambda_{n-1}} & 0 \\ 0 & \cdots & 0 & e^{\lambda_n} \end{pmatrix} U. \tag{13.51}$$

This last equation holds because we chose the matrix $UMU^{\dagger}$ to be diagonal

$$UMU^{\dagger} = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & \lambda_{n-1} & 0 \\ 0 & \cdots & 0 & \lambda_n \end{pmatrix} \tag{13.52}$$

and the power series gives the exponential of each diagonal term. This approach can obviously be extended to all functional forms, so one can always construct matrix-valued functions in this fashion. The general result is

$$f(M) = U^{\dagger} \begin{pmatrix} f(\lambda_1) & 0 & \cdots & 0 \\ 0 & f(\lambda_2) & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & f(\lambda_{n-1}) & 0 \\ 0 & \cdots & 0 & f(\lambda_n) \end{pmatrix} U. \tag{13.53}$$

Section 13.4 Problems:

13.4.1 Functions of Matrices That Are Not Diagonalizable

Some matrices cannot be diagonalized, in the sense that they do not have n linearly independent eigenvectors. Write a short essay that describes how one can define functions for these types of matrices.

Hint: Only one of our approaches works here.

13.4.2 Matrix Square Root

One of the most important functions is the matrix square root. We will work with a Hermitian matrix. Show how $\sqrt{M^{\dagger}M} = |M|$, taken in the matrix sense, where

$$\sqrt{M^\dagger M} = |M| = U^\dagger \begin{pmatrix} |\lambda_1| & 0 & \cdots & 0 \\ .0 & |\lambda_2| & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & |\lambda_{n-1}| & 0 \\ 0 & \cdots & 0 & |\lambda_n| \end{pmatrix} U. \tag{13.54}$$

Note that one could choose other square roots, since each diagonal term can have its sign changed. But, the standard definition of the matrix square root is with all of the positive roots.

13.4.3 Matrix Square Root II

Suppose we look at the 2^n different possible matrix square roots. Unlike the situation with complex numbers, where every real number has two square roots, with matrices, we have 2^n square roots. Write an essay that brainstorms about applications that one might have that use more than just the standard square root, discussed in the previous problem.

13.4.4 Scaling and Squaring

Sometimes, computing the exponential of a matrix by summing the power series is too long of a calculation to carry out. There is an alternative way we can calculate exponentials, called the scaling and squaring method. In this approach, we note the identity that

$$e^M = \left(e^{\frac{1}{2}M}\right)\left(e^{\frac{1}{2}M}\right), \tag{13.55}$$

because M commutes with itself. Repeat this procedure $n-1$ more times to show that

$$e^M = \left(e^{\frac{1}{2^n}M}\right)^{2^n}. \tag{13.56}$$

Now, the way the algorithm works is we pick an n large enough, so that $\exp(\frac{1}{2^n}M) \approx \mathbb{I} + \frac{1}{2^n}M$. Then we square that matrix n times to compute the exponential. Write a brief discussion explaining how to implement such an approach and how we would pick the correct n to use.

13.5 Matrix Absolute Value and Positive Semidefinite Matrices

The absolute value of a matrix (or, if you like, the square root of the square of a matrix) can be defined in terms of the matrix in diagonal form, even if there is no power-series expansion for the absolute value. For the absolute value of a matrix, we choose the unique square root that has all nonnegative eigenvalues. It is given by

$$|M| = U^\dagger \begin{pmatrix} |\lambda_1| & 0 & \cdots & 0 \\ .0 & |\lambda_2| & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & |\lambda_{n-1}| & 0 \\ 0 & \cdots & 0 & |\lambda_n| \end{pmatrix} U. \tag{13.57}$$

Note that, in general after multiplying by U and $U^\dagger$, we have that the ij element of $|M|_{ij}$ is often not equal to $|M_{ij}|$.

One important property of the matrix absolute value is that it is a positive semidefinite matrix. This means for any vector ψ, we have that

$$\psi^* |M| \psi = \sum_{ij} \psi_i^* |M|_{ij} \psi_j = \sum_{ijk} \psi_i^* U_{ji}^* |\lambda_j| U_{jk} \psi_k \geq 0. \tag{13.58}$$

It might not be completely obvious why this is nonnegative until you recognize that

$$\phi_j = \sum_i U_{ji} \psi_i \tag{13.59}$$

has a complex conjugate given by

$$\phi_j^* = \left(\sum_i U_{ji} \psi_i \right)^* = \sum_i \psi_i^* U_{ji}^* \tag{13.60}$$

and the summation for the expectation value of $|M|$ in the state ψ becomes $\sum_j |\phi_j|^2 |\lambda_j|$, which is clearly nonnegative. If all of the eigenvalues of the matrix are positive, and never zero, then it is a *positive definite* matrix.

Section 13.5 Problems:

13.5.1 Matrix Absolute Value of the Pauli Spin Matrices

Show that the matrix absolute value of each Pauli matrix is the 2×2 identity matrix $\mathbb{I}_2$.

13.5.2 Structure of a Positive Semidefinite Matrix

If M is a Hermitian matrix that is not diagonal, explain what the matrix absolute value might look like. Is it the same as $|M_{ij}|$? Can $|M|$ have negative matrix elements? Explain in a few sentences or with some concrete examples.

13.6 Trace and Partial Trace

The *trace* of a matrix is a simple object to define—it is the sum of the diagonal elements of the matrix, so we have

$$\mathrm{Tr}\, M = \sum_{i=1}^{n} M_{ii}. \tag{13.61}$$

This is the second important summary number for a matrix (after the determinant). When we worked with the determinant, we found that $\det(AB) = \det A \cdot \det B$. What about for the trace? If we compute the trace of AB, we find it satisfies

$$\mathrm{Tr}(AB) = \sum_{ij=1}^{n} A_{ij} B_{ji} = \sum_{ij=1}^{N} B_{ji} A_{ij} = \mathrm{Tr}(BA). \tag{13.62}$$

So, we do not have the trace of a product is equal to the product of the traces, but instead we have that the trace is invariant under the reordering of its factors $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$. This property ends up being very important and is often used in quantum statistical mechanics to simplify expressions by reordering the terms in a product under the trace. The trace is invariant under any cyclic reordering, so we have

$$\mathrm{Tr}(ABC) = \mathrm{Tr}(BCA) = \mathrm{Tr}(CAB) \tag{13.63}$$

and its generalization to larger number of products of terms.

This invariance of the trace of a product of matrices with a cyclic permutation immediately tells us that the trace is independent of the basis that the matrix is calculated in, because

$$\mathrm{Tr}(SMS^{-1}) = \mathrm{Tr}(MS^{-1}S) = \mathrm{Tr}(M\mathbb{I}) = \mathrm{Tr}M. \tag{13.64}$$

Hence, the trace is unchanged under a similarity transformation. So, for a Hermitian matrix, the trace is given by the sum of the eigenvalues (similar to the determinant is given by the product of the eigenvalues), which is found by calculating the trace in the eigenbasis of the matrix.

There is another interesting identity that relates the trace and the determinant. It also uses the functions of a matrix we discussed previously. The relation is that

$$\det M = \exp\big(\mathrm{Tr}(\ln M)\big). \tag{13.65}$$

This result should be obvious to you if you look in the diagonal basis of the Hermitian matrix M. It is discussed further in Problem 13.6.5. Note that this identity will also hold for infinite matrices, so it becomes a useful formula for more advanced quantum calculations. That is it for the identities about the trace. It may seem like a simple concept, but it often is used in quite advanced calculations, so it is a concept that you need to know.

When we work with tensor-product matrices, we often discuss the partial trace of a tensor-product matrix. Consider the tensor product $C = A \otimes B$ of 2×2 matrices. It is formed by placing the B matrix where each element of the A matrix is and multiplying out the terms. So, we have that

$$C = \begin{pmatrix} a_{11}\begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} & a_{12}\begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \\ a_{21}\begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} & a_{22}\begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \end{pmatrix} = \begin{pmatrix} a_{11}b_{11} & a_{11}b_{12} & a_{12}b_{11} & a_{12}b_{12} \\ a_{11}b_{21} & a_{11}b_{22} & a_{12}b_{21} & a_{12}b_{22} \\ a_{21}b_{11} & a_{21}b_{12} & a_{22}b_{11} & a_{22}b_{12} \\ a_{21}b_{21} & a_{21}b_{22} & a_{22}b_{21} & a_{22}b_{22} \end{pmatrix}. \tag{13.66}$$

The last form motivates thinking of the C matrix in terms of a complex index as

$$c_{i\alpha,j\beta} = a_{ij}b_{\alpha\beta}. \tag{13.67}$$

Indeed, many operations can be treated more efficiently in terms of this compound index. For example, the trace of a tensor-product matrix is given by

$$\mathrm{Tr}C = \sum_{i\alpha} c_{i\alpha,i\alpha} = \sum_{i}\sum_{\alpha} a_{ii}b_{\alpha\alpha} = \mathrm{Tr}A\mathrm{Tr}B. \tag{13.68}$$

Be sure that you can follow each step in that derivation. The partial trace is used heavily in quantum information. It involves the sum over one of the compound indices only. We usually use a subscript to indicate the matrix of the tensor product that we are tracing over. For example, we have

$$\mathrm{Tr}_B(C) = \sum_{\alpha} c_{i\alpha,j\alpha} = \sum_{\alpha} a_{ij}b_{\alpha\alpha} = (\mathrm{Tr}B)A. \tag{13.69}$$

The partial trace over A becomes

$$\mathrm{Tr}_A(C) = \sum_{i} c_{i\alpha,i\beta} = \sum_{i} a_{ii}b_{\alpha\beta} = (\mathrm{Tr}A)B. \tag{13.70}$$

Note that the partial trace is a matrix, whose dimensions are smaller than the dimensions of the original tensor-product matrix. Note further that while these expressions look simple when expressed in terms of the compound indices, they are more complicated to sort out in terms of the ordinary matrix indices of the matrix C.

Section 13.6 Problems:

13.6.1 Trace of a Commutator

For any finite-dimensional matrices, the trace of the commutator vanishes. Prove this by using the invariance of the trace with respect to the ordering of two matrices in a product.

Look back at examples of this from the Pauli spin matrices and the angular momentum matrices with $J = 1$, and verify that the rule holds.

Surprisingly, if we look at the matrices that represent $\hat{p}$ and $\hat{x}$ for the harmonic oscillator in one dimension, we find that these matrices are infinite dimensional, and the identity no longer holds. Explain why. If you want to, you can compute the matrices for position and momentum in the energy eigenbasis. You will find they each have nonzero terms only on the first left and right subdiagonals.

13.6.2 Consequences of Zero Trace

Show that any 2×2 matrix with zero trace squares to a diagonal matrix.

13.6.3 Traces of Products and Tensor Products

Explain why, for general matrices, we have $\mathrm{Tr}(AB) \neq \mathrm{Tr}A \cdot \mathrm{Tr}B$, but for a tensor-product matrix, we have $\mathrm{Tr}\big(A \otimes B\big) = \mathrm{Tr}A \cdot \mathrm{Tr}B$.

13.6.4 Trace of Transpose and Hermitian Conjugate Matrices

Show that $\mathrm{Tr}A^T = \mathrm{Tr}A$; that is, the trace of a matrix is equal to the trace of its transpose. Under what circumstances is it true that the trace of the Hermitian conjugate of a matrix is equal to the trace of the original matrix? What is the relationship between the trace of the Hermitian conjugate of a matrix and the original matrix?

13.6.5 Relationship Between the Determinant and the Trace

In the main text, we learned that for Hermitian matrices $\det M = \exp\big(\mathrm{Tr}(\ln M)\big)$. The identity does require that M is nonsingular ($\det M \neq 0$), so that the logarithm is well-defined. It does not require the matrix to be Hermitian, but it is simpler to work with.

There is an equivalent form, which does not use the logarithm of a matrix, so it holds more generally. Replace $M \to \exp(M)$. Then you find that

$$\det\left(e^{M}\right) = \exp\big(\mathrm{Tr}M\big). \tag{13.71}$$

Verify that this is a correct identity. Note that e^{M} is a nonsingular matrix, even if M is singular.

Taking the logarithm of both sides of the original identity gives

$$\ln\big(\det M\big) = \mathrm{Tr}\big(\ln M\big). \tag{13.72}$$

What are the conditions where this equation is valid?

13.6.6 Partial Trace in Pictures

We can think of the partial trace with respect to the second matrix of a tensor product to be expressed graphically as

$$\mathrm{Tr}_B C = \begin{pmatrix} a_{11}\mathrm{Tr}\begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} & a_{12}\mathrm{Tr}\begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \\ a_{21}\mathrm{Tr}\begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} & a_{22}\mathrm{Tr}\begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \end{pmatrix} = (\mathrm{Tr}B)\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}. \tag{13.73}$$

Can you come up with a picture for the partial trace with respect to A?

13.6.7 Alternative Formula for the Partial Trace

Consider the expression for the tensor product of two 2×2 matrices A and B to make a 4×4 matrix C, with the indices c_{ij} running from 1 to 4. Write a concrete expression in terms of the matrix elements of C using the regular indices i and j for the partial trace with respect to B and for the partial trace with respect to A.

13.6.8 Dirac Matrices as Tensor Products

When Dirac worked out an equation for the relativistic quantum mechanics of fermions, he employed four matrices constructed out of tensor products of the Pauli matrices. He used

$$\alpha_x = \sigma_x \otimes \sigma_x, \quad \alpha_y = \sigma_x \otimes \sigma_y, \quad \alpha_z = \sigma_x \otimes \sigma_z, \quad \text{and} \quad \beta = \sigma_z \otimes \mathbb{I}_2. \tag{13.74}$$

Determine all four matrices concretely and then compute their partial traces with respect to A and B.

We will be explaining why these matrices were chosen in this form later when we discuss relativistic quantum mechanics and the Dirac equation.

13.7 Completeness and Orthonormality

The spectral theorem tells us that the eigenvectors of a Hermitian matrix form a complete orthonormal set. This means they are all unit norm, they are mutually orthogonal, and they span the entire space. This means the eigenvectors can be used as a basis for the space. The spectral theorem holds for finite and infinite vector spaces, with a number of technical issues needed for the infinite-dimensional case—the operator must be self-adjoint; Hermitian is insufficient, for example. The spectral theorem is quite an advanced theorem, especially in the infinite-dimensional limit. So, we will not prove it in its most general form. Instead, since we have already been working with it with respect to eigenvectors of operators, we simply summarize how it works for finite-dimensional matrices, providing heuristic arguments, or proofs, where appropriate.

Orthonormality is two properties in one. First, it says all eigenvectors are unit norm. Second, it says that the eigenvectors are mutually orthogonal. Both properties hold for any Hermitian matrix. The proof for the situation where the eigenvalues are nondegenerate is simple. Let $\{\psi^\lambda\}$ be the set of eigenvectors. Then we can evaluate the sum $\sum_{ij}(\psi_i^{\lambda'})^* M_{ij}\psi_j^\lambda$ in two ways. The first way is to let the matrix act to the right, where it becomes $\lambda\psi_i^\lambda$, and to the left, where it becomes $\lambda'(\psi_j^{\lambda'})^*$. Because these two results are equal, we find that

$$(\lambda - \lambda')\sum_i (\psi_i^{\lambda'})^*\psi_i^\lambda = 0. \tag{13.75}$$

Hence, for nondegenerate eigenvalues, where $\lambda \neq \lambda'$, we must have $\sum_i (\psi_i^{\lambda'})^*\psi_i^\lambda = 0$. We have said in the past that for degenerate eigenvectors, we can show that they

can be taken to be mutually orthogonal as well. We will show this later in this section when we discuss Gram-Schmidt orthogonalization. So, this concludes the argument that the eigenvectors must be orthonormal.

This was the easy part of the argument, that the set of eigenvectors is complete is a bit more subtle to argue. First off, if we think physically, we have a n-dimensional vector space. A Hermitian matrix will be an $n \times n$ matrix, with n real eigenvalues, which can be the result of a measurement corresponding to the property of the matrix. Since we have an n-dimensional vector space, and we have n mutually orthogonal eigenvectors, they must span the space. Hence, we must have the completeness relation, that is,

$$\sum_{\lambda} \psi_i^{\lambda} (\psi_j^{\lambda})^* = \delta_{ij}. \tag{13.76}$$

The

$$e_{\text{un}}^2 = \psi^2 - (e^1 \cdot \psi^2) e^1, \tag{13.77}$$

and then normalize it. Note that for a complex vector space, the first vector in the dot product must be complex conjugated. Again, we will use the symbol e^2 to denote the normalized basis vector. This vector is clearly orthogonal to our first vector because we subtracted the component of ψ^2 that lay along the direction of e^1. To go further, we just repeat the process, so

$$e_{\text{un}}^3 = \psi^3 - (e^1 \cdot \psi^3) e^1 - (e^2 \cdot \psi^3) e^2. \tag{13.78}$$

The pattern for how to complete the approach should now be clear. Note that when we do this numerically, if the new basis vector has a very small norm, because most of the corresponding ψ vector lay along the directions of the basis vectors we already determined, the algorithm can lose accuracy. In addition, the algorithm has two ways it can be implemented numerically—one leads to huge errors, while the other is stable. Look up the stable algorithm if you want to implement this yourself. The difference in the algorithms is just the order in which we subtract the projections. So, how do we find the orthonormal eigenvectors when we have degeneracy? For each degenerate subspace, we simply apply the Gram-Schmidt procedure and add the orthonormal vectors we find to the vectors already found for the other eigenvalues. It works like a charm. We have a video https://tinyurl.com/y8aj66w4 discussing more complex ways to look at Gram-Schmidt orthogonalization.

Section 13.7 Problems:

13.7.1 Practice with Gram-Schmidt Orthogonalization I

Use the Gram-Schmidt orthogonalization process on the following vectors:

$$\psi^1 = (1, 1, 0, 0), \quad \psi^2 = (1, 1, 1, -1), \quad \psi^3 = (1, -1, 1, -1), \quad \text{and}$$
$$\psi^4 = (1, 0, 1, -1). \tag{13.79}$$

Note that if there are only three linearly independent vectors, you will find one of your basis vectors is 0 in the Gram-Schmidt process. This means you had more vectors than the span of the vectors.

13.7.2 Practice with Gram-Schmidt Orthogonalization II

Use the Gram-Schmidt process on the following vectors:

$$\psi^1 = (1, 1, i), \quad \psi^2 = (1, 1, -2i), \quad \text{and} \quad \psi^3 = (1, -1, 1). \tag{13.80}$$

Note that these are complex vectors, so do not forget where and when to include a complex conjugation in your calculations.

13.8 Summations of Exponentials, Cosines, Sines, and Powers

In this section, we will be using the geometric series and the sums of powers of integers to determine some powerful identities we need to evaluate time evolution in quantum systems. These identities do lead to the results for the integrals of exponentials, cosines, sines, and powers, but we can get these results without requiring all of the calculus background. For us, it will always look like we are trying to sum up these special series. The only subtlety for what we do is that we will need to evaluate these exact results using some limits at the end. We have discussed the concept of limits before and have used them in a number of different places. We will be careful to proceed slowly through the material so it makes sense to you, even if you do not feel so comfortable with evaluating limits. As you will see from the work we do here, using the limits actually simplifies our results and makes things easier, not harder. I hope you will agree.

Previously, we worked out the geometric series showing that

$$\sum_{j=0}^{n} x^j = \frac{1 - x^{n+1}}{1 - x}. \tag{13.81}$$

The derivation followed by taking the sum, multiplying it by $1 - x$, and seeing that most terms in the sum cancel, except those on the "boundary." Now, we let $x = e^{\alpha \Delta t}$, with α a (possibly complex) number and we take Δt to be a small constant, which we will eventually take the limit as it goes to zero. We find that the geometric series gives us an exact result for the following summation:

$$\Delta t \sum_{j=0}^{n} e^{\alpha j \Delta t} = \Delta t \frac{1 - e^{\alpha (n+1) \Delta t}}{1 - e^{\alpha \Delta t}}. \tag{13.82}$$

This result is exact, but it is messy. We now are going to take our limit. We assume the integer n and the interval Δt vary in such a way that $n \Delta t = t$ is fixed. We can think of this as dividing the interval between 0 and t into n steps, each of length $\Delta t = \frac{t}{n}$. Then, one can see that $n \Delta t = t$ always, for all n. But, as we take n to be very large, then Δt becomes very small. When Δt is very small, we can use the power-series expansion for the exponential to evaluate $e^{\alpha \Delta t} \approx 1 - \alpha \Delta t + \frac{1}{2} \alpha^2 (\Delta t)^2 + \cdots$. If we substitute this into the denominator of Eq. (13.82), we find

$$\Delta t \sum_{n=0}^{n} e^{\alpha j \Delta t} = \frac{\Delta t}{1 - 1 - \alpha \Delta t - \frac{1}{2} \alpha^2 (\Delta t)^2 - \cdots} \left(1 - e^{\alpha (t + \Delta t)}\right). \tag{13.83}$$

Here, we used the fact that $n \Delta t = t$ in the exponent of the last term. Now, you should see that the ones cancel in the denominator, and if we factor a Δt out of the denominator and cancel it with the numerator, we find that

$$\Delta t \sum_{j=0}^{n} e^{\alpha j \Delta t} = \frac{1}{\alpha (1 + \frac{1}{2} \alpha \Delta t + \cdots)} \left(e^{\alpha t} (1 + \alpha \Delta t + \cdots) - 1\right). \tag{13.84}$$

So far, everything we have done is exact. Now we have one more step to work out—we want to take the limit where $n \to \infty$, so that $\Delta t \to 0$. Remember, that when we do this limit, we hold fixed $n \Delta t = t$. To evaluate this limit, any Δt term on the right-hand side that is not multiplied by n is set equal to zero. But, on the left-hand side, we keep all of those terms, just put a limit sign out front. So, our final result is

$$\lim_{n \to \infty} \Delta t \sum_{j=0}^{n} e^{\alpha j \Delta t} = \frac{1}{\alpha} \left(e^{\alpha t} - 1\right), \tag{13.85}$$

which is indeed simpler than the exact result. If you have taken calculus, you will recognize this as the integral of an exponential from 0 to t. But, you do not need to know that jargon at all. In our work, we will simply need to evaluate summations like this and then take the same limit as a last step. Now, we know what the final result is for this procedure!

So, why did we multiply the sum by Δt? You can see that if we double n, the number of terms in the sum doubles. But Δt shrinks in half. So the product of the two is approximately the same. This allows the sum with the extra factor of Δt to be a well-defined number in the limit as $n \to \infty$. If it wasn't there, the result of taking $n \to \infty$ would give us infinity for the sum. This is a subtle point, but it is easy to follow once it has been pointed out to you. These types of issues often arise when we discuss limits.

Having completed the summation for the exponential, the summation for cosines or sines is simple. We let α be imaginary $\alpha \to i\alpha$, and using the Euler formula, we take the real part for a cosine and an imaginary part for the sine summations. Hence,

$$\lim_{n\to\infty} \sum_{j=0}^{n} \cos(\alpha j \Delta t) = \text{Re}\left[\frac{1}{i\alpha}\left(e^{i\alpha t} - 1\right)\right] = \frac{1}{\alpha}\sin(\alpha t), \tag{13.86}$$

and

$$\lim_{n\to\infty} \sum_{j=0}^{n} \sin(\alpha j \Delta t) = \text{Im}\left[\frac{1}{i\alpha}\left(e^{i\alpha t} - 1\right)\right] = \frac{1}{\alpha}\big(1 - \cos(\alpha t)\big). \tag{13.87}$$

It is as simple as that!

The summation of powers is a bit more complicated. I have to mention that in the early 1990s there was a rap song by the group Snap! called *The Power*. It had a haunting chorus by singer Penny Ford who kept repeating "I've got the power" to a beat that everyone wanted to exercise to. So, for years, when I did step aerobics, I would hear "I've got the power." As we move here to calculate the sums of powers, perhaps you should get into the mood, by listening to their song.

There is a story that when Gauss was a young child, his teacher gave the class a problem to sum all of the integers from 1 to 100. Thinking that they would do it the long way, the task was expected to take hours to complete. Young Gauss figured out a neat way to do it that gave him the answer in seconds. When I was in elementary school, my teacher told us this story. I was highly motivated to figure it out, and I did. Now I will remind you how it is done, because we have seen it before.

Define

$$\text{sum}_k(n) = \sum_{j=1}^{n} j^k = \text{ sum of first n integers raised to the k power.} \tag{13.88}$$

The simplest case is $k = 1$.

$$\text{sum}_1(n) = 1 + 2 + 3 + ... + n. \tag{13.89}$$

We can write $\text{sum}_1(n)$ in reverse order underneath, and add down the columns:

$$\begin{aligned} \text{sum}_1(n) &= 1 + \quad 2 \quad + \quad 3 \quad + \ldots + n \\ \text{sum}_1(n) &= n \; + \; (n-1) \; + \; (n-2) + \ldots + 1 \end{aligned}$$

$$2 \times \text{sum}_1(n) = \underbrace{(n+1) + (n+1) + (n+1) + ... + (n+1)}_{n \text{ terms}} \tag{13.90}$$

There are n terms of $(n+1)$, so

$$\text{sum}_1(n) = \frac{n(n+1)}{2}. \tag{13.91}$$

This result is very direct, simple, and neat. The trick is to recognize that when we add the series of terms in opposite orders, each sum is the same, so we need to only determine how many of those combinations we have.

Unfortunately, this approach is not easily generalized to other k powers. So we have to proceed a different way. We do so by computing differences of finite sums in two different ways. This is motivated by how we found the geometric series by multiplying by $1 - x$ and having most terms in the summation cancel except the "boundary terms." Consider

$$\sum_{j=1}^{n} \left[(j+1)^2 - j^2 \right] = (n+1)^2 - 1 = n^2 + 2n, \tag{13.92}$$

which follows since many of the terms (all except the first term of the second sum and the last term of the first sum) cancel when the sums are subtracted from each other. Now, we compute a second way, by expanding the terms in the sums (this is mathematically allowed because all sums are *finite*). We find

$$\sum_{j=1}^{n} \left[(j+1)^2 - j^2 \right] = \sum_{j=1}^{n} \left(\cancel{j^2} + 2j + 1 - \cancel{j^2} \right) = \sum_{j=1}^{n} 2j + \sum_{j=1}^{n} 1. \tag{13.93}$$

We realize that

$$\sum_{j=1}^{n} 2j = 2 \sum_{j=1}^{n} j \tag{13.94}$$

and

$$\sum_{j=1}^{n} 1 = n. \tag{13.95}$$

So we immediately discover (after equating the two ways to evaluate the summations) that

$$2\,\mathrm{sum}_1(n) = n^2 + 2n - n = n(n+1) \tag{13.96}$$

so

$$\mathrm{sum}_1(n) = \frac{n(n+1)}{2} \tag{13.97}$$

as before.

But here is the beauty—this strategy can be generalized to higher powers of k.

We can check the next case where we work with cubes instead of squares. We have

$$\sum_{j=1}^{n} \left[(j+1)^3 - j^3 \right] = (n+1)^3 - 1 = n^3 + 3n^2 + 3n \tag{13.98}$$

and

$$\sum_{j=1}^{n} \left[(j+1)^3 - j^3 \right] = \sum_{j=1}^{n} \left(\cancel{j^3} + 3j^2 + 3j + 1 - \cancel{j^3} \right) \tag{13.99}$$

$$= 3\,\mathrm{sum}_2(n) + 3\,\mathrm{sum}_1(n) + \mathrm{sum}_0(n) \tag{13.100}$$

$$= n^3 + 3n^2 + 3n. \tag{13.101}$$

So

$$\mathrm{sum}_2(n) = \frac{1}{3}\left(\underbrace{n^3 + 3n^2 + 3n}_{\text{sum first way}} \underbrace{-\frac{3}{2}n^2 - \frac{3}{2}n}_{-3\,\mathrm{sum}_1(n)} \underbrace{-n}_{-\mathrm{sum}_0(n)} \right) \tag{13.102}$$

$$= \frac{1}{3}\left(n^3 + \frac{3}{2}n^2 + \frac{1}{2}n \right) \tag{13.103}$$

$$= \frac{1}{6}n\left(2n^2 + 3n + 1 \right) \tag{13.104}$$

$$= \frac{1}{6} n\,(2n+1)\,(n+1)\,. \tag{13.105}$$

A mathematician named Cavalieri extended this to all the way to $k = 10$. We need to use the results for all previous powers to obtain the final answer for a given k. Note that there does not seem to be any simple pattern to these final answers, so we do not try to establish them by induction. We have a video discussing Summation of Powers https://tinyurl.com/2ccak38c.

So far, we have not used any limits. That enters if we wish to sum

$$\Delta t \sum_{j=1}^{n} (j\,\Delta t)^k, \tag{13.106}$$

which can be thought of as the sum of time values raised to a power for a discretization of the time axis into n pieces. The answer is, of course, $(\Delta t)^{k+1}\text{sum}_k$. We have found that sum_k is a polynomial in n of degree $k+1$: $\text{sum}_k = \sum_{j=0}^{k+1} c_j n^j$. But, if we multiply this by $(\Delta t)^{k+1}$ and take the limit as $n \to \infty$, we find only c_{k+1} enters into the answer. This will be worked out in detail in the problems.

The results we have found here will be quite helpful when we get to time evolution of quantum systems in the next chapter.

Section 13.8 Problems:

13.8.1 Exponential Sum over Negative and Positive Times

In the text, we derived a summation for the exponentials that ranged over values of time ranging from 0 to t. Extend this result to perform the sum to extend for times from $-t$ to t.

Hint: The case from $-t$ to zero can be expressed in terms of the case from 0 to t, with $\alpha \to -\alpha$. Be careful to track your signs in the calculation.

Ans. $\frac{1}{\alpha}\left(e^{\alpha t} - e^{-\alpha t}\right)$.

13.8.2 Summation of the Product of Cosine and Sine

Perform the following summations. Take the limit as n goes to infinity with $n\Delta t = t$, as before.

$$\text{(a)} \quad \Delta t \sum_{j=0}^{n} \cos(\alpha j\,\Delta t)\,\sin(\alpha j\,\Delta t), \tag{13.107}$$

$$\text{(b)} \quad \Delta t \sum_{j=0}^{n} \cos^2(\alpha j \Delta t), \tag{13.108}$$

$$\text{(c)} \quad \Delta t \sum_{j=0}^{n} \sin^2(\alpha j \Delta t). \tag{13.109}$$

Hint: The trigonometric double-angle formulas will come in handy here.

13.8.3 Finite Summation of the Third and Fourth Power of an Integer

Following the general procedure outlined in the text, compute

$$\text{sum}_3 = \sum_{j=1}^{n} j^3 \text{ and } \text{sum}_4 = \sum_{j=1}^{n} j^4. \tag{13.110}$$

13.8.4 Taking the Limit of Sums of Powers

Show that

$$\lim_{n\to\infty} \Delta \sum_{j=1}^{n} (j\Delta t)^k = c_k t^{k+1}. \tag{13.111}$$

Determine c_k for the general case.

13.9 Counting Approach to the Born Rule

Earlier in this book, when we were developing quantum mechanics for particles, we made some heuristic arguments to justify why $[\hat{x}, \hat{p}] = i\hbar$. It wasn't an ironclad derivation, but it was a pretty strong argument for why the canonical commutation relation is what it is. Similarly, when we discussed how we perform measurements, in particular projective measurements, we used the Born rule, which assigns a probability for a particular eigenstate to be measured to be given by the modulus squared of the amplitude of that eigenstate in the current quantum state. At that time, we took the Born rule as a postulate. Here, we provide another heuristic argument based on counting that "derives" the Born rule.

Back as early as the 1950s, a theorem, now know as *Gleason's theorem*, was proved. This theorem showed that if you want to assign probabilities to unit vectors, within the framework of quantum mechanics, then you must do so using the Born rule. It requires that the theory be *noncontextual*, which means the results of measurements should depend only on the eigenvectors being projected on. The theorem works for all quantum systems with three or more degrees of freedom. But, this theorem is difficult to state, let alone prove. It does not provide much insight into why we have the Born rule in the first place. Instead, we proceed here using what is essentially a counting argument to derive the Born rule. Again, the arguments we will provide appear sound, but whether this is a true derivation or just a heuristic argument is currently a matter of debate in the community. The argument is simple and clear, so let's get on with it.

The key assumption is that we must work with unit vectors. Let's suppose I have a Hermitian operator $\hat{O}$ that I want to measure. It has a complete set of eigenvectors that form an orthonormal basis. Let's assume that the state is given by

$$|\psi\rangle = \frac{1}{\sqrt{2}}|\phi_1\rangle + \frac{1}{\sqrt{2}}|\phi_2\rangle, \tag{13.112}$$

a state where either eigenstate $|\phi_1\rangle$ or $|\phi_2\rangle$ is equally likely. The Born rule says we take the coefficient of each state and square it, and that yields the corresponding probabilities, which in this case are $\frac{1}{2}$ for each. The counting argument, instead, says we have two *equally likely* possibilities, so we must have equal probability for each. Hence, the probability is 50% for each.

You might be saying to yourself that this seems to be completely obvious, as we said the probability was equal for both from the beginning. So, to continue the argument, consider the state

$$|\psi\rangle = \frac{1}{\sqrt{3}}|\phi_1\rangle + \frac{\sqrt{2}}{\sqrt{3}}|\phi_2\rangle. \tag{13.113}$$

The Born rule says the first state has probability $\frac{1}{3}$ and the second has probability $\frac{2}{3}$. How do we make this argument with the counting approach? Well, the counting approach assumes that when the coefficients of each state in the expansion are the same, the probability for each state is the same. But, we do not have that situation in this case, so what do we do? We convert this case into that situation. So, we create a fictitious state $\frac{1}{\sqrt{2}}|\chi_1\rangle + \frac{1}{\sqrt{2}}|\chi_2\rangle$ and we replace the unit norm state $|\phi_2\rangle$ by this fictitious state. We then find that our state can be written as

$$|\psi\rangle = \frac{1}{\sqrt{3}}(|\phi_1\rangle + |\chi_1\rangle + |\chi_2\rangle). \tag{13.114}$$

Now, we have an expansion in terms of equal amplitudes, so the probability for each state must be the same, namely, $\frac{1}{3}$. But, to compute the probability for $|\phi_2\rangle$, we must add the probabilities of each fictitious state used in the expansion. This gives us a

probability of $\frac{1}{3} + \frac{1}{3} = \frac{2}{3}$ for $|\phi_2\rangle$. And, once again, we have derived the Born rule by counting. This rule can be generalized for arbitrary rational probabilities, as you will explain in the problems below.

What about cases where the probabilities are not rational numbers? In this case, we simply insist that our probability mapping be continuous. Since the rational numbers form a dense set in the real numbers, the Born rule for all cases follows by the continuity requirement.

This is pretty cool. We have managed to boil down the association of a probability as the modulus squared of the probability amplitude simply by requiring the states to be unit vectors, requiring all states to have equal probability (including the fictitious ones we introduce), and then counting. But this derivation still leaves us feeling somewhat flat, as it does not explain why the states should be unit vectors, and why all states have equal probability to be measured. In other words, it provides no intuition as to why we have the Born rule. But, as a mathematical consequence of our theory, it is a pretty neat way to justify the Born rule via a simple egalitarian counting argument.

Section 13.9 Problems:

13.9.1 Generalizing Our Arguments to the Arbitrary Rational Number Case

Consider the general case

$$|\psi\rangle = \sqrt{\frac{p}{q}}|\phi_1\rangle + \sqrt{\frac{q-p}{q}}|\phi_2\rangle. \tag{13.115}$$

Use the counting argument, by replacing both states by the appropriate fictitious states, to prove the Born rule holds for this expansion. Discuss how one can generalize this argument to an expansion of the state $|\psi\rangle$ in terms of an arbitrary number of quantum states, as long as the coefficients of each state are given by the square root of rational fractions.

13.9.2 A Subtlety of the Argument

When we replace a state $|\psi_2\rangle$, by two states $|\psi_2\rangle = \frac{1}{\sqrt{2}}(|\chi_1\rangle + |\chi_2\rangle)$, we implicitly assumed that we can find states in the final expansion that are all mutually orthogonal. Write an essay discussing this issue and thinking about whether it is likely to hold as $q \to \infty$ for the general case of rational numbers. Note that this is part of the reason why the argument is more heuristic than rigorous.

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Chapter 14
Time Evolution

Everybody has to learn that the world works in very specific ways, and you're not in charge of it.

Samuel L. Jackson, Actor

Until now, we have been working with time-independent states. The real world is dynamic and it changes from one time to another. The quantum world provides a basis for the real world, so it has dynamics too. In this chapter, we discuss the time evolution of quantum states and a summary is available at https://tinyurl.com/mus52buy. It may surprise you to learn that many research problems in quantum mechanics involve trying to understand the time evolution of complex quantum states. We still do not have an efficient way to determine the dynamical properties of complicated quantum-mechanical systems. You will get your first taste of how we describe time variation of quantum states in this chapter and why the analysis can get complicated.

14.1 Evolution Operator for Time-Independent Hamiltonians

In many treatments of quantum mechanics, the time-dependent Schrödinger equation is postulated. It is a first-order differential equation, which describes how quantum systems change as a function of time. But, we are presenting quantum mechanics without calculus, so we certainly want to avoid working with a differential equation. So, what do we do? Our approach is to first outline a number of requirements that time evolution must have, and then from those requirements, we will be able to propose how to determine the time evolution of a system. We will do this without needing to discuss any differential equations.

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The first requirement that we have is that time evolution preserves the total probability of the quantum state. If you recall, the fact that something must happen, and that the probability for all possible occurrences must add up to one, we find that the normalization of a quantum state must not change during time evolution. So, if we have a quantum state at time t_0, which we denote as $|\psi(t_0)\rangle$, then if the time-evolution operator is denoted by $\hat{U}(t, t_0)$, and it carries the quantum state from time t_0 to time t via

$$|\psi(t)\rangle = \hat{U}(t, t_0)|\psi(t_0)\rangle, \tag{14.1}$$

we must have

$$\langle\psi(t)|\psi(t)\rangle = \langle\psi(t_0)|\hat{U}^\dagger(t, t_0)\hat{U}(t, t_0)|\psi(t_0)\rangle = 1. \tag{14.2}$$

Since this result must hold for all possible initial states $|\psi(t_0)\rangle$, we must have $\hat{U}^\dagger(t, t_0)\hat{U}(t, t_0) = \hat{\mathbb{I}}$. As we recall, unitary operators satisfy this property, so we require the time-evolution operator to be unitary.

Another property of time evolution is encapsulated in what is called the *semigroup rule*. While this sounds formidable, the concept is quite simple. We simply say that if we time evolve from t_0 to t_1 that must be the same as time evolving from t_0 to t_2 and then from t_2 to t_1. This must hold for all values of t_2, even those that do not lie in between t_0 and t_1. This condition is completely reasonable, and we should require it, but it is not so simple to see how to guarantee it. To get more concrete, we assume that we can write our time-evolution operator as the exponential of a Hermitian operator in the form $\hat{U}(t_1, t_0) = e^{-i(t_1-t_0)\hat{O}}$. Then, the Hermiticity of $\hat{O}$ guarantees that the operator is unitary, while the linear dependence on time in the exponential guarantees that we satisfy the semigroup rule. Note that this is not the most general way that we can satisfy the semigroup rule, but we will see this is general enough to let us determine the time-evolution operator. If you think carefully about this, a similar argument could be given for spatial or momentum translations, with the same result, implying that you have seen this concept already. So, what operator should go in the exponent? If we use the correspondence principle, the dynamics of a classical system is determined from its Hamiltonian. So, we require this to be the same for the quantum system. But, the argument of the exponential must be dimensionless, so we quickly see that $\hat{O} = \frac{1}{\hbar}\hat{H}$ is our requirement as well. What about the sign in the exponent? It is chosen, by convention, to be a minus sign. This is motivated by the fact that a wave with a wavenumber (or momentum to the left) will actually move to the left under time evolution, as we want it to. This simple form we have "derived" will only work for a time-independent Hamiltonian, so we have our postulate for time evolution of a system with a time-independent Hamiltonian to be

$$\hat{U}(t, t_0) = e^{-\frac{i}{\hbar}(t-t_0)\hat{H}}. \tag{14.3}$$

This evolution operator is unitary, it is governed by the Hamiltonian operator, and it satisfies the semigroup rule. So, it has all of the elements that we require. As

before with our "derivation" of the canonical commutation relation, the derivation we provided here for the evolution operator is really a motivation for the final result. But, this is no different than postulating the time-dependent Schrödinger equation, which then is solved with the above time-evolution operator for a time-independent Hamiltonian.

You may have heard that quantum mechanics requires us to use complex numbers. But, for all that we worked on up to now, with a few exceptions (like spin states in the y-direction), we were able to get away with working with real numbers only. Once we get to time evolution, this can no longer be done. In fact complex numbers are absolutely required to describe the dynamics of quantum states. The reason why is simple. To get a unitary operator, I must exponentiate i times a Hermitian operator. Then, I will have complex numbers occurring everywhere.

Now that we have determined the time-evolution operator for a time-independent Hamiltonian, how do we extend it to a time-dependent Hamiltonian? The idea is to take our time interval $t - t_0$ and divide it into N equally spaced points, with a size $\Delta t = \frac{t-t_0}{N}$. Then, we approximate a time-dependent Hamiltonian $\hat{H}(t)$ as the piecewise constant Hamiltonian

$$\hat{H}(t) \approx \hat{H}\left(\tfrac{t_i+t_{i-1}}{2}\right) \quad \text{for } t_i < t < t_{i-1}. \tag{14.4}$$

Then for each "piece," we can use the time-independent result for the time-evolution operator to create the *Trotter product formula*, or the *time-ordered product*, as

$$\begin{aligned}\hat{U}(t,t_0) &= U(t,t-\Delta t)\hat{U}(t-\Delta t,t-2\Delta t)\cdots \\ &\quad \hat{U}(t,t_0+2\Delta t,t_0+\Delta t)\hat{U}(t+\Delta t,t_0) \\ &= e^{-\frac{i\Delta t}{\hbar}\hat{H}\left(t-\frac{\Delta t}{2}\right)}e^{-\frac{i\Delta t}{\hbar}\hat{H}\left(t-\frac{3\Delta t}{2}\right)}\cdots e^{-\frac{i\Delta t}{\hbar}\hat{H}\left(t_0+\frac{3\Delta t}{2}\right)}e^{-\frac{i\Delta t}{\hbar}\hat{H}\left(t_0+\frac{\Delta t}{2}\right)}.\end{aligned} \tag{14.5}$$

The mnemonic to remember the time-ordered product is that "later times are placed to the left." In this case, we have evaluated the Hamiltonian with its value at the center of each time interval. We might want to change this to using the right endpoint, or the left endpoint. We can call these different approximations the midpoint Trotter formula, the rightpoint Trotter formula, or the leftpoint Trotter formula. This generalization is easy to work out. Note that in the limit as $\Delta t \to 0$, all different discretizations will yield the same final result. This formula for the time-ordered product was first worked out by Hale Trotter, a Professor of Mathematics at Princeton University, in the mid-1950s (see Fig. 14.1) in the context of products of Lie group elements in their exponential form. The Trotter Product Formula video https://tinyurl.com/42zbunwe discusses some more ideas on this.

The result we have worked out might appear straightforward to you, but it is a highly nontrivial result. This is because the time-evolution operator is an operator that transforms any quantum state at one time to another time. This is an incredibly powerful operator. It will take us quite a bit of time to become familiar with it and to be comfortable using it to analyze the dynamics of quantum systems. We start by examining how it behaves on energy eigenstates of a time-independent Hamiltonian.

Fig. 14.1 Hale Trotter in Berkeley, California, in 1978. [Photograph taken by George M. Bergmann, in the Oberwolfach Photo Collection, Creative Commons License Attribution-Share Alike 2.0 Germany. Converted to look like a sketch by James Freericks.]

We will then examine time-dependent problems that map onto time-independent ones before we move into the general time-dependent case. Time evolution remains a complicated problem that is an active research area, especially for systems of many quantum particles that interact strongly with each other.

We begin by discussing how energy eigenstates and then superpositions of energy eigenstates evolve under time. Energy eigenstates are only well defined for time-independent Hamiltonians, so we will restrict ourselves to this special case first. Suppose I have a time-independent Hamiltonian $\hat{H}$, with energy eigenstates that satisfy $\hat{H}|n\rangle = E_n|n\rangle$, with E_n the energy eigenvalue for the energy eigenstate $|n\rangle$. A general state $|\psi\rangle$ can be expanded in the energy eigenstate basis (because it is complete) via

$$|\psi\rangle = \sum_{n=0}^{\infty} c_n|n\rangle \text{ with } c_n = \langle n|\psi\rangle. \tag{14.6}$$

This result follows from just using the completeness relation for the energy eigenstates. It does assume that all of the bound states form a complete set; there are some potentials which have bound and continuum energy eigenstates, but we will not worry about that complication here.

To determine the time-evolved state, we simply operate the time-evolution operator onto the state $|\psi\rangle$. For simplicity in our formulas, we will set $t_0 = 0$. Suppose, at first, that the state is a single energy eigenstate. Then

$$|\psi_n(t)\rangle = e^{-\frac{i}{\hbar}\hat{H}t}|n\rangle = e^{-\frac{i}{\hbar}E_n t}|n\rangle. \tag{14.7}$$

An energy eigenstate evolves by having its complex phase change as a function of time. But, because the global phase of a wavefunction does not change any properties of the wavefunction, these states act the same for all times. This motivates our calling them *stationary states*. Because a bra is the Hermitian conjugate of a ket, the expectation value of any operator with respect to this state is time independent. For we have that

$$\langle\psi_n(t)|\hat{O}|\psi_n(t)\rangle = \langle n|e^{\frac{i}{\hbar}E_n t}\hat{O}e^{-\frac{i}{\hbar}E_n t}|n\rangle = \langle n|\hat{O}|n\rangle. \tag{14.8}$$

The expectation value of any operator in a stationary state does not depend on time!

Things become more interesting when we look at a state $|\psi\rangle$ which is a superposition of energy eigenstates. Then, we find the state evolves as a function of time in a nontrivial way via

$$|\psi(t)\rangle = e^{-\frac{i}{\hbar}\hat{H}t}|\psi\rangle = \sum_{n=0}^{\infty} c_n e^{-\frac{i}{\hbar}E_n t}|n\rangle, \tag{14.9}$$

where we used the fact that $\hat{H}|n\rangle = E_n|n\rangle$. Note that in this case, the time evolution is more complicated than just multiplying by a complex phase when we have at least two c_n's that are nonzero and that have different energy eigenvalues. If we now look at the expectation value of an operator, we find a nontrivial time dependence:

$$\langle\psi(t)|\hat{O}|\psi(t)\rangle = \sum_{m=0}^{\infty}\sum_{n=0}^{\infty} c_m^* c_n e^{\frac{i}{\hbar}(E_m - E_n)t}\langle m|\hat{O}|n\rangle. \tag{14.10}$$

In this formula, you can see it is the differences in the energy eigenvalues that determine the time dependence for the time evolution of the expectation value of an operator.

The formula for time evolution is complicated and opaque. To get a sense for how it works, we will consider a simple example from the particle in a box. We learned that the wavefunctions (and energy eigenvalues) for a particle in a box of length L were given by

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{\pi n x}{L}\right) \quad \text{and} \quad E_n = \frac{\hbar^2\pi^2 n^2}{2ML^2}, \tag{14.11}$$

with $n = 1, 2, \cdots$. This is the case where the box ranges from $0 \le x \le L$. We will consider a superposition consisting of equal amounts of the ground state and the first excited state, so

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}}e^{-\frac{i}{\hbar}E_1 t}\left(|1\rangle + e^{\frac{i}{\hbar}(E_1 - E_2)t}|2\rangle\right), \tag{14.12}$$

and the time-evolved wavefunction becomes

$$\begin{aligned}\psi(x,t) = \langle x|\psi(t)\rangle = \frac{1}{\sqrt{L}}\exp\left(-\frac{i\hbar\pi^2 t}{2ML^2}\right)\\ \times\left(\sin\left(\frac{\pi x}{L}\right) + \exp\left(-\frac{3i\hbar\pi^2 t}{2ML^2}\right)\sin\left(\frac{2\pi x}{L}\right)\right).\end{aligned} \tag{14.13}$$

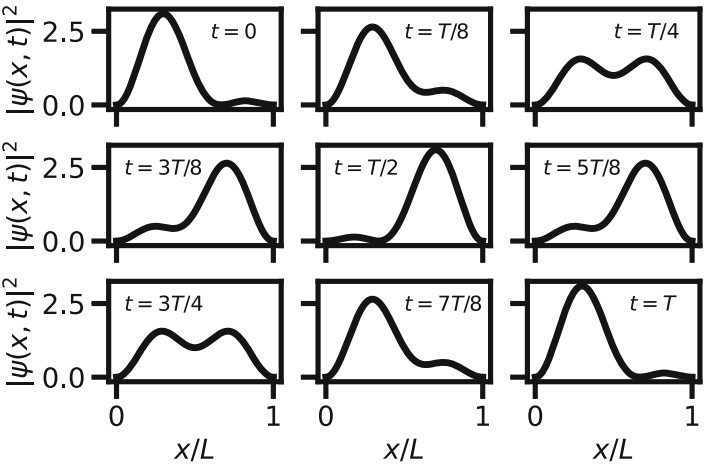

Fig. 14.2 Snapshots of the probability distribution for the particle in a box as a function of time over one period T of the motion using the superposition discussed in the text. Note how the main peak in the probability sloshes from the left to the right and back again

The time-dependent wavefunction is complex valued as we mentioned above. To better understand how it behaves, we plot the probability distribution $|\psi(x,t)|^2$ for different times in Fig. 14.2. You can see that the motion is periodic, with a period given by $T = \frac{4ML^2}{3\hbar\pi}$. You also can see that the probability distribution sloshes back and forth as a function of time. This is perhaps how you might envision motion in a particle in a box to behave, as the particle can never escape the box, so its motion must move back and forth. Note that this superposition describes the motion of a state whose energy is indeterminate. This sloshing motion disappears for definite energy states, such as the ground state, because they are stationary. But stationary states still have nonzero expectation values for the kinetic energy. This means we cannot sense motion of a quantum particle just by looking at how probability distributions change with time. It is more complicated than that, and not easy to visualize. This has led people to say quanta are fluctuating fuzzy blobs, but that is not a great image either. As with many quantum phenomena, it is difficult to have a clear mental image for how quanta behave. We just have to live with this.

Section 14.1 Problems:

14.1.1 Matrices Representing the Time-Evolution Operator Are Unitary

In the main text, we argued that the operator for time evolution must satisfy $\hat{U}^\dagger \hat{U} = \hat{\mathbb{I}}$. Represent this operator equation as a matrix equation using a complete set of states as the basis, and show that the matrices satisfy $U^\dagger U = \mathbb{I}$ as a matrix

multiplication equation. This requires the matrices that represent the time-evolution operators to be unitary.

14.1.2 Functional Forms That Satisfy the Semigroup Rule

Verify the semigroup rule holds for the form $e^{i(t-t')\hat{O}}$. Verify that it holds for the Trotter formula for time-varying Hamiltonians, even though we might have $[\hat{H}(t), \hat{H}(t')] \neq 0$. Can you think of any other functional form that satisfies the semigroup rule?

14.1.3 Inner Products Between Time Evolved States

Show that the inner product between two time-evolving states in the vector space are independent of time because the time-evolution operator is unitary. Discuss how this occurs and what it means for completeness and orthonormality of a complete basis as a function of time.

14.1.4 Time Dependence of the Expectation Value of Operators

The time evolution of an energy eigenstate involves just multiplication by a global complex phase, so expectation values of any operator with respect to a time-varying energy eigenstate will be independent of time. This is why they are called stationary states. But superpositions of energy eigenstates will have nontrivial time dependence. Write a short essay that discusses this phenomena.

14.1.5 Time Evolution of the Simple Harmonic Oscillator

Consider an arbitrary superposition of simple-harmonic-oscillator energy eigenstates. Show that the motion is always periodic, with a period of $T = \frac{2\pi}{\omega}$. Under what circumstances can the period of the motion be even shorter?

14.1.6 Time Evolution of the Particle in a Box

For a general superposition of particle in a box energy eigenstates, the motion will not be periodic. But, if the superposition includes only a finite number of terms, it will be periodic. Determine the period for such a finite superposition.

Hint: You will need to use the concept of the *least common multiple*.

14.1.7 Taking the Limit of $\Delta t \to 0$ in the Trotter Formula

The Trotter product formula becomes more and more accurate as we increase the number of steps for a fixed time interval. Write an essay that describes how we should think of the Trotter product formula in the limit as $\Delta t \to 0$. Can you explain in words what the limit means and how we could work with such an abstract object. Under what circumstances could we actually take the limit explicitly?

14.2 Magnetic Resonance

The problem of *magnetic resonance* is one of the simplest time-dependent problems that can be solved exactly. In magnetic resonance, we have a spin (which can be an electron spin or a nuclear spin) that precesses in a large uniform magnetic field. Then a small time-dependent field that rotates in the plane perpendicular to the large field is added to the system, and when the rate of rotation of the small field is equal to the rate of precession of the spin, then in the rotating frame of reference, we see two static fields—the large uniform one and the small one perpendicular to the large one. In the rotating frame, we have no precession around the large field anymore. The precession around the small field will have the spin flip from up to down, then back to up, and so on. This flipping of the spin is called Rabi flopping after Isidor Rabi, a Nobel Prize-winning physicist, who developed the field of atomic beam physics, which extended the Stern-Gerlach experiment broadly. Isidor Rabi is shown in Fig. 14.3. In this section, we solve a mathematical model for magnetic resonance.

But, before we jump into the full problem, we start by looking at the simpler problem of precession in a large uniform field. Recall that the Hamiltonian of a magnetic dipole in a magnetic field is given by minus the dot product of the magnetic dipole with the magnetic field. The magnetic dipole moment is proportional to the intrinsic spin of the particle and is given by

$$\hat{\vec{\mu}} = \frac{gq\hbar}{2M} \frac{\hat{\vec{S}}}{\hbar}, \tag{14.14}$$

which is expressed in terms of a dimensionless spin operator that we have split off in the fraction to the right. The term g is the g-factor, which is approximately 2 for an electron (we will see the Dirac equation derives it to be exactly 2, but it is modified by quantum electrodynamics corrections). For the proton, it is 5.586, while for the neutron, it is -3.826. The symbol q is the charge of the particle and M is its mass (for the neutron we use $q = e$ even though it is charge neutral). We will focus on the general magnetic resonance of quantum particle in this section and will discuss nuclear magnetic resonance imaging in the next section.

Fig. 14.3 Isidor Rabi (Nobel Prize image, 1944). [Public domain. Converted to look like a sketch by James Freericks]

So, the Hamiltonian for precession in a uniform field, oriented along the z-direction is

$$\hat{H}_{\text{prec}} = -\hat{\vec{\mu}} \cdot \vec{B} = -\frac{ge\hbar}{2M} \frac{\hat{S}_z}{\hbar} B_0 = -\gamma B_0 \frac{\hat{S}_z}{\hbar} = -\Omega \hat{S}_z. \tag{14.15}$$

There are a number of important symbols defined here. The symbol $\gamma = \frac{ge\hbar}{2M}$ is called the *gyromagnetic ratio*, while $\Omega = \frac{\gamma B_0}{\hbar}$ is called the Larmor frequency. Note that there is a dependence on the inverse mass of the particle in the magnetic dipole moment. This means that the magnetic dipole of a proton is approximately one thousand times smaller than that of an electron. The strength of the dipole moment of the electron is summarized in terms of the *Bohr magneton* μ_B, which satisfies $\mu_B = \frac{e\hbar}{2M_e} \approx 5.79 \times 10^{-5} \frac{\text{ev}}{\text{Tesla}}$. This tells you that large magnetic fields are needed to significantly change the energy of an electron (and substantially larger fields for nuclei). We will describe all of the results in terms of the Larmor frequency in this section. If the quantum particle is negatively charged, or has an effective negative magnetic moment, the Larmor frequency will be negative to work with the formulas developed here.

Now, when we describe a spin s particle, the states of definite spin and definite z-component of spin are denoted $|s, m\rangle$; when we add in time dependence, we will denote them as $|s, m; t\rangle$. These states are eigenstates of the Hamiltonian, with $E_m = -\hbar\Omega m$. This implies the time evolution of these stationary states is given by

$$|s, m; t\rangle = e^{im\Omega t} |s, m; t{=}0\rangle. \tag{14.16}$$

This describes the precession of spin about the axis of the magnetic field.

Now we are ready to describe magnetic resonance. The precession of the spin corresponds to a complex phase that rotates in the clockwise direction. To create a resonance effect, where the rotating magnetic field rotates at the same rate as the precession occurs, we need to rotate the time-dependent field in the clockwise direction as well. So, the Hamiltonian for magnetic resonance is given by

$$\hat{H} = -\Omega \hat{S}_z - \gamma B_1 \left(\cos(\omega t) \frac{\hat{S}_x}{\hbar} - \sin(\omega t) \frac{\hat{S}_y}{\hbar} \right). \tag{14.17}$$

Here, B_1 is the magnitude of the time-dependent field, which rotates in the $x - y$ plane. The time-dependent term can be rewritten in terms of the Hadamard lemma, as we described in Sect. 9.2. The Hamiltonian then becomes

$$\hat{H} = -\Omega \hat{S}_z - \frac{\gamma B_1}{\hbar} e^{\frac{i}{\hbar}\omega t \hat{S}_z} \hat{S}_x e^{-\frac{i}{\hbar}\omega t \hat{S}_z}. \tag{14.18}$$

Because the Hadamard lemma is using the z-component of spin, we can also apply it to the first term in the Hamiltonian. Then we find that we have the identity

$$\hat{H} = -e^{\frac{i}{\hbar}\omega t \hat{S}_z} \left(\Omega \hat{S}_z + \frac{\gamma B_1}{\hbar} \hat{S}_x \right) e^{-\frac{i}{\hbar}\omega t \hat{S}_z}. \tag{14.19}$$

We are now ready to determine the time evolution for the magnetic resonance problem. We will work with the Trotter factors of the form

$$\begin{aligned} e^{-\frac{i}{\hbar}\hat{H}(t)\Delta t} &= \exp\left(\frac{i}{\hbar} e^{\frac{i}{\hbar}\omega t \hat{S}_z} \left(\Omega \hat{S}_z + \frac{\gamma B_1}{\hbar} \hat{S}_x \right) e^{-\frac{i}{\hbar}\omega t \hat{S}_z} \Delta t \right) \\ &= e^{\frac{i}{\hbar}\omega t \hat{S}_z} \exp\left(\frac{i}{\hbar} \left(\Omega \hat{S}_z + \frac{\gamma B_1}{\hbar} \hat{S}_x \right) \Delta t \right) e^{-\frac{i}{\hbar}\omega t \hat{S}_z}, \end{aligned} \tag{14.20}$$

where we used the braiding relation to move the Hadamard lemma outside the exponential. Each Trotter factor depends on time, but, as you can see by the above result, the time only enters into the first and third exponential factor. Now, when we multiply the third Trotter factor for t and the first Trotter factor for $t - \Delta t$ together, the middle exponential factors partially cancel, replacing the time in the exponent by Δt. Hence, we have

$$\begin{aligned} \hat{U}(t) = e^{\frac{i}{\hbar}\omega t \hat{S}_z} e^{\frac{i}{\hbar}\left(\Omega \hat{S}_z + \frac{\gamma B_1}{\hbar}\hat{S}_x\right)\Delta t} e^{-\frac{i}{\hbar}\omega \hat{S}_z \Delta t} e^{\frac{i}{\hbar}\left(\Omega \hat{S}_z + \frac{\gamma B_1}{\hbar}\hat{S}_x\right)\Delta t} e^{-\frac{i}{\hbar}\omega \hat{S}_z \Delta t} \cdots \\ \times \underbrace{e^{\frac{i}{\hbar}\left(\Omega \hat{S}_z + \frac{\gamma B_1}{\hbar}\hat{S}_x\right)\Delta t} e^{-\frac{i}{\hbar}\omega \hat{S}_z \Delta t}}_{\text{repeated pair}}. \end{aligned} \tag{14.21}$$

We have set $t_0 = 0$ for simplicity again. You can see that after the leftmost term, all the remaining terms in the product are paired and repeated. We use the Baker-Campbell-Hausdorff formula on each pair to obtain

$$e^{\frac{i}{\hbar}\left(\Omega\hat{S}_z+\frac{\gamma B_1}{\hbar}\hat{S}_x\right)\Delta t}e^{-\frac{i}{\hbar}\omega\hat{S}_z\Delta t} = e^{\frac{i}{\hbar}\left((\Omega-\omega)\hat{S}_z+\frac{\gamma B_1}{\hbar}\hat{S}_x\right)\Delta t+\mathcal{O}\left((\Delta t)^2\right)}. \tag{14.22}$$

In Problem 14.2.1, you will compute the operator term that multiplies $(\Delta t)^2$. But, because the term is the same for every factor, we can combine the results of all of the pairs together, and using the fact that $N\Delta t = t$, we find that

$$\hat{U}(t) = e^{\frac{i}{\hbar}\omega t\hat{S}_z}e^{\frac{i}{\hbar}\left((\Omega-\omega)\hat{S}_z+\frac{\gamma B_1}{\hbar}\hat{S}_x\right)t+t\Delta t\times\text{term}}, \tag{14.23}$$

where "term" is the lowest-order commutator term correction from the BCH formula; it is explicitly computed in Problem 14.2.1. One can immediately see that the correction term vanishes when we take the limit $\Delta t \to 0$. This is an important result of the Trotter formula—we can combine two exponential factors with exponents proportional to Δt and find that the error in doing this is always of order $(\Delta t)^2$. This means we have a simple factorized form for the time-evolution operator in the limit $\Delta t \to 0$. It is

$$\hat{U}(t) = e^{\frac{i}{\hbar}\omega t\hat{S}_z}e^{\frac{i}{\hbar}\left((\Omega-\omega)\hat{S}_z+\frac{\gamma B_1}{\hbar}\hat{S}_x\right)t}. \tag{14.24}$$

The time-evolved state is then $|\psi(t)\rangle = \hat{U}(t)|\psi(0)\rangle$. This time-evolution operator is remarkable. We have found an exact factorized form of the time evolution by recognizing the Hamiltonian can be expressed in terms of the Hadamard lemma, using the braiding relation, the BCH formula, and finally, taking the limit as $\Delta t \to 0$. It has some interesting properties. One can think of the left term in the product as converting from the rotating frame to the lab frame, while the right factor can be thought of as the time evolution of a time-independent problem in the rotating frame. Notice how, when we are on resonance, with $\Omega = \omega$, the coefficient of the term with $\hat{S}_z$ vanishes. Note further that if we ran the time-dependent field in the opposite direction (counter-clockwise rotation), we would never achieve resonance, and the modification of the time evolution from simple precession would always be small. Resonance requires both a matching of the frequency and a rotation in the correct orientation.

We worked hard to find an exact expression for the time-evolution operator. It was made easier because the time-dependent Hamiltonian was related to a time-independent Hamiltonian (as we expected it to be in the rotating frame). Now that we have this nice result, what do we use it for? In many time-dependent problems, we are interested in how the time-dependent Hamiltonian causes transitions between different states. Time-independent Hamiltonians have stationary states that do not transform between one another, but time-dependent Hamiltonians can drive transitions between states. So, let's do this for the problem we are working on.

Suppose we start in the state $|s, m; t{=}0\rangle$ at the initial time. Then as the state evolves in time, what is the probability to find it in the state $|s, m'\rangle$? The answer comes from simply determining how much of the state $|s, m'\rangle$ is in $|s, m; t\rangle$ and taking the modulus squared of the amplitude. Hence,

$$P_{m' \leftarrow m}(t) = \left| \langle s, m' | e^{\frac{i}{\hbar}\omega t \hat{S}_z} e^{\frac{i}{\hbar}\left((\Omega-\omega)\hat{S}_z + \frac{\gamma B_1}{\hbar}\hat{S}_x\right)t} |s, m\rangle \right|^2 . \tag{14.25}$$

The state $\langle s, m'|$ is an eigenstate of $\hat{S}_z$, so when the operator $e^{\frac{i}{\hbar}\omega t \hat{S}_z}$ acts on it, it gives the complex phase $e^{\frac{i}{2}m'\omega t}$. This is a complex phase that has modulus 1, so we only need to compute

$$P_{m' \leftarrow m}(t) = \left| \langle s, m' | e^{\frac{i}{\hbar}\left((\Omega-\omega)\hat{S}_z + \frac{\gamma B_1}{\hbar}\hat{S}_x\right)t} |s, m\rangle \right|^2 . \tag{14.26}$$

To evaluate this in the general case, we can use the exponential disentangling identity to convert the exponential into an exponential product of $\hat{S}_+$, $\hat{S}_-$, and $\hat{S}_z$. Then, just as we did when we computed the spherical harmonics, we can evaluate the action of each term on the state and compute the amplitude for each term. We then take the modulus squared, and we have our probability. You will work this out explicitly for the general case in Problem 14.2.5.

For special cases, like spin one-half, or spin one, we can use our matrix representations and the generalized Euler identity to compute the probability directly. We illustrate for you how to do this for spin one-half and leave the case of spin one to Problem 14.2.4. We use the generalized Euler relation to simplify the operator. Since $e^{i\vec{v}\cdot\vec{\sigma}} = \cos|\vec{v}|\,\mathbb{I} + i\sin|\vec{v}|\,\frac{\vec{v}\cdot\vec{\sigma}}{|\vec{v}|}$ and

$$\vec{v} = \left(\frac{\gamma B_1 t}{2\hbar},\ 0,\ \frac{(\Omega-\omega)t}{2}\right), \tag{14.27}$$

we find the operator is

$$\begin{aligned} e^{\frac{i}{\hbar}\left((\Omega-\omega)\hat{S}_z + \frac{\gamma B_1}{\hbar}\hat{S}_x\right)t} &= \cos\left(\tfrac{t}{2}\Delta\right)\mathbb{I} + i\sin\left(\tfrac{t}{2}\Delta\right)\left(\frac{\gamma B_1}{\hbar\Delta}\sigma_x + \frac{\Omega-\omega}{\Delta}\sigma_z\right) \\ \text{with } |\vec{v}| &= \frac{t}{2}\sqrt{\frac{(\gamma B_1)^2}{\hbar^2} + (\Omega-\omega)^2} = \frac{t}{2}\Delta . \end{aligned} \tag{14.28}$$

We first calculate the probability to flip the spin. It is

$$P_{\downarrow \leftarrow \uparrow}(t) = \left| \langle \downarrow | \left[\cos\left(\tfrac{t}{2}\Delta\right)\mathbb{I} + i\sin\left(\tfrac{t}{2}\Delta\right)\left(\frac{\gamma B_1}{\hbar\Delta}\sigma_x + \frac{\Omega-\omega}{\Delta}\sigma_z\right)\right] | \uparrow \rangle \right|^2$$

$$= \frac{(\gamma B_1)^2}{(\hbar\Delta)^2} \sin^2\left(\tfrac{t\Delta}{2}\right). \tag{14.29}$$

We expect this probability to vanish as $t \to 0$, because we have not evolved the state yet in that limit. We also note that it is a simple oscillating function, which is what is expected for Rabi flopping—the spin flips, then flips back, and repeats. To confirm everything is working properly, let us also calculate the probability for it to stay in the same state:

$$\begin{aligned} P_{\uparrow\leftarrow\uparrow}(t) &= \left| \langle\uparrow | \left[\cos\left(\tfrac{t}{2}\Delta\right)\mathbb{I} + i \sin\left(\tfrac{t}{2}\Delta\right)\left(\frac{\gamma B_1}{\hbar\Delta}\sigma_x + \frac{\Omega-\omega}{\Delta}\sigma_z\right)\right] | \uparrow\rangle \right|^2 \\ &= \cos^2\left(\tfrac{t\Delta}{2}\right) + \frac{(\Omega-\omega)^2}{\Delta^2} \sin^2\left(\tfrac{t\Delta}{2}\right) \\ &= 1 - \frac{(\gamma B_1)^2}{(\hbar\Delta)^2} \sin^2\left(\tfrac{t\Delta}{2}\right). \end{aligned} \tag{14.30}$$

One can see the sum of the two probabilities is equal to 1, as it must be; the last line follows by replacing $\cos^2 = 1 - \sin^2$ and using the definition of Δ. These results show the typical behavior that we see with Rabi flopping—the probability to flip goes to zero as $t \to 0$, and it oscillates with the sum of all probabilities equal to 1 at all times. We have a video https://tinyurl.com/ms2xnfph that discusses the algebraic details.

We end this chapter with a discussion of the *geometric phase*, as introduced by Michael Berry (1984). We pick special values of the magnetic fields and focus on spin one-half, where we can write

$$\begin{aligned} \hat{H}(t) &= \cos\alpha\,\sigma_z + \sin\alpha(\cos\omega t\,\sigma_x - \sin\omega t\,\sigma_y) \\ &= \begin{pmatrix} \cos\alpha & e^{i\omega t}\sin\alpha \\ e^{-i\omega t}\sin\alpha & -\cos\alpha \end{pmatrix} = e^{\frac{i}{2}\omega t\,\sigma_z}(\cos\alpha\,\sigma_z + \sin\alpha\,\sigma_x)e^{-\frac{i}{2}\omega t\,\sigma_z}. \end{aligned} \tag{14.31}$$

Note how the Hamiltonian returns to itself after a time $T = \frac{2\pi}{\omega}$. You should also be able to see that the instantaneous eigenvalues of the Hamiltonian are given by ± 1, regardless of the value of α and t. The time-evolution operator becomes

$$\hat{U}(t) = e^{\frac{i}{2}\omega t\,\sigma_z} e^{it\left((\cos\alpha - \frac{\omega}{2})\,\sigma_z + \sin\alpha\,\sigma_x\right)} = \begin{pmatrix} e^{\frac{i}{2}\omega t} & 0 \\ 0 & e^{-\frac{i}{2}\omega t} \end{pmatrix}$$

$$\times \begin{pmatrix} \cos R(t) + \frac{it(\cos\alpha - \frac{\omega}{2})}{R(t)} \sin R(t) & \frac{it\sin\alpha}{R(t)} \sin R(t) \\ \frac{it\sin\alpha}{R(t)} \sin R(t) & \cos R(t) - \frac{it(\cos\alpha - \frac{\omega}{2})}{R(t)} \sin R(t) \end{pmatrix}$$

$$= \begin{pmatrix} e^{\frac{i\pi t}{T}} \left(\cos\left(\frac{tR(T)}{T}\right) + \frac{i(T\cos\alpha - \pi)}{R(T)} \sin\left(\frac{tR(T)}{T}\right)\right) & e^{\frac{i\pi t}{T}} \frac{iT\sin\alpha}{R(T)} \sin\left(\frac{tR(T)}{T}\right) \\ e^{-\frac{i\pi t}{T}} \frac{iT\sin\alpha}{R(T)} \sin\left(\frac{tR(T)}{T}\right) & e^{-\frac{i\pi t}{T}} \left(\cos\left(\frac{tR(T)}{T}\right) - \frac{i(T\cos\alpha - \pi)}{R(T)} \sin\left(\frac{tR(T)}{T}\right)\right) \end{pmatrix}, \tag{14.32}$$

with $R(t) = t\sqrt{1 - \omega\cos\alpha + \frac{\omega^2}{4}}$ and $R(T) = \sqrt{T^2 - 2T\pi\cos\alpha + \pi^2}$; recall $T\omega = 2\pi$. As we evolve an initial eigenstate of $\hat{H}$ around one period, it returns to itself with a change in phase. The phase is given by the sum of the dynamical phase over the evolved time, or $-\Delta t \sum_{j=1}^{N} E(j\Delta t) = \mp T$ for this problem, and a geometric phase, which is determined by the path the Hamiltonian takes in its "phase space." Technically speaking the geometric phase is determined when the time evolution is very slow, so in the limit where $T \to \infty$. This geometric phase is independent of the period of the evolution and depends only on the geometry of the phase space, which we describe next.

If we look at the instantaneous magnetic field at time t, which is

$$\vec{B}(t) = -(\sin\alpha\ \cos\omega t, -\sin\alpha\ \sin\omega t, \cos\alpha), \tag{14.33}$$

we see it traces out a cone on the surface of the unit sphere. The surface area enclosed by the spherical cap is $2\pi(1 - \cos\alpha)$; as $\alpha \to 0$ it vanishes and as $\alpha \to \pi$ it becomes the entire sphere, or 4π. We use the sphere as the "phase space" because as α and ωt change, we trace out the surface of a unit sphere. We already stated that the instantaneous eigenvalues of the Hamiltonian are ± 1 for all t. The instantaneous eigenstate with energy 1 is

$$\begin{pmatrix} e^{-\frac{2i\pi t}{T}} \cos\frac{\alpha}{2} \\ \sin\frac{\alpha}{2} \end{pmatrix}. \tag{14.34}$$

Note that the instantaneous eigenstates do return to themselves after one period. After one period, the dynamical phase is equal to $-T$. To compute the geometric phase, we need to find the total phase change of the eigenstate with $E = 1$ after evolving it for one period and subtract from that the dynamical phase. This is a problem for you below.

Section 14.2 Problems:

14.2.1 Technical Details of the Baker-Campbell-Hausdorff Formula Applied to the Magnetic Resonance Problem

The BCH formula tells us that $e^{\hat{A}}e^{\hat{B}} = e^{\hat{A}+\hat{B}+\frac{1}{2}[\hat{A},\hat{B}]+\cdots}$, where the dots denote higher-order nested commutators. Taking the pair of two repeated operators in the

Trotter product for the time evolution, show that the higher-order commutators are of order $(\Delta t)^3$ or higher. Next, compute the first commutator and concretely determine the term proportional to $(\Delta t)^2$. Verify that the exponent for each pair, including the BCH commutator corrections, are identical for each term, so they can all be added together (which corresponds to multiplying them by N). Finally, using $N\Delta t = t$ before taking the limit of $\Delta t \to 0$, show that the result discussed in the main text for the time evolution is indeed correct.

14.2.2 Magnetic Resonance for $|\downarrow\rangle$

Repeat the calculation for the magnetic resonance, this time starting from the down state. Compute $P_{\uparrow\leftarrow\downarrow}(t)$ and $P_{\downarrow\leftarrow\downarrow}(t)$. Verify that they have the proper limit as $t \to 0$ and that they add to 1 for all times.

14.2.3 Magnetic Resonance for a Counter-Clockwise Rotation

Work out the time-evolution operator for the following Hamiltonian:

$$\hat{H} = \Omega \hat{S}_z - \frac{\gamma B_1}{\hbar}\left(\cos(\omega t)\hat{S}_x + \sin(\omega t)\hat{S}_y\right). \tag{14.35}$$

Use a similar Hadamard lemma identity in your derivation.

Take your results for the time-evolution operator and compute $P_{\uparrow\leftarrow\uparrow}(t)$ and $P_{\downarrow\leftarrow\uparrow}(t)$. Verify they behave properly for $t \to 0$ and that the sum of the probabilities is 1 for all times.

14.2.4 Magnetic Resonance for Spin One

Using Eq. (14.26) for $s = 1$, compute $P_{1\leftarrow 1}(t)$, $P_{0\leftarrow 1}(t)$, and $P_{-1\leftarrow 1}(t)$. You will find it easiest to use the generalized Euler identity that we derived for the $s = 1$ case. Verify that it has the right behavior as $t \to 0$ and that the sum of all probabilities sum to one for each time.

14.2.5 Magnetic Resonance for Spin s

In this problem, we develop the general formula for spin s. To do this, we need to find an exponential disentangling identity for $e^{i\alpha\sigma_z + i\beta\sigma_x}$. We proceed as we did

before. First, compute the exponential exactly using the generalized Euler identity for spin one-half. Next, compute the product $e^{a\sigma_-}e^{b\sigma_z}e^{c\sigma_+}$ and equate the two to find a, b, and c in terms of α and β. Use the result you have found for the general spin case to rewrite the time-evolution operator in its disentangled form.

Next, we compute the probability for the transition $P_{m\leftarrow s}(t)$ to be found in any spin state, if we start in the maximal spin state ($m = s$) at $t = 0$. You should find the first exponential in the disentangled form can be replaced by 1, the second one can be exactly evaluated since the state is an eigenvector of it, and everything boils down to computing the final exponential, which involves just a polynomial of terms in its expansion. Using the general formula for how the lowering operator acts on a general spin state, compute each of the probabilities. Verify the result is correct for $t = 0$ and that the sum of all probabilities is equal to 1.

14.2.6 Ramsey Interference

Ramsey interferometry uses a waiting time to measure the frequency Ω in our magnetic resonance Hamiltonian. You might ask why would we want to do that, don't we know the field we applied? The reason why is we can map this same Hamiltonian onto an atomic problem, where the up state is the ground state, the down state is the excited state, and Ω is the energy splitting between these states. Then a Ramsey interferometry experiment will measure the energy splitting in the atom.

How does it work? We apply a $\frac{\pi}{2}$ pulse $e^{i\frac{\pi}{2}\frac{\sigma_x}{2}} = e^{i\frac{\pi}{4}\sigma_x}$. This transformation maps the up spin to a spin in the y-direction. Then we wait a time T, and the spin precesses in the x-y plane for that period of time, advancing an angle of $\Omega T/2$. We then apply a $-\frac{\pi}{2}$ pulse to bring the spin back to the $+z$ direction. We then measure the probability that it is in the $|\uparrow\rangle_z$ state. When this probability is maximal, we have determined Ω by measuring T.

Go through this analysis, determine the probability to be up as a function of T, and explain how this approach is used to measure the frequency Ω. Variants of this approach are used in atomic clocks to define the second.

14.2.7 Geometric Phase

We start in the state that is up along the α axis in the x-z plane, or $(\cos\frac{\alpha}{2}, \sin\frac{\alpha}{2})$. Evolve it for a time t, and verify that the time evolved state is given by

$$\begin{pmatrix} e^{\frac{i\pi t}{T}}\cos\frac{\alpha}{2}\left(\cos\left(\frac{tR(T)}{T}\right)+\frac{i(T-\pi)}{R(T)}\sin\left(\frac{tR(T)}{T}\right)\right) \\ e^{-\frac{i\pi t}{T}}\sin\frac{\alpha}{2}\left(\cos\left(\frac{tR(T)}{T}\right)+\frac{i(T+\pi)}{R(T)}\sin\left(\frac{tR(T)}{T}\right)\right) \end{pmatrix}. \tag{14.36}$$

Note that this is not a phase multiplied by the initial state unless we take $T \to \infty$ (recall that diabatic time evolution always has some tunneling out of an energy eigenstate); in that limit, the two terms in parenthesis to the right are the same in each row and equal to a complex phase. Nevertheless, we can still define a phase associated with this evolution by computing the phase of the inner product of the initial state with the time evolved state at $t = T$. Show this satisfies

$$\text{phase} = \arg\left(-\cos R(T) - \frac{i(T-\pi\cos\alpha)}{R(T)}\sin R(T)\right). \tag{14.37}$$

This form is difficult to work with, so we proceed as follows. First define a new function

$$1 - F(T) = -\frac{(T-\pi\cos\alpha)}{R(T)}. \tag{14.38}$$

Using this, show that the total phase minus the dynamical phase is equal to

$$\text{geometric phase} = T - \pi - R(T) + \arctan\left(\frac{F(T)\sin\big(2R(T)\big)}{2\Big(1 - F(T)\sin^2\big(R(T)\big)\Big)}\right). \tag{14.39}$$

It is easiest to do this by writing $-1 = e^{-i\pi}$ and factoring out a factor of $e^{-iR(T)}$. Now, if we take the limit as $T \to \infty$, we find the geometric phase is given by $-\pi(1-\cos\alpha)$ because $\sqrt{1+\delta} \approx 1 + \frac{\delta}{2}$.

One can analyze the corrections for finite T, and there are interesting results to see from that analysis. Consult (Berry, 2023) for further details.

14.3 Nuclear Magnetic Resonance and Magnetic Resonance Imaging

This section will be primarily descriptive. We have discussed nuclear magnetic resonance a number of times before, including the quantitative description in the previous section. Now, we want to describe how it is used in the application of magnetic resonance imaging.

One key result is that if the large magnetic field is uniform, then all of the spins in the sample precess at the same rate. This is great for having a strong signal, arising from many spins, but is not useful for imaging. For imaging, we want the rates to change with position. So, we add a gradient to the uniform magnetic field, so that it varies in a linear fashion, such as $B_z(z) = B_0 + \alpha z$. Then, the rate of precession is correlated with z position in the sample. If we could examine the Rabi flopping directly, we would use this to simply measure the resonance condition as a function of position and thereby find the concentration of spins as a function of position. Since the spin is a hydrogen nucleus, this then maps to the concentration of water in the sample as a function of position.

But this is not how magnetic resonance imaging works. In the real world, the spins *relax*. This means that if we flip a spin to lie opposite to the field and wait, it will eventually align itself back with the field again. Our quantum mechanics formalism does not yet describe this relaxation process. Nevertheless, we can still describe how it works. The experiment would simply flip the spin from along the field to opposite the field and then measure the so-called T_1 relaxation, by fitting the time dependence to the functional form $B_z(t) = B_0(1 - 2e^{-t/T_1})$. The exponential decay time is called the longitudinal relaxation time, or also the spin-lattice relaxation time. This relaxation time measures how long it takes for the spin to go back to the state it was before we flipped it. It is always the longer of the two relaxation times.

The second relaxation time is called the T_2 relaxation time, the transverse relaxation time, or the spin-spin relaxation time. If you recall our discussion of spin echo in Sect. 2.4, we described how the presence of different local fields caused the spins to splay out from each other as they precessed. This splaying out occurred from the different inhomogeneous local fields in the sample. We could erase their effect by reversing the spins and essentially running the time evolution backward, until the spins all merge again creating an echo pulse. This analysis ignored the relaxation effects leading to the T_2 time. We can think of this as an irreversible effect of neighboring spins on the motion. This effect also causes splaying, but does not reverse when we reverse the spins to perform the spin echo. So, these effects lead to a splaying that cannot be reversed. As a result, the magnitude of the echo decreases over time. It does so also exponentially, with a time constant given by the T_2 relaxation time. In solids, the T_1 relaxation time is much, much larger than the T_2 relaxation time. In liquids, the two relaxation times are nearly identical.

In magnetic resonance imaging, different tissues are more visible when measured via T_1 relaxation or via T_2 relaxation. The technical term for this is *contrast*. Depending on what is being imaged, an MRI image might be looked at in T_1 contrast or T_2 contrast. If you really want to impress your MRI tech the next time you need an image, ask them which contrast they are using and why! You probably know more about the physics and quantum mechanics of the device than they do.

So, how do we add in these relaxation mechanisms? They are generically called decoherence. There is an entire field of quantum mechanics devoted to describing relaxation. It uses cool sounding things like the "master equation." What could be better than that! We will examine some simple models of decoherence later in Chap. 22 and we will revisit magnetic resonance there again.

Section 14.3 Problems:

14.3.1 Measuring Relaxation

Using your own words, describe what the two different relaxation times are and how we measure them.

14.3.2 T_2^*

If the transverse relaxation time is not measured with a spin-echo experiment, it includes the intrinsic spin-spin relaxation mechanisms plus the inhomogeneous field relaxation. Do you expect T_2^* to be larger or smaller than T_2? Explain.

14.3.3 T_1 Versus T_2

In the text, we said that T_1 is always larger than T_2. Find some other sources on NMR/MRI and explain more concretely why this must be so. Also explain why the two relaxation times are nearly the same in liquids.

14.4 Trotter Formula for Time-Dependent Hamiltonians

In this section, we focus on time evolution for time-dependent Hamiltonians. Recall that for a general time-dependent Hamiltonian, we used the fact that time evolution is unitary, satisfies the semigroup rule, is governed by the Hamiltonian, and can be approximated by piecewise constant Hamiltonians to approximate the time evolution via

$$\hat{U}(t) = e^{-\frac{i}{\hbar}\hat{H}(t)\Delta t} e^{-\frac{i}{\hbar}\hat{H}(t-\Delta t)\Delta t} \cdots e^{-\frac{i}{\hbar}\hat{H}(\Delta t)\Delta t}, \tag{14.40}$$

where we used a left-point rule for evaluating the piecewise constant Hamiltonians. If we have a situation where the Hamiltonians at different times commute, then we can put all Hamiltonians into the same exponent, and we have the summation over all of them, which those of you who have studied calculus will recognize as a discrete approximation to an integral. As we discussed in Sect. 13.8, such discrete summations can be exactly summed for exponentials, sines, cosines, or powers relatively easily, and this will suffice for our work.

But, in most cases, the Hamiltonian does not commute with itself at two different times, namely, $[\hat{H}(t), \hat{H}(t')] \neq 0$. In such cases, it can be challenging to simplify the time-ordered product. We saw an example of how this works for the magnetic resonance calculation that we did in Sect. 14.2. That case can be generalized to the case where the time evolution is governed by repeated pairs of operators in the form $e^{-i\hat{A}\Delta t} e^{-i\hat{B}\Delta t}$, which can be combined and summed, as we discuss in Problem 14.4.1 below. If we think in terms of our operator identities, the braiding operation can be used to move Hadamard lemmas into and out of the exponent. The Baker-Campbell-Hausdorff formula can be used to combine exponential factors, and in the cases where both operators are multiplied by Δt, the commutator correction terms can be neglected. We can also use the exponential disentangling identity, in

some cases, to re-express products of operators in different forms. Finally, in the cases where we work with finite-size representations in terms of matrices (Pauli spin matrices or spin-1 matrices), the generalized Euler formula can be helpful in simplifying expressions. Often, what we are looking for is a way to factorize the time-evolution operator into two (or more) independent terms. This was also done in the magnetic resonance example in Sect. 14.2. In the remainder of this section, we will discuss some of these different approaches and leave examples for how to do this to the problems.

One important point to reiterate is that because the time-ordered product puts later times to the left, all of the action at the latest time is happening at the left-hand side of the operator. Everything to the right of the last term remains the same. This implies that we can build up moving to later and later times by simply adding just one new Trotter factor to the left. While this sounds like a nice simplification for time evolution, in practice, it is difficult to work with the full time-evolution operator, because it is too large of an object to store into a computer memory that we often have to use other techniques when we do time evolution. Let us elaborate on this issue. The time-evolution operator $\hat{U}(t)$ is a very versatile operator, because it will evolve *any state* forward in time. But, for many applications, we care only about evolving *one specific state* forward in time. In that case, because an operator maps a vector to a vector, if we can operate a single Trotter factor onto a state efficiently, then an alternative for how we can perform time evolution is to evolve the state one Trotter factor at a time, storing only the vector in memory and not needing to store the operator (typically a matrix). You might wonder why this could be a savings, because we still have to operate the Trotter factor, which can be a very large operator onto our vector. How do we store that object? The answer is that, in many cases, we don't need to store it. We either construct the action of that operator onto the vector "on the fly" or use sparse matrix methods to compute it in a power series, using a small number of terms because Δt is small. As you can see, there are a lot of innovative ideas that have been developed to allow us to determine the time evolution of quantum systems.

Before ending this section, we will discuss two perspectives for how to view time evolution of quantum systems called the *Schrödinger representation* and the *Heisenberg representation*. To describe both of these approaches, we first need to remind you that many calculations in quantum mechanics involve calculating matrix elements of the form $\langle\phi|\hat{O}|\psi\rangle$, where we take the expectation value of an operator sandwiched between two quantum states. If these states are time dependent, we can express them in terms of the time-evolution operators via

$$\langle\phi(t_0)|\hat{U}^\dagger(t,t_0)\hat{O}\hat{U}(t,t_0)|\psi(t_0)\rangle. \tag{14.41}$$

There are two ways to think of this formula, as summarized in the next two equations:

$$\underbrace{\langle\phi(t_0)|\hat{U}^\dagger(t,t_0)}_{\text{state}}\,\hat{O}\,\underbrace{\hat{U}(t,t_0)|\psi(t_0)\rangle}_{\text{state}}\quad\text{or}\quad\langle\phi(t_0)|\,\underbrace{\hat{U}^\dagger(t,t_0)\hat{O}\hat{U}(t,t_0)}_{\text{operator}}\,|\psi(t_0)\rangle. \tag{14.42}$$

The results are of course identical, but in the expression on the left-hand side, we emphasize the time evolution as coupled to the states. This is what is called the Schrödinger representation. The time evolution is always in the states, and the operators have no time evolution, but they can have an intrinsic time dependence, if they are a time-dependent operator. The second expression on the right emphasizes the time evolution with respect to the operators, leaving the states time independent (and evaluated at their initial time t_0). This is the Heisenberg representation, where all time dependence is in the operators and the states remain time independent. While the Schrödinger representation looks more familiar to you, given how we have developed time evolution, the Heisenberg representation is often used in many applications. One reason you can see why is that the time-evolution operator acts like a similarity transformation on the operator, and it has the potential to be evaluated via the Hadamard lemma. Indeed, we will see this often is the case for interesting problems.

The problems below are somewhat technical, showing different strategies that can be used to simplify time-ordered products. Some of the manipulations we perform below will be revisited when we discuss other applications of time evolution later in this chapter.

Section 14.4 Problems:

14.4.1 Simplification of Time-Ordered Products That Have Repeated Pairs of Operators

Suppose a Trotter factor takes the form

$$e^{-\frac{i\Delta t}{\hbar}\hat{A}_N}e^{-\frac{i\Delta t}{\hbar}\hat{B}_N}e^{-\frac{i\Delta t}{\hbar}\hat{A}_{N-1}}e^{-\frac{i\Delta t}{\hbar}\hat{B}_{N-1}}\cdots e^{-\frac{i\Delta t}{\hbar}\hat{A}_1}e^{-\frac{i\Delta t}{\hbar}\hat{B}_1}. \tag{14.43}$$

The A operators commute $[\hat{A}_i,\hat{A}_j]=0$, as do the B operators, but $[\hat{A}_i,\hat{B}_j]\neq 0$. Show that the time-evolution operator, after putting all operator terms in the same exponent and neglecting terms of order $(\Delta t)^3$, becomes

$$\exp\left(-\frac{i\Delta t}{\hbar}\sum_{i=1}^{N}(\hat{A}_i+\hat{B}_i)\right.$$
$$\left.-\frac{(\Delta t)^2}{2\hbar^2}\sum_{i=1}^{N}\left([\hat{A}_i,\hat{B}_i]+\left[\hat{A}_i,\sum_{j=1}^{i-1}\hat{B}_j\right]-\left[\sum_{j=1}^{i-1}\hat{A}_j,\hat{B}_i\right]\right)\right). \tag{14.44}$$

Show that in the case of magnetic resonance, where all A operators are the same, as are all B operators, the contribution of the last two terms proportional to $(\Delta t)^2$ vanishes; hence, the total contribution goes like $N(\Delta t)^2$, which vanishes as $\Delta t \to 0$. But, if the operators are all different, there can be a contribution from those terms, because the number of those terms is approximately N^2 and $N^2(\Delta t)^2 = t^2$ has a nonvanishing limit as $\Delta t \to 0$. This problem shows that, in general, putting all operator terms into a single exponent can lead to complicated expressions. Only in special cases do they simplify.

14.4.2 Simplification of Time-Evolution Operators for Which the Weyl Form of the BCH Formula Is Exact

In this problem, we have the Trotter factors all of the form

$$e^{-\frac{i\Delta t}{\hbar}(\hat{A}_N+\hat{B}_N)}e^{-\frac{i\Delta t}{\hbar}(\hat{A}_{N-1}+\hat{B}_{N-1})}\cdots e^{-\frac{i\Delta t}{\hbar}(\hat{A}_1+\hat{B}_1)} \tag{14.45}$$

and $[\hat{A}_i, \hat{A}_j] = [\hat{B}_i, \hat{B}_j] = 0$. We also have $[\hat{A}_i, \hat{B}_j] \neq 0$, but all double-nested commutators vanish.

The first step of this problem is to use the BCH formula in the Weyl form and separate the Trotter terms into exponentials of A, exponentials of B, and exponentials of their commutator. The commutator terms can be all moved to the right because they commute with everything. Our next step is to move all of the A terms to the far left. There they can be combined together and all put into one exponent, because they commute. Similarly, all the commutator terms can be moved to the far right, and all B terms can be collected together and put into one exponent in the middle. In this way, we have factorized the time-evolution operator into the A terms first, the B terms second, and all commutator terms third. Because the commutator terms commute with everything, we can also put all of the commutator terms into the exponent with the A operators, or with the B operators.

Hint: This problem is straightforward, but requires careful book-keeping to get it correct, as there are many commutator terms to collect. It is definitely a French cooking problem.

14.4.3 Compression Algorithm for SU(2)

In quantum computing, one often applies Pauli spin rotations along different axes to a qubit (which can be thought of as a two-state system described by a spin degree of freedom). Since each operation on a quantum computer is expensive to apply, researchers have looked into circuit compression techniques to reduce the number of circuit elements that need to be applied. Suppose that we have a sequence of exponential operators that alternate rotations along the x- and z-axes. This would look like

$$e^{-\frac{i\alpha_1}{2}\sigma_x}e^{-i\frac{\beta_1}{2}\sigma_z}e^{-\frac{i\alpha_2}{2}\sigma_x}e^{-\frac{i\beta_2}{2}\sigma_z}\cdots e^{-\frac{i\alpha_n}{2}\sigma_x}e^{-\frac{i\beta_n}{2}\sigma_z}. \tag{14.46}$$

Schematically, we can refer to such a chain of exponentials as $XZXZ\cdots XZ$. To perform a compression, we want to reduce the number of rotations that we apply to a qubit. One way to do this is to convert the spin rotation products from one form to another. Again schematically, we can write any three rotations of the form XZX in the alternate form ZXZ. This identity is an exact operator relation. If we do this, we can convert $XZXZXZ\cdots$ into $ZXZZXZ\cdot$ (by converting the leftmost XZX term to a ZXZ term) and now the two adjacent Z factors can be directly combined, reducing the chain by one exponential factor. Then it has the form $ZXZXZ\cdots$, and we can repeat the procedure, taking the first XZX factor, replacing it by ZXZ, so we have $ZZXZZX\cdots$, where we can reduce the two adjacent ZZ factors again. The process can be repeated until we have only three factors left. In this fashion, we have compressed the circuit. More sophisticated implementations use properties of Lie algebras to perform even more complex circuit compressions.

In this problem, we will derive the identity $XZX = ZXZ$ used in the compression algorithm by computing

$$e^{-\frac{i\alpha}{2}\sigma_x}e^{-\frac{i\beta}{2}\sigma_z}e^{-\frac{i\gamma}{2}\sigma_x} = e^{-\frac{ia}{2}\sigma_z}e^{-\frac{ib}{2}\sigma_x}e^{-\frac{ic}{2}\sigma_z}, \tag{14.47}$$

where you will compute a, b, and c from the known coefficients α, β, and γ. Use the generalized Euler identity to convert the left and the right sides of the above equation into 2×2 matrices and compute the relationships between the coefficients by equating the four matrix elements of the final products. Show that you obtain the exact inverse trigonometric relations given by the following:

$$a = \arctan\left[\tan\left(\frac{\beta}{2}\right)\frac{\cos\left(\frac{\alpha-\gamma}{2}\right)}{\cos\left(\frac{\alpha+\gamma}{2}\right)}\right] - \arctan\left[\tan\left(\frac{\beta}{2}\right)\frac{\sin\left(\frac{\alpha-\gamma}{2}\right)}{\sin\left(\frac{\alpha+\gamma}{2}\right)}\right], \tag{14.48}$$

$$\frac{b}{2} = \arctan\left[\sqrt{\frac{\sin^2\left(\frac{\alpha+\gamma}{2}\right)+\tan^2\left(\frac{\beta}{2}\right)\sin^2\left(\frac{\alpha-\gamma}{2}\right)}{\cos^2\left(\frac{\alpha+\gamma}{2}\right)+\tan^2\left(\frac{\beta}{2}\right)\cos^2\left(\frac{\alpha-\gamma}{2}\right)}}\right], \tag{14.49}$$

and

$$c = \arctan\left[\tan\left(\frac{\beta}{2}\right)\frac{\cos\left(\frac{\alpha-\gamma}{2}\right)}{\cos\left(\frac{\alpha+\gamma}{2}\right)}\right] + \arctan\left[\tan\left(\frac{\beta}{2}\right)\frac{\sin\left(\frac{\alpha-\gamma}{2}\right)}{\sin\left(\frac{\alpha+\gamma}{2}\right)}\right]. \tag{14.50}$$

Describe in words how one performs the update to convert from XZX to ZXZ and compress the circuit. Note that it does not require any matrix multiplications, just working with the different coefficients.

14.4.4 Compression Algorithm from the Generalized Euler Formula

In Sect. 14.8, we will examine a famous problem of diabatic passage. In the solution to that problem, we need to evaluate Trotter products that include Trotter factors of the form

$$U(t, t - \Delta t) = e^{-\frac{i}{\hbar}(vt\sigma_z + \delta\sigma_x)\Delta t}. \tag{14.51}$$

This product of factors can be compressed by using the result we worked out for the closure of a Lie group in Problem 6.3.11. Let $\vec{v} = \frac{\Delta t}{\hbar}(\delta,\ 0,\ vt)$ be the vector representation of the Trotter factor $e^{-i\vec{v}\cdot\vec{\sigma}}$. Then, we have that the product of two exponentials $e^{i\vec{v}\cdot\vec{\sigma}}e^{i\vec{v}'\cdot\vec{\sigma}} = e^{i\vec{v}''\cdot\vec{\sigma}}$ can be expressed in terms of a single vector $\vec{v}''$. Verify that the new vector is found from

$$\cos|\vec{v}''| = \cos|\vec{v}|\cos|\vec{v}'| - \sin|\vec{v}|\sin|\vec{v}'|\frac{\vec{v}\cdot\vec{v}'}{|\vec{v}||\vec{v}'|}, \tag{14.52}$$

and

$$\sin|\vec{v}''|\frac{\vec{v}''}{|\vec{v}''|} = \left(\cos|\vec{v}|\sin|\vec{v}'|\frac{\vec{v}'}{|\vec{v}'|} + \sin|\vec{v}|\cos|\vec{v}'|\frac{\vec{v}}{|\vec{v}|} - \sin|\vec{v}|\sin|\vec{v}'|\frac{\vec{v}\times\vec{v}'}{|\vec{v}||\vec{v}'|}\right). \tag{14.53}$$

Unfortunately, there does not appear to be any simple way to write down this result in a way that we can determine the final compressed expression by induction. Instead, explain how, if we have 2^n Trotter factors in our time-evolution operator, we can compress it to one factor in n steps by pairing adjacent operators together at each step and replacing each pair by a single factor using our exact identity and then repeating. Also explain how we efficiently perform the update to determine $\vec{v}''$ from the vectors for each factor in the pair being compressed. Do we need to compute trigonometric functions (or inverse trigonometric functions), or can it be done in an alternative fashion?

14.4.5 Heisenberg Representation of Position and Momentum for a Simple Harmonic Oscillator

Consider a simple harmonic oscillator, where the time-evolution operator is given by

$$\hat{U}(t) = e^{-\frac{it}{\hbar}\left(\frac{\hat{p}^2}{2M} + \frac{1}{2}M\omega^2\hat{x}^2\right)}. \tag{14.54}$$

Use the Hadamard lemma to compute the Heisenberg operators $\hat{x}_H(t) = \hat{U}^\dagger(t)\hat{x}\hat{U}(t)$ and $\hat{p}_H(t) = \hat{U}^\dagger(t)\hat{p}\hat{U}(t)$.

Hint: When you determine the nested commutators, you should find they alternate between position and momentum operators. By collecting all position operator terms and all momentum operator terms separately, you should find the series for the sine and cosine, which will allow you to simplify your result.

14.4.6 Loschmidt Echo

The Loschmidt echo is used to answer the following question: If I start in a state $|\psi\rangle$ and evolve the state for a time t under a Hamiltonian $\hat{H}_1$, and I evolve the same state for the same time under a different Hamiltonian $\hat{H}_2$, how different are the two states? This is measured by the fidelity, given by

$$\mathcal{F} = |\langle \psi(t);\, \hat{H}_2 | \psi(t);\, \hat{H}_1 \rangle|^2. \tag{14.55}$$

The symbols with the Hamiltonians indicate which Hamiltonian is governing the time evolution. If the two states are the same, the fidelity is 1.

Usually, we do not evolve the two states forward in time under two different Hamiltonians and compare them. Show instead that the fidelity satisfies

$$\mathcal{F} = |\langle \psi | e^{\frac{i}{\hbar}\hat{H}_2 t} e^{-\frac{i}{\hbar}\hat{H}_1 t} | \psi \rangle|^2. \tag{14.56}$$

One way to perform the experiment is to time evolve under $\hat{H}_1$ for time t, then evolve further in time under the Hamiltonian $-\hat{H}_2$, and then measure the fidelity of the time-evolved state with the original state.

These types of calculations can be done rather directly on quantum computers. They also arise in nonequilibrium situations where we evolve with time-dependent Hamiltonians. Explain how that changes the analysis.

14.4.7 Improving the Accuracy of the Trotter Formula

In this problem, we will consider a Trotter product given by a repeating sequence of terms corresponding to operators $\hat{A}$ and $\hat{B}$ as an approximation to the exponential of the sum, so that

$$e^{i\alpha\hat{A}+i\beta\hat{B}} \approx \left(e^{i\frac{\alpha}{N}\hat{A}} e^{i\frac{\beta}{N}\hat{B}}\right)^N. \tag{14.57}$$

Why do we want to do this? Sometimes we need to evaluate the exponential of a sum of operators with large coefficients, but they do not commute with each other.

In some numerical calculations, it is easier to calculate the exponentials of each operator separately, but not both in the same exponent. This situation often occurs in quantum computing, for example, where it is often not possible to implement the exponential of the sum of two noncommuting operators. So, how accurate is this approximation? This is the question we ask here.

Compute the right-hand side of Eq. (14.57) by first using the Baker-Campbell-Hausdorff identity to bring the product of the exponentials $e^{i\frac{\alpha}{N}\hat{A}}e^{i\frac{\beta}{N}\hat{B}}$ into one exponent. Collect only the first commutator correction term in the exponent. Then you can raise it to the Nth power easily, because each factor is identical. You should find you have an error that goes like $\frac{1}{N}$.

Now, instead, compute the following product:

$$e^{-i\frac{\alpha}{2N}\hat{A}}\left(e^{i\frac{\alpha}{N}\hat{A}}e^{i\frac{\beta}{N}\hat{B}}\right)^N e^{i\frac{\alpha}{2N}\hat{A}} = e^{i\frac{\alpha}{2N}\hat{A}}e^{i\frac{\beta}{N}\hat{B}}\left(e^{i\frac{\alpha}{N}\hat{A}}e^{i\frac{\beta}{N}\hat{B}}\right)^{N-1} e^{i\frac{\alpha}{2N}\hat{A}}, \tag{14.58}$$

which can be thought of as splitting off the leftmost factor $e^{i\frac{\alpha}{N}\hat{A}}$ and keeping the square root of it on the left and adding the square root of it onto the right. In this case, you should find the error goes like $\frac{1}{N^2}$.

Hint: The first form suggests using the Hadamard lemma after one uses BCH on all of the factors. Use it in the braiding relation by bringing it into the argument of the final exponential.

As a bonus problem, determine precisely the $\frac{1}{N^2}$ corrections in terms of commutators of operators. This requires including the double-nested commutators when working out the BCH formula, rather than just the single commutator term.

14.5 Time Evolution of the Harmonic Oscillator

The time-evolution operator for the simple harmonic oscillator can be written in two equivalent ways. The first way emphasizes the position and momentum operators and is given by

$$\hat{U}(t,t_0) = e^{-\frac{i}{\hbar}(t-t_0)\left(\frac{\hat{p}^2}{2M}+\frac{1}{2}M\omega^2\hat{x}^2\right)}, \tag{14.59}$$

while the second way emphasizes the raising and lowering operators and is given by

$$\hat{U}(t,t_0) = e^{-i\omega(t-t_0)\hat{a}^\dagger\hat{a}}e^{-\frac{i}{2}\omega(t-t_0)}. \tag{14.60}$$

The second form is particularly useful for determining the Heisenberg representation of the raising and lowering operators. The result follows from a simple calculation using the Hadamard lemma:

$$\begin{aligned}\hat{a}_H(t) &= e^{i\omega t\hat{a}^\dagger\hat{a}}\hat{a}e^{-i\omega t\hat{a}^\dagger\hat{a}} \\ &= \hat{a} + i\omega t[\hat{a}^\dagger\hat{a},\hat{a}] + \frac{(i\omega t)^2}{2}[\hat{a}^\dagger\hat{a},[\hat{a}^\dagger\hat{a},\hat{a}]] + \cdots \\ &= \sum_{n=0}^{\infty}\frac{(i\omega t)^n}{n!}(-1)^n\hat{a} = e^{-i\omega t}\hat{a}. \end{aligned} \tag{14.61}$$

By taking the Hermitian conjugate, we find that

$$\hat{a}_H^\dagger(t) = e^{i\omega t}\hat{a}^\dagger. \tag{14.62}$$

We have set $t_0 = 0$ again for simplicity. Having the Heisenberg representation for the ladder operators allows us to immediately find the Heisenberg representation of any function of the ladder operators, because the Heisenberg representation of the operator is given by a Hadamard lemma, and this goes into functions via the braiding operation. This makes time dependence calculations for the simple harmonic oscillator particularly easy to work out.

We already know the time dependence of the energy eigenstates—they are given by $|n(t)\rangle = e^{-i\omega t(n+\frac{1}{2})}|n\rangle$. But, we know a wide array of other states associated with the simple harmonic oscillator. For example, we have the coherent state, given by

$$|\alpha\rangle = \hat{D}(\alpha)|0\rangle = e^{-\alpha^*\hat{a}+\alpha\hat{a}^\dagger}|0\rangle. \tag{14.63}$$

You may want to review the coherent states in Sect. 7.11 to refresh yourself on what these states are. To find the time dependence of the coherent state, we need to construct the Hadamard lemma, and we do this by introducing a $\hat{U}^\dagger\hat{U}$. So, we have

$$|\alpha(t)\rangle = \hat{U}(t)|\alpha\rangle = \hat{U}(t)\hat{D}(\alpha)\hat{U}^\dagger(t)\hat{U}(t)|0\rangle. \tag{14.64}$$

The braiding relation brings the Hadamard transformation into the exponent of the operator, so

$$|\alpha(t)\rangle = e^{-\alpha^*\hat{U}(t)\hat{a}\hat{U}^\dagger(t)+\alpha\hat{U}(t)\hat{a}^\dagger\hat{U}^\dagger(t)}\hat{U}(t)|0\rangle. \tag{14.65}$$

Now, we can evaluate the transformed ladder operators, but we need to note that the order of the time-evolution operators is swapped from the result we calculated above, which is the same as taking $t \to -t$, so we have that

$$|\alpha(t)\rangle = e^{-\alpha^*e^{i\omega t}\hat{a}+\alpha e^{-i\omega t}\hat{a}^\dagger}e^{-\frac{i}{2}\omega t}|0\rangle = e^{-\frac{i}{2}\omega t}\hat{D}\big(\alpha e^{-i\omega t}\big)|0\rangle = e^{-\frac{i}{2}\omega t}|\alpha e^{-i\omega t}\rangle. \tag{14.66}$$

Hence, the time evolution of a coherent state is summarized simply as a time evolution of the α parameter that determines the coherent state. One can see

that $\alpha(t)$ periodically varies, returning to its original value each period of the harmonic oscillator, so the motion of the coherent state is periodic. The shape of the wavefunction does not change, since a coherent state does not change its shape as α varies—instead, it simply translates the ground state in position and multiplies it by a complex phase that depends on x. The motion of the probability density of a coherent state as a function of time is simply that of the Gaussian profile that moves back and forth with the period of the spring and does not change its shape. This is the closest we get to the motion of a mass on a spring in quantum mechanics. It has a remarkably classical nature to its behavior. Note that the time-varying α will change the average position and momentum of the oscillator, as we know from our previous work, and as we expect for an oscillating mass on a spring.

In Fig. 14.4, we use a three-dimensional representation of a time-varying coherent state. The complex number is plotted perpendicular to the plane, with the real part vertical and the imaginary part horizontal (projections on the back plane and bottom plane are shown in the image as well). A surface that envelopes the wavefunction is the probability distribution, so that the volume enclosed by a slice of width Δx is proportional to the probability to be measured between x and $x+\Delta x$. In Fig. 14.5, we plot the 3d surface along with the wavefunction (solid line). Note

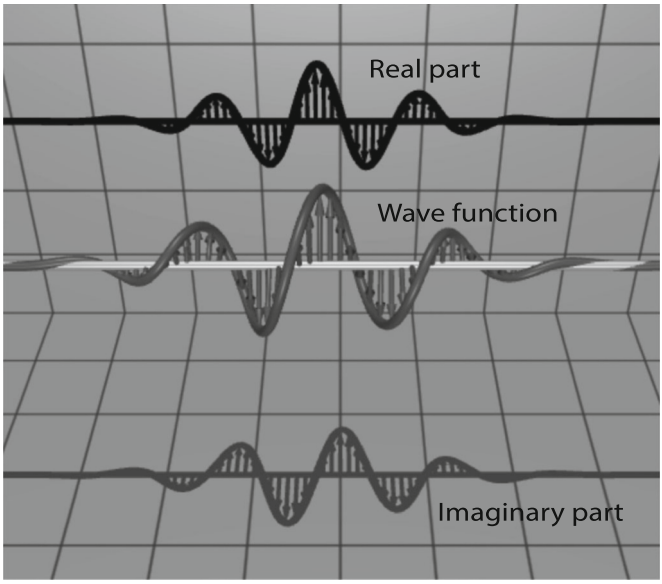

Fig. 14.4 Three-dimensional representation of a time-dependent wavefunction I. At each position x, we plot a snapshot of the wavefunction $\psi(x,t)$ as a complex number in the plane perpendicular to the x-axis (arrows). This is the middle part of the image where we depict the wavefunction in 3d. The projection onto the plane behind the image (black curve) is the real part of the wavefunction, while the projection on the plane below (gray image) is the imaginary part. The probability density over a width Δx is given by the volume of a cylindrical slice of width Δx. Hence, a 3d surface enveloping the wavefunction represents the probability density associated with the wavefunction

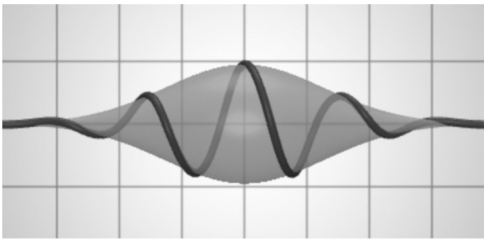

Fig. 14.5 Three-dimensional representation of a time-dependent wavefunction II. The solid line is the wavefunction (the same wavefunction as in the previous figure), while the gray surface is the envelope of the wavefunction, whose volume gives the probability distribution

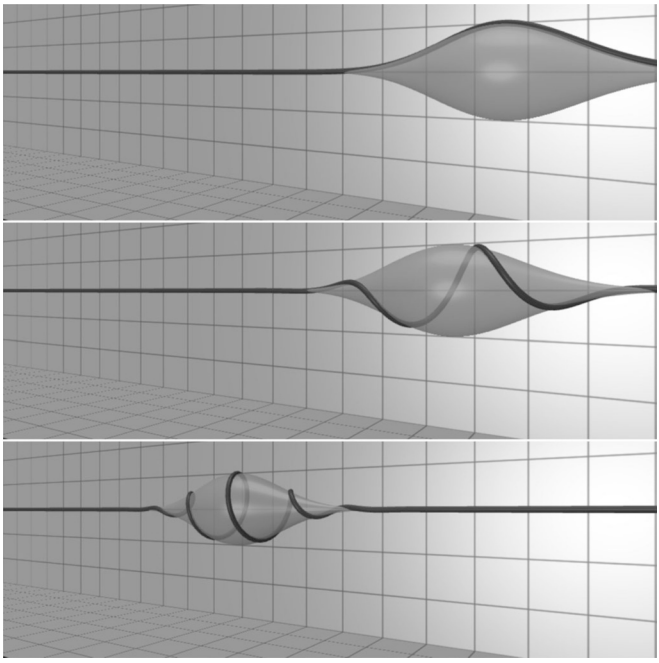

Fig. 14.6 Three-dimensional representation of a time-dependent wavefunction for the coherent state during one-fourth of the period of the oscillation. Top: The coherent state at the maximum extension of the spring. Middle: The coherent state starting to move toward the center. Bottom: The coherent state at the origin

that this wavefunction has momentum and is moving because it is complex valued. In fact, the tightness of the spiral determines how fast it is moving—a tighter spiral moves faster.

In Fig. 14.6, we show snapshots of the coherent state over a period of motion of the simple harmonic oscillator. Note how in the center of the motion, it is the most tightly wound spiral, indicating it is moving the fastest, and how it has no spiral at

the maximum, because that is where it is instantly motionless. You can also see how the shape of the probability distribution remains the same throughout the motion. Note that it is challenging to depict the motion of a time-varying quantum state using just still images, even if they are snapshots. Movies depict the motion much better, but we cannot show them in books.

How do we create such coherent states? We first create a ground state, which we describe how to do below via a cooling protocol. Then we have to provide either a push or a pull to it, by giving it an impetus. This is called "impulsing the oscillator." Once we do this, we will then set the oscillator in motion, and if there is no friction, it will oscillate forever. We will describe below some clever ways to impulse a quantum oscillator.

Let us examine the expectation values for position and momentum. We can calculate them in two ways. First we will use the properties of the coherent state that $\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$ and $\langle\alpha|\hat{a}^\dagger = \langle\alpha|\alpha^*$. So, we have

$$\begin{aligned}\langle\alpha(t)|\hat{x}|\alpha(t)\rangle &= \sqrt{\frac{\hbar}{2m\omega}}\langle\alpha(t)|(\hat{a}+\hat{a}^\dagger)|\alpha(t)\rangle = \sqrt{\frac{\hbar}{2m\omega}}\left(\alpha e^{-i\omega t}+\alpha^* e^{i\omega t}\right)\\ &= \sqrt{\frac{\hbar}{2m\omega}}\left(\left(\sqrt{\frac{m\omega}{2\hbar}}x_0+\frac{i}{\sqrt{2\hbar m\omega}}p_0\right)e^{-i\omega t}+\left(\sqrt{\frac{m\omega}{2\hbar}}x_0-\frac{i}{\sqrt{2\hbar m\omega}}p_0\right)e^{i\omega t}\right)\\ &= x_0\cos\omega t+\frac{p_0}{m\omega}\sin\omega t. \end{aligned} \tag{14.67}$$

In this derivation, we used the expression for the α parameter in terms of the initial position x_0 and initial momentum p_0. It can be rewritten in terms of the initial velocity as $\langle x(t)\rangle_\alpha = x_0\cos\omega t + \frac{v_0}{\omega}\sin\omega t$. A similar calculation of the expectation value of the momentum gives $\langle p(t)\rangle_\alpha = -m\omega x_0\sin\omega t + p_0\cos\omega t$, which you should be sure to verify. In this derivation, we used the eigenvalue property of the coherent state. But, we could also have performed the calculation with the Hadamard lemma. This latter approach works better for performing similar calculations with squeezed states. The way it works is by using the Heisenberg operator for position or momentum, by moving the time-evolution operators from the states to the operators. So, we have

$$\begin{aligned}\langle\alpha(t)|\hat{x}|\alpha(t)\rangle &= \langle\alpha|\hat{U}^\dagger(t)\hat{x}\hat{U}(t)|\alpha\rangle = \sqrt{\frac{\hbar}{2m\omega}}\langle\alpha|e^{i\omega t\hat{a}^\dagger\hat{a}}(\hat{a}+\hat{a}^\dagger)e^{-i\omega t\hat{a}^\dagger\hat{a}}|\alpha\rangle\\ &= \sqrt{\frac{\hbar}{2m\omega}}\langle\alpha|\left(\hat{a}e^{-i\omega t}+\hat{a}^\dagger e^{i\omega t}\right)|\alpha\rangle\\ &= \sqrt{\frac{\hbar}{2m\omega}}\left(\alpha e^{-i\omega t}+\alpha^* e^{i\omega t}\right)\\ &= x_0\cos\omega t+\frac{p_0}{m\omega}\sin\omega t, \end{aligned} \tag{14.68}$$

which is the same result as before. The momentum gives the same result as before too. These results for the time variation of the expectation values of the position and momentum in a coherent state agree exactly with the classical expressions for a mass moving on a spring with a given initial position and momentum. This is one of the reasons why the coherent states are called the most classical states of the harmonic oscillator. But, in any case, the result that the expectation value of the position and momentum as a function of time agrees with the classical equation of motion is a general result for quantum systems, which is called the *Ehrenfest theorem* after Paul Ehrenfest.

Now we move on to look at squeezed states. Recall a squeezed coherent state is described by

$$\hat{D}(\alpha)\hat{S}(\xi,\eta)|0\rangle = e^{-\alpha^*\hat{a}+\alpha\hat{a}^\dagger}e^{-\frac{\xi}{2}\left(\hat{a}^\dagger\right)^2+i\frac{\eta}{2}\left(\hat{a}^\dagger\hat{a}+\hat{a}\hat{a}^\dagger\right)+\frac{\xi^*}{2}\hat{a}^2}|0\rangle. \tag{14.69}$$

To determine the time evolution, we must multiply by $\hat{U}(t)$ from the left. We can create Hadamards by introducing $\hat{U}^\dagger(t)\hat{U}(t)$ in between the exponential factors and to the left of the ground state. Then the braiding operation is used to bring the Hadamard into the exponents. We evaluate them and find that the time-dependent coherent-squeezed state is given by

$$\hat{D}(\alpha e^{-i\omega t})\hat{S}(\xi e^{2i\omega t},\eta)|0\rangle. \tag{14.70}$$

This is a complicated result. The α parameter in the displacement operator still oscillates with the period of the harmonic oscillator. The squeezing parameter ξ oscillates twice as fast, while the η parameter does not change with time. The α parameter governs the spiraling of the wavefunction. The ξ parameter determines the shape change of the wavefunction—squeezing it for part of the motion (near the origin or near the maximal extension, depending on the parameters) and expanding it for the other part of the motion (the vice versa of where it is squeezed). Because the squeezed state squeezes and expands during the motion, its period is half that of the harmonic oscillator. It is difficult to visualize this motion with a sequence of still images, so we do not even try. Instead, we recommend you look up some videos of the time dependence of these states on the Web. There are a lot of nice videos to choose from. Go explore.

We discussed Wavefunction@Wave-particle duality earlier in the book, when we discussed how a single-photon light source works with a two-slit experiment. The behavior there sometimes leads people to say that quanta "travel like waves, but are observed as particles." In my opinion, this is not an accurate way to describe Wavefunction@Wave-particle duality. Coherent states provide a nice way to further elaborate on Wavefunction@Wave-particle duality and understand better its nuances. You see, in the light example, we clearly see particles hitting the screen and only over time do we learn that they also have an underlying wave pattern. But in the coherent-state example, one look at Fig. 14.6 beautifully illustrates another aspect of Wavefunction@Wave-particle duality—that the wave nature and particle

nature are intertwined and inseparable. For in that figure you can clearly see the wave aspect by looking at the wavefunction, which has an oscillating pattern as it winds around (if unsure look at the real and imaginary parts in Fig. 14.4 which clearly have a wave character to them). But, because our detectors detect according to the modulus squared of the wavefunction in the interval Δx corresponding to their width, we always are probing the particle-like aspect, because the probability distribution varies more slowly and shows no oscillations. To be clear, *one should never think of Wavefunction@Wave-particle duality as a quanta deciding whether it is a wave or a particle*, or even that it *travels as a wave but is detected as a particle*. Instead, one should think of duality as a quanta *always having wave and particle aspects simultaneously and in an inseparable fashion.* It is how we perform an experiment, and what the experiment is sensitive to, that determines whether we are probing the wave or particle aspect. Here, a particle detector, sensitive to the probability distribution generated from the wavefunction, is insensitive to the wave characteristics. Usually to see the wave properties we must find a way to create interference patterns, which then clearly show the wavelike behavior. We describe how to do this with a coherent state next.

There is an interesting harmonic oscillator state that we can create called a *Schrödinger cat state* that directly illustrates the wave nature of the coherent state. Such states have been created in ion traps, which we will describe for you below. But first we discuss just what the cat state is—it is a superposition of two coherent states—one with momentum p_0 and the other with momentum $-p_0$. Hence, it is of the form $\frac{1}{\sqrt{2}}(|\alpha\rangle + |\alpha^*\rangle)$. The state acts like two independent coherent states oscillating back and forth in the harmonic oscillator. However, when they meet in the center of the oscillation, they interfere, and the wave pattern becomes apparent (see Figs. 14.7 and 14.8). If we have particle detectors smaller than the wavelength of the oscillation, we can measure them by just counting particles. Usually we do not have so small detectors, so we need to first magnify the probability distribution before measuring it. This can be done with, for example, a time-of-flight experiment. The requirement to magnifying the quantum behavior before it can be directly observed is a common feature in quantum measurement. How such a state is created, where we impulse the system in two different directions at the same time is described below after we discuss ion and atom traps. So, the wave and particle aspects are always there. It is just how we set up an experiment that determines which aspect (or both) we are sensitive to.

The field of single ion and single atom trapping is about 50 years old, yet many of us are not familiar with how it works. We will start with an ion trap, namely, a Paul trap, since they are perhaps the easiest to understand. The challenge with trapping single ions is that they cannot be trapped by static fields. There is a theorem in electromagnetism called Earnshaw's theorem that forbids this. So, one sets up a saddlelike potential and rotates it, so that when the particle deviates to the left, the saddle pushes it to the right, and when it deviates to the right, the saddle has rotated just the right amount to push it to the left. This may sound hard to do, but if you know the period of the oscillation, it is not so difficult to get the timing

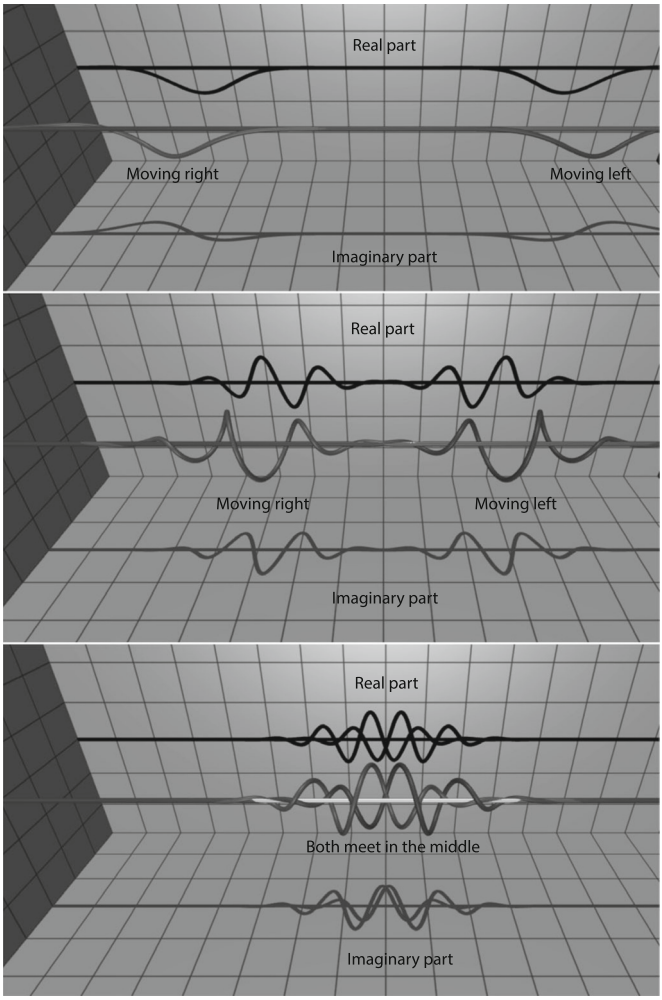

Fig. 14.7 Schrödinger cat state with the two superpositions identified by different grayscales. You can clearly see the wavelike features in the real and imaginary parts, and there is clear interference when the two parts of the superposition meet in the center. Recall that as the magnitude of the momentum increases, the wavefunction coils more

right. Then, we just need to bring in low-energy ions that will get trapped. This is done by taking ions from an oven which are hot and cooling them. Usually cooling is performed by magnetic and optical fields in a very special fashion to slow the particles down. The trapping of single atoms is similar, except we do not use a rotating trap; usually an optical trap is formed to contain the atoms. The atoms are trapped by their polarization and how it interacts with a light field. Once trapped, the ion or atom needs to be cooled. The simplest way to do this, at first, is via Doppler cooling. One shines light on the ion so that it is detuned to the red from an atomic

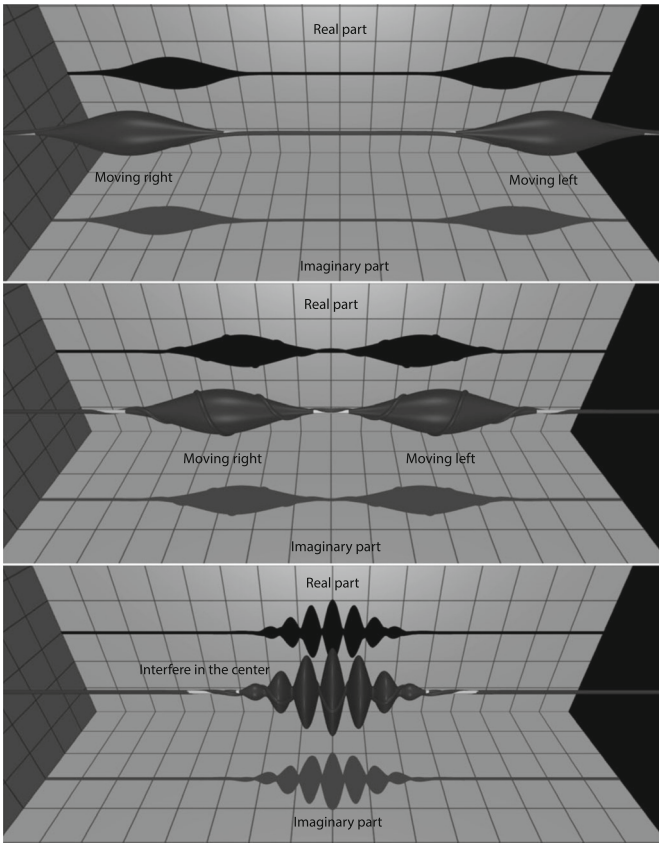

Fig. 14.8 Schrödinger cat state with the two superpositions indistinguishable and using the envelope to emphasize the probability distribution, which is what the detector will measure. Note the large oscillations when the two parts of the superposition meet in the center

transition (meaning it has too little energy for the transition). Then, if the atom moves toward the light, it sees a blue-shifted light from the Doppler effect, which transforms it to the right frequency to be absorbed. This kicks the atom in a direction opposite to the direction it is moving, because it absorbs the light only when the photon is moving toward it for a "head-on collision." One can lower the temperature significantly using this approach of selective scattering of light. Finally, for ions, we use sideband cooling to cool them down to the ground state of the motion. This is achieved by shining light that is red shifted by precisely the vibrational frequency of the harmonic oscillator from an atomic transition. This requires the ion to lose one quanta of vibration when it absorbs the light to conserve energy. Hence, we reduce the average quantum number by one unit with each absorption. Eventually we can get very close to the ground state, with the average of $\hat{n} = \hat{a}^{\dagger}\hat{a}$ being less than 0.1. At this stage, we are essentially in the motional ground state inside the trap.

To prepare a coherent state from the ground state, we need to impulse the atom in a specific direction. This can be done by applying an optical force to the ion that pushes it. But, a new idea is used in quantum engineering ions in a trap. This is called a *spin-dependent force*. Here, the light is tuned, so that it pushes to the left if the ion is in one quantum state and to the right if it is in another state. This can be easily done if the two states have different energies and we detune the light the same amount to the blue for one transition and to the red for the other (the positive or negative detuning determines the sign of the force). Finally, to make a cat state, we simply create an equal superposition of the two quantum states of the ion and then apply the spin-dependent force. This makes the entangled state $\frac{1}{\sqrt{2}}(|GS\rangle \otimes |\alpha\rangle + |ES\rangle \otimes |\alpha^*\rangle)$, which is almost the Schrödinger cat state. Here, GS is the ground state and ES is the excited state of the ion. To complete the state preparation, we need to make a product state, by making the two ionic states the same. This can be done by a selective transition that takes the excited state to the ground state but leaves the ground state as is (sometimes called optical pumping).

In this discussion, we described how one can make the cat state and also how one can make a general coherent state—just cool to the ground state and impulse the system appropriately. How would we make a squeezed state? Well, if we control the trap potential, we can suddenly change the trap potential from one curvature to another curvature. One has to be careful to make the change small; otherwise, the ion may no longer be trapped, so there isn't likely to be a large amount of squeezing. But now, we have quenched the ground state with one frequency into a trap with another frequency, which creates a squeezed state. What about Fock states? How can they be created? The ground state is the simplest Fock state and we can make that by simply cooling the system down. The higher Fock states are actually quite difficult to create. Cindy Regal's lab has developed an ingenious way to do this with trapped atoms and a double-well potential (Brown et al., 2023). They trap the atom in a double-well potential that is asymmetric, so one minimum is much lower than the other minimum. They cool to the ground state in the left half of the well (higher minimum) and then transfer to the right half (lower minimum). By tuning the right half excited Fock energy to the ground-state energy in the left half, the transfer can be quite high fidelity. Then they split the double well into a single well, with the oscillator in a higher-energy Fock state. We will discuss this process (and the experiment) more in the next section.

Section 14.5 Problems:

14.5.1 Expectation Value of Position and Momentum in a Squeezed State

Compute the expectation value of position and momentum in a general squeezed state as a function of time. If you are bold, repeat this for a squeezed coherent state.

We know the squeezed state has the width of the state oscillate at half the period of the harmonic oscillator. Do the expectation values of position and momentum have a dependence on half the period as well?

14.5.2 Time-Dependent Uncertainty in a Squeezed State

Go back and look at the calculation we did for the uncertainty of a coherent state in Sect. 7.11. This uncertainty is independent of time for a time-varying coherent state, because the shape of the state does not depend on time. But, this is not true for a squeezed state. Compute the uncertainty in position and in momentum as a function of time for a squeezed state. Determine when the uncertainty for position is smaller than the corresponding coherent state uncertainty, and similarly for momentum. What is the smallest the position uncertainty becomes as a function of time?

The results of this problem show that in order to get an accuracy boost from quantum mechanics in an experiment using squeezed states, one must properly time the measurement so that there actually is an advantage.

Hint: Because the previous calculation worked with a complex ξ, you should be able to write down the answer based on how ξ varies with time. Of course, you can repeat the calculation if you want to as well.

14.5.3 Wavefunction@Wave-particle duality in Your Own Words

Write an essay about the Schrödinger cat state and what Wavefunction@Wave-particle duality really means in your own words. Do you agree or disagree with the commonly used statement "quanta travel like waves, but are detected like particles?"

14.6 Free Expansion of a Gaussian Wavepacket

In this section, we treat an important problem—how a wavepacket expands as a function of time, even if it has no forces acting on it. We do so with a Gaussian wavepacket, which is just the well-known coherent state of a simple harmonic oscillator. If we recall that the coherent state does not expand or contract in the harmonic oscillator potential, then you might be surprised to learn that it does expand in free space. This is related to the phenomenon of squeezing, and our discussion will be within that context. The results presented here follow closely

the discussion in Orjuela and Freericks (2023), which showed how to solve this problem using an operator formalism.

We examine the free expansion of a Gaussian wavepacket starting from a state with variance σ and zero average momentum, which is given by the following form at $t = 0$:

$$\psi(x, t{=}0) = \frac{1}{(2\pi\sigma^2)^{\frac{1}{4}}} e^{-x^2/4\sigma^2}. \tag{14.71}$$

The application of a squeezing operator for the free expansion of a Gaussian comes from the facts that the ground state of a harmonic oscillator is also a Gaussian, and that the free time-evolution operator is a squeezing operator (for the harmonic oscillator system). Hence, the time evolution of a Gaussian can be solved by simply applying a squeezing operator to the harmonic oscillator ground state!

The ground-state wavefunction (in position space) of a simple harmonic oscillator, with mass m and frequency ω is given by

$$\psi_{gs}(x) = \langle x|0\rangle = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} e^{-m\omega x^2/2\hbar}. \tag{14.72}$$

The variance of σ^2 of the initial Gaussian is then set by choosing $\omega = \hbar/2m\sigma^2$. Next, we can use a displacement operator to displace the Gaussian in both position space and momentum space. The displacement by x_0 and p_0, respectively, is given by

$$\hat{D}(x_0, y_0) = e^{-\frac{i}{\hbar}(x_0\hat{p} - p_0\hat{x})}. \tag{14.73}$$

The displaced Gaussian state can then be represented by

$$|\psi_{gs}; x_0, p_0\rangle = \hat{D}(x_0, p_0)|0\rangle; \tag{14.74}$$

our choice for writing the wavepacket in this fashion fixes its initial global phase.

The free particle evolves according to a free Hamiltonian, given by $\hat{\mathcal{H}}_{\text{free}} = \hat{p}^2/2m$, so the time evolution of the initial Gaussian state becomes

$$|\psi_{gs}(t); x_0, p_0\rangle = e^{-\frac{i}{\hbar}\hat{\mathcal{H}}_{\text{free}}t}\hat{D}(x_0, p_0)|0\rangle, \tag{14.75}$$

and the spreading Gaussian wavepacket is given by

$$\psi_{gs}(x, t; x_0, p_0) = \langle x|e^{-\frac{i}{\hbar}\hat{\mathcal{H}}_{\text{free}}t}\hat{D}(x_0, p_0)|0\rangle. \tag{14.76}$$

This is in the form of a squeezed and displaced simple-harmonic-oscillator wavefunction! The free-particle Hamiltonian is expressed as

$$\frac{\hat{p}^2}{2m} = -\frac{\hbar\omega}{4}\left((\hat{a}^\dagger)^2 - \hat{a}^\dagger\hat{a} - \hat{a}\hat{a}^\dagger + \hat{a}^2\right), \tag{14.77}$$

in terms of the ladder operators, so the time-evolution operator $e^{-i\hat{p}^2 t/2m\hbar}$ is simply the squeezing operator with

$$\xi = -\frac{i\omega t}{2} \text{ and } \eta = -\frac{\omega t}{2}. \tag{14.78}$$

Hence, the most general spreading Gaussian is a displaced-squeezed state, with the magnitudes of both ξ and η increasing linearly in time. Note that this squeezing operator is a special case, in which $\xi = i\eta$. In this case, we find it is clearer to express the simplified squeezing operator in the form

$$\hat{S}_\pm(k) = e^{ik(\hat{a}^\dagger \pm \hat{a})^2}, \tag{14.79}$$

where $k = \omega t/4$. Here, the operator we work with for the time evolution is $\hat{S}_-\left(\frac{\omega t}{4}\right)$, because it involves the exponential of the square of the momentum operator; we will find we also need to work with the exponential of a constant times the square of the position operator, which corresponds to $\hat{S}_+(k')$ for a suitably chosen k'.

We start with a coherent state $|\alpha\rangle$ with initial position $x_0 = 0$ and some arbitrary initial momentum p_0. Then, using the standard notation for coherent states, we have that $\alpha = ip_0\frac{1}{\sqrt{2\hbar m\omega}}$ for this special case. We use the free-particle Hamiltonian, with m and ω fixed in our calculations, and we have that the initial variance of the probability distribution is given by $\sigma_0^2 = \Delta_x^2 = \hbar/2m\omega$, since it is a coherent state.

To obtain the time-dependent wavefunction, we take the overlap of a position bra with the time-evolved initial coherent state;

$$\psi(x,t) = \langle x|e^{-\frac{i}{\hbar}\hat{\mathcal{H}}_{\text{free}}t}|\alpha\rangle = \langle x|e^{-\frac{it}{2\hbar m}\hat{p}^2}e^{\frac{ip_0}{\hbar}\hat{x}}|0\rangle. \tag{14.80}$$

Now, we begin our work to simplify Eq. (14.80). We use a multiply by 1 to create a Hadamard lemma expression, which, after evaluating the Hadamard lemma, effectively moves the $\exp(\frac{ip_0}{\hbar}\hat{x})$ term from the right of the time-evolution operator to its left.

$$\psi(x,t) = \langle x|\underbrace{e^{\frac{ip_0}{\hbar}\hat{x}}e^{-\frac{ip_0}{\hbar}\hat{x}}}_{=1}e^{-\frac{it}{2\hbar m}\hat{p}^2}e^{\frac{ip_0}{\hbar}\hat{x}}|0\rangle = \langle x|e^{\frac{ip_0}{\hbar}\hat{x}}\underbrace{e^{-\frac{ip_0}{\hbar}\hat{x}}e^{-\frac{it}{2\hbar m}\hat{p}^2}e^{\frac{ip_0}{\hbar}\hat{x}}}_{\text{Hadamard}}|0\rangle$$

$$= \langle x|e^{\frac{ip_0}{\hbar}\hat{x}}e^{-\frac{it}{2\hbar m}e^{-\frac{ip_0}{\hbar}\hat{x}}\hat{p}^2 e^{\frac{ip_0}{\hbar}\hat{x}}}|0\rangle = e^{\frac{ixp_0}{\hbar}}\langle x|e^{-\frac{it}{2\hbar m}(\hat{p}+p_0)^2}|0\rangle. \tag{14.81}$$

Here, we used the fact that $\langle x|\hat{x} = \langle x|x$ to replace $\langle x|\exp(\frac{ip_0}{\hbar}\hat{x})$ by $\langle x|\exp(\frac{ip_0}{\hbar}x)$ and then moved the number out of the matrix element. Evaluating the Hadamard lemma requires use of the braiding relation to move it into the exponent, where it

truncates after just the first commutator because it is a number, which subsequently commutes with all operators.

Next, we expand the squared exponent in Eq. (14.81), pull out the constant terms, let the terms with a linear exponent in $\hat{p}$ act on the bra to translate the position state, and then replace the momentum operator with its expression in terms of ladder operators in the quadratic exponent to give us

$$\psi(x,t) = e^{\frac{ip_0x}{\hbar} - \frac{itp_0^2}{2\hbar m}} \langle x - \tfrac{tp_0}{m} | e^{\frac{i\omega t}{4}(\hat{a}^\dagger - \hat{a})^2} |0\rangle. \tag{14.82}$$

We need to motivate our next step. The exponent of the operator is proportional to the difference of the ladder operators, and we do not know how such an operator acts on the position bra to the left. But, if we could find a way to change the difference of ladder operators into a sum in the exponent, then the operator in the exponent would be proportional to $\hat{x}^2$, which can be readily applied onto the position eigenbra to its left; this is the same strategy we employed when calculating the wavefunction of the simple harmonic oscillator back in Sect. 7.10. This then leaves us with the ground-state wavefunction evaluated at $x - \frac{tp_0}{m}$. In order to accomplish this sign change, we first use the exponential disentangling identity acting on the simple-harmonic-oscillator ground state and apply it to the operator with a quadratic exponent as given in Problem 14.6.1 below. We use the case with a difference of the ladder operators and $k = \omega t/4$ to yield

$$\psi(x,t) = \frac{e^{\frac{ip_0x}{\hbar} - \frac{itp_0^2}{2\hbar m}}}{\sqrt{1 + \frac{i\omega t}{2}}} \langle x - \tfrac{tp_0}{m} | e^{\frac{i\omega t}{4+2i\omega t}(\hat{a}^\dagger)^2} |0\rangle, \tag{14.83}$$

where two of the exponential factors in the exponential disentangling identity have been replaced by 1 because $\hat{a}|0\rangle = 0$. Next, we use the identity again, but now in reverse for the sum of ladder operators and with $k' = \frac{\omega t}{4(1+i\omega t)}$. In addition, we start from the expression on the right and replace it by the equivalent expression on the left. Some straightforward algebra then results in

$$\frac{ik'}{1-2ik'} = \frac{i\omega t}{4(1+i\omega t)\left(1 - \frac{2i\omega t}{4(1+i\omega t)}\right)} = \frac{i\omega t}{4+2i\omega t}. \tag{14.84}$$

So, we find

$$\begin{aligned}\psi(x,t) &= \frac{e^{\frac{ip_0x}{\hbar} - \frac{itp_0^2}{2\hbar m}}}{\sqrt{1 + \frac{i\omega t}{2}}} \langle x - \tfrac{tp_0}{m} | \sqrt{\tfrac{1+\frac{i\omega t}{2}}{1+i\omega t}}\, e^{\frac{i\omega t}{4(1+i\omega t)}(\hat{a}^\dagger + \hat{a})^2} |0\rangle \\ &= \frac{e^{\frac{ip_0x}{\hbar} - \frac{itp_0^2}{2\hbar m}}}{\sqrt{1 + i\omega t}} \langle x - \tfrac{tp_0}{m} | e^{\frac{im\omega^2 t}{2\hbar(1+i\omega t)}\hat{x}^2} |0\rangle\end{aligned}$$

$$= \frac{e^{\frac{ip_0x}{\hbar} - \frac{itp_0^2}{2\hbar m}}}{\sqrt{1+i\omega t}} e^{\frac{im\omega^2 t}{2\hbar(1+i\omega t)}\left(x - \frac{tp_0}{m}\right)^2} \langle x - \frac{tp_0}{m}|0\rangle$$

$$= \frac{e^{\frac{ip_0x}{\hbar} - \frac{itp_0^2}{2\hbar m}}}{\sqrt{1+i\omega t}} e^{\frac{im\omega^2 t}{2\hbar(1+i\omega t)}\left(x - \frac{tp_0}{m}\right)^2} \psi_{gs}\left(x - \frac{tp_0}{m}\right). \tag{14.85}$$

Substituting in the ground-state wavefunction from Eq. (14.72) yields our final result

$$\psi(x,t) = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} \frac{1}{\sqrt{1+i\omega t}} e^{-\frac{m\omega}{2\hbar(1+i\omega t)}\left(x - \frac{tp_0}{m}\right)^2 + \frac{ip_0x}{\hbar} - \frac{itp_0^2}{2\hbar m}}. \tag{14.86}$$

The probability density becomes

$$|\psi(x,t)|^2 = \sqrt{\frac{m\omega}{\pi\hbar(1+\omega^2t^2)}} e^{-\frac{m\omega}{\hbar(1+\omega^2t^2)}\left(x - \frac{tp_0}{m}\right)^2}. \tag{14.87}$$

Now we see the variance is increasing approximately quadratically in time as

$$\sigma^2(t) = \Delta_x^2(t) = \frac{\hbar}{2m\omega}(1+\omega^2t^2) = \sigma^2(0)(1+\omega^2t^2). \tag{14.88}$$

What is remarkable is that this result can be directly related to recent experiments in trapped atoms, where they measure the shape of the harmonic oscillator wavefunction by performing a time-of-flight experiment, which, for the ground state, is just the free expansion of a Gaussian wavepacket! The recent experimental work has (i) created the ground state, the first excited state, and the second excited state and (ii) then used time-of-flight spectroscopy to measure the momentum distribution of the atom when it is in these different Fock states Brown et al. (2023). This can then be related to the position-space wavefunctions, because the shape of the wavefunctions are the same in position and momentum space for the harmonic oscillator.

In a time-of-flight experiment, one prepares a reasonably localized state (in Regal's experiment the state is localized on the order of $0.3 - 0.5\,\mu$m). Then at a specific time, the harmonic oscillator trapping potential is shut off (a quench to $\omega \to 0$, just like we analyzed above). We start a clock at this time and let the atom expand in its wavepacket for some time before measuring its position (the time of flight is about $10\,\mu$s). At the moment of the measurement, we know the position to the accuracy of the position detector—but, we also know its momentum, which is inferred from the fact that it traveled as a free particle from its localized origin to where it was detected in the time of flight. So, at the moment of the measurement, we know the position and momentum of the quantum particle for this one experimental shot! More details about the experiment and its results are shown in Fig. 14.9.

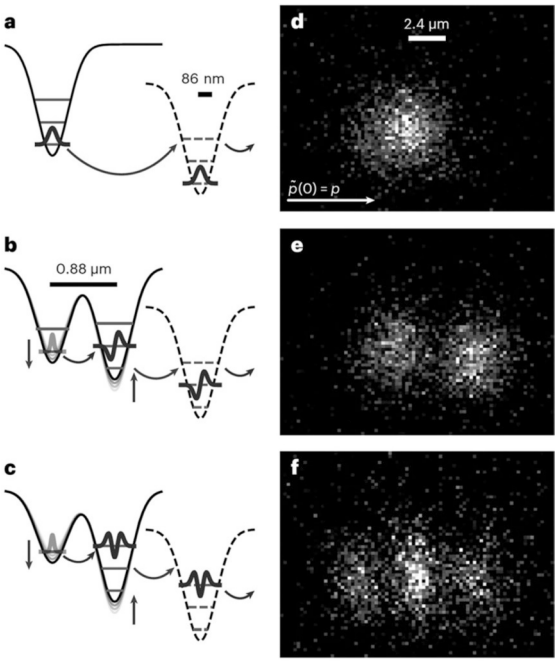

Fig. 14.9 Single-atom harmonic oscillator Fock-state preparation and imaging from the Regal experiment. Panels (**a**)–(**c**): Illustration of motional state preparation of $n = 0,\ 1$, and 2 states, respectively. Panels (**d**)–(**f**): Time-of-flight momentum distributions after allowing the atom to expand when the trap was dropped. You can clearly see the shapes of the probability distributions for the ground and first two excited states. This is one of the ways we can indirectly image a wavefunction (since this wavefunction can be taken to be real, it is equal to the square root of the probability distribution). [Figure reprinted from Brown et al. (2023) with permission © Nature Publishing Group]

Does this violate uncertainty? Of course not. The uncertainty principle applies to the spread in the results if we repeat the measurement many times (which will actually determine the original momentum distribution of the quantum state), because the momentum distribution does not spread in time. And what about the original position uncertainty? Unfortunately, we cannot determine that, because this measurement is not sensitive to the initial position. But the uncertainty in knowing just what the initial position is (due to the initial wavepacket spread) does lead to an intrinsic uncertainty in the momentum measurement in the time-of-flight experiment. We can make that uncertainty as small as desired by simply increasing the time for the time of flight (which reduces the relative error for the momentum measurement).

Note that one cannot create a quantum state with definite position and momentum, but this does not mean that we cannot *measure* position and momentum at the same time. We can as described above.

In the measurements, they examine not just the momentum distribution of the ground state, but of the two lowest excited states as well. We can easily generalize the techniques we developed above to determine the momentum distributions for these different Fock states too. The calculation is quite simple because all harmonic oscillator Fock states maintain their shape (in momentum space) during time of flight. So, if we start in the nth Fock state, given by $\frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle$, then the momentum distribution after a time of flight of duration t is simply given by

$$\phi_n(p,t) = \langle p|e^{-\frac{i}{\hbar}\hat{\mathcal{H}}_{free}t}\frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle, \tag{14.89}$$

which can be immediately solved to be

$$\phi_n(p,t) = \frac{1}{(\pi\hbar m\omega)^{\frac{1}{4}}}\frac{1}{\sqrt{n!2^n}}H_n\left(\frac{p}{\sqrt{\hbar m\omega}}\right)e^{-\frac{p^2}{2\hbar m\omega}-\frac{ip^2t}{2\hbar m}}. \tag{14.90}$$

One might say that because we directly measure the position in the time of flight, we should really calculate the position distribution at time t. One can do this as well for the Fock states, but it takes a few more steps. We leave this to Problem 14.6.4. I hope you will agree the free expansion of a Gaussian is a fascinating problem, which is even more exciting because it has been actually measured with trapped atoms. I have given a talk encouraging others to teach this material in a video https://tinyurl.com/mv8rmx7v.

Section 14.6 Problems:

14.6.1 Disentangling Identities of the Symplectic Group

In this problem, we will work out disentangled expressions for the following two operators:

$$e^{ik\left(\hat{a}^\dagger\pm\hat{a}\right)^2} = e^{ik\left((\hat{a}^\dagger)^2\pm(\hat{a}^\dagger\hat{a}+\hat{a}\hat{a}^\dagger)+\hat{a}^2\right)}, \tag{14.91}$$

with k a complex number.

We start by working with the generators of the symplectic Lie algebra:

$$\hat{K}_0 = \frac{1}{4}(\hat{a}^\dagger\hat{a}+\hat{a}\hat{a}^\dagger),\ \hat{K}_+ = \frac{1}{2}(\hat{a}^\dagger)^2,\ \text{and}\ \hat{K}_- = \frac{1}{2}\hat{a}^2, \tag{14.92}$$

which satisfy the commutation relations

$$\left[\hat{K}_0,\hat{K}_\pm\right] = \pm\hat{K}_\pm \text{ and } \left[\hat{K}_+,\hat{K}_-\right] = -2\hat{K}_0. \tag{14.93}$$

Then, we have

$$e^{ik(\hat{a}^\dagger \pm \hat{a})^2} = e^{2ik\left(\hat{K}_+ \pm 2\hat{K}_0 + \hat{K}_-\right)}, \tag{14.94}$$

and our goal is to find a, b, and c such that

$$e^{2ik(\hat{K}_+ \pm 2\hat{K}_0 + \hat{K}_-)} = e^{a\hat{K}_+} e^{b\hat{K}_0} e^{c\hat{K}_-}. \tag{14.95}$$

Use the 2×2 matrix representation of the symplectic Lie algebra:

$$\hat{K}_0 \leftrightarrow \frac{1}{2}\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \ \hat{K}_+ \leftrightarrow \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix}, \text{ and } \hat{K}_- \leftrightarrow \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}. \tag{14.96}$$

You should verify this representation satisfies the commutation relations before proving the disentangling identities. Verify that you obtain

$$e^{ik(\hat{a}^\dagger \pm \hat{a})^2} = e^{\frac{ik}{1 \mp 2ik}(\hat{a}^\dagger)^2} e^{-\frac{1}{2}\ln(1 \mp 2ik)(\hat{a}^\dagger \hat{a} + \hat{a}\hat{a}^\dagger)} e^{\frac{ik}{1 \mp 2ik}\hat{a}^2}. \tag{14.97}$$

Act this on the simple-harmonic-oscillator ground state to show that

$$e^{ik(\hat{a}^\dagger \pm \hat{a})^2}|0\rangle = \frac{1}{\sqrt{1 \mp 2ik}} e^{\frac{ik}{1 \mp 2ik}(\hat{a}^\dagger)^2}|0\rangle. \tag{14.98}$$

This last identity is used in our derivation with two different k values, $k = \frac{\omega t}{4}$ and $k' = \frac{\omega t}{4(1 + i\omega t)}$.

14.6.2 Momentum-Space Evolution of a Gaussian Wavepacket

Start from

$$\phi(p, t) = \langle p|e^{-\frac{i}{\hbar}\hat{\mathcal{H}}_{\text{free}} t}|\alpha\rangle = \langle p|e^{-\frac{it}{2\hbar m}\hat{p}^2} e^{\frac{ip_0}{\hbar}\hat{x}}|0\rangle. \tag{14.99}$$

Simplify to

$$\phi(p, t) = e^{-\frac{ip^2 t}{2m\hbar}}\langle p - p_0|0\rangle. \tag{14.100}$$

Use the ground-state wavefunction of the harmonic oscillator in momentum space to further simplify to

$$\phi(p, t) = \frac{1}{(\pi\hbar m\omega)^{\frac{1}{4}}} e^{-\frac{(p-p_0)^2}{2\hbar m\omega} - \frac{ip^2 t}{2m\hbar}}. \tag{14.101}$$

14.6.3 Expansion or No Expansion

Why does the width of the wavepacket expand in position space, but not in momentum space? Write a persuasive essay to explain how this works.

14.6.4 Free Evolution of a Fock State

In this problem, we will examine the free evolution of excited states of the harmonic oscillator. The only difference in the derivation is at the point when we use the exponential disentangling identity. In this case, the exponentials of $\hat{a}^2$ and of $\hat{a}^\dagger\hat{a}$ cannot immediately act on the ground state because the $\left(\hat{a}^\dagger\right)^n$ term is in the way. Move the exponentials past that operator by using a "multiply by one" followed by the Hadamard lemma. Do this twice—once when you use the disentangling identity on the square of the momentum operator and once when you bring the required factors needed to convert the exponential to the square of the position operator. Verify that you obtain

$$\psi_n(x,t) = \frac{1}{\sqrt{1+i\omega t}} e^{\frac{im\omega^2 t}{2\hbar(1+i\omega t)}x^2} \frac{1}{\sqrt{n!}(1+i\omega t)^n} \times \langle x| \left(\hat{a}^\dagger + (i\omega t)^2\hat{a}\right)^n |0\rangle. \tag{14.102}$$

If $n = 0$, this is the same answer as we had before. Explicitly work out the result for $n = 1$. Explain why an experiment measuring a particle prepared in this state will have the shape of the probability distribution of the first excited state of the harmonic oscillator.

14.6.5 Time-of-Flight Experiment

Write an essay about time-of-flight experiment for free expansion of a Gaussian wavepacket and explain why it measures the initial momentum distribution of the quantum particle in the ground state of the harmonic oscillator. In the experiment from the Regal lab, they let the atoms expand and then measure them repeatedly in quick succession so that enough light is emitted to detect the atomic positions. Explain in your essay what this implies with regard to whether we are measuring position or momentum and the state the system collapses into after each measurement.

14.7 Factorization of the Evolution Operator and the Interaction Representation

We discussed two different ways to represent time-dependent quantum mechanics—one was the Schrödinger picture, which put all time dependence in the states—and one was the Heisenberg picture, which put all the time dependence into the operators. Dirac devised a third picture, called *the interaction representation*, which can loosely be thought of as "halfway between the other two." This representation is used to create a perturbation theory expansion for the time-evolution operator by factorizing the evolution operator and then expanding one of the factors in a power series. We will examine how this works in this section.

The idea behind the interaction representation is that we decompose the Hamiltonian into two pieces, $\hat{H} = \hat{H}_0 + \hat{V}$, and then factorize the evolution operator into a piece that depends only on the time evolution with respect to $\hat{H}_0$ and whatever remains (which depends on both $\hat{V}$ and $\hat{H}_0$). One might ask, "why would you want to do that?" The answer is that some problems have Hamiltonians that can be decomposed into two pieces. The first piece $\hat{H}_0$ has an evolution operator that can be exactly determined, and the second piece is small in some sense. The factorization does not require the smallness, but in the most used applications, the $\hat{V}$ term is small, so we can further expand the second factor in a power series in $\hat{V}$. This process is called perturbation theory, more specifically *time-dependent perturbation theory*. And we will explain how we use it and how the theory is developed.

The derivation is straightforward and just uses ideas we have already developed. We need the Trotter formula, the Baker-Campbell-Hausdorff formula, and the braiding relation. But, I want to point out that the derivation has a number of steps to it, so it is easy to get lost. We will go through it slowly, in order to keep you on track. Once you see the pattern of what we are doing, it will all become clear to you. We start by writing the time-evolution operator in the standard Trotter form:

$$\hat{U}(t) = e^{-\frac{i\Delta t}{\hbar}\left(\hat{H}_0(t)+\hat{V}(t)\right)} e^{-\frac{i\Delta t}{\hbar}\left(\hat{H}_0(t-\Delta t)+\hat{V}(t-\Delta t)\right)} \cdots e^{-\frac{i\Delta t}{\hbar}\left(\hat{H}_0(\Delta t)+\hat{V}(\Delta t)\right)}. \tag{14.103}$$

Now, we know from the earlier work that we can take each Trotter factor and replace it by the product of an exponential of each term in the exponent multiplied by an error of order $(\Delta t)^2$. But, in the limit where we take $\Delta t \to 0$, we can ignore the correction terms. So, we have that

$$\hat{U}(t) = e^{-\frac{i\Delta t}{\hbar}\hat{V}(t)} e^{-\frac{i\Delta t}{\hbar}\hat{H}_0(t)} e^{-\frac{i\Delta t}{\hbar}\hat{V}(t-\Delta t)} e^{-\frac{i\Delta t}{\hbar}\hat{H}_0(t-\Delta t)} \cdots e^{-\frac{i\Delta t}{\hbar}\hat{V}(\Delta t)} e^{-\frac{i\Delta t}{\hbar}\hat{H}_0(\Delta t)}. \tag{14.104}$$

Note that we could have chosen either order for the exponential factors corresponding to each Trotter factor—with the $\hat{V}$ first or last. This choice with the $\hat{V}$ first makes for a clearer derivation. Now, we do an exponential reordering. This is done by a multiply by one followed by a braiding operation. So, we take

$$e^{-\frac{i\Delta t}{\hbar}\hat{V}(t)}e^{-\frac{i\Delta t}{\hbar}\hat{H}_0(t)} = \underbrace{e^{-\frac{i\Delta t}{\hbar}\hat{H}_0(t)}e^{\frac{i\Delta t}{\hbar}\hat{H}_0(t)}}_{=1} e^{-\frac{i\Delta t}{\hbar}\hat{V}(t)}e^{-\frac{i\Delta t}{\hbar}\hat{H}_0(t)}$$

$$= e^{-\frac{i\Delta t}{\hbar}\hat{H}_0(t)} \underbrace{e^{-\frac{i\Delta t}{\hbar}e^{\frac{i\Delta t}{\hbar}\hat{H}_0(t)}\hat{V}(t)e^{-\frac{i\Delta t}{\hbar}\hat{H}_0(t)}}}_{\text{braiding}}. \tag{14.105}$$

We do this transformation for every Trotter factor, moving the exponential of the corresponding $\hat{H}_0$ term to the left one step followed by a braiding operation. This gives

$$\begin{aligned}\hat{U}(t) = {} & e^{-\frac{i\Delta t}{\hbar}\hat{H}_0(t)}e^{-\frac{i\Delta t}{\hbar}e^{\frac{i\Delta t}{\hbar}\hat{H}_0(t)}\hat{V}(t)e^{-\frac{i\Delta t}{\hbar}\hat{H}_0(t)}} \\ & \times e^{-\frac{i\Delta t}{\hbar}\hat{H}_0(t-\Delta t)}e^{-\frac{i\Delta t}{\hbar}e^{\frac{i\Delta t}{\hbar}\hat{H}_0(t-\Delta t)}\hat{V}(t-\Delta t)e^{-\frac{i\Delta t}{\hbar}\hat{H}_0(t-\Delta t)}} \\ & \times \cdots e^{-\frac{i\Delta t}{\hbar}\hat{H}_0(\Delta t)}e^{-\frac{i\Delta t}{\hbar}e^{\frac{i\Delta t}{\hbar}\hat{H}_0(\Delta t)}\hat{V}(\Delta t)e^{-\frac{i\Delta t}{\hbar}\hat{H}_0(\Delta t)}}.\end{aligned} \tag{14.106}$$

We now move each of the exponential of the $\hat{H}_0$ factors to the left one more step. This will give another braiding operation to every $\hat{V}$ term, except the rightmost one. Note that this is building up the time-evolution operator in the exponentials that have the $\hat{V}$ terms because the time-evolution operators are being accumulated in the proper order for the time-ordered product (or its Hermitian conjugate). So this next step becomes

$$\begin{aligned}\hat{U}(t) = {} & e^{-\frac{i\Delta t}{\hbar}\hat{H}_0(t)}e^{-\frac{i\Delta t}{\hbar}\hat{H}_0(t-\Delta t)}e^{-\frac{i\Delta t}{\hbar}e^{\frac{i}{\hbar}\hat{H}_0(t-\Delta t)}e^{\frac{i\Delta t}{\hbar}\hat{H}_0(t)}\hat{V}(t)e^{-\frac{i\Delta t}{\hbar}\hat{H}_0(t)}e^{-\frac{i\Delta t}{\hbar}\hat{H}_0(t-\Delta t)}} \\ & \times e^{-\frac{i\Delta t}{\hbar}\hat{H}_0(t-2\Delta t)}e^{-\frac{i\Delta t}{\hbar}e^{\frac{i\Delta t}{\hbar}\hat{H}_0(t-2\Delta t)}e^{\frac{i\Delta t}{\hbar}\hat{H}_0(t-\Delta t)}\hat{V}(t-\Delta t)e^{-\frac{i\Delta t}{\hbar}\hat{H}_0(t-\Delta t)}e^{-\frac{i\Delta t}{\hbar}\hat{H}_0(t-2\Delta t)}} \\ & \cdots e^{-\frac{i\Delta t}{\hbar}e^{\frac{i\Delta t}{\hbar}\hat{H}_0(\Delta t)}\hat{V}(\Delta t)e^{-\frac{i\Delta t}{\hbar}\hat{H}_0(\Delta t)}}.\end{aligned} \tag{14.107}$$

We repeat this $N-2$ more times. You can see that we will then have moved all of the $\hat{H}_0$ exponential terms to the left. Furthermore, the exponentials with all of the $\hat{V}$ terms will have a similarity transformation with the evolution operator with respect to the $\hat{H}_0$ terms only. We need to define the time-evolution operator with respect to $\hat{H}_0(t)$ only

$$\hat{U}_0(t) = e^{-\frac{i\Delta t}{\hbar}\hat{H}_0(t)}e^{-\frac{i\Delta t}{\hbar}\hat{H}_0(t-\Delta t)}\cdots e^{-\frac{i\Delta t}{\hbar}\hat{H}_0(\Delta t)}, \tag{14.108}$$

and the interaction representation "potential"

$$\hat{V}_I(t) = \hat{U}_0^{\dagger}(t)\hat{V}(t)\hat{U}_0(t). \tag{14.109}$$

Then, we can write the time-evolution operator in the following factorized form:

$$\hat{U}(t) = \hat{U}_0(t) e^{-\frac{i\Delta t}{\hbar}\hat{V}_I(t)} e^{-\frac{i\Delta t}{\hbar}\hat{V}_I(t-\Delta t)} \cdots e^{-\frac{i\Delta t}{\hbar}\hat{V}_I(\Delta t)}. \tag{14.110}$$

This is the final result for the interaction representation. We have factorized the time-evolution operator into the evolution operator with respect to $\hat{H}_0$ only, followed by a time-ordered product of the interaction representation "potential," which can be thought of as time evolution with respect to $\hat{V}_I$. It may surprise you that a time evolution with respect to the full Hamiltonian can be separated in this way.

There are some special cases of this formula that are important to analyze. The first one is the case where $\hat{H}_0$ is independent of time. In this situation, the factorized formula becomes

$$\begin{aligned}\hat{U}(t) = e^{-\frac{it}{\hbar}\hat{H}_0} e^{-\frac{i\Delta t}{\hbar} e^{\frac{it}{\hbar}\hat{H}_0}\hat{V}(t)e^{-\frac{it}{\hbar}\hat{H}_0}} e^{-\frac{i\Delta t}{\hbar} e^{\frac{i(t-\Delta t)}{\hbar}\hat{H}_0}\hat{V}(t-\Delta t)e^{-\frac{i(t-\Delta t)}{\hbar}\hat{H}_0}} \cdots \\ \times e^{-\frac{i\Delta t}{\hbar} e^{\frac{i\Delta t}{\hbar}\hat{H}_0}\hat{V}(\Delta t)e^{-\frac{i\Delta t}{\hbar}\hat{H}_0}}.\end{aligned} \tag{14.111}$$

Usually, when we work with this expression, the Hadamard lemma computation to determine $\hat{V}_I$ is able to be carried out concretely as well, but we cannot show how that works without choosing explicit examples for the terms in the original Hamiltonian. A second case, which might not seem so obvious at first, is the case where $[\hat{H}_0(t), \hat{H}_0(t')] = 0$, that is, the unperturbed Hamiltonian commutes with itself at two different times. There are a number of different ways how this can occur, but the simplest way is if $\hat{H}_0(t) = f(t)\hat{O}$, a function of time multiplied by an operator. In this case, we have

$$\hat{U}_0(t) = e^{-\frac{i\Delta t}{\hbar}f(t)\hat{O}} e^{-\frac{i\Delta t}{\hbar}f(t-\Delta t)\hat{O}} \cdots e^{-\frac{i\Delta t}{\hbar}f(\Delta t)\hat{O}} = e^{-\frac{i\Delta t}{\hbar}\sum_{n=1}^{N} f(n\Delta t)\hat{O}}. \tag{14.112}$$

This is because $\hat{O}$ commutes with itself, so the product of terms in the first equation can be consolidated into one single exponent. We have already seen that summations of the form $\Delta t \sum_{n=1}^{N} f(n\Delta t)$ can be explicitly evaluated if f is an exponential, a sine or cosine, or a power. We will only analyze such cases. Once $\hat{U}_0$ is determined, then the interaction representation potential is found from the Hadamard lemma, and often that calculation can also be evaluated exactly.

One of the main reasons to perform this factorization is to isolate the time evolution with respect to $\hat{H}_0$ from that of $\hat{V}_I$. This is because in the case where $\hat{H}_0$ is large and $\hat{V}_I$ is small, we can now perform an expansion with respect to the small object. If we had not separated the two terms and we did an expansion, there is no clear justification for it, as we are expanding large and small terms in the power series. Problem 14.7.2 below investigates this issue more concretely. Here, we will follow the expansion when the potential is small. Then we replace each Trotter factor in $\hat{V}_I$ by a power series expansion that is truncated to a chosen order. We then collect all terms in the expansion of the product to that

order and this final expression is called the nth-order Born approximation, or the nth-order time-dependent perturbation theory expansion. Let's work out the first Born approximation. The second Born approximation is left for Problem 14.7.3. In most applications people stop after the first Born approximation, not because the higher-order terms are unimportant, but because they are too difficult to work with (although sometimes they really are unimportant). So watch carefully when someone else does an expansion to understand why they stopped at the order they stopped at.

We now show how to perform the expansion for the first Born approximation. We start by expanding the exponential containing each $\hat{V}_I$ term in a power series, keeping only the first two terms. Then Eq. (14.110) becomes

$$\hat{U}(t) \approx \hat{U}_0(t)\left(1 - \frac{i\Delta t}{\hbar}\hat{V}_I(t)\right)\left(1 - \frac{i\Delta t}{\hbar}\hat{V}_I(t-\Delta t)\right)\cdots\left(1 - \frac{i\Delta t}{\hbar}\hat{V}_I(\Delta t)\right). \tag{14.113}$$

Our next step is to collect the terms of zeroth order and first order in $\hat{V}_I$ only. One can see the zeroth-order term is just 1, while the first-order terms become the sum over all $\hat{V}_I$ terms. So we obtain

$$\hat{U}(t) \approx \hat{U}_0(t)\left(1 - \frac{i\Delta t}{\hbar}\sum_{n=1}^{N}\hat{V}_I(n\Delta t)\right). \tag{14.114}$$

Interestingly, the first Born approximation no longer has any time ordering for the interaction representation potential terms, because they appear only to first order. You will see in the problems that time ordering does enter for the higher-order expansions. The general approach proceeds in the same fashion. We truncate the exponential at a given order and collect all terms of that order after evaluating all the products. It is straightforward, but cumbersome to work out higher-order contributions.

As we mentioned before, one of the important quantities to calculate is the probability to make a transition from a state $|m\rangle$ to a state $|n\rangle$ due to the time dependence of the system. We examine the simplified case, where $\hat{H}_0$ is time independent, so it has energy eigenstates, which we denote with a subscript 0, so $\hat{H}_0|m\rangle_0 = E_m|m\rangle_0$. Then, we find that

$$P_{n\leftarrow m}(t) = \left|{}_0\langle n|e^{-\frac{it}{\hbar}\hat{H}_0}\left(1 - \frac{i\Delta t}{\hbar}\sum_{j=1}^{N}\hat{V}_I(j\Delta t)\right)|m\rangle_0\right|^2, \tag{14.115}$$

in the first Born approximation. The first factor in the evolution operator can be evaluated against the energy eigenstate bra, giving the factor $e^{-\frac{i}{\hbar}E_n t}$, which is a complex phase whose modulus is 1 and can be removed. Similarly, because $\hat{V}_I(t) =$

$e^{\frac{it}{\hbar}\hat{H}_0}\hat{V}(t)e^{-\frac{it}{\hbar}\hat{H}_0}$, we can evaluate the exponential factors against the eigenstates as well. Assuming $m \neq n$, we find that

$$P_{\substack{n\leftarrow m \\ m\neq n}}(t) = \left| \frac{i\Delta t}{\hbar} \sum_{j=1}^{N} e^{\frac{ij\Delta t}{\hbar}(E_n - E_m)}\, {}_0\langle n|\hat{V}(j\Delta t)|m\rangle_0 \right|^2, \tag{14.116}$$

or

$$P_{\substack{n\leftarrow m \\ m\neq n}}(t) = \frac{1}{\hbar^2} \left| \Delta t \sum_{j=1}^{N} e^{i\omega_{nm} j\Delta t}\, {}_0\langle n|\hat{V}(j\Delta t)|m\rangle_0 \right|^2. \tag{14.117}$$

Here, we set $\frac{1}{\hbar}(E_n - E_m) = \omega_{nm}$. For an arbitrary time-varying operator, the matrix element is complicated and we cannot simplify it further.

Let's apply the general formula to the magnetic resonance problem with spin one-half. Then $\hat{H}_0 = -\Omega\hat{S}_z$ and $\hat{V}(t) = -\frac{\gamma B_1}{\hbar} e^{i\omega\hat{S}_z}\hat{S}_x e^{-i\omega\hat{S}_z}$. Plugging into the transition formula, we have

$$P_{\downarrow\leftarrow\uparrow}(t) = \frac{\gamma^2 B_1^2}{\hbar^4} \left| \Delta t \sum_{j=1}^{N} e^{i\Omega j\Delta t}\, {}_z\langle\downarrow| e^{\frac{i}{\hbar}\omega j\Delta t\hat{S}_z}\hat{S}_x e^{-\frac{i}{\hbar}\omega j\Delta t\hat{S}_z}|\uparrow\rangle_z \right|^2. \tag{14.118}$$

The exponential factors of $\hat{S}_z$ can be directly evaluated against the states to the right or to the left, because they are eigenstates of that operator. Then, the matrix element of $\hat{S}_x$ can be computed. We find

$$P_{\downarrow\leftarrow\uparrow}(t) = \frac{\gamma^2 B_1^2}{4\hbar^2} \left| \Delta t \sum_{j=1}^{N} e^{i(\Omega-\omega)j\Delta t} \right|^2. \tag{14.119}$$

The sum is just a geometric sum, which we worked out earlier (we need to factor out a factor of $e^{i(\Omega-\omega)\Delta t}$). Using Eq. (13.82), we find it is equal to

$$P_{\downarrow\leftarrow\uparrow}(t) = \frac{\gamma^2 B_1^2}{4\hbar^2} \left| e^{i(\Omega-\omega)\Delta t}\Delta t \frac{1 - e^{i(\Omega-\omega)N\Delta t}}{1 - e^{i(\Omega-\omega)\Delta t}} \right|^2. \tag{14.120}$$

Then, taking the limit as $\Delta t \to 0$, we find

$$P_{\downarrow\leftarrow\uparrow}(t) = \left(\frac{\gamma B_1}{2\hbar(\Omega-\omega)} \right)^2 \left| 1 - e^{i(\Omega-\omega)t} \right|^2 \tag{14.121}$$

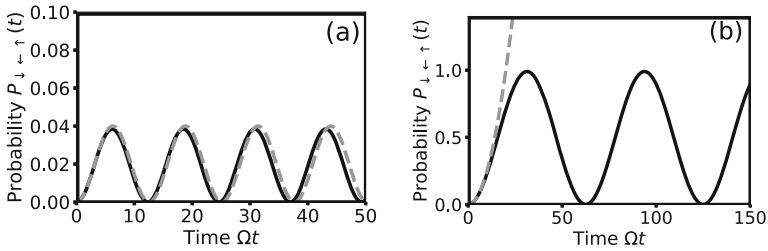

Fig. 14.10 Plot of the probability to flip a spin in a magnetic resonance experiment. Panel (**a**) is off-resonant $\omega = 0.5\Omega$ and panel (**b**) is near-resonant $\omega = 0.99\Omega$. The solid black line is the exact solution and the dashed gray line is the perturbation theory. Note how both results are quite accurate for short times, but become more and more inaccurate for long times. This occurs faster for the near-resonant case. In both calculations, we choose $\frac{\gamma B_1}{\hbar\Omega} = 0.1$. Note that the scales are different in the two figures

or

$$P_{\downarrow\leftarrow\uparrow}(t) = \left(\frac{\gamma B_1}{2\hbar(\Omega-\omega)}\right)^2 2\big(1-\cos(\Omega-\omega)t\big)$$
$$= \left(\frac{\gamma B_1}{\hbar(\Omega-\omega)}\right)^2 \sin^2\left(\frac{(\Omega-\omega)t}{2}\right). \tag{14.122}$$

This result resembles the exact result in Eq. (14.29) very closely. But there is an important difference. In the exact formula, the denominator never vanishes, but here, it does when $\Omega = \omega$. This condition is called *resonance*, and the perturbation theory result is often inaccurate near resonance. When we are on resonance, one can see that the result becomes $P_{\downarrow\leftarrow\uparrow}(t) = \left(\frac{\gamma B_1 t}{2\hbar}\right)^2$. This is accurate for short times, but as t grows, it will eventually become larger than 1, which is nonsense. This is a well-known issue with perturbation theory. It is an expansion valid for short times only. Here, the criterion is that it is good as long as the probability to flip the spin remains very small. We compare the exact and the perturbation theory result away from resonance and on resonance in Fig. 14.10.

For a class of operators called *harmonic perturbations*, we can simplify the general probability for the transition formula even more. A harmonic perturbation is defined in terms of

$$\hat{V}(t) = e^{i\Omega t}\hat{v}^\dagger + e^{-i\Omega t}\hat{v}; \tag{14.123}$$

that is, it is the sum of two terms that are Hermitian conjugates of each other with $\hat{v}$ any operator. We pick $\Omega > 0$ for concreteness. We also assume that both $v_{nm} = {}_0\langle n|\hat{v}|m\rangle_0$ and $v^*_{mn} = {}_0\langle n|\hat{v}^\dagger|m\rangle_0$ are nonzero; note that because the operators are not Hermitian, the matrix elements need not be directly related to each other. The probability for a transition then becomes

$$
\begin{aligned}
P_{\substack{n\leftarrow m\\ m\neq n}}(t) &= \frac{1}{\hbar^2}\left|\Delta t\sum_{j=1}^{N} e^{i\omega_{nm}j\Delta t}\left(e^{i\Omega j\Delta t}v_{mn}^{*}+e^{-i\Omega j\Delta t}v_{nm}\right)\right|^2\\
&= \frac{1}{\hbar^2}\left|\frac{v_{mn}^{*}}{i(\omega_{mn}-\Omega)}\left(e^{i(\omega_{nm}+\Omega)t}-1\right)+\frac{v_{nm}}{i(\omega_{nm}-\Omega)}\left(e^{i(\omega_{nm}-\Omega)t}-1\right)\right|^2\\
&= \frac{4}{\hbar^2}\left(\frac{|v_{mn}|^2}{(\omega_{mn}-\Omega)^2}\sin^2(\omega_{mn}-\Omega)\tfrac{t}{2}+\frac{|v_{nm}|^2}{(\omega_{nm}-\Omega)^2}\sin^2(\omega_{nm}-\Omega)\tfrac{t}{2}\right.\\
&\qquad \left.+\ \text{cross terms}\right). \qquad (14.124)
\end{aligned}
$$

We summed the summation using the geometric series, took the limit of $\Delta t \to 0$, and then expanded the modulus squared. We switched $\omega_{nm} \to -\omega_{mn}$ in one term, and we did not write out explicitly the cross terms, as they are more complicated (they are left as a problem for you). The system is on resonance when the denominators vanish. If $\omega_{nm} > 0$, then this occurs when $\omega_{nm} = \Omega$. If, on the other hand, $\omega_{mn} > 0$, then resonance occurs when $\omega_{mn} = \Omega$. Hence, we have resonance whenever $|\omega_{nm}| = \Omega$. When $E_n = E_m + \hbar\Omega$, we have *stimulated absorption* (a transition to a higher-energy state), while when $E_n = E_m - \hbar\Omega$, we have *stimulated emission* (a transition to a lower-energy state). We use the word stimulated because the transitions are driven by the field we have applied. The transitions occur most rapidly when we are on resonance.

One of the problems with the first Born approximation is that it neglects the phenomena of *depletion* and *return*. Depletion occurs when we have had a lot of transitions from the m state. Then there is a lower probability to be in the m state and then to transition out of it. The approximation we evaluated does not include this fact, as it is a higher-order effect. Return also occurs when we have had a high probability to transition out of the m state. Then it is possible to return from another state back to the m state. This is also a higher-order process that is neglected. In some cases return does not occur, such as in a transition that emits a photon from an atom in free space. Then the photon travels away from the atom and it is so far away it cannot return to cause the reverse transition. In such cases, the first Born approximation is more accurate.

Section 14.7 Problems:

14.7.1 Completing the Argument Leading to the Factorized Form of the Interaction Representation

When we factorized the evolution operator into the two terms, we jumped from the second step to the final result. Carefully fill in the details to show how the argument proceeds to the final answer.

14.7.2 Expanding U Versus Expanding the Interaction Representation

Starting from the Trotter formula in Eq. (14.103), expand the time-evolution operator through second order in $\hat{H} = \hat{H}_0 + \hat{V}$. Next, take the final result for the interaction representation in Eq. (14.110) and expand only the terms with $\hat{V}_I$ to second order (if having trouble, look at the next problem).

Compare the two expansions and explain why it is better to expand the interaction representation with respect to $\hat{V}_I$ instead of the ordinary Trotter formula with respect to $\hat{H}$ if $\hat{V}_I$ is small.

14.7.3 The Second Born Approximation

Expand each of the Trotter factors with the $\hat{V}_I$ terms to second order in $\hat{V}_I$. Then collect all terms up to the quadratic order. You will find the second-order answer involves double summations of products of the $\hat{V}_I$ terms at two different times. You should find all products are time ordered, with late times to the left in the products.

You should find that

$$\hat{U}_I(t) = 1 - \frac{i\Delta t}{\hbar}\sum_{j=1}^{N}\hat{V}_I(j\Delta t) - \frac{(\Delta t)^2}{\hbar^2}\sum_{j=1}^{N}\sum_{k=1}^{j-1}\hat{V}_I(j\Delta t)\hat{V}_I(k\Delta t) - \frac{1}{2\hbar^2}\sum_{j=1}^{N}\Big(\hat{V}_I(j\Delta t)\Big)^2. \tag{14.125}$$

But, when we take the limit as $\Delta t \to 0$, the last term vanishes because we only have N terms, but it is multiplied by $(\Delta t)^2$. Note that the k index is always smaller than the j index.

14.7.4 Second Born Approximation for Magnetic Resonance

Taking the second-order expansion results from the previous problem, work out the results for the magnetic resonance problem with spin one-half to find the probability to flip the spin as a function of time in the second Born approximation. Compare your result to the exact answer.

14.7.5 Harmonic Perturbation

Show how the magnetic resonance problem is a harmonic perturbation, but for spin one-half we had just one term, not two (ignoring cross terms, which also are not there for this case). Explain why.

Hint: $\hat{S}_x = \frac{1}{2}(\hat{S}_+ + \hat{S}_-)$.

14.7.6 Cross Terms for the Harmonic Perturbation

Determine the cross terms in the transition probability for the harmonic perturbation and verify that the terms included in the main text are the dominant ones on resonance. Do the cross terms have any other condition for resonance?

14.7.7 Driving a Time-Independent Problem on Resonance

A spin is initialized in the $|\uparrow\rangle_z$ state. At time $t = 0$, a magnetic field is applied in the θ-ϕ direction. The Hamiltonian for $t > 0$ is then $-\gamma \hat{\vec{S}} \cdot \vec{B}$. Compute the probability for the spin to flip to the down state as a function of time in the first Born approximation. You should find the probability is equal to $\frac{\gamma^2 B^2 \sin^2\theta}{4\hbar^2} t^2$, with B the magnitude of the magnetic field applied at $t = 0$.

Compute the probability for the spin to flip to the down state as a function of time for the exact solution. You should find the probability is given by $\sin^2 \frac{\gamma B}{2\hbar} \times \sin^2\theta$.

Compare these two expressions and discuss under what circumstances the perturbation theory is accurate.

14.7.8 A Time-Dependent Spin Model

Consider a two-spin antiferromagnet placed in its ground state

$$|\psi\rangle = \frac{1}{\sqrt{2}} \left(|\uparrow\rangle_z \otimes |\downarrow\rangle_z - |\downarrow\rangle_z \otimes |\uparrow\rangle_z\right). \tag{14.126}$$

At time $t = 0$, a magnetic field of magnitude B is turned on in the x-direction. The Hamiltonian after $t = 0$ is given by

$$\hat{H} = J \hat{\vec{S}}_1 \cdot \hat{\vec{S}}_2 - \frac{\gamma B}{\hbar}(\hat{S}_1^x + \hat{S}_2^x), \tag{14.127}$$

with $J > 0$. What is the probability as a function of time for the system to be found in the $|\uparrow\rangle_z \otimes |\uparrow\rangle_z$ state?

You can either solve the problem exactly or in the first Born approximation.

14.8 The Landau-Zener Problem

The Landau-Zener problem is an exactly solvable time-dependent problem that describes how a quantum particle tunnels between two states as a function of the rate that it traverses a minimal energy gap. The Landau-Zener problem was originally solved in 1932 by Landau (1932), Zener (1932), Stueckelberg (1932), and Majorana (1932) (see Fig. 14.11). The exact solution involves mapping the time-dependent Schrödinger equation onto the so-called Weber equation, which is solved with parabolic cylinder functions. We will not describe how to solve the problem that way. Instead, we will focus on the Trotter product formula as a means to solve the problem numerically. The work in this section is based heavily on recent work that shows how to solve this problem efficiently using the Trotter formula Guttieres et al. (2023).

The Landau-Zener problem seeks to find the probability for a transition from the ground state to the excited state after the two states approach each other, reach a minimal energy difference, and then depart from each other. The Landau-Zener system is described by the Hamiltonian

$$\hat{H}(t) = \begin{pmatrix} vt & \delta \\ \delta & -vt \end{pmatrix} = vt\sigma_z + \delta\sigma_x, \tag{14.128}$$

where t is time, v is the rate at which the two levels approach each other, and δ is the coupling between them that determines the minimal energy gap (2δ) occurring at time $t = 0$. Both v and δ are real numbers with units of energy/time and energy, respectively. Initially, when $t = -\infty$, the Hamiltonian is dominated by the σ_z term, and the two (instantaneous) energy eigenvectors $|\psi_\pm\rangle$ are given by

Fig. 14.11 Lev Landau, Klaus Zener, Ernst Stueckelberg, and Ettore Majorana. [Landau image: original drawing by Peter Geoffrey Cook, courtesy AIP Emilio Segrè Visual Archives, Weber Collection; Zener image: AIP Emilio Segrè Visual Archives, Physics Today Collection; Stueckelberg image: AIP Emilio Segré Visual Archives; and Majorana image: © E. Recami and Maria Majorana, courtesy of AIP Emilio Segré Visual Archives, E. Recami and Fabio Majorana Collection. All images converted to look like similar sketches by James Freericks.]

$$|\psi_+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } |\psi_-\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \tag{14.129}$$

which are infinitely separated in energy; the initial ground state is $|\psi(-\infty)\rangle = |\psi_+\rangle$. As the system evolves with time, the two levels $E_+(t)$ (the upper instantaneous energy level) and $E_-(t)$ (the lower instantaneous energy level) approach each other as $t \to 0$ and then move apart as $t \to \infty$. Because the ground state smoothly changes from $|\psi_+\rangle$ as $t \to -\infty$ to $|\psi_-\rangle$ as $t \to \infty$, the probability to remain in the ground state for large positive times is given by $P_-(t) = |\langle\psi_-|\psi(t)\rangle|^2$; note that $|\psi_\pm\rangle$ are not the instantaneous eigenstates for all t just the initial and final times. Similarly, the probability to end in the excited state at long times is given by $P_+(t) = 1 - P_-(t) = |\langle\psi_+|\psi(t)\rangle|^2$. We want to determine both of these probabilities when $t \to \infty$.

As shown in Fig. 14.12a, to transition from the lower-energy state to the higher-energy state, the system has to tunnel through a gap of size at least 2δ. If the system evolves adiabatically, that is, extremely slowly, it will always remain in the lower-energy state (the lower band); that is, it will make perfect transitions from one instantaneous ground state to another along its time evolution. This means that around $t = 0$ there will be a slow and smooth transition from $|\psi_+\rangle$ to $|\psi_-\rangle$. If we let the system evolve diabatically (fast), it always tunnels from the lower to the upper band. For other rates, we need to determine the transition probability numerically.

Zener found the exact solution to be $\mathcal{P}_+(\infty) = \exp\left(-\pi\delta^2/\hbar v\right)$; we use a new symbol to distinguish the exact final probability from approximate ones. Our goal is to show how to get the probability by directly determining the time-evolution operator. The first issue that arises when considering this problem from a numerical perspective is how to deal with the infinite times. Numerical simulations work with finite times, so how do we effectively start with a state at $t = -\infty$ and obtain a result at $t = +\infty$ on a computer? It is usually assumed that starting in the state $|\psi_+\rangle$ at some sufficiently large (but finite) negative time is justified and will lead to an accurate numerical solution. But, the Landau-Zener problem is known for its slowly decaying oscillations of the transition probability $P_+(t)$ with time, which are illustrated on the right-hand side of Fig. 14.12b and in Fig. 14.12c. Although the time evolution of a two-level system is a simple problem to solve numerically and is not computationally demanding, the persistence of these slowly decaying oscillations presents a serious problem in accurately determining the transition probability as $t \to \infty$. We begin by dividing time evolution into three parts, where propagation from $-\infty$ to some cutoff time and from another cutoff time to $+\infty$ is found with time-dependent perturbation theory (see the gray areas in Fig. 14.12a), whereas the evolution on the finite time interval between the cutoffs is computed using the Trotter product formula.

We write the evolution operator $\hat{U}(T_{\max}, -T_{\max})$ as a time-ordered product that employs the following for each Trotter factor. It is simplified via

$$\hat{U}(t + \Delta t, t) = e^{-\frac{i}{\hbar}(vt\sigma_z + \delta\sigma_x)\Delta t} = \cos\left(\sqrt{v^2t^2 + \delta^2}\,\frac{\Delta t}{\hbar}\right)\mathbb{I}$$

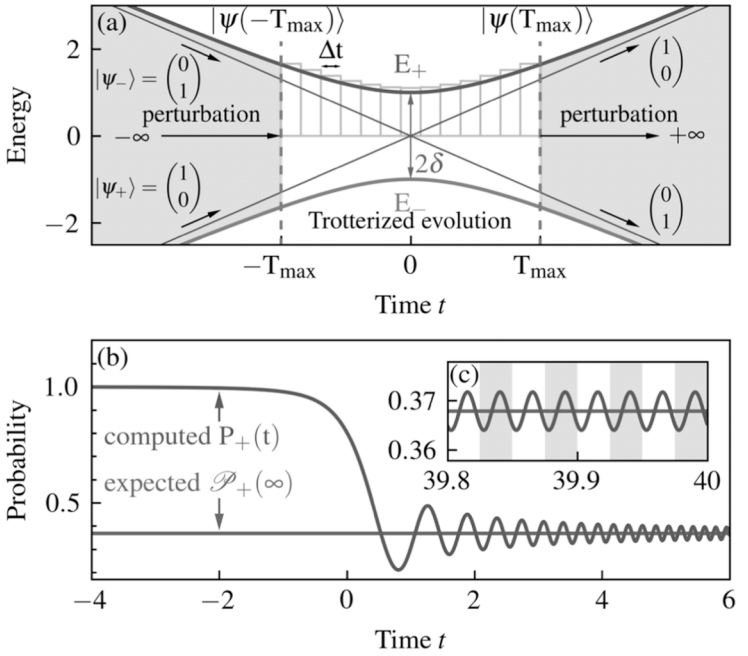

Fig. 14.12 Landau-Zener system: (**a**) Time evolution of the two instantaneous eigenenergy levels $E_\pm(t) = \pm\sqrt{v^2t^2 + \delta^2}$. At time $t = 0$, the two levels are separated by the minimal energy gap of $E_+(0) - E_-(0) = 2\delta$. The straight lines show the crossing of the ($\delta = 0$) energy levels $\pm vt$ at time $t = 0$ in a system that has no coupling between the levels. The time-evolution algorithm that we use is divided into three parts. The initial state $|\psi(-T_{max})\rangle$ is obtained either using perturbation theory applied to $|\psi_+\rangle$ at $t = -\infty$ or is set to $|\psi_+\rangle$ at $-T_{max}$. This initial state is then propagated toward $|\psi(T_{max})\rangle$ using an evolution operator in the Trotterized form with a time step of Δt. An additional perturbation is applied to $|\psi(T_{max})\rangle$ to obtain the final state at $t = +\infty$, or $|\psi(T_{max})\rangle$ is considered to be the final state (for the two different types of calculations). (**b**) Time evolution of the computed transition probability $P_+(t) = |\langle\psi_+|\psi(t)\rangle|^2$ compared to the exact probability obtained from the analytical expression for the Landau-Zener transition. The rate is $v = \pi$ and $\delta = 1$; we use these same parameters for all of the numerical calculations. The inset in panel (**c**) shows the fast oscillations of the transition probability $P_+(t)$ with time on a backdrop of width $vt^2 = 4N\pi$ (gray and white background), showing that these oscillations have a period proportional to $\sim t^2$. The amplitude of these oscillations for the interval $[-T_{max}, T_{max}]$ decays quite slowly with T_{max} (as a power law) unless corrected by the time-dependent perturbation theory. [Figure reprinted from Guttieres et al. (2023) with permission © American Institute of Physics]

$$-i\frac{\sin\left(\sqrt{v^2t^2+\delta^2}\,\frac{\Delta t}{\hbar}\right)}{\sqrt{v^2t^2+\delta^2}}\,(vt\sigma_z + \delta\sigma_x)\,, \tag{14.130}$$

after using the generalized Euler identity for the exponentials of the Pauli matrices. Each of the exponential Trotter factors $U(t + \Delta t, t)$ is now a concrete 2×2 matrix that acts on the state at a given time. Here, we use the exponentiated linear

superposition of Pauli matrices with the vector $\vec{\gamma} = -\frac{\Delta t}{\hbar}(vt, 0, \delta)$ (which is a time-dependent vector) for each Trotter factor. This is called the *exact form* of the Trotter product. We can also approximate a Trotter factor via

$$\hat{U}(t + \Delta t, t) \approx e^{-\frac{i}{\hbar} vt\sigma_z \Delta t} e^{-\frac{i}{\hbar}\delta\sigma_x \Delta t}, \tag{14.131}$$

which we call the *split form* of the Trotter product.

The time evolution can be implemented in two different ways. One can write a function that computes the time-evolution operator $\hat{U}(t + \Delta t, t)$ at every time step and applies it to the state $|\psi(t)\rangle$ to obtain $|\psi(t + \Delta t)\rangle$ (propagating the state), or one can multiply this time-evolution operator with the accumulated time-evolution operator for all previous times $\hat{U}(t + \Delta t, -T_{\max}) = \hat{U}(t + \Delta t, t)\hat{U}(t, -T_{\max})$ (propagating the operator). Then the state is obtained from $|\psi(t + \Delta t)\rangle = \hat{U}(t + \Delta t, -T_{\max})|\psi(-T_{\max})\rangle$. Since each Trotter factor is a concrete 2×2 matrix, either is easy to program. We will see the challenge is in properly determining the initial state to work with at each time cutoff.

For large positive and negative times, we must use time-dependent perturbation theory. We split the Landau-Zener Hamiltonian into two parts: the main time-dependent Hamiltonian $\hat{H}_0(t) = vt\sigma_z$ and the time-independent perturbation $\hat{V} = \delta\sigma_x$. The evolution operator is then constructed in the interaction picture via

$$\begin{aligned}\hat{U}(t, t_0) &= \hat{U}_0(t, t_0)\hat{U}_I(t, t_0) \\ &= \hat{U}_0(t, t_0) e^{-\frac{i\Delta t}{\hbar}\hat{V}_I(t)} e^{-\frac{i\Delta t}{\hbar}\hat{V}_I(t-\Delta t)} \cdots e^{-\frac{i\Delta t}{\hbar}\hat{V}_I(t_0+\Delta t)},\end{aligned} \tag{14.132}$$

with $\hat{V}_I(t) = \hat{U}_0^\dagger(t, t_0)\,\hat{V}\,\hat{U}_0(t, t_0)$; recall $\hat{V}$ is independent of time.

The evolution operator for the unperturbed Hamiltonian $\hat{H}_0(t)$ can be computed analytically because it commutes with itself for all times. Hence, we can just sum the unperturbed piece to find $\hat{U}_0(t, t_0) = e^{-\frac{i}{\hbar}\frac{1}{2}v(t^2 - t_0^2)\sigma_z}$. This sum can be completed because it is a power of t as we worked out before. The time t_0 will be taken to $-\infty$, but we do not do that until later. We next use this exact result to determine the perturbation in the interaction picture (which is a rotation of the σ_x matrix about the z-axis). It becomes

$$\begin{aligned}\hat{V}_I(t) &= e^{\frac{i}{\hbar}\frac{1}{2}v(t^2 - t_0^2)\sigma_z}\delta\sigma_x e^{-\frac{i}{\hbar}\frac{1}{2}v(t^2 - t_0^2)\sigma_z} \\ &= \delta\left(\sigma_x \cos\frac{v(t^2 - t_0^2)}{\hbar} - \sigma_y \sin\frac{v(t^2 - t_0^2)}{\hbar}\right),\end{aligned} \tag{14.133}$$

which can be found by using the generalized Euler identity to determine each exponential factor and then multiplying the three matrices together. The strategy of perturbation theory is to approximate the time-ordered product for $\hat{U}_I(t, t_0)$ in the evolution operator by

$$\hat{U}(t,t_0) \approx \hat{U}_0(t,t_0)\left(\mathbb{I} - \frac{i\Delta t}{\hbar}\sum_{j=1}^{N}\hat{V}_I(t_0+j\Delta t)\right), \tag{14.134}$$

which is accurate for t close to t_0, or when $\hat{V}_I$ is "small." Note that we have $t = t_0 + N\Delta t$ here.

Directly computing the $\hat{V}_{\mathrm{I}}(t)$ operator from Eq. (14.133) and then factoring the result in terms of Pauli matrices gives

$$\begin{aligned}\hat{V}_I(t') &= \begin{pmatrix} 0 & e^{\frac{i}{\hbar}v(t'^2-t_0^2)} \\ e^{-\frac{i}{\hbar}v(t'^2-t_0^2)} & 0 \end{pmatrix} = e^{-\frac{i}{\hbar}\frac{1}{2}vt_0^2\sigma_z}\begin{pmatrix} 0 & e^{\frac{i}{\hbar}vt'^2} \\ e^{-\frac{i}{\hbar}vt'^2} & 0 \end{pmatrix} e^{\frac{i}{\hbar}\frac{1}{2}vt_0^2\sigma_z} \\ &= \hat{Z}(t_0)\,\hat{V}_I'(t')\,\hat{Z}^{-1}(t_0),\end{aligned} \tag{14.135}$$

where $\hat{Z}(t) = e^{-\frac{i}{\hbar}\frac{1}{2}vt^2\sigma_z}$ and $V_I'(t')$ has the t_0 dependence removed. The unit matrix can also be written as $\mathbb{I} = \hat{Z}(t_0)\,\mathbb{I}\,\hat{Z}^{-1}(t_0)$ and $\hat{U}_0(t,t_0) = \hat{Z}(t)\hat{Z}^{-1}(t_0)$, so the evolution operator becomes

$$\begin{aligned}\hat{U}(t,t_0) &\approx \hat{Z}(t)\hat{Z}^{-1}(t_0)\left[\hat{Z}(t_0)\mathbb{I}\hat{Z}^{-1}(t_0) - \frac{i\Delta t}{\hbar}\hat{Z}(t_0)\sum_{j=1}^{N}\hat{V}_I'(j\Delta t)Z^{-1}(t_0)\right] \\ &= \hat{Z}(t)\left[\mathbb{I} - \frac{i\Delta t}{\hbar}\sum_{j=1}^{N}\hat{V}_I'(j\Delta t)\right]Z^{-1}(t_0).\end{aligned} \tag{14.136}$$

Equation (14.136) yields an approximate formula for the evolution operator in the interaction picture for any two times. We now apply it to compute $\hat{U}(-T_{\max},-\infty)$ and $\hat{U}(+\infty,T_{\max})$. In the first case, the operator $\hat{Z}^{-1}(t_0 = -\infty)$ on the right side of Eq. (14.136) can be neglected because we start from an eigenstate of the σ_z operator (the $|\psi_+\rangle$ state) and acting with $\hat{Z}^{-1}$ will just produce a global complex phase, which does not affect the probabilities. In the second case, when computing $\hat{U}(+\infty,T_{\max})$ the operator on the left side of Eq. (14.136) (i.e. $\hat{Z}(t=\infty)$) can be neglected because we are interested only in the probability $P_+(\infty)$ and that term also just contributes a complex phase, which cancels out. The summations in Eq. (14.136) can be analytically computed (in terms of so-called error functions) in both cases, but they go beyond the types of summations we have worked on previously. We will not describe how to do it; instead, we will just write down the result. For $\hat{U}(-T_{\max},-\infty)$ we find

$$\mathbb{I} - \frac{i\Delta t}{\hbar}\sum_{j}\hat{V}_I'(j\Delta t) = \begin{pmatrix} 1 & -\frac{i}{\hbar}\xi\,(T_{\max}) \\ -\frac{i}{\hbar}\eta\,(T_{\max}) & 1 \end{pmatrix}, \tag{14.137}$$

where

$$\eta(t) = \sqrt{\frac{\pi}{2v}}\frac{(i-1)}{2}\left[1 - \text{erf}\left(\sqrt{\frac{v}{2}}(i+1)t\right)\right], \tag{14.138}$$

$$\xi(t) = \sqrt{\frac{\pi}{2v}}\frac{(i+1)}{2}\left[1 + \text{erf}\left(\sqrt{\frac{v}{2}}(i-1)t\right)\right], \tag{14.139}$$

and the summation goes over discretized time values from $-\infty$ to $-T_{\max}$. Both off-diagonal components of the perturbation matrix, $\eta(t)$ and $\xi(t)$, are expressed in terms of the error function with a complex argument. The evolution operator obtained through perturbation theory is approximate and not necessarily unitary. This means the quantum state must be normalized "by hand" after applying the approximate evolution operator. In other words, we renormalize both at time $t = -T_{\max}$ and at $t = \infty$. Note that the central summation in Eq. (14.136) does not change for $\hat{U}(+\infty, T_{\max})$, but the $\hat{Z}(t)$ and $\hat{Z}^{-1}(t_0)$ operators do change.

The improvement in accuracy for the calculation that includes the time-dependent perturbation theory arises from its use of a more accurate initial state at $-T_{\max}$. This improvement is shown in Fig. 14.13. In Fig. 14.13a, we compare

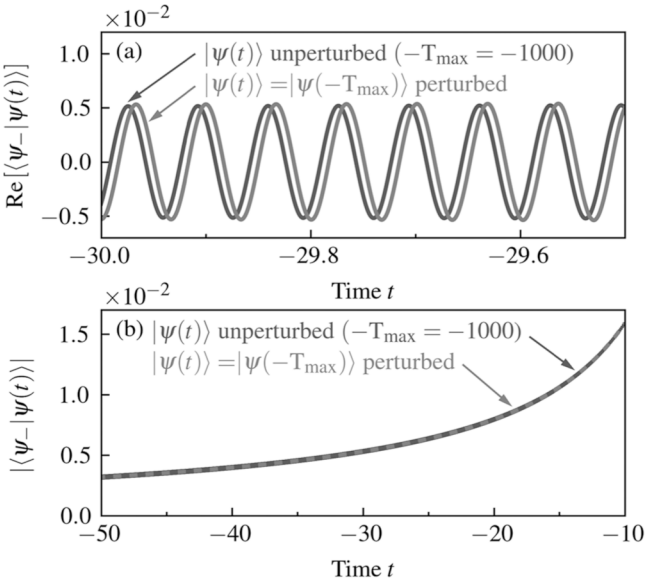

Fig. 14.13 State evolution for different starting conditions: (**a**) the real part of the projected wave function $\text{Re}[\langle\psi_-|\psi(t)\rangle]$ for the initial state computed as a function of time using perturbation theory $|\psi(t)\rangle = |\psi(-T_{\max})\rangle$ versus the unperturbed state $|\psi(t)\rangle$ computed by the time evolution from $|\psi_+\rangle$ starting at $-T_{\max} = -1000$. (**b**) Time evolution of the modulus of the state projection $|\langle\psi_-|\psi(t)\rangle|$ computed for the same two starting conditions as in panel (**a**). You can see the modulus is essentially unchanged by the perturbation theory, just the phase is shifted. [Figure reprinted from Guttieres et al. (2023) with permission © American Institute of Physics]

the phase of the state $|\psi(t)\rangle$ obtained by propagating $|\psi_+\rangle$ from a distant time $-T_{\max} = -1000$ (blue curve) with the one obtained by perturbation theory as functions of time ($t = T_{\max}$, the orange curve). In general, the amplitude $|\langle\psi_-|\psi(t)\rangle|$ in Fig. 14.13b matches the perturbed and unperturbed state, but as shown in Fig. 14.13a they differ by a phase. This initial phase difference introduces an error for later times that propagates through the Trotter product. The advantage of the perturbation theory is that it reduces the computational demand by reducing the time range $[-T_{\max}, T_{\max}]$ needed for the numerical calculation. Instead of propagating the state from a very distant past, one can obtain a precise result for a much smaller $T_{\max}$. The perturbation that brings the state to $t = +\infty$ is even more important, because it removes the small but persistent oscillations of the transition probability. I have a video https://tinyurl.com/4jxtszn7 that discusses these ideas from a different perspective using more advanced math.

Section 14.8 Problems:

14.8.1 Numerical Implementation of the Landau-Zener Problem over a Finite Time Interval

When performing numerical calculations, it is important to work in dimensionless units, so we choose v, δ, t, and $\hbar$ to be dimensionless. We set $\hbar = 1$, and use $v = \pi$ and $\delta = 1$ in the calculations—if needed true dimensions can be restored later—but, we won't need them. Compute the time evolution using the exact form for the Trotter factor and an initial state at $-T_{\max}$ given by $(1, 0)$. Choose how big you want to take $T_{\max}$ and Δt in your calculations, and vary them until you are satisfied with your answer. How close can you get to the exact result for the probability to stay in the ground state? If you are able to, plot the probability as a function of time and comment on how accurate your result is.

14.8.2 Numerical Implementation of the Landau-Zener Problem over the Infinite Time Interval

Repeat the numerical calculation of the previous problem, but append the perturbation theory result for large negative times and large positive times, as described in the text. You will need to find a package that will calculate the error function of a complex argument to carry this out (e.g., NumPy has such packages). Compare your final result with the exact answer and the results in Fig. 14.14. We include numerical results for the split form there as well.

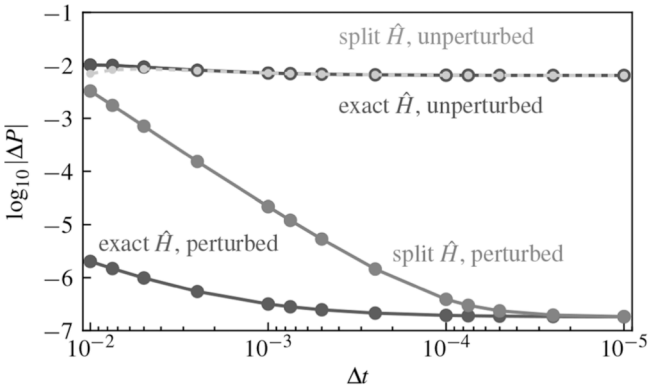

Fig. 14.14 The accuracy of the computed transition probability as a function of the time step Δt. In the case of perturbed results, we compute the difference as $\Delta P = P_+(\infty) - \mathcal{P}_+(\infty)$, whereas in the case of unperturbed results we compute $\Delta P = P_+(T_{\max}) - \mathcal{P}_+(\infty)$. Here, $\mathcal{P}_+(\infty)$ is the probability obtained by the exact Landau-Zener formula $\exp\left(-\frac{\pi\delta^2}{\hbar v}\right)$, while $P_+(\infty)$ and $P_+(T_{\max})$ are the probabilities computed using the Trotterized evolution, with or without the perturbation, respectively. We also compare the accuracy when the Trotter step exponential is evaluated exactly, or when the Hamiltonian is split between σ_z and σ_x terms in each Trotter step. In all cases $T_{\max} = 30$. High accuracy can only be attained for this value of $T_{\max}$ when the infinite tails are included perturbatively. [Figure reprinted from Guttieres et al. (2023) with permission © American Institute of Physics]

14.8.3 Subtle Properties of the Split Form of the Trotter Formula

In the previous two problems, we examined the time evolution using the exact Trotter formula. In this problem, we analyze what happens with the split form. Note that the exponentials of the σ_x matrix $\exp(-\frac{i}{\hbar}\delta\sigma_x\Delta t)$ in the Trotter product sequence are constant for a fixed time step Δt, whereas the σ_z terms $\exp\left(-\frac{i}{\hbar}vt\sigma_z\Delta t\right)$ depend on time t. At special times, $T_{\rm n}$, the coefficient of the σ_z term satisfies $vT_{\rm n}\Delta t/\hbar = 2n\pi$. At these special times, the given Trotter component with σ_z (in the split form) is a unit matrix. The next σ_z component in the Trotter product will have the exponent $v(T_{\rm n} + \Delta t)\Delta t/\hbar$ which is the same as the earlier term $v(T_{\rm n-1} + \Delta t)\Delta t/\hbar$ and the same as the even earlier time $v\Delta t\Delta t/\hbar$, and so on. This means that in the split form, the Trotter product repeats with period $T = 2\pi\hbar/(v\Delta t)$. But we know that the exact time-evolution operator is not periodic. So this periodicity is an artifact of using the split form.

Write an essay about how you can deal with this issue in a numerical calculation that uses the split form instead of the exact form. In particular, discuss how Δt must be related to $T_{\max}$ in order for the result to be accurate.

14.8.4 Application of Compression to the Landau-Zener Problem

In Problems 14.4.3 and 14.4.4, we discussed two different types of compression algorithms. Pick one of them and discuss how such an algorithm can be employed to compress the numerical evaluation of the time-evolution operator. Does this approach make the code more efficient, less efficient, or about the same efficiency? Discuss circumstances where it is useful to employ compression.

14.9 Driven Harmonic Oscillator

In this section, we solve another exactly solvable time-evolution problem—the driven simple harmonic oscillator. This can be thought of as a harmonic oscillator, with the mass at the end of the spring pushed or pulled periodically as a function of time. The path to the exact solution is complicated, and we will carefully describe how we proceed. It turns out we need to factorize the time-evolution operator into three distinct factors to be able to solve it.

We start by writing down the time-dependent Hamiltonian, which involves a harmonic perturbation:

$$\hat{H} = \underbrace{\hbar\omega\left(\hat{a}^\dagger\hat{a} + \tfrac{1}{2}\right)}_{\hat{H}_0} + \underbrace{\hbar C\left(e^{i\Omega t}\hat{a}^\dagger + e^{-i\Omega t}\hat{a}\right)}_{\hat{V}(t)}. \tag{14.140}$$

We have indicated the breakup into a time-independent $\hat{H}_0$ and a time-dependent $\hat{V}(t)$. The number C is a real number with dimensions of frequency, and Ω is the driving frequency that need not be related to the harmonic oscillator frequency ω; Ω can be positive or negative.

Before we go too far, we should see what kind of approaches we can try to solve this. The problem appears to be similar to the magnetic resonance problem. If we recall the Hadamard lemma in the form

$$e^{i\alpha t\hat{a}^\dagger\hat{a}}\hat{a}e^{-i\alpha t\hat{a}^\dagger\hat{a}} = e^{-i\alpha t}\hat{a} \tag{14.141}$$

and the fact that the number operator commutes with itself, we can rewrite the Hamiltonian as

$$e^{i\Omega t\hat{a}^\dagger\hat{a}}\left(\hbar\omega\left(\hat{a}^\dagger\hat{a} + \tfrac{1}{2}\right) + \hbar C(\hat{a}^\dagger + \hat{a})\right)e^{-i\Omega t\hat{a}^\dagger\hat{a}}, \tag{14.142}$$

and we can construct the time-evolution operator the same way as we did in the magnetic resonance problem. The time-dependent Hamiltonian is a Hadamard

lemma of a time-independent Hamiltonian. Then, the time-evolution operator becomes

$$\hat{U}(t) = e^{i\Omega t\hat{a}^\dagger\hat{a}} e^{-i\Delta t\left(\omega\left(\hat{a}^\dagger\hat{a}+\frac{1}{2}\right)+C(\hat{a}^\dagger+\hat{a})\right)} e^{-i\Omega\Delta t\hat{a}^\dagger\hat{a}} e^{-i\Delta t\left(\omega\left(\hat{a}^\dagger\hat{a}+\frac{1}{2}\right)+C(\hat{a}^\dagger+\hat{a})\right)} \cdots$$
$$\times\, e^{-i\Delta t\left(\omega\left(\hat{a}^\dagger\hat{a}+\frac{1}{2}\right)+C(\hat{a}^\dagger+\hat{a})\right)} e^{-i\Omega\Delta t\hat{a}^\dagger\hat{a}}. \tag{14.143}$$

We now can factor out the constant term $e^{-\frac{i\omega t}{2}}$, combine the neighboring terms into one exponential, and take the limit as $\Delta t \to 0$, just as we did in the magnetic resonance problem (the terms proportional to the commutator, with $(\Delta t)^2$ vanish in the limit $\Delta t \to 0$, just as they did before, because they are all the same). We find the final result

$$\hat{U}(t) = e^{i\Omega t\hat{a}^\dagger\hat{a}} e^{-it\left((\Omega+\omega)\hat{a}^\dagger\hat{a}+C(\hat{a}^\dagger+\hat{a})\right)} e^{-\frac{i\omega t}{2}}. \tag{14.144}$$

This is an exact result for the time-evolution operator. We will manipulate it more in the problems.

We have another way to compute an exact result for this operator. If we look at the commutator of the Hamiltonian with itself at two different times, we find

$$[\hat{H}(t), \hat{H}(t')] = \hbar^2 C\omega\left(e^{i\Omega t'}\hat{a}^\dagger - e^{-i\Omega t'}\hat{a}\right) - \hbar^2 C\omega\left(e^{i\Omega t}\hat{a}^\dagger - e^{-i\Omega t}\hat{a}\right)$$
$$+ \hbar^2 C^2\left(e^{-i\Omega(t-t')} - e^{i\Omega(t-t')}\right). \tag{14.145}$$

This result does not commute with everything, so it is not one of the cases that we can use for further simplification. Note, though, that it does vanish for $t = t'$ as it must.

The situation changes after going to the interaction representation. Using the harmonic oscillator Hamiltonian as the $\hat{H}_0$, we have

$$\hat{V}_I(t) = \hbar C e^{i\omega t\hat{a}^\dagger\hat{a}}\left(e^{i\Omega t}\hat{a}^\dagger + e^{-i\Omega t}\hat{a}\right)e^{-i\omega t\hat{a}^\dagger\hat{a}} = \hbar C\left(e^{i(\Omega+\omega)t}\hat{a}^\dagger + e^{-i(\Omega+\omega)t}\hat{a}\right). \tag{14.146}$$

Now, we have

$$\left[\hat{V}_I(t), \hat{V}_I(t')\right] = \hbar^2 C^2\left(e^{-i(\Omega+\omega)(t-t')} - e^{i(\Omega+\omega)(t-t')}\right)$$
$$= -2i\hbar^2 C^2 \sin\left((\Omega+\omega)(t-t')\right). \tag{14.147}$$

This is a number, which commutes with everything, so it is likely we can use this fact to simplify the time-evolution operator. Indeed, we can.

The situation is identical to the case worked out in Problem 14.4.2. We will work out the details here. The interaction part of the time-evolution operator is given by

$$\hat{U}_I(t) = e^{-iC\Delta t\left(e^{i(\Omega+\omega)t}\hat{a}^\dagger + e^{-i(\Omega+\omega)t}\hat{a}\right)} e^{-iC\Delta t\left(e^{i(\Omega+\omega)(t-\Delta t)}\hat{a}^\dagger + e^{-i(\Omega+\omega)(t-\Delta t)}\hat{a}\right)} \cdots$$
$$\times\, e^{-iC\Delta t\left(e^{i(\Omega+\omega)\Delta t}\hat{a}^\dagger + e^{-i(\Omega+\omega)\Delta t}\hat{a}\right)}. \tag{14.148}$$

Now, we combine the product of Trotter factors one by one. When we do this, there is a commutator we need to evaluate for the Weyl form of the Baker-Campbell-Hausdorff formula (recall $e^{\hat{A}}e^{\hat{B}} = e^{A+B+\frac{1}{2}[\hat{A},\hat{B}]}$). Those commutators are numbers, so they can be collected to the right in the product. Each time we add a new factor from the right to the exponent, it needs to be commuted with all of the $\hat{V}_I$ terms that are already in the exponent. So the number of Weyl-form correction terms (i.e., commutators) becomes large (growing like N^2). The final result is

$$\begin{aligned}\hat{U}(t) &= e^{-i\omega t\left(\hat{a}^\dagger\hat{a}+\frac{1}{2}\right)} e^{-iC\Delta t\sum_{j=1}^{N}\left(e^{i(\Omega+\omega)j\Delta t}\hat{a}^\dagger + e^{-i(\Omega+\omega)j\Delta t}\hat{a}\right)} \\ &\quad\times e^{iC^2(\Delta t)^2\sum_{j=1}^{N}\sum_{k=1}^{j-1}\sin\left((\Omega+\omega)(j-k)\Delta t\right)} \\ &= e^{-i\omega t\left(\hat{a}^\dagger\hat{a}+\frac{1}{2}\right)} e^{-\frac{C}{\Omega+\omega}\left(\left(e^{i(\Omega+\omega)t}-1\right)\hat{a}^\dagger - \left(e^{-i(\Omega+\omega)t}-1\right)\hat{a}\right)} \\ &\quad\times e^{\frac{iC^2}{(\Omega+\omega)^2}\left((\Omega+\omega)t-\sin\left((\Omega+\omega)t\right)\right)}\end{aligned} \tag{14.149}$$

after evaluating the different summations (details are given in Problem 14.9.1). To work with this, we would also want to separate the exponential term with sum of the two ladder operators using the Baker-Campbell-Hausdorff formula in the Weyl form again, as it would then be in a form (exponentials of raising or lowering operators) that it could act on harmonic oscillator states. We show how this works in Problem 14.9.2. Surprisingly, this result is identical to the result found in Eq. (14.144), although it certainly does not look like it. You will work through the steps to show the two are indeed equivalent in Problem 14.9.2. When everything is put into the same exponent, this is called the *Magnus expansion*. It is straightforward, but tedious, to continue the expansion when the triple and higher-order nested commutators do not vanish. One can clearly see the relationship with the Baker-Campbell-Hausdorff expansion.

You might ask, why do we go through rewriting the evolution operator in a factorized form? The answer is not so obvious at first. The formulas often look more, rather than less complicated. But, by isolating operators into individual exponentials, we often can have them act on states with definite results, so this approach allows us to perform calculations more easily than in the full time-evolution operator form, where we just do not know how the operator acts on states. For example, we will use this result to evaluate the probability to make a transition from the ground state to an excited state in the harmonic oscillator as a function of time when the quantum oscillator is driven.

Section 14.9 Problems:

14.9.1 Technical Details for the Derivation of the Second Factorized Form

Verify the first line of Eq. (14.149).

In the rest of this problem, we work out the summations. Show that

$$\sum_{j=1}^{N} e^{i(\Omega+\omega)j\Delta t} = \frac{1 - e^{(\Omega+\omega)N\Delta t}}{1 - e^{i(\Omega+\omega)\Delta t}}. \tag{14.150}$$

Take the limit as $\Delta t \to 0$ with $N\Delta t = t$ to find that the sum becomes $\left(e^{i(\Omega+\omega)t} - 1\right)/i(\Omega+\omega)$.

Complete the double summation using the following methods:

$$\begin{aligned}(\Delta t)^2 \sum_{j=1}^{k}\sum_{k=1}^{j-1} \sin\left((\Omega+\omega)(j-k)\Delta t\right) \\ = \text{Im}\left(\Delta t \sum_{j=1}^{N} e^{i(\Omega+\omega)j\Delta t}\, \Delta t \sum_{k=1}^{j-1} e^{-i(\Omega+\omega)k\Delta t}\right),\end{aligned} \tag{14.151}$$

which uses the Euler relation to convert the sum over a sine to a sum over an exponential. The sum over k is done first, followed by the sum over j. Show you obtain

$$\begin{aligned}&\text{Im}\left(\Delta t \sum_{j=1}^{N} e^{i(\Omega+\omega)j\Delta t}\, \Delta t \sum_{k=1}^{j-1} e^{-i(\Omega+\omega)k\Delta t}\right) \\ &= \text{Im}\left(\Delta t \sum_{j=1}^{N} \frac{e^{i(\Omega+\omega)(j-1)\Delta t} - 1}{i(\Omega+\omega)}\right) \\ &= \text{Im}\left(\frac{1 - e^{i(\Omega+\omega)t}}{(\Omega+\omega)^2} - \frac{t}{i(\Omega+\omega)}\right) \\ &= \frac{1}{(\Omega+\omega)^2}\left((\Omega+\omega)t - \sin\left((\Omega+\omega)t\right)\right).\end{aligned} \tag{14.152}$$

14.9.2 Verification That the Two Factorized Forms of the Time-Evolution Operator Are the Same

First verify the "shift" relation

$$e^{\alpha(\hat{a}-\hat{a}^\dagger)}\hat{a}e^{-\alpha(\hat{a}-\hat{a}^\dagger)} = \hat{a} + \alpha \tag{14.153}$$

and its Hermitian conjugate. Use these relations with $\alpha = \frac{C}{\Omega+\omega}$ to rewrite Eq. (14.144) as

$$e^{i\Omega t\hat{a}^\dagger\hat{a}}e^{\frac{C}{\Omega+\omega}(\hat{a}-\hat{a}^\dagger)}e^{-it(\Omega+\omega)\hat{a}^\dagger\hat{a}}e^{-\frac{C}{\Omega+\omega}(\hat{a}-\hat{a}^\dagger)}e^{\frac{i\omega t}{2}}e^{\frac{itC^2}{\Omega+\omega}} \tag{14.154}$$

after using the braiding relation (the constant term on the far right comes from completing the square). Use a multiply by one and a Hadamard lemma to simplify it to

$$e^{-i\omega t\left(\hat{a}^\dagger\hat{a}+\frac{1}{2}\right)}e^{\frac{C}{\Omega+\omega}\left(e^{-i(\Omega+\omega)t}\hat{a}-e^{i(\Omega+\omega)t}\hat{a}^\dagger\right)}e^{-\frac{C}{\Omega+\omega}(\hat{a}-\hat{a}^\dagger)}e^{\frac{i\omega t}{2}}e^{\frac{itC^2}{\Omega+\omega}}. \tag{14.155}$$

The next step is to use the Weyl form of the Baker-Campbell-Hausdorff formula to combine the product of the middle two exponential factors (with the sum of the ladder operators) into a single exponential factor. The Weyl form of the Baker-Campbell-Hausdorff identity applies here because the commutator of the two exponents is a number which commutes with everything. After doing this step, you should find that you obtain Eq. (14.149). This requires you to carefully determine the constant correction term from the commutator.

You have now established that the two forms of the evolution operator are indeed the same.

14.9.3 Exact Calculation of Transitions from the Ground State to an Excited State in the Driven Harmonic Oscillator

Suppose we start in the ground state of the harmonic oscillator $|0\rangle$, and we turn on the driving of the harmonic oscillator at $t = 0$. What is the probability to make a transition to the mth excited state as a function of time $P_{m\leftarrow 0}(t)$?

It is best to start from Eq. (14.149). After removing the complex phase factors that have modulus 1, you should find that

$$P_{m\leftarrow 0}(t) = \left|\langle m|e^{-\frac{C}{\Omega+\omega}\left(\left(e^{i(\Omega+\omega)t}-1\right)\hat{a}^\dagger-\left(e^{-i(\Omega+\omega)t}-1\right)\hat{a}\right)}|0\rangle\right|^2. \tag{14.156}$$

You can solve this by using the Baker-Campbell-Hausdorff formula in the Weyl form to put the exponential of the raising operator to the left of the exponential of the lowering operator. Then, one can evaluate the matrix element. You should find that the probability is given by

$$P_{m\leftarrow 0}(t)=\frac{1}{m!}\left(\frac{2C^2}{(\Omega+\omega)^2}\right)^m\left(1-\cos\left((\Omega+\omega)t\right)\right)^m e^{-\frac{2C^2}{(\Omega+\omega)^2}\left(1-\cos((\Omega+\omega)t)\right)}. \tag{14.157}$$

Verify that the sum of the probability to go to any state is equal to one, as it must be ($\sum_{m=0}^{\infty} P_{m\leftarrow 0}(t)=1$).

14.9.4 Comparison of the Exact Formula for Transitions with Time-Dependent Perturbation Theory

Use first-order time-dependent perturbation theory and show that it gives the correct result for $P_{1\leftarrow 0}(t)$, but sets all other probabilities to zero. Go to second-order time-dependent perturbation theory and show it properly gives $P_{2\leftarrow 0}(t)$. In general we need mth-order perturbation theory to get the transition to the mth excited state.

14.10 Photoproduction of Hydrogen

One of the most important time-dependent problems is the spontaneous emission of an electron in an atom from a higher-energy state to a lower-energy state (eventually ending in the ground state). The transitions occur even if we are not driving the atom with an electric field. This is called *spontaneous emission* and was discovered by Einstein. One can think of the spontaneous emission as being driven by the vacuum fluctuations of the electric field. We will discuss this more below. For now, we will examine a hydrogen atom in the presence of real electric and magnetic fields as a paradigm for this phenomena.

It is not assumed that you have seen the *vector potential* before. It is a function from which one can find the electric and magnetic fields. In classical mechanics, this object is an abstract construct that facilitates determining the electric and magnetic fields—only the electric and magnetic fields are physical. This was believed to also hold in the quantum realm until Aharonov and Bohm showed that one can set up an experiment to determine the vector potential even in regions where the magnetic field vanishes. This led us to conclude that it is actually the vector potential that is the fundamental object. So, in quantum mechanics, the Hamiltonian for an atom interacting with electric and magnetic fields is written in terms of the vector potential. We cannot motivate the form of the Hamiltonian without resorting to the correspondence principle, so, please take the following Hamiltonian as a postulate

for how an atom interacts with electric and magnetic fields; it isn't really a postulate, but instead follows from comparing to classical electrodynamics, but we do not assume that you know the corresponding classical electrodynamics.

The Hamiltonian takes the following form:

$$\hat{H} = \frac{1}{2M}\left(\hat{\vec{p}} + \frac{|e|}{c}\vec{A}(\hat{\vec{r}}, t)\right)^2 - \frac{e^2}{4\pi\epsilon_0 \hat{r}} + \frac{g|e|}{2Mc}\hat{\vec{S}} \cdot \vec{B}(\hat{\vec{r}}, t). \tag{14.158}$$

The first term is the kinetic energy, modified by the presence of an electromagnetic field, the second term is the Coulomb attraction between the electron and the proton, and the third term is the direct magnetic coupling of the spin to the magnetic field. The magnetic field is written in terms of the vector potential as

$$\vec{B}(\hat{\vec{r}}, t) = \frac{i}{\hbar}\sum_{ijk} \epsilon_{ijk}\left[\hat{p}_i, A_j(\hat{\vec{r}}, t)\right]\vec{e}_k, \tag{14.159}$$

which you can view as a definition of the magnetic field. In electromagnetism, the vector potential (and a scalar potential) have a gauge freedom associated with them. This essentially means multiple potentials lead to the same fields. We use this freedom to make a constraint on the vector potential. It is given by

$$\sum_i \left[\hat{p}_i, A_i(\hat{\vec{r}}, t)\right] = 0, \tag{14.160}$$

which is called the *Coulomb gauge*. When we expand the square of the operators in the kinetic energy, we will get terms that go like $\hat{\vec{p}} \cdot \vec{A} + \vec{A} \cdot \hat{\vec{p}}$, but the Coulomb gauge guarantees that these two operators commute, so we do not need to worry about ordering issues with respect to the momentum and the vector potential. This is the main benefit of using the Coulomb gauge.

So, expanding the Hamiltonian gives

$$\hat{H} = \frac{\hat{\vec{p}}^2}{2M} + \frac{|e|}{Mc}\vec{A}(\hat{\vec{r}}, t) \cdot \hat{\vec{p}} + \underbrace{\frac{e^2}{2Mc^2}\hat{A}(\hat{\vec{r}}, t) \cdot \vec{A}(\hat{\vec{r}}, t)}_{\text{neglect since small}} - \frac{e^2}{4\pi\epsilon_0 \hat{r}} + \frac{g|e|}{2mc}\hat{\vec{S}} \cdot \vec{B}(\hat{\vec{r}}, t). \tag{14.161}$$

We neglect the term that goes like the square of the vector potential because it is much smaller than the remaining terms. Then, we separate the Hamiltonian into two terms for performing time-dependent perturbation theory. Before doing that though, we introduce a concrete form for the vector potential. We write it as

$$\vec{A}(\hat{\vec{r}}, t) = \vec{A}_0 e^{i(\vec{k}\cdot\hat{\vec{r}} - \omega t)} + \vec{A}_0^* e^{-i(\vec{k}\cdot\hat{\vec{r}} - \omega t)}. \tag{14.162}$$

Here, $\vec{A}_0$ is a vector of complex numbers, $\vec{k}$ is a real-valued wavevector, and ω is a real frequency. Then, we find

$$\hat{H} = \underbrace{\frac{\hat{\vec{p}}^2}{2M} - \frac{e^2}{4\pi\epsilon_0 \hat{r}}}_{\hat{H}_0} \times + \underbrace{\frac{|e|}{Mc}\left(\vec{A}_0 e^{i(\vec{k}\cdot\hat{\vec{r}})} + \vec{A}_0^* e^{-i(\vec{k}\cdot\hat{\vec{r}}-\omega t)}\right)\cdot\hat{\vec{p}} + \frac{ig|e|}{2mc}\hat{\vec{S}}\cdot\left(\vec{k}\times\vec{A}_0 - \vec{k}\times\vec{A}_0^*\right)}_{\hat{V}(t)}. \tag{14.163}$$

This is a harmonic perturbation with $\hat{V}(t) = \hat{V}_\omega e^{i\omega t} + \hat{V}_{-\omega} e^{-i\omega t}$, $\hat{V}_{-\omega} = \hat{V}_\omega^\dagger$ and

$$\hat{V}_{-\omega} = \frac{|e|}{Mc} e^{i\vec{k}\cdot\hat{\vec{r}}}\left(\vec{A}_0\cdot\hat{\vec{p}} + i\hat{\vec{S}}\cdot\left(\vec{k}\times\vec{A}_0\right)\right). \tag{14.164}$$

Note that treating the electric and magnetic fields as functions of time is a so-called *semiclassical treatment* of the transitions. This is because we are neglecting the granularity associated with individual photons (to add this in requires us to quantize light, which we do in Chap. 20). This semiclassical approach is very accurate in the limit when we have a lot of photons around, such as when we illuminate an atom by an intense laser. It turns out that the results we find here also describe spontaneous emission, although this is not immediately obvious.

Before we proceed any further, we need to look at the size of different terms in the Hamiltonian. We know the size of the atom is on the order of a_0. The energy scale for transitions is on the order of eV. Light that has an energy of eV has a wavelength on the order of a micron. This means the wavelength of the light is about 1000 times larger than the size of the atom. So, over the length scale of the atom, we can assume the light is approximately constant in space. This motivates expanding $e^{i\vec{k}\cdot\hat{\vec{r}}}$ in a power series, because the exponent is approximately ka_0, which is much smaller than 1. The harmonic perturbation then becomes

$$\hat{V}_{-\omega} = \frac{|e|}{Mc}\left(1 + i\vec{k}\cdot\hat{\vec{r}} - \tfrac{1}{2}(\vec{k}\cdot\hat{\vec{r}})^2 + \cdots\right)\left(\vec{A}_0\cdot\hat{\vec{p}} + i\hat{\vec{S}}\cdot\left(\vec{k}\times\vec{A}_0\right)\right) \tag{14.165}$$

or

$$\hat{V}_{-\omega} = \frac{|e|}{Mc}\left(\vec{A}_0\cdot\hat{\vec{p}} + i\hat{\vec{S}}\cdot\left(\vec{k}\times\vec{A}_0\right) + i\vec{k}\cdot\hat{\vec{r}}\vec{A}_0\cdot\hat{\vec{p}} + O(k^2)\right). \tag{14.166}$$

This can be rearranged by using add zero, with the following result:

$$\begin{aligned}\vec{k}\cdot\hat{\vec{r}}\vec{A}_0\cdot\hat{\vec{p}} &= \sum_i\sum_j k_i\hat{r}_i A_{0j}\hat{p}_j \\ &= \tfrac{1}{2}\sum_i\sum_j k_i A_{0j}\left(\hat{r}_i\hat{p}_j+\hat{r}_i\hat{p}_j+\hat{p}_i\hat{r}_j-\hat{p}_i\hat{r}_j+\hat{p}_j\hat{r}_i-\hat{p}_j\hat{r}_i\right) \\ &= \tfrac{1}{2}\sum_i\sum_j k_i A_{0j}\Bigg(\hat{r}_i\hat{p}_j+\hat{p}_i\hat{r}_j+\underbrace{\hat{r}_i\hat{p}_j-\hat{p}_j\hat{r}_i}_{=i\hbar\delta_{ij}}+\underbrace{\hat{p}_j\hat{r}_i-\hat{p}_i\hat{r}_j}_{=\sum_k\epsilon_{ijk}\hat{L}_k}\Bigg).\end{aligned} \tag{14.167}$$

If we note that $\sum_i k_i A_{0i} = 0$ from the Coulomb gauge, we see the commutator term vanishes, and we are left with

$$\hat{V}_{-\omega} = \frac{|e|}{Mc}\left(\vec{A}_0\cdot\hat{\vec{p}}+\tfrac{i}{2}(\hat{\vec{L}}+g\hat{\vec{S}})\cdot\left(\vec{k}\times\vec{A}_0\right)+\tfrac{i}{2}\sum_i\sum_j k_i A_{0j}(\hat{r}_i\hat{p}_j+\hat{p}_i\hat{r}_j)\right). \tag{14.168}$$

This is the form we will use in the rest of our calculations (with $g = 2$ for the electron).

We will need to know the matrix elements of the harmonic perturbation in eigenstates of $\hat{H}_0$, which satisfy $\hat{H}_0|n\rangle = E_n|n\rangle$. The expectation value of the momentum vector $\hat{\vec{p}} = \sum_\alpha \hat{p}_\alpha\vec{e}_\alpha$ is more conveniently expressed in terms of the commutator identity that we used when we derived the canonical commutation relation

$$\hat{\vec{p}} = \frac{iM}{\hbar}[\hat{H}_0,\hat{\vec{r}}], \tag{14.169}$$

with $\hat{\vec{r}} = \sum_\alpha \hat{r}_\alpha\vec{e}_\alpha$. We then have

$$\langle n|\hat{\vec{p}}|m\rangle = \frac{iM}{\hbar}\langle n|[\hat{H}_0,\hat{\vec{r}}]|m\rangle = iM\omega_{nm}\langle n|\hat{\vec{r}}|m\rangle, \tag{14.170}$$

with $\omega_{nm} = (E_n - E_m)/\hbar$. Similarly, we have

$$\hat{r}_i\hat{p}_j+\hat{p}_i\hat{r}_j = \frac{iM}{\hbar}\left(\hat{r}_i[\hat{H}_0,\hat{r}_j]+[\hat{H}_0,\hat{r}_i]\hat{r}_j\right) = \frac{iM}{\hbar}[\hat{H}_0,\hat{r}_i\hat{r}_j], \tag{14.171}$$

so that

$$\langle n|\hat{r}_i\hat{p}_j+\hat{p}_i\hat{r}_j|m\rangle = iM\omega_{nm}\langle n|\hat{r}_i\hat{r}_j|m\rangle. \tag{14.172}$$

We use this to compute the matrix element of the harmonic perturbation. We rewrite $3\sum_i\sum_j k_i A_{0j}\hat{r}_i\hat{r}_j = \sum_i\sum_j k_i A_{0j}\left(3\hat{r}_i\hat{r}_j-\delta_{ij}\hat{r}^2\right)$, which holds because $\vec{k}\cdot\vec{A}_0 = 0$. Hence, we find that

$$\langle n|\hat{V}_{-\omega}|m\rangle = -i\frac{\omega_{nm}}{c}\vec{A}_o\cdot\langle n|(-|e|\hat{\vec{r}})|m\rangle - i(\vec{k}\times\vec{A}_0)\cdot\langle n|\Big(-\tfrac{|e|}{2Mc}\Big)\big(\hat{\vec{L}}+2\hat{\vec{S}}\big)|m\rangle$$
$$+\frac{\omega_{nm}}{6c}\sum_i\sum_j k_i A_{0j}\langle n|(3\hat{r}_i\hat{r}_j-\delta_{ij}\hat{r}^2)|m\rangle + O(k^2). \qquad (14.173)$$

We define the *electric dipole operator* via $\hat{\vec{D}} = -|e|\hat{\vec{r}}$, the *magnetic dipole operator* via $\hat{\vec{\mu}} = \frac{|e|}{2Mc}\big(\hat{\vec{L}}+2\hat{\vec{S}}\big)$, and the *electric quadrupole moment operator* via $\hat{Q}_{ij} = -|e|\big(3\hat{r}_i\hat{r}_j-\delta_{ij}\hat{r}^2\big)$. Our final result for the harmonic perturbation matrix element is

$$\Big(\hat{V}_{-\omega}\Big)_{nm} = -i\frac{\omega_{nm}}{c}\hat{A}_0\cdot\big(\vec{D}\big)_{nm} - i\big(\vec{k}\times\vec{A}_0\big)\cdot\big(\vec{\mu}\big)_{nm} + \frac{\omega_{nm}}{6c}\sum_i\sum_j k_i A_{0j}\big(Q_{ij}\big)_{nm}. \qquad (14.174)$$

This general form holds for all atoms, not just hydrogen, although the operator definitions are more complicated for multielectron atoms.

The electric dipole transitions, abbreviated as $E1$, are *allowed transitions*, while the magnetic dipole, abbreviated $M1$, and electric quadrupole, abbreviated $E2$, are called *forbidden transitions*. The allowed transitions occur much more often than the forbidden transitions, so when allowed transitions are possible, the forbidden transitions can be neglected; but, when the allowed transitions cannot occur, then the forbidden transitions are often the important transitions. Note that allowed and forbidden are poor word choices, as both transitions can, and do, occur. The transitions have specific *selection rules*, which govern when the corresponding matrix elements are nonzero. To understand how to derive the selection rules, we need to realize that there are conditions under which $\langle n|\hat{\vec{r}}|m\rangle \neq 0$. The first condition comes from orbital angular momentum. We know the matrix element between two states with definite angular momentum vanishes unless the angular momentum is the same. When we take a state $|m\rangle$ and let $\hat{\vec{r}}$ operate on it, it is as if we are adding one unit of angular momentum to the angular momentum of state $|m\rangle$ because the operators $\hat{r}_x$, $\hat{r}_y$, and $\hat{r}_z$ carry angular momentum of 1 because they are vector operators. So, if the state $|m\rangle$ has angular momentum j, then the state $|n\rangle$ must have angular momentum given by the total angular momentum found when combining j with 1, or $\Delta J = \pm 1,\ 0$. But, when we examine the matrix element, we must respect the parity of the state, and since $\hat{\vec{r}}$ is an odd parity operator, the parity must also change. This means $\Delta L = \pm 1$ and then $\Delta S = 0$. Finally, when we have both s-wave states, we always have $\langle s|\hat{\vec{r}}|s\rangle = 0$. So there can be no $0 \to 0$ transition. The analysis for the $M1$ transition is similar, because angular momentum is a vector operator. We have again $\Delta J = \pm 1,\ 0$, $\Delta L = \Delta S = 0$ and there is no $0 \to 0$ transition allowed. Finally, for the $E2$ transition, which is a tensor operator, we have $J_n + J_m \geq 2 \geq |J_n - J_m|$, $\Delta S = 0$ and no $0 \to 0$ transition allowed. We will use these selection rules when analyzing different transitions.

So far we have determined the Hamiltonian that can drive transitions between different energy eigenstates of hydrogen and we have partially evaluated the matrix elements of the time-dependent perturbation between the energy eigenstates of the

unperturbed Hamiltonian. We need to re-examine the formula for the transition matrix elements. Our original derivation was between discrete energy eigenstates, but now, the transition is to a continuum of states because a photon is involved, which has a definite energy, but can have a wavenumber that points in any direction (and similarly a polarization that can point anywhere, unless the light source is polarized). So we need to go back to re-examine the transition matrix element we have been looking at. In time-dependent perturbation theory, we found the probability to make a transition between two different eigenstates in first-order perturbation theory is given by

$$P_{n\leftarrow m}(t) = \frac{1}{\hbar^2}\left|\Delta t \sum_{j=1}^{N} e^{\frac{i}{\hbar}E_{nm}(j\Delta t - t_0)} v_{nm}\right|^2 \tag{14.175}$$

for a time-independent perturbation. The summation gives

$$\begin{aligned}\Delta t \sum_{j=1}^{N} e^{\frac{i}{\hbar}E_{nm}(j\Delta t - t_0)} &= \hbar\left(1 - \cos\left(\tfrac{i}{\hbar}E_{nm}(t-t_0)\right)\right)/iE_{nm} \\ &= 2\hbar e^{\frac{i}{2\hbar}E_{nm}t}\sin\left(\tfrac{1}{2\hbar}E_{nm}(t-t_0)\right)/E_{nm},\end{aligned} \tag{14.176}$$

and the probability becomes

$$P_{n\leftarrow m}(t) = \frac{4\sin^2\left(\frac{1}{2\hbar}E_{nm}(t-t_0)\right)}{E_{nm}^2}|v_{nm}|^2. \tag{14.177}$$

This is the result for the transition between two states with a specific energy difference. If, instead, we have a transition to states in a continuum, we can continuously vary the energy difference, and write it as just E. The plot is given in Fig. 14.15. One can see that as $t \to \infty$, the curve becomes a sharp peak around $E = 0$, with an area given by $\frac{2\pi t}{\hbar}$. Because it gets so narrow, in the long-time limit, we have that the energy is conserved, in the sense that $E_{nm} = 0$ implies $E_n = E_m$. Then, in the limit as t is large, we have

$$P_{n\leftarrow m}(t) = \frac{2\pi}{\hbar}t|v_{nm}|^2\Big|_{E_n=E_m}. \tag{14.178}$$

The condition that the initial and final energies are the same is mathematically taken into account with a Dirac delta function, but we cannot discuss that without calculus. We will find remembering the constraint, which is what the vertical line does, is all we will need for these calculations. As you can see the probability increases linearly in time, and we are considering the large time limit. This becomes unphysical in that realm. So, we instead focus on the rate that the probability changes, which is the slope of the linear curve. This then gives us the Fermi golden rule:

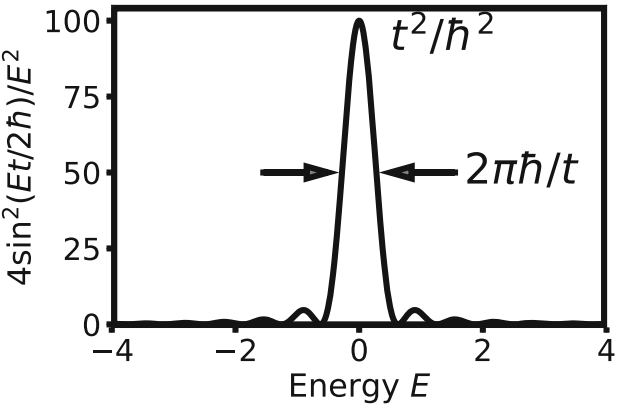

Fig. 14.15 Plot of $4\sin^2\frac{Et}{2\hbar}/E^2$ versus E for large t. The plot is always nonnegative and the first zero occurs at $E = \pm\frac{2\pi\hbar}{t}$, so the half width of the peak is approximately $\frac{2\pi\hbar}{t}$. The height is approximately $\frac{t^2}{\hbar^2}$, so the area under the curve is approximately $\frac{2\pi t}{\hbar}$. Here we chose $t = 10$

$$\text{rate of probability increase} = \frac{2\pi}{\hbar}|v_{nm}|^2\Big|_{E_n=E_m}. \tag{14.179}$$

You might ask how do we use such a result? The way to think of it is to think in terms of a large ensemble of atoms, where we shine a laser onto them. Then, we will see the transitions occurring at a constant rate, which is given by the Fermi golden rule. In experiments, we often measure this steady state of the ensemble. That is, we directly measure the rates. The key in our calculations is that we need to sum up all of the contributions from all possible transitions (as we mentioned before all directions of momentum and all directions of polarizations). Finally, note that one can relax the condition that the perturbation becomes time independent, because the sharp peak "squeezes" the contributions to being at a single time regardless.

As we mentioned above, we are working in a semiclassical approach, where we have not quantized the electric field into the elementary photons. Nevertheless, we can express our results in the language of the quantized photons. So, if we have n photons present, we use the fact that photons behave like the simple harmonic oscillator to write the vector potential as

$$\vec{A}_0 = \vec{\epsilon}\sqrt{\frac{\hbar c^2}{2\epsilon_0\omega_k}} \times \begin{cases} \sqrt{n+1} & \text{emission} \\ \sqrt{n} & \text{absorption} \end{cases}. \tag{14.180}$$

Here, $\vec{\epsilon}$ is the polarization vector, which points in the direction of the electric field, and the factors $\sqrt{n+1}$ and $\sqrt{n}$ come from the harmonic oscillator formulas for increasing or decreasing the number of excitations. The symbol $\omega_k = ck$ is the frequency associated with a light wave of wavevector k. Note that the polarization

vector is a unit vector perpendicular to the wavevector $\vec{k}$. There are two independent directions perpendicular to $\vec{k}$, so there are two polarization vectors, which we label with α when necessary. Since there are two polarizations for each wavevector, and the wavevector directions are distributed uniformly over the unit sphere, we describe the rate in terms of a small region of solid angle, which just means a patch of small area on the unit sphere, which we denote as $\Delta\Omega_\gamma$ since γ is the symbol reserved for a photon. So, for the $E1$ transition, we have

$$\text{rate per solid angle} = \frac{2\pi}{\hbar}\rho_\gamma(k)\sum_\alpha \left|\sqrt{\frac{\hbar\omega_k}{2\epsilon_0}}\,\vec{\epsilon}_\alpha\cdot\left(D_{fi}\right)\right|^2 \Big|_{E_f+\hbar\omega_k=E_i}\Delta\Omega_\gamma, \tag{14.181}$$

for spontaneous emission, where $n = 0$. The symbols i and f stand for the initial and final states of the transition. We sum the squares of the matrix elements for the different polarizations, because each polarization carries its own "which-way" information, and so they cannot "move in superposition." This is called *adding results by quadrature*. The energy conservation says that the energy of the initial state of the atom E_i must equal the sum of the energy of the final state of the atom plus the energy of the photon $E_f + \hbar\omega_k$. The symbol $\rho_\gamma(k)$ is the photon density of states, given by

$$\rho_\gamma(k) = \frac{1}{(2\pi)^3}\,\frac{\omega_k^2}{\hbar c^3}. \tag{14.182}$$

It measures the number of photons available that have the energy needed for the transition, as a function of the photon frequency. This number arises from the fact that we are looking at the directions of the photon along a unit sphere, but the wavevector is distributed along a sphere whose radius is $|\vec{k}|$. Simplifying the rate per solid angle, we have

$$\text{rate per solid angle} = \frac{\omega_k^3}{8\pi^2\epsilon_0\hbar c^3}\sum_\alpha \left|\vec{\epsilon}_\alpha\cdot\left(\vec{D}_{fi}\right)\right|^2 \Big|_{E_f+\hbar\omega_k=E_i}\Delta\Omega_k. \tag{14.183}$$

To get the total rate, Γ_{fi}, we need to add up all the results over the entire unit sphere, so

$$\Gamma_{fi} = \sum_{\text{surface of unit sphere}} \frac{\omega_k^3}{8\pi^2\epsilon_0\hbar c^3}\sum_\alpha \left|\vec{\epsilon}_\alpha\cdot\left(\vec{D}_{fi}\right)\right|^2 \Big|_{E_f+\hbar\omega_k=E_i}\Delta\Omega_k. \tag{14.184}$$

For the $E1$ transitions, we then have

$$\Gamma_{fi}^{E1} = \frac{\omega_k^3 e^2}{8\pi^2\epsilon_0\hbar c^3}\sum_{\text{surface of unit sphere}}\sum_\alpha\sum_i\sum_j \vec{\epsilon}_{\alpha i}\langle f|\hat{r}_i|i\rangle\vec{\epsilon}_{\alpha j}\langle i|\hat{r}_j|f\rangle \Big|_{E_f+\hbar\omega_k=E_i}\Delta\Omega_k. \tag{14.185}$$

The sum over polarizations can be performed using completeness for the three-dimensional space spanned by each polarization vector, and the unit vector in the direction of $\vec{k}$, that is the three unit vectors that span the three-dimensional space. We know that

$$\delta_{ij} = \sum_{\alpha} \vec{\epsilon}_{\alpha i} \vec{\epsilon}_{\alpha j} + \frac{k_i k_j}{|\vec{k}|^2}. \tag{14.186}$$

Solving for the sum, we find that

$$\sum_{\alpha} \vec{\epsilon}_{\alpha i} \vec{\epsilon}_{\alpha j} = \delta_{ij} - \frac{k_i k_j}{|\vec{k}|^2}. \tag{14.187}$$

We will substitute this into the equation for the total rate. The summation over the unit sphere can be easily done now. The sum of $\delta_{ij} \Delta\Omega_k = 4\pi\delta_{ij}$, because the summations do not depend on where we are on the sphere, and the total solid angle is 4π (recall the total area of a sphere of radius R is $4\pi R^2$). The sum of $\frac{k_i k_j}{|\vec{k}|^2}$ is determined as follows. First, when $i \neq j$, the sum is zero, because every term with a specific value of k_i has another term exactly equal to $-k_i$, so they cancel in pairs unless $i = j$, where all those terms become positive. But because the three spatial directions are indistinguishable on a unit sphere, they all must contribute the same amount, namely, $\frac{4\pi}{3}$. This means we have

$$\sum_{\text{surface of unit sphere}} \sum_{\alpha} \vec{\epsilon}_{\alpha i} \vec{\epsilon}_{\alpha j} = \frac{8\pi}{3}\delta_{ij}, \tag{14.188}$$

and the total rate becomes

$$\begin{aligned} \Gamma_{fi}^{E1} &= \frac{\omega_k^3 e^2}{3\pi\epsilon_0 \hbar c^3} \sum_i \langle f|\hat{r}_i|i\rangle\langle i|\hat{r}_i|f\rangle \Big|_{E_f+\hbar\omega_k=E_i} \\ &= \frac{\omega_k^3 e^2}{3\pi\epsilon_0 \hbar c^3} |\langle f|\hat{\vec{r}}|i\rangle|^2 \Big|_{E_f+\hbar\omega_k=E_i}. \end{aligned} \tag{14.189}$$

The lifetime is the inverse of the rate. So, the partial lifetime to make a transition to state f is given by $\tau_{f\leftarrow i} = 1/\Gamma_{f\leftarrow i}$. The total lifetime of an initial state is then $\tau_i = 1/\sum_f \Gamma_{f\leftarrow i}$.

Before doing any calculations, let us estimate the size of the different terms in this expression to approximate the lifetimes. It is convenient to use the *fine structure constant* given by

$$\frac{e^2}{4\pi\epsilon_0 \hbar c} = \alpha \approx \frac{1}{137.036}. \tag{14.190}$$

The frequency of light for an atomic transition is approximately given by the bound-state energies or

$$\omega_k \approx \frac{e^2}{4\pi\epsilon_0 \hbar a_0}, \tag{14.191}$$

and then $ka_0 = \omega a_0/c \approx \alpha$, which confirms assuming it is small in the treatment of the photoproduction of hydrogen. Since the matrix element $\langle f|\hat{r}|i\rangle \approx a_0$, we then have for the rate

$$\Gamma_{f\leftarrow i} \approx \frac{\omega_k^3}{4\pi\epsilon_0\hbar c^3}\, a_0^2 \approx \alpha^4 \frac{c}{a_0} \approx \left(\frac{1}{137}\right)^4 \frac{3\times 10^8\,\text{m/s}}{0.0529\times 10^{-9}\,\text{m}} \approx 10^{10}/\text{s}, \tag{14.192}$$

or lifetimes on the order of one-tenth of a nanosecond. The forbidden transitions $M1$ and $E2$ have lifetimes $1/\alpha^2$ larger or on the order of a microsecond. Other transitions can be even longer. The calculation of the specific lifetimes is examined in Problem 14.10.1 and is one of the important quantities we can calculate to compare with experiment. In Chap. 20, we will discuss experiments that measure the lifetime of a photon coming from an atomic transition.

The standard description for why we have spontaneous emission, in the absence of any electromagnetic fields is that the vacuum has an average field that vanishes, but the fluctuations of the field are nonzero. So, for a short period of time, the vacuum has an electric field. This field can then stimulate the emission of the electron in the atom from a higher-energy state to the ground state, although the precise process for doing this is a bit vague. I think there is a better way to look at this in terms of energy eigenstates. If we were to quantize light (which we will do later in the book), then, because light and matter interact, if we have both atoms and light present, the energy eigenstates are not just energy eigenstates of atoms, but complicated superpositions of atoms with photons. Then, if the excited energy state of the atom with no photon is not an energy eigenstate, it will oscillate in time with respect to the energy eigenstates needed to describe the original state as a superposition. So, the amplitudes of the system will oscillate, with the amplitude of the ground state growing at the expense of the excited state. The ground state must also have a photon from energy conservation. Once the photon is created, it moves away from the atom and cannot come back to be absorbed (in most cases unless the atom is inside a highly reflective cavity which keeps the photon around). So, once the ground-state amplitude becomes large, the photon disappears because it travels away from the atom, and then it cannot oscillate back to the excited state plus no photon, because it is no longer energetically allowed. While not perfect, this is a reasonable way to understand why the energy eigenstates of an atom become unstable and decay to the ground state when the atom interacts with light (even if it is just the vacuum). Without this additional interaction, we would expect higher-energy states to be stable. But, this rarely occurs. In other contexts, we use the word

decoherence to describe the decay to the ground state when the system interacts with some kind of environment.

We have estimated that the $2p$ energy eigenstates have a lifetime of about 1 ns. But the $2s$ state cannot easily decay to the $1s$ state. This is because it is a $0 \leftarrow 0$ transition, and these do not occur. Hence, it is a forbidden transition that does not even have transitions via $M1$ or $E2$. So we expect the lifetime to be quite long. In fact, this state only decays via a nonlinear two-photon transition and it is used as one of the measurements of the proton charge radius. The lifetime is about 0.1 s, or eight orders of magnitude longer than the $2p$ states. Do other s states have the same protection? The answer is no, and you will have a chance to explore this further in Problem 14.10.2. We also have a video https://tinyurl.com/548xh9xp that discusses photoproduction of hydrogen from a more advanced perspective.

Section 14.10 Problems:

14.10.1 Transition from the $n = 2$ Shell to the $n = 1$ Shell in Hydrogen

In this problem, we will use our formulas to compute the lifetimes of the $n = 2$ states (ignoring fine structure effects). Assume one is in the $|n{=}2, l{=}1, m\rangle$ state or the $|n{=}2, l{=}0, m{=}0\rangle$ state and compute the partial lifetime to decay to the $1s$ state (with $l = 0$ and $m = 0$).

You will need a number of matrix elements. While, in principle, they can be calculated using methods similar to what we used for the Kramers-Pasternack recurrence relations, the derivation of these formulas is cumbersome. So, we just give you the nonzero results, which are as follows:

$$\begin{aligned}
\langle n{=}1, l{=}0, m{=}0|\hat{r}_x|n{=}2, l{=}1, m{=}\pm 1\rangle &= \pm\frac{2^7}{3^5}a_0 \\
\langle n{=}1, l{=}0, m{=}0|\hat{r}_y|n{=}2, l{=}1, m{=}\pm 1\rangle &= i\frac{2^7}{3^5}a_0 \\
\langle n{=}1, l{=}0, m{=}0|\hat{r}_z|n{=}2, l{=}1, m{=}0\rangle &= \sqrt{2}\,\frac{2^7}{3^5}a_0.
\end{aligned} \tag{14.193}$$

In particular, $\langle n{=}1, l{=}0, m{=}0|\,\hat{\vec{r}}\,|n{=}2, l{=}0, m{=}0\rangle = 0$. Use these results to find the lifetimes for $2p$ and $2s$ states.

Ans: you should find the $2p$ states have a lifetime of about 1.6 ns, while the lifetime of the $2s$ state is infinite. Explain why the $2s$ state having an infinite lifetime is incorrect, because that state is unstable and does decay to the $1s$ state.

14.10.2 *Lifetime of 3s States Versus 2s States*

In the main text, we described why the $2s$ state has a very long lifetime. What about higher s states, like the $3s$ or $4s$ states? Write a short essay that describes how they decay, and whether we think their lifetime is likely to be long or short? Be sure to contrast what you describe here to what happens for the $2s$ states. Also, be as explicit as possible in describing the chain of transitions that take us from the original state down to the $1s$ state.

Hint: It requires more than one transition to get to the $1s$ state.

14.10.3 *Why Is the Lifetime of Rydberg States So Long?*

Recall that Rydberg states have large n and often (but not always) large l. If we look at an estimate of $\langle f|\hat{r}|i\rangle$, we find it is on the order of $n^2 a_0$ when both states have large n. The frequency goes like $1/n^3$ in this case. Show that this leads to a much longer lifetime than for the $n = 2$ states. Use $n = 100$ for the Rydberg states to be concrete.

If we instead have small l, then the overlap with states that have small n is suppressed because the large n wavefunctions oscillate a lot in the region close to the $1s$ shell to be orthogonal to them. This means we estimate $\langle f|\hat{r}|i\rangle$ to be much smaller than a_0, even though the frequency is on the order of eV now. Discuss how much smaller it needs to be in order to have long lifetimes for these states as well. Discuss what scenario is more likely to have long-lived states for Rydberg atoms—large n and large l or large n and small l, or does it not matter.

14.11 XY Model as a Stroboscopic Ising Model in the Rotating-Wave Approximation

Another important approximation for time-dependent systems is the *rotating-wave approximation*, where we neglect terms in the Hamiltonian that oscillate rapidly, even if they have a large amplitude. We already saw an example of this in magnetic resonance. When ω was far from resonance (hence the field rotates fast in the rotating frame), the probability to flip the spin was always small and oscillated rapidly. For me, this approximation always appears counterintuitive, and I often have a hard time believing it really holds. But, it does, and we need to understand it better. In this section, we will describe how the rotating-wave approximation works for the Ising model in a large field, where the system approximates the XY model, but only if we interrogate it at specific times. Hence, it requires looking at the system stroboscopically. This section covers some more advanced ideas from time

evolution in quantum mechanics. They have been used in real experiments, so there are practical applications of them.

We focus on employing the rotating-wave approximation to go from the transverse-field Ising model, given by

$$\hat{H}_{TFI} = \sum_{j<j'} J_{jj'} \sigma_j^x \sigma_{j'}^x - B \sum_j \sigma_j^z \tag{14.194}$$

to the XY model, which is given by the following Hamiltonian:

$$\hat{H}_{XY} = \frac{1}{2} \sum_{j<j'} J_{jj'} \left(\sigma_j^x \sigma_{j'}^x + \sigma_j^y \sigma_{j'}^y \right) = \frac{1}{4} \sum_{j<j'} J_{jj'} \left(\sigma_j^+ \sigma_{j'}^- + \sigma_j^- \sigma_{j'}^+ \right), \tag{14.195}$$

where we used the spin raising and lowering operators via $\sigma^{\pm} = \sigma^x \pm i\sigma^y$ in the second equation. The relationship between the transverse-field Ising model and the XY model is then found by substituting the raising and lowering operators into the transverse-field Ising model which yields

$$\hat{H}_{TFI} = \frac{1}{4} \sum_{j<j'} J_{jj'} (\underbrace{\sigma_j^+ \sigma_{j'}^+}_{\text{extra}} + \underbrace{\sigma_j^+ \sigma_{j'}^- + \sigma_j^- \sigma_{j'}^+}_{\text{XY piece}} + \underbrace{\sigma_j^- \sigma_{j'}^-}_{\text{extra}}) - B \sum_j \sigma_j^z. \tag{14.196}$$

Note how two of the spin-exchange terms are the same as those in the XY model, but there are two extra operators which involve either raising the spins twice or lowering them twice. If the magnetic field is large, there will be a large energy cost for those double spin flips, as opposed to having no energy cost for the XY terms, which flip one spin up and the other spin down. This provides a hint that there should be a relationship between these two models in a large magnetic field.

To make the mapping more precise, we invoke the rotating-wave approximation, which requires us to go to the interaction representation with respect to the magnetic field piece of the Hamiltonian. We then transform the spin-exchange part of the Hamiltonian via $\hat{H}_{TFI}^{SE} \rightarrow \exp[i\hat{H}_{TFI}^{B} t] \hat{H}_{TFI}^{SE} \exp[-i\hat{H}_{TFI}^{B} t]$ to give us the interaction representation of $\hat{V}_I$. In the rotating frame, the magnetic-field piece of the Hamiltonian is accounted for in the time dependence under $\hat{H}_0$, so the "interaction piece" of the transverse-field Ising model becomes

$$\hat{H}_{TFI} \rightarrow \frac{1}{4} \sum_{j<j'} J_{jj'} \left(\sigma_j^+ \sigma_{j'}^+ e^{4iBt} + \sigma_j^+ \sigma_{j'}^- + \sigma_j^- \sigma_{j'}^+ + \sigma_j^- \sigma_{j'}^- e^{-4iBt} \right). \tag{14.197}$$

When the magnetic field, B, is large, the rotating terms are rapidly oscillating and average to zero. The rotating-wave approximation then says we can ignore them. So, in this large-field limit of the Ising model, it should act like an XY model. To fully understand this mapping, though, we need to explore in detail how the energy eigenvalues relate as well as the dynamics of the wavefunctions. We will

find the mapping holds stroboscopically in time because the frame is rotating at the Larmor frequency, and hence the wavefunctions have an oscillating phase which approximately returns to a multiple of 2π every Larmor period. We will need to balance the improved accuracy given from a larger field with the difficulty in properly timing the stroboscopic measurements when the complex phase factors oscillate too rapidly in order to determine the optimal mapping between the two models.

We work with the J_{ij} given by a simple power law

$$J_{jj'} \approx \frac{J_0}{|j - j'|^\alpha}, \tag{14.198}$$

where α varies from 0 to 3 because this is the situation that can be examined in ion-trap experiments. We will see later that there is an exact formula for the J_{ij} coefficients, but here, we just assume they have a fixed power law. We work with an α varying between 0.63 and 1.19 and with $J_0 \approx 500\,\text{Hz}$ (note that Hz denotes a regular frequency as opposed to an angular frequency—they are related by a factor of 2π). Both the transverse-field Ising model and the XY model are time independent. The evolution operator is then given by $\hat{U}(t) = e^{-i\hat{H}t}$.

Our first illustration of the mapping between these models involves a comparison of their energy levels. The transverse-field Ising model energy levels in a strong transverse field are approximately shifted by $-2\,mB$, where $2\,m$ is the eigenvalue of the $\sigma^z_{tot} = \sum_j \sigma^z_j$ operator. The shift is approximate because σ^z_{tot} does not commute with the transverse-field Ising Hamiltonian. We identify approximate σ^z_{tot} blocks in the transverse-field Ising energy levels in the limit of a large transverse field; that is, the energy levels will split based on the approximate value of σ^z_{tot} acting on the corresponding eigenstate. Figure 14.16 shows a comparison of Ising states with $S^z_{tot} \approx 0$ for $B/J_0 = 10$ in a six-site lattice with $\alpha \approx 0.63$. The XY Hamiltonian commutes with the σ^z_{tot} operator, so we can compare the energy states of both models on the basis of their σ^z_{tot} value (approximate for the transverse-field Ising model and exact for the XY model). Note how the XY model has some states with $S^z_{tot} \neq 0$ intermixed in the energy levels with $S^s_{tot} = 0$. These intermixed energy levels are not in the corresponding Ising model. In the right panel, we show the energy differences between two comparative energy levels: (i) the ground state and the second excited state. The energy levels differ by the order of 10^{-5} here, with higher energy levels having larger deviations. This is a small error, indicating how good the rotating-wave approximation is.

A numerical evaluation of the modulus squared of the overlap between the XY and Ising states as a function of time is shown in Fig. 14.17 for a five-ion chain with $\alpha \approx 0.63$ and field strengths of $B/J_0 = 5$, 10, 15, and 20. The initial state for these calculations is the state with all spins oriented in the $-\hat{y}$ direction, which is represented in the z-basis as the direct-product state $|\Psi\rangle = (|\uparrow\rangle - i|\downarrow\rangle)_1 \bigotimes (|\uparrow\rangle - i|\downarrow\rangle)_2 \bigotimes \ldots \bigotimes (|\uparrow\rangle - i|\downarrow\rangle)_N$. This state is not an energy eigenstate, so it varies in time. The black dots are placed at $2\pi n/\omega_L$, for

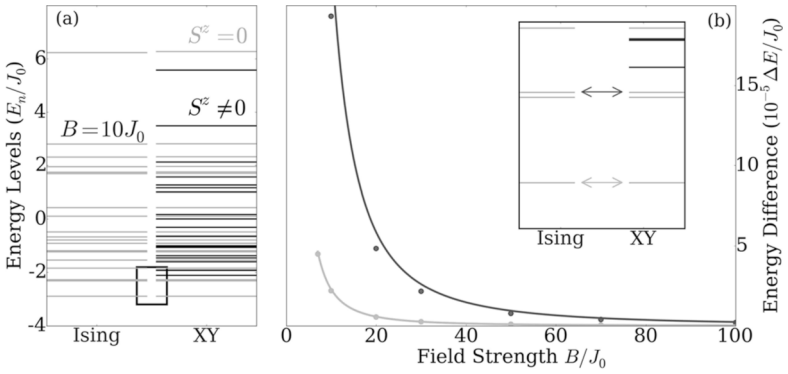

Fig. 14.16 Transverse-field Ising model energy levels for $B/J_0 = 10$ and XY model energy levels for a chain of six ions with $\alpha \approx 0.63$. Levels in the $S^z_{tot} \approx 0$ block of the Ising model and the $S^z_{tot} = 0$ block of the XY model are colored in gray. (**b**) Difference between the corresponding XY and transverse-field Ising model energy levels plotted as a function of field strength for $B/J_0 = 7$, 10, 20, 30, 50, 70, and 100. The particular levels used to measure the difference are shown in the inset with the rows. You can see the corresponding energy levels are quite close to one another, but there are some energy levels in the XY model that have different S^z_{tot} values, and these energy levels are ignored in this mapping. [Figure reprinted from Kiely and Freericks (2018). © American Physical Society.]

integer n with $\omega_L = 4\pi B$, which corresponds multiples of the Larmor period. The light gray dots are placed according to an optimized frequency, which is found by modifying the Larmor frequency from $4\pi B$ to $4\pi\sqrt{B^2 + (aJ_0)^2}$ and varying a until the combined sum of all modulus squares of the overlaps for a given range of times reaches a local maximum at integer multiples of the modified period. The form of the correction assumes that the spin-exchange interaction can be treated as an approximate mean field in the x-direction and that the frequency of the oscillations depends on the resultant magnitude of the total field; in other words, we assume a specific form, similar to what we had in magnetic resonance (with a perpendicular field proportional to J_0) and find the best fit. The corrected frequency in Fig. 14.17 corresponds to $a = 1.67$, which was determined by optimizing the sum of all plotted points between $tJ_0 = 0$ and $tJ_0 = 1$. In general, we found that a depends on the lattice size and on the initial state of the system, so it is not easy to know what it would be without solving the problem *a priori*. We went through this exercise to try to extend the period in time where the two models had quantum states that could be identified with each other stroboscopically. In general, however, if we don't have more accurate information available to us, we simply have to use the Larmor period, which breaks down quite a bit sooner than the corrected period. It should also be clear from the figure that as the magnetic field is made larger, the approximation works better, but the stroboscopic times become closer and closer together and hence harder to accurately make measurements at.

The squared overlap oscillates between 1 and 0 at the Larmor frequency, while the envelope of the amplitude decays with time. The importance of the mean-field

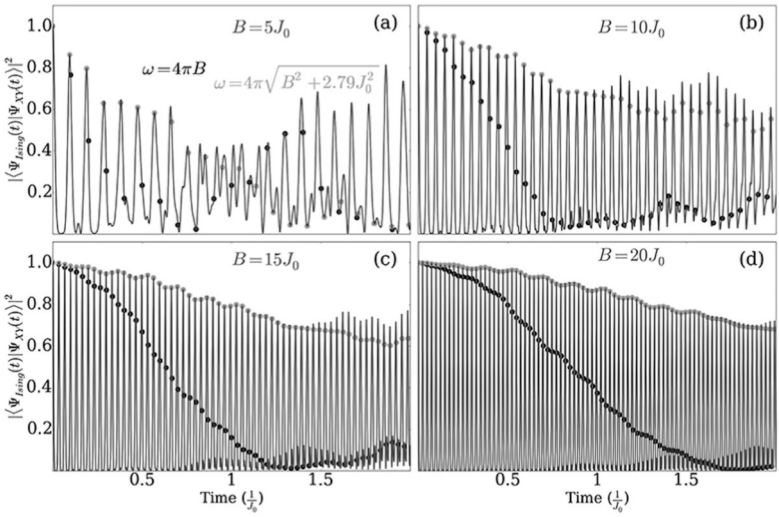

Fig. 14.17 Plot of the modulus squared of the overlap of the time-evolved XY and transverse-field Ising state vectors, $\langle\Psi_{Ising}(t)|\Psi_{XY}(t)\rangle$, for a five-ion chain with $\alpha \approx 0.63$ and various transverse-field strengths. Values close to 1 indicate the rotating-wave approximation is good. Panels (**a**)–(**d**) plot the squared overlap between $tJ_0 = 0$ and $tJ_0 = 2$ for $B/J_0 = 5$, 10, 15, and 20, respectively. The black dots are plotted at the Larmor frequency, $\omega_L = 4\pi$. The light gray dots are plotted at a numerically optimized frequency, given by $\omega_{opt} = 4\pi\sqrt{B^2 + (1.67J_0)^2}$. What one can see is the fidelity of the ground state is large for a much longer time when we use the optimized frequency versus the Larmor frequency, where the fidelity rapidly decreases to very small values, indicating the system is not in the ground state at those stroboscopic times. The fidelity measurement automatically ignores any global phase differences between the two states. [Figure reprinted from Kiely and Freericks (2018). ©American Physical Society.]

correction to the measurement frequency is shown clearly by the rapid rate of decay of the black dots relative to the light gray dots, especially for larger magnetic fields. Even for $B/J_0 = 20$, measurements taken with $\omega_L = 4\pi B$ will fall so far out of phase by $tJ_0 = 1$ that $|\langle\Phi_{Ising}|\Phi_{XY}\rangle|^2 \approx 0.5$, even though the state vectors are still coming into a maximum alignment of $|\langle\Psi_{Ising}|\Psi_{XY}\rangle|^2 \approx 0.9$ at slightly different stroboscopic times. This result tells us two things. First, getting the period right for the stroboscopic behavior is important, and it is corrected from the simple Larmor frequency, and second, the rotating-wave approximation is just that—an approximation—and so we expect it to hold only to a specific level of accuracy.

In studying the dynamics of this spin system, it is useful to describe how spin excitations evolve as a function of time. One way to do this is to flip a spin at time t_0 and site i and flip it back at time t and site j and ask how much has the system changed in the time interval $t - t_0$? In practice, we examine both possible flips simultaneously by using a σ_i^x operator. Such an object is called a *response function* or more commonly a *Green's function*. The "pure-wavefunction" retarded spin-spin Green's function is then defined by

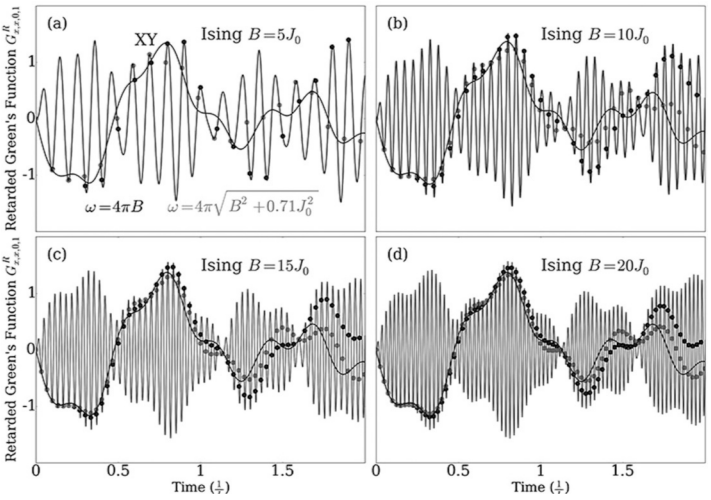

Fig. 14.18 Plot of the pure-wavefunction retarded Green's function of the transverse-field Ising Hamiltonian, in black with fast oscillations, and of the XY Hamiltonian, in gray (dashed line) with slow oscillations, for a chain of seven ions with $\alpha \approx 1$ and various transverse-field strengths. Panels (**a**)–(**d**) plot the Green's functions between $tJ_0 = 0$ and $tJ_0 = 2$ for $B/J_0 = 5$, 10, 15, and 20, respectively. The black dots are plotted at the Larmor frequency, $\omega_L = 4\pi$, and the light gray dots are plotted at a $\omega_{opt} = 4\pi\sqrt{B^2 + (0.84J_0)^2}$. [Figure reprinted from Kiely and Freericks (2018). © American Physical Society.]

$$G^R_{\alpha,\beta,j,j'}(t,t_0) = i\theta(t-t_0)\langle\Psi_0|[\sigma^\alpha_j(t),\sigma^\beta_{j'}(t_0)]|\Psi_0\rangle \tag{14.199}$$

where $\sigma^\alpha_j(t) = \hat{U}^\dagger(t)\sigma^\alpha_j\hat{U}(t)$ is a Pauli matrix in the Heisenberg picture. The word retarded is added because the unit step function requires $t \geq t_0$. The commutator may seem strange to you, but this is included because the commutator arises when we examine the linear response of a strongly interacting system (we do not have the time to properly motivate why one would include this here). We examine the $G^R_{x,x,j,j'}$ components of the Green's function because they can be measured experimentally with Ramsey spectroscopy. In Fig. 14.18, we show the numerical evaluation of the $G^R_{x,x,0,1}(t,0)$ for a seven-ion chain in the XY model and the transverse-field Ising model with $B/J_0 = 5$, 10, 15, and 20 and $\alpha \approx 1$. The pure state used in the Green's function calculation is defined by $|\Phi_T\rangle = \sum_n \sqrt{\exp[-\beta E_n]/Z}|n\rangle$, where $Z = \sum_n \exp[-\beta E_n]$ and $\beta = 2/J_0$. Note that this wavefunction is a linear superposition of the eigenstates with the amplitudes of each state chosen with a specific functional form (this choice is made to resemble the behavior of the system at a specific temperature). Dots indicate measurements of the transverse-field Ising model Green's function at the particular stroboscopic times, which correspond to the simulation of the XY model Green's function. The light gray dots correspond to a mean-field correction of $0.84J_0$ ($a = 0.84$) to the Larmor frequency (as discussed above), which was determined by optimizing the modulus

squared of the overlap between XY and the transverse-field Ising evolutions of $|\Psi_T\rangle$ between $tJ_0 = 0$ and $tJ_0 = 1$.

The XY Green's function initially traces the envelope of the fast-oscillating Ising Green's function, but this relationship breaks down at around $tJ_0 = 0.9$. The dots do not reliably track the XY Green's function until $B/J_0 = 15$, but even for $B/J_0 = 20$, the mapping falls off around $tJ_0 = 1$. Further, it is important to note that the instantaneous slope of the transverse-field Ising model Green's function at measurement times increases with B because the XY curve does not simply follow its envelope. This means that experimental error will be amplified considerably in the presence of a large transverse field due to the difficulty in measuring the required points at the precisely needed stroboscopic times.

The plots of the Green's function and of the wavefunction overlap indicate that there is an experimentally optimal field strength that would produce the most accurate simulation of the XY model for a given experimental error in data collection times. This optimal value is important because the slope of the oscillations of the transverse-field Ising model data at the times where data is collected can be huge.

For fields of equal or lower magnitude than the optimal field strength, there are also maximum dephasing times, after which the overlap of the transverse-field Ising model evolved state and the XY evolved state will be too small to say that the two results are equivalent. Note also that the value of the overlap will differ depending on whether a simple Larmor frequency is used or whether a correction factor is included. Table 14.1 summarizes this dephasing time for the modulus squared of the overlap, defining the dephasing time as the time after which the squared overlap is less than 0.7.

In this section, we examined a counterintuitive result—when terms in a Hamiltonian oscillate rapidly, they can be ignored, even if they have a large amplitude. This is an approximation, so it has errors associated with it. When we applied this to the time evolution of the transverse-field Ising model in an external field, we find that the system behaves more like the XY model when the magnetic field is large. But, it requires care to determine the behavior of the XY model, because we

Table 14.1 Dephasing time of the modulus squared of the overlap of the evolved XY and transverse-field Ising states, defined as the time at which $|\langle\Phi_{XY}(t)|\Phi_{Ising}(t)\rangle|^2 \leq 0.7$ compared to the strength of the transverse field. The data used to determine these values is the same as that presented in Fig. 14.17. The Larmor dephasing time corresponds to the black dots in Fig. 14.17, which are placed at a frequency of $\omega_L = 4\pi B$, and the corrected dephasing time corresponds to the gray dots, placed at $\omega_{opt} = 4\pi\sqrt{B^2 + (1.67J_0)^2}$. One can clearly see that using the corrected frequency significantly improves the accuracy for short times

Field strength (B/J_0)	Larmor dephasing time ($1/J_0$)	Optimal dephasing time ($1/J_0$)
5	0.20	0.28
10	0.35	0.79
15	0.50	1.29
20	0.63	1.79

have to interrogate the Ising model at specific stroboscopic times. Nevertheless, we found that the energy eigenvalues are quite similar, the wavefunctions have large overlap (especially if we use an optimized frequency), and correlation functions involving flipping spins behave similarly, at least for short times. These ideas have been examined in experimental settings, such as in ion-trap experiments. We won't describe the detail of those experiments here, but we do cover ion traps in more detail in Chap. 21.

Section 14.11 Problems:

14.11.1 Rotating Frame

The time-evolution operator, in its matrix form, is expressed as

$$U(t) = e^{it\sum_j \sigma_j^z} e^{-i\Delta t V_I(t)} e^{-i\Delta t V_I(t-\Delta t)} \cdots e^{-i\Delta t V_I(\Delta t)}, \tag{14.200}$$

where V_I is the spin-spin interaction of the transverse-field Ising model, in the rotating frame. Note how the first term returns to the identity matrix when we evolve by the Larmor frequency. Explain how this does not guarantee that the wavefunction has evolved back to itself at that period and why the proper period can be perturbed from this. Note that it is this factorization that is used for what we call the rotating frame, because of the leftmost term on the factorization, which transforms to this rotating frame.

14.11.2 Modifying the Larmor Frequency

In the magnetic resonance experiment, we drove the spin in a large magnetic field so that it precessed at the Larmor frequency, but the oscillation frequency of the probabilities to flip were given by the shifted frequency, proportional to the magnitude of the full magnetic field, as opposed to just its z-component. A similar thing occurs here, if we think of the system as having an approximate B_x given by $J_0\langle\sigma^x\rangle$. Explain how this is just an intuitive picture, so we do not know the precise coefficient to use in the modified Larmor frequency given by $4\pi\sqrt{B^2 + (aJ_0)^2}$. Also discuss some strategies to determine the coefficient a.

14.11.3 Pros and Cons of a Stroboscopic Approach

We learned that the rotating-wave approximation can be employed to map the transverse-field Ising model onto an XY model, but only at specific stroboscopic

times. Write a short essay that describes the benefits and the pitfalls of such a stroboscopic approach, especially since one needs to measure at the precise times (as opposed to just looking at the envelope of the oscillations).

14.11.4 Plane-Polarized Light in Magnetic Resonance

While we have not explicitly discussed circularly polarized light, as you can imagine, it corresponds to light whose polarization rotates in a circular fashion. If we take equal amounts of left circularly polarized light and right circularly polarized light, we obtain plane-polarized light, whose polarization points in just one direction. Knowing this fact, use the rotating-wave approximation to explain why one can use plane-polarized light instead of circularly polarized light to find the resonance. Because plane-polarized light is easier to make than circularly polarized light, this trick is often used in magnetic resonance experiments.

Hint: When the plane-polarized light is broken into its two circular components, one component can be made to be in resonance with the precession of the spin, while the other never can, because it rotates in the wrong orientation.

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Chapter 15
Exchange and the Heisenberg Model

Be prepared, work hard, and hope for a little luck.

Edward Bradley, Journalist

One of the most important quantum-mechanical properties is the property of exchange. The ideas behind this arise from the fact that elementary quanta are *indistinguishable* in the real word, but they are completely distinguishable in our theory, where we use different labels for the different particles. The procedure by which we make our theory indistinguishable as well leads to a new "force," called quantum-mechanical exchange. In this chapter, we will develop just what exchange is, and we will solve a highly nontrivial model of exchange, which opened up an entire industry on how one can solve one-dimensional models that interact strongly. There are going to be a lot of exciting new ideas presented here and you can see what they are at https://tinyurl.com/2zwtczbb. You will find it amazing to learn the quantum origin of magnetism and details about how we work with quantum systems that have many particles. Much of the notions presented here come from Daniel Mattis' fantastic book on quantum magnetism, which emphasizes these ideas and how they lead to remarkable collective behavior (Mattis, 1981). For those interested in a more mathematical treatment, Hal Tasaki has another outstanding book (Tasaki, 2020) that describes how one can rigorously establish many of these results. Our treatment will be much more compact than either of those treatments; interested readers can consult them later for more advanced work. In any case, always remember that permanent magnets are one of the best examples of macroscopic quantum mechanics.

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15.1 Quantum-Mechanical Exchange

Even before quantum mechanics was fully discovered, it was known that classical mechanics cannot explain permanent magnets. First, Bohr (1911) and then van Leeuwen (1921) proved that one cannot have permanent magnets at nonzero temperatures in a classical theory. The proof is quite straightforward, but requires knowledge of statistical mechanics and calculus, so we will not show it to you here. The basic idea behind it is that in the theory the field enters via the vector potential, which shifts the momentum (as we used for the photoproduction of hydrogen in Sect. 14.10). But if one works with velocity, instead of momentum, then one can easily show that the magnetism vanishes in a vanishing magnetic field, so there is no permanent magnetism. This means permanent magnets can only be described with quantum mechanics. Just think of it, when you were playing with magnets on your refrigerator, you were working with quantum mechanics and didn't even know it. Cool!

The origin of the quantum theory begins with our old friends Heisenberg (1926) and Dirac (1926). We start by examining the idea of exchange in a treatment with just two particles; for concreteness think of them as electrons. A general quantum state is represented with a tensor product as

$$|\psi\rangle = \sum_{ij} \sum_{\sigma_1 \sigma_2} c_{ij\sigma_1\sigma_2} \underbrace{|\phi_i, \sigma_1\rangle}_{\text{particle 1}} \otimes \underbrace{|\phi_j, \sigma_2\rangle}_{\text{particle 2}}, \tag{15.1}$$

with a complete basis of single-particle states $\{|\phi_i\rangle\}$, spin states described by σ (we have not specified the axis yet) and complex coefficients $c_{ij\sigma_1\sigma_2}$. The first entry in the tensor product is for the first particle and the second entry for the second one. Note how we clearly distinguish the two particles in our representation of the state. They are not indistinguishable because for any $c_{ij\sigma_1\sigma_2} \neq 0$, we can say electron 1 is in state $|\phi_i\rangle$ with spin σ_1, while electron 2 is in state $|\phi_j\rangle$ with spin σ_2. The components that have $i = j$ and $\sigma_1 = \sigma_2$ would be indistinguishable, but they are forbidden by the Pauli exclusion principle, which does not allow any two fermions to occupy the same quantum state (same in both spatial degrees of freedom *and* spin degrees of freedom).

To make the particles indistinguishable, we need to write the quantum state as a superposition of states, where we can no longer say which particle is which. In addition, for fermions, the quantum state must be *totally antisymmetric*, which means the state changes sign when we interchange two particles (interchanging both the spatial and spin states). The antisymmetry property is explained fully in Chap. 17—we will simply assume it to be true here. So, we can guarantee the antisymmetry if we always have

$$c_{ij\sigma_1\sigma_2} = -c_{ji\sigma_2\sigma_1} \tag{15.2}$$

for the nonzero coefficients (this result trivially holds when the coefficients are zero). Within this requirement, we can still have a wide variety of different states—we can have states that factorize between the spatial and spin degrees of freedom, and have

symmetric spatial states and antisymmetric spin states, or vice versa, or we can have entangled states that do not factorize, but are antisymmetric. It is easier to work with these systems if we can characterize the different permutations of the degrees of freedom in the state. We go through this characterization next.

There are three types of permutations: (i) permutations of just the spin degrees of freedom (called Bartlett permutations), (ii) permutations of just the spatial degrees of freedom (called Majorana permutations), and (iii) permutations of both the spatial and spin degrees of freedom (called Heisenberg permutations). Clearly the product of the first two permutations is equal to the third, and because the Heisenberg permutation must return the wavefunction multiplied by -1 (because of the antisymmetry requirement), the other two permutations are essentially inverses of each other (with the modification of the global phase of -1). In equations, we have

$$\begin{aligned}
\hat{P}_{12}^{B}|\psi\rangle &= \sum_{ij}\sum_{\sigma_1\sigma_2} c_{ij\sigma_1\sigma_2}|\phi_i,\sigma_2\rangle\otimes|\phi_j,\sigma_1\rangle \\
\hat{P}_{12}^{M}|\psi\rangle &= \sum_{ij}\sum_{\sigma_1\sigma_2} c_{ij\sigma_1\sigma_2}|\phi_j\sigma_1\rangle\otimes|\phi_i\sigma_2\rangle \\
\hat{P}_{12}^{H}|\psi\rangle &= \sum_{ij}\sum_{\sigma_1\sigma_2} c_{ij\sigma_1\sigma_2}|\phi_j\sigma_2\rangle\otimes|\phi_i\sigma_1\rangle = -|\psi\rangle,
\end{aligned} \tag{15.3}$$

for the three different permutations; the last expression is the antisymmetrization requirement and follows from the condition we imposed on the coefficients. Note that the position of the ket in the tensor product determines which particle is which, so when we exchange the kets on different sides of the tensor product, we are exchanging the states for particle 1 and particle 2. Since the exchange of the spatial basis states is challenging to work out, we focus on finding the operator that exchanges the spin degrees of freedom (the Bartlett permutation).

Let's look at the different spin states for two particles. We have $|\uparrow\rangle\otimes|\uparrow\rangle$, $|\uparrow\rangle\otimes|\downarrow\rangle$, $|\downarrow\rangle\otimes|\uparrow\rangle$, and $|\downarrow\rangle\otimes|\downarrow\rangle$. The two states with the same spin states for each particle have the identity operator as the exchange operator, because the states are the same after exchanging the kets. The states where they are opposite map to each other under the exchange operation. We know how to do the latter; we simply apply

$$\frac{1}{\hbar^2}\left(\hat{S}_1^+\otimes\hat{S}_2^- + \hat{S}_1^-\otimes\hat{S}_2^+\right) \tag{15.4}$$

to create an operator that maps $|\uparrow\rangle\otimes|\downarrow\rangle \rightarrow |\downarrow\rangle\otimes|\uparrow\rangle$ and vice versa. But this operator annihilates the states with the same spin projection for each particle. So we need to add an operator that acts like the identity on these states and annihilates the states with opposite projections. After a little thought, we find that

$$\hat{P}_{12}^{B} = \frac{2}{\hbar^2}\left(\frac{\hbar^2}{4}\hat{\mathcal{I}} + \frac{1}{2}\left(\hat{S}_1^+\otimes\hat{S}_2^- + \hat{S}_1^-\otimes\hat{S}_2^+\right) + \hat{S}_1^z\otimes\hat{S}_2^z\right) = \frac{2}{\hbar^2}\left(\frac{\hbar^2}{4}\hat{\mathcal{I}} + \hat{\vec{S}}_1\cdot\hat{\vec{S}}_2\right), \tag{15.5}$$

where we dropped the tensor product in the rightmost equation. We are familiar with the second operator term (the identity term just gives a constant term when operated on states and can be dropped). It is the sum of the XY-model interaction plus a perpendicular Ising model interaction. When the interaction is isotropic, as it is here, being represented in terms of the inner product of the spin operators, we call this operator the *Heisenberg spin-exchange operator*. To make it into a Hamiltonian, we need to multiply it by a constant with dimensions of energy. It is common to write it in the form

$$-\frac{4J}{\hbar^2}\hat{\vec{S}}_1 \cdot \hat{\vec{S}}_2 = -J\vec{\sigma}_1 \cdot \vec{\sigma}_2, \tag{15.6}$$

with J being the *exchange constant*. Then, when $J > 0$, the ferromagnetic states (highest spin) are lowest in energy, while if $J < 0$, antiferromagnetic states (usually the lowest spin) are lowest in energy. We will cover these distinctions in full detail in the next section.

It is important to remember that the exchange interaction *is not a direct spin-spin interaction between the spins of the electrons*. Rather, it results from the fact that the Coulomb interaction between two electrons prefers certain spin arrangements to give it the lowest energy. How this works is subtle, so we explain it carefully. The idea is a combination of the exclusion principle and Coulomb repulsion. We work in a basis that involves quantum states that localize electrons in some spatial regions. If two electrons are localized in the same region, we pay a large Coulomb energy cost. But, if they are required to be in different regions, the cost is much less, because the Coulomb repulsion becomes weaker at longer distances. The exclusion principle tells us if both spins are the same, then they cannot be in the same spatial region. So their energy is automatically lowered relative to opposite spins, which can be in the same region. Hence, the electrostatic Coulomb repulsion plus the exclusion principle favors states of specific total spin. If you studied Hund's rules in a chemistry class, this is the origin of the rule that asked you to maximize the spin in a partially filled shell. This implies that the exchange interaction arises from the mutual Coulomb repulsion of two electrons and the Pauli exclusion principle. In contrast, the direct dipole-dipole coupling of spins is much too small a force to lead to magnetic ordering. It is quite remarkable how this works!

So why isn't everything magnetic? There is a balance between the kinetic and potential energies and quantum exchange. Usually minimizing the kinetic energy via the bathtub principle also moves the electrons far apart from each other, so the exchange interaction is not needed to reduce the energy further. Only in cases where this does not happen can the exchange interaction do its magic. This typically occurs when the possible electronic states are *degenerate* or nearly degenerate, such as d-electrons in an atom. Then, the kinetic energy does not spread them out enough, so the exchange interaction can make them ferromagnetic. This is believed to be the basic reason why iron is magnetic, due to its partially filled d-electrons, and why copper, with its filled d-electrons, is not.

For exchange to be active, the electrons must occupy quantum states that *overlap* with each other. We describe what this means with a very simple model, due to Elliott Lieb and Daniel Mattis. Consider our two-electron model with each electron forced to sit in one of two adjacent boxes. The electrons interact by the Coulomb interaction between the boxes $e^2/4\pi\epsilon_0|\hat{x}_1 - \hat{x}_2|$, which is symmetric under the interchange $1 \leftrightarrow 2$. The wavefunction can be written as in Eq. (15.1) with $|\phi_i\rangle$ chosen in the left box and $|\phi_j\rangle$ chosen from the right box. Now, consider an expectation value $\langle\psi|O(\hat{x}_1, \hat{x}_2)|\psi\rangle$ of an operator that depends on the position operators of each particle, but not on their spin degrees of freedom. Now $\hat{P}_{12}^B$ commutes with O, because it has no spin operator dependence. But, if $O(\hat{x}_1, \hat{x}_2) = O(\hat{x}_2, \hat{x}_1)$, then O commutes with $\hat{P}_{12}^M$ as well. This is a bit more subtle, because naively, we do not think of the parity operator as having any effect on this operator. Now, if we write down the matrix element

$$\langle x_1| \otimes \langle x_2|O(\hat{x}_1, \hat{x}_2)|\psi\rangle = O(x_1, x_2)\psi(x_1, x_2), \tag{15.7}$$

we see that the permutation of $|\phi_i\rangle \otimes |\phi_j\rangle \leftrightarrow |\phi_j\rangle \otimes |\phi_i\rangle$ is identical to the permutation of $x_1 \leftrightarrow x_2$ for the matrix element. Next, we look at the expectation value of O in the product states $|\psi_i\rangle \otimes |\sigma_1\rangle \otimes |\phi_j\rangle \otimes |\sigma_2\rangle$. We can write the diagonal elements schematically as

$$\begin{aligned}&\sum_{x_1}\sum_{x_2}\langle\phi_1|x_1\rangle\langle\sigma_1|\sigma_1\rangle O(x_1, x_2)\langle\phi_2|x_2\rangle\langle\sigma_2|\sigma_2\rangle\langle x_1|\phi_i\rangle\langle x_2|\phi_j\rangle\Delta x_1\Delta x_2\\ &\quad= \sum_{x_1}\sum_{x_2}|\phi_i(x_1)|^2|\phi_j(x_2)|^2 O(x_1, x_2)\Delta x_1\Delta x_2,\end{aligned} \tag{15.8}$$

and the off-diagonal elements become

$$\sum_{x_1}\sum_{x_2}\phi_i^*(x_1)\phi_{i'}(x_1)\phi_j^*(x_2)\phi_{j'}(x_2)O(x_1, x_2)\langle\sigma_1|\sigma_1'\rangle\,\langle\sigma_2|\sigma_2'\rangle\Delta x_1\Delta x_2, \tag{15.9}$$

where the primes indicate the product state on the right of the matrix element. Now, one can clearly see if i and i' are in different boxes, the product is always zero, because the x values where $\phi_i(x) \neq 0$ are precisely the x values for which $\phi_{i'}(x) = 0$, because there is no overlap due to the separate boxes. Hence, every term in the product vanishes when either i and i' are in different boxes, or j or j' are in different boxes. This is no longer true when we do have overlap. Furthermore, the diagonal and off-diagonal matrix elements are independent of the spin state (when they are nonzero).

Now suppose $|\psi\rangle$ is a ground-state wave function where the first particle is in box one and the second in box two. Then $|\psi'\rangle = \hat{P}_{12}^M|\psi\rangle$ is also a ground state, because the Hamiltonian operator, being symmetric in the operators in the different boxes, commutes with the Majorana permutation operator. For N particles, the $N!$ states, corresponding to all possible permutations, are also degenerate. Now, not

all of these states are allowed to be quantum states for fermions. Only $2^N < N!$ are allowed, and they are formed from totally antisymmetric linear combinations of these $N!$ states. But, the key point for us is that the states are degenerate. Now, because the Hamiltonian is independent of spin, the different total spin operators commute with it, so we can formulate our states in terms of definite total spin and definite total z-component of spin eigenvalues. Then, the eigenstates have definite spin and z-component of spin. But, they are also degenerate, as we argued above. So, this means that the ground state (and indeed all energy eigenstates) is 2^N-fold degenerate. Now, when we discuss exchange, it relates to states that have different energy for different spins. Since this does not occur when the states do not overlap, it is clear that we must have overlap in order to have spin states with different energy. An explicit example of this is given for the $N = 2$ case in Problem 15.1.3, where we solve the problem with overlap (and see the degeneracy lifted), but also see how it can be restored when the overlap goes away.

We have one other point we need to make. In the case with two particles, the possible spin states are the $S = 0$ singlet and the $S = 1$ triplet. In this case, the totally antisymmetric state is formed from a spatially symmetric state multiplied by an antisymmetric spin state or vice versa. Note that it is only the case when $N = 2$ where it occurs this way. For $N = 3$ and higher, we cannot separate all states in this fashion. They just are more complicated. This point is also illustrated in Problem 15.1.2.

Section 15.1 Problems:

15.1.1 Antisymmetrization of Fermionic States

Use induction to show that the number of permutations of N elements is $N!$. You may want to start by showing $P(N) = NP(N-1)$.

Count the number of spin states available for a system with N spins.

Explain why it must be that some spin states found by permuting spins are not consistent with the antisymmetry of a fermionic system once $N \geq 3$. Use only a counting argument—do not determine states explicitly.

15.1.2 Fermionic States Are Not Simply Products of Symmetric with Antisymmetric States

Let's determine the number of fully antisymmetric combinations of N spins. Start with any product state of spins and then add to it its permutation multiplied by 1 or -1 depending on the parity of the permutation. Continue until you have gone through including all permutations and have a total of $N!$ terms. When you simplify,

some terms may cancel, depending on the state you started from. You also may find all terms cancel. Continue with different starting states.

You should find that in general if you start from any state that is a component of a fully antisymmetric state, then it will produce the same fully antisymmetric state (modulo a global phase of ± 1).

Now get concrete and try the $N = 2$ states, starting from all four possible product states. How many unique fully antisymmetric ones do you have? You should find exactly one. In this case we can factorize into symmetric/antisymmetric pairs. Repeat for $N = 3$ trying all eight product states and also $N = 4$. Can you determine the general result?

Take the $N = 3$ example. It has one spin $\frac{3}{2}$ state and two spin $\frac{1}{2}$ states. How many fully antisymmetric states did you find? If it is less than four, then you cannot have the factorization into symmetric/antisymmetric states. Explain why.

Repeat the above analysis for $N = 4$. Are there fully antisymmetric states for $N = 4$? Can you generalize your result for arbitrary N?

15.1.3 Overlap and Exchange for the Simplest $N = 2$ case

In this problem, we will examine a common method in quantum chemistry called the *configuration interaction* method. In this approach, we use a finite basis of many-body states, typically formed from many-body product states, and determine the matrix for the Hamiltonian in this basis. Then we find the eigenvalues and eigenvectors in that basis. This forms an approximation to the ground-state energy and eigenstate of the full system. In a typical calculation, one works with as large a matrix as the computer can handle to get the most accurate solution. We will explore this method more fully in Chap. 19. Here, however, we will work with a basis of just two states.

We consider a problem where we have two "boxes" separated by a distance R in one dimension. We put the origin in the center, and then the Hamiltonian becomes

$$\hat{H} = \underbrace{\frac{\hat{p}_1^2}{2m} + V\left(\hat{x}_1 + \tfrac{R}{2}\right)}_{\text{particle 1}} + \underbrace{\frac{\hat{p}_2^2}{2m} + V\left(\hat{x}_2 - \tfrac{R}{2}\right)}_{\text{particle 2}} + V\left(\hat{x}_1 - \tfrac{R}{2}\right) + V\left(\hat{x}_2 + \tfrac{R}{2}\right) + V_{int}(\hat{x}_1, \hat{x}_2), \tag{15.10}$$

which includes the Hamiltonian for particle 1 (centered at $x = -\frac{R}{2}$), for particle 2 (centered at $x = \frac{R}{2}$), and the interaction between them. We include potentials for particle 1 in box 2 and for particle 2 in box 1 as part of the "interaction," because it treats both particles symmetrically (which is certainly needed for indistinguishability). At this stage, we need not specify what the potentials for the "boxes" are, nor do we need to specify the interaction potential—we will discuss these terms in detail later. But we do assume that $V_{int}(\hat{x}_1, \hat{x}_2) = V_{int}(\hat{x}_2, \hat{x}_1)$ is

symmetric under an interchange of the particle labels. With these assumptions, we have that the Hamiltonian is symmetric under and interchange of the 1 and 2 labels.

We first determine the single-particle basis composed of solutions of the single-particle Hamiltonians (each underbraced term in the Hamiltonian) via $\hat{H}_L|\psi_n^L\rangle = E_n|\psi_n^L\rangle$ and $\hat{H}_R|\psi_n^R\rangle = E_n|\psi_n^R\rangle$, where we are assuming the potential has the same functional form for each "box." It makes sense to try to form the two-particle state from the ground states for each particle. The most likely choice is then $|\psi\rangle = |\psi_0^L\rangle \otimes |\psi_0^R\rangle$ for the two particles. But, we know that the requirement of antisymmetry means we must also consider the state $|\psi'\rangle = |\psi_0^R\rangle \otimes |\psi_0^L\rangle$. These two states will form the basis we will work in for this problem.

We assume the two ground states overlap, meaning $\langle\psi_0^L|\psi_0^R\rangle = L \neq 0$. If we introduce a complete set of position states, this means

$$\sum_i \langle\psi_0^L|x_i\rangle\,\langle x_i|\psi_0^R\rangle\Delta x = \sum_i \psi_0^{L*}(x_i)\psi_0^R(x_i)\Delta x \neq 0. \tag{15.11}$$

We can see that we are lifting the requirement of no overlap, which we discussed before, by allowing $\psi_0^{L*}(x)\psi_0^R(x) \neq 0$ for some x values.

We seek the ground state in this restricted basis, which we denote as $|\phi\rangle = \alpha|\psi\rangle + \beta|\psi'\rangle$, where α and β can be complex numbers. The energy eigenvalue problem can be reformulated as a minimization of the variational energy, in terms of the α and β coefficients. In other words, we seek to minimize

$$E(\alpha, \beta) = \frac{\langle\phi|\hat{H}|\phi\rangle}{\langle\phi|\phi\rangle}, \tag{15.12}$$

with respect to α and β. This can be rearranged into the form

$$\langle\phi|\hat{H}|\phi\rangle - E\langle\phi|\phi\rangle = 0. \tag{15.13}$$

Using the coefficients of the wavefunction, this can be rewritten as

$$(\alpha^*, \beta^*)\begin{pmatrix} V & U \\ U^* & V \end{pmatrix}\begin{pmatrix}\alpha \\ \beta\end{pmatrix} - \big(E(\alpha^*, \beta^*) - 2E_0\big)\begin{pmatrix} 1 & L^2 \\ L^2 & 1 \end{pmatrix}\begin{pmatrix}\alpha \\ \beta\end{pmatrix} = 0. \tag{15.14}$$

Here, we have $V = \langle\psi|V(\hat{x}_1 - \frac{R}{2}) + V(\hat{x}_2 + \frac{R}{2}) + V_{int}(\hat{x}_1, \hat{x}_2)|\psi\rangle$ and $U = \langle\psi|V(\hat{x}_1 - \frac{R}{2}) + V(\hat{x}_2 + \frac{R}{2}) + V_{int}(x_1, x_2)|\psi'\rangle$. Verify that this equation holds. It is easiest to note the symmetry of $\hat{H}$ under the interchange of the labels 1 and 2.

We can always find α and β that solve this equation. In fact, it should be clear that if we can solve

$$\begin{pmatrix} V & U \\ U^* & V \end{pmatrix}\begin{pmatrix}\alpha \\ \beta\end{pmatrix} = (E - 2E_0)\begin{pmatrix} 1 & L^2 \\ L^2 & 1 \end{pmatrix}\begin{pmatrix}\alpha \\ \beta\end{pmatrix}, \tag{15.15}$$

then we have solved the problem. This looks like an eigenvalue problem. It is called a *generalized eigenvalue problem*, and it takes this form because we are working with a basis that is not orthogonal. Note that we could orthogonalize the basis using a Gram-Schmidt process, but it is often simpler to instead solve the generalized problem.

The eigenvalues solve the equation

$$\det\begin{pmatrix} V+2E_0-E & U+2E_0L^2-EL^2 \\ U^*+2E_0L^2-EL^2 & V+2E_0-E \end{pmatrix}=0, \tag{15.16}$$

which is a quadratic equation for E. Rather than solving the quadratic equation the normal way, it is much simpler to note that the equation has the form of $x^2-y^2=0$, which means $x=\pm y$, which you can use to solve for the two solutions of E.

Solve for the eigenvalues and show that the difference between the solutions is given by

$$\Delta E = 2\frac{VL^2-U}{1-L^4}. \tag{15.17}$$

You should find that the eigenfunctions are symmetric and antisymmetric combinations of the spatial states $|\psi\rangle$ and $|\psi'\rangle$. This makes sense for a two-particle problem because the spin states are symmetric and antisymmetric combinations as well.

Note how the energy splitting goes to zero as $L\to 0$ (recall $U\to 0$ if $L\to 0$). This is precisely the result we found earlier that the exchange interaction requires overlap.

Rewrite the energy difference in terms of the corresponding exchange integral J. Note how the spin singlet state is lower in energy for one sign of J and the triplet is lower in energy for the other sign of J. For this problem, is the ground state the singlet state or the triplet state?

This procedure also illustrates how one can extract the exchange integral from an exact solution in a truncated basis. As you can see, it is not a simple procedure!

15.2 Heisenberg Model Properties

In Problem 15.1.3, we solved for the energy eigenstates of the simplest many-body problem—two particles and a basis of just two many-body states. If you have not yet solved this problem, please go back and do so. The results are critical for our work in this section. At the end of the problem, we found we could fit the Heisenberg exchange integral J to give us the energy splitting between the singlet and triplet states. In general, this procedure can be repeated for other many-body problems, but there are a few points to make about that. In general, the energy levels for a many-body problem do not all map onto the energy levels of the

Heisenberg exchange Hamiltonian. Don't be surprised by that. We did not say that all many-body problems reduce to a Heisenberg exchange model, just that it was an important component in understanding the energy levels. In addition, there can be many different exchange integrals, corresponding to different ranges of the spin-spin coupling. This has led many researchers to study Heisenberg exchange in general, rather than trying to find the best fit exchange model for a given many-body problem. But, we still expect that in certain limits, many different many-body problems will reduce to the Heisenberg model. We will see later in Chap. 18, for example, that when the electron-electron interaction is strong, the Hubbard model does reduce to a Heisenberg model at a certain concentration of electrons.

So, now that we have the freedom to consider the Heisenberg model in general, let us discuss some of the different options that we have for the model. Just like the Ising and XY models, in the Heisenberg model, we usually think of placing the spin degrees of freedom onto a lattice. This can be a one-dimensional lattice, a two-dimensional square or triangular lattice, a three-dimensional cubic, body-centered cubic, or face-centered cubic lattice, or many other options. In most cases, the interaction is between the nearest neighbors, but we can extend it to include additional interactions with second nearest neighbors or even further neighbors. Hence, a general Heisenberg model takes the following form:

$$\hat{H} = -\frac{4}{\hbar^2} \sum_{ij \in \Lambda} J_{ij} \hat{\vec{S}}_i \cdot \hat{\vec{S}}_j = -\sum_{ij \in \Lambda} J_{ij} \vec{\sigma}_i \cdot \vec{\sigma}_j, \tag{15.18}$$

where Λ is a collection of $|\Lambda|$ points. The J_{ij} matrix is real and symmetric with $J_{ii} = 0$—in some cases, it is common to sum over the pairs ij and ji once and only once. This is usually stated clearly when the Hamiltonian is written down—it just renormalizes the J_{ij}.

The case where $J_{ij} \geq 0$ is the ferromagnetic Heisenberg model. This is because the ferromagnetic state is the ground state of this system. The case where the exchange is just between nearest neighbors on a *bipartite* lattice (where the lattice is broken into two distinct sublattices and $J_{ij} \neq 0$ only for i and j elements from two different sublattices), and $J_{ij} \leq 0$, is the antiferromagnetic model, which usually has a state that is ferromagnetic on each sublattice, but with opposite orientations for the two different sublattices as the ground state; quantum fluctuations will significantly change this state. Some cases are neither ferromagnetic nor antiferromagnetic, but have exchange integrals that can change sign. We can even consider random J_{ij}. Finally, one can have anisotropic interactions (the x and y interactions different from the z) or interactions that involve three or more spins. The possibilities are nearly endless! But here we will focus just on the simpler Heisenberg model.

We need to note that when we work solely with abstract spins, we no longer have the full fermionic degrees of freedom around. Hence, we no longer require

antisymmetry of the quantum state. Instead, we usually work with states of definite total spin and definite z-component of spin. The reason why we do this becomes clear next, after we discuss the symmetries of the model.

We know that the XY model terms involve flips of connected spins. This does not change the number of up and down spins, just rearranges them on the lattice. Furthermore, the z interaction terms also commute with the z-component of spin. So, we expect the total z-component of spin to be a good quantum number in this case. It must be the total spin and not just the spin on any given site, because the XY terms can rearrange the spins locally, just not change the total number of ups and downs. To check this, we need to evaluate the commutator $[\hat{S}_z, \hat{H}]$ and see whether it vanishes. Indeed, we can compute

$$\begin{aligned}
\left[\sum_{k\in\Lambda}\hat{S}_k^z, -\frac{4}{\hbar^2}\sum_{ij\in\Lambda}J_{ij}\hat{\vec{S}}_i\cdot\hat{\vec{S}}_j\right] &= -\frac{2}{\hbar^2}\sum_{ijk\in\Lambda}J_{ij}[\hat{S}_k^z, \hat{S}_i^+\hat{S}_j^- + \hat{S}_i^-\hat{S}_j^+] \\
&= -\frac{2}{\hbar^2}\sum_{ijk\in\Lambda}J_{ij}\left(\hat{S}_i^+[\hat{S}_k^z, \hat{S}_j^-] + [\hat{S}_k^z, \hat{S}_i^+]\hat{S}_j^- + \hat{S}_i^-[\hat{S}_k^z, \hat{S}_j^+] + [\hat{S}_k^z, \hat{S}_i^-]\hat{S}_j^+\right) \\
&= -\frac{2i}{\hbar}\sum_{ijk\in\Lambda}J_{ij}\left(-\hat{S}_i^+\hat{S}_j^-\delta_{jk} + \hat{S}_i^+\hat{S}_j^-\delta_{ik} + \hat{S}_i^-\hat{S}_j^+\delta_{jk} - \hat{S}_i^-\hat{S}_j^+\delta_{ik}\right) \\
&= -\frac{2i}{\hbar}\sum_{ij\in\Lambda}J_{ij}\left(-\hat{S}_i^+\hat{S}_j^- + \hat{S}_i^+\hat{S}_j^- + \hat{S}_i^-\hat{S}_j^+ - \hat{S}_i^-\hat{S}_j^+\right) = 0. \qquad (15.19)
\end{aligned}$$

So, the total z-component of spin does commute with the Heisenberg model. Now, because the coupling is of the form $\hat{S}_i^x\hat{S}_j^x + \hat{S}_i^y\hat{S}_j^y + \hat{S}_i^z\hat{S}_j^z$, which is symmetric under any permutation of xyz, one can see that we must also have the total x and y components of spin also commute with $\hat{H}$. If all components commute, then the total spin $\sum_{ij\in\Lambda}\hat{\vec{S}}_i\cdot\hat{\vec{S}}_j$ commutes with $\hat{H}$ as well. This means we can organize our states in terms of quantum numbers with total spin s and total z component of spin m.

Section 15.2 Problems:

15.2.1 Verify the Symmetries

Explicitly verify that the total x and total y components of spin commute with $\hat{H}$. Explain why this means the total squared spin operator also commutes with $\hat{H}$.

15.2.2 Toy Model on a Bipartite Lattice with Equally Sized Sublattices

Consider a Heisenberg model on a bipartite lattice with all interactions the same magnitude and the coupling between every pair of sites on the different sublattices. If we let the number of lattice sites be $|\Lambda|$, then the number on the A sublattice $|A|$ and on the B sublattice $|B|$ are equal and satisfy $|A| = |B| = \frac{|\Lambda|}{2}$. The Hamiltonian is

$$\hat{H} = -\frac{4J}{\hbar^2}\sum_{i\in A}\sum_{j\in B}\hat{\vec{S}}_i \cdot \hat{\vec{S}}_j - \frac{4J}{\hbar^2}\sum_{i\in B}\sum_{j\in A}\hat{\vec{S}}_i \cdot \hat{\vec{S}}_j. \tag{15.20}$$

Use our completing the square trick to rewrite this as

$$\hat{H} = -\frac{4J}{\hbar^2}\left\{\left(\sum_{i\in\Lambda}\hat{\vec{S}}_i\right)\cdot\left(\sum_{j\in\Lambda}\hat{\vec{S}}_j\right) - \left(\sum_{i\in A}\hat{\vec{S}}_i\right)\cdot\left(\sum_{j\in A}\hat{\vec{S}}_j\right) - \left(\sum_{i\in B}\hat{\vec{S}}_i\right)\cdot\left(\sum_{j\in B}\hat{\vec{S}}_j\right)\right\}. \tag{15.21}$$

Hence, the eigenvalues of $\hat{H}$ are

$$E(s, s_A, s_B) = -4J\big(s_{tot}(s_{tot}+1) - s_A(s_A+1) - s_B(s_B+1)\big), \tag{15.22}$$

where s_{tot} is the total spin and s_A and s_B are the total spin on the A and B sublattices, respectively. Use the results for how we combine angular momentum to determine the allowed ranges for s_{tot}, s_A, and s_B and then determine the ground-state energy and the ground-state spin for $J > 0$ and $J < 0$.

Ans.: For $J < 0$ you should find $s_{tot} = 0$, $s_A = \frac{|\Lambda|}{4} = s_B$, and $E = -\frac{J}{2}|\Lambda|(|\Lambda|+4)$.

15.2.3 Toy Model on a Bipartite Lattice with Different Sized Sublattices

Repeat the above problem with the two sublattices having different sizes ($|A| \neq |B|$ and $|A| + |B| = |\Lambda|$).

Ans: For $J < 0$, you should find $s_{tot} = \frac{1}{2}\big||A| - |B|\big|$, $s_A = \frac{|A|}{2}$, and $s_B = \frac{|B|}{2}$. Verify this is consistent with the previous problem when the two sublattices are the same size.

15.3 Ferromagnetic Versus Antiferromagnetic Coupling

In this section, we will show how to solve for the energy eigenvalues and eigenstates of the Heisenberg model for small systems—those with 2, 3, and 4 lattice sites. This approach is called *full configuration interaction* in the chemistry community and *exact diagonalization* in the physics community. Because the Hilbert space grows as $2^{|\Lambda|}$, it rapidly becomes impossible to solve this problem on a computer because one cannot even store the eigenstate in the computers memory once $|\Lambda|$ is too large (typically in the 40–50 range after using many tricks to be able to efficiently solve the problem). We won't work on those large problems here, just the smallest nontrivial ones.

We start with the two-site system ($|\Lambda| = 2$). The Hamiltonian is

$$\hat{H}_2 = -\frac{J}{2}\left(\sigma_1^+ \otimes \sigma_2^- + \sigma_1^- \otimes \sigma_2^+\right) - J\sigma_1^z \otimes \sigma_2^z; \tag{15.23}$$

note how we are being explicit here to avoid any double counting of the pairs, which is commonly, but not always done. The four states in our Hilbert space are spanned by the states $|\uparrow\rangle_z \otimes |\uparrow\rangle_z$, $|\uparrow\rangle_z \otimes |\downarrow\rangle_z$, $|\downarrow\rangle_z \otimes |\uparrow\rangle_z$, and $|\downarrow\rangle_z \otimes |\downarrow\rangle_z$. To find the matrix, we take each of the four states, act $\hat{H}$ on them, and expand the final vector in the original basis. So

$$\begin{aligned}
\hat{H}_2|\uparrow\rangle_z \otimes |\uparrow\rangle_z &= -J|\uparrow\rangle_z \otimes |\uparrow\rangle_z \\
\hat{H}_2|\uparrow\rangle_z \otimes |\downarrow\rangle_z &= -2J|\downarrow\rangle_z \otimes |\uparrow\rangle_z + J|\uparrow\rangle_z \otimes |\downarrow\rangle_z \\
\hat{H}_2|\downarrow\rangle_z \otimes |\uparrow\rangle_z &= -2J|\uparrow\rangle_z \otimes |\downarrow\rangle_z + J|\downarrow\rangle_z \otimes |\uparrow\rangle_z \\
\hat{H}_2|\downarrow\rangle_z \otimes |\downarrow\rangle_z &= -J|\downarrow\rangle_z \otimes |\downarrow\rangle_z,
\end{aligned} \tag{15.24}$$

where we recall the nonzero off-diagonal element of the Pauli raising or lowering matrices is equal to 2. So, we can write down the Hamiltonian matrix as

$$H_2 = \begin{pmatrix} -J & 0 & 0 & 0 \\ 0 & J & -2J & 0 \\ 0 & -2J & J & 0 \\ 0 & 0 & 0 & -J \end{pmatrix}. \tag{15.25}$$

Two eigenvalues are $-J$. We need to solve the quadratic equation $E^2 - 2JE - 3J^2 = 0$ to find the other two. They are $E = -J$ and $3J$. So, we find that three eigenvalues are $-J$ and one is $3J$. We expect this degeneracy pattern, because the spin triplet state should have three degenerate states, and the spin singlet state should be nondegenerate. So, everything seems fine. If $J > 0$, then the ferromagnetic state ($s = 1$) is the ground state, while if $J < 0$, the antiferromagnetic state ($s = 0$) is the ground state.

Let's think a bit more analytically about what we have just done. If we look at the matrix, we find the states with $m = \pm 1$ had no off-diagonal terms, so they are eigenstates and the eigenvalues can be read off from the diagonal elements. The states with $m = 0$ include a singlet and a triplet state, but when we work with the product state basis, they are mixed together and we have a 2×2 eigenvalue problem to solve. Can we do better? Yes, if we take instead the following basis states $\frac{1}{\sqrt{2}}(|\uparrow\rangle_z \otimes |\downarrow\rangle_z \pm |\downarrow\rangle_z \otimes |\uparrow\rangle_z)$ for the $m = 0$ sector, we find the Hamiltonian matrix becomes

$$H_2' = \begin{pmatrix} -J & 0 & 0 & 0 \\ 0 & -J & 0 & 0 \\ 0 & 0 & 3J & 0 \\ 0 & 0 & 0 & -J \end{pmatrix}, \tag{15.26}$$

with the prime indicating that this matrix uses a different set of basis states. Be sure you can do this calculation yourself. You simply take the results we found in the product states for the action of $\hat{H}$ add or subtract them appropriately and re-express the final result in the new basis.

So, we have learned that by thinking of conserved quantum numbers and writing down states with conserved quantum numbers, we "block diagonalize" the Hamiltonian matrix, making it easier to solve the eigenvalue problem. Furthermore, because the higher spin multiplets are all degenerate in energy, we can perform our calculations in the set of states with minimal magnitude of the z-component of spin ($m = 0$ for an even number of spins and $m = \frac{1}{2}$ for an odd number of spins).

We now illustrate this for the $|\Lambda| = 3$ case. If you recall how we combine three spins together, we find one $s = \frac{3}{2}$ multiplet with four states and two $s = \frac{1}{2}$ multiplets with two states each, for a total of $2^3 = 8$ states. Working in the basis with $m = \frac{1}{2}$ and recalling that the state with all up spins has $m = \frac{3}{2}$ allows us to find the $s = \frac{3}{2}$ state with $m = \frac{1}{2}$ simply by applying the total spin lowering operator and normalizing the state. So, we have our first basis state is

$$|1\rangle = \frac{1}{\sqrt{3}}\left(|\uparrow\uparrow\downarrow\rangle_z + |\uparrow\downarrow\uparrow\rangle_z + |\downarrow\uparrow\uparrow\rangle_z\right), \tag{15.27}$$

where we are now using our condensed notation omitting the tensor products to make the formulas less cumbersome. The other two basis states must be perpendicular to each other and to this state. This requires us to use the so-called SU(N) unit vectors, which are discussed in Problem 15.3.1. This then gives us the other two basis states

$$\begin{aligned} |2\rangle &= \frac{1}{\sqrt{2}}\left(|\uparrow\uparrow\downarrow\rangle_z - |\downarrow\uparrow\uparrow\rangle_z\right) \\ |3\rangle &= \frac{1}{\sqrt{6}}\left(|\uparrow\uparrow\downarrow\rangle_z - 2|\uparrow\downarrow\uparrow\rangle_z + |\downarrow\uparrow\uparrow\rangle_z\right). \end{aligned} \tag{15.28}$$

Now, we compute the Hamiltonian matrix. The Hamiltonian is

$$\hat{H}_3 = -\frac{J}{2}\left(\sigma_1^+\sigma_2^- + \sigma_1^-\sigma_2^+ + \sigma_2^+\sigma_3^- + \sigma_2^-\sigma_3^+ + \sigma_3^+\sigma_1^- + \sigma_3^-\sigma_1^+\right) - J\left(\sigma_1^z\sigma_2^z + \sigma_2^z\sigma_3^z + \sigma_3^z\sigma_1^z\right). \tag{15.29}$$

where we also dropped the tensor-product symbols. Note that the interaction is for spins on an equilateral triangle, interacting with their nearest neighbors. We know the first state should be an eigenvector and the second and third states should form a 2×2 matrix. Work carefully to verify that

$$\hat{H}_3|1\rangle = -3J|1\rangle. \tag{15.30}$$

In this calculation, you should find the raising and lowering operator terms give a $-4J$ contribution, while the z-component terms give a $+J$ contribution. It is an eigenvector already, as we expected. Now, we need to repeat for the calculation of the other two states. This requires some careful work. After a bit of algebra, you should find that

$$\hat{H}_3|2\rangle = 3J|2\rangle \quad \text{and} \quad \hat{H}_3|3\rangle = 3J|3\rangle. \tag{15.31}$$

These states are eigenstates as well. The reason why is that the Hamiltonian also has spatial symmetries that we have not discussed, but it turns out the states we chose have different spatial symmetries, so they cannot mix. Hence, they are eigenstates already. So, we have diagonalized the $|\Lambda| = 3$ Hamiltonian as well. It is amazing that by using some simple ideas, we reduce from working with an 8×8 matrix down to an immediate diagonalization of the problem. This is a *huge* simplification! Again, when $J > 0$, the $s = \frac{3}{2}$ state is the ground state, while for $J < 0$, both $s = \frac{1}{2}$ states are the ground state and we have degeneracy. The degenerate ground state arises from frustration of the antiferromagnetic state. We discussed this earlier in Chap. 10, where we showed how one cannot obtain an antiferromagnetic arrangement with three spins on a triangle. Here, in a fully quantum model, this is evidenced by having degeneracy in the ground state. In fact, we often find frustrated spin systems either have degenerate or nearly degenerate ground states, and it is a challenging problem to try to determine precisely what the energy eigenvalues are for the ground state and the low-lying excited states.

We will solve one more problem here, the $|\Lambda| = 4$ case. Here, we have $2^4 = 16$ states, and we form one $s = 2$ manifold with five states, three different $s = 1$ triplets, accounting for nine states, and two $s = 0$ spin singlets. The total number of states is 16, as it must be. But, by working in the $m = 0$ subspace, and using states with definite spin, we can reduce the 16×16 problem down to a 1×1 problem, a 3×3 problem, and a 2×2 problem. We also know that some of these smaller matrix blocks may be even smaller due to spatial symmetries, as we saw in the $|\Lambda| = 3$ example. The Hamiltonian is

$$\hat{H}_4 = -\frac{J}{2}\Big(\sigma_1^+\sigma_2^- + \sigma_1^-\sigma_2^+ + \sigma_2^+\sigma_3^- + \sigma_2^-\sigma_3^+ + \sigma_3^+\sigma_4^- + \sigma_3^-\sigma_4^+ + \sigma_4^+\sigma_1^- + \sigma_4^-\sigma_1^+\Big) - J\Big(\sigma_1^z\sigma_2^z + \sigma_2^z\sigma_3^z + \sigma_3^z\sigma_4^z + \sigma_4^z\sigma_1^z\Big). \tag{15.32}$$

The $s = 2$ multiplet starts with the all up state, and then we hit it with the lowering operator twice, and normalize. This gives

$$|1\rangle = \frac{1}{\sqrt{6}}\Big(|\uparrow\uparrow\downarrow\downarrow\rangle_z + |\uparrow\downarrow\uparrow\downarrow\rangle_z + |\downarrow\downarrow\uparrow\uparrow\rangle_z + |\downarrow\uparrow\uparrow\downarrow\rangle_z + |\uparrow\downarrow\uparrow\downarrow\rangle_z + |\downarrow\uparrow\downarrow\uparrow\rangle_z\Big). \tag{15.33}$$

To find the $s = 1$ states, we work in the $m = 1$ sector finding three states perpendicular to the $s = 2$ state. These states are

$$\begin{aligned}
|2'\rangle &= \frac{1}{2}\Big(|\uparrow\uparrow\uparrow\downarrow\rangle_z + |\uparrow\uparrow\downarrow\uparrow\rangle_z - |\uparrow\downarrow\uparrow\uparrow\rangle_z - |\downarrow\uparrow\uparrow\uparrow\rangle_z\Big),\\
|3'\rangle &= \frac{1}{2}\Big(|\uparrow\uparrow\uparrow\downarrow\rangle_z - |\uparrow\uparrow\downarrow\uparrow\rangle_z + |\uparrow\downarrow\uparrow\uparrow\rangle_z - |\downarrow\uparrow\uparrow\uparrow\rangle_z\Big),\\
|4'\rangle &= \frac{1}{2}\Big(|\uparrow\uparrow\uparrow\downarrow\rangle_z - |\uparrow\uparrow\downarrow\uparrow\rangle_z - |\uparrow\downarrow\uparrow\uparrow\rangle_z + |\downarrow\uparrow\uparrow\uparrow\rangle_z\Big).
\end{aligned} \tag{15.34}$$

You can easily see that these three states are orthogonal to each other and to the $s = 2$ state with $m = 1$ (this strategy of using plus and minus signs works in even dimensions for finding orthogonal vectors and is well worth remembering). Now, we hit each of these with the lowering operator to find the $s = 1$ states with $m = 0$. A quick calculation then gives

$$\begin{aligned}
|2\rangle &= \frac{1}{\sqrt{2}}\Big(|\uparrow\uparrow\downarrow\downarrow\rangle_z - |\downarrow\downarrow\uparrow\uparrow\rangle_z\Big),\\
|3\rangle &= \frac{1}{\sqrt{2}}\Big(|\uparrow\downarrow\uparrow\downarrow\rangle_z - |\downarrow\uparrow\downarrow\uparrow\rangle_z\Big),\\
|4\rangle &= \frac{1}{\sqrt{2}}\Big(|\uparrow\downarrow\downarrow\uparrow\rangle_z - |\downarrow\uparrow\uparrow\downarrow\rangle_z\Big).
\end{aligned} \tag{15.35}$$

Finally, the two $s = 0$ states must be perpendicular to these four states and each other. They are

$$\begin{aligned}
|5\rangle &= \frac{1}{2}\Big(|\uparrow\uparrow\downarrow\downarrow\rangle_z - |\uparrow\downarrow\downarrow\uparrow\rangle_z + |\downarrow\downarrow\uparrow\uparrow\rangle_z - |\downarrow\uparrow\uparrow\downarrow\rangle_z\Big)\\
|6\rangle &= \frac{1}{2\sqrt{3}}\Big(|\uparrow\uparrow\downarrow\downarrow\rangle_z + |\uparrow\downarrow\downarrow\uparrow\rangle_z + |\downarrow\downarrow\uparrow\uparrow\rangle_z + |\downarrow\uparrow\uparrow\downarrow\rangle_z - 2|\uparrow\downarrow\uparrow\downarrow\rangle_z
\end{aligned}$$

$$-2|\downarrow\uparrow\downarrow\uparrow\rangle_z\Big). \tag{15.36}$$

One can easily verify that these vectors are normalized and orthogonal to each other. Now we need to go through and compute the Hamiltonian matrix. This is again a bit of algebra in our French cooking style. It is still reasonable to do it, but it is getting a bit tedious, and it is clear we do not want to go further than this case by hand. It does seem like the basis states we have chosen also have additional spatial symmetries, so we might expect some additional matrix elements to be 0 due to this symmetry, which we have not explicitly taken into account. We find the $s = 2$ state satisfies

$$\hat{H}|1\rangle = -4J|1\rangle; \tag{15.37}$$

The terms with two up and two down arrows adjacent have a $-8J$ contribution multiplying the two Néel states from the spin-flip terms, and no contribution from the z terms, while the Néel terms produce $-4J$ times each of the two up and two down terms, and a $+4J$ times the Néel terms from the z terms. Recall that the Néel states are $|\uparrow\downarrow\uparrow\downarrow\rangle$ and $|\downarrow\uparrow\downarrow\uparrow\rangle$. So the net effect is multiplying the state by $-4J$.

The $s = 1$ states give

$$\begin{aligned}\hat{H}|2\rangle &= 0\\ \hat{H}|3\rangle &= 4J|3\rangle\\ \hat{H}|4\rangle &= 0,\end{aligned} \tag{15.38}$$

showing they are all eigenstates as well. Finally, the $s = 0$ states satisfy

$$\begin{aligned}\hat{H}|5\rangle &= 0\\ \hat{H}|6\rangle &= 8J|6\rangle,\end{aligned} \tag{15.39}$$

which are both eigenstates as well. One can see for $J > 0$, the ground state is ferromagnetic, with $s = 2$, while for $J < 0$, the ground state is antiferromagnetic, with $s = 0$. But, note that it does not include just the Néel states, but has contributions from all other states as well.

This section examined the eigenstates of the Heisenberg model for the smallest nontrivial cases of $|\Lambda| = 2$, 3 and four spins on a one-dimensional circle, with nearest-neighbor interactions. We found that when $J > 0$, the ferromagnetic state (including all of its multiplets) is the ground state, while for $J < 0$, we found that the state always has the minimal allowed spin. We do find energy levels that are degenerate in some cases and described how this could be due to frustration, but can have other causes as well. It turns out there is a hidden positivity relation to the antiferromagnetic states. This is called the *Marshall sign rule*, and we describe it in detail in the next section as well as having Problem 15.3.2 devoted to it below.

Section 15.3 Problems:

15.3.1 SU(N) Unit vectors

Show that the following set of $N+1$ vectors are orthonormal:

$$\begin{aligned}
&\frac{1}{\sqrt{N}}\underbrace{(1,1,1,1,\cdots,1,1)}_{N \text{ entries}}\\
&\frac{1}{\sqrt{2}}(1,-1,0,0,\cdots,0,0)\\
&\frac{1}{\sqrt{6}}(1,1,-2,0,\cdots,0,0)\\
&\frac{1}{\sqrt{12}}(1,1,1,-3,\cdots,0,0)\\
&\qquad\qquad\vdots\\
&\frac{1}{\sqrt{(N-2)(N-1)}}(1,1,1,1,\cdots,-N+2,0)\\
&\frac{1}{\sqrt{(N-1)N}}(1,1,1,1,\cdots 1,-N+1).
\end{aligned} \tag{15.40}$$

Explain in words how we construct these basis vectors. Is it possible to change the filling pattern for which entries are nonzero? Explain how, or explain why it cannot be done.

15.3.2 Marshall Sign Rule

In this problem, we will work with the Marshall sign rule, which is covered in much more detail in the next section. To work with the Marshall sign rule, we must divide our lattice into two sublattices, with the exchange interactions only between sites on different sublattices. We assume the two sublattices are the same size here. A quick check of the three Hamiltonians we just looked at tells us that the $|\Lambda| = 2$ and $|\Lambda| = 4$ cases are on bipartite lattices. Let us focus on the four-site case. Take the B sublattice to be sites 2 and 4. We assign a minus sign to each down spin in the product state that is on sublattice B. Then, the Marshall sign rule tells us that the ground state can be written as

$$|\psi_{gs}\rangle = \sum_{\sigma_1,\sigma_2,\sigma_3,\sigma_4|\sum_i \sigma_i=0} (-1)^{N_\downarrow(B)} c_{\sigma_1,\sigma_2,\sigma_3,\sigma_4} |\sigma_1\rangle_z \otimes |\sigma_2\rangle_z \otimes |\sigma_3\rangle_z \otimes |\sigma_4\rangle_z, \tag{15.41}$$

where all $c_{\vec{\sigma}} > 0$ or all $c_{\vec{\sigma}} < 0$. Here, $\vec{\sigma} = \sigma_1, \sigma_2, \sigma_3, \sigma_4$ and $N_\downarrow(B)$ counts the number of lattice sites with the state $|\downarrow\rangle_z$ on the B sublattice in the given product state. Note that this means that *all* coefficients are nonzero.

Does the ground state we found for the four-site Hamiltonian when $J < 0$ satisfy the Marshall sign rule?

Verify whether the ground state for the two-site model satisfies the Marshall sign rule for $J < 0$. You will need to carefully formulate the sign rule for this state as well.

15.3.3 Frustration with $|\Lambda| = 3$

Add the additional exchange term

$$-\frac{J'}{2}\left(\sigma_1^+\sigma_3^- + \sigma_1^-\sigma_3^+\right) - J'\sigma_1^z\sigma_3^z \tag{15.42}$$

to $\hat{H}_3$ and find the energy eigenvalues for this Hamiltonian. Note that you already know the action of the other terms in the Hamiltonian on the basis states. You need only *add* the action of this extra term.

Fix $J = 1$ and vary J' over positive and negative values. Does the spin of the ground state change with J'? Does J' lift the ground-state degeneracy?

15.3.4 Level Crossings with $|\Lambda| = 4$

Add the second-neighbor exchange term

$$-\frac{J'}{2}\left(\sigma_1^+\sigma_3^- + \sigma_1^-\sigma_3^+ + \sigma_2^+\sigma_4^- + \sigma_2^-\sigma_4^+\right) - J'\left(\sigma_1^z\sigma_3^z + \sigma_2^z\sigma_4^z\right) \tag{15.43}$$

to $\hat{H}_4$ and find the energy eigenvalues for this Hamiltonian. Note that you already know the action of the other terms in the Hamiltonian on the basis states. You need only add the action of this extra term.

Set $J = 1$ and vary J' over all possible positive and negative values. Does the spin of the ground state change?

15.3.5 Algorithm to Solve the Heisenberg Model for Larger $|\Lambda|$ with a Computer

In working through this section, we had to perform a lot of tedious calculations in order to determine the action of the exchange Hamiltonian on different states. Since each lattice site has just two states, we can map $\uparrow \rightarrow 0$ and $\downarrow \rightarrow 1$ and then represent a quantum product state as a binary integer, with 0's and 1's for each lattice site to represent the up and down spins. Explain how one can act terms of the Hamiltonian onto a product state in this binary form. Note that the bitwise NOT operation takes $0 \rightarrow 1$ and $1 \rightarrow 0$ and bitwise binary addition modulo 2 satisfies $0 + 0 = 0$, $1 + 1 = 0$, $0 + 1 = 1$, and $1 + 0 = 1$. Describe how one can create product states in this binary format, act the Hamiltonian elements onto it in this binary format, and eventually determine the Hamiltonian matrix. Then if the matrix is small enough in size, it can be diagonalized on the computer to find all energy eigenvalues and energy eigenstates. Discuss how one can work on the $m = 0$ sector for even sized lattices and how one can work with states of definite s for the basis states.

These types of techniques are used in exact diagonalization studies of the Heisenberg model, along with more efficient methods that take into account the sparsity of these matrices to allow even larger systems to be diagonalized. We won't discuss those approaches here.

15.4 Marshall Sign Rule

The Marshall sign rule was discovered in 1955 for the one-dimensional lattice (Marshall, 1955) and was extended to a more general case by Lieb and Mattis (1962a). Our discussion here is a simplified version of the discussion presented in Tasaki's book (2020) for spin one-half. The Marshall sign rule is important because it establishes the spin of the ground state and its nondegeneracy and uniqueness. It does not tell us about the long-range order. We will not discuss the ordering problem in detail, but we will summarize some of the results at the end of the section.

We begin by discussing the ferromagnetic case, where $J > 0$. We consider the Heisenberg model on a general graph with $|\Lambda|$ lattice sites and N_B bonds (which are pairs of lattice sites where $J_{ij} \neq 0$). We also assume the Heisenberg exchange is the same for each bond, so $J_{ij} = J$ for all nonzero bonds (it will turn out this restriction is not needed, just that they are all the same sign, but it is easier to think in terms of a constant J). Then, if we let $|\tilde{\boldsymbol{\sigma}}\rangle_z = |\sigma_1\rangle_z \otimes |\sigma_2\rangle_z \otimes \cdots \otimes |\sigma_{|\Lambda|}\rangle_z$ denote the $2^{|\Lambda|}$ many-body basis states for the spin Hilbert space, we can examine the energy expectation value for the fully polarized state $|\psi_{fm}\rangle = |\uparrow\uparrow \cdots \uparrow\rangle_z$ to find that

$$\langle \psi_{fm}|\hat{H}|\psi_{fm}\rangle = \langle \uparrow\uparrow \cdots \uparrow |\hat{H}| \uparrow\uparrow \cdots \uparrow\rangle = -JN_B, \tag{15.44}$$

because $\sigma_i^+|\psi_{fm}\rangle = 0$. This is the ground state. It is not immediately obvious that one could not find a lower energy state. The proof that this is the ground state of the ferromagnetic Heisenberg exchange model is given in Problem 15.4.1 below.

Now, we move on to the main event of investigating the Marshall sign rule for the antiferromagnetic case. We again have a graph with $|\Lambda|$ lattice sites and N_B bonds. We assume the lattice is bipartite, which means it can be broken into two sublattices A and B, such that $J_{ij} \neq 0$ only when $i \in A$ and $j \in B$ or $i \in B$ and $j \in A$. In other words, a bond, corresponding to two sites where $J_{ij} \neq 0$, occurs only between lattice sites on *different* sublattices. Note that it does not say that when i and j are from different sublattices, then $J_{ij} \neq 0$, but instead it says whenever $J_{ij} \neq 0$, we must have i and j on different sublattices. We assume further that the two sublattices are the same size, so $|A| = |B| = |\Lambda|/2$. Hence, $|\Lambda|$ is a positive and even integer. Because we have an even number of lattice sites, the minimal total spin we can have is $s = 0$, so we can work with a subblock of the Hamiltonian matrix restricted to the states with $m = 0$, because all states with integer spin have a representative in that subspace. The many-body product states that we work with are then all of the form

$$|\boldsymbol{\sigma}\rangle = (-1)^{N_\downarrow(B)}|\sigma_1\rangle_z \otimes |\sigma_2\rangle_z \otimes \cdots \otimes |\sigma_{|\Lambda|}\rangle_z, \tag{15.45}$$

where we have the constraint that $\sum_{i=1}^{|\Lambda|} \sigma_i = 0$, and the symbol $N_\downarrow(B)$ denotes the total number of down spins on the B sublattice. We use a bold symbol $\boldsymbol{\sigma}$ to denote the $|\Lambda|$-dimensional vector $(\sigma_1, \sigma_2, \cdots, \sigma_{|\Lambda|})$ that specifies the product state used for the many-body basis states. The specific sign that we chose for the basis states will become clear next.

Our strategy for establishing the Marshall sign rule for an antiferromagnet is to use the Perron-Frobenius theorem, which we established in Sect. 13.3. Recall that for this theorem we just require that the matrix is real-valued and all off-diagonal matrix elements are negative. Since J is a real number, it should be clear that $\langle\boldsymbol{\sigma}|\hat{H}|\boldsymbol{\sigma'}\rangle$ is also a real number. Furthermore, the off-diagonal matrix elements arise only from the terms in the Hamiltonian of the form $\frac{|J|}{2}(\sigma_i^+\sigma_j^- + \sigma_i^-\sigma_j^+)$ (where we use $|J|$ because the antiferromagnetic case corresponds to $J < 0$). These matrix elements are nonzero only if $|\boldsymbol{\sigma}\rangle$ has the same spin projections on each lattice site as $|\boldsymbol{\sigma'}\rangle$ except for the sites i and j, where the spins are opposite for those two lattice sites. Then, because the operator either adds or removes one down spin from the state (on the B sublattice), we get a minus sign from the $(-1)^{N_\downarrow(B)}$ term, and hence the matrix element is $-|J|$ for these states.

This means that the $m = 0$ block of the Hamiltonian matrix has all of its off-diagonal matrix elements equal to zero or negative—that is, they are nonpositive. This is precisely the requirement of the Perron-Frobenius theorem, which immediately tells us the ground state, given by

$$|\psi_{GS}\rangle = \sum_{\boldsymbol{\sigma}|\sum_i \sigma_i = 0} c_{\boldsymbol{\sigma}}|\boldsymbol{\sigma}\rangle_z \tag{15.46}$$

has all coefficients satisfy $c_{\boldsymbol{\sigma}} \geq 0$. Furthermore, if the graph is connected by the bonds generated by the J_{ij} matrix, then the ground state is also unique and furthermore, all $c_{\boldsymbol{\sigma}} > 0$, implying every many-body basis state with $m = 0$ has some amplitude in the ground state. You can see this is exactly what we had with our $|\Lambda| = 2$ and $|\Lambda| = 4$ solutions that we found in the previous section. The proof that the Hamiltonian matrix is connected in the many-body product space if the Heisenberg exchange matrix J_{ij} is connected on the graph is given in Problem 15.4.2 below.

Now, we have to determine the total spin of this ground state. To do this, we use an argument from the Lieb and Mattis work (Lieb & Mattis, 1962a), where they compare the ground state of the antiferromagnetic Heisenberg exchange model on the original graph with the ground state of the antiferromagnetic Heisenberg exchange model on a toy graph, whose spin is easy to determine. This question was given in Problem 15.2.2. We examine the solution in detail now. The Hamiltonian is chosen so that $J_{ij} = \frac{J}{2}$ for $i \in A$ and $j \in B$ or $i \in B$ and $j \in A$. In other words, the coupling is from every site in the A sublattice to every site in the B sublattice. Using the result for competing the square, one can show that the energy is given by the eigenvalues of $\frac{|J|}{8\hbar^2}\big((\hat{\vec{S}}_A + \hat{\vec{S}}_B)^2 - (\hat{\vec{S}}_A)^2 - (\hat{\vec{S}}_B)^2\big)$. This eigenvalue is minimized when we maximize the spin on the A and B sublattices and minimize the total spin. When $|A| = |B|$, the minimal spin is $s = 0$ and the spin on each sublattice is $\frac{|\Lambda|}{4}$. The key point is that the toy model also satisfies the assumptions of our theorem—the lattice is bipartite and connected, the interactions are all nonpositive, and $m = 0$. Hence, all of the coefficients in the expansion for the wavefunction, using the same basis as we did for the original problem, can be chosen to be positive or $c_{\boldsymbol{\sigma}}^{toy} > 0$. In the problems, we prove that two eigenstates that are eigenstates of an operator with discrete eigenvalues must have the same eigenvalue if they have a nonzero overlap. Since the overlap of the two ground states satisfies

$$\langle \psi_{GS} | \psi_{GS}^{toy} \rangle = \sum_{\boldsymbol{\sigma}} c_{\boldsymbol{\sigma}} c_{\boldsymbol{\sigma}'} > 0, \tag{15.47}$$

this implies that the ground state of the original Hamiltonian is also a spin singlet state, which completes our proof.

Up to now, we assumed that the size of the two sublattices was the same, that is, $|A| = |B|$. But, this requirement is not necessary. If $|A| \neq |B|$ everything we discussed above still holds, except for the toy model, we can only minimize the total spin with a total spin eigenvalue given by $s = \frac{1}{2}\big||A| - |B|\big|$, which follows from the standard theory of combining angular momenta. Again, we minimize all three terms simultaneously for the toy Hamiltonian. Everything else follows as before, so the spin eigenvalue of the original Hamiltonian is the same as that of the toy model, or $s = \frac{1}{2}\big||A| - |B|\big|$. It is quite remarkable that we find such a powerful result—we determined the total spin eigenvalue of a wide class of antiferromagnetic Hamiltonians on a huge variety of different lattices (more precisely graphs, as they

need not even be periodic). It is amazing how powerful the Marshall sign rule is and the Perron-Frobenius theorem.

It is important to discuss what this theorem does not show—it says nothing about the spatial pattern of the spins in the antiferromagnet. As we know, the ground state involves a superposition of all $m = 0$ product states, so to see long-range order, we need to look at correlation functions. This is done in one of the problems. In the next subsection, however, we will explicitly find the ground state for the Heisenberg exchange model on a one-dimensional lattice with nearest-neighbor spin-spin interactions only. You will see this is quite challenging to work out, which tells you why understanding frustrated antiferromagnets remains an active area of research. It also has no long-range order at a nonzero temperature, just as the Ising model had none in one dimension.

We will not attempt to work out theorems that prove long-range order, but we will summarize a bit of what is known. First off, the proofs about the ground state for the Heisenberg model (both ferromagnetic and antiferromagnetic) can be extended from spin-one-half on each lattice site to arbitrary spin. The techniques to do so are covered in Tasaki's book and follow closely the ideas we covered here. In two dimensions, there is a famous theorem, called the Mermin-Wagner theorem, that says that there can be no long-range order in spin-one-half Heisenberg models on two-dimensional lattices at any nonzero temperature. This is very different from the Ising model case, where the system does have long-range order at low enough temperatures. Long-range order does occur for dimensions greater or equal to three and all nonzero spin values. It also occurs in two dimensions for spins greater than or equal to one. Clearly this indicates the problem is complicated and subtle!

Section 15.4 Problems:

15.4.1 Proof That the Polarized State Is the Ground State of the Ferromagnetic Heisenberg Exchange Model

We first focus on an identity of the spin operators and then convert it to the Pauli matrices. We have seen before that

$$\hat{\vec{S}}_i \cdot \hat{\vec{S}}_j = \frac{1}{2}\Big(\Big(\hat{\vec{S}}_i + \hat{\vec{S}}_j\Big)^2 - \Big(\hat{\vec{S}}_i\Big)^2 - \Big(\hat{\vec{S}}_j\Big)^2 \Big). \tag{15.48}$$

For spin one-half, we know the sum of the two spins can give a total spin of 0 or 1, while the square of each spin has an eigenvalue of $\frac{3}{4}\hbar^2$. So, the maximum eigenvalue for $\hat{\vec{S}}_i \cdot \hat{\vec{S}}_j$ is $\frac{1}{2}\left(2 - \frac{3}{2}\right)\hbar^2 = \frac{1}{4}\hbar^2$. Converting this to the Pauli matrices, we find the maximal eigenvalue for $\vec{\sigma}_i \cdot \vec{\sigma}_j$ is 1. First, verify these derivations, then use them to argue that one cannot have an eigenvalue less than $-JN_B$ for the ferromagnetic

Heisenberg model, so the fully polarized state (which saturates the bound) is the ground state.

This state is $|\Lambda| + 1$-fold degenerate, corresponding to the states with different eigenvalues of $\hat{S}^z_{tot}$. Construct the state $\hat{S}^-_{tot}|\psi_{fm}\rangle$ and compute explicitly $\hat{H}\hat{S}^-_{tot}|\psi_{fm}\rangle$. Verify that it is an eigenstate with the eigenvalue $-JN_B$, so it is also a ground state. This can be repeated for all of the remaining states, but we do not ask you to do that.

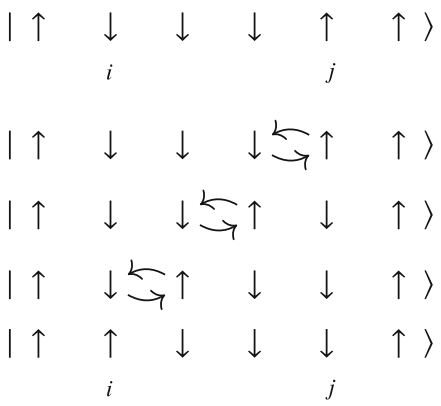
15.4.2 *Proof That the Many-Body Product Space States Are Connected if J_{ij} Is Connected on the Graph*

We sketch the idea of the proof for you and allow you to fill in the rest of the details.

Start with two states $|\boldsymbol{\sigma}\rangle$ and $|\boldsymbol{\sigma}'\rangle$, both with $m = 0$. The only difference between the two states is the ordering of the spins on each lattice site. The total number of up spins and down spins in each state is the same.

The proof starts by finding the first spin state (from the left) in $|\boldsymbol{\sigma}'\rangle$ that is different from the spin state in $|\boldsymbol{\sigma}\rangle$. Say it is at the ith lattice site and the state is $\uparrow$ for concreteness. Skipping the ith lattice site, find the next spin state of the projection opposite to that at the ith site that is different (i.e., the first $\downarrow$ site that is different). Say it is the jth site. Such a site always exists, because if all the up spin positions in $|\boldsymbol{\sigma}\rangle$ are the same as all the up spin positions in $|\boldsymbol{\sigma}'\rangle$ then all the down spin positions are the same as well (because each site has only one of two states). Now, sites i and j are connected by a chain of nonzero J values in a chain, such as $J_{in_1}J_{n_1n_2}\cdots J_{n_mj}$ of $m+1$ factors. Consider the sites $n_1, n_2, \cdots, n_m$ and see whether any states along that path is an $\uparrow$. If so, take the rightmost of those sites and mark it, and we will use the chain from that site to site j. Apply the spin-flip part of the Hamiltonian along that chain, in turn, and it will move the up state from the marked site to site j and a down state from site j to the marked state.

After completing this, we have put an up arrow in the proper location (the jth site), and we have removed one of the up states from the path. The next step is to move the next rightmost up state in the path to the marked site using a subset of the chain. Continue in this fashion until you have rearranged the states such that the ith state is now a down state and the jth state is now an up arrow. This has fixed the states on the ith and jth sites without changing the states on any other site. Note that we did not necessarily move the spin from i to j, but we achieved the transformation anyway. An example (where the first mismatched state is now a $\downarrow$ state) is summarized below, with $|\boldsymbol{\sigma}\rangle = |\uparrow\uparrow\uparrow\downarrow\downarrow\downarrow\rangle$ and $|\boldsymbol{\sigma}'\rangle = |\uparrow\downarrow\downarrow\downarrow\uparrow\uparrow\rangle$. Then, the algorithm goes as follows (with neighboring spins connected):



(15.49)

You can see, we have managed to swap the spin states at positions i and j, but we did not move the spin from i to the j location (although we did move the spin at j to the i location).

Put together this basic algorithm to determine how the two states are connected. For example, in the above concrete case, one needs to perform the algorithm one more time. Your job is to clearly write out how the general case works.

15.4.3 Quantum States with All Positive Coefficients Share the Same Quantum Numbers

In this problem, consider two states $|\psi\rangle$ and $|\phi\rangle$ that are eigenstates of a Hermitian operator $\hat{A}$, with eigenvalues a_ψ and a_ϕ, respectively. If both eigenvalues are nonzero, show that if $\langle\psi|\phi\rangle \neq 0$, then we must have $a_\psi = a_\phi$.

How can this argument be extended if the eigenvalue is zero?

If one state has an expansion solely in terms of positive coefficients and the other state has an expansion solely in terms of nonnegative coefficients (in the same orthonormal basis), show that $\langle\psi|\phi\rangle \neq 0$.

Hint: Introduce a multiply by one inside the matrix element in the form $\hat{A}/a_\psi$ or $\hat{A}/a_\phi$ and then allow $\hat{A}$ to operate on the other state.

15.4.4 Antiferromagnetism on Nonbipartite Lattices

Suppose the lattice you are considering is not bipartite. Then the Perron-Frobenius theorem does not apply. But if the degree of "nonbipartiteness" is small (say adding a second-neighbor hopping with $|J_2| \ll |J_1|$), then by continuity, we anticipate that the results of the Perron-Frobenius theorem continue to hold until $|J_2|$ is large

enough. Write an essay that discusses how the Perron-Frobenius theorem can break down and what that implies about the total spin of the ground state.

15.4.5 Correlation Functions

If we think of the two Néel states, they are opposites of each other, so if we average simple properties over these two states, we expect them to give zero. But, if we look at the product of the z-components of spin between neighboring lattice sites, we find for both states they are negative. Hence, looking at averages of the correlations between spins can provide us with useful information about how the order appears in the system. This motivates us to study so-called correlation functions of spins.

Consider the following static correlation function:

$$\langle \psi_{GS} | \hat{\vec{S}}_i \cdot \hat{\vec{S}}_j | \psi_{GS} \rangle \tag{15.50}$$

for the spin-one-half Heisenberg antiferromagnet on a bipartite lattice with $|A| = |B|$. In this problem, we will show the correlation function is positive when i and j are on the same sublattice and negative when i and j are on different sublattices.

To start, we need to remember that the ground state is a spin singlet. Spin singlet states have $\hat{S}^{\alpha}_{tot} |\psi_{GS}\rangle = 0$ for $\alpha = x$, y, and z. The z-component is clear, because $m = 0$. The x and y components become clear, because a spin-singlet state cannot have the z components raised or lowered (if troubled by this, try it on the $|\Lambda| = 2$ spin singlet state). Now, this means that the ground state is invariant under any global rotation (that uses total spin operators). This then means that

$$\begin{aligned} \langle \psi_{GS} | \hat{S}^{\alpha}_i \hat{S}^{\alpha}_j | \psi_{GS} \rangle &= \langle \psi_{GS} | \hat{U}^{\dagger}_{rot} \hat{U}_{rot} \hat{S}^{\alpha}_i \hat{U}^{\dagger}_{rot} \hat{U}_{rot} \hat{S}^{\alpha}_j \hat{U}^{\dagger}_{rot} \hat{U}_{rot} | \psi_{GS} \rangle \\ &= \langle \psi_{GS} | \underbrace{\hat{U}_{rot} \hat{S}^{\alpha}_i \hat{U}^{\dagger}_{rot}}_{\text{rotated operator}} \underbrace{\hat{U}_{rot} \hat{S}^{\alpha}_j \hat{U}^{\dagger}_{rot}}_{\text{rotated operator}} | \psi_{GS} \rangle . \end{aligned} \tag{15.51}$$

Here, the first equality is just using three multiply-by-ones, while the second line is using the fact that the ground state is invariant under any rotation. Now, by choosing an appropriate rotation, we can rotate the spin operator in the α direction to any other direction. This implies that $\langle \psi_{GS} | \hat{S}^{\alpha}_i \hat{S}^{\alpha}_j | \psi_{GS} \rangle$ does not depend on α, and the result is the same for the x-, y-, and z-directions. Use this fact to show that

$$\langle \psi_{GS} | \hat{\vec{S}}_i \cdot \hat{\vec{S}}_j | \psi_{GS} \rangle = \frac{3}{2} \mathrm{Re} \left\{ \langle \psi_{GS} | \hat{S}^{+}_i \hat{S}^{-}_j | \psi_{GS} \rangle \right\} . \tag{15.52}$$

To complete the proof, we use the unitary operator $\hat{U} = (-1)^{N_{\downarrow}(B)}$, which also satisfies $\hat{U}^2 = \mathcal{I}$. Define the symbol $\epsilon(i, j)$ to be 0 if i and j are from the same sublattice and to be 1 if i and j are from different sublattices. Next show that

$$(-1)^{\epsilon(i,j)}\hat{S}_i^+\hat{S}_j^- = \hat{U}\hat{S}_i^+\hat{S}_j^-\hat{U}. \tag{15.53}$$

Then show that

$$\begin{aligned}(-1)^{\epsilon(i,j)}\langle\psi_{GS}|\hat{S}_i^+\hat{S}_j^-|\psi_{GS}\rangle &= \langle\psi_{GS}|\hat{U}\hat{S}_i^+\hat{S}_j^-\hat{U}|\psi_{GS}\rangle \\ &= \sum_{\boldsymbol{\sigma}\boldsymbol{\sigma}'|m=0} c_{\boldsymbol{\sigma}}c_{\boldsymbol{\sigma}'}\langle\tilde{\boldsymbol{\sigma}}|\hat{S}_i^+\hat{S}_j^-|\tilde{\boldsymbol{\sigma}}'\rangle.\end{aligned} \tag{15.54}$$

The right-hand side can be shown to be explicitly positive, because the matrix element is equal to 0 or 1 only (be sure to explain why). Note that the tilde means that these states *do not* have the $(-1)^{N_\downarrow(B)}$ factors in them. Use this to establish the desired result. In particular, be sure to explain why the right-hand side is never equal to zero.

While it might seem like this shows long-range order, that requires the absolute value of the correlation function to be larger than some constant as the distance between i and j gets larger and larger. That is clearly more difficult to establish.

15.5 One-Dimensional Bethe Ansatz Solution

The last topic we will cover is to find the exact solution of the spin-one-half Heisenberg exchange model on a one-dimensional lattice with nearest-neighbor interactions. This problem was solved back in 1931 by Hans Bethe (1931), who won the Nobel prize in 1967 primarily for his work in nuclear physics. A picture of Bethe in the 1930s is shown in Fig. 15.1.

The method we will use to solve this problem is called *the Bethe ansatz*, and it is the technique used to solve a wide variety of different one-dimensional problems, ranging from interacting bosons to interacting fermions, to the Hubbard model. In general, the approach is quite technical and complicated to implement. Fortunately for us, the Heisenberg model solution is probably the simplest application of the Bethe ansatz. Nevertheless, we will find the approach can get quite involved. So, we have some French cooking ahead of us!

Our goal is to find all eigenstates, so we start with the simplest one, which is the fully polarized ferromagnetic state and then we work on finding the eigenstates with one spin flip, then two, then three, and so on. We will find it is more difficult to determine the eigenstates with one spin flip, even more difficult for two spin flips, but then for three or more, we can just use our previous work to find them. This is the magic of the Bethe ansatz—once the two-particle solution is found all the rest can be found straightforwardly. It requires many special properties of the Hamiltonian to hold, which usually relies on it being a one-dimensional chain with just nearest-neighbor connections. But, we won't go into full detail on how to determine problems solvable by the Bethe ansatz here. We will instead focus on solving the problem at hand.

Fig. 15.1 Hans Bethe in the 1930s. [Courtesy of AIP Emilio Segrè Visual Archives, Physics Today Collection, Fermi Film Collection. Converted to look like a sketch by James Freericks.]

Recall that the Hamiltonian for a one-dimensional chain on a ring is given by

$$\hat{H} = -\frac{J}{4}\sum_{j=1}^{|\Lambda|}\left[\frac{1}{2}\left(\hat{\sigma}_j^+\hat{\sigma}_{j+1}^- + \hat{\sigma}_j^-\hat{\sigma}_{j+1}^+\right) + \hat{\sigma}_j^z\hat{\sigma}_{j+1}^z\right], \tag{15.55}$$

with the identification of site $j = |\Lambda| + 1$ with site $j = 1$. Note that we have divided by a factor of four here, as this is the conventional normalization used by others. The starting point is the case where all spins are up. The state $|\uparrow\uparrow\cdots\uparrow\rangle_z$ is an eigenstate of $\hat{H}$ with eigenvalue $-J|\Lambda|/4$, because the spin-flip terms annihilate the state, and the number of bonds is $|\Lambda|$.

Now we move on to the single spin-flip sector. There are $|\Lambda|$ states. We might want to try a single spin-flip state as an eigenstate, but we can immediately see that the spin-flip terms in the Hamiltonian move the spin flip to its neighboring sites, so it cannot be an eigenstate. To find an energy eigenstate, we must have a superposition of *all* single spin-flip states to be able to have the Hamiltonian acting on the state reproduce it times a number.

We note that our system is periodic on the ring with what we call *translation invariance*—each site of the ring looks like every other site. So, when we want to come up with an ansatz for such a superposition, we want the amplitude of each term to be the same (otherwise the probability would be favored at one site versus another). So, if we work with an unnormalized state, we can set the amplitude of each term to a complex phase. We also want the phase difference between the sites to be uniform due to translation invariance as well, which means we will try an

ansatz of the form

$$|\psi_1(p)\rangle = \sum_{j=1}^{|\Lambda|} e^{ipj} |\uparrow\uparrow\cdots\uparrow\downarrow\uparrow\cdots\uparrow\rangle_z, \tag{15.56}$$

where the down arrow is in the jth position. One can also write this as

$$|\psi_1(p)\rangle = \frac{1}{2}\sum_{j=1}^{|\Lambda|} e^{ipj}\hat{\sigma}_j^- |\uparrow\uparrow\cdots\uparrow\rangle_z, \tag{15.57}$$

if desired. We need to operate the Hamiltonian operator onto this state to see if it is an eigenstate.

Acting the Hamiltonian onto the ansatz state gives

$$\begin{aligned}\hat{H}|\psi_1(p)\rangle &= -\frac{J}{4}\frac{1}{2}\sum_{j=1}^{|\Lambda|} e^{ipj}\left(2\hat{\sigma}_{j+1}^- + 2\hat{\sigma}_{j-1}^- + (|\Lambda|-4)\hat{\sigma}_j^-\right)|\uparrow\uparrow\cdots\uparrow\rangle_z \\ &= -\frac{J}{4}(4\cos p + |\Lambda| - 4)|\psi_1(p)\rangle, \end{aligned} \tag{15.58}$$

because only two terms from the raising and lowering operators contribute nonzero solutions (recall the raising and lowering operators in the Pauli matrix representation have 2's on their off diagonals), but all of the z operators contribute ($|\Lambda|-2$ are $+1$ and 2 are -1). Note that we must use a multiply by one in the form $e^{ip}e^{-ip}$ for the first two terms and then use an index shift in the summation to rearrange those terms to look like the ansatz. Be sure you can workout those details yourself. So, it is an eigenstate! The single spin case is pretty easy to accomplish. But we have one issue—what are the allowed values of p? To find these, we must use our periodicity condition. If we identify site $|\Lambda|+1$ with site 1, then the phases must be the same, so

$$e^{ip(|\Lambda|+1)} = e^{ip}, \tag{15.59}$$

or $p|\Lambda| = 2\pi n$ for some integer n. Hence,

$$p = \frac{2\pi n}{|\Lambda|} \text{ for } 1 \le n \le |\Lambda|. \tag{15.60}$$

We have exactly $|\Lambda|$ solutions as required. This completes the single-spin case. The energies run all the way down to the energy of the ferromagnetic case (when $\cos p = 1$) as expected, since there is a spin-s state among the single-spin flip ansatz states (in fact, it is the case with $p = 2\pi$). If we look at the difference in energy $E_1(p) - E_0$, we find it is equal to $-J(\cos p - 1)$. This excitation energy (it is larger

than 0 for $J > 0$) can be thought of as the *single magnon* excitation energy of the ferromagnet. These excitations are called *spin waves* and there is a large effort that goes into understanding how to describe spin-wave excitations in magnetic systems. The term magnon connotes the idea of a particle—indeed, one can think of these excitations as akin to particle-like excitations of the ferromagnet; then p is the *momentum* associated with the magnon. The magnon energy is important, so we will use a new symbol to describe it—$\varepsilon(p) = -J(\cos p - 1)$.

The two-spin-flip case is more challenging, as we have two types of product states in our expansion—those where the two spin flips are separated by at least one unflipped spin, and those where the two spin flips are adjacent. This complicates the analysis, as we will soon see. Nevertheless, if the two magnons are widely separated, we expect that they do not really see each other and so their energy should just be the sum of two magnon energies. This is essentially what we will find, except the momenta p can change a bit from the values we found for the one-magnon case.

The number of states with two spin flips is $\frac{1}{2}|\Lambda|(|\Lambda| - 1)$, so we will be looking for that many solutions. Since we anticipate two well-separated magnons to be essentially two single magnons, we write the two-spin-flip wavefunction in the form

$$|\psi_2(p_1, p_2)\rangle = \frac{1}{4} \sum_{n_1 < n_2} C(p_1, p_2; n_1, n_2) \hat{\sigma}^-_{n_1} \hat{\sigma}^-_{n_2} | \uparrow\uparrow \cdots \uparrow\rangle_z, \tag{15.61}$$

with

$$C(p_1, p_2; n_1, n_2) = A(p_1, p_2) e^{i(p_1 n_1 + p_2 n_2)} + A(p_2, p_1) e^{i(p_2 n_1 + p_1 n_2)}. \tag{15.62}$$

Note that this summation has $\frac{1}{2}N(N-1)$ terms and we have explicitly included the amplitudes corresponding to the single magnons. Because the two spin excitations must be indistinguishable, we must symmetrize the result, which explains the sum over the two terms in the C coefficients. The coefficient C is well-defined when $n_1 = n_2$ even though there is no corresponding spin-flip state. This allows us to work with the "unphysical" coefficient, even though it does not appear in the expansion of one of our quantum states. We expect the amplitudes A to be complex phases (ignoring overall normalization), and only their ratio, given by

$$S(p_1, p_2) = \frac{A(p_2, p_1)}{A(p_1, p_2)}, \tag{15.63}$$

needs to be determined. This ratio is called the *scattering matrix* of two magnons with momenta p_1 and p_2, respectively. To find the solution to the eigenvalue problem, we need to adjust the scattering matrix until we have an eigenvalue-eigenvector relationship, which then determines the energy $E_2(p_1, p_2)$. We denote $\frac{1}{4}\hat{\sigma}^-_{n_1}\hat{\sigma}^-_{n_2}|\uparrow\uparrow\cdots\uparrow\rangle_z = |n_1, n_2\rangle$, which labels the two sites where we have flipped the spins. Then we have

$$\frac{1}{4}\sum_{j=1}^{|\Lambda|}\hat{\sigma}_j^+\hat{\sigma}_{j+1}^-|n_1,n_2\rangle = |n_1+1,n_2\rangle + |n_1,n_2+1\rangle, \tag{15.64}$$

and

$$\frac{1}{4}\sum_{j=1}^{|\Lambda|}\hat{\sigma}_j^-\hat{\sigma}_{j+1}^+|n_1,n_2\rangle = |n_1-1,n_2\rangle + |n_1,n_2-1\rangle. \tag{15.65}$$

These two results simply follow from the fact that these terms in the Hamiltonian flip neighboring spins. Because $n_2 > n_1+1$, we can always achieve these spin flips. Similarly, we have that

$$\frac{1}{4}\sum_{j=1}^{|\Lambda|}\hat{\sigma}_j^z\hat{\sigma}_{j+1}^z|n_1,n_2\rangle = \frac{|\Lambda|-8}{4}|n_1,n_2\rangle, \tag{15.66}$$

because we have $|\Lambda|-4$ pairs with the same spin projection and exactly four pairs with opposite spin projections.

The strategy is to break the eigenvalue-eigenvector relationship into two cases: (i) the case with $n_2 > n_1+1$, where the two spin excitations are separated by at least one up spin, and (ii) the case where $n_2 = n_1+1$, where the two spin excitations are adjacent. We start with the first case. When $n_2 > n_1+1$, applying $\hat{H}$ onto the state, and looking at the coefficient of the $|n_1,n_2\rangle$ state yields

$$\begin{aligned}
&E_2(p_1,p_2)C(p_1,p_2;n_1,n_2)\\
&\quad = -\frac{J}{2}C(p_1,p_2;n_1-1,n_2) - \frac{J}{2}C(p_1,p_2;n_1,n_2-1)\\
&\qquad - \frac{J}{2}C(p_1,p_2;n_1+1,n_2) - \frac{J}{2}C(p_1,p_2;n_1,n_2+1)\\
&\qquad - J\,\frac{|\Lambda|-8}{4}C(p_1,p_2;n_1,n_2).
\end{aligned} \tag{15.67}$$

We use the explicit form for the C coefficients in Eq. (15.62) to express all of the coefficients in Eq. (15.67) in terms of $C(p_1,p_2;n_1,n_2)$ only. Using

$$C(p_1,p_2;n_1+1,n_2) + C(p_1,p_2;n_1,n_2+1) = (e^{ip_1}+e^{ip_2})C(p_1,p_2;n_1,n_2) \tag{15.68}$$

and

$$C(p_1,p_2;n_1-1,n_2) + C(p_1,p_2;n_1,n_2-1) = (e^{-ip_1}+e^{-ip_2})C(p_1,p_2;n_1,n_2) \tag{15.69}$$

then gives us

$$E_2(p_1, p_2)C(p_1, p_2; n_1, n_2)$$
$$-= \frac{J}{2}\left(e^{-ip_1} + e^{-ip_2} + e^{ip_1} + e^{ip_2} + \frac{|\Lambda| - 8}{2}\right) C(p_1, p_2; n_1, n_2). \tag{15.70}$$

From this, we can read off the energy eigenvalue as

$$E_2(p_1, p_2) = -J(\cos p_1 + \cos p_2 - 2) - J\frac{|\Lambda|}{4} = \varepsilon(p_1) + \varepsilon(p_2) - J\frac{|\Lambda|}{4}. \tag{15.71}$$

To complete the solution, we need to consider the case where the two spin flips are adjacent. One can directly establish that

$$\frac{1}{4}\sum_{j=1}^{|\Lambda|} \hat{\sigma}_j^+ \hat{\sigma}_{j+1}^- |n_1, n_1 + 1\rangle = |n_1, n_1 + 2\rangle,$$
$$\frac{1}{4}\sum_{j=1}^{|\Lambda|} \hat{\sigma}_j^- \hat{\sigma}_{j+1}^+ |n_1, n_1 + 1\rangle = |n_1 - 1, n_1 + 1\rangle, \tag{15.72}$$

and

$$\frac{1}{4}\sum_{j=1}^{|\Lambda|} \hat{\sigma}_j^z \hat{\sigma}_{j+1}^z |n_1, n_1 + 1\rangle = \frac{|\Lambda| - 4}{4} |n_1, n_1 + 1\rangle. \tag{15.73}$$

Again, we look at the $|n_1, n_1 + 1\rangle$ coefficient. We find

$$E_2(p_1, p_2)C(p_1, p_2; n_1, n_1 + 1)$$
$$= -\frac{J}{2}C(p_1, p_2; n_1, n_1 + 2) - \frac{J}{2}C(p_1, p_2; n_1 - 1, n_1 + 1)$$
$$- J\,\frac{|\Lambda| - 4}{4}C(p_1, p_2; n_1, n_1 + 1). \tag{15.74}$$

Next, we take the expression in Eq. (15.67) with $n_2 = n_1 + 1$, which is proven to hold in Problem 15.5.1:

$$E_2(p_1, p_2)C(p_1, p_2; n_1, n_1 + 1)$$
$$= -\frac{J}{2}C(p_1, p_2; n_1 - 1, n_1 + 1) - \frac{J}{2}C(p_1, p_2; n_1, n_1)$$
$$- \frac{J}{2}C(p_1, p_2; n_1 + 1, n_1 + 1) - \frac{J}{2}C(p_1, p_2; n_1, n_1 + 2)$$

$$- J \frac{|\Lambda| - 8}{4} C(p_1, p_2; n_1, n_1 + 1), \tag{15.75}$$

and subtract it from Eq. (15.74). This gives the following constraint:

$$\frac{1}{2} C(p_1, p_2; n_1, n_1) + \frac{1}{2} C(p_1, p_2; n_1 + 1, n_1 + 1) - C(p_1, p_2; n_1, n_1 + 1) = 0, \tag{15.76}$$

in order for the eigenvalue-eigenvector equation to hold. Substituting the explicit form of the coefficient in and collecting terms that multiply each amplitude then gives us

$$A(p_1, p_2) \left(\tfrac{1}{2} + \tfrac{1}{2} e^{ip_1 + ip_2} - e^{ip_2} \right) = A(p_2, p_1) \left(-\tfrac{1}{2} - \tfrac{1}{2} e^{ip_1 + ip_2} + e^{ip_1} \right). \tag{15.77}$$

This allows us to solve for the scattering matrix, which gives us

$$S(p_1, p_2) = \frac{A(p_2, p_1)}{A(p_1, p_2)} = -\frac{1 + e^{ip_1 + ip_2} - 2e^{ip_2}}{1 + e^{ip_1 + ip_2} - 2e^{ip_1}}. \tag{15.78}$$

This is the second quantity we needed to determine. It can be re-expressed as

$$S(p_1, p_2) = \frac{\frac{1}{2} \cot \frac{p_1}{2} - \frac{1}{2} \cot \frac{p_2}{2} - i}{\frac{1}{2} \cot \frac{p_1}{2} - \frac{1}{2} \cot \frac{p_2}{2} + i} = e^{-i\alpha}, \tag{15.79}$$

which you will work out in Problem 15.5.2; the final equality defines the complex phase α. In the same problem, you will also show the amazingly simple formula

$$2 \cot \tfrac{\alpha}{2} = \cot \tfrac{p_1}{2} - \cot \tfrac{p_2}{2}, \tag{15.80}$$

which also follows from the constraint.

We have not yet used the periodicity condition. Clearly, what we must have, since the site $|\Lambda| + 1 = 1$, is that

$$C(p_1, p_2; 1, n_2) = C(p_1, p_2; n_2, |\Lambda| + 1), \tag{15.81}$$

which can be rewritten as both

$$e^{-i\alpha} = S(p_1, p_2) = e^{-ip_1 |\Lambda|} = e^{ip_2 |\Lambda|}. \tag{15.82}$$

Hence, we have

$$p_1 = \frac{2\pi m + \alpha}{|\Lambda|} \quad \text{and} \quad p_2 = \frac{2\pi m' - \alpha}{|\Lambda|}, \tag{15.83}$$

with m and m' integers that range from 1 to $|\Lambda|$. This result shows us that the two magnon solution is similar in form to two one-magnon solutions but with a phase shift that changes the energy. This phase shift $\pm\alpha/|\Lambda|$ is solely due to the effect of the interaction between the magnons. Note that if we compare the coefficients at n_1 and n_2 to those at n_1+1 and n_2+1, the coefficients are shifted by the phase $e^{ip_1+ip_2}$. This tells us that the total momentum of the two-magnon state is given by $p_1 + p_2$. The total momentum is given by $\frac{2\pi(m+m')}{|\Lambda|}$ and is independent of the phase shifts! This is similar to the breakup we had when we solved for the motion of hydrogen—the center of mass motion was independent of the Coulomb interaction, while the relative motion depended on it. Here, the equivalent to center-of-mass motion is independent of the phase shifts while the total energy depends on it.

Now we want to carefully solve for the phase shifts of the two-magnon problem and verify we have found all two spin-flip states. This proceeds in a number of steps because there are a number of different cases to go through and we must do this carefully. We follow closely (Karbach & Müller, 1997) for this analysis. Initially looking at Eq. (15.80) and combining with Eq. (15.83), we obtain

$$2\cot\tfrac{\alpha}{2} = \cot\left(\frac{2\pi m+\alpha}{2|\Lambda|}\right) - \cot\left(\frac{2\pi m'-\alpha}{2|\Lambda|}\right). \tag{15.84}$$

Naively, we have $|\Lambda|^2$ possible values of m and m', but we pick $m \leq m'$, because the wave function is symmetric under interchange of the momenta. This reduces the possible solutions to $|\Lambda|(|\Lambda|+1)/2$, which is still larger than the total number of two magnon states, so not all solutions can work. We need to analyze them more carefully.

We start with searching for the solutions with $\alpha = 0$. These solutions must exist because that is the energy of the $s = |\Lambda|/2$ and $s = |\Lambda|/2-1$ that we found with the zero magnon case ($m = m' = |\Lambda|$) and the one magnon case $m \neq |\Lambda|$, $m' = |\Lambda|$). We cannot use Eq. (15.78) because the left-hand side is infinity if $\alpha = 0$. So, we go back to Eq. (15.78) and set $p_2 = 2\pi$ ($m' = |\Lambda|$). Then $S(p_1, p_2 = 2\pi) = 1$, implying $\alpha = 0$. This holds for all $m < |\Lambda|$ and for the limit as $m \to |\Lambda|$. This gives us the $|\Lambda|$ solutions from the 0 and 1 magnon solutions that have been lowered to the two magnon states. These energies are identical to the 0 and 1 magnon energies because the energy is zero when $\alpha = 0$ and m or $m' = |\Lambda|$.

Now, we want to find the nontrivial solutions to Eq. (15.83). We have $m' \neq |\Lambda|$ and $m' > m$. We first examine the case $m' = m+1 < |\Lambda|$. Here, we can immediately see that the solution corresponds to $\alpha = \pi$. It may take a little trial and error to verify this is the solution—it is easiest to just plug in and see it works. But, there is a problem with this solution. Since $\alpha = \pi$, we have $S(p_1, p_2) = -1$ and in this case we have $p_1 = p_2$. In such a situation, $C(p_1, p_2; n_1, n_2) = 0$, so there is no state! Hence, we must have $m' > m + 1$. In this case, Eq. (15.83) can be solved for all possible m and m'. This yields $(|\Lambda|-2)(|\Lambda|-3)/2$ new solutions. Combined with the previous $|\Lambda|$ solutions we found, we have $|\Lambda|-3$ solutions remaining to find. If we set $m = m' < |\Lambda|$, we have $|\Lambda|-1$ possibilities, and if we set $m' = m+1 < |\Lambda|$, we have $|\Lambda| - 2$ possibilities. This yields $2|\Lambda| - 3$ possible solutions, but we only

need $|\Lambda| - 3$, so not all equations will yield valid solutions. Indeed, we will find only $|\Lambda| - 3$ solutions, which is exactly what we need. These remaining solutions are called *bound-state* solutions.

The analysis is complicated, so we have to go through it very carefully. The key point is that α becomes *imaginary*. Let's start with the case where $m = m'$ and write $\alpha = i\gamma$. We go back to Eq. (15.78). Choosing $\alpha = i\gamma$, so that $p_1 = \frac{2\pi m}{|\Lambda|} + \frac{i\gamma}{|\Lambda|}$ and $p_2 = \frac{2\pi m}{|\Lambda|} - \frac{i\gamma}{|\Lambda|}$, we find

$$\begin{aligned} e^{\gamma} &= -\frac{1 + e^{\frac{4i\pi m}{|\Lambda|}} - 2e^{\frac{2i\pi m}{|\Lambda|} + \frac{\gamma}{|\Lambda|}}}{1 + e^{\frac{4i\pi m}{|\Lambda|}} - 2e^{\frac{2i\pi m}{|\Lambda|} - \frac{\gamma}{|\Lambda|}}} \\ &= -\frac{e^{\frac{-2i\pi m}{|\Lambda|}} + e^{\frac{2i\pi m}{|\Lambda|}} - 2e^{\frac{\gamma}{|\Lambda|}}}{e^{\frac{-2i\pi m}{|\Lambda|}} + e^{\frac{2i\pi m}{|\Lambda|}} - 2e^{-\frac{\gamma}{|\Lambda|}}} \\ &= -\frac{\cos\frac{2\pi m}{|\Lambda|} - e^{\frac{\gamma}{|\Lambda|}}}{\cos\frac{2\pi m}{|\Lambda|} - e^{-\frac{\gamma}{|\Lambda|}}}. \end{aligned} \tag{15.85}$$

Rearranging gives

$$\cos\frac{2\pi m}{|\Lambda|} = \frac{\cosh\left(\frac{\gamma}{2} - \frac{\gamma}{|\Lambda|}\right)}{\cosh\frac{\gamma}{2}}. \tag{15.86}$$

Since the hyperbolic cosine is positive, this can only have a solution with $\gamma > 0$ when the cosine is positive. Hence, we can solve it when $m = 1, 2, \cdots \frac{|\Lambda|}{4} - 1$ and $m = \frac{3|\Lambda|}{4} + 1, \cdots, |\Lambda| - 1$. This accounts for $\frac{|\Lambda|}{2} - 2$ solutions.

The next case is when $m' = m + 1$ and the real part of α is π, so that p_1 and p_2 are identical again. Repeating the analysis above for $\alpha = \pi + i\gamma$ yields

$$\cos\frac{\pi(2m+1)}{|\Lambda|} = \frac{\sinh\left(\frac{\gamma}{2} - \frac{\gamma}{|\Lambda|}\right)}{\sinh\frac{\gamma}{2}}. \tag{15.87}$$

Again, we must have the cosine positive, so we have an additional $\frac{|\Lambda|}{2} - 2$ solutions. The only subtlety is some of these solutions have real α, which requires one to go back and solve Eq. (15.79) with $\alpha \neq \pi$. The crossover occurs when the cosine is larger than the value of the right-hand side when $\gamma = 0$, which is $1 - \frac{2}{|\Lambda|}$, as we can evaluate by taking the limit and using $\sinh x \approx x$ for small x. This is because the right-hand side decreases as γ increases.

There is one more solution we need to determine all possible solutions. Here, we set $m = m' = \frac{|\Lambda|}{4}$ and $\alpha = i\gamma$. Substituting into Eq. (15.78) then gives us

$$-e^{-\frac{2\gamma}{|\Lambda|}} = e^{-\gamma}. \tag{15.88}$$

Naively, this does not look like it has any possible solution, but if we let $\gamma \to \infty$, we get $0 = 0$ and it works! This is our final solution.

The energy of the state when we have a complex solution is given by

$$\begin{aligned} E_2(\text{bound}) &= -J(\cos p_1 + \cos p_2 - 2) - J\frac{|\Lambda|}{4} \\ &= -J\left(2\cos(\text{Re}(p_1))\cosh\frac{\gamma}{|\Lambda|} - 2\right) - J\frac{|\Lambda|}{4}. \end{aligned} \tag{15.89}$$

Here, we used the fact that $\text{Re}(p_1) = \text{Re}(p_2)$ for all the cases we considered. The case with infinite imaginary part is more complicated to evaluate because the cosine goes to zero while the hyperbolic cosine goes to infinity, so it must be evaluated properly as a limit. It has an energy of $J - J\frac{|\Lambda|}{4}$.

Now, we move on to determining the results for more magnons. We will start with three magnons and then rapidly move to the arbitrary number of magnons case. One of the key issues in discussing this material is that the notation gets cumbersome even if the ideas are straightforward. There is not much we can do about this. We won't go through all of the complex algebra here though, but instead explain how the calculations work, including the relevant formulas. In other words, we will tell you results without fully deriving them.

The three-magnon state is expressed as

$$|\psi_3(p_1, p_2, p_3)\rangle = \frac{1}{8}\sum_{n_1<n_2<n_3} C(p_1, p_2, p_3; n_1, n_2, n_3)\hat{\sigma}^-_{n_1}\hat{\sigma}^-_{n_2}\hat{\sigma}^-_{n_3}|\uparrow\uparrow\cdots\uparrow\rangle_z. \tag{15.90}$$

Here, the coefficient involves a sum over the six permutations of the three momenta:

$$\begin{aligned} C(p_1, p_2, p_3; n_1, n_2, n_3) &= A(p_1, p_2, p_3)e^{i(p_1n_1+p_2n_2+p_3n_3)} \\ &+ A(p_2, p_3, p_1)e^{i(p_2n_1+p_3n_2+p_1n_3)} + A(p_3, p_1, p_2)e^{i(p_3n_1+p_1n_2+p_2n_3)} \\ &+ A(p_2, p_1, p_3)e^{i(p_2n_1+p_1n_2+p_3n_3)} + A(p_1, p_3, p_2)e^{i(p_1n_1+p_3n_2+p_2n_3)} \\ &+ A(p_3, p_2, p_1)e^{i(p_3n_1+p_2n_2+p_1n_3)}. \end{aligned} \tag{15.91}$$

Now, we have six objects to determine: the energy $E_3(p_1, p_2, p_3)$ and the ratio of the amplitudes relative to $A(p_1, p_2, p_3)$. This is clearly a challenging thing to do. The miracle of the Bethe ansatz is that the energies have the same form as we had earlier:

$$E_3(p_1, p_2, p_3) = \varepsilon(p_1) + \varepsilon(p_2) + \varepsilon(p_3) - J\frac{|\Lambda|}{4}. \tag{15.92}$$

Furthermore, the ratio of amplitudes factorizes into a product of scattering matrices. The rule is that each time we permute a pair of neighboring momenta in an amplitude, we get a scattering matrix of the lower index momentum followed by

the higher index. So, for example,

$$A(p_3, p_1, p_2) = S(p_1, p_3)S(p_1, p_2)A(p_1, p_2, p_3), \tag{15.93}$$

which allows us to find the ratio of the amplitudes. Of course one has to explicitly prove that this holds—the algebra to do this is tedious, and we do not go through the details here. One also has to work out the periodicity requirements as well. Finally, there will be a large set of phase shifts found by formulae similar to the one we used for the two-magnon case, as well as additional "bound-state" solutions to find as well. In addition, every solution for the two-magnon case must also be a solution for the three-magnon case because all of those states can be lowered. Clearly this is a lot of work.

We just state the results for the general case of n flips. We let $\Pi(i)$ be the permutation of the ith element of the set $1, 2, \cdots, n$. Then the wavefunction is expressed as

$$\begin{aligned}|\psi_n(p_1, p_2, \cdots, p_n)\rangle = \frac{1}{2^n} &\sum_{n_1<n_2<\cdots<n_n} A(p_1, p_2, \cdots, p_n) \\ &\times \sum_{\text{perm } \Pi} \frac{A(p_{\Pi(1)}, p_{\Pi(2)}, \cdots, p_{\Pi(n)})}{A(p_1, p_2, \cdots, p_n)} \\ &\times e^{i(p_{\Pi(1)}n_1 + p_{\Pi(2)}n_2 + \cdots + p_{\Pi(n)}n_n)} \hat{\sigma}^-_{n_1}\hat{\sigma}^-_{n_2}\cdots\hat{\sigma}^-_{n_n}|\uparrow\uparrow\cdots\uparrow\rangle_z.\end{aligned} \tag{15.94}$$

The scattering amplitudes factorize, as we discussed before and satisfy the periodicity requirement. The energy takes the form

$$E_n(p_1, p_2, \cdots, p_n) = \varepsilon(p_1) + \varepsilon(p_2) + \cdots \varepsilon(p_n) - J\frac{|\Lambda|}{4}. \tag{15.95}$$

The momenta are given by

$$p_j = \frac{2\pi m_j + \sum_{j'\neq j}\alpha_{jj'}}{|\Lambda|} \tag{15.96}$$

with phase shifts that satisfy

$$2\cot\frac{\alpha_{jj'}}{2} = \cot\frac{p_j}{2} - \cot\frac{p_{j'}}{2} \tag{15.97}$$

and

$$e^{-i\alpha_{jj'}} = -\frac{1 + e^{ip_j + ip_{j'}} - 2e^{ip_{j'}}}{1 + e^{ip_j + ip_{j'}} - 2e^{ip_{j'}}}. \tag{15.98}$$

Again, one also has special values of the phase shifts when one of the indices is $|\Lambda|$ and for the bound states, where m_j values are the same or differ by 1. We do not describe those in detail here.

So far, we have focused on the ferromagnetic case, where the spin-flip states have a higher energy than the ground state. But, now we turn our attention to the antiferromagnetic case, where $J < 0$, and the ground state on a lattice with $|\Lambda|$ sites has $|\Lambda|/2$ spin flips. Although we did not state it earlier, the regular solutions still require neighboring m_j values to differ by at least 2. For $|\Lambda|/2$ spin flips, we must have $m_1 = 1, m_2 = 3, m_3 = 5, \cdots m_{|\Lambda|/2} = |\Lambda| - 1$. It turns out this is the ground state. The ground state energy is the sum of $\varepsilon(p_j)$ for the allowed p_j values (which include the phase shifts). What is amazing is that the summation can be done exactly in the limit as $|\Lambda| \to \infty$. Hulthén was the first to work this out (Hulthén, 1938). The result is

$$\frac{E_{GS}(AF) - E(FM)}{|\Lambda||J|} = -\ln 2 \approx -0.693147. \tag{15.99}$$

You can explore what the value is for this for the $|\Lambda| = 6$ case in the problems.

We end this section with a discussion of the anisotropic Heisenberg model. In this model, the z-exchange coefficient changes from $-J$ to $-gJ$. The isotropic Heisenberg model corresponds to $g = 1$. One expects for $g \approx 1$ the system should behave similarly by continuity. What is more surprising is that the strategy we used for the Bethe ansatz solution of the Heisenberg model also works for this anisotropic XXZ model. The changes are rather minor, and we encourage you to examine them in Problem 15.5.6.

Section 15.5 Problems:

15.5.1 Eigenvalue Equation Identity

Verify that Eq. (15.67) holds even if $n_2 = n_1 + 1$ by explicitly working out the result after substituting in $E_2(p_1, p_2) = -J(\cos p_1 + \cos p_2 - 2) - \frac{1}{4}J|\Lambda|$, and rewriting all of the coefficients in terms of $C(p_1, p_2; n_1, n_1 + 1)$.

15.5.2 Derive Eq. (15.79) and Other Identities

Starting from Eq. (15.78), factor out $-4ie^{\frac{i}{2}(p_1+p_2)}$ from the numerator and denominator and rearrange it to the form given in Eq. (15.79).

One can immediately see that $S(p_1, p_2)$ is a complex phase by showing the norm of the numerator and the denominator is the same. Do this. Then set the phase equal to $e^{-i\alpha}$ and determine a quadratic equation in $x = \cot\frac{p_1}{2} - \cot\frac{p_2}{2}$, which is in the

form $x^2 - 4\cot\alpha - 4 = 0$. Solve the quadratic equation, and use half-angle formulas and other trig identities to show that the solution is either

$$2\cot\tfrac{\alpha}{2} = \cot\tfrac{p_1}{2} - \cot\tfrac{p_2}{2} \quad \text{or} \quad 2\tan\tfrac{\alpha}{2} = \cot\tfrac{p_2}{2} - \cot\tfrac{p_1}{2}. \tag{15.100}$$

Verify that if we substitute the solution on the left into the original expression, it gives $e^{-i\alpha}$, while the one on the right gives $-e^{-i\alpha}$, so the one on the left is the correct solution.

15.5.3 How to Compute the Phase Shifts for the Bethe Ansatz

Determine a procedure to calculate the solutions to Eq. (15.79) when $m' > m+1$. If you want to be concrete, find all of the phase shifts and energies for these solutions with $|\Lambda| = 8$. You do not need to solve for the bound-state solutions. If you have never solved one-dimensional roots before, Brent's one-dimensional solver is a great place to start.

15.5.4 The $|\Lambda| = 4$ Case by Bethe Ansatz

Previously, we worked out all of the eigenvalues and eigenvectors for the four-site case. Repeat, but now using the Bethe ansatz. You should be able to find all energies without requiring any root solver for the zero, one, and two spin-flip sectors. Recall that the energy with four spin flips is the same as with zero and the energy with three flips is the same as with one. Note that because of our change in normalization of the Hamiltonian, all energies we calculated previously by exact diagonalization need to be divided by four to compare with the work here.

Hint: The six (m, m') pairs you need to use are $(1, 4)$, $(2, 4)$, $(3, 4)$, $(4, 4)$, $(1, 1)$, and $(1, 3)$. Only $(1, 1)$ is the special case with infinite imaginary part. You only need to solve a transcendental equation for $(1, 3)$, but it should be easy to solve, even analytically.

15.5.5 The Antiferromagnetic Ground-State Energy for $|\Lambda| = 6$

Using $m_1 = 1$, $m_2 = 3$ and $m_3 = 5$, compute the antiferromagnetic ground state energy for $|\Lambda| = 6$. Form $(E_{GS}(AF) - E(FM))/6|J|$ and compare it to $-\ln 2$. You will need to numerically solve for the phase shifts.

Ans. -0.71713.

15.5.6 The XXZ Model

In the XXZ model, the exchange constant J remains the same for the raising and lowering operator terms, but changes for the z interactions with $J \rightarrow gJ$. Revisit our derivations and see what changes for $g \neq 1$. Write a short essay discussing what occurs, without explicitly reworking formulas. The same procedure we worked on to find eigenvalues and eigenvectors works for this case too, but the details are different.

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Chapter 16
Relativity and the Dirac Equation

If you can think it, you can do it.

Dionne Warwick, Singer, 2011

In this chapter, we discuss relativistic quantum mechanics. See a video tour at https://tinyurl.com/2p8ewsj9. When it was discovered by Dirac in 1927, it had a tremendous impact—it explained why the anomalous g factor for electrons is almost 2, it predicted antiparticles (and the positron was discovered in 1932 by Carl Anderson), and it almost got the spectra for hydrogen correct; it took another 20 years for the Lamb shift to be accurately measured and then explained. But, it turns out that relativistic quantum mechanics is not a complete theory. This is because the mass-energy equivalence in relativity means that we can create particle-antiparticle pairs and relativistic quantum mechanics does not have this phenomenon in it. One had to wait until quantum electrodynamics was fully developed in the late 1940s before we had the proper quantum field theory description of relativistic quantum mechanics. Do not fret though. Relativistic quantum mechanics is still very good at determining relativistic corrections when they are small and when we are not close to the situation where particle-antiparticle pairs can be created. So, it plays an important role for hydrogen and is useful to know. Paul Dirac's image is shown in Fig. 16.1 from the period when he developed this theory.

16.1 Deriving the Dirac Equation

Back in Sect. 14.10, we discussed how to describe the Hamiltonian of a system in an electric and magnetic field. It required using what is called the *minimal coupling* of electric and magnetic fields to the vector potential. We shifted $\hat{\vec{p}} \to \hat{\vec{p}} - \frac{q}{c}\vec{A}(\hat{\vec{r}}, t)$

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Fig. 16.1 Paul Dirac in the 1930s. [Photograph by A. Bortzells Tryckeri, courtesy AIP Emilio Segrè Visual Archives, E. Scott Barr Collection, Weber Collection. Converted to look like a sketch by James Freericks]

in the kinetic energy and added a scalar electric potential $q\Phi(\hat{\vec{r}}, t)$ to it. Back in that section, we only showed the specific case of an electron in hydrogen, where $q = -|e|$ and $\Phi = e/4\pi\epsilon_0\hat{r}$. We also added in the explicit coupling of spin to the magnetic field, including having to introduce a g factor, so that we could obtain results close to the experiment.

Pauli was the first to notice that the minimal substitution was ambiguous, because there were multiple ways one can obtain $\hat{\vec{p}}^2/2M$ and hence multiple minimal substitutions. He instead advocated that the correct Hamiltonian for a free particle is

$$\hat{H}_{\text{free}} = \frac{\left(\vec{\sigma}\cdot\hat{\vec{p}}\right)^2}{2M}, \tag{16.1}$$

which we verify by using the identity $\left(\vec{\sigma}\cdot\vec{A}\right)\left(\vec{\sigma}\cdot\vec{B}\right) = \vec{A}\cdot\vec{B} + i\vec{\sigma}\cdot(\vec{A}\times\vec{B})$ that was proven back when we were discussing Pauli spin matrices. Since $\vec{A} = \vec{B}$ and they commute with themselves, we find that $\hat{\vec{p}}\times\hat{\vec{p}} = 0$ and $\hat{H}_{\text{free}} = \hat{\vec{p}}^2/2M$. This is also our free-particle Hamiltonian, so indeed, there is just such an ambiguity.

Next, we simply use Pauli's version of the free-particle Hamiltonian and make the minimal substitution. We find

$$\hat{H}_{\text{Pauli}} = \frac{\vec{\sigma}\cdot\left(\hat{\vec{p}} - \frac{q}{c}\vec{A}(\hat{\vec{r}}, t)\right)\vec{\sigma}\cdot\left(\hat{\vec{p}} - \frac{q}{c}\vec{A}(\hat{\vec{r}}, t)\right)}{2M} + q\Phi(\hat{\vec{r}}, t). \tag{16.2}$$

Now, because

$$i\vec{\sigma}\cdot\left\{\left(\hat{\vec{p}}-\frac{q}{c}\vec{A}(\hat{\vec{r}},t)\right)\times\left(\hat{\vec{p}}-\frac{q}{c}\vec{A}(\hat{\vec{r}},t)\right)\right\}=-i\frac{q}{c}\sum_{ijk}\epsilon_{ijk}\sigma_i[\hat{p}_j,A_k(\hat{\vec{r}},t)], \tag{16.3}$$

we can use the definition of the magnetic field in Eq. (14.159), to find that the Pauli Hamiltonian becomes

$$\hat{H}_{\text{Pauli}}=\frac{\left(\hat{\vec{p}}-\frac{q}{c}\vec{A}(\hat{\vec{r}},t)\right)\cdot\left(\hat{\vec{p}}-\frac{q}{c}\vec{A}(\hat{\vec{r}},t)\right)}{2M}+q\Phi(\hat{\vec{r}},t)-\frac{q}{2Mc}\hbar\vec{\sigma}\cdot\vec{B}(\hat{\vec{r}},t). \tag{16.4}$$

Since $\hat{\vec{S}}=\frac{\hbar}{2}\vec{\sigma}$, we find that the Pauli Hamiltonian, with the minimal substitution, is given by

$$\hat{H}_{\text{Pauli}}=\frac{\left(\hat{\vec{p}}-\frac{q}{c}\vec{A}(\hat{\vec{r}},t)\right)\cdot\left(\hat{\vec{p}}-\frac{q}{c}\vec{A}(\hat{\vec{r}},t)\right)}{2M}+q\Phi(\hat{\vec{r}},t)-\frac{q}{Mc}\hat{\vec{S}}\cdot\vec{B}(\hat{\vec{r}},t). \tag{16.5}$$

This includes the coupling of spin to the magnetic field with $g=2$! Unfortunately the Pauli Hamiltonian does not give a full relativistic treatment. Note that it has a 2×2 matrix structure to it from the spin operators. This makes it a bit more complicated to solve than a scalar equation.

In order to develop the proper relativistic formalism, we need to have a lightning review of energy in a relativistic system. The key result we need to use is the relationship between energy and momentum, given by

$$E^2=\vec{p}^2c^2+M^2c^4, \tag{16.6}$$

which holds for a classical relativistic system. When $\vec{p}=0$, it has hidden in it the famous $E=mc^2$ equation that is associated with Einstein's achievements. Our treatment of relativity will be brief and dense. Readers who want to get some more depth should consult other excellent sources for special relativity such as Epstein's book (1983) or Taylor and Wheeler's book (1992), which is a bit more advanced.

In relativity, we need to be careful to define what an *event* is, which is something that occurs at a specific location at a specific time in a framework of rulers and clocks that have been synchronized to take into account the finite speed of the speed of light. This is done by starting with a reference clock at the origin, setting each clock forward by a time given by how long it takes light to travel from the origin to the clock and then having the clock at the origin send out a reference pulse of light that starts each clock in the framework once the light pulse hits it. If we measure time and distances with this framework, then we find the spacetime interval, given by

$$(\Delta\tau)^2=(\Delta t)^2-\left(\frac{\Delta x}{c}\right)^2-\left(\frac{\Delta y}{c}\right)^2-\left(\frac{\Delta z}{c}\right)^2, \tag{16.7}$$

is measured to be the same in any frame of reference moving with a constant velocity relative to our frame. The symbol Δ means the difference of the results for two events. So $\Delta\tau = \tau_2 - \tau_1$, $\Delta t = t_2 - t_1$, $\Delta x = x_2 - x_1$, and so on, and the first event is at (t_1, x_1, y_1, z_1), while the second is at (t_2, x_2, y_2, z_2).

Just as an event has position and time incorporated into it, energy and momentum are related in relativity as well. We first need to determine how to find them. In nonrelativistic systems, the velocity is just how far we traveled divided by the time interval. To make it more and more accurate, we pick very small time intervals to determine an instantaneous velocity. In a relativistic system, we use the spacetime interval instead of time, so that $p_x = M\frac{\Delta x}{\Delta\tau}$ and similarly for the y- and z-directions. The definition for the energy will seem odd to you. We take it to be $E = Mc^2\frac{\Delta t}{\Delta\tau}$. You will verify in the problems that this gives us the usual kinetic energy in the nonrelativistic limit, which helps verify we are using the right result. Note that the definition of relativistic momentum and energy can be made much more rigorous than what we are doing here, but it takes a bit more time to fully develop the theory. Taylor and Wheeler's book does this very well if you want further information. Now, we take our invariant interval relation, multiply it by $M^2c^4/(\Delta\tau)^2$. Using our definitions, then gives us

$$M^2c^4 = E^2 - \vec{p}^2c^2 \text{ or } E^2 = \vec{p}^2c^2 + M^2c^4. \tag{16.8}$$

This then has derived the relativistic relationship between energy and momentum from the invariance of the spacetime interval.

Dirac sought out a linear equation for the relativistic generalization of the Schrödinger equation. Recall, our nonrelativistic Schrödinger equation for a state with definite energy is

$$\left(\frac{\hat{\vec{p}}^2}{2M} + V(\hat{\vec{r}})\right)|\psi\rangle = E|\psi\rangle, \tag{16.9}$$

which is a linear eigenvalue equation. The linear equation means that we have linear superposition, and so forth, a very important property of quantum mechanics. The relativistic equation is a quadratic equation for the energy—so, we want to find a way to make it into a linear equation so we maintain the same type of linear superposition. It is by no means obvious how to do this and Dirac struggled to figure it out. Inspired by the Pauli equation, Dirac tried to find objects α_i and β such that

$$\left(\vec{\alpha}\cdot\hat{\vec{p}}c + \beta Mc^2\right)^2 = \vec{p}^2c^2 + M^2c^4. \tag{16.10}$$

Then, the linear equation for the energy eigenvalues of a relativistic free particle would be

$$\left(\vec{\alpha}\cdot\hat{\vec{p}}c + \beta Mc^2\right)|\Psi\rangle = E|\Psi\rangle. \tag{16.11}$$

For this to work, we need $\alpha_i^2 = \beta^2 = 1$, $\alpha_i\alpha_j + \alpha_j\alpha_i = 0$, and $\alpha_i\beta + \beta\alpha_i = 0$. We have seen properties like this for the Pauli matrices, but one cannot use 2×2 Pauli matrices to construct the α_i's and β. This follows from a simple counting argument because the matrices must be independent of each other. We need four linearly independent matrices that anticommute with each other. But, we only have three with the Pauli matrices σ_x, σ_y, and σ_z. Dirac's matrices must have zero trace, by the following argument:

$$\begin{aligned} \alpha_i\beta + \beta\alpha_i &= 0 \\ \beta\alpha_i\beta + \beta^2\alpha_i &= 0 \\ \mathrm{Tr}(\beta\alpha_i\beta) + \mathrm{Tr}(\mathbb{I}\alpha_i) &= 0 \\ \mathrm{Tr}(\beta^2\alpha_i) + \mathrm{Tr}(\alpha_i) &= 0, \end{aligned} \tag{16.12}$$

or $\mathrm{Tr}\alpha_i = 0$. The argument starts from the anticommutation relation. We multiply by β from the left, use the fact that $\beta^2 = \mathbb{I}$, use the invariance of the trace $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$, and use the fact that $\beta^2 = \mathbb{I}$ again. Because these matrices square to the unit matrix, their eigenvalues are ± 1. But, they are also traceless. So, we cannot construct such a matrix in three dimensions. They must be 4×4 matrices. It turns out we can construct them from the Pauli matrices, which we do next.

The procedure to construct them is to work with tensor-product matrices formed out of the tensor product of two matrices chosen from the Pauli matrices or the identity matrix. Recall to make a tensor-product matrix, we place the second matrix at the location of each matrix element of the first matrix, and multiply the first matrix element by the second matrix. So, for example, we have

$$\sigma_x \otimes \sigma_z = \begin{pmatrix} 0 & \sigma_z \\ \sigma_z & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}. \tag{16.13}$$

When we multiply two matrices written in tensor-product form, we multiply the tensor product components, so $A \otimes B$ multiplied by $C \otimes D$ is $(AC) \otimes (BD)$. These properties, plus the properties of the Pauli matrices—that they anticommute with each other and square to the unit matrix—will allow us to construct matrices that satisfy the requirements of the Dirac equation. Namely, we find that the following construction works:

$$\alpha_x = \sigma_x \otimes \sigma_x = \begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}, \quad \alpha_y = \sigma_x \otimes \sigma_y = \begin{pmatrix} 0 & \sigma_y \\ \sigma_y & 0 \end{pmatrix}, \quad \alpha_z = \sigma_x \otimes \sigma_z = \begin{pmatrix} 0 & \sigma_z \\ \sigma_z & 0 \end{pmatrix}, \tag{16.14}$$

and

$$\beta = \sigma_z \otimes \mathbb{I}_2 = \begin{pmatrix} \mathbb{I}_2 & 0 \\ 0 & -\mathbb{I}_2 \end{pmatrix}. \tag{16.15}$$

You will verify that these matrices satisfy the required relations in their 4×4 matrix format in the problems. Here, we will argue why they satisfy them using the tensor-product notation. It is clear that each of these four matrices square to one—$(\sigma_x \otimes \sigma_i)\,(\sigma_x \otimes \sigma_i) = \sigma_x^2 \otimes \sigma_i^2 = \mathbb{I}_2 \otimes \mathbb{I}_2 = \mathbb{I}_4$ and similarly for the β matrix. Anticommutation for $i \neq j$ follows from

$$\alpha_i \alpha_j + \alpha_j \alpha_i = \sigma_x^2 \otimes (\sigma_i \sigma_j + \sigma_j \sigma_i) = 0, \tag{16.16}$$

because different Pauli matrices anticommute. Similarly, for the β matrix, we have

$$\alpha_i \beta + \beta \alpha_i = (\sigma_x \sigma_z + \sigma_z \sigma_x) \otimes \sigma_i = 0, \tag{16.17}$$

as well. This then shows what we needed to verify.

To obtain the full Dirac equation, we need to add the potential and also to perform the minimal substitution of the electromagnetic potentials (scalar and vector) to describe the full Hamiltonian in the presence of electric and magnetic fields. For an energy eigenstate, we find that we have

$$\left(c\vec{\alpha} \cdot \left(\hat{\vec{p}} - \tfrac{q}{c} \vec{A}(\hat{\vec{r}}, t) \right) + q\Phi(\hat{\vec{r}}, t) + V(\hat{\vec{r}}) + \beta M c^2 \right) |\Psi\rangle = E|\Psi\rangle. \tag{16.18}$$

The energy eigenstate $|\Psi\rangle$ has four components to it. We usually let $\psi = (\psi_1, \psi_2)$ denote the so-called "large" components and $\chi = (\chi_1, \chi_2)$ the so-called "small" components, with the four-component vector written as $(\psi, \chi) = (\psi_1, \psi_2, \chi_1, \chi_2)$. Then we have

$$\begin{pmatrix} q\Phi(\hat{\vec{r}}, t) + V(\hat{\vec{r}}) + Mc^2 & c\vec{\sigma} \cdot \left(\hat{\vec{p}} - \frac{q}{c} \vec{A}(\hat{\vec{r}}, t) \right) \\ c\vec{\sigma} \cdot \left(\hat{\vec{p}} - \frac{q}{c} \vec{A}(\hat{\vec{r}}, t) \right) & q\Phi(\hat{\vec{r}}, t) + V(\hat{\vec{r}}) - Mc^2 \end{pmatrix} \begin{pmatrix} \psi \\ \chi \end{pmatrix} = E \begin{pmatrix} \psi \\ \chi \end{pmatrix}. \tag{16.19}$$

This is a form that we will often work with, sometimes separating out into two equations for each of the two-component vectors ψ and χ.

There are many things that often are done with the Dirac equation that we will not do here. This includes showing that it is relativistically covariant. In addition, the modern notation used for the Dirac equation is different from Dirac's original form. In modern notation, we work with the so-called gamma matrices, given by $\gamma_i = -i\beta\alpha_i$ and $\gamma_0 = \beta$. The Dirac equation can be written quite compactly with this notation, but, we will not pursue that further here.

Section 16.1 Problems:

16.1.1 Relativistic Energy

Using $\Delta\tau/\Delta t = \sqrt{1-\frac{\vec{v}^2}{c^2}}$ show that $E = Mc^2\Delta t/\Delta\tau$ can be used with $E^2 = \vec{p}^2c^2 + M^2c^4$ to find that $\vec{p}^2 = M^2\vec{v}^2/(1-\frac{\vec{v}^2}{c^2})$ for relativistic momentum. Verify that this takes the usual form in the nonrelativistic limit when we have $\frac{v}{c} \ll 1$.

16.1.2 Relativistic Kinetic Energy

The relativistic kinetic energy is defined as $E - Mc^2$. Using the positive square root for the energy, we find that

$$E = \sqrt{\vec{p}^2c^2 + M^2c^4} = Mc^2\sqrt{1+\frac{\vec{p}^2}{M^2c^2}}. \tag{16.20}$$

Using the expansion for the square root, given by $\sqrt{1+\delta} = 1 + \frac{\delta}{2} - \frac{1}{8}\delta^2 + \cdots$ for $\delta \ll 1$, expand the kinetic energy to show that it takes the ordinary form, plus a relativistic correction, which is small if $|\vec{v}|/c \ll 1$.

16.1.3 Direct Verification of $\vec{\alpha}$ and β Matrices

Write out the $\vec{\alpha}$ and β matrices as 4×4 matrices, and verify that $\alpha_i^2 = \beta^2 = \mathbb{I}_4$ by brute force matrix multiplication of the 4×4 matrices. Also show that $\alpha_i\alpha_j + \alpha_j\alpha_i = 0$ for $i \neq j$ and $\alpha_i\beta + \beta\alpha_i = 0$ using matrix multiplication.

16.1.4 Operator Square Root

Previously, when we discussed an operator square root, we described how we diagonalize the operator, taking the square root of each the diagonal matrix element, leaving a choice of plus or minus signs for each diagonal element (yielding 2^n possible choices for the square root) for an $n \times n$ matrix in the eigenvector basis. But here, we instead took "scalar" operators and added an additional matrix structure to them to compute the square root. Discuss, in a short essay, what the differences are from our earlier way to compute the square root, and why we did it this way for the relativistic theory.

16.1.5 Gamma Matrices

The gamma matrices are given by $\gamma_i = -i\beta\alpha_i$ and $\gamma_0 = \beta$. Compute the three gamma matrices for $i = 1$, 2, and 3.

16.1.6 Noninteracting Problem

Set $\vec{A} = \Phi = V = 0$. An eigenstate of $\hat{\vec{p}}$ will also be an energy eigenstate of the noninteracting problem. Consider the eigenstates corresponding to $E = \sqrt{\vec{p}^2c^2 + M^2c^4}$. Find the eigenvectors ψ and χ corresponding to this eigenvalue. Is the state degenerate or nondegenerate?

16.1.7 Nonrelativistic Limit

When we are in the nonrelativistic limit, with $v/c = |\vec{v}|/c \ll 1$, it turns out that the Dirac equation maps onto the Pauli equation, with the ψ components being large and the χ components being small. Separating out the two components of the equation and introducing the Dirac ket for the two component pieces, we have

$$\left(q\Phi(\hat{\vec{r}}, t) + V(\hat{\vec{r}})\right)|\psi\rangle + c\vec{\sigma} \cdot \left(\hat{\vec{p}} - \tfrac{q}{c}\vec{A}(\hat{\vec{r}}, t)\right)|\chi\rangle = (E - Mc^2)|\psi\rangle \tag{16.21}$$

$$\left(q\Phi(\hat{\vec{r}}, t) + V(\hat{\vec{r}})\right)|\chi\rangle + c\vec{\sigma} \cdot \left(\hat{\vec{p}} - \tfrac{q}{c}\vec{A}(\hat{\vec{r}}, t)\right)|\psi\rangle = (E + Mc^2)|\chi\rangle. \tag{16.22}$$

For nonrelativistic motion, $E/Mc^2 - 1 \ll 1$. We assume that $|\psi\rangle \approx 1$. Under the assumption that the potentials and momentum are all small, show that we can approximate

$$|\chi\rangle \approx \frac{1}{2Mc}\vec{\sigma} \cdot \left(\hat{\vec{p}} - \tfrac{q}{c}\vec{A}(\hat{\vec{r}}, t)\right)|\psi\rangle \tag{16.23}$$

and then we have

$$\left(\frac{\left(\vec{\sigma} \cdot \left(\hat{\vec{p}} - \frac{q}{c}\vec{A}(\hat{\vec{r}}, t)\right)\right)^2}{2M} + q\Phi(\hat{\vec{r}}, t) + V(\hat{\vec{r}})\right)|\psi\rangle \approx \left(E - Mc^2\right)|\psi\rangle, \tag{16.24}$$

which is the Pauli equation after we recognize that Pauli's energy is the energy relative to the rest-mass energy, which is a small number. So, this analysis shows, for nonrelativistic motion, that the Dirac equation reduces to the Pauli equation. Just remember that when we expand out the first term, we obtain both the minimal substituted kinetic energy and the coupling of a spin-one-half particle to a magnetic field with a g factor equal to 2. This is the first clue that the Dirac equation describes spin-one-half particles.

16.2 Central Force Problems

Our main focus in this chapter is to solve the relativistic hydrogen problem. This is a central-force problem, so we need to look into how we separate variables into radial and angular motion and how angular momentum is described in the relativistic Dirac equation.

In central force problems, orbital angular momentum is always conserved (because the force is always radially directed). We expect the total angular momentum to commute with the relativistic Hamiltonian with a central force. But, because we know that the Dirac equation (and the Pauli equation) have an intrinsic coupling to an "internal" spin degree of freedom, this should somehow enter into the total angular momentum. Let' s see how we construct the total angular momentum.

We first form the orbital angular momentum, given by $\hat{\vec{L}} = \hat{\vec{r}} \times \hat{\vec{p}}\,\mathbb{I}_4$, where we include the explicit 4×4 identity matrix expression. We take the Dirac Hamiltonian in the absence of a field, so $\hat{H} = c\vec{\alpha} \cdot \hat{\vec{p}} + \beta Mc^2 + V(\hat{r})\,\mathbb{I}$. We know that $[\hat{\vec{L}}, \hat{r}] = 0$, so the orbital angular momentum does commute with the central force potential. It also commutes with the βMc^2 term, because the identity matrix commutes with everything. But, the momentum operator is a vector operator, so, we have $[\hat{L}_i, \hat{p}_j] = i\hbar \sum_k \epsilon_{ijk} \hat{p}_k$. This then gives

$$[\hat{\vec{L}}, \hat{H}] = i\hbar c\vec{\alpha} \times \hat{\vec{p}} \neq 0. \tag{16.25}$$

Hence, the components of the orbital angular momentum *do not commute* with a central-force Hamiltonian! For the nonrelativistic case, the Hamiltonian depended on $\hat{\vec{p}}^2$, which does commute with $\hat{\vec{L}}$. But here, we have a linear dependence on momentum, so the angular momentum no longer commutes with the Hamiltonian.

In essence, this is a result we expected. Because the Dirac equation has an intrinsic spin incorporated in it, it should have that spin incorporated into the total angular momentum, and we have not done that yet. The spin operator is represented by the following:

$$\hat{\vec{S}} = \frac{\hbar}{2}\mathbb{I}_2 \otimes \vec{\sigma} = \frac{\hbar}{2}\begin{pmatrix} \vec{\sigma} & 0 \\ 0 & \vec{\sigma} \end{pmatrix}, \tag{16.26}$$

in the Dirac four-spinor space—it acts as the conventional spin operator on both the upper and lower components of the Dirac four-spinor. Let us compute the commutator of $\hat{\vec{S}}$ with $\hat{H}$. It is easiest to do this using the tensor-product notation for the matrices. We have

$$[\hat{S}_i, \alpha_j] = \tfrac{\hbar}{2}[\mathbb{I}_2 \otimes \sigma_i, \sigma_x \otimes \sigma_j] = \tfrac{\hbar}{2}\sigma_x \otimes (\sigma_i\sigma_j - \sigma_j\sigma_i) = i\hbar \sum_k \epsilon_{ijk}\sigma_x \otimes \sigma_k, \tag{16.27}$$

which is $i\hbar \sum_k \epsilon_{ijk}\alpha_k$. This means that

$$[\hat{\vec{S}}, \hat{H}] = -i\hbar c\vec{\alpha} \times \hat{\vec{p}}. \tag{16.28}$$

Hence, the Dirac Hamiltonian does commute with the total (orbital plus spin) angular momentum. We define $\hat{\vec{J}} = \hat{\vec{L}} + \hat{\vec{S}}$, and then $[\hat{J}, \hat{H}] = 0$. We have found the proper angular momentum that is conserved in the relativistic central-force problem. This means we can classify states according to the total angular momentum j and the z component of the total angular momentum m_z. Note that this derivation makes it seem like we can easily extract the orbital and spin components of the total angular momentum. But, this is not quite as obvious as it may seem. In the nonrelativistic limit, the decomposition is clear, but in the relativistic case, we could have corrections to each that add to zero. We will not pursue this further here, but just keep in mind that the relativistic spin of the physical particle might be a more complicated object than just $\hat{\vec{S}}$.

Because orbital angular momentum has the eigenvalue of $\hat{L}^2$ to be $\hbar^2 l(l+1)$ with integer l and the eigenvalue of $\hat{S}^2$ is $\frac{3}{4}\hbar^2$, the total angular momentum for the relativistic system corresponds to half-odd integer j, which is equal to either $j = l + \frac{1}{2}$ or $j = l - \frac{1}{2}$. In the problems you will show that $[\hat{L}^2, \hat{H}] \neq 0$ and $[\hat{S}^2, \hat{H}] = 0$. Even though $\hat{\vec{L}}^2$ does not commute with $\hat{H}$, we can still use the coupled basis $|j, m_z, l, s\rangle$ to describe our states. But, we will find that while our states have definite j, they will be constructed from linear combinations of the two different l values. It also turns out that we need to work with another quantity to simplify the Dirac equation for energy eigenstates in central potentials. We follow the ideas of Martin and Glauber from the 1950s (Glauber & Martin, 1958), as implemented by Green (1965) to determine this new operator, which is needed specifically for the hydrogen problem. We start by examining the eigenvalues of $\hat{\vec{L}} \cdot \hat{\vec{S}}$ using the fact that we can write $2\hat{\vec{L}} \cdot \hat{\vec{S}} = \hat{J}^2 - \hat{L}^2 - \hat{S}^2$. Since $j = l \pm \frac{1}{2}$, we have that the eigenvalues of $\hat{\vec{L}} \cdot \hat{\vec{S}}$ are given by

$$\frac{\hbar^2}{2}\left(\left(l \pm \tfrac{1}{2}\right)\left(l + 1 \pm \tfrac{1}{2}\right) - l(l+1) - \tfrac{3}{4}\right) = \pm\frac{\hbar^2}{2}\left(l + \tfrac{1}{2} \mp \tfrac{1}{2}\right) = \begin{cases} \frac{\hbar^2}{2} l \\ -\frac{\hbar^2}{2}(l+1). \end{cases} \tag{16.29}$$

Note that to determine these eigenvalues, we need to operate on states of definite j, m_z, l, and s, as we did when we examined combining angular momentum. This is allowed because we can form angular momentum basis states from such states.

Green next suggests that we examine an operator of the form (our operator is slightly different from his)

$$\hat{\Lambda} = -\frac{2\hat{\vec{L}} \cdot \hat{\vec{S}} + \hbar^2 \mathbb{I}_4}{\hbar} - 2i\bar{\lambda}\hat{\vec{e}}_r \cdot \hat{\vec{S}}\,\sigma_x \otimes \mathbb{I}_2, \tag{16.30}$$

with $\bar{\lambda}$ an arbitrary dimensionless number. Note that this operator is not Hermitian for real λ and that it incorporates the unit vector operator in the radial direction $\hat{\vec{e}}_r = \frac{\hat{\vec{r}}}{\hat{r}}$, which we have worked with before. We then compute

$$\begin{aligned}\hat{\Lambda}^2 &= \frac{(2\hat{\vec{L}}\cdot\hat{\vec{S}}+\hbar^2\mathbb{I}_4)(2\hat{\vec{L}}\cdot\hat{\vec{S}}+\hbar^2\mathbb{I}_4)}{\hbar^2} + \frac{(2\hat{\vec{L}}\cdot\hat{\vec{S}}+\hbar^2\mathbb{I}_4)}{\hbar}2i\bar{\lambda}\hat{\vec{e}}_r\cdot\hat{\vec{S}}\,\sigma_x\otimes\mathbb{I}_2 \\ &\quad + 2i\bar{\lambda}\hat{\vec{e}}_r\cdot\hat{\vec{S}}\,\sigma_x\otimes\mathbb{I}_2\frac{(2\hat{\vec{L}}\cdot\hat{\vec{S}}+\hbar^2\mathbb{I}_4)}{\hbar} - 4\bar{\lambda}^2(\hat{\vec{e}}_r\cdot\hat{\vec{S}})\,\sigma_x\otimes\mathbb{I}_2(\hat{\vec{e}}_r\cdot\hat{\vec{S}})\,\sigma_x\otimes\mathbb{I}_2 \\ &= \frac{(2\hat{\vec{L}}\cdot\hat{\vec{S}}+\hbar^2\mathbb{I}_4)^2}{\hbar^2} - \hbar^2\bar{\lambda}^2\mathbb{I}_4. \end{aligned} \tag{16.31}$$

The derivation of this expression is assigned as Problem 16.2.2 below. Using this result, and continuity, as $\bar{\lambda}\to 0$, we find that the eigenvalues of $\hat{\Lambda}$ are

$$\Lambda_+ = -\hbar\sqrt{(l+1)^2-\bar{\lambda}^2} \text{ and } \Lambda_- = \hbar\sqrt{l^2-\bar{\lambda}^2}, \tag{16.32}$$

after recalling the minus sign in the definition of $\hat{\Lambda}$; the plus sign (in the subscript) is for $j = l+\frac{1}{2}$ and the minus sign (in the subscript) is for $j = l-\frac{1}{2}$. We can summarize both cases as $\Lambda_\pm = \mp\hbar\sqrt{(j+1/2)^2-\bar{\lambda}^2}$.

Our next step is to perform the separation of variables to the Dirac equation, separating into "angular" motion and radial motion ("angular" is in quotes, because the total angular momentum depends on spin too). In the process, we will find the effective Hamiltonian for fixed j and m_z that we need to solve to determine the energy eigenvalues (and later the wavefunctions). The standard approach is to "square" the Dirac Hamiltonian, so we have a quadratic momentum term as before, which we know how to separate variables with. But, one needs some caution here, because the solution we seek is to the Dirac equation, not its "squared" cousin. So we will need to do some additional work later when we determine the wavefunction for the original Dirac equation. This approach then has us take an energy eigenstate $|\Psi\rangle$ and represent it in terms of another state $|\Phi\rangle$ via

$$|\Psi\rangle = \beta\left(c\vec{\alpha}\cdot\hat{\vec{p}} - \beta Mc^2 + V(\hat{r}) - E\right)|\Phi\rangle. \tag{16.33}$$

Then we have

$$\begin{aligned}\beta(\hat{H}-E)|\Psi\rangle &= \beta\left(c\vec{\alpha}\cdot\hat{\vec{p}} + \beta Mc^2 + V(\hat{r}) - E\right) \\ &\quad\times\beta\left(c\vec{\alpha}\cdot\hat{\vec{p}} - \beta Mc^2 + V(\hat{r}) - E\right)|\Phi\rangle \\ &= 0, \end{aligned} \tag{16.34}$$

because $|\Psi\rangle$ is an energy eigenstate of $\hat{H}$ with eigenvalue E. This result can be rewritten as

$$\left(E - V(\hat{r})\right)^2 \mathbb{I}_4 |\Phi\rangle = \left(M^2c^4\,\mathbb{I}_4 + c^2\hat{p}^2\,\mathbb{I}_4 - c\vec{\alpha}\cdot[V(\hat{r}),\hat{\vec{p}}]\right)|\Phi\rangle. \tag{16.35}$$

Now, we substitute in $V(\hat{r}) = -\frac{e^2}{4\pi\epsilon_0\hat{r}} = -\frac{\alpha_F\hbar c}{\hat{r}}$, for the hydrogen problem, with $\alpha_F = \frac{e^2}{4\pi\epsilon_0\hbar c} \approx \frac{1}{137}$ the fine-structure constant. Then, we have

$$\left(\frac{E}{c} + \frac{\alpha_F\hbar}{\hat{r}}\right)^2 \mathbb{I}_4|\Phi\rangle = \left(M^2c^2\,\mathbb{I}_4 + \hat{p}^2\,\mathbb{I}_4 + \alpha_F\hbar\vec{\alpha}\cdot\left[\frac{1}{\hat{r}},\hat{\vec{p}}\right]\right)|\Phi\rangle. \tag{16.36}$$

Rearranging, we find

$$\left(\frac{E^2}{c^2} - M^2c^2\right)\mathbb{I}_4|\Phi\rangle = \left(\hat{p}^2\,\mathbb{I}_4 - \frac{2\alpha_F\hbar E}{c\hat{r}}\,\mathbb{I}_4 - \frac{\alpha_F^2\hbar^2}{\hat{r}^2}\,\mathbb{I}_4 - i\alpha_F\hbar^2\frac{\vec{\alpha}\cdot\hat{\vec{r}}}{\hat{r}^3}\right)|\Phi\rangle. \tag{16.37}$$

The next step uses the fact that $\hat{p}^2 = \hat{p}_r^2 + \frac{\hat{L}^2}{\hat{r}^2}$. We want to relate this to the operator $\hat{\Lambda}$, so we use the fact that

$$\hat{L}^2 = (\hat{\vec{L}}\cdot\vec{\sigma})(\hat{\vec{L}}\cdot\vec{\sigma}) - i\vec{\sigma}\cdot\hat{\vec{L}}\times\hat{\vec{L}} = (\hat{\vec{L}}\cdot\vec{\sigma})(\hat{\vec{L}}\cdot\vec{\sigma}) + \hbar\hat{\vec{L}}\cdot\vec{\sigma}. \tag{16.38}$$

This can be rewritten as

$$\hat{L}^2\hbar^2\,\mathbb{I}_4 = \left(2\hat{\vec{L}}\cdot\hat{\vec{S}}\right)\left(2\hat{\vec{L}}\cdot\hat{\vec{S}} + \hbar^2\,\mathbb{I}_4\right). \tag{16.39}$$

The other reorganization we need is that $\vec{\alpha} = \sigma_x\otimes\vec{\sigma} = (\sigma_x\otimes\mathbb{I}_2)(\mathbb{I}_2\otimes\vec{\sigma})$, so that $\vec{\alpha}\cdot\hat{\vec{r}} = \frac{2}{\hbar}(\hat{\vec{r}}\cdot\hat{\vec{S}})\,\sigma_x\otimes\mathbb{I}_2$. Putting this together yields

$$\begin{aligned}
&\left(\frac{E^2}{c^2} - M^2c^2\right)\mathbb{I}_4|\Phi\rangle \\
&\quad= \left(\hat{p}_r^2\,\mathbb{I}_4 - \frac{2\alpha_F\hbar E}{c\hat{r}}\,\mathbb{I}_4\right. \\
&\qquad\left. + \frac{(2\hat{\vec{L}}\cdot\hat{\vec{S}})(2\hat{\vec{L}}\cdot\hat{\vec{S}} + \hbar^2\,\mathbb{I}_4) - \alpha_F^2\hbar^4\,\mathbb{I}_4 - 2i\alpha_F\hbar^3\hat{\vec{e}}_r\cdot\hat{\vec{S}}}{\hbar^2\hat{r}^2}\right)|\Phi\rangle. \\
&\quad= \left(\hat{p}_r^2\,\mathbb{I}_4 - \frac{2\alpha_F\hbar E}{c\hat{r}}\,\mathbb{I}_4 + \frac{\hat{\Lambda}(\hat{\Lambda}+\hbar)}{\hat{r}^2}\right)|\Phi\rangle,
\end{aligned} \tag{16.40}$$

with $\bar{\lambda} = \alpha_F$ The last line is verified in Problem 16.2.3. Note that this equation is different in form from what we have worked with for definite energy states before.

The left hand side is a number multiplying a vector, and the right hand side is an operator acting on a vector, which looks like an eigenvalue-eigenvector relationship, but the eigenvalue we seek is E, *which appears on both sides of the equation*. This will play a role in determining how we solve the problem using the factorization method in the next section. It arose simply from the fact that we "squared" the equation.

Section 16.2 Problems:

16.2.1 Commutator of the Dirac Hamiltonian with the Square of Orbital and Spin Angular Momentum

Use the commutation relations of the components of the orbital and spin angular momentum with $\hat{H}$ to show that $\hat{L}^2$ does not commute with $\hat{H}$ and that $\hat{S}^2$ does commute with $\hat{H}$. Note that you will need to use the Leibniz rule and carefully think about the different cases brought on by the Levi-Civita tensor to verify the results.

16.2.2 Derivation of $\hat{\mathbf{\Lambda}}^2$

Show that

$$(2\hat{\vec{L}}\cdot\hat{\vec{S}}+\hbar^2)\,\mathbb{I}_4\,(\hat{\vec{r}}\cdot\hat{\vec{S}})\,\sigma_x\otimes\mathbb{I}_2+(\hat{\vec{r}}\cdot\hat{\vec{S}})\,\sigma_x\otimes\mathbb{I}_2\,(2\hat{\vec{L}}\cdot\hat{\vec{S}}+\hbar^2)\,\mathbb{I}_4=0, \tag{16.41}$$

by expanding out each term and simplifying. You will need to show that $\hat{\vec{r}}\cdot\hat{\vec{L}}=0$ in your derivation. It is easiest to do this by expanding out each component of the dot product. Using $\hat{\vec{S}}=\frac{\hbar}{2}\mathbb{I}_2\otimes\vec{\sigma}$, show that

$$4(\hat{\vec{r}}\cdot\hat{\vec{S}})\,\sigma_x\otimes\mathbb{I}_2\,(\hat{\vec{r}}\cdot\hat{\vec{S}})\,\sigma_x\otimes\mathbb{I}_2=\hbar^2\hat{r}^2\mathbb{I}_4. \tag{16.42}$$

These two relations establish Eq. (16.31).

16.2.3 Verify Eq. (16.40)

Verify the last line of Eq. (16.40).

16.2.4 The Free-Particle in Spherical Coordinates

Determine the eigenvalue equation analogous to Eq. (16.40) for the free particle with $V = 0$. Can this "Hamiltonian" be solved by factorization?

16.2.5 Isotropic Oscillator

Determine the eigenvalue equation analogous to Eq. (16.40) for the case where $V = \frac{1}{2}M\omega^2\hat{r}^2$. Can this "Hamiltonian" be solved by factorization?

16.3 Relativistic Hydrogen Energy Eigenstates

As you can see from the work in the previous section, the energy eigenvalue relation is more complicated in the relativistic case. The energy eigenvalue E appears on both sides of the equation! Nevertheless, if we can determine the eigenvalue of the operator, it will lead us to the energy eigenvalue of the relativistic problem. We just have to solve an additional equation to determine it. We will show the details for how to do this shortly.

The first thing to notice about Eq. (16.40) is that the bound state energy for hydrogen ($-$ 13.6 eV) is small compared to the rest mass energy (500,000 eV). So, we expect the E on the right hand side of the energy eigenvalue equation to be very close to the rest mass energy Mc^2. If we call the operator on the right-hand-side of the equation $\hat{O}$ with eigenvalues $\varepsilon(E)$, which depend on E, after we find an eigenvalue, we must solve $\frac{E^2}{c^2} - M^2c^2 = \varepsilon(E)$ to determine E. Because we have solved energy eigenvalue equations involving operators that look just like $\hat{O}$ (with $\hat{\Lambda} \rightarrow \hbar l$), we know the energy ε is proportional to the coefficient of the $\frac{1}{\hat{r}}$ term. The left hand side is approximately proportional to it, because we can factorize the difference of squares, and we find it is approximately equal to $\left(\frac{E}{c} - Mc\right)2Mc$, which means the corrections of the energy from the nonrelativistic case should be small. Again, this is not surprising, because we estimated the electron's average speed was on the order of one one-hundredth of the speed of light. In such a case relativistic corrections will be small.

Our next step is to finish the separation of variables so we can solve the factorization problem. We will find that we have already solved this problem previously, so the factorization method will be able to be applied easily to this problem. We schematically write our state as $|\Phi\rangle = |\phi\rangle \otimes |j, m\rangle$, where $j = l \pm \frac{1}{2}$ and l is an integer (we will find in the next section how to construct this state and find its form is a bit more complicated than what we assumed here, but the eigenvalue remains unchanged). We can definitely separate the radial and angular (and spin) contributions because $[\hat{p}_r, \hat{\Lambda}] = 0$, which is easy to verify. Now, acting

the $\hat{\Lambda}$ operator onto this state, has it act only on the second factor of the tensor-product state, where it produces the eigenvalues we found previously $\Lambda_\pm = \hbar\lambda_\pm$ in Eq. (16.32), where the plus sign is when $j = l + \frac{1}{2}$ and the minus sign when $j = l - \frac{1}{2}$. Our eigenvalue equation becomes

$$\left(\hat{p}_r^2 + \frac{\hbar^2\Lambda_\pm(\Lambda_\pm + 1)}{\hat{r}^2} - \frac{2\alpha_F\hbar E}{c\hat{r}}\right)|\phi\rangle = \varepsilon(E)|\phi\rangle. \tag{16.43}$$

Aside from the fact that this equation is missing a $2M$ factor in the denominator, it has the exact same form as the types of equations we factorized when we solved for hydrogen (especially with Problem 11.8.7), where we added a perturbation to the inverse $\hat{r}$ square term to the hydrogen Hamiltonian and saw how the energy levels changed. This implies that we factorize as follows:

$$\hat{O} = \hat{O}_\lambda = \hat{B}_\lambda^\dagger \hat{B}_\lambda + \varepsilon_\lambda, \tag{16.44}$$

with

$$\hat{B}_\lambda = \hat{p}_r - i\hbar\left(\frac{\alpha_\lambda}{\bar{a}_0} + \frac{\beta_\lambda}{\hat{r}}\right), \tag{16.45}$$

and $\bar{a}_0 = \frac{\hbar c}{\alpha_F E}$. Note that this definition of the Bohr radius reduces to our old result if $E = Mc^2$. We dropped the $\pm$ subscript and use the generic λ index for simplicity. We find that

$$\hat{B}_\lambda^\dagger \hat{B}_\lambda = \hat{p}_r^2 + \frac{\hbar^2\beta_\lambda}{\hat{r}^2} + \frac{\hbar^2\alpha_\lambda^2}{\bar{a}_0^2} + \frac{2\hbar^2\alpha_\lambda\beta_\lambda}{\bar{a}_0\hat{r}} + \frac{\hbar^2\beta_\lambda^2}{\hat{r}^2}. \tag{16.46}$$

Hence, $-\alpha_\lambda\beta_\lambda = 1$ and $\beta_\lambda(\beta_\lambda + 1) = \lambda(\lambda + 1)$. This implies $\alpha_\lambda = -\frac{1}{\beta_\lambda}$, and $\beta_\lambda = \lambda$ or $\beta_\lambda = -(\lambda + 1)$. Finally, we have that $\varepsilon_\lambda = -\frac{\hbar^2\alpha_\lambda^2}{\bar{a}_0^2}$.

Now, we need to decide what value of β_λ is the correct one to use. We have a number of different cases to go through, so be patient. We start with the case where $j = l + \frac{1}{2}$, where $l = 0,\ 1,\ \cdots$. In this case, $\lambda = -\sqrt{(l+1)^2 - \alpha_F^2}$. This is well defined for all allowed l, because $\alpha_F^2 < 1$. To have a well-defined factorization, with a decaying wavefunction as $r \to \infty$, we must have the superpotential be positive as $r \to \infty$. Hence, $\alpha_\lambda > 0$ and $\beta_l < 0$. Since $\lambda_+ < 0$, we must choose $\beta_\lambda = \lambda_+ = -\sqrt{(l+1)^2 - \alpha_F^2}$, and then $\alpha_\lambda = 1/\sqrt{(l+1)^2 - \alpha_F^2}$. The next case is $j = l - \frac{1}{2}$, with $l = 1,\ 2,\ \cdots$. The case $l = 0$ is not allowed because $j > 0$. Now, we have $\lambda_- = \sqrt{l^2 - \alpha_F^2}$, which is also well defined for $l \geq 1$. Then we find that $\beta_\lambda = -1 - \lambda_- = -1 - \sqrt{l^2 - \alpha_F^2}$ and $\alpha_\lambda = 1/\left(1 + \sqrt{l^2 - \alpha_F^2}\right)$. There is another case that is not physical and is discussed in Problem 16.3.6.

Having determined the factorization for both physical cases, we now will determine the energy. Recall that

$$\varepsilon_\lambda(E) = -\frac{\hbar^2\alpha_\lambda^2}{\bar{a}_0^2} = -\frac{\alpha_F^2 E^2}{c^2\beta_\lambda^2} = \frac{E^2}{c^2} - M^2c^2. \tag{16.47}$$

Solving for the relativistic energy E gives us

$$E = \frac{Mc^2}{\sqrt{1+\frac{\alpha_F^2}{\beta_\lambda^2}}}, \tag{16.48}$$

where we choose the positive sign, because the energy must be close to the rest-mass energy. One can see the energy is slightly less than the rest-mass energy, implying the state is a bound state. In fact, if we expand the result for small α_F, we find

$$E \approx Mc^2 - \frac{\alpha_F^2 Mc^2}{2\beta_\lambda^2} \approx Mc^2 - \frac{\alpha_F^2 Mc^2}{2(l+1)^2} \approx Mc^2 - \frac{e^2}{8\pi\epsilon_0 a_0 (l+1)^2}, \tag{16.49}$$

which is the correct result for the bound-state energy in the nonrelativistic case, for the ground state with angular momentum l (we have to subtract the rest-mass energy to find the bound-state energy in the nonrelativistic limit and we now are using the usual Bohr radius).

It is common to express the results in terms of j rather than l. Doing this for the case $j = l + \frac{1}{2}$ gives us

$$E_+ = \frac{Mc^2}{\sqrt{1+\frac{\alpha_F^2}{(j+\frac{1}{2})^2-\alpha_F^2}}}. \tag{16.50}$$

For the case $j = l - \frac{1}{2}$, we have

$$E_- = \frac{Mc^2}{\sqrt{1+\frac{\alpha_F^2}{\left(1+\sqrt{(j+\frac{1}{2})^2-\alpha_F^2}\right)^2}}}. \tag{16.51}$$

To determine the bound-state energy, we subtract Mc^2 from these results. A key fact is that the energy is just a function of j and is independent of l (even though we use a slightly different form for the $+$ and $-$ cases). Note that these results are only for the ground states for each value of l, because we have not yet constructed the factorization chain.

The way we find the higher-energy states is to follow the standard procedure. We compute $\hat{B}_\lambda \hat{B}_\lambda^\dagger + \varepsilon_\lambda$ and refactorize to determine the next set of α_λ and β_λ values.

We find

$$\hat{B}_\lambda \hat{B}_\lambda^\dagger + \varepsilon_\lambda = \hat{p}_r^2 + \frac{2\hbar^2 \alpha_\lambda \beta_\lambda}{\bar{a}_0 \hat{r}} + \frac{\hbar^2 \beta_\lambda (\beta_\lambda - 1)}{\hat{r}^2}, \tag{16.52}$$

which shifts the value for β_λ that we had for the previous case by -1 for each case ($j = l \pm \frac{1}{2}$) because β_λ is negative. This allows us to write down the energies for the general case immediately, with one formula for both cases. The energies become

$$E(n') = \frac{Mc^2}{\sqrt{1 + \frac{\alpha_F^2}{\left(n' + \sqrt{(j+\frac{1}{2})^2 - \alpha_F^2}\right)^2}}}, \tag{16.53}$$

where for the $+$ case we start $n' = 0,\ 1,\ ,\cdots$, while for the $-$ case, we start with $n' = 1,\ 2,\ \cdots$. Then, the $j = l + \frac{1}{2}$ eigenstate is schematically given by

$$\begin{aligned} &|n', l, j = l + \tfrac{1}{2}\rangle \\ &= \frac{1}{\sqrt{\left(\varepsilon_{n'}^+(E) - \varepsilon_{n'-1}^+(E)\right)\left(\varepsilon_{n'}^+(E) - \varepsilon_{n'-2}^+(E)\right)\cdots\left(\varepsilon_{n'}^+(E) - \varepsilon_0^+(E)\right)}} \\ &\times \hat{B}_{\lambda_+(0,l)}^\dagger \hat{B}_{\lambda_+(1,l)}^\dagger \cdots \hat{B}_{\lambda_+(n'-1,l)}^\dagger |\lambda_+(n', l)\rangle \end{aligned} \tag{16.54}$$

with $\hat{B}_{\lambda_+(n',l)}|\lambda_+(n', l)\rangle = 0$, $\hat{\Lambda}|\lambda_+(n', l)\rangle = \hbar\lambda_+(n', l)|\lambda_+(n', l)\rangle$, and $\lambda_+(n', l) = -n' - \sqrt{(l+1)^2 - \alpha_F^2}$. Note that the energies in the normalization factor are the ε energies, not the E energies. The normalization factor is

$$C_{n',l}^+ = \left(\frac{\hbar}{\bar{a}_0\sqrt{n' + \sqrt{(l+1)^2 - \alpha_F^2}}}\right)^{n'} \sqrt{\frac{n'!\Gamma(n'+1+2\sqrt{(l+1)^2-\alpha_F^2})\Gamma(1+\sqrt{(l+1)^2-\alpha_F^2})}{\Gamma(1+2\sqrt{(l+1)^2-\alpha_F^2})\Gamma(n'+1+\sqrt{(l+1)^2-\alpha_F^2})}}, \tag{16.55}$$

where $\Gamma(z)$ is called the Gamma function and is a generalization of the factorial, satisfying $\Gamma(z+1) = z\Gamma(z)$. Here, we have $n' = 1,\ 2,\ \cdots$; the state with $n' = 0$ is just $|\lambda_+(0, l)\rangle$. Note that these relations are schematic, because we have not explicitly worked out the Dirac four-spinor structure or the tensor-product structure of these states (which we will do in the next section). For the case with $j = l - \frac{1}{2}$, we have

$$\begin{aligned} &|n', l, j = l - \tfrac{1}{2}\rangle \\ &= \frac{1}{\sqrt{\left(\varepsilon_{n'}^-(E) - \varepsilon_{n'-1}^-(E)\right)\left(\varepsilon_{n'}^-(E) - \varepsilon_{n'-2}^-(E)\right)\cdots\left(\varepsilon_{n'}^-(E) - \varepsilon_1^-(E)\right)}} \\ &\times \hat{B}_{\lambda_-(1,l)}^\dagger \hat{B}_{\lambda_-(2,l)}^\dagger \cdots \hat{B}_{\lambda_-(n'-1,l)}^\dagger |\lambda_-(n', l)\rangle \end{aligned} \tag{16.56}$$

with $\hat{B}_{\lambda_-(n',l)}|\lambda_-(n',l)\rangle = 0$, $\hat{\Lambda}|\lambda_-(n',l)\rangle = \hbar\lambda_-(n',l)|\lambda_-(n',l)\rangle$, and $\lambda_-(n',l) = n' - 1 + \sqrt{l^2 - \alpha_F^2}$. Here, we have $n' = 2,\ 3,\ \cdots$; the state with $n' = 1$ is just $|\lambda_-(1,l)\rangle$. The similar normalization factor can be worked out as well. Note that all states $|\lambda\rangle$ are assumed to be normalized $\langle\lambda|\lambda\rangle = 1$.

Let's look at the lowest energy states. We start with $l = 0$, where we have only $j = \frac{1}{2}$. The energy of this state is

$$E\left(1S_{\frac{1}{2}}\right) = \frac{Mc^2}{\sqrt{1 + \frac{\alpha_F^2}{1-\alpha_F^2}}} = Mc^2\sqrt{1-\alpha_F^2}. \tag{16.57}$$

While this state is twofold degenerate with respect to the spin multiplet degeneracy (for $m = \pm\frac{1}{2}$), we usually refer to it as a *nondegenerate* state, because it is not degenerate with any of the other term symbols. This corresponds to the $n = 1$ shell of the nonrelativistic system. Next, we consider the states with $n = 2$. They are made from the state with $l = 0$, $j = \frac{1}{2}$ (the $2S_{\frac{1}{2}}$ state), and $n' = 1$, the state with $l = 1$, $j = \frac{3}{2}$, and $n' = 0$ (the $2P_{\frac{3}{2}}$ state), and the state with $l = 1$, $j = \frac{1}{2}$, and $n' = 1$ (the $2P_{\frac{1}{2}}$ state). You can see that the $2S_{\frac{1}{2}}$ and $2P_{\frac{1}{2}}$ states are degenerate with energy given by

$$E\left(2S_{\frac{1}{2}}\right) = E\left(2P_{\frac{1}{2}}\right) = \frac{Mc^2}{\sqrt{1 + \frac{\alpha_F^2}{(1+\sqrt{1-\alpha_F^2})^2}}} = Mc^2\sqrt{\frac{1+\sqrt{1-\alpha_F^2}}{2}}. \tag{16.58}$$

The energy of the $2P_{\frac{3}{2}}$ state is

$$E\left(2P_{\frac{3}{2}}\right) = \frac{Mc^2}{\sqrt{1 + \frac{\alpha_F^2}{4-\alpha_F^2}}} = Mc^2\frac{\sqrt{4-\alpha_F^2}}{2}. \tag{16.59}$$

You will examine higher shells in the problems. Note that we lifted some degeneracy here due to relativistic effects. In the nonrelativistic solution, all eight states with $n = 2$ were degenerate. Here, they are reduced to two sets of fourfold degenerate states. Note that we have to be careful not to confuse n with n' when working with the relativistic problem. This twofold degeneracy (between different term symbols) continues for all states except for the ground state for each l with $j = l + \frac{1}{2}$, which is nondegenerate. This situation occurs due to a supersymmetry and is discussed in a problem in the next section. Note that we usually do not discuss the additional angular momentum multiplet degeneracy associated with the total angular momentum j.

Section 16.3 Problems:

16.3.1 Energy splitting for $n = 2$

Using the results given above for the energies of the $n = 2$ shell, compute the energy difference, and express it in eV. Use $\alpha_F \approx \frac{1}{137.036}$ and $Mc^2 = 510,999 \times 0.99946\,\text{eV}$ for the reduced mass of the electron in hydrogen (with a proton in the nucleus). Your calculation should use all dimensionless constants except for the Mc^2 term.

16.3.2 Energies of the Eigenstates with $n = 3$

For $n = 3$, we have the following cases: (i) $3S_{\frac{1}{2}}$ with $n' = 2$, $l = 0$, and $j = \frac{1}{2}$; (ii) $3P_{\frac{1}{2}}$ with $n' = 2$, $l = 1$, and $j = \frac{1}{2}$; (iii) $3P_{\frac{3}{2}}$ with $n' = 1$, $l = 1$, and $j = \frac{3}{2}$; (iv) $3D_{\frac{3}{2}}$ with $n' = 1$, $l = 2$, and $j = \frac{3}{2}$; and (v) $3D_{\frac{5}{2}}$ with $n' = 0$, $l = 2$, and $j = \frac{5}{2}$. Determine the formulas for the different energy levels, and plot an energy-level diagram showing the degeneracies and the ordering of the energy levels. Which energy levels are degenerate?

16.3.3 Energies of the Eigenstates with $n = 4$

Repeat the previous problem, but now for $n = 4$.

16.3.4 Relate n' to n

The principal quantum number that we used in the nonrelativistic case can be related to the quantum number n' via $n' = n - j - \frac{1}{2}$. Show how this result is derived.

16.3.5 Details of the Relativistic Factorization Chain

In the main text, we quickly described how to form the factorization chain for the relativistic problem. In this problem, you are asked to flesh out the details and explain carefully how the factorization chain is formed, what the auxiliary

Hamiltonians are, and so on. You should also schematically plot the energies for each factorization chain. You should find there is a degeneracy between two states for each principal quantum number, except the state that is the lowest energy state for a given l value.

16.3.6 Ruling Out the Set of Extra Solutions

When $l = 0$ and $j = l + \frac{1}{2}$, we could have chosen $\beta_\lambda = -1 - \lambda = -1 + \sqrt{1 - \alpha^2} < 0$. This choice only works for $l = 0$, because higher l values do not have $\beta_\lambda < 0$. Expand the energy out for small α_F^2 to show that there are no nonrelativistic states near this energy, ruling out this set of states by comparison with the nonrelativistic limit.

Ans.: You should find that this predicts a bound state at the order of $-100{,}000\,\text{eV}$. Curiously, the wavefunction for this state is well behaved for large and small r, but the state cannot be physical, nevertheless. Indeed, it does suffer from a number of pathological issues when calculating the probability current or the average kinetic energy.

16.4 Relativistic Hydrogen Wavefunctions

Previously, we used a trick of "effectively" squaring the Hamiltonian and introduced the operator $\hat{\Lambda}$, which acted similarly to angular momentum, to determine the relevant physically meaningful eigenvalues of the relativistic problem. In this section, we determine the wavefunctions of hydrogen. To do so requires a number of steps, which we outline next.

The key issue is we have to work out the "shape" or "direction" of the Dirac four-spinor for a given energy eigenstate. This is done by first separating the spinor into its even and odd components (as opposed to working with its large and small components, as we did in the weak relativistic limit). When we do this, we learn that the odd component can be constructed from the even one. Next, we use the even component, to reduce the problem to a two-dimensional Pauli spinor problem. Using states with definite total angular momentum and definite z-component of the total angular momentum, we first find the eigenvectors of $\hat{\Lambda}$ and then reduce the problem to a scalar problem for the energy eigenstate. This last problem is then solved using factorization and the operator form of the Rodrigues formula to determine the wavefunction, as we have done before. In fact, this is closely related to problems that have been solved previously. Finally, we put everything back together and construct the final wavefunction as a Dirac spinor. We have a long job ahead of us. Let us get started.

The first step is to project onto the even and odd components corresponding to the large and small components of the Dirac four-spinor. The way this goes is to work

with the matrix $\gamma_5 = \sigma_x \otimes \mathbb{I}_2$ that we introduced when we discussed the alternate Dirac matrices; it also was a part of the $\hat{\Lambda}$ operator. We create projection operators

$$P_\pm = \frac{1}{2}(\mathbb{I}_4 \pm \gamma_5) = \frac{1}{2}\begin{pmatrix} \mathbb{I}_2 & \pm\mathbb{I}_2 \\ \pm\mathbb{I}_2 & \mathbb{I}_2 \end{pmatrix}. \tag{16.60}$$

Because $\gamma_5^2 = \mathbb{I}_4$, we have $P_\pm^2 = P_\pm$, $P_\pm P_\mp = 0$, and $P_+ + P_- = \mathbb{I}_4$. This means these two operators are projection operators. We can use these operators to reduce the problem from solving for Dirac four-spinors to solving for Pauli two-spinors. We take the Dirac equation for an energy eigenstate $|\Psi\rangle$ (and multiplied by β from the left) and move the mass term to the right hand side. It is given by

$$\Big(c\beta\vec{\alpha}\cdot\hat{\vec{p}} + \beta\big(V(\hat{r}) - E\big)\Big)|\Psi\rangle = -Mc^2|\Psi\rangle. \tag{16.61}$$

Now, using the fact that $\gamma_5\beta = -\beta\gamma_5$ and $\gamma_5\vec{\alpha} = \vec{\alpha}\gamma_5$ (these are most easily verified in the tensor-product form for the matrices), we find that $P_\pm\beta = \beta P_\mp$ and $P_\pm\vec{\alpha} = \vec{\alpha}P_\pm$. Then

$$\begin{aligned} P_-\Big(c\beta\vec{\alpha}\cdot\hat{\vec{p}} + \beta\big(V(\hat{r}) - E\big)\Big)|\Psi\rangle &= \Big(c\beta\vec{\alpha}\cdot\hat{\vec{p}} + \beta\big(V(\hat{r}) - E\big)\Big)P_+|\Psi\rangle \\ &= -Mc^2P_-|\Psi\rangle, \end{aligned} \tag{16.62}$$

or

$$P_-|\Psi\rangle = -\frac{1}{Mc^2}\Big(c\beta\vec{\alpha}\cdot\hat{\vec{p}} + \beta\big(V(\hat{r}) - E\big)\Big)P_+|\Psi\rangle. \tag{16.63}$$

Now, because $P_+ + P_- = \mathbb{I}_4$, we have $|\Psi\rangle = P_+|\Psi\rangle + P_-|\Psi\rangle$ or

$$|\Psi\rangle = \left(1 - \frac{1}{Mc^2}\Big(c\beta\vec{\alpha}\cdot\hat{\vec{p}} + \beta\big(V(\hat{r}) - E\big)\Big)\right)P_+|\Psi\rangle. \tag{16.64}$$

Multiplying by $-Mc^2$ and using the fact that the Dirac equation says that

$$\Big(c\beta\vec{\alpha}\cdot\hat{\vec{p}} + \beta\big(V(\hat{r}) - E\big) + Mc^2\Big)|\Psi\rangle = 0, \tag{16.65}$$

then gives us

$$\Big(c\beta\vec{\alpha}\cdot\hat{\vec{p}} + \beta\big(V(\hat{r}) - E\big) + Mc^2\Big)\Big(c\beta\vec{\alpha}\cdot\hat{\vec{p}} + \beta\big(V(\hat{r}) - E\big) - Mc^2\Big)P_+|\Psi\rangle = 0. \tag{16.66}$$

If we denote the state $|\Psi\rangle$ in terms of the two Pauli spinors ψ and χ, then we have

$$|\Psi\rangle = \begin{pmatrix} \psi \\ \chi \end{pmatrix}, \quad P_+|\Psi\rangle = \begin{pmatrix} \frac{1}{2}(\psi+\chi) \\ \frac{1}{2}(\psi+\chi) \end{pmatrix}, \quad \text{and} \quad P_-|\Psi\rangle = \begin{pmatrix} \frac{1}{2}(\psi-\chi) \\ \frac{1}{2}(-\psi+\chi) \end{pmatrix}. \tag{16.67}$$

In this fashion, we have converted the problem from solving for the Dirac four-spinor to solving for the Pauli two-spinor $\frac{1}{2}(\psi+\chi)$. Once it is found, we use Eq. (16.64) to construct $|\Psi\rangle$. We next solve for Pauli two-spinor component of $P_+|\Psi\rangle$.

If we compare Eqs. (16.66) to (16.34), we see that they are identical, except now, knowing the four-spinor structure of $P_+|\Psi\rangle$ allows us to interpret the equation as limited to a Pauli two-spinor. Hence, after performing a similar simplification as we did before, we now have

$$\left(\frac{E^2}{c^2} - M^2c^2\right)\mathbb{I}_2|\Psi_2\rangle = \left(\hat{p}_r^2\mathbb{I}_2 - \frac{2\alpha_F\hbar E}{c\hat{r}}\mathbb{I}_2 + \frac{\hat{\Lambda}_2(\hat{\Lambda}_2+\hbar)}{\hat{r}^2}\right)|\Psi_2\rangle, \tag{16.68}$$

where

$$\hat{\Lambda}_2 = -\frac{2\hat{\vec{L}}\cdot\hat{\vec{S}}+\hbar^2\mathbb{I}_2}{\hbar} - i\alpha_F\hbar\hat{\vec{e}}_r\cdot\hat{\vec{\sigma}}, \tag{16.69}$$

is the 2×2 truncation of the Λ operator, when acting on symmetric states, and $|\Psi_2\rangle$ is the upper Pauli spinor of the symmetric Dirac four-spinor. Note that here, the spin operator is now represented in the ordinary way as $\hat{\vec{S}} = \frac{\hbar}{2}\vec{\sigma}$. Every operator in the energy eigenvalue equation is a diagonal operator in the two-dimensional Pauli spinor space when it acts on states of definite j, m_z, l, and s except for the operator $\hat{\vec{e}}_r\cdot\vec{\sigma}$. Hence, we need to study this operator in detail to determine how to reduce the Pauli two-spinor equation to a scalar equation.

The operator $\hat{\vec{e}}_r\cdot\vec{\sigma}$ satisfies the following properties when it acts on a state of definite j and l: (i) its eigenvalues are ± 1; (ii) it changes the value of l by ± 1; and (iii) it does not change the value of j. Our first goal is to establish these three properties. The first is rather simple, because the operator is Hermitian and $(\hat{\vec{e}}_r\cdot\vec{\sigma})^2 = \mathbb{I}_2$, which means the eigenvalues of $\hat{\vec{e}}_r\cdot\vec{\sigma}$ must be ± 1. As for the second property, this follows if we write $\hat{\vec{e}}_r = \frac{\hat{\vec{r}}}{\hat{r}}$ and use the fact that $\hat{\vec{r}}$ is a vector operator and $[\hat{\vec{L}}, \hat{r}] = 0$. A vector operator carries orbital angular momentum of $l = 1$. When we combine this with a state of definite l, we will obtain a linear combination of states with $l+1$, l, and $l-1$. But $\hat{\vec{r}}$ is odd under the spatial parity operation of $\hat{\vec{r}} \to -\hat{\vec{r}}$, so it changes the parity of the state. This means there can be no l component, only components with $l+1$ and $l-1$, each which has opposite parity from l. For the last one, we have to show that $[\hat{\vec{J}}, \hat{\vec{e}}_r\cdot\vec{\sigma}] = 0$, because commuting operators can have eigenstates with definite eigenvalues for both operators. This requires a direct calculation to verify it. We find

$$[\hat{J}_i, \hat{\vec{e}}_r \cdot \vec{\sigma}] = \sum_j \left(\frac{1}{\hat{r}} [\hat{L}_i, \hat{r}_j] \sigma_j + \frac{\hat{r}_j}{\hat{r}} [\hat{S}_i, \sigma_j] \right)$$
$$= i\hbar \sum_j \sum_k \epsilon_{ijk} \left(\frac{\hat{r}_k}{\hat{r}} \sigma_j + \frac{\hat{r}_j}{\hat{r}} \sigma_k \right) = 0, \tag{16.70}$$

since the term in the parenthesis is symmetric under $j \leftrightarrow k$, while the Levi-Civita tensor is antisymmetric in this interchange. Because they commute, the operator does not change j. We have to think a bit to understand how this can work. If I have a state of definite j, then it has $l = j \pm \frac{1}{2}$. What these results tell us is that

$$\hat{\vec{e}}_r \cdot \vec{\sigma} |j, m, l = j + \tfrac{1}{2}, s\rangle = e^{i\theta} |j, m, l = j - \tfrac{1}{2}, s\rangle, \tag{16.71}$$

and with the factor $e^{-i\theta}$ times the $l = j + \frac{1}{2}$ state, when operator operates on the $l = j - \frac{1}{2}$ state. We can always pick $\theta = \pi$, as discussed in Problem 16.4.1, which we do here. See how that problem explicitly verifies this result below using harmonic polynomials for $j = \frac{1}{2}$.

We are now ready to find the eigenvectors of $\hat{\Lambda}_2$. We first recall that the eigenvalue of $2\hat{\vec{L}} \cdot \hat{\vec{S}} + \hbar^2$ is given by $\hbar^2(l+1)$ for $j = l + \frac{1}{2}$ $\left(\Lambda_+ = -\hbar\sqrt{(l+1)^2 - \alpha_F^2}\right)$ and $-\hbar^2 l$ for $j = l - \frac{1}{2}$ $\left(\Lambda_- = \hbar\sqrt{l^2 - \alpha_F^2}\right)$. Now, the eigenvectors of $\hat{\Lambda}_2$ are formed by linear combinations of the following expressions:

$$|\Lambda_+\rangle = \alpha_+ |j{=}l + \tfrac{1}{2}, m, l, s\rangle + \beta_+ |j{=}l + \tfrac{1}{2}, m, l+1, s\rangle \quad l = 0, 1, \cdots \tag{16.72}$$

with $\hat{\Lambda}_2 = -\hbar(l+1)\sigma_z + i\hbar\alpha_F\sigma_x$ and

$$|\Lambda_-\rangle = \alpha_- |j{=}l - \tfrac{1}{2}, m, l, s\rangle + \beta_- |j{=}l - \tfrac{1}{2}, m, l-1, s\rangle \quad l = 1, 2, \cdots \tag{16.73}$$

with $\hat{\Lambda}_2 = \hbar l\sigma_z + i\hbar\alpha_F\sigma_x$. In order for these results to properly reproduce the nonrelativistic limit, where the states have angular momentum l, we must have the $\beta_\pm$ coefficients approach 0 when the relativistic corrections are ignored. At this stage, we simply act $\hat{\Lambda}_2$ onto these states and force them to work. We find

$$\frac{\alpha_+}{\beta_+} = \frac{i\alpha_F}{l + 1 - \sqrt{(l+1)^2 - \alpha_F^2}} \quad \text{and} \quad \frac{\alpha_-}{\beta_-} = -\frac{i\alpha_F}{l - \sqrt{l^2 - \alpha_F^2}}. \tag{16.74}$$

This might not look like β is small when $\alpha_F \to 0$, but because the denominator approaches $O(\alpha_F^2)$ for small α_F, the β coefficients are indeed small compared to the α coefficients as we require in that limit. The last step is to produce the normalized constants. This gives

$$\alpha_+ = \frac{i\alpha_F}{\sqrt{2(l+1)^2-2(l+1)\sqrt{(l+1)^2-\alpha_F^2}}} \text{ and } \beta_+ \quad = \frac{l+1-\sqrt{(l+1)^2-\alpha_F^2}}{\sqrt{2(l+1)^2-2(l+1)\sqrt{(l+1)^2-\alpha_F^2}}}, \tag{16.75}$$

and

$$\alpha_- = -\frac{i\alpha_F}{\sqrt{2l^2-2l\sqrt{l^2-\alpha_F^2}}} \text{ and } \beta_- = \frac{l-\sqrt{l^2-\alpha_F^2}}{\sqrt{2l^2-2l\sqrt{l^2-\alpha_F^2}}}. \tag{16.76}$$

This leaves us with two radial scalar equations we have to solve due to the schematic tensor-product structure of these states in the form $|\psi_r\rangle \otimes |\Lambda\rangle$ with the state $|\Lambda\rangle$ being a Pauli two-spinor and $|\psi_r\rangle$ being a scalar state. The two radial equations are given by

$$\left(\frac{E^2}{c^2} - M^2c^2\right)|\psi_r^+\rangle = \left(\hat{p}_r^2 - \frac{2\alpha_F\hbar E}{c\hat{r}} + \frac{\Lambda_+(\Lambda_+ + \hbar)}{\hat{r}^2}\right)|\psi_r^+\rangle \tag{16.77}$$

for $j = l + \frac{1}{2}$ and

$$\left(\frac{E^2}{c^2} - M^2c^2\right)|\psi_r^-\rangle = \left(\hat{p}_r^2 - \frac{2\alpha_F\hbar E}{c\hat{r}} + \frac{\Lambda_-(\Lambda_- + \hbar)}{\hat{r}^2}\right)|\psi_r^-\rangle \tag{16.78}$$

for $j = l - \frac{1}{2}$. Both of these equations have an identical form and we already know how to factorize them. We do this next, and then we determine the radial wavefunctions.

As we saw in the previous section, we use raising operators of the same form as we had for hydrogen, but the parameters in the raising operator are now different and given by $\Lambda_\pm(n', l)$. The raising operators take the form (do not conflate $\alpha_\pm$ with α_λ and similarly with β and β_λ)

$$\hat{B}_\lambda^\dagger = \hat{p}_r + i\hbar\left(\frac{\alpha_\lambda}{\bar{a}_0} + \frac{\beta_\lambda}{\hat{r}}\right). \tag{16.79}$$

Note that for the relativistic case, we no longer have the $\frac{1}{\sqrt{2M}}$ factor, and the form is identical to what we had before with λ essentially taking the role of l. Here, we have for the case $j = l + \frac{1}{2}$ that $\lambda_+(n', l) = -n' - \sqrt{(l+1)^2 - \alpha_F^2}$ (with $n' = 0, 1, \cdots$) and then $\beta_\lambda = \lambda_+(n', l)$ and $\alpha_l = -\frac{1}{\lambda_+(n',l)}$. While for $j = l - \frac{1}{2}$, we have $\lambda_-(n', l) = n' - 1 + \sqrt{l^2 - \alpha_F^2}$ (with $n' = 1, 2, \cdots$) and then $\beta_\lambda = -1 - \lambda_-(n', l)$ and $\alpha_l = \frac{1}{1+\lambda_-(n',l)}$. In both cases, we can determine the similarity transformation for the raising operator as

$$\hat{B}_{\lambda}^{\dagger} = \hat{r}^{\beta_\lambda} e^{\frac{\alpha_\lambda \hat{r}}{\bar{a}_0}} \hat{p}_r e^{-\frac{\alpha_\lambda \hat{r}}{\bar{a}_0}} \hat{r}^{-\beta_\lambda} \tag{16.80}$$

and then we have the state annihilated by the momentum operator is given by

$$\hat{p}_r e^{\frac{\alpha_\lambda \hat{r}}{\bar{a}_0}} \hat{r}^{\beta_\lambda} |\phi_r\rangle = 0, \tag{16.81}$$

where the state $|\phi_r\rangle$ is the appropriate auxiliary ground state for the given case being considered. We use the label ϕ, because these states are auxiliary ground states in the factorization chain once we start forming it. It is only the leftmost element of the factorization chain which produces an energy eigenstate of the original Hamiltonian.

We now work out the case of $j = l + \frac{1}{2}$. We have

$$\begin{aligned}
&|\psi_{n'}; l, j = l + \tfrac{1}{2}\rangle \\
&\quad = C^{+}_{n',l} \hat{r}^{-\sqrt{(l+1)^2-\alpha_F^2}} e^{\frac{\hat{r}}{\sqrt{(l+1)^2-\alpha_F^2}\bar{a}_0}} \hat{p}_r e^{-\frac{\hat{r}}{\left(1+\sqrt{(l+1)^2-\alpha_F^2}\right)\sqrt{(l+1)^2-\alpha_F^2}\bar{a}_0}} \frac{1}{\hat{r}} \\
&\qquad \times \hat{p}_r e^{\frac{\hat{r}}{\left(2+\sqrt{(l+1)^2-\alpha_F^2}\right)\left(1+\sqrt{(l+1)^2-\alpha_F^2}\right)\bar{a}_0}} \frac{1}{\hat{r}} \cdots \\
&\qquad \times \hat{p}_r e^{-\frac{\hat{r}}{\left(n'-1+\sqrt{(l+1)^2-\alpha_F^2}\right)\bar{a}_0}} \hat{r}^{n'-1+\sqrt{(l+1)^2-\alpha_F^2}} |\phi^{+}_{n'}; l\rangle.
\end{aligned} \tag{16.82}$$

Using the state annihilated by $\hat{p}_r$, we can re-express this in terms of nested commutators via

$$\begin{aligned}
|\psi_{n'}; l, j = l + \tfrac{1}{2}\rangle &= C^{+}_{n',l} \hat{r}^{-n'+1-2\sqrt{(l+1)^2-\alpha_F^2}} e^{\frac{\left(n'+2\sqrt{(l+1)^2-\alpha_F^2}\right)\hat{r}}{\left(n'+\sqrt{(l+1)^2-\alpha_F^2}\right)\sqrt{(l+1)^2-\alpha_F^2}\bar{a}_0}} \\
&\times \left[\frac{1}{\hat{r}}\hat{p}_r, e^{-\frac{\hat{r}}{\left(1+\sqrt{(l+1)^2-\alpha_F^2}\right)\sqrt{(l+1)^2-\alpha_F^2}\bar{a}_0}} \left[\frac{1}{\hat{r}}\hat{p}_r, e^{-\frac{\hat{r}}{\left(2+\sqrt{(l+1)^2-\alpha_F^2}\right)\left(1+\sqrt{(l+1)^2-\alpha_F^2}\right)\bar{a}_0}},\right.\right. \\
&\times \left[\frac{1}{\hat{r}}\hat{p}_r, \ldots\ldots, e^{-\frac{\hat{r}}{\left(n'-1+\sqrt{(l+1)^2-\alpha_F^2}\right)\left(n'-2+\sqrt{(l+1)^2-\alpha_F^2}\right)\bar{a}_0}}\right. \\
&\times \left.\left.\left.\left[\frac{1}{\hat{r}}\hat{p}_r, e^{-\frac{\left(2n'-1+2\sqrt{(l+1)^2-\alpha_F^2}\right)\hat{r}}{\left(n'+\sqrt{(l+1)^2-\alpha_F^2}\right)\left(n'-1+\sqrt{(l+1)^2-\alpha_F^2}\right)\bar{a}_0}} \hat{r}^{2n'-1+2\sqrt{(l+1)^2-\alpha_F^2}}\right] \cdots\right]\right]\right]_{n'} \\
&|\phi^{+}_{n'}; l\rangle.
\end{aligned} \tag{16.83}$$

This expression agrees with what we had before for hydrogen, if we take $n{=}n'{+}\sqrt{(l{+}1)^2{-}\alpha_F^2}$ in the exponents. The next step is to establish the Laguerre polynomial that this corresponds to.

When $n' = 0$, the result is trivial and $|n' = 0, l,\, j = l + \frac{1}{2}\rangle = |\phi^+_{n'=0}; l\rangle$. For $n' = 1$, we have

$$
\begin{aligned}
&|\psi_{n'=1}; l,\, j{=}l + \tfrac{1}{2}\rangle \\
&\quad = C^+_{n'=1,l}\frac{-i\hbar}{2}\frac{1}{\hat r}\frac{1{+}2\sqrt{(l{+}1)^2{-}\alpha_F^2}}{\sqrt{(l{+}1)^2{-}\alpha_F^2}}\left(2\sqrt{(l{+}1)^2{-}\alpha_F^2}-\frac{2\hat r}{\left(1{+}\sqrt{(l{+}1)^2{-}\alpha_F^2}\right)\tilde a_0}\right)|\phi^+_{n'}; l\rangle.
\end{aligned}
\tag{16.84}
$$

The case with $n' = 2$ is a bit more complicated. It gives us

$$
\begin{aligned}
&|\psi_{n'=2}; l,\, j{=}l + \tfrac{1}{2}\rangle = C^+_{n'=2,l}\left(\frac{-i\hbar}{2}\right)^2\frac{1}{\hat r^2}\frac{\left(3{+}2\sqrt{(l{+}1)^2{-}\alpha_F^2}\right)\left(2{+}2\sqrt{(l{+}1)^2{-}\alpha_F^2}\right)2}{\left(1{+}\sqrt{(l{+}1)^2{-}\alpha_F^2}\right)\sqrt{(l{+}1)^2)-\alpha_F^2}} \\
&\quad\times\left(\frac{\left(1{+}2\sqrt{(l{+}1)^2{-}\alpha_F^2}\right)2\sqrt{(l{+}1)^2{-}\alpha_F^2}}{2}\right. \\
&\quad\left.-\left(1+2\sqrt{(l+1)^2-\alpha_F^2}\right)\frac{2\hat r}{\left(2{+}\sqrt{(l{+}1)^2{-}\alpha_F^2}\right)a_0}+\frac{1}{2}\left(\frac{2\hat r}{\left(2{+}\sqrt{(l{+}1)^2{-}\alpha_F^2}\right)a_0}\right)^2\right)|\phi^+_{n'}; l\rangle.
\end{aligned}
\tag{16.85}
$$

This then allows us to determine the general ansatz for the nested commutators as

$$
\begin{aligned}
&\hat r^{-n'+1-2\sqrt{(l+1)^2-\alpha_F^2}}e^{\frac{\left(n'+2\sqrt{(l+1)^2-\alpha_F^2}\right)\hat r}{\left(n'+\sqrt{(l+1)^2-\alpha_F^2}\right)\sqrt{(l+1)^2-\alpha_F^2}\tilde a_0}} \\
&\quad\times\left[\frac{1}{\hat r}\hat p_r, e^{-\frac{\hat r}{\left(1+\sqrt{(l+1)^2-\alpha_F^2}\right)\sqrt{(l+1)^2-\alpha_F^2}\tilde a_0}}\right]\left[\frac{1}{\hat r}\hat p_r, e^{-\frac{\hat r}{\left(2+\sqrt{(l+1)^2-\alpha_F^2}\right)\left(1+\sqrt{(l+1)^2-\alpha_F^2}\right)\tilde a_0}}\right], \\
&\quad\left[\frac{1}{\hat r}\hat p_r,\cdots\right. \\
&\quad\left.\cdots, e^{-\frac{\hat r}{\left(n'-1+\sqrt{(l+1)^2-\alpha_F^2}\right)\left(n'-2+\sqrt{(l+1)^2-\alpha_F^2}\right)\tilde a_0}}\right. \\
&\quad\left.\times\left[\frac{1}{\hat r}\hat p_r, e^{-\frac{\left(2n'-1+2\sqrt{(l+1)^2-\alpha_F^2}\right)\hat r}{\left(n'+\sqrt{(l+1)^2-\alpha_F^2}\right)\left(n'-1+\sqrt{(l+1)^2-\alpha_F^2}\right)\tilde a_0}}\hat r^{2n'-1+2\sqrt{(l+1)^2-\alpha_F^2}}\right]\cdots\right]\Bigg]_{n'} \\
&= \left(\frac{-i\hbar}{2}\right)^{n'}\left(2n'-1+2\sqrt{(l+1)^2-\alpha_F^2}\right)\left(2n'-2+2\sqrt{(l+1)^2-\alpha_F^2}\right)\cdots\left(n'+2\sqrt{(l+1)^2-\alpha_F^2}\right)\times
\end{aligned}
$$

$$\frac{n'!}{\left(n'-1+\sqrt{(l+1)^2-\alpha_F^2}\right)\left(n'-2+\sqrt{(l+1)^2-\alpha_F^2}\right)\cdots\sqrt{(l+1)^2-\alpha_F^2}}$$
$$\times L_{n'}^{\left(2\sqrt{(l+1)^2-\alpha_F^2}-1\right)}\left(\frac{2\hat{r}}{\left(n'+\sqrt{(l+1)^2-\alpha_F^2}\right)\bar{a}_0}\right). \tag{16.86}$$

The proof follows the same strategy as we used before. It is just that here we have more complicated exponents to work with. You verify this result in the problems.

We still need to determine the auxiliary ground-state radial wavefunction given by $\langle r|\phi_{n'}^+; l\rangle$. To find this, we recall that it satisfies $\hat{B}_\lambda|\phi_{n'}^+; l\rangle = 0$, or

$$\hat{p}_r|\phi_{n'}^+; l\rangle = i\hbar\left(\frac{1}{\left(n'+\sqrt{(l+1)^2-\alpha_F^2}\right)\bar{a}_0} - \frac{n'+\sqrt{(l+1)^2-\alpha_F^2}}{\hat{r}}\right)|\phi_{n'}^+; l\rangle. \tag{16.87}$$

We add $\frac{i\hbar}{\hat{r}}|\phi_{n'}^+; l\rangle$ to both sides of the equation and then find the state annihilated by this operator, which is

$$\left(\hat{p}_r + \frac{i\hbar}{\hat{r}}\right) r^{-n'+1-\sqrt{(l+1)^2-\alpha_F^2}} e^{\frac{\hat{r}}{\left(n'+\sqrt{(l+1)^2-\alpha_F^2}\right)\bar{a}_0}}|\phi_{n'}^+; l\rangle = 0. \tag{16.88}$$

Then by using our multiply-by-one trick and introducing the translation operator in spherical coordinates, we find that

$$\langle r|\phi_{n'}^+; l\rangle = \langle 0|\frac{1}{\hat{r}^{n'-1+\sqrt{(l+1)^2-\alpha_F^2}}}|\phi_{n'}^+; l\rangle r^{n'-1+\sqrt{(l+1)^2-\alpha_F^2}} e^{-\frac{r}{\left(n'+\sqrt{(l+1)^2-\alpha_F^2}\right)\bar{a}_0}}, \tag{16.89}$$

where the first factor on the right hand side is the normalization constant. You can immediately see that for the s-wave case, with $l = 0$ and $n' = 0$, that the wavefunction diverges as a small power law as $r \to 0$. This divergence is rather weak, but will be important in our analysis of the wavefunctions themselves. Note that the state remains normalizable in spite of this divergence.

The normalization constant is complicated to work out for the relativistic problems, and we will not state explicitly what it is here. In the end, the normalization must be worked out for the final four-vector form of the wavefunction, as opposed to the two-spinor result. We now summarize the results we have found for the case $j = l + \frac{1}{2}$. Using our schematic notation, we have

$$\psi_{n',l,j=l+\frac{1}{2}}(r) = CL_{n'}^{\left(2\sqrt{(l+1)^2-\alpha_F^2}-1\right)}\left(\frac{2\hat{r}}{\left(n'+\sqrt{(l+1)^2-\alpha_F^2}\right)\bar{a}_0}\right)$$
$$\times r^{n'-1+\sqrt{(l+1)^2-\alpha_F^2}} e^{-\frac{r}{\left(n'+\sqrt{(l+1)^2-\alpha_F^2}\right)\bar{a}_0}} \otimes |\Lambda_+\rangle, \tag{16.90}$$

in a "tensor product" with the $|\Lambda_+\rangle$ state. It is the form of the Lambda state that determines the two-spinor we need to then apply the projection operator onto to determine the four-spinor. To determine the $|\Lambda_+\rangle$ state, we first need to express the total angular momentum states in terms of the orbital and spin angular momentum. Doing this, shows us that the $|\Lambda_+\rangle$ state is expressed as

$$|\Lambda_+\rangle = \alpha_+ \begin{pmatrix} \sqrt{\frac{l+m+\frac{1}{2}}{2l+1}}|l, m-\frac{1}{2}\rangle \\ \sqrt{\frac{l-m+\frac{1}{2}}{2l+1}}|l, m+\frac{1}{2}\rangle \end{pmatrix} + \beta_+ \begin{pmatrix} -\sqrt{\frac{l+1-m+\frac{1}{2}}{2l+3}}|l+1, m-\frac{1}{2}\rangle \\ \sqrt{\frac{l+1+m+\frac{1}{2}}{2l+3}}|l+1, m+\frac{1}{2}\rangle \end{pmatrix}, \tag{16.91}$$

in the two-spinor notation (recall that the upper component of the spinor is $|\uparrow\rangle_z$ and the bottom component is $|\downarrow\rangle_z$). The case with $j = l - \frac{1}{2}$ is discussed in Problem 16.4.7.

We now extend this to a four-spinor, by copying the two-spinor twice—once in the upper two components and once in the lower two components. We next need to apply the operator in Eq. (16.64) to the state $\hat{P}_+|\Psi\rangle$, which we have already determined. Everything is easy to apply, except for the $\vec{\alpha}\cdot\hat{\vec{p}}$ operator. So, we work on that operator next.

Our energy eigenstate is determined by the action of the radial momentum onto it, so we do not know what the action of the Cartesian coordinates of momentum are on the energy eigenstate. The way around this issue is to rewrite $\vec{\alpha}\cdot\hat{\vec{p}}$ as $\vec{\alpha}\cdot\hat{\vec{e}}_r\vec{\alpha}\cdot\hat{\vec{e}}_r\vec{\alpha}\cdot\hat{\vec{p}}$, because the product of the first two terms is given by $\mathbb{I}_4$. Next, we instead multiply the second and third terms together, which gives us

$$\vec{\alpha}\cdot\hat{\vec{e}}_r\,\vec{\alpha}\cdot\hat{\vec{p}} = \mathbb{I}_2 \otimes \frac{1}{\hat{r}}\left(\hat{\vec{r}}\cdot\hat{\vec{p}} + i\vec{\sigma}\cdot\hat{\vec{L}}\right) = \mathbb{I}_2 \otimes \left(\hat{p}_r + \frac{i\hbar}{\hat{r}} + \frac{i\vec{\sigma}\cdot\hat{\vec{L}}}{\hat{r}}\right). \tag{16.92}$$

Now, we have our final formula, which is

$$\vec{\alpha}\cdot\hat{\vec{p}} = \sigma_x \otimes \vec{\sigma}\cdot\hat{\vec{e}}_r\left(\hat{p}_r + \frac{i\hbar}{\hat{r}} + \frac{i\vec{\sigma}\cdot\hat{\vec{L}}}{\hat{r}}\right) = \vec{\alpha}\cdot\hat{\vec{e}}_r\left(\hat{p}_r + \frac{i\hbar}{\hat{r}} + \frac{i\vec{\sigma}\cdot\hat{\vec{L}}}{\hat{r}}\right). \tag{16.93}$$

This is now in terms of operators that we know how they act on the different states. The operators $\vec{\sigma}\cdot\hat{\vec{e}}_r$ and $\vec{\sigma}\cdot\hat{\vec{L}}$ act only on the angular momentum states, while $\hat{r}$ and $\hat{p}_r$ both act only on the radial state. Next, we use the fact that $(\vec{\alpha}\cdot\hat{\vec{e}}_r)^2 = \mathbb{I}_4$ to re-express $\vec{\alpha}\cdot\hat{\vec{p}} - \frac{\alpha_F\hbar}{\hat{r}}$ as

$$\vec{\alpha}\cdot\hat{\vec{p}} - \frac{\alpha_F\hbar}{\hat{r}} = \vec{\alpha}\cdot\hat{\vec{e}}_r\left(\hat{p}_r + \frac{i\hbar + i\vec{\sigma}\cdot\hat{\vec{L}} - \vec{\alpha}\cdot\hat{\vec{e}}_r\alpha_F\hbar}{\hat{r}}\right) = \vec{\alpha}\cdot\hat{\vec{e}}_r\left(\hat{p}_r - \frac{i\hat{\Lambda}}{\hat{r}}\right). \tag{16.94}$$

Using this result, we find that

$$Mc^2 - c\beta\vec{\alpha}\cdot\hat{\vec{p}} + \beta\left(\frac{\alpha_F\hbar c}{\hat{r}} + E\right) = Mc^2 - c\beta\vec{\alpha}\cdot\hat{\vec{e}}_r\left(\hat{p}_r - \frac{i\hat{\Lambda}}{\hat{r}}\right) + \beta E. \quad (16.95)$$

Because the $\hat{\Lambda}$ operator commutes with $\hat{p}_r$ and $\hat{r}$, it can move through all of the $\hat{B}$ operators and act directly on the state, where it is replaced by $\hbar\lambda_+(n', l)$. Then using the definition of the $\hat{B}$ operators, we find that

$$\begin{aligned}
&|\Psi_{n'}; l, j = l + \tfrac{1}{2}\rangle \\
&\quad = \frac{1}{Mc^2}\left[Mc^2 - \beta\vec{\alpha}\cdot\hat{\vec{e}}_r\left(c\hat{B}_{\lambda_+(0,l)} + \frac{i\hbar cn'}{\hat{r}} - \frac{i\hbar c}{\lambda_+(n', l)\bar{a}_0}\right) + \beta E\right] \\
&\qquad \times \begin{pmatrix} |\psi_{n',l,j+\frac{1}{2}}\rangle \otimes |\lambda_+(n'; l, j + \frac{1}{2})\rangle \\ |\psi_{n',l,j+\frac{1}{2}}\rangle \otimes |\lambda_+(n'; l, j + \frac{1}{2})\rangle \end{pmatrix},
\end{aligned} \quad (16.96)$$

with the $|\lambda_+\rangle$ state expressed as a two-spinor. Now, we have a string of $\hat{B}^\dagger$ operators acting on $|\phi^+_{n'}; l\rangle$, but the $\hat{B}$ operator can be combined with its matching $\hat{B}^\dagger$ operator that can be rewritten as $\hat{H}_1^{(\text{aux})} - \varepsilon_0(E)$ and then with the intertwining operation moved through until it hits the state at the right, where it gives $\varepsilon_{n'}(E) - \varepsilon_0(E)$. But, this operation has also removed the rightmost operator $\hat{B}^\dagger_{\lambda_+(0,l)}$ from the quantum state. This requires us to repeat the operator Rodrigues formula derivation without the last nested commutator. While this calculation is messy, it can be worked out and we leave the further details to the problems.

We will work out the simplest case though, which corresponds to the case of $n' = 0$. In this case, we have $E = Mc^2\frac{\sqrt{(l+1)^2-\alpha_F^2}}{l+1}$, and after substituting in for $\bar{a}_0$, we find that

$$\begin{aligned}
&|\Psi_0; l, j = l + \tfrac{1}{2}\rangle \\
&= \frac{1}{l+1}\begin{pmatrix} \left(l + 1 - \sqrt{(l+1)^2 - \alpha_F^2} + i\alpha_F\vec{\alpha}\cdot\hat{\vec{e}}_r\right)|\psi_{0,l,j+\frac{1}{2}}\rangle \otimes |\lambda_+(0; l, j + \frac{1}{2})\rangle \\ \left(l + 1 + \sqrt{(l+1)^2 - \alpha_F^2} - i\alpha_F\vec{\alpha}\cdot\hat{\vec{e}}_r\right)|\psi_{0,l,j+\frac{1}{2}}\rangle \otimes |\lambda_+(0; l, j + \frac{1}{2})\rangle \end{pmatrix}.
\end{aligned} \quad (16.97)$$

Re-expressing the $|\Lambda_+\rangle$ state in terms of the angular momentum states and simplifying with the α_+ and β_+ coefficients gives us

$$\begin{aligned}
&|\Psi_0; l, j = l + \tfrac{1}{2}\rangle \\
&\quad = \sqrt{\frac{2}{l+1}}\frac{i}{\sqrt{l+1+\sqrt{(l+1)^2-\alpha_F^2}}}\begin{pmatrix} \sqrt{l+1+\sqrt{(l+1)^2-\alpha_F^2}}|\psi_{0,l,j+\frac{1}{2}}\rangle\otimes|j=l+\frac{1}{2},m,l,s\rangle \\ -i\alpha_F|\psi_{0,l,j+\frac{1}{2}}\rangle\otimes|j=l+\frac{1}{2},m,l+1,s\rangle \end{pmatrix},
\end{aligned} \quad (16.98)$$

which has the "l" state in the upper spinor and the "$l+1$" state in the lower spinor. This only occurs for the case $n'=0$; we multiplied the numerator and denominator of both terms by $\sqrt{l+1-\sqrt{(l+1)^2-\alpha_F^2}}$ to put them into a form where it is easier to see what are the large and small components. While it is possible to work out all of the components for the general case (the Wikipedia page on hydrogen-like atoms has an excellent summary of the exact formulas for all cases), it is a challenge to get all of the details right and we will not do that here.

We now will discuss the lowest energy states corresponding to $1S_{\frac{1}{2}}$, $2S_{\frac{1}{2}}$, $2P_{\frac{1}{2}}$, and $2P_{\frac{3}{2}}$. Two of these states are the $n'=0$ and $j=l+\frac{1}{2}$ cases we worked out above, while one is a $n'=1$ case of $j=l+\frac{1}{2}$. The remaining one is a $n'=1$ case for $j=l-\frac{1}{2}$. Ignoring the overall normalization constant, we show the shape of the two $n'=0$ cases and then compare the radial wavefunctions of the large components of the Dirac four-spinor with the nonrelativistic results we used before (here with a normalization that is the same for both). We have the wavefunction spinor is composed of factors that include $\psi_{n',l,j\pm\frac{1}{2}}(r)=\langle r|\psi_{n',l,j\pm\frac{1}{2}}\rangle$ and $Y_{lm}(\theta,\phi)=\langle\theta,\phi|l,m\rangle$. We start with the $1S_{\frac{1}{2}}$ state which has $m=\frac{1}{2}$ and $m=-\frac{1}{2}$:

$$\Psi\left(\vec{r},m{=}\tfrac{1}{2};1S_{\frac{1}{2}}\right)=C\begin{pmatrix}\sqrt{1+\sqrt{1-\alpha_F^2}}\frac{1}{\sqrt{4\pi}}\\0\\i\alpha_F\frac{1}{\sqrt{4\pi}}\cos\theta\\i\alpha_F\frac{1}{\sqrt{4\pi}}\sin\theta e^{i\phi}\end{pmatrix}r^{\sqrt{1-\alpha_F^2}-1}e^{-\frac{r}{a_0}},\tag{16.99}$$

$$\Psi\left(\vec{r},m{=}-\tfrac{1}{2};1S_{\frac{1}{2}}\right)=C\begin{pmatrix}0\\\sqrt{1+\sqrt{1-\alpha_F^2}}\frac{1}{\sqrt{4\pi}}\\-i\alpha_F\frac{1}{\sqrt{4\pi}}\sin\theta e^{-i\phi}\\i\alpha_F\frac{1}{\sqrt{4\pi}}\cos\theta\end{pmatrix}r^{\sqrt{1-\alpha_F^2}-1}e^{-\frac{r}{a_0}},\tag{16.100}$$

with C a normalization constant, which satisfies $C=\left(\frac{2}{a_0}\right)^{\frac{3}{2}}/\sqrt{2\Gamma(1+2\sqrt{1-\alpha_F^2})}$. Note how one of the components vanishes in each spinor. The $2P_{\frac{3}{2}}$ is similar. Two of its components are

$$\Psi\left(\vec{r},m{=}\tfrac{3}{2};1P_{\frac{3}{2}}\right)=C'\begin{pmatrix}-\sqrt{2+\sqrt{2-\alpha_F^2}}\sqrt{\frac{3}{8\pi}}\sin\theta e^{i\phi}\\0\\-i\alpha_F\sqrt{\frac{3}{8\pi}}\cos\theta\sin\theta e^{i\phi}\\-i\alpha_F\sqrt{\frac{5}{16\pi}}\sin^2\theta e^{2i\phi}\end{pmatrix}r^{\sqrt{2-\alpha_F^2}-1}e^{-\frac{r}{2a_0}},\tag{16.101}$$

$$\Psi\left(\vec{r}, m{=}\tfrac{1}{2}; 1P_{\frac{3}{2}}\right) = C'' \begin{pmatrix} \sqrt{2{+}\sqrt{2{-}\alpha_F^2}}\sqrt{\frac{1}{2\pi}}\cos\theta \\ -\sqrt{2{+}\sqrt{2{-}\alpha_F^2}}\sqrt{\frac{1}{8\pi}}\sin\theta e^{i\phi} \\ -i\alpha_F\sqrt{\frac{1}{8\pi}}(3\cos^2\theta - 1) \\ -i\alpha_F\sqrt{\frac{9}{8\pi}}\cos\theta\sin\theta e^{i\phi} \end{pmatrix} r^{\sqrt{2-\alpha_F^2}-1} e^{-\frac{r}{2a_0}}, \tag{16.102}$$

with C' and C'' normalization constants that we won't explicitly specify. We won't show the other two components, which are similar. For the $2S_{\frac{1}{2}}$ state, the radial function behaves as

$$\begin{aligned} & L_1^{\left(2\sqrt{1-\alpha_F^2}\right)}\left(\frac{2r}{\sqrt{2\left(1+\sqrt{1-\alpha_F^2}\right)}}\right)\left(\frac{2r}{\sqrt{2\left(1+\sqrt{1-\alpha_F^2}\right)}}\right)^{\sqrt{1-\alpha_F^2}} e^{-\frac{r}{\sqrt{2\left(1+\sqrt{1-\alpha_F^2}\right)}}} \\ &= e^{-\frac{r}{\sqrt{2\left(1+\sqrt{1-\alpha_F^2}\right)}}} \\ &\quad \times \left(\sqrt{2\left(1+\sqrt{1-\alpha_F^2}\right)}\left(\frac{2r}{\sqrt{2\left(1+\sqrt{1-\alpha_F^2}\right)}}\right)^{\sqrt{1-\alpha_F^2}-1}\right. \\ &\quad \left. -\frac{1+\sqrt{2\left(1+\sqrt{1-\alpha_F^2}\right)}}{1+2\sqrt{1-\alpha_F^2}}\left(\frac{2r}{\sqrt{2\left(1+\sqrt{1-\alpha_F^2}\right)}}\right)^{\sqrt{1-\alpha_F^2}}\right). \end{aligned} \tag{16.103}$$

While for the $2P_{\frac{1}{2}}$ state, we have

$$\begin{aligned} & \left(\left(\sqrt{2\left(1+\sqrt{1-\alpha_F^2}\right)}-2\right)\left(\frac{2r}{\sqrt{2\left(1+\sqrt{1-\alpha_F^2}\right)}}\right)^{\sqrt{1-\alpha_F^2}-1}\right. \\ &\quad \left. +\frac{1-\sqrt{2\left(1+\sqrt{1-\alpha_F^2}\right)}}{1+2\sqrt{1-\alpha_F^2}}\left(\frac{2r}{\sqrt{2\left(1+\sqrt{1-\alpha_F^2}\right)}}\right)^{\sqrt{1-\alpha_F^2}}\right) e^{-\frac{r}{\sqrt{2\left(1+\sqrt{1-\alpha_F^2}\right)}}}. \end{aligned} \tag{16.104}$$

There are a few notable points to mention about these wavefunctions. The exponential decay for the $1S_{\frac{1}{2}}$ and $2P_{\frac{3}{2}}$ are the same as the nonrelativistic case—the exponential functions decay a bit faster for the $2S_{\frac{1}{2}}$ and $2P_{\frac{1}{2}}$ states, because $\sqrt{2\left(1+\sqrt{1-\alpha_F^2}\right)}$ is slightly smaller than 2. Furthermore, since $\sqrt{1-\alpha_F^2}-1<0$, we see

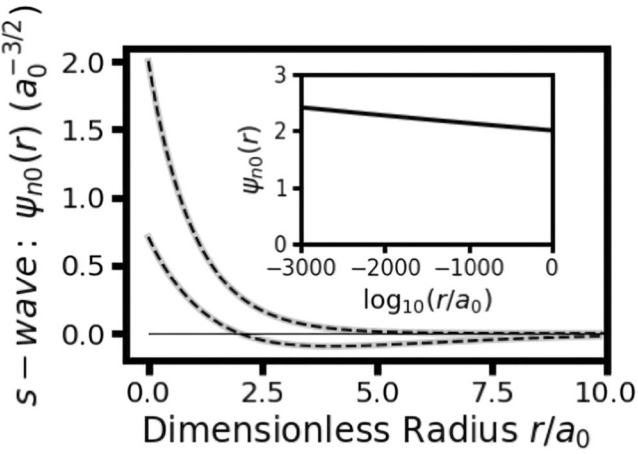

Fig. 16.2 Comparison of the nonrelativistic results (solid gray lines) for $1S_{\frac{1}{2}}$ (no node) and $2S_{\frac{1}{2}}$ (one node) to the relativistic results (dashed lines). You can see the two curves essentially lie on top of each other. Inset, we show the $1S_{\frac{1}{2}}$ state for very small radius where you can start to see the divergence as $r \to 0$. The divergence is extremely weak

divergences as $r \to 0$ in both the s states and the p state with $j = \frac{1}{2}$. For hydrogen, these divergences are minor and can only be seen at very small values of r, hence the logarithmic plot in the figure.

In Fig. 16.2, we plot the relativistic wavefunctions for the large component of the spinor. We see that the two curves essentially lie on top of each other with no discernible difference (indeed you have to look close to even see the two curves). Inset, we show the behavior for very small r/a_0. There, you can see the divergence at the origin is exceedingly slow. Even at the radial size of the nucleus (near -5 on the horizontal axis), the relativistic effects are indistinguishable from the nonrelativistic result. It is only if we are looking for high precision results, or we are studying heavier atomic nuclei, will the relativistic effects be large enough to see them. We discuss the latter effect in the next section.

Section 16.4 Problems:

16.4.1 Properties of $\hat{\vec{e}}_r \cdot \vec{\sigma}$ via Harmonic Polynomials

Our first step is to examine $|l, m\rangle$ states with harmonic polynomials. Recall harmonic polynomials for states with definite l are polynomials that have the same total degree for each term and which satisfy the commutator form of Laplace's equation. When we examine states with definite l and m, we have to add in

additional restrictions. We will not describe the general case here, but we will examine the first few solutions.

We start with the state $|l{=}0, m{=}0\rangle = |0\rangle$, which satisfies $\hat{L}_\alpha|0\rangle = 0$ for $\alpha = x$, y, and z. It should be obvious that this state is an eigenstate of $\hat{L}^2$ and $\hat{L}_z$ with eigenvalues of 0 for both.

Next, we examine the states with $l = 1$. Our claim is that

$$|l{=}1, m{=}1\rangle = -\sqrt{\tfrac{3}{2}}\,\frac{\hat{r}_x + i\hat{r}_y}{\hat{r}}|0\rangle, \quad |l{=}1, m{=}0\rangle = \sqrt{3}\,\frac{\hat{r}_z}{\hat{r}}|0\rangle, \quad \text{and}$$
$$|l{=}1, m{=}-1\rangle = \sqrt{\tfrac{3}{2}}\,\frac{\hat{r}_x - i\hat{r}_y}{\hat{r}}|0\rangle. \tag{16.105}$$

Verify this, by showing that $\hat{L}_z|l, m\rangle = \hbar m|l, m\rangle$ and $\hat{L}^2|l, m\rangle = 2\hbar^2|l, m\rangle$. Be sure to use the fact that $\hat{L}_\alpha|0\rangle = 0$ to allow you to use the add-zero trick to create relevant commutators. Note that the numerical factors out front are normalization factors with $\langle 0|0\rangle = 1$. Also verify that $\hat{L}_+|l{=}1, m\rangle = \sqrt{2}\hbar|l{=}1, m+1\rangle$ and that it annihilates the highest state. Do the same for $\hat{L}_-$.

We examine the action of $\hat{\vec{e}}_r \cdot \vec{\sigma}$ on the states with $j = \frac{1}{2}$ and $m = \frac{\hbar}{2}$. We have

$$\hat{\vec{e}}_r \cdot \vec{\sigma} = \tfrac{1}{2}\frac{\hat{r}_x + i\hat{r}_y}{\hat{r}}\sigma_- + \tfrac{1}{2}\frac{\hat{r}_x - i\hat{r}_y}{\hat{r}}\sigma_+ + \frac{\hat{r}_z}{\hat{r}}\sigma_z, \tag{16.106}$$

$$|j{=}\tfrac{1}{2}, m{=}\tfrac{1}{2}, l{=}0, s{=}\tfrac{1}{2}\rangle = |0\rangle \otimes |\uparrow\rangle_z, \tag{16.107}$$

and

$$|j{=}\tfrac{1}{2}, m{=}\tfrac{1}{2}, l{=}1, s{=}\tfrac{1}{2}\rangle = -\frac{\hat{r}_x + i\hat{r}_y}{\hat{r}}|0\rangle \otimes |\downarrow\rangle_z - \frac{\hat{r}_z}{\hat{r}}|0\rangle \otimes |\uparrow\rangle_z. \tag{16.108}$$

Now apply $\hat{\vec{e}}_r \cdot \vec{\sigma}$ onto the state $|j{=}\frac{1}{2}, m{=}\frac{1}{2}, l{=}0, s{=}\frac{1}{2}\rangle$ and verify it equals $-|j{=}\frac{1}{2}, m{=}\frac{1}{2}, l{=}1, s{=}\frac{1}{2}\rangle$ and vice versa.

To generalize this to all $|l, m\rangle$ states, we need to know that $|l, m\rangle$ is a polynomial with terms that have factors of $(\hat{r}_x \pm i\hat{r}_y)/\hat{r}$ and $\hat{r}_z/\hat{r}$ and satisfy the harmonic polynomial requirements plus $P_l(\hat{r}_x, \hat{r}_y, \hat{r}_z)/\hat{r}^l|0\rangle$ is an eigenstate of $\hat{L}_z$ with eigenvalue $\hbar m$. Describe how the calculation of $\hat{\vec{e}}_r \cdot \vec{\sigma}$ acting on the $|j, m, l, s\rangle$ state works, but do not try to work out the general case here.

16.4.2 Laguerre Polynomial Calculation for $j = l + \frac{1}{2}$

Fill in the details for the Laguerre polynomial derivation in the case where we have $j = l + \frac{1}{2}$.

16.4.3 Laguerre Polynomial for the $j = l - \frac{1}{2}$ Case

Redo the derivation for the Laguerre polynomials in the case where $j = l - \frac{1}{2}$. You will need to verify the product of ladder operators needed for the state, and work out and solve the recurrence relation after determining the state annihilated by the radial momentum.

16.4.4 Supersymmetry in Relativistic Hydrogen

Supersymmetric quantum mechanics is closely related to the factorization method. It was discovered in the mid-1980s by Ed Witten. For our purposes, we think of it as the first step of the factorization chain—we have the original Hamiltonian $\hat{H}_0$ and the first auxiliary Hamiltonian $\hat{H}_1^{(\text{aux})}$. All of the eigenvalues of $\hat{H}_1^{(\text{aux})}$ are shared with those of $\hat{H}_0$, but $\hat{H}_0$ has one more lower energy eigenvalue, corresponding to its ground state. If we look at the energy eigenvalues for a fixed value of j, we can see that there is one more state for the $j = l + \frac{1}{2}$ ($n' \geq 0$) than for the $j = (l+1) - \frac{1}{2}$ ($n' \geq 1$). For example, $n\mathrm{S}_{\frac{1}{2}}$ and $n\mathrm{P}_{\frac{1}{2}}$ share all energy levels for all n except $n = 1$. One can find similar pairings for $j = \frac{3}{2}$ and so on. Please verify these results for yourself. Note that we are working with the scalar quadratic energy eigenvalue problem here, not the Dirac equation.

The fundamental property of supersymmetric quantum mechanics is that there exist two Hermitian conjugate operators $\hat{B}_\gamma^\dagger$ and $\hat{B}_\gamma$ which relate the energy eigenstates of $\hat{H}_0$ (given by $|\psi_n\rangle$ with energy ε_n for $n \geq 0$) to the energy eigenstates of $\hat{H}_1^{(\text{aux})}$ (given by $|\phi_0^{(n)}\rangle$ with energy ε_n for $n \geq 1$). The relation is

$$\hat{B}_\gamma^\dagger |\phi_0^{(n)}\rangle \propto |\psi_n\rangle \text{ and } \hat{B}_\gamma |\psi_n\rangle \propto |\phi_0^{(n)}\rangle, \tag{16.109}$$

for $n \geq 1$ and $\hat{B}_\gamma |\psi_0\rangle = 0$. Find the parameter γ and verify that these results hold for the $j = \frac{1}{2}$ supersymmetric family. Can you generalize it to arbitrary j? Note that here we are working with the ε eigenvalues, not the E eigenvalues, although the degeneracy patterns holds for the E values as well.

16.4.5 Rodrigues Formula for the Second Term in the Wavefunction

In the case where $j = l + \frac{1}{2}$, we need to evaluate the operation of a $\hat{B}$ operator indexed by $n' = 0$ onto the Laguerre polynomial. There are two ways one can do this. One can compute directly the commutator of $\hat{p}_r$ with the polynomial and add on the action of the radial momentum on the state (and then combine the other

terms) to determine the resulting Laguerre polynomial. Or, one can use the $\hat{B}\hat{B}^\dagger$ identity to replace that by a Hamiltonian operator and a constant term and then shift it through the string of raising operators (using intertwining) until it acts on the state at the end. Then you have to evaluate the Laguerre polynomial that results after removing the raising operator with the $n' = 0$ index as the leftmost factor using the Rodrigues formula approach. Whichever approach you take, determine the Laguerre polynomial for this case.

16.4.6 Concrete Dirac Spinors for $2S_{\frac{1}{2}}$ and $2P_{\frac{1}{2}}$

Derive the shape of the $2P_{\frac{1}{2}}$ and $2S_{\frac{1}{2}}$ states as Dirac four-spinors.

16.4.7 Dirac Four-Spinors for the $j = l - \frac{1}{2}$ Case with $n' = 1$

Work out the spinor structure for the case $n' = 1$ only.

16.4.8 General Result for the Dirac Spinor Form of the Wavefunctions

This problem is more of a project than a homework problem. Work out the general results for all n' for the case $j = l - \frac{1}{2}$ up to the normalization factors. The Wikipedia page on hydrogenic atoms can be helpful in checking your work.

16.4.9 Hyperfine Structure with Divergent Wavefunctions at $r = 0$

When we worked out the calculations for the hyperfine structure in Sect. 11.14, we needed to use the value of the radial wavefunction at the origin. But now, every case with $j = \frac{1}{2}$ diverges at the origin. Write an essay that discusses how the hyperfine structure results will change, if they do change, due to the relativistic theory. Be sure to review the approximations used to determine the hyperfine structure and rethink how they can be adjusted. You may find reviewing how we analyzed the proton charge radius problem to be helpful in this as well.

16.5 Fine Structure, the Lamb Shift, and Experimental Implications

We have solved the Dirac equation, which is an amazing achievement, so it may seem as a let down to learn that there remains important physics beyond the Dirac equation. You see, the Dirac equation does not have particle-antiparticle pair creation in it, so it is not a fully self-consistent relativistic theory. To develop that, one needs to create quantum field theory, with the simplest version being quantum electrodynamics, which describes the field theory of electric and magnetic fields encompassing the electromagnetism of quantum particles and the photon. We have already shown a simplified theory to quantize the electromagnetic field inside a box, but we did not include in a self-consistent fashion the interaction of the photons with matter and the matter field theory. We won't be able to do that in this book, but we can describe some of the effects of the more correct theories.

We have worked hard to solve for the energy levels of the Dirac equation, which provide relativistic shifts to the energy levels that partially break the degeneracy we had in the nonrelativistic theory. But, because the Dirac equation is not a fully relativistic quantum field theory, as described above, it is not the final story. The quantum electrodynamic effects cause the anomalous g factor to be slightly more than 2, being approximately 2.002319 for the electron and there are additional energy splittings from the Lamb shift, which involves an interaction of the hydrogen atom with the vacuum fluctuations of the electromagnetic field and cause a splitting of approximately 1060 MHz between the $2P_{\frac{1}{2}}$ and $2S_{\frac{1}{2}}$ states with the p-state lowest in energy. Finally, the spin of the electron interacts with the spin of the nucleus, and this causes additional small splittings called the hyperfine structure. These splittings are typically below about 100 MHz. By performing ultra high precision measurements of the energy splittings, we can learn important results about quantum field theory effects. See Fig. 16.3 for a graphical illustration of these different spectral lines.

If we modify our theory from hydrogen to hydrogenic ions, which still have only one electron but the nuclear charge Z is now larger than 1, it requires a simple change of $\alpha_F \rightarrow Z\alpha_F$ in most of our formulas. If we look at the expression that came up for the 1S calculation of hydrogen, it involved $\sqrt{1-\alpha_F^2}$ which now becomes $\sqrt{1-Z^2\alpha_F^2}$. Since $\alpha_F \approx \frac{1}{137}$, we find that we cannot have atoms with $Z > 137$ and be able to describe them with this theory. Hence, the relativistic theory provides a limit on how large the nucleus of an atom can be. This is an unexpected result, but it is certainly consistent with the known atoms. This result also implies that when we have a case with a large Z value (such as uranium with $Z = 92$), the relativistic effects of the Dirac equation are much larger and much more important.

So far, we have ignored the negative energy solutions of the Dirac equation. These are the solutions for which the lower two elements of the Dirac four-spinor are the largest components. We neglected these solutions because they were

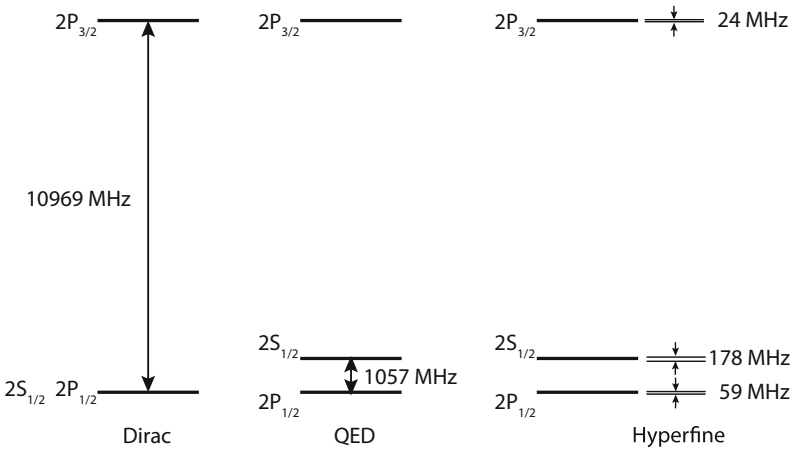

Fig. 16.3 Dirac, quantum electrodynamics, and hyperfine structure effects on the $n = 2$ shell of hydrogen. In the Dirac equation, the $2P_{\frac{3}{2}}$ state has a different energy than the $2P_{\frac{1}{2}}$ and $2S_{\frac{1}{2}}$, which are degenerate. Quantum electrodynamics has two effects—the first is to change g from 2 to slightly larger than 2 and the second is to split the $2P_{\frac{1}{2}}$ and $2S_{\frac{1}{2}}$ states via the Lamb shift. Finally, the hyperfine interaction (between the electron spin and the spin of the nucleus) causes an additional small splitting of the states. Note that we are just reporting approximate numbers for these splittings—they are known much more accurately than what is shown here

unphysical—we have never seen such states. Dirac argued that we can get rid of worrying about them if we envision all of the negative energy states to be filled in the physical world, so only the positive energy states are available for the electrons. But, this also allows us to consider exciting a filled negative energy state to a hole in the filled sea of negative energy states and a particle in the positive energy states. This would be the particle-antiparticle pair creation, and it predicts the antiparticle with positive charge, the antielectron, as the hole in the filled sea of negative energy states. Indeed, the antielectron (or what we usually call the positron) was discovered a few years after the Dirac equation was discovered, providing extra support for its correctness. Modern quantum field theory no longer reasons this way and we regard the negative energy solutions as just another problematic part of the Dirac equation because it is not a proper field theory.

The two most observable quantum field theory effects are the Lamb shift and the anomalous value for the g-factor. These are small effects on the order of α_F^2. The Lamb shift arises from the fact that the electromagnetic field vacuum has zero average field but has nonzero fluctuations, indicating virtual photons are created and destroyed in short time intervals. These fluctuations can interact with the electron in hydrogen and cause its energy levels to shift. The effect is smaller than the typical relativistic effects and it creates an additional splitting between levels that is not present in the Dirac equation. Its discovery was instrumental in helping verify that quantum electrodynamics was properly calculating the right physical results. The other effect is that these vacuum fluctuations also modify the anomalous g-factor as

well. The shift is small, but it has been measured to incredibly high accuracy using a Penning trap for the electron and in an experiment currently underway at Fermilab for the muon. The experiments are expected to be accurate enough that we need to include effects from the weak interactions and also the strong interactions to correct for all modifications of the g factor from 2. Current calculations are not yet accurate enough to be able to account for the anticipated accuracy of the newest experiments, so hard work is being undertaken to determine all of these contributions. This again points to the importance of high-precision measurements and how they can lead to discovering new results not expected from the currently accepted theories.

One of the main effects of these experimental observations from high-precision measurements is that they show us that we must go beyond quantum mechanics, as described by the Schrödinger or Dirac equations into more complicated theories such as quantum electrodynamics and then the standard model of particle physics. These theories require sophisticated mathematics to understand them primarily because there are many infinities that arise that have to be removed. The removal process is called *regularization*. After the theory is regularized, it then often requires *renormalization* to properly determine the interactions between the different particles. We will not be able to go into any more detail into these fascinating ideas here.

We end with a brief discussion of the physical principles behind the relativistic corrections that are in the Dirac equation. Previously, we saw that if we expand the relativistic kinetic energy, defined by

$$\begin{aligned}\sqrt{p^2c^2+M^2c^4}-Mc^2 &\approx Mc^2\left(1+\tfrac{1}{2}\frac{p^2c^2}{M^2c^4}-\tfrac{1}{8}\frac{p^4c^4}{M^4c^8}+\cdots\right)\\ &\approx \frac{p^2}{2M}-\frac{p^4}{8M^3c^2}+\cdots,\end{aligned} \tag{16.110}$$

we find the nonrelativistic kinetic energy in the first term and the relativistic correction in the second term. So, the kinetic energy is modified by relativistic effects. Another correction is the so-called spin-orbit interaction. The idea behind this is that a moving charge creates a magnetic field and that magnetic field interacts with the spin of the electron. The spin-orbit interaction is proportional to $\hat{\vec{L}}\cdot\hat{\vec{S}}/\hat{r}^3$, so it vanishes in a spin-singlet state. There are complications to deriving this result, because the electron is accelerating, leading to an additional factor of 2 called the Thomas precession factor. The final term is the so-called Darwin term, which only affects s-wave states, and it is proportional to the mod-square of the radial wavefunction at the origin. This term, in effect, provides a result that looks just like the spin-orbit interaction term for the s-wave states. One can find these terms by approximating the small parts of the Dirac spinor, as we did previously, and then expanding out the effective Hamiltonian for them to one order higher than what was done before, where we discovered the Pauli equation. We will not do that analysis here.

Section 16.5 Problems:

16.5.1 Modern High-Precision Measurements for Hydrogen

Write an essay on high-precision measurements and what they can test when performed on hydrogen. Be sure to describe both the experiment itself and also what it can test.

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Chapter 17
Bosons and Fermions

Reason well from the beginning and then there will never be any need to look back with confusion and doubt.

14th Dalai Lama, Spiritual Leader, Nobel Laureate, 1994

In this chapter, we discuss the property of indistinguishability which is summarized in a short video https://tinyurl.com/yeejx2uf. When you work with the smallest known objects, it is difficult to tag them, unless they have an internal property (such as polarization for a photon or spin for an electron). But because those tags are discrete, we run out of options pretty quickly. If we have a photon of a single color and with a definite polarization and then we look at a second photon with the same color and polarization, is there any way to tell them apart? The answer is no! They are indistinguishable. We need to ensure when we work out a theory that it is indistinguishable too. This chapter describes how we do that.

17.1 Bosons and Symmetric States

The key issue with indistinguishability is that nature requires it, but our formalism is usually manifestly distinguishable. When we write out a tensor-product state of two particles, the formalism says the left factor in the product is the state corresponding to particle 1, while the right factor corresponds to particle 2. So, our representation is clearly distinguishable (until we fix it to make it indistinguishable). We already encountered this in Chap. 15, when we discussed Heisenberg exchange. The key to making a quantum state indistinguishable is to swap the two particles and create a superposition with the two states, which has the same (or the opposite) coefficients. We have these two choices, because when we swap twice, that is the same as doing nothing, so the permutation operator for a pair swap squares to 1. Hence, its

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eigenvalues are ± 1. When it is $+1$ we have symmetric states, and when it is -1, we have antisymmetric states. Because all of the more complicated permutations can be constructed just out of pair swaps, understanding the behavior under pair swaps is sufficient for understanding the whole problem.

Let's get more formal, first focusing on two particle states. The two-particle state can be expanded in terms of tensor product states of a complete set of single-particle states for each particle (we assume they are the same sets here, because we consider a two-particle state constructed from the same particles). The expansion is

$$|\psi\rangle = \sum_{ij} c_{ij} |\phi_i\rangle \otimes |\phi_j\rangle, \tag{17.1}$$

with coefficients c_{ij}. In this expansion, the first factor is the first particle, and the second factor is the second. In general, this wavefunction is distinguishable if it contains $c_{ij} \neq 0$ for $i \neq j$. Because the two-particle state must be indistinguishable, when we swap two particles, it cannot change the state. This implies that we must have $\hat{\Pi}_{\text{swap}}|\psi\rangle = \pm|\psi\rangle$, with $\hat{\Pi}_{\text{swap}}$ the swap operator. This means

$$\sum_{ij} c_{ij} |\phi_i\rangle \otimes |\phi_j\rangle = \pm \sum_{ij} c_{ij} |\phi_j\rangle \otimes |\phi_i\rangle. \tag{17.2}$$

The identity holds when we have $c_{ij} = \pm c_{ji}$ for all i and j that are nonzero, with the same sign for all of these nonzero cases. This is the requirement for indistinguishability of a two-particle state. Note that for symmetric states, we can have $c_{ii} \neq 0$, while for antisymmetric states, we must have $c_{ii} = 0$. One can see that antisymmetric states automatically satisfy the Pauli exclusion principle, which says we cannot have two particles in the same quantum state.

In this section, we will focus on the symmetric states. There is an important theorem, called the CPT theorem, which is often proven in a quantum field theory class. It says that fermions, which are antisymmetric, must have half-odd integer spin, while bosons, which are symmetric, must have integer spins. So, as a rule, if we have symmetric states, it corresponds to bosonic particles with integer spin, while if we have antisymmetric states, it corresponds to fermions with half-odd integer spins. We will discuss bosonic systems first and then fermionic ones. Bosons are named after Satyendra Bose, who, along with Einstein, first described this class of elementary particles. Bose is shown in Fig. 17.1.

What about an arbitrary number of particles? The expansion for an arbitrary state takes the form

$$|\psi_N\rangle = \sum_{i_1 i_2 \cdots i_N} c_{i_1 i_2 \cdots i_N} |\phi_{i_1}\rangle \otimes |\phi_{i_2}\rangle \otimes \cdots \otimes |\phi_{i_N}\rangle, \tag{17.3}$$

and the symmetrization requirement is that

$$c_{P i_1 P i_2 \cdots P i_N} = c_{i_1 i_2 \cdots i_N}, \tag{17.4}$$

Fig. 17.1 Satyendra Bose in the 1930s. [Falguni Sarkar, courtesy AIP Emilio Segrè Visual Archives. Converted to look like a sketch by James Freericks]

where P is one of the $N!$ permutations of the indices. This is a straightforward operation, but it is tedious to work out for large N because the number of terms grows very quickly. Note that there are many terms in the superposition, but each one of the product states is orthogonal to every other product state, so the normalization procedure is the same as what we have done before. It just is a long exercise to compute all of the contributions.

In addition to the symmetrized quantum state, we can also compute the symmetrized wavefunction. To compute it, we multiply from the left with the position bra $\langle x_1| \otimes \langle x_2| \otimes \cdots \otimes |\langle x_N|$ to find

$$\begin{aligned}\langle \vec{x}|\psi_N\rangle &= \sum_{i_1 i_2 \cdots i_N} c_{i_1 i_2 \cdots i_N} \langle x_1|\phi_{i_1}\rangle \langle x_2|\phi_{i_2}\rangle \cdots \langle x_N|\phi_{i_N}\rangle \\ &= \sum_{i_1 i_2 \cdots i_N} c_{i_1 i_2 \cdots i_N} \phi_{i_1}(x_1)\phi_{i_2}(x_2)\cdots\phi_{i_N}(x_N). \end{aligned} \tag{17.5}$$

If we group all the terms with a given set of indices $i_1, i_2, \cdots, i_N$ and all of its permutations, that term can be represented in terms of a *permanent* (we have all i_j distinct in this case, although this is not technically needed for a permanent). The permanent is just like a determinant, but we have no minus signs. In other words, for the example of a three-particle state, we have

$$\begin{aligned}
&\text{perm}\begin{pmatrix}\phi_{i_1}(x_1)\ \phi_{i_2}(x_1)\ \phi_{i_3}(x_1)\\ \phi_{i_1}(x_2)\ \phi_{i_2}(x_2)\ \phi_{i_3}(x_2)\\ \phi_{i_1}(x_3)\ \phi_{i_2}(x_3)\ \phi_{i_3}(x_3)\end{pmatrix}\\
&\quad=\phi_{i_1}(x_1)\phi_{i_2}(x_2)\phi_{i_3}(x_3)+\phi_{i_2}(x_1)\phi_{i_1}(x_2)\phi_{i_3}(x_3)\\
&\qquad+\phi_{i_2}(x_1)\phi_{i_3}(x_2)\phi_{i_1}(x_3)+\phi_{i_1}(x_1)\phi_{i_3}(x_2)\phi_{i_2}(x_3)\\
&\qquad+\phi_{i_3}(x_1)\phi_{i_1}(x_2)\phi_{i_2}(x_3)+\phi_{i_3}(x_1)\phi_{i_2}(x_2)\phi_{i_1}(x_3).
\end{aligned}\tag{17.6}$$

This has a clear generalization to higher numbers of particles. Note that in general, we cannot rewrite the superposition for $|\psi_N\rangle$ in terms of just a single permanent, but we would require a number of permanents in the superposition. The challenge with this representation, even though it is compact, is that it is difficult to calculate permanents with algorithms that scale like a power of the size of the matrix N. Instead, the most direct algorithm scales with the factorial of N. Such scaling is not practical for calculations with large N.

Fortunately, there is another way to formulate these many-particle problems. This approach does not require such complicated expressions; it automatically enforces the symmetry of the state and its indistinguishability. The method is called *second quantization*. It is the most common way to work with bosonic states with more than one particle. Loosely speaking, the way second quantization works is similar to the way the physical world works—we simply state the number of bosons we have and which states they occupy, and it automatically ensures that the states are properly symmetrized. This is similar to the physical world, because that is what we do there as well—we simply decide how many particles we have in a state and what states they occupy, without introducing particle labels. This sounds like a great thing to do. Let's look carefully at how it might work. The formalism ends up looking a lot like how we worked with harmonic oscillator states.

We start with the vacuum state $|0\rangle$, which contains no particles. Then we use creation and annihilation operators, denoted $\hat{a}_i^\dagger$ and $\hat{a}_i$, respectively. Here, the index i denotes that a boson is created in the state $|\phi_i\rangle$. Note that we do not specify *which* boson is created in that state, just that a boson is created in that state. Then, by specifying the number of bosons in each state, we can create such a state using these operators. For example, a normalized state with m_1 bosons in $|\phi_1\rangle$, m_2 bosons in $|\phi_2\rangle$ and so on, is given by

$$|m_1, m_2, \cdots\rangle = \frac{(\hat{a}_1^\dagger)^{m_1}}{\sqrt{m_1!}}\frac{(\hat{a}_2^\dagger)^{m_2}}{\sqrt{m_2!}}\cdots|0\rangle. \tag{17.7}$$

This is reminiscent of our use of ladder operators in the harmonic oscillator, especially with regard to the normalization factors, which you will verify in the problems. Indeed, we require the following commutation relations of the creation and annihilation operators:

$$\left[\hat{a}_i^\dagger, \hat{a}_j^\dagger\right] = 0, \quad \left[\hat{a}_i, \hat{a}_j\right] = 0, \quad \text{and} \quad \left[\hat{a}_i, \hat{a}_j^\dagger\right] = \delta_{ij}. \tag{17.8}$$

You can see we do not specify which boson is which—there are no particle labels on the bosons at all. In fact, if you wanted to think of a specific boson as being determined by where the operator is in the product of factors that are acting on the vacuum, this will not work, because the fact that two creation operators commute means you can move the factor anywhere in the product and the result remains the same. In fact, this property is what ensures the system is automatically symmetric with respect to particle interchange. The expression is totally symmetric in the operators! It may seem strange to work with quantum mechanics problems in this fashion. Up to now, we have only described how to construct bosonic states. In order to calculate something, we need to represent operators like the Hamiltonian in terms of the creation and annihilation operators. This is straightforward to do, but we will explore it in more detail when we perform second quantization with fermions later in the chapter. Note that it is critically important not to conflate the label, which determines what state we are occupying from the boson particle label, which says which boson occupies it. In the second-quantization formalism, we have no boson particle labels, only state labels.

While it is possible to talk about wavefunctions, when we work in second quantization, we do not usually do it. To construct the wavefunction, we simply multiply from the right with the position-space or momentum-space bra's for the different particles. The evaluation of the wavefunction results in the sum of the fully symmetric wavefunction with the huge number of terms, just as what we did before. But, in general, it is not useful to do this. You see, the operators condense the state so much that it becomes manageable to work with (it is a factorial-times compression). Why make it unmanageable again?

We end this section with a discussion of black body radiation and the Planck formula. We will not discuss the long history behind this, just describe how one determines the proper result. First off, what is a black body anyway? The name comes from the observation that when we heat objects to very hot temperatures, they first start to glow red-orange and then they become yellower and eventually become white. This progression is independent of the material that the object is made from. It is a *universal* response. The question we will address here is why? What is the reason why everything that is heated up to some temperature glows with a specific color? The answer lies in quantum mechanics. Indeed, solving this problem signaled the dawn of quantum mechanics.

We start by modeling the black body by a box that is filled with photons. The photons must fit inside the box. If we take it to be a cube of length L on each side, then this problem of a particle in a three-dimensional box was already solved in Problem 12.5.4. The wavefunction takes the form

$$\psi(x, y, z) = C \sin(k_x L) \sin(k_y L) \sin(k_z L), \tag{17.9}$$

where $k_\alpha = n_\alpha \pi / L$ with n_α an integer. The allowed wavenumber values fall on a cubic lattice, indexed by the integers n_x, n_y, and n_z, with a spacing between adjacent points of π/L in each direction. The relationship between the frequency and the wavenumber is that

$$\frac{\omega}{c} = |\vec{k}| = \frac{\pi}{L}\sqrt{n_x^2 + n_y^2 + n_z^2}. \tag{17.10}$$

This formula tells us that the states with constant energy lie on the surface of a sphere, whose radius is given by the wavenumber. Our goal now is to count the number of modes. We can express the frequency f in terms of the angular frequency ω, but we want to know how many different wavenumbers, or more correctly wavevectors, have a given value of frequency. This is the equivalent to changing variables from a Cartesian coordinate system n_x, n_y, and n_z to a spherical coordinate system r, θ, and ϕ, except that here, the system is spherically symmetric, so we only have radial dependence and no dependence on the angles. The procedure to count the states comes from counting the number of grid points that lie inside a thin shell of a fixed radius. Our procedure is then as follows: The frequency satisfies

$$f = \frac{\omega}{2\pi} = \frac{c}{2L}\sqrt{n_x^2 + n_y^2 + n_z^2}, \tag{17.11}$$

while the number of points in between the two spheres of radius r and $r + \Delta r$ is proportional to the volume in between two spheres with slightly different radii, Which is just the product of the thickness of the shell Δr and surface area of the sphere $4\pi r^2$, with $r = \frac{kL}{\pi}$. The density of points is uniform and given by $1/L^3 = 1/V$. In addition, we choose all of the integers $n_\alpha \geq 0$, so they all sit in the first octant only (which reduces the volume between the spheres by a factor of 8). Finally, we use the relationship between the wavevector and the frequency given by $k = \frac{2\pi f}{c}$ in our final formula. Putting all of this together, we find that the density of states as a function of the radius $N(r)$ satisfies

$$\Delta N(r) = 4\pi V \frac{f^2}{c^3} \Delta f. \tag{17.12}$$

The total density of states $n(f)$ is defined to be

$$n(f) = \sum_{\text{pol.}} \frac{1}{V} \frac{\Delta N(r)}{\Delta f} = 8\pi \frac{f^2}{c^3}. \tag{17.13}$$

You may have heard that electromagnetic waves act like harmonic oscillators. This is by no means obvious except for the fact that the motion, given by sines or cosines, is the same as the motion of a simple harmonic oscillator. It turns out that mathematically, they are *identical* (which we will see when we quantize light in Chap. 20). When you study statistical mechanics, you will learn that the average

energy $\bar{E}$ of a harmonic oscillator at temperature T is given by $\bar{E} = k_B T$, where k_B is the Boltzmann constant. There is no simple way for me to argue about why this is correct. Energy and temperature are related to each other, which you probably already know and the Boltzmann constant is used to relate them. Furthermore, as something gets hotter, corresponding to a higher temperature, it has more energy in it, so having the average energy proportional to the temperature also makes sense. The fact that the proportionality constant is exactly k_B and the relationship is linear requires a classical statistical mechanics calculation which is beyond this work. I hope you will be comfortable with this result as being correct, nevertheless. Using this result, we compute the average energy for the electromagnetic waves at a temperature T by simply taking the density of states for the photons as a function of frequency ($n(f)$ calculated above) and multiply it by the energy for the light at that frequency. Here, the latter is just $\bar{E}$, which depends only on temperature and not on frequency. We let $u(f)$ denote the *energy density* as a function of frequency. It then satisfies

$$u(f) = \frac{8\pi f^2 \bar{E}}{c^3} = \frac{8\pi f^2 k_B T}{c^3}. \tag{17.14}$$

which is called the *Rayleigh-Jeans* law. Even though we calculated the density of states with a reference to quantum mechanics, it turns out that this result is what a classical calculation of the black-body radiation also gives us. All seems well and good until we ask the question, what is the total energy stored inside the box? Well, to get that, we need to sum up the contributions for all frequencies. But the energy density grows like f^2, so as we add in results for higher and higher frequencies, it is easy to see that this must diverge. This is called the *ultraviolet catastrophe*, because it says the total energy of the photons trapped in every box is infinite at any nonzero temperature! Clearly this seems like it must be either a complete failure of the classical theory, or it involves something very subtle about issues related to what infinite energy means. It turns out the former is the case.

So, this situation was very confusing around the turn of the 1900s. Enter Max Planck (see Fig. 17.2), who postulated that light in a black body is not to be treated as a classical wave, but as a "particle" whose energy is equal to $E = hf$, with a constant h, now called the Planck constant (note that $h = 2\pi\hbar$). If I have n photons, then the energy will be $E_n = nhf$, since the photons are independent of each other (i.e., do not interact) and the energy of each photon is determined by its color. Note that this was really just a guess. There was no solid evidence that this should be true. In addition, it was only after Einstein described the photoelectric effect with this same idea that one had some evidence that it must be true. But Planck did his work 5 years earlier. In fact, he viewed this more as a fictitious idea to motivate the formulas he found for the blackbody radiation than the discovery of the particle nature of a photon.

The first step that we need is to calculate the average energy of these photons. To start, we need to compute how many photons will be present with frequency f if the temperature is T. This requires us to do some statistical mechanics. Once again,

Fig. 17.2 Max Planck. [Drawing by William J. Numeroff, courtesy of AIP Emilio Segrè Visual Archives, Physics Today Collection]

you probably have not seen this before, so bear with me. Everything is correct, but I won't be able to derive all the formulas in full. The probability to have n photons of frequency f present if the temperature is T is proportional to e^{-E_n/k_BT}. A higher temperature should emit a higher energy, and this is consistent with that notion. It will be fully derived for you when you study statistical mechanics. We also know that the sum of the probabilities for there to be all photons present must add to 1, so the probability to have n photons is given by

$$P(n) = \frac{e^{-\frac{E_n}{k_BT}}}{\sum_{n=0}^{\infty} e^{-\frac{E_n}{k_BT}}}, \tag{17.15}$$

with $E_n = nhf$. The sum in the denominator is a geometric series for $x = e^{-\frac{E_n}{k_BT}}$, so we can immediately sum it to be $1/(1-x)$.

We are more interested in the average energy of the photons with frequency f at temperature T, denoted $\bar{E}(f,T)$. This is calculated in Problem 17.1.5 and it is given by

$$\bar{E}(f,T) = \frac{hf}{e^{\frac{hf}{k_BT}} - 1}. \tag{17.16}$$

This result then tells us the energy density $u(f)$ which satisfies

$$u(f) = \frac{8\pi f^2 \bar{E}(f,T)}{c^3} = \frac{8\pi h f^3}{c^3\left(e^{\frac{hf}{k_B T}} - 1\right)}. \tag{17.17}$$

This is called the *Planck radiation law* and it agrees well with the experimental distribution as a function of frequency.

By analyzing the color spectrum of the black-body radiation, we can determine the temperature of an object. This is done by determining where it peaks and noting that we see a different peak for different temperatures. For example, the sun appears yellow to us, suggesting, without doing any measurements, that the black body radiation curve is peaked around the yellow part of the optical spectrum. This means the temperature is on the order of $5000^{\text{o}} - 6000^{\text{o}}$ K. And we get this simply by understanding black body radiation and making an observation. The more accurate temperature for the surface of the sun is approximately 5700^{o} K. Similarly, when we see a red star, it is at a lower temperature, while a blue star is at a hotter temperature. It is remarkable we can measure the temperature of something that can be many, many light years away from us by using quantum mechanics!

There are many other phenomena in many-body physics that involve bosons. They include superfluidity, Bose-Einstein condensation, superfluid-insulator transitions and so on. Most of these phenomena require a complete development of quantum statistical mechanics to fully understand them, and we do not have the space to pursue them here. We have, however, prepared you well for going into those topics on your own if you so desire.

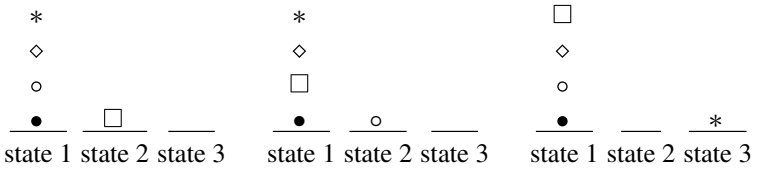
Section 17.1 Problems:

17.1.1 Counting Bosons

Our problem here is going to be quite concrete. We have five particles that we will denote by symbols. We will consider the case where the different symbols correspond to classical distinguishable particles to bosons and to fermions (here, we will use spin to denote the particles, because fermions always have spin). The first and second cases will turn out to be relatively simple to work out. We will find the boson case is much harder. We are going to work on a simple counting problem. We want to distribute these five particles among three different quantum states. It doesn't matter what these states are, but if you need a picture in your head, you can think of them as states with three different energies. We will use lines to denote the different states (you will see what we are doing is reminiscent of what you may have done with energy-level diagrams in chemistry). The counting problem asks us to count the different allowed configurations that we have for distributing the particles. You may ask why would we want to do such a thing. It turns out that all

of statistical mechanics boils down to proper counting. So we need to develop the skill to do such counting.

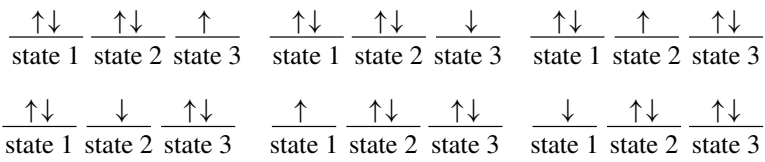
In the classical particle case, it is fairly clear how to count. We have no restrictions. The symbols represent the different distinguishable particles, so placing an open circle in one state is different from placing an asterisk in that state. The different configurations look like the following:



and many more.

For the classical case, the first two configurations are different because of the different shapes in the different states. So they must *each* be counted. Essentially in this classical case, there are no restrictions at all, and all possibilities must be considered. There are five particles, and each particle has one of three possible states. So one particle has three choices, the second has three choices, and so on. The pattern is clear. The total number of choices is $3 \times 3 \times 3 \times 3 \times 3 = 3^5 = 243$. Obviously, we will not be listing all of them here.

The fermion case is the next simplest case. It allows no more than one particle in each state (due to the Pauli exclusion principle). Fermions have spin, so we will denote them with up or down arrows, as we have done previously (we consider only spin-one-half fermions, for now). Here, because we have three states and two choices per state, there are a total of six possible places to put five objects. If you know how to do combinatorics, you will easily verify there are in total only six fermionic states. This is a huge reduction from the 243 that we had in the classical case. We can show all six, so we do so now.



You can see that we have one "hole" in this system that we place in one of the available states (up or down spin, and state 1, 2, or 3). Many fermionic problems are easier to solve when we think of them in this "hole" picture. We will have much more experience with this later in the book.

Now, on to the main event—bosons. You may want to review the binomial coefficient and how it is used to count collections to help you with your work. The problem has a bit of a trick to it. If you see it, solving the problem is easy. If you

do not, it can be very hard. Feel free to depict all possible bosonic states and recall you must invoke indistinguishability (use the same symbol for each particle). We are solving the five bosons in three states problem here.

Ans. You should find 21 states for this case. The general formula for N bosons and s states is

$$\binom{N+s-1}{N} = \frac{(N+s-1)!}{N!(s-1)!}. \tag{17.18}$$

If you want to learn more about how to think this way, search for the "stars and bars" method in combinatorics.

17.1.2 Normalizing the Second-Quantized Representation of a State

We claimed that the normalization factor for the bosonic states that have m_1 bosons in state 1, m_2 bosons in state 2, and so on is given by a factor of $1/\sqrt{m_j!}$ for each state j. This looks just like the normalization factor for the simple harmonic oscillator states.

Use the fact that $\hat{a}_j|0\rangle = \langle 0|\hat{a}_j^\dagger = 0$ to verify the normalization factor. First work with a bosonic state where you have m bosons in one state and then generalize to the full case.

17.1.3 Why Permanents Are Difficult to Calculate

The formula for a permanent is very clear. It involves $N!$ terms when we have N particles. This becomes very large very quickly. Then it can take "forever" to calculate with it.

One might have thought the same thing is true about determinants, but they can be calculated much more easily using an approach called row reduction.

Look up how to calculate determinants efficiently and write a short essay describing the differences between determinants and permanents, and why the permanents are much more difficult to calculate with.

17.1.4 Boson Sampling

Scott Aaronson introduced the idea of boson sampling as a potential application of quantum simulators that will always do better than conventional computers.

The idea of boson sampling is that we have N beam splitters and M identical photons. We send the photons into the device, and they have 2^N different paths they can take. What is the probability distribution of the bosons for each of those paths. Note the paths need not be equivalent or equal probability.

Do some more research about the boson sampling problem and write a short essay describing why it is an interesting problem to solve with quantum devices.

17.1.5 Computing the Average Energy of Light

In this problem, we will compute the average energy of light, given by

$$\bar{E}(f,T) = \frac{\sum_{n=0}^{\infty} nhf e^{-\frac{nhf}{k_B T}}}{\sum_{n=0}^{\infty} e^{-\frac{nhf}{k_B T}}}. \tag{17.19}$$

The summation in the numerator is of the form $\sum nx^n$, with a suitably chosen x. This looks like the geometric series, but is different. Can we sum it? Yes, we can. We describe the procedure how to do it next.

The trick is to see that we have a chain of repeated summations, where each summation is related to the geometric series. Concretely, we have that

$$\begin{aligned}
\sum_{n=0}^{\infty} nx^n = {} & x + x^2 + x^3 + x^4 + \cdots \\
& + \quad x^2 + x^3 + x^4 + \cdots && (17.20)\\
& + \quad x^3 + x^4 + \cdots && (17.21)\\
& + \quad x^4 + \cdots && (17.22)\\
& \quad \vdots && (17.23)
\end{aligned}$$

where the extra summation shifts to the left one spot with each new line to properly count the n repeated terms in the summation. This extends an infinite number of lines, with it being shifted to the right with each new line. We can rewrite this summation as

$$\sum_{n=0}^{\infty} nx^n = (1 + x + x^2 + x^3 + \cdots)(x + x^2 + x^3 + x^4 + \cdots). \tag{17.24}$$

Re-express the product of these two summations in terms of the geometric series and show that we have

$$\sum_{n=0}^{\infty} n x^n = \frac{x}{(1-x)^2}. \tag{17.25}$$

Use this identity to complete the computation of the average energy of photons of frequency f at temperature T.

17.1.6 Limiting Behavior of the Planck Radiation Law at High and Low T

At high temperature, the Planck radiation law can be simplified and evaluated approximately. Use the power series expansion for the exponential to compute the lowest-order approximation to the Planck distribution law at high temperature. Compare it to the Rayleigh-Jeans distribution.

At low temperature, the exponential term dominates over the -1 term in the denominator of the Planck radiation law. Approximate the distribution in this case as well.

Taking those two results, now think in terms of a fixed temperature, but vary the frequency. What is the limiting behavior for low and high frequency?

17.2 Hong-Ou-Mandel Experiment and Photon Bunching Revisited

In this section, we will discuss a "baby" version of quantum optics, which will become much more precise in Chap. 20, when we fully quantize light and develop the subject in its entirety. We focus on three topics here: (i) the Hanbury-Brown and Twiss experiment, (ii) photon sampling with a single (unbalanced) beam splitter, and (iii) the Hong-Ou-Mandel experiment. We ignore the polarization of light in the discussion.

We start with the Hanbury-Brown and Twiss experiment. As you may recall, this experiment can be thought of as two monochromatic light sources (A and B), separated by a distance r that are a distance L away from the detectors (1 and 2). We place the two detectors a distance R apart and study the correlation between the intensity of the light detected in each detector $\langle I_1 I_2 \rangle$, where the angle brackets denote the quantum average of the intensities. See Fig. 17.3 for a schematic of the experiment. The frequency of the light is given by ω, while the wavevectors of the light from the two sources are slightly different, given by $\vec{k}$ and $\vec{k}'$, with equal norms, but the angle between them is given by $\phi = r/L$.

We examine the situation with precisely two photons emitted and detected. Since we are looking at the correlations of the amplitudes in the two detectors, we must

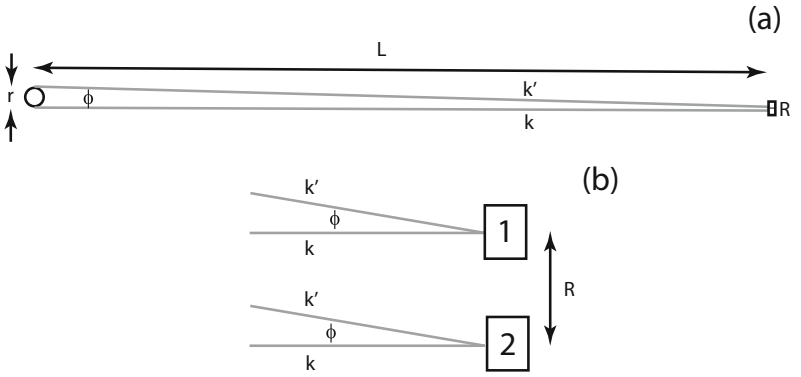


Fig. 17.3 Schematic of the Hanbury-Brown and Twiss experiment. The separation of the sources is r, their distance is L and the separation of the detectors is R. We describe the photons with the same frequency ω, but slightly different wavevectors $\vec{k}$ and $\vec{k}'$. The angle between them is $\phi = r/L$

have them each detect one photon. There are four ways this can occur—(i) two photons leave source A and are detected in 1 and 2; (ii) two photons leave source B and are detected in detectors 1 and 2; (iii) a photon leaves A and is detected in 1, while another leaves B and is detected in 2; and (iv) a photon leaves A and is detected in 2, while another leaves B and is detected in 1. The first two processes are different events, because the source and detector conditions are different. Each occurs one alternative way, so there is no interference. The last two are the same event, with two alternative ways to happen, so there is interference. We will use creation and annihilation operators, labeled by the wavevector, to describe the different photons. We know the photons have a phase determined by how far they travel. Previously, we described them with clocks and arrows. Now, we use complex exponentials, so that the complex phase associated with the creation operator is given by $e^{-i(\vec{k}\cdot\vec{r}-\omega t)}$, where $\vec{r}$ is the location we are detecting at and t is the time we are detecting at.

Because we do not know where the photon came from, we describe it in a superposition state. The operator used to describe the photon at $\vec{r}$ and at time t is then given by

$$\left(\hat{a}^\dagger_{\vec{k}} e^{-i(\vec{k}\cdot\vec{r}-\omega t)} + \hat{a}^\dagger_{\vec{k}'} e^{-i(\vec{k}'\cdot\vec{r}-\omega t)}\right). \tag{17.26}$$

Since we are detecting one photon at $\vec{r}_1$ and one photon at $\vec{r}_2$, both occurring at time t, the state at the detectors is given by

$$|\psi_d\rangle = \left(\hat{a}^\dagger_{\vec{k}} e^{-i(\vec{k}\cdot\vec{r}_2-\omega t)} + \hat{a}^\dagger_{\vec{k}'} e^{-i(\vec{k}'\cdot\vec{r}_2-\omega t)}\right)\left(\hat{a}^\dagger_{\vec{k}} e^{-i(\vec{k}\cdot\vec{r}_1-\omega t)} + \hat{a}^\dagger_{\vec{k}'} e^{-i(\vec{k}'\cdot\vec{r}_1-\omega t)}\right)|0\rangle, \tag{17.27}$$

where $\vec{r}_i$ is the location of detector i. The intensity correlations are then given by the norm squared of this state. We will describe how it includes all of the four different processes as we evaluate the expression. The norm-squared becomes

$$\begin{aligned}\langle I_1 I_2\rangle &= \langle 0|\left((\hat{a}_{\vec{k}})^2 e^{i\vec{k}\cdot(\vec{r}_1+\vec{r}_2)} + \hat{a}_{\vec{k}}\hat{a}_{\vec{k}'}\left(e^{i(\vec{k}\cdot\vec{r}_1+\vec{k}'\cdot\vec{r}_2)} + e^{i(\vec{k}\cdot\vec{r}_2+\vec{k}'\cdot\vec{r}_1)}\right)\right.\\ &\quad\left.+(\hat{a}_{\vec{k}'})^2 e^{i\vec{k}'\cdot(\vec{r}_1+\vec{r}_2)}\right)\\ &\quad\times\left((\hat{a}^\dagger_{\vec{k}})^2 e^{-i\vec{k}\cdot(\vec{r}_1+\vec{r}_2)} + \hat{a}^\dagger_{\vec{k}}\hat{a}^\dagger_{\vec{k}'}\left(e^{-i(\vec{k}\cdot\vec{r}_1+\vec{k}'\cdot\vec{r}_2)} + e^{-i(\vec{k}\cdot\vec{r}_2+\vec{k}'\cdot\vec{r}_1)}\right)\right.\\ &\quad\left.+(\hat{a}^\dagger_{\vec{k}'})^2 e^{-i\vec{k}'\cdot(\vec{r}_1+\vec{r}_2)}\right)|0\rangle\\ &= \underbrace{\langle 0|(\hat{a}_{\vec{k}})^2(\hat{a}^\dagger_{\vec{k}})^2|0\rangle}_{2\text{ from }A} + \underbrace{\langle 0|(\hat{a}_{\vec{k}'})^2(\hat{a}^\dagger_{\vec{k}'})^2|0\rangle}_{2\text{ from }B}\\ &\quad+\underbrace{\langle 0|\hat{a}_{\vec{k}}\hat{a}_{\vec{k}'}\hat{a}^\dagger_{\vec{k}}\hat{a}^\dagger_{\vec{k}'}|0\rangle 2\Big(1+\cos\left((\vec{k}-\vec{k}')\cdot(\vec{r}_1-\vec{r}_2)\right)\Big)}_{1\text{ from }A\text{ and }1\text{ from }B}.\end{aligned} \tag{17.28}$$

You can see it includes the terms with no interference plus the contributions with interference. We can simplify the argument of the cosine $(\vec{k}-\vec{k}')\cdot(\vec{r}_1-\vec{r}_2)$ to make the expression clearer. Going back to the original Hanbury-Brown and Twiss geometry, we think of the light with $\vec{k}$ perpendicular to $\vec{r}_1-\vec{r}_2$ and $\vec{k}'$ oriented at an angle ϕ from the vertical. Then $(\vec{k}-\vec{k}')\cdot(\vec{r}_1-\vec{r}_2) = |\vec{k}||\vec{r}_1-\vec{r}_2|\phi$. Since $\vec{k}$ is known, we vary the distance between the detectors and use that to determine ϕ, the angular width of the two sources. You can see that once we have determined the operators corresponding to the superposition of the two sources, the calculation itself is straightforward and simple to evaluate, even though it has many terms. Note further that we evaluated these results for a total of two photons, but if we replace the vacuum state with another photon state (say from a thermal source), we can still use the same formula as written above with the only modification being changing the states that we compute the averages with respect to.

We now move on to describe boson sampling. We will work just with a single beam splitter, but we will make it general, in the sense that it is not necessarily a 50–50 beam splitter. We assume the beam splitter transmits a probability of $T = t^2$ photons and reflects $R = r^2 = 1-t^2$ photons (see Fig. 17.4). Here, t and r are both real numbers. Previously, we worked with $t = r = \frac{1}{\sqrt{2}}$. But now we allow them to vary from 0 to 1. The beam splitter has two input ports (1 and 2) and two output ports (3 and 4). The reflection from port 1 has the minus sign, so we can relate the operators as follows:

$$\begin{aligned}\hat{a}^\dagger_3 &= -\sqrt{1-t^2}\,\hat{a}^\dagger_1 + t\hat{a}^\dagger_2 \text{ and } \hat{a}^\dagger_4 = t\hat{a}^\dagger_1 + \sqrt{1-t^2}\,\hat{a}^\dagger_2\\ \hat{a}^\dagger_1 &= -\sqrt{1-t^2}\,\hat{a}^\dagger_3 + t\hat{a}^\dagger_4 \text{ and } \hat{a}^\dagger_2 = t\hat{a}^\dagger_3 + \sqrt{1-t^2}\,\hat{a}^\dagger_2.\end{aligned} \tag{17.29}$$

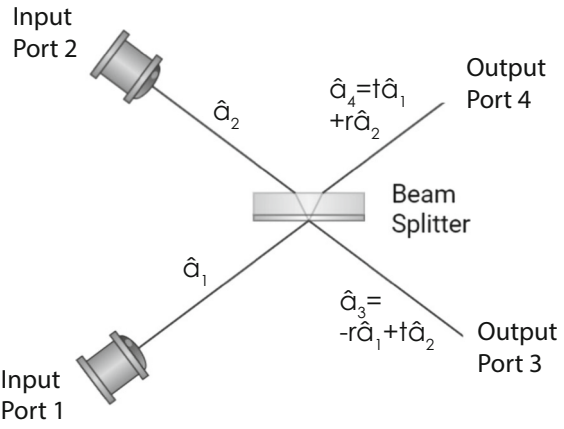


Fig. 17.4 Schematic of a general beam splitter. The connection between the creation operators in the input and output ports is as follows: (i) $\hat{a}_3^\dagger = -\sqrt{1-t^2}\hat{a}_1^\dagger + t\hat{a}_2^\dagger$ and (ii) $\hat{a}_4^\dagger = t\hat{a}_1^\dagger + \sqrt{1-t^2}\hat{a}_2^\dagger$. Note that $r = \sqrt{1-t^2}$

Now, we start with a state with n photons in port 1 and m photons in port 2. It is given by

$$|\psi\rangle = \frac{(\hat{a}_1^\dagger)^n}{\sqrt{n!}} \frac{(\hat{a}_2^\dagger)^m}{\sqrt{m!}} |0\rangle_1 \otimes |0\rangle_2. \tag{17.30}$$

The boson sampling problem asks for the probability to have m' bosons in port 3 and $n-m'$ bosons in port 4. To compute this, we replace the creation operators with the operators corresponding to the output states. This gives us

$$\begin{aligned}
|\psi\rangle &= \frac{(-\sqrt{1-t^2}\hat{a}_3^\dagger + t\hat{a}_4^\dagger)^n}{\sqrt{n!}} \frac{(t\hat{a}_3^\dagger + \sqrt{1-t^2}\hat{a}_2^\dagger)^m}{\sqrt{m!}} |0\rangle_3 \otimes |0\rangle_4 \\
&= \frac{1}{\sqrt{n!}} \left(\sum_{k=0}^{n} \frac{n!}{k!(n-k)!} (-1)^k (\sqrt{1-t^2})^k t^{n-k} (\hat{a}_3^\dagger)^k (\hat{a}_4^\dagger)^{n-k} \right) \\
&\quad \times \frac{1}{\sqrt{m!}} \left(\sum_{l=0}^{m} \frac{m!}{l!(m-l)!} t^l (\sqrt{1-t^2})^{m-l} (\hat{a}_3^\dagger)^l (\hat{a}_4^\dagger)^{m-l} \right) |0\rangle_3 \otimes |0\rangle_4 \\
&= \sum_{k=0}^{n} \sum_{l=0}^{m} \frac{\sqrt{n!m!(l+k)!(n+m-l-k)!}}{k!l!(n-k)!(m-l)!} (-1)^k (\sqrt{1-t^2})^{m+k-l} t^{n+l-k} \\
&\quad \times |k+l\rangle_3 \otimes |2n-l-k\rangle_4,
\end{aligned} \tag{17.31}$$

where we used the binary theorem and simplified the results. This can now be used to determine the probability to find j photons in port 3 and $n+m-j$

photons in port 4. To determine this, we restrict the summations to the terms where $k + l = j$ and then take the modulus squared of the coefficient. One can clearly see that this is a complicated mess, and we only used one beam splitter here. In principle, generalizing to more beam splitters is not so complicated, but it becomes very tedious. We will not do that here.

This result can be directly applied to the Hong-Ou-Mandel experiment. We take $t = \frac{1}{\sqrt{2}}$ and $n = m = 1$. We then examine the three cases: $j = 0$, $j = 1$, and $j = 2$. We find for $j = 0$, the coefficient is $\frac{1}{\sqrt{2}}$ for $|0\rangle_3 \otimes |2\rangle_4$. For $j = 1$, the amplitude is 0 for the state $|1\rangle_3 \otimes |1\rangle_4$ (due to an exact cancellation of the two alternative way contributions). Finally, for $j = 2$, we have the amplitude is $\frac{1}{\sqrt{2}}$ for the state $|2\rangle_3 \otimes |0\rangle_4$. This is precisely what we had before for the Hong-Ou-Mandel experiment.

Section 17.2 Problems:

17.2.1 Hanbury-Brown and Twiss experiment

Write an essay describing how one uses the Hanbury-Brown and Twiss experiment to measure the angular size of stars.

17.2.2 Technical Details of the Hanbury-Brown and Twiss Experiment

In the main text, we showed that with a specific geometry $(\vec{k}-\vec{k}')\cdot(\vec{r}_1-\vec{r}_2) = |\vec{k}|R\phi$, where R is the distance between the two detectors and ϕ is the angle between $\vec{k}$ and $\vec{k}'$ (and this result neglects higher-order contributions in ϕ). Verify that $|\vec{k} - \vec{k}'| = |\vec{k}|\phi$, so the result quoted in the text is the largest possible result for the dot product between these vectors.

17.2.3 Hong-Ou-Mandel Experiment with a General Beam Splitter

Compute the probabilities for each of the three states in the Hong-Ou-Mandel experiment for general t. Verify that it gives the standard results for the 50–50 beam splitter.

17.2.4 Clarifying a Small Swindle We Had Before in the Hong-Ou-Mandel Experiment

Write an essay that describes the swindle we did when discussing the Hong-Ou-Mandel experiment using the Feynman methodology. As you may recall from that discussion, we did not multiply by the shrink factor a second time for the cases with two photons in one detector. Instead, we said we know where the photon goes, so there is no additional shrinkage of the arrow. But, such an approach gives the wrong result for the general case, as you can verify with the above problem when $r \neq t$. Explain how we corrected this swindle using the general formula derived in this section when the beam splitter is not balanced.

17.3 Fermions and Antisymmetric States

Fermions are named after Enrico Fermi, a Nobel Prize-winning physicist, who was one of the twentieth centuries greatest experimental minds. The basic ideas were developed by others as well, especially Dirac. Nevertheless, we honor Fermi with the name of the fermion. You can see a sketch of him from this era where fermions were first described in Fig. 17.5.

In the fermionic case, the expansion of an arbitrary state N-particle state is given by

Fig. 17.5 Enrico Fermi in 1928. [Photo by G.C. Trabacchi, courtesy AIP Emilio Segrè Visual Archives Converted to look like a sketch by James Freericks]

$$|\psi_N\rangle = \sum_{i_1 i_2 \cdots i_N} c_{i_1 i_2 \cdots i_N} |\phi_{i_1}\rangle \otimes |\phi_{i_2}\rangle \otimes \cdots \otimes |\phi_{i_N}\rangle, \tag{17.32}$$

with the antisymmetrization requirement being

$$c_{Pi_1 Pi_2 \cdots Pi_N} = (-1)^P c_{i_1 i_2 \cdots i_N}. \tag{17.33}$$

Here, P is one of the $N!$ permutations of the indices and $(-1)^P$ is 1 for an even permutation and -1 for an odd permutation. While this looks similar to what we did for bosons, the presence of the minus signs has significant consequences. First off, we have no contributions from terms where two or more of the $|\phi_i\rangle$ are the same. While this arises naturally from the minus signs (because the state is unchanged when we swap the two identical ϕ's, but the coefficient must change sign, and the only coefficient that does this is zero). Hence, the antisymmetrization automatically enforces the Pauli exclusion principle.

Similar to what we did before, we can also compute the antisymmetrized wavefunction. We simply multiply from the left with the position bra $\langle x_1| \otimes \langle x_2| \otimes \cdots \otimes \langle x_N|$ to find

$$\begin{aligned} \langle \vec{x}|\psi_N\rangle &= \sum_{i_1 i_2 \cdots i_N} c_{i_1 i_2 \cdots i_N} \langle x_1|\phi_{i_1}\rangle \langle x_2|\phi_{i_2}\rangle \cdots \langle x_N|\phi_{i_N}\rangle \\ &= \sum_{i_1 i_2 \cdots i_N} c_{i_1 i_2 \cdots i_N} \phi_{i_1}(x_1)\phi_{i_2}(x_2)\cdots\phi_{i_N}(x_N), \end{aligned} \tag{17.34}$$

with the coefficients satisfying the antisymmetrization requirement. If we group all the terms with a given set of indices $i_1, i_2, \cdots, i_N$ and all of its permutations, that term can be represented in terms of a *determinant* now. The determinant has extra minus signs in it from the permanent that we discussed earlier. As an example, a three-particle state (unnormalized) looks like

$$\begin{aligned} \det \begin{pmatrix} \phi_{i_1}(x_1) & \phi_{i_2}(x_1) & \phi_{i_3}(x_1) \\ \phi_{i_1}(x_2) & \phi_{i_2}(x_2) & \phi_{i_3}(x_2) \\ \phi_{i_1}(x_3) & \phi_{i_2}(x_3) & \phi_{i_3}(x_3) \end{pmatrix} &= \phi_{i_1}(x_1)\phi_{i_2}(x_2)\phi_{i_3}(x_3) - \phi_{i_2}(x_1)\phi_{i_1}(x_2)\phi_{i_3}(x_3) \\ &\quad + \phi_{i_2}(x_1)\phi_{i_3}(x_2)\phi_{i_1}(x_3) - \phi_{i_1}(x_1)\phi_{i_3}(x_2)\phi_{i_2}(x_3) \\ &\quad + \phi_{i_3}(x_1)\phi_{i_1}(x_2)\phi_{i_2}(x_3) - \phi_{i_3}(x_1)\phi_{i_2}(x_2)\phi_{i_1}(x_3). \end{aligned} \tag{17.35}$$

But, unlike the issue with bosonic states, there is a simple and efficient way to compute determinants via a procedure called row-reduction. If you are interested in how this works, please consult a linear-algebra textbook that describes the algorithm (and why it works), as the specific details are not critical for our work here.

John Slater introduced the use of these determinants in computation, and he is honored by our calling this representation of states *Slater determinants*, which is the

antisymmetrization of an N-particle product state consisting of placing N fermions into N different states $\{|\phi_i\rangle\}$. There is a subtle point about these states. We often call them "product states" even though the antisymmetrized state is clearly not a product state, using our original definition. So, why do we call them product states? Primarily because they are product states when represented in second quantization, in terms of the creation operators acting on the vacuum. We will discuss second quantization in the next section of the chapter. Even in first quantization, the state is a product state in the sense that it is a product state plus antisymmetrization of the state to make it a set of N indistinguishable fermions. This simple fact allows for a number of simplifications when computing matrix elements of operators in single Slater determinant states, and because of this simplification, calling them by a specific name makes sense. In the remainder of the section, we will discuss applications of these ideas to the degenerate Fermi gas, the Fermi-Dirac distribution function, and neutron stars.

Quantum effects are most pronounced at low temperatures. This is because thermal fluctuations, which are stronger at higher temperatures, are deleterious to quantum effects, especially superposition and entanglement. So, how low a temperature do we need to get to in order to see these quantum effects? For fermions, we say this occurs when we are in a degenerate situation, which means we occupy the lowest energy levels, consistent with the Pauli exclusion principle, and the only place where we have excitations from the ground state are for the highest-energy states, where we can excite electrons from occupied to unoccupied states without having any issue with the Pauli exclusion principle. We study the degenerate Fermi gas to make these points clear. What is amazing is the same behavior governs neutral atoms cooled to 10^{-9} K, metals at room temperature, and neutron stars at 10^6 K; for these diverse systems, the notion of what is a low-energy excitation differs by at least 15 orders of magnitude! The concept of universal behavior has incredible applications to many different physical systems with widely different energy scales, but all of them being quantum.

It is common to start a discussion of many fermions with the case of noninteracting particles. It is even simpler, if we put them in free space, so they do not interact with a potential either. Hence, we have the Fermi gas problem. To start, we think of the fermions as being placed in a large cubic box of side length L and just apply the Pauli exclusion principle. We worked with a similar situation when we discussed black-body radiation, and just as we did there, we need to find the set of allowed wavefunctions that fit inside the box. For black-body radiation, we used sine waves, but for electrons it is more common to use periodic boundary conditions and plane waves, given by complex exponentials. Hence we have our $|\phi\rangle$ states being given by momentum eigenstates (which we label by the wavenumber $\vec{k} = \vec{p}/\hbar$). The periodic boundary conditions require $e^{ik_\alpha L} = 1$ for $\alpha = x$, y, and z. Hence, $k_\alpha = \frac{2\pi n_\alpha}{L}$, with n_α an integer, that we typically take to run from the integer closest to $-\frac{L}{2}$ to the integer closest to $\frac{L}{2}$. The Pauli exclusion principle will require us to have the allowed $\vec{k}$ values be different for each electron, except for the cases where we place an up spin electron and a down spin electron into the same $\vec{k}$ state. We use the bathtub

principle to fill in the lowest energy states. Since the energy is given by $\epsilon_{\vec{k}} = \frac{\hbar^2 k^2}{2M}$, we again find constant energy surfaces to be spheres. So, a sphere, whose radius is chosen to give us the correct total number of electrons, encloses all of the occupied states. Solving the degenerate Fermi gas problem simply requires us to properly determine the radius and thereby the occupied states of the system. Then we can use it to further determine other physical properties. Things become more complicated when we want to examine the properties at a low, but nonzero temperature. We will discuss some aspects of this, but we will not go through all of the details here.

Our next step is to solve for the number of states. This is just a counting exercise. We know the volume of a sphere is given by $\frac{4}{3}\pi k^3$ and the spacing between the different wavevectors is $\frac{2\pi}{L}$ in each dimension. This means we can think of each wavevector that designates a state as occupying a volume given by $\left(\frac{2\pi}{L}\right)^3$. So the number of electronic states (including spin) in a sphere of size k is

$$N_e = \underbrace{2}_{\text{spin}} \times \frac{\frac{4}{3}\pi k^3}{\left(\frac{2\pi}{L}\right)^3} = \frac{(kL)^3}{3\pi^2}, \tag{17.36}$$

or the density of electrons $\rho_e = \frac{N_e}{L^3}$ is given by

$$\rho_e = \frac{k_F^3}{3\pi^2}, \tag{17.37}$$

where, k_F, the radius of the sphere is called the Fermi wavevector. We also will want to know the ground-state energy. This is worked out in Problem 17.3.1. Sometimes, we prefer to solve for the Fermi wavevector in terms of the electron density. It is given by

$$k_F = \left(3\pi^2 \rho_e\right)^{\frac{1}{3}}. \tag{17.38}$$

It is in this fashion that we filled the lowest energy levels using the bathtub principle and found the configuration of electrons as a function of their density.

This degenerate Fermi gas is a strongly "locked" quantum state. It costs a lot of energy to excite an electron in a state with $k \ll k_F$, because we must excite it to unoccupied states outside the Fermi sphere. The electrons in states very close to the surface of the Fermi sphere can be easily excited, with low energies. But there are much fewer of these electrons than the total available. This means low-temperature properties of electrons in a degenerate Fermi gas act as if there is a much smaller number of electrons available, because the remaining electrons are "locked" in the low-lying states and cannot be easily excited. Many of the properties of electrons in solids can be understood from this simple fact. So, how high in temperature do we need to be in order for the fermions not to be degenerate? This is larger than the

Fermi temperature, which usually is a very large number (typically 10,000° K or more for metals). So, for most solids, the electrons are always in a degenerate state.

While we will not discuss statistical mechanics in great detail here, we will discuss the Fermi-Dirac distribution. It answers the question: If I am at a temperature T, what is the probability that I have an electron in a state of energy ϵ? Again, we use the Boltzmann factor, given by e^{-E/k_BT} for a state with energy E. Noting that for a fermion that we will put in a state with energy ϵ, we have $E = 0$ when we have no electron and $E = \epsilon$ when we have one. The probability for the state to be occupied is given by the ratio of the Boltzmann factor to the partition function (sum of Boltzmann weights for both states), which is

$$P(\epsilon) = \frac{e^{-\epsilon/k_BT}}{1 + e^{-\epsilon/k_BT}} = \frac{1}{1 + e^{\epsilon/k_BT}}. \tag{17.39}$$

The rightmost form is what we call the *Fermi-Dirac distribution* for an electron with energy ϵ. One of the interesting applications of the Fermi-Dirac distribution is that it is employed in the training of deep neural networks. Before this idea was introduced, linear functions were used in the mappings of inputs to outputs in the network. But, one could prove that they cannot map all possibilities. So, the Fermi-Dirac distribution was introduced, and this allowed for the training of deep neural networks and the boom in artificial intelligence and big data.

We end with a brief discussion about neutron stars, which is one of the most interesting examples of a degenerate Fermi gas. One can think of a neutron star as a hydrogen star that has been compressed so much so that the electron is combined with the proton to make a neutron and the entire object is so dense it appears similar to a huge nucleus, now consisting of an almost unthinkable large number of neutrons. The neutrons are fermions, so they satisfy the Pauli exclusion principle and fill in the lowest energy states as in the bathtub principle. The Fermi temperature of this system is tremendous, and the pressure arising from the exclusion principle is the only thing that keeps the star from collapsing into a black hole. One way to think of how an ordinary star evolved into this state is as follows. During the normal lifetime, the star will burn nuclear fuel. This fuel will provide an outward pressure that is balanced by the gravitational pressure trying to collapse the star inward. Eventually, even the Fermi pressure of the hydrogen is not enough to keep back this gravitational pull once the nuclear cycle has run out of fuel, and after a supernova, the remnants of the star consists of just nucleons packed tightly to seem as if it is a giant nucleus. Here, the Fermi pressure from the neutrons balances the general relativistic gravitational forces and the system remains stable. One can understand how this works with a fairly straightforward analysis, but it does require ideas from general relativity, which go beyond what we can do here.

The neutron stars are quite interesting beasts. They typically have a huge magnetic field that they sweep around like a lighthouse, as the star rotates about an axis. The sweeps are so regular they can be used as astronomical clocks and can also be used to observe the loss of energy from gravitational waves in a binary pulsar. The superconducting temperature for this system is also enormous, and even though

the star is hot, it is believed to also exhibit superconductivity! Finally, neutron star mergers are supposed to be responsible for all of the heavy metal formation in the universe (atoms with atomic number higher than iron). There is so much fascinating behavior in neutron stars. And to think it all comes from the fact that they behave like a huge collection of fermions!

Section 17.3 Problems:

17.3.1 Ground-State Energy of a Degenerate Fermi Gas

In the main text, we saw how a noninteracting Fermi gas in a box of volume L^3 is described by occupying the lowest N_e free-particle energy levels, resulting in a Fermi wavevector of $k_F = (3\pi^2\rho_e)^{\frac{1}{3}}$. Here, $\rho_e = \frac{N_e}{V}$ is the electron density. To find the energy of the ground state, we need to add up the energies of all of the filled energy levels.

We start by observing that because the energy is given by $\frac{\hbar^2 k^2}{2M}$ for a state with wavevector $\vec{k}$, all states on the surface of a sphere of radius k have the same energy. The number of electrons of each spin in a thin shell of width Δk is given by the volume of the shell (for each spin) divided by the volume for each state, which is $\left(\frac{2\pi}{L}\right)^3$. Determine the number of electrons in this thin shell. Do not forget to include spin in your analysis.

To find the total energy, we multiply this number by the energy of the state and add up over all filled states. You should find your result takes the form

$$E_{\text{gs}} = 2\sum_{\text{shells}} \frac{\hbar^2 k^2}{2M} \frac{4\pi k^2 \Delta k V}{(2\pi)^3}. \tag{17.40}$$

Be sure you understand where each term comes from.

How do we sum over the shells? We use our result for the sums of powers of integers. We let the nth shell have a radius $k_n = n\Delta k$ and the sum becomes

$$E_{\text{gs}} = \frac{\hbar^2 V}{2M\pi^2} \sum_{n=0}^{N} n^4 (\Delta k)^5. \tag{17.41}$$

N is determined by the fact that the largest shell has a radius given by k_F. Use this to find N.

Next, use our methods to perform the sum, replace $N\Delta k$ by k_F, and then take the limit as $\Delta k \to 0$. This final result is the ground-state energy for a degenerate Fermi gas.

Ans: $E_{\text{gs}} = \frac{3}{5} E_F N_e$.

17.3.2 Fermi Temperature and the Fermi-Dirac Distribution

The Fermi temperature is defined to be the temperature that is equal to the Fermi energy, so

$$k_B T_F = \frac{\hbar^2 k_F^2}{2M}. \tag{17.42}$$

It is common to rewrite the energy as being measured relative to the Fermi energy, so $\epsilon \to \epsilon' = \epsilon + E_F$. Use the Fermi temperature to rewrite the Fermi-Dirac distribution in the form

$$f_{\text{FD}}(\epsilon') = \frac{1}{1 + e^{\frac{\epsilon'}{k_B T} - \frac{T_F}{T}}}. \tag{17.43}$$

Using $\epsilon'/k_B T_F$ as the horizontal axis, plot the result for four different temperatures on the same plot: ($T/T_F = 0$, 0.25, 1, and 4).

The properties of a degenerate Fermi gas are determined by the ratio of the temperature to the Fermi temperature. The Fermi temperature varies widely for different systems. Find out what it is for ultracold atoms, electrons in metals, nucleons in the nucleus of an atom, and neutron stars, to get a sense of this variance. Use a common scale of eV/k_B for your temperatures.

17.3.3 Fermi Pressure

The first law of thermodynamics tells us that we can relate the pressure to changes in the energy with respect to volume. The precise relation for the degenerate Fermi gas is

$$P = -\frac{\Delta E_{\text{gs}}}{\Delta V}. \tag{17.44}$$

What this means is we take our formula for the ground-state energy and imagine we increase the volume from V to $V + \Delta V$. Then we find the change in the energy $E_{\text{gs}} \to E_{\text{gs}} + \Delta E_{\text{gs}}$ and divide ΔE_{gs} by ΔV. If you have never seen something like this, it might look a little scary, but the calculation is straightforward. Take the formula for the ground-state energy solved above and locate its volume dependence from the volume dependence of k_F. Then compute how the energy changes by using the binomial theorem after replacing $V \to V + \Delta V$, but only keep the *linear terms* in ΔV. Finally use the relation for the pressure to determine the degeneracy pressure of a Fermi gas. You should find $P = \frac{2}{5}\frac{N_e}{V} E_F = \frac{2}{5}\rho_e E_F$. It is this pressure that keeps a white dwarf star from collapsing if the degeneracy pressure is from electrons, and it is what keeps a neutron star from collapsing if it is from neutrons. Determining

the properties of the white dwarf requires adding in the forces due to gravity. But for a neutron star, we need to use a general relativistic approach to gravity to get the results correct.

17.3.4 Why Nonzero Temperature Is More Difficult Than Zero Temperature

When we calculate a ground-state property, we only need to work with the lowest-energy eigenstate in our calculations. But when we want to calculate at any nonzero temperature, we must average over all energy eigenstates, averaged by the Boltzmann factor divided by the partition function. Write a short essay explaining the challenges in working at nonzero temperature. Are their situations where nonzero temperature results simplify and become easier to work with? Think about extreme cases for the temperature.

17.4 Second Quantization for Fermions

We describe second quantization of fermions in more detail here. Recall, we start with a set of single-particle states, $\{\phi_n(\vec{r})\}$ or $\{|\phi_n\rangle\}$. The N-particle wavefunction can be expanded in terms of "product-state" Slater determinants, with each determinant given by

$$\psi_{i_1,i_2,\cdots,i_N}(\vec{r}_1,\vec{r}_2,\cdots,\vec{r}_N) = \frac{1}{\sqrt{N!}}\sum_P (-1)^P \phi_{i_1}(P\vec{r}_1)\phi_{i_2}(P\vec{r}_2)\cdots\phi_{i_N}(P\vec{r}_{i_N}), \tag{17.45}$$

and the sum over all $N!$ permutations of N objects. The full many-body wavefunction is then given by

$$\psi(\vec{r}_1,\vec{r}_2,\cdots,\vec{r}_N) = \sum_{i_1,i_2,\cdots,i_N} c_{i_1,i_2,\cdots,i_N}\psi_{i_1,i_2,\cdots,i_N}(\vec{r}_1,\vec{r}_2,\cdots,\vec{r}_N). \tag{17.46}$$

It is quite challenging to work with Slater determinants in this fashion. But, because the Slater determinants are determined just by specifying the states that are in the determinant, we can simplify the notation by using the *occupation-number representation*. We just specify only the states included in the Slater determinant. So, for example, we have $|1,0,\cdots\rangle = \phi_1(\vec{r}_1)$ and $|0,1,1,0,\cdots\rangle = \frac{1}{\sqrt{2}}\big(\phi_2(\vec{r}_1)\phi_3(\vec{r}_2) - \phi_2(\vec{r}_2)\phi_3(\vec{r}_1)\big)$, and so on. We can simplify it further by only including the nonzero terms in the Dirac notation.

We introduce operators in this number-operator space to create, annihilate, and count the individual states in any given Slater determinant. Formally, we say that

Fig. 17.6 Vladimir Fock in the 1930s. [Converted to look like a sketch by James Freericks]

$$\hat{c}_k^\dagger |n_1, n_2, \cdots, n_k, \cdots\rangle = \pm |n_1, n_2, \cdots, n_k + 1, \cdots\rangle, \tag{17.47}$$

creates a state at position k and

$$\hat{c}_k |n_1, n_2, \cdots, n_k, \cdots\rangle = \pm |n_1, n_2, \cdots, n_k - 1, \cdots\rangle, \tag{17.48}$$

annihilates a state at position k. We have included plus or minus signs, because the actual sign after operating the operator on the state depends on which other states are occupied or not, as we will discuss later. Vladimir Fock, a Soviet physicist, was instrumental in developing the occupation-number representation, which is often called *Fock space*. We have a picture of him from the 1930s in Fig. 17.6.

Now, the Pauli exclusion principle tells us that $n_k = 0$ or $n_k = 1$ only, so we must have $(\hat{c}_k^\dagger)^2 = 0$ and $\hat{c}_k^2 = 0$. We define the vacuum state to be the state with all $n_k = 0$, or $|0\rangle = |0, 0, 0, \cdots\rangle$. Then $\langle \vec{r}_1 | \hat{c}_k^\dagger |0\rangle = \phi_k(\vec{r}_1)$ and

$$\langle \vec{r}_1, \vec{r}_2 | \hat{c}_{k'}^\dagger \hat{c}_k^\dagger |0\rangle = \frac{1}{\sqrt{2}} \left(\phi_k(\vec{r}_1)\phi_{k'}(\vec{r}_2) - \phi_k(\vec{r}_2)\phi_{k'}(\vec{r}_1) \right), \tag{17.49}$$

and

$$\langle \vec{r}_1, \vec{r}_2 | \hat{c}_k^\dagger \hat{c}_{k'}^\dagger |0\rangle = \frac{1}{\sqrt{2}} \left(\phi_{k'}(\vec{r}_1)\phi_k(\vec{r}_2) - \phi_{k'}(\vec{r}_2)\phi_k(\vec{r}_1) \right). \tag{17.50}$$

This implies we must have

$$\hat{c}_k^\dagger \hat{c}_{k'}^\dagger + \hat{c}_{k'}^\dagger \hat{c}_k^\dagger = 0. \tag{17.51}$$

The *anticommutator* of any two creation operators equals zero. By taking the Hermitian conjugate, we find the anticommutator of any two annihilation operators is also zero: $\hat{c}_k \hat{c}_{k'} + \hat{c}_{k'} \hat{c}_k = 0$. For the anticommutator between the creation and annihilation operators, we must first work out how the operators act on the states. Assuming we have at most one electron in our state (to avoid dealing with the plus or minus signs for the moment), we find that

$$\hat{c}_k^\dagger |0_k\rangle = |1_k\rangle, \quad \hat{c}_k^\dagger |1_k\rangle = 0, \quad \hat{c}_k |0_k\rangle = 0, \text{ and } \hat{c}_k |1_k\rangle = |0_k\rangle. \tag{17.52}$$

From this, we can easily see that we must have

$$\hat{c}_k^\dagger \hat{c}_{k'} + \hat{c}_{k'} \hat{c}_k^\dagger = \delta_{kk'}. \tag{17.53}$$

Here, we require the creation and annihilation operators for different states to also anticommute, although this is not a strict requirement of the formalism. Now, when we act the creation and annihilation operators on states with one or more electrons already in them, we will pick up another $(-1)^M$ factor if we have to interchange the operator with M fermionic creation operators that are blocking where the current operator is supposed to go (i.e., M is the number of creation operators in the state already with index smaller than the index of the current operator). We can see that a single Slater determinant is now written as $\hat{c}_{i_1}^\dagger \hat{c}_{i_2}^\dagger \cdots \hat{c}_{i_N}^\dagger |0\rangle$, which motivates why we call it a product state (even though it is a complicated and entangled sum over many terms when we construct its wavefunction—it is because it maintains the product representation in terms of operators, which we will see comes in handy for calculations).

We can also define the *number operator*, which counts the number of fermions of the given index in a state. Using Eq. (17.53), we immediately see that

$$\hat{c}_k^\dagger \hat{c}_k |0_k\rangle = 0 \text{ and } \hat{c}_k^\dagger \hat{c}_k |1_k\rangle = |1_k\rangle, \tag{17.54}$$

precisely does this counting for us. We define $\hat{n}_k = \hat{c}_k^\dagger \hat{c}_k$ as the number operator. Then $\hat{N} = \sum_k \hat{n}_k$ is the total number operator. One can directly verify the commutation relations

$$[\hat{n}_k, \hat{c}_k^\dagger] = \hat{c}_k^\dagger \text{ and } [\hat{n}_k, \hat{c}_k] = -\hat{c}_k. \tag{17.55}$$

These results are very similar to what we had for the ladder and number operators of the harmonic oscillator.

We have discussed the states. Now, we must discuss how we represent operators in the second-quantized formalism. Operators are characterized in terms of the

number of fermions they affect—single-particle operators depend on a single particle, two-particle operators on two, and so forth. Most terms in a Hamiltonian involve only single and two-particle terms.

We denote single-particle operators as $\hat{O}_1$. They take the form of a matrix element between single-fermion states $\langle k\sigma|\hat{O}|k'\sigma'\rangle$, where we now explicitly include the spin labels. In second quantization, we have

$$\hat{O}_1 \leftrightarrow \sum_{k\sigma k'\sigma'} \langle k\sigma|\hat{O}_1|k'\sigma'\rangle \hat{c}^\dagger_{k\sigma}\hat{c}_{k'\sigma'}. \tag{17.56}$$

Here k labels the spatial degree of freedom and σ labels the spin degree of freedom. An example is the momentum operator. If we choose the set of $\{|\phi_i\rangle\}$ to be the momentum states in a box with periodic boundary conditions, the allowed momenta are discrete points uniformly distributed in momentum space. We have the momentum expectation value satisfies $\langle\vec{p}'|\hat{\vec{p}}|\vec{p}''\rangle = \vec{p}'\delta_{\vec{p}'\vec{p}''}$, so in second quantization, the momentum operator becomes

$$\hat{\vec{p}} \leftrightarrow \sum_{\vec{p}'\sigma} \vec{p}'\hat{c}^\dagger_{\vec{p}'\sigma}\hat{c}_{\vec{p}'\sigma} = \sum_{\vec{p}'\sigma} \vec{p}'\hat{n}_{\vec{p}'\sigma}. \tag{17.57}$$

Similarly, the spin operators become

$$\hat{S}_z \leftrightarrow \frac{\hbar}{2}\sum_k \left(\hat{c}^\dagger_{k\uparrow}\hat{c}_{k\uparrow} - \hat{c}^\dagger_{k\downarrow}\hat{c}_{k\downarrow}\right), \quad \hat{S}_+ \leftrightarrow \hbar\sum_k \hat{c}^\dagger_{k\uparrow}\hat{c}_{k\downarrow}, \quad \text{and} \quad \hat{S}_- \leftrightarrow \hbar\sum_k \hat{c}^\dagger_{k\downarrow}\hat{c}_{k\uparrow}. \tag{17.58}$$

This representation of total spin operators should be clear. For the z-component, we count the number up minus the number down and multiply by $\frac{\hbar}{2}$. The raising operator removes a down spin and creates an up spin, and so on. Note that it is very common when working with model systems to drop the $\hbar$ and work with dimensionless total spin operators. We will, in fact, do this in the next chapter.

Things get a bit more complicated when we work with two-particle operators, such as the Coulomb repulsion between electrons. In this case, we have

$$\begin{aligned}\hat{O}_2 &= \frac{e^2}{4\pi\epsilon_0|\hat{\vec{r}}_1 \otimes \mathbb{I} - \mathbb{I} \otimes \hat{\vec{r}}_2|} \\ &\leftrightarrow \sum_{\substack{k_1k_2k_3k_4 \\ \sigma_1\sigma_2\sigma_3\sigma_4}} \langle k_1\sigma_1| \otimes \langle k_2\sigma_2|\hat{O}_2|k_4\sigma_4\rangle \otimes |k_3\sigma_3\rangle \hat{c}^\dagger_{k_1\sigma_1}\hat{c}^\dagger_{k_2\sigma_2}\hat{c}_{k_3\sigma_3}\hat{c}_{k_4\sigma_4}.\end{aligned} \tag{17.59}$$

Here $\hat{\vec{r}}_1$ is associated with the states $\langle k_1\sigma_1|$ and $|k_4\sigma_4\rangle$ and $\hat{\vec{r}}_2$ is associated with $\langle k_2\sigma_2|$ and $|k_3\sigma_3\rangle$. The Coulomb interaction is one of the most important operators to get an explicit formula for. Unfortunately, the derivation requires calculus to evaluate all of the matrix elements. So, we will just state the result. The Coulomb interaction, when we work in a momentum-space basis, is given by

$$\frac{1}{2}\sum_{\vec{k}\vec{k}'\vec{q}\sigma\sigma'}\frac{e^2}{V\epsilon_0 q^2}\hat{c}^\dagger_{\vec{k}+\vec{q}\sigma}\hat{c}^\dagger_{\vec{k}'-\vec{q}\sigma'}\hat{c}_{\vec{k}'\sigma'}\hat{c}_{\vec{k}\sigma}. \tag{17.60}$$

The factor of $\frac{1}{2}$ in front is introduced to compensate for the so-called double counting of the interaction and here we are writing the momentum as $\vec{p} = \hbar\vec{k}$, and so on. The double counting correction is subtle. If we go back to the original form of the Coulomb interaction in position space, we see it has a denominator that goes like $|\vec{r}_1 - \vec{r}_2|$. If we sum freely over all $\vec{r}_1$ and $\vec{r}_2$, we count the case where one electron is at a position and the other is at another position twice, because both of the position vectors run over all possible position values. This is also similar to the issue we had when discussing interactions on a lattice and we ran over each nearest-neighbor pair once, while if we summed over all sites, we would include each nearest-neighbor pair twice. Going to momentum space does not correct the double counting. So, we must multiply the result by $\frac{1}{2}$. We have a video Fermionic Creation and Annihilation Operators https://tinyurl.com/mt39dwva that covers some different perspectives on this.

In the remainder of this section, we will discuss how to work with operators and states and we will look carefully at a simplified model to describe electrons in metals called the jellium model. This is a model where electrons repel each other with the Coulomb interaction, but they interact with the ions in an approximate way, where the positive charge of the ion cores in a metal are smeared out into a uniform positive charge. The Hamiltonian, when written in a momentum-space basis, takes the form

$$\hat{H}_{\text{jellium}} = \sum_{\vec{k}\sigma}\frac{\hbar^2 k^2}{2M}\hat{c}^\dagger_{\vec{k}\sigma}\hat{c}_{\vec{k}\sigma} + \frac{e^2}{2\epsilon_0 V}\sum_{\substack{\vec{k}\vec{k}'\vec{q}\sigma\sigma' \\ \vec{q}\neq 0}}\frac{1}{q^2}\hat{c}^\dagger_{\vec{k}+\vec{q}\sigma}\hat{c}^\dagger_{\vec{k}'-\vec{q}\sigma'}\hat{c}_{\vec{k}'\sigma'}\hat{c}_{\vec{k}\sigma}. \tag{17.61}$$

The first term is the kinetic energy for the electrons, and the second term is the potential energy. We do not include the term with $\vec{q} = 0$, because that term is cancelled by the uniform positive background charge. This avoids us having to deal with an infinity. To approximately solve the model, we use the bathtub principle to minimize the kinetic energy and then simply calculate the potential energy contribution. The Fermi wavevector is the same as before ($k_F = (3\pi\rho_e)^{1/3}$), as is the kinetic energy, which is given by $\frac{3}{5}E_F\rho_e V$.

To calculate the different expectation values in second quantization, we will use Wick's theorem, which tells us that we can replace averages of products of fermionic operators by products of averages of pairs of fermionic operators when the average is taken in a state with a single Slater determinant. You will prove the theorem in Problem 17.4.2. Here, we examine it for the expectation value for the kinetic energy and for the potential energy. For the kinetic energy, we have that it is equal to

$$\sum_{\vec{k}\sigma} \frac{\hbar^2 k^2}{2M} \langle \psi | \hat{c}^\dagger_{\vec{k}\sigma} \hat{c}_{\vec{k}\sigma} | \psi \rangle, \tag{17.62}$$

where

$$|\psi\rangle = \prod_{\substack{\vec{k}\sigma \\ |\vec{k}| \le k_F}} \hat{c}^\dagger_{\vec{k}\sigma} |0\rangle, \tag{17.63}$$

which occupies the N_e lowest energy levels, each occupied with an up spin and a down spin until all of the electrons are exhausted via the bathtub principle. The operator in the expectation value is the number operator, so the expectation value is equal to 1 if $\vec{k}$ is occupied and is equal to 0 if it is not. This then leads to the result for the kinetic energy that we quoted above. Now, for the potential energy, we have

$$\frac{e^2}{2\epsilon_0 V} \sum_{\substack{\vec{k}\vec{k}'\vec{q}\sigma\sigma' \\ \vec{q}\neq 0}} \frac{1}{q^2} \langle \psi | \hat{c}^\dagger_{\vec{k}+\vec{q}\sigma} \hat{c}^\dagger_{\vec{k}'-\vec{q}\sigma'} \hat{c}_{\vec{k}'\sigma'} \hat{c}_{\vec{k}\sigma} | \psi \rangle. \tag{17.64}$$

Because we have annihilation operators to the right in the expectation value, it will yield 0 unless the state is occupied in $|\psi\rangle$. Once a state in $|\psi\rangle$ is annihilated, we must create it again so that we reconstruct $|\psi\rangle$, so we can take its inner product with $\langle\psi|$. There are two ways to do this: (i) $\vec{k} + \vec{q} = \vec{k}$ and $\vec{k}' - \vec{q} = \vec{k}'$ (which requires $\vec{q} = 0$ or (ii) $\vec{k} + \vec{q} = \vec{k}'$, $\vec{k}' - \vec{q} = \vec{k}$, and $\sigma = \sigma'$. In case (i), we must have $\vec{q} = 0$. But there is no term with $\vec{q} = 0$ in the summation, so that term does not contribute. For case (ii), we have $\vec{q} = \vec{k}' - \vec{k}$ solves the problem. To evaluate the averages as averages of number operators, we must interchange one creation and one annihilation operator, which gives us a minus sign because they anticommute. Hence, we have the potential energy becomes

$$-\frac{1}{2V} \sum_{\vec{k}\neq\vec{k}'\sigma} \frac{e^2}{\epsilon_0 |\vec{k}' - \vec{k}|} \theta(|\vec{k}| < k_F)\theta(|\vec{k}'| < k_F). \tag{17.65}$$

Remarkably, this sum can be done exactly. It requires calculus, so we just state the result

$$-\frac{V e^2}{16\pi^4 \epsilon_0} k_F^4. \tag{17.66}$$

We will re-express this result in a different way that is commonly used for analysis in Problem 17.4.3.

This calculation of the energy in the jellium model is a variational calculation. We make a guess for the wavefunction (here chosen by minimizing the kinetic energy) and compute the total energy by computing the expectation value of the Hamiltonian

in this state. It is a rather simplistic model for a metal, because it does not include the fact that metals have the ion cores organized on a periodic lattice. Nevertheless, it is a quantitative model that can be evaluated and examined to see what properties it has. Here, we considered the simplest approach, by using the state that minimizes the kinetic energy. Improving on this result becomes increasingly difficult. You can study how we do that and how we include the periodic lattice in your favorite solid-state physics textbook.

Section 17.4 Problems:

17.4.1 Operator Gymnastics with Spin and Pseudospin

Consider a two-level system described by fermionic creation and annihilation operators that satisfy the following anticommutation relations:

$$\{\hat{c}^\dagger_{i\sigma}, \hat{c}_{j\sigma'}\} = \delta_{ij}\delta_{\sigma\sigma'}, \quad \{\hat{c}^\dagger_{i\sigma}, \hat{c}^\dagger_{j\sigma'}\} = 0, \quad \text{and} \quad \{\hat{c}_{i\sigma}, \hat{c}_{j\sigma'}\} = 0, \tag{17.67}$$

for $i, j = 1, 2$.

Construct the total spin operators $\hat{S}^+$, $\hat{S}^-$, and $\hat{S}^z$ (raising, lowering, and z-component). Verify, using the anticommutation relations of the second-quantized fermionic operators, that

$$[\hat{S}^+, \hat{S}^-] = 2\hbar\hat{S}^z \quad \text{and} \quad [\hat{S}^z, \hat{S}^\pm] = \pm\hbar\hat{S}^\pm. \tag{17.68}$$

The pseudospin operators are defined by

$$\begin{aligned}
\hat{J}^+ &= \hat{c}^\dagger_{1\uparrow}\hat{c}^\dagger_{1\downarrow} - \hat{c}^\dagger_{2\uparrow}\hat{c}^\dagger_{2\downarrow}, \\
\hat{J}^- &= \hat{c}_{1\downarrow}\hat{c}_{1\uparrow} - \hat{c}_{2\downarrow}\hat{c}_{2\uparrow}, \\
\hat{J}^z &= \frac{1}{2}\left(\hat{c}^\dagger_{1\uparrow}\hat{c}_{1\uparrow} + \hat{c}^\dagger_{1\downarrow}\hat{c}_{1\downarrow} + \hat{c}^\dagger_{2\uparrow}\hat{c}_{2\uparrow} + \hat{c}^\dagger_{2\downarrow}\hat{c}_{2\downarrow}\right) - 1.
\end{aligned} \tag{17.69}$$

Show that the pseudospin operators satisfy an SU(2) Lie algebra:

$$[\hat{J}^+, \hat{J}^-] = 2\hat{J}^z \quad \text{and} \quad [\hat{J}^z, \hat{J}^\pm] = \pm\hat{J}^\pm. \tag{17.70}$$

Also show that the spin and pseudospin operators commute:

$$[\hat{J}^\pm, \hat{S}^\pm] = [\hat{J}^\pm, \hat{S}^z] = [\hat{J}^z, \hat{S}^\pm] = [\hat{J}^z, \hat{S}^z] = 0. \tag{17.71}$$

In this system, the total J (and total S) value can be 0, $\frac{1}{2}$, or 1, the m_J value ranges from $-J$ to J (and the spin m_S from $-S$ to S). Find J, m_J, S, and m_S for

the following states:

$$\begin{aligned}
&\hat{c}^\dagger_{1\uparrow}|0\rangle \\
&\hat{c}^\dagger_{1\uparrow}\hat{c}^\dagger_{2\uparrow}|0\rangle \\
&\frac{1}{\sqrt{2}}\left(\hat{c}^\dagger_{1\uparrow}\hat{c}^\dagger_{2\downarrow} \pm \hat{c}^\dagger_{1\downarrow}\hat{c}^\dagger_{2\uparrow}\right)|0\rangle \\
&\frac{1}{\sqrt{2}}\left(\hat{c}^\dagger_{1\uparrow}\hat{c}^\dagger_{1\downarrow} - \hat{c}^\dagger_{2\uparrow}\hat{c}^\dagger_{2\downarrow}\right)|0\rangle.
\end{aligned} \tag{17.72}$$

The calculation can be done either in a brute-force way working with the operators or in a more clever way by thinking about the structure of the two SU(2) operators and how the raising and lowering operators carry us between different states.

17.4.2 Wick's Theorem

We work with a general $|\psi\rangle$ given by

$$\prod_{n=1}^{N} \hat{c}^\dagger_{i_n\sigma_n}|0\rangle \tag{17.73}$$

for a N fermion state. Consider the expectation value

$$\langle\psi|\hat{c}^\dagger_{k\sigma}\hat{c}_{k'\sigma'}|\psi\rangle. \tag{17.74}$$

Show that it vanishes unless $k = k'$ and $\sigma = \sigma'$ and the state k, σ is in the set $\{i_n, \sigma_n | n = 1, 2, \cdots, N\}$. In other words, the expectation value vanishes unless it is a number operator, and in that case, it is nonvanishing only if the state it counts is a state in $|\psi\rangle$. Verify as well that the expectation value of two creation operators or two annihilation operators always vanishes.

The idea of Wick's theorem is that the only nonzero averages arise when we can create number operators by our string of creation and annihilation operators. When we have more than two operators, we can construct number operators in several different ways. Each way is called a *contraction*. We will now study this for the case with four operators. We must have two creation operators and two annihilation operators. We consider the following case (which is not the most general case):

$$\langle\psi|\hat{c}^\dagger_{k_1\sigma}\hat{c}^\dagger_{k_2\sigma'}\hat{c}_{k_3\sigma'}\hat{c}_{k_4\sigma}|\psi\rangle. \tag{17.75}$$

Note first that if $k_1 = k_2 = k_3 = k_4$ and $\sigma = \sigma'$ that the expectation value vanishes, because the square of a creation or annihilation operator vanishes. There are two

ways to pair the indices to make number operators. One of them requires a change of sign due to anticommutation. Verify that

$$\langle\psi|\hat{c}^\dagger_{k_1\sigma}\hat{c}^\dagger_{k_2\sigma'}\hat{c}_{k_3\sigma'}\hat{c}_{k_4\sigma}|\psi\rangle = \delta_{k_1k_4}\delta_{k_2k_3}\langle\psi|\hat{c}^\dagger_{k_1\sigma}\hat{c}_{k_1\sigma}\hat{c}^\dagger_{k_2\sigma'}\hat{c}_{k_2\sigma'}|\psi\rangle - \delta_{k_1k_3}\delta_{k_2k_4}\delta_{\sigma\sigma'}\langle\psi|\hat{c}^\dagger_{k_1\sigma}\hat{c}_{k_1\sigma}\hat{c}^\dagger_{k_2\sigma}\hat{c}_{k_2\sigma}|\psi\rangle. \tag{17.76}$$

Note that when $k_1 = k_2 = k_3 = k_4$ and $\sigma = \sigma'$ the first term cancels the second, so it is zero, as claimed. This means we can assume that case does not occur when computing the averages. This allows them to be independent of each other. Verify that

$$\langle\psi|\hat{c}^\dagger_{k_1\sigma}\hat{c}^\dagger_{k_2\sigma'}\hat{c}_{k_3\sigma'}\hat{c}_{k_4\sigma}|\psi\rangle = \langle\psi|\hat{c}^\dagger_{k_1\sigma}\hat{c}_{k_4\sigma}|\psi\rangle\,\langle\psi|\hat{c}^\dagger_{k_2\sigma'}\hat{c}_{k_3\sigma'}|\psi\rangle - \langle\psi|\hat{c}^\dagger_{k_1\sigma}\hat{c}_{k_3\sigma}|\psi\rangle\,\langle\psi|\hat{c}^\dagger_{k_2\sigma}\hat{c}_{k_4\sigma}|\psi\rangle\delta_{\sigma\sigma'}. \tag{17.77}$$

It is this last result which is called Wick's theorem. We replace the averages of a product of operators by the product of the averages of pairs of operators, with each pair chosen by the different contractions we can make. It only holds for Slater determinant states individually.

Describe how the result will generalize for expectation values of larger number of products of operators. You do not need to prove it for the general case.

17.4.3 Re-expressing the Jellium Energy in Terms of r_s

In the main text, we found the variational energy per unit volume was given by

$$\frac{E}{V} = \frac{\hbar^2}{2M}\frac{k_F^5}{5\pi^2} - \frac{e^2}{4\pi\epsilon_0}\frac{k_F^4}{4\pi^3}. \tag{17.78}$$

It is more common to express this in terms of the energy per particle. Using $N_e = V\rho_e = Vk_F^3/3\pi^2$ and the Bohr radius $a_0 = 4\pi\epsilon_0\hbar^2/Me^2$, show that the energy per particle can be written as

$$\frac{E}{N_e} = \frac{e^2}{8\pi\epsilon_0 a_0}\left(\frac{3}{5}(k_F a_0)^2 - \frac{3}{2\pi}(k_F a_0)\right). \tag{17.79}$$

It is more common to write the result in terms of r_S which is the ratio of the radius of a sphere that contains one electron to the Bohr radius. That is, we have $\frac{4}{3}\pi r^3 = \frac{1}{\rho_e}$ and $r_s = \frac{r}{a_0}$. Express the energy per particle in terms of r_s. You should find that

$$\frac{E}{N_e} = \left(\frac{2.210}{r_s^2} - \frac{0.916}{r_s} \right) \text{ Ry}, \tag{17.80}$$

where Ry is a Rydberg and is equal to 13.6 eV.

A more detailed analysis, requiring a lot more work, shows that the next terms in the expansion are $0.0622 \ln r_s - 0.142$ in Rydbergs. This expansion is accurate for small r_s. Unfortunately, in metals one typically has r_s ranging from 1 to 10. So, this approximation does not work so well there.

17.4.4 Hopping of Electrons on a Lattice and the Tight-Binding Bandstructure

Most of the solid state condenses into periodic lattices. In metals, we have the core electrons tightly bound to the nucleus forming what is called the ion core, which has a net positive charge and is localized at each lattice site. The remaining electrons are free to move throughout the lattice and one of the simplest pictures to describe this is that they tunnel from one lattice site to a neighboring lattice site with an energy that governs the rate of tunneling. In second-quantization the model becomes a simplified model that is called the hopping model. The Hamiltonian is given by

$$\hat{H}_{\text{hopping}} = -\sum_{ij} \sum_{\sigma} t_{ij} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma}. \tag{17.81}$$

You can see the operator removes an electron of spin σ from site j and places it at site i without changing the spin. These lattice sites are located at the positions $\vec{R}_i$, which sit at equally spaced locations separated by the distance a in the x or y directions for a square lattice. The energy $-t_{ij}$ is the *hopping matrix*. For our work, we will take it to be a real symmetric matrix. If we only have nearest-neighbor hopping, then $-t_{ij} \neq 0$ only when i and j are nearest neighbors. Note that the hopping moves the electrons of each spin around on the lattice but never converts one spin into the other. For this problem, on a square lattice, we take $-t_{ij} = -t$ if $\vec{R}_i - \vec{R}_j = \pm a\vec{e}_x$ or $\pm a\vec{e}_y$ only.

Surprisingly, this Hamiltonian can be exactly solved if the lattice has periodic boundary conditions. For concreteness, we will work with a square lattice of length aL. Periodic boundary conditions means that the lattice site i is equivalent to the lattice site translated by aL in either the x or the y direction.

We define new fermionic operators in terms of the original ones via

$$\hat{a}^{\dagger}_{\vec{k}\sigma} = \frac{1}{L} \sum_{j} e^{i\vec{k}\cdot\vec{R}_j} \hat{c}^{\dagger}_{j\sigma} \text{ and } \hat{a}_{\vec{k}\sigma} = \frac{1}{L} \sum_{j} e^{-i\vec{k}\cdot\vec{R}_j} \hat{c}_{j\sigma}. \tag{17.82}$$

Show that the anticommutator of two of these new operators is given by

$$\left\{\hat{a}^{\dagger}_{\vec{k}\sigma}, \hat{a}_{\vec{k}'\sigma'}\right\} = \frac{1}{L^2}\sum_j e^{i(\vec{k}-\vec{k}')\cdot\vec{R}_j}\delta_{\sigma\sigma'}. \tag{17.83}$$

One can immediately see that if $\vec{k} = \vec{k}'$ then the sum is equal to L^2 which cancels the L^2 factor in the denominator and the net result is equal to one. Using the fact that the periodic boundary conditions requires the wavevector to take the form

$$\vec{k} = \frac{2\pi}{aL}(m\vec{e}_x + n\vec{e}_y), \tag{17.84}$$

with $0 \leq m, n \leq L-1$, show that when $\vec{k} \neq \vec{k}'$ that the sum equals zero. To get started, rewrite the sum over j as the sum over the x and y lattice site directions separately and note that $\vec{k} - \vec{k}'$ is a nonzero wavevector of the form $(2\pi/aL)(m\vec{e}_x + n\vec{e}_y)$ for some integers m and n (and both cannot be equal to zero). Then, we need to evaluate

$$\sum_{j=0}^{L-1}\sum_{j'=0}^{L-1} e^{i\frac{2\pi}{L}(mj+nj')}. \tag{17.85}$$

Show this vanishes by using results from our analysis of the geometric series as long as at least one of m or n is nonzero. This establishes that

$$\frac{1}{L^2}\sum_j e^{i(\vec{k}-\vec{k}')\cdot\vec{R}_j} = \delta_{\vec{k}\vec{k}'}. \tag{17.86}$$

It also shows that the $\hat{a}$ and $\hat{a}^{\dagger}$ operators are fermionic creation and annihilation operators.

Use this relation to verify the inverse relations

$$\hat{c}^{\dagger}_{j\sigma} = \frac{1}{L}\sum_{\vec{k}} e^{-i\vec{k}\cdot\vec{R}_j}\hat{a}^{\dagger}_{\vec{k}\sigma} \quad \text{and} \quad \hat{c}_{j\sigma} = \frac{1}{L}\sum_{\vec{k}} e^{i\vec{k}\cdot\vec{R}_j}\hat{a}_{\vec{k}\sigma}. \tag{17.87}$$

Use the fact that the $\hat{c}$ operator must be unchanged when we translate $\vec{R}_j \rightarrow \vec{R}_j + aL\vec{e}_x$ and by $\vec{R}_j \rightarrow \vec{R}_j + aL\vec{e}_y$; show that the $\vec{k}$ values must be given by the form we stated above.

Next, substitute these expressions for the $\hat{c}$ and $\hat{c}^{\dagger}$ operators into the Hamiltonian and re-express it in terms of the $\hat{a}$ and $\hat{a}^{\dagger}$ operators. You should find that

$$\hat{H}_{\text{hopping}} = \sum_{\vec{k}}\sum_{\sigma} \epsilon(\vec{k})\hat{a}^{\dagger}_{\vec{k}\sigma}\hat{a}_{\vec{k}\sigma} \tag{17.88}$$

with

$$\epsilon(\vec{k}) = -2t\cos(\vec{k}\cdot\vec{e}_x a) - 2t\cos(\vec{k}\cdot\vec{e}_y a) = -2t\cos(k_x a) - 2t\cos(k_y a). \quad (17.89)$$

The Hamiltonian is now in a diagonal form, with energies given by the bandstructure for each $\vec{k}$ value. We will explore this further later in the book. What we have done here is transform the problem from a position-space representation to a momentum-space representation. But, we did it with the operators rather than the wavefunctions, because we are working with second quantization.

17.5 Working with Fermionic States

As you can see, it is very tedious to work with fermionic states and to get the signs correct. Every time we interchange two fermionic operators, we get a sign change if they anticommute (we also get an extra term if the anticommutator is not zero). This becomes quite difficult to keep straight with pen-and-paper calculations. If you want to get serious in working with fermionic operators, you eventually have to do things numerically. This works well, because computers are perfectly content doing tedious and repetitive tasks for you. But, to get computers to work well, we need to understand a bit about how we would work with these systems on a computer. In this section, we will describe some of the strategies to work with fermionic operators and states in second quantization.

The first thing we need to decide is the protocol we will use for storing the states. Because each state can be occupied by a spin-up or a spin-down fermion, we have two obvious protocols to use. Both work, but you need to stick to one and only one. The first way to do it is to order the states as $1\uparrow, 1\downarrow, 2\uparrow, 2\downarrow$, and so on. Here, we include the spatial states in order with both possibilities of spin. The second way is to separate all up spins first and then all down spins. This looks like $1\uparrow, 2\uparrow, \cdots, N\uparrow, 1\downarrow, 2\downarrow$, and so on. How this works is that when we apply a $\hat{c}^\dagger_{j\sigma}$ operator onto a state, we must first move it to the location where it is supposed to sit. This may require interchanging it with some number of other creation operators, all of which it anticommutes with. We count the number of interchanges and if it is odd, we have a minus sign, if even, a plus sign.

Storing and working with the state can be done very efficiently. Once the mapping of each position for the different creation operators has been fixed, the only information we need for a given state is whether such an operator is present or not. This is a binary choice, and we usually assign 0 for not there and 1 for is there. Hence, if I have $2N$ possible spin states, each Slater determinant can be stored with a $2N$-bit long binary integer that has precisely N_e binary 1 digits in it. This can be a very efficient way to store the states. It also can usually be done with unsigned 64 bit integers, because we rarely work with more than 32 spatial states in a calculation. Languages that allow larger unsigned integers can handle those larger systems as well.

The full state is a linear superposition of many different Slater determinants, each with a different complex-valued coefficient (sometimes the properties of the problem restrict the coefficients to always being real). A clever way of storing a state is either with an array or with a dictionary. If we use the integer number that represents the state in binary form to be the index of the array (or the keys for a dictionary), then the value is the coefficient. This is by far the most efficient way to encode the data. But, it can run into memory issues if the number of entries is too large. In this case, if one expects the total number of nonzero coefficients to be a small subset of the total number of allowed Slater determinants, one can use pointers for the states as a way to compress the storage further. In any case, if one chooses a binary encoding, there still is the issue of how to work with such a state. While one can use bit operations to perform tasks on the states, it usually is much easier to use a conversion routine that will convert the binary encoding into a more human readable encoding, such as storing the respective spin orbitals that are occupied in the given Slater determinant as an array. Then, we work with the states in this more manageable form and convert back to the binary form only when we need to store them.

The key issue is how do we act operators onto these states? We know that the second-quantized operators are usually formed as sums over products of operators with numerical coefficients. So, we focus on just applying a product of creation and annihilation operators to a state. Suppose, we wish to apply a single-particle operator, of the form $\hat{c}^{\dagger}_{j\sigma}\hat{c}_{j'\sigma'}$ onto a state $|\psi\rangle$. Rather than write out a general result, it is easier to consider a concrete example to see how this works. So, we will apply $\hat{c}^{\dagger}_{1\uparrow}\hat{c}_{1\downarrow}$ to the state (stored in the up-spin-first format) $|\psi\rangle = \hat{c}^{\dagger}_{2\uparrow}\hat{c}^{\dagger}_{1\downarrow}|0\rangle$. How does it work? The simplest strategy to take is to move the pair of operators as a unit to the location where the creation operator partner to the annihilation operator in the operator is. In this case, we wish to move the operator $\hat{c}^{\dagger}_{1\uparrow}\hat{c}_{1\downarrow}$ to the location just to the left of the $\hat{c}^{\dagger}_{1\downarrow}$ operator in the state. For this move, there are no sign changes, because when the pair of operators moves through any single fermionic operator, we always have two sign changes. Note that if the $1\uparrow$ state is already occupied, then this operator will annihilate the state, because we would have a $(\hat{c}^{\dagger}_{1\uparrow})^2$ operator, which vanishes. Similarly, if the $1\downarrow$ state is not occupied, this operator will annihilate the state as well, because $\hat{c}_{1\downarrow}|0\rangle = 0$. One can easily check these two conditions before going through the action of the operator on the state, to make your code more efficient. Now, after moving the pair to its proper spot, we have

$$\hat{c}^{\dagger}_{1\uparrow}\hat{c}_{1\downarrow}|\psi\rangle = \hat{c}^{\dagger}_{2\uparrow}\hat{c}^{\dagger}_{1\uparrow}\hat{c}_{1\downarrow}\hat{c}^{\dagger}_{1\downarrow}|0\rangle, \tag{17.90}$$

with no sign change. Now, we use our add zero trick (which works even if there are additional operators to the right of the operator we are working with) to replace $\hat{c}_{1\downarrow}\hat{c}^{\dagger}_{1\downarrow}|0\rangle$ with $\{\hat{c}_{1\downarrow}, \hat{c}^{\dagger}_{1\downarrow}\}|0\rangle = |0\rangle$, because the anticommutator is equal to 1. Now, we have the state

$$\hat{c}^{\dagger}_{1\uparrow}\hat{c}_{1\downarrow}|\psi\rangle = \hat{c}^{\dagger}_{2\uparrow}\hat{c}^{\dagger}_{1\uparrow}|0\rangle. \tag{17.91}$$

But, it is in the wrong order. To put it into its canonical form, we need to move the $\hat{c}^\dagger_{1\uparrow}$ operator to its canonical location. This then gives us our final result:

$$\hat{c}^\dagger_{1\uparrow}\hat{c}_{1\downarrow}|\psi\rangle = -\hat{c}^\dagger_{1\uparrow}\hat{c}^\dagger_{2\uparrow}|0\rangle. \tag{17.92}$$

Note that in this case we do have a minus sign when the operator operates on the state. Working out the general case is tedious, but straightforward, and hopefully this example makes clear how it must work. In Problem 17.5.2, you will explore how to do this for a two-particle operator.

Sometimes we have to store operators in the computer as matrices, and this can exhaust all of the computer memory very quickly. Usually, the operators that need to be stored are sparse operators, which have many entries that are 0 in their matrix representation. Then, we can use sparse matrix methods to store only the nonzero elements. This usually requires storing the nonzero values of the matrix element and the two indices of the matrix element.

Another question we have to answer is which basis do we use? This is much more than do I work in position space or momentum space. As we saw, the translational invariance of a finite-sized lattice with periodic boundary conditions makes working in momentum space appear attractive, but this is not always a savings in our work, as some operators might be local in position space and thereby sparse, but are much less sparse in momentum space. Only after thinking through all of the details for a given problem should you make your decision.

Finally, we know matrices that represent operators will take a block-diagonal form when the states have definite quantum numbers, because two states with definite quantum numbers will have a vanishing matrix element for an operator that has the symmetry corresponding to those quantum numbers. For example, if the Hamiltonian commutes with the total spin and z-component of spin operators, then we can arrange our states into total spin multiplets, and matrix elements of the Hamiltonian between states with different quantum numbers will be zero. Furthermore, the energy eigenvalues for any state in a spin multiplet will be the same, so we need only find one representative to compute the energy of that multiplet. As an example, assume I have an even number of electrons; then, if I work in the subspace with $m_S = 0$ (equal numbers of up spins as down spins), I will have a representative for every total spin state. But, the Hamiltonian matrix will only become block diagonal if I also construct states that have definite total spin. There is a trade-off to doing this. If I work with Slater determinants as my basis vectors, then each time an operator acts on a state, it acts on a single Slater determinant, which is simple. But, if I work with states of definite total spin, they will involve linear superpositions of many Slater determinants, which means when an operator acts on the state, it acts on many Slater determinants, and then inner products with other states in this basis also involve many Slater determinants as well. Even though this sounds more complicated, usually it is better to use the symmetry to block-diagonalize the matrices. And the computer has no problem doing the work for you. The way to find the different total spin states is usually to construct them starting

from a single spin. One can also construct the maximal m_S spin states for a given S and then hit them with the appropriate number of $\hat{S}^-$ operators to get the state with $m_S = 0$. In any case, it takes some work to generate these states.

There are other symmetries as well. In a model that has pseudospin symmetry, we can use that symmetry. Often there are spatial symmetries such as reflections or rotations. These symmetries usually form a finite group, and then one has to use more sophisticated analyses to form states that have definite symmetries according to all of the different representations of the group. Describing the specific procedure for doing this takes us beyond what we will do here. But for reflections, or 180^o rotations, the symmetries are essentially even-odd symmetries which are fairly easy to implement, as we will see for some examples to come.

We are not going to write our own code to do this computer algebra in this book. We just wanted to give you a taste for how these things work. In many cases, you might not need to write your own code either, because many different packages exist that allow you to work with these operators according to the specific protocols that they have implemented in their software. In any case, understanding how such codes actually work often helps you better use the codes, even if you are not writing them yourself.

Section 17.5 Problems:

17.5.1 Storing States in Binary Form

We will work with a lattice of two sites, so we have a total of four possible single-particle states $1\uparrow$, $2\uparrow$, $1\downarrow$, and $2\downarrow$. We will use this ordering for our states, so the binary encoding is of the form

$$\underbrace{a\ \ b}_{\text{spin}\uparrow}\ \ \underbrace{c\ \ d}_{\text{spin}\downarrow}, \tag{17.93}$$

with a, b, c, and d equal to 0 or 1. For example, the state $|1\uparrow 2\uparrow 1\downarrow\rangle$ is represented by $1110 = 14$.

Work out the representation for the four two-particle states with $m_S = 0$. Report the binary and Arabic numbers for each of the states.

17.5.2 Operating a Two-Particle Operator onto a State

Determine the operation of $\hat{c}^\dagger_{1\uparrow}\hat{c}^\dagger_{3\uparrow}\hat{c}_{2\downarrow}\hat{c}_{4\uparrow}$ onto the state $\hat{c}^\dagger_{2\uparrow}\hat{c}^\dagger_{4\uparrow}\hat{c}^\dagger_{2\downarrow}|0\rangle$. Explain carefully how the final sign is determined for the state after the operator is applied to it.

Hint: You may find it simpler to first rewrite the operator as $\hat{c}^\dagger\hat{c}\hat{c}^\dagger\hat{c}$ (taking into account the sign change, if any) and then operate each $\hat{c}^\dagger\hat{c}$ term in turn.

When we operated a two-particle operator on $|\psi\rangle$ and then took the expectation value with $\langle\psi|$, we found we had two contributions, but when we evaluate the action of a single two-particle operator onto $|\psi\rangle$ it produces one result. Can you explain why both statements are correct?

17.5.3 Ground-State Energy of the Hopping Model

Consider the hopping Hamiltonian on a four-site square lattice. Note that the periodic boundary conditions show that if we number the lattice sites so the neighbors of even numbers are odd numbers and vice versa, the four neighbors of site 1 are site 2, 4, and then 2, and 4 again. The allowed wavenumbers are $\frac{\pi}{a}(m\vec{e}_x + n\vec{e}_y)$, with $m, n = 0, 1$.

There are four energy levels in this bandstructure. You should find two of them are degenerate. Determine the ground-state energy for four electrons and count how many degenerate states there are with the same energy. Note that for the hopping model, we occupy the electrons according to the bath-tub principle, but if there is a degeneracy in the bandstructure, we have extra possibilities for how the electrons energy levels are occupied in the many-body state.

17.5.4 Symmetries for a Two-Site Model with Translational Invariance

We consider a two-site model, with $j = 1, 2$. The number of two-particle states is 6: $|1\uparrow 1\downarrow\rangle$, $|2\uparrow 2\downarrow\rangle$, $|1\uparrow 2\uparrow\rangle$, $|1\uparrow 2\downarrow\rangle$, $|1\downarrow 2\uparrow\rangle$, and $|1\downarrow 2\downarrow\rangle$. There is one spin triplet state and three spin singlet states. Using the pseudospin symmetry, there is one triplet pseudospin state and five singlet pseudospin states. Finally, if the system is periodic, we can construct even or odd states with respect to the interchange of labels $1 \leftrightarrow 2$. Construct the six symmetrized states that are classified as follows: (i) $S = 1, m_S = 1, J = 0, m_J = 0$, odd spatial symmetry; (ii) $S = 1, m_S = 0, J = 0, m_J = 0$, odd spatial symmetry; (iii) $S = 1, m_S = -1, J = 0, m_J = 0$, odd spatial symmetry; (iv) $S = 0, m_S = 0, J = 1, m_J = 1$, odd spatial symmetry; and (v) and (vi) $S = 0, m_S = 0, J = 0, m_J = 0$, even spatial symmetry (two linearly independent states). This classification of the symmetry tells us that the Hamiltonian matrix will have four 1×1 blocks and one 2×2 block for a general Hamiltonian that conserves spin, pseudospin, and is symmetric under spatial reflection. We will discuss just such a model in the next Chapter.

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Chapter 18
Hubbard Model

The only limit to the height of your achievements is the reach of your dreams and your willingness to work hard for them.

Michelle Obama, Former First Lady of the United States, 2008

The Hubbard model is one of the most important models of many-body physics. It is a simplified model for how electrons move and interact in a crystal. It consists of a hopping term, where electrons can move between neighboring sites (sometimes more distant neighbors are allowed). When two electrons (of opposite spin) occupy the same lattice site, they feel a Coulomb repulsion. This model can be exactly solved only in one dimension, in the limit of infinite dimensions, or for small systems via exact diagonalization. It has been studied by thousands of physicists over the years. While much is known about the model, even more remains unknown, so it is still a great frontier to examine further. In this chapter, we focus on the known solutions and provide concrete examples and rigorous theorems about the behavior of the model, which are all summarized neatly for you in a video https://tinyurl.com/bdj8whnp.

18.1 Introduction to the Model

In the previous chapter, we introduced the tight-binding model, which had a bandstructure (for the square lattice, we found it to be $\epsilon(\vec{k}) = -2t\big(\cos(k_x a) + \cos(k_y a)\big)$. While we solved the problem on a finite-sized lattice, one can easily generalize the problem from a finite system to an infinite system by taking the spacing between the wavevectors to go to zero (they still remain inside a finite region of size approximately $\frac{2\pi}{a}$ in each dimension). Then, if we fill the electrons according to some electron density (such as one electron per lattice site), we find we

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fill up exactly half of the energy levels. The remaining energy levels are available to be filled at any energy (at low energy). This means the system is a metal. While we won't solve the problem here, if we changed the square lattice to a checkerboard lattice, where we added a local site energy $\Delta \hat{c}^{\dagger}_{j\sigma} \hat{c}_{j\sigma}$ to the sites corresponding to the black squares on the lattice, we would instead find a gap opened in the center of the energy band, right at the point where we have put in one electron per lattice site. In this case, there is a gap to excitations, and the system would be an insulator if the gap was large or a semiconductor if the gap was small. In this way, tight-binding models can describe metals, semiconductors, and insulators. Modern work on this problem is much more sophisticated, using real atomic systems in the calculations, than simple model ones, but the conclusions are the same. If the bandstructure has no gap, we are in a metal, while if it has a gap, we are in a semiconductor or insulator. In most solids, the lattice constant a is a bit larger than the size of an atom, typically $1 - 3$ Å.

This approach would say metals remain metals even if we expand the lattice constant a to be much larger than what it is in crystals. If we could make a as large as a meter, would we still expect the system to be a metal? The simple tight-binding bandstructure says yes, but we know it cannot be. We would instead have the electrons localized around each ion core and would have a collection of neutral atoms rather than a metallic solid. So, how does this transition take place? And what does the tight-binding model miss? This was the question taken up by Neville Mott, a Nobel Prize-winning physicist from England. Mott decided to examine the simplest possible model of this in a thought experiment. Take a collection of hydrogen atoms in a one-dimensional chain. Squeeze them together to reduce the lattice spacing a until the electrons in neighboring atoms start to overlap. At this point, the electrons become free to tunnel to neighboring sites, and the system becomes metallic. The transition from metallic to insulating behavior occurs when the electrons go from being delocalized through the lattice to being localized at the atomic cores. The transition is expected to be very sharp in a (in other words, very suddenly, so as a changes, we will see a transition from one state to the other over a very small range of a values). Well, what is tight-binding missing? The repulsion between the electrons! In most quantum problems, the energy is balanced between the kinetic and potential energies. But the tight-binding model only has kinetic energy. Something must be done to modify this to be able to describe the metal-insulator transition. We honor Mott for this original work by calling such a transition the Mott metal-insulator transition, or sometimes just the Mott transition.

John Hubbard then enters the scene (see Fig. 18.1). He argued that the Coulomb repulsion between electrons in a lattice, especially if the system is metallic, should be short-ranged, because the electrons are mobile and can screen out the charge of other electrons. The simplest way to do this in a model is to have the Coulomb repulsion limited to just the case where two electrons sit on the same lattice site. (Note that when we say sit on a lattice site, we mean closer to the ion core located at that lattice site than to any other ion core in the lattice, so the electron is not completely localized, just localized to the region of one ion core.) This model, which was introduced in 1963 (Hubbard, 1963), is now called the Hubbard model. It is one of the most important models in solid-state physics. It includes the kinetic energy

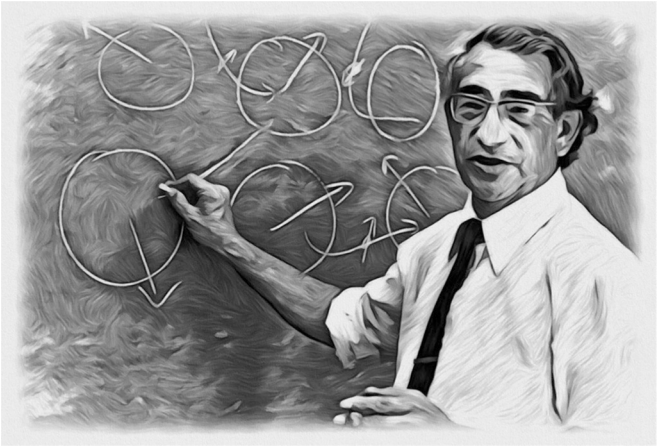

Fig. 18.1 John Hubbard in the 1970s. [Courtesy of AIP Emilio Segrè Visual Archives. Converted to look like a sketch by James Freericks]

term from the hopping and the potential energy term from the on-site repulsion. We write it as

$$\hat{H}_{\text{Hubbard}} = -\sum_{ij\in\Lambda}\sum_{\sigma} t_{ij}\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma} + U\sum_{i\in\Lambda}\hat{c}^{\dagger}_{i\uparrow}\hat{c}_{i\uparrow}\hat{c}^{\dagger}_{i\downarrow}\hat{c}_{i\downarrow}. \tag{18.1}$$

Here, Λ is a graph with points corresponding to lattice sites and connections between lattice sites determined by the sites where $t_{ij} \neq 0$. The number of lattice sites in Λ is equal to $|\Lambda|$. The first term is the hopping term. It is described by a real symmetric matrix $t_{ij} = t_{ji}$. The second term is the interaction term, given by the strength U multiplied by an operator that counts which sites are *doubly occupied*. While this looks like a relatively mild modification of the hopping model, it is notoriously difficult to solve. The Hubbard model, when we have a single hopping integral $-t$ between nearest neighbors, is often written in the form

$$\hat{H}_{\text{Hubbard}} = -t\sum_{\langle ij\rangle}\sum_{\sigma}\left(\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma} + \hat{c}^{\dagger}_{j\sigma}\hat{c}_{i\sigma}\right) + U\sum_{i}\hat{n}_{i\uparrow}\hat{n}_{i\downarrow}, \tag{18.2}$$

where we use the number operator for the interaction term. Here, the symbol $\langle ij\rangle$ means sum over nearest-neighbor pairs only once. Usually, in this case, we are on a periodic lattice, and we use $\vec{R}_i$ to denote the spatial locations of the different lattice sites. We will spend the rest of this chapter discussing the Hubbard model and its properties.

Our next step is to rewrite the Hamiltonian in momentum space. This step is only possible if Λ is a periodic lattice in space. We have already seen these operators, we define

$$\hat{a}^\dagger_{\vec{k}\sigma} = \frac{1}{\sqrt{|\Lambda|}} \sum_{j\in\Lambda} e^{i\vec{R}_j\cdot\vec{k}} \hat{c}^\dagger_{j\sigma} \text{ and } \hat{a}_{\vec{k}\sigma} = \frac{1}{\sqrt{|\Lambda|}} \sum_{j\in\Lambda} e^{-i\vec{R}_j\cdot\vec{k}} \hat{c}_{j\sigma}. \tag{18.3}$$

Then, using the identity

$$\frac{1}{|\Lambda|} \sum_{\vec{k}} e^{i(\vec{R}_i-\vec{R}_j)\cdot\vec{k}} = \delta_{ij}, \tag{18.4}$$

we find that

$$\hat{c}^\dagger_{i\sigma} = \frac{1}{\sqrt{|\Lambda|}} \sum_{\vec{k}} e^{-i\vec{R}_i\cdot\vec{k}} \hat{a}^\dagger_{\vec{k}\sigma} \text{ and } \hat{c}_{i\sigma} = \frac{1}{\sqrt{|\Lambda|}} \sum_{\vec{k}} e^{i\vec{R}_i\cdot\vec{k}} \hat{a}_{\vec{k}\sigma}. \tag{18.5}$$

The identity in Eq. (18.4) is straightforward to derive. When $\vec{R}_i = \vec{R}_j$, the sum is obviously equal to one, because each term in the summand is 1 and there are $|\Lambda|$ of them. When $\vec{R}_i \neq \vec{R}_j$, the sum involves complex numbers equally spaced around the unit circle, whose sum is zero. Alternatively, you can use the geometric series and the periodic boundary conditions to show it vanishes as well. The next step we have is to substitute in for the position space creation and annihilation operators in terms of the momentum space ones from Eq. (18.5) into the Hamiltonian. We note that the periodicity allows us to rewrite the summation over nearest-neighbor pairs as a sum over all sites for i, the nearest-neighbor translation vectors $\vec{\delta}$ for j, with $\vec{R}_j = \vec{R}_i + \vec{\delta}$. The nearest-neighbor translation vectors are the same set for each lattice site due to the periodicity. Since this resummation counts each pair twice, we multiply the sum by $\frac{1}{2}$ to avoid double counting. The Hamiltonian becomes

$$\begin{aligned}\hat{H} = &-\frac{t}{2|\Lambda|} \sum_{i\in\Lambda} \sum_{\vec{\delta}} \sum_{\sigma} \sum_{\vec{k}\vec{k}'} \left(e^{-i(\vec{R}_i\cdot\vec{k}-\vec{R}_i\cdot\vec{k}'-\vec{\delta}\cdot\vec{k}')} + e^{i(\vec{R}_i\cdot\vec{k}'-\vec{R}_i\cdot\vec{k}-\vec{\delta}\cdot\vec{k})} \right) \hat{a}^\dagger_{\vec{k}\sigma} \hat{a}_{\vec{k}'\sigma} \\ &+ \frac{U}{|\Lambda|^2} \sum_{i\in\Lambda} \sum_{\vec{k}\vec{k}'\vec{k}''\vec{k}'''} e^{-i\vec{R}_i\cdot(\vec{k}-\vec{k}'+\vec{k}''-\vec{k}''')} \hat{a}^\dagger_{\vec{k}\uparrow} \hat{a}_{\vec{k}'\uparrow} \hat{a}^\dagger_{\vec{k}''\downarrow} \hat{a}_{\vec{k}'''\downarrow}. \end{aligned} \tag{18.6}$$

In the terms on the first line, the sum over i will set $\vec{k} = \vec{k}'$, so we can perform the sum over $\vec{k}'$ too. For the second line, the sum over i will set $\vec{k} - \vec{k}' + \vec{k}'' - \vec{k}''' = 0$, which allows us to do the sum over $\vec{k}'''$. We also let $\vec{k}' = \vec{k} + \vec{q}$ to make the formula look nicer. This gives us

$$\hat{H} = \sum_{\vec{k}} \sum_{\sigma} \left(-\frac{t}{2} \sum_{\vec{\delta}} \left(e^{i\vec{k}\cdot\vec{\delta}} + e^{-i\vec{k}\cdot\vec{\delta}} \right) \right) \hat{a}^\dagger_{\vec{k}\sigma} \hat{a}_{\vec{k}\sigma} + \frac{U}{|\Lambda|} \sum_{\vec{k}\vec{k}'\vec{q}} \hat{a}^\dagger_{\vec{k}\uparrow} \hat{a}_{\vec{k}+\vec{q}\uparrow} \hat{a}^\dagger_{\vec{k}'\downarrow} \hat{a}_{\vec{k}'-\vec{q}\downarrow}. \tag{18.7}$$

The term

$$\epsilon(\vec{k}) = -\frac{t}{2}\sum_{\vec{\delta}}\left(e^{i\vec{k}\cdot\vec{\delta}} + e^{-i\vec{k}\cdot\vec{\delta}}\right) = -t\sum_{\vec{\delta}}\cos(\vec{k}\cdot\vec{\delta}) \tag{18.8}$$

is called the bandstructure, and with it, we obtain our final form for the Hamiltonian in momentum space

$$\hat{H} = \sum_{\vec{k}}\sum_{\sigma}\epsilon(\vec{k})\hat{a}^{\dagger}_{\vec{k}\sigma}\hat{a}_{\vec{k}\sigma} + \frac{U}{|\Lambda|}\sum_{\vec{k}\vec{k}'\vec{q}}\hat{a}^{\dagger}_{\vec{k}\uparrow}\hat{a}_{\vec{k}+\vec{q}\uparrow}\hat{a}^{\dagger}_{\vec{k}'\downarrow}\hat{a}_{\vec{k}'-\vec{q}\downarrow}. \tag{18.9}$$

When we compare the position-space version of the Hubbard model to the momentum-space version, we see that in position space, the interaction is diagonal, while in momentum space the hopping term is diagonal. We often prefer to work in real space, because the Hamiltonian is sparser there, especially if the hopping is only between nearest neighbors.

We end this section with a discussion of some of the symmetries of the model, leaving a more thorough discussion of symmetry to the next section. The first thing is to recall what a bipartite lattice is. We say Λ is bipartite if it can be separated into two sublattices A and B, such that $t_{ij} \neq 0$ only when $i \in A$ and $j \in B$ or $i \in B$ and $j \in A$. When the Hubbard model is solved on bipartite lattice, the energy eigenvalue spectrum (the set of the energy eigenvalues) is the same if we transform $t \to -t$. The proof follows by performing a unitary transformation that changes the sign of the hopping. Because the set of energy eigenvalues are not changed in a unitary transformation (proved in Problem 18.1.1), the set of energy eigenvalues is unchanged when $t_{ij} \to -t_{ij}$. The transformation we have to use is

$$\hat{O} = e^{i\pi\sum_{j\in A}(\hat{n}_{j\uparrow}+\hat{n}_{j\downarrow})}. \tag{18.10}$$

Clearly, we have

$$\hat{\tilde{c}}_{j\sigma} = \hat{O}\hat{c}_{j\sigma}\hat{O}^{\dagger} = \begin{cases} -\hat{c}_{j\sigma} & \text{if } j \in A \\ \hat{c}_{j\sigma} & \text{if } j \in B \end{cases}, \tag{18.11}$$

and similarly for the creation operator. The proof of this is also given in Problem 18.1.1 and simply uses the Hadamard lemma. Now, if we apply this to the Hubbard model, the interaction term remains the same, because number operators always have a creation and an annihilation operator from the same lattice site, so there is no sign change, while the hopping term changes sign, because it always has one index from the A sublattice and one from the B sublattice, and then we always get a net minus sign. This then changes the sign of the hopping matrix.

The last symmetry we examine here is the *partial particle-hole symmetry* again for bipartite lattices. In this symmetry, we leave the down-spin particles unchanged, but we interchange creation and annihilation operators (with a sign change if on the A sublattice) for the up-spin particles. In other words, we have

$$\hat{c}^\dagger_{j\uparrow} \to (-1)^{\Pi(j)}\hat{d}_{j\uparrow},\ \ \hat{c}_{j\uparrow} \to (-1)^{\Pi(j)}\hat{d}^\dagger_{j\uparrow},\ \ \hat{c}^\dagger_{j\downarrow} \to \hat{d}^\dagger_{j\downarrow},\ \text{ and } \hat{c}_{j\downarrow} \to \hat{d}_{j\downarrow}. \tag{18.12}$$

Here, $\Pi(j) = 1$ if $j \in A$ and $\Pi(j) = 0$ if $j \in B$. After re-expressing the Hamiltonian in terms of the $\hat{d}$ operators, and using the anticommutation relations, we find that

$$\hat{H} \to -\sum_{ij\in\Lambda}\sum_{\sigma} t_{ij}\hat{d}^\dagger_{i\sigma}\hat{d}_{j\sigma} - U\sum_{i\in\Lambda}\hat{d}^\dagger_{i\uparrow}\hat{d}_{i\uparrow}\hat{d}^\dagger_{i\downarrow}\hat{d}_{i\downarrow} + U\sum_{i\in\Lambda}\hat{d}^\dagger_{i\downarrow}\hat{d}_{i\downarrow}. \tag{18.13}$$

If we let $E(U, N_\uparrow, N_\downarrow)$ be an energy eigenvalue for the case with $N_\uparrow$ spin up electrons and $N_\downarrow$ spin-down electrons, then this transformation tells us that

$$E(U, N_\uparrow, N_\downarrow) = E(-U, |\Lambda| - N_\uparrow, N_\downarrow) + UN_\downarrow. \tag{18.14}$$

The identity is most interesting at half-filling, where $N_\uparrow = N_\downarrow = \frac{|\Lambda|}{2}$. Then, we have that

$$E\left(U, \tfrac{|\Lambda|}{2}, \tfrac{|\Lambda|}{2}\right) = E\left(-U, \tfrac{|\Lambda|}{2}, \tfrac{|\Lambda|}{2}\right) + U\tfrac{|\Lambda|}{2}. \tag{18.15}$$

This says the energy eigenvalues are even in U up to a constant shift. We have a video The Hubbard model https://tinyurl.com/2s46t72c that assumes an introductory background in solid-state physics, but you may still find it rewarding.

Section 18.1 Problems:

18.1.1 Unitary Transformations Do Not Change Eigenvalues

Consider the eigenvalue-eigenvector relationship for a Hamiltonian, given by $\hat{H}|\psi\rangle = E|\psi\rangle$. We will now transform the Hamiltonian via a unitary transformation, where $\hat{H} \to \hat{U}\hat{H}\hat{U}^\dagger = \hat{H}'$ and transform the states via $|\psi\rangle \to \hat{U}|\psi\rangle = |\psi'\rangle$. Show that the unitary transformation does not change eigenvalue-eigenvector relationship. Namely, show that $\hat{H}'|\psi'\rangle = E|\psi'\rangle$.

Consider the transformation in Eq. (18.10). Show that $\hat{\tilde{c}}_{j\sigma} = \pm\hat{c}_{j\sigma}$ depending on which sublattice the annihilation operator is in. Repeat for the creation operators. Verify that the transformation is indeed unitary.

18.1.2 Chemical Potential and Particle-Hole Symmetric Interactions

We often add a term $-\mu\sum_{i\in\Lambda}\sum_\sigma \hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma}$ to the Hamiltonian. It is called the chemical potential term and μ is the chemical potential. When we solve the Hubbard

model, we can either fix the number of electrons in our solution (called working in the *canonical ensemble*), or we can solve for all possible number of electrons, but include a chemical potential, which is adjusted to yield the correct average electron filling (called the *grand canonical ensemble*). While this may seem silly when doing an exact diagonalization calculation, because working with a fixed number of electrons is much simpler than working with all numbers of electrons, when one wants to go to the infinite size limit, working with a chemical potential becomes very convenient.

Show that the chemical potential term can be absorbed into the interaction term by shifting it via

$$U\sum_{i\in\Lambda}(\hat{n}_{i\uparrow}+\hat{n}_{i\downarrow})-\mu\sum_{i\in\Lambda}(\hat{n}_{i\uparrow}+\hat{n}_{i\downarrow}) = U\sum_{i\in\Lambda}\left(\hat{n}_{i\uparrow}-\tfrac{\mu}{U}\right)\left(\hat{n}_{i\downarrow}-\tfrac{\mu}{U}\right)+C. \tag{18.16}$$

Determine C. When $\mu = \frac{U}{2}$, the interaction is said to be in its particle-hole symmetric form if we ignore the constant term C. Show in that case that the interaction energy at each site takes on the values $\pm\frac{U}{4}$ depending on whether there is zero, one, or two electrons on the site.

18.1.3 Full Particle-Hole Symmetry and Half-Filling

Consider the full particle-hole symmetry, given by

$$\begin{aligned}&\hat{c}^{\dagger}_{j\uparrow}\to(-1)^{\Pi(j)}\hat{d}_{j\uparrow},\ \ \hat{c}_{j\uparrow}\to(-1)^{\Pi(j)}\hat{d}^{\dagger}_{j\uparrow},\\ &\hat{c}^{\dagger}_{j\downarrow}\to(-1)^{\Pi(j)}\hat{d}_{j\downarrow},\ \ \text{and}\ \ \hat{c}_{j\downarrow}\to(-1)^{\Pi(j)}\hat{d}^{\dagger}_{j\downarrow}.\end{aligned} \tag{18.17}$$

Show that under this transformation we have

$$\begin{aligned}E(U,N_{\uparrow},N_{\downarrow})-\mu(N_{\uparrow}+N_{\downarrow})\to E(U,|\Lambda|-N_{\uparrow},|\Lambda|-N_{\downarrow})\\ +U(|\Lambda|-N_{\uparrow}-N_{\downarrow})-\mu(|\Lambda|-N_{\uparrow}+|\Lambda|-N_{\downarrow}),\end{aligned} \tag{18.18}$$

when we include the chemical potential. At half-filling, we have $N_{\uparrow}=N_{\downarrow}=\frac{|\Lambda|}{2}$. Show that we must have $\mu=\frac{U}{2}$ for the energies to be unchanged after this transformation. This means $\mu=\frac{U}{2}$ corresponds to the particle-symmetric point of the Hubbard model, which is at half filling. Note this only holds for bipartite lattices.

18.1.4 Limiting Behavior of the Hubbard Model

When $U = 0$, we have just the hopping model. Explain how it is solved in momentum space using the bathtub principle.

When $U = \infty$, we cannot have two electrons on the same site. Show that at half-filling this means we have one electron per site (with a spin degree of freedom), but the electrons cannot move, so it is an insulator.

When $U = -\infty$, we must have two electrons on each site. Show that the system is an insulator for all even electron numbers in this case, because we cannot break the strongly bound pair.

Hence, at half-filling, there must be a metal insulator transition as a function of U. Numerical calculations show this occurs at a critical value of U in infinite dimensions, while the Bethe ansatz solution shows it occurs immediately once $U > 0$ in one dimension.

18.2 Spin, Pseudospin, and Number Operators

The Hubbard model has a number of good quantum numbers associated with it. This occurs because we can find operators that commute with the Hubbard Hamiltonian. Whenever we find an operator that commutes with $\hat{H}_{\text{Hubbard}}$, we call that operator a symmetry of the Hamiltonian. Our goal in this section is to determine the maximal set of commuting operators of the Hubbard Hamiltonian. We will actually see that not all of our symmetry operators commute with the Hamiltonian. In this case we have what we would call a *broken* symmetry. We will find that even in this case, we can still find good quantum numbers associated with the broken symmetry.

Our first symmetry operators are the total number operators of each spin. Because the Hamiltonian can be described as placing electrons of definite spin onto lattice sites, letting them hop from site to site, obeying the Pauli exclusion principle, and paying an extra cost in energy equal to U when two electrons are on the same site, one can see that because the hopping term does not flip spins and neither does the interaction term, we must have the total number of up spin electrons and the total number of down spin electrons conserved in the model. Let's check. These number operators are given by

$$\hat{N}_\sigma = \sum_{i \in \Lambda} \hat{n}_{i\sigma}. \tag{18.19}$$

If they are conserved quantities, they must commute with the Hamiltonian. Since number operators commute with themselves, the potential-energy term commutes with both number operators. For the kinetic energy, we find that

$$[\hat{H}, \hat{N}_\sigma] = \sum_{ijk\in\Lambda}\sum_{\sigma'}(-t_{ij})[\hat{c}^\dagger_{i\sigma'}\hat{c}_{j\sigma'}, \hat{n}_{k\sigma}] = -\sum_{ijk\in\Lambda} t_{ij}\left(\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\delta_{jk} - \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\delta_{ik}\right)$$
$$= -\sum_{ij\in\Lambda}(t_{ij} - t_{ij})\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma} = 0. \tag{18.20}$$

So it is a symmetry. But of course, it is only a symmetry after we sum over all terms in the lattice, because on any given site, the hopping can change the number of electrons on that site. It is convenient to rewrite these conserved symmetries in a different way. We define the z-component of spin to be

$$\hat{S}^z = \frac{1}{2}\sum_{i\in\Lambda}\left(\hat{n}_{i\uparrow} - \hat{n}_{i\downarrow}\right), \tag{18.21}$$

where we now work with dimensionless spin, dropping the usual $\hbar$ factor. In addition, we define the z-component of pseudospin via

$$\hat{J}^z = \frac{1}{2}\sum_{i\in\Lambda}\left(\hat{n}_{i\uparrow} + \hat{n}_{i\downarrow} - 1\right). \tag{18.22}$$

While the z-component of pseudospin can be defined on any lattice, the full pseudospin symmetry is only defined on bipartite lattices, as we will see below. At this stage, we find that we can work with states that have definite total number of electrons and have definite z-component of spin in addition to the definite energy of an energy eigenstate.

The spin raising operator is $\hat{S}^+ = \sum_{i\in\Lambda}\hat{c}^\dagger_{i\uparrow}\hat{c}_{i\downarrow}$. It should be clear that this operator flips a spin from down to up. We next show that it commutes with the Hamiltonian. We have

$$[\hat{H}, \hat{S}^+] = \sum_{ijk\in\Lambda}\sum_{\sigma}(-t_{ij})[\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}, \hat{c}^\dagger_{k\uparrow}\hat{c}_{k\downarrow}] + U\sum_{ij\in\Lambda}[\hat{n}_{i\uparrow}\hat{n}_{i\downarrow}, \hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow}]$$
$$= -\sum_{ijk\in\Lambda} t_{ij}\left(\hat{c}^\dagger_{i\uparrow}\hat{c}_{k\downarrow}\delta_{jk} - \hat{c}^\dagger_{k\uparrow}\hat{c}_{j\downarrow}\,\delta_{ik}\right)$$
$$+ U\sum_{ij\in\Lambda}\left(-\hat{n}_{i\uparrow}\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow} + \hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow}\hat{n}_{i\downarrow}\right)\delta_{ij}$$
$$= -\sum_{ij\in\Lambda} t_{ij}\left(\hat{c}^\dagger_{i\uparrow}\hat{c}_{j\downarrow} - \hat{c}^\dagger_{i\uparrow}\hat{c}_{j\downarrow}\right) + U\sum_{i\in\Lambda}\left(-\hat{c}^\dagger_{i\uparrow}\hat{c}_{i\downarrow} + \hat{c}^\dagger_{i\uparrow}\hat{c}_{i\downarrow}\right) = 0. \tag{18.23}$$

Taking the Hermitian conjugate of this equation shows us that $[\hat{H}, \hat{S}^-] = 0$ as well. Here, $\hat{S}^- = \sum_{i\in\Lambda}\hat{c}^\dagger_{i\downarrow}\hat{c}_{i\uparrow}$. So, we have shown total spin is also a symmetry of

the Hamiltonian. We verify that the spin operators satisfy an SU(2) algebra in the problems.

When we are on a bipartite lattice, we have pseudospin raising and lowering operators given by

$$\hat{J}^+ = \sum_{i\in\Lambda} \hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{i\downarrow}(-1)^{\Pi(i)} \text{ and } \hat{J}^- = \sum_{i\in\Lambda} \hat{c}_{i\downarrow}\hat{c}_{i\uparrow}(-1)^{\Pi(i)}. \tag{18.24}$$

Recall that $\Pi(i) = 1$ for $i \in A$ and 0 for $i \in B$. We compute the commutator again:

$$\begin{aligned}
[\hat{H}, \hat{J}_+] &= \sum_{ijk\in\Lambda}\sum_{\sigma}(-t_{ij})[\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}, \hat{c}^\dagger_{k\uparrow}\hat{c}^\dagger_{k\downarrow}](-1)^{\Pi(k)} + U\sum_{ij\in\Lambda}[\hat{n}_{i\uparrow}\hat{n}_{i\downarrow}, \hat{c}^\dagger_{j\uparrow}\hat{c}^\dagger_{j\downarrow}](-1)^{\Pi(j)} \\
&= -\sum_{ijk\in\Lambda} t_{ij}\left(\hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{k\downarrow}(-1)^{\Pi(k)}\delta_{jk} + \hat{c}^\dagger_{k\uparrow}\hat{c}^\dagger_{i\downarrow}(-1)^{\Pi(k)}\delta_{jk}\right) \\
&\quad + U\sum_{ij\in\Lambda}\left(\hat{n}_{i\uparrow}\hat{c}^\dagger_{j\uparrow}\hat{c}^\dagger_{j\downarrow}\delta_{ij} + \hat{c}^\dagger_{j\uparrow}\hat{c}^\dagger_{j\downarrow}\hat{n}_{i\downarrow}\delta_{ij}\right)(-1)^{\Pi(j)} \\
&= -\sum_{ij\in\Lambda} t_{ij}(-1)^{\Pi(j)}\left(\hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{j\downarrow} + \hat{c}^\dagger_{j\uparrow}\hat{c}^\dagger_{i\downarrow}\right) + U\sum_{i\in\Lambda}\hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{i\downarrow}(-1)^{\Pi(i)} \\
&= U\hat{J}^+.
\end{aligned} \tag{18.25}$$

This result follows because in the next to last line, we switch $i \leftrightarrow j$ in the second term with the hopping matrix, using the fact that $t_{ij}(-1)^{\Pi(j)} + t_{ji}(-1)^{\Pi(i)} = 0$ because i and j are on different sublattices and t_{ij} is a symmetric matrix. This makes the hopping term vanish. For the term proportional to U, we use the fact that $\hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{i\downarrow}\hat{n}_{i\downarrow} = 0$ because it has the square of a creation operator and $\hat{n}_{i\uparrow}\hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{i\downarrow} = \hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{i\downarrow}$ by using an add zero to make an anticommutator with the middle two creation and annihilation operators in the product of four fermionic operators. This shows that $\hat{J}^+$ is a raising operator for $\hat{H}$, increasing the energy eigenvalues by U. Similarly, by taking the Hermitian conjugate, we learn that $\hat{J}^-$ is a lowering operator, reducing the energy by U. One might think this rules out these operators as being a symmetry of the problem, because they do not commute with the Hamiltonian, but the pseudospin quantum numbers are still good quantum numbers for this problem; it is just the raising and lowering feature which means terms in a pseudospin multiplet are not degenerate, but have different energies if $U \neq 0$. We will carefully explain this below as we use the eigenstates of these operators to construct different states. In the problems, you will verify that pseudospin indeed has the SU(2) Lie algebra, so it can be classified in terms of the total J and the z-component m_J.

We end this section with a discussion of the allowed eigenvalues for a given number of up and down electrons. The z-component of spin is equal to $\frac{1}{2}(N_\uparrow - N_\downarrow)$, so S runs from $\frac{1}{2}(N_\uparrow - N_\downarrow)$ to $\frac{1}{2}(N_\uparrow + N_\downarrow)$. The z-component of pseudospin is

$\frac{1}{2}(N_\uparrow + N_\downarrow - |\Lambda|)$, so the J value runs from $\frac{1}{2}(N_\uparrow + N_\downarrow - |\Lambda|)$ to $\frac{|\Lambda|}{2}$. It takes some time to get used to the allowed eigenvalues of these operators, but with some practice, you can master it.

Section 18.2 Problems:

18.2.1 SU(2) Algebra for Spin and Pseudospin

Using the fermionic representation of the operators, show that

$$[\hat{S}^+, \hat{S}^-] = 2\hat{S}^z \text{ and } [\hat{S}^z, \hat{S}^\pm] = \pm\hat{S}^\pm \tag{18.26}$$

and show as well that

$$[\hat{J}^+, \hat{J}^-] = 2\hat{J}^z \text{ and } [\hat{J}^z, \hat{J}^\pm] = \pm\hat{J}^\pm. \tag{18.27}$$

This shows that spin and pseudospin each generate an SU(2) Lie group.

18.2.2 Spin and Pseudospin Commute

Using the fermionic representation of the operators, show that

$$[\hat{S}^z, \hat{J}^z] = [\hat{S}^z, \hat{J}^\pm] = [\hat{S}^\pm, \hat{J}^z] = [\hat{S}^\pm, \hat{J}^\pm] = 0. \tag{18.28}$$

This shows the two SU(2) Lie groups are independent of each other.

18.2.3 Quantum Numbers for a Bipartite Lattice with $|\Lambda| = 2$

When $|\Lambda| = 2$, we have a total of $4^2 = 16$ states. We describe these states in a localized product state basis and discuss the different quantum numbers of the states.

The number of electrons we can have ranges from 0 to 4. The number of states is one for 0 and 4, four for 1 and 3, and six for 2. We list them below as the eigenstates of spin and pseudospin. You job is to determine what the spin, z-component of spin, pseudospin, and z-component of pseudospin are for these different states.

For $N = 0$, we have $|0\rangle$. For $N = 1$, we have $|1\uparrow\rangle$, $|1\downarrow\rangle$, $|2\uparrow\rangle$, and $|2\downarrow\rangle$. For $N = 2$ we have $|1\uparrow 2\uparrow\rangle$, $\frac{1}{\sqrt{2}}(|1\uparrow 1\downarrow\rangle \pm |2\uparrow 2\downarrow\rangle)$ (two states), $\frac{1}{\sqrt{2}}(|1\uparrow 2\downarrow\rangle \pm |1\downarrow 2\uparrow\rangle)$ (two states), and $|1\downarrow 2\downarrow\rangle$. For $N = 3$, we have $|1\uparrow 1\downarrow 2\uparrow\rangle$,

$|1\uparrow 1\downarrow 2\downarrow\rangle$, $|1\uparrow 2\uparrow 2\downarrow\rangle$, and $|1\downarrow 2\uparrow 2\downarrow\rangle$. Finally, for $N=4$, we have $|1\uparrow 1\downarrow 2\uparrow 2\downarrow\rangle$.

Note that with careful thought, and a process of elimination, this should not require significant computations.

18.3 Two-Site Hubbard Model

The many-body problem is subtle and complicated. We find insight only by examining many different cases. This section is our first nontrivial one for electrons. We will solve for all of the energy levels of the two-site Hubbard model, given by

$$\hat{H} = -t\sum_{\sigma}(\hat{c}^{\dagger}_{1\sigma}\hat{c}_{2\sigma} + \hat{c}^{\dagger}_{2\sigma}\hat{c}_{1\sigma}) + U(\hat{n}_{1\uparrow}\hat{n}_{1\downarrow} + \hat{n}_{2\uparrow}\hat{n}_{2\downarrow}). \tag{18.29}$$

We will be able to solve this problem analytically.

This problem has a total of $4^2 = 16$ states, so a brute force approach to solving it would require us to calculate $16\times 16 = 256$ matrix elements and then diagonalize the 16×16 matrix to find its eigenvalues and eigenvectors. Fortunately, if we analyze the problem before going the brute-force way and think about the implications of symmetry, we can reduce the problem to 14 1×1 blocks, where we just get the energy eigenvalue immediately and one 2×2 block that needs to be diagonalized. This is a remarkable savings of effort—it pays to think before you calculate!

In Problem 18.2.3, we discussed the different states. We now take each state and work out the eigenstates and eigenvalues. Note that in that problem, we just worked on states with definite spin and pseudospin. We mentioned previously that there also can be spatial symmetries. Here they are just that we want to construct states that are even or odd under the interchange of sites 1 with 2. We now go about solving for each set of eigenvalues according to the number of electrons.

We start with $N=0$. The state is the vacuum state. Since annihilation operators annihilate this state, we immediately find $\hat{H}|0\rangle = 0$, so the energy for $N=0$ is 0. The quantum numbers are, of course, $S=0$, $m_S=0$, $J=1$, and $m_J=-1$. You might be surprised that the pseudospin value is $J=1$, but because the $m_J=-1$, this is the only value it can have, because $J\le\frac{|\Lambda|}{2}$.

For $N=1$, we construct even and odd states under the interchange of the site label via $\frac{1}{\sqrt{2}}(|1\uparrow\rangle \pm |2\uparrow\rangle)$ and $\frac{1}{\sqrt{2}}(|1\downarrow\rangle \pm |2\downarrow\rangle)$. The potential energy vanishes when acting on this state, because it does not have two electrons on any site. The hopping term gives $-t$ for the $+$ sign and t for the $-$ sign. These states have $S=\frac{1}{2}$, $m_S=\pm\frac{1}{2}$, $J=\frac{1}{2}$ and $m_J=-\frac{1}{2}$. They are each twofold degenerate with respect to the m_S eigenvalues. This is expected, because the spin multiplets should each have the same energy.

Now, we move on to $N=2$. There are six states. Rather than list them all at the outset, we will work through them carefully using what we already know. We start

with the fact that we have a state with $N = 0$ that has $J = 1$ and $m_J = -1$. If we operate J^+ onto that state, it will be $J = 1$ and $m_J = 0$. The state is

$$\tfrac{1}{\sqrt{2}}\hat{J}^+|0\rangle = \tfrac{1}{\sqrt{2}}(-|1\uparrow 1\downarrow\rangle + |2\uparrow 2\downarrow\rangle). \tag{18.30}$$

The $\frac{1}{\sqrt{2}}$ is introduced to normalize the state. This state also has $S = 0$, $m_S = 0$, and $E = U$. All other states with $N = 2$ must have $J = 0$ and $m_J = 0$. We start with the spin triplet state with $S = 1$ and $m_S = -1$. This state is $|1\downarrow 2\downarrow\rangle$. The hopping term annihilates this state, and the potential term annihilates it as well, so we have $E = 0$. Clearly, we get the same result for the other states in the multiplet with $m_S = 0$ and $m_S = -1$. The latter state is $|1\downarrow 2\downarrow\rangle$. To determine the state with $m_S = 0$, we simply compute $\hat{S}^-|1\uparrow 2\uparrow\rangle$. But to do this, we need to specify what the rules we are using for the ordering of the creation operators in the states. We pick the ordering by using all up spins first followed by all down spins. Then we have the state with $m_S = 0$ is given by

$$\begin{aligned}\tfrac{1}{\sqrt{2}}\hat{S}^-\hat{c}^\dagger_{1\uparrow}\hat{c}^\dagger_{2\uparrow}|0\rangle &= \tfrac{1}{\sqrt{2}}(\hat{c}^\dagger_{1\downarrow}\hat{c}_{1\uparrow} + \hat{c}^\dagger_{2\downarrow}\hat{c}_{2\uparrow})\hat{c}^\dagger_{1\uparrow}\hat{c}^\dagger_{2\uparrow}|0\rangle \\ &= \tfrac{1}{\sqrt{2}}(-\hat{c}^\dagger_{2\uparrow}\hat{c}^\dagger_{1\downarrow}|0\rangle + \hat{c}^\dagger_{1\uparrow}\hat{c}^\dagger_{2\downarrow}|0\rangle) \\ &= \frac{1}{\sqrt{2}}(-|1\downarrow 2\uparrow\rangle + |1\uparrow 2\downarrow\rangle).\end{aligned} \tag{18.31}$$

Note that our rule for where the creation operators go determines the sign of the states, regardless of how we write the label in the ket. There are two states left. They both have $S = 0$ and $m_S = 0$. They are the states orthogonal to the other $m_S = 0$ states we have found. They are

$$|a\rangle = \tfrac{1}{\sqrt{2}}(|1\uparrow 1\downarrow\rangle + |2\uparrow 2\downarrow\rangle) \text{ and } \tfrac{1}{\sqrt{2}}(|1\uparrow 2\downarrow\rangle + |1\downarrow 2\uparrow\rangle). \tag{18.32}$$

Since these states are both even under the interchange of sites 1 and 2 (best seen by looking at the states in terms of the creation operators—the interchange of $1 \leftrightarrow 2$ still has the $\uparrow$ operator to the left of the $\downarrow$ operator). Since they have the same symmetry, they should be coupled together by the Hamiltonian forming a 2×2 block. To find this, we simply compute

$$\langle a|\hat{H}|a\rangle = U, \quad \langle a|\hat{H}|b\rangle = \langle b|\hat{H}|a\rangle = -2t, \quad \text{and} \quad \langle b|\hat{H}|b\rangle = 0. \tag{18.33}$$

To find the eigenvalues, we must diagonalize

$$\begin{pmatrix} U & -2t \\ -2t & 0 \end{pmatrix}. \tag{18.34}$$

Following the standard procedure gives us $E^2 - UE - 4t^2 = 0$, which has solutions given by

$$E_{\pm} = \frac{U}{2} \pm \frac{1}{2}\sqrt{U^2 + 16t^2}. \tag{18.35}$$

This completes the $N = 2$ case.

Now, we move on to $N = 3$. We again form the states that have even and odd spatial symmetry. They are

$$\frac{1}{\sqrt{2}}(|1\uparrow 1\downarrow 2\uparrow\rangle \pm |1\uparrow 2\uparrow 2\downarrow\rangle) \text{ and } \frac{1}{\sqrt{2}}(|1\uparrow 1\downarrow 2\downarrow\rangle \pm |1\downarrow 2\uparrow 2\downarrow\rangle). \tag{18.36}$$

The eigenvalues are $-t + U$ and $t + U$, each twofold degenerate. We can get this either by acting the Hamiltonian on each state or by using the raising operator $\hat{J}^+$ on the states with $N = 1$. The quantum numbers are $S = \frac{1}{2}$, $m_S = \pm\frac{1}{2}$, $J = \frac{1}{2}$, and $m_J = \frac{1}{2}$.

Finally, we have the state with $N = 4$, given by $|1\uparrow 1\downarrow 2\uparrow 2\downarrow\rangle$. This state has $S = 0$, $m_S = 0$, $J = 1$, $m_J = 1$, and $E = 2U$.

We summarize these results in Table 18.1 listing the 16 different states, their energies, and their quantum numbers. One can see that for $N = 1$, the ground state has energy $-t$ and is twofold degenerate. The quantum numbers are $S = \frac{1}{2}$ and $J = \frac{1}{2}$. For $N = 2$, the ground state has energy $\frac{U}{2} - \frac{1}{2}\sqrt{U^2 + 16t^2}$, is nondegenerate, and has $S = 0$ and $J = 0$. The ground state for $N = 3$ has energy $-t + U$, is twofold degenerate, and has $S = \frac{1}{2}$ and $J = \frac{1}{2}$. It is not a coincidence that the ground state at half filling $N = 2$ has both spin and pseudospin equal to 0. We will prove this result later in the chapter for a wide range of lattices. We saw in both cases where the spin of the ground state was $\frac{1}{2}$ we find that the ground-state is degenerate. This is a general result called *Kramers degeneracy*. When we have half-odd integer spin and no magnetic fields, the ground state must be at least doubly degenerate. We see this with the solutions here. We will not prove why Kramers degeneracy occurs in this book; we simply mention that it does occur. The proof requires working with time-reversal symmetry and can be found in comprehensive solid-state physics texts.

It is useful to discuss some of the properties of the ground state as well. In the problems, you will solve for α and β. The results are

$$\alpha = -\frac{\sqrt{U^2 + U\sqrt{U^2 + 16t^2} + 16t^2}}{\sqrt{2}\sqrt{U^2 + 16t^2}} \text{ and } \beta = \frac{2\sqrt{2}t}{\sqrt{U^2 + U\sqrt{U^2 + 16t^2} + 16t^2}}. \tag{18.37}$$

We will examine the behavior in the limiting cases. First, let's examine what happens when $U = 0$. Then, we find that $\alpha = -\frac{1}{\sqrt{2}}$ and $\beta = \frac{1}{\sqrt{2}}$. Substituting this into the wavefunction, one finds this corresponds to the product state made by filling

Table 18.1 States, quantum numbers, and energies of the different energy eigenstates of the Hubbard model with $|\Lambda| = 2$. The coefficients α and β satisfy $\alpha^2 + \beta^2 = 1$ and need to be determined by diagonalizing the 2×2 matrix. They are given in the main text, and there is also a problem to solve for them

N	state	S	m_S	J	m_J	E
0	$\|0\rangle$	0	0	1	-1	0
1	$\frac{1}{\sqrt{2}}(\|1\uparrow\rangle + \|2\uparrow\rangle)$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$-t$
	$\frac{1}{\sqrt{2}}(\|1\downarrow\rangle + \|2\downarrow\rangle)$	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$-t$
	$\frac{1}{\sqrt{2}}(\|1\uparrow\rangle - \|2\uparrow\rangle)$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	t
	$\frac{1}{\sqrt{2}}(\|1\downarrow\rangle - \|2\downarrow\rangle)$	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	t
2	$\frac{1}{\sqrt{2}}(-\|1\uparrow 1\downarrow\rangle + \|2\uparrow 2\downarrow\rangle)$	0	0	1	0	U
	$\|1\uparrow 2\uparrow\rangle$	1	1	0	0	0
	$\frac{1}{\sqrt{2}}(-\|1\downarrow 2\uparrow\rangle + \|1\uparrow 2\downarrow\rangle)$	1	0	0	0	0
	$\|1\downarrow 2\downarrow\rangle$	1	-1	0	0	0
	$\frac{\alpha}{\sqrt{2}}(\|1\uparrow 1\downarrow\rangle + \|2\uparrow 2\downarrow\rangle)$ $+ \frac{\beta}{\sqrt{2}}(\|1\uparrow 2\downarrow\rangle + \|1\downarrow 2\uparrow\rangle)$	0	0	0	0	$\frac{U}{2} + \frac{1}{2}\sqrt{U^2 + 16t^2}$
	$\frac{\beta}{\sqrt{2}}(\|1\uparrow 1\downarrow\rangle + \|2\uparrow 2\downarrow\rangle)$ $- \frac{\alpha}{\sqrt{2}}(\|1\uparrow 2\downarrow\rangle + \|1\downarrow 2\uparrow\rangle)$	0	0	0	0	$\frac{U}{2} - \frac{1}{2}\sqrt{U^2 + 16t^2}$
3	$\frac{1}{\sqrt{2}}(\|1\uparrow 1\downarrow 2\uparrow\rangle + \|1\uparrow 2\uparrow 2\downarrow\rangle)$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$-t + U$
	$\frac{1}{\sqrt{2}}(\|1\uparrow 1\downarrow 2\downarrow\rangle + \|1\downarrow 2\uparrow 2\downarrow\rangle)$	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$-t + U$
	$\frac{1}{\sqrt{2}}(\|1\uparrow 1\downarrow 2\uparrow\rangle - \|1\uparrow 2\uparrow 2\downarrow\rangle)$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$t + U$
	$\frac{1}{\sqrt{2}}(\|1\uparrow 1\downarrow 2\downarrow\rangle - \|1\downarrow 2\uparrow 2\downarrow\rangle)$	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$t + U$
4	$\|1\uparrow 1\downarrow 2\uparrow 2\downarrow\rangle$	0	0	1	1	$2U$

the lowest two single particle energy levels, as it should. When we let $U \to \infty$, the coefficients become $\alpha = -1$ and $\beta = 0$, which tells us we have no double occupancies. Finally, as $U \to -\infty$, we find that $\alpha = 0$ and $\beta = 1$. This shows the state has only double occupancies in it.

In the general case, we find for half-filling that as $U \to \infty$, all states with no double occupancies are degenerate, but for U slightly less than ∞, the state with $S = 0$ will be the ground state when we are on a bipartite lattice with equal sized sublattices. We will prove this later in the chapter. We also find the ground state is nondegenerate for $0 < U < \infty$ and has $J = 0$ as well. For the attractive case, and all even N fillings, the ground state will have $S = 0$ and be nondegenerate for $-\infty < U < 0$. You will see how we establish these remarkable results later. It is amazing that such general results can be established for this model. We have a video https://tinyurl.com/557stwc7 discussing this solution.

Section 18.3 Problems:

18.3.1 Compute the Ground State for $N = 2$

Compute α and β for the ground state by solving for the eigenvectors of the 2×2 matrix. Recall the ground state must be normalized.

18.3.2 Ground State Properties for All U at Half Filling

Write an essay that discusses the evolution of the ground state for all U for $N = 2$ in terms of how the state changes and what the energy is. Discuss what the quantum numbers are. Do you see any interesting trends with the quantum numbers?

18.4 Constructing States of Definite Spin, Pseudospin, and Spatial Symmetry

In the solution of the two-site Hubbard model, we saw the importance of finding states with definite quantum numbers of spin and pseudospin as well as having definite spatial symmetry. Using symmetry took us from having to find the eigenvalues and eigenvectors of a 16×16 matrix to needing to solve just one 2×2 matrix. This example shows that finding states with definite quantum numbers is quite helpful in simplifying the energy eigenvalue problem. While we just worked out the examples directly for $|\Lambda| = 2$, for the general case, we need to approach the problem in a more thoughtful manner. In this section, we go through precisely how one does that. Because the energy levels for more than half filling are easily related to those at less than half filling, we focus on building the states just up to half filling.

We start with the spin symmetry. Back in Problem 9.4.3, we sketched out how one can create states with definite spin. One does not try to do it "all at once" by working with the states of N spins, but instead, we build up the states starting from one spin and increasing one spin at a time. This can be a bit tedious, but we are thinking of a computer doing the algebra for us, so that it can be done painlessly. So, if we recall the problem, with two spins, we have four states, one $S = 1$ triplet and one singlet. Then, for $N = 3$, we have one $S = \frac{3}{2}$ quartet and two $S = \frac{1}{2}$ states, to give us the total of eight possible states. To find both of the $S = \frac{1}{2}$ states, we need to actively construct them, one from the singlet and one from the triplet. And then we just keep on going, adding one spin one-half to each of the states we have already made. This will create two classes of states, with $S \pm \frac{1}{2}$, except for the case

where $S = 0$, where we make just a spin-one-half state. This approach works for eventually getting us to all of the states with N spins. But for fermionic problems, things are a bit harder. You see, we have to first decide which *different* sites we are placing the electrons at. If $N < |\Lambda|$ there will be $|\Lambda|!/[(|\Lambda| - N)!N!]$ possible choices for the different electrons. We need to place the electrons in each of these different choices, and each one has all of the different possible spin states that we need to construct. Still straightforward, but tedious.

But, we are not quite done. We also have states where there are two electrons on a site. Every time we place two electrons on a site, they must be in a spin-singlet state, because they are in the same spatial state and must be antisymmetric. Hence, once we have finished forming all of the states with definite spin that have no double occupancies, we form the ones with one double occupancy. To do this, we pick the site with the double occupancy, and we place the remaining electrons on $N-2$ different locations from where the double occupancy lies. Then we repeat for all remaining choices for where to place the double occupancy and the remaining electrons. Finally, we move onto the case with two double occupancies and so on, until all are exhausted. Each case proceeds in the same way. Fill the double occupancies first and then fill in the remaining electrons with definite spin states in the remaining sites.

As an example, consider the case with $N = 3$ and $|\Lambda| = 4$. First, let's count how many states we have. There are many different ways to do this. If all electrons are on different sites, then we have two choices for each site (up or down), and we can place three electrons on four sites in four different ways. So this is a total of 32 states. Then for the ones with one double occupancy, we have four choices for where the double occupancy goes and three choices for the remaining electron, which can be up or down. So, there are 24 states. The total is then 56 states. This will be split into 4 $S = \frac{3}{2}$ states, 8 $S = \frac{1}{2}$ states with no double occupancy, and 12 states with $S = \frac{1}{2}$ with one double occupancy. We now describe how to find these states. We will describe how to form the states with maximal m_S, as the remaining states can be found by just applying the lowering operator onto them.

The states with $S = \frac{3}{2}$ are $|1 \uparrow 2 \uparrow 3 \uparrow\rangle$ and the three different permutations $|1 \uparrow 2 \uparrow 4 \uparrow\rangle$, $|1 \uparrow 3 \uparrow 4 \uparrow\rangle$, $|2 \uparrow 3 \uparrow 4 \uparrow\rangle$. Constructing the spin-one-half states with no double occupancies is direct. We apply $\hat{S}^-$ onto the first spin-three-half state to obtain

$$\frac{1}{\sqrt{3}}(|1 \uparrow 2 \uparrow 3 \downarrow\rangle + |1 \uparrow 2 \downarrow 3 \uparrow\rangle + |1 \downarrow 2 \uparrow 3 \uparrow\rangle) \tag{18.38}$$

(recall for these states the raising operator for the spin-down particle is always the rightmost operator in each term regardless of the order it is given in the Dirac ket label). The two spin-one-half states are orthogonal to this. To construct them we need to use the SU(3) unit vectors from Problem 15.3.1. The two orthogonal vectors leading to these states are

$$\frac{1}{\sqrt{2}}(|1\uparrow 2\downarrow 3\uparrow\rangle - |1\downarrow 2\uparrow 3\uparrow\rangle) \text{ and}$$
$$\frac{1}{\sqrt{6}}(|1\uparrow 2\downarrow 3\uparrow\rangle + |1\downarrow 2\uparrow 3\uparrow\rangle - 2|1\uparrow 2\uparrow 3\downarrow\rangle). \tag{18.39}$$

The other spin-one-half states are found by permuting the spatial indices to include the state on site 4, just as we did for the spin-three-half states. We have just the doubly occupied states. They are easy to make. We have $|1\uparrow 1\downarrow 2\uparrow\rangle$ and all of the permutations of the site indices.

Having finished with spin symmetry, we now move on to pseudospin symmetry. This is also straightforward to construct, but it requires us to work with states that have electron numbers less than the current state. We then apply $\hat{J}^+$ to the states to bring their electron number up to the value we are working with. We have to apply the raising operator to states with smaller number of electrons that also have definite spin to preserve the spin state quantum numbers. This means we will need to re-orthogonalize our states afterward, using the Gram-Schmidt procedure if necessary. Then we will have states of definite spin and pseudospin.

We illustrate this as well with our $N = 3$ example. We start with the states that have $N = 1$, since the raising operator of pseudospin adds in two electrons. We work with the state $|3\uparrow\rangle$ and then apply $\hat{J}^+$ to it. It gives the state (being careful to get the signs right)

$$\frac{1}{\sqrt{3}}\hat{J}^+|3\uparrow\rangle = \frac{1}{\sqrt{3}}(|1\uparrow 1\downarrow 3\uparrow\rangle - |2\uparrow 2\downarrow 3\uparrow\rangle + |3\uparrow 4\uparrow 4\downarrow\rangle), \tag{18.40}$$

where the odd-numbered sites are the A sublattice and the even numbers the B sublattice. Since the down state creation operator must be last, the first state has one sign change, so does the second, but the third state has none. Remaining states are found by permuting the labels of the states and being careful about signs, if some states need to be re-ordered to put them into our canonical form. One needs to find the states orthogonal to these within the subspaces that have definite spin, to find the states with $J = \frac{1}{2}$. But this is just a Gram-Schmidt orthogonalization process.

Finally, we need to work with the spatial symmetry. As we stated before, in most cases this involves constructing even and odd states with respect to the permutation of the spatial site indices. But, in some instances, it requires a more complicated procedure. The way to properly analyze this uses group theory for finite groups. Each group has a series of representations of the group that agree with the group multiplication table. Most of these representations are not faithful representations (such as replacing every group element by the identity element). Nevertheless, each representation can be employed to construct states that have definite symmetry with respect to the group (similar to even and odd states, but perhaps more complex). We will not develop the theory for how to do this in any great detail. One needs to look into a group theory book for the details if you choose to do this. We will, however, provide a concrete example for how this is done when we work on exact solutions with three and four sites, so that you can see the procedure once.

One of the key elements to keep in mind is that we need to use both orthogonality and counting to help us in generating all of the states that have definite quantum numbers. As you have seen, we have different strategies to create states with definite quantum numbers, but we have to do it one at a time. Because the states with definite quantum numbers are always degenerate, we then form the proper linear combinations of these states to find the states that have definite values for multiple quantum numbers. Then, at the end, we use counting and orthogonality to verify we have found and classified all of the states. The counting tells us we have found all of them, while the orthogonality ensures that the states are linearly independent of each other. It is tedious, but relatively simple for a computer to handle this.

Is this all of the symmetries? Well, yes and no. There can actually be other symmetries. A nice paper that discusses this is by Heilmann and Lieb on the six-site benzene molecule, as modeled by the six-site Hubbard model on a periodic ring (Heilmann & Lieb, 1970). In that work, in addition to the spin, pseudospin, and spatial symmetries, they found both extra parity symmetries and dynamical symmetries (where the symmetry operator changes with the interaction U). So, extra symmetries may be there in the Hubbard model. But, it is usually quite difficult to work with these symmetries, so they are often not incorporated into an analysis.

John Wilkins, a well-respected condensed-matter theorist, often commented that we never forget our Ph.D. thesis. Indeed, the type of work described in this chapter made up a large part of my own Ph.D. thesis. In that work, I wrote a computer program (in Fortran, no less) that performed all of this algebra, constructed the states and then the individual blocks of the Hamiltonian matrix, and then diagonalized them to find the different energy eigenstates. The computer code actually wrote Fortran code to evaluate the different matrix blocks as functions, so they could be diagonalized. I loved the work, but the debugging was extraordinarily painful! Doing similar things now is much easier.

In the problems, we will construct the symmetry states for a number of examples that we will be working with later in the chapter, when we diagonalize a number of these small clusters.

Section 18.4 Problems:

18.4.1 Symmetry States for a Triangle

In the main text, we discussed how you can construct the states for $N = 3$ on a small cluster with $|\Lambda| = 4$. In this problem, we do the same for the case with $|\Lambda| = 3$. This is a subset of the states we found before. There is one $S = \frac{3}{2}$ multiplet (4 states), two $S = \frac{1}{2}$ multiplets with no double occupancy (4 states), and six $S = \frac{1}{2}$ multiplets with one double occupancy (12 states). There is a total of 20 states.

First, find all the states of definite spin and the maximal m_S. Next, find the linear combinations that have definite pseudospin. We will not have you work out the spatial symmetry.

18.4.2 Symmetry States for $N = 2$ and $N = 3$ for a Square

In the main text, we worked out the states for $N = 3$ and $|\Lambda| = 4$, without regard for the spatial structure of the lattice. Now, consider a square, where even numbered sites connect to odd numbered sites and vice versa. Carefully summarize the states, especially in terms of their pseudospin, which is a definite symmetry on the square. Also determine the $N = 2$ states given by six $S = 1$ multiplets (18 states), six $S = 0$ multiplets (6 states) with no double occupancy, and four $S = 0$ multiplets with a double occupancy (4 states). The total is 28 states.

In both cases, find only the states with maximal m_S. Identify the pseudospin quantum numbers as well.

18.4.3 Symmetry States for $N = 4$ and $|\Lambda| = 4$ on a Square

There are 70 states: one $S = 2$ (five states), three $S = 1$ with no double occupancy (nine states), and two $S = 0$ with no double occupancy (two states). There are 12 $S = 1$ states with one double occupancy (36 states) and 12 $S = 0$ states with 1 double occupancy (12 states). There are six $S = 0$ states with two double occupancies (6 states).

Be sure to classify them in terms of pseudospin as well. There is 1 $J = 2$ state, 27 $J = 1$ states, and 42 $J = 0$ states.

Note that you do not have to write out all possible states; instead, you can write the general form and then how things change when one permutes site indices.

Note further that the spin states themselves will also be valid for nonbipartite structures, like a tetrahedron.

18.5 Nagaoka Theorem and Ferromagnetism

The Nagaoka theorem is a remarkable result. It says that when U is large enough, the ground state for $N = |\Lambda| - 1$ electrons (one hole in the half-filled band) is ferromagnetic. It was originally proven in 1965–1966 (Nagaoka, 1965, 1966), very shortly after the Hubbard model itself was introduced. The work in this section follows closely Tasaki's modernized proof of the theorem (Tasaki, 2020). We prove it two different ways—one is a direct proof, the other uses the Perron-Frobenius theorem.

We take the Hubbard model on the graph Λ with $U \to \infty$. In this case, we have no double occupancies. We choose an ordering for the lattice sites, labeling them from 1 to $|\Lambda|$. The states with one hole in a half-filled band have a lattice site i where there is no electron and has a spin background $\{\sigma_1, \sigma_2, \cdots, \sigma_{i-1}, \sigma_{i+1}, \cdots, \sigma_{|\Lambda|}\}$ that the hole lives in. We use the following set of states as our basis states. Note that this *is not* the same as the canonical choices for ordering the operators that we used before. This scheme is chosen just for the Nagaoka problem. The states are

$$|i, \{\vec{\sigma}\}\rangle = (-1)^i \hat{c}^\dagger_{1\sigma_1} \hat{c}^\dagger_{2\sigma_2} \cdots \hat{c}^\dagger_{i-1\sigma_{i-1}} \hat{c}^\dagger_{i+1\sigma_{i+1}} \cdots \hat{c}^\dagger_{|\Lambda|\sigma_{|\Lambda|}} |0\rangle. \tag{18.41}$$

Note the sign $(-1)^i$ which depends on where the hole is located. This choice of sign is critical for the theorem. Similarly, the theorem requires $-t_{ij} \geq 0$ for all i and j (and that $-t_{ij}$ is a real and symmetric matrix). For the case of a bipartite lattice, we can change the sign of $-t_{ij}$ by the partial particle-hole transformation, so the sign condition only becomes important for nonbipartite lattices.

Since the Hamiltonian consists of the hopping term plus the no double-occupancy condition, we focus on what the hopping term does to the states. We say a state $|i, \{\vec{\sigma}\}\rangle$ is connected to a state $|j, \{\vec{\tau}\}\rangle$ if

$$\langle j, \{\vec{\tau}\}| \left(\hat{c}^\dagger_{i\uparrow} \hat{c}_{i\uparrow} + \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\downarrow} \right) |i, \{\vec{\sigma}\}\rangle \neq 0. \tag{18.42}$$

If the state $|i, \{\vec{\sigma}\}\rangle$ is connected to $|j, \{\vec{\tau}\}\rangle$, then we must have $\sigma_\alpha = \tau_\alpha$ for all indices except for $\alpha = i$ and $\alpha = j$. Then, we must have $\sigma_i = \tau_j$. For two connected states, if we move the operator $\hat{c}^\dagger_{i\sigma_j} \hat{c}_{j\sigma_j}$ to the j location in $|i, \{\vec{\sigma}\}\rangle$, the pair of operators moves through all of the operators "in the way" without requiring any sign changes. Then, we can replace $\hat{c}_{i\sigma_j} \hat{c}^\dagger_{i\sigma_j}$ by its anticommutator, because it acts on the vacuum state. This gets rid of those operators. Now, we have to move the $\hat{c}^\dagger_{i\sigma_j}$ operator from the j location to the i location. This requires a sign change given by $(-1)^{i-j-1}$. But, the sign convention of the state produces another $(-1)^{j-i}$ sign change. The net effect is -1. Hence, we have that

$$\langle j, \{\vec{\tau}\}| \sum_{ij \in \Lambda} (-t_{ij}) \left(\hat{c}^\dagger_{i\uparrow} \hat{c}_{i\uparrow} + \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\downarrow} \right) |i, \{\vec{\sigma}\}\rangle = t_{ij} \leq 0 \tag{18.43}$$

when the two states are connected.

We say that a lattice Λ satisfies the *connectivity condition* if for any two states $|i, \{\vec{\sigma}\}\rangle$ and $|j, \{\vec{\tau}\}\rangle$, we can find a chain of sites

$$|i, \{\vec{\sigma}\}\rangle \leftrightarrow |i_2, \{\vec{\sigma}_2\}\rangle \leftrightarrow |i_3, \{\vec{\sigma}_3\}\rangle \leftrightarrow \cdots \leftrightarrow |i_M, \{\vec{\sigma}_M\}\rangle \leftrightarrow |j, \{\vec{\tau}\}\rangle, \tag{18.44}$$

where each of the states in the chain is connected to its neighbor on either side. It is not simple to prove what graphs satisfy this condition, but if we work on a periodic lattice, then as long as the lattice has either triangles (labeled by i, j, and k) such that $t_{ij} t_{jk} t_{ki} \neq 0$ or squares (labeled by i, j, k, and l) such that $t_{ij} t_{jk} t_{kl} t_{ki} \neq 0$ and there is at least one other site than site i such that every site on the lattice is connected to it via a path that does not go through site i, then the states with one hole will satisfy the connectivity condition. It turns out that essentially all lattices, with one exception, satisfy this rule. So, the square lattice, triangular lattice, simple cubic lattice, body-centered cubic lattice, and face-centered cubic lattice all satisfy this condition. The one-dimensional lattice with nearest-neighbor hopping only does not satisfy this.

Now, we move onto the proof that among all of the possible ground states, at least one of them has maximal spin $S = S_{\max} = \frac{|\Lambda|-1}{2}$. The proof starts with a ground state $|\psi\rangle$, which satisfies $|\psi\rangle = \sum_{i,\{\vec{\sigma}\}} \psi_{i,\vec{\sigma}} |i, \{\vec{\sigma}\}\rangle$. Then, we form the state $|\phi\rangle$ which has maximal spin, given by $|\phi\rangle = \sum_i \phi_i |i, \{\uparrow\uparrow \cdots \uparrow\}\rangle$, which is a linear combination of all states with a hole at i and the spin background being all up states. The state ϕ has maximal spin. The coefficients ϕ_i are given by

$$\phi_i = \sqrt{\sum_{\{\vec{\sigma}\}} |\psi_{i,\{\vec{\sigma}\}}|^2}. \tag{18.45}$$

Note that these coefficients are all real and nonnegative. Next, we compute the energy in the $|\psi\rangle$ state and the $|\phi\rangle$ state. This gives us the inequality

$$\begin{aligned}\langle\psi| \sum_{ij\in\Lambda} (-t_{ij}) \left(\hat{c}^\dagger_{i\uparrow}\hat{c}_{i\uparrow} + \hat{c}^\dagger_{i\downarrow}\hat{c}_{i\downarrow}\right) |\psi\rangle &= \sum_{\vec{\sigma},\vec{\tau}} \sum_{ij} t_{ij} \psi^*_{j\vec{\tau}} \psi_{i\vec{\sigma}} \\ \geq \sum_{ij} t_{ij}\phi_j\phi_i = \langle\phi| \sum_{ij\in\Lambda} (-t_{ij}) \left(\hat{c}^\dagger_{i\uparrow}\hat{c}_{i\uparrow} + \hat{c}^\dagger_{i\downarrow}\hat{c}_{i\downarrow}\right) |\phi\rangle. &\end{aligned} \tag{18.46}$$

In the first summation, the state $|i, \{\vec{\sigma}\}\rangle$ is connected to the state $|j, \{\vec{\tau}\}\rangle$. The inequality comes from the Schwartz inequality, which we proved earlier in Sect. 7.7. The Schwartz inequality says that $\langle a|b\rangle \leq \|a\|\,\|b\|$. Here, we apply it $|\Lambda|$ times. We choose $b_\alpha = \psi_{i,\vec{\sigma}}$, where i is fixed, and α corresponds to the different $\vec{\sigma}$ values. Then $a_\alpha = \psi_{j,\vec{\tau}}$, where the i and $\vec{\sigma}$ are fixed and j and $\vec{\tau}$ are the one state connected to i and $\vec{\sigma}$. Then considering all possible $\vec{\sigma}$ terms in the state $|\psi\rangle$, we see that $\|b\| = \phi_i$ and $\|a\| = \phi_j$. We then apply the Schwartz inequality for each i and its partner j. This result shows that for any $|\psi\rangle$, we can construct a maximal spin state $|\phi\rangle$ that has an energy that is equal to or less than the energy of the state $|\psi\rangle$. But, if $|\psi\rangle$ is a ground state, then the energy must be equal, so that among the different ground states, there is at least one that has maximal spin. This means the system has a ferromagnetic ground state.

But, we can go further than this. We can show that the ground state is unique, which implies it is only a ferromagnetic ground state, with the degeneracy given by the number of states in the spin multiplet $|\Lambda|$. It is easiest to prove this with the Perron-Frobenius theorem, which we proved in Sect. 13.3. Just consider the matrix $M = -H$, written in the basis we have been using with a fixed value of m_S. The matrix elements are then all nonnegative. Furthermore, the connectivity condition implies that the matrix is connected. Then the Perron-Frobenius theorem tells us the ground state has all elements the same sign and it is unique. The states with maximal S also have all coefficients positive (recall how the lowering operator works), so this shows, within each sector of m_S, the ground state has maximal S. This is of course what it must be when we have the degeneracy given by the degeneracy of the ferromagnetic state.

Since the state is ferromagnetic for $U = \infty$, it must be ferromagnetic for finite U as well due to continuity of the ground-state energy with U. Eventually, there must be a transition to a lower-spin state, because the ground state with $U = 0$ is not ferromagnetic. It turns out the transition occurs for quite large values of U, as we will investigate in the next section. Furthermore, if we look at two holes in the half-filled band, we usually find a spin-singlet state as the ground state. So, there is not a finite region of electron density where the system is ferromagnetic for large U. We have a video https://tinyurl.com/7e7e9hj4 on this too.

Section 18.5 Problems:

18.5.1 Bipartite or Nonbipartite

Write a short essay discussing what the results of the Nagaoka theorem say for a bipartite and for a nonbipartite lattice that satisfies the connectivity condition.

18.6 Three- and Four-Site Hubbard Model

In this section, we will solve the three-site and four-site Hubbard models on fully connected clusters—a triangle for $|\Lambda| = 3$ and a tetrahedron for $|\Lambda| = 4$. These geometric structures have the most symmetry, so by using states with definite symmetry, we will be able to reduce the Hamiltonian matrix blocks to be as small as possible. But, because neither is bipartite, there will be no pseudospin symmetry in these systems. While the algebra to go through these solutions is long, we do learn some interesting things about how electron correlations work and about the theorems we will prove about these systems. So, the effort is certainly worthwhile.

The three-site Hubbard model is given by

$$\hat{H} = -t \sum_{\sigma} (\hat{c}^{\dagger}_{1\sigma} c_{2\sigma} + \hat{c}^{\dagger}_{2\sigma} c_{3\sigma} + \hat{c}^{\dagger}_{3\sigma} c_{1\sigma} + \hat{c}^{\dagger}_{2\sigma} c_{1\sigma} + \hat{c}^{\dagger}_{3\sigma} c_{2\sigma} + \hat{c}^{\dagger}_{1\sigma} c_{3\sigma}) + U \sum_{i=1}^{3} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}. \tag{18.47}$$

One can see a schematic of this model in Fig. 18.2, along with the three energy levels we will solve for in the $N = 1$ case, which illustrates the bandstructure of the cluster—it changes character as the sign of the hopping is changed.

The case with $N = 0$, is trivial, and we do not discuss it further. For $N = 1$, we can proceed in two ways, and we show both. We can use the localized basis of three sites and construct a 3×3 matrix that we can diagonalize. Or, we can focus on the symmetry of the model to find the symmetric states first and just work with the different symmetry blocks. Before jumping into this, though, we want to discuss a bit about the symmetry of the triangle. The symmetry group which involves 120^{o}

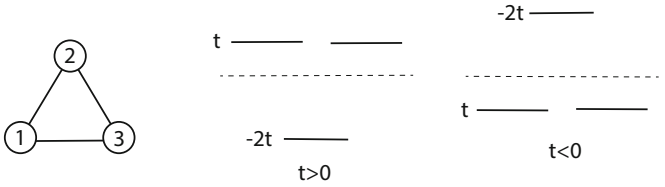

Fig. 18.2 (Left) Schematic of the Hubbard model on a triangle, where the electrons can sit on any of the three different lattice sites. The solid lines indicate bonds, where the electrons can hop from one site to another. (Center and Right) Bandstructure of the three-site model (the dashed line is at zero energy). We show both the case with $t > 0$, which has one state with energy $-2t$ below two degenerate states with t, while for $t < 0$, the situation is reversed

rotations and reflections about three mirrors that each go through a vertex and the middle of the opposite side has two one-dimensional representations and one non-trivial two-dimensional representation. The two-dimensional representation means that every quantum state that has the symmetry associated with that representation will have states that are twofold degenerate. We can work with just one of these two states if we know how to generate the state with a specific symmetry and not even worry about finding the degenerate partner state, unless we need that state for calculating some specific result (such as calculating a specific matrix element of some operator). Okay, back to the calculation. We will use a simplified notation, where we use 1σ to denote $\hat{c}^{\dagger}_{1\sigma}|0\rangle$ and so on. The symbols label where the creation operators sit, which is different from how we did it before with the kets, where the labels could be put in in any order, but the creation operator order was determined according to a specific canonical ordering scheme. This new representation is more compact and makes determining the signs easier, because it is as if we were working directly with the operators. Because spin is preserved, we work with the up spin states only (the down spin states are degenerate). Our basis states are then $1\uparrow$, $2\uparrow$, and $3\uparrow$. Since each site hops to the other two sites, it is not too difficult to find that the Hamiltonian matrix for these three states is given by

$$\hat{H}_{N=1} = \begin{pmatrix} 0 & -t & -t \\ -t & 0 & -t \\ -t & -t & 0 \end{pmatrix}. \tag{18.48}$$

We subtract $E\mathbb{I}_3$ from it and calculate the determinant to find the characteristic equation $E^3 - 3Et^2 + 2t^3 = 0$. One can immediately see that the roots are $E = -2t$ (once) and $E = t$ (twice). These energy levels are the bandstructure. Note how they all add to zero. This is not a coincidence. Recall that the trace of a matrix is given by the sum of all of its eigenvalues. It also is given by the sum of all of the diagonal matrix elements in any basis. But, because the hopping matrix has all of its diagonal elements zero, its trace is always zero. Hence, the sum of the eigenvalues of the bandstructure always vanishes.

Now, we solve it using symmetrized states. We first think about what is the analog of an even state. It is a state with the full symmetry of the lattice—it does not change if we rotate it or apply a mirror plane. The rotations are given by $123 \rightarrow 231$ or 312, while the mirror planes are $123 \rightarrow 132$, 321, or 213. So, we want a state unchanged by all of these different operations. It is clearly the state

$$\frac{1}{\sqrt{3}}(1\uparrow + 2\uparrow + 3\uparrow). \tag{18.49}$$

If we apply $\hat{H}$ on it, we find it is an eigenvector, with $-2t$ as the eigenvalue. The other two eigenvectors must be perpendicular to it. We will find they are degenerate. Using our SU(3) vectors, we find they are

$$\frac{1}{\sqrt{2}}(1\uparrow - 2\uparrow) \text{ and } \frac{1}{\sqrt{6}}(1\uparrow + 2\uparrow - (2)\cdot 3\downarrow), \tag{18.50}$$

with the parenthesis and the $\cdot$ denoting multiplication of the number two times the state $3\uparrow$. Operating the Hamiltonian on these two states, we do indeed find that they are eigenstates with eigenvalues of t. This then gives the bandstructure shown in Fig. 18.2. Personally, I do think working with the symmetrized states is better for analytical work. But, it does take an effort to generate them, which becomes increasingly more difficult as we have more electrons in the state.

Now, we move on to the $N = 2$ case. There are 15 total states. We count this by looking at the total number of spin orbitals ($3 \times 2 = 6$) and determining how many different ways we can choose two of them. This is $6 \times 5/2 = 15$. Nine of these states are $S = 1$ (3 triplet states). We will examine these states in the two ways we did before. The three states in a localized basis are

$$1\uparrow 2\uparrow, \quad 1\uparrow 3\uparrow, \quad \text{and} \quad 2\uparrow 3\uparrow . \tag{18.51}$$

There are no double occupancies, while the hopping involves either of the electrons in the state hopping to the empty site, so the Hamiltonian block becomes

$$\hat{H}_{N=2,\ S=1} = \begin{pmatrix} 0 & -t & t \\ -t & 0 & -t \\ t & -t & 0 \end{pmatrix}. \tag{18.52}$$

The sign change in the 13 and 31 matrix elements arises from having to reorder the creation operators in the state, which changes the sign. The characteristic equation becomes $E^3 - 3Et^2 - 2t^3$. The roots are $2t$, $-t$, and $-t$. Note that they are the opposite sign of the energies we found for $N = 1$. This is because when we look at the $S = 1$ states, we can think of them as two filled electrons, or as one hole, corresponding to the empty site. It turns out the energies for holes are the negative of the energies for electrons, so the result confirms this rule. Again, if we want to look at the symmetrized states, they are

$$\tfrac{1}{\sqrt{3}}(1\uparrow 2\uparrow -1\uparrow 3\uparrow +2\uparrow 3\uparrow),\quad \frac{1}{\sqrt{2}}(1\uparrow 2\uparrow +1\uparrow 3\uparrow),$$
$$\text{and}\quad \frac{1}{\sqrt{6}}(1\uparrow 2\uparrow -1\uparrow 3\uparrow -(2)\cdot 2\uparrow 3\uparrow). \tag{18.53}$$

You should be able to directly verify these states are eigenvectors with the stated eigenvalues ($2t$, $-t$, and $-t$). Note that there are sign changes from what we had before arising from how we chose the ordering of the creation operators in the basis states. Next up are the six spin-singlet states. These states have one double occupancy or no double occupancy. We will use the symmetrized states. The singly degenerate states are given by

$$|1\rangle = \frac{1}{\sqrt{3}}(1\uparrow 1\downarrow +2\uparrow 2\downarrow +3\uparrow 3\downarrow)\quad \text{and}$$
$$|2\rangle = \frac{1}{\sqrt{6}}(1\uparrow 2\uparrow +1\uparrow 3\uparrow +2\uparrow 1\downarrow +2\uparrow 3\downarrow +3\uparrow 1\downarrow +3\uparrow 2\downarrow). \tag{18.54}$$

Operating the Hamiltonian on these states, and paying attention to the normalizations, gives us $\hat{H}|1\rangle = U|1\rangle - 2\sqrt{2}t|2\rangle$ and $\hat{H}|2\rangle = -2\sqrt{2}t|1\rangle - 2t|2\rangle$. This gives a 2×2 matrix

$$\hat{H}_{N=2,\ S=0} = \begin{pmatrix} U & -2\sqrt{2}t \\ -2\sqrt{2}t & -2t \end{pmatrix}. \tag{18.55}$$

The energy eigenvalues then become

$$E = -t + \frac{U}{2} \pm \frac{1}{2}\sqrt{(2t+U)^2 + 32t^2}. \tag{18.56}$$

The remaining four states can be divided into two sets of states that are degenerate with each other, but not directly coupled in the Hamiltonian. We will work with one set of the states here. The other will be examined in Problem 18.6.1 below. We have

$$|3\rangle = \tfrac{1}{\sqrt{2}}(1\uparrow 1\downarrow -2\uparrow 2\downarrow)\ \text{and}\ |4\rangle = \frac{1}{2}(1\uparrow 3\downarrow -2\uparrow 3\downarrow +3\uparrow 1\downarrow -3\uparrow 2\downarrow). \tag{18.57}$$

You then find that $\hat{H}|3\rangle = U|3\rangle - \sqrt{2}t|4\rangle$ and $\hat{H}|4\rangle = -\sqrt{2}t|3\rangle + t|4\rangle$. After constructing the matrix and diagonalizing it, we find that the energies are

$$E = \frac{t}{2} + \frac{U}{2} \pm \frac{1}{2}\sqrt{(t-U)^2 + 8t^2}. \tag{18.58}$$

Each of these energies is twofold degenerate.

To find the ground state, we need to compare all of the different energy eigenvalues and find the lowest ones. For $t > 0$, we find the eigenvalue $E = -t + \frac{U}{2} - \frac{1}{2}\sqrt{(2t+U)^2 + 32t^2}$ is the lowest eigenvalue for all U. Hence, the ground state is $S = 0$. While for $t < 0$, we find that the same spin singlet is the ground state for $U < 0$, but for $U > 0$, the state $E = 2t$ is the ground state for all $U > 0$. Note that this is the Nagaoka theorem result that the ground state must be ferromagnetic for $U \to \infty$ (when $t < 0$). Here, we find the transition extends all the way to $U = 0$.

Our last case is half-filling, with $N = 3$. We have one $S = \frac{3}{2}$ state and eight $S = \frac{1}{2}$, which gives a total of 20 states. The $S = \frac{3}{2}$ state fills all of the energy levels in the bandstructure with spin up electrons (and the other three states in the multiplet). The total energy of this state is $E = 0$. For the $S = \frac{1}{2}$ states, we have six states that have one double occupancy and two states with no double occupancy. We work with the symmetrized states only.

We use group theory to show how to symmetrize the states. The approach is a bit tedious, but again, for larger systems, one can automate this on a computer. As we mentioned before, the group we are working with is the permutation group of three elements, with group elements 123, 231, 312, 213, 132, and 321. The first element is the identity, the next two are the rotations of the triangle, and the last three are the three reflections. To properly work out the group theory, we first want to write down the multiplication table for this group, for which, we use the identity I, rotation R_c and R_{cc} (for clockwise and counterclockwise) and reflection symbols r_1, r_2, and r_3 with the index labeling the vertex through which the mirror plane goes. The multiplication table is given in Table 18.2.

To find the symmetrized states, we must find all independent group representations. A *representation* of a group is a mapping of the group elements to numbers of matrices, which satisfy the group multiplication table. The mapping of every group element to the number 1 is just such a mapping. This exists for every group and is called the trivial representation. Here, we have another one-dimensional representation, because there is an even and odd property with reflections versus rotations. If I take rotations as even and reflections as odd, then you can see rotation

Table 18.2 Multiplication table for the symmetries of the triangle (equivalent to the permutation group of three elements). Looking at the upper left 3 × 3 block, you can see that the rotations form a subgroup of the group, as they are closed under multiplication as well. Our convention for the reflection planes is that they always point to the same vertex, even if that vertex has been renumbered (in other words r_1 has its vertex in the lower left corner always—it does not move to where the 1 site currently is)

×	I	R_{cc}	R_c	r_1	r_2	r_3
I	I	R_{cc}	R_c	r_1	r_2	r_3
R_{cc}	R_{cc}	R_c	I	r_3	r_1	r_2
R_c	R_c	I	R_{cc}	r_2	r_3	r_2
r_1	r_1	r_2	r_3	I	R_{cc}	R_c
r_2	r_2	r_3	r_1	R_c	I	R_{cc}
r_3	r_3	r_1	r_2	R_{cc}	R_c	I

times rotation is even, reflection times reflection is even, but rotation times reflection is odd. So the mapping I, R_c and R_{cc} to 1 and r_1, r_2, and r_3 to -1 is another representation of the group. The final representation is a two-dimensional one, which we can construct with Pauli matrices. Using the generalized Euler identity for rotations, we see the angles must be $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$. We choose the Pauli matrix to be σ_y to have a real-valued matrix. We find

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad R_{cc} = \frac{1}{2}\begin{pmatrix} -1 & \sqrt{3} \\ -\sqrt{3} & -1 \end{pmatrix}, \quad \text{and} \quad R_c = \frac{1}{2}\begin{pmatrix} -1 & -\sqrt{3} \\ \sqrt{3} & -1 \end{pmatrix}. \tag{18.59}$$

We cannot construct the reflections from matrices that are exponentials of Pauli matrices, because those matrices all have determinant 1 and the reflections must have determinant -1. The strategy is to pick the simplest matrix with determinant -1, assign it to the r_1 reflection, and use the multiplication table to find the other two rotations. This gives

$$r_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad r_2 = \frac{1}{2}\begin{pmatrix} -1 & \sqrt{3} \\ \sqrt{3} & 1 \end{pmatrix}, \quad \text{and} \quad r_3 = \frac{1}{2}\begin{pmatrix} -1 & -\sqrt{3} \\ -\sqrt{3} & 1 \end{pmatrix}. \tag{18.60}$$

You are asked to verify that this is a *faithful representation* in Problem 18.6.2. This means we have a unique assignment of each group element to a matrix and it satisfies the multiplication table of the group. Note that the multiplication table is guaranteed to work from the way we constructed these matrices, but it is a useful exercise to check it anyway.

Now that we have the group representation, we can construct our states. Recall, we have eight possible states. Six that have one double occupancy and two that have no double occupancy. We start with the ones with a double occupancy. They are

$$\begin{aligned} |1\rangle &= 1\uparrow 2\uparrow 1\downarrow, \quad |2\rangle = 1\uparrow 3\uparrow 1\downarrow, \quad |3\rangle = 1\uparrow 2\uparrow 2\downarrow \\ |4\rangle &= 2\uparrow 3\uparrow 2\downarrow, \quad |5\rangle = 1\uparrow 3\uparrow 3\downarrow, \quad |6\rangle = 2\uparrow 3\uparrow 3\downarrow . \end{aligned} \tag{18.61}$$

The way that we find the symmetrized states is that we start with one of the states, say $|1\rangle$. We perform the permutation of the site labels in the states according to each of the six group elements, multiply the end result by the value of the group representation (if a one-dimensional representation) or the 1,1 matrix element of the representation (if it is a higher dimensional representation), and add the results for all six operations together. One must be careful to add in minus signs if needed, when the transformed state is not in our canonical form. We work with the one-dimensional representations first. Starting from the trivial representation, we find, after acting on state $|1\rangle$ that we obtain

$$|a\rangle = \frac{1}{2\sqrt{2}}(1\uparrow 2\uparrow 1\downarrow + 1\uparrow 3\uparrow 1\downarrow - 1\uparrow 2\uparrow 2\downarrow$$

$$+2\uparrow 3\uparrow 2\downarrow -1\uparrow 3\uparrow 3\downarrow -2\uparrow 3\uparrow 3\downarrow). \tag{18.62}$$

The minus signs come from our convention for how we write our states. The representation that treats rotations as even and reflections as odd gives the state

$$\begin{aligned}|b\rangle = \frac{1}{2\sqrt{2}}(&1\uparrow 2\uparrow 1\downarrow -1\uparrow 3\uparrow 1\downarrow +1\uparrow 2\uparrow 2\downarrow\\ &+2\uparrow 3\uparrow 2\downarrow -1\uparrow 3\uparrow 3\downarrow +2\uparrow 3\uparrow 3\downarrow).\end{aligned} \tag{18.63}$$

You can see these two states are orthogonal to each other. Now, for the two-dimensional representations, we do the same, using the 1,1 and the 2,2 matrix elements of the representation. This gives the states

$$\begin{aligned}|c\rangle = \frac{1}{2\sqrt{3}}(&-(2)\cdot 1\uparrow 2\uparrow 1\downarrow -(2)\cdot 1\uparrow 3\uparrow 1\downarrow -1\uparrow 2\uparrow 2\downarrow\\ &+2\uparrow 3\uparrow 2\downarrow -1\uparrow 3\uparrow 3\downarrow -2\uparrow 3\uparrow 3\downarrow)\\ |d\rangle = \frac{1}{2\sqrt{3}}(&-(2)\cdot 1\uparrow 2\uparrow 1\downarrow +(2)\cdot 1\uparrow 3\uparrow 1\downarrow +1\uparrow 2\uparrow 2\downarrow\\ &+2\uparrow 3\uparrow 2\downarrow -1\uparrow 3\uparrow 3\downarrow +2\uparrow 3\uparrow 3\downarrow).\end{aligned} \tag{18.64}$$

We can find two states orthogonal to these four, which are the rest of the symmetrized states. They are

$$\begin{aligned}|e\rangle &= \frac{1}{2}(1\uparrow 2\uparrow 2\downarrow -2\uparrow 3\uparrow 2\downarrow -1\uparrow 3\uparrow 3\downarrow -2\uparrow 3\uparrow 3\downarrow)\\ |f\rangle &= \frac{1}{2}(-1\uparrow 2\uparrow 2\downarrow -2\uparrow 3\uparrow 2\downarrow -1\uparrow 3\uparrow 3\downarrow +2\uparrow 3\uparrow 3\downarrow).\end{aligned} \tag{18.65}$$

We have two more states to work with. By using the spin lowering operator, we find that the $S = \frac{3}{2}$ state with $m_S = \frac{1}{2}$ is

$$\frac{1}{\sqrt{3}}(1\uparrow 2\uparrow 3\downarrow -1\uparrow 3\uparrow 2\downarrow +2\uparrow 3\uparrow 1\downarrow). \tag{18.66}$$

The states we want are orthogonal to this state. Using our symmetrization procedure, we find the first state and orthogonality gives us the second. We have

$$\begin{aligned}|g\rangle &= \frac{1}{\sqrt{2}}(1\uparrow 2\uparrow 3\downarrow +1\uparrow 3\uparrow 2\downarrow)\\ |h\rangle &= \frac{1}{\sqrt{6}}(1\uparrow 2\uparrow 3\downarrow -1\uparrow 3\uparrow 2\downarrow -(2)\cdot 2\uparrow 3\uparrow 1\downarrow).\end{aligned} \tag{18.67}$$

We have all of our states, now we have to apply the Hamiltonian on them. We start with state $|a\rangle$. It has one double occupancy. The hopping term annihilates the state, so it is an eigenstate with $E = U$. Note, there is no simple way to see that the hopping annihilates the state. One has to do the calculation. Next, we look at the state $|b\rangle$. It also has one double occupancy and is annihilated by the hopping, so it also has $E = U$. The degenerate representations will form a 3×3 block. We have

$$\begin{aligned}
\hat{H}|c\rangle &= \left(-\frac{3}{2}t + U\right)|c\rangle - \frac{\sqrt{3}}{2}t|f\rangle + \sqrt{\frac{3}{2}}t|g\rangle \\
\hat{H}|f\rangle &= -\frac{\sqrt{3}}{2}t|c\rangle + (3t + U)|f\rangle + \frac{3}{\sqrt{2}}t|g\rangle \\
\hat{H}|g\rangle &= \sqrt{\frac{3}{2}}t|c\rangle + \frac{3}{\sqrt{2}}t|f\rangle.
\end{aligned} \tag{18.68}$$

So

$$H_{N=3\ S=\frac{1}{2}} = \begin{pmatrix} -\frac{3}{2}t + U & -\frac{\sqrt{3}}{2}t & \sqrt{\frac{3}{2}}t \\ -\frac{\sqrt{3}}{2}t & 3t + U & \frac{3}{\sqrt{2}}t \\ \sqrt{\frac{3}{2}}t & \frac{3}{\sqrt{2}}t & 0 \end{pmatrix}. \tag{18.69}$$

The characteristic equation becomes

$$E^3 - \left(\tfrac{3}{2}t + 2U\right)E^2 + \left(-\tfrac{45}{4}t^2 + \tfrac{3}{2}tU + U^2\right)E + \tfrac{9}{4}t^3 + 6t^2U = 0. \tag{18.70}$$

Each root of the cubic equation represents a doubly degenerate energy because these states transform as a two-dimensional representation.

Note that these calculations can be tedious. Often, once one has the symmetry analysis done, you do not have to determine how the Hamiltonian acts on each state, but you can pick specific states and use them to determine the weight of the coefficients of the different states after $\hat{H}$ acts on the state (this is because we are guaranteed to map to one of our symmetry states). There are a few checks on your algebra as well. The Hamiltonian blocks are real symmetric matrices, so corresponding off-diagonal terms must be equal. But, even with this, these calculations are challenging to carry out by hand, pushing what can be done to the limit. For this reason, do not feel bad if you cannot complete these calculations fully yourself. If you work carefully and take enough time, you should be able to do it, but some of you might not have enough time to do so. There is a tremendous satisfaction in being able to carry out these calculations to the end though. It also builds confidence that you can do long and complex calculations when you get involved in research.

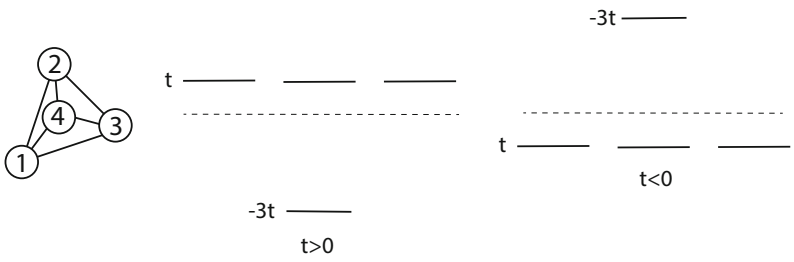

Fig. 18.3 (Left) Schematic of the Hubbard model on a tetrahedron, where the electrons can sit on any of the four different lattice sites. (Right) Bandstructure of the four-site model. We show both the case with $t > 0$, which has one state with energy $-3t$ below three degenerate states with t, while for $t < 0$, the situation is reversed]

We end this section with a discussion of the Hubbard model on a tetrahedron. We show all of the results for you, rather than deriving them. These results were worked out using computer algebra, although it is possible to determine the Hamiltonian matrix blocks using the same techniques we have used before and should not be too complicated for smaller numbers of electrons. The half-filled case is challenging to try to complete though. In Problem 18.8.1, we will give you the opportunity to try solve these problems for yourself with the four-site square model. The tetrahedron is depicted in Fig. 18.3, and the Hubbard model is given by

$$\hat{H} = -t \sum_{\sigma} (\hat{c}^{\dagger}_{1\sigma}\hat{c}_{2\sigma} + \hat{c}^{\dagger}_{1\sigma}\hat{c}_{3\sigma} + \hat{c}^{\dagger}_{1\sigma}\hat{c}_{4\sigma} + \hat{c}^{\dagger}_{2\sigma}\hat{c}_{3\sigma} + \hat{c}^{\dagger}_{2\sigma}\hat{c}_{4\sigma} + \hat{c}^{\dagger}_{3\sigma}\hat{c}_{4\sigma} + \text{h.c.}) + U(\hat{n}_{1\uparrow}\hat{n}_{1\downarrow} + \hat{n}_{2\uparrow}\hat{n}_{2\downarrow} + \hat{n}_{3\uparrow}\hat{n}_{3\downarrow} + \hat{n}_{4\uparrow}\hat{n}_{4\downarrow}). \tag{18.71}$$

Here, "h.c." stands for hermitian conjugate and what it means is that we must add the hermitian conjugate of each of the terms in the parenthesis.

We start by counting the states. We will again only work out the results below half filling. For $N = 0$, there is 1 state; for $N = 1$, 8 states; for $N = 2$, 28 states; for $N = 3$, 56 states; and for $N = 4$, 70 states. If we add up these results (doubling the results for $N \neq 4$ to count the states with $N > 4$ as well), we find a total of $4^4 = 256$ states, which is correct. Within each fixed N values, we have to count the states with different spin values. In addition, the tetrahedron has a lot of symmetry operations (they are again reflections and rotations). There are two one-dimensional representations—one which has the full symmetry of the system (trivial representation) and one that is odd under the reflections. There is one two-dimensional representation and two three-dimensional representations. We will not go into detail about which representation is which here, because such discussions might not be too relevant to you. Furthermore, we will only show one representative of the symmetric states and simply keep a count of the total number of states taking into account the spin and the spatial degeneracies.

We start with $N = 0$. This is one state, with $S = 0$ and $E = 0$. For $N = 1$, we have the states corresponding to the bandstructure and a degeneracy of two due to spin. One of the spatial representations is a one-dimensional, and the other is three-dimensional. The former one has the state

$$|1\rangle_{N=1\ S=\frac{1}{2}} = \frac{1}{2}(1\uparrow + 2\uparrow + 3\uparrow + 4\uparrow) \tag{18.72}$$

and its energy is $E = -3t$. The latter is represented by

$$|2\rangle_{N=1\ S=\frac{1}{2}} = \frac{1}{2}(1\uparrow + 2\uparrow - 3\uparrow - 4\uparrow) \tag{18.73}$$

and its two partner states; it has energy $E = t$. One can see the sum of the energy eigenvalues gives zero, after we account for the degeneracy. The state $E = -3t$ has twofold degeneracy and the state $E = t$ has six-fold degeneracy.

Now we move on to $N = 2$. We have two possible spin states $S = 1$ (which has a threefold degeneracy for the spin) and $S = 0$, which is nondegenerate. There are six $S = 1$ states composed of the two different three-dimensional spatial symmetries (so each state is ninefold degenerate). There are then ten $S = 0$ states, divided into two one-dimensional states (which are coupled together), one two-dimensional representation, and two three-dimensional representations (coupled). The two representative spin triplet states of definite spatial symmetry are

$$\begin{aligned} |1\rangle_{N=2\ S=1} &= \frac{1}{2}(1\uparrow 3\uparrow + 1\uparrow 4\uparrow + 2\uparrow 3\uparrow + 2\uparrow 4\uparrow) \\ |2\rangle_{N=2\ S=1} &= \frac{1}{2}(1\uparrow 2\uparrow - 1\uparrow 4\uparrow + 2\uparrow 3\uparrow + 3\uparrow 4\uparrow). \end{aligned} \tag{18.74}$$

State $|1\rangle$ has energy $-2t$, while state $|2\rangle$ has energy $2t$. Each is ninefold degenerate. One can easily see the possible energies by looking at how one can fill the different bandstructure states with up spin particles because these states have no interactions due to the Pauli exclusion principle.

There are five representative spin singlet states of definite spatial symmetry. They are

$$\begin{aligned} |1\rangle_{N=2\ S=0} &= \frac{1}{2}(1\uparrow 1\downarrow + 2\uparrow 2\downarrow + 3\uparrow 3\downarrow + 4\uparrow 4\downarrow) \\ |2\rangle_{N=2\ S=0} &= \frac{1}{2\sqrt{3}}(1\uparrow 2\downarrow + 1\uparrow 3\downarrow + 1\uparrow 4\downarrow + 2\uparrow 1\downarrow + 2\uparrow 3\downarrow + 2\uparrow 4\downarrow \\ &\quad + 3\uparrow 1\downarrow + 3\uparrow 2\downarrow + 3\uparrow 4\downarrow + 4\uparrow 1\downarrow + 4\uparrow 2\downarrow + 4\uparrow 3\downarrow) \\ |3\rangle_{N=2\ S=0} &= \frac{1}{2\sqrt{2}}(1\uparrow 2\downarrow - 1\uparrow 3\downarrow + 2\uparrow 1\downarrow - 2\uparrow 4\downarrow - 3\uparrow 1\downarrow + 3\uparrow 4\downarrow \\ &\quad - 4\uparrow 2\downarrow + 4\uparrow 3\downarrow) \end{aligned}$$

$$|4\rangle_{N=2\ S=0} = \frac{1}{2}(1\uparrow 1\downarrow +2\uparrow 2\downarrow -3\uparrow 3\downarrow -4\uparrow 4\downarrow)$$
$$|5\rangle_{N=2\ S=0} = \frac{1}{2}(1\uparrow 2\downarrow +2\uparrow 1\downarrow -3\uparrow 4\downarrow -4\uparrow 3\downarrow). \tag{18.75}$$

States $|1\rangle_{N=2\ S=0}$ and $|2\rangle_{N=2\ S=0}$ are nondegenerate and have the energy eigenvalues determined by diagonalizing the 2×2 Hamiltonian block

$$\begin{pmatrix} U & -2\sqrt{3}t \\ -2\sqrt{3}t & -2t \end{pmatrix}. \tag{18.76}$$

State $|3\rangle_{N=2\ S=0}$ is twofold degenerate and has energy $E = 2t$, while states $|4\rangle_{N=2\ S=0}$ and $|5\rangle_{N=2\ S=0}$ are each threefold degenerate and have the Hamiltonian block

$$\begin{pmatrix} U & -2t \\ -2t & 0 \end{pmatrix}. \tag{18.77}$$

You should see whether you can verify these energies and the Hamiltonian matrix blocks on your own.

We now move on to $N = 3$. There are 16 $S = \frac{3}{2}$ states and 40 $S = \frac{1}{2}$ states. For $S = \frac{3}{2}$, we have four spatial symmetry states—one is a one-dimensional representation and the other is three dimensional—so, including spin, one state is 4-fold degenerate and the other is 12-fold degenerate. Each state is an eigenstate of the Hamiltonian. The 4-fold degenerate state is

$$|1\rangle_{N=3\ S=\frac{3}{2}} = \frac{1}{2}(1\uparrow 2\uparrow 3\uparrow -1\uparrow 2\uparrow 4\uparrow +1\uparrow 3\uparrow 4\uparrow -2\uparrow 3\uparrow 4\uparrow), \tag{18.78}$$

which has energy $E = 3t$, while the 12-fold degenerate state has the following representative state:

$$|2\rangle_{N=3\ S=\frac{3}{2}} = \frac{1}{2}(1\uparrow 2\uparrow 3\uparrow +1\uparrow 2\uparrow 4\uparrow +1\uparrow 3\uparrow 4\uparrow +2\uparrow 3\uparrow 4\uparrow), \tag{18.79}$$

which has energy $E = -t$. The $S = \frac{1}{2}$ states have a twofold spin degeneracy. The spatial degeneracy has one one-dimensional, two two-dimensional, and five three-dimensional representations. This is 20 spatial states, which gives the required 40 $S = \frac{1}{2}$ states after we include the spin degeneracy. The eight different representative states will be listed next. We start with the one-dimensional state

$$|1\rangle_{N=3\ S=\frac{1}{2}} = \frac{1}{2\sqrt{3}}(1\uparrow 2\uparrow 1\downarrow -1\uparrow 2\uparrow 2\downarrow +1\uparrow 3\uparrow 1\downarrow -1\uparrow 3\uparrow 3\downarrow$$
$$+1\uparrow 4\uparrow 1\downarrow -1\uparrow 4\uparrow 4\downarrow +2\uparrow 3\uparrow 2\downarrow -2\uparrow 3\uparrow 3\downarrow$$

$$+2\uparrow 4\uparrow 2\downarrow -2\uparrow 4\uparrow 4\downarrow +3\uparrow 4\uparrow 3\downarrow -3\uparrow 4\uparrow 4\downarrow). \tag{18.80}$$

State $|1\rangle_{N=3\ S=\frac{1}{2}}$ is doubly degenerate and has energy $E=-t+U$. Next, we look at the two two-dimensional states given by

$$\begin{aligned}
&|2\rangle_{N=3\ S=\frac{1}{2}} \\
&\quad=\frac{1}{2\sqrt{2}}(1\uparrow 2\uparrow 1\downarrow -1\uparrow 2\uparrow 2\downarrow -1\uparrow 3\uparrow 1\downarrow +1\uparrow 3\uparrow 3\downarrow \\
&\qquad -2\uparrow 4\uparrow 2\downarrow +2\uparrow 4\uparrow 4\downarrow +3\uparrow 4\uparrow 3\downarrow -3\uparrow 4\uparrow 4\downarrow) \\
&|3\rangle_{N=3\ S=\frac{1}{2}} \\
&\quad=\frac{1}{2\sqrt{6}}(1\uparrow 2\uparrow 3\downarrow -1\uparrow 2\uparrow 4\downarrow -1\uparrow 3\uparrow 2\downarrow +1\uparrow 3\uparrow 4\downarrow \\
&\qquad -(2)1\uparrow 4\uparrow 2\downarrow +(2)1\uparrow 4\uparrow 3\downarrow -(2)2\uparrow 3\uparrow 1\downarrow \\
&\quad +(2)2\uparrow 3\uparrow 4\downarrow \\
&\qquad -2\uparrow 4\uparrow 1\downarrow +2\uparrow 4\uparrow 3\downarrow +3\uparrow 4\uparrow 1\downarrow -3\uparrow 4\uparrow 2\downarrow),
\end{aligned} \tag{18.81}$$

where the numerical coefficients (2) are put inside parentheses. States $|2\rangle_{N=3\ S=\frac{1}{2}}$ and $|3\rangle_{N=3\ S=\frac{1}{2}}$ are each fourfold degenerate, and the energy is found from diagonalizing the 2×2 block

$$\begin{pmatrix} 2t+U & \sqrt{3}t \\ \sqrt{3}t & 0 \end{pmatrix}. \tag{18.82}$$

Next, we have three three-dimensional representations with the same spatial symmetry. They are

$$\begin{aligned}
|4\rangle_{N=3\ S=\frac{1}{2}} &= \frac{1}{2}(1\uparrow 2\uparrow 1\downarrow -1\uparrow 2\uparrow 2\downarrow -3\uparrow 4\uparrow 3\downarrow +3\uparrow 4\uparrow 4\downarrow) \\
|5\rangle_{N=3\ S=\frac{1}{2}} &= \frac{1}{2\sqrt{2}}(1\uparrow 3\uparrow 1\downarrow +1\uparrow 3\uparrow 3\downarrow +1\uparrow 4\uparrow 1\downarrow +1\uparrow 4\uparrow 4\downarrow \\
&\qquad +2\uparrow 3\uparrow 2\downarrow +2\uparrow 3\uparrow 3\downarrow +2\uparrow 4\uparrow 2\downarrow +2\uparrow 4\uparrow 4\downarrow) \\
|6\rangle_{N=3\ S=\frac{1}{2}} &= \frac{1}{2\sqrt{2}}(1\uparrow 3\uparrow 2\downarrow +1\uparrow 3\uparrow 4\downarrow +1\uparrow 4\uparrow 2\downarrow +1\uparrow 4\uparrow 3\downarrow \\
&\qquad +2\uparrow 3\uparrow 1\downarrow +2\uparrow 3\uparrow 4\downarrow +2\uparrow 4\uparrow 1\downarrow +2\uparrow 4\uparrow 3\downarrow).
\end{aligned} \tag{18.83}$$

The states $|4\rangle_{N=3\ S=\frac{1}{2}}$, $|5\rangle_{N=3\ S=\frac{1}{2}}$, and $|6\rangle_{N=3\ S=\frac{1}{2}}$ are each sixfold degenerate and the eigenvalues are found from diagonalizing the following 3×3 block:

$$\begin{pmatrix} t+U & -\sqrt{2}t & \sqrt{2}t \\ -\sqrt{2}t & -2t+U & -3t \\ \sqrt{2}t & -3t & -2t \end{pmatrix}. \tag{18.84}$$

And last, but not least, we have the two three-dimensional representations of the other spatial symmetry:

$$\begin{aligned} |7\rangle_{N=3\ S=\frac{1}{2}} &= \frac{1}{2\sqrt{2}}(1\uparrow 2\uparrow 1\downarrow +1\uparrow 2\uparrow 2\downarrow -1\uparrow 4\uparrow 1\downarrow -1\uparrow 4\uparrow 4\downarrow \\ &\quad +2\uparrow 3\uparrow 2\downarrow +2\uparrow 3\uparrow 3\downarrow +3\uparrow 4\uparrow 3\downarrow +3\uparrow 4\uparrow 4\downarrow) \\ |8\rangle_{N=3\ S=\frac{1}{2}} &= \frac{1}{2\sqrt{6}}(1\uparrow 2\uparrow 3\downarrow +1\uparrow 2\uparrow 4\downarrow +(2)1\uparrow 3\uparrow 2\downarrow -(2)1\uparrow 3\uparrow 4\downarrow \\ &\quad 1\uparrow 4\uparrow 2\downarrow -1\uparrow 4\uparrow 3\downarrow +2\uparrow 3\uparrow 1\downarrow +2\uparrow 3\uparrow 4\downarrow \\ &\quad (2)2\uparrow 4\uparrow 1\downarrow +(2)2\uparrow 4\uparrow 3\downarrow +3\uparrow 4\uparrow 1\downarrow +3\uparrow 4\uparrow 2\downarrow). \end{aligned} \tag{18.85}$$

The states $|7\rangle_{N=3\ S=\frac{1}{2}}$ and $|8\rangle_{N=3\ S=\frac{1}{2}}$ are each sixfold degenerate, and the energy eigenvalues are found from diagonalizing the following 2×2 matrix:

$$\begin{pmatrix} U & -\sqrt{3}t \\ -\sqrt{3}t & 2t \end{pmatrix}. \tag{18.86}$$

This completes the $N = 3$ case.

Next up is the $N = 4$ case, which is the most difficult one. In terms of counting, we have 1 state with $S = 2$ (fivefold degenerate), 15 $S = 1$ states (threefold degenerate), and 20 $S = 0$ states. This accounts for all of the 70 states that we have. The $S = 2$ state is given by $1\uparrow 2\uparrow 3\uparrow 4\uparrow$, is fivefold degenerate, and has energy $E = 0$. Next, the $S = 1$ states, separated according to their spatial symmetry. The first one is a one-dimensional representation:

$$\begin{aligned} |1\rangle_{N=4\ S=1} = \frac{1}{2\sqrt{3}}&(1\uparrow 2\uparrow 3\uparrow 1\downarrow +1\uparrow 2\uparrow 3\uparrow 2\downarrow +1\uparrow 2\uparrow 3\uparrow 3\downarrow \\ &-1\uparrow 2\uparrow 4\uparrow 1\downarrow -1\uparrow 2\uparrow 4\uparrow 2\downarrow -1\uparrow 2\uparrow 4\uparrow 4\downarrow \\ &+1\uparrow 3\uparrow 4\uparrow 1\downarrow +1\uparrow 3\uparrow 4\uparrow 3\downarrow +1\uparrow 3\uparrow 4\uparrow 4\downarrow \\ &-2\uparrow 3\uparrow 4\uparrow 2\downarrow -2\uparrow 3\uparrow 4\uparrow 3\downarrow -2\uparrow 3\uparrow 4\uparrow 4\downarrow). \end{aligned} \tag{18.87}$$

The energy of this state is $E = U$. It is threefold degenerate. Next, we have one two-dimensional representation, given by

$$
\begin{aligned}
|2\rangle_{N=4\ S=1} = \frac{1}{2\sqrt{6}} \times \big(& (2)1\uparrow 2\uparrow 3\uparrow 1\downarrow \\
& -1\uparrow 2\uparrow 3\uparrow 2\downarrow -1\uparrow 2\uparrow 3\uparrow 3\downarrow +1\uparrow 2\uparrow 4\uparrow 1\downarrow \\
& -(2)1\uparrow 2\uparrow 4\uparrow 2\downarrow +1\uparrow 2\uparrow 4\uparrow 4\downarrow -1\uparrow 3\uparrow 4\uparrow 1\downarrow \\
& +(2)1\uparrow 3\uparrow 4\uparrow 3\downarrow -1\uparrow 3\uparrow 4\uparrow 4\downarrow +2\uparrow 3\uparrow 4\uparrow 2\downarrow \\
& +2\uparrow 3\uparrow 4\uparrow 3\downarrow -(2)2\uparrow 3\uparrow 4\uparrow 4\downarrow \big).
\end{aligned}
\tag{18.88}
$$

This state also has $E = U$ and is sixfold degenerate. Next, we have the ninefold degenerate state, represented by

$$
\begin{aligned}
& |3\rangle_{N=4\ S=1} \\
& = \frac{1}{2\sqrt{2}}(1\uparrow 2\uparrow 3\uparrow 1\downarrow -1\uparrow 2\uparrow 3\uparrow 2\downarrow +1\uparrow 2\uparrow 4\uparrow 1 - 1\uparrow 2\uparrow 4\uparrow 2\downarrow \\
& \quad -1\uparrow 3\uparrow 4\uparrow 3\downarrow +1\uparrow 3\uparrow 4\uparrow 4\downarrow -2\uparrow 3\uparrow 4\uparrow 3\downarrow +2\uparrow 3\uparrow 4\uparrow 4\downarrow).
\end{aligned}
\tag{18.89}
$$

This state also has energy $E = U$. The last spatial symmetry has three different ninefold degenerate states represented by

$$
\begin{aligned}
& |4\rangle_{N=4\ S=1} \\
& = \frac{1}{2\sqrt{2}}(1\uparrow 2\uparrow 3\uparrow 1\downarrow +1\uparrow 2\uparrow 3\uparrow 3\downarrow +1\uparrow 2\uparrow 4\uparrow 2\downarrow +1\uparrow 2\uparrow 4\uparrow 4\downarrow \\
& \quad +1\uparrow 3\uparrow 4\uparrow 1\downarrow +1\uparrow 3\uparrow 4\uparrow 3\downarrow +2\uparrow 3\uparrow 4\uparrow 2\downarrow +2\uparrow 3\uparrow 4\uparrow 4\downarrow) \\
& |5\rangle_{N=4\ S=1} \\
& = \frac{1}{2}(1\uparrow 2\uparrow 3\uparrow 2\downarrow +1\uparrow 2\uparrow 4\uparrow 1\downarrow +1\uparrow 3\uparrow 4\uparrow 4\downarrow +2\uparrow 3\uparrow 4\uparrow 3\downarrow) \\
& |6\rangle_{N=4\ S=1} \\
& = \frac{1}{2}(1\uparrow 2\uparrow 3\uparrow 4\downarrow +1\uparrow 2\uparrow 4\uparrow 3\downarrow +1\uparrow 3\uparrow 4\uparrow 2\downarrow +2\uparrow 3\uparrow 4\uparrow 1\downarrow).
\end{aligned}
\tag{18.90}
$$

These three states (each nine-fold degenerate) have energy levels given by the eigenvalues of the 3×3 block given by

$$\begin{pmatrix} U & -2\sqrt{2}t & -2\sqrt{2}t \\ -2\sqrt{2}t & U & 0 \\ -2\sqrt{2}t & 0 & 0 \end{pmatrix}. \tag{18.91}$$

All we have left are the $S = 0$ states. There are nine representative states. Two one-dimensional representatives are coupled together with the same symmetry, They are

$$\begin{aligned} |1\rangle_{N=4\ S=0} &= \frac{1}{\sqrt{6}}(1\uparrow 2\uparrow 1\downarrow 2\downarrow + 1\uparrow 3\uparrow 1\downarrow 3\downarrow + 1\uparrow 4\uparrow 1\downarrow 4\downarrow \\ &\quad + 2\uparrow 3\uparrow 2\downarrow 3\downarrow + 2\uparrow 4\uparrow 2\downarrow 4\downarrow + 3\uparrow 4\uparrow 3\downarrow 4\downarrow) \\ |2\rangle_{N=4\ S=0} &= \frac{1}{2\sqrt{6}}(1\uparrow 2\uparrow 1\downarrow 3\downarrow + 1\uparrow 2\uparrow 1\downarrow 4\downarrow - 1\uparrow 2\uparrow 2\downarrow 3\downarrow - 1\uparrow 2\uparrow 2\downarrow 4\downarrow \\ &\quad + 1\uparrow 3\uparrow 1\downarrow 2\downarrow + 1\uparrow 3\uparrow 1\downarrow 4\downarrow + 1\uparrow 3\uparrow 2\downarrow 3\downarrow - 1\uparrow 3\uparrow 3\downarrow 4\downarrow \\ &\quad + 1\uparrow 4\uparrow 1\downarrow 2\downarrow + 1\uparrow 4\uparrow 1\downarrow 3\downarrow + 1\uparrow 4\uparrow 2\downarrow 4\downarrow + 1\uparrow 4\uparrow 3\downarrow 4\downarrow \\ &\quad - 2\uparrow 3\uparrow 1\downarrow 2\downarrow + 2\uparrow 3\uparrow 1\downarrow 3\downarrow + 2\uparrow 3\uparrow 2\downarrow 4\downarrow - 2\uparrow 3\uparrow 3\downarrow 4\downarrow \\ &\quad - 2\uparrow 4\uparrow 1\downarrow 2\downarrow + 2\uparrow 4\uparrow 1\downarrow 4\downarrow + 2\uparrow 4\uparrow 2\downarrow 3\downarrow + 2\uparrow 4\uparrow 3\downarrow 4\downarrow \\ &\quad - 3\uparrow 4\uparrow 1\downarrow 3\downarrow + 3\uparrow 4\uparrow 1\downarrow 4\downarrow - 3\uparrow 4\uparrow 2\downarrow 3\downarrow + 3\uparrow 4\uparrow 2\downarrow 4\downarrow). \end{aligned} \tag{18.92}$$

The 2×2 Hamiltonian block that needs to be diagonalized is

$$\begin{pmatrix} 2U & -4t \\ -4t & U \end{pmatrix}. \tag{18.93}$$

Each state is nondegenerate. The next symmetry has three coupled states, given by

$$\begin{aligned} |3\rangle_{N=4\ S=0} &= \frac{1}{2}(1\uparrow 2\uparrow 1\downarrow 2\downarrow - 1\uparrow 3\uparrow 1\downarrow 3\downarrow - 2\uparrow 4\uparrow 2\downarrow 4\downarrow + 3\uparrow 4\uparrow 3\downarrow 4\downarrow) \\ |4\rangle_{N=4\ S=0} &= \frac{1}{4}(1\uparrow 2\uparrow 1\downarrow 4\downarrow - 1\uparrow 2\uparrow 2\downarrow 3\downarrow - 1\uparrow 3\uparrow 1\downarrow 4\downarrow - 1\uparrow 3\uparrow 2\downarrow 3\downarrow \\ &\quad + 1\uparrow 4\uparrow 1\downarrow 2\downarrow - 1\uparrow 4\uparrow 1\downarrow 3\downarrow - 1\uparrow 4\uparrow 2\downarrow 4\downarrow + 1\uparrow 4\uparrow 3\downarrow 4\downarrow \\ &\quad - 2\uparrow 3\uparrow 1\downarrow 2\downarrow - 2\uparrow 3\uparrow 1\downarrow 3\downarrow - 2\uparrow 3\uparrow 2\downarrow 4\downarrow - 2\uparrow 3\uparrow 3\downarrow 4\downarrow \\ &\quad - 2\uparrow 4\uparrow 1\downarrow 4\downarrow - 2\uparrow 4\uparrow 2\downarrow 3\downarrow + 3\uparrow 4\uparrow 1\downarrow 4\downarrow - 3\uparrow 4\uparrow 2\downarrow 3\downarrow) \end{aligned}$$

$$
\begin{aligned}
&|5\rangle_{N=4\ S=0} \\
&= \frac{1}{2\sqrt{3}}(1\uparrow 2\uparrow 3\downarrow 4\downarrow - 1\uparrow 3\uparrow 2\downarrow 4\downarrow - (2)1\uparrow 4\uparrow 2\downarrow 3\downarrow \\
&\quad - (2)2\uparrow 3\uparrow 1\downarrow 4\downarrow - 2\uparrow 4\uparrow 1\downarrow 3\downarrow + 3\uparrow 4\uparrow 1\downarrow 2\downarrow). \qquad (18.94)
\end{aligned}
$$

These states are all twofold degenerate, and the energy eigenvalues are found by diagonalizing the following 3×3 block:

$$
\begin{pmatrix} 2U & 2t & 0 \\ 2t & U & 2\sqrt{3}t \\ 0 & 2\sqrt{3}t & 0 \end{pmatrix}. \qquad (18.95)
$$

We have two groups of three-dimensional representations. The first one is

$$
\begin{aligned}
&|5\rangle_{N=4\ S=0} \\
&= \frac{1}{\sqrt{2}}(1\uparrow 2\uparrow 1\downarrow 2\downarrow - 3\uparrow 4\uparrow 3\downarrow 4\downarrow) \\
&|6\rangle_{N=4\ S=0} \\
&= \frac{1}{4}(1\uparrow 2\uparrow 1\downarrow 3\downarrow + 1\uparrow 2\uparrow 1\downarrow 4\downarrow - 1\uparrow 2\uparrow 2\downarrow 3\downarrow - 1\uparrow 2\uparrow 2\downarrow 4\downarrow \\
&\quad + 1\uparrow 3\uparrow 1\downarrow 2\downarrow + 1\uparrow 3\uparrow 3\downarrow 4\downarrow + 1\uparrow 4\uparrow 1\downarrow 2\downarrow - 1\uparrow 4\uparrow 3\downarrow 4\downarrow \\
&\quad - 2\uparrow 3\uparrow 1\downarrow 2\downarrow + 2\uparrow 3\uparrow 3\downarrow 4\downarrow - 2\uparrow 4\uparrow 1\downarrow 2\downarrow - 2\uparrow 4\uparrow 3\downarrow 4\downarrow \\
&\quad + 3\uparrow 4\uparrow 1\downarrow 3\downarrow - 3\uparrow 4\uparrow 1\downarrow 4\downarrow + 3\uparrow 4\uparrow 2\downarrow 3\downarrow - 3\uparrow 4\uparrow 2\downarrow 4\downarrow) \\
&|7\rangle_{N=4\ S=0} \\
&= \frac{1}{2\sqrt{2}}(1\uparrow 3\uparrow 1\downarrow 4\downarrow - 1\uparrow 3\uparrow 2\downarrow 3\downarrow + 1\uparrow 4\uparrow 1\downarrow 3\downarrow - 1\uparrow 4\uparrow 2\downarrow 4\downarrow \\
&\quad - 2\uparrow 3\uparrow 1\downarrow 3\downarrow + 2\uparrow 3\uparrow 2\downarrow 4\downarrow - 2\uparrow 4\uparrow 1\downarrow 4\downarrow + 2\uparrow 4\uparrow 2\downarrow 3\downarrow), \qquad (18.96)
\end{aligned}
$$

with the 3×3 matrix that determines the energy eigenvalues being

$$
\begin{pmatrix} 2U & -2\sqrt{2}t & 0 \\ -2\sqrt{2}t & U & 2\sqrt{2}t \\ 0 & 2\sqrt{2}t & U \end{pmatrix}. \qquad (18.97)
$$

Each state is threefold degenerate. We have one final state

$$
|8\rangle_{N=4\ S=0}
$$

$$= \frac{1}{4}(1\uparrow 2\uparrow 1\downarrow 3\downarrow +1\uparrow 2\uparrow 2\downarrow 4\downarrow +1\uparrow 3\uparrow 1\downarrow 2\downarrow -1\uparrow 3\uparrow 1\downarrow 4\downarrow$$
$$-1\uparrow 3\uparrow 2\downarrow 3\downarrow -1\uparrow 3\uparrow 3\downarrow 4\downarrow -1\uparrow 4\uparrow 1\downarrow 3\downarrow +1\uparrow 4\uparrow 2\downarrow 4\downarrow$$
$$-2\uparrow 3\uparrow 1\downarrow 3\downarrow +2\uparrow 3\uparrow 2\downarrow 4\downarrow +2\uparrow 4\uparrow 1\downarrow 2\downarrow +2\uparrow 4\uparrow 1\downarrow 4\downarrow$$
$$+2\uparrow 4\uparrow 2\downarrow 3\downarrow -2\uparrow 4\uparrow 3\downarrow 4\downarrow -3\uparrow 4\uparrow 1\downarrow 3\downarrow -3\uparrow 4\uparrow 2\downarrow 4\downarrow). \tag{18.98}$$

The energy of this threefold degenerate state is $E = U$.

This was a huge effort. Look back at what we achieved—we took a 256×256 matrix that described the Hubbard model on the four-site tetrahedron, and we reduced it to Hamiltonian blocks that were no larger than 3×3 blocks. Because one can, in principle, analytically find the roots of cubic equations, we have essentially analytically solved for all of the energy levels of the model! This is quite amazing and points to the benefits of performing a symmetry analysis for these types of problems, even if using them can be a tedious operation.

While we have essentially solved these models, we actually need to find the different energy eigenvalues and determine which states (and what their quantum numbers are) that correspond to the ground state. Some of these questions will be explored in the problems below.

Section 18.6 Problems:

18.6.1 Completing the Symmetry Analysis for $N = 2$ on the Triangle

Find the other set of states for the $N = 2$ case on the triangle that completes the states of the two-dimensional representation. Verify it is correct by computing the Hamiltonian block and seeing that it gives the same energy eigenvalues.

18.6.2 Verifying the Multiplication Table for the 2×2 Matrix Representation of the Group

Verify the multiplication table for the 2×2 representation of permutation group of three elements using the matrices developed in the main text.

18.6.3 Verifying the Hamiltonian Matrix Block for the Other Symmetry Sector on the Triangle

In the main text, we worked with the states $|c\rangle$, $|f\rangle$, and $|g\rangle$ for the half-filled case on the triangle. Compute the Hamiltonian matrix block using states $|d\rangle$, $|e\rangle$ and $|h\rangle$, and verify that they give the same energy eigenvalues.

18.6.4 Nagaoka Theorem for the Tetrahedron

Find the critical value of U for the Nagaoka theorem where the $S = \frac{1}{2}$ ground state transitions to the $S = \frac{3}{2}$ ground state by using the energies we found for the exact diagonalization on the tetrahedron with $N = 3$. Note that the ferromagnetism only holds for one sign of t. Which one?

18.6.5 Ground-State Quantum Numbers

Using the Hamiltonian subblocks for the case of half-filling on the tetrahedron, determine the ground state quantum number for total spin S, by finding all energy eigenvalues as a function of U and determining which one is the lowest as a function of U. Is there any transition in the ground state from one state to another? Consider both signs of the hopping t and both signs of the interaction U. Note that this is a brute-force problem. For a given U value, compute all of the energy levels, find the one that is the lowest, and record its quantum number; then change U and repeat. Work with a sufficient number of U values that it is clear you have determined there are no transitions or a transition has occurred. These transitions, when they happen, are called *level crossings*, and they can only occur between two states with different symmetries.

18.6.6 How to Find Quantum Numbers Without Operating the Operator onto a State

In this section, we worked with nonbipartite lattices. In this case, there are no pseudospin quantum numbers. Suppose that we instead were working with a lattice that had pseudospin symmetry. Just by considering the energy levels (and other quantum numbers), explain how you can determine the total pseudospin quantum numbers for each state. You may assume you have the energy eigenvalues for all values of electron number.

18.7 Lieb Theorems and Spin-Reflection Positivity

In this section, we discuss an important idea called *spin-reflection positivity* formulated by Elliott Lieb in the late 1980s (Lieb, 1989; Freericks & Lieb, 1995; Freericks, 2022a). This idea, allows us to prove three important theorems about the Hubbard model which relate to the eigenvalues of the ground state. This section uses many of the ideas we have already developed in this chapter, so feel free to go back and review them if you find you need to. It is remarkable how we can establish these rigorous results just by approaching the problem from the right perspective.

The Hubbard Hamiltonian we will work with is slightly more general from the initial one we had. We still take a hopping matrix and an interaction term, but here we restrict the hopping matrix to be a real and symmetric matrix rather than a Hermitian matrix and we allow the interaction U_i to depend on the lattice site i. The Hamiltonian then becomes

$$\hat{\mathcal{H}} = -\sum_{\sigma} \sum_{ij\in\Lambda} t_{ij} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + \sum_{i\in\Lambda} U_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}. \tag{18.99}$$

The first theorem is about the existence of a ground state that has $S = 0$, for the case where the interactions U_i are all chosen to satisfy $U_i \leq 0$ for all $i \in \Lambda$. This is what we call the attractive case of the model, because the two particles are attracted together. While we know that, in general, electrons repel each other due to their like charge, it is common to generalize models to different cases—we will return to the more physical repulsive case at the end of the section.

The first requirement we have for this proof is that the number of electrons $N_e = 2N$ is even and that the number of lattice sites $|\Lambda|$ is finite. Next, we restrict the many-body states to $m_S = 0$, because states with every possible S eigenvalue have a representative in this subspace. Then, we have the same number of up-spin and down-spin electrons, which we denote N and hence $N_e = N_\uparrow + N_\downarrow = N + N = 2N$. The fact that the Hamiltonian has all real parameters in it will become important a bit later in the proof.

Theorem 1 (Existence of a Spin-singlet Ground State for the Attractive Hubbard Model) *If the interactions are all attractive, so that $U_i \leq 0 \; \forall \; i \in \Lambda$, then among all of the possible ground states of the Hubbard model with an even number of electrons $N_e = 2N$, there is at least one state that is a spin-singlet state with $S = 0$.*

Note that we have $U_i \leq 0, \; \forall \; i \in \Lambda$, so that we can write them as $U_i = -|U_i|$. We let $I = (i_1, i_2, \cdots, i_N)$ be the N-tuple of labels for the locations of up-spin electrons on Λ and $J = (j_1, j_2, \cdots, j_N)$ be the corresponding down-spin labels. The basis states, labeled by I and J, is given by

$$\hat{c}^{\dagger}_{i_1\uparrow} \hat{c}^{\dagger}_{i_2\uparrow} \cdots \hat{c}^{\dagger}_{i_N\uparrow} |0\rangle \otimes \hat{c}^{\dagger}_{j_1\downarrow} \hat{c}^{\dagger}_{j_2\downarrow} \cdots \hat{c}^{\dagger}_{j_N\downarrow} |0\rangle, \tag{18.100}$$

and the coefficients are given by $\Psi(I, J)$ in the expansion of the many-body state ($|\psi\rangle = \sum_{IJ} \psi(I, J)|I\rangle\otimes|J\rangle$ with the first factor in the tensor product being spin up and the second spin down). The antisymmetry of the wavefunction is guaranteed by the anticommutation relations of the fermionic creation operators. We often find it convenient to discuss the quantum state in terms of these coefficients. For example, because $\Psi(J, I)$, $\Psi^*(I, J)$, and $\Psi(J, I)^*$ are also eigenstates with the same energy, we can, without loss of generality, assume that $\Psi(I, J) = \Psi^*(J, I)$; that is, the matrix of coefficients is Hermitian. The proof of this is in Problem 18.7.1 below. Note that this assumption *does not imply* that every $\Psi(I, J)$ is real valued. This matrix Ψ is a $d \times d$ matrix, with $d = \binom{|\Lambda|}{N}$ being the number of single-spin fermionic states with N electrons on the lattice. Any finite-dimensional Hermitian matrix can be expanded in an eigenfunction basis $\{\phi_\alpha \mid \sum_J \Psi(I, J)\phi_\alpha(J) = w_\alpha\phi_\alpha(I)\}$, with real eigenvalues w_α; recall we are simply multiplying the matrix Ψ by a vector ϕ_α and finding it is equal to a number w_α times the eigenvector ψ_α. Note that each ϕ_α is orthonormal, with $\sum_I \phi_\alpha^*(I)\phi_\beta(I) = \delta_{\alpha\beta}$. The expansion for $\Psi(I, J)$ becomes

$$\Psi(I, J) = \sum_{\alpha=1}^{d} w_\alpha \phi_\alpha(I)\phi_\alpha^*(J). \tag{18.101}$$

This is the standard way to express a matrix in terms of its eigenvalues and eigenvectors. Again, we want to reiterate that this matrix structure is essentially a hidden property of the many-body state that emerges only when we treat the many-body state in this tensor-product fashion.

We have introduced a number of abstract ideas here, so we pause for a moment to be sure everything is clear. First off, instead of thinking of the quantum state as a sum of coefficients multiplied by each orthonormal basis vector of the Hilbert space, we instead are using a matrix structure for the coefficients that uses the tensor-product structure of the many-body states and is a compact way of storing the coefficients of the expansion. Once we have done this, we learn that due to the fact that the parameters in the Hamiltonian are all real, this matrix of coefficients is a Hermitian matrix, which means it has a complete set of eigenvectors and real eigenvalues and we can use them to simplify calculations. Note that the eigenvalues of the matrix Ψ are not directly related to the eigenvalues of the Hamiltonian. *These are two very different objects.* It is important you keep these different concepts straight.

Our approach for how we will proceed is to use a variational method to prove the theorem. This means we need to first compute the expectation value of the energy, in the state Ψ, we use the spectral decomposition in Eq. (18.101). The kinetic-energy piece is given by

$$\langle\Psi|\hat{K}|\Psi\rangle = \sum_{\alpha=1}^{d}\sum_{\beta=1}^{d} w_\alpha w_\beta \sum_I \sum_J \sum_K \Big(\phi_\alpha^*(I)K_\uparrow(I, J)\phi_\beta(J)\phi_\alpha(K)\phi_\beta^*(K)$$

$$+ \phi_\alpha(I) K_\downarrow(I, J) \phi^*_\beta(J) \phi^*_\alpha(K) \phi_\beta(K) \Big). \tag{18.102}$$

Here, $\hat{K}_\sigma(I, J)$ is the matrix element of the kinetic-energy operator $-\sum_{i,j\in\Lambda} t_{ij} \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma}$, expressed in the product-space basis we used to expand the wavefunctions. When we apply the hopping operator onto a spinless fermion state denoted by I, we allow each fermionic creation operator to hop to the sites connected by t_{ij}. For each state J that it hops to, we accumulate a $\pm t_{ij}$, depending on the signs needed to rearrange the states into the canonical form. Using the facts that the $\{\phi_\alpha\}$ are an orthonormal set (recall they are eigenvectors of a Hermitian matrix), that K_σ is independent of spin, and that the t_{ij} are real yields

$$\langle \Psi | \hat{K} | \Psi \rangle = 2 \sum_{\alpha=1}^{d} w_\alpha^2 \langle \phi_\alpha | \hat{K} | \phi_\alpha \rangle. \tag{18.103}$$

Here, we have used a Dirac notation for these eigenvectors and the K matrix, which should be clear. Note that the ϕ_α here can only be found from the eigenvectors of the matrix Ψ and hence have no clear physical interpretation. But, we will see they are very useful in computations. Similarly, we have that the potential energy piece satisfies

$$\langle \Psi | \hat{L} | \Psi \rangle = -\sum_{i\in\Lambda} |U_i| \sum_{\alpha=1}^{d} \sum_{\beta=1}^{d} w_\alpha w_\beta \sum_I \sum_J \sum_K \sum_{K'} \phi^*_\alpha(I) L_i(I, J) \phi_\beta(J) \times \phi_\alpha(K) L_i(K, K') \phi^*_\beta(K'), \tag{18.104}$$

where $L_i(I, J)$ is the matrix representation of $n_{i\sigma}$ (in the same basis we have been using), which is independent of σ. Using the fact that $L^*_i(I, J) = L_i(J, I)$ then yields

$$\langle \Psi | \hat{L} | \Psi \rangle = -\sum_{i\in\Lambda} |U_i| \sum_{\alpha=1}^{d} \sum_{\beta=1}^{d} w_\alpha w_\beta |\langle \phi_\alpha | \hat{L}_i | \phi_\beta \rangle|^2, \tag{18.105}$$

where, again, we use this Dirac notation for the ϕ's. Finally, the overlap of the wavefunction with itself is

$$\langle \Psi | \Psi \rangle = \sum_{\alpha=1}^{d} \sum_{\beta=1}^{d} \sum_I \sum_J \phi^*_\alpha(I) \phi_\beta(I) \phi_\alpha(J) \phi^*_\beta(J) = \sum_{\alpha=1}^{d} w_\alpha^2, \tag{18.106}$$

because the eigenvectors are orthonormal. This leads to the energy expectation value

$$E(\Psi) = \frac{2\sum_{\alpha=1}^{d} w_\alpha^2 \langle\phi_\alpha|\hat{K}|\phi_\alpha\rangle - \sum_{i\in\Lambda} |U_i| \sum_{\alpha=1}^{d}\sum_{\beta=1}^{d} w_\alpha w_\beta |\langle\phi_\alpha|\hat{L}_i|\phi_\beta\rangle|^2}{\sum_{\alpha=1}^{d} w_\alpha^2}. \tag{18.107}$$

One can immediately see that replacing w_α by $|w_\alpha|$ can only make the energy smaller. It does not change the denominator or the kinetic energy terms, because they only depend on $w_\alpha^2 = |w_\alpha|^2$ (recall the w_α are real). But, because the coefficients of $w_\alpha w_\beta$ are net negative, replacing by $|w_\alpha|\,|w_\beta|$ only lowers the expectation value. So, because the ground state is the lowest energy state, we can assume that the ground state includes a state that satisfies $\Psi = |\Psi|$ in a matrix sense, where $|\Psi| = \sqrt{\Psi^2}$; note that this is usually not the same as $\Psi(I,J) \to |\Psi(I,J)|$, as that only holds in the diagonal basis. This means that $\mathrm{Tr}\,\Psi = \sum_{\alpha=1}^{d} |w_\alpha| > 0$, which implies that $\Psi(\bar{I},\bar{I}) \neq 0$ for some $\bar{I}$. Then the up spin particles are at the same locations as the down spin particles—this requires them to be in a spin-singlet state. So, among the possible ground states, there is at least one ground state that is a spin singlet.

The existence proof above illustrates the concept of *spin reflection positivity*. When we wrote the wavefunction in the square matrix form (possible because we are in the sector with $m_S = 0$), we made manifest the spin-reflection symmetry, which switches the up spins and the down spins and vice versa, and showed that a positivity criterion for the wavefunction does not raise the variational energy; hence the ground state must be in this positive semidefinite form. From this, we learn that the ground state has a $S = 0$ component to it.

The uniqueness proof is both technically more complicated and requires additional restrictions on the Hamiltonian. We will not prove it completely here, but we will sketch how the argument goes for the theorem, and additional details will be examined in Problem 18.7.4 below. Key to this work is the idea of a lattice structure in the many-body configuration space of the wavefunction. If this configuration space is fully connected, then the ground-state is unique (similar in many respects to the methodology used to prove the uniqueness in the Perron-Frobenius theorem, which also relies on positivity and connectivity arguments). We have hinted at this lattice structure already, when we described how we compute the K matrix, which is a matrix (or operator) acting in the many-body space of states.

The additional restrictions for uniqueness are not too severe—we must have the hopping be connected on the graph Λ, and we must require all U_i to be strictly negative $U_i < 0$. The concept of *connected* on the lattice Λ is defined as follows: every two lattice sites $i, j \in \Lambda$ are said to have a bond between them if $t_{ij} \neq 0$. Then, the lattice is connected, if for every pair $i \neq j \in \Lambda$, there is a chain of bonds along a path that connects i to j. This path (with the chain of bonds) is defined as $\{i_m \mid i_m \in \Lambda, i_m \neq i_n \text{ if } m \neq n,\ i = i_1, i_2, \cdots, i_{n-1}, i_n = j, \text{ and } t_{i_1 i_2} t_{i_2 i_3} \cdots t_{i_{n-1} i_n} \neq 0\}$. Hence, we say for every pair of lattice points i and j there is a path from i to j connected by a chain of bonds. This is the conventional condition for connectedness of a lattice via its bonds, which we have seen before.

The lemma we will need to prove first is that if the lattice Λ is connected, then the single-spin many-body configuration space is also connected, when thought of as an

abstract lattice in the many-body space. The single-spin many-body configuration space is the collection of all many-body states with N electrons of the form

$$\{\hat{c}_{i_1}^\dagger \cdots \hat{c}_{i_N}^\dagger |0\rangle \mid i_m \in \Lambda \forall\, m = 1, \cdots, N; i_m \neq i_n \forall\, m, n = 1, \cdots, N \text{ when } m \neq n\}, \tag{18.108}$$

in the second-quantization language. Note that we have suppressed the spin label here. You can think of these as exclusively spin-up or exclusively spin down electron states. We can collect all of the many-body states into a many-body graph, denoted $\tilde{\Lambda}$, defined by the collection of all of the many-body states, as defined in Eq. (18.108), being the nodes, and the bonds being the connections generated by K. We index these states with the same N-tuple as we used before, $I = (i_1, i_2, \cdots, i_N)$, with all $i_m \in \Lambda$. We say that two many-body states are connected by a many-body bond on $\tilde{\Lambda}$ if the two graph states I and J satisfy

$$I = (i_1, i_2, \cdots, i, \cdots, i_N) \text{ and } J = (i_1, i_2, \cdots, j, \cdots, i_N) \tag{18.109}$$

where the two N-tuples are identical except for one element—I contains the element i, while J contains the element j with all other indices identical—and $t_{ij} \neq 0$. Heuristically, this means the many-body state J can be reached from the many-body state I by acting the kinetic-energy operator $\hat{K}$ onto it. This is essentially how we described how we compute the K matrix—we allow each creation operator in the state labeled by I to hop to all possible states that are able to be reached by a single hopping governed by the nonzero matrix elements of t_{ij} and where the electrons are located in the state I. Note that it is convenient when representing these states to not worry about ordering of the labels inside the N-tuples I and J, but instead thinking of the equivalence class of all possible orderings being the same state, because our canonical ordering scheme chooses the products of the creation operators to always be ordered according to their site index. We do have to be careful though that no two indices are the same, for then the state is annihilated.

We also need to define the one-dimensional projector $\hat{\Pi}^I = \hat{n}_{i_1}\hat{n}_{i_2}\cdots\hat{n}_{i_N}$ as the projector onto the many-body state, where electrons are placed at the N-lattice sites (in Λ) and labeled by the N-tuple I. It is easy to see that this is a Hermitian operator and that it is a one-dimensional projector (because $\hat{n}_i^2 = \hat{n}_i$). Note that because number operators commute with each other, the projector is identical for two states I and I' that are just a permutation of the respective indices. Indeed, this is what is meant when we say we focus on the equivalence classes of the projectors, with respect to the permutation operation. In essence, this is the same as saying that we should just work with the creation operators themselves in canonical order.

Lemma 1 (Many-body Connectedness of the Single-spin Configuration Space Lattice $\tilde{\Lambda}$) *If the Λ lattice is connected, such that any two points $i, j \in \Lambda$ can be connected by a path in Λ such that a bond connects each step along the path with a nonzero hopping matrix element, then the many-body lattice $\tilde{\Lambda}$ is also connected with respect to the kinetic energy operator, in the sense that for any $I, J \in \tilde{\Lambda}$, there exists a set of elements $I_m \in \tilde{\Lambda}$ with $I = I_1, I_2, \cdots, I_{n-1}, I_n = J$ such that*

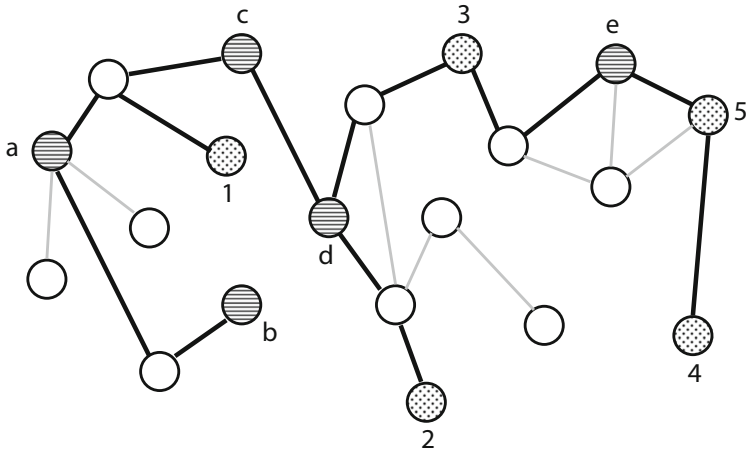

Fig. 18.4 Schematic for the proof of the connectivity of $\tilde{\Lambda}$. The figure illustrates an initial state I (circles with a dot pattern) and a final state J (circles with a horizontal line pattern), which are elements of $\tilde{\Lambda}$. Here, the circles that are filled with a pattern illustrate the initial state (with electrons on sites 1–5) and the final state (with electrons on sites a–e). The lines illustrate the bonds between the sites on the original lattice Λ. The solid black lines indicate the bonds in Λ over which we will be moving electrons from one site to another. Each intermediate state in this figure corresponds to a particular state $\tilde{I}$ on the many-body configuration lattice $\tilde{\Lambda}$. The lattice $\tilde{\Lambda}$ is not illustrated here

$I_m \neq I_{m'}$ for all $m \neq m'$ and the product of matrix elements, determined by the operator chain below satisfies

$$\hat{\Pi}^{I_n} \hat{K} \hat{\Pi}^{I_{n-1}} \cdots \hat{\Pi}^{I_2} \hat{K} \hat{\Pi}^{I_1} \neq 0. \tag{18.110}$$

The proof begins with solving a geometrical problem, which is visualized concretely in Fig. 18.4. We put unlabeled markers on the lattice Λ at the N locations of the initial configuration $I = (i_1, \cdots, i_N) \in \tilde{\Lambda}$. Our goal is to move those markers, via bonds in Λ (determined by nonzero hopping matrix elements connecting two lattice sites), from the sites corresponding to configuration I to the sites corresponding to the configuration J. This is illustrated in the figure as follows: (i) sites corresponding to I are the circles with a dotted pattern, with the labels indicated by numerals from 1 to 5; (ii) sites corresponding to J are the circles with a horizontal line pattern, with the labels indicated by letters from a to e; and (iii) solid dark lines indicate the bonds we will use to move the markers during our algorithm. We first describe how it works for the concrete figure and then describe the general case.

In our first step, we will move a marker from site 1 to site a, along the black-line bonds, to end in a state with one marker on a and four markers on 2–5. This first step is simple, as we have a direct path from 1 to a with no other markers along the path, so we make the move. Next, we want to move the marker from 2 to b. But here, the

marker on a blocks our ability to move the marker from 2 to b. So, we move the marker from 2 to the empty circle adjacent to a, then move the marker from a to b (which has a direct path), and then finally move the marker from the site adjacent to a to a. Now we have markers on a and b and 3–5. Moving the marker from 3 to c is direct, with no obstructions, so we do it. Moving from 4 to d is blocked on the first attempt. Instead, we move from 5 to d and then from 4 to 5. Finally, in the last step, we move from 5 to e. As you can see, when we move the markers from 1–5 to a–e, we can always succeed, but we do not always move the marker originally sitting on 1 to site a and similarly for the remaining markers—when there is blockage, we must move them in tandem—but this is always possible due to the connectivity of the original lattice Λ. This shows I and J are connected on $\tilde{\Lambda}$. Note that when we determine these states, they may need to be multiplied by a -1 depending on the number of reorderings of the creation operators. Of course, the projection operators have no such sign change, because number operators commute.

For the general case, we proceed in essentially the same way. Start by placing markers on the N sites given by $I = (i_1, i_2, \cdots, i_N)$. Next, find a path from the site i_1 to the site j_1. Such a path is guaranteed by the connectivity of Λ. Identify whether any other markers are on the path. If none, move the marker directly from i_1 to j_1. If there are markers along the path, then move the marker that has a direct path to j_1 from its lattice site to j_1. Then move the next marker down the path (marching backwards from j_1 towards i_1) from where it sits to where the marker originally was that we moved to j_1. Repeat this process for all remaining markers on the path. In this fashion, we have moved a marker from i_1 to j_1 and have left unchanged the markers on sites i_2 to i_N. Moving the marker from i_2 to j_2 proceeds in a similar fashion. Note that markers along the path could be on an I site or on a J site. It does not matter, the algorithm works for either scenario, as you can readily check.

Now, with the geometrical problem solved, we select the n I configurations $I_1, \cdots, I_n$ by choosing every configuration of the markers that were used in the rearrangement to move the markers from I to J. Because $\hat{\Pi}^{I_m}$ is a projection operator, so $\hat{\Pi}^{I_m} = \left(\hat{\Pi}^{I_m}\right)^2$, we can rewrite

$$\hat{\Pi}^{I_n}\hat{K}\hat{\Pi}^{I_{n-1}}\cdots\hat{\Pi}^{I_2}\hat{K}\hat{\Pi}^{I_1} = \left(\hat{\Pi}^{I_n}\hat{K}\hat{\Pi}^{I_{n-1}}\right)\left(\hat{\Pi}^{I_{n-1}}\hat{K}\hat{\Pi}^{I_{n-1}}\right)\cdots\left(\hat{\Pi}^{I_2}\hat{K}\hat{\Pi}^{I_1}\right). \tag{18.111}$$

We now introduce complete sets of states in between each projection operator and to the far left and to the far right. The states I_m and I_{m+1} in $\tilde{\Lambda}$ agree on all but two of their index values, which we label i and j (recall that we work with the equivalence class due to permutations of each I state). This means we have $I_m = (i_1, i_2, \cdots, i, \cdots, i_N)$ and $I_{m+1} = (i_1, i_2, \cdots, j, \cdots, i_N)$. Then the matrix element satisfies

$$\langle I_{m+1}|\hat{\Pi}^{I_{m+1}}\hat{K}\hat{\Pi}^{I_m}|I_m\rangle = \pm t_{ij}. \tag{18.112}$$

In this basis, only one state $\langle I_{m+1}|$ will yield a nonzero matrix element for a given $|I_m\rangle$, for the operator $\hat{\Pi}^{I_{m+1}}\hat{K}\hat{\Pi}^{I_m}$, and for the specific i and j. Then, because of the connectivity of the original lattice Λ, we have that the product of matrix elements is nonzero, as we claimed. Hence, the state I and J in $\tilde{\Lambda}$ are connected. We are now ready for the proof.

Theorem 2 (Uniqueness of the Hubbard Model When Λ is Connected and $U_i < 0\,\forall\, i \in \Lambda$) *The ground state of the Hubbard model when the interactions are strictly attractive and the lattice is connected is unique. Since we always have at least one ground state with $S = 0$, the ground state must have $S = 0$ as well.*

We sketch the uniqueness proof. The proof is by contradiction. Assume two solutions exist Ψ_1 and Ψ_2. Then the state $\Psi(\lambda) = \Psi_1 + \lambda\Psi_2$ is also a ground state for all real λ. For some value of λ, the matrix $\Psi(\lambda)$ is neither positive semidefinite nor negative semidefinite. Fix λ for this value, then since $|\Psi(\lambda)|$ is also a ground state, the two matrices $\Psi_\pm = \frac{1}{2}(|\Psi(\lambda)| \pm \Psi(\lambda))$ are also ground states—the state Ψ_+ is a positive semidefinite ground state. Ψ_+ satisfies the following Schrödinger equation:

$$\hat{K}\Psi_+ + \Psi_+\hat{K} - \sum_{i\in\Lambda} |U_x|\hat{L}_i\Psi_+\hat{L}_i = E\Psi_+. \tag{18.113}$$

Define H_+ as the range of Ψ_+ and $H_\perp$ as the orthogonal complement of H_+ in the Hilbert space. If there are two ground states, then both H_+ and $H_\perp$ are nontrivial subspaces of the Hilbert space. If we multiply Eq. (18.113) on the right and the left by $\hat{\Pi}_\perp$, the projector onto $H_\perp$, we find

$$\Psi_+\hat{L}_i\hat{\Pi}_\perp = 0. \tag{18.114}$$

This result follows because each $U_i \neq 0$, $\Psi_+\hat{\Pi}_\perp = \hat{\Pi}_\perp\Psi_+ = 0$, and Ψ_+ is a positive semidefinite matrix. This says that $\hat{L}_i$ maps H_+ to H_+. Keeping in mind Eq. (18.114), we multiply the Schrödinger equation in Eq. (18.113) on the right by $\hat{\Pi}_\perp$ and find that

$$\Psi_+\hat{K}\hat{\Pi}_\perp = 0, \tag{18.115}$$

as well. This means $\hat{K}$ also maps H_+ to H_+. Hence, the subspaces H_+ and $H_\perp$ are left invariant by $\hat{K}$ and $\hat{\Pi}_\perp$.

The sum of the projectors, over one element of each equivalence class, is the identity element on the Hilbert space, so there is some $\bar{I} \in \tilde{\Lambda}$ such that $\hat{\Pi}^{\bar{I}}\Psi_+ \neq 0$ because H_+ is nontrivial. Furthermore, for some element $\psi_\perp \in H_\perp$, there exists a $\bar{J} \in \tilde{\Lambda}$ such that $\hat{\Pi}^{\bar{J}}\psi_\perp \neq 0$. Now, choose the operator in Eq. (18.110) with $I = \bar{I}$ and $J = \bar{J}$. We must have the matrix element between $\langle\psi_\perp|$ and $|\Psi_+\rangle$ is nonzero. But this operator keeps the subspaces H_+ and $H_\perp$ invariant, because it is constructed

from $\hat{K}$ and $\hat{\Pi}^X$ operators. Hence, the matrix element cannot be nonzero. This is a contradiction. So, the ground state must be unique. Since it is unique, it must have $S = 0$.

We can relate these results for the attractive case to the repulsive case when we are also on a bipartite lattice and $U_i = U$ for all $i \in \Lambda$. The partial particle-hole transformation is a particle-hole transformation on the up-spin particle, leaving the down spin particles unchanged. Recall that, in this transformation, we have

$$\hat{c}_{i\uparrow} \to \hat{d}^\dagger_{i\uparrow}(-1)^{\Pi(i)} \text{ and } \hat{c}_{i\downarrow} \to \hat{d}_{i\downarrow}. \tag{18.116}$$

In this case, the number operators transform as $\hat{n}_{i\uparrow} \to (1 - n_{i\uparrow})$ and $\hat{n}_{i\downarrow} \to \hat{n}_{i\downarrow}$. Hence, the filling changes by $N_\uparrow \to |\Lambda| - N_\uparrow$ and $N_\downarrow \to N_\downarrow$. This means all even fillings are transformed to half-filling $N_\uparrow + N_\downarrow = |\Lambda|$. Furthermore, the interaction changes sign (plus a shift in energy given by $UN_\downarrow$, which is unimportant for our work), so the attractive model becomes the repulsive model (the hopping term remains in tact, if the lattice is bipartite). Because it is a unitary transformation, the energies remain the same, in particular, the repulsive Hubbard model, after the constant shift, has the same ordering of the energy levels as in the attractive model. But the spin operators transform to the pseudospin operators and the pseudospin operators transform to the spin operators. Key is the fact that the ground state is unique, so it has no level crossings. This means its spin quantum number, which is heretofore unknown, cannot change with U. But, in the large-U limit, the Hubbard model maps to a Heisenberg antiferromagnet; note that we just state this here as we have not shown it, but it is true, nevertheless. So, it must have the same spin quantum number as the antiferromagnet. But, this is known to be $\Big||\Lambda_A| - |\Lambda_B|\Big|/2$, as we saw when we worked with the Marshall sign rule in Sect. 15.4. So, in the case where the two sublattices of the bipartite lattice have different numbers of lattice sites, the system is ferrimagnetic (no long-range order was proven, just that the spin quantum number is in between the saturated ferromagnet and the spin singlet). Most bipartite lattices that people are familiar with have $|\Lambda_A| = |\Lambda_B|$. Lieb pointed out that by removing bonds from the face-centered-cubic lattice, one could create a bipartite lattice with one sublattice having three-times as many sites as the other sublattice. These lattices are special. Now bipartite lattices with different numbers of sites on each lattice are called *Lieb lattices*. They are characterized by having a macroscopic number of zero eigenvalues in the hopping matrix and are often called *flat band models*.

So, what does this mean? For any Hubbard model on a bipartite lattice, the ground state at half filling is unique for $0 < U < \infty$, and it has spin given by $S = \Big||\Lambda_A| - |\Lambda_B|\Big|$. It is pretty amazing that we can arrive at such a conclusion with the only required property being that the lattice is bipartite! We have a video https://tinyurl.com/58tbaaf3 summarizing these ideas.

Section 18.7 Problems:

18.7.1 The Matrix Ψ Is a Hermitian Matrix

In this problem, we will verify that the matrix Ψ can be chosen to be Hermitian. First, the Schrödinger equation for Ψ can be written as

$$K\Psi + \Psi K - \sum_{i\in\Lambda} |U_i| L_i \Psi L_i = E\Psi, \tag{18.117}$$

where K, L_i, and Ψ are all $d \times d$ matrices. K and L are real and symmetric matrices because the hopping is real. Introduce the summations corresponding to the matrix multiplications and show that if we replace $\Psi(I, J) \rightarrow \Psi(J, I)$, it corresponds to the exact same Schrödinger equation, and hence it must be a solution as well. Repeat for $\Psi(I, J)^*$ and $\Psi(J, I)^*$; you must use the reality of t_{ij} for this. All of these solutions have identical energies E.

Now, form the matrices $\Psi(I, J) + \Psi^*(J, I)$ and $i(\Psi(I, J) - \Psi^*(J, I))$. These matrices are also solutions with the same energy E. But, both matrices are manifestly Hermitian, and we cannot have both matrices vanish and also have a normalized Ψ. This means, we can assume Ψ is a Hermitian matrix without any loss of generality. For if it was not, we could construct a Hermitian matrix from it.

18.7.2 Relation Between the Hubbard Model and the Heisenberg Antiferromagnet

In Fig. 18.5, we illustrate the spins on two neighboring lattice sites in a ferromagnetic (parallel) and antiferromagnetic (antiparallel) arrangement when $U \rightarrow \infty$. Note that the ferromagnetic arrangement cannot have the electrons hop to the neighboring site, because the Pauli exclusion principle does not allow two like

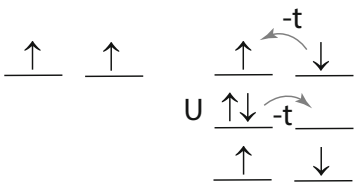

Fig. 18.5 Strong-coupling limit of the Hubbard model illustrated by (left) parallel and (right) antiparallel spins and their approximate energies. The left configuration cannot hop to the neighboring site due to the Pauli principle. The right one can, with an increase of energy given by U. The spins can hop back to the same or the opposite spin arrangement depending on which spin hops back

spins on the same site. But the antiparallel arrangement can have an electron virtually hop to a neighboring site and back. The matrix element for the hop is $\pm t_{ij}$, while the energy increases by U due to the double occupancy in the intermediate state. The energy of the antiparallel state is lowered by $|t_{ij}|^2/U$ relative to the parallel arrangement due to it having the possibility to have these quantum fluctuations. Note that this conclusion comes from applying second-order perturbation theory to the problem, which we have not covered in this book, so trust us that this is a correct statement. Since this interaction is different for the $S = 1$ arrangement versus the $S = 0$ arrangement, it can be described by a Heisenberg exchange interaction. Because the opposite spin arrangement is lower, it is an antiferromagnetic interaction.

We illustrate this behavior using our exact diagonalization solutions from the previous section. Consider the Hamiltonian block

$$\begin{pmatrix} U & -2\sqrt{2}t & -2\sqrt{2}t \\ -2\sqrt{2}t & U & 0 \\ -2\sqrt{2}t & 0 & 0 \end{pmatrix}. \tag{18.118}$$

Compute its characteristic equation used to find the three energy eigenvalues. Check that if you set $t = 0$, the three roots are 0, U, and U. Expand the exact characteristic equation for nonzero t in the case where U is large and show that the energy eigenvalue close to 0 will behave like $E \approx ct^2/U$. Find c. This shows that the corrections to the energies in the large-U limit are given by terms proportional to the Heisenberg exchange integral $J \approx 4t^2/U$, which supports our conclusion of the mapping to a Heisenberg antiferromagnet at half filling for large U. One can perform the analysis completely to derive the Heisenberg Hamiltonian by using a canonical transformation that removes double occupancies from the system, but this is beyond what we do here.

18.7.3 Concrete Discussion of Connectedness for a Four-Site Square Lattice

Consider the four-site square lattice, shown in Fig. 18.6, with bonds drawn by solid lines connecting the lattice sites. Here, we have $t_{ij} \neq 0$ for $i, j = 1, 2; 2,3; 3,4; 4,1$ and the same pairs in opposite order.

First show that for any i and j on the lattice that you can find a connected path using only bonds that create a connected path from i to j. This establishes that the lattice is connected.

Next, consider the many-body state constructed from the 36 tensor product states formed from the six pairs 12, 13, 14, 23, 24, and 34 used for each spin. We have the 36 states being $I \otimes J$ or $i_1 \uparrow i_2 \uparrow j_1 \downarrow j_2 \downarrow$, with the i_1, i_2 and j_1, j_2 pairs chosen from the above six choices. We want to show the graph containing these 36

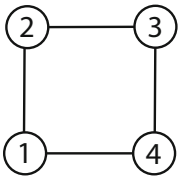

Fig. 18.6 Four-site square lattice with nearest-neighbor hopping. The solid lines indicate the bonds, which are sites connected by nonzero hopping. This is a bipartite lattice

nodes is also connected because the t_{ij} was connected. We will do this concretely. Fortunately we need to do this only for each spin, so we have only six states here. Construct a graph with six nodes and draw the connections between the states with lines, annotated with $\pm t_{ij}$, to illustrate the graph of $\tilde{\Lambda}$. Once the bonds have been drawn in, argue that $\tilde{\Lambda}$ is a connected graph. As an example, the state 12 is connected to the state 13 by the hopping matrix element $-t_{23}$.

18.7.4 Additional Details of the Uniqueness Proof

In this problem, we provide some additional details about the uniqueness proof. First, verify the statement about Eq. (18.112) that for any state I_m only one state I_{m+1} can have a nonzero matrix element for a specific i and j. Next, verify that Ψ_+ is a positive semidefinite matrix. Explain how L_i and K leave the H_+ invariant. Finally, write a short explanation how these results complete the proof and show that the ground state is unique.

18.8 Spin Tower and Minimal Pseudospin

Another interesting area where significant work has been done is in the description of spin and pseudospin towers. A spin tower says the lowest energy eigenvalue with total spin quantum number $S + 1$ is greater than the lowest energy eigenvalue with total spin quantum number S. Or, in other words, the ground state is 0; then there is an $S = 1$ excited state before an $S = 2$ and so on, all the way up to the ferromagnetic state. One way to think of how this occurs is for $U \to -\infty$. There, the ground state has all states doubly occupied and is a spin singlet. To get higher spin, we must break the double occupancy, and such a spin tower becomes clear. A spin tower was proven to exist in one dimension in the original work by Lieb and Mattis in the 1960s (Lieb & Mattis, 1962b). But, it can only be proven in one dimension and with finite-strength potentials. Lieb's original proof for the Hubbard model shows that the ground-state for the attractive model has $S = 0$ whenever the conditions for uniqueness hold. Shen extended Lieb's proof, via an extension of the

uniqueness proof to also show that $|\Psi|$ is positive definite, as opposed to positive semidefinite (Shen, 1996, 1998). This allows us to determine quantum numbers of the ground state if we can find a positive semidefinite state with a specific quantum number—such a state has nonzero overlap with the positive-definite state and so they must share the same eigenvalue. This approach can be used to show that the ground states of the attractive Hubbard model have minimal pseudospin for $N \leq 2\min\{|\Lambda_A|, |\Lambda_B|\}$; then the pseudospin is fixed at $\big||\Lambda_A| - |\Lambda_B|\big|/2$ until $N \geq 2\max\{|\Lambda_A|, |\Lambda_B|\}$, where it becomes minimal again. To do this, we construct the pseudospin states concretely, following Boretsky et al. (2018).

After establishing such a tower, we can perform a partial particle-hole transformation and then learn that the half-filled repulsive Hubbard model has a partial or a full spin tower—the tower forms for spin values that satisfy $S \geq \big||\Lambda_A| - |\Lambda_B|\big|/2$, because the ground state has $S = \big||\Lambda_A| - |\Lambda_B|\big|/2$. We do not learn about the relative ordering of lower spin values, except that they are above the ground state in energy. Of course, for a bipartite lattice with equally sized sublattices, we have a full spin tower.

What remains unproven is whether there exists a spin tower for the attractive Hubbard model (even though this case seems to be the most obvious one as $U \to -\infty$). If it can be proven, then it would imply the ground state of the Hubbard model with even numbers of electrons always has minimal pseudospin—a pseudospin tower. Perhaps one of you has just the right idea to establish this result.

To begin, we show *Shen's lemma*—if one wavefunction is positive definite and the other is positive semidefinite, then their overlap is nonzero. Let W be the positive definite wavefunction and W' the positive semidefinite wavefunction. Introduce the unitary matrix U which diagonalizes W via $W = U^\dagger D U$, where D is a diagonal matrix whose diagonal elements all satisfy $|w_\alpha| > 0$. Then, we can immediately compute the overlap

$$\mathrm{Tr} W'^\dagger W = \mathrm{Tr} W'^\dagger U^\dagger D U = \mathrm{Tr} U W'^\dagger U^\dagger D = \sum_{\alpha\beta\gamma} U_{\alpha\beta} W'_{\beta\gamma} U^*_{\alpha\gamma} |w_\alpha|. \tag{18.119}$$

But $\sum_{\beta\gamma} U_{\alpha\beta} W'_{\beta\gamma} U^*_{\alpha\gamma} \geq 0$ for every α because W' is positive semidefinite; to see this fix α, then this looks like we multiply by a vector from the right and its conjugate from the left, which must be larger than zero if the "vector" is nonzero. Not all can be zero or $W' = 0$, which is not possible if $\mathrm{Tr} W'^\dagger W' = 1$. Hence, the summation includes some nonzero terms and all nonzero terms are positive. Therefore, $\mathrm{Tr} W'^\dagger W > 0$.

A full pseudospin tower occurs for a fixed filling of electrons $N_e = 2N$, when the minimal energy state for a given pseudospin is ordered according to the pseudospin values from the lowest allowed pseudospin $J = (|\Lambda_B| + |\Lambda_A| - 2N)/2$ up to the maximal pseudospin $J = (|\Lambda_B| + |\Lambda_A|)/2$; that is, $E_{min}(J+1) > E_{min}(J)$ for all $J \geq (|\Lambda_B| + |\Lambda_A| - 2N)/2$. Here, the subscript min denotes the minimal energy state with the given value of pseudospin. A partial pseudospin tower has the

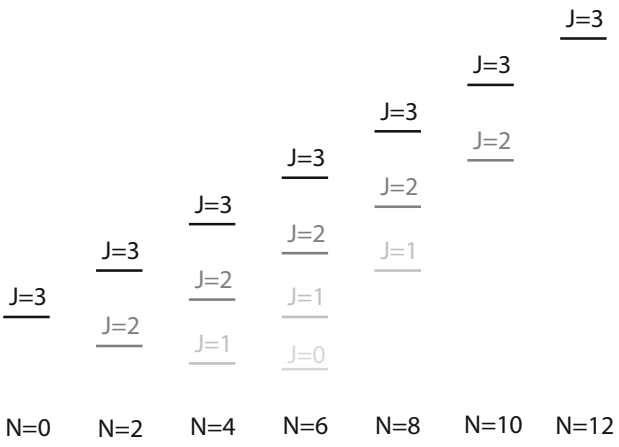

Fig. 18.7 Schematic of a pseudospin tower for the attractive Hubbard model for $|\Lambda| = 6$ and equal sized sublattices $|\Lambda_A| = |\Lambda_B| = 3$. We only show the lowest state with each pseudospin value

minimal energy ordering down to a given pseudospin J_0, and for pseudospins with lower quantum numbers, the minimal energy states are all larger than the energy of the minimal state with pseudospin J_0. These different cases are illustrated in Fig. 18.7. For the attractive Hubbard model, we have a full pseudospin tower for the case where $|\Lambda_A| = |\Lambda_B|$, while we have $J_0 = (|\Lambda_B| - |\Lambda_A|)/2$ for a Lieb lattice. Next, we develop how one can prove the existence of a pseudospin tower.

We use Shen's lemma to find the quantum number of the ground state of the attractive Hubbard model by finding any positive semidefinite state that has a definite value of the particular quantum number. Then since the overlap of the ground state with the positive semidefinite state is nonzero, it must share the same eigenvalue with the ground state. Shen originally did this by finding a specific pseudospin model that had positive semidefinite ground states, from which one can infer the overall quantum number of the ground state (the model was a pseudospin version of a Heisenberg model). We adopt the same methodology for determining the pseudospin of the ground state, but we instead construct the positive semidefinite state directly rather than inferring it from the ground state of a fictitious pseudospin Hamiltonian.

The states we will consider consist of linear combinations of product states that involve pairs of up and down spin electrons created on specific lattice sites. When expressed in the matrix form using our $|I\rangle \otimes |J\rangle$ basis, these states all are represented by diagonal matrices. If the diagonal elements are all nonnegative coefficients, then the state is a positive semidefinite state, which will have a nonzero overlap with the ground-state wavefunction.

We begin with the case $N = 0$ of no electrons. The zero-electron state has $J = |\Lambda_A| + |\Lambda_B| = |\Lambda|$. If we apply J^+ onto this state, then $J^+|0\rangle$ has coefficients that are negative and positive. The operator creates a linear combination of single paired

electrons on each site—those on the A sublattice are multiplied by -1 and those on the B sublattice are multiplied by $+1$. Hence this state is not positive semidefinite. We can form an (unnormalized) positive semidefinite state by choosing every paired state on the A sublattice to have the coefficient $|\Lambda_B|$ and every paired state on the B sublattice to have the coefficient $|\Lambda_A|$

$$|\psi_+(N_e=2)\rangle = |\Lambda_B| \sum_{i\in\Lambda_A} c^\dagger_{i\uparrow} c^\dagger_{i\downarrow}|0\rangle + |\Lambda_A| \sum_{j\in\Lambda_B} c^\dagger_{j\uparrow} c^\dagger_{j\downarrow}|0\rangle. \tag{18.120}$$

One can immediately show that $J^-|\psi_+(N=2)\rangle = 0$, so that $J = (|\Lambda_A|+|\Lambda_B|-1)$ for this state, and hence the $N_e = 2$ ground state has minimal pseudospin.

We can keep continuing in this fashion. The state with $N_e = 2N$ electrons is composed of N sites of paired electrons. We form the positive semidefinite state with the same coefficient for product states that have m_A pairs on the A sublattice and m_B pairs on the B sublattice ($N = m_A + m_B$), so they have the form

$$|\psi_+(N_e=2N)\rangle = \sum_{m_A=0}^{N} c_{m_A,N-m_A} \sum_{\substack{\text{all states with}\\ m_A \text{ pairs in } \Lambda_A \text{ and}\\ N-m_A \text{ pairs in } \Lambda_B}} |m_A; N-m_A\rangle, \tag{18.121}$$

with all $c_{m_A,N-m_A} \geq 0$; in other words, each state of the same type has the same coefficient in the linear combination, but different types have different coefficients. The coefficients are chosen such that $J^-|\psi_+(N_e=2N)\rangle = 0$, so that the state has $J = (|\Lambda_A| + |\Lambda_B| - N)$. Finding these coefficients is a simple counting exercise. For a given m_A, we have $|\Lambda_A|!/[m_A!(|\Lambda_A| - m_A)!] \times |\Lambda_B|!/[(N - m_A)!(|\Lambda_B| - N + m_A)!]$ different states. When J^- acts on this state, each term on the A sublattice will create m_A new terms of the form $-|m_A - 1; M - m_A\rangle$ and $N - m_A$ terms of the form $|m_A, N - m_A - 1\rangle$. We need the negative terms with the same numbers of pairs on each sublattice to cancel against the positive terms. The total number of negative terms is $m_A|\Lambda_A|!/[m_A!(|\Lambda_A| - m_A)!] \times |\Lambda_B|!/[(N - m_A)!(|\Lambda_B| - N + m_A)!]$. This is equal to $(|\Lambda_A| - m_A + 1)$ copies of the $|\Lambda_A|!/[(m_A - 1)!(|\Lambda_A| - m + 1)!] \times |\Lambda_B|!/[(N - m_A)!(|\Lambda_B| - N + m_A)!]$ terms in the $|m_A - 1; N - m_A\rangle$ sector. Since each individual term in the set of $|m_A - 1; N - m_A\rangle$ set of states is created from $|\Lambda_A| - m_a + 1$ possible "father" states when a pair is removed from the A sublattice and $|\Lambda_B| - N + m_A$ "father" states when a pair is removed from the B sublattice, all terms appear the same number of times. This implies

$$-(|\Lambda_A| - m_A + 1)c_{m_A,N-m_A} + (|\Lambda_B| - N + m_A)c_{m_A-1,N-m_A+1} = 0, \tag{18.122}$$

for $1 \leq m_A \leq N$ when $N < |\Lambda_A|$. Start with $c_{0,N}$. Then $c_{1,N-1} = (|\Lambda_B| - N + 1)c_{0,N}/|\Lambda_A|$, $c_{2,N-2} = (|\Lambda_B| - N + 1)(|\Lambda_B| - N + 2)c_{0,N}/[|\Lambda_A|(|\Lambda_A| - 1)]$, $\cdots$,

$$c_{m,N-m} = \frac{(|\Lambda_B| - N + m)!}{(|\Lambda_B| - N)!} \frac{(|\Lambda_A| - m)!}{|\Lambda_A|!} c_{0,N}. \tag{18.123}$$

As long as $N < |\Lambda_A|$, then this state $|\psi_+(N_e = 2N)\rangle$ does not remain positive semidefinite when J^+ is applied to it, because all of the states with $N + 1$ pairs on the A sublattice and none on the B sublattice have negative coefficients coming from the terms in J^+ which create pairs on the A sublattice and acted on the states of the form $|N; 0\rangle$. Hence, as long as $N < |\Lambda_A|$, the ground state has minimal pseudospin.

When $N = |\Lambda_A|$, we cannot add any new pairs to the A sublattice from the $||\Lambda_A| = N, 0\rangle$ state, so it may be possible now that applying J^+ to this state can lead to a positive semidefinite state. In fact, by restricting the coefficients in Eq. (18.123) to run only over $1 \leq m_A \leq |\Lambda_A|$ for $|\Lambda_A| \leq N \leq |\Lambda_B|$, we find that the J^+ operator can be applied $|\Lambda_B| - |\Lambda_A|$ times and the coefficients remain all positive; they are multiplied by $|\Lambda_B| - N$ for $|\Lambda_A| \leq N \leq |\Lambda_B|$. So the states with $2|\Lambda_A| \leq |\Lambda| \leq 2|\Lambda_B|$ all have $J = |\Lambda_B| - |\Lambda_A|$ and the ground-state pseudospin is no longer minimal.

Finally, as N runs from $|\Lambda_B|$ to $|\Lambda_A| + |\Lambda_B|$, the ground state is minimal pseudospin again. This can be seen because the positive-definite state with a definite J quantum number no longer remains positive definite when J^+ is applied to it. This occurs because the state in the sector $|N - |\Lambda_B|; |\Lambda_B|\rangle$ has negative coefficients for all the $|N - |\Lambda_B| + 1; |\Lambda_B|\rangle$ states, and since they all come from just the $|N - |\Lambda_B|; |\Lambda_B|\rangle$ sector, the states are not positive semidefinite. This implies that any positive semidefinite state with a definite J value cannot be raised and still remain positive semidefinite. Hence the positive semidefinite state must be annihilated when J^+ is applied, implying the pseudospin is minimal.

This establishes the proof about the pseudospin quantum number for the ground state of the attractive Hubbard model. This proof employed a constructive method which created positive-semidefinite states with definite J which have nonzero overlap with the ground state and hence share the same quantum number.

All of the work we have done so far was for the attractive case with $U < 0$. The repulsive case can be connected to the attractive one via a partial particle-hole transformation. Then we find that the spin operators transform to the pseudospin operators and vice versa. The up spin filling is transformed from N electrons to $|\Lambda| - N$ electrons. Finally, the interaction is changed via $U \to -U$.

Let's examine what happens to the $m_S = 0$ states with $N_e = 2N$ electrons (N up spin and N down spin). The electron filling changes to $|\Lambda| - N$ up spins and N down spins, so the total filling is $N_e = |\Lambda|$, which is what we call half-filling. The z component of the spin is now $|\Lambda|/2 - N$, so as N runs from 0 to $|\Lambda|$, m_S runs from $-|\Lambda|/2$ to $|\Lambda|/2$.

If we plot the energy levels for half-filling as a function of m_S, a full spin tower implies that $E(|m_S|) > E(|m'_S|)$ whenever $|m_S| > |m'_S|$. A partial spin tower is a spin tower for all $|m_S| > S_0$ and then $E(|m_S|) > E(S_0)$ for $|m_S| < S_0$. This implies that the ground state has spin S_0 and a spin tower exists for spins larger than S_0. It should be clear that we can now prove that on a bipartite lattice with $|\Lambda_A| = |\Lambda_B|$, we have a full spin tower, and the ground state is a spin singlet at half-filling for the repulsive case when $U < \infty$. This result generalizes the Lieb-Mattis tower, which was proved for attractive and repulsive interactions for one-dimensional systems. In addition, we can now establish that $S_0 = (|\Lambda_B| - |\Lambda_A|)/2$ for Lieb lattices. The situation is shown schematically in Fig. 18.8.

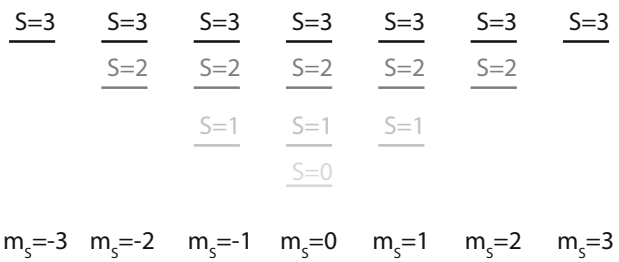

Fig. 18.8 Schematic of a spin tower at half-filling for the repulsive Hubbard model for $|\Lambda| = 6$ on a lattice with $|\Lambda_A| = |\Lambda_B|$ The energy levels are plotted on the vertical axis and the m_S value of the electron spin on the horizontal. Only the lowest energy state in each sector is plotted

The proof is rather straightforward. We first need to relate the energies of the attractive and repulsive cases. In the current Hamiltonian, we have the interaction energy is $U \sum_{i\in\Lambda} c^\dagger_{i\uparrow} c_{i\uparrow} c^\dagger_{i\downarrow} c_{i\downarrow}$, which clearly gives a different result if we have zero, one, or two electrons at a site. But if we instead write it in the particle-hole symmetric form as $U \sum_{i\in\Lambda} (c^\dagger_{i\uparrow} c_{i\uparrow} - \frac{1}{2})(c^\dagger_{i\downarrow} c_{i\downarrow} - \frac{1}{2})$, then the interaction is the same when there are zero or two electrons on a site, while the case with one electron on a site gives the negative value. Since the partial particle-hole transformation changes the up filling from one to zero as well as changing the sign of U, it actually does not change the potential energy when written in this form. This means the attractive and repulsive cases have the same potential energy. The kinetic energy is also unchanged for the half-filled case, because the kinetic energy for the down spins are unchanged in the transformation. For the up spins, the filling goes from N to $|\Lambda| - N$, but because of the bipartite nature of the hopping, the kinetic energy with N particles and with N holes (or $|\Lambda| - N$ particles) is identical. Hence the energies are the same for the modified interaction after the particle-hole transformation.

Now the conversion from the original interaction and the particle-hole symmetric one requires us to add $-U \sum_{i\in\Lambda} (c^\dagger_{i\uparrow} c_{i\uparrow} + c^\dagger_{i\downarrow} c_{i\downarrow}) + U|\Lambda|/4$. All of our previous energy eigenstates are eigenstates of this additional operator as well, so the shift of the eigenvalues can be immediately computed. More importantly, the shift has no effect on the pseudospin eigenvalues or the spin eigenvalues.

So, at this stage we know the following: (1) the $m_S = 0$ attractive eigenstates map to $m_J = 0$ repulsive eigenstates corresponding to half-filling; (2) for a given m_J value for the attractive case, we know the J value of the minimal energy eigenstate, so for the repulsive case at half-filling we know the S eigenvalue of the minimal energy state for fixed m_S; and (3) the energy eigenvalues for the repulsive and attractive cases are the same.

Now focus on the repulsive case at half-filling. We have that $S_0 = (|\Lambda_B| - |\Lambda_A|)/2$. If $m_S > S_0$, then we know the minimal energy states for $|m_S| > |m'_S|$ must satisfy $E(|m_S|) > E(|m'_S|)$. This follows because in the attractive case, when we apply J^+ to the minimal energy state with a given $m_J < 0$ value, the energy changes by $U - U = 0$ for the modified potential. But the minimal energy state with $m_J + 1$ must be lower in energy. Performing the partial particle-hole transformation

then yields the desired result. If $|m_S| < S_0$, then $E(|m_S|) > E(S_0)$ follows because the minimal energy state is a pseudospin multiplet for the attractive case. This then establishes the (partial) spin tower. Since we are at half-filling, the term we added to the Hamiltonian is just a constant, so we can remove it and go back to the original Hamiltonian, and the result remains the same—the repulsive Hubbard model has the same spin tower at half-filling.

Note that this shows that the ground state is ferrimagnetic (with $S = ||\Lambda_B| - |\Lambda_A||/2$) on a Lieb lattice. In particular, it agrees with the known spin of the ground state of an antiferromagnetic Heisenberg model, to which the repulsive Hubbard model maps when U is positive and large in magnitude, as we discussed earlier.

What remains to be proven is that the attractive case has a spin tower for all even fillings. If true, then the particle-hole transformation would show that the pseudospin quantum number of the repulsive model with even numbers of electrons has minimal pseudospin. But, it is difficult to use any of the similar strategies that worked with the above proofs for these cases because $N_\uparrow \neq N_\downarrow$ implies that the matrix Ψ is not square. Furthermore, because the up spin and down spin basis functions are different, we can no longer show that Ψ is Hermitian. The situation of nonzero m_S and half-filling, is, however, unique in that this case does have square matrices. Furthermore, if one can establish a spin tower here, one can establish it at all other fillings too by using pseudospin raising and lowering operators. But one cannot show that Ψ is Hermitian anymore, which makes the general approach to solving the problem difficult (i.e., spin-reflection positivity cannot be easily employed in this case).

Perhaps one of you will figure out how to do this!

Section 18.8 Problems:

18.8.1 *Half-Filling for the* $|\Lambda| = 4$ *Square Lattice Using the Lieb Formalism*

In this problem, we will investigate further the four-site Hubbard model on a square at half-filling by using the Lieb form of the matrix representation for the different energy eigenstates. We will focus on the ground state and will determine it in a *strong-coupling expansion* in the limit of large U. I hope that you find working concretely with these matrices helps you understand better how the approach works. (If you have some experience in quantum mechanics outside of this book, you may realize that we are trying to solve a degenerate perturbation theory problem here—the simplification comes from the fact that Lieb's theorem tells us the direction in the degenerate space that the lowest energy state lies, so we do not need to actually do an expansion in a degenerate space.)

While we did not solve this problem when we looked at exact diagonalization, one can do so. After performing the same symmetry analysis, looking at all energy levels and so forth, we find that the ground state is constructed from the following states:

$$
\begin{aligned}
&|1\rangle_{N=4\ S=0} \\
&= \frac{1}{2\sqrt{3}}\Big(1\uparrow 2\uparrow 1\downarrow 2\downarrow + 1\uparrow 4\uparrow 1\downarrow 4\downarrow + 2\uparrow 3\uparrow 2\downarrow 3\downarrow + 3\uparrow 4\uparrow 3\downarrow 4\downarrow \\
&+ (2)1\uparrow 3\uparrow 1\downarrow 3\downarrow) + (2)2\uparrow 4\uparrow 2\downarrow 4\downarrow \Big) \qquad (18.124)
\end{aligned}
$$

$$
\begin{aligned}
&|2\rangle_{N=4\ S=0} \\
&= \frac{1}{4}\Big(1\uparrow 2\uparrow 1\downarrow 3\downarrow - 1\uparrow 2\uparrow 2\downarrow 4\downarrow + 1\uparrow 3\uparrow 1\downarrow 2\downarrow + 1\uparrow 3\uparrow 1\downarrow 4\downarrow \\
&+ 1\uparrow 3\uparrow 2\downarrow 3\downarrow - 1\uparrow 3\uparrow 3\downarrow 4\downarrow + 1\uparrow 4\uparrow 1\downarrow 3\downarrow + 1\uparrow 4\uparrow 2\downarrow 4\downarrow \\
&+ 2\uparrow 3\uparrow 1\downarrow 3\downarrow + 2\uparrow 3\uparrow 2\downarrow 4\downarrow - 2\uparrow 4\uparrow 1\downarrow 2\downarrow + 2\uparrow 4\uparrow 1\downarrow 4\downarrow \\
&+ 2\uparrow 4\uparrow 2\downarrow 3\downarrow + 2\uparrow 4\uparrow 3\downarrow 4\downarrow - 3\uparrow 4\uparrow 1\downarrow 3\downarrow + 3\uparrow 4\uparrow 2\downarrow 4\downarrow \Big) \qquad (18.125)
\end{aligned}
$$

$$
\begin{aligned}
&|3\rangle_{N=4\ S=0} \\
&= \frac{1}{2}\Big(1\uparrow 2\uparrow 3\downarrow 4\downarrow - 1\uparrow 4\uparrow 2\downarrow 3\downarrow - 2\uparrow 3\uparrow 1\downarrow 4\downarrow + 3\uparrow 4\uparrow 1\downarrow 2\downarrow \Big). \qquad (18.126)
\end{aligned}
$$

In this basis, the Hamiltonian matrix block is given by

$$
H = \begin{pmatrix} 2U & -2\sqrt{3}t & 0 \\ -2\sqrt{3}t & U & 2t \\ 0 & 2t & 0 \end{pmatrix}. \qquad (18.127)
$$

The secular equation for the energy eigenvalues is then the cubic equation

$$
E^3 - 3UE^2 + 2(U^2 - 8t^2)E + 8t^2U = 0. \qquad (18.128)
$$

The minimum energy solution for the attractive model is $E = 2U = -2|U|$ for $|U| \to \infty$. The correction to the energy is found by substituting $E = 2U + \delta$ into the equation and solving for δ, assuming t is small. This gives

$$
\delta = 12\frac{t^2}{U} - \frac{3\delta^2}{4U} + \frac{8t^2\delta}{U^2} - \frac{\delta^3}{2U^2}. \qquad (18.129)
$$

Expanding the solution for small δ gives us

$$
E = -2|U| - 12\frac{t^2}{|U|} + 120\frac{t^4}{|U|^3} + \cdots. \qquad (18.130)
$$

Your first question is to verify the expansion of the energy.

Note that there are 36 localized product states basis vectors for this problem, but only 26 appear in the ground-state expansion. This explicitly shows us that some of the coefficients vanish in the Lieb theorem, so it is not the same type of proof as the Perron-Frobenius theorem, where no coefficient would be zero.

The six up spin and down spin states were worked out in Problem 18.7.3, and we use that work here. Show that the hopping matrix K and the number operator matrices L_i satisfy

$$K = \begin{pmatrix} 0 & -t & 0 & 0 & t & 0 \\ -t & 0 & -t & -t & 0 & t \\ 0 & -t & 0 & 0 & -t & 0 \\ 0 & -t & 0 & 0 & -t & 0 \\ t & 0 & -t & -t & 0 & -t \\ 0 & t & 0 & 0 & -t & 0 \end{pmatrix}, \tag{18.131}$$

$$L_1 = \begin{pmatrix} 1&0&0&0&0&0 \\ 0&1&0&0&0&0 \\ 0&0&1&0&0&0 \\ 0&0&0&0&0&0 \\ 0&0&0&0&0&0 \\ 0&0&0&0&0&0 \end{pmatrix}, \quad L_2 = \begin{pmatrix} 1&0&0&0&0&0 \\ 0&0&0&0&0&0 \\ 0&0&0&0&0&0 \\ 0&0&0&1&0&0 \\ 0&0&0&0&1&0 \\ 0&0&0&0&0&0 \end{pmatrix}, \quad L_3 = \begin{pmatrix} 0&0&0&0&0&0 \\ 0&1&0&0&0&0 \\ 0&0&0&0&0&0 \\ 0&0&0&1&0&0 \\ 0&0&0&0&0&0 \\ 0&0&0&0&0&1 \end{pmatrix},$$

$$L_4 = \begin{pmatrix} 0&0&0&0&0&0 \\ 0&0&0&0&0&0 \\ 0&0&1&0&0&0 \\ 0&0&0&0&0&0 \\ 0&0&0&0&1&0 \\ 0&0&0&0&0&1 \end{pmatrix}. \tag{18.132}$$

You can immediately see that $\sum_{x\in\Lambda} L_x = 2\mathbb{I}$, as it must, because it counts the total number of electrons of one spin. It is also useful to calculate powers of K. In particular, show that

$$K^2 = 2t^2 \begin{pmatrix} 1&0&0&0&0&-1 \\ 0&2&0&0&0&0 \\ 0&0&1&1&0&0 \\ 0&0&1&1&0&0 \\ 0&0&0&0&2&0 \\ -1&0&0&0&0&1 \end{pmatrix} \tag{18.133}$$

and $K^3 = 4t^2 K$.

In this problem, we will solve the Schrödinger equation, as written by Lieb in his proof, using an expansion about the $U \to -\infty$ limit solution for the four-site square lattice; the order of the expansion denotes the powers of t. Recall that the combination of Lieb's proof and Shen's lemma says that the ground state Ψ is a positive definite matrix for all U. In the limit as $U \to -\infty$, the ground state includes only states that have maximal number of double occupancies. These states have coefficients only on the diagonal (in the localized basis), and all of them are nonzero and positive. We start with the solution in this limit. At zeroth order (neglecting the hopping terms), we have

$$-|U| \sum_{i \in \Lambda} L_i \psi_0 L_i = E_0 \psi_0. \tag{18.134}$$

We take the trace of this equation to find

$$E_0 \mathrm{Tr}\psi_0 = -|U| \sum_{i \in \Lambda} \mathrm{Tr} L_i \psi_0 L_i = -|U| \sum_{i \in \Lambda} \mathrm{Tr} L_i \psi_0 = -2|U| \mathrm{Tr}\psi_0, \tag{18.135}$$

because $L_i^2 = L_i$ and $\sum_{i \in \Lambda} L_i$ is proportional to the identity matrix and counts the total number of electrons of the same spin, or $\sum_{i \in \Lambda} L_i \psi_0 = N_\uparrow \mathbb{I} \psi_0$ and $\sum_{i \in \Lambda} \psi_0 L_i = \psi_0 N_\downarrow \mathbb{I}$. Since $N_\uparrow = N_\downarrow$ for all cases we consider, we can use either one of them in the summarizing formula. Because ψ_0 is full rank and positive definite, $\mathrm{Tr}\psi_0 \neq 0$, so we learn that

$$E_0 = -2|U|. \tag{18.136}$$

Note that at zeroth order, we do not yet have enough information to determine ψ_0. It will actually be found at second order.

At first order we have

$$K\psi_0 + \psi_0 K - |U| \sum_{i \in \Lambda} L_i \psi_1 L_i = E_0 \psi_1 + E_1 \psi_0. \tag{18.137}$$

Again, we take the trace of the equation. Noting that the K matrices are off-diagonal only and ψ_0 is diagonal, we have that $\mathrm{Tr} K\psi_0 = \mathrm{Tr}\psi_0 K = 0$. Furthermore, $\sum_i \mathrm{Tr} L_i \psi_1 L_i = 2\mathrm{Tr}\psi_1$. So, we learn that $E_1 = 0$. Next, we need another operator identity. For a general matrix representation of the quantum state ψ, we have that

$$\begin{aligned} N_\sigma \psi - \sum_{i \in \Lambda} L_i \psi L_i &= \frac{1}{2} \sum_{i \in \Lambda} (L_i \psi + \psi L_i - 2 L_i \psi L_i) \\ &= \frac{1}{2} \sum_{i \in \Lambda} (L_i \psi (1 - L_i) + (1 - L_i) \psi L_i) \\ &= \frac{1}{2} \sum_{i \in \Lambda} \left(\psi - \underbrace{L_i \psi L_i}_{\text{double occupancies}} - \underbrace{(1 - L_i) \psi (1 - L_i)}_{\text{empty sites}} \right). \end{aligned} \tag{18.138}$$

In the last line, we see we subtract the number of double occupancies plus the number of empty sites. On the diagonal, the number of double occupancies is given by N_σ and of empty sites $|\Lambda| - N_\sigma$. Hence, the matrix in the last line above always has a vanishing diagonal! Using this identity, we can determine the off-diagonal contribution to ψ_1. It is given by

$$(\psi_1)_{\text{off-diag}} = -\frac{K\psi_0 + \psi_0 K}{|U|}. \tag{18.139}$$

We do not expect diagonal contributions to ψ_1, but we must go to higher order in the perturbation theory to determine whether there are such contributions.

We now move to second order. The Schrödinger equation says

$$K\psi_1 + \psi_1 K - |U| \sum_{i\in\Lambda} L_i \psi_2 L_i = E_0\psi_2 + E_1\psi_1 + E_2\psi_0. \tag{18.140}$$

Recalling that $E_1 = 0$ and assuming the diagonal contribution to ψ_1 vanishes, we have

$$-\frac{K^2\psi_0 + 2K\psi_0 K + \psi_0 K^2}{|U|} - |U| \sum_{i\in\Lambda} L_i \psi_2 L_i = 2|U|\psi_2 + E_2\psi_0. \tag{18.141}$$

Now, we take the trace and find the result that

$$E_2 = -\frac{4\text{Tr}K^2\psi_0}{|U|\text{Tr}\psi_0}, \tag{18.142}$$

which holds, even if ψ_1 has some nonvanishing diagonal part. Next, focusing on the diagonal part of the Schrödinger equation at this order, we find that

$$-\frac{\text{diag}(K^2\psi_0 + 2K\psi_0 K + \psi_0 K^2)}{|U|} = E_2\,\text{diag}(\psi_0), \tag{18.143}$$

where "diag" denotes the diagonal part of the matrix. This is an equation that is similar to an eigenvalue-eigenvector equation for ψ_0, which we still have not yet determined. The equation can be rewritten as

$$\frac{\text{diag}(\psi_0)}{\text{Tr}\psi_0} = -\frac{\text{diag}(K^2\psi_0 + 2K\psi_0 K + \psi_0 K^2)}{4\text{Tr}K^2\psi_0}. \tag{18.144}$$

You will directly solve this for the square below.

At second order, we have three different types of states that enter into the perturbation theory. At first-order in the hopping, all we can do is hop one electron from a doubly occupied site to an empty site. This will leads to our having $N_\sigma - 1$ doubly occupied sites, two singly occupied sites, and $N_\sigma - 1$ empty sites. Adding one more hopping leaves us with three different possibilities: (i) we hop the singly occupied electron back (or hop the opposite spin electron to where the first electron hopped) and have N_σ doubly occupied sites and N_σ empty sites; (ii) we can hop

the single occupied site to another empty site and continue to have $N_\sigma - 1$ doubly occupied sites, two singly occupied sites, and $N_\sigma - 1$ empty sites; or (iii) we can hop another electron from a doubly occupied site to an empty site giving us $N_\sigma - 2$ doubly occupied sites, four singly occupied sites, and $N_\sigma - 2$ empty sites. In this analysis, we are assuming the lattice is large enough in size to accommodate all of these different states. The processes in category (i) involve the diagonal terms in ψ, but those terms are projected out of the Schrödinger equation at this order, and hence they must enter at higher order. Processes (ii) and (iii) correspond to off-diagonal terms. The projection in Eq. (18.138) gives a factor of 1 for process (ii) and a factor of 2 for process (iii). This then gives us

$$(\psi_2)_{\text{off diag}} = \frac{2(K^2\psi_0 + 2K\psi_0 K + \psi_0 K^2)_{\text{off diag}}}{|U|^2\langle n_{\text{sing}}\rangle}, \tag{18.145}$$

where $\langle n_{\text{sing}}\rangle$ counts the number of singly occupied sites in the states in the off-diagonal elements (it is equal to 2 for process (ii) and 4 for process (iii)). It is the presence of this normalization factor $\langle n_{\text{sing}}\rangle$ that makes writing down general formulas a bit complicated. It turns out that we can fix the normalization such that $\text{Tr}(\psi_0\psi_k) = 0$ and then the diagonal of all ψ_n vanishes for all $n \geq 1$. We do not show the details here. Instead, we discuss an alternative approach to show how to find that they vanish. But for the calculations of the energy below, you can assume the diagonals vanish.

At third order, we have

$$K\psi_2 + \psi_2 K - |U|\sum_{x\in\Lambda} L_x\psi_3 L_x = E_0\psi_3 + E_2\psi_1 + E_3\psi_0. \tag{18.146}$$

again, we take the trace. Assuming ψ_1 has zeros on the diagonal, which we verify below, we have $2\text{Tr}K\psi_2 = E_3\text{Tr}\psi_0$. Using the form for ψ_2 and recognizing that the diagonal piece does not contribute, then gives $E_3 \propto \text{Tr}K^3\psi_0 = 0$, which vanishes because K^3 has no diagonal elements, so there is no third-order correction to the energy. For the eigenstate, we have

$$(\psi_3)_{\text{off diag}} = -\frac{2(K^3\psi_0 + 3K^2\psi_0 K + 3K\psi_0 K^2 + \psi_0 K^3 - E_2\psi_1)}{|U|^3\langle n_{\text{sing}}\rangle}. \tag{18.147}$$

Here, if we look at the diagonal component of the eigenvalue equation, because $K\psi_2 + \psi_2 K$ has no diagonal component, we now confirm that $(\psi_1)_{\text{diag}} = 0$.

At fourth order, we have that

$$K\psi_3 + \psi_3 K - |U|\sum_{x\in\Lambda} L_x\psi_4 L_x = E_0\psi_4 + E_2\psi_2 + E_4\psi_0. \tag{18.148}$$

Here we find the following results: first, taking the trace gives

$$E_4 = \frac{2\text{Tr}K\psi_3 - E_2\text{Tr}\psi_2}{\text{Tr}\psi_0}. \tag{18.149}$$

The diagonal component is

$$\operatorname{diag}(K\psi_3 + \psi_3 K) = E_2 \operatorname{diag}(\psi_2) + E_4 \operatorname{diag}(\psi_0). \tag{18.150}$$

Again, this requires a complicated nonlinear equation to solve to determine both E_4 and the diagonal part of ψ_2. The off-diagonal component of ψ_4 is then

$$(\psi_4)_{\text{off diag}} = (K\psi_3 + \psi_3 K)_{\text{off diag}} - E_2(\psi_2)_{\text{off diag}}. \tag{18.151}$$

OK. That is a lot of equations. Now, we apply it to the square.
Show that we obtain

$$\psi_0 = \frac{1}{2\sqrt{3}} \begin{pmatrix} 1&0&0&0&0&0\\ 0&2&0&0&0&0\\ 0&0&1&0&0&0\\ 0&0&0&1&0&0\\ 0&0&0&0&2&0\\ 0&0&0&0&0&1 \end{pmatrix}, \quad \psi_1 = \frac{\sqrt{3}t}{2|U|} \begin{pmatrix} 0&1&0&0&-1&0\\ 1&0&1&1&0&-1\\ 0&1&0&0&1&0\\ 0&1&0&0&1&0\\ -1&0&1&1&0&1\\ 0&-1&0&0&1&0 \end{pmatrix}, \tag{18.152}$$

$$\psi_2 = \frac{\sqrt{3}t^2}{|U|^2} \begin{pmatrix} 0&0&0&0&0&-1\\ 0&0&0&0&0&0\\ 0&0&0&1&0&0\\ 0&0&1&0&0&0\\ 0&0&0&0&0&0\\ -1&0&0&0&0&0 \end{pmatrix}, \quad \text{and} \quad \psi_3 = -\frac{8\sqrt{3}t^3}{2|U|^3} \begin{pmatrix} 0&1&0&0&-1&0\\ 1&0&1&1&0&-1\\ 0&1&0&0&1&0\\ 0&1&0&0&1&0\\ -1&0&1&1&0&1\\ 0&-1&0&0&1&0 \end{pmatrix}$$

$$\psi_3 = -\frac{16t^2}{|U|^2}(\psi_1)_{\text{off diag}}, \tag{18.153}$$

where the question marks denote terms we have not found yet. Verify that the energy comes out correct through fourth order as well.

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Chapter 19
Applications to Chemistry

The one who finds something no matter what it might be, even if his intention were not to search for it, at least arouses our curiosity, if not our admiration.

Pablo Picasso, Artist, 1923

Quantum mechanics plays a huge role in the field of chemistry. The chemical bond is quantum mechanical in origin, reflecting how electrons rearrange themselves when atoms are brought close together to form molecules. The electron bonds are "squishy" in the sense that we can compress or pull the molecules. This leads to molecular vibrations, which will be our first topic in this chapter. The molecules also have a three-dimensional character to them, so they can be rotated about different axes, giving rise to rotational spectroscopy. Next, we discuss how two hydrogen atoms bond together to form a H_2 molecule, the simplest of all molecules. We will describe this problem in a simplified fashion concretely, and then in the following section, we will discuss more thoroughly how the Hartree-Fock solution works to describe the "electron glue" that holds molecules together. Then we discuss more accurate approaches that go beyond mean field, including the coupled-cluster approach, the unitary coupled-cluster approach, and how it is used in quantum computation, and finally we end with a section discussing how coupled cluster and unitary coupled cluster inter-relate. You will see that many of our fundamental operator identities play a large role in working with quantum chemistry. Quantum chemistry is a remarkable field, where very clever chemists have found incredibly efficient ways for determining ground states, including finding a ground-state energy eigenstate without ever actually constructing it! There is some amazing material in this chapter. Our focus is on describing how a number of different approaches work, not the full details to carry out calculations yourself, nor the qualitative features of bonding and hybridization, which are covered elsewhere. See the full story of Chap. 19 at https://tinyurl.com/27ws5hde.

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19.1 Morse Potential and Vibrational Spectra

Molecules are formed when the outer electrons of atoms interact with each other and find a stable arrangement (with lower energy than the isolated atoms). The arrangement is often called a *chemical bond*, and it can involve a sharing of electrons (covalent bond) or a transfer of electrons (an ionic bond). We will discuss this phenomenon of bonding and electronic structure later in this chapter. Here, we wish to focus on the vibrations that a molecule exhibits because these bonds are not rigid, but are "squishy." We think of the atoms and the bonds in the molecule loosely as balls and springs. We will later see how we can determine the potential surface of the "spring," but for now we simply assume that such a structure exists and we wish to analyze its behavior.

The first thing we want to do is to estimate the energy of a molecular vibration. Since, if we push atoms too close together, they eventually will have their nuclei overlap, we know the potential must be strongly repulsive at short distances (because the atomic cores are positively charged and strongly repel each other). But, as we pull the atoms farther and farther apart, eventually it will be more favorable for the electrons to stick to the individual atoms than to stay as the glue between them in the collective molecule. So, at large distances, the potential approaches zero. A measure of the depth of the potential is the dissociation energy—the electronic energy needed to break the molecule into pieces (and called D_e). This energy is typically on the order of a few eV (let's say about 1 eV to be concrete). The bond length in equilibrium is on the order of a few angstroms (let's say 1 Å to be concrete). This should give us approximately a spring constant of 1 eV/Å^2. The frequency is then

$$\omega_{\text{vib}} \approx \sqrt{\frac{k}{M}} = \sqrt{\frac{1\text{ eV } c^2}{Mc^2\text{Å}^2}} = \sqrt{\frac{1\text{ eV } (3\times 10^{18}\text{Å/s})^2}{10\times 10^9\text{ eV Å}^2}} \approx 3\times 10^{13}\text{s}^{-1}. \tag{19.1}$$

This is about 100 times less energy than optical light. The estimate turns out to be a little low. We usually report the frequencies in inverse cm for molecular spectroscopy and 8071 cm^{-1} is about 1 eV, so this estimate is about 150 cm^{-1}, and the true spectra is usually in the range of 300−3000 cm^{-1}. This energy unit might be unfamiliar to you at first, but it has become the unit everyone who works in this area uses, so we need to become comfortable with it. You can feel free to convert them to eV at first to give you a better feeling for what they are; eventually you will become comfortable with using them.

In order for us to be able to directly observe the molecular vibrations, the molecule must have a dipole moment that couples to electromagnetic waves as it oscillates, so that light can be absorbed or emitted. This coupling is similar to what we saw when we discussed the photoproduction of hydrogen in Sect. 14.10. For a binary molecule, we expect the dipole moment to oscillate with the vibration, as long as the two species that make up the molecule are different, such as in carbon

monoxide CO. A homogeneous molecule like H_2 will not have any observable vibrational spectroscopy because it has no permanent dipole moment.

Because nearly all potentials have quadratic behavior near their minimum, our first approximation to molecular vibrations is via a harmonic potential, with the minimum potential energy approximately at the negative of the dissociation energy of the molecule. Let's discuss some of the behavior of such a description. First, we expect it to be accurate only at low energies. This is because the harmonic oscillator potential has bound states at any energy. But we know the molecule will eventually break if it is pulled far enough apart. So eventually, the potential must go to zero at long distances. We will have more to say about that later in the section. Second, the selection rule for a dipole-allowed transition involves a matrix element of $\hat{r}_x$ between the two oscillator states (the x-direction lying along the molecular bond). The way to see this is that we have a dipole moment that is equal to μ_x at the equilibrium separation and it varies linearly with position away from the equilibrium separation as

$$\mu_x(\hat{r}_x) = \mu_x(r_e) + \Delta\mu_x(\hat{r}_x - r_e) + \cdots \tag{19.2}$$

where we are using a linear variation away from the equilibrium value (usually when we approximate functions about equilibrium they vary quadratically about their minimum, but here the dipole moment is not minimal at the equilibrium separation, so it will vary linearly away from equilibrium—only the potential varies quadratically about the equilibrium position). Now, the probability for the transition will be proportional to the square of the matrix element of the dipole moment (similar to what we had for the photoproduction of hydrogen). Hence, we must compute $\langle m|\mu_x(\hat{r}_x)|n\rangle$ with $m \neq n$. For the harmonic oscillator, we know that $\langle m|n\rangle = 0$ for $m \neq n$, and $\langle m|\hat{r}_x|n\rangle = 0$, unless $m = n \pm 1$. Hence, the selection rule for vibrational transitions becomes $m = n \pm 1$. But, because the true potential is not precisely harmonic, this selection rule only holds approximately, as the true eigenstates are not harmonic oscillator eigenstates, but eigenstates of an anharmonic potential. Note that because the potential looks more harmonic toward its minimum, the selection rule holds better for the low-lying states than the higher ones. We will see the consequences of this more later in the section.

The correct molecular potential of a diatomic molecule is highly repulsive when the two atoms are close to each other, has a nearly symmetric minimum at an intermediate length (which will be on the order of an Å), and then will go to zero for large distances. We have already encountered a good approximate potential for molecular vibrations that satisfies this behavior. It is the Morse potential, which we saw earlier when we were discussing the application of the harmonic oscillator in Sect. 7.14. The Morse potential is

$$V_{Morse}(\hat{r}_x) = D_e\left(e^{-2a(\hat{r}_x - r_e)} - 2e^{-a(\hat{r}_x - r_e)}\right) = D_e\left(1 - e^{-a(\hat{r}_x - r_e)}\right)^2 - D_e, \tag{19.3}$$

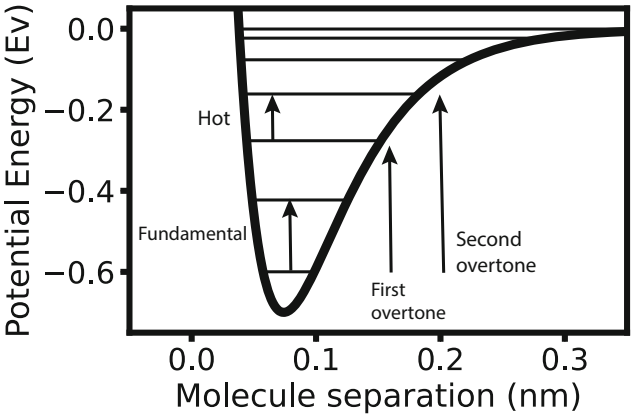

Fig. 19.1 Schematic of the energy levels of the Morse potential (horizontal lines) and the different types of spectroscopic lines—fundamental ($\Delta n = \pm 1$), overtones ($\Delta n \neq \pm 1$), and hot spectroscopy (initial state is not the ground state)

with D_e the electronic dissociation energy. See Fig. 19.1. Previously, we said the energy eigenvalues can be written as

$$E_n = \hbar\omega\left(n + \tfrac{1}{2}\right) - \chi\hbar\omega\left(n + \tfrac{1}{2}\right)^2 - D_e, \tag{19.4}$$

with $\omega = a\sqrt{2D_e/M}$ and $\chi = \hbar\omega/(4D_e)$; we use M for μ_{RED} for simplicity. You are asked to solve for these energy eigenvalues using the factorization method in Problem 19.1.1. On important point to note is that the Morse potential has a *finite number* of bound states (these correspond to the states where E_n is increasing as n increases). Once we reach the point where $E_{n+1} \leq E_n$, then we have found the last bound state (corresponding to $|n\rangle$). The number of bound states is determined by D_e and ω.

As we already mentioned, the selection rules are more relaxed for the Morse potential. One can have so-called overtone transitions. Suppose the molecule is in its ground state $|0\rangle$, then the transition corresponding to the harmonic oscillator sum rule ($m = n \pm 1$) is given by the transition to the state $|1\rangle$, and the energy difference is equal to $E_1 - E_0$, which is equal to $h\nu$ with ν the observed frequency in the spectroscopy (note we use Planck's constant, not the reduced Planck's constant because we are working with frequency, not angular frequency). The overtones correspond to transitions to higher energy states. The first overtone is to the state $|2\rangle$, the second to the state $|3\rangle$, and so on. Typically these overtone transitions are much smaller amplitude than the fundamental, but they can be observed in spectroscopy. They provide additional information about the energy levels and, in particular, can tell us whether there are deviations from the levels predicted by the Morse potential.

We start by looking at specific values for the fundamental and the overtones. In the molecule HCl (hydrogen chloride), we find that the fundamentals and overtones lie close to multiples of each other, but are different from that. We show the frequencies, the differences of the frequencies, and the second-order differences:

$$
\begin{aligned}
E_0 - E_0 &= 0 \\
&\qquad\qquad \Delta G_{\frac{1}{2}} = 2885.90\ \text{cm}^{-1} \\
E_1 - E_0 &= 2885.90\ \text{cm}^{-1} \qquad\qquad \Delta G_1^2 = -103.7\ \text{cm}^{-1} \\
&\qquad\qquad \Delta G_{\frac{3}{2}} = 2782.15\ \text{cm}^{-1} \\
E_2 - E_0 &= 5668.05\ \text{cm}^{-1} \qquad\qquad \Delta G_2^2 = -103.22\text{cm}^{-1} \\
&\qquad\qquad \Delta G_{\frac{5}{2}} = 2678.93\ \text{cm}^{-1} \\
E_3 - E_0 &= 8346.98\ \text{cm}^{-1} \qquad\qquad \Delta G_3^2 = -102.80\ \text{cm}^{-1} \\
&\qquad\qquad \Delta G_{\frac{7}{2}} = 2576.13\ \text{cm}^{-1} \\
E_4 - E_0 &= 10923.11\ \text{cm}^{-1}.
\end{aligned}
\tag{19.5}
$$

This data is isotopically pure $H^{35}Cl$ and is from Yamanouchi (2012). The subscript on the ΔG (and ΔG^2) is given by the average of the two indices that one takes the differences of. A perfect description by a Morse potential would have the ΔG^2 terms all equal to each other, which we discuss now. Consider $E_n - E_0 = \hbar\omega n - \chi\hbar\omega n(n+1)$ and $E_{n+1} - E_0 = \hbar\omega(n+1) - \chi\hbar\omega(n+1)(n+2)$. This gives us $\Delta G_{n+\frac{1}{2}} = \hbar\omega - 2\chi\hbar\omega(n+1)$ and $\Delta G_n^2 = -2\chi\hbar\omega$. This is why the second differences should be a constant and why their deviation tells us about how well the vibrations are described by the Morse potential. Furthermore, one can use this data to extract the parameters ω and χ or D_e and a for the Morse potential from this data. You are asked to do this for HCl in Problem 19.1.5.

In Fig. 19.2, we plot the spectra for the HCl molecule as found in nature. The figure is modified with permission from Bigham et al. (2018). The top plot is the fundamental spectral, while the lower plot is the overtone spectra. Here, we have high-precision spectra, and we can immediately see that the spectra is not really what we would expect based on the discussion so far. What happens in these systems is we have so-called *rovibrational* spectra, which incorporates effects of both the vibrations and the rotations. Our job now is to understand how such a picture modifies what we have discussed so far, and how we can make sense of all of the details in these spectra. To start, we will make a simplifying assumption that separates the rotational degrees of freedom from the vibrational degrees of freedom, so the wavefunction will be represented as $|\psi\rangle = |\psi_o\rangle \otimes |J, m_J\rangle$, where the subscript o denotes oscillator and the rotational degrees of freedom correspond to rotational motion with definite angular momentum and z-component of angular

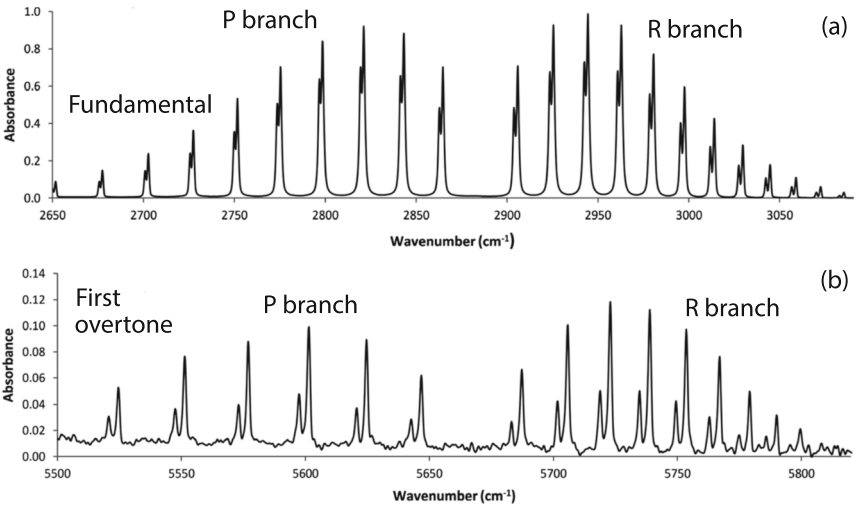

Fig. 19.2 (**a**) Fundamental and (**b**) first overtone spectroscopy of HCl. This sample has a natural abundance of Cl, which is approximately 76% ^{35}Cl and 24% ^{37}Cl (Figure modified with permission from Figure 2 of Bigham et al. (2018) © American Chemical Society (2018))

momentum. This approximation ignores the possibility that the moment of inertia can change for different vibrational states and that the equilibrium separation can change for higher angular momentum states. We already know that diatomic molecules are symmetric rotors, so we cannot have the m_J quantum number change, and we can ignore that quantum number (if $\Delta m_J \neq 1$, the change in energy is huge, if not infinite, because the moment of inertia for rotations about the axis of the molecule is essentially equal to 0). Hence, we have $\hat{H} = \hat{H}_o + \frac{\hat{J}^2 - \hat{J}_z^2}{2I}$, so that the energy levels are

$$E_{nJ} = \hbar\omega\left(n + \tfrac{1}{2}\right) - \chi\hbar\omega\left(n + \tfrac{1}{2}\right)^2 + \frac{\hbar^2 J(J+1) - \hbar^2 m_j^2}{2I}. \tag{19.6}$$

We already know the selection rule for the vibrational spectra—$\Delta n = \pm 1$ for the harmonic case, but for an anharmonic potential, we can see additional changes in n. We also need the selection rule for the rotational degrees of freedom. This requires us to compute the expectation value of the $\hat{\vec{r}}$ operator between different $|J, m_J\rangle$ states. The position operator is a vector operator that acts essentially like an angular momentum equal to one object plus being odd in parity. The former means we can couple J to $J' = J - 1$, J, and $J + 1$. The parity condition says that there is no $J \leftrightarrow J$ coupling with $\Delta J = 0$. Hence, we have $\Delta J = \pm 1$. Because the rotational levels lie at or above the vibrational levels, transitions with $\Delta J = 1$ have frequencies larger than the vibrational frequency (called the R branch), while those with $\Delta J = -1$ have energies less than the vibrational frequency (called the

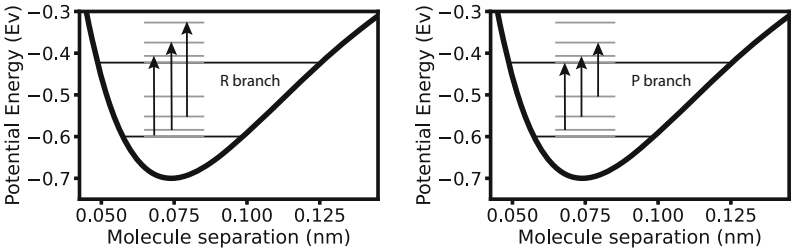

Fig. 19.3 Schematic of rovibrational spectra of a molecule. The $J = 0$ rotational level lies at the vibrational state energy. The $J = 1$ state is $\hbar^2/I$ higher, the $J = 2$ state is $3\hbar^2/I$ higher, and the $J = 3$ state is $6/\hbar^2/I$ higher. We do not include higher J levels in the plot. The R branch has $\Delta J = 1$ (left), while the P branch has $\Delta J = -1$ (right). The vibrational levels are in black, while the rovibrational levels are in gray

P branch). Note that the requirement that there are no transitions with $\Delta J = 0$ means we have no absorption line at the vibrational frequency! Finally recall that the constraint that $\Delta m_j = 0$ was used throughout this discussion and allows us to drop the term proportional to m_j^2 from the analysis. This is all illustrated for you schematically in Fig. 19.3.

The next thing to focus on in the experimental spectra is the near doubling of the lines. This arises from the two different stable Cl isotopes. The different mass leads to a different moment of inertia, for the different species, and the fact that one is three times more abundant than the other leads to the behavior shown there. In addition, we saw before that the separation of the spectral lines had uniform spacing for rotational spectra, but here, we can clearly see the spacing gets wider as we move to the left in the P branch and more compressed as we move to the right in the R branch. Phenomenologically, we can express this, similar to how we expressed the behavior for the vibrational states due to anharmonicity, as (we drop the m_j dependence which is always a constant)

$$E_{nJ} = \hbar\omega\left(n + \tfrac{1}{2}\right) - \chi\hbar\omega\left(n + \tfrac{1}{2}\right)^2 + \frac{\hbar^2 J(J+1)}{2I} - D\frac{\hbar^2 J^2 (J+1)^2}{2I}. \tag{19.7}$$

Using this modified form, one can immediately reproduce the results seen in the spectra qualitatively.

If we compare the fundamental line to the overtone line, the behavior is quite similar. Note that in Fig. 19.2, the two panels have different horizontal scales, so be careful when comparing them. In general, the overtone spectra is much weaker than the fundamental, so one cannot get as good data for that spectroscopy.

When we discussed spectroscopy, so far, we have assumed that the vibrational state is in the ground state, but we have allowed the rotational state to be in excited energy levels. Why? Well, room temperature corresponds to an energy of approximately 200 cm^{-1}. In quantum statistical mechanics, the population of a state is proportional to $e^{-E/k_B T}$, so we need an energy on the order of $2k_B T$ or lower to

have significant occupation at room temperature. Because vibrational spectra are usually in the thousands of cm^{-1}, only the ground state is available as the initial state. But rotational levels are much lower in energy, so they can be occupied at room temperature. To get the higher vibrational energy levels occupied, we need to raise the temperature—often this is raised to thousands of degrees for hot spectroscopy from an excited vibrational state. What about very low temperatures? The theory we have worked out is for isolated molecules, so if they are in a dilute gas phase, everything is fine. When we cool them, they eventually liquify and then solidify. Then the theory gets more complicated. We won't discuss that further here. As you can see, there are many options available to experimentalists!

We have only touched the surface of molecular spectroscopy, because we have only considered diatomic molecules. Many other polyatomic molecules exist, and they have additional vibrational degrees of freedom available to them. We cannot discuss these in detail here, but instead just mention briefly how this works. The key is we try to find the oscillations where all atoms that move at the same frequency. To solve this, we need to find the eigenvalues of a matrix that gives rise to the forces. The eigenvalue tells us what the frequency is for the oscillation, while the eigenvector tells us how the atoms move in this motion. This analysis is called *finding the normal modes* of the molecule. We will solve a simpler version of this problem when we look at ions in a trap. But we will not go through the details of how this works for the general case of polyatomic molecules.

We end this section with a brief discussion of the very interesting problem of the vibrational motion of ammonia (NH_3). The structure of an ammonia molecule is that of a flattened trigonal pyramid. The three hydrogens form the base of the pyramid, with the nitrogen above, or below. It turns out both possibilities are equally likely and the nitrogen oscillates back and forth between these two configurations. The potential that the nitrogen sits in is a double-well potential. It has two equivalent minima in it, corresponding to the nitrogen being on the top or on the bottom. This is a classic problem in quantum mechanics, where the ground state is a superposition of the nitrogen being in the two minima (the even combination has no nodes and so must be the ground state). Hence, if we start with it sitting above, or below, then it will oscillate back and forth because it is not in an eigenstate. You will get a chance to analyze this in detail in Problem 19.1.8 below.

Much of this section is based on material from Yamanouchi's wonderful book on molecules (Yamanouchi, 2012) and you are encouraged to look there for further reading.

Section 19.1 Problems:

19.1.1 Use the Factorization Method to Find the Eigenvalues of the Morse Potential

The Morse potential is written in Eq. (19.3). Use the factorization method to find the eigenvalues. This is a shape-invariant potential, and a superpotential of the form

$\hbar k(-e^{-a(\hat{r}_x - r_e)} + \lambda)$ will allow you to do the factorization; this form has the right behavior of the superpotential as $x \to \pm\infty$ as long as $\lambda > 0$. Note that you can calculate the commutator with an exponential by first using the Hadamard lemma and then multiplying by an exponential from the left and rearranging terms.

You may find it clearer to work in terms of a and ω instead of a and D_e. You should find that

$$\hat{A}_n = \frac{1}{\sqrt{2M}} \left\{ \hat{p}_x - \frac{iM\omega}{a} \left(e^{-a(\hat{r}_x - r_e)} + 1 - (2n+1)\chi \right) \right\}. \tag{19.8}$$

19.1.2 Selection Rules for the Morse Potential

We will work out the selection rules for the fundamental and the overtones starting from the ground state.

The fundamental matrix element is given by

$$\Delta\mu_x \frac{1}{\sqrt{E_1 - E_0}} \langle \phi_1 | \hat{A}_0 \hat{r}_x | \psi_0 \rangle, \tag{19.9}$$

while the first overtone is given by

$$\Delta\mu_x \frac{1}{\sqrt{(E_2 - E_1)(E_2 - E_0)}} \langle \phi_2 | \hat{A}_0 \hat{A}_1 \hat{r}_x | \psi_0 \rangle, \tag{19.10}$$

and the second overtone is given by

$$\Delta\mu_x \frac{1}{\sqrt{(E_3 - E_2)(E_3 - E_1)(E_3 - E_0)}} \langle \phi_3 | \hat{A}_0 \hat{A}_1 \hat{A}_2 \hat{r}_x | \psi_0 \rangle, \tag{19.11}$$

where $|\psi_0\rangle$ is the ground state of the original Morse potential and $|\phi_i\rangle$ is the ground state of the ith auxiliary Hamiltonian coming from the factorization chain.

Our goal is to compute these results using operator methods. The key facts we will use are the subsidiary condition $\hat{A}_0|0\rangle = 0$ and the relationship of the ladder operators—$\hat{A}_n = \hat{A}_0 + \frac{iM\omega\chi n}{a\sqrt{2M}}$. The subsidiary condition allows you to replace $\hat{A}_0 \hat{r}_x |0\rangle$ by $[\hat{A}_0, \hat{r}_x]|0\rangle$, and the other condition allows you to rewrite ladder operators in terms of $\hat{A}_0$ and constants. Using these results, you should be able to show that the matrix element for the fundamental is

$$-i \frac{\Delta\mu_x}{a} \sqrt{\frac{\hbar a^2}{2M\omega}} \frac{1}{\sqrt{1 - 2\chi}} \langle \phi_1 | \psi_0 \rangle, \tag{19.12}$$

for the first overtone is

$$\frac{\Delta\mu_x}{a}\frac{\chi}{2\sqrt{2(1-3\chi)(1-4\chi)}}\langle\phi_2|\psi_0\rangle, \tag{19.13}$$

and for the second overtone is

$$i\frac{\Delta\mu_x}{a}\sqrt{\frac{2M\omega}{\hbar a^2}}\frac{\chi^2}{2\sqrt{6(1-4\chi)(1-5\chi)(1-6\chi)}}\langle\phi_3|\psi_0\rangle. \tag{19.14}$$

You can see that these results become the simple harmonic oscillator results when $\chi = 0$ and are increasingly reduced, the farther we go from the harmonic oscillator selection rules. Hence, anharmonic effects are typically small. Note that the overlaps $\langle\phi_i|\psi_0\rangle$ are of the order of 1, unless i is large, where they will start to decrease. As we approach the harmonic oscillator limit, we have $|\phi_i\rangle = |\psi_0\rangle = |0\rangle$ is the ground state of the harmonic oscillator.

Pick the parameters to be $\hbar^2a^2/2M = 10\ \text{cm}^{-1}$, $\hbar\omega = 1000\ \text{cm}^{-1}$, and $\chi = 0.1$ to numerically compute the ratios of the probabilities to have a first or second overtone transition versus the probability of the fundamental transition.

19.1.3 Why Does the Morse Potential Have a Finite Number of Bound States?

Write a short essay that explains why the Morse potential has a finite number of bound states from rules of factorization.

19.1.4 Fitting Real Data to the Morse Potential

Use the data given in Eq. (19.5) to fit both ω and χ, as well as D_e and a for the Morse potential. How does one fit r_e?

19.1.5 Continuum States

Write an essay explaining why the Morse potential has continuum states and what the continuum states mean physically.

19.1.6 D_e Versus D_o

In the Morse potential, we said D_e was the electronic dissociation energy. But, the true dissociation energy is a bit lower than that, because the vibrational state cannot

have lower energy than its ground state. We call D_o the vibrational dissociation energy. Explain how one can try to use experimentally determined dissociation energies D_o to determine what D_e must be for the Morse potential.

19.1.7 Bound-State Wavefunctions for the Morse Potential

Use the operator form of the Rodrigues formula approach to determine the wavefunctions of the Morse potential. Some highlights are that we have

$$\hat{A}_n^\dagger = \exp\left[\frac{M\omega}{\hbar a}\left(\frac{1}{a}e^{-a(\hat{r}_x-r_e)} + \big(1-(2n+1)\chi\big)\hat{r}_x\right)\right] \times \frac{\hat{p}_x}{\sqrt{2M}}\exp\left[-\frac{M\omega}{\hbar a}\left(\frac{1}{a}e^{-a(\hat{r}_x-r_e)} + \big(1-(2n+1)\chi\big)\hat{r}_x\right)\right] \tag{19.15}$$

for the similarity transformation, the nth auxiliary ground state annihilated by $\hat{p}_x$ satisfies $\hat{A}_n|\phi_n\rangle = 0$ and

$$\hat{p}_x \exp\left[\frac{M\omega}{\hbar a}\left(\frac{1}{a}e^{-a(\hat{r}_x-r_e)} + \big(1-(2n+1)\chi\hat{r}_x\big)\right)\right]|\phi_n\rangle = 0, \tag{19.16}$$

and the final result is of the form

$$|n\rangle = \frac{1}{\sqrt{(E_n-E_0)\cdots(E_n-E_{n-1})}}\exp\left\{\tfrac{2M\omega}{\hbar a}\left(\tfrac{1}{a}e^{-a(\hat{r}_x-r_e)} + \big(1-\chi\big)\hat{r}_x\right)\right\} \times \left[\hat{p}_x, e^{-\frac{2\chi M\omega}{\hbar a}\hat{r}_x}\left[\hat{p}_x, e^{-\frac{2\chi M\omega}{\hbar a}\hat{r}_x}\left[\hat{p}_x, \cdots\right.\right.\right. \left.\left.\left.\left[\hat{p}_x, \exp\left\{-\tfrac{2M\omega}{\hbar a}\left(\tfrac{1}{a}e^{-a(\hat{r}_x-r_e)} + \big(1-n\chi\big)\hat{r}_x\right)\right\}\right]\cdots\right]\right]\right]|\phi_n\rangle. \tag{19.17}$$

Because the Morse potential is asymmetric, with a long tail to the right, we expect the probability distribution for large n states to be skewed more to the right than what we see in a harmonic oscillator wavefunction. Indeed, this is what is found, as one can see in Fig. 19.4. For further details, see Li et al. (2025).

19.1.8 Ammonia

The ammonia molecule is a flattened trigonal pyramid, with the N atom lying above or below the plane of the pyramid. The H atoms are far enough apart that the N atom can tunnel between these two positions. Since both positions are equivalent,

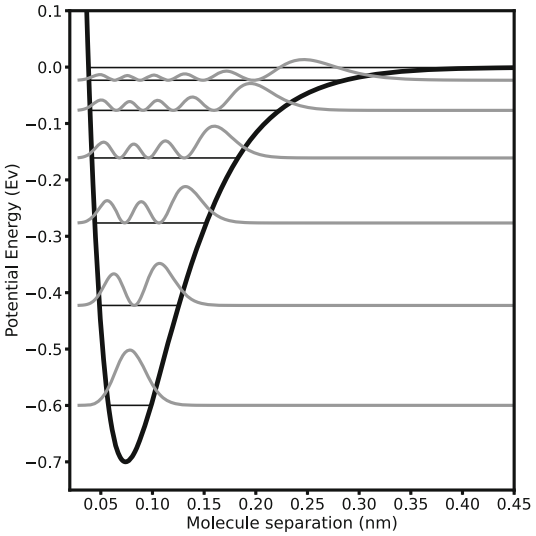

Fig. 19.4 Probability density for the ground state and the first five excited states in the Morse potential that has seven bound states. You can see how the probability asymmetrically shifts to the right for the higher-energy states

they have the same energy, and the Hamiltonian can be written as

$$\hat{H}|\text{above}\rangle = E_0|\text{above}\rangle - t|\text{below}\rangle \quad \text{and} \quad \hat{H}|\text{below}\rangle = -t|\text{above}\rangle + E_0|\text{below}\rangle. \tag{19.18}$$

Hence, it can be written as a 2×2 matrix, with $|\text{above}\rangle \leftrightarrow (1, 0)$ and $|\text{below}\rangle \leftrightarrow (0, 1)$. We have

$$\hat{H} \leftrightarrow \begin{pmatrix} E_0 & -t \\ -t & E_0 \end{pmatrix}. \tag{19.19}$$

The energy eigenstates are given by $E_\pm = E_0 \pm t$, and the eigenstates are the even and odd combinations—the minus state is $\frac{1}{\sqrt{2}}(1, 1)$ and the plus state is $\frac{1}{\sqrt{2}}(1, -1)$. What this means is the state $|\text{above}\rangle$ is not an energy eigenstate, and neither is $|\text{below}\rangle$. So, if we start in one of them, the system will oscillate between the two. Using the fact that energy eigenstates evolve in time as $|\psi(t)\rangle = e^{-iEt/\hbar}|\psi(0)\rangle$, determine the time dependence of the quantum state that starts in the state $|\text{above}\rangle$. How long does it take before the probability to be in $|\text{below}\rangle$ is one? How long until the probability to be in $|\text{above}\rangle$ is one again? Determine the period of this oscillation in terms of E_0 and $-t$. If the energy difference is 0.79 cm^{-1}, how long is the period of the oscillation?

19.2 Molecular Rotations

One of the most important measurements of molecules is their rotational spectroscopy. If the molecule has an asymmetry in its charge distribution, then, as

it rotates, it will act like a time-varying dipole (among other more complicated multipoles), and as a result, it will couple to light and have transitions where light is absorbed and, if in an excited state, where light is emitted. The treatment of the rotation of a three-dimensional "rigid-body" molecule is more complicated than just working with angular momentum as we have before. This is because the rigid body (which means the lengths between the ions in the molecule and the angles between them are "rigid" and do not change) has three angular degrees of freedom called the Euler angles. They correspond to two angles which orient the axis the molecule rotates about and the third angle being how much it rotates about that axis. The angular momentum of an atom involved just two degrees of freedom and hence had two conserved quantum numbers. Now, with three degrees of freedom, we expect three conserved quantum numbers. This is indeed just what we will find.

The original problem of molecular rotations was solved by Casimir in 1931 as his Ph.D. thesis. Most treatments since then closely follow the strategy that he used. It involves working with differential operators for momentum in the position representation and determining complex projections of these different operators in the lab frame and in a frame fixed to the rigid body, called the *body-fixed frame*. In physics, the problem is often not discussed in any detail. Two exceptions are the textbooks by Landau and Lifshitz (1977) and the second edition of Weinberg (2012). Much of our notation will follow that of Yamanouchi's comprehensive text (Yamanouchi, 2012). Our treatment works in a representation independent formalism and requires no reference to Euler angles. In this sense, it is quite different from the conventional approach (Toprak et al., 2025).

We consider a molecule that consists of N ion cores, surrounded by electrons. Because the ion cores are significantly more massive than the electrons, we ignore the electrons in the analysis. We further assume the system is a *rigid body*, meaning the distances between any two ions and the angles between any three ions are fixed and do not change during the motion. This is a simplification, but it is often used as the first description of rotational motion. If one relaxes this condition, then we have *ro-vibrational motion*, which we will not discuss this in detail here (it was covered for diatomics in the previous section). Hence, we will use the position operators $\{\hat{\vec{r}}_1, \hat{\vec{r}}_2, \cdots, \hat{\vec{r}}_N\}$ for the positions of each ion in the molecule and $\{\hat{\vec{p}}_1, \hat{\vec{p}}_2, \cdots, \hat{\vec{p}}_N\}$ for their momentum. We also let $\{m_1, m_2, \cdots, m_N\}$ denote the masses of each of the ions. The operators for different ions obviously commute with each other, so we have

$$[\hat{r}_{n\alpha}, \hat{p}_{n'\beta}] = i\hbar\delta_{nn'}\delta_{\alpha\beta}, \tag{19.20}$$

where we use α and β to denote the Cartesian coordinates of the operators in the lab frame; that is, $\hat{\vec{r}}_n \cdot \vec{e}_\alpha = \hat{r}_{n\alpha}$, where $\vec{e}_\alpha$ are the standard Cartesian unit vectors (for $\alpha = x$, y, and z). The unit vectors are numbers not operators. We will later discuss the unit vectors in the body-fixed frame, which are operators, because they rotate with the rigid body.

When we discussed the two-body problem, we found it convenient to separate into the average and relative coordinates. The situation for a rigid body is similar. We work with the center-of-mass position operator $\hat{\vec{R}}$ and the total momentum operator $\hat{\vec{P}}$. We have

$$\hat{\vec{R}} = \frac{1}{M}\sum_{n=1}^{N} m_n \hat{\vec{r}}_n \quad \text{and} \quad \hat{\vec{P}} = \sum_{n=1}^{N} \hat{\vec{p}}_n, \tag{19.21}$$

where $M = \sum_{n=1}^{N} m_n$ is the total mass of the molecule. These two operators satisfy the canonical commutation relation, which can be immediately verified as follows:

$$\begin{aligned}[\hat{R}_\alpha, \hat{P}_\beta] &= \frac{1}{M}\sum_{n=1}^{N}\sum_{n'=1}^{N} m_n [\hat{r}_{n\alpha}, p_{n'\beta}] = \frac{1}{M}\sum_{n=1}^{N}\sum_{n'=1}^{N} m_n i\hbar\delta_{nn'}\delta_{\alpha\beta} \\ &= \frac{1}{M}\sum_{n=1}^{N} m_n i\hbar\delta_{\alpha\beta} = i\hbar\delta_{\alpha\beta}.\end{aligned} \tag{19.22}$$

You may be surprised that we included mass factors in the definition of $\hat{\vec{R}}$ but not $\hat{\vec{P}}$ as this appears to be asymmetric. But when we recall that momentum is the mass multiplied by the velocity, we see it actually is symmetric. Indeed, it is only this combination that gives the canonical commutation relation between the two operators. Now, using these operators, we define the operators relative to $\hat{\vec{R}}$ and $\hat{\vec{P}}$. They are defined via the relations

$$\hat{\vec{r}}_n = \hat{\vec{R}} + \Delta\hat{\vec{r}}_n \quad \text{and} \quad \hat{\vec{p}}_n = \frac{m_n}{M}\hat{\vec{P}} + \Delta\hat{\vec{p}}_n. \tag{19.23}$$

The reason for the m_n/M factor in the second relation is that a more massive particle should have a larger share of the total momentum when determining the relative momentum. Note that these relative position and momentum operators do not satisfy the canonical commutation relation. In fact, we have

$$\begin{aligned}[\Delta\hat{r}_{n\alpha}, \Delta\hat{p}_{n'\beta}] &= \left[\hat{r}_{n\alpha} - \hat{R}_\alpha, \hat{p}_{n'\beta} - \frac{m_{n'}}{M}\hat{P}_\beta\right] \\ &= [\hat{r}_{n\alpha}, \hat{p}_{n'\beta}] - [\hat{R}_\alpha, \hat{p}_{n'\beta}] - \frac{m_{n'}}{M}[\hat{r}_{n\alpha}, \hat{P}_\beta] + \frac{m_{n'}}{M}[\hat{R}_\alpha, \hat{P}_\beta] \\ &= i\hbar\delta_{nn'}\delta_{\alpha\beta} - \frac{m_{n'}}{M}i\hbar\delta_{\alpha\beta} - \frac{m_{n'}}{M}i\hbar\delta_{\alpha\beta} + \frac{m_{n'}}{M}i\hbar\delta_{\alpha\beta} \\ &= i\hbar\delta_{nn'}\delta_{\alpha\beta} - i\hbar\frac{m_{n'}}{M}\delta_{\alpha\beta},\end{aligned} \tag{19.24}$$

which is not the canonical commutation relation. Hence it is often more convenient to work with the $\hat{\vec{r}}_n$, $\hat{\vec{p}}_n$, $\hat{R}$, and $\hat{P}$ operators when we compute commutators. However, we do have that the "average" operators do commute with the relative ones, since

$$[\hat{R}_\alpha, \Delta p_{n'\beta}] = \left[\hat{R}_\alpha, \hat{p}_{n'\beta} - \frac{m_{n'}}{M}\hat{P}_\beta\right] = \frac{m_{n'}}{M} i\hbar\delta_{\alpha\beta} - \frac{m_{n'}}{M} i\hbar\delta_{\alpha\beta} = 0, \tag{19.25}$$

and similarly we have that $[\Delta\hat{r}_{n\alpha}, \hat{P}_\beta] = 0$. Even though the relative operators do not satisfy the canonical commutation relations, we will show that they can be used to define the relative angular momentum and that the relative angular momentum does satisfy the conventional angular momentum commutation relations in the lab frame.

We compute the total angular momentum by summing the contributions of each ion via

$$\hat{\vec{L}} = \sum_{n=1}^{N} \hat{\vec{r}}_n \times \hat{\vec{p}}_n. \tag{19.26}$$

Then, rewriting in terms of the average and relative operators, we find that

$$\begin{aligned}
\hat{\vec{L}} &= \sum_{n=1}^{N} \left(\hat{\vec{R}} + \Delta\hat{\vec{r}}_n\right) \times \left(\frac{m_n}{M}\hat{\vec{P}} + \Delta\hat{\vec{p}}_n\right) \\
&= \sum_{n=1}^{N} \frac{m_n}{M} \hat{\vec{R}} \times \hat{\vec{P}} + \hat{\vec{R}} \times \sum_{n=1}^{N} \Delta\hat{\vec{p}}_n + \sum_{n=1}^{N} \frac{m_n}{M} \Delta\hat{\vec{r}}_n \times \hat{\vec{P}} + \sum_{n=1}^{N} \Delta\hat{\vec{r}}_n \times \Delta\hat{\vec{p}}_n \\
&= \hat{\vec{R}} \times \hat{\vec{P}} + \sum_{n=1}^{N} \Delta\hat{\vec{r}}_n \times \Delta\hat{\vec{p}}_n.
\end{aligned} \tag{19.27}$$

The middle two terms vanish because $\sum_n m_n \Delta\hat{\vec{r}}_n = 0$ and $\sum_n \Delta\hat{\vec{p}}_n = 0$. For example, since $\hat{\vec{P}} = \sum_n \hat{\vec{p}}_n$, we find that $\sum_n \Delta\hat{\vec{p}}_n = \sum_n \left(\hat{\vec{p}}_n - \frac{m_n}{M}\hat{\vec{P}}\right) = \sum_n \hat{\vec{p}}_n - \hat{\vec{P}} = 0$. One can similarly show that $\sum_n \frac{m_n}{M} \Delta\hat{\vec{r}}_n = 0$. Hence, we have that the total angular momentum separates into the angular momentum of the center-of-mass and the angular momentum about the center of mass. Given that $\hat{\vec{R}}$ and $\hat{\vec{P}}$ satisfy the canonical commutation relations, then $\hat{\vec{R}} \times \hat{\vec{P}}$ satisfies the commutation relations of an angular momentum. We need to show that the second term in the last line of Eq. (19.27) is also an angular momentum. This follows because $\hat{\vec{L}}$ is an angular momentum, as is $\hat{\vec{R}} \times \hat{\vec{P}}$, then so must the difference $\sum_n \Delta\hat{\vec{r}}_n \times \Delta\hat{\vec{p}}_n$. In any case, we will show this directly next, just to confirm it.

We have

$$\begin{aligned}
&\left[\sum_{n=1}^{N}\left(\Delta\hat{\vec{r}}_n\times\Delta\hat{\vec{p}}_n\right)_\alpha,\sum_{n'=1}^{N}\left(\Delta\hat{\vec{r}}_{n'}\times\Delta\hat{\vec{p}}_{n'}\right)_\beta\right]\\
&=\sum_{n,n'=1}^{N}\sum_{\gamma\delta\gamma'\delta'}\epsilon_{\alpha\gamma\delta}\epsilon_{\beta\gamma'\delta'}\left(\Delta\hat{r}_{n'\gamma'}[\Delta\hat{r}_{n\gamma},\Delta\hat{p}_{n'\delta'}]\Delta\hat{p}_{n\delta}\right.\\
&\quad\left.+\Delta\hat{r}_{n\gamma}[\Delta\hat{p}_{n\delta},\Delta\hat{r}_{n'\gamma'}]\Delta\hat{p}_{n'\delta'}\right)\\
&=i\hbar\sum_{n,n'=1}^{N}\sum_{\gamma\delta\gamma'\delta'}\epsilon_{\alpha\gamma\delta}\epsilon_{\beta\gamma'\delta'}\left(\Delta\hat{r}_{n'\gamma'}\delta_{\gamma\delta'}(\delta_{nn'}-\tfrac{m_{n'}}{M})\Delta\hat{p}_{n\delta}\right.\\
&\quad\left.-\Delta\hat{r}_{n\gamma}\delta_{\gamma'\delta}(\delta_{nn'}-\tfrac{m_n}{M})\Delta\hat{p}_{n'\delta'}\right)
\end{aligned}\tag{19.28}$$

after using the commutation relation between the relative operators. The terms with the fractions of the masses in them vanish, because $\sum_n m_n\Delta\hat{\vec{r}}_n=0$. As for the remaining terms, the Kronnecker delta function from the commutator makes one of the indices for the product of the Levi-Cevita tensors to be the same, allowing the sum over that index to be performed. Then we obtain a number of additional Kronnecker delta functions, which can also be used to eliminate more summations. In the end, we find that

$$\left[\sum_{n=1}^{N}\left(\Delta\hat{\vec{r}}_n\times\Delta\hat{\vec{p}}_n\right)_\alpha,\sum_{n'=1}^{N}\left(\Delta\hat{\vec{r}}_{n'}\times\Delta\hat{\vec{p}}_{n'}\right)_\beta\right]=i\hbar\sum_\gamma\epsilon_{\alpha\beta\gamma}\sum_{n=1}^{N}\left(\Delta\hat{\vec{r}}_n\times\Delta\hat{\vec{p}}_n\right)_\gamma,\tag{19.29}$$

which is the conventional commutation relation for angular momentum. You should verify these steps yourself. This establishes the separation of the rotational motion to the motion of the center of mass and the motion about the center of mass.

In addition to working in the lab frame, we must also work in the body-fixed frame. This corresponds to a set of three orthonormal unit vector operators that form a right-handed frame ($\hat{\vec{f}}_a\times\hat{\vec{f}}_b=\hat{\vec{f}}_c$) and rotate with the rigid body. Just like in spherical coordinates, where the unit vectors were operators because they change direction when the angles in spherical coordinates change, a similar thing happens with the body-fixed frame, where the unit vectors must move in the lab frame to remain fixed to the rigid body. We construct these vectors by taking linear combinations of the $\Delta\hat{\vec{r}}_n$ operators. This guarantees that they rotate with the rigid body, because the ions in the molecule do just that. Hence, we have

$$\hat{\vec{f}}_a=\sum_{n=1}^{N}C_{na}\Delta\hat{\vec{r}}_n.\tag{19.30}$$

The coefficients C_{na} are numbers, not operators, which is possible because the molecule is assumed to be a rigid body. For example, if we had a molecule made of ions that lie a specific distance down the x-, y-, and z-axes from the center-of-mass position, we would form the unit vectors by taking the operators corresponding to the ions in the positive direction along each axis and we would divide them by the distance to the origin. In the general case, we will need to take linear combinations of many different relative position operators of the ions to form these unit vectors, and the procedure is not unique—many different choices are possible for this. Note that is important that you do not form $\hat{\vec{f}}_c = \hat{\vec{f}}_a \times \hat{\vec{f}}_b$ from the solution found for the first two unit vectors, because that would produce an expression for the last unit vector that is quadratic in the relative position operators, and we want them to all be linear functions of the relative position operators. Now, because the relative position operators are all vector operators, we can immediately find the commutation relation with the angular momentum. It is

$$[\hat{L}_\alpha, \hat{\vec{f}}_a \cdot \vec{e}_\beta] = \sum_{n=1}^{N} C_{na} [\hat{L}_\alpha, \Delta\hat{\vec{r}}_{n\beta}] = i\hbar \sum_n C_{na} \sum_\gamma \epsilon_{\alpha\beta\gamma} \Delta\hat{\vec{r}}_{n\gamma} = i\hbar \sum_\gamma \epsilon_{\alpha\beta\gamma} \hat{\vec{f}}_a \cdot \vec{e}_\gamma. \tag{19.31}$$

This says that the Cartesian components of the unit vectors in the body frame transform as vector operators in the lab frame.

Now we are ready to define the lab-frame and body-frame angular momenta of the molecules. We assume that the angular momentum state for the center-of-mass and total momentum operators is given by an s-wave state with vanishing angular momentum, so the $\hat{\vec{R}} \times \hat{\vec{P}}$ operator does not enter into our analysis, and we work solely with the angular momentum relative to the center-of-mass. It is customary to define the components in the lab frame via

$$\hat{J}_\alpha = \sum_{n=1}^{N} \left(\Delta\hat{\vec{r}}_n \times \Delta\hat{\vec{p}}_n \right) \cdot \vec{e}_\alpha. \tag{19.32}$$

We have already shown that $[\hat{J}_\alpha, \hat{J}_\beta] = i \sum_\gamma \epsilon_{\alpha\beta\gamma} \hat{J}_\gamma$. We use the same operator to define the components in the body frame, so we have

$$\hat{\mathcal{J}}_a = \sum_{n=1}^{N} \left(\Delta\hat{\vec{r}}_n \times \Delta\hat{\vec{p}}_n \right) \cdot \hat{\vec{f}}_a. \tag{19.33}$$

We will work out the commutation relations of these operators shortly.

The first thing to note is that $\sum_\alpha \hat{J}_\alpha^2 = \sum_a \hat{\mathcal{J}}_a^2$, because the norm squared of a vector operator is independent of the coordinate system used to determine the components in a specific basis, even if one frame is rotating. The second thing to note is that $[\hat{J}_\alpha, \hat{\mathcal{J}}_a] = 0$; that is, the components of the angular momentum in the lab frame commute with the components of the angular momentum in the body

frame. I know this sounds very strange, but it comes from the fact that the body frame is rotating, so that our intuition is flawed in trying to determine the behavior of these commutators without explicitly calculating them. So we calculate them right now. To do this, we first introduce a projection onto the Cartesian unit vectors for the body frame. We have

$$\begin{aligned}[\hat{J}_\alpha, \hat{\mathcal{J}}_a] &= \sum_\beta [\hat{J}_\alpha, \hat{J}_\beta \hat{\vec{f}}_a \cdot \vec{e}_\beta] \\ &= \sum_\beta \left([\hat{J}_\alpha, \hat{J}_\beta] \hat{\vec{f}}_a \cdot \vec{e}_\beta + \hat{J}_\beta [\hat{J}_\alpha, \hat{\vec{f}}_a \cdot \vec{e}_\beta] \right) \\ &= \sum_{\beta\gamma} \left(\epsilon_{\alpha\beta\gamma} \hat{J}_\gamma \hat{\vec{f}}_a \cdot \vec{e}_\beta + \epsilon_{\alpha\beta\gamma} \hat{J}_\beta \hat{\vec{f}}_a \cdot \vec{e}_\gamma \right) = 0, \end{aligned} \tag{19.34}$$

after we interchange $\beta \leftrightarrow \gamma$ in the second term and switch the order of the β and γ indices of the Levi-Cevita tensor. So, indeed, we see that the components of angular momentum in the lab and in the body frames do commute. This means that we have three operators that mutually commute—$\hat{\vec{J}}^2$, $\hat{J}_z$, and $\hat{\mathcal{J}}_c$ (on occasion $\hat{\mathcal{J}}_a$ is quantized instead). We will use the eigenvectors of these three operators to describe the states we will be working with. But before we can do that, we need to compute the commutator between two angular-momentum operators in the body frame. We have

$$\begin{aligned}[\hat{\mathcal{J}}_a, \hat{\mathcal{J}}_b] &= \sum_{\alpha\beta} [\hat{J}_\alpha \hat{\vec{f}}_a \cdot \vec{e}_\alpha, \hat{J}_\beta \hat{\vec{f}}_b \cdot \vec{e}_\beta] \\ &= \sum_{\alpha\beta} \left([\hat{J}_\alpha, \hat{J}_\beta] \hat{\vec{f}}_a \cdot \vec{e}_\alpha \hat{\vec{f}}_b \cdot \vec{e}_\beta + \hat{J}_\beta [\hat{J}_\alpha, \hat{\vec{f}}_b \cdot \vec{e}_\beta] \hat{\vec{f}}_a \cdot \vec{e}_\alpha + \hat{J}_\alpha [\hat{\vec{f}}_a \cdot \vec{e}_\alpha, \hat{J}_\beta] \hat{\vec{f}}_b \cdot \vec{e}_\beta \right) \\ &= i\hbar \sum_{\alpha\beta\gamma} \epsilon_{\alpha\beta\gamma} \left(\hat{J}_\gamma \hat{\vec{f}}_a \cdot \vec{e}_\alpha \hat{\vec{f}}_b \cdot \vec{e}_\beta + \hat{J}_\beta \hat{\vec{f}}_a \cdot \vec{e}_\alpha \hat{\vec{f}}_b \cdot \vec{e}_\gamma + \hat{J}_\alpha \hat{\vec{f}}_a \cdot \vec{e}_\gamma \hat{\vec{f}}_b \cdot \vec{e}_\beta \right) \\ &= i\hbar \sum_{\alpha\beta\gamma} \epsilon_{\alpha\beta\gamma} \hat{J}_\alpha \hat{\vec{f}}_a \cdot \vec{e}_\gamma \hat{\vec{f}}_b \cdot \vec{e}_\beta, \end{aligned} \tag{19.35}$$

where the first two terms in the third line cancel, as can be seen by $\beta \leftrightarrow \gamma$ in the second term. Using the representation of the vector triple product in terms of the Levi-Cevita tensor then tells us that

$$[\hat{\mathcal{J}}_a, \hat{\mathcal{J}}_b] = -i\hbar \sum_\alpha \hat{J}_\alpha \big(\hat{\vec{f}}_a \times \hat{\vec{f}}_b \big)_\alpha = -i\hbar \sum_\alpha \hat{J}_\alpha \big(\hat{\vec{f}}_c \big)_\alpha = -i\hbar \sum_c \epsilon_{abc} \hat{\mathcal{J}}_c, \tag{19.36}$$

which is the standard angular momentum commutation relation with one exception—there is an extra minus sign. This is a very odd result, but we already

predicted it above due to the fact that the unit vectors in the body frame are rotating. The extra minus sign affects a few things in our analysis below, especially how we generate the different eigenstates.

We will be working with states that are eigenfunctions of $\hat{\vec{J}}^2$, $\hat{J}_z$, and $\hat{\mathcal{J}}_c$, with eigenvalues $\hbar^2 j(j+1)$, $\hbar m$, and $\hbar k$, respectively. The eigenfunctions for the m eigenvalue act like regular angular momentum, so that $\hat{J}_+ = \hat{J}_x + i\hat{J}_y$ is the raising operator for the m eigenvalue and $\hat{J}_- = \hat{J}_x - i\hat{J}_y$ is the lowering operator for the m eigenvalue. We worked out previously that

$$\hat{J}_\pm |j, m, k\rangle = \hbar\sqrt{j(j+1) - m(m\pm 1)}\,|j, m\pm 1, k\rangle. \tag{19.37}$$

In the body frame, however, $\hat{\mathcal{J}}_+ = \hat{\mathcal{J}}_x + i\hat{\mathcal{J}}_y$ is the lowering operator and $\hat{\mathcal{J}}_- = \hat{\mathcal{J}}_x - i\hat{\mathcal{J}}_y$ is the raising operator, so that

$$\hat{\mathcal{J}}_\pm |j, m, k\rangle = \hbar\sqrt{j(j+1) - k(k\mp 1)}\,|j, m, k\mp 1\rangle. \tag{19.38}$$

You will establish these results in Problem 19.2.2.

Because the position vectors in the body frame are fixed to the body, they do not move, but the velocity of the ions is given by $\vec{\omega} \times \Delta\vec{r}_n$, where $\vec{\omega}$ is a vector pointing in the direction of the axis of rotation, with the magnitude given by the rate of rotation of the rigid body. Now, it may seem like we cannot have a velocity for an ion in the body frame, because everything is fixed to the rigid body in the body frame, but because this is a rotating frame, it turns out that there is a velocity in this frame, determined as we stated above. Since we work in the body frame, we will denote $\Delta\vec{r}_n = (\Delta r_{na}, \Delta r_{nb}, \Delta r_{nc})$. Then, we need to work with some vector identities. We have

$$\begin{aligned}
(\vec{\omega}\times\Delta\vec{r}_n)\cdot(\vec{\omega}\times\Delta\vec{r}_n) &= \vec{\omega}^2\Delta\vec{r}_n^{\,2}\sin^2\theta = \vec{\omega}^2\Delta\vec{r}_n^{\,2}(1-\cos^2\theta)\\
&= \vec{\omega}^2\Delta\vec{r}_n^{\,2} - (\vec{\omega}\cdot\Delta\vec{r}_n)^2\\
&= (\omega_a^2+\omega_b^2+\omega_c^2)(\Delta r_{na}^2+\Delta r_{nb}^2+\Delta r_{nc}^2) - (\omega_a\Delta r_{na}+\omega_b\Delta r_{nb}+\omega_c\Delta r_{nc})^2\\
&= \omega_a^2(\Delta r_{nb}^2+\Delta r_{nc}^2)+\omega_b^2(\Delta r_{na}^2+\Delta r_{nc}^2)+\omega_c^2(\Delta r_{na}^2+\Delta r_{nc}^2)\\
&\quad - 2\omega_a\omega_b\Delta r_{na}\Delta r_{nb} - 2\omega_b\omega_c\Delta r_{nb}\Delta r_{nc} - 2\omega_c\omega_a\Delta r_{nc}\Delta r_{na},
\end{aligned} \tag{19.39}$$

where θ is the angle between $\vec{\omega}$ and $\Delta\vec{r}_n$. Note that we have not put any hats on the position variables here, because we will work out the kinetic energy of the rotation classically first, and then carry over to the quantum result. Summing over all ion positions, we can write the kinetic energy as

$$\begin{aligned}
\text{KE} = {}& \tfrac{1}{2}\omega_a I_{aa}\omega_a + \tfrac{1}{2}\omega_b I_{bb}\omega_b + \tfrac{1}{2}\omega_c I_{cc}\omega_c\\
&+ \tfrac{1}{2}\omega_a I_{ab}\omega_b + \tfrac{1}{2}\omega_a I_{ac}\omega_c + \tfrac{1}{2}\omega_b I_{ba}\omega_a\\
&+ \tfrac{1}{2}\omega_b I_{bc}\omega_c + \tfrac{1}{2}\omega_c I_{ca}\omega_a + \tfrac{1}{2}\omega_c I_{cb}\omega_b.
\end{aligned} \tag{19.40}$$

Here, we have defined the moments of inertia

$$I_{aa} = \sum_{i=1}^{N} m_n(\Delta r_{nb}^2 + \Delta r_{nc}^2), \quad I_{bb} = \sum_{i=1}^{N} m_n(\Delta r_{na}^2 + \Delta r_{nc}^2),$$

$$\text{and } I_{cc} = \sum_{i=1}^{N} m_n(\Delta r_{na}^2 + \Delta r_{nb}^2), \tag{19.41}$$

and the products of inertia to be

$$I_{ab} = I_{ba} = -\sum_{n=1}^{N} m_n \Delta r_{na} \Delta r_{nb}, \quad I_{bc} = I_{cb} = -\sum_{n=1}^{N} m_n \Delta r_{nb} \Delta r_{nc},$$

$$\text{and } I_{ca} = I_{ac} = -\sum_{n=1}^{N} m_n \Delta r_{nc} \Delta r_{na}. \tag{19.42}$$

Now, the moment-of-inertia matrix is a real symmetric matrix, so it can be diagonalized, and when we do that, we say we are working with a basis called the *principal axes*, with the diagonals of the moment-of-inertia matrix called the *principal moments of inertia* (I_a, I_b, and I_c). In this coordinate frame, we have that

$$\text{KE} = \tfrac{1}{2}\omega_a I_a \omega_a + \tfrac{1}{2}\omega_b I_b \omega_b + \tfrac{1}{2}\omega_c I_c \omega_c. \tag{19.43}$$

It is more convenient to write this in terms of the components of the angular momentum, which are given by $L_a = I_a\omega_a$, $L_b = I_b\omega_b$, and $L_c = I_c\omega_c$, where we have

$$\text{KE} = \frac{L_a^2}{2I_a} + \frac{L_b^2}{2I_b} + \frac{L_c^2}{2I_c}. \tag{19.44}$$

To go to the quantum case, we treat the principal moments of inertia as numbers and the angular momentum as operators (now the angular momentum components in the body frame), so that

$$\hat{H} = \frac{\hat{\mathcal{J}}_a^2}{2I_a} + \frac{\hat{\mathcal{J}}_b^2}{2I_b} + \frac{\hat{\mathcal{J}}_c^2}{2I_c}. \tag{19.45}$$

Once we have determined the moments of inertia, we can classify the four different types of molecules that we can have: (i) linear; (ii) spherical; (iii) symmetric; and (iv) asymmetric. In a linear molecule, all the ions are lined up along a single axis. In this case, one of the moments of inertia vanishes, so $I_c = 0$ and

$I_a = I_b$. For a spherical top molecule, we have $I_a = I_b = I_c$. The symmetric top has $I_a = I_b \neq I_c$. If $I_c < I_a = I_b$, it is called a *prolate symmetric top* (which looks more cylindrical, like a cucumber), and if $I_c > I_a = I_b$, it is called a *oblate symmetric top* (which looks more flattened, like a pancake); sometimes for the prolate top, we quantize on the a rather than c axis, which requires some relabeling of axes. Finally if we have $I_a \neq I_b \neq I_c$, it is called an *asymmetric top* molecule.

Our next step is to solve for the energy levels of the different types of molecules. The Hamiltonian does not depend on $\hat{J}_z$, so the energies are independent of that quantum number, providing many degeneracies. For the linear molecule, we remove the term in the Hamiltonian where $I_c = 0$, so we have $\hat{H}_{\text{linear}} = (\hat{\vec{J}}^2 - \hat{\mathcal{J}}_c^2)/2I$, whose energy eigenvalues are $\hbar^2\big(j(j+1) - k^2\big)/2I$. Next, we have the spherical top molecule, where the Hamiltonian is $\hat{H}_{\text{spherical}} = \hat{\vec{J}}^2/2I$, and the energy eigenvalues are $\hbar^2 j(j+1)/2I$. Next is the symmetrical top, where $\hat{H}_{\text{symmetrical}} = (\hat{\vec{J}}^2 - \hat{\mathcal{J}}_c^2)/2I_a + \hat{\mathcal{J}}_c^2/2I_c$, and the energies are $\hbar^2\big(j(j+1) - k^2\big)/2I_a + \hbar^2 k^2/2I_c$. The asymmetric top molecule is more complicated and has to be worked out carefully.

Since

$$\hat{\mathcal{J}}_a^2 = \frac{1}{4}\left(\hat{\mathcal{J}}_+^2 + \hat{\mathcal{J}}_+\hat{\mathcal{J}}_- + \hat{\mathcal{J}}_-\hat{\mathcal{J}}_+ + \hat{\mathcal{J}}_-^2\right) \tag{19.46}$$

and

$$\hat{\mathcal{J}}_b^2 = -\frac{1}{4}\left(\hat{\mathcal{J}}_+^2 - \hat{\mathcal{J}}_+\hat{\mathcal{J}}_- - \hat{\mathcal{J}}_-\hat{\mathcal{J}}_+ + \hat{\mathcal{J}}_-^2\right), \tag{19.47}$$

we can see that the Hamiltonian connects states with eigenvalue k to states with $k \pm 2$. Furthermore, because the Hamiltonian is unchanged under the transformation $\hat{\mathcal{J}}_a \to -\hat{\mathcal{J}}_a$, $\hat{\mathcal{J}}_b \to \hat{\mathcal{J}}_b$, and $\hat{\mathcal{J}}_c \to -\hat{\mathcal{J}}_c$, there is a rotational-parity symmetry, which says we can classify energy eigenstates according to their parity under $k \to -k$. We use these facts to determine how to find the energy eigenvalues for different values of j.

There is no general formula for the energy eigenvalues for arbitrary j, because the states are coupled in a way that cannot be analytically solved. So, once j is large enough, we must solve a set of matrix eigenvalue equations to find the energy eigenvalues. The total angular momentum quantum number j takes on integer values, so we have the cases $j = 0, 1, 2, \cdots$. For $j = 0$, all operators annihilate against the state so $E = 0$. For $j = 1$, the state with $k = 0$ does not couple to anything, so it is an eigenstate, while using parity, we have that $(|1, m, 1\rangle \pm |1, m, -1\rangle)/\sqrt{2}$ must be energy eigenstates. So the energies can be easily found analytically. We have that the $k = 0$ state has

$$E_{k=0} = \hbar^2\left(\frac{1}{I_a} + \frac{1}{I_b}\right), \tag{19.48}$$

while the $k \neq 0$ states have

$$E_{|k|=1,+} = \hbar^2 \left(\frac{1}{I_a} + \frac{1}{I_c} \right) \text{ and } E_{|k|=1,-} = \hbar^2 \left(\frac{1}{I_b} + \frac{1}{I_c} \right). \tag{19.49}$$

You should carefully work out these results yourself. When solving for higher j values, it remains useful to work with the states of definite parity, as you will see in the problems.

The selection rules involve situations where the electric dipole matrix element is nonzero between two different rotational states. The way to think of this is that the dipole moment operator, as a vector operator, carries angular momentum $l = 1$. Hence, when we combine it with a state that has angular momentum j, we find the transitions will be allowed to $j + 1$, j, and $j - 1$. Hence $\Delta j = \pm 1,\ 0$. We examine the behavior in the lab frame, where again, using the conservation of total angular momentum idea, we find that $\Delta m = \pm 1,\ 0$ as well. Of course, there are no $0 \rightarrow 0$ transitions. What about the quantum number k? For symmetric top molecules, it turns out it does not change, so $\Delta k = 0$. The way to reason this out is that the dipole moment will point along the same axis as we use for quantizing $\hat{\mathcal{J}}_c$, so rotations about the dipole moment do not affect k and hence $\Delta k = 0$. The situation is more complicated for allowed Δk values for asymmetric top molecules, because the energy eigenstates typically are superpositions of different k values, so we do not discuss that case in detail here. A more extensive discussion of how this works is given in Yamanouchi (2012).

As a molecule spins rapidly, the bonds become larger due to the centripetal force pushing the ions away from the center. As a result, the rotational motion can excite vibrational motion. Such spectroscopy is called ro-vibrational spectroscopy, and it requires us to no longer work with a rigid body. Discussing the details for how to work with this is beyond what we will do here. If you are interested in further details, Yamanouchi (2012) is a good place to start.

Section 19.2 Problems:

19.2.1 Comparison with Average and Relative Coordinates for $N = 2$

Take the case $N = 2$, and compare the coordinates defined here with the average and relative coordinates used before in Chap. 11. You should notice some differences, which give rise to the different commutation relations we have here versus before. Explain carefully why we need to proceed in the way described in the text when generalizing to $N > 2$.

19.2.2 Eigenstates in the Body-Fixed Frame

Following the procedures we used in Chap. 9, show the details for how to derive Eq. (19.38).

19.2.3 Calculate Moments of Inertia of the Water Molecule

Consider a water molecule, with a bond angle of 109.45^{o} and a H-O bond distance given by r. Determine the moments of inertia for this molecule in the principal axis basis (by solving an eigenvalue problem if you have to). Is the water molecule a symmetric or asymmetric top?

19.2.4 Oblate, Prolate, or Spherical Molecules

Find images of the following molecules and classify them, by looking at them as to whether they are prolate or oblate symmetric tops or spherical tops: (i) CH_3Cl; (ii) methane CH_4; (iii) benzene C_6H_6; (iv) SF_6; (v) fullerene C_{60}; and (vi) ammonia NH_3. Note that this involves no calculations.

19.2.5 Essay About Why the Commutation Relations in the Two Different Frames Behave This Way

Write an essay that carefully explains how the components of the same angular momentum operator in the lab and body frames commute and also why the components of the angular momentum in the body frame have an extra minus sign in their commutation relations.

19.2.6 Asymmetric Top with $J = 2$

Consider the asymmetric top case with $j = 2$ and find its energy eigenvalues. You should find $k = \pm 1$ are coupled, as are $k = 0, \ \pm 2$. By using the spin parity, you should find three 1×1 blocks and one 2×2 block to solve. You should be able to find analytic formulas for all of the energy eigenvalues, although they will be a bit complicated.

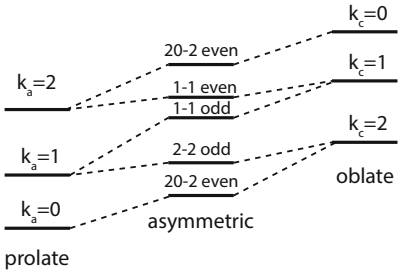

Fig. 19.5 Energy levels for the prolate and oblate symmetrical top and how they are connected through the asymmetric top energy levels. We label the asymmetric top energy levels via the k_c values and whether they are even or odd. For the prolate symmetric top, we quantize k_a, not k_c

If you organize your energy eigenvalues in order, you should find they are connected to the prolate and oblate cases, as we schematically take I_b from equal to I_c to equal to I_a with $I_c < I_a$ we make the transition from a prolate spherical top (quantize along a-axis) to an oblate spherical top (quantize about c-axis). An energy level diagram showing this connection is given in Fig. 19.5, which you should verify against your results.

19.2.7 Determining Structure of a Molecule from Rotational Spectroscopy

We will work out the bond-bond distance and the angle of the NF_3 (XY_3) molecule using rotational spectroscopy. First, we must find the moments of inertia. Verify that

$$I_a = I_b = 2m_Y r^2 \sin^2\tfrac{\theta}{2} + \frac{m_X m_Y}{m_X + 3m_Y} r^2\left(3 - 4\sin^2\tfrac{\theta}{2}\right) \quad \text{and} \quad I_c = 4m_Y r^2 \sin^2\tfrac{\theta}{2}. \tag{19.50}$$

Assuming the structure is the same for $^{14}NF_3$ and $^{15}NF_3$, the frequency for the transition between $j = 0$ to $j = 1$ for the ^{14}N isotope is 21362.18 MHz. Use this to determine I_b in MHz. Repeat for the ^{15}N isotope, with the transition frequency being 21258.92 MHz. Use these two results to determine r and θ. In your calculations, use $m(\text{F}) = 18.998405$ amu, $m(^{14}\text{N}) = 14.003074$ amu, and $m(^{15}N) = 15.000108$ amu. This is one of the most accurate ways to determine the inter-ion distances in a molecule (See Fig. 19.6).

Hint: first find I_b for both cases and then use the formula for I_b to isolate r and θ.

Ans. $r = 1.0156$ Å and $\theta = 107.28°$.

This problem is based on discussions in Yamanouchi (2012).

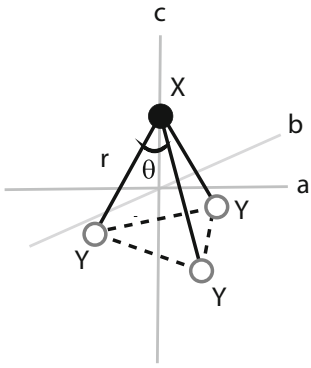

Fig. 19.6 Structural positions of the ions in a XY_3 type molecule, which is an oblate symmetric top. Note the bond distance is r from X to Y, and the bond angle at the X vertex is θ. The axes of the body frame go through the center of mass

19.3 Hydrogen Molecule

In this section, we investigate the simplest possible molecule, H_2, using the minimal set of basis functions. This makes the problem particularly simple, but it also allows us to introduce many of the ideas used to describe the electronic structure of more complex molecules. Some of the tools needed here involve calculus in order to explicitly compute the numerical results. But, we will see in the next section that the standard approach used in quantum chemistry calculations employs a somewhat different methodology based on algebraic methods. This allows us to avoid requiring calculus to describe the behavior of H_2. It turns out not to be so critical for these problems, as you will soon see.

One important approximation that is used for molecules is the *Born-Oppenheimer approximation*. This uses the fact that the mass of the ion cores in molecules is much higher than that of electrons (many thousands of times larger), so the strategy to solve the problem is broken into two separate pieces. First, we assume the masses of the ion cores are infinite, fix their positions, and compute the electronic energy of the molecule. Then we change the locations of the ion cores and repeat. By doing this many, many times, we can construct a potential-energy surface for the molecule based on the changes in the coordinates of the ion-core positions. For large polyatomic molecules, this gets challenging to work with rather quickly, but it is a simple one-dimensional potential for diatomics if we focus on just the stretching of the bond (things can be a bit more different, especially for linear molecules with more atoms, as three-dimensional bending modes become important as well). Then, once we have found the potential energy surface, we use that in a quantum-mechanical problem for the ions only to find their vibrational motion. This fits perfectly with our treatment of vibrations if the potential energy surface of the diatomic was well described by the Morse potential. In this approximation, we are assuming that the electrons respond instantaneously to the changes in the ion-core positions. This is an accurate approximation in essentially all molecules

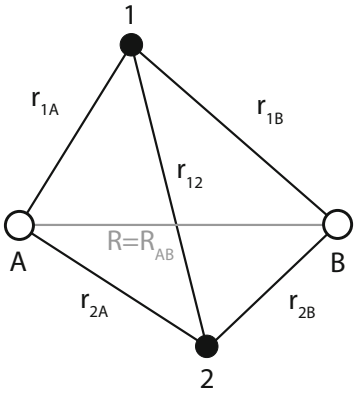

Fig. 19.7 Geometry of the hydrogen molecule. The open circles represent the atomic positions, while the solid circles represent the electron positions. The lines represent the Coulomb interactions

We now consider the first step of the Born-Oppenheimer approach. We neglect the kinetic energy of the ion cores. Then the Hamiltonian for a H_2 molecule is given by

$$\hat{H} = \underbrace{\frac{\hat{\vec{p}}_1^2}{2M} - \frac{e^2}{4\pi\epsilon_0 \hat{r}_{1A}}}_{\text{atom A}} + \underbrace{\frac{\hat{\vec{p}}_2^2}{2M} - \frac{e^2}{4\pi\epsilon_0 \hat{r}_{2B}}}_{\text{atom B}} \underbrace{- \frac{e^2}{4\pi\epsilon_0 \hat{r}_{1B}} - \frac{e^2}{4\pi\epsilon_0 \hat{r}_{2A}} + \frac{e^2}{4\pi\epsilon_0 r_{AB}} + \frac{e^2}{4\pi\epsilon_0 \hat{r}_{12}}}_{\text{additional Coulomb}} . \tag{19.51}$$

Here, 1 and 2 denote the electron positions, A and B denote the nuclear positions, M is the electron mass, and $\hat{r}_{ij} = |\hat{\vec{r}}_i - \hat{\vec{r}}_j|$; when the label refers to an atom, $\vec{r}_{A,B}$ is a fixed position, and not an operator. If we initially associate electron 1 with atom A and electron 2 with atom B (which is possible if they are far apart), then the first two sets of terms are the Hamiltonians for the two different hydrogen atoms, with distinguishable electrons. The additional Coulomb interactions include the Coulomb interaction of each electron with the other atom (which makes them indistinguishable in the Hamiltonian), the repulsion of the two nuclei, and the repulsion of the two electrons. The details of the Hamiltonian are depicted in Fig. 19.7. It is the last term that makes this problem difficult to solve, and our treatment here will only be an approximate treatment. We also just ensure that the system treats the electrons indistinguishably in the quantum state. Finally, by varying r_{AB}, one can compute the potential energy surface by finding the total energy as a function of r_{AB}.

We need to solve this problem in the two-electron many-body Fock space. The simplest thing to do is to employ a *configuration interaction* approach, which is to solve the many-body problem in a truncated many-body space of states. The simplest one can pick for this many-body space is a basis of two spatial orbitals.

The best choice are the orbitals that solve the different atomic problems, so the 1s orbital centered on nucleus A and the 1s orbital centered on the nucleus B. We call these states $|\phi_A^{1s}\rangle$ and $|\phi_B^{1s}\rangle$. Here, the superscript says the type of orbital, and the subscript says which nuclear site the orbital is centered at. We also need to specify the spin state. While we usually do this with up or down arrows, in chemistry, they choose to do it with α for the up spin state and β for the down spin state. So, the four *spin orbitals* we have are $|\phi_A^{1s}\alpha\rangle$, $|\phi_A^{1s}\beta\rangle$, $|\phi_B^{1s}\alpha\rangle$, and $|\phi_B^{1s}\beta\rangle$. The four two-electron states that are antisymmetric include a spatially symmetric and spin antisymmetric singlet state and three spatially antisymmetric and spin symmetric triplet states. The singlet state is

$$\frac{1}{2}\big(|\phi_A^{1s}\alpha\rangle\otimes|\phi_B^{1s}\beta\rangle-|\phi_A^{1s}\beta\rangle\otimes|\phi_B^{1s}\alpha\rangle+|\phi_B^{1s}\alpha\rangle\otimes|\phi_A^{1s}\beta\rangle-|\phi_B^{1s}\beta\rangle\otimes|\phi_A^{1s}\alpha\rangle\big), \tag{19.52}$$

which can be written schematically as

$$\frac{1}{2}\big(|\phi_A^{1s}\rangle\otimes|\phi_B^{1s}\rangle+|\phi_B^{1s}\rangle\otimes|\phi_A^{1s}\rangle\big)\big(|\alpha\rangle\otimes|\beta\rangle-|\beta\rangle\otimes|\alpha\rangle\big), \tag{19.53}$$

as a factorized form in space and spin (we still use the convention where the first electron corresponds to the first member of each tensor product and the second electron as the second one). The spin triplet state is given by the two forms

$$\frac{1}{\sqrt{2}}\big(|\phi_A^{1s}\alpha\rangle\otimes|\phi_B^{1s}\alpha\rangle-|\phi_B^{1s}\alpha\rangle\otimes|\phi_B^{1s}\alpha\rangle\big) \tag{19.54}$$

and

$$\frac{1}{\sqrt{2}}\big(|\phi_A^{1s}\rangle\otimes|\phi_B^{1s}\rangle-|\phi_B^{1s}\rangle\otimes|\phi_A^{1s}\rangle\big)\big(|\alpha\rangle\otimes|\alpha\rangle\big); \tag{19.55}$$

the other two spin triplet states can be formed by applying the lowering operator onto the spin part of the quantum state.

Because these states factorize, and the Hamiltonian is independent of the spin degrees of freedom, we can compute the energy eigenvalues for the singlet and triplet states using just the spatial part of the quantum states. Note, that this simplification does not hold for generic states, but it does hold here, so we will use it.

The first step is to compute the matrix elements of the Hamiltonian. The atomic states satisfy

$$\left(\frac{\hat{\vec{p}}_1^2}{2M}-\frac{e^2}{4\pi\epsilon_0\hat{r}_{1A}}\right)|\phi_A^{1s}\rangle=E_H|\phi_A^{1s}\rangle \text{ and } \left(\frac{\hat{\vec{p}}_2^2}{2M}-\frac{e^2}{4\pi\epsilon_0\hat{r}_{2B}}\right)|\phi_B^{1s}\rangle=E_H|\phi_B^{1s}\rangle. \tag{19.56}$$

Here, $E_H = -\frac{e^2}{8\pi\epsilon_0 a_0}$ is the atomic hydrogen ground-state energy. We also split off the repulsion energy of the two nuclei, because it is independent of the electronic wavefunctions (we use $r_{AB} = R$). The second step is to compute the direct interaction terms, which can be thought of as the "diagonal" terms

$$\begin{aligned} J &= \langle\phi_A^{1s}| \otimes \langle\phi_B^{1s}| \left(-\frac{e^2}{4\pi\epsilon_0\hat{r}_{1B}} - \frac{e^2}{4\pi\epsilon_0\hat{r}_{2A}} + \frac{e^2}{4\pi\epsilon_0\hat{r}_{12}}\right) |\phi_A^{1s}\rangle \otimes |\phi_B^{1s}\rangle \\ &= \langle\phi_B^{1s}| \otimes \langle\phi_A^{1s}| \left(-\frac{e^2}{4\pi\epsilon_0\hat{r}_{1B}} - \frac{e^2}{4\pi\epsilon_0\hat{r}_{2A}} + \frac{e^2}{4\pi\epsilon_0\hat{r}_{12}}\right) |\phi_B^{1s}\rangle \otimes |\phi_A^{1s}\rangle. \end{aligned} \tag{19.57}$$

You should verify explicitly why these two lines are equal to each other, as it is not immediately obvious that this should be so. Next, we have the exchange interaction terms, which can be thought of as the "off-diagonal" terms:

$$\begin{aligned} K &= \langle\phi_A^{1s}| \otimes \langle\phi_B^{1s}| \left(-\frac{e^2}{4\pi\epsilon_0\hat{r}_{1B}} - \frac{e^2}{4\pi\epsilon_0\hat{r}_{2A}} + \frac{e^2}{4\pi\epsilon_0\hat{r}_{12}}\right) |\phi_B^{1s}\rangle \otimes |\phi_A^{1s}\rangle \\ K^* &= \langle\phi_B^{1s}| \otimes \langle\phi_A^{1s}| \left(-\frac{e^2}{4\pi\epsilon_0\hat{r}_{1B}} - \frac{e^2}{4\pi\epsilon_0\hat{r}_{2A}} + \frac{e^2}{4\pi\epsilon_0\hat{r}_{12}}\right) |\phi_A^{1s}\rangle \otimes |\phi_B^{1s}\rangle. \end{aligned} \tag{19.58}$$

There is one additional issue that we have to deal with which is that this basis we are using is not orthogonal. This is because $\langle\phi_A^{1s}|\phi_B^{1s}\rangle = S \neq 0$. We will discuss the properties of this overlap matrix element below; note that we will assume the overlap can be complex, even though for this case the wavefunctions are real valued. Right now, however, we work out the matrix form of the eigenvalue equation using the basis $|\phi_A^{1s}\rangle \otimes |\phi_B^{1s}\rangle$ and $|\phi_B^{1s}\rangle \otimes |\phi_A^{1s}\rangle$. The energy eigenvalue equation becomes

$$\begin{pmatrix} J & K \\ K^* & J \end{pmatrix} = \left(E - 2E_H - \frac{e^2}{4\pi\epsilon_0 R}\right) \begin{pmatrix} 1 & S^2 \\ S^{*2} & 1 \end{pmatrix}. \tag{19.59}$$

This is what is called a generalized eigenvalue equation, because it is working with a nonorthogonal basis and the equation is more complicated than the ordinary eigenvalue equation. To solve it, we still subtract the right-hand side from the left-hand side and set the determinant of the corresponding matrix to zero to determine the energy eigenvalues. Fortunately, we do not need to do that here, because we know the states of definite spin are energy eigenstates; hence, the eigenstates must be $(1, 1)/\sqrt{2}$ and $(1, -1)/\sqrt{2}$. Note that as R changes, the Hamiltonian matrix elements all change (as well as the overlap S), so to find the equilibrium extension of the molecule, we must repeat the calculation many times to find the minimum of the energy.

Before discussing the energy eigenvalues and the properties of the solutions, we go back to the question of the overlap matrix elements. Recall that this is the inner

product of 1s hydrogen wavefunctions, with one shifted relative to the other by a distance of R, which is the fixed distance between the two nuclei. Taking the z-axis along the molecule axis, we have

$$S = \left\langle \phi_A^{1s} | \phi_B^{1s} \right\rangle = \left\langle \phi_A^{1s} | e^{-\frac{i}{\hbar} R \hat{p}_z} | \phi_A^{1s} \right\rangle. \tag{19.60}$$

That is, we translate one of the wavefunctions a distance R along the molecular axis before computing the inner product. Clearly, this approaches 1 as $R \to 0$, because the hydrogen states are normalized, and it approaches 0 as $R \to \infty$, because the wavefunctions decrease exponentially for large distances.

We would like to compute this matrix element of the translation operator. Indeed, while this was already done when we computed the overlap of two coherent states with different parameters in Sect. 7.11, the approach used there, where we express the momentum operator in terms of the ladder operators, then separate the single exponential factor using the Weyl form of the Baker-Campbell-Hausdorff formula, and finally evaluate the remaining exponential operators to the left and to the right (where they are replaced by 1) cannot be used here. It is not possible with Coulomb states for reasons described next. Using the Cartesian factorization, we can certainly express the momentum operator in the z-direction in terms of the ladder operators, but the Baker-Campbell-Hausdorff formula is not simple anymore and involves an infinite number of terms, which makes evaluating this impossible with the same approach. It is here where calculus can come to save the day, as one can introduce a complete set of position states into the inner product and then convert the expression into an integral, which can be directly evaluated. We will not go into those details here, because it turns out that in quantum chemistry, we instead approximate the atomic wavefunctions by weighted sums over Gaussians. Then, the translation operator can be immediately evaluated, because it looks just like the coherent state calculation. The direct and exchange interaction terms can also be evaluated for Gaussians, which then allows for the entire set of terms in the Hamiltonian to be determined algebraically. This algebraic computational framework allows for the Hamiltonian matrix elements to be computed much more efficiently than any approach that uses integrations and calculus. We will have more to say about this issue in the next section, as this result is far from obvious for the interaction terms. The approximation of representing Coulomb wavefunctions in terms of linear combinations of Gaussians will be explored further in the next section.

Now, we come back to the energy calculation. You will compute the energies directly using the even and odd combinations, which are the two eigenstates, in Problem 19.3.4. The energies are given by

$$E_\pm = 2E_H + \frac{e^2}{4\pi\epsilon_0 R} + \frac{J \pm K}{1 \pm S^2}. \tag{19.61}$$

In order for a molecule to be formed, we must have at least one of these energies to be less than $2E_H$, which is the total energy of the isolated atoms. The singlet state is

always lower in energy than the triplet state. The splitting between the energy levels is computed in Problem 19.3.4.

We will discuss, to a limited extent, how this methodology is generalized for more complicated models in the next section. Note, however, that the calculation given here, while exact in this limited basis, remains an approximation to the exact result, because we worked with a very small (incomplete) basis of many-body states. This ends up being the case for nearly all quantum chemistry calculations, although some of the more advanced approaches do include small contributions from large numbers of different many-body states, which greatly improves the overall result.

Section 19.3 Problems:

19.3.1 Direct and Exchange Matrix Element Properties

When we computed the ground-state wavefunction of hydrogen, we found it had an exponential form ce^{-r/a_0}. We take c to be a positive real number; then the wavefunction is real and positive (be sure you check what the spherical harmonic factor is to verify this). Use this behavior to describe what we expect the overlap S to behave like for all R and similarly what J and K behave like for all R (this is a qualitative discussion, not a quantitative one, except in the extreme limits of $R \to 0$ and $R \to \infty$). When R is large, do these results tend to exponentially decrease, power-law decrease, or go to a constant? You may find reviewing the virial theorem for hydrogen will be helpful for you.
Ans.: $S \to 1$, and $J, K \to -e^2/4\pi\epsilon_0 a_0$ for $R \to 0$ and all approach zero as $R \to \infty$.

19.3.2 Relationship Between Hamiltonian Matrix Elements and the Translation Operator

Using the fact that $|\phi_B^{1s}\rangle$ is translated a distance of R along the z-axis away from $|\phi_A^{1s}\rangle$, express S, J, and K in terms of just the bras and kets at A and translation operators.

19.3.3 Improving the Calculation with Additional Variational Freedom

The current way we are doing the calculation for H_2 is by using an unaltered 1s wavefunction of H, just shifting their origin. But, if we wanted to go one step further, we could allow the exponent be $e^{-\alpha r/a_0}$ and minimize with respect to α as well.

Explain how such an optimization approach would work. No need to explain it in the most efficient way, just explain it in any way that will actually work.

Such an approach allows one to take into account the effect of many excited states of H, but even with this, one does not obtain an exact result. Can you explain what remains that is missing?

19.3.4 Energy Eigenvalues for H_2

Compute the energy eigenvalues $E_\pm$ and their difference $\Delta E = E_- - E_+$ for the exact Hamiltonian we worked with. It is easiest to just let $\hat{H}$ act on the known eigenstates to find the energies.

What changes in this calculation as R changes? (This result is special to H_2—in more complicated molecules more changes.)

19.3.5 Algebraic Approaches to Computing Overlap Matrix Elements

Write an essay describing how we can compute the overlap S if the states are harmonic oscillator ground states (products of Gaussians in each Cartesian direction). Explain why it is challenging to do it with Coulomb wavefunctions (although in principle, it can still be calculated; it is much more challenging).

19.3.6 Ortho and Para Hydrogen

Since the electrons lie in a spin singlet state for the lowest-energy state, we have the nuclear spins available on each nucleus. They will interact with each other and can form singlet (parahydrogen) or triplet (orthohydrogen) states. The energy splitting is very small. At room temperature, each state is equally likely, so we see three times as much ortho-H_2 than para-H_2. When we get down to close 60° K, the probability of ortho and para hydrogen are about the same. In the liquid state, it is essentially all parahydrogen. Write a short essay explaining how this works, estimate the energy splitting, and explain why it may be difficult to liquify hydrogen if it is in a mixture of ortho and para hydrogen, if the conversion from ortho to para costs too much energy for each molecule.

19.4 Hartree-Fock Approximation

In the previous section, we showed how one calculates the approximate ground-state energy of the H_2 molecule in a minimal basis. We interpret that calculation

as a configuration-interaction calculation since it is a linear combination of Slater determinants. The Hartree-Fock method approximates the ground-state by using a *single* Slater determinant (or a single product state when expressed in a second-quantized basis). This is an approximate result, because the true ground state will generically be a superposition of many Slater determinants. We also know that when we work with a single Slater determinant, Wick's theorem holds, which allows us to determine expectation values of products of operators in terms of products of expectation values of pairs of operators. But, this is an approximation when compared to what happens when we have a superposition of Slater determinants. In this section, we describe some of the additional complexities that arise in carrying out a Hartree-Fock calculation for more complex molecules.

The first step is to choose a basis set. As we mentioned before, this is usually done with a weighted sum of Gaussians to describe each "atomic" basis set. The reason why is that with these basis sets, the different Coulomb integrals in the matrix representation of the Hamiltonian can be calculated efficiently. It has two problems associated with it: (i) the Gaussian is flat at zero argument, while the atomic wavefunctions have a nonzero slope at the origin, and (ii) the Gaussians decay too rapidly at long distances. The second issue can be dealt with to some extent by including more Gaussians in the superposition at farther distances, but is a problem with this choice. However, there is no other way.

The goal of the Hartree-Fock approach is to find the optimal Slater determinant. Within this approach, there are two different ways we can proceed. The first is called an unrestricted Hartree-Fock approximation, where one can fill the Hartree-Fock energy levels in any fashion, and the restricted Hartree-Fock approximation, where we always fill each level with both up and down spin electrons. This latter case allows for a number of simplifications, and from the bathtub principle, it should give the lowest-energy state for closed-shell systems.

As we saw in the H_2 example, we often need to make a linear combination of the basis states we work with to find the best choice for the Hartree-Fock orbitals that we wish to fill. Because energy eigenstates are orthogonal, we will find the Hartree-Fock orbitals must be orthogonal as well, even if the basis states have nonzero overlap.

There are many ways to perform Hartree-Fock calculations; we will describe one approach.

1. The first step is to choose the spin-orbital basis set $\{|\phi_j\rangle : j = 1, \cdots, N\}$ for the calculation. This is usually larger than the minimal basis set, which often is not very accurate because it has too few states.
2. The second step is to compute all of the different matrix elements in the Hamiltonian. This can be a large number—with N orbitals, the number of matrix elements grows like N^4.
3. The third step is the Löwdin orthogonalization. Compute the transformation matrix M that diagonalizes the overlap matrix $S_{ij} = \langle\phi_i|\phi_j\rangle$ ($M^T S M$ is a diagonal matrix).

4. The fourth step is to re-express the matrix elements in the new orthogonal basis, which is done by contracting each index in the single-particle matrices and the two-particle tensors with M if it is a ket index and M^T if it is a bra index.
5. The fifth step is to express the Hamiltonian in this general form, using $\hat{c}_i$ to denote the second-quantized operators for the orthonormal basis $|\psi_i\rangle = \sum_j M_{ij}|\phi_j\rangle$. The final form is

$$\hat{H} = \sum_{ij} C_{ij}\hat{c}_i^\dagger \hat{c}_i + \sum_{ijkl} D_{ijkl}\hat{c}_i^\dagger \hat{c}_j^\dagger \hat{c}_k \hat{c}_l \,. \tag{19.62}$$

6. The sixth step is to use the fact that we will be using a single Slater determinant state $|\Psi\rangle$, so the expectation values of the two-particle terms factorize in terms of the two possible contractions, so

$$\begin{aligned}\hat{H} \to \sum_{ij} C_{ij}\hat{c}_i^\dagger \hat{c}_i + \frac{1}{2}\sum_{ijkl} D_{ijkl}\big(&\hat{c}_i^\dagger \hat{c}_l \,\langle\Psi|\hat{c}_j^\dagger \hat{c}_k|\Psi\rangle + \hat{c}_j^\dagger \hat{c}_k \langle\Psi|\hat{c}_i^\dagger \hat{c}_l\,|\Psi\rangle \\ &- \hat{c}_i^\dagger \hat{c}_k \langle\Psi|\hat{c}_j^\dagger \hat{c}_l\,|\Psi\rangle - \hat{c}_j^\dagger \hat{c}_l\,\langle\Psi|\hat{c}_i^\dagger \hat{c}_k|\Psi\rangle\big),\end{aligned} \tag{19.63}$$

   with the factor of $\frac{1}{2}$ to prevent double counting. This form of the Hamiltonian is now quadratic in the fermionic operators, so it is of the form $\sum_{ij} O_{ij}\hat{c}_i^\dagger \hat{c}_j$. To find the ground state of this, we diagonalize the matrix O, find the different energy levels, and use the bathtub principle to fill in the lowest levels.
7. The seventh step has us deal with the fact that the state $|\Psi\rangle$ changes after the calculation is finished versus what it was before, because the O_{ij} are determined from $|\Psi\rangle$. So, we repeat again, until we get to the point where $|\Psi\rangle$ does not change from one iteration to the next. This creates what is called the self-consistent field solution to the Hartree-Fock problem. After we have found the converged final result, we work within the Hartree-Fock basis to then improve the solution by adding many-body correlations to the quantum state. The Hartree-Fock basis is the basis that diagonalized the final matrix O.

As we mentioned previously, the Hartree-Fock calculation is an approximate calculation. We define the correlation energy as the total energy (computed exactly) minus the Hartree-Fock energy. It measures how important the interactions are for a molecule. For many molecules, the Hartree-Fock energy (for a sufficiently large basis set) is often more than 90% of the total energy—the correlation energy is less than 10%. Unfortunately the Hartree-Fock calculations are not accurate enough for quantum chemistry applications (where we need an accuracy of 0.05 eV or better), so much effort is spent on trying to get that remaining 10% of the energy.

In general, the Hartree-Fock approximation is fairly accurate near the equilibrium distances for the molecule, but it gets worse as the molecule is stretched toward dissociation. This is because the stretched molecule is more strongly correlated. This often sounds like it is the opposite of what one might expect, so it is useful to explain

why. As we stretch the molecule, the single-particle terms in the Hamiltonian decrease faster than the Coulomb terms, so we eventually get into the regime where the interactions are much larger than the kinetic energy. You can envision this because the single-particle terms depend exponentially on the distance between the nuclei (when they are far enough apart), while the Coulomb terms change with a power law. Hence, correlations become more important if you want to determine the potential-energy surface all the way out to dissociation. In addition, when we have correlated quantum states, they cannot be accurately represented in terms of a product state.

So, how do we go beyond the Hartree-Fock approximation? We need to go from working with a single Slater determinant to a superposition of many of them. But, even with this superposition, the Hartree-Fock term is by far the dominant contribution in single-reference problems. So, the problem we have is we need to find a way to include correlations accurately, but also in a way where we can compute their contribution efficiently. We describe one very clever approach to do this in the next section.

Section 19.4 Problems:

19.4.1 Orthogonality of H_2 Eigenstates

Remembering that $\langle \phi_A^{1s} | \phi_B^{1s} \rangle$ is nonzero, compute the inner product between the two states $\frac{1}{\sqrt{2}}(|\phi_A^{1s}\rangle \pm |\phi_B^{1s}\rangle$ and verify that it is zero.

19.4.2 Gaussians Versus Exponentials

Plot the exponential e^{-x} and the Gaussian e^{-x^2} and compare the two. For small x one is flat, while the other has a nonzero slope. For large x, one decays much faster than the other. Explain how we can obtain accurate results if we use Gaussians instead of exponentials for the basis functions. What does it say about the x values in equilibrium?

19.4.3 Calculating the Interaction Term with Gaussian States

In this problem, we will focus on the interaction terms of the electron with a nucleus. We assume that one can calculate the matrix element

$$\langle 0|\frac{1}{|\hat{\vec{r}}-\vec{R}|}|0\rangle, \tag{19.64}$$

where $|0\rangle$ is the ground state of a harmonic oscillator in three-dimensions (the frequencies need not be the same in each direction). The generic interaction we need to calculate is then of the form

$$\langle 0|\frac{1}{|\hat{\vec{r}}-\vec{R}|}e^{-\frac{i}{\hbar}\vec{R}'\cdot\hat{\vec{P}}}|0\rangle. \tag{19.65}$$

To show this can be calculated, we need to reduce it to the form in Eq. (19.64). We do this using the following steps: (i) introduce the ladder operators for each dimension to re-express the momentum operator in terms of ladder operators; (ii) use the Weyl form of the Baker-Campbell-Hausdorff formula to rewrite these terms with the raising operators to the left of the lowering operators (and a correction factor); (iii) operate the lowering operator term on the vacuum to get rid of the exponential term; (iv) re-express the raising operator in terms of the corresponding position and momentum operators; (v) use a multiply by one to make a Hadamard lemma with respect to the interaction term operator; and (vi) evaluate the Hadamard lemma. At this stage, you should be able to show that the desired interaction term is of the form we already know how to calculate and an additional multiplicative factor. This shows how knowing how to calculate the simple matrix element allows us to determine these more complicated matrix elements. This is the reason why the linear combination of Gaussians approach works so well.

19.4.4 Why Is a Mean-Field Solution Not an Exact Solution?

Write an essay explaining why the mean-field approximation is not an exact solution. What is the mean-field solution missing?

19.4.5 Is the Hartree-Fock Approximation a Variational Approximation?

In the way we described how the Hartree-Fock calculation is determined, it does not seem like it is a variational calculation. But, it turns out that it is. Write a short essay explaining why it is a variational calculation.
Hint: Calculate the expectation value of the Hamiltonian in the state $|\Psi\rangle$ and compare that to the Hartree-Fock energy (which also is calculated in the state $|\Psi\rangle$).

19.5 Coupled-Cluster Approximation

The coupled-cluster method is one of the most ingenious approaches to solving the many-body problem—it calculates the energy of a quantum state without ever generating the state, and unlike the factorization method, which is for single-particle quantum mechanics, this is for the many-body problem. The key idea is that a *low-rank* quantum state will be an accurate approximation to the exact state. This means, the expansion for the ground-state in terms of basis states $|\psi\rangle = \sum_n c_n |\psi_n\rangle$ uses a much smaller number of parameters to determine the coefficients than the total number of coefficients. This will become clearer as we describe the approach further below.

Before jumping into the theory, it is interesting to discuss a bit of the history. The story begins in the mid-1950s, when Fritz Coester and Hermann Kümmel introduced the idea into nuclear physics. The idea did not get much traction, and it was a decade later when Jirí Čížek introduced the idea into chemistry (Čižek, 1966) with a calculation on benzene. Cížek collaborated with Jozef Paldus on much of the initial development of coupled-cluster theory. I am indebted to Rodney Bartlett and Dominika Zgid for clarifying many of the coupled-cluster ideas for me.

As we have already mentioned, the key to the coupled-cluster quantum state is the low-rank representation—we use a small number of parameters, but have a large number of terms in the state. This implies that many of the coefficients in the expansion are constructed from expressions derived from the smaller number of parameters that go into the definition of the state. This means that we are not computing with a full variational freedom in the wavefunctions—indeed coupled-cluster theory is not a variational technique, as we will describe in more detail below. One of the most important properties is called *size consistency* of the coupled-cluster approach. This aspect allows a dissociated molecule to properly separate into its atomic constituents within the theory. This is a nontrivial issue, as one can see with a simple example from configuration interaction theory. Suppose our simplified atom has a form $|\psi_{\text{atom}}\rangle = \alpha|\psi_1\rangle + \beta|\psi_2\rangle$. Then, when the molecule is stretched into a separated atomic state R units away, the full quantum state is given by

$$|\psi\rangle = \big(\alpha|\psi_1\rangle + \beta|\psi_2\rangle\big) \otimes e^{-\frac{i}{\hbar}R\hat{p}_z}\big(\alpha|\psi_1\rangle + \beta|\psi_2\rangle\big). \tag{19.66}$$

But, if the configuration interaction theory only used the states $|\psi_1\rangle \otimes e^{-\frac{i}{\hbar}R\hat{p}_z}|\psi_1\rangle$, $|\psi_2\rangle \otimes e^{-\frac{i}{\hbar}R\hat{p}_z}|\psi_2\rangle$ and $|\psi_2\rangle \otimes e^{-\frac{i}{\hbar}R\hat{p}_z}|\psi_1\rangle$, it cannot represent the factorized quantum state. While this simple example is contrived, such phenomena usually occur with configuration-interaction calculations, so it becomes a pervasive problem in chemical calculations. The coupled-cluster approach, however, uses an exponential ansatz for the quantum state. This always factorizes and can always produce the correct dissociated states. Hence, it should produce more accurate potential-energy surfaces, especially as we get closer to dissociation than a configuration interaction calculation will produce.

We express the coupled-cluster quantum state in the form

$$e^{T_1+T_2+T_3+T_4+\cdots}|\psi_{\rm HF}\rangle, \tag{19.67}$$

where $|\psi_{\rm HF}\rangle$ is the Hartree-Fock quantum state (a single reference state) and the operators T_n are rank-n excitation operators of the following forms

$$\begin{aligned}
\hat{T}_1 &= \sum_{i}^{\rm occ}\sum_{a}^{\rm vir}\alpha_i^a\hat{a}_a^\dagger\hat{a}_i\,,\\
\hat{T}_2 &= \frac{1}{2!}\sum_{ij}^{\rm occ}\sum_{ab}^{\rm vir}\alpha_{ij}^{ab}\hat{a}_a^\dagger\hat{a}_b^\dagger\hat{a}_i\,\hat{a}_j\,,\\
\hat{T}_3 &= \frac{1}{3!}\sum_{ijk}^{\rm occ}\sum_{abc}^{\rm vir}\alpha_{ijk}^{abc}\hat{a}_a^\dagger\hat{a}_b^\dagger\hat{a}_c^\dagger\hat{a}_k\hat{a}_j\hat{a}_i\,,\cdots
\end{aligned} \tag{19.68}$$

where the $i,\ j,\ k,\cdots$ indices are for the single-particle states occupied in the Hartree-Fock state (from the middle of the alphabet), and $a,\ b,\ c,\cdots$ are indices for the virtual (unoccupied) orbitals (from the beginning of the alphabet). The operators $\hat{a}_i$ are destruction operators for real orbitals (orbitals used in $|\psi_{\rm HF}\rangle$), and $\hat{a}_a^\dagger$ are creation operators for virtual orbitals (orbitals in the basis set, but not in $|\psi_{\rm HF}\rangle$). They satisfy the conventional fermionic anticommutation relations. The numbers $\alpha_{ijk\cdots}^{abc\cdots}$ are the amplitudes. The standard coupled-cluster ansatz will include all possible singles (S) and doubles (D). It then usually selects the most important triples (T) and quadruples (Q). This leads to notations such as CCSD for a coupled-cluster ansatz with singles and doubles, or CCSDTQ, which includes all rank operators up to quadruples (although usually the triples and quadruples are selectively included, because there are so many of them). All orbitals discussed here are spin orbitals, so we neglect the spin index on all of them. The amplitudes α are all taken to be real, for problems in the absence of an external magnetic field, which is what we assume here.

We can now describe how the coupled-cluster ansatz is constructed. We let $\hat{T} = \hat{T}_1 + \hat{T}_2 + \cdots$ and then we want to solve the eigenvalue problem

$$\hat{H}e^{\hat{T}}|\psi_{\rm HF}\rangle = E_{\rm gs}e^{\hat{T}}|\psi_{\rm HF}\rangle \tag{19.69}$$

for the ground-state energy and ground state. The first thing you might think is let's just calculate the variational energy and minimize it by choosing the optimal parameters. This would lead to working with an expression of the form

$$\langle\psi_{\rm HF}|e^{\hat{T}^\dagger}\hat{H}e^{\hat{T}}|\psi_{\rm HF}\rangle/\langle\psi_{\rm HF}|e^{\hat{T}^\dagger}e^{\hat{T}}|\psi_{\rm HF}\rangle. \tag{19.70}$$

The challenge is that we have no easy way to work with this expression. We would simply need to compute the quantum state $e^{\hat{T}}|\psi_{\text{HF}}\rangle$ and then the variational energy, followed by a complex optimization scheme. This would rapidly become too large of a problem for a computer to handle. The ingenious idea of coupled cluster is to instead perform a similarity transformation, by multiplying the eigenvalue-eigenstate equation from the left by $e^{-\hat{T}}$. Then, the Hamiltonian has a similarity transformation applied to it $\hat{H} \to e^{-\hat{T}}\hat{H}e^{\hat{T}}$. If we recall, a similarity transformation does not change eigenvalues. But, note, this similarity transformation is *not* a unitary transformation. This is because $\left(e^{\hat{T}}\right)^{\dagger} = e^{\hat{T}^{\dagger}} \neq e^{-\hat{T}}$.

Okay, we have performed a similarity transformation, how do we find the amplitudes α, if we do not do so variationally? The answer is that after the similarity transformation, we require the leftmost column of the transformed matrix to have the form $(E_{\text{gs}}, 0, 0, \cdots, 0)$, so the transformed matrix is diagonal, at least for the ground state. Using a numbering scheme, where $|\psi_0\rangle = |\psi_{\text{HF}}\rangle$, the first set of states are the singles excitations $|\psi_I\rangle = \hat{a}_a^{\dagger}\hat{a}_i|\psi_{\text{HF}}\rangle$ for $I = 1, \cdots, N_S$, and then for doubles, via $|\psi_I\rangle = \hat{a}_a^{\dagger}\hat{a}_b^{\dagger}\hat{a}_i\,\hat{a}_j|\psi_{\text{HF}}\rangle$, for $I = N_S + 1, \cdots, N_S + N_D$, and so on. Here N_S is the number of singles excitations and N_D is the number of doubles excitations. Then, the coupled-cluster amplitude equations are

$$\begin{aligned} E_{\text{gs}} &= \langle\psi_0|e^{-\hat{T}}\hat{H}e^{\hat{T}}|\psi_0\rangle, \\ 0 &= \langle\psi_1|e^{-\hat{T}}\hat{H}e^{\hat{T}}|\psi_0\rangle, \\ 0 &= \langle\psi_2|e^{-\hat{T}}\hat{H}e^{\hat{T}}|\psi_0\rangle, \\ &\vdots \end{aligned} \tag{19.71}$$

We will not describe the strategies used to solve these equations in detail.

Nevertheless, we still have not shown any simplicity—it seems like now we need to calculate not one matrix element of the Hamiltonian in the complicated quantum state, but a large number of them! Here is the secret idea—the transformation $\hat{H} \to e^{-\hat{T}}\hat{H}e^{\hat{T}}$ corresponds to a Hadamard lemma! Even more powerful is the fact that the Hadamard lemma must truncate after no more than four nested commutators (which you will prove in Problem 19.5.1, as it is not obvious). This allows us to solve the coupled-cluster problem *without ever determining the quantum state* $e^{\hat{T}}|\psi_{\text{HF}}\rangle$. You should be able to see that this approach is not a variational approach. We are not calculating the energy and minimizing it. We instead use the coupled-cluster amplitude equations to solve the problem. These equations are nonlinear equations which are solved iteratively in most cases. Because we are not explicitly determining the trial quantum state, which includes an exponentially large number of terms, the coupled cluster approximation has an intrinsic exponential speed-up compared to all other algorithms. This is why it is so commonly used in quantum chemistry calculations.

We want to emphasize another point of the ansatz. In the operator $\hat{T}$, all excitation operators in the expansion commute with all other operators. This is because the set of real spin orbitals is orthogonal to the set of virtual orbitals and we only use creation operators for the virtual orbitals and destruction operators for the real orbitals, so these individual fermionic operators always anticommute with each other. This then implies that all of the low-rank excitation operators also commute with each other. This result implies that we can put all of the exponential factors into one exponent, as we have done, or we can put them into individual factors, broken up according to each excitation operator. It is also because of this separation into different categories for the creation and annihilation operators in $\hat{T}$ that the Hadamard lemma truncates at the fourth-order term at the worst case.

There is another interesting result for the singles excitations—all they do is change the single-particle basis we are working with. We will show why, in just a moment. Before doing that, though, we want to comment why one might want to do this. After all, the Hartree-Fock basis is the optimal basis for the Hartree-Fock solution. It turns out that when we go from the single determinant of the Hartree-Fock state to a sum over many determinants for a general many-body state, the optimal single-particle basis might change. In chemistry, this is expressed in terms of working with a diagonal single-particle density matrix, given by $M_{rs} = \langle\psi|\hat{a}_r^\dagger \hat{a}_s|\psi\rangle$, where the indices r and s run over all possible (real or virtual) indices. One can see this matrix is diagonal, if we take the state to be $|\psi_{\text{HF}}\rangle$, but that it might not be diagonal once we move to the many-body state. Now, how do we see this result? We simply use a multiply by one many times. Then, we have

$$e^{T_1}\hat{a}_1^\dagger\hat{a}_2^\dagger\cdots\hat{a}_m^\dagger|0\rangle = \left(e^{\hat{T}_1}\hat{a}_1^\dagger e^{-\hat{T}_1}\right)\left(e^{\hat{T}_1}\hat{a}_2^\dagger e^{-\hat{T}_1}\right)\cdots\left(e^{\hat{T}_1}\hat{a}_m^\dagger e^{-\hat{T}_1}\right)|0\rangle. \tag{19.72}$$

You will examine further the transformation of each creation operator for a singles only ansatz in Problem 19.5.5 and show that it corresponds to a linear transformation on these operators.

Unlike the singles, which change the basis we are working with, the doubles operators add entanglement and couple the system to all higher-order terms. One can see this schematically, because evaluating the Hadamard lemma for $e^{\hat{T}_2}\hat{a}_i^\dagger e^{-T_2}$ will continue for many, many terms before it eventually truncates, and along the way, it will include an increasing number of fermionic operators for each subsequent nested commutator. This also provides a clear picture about how much more complex the higher-rank operator are when they act on a given state.

In summary, this coupled-cluster approach produces an efficient way to determine optimal low-rank approximations to the many-body quantum state. In cases where there is one leading contribution, and subsequent contributions involve excitations relative to the single contribution (so-called single-reference cases), the coupled-cluster approach is both accurate and efficient in determining a wide variety of ground-state properties. But, in cases where there are two or more quite different determinants with substantial weight in the many-body state, and the remaining important terms are excitations of both of these references, then we have a much more complex problem. Essentially, we want a different excitation operator $\hat{T}$ for

each reference state in order to have a low-rank representation, and this greatly complicates the calculation. We won't discuss such situations further here.

Section 19.5 Problems:

19.5.1 Why Does the Hadamard Lemma Truncate at Fourth Order?

Recall that the chemical Hamiltonian can be written as

$$\hat{H} = \sum_{rs} C^{(1)}_{rs} \hat{a}^\dagger_r \hat{a}_s + \sum_{rstu} C^{(2)}_{rstu} \hat{a}^\dagger_r \hat{a}^\dagger_s \hat{a}_t \hat{a}_u, \tag{19.73}$$

where the summation is over both real and virtual states, and the terms $C^{(1)}$ and $C^{(2)}$ are numbers.

Now, just consider what happens with each nested commutator of the Hadamard lemma acting on each of the terms in the Hamiltonian. Keep track of the largest number of $\hat{a}^\dagger$ terms that correspond to real orbitals and how many $\hat{a}$ terms correspond to virtual orbitals. Think about why only those cases are the important ones to keep track of. Use this to show the Hadamard lemma must truncate at fourth order.

Does this continue to truncate for all ranks of excitation operators?

19.5.2 How Does the Exponential Nature Used in Creating the Quantum State Allow It to Factorize into Two Neutral Atoms for a Stretched Diatomic Molecule

Write an essay explaining how the exponential form of the coupled-cluster ansatz allows for the many-body wavefunction to factorize into neutral atomic results on each atom of a diatomic molecule when it is stretched. You can assume the Hartree-Fock orbitals become more and more localized about each atom as the system is stretched.

19.5.3 Counting

Suppose we have N real spin orbitals, and M virtual spin orbitals. How many singles operators are there? How many doubles operators? How many triples? How many quadruples? If $N = 10$ and $M = 20$, you can see it the growth of the number of terms in each rank is so fast and that it rapidly becomes too many to be able to treat them all.

19.5.4 Explain Why the Coupled-Cluster Theory Is Not Variational

Write an essay about the coupled-cluster approximation and explain why the approach used to find and solve the amplitude equations cannot be thought of as a variational calculation. Can you identify what would need to be changed to make it variational?

19.5.5 Explain the Difference Between Singles Excitations and All Higher-Rank Excitations

Singles excitations act differently than all higher-rank excitations. Using the discussion in the main text, show how the singles-only ansatz will just change the basis used for the calculation, and keep the system in a single determinant, while using second-order or higher rank operators always creates more complicated states.

19.5.6 Discuss How One Can Calculate Energies Without Wavefunctions

Describe how one calculates energies without quantum states in your own words. Can you describe how it is a different procedure here than what we did for factorization?

19.5.7 How Does Coupled-Cluster Fail?

Eventually, when the many-body wavefunction becomes too strongly correlated, the coupled-cluster approach will fail. Write an essay explaining what is happening that is making it fail. You may want to consider the different assumptions that went into coupled-cluster theory and determine how they change when you are in the strongly-correlated limit.

19.5.8 When Are Multireference Initial States Needed

We have assumed in our work that we can use a single reference (determinant) and a low-rank representation of the quantum state in performing a quantum chemistry

calculation. But, there are some situations that require multiple references. Can you describe some circumstances that require multireference? Why is this case more complicated than the single-reference case?

Hint: Multireference situations often arise when one has strong correlations.

19.6 Unitary Coupled-Cluster Approximation and Quantum Computing

In the previous section, we learned about the coupled-cluster approximation, which is currently viewed as the "gold-standard" for quantum chemistry problems. This is not because it solves all problems, but because it usually is accurate enough, and efficient enough, that it can be applied to most molecules and will produce sufficiently accurate results and that it is all that is needed to complete a given calculation. But, it was known that coupled-cluster theory does not solve everything. In particular, it is poor for molecules that have some elements, such as transition metals, which have strong correlations, and other atoms, like oxygen, carbon, or hydrogen, which are not so strongly correlated. So chemists had to look for alternatives. The case we consider here is called *unitary coupled-cluster theory*, where we apply a unitary operator to the Hartree-Fock state in order to make the calculation variational, which can cure some of the challenges with coupled-cluster theory.

One of the suggested improvements is unitary coupled-cluster theory, first proposed by Kutzelnigg (1977) in the mid-1970s. The basic idea for this is rather simple. We know that the exponential of anti-Hermitian operators are unitary, so we must change $\hat{T}$ to an anti-Hermitian operator. This is achieved by taking $\hat{T} \rightarrow \hat{T} - \hat{T}^\dagger$. The operator is chosen in this form, instead of using operators multiplied by i, because this allows you to always work with real valued variables on the computer—but it is not clear whether such a restriction limits the states that can be represented in the theory.

While this seems like a simple change to the system, it is a profound one. We can no longer easily calculate the Hadamard lemma anymore, because the Hadamard lemma no longer truncates at fourth order (recall now that we have both creation and annihilation operators for both real and virtual orbitals in the unitary coupled-cluster theory). This means the only way one can do the calculation is variationally. To do this, we need to compute the quantum state. This makes these types of calculations much more difficult than conventional coupled-cluster theory.

This approach might have been just a mathematical curiosity studied by a handful of experts, but as quantum computers started to be built, it was recognized that only unitary coupled-cluster theory could be applied to these systems. This is because quantum computers can only apply unitary operations into their qubits (unless they are using mid-circuit resets). So, to work on a quantum computer, one is forced

into working with the unitary coupled-cluster theory. Early on, it was recognized that a variational quantum eigensolver approach (Peruzzo et al., 2014) could be employed here. This essentially applies the operator $e^{\hat{T}-\hat{T}^\dagger}$ onto the Hartree-Fock state, computes the energy of the Hamiltonian, and then optimizes the parameters to find the lowest energy. A quantum computer has even more constraints to it. In each clock cycle, we can apply parallel operations on all of the qubits, but they must be in terms of quantum operators that commute with each other. This condition requires us to work with the factorized form of the unitary coupled-cluster approximation, which takes the form

$$|\psi\rangle = \prod_{i}^{\text{real}} \prod_{a}^{\text{virtual}} e^{\theta_i^a \left(\hat{a}_a^\dagger \hat{a}_i - \hat{a}_i^\dagger \hat{a}_a\right)} \times \prod_{i}^{\text{real}} \prod_{a}^{\text{virtual}} e^{\theta_{ij}^{ab} \left(\hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_j \hat{a}_i - \hat{a}_i^\dagger \hat{a}_j^\dagger \hat{a}_b \hat{a}_a\right)} |\psi_{\text{HF}}\rangle. \tag{19.74}$$

One modification that we have to deal with is the individual excitation and de-excitation operators do not commute with each other. So, the operator with all terms in one exponent, and the operator with terms in individual factors, are no longer the same. We know that we can use a Trotter product formula to approximate the exponential with all terms in one exponent, and one can view the product of each factor once and only once as the first-order Trotter approximation. While this ordering problem sounds like it might be serious, we have to recall that we will be solving the problem variationally. This means that we might be able to trade off some of the problems associated with ordering issues with the variational principle. It is customary to try to determine which terms are the most important (for doubles, this can be done with a perturbation theory calculation) and apply them first and then apply the subsequent ones. For quantum computing, we have to use the factorized form, so this factorized form has been investigated intensively ever since quantum computers became a reality. The alternative to deal with the ordering problem is to introduce copies of particular single factors multiple times in the chain of operators that are used. In most cases, the system does not have any issues associated with the ordering problem—no matter what order is chosen, you can still optimize the final result.

One of the interesting discoveries in this problem is that there is a hidden SU(2) symmetry in the different factorized unitary coupled-cluster factors. We illustrate the identity directly here for rank-one operators following (Chen et al., 2021), and you will prove the result in general for all higher rank cases in the problems.

First define the "hidden" spin operators as

$$\hat{S}_+ = i\hat{a}_a^\dagger \hat{a}_i; \quad \hat{S}_- = (\hat{S}_+)^\dagger = -i\hat{a}_i^+ \hat{a}_a; \quad \hat{S}_z = \frac{1}{2}[\hat{S}_+, \hat{S}_-] = \frac{1}{2}(\hat{a}_a^\dagger \hat{a}_a - \hat{a}_i^\dagger \hat{a}_i). \tag{19.75}$$

You can see that this is the conventional representation of spin that we saw in the Hubbard model, if you map the virtual spin-orbital a to spin-up and the real spin-

orbital i to spin down. The commutation relations of these operators can then be immediately verified to be

$$[\hat{S}_+, \hat{S}_-] = 2\hat{S}_z, \qquad [\hat{S}_z, \hat{S}_+] = \hat{S}_+, \quad \text{and} \quad [\hat{S}_z, \hat{S}_-] = -\hat{S}_-. \tag{19.76}$$

This is the conventional $SU(2)$ algebra.

There are some subtleties to this, as these operators are not an independent $SU(2)$ algebra. Instead they arise as a subgroup of the permutation symmetry of all the generators of the allowed unitary coupled-cluster factors. Hence, they represent a direct sum of $S = 0$ and $S = 1/2$ representations when acting on any product state in the Hilbert space. We see this when we examine some additional operator identities given by

$$\begin{aligned} \hat{S}_+^2 = \hat{S}_-^2 = 0; \quad (\hat{S}_+ + \hat{S}_-)^2 = \hat{S}_+\hat{S}_- + \hat{S}_-\hat{S}_+; \\ (\hat{S}_+ + \hat{S}_-)^3 = \hat{S}_+\hat{S}_-\hat{S}_+ + \hat{S}_-\hat{S}_+\hat{S}_- = 2\hat{S}_z\hat{S}_+ - 2\hat{S}_z\hat{S}_- = \hat{S}_+ + \hat{S}_-, \end{aligned} \tag{19.77}$$

which are not general operator identities of the $SU(2)$ algebra, but are specific to this direct-sum space. These identities immediately imply that odd powers of $\hat{S}_+ + \hat{S}_-$ are equal to $\hat{S}_+ + \hat{S}_-$ and even powers are equal to $\hat{S}_+\hat{S}_- + \hat{S}_-\hat{S}_+$. These identities are similar to a spin-1 representation that we examined earlier, where the cube of the Cartesian angular momentum operators are equal to the Cartesian angular momentum operators. Indeed, this same identity allows us to evaluate the exponentials exactly—simply expand the exponential of the corresponding factor in a power series and use the fact that all nonzero even powers of the operator are the same and all odd powers of the operator are the same. Then the numerical factors can be immediately resummed to yield

$$\begin{aligned} \exp\left[-i\theta_i^a(\hat{S}_+ + \hat{S}_-)\right] &= 1 - i\theta_i^a(\hat{S}_+ + \hat{S}_-) + \frac{(i\theta_i^a)^2}{2}(\hat{S}_+ + \hat{S}_-)^2 \\ &\quad - \frac{(i\theta_i^a)^3}{3!}(\hat{S}_+ + \hat{S}_-)^3 + \cdots \\ &= 1 - i\sin\theta_i^a(\hat{S}_+ + \hat{S}_-) + (\cos\theta_i^a - 1)(\hat{S}_+\hat{S}_- + \hat{S}_-\hat{S}_+). \end{aligned} \tag{19.78}$$

For this single UCC excitation, inserting Eq. (19.75) into Eq. (19.78) yields

$$\exp\left[\theta_i^a(\hat{a}_i^a - \hat{a}_a^i)\right] = 1 + \sin\theta_i^a(\hat{a}_i^a - \hat{a}_a^i) + (\cos\theta_i^a - 1)(\hat{n}_a + \hat{n}_i - 2\hat{n}_a\hat{n}_i), \tag{19.79}$$

where $\hat{n}_\gamma = \hat{a}_\gamma^\dagger\hat{a}_\gamma$ is the number operator for $\gamma = a$ or i. In Problem 19.6.3, you generalize this to arbitrary rank excitation and de-excitation operators. This then

allows us to use these identities when we calculate the quantum state for the unitary coupled-cluster ansatz.

The strategy for how to carry out a calculation are now clear. We do the following process:

1. Determine the ordering of the factors to be used. For doubles, this is done by comparing to a perturbation theory calculation, for singles, one just picks any order.
2. For each factor, we scroll through each term in the quantum state already, for each term, we determine whether it can be excited or de-excited. If it can, we use the SU(2) formula to create two terms from the initial term. If it cannot, we do nothing.
3. Repeat for each factor in the ansatz.
4. Once the quantum state has been constructed, calculate the expectation value of the Hamiltonian.
5. Optimize the parameters until the solution is accurate enough to stop.

There are a few subtleties to this. In a worst-case scenario, the quantum state doubles in size with each new factor. This rapidly creates a state that is too large to store on a conventional computer. We deal with this in three ways. First, not all states double, so the true multiplier is less than 2. Second, some of the states created are duplicates of ones already there. When this happens, we can combine the results and save on the storage. Third, if the amplitudes are too small, we can neglect them. The other issue is that we need to optimize the solutions. This can be a time-consuming step, and we do not describe the details of how to do that here.

The main point of this work is that on a quantum computer, we do not have any issues with the memory troubles on a conventional computer. Quantum computers have immense memory and are very efficient in storing quantum states. This is why using a quantum computer is viewed positively for these types of calculations. But, current hardware is not good enough yet to be competitive with conventional hardware. Someday it will be.

Now, we illustrate how this approach works on a nontrivial example given by the H_{10} chain in Fig. 19.8. This system is a useful test case, because it can produce accurate results with a minimal basis and it becomes strongly correlated when it is stretched. The main result to take away from this figure is that the unitary coupled-cluster method is competitive with the coupled-cluster method, but it works for much longer (well after the coupled-cluster method fails due to the correlations becoming too strong).

The other issue we wish to discuss is about the angles in the optimal ansatz. As we discussed earlier, in chemistry, we often have situations where there are a large number of small contributions, which accumulate to a large contribution to the total energy. This would mean that we would expect many of the angles in the ansatz to be small. Then, as the correlations increase, the angles should become larger. This result is exactly what we see in Fig. 19.9. The fact that most of the larger angles are to the left in the figure (except for the singles on the far right) is because the Møller-Plesset perturbation theory is indeed predicting the ordering of the factors

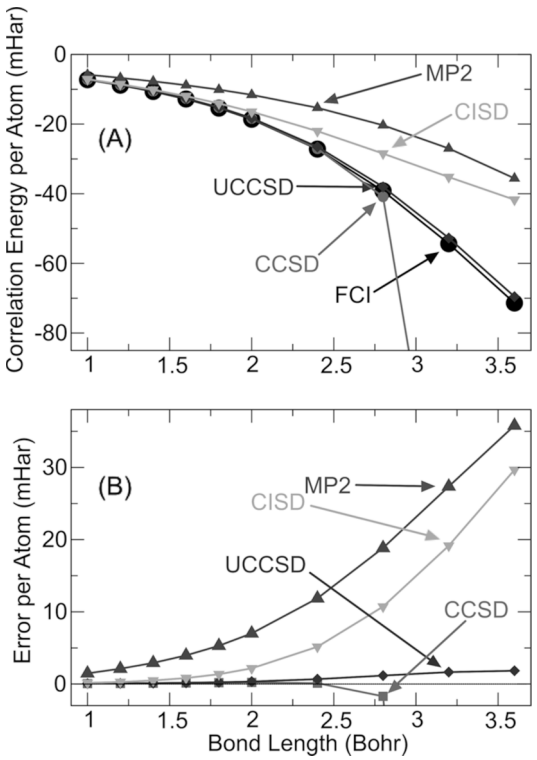

Fig. 19.8 (**a**) Comparison of different computational methods for a linear hydrogen chain of ten atoms as it is uniformly stretched. (**b**) Difference in the energy of the different methods from the exact result. The different methods are Møller-Plesset second-order perturbation theory (MP2), configuration interaction (CI), coupled-cluster singles and doubles (CCSD), unitary coupled-cluster singles and doubles (UCCSD), and full configuration interaction (FCI) or exact diagonalization. The units are in terms of Bohr radii a_0 for the length and milliHartrees (0.0272 eV) for the energies. One can see the UCCSD is competitive with the CCSD for short lengths and then becomes better after CCSD fails for longer-length scales (This figure is used with permission from Chen et al. (2021) © American Chemical Society. It has been modified to grayscale)

properly. When the angles are very small, one can try to approximate their effects using perturbation theory methods, and this is an active area of research.

The unitary coupled-cluster approximation is definitely different from the conventional coupled-cluster method. But, is it possible to find an exact relationship between the two? The fact that there is a hidden SU(2) symmetry suggests that we might be able to do so. There should be similar disentangling identities that will allow us to disentangle the unitary coupled-cluster factors and convert them to the conventional coupled-cluster factors. Indeed, this is possible, and we describe how it works in the next section.

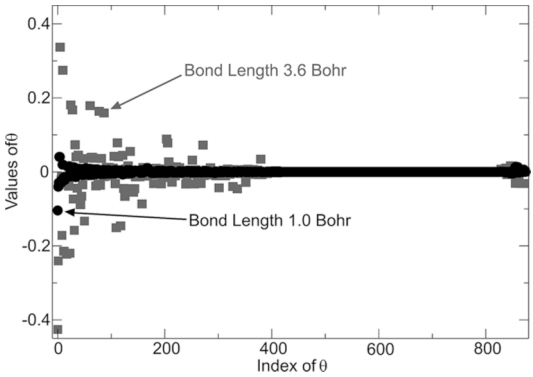

Fig. 19.9 Angles versus index of factor in the ansatz. The black circles are for small H-H spacing, and the gray squares are from the stretched molecule, which is strongly correlated. One can clearly see the angle values are much larger for the gray symbols than the black symbols. But even in the gray case, many angles remain small (This figure is used with permission from Chen et al. (2021) © American Chemical Society. It has been modified to grayscale)

Section 19.6 Problems:

19.6.1 Why Does the Hadmard Lemma No Longer Truncate?

In the conventional coupled-cluster approximation, we could show that the Hadamard lemma truncated after at most the fourth order. Show that for the unitary coupled-cluster approximation, the Hadamard lemma never truncates.

19.6.2 Verify the Commutation Relations for the Rank-One SU(2) Algebra

Verify the SU(2) commutation relations for the rank-one unitary coupled-cluster operators, as described in the text.

19.6.3 Determine the Hidden SU(2) Symmetry for Arbitrary Ranks

Verify the SU(2) symmetry and the generalized Euler identity for an arbitrary rank unitary coupled-cluster factor. Use

$$\hat{S}_+ = i\hat{a}^{a_1 a_2 \cdots a_n}_{i_1 i_2 \cdots i_n}, \quad \hat{S}_- = -i\hat{a}^{i_1 i_2 \cdots i_n}_{a_1 a_2 \cdots a_n}, \quad \text{and}$$
$$\hat{S}_z = \frac{1}{2}\left(\hat{n}_{a_1} \ldots \hat{n}_{a_n}(1-\hat{n}_{i_1}) \cdots (1-\hat{n}_{i_n}) - (1-\hat{n}_{a_1}) \ldots (1-\hat{n}_{a_n})\hat{n}_{i_1} \cdots \hat{n}_{i_n}\right). \tag{19.80}$$

Show that these spin operators satisfy the $SU(2)$ commutation relations in Eq. (19.76) and also satisfy the additional operator identities in Eq. (19.77).

Use these results to show that the arbitrary rank operator satisfies the final exact operator identity:

$$\begin{aligned} U^{a_1 \ldots a_n}_{i_1 \ldots i_n} = \exp\left[\theta\left(\hat{a}^{a_1 \ldots a_n}_{i_1 \ldots i_n} - \hat{a}^{i_1 \ldots i_n}_{a_1 \ldots a_n}\right)\right] &= 1 + \sin\theta\left(\hat{a}^{a_1 \ldots a_n}_{i_1 \ldots i_n} - \hat{a}^{i_1 \ldots i_n}_{a_1 \ldots a_n}\right) \\ &+ (\cos\theta - 1)[\hat{n}_{a_1} \ldots \hat{n}_{a_n}(1-\hat{n}_{i_1}) \ldots (1-\hat{n}_{i_n}) \\ &+ (1-\hat{n}_{a_1}) \ldots (1-\hat{n}_{a_n})\hat{n}_{i_1} \ldots \hat{n}_{i_n}]. \end{aligned} \tag{19.81}$$

This identity also acts in a direct sum space of $S = 0$ and $S = 1/2$: when $S = 0$, which happens when no excitation or de-excitation is possible, the operator acts as the identity, but when $S = 1/2$, which happens when an excitation or de-excitation is possible, the operator acts analogous to a spin-one-half state, which is rotated by the angle θ.

19.6.4 Using a Tree Structure for the Calculation of the Unitary Coupled-Cluster Calculation on a Conventional Computer

The exact operator identity takes a term in the current quantum state and either leaves it alone or takes the state, multiplies it by $\cos\theta$, and multiplies the excited or de-excited state by $\sin\theta$, and both are added to the new state. This implies that we can map the action of a unitary coupled-cluster factor onto a many-body state via a tree-like structure, where a node, corresponding to the current state, is either translated down unchanged to the next level, or it bifurcates into two leaves, one weighted by the cosine and the other by the sine.

Write an essay describing the rules for how you can create this tree structure. Consider as well how you can merge leaves that are identical.

19.6.5 Details for How to Apply a Unitary Coupled-Cluster Factor on a Quantum Computer

We discuss one way to calculate the unitary coupled-cluster factor on a quantum computer. We will not go into the full details, just schematically show how it works.

In a quantum computer, we map $\hat{a}_i^\dagger \rightarrow \sigma_i^x - i\sigma_i^y$ and $\hat{a}_i \rightarrow \sigma_i^x + i\sigma_i^y$. Using this mapping, show that

$$\hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_j \hat{a}_i \rightarrow 2i \left(\sigma_a^y \sigma_b^x \sigma_j^x \sigma_i^x + \sigma_a^x \sigma_b^y \sigma_j^x \sigma_i^x - \sigma_a^x \sigma_b^x \sigma_j^y \sigma_i^x - \sigma_a^x \sigma_b^x \sigma_j^x \sigma_i^y \cdots \right), \tag{19.82}$$

where the dots indicate the terms with three σ^y factors and one σ^x factor.

Show that each of the individual strings of Pauli matrices commute with each other. Do this by showing different Pauli matrices on the same site anticommute. If we have an even number of anticommuting terms, then the whole product commutes. Because they commute, we can evaluate each factor separately. For a rank-n operator, how many factors do we need to evaluate?

A quantum computer can evaluate the operator $e^{2i\theta\sigma_a^z\sigma_b^z\sigma_j^z\sigma_i^z}$. It does this by using strings of control-not operations, but we need not go into those details here. Instead, we focus on how to convert the strings of x and y Pauli matrices into a string of z Paulis. We can do this by multiplying on the left and the right by opposite exponential factors of the right Pauli matrix with the right angle and then use the braiding relation and the Hadamard lemma to determine the proper operators. Find the operators to convert σ^x to σ^z and σ^y to σ^z.

Schematically describe how such a calculation is done on a quantum computer. How much harder is a rank-n calculation compared to a rank-$n+1$ calculation?

Note that there are more efficient ways to do this calculation on a quantum computer now, but they go beyond what we will discuss here.

19.6.6 *Using the Unitary Coupled-Cluster Approximation to Create the Exact Ground State of a Four-Site Hubbard Model on a Square Lattice*

We consider the four-site Hubbard model on a square lattice. We work in momentum space, so that the states are labeled by the momentum $\frac{j\pi}{2}$ for $j = 0,\ 1,\ 2,\ 3$. The ground state is given by the following form:

$$\begin{aligned} |\psi_{\mathrm{GS}}\rangle = {} & \alpha(|0\uparrow 1\uparrow 0\downarrow 1\downarrow\rangle - |0\uparrow 3\uparrow 0\downarrow 3\downarrow\rangle) + \beta(|0\uparrow 1\uparrow 2\downarrow 3\downarrow\rangle \\ & + |0\uparrow 3\uparrow 1\downarrow 2\downarrow\rangle + |1\uparrow 2\uparrow 0\downarrow 3\downarrow\rangle + |2\uparrow 3\uparrow 0\downarrow 1\downarrow\rangle \\ & + (2)|0\uparrow 2\uparrow 1\downarrow 3\downarrow\rangle + (2)|1\uparrow 3\uparrow 0\downarrow 2\downarrow\rangle) \\ & + \gamma(|2\uparrow 3\uparrow 2\downarrow 3\downarrow\rangle - |1\uparrow 2\uparrow 1\downarrow 2\downarrow\rangle). \end{aligned} \tag{19.83}$$

Note how the states with β and γ coefficients are all created from the states with an α coefficient.

Use the following strategy to create the final state:

1. Start from the state $|0\uparrow 1\uparrow 0\downarrow 1\downarrow\rangle$ and apply two doubles operators to create the states $|0\uparrow 2\uparrow 1\downarrow 3\downarrow\rangle$ and $|1\uparrow 3\uparrow 0\downarrow 2\downarrow\rangle$.
2. You will find that you have also created the state $|2\uparrow 3\uparrow 2\downarrow 3\downarrow\rangle$. Use a quad term to remove this state, without affecting the other two.
3. Create the partner of the state with the α coefficient.
4. Create the remaining states, by using doubles operators that work on the states multiplied by α, applying doubles to each of them, in turn, twice, to finish the calculation.

This problem shows you that proper use of the unitary coupled-cluster approach can prepare exact states if applied in the right order and with the right operators.

19.7 Relationship Between Coupled Cluster and Unitary Coupled Cluster

In this section, we will work out the relationship between the unitary coupled cluster and the conventional coupled cluster. We will find that a low-rank representation of unitary coupled cluster does not usually correspond to a low-rank representation of the coupled-cluster approximation. This is the main result of this section, as it clearly shows the two approaches must be different from each other. They only start to resemble each other when all of the angles are small. Then the higher-order terms become less important, and the differences can often be neglected.

If all we care about is the relationship between the ansatzes, there is a simple procedure to convert from a unitary coupled-cluster ansatz to a coupled-cluster ansatz, but not the other way around. One first applies all unitary coupled-cluster factors to the reference state and expands out the entire quantum state. Next, we use the coupled cluster and the excitation corresponding to the component in each singles operator in $\hat{T}_1$ to identify the amplitude of each singles excitation. Then, we subtract $e^{\hat{T}_1}|\psi_{\text{HF}}\rangle$ from the unitary ansatz. This will modify the coefficients of some of the doubles and higher excitations. We continue to do the same thing with the doubles excitations and then the triples and so on, until we have identified every term in the coupled-cluster expansion. This is a straightforward but tedious procedure, and it provides us with no information about the operator relationship between the two. So, we now formulate how to do this via an operator identity. The discussion closely follows Freericks (2022b).

We start by defining the hidden "effective" spin operators for a rank-n unitary coupled-cluster factor via

$$\hat{S}_+ = i\underbrace{\hat{a}_a^\dagger\hat{a}_b^\dagger\cdots}_{n\text{ terms}}\underbrace{\cdots\hat{a}_j\hat{a}_i}_{n\text{ terms}} \quad\text{and}\quad \hat{S}_- = \hat{S}_+^\dagger = -i\underbrace{\hat{a}_i^\dagger\hat{a}_j^\dagger\cdots}_{n\text{ terms}}\underbrace{\cdots\hat{a}_b\,\hat{a}_a}_{n\text{ terms}}. \tag{19.84}$$

We then define $\hat{S}_z$ via $[\hat{S}_+, \hat{S}_-] = 2\hat{S}_z$, which gives us

$$\hat{S}_z = \frac{1}{2}\Big(\underbrace{\hat{n}_a \hat{n}_b \ \cdots}_{n \text{ terms}} \ \underbrace{\cdots (1-\hat{n}_j)(1-\hat{n}_i)}_{n \text{ terms}} - \underbrace{\hat{n}_i \hat{n}_j \cdots}_{n \text{ terms}} \ \underbrace{\cdots (1-\hat{n}_b\ )(1-\hat{n}_a)}_{n \text{ terms}} \Big). \tag{19.85}$$

Here the number operator is schematically given by $\hat{n} = \hat{a}^\dagger \hat{a}$ and the z-component of the effective spin is the difference of two projection operators—one onto states that can be excited and one onto states that can be de-excited. We also have that $[\hat{S}_z, \hat{S}_\pm] = \pm \hat{S}_\pm$, which establishes the SU(2) symmetry for these operators.

The Pauli exclusion principle requires $\hat{a}^2 = (\hat{a}^\dagger)^2 = 0$, which produces additional identities given by

$$(\hat{S}_\pm)^2 = 0, \quad \hat{S}_+ \hat{S}_- \hat{S}_+ = \hat{S}_+, \quad \text{and} \quad \hat{S}_- \hat{S}_+ \hat{S}_- = \hat{S}_-. \tag{19.86}$$

Any unitary coupled-cluster factor can then be written as

$$e^{-i\theta^{abc\cdots}_{ijk\cdots}(\hat{S}_+ + \hat{S}_-)} = \sum_{n=0}^{\infty} \frac{\left(-i\theta^{abc\cdots}_{ijk\cdots}\right)^n}{n!} (\hat{S}_+ + \hat{S}_-)^n. \tag{19.87}$$

The identities in Eq. (19.86) then tell us that $(\hat{S}_+ + \hat{S}_-)^2 = (\hat{S}_+)^2 + \hat{S}_+\hat{S}_- + \hat{S}_-\hat{S}_+ + (\hat{S}_-)^2$, which is equal to $\hat{S}_+\hat{S}_- + \hat{S}_-\hat{S}_+$. Hence, $(\hat{S}_+ + \hat{S}_-)^3 = (\hat{S}_+\hat{S}_- + \hat{S}_-\hat{S}_+)(\hat{S}_+ + \hat{S}_+)$, which is $\hat{S}_+ + \hat{S}_-$. So, we have discovered that

$$(\hat{S}_+ + \hat{S}_-)^n = \begin{cases} \mathbb{I}, & \text{if } n = 0 \\ \hat{S}_+ + \hat{S}_-, & \text{if } n = \text{odd} \\ \hat{S}_+\hat{S}_- + \hat{S}_-\hat{S}_+, & \text{if } n = \text{even and positive.} \end{cases} \tag{19.88}$$

This allows us to explicitly sum the series, giving us

$$e^{-i\theta^{abc\cdots}_{ijk\cdots}(\hat{S}_+ + \hat{S}_-)} = \hat{\mathbb{I}} - i \sin\theta^{abc\cdots}_{ijk\cdots}(\hat{S}_+ + \hat{S}_-) + \left(\cos\theta^{abc\cdots}_{ijk\cdots} - 1\right)(\hat{S}_+\hat{S}_- + \hat{S}_-\hat{S}_+). \tag{19.89}$$

Because these operators satisfy an SU(2) algebra, we already know that we also have an exponential disentangling identity, given by

$$\begin{aligned} e^{-i\theta^{abc\cdots}_{ijk\cdots}(\hat{S}_+ + \hat{S}_-)} &= e^{-i\tan\theta^{abc\cdots}_{ijk\cdots}\hat{S}_+} e^{-2\ln\left(\cos\theta^{abc\cdots}_{ijk\cdots}\right)\hat{S}_z} e^{-i\tan\theta^{abc\cdots}_{ijk\cdots}\hat{S}_-} \\ &= e^{-i\tan\theta^{abc\cdots}_{ijk\cdots}\hat{S}_+} e^{-\ln\left(\cos\theta^{abc\cdots}_{ijk\cdots}\right)(\hat{S}_+\hat{S}_- - \hat{S}_-\hat{S}_+)} e^{-i\tan\theta^{abc\cdots}_{ijk\cdots}\hat{S}_-}. \end{aligned} \tag{19.90}$$

This separates out the excitation operators to the left, the difference of the two projection operators in the center, and the de-excitation operators to the right. You will prove this identity using just the effective spin operators and their properties in Problem 19.7.1.

The case $\theta = \pm\frac{\pi}{2}$ can present an issue for the disentangling identity due to the divergence of the tangent at half-odd integer multiples of π. But, in general, we do not expect there to ever be an angle that satisfies $\theta = \frac{\pi}{2}$, because in such a case, the reference state is *removed* from the wavefunction, when this term is applied ($\cos\theta = 0$ implies no reference state remains, as shown by the generalized Euler formula). Since the reference is supposed to be a large amplitude term in the final wavefunction, we would never use such a large angle. Indeed the magnitude of the angles are usually not larger than $\frac{\pi}{4}$, so we simply assume the case of $\pm\frac{\pi}{2}$ does not occur.

We want to make one additional comment about the disentangling identity. Note, how the angles for the unitary coupled-cluster approximation are restricted to lie between $-\pi/2 \leq \theta \leq \pi/2$, whereas the coupled-cluster amplitudes can have any real value. This is exhibited clearly in the disentangling identity, as an angle (which becomes a cosine or a sine using the $SU(2)$ identity) is replaced by a $\tan\theta$ in the disentangling identity, *showing it can take any real value*. We want to stress this observation, because it makes sense as to why the identity has the form it has. Finally, because we anticipate most angles have their magnitude bound by $\pi/4$ instead of $\pi/2$, this indicates that in a coupled-cluster approximation, most amplitudes should be less than 1 in magnitude.

The way the disentangling identity is used to simplify the factorized form of the unitary coupled-cluster ansatz is that we replace each factor of the form $e^{-i\theta(\hat{S}_+ + \hat{S}_-)}$, with the first line of Eq. (19.90). This separates the excitation operators from the de-excitation operators, but it is not yet in the normal-ordered form (with all $\hat{S}_+$ to the left and all $\hat{S}_-$ to the right; we move the projection operator to the right, to the extent that we can, as well). In this normal-ordered form, we have only the coupled-cluster operators left. Note that all other operators will annihilate against $|\psi_{\mathrm{HF}}\rangle$ or can be immediately evaluated as numbers. Recall as well that the effective spin operators depend on each specific excitation and de-excitation operator used in a particular factor (we suppressed this notation with indices corresponding to the virtual and real spin orbitals involved above), so one must convert from the effective spin notation back to the fermionic creation and destruction operators (including all relevant indices) before putting the product of operators into normal-ordered form.

Our next step is to determine how to re-order the different exponential factors. This is, of course, done with the Hadamard lemma and the braiding relation. In principle, it is straightforward now to evaluate all of the re-orderings, but it is quite cumbersome to do so. The notation needed to describe these commutations is also cumbersome, so we need to develop new notation in order to sketch how we carry out the re-orderings. Every term that we work with has the form of some number of creation operator factors and some number of destruction operator factors, always organized so the creation operators are to the left of the destruction operators, except

in the projection operators, where half of the terms are in the opposite order. But unlike the original operators where the raising and lowering operators are always grouped into virtual or real spin-orbitals, after we have re-ordered exponentials, we will generate operators where the indices are *mixed* between the two groups. This is why we need a more general notation to describe the re-orderings.

We use the notation

$$\hat{A}(a_1, \cdots, a_n; b_1, \cdots, b_n | c_1, \cdots, c_m; d_1, \cdots, d_{m'}) = \underbrace{\hat{a}^\dagger_{a_1} \cdots \hat{a}^\dagger_{a_n}}_{n \text{ terms}} \underbrace{\hat{a}_{b_n} \cdots \hat{a}_{b_1}}_{n \text{ terms}} \underbrace{\hat{n}_{c_1} \cdots \hat{n}_{c_m}}_{m \text{ terms}} \underbrace{(1 - \hat{n}_{d_1}) \cdots (1 - \hat{n}_{d_{m'}})}_{m' \text{ terms}}, \tag{19.91}$$

to describe the operators that we need to work with. The general operator is broken into two halves. The left half includes the isolated creation and annihilation operators, while the right half includes the projection operators—the number of creation operators is always the same as the number of destruction operators—hence the total number of creation plus destruction operators in each product string is even; the ordering of the destruction operators is reversed, just as is done in the excitation and de-excitation operators. Note that it is best to represent $\hat{n}$ as $\hat{a}^\dagger \hat{a}$ and $1 - \hat{n}$ as $\hat{a}\hat{a}^\dagger$, so that each operator $\hat{A}$ is a product of creation and destruction operators, but it is often not normal ordered (e.g., when $n \neq 0$ and m or m' are nonzero).

This operator notation can describe a *pure excitation* when $m = m' = 0$ and $\{a_1, \cdots, a_n\}$ are all virtual spin orbitals and $\{b_1, \cdots, b_n\}$ are all real spin orbitals. It is a *pure de-excitation* when $m = m' = 0$ and $\{a_1, \cdots, a_n\}$ are all real spin orbitals and $\{b_1, \cdots, b_n\}$ are all virtual spin orbitals. Similarly, when $n = 0$ and $m \neq 0$ and/or $m' \neq 0$, the operator is a *pure projection operator*. If the operator is not a pure operator, it is a *mixed operator*. Such a mixed operator mixes excitation and de-excitation between different spin orbitals in the same operator. In cases when $m \neq 0$ and/or $m' \neq 0$, the operator is said to be *with projection*. If $m = m' = 0$ it is *without projection*.

It is useful to describe some rules about these operators. If any $a_i = a_j$, any $b_i = b_j$, any $a_i = c_j$, any $b_i = d_j$, or any $c_i = d_j$, then the operator vanishes because it has the square of a creation or destruction operator or a product $\hat{n}(1 - \hat{n}) = 0$. If any c or d index repeats, the repeating index can be removed. If any $a_i = d_j$, the d_j index can be removed (because $\hat{a}^\dagger_i (1 - \hat{n}_i) = \hat{a}^\dagger_i$) and if any $b_i = c_j$, the c_j index can be removed (because $\hat{a}_i \hat{n}_i = \hat{a}_i$). Finally, if any $a_i = b_j$, we move it into the c indices. We say the operator is in *canonical form* if all of these "contractions" have been applied to the operator. In this case, all indices that are in the operator are different. In addition, when in canonical form, all of the indices in each grouping a, b, c, and d has the index values in each subgroup ordered, so that $a_1 < a_2 < a_3 \cdots$ and similarly for the other three sets of indices.

Before we start re-ordering the exponential factors to place the products in an "excitation-only" form, every operator in the exponential is a pure excitation without projection, a pure de-excitation without projection, or a pure projection,

because this is the form of the exponential disentangling identity for each unitary coupled-cluster factor. As we re-order operators, this changes, and many operators become mixed and with projection. But, there are some simple rules for how the re-ordering changes terms. These rules are rather tedious to carry out "by hand," but are straightforward to implement on a computer.

The re-ordering of exponential terms always involves interchanging the order of an $\hat{A}$ operator and an $\hat{A}'$ operator, in the form

$$e^{\alpha\hat{A}}e^{\alpha'\hat{A}'} = e^{e^{\alpha\hat{A}}\alpha'\hat{A}'e^{-\alpha\hat{A}}}e^{\alpha\hat{A}}, \tag{19.92}$$

where the operators $\hat{A}$ and $\hat{A}'$ can both be assumed to be in canonical form. In most cases, the Hadamard lemma truncates after a finite number of nested commutators, for the following reasons. First, the commutator vanishes if $\{a_i \mid i = 1, \cdots, n\} \cap \{a'_i \mid i = 1, \cdots, n'\} \neq 0$ or $\{b_i \mid i = 1, \cdots, n\} \cap \{b'_i \mid i = 1, \cdots, n'\} \neq 0$ and no a_i index is in $\{b'_i\}$ and no b_i is in $\{a'_i\}$, because if there is a common index in both operators, the canonical form guarantees there is no c or d index that is the same, so both $\hat{A}$ and $\hat{A}'$ have the same creation operator in them, or the same destruction operator in them, but not both. This means their product vanishes in either order (because the square of a fermionic operator is zero). So, we assume these two sets have no intersection. Then the commutator $[\hat{A}, \hat{A}'] \neq 0$ if and only if there is at least one of the following:

$$\exists\ a_i = b'_j \ \text{ or } \ a_i = c'_j \ \text{ or } \ a_i = d'_j \tag{19.93}$$

$$\exists\ b_i = a'_j \ \text{ or } \ b_i = c'_j \ \text{ or } \ b_i = d'_j. \tag{19.94}$$

$$\exists\ c_i = a'_j \ \text{ or } \ c_i = b'_j \tag{19.95}$$

$$\exists\ d_i = a'_j \ \text{ or } \ d_i = b'_j \tag{19.96}$$

Because the operators are in canonical form, only one of the possibilities (for any specific i) in any line can occur, but there can be more than one i that satisfies this condition for any line and we can have the condition satisfied on more than one line.

Let us carefully look at the cases. If Eq. (19.93) holds, then if the match is with c'_j or d'_j, every term in the commutator has an $a^\dagger_{a_i}$ factor in the product of operators (after putting the commutator into canonical form), which annihilates with $\hat{A}$ when multiplied on either side. This means only the first commutator is nonzero and all higher-order ones vanish. The same is true for Eq. (19.94) if the match is with c'_j or d'_j. If Eq. (19.95) holds we have two different possible behaviors. First, if $\{a_i \mid i = 1, \cdots, n\} \subset \{a'_i \mid i = 1, \cdots, n'\}$ and $\{b_i \mid i = 1, \cdots, n\} \subset \{b'_i \mid i = 1, \cdots, n'\}$, then in each term in the commutator, all of the unpaired creation and annihilation operators in $\hat{A}$ are paired into $\hat{n}$ or $1 - \hat{n}$ factors. In this case, the nested commutators go on for an infinite number of terms, but they can all be summed into an exponential factor that multiplies the result. But, if both sets for the indices of the creation and destruction operators of $\hat{A}$ are not also in the labels for $\hat{A}'$, then

we always have at least one creation or annihilation operator left over in every term in the commutator—in this situation, $\hat{A}$ annihilates the operator when multiplied to the left or the right, and the expansion truncates after the first commutator. We have a similar result for Eq. (19.96). The remaining case occurs when the match is only between the creation and annihilation operators in Eqs. (19.93) and (19.94).

In this case, the number operators commute with everything, so we need not worry about them anymore. Then we have at most two nested commutators. Consider the first index i, such that $a_i = b'_j$. After the first commutator, one term in the commutator has no $a^\dagger_{a_i}$ and $\hat{a}_{a_i}$ terms in it, while the rest have either a $\hat{n}_{a_i}$ term or a $1 - \hat{n}_{a_i}$ term. After the second nested commutator, all terms will have a $\hat{a}_{a_i}$ term in it, which then vanishes when multiplied by $\hat{A}$ on either side for the third nested commutator. We have a similar argument if the only indices that are the same are $b_i = a'_j$. The second commutator only enters in situations where both $\{a_i \mid i = 1, \cdots, n\} \subset \{b'_i \mid i = 1, \cdots, n'\}$ and $\{b_i \mid i = 1, \cdots, n\} \subset \{a'_i \mid i = 1, \cdots, n'\}$, as before, since, otherwise, we have a lone fermionic operator remaining in all terms of the commutator, which then vanishes when multiplied by $\hat{A}$ on the right or on the left. So, the reordering is always of the form

$$e^{\alpha \hat{A}} e^{\alpha' \hat{A}'} = e^{\alpha' \hat{A}' + \alpha\alpha' [\hat{A}, \hat{A}']} e^{\alpha \hat{A}}, \tag{19.97}$$

when the a and b indices of the operator $\hat{A}$ do not entirely lie within the b' and a' indices of $\hat{A}'$. We use the following terminology: when the a and b indices of $\hat{A}$ are a subset of the b' and a' indices of $\hat{A}'$, we say $\hat{A}$ *matches* $\hat{A}'$; if they are not both subsets, we say $\hat{A}$ *does not match* $\hat{A}'$.

If $\hat{A}'$ does not match $\hat{A}$, then the commutator $[\hat{A}, \hat{A}']$ has lone creation or annihilation operators from $\hat{A}'$ in it, meaning $\hat{A}'$ annihilates it when multiplied to the right or to the left (after putting the commutator into canonical form). Hence, $\hat{A}'$ and $[\hat{A}, \hat{A}']$ commute and we can separate the factors into

$$e^{\alpha \hat{A}} e^{\alpha' \hat{A}'} = e^{\alpha' \hat{A}'} e^{\alpha\alpha' [\hat{A}, \hat{A}']} e^{\alpha \hat{A}}. \tag{19.98}$$

The re-ordering has now been put into a form of exponentials of individual A-form operators. If $\hat{A}'$ matches $\hat{A}$, then the commutator $[\hat{A}', [\hat{A}, \hat{A}']]$ has lone creation or annihilation operators in it (following similar arguments as given above), so it commutes with $\hat{A}'$, but it does not necessarily commute with $[\hat{A}, \hat{A}']$ (which has no lone fermionic operators from $\hat{A}'$). In this case, there is no simple way to separate the terms, as nested commutators of arbitrary order will be nonzero. But, of course, it can be put into an infinite product of exponentials of individual A-form operators, by use of the Zassenhaus formula (and many of the nested commutators in that formula will vanish). Because the commutator of a projection operator with a fermionic operator is proportional to the fermionic operator, the nested commutators will repeat in form, and in the end, these infinite factors can all be combined with a new numerical factor multiplying the corresponding operator. We try to separate the two terms in Eq. (19.98) when $\hat{A}'$ matches $\hat{A}$. If we let $\hat{A}' \to \hat{A}$ and $[\hat{A}, \hat{A}'] \to \hat{B}$

in the Baker-Campbell-Hausdorff formula, then, because $[\hat{A}, [\hat{A}, \hat{B}]] = 0$, all terms including this nested commutator vanish in the expansion. This will simplify the expression, but it is not enough for us to determine the required Zassenhaus factorization of the expression, without significant additional computation.

Now, if $\hat{A}$ does match $\hat{A}'$, then the exponential re-ordering identity takes the form

$$e^{\alpha\hat{A}}e^{\alpha'\hat{A}'} = e^{\alpha'\hat{A}'+\alpha\alpha'[\hat{A},\hat{A}']+\frac{1}{2}\alpha^2\alpha'[\hat{A},[\hat{A},\hat{A}']]}e^{\alpha\hat{A}}. \tag{19.99}$$

If $\hat{A}'$ does not match $\hat{A}$, then we have lone fermionic operators from $\hat{A}'$ in each of the commutators, which means the three separate operator terms in the exponent, all mutually commute, and we can re-write the exponential re-ordering identity as

$$e^{\alpha\hat{A}}e^{\alpha'\hat{A}'} = e^{\alpha'\hat{A}'}e^{\alpha\alpha'[\hat{A},\hat{A}']}e^{\frac{1}{2}\alpha^2\alpha'[\hat{A},[\hat{A},\hat{A}']]}e^{\alpha\hat{A}}. \tag{19.100}$$

The case where $\hat{A}'$ also matches $\hat{A}$ is again complicated and we have to use the infinite-order Zassenhaus formula again, perhaps recursively, because there are three terms in the exponent. In this case, the operators may not reduce to a simple form.

Finally, we still have to work out the cases from Eqs. (19.95) and (19.96) when $\hat{A}$ matches $\hat{A}'$. To do this, we first need to determine the general similarity transformation

$$\begin{aligned}\hat{\mathcal{O}}(\alpha) = \exp\left(\alpha\prod_{i=1}^{m+n}\hat{n}_i\prod_{i=m+n+1}^{m+n+m'+n'}(1-\hat{n}_i)\right)\\ \left(\prod_{i=1}^{m}\hat{a}_i^\dagger\prod_{i=m+1}^{m+n}\hat{a}_i\prod_{i=m+n+1}^{m+n+m'}\hat{a}_i^\dagger\prod_{i=m+n+m'+1}^{m+n+m'+n'}\hat{a}_i\right)\\ \times\exp\left(-\alpha\prod_{i=1}^{m+n}\hat{n}_i\prod_{i=m+n+1}^{m+n+m'+n'}(1-\hat{n}_i)\right).\end{aligned} \tag{19.101}$$

An analysis based on calculus then tells us that $\hat{\mathcal{O}}(\alpha)$ is either unchanged or multiplied by a factor of $e^{\pm\alpha}$ after the re-ordering of the operators. This correction typically changes an amplitude by a factor of a secant of another amplitude; in many cases the secant is raised to an operator exponent.

It is time to summarize this complex algebra. What we have found is that in cases where neither $\hat{A}$ matches $\hat{A}'$ nor $\hat{A}'$ matches $\hat{A}$, the exponential re-ordering is simple because it involves evaluating one commutator and requires no additional operator manipulations. If $\hat{A}$ matches $\hat{A}'$, but $\hat{A}'$ does not match $\hat{A}$, then we have two commutators to evaluate. But if both operators match in both directions, the formula is complicated and might not even be able to be written down analytically. This raises the question, can such a situation occur? In general, it seems like it is a

rare occurrence, if it happens at all, but without looking at some examples, we will not be able to tell. In cases where this does not occur, we have established the rules needed to perform these algebraic manipulations, but they are clearly complicated to try to carry out "by hand." However, they are straightforward to implement using computer algebra—but, this will not be done here.

We now discuss why this process should work, even though we have not shown how it works in the most general case. The reason is rather simple. We can take the factorized form of the unitary coupled-cluster approximation and use the generalized Euler-identity like relation, which breaks each factor into either an identity operator or a term that has a cosine multiplying one term and a sine multiplying another. This means when we have created an intermediate ansatz wavefunction, by acting a number of these factors onto the reference, we find that the next factor will either leave a determinant in the superposition unchanged or will multiply it by a cosine and add a second determinant that is multiplied by a sine. This means all terms in the expansion for the wavefunction involve polynomials in sines and cosines of the different angles. As we saw in our examples, we expect the transformation from unitary coupled cluster to conventional coupled cluster to have a prefactor of a cosine of the angle for each factor. This will change many of the cosine factors in the polynomial coefficients to ones and will change sines to tangents (it may also introduce some secant terms). But there are no other types of terms that can be created. Furthermore, all of the determinants are excitations from the single reference. This implies that either we never run into any of the "edge" cases where separating out the different exponents after reordering two operators is one of the complicated cases (such as when the two operators match in each direction) or the separation can be performed, yielding a simple final result. While not a rigorous proof, this is a strong indication that the above algorithm will always work to perform this conversion, even if edge cases do occur. But note, this argument does not preclude that the amplitudes may be raised to operator-valued integer powers.

Another point we have found in our analysis is that the conversion often requires additional terms than those that were in the original unitary coupled-cluster ansatz. It is possible that these new terms could be the same rank as the terms that were being re-ordered (if they have a definite rank), or it can be a higher rank, although creating a higher-rank term that does not annihilate against the reference will require terms arising from many re-orderings. In addition, many of the terms we create as we re-order will annihilate against the reference; this keeps the identity (which is an exact operator identity) from becoming too large, which it easily can if we do not apply to the reference. We also expect that a unitary coupled-cluster ansatz that is low rank, involving excitation-de-excitation operators all less than some specific rank, will likely map to a coupled-cluster ansatz that has many higher-rank terms. In fact, it is likely that the number of amplitudes that we find in the mapped coupled-cluster ansatz are as many as the determinants in the ansatz wavefunction, because the exponential form that arises from the coupled-cluster ansatz does not produce the same amplitudes for higher-rank determinants as the unitary coupled-cluster

does—then an additional higher-rank amplitude is needed to produce the same ansatz wavefunction. Finally, because some of the amplitudes determined in this operator identity depend on the determinants that they operate on, we have to use a factorized form for the coupled-cluster ansatz, because not all factors commute with each other anymore. This is able to be fixed by employing additional higher rank factors, but we do not pursue that more thoroughly here.

The best way to carry out this algorithm is on a computer following the rules we developed above. Because the unitary coupled-cluster ansatz in its factorized form can involve an exponentially large number of terms, we might be limited to smaller-size systems. While we have not provided an analytical formula connecting the amplitudes of the two ansatzes, the approach outlined here helps us understand how the two ansatzes relate to each other. They typically require going from a low-rank representations to a high-rank one, but, for small angles, this may not be much of an issue.

Section 19.7 Problems:

19.7.1 Alternative Operator Proof of the Disentangling Identity

Start from the fact that $e^{-i\tan\theta\hat{S}_\pm} = 1 - i\tan\theta\hat{S}_\pm$. Then expand

$$e^{-2\ln(\cos\theta)\hat{S}_z} = \sum_{n=0}^{\infty} \frac{\Big(-2\ln(\cos\theta)\Big)^n}{n!}\Big(\hat{S}_z\Big)^n \tag{19.102}$$

and note that

$$\Big(2\hat{S}_z\Big)^2 = (\hat{S}_+\hat{S}_- - \hat{S}_-\hat{S}_+)^2 = \hat{S}_+\hat{S}_- + \hat{S}_+\hat{S}_-, \tag{19.103}$$

so we have

$$\begin{aligned} e^{-2\ln(\cos\theta)\hat{S}_z} &= \hat{\mathbb{I}} + \Big(\cosh\Big(\ln(\cos\theta)\Big) - 1\Big)(\hat{S}_+\hat{S}_- + \hat{S}_-\hat{S}_+) \\ &\quad - \sinh\Big(\ln(\cos\theta)\Big)(\hat{S}_+\hat{S}_- - \hat{S}_-\hat{S}_+). \end{aligned} \tag{19.104}$$

Now, we substitute into the disentangling identity in Eq. (19.90) and use all of the different effective spin operator identities to show that

$$e^{-i\theta(\hat{S}_+ + \hat{S}_-)} = \hat{\mathbb{I}} - i\sin\theta(\hat{S}_+ + \hat{S}_-) + (\cos\theta - 1)(\hat{S}_+\hat{S}_- + \hat{S}_-\hat{S}_+), \tag{19.105}$$

and establish the identity directly in terms of the operators themselves.

19.7.2 Direct Way to Relate Unitary Coupled Cluster to Conventional Coupled Cluster

Write an essay that describes how we can directly read off the differences once the unitary coupled-cluster ansatz has been constructed. Also discuss why trying to do this from coupled cluster to unitary coupled cluster is much more challenging.

19.7.3 Example of Re-ordering for the Doubles Case

Consider the case with just two doubles factors. First, we use the exponential disentangling identity to separate the terms in each:

$$\begin{aligned} & e^{\theta_{kl}^{cd}\left(\hat{a}_c^\dagger \hat{a}_d^\dagger \hat{a}_l \hat{a}_k - \hat{a}_k^\dagger \hat{a}_l^\dagger \hat{a}_d \hat{a}_c\right)} e^{\theta_{ij}^{ab}\left(\hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_j \hat{a}_i - \hat{a}_i^\dagger \hat{a}_j^\dagger \hat{a}_b \hat{a}_a\right)} |\Psi_0\rangle \\ & \quad = e^{\theta_{kl}^{cd}\left(\hat{A}(cd;kl|) - \hat{A}(kl;cd|)\right)} e^{\theta_{ij}^{ab}\left(\hat{A}(ab;ij|) - \hat{A}(ij;ab|)\right)} |\Psi_0\rangle \\ & \quad = e^{\tan\theta_{kl}^{cd} \hat{A}(cd;kl|)} e^{-\ln\left(\cos\theta_{kl}^{cd}\right)\left(\hat{A}(|cd;kl) - \hat{A}(|kl;cd)\right)} e^{-\tan\theta_{kl}^{cd}\left(\hat{A}(kl;cd)\right)} \\ & \quad \times e^{\tan\theta_{ij}^{ab} \hat{A}(ab;ij|)} e^{-\ln\left(\cos\theta_{ij}^{ab}\right)\left(\hat{A}(|ab;ij) - \hat{A}(|ij;ab)\right)} e^{-\tan\theta_{ij}^{ab}\left(\hat{A}(ij;ab)\right)} |\Psi_{\mathrm{HF}}\rangle. \end{aligned} \tag{19.106}$$

The rightmost term has an exponent that annihilates against the reference and can be removed. The second rightmost term will yield a cosine when it acts on the reference. Our first re-ordering involves the next two terms. Note that we cannot have $\hat{A}$ match $\hat{A}'$, because the two doubles operators would be identical if they did. The re-ordering then requires one commutator

$$\begin{aligned} e^{-\tan\theta_{kl}^{cd} \hat{A}(kl;cd)} e^{\tan\theta_{ij}^{ab} \hat{A}(ab;ij|)} &= e^{\tan\theta_{ij}^{ab} \hat{A}(ab;ij|) - \tan\theta_{ij}^{ab} \tan\theta_{kl}^{cd} [\hat{A}(kl;cd), \hat{A}(ab;ij)]} \\ &\quad \times e^{-\tan\theta_{kl}^{cd} \hat{A}(kl;cd)}, \end{aligned} \tag{19.107}$$

This calculation can get quite lengthy, as there are eight possible terms that can contribute in this first re-ordering. But because $a \neq b$, $c \neq d$, $i \neq j$, and $k \neq l$, at most three terms can contribute. Even so, it is still cumbersome to include the most general situation, even for doubles. So, instead, we will consider some specific cases.

The first one is where one index is in common, say $i = k$. Compute the result in this case. Does the rank of the operators remain the same, decrease, or increase? You should find that the rightmost two terms annihilate against the reference state.

The second re-ordering is

$$
\begin{aligned}
& e^{-\ln\left(\cos\theta_{il}^{cd}\right)\left(\hat{A}(|cd;il)-\hat{A}(|il;cd)\right)} e^{\tan\theta_{ij}^{ab}\hat{A}(ab;ij|)} \\
& = e^{\tan\theta_{ij}^{ab}\left(\cos\theta_{il}^{cd}\right)^{-\hat{n}_l-\hat{n}_c+\hat{n}_l(\hat{n}_c+\hat{n}_d)}\hat{A}(ab;ij|)} \\
& \times e^{-\ln\left(\cos\theta_{il}^{cd}\right)\left(\hat{A}(|cd;il)-\hat{A}(|il;cd)\right)},
\end{aligned}
\tag{19.108}
$$

which requires the full Hadamard, just like before; except here, because the number operator that matches with the lowering operator (index i) is multiplied by number operators, they enter into the exponent. Work out this term carefully.

Verify that you obtain

$$
\begin{aligned}
& e^{\theta_{kl}^{cd}\left(\hat{a}_c^\dagger\hat{a}_d^\dagger\hat{a}_l\hat{a}_k-\hat{a}_k^\dagger\hat{a}_l^\dagger\hat{a}_d\hat{a}_c\right)} e^{\theta_{ij}^{ab}\left(\hat{a}_a^\dagger\hat{a}_b^\dagger\hat{a}_j\hat{a}_i-\hat{a}_i^\dagger\hat{a}_j^\dagger\hat{a}_b\hat{a}_a\right)}|\Psi_{\mathrm{HF}}\rangle \\
& \quad = \cos\theta_{ij}^{ab}\cos\theta_{il}^{cd} e^{\tan\theta_{il}^{cd}\hat{a}_c^\dagger\hat{a}_d^\dagger\hat{a}_l\hat{a}_i+\tan\theta_{ij}^{ab}\sec\theta_{il}^{cd}\hat{a}_a^\dagger\hat{a}_b^\dagger\hat{a}_j\hat{a}_i}|\Psi_{\mathrm{HF}}\rangle.
\end{aligned}
\tag{19.109}
$$

Note, that this is in the standard CC form because we could remove the operator exponent on the cosine factor in the amplitude by acting on the reference state before putting all excitation terms in the same exponent.

19.7.4 Direct Way to Relate Unitary Coupled Cluster to Conventional Coupled Cluster

Directly calculate the unitary coupled-cluster form and verify that it gives the same result as in the problem above. Use the generalized Euler identity for each factor and operate them on the state. Then compare to the coupled-cluster form.

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Chapter 20
Quantizing Light

I've seen it all, done it all, known it all.

Israel Kamakawiwo'ole, Singer

James Maxwell is one of the most important physicists of all time (see Fig. 20.1). He did landmark work in unifying electricity and magnetism and also in describing the two with a coherent mathematical framework. In the process, he discovered that light is an electromagnetic wave that can oscillate in a vacuum. He also played critical roles in the development of statistical mechanics, which is why we call e^{-E/k_BT} the Maxwell-Boltzmann distribution. Unfortunately, to fully work with his mathematical formulation of electromagnetism, via the so-called Maxwell equations, requires one to master vector calculus. So, in this chapter, we will not cover the complete derivation of his work, but instead show how it applies directly to light and from that we will then be able to jump into quantum optics. The goal of this chapter is to precisely explain and illustrate for you what a photon really is. Not just a particle of light, nor just a wavepacket, but a specific quantum structure that has the property that it can be measured once and only once. Then we put together all of these new insights to explain how one measures gravitational waves with quantum mechanics. Much of the material presented here was shaped by Alain Aspect's wonderful courses on single and multimode photons presented on coursera. See our overview of the photon at https://tinyurl.com/5dhfhynj.

20.1 Maxwell's Equations and Light Confined to a Box

As you have certainly learned previously, we can describe waves via complex valued functions, and the most common three-dimensional plane wave takes the form $e^{i(\vec{k}\cdot\vec{r}-\omega t)}$. We will assume that plane-wave light takes this form, and from it, we

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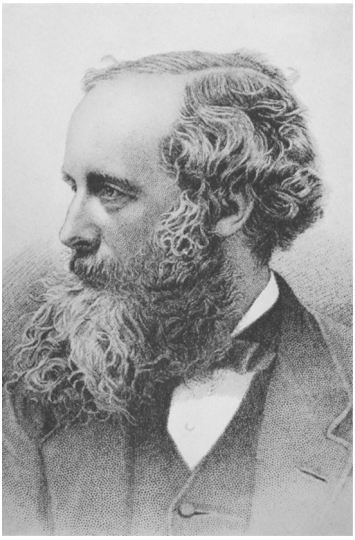

Fig. 20.1 James Maxwell. [Engraved by G. J. Stodart, courtesy of AIP Emilio Segrè Visual Archives]

will discuss Maxwell's four equations of electromagnetism and how they are used to describe light.

The first question is what is meant by an equation for light? We will find that Maxwell's equations show us that light involves time-varying electric and magnetic fields that vary in sync, in such a way that they become a traveling wave. As we also recall from our earlier work, the electric and magnetic fields tell us about the forces on charged particles, so by measuring these forces, we can infer these fields—in other words, they are measurable, or observable quantities. We also recall that light has a property called polarization (the direction of the electric field). This motivates for us a form for light that has a time and space varying electric field given by

$$\vec{E}(\vec{r},t) = \sum_l \left(\vec{\varepsilon}_l E_l e^{i(\vec{k}_l \cdot \vec{r} - \omega_l t)} + \vec{\varepsilon}_l^* E_l^* e^{-i(\vec{k}_l \cdot \vec{r} - \omega_l t)} \right), \tag{20.1}$$

where l is the label for the component of the superposition and $\vec{\varepsilon}_l$ is a unit vector, called the polarization vector. It can be complex, but in our work, it will always be real. E_l is the amplitude of the electric field component (which can also be complex), $\vec{k}_l$ is its wavevector, and ω_l is its frequency. One of the things we will do is use Maxwell's equations to help us determine relationships between these different parameters, and the form of the magnetic field (although because it should be in sync, it will also have a factor of $e^{\pm i(\vec{k}_l \cdot \vec{r} - \omega_l t)}$).

There are four Maxwell equations. The first one is called *Gauss' law*, and it tells us that $\vec{k}_l \cdot \vec{\varepsilon}_l = 0$ for each component. This tells us the wavevector $\vec{k}_l$ is perpendicular to the polarization vector $\vec{\varepsilon}_l$ for each component in the superposition. Next, we consider *Faraday's law of induction*. If we write the magnetic field as

$$\vec{B}(\vec{r},t) = \sum_l \left(\vec{B}_l e^{i(\vec{k}_l \cdot \vec{r} - \omega_l t)} + \vec{B}_l^* e^{-i(\vec{k}_l \cdot \vec{r} - \omega_l t)} \right), \tag{20.2}$$

then Faraday's law of induction tells us that $(\vec{k}_l \times \vec{\varepsilon}_l)E_l = \omega_l \vec{B}_l$. This says the direction of the magnetic field is perpendicular to both the polarization vector of the electric field and the wavevector. The third law is *Gauss' law for magnetism*, which says $\vec{k}_l \cdot \vec{B}_l = 0$. But, this is already satisfied because $\vec{B}_l = (\vec{k}_l \times \vec{\varepsilon}_l)E_l/\omega_l$, which is already perpendicular to the wavevector. Finally, *Ampere's circuital law* tells us that $c^2\vec{k}_l \times \vec{B}_l = -\omega_l \vec{\varepsilon}_l E_l$. Substituting in the result for $\vec{B}_l$, we find that $c^2\vec{k}_l \times \left(\vec{k}_l \times \vec{\varepsilon}_l\right) = -\omega_l^2 \vec{\varepsilon}_l$. In Problem 20.1.1, we show that $\vec{k}_l \times \left(\vec{k}_l \times \vec{\varepsilon}_l\right) = (\vec{k}_l \cdot \vec{\varepsilon}_l)\vec{k}_l - k_l^2 \vec{\varepsilon}_l$, which becomes $\vec{k}_l \times \left(\vec{k}_l \times \vec{\varepsilon}_l\right) = -k_l^2 \vec{\varepsilon}_l$, because the wavevector and the polarization vector are perpendicular to each other. Then, we learn that $\omega_l = c|\vec{k}_l|$. Putting this all together, we have the electric field is given by Eq. (20.1), while the magnetic field is given by

$$\vec{B}(\vec{r}, t) = \sum_l \left(\frac{\vec{k}_l \times \vec{\varepsilon}_l}{\omega_l} E_l e^{i(\vec{k}_l \cdot \vec{r} - \omega_l t)} + \frac{\vec{k}_l \times \vec{\varepsilon}_l}{\omega_l} E_l^* e^{-i(\vec{k}_l \cdot \vec{r} - \omega_l t)} \right). \tag{20.3}$$

Here, we have $\vec{k}_l \cdot \vec{\varepsilon}_l = 0$, $\omega_l = c|\vec{k}_l|$, and the vectors $\vec{k}_l$, $\vec{\varepsilon}_l$, and $\vec{k}_l \times \vec{\varepsilon}_l$ form a triad of vectors for the coordinate system of each mode (recall that we will be working with real valued polarization vectors).

We have just used the four Maxwell equations (as they act on our electromagnetic wave ansatz) to determine the relationship between the electric and magnetic fields in a classical light wave. We did not provide any deep context for why the Maxwell equations were what they were, or how they specifically give rise to the phenomena of light as an electromagnetic wave. This is the best we can do without using calculus. The important take away from this work is that the electric and magnetic fields have the forms given above in terms of the wavevector, polarization vector, and frequency. Given these results, our next goal is to quantize light. This will require us to convert the classical system into a quantum one. There are a number of technical problems associated with trying to quantize light in an infinite-sized space (you can take a course in relativistic quantum field theory later to learn all of these details). Instead, here, we assume that we quantize light in a large box, and to reduce the finite-size-effects of the box, we give it periodic boundary conditions (like our particle on a circle). The main effect of this technical trick is to make the number of photon modes countably infinite instead of a continuum. This makes it much easier to work with the formalism. It even is correct physically, because we can think of our universe as being a very large, but finite box, so we should not even require quantizing in an infinite-sized space!

Suppose the box is a cube of edge L and volume $V = L^3$. Then the periodic boundary conditions tell us that $e^{i\vec{k}_l \cdot \vec{r}} = e^{i\vec{k}_l \cdot \vec{r} + ik_{l\alpha}L}$ or $e^{ik_{l\alpha}L} = 1$, where $\alpha = x$, y, and z and denote the three different spatial components of the wavevector. This then means that we can associate each mode l with the integers n_x, n_y and n_z, via $k_l = \frac{2\pi}{L}(n_x, n_y, n_z)$. We are then constrained to choose $\vec{\varepsilon}_l \perp \vec{k}_l$. Note that there are two linearly independent polarizations for each wavevector. We will find

it convenient to choose the polarization for the state $\vec{k}_{-l}$ to be $\vec{\varepsilon}_{-l} = \vec{\varepsilon}_l$, where $\vec{k}_{-l} = -\vec{k}_l$ when we are doing summations over modes. We will still use the label l to label a specific mode, but we now know that the mode is determined by the three numbers $\{n_x, n_y, n_z\}$ and the polarization $\vec{\varepsilon}_l$; there are two polarizations for each set of integers; hence, there are two modes for each set of integers. In addition, the wavevector $\vec{k}_l$ takes discrete values, with a spacing given by $\frac{2\pi}{L}$ in each dimension. Note how the presence of the box quantizes the allowed states of electromagnetic radiation.

We now introduce some notation that will make it easier for us to make contact with the quantum world. We define the single-photon amplitude in mode l to be $\mathcal{E}_l^{(1)} = \sqrt{\frac{\hbar\omega_l}{2\epsilon_0 V}}$ where $V = L^3$ is the "quantization volume." The total amplitude of the field component is E_l (which can be complex). Absorbing the time variation of the wave into the amplitude, it becomes $E_l(t) = E_l e^{-i\omega_l t}$. Using the single-photon amplitude, we instead write it as

$$E_l(t) = i\mathcal{E}_l^{(1)}\alpha_l(t), \tag{20.4}$$

with $\alpha_l(t)$ complex. This notation is designed to make contact with coherent states, as you will later see. It may not seem obvious to you as to why we chose it this way. Indeed, it is completely opaque at this stage. Just trust that doing it this way, which can be viewed as the definition of $\alpha_l(t)$, makes our notation much simpler later. It is another case of 20–20 hindsight. Using this notation, we now define the two quadrature parameters $Q_l(t) = \sqrt{\frac{\hbar}{2}}\left(\alpha_l(t)+\alpha_l^*(t)\right)$ and $P_l(t) = \sqrt{\frac{\hbar}{2}}\frac{1}{i}\left(\alpha_l(t)-\alpha_l^*(t)\right)$, which you can see isolate the real and imaginary parts of the amplitude of the wave. As you know, when working with complex numbers, we can do so via the modulus and phase, or the real and imaginary parts. The real and imaginary parts tend to be the preferred way to work for quantized light. We use labels similar to position and momentum, because we will find they appear in the Hamiltonian in this form, which then will look like a harmonic oscillator. It is in this way, using the relation of the modulus squared of a complex number to the sum of the squares of the real and imaginary parts that is where the relationship of quantized light to the harmonic oscillator occurs. It is not, because we are physically modeling some oscillation of something as a harmonic oscillator by analogy.

Now, we want to compute the energy. You might be tempted to say is it something like the sum of $\hbar^2 k_l^2/2m$, as we have seen earlier in the class for free particles. But this must be wrong—photons have no mass! Instead, we rely on the classical notion of the energy in a field and sum the square of the electric and magnetic fields via

$$\varepsilon(\vec{r}, t) = \frac{\epsilon_0}{2}\left(\vec{E}^2(\vec{r}, t) + c^2\vec{B}^2(\vec{r}, t)\right), \tag{20.5}$$

which supports the notion that the energy of a wave is proportional to the square of its amplitude. This is the energy of the field localized at $\vec{r}$ as a function of time.

What we really want is the total energy in the electromagnetic wave inside the entire box. This requires us to sum up these results over all spatial positions. Officially speaking, this must be done with an integral. But, as we describe below, this can be easily worked out by simply breaking the big box into smaller boxes and summing the value in each little box using the geometric series. As you might suspect, the total energy is independent of time, because the oscillating wave oscillates similarly in time and space, so accumulating its contributions over all space is the same as averaging over a period in time.

The key object we need to sum over the box is $e^{i\vec{k}\cdot\vec{r}} = e^{ik_x r_x} e^{ik_y r_y} e^{ik_z r_z}$. We do the same thing we have done before. We break the big box into N boxes along each dimension, so they each have a side length of $\Delta = \frac{L}{N}$. We need to evaluate

$$\sum_{\text{boxes}} e^{ik_x r_x} e^{ik_y r_y} e^{ik_z r_z} \Delta^3 = \sum_{j=0}^{N-1} e^{ik_x \Delta j} \sum_{k=0}^{N-1} e^{ik_y \Delta k} \sum_{l=0}^{N-1} e^{ik_z \Delta l} \Delta^3. \tag{20.6}$$

As you can see, each sum is a geometric series. Whenever $k_\alpha \neq 0$, we obtain

$$\sum_{\text{boxes}} e^{ik_x r_x} e^{ik_y r_y} e^{ik_z r_z} \Delta^3 = \frac{1 - e^{ik_x \Delta N}}{1 - e^{ik_x \Delta}} \frac{1 - e^{ik_y \Delta N}}{1 - e^{ik_y \Delta}} \frac{1 - e^{ik_z \Delta N}}{1 - e^{ik_z \Delta}} \Delta^3. \tag{20.7}$$

Now, because $e^{ik_\alpha \Delta N} = e^{ik_\alpha L} = 1$, all of the numerators vanish, and so does the sum. In fact if *any* of the $k_\alpha \neq 0$, then the sum vanishes, because one of the factors is zero. So, the only way to obtain a nonzero result is for all $k_\alpha = 0$. Then going back to the original sum, each term is 1, there are N^3 of them, so we get the sum is equal to $(\Delta N)^3 = L^3 = V$. In summary, we have

$$\sum_{\text{boxes}} e^{ik_x r_x} e^{ik_y r_y} e^{ik_z r_z} \Delta^3 = V \delta_{n_x,0} \delta_{n_y,0} \delta_{n_z,0}, \tag{20.8}$$

where we have used the fact that $k_\alpha = \frac{2\pi n_\alpha}{L}$. We will now use this result to compute the total energy in the box. We will do the electric field contribution here. You will complete the calculation with the magnetic field term in the problems.

So, we need to compute

$$\varepsilon^{\text{el}}_{\text{box}}(t) = \sum_{\text{boxes}} \varepsilon^{\text{el}}(\vec{r}, t) = \frac{\epsilon_0}{2} \sum_{\text{boxes}} \vec{E}^2(\vec{r}, t). \tag{20.9}$$

Now, we work out the summation for each box. There are a lot of terms because of the products. We first use the shortened mode label l to make the equations more compact. Because we are squaring the electric field, and it involves a sum over modes, the squared field has two summations over modes. This will become three sums when we add the sum over the boxes. First, we compute the square of the field:

$$\vec{E}^2(\vec{r},t) = \sum_{ll'} \vec{\varepsilon}_l \cdot \vec{\varepsilon}_{l'} \Big(E_l E_{l'} e^{i\left((\vec{k}_l+\vec{k}_{l'})\cdot\vec{r}-(\omega_l+\omega_{l'})t\right)} + E_l E_{l'}^* e^{i\left((\vec{k}_l-\vec{k}_{l'})\cdot\vec{r}-(\omega_l-\omega_{l'})t\right)} + E_l^* E_{l'} e^{i\left((-\vec{k}_l+\vec{k}_{l'})\cdot\vec{r}-(-\omega_l+\omega_{l'})t\right)} + E_l^* E_{l'}^* e^{i\left((-\vec{k}_l-\vec{k}_{l'})\cdot\vec{r}-(-\omega_l-\omega_{l'})t\right)} \Big). \tag{20.10}$$

When we do the summation over the boxes, we get zero unless the coefficient of $\vec{r}$ in each exponential is equal to zero. This gives two cases where $\vec{k}_l + \vec{k}_{l'} = 0$ and two cases where $\vec{k}_l - \vec{k}_{l'} = 0$. In the first case, we have $\vec{k}_{l'} = -\vec{k}_l = \vec{k}_{-l}$ using our notation discussed above. In the second case, we have $\vec{k}_{l'} = \vec{k}_l$. Since ω is always a positive number, the case where $\vec{k}_l - \vec{k}_{l'} = 0$, the two frequency terms cancel, while they add together when $\vec{k}_l + \vec{k}_{l'} = 0$. In both cases, we have $\vec{\varepsilon}_{l'} = \vec{\varepsilon}_l$ according to our notation. This then gives us

$$\varepsilon_{\text{box}}^{\text{el}}(t) = \frac{\epsilon_0 V}{2} \sum_l \left(E_l E_{-l} e^{-2i\omega_l t} + 2|E_l|^2 + E_l^* E_{-l}^* e^{2i\omega_l t} \right). \tag{20.11}$$

You will work out the contribution from the magnetic field in Problem 20.1.2. After adding them together, we arrive at the total for the energy, which is equal to

$$\varepsilon_{\text{box}}(t) = 2\epsilon_0 V \sum_l |E_l|^2, \tag{20.12}$$

which is independent of time, as we claimed it must be. If we reexpress this in terms of $\alpha_l(t)$, or the quadrature parameters, we find it gives

$$\varepsilon_{\text{box}}(t) = \sum_l \hbar\omega_l |\alpha(t)|^2 = \sum_l \frac{\omega_l}{2} \left(Q_l^2(t) + P_l^2(t) \right), \tag{20.13}$$

which looks very similar to the harmonic oscillator results, if we think of the α as relating to the lowering operator in the first form, and we rescale with respect to effective mass and spring constants in the second form. Note as well that the result for the total energy in the box is not time dependent, because the time dependence of $\alpha(t)$ is via a complex phase. When we quantize light in the next section, we will use this result to aid us in determining the Hamiltonian for light.

Section 20.1 Problems:

20.1.1 Evaluate the Vector Triple Product

Show that

$$\vec{A} \times (\vec{A} \times \vec{B}) = (\vec{A} \cdot \vec{B})\vec{A} - (\vec{A} \cdot \vec{A})\vec{B}, \tag{20.14}$$

and verify it was implemented properly when we were using Maxwell's equations to verify the form of the electromagnetic waves. You might find the identity we used for the Levi-Cevita tensor to come in handy with your proof.

20.1.2 Determine the Magnetic Field Contribution to the Total Energy in the Box

Calculate the contribution to the total energy due to the magnetic field, given by

$$\varepsilon_{\text{box}}^{\text{mag}}(t) = \frac{\epsilon_0 c^2}{2} \sum_{\text{box}} \vec{B}^2(\vec{r}, t). \tag{20.15}$$

Verify that when we add it to the electric field contribution, we obtain the result stated in the text.

20.2 Quantization

We have laid out all of the required ground work. Building off of this work, we now find that quantizing light is completely straightforward (but we will see some surprises anyway). We start from the total energy in Eq. (20.13), which we want to quantize. Taking the rightmost form, as $\varepsilon_{\text{box}} = \sum_l \frac{\omega_l}{2}\left(Q_l^2 + P_l^2\right)$, because the energy is time independent, we will elevate it to a Hamiltonian, with $\hat{Q}_l$ and $\hat{P}_l$ the position and momentum quadrature operators for light, which satisfy the commutation relation $[\hat{Q}_l, \hat{P}_{l'}] = i\hbar\delta_{ll'}$. These quadrature operators were constructed, so that they had dimensions of $\sqrt{\hbar}$ for each of them. Now, if we define ladder operators for each mode via

$$\hat{a}_l = \frac{1}{\sqrt{2\hbar}}\left(\hat{Q}_l + i\,\hat{P}_l\right) \quad \text{and} \quad \hat{a}_l^\dagger = \frac{1}{\sqrt{2\hbar}}\left(\hat{Q}_l - i\,\hat{P}_l\right), \tag{20.16}$$

then we immediately see that the Hamiltonian becomes

$$\hat{H}_{\text{light}} = \sum_l \frac{\omega_l}{2}\left(\hat{Q}_l^2 + \hat{P}_l^2\right) = \sum_l \hbar\omega_l \left(\hat{a}_l^\dagger \hat{a}_l + \frac{1}{2}\right), \tag{20.17}$$

by simply following the same steps as we had when we worked with the operators for the simple harmonic oscillator. We also have an operator for the electric field in the Schrödinger picture at the position $\vec{r}$, which is

$$\hat{\vec{E}}(\vec{r}) = \sum_l i\mathcal{E}_l^{(1)} \left(\vec{\varepsilon}_l \hat{a}_l \, e^{i\vec{k}_l \cdot \vec{r}} - \vec{\varepsilon}_l^* \hat{a}_l^\dagger e^{-i\vec{k}_l \cdot \vec{r}} \right) = \hat{\vec{E}}^{(+)}(\vec{r}) + \hat{\vec{E}}^{(-)}(\vec{r}) \text{ with}$$

$$\hat{\vec{E}}^{(+)}(\vec{r}) = i \sum_l \mathcal{E}_l^{(1)} \vec{\varepsilon}_l \hat{a}_l \, e^{i\vec{k}_l \cdot \vec{r}} \text{ and } \hat{\vec{E}}^{(-)}(\vec{r}) = -i \sum_l \mathcal{E}_l^{(1)} \vec{\varepsilon}_l^* \hat{a}_l^\dagger e^{-i\vec{k}_l \cdot \vec{r}}. \quad (20.18)$$

Be careful to focus on the notation used here. The "+" component of the electric field operator is given by the lowering operator contributions, while the "−" component is given by the raising operator contributions. Note that because we will think of these operators as creating or destroying photons, we will tend to call them creation and annihilation operators rather than raising and lowering operators.

As we discussed before when we were working with general bosons, we will use a number operator representation for the states. We think of this as organizing all of the modes into an ordered list in terms of the mode index l, and then a quantum state is given by

$$|n_1, n_2, n_3, n_4, \cdots\rangle = \frac{\left(\hat{a}_1^\dagger\right)^{n_1}}{\sqrt{n_1!}} \frac{\left(\hat{a}_2^\dagger\right)^{n_2}}{\sqrt{n_2!}} \frac{\left(\hat{a}_3^\dagger\right)^{n_3}}{\sqrt{n_3!}} \frac{\left(\hat{a}_4^\dagger\right)^{n_4}}{\sqrt{n_4!}} \cdots |0\rangle, \quad (20.19)$$

with the state the operators act on being the vacuum state which has zero photons in every mode. The energy of the state is given by

$$E_{n_1,n_2,n_3,n_4,\cdots} = \sum_{l=1}^{\infty} \hbar\omega_l \left(n_l + \frac{1}{2} \right). \quad (20.20)$$

We can see we have run into our first real problem with this quantization of the system. Because of the zero-point energy, and the fact that number of modes is infinite in every box—this sum is always infinity! Even the vacuum has infinite energy. So, we deal with this by computing the difference in energies between the state and the vacuum. In other words, we always work with $E_{n_1,n_2,n_3,\cdots} - E_{\text{vac}}$, which is always finite. Because it is a pain to continue to write $- E_{\text{vac}}$, everywhere, we drop the zero-point energy from the Hamiltonian and work then with energies that are automatically defined relative to the vacuum energy.

Our next task is to examine the expectation values of the field and the square of the field in a single-mode state with n_l single-mode photons. The state is then

$$|n_l\rangle = \frac{\left(\hat{a}_l^\dagger\right)^{n_l}}{\sqrt{n_l!}} |0\rangle. \quad (20.21)$$

For the field operator, we want the expectation value to have time dependence, so we must first compute the operators in the Heisenberg representation at time t. Recall, this takes the form

$$\hat{a}_l^\dagger(t) = \hat{U}^\dagger(t) \hat{a}_l^\dagger \hat{U}(t) = e^{\frac{i}{\hbar}\hat{H}t} \hat{a}_l^\dagger e^{-\frac{i}{\hbar}\hat{H}t} = \hat{a}_l^\dagger e^{i\omega_l t}, \quad (20.22)$$

and we have $\hat{a}_l(t) = \hat{a}_l e^{-i\omega_l t}$. In both cases, the derivation goes similar to what we did before—we have a Hadamard lemma, and it produces a repeating pattern, which then can be summed exactly. You will prove these two results in Problem 20.2.1.

Now we will compute the expectation value of the electric field and the fluctuations of the field for a single-mode photon state with precisely n_l photons. The results are surprising. We have

$$\langle n_l|\hat{\vec{E}}(\vec{r},t)|n_l\rangle = i\mathcal{E}_l^{(1)}\langle n_l|\left(\vec{\varepsilon}_l\hat{a}_l\, e^{i\vec{k}_l\cdot\vec{r}-i\omega_l t} - \vec{\varepsilon}_l^{\,*}\hat{a}_l^\dagger e^{-i\vec{k}\cdot\vec{r}+i\omega_l t}\right)|n_l\rangle = 0. \tag{20.23}$$

No matter how many photons we have in the state, the electric field is measured to be zero, everywhere! A single-mode photon state with a specific number of photons has a vanishing electric field. This arises because the number of photons takes on just one value, and it is quite surprising of a result. The fluctuations, however, are not zero. We compute

$$\begin{aligned}\langle n_l|\hat{\vec{E}}(\vec{r},t)\cdot\hat{\vec{E}}(\vec{r},t)|n_l\rangle &= -\left(\mathcal{E}_l^{(1)}\right)^2\langle n_l|\left(\hat{a}_l^\dagger\hat{a}_l^\dagger e^{-2i\vec{k}_l\cdot\vec{r}+2i\omega_l t} - \hat{a}_l^\dagger\hat{a}_l - \hat{a}_l\,\hat{a}_l^\dagger\right.\\ &\qquad\left.+\hat{a}_l\,\hat{a}_l\, e^{2i\vec{k}\cdot\vec{r}-2i\omega_l t}\right)|n_l\rangle\\ &= \left(\mathcal{E}_l^{(1)}\right)^2(2n_l+1)\end{aligned} \tag{20.24}$$

where we assumed the polarization is real and a unit vector. The fluctuations are nonzero, even in the vacuum state with $n_l = 0$. In particular, we have $\Delta E_l = \mathcal{E}_l^{(1)}\sqrt{2n_l+1}$.

The fluctuations of the vacuum are important. They lead to the Lamb shift, spontaneous emission, the Casimir effect, and more. So, even in the case where we measure a vanishing electric field, the field itself is full of quantum fluctuations that can interact with a physical system. This is already a surprising set of results, but there is more to come.

Congratulations, we have now successfully quantized light. Our next step is to determine how to measure individual quanta of light. That is, how to detect the presence of a single photon. We have a video https://tinyurl.com/yc72czpx of a lecture showing how to quantize light as well.

Section 20.2 Problems:

20.2.1 The Heisenberg Representation for the Creation and Annihilation Operators

Verify the Heisenberg representation for a creation operator of a single mode photon by explicitly computing the Hadamard lemma in Eq. (20.22) and repeat for the corresponding annihilation operator.

20.2.2 Average Value of the Field Versus Its Fluctuations

We saw in our explicit calculations that the average value of the electric field vanishes, but the fluctuations do not for states that have a fixed specific number of photons in each mode. Write an essay that discusses these different points. In particular, why is the electric field vanishing for a state with n_l photons, and how is it that it can have fluctuations if its expectation value is zero? You can describe why this happens mathematically but also discuss some of the physical reasons behind this behavior.

20.2.3 Fluctuations of Quadrature Operators

Using the quadrature operators, compute the average values $\langle n_l|\hat{Q}_l(t)|n_l\rangle$ and $\langle n_l|\hat{P}_l(t)|n_l\rangle$, and then the corresponding fluctuations. Show that the vacuum is a minimal uncertainty state.

20.2.4 Single-Mode Versus Multimodal Photons in Single-Photon States

A single-mode quantum state is given by $\hat{a}_l^\dagger|0\rangle$. A multimode single-photon state is given by

$$|\psi\rangle = \sum_l c_l \hat{a}_l^\dagger |0\rangle, \tag{20.25}$$

with $\sum_l |c_l|^2 = 1$. We define the total number operator to be

$$\hat{N} = \sum_l \hat{a}_l^\dagger \hat{a}_l \,. \tag{20.26}$$

Show that the single-mode photon state and the multimode photon state both have the expectation value of $\hat{N}$ and $\hat{N}^2$ to be equal to one, so both states have precisely one photon and no fluctuations. Write a short discussion about what this means for the multimode photon state.

20.3 How a Single-Photon Detector Works

In order to understand what a photon is, we first need to understand how we can detect a single photon, because once we know how to observe individual photons,

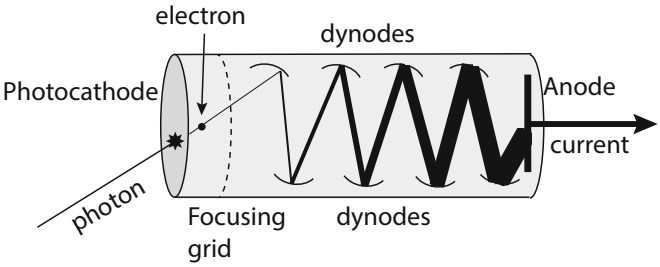

Fig. 20.2 Schematic of a photomultiplier tube. On the far left, a photon impinges on a metal (photocathode) with a low work function. A photon with an energy higher than the work function emits an electron. This emitted electron is focused and directed to a dynode with a large potential difference. The accelerated electron has enough energy to eject two to five electrons from the dynode surface, which then head to the next dynode and repeat. After 20–30 sets of dynodes, we have a large enough bunch of electrons that we can measure the current of the pulse with a sensitive ammeter. This is how the photomultiplier tube operates

we can then plan and carry out experiments that use single photons and see what they tell us about quantum mechanics. Back in the beginning of this book, when we asked how can we separate atoms, we found that by finding a state-selective force (there, an inhomogeneous magnetic field provides a force proportional to the spin projection of the atom), we could let it act for some period of time and then let the atoms move over a larger distance to amplify the separation and finally measure the position to infer the spin projection. These three key concepts—create a signal that differentiates what you would like to measure, amplify the signal, and then detect it—underlie much of quantum measurement. Here, our focus is on measuring single photons, which we can think of most clearly in terms of a counting experiment. How can I set up a detector that can count individual photons? We need to find a way to be sensitive to an individual photon, to amplify the signal so we can detect it, and then detect it. These three criteria are components of how we measure single photons with a photomultiplier tube.

In Fig. 20.2, we schematically depict a photomultiplier tube, which can be used to detect individual photons. The first step if the photomultiplier tube is the photon transduction step—we convert a photon to an electron via the photoelectric effect with a thin sheet of metal that has a low work function and is called the *photocathode*. In the photoelectric effect, energy is transferred from the light to the electrons in the metal. A single photon must have enough energy to "lift" the electron above the work function of the metal. Electrons with energy this high are free to leave the surface of the metal. The energy of the photon is determined by the frequency of light not the intensity of light (which tells us how many photons hit the metal per unit time and unit area). But, in general, a single photon will emit a single electron. We are still left with the issue of how do we detect individual quanta. Of course, an electron is a charged particle, and charged particles are easier to manipulate than neutral particles, so the photon transduction does help a lot in being able to construct a single-photon detector.

The next step is to amplify the signal. We do this by accelerating the electron into a series of dynodes. If the electron energy is high enough when it hits the dynode, then it emits two to five electrons from the surface of the metal. We continue this process through many dynodes until we have a bunch of electrons with a significant number, large enough that they can be measured using classical meters, such as an ammeter. Then, by detecting the current, we associate that with the detection of a single photon. Note that this approach has a great sensitivity issue. You can see that it is like an avalanche, just ready to fall. The photon comes in and "tickles the tail of the dragon," leading to a large response. It is a very clever design. These detectors are both efficient and fast. This is needed for quantum optics experiments, because we often use coincidence measurements to discriminate between different phenomena.

Because a single electron can set off the avalanche that then gets detected by the ammeter, if we have any stray electrons around, they can set off the photomultiplier tube. Even in the best experiments, we cannot prevent this, so photomultiplier tubes always have a certain level of *dark current* determined by the rate of false counts that occur even if there are no photons impinging on the photomultiplier tube. Of course, it is fairly simple to measure this dark current—just turn the photomultiplier tube on and cover the front end so no photons can reach it. Hence, in any given experiment, we must ensure the signal we wish to measure is larger than the dark current (or we need to amplify it even more over the dark current) in order for us to be able to reliably measure it.

There are other types of single-photon detectors as well. Let's briefly discuss two of them. The first is the single-photon avalanche detector, sometimes called a SPAD. It operates in a very similar fashion to the photomultiplier tube, but it is constructed out of a semiconductor material, so it can be significantly more compact. Its avalanche comes from a process called impact ionization, where an energetic electron moving through a solid can cause multiple electrons to be emitted (similar, in many respects to the e-2e experiment we described earlier when discussing how to visualize what the hydrogen atom looks like). Another type of detector is a superconducting transition-edge detector. These detectors use superconducting wires tuned close to the superconducting-normal state transition, where very small changes in temperature cause large changes in the current-voltage relationship of the wire. These detectors detect photons by measuring the heat absorbed by the wire that is deposited by the photon as it is absorbed. They are so sensitive, that they can even be used to detect multiple photons at the same time and also tell us about the energy of each photon being detected. They tend to be a bit slower to recover after detection though, as they must be brought back to their starting temperature to be reset.

There is one final type of single-photon detection, but it is of a quite different character than the ones we have been discussing. So far, the photon detectors have destroyed the photon in order to detect them. Is this always needed? The surprising answer is no, it is not. There are systems called optical cavities, which are boxes with highly polished mirrors on the walls. If we put a number of photons inside the cavity, they will live for a long time bouncing off the walls. It turns out the resonant

frequency of the cavity is affected by the number of photons inside the cavity. So, we can use a measurement of the resonant frequency to count the number of photons in the cavity. This type of detector is used for many advanced quantum experiments, especially with atoms interacting with photons. Such systems are called cavity quantum electrodynamics systems or cavity QED. They are fascinating systems.

When detecting single photons, one has to worry about noise. We mentioned this earlier, but let's get more concrete about where the noise can come from. First, it it not possible to create an experiment where all environmental photons are blocked from the device. Hence, we can have stray photons enter the detector and be measured. This noise source can be reduced by removing as much extraneous light as possible. Second, the photocathode only weakly binds the electrons to its surface. They can be spontaneously "boiled off" the surface. This noise can be reduced by cooling the detector, which greatly reduces the electron emission rate. Third, other ionized matter can also enter the tube, sometimes coming off the walls, or from cosmic rays and muons. Anything that hits the first dynode will launch the cascade. Because the dynode is at a high potential, it is easy for charged particles to get accelerated toward the dynode. Hence, the fact that the photomultiplier tube is sitting on the edge of the precipice allows it to be susceptible to noise that generates false-positive detections of photons. This is called the dark current of the photodetector, and it typically occurs with a constant rate during the course of an experiment. Sometimes it is hard to shield from all sources of dark current. Another issue to worry about is the efficiency of the detector. It won't measure every single photon that impinges on it. Some fail to emit an electron, or the electron does not focus onto a dynode. Hence, there is always an efficiency level for photon detection that we will also need to contend with.

Section 20.3 Problems:

20.3.1 Essay on How We Measure Single Quanta

We have discussed the characteristics needed to be able to measure single quanta. Write a brief essay that explains, in your own words, the process needed to measure individual quanta. Be sure to discuss how we need to implement the steps of single quantum sensitivity, amplification, and detection to make the detector work. Also comment on the issue of noise. Feel free to bring outside sources into your discussion.

20.3.2 Essay on How to Measure Energy and Number of Photons at the Same Time

We described one method to detect the number of photons and their energy at the same time. Are there others? Can you discover more details about how these types

of detectors work? Write a short essay explaining how we can measure both the energy and count the number of photons at the same time.

20.4 How a Single-Photon Source Works

We have described how to detect single photons. But, in order to work with them in an experiment, we must be able to have a single-photon source available. This is not as simple to achieve as you might initially think it is. We go into the details of how to create and verify single-photon sources. No, they do not correspond to just taking any light source and making it very dim.

In 1909, Geoffrey Taylor performed an experiment where he used dimmer and dimmer light to generate the diffraction pattern of a needle (very similar to a two-slit experiment) (Taylor, 1909). One of the exposures was so weak; it took 3 months to develop the image. It was believed at that time that for the dimmest light, there was on average never more than one photon in the apparatus at any given time. This is an argument often used to justify dim light as being a single-photon light source. But, as we will see in this section, it is not. Any form of classical light, be it from a light bulb (where it is incoherent) or a laser (where it is coherent), is never a single-photon light source. It always has statistics that bunch photons, so there always will be times where two (or more) photons are in the apparatus at the same time. We will show precisely why this is so later in this section.

If dim light is never a single-photon light source, how do I make a single-photon light source? The calcium radiative cascade source is the single-photon source used in many of the early true single-photon light experiments (Aspect et al., 1981). This is an atom that has a two-photon excitation from the ground state (see Fig. 20.3). The excited state is metastable with a moderate lifetime, which then decays back to the ground state through a two-photon decay. The intermediate state ($4s4p\ ^1P_1$) is short lived. So, by shining a laser onto the calcium source (typically a dilute

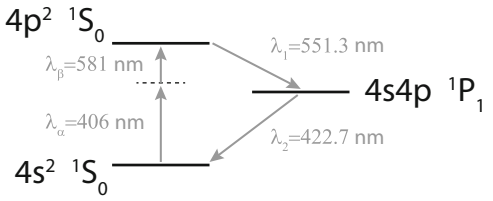

Fig. 20.3 Energy diagram of the calcium cascade single-photon light source. The atom is excited by a combination of two lasers, with the wavelengths of 406 nm and 581 nm. The excited state is moderately stable and then randomly decays to the intermediate state, emitting a photon of wavelength 551.3 nm. Then within about 5 ns, a second photon is emitted with a wavelength of 422.7 nm

gas source), we can excite a large number of calcium atoms. Then we wait until an atom spontaneously decays. We can detect the first photon released, which then "heralds" the release of the second photon. In this fashion, we can use coincidence detection of the two photons as a way to determine when a single photon has entered the experiment. Note that in the experiment, the two photons must be emitted in opposite directions for us to detect both of them. When this happens, we know that a single photon is imminent on our device. But this is not a single-photon source "on demand." We cannot tell exactly when the photon will be emitted, because the decay of the metastable state is random.

The more modern single-photon source is a parametric down conversion single-photon source. This is not an on-demand photon source either. It uses a laser that shines light onto a special nonlinear crystal made of β-barium borate or lithium niobate. This crystal converts a single high-frequency photon into two photons of approximately half the frequency with perpendicular polarizations. The photons are emitted in an entangled state that has indeterminate polarizations, either horizontal on the left and vertical on the right or vice versa. But they are entangled, not one or the other. One can create other forms of entangled polarizations as well. The energy of the two photons can vary somewhat, but the sum of the two emitted photons is very closely equal to the energy of the initial photon that it came from by energy conservation. These states are often called *hyperentangled states*, because they essentially have entangled polarizations, energies, and the directions that they emerge from the crystal. In this setup, because the photons come off with very precise angles between them, one can measure one of the photons, which then heralds the other. Because of the entanglement, one can know the polarization of the heralded photon as well. Alternatively, one can work with experiments on entangled pairs of photons too. The only reason why this source is not a single photon on demand source is that the odds of a single photon undergoing the down conversion is very low, so we use intense lasers shone onto the crystal, and then the entangled pairs emerge at random times.

More recently, there has been a push to try to develop on-demand single-photon light sources. One of the most promising approaches is with quantum dots. They work in the following way: (i) a pulse of light creates an exciton in the quantum dot, and (ii) the dot is in an optical cavity which promotes the emission of a single photon within a short time frame. One can establish that photons from this source, with the same polarization, are indistinguishable via the Hong-Ou-Mandel experiment. These are unheralded light single-photon light sources, but they can emit light within a rather definite time window.

Once we have created a single-photon source, we need to verify that it really is a single-photon light source. We do this by performing some experiments to clearly demonstrate the single-photon character of the light.

We now discuss how to detect a heralded photon from a Ca cascade source, as illustrated in Fig. 20.4. The single-photon source is to the left of a cylindrical waveguide of total cross-sectional area S. The photon detection rate is proportional to the electric field at the detector. So, a state $|\psi\rangle$ has a detection probability proportional to $\left\| \hat{\vec{E}}^{(+)}(\vec{r}, t)|\psi\rangle \right\|^2$, because we can only detect a photon if one

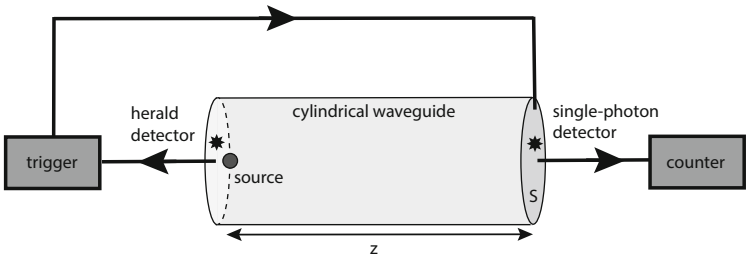

Fig. 20.4 Setup for a single-photon detection experiment from a heralded Ca cascade source. The source lies on the far left of the cylindrical waveguide. It is a gas of atoms illuminated by two lasers to excite the transition. When a photon from the first transition is detected in the left photodetector, it triggers the right detector to be activated for a period of 10–20 ns. When the second photon emerges, it is detected in the right detector with a delay time of approximately z/c

already exists in the state $|\psi\rangle$ and when we detect a photon, we remove it with a lowering operator. The detection rate is proportional to the cross-sectional area S and to the rate photons are incident from the left. So, we have the probability to detect one photon in a time interval Δt is

$$\Delta P(\vec{r}, t) = W^{(1)}(\vec{r}, t) S \Delta t, \tag{20.27}$$

with the single-photon detection probability $W^{(1)}$ that satisfies

$$W^{(1)}(\vec{r}, t) = s \left\| \hat{\vec{E}}^{(+)}(\vec{r}, t)|\psi\rangle \right\|^2, \tag{20.28}$$

where s is the sensitivity of the detector, described in more detail below.

This formula should seem reasonable to you as we will soon see when we perform the calculation. Assume $|\psi\rangle$ is a single-photon state in mode l. Then, we have

$$W^{(1)}(\vec{r}, t) = s \left\| \sum_{l'} i\vec{\varepsilon}_{l'} \mathcal{E}_{l'}^{(1)} e^{i(\vec{k}_{l'} \cdot \vec{r} - \omega_{l'} t)} \hat{a}_{l'} \hat{a}_l^\dagger |0\rangle \right\|^2. \tag{20.29}$$

The sum contains only one nonzero term, given by $l' = l$. We evaluate $\hat{a}_l \, \hat{a}_l^\dagger |0\rangle$, after using our add-zero trick to make the commutator $[\hat{a}_l \, , \hat{a}_l^\dagger]|0\rangle = |0\rangle$. So, we find that

$$W^{(1)}(\vec{r}, t) = s \left| \mathcal{E}_l^{(1)} \right|^2. \tag{20.30}$$

Now, we multiply the probability by the cross-sectional area S, and we use the interval $\Delta t = T$ corresponding to the lifetime of the excited state that emits our single photon. Then we find the probability for detection is

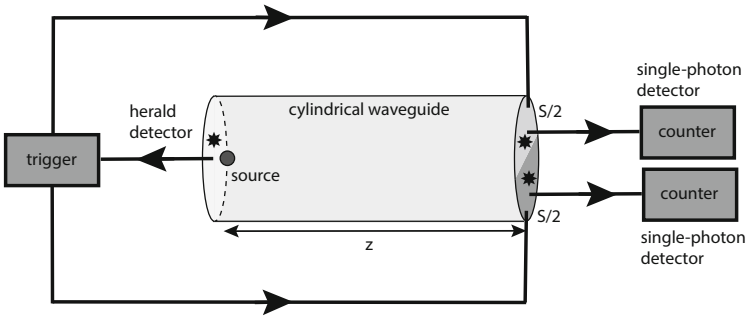

Fig. 20.5 Setup for a double-photon detection experiment from a heralded Ca cascade source. The source lies on the far left of the cylindrical waveguide. When a photon from the first transition is detected in the left photodetector, it triggers both the right detectors to be activated. These detectors detect over half of the circular area now. We will never see two photons detected in the time window when there is just one heralded photon, because a photon can only be detected once

$$P^{(1)} = s\left|\mathcal{E}_l^{(1)}\right|^2 ST = s\frac{\hbar\omega_l}{2\epsilon_0 V}ST = s\frac{\hbar\omega_l}{2\epsilon_0 c}, \tag{20.31}$$

because the volume the photon lives in is $V = ScT$. The probability for detecting a single photon equals one in the perfect case, so $s_{\text{perfect}} = 2\epsilon_0 c/\hbar\omega_l$ is the perfect quantum detection efficiency. We define the quantum efficiency as $s = \eta s_{\text{perfect}}$ with $0 < \eta \leq 1$. Then, the single-photon quantum amplitude for photons in the waveguide can be written as $\mathcal{E}_l^{(1)} = \frac{\eta}{ST}$.

The two-photon detection probability (at the same time) $W^{(2)}(\vec{r}, t, \vec{r}', t)$ satisfies

$$W^{(2)}(\vec{r}, t, \vec{r}', t) = s^2 \|\hat{\vec{E}}^{(+)}(\vec{r}, t)\hat{\vec{E}}^{(+)}(\vec{r}', t)|\psi\rangle\|^2. \tag{20.32}$$

We have the square of the efficiency, because there are two photon detectors, and the measurements are required to be coincident, or occurring within the same time window. What is the probability for detection of two photons in a one photon state as illustrated in Fig. 20.5? Because detection in one detector destroys the photon, and we only started with one photon, the joint detection must be zero! This is how we can verify that a single-photon source emits one and only one photon at a time. Note, however, that because these detectors have dark counts, we will sometimes see coincidences in the detectors, so a more sophisticated approach is needed to do this properly, and this is what we will explore in more detail below.

We get started by properly describing the photon state emerging from the source, because it cannot be a single-mode photon. This is because the excited state of an atom has a finite lifetime, and a single mode photon exists for all time and throughout all of space—it is an idealization that can never be created. Instead, we must work with multimode photons. For a photon emitted by an atomic decay, we have that

$$|\psi(t_0)\rangle = \sum_l c_l |1_l\rangle = \sqrt{\frac{c\Gamma}{L}} \sum_l \frac{e^{i\omega_l t_0}}{\omega_l - \omega_0 + i\frac{\Gamma}{2}} \hat{a}_l^\dagger |0\rangle. \tag{20.33}$$

This state is a linear superposition of single-photon states, centered about $\omega_l = \omega_0$ and with a lifetime determined by the linewidth $\Gamma > 0$. This functional form is called a *Lorentzian*. The state is normalized in the sense that $\sum_l |c_l|^2 = 1$, but we will not show the details of how to compute this normalization.

We have the state. Our next task is to compute the probability to measure it at position z if it was emitted at time t_0. We have

$$\begin{aligned} W^{(1)}(z,t) &= s \left\| \hat{\vec{E}}^{(+)}(z,t)|\psi(t_0)\rangle \right\|^2 \\ &= s \left\| \sum_{l'} i\vec{\varepsilon}_l \mathcal{E}_{l'}^{(1)} e^{ik_{l'}z - i\omega_{l'}t} \hat{a}_{l'} \sum_l \sqrt{\frac{c\Gamma}{L}} \frac{e^{i\omega_l t_0}}{\omega_l - \omega_0 + i\frac{\Gamma}{2}} \hat{a}_l^\dagger |0\rangle \right\|^2 \\ &= \frac{c\Gamma}{L} \left\| \sum_l \mathcal{E}_l^{(1)} \frac{e^{-i\omega_l(t - t_0 - \frac{z}{c})}}{\omega_l - \omega_0 + i\frac{\Gamma}{2}} \right\|^2 . \end{aligned} \tag{20.34}$$

We have to do the summation. Unfortunately, the only way to do this is to convert the summation to an integral and evaluate it by the method of residues. We won't do that here. Instead we will just state the result. But this summation is called a Fourier transform when it is converted to an integral, and Fourier transforms come in pairs—if the Fourier transform of f is g, then the inverse Fourier transform of g is f. We can do the inverse transformation by a simple summation, and that is given in Problem 20.4.1 below. So, while we cannot do the summation without calculus here, we can show you that the answer we obtained is a valid one in the problems. The result is

$$W^{(1)}(z,t) = s \frac{\hbar\omega_0}{2\epsilon_0 V} \frac{L\Gamma}{c} e^{-\Gamma(t - t_0 - \frac{z}{c})} \theta(t - t_0 - \tfrac{z}{c}), \tag{20.35}$$

where we evaluated $\mathcal{E}_l^{(1)}$ at $\omega_l = \omega_0$ because the multimode superposition is strongly peaked about ω_0 and $\theta(t)$ is the unit step function, which equals 0 for $t < 0$ and equals 1 for $t > 0$.

The experiment is schematically illustrated in Fig. 20.4. Once a photon has been heralded, we wait the time $\frac{z}{c}$ for the second photon to reach our detector and then trigger the detector to detect over a short window of a few nanoseconds. The actual experiment has a somewhat different geometry, where we use a parabolic reflector to reflect all of our single photons down the cylindrical waveguide from the left. What does the probability for detection look like? It is plotted for you in Fig. 20.6. You can see we get nothing until there has been enough time for the photon to reach the detector, then the probability becomes very high very quickly, and then it decays with an exponential decay determined by the linewidth Γ. You can think of this as telling us what a photon is. The particle nature arises, because we have to wait the

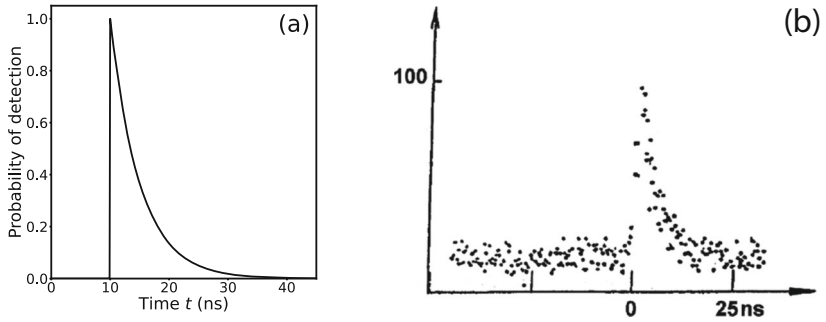

Fig. 20.6 Simulated (left) and experimental (right) signals from a heralded single-photon Ca-cascade source. You can see in the experimental data that there are extra spontaneous emissions and dark counts, so the signal does not go to zero. The experimental figure is courtesy of Alain Aspect from Aspect (2002)

time for it to travel a distance z at speed c before we can detect it. But, it has some wavelike character, because they are not all instantaneously detected, but instead they are spread out in a wavepacket with different probabilities to be detected (the superposition is an indeterminant state). There is a decaying probability to find the photon over a period of time after the arrival time. Also, once measured, we cannot measure it again. The photon can only be measured once. One can also emphasize a particle perspective only for this experiment—in this case we would say that we have to wait over a period of time for the photon to decay at some time during a few lifetimes, but once it is created, it travels at speed c a distance z to the detector and is immediately detected. In this experiment, we cannot tell the difference between the two different ways to describe the system. Note that for the Ca-cascade source, the lifetime is about 5 ns ($\Gamma \approx 0.2\ \mathrm{s}^{-1}$), and the collection window for the detector would be about 20 ns after the time needed for the photon to reach the detector.

We have shown how we can make a single-photon source. But have we really verified that it is a single photon? There is a different experiment that is used to verify the quality of the single-photon light source. It starts by shining light from a single-photon source onto a beam splitter as shown in Fig. 20.7. The photon coming in from port 1 can be reflected or transmitted. This is the type of experiment that we can describe using the Feynman model with the arrows, and we can think of the arrows as loosely being the electric fields. Now, the input arrow into port 1 will be split into a reflected arrow and a transmitted arrow, and the same for a photon input into port 2. One of the reflections gets a minus sign, because it reflects from glass. So, we have

$$E_3 = rE_1 + tE_2 \ \text{ and } \ E_4 = tE_1 - rE_2, \tag{20.36}$$

and we have $r = t = \frac{1}{\sqrt{2}}$ for a 50–50 beam splitter. The probability for single-photon detection in detector 3 and detector 4 then are

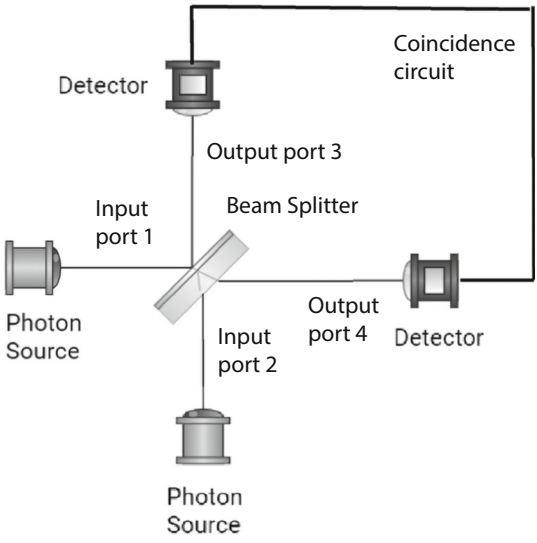

Fig. 20.7 Experiment to determine the single-photon character of a single-photon light source. A single-photon light source beam is shot at one of the input ports of the beam splitter, and then we have detectors along both output ports. Note that we show two photon sources here, but the experiment uses *only one* of them. There is a coincidence circuit to determine whether the two detectors fire within a short time window (within a few ns)

$$W^{(1)}(\vec{r}_3,t) = s\left\|\hat{E}_3^{(+)}(\vec{r}_3,t)|\psi_{\text{input}}\rangle\right\|^2 \text{ and } W^{(1)}(\vec{r}_4,t) = s\left\|\hat{E}_4^{(+)}(\vec{r}_4,t)|\psi_{\text{input}}\rangle\right\|^2, \tag{20.37}$$

where $\hat{E}_3^{(+)} = r\hat{E}_1^{(+)} + t\hat{E}_2^{(+)}$ and $\hat{E}_4^{(+)} = t\hat{E}_1^{(+)} - r\hat{E}_2^{(+)}$. $|\psi_{\text{input}}\rangle$ is the input photon state into the beam splitter. If the input state is a single-photon state, we should not see any double counts (both detectors fire) within the measurement window of a few ns. The real experiment has some issues with it that we have to take into account. First, not all heralded photons will be detected. Let's say the probability to detect the heralded photon is $|\gamma|^2$. This makes the input state equal to

$$\gamma|1\rangle_1 \otimes |0\rangle_2 + \sqrt{1-|\gamma|^2}|0\rangle_1 \otimes |0\rangle_2. \tag{20.38}$$

Then, if we recall the quantum efficiency, we find that the probability for detection in port 3 is

$$W^{(1)}(\vec{r}_3,t_3) = \eta_3|\gamma|^2|r|^2\theta(t_3 - t_0 - \tfrac{z_3}{c})e^{-\Gamma(t_3-t_0-\frac{z_3}{c})} \tag{20.39}$$

and in port 4, we have

$$W^{(1)}(\vec{r}_4,t_4) = \eta_4|\gamma|^2|t|^2\theta(t_4 - t_0 - \tfrac{z_4}{c})e^{-\Gamma(t_4-t_0-\frac{z_4}{c})}. \tag{20.40}$$

If we wait a few inverse Γs, we will be guaranteed to collect the photon in each channel, so the number of photons found in the two output ports is

$$N_3 = \eta_3|\gamma|^2|r|^2, \quad N_4 = \eta_4|\gamma|^2|t|^2, \text{ and } N_c = 0. \tag{20.41}$$

Here, N_c are the coincidences, where both detectors fire. But, in a real experiment, there will be some coincidences due to dark photons and due to multiple spontaneous emissions occurring in one measurement window. The atomic source is dilute enough that we do not have too many spontaneous emissions, but dense enough that we have enough single photons that we can carry out the experiment. Then the probabilities to measure in ports 3 and 4 and of coincidences are given by

$$P_3 = \frac{N_3}{N_H}, \quad P_4 = \frac{N_4}{N_H}, \quad \text{and} \quad P_c = \frac{N_c}{N_H}, \tag{20.42}$$

with N_H the number of heralded photons. We define the ratio α by

$$\alpha = \frac{P_c}{P_3 P_4} = \frac{N_c N_H}{N_3 N_4}. \tag{20.43}$$

A perfect single-photon source has $\alpha = 0$, while classical light sources always have $\alpha = 1$. This allows us to use this parameter to test for a single-photon source. For a classical source, we have $W^{(2)} = \left(W^{(1)}\right)^2$, which gives $\alpha = 1$ (you will prove this in Problem 20.4.3 for coherent states of light). Now, when we add in dark currents and spontaneous emissions, we find $\alpha_{\text{classical}} \geq 1$, so we can truly verify a single-photon source whenever $\alpha < 1$. The first experiment showing this is in Fig. 20.8.

We consider the effect of extra spontaneous emissions. Say within the measurement window I have $\nu < 1$ extra spontaneous emissions. Then, we would have $P_3 = \epsilon_3(1 + \nu)$ and $P_4 = \epsilon_4(1 + \nu)$, where ϵ_3 and ϵ_4 are the probabilities to be measured in ports 3 and 4 when there are no spontaneous emissions. The probability for a coincidence is then

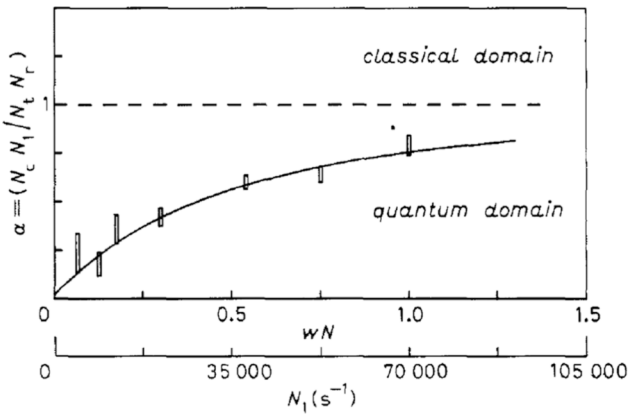

Fig. 20.8 Original verification of a single-photon light source by measuring α on a beam splitter. Image from Grangier et al. (1986) and used with permission © Institute of Physics

$$N_c = \epsilon_3(1+\nu)\epsilon_4(1+\nu) - \epsilon_3\epsilon_4 = \epsilon_3\epsilon_4(2\nu+\nu^2). \tag{20.44}$$

This arises, because it must be proportional to the product of the measurements in each port, but when $\nu = 0$, there cannot be any coincidences. Using this form, we then find that

$$\alpha = \frac{2\nu+\nu^2}{(1+\nu)^2} = 1 - \frac{1}{(1+\nu)^2} < 1. \tag{20.45}$$

This is why we say for a single-photon source, we must have $\alpha < 1$. In one of the early experiments, they were able to achieve an $\alpha \approx 0.18$, corresponding to $\nu \approx 0.1$. This experiment is often called a g2 experiment.

Let's summarize what a single-photon source is. The single-photon source creates a superposition of single-photon states that is an eigenstate of the total number operator $\hat{N} = \sum_l \hat{a}_l^\dagger \hat{a}_l$ but is not an eigenstate of the Hamiltonian $\sum_l \hbar\omega_l \hat{a}_l^\dagger \hat{a}_l$. So, it has definite particle number but does not have definite energy. When we measure a multimode single-photon state, it can be measured once and only once. This can be verified by looking at coincidences when the single-photon light source is shone onto a beam splitter with a detector on each output port. Note how this is much more nuanced than just a photon is a particle of light. In fact, these are the critical ideas needed to understand properly what a photon really is. We have a video https://tinyurl.com/cuvmey2x describing single-photon sources and detectors.

Section 20.4 Problems:

20.4.1 Inverse Fourier Transform of the Integral Needed in the Text

In this problem, we perform the partner summation for the one needed in the main text. We take as the function we will sum the square root of the norm squared summation result we had in the main text. We also must weigh the summation with an exponential phase factor. That is, we will sum

$$\sum_{j=0}^{N} e^{i\omega j\Delta t - \frac{\Gamma}{2} j\Delta t}\Delta t, \tag{20.46}$$

where the unit step function is taken into account by the summation running from 0 to N. Show that this is equal to

$$\frac{\Delta t}{1 - e^{i\omega\Delta t - \Gamma\Delta t}}, \tag{20.47}$$

in the limit as $N \to \infty$ when $\Gamma > 0$. If we also take the limit as Δt to 0, we obtain

$$\frac{i}{\omega + i\frac{\Gamma}{2}}. \tag{20.48}$$

Using the theorem of Fourier transforms, which we will not attempt to prove here, the summation corresponding to the Fourier transform is then

$$i \sum_{j=0}^{N} \Delta\omega \frac{e^{-ij\Delta\omega t}}{j\Delta\omega + i\frac{\Gamma}{2}}. \tag{20.49}$$

We let $j\Delta\omega \to \omega_l - \omega_0$ and $t \to t - t_0 - \frac{z}{c}$, and you can see it becomes the summation we needed to work out. Because of this partner property, we have established the summation needed for the calculation in the main text.

20.4.2 How Do We Detect Photon Bunching?

Write a short essay explaining how you can set up an experiment to detect photon bunching of a classical light source? Think about what the probability is to find two photons together, three photons together, and so on.

20.4.3 The g2 Experiment with Coherent States

Compute the g2 probability for a coherent state. Recall that a coherent state $|\alpha\rangle$ satisfies $\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$. You can do this with a single mode coherent state. Show that for this state, we have $W^{(2)} = (W^{(1)})^2$ and that $\alpha = 1$.

Alain Aspect provided some experimental numbers for coherent states in an experiment done with a light-emitting diode. There, they had $N_H = 9.55 \times 10^8$, $N_A = 8.32 \times 10^5$, $N_B = 9.14 \times 10^5$, and $N_c = 840$. Use these numbers to compute α. You should find it is larger than 1, because this is a classical source of light.

You may have been surprised that only a very small fraction of heralded photons are subsequently measured. You might want to think about why this is so.

20.4.4 Essay on the Effect of Dark Counts on the Experiment

Write an essay explaining how the presence of dark counts affects the determination of α.

20.4.5 Compute g2 with Real Experimental Data

Prof. Kiko Galvez shared data from a modified g2 experiment he conducted with his undergraduate students at Colgate University. In this experiment, a parametric down conversion source is used. One of the entangled photons emitted from the nonlinear crystal goes straight to a detector, which measures the number of heralded photons we should see in the system.

The other photon is sent into a Mach-Zehnder interferometer that has control over the path length of the reference path. Detectors are then set up at the bright and dark detector locations. This modification is put in, so that they can test the effect of varying r and t for the beam splitter without having to set up the experiment again. Recall by adding a phase to the reference path (or changing the length of the reference path) we can change the probabilities at the bright and dark detectors.

The results of the experiment are as follows for a total collection time of 45 s: $N_H = 2\ 914\ 321$, $N_{\text{bright}} = 108\ 972$, $N_{\text{dark}} = 120\ 788$, and $N_{\text{coincident}} = 128$. Note that the detection in the bright and dark detectors is for the heralded photons, where the partner photon is detected in the same time window as the bright or dark detector. Use this data to compute g2.

You might be surprised that the losses of the heralded photon are so large—about 90% of the photons that should have been detected are lost. This occurs due to alignment issues, detector efficiency, and other experimental errors. Describe what the different sources of these losses could be.

It is amazing that one can perform reliable single-photon experiments in cases where there are such high losses. The quality of this single-photon source is excellent, with $\alpha \ll 1$.

20.5 Heterodyne and Homodyne Detection

In principle, we should be able to directly measure the electric field due to photons, because the electric field is an observable. But for optical light, the field oscillates too rapidly to be directly measured. Oscillating electric signals can be measured on an oscilloscope, with the fastest oscilloscope operating at frequencies up to 100 GHz. These devices are a tour de force of engineering that requires using superconducting electronics. Most oscilloscopes can measure signals that are less than a few GHz. But light oscillates at $10^{14}-10^{15}$ Hz (five to six orders of magnitude faster than a standard oscilloscope). It is impossible to observe such fast oscillations with any known electronics. So, if the electric field oscillates too quickly, we cannot easily measure it. Note that the photomultiplier tube did not detect the electric field by observing its oscillations. Instead, it detected photons that were energetic enough via the photoelectric effect and an electron avalanche that created a current pulse. In this section, we discuss strategies that one can use to measure the electric field using clever ideas. First, we discuss heterodyne detection. It is used to amplify a

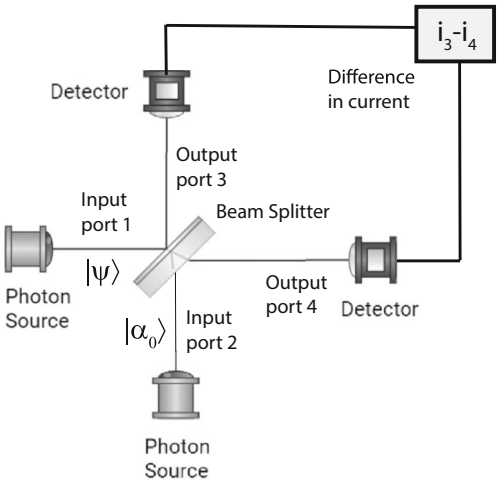

Fig. 20.9 Setup for a heterodyne detection experiment with a weak photon source $|\psi\rangle$ input into port 1 and a strong local oscillator signal as a coherent state $|\alpha_0\rangle$ with $|\alpha_0| \gg 1$ on port 2 and a frequency that closely matches the frequency we are trying to measure. The experiment measures the difference in the photon counts between the two detectors

weak signal and also to reduce the oscillation rate by using the phenomenon of beats.

Figure 20.9 shows the setup, which works with a beam splitter, but now with light incident on both inputs. The first port is used for the weak signal we are trying to measure. The second port has a large amplitude local oscillator signal at a given frequency that is very close to the frequency we are trying to measure. We assume, for now, that both input signals are coherent states. We then detect the difference of the photocurrents on the two arms of the beam splitter. Note that this is different from counting individual photons but instead is accumulating the total number of photons incident over an extended time window and then looking at the difference between these numbers through the different ports. Our initial state is

$$|\psi_{in}\rangle = |\alpha_1(t)\rangle \otimes |\alpha_2(t)\rangle, \tag{20.50}$$

the tensor product of two coherent states incident on each port; we use α_1 for the weak signal coherent state incident at port 1 and α_2 for the strong local oscillator coherent state incident at port 2. The electrical current is the rate of change of the probability per unit time multiplied by the electric charge, so we find

$$i_4(t) = eSW^{(1)}(\vec{r}_4, t_4) = eSs\|E_4^{(+)}(\vec{r}_4, t_4)|\psi_{in}\rangle\|^2, \tag{20.51}$$

where the S is the cross-sectional area of the photomultiplier tube. Using the facts that $E_4 = tE_1 - rE_2$, $\alpha(t) = \alpha e^{-i\omega t}$, and that coherent states are eigenstates of the annihilation operator, then yields

$$i_4(t) = eSs\left(\mathcal{E}_\omega^{(1)}\right)^2 |t\alpha_1 e^{-i\omega_1 t} - r\alpha_2 e^{-i\omega_2 t}|^2. \tag{20.52}$$

We write $\alpha_1 = |\alpha_1|e^{i\phi_1} = \sqrt{N_1}e^{i\phi_1}$, $\alpha_2 = |\alpha_2|e^{i\phi_2} = \sqrt{N_2}e^{i\phi_2}$, $r = |r|$, and $t = |t|$ (both reflection and transmission coefficients are real); N_α is the number of photons that are, on average, in the respective coherent state because $\langle\alpha|\hat{N}|\alpha\rangle = |\alpha|^2$. Then we recall that $s\left(\mathcal{E}_\omega^{(1)}\right)^2 = \frac{\eta\hbar\omega}{ST}$, with T the time interval over which the photons travel before measurement. Simplifying our expression gives us that

$$i_4(t) = \frac{e\eta\hbar\omega}{T}\left(|t|^2N_1 + |r|^2N_2 - 2|r||t|\sqrt{N_1N_2}\cos\left((\omega_1-\omega_2)t - \phi_1 + \phi_2\right)\right). \tag{20.53}$$

The number of photons in the signal depends on the time interval of the measurement (in other words, we are calculating the rate of measuring photons in the detector). We define the photon flux Φ^{phot} to be the energy flux divided by $\hbar\omega$, or Energy density $= \Phi^{\text{phot}}\hbar\omega = \frac{|\alpha|^2\hbar\omega}{T}$. Then we find the current becomes

$$\begin{aligned} i_4(t) = e\eta\Bigg(&|t|^2\Phi_1^{\text{phot}} + |r|^2\Phi_2^{\text{phot}} - 2|r||t|\sqrt{\Phi_1^{\text{phot}}\Phi_2^{\text{phot}}} \\ &\times\cos\left((\omega_1-\omega_2)t - \phi_1 + \phi_2\right)\Bigg). \end{aligned} \tag{20.54}$$

If we can adjust the setup so that $|t|^2\Phi_1^{\text{phot}} = |r|^2\Phi_2^{\text{phot}}$, then we find we have the signal becomes

$$i_4(t) = 2|t|^2e\eta\Phi_1^{\text{phot}}\big(1 - \cos((\omega_1-\omega_2)t - \phi_1 + \phi_2)\big) \tag{20.55}$$

which has a maximal visibility, oscillating between 0 and a maximal value. If the frequencies agree to 4–5 decimal points, then we can measure the oscillation on an oscilloscope, and it is proportional to the weak signal amplitude. If, instead, we do not match the amplitudes as proposed above, we then measure an oscillation with an amplitude proportional to $\sqrt{\Phi_1^{\text{phot}}\Phi_2^{\text{phot}}}$, which can be much larger if the local oscillator flux is very large compared to the signal we are trying to measure. Note that the cleverness of this approach is that we take a small constant signal Φ_1^{phot}, which is easily lost in a noisy background, and boost it to an oscillating current whose amplitude is easier to measure and is clearly distinguished from the background. If we calculate $i_3(t)$, we find it is the same as $i_4(t)$ except the sign of the cross term is opposite. Then, by measuring $i_3(t) - i_4(t)$, we can isolate the oscillating term. In heterodyne detection, we can either measure at one output port or take the difference at the two output ports.

But do we really win? If one calculates the signal-to-noise ratio for each proposed measurement, we find they are the same (this is Problem 20.5.2 for you below). So why bother? Because the dark current is independent of the amplitude of the signals. By using heterodyne detection, we can reduce the noise from the dark current, relative to the signal we are measuring, which makes the net errors smaller. So heterodyne detection is a plus and is often used.

Homodyning is very similar to heterodyning, but we pick $\omega_1 = \omega_2$, $\vec{\epsilon}_1 = \vec{\epsilon}_2$, and $r = t = \frac{1}{\sqrt{2}}$. You might ask, how do we do this—usually we use the same laser source for the local oscillator and for the signal. Then by splitting the initial beam, we have a source that oscillates at essentially the same frequency (although we know lasers have a range of frequencies they oscillate at because they cannot be a pure monochromatic source). Good quality lasers typically have spreads in their frequency on the order of MHz, but high-quality lasers can go much lower—spreads as low as 1 Hz are possible. Moreover, fluctuations in the laser lead to the same fluctuations, at the same time in the signal beam. Again, we measure $i_3(t) - i_4(t)$, which is not measuring just coincidences, because we add the difference over a long time interval. Recall that

$$i_{3,4}(t) = \frac{e\eta}{T}\langle\psi_{in}|\hat{N}_{3,4}|\psi_{in}\rangle. \tag{20.56}$$

To get the number operators at the output ports, we simply recall that the raising and lowering operators transform as the field transforms, so

$$\hat{a}_3 = r\hat{a}_1 + t\hat{a}_2 \text{ and } \hat{a}_4 = t\hat{a}_1 - r\hat{a}_2. \tag{20.57}$$

This then gives

$$\hat{N}_3 = \hat{a}_3^\dagger\hat{a}_3 = |r|^2\hat{N}_1 + |t|^2\hat{N}_2 + r^*t\hat{a}_1^\dagger\hat{a}_2 + rt^*\hat{a}_2^\dagger\hat{a}_1 = \frac{1}{2}(\hat{N}_1 + \hat{N}_2 + \hat{a}_1^\dagger\hat{a}_2 + \hat{a}_2^\dagger\hat{a}_1), \tag{20.58}$$

since $r = t = \frac{1}{\sqrt{2}}$. In Problem 20.5.3, you will determine $\hat{N}_4$.

One of the main uses of homodyne detection is to measure the quadrature expectation values $\hat{Q}$ and $\hat{P}$. Let us show you how. Note that $i_3 - i_4 \propto \langle\hat{N}_3 - \hat{N}_4\rangle = \langle\hat{a}_2^\dagger\hat{a}_1 + \hat{a}_1^\dagger\hat{a}_2\rangle$, so we find that

$$i_3(t) - i_4(t) = \frac{e\eta}{T}\langle\psi_{in}|\hat{a}_2^\dagger\hat{a}_1 + \hat{a}_1^\dagger\hat{a}_2|\psi_{in}\rangle \tag{20.59}$$

Our input state is $|\psi_{in}\rangle = |\psi_1\rangle \otimes |\psi_2\rangle$, and we let $|\psi_2\rangle = |\alpha_{LO}\rangle$ with $\alpha_{LO} = |\alpha_{LO}|e^{i\phi_{LO}}$. Here LO denotes the *local oscillator*. Then, because the local oscillator state is a coherent state, we find that

$$i_3(t) - i_4(t) = \frac{e\eta}{T}|\alpha_{LO}|(\langle\psi_1|\hat{a}_1|\psi_1\rangle e^{-i\phi_{LO}} + \langle\psi_1|\hat{a}_1^\dagger|\psi_1\rangle e^{i\phi_{LO}}) \tag{20.60}$$

Using the Euler identity, we find it can be rewritten as

$$i_3(t) - i_4(t) = \frac{e\eta}{T}|\alpha_{LO}|(\langle\psi_1|\hat{a}_1 + \hat{a}_1^\dagger|\psi_1\rangle\cos\phi_{LO} - i\langle\psi_1|\hat{a}_1 - \hat{a}_1^\dagger|\psi_1\rangle\sin\phi_{LO}). \tag{20.61}$$

If we recall that $\hat{a}+\hat{a}^\dagger \propto \hat{Q}$ and $-i\hat{a}+i\hat{a}^\dagger \propto \hat{P}$, we see that by adjusting the phase of the local oscillator, we can measure either the Q-quadrature or the P-quadrature. So balanced homodyne detection is used to measure the quadrature operators as a function of the local oscillator phase!

Now, we need to calculate the fluctuations. We get them by calculating

$$(\hat{N}_3-\hat{N}_4)^2=(\hat{a}_2^\dagger\hat{a}_1+\hat{a}_1^\dagger\hat{a}_2)^2=(\hat{a}_2^\dagger)^2\hat{a}_1^2+(\hat{a}_1^\dagger)^2\hat{a}_2^2+\hat{a}_2^\dagger\hat{a}_1\hat{a}_1^\dagger\hat{a}_2+\hat{a}_1^\dagger\hat{a}_2\hat{a}_2^\dagger\hat{a}_1. \tag{20.62}$$

We must normal order the result above, bringing all raising operators to the left and lowering operators to the right. This gives us

$$(\hat{N}_3-\hat{N}_4)^2=(\hat{a}_2^\dagger)^2\hat{a}_1^2+(\hat{a}_1^\dagger)^2\hat{a}_2^2+2\hat{a}_1^\dagger\hat{a}_2^\dagger\hat{a}_1\hat{a}_2+\hat{a}_1^\dagger\hat{a}_1+\hat{a}_2^\dagger\hat{a}_2 \tag{20.63}$$

Now, we take the average with respect to the in state $|\psi_{in}\rangle$. We need to recall that the local oscillator state has a lot of photons in it, so $\frac{|\alpha_{LO}|^2}{T}\gg 1$, while the expectation value $\langle\psi_1|\hat{a}_1^\dagger\hat{a}_1|\psi_{in}\rangle \ll |\alpha_{LO}|^2$; this means we can drop the term $\langle\psi_1|\hat{a}_1^\dagger\hat{a}_1|\psi_1\rangle$ relative to the $\langle\alpha_{LO}|\hat{a}_2^\dagger\hat{a}_2|\alpha_{LO}\rangle$. So, we find that

$$\begin{aligned}\langle\psi_{in}|(\hat{N}_3-\hat{N}_4)^2|\psi_{in}\rangle &= |\alpha_{LO}|^2\big(\langle\psi_1|\hat{a}_1^2|\phi_1\rangle e^{-2i\phi_{LO}}+\langle\psi_1|(\hat{a}_1^\dagger)^2|\psi_1\rangle e^{2i\phi_{LO}}\\ &\quad+2\langle\psi_1|\hat{a}_1^\dagger\hat{a}_1|\psi_1\rangle+1\big)\\ &= |\alpha_{LO}|^2\langle\psi_1|\big(\hat{a}_1^\dagger e^{i\phi_{LO}}+\hat{a}_1e^{-i\phi_{LO}}\big)^2|\psi_1\rangle\end{aligned} \tag{20.64}$$

where the last result uses the fact that $[\hat{a}_1,\hat{a}_1^\dagger]=1$. Recalling that $\langle\psi_{in}|(\hat{N}_3-\hat{N}_4)|\psi_{in}\rangle=|\alpha_{LO}|\langle\psi_1|\hat{a}_1^\dagger e^{i\phi_{LO}}+\hat{a}_1e^{-i\phi_{LO}}|\psi_1\rangle$, we find that

$$\begin{aligned}\Big(\Delta(\hat{N}_3-\hat{N}_4)_{|\psi_{in}\rangle}\Big)^2 &= |\alpha_{LO}|^2\big(\langle\psi_1|(\hat{a}_1^\dagger e^{i\phi_{LO}}+\hat{a}_1e^{-i\phi_{LO}})^2|\psi_1\rangle\\ &\quad-\langle\psi_1|(\hat{a}_1^\dagger e^{i\phi_{LO}}+\hat{a}_1e^{-i\phi_{LO}})|\psi_1\rangle^2\big)\\ &= |\alpha_{LO}|^2\left(\Delta(\hat{a}_1^\dagger e^{i\phi_{LO}}+\hat{a}_1e^{-i\phi_{LO}})_{|\psi_1\rangle}\right)^2\end{aligned} \tag{20.65}$$

These results are related to the quadrature operators. Recall that $\hat{Q}_1=\sqrt{\frac{\hbar}{2}}(\hat{a}_1+\hat{a}_1^\dagger)$ and $\hat{P}_1=\sqrt{\frac{\hbar}{2}}\frac{1}{i}(\hat{a}_1-\hat{a}_1^\dagger)$. We define a generalized quadrature operator to be

$$\hat{Q}_1(\phi_{LO})=\sqrt{\frac{\hbar}{2}}(\hat{a}_1e^{-i\phi_{LO}}+\hat{a}_1^\dagger e^{i\phi_{LO}}). \tag{20.66}$$

Then $\hat{Q}_1=\hat{Q}_1(\phi_{LO}{=}0)$ and $\hat{P}_1=\hat{Q}_1(\phi_{LO}{=}\frac{\pi}{2})$. Using the generalized quadratures $\hat{Q}_1(\theta)=\sqrt{\frac{\hbar}{2}}(\hat{a}_1^\dagger e^{i\theta}+\hat{a}_1e^{-i\theta})$ and $\hat{P}_1(\theta)=\hat{Q}_1(\theta+\frac{\pi}{2})$, you will compute

the generalized quadratures commutator in Problem 20.5.4. Homodyne detection can be employed to measure these generalized quadrature operators, which act like position and momentum when the phase is changed by $\frac{\pi}{2}$.

We have learned a few important ideas about measuring photons. Heterodyne detection can help boost a weak signal (and make it possible to measure in the first place) by using the phenomenon of beats. Homodyning is used for measuring the quadrature operators, which are tuned by adjusting the phase of the local oscillator state with respect to the weak signal state. These practical ideas greatly help our ability to perform experiments with light.

Section 20.5 Problems:

20.5.1 The Phenomenon of Beats

When two oscillating signals with frequencies that are close to each other are in superposition, we see the phenomenon of beats. Take the sum of two oscillations with the same amplitude $e^{i\omega_1 t} + e^{i\omega_2 t}$ and reexpress it in terms of the average frequency $\omega = \frac{1}{2}(\omega_1 + \omega_2)$ and the frequency difference $\Delta\omega = \omega_1 - \omega_2$. You should be able to factor out an oscillation term at the average frequency and have left over a cosine of the frequency difference. This latter factor oscillates very slowly relative to the oscillations at the average frequency and is easier to measure because of its slow variation in time.

Explain how this phenomenon of beats can be employed to measure signals containing frequencies that vary too fast to be measured with available electronics.

20.5.2 Signal to Noise for Heterodyne Detection

In this problem, we discuss how to determine the signal-to-noise ratio for measurements of photocurrents. Recall the measurement of a photocurrent requires us to measure over a long time interval T. Then the current is given by $i = eN(T)/T$, where $N(T) \propto \langle\psi|\hat{N}|\psi\rangle$ is the total number of photons counted over the time window T, or equivalently, the number of photons in the quantum volume ScT.

In a coherent state, we have $\langle\alpha|\hat{a}^\dagger\hat{a}|\alpha\rangle = |\alpha|^2$, and $\langle|\hat{a}^\dagger\hat{a}\hat{a}^\dagger\hat{a}|\psi\rangle = |\alpha|^4 + |\alpha|^2$, so $(\Delta N)_\alpha = |\alpha|$.

Let Φ^e be the photon energy flux. Then, we showed in the main text that $|\alpha|^2 = \Phi^e T/\hbar\omega$. Use this to show that

$$\frac{N(T)}{\Delta N(T)} = \sqrt{\frac{\Phi^e T}{\hbar\omega}} \tag{20.67}$$

and that the fluctuations in the photocurrent are given by

$$\Delta i = e\frac{\Delta N(T)}{T} = e\sqrt{\frac{\Phi^e}{\hbar\omega T}} = e\sqrt{\frac{\Phi^e \Delta\omega}{2\pi\hbar\omega}}, \tag{20.68}$$

where $\Delta\omega = 2\pi/T$ is the frequency bandwidth associated with the collection time. This is called the *Shot noise* associated with the current. Note that these calculations assume perfect quantum efficiency, or $\eta = 1$.

Now, back to the heterodyne measurement. Suppose we measure only in output port 4. If we do not heterodyne, and we measure the weak signal only, we have $i_{\text{direct}} = \eta e t^2 \Phi_1^{\text{phot}}$, and if we measure it with heterodyning, we have $i_{\text{het}} = \eta e r t \sqrt{\Phi_1^{\text{phot}} \Phi_2^{\text{phot}}}$. Show that

$$\frac{i_{\text{het}}}{i_{\text{direct}}} = \frac{r}{t}\sqrt{\frac{\Phi_2^{\text{phot}}}{\Phi_1^{\text{phot}}}}. \tag{20.69}$$

As we saw before, this boosts the signal if the local oscillator amplitude is large.

The signal-to-noise ratio is given by the ratio of the current to its fluctuations, so

$$\frac{S}{N} = \frac{i}{\Delta i}. \tag{20.70}$$

Show that the signal-to-noise ratio for the direct measurement and for the heterodyned measurement is essentially the same if we ignore dark current. Of course, including effects of dark current show that heterodyning is always better.

20.5.3 Compute N_4 for Homodyne Detection

Using the operators in Eq. (20.57) to compute $\hat{N}_4$.

20.5.4 Compute the Generalized Quadratures Commutator

Compute the generalized quadrature operators commutator $[\hat{Q}_1(\theta), \hat{P}_1(\theta)]$ using $\hat{Q}_1(\theta) = \sqrt{\frac{\hbar}{2}}(\hat{a}_1^\dagger e^{i\theta} + \hat{a}_1 e^{-i\theta})$ and $\hat{P}_1(\theta) = \hat{Q}_1(\theta + \frac{\pi}{2})$.

20.5.5 Measuring the Electric Field

Recall that the electric field is given by

$$\vec{E}(\vec{r},t) = \sum_l \vec{\epsilon}_l \mathcal{E}_l^{(1)} \sqrt{\frac{2}{\hbar}} \left(-\hat{Q}_l \sin(\vec{k}_l \cdot \vec{r} - \omega_l t) - \hat{P}_l \cos(\vec{k}_l \cdot \vec{r} - \omega_l t) \right); \tag{20.71}$$

$\hat{Q}$ is the real part of the field and $\hat{P}$ is the imaginary part of the field. Take the state to be $|\psi_1\rangle = |\alpha_1\rangle$ with $\alpha_1 = |\alpha_1| e^{i\phi_1}$. Show that

$$\langle \psi_1 | \hat{Q}(\theta) | \psi_1 \rangle = \sqrt{\frac{\hbar}{2}} \langle \psi_1 | \hat{a}_1 e^{-i\theta} + \hat{a}_1^\dagger e^{i\theta} | \psi_1 \rangle = \sqrt{2\hbar} |\alpha_1| \cos(\theta - \phi_1) \tag{20.72}$$

and

$$\langle \psi_1 | \hat{P}(\theta) | \psi_1 \rangle = \sqrt{\frac{\hbar}{2}} \langle \psi_1 | - i\hat{a}_1 e^{-i\theta} + i\hat{a}_1^\dagger e^{i\theta} | \psi_1 \rangle = -\sqrt{2\hbar} |\alpha_1| \sin(\theta - \phi_1). \tag{20.73}$$

Next, we compute the variances. Show that

$$\begin{aligned} \langle \psi_1 | \hat{Q}^2(\theta) | \psi_1 \rangle &= 2\hbar |\alpha_1|^2 \cos^2(\theta - \phi_1) + \frac{\hbar}{2} \\ \langle \psi_1 | \hat{P}^2(\theta) | \psi_1 \rangle &= 2\hbar |\alpha_1|^2 \sin^2(\theta - \phi_1) + \frac{\hbar}{2}. \end{aligned} \tag{20.74}$$

You will need to use some trigonometric identities to get it into this form. This then gives us the variances

$$(\Delta Q(\theta))_{|\psi_1\rangle} = \sqrt{\frac{\hbar}{2}} = (\Delta P(\theta))_{|\psi_1\rangle}. \tag{20.75}$$

Hence all coherent states of light are minimal uncertainty states! We will see an experimental image of these fluctuations later in the chapter.

20.6 Mach-Zehnder Interferometer

The Mach-Zehnder interferometer is illustrated in Fig. 20.10. We wish to analyze it using our operator formalism. The standard way to analyze a quantum optics experiment in terms of the operators is to relate all subsequent operators to the initial operators as we move through each piece of the apparatus. The initial operators are $\hat{a}_1$ and $\hat{a}_2$. We work with the lowering operators, because they will be put into the corresponding $\hat{E}^{(+)}$ operator to determine the final measurement probability. When we pass through the first beam splitter, which is a 50–50 beam splitter (so $r = t = \frac{1}{\sqrt{2}}$), we have

$$\hat{a}_3 = \frac{1}{\sqrt{2}} \left(\hat{a}_1 + \hat{a}_2 \right) \text{ and } \hat{a}_4 = \frac{1}{\sqrt{2}} \left(\hat{a}_1 - \hat{a}_2 \right), \tag{20.76}$$

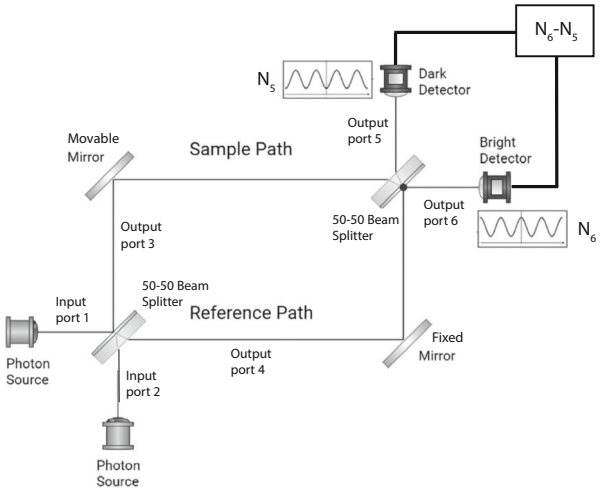

Fig. 20.10 Schematic of a Mach-Zehnder interferometer, detailing the different ports that will be used in our calculations. This has two input ports and two output ports

with the minus sign coming from the reflection from inside glass. Note how similar working with the operators is to working with the Feynman arrows. We let L_3 be the length along the sample path and L_4 the length along the reference path; the movable mirror allows us to change L_3 in the experiment, while L_4 is fixed. Recall, to determine the phase that goes with the operator, we need the time, which is $\frac{L_3}{c}$ for the sample path and $\frac{L_4}{c}$ for the reference path. Then, using the fact that $\frac{\omega}{c} = k$, we find that just before getting to the second beam splitter, we have the two operators which are

$$\hat{a}_3 = \frac{e^{-ikL_3}}{\sqrt{2}}\left(\hat{a}_1 + \hat{a}_2\right) \text{ and } \hat{a}_4 = \frac{e^{-ikL_4}}{\sqrt{2}}\left(\hat{a}_1 - \hat{a}_2\right). \tag{20.77}$$

We now move through the beam splitter and find we have

$$\begin{aligned}\hat{a}_5 &= \frac{e^{-ikL_3}}{2}\left(-\hat{a}_1 - \hat{a}_2\right) + \frac{e^{-ikL_4}}{2}\left(\hat{a}_1 - \hat{a}_2\right) \text{ and} \\ \hat{a}_6 &= \frac{e^{-ikL_3}}{2}\left(\hat{a}_1 + \hat{a}_2\right) + \frac{e^{-ikL_4}}{2}\left(\hat{a}_1 - \hat{a}_2\right).\end{aligned} \tag{20.78}$$

We factor out $e^{-\frac{i}{2}k(L_3+l_4)}$ out of the terms, then using $\delta = k(L_3 - L_4)$, we find

$$\begin{aligned}\hat{a}_5 &= e^{-\frac{i}{2}k(L_3+L_4)}\left(i\sin\tfrac{\delta}{2}\hat{a}_1 - \cos\tfrac{\delta}{2}\hat{a}_2\right) \text{ and} \\ \hat{a}_6 &= e^{\frac{i}{2}k(L_3+L_4)}\left(\cos\tfrac{\delta}{2}\hat{a}_1 - i\sin\tfrac{\delta}{2}\hat{a}_2\right).\end{aligned} \tag{20.79}$$

We examine the same case as we did before—we input a coherent state $|\alpha\rangle$ at port 1 and a vacuum state at port 2. So the initial state is $|\psi_{\text{in}}\rangle = |\alpha\rangle \otimes |0\rangle$. Then, we find the results at the bright detector are $\langle\psi_{\text{in}}|\hat{a}_6^\dagger \hat{a}_6|\psi_{\text{in}}\rangle = |\alpha|^2 \cos^2 \frac{\delta}{2}$, and at the dark detector are $\langle\psi_{\text{in}}|\hat{a}_5^\dagger \hat{a}_5|\psi_{\text{in}}\rangle = |\alpha|^2 \sin^2 \frac{\delta}{2}$, as expected. But, the measurement we want to focus on is the *difference* of the number of photons in each detector. We find that

$$\langle\psi_{\text{in}}|(\hat{N}_6 - \hat{N}_5)|\psi_{\text{in}}\rangle = |\alpha|^2 \left(\cos^2 \tfrac{\delta}{2} - \sin^2 \tfrac{\delta}{2}\right) = |\alpha|^2 \cos\delta. \tag{20.80}$$

One typically works near $\delta = \frac{\pi}{2}$, because it is easier to measure near zero photons than to try to determine the maximal number of photons that hit, on average, in a given collection time interval.

Computing the operator $\hat{N}_6 - \hat{N}_5$ gives us

$$\hat{N}_6 - \hat{N}_5 = \cos\delta \left(\hat{a}_1^\dagger \hat{a}_1 - \hat{a}_2^\dagger \hat{a}_2\right) - i \sin\delta \left(\hat{a}_1^\dagger \hat{a}_2 - \hat{a}_2^\dagger \hat{a}_1\right). \tag{20.81}$$

If we work at $\delta = \frac{\pi}{2}$, we find for a state $|\psi_{\text{in}}\rangle = |\alpha\rangle \otimes |\psi_2\rangle$, with $\alpha = |\alpha| e^{i\phi}$ that

$$\langle\psi_{\text{in}}|(\hat{N}_6 - \hat{N}_5)|\psi_{\text{in}}\rangle = i|\alpha|\langle\psi_2| \left(\hat{a}_2^\dagger e^{i\phi} - \hat{a}_2 e^{-i\phi}\right)|\psi_2\rangle. \tag{20.82}$$

One can see for $\phi = 0$, this measures the P quadrature and for $\phi = \frac{\pi}{2}$, we have the Q quadrature; note that ϕ is different from the θ we used before for the generalized quadratures. The fluctuations would then be the corresponding P or Q quadrature fluctuations. If we use a vacuum state for $|\psi_2\rangle$, we will measure the vacuum fluctuations multiplied by $|\alpha|$, while if we use a squeezed state, we can reduce the fluctuations. It is quite remarkable that what we do to the port corresponding to the state we are not measuring affects the fluctuations in our measurement of the signal sent into port 1. We will use this fact to our advantage when we discuss the LIGO experiment later in the chapter.

It is important to have a brief discussion about language and how these calculations differ from our previous ones that used arrows or Dirac notation. We would refer to the states inside the interferometer as a superposition of the photon traveling in indeterminate states determined by the paths. In the quantum optics language, we would instead say the superposition is in terms of the states with one photon in the sample path and zero in the reference path and the state with zero photons in the sample path and one in the reference path. In the way we organize the calculations here, we work by propagating the *operators* and not the states, so that language does not directly enter into the analysis. This note is just a caution about the need for precision in language, especially as the formalism becomes more precise.

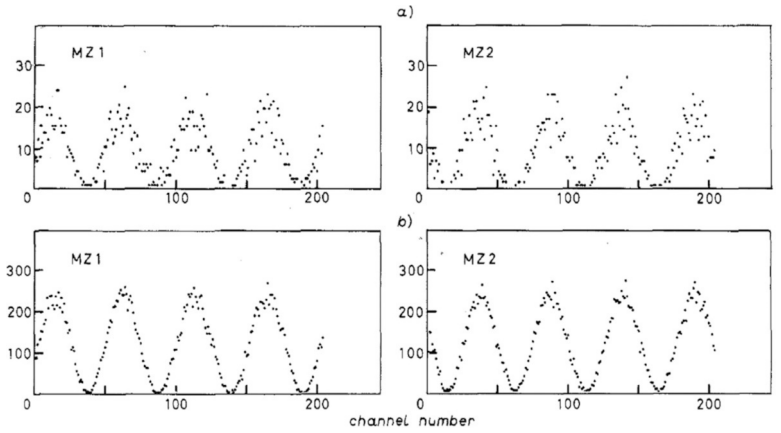

Fig. 20.11 Interference fringes for a Mach-Zehnder interferometer with a single-photon light source: MZ1 is the bright detector and MZ2 is the dark detector. The collection times are 1 s for the top two images and 15 s for the bottom two. The horizontal axes are the parameter δ for the phase difference between the 2 paths with 50 channels corresponding to 1 wavelength of the light. You can see that the interference patterns become sharper over time. Figure courtesy of Alain Aspect from Grangier et al. (1986) and used with permission (© Institute of Physics)

We conclude this section with a discussion of single-photon interference and Wavefunction@Wave-particle duality. These are some of the most interesting experiments done with single-photon sources that verify claims we made earlier in this book. In essence, when we use photon sources with lots of photons, like a laser, we see strong interference patterns in the Mach-Zehnder interferometer. The patterns in one detector look like concentric rings, as we showed in Fig. 4.6, which are complimentary to each other for the bright and dark detectors. By adjusting δ (with the movable mirror), we can scan through the fringes. Now, here is the really cool thing—when we use a single-photon light source, we get the same pattern. It just needs to build up by collecting enough photons. This has at the heart of it Wavefunction@Wave-particle duality, where particle represented by the individual photons has wave properties that show interference when enough of them are collected. Alain Aspect illustrates this with a single-photon experiment shown in Fig. 20.11 (Grangier et al., 1986).

Section 20.6 Problems:

20.6.1 Mach-Zehnder Interferometer with a Polarizing Beam Splitter

In Fig. 20.12, we show a Mach-Zehnder interferometer, with a polarized single-photon light source and a 50–50 polarizing beam splitter which transmits V

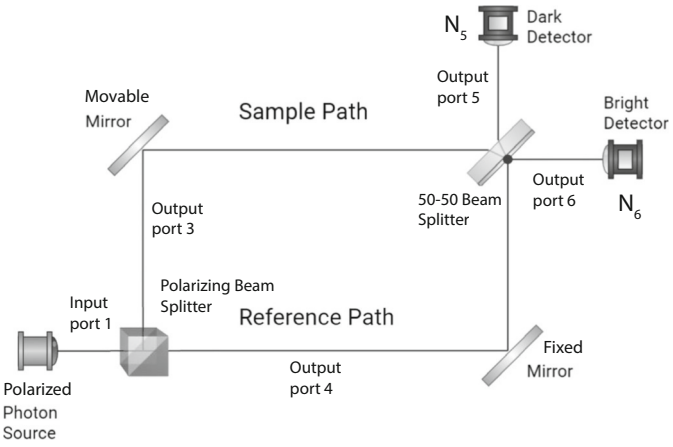

Fig. 20.12 Mach-Zehnder interferometer with one (polarized) input light source and a polarizing beam splitter. We examine what happens when light reaches the bright and dark detectors and what happens if a polarizer is placed before either detector, or both

polarized light and reflects H polarized light. Using a notation that includes the polarization of the photon mode in the creation and annihilation operator, we start in a diagonal polarized state, given by $\frac{1}{\sqrt{2}}(\hat{a}_{1V}^{\dagger} + \hat{a}_{1H}^{\dagger})|0\rangle$, which is input into port 1 of the interferometer (with nothing input into port 2 of the interferometer). Develop the rules for the output operators, and determine the probability for detection in the bright and dark detectors. You should find it is independent of δ for both of them.

Now, introduce additional polarizers after the second beam splitter and in front of the bright and dark detectors. As functions of the angles θ_{bright} and θ_{dark}, for each of the polarizers, with respect to vertical, describe what the probability for measurement in the detectors is as a function of the phase difference δ.

20.6.2 Fluctuations of the General Mach-Zehnder Interferometer

In this problem, we will determine the operator for fluctuations in the P-quadrature measurement, $(\hat{N}_6 - \hat{N}_5)^2$, and express it in normal-ordered form in terms of the input operators $\hat{a}_1$, $\hat{a}_1^{\dagger}$, $\hat{a}_2$, and $\hat{a}_2^{\dagger}$. The one thing that we change from a standard beam splitter in a Mach-Zehnder interferometer is that here, we have general reflection r and transmission t amplitudes instead of a 50–50 beam splitter.

Using the principles we developed, determine $(\hat{N}_6 - \hat{N}_5)^2$ in terms of the input operators, and put them into normal-ordered form, where all raising operators are to the left of all lowering operators. Assume the reflection and transmission coefficients are both real and assume the interferometer is "balanced" for the incident light,

Fig. 20.13 Jim Franson, the inventor of the Franson interferometer at the time of the invention. [Photo courtesy of Jim Franson. Converted to look like a sketch by James Freericks]

so that there is no phase difference between the path lengths on either arm of the interferometer ($\delta = 0$). Better to work with the sums of factorized terms as coefficients and not expand them out, or your expressions will become very lengthy.

20.7 Franson Interferometer

In quantum mechanics, we have studied entanglement of a single object (a photon's position and its polarization or an electron's position and spin), and we have studied entanglement of two particles (EPR experiment or Bohm's variant with spin). In both of these situations, we entangle the quanta's position with some other degree of freedom. The Franson interferometer does something quite different. It entangles quanta in *time*. Entanglement itself is hard to understand. Entanglement in time really makes your head spin. In this section, we carefully describe just how this entanglement in time works. An image of Jim Franson, the inventor of the Franson interferometer, is shown in Fig. 20.13.

The Franson interferometer is illustrated schematically in Fig. 20.14 (Franson, 1989). It consists of a two-photon light source, where the two photons must be created at the same time. The most common source for this is a parametric down conversion source. Then, each photon is taken on a different path to a modified Mach-Zehnder interferometer (note how we switched the positions of a beam splitter and a mirror in the device). This creates a long path and a short path for each photon to move on. The long path length can be slightly adjusted with a movable mirror, which allows the phase to be changed of the light. The long path is sufficiently longer than the size of the photon, so that there are no superposition or interference effects in a single interferometer. The light that comes out can be detected in one of four detectors, although in a typical experiment, we look at only two of them (here

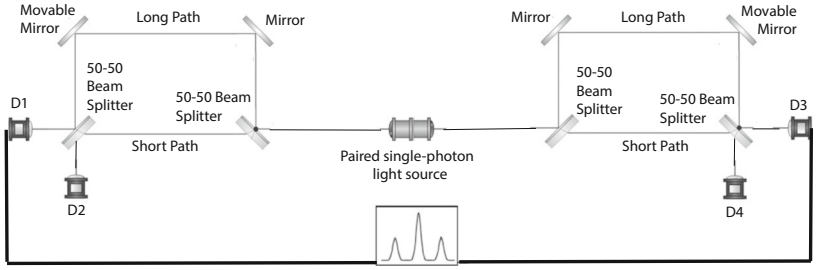

Fig. 20.14 Schematic of a Franson interferometer. It is composed of a paired photon light source (such as a parametric down conversion source that creates pairs of photons at the same time) and two modified Mach-Zehnder interferometers. The interferometers have a large difference in time between the time to travel the long path and the short path (longer than the size of the photon). Two detectors are chosen to examine for coincidences in the light detection, here D1 and D3

D1 and D3). The experiment looks for coincidences. If we plot the photons detected as a function of the time difference for when they were detected $t_1 - t_3$, we have three possibilities (and hence three peaks): The photon arrives in D1 before D3; the photon arrives in D1 after D3; or the photons arrive at the same time in D1 and D3. The first case corresponds to a short path to 1 and a long path to 3. The second a long path to 1 and a short path to 3. The third is either a short path to 1 and a short path to 3 or a long path to 1 and a long path to 3. These last two options are indistinguishable, because they both produce coincidences, so they interfere with each other, as we will see. What is so mind blowing about this is it seems to be impossible to have objects that arrive at different times interfere with each other. Yet, as we will see, they most definitely do. In fact, when this idea was first proposed, many experts in quantum optics did not believe it would work. These ideas were all dispelled when Franson illustrated a working device.

Let us first analyze this experiment in the Feynman way using our simple rules with clocks and arrows. The first thing we must do is identify each event, then the alternative way each event occurs, and finally the arrows that describe each alternative way. Then we sum the arrows if there is more than one way, square the final arrows, and from that determine the probability for each event to occur. But, rather than work with the times on the clocks, we will work with complex numbers, because that is what the arrows represented anyway. Now that we know so much more quantum mechanics than at the beginning of the book, this small change is completely straightforward to implement.

We start with our first event of detection where $t_1 - t_3 < 0$—the left photon goes on a short path and the right photon goes on a long path. There is only one way this event can occur. We have a complex number associated with each photon, which we will multiply together for the composite arrow of the event. So, let's start at the source, with the left photon described by the number 1. As it moves on the short path, which takes time t_1, the arrow rotates to become $e^{-i\omega t_1}$. We also move through two 50–50 beam splitters, which each shrink the arrow by factors of $\frac{1}{\sqrt{2}}$,

for a total shrinkage of $\frac{1}{2}$; both paths reflect from air, so there is no change of sign. So the arrow for this photon is $\frac{1}{2}e^{-i\omega t_1}$; we use a lower-case t for the short path. Now, we must work out the same arrow for the photon on the right. It goes on a long path, so its arrow is $\frac{1}{2}e^{-i\omega T_3}$, where T_3 is the time it takes to travel the long path (upper case for the long path). Because this is a two photon state and we measure both photons, we must multiply these two arrows together to get the final arrow. It is $\frac{1}{4}e^{-i\omega(t_1+T_3)}$. When we square it to get the probability, we find it is equal to $\frac{1}{16}$. The procedure for the $t_1 - t_3 > 0$ case (the L to the left and S to the right) is similar, giving us a final arrow of $\frac{1}{4}e^{-i\omega(T_1+t_3)}$ and a final probability of $\frac{1}{16}$. Finally, for the short-short path, we have a final arrow given by $\frac{1}{4}e^{-i\omega(t_1+t_3)}$ and for the long-long path, it is $\frac{1}{4}e^{-i\omega(T_1+T_3)}$. Combining these two arrow together gives

$$\frac{1}{2}e^{-\frac{i}{2}\omega(t_1+t_3+T_1+T_3)}\cos\left(\tfrac{1}{2}\omega(t_1+t_3-T_1-T_3)\right), \tag{20.83}$$

yielding a probability for coincidence of

$$P_c = \frac{1}{8}\big(1+\cos\omega(t_1+t_3-T_1-T_3)\big). \tag{20.84}$$

Now, here is an interesting question for you. If I move one of the mirrors to change the path length (or time) and thereby vary the argument of the cosine over the range of 2π, the probability varies from 0 to $\frac{1}{4}$. Where does the rest of the probability go? You will answer that in Problem 20.7.2.

It is quite common to use a Franson interferometer with only two detectors, but we can have up to four of them, and then we have a lot of choices. Which two pairs of detectors from each side do we pick to look for coincidences and how many detectors do I use? You can see there are many, many options here. You will explore this further in the problems.

We can also analyze these experiments using our quantized operators. As before, we need to start from the input operators and convert all output operators into expressions involving the input operators. There are a total of four output operators corresponding to the operators describing the photons just before they enter any of the four detectors. Such an analysis is a little tedious but quite straightforward to carry out. You will have a chance to do so in Problem 20.7.1. But, because this experiment entangles in time, to analyze it properly, we need to use wavepackets of light in each arm. This analysis then becomes complicated, because we have to follow the wavepackets through each piece of the apparatus, where they split and rejoin. As we saw earlier, when we work with wavepackets, we typically have to work with Fourier transforms, which requires us to use calculus. There is no way to get out of that issue here, so we will not go any further with the analysis. But, you might want to think about what it would entail and how it could be performed. This will take you to an advanced level of quantum mechanics if you can do it.

Section 20.7 Problems:

20.7.1 Quantum Optics Calculation of the Franson Interferometer

Reanalyze the experiment we analyzed in the main text, with detectors D1 and D3 only, using operators and quantum optics, as described in the text. You will need to determine precisely what you will measure (recall the detectors count photons) and how it can be used to make the plot seen in the figure.

20.7.2 Conservation of Probability

Consider the arrangement where we measure the photons in all four detectors and we assume there is no loss, so every photon in is measured in a detector. Now, we always will have two detectors click, and they can be simultaneous clicking, or one before the other. Consider all possibilities and be careful to identify all events determined by the final states of the four detectors. Show that the sum of the probabilities for all events is always equal to 1, so there is no issue with probability conservation and that the simultaneous detection probabilities are correlated between different detectors depending on the times for each of the paths.

20.7.3 Partial Coincidence Circuits

Suppose we set up the Franson interferometer and measure in all four detectors but only look at coincidences that occur in D1 and D3 and in D2 and D4. Describe what this experiment will measure, why it appears to lose probability, and how it can be understood.

20.7.4 Essay on Entanglement in Time

Entanglement in time is as eerie and spooky a behavior that we have ever had. Note that we are measuring coincidences in the detectors, and because we do not know when the light was emitted by the source, we have no "which-time" information, which is why the two results for coincident time detection must interfere. Write an essay discussing this entanglement in time and how it differs from entanglement

in space. Is there a way to obtain "which-time" information? How would you do it? Can one make a delayed choice experiment with this device? Explore different interesting options on extending the types of experiments that can be done with states that are entangled in time.

20.7.5 Wavepackets and the Franson Interferometer

Write an essay that discusses the way in which a calculation with wavepackets would work and how it will properly describe the arrival times at the different detectors (at least within the measurement time windows). Why is it important for the difference in time between the long path and the short path be much longer than the extent of the photon wavepacket in time?

20.8 Squeezed Light and LIGO

The laser interferometry gravitational observatory (LIGO) is one of the most incredible achievements of humankind. It can measure differences in length to better than one part in 10^{21}, which is a truly remarkable precision. This extreme precision is required in order to measure gravitational waves that appear in a frequency range near 100 Hz. Obviously, this requires careful engineering. It is so sensitive that quantum mechanical uncertainty actually limits its accuracy. It is the ultimate quantum sensing device and should be seen by every quantum mechanics student. We describe both how it works and how the accuracy is improved by using the quantum mechanics of the squeezed vacuum. This section follows closely many of the ideas in Saulson's book (Saulson, 2017), often viewed as the "bible" for the theory behind how LIGO works. In Fig. 20.15, you can see the three intellectual leaders of the LIGO project who were awarded a Nobel prize for their work.

Gravitational waves are angular momentum equal to two excitations, so they have the shape of a d-orbital, with four lobes that oscillate out of sync with each other. This is illustrated for you in Fig. 20.16. You can think of the waves as pulling in one direction and pushing in a perpendicular direction and then reversing the two. The two directions can be along coordinate axes, or the diagonals, and more. It is very different from a breathing mode (similar to blowing air into a balloon), which is pushed out and then in in all directions. Here, we have this more complicated motion.

The interferometer itself is built in the same design used by Michelson and Morley and illustrated schematically in Fig. 20.17. Each arm includes a Fabry-Perot cavity, where the light reflects back and forth about 300 times, making the 4 Km long instrument appear 1200 Km long. Such a long effective length in the arms is required to observe gravitational waves but cannot be constructed on earth due to earth's curvature. It uses an extreme high power laser (kilowatts of effective power in the

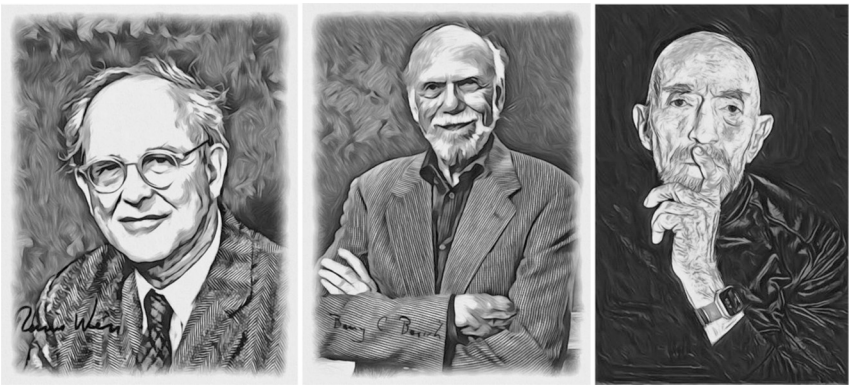

Fig. 20.15 2017 Nobel prize winners for the creation of LIGO and the observation of gravitational waves. Left Rainer Weiss; Center, Bary Barish; and Right, Kip Thorne. [Rainer Weiss image: AIP Emilio Segrè Visual Archives, Gift of Rainer Weiss, W.F. Meggers Gallery of Nobel Laureates. Barry Barish image: AIP Emilio Segrè Visual Archives, Gift of Barry Barish, W.F. Meggers Gallery of Nobel Laureates. Kip Thorne image: from Christopher Michel, Creative Commons Attribution-Share Alike 4.0 International. All images converted to look like sketches by James Freericks]

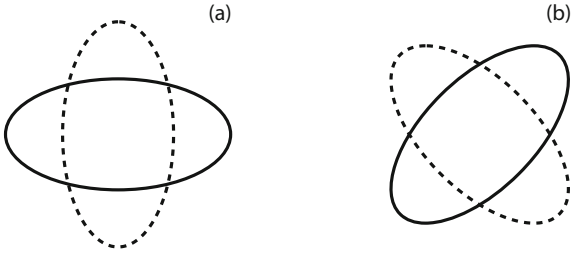

Fig. 20.16 Schematic of how gravitational waves are polarized. You should think of this as an elliptical balloon that is squashed to look elliptical in the other direction and repeated. The two different polarizations shown here are orthogonal polarizations (there are three other ones as well)

cavities). The mirrors are suspended on thin silicon cables, weigh about 100 pounds, and have active noise cancellation and vibration reduction. The laser is unbelievably stable. It is stable to one part in 10^{15}. This is the equivalent of about 1 Hz. Excellent lasers are stable to KHz scales—this one is many orders of magnitude more stable! This means the wavelength of the light is exceedingly accurate, which is needed in order to measure its differences to such high precision. Why does it use this "L" shape? Because gravitational waves have a quadrupolar shape. This is like an $l = 2$ spherical harmonic, or a d-orbital from chemistry.

Noise is obviously a big issue. They have a number of ways to isolate noise. The instrument is most sensitive near 100 Hz, which is the expected frequency for most gravitational waves and which is also in the audible range, so the signals can be converted to sound. This isolation looks for the signals in an area where no other

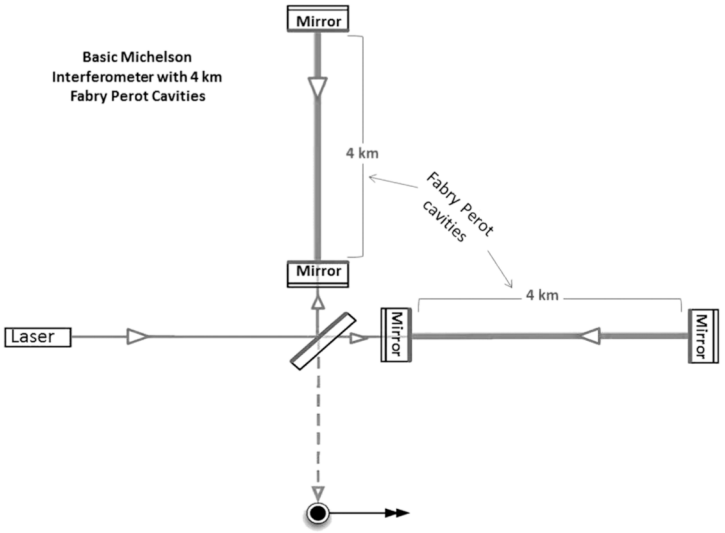

Fig. 20.17 Schematic of a LIGO observatory in the form of a Michelson-Morley interferometer with a Fabry-Perot cavity on each arm. The cavities are each 4 Km long, and the Fabry-Perot cavities act to increase the apparent length by a factor of about 300. Note how the L-shape is useful for detecting gravitational waves with contraction in one arm and expansion in the other [Image courtesy of Caltech/MIT/LIGO Lab]

resonant frequencies due to vibrations exist in the apparatus (or seismically). The requirement for the signal to be measured simultaneously at both sites also removes the noise from any local events in the vicinity of the instrument; the signals must be seen with a similar amplitude in both places in order to be due to gravitational waves. In many respects, it is one of the most amazing feats of physics and of engineering to successfully carry this out.

An interferometer can measure a distance on the order of one one hundredth of the wavelength. Since the wavelength of the laser light is 1064 nm, this means one can measure a length difference on the order of 10 nm. Since the arms have a length of 4000 m and the resonant cavities make them effectively 1200000 m long, one can compute what the relative distance is that can be measured. It is about one part in 10^{14}. This is seven orders of magnitude less than what the true sensitivity is. Where does the rest come from? It essentially comes from the high power of the laser. If we sit on a region where the interferometer sends 0 photons to the detector and the light is intense, it rises from zero very quickly. So by counting the rise in photons, we can correlate that with a small change in distance and this allows us to better discriminate the dark fringes. In the experiment, we measure the position where the light beam goes through zero via a P quadrature (which is easier to discriminate than the place where the beam is a maximum via a Q quadrature). Locating the zero is also easier if the curve has a near vertical slope, which is where the high

power comes in. But the true way in which this works is to create an oscillating signal (easier to measure than a static one) and operate the detector as a null device, where adjustments are made to keep the signal through the device constant and the adjustments are the signal that determines the measurement. The full details for these last seven orders of magnitude of sensitivity are complicated and go beyond what we can discuss here.

There is one more element that adds into the precision, and this is the uncertainty principle and squeezing the light. In Problem 14.5.2, you were asked to calculate the time-dependent uncertainties for position and momentum in a squeezed coherent state. The width oscillates between maximal and minimal spreads as a function of time, and hence the uncertainty changes with time. When it is minimal, it is actually less than the uncertainty we find in a coherent state (which does not change with time; recall how the coherent state evolution had a Gaussian that was the same width always). This means, to get high accuracy, we want to time our measurements to precisely the points in time where the uncertainty is minimal. Then we can get the benefit of a higher accuracy than what we would obtain otherwise.

The situation with light is identical, because light maps onto the same operator structure and has the same uncertainty relations, in this case, with regard to the quadrature operators Q and P. The Q quadrature measures light near its maximum amplitude and the P quadrature near its zero. It is the latter that is used in LIGO. To see how squeezing light changes the fluctuations versus coherent states, see Fig. 20.18 reprinted from Breitenbach et al. (1997).

So, if we use a squeezed state, the uncertainty is "squeezed" in one direction at a given time and in another direction at a later time, so we must time the measurements properly to improve their accuracy. In LIGO, this means we want to have a squeezed vacuum input into the unused port rather than the ordinary vacuum. Doing so provides a 30% or so improvement in the precision, which leads to the instrument being sensitive to two times as large a region of the universe than without using squeezing. So, it is a huge effect. Advanced LIGO now uses the squeezed vacuum

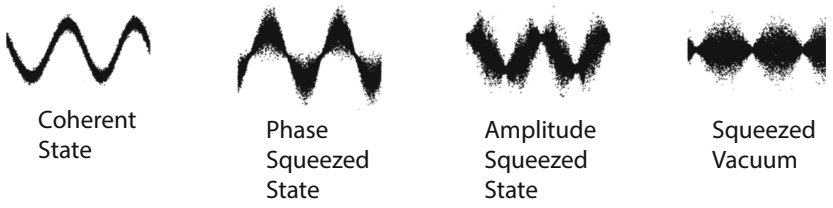

Fig. 20.18 Quadrature measurements of squeezed light with the same wavelength as the light used in LIGO. From left to right, we have a coherent state, with equal fluctuations everywhere, phase squeezed states, with low fluctuations near the zero of the field, amplitude squeezed states with low fluctuations near the maximum of the field, and a squeezed vacuum, which is used in LIGO [Images courtesy of Breitenbach et al. (1997) and adapted into a new format]

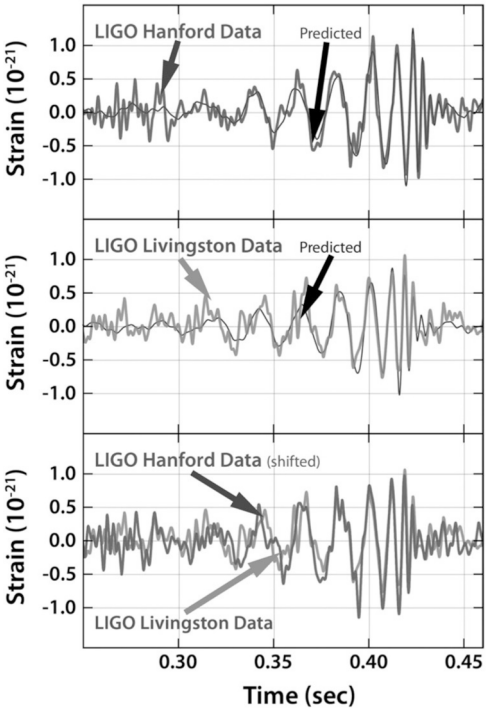

Fig. 20.19 Original measurement of the black-hole merger in 2015 as detected by LIGO (Abbott et al., 2016). The two signals are shown on the upper and middle panels compared to simulations of the events. The two signals, shifted so that they start at a common time, appears in the lowest panel. This image shows the gravitational signal of two moderately sized black holes spiraling into each other and merging together. You can see that the signal is on the order of one part in 10^{21} [Figure courtesy of Caltech/MIT/LIGO Lab and converted to grayscale with arrows added to indicate the different curves]

as an input into the unused port of the beam splitter. The idea came from Carlton Caves back in the early 1980s.

The LIGO project has been active for many decades (the idea originated in the 1980s). After many years of null signals, in 2015, LIGO measured a black hole merger. Within a year after, a neutron-star merger was measured. Converting the electrical signals into sounds, the gravitational wave increases its frequency, going from a low frequency hum to a higher frequency louder signal, before stopping once the merger is complete.

You can see in Fig. 20.19 how the simultaneous measurement at the two sites works quite well at being able to determine a gravitational wave has been observed.

This is just some of the coolest stuff there is. We have the most sensitive instrument ever built. It combines effects of quantum mechanics and general relativity on many different levels to design the measurement apparatus. And the results tell us about some of the rarest, most mind-boggling events in the universe. I hope that this book has helped you be able to understand some parts of how this measurement works.

This is why we study physics!

I want to end this chapter with a summary of what a photon actually is, as this is the main goal of the chapter. You should no longer think of a photon as just a particle of light. Instead, note the following five facts about photons:

1. They have definite particle number, but not definite energy or frequency;
2. They have a finite extent in time, and this is determined by how the photon was created;
3. They can be detected only once in destructive measurements;
4. They are detected as a discrete lump, always of the same "size" when discretely counted; and
5. They can show interference patterns after many are collected.

These facts about the photon should now be familiar to you and provide a much richer picture of what a photon really is. You have also learned how to perform calculations with these photons and understand many complex physics ideas that use them.

Section 20.8 Problems:

20.8.1 Essay: Describe the Squeezed Vacuum State

When we use a conventional vacuum state on the unused import port, we do nothing. But a squeezed vacuum requires careful preparation. Write a short essay explaining what a squeezed vacuum is. In particular, does the squeezed vacuum have photons in it?

20.8.2 Timing of Experiments

Look at the uncertainty relation for the squeezed state as a function of time. If I am measuring a P quadrature, what fraction of the period has a P quadrature with fluctuation smaller than what a coherent state has for the P quadrature.

20.8.3 Null Instrument

Research more about what a null instrument is and how they work. Write a short essay that discusses why null instruments are well-suited for measuring small signals and how they measure signals at a nonzero frequency, so they can be more easily distinguished from the noise. How does a null instrument relate to a lock-in amplifier?

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Chapter 21
Quantum Simulation with Trapped Ions

As long as one keeps searching, the answers will come.

Joan Baez, Singer and Songwriter

In this chapter, we examine ion-trap-based quantum simulators, and the overview is available at https://tinyurl.com/3tvk3a6r. This is the same type of hardware that is employed for ion-trap quantum computers, but we will not be describing gated quantum computing in this book. Instead, we focus on examining the complex quantum dynamics in systems where individual ions can be trapped and manipulated. These systems are one of the hallmarks of the second quantum revolution, where we can manipulate quantum systems at the single-quantum level. We have a video https://tinyurl.com/5yc7ruev that is an introduction to the physics of ion traps.

21.1 Normal Modes in an Ion Trap

We start with the trapping of the ions themselves. One of the benefits of trapping ions—that is, atoms with at least one electron removed from them—instead of atoms is that we can manipulate an ion with electromagnetic fields, because they have a net charge, whereas the manipulation of neutral atoms is more complicated. Ionizing the atoms is fairly simple—if we hit them with a laser of high enough energy, it can kick an electron out of the ion. One can also do this with a beam of electrons hitting the atoms. The material in this chapter follows closely the ideas of James, who first worked out the classical mechanics of these one-dimensional traps (James, 1998; Marquet et al., 2003).

But manipulating an ion is not the same as trapping the ion. Indeed, there is a theorem from electromagnetism called *Earnshaw's theorem* that says one cannot

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J. K. Freericks, *Quantum Mechanics Done Right*,
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stably trap any charged particle with static electric and magnetic fields. So, we have to use a time-dependent field. The way this works is really odd. The field creates a potential shaped like a saddle, so in one direction, it pushes particles towards the center, while in the other direction, it pushes them away from the center (think of gently rolling a marble on a saddle to get a clear picture of this in your head). The ingenious trick is to rotate the saddle about an axis perpendicular to it and running through the center. Then, if things are timed right, the marble rolls up the saddle at one moment of time, heads down to the center, and then the saddle has rotated to roll the marble back down the other side when it gets there. We won't discuss this any more detail here, because even though the trapping is a dynamical process requiring this careful tuning of the rate of rotation of the saddle, the final analysis for how the ion is trapped can be done using a *static pseudopotential*, which is what we do now.

Hence, we assume that the ions sit in a trap that satisfies

$$V(\vec{r}) = \frac{1}{2}m\omega_x^2 r_x^2 + \frac{1}{2}m\omega_y^2 r_y^2 + \frac{1}{2}m\omega_z^2 r_z^2, \tag{21.1}$$

where we have no hats, because this is a purely classical analysis for the moment. The frequencies need not be the same. For one-dimensional traps, the frequency in the z direction is much smaller than the frequency in the x and y directions, so that the ions have their motion primarily in the z-direction. Now, if we put a set of ions into such a trap, they would all go to the center of the trap if they had no electronic charge on them. But the ions are each identically charged, and so they repel each other by the Coulomb repulsion. Then, if I use a label of i for each of the N trapped ions, the total energy of the system is given by

$$E = \sum_{i=1}^{N} \frac{p_i^2}{2m} + \sum_{i=1}^{N}\left(\frac{1}{2}m\omega_x^2 r_{ix}^2 + \frac{1}{2}m\omega_y^2 r_{iy}^2 + \frac{1}{2}m\omega_z^2 r_{iz}^2\right) + \sum_{i=1}^{N}\sum_{\substack{j=2\\ j>i}}^{N} \frac{e^2}{4\pi\epsilon_0|\vec{r}_i - \vec{r}_j|}, \tag{21.2}$$

where the Coulomb interaction is between ions that are assumed to all have a charge of $+|e|$.

We start by finding the equilibrium positions of the ions, followed by the determination of the effective harmonic oscillator about the equilibrium position, which is used for our further analysis. The equilibrium positions are found at the points where the forces on the ions vanish. The force of a harmonic oscillator is given by $-m\omega^2\vec{r}$, while the force of the Coulomb interaction is given by $\frac{e^2}{4\pi\epsilon(\vec{r}-\vec{r}')^2}$, directed from the position of the particle toward that of the one it interacts with, with the sign determined by the direction of a repulsive interaction. Because we know the ions will be trapped along a one-dimensional chain (with $r_x = r_y = 0$ for all ions at equilibrium), we find the equation for zero force on the ith ion is given by

$$-m\omega_z^2 r_{iz} + \sum_{j<i} \frac{e^2}{4\pi\epsilon_0(r_{iz} - r_{jz})^2} - \sum_{j>i} \frac{e^2}{4\pi\epsilon_0(r_{iz} - r_{jz})^2} = 0 \tag{21.3}$$

where we have the ions ordered along the chain according to their index. This equation can be solved analytically for $N = 2$ and for $N = 3$. For higher N values, one has to solve the equations numerically. We will first show explicitly how the small N solutions are found, and then for the remaining analysis, we will assume that we have found the numerical solutions for the other cases.

Because of the symmetry in this problem, when $N = 2$, we must have $r_{1z} = -r_{2z} = z$; that is, the two positions lie equal distances away from the origin. Then, the equation for equilibrium is

$$m\omega_z^2 z = \frac{e^2}{4\pi\epsilon_0(2z)^2}, \tag{21.4}$$

or

$$z(N{=}2) = \left(\frac{e^2}{16\pi\epsilon_0 m\omega_z^2}\right)^{\frac{1}{3}} = \left(\frac{1}{2}\right)^{\frac{2}{3}} l_0, \tag{21.5}$$

with $l_0 = (e^2/4\pi\epsilon_0 m\omega_z^2)^{\frac{1}{3}}$ the length scale in this problem. When $N = 3$, the three positions are $-z$, 0, and z, so we have

$$m\omega_z^2 z = \frac{e^2}{4\pi\epsilon_0 z^2}\left(1 + \frac{1}{4}\right), \tag{21.6}$$

which is solved by $z(N{=}3) = (5/4)^{\frac{1}{3}} l_0$. The other cases can be solved as well, and we plot the equilibrium positions in units of l_0 for $2 \leq N \leq 10$ in Fig. 21.1.

Our next step is to find the oscillation frequencies about equilibrium. This also allows us to determine the so-called *normal modes* of motion in the ion trap. A normal mode is a motion of the ions where all ions that move do so at exactly the same oscillation frequency. In order to solve this problem, we must expand the potential energy about the equilibrium positions through quadratic order in the deviations from equilibrium. This determines an effective simple harmonic oscillator, but the motion of each ion is coupled with all other ions. We need to then extract from the expansion the normal modes and their frequencies. This is a complicated procedure, so we describe it carefully.

The first step is to focus on the longitudinal motion (motion along the one-dimensional axis of the trapped ions). Then we will tackle the transverse motion (perpendicular to the chain). Fortunately, the system has the motion along each coordinate axis decoupled from each other, so we can treat them separately. The expansion is straightforward, but tedious. We let $r_{iz}^{(0)}$ denote the equilibrium position of each of the ions. They satisfy

$$r_{iz}^{(0)} = \sum_{j<i} \frac{1}{(r_{iz}^{(0)} - r_{jz}^{(0)})^2} - \sum_{j>i} \frac{1}{(r_{iz}^{(0)} - r_{jz}^{(0)})^2}, \tag{21.7}$$

where all distances are measured in units of l_0. Then we need to find an expansion for the potential to second order in the deviations from equilibrium. The expansion has us write $r_{iz} = r_{iz}^{(0)} + \delta r_{iz}$. We find, using the geometric series, that

$$\frac{1}{\left|r_{iz}^{(0)} + \delta r_{iz} - r_{jz}^{(0)} - \delta r_{jz}\right|} = \frac{1}{\left|r_{iz}^{(0)} - r_{jz}^{(0)}\right|} - \frac{\left(\delta r_{zi} - \delta r_{zj}\right)}{\left(r_{iz}^{(0)} - r_{jz}^{(0)}\right)^2} + \frac{\left(\delta r_{zi} - \delta r_{zj}\right)^2}{\left|r_{iz}^{(0)} - r_{jz}^{(0)}\right|^3} + \cdots . \tag{21.8}$$

Plugging this into the total potential energy yields

$$V(\vec{r}_1, \vec{r}_2, \cdots, \vec{r}_N) = V_0 + \frac{1}{2} m\omega_z^2 l_0^2 \sum_{i,j=1}^{N} K_{ij}^{(l)} \delta r_{iz} \delta r_{jz} \tag{21.9}$$

where the constant term is given by the full potential, evaluated at $\vec{r}_i = \vec{r}_i^{(0)}$ and the quadratic term is given by

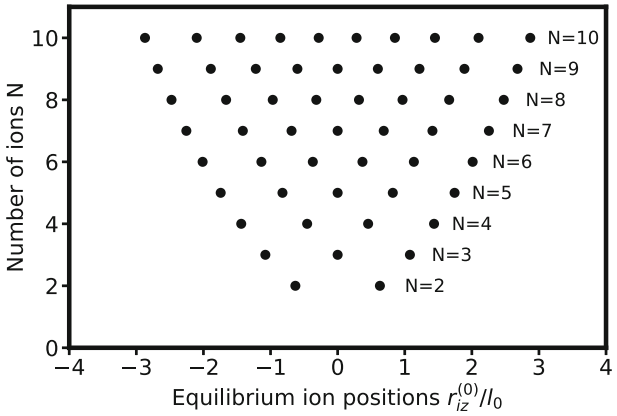

Fig. 21.1 Equilibrium positions of the ions in a one-dimensional trap with $2 \leq N \leq 10$. These positions are found by solving the equilibrium equations for vanishing force on each ion. Note how the spacing becomes uneven, and how the spacing is more compressed toward the center as the number of ions increases. This means the total length of the chain grows more slowly than just being proportional to N. Eventually the ion chain will buckle into a different equilibrium configuration. We analyze when that happens below

$$K^{(l)} = m\omega_z^2 l_0^2 \begin{cases} -\frac{2}{\left|r_{iz}^{(0)} - r_{jz}^{(0)}\right|^3} & i \neq j, \\ 1 + \sum_{\substack{j=1 \\ j\neq i}}^{N} \frac{2}{\left|r_{iz}^{(0)} - r_{jz}^{(0)}\right|^3} & i = j, \end{cases} \tag{21.10}$$

with the positions all measured in units of l_0. The terms linear in the deviations from equilibrium cancel, because the force at equilibrium vanishes at each equilibrium ion site. You will show this derivation in Problem 21.1.2. Note that we freely sum over all i and j here—one has to work carefully to get the factors of 2 correct.

The equation of motion is given by Newton's second law. We assume that all of the positions δr_{jz} oscillate with the same frequency, written as $\delta r_{jz} = b_j^{(\alpha)} e^{i\omega_l^{(\alpha)} t}$ for the dimensionless position, and then $-\left(\omega_l^{(\alpha)}\right)^2 l_0 b_j^{(\alpha)} e^{i\omega_l^{(\alpha)} t}$ for the dimensionful acceleration. Newton then says that

$$m\omega_z^2 l_0 \frac{\left(\omega_l^{(\alpha)}\right)^2}{\omega_z^2} b_i^{(\alpha)} = m\omega_z^2 l_0 \sum_j K_{ij}^{(l)} b_j^{(\alpha)}. \tag{21.11}$$

Dividing both sides by $m\omega_z^2 l_0$, we find a matrix eigenvalue equation, where $K^{(l)}\vec{b}^{(\alpha)} = \left(\omega_l^{(\alpha)}\right)^2 \vec{b}^{(\alpha)}/\omega_z^2$. This is just like eigenvalue equations we have dealt with in the past, except the frequency of the normal mode, measured in units of ω_z, is determined by the *square root* of the eigenvalue. This implies that the eigenvalue must be larger than zero for the oscillation to be a stable oscillation! While it is straightforward to set up these equations and solve for the eigenvalues, it is so tedious, that it must be done by computer. But, once solved for, then the dimensionless eigenvector, labeled by α is $b_j^{(\alpha)}$ and the frequency, in units of ω_z, is given by $\omega_l^{(\alpha)}$. Interestingly, the center of mass frequency can be found immediately. The eigenvector is given by $b_j^{\text{COM}} = 1/\sqrt{N}$, which gives $\omega^{\text{COM}} = 1$, so the center of mass frequency is always given by the trap frequency ω_z (because these dimensionless frequencies are measured in units of ω_z). All other frequencies are higher in energy. We illustrate this in Fig. 21.2 for ion traps with $2 \leq N \leq 10$. Note that because all frequencies are larger than ω_z, no longitudinal mode ever becomes unstable.

The next step is to evaluate the transverse phonon modes. We need another expansion. First note that when we have a displacement along the x-axis, we have

$$|\vec{r}_i^{(0)} + \delta\vec{r}_i - \vec{r}_j^{(0)} - \delta\vec{r}_j| = \sqrt{\left(r_{iz}^{(0)}\right)^2 - \left(r_{iz}^{(0)}\right)^2 + \left(\delta r_{ix} - \delta r_{jx}\right)^2}, \tag{21.12}$$

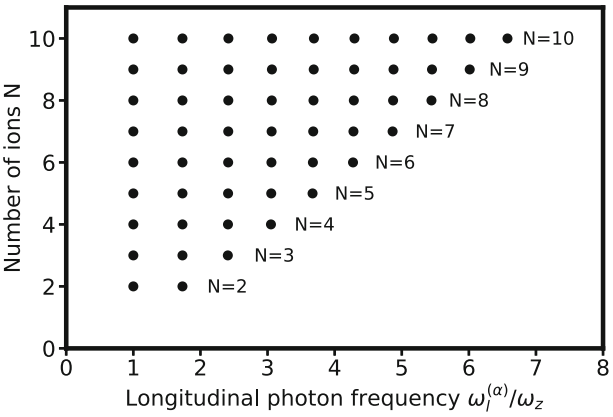

Fig. 21.2 Longitudinal normal mode frequencies in units of ω_z for $2 \leq N \leq 10$. Note how all frequencies are larger than the center of mass frequency, which is equal to 1, corresponding to the trap frequency ω_z. Furthermore, we can see that the frequencies for $N + 1$ are nearly identical to those for N plus one additional frequency at higher energy. The longitudinal frequencies are independent of the anisotropy of the trap

which can be expanded into

$$|\vec{r}_i^{(0)} + \delta\vec{r}_i - \vec{r}_j^{(0)} - \delta\vec{r}_j| = |r_{iz}^{(0)} - r_{jz}^{(0)}| \left(1 + \frac{1}{2}\frac{\left(\delta r_{ix} - \delta r_{jx}\right)^2}{\left(r_{iz}^{(0)} - r_{jz}^{(0)}\right)^2} + \cdots\right), \tag{21.13}$$

to second order in the deviations. We use the first two terms of the geometric series to perform the expansion of the inverse distance. If the two transverse frequencies are the same, which we assume to be true here, then we find the potential energy for these deviations becomes

$$V(\vec{r}_1, \vec{r}_2, \cdots, \vec{r}_N) = V_0 + \frac{1}{2} m\omega_z^2 l_0^2 \sum_{i,j=1}^{N} K_{ij}^{(t)} \delta r_{ix} \delta r_{jx} \tag{21.14}$$

where the constant term is given by the full potential, evaluated at $\vec{r}_i = \vec{r}_i^{(0)}$ and the quadratic term is given by

$$K^{(t)} = m\omega_z^2 l_0^2 \begin{cases} \frac{1}{\left|r_{iz}^{(0)} - r_{jz}^{(0)}\right|^3} & i \neq j, \\ \frac{\omega_x^2}{\omega_z^2} - \sum_{\substack{j=1 \\ j\neq i}}^{N} \frac{1}{\left|r_{iz}^{(0)} - r_{jz}^{(0)}\right|^3} & i = j. \end{cases} \tag{21.15}$$

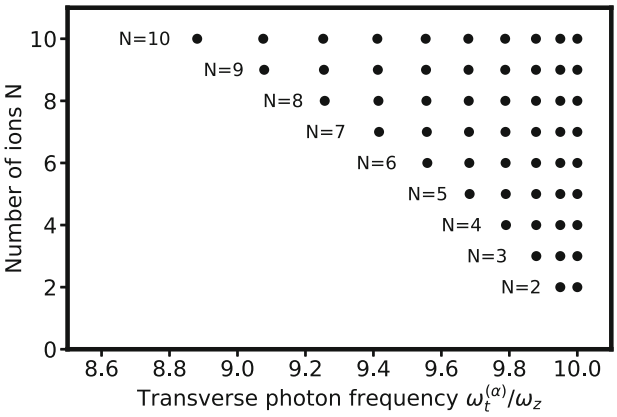

Fig. 21.3 Transverse normal mode frequencies in units of ω_z for $2 \leq N \leq 10$. Note how all frequencies are smaller than the center of mass frequency, which satisfies $\omega_x = 10\omega_z$ for both x and y. Again, the frequencies nearly line up together for the $N + 1$ case relative to the N case, with one additional frequency added in at the low end as N increases. If we continue plotting these frequencies, the lowest one becomes unstable at $N = 25$

The main differences here are that there is a factor of $-1/2$ in the expansion of the distances that is different in the transverse case versus the longitudinal case. In addition, the constant term is different due to the different trap frequency (and the fact that we continue to measure frequencies in units of ω_z).

The eigenvalue equation has the same form as before, and we continue to need to take the square root of the eigenvalue to get the frequency of the normal mode. This requires the eigenvalue to be larger than zero for a stable equilibrium configuration. One can show again that the center of mass normal mode has a frequency given by the trap frequency in the x direction, or ω_x. The main difference here is that all other frequencies lie *below* the trap frequency (see Fig. 21.3). This means they can go all the way to zero and become unstable. This does actually occur and is called the *zig-zag transition* when it happens. You can easily illustrate this behavior yourself by pushing on a spring, and seeing it bows out, away from the one-dimensional axis, as you increasingly push on it. The same thing happens with trapped ions. We will be working only with cases where the ions have stable normal mode frequencies, and we are not in the zig-zag phase in this book.

This completes our study of the classical mechanics of the trapped-ion chain. We will next explore its quantum mechanics, including quantizing the normal mode motion, and describing what happens when the system is driven by specifically tuned laser light.

Section 21.1 Problems:

21.1.1 Numerical Solution of the Equilibrium Positions

This problem is a numerical problem. Only try it if you are comfortable with programming and numerics.

Using the forces applied to each ion, devise a scheme, similar to a Newton's method root finder, that will iteratively update the ion positions until they have converged to their equilibrium values. If you do not want to use Newton's method, there are other strategies you can use such as bisection to find the roots for each i and then repeat until they stop changing. Compare a plot of your positions with those shown in Fig. 21.1.

21.1.2 Derivation of the Longitudinal $K^{(l)}$ Matrix

Derive the longitudinal spring-constant matrix in Eq. (21.10). Pay particular attention to the constants and the "double counting" for the off diagonal terms. Note that in the potential energy, we summed over $j > i$ always, but for the spring constant, we sum freely over i and j. Be sure to show that the linear term vanishes due to the equilibrium condition.

21.1.3 Numerical Solution of the Longitudinal Frequencies

This problem is a numerical problem. Only try it if you are comfortable with programming and numerics.

This problem requires you to have already solved for the equilibrium positions. Use those equilibrium positions to construct the longitudinal spring-constant matrix and diagonalize it to find the frequencies of the normal modes. Don't forget to take the square root! Compare your results to those in Fig. 21.2.

21.1.4 Exact Frequency of the Longitudinal Breathing Mode

In the text, we showed that the center of mass mode always had a frequency of ω_z. The longitudinal breathing mode, given by $b_j \propto r_{iz}^{(0)}$ also has a frequency that can be found exactly. Show that this frequency is given by $\sqrt{3}\omega_z$.

Hint: you will need to use the Eq. (21.7) in your derivation. Be sure to be careful with regard to absolute values in your derivation.

21.1.5 Derivation of the Transverse $K^{(t)}$ Matrix

Derive the transverse spring-constant matrix in Eq. (21.15). Pay particular attention to the constants and the "double counting" for the off diagonal terms. Note that in the potential energy, we summed over $j > i$ always, but for the spring constant, we sum freely over i and j. Be sure to properly identify the factor of $-\frac{1}{2}$ when compared to the longitudinal normal mode derivation.

21.1.6 Numerical Solution of the Transverse Frequencies

This problem is a numerical problem. Only try it if you are comfortable with programming and numerics.

This problem requires you to have already solved for the equilibrium positions. Use those equilibrium positions to construct the transverse spring-constant matrix and diagonalize it to find the frequencies of the normal modes. Use $\omega_x/\omega_z = 10$ (and the same for y). Don't forget to take the square root! Compare your results to those in Fig. 21.3.

As a bonus, continue increasing N until you hit the zig-zag transition. Determine which value of N for these trap frequencies is the largest value that is still stable.

21.1.7 Exact Frequency of the Transverse Tilt Mode

In the text, we showed that the center of mass mode always had a frequency of ω_x. The transverse tilt mode, given by $b_j \propto r_{iz}^{(0)}$, also has a frequency that can be found exactly. Show that this frequency is given by

$$\omega_t^{\text{tilt}} = \omega_z \sqrt{\left(\frac{\omega_x}{\omega_z}\right)^2 - 1}. \tag{21.16}$$

Note that the tilt mode has a displacement proportional to the equilibrium position in the z direction, even though it is moving in the transverse direction.

Hint: you will need to use the Eq. (21.7) in your derivation. Be sure to be careful with regard to absolute values in your derivation.

21.1.8 Physical Size of a Real Trapped Ion Chain

$^{171}\text{Yb}^+$ is a common ion used in trapped ion systems. For example, it is used in ionQ quantum computers. Assuming trap frequencies of $\omega_z = 2\pi \times 10\,\text{MHz}$, $2\pi \times 1\,\text{MHz}$, and $2\pi \times 100\,\text{KHz}$, compute l_0 and determine the average spacing between ions, as

well as the minimal spacing between two ions near the center of the trap. How tightly focused would a light beam need to be to address an individual ion? How wide would a beam need to be to address all ions equally?

21.1.9 Analysis of the Equilibrium Positions and Phonon Frequencies

Write an essay discussing the facts we have found about ion traps. This includes that the relative equilibrium ion positions, in units of l_0, do not depend on any of the trap frequencies, as shown in Fig. 21.1. Interestingly, the stability of these equilibrium configurations does depend on the trap asymmetry ω_x/ω_z. Explain how this works to eventually make the chain unstable if N becomes too large for a given asymmetry.

21.2 Raman Transitions and Rabi Oscillations

In this section, we focus on how trapped ions can be used for quantum information science. Many different ions have been used by different groups, and while all of the techniques are similar in spirit, the details matter very much, so we will focus on the $^{171}\mathrm{Yb}^{+}$ ion which is used commercially in ion trap quantum computers.

Much of atomic physics boils down to examining phenomena closely related to what we examined for the hydrogen atom. In the Yb ion, the ground state is a $S_{\frac{1}{2}}$ state, with a degeneracy of two. The $P_{\frac{1}{2}}$ excited state is the most important one for our work (there are many other excited states too). Because we are working with high-precision physics, we need to take into account the hyperfine coupling, as the $^{171}\mathrm{Yb}$ ion has a nucleus that also has spin $\frac{1}{2}$. It is common to denote the total angular momentum of the electron and the nucleus with the symbol F. Here, we will have a singlet state, which is lowest in energy, and a triplet state lying above it, with a splitting in the GHz range. Two of the states have $m_F = 0$, which is insensitive to magnetic fields (to linear order in B) and are called the *clock states*, because they are used in atomic clocks based on Yb. These two states are the two qubits that will be used for an effective spin-up and an effective spin down state of the qubit. They have an energy splitting of about 12.6 GHz, which will come in handy for measuring the state the ion is in, as we will see below. The system is placed in a uniform magnetic field to split the energies of the triplet state (recall the energies in a magnetic field split according to $g\mu_B m_F B$ due to the Zeeman effect). A schematic of these low-energy states is given in Fig. 21.4.

In addition to the low-energy states, we focus on the $P_{\frac{1}{2}}$ manifold, which also becomes a spin singlet and three spin-triplet states when the hyperfine interaction is taken into account. These states are used for many different things in determining

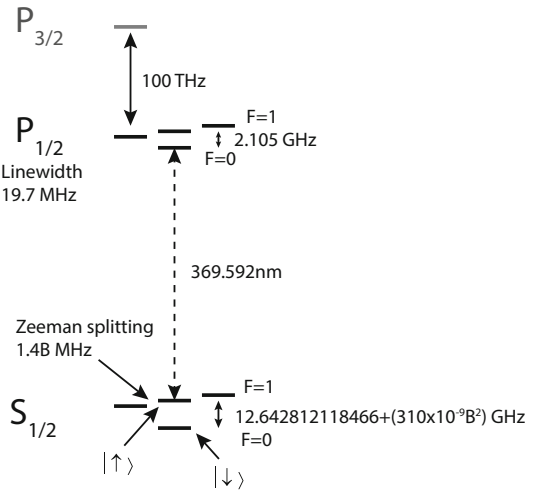

Fig. 21.4 Energy level diagram for $^{171}Yb^{+}$, illustrating the low-energy hyperfine states and the higher-energy hyperfine states that are used for the stimulated Raman transitions. The low lying states split into $F = 0$ and $F = 1$ manifolds, as do the upper states. The clock states, with $m_F = 0$, have a splitting given by $12.64281218466 + 3.1 \times 10^{-7} B^2$ GHz (with B measured in Gauss) and the Zeeman splitting of the upper states is given by $1.4B$ MHz (with B in Gauss). The excitation between the upper clock state and the lowest clock state in the upper manifold is at 369.529 nm, while the upper manifold of states is separated by 2.105 GHz when $B = 0$. A nonzero magnetic field is used to split the $F = 1$ states

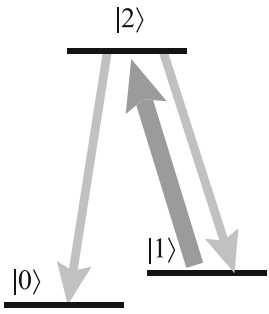

Fig. 21.5 Energy level diagram for a simplified optical pumping experiment. The applied light can excite from $|1\rangle$ to $|2\rangle$ only (thick dark line). But $|2\rangle$ decays to both $|0\rangle$ and $|1\rangle$ (thin gray lines), eventually putting all population into the dark state $|0\rangle$

the dynamics of the qubit states. First, they are used to initialize all of the ions in one of the two qubit states, second they are used in detecting which state the qubit is in, and third, they are used to create a spin (or state) dependent force on the Yb ions, which allows for quantum computing protocols.

We begin with a discussion about how optical pumping works in general with a three-state model. For the Yb case, the lower two states are the clock states, and the upper state is one of the P states. The diagram for the three-state model for optical pumping is given schematically in Fig. 21.5. We have three states in the model—two lower energy states, denoted state 0 and 1, and one high energy state, denoted

state 2. In the absence of any external electromagnetic fields, the atom Hamiltonian is diagonal in the three energy states and is written as

$$\hat{H}_0 = \hbar\omega_0|0\rangle\langle 0| + \hbar\omega_1|1\rangle\langle 1| + \hbar\omega_2|2\rangle\langle 2|. \tag{21.17}$$

Here, the energies of each of the atomic states are given by $E_i = \hbar\omega_i$. The key to optical pumping is that when the atom is excited to state 2, it has a nonzero probability to decay to either of the low-energy states. Then, if I apply light with a frequency given by $\omega_2 - \omega_1$, and the energy difference $\omega_1 - \omega_0$ is much larger than the line-width of the transition, we can only excite electrons from state 1 to state 2 with this specific choice of frequency for the light. Then the process goes as follows: (i) any atom with the electron in state 0 stays in state 0; (ii) any atom with an electron in state 1, will be excited into state 2, and then decay into states 0 or 1, with some specific probability for each; (iii) as the light is shone on the system, it will have state 1 depleted and state 0 will have its probability increased; and (iv) after we wait for a long enough time, essentially all atoms will be in state 0. The key to optical pumping is that we can only excite one way, but we can decay numerous ways, so we end up in the state that cannot be excited.

The state that we pump into is called the *dark state*. It is called this, because it cannot absorb light. If we apply optical pumping to a gas of atoms, then what we find is initially, the light is absorbed and scattered by the atoms, and they are not transparent to it. But once the dark state is fully occupied, then the light transmits completely through the gas. This is called an *induced transparency* of the system, which arises from the depletion of all states that can absorb light. In addition to the dark state, there is a *bright state* which can absorb the light.

Another interesting state that can be created in this system is called a *coherent dark state*. Working out the theory for how to determine this state gives us some practice in working out the mathematics of these systems. In addition to the atomic Hamiltonian $\hat{H}_0$, we also have an interaction Hamiltonian induced by the dipole transitions from the applied light field. We focus here on linearly polarized light, which corresponds to a real-valued field. The coupling is given by a matrix element equal to the dipole matrix element in the direction of the light polarization (the light is treated classically here, because it is from an intense laser, so the discreteness of the photon excitations does not play a significant role). Recall, the dipole matrix element is defined by

$$-|e|\langle\psi_f|\hat{\vec{r}}|\psi_i\rangle, \tag{21.18}$$

with $|\psi_i\rangle$ the initial and $|\psi_f\rangle$ the final state. The electric field is given by $\vec{E}(t) = \vec{\epsilon}E_0\cos(\nu t)$, where $\vec{\epsilon}$ is the real polarization vector for the linear polarization we will consider here, E_0 is the amplitude of the field, and ν is the frequency. We define $\mu_{fi} = |e|\langle\psi_f|\vec{\epsilon}\cdot\hat{\vec{r}}|\psi_i\rangle$ and then the interaction Hamiltonian for a given state takes the form $-\mu_{fi}E_0\cos(\nu t)\big(|\psi_f\rangle\langle\psi_i| + |\psi_i\rangle\langle\psi_f|\big)$, where we assumed that the polarization, the field, and the matrix elements are all real (which occurs if we use

the Cartesian basis for the atomic states—if one uses the angular-momentum basis, then one will have complex conjugates to take into account). Using these results, we have that the interaction Hamiltonian, for the interaction of the three-state atoms with two electric fields of frequency ν_0 and ν_1 becomes

$$\hat{V}(t) = -\mu_{20} E_0 \cos(\nu_0 t)\big(|2\rangle\langle 0| + |0\rangle\langle 2|\big) - \mu_{21} E_0 \cos(\nu_1 t)\big(|2\rangle\langle 1| + |1\rangle\langle 2|\big). \tag{21.19}$$

The frequencies are chosen to equal the energy differences of the states, so that $\nu_0 = \omega_2 - \omega_0$ and $\nu_1 = \omega_2 - \omega_1$, which justifies the transitions between the above states in the Hamiltonian.

The full Hamiltonian is $\hat{H}(t) = \hat{H}_0 + \hat{V}(t)$. We are interested in the time dependence of an arbitrary initial quantum state $|\psi\rangle = c_0|0\rangle + c_1|1\rangle + c_2|2\rangle$. We will use the interaction representation to work this out. We find

$$\begin{aligned}
\hat{V}_I(t) &= e^{\frac{i}{\hbar}\hat{H}_0 t}\hat{V}(t)e^{-\frac{i}{\hbar}\hat{H}_0 t} \\
&= -\frac{\mu_{20}E_0}{2}\Big(\big(e^{i(\nu_0+\omega_2-\omega_0)t} + e^{i(-\nu_0+\omega_2-\omega_0)t}\big)|2\rangle\langle 0| \\
&\qquad + \big(e^{i(\nu_0-\omega_2+\omega_0)t} + e^{i(-\nu_0-\omega_2+\omega_0)t}\big)|0\rangle\langle 2|\Big) \\
&\quad - \mu_{21}E_0\Big(\Big(\big(e^{i(\nu_1+\omega_2-\omega_1)t} + e^{i(-\nu_1+\omega_2-\omega_1)t}\big)|2\rangle\langle 1| \\
&\qquad + \big(e^{i(\nu_1-\omega_2+\omega_1)t} + e^{i(-\nu_1-\omega_2+\omega_1)t}\big)|1\rangle\langle 2|\Big),
\end{aligned} \tag{21.20}$$

by directly acting the exponentials on the states—that is, no Hadamard lemma is needed to determine this result. This Hamiltonian has eight oscillating terms in it. If we tune the two lasers so that $\nu_0 = \omega_2 - \omega_0$ and $\nu_1 = \omega_2 - \omega_1$, we have four terms that are constant, and four terms that oscillate rapidly. In the *rotating wave approximation*, we simply drop the fast oscillating terms, because their effects tend to average to zero over long times. The interaction potential V_I then becomes a constant in time

$$\hat{V}_I = -\frac{\mu_{20}E_0}{2}\big(|2\rangle\langle 0| + |0\rangle\langle 2|\big) - \frac{\mu_{21}E_0}{2}\big(|2\rangle\langle 1| + |1\rangle\langle 2|\big). \tag{21.21}$$

Because the interaction representation potential is a constant, we can immediately write down the time evolution operator

$$\hat{U}(t) = \begin{pmatrix} e^{-i\omega_0 t} & 0 & 0 \\ 0 & e^{-i\omega_1 t} & 0 \\ 0 & 0 & e^{-i\omega_2 t} \end{pmatrix} \times \exp\left[\frac{iE_0 t}{2}\begin{pmatrix} 0 & 0 & \mu_{20} \\ 0 & 0 & \mu_{21} \\ \mu_{20} & \mu_{21} & 0 \end{pmatrix}\right]. \tag{21.22}$$

The exact time evolution operator is calculated in Problem 21.2.1 as well as the determination of the coherent dark state.

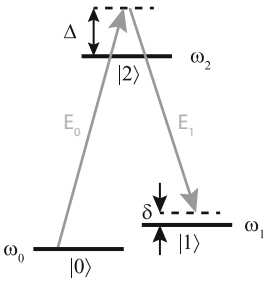

Fig. 21.6 Simplified energy level diagram for the stimulated Raman transitions. The three levels have two light beams of amplitude E_0 and E_1 applied to the system with detunings Δ and δ

So far, we have focused on resonant transitions. But resonant transitions often have a real occupation of the excited state. Then, it has two pathways to decay to a lower-energy state—it can decay coherently via stimulated emission or incoherently via spontaneous emission. The latter leads to random changes in the phase of the quantum state and often interfere with the coherent time evolution desired of the qubit. This problem is resolved by driving the excited state off of resonance. Such a situation is called a *stimulated Raman transition*, where the excited always has $c_2(t) \approx 0$, but its presence affects the dynamics. This is the next topic that we will examine here.

In Fig. 21.6, we plot the energy level diagram and the photon energies for off resonant driving. The symbol Δ is the detuning of the light excitation from state $|0\rangle$ to state $|2\rangle$, while δ is the detuning of the transition back to the state $|1\rangle$. The Raman process involves two photons—one absorbed, and one emitted (via stimulated emission), and hence it is a two-photon process. The $\hat{H}_0$ of the system is the same as before, but we will instead define a H_0' via

$$\hat{H}_0' = \hbar\omega_0|0\rangle\langle 0| + \hbar\omega_1|1\rangle\langle 1| + \hbar(\omega_2 + \Delta)|2\rangle\langle 2|, \tag{21.23}$$

and compute the interaction representation with respect to $\hat{H}_0'$, as this is needed to make the interaction potential have as little time dependence as we can. The shifted interaction, in the rotating wave approximation, is

$$\begin{aligned}\hat{V}(t) &= -\frac{E_0\mu_{20}}{2}\left(e^{-i\nu_0 t - i\phi_0}|2\rangle\langle 0| + e^{i\nu_0 t + i\phi_0}|0\rangle\langle 2|\right) \\ &\quad - \frac{E_1\mu_{21}}{2}\left(e^{-i\nu_1 t - i\phi_1}|2\rangle\langle 1| + e^{i\nu_1 t + i\phi_1}|1\rangle\langle 2|\right) - \Delta|2\rangle\langle 2| \\ &= -\frac{E_0\mu_{20}}{2}\left(e^{-i(\omega_{20}+\Delta)t - i\phi_0}|2\rangle\langle 0| + e^{i(\omega_{20}+\Delta)t + i\phi_0}|0\rangle\langle 2|\right) \\ &\quad - \frac{E_1\mu_{21}}{2}\left(e^{-i(\omega_{21}+\Delta-\delta)t - i\phi_1}|2\rangle\langle 1| + e^{i(\omega_{21}+\Delta-\delta)t + i\phi_1}|1\rangle\langle 2|\right) - \hbar\Delta|2\rangle\langle 2|,\end{aligned} \tag{21.24}$$

where we have added a constant phase to the applied electric fields (ϕ_0 and ϕ_1), and we have replaced the light frequencies according to their equivalent energies from the energy diagram. We also have subtracted the energy shift of the excited state by Δ. We next use $\hat{H}_0'$ to find the interaction potential

$$\begin{aligned}\hat{V}_I'(t) &= e^{\frac{i}{\hbar}\hat{H}_0' t}\hat{V}(t)e^{-\frac{i}{\hbar}\hat{H}_0' t} \\ &= -\frac{E_0\mu_{20}}{2}\left(e^{-i\phi_0}|2\rangle\langle 0| + e^{i\phi_0}|0\rangle\langle 2|\right) \\ &\quad - \frac{E_1\mu_{21}}{2}\left(e^{i\delta t - i\phi_1}|2\rangle\langle 1| + e^{-i\delta t + i\phi_1}|1\rangle\langle 2|\right) - \hbar\Delta|2\rangle\langle 2| \qquad (21.25)\end{aligned}$$

In the limit where the detuning Δ is large, it is not easy to excite from a low-energy state to the excited state. Hence, we anticipate that the population of the excited state is very low. In this limit, we can adiabatically eliminate the excited state and work on a smaller problem involving only the lower-energy states. When we construct the Trotter product formula, it takes the form $e^{-\frac{i}{\hbar}\hat{H}_0'\Delta t}(1 - \frac{i}{\hbar}V_I'(t)\Delta t)$ for the factors that will be repeated in the Trotter product formula. In order for the $c_2(t)$ coefficient not to grow in time, we must have the "2" component of the state vector to vanish when $\hat{V}_I'(t)$ acts on $|\psi\rangle$. This requires

$$c_2(t) = -\frac{E_0\mu_{20}}{2\hbar\Delta}e^{-i\phi_0}c_0(t) - \frac{E_1\mu_{21}}{2\hbar\Delta}e^{i\delta t - i\phi_1}c_1(t). \qquad (21.26)$$

We then use this relation to remove the excited state from the problem and find the reduced "noninteracting" Hamiltonian becomes

$$\hat{\tilde{H}}_0' = \hbar\omega_0|0\rangle\langle 0| + \hbar\omega_1|1\rangle\langle 1| \qquad (21.27)$$

and

$$\hat{\tilde{V}}_I'(t) = \frac{E_0^2\mu_{20}^2}{4\hbar\Delta}|0\rangle\langle 0| + \frac{E_0E_1\mu_{20}\mu_{21}}{4\hbar\Delta}e^{i\delta t + i\phi_0 - i\phi_1}|0\rangle\langle 1| \qquad (21.28)$$

$$+ \frac{E_0E_1\mu_{20}\mu_{21}}{4\hbar\Delta}e^{-i\delta t - i\phi_0 + i\phi_1}|1\rangle\langle 0| + \frac{E_1^2\mu_{21}^2}{4\hbar\Delta}|1\rangle\langle 1|. \qquad (21.29)$$

It is customary to define the two frequencies via $\hbar\Omega_0 = E_0\mu_{20}$ and $\hbar\Omega_1 = E_1\mu_{21}$. Then the Rabi frequency is defined by $\Omega = \Omega_0\Omega_1/(2\Delta)$ and the AC stark shifts (or light shifts) are defined by $\delta_i = \Omega_i^2/(4\Delta)$. Note how these values can be tuned by the amplitude and the frequencies of the applied light. In particular, by increasing the power (E_0 and E_1), we can work at a larger detuning Δ and have the same Rabi frequency. As the detuning increases, the chance for spontaneous

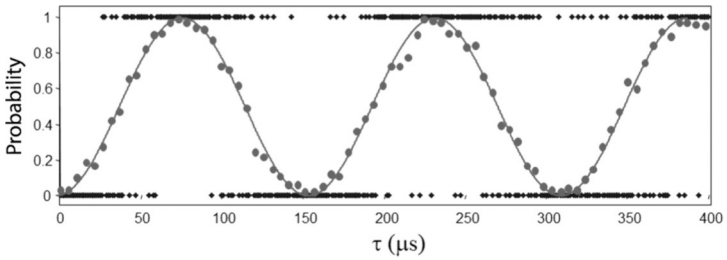

Fig. 21.7 Rabi oscillations for the qubit states of an ion in a trap. Note how each measurement is either in the 0 state or the 1 state (black diamonds at top and bottom, with the frequency of the measurement determining the probability), but the averages of the measurements (gray circles) follow the oscillating curve (which is the exact result). The symbol τ is the time of evolution of the Rabi oscillation before measurement. This experiment was tuned with a rather small Rabi frequency compared to typical values used (Data shared by Chris Monroe's ion trapping group in 2011)

emission decreases, because the excited state is only virtually occupied. This makes the dynamics of the qubits cleaner.

When the detuning to the state $|1\rangle$ vanishes, $\delta = 0$, the interaction frame potential becomes time independent, so that the time evolution operator could be immediately calculated by using the generalized Euler identity. But it is customary to use yet another new interaction picture for this 2×2 Hamiltonian for all δ via

$$\hat{\tilde{H}}_0'' = \begin{pmatrix} \delta_0 & 0 \\ 0 & \delta_1 \end{pmatrix} \text{ and } \hat{\tilde{V}}_I''(t) = \begin{pmatrix} 0 & \frac{\Omega}{2} e^{i\delta_0 t - i\delta_1 t + i\delta t + i\phi_0 - i\phi_1} \\ \frac{\Omega}{2} e^{-i\delta_0 t + i\delta_1 t - i\delta t - i\phi_0 + i\phi_1} & 0 \end{pmatrix}. \tag{21.30}$$

The time evolution in this case is identical to what we studied with magnetic resonance back in Sect. 14.2. The formulas are a bit cumbersome to write out explicitly, but they are straightforward to work out, if you want to. When $\delta = 0$ the oscillations are called Rabi oscillations. An example of these oscillations can be seen in Fig. 21.7 from Chris Monroe's laboratory. One can think of this as simply evolving the qubit with a σ_x operator, at least in a frame that rotates about the z-axis at the rate $\delta_0 - \delta_1 + \delta$, when $\phi_0 = \phi_1$. The interaction becomes time-independent when the rate vanishes.

When $\delta_0 - \delta_1 + \delta = 0$, the second interaction Hamiltonian becomes time-independent, and the phase $\phi_0 - \phi_1$ determines the direction of the Pauli spin matrix in the $x - y$ plane. This transition is called the *hyperfine carrier transition*—it creates transitions between the two clock states of the Yb ion. This causes the populations of the two hyperfine states to oscillate with the frequency Ω, because the probabilities (or populations) are found from the square of the amplitudes. This behavior is essentially the behavior of an external magnetic field acting on the qubits with the direction of the field determined by the phase $\phi_0 - \phi_1$. The typical values of these parameters are Ω_0 and Ω_1 are $2\pi \times 1$ GHz (set by the amplitudes of the fields), and the detuning Δ ranging from $2\pi \times 10$ GHz to $2\pi \times 10$ THz. The detuning is the

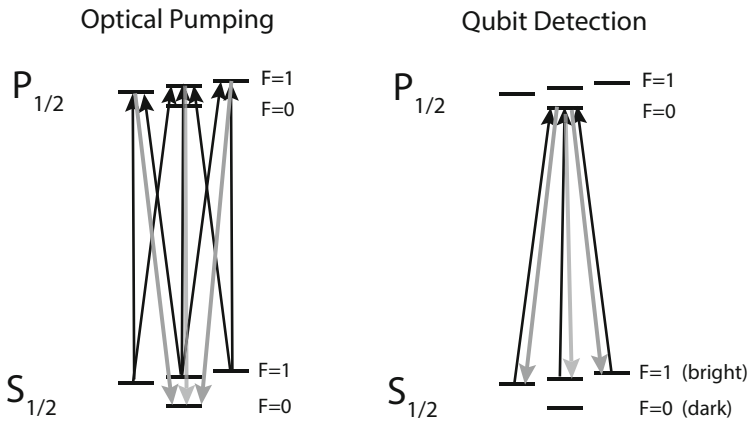

Fig. 21.8 Optical pumping (left) and state detection (right) for the $^{171}Yb^{+}$ ion. On the left, we only show the transitions leading back to the dark state. The same laser is often used for both processes. For the optical pumping, it is put through an electro-optical modulator at 2.105 GHz to create the higher-frequency light

simplest parameter to change in determining the Rabi frequency. A typical value of Ω is then on the order of $2\pi \times 1$ MHz.

We end this section with a discussion of how the optical pumping is modified for the Yb case, and we also describe how the qubit states are detected (see Fig. 21.8). In the optical pumping case, we tune the lasers, so they will excite the atoms from the $F = 1$ manifold for $S_{\frac{1}{2}}$ to the $F = 1$ manifold for $P_{\frac{1}{2}}$. They can then decay to any of the states in the low-energy manifold. Those that decay to the $F = 0$ state are stuck there and cannot be excited out. This is how the optical pumping is carried out. For the state detection, we instead excite from the $F = 1$ manifold for $S_{\frac{1}{2}}$ to the $F = 0$, $P_{\frac{1}{2}}$ state. Because there are no $0 \rightarrow 0$ transitions, when the atoms decay, they decay back to the $F = 1$, $S_{\frac{1}{2}}$ states. This process is repeated many times. If an ion appears bright, it was in state $|1\rangle$, while if it is dark, it is in state $|0\rangle$. This method of detection is quite efficient and accurate.

Section 21.2 Problems:

21.2.1 Time Evolution Operator for Coherent Dark States

Using the facts that

$$\begin{pmatrix} 0 & 0 & \mu_{20} \\ 0 & 0 & \mu_{21} \\ \mu_{20} & \mu_{21} & 0 \end{pmatrix}^2 = \begin{pmatrix} \mu_{20}\mu_{20} & \mu_{20}\mu_{21} & 0 \\ \mu_{20}\mu_{21} & \mu_{21}\mu_{21} & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{21.31}$$

and

$$\begin{pmatrix} 0 & 0 & \mu_{20} \\ 0 & 0 & \mu_{21} \\ \mu_{20} & \mu_{21} & 0 \end{pmatrix}^3 = (\mu_{20}^2 + \mu_{21}^2) \begin{pmatrix} 0 & 0 & \mu_{20} \\ 0 & 0 & \mu_{21} \\ \mu_{20} & \mu_{21} & 0 \end{pmatrix} \tag{21.32}$$

you can sum the exponential series and determine the time evolution operator. Show that

$$\begin{aligned} \hat{U}(t) = & \begin{pmatrix} e^{-i\omega_0 t} & 0 & 0 \\ 0 & e^{-i\omega_1 t} & 0 \\ 0 & 0 & e^{-i\omega_2 t} \end{pmatrix} \times \left[\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right. \\ & + \frac{\cos\left(\frac{1}{2} E_0 t \sqrt{\mu_{20}^2 + \mu_{21}^2}\right) - 1}{\frac{1}{4} E_0^2 t^2 (\mu_{20}^2 + \mu_{21}^2)} \begin{pmatrix} \mu_{20}\mu_{20} & \mu_{20}\mu_{21} & 0 \\ \mu_{20}\mu_{21} & \mu_{21}\mu_{21} & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ & \left. + \frac{i \sin \frac{1}{2} E_0 t \sqrt{\mu_{20}^2 + \mu_{21}^2}}{\frac{1}{2} E_0 t \sqrt{\mu_{20}^2 + \mu_{21}^2}} \begin{pmatrix} 0 & 0 & \mu_{20} \\ 0 & 0 & \mu_{21} \\ \mu_{20} & \mu_{21} & 0 \end{pmatrix} \right]. \end{aligned} \tag{21.33}$$

Starting from the state $|\psi\rangle = c_0|0\rangle + c_1|1\rangle + c_2|2\rangle$, we have that the dark state corresponds to the case where $c_2(t) = 0$ for all time, because in this case, we do not occupy the excited state. Show that we need

$$c_0 = \frac{\mu_{21}}{\sqrt{\mu_{20}^2 + \mu_{21}^2}}, \quad c_1 = -\frac{\mu_{20}}{\sqrt{\mu_{20}^2 + \mu_{21}^2}}, \quad \text{and} \quad c_2 = 0 \tag{21.34}$$

for the dark state, and when we choose these values for the initial state, the state $|\psi\rangle$ does not change with time.

The low-energy state orthogonal to the coherent dark state is called the *coherent bright state*. Determine it by finding another state that also has $c_2(0) = 0$. How does the coherent bright state vary with time?

21.2.2 Rabi Oscillation

Compute the time evolution operator for the system in Eq. (21.30) for arbitrary detuning. Assume that the system starts at time $t = 0$ in state $|0\rangle$. What is the probability to remain in that state as a function of time? How does it change as the detuning changes?

21.2.3 Direction of the Pauli Spin Operator

Show that

$$\begin{pmatrix} 0 & e^{i\phi} \\ e^{-i\phi} & 0 \end{pmatrix} = \cos\phi\sigma_x - \sin\phi\sigma_y = \sigma_\phi, \tag{21.35}$$

which can be thought of as the Pauli spin matrix σ_x rotated by ϕ about the z-axis. These Pauli matrices enter into the description of how lasers drive the ions.

21.3 Atomic Physics of Spin-Dependent Forces

So far, we have ignored the spatial profile of the electric field from the laser. But because the ions are spaced on the order of microns, the laser light varies on the same scale, so we need to take this into account. Furthermore, because the ions will be cooled down to almost their motional ground state, the creation or destruction of even a single phonon needs to be kept track of. This means we must use a quantum description for the phonons.

The procedure is similar to what we did classically. We still describe the position of each ion in terms of the equilibrium, plus the deviations about the equilibrium, but now the latter are quantized. So, we have

$$\begin{aligned}\hat{\vec{r}}_i = \vec{r}_i^{(0)} &+ \sum_{\alpha=1}^{N} b_i^{(\alpha)} \sqrt{\frac{\hbar}{2M\omega_l^{(\alpha)}}} (\hat{a}_{\alpha z} + \hat{a}_{\alpha z}^\dagger)\vec{e}_z + \sum_{\alpha=1}^{N} b_i^{(\alpha)} \sqrt{\frac{\hbar}{2M\omega_t^{(\alpha)}}} (\hat{a}_{\alpha x} + \hat{a}_{\alpha x}^\dagger)\vec{e}_x \\ &+ \sum_{\alpha=1}^{N} b_i^{(\alpha)} \sqrt{\frac{\hbar}{2M\omega_t^{(\alpha)}}} (\hat{a}_{\alpha y} + \hat{a}_{\alpha y}^\dagger)\vec{e}_y. \end{aligned} \tag{21.36}$$

The normal modes are indexed by α and correspond to longitudinal modes along the z direction and transverse modes along the x and y directions (we assume the transverse trap frequencies are identical here). The normal mode frequencies are $\omega_l^{(\alpha)}$ for the longitudinal modes and $\omega_t^{(\alpha)}$ for the transverse modes. The normal mode eigenvectors $b_i^{(\alpha)}$ tell us how much the ith ion is displaced in the αth mode along the corresponding spatial direction. M is the mass of the ion. Note how this is similar to the expansions we used before, but now we have an expansion for each spatial direction, and a weighting by the normal mode "pattern" in each expansion. The Hamiltonian for the phonons is simply

$$\hat{H}_{\text{phon}} = \sum_{\alpha} \hbar\omega_l^{(\alpha)} \left(\hat{a}_{\alpha z}^{\dagger}\hat{a}_{\alpha z} + \tfrac{1}{2}\right) + \sum_{\alpha} \hbar\omega_t^{(\alpha)} \left(\hat{a}_{\alpha x}^{\dagger}\hat{a}_{\alpha x} + \tfrac{1}{2}\right) + \sum_{\alpha} \hbar\omega_t^{(\alpha)} \left(\hat{a}_{\alpha y}^{\dagger}\hat{a}_{\alpha y} + \tfrac{1}{2}\right). \tag{21.37}$$

We will use these operator expressions, where needed, to describe the quantum dynamics of the ions.

In the previous section, we discussed the carrier transition, which enabled us to simulate an effective magnetic field, in an arbitrary direction in the $x - y$ plane that acts on each qubit. Now, we discuss the sideband transitions, which involve the additional creation or annihilation of a phonon. We focus here on sideband transitions with respect to the transverse modes. We apply the two laser beams such that their difference in wavevector $\Delta\vec{k}$ lies in the transverse direction (say in the x direction). We choose their frequency difference such that $\nu_0 - \nu_1 = \omega_{\text{phon}} + \omega_1 - \omega_0$, for a specific phonon that we wish to excite or de-excite; one could include a detuning δ as well, but for now we just consider the transitions that are on resonance. The interacting Hamiltonian, ignoring the diagonal AC Stark shift terms, becomes

$$\hat{\tilde{V}}_I'(t) = \frac{\hbar\Omega}{2} e^{-i\Delta\vec{k}\cdot\hat{\vec{r}}_I + i\omega_{\text{phon}}t + i\phi_0 - i\phi_1} |0\rangle\langle 1| + \frac{\hbar\Omega}{2} e^{i\Delta\vec{k}\cdot\hat{\vec{r}}_I - i\omega_{\text{phon}}t - i\phi_0 + i\phi_1} |1\rangle\langle 0|, \tag{21.38}$$

after adiabatically eliminating the excited state and including the phase contribution from where the ion is located. The position operator can be thought of as the position for a specific trapped ion, but we are being schematic here and will become more precise later. The position operator is then expanded as

$$\hat{r}_x = x^{(0)} + x_{\text{phon}}(\hat{a} + \hat{a}^{\dagger}), \tag{21.39}$$

with $x_{\text{phon}} = \sqrt{\hbar/2M\omega_{\text{phon}}}$. We write $\Delta\vec{k} = \Delta k\vec{e}_x$, so that $\Delta\vec{k}\cdot\hat{\vec{r}} = \Delta k x^{(0)} + \Delta k x_{\text{phon}}(\hat{a} + \hat{a}^{\dagger})$. Converting to the interaction representation, with respect to the phonon Hamiltonian requires $\hat{a} \to \hat{a}e^{-i\omega_{\text{phon}}}$ and the Hermitian conjugate relation for $\hat{a}^{\dagger}$. We assume the oscillations from the phonons are small and expand the exponential in a power series for the phonon terms. It becomes

$$\hat{\tilde{V}}_I'(t) = \frac{\hbar\Omega}{2}\left(1 - i\eta(\hat{a}e^{-i\omega_{\text{phon}}t} + \hat{a}^{\dagger}e^{i\omega_{\text{phon}}t}) + \cdots\right) e^{i\omega_{\text{phon}}t + i\phi_0 - i\phi_1} |0\rangle\langle 1| + \frac{\hbar\Omega}{2}\left(1 + i\eta(\hat{a}e^{-i\omega_{\text{phon}}t} + \hat{a}^{\dagger}e^{i\omega_{\text{phon}}t}) + \cdots\right) e^{-i\omega_{\text{phon}}t - i\phi_0 + i\phi_1} |1\rangle\langle 0|, \tag{21.40}$$

where $\eta = \Delta k x_0 \ll 1$ is the Lamb-Dicke parameter, which is small in the regime we are working in. Next, we use the rotating wave approximation to remove terms that oscillate fast, to find

$$\hat{\tilde{V}}_I'(t) = -\frac{i\eta\hbar\Omega}{2}\hat{a}e^{i\phi_0 - i\phi_1}|0\rangle\langle 1| + \frac{i\eta\hbar\Omega}{2}\hat{a}^\dagger e^{-i\phi_0 + i\phi_1}|1\rangle\langle 0|. \tag{21.41}$$

This makes the transition from $|0\rangle\otimes|n\rangle$ to $|1\rangle\otimes|n+1\rangle$ and *vice versa*. This transition is higher in energy than the carrier transition and is called the *blue sideband*. The full name of the transition, on resonance, is called the *resonant stimulated Raman blue sideband transition*. Similarly, if we set $\nu_0 - \nu_1 = -\omega_{\text{phon}} + \omega_1 - \omega_0$, the interaction Hamiltonian becomes

$$\hat{\tilde{V}}_I'(t) = -\frac{i\eta\hbar\Omega}{2}\hat{a}^\dagger e^{i\phi_0 - i\phi_1}|0\rangle\langle 1| + \frac{i\eta\hbar\Omega}{2}\hat{a}e^{-i\phi_0 + i\phi_1}|1\rangle\langle 0|. \tag{21.42}$$

This is called the *resonant stimulated Raman red sideband transition*. It makes the transition from $|0\rangle\otimes|n\rangle$ to $|1\rangle\otimes|n-1\rangle$ and *vice versa*. The sideband transitions can be used to cool the system, but we will not describe how to do this in detail here. Instead, we see Problem 21.3.1 below.

Now, we will describe the full laser-ion interaction that is used in quantum simulation with trapped ions. The setup includes the carrier transition, plus a red-detuned and blue-detuned transition. This requires us to shine the laser beam with frequency $\nu_0 = \omega_2 - \omega_0 + \Delta$, and then three additional beams: (i) the carrier beam $\nu_1 = \omega_2 - \omega_1 + \Delta$; (ii) the blue-detuned beam $\nu_1 = \omega_2 - \omega_1 + \Delta - \mu$; and (iii) the red-detuned beam $\nu_1 = \omega_2 - \omega_1 + \Delta + \mu$. The detuning μ will be on the order of magnitude of the phonon frequencies, and we are ignoring the AC stark shifts for simplicity. The carrier interaction Hamiltonian in the interaction picture is

$$\hat{H}_{cI} = \sum_{j=1}^{N}\frac{\hbar\Omega_j}{2}\left(\sigma_j^+ e^{i\Delta\vec{k}\cdot\hat{\vec{r}}(t) - i\delta_c t - i\phi_c} + \sigma_j^- e^{-i\Delta\vec{k}\cdot\hat{\vec{r}}(t) + i\delta t + i\phi_c}\right), \tag{21.43}$$

where Ω_j is the effective Rabi frequency at the ionic equilibrium site j, δ_c is the detuning $\nu_0 - \nu_1 - \omega_1 + \omega_0$, and ϕ_c is the carrier phase shift, given by $\phi_1 - \phi_0$. We will see below that the carrier transition is used to create a magnetic field. The blue-detuned interaction Hamiltonian in the interaction picture is

$$\hat{H}_{bI} = \sum_{j=1}^{N}\frac{\hbar\Omega_j}{2}\left(\sigma_j^+ e^{i\Delta\vec{k}\cdot\hat{\vec{r}}(t) - i\mu t + i\phi_b} + \sigma_j^- e^{-i\Delta\vec{k}\cdot\hat{\vec{r}}(t) + i\mu t - i\phi_b}\right), \tag{21.44}$$

and the red-detuned interaction Hamiltonian in the interaction picture is

$$\hat{H}_{rI} = \sum_{j=1}^{N}\frac{\hbar\Omega_j}{2}\left(\sigma_j^+ e^{i\Delta\vec{k}\cdot\hat{\vec{r}}(t) + i\mu t + i\phi_r} + \sigma_j^- e^{-i\Delta\vec{k}\cdot\hat{\vec{r}}(t) - i\mu t - i\phi_r}\right), \tag{21.45}$$

with the detuning $\mu > 0$. In the experiment we will describe, we only use the red- and blue-detuned interaction Hamiltonians to create effective spin-spin interactions.

We study the effective spin-spin interactions first. Adding the red and blue sideband terms together, we find

$$\hat{H}_{bI} + \hat{H}_{rI} = \sum_j \hbar\Omega_j \cos(\mu t + \phi_M) \left(e^{i\Delta\vec{k}\cdot\delta\hat{\vec{r}}_j(t)} \sigma_j^+ e^{-i\phi_{Sj}} + e^{-i\Delta\vec{k}\cdot\delta\hat{\vec{r}}_j(t)} \sigma_j^- e^{i\phi_{Sj}} \right), \tag{21.46}$$

where $\phi_M = (\phi_r - \phi_b)/2$ and $\phi_{Sj} = -\Delta\vec{k}\cdot\vec{r}_j^{(0)} - (\phi_b + \phi_r)/2$. In the Lamb-Dicke limit, we approximate $e^{i\Delta\vec{k}\cdot\delta\hat{\vec{r}}(t)} \approx 1 + i\Delta\vec{k}\cdot\delta\hat{\vec{r}}_j(t)$. The final form is

$$\hat{H}_{bI} + \hat{H}_{rI} = \sum_j \hbar\Omega_j \cos(\mu t + \phi_M) \left(e^{-i\phi_{Sj}} \sigma_j^+ (1 + i\Delta\vec{k}\cdot\delta\hat{\vec{r}}_j(t)) \right.$$
$$\left. + e^{i\phi_{Sj}} \sigma_j^- (1 - i\Delta\vec{k}\cdot\delta\hat{\vec{r}}_j(t)) \right). \tag{21.47}$$

The term with the 1 in it is a carrier transition, which creates no phonons. It has fast oscillations with a low amplitude, so we neglect it. The sideband transitions take the form

$$\hat{H}_I^{SB} = \sum_j \hbar\Omega_j \cos(\mu t + \phi_M) \sigma_j^{-\phi_M - \frac{\pi}{2}} \Delta k \cdot \delta\hat{\vec{r}}_j(t). \tag{21.48}$$

If we drive the transverse phonons, then $\Delta\vec{k}\cdot\delta\vec{r}_j^{(0)} = 0$, because they are perpendicular. Then, we fix the phases such that $-\phi_b = \phi_r = \frac{\pi}{2}$ (or $\phi_M = \phi_S = \frac{\pi}{2}$), and we have

$$\hat{H}_I^{SB} = -\sum_{j=1}^{N} \sum_{\alpha=1}^{N} \hbar\Omega_j \sin(\mu t) \sigma_j^x \eta_{jx}^{\alpha} b_j^{(\alpha)} \left(\hat{a}_{\alpha x} e^{-i\omega_t^{(\alpha)} t} + \hat{a}_{\alpha x}^{\dagger} e^{i\omega_t^{(\alpha)} t} \right), \tag{21.49}$$

with

$$\eta_{jx}^{\alpha} = \Delta k \sqrt{\frac{\hbar}{2M\omega_t^{(\alpha)}}}. \tag{21.50}$$

In the next section, we show how this laser-ion interaction leads to an effective spin-spin coupling between the qubits.

In addition to the interaction that will lead to an effective spin-spin interaction, we also need to determine how to create a transverse magnetic field in a direction perpendicular to the spin-spin interactions. This then will allow us to create the transverse-field Ising model, which will be the main model we examine with the trapped ions. In this case, we keep $\mu = 0$, so we have just the carrier transition, and

the sidebands vanish. Then, we pick the phases on the two laser beams to have a phase difference of π, and then the Pauli matrix generated by the carrier transition will be in the y-direction. This allows us to separately control the transverse field in a direction perpendicular to the spin-spin interactions.

Section 21.3 Problems:

21.3.1 Sideband Cooling

Write an essay describing how sideband cooling works. The basic idea is we start with the system optically pumped into the $|0\rangle \otimes |n\rangle$ state which has n phonons excited. Applying a red sideband transition flips the spin to $|1\rangle$ and removes a phonon. Then optical pumping takes us to the state $|0\rangle \otimes |n-1\rangle$. We repeat to cool down to the ground state.

Explain how this works if we have a superposition of phonon states? What about the situation where we have many ions, and hence, many phonon modes? Does the frequency of the red sideband transition depend on the number of phonons excited? Discuss these ideas as well in your essay.

21.3.2 Describe the Different Approximations That Go into the Derivation of the Laser-Ion Interaction Hamiltonians

Write an essay discussing the different approximations we used in the derivation—the rotating wave approximation and the Lamb-Dicke approximation. Under what circumstances will these approximations break down? What are the implications for experiments?

21.4 Effective Spin Model

As we saw in the previous section, we can use the laser-ion coupling to do two things: (i) create an effective magnetic field of arbitrary strength in any direction in the $x-y$ plane and (ii) create an entangling operation between the spin degree of freedom at each lattice site and the phonons of the ions in the trap. When we only virtually excite the phonons, they can be adiabatically removed from the system, leaving behind an effective spin-spin interaction. When the magnetic field lies in the same direction as the spin-spin interaction (the so-called Ising quantization axis), then one can find an exact mapping that removes the phonons. When the magnetic field is in another direction (transverse-field Ising model), the mapping is

approximate. We will first work with no magnetic field and only add it in toward the end of the section. The original ideas behind this construction come from Cirac and Zoller (1995), but our approach here is a bit different from how they developed the theory and follows Wang and Freericks (2012) more closely. It is best to revisit Sect. 14.9 to review the theory we will apply here and Sect. 10.6 for the effective spin model we will work with.

In the previous section, we worked with the three-state model and determined the effective interaction Hamiltonian in the two-state model. Summarizing those results, and using the spin representation such that $|0\rangle = |\downarrow\rangle_z$ and $|1\rangle = |\uparrow\rangle_z$, we find the Hamiltonian can be described in terms of the "bare Hamiltonian"

$$\hat{H}_0 = \sum_j \left(\frac{\hbar(\omega_1 - \omega_0)}{2} \sigma_j^z + \frac{\hbar(\omega_0 + \omega_1)}{2} \mathbb{I}_j \right), \tag{21.51}$$

and the interaction Hamiltonian (with driving in the x-direction)

$$\hat{V}_I(t) = \sum_{\alpha=1}^{N} \hbar\omega_t^{(\alpha)} \left(\hat{a}_{\alpha x}^\dagger \hat{a}_{\alpha x} + \frac{1}{2} \right) - \sin(\mu t) \sum_{j=1}^{N} \hbar\Omega_j \sigma_j^x \sum_{\alpha=1}^{N} \eta_{jx}^\alpha b_j^{(\alpha)} \left(\hat{a}_{\alpha x} + \hat{a}_{\alpha x}^\dagger \right), \tag{21.52}$$

arising from the red and blue sidebands in the interaction representation plus the bare phonon Hamiltonian, which is not part of the "bare" Hamiltonian used for the interaction representation. For now, we do not add in the extra magnetic field term from the carrier transition.

It turns out that we can neglect the $e^{-\frac{i}{\hbar}\hat{H}_0 t}$ term in all of our work. This is because we always measure the system in the spin-z basis. Hence, when we multiply by a bra from the left, which is a product state eigenstate of $\hat{H}_0$, the bare Hamiltonian contribution is just a complex phase, which does not enter into the probability to measure that particular state, and hence, we can ignore it. So, the focus of all of the calculations is on evaluating the time-ordered product for the interaction Hamiltonian. Note that this is a classic "French cooking" problem, with lots of steps to it, so follow closely.

Our strategy for working with the interaction Hamiltonian is to use the interaction representation again to further factorize the Hamiltonian, with $\hat{H}_0' = \sum_{j=1}^{N} \hbar\omega_t^{(\alpha)} \left(\hat{a}_{\alpha x}^\dagger \hat{a}_{\alpha x} + \frac{1}{2} \right)$, and $\hat{V}_I'(t) = e^{\frac{i}{\hbar}\hat{H}_0' t} \hat{V}_I(t) e^{-\frac{i}{\hbar}\hat{H}_0' t}$ or

$$\hat{V}_I'(t) = -\frac{1}{2i} \sum_{j=1}^{N} \hbar\Omega_j \sigma_j^x \sum_{\alpha=1}^{N} \eta_{jx}^\alpha b_j^{(\alpha)} \left\{ \hat{a}_{\alpha x} \left(e^{i\left(-\omega_t^{(\alpha)} + \mu\right)t} - e^{i\left(-\omega_t^{(\alpha)} - \mu\right)t} \right) + \hat{a}_{\alpha x}^\dagger \left(e^{i\left(\omega_t^{(\alpha)} + \mu\right)t} - e^{i\left(\omega_t^{(\alpha)} - \mu\right)t} \right) \right\}. \tag{21.53}$$

Previously, we would take the two contributions and break them into the slowly varying ones (corresponding to the difference of the detuning and the phonon frequencies) and the fast oscillating terms (corresponding to the sum of the two). But we can actually handle both terms exactly, as we now show through our derivation.

We start with the calculation of the time evolution operator as

$$\hat{U}'(t) = e^{-i\sum_{\alpha=1}^{N}\omega_t^{(\alpha)}t\left(\hat{a}_{\alpha x}^{\dagger}\hat{a}_{\alpha x}+\frac{1}{2}\right)}e^{-\frac{i}{\hbar}\hat{V}_I'(t)\Delta t}e^{-\frac{i}{\hbar}\hat{V}_I'(t-\Delta t)\Delta t}\cdots e^{-\frac{i}{\hbar}\hat{V}_I'(\Delta t)\Delta t}. \tag{21.54}$$

The key point is that the commutator

$$\begin{aligned}[\hat{V}_I'(t),\hat{V}_I'(t')] = -\frac{1}{4}\sum_{j=1}^{N}\sum_{j'=1}^{N}\hbar^2\Omega_j\Omega_{j'}\sigma_j^x\sigma_{j'}^x\sum_{\alpha=1}^{N}\sum_{\alpha'=1}^{N}\eta_{jx}^{\alpha}\eta_{j'x}^{\alpha'}b_j^{(\alpha)}b_{j'}^{(\alpha')}\\ \times\Big(e^{-i\omega_t^{(\alpha)}t+i\omega_t^{(\alpha')}t'+i\mu(t+t')} - e^{-i\omega_t^{(\alpha)}t+i\omega_t^{(\alpha')}t'+i\mu(t-t')}\\ - e^{-i\omega_t^{(\alpha)}t+i\omega_t^{(\alpha')}t'-i\mu(t-t')} + e^{-i\omega_t^{(\alpha)}t+i\omega_t^{(\alpha')}t'-i\mu(t+t')}\\ - e^{i\omega_t^{(\alpha)}t-i\omega_t^{(\alpha')}t'+i\mu(t+t')} + e^{i\omega_t^{(\alpha)}t-i\omega_t^{(\alpha')}t'+i\mu(t-t')}\\ + e^{i\omega_t^{(\alpha)}t-i\omega_t^{(\alpha')}t'-i\mu(t-t')} - e^{i\omega_t^{(\alpha)}t-i\omega_t^{(\alpha')}t'-i\mu(t+t')}\Big)\end{aligned} \tag{21.55}$$

is a number, not an operator, so we can put all the terms in the Trotter product formula into a single exponent, after we account for all of the correction terms from the commutators. The precise formula is that we sum all $\hat{V}_I'(k\Delta t)$ terms with $k\Delta t < t$ and then add on

$$\frac{1}{2}\sum_{k=1}^{t/\Delta t}\sum_{k'=1}^{k}[\hat{V}_I(k\Delta t),\hat{V}_I(k'\Delta t)], \tag{21.56}$$

with $k'\Delta t \le k\Delta t \le t$. Note that we have to accumulate a large number of terms, but the summations can all be done by using the geometric series and then taking the limit as $\Delta t \to 0$. It is a bit tedious to sum all of the terms. Finally, we obtain the factorized form of the evolution operator as

$$\begin{aligned}\hat{U}(t) = &\underbrace{e^{-i\sum_{\alpha=1}^{N}\left(\hat{a}_{\alpha x}^{\dagger}\hat{a}_{\alpha x}+\frac{1}{2}\right)\omega_t^{(\alpha)}t}}_{\text{phonon}}\underbrace{e^{-i\sum_{j=1}^{N}\sum_{\alpha=1}^{N}\sigma_j^x\left(\Gamma_j^{(\alpha)}(t)\hat{a}_{\alpha x}+\Gamma_j^{(\alpha)*}(t)\hat{a}_{\alpha x}^{\dagger}\right)}}_{\text{phonon-spin entanglement}}\\ &\times\underbrace{e^{-i\sum_{j=1}^{N}\sum_{j'=1}^{N}\Phi_{jj'}(t)\sigma_j^x\sigma_{j'}^x}}_{\text{spin-spin interaction}}\end{aligned} \tag{21.57}$$

with

$$\Gamma_j^{(\alpha)}(t) = \frac{\Omega_j \eta_{jx}^{\alpha} b_j^{(\alpha)}}{(\omega_t^{(\alpha)})^2 - \mu^2} \left(-i\omega_t^{(\alpha)} e^{-i\omega_t^{(\alpha)} t} \sin(\mu t) - \mu e^{-i\omega_t^{(\alpha)} t} \cos(\mu t) + \mu \right), \tag{21.58}$$

and

$$\Phi_{jj'}(t) = -\frac{1}{2}\Omega_j \Omega_{j'} \sum_{\alpha=1}^{N} \frac{\eta_{jx}^{\alpha} \eta_{j'x}^{\alpha} b_j^{(\alpha)} b_{j'}^{(\alpha)}}{\mu^2 - (\omega_t^{(\alpha)})^2} \times \left(\omega_t^{(\alpha)} t - \frac{\omega_t^{(\alpha)}}{2\mu} \sin(2\mu t) + \frac{\mu}{\omega_t^{(\alpha)} + \mu} \sin\left((\omega_t^{(\alpha)} + \mu)t\right) - \frac{\mu}{\omega_t^{(\alpha)} - \mu} \sin\left((\omega_t^{(\alpha)} - \mu)t\right) \right). \tag{21.59}$$

You are asked to verify these formulas in Problem 21.4.2. One can rewrite the spin-spin interaction term as a time-ordered product of an Ising spin model, which has the form

$$\hat{U}_{\text{spin-spin}}(t) = \prod_{k=1}^{M} e^{-\frac{i}{\hbar} \sum_{j=1}^{N} \sum_{j'=1}^{N} J_{jj'}(k\Delta t) \sigma_j^x \sigma_{j'}^x \Delta t}, \tag{21.60}$$

with the product going from early times to late times as we go from right to left and with

$$J_{jj'}(t) = -\frac{\hbar}{2}\Omega_j \Omega_{j'} \sum_{\alpha=1}^{N} \frac{\eta_{jx}^{\alpha} \eta_{j'x}^{\alpha} b_j^{(\alpha)} b_{j'}^{(\alpha)}}{\mu^2 - \left(\omega_t^{(\alpha)}\right)^2} \times \left(\omega_t^{(\alpha)} - \omega_t^{(\alpha)} \cos(2\mu t) - 2\mu \sin(\omega_t^{(\alpha)} t) \sin(\mu t) \right) \tag{21.61}$$

Note how the spin-exchange terms have a strong time dependence to them. In the absence of a transverse field, the time-ordered product needs no time ordering, and all terms can be put into the same exponent and summed. But we will see this is not the case when we have a transverse field present.

The form of the different terms is interesting. The phonon piece is just the evolution operator for noninteracting phonons. The spin-phonon entangling piece actually creates phonon coherent states, with the effective position and momentum for each phonon normal mode determined by the particular spin state. The spin-spin interaction term is an Ising model, just with the Ising axis being the x-axis rather than the conventional z-axis.

So far, we have worked with spin operators that commute. But, if we add in a transverse magnetic field $\hat{H}_B(t) = -\sum_j B_j(t)\sigma_j^y$, then the manipulations we have done are not all valid. The way we organize them is to still move all of the phonon-spin entanglement terms to left and to have a time-ordered product of the spin-spin terms plus the magnetic field. But because $[\sigma_j^x, \sigma_j^y] = 2i\sigma_j^z$, there will be additional

correction terms that need to be taken into account. Our strategy is to continue to place the time-ordered product of the spin-only interactions as the third factor as the time-ordered product of the spin-only Hamiltonian, given by

$$\hat{H}_{\text{spin-spin}}(t) = \sum_{j=1}^{N} \sum_{j'=1}^{N} J_{jj'}(t) \sigma_j^x \sigma_{j'}^x - B(t) \sum_{j=1}^{N} \sigma_j^y, \tag{21.62}$$

where we assumed the transverse field was the same for each site, and we used the conventional sign of the magnetic field interaction (many papers on simulations in ion traps change the sign of the field; we will have to also in order to compare to their results). We then accumulate all other correction terms in a fourth factor to the rightmost. These terms are complicated to determine, and there are many of them. You will describe them in an essay question in Problem 21.4.6. It is common to neglect them, even if it is not completely clear that they are all small. We will comment on the effect of these terms later.

There is an interesting result that we can show, if we neglect the additional corrections to the transverse magnetic field (or for any case with a longitudinal field or no field), where measurements of the populations of the different spin states in the x-basis are independent of the phonons that are created, if we start the system in the phonon ground state. This tells us that when the spin-phonon entanglement occurs, it does not affect the measurement of the spin-state populations, and hence we can ignore that term in the time evolution and think of the system as a spin-only system. This makes the overall analysis much simpler.

If we ignore the corrections to the transverse field, start in the phonon ground state, and in the spin state for a large magnetic field, then our initial state is

$$|\psi(0)\rangle = \underbrace{|\uparrow\rangle_{1y} \otimes |\uparrow\rangle_{2y} \otimes \cdots \otimes |\uparrow\rangle_{Ny}}_{\text{spin part}} \otimes \underbrace{|0\rangle_1 \otimes |0\rangle_2 \otimes \cdots \otimes |0\rangle_N}_{\text{phonon part}}. \tag{21.63}$$

The spin states are up along the y-axis for all spins, and the phonon states are all in the ground state (the phonons are described in their normal mode Fock states). When the spin-spin interaction acts on this state, it takes us to a different spin state, which we expand in a set of coefficients c in terms of the x-basis, given by

$$|\psi_{\text{spin}}\rangle = \sum_{m_1} \cdots \sum_{m_N} c_{m_1 m_2 \cdots m_N} |m_1\rangle_{1x} \otimes |m_2\rangle_{2x} \otimes \cdots \otimes |m_N\rangle_{Nx}, \tag{21.64}$$

where $m_i = \uparrow$ or $\downarrow$ along the x-direction for each site ($m_i = \pm 1$). This is then tensored with the phonon state, which still has all phonons in the ground state, because the spin-spin evolution operator does not affect the phonon states.

Next, the spin-phonon entanglement piece acts on the state. It creates an entangled state between the spins and where each phonon state is made into a coherent state, with the position and momentum of the state determined by the spin state. In particular, we use the Weyl form of the Baker-Campbell-Hausdorff formula

to show that

$$\hat{U}_{\text{spin-phonon}}(t) = \prod_{\alpha=1}^{N} \left(e^{-i\sum_{j=1}^{N}\sigma_j^x \Gamma_j^{(\alpha)*}(t)\hat{a}_{\alpha x}^\dagger} e^{-i\sum_{j=1}^{N}\sigma_j^x \Gamma_j^{(\alpha)}(t)\hat{a}_{\alpha x}} e^{-\frac{1}{2}\sum_{j=1}^{N}\sum_{j'=1}^{N}\sigma_j^x\sigma_{j'}^x\left|\Gamma_j^{(\alpha)}\right|^2} \right) \tag{21.65}$$

which takes our state to

$$|\psi_{\text{spin-phonon}}\rangle = \sum_{m_1}\cdots\sum_{m_N} c_{m_1 m_2 \cdots m_N} |m_1\rangle_{1x} \otimes |m_2\rangle_{2x} \otimes \cdots \otimes |m_N\rangle_{Nx} \otimes \left|-i\sum_j m_j \Gamma_j^{(\alpha=1)*}(t)\right\rangle_1 \otimes \cdots \otimes \left|-i\sum_j m_j \Gamma_j^{(\alpha=N)*}(t)\right\rangle_N \tag{21.66}$$

where we label each coherent state by its "α" label. Now, when we measure, because we do not measure the phonon states, we multiply by the bra corresponding to the spin-state measured ${}_x\langle m_1| \otimes {}_x\langle m_2| \otimes \cdots \otimes {}_x\langle m_N|$, and we multiply by a bra corresponding to a specific Fock state for all phonons ${}_1\langle n_1| \otimes {}_2\langle n_2| \otimes \cdots \otimes {}_N\langle n_N|$. Then, we square the matrix element and sum over all of the phonon states. Using the fact that

$$\sum_{n=0}^{\infty} |\langle n|\alpha\rangle|^2 = \langle\alpha|\alpha\rangle = 1, \tag{21.67}$$

with the "α" label being the coherent state label, we find that the probability to measure the particular state is given by

$$P_{m_1,m_2,\cdots,m_N} = |{}_x\langle m_1| \otimes {}_x\langle m_2| \otimes \cdots \otimes {}_x\langle m_N||\psi_{\text{spin}}\rangle|^2 = |c_{m_1,m_2,\cdots,m_N}|^2. \tag{21.68}$$

This is what we would have gotten had we ignored the spin-phonon entanglement! In Problem 21.4.5, you will explain what happens if we measure a spin state in another direction.

In a specific experiment, we choose a detuning, which determines the different spin exchange terms. If we detune far to the blue of the center-of-mass mode of the transverse phonons, we excite all of the phonons weakly. This leads to (time-averaged) spin interactions that decay, approximately as $1/|i-j|^3$. As we move the detuning closer to the center-of-mass mode, the behavior is still approximately power law, but it decreases all the way down to a power of zero, when we drive the center-of-mass mode nearly on resonance. This is because the center-of-mass phonon has the same amplitude for each ion. If we push past the center-of-mass mode and detune in between phonon modes, then we find that the spin exchange

interactions have more complex behavior, varying both in magnitude and sign in somewhat unpredictable ways. Of course, we can calculate explicitly what they are for any given case. What is particularly nice is we can measure each of these spin-exchange constants individually, and they agree with the theory very well (there are no adjustable parameters in the theory).

We have seen that the spin exchange values can have a strong time dependence, yet most analyses assume they are given by their time-averaged values. Why? Well, one reason is the rotating wave approximation, because the oscillations are much faster than the spin dynamics (typically MHz versus KHz), but if we solve the problem including the phonons, instead of adiabatically eliminating them, we find that the presence of the phonons damps out the oscillations and makes the system behave more and more like it was described just in terms of the time-averaged spin-exchange terms. For this reason, the time dependence of $J_{jj'}$ is neglected for most theoretical treatments. We have a video https://tinyurl.com/y86bdtjk that discusses many of the themes in the past three sections.

Section 21.4 Problems:

21.4.1 Energy Scales for the Spin-Spin Interactions

Pick a Lamb-Dicke parameter of $\eta = 0.05$, a Rabi frequency of $\Omega = 2\pi \times 1\,\text{MHz}$, the transverse trap frequency as varying from $\omega_x = 2\pi \times 10\,\text{MHz}$ to $\omega_x = 2\pi \times 100\,\text{KHz}$, and a detuning of $\mu = \omega_x + 50\,\text{KHz}$. Use these values to estimate the time-averaged spin-exchange couplings.

21.4.2 Geometric Summations Needed for the Evaluation of the Time-Ordered Product

Previously, we found that we can use the geometric summation to show that

$$\lim_{\Delta t\to 0}\sum_{k=0}^{M-1} e^{i\omega k\Delta t} = \lim_{\Delta t\to 0}\frac{1-e^{i\omega t}}{1-e^{i\omega\Delta t}} = \frac{1}{i\omega}\left(e^{i\omega t}-1\right), \tag{21.69}$$

where $t = M\Delta t$. Note that this holds as well even if we shift the limits to sum from $k = 1$ to $k = M$, because $\lim_{\Delta t\to 0} e^{i\omega\Delta t} = 1$. Use this identity to show that

$$\sum_{k=1}^{M}\hat{V}_I'(k\Delta t)\Delta t = \hbar\sum_{j=1}^{N}\sum_{\alpha=1}^{N}\sigma_j^x\left(\Gamma_j^{(\alpha)}(t)\hat{a}_{\alpha x}+\Gamma_j^{(\alpha)*}(t)\hat{a}_{\alpha x}\right), \tag{21.70}$$

where $\Gamma_j^{(\alpha)}(t)$ is given in Eq. (21.58) and $t = M\Delta k$. You will need to take the limit as $\Delta t \to 0$ to obtain this result.

To find the $J_{jj'}(t)$, we need to compute

$$\frac{\Delta t}{2}\left[\hat{V}_I'(t), \sum_{k=1}^{M-1} \hat{V}_I'(k\Delta t)\right] = -i\hbar \sum_{j=1}^{N}\sum_{j'=1}^{N} J_{jj'}(t)\sigma_j^x \sigma_{j'}^x. \tag{21.71}$$

Use the fact that we already computed the summation above, and evaluate the commutator to find the spin-spin interaction coefficients.

The exponent in the spin-spin time evolution operator can also be explicitly summed. Use the fact that Ising spins commute with each other to evaluate the coefficients $\Phi_{jj'}(t)$.

21.4.3 Simplification of the Ion-Spin Time Evolution Operator When We Use the Rotating Wave Approximation

If we perform the rotating wave approximation on the interaction Hamiltonian, we have

$$\hat{V}_I'(t) = -\frac{1}{2i}\sum_{j=1}^{N} \hbar\Omega_j \sigma_j^x \sum_{\alpha=1}^{N} \eta_{jx}^{\alpha} b_j^{(\alpha)} \left\{\hat{a}_{\alpha x} e^{i\left(-\omega_t^{(\alpha)}+\mu\right)t} - \hat{a}_{\alpha x}^{\dagger} e^{i\left(\omega_t^{(\alpha)}-\mu\right)t}\right\}, \tag{21.72}$$

with $\hat{H}_0$ given by the phonon Hamiltonian. This is a harmonic perturbation, and it can be rewritten as the Hadamard lemma applied to a time-independent Hamiltonian. Do this, and then compute the time-ordered product for the case with no transverse field. Put all terms into one exponent, computing corrections that survive as $\Delta t \to 0$. Then use the displacement Hadamard to write the final term as a displacement of the $\hat{a}^{\dagger}\hat{a}$ terms. Finally move all number operator terms to the left, and simplify the final result so that it takes the form of the three factors we had before for the time-ordered product. Compare these two expressions. It is best to do this by keeping the spin-spin evolution operator written as a time-ordered product with the $J_{jj'}$ instead of using the Φ.

21.4.4 The Møllmer-Sørensen Gate

The Møllmer-Sørensen gate (Sørensen & Møllmer, 1999) is an interesting construction, because it entangles spins together but with no spin-phonon entanglement, even in cases where there are phonons excited in the system. It uses phonon excitations during the execution of the gate but then precisely removes them at the end.

In this problem, we investigate the gate in the case where we excite the center-of-mass mode only (achieved by detuning close to the center-of-mass frequency). Show that the spin-phonon time evolution operator is given by

$$e^{-\frac{i\Omega\eta_{\mathrm{CM}}}{2\hbar\sqrt{N}(\omega_t^{(\alpha)}-\mu)}\left((1-e^{-i(\omega_t^{(\alpha)}-\mu)t})\hat{a}_{\mathrm{CM}}+(1-e^{i(\omega_t^{(\alpha)}-\mu)t})\hat{a}_{\mathrm{CM}}^{\dagger}\right)\sum_{j=1}^{N}\sigma_j^x}, \tag{21.73}$$

for time evolution from 0 to t, where we assume the Lamb-Dicke parameter and the Rabi frequency do not depend on the lattice site. You will need the center-of-mass phonon eigenvector (normalized!) in the derivation.

You can see that the exponent vanishes at $t = 0$, as expected. For what other times does it vanish? The Møllmer-Sørensen gate is run for a time long enough that the spin-phonon entanglement evolution operator exponent vanishes. This is possible when one phonon frequency dominates the dynamics. Explain how phonons are entangled with spins for intermediate times but become disentangled at these specific times. The spin part of the quantum state also evolves under the spin Hamiltonian during this time, and it creates a spin-only modification of the state when we stop at these special times. Explain how this changes when we include all phonon modes, so one cannot evolve for a single time where the spin-phonon entanglement vanishes.

21.4.5 What Happens to the Measurement in a Direction Different from the Ising Spin Direction?

Write an essay describing what happens to the measurements of spin states if they are measured in a different direction than the Ising direction. Think about how the argument made in the text changes in this case and what the implications are. Also discuss what occurs when we measure in the x-direction when we do have a transverse field in the y-direction. Under what circumstances are the phonon effects small in both cases?

21.4.6 Extra Terms That Enter When We Have a Transverse Field

Write an essay that describes how we include the transverse magnetic field into the analysis and how we would determine the extra terms that arise. Under what circumstances are these extra terms small and can be neglected?

21.4.7 Spin-Spin Interactions for the Three-Site Model

For an $N = 3$ ion chain, there are two independent spin-exchange terms $J_{12} = J_{23} = J$ and $J_{13} = J'$. Assume $J = 2\pi \times 800$ Hz. Find J' in the same units, using

$$\frac{\omega_{\text{tilt}}}{\omega_{\text{COM}}} = \frac{2.5474}{2.7367}, \quad \frac{\omega_{\text{zig-zag}}}{\omega_{\text{COM}}} = \frac{2.2560}{2.7367}, \quad \text{and} \quad \mu = 0.874\omega_{\text{COM}}. \tag{21.74}$$

We are also assuming the Rabi frequency and Lamb-Dicke parameters are independent of j. You will need to determine the phonon eigenvectors, which were discussed in Sect. 21.1.

21.4.8 Plotting the Spin-Spin Interactions as a Function of Time

Using the parameters from the previous problem, plot the spin-spin interactions as a function of t. Explain why the rotating wave approximation still works, so that we can use just the time-averaged coefficients.

21.5 Adiabatic State Preparation and Shortcuts to Adiabaticity

The initial experiments to use ion traps for quantum simulation and computation were born in the 1990s at the National Institute of Standards and Technology and led by Dave Wineland and Chris Monroe (see Fig. 21.9). Chris Monroe later co-founded ionQ, a company that has commercialized ion-trap-based quantum computers. In this section, we will discuss some of the first simulation experiments which used the ion trap, driven by lasers, to prepare a nontrivial entangled ground state of a quantum spin system. I was personally involved in a number of these projects in the 2010s, and this section will focus on that work, which tells the story about how one can prepare specific quantum states.

The adiabatic theorem (Born & Fock, 1928) states that for any quantum system with an energy gap between the ground and excited state, the ground state can be reached from an easy to prepare state by slowly changing the Hamiltonian from the easy Hamiltonian to the complex Hamiltonian, as long as there is always a gap for all parameters chosen in the path from the initial to final Hamiltonian. While this sounds like a great way to prepare difficult ground states—start from an easy to prepare state of a simple Hamiltonian and then just evolve it slowly enough to the complicated Hamiltonian—the problem is that "slow enough" is often excruciatingly slow. As a result, it is difficult to evolve from one state to another

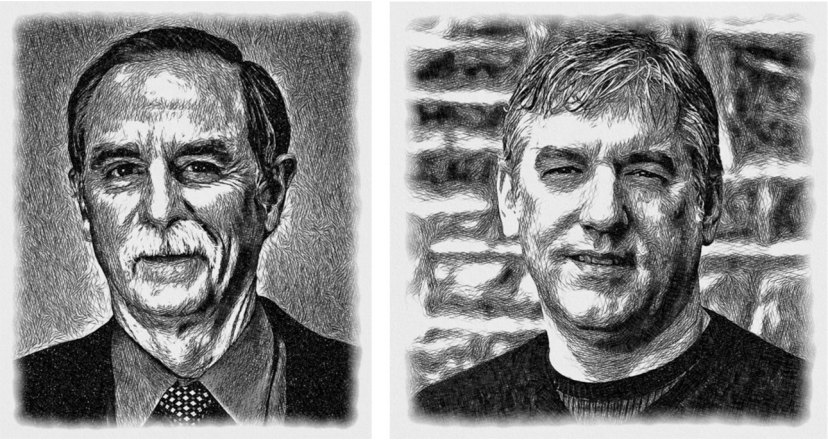

Fig. 21.9 (Left) Dave Wineland, 2012 Nobel laureate, and (right) Chris Monroe, co-founder of ionQ. In the 1990s, these two developed much of the initial experimental work on how to manipulate trapped ions for quantum simulation and quantum computation (Wineland photograph by James Burrus/NIST, courtesy of AIP Emilio Segrè Visual Archives, W. F. Meggers Gallery of Nobel Laureates Collection; modified to look like a sketch by James Freericks. Monroe photograph from his website and used by permission; modified to look like a sketch by James Freericks)

without creating diabatic excitations out of the ground state. We already saw this when we studied the Landau-Zener problem in Sect. 14.8, where the amount of excitation depended both on the minimal energy gap, and the speed at which the minimal gap is traversed.

So, for any reasonable time frame, when one tries to engage in adiabatic state preparation, there will be some level of excitation out of the ground state. The probability to remain in the ground state is called the *fidelity*, and it is given by $|\langle\psi_{gs}|\psi(t_f)\rangle|^2$ where $|\psi(t_f)\rangle$ is the time-evolved quantum state along the adiabatic path at the final time. The next best thing to a full adiabatic time evolution is what is called the *local adiabatic time evolution*, which finds the optimal path in parameter space to evolve the system to maximize the final fidelity over a predetermined time interval. This is done by making the diabatic excitation out of the ground state to be the same in each time interval. As we learned from the Landau-Zener problem, this requires us to evolve quickly when the gap is large and slowly when it is small. Doing precisely this builds what is called the *local adiabatic field*, which is the optimal field to reduce the excitation away from the ground state.

Let us now get concrete to discuss these issues within the context of the transverse-field Ising model, which can be simulated with trapped ions and the proper driving laser fields. Since we want to start the system in an easy to prepare state at $t = 0$, we choose the ground state of $\hat{H}_B$, which is the field-only part of the Hamiltonian. Its ground state simply has every ion state in the $|\uparrow\rangle_y$ state, because the up spin is the ground state of the magnetic field. Then, we slowly turn on the spin-spin interactions and reduce the magnetic field until at t_f, where the magnetic

field is equal to the final value for where we seek the ground state of the system (in most cases, we pick $B(t_F) = 0$). In a realistic experiment, we cannot start the system with $J_{jj'}(0) = 0$, so we instead pick a large B field, with a constant $J_{jj'}$, and ramp from $B_{\max}$ to $B_{\min}$. When we do this, we still start in the easy to prepare state, so we have some diabatic excitation right from the start. As we evolve the system further, the amount of diabatic excitation just increases. Nevertheless, the goal is still to achieve the highest possible fidelity at the end of the time evolution. But you can see it is becoming more difficult to carry out.

There are typically two symmetries in the transverse-field Ising model. The first is a spin-parity symmetry. If we map

$$\sigma_j^x \to -\sigma_j^x, \quad \sigma_j^y \to \sigma_j^y, \quad \text{and} \quad \sigma_j^z \to -\sigma_j^z, \tag{21.75}$$

then the Hamiltonian maps to itself (because the x terms are quadratic and there are no z terms) and all of the Pauli matrix commutation relations are preserved (because we change the sign of only two of the three matrices). This symmetry is present in all transverse-field Ising models. But, if we look at the spin-phonon entangling terms, they are linear in σ^x, so the exponent changes sign. This means it is not a symmetry of the system when we include the effects of the phonons. But in cases where the phonons do not affect the measurement results, it will still be a symmetry for the system. The other symmetry we have is a spatial reflection symmetry. This happens if the system looks the same when reflected in a mirror about the center. In equations, it means we must have

$$B_j(t) = B_{N+1-j}(t) \quad \text{and} \quad J_{jj'} = J_{N+1-j\,N+1-j'}. \tag{21.76}$$

In order for this to occur, when we have a uniform magnetic field, uniform Rabi frequency, and uniform Lamb-Dicke parameters, is that the normal-mode eigenvectors have the symmetry $b_j^{(\alpha)} = b_{N+1-j}^{(\alpha)}$, which occurs when the applied pseudopotential is an even function of position. So, it will be a symmetry for the problems we consider.

In both cases, these are parity symmetries. So when we apply the symmetry to a state, it will return $\pm$ the state. This gives us four symmetry classifications—even or odd with respect to each parity. Since the Hamiltonian shares this symmetry, it never changes the parity of the state and as a result, whatever symmetry sector we start in, that is the sector we stay in during the time evolution (with the caveat that the phonons do not affect the results). Knowing this allows us to prepare states with specific symmetries, and then because we have a discrete quantum number (the parity), that quantum number cannot change. This can make the computation more robust, a fact we use more heavily in the next section.

As an example, let us determine the parity symmetry of the initial state in a magnetic field. There is a trick to working with these systems. Because of the cyclic symmetry, we can shift $x \to z$, $y \to x$, and $z \to y$ in the Pauli matrices. Then the system looks like our standard Pauli matrices and the standard transverse-field Ising

model with the Ising interaction in the z direction and the field in the x direction. Then, when we create a state with all spins in the $|\uparrow\rangle_x$ direction, it is given by

$$|\psi(0)\rangle = \frac{1}{\sqrt{2^N}}(|\uparrow\uparrow\cdots\uparrow\rangle_z + |\downarrow\uparrow\cdots\uparrow\rangle_z + \cdots + |\downarrow\downarrow\cdots\downarrow\rangle_z), \tag{21.77}$$

where we include a linear superposition of *every* product state in the Hilbert space with the same coefficient for each state in the expansion. By performing the symmetry transformations, it is clear this is in the even spin and even space parity sector. If, as is usually done, the magnetic field term enters with a plus sign instead of a minus sign, we must use the tensor product of the $|\downarrow\rangle_x$ states, which will have a minus sign in front of each term with an odd number of $\downarrow$ states in the product state. Then, the spatial parity remains even, but the spin parity becomes odd. Once the symmetry of the initial state is determined, then the system remains in that state for all of the time evolution. So, we must pick the right symmetry to begin with when doing the adiabatic state preparation.

In order to perform the adiabatic state preparation, it is best to construct the local adiabatic field. Before people tried this, they tried a simple linear ramp, but that does not work so well. Then an exponential ramp was tried, because it changes faster at earlier times than later times. It worked better. But the best way is with a local adiabatic field. Constructing it uses calculus. Nevertheless, it is easy to describe how one constructs it. We have to compute the energy gap to the lowest-energy excited state with the same symmetry. We do this as a function of magnetic field and call it $\Delta E(B)$. The Landau-Zener problem tells us that the speed we can traverse a gap is proportional to the square of the gap. This tells us that as we decrement the field by ΔB, then the time must change by $\Delta t \propto \frac{\Delta B}{(\Delta E(B))^2}$. To find the field profile, we have two more steps. We choose a B grid in steps of ΔB and compute the time t by summing the different Δt values, computed to allow for the same excitation in each magnetic field step. The total time is determined by the constant scaling factor that relates the time to the change in the magnetic field divided by the square of the energy gap. Then, we have a function $t(B)$ determined on this grid. We invert it to find $B(t)$. This result is plotted in Fig. 21.10 for an example case. One of the drawbacks of using this approach is that you need *a priori* knowledge about your system—namely, the same symmetry excitation gap along the adiabatic path. Often this is not known for hard problems that have not been solved yet.

We now summarize some of the earliest experiments with adiabatic state preparation in trapped ions (Kim et al., 2010; Islam et al., 2011; Kim et al., 2011). As mentioned before, these experiments use a Hamiltonian with a $+B$ term rather than the customary $-B$ term. The first thing that needs to be verified experimentally is that the exchange couplings are what they are predicted to be by theory. For a three-site ion chain, we only have the nearest-neighbor spin exchange J_1 and the second-neighbor spin exchange J_2. To measure them, we simply create a state that has a superposition of some of the states (if we work with states of definite parity, then each sector has one or three energy states). Then, because a general state, with given symmetry, can be expanded in terms of the energy eigenstates as

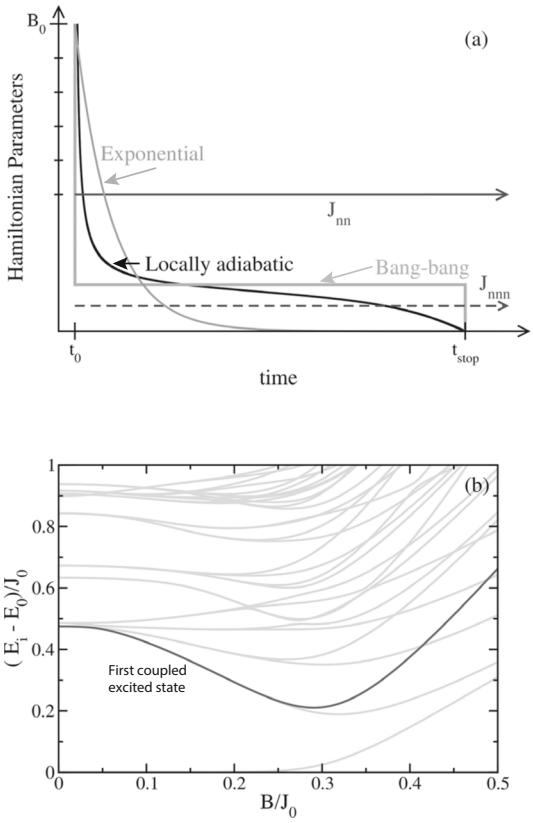

Fig. 21.10 (**a**) Exponential ramp, local adiabatic field, and bang-bang approximation for the transverse-field Ising model with $N = 10$ and an approximate power law of 1.05 for the decay of the exchange coefficients with distance. Note how the local adiabatic field decreases rapidly for large fields, decreases the slowest near the minimal energy gap, and then decreases more rapidly as the gap grows again. (**b**) Low-energy spectrum for this same system (Both figures from Balasubramanian et al. (2018) © American Physical Society)

$|\psi(t)\rangle = \sum_n c_n e^{-i\omega_n t}|n\rangle$, where $\hbar\omega_n = E_n$, then we can see that the measurement of any expectation value, such as the average magnetization in the x-direction or the probability to be in the all down state, will oscillate at the energy differences. So, by preparing the same state every time (such as the all up in y or all down in y states), and then measuring the expectation values at different times after evolving the system with the Ising interaction term only, we find an oscillating time trace. By performing an operation called a Fourier transformation, we can find the frequencies in the oscillation, and from that the energy differences. When enough differences are measured, we can then extract J_1 and J_2. This procedure is illustrated in Fig. 21.11. One can see that the theory and the experiment agree well, with no adjustable parameters.

Now that we are assured that we know how to determine the terms in the Hamiltonian, it is time to try to perform an adiabatic state preparation. For the three-site system, the schematic of what will be run is shown in Fig. 21.12. The system is started in the all down state in the y-direction $|\downarrow\downarrow\downarrow\rangle_y$ (because the magnetic field used is a $+B$ term). Next, we turn on the Ising interaction and a large field, then slowly reduce the field to its minimal value, and finally measure the state after the

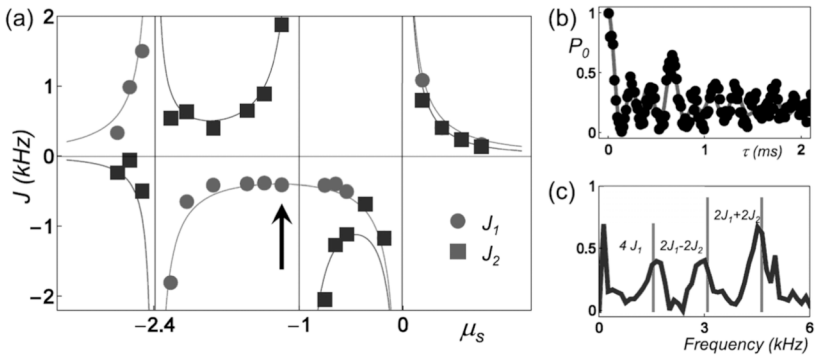

Fig. 21.11 (**a**) Measured couplings $J_1 = J_{12} = J_{23}$ and $J_2 = J_{13}$ for three ions as a function of detuning μ (symbols) compared to theory (lines). The plot uses a scaled detuning $\mu_s = (\mu^2 - \omega_t^{(\mathrm{COM})})/\omega_l^{(\mathrm{COM})}$. In these units, the center of mass mode is at 0, the tilt mode is at -1 and the zig-zag mode is at -2.4. The arrow indicates the detuning used for the results in the other two panels (-50 KHz from the tilt mode). (**b**) Time evolution of the probability P_{down} of $|\downarrow\downarrow\downarrow\rangle_x$ under the Ising-only interaction. The solid line is a fit with an empirical exponential decay. (**c**) The Fourier transform of the experimental curve shown in (**b**), where three peaks originate from the frequency components of $4J_1$, $2J_1 - 2J_2$ and $2J_1 + 2J_2$. The vertical lines are the calculated values (Figure is adapted from Kim et al. (2011) © Institute of Physics)

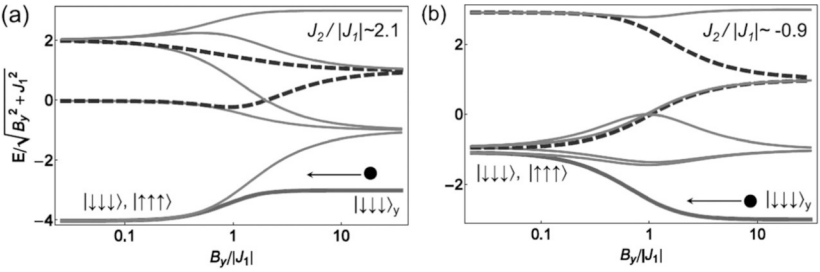

Fig. 21.12 Energy level diagrams for the transverse-field Ising model with two different types of spin–spin interactions. For both panels, the nearest-neighbor interactions are ferromagnetic. (**a**) The second-neighbor interaction is FM with $|J_2|/|J_1| \approx 2$ and (**b**) antiferromagnetic with $|J_2|/|J_1| \approx 0.9$. The adiabatic path moves in the direction given by the arrow with $B_y/|J_1| \approx 10$ at $t = 0$ (the initial state $|\downarrow\downarrow\downarrow\rangle_y$ is only an approximate ground state of the system). Note how the ground state (thick solid line) has no level crossings with any excited state. The dotted lines represent coupled excited states. In both (**a**) and (**b**), the final ground state is ferromagnetic in the x-direction and becomes degenerate at $B_y = 0$. The adiabatic state preparation takes the system to a specific linear combination of the two ferromagnetic states determined by the spin parity (Figure is adapted from Kim et al. (2011) © Institute of Physics)

diabatic time evolution has been completed. In a typical experiment, the exchange terms are on the order of 1–10 KHz, and the total time that can be evolved is a few ms (the earliest experiments were about one half of a ms). So the time evolution is rather rapid in terms of the number of inverse exchange interactions before the experiment is over. The ferromagnetic case, where the exchange integrals are all

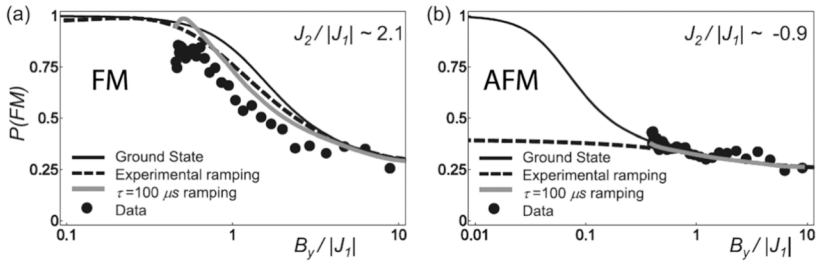

Fig. 21.13 (**a**) Ferromagnetic case. The solid gray line is the theoretical prediction for a 35 μs exponential ramp time constant, the dashed line is for a 100 μs exponential ramp, while the solid black line is for the adiabatic limit. The solid dots are the experimental data, which show a reasonable agreement for the population. (**b**) Antiferromagnetic case. Here, the ramp does not go as far, because the excitation gap becomes too small to try to move past it rapidly. As a result, we do not get so close to the $B_y \to 0$ limit (Figure is adapted from Kim et al. (2011) © Institute of Physics)

negative, has a larger gap of the same symmetry than the antiferromagnetic case has. So, one can evolve the field to lower values in that case and achieve a higher fidelity for the zero-field state. An example of an experimental run is shown in Fig. 21.13.

The current discussion has focused on the probability to find the system in the ferromagnetic state: (i) $|\uparrow\uparrow\uparrow\rangle_x$ or (ii) $|\downarrow\downarrow\downarrow\rangle_x$. The results shown in Fig. 21.13 are calculating sum of the probabilities to be in these two states, which are technically called the population to be in the ferromagnetic state. But the parity symmetry tells us the proper state we go to will be the $(|\uparrow\uparrow\uparrow\rangle_x - |\downarrow\downarrow\downarrow\rangle_x)/\sqrt{2}$. Note that a state that is half the time in the all up state and half the time in the all down state will have the same populations as the superposition with odd spin reflection parity. To tell we really have the superposition state, we need to be able to detect the superposition itself. This can be done by performing state tomography, but that is often a very expensive procedure. Instead, one can also determine whether one is in a superposition by using an *entanglement witness*. These are operators that when we measure their expectation value, if they are negative, then the system is in an entangled superposition. The theory for them is complex, and we will not develop them here. But, they were used in the original experiment (Kim et al., 2010) to verify that the state preparation really is creating a superposition of states.

Finally, in Fig. 21.14, we show the results for what happens as we scale the number of spins from $N = 2$ to $N = 9$ (Islam et al., 2011). In this case, we see that the transition of the state from a superposition of all states to the entangled superposition of the all up and all down states occurs with a steeper and steeper slope in the magnetic field as N increases. This indicates as N is made larger, the transition becomes sharper, which is what is expected in the theory of phase transitions. While we cannot get close to the true phase transition, this behavior is indicative that the experiments are behaving as they should as N is made larger. Another result that came from this work is that if one wants high fidelity of the final ground state, then one needs to investigate alternatives for how to create the ground

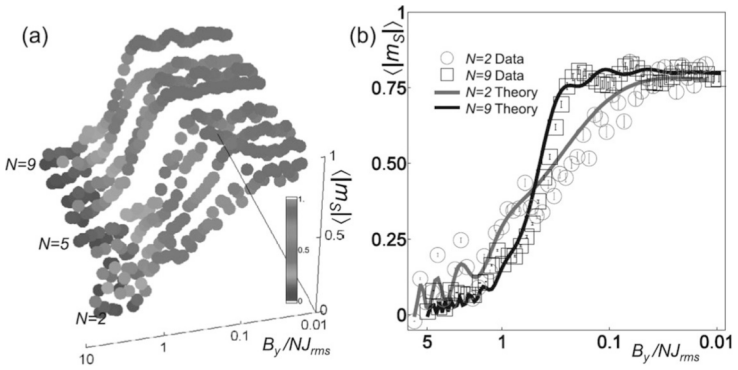

Fig. 21.14 (**a**) Scaled average absolute magnetization per site, versus $B/(NJ_{\rm rms})$ plotted for $N = 2 - 9$ spins. Note that the magnetic field is on a logarithmic scale going from large to small as we go from left to right. As the field is lowered, the spin ordering undergoes a crossover from a paramagnetic to ferromagnetic phase. The crossover curves sharpen as the system size is increased. The oscillations in the data arise due to imperfect initial state preparation and non-adiabaticity due to finite ramping time. (**b**) Magnetization data for $N = 2$ spins (circles) and $N = 9$ spins (squares). The disagreement between theory and experiment is predominantly due to spontaneous emissions and intensity fluctuations of Raman laser beams (Figure is adapted from Kim et al. (2011) © Institute of Physics)

state, because the ramps are always too fast, and significant diabatic excitation tends to occur in these systems.

These conclusions require us to think about different ways to try to create the ground state. One approach is called a *shortcut to adiabatic state preparation*, which strives to complete the creation of the ground state in a much shorter time than required by the adiabatic theorem. The original idea was to find a Hamiltonian that could evolve the system in such a way that it always stayed in the ground state of the Hamiltonian along the adiabatic path, but over a time interval as short as desired. Unfortunately, the Hamiltonian needed to do this is horribly complicated and cannot be used in any realistic experiment. Simple alternatives exist as well. One is called the bang-bang protocol, which is the way we operate our thermostats for heating or cooling. If the room is being heated and it falls to too low a temperature, we turn the system on at full strength until the temperature is raised to an upper threshold, where the system is turned off. In spite of its simplicity, this approach is often the most efficient way to control the temperature in a house. The way we use this approach in the transverse-field Ising model is to start the system in the ground state of the $B_y \to \infty$ limit and turn both the Ising term and magnetic field term on with a moderate field value that is held constant for a given time and then turned off. The time it is held on and its value are adjusted until we obtain the highest fidelity of the final state. In theoretical calculations (Balasubramanian et al., 2018), it is shown that usually the local adiabatic field does better, especially as the power law for the exchange interactions gets larger. But when the interactions become more uniform, the bang-bang optimization can sometimes work better than the local adiabatic field, when run over the same time interval.

There are further alternatives to shortcuts to adiabaticity, which are focused on using the variational principle to reduce the number of steps needed to get to the ground state with a certain fidelity. The first general approach to this is called the *quantum approximate optimization algorithm* or QAOA (Farhi et al., 2014). It is a very simple idea. Fix the number of steps (which practitioners prefer to call layers) of a Trotterized time evolution, where the coefficients of the exponent in each Trotter factor are chosen to be a variational parameter. The form for the approximation is then

$$|\psi_f\rangle = \underbrace{e^{-i\beta_p \hat{H}_B} e^{-i\gamma_p \hat{H}_I}}_{\text{single layer}} e^{-i\beta_{p-1}\hat{H}_B} e^{-i\gamma_{p-1}\hat{H}_I} \cdots e^{-i\beta_1 \hat{H}_B} e^{-i\gamma_1 \hat{H}_I} |\psi_0\rangle, \tag{21.78}$$

where we alternate the magnetic field term and the Ising term, with "angles" β and γ at each layer of the ansatz. If a local adiabatic approximation takes 1000 time steps to reach a high fidelity for the ground state, a QAOA approximation can reach the same fidelity in about 30 layers. This is a huge savings! It arises from the use of the variational principle in optimizing the parameters, which allows us to take large steps for each layer and still achieve accurate results. One of the main benefits of the variational principle is that it allows us to engineer a return to the ground state, which can easily overtake the diabatic loss from the ground state, resulting in a higher fidelity with fewer layers. One can also achieve a similar approach by using what we call *modulated time evolution*. Here, we approximately use a local adiabatic evolution for the field but multiply the Hamiltonian by a scaling factor $\lambda(t)$ at each time step and employ a variational principle to determine $\lambda(t)$. This can do even better than QAOA, if we include both factors for a single layer of the QAOA into one exponent of a single Trotter time step.

The final topic we will talk about in this section is *diabatic spectroscopy*. This uses the fact that the state preparation has created some excitation in the system to measure different properties. There are two important observables that can be measured with this: (i) the fidelity of the ground state and (ii) the excitation energy to other excited states. We have already touched on these ideas earlier in the section. It all starts from expanding the current state in terms of the instantaneous energy eigenstates as $|\psi(t)\rangle = \sum_n c_n e^{-i\omega_n t}|n\rangle$. Then, we simply measure some operator and find that

$$\langle\psi(t)|\hat{O}|\psi(t)\rangle = \sum_{mn} c_m^* c_n e^{i(\omega_m - \omega_n)t} \langle m|\hat{O}|n\rangle. \tag{21.79}$$

If we take the time traces and extract the frequency differences, which can be done via Fourier transformation, or compressive sensing, then we learn about the excited states. By instead fitting the amplitudes of the oscillations, we can learn about $|c_n|$, which gives us information about the populations of the different states, especially it tells us about the fidelity of the ground state. There are many ways to make such measurements efficient for specific observables.

Section 21.5 Problems:

21.5.1 Time Evolution of the Three-Site Model

The following is a numerical problem. Using the same parameters as discussed in Problem 21.4.7, pick $B_0 = 2\pi \times 10\,\text{KHz}$ and $\tau = 0.0001\,\text{s}$. Take the exponential ramp $B(t) = B_0 e^{-t/\tau}$, and use the $+B$ form of the Hamiltonian. Use the Trotter product formula to determine time evolution of the system numerically, and find the fidelity to be in the ground state at $t = 0.5\,\text{ms}$ and $t = 1\,\text{ms}$. Plot $P(|\uparrow\uparrow\uparrow\rangle_x) + P(|\downarrow\downarrow\downarrow\rangle_x)$, $P(|\uparrow\downarrow\uparrow\rangle_x) + P(|\downarrow\uparrow\downarrow\rangle_x)$, and $P(|\uparrow\uparrow\downarrow\rangle_x) + P(|\downarrow\downarrow\uparrow\rangle_x) + P(|\downarrow\uparrow\uparrow\rangle_x) + P(|\uparrow\downarrow\downarrow\rangle_x)$ as three different curves. Discuss how one can measure the probability to be in the states $(|\uparrow\uparrow\uparrow\rangle_x \pm |\downarrow\downarrow\downarrow\rangle_x)/\sqrt{2}$, and plot what those probabilities are versus time as well.

21.5.2 How Do We Measure the Final State?

In the experiments described here, we are measuring the populations of the different states in the superposition, which gives us information about the amplitudes in the superposition. How can we obtain information about the relative phases of the different terms?

21.5.3 How Does One Carry Out the Bang-Bang Protocol?

Explain how one might concretely carry out the bang-bang protocol. Is it possible to measure the fidelity at the end of the calculation? If not, what else could be used to optimize to find the best protocol?

21.5.4 Diabatic Spectroscopy

Write a short essay that explains details about how diabatic spectroscopy works.

21.6 Lipkin-Meshkov-Glick and Dicke Models

So far, we have focused on the case where the detuning is sufficiently far from any individual phonon mode that the phonons are only virtually excited, resulting in an

effective spin model only. As we detune closer and closer to a specific phonon mode, then we resonantly excite phonons during the laser driving, and the description of the system must involve both the spin and phonon degrees of freedom. In this section, we will discuss such near-resonant driving of the center-of-mass mode and explain the type of behavior that occurs in such a system. It is described by the Dicke model, which we encountered earlier in Sect. 10.8.

Recent experiments illustrating the behavior of the Dicke model were performed in a Penning trap, which is somewhat different from the linear Paul trap we have been discussing up until now. In the Penning trap, the ions are placed in a large external magnetic field, which makes them rotate in cyclotron orbits. If a trap is added in, and we take into account the Coulomb repulsion, the system forms a lattice-like structure which is nearly perfect in the center and then often has a curved shape for the boundary. The lattice rotates for stability, which can create complications for imaging and other types of experiments. In a 4.24 T magnetic field, the $^9Be^+$ ions have a splitting of 124 GHz between their spin up and spin down states. These are the qubit states used for the system. The optical dipole force is applied in a similar way; except here, the carrier transition is usually directly applied by microwaves rather than using Raman beams.

We consider the case where we detune close to the center of mass mode and slightly to the red of it, so that $\mu - \omega^{(\mathrm{COM})} = \delta < 0$. The bare Hamiltonian with the laser driving takes the form

$$\hat{H}(t) = \hbar\omega^{(\mathrm{COM})}\hat{a}^\dagger_{\mathrm{COM}}\hat{a}_{\mathrm{COM}} - \frac{\hbar\Omega\eta_{\mathrm{COM}}}{\sqrt{N}}(\hat{a}_{\mathrm{COM}} + \hat{a}_{\mathrm{COM}})\hat{S}^z\cos(\mu t) - B_x(t)\hat{S}^x, \tag{21.80}$$

which is the same as Eq. (10.177). Note that we have applied a shift of the phase by $\frac{\pi}{2}$ in the laser drive, we have used the fact that the center of mass phonon has $b_j^{(\mathrm{COM})} = \frac{1}{\sqrt{N}}$, we use the total spin operators $\hat{S}^\alpha = \hbar\sum_{j=1}^N \sigma_j^\alpha/2$, and we have permuted the spin directions, so that the z-direction is the Ising direction and the transverse field is in the x direction. It is common to set $g = 2\Omega\eta_{\mathrm{COM}}$. To go to the rotating system, we use the interaction representation, where we take $\hat{H}_0 = \hbar\mu\hat{a}^\dagger_{\mathrm{COM}}\hat{a}_{\mathrm{COM}}$ to correspond to the laser driving frequency. Then, if we look at the interaction term in the interaction representation, it is given by

$$\hat{V}'_I(t) = -\hbar\delta\hat{a}^\dagger_{\mathrm{COM}}\hat{a}_{\mathrm{COM}} - \frac{\hbar g}{\sqrt{N}}\left\{\hat{a}_{\mathrm{COM}}\left(1 + e^{-2i\mu t}\right) + \hat{a}^\dagger_{\mathrm{COM}}\left(1 + e^{2i\mu t}\right)\right\}\hat{S}^z - B_x(t)\hat{S}^x. \tag{21.81}$$

In the rotating wave approximation, we drop the fast oscillating terms $e^{\pm 2i\mu t}$.

Again, we have a spin-phonon reflection symmetry, where $\hat{S}^x \to \hat{S}^x$, $\hat{S}^y \to -\hat{S}^y$, $\hat{S}^z \to -\hat{S}^z$, $\hat{a} \to -\hat{a}$, and $\hat{a}^\dagger \to -\hat{a}^\dagger$, which maps the Hamiltonian to itself. This is again a parity symmetry, and we can classify states according to whether they are positive or negative parity with respect to this symmetry.

As a first approximation, we can try to adiabatically eliminate the phonons. This is easily done here by defining new phonon ladder operators such that $\hat{A} = \hat{a}_{\text{COM}} + g\hat{S}^z/(\hbar\delta\sqrt{N})$ and $\hat{A}^\dagger = \hat{a}^\dagger_{\text{COM}} + g\hat{S}^z/(\hbar\delta\sqrt{N})$. The commutation relation of the ladder operators and the spin operators are the same (but the ladder operators and spin operators no longer commute), and

$$\hat{V}_I'(t) = -\hbar\delta\hat{A}^\dagger\hat{A} + \frac{g^2}{\hbar\delta N}\left(\hat{S}^z\right)^2 - B_x(t)\hat{S}^x. \tag{21.82}$$

This looks like independent Lipkin-Meshkov-Glick and phonon models, but because $[\hat{A}, \hat{S}^x] \neq 0$, the phonon and spin systems are not independent, and one cannot construct states as tensor products of two independent spaces. Furthermore, because we typically are working close to resonance, we may have that δ is smaller than either B_x or $g^2/\hbar\delta N$, we can have a situation where the low-energy excitations are phonon creations, not spin excitations. In such a situation, we need to solve the problem with both degrees of freedom treated together. Hence, although it looks like we can create the Lipkin-Meshkov-Glick model by tuning near resonance to the center-of-mass mode, we cannot. We always have a Dicke model, where the spin and phonon degrees of freedom will be entangled by the Hamiltonian dynamics.

We have already discussed many of the properties of the Dicke model in Sect. 10.8, and we suggest you review those results before we discuss how they are implemented experimentally.

We now discuss the results of two experiments in Safavi-Naini et al. (2018), where an exponential and linear ramp were used and Cohn et al. (2018), where the bang-bang approach was used. The Penning trap held approximately 70 ions in it, and the system had two systematic experimental errors that can cause errors in the simulation—the first is decoherence, which reduces the z spin projection and an additional longitudinal magnetic field, which causes the spin states to precess in the $x-y$ plane. The expected behavior is that starting with a large field, the ground state, which has definite $\hat{\vec{S}}^2$, will be oriented along the x axis, with the expectation value near $\hbar N/2$. As the field is reduced, the ground state becomes a linear combination of the all up state and the all down state in the z direction, with the all up state having the phonons pushed up and the all down state has the phonons pushed down. This state is called a *spin-cat state*. The average z-component of spin approaches zero, but the average absolute value of the spin approaches $\hbar N/2$, while the average x-component approaches zero. So this is the expected behavior (we do not measure the phonons directly, but we do measure a spin-phonon correlation function that is discussed in the problems). Examining the results with the exponential ramp, and comparing them to theoretical results (which approximate the decoherence effects), show remarkably good agreement in Fig. 21.15. This is even more remarkable, because the center-of-mass mode was only cooled down to an average number of phonons of about $\langle\hat{n}_{\text{COM}}\rangle \approx 6$, which is rather far from the ground state. In addition, these results show clearly that the Lipkin-Meshkov-Glick model is not accurate for describing the system.

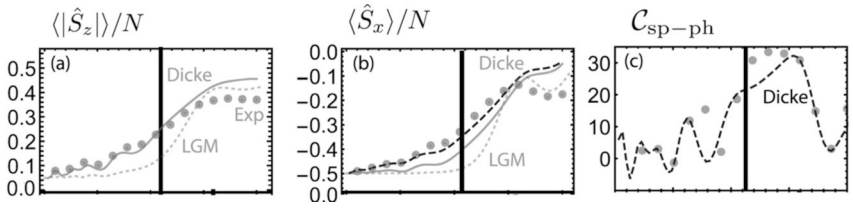

Fig. 21.15 (**a**) Instantaneous magnetization in the z direction, (**b**) in the x direction, and (**c**) the spin-phonon correlation function. The filled circles are experimental measurements (statistical error is on the order of marker size), the solid and black dashed lines are the theory results without and with dephasing (dephasing is absent in panel (**a**) because the z magnetization is less sensitive to this decoherence) and the dotted lines are the Lipkin-Meshkov-Glick model results. We indicate the time at which B_c is reached in each ramp by a vertical line. The initial field is $B(t{=}0)/2\pi \approx 7.1\,\mathrm{KHz}$, $g/2\pi \approx 1.32\,\mathrm{KHz}$, $\delta/2\pi \approx -1\,\mathrm{KHz}$, and $J/2\pi = g^2/(2\pi\delta) \approx 1.75\,\mathrm{KHz}$. This experiment has $N = 68$ (Figure is adapted from Safavi-Naini et al. (2018) © American Physical Society)

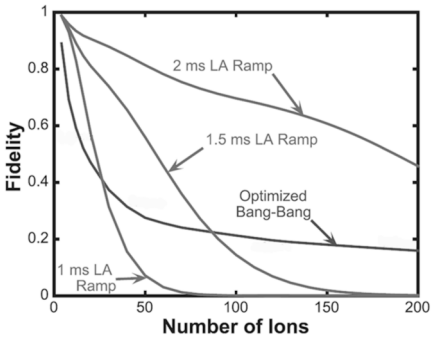

Fig. 21.16 Ground-state fidelity of the Dicke Hamiltonian as a function of system size for the bang-bang and LA ramps of 1.0, 1.5 and 2.0 ms. The bang-bang approach outperforms the LA ramp of 1.5 ms at approximately 90 ions (Figure is adapted from Cohn et al. (2018) CC BY license)

In addition, the bang-bang shortcut to adiabaticity was compared to the local adiabatic ramp with about 75 ions (Cohn et al., 2018). For ramps up to about 1.5 ms, the local adiabatic ramp performed better than the bang-bang protocol. But, surprisingly, as we go to higher and higher numbers of ions, all of sudden the bang-bang approach works better. These latter experiments have not been done yet, but the theory was calculated and clearly shows these results in Fig. 21.16.

There is one more important phenomena that we need to discuss about these systems related to entanglement and the conserved parity quantum number. Part of the reason why the results work so well is because the parity quantum number protects the calculations from errors. The only thing that can erode the parity quantum number is perturbations that are not symmetric under the spin-phonon reflection symmetry. An example of that is the longitudinal field, which breaks this symmetry. But, there is more to this system than just protecting the parity. There is a simple protocol that can evolve both the positive distorted phonon and the negative distorted phonon to both go back to the origin at the same time. It simply

requires changing the sign of the detuning. If we allow the system to evolve this way, then after the system has had both phonon distortions transferred to the origin, we have changed the system from an entangled state to a product state with respect to the spin and the phonons. Yet it remains an entangled spin state! In other words, we have transferred the entanglement from the spin-phonon system to the spin-only system. This then opens up a new avenue in *reservoir engineering*, where we engineer entanglement with the reservoir to protect the calculation and then remove the entanglement at the end to be able to work with the pure spin state. It is a neat idea with great potential for how one can use these quantum simulators to solve different types of problems.

Section 21.6 Problems:

21.6.1 Difference in the Behavior for Red-Detuned Versus Blue-Detuned Lasers

In the mapping, we had a red-detuned laser driving, which created a negative term for the phonons in the interaction Hamiltonian. If, on the other hand, we were blue detuned, then the phonons have a positive term in front of them. Discuss in an essay what the differences are for these two cases. In particular, is it even possible to do the rotating wave approximation for the blue detuned case?

21.6.2 Spin-Phonon Correlation Function

The spin-phonon correlation function plotted in panel (c) of Fig. 21.15 is given by $\langle\psi(t)|(\hat{a}_{\text{COM}} + \hat{a}^{\dagger}_{\text{COM}})\hat{S}^{y}|\psi(t)\rangle$. If we consider the difference of the expectation value of $\hat{S}^{x}$ at time $t + \Delta t$ and time t, we find that

$$\begin{aligned}\langle\psi(t+\Delta t)|\hat{S}^{x}|\psi(t+\Delta t)\rangle &= \langle\psi(t)|e^{\frac{i}{\hbar}\hat{V}'_{I}(t)\Delta t}\hat{S}^{x}e^{-\frac{i}{\hbar}\hat{V}'_{I}(t)\Delta t}|\psi(t)\rangle \\ &\approx \langle\psi(t)|\big(1+\tfrac{i}{\hbar}\hat{V}'_{I}(t)\Delta t\big)\hat{S}^{x}\big(1-\tfrac{i}{\hbar}\hat{V}'_{I}(t)\Delta t\big)|\psi(t)\rangle \\ &\approx \langle\psi(t)|\hat{S}^{x}|\psi(t)\rangle + \tfrac{i}{\hbar}\langle\psi(t)|[\hat{V}'_{I}(t),\hat{S}^{x}]|\psi(t)\rangle\Delta t \\ &\quad + O\big((\Delta t)^{2}\big). \end{aligned} \tag{21.83}$$

By evaluating the commutator, find a formula for the correlation function in terms of the rate that the expectation value for $\hat{S}^{x}$ changes with time. In the experiment, the system also has decoherence, which adds an extra term to the determination of the correlation function, but we do not discuss that contribution here.

21.6.3 The Role of Parity Symmetry to Protect Entanglement and to Transfer Entanglement

Write an essay filling in the details for how the spin-phonon entanglement can be removed at the end of the calculation. If having trouble, consult the two experimental papers that discuss the approach needed to remove the entanglement (especially the supplemental information in Safavi-Naini et al., 2018). Think about circumstances where this procedure is helpful for a quantum simulation.

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Chapter 22
Decoherence and Squeezed States

Pattern finding is the purpose of the mind and the construct of the universe.

Agnes Denes, Artist

Quantum systems are rarely isolated and often interact with a large number of degrees of freedom that we generically call *the environment*. As the system becomes entangled with the environment, it effectively loses the interference within the system because it transforms from a product state superposition to an entangled state superposition. Thus, when the system is measured, the interference effects within the system are reduced, because we do not measure the entangled environment degrees of freedom. The details for how we describe the behavior of this so-called decoherence are complicated. In this chapter, we investigate how to formulate descriptions of decoherence, we apply this formalism to real systems such as nuclear magnetic resonance, and then we examine the interplay between squeezing and decoherence in trapped ions and the implications for precision measurements. Get a jump on entanglement and decoherence with the summary at https://tinyurl.com/yenuahu2.

22.1 Mixed States and Density Matrices

When we described both the Stern-Gerlach experiment and the two-slit experiment with polarizers, we determined the quantum states, just before measurement, as entangled superpositions—in the Stern-Gerlach experiment, it was $(|\uparrow\rangle \otimes |\text{up}\rangle + |\downarrow\rangle \otimes |\text{down}\rangle)/\sqrt{2}$, while for the two slit experiment, it was $(|\text{left}\rangle \otimes |V\rangle + |\text{right}\rangle \otimes |H\rangle)/\sqrt{2}$. In both cases, the entangled states imply we lose interference effects, even if we do not measure one of the entangled degrees of freedom (spin projection for Stern-Gerlach and polarization for the two-slit experiment). For concreteness, we

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focus our discussion on the two-slit experiment. When we measure the entangled state on the screen, we find no interference—it is as if half of the time the photon went through the left slit and half of the time through the right slit. In general, we don't have a clear formalism to describe such situations, even though they are easy to describe in words. To do this properly, we need to represent quantum states in terms of operators, not vectors. This then allows us to describe situations when the quantum system is not described by a single, pure quantum state. It turns out that all density matrices can be thought of as pure entangled states where we do not measure at least one of the entangled degrees of freedom, which will be discussed more thoroughly below.

Our first step is to represent the pure quantum state in terms of an operator. If I have a state $|\psi\rangle$, the most natural operator to work with is the projection operator onto that pure state: $|\psi\rangle\langle\psi|$. We can do a number of different calculations with the projection operator. The most important is determining the expectation value of a Hermitian operator $\hat{O}$, with eigenvalues o_n and eigenstates $|o_n\rangle$. To do this from the operator viewpoint, we work in the eigenbasis $\{|o_n\rangle\}$ of the operator $\hat{O} = \sum_n o_n |o_n\rangle\langle o_n|$. We know the probability to measure o_n satisfies $P(o_n) = |\langle o_n|\psi\rangle|^2$, and the expectation value of $\hat{O}$ satisfies

$$\langle\psi|\hat{O}|\psi\rangle = \sum_n o_n P(o_n) = \sum_n o_n |\langle o_n|\psi\rangle|^2 = \sum_n o_n \langle o_n|\psi\rangle\langle\psi|o_n\rangle. \tag{22.1}$$

From the operator perspective, we can reexpress these results as

$$\langle\psi|\hat{O}|\psi\rangle = \sum_n \langle o_n|\hat{O}|\psi\rangle\langle\psi|o_n\rangle = \mathrm{Tr}\big(\hat{O}|\psi\rangle\langle\psi|\big), \tag{22.2}$$

where the invariance of the trace means we can take it in any complete orthonormal set of basis vectors, even though the derivation explicitly used the eigenbasis of the operator $\hat{O}$. We define the *density operator* $\hat{\rho}$ of a pure state via the projection operator $\hat{\rho} = |\psi\rangle\langle\psi|$ and the *density matrix* via the representation of the density operator in a specific basis $\rho_{mn} = \langle m|\hat{\rho}|n\rangle = \langle m|\psi\rangle\langle\psi|n\rangle$. Oftentimes, the name density operator and density matrix are conflated, so we will not try to stick to proper language and typically use the more common density matrix moniker for both throughout. It should be clear whether the expression is an operator or a matrix by the context.

Our formalism now allows us to go beyond pure states. If we have a set of N nonnegative numbers, $0 \leq p_i \leq 1$, that sum to 1, $\sum_{i=1}^N p_i = 1$, then we can define a density matrix for a *mixed state* via

$$\hat{\rho} = \sum_{i=1}^N p_i |\psi_i\rangle\langle\psi_i|, \tag{22.3}$$

when there are at least two terms with $p_i > 0$ in the summation (a pure state has only one term with $p_i > 0$ in the summation; in fact $p_i = 1$). Note that the mixed state is what we were seeking—a probabilistic collection of pure states. Note further

that nowhere do we require that the states in the mixture be orthogonal. They need not be. There are many equivalent ways of expressing the same density matrix. The density matrix is often called a *quantum state*, whether it is a pure state or a mixed state—it is the most general form of a quantum state.

Now that we have determined the definition of the density matrix, we need to examine some of its properties. We just go through the main properties in order.

1. *Hermiticity*: The density matrix is a Hermitian operator. This follows because all projection operators are Hermitian and probabilities are real—$(|\psi\rangle\langle\psi|)^\dagger = |\psi\rangle\langle\psi|$.
2. *Unit trace*, $\mathrm{Tr}\hat{\rho} = 1$: Take a complete set of states $\{|n\rangle\}$. Then $\mathrm{Tr}\hat{\rho} = \sum_n \sum_i p_i |\langle n|\psi_i\rangle|^2$. Reorder to do the sum over n first, which gives $\sum_i \sum_n p_i \langle\psi_i|n\rangle\langle n|\psi_i\rangle = \sum_i p_i \langle\psi_i|\psi_i\rangle = \sum_i p_i = 1$. Note that we only use the fact that the $\{|\psi_i\rangle\}$ are normalized in this derivation.
3. *$\hat{\rho}$ is a positive semidefinite operator*: This means that $\langle\psi|\hat{\rho}|\psi\rangle \geq 0$ for all $|\psi\rangle$. We have $\langle\psi|\hat{\rho}|\psi\rangle = \sum_i p_i |\langle\psi|\psi_i\rangle|^2 \geq 0$, because the mod square of a matrix element is nonegative as are the probabilities p_i.
4. *Expectation value of $\hat{O}$ in the quantum state $\hat{\rho}$ is* $\mathrm{Tr}(\hat{O}\hat{\rho})$: We have $\mathrm{Tr}(\hat{O}\hat{\rho}) = \sum_n \sum_i p_i o_n |\langle o_n|\psi_i\rangle|^2 = \sum_i p_i \langle\psi_i|\hat{O}|\psi_i\rangle$, which is the sum of the average value for each state represented in the density matrix, and weighted by the probability of that state to appear. This is clearly the expectation value for the quantum state $\hat{\rho}$.
5. $\mathrm{Tr}\hat{\rho}^2 \leq 1$: We have $\mathrm{Tr}\hat{\rho}^2 = \mathrm{Tr}\sum_i \sum_j p_i p_j |\langle\psi_j|\psi_i\rangle|^2 \leq \sum_i p_i \langle\psi_i|\psi_i\rangle \sum_j p_j \langle\psi_j|\psi_j\rangle = 1$. We used the Schwartz inequality $|\langle\psi_i|\psi_j\rangle|^2 \leq \langle\psi_i|\psi_i\rangle\langle\psi_j|\psi_j\rangle$ for the inequality step.

The last property is a test to see whether we have a pure state or a mixed state. The quantity $\mathrm{Tr}\hat{\rho}^2$ is called the *purity*. When it is 1, we have a pure state. When it is less than 1, we have a mixed state. In Problem 22.1.2, we will show it is always larger than $\frac{1}{d}$, where d is the dimensionality of the Hilbert space for the quantum system.

We now discuss a simple example of a density matrix corresponding to the description of a random source of Stern-Gerlach atoms. The cleanest way to describe this is to create a random probability for the atom to have a spin projection of σ in the θ and ϕ directions. So

$$\hat{\rho}_{\text{oven}} = \sum_\theta \sum_\phi \sum_\sigma P(\theta, \phi, \sigma) |\sigma\rangle_{\theta,\phi\ \theta,\phi}\langle\sigma|, \tag{22.4}$$

where we must pick the probabilities, so that we have a completely random result. This means the probabilities should have no bias to them. To do this on a sphere, we pick uniformly with respect to $\cos\theta$ and ϕ and uniformly in σ. Picking a distribution uniformly in $\cos\theta$ may seem like an odd thing to do but recall if we look at the sphere, the relative area is largest at the equator and is smallest at the poles. Recall $|\uparrow\rangle_{\theta,\phi} = \cos\frac{\theta}{2}|\uparrow\rangle_z + e^{i\phi}\sin\frac{\theta}{2}|\downarrow\rangle_z$. We will compute the density matrix for $\sigma = \uparrow$ only and complete the calculation in Problem 22.1.3. We have

$$\hat{\rho}_{\text{oven}} = \sum_{\cos\theta}\sum_{\phi}\Big(\cos^2\tfrac{\theta}{2}|\uparrow\rangle_{z\,z}\langle\uparrow| + e^{i\phi}\cos\tfrac{\theta}{2}\sin\tfrac{\theta}{2}|\uparrow\rangle_{z\,z}\langle\downarrow| + e^{-i\phi}\cos\tfrac{\theta}{2}\sin\tfrac{\theta}{2}|\downarrow\rangle_{z\,z}\langle\uparrow| + \sin^2\tfrac{\theta}{2}|\downarrow\rangle_{z\,z}\langle\downarrow|\Big) \tag{22.5}$$

We have done these types of sums before. Recall that

$$\sum_{j=0}^{N-1} e^{i\frac{2\pi}{N}j} = \frac{1-e^{i2\pi}}{1-e^{i\frac{2\pi}{N}}} = 0, \tag{22.6}$$

and similarly for the case with $e^{-i\phi}$. So the cross terms both vanish. For the other terms, we note that $\cos^2\frac{\theta}{2} = \frac{1}{2}(1+\cos\theta)$ and $\sin^2\frac{\theta}{2} = \frac{1}{2}(1-\cos\theta)$, so the summations we need to evaluate involve $1 \pm \cos\theta$, and we sample uniformly in $-1 \leq \cos\theta \leq 1$. The constant term will sum to 1, because the distribution is uniform, while the cos term sums to zero, because it is odd. Hence, we find that

$$\hat{\rho}_{\text{oven}} = \tfrac{1}{2}|\uparrow\rangle_{z\,z}\langle\uparrow| + \tfrac{1}{2}|\downarrow\rangle_{z\,z}\langle\downarrow|. \tag{22.7}$$

Note that we obtain the maximally mixed state even when we only sum over the $\sigma = \uparrow$ cases. Can you explain why this is the case?

One of the most notable results of the previous calculation was that while the scheme for averaging over all angles uniformly and both projections of the spins does indeed produce the right final density matrix, we also saw that when we did the sum over all directions and just one projection of the spin, it also yielded the same result. So, summing over both cases was redundant. How much redundancy do we have? Let's focus on the maximally mixed state and ask how many pure states do I need to sum over to create the density matrix of the maximally mixed state? Since this density matrix is given by $\frac{1}{d}\mathbb{I}$ in any basis, we immediately see that by using the resolution of unity, where $\hat{\mathbb{I}} = \sum_{n=1}^{d}|n\rangle\langle n|$, we have that the minimal number of states needed to represent this is the d orthonormal basis states, which are summed over with equal weight. It is clear this cannot be done with any fewer. But of course, the density matrix of a pure state can be represented by just one term in the weighted sum of projection operators. Hence, there is no simple way to know how to fully remove all redundancies, but it is clear the total number of needed pure states that we must sum over is less than or equal to d. A more complete analysis can be done of this, but we will not go into any further details here.

Using the above argument, we find that for the oven source of the Stern-Gerlach experiment, we can determine the maximally mixed state with just a sum over two states. In fact, it is given by $\frac{1}{2}(|\uparrow\rangle\langle\uparrow| + |\downarrow\rangle\langle\downarrow|)$ with the quantization axis being in any direction. So, the averaging over all angles is not really needed. This is somewhat mind-boggling, but it comes from the fact that the Stern-Gerlach experiment has only two results—up or down projections—and in

quantum mechanics, this means we only need two terms to represent the density matrix.

When we work with the density matrix in a specific basis $\rho_{mn} = \langle m|\hat{\rho}|n\rangle$, we call the diagonal elements the *populations*, while the off-diagonal elements are called the *coherences*. How do these two objects evolve in time? If our density matrix is represented as $\hat{\rho} = \sum_i p_i |\psi_i\rangle \langle\psi_i|$, then the time-dependent density matrix should be given by the time-evolved states, or $\hat{\rho}(t) = \sum_i p_i |\psi_i(t)\rangle \langle\psi_i(t)|$. This means

$$\hat{\rho}(t) = \hat{U}(t) \left(\sum_i p_i |\psi_i\rangle \langle\psi_i| \right) \hat{U}^\dagger(t) = \hat{U}(t)\hat{\rho}\hat{U}^\dagger(t). \tag{22.8}$$

Now, back to the populations. Consider the density matrix evaluated in the energy eigenbasis of the time-independent Hamiltonian for the given problem. Then the density matrix elements satisfy

$$\rho_{mn}(t) = \sum_i p_i \langle m|\hat{U}(t)|\psi_i\rangle \langle\psi_i|\hat{U}^\dagger(t)|n\rangle = e^{-i\omega_{mn}t}\rho_{mn}(0). \tag{22.9}$$

Here, we have $\omega_{mn} = (E_m - E_n)/\hbar$. Note that one has to be careful to get the sign of the exponent correct, because the operator acting on the bra state is the Hermitian conjugate of what we have in time evolution of the energy eigenstate ket. This means the populations are time independent in this basis, while the coherences oscillate at the difference in the frequencies associated with each state in the basis.

There is a special representation for the density matrix of a spin-one-half degree of freedom that is called the *Bloch representation*. It is constructed by simply taking the density matrix and reexpressing it as a linear combination of the identity matrix and the three Pauli spin matrices. It takes the form

$$\rho^{(2)}_{mn} = \begin{pmatrix} \rho_{\uparrow\uparrow} & \rho_{\uparrow\downarrow} \\ \rho_{\downarrow\uparrow} & \rho_{\downarrow\downarrow} \end{pmatrix} = \tfrac{1}{2}\mathbb{I}_2 + \tfrac{1}{2}\vec{v}\cdot\vec{\sigma}. \tag{22.10}$$

The vector $\vec{v}$ is called the *Bloch vector*. Writing the quantum state in terms of the Bloch vector, we find

$$\rho^{(2)}_{mn} = \frac{1}{2}\begin{pmatrix} 1+v_z & v_x - iv_y \\ v_x + iv_y & 1-v_z \end{pmatrix}. \tag{22.11}$$

Taking the determinant, we find $\det\rho^{(2)} = \frac{1}{4}(1 - |\vec{v}|^2)$. So we can determine the magnitude of the Bloch vector by calculating the determinant of the density matrix. But there is more. Because the density matrix is positive semidefinite, the determinant, which is the product of the two real eigenvalues of the matrix, must be nonnegative, so $|\vec{v}|^2 \leq 1$. When we have a pure state, we know the trace of $\hat{\rho}^2$ is equal to 1. Computing the square of the density matrix in the Bloch form shows its

trace is $\frac{1}{2}(1+|\vec{v}|^2)$. Hence $|\vec{v}|=1$ for a pure state. Then we see the determinant vanishes for a pure state as well.

In many cases, it is preferable to work with density matrices when describing composite systems, especially when one of the components will not be explicitly measured. The challenge is that the notation gets cumbersome. The way we construct the density matrix follows simply the way we represent the quantum state. Suppose we have two Hamiltonians $\hat{H}_A$ and $\hat{H}_B$, which need not be the same, or even similar at all, and we work on a tensor product space where the tensor-product Hamiltonian is $\hat{H}_A \otimes \hat{H}_B$. Let $|\psi_i\rangle_A$ be a basis for the A problem and $|\phi_j\rangle_B$ be the basis for the B problem. Then any state in the composite system can be written as

$$|\psi\rangle = \sum_{i=1}^{d_A}\sum_{j=1}^{d_B} c_{ij}|\psi_i\rangle_A \otimes |\phi_j\rangle_B, \tag{22.12}$$

where the Hilbert-space dimension of $\hat{H}_A$ is d_A and of $\hat{H}_B$ is d_B; note we have seen this form before when we discussed the Hubbard model. Now to form the density operator for the pure state, we have

$$\begin{aligned}\hat{\rho} &= \sum_{i=1}^{d_A}\sum_{i'=1}^{d_A}\sum_{j=1}^{d_B}\sum_{j'=1}^{d_B} c_{ij}c^*_{i'j'}|\psi_i\rangle_A \otimes |\phi_j\rangle_B\, {}_A\langle\psi_{i'}| \otimes {}_B\langle\phi_{j'}| \\ &= \sum_{i=1}^{d_A}\sum_{i'=1}^{d_A}\sum_{j=1}^{d_B}\sum_{j'=1}^{d_B} c_{ij}c^*_{i'j'}|\psi_i\rangle_A\, {}_A\langle\psi_{i'}| \otimes |\phi_j\rangle_B\, {}_B\langle\phi_{j'}|,\end{aligned} \tag{22.13}$$

where in the second line we write the density operator as a sum of tensor-producted operators. While the idea here is simple, you can see it becomes cumbersome. If we compute the density matrix in this basis, it is given by the product of the two coefficients, but you see it has twice the number of indices. This is because, as a matrix, it also has a tensor-product structure to it. Now, to form the general density matrix, we label each state with the index α and weight them with the probability p_α, so that

$$\rho_{ij,i'j'} = \sum_{\alpha=1}^{N} p_\alpha c_{ij}^{(\alpha)}\big(c_{i'j'}^{(\alpha)}\big)^* \tag{22.14}$$

for a mixed state with N components. Everything is straightforward as long as you keep track of the indices correctly.

Let us describe a situation where we would want to use this formalism. Consider the ion-trap system. We saw previously, that as we evolve the system, we create an entangled state between the spins and the phonons. But, if we do not measure the phonons, the final results will ultimately take place only within the spin space. While

it seems to be a perfectly fine exercise to just treat the whole system and evaluate as we always do, there is a benefit to thinking of it as a system (the spins) entangled with a reservoir (the phonons). Then, it is very convenient to map all of the work to the system only, especially since the environment can contain very many degrees of freedom, and we might not even know how to properly describe them. So, when we calculate an expectation value via $\langle\psi|\hat{O}|\psi\rangle = \mathrm{Tr}(\hat{O}\hat{\rho})$, we sum over the indices of each subsystem to perform the trace. We can reexpress this as

$$\sum_{i=1}^{d_A}\sum_{j=1}^{d_B} \rho_{ij,ij}\, {}_A\langle\psi_i| \otimes {}_B\langle\phi_j|\hat{O}|\psi_i\rangle_A \otimes |\phi_j\rangle_B, \tag{22.15}$$

and then perform the sum over the B system if the operator $\hat{O}$ acts only on the A subsystem. We rewrite this as $\langle\psi|\hat{O}_A|\psi\rangle = \mathrm{Tr}_A(\hat{O}_A\hat{\rho}_A^{\mathrm{red}})$, with the *reduced density matrix* defined in terms of the partial trace—$\hat{\rho}_A^{\mathrm{red}} = \mathrm{Tr}_B\hat{\rho}$—or

$$\hat{\rho}^{\mathrm{red}}_{Aii'} = \sum_{j=1}^{d_B} \rho_{ij,i'j}. \tag{22.16}$$

One can see that there is a loss of information when we perform a partial trace, and, in general, if the quantum state is an entangled pure state, then the partial trace will create a mixed state, as we illustrate with a simple example next.

Consider a system prepared in a spin singlet state via $|\psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle - |\downarrow\rangle \otimes |\uparrow\rangle)$. Its density matrix is given by

$$\rho_{mn} = \left(\begin{array}{cc|cc} \frac{1}{2} & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \\ -\frac{1}{2} & 0 & 0 & \frac{1}{2} \end{array}\right), \tag{22.17}$$

where we used dashes to illustrate the 2×2 structure of 2×2 blocks. You should be able to verify that $\mathrm{Tr}\hat{\rho} = 1$ and $\mathrm{Tr}\hat{\rho}^2 = 1$. To perform the partial trace, we construct a matrix where we perform the trace of each of the four 2×2 blocks. This gives

$$\rho^{\mathrm{red}}_{mn} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix}. \tag{22.18}$$

You should recognize this is the density matrix of the maximally mixed state. What this says is that if we trace out one degree of freedom of an entangled state, we find the remaining degree of freedom becomes a mixed state. Physically, you know this.

If I have a singlet state and I measure only one spin, then I will find half of the time it is in the up state and half of the time in the down state.

There is a procedure called *purification* of a density matrix. It says, for any density matrix, whether a pure state or a mixed state, it can be written as the partial trace of the density matrix of a pure state of a composite system. We will illustrate this for a simple representation of the density matrix. While we do not prove it, this simple representation is a form that every density matrix can be represented in. Suppose $\hat{\rho} = \sum_{i=1}^{d_A} p_i |\psi_i\rangle \langle\psi_i|$. Then we form a new density matrix of a composite system via

$$\hat{\rho}^{\text{pure}} = \sum_{i=1}^{d_A} \sum_{j=1}^{d_A} \sqrt{p_i p_j} |\psi_i\rangle_A \otimes |\psi_i\rangle_B \ {}_A\langle\psi_j| \otimes_B \langle\psi_j|. \tag{22.19}$$

This density matrix is formed in a doubled tensor-product space, from the pure state

$$|\psi\rangle = \sum_{i=1}^{d_A} \sqrt{p_i} |\psi_i\rangle_A \otimes |\psi_i\rangle_B. \tag{22.20}$$

You should be able to verify that $\text{Tr}_B(\hat{\rho}^{\text{pure}}) = \hat{\rho}$.

The last topic we will discuss is the topic of *tomography*. It is a way of determining the quantum state by measurement. While we have said before the quantum state is not directly measurable, we can construct a sequence of measurements, from which we can learn precisely what the quantum state is. We know that if we measure the probability of a quantum state to be in each of the orthonormal basis vectors of the Hilbert space, then we obtain the probability of each basis vector (and from the square root, the amplitude), while interference experiments can tell us about the relative phases between the different terms in the expansion. We illustrate how this works for a single qubit state represented by the density matrix $\hat{\rho} = \frac{1}{2}(\mathbb{I} + \vec{r} \cdot \vec{\sigma})$. This is the most general quantum state for spin one-half. Now, using the formula for the average value of an operator that is being measured, we immediately see that $r_i = \text{Tr}(\sigma_i \hat{\rho})$. So we need to measure each Pauli matrix. But how do we really do that? For σ_z, it is easy. If we prepare the state and measure what eigenstate it is in the z-basis, we would find it is given by the populations of the up state and of the down state. To determine the measurement of σ_z, we then simply take the difference of the up minus the down populations. You can see this immediately gives r_z. What about the other directions? For the x direction, we must rotate the x-direction to the z-direction and then measure σ_z. Similar for the y-direction. The details for how to do this with the density matrix are worked out in Problem 22.1.4.

Section 22.1 Problems:

22.1.1 Phase Ambiguity

For a conventional pure quantum state, written as a state vector $|\psi\rangle$, we have a phase ambiguity—we can let $|\psi\rangle \rightarrow e^{i\alpha}|\psi\rangle$, for any real α, and no properties of the quantum state are changed. Show that when we write the quantum state as a density matrix that it has no phase ambiguity when in a pure state or a mixed state. Hence, the quantum state, written in a density matrix form, has no complex phase freedom to it.

22.1.2 Maximally Mixed State

The maximally mixed state is the state with the lowest possible purity. When we consider a density matrix, the restriction of the density matrix with vanishing off-diagonal matrix elements always has a lower purity than a density matrix that has the same diagonal elements and nonzero off-diagonals. Prove that this is true by showing that setting the off diagonal elements to zero preserves all of the properties of a density matrix.

Then, to find the smallest purity, we assume the diagonal elements are of the form p_i, with $\sum_{i=1}^{d} p_i = 1$. We show that $\sum_{i=1}^{d} p_i^2$ is minimized by choosing $p_i = \frac{1}{d}$. The proof comes from the Schwarz inequality. Consider two d-dimensional vectors: $\vec{x} = \frac{1}{d}(1, 1, \cdots, 1)$ and $\vec{y} = (p_1, p_2, \cdots, p_d)$. Massage $\vec{x} \cdot \vec{y} \leq |\vec{x}|\,|\vec{y}|$ to prove that

$$\frac{1}{d} \leq \sum_{i=1}^{d} p_i^2. \tag{22.21}$$

Hint: You will need to use the fact that $\sum_{i=1}^{d} p_i = 1$.

22.1.3 Stern-Gerlach Oven Source

Show that the density matrix contributions from $\sigma = \downarrow$ give the same result as what we found in the main text for $\sigma = \uparrow$.

22.1.4 Details of Tomography

We work out the details of the tomography measurement. Recall that $\sigma_x = (i\sigma_y)\sigma_z(-i\sigma_y)$. Use this fact with the invariance of the trace upon a cyclic rearrangement of matrix factors to show that one can reexpress the measurement of σ_x in the state $\hat{\rho}$ as the measurement of σ_z in a state that has been rotated by $\pi/2$ about the y-axis. Recall the generalized Euler identity and how states are rotated, when determining how a density matrix is rotated. Repeat for the measurement of σ_y.

In this fashion, you see, we only need to know how to measure in the z-direction and how to transform states before measurement, in order to measure all properties of a qubit. This very strategy is used in all forms of gated quantum computing.

22.2 Thermal Distributions

The most important mixed state is the thermal state, which represents what a quantum system looks like at a fixed temperature. It is most concretely represented by working in the energy eigenbasis and using probabilities given by Maxwell-Boltzmann factors as

$$\hat{\rho}_{\text{th}} = \sum_n \frac{e^{-\beta E_n}}{Z} |n\rangle \langle n| = \frac{e^{-\beta \hat{H}}}{Z}, \tag{22.22}$$

where $\beta = 1/k_B T$ is the inverse temperature in units of energy (k_B is the Boltzmann constant which converts temperature to energy). The factor Z is the partition function given by $Z = \text{Tr}\exp(-\beta \hat{H}) = \sum_n e^{-\beta E_n}$.

One might ask, why do we choose the probabilities as negative exponentials of the ratios of the energy to the temperature? If we look at the properties of this, the choice seems reasonable—if T is low, only low-energy states have a high probability, whereas if T is high, all states have similar probabilities. The technical reason is that one can prove that this choice maximizes the entropy for a fixed average energy, although we will not show the details of that here, as the proof requires an optimization using calculus. This particular mixed state is so useful, because most situations where we wish to perform a calculation correspond to working with systems that are in equilibrium. So, this is the state we would use for such work.

We will discuss the simplest thermal state example of a spin-one-half degree of freedom in a magnetic field. The Hamiltonian is $\hat{H} = -g\mu_B B \hat{S}_z/\hbar$, with energy levels of $\mp g\mu_B B = \mp\hbar\Omega$ in terms of the Larmor frequency. Now, the first step is to determine the partition function. It is given by

$$Z = \sum_{\pm} e^{-\beta\hbar\Omega} = 2\cosh\beta\hbar\Omega. \tag{22.23}$$

The density matrix is

$$\hat{\rho} = \frac{1}{2\cosh\beta\hbar\Omega}\left(e^{\beta\hbar\Omega}|\uparrow\rangle\langle\uparrow| + e^{-\beta\hbar\Omega}|\downarrow\rangle\langle\downarrow|\right), \tag{22.24}$$

in its operator form. Finally, we can examine things like the average magnetization, given by $m(T) = \frac{\hbar}{2}\mathrm{Tr}(\sigma_z\hat{\rho}) = \tanh\beta\hbar\Omega$. Since $\beta \to \infty$ at low temperature and $\beta \to 0$ at high temperature, we can see the magnetization starts with a polarized state in the $\uparrow$ direction at low temperature, to a state that is equally likely up or down at high temperature, with a smooth crossover in between.

Let us do a similar analysis for the harmonic oscillator, with $\hat{H} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)$. We find that

$$Z = \sum_{n=0}^{\infty} e^{-\beta\hbar\omega\left(n+\frac{1}{2}\right)} = \frac{e^{-\frac{1}{2}\beta\hbar\omega}}{1 - e^{-\beta\hbar\omega}} = \frac{1}{2\sinh\frac{1}{2}\beta\hbar\omega}, \tag{22.25}$$

which is obtained by using the geometric series (because the exponential factor is always less than 1). The density matrix then becomes

$$\hat{\rho}_{\mathrm{th}} = \left(1 - e^{-\beta\hbar\omega}\right)\sum_{n=0}^{\infty} e^{-\beta\hbar\omega n}|n\rangle\langle n|. \tag{22.26}$$

With the density matrix, we can calculate different thermal averages. For example, the average density of excitations of the harmonic oscillator is given by

$$\langle\hat{n}\rangle_{\mathrm{th}} = \left(1 - e^{-\beta\hbar\omega}\right)\sum_{n=0}^{\infty} e^{-\beta\hbar\omega n}\langle n|\hat{n}|n\rangle = \left(1 - e^{-\beta\hbar\omega}\right)\sum_{n=0}^{\infty} e^{-\beta\hbar\omega n} n. \tag{22.27}$$

Recalling that $\sum_{n=0}^{\infty} n x^n = \frac{x}{(1-x)^2}$, we find that

$$\langle n\rangle_{\mathrm{th}} = \frac{1}{e^{\beta\hbar\omega} - 1}, \tag{22.28}$$

which is called the *Bose-Einstein distribution function*. Note that this is exactly the functional form that Planck used when he solved the black-body radiation problem. As $T \to 0$ ($\beta \to \infty$), we find the average density of excitations goes to zero, since the system must be in the ground state. While, as $T \to \infty$ ($\beta \to 0$), we have that average density of excitations diverges!

Section 22.2 Problems:

22.2.1 Determine the Bloch Form for the Spin Thermal Density Matrix

Using a basis in the z-direction, determine the Bloch form of the spin thermal-density matrix. Repeat this for a basis in the x-direction.

22.2.2 Thermal State of the Two-Site Heisenberg Spin Model

Consider a two-site Heisenberg model given by $\hat{H} = -\frac{J}{\hbar}\hat{\vec{S}}_1 \cdot \hat{\vec{S}}_2$. Determine the four energy levels, the partition function, the density matrix, and the average energy as a function of the temperature. Consider both signs of the exchange integral J. Discuss the behavior of this system.

22.2.3 Average Energy of the Thermal State of the Harmonic Oscillator

Compute the average energy of the simple harmonic oscillator in the thermal state. Plot this as a function of temperature over a range from $0 \leq k_B T \leq 5\hbar\omega$. You should find at high temperature that the average energy satisfies the equipartition theorem, where $\langle \hat{H} \rangle_{\text{th}} \approx k_B T$.

These results can be used in the solution of the blackbody radiation problem.

22.3 Quantum Channels

In this section, we discuss the time evolution of quantum states described by density matrices. If our quantum state is a weighted average of pure states, such as $\hat{\rho} = \sum_{i=1}^{N} p_i |\psi_i\rangle \langle\psi_i|$, then the density matrix clearly evolves via $\hat{\rho}(t) = \hat{U}(t)\hat{\rho}(0)\hat{U}^{\dagger}(t)$. But is this the only way a quantum state can evolve in time? The answer is surprisingly no, because the only critical condition for the time evolution is that we map the quantum state to a quantum state. This requires the map to be linear, preserve the trace of the state, preserve the Hermiticity of the state, and preserve the positivity of the state. It turns out the last point is a bit subtle, because we need to maintain positivity for all partial traces of composite systems as well, which is a

requirement called *complete positivity*. We will not delve into the technical details of the difference here in this brief treatment, but it is important for you to know the correct requirements should you pursue this line yourself later. The general name for these maps between quantum states is called a *quantum channel* from its relationship to classical communication channels. A map that satisfies these properties is called a *completely positive trace preserving map* or CPTP map. The work we develop in this section is influenced by Manenti and Motta's wonderful book on quantum information (Manenti & Motta, 2023).

If the map that evolves the quantum state forward in time is unitary, it preserves the information content of the quantum state; here we are equating purity with information, but there are better measures of information content based on entropy. Such maps are invertible by simply applying the inverse unitary transformation. But quantum states can be mapped in a fashion that is not invertible. In this case, one typically loses information as we go forward in time, and this loss of information is often chosen to be synonymous with the phenomenon of decoherence. Our goal in this section is to illustrate different types of CPTP maps concretely on spin one-half systems (which can also be thought of as maps acting on a single qubit).

We start with the so-called *depolarizing channel*. This is a quantum channel that loses information as a function of time, eventually mapping the system onto the maximally mixed state, which has no information content at all. The map is simply to write down:

$$\hat{\rho}_{\text{new}} = (1-\gamma)\hat{\rho}_{\text{old}} + \gamma \tfrac{1}{2}\mathbb{I}_2. \tag{22.29}$$

What it does is with probability $1-\gamma$, the state stays the same, while with probability γ, the state becomes the maximally mixed state. Suppose we think of this map as a single step of a time evolution, so that

$$\hat{\rho}(t+\Delta t) = (1-\gamma)\hat{\rho}(t) + \gamma \tfrac{1}{2}\mathbb{I}_2. \tag{22.30}$$

Then

$$\hat{\rho}(t+2\Delta t) = (1-\gamma)^2\hat{\rho}(t) + (2\gamma-\gamma^2)\tfrac{1}{2}\mathbb{I}_2. \tag{22.31}$$

Repeating N times then gives us

$$\hat{\rho}(t+N\Delta t) = (1-\gamma)^N\hat{\rho}(t) + [1-(1-\gamma)^N]\tfrac{1}{2}\mathbb{I}_2. \tag{22.32}$$

Now, if we let $t = N\Delta t$ and we write $\gamma = \Gamma\Delta t$, we find, in the limit where $\Delta t \to 0$, that the map, as a function of time, becomes

$$\hat{\rho}(t) = e^{-\Gamma t}\hat{\rho}(0) + \left(1-e^{-\Gamma t}\right)\tfrac{1}{2}\mathbb{I}_2. \tag{22.33}$$

In this form, we clearly see that the off-diagonal terms in the quantum state decrease to zero exponentially fast, while the diagonal terms approach the maximally mixed state values exponentially fast as well. It is also clear that no matter what state we start from, we always end in the maximally mixed state, which again shows us that this depolarizing channel loses information as a function of time, because the maximally mixed state has no information, being completely random with respect to which state we are in (and having the lowest possible purity). Finally, due to this last fact, the maximally mixed state is mapped to the maximally mixed state by this map. This means the map is *unital*. Unital maps can be shown to always lose information—the purity decreases with each step forward in time. Note that with any map that loses information, it is not surprising that it can be represented in multiple ways. Some examples of other ways to represent the depolarizing channel are given in the problems. These depolarizing errors in a quantum computer are often the most difficult to remove via error correction.

One general form for representing a quantum channel is the so-called *Kraus map*. It represents the quantum channel in the general form

$$\hat{\rho} \to \sum_{i=1}^{N} \hat{K}_i \, \hat{\rho} \hat{K}_i^\dagger, \tag{22.34}$$

which is the sum of a sequence of maps that are guaranteed to map a quantum state to a quantum state when they satisfy a specific normalization condition that we will discuss more thoroughly in the next section. We represent the amplitude damping channel in this form, using

$$\begin{aligned} \hat{K}_1 &= \begin{pmatrix} 0 & \sqrt{\Gamma \Delta t} \\ 0 & 0 \end{pmatrix} = \tfrac{1}{2}(\sigma_x + i\sigma_y)\sqrt{\Gamma \Delta t} \text{ and} \\ \hat{K}_2 &= \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\Gamma \Delta t} \end{pmatrix} = \tfrac{1}{2}(1+\sqrt{1-\Gamma \Delta t})\mathbb{I}_2 + \tfrac{1}{2}(1-\sqrt{1-\Gamma \Delta t})\sigma_z, \end{aligned} \tag{22.35}$$

for a single time step of size Δt. Let us look at the first term. One finds that

$$\hat{K}_1 \sigma_x \hat{K}_1^\dagger = \tfrac{\Gamma \Delta t}{4}(\sigma_x + i\sigma_y)\sigma_x(\sigma_x - i\sigma_y) = 0, \tag{22.36}$$

and the same result for σ_y, while $\hat{K}_1 \sigma_z \hat{K}_1^\dagger = -\frac{\Gamma \Delta t}{2}(\mathbb{I}_2 + \sigma_z)$. We also have $\hat{K}_1 \mathbb{I}_2 \hat{K}_1^\dagger = \frac{\Gamma \Delta t}{2}(\mathbb{I}_2 + \sigma_z)$. Similarly, we find that

$$\begin{aligned} \hat{K}_2 \sigma_x \hat{K}_2^\dagger = \sqrt{1-\Gamma \Delta t}\sigma_x, \quad \hat{K}_2 \sigma_y \hat{K}_2^\dagger = \sqrt{1-\Gamma \Delta t}\sigma_y, \quad \text{and} \\ \hat{K}_2 \sigma_z \hat{K}_2^\dagger = \tfrac{\Gamma \Delta t}{2}\mathbb{I}_2 + \big(1 - \tfrac{\Gamma \Delta t}{2}\big)\sigma_z. \end{aligned} \tag{22.37}$$

We also have $\hat{K}_2 \mathbb{I}_2 \hat{K}_2^\dagger = \left(1 - \frac{\Gamma \Delta t}{2}\right)\mathbb{I}_2 + \frac{\Gamma \Delta t}{2}\sigma_z$. Combining these results together, we see that the Bloch form for the quantum state evolves as

$$\vec{r}(t + \Delta t) = \vec{r}(t) + \left(\left(\sqrt{1 - \Gamma \Delta t} - 1\right)r_x, \left(\sqrt{1 - \Gamma \Delta t} - 1\right)r_y, \Gamma \Delta t (1 - r_z)\right). \tag{22.38}$$

After N steps, we have

$$\vec{r}(t + N\Delta t) = \left((1 - \Gamma \Delta t)^{\frac{N}{2}} r_x, (1 - \Gamma \Delta t)^{\frac{N}{2}} r_y, r_z + \left(1 - (1 - \Gamma \Delta t)^N\right)(1 - r_z)\right). \tag{22.39}$$

Taking the limit of $\Delta t \to 0$ with $N \Delta t = t$, we obtain

$$\vec{r}(t) = \left(e^{-\frac{\Gamma t}{2}} r_x(0), e^{-\frac{\Gamma t}{2}} r_y(0), r_z(0) + (1 - e^{-\Gamma t})\left(1 - r_z(0)\right)\right). \tag{22.40}$$

Be sure you can go through these steps and verify this result. You can see that as $t \to \infty$, we find $\vec{r}(t) \to (0, 0, 1)$, which means the system goes to the ground state, a pure state! One can check that this map is not unital and it can increase the purity of a system. In fact, this map maps any initial state to the ground state. We can think of it as a model for describing how spontaneous emission eventually sends any excited system to the ground state (if the ground state is coupled to all other excited states). The term Γ is the relaxation rate toward the ground state. Its inverse $T_1 = \frac{1}{\Gamma}$ is called the relaxation time. This type of relaxation typically occurs in superconducting-based quantum computers and is a cause of error as the excited states decay to the ground state. In general, we want this relaxation time as long as possible, so simply waiting for the system to go to the ground state is not a practical state initialization protocol. There are more rapid ways to perform a reset of a qubit—the simplest being to measure the qubit. If the state is up, do nothing, if down, hit is with a σ_x to transform it to up. If in doubt, repeat the process. Such a strategy is called an *active reset*.

The last channel we consider is the *dephasing channel*. This relaxation occurs when the Larmor frequency varies randomly as a function of time, typically arising from a fluctuating magnetic field. This yields the following map:

$$\hat{\rho}(t + \Delta t) = \hat{\rho}(t) + \Gamma \Delta t \left(\sigma_z \hat{\rho}(t) \sigma_z - \hat{\rho}(t)\right). \tag{22.41}$$

Note that the diagonal components of the density matrix are unchanged by this map, so we have

$$\vec{r}(t + \Delta t) = \vec{r}(t) - 2\Gamma \Delta t (r_x(t), r_y(t), 0). \tag{22.42}$$

Or

$$\vec{r}(t + N\Delta t) = \left((1 - 2\Gamma \Delta t)^N r_x(t), (1 - 2\Gamma \Delta t)^N r_y(t), r_z(t)\right). \tag{22.43}$$

This can be rewritten as

$$\vec{r}(t) = \left(e^{-2\Gamma t} r_x(0), e^{-2\Gamma t} r_y(0), r_z(0)\right). \tag{22.44}$$

We can see that the off-diagonal pieces relax to zero, while the diagonal pieces remain constant on time. The time $t_\phi = 1/2\Gamma$ is called the *dephasing time*.

One interesting fact about these last two examples is that they operate independent of each other. If we apply an amplitude damping channel with parameter Γ and a dephasing channel with amplitude Γ', then you should be able to establish that these two relaxation mechanics commute with each other so that the combined map is given by

$$\vec{r}(t) = \left(e^{-(\Gamma+2\Gamma')t} r_x(0), e^{-(\Gamma+2\Gamma')t} r_y(0), r_z(0) + (1 - e^{-\Gamma t})\big(1 - r_z(0)\big)\right). \tag{22.45}$$

In this case, the net decoherence time is determined by a combination of the dephasing and T_1 relaxation times via $\frac{1}{\Gamma} + \frac{1}{2\Gamma'}$.

The final topic we discuss here is that of a *system plus reservoir*. Many systems that dissipate energy, or are in thermal equilibrium, do so by having a small system in contact with a large reservoir. We also have some intrinsic problems we have studied of this form, such as the ion-trap problem, where we can think of the spins as the system and the phonons as the reservoir. If we simulate the entire system plus reservoir, it evolves with unitary time evolution, because it is a pure state that evolves under the rules of quantum mechanics governed by a Hamiltonian. What is more interesting is to restrict ourselves to the system only, by removing the degrees of freedom of the reservoir. We must do this for practical reasons if we do not know what the reservoir is—even if we do know, the size of the problem we must simulate is drastically smaller if we work with the system only instead of the system plus environment. This can be achieved by a partial tracing out of the reservoir degrees of freedom. Now, physically speaking, we can think of an excitation of the system as entering the reservoir at some time t, only to come back to the system again at a later time t', because it moves around in the reservoir for a time $t' - t$ before coming back. So, the net interactions within the system can become complicated, as we saw when we removed the effects of the phonons from the ion-trap system earlier. It turned out that some special circumstances of that problem made the analysis simpler there than it is in the general case. But the basic idea remains similar to what we did before—working just with the system saves a huge amount of labor, because the Hilbert space of the system is substantially smaller than that of the system plus reservoir. This simplicity comes at a price—the time evolution of the system, after removing the reservoir degrees of freedom, causes the interaction of excitations within the system to be complicated—in the most general case they are described by quantum maps. We will consider the general form of this time evolution in the next section.

Section 22.3 Problems:

22.3.1 Equivalent Form of the Depolarizing Channel

In this problem, we will investigate another representation of the depolarizing channel. The map $\hat{\rho} \rightarrow \sigma_x \hat{\rho} \sigma_x$ is called a bit-flip operation. The map $\hat{\rho} \rightarrow \sigma_z \hat{\rho} \sigma_z$ is a phase flip operation. And the map $\hat{\rho} \rightarrow \sigma_y \hat{\rho} \sigma_y$ is both a phase and bit flip operation. Consider the map that includes all three with equal probability:

$$\hat{\rho} \rightarrow (1-\beta)\hat{\rho} + \frac{\beta}{3}(\sigma_x \hat{\rho} \sigma_x + \sigma_y \hat{\rho} \sigma_y + \sigma_z \hat{\rho} \sigma_z). \tag{22.46}$$

Show that this is a depolarizing channel by finding the relationship between β and γ.

Hint: the identity

$$\sigma_x \hat{\rho} \sigma_x + \sigma_y \hat{\rho} \sigma_y + \sigma_z \hat{\rho} \sigma_z = 2\mathbb{I}_2 - \hat{\rho}, \tag{22.47}$$

which you should prove, will come in handy in your analysis.

22.3.2 Random Rotations Are Equivalent to a Depolarizing Channel

Recall that a rotation by an angle α about an axis $\vec{e}_n = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$ is given by a transformation $\sigma_i \rightarrow R\sigma_i R^\dagger$, with $R = \cos\alpha \mathbb{I}_2 + i \sin\alpha \vec{e}_n \cdot \vec{\sigma}$. The idea of the random rotation is that we look at the transformation of the quantum state via a series of random rotations, with α distributed uniformly from 0 to 2π, $\cos\theta$ uniformly distributed from -1 to 1, and ϕ uniformly distributed from 0 to 2π.

First, verify the following identities:

$$\sum_{\cos\theta}\sum_{\phi} e_{ni} = \sum_{\cos\theta}\sum_{\phi} \begin{cases} \sin\theta\cos\phi \\ \sin\theta\sin\phi \\ \cos\theta \end{cases} = 0 \tag{22.48}$$

and

$$\sum_{\cos\theta}\sum_{\phi} e_{ni} e_{nj} = \frac{\delta_{ij}}{3}. \tag{22.49}$$

In both cases, the summations indicate the uniform distribution in the summand as we have used previously.

Hint: It is easiest to show it assumes the form of the previous problem rather than the original form of the depolarizing channel.

22.3.3 Example of an Amplitude Damping Channel That Increases Purity

Act the amplitude damping quantum channel onto the maximally mixed state, with $\vec{r}(0) = 0$. Compute the time dependence of the purity $\mathrm{Tr}(\hat{\rho}^2)$, and plot it versus Γt. Comment on the results.

22.3.4 Relaxation to a Thermal State at Temperature T

The amplitude damping example we looked at in the main text showed how to relax to the ground state. Are there similar amplitude damping channels that can relax to a thermal state? The answer is yes, and we illustrate it in this problem.

The map is of the form $\hat{\rho} \to \sum_{i=1}^{4} \hat{K}_i \, \hat{\rho} \hat{K}_i^\dagger$, with the following Kraus operators:

$$\hat{K}_1 = \frac{1}{\sqrt{1+e^{-\beta\hbar\Omega}}} \begin{pmatrix} 0 & \sqrt{\Gamma\Delta t} \\ 0 & 0 \end{pmatrix}, \quad \hat{K}_2 = \frac{1}{\sqrt{1+e^{-\beta\hbar\Omega}}} \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\Gamma\Delta t} \end{pmatrix}$$
$$\hat{K}_3 = \frac{1}{\sqrt{1+e^{\beta\hbar\Omega}}} \begin{pmatrix} 0 & 0 \\ \sqrt{\Gamma\Delta t} & 0 \end{pmatrix}, \quad \text{and} \quad \hat{K}_4 = \frac{1}{\sqrt{1+e^{\beta\hbar\Omega}}} \begin{pmatrix} \sqrt{1-\Gamma\Delta t} & 0 \\ 0 & 1 \end{pmatrix}. \tag{22.50}$$

Compute $\hat{\rho}(t)$ under the assumption that $N\Delta t = t$ in the limit $\Delta t \to 0$. Verify that the relaxation rate to the thermal state is determined by Γ, while the final thermal state itself is determined by the β in the prefactors of the matrices. The limit $T \to 0$ reduces to the result considered in the main text. Note that this is a long problem but straightforward to complete. Work carefully; it is easy to make a mistake.

22.3.5 Alternative Form for the Dephasing Channel

Show that the following map is the same as the dephasing map: $\hat{\rho} \to \hat{K}_1 \hat{\rho} \hat{K}_1^\dagger + \hat{K}_2 \hat{\rho} \hat{K}_2^\dagger$, with

$$\hat{K}_1 = \sqrt{1-\Gamma\Delta t}\,\mathbb{I}_2, \quad \text{and} \quad \hat{K}_2 = \sqrt{\Gamma\Delta t}\,\sigma_z. \tag{22.51}$$

22.3.6 Dephasing Channel as Random z-Rotations

Similar to the depolarizing channel, the dephasing channel can be shown to be the same as a random distribution of rotations about the z-axis. Perform the same analysis as in Problem 22.3.2, except limit the axis to only the z-axis. Show that this yields the dephasing channel.

22.3.7 Classical Reservoir

So far, we have described the system plus reservoir as a full quantum system, but we can instead consider a classical reservoir, which, in its simplest form, has the particle enter the reservoir and then immediately come back from it and just corresponds to taking the original Hamiltonian and adding in an arbitrary selection of time-dependent operators to it. These operators can do many things—they can add or remove energy from the system or cause states within the Hamiltonian to be coupled together. The key is that the Hilbert space of the system is not enlargened by this construction, as the classical reservoir has no quantum degrees of freedom. This means a classical reservoir never entangles the system with something outside of the system.

Write an essay describing the difference between a classical reservoir and a quantum reservoir.

22.3.8 Stinespring Dilation

The problem of working with a system plus reservoir and how it connects to a quantum channel is called the Stinespring dilation. Write a short essay that explains the pros and cons of working with the whole system plus reservoir, versus working with the system only, but using a more complicated quantum channel for the time evolution.

22.3.9 Meaning of Decoherence

We have seen in many cases when we do not have a unitary quantum channel that the system loses information, as can be tracked by seeing the purity of the system decrease in time. Write a short essay that explains the relationship between entanglement of the system with degrees of freedom outside of the system and this loss of information, or decoherence.

22.4 Kraus Operators and the Lindblad Equation

In this section, we discuss in more detail the general formulation of the time evolution of an open quantum system. The work in this section is motivated by Bertlmann and Friis' recent book (Bertlmann and Friis, 2023).

Suppose we have a composite system plus reservoir that starts in a system state $\hat{\rho}$, and the environment is in a pure state given by $|\phi_i\rangle\langle\phi_i|$, so that our initial density matrix is given by $\hat{\rho}\otimes|\phi_i\rangle\langle\phi_i|$. The time evolution is given by a unitary time-ordered product that we call $\hat{U}(t,t_0)$ that acts on both the system and reservoir. The time evolved density matrix is then given by $\hat{U}(t,t_o)\big(\hat{\rho}\otimes|\phi_i\rangle\langle\phi_i|\big)\hat{U}^\dagger(t,t_0)$. This is how we describe the evolution of the system plus reservoir together. But it can be much more efficient to describe the evolution of the system only. To do this, we must take a partial trace over the reservoir. This gives us

$$\hat{\rho}_{\text{sys}}(t)=\text{Tr}_{\text{res}}\left(\hat{U}(t,t_o)\big(\hat{\rho}_{\text{sys}}\otimes|\phi_i\rangle\langle\phi_i|\big)\hat{U}^\dagger(t,t_0)\right), \tag{22.52}$$

which is formally an exact equation. The challenge is we need to know the complicated dynamics of the two systems to have any chance to determine this result exactly. Fortunately, we are interested in the form of the time evolution for the system only, so we can still make progress.

Let $\{|\phi_n\rangle\}$ be the set of basis state vectors that describe the reservoir by itself. Then the trace over the reservoir can be easily done via

$$\hat{\rho}_{\text{sys}}(t)=\sum_{n=1}^{d_R}\langle\phi_n|\hat{U}(t,t_0)|\phi_i\rangle\hat{\rho}_{\text{sys}}\langle\phi_i|\hat{U}^\dagger(t,t_0)|\phi_n\rangle, \tag{22.53}$$

where the matrix elements $\langle\phi_n|\hat{U}(t,t_0)|\phi_i\rangle$ are *operators* defined in the system subspace only (because the operator part in the reservoir subspace has been removed by the matrix element). Note that it is easy to forget that this is an operator and not a number, so it requires diligence to be able to follow these types of derivations. Now, we define the *Kraus operator* via

$$\hat{K}_n(t,t_0)=\langle\phi_n|\hat{U}(t,t_0)|\phi_i\rangle, \tag{22.54}$$

which satisfy $\sum_{n=1}^{d_R}\hat{K}_n^\dagger(t,t_0)\hat{K}_n(t,t_0)=\mathbb{I}$. This last identity is the required normalization condition of Kraus operators that we mentioned previously. Now, using these Kraus operators, we find the quantum map takes the form

$$\hat{\rho}_{\text{sys}}(t)=\sum_{n=1}^{d_R}\hat{K}_n(t,t_0)\hat{\rho}_{\text{sys}}(t_0)\hat{K}_n^\dagger(t,t_0), \tag{22.55}$$

which is called the *Kraus form* of the map. One can see it is automatically Hermitian, and the normalization condition makes it trace preserving. While not also quite so obvious, it is also a completely positive map. Note we have specified no property of the $\hat{K}$ operators—they need not be Hermitian or unitary. If the reservoir is not initially in a pure state, we can define additional Kraus operators for all of the states in the expansion of the density matrix of the reservoir (which increases the total number from d_R to d_R^2). But we will not go into this complication further here.

While we have achieved a nice simplification, where we have the quantum map acting solely on the system, computing the individual Kraus operators appears to be a highly complicated procedure. While this can be done for some simple examples, in general, it is so complicated that it rapidly becomes intractable. Nevertheless, because we know the exact form of the Kraus map, we can instead examine what types of maps we have by choosing the Kraus operators wisely. You can see many of our single qubit examples from the previous section were expressed in this Kraus representation form. There is another issue, however. This map carries us over a finite time interval $t - t_0$. Can we figure out how to evolve the system over a single time step instead? This is the essence of the so-called *master equation*, which we examine next.

To begin our discussion of the master equation, we must first start with the time evolution of a system alone that evolves unitarily. Then we have

$$\hat{\rho}_{\text{sys}}(t) = \hat{U}(t, t_0)\hat{\rho}_{\text{sys}}(t_0)\hat{U}^{\dagger}(t, t_0), \tag{22.56}$$

which can be rewritten in terms of the Trotter formula into the single-step time evolution via

$$\begin{aligned}\hat{\rho}_{\text{sys}}(t + \Delta t) &= e^{-\frac{i}{\hbar}\hat{H}(t)\Delta t}\hat{\rho}_{\text{sys}}(t)e^{\frac{i}{\hbar}\hat{H}(t)\Delta t} \\ &= \hat{\rho}_{\text{sys}}(t) - \frac{i\Delta t}{\hbar}[\hat{H}(t), \hat{\rho}_{\text{sys}}(t)] + O\big((\Delta t)^2\big).\end{aligned} \tag{22.57}$$

This shows that the single time step for unitary time evolution involves a commutator of the quantum state with the Hamiltonian. This result is only part of the time evolution when the system evolves via a quantum map. We will not develop the full time evolution of the quantum map directly but instead just state the result. The Lindblad form of the master equation is given by

$$\begin{aligned}\hat{\rho}_{\text{sys}}(t + \Delta t) = \hat{\rho}_{\text{sys}}(t) &- \frac{i\Delta t}{\hbar}[\hat{H}(t), \hat{\rho}_{\text{sys}}(t)] \\ &+ \Delta t \sum_n \left(\hat{L}_n\hat{\rho}_{\text{sys}}(t)\hat{L}_n^{\dagger} - \frac{1}{2}\hat{\rho}_{\text{sys}}(t)\hat{L}_n^{\dagger}\hat{L}_n - \frac{1}{2}\hat{L}_n^{\dagger}\hat{L}_n\hat{\rho}_{\text{sys}}(t)\right),\end{aligned} \tag{22.58}$$

where the second line is the dissipator in the Lindblad form. The operators $\hat{L}_n$ are called the Lindblad operators. This form of dissipator is the only form that preserves the trace, is Hermitian, and completely positive. One can derive it in a straightforward fashion from the Kraus operator form, but it is not too enlightening. This form, however, is not the most general form of a master equation. It assumes a Markovian process, where the system has no memory along the time evolution, so the result at the next time step is determined entirely from the previous time step. As we described before, if an excitation enters the reservoir at time t and emerges at time t', we expect the behavior of the system depends on everything happening between time t and t'. So the Markovian approximation can be a significant one. Nevertheless, we examine some examples of how this works next.

As a first example, we will revisit the amplitude-damping channel in the Lindblad form. We work with a 2×2 density matrix, with $\hat{H} = -\frac{\hbar\Omega}{2}\sigma_z$, and we need just one Lindblad operator given by $\hat{L}_1 = \frac{\sqrt{\Gamma}}{2}(\sigma_x + i\sigma_y)$. Using

$$\hat{\rho} = \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix}, \tag{22.59}$$

we can compute the different terms in the master equation. We find

$$[\hat{H}, \hat{\rho}] = -\hbar\Omega \begin{pmatrix} 0 & \rho_{01} \\ -\rho_{10} & 0 \end{pmatrix}, \quad \hat{L}_1 \hat{\rho}_{\text{sys}} \hat{L}_1^\dagger = \Gamma \begin{pmatrix} \rho_{11} & 0 \\ 0 & 0 \end{pmatrix},$$
$$\text{and} \quad \hat{\rho}_{\text{sys}}(t)\hat{L}_1^\dagger \hat{L}_1 + \hat{L}_1^\dagger \hat{L}_1 \hat{\rho}_{\text{sys}}(t) = \Gamma \begin{pmatrix} 0 & \rho_{01} \\ \rho_{10} & 2\rho_{11} \end{pmatrix}. \tag{22.60}$$

Plugging into the Lindblad equation, we find that

$$\hat{\rho}(t + \Delta t) = \hat{\rho}(t) + i\Omega\Delta t \begin{pmatrix} 0 & \rho_{01} \\ -\rho_{10} & 0 \end{pmatrix} + \Gamma\Delta t \begin{pmatrix} \rho_{11} & -\frac{1}{2}\rho_{01} \\ -\frac{1}{2}\rho_{10} & -\rho_{11} \end{pmatrix}. \tag{22.61}$$

Repeating N times gives

$$\hat{\rho}(t + N\Delta t) = \begin{pmatrix} \rho_{00} + (1 - (1 - \Gamma\Delta t)^N)\rho_{11} & (1 + i\Omega\Delta t - \frac{1}{2}\Gamma\Delta t)^N \rho_{01} \\ (1 - i\Omega\Delta t - \frac{1}{2}\Gamma\Delta t)^N \rho_{10} & (1 - \Gamma\Delta t)^N \rho_{11} \end{pmatrix}. \tag{22.62}$$

Note that we used $\sum_{n=0}^{N-1} x^n = (1 - x^N)/(1 - x)$, with $x = 1 - \Gamma\Delta t$ in the derivation. After taking the limit of $\Delta t \to 0$ with $N\Delta t = t$, we find

$$\hat{\rho}(t) = \begin{pmatrix} \rho_{00} + (1 - e^{-\Gamma t})\rho_{11} & e^{i\Omega t} e^{-\frac{1}{2}\Gamma t} \rho_{01} \\ e^{-i\Omega t} e^{-\frac{1}{2}\Gamma t} \rho_{10} & e^{-\Gamma t} \rho_{11} \end{pmatrix}. \tag{22.63}$$

This is identical behavior to what we saw with amplitude damping, previously. It is only that here, we derived the result using the Lindblad form of the master equation. Note again how the dephasing (off-diagonal terms) decay more slowly than the diagonal ones.

In the previous example, we showed what happens when a qubit can spontaneously emit to decay to its ground state. Unlike the closed system situation, where we had to posit a transition, here, by simply having the system interact with a bath, we find it will decay to the ground state with a characteristic time given by $\frac{1}{\Gamma}$. Our bath had no specific properties to it, just an effective coupling that allows the transition. But, what if we had a bath of photons at a temperature T. Then, in addition to spontaneous emission, we should have stimulated emission and stimulated absorption from the bath. We will now modify our master equation to allow for these additional processes. The average number of photons (of frequency Ω) present in the reservoir at temperature T satisfies

$$n_{\text{ave}} = \frac{1}{e^{\beta\hbar\Omega} - 1}. \tag{22.64}$$

We only care about these photons, because they are the ones that can be absorbed or emitted by the qubit. We already worked out the master equation for spontaneous emission, and it is independent of n_{ave}. The spontaneous emission has an identical form and is multiplied by n_{ave}, so we simply multiply the dissipator by $(1 + n_{\text{ave}})$ to account for spontaneous and stimulated emissions. What about stimulated absorption? It is obviously proportional to n_{ave}, and being an absorption, it corresponds to the interchange of the Lindblad operators via $\hat{L}_2 \leftrightarrow \hat{L}_1^\dagger$. This results in the Lindblad equation of the form

$$\begin{aligned}\hat{\rho}(t + \Delta t) = \hat{\rho}(t) &+ i\Omega\Delta t \begin{pmatrix} 0 & \rho_{01} \\ -\rho_{10} & 0 \end{pmatrix} + (1 + n_{\text{ave}})\Gamma\Delta t \begin{pmatrix} \rho_{11} & -\frac{1}{2}\rho_{01} \\ -\frac{1}{2}\rho_{10} & -\rho_{11} \end{pmatrix} \\ &+ n_{\text{ave}}\Gamma\Delta t \begin{pmatrix} -\rho_{00} & -\frac{1}{2}\rho_{01} \\ -\frac{1}{2}\rho_{10} & \rho_{00} \end{pmatrix}.\end{aligned} \tag{22.65}$$

We first set $\rho_{11} = 1 - \rho_{00}$ for one step and then show how N steps evolve,

$$\begin{aligned}\rho_{00}(t + N\Delta t) &= \big(1 - (1 + 2n_{\text{ave}})\Gamma\Delta t\big)^N \rho_{00} \\ &\quad + \frac{1 + n_{\text{ave}}}{1 + 2n_{\text{ave}}}\big(1 - \big(1 - (1 + 2n_{\text{ave}})\Gamma\Delta t\big)^N\big) \\ \rho_{01}(t + N\Delta t) &= (1 + i\Omega\Delta t - (\tfrac{1}{2} + n_{\text{ave}})\Gamma\Delta t)^N \rho_{01} \\ \rho_{10}(t + N\Delta t) &= (1 - i\Omega\Delta t - (\tfrac{1}{2} + n_{\text{ave}})\Gamma\Delta t)^N \rho_{10},\end{aligned} \tag{22.66}$$

where, again, we use the ideas from the geometric series. The off-diagonal terms are easy to determine in the limit of $\Delta t \to 0$:

$$\rho_{01}(t) = e^{i\Omega t - \left(\frac{1}{2}+n_{\text{ave}}\right)\Gamma t}\rho_{01},$$
$$\rho_{10}(t) = e^{-i\Omega t - \left(\frac{1}{2}+n_{\text{ave}}\right)\Gamma t}\rho_{10}. \tag{22.67}$$

The diagonal terms clearly go to

$$\rho_{00}(t) = e^{-(1+2n_{\text{ave}})\Gamma t}\rho_{00} + \frac{1+n_{\text{ave}}}{1+2n_{\text{ave}}}\left(1 - e^{-(1+2n_{\text{ave}})\Gamma t}\right) \tag{22.68}$$

We can see that as $t \to \infty$, the population in the ground state goes to

$$\frac{1+n_{\text{ave}}}{1+2n_{\text{ave}}} = \frac{1+\frac{1}{e^{\beta\hbar\Omega}-1}}{1+\frac{2}{e^{\beta\hbar\Omega}-1}} = \frac{1}{1+e^{-\beta\hbar\Omega}}, \tag{22.69}$$

which is the thermal occupancy of the ground state at temperature T. Hence, the equilibration of the qubit with the bath of photons at a fixed temperature produces a qubit at that same temperature! This is a beautiful example of how a system can thermalize. The general way to thermalize a system is often more complicated than this, but it remains a nice instructive example of how we can work with master equations.

Section 22.4 Problems:

22.4.1 Trace-Preserving Nature of the Master Equation

Take the trace of both sides of Eq. (22.58) and show that $\text{Tr}\left(\hat{\rho}_{\text{sys}}(t+\Delta t)\right) = \text{Tr}\left(\hat{\rho}_{\text{sys}}(t)\right)$. This helps explain what the Lindblad form is the way that it is.

22.4.2 Verify the Derivation of Eq. (22.65)

Using the form described for the stimulated absorption, derive the master equation in Eq. (22.65).

22.4.3 Implementing an Approximate Kraus Map on a Quantum Computer

Quantum computers implement unitary operators and resets. How can we implement a Kraus map in such a circumstance? In this problem, we explore such a

construction. The strategy is to use a purified unitary version of the Kraus map and then trace out the ancilla degree of freedom to create the quantum channel. The map we will look at will produce proper diagonal elements but not the off-diagonal ones. It can be used when only the diagonal elements are needed in the calculation.

We work with some simple Kraus operators that act on a single qubit state. They are

$$\hat{K}_1 = \sqrt{1-\alpha}|0\rangle\langle 0|, \quad \hat{K}_2 = \sqrt{\alpha}|1\rangle\langle 0|, \quad \hat{K}_3 = \sqrt{1-\gamma}|1\rangle\langle 1|, \quad \text{and}$$
$$\hat{K}_4 = \sqrt{\gamma}|0\rangle\langle 1|. \tag{22.70}$$

We can describe this Kraus map in words as follow: if the system is in the state $|0\rangle$ with probability $1-\alpha$, do nothing, and with probability α, flip the qubit and if the system is in state $|1\rangle$ with probability $1-\gamma$, do nothing, and with probability γ, flip the qubit.

Once we have the description in words, the procedure is straightforward. First, we rotate the ancilla qubit, so it is in the state $\sqrt{1-\alpha}|0\rangle + \sqrt{\alpha}|1\rangle$. Then we use a control operation on the system qubit to rotate the ancilla qubit from its current state to $\sqrt{1-\gamma}|0\rangle + \sqrt{\gamma}|1\rangle$ if the system is in state $|1\rangle$. Next we perform a CNOT, which flips the system qubit if the ancilla qubit is in the $|1\rangle$ state. Finally, we measure the ancilla to lock in one of the four possible results and then get ready to perform the map again.

Write an essay that fills in the details by determining the specific rotations needed, by showing us the different states after each operation is applied to the ancilla and qubit, and then by explicitly verifying that this quantum computation will faithfully reproduce the Kraus map described above at each time step.

Note that this Kraus map only maps to a mixed state with no off-diagonal elements. It is a simplified implementation that has the proper diagonal elements but does not preserve the correct coherences. One can extend the map to include the proper determination of the off-diagonal terms as well. It is a bit more complicated to derive this, so we do not discuss it further here.

22.5 Nuclear Magnetic Resonance of a Single Spin

In this section, we examine a number of experiments in nuclear magnetic resonance in the case where the spin interacts with a dissipator described by the following three Lindbladian operators:

$$\hat{L}_1 = \frac{\sqrt{\alpha}}{2}(\sigma_x + i\sigma_y), \quad \hat{L}_2 = \frac{\sqrt{\gamma}}{2}(\sigma_x - i\sigma_y), \quad \text{and} \quad \hat{L}_3 = \sqrt{\delta}\sigma_z. \tag{22.71}$$

The first two operators were used with the last example in the previous section, while the third one is new. It is easy to evaluate because $\sigma_z\sigma_z\sigma_z = \sigma_z$, while there is a sign

change for σ_x and σ_y, furthermore $\sigma_z^2 = \mathbb{I}_2$ which also makes the calculation of the dissipator terms straightforward. Plugging into the time evolution of the Lindblad master equation yields

$$\rho_{00}(t+\Delta t) = (1-\gamma\Delta t)\rho_{00}(t) + \alpha\Delta t\rho_{11}(t) \tag{22.72}$$

$$\begin{aligned}
\rho_{01}(t+\Delta t) &= (1+i\Omega\Delta t - \tfrac{\alpha\Delta t}{2} - \tfrac{\gamma\Delta t}{2} - 2\delta\Delta t)\rho_{01}(t)\\
\rho_{10}(t+\Delta t) &= (1-i\Omega\Delta t - \tfrac{\alpha\Delta t}{2} - \tfrac{\gamma\Delta t}{2} - 2\delta\Delta t)\rho_{10}(t)\\
\rho_{11}(t+\Delta t) &= (1-\alpha\Delta t)\rho_{11}(t) + \gamma\Delta t\rho_{00}(t).
\end{aligned} \tag{22.73}$$

The off-diagonal terms can be directly solved in the same way as before to yield

$$\rho_{01}(t) = \rho_{01}(0)e^{i\Omega t - \left(\frac{1}{2}\alpha+\frac{1}{2}\gamma+2\delta\right)t} \text{ and } \rho_{10}(t) = \rho_{10}(0)e^{-i\Omega t - \left(\frac{1}{2}\alpha+\frac{1}{2}\gamma+2\delta\right)t}. \tag{22.74}$$

For the diagonal ones, we note that $\rho_{11} = 1 - \rho_{00}$ producing

$$\rho_{00}(t+N\Delta t) = (1-(\alpha+\gamma)\Delta t)^N\rho_{00} + \frac{\alpha}{\alpha+\gamma}(1-(1-(\alpha+\gamma)\Delta t)^N), \tag{22.75}$$

which is solved by

$$\rho_{00}(t) = \rho_{00}(0)e^{-(\alpha+\gamma)t} + \frac{\alpha}{\alpha+\gamma}(1-e^{-(\alpha+\gamma)t}). \tag{22.76}$$

The only change from before is that now, the off-diagonal terms have a different relaxation time. Note that the constants α and γ need to be chosen in such a way that the system relaxes to the proper final state. We can see that by choosing $\frac{\alpha}{\gamma} = e^{\beta\hbar\Omega}$, the system relaxes to a thermal state.

The simplified version of an NMR experiment is to start the system aligned along the z-axis with a positive projection, flip the spin by $\frac{\pi}{2}$ to place it in the $x-y$-plane, let it evolve over time, and measure how the diagonal and off-diagonal terms decay to extract the different decay constants. One can see that the longitudinal decay, which governs the diagonal elements, is given by $T_1 = 1/(\alpha+\gamma)$, while the off-diagonal decay is given by $T_2 = 1/(\frac{1}{2}\alpha + \frac{1}{2}\gamma + 2\delta)$, which is a shorter decay time if $2\delta > \frac{1}{2}\alpha + \frac{1}{2}\gamma$, which is typically the case. Now, we look more carefully into how the different measurements are actually done.

The easiest way to measure the longitudinal relaxation time is to place qubit (or spin) into the excited state and fit the decay to an exponential decay. This is done as follows: (i) Place all spins in the ground state; (ii) flip all spins to the excited state; (iii) wait a time t; (iv) measure the populations of the spins; and (v) fit the curve of the populations as a function of time to exponentials to extract T_1. This corresponds to our above time evolution, with $\rho_{00}(0) = \rho_{01}(0) = \rho_{10}(0) = 0$, so that

$$\rho_{00}(t) = \frac{\alpha}{\alpha+\gamma}\left(1 - e^{-(\alpha+\gamma)t}\right) \text{ and } \rho_{11}(t) = \frac{\gamma}{\alpha+\gamma} + \frac{\alpha}{\alpha+\gamma}e^{-(\alpha+\gamma)t}. \tag{22.77}$$

These diagonal values give us the populations, which have an exponential decay that is fit as a function of t to determine the relaxation time $T_1 = 1/(\alpha+\gamma)$.

The T_2 relaxation time is determined, most commonly, via a Ramsey experiment (technically speaking when determined by the Ramsey experiment, it is often called the T_2^* relaxation time). The experiment is relatively simple. We start in the ground state, flip the spins to the $x-y$-plane via a rotation by $\frac{\pi}{2}$ about the y-axis, let them evolve a time t, flip back to the z-axis with a second rotation by $\frac{\pi}{2}$ about the y-axis, and measure the populations. To analyze this, we start with the spins up and then flip them to the $|\uparrow\rangle_x$ state, so we have $\rho_{ij}(0) = \frac{1}{2}$. Now, we evolve them in time, using our general result, arriving at

$$\begin{aligned}
&\rho_{00}(t) = \frac{\alpha}{\alpha+\gamma} + \frac{-\alpha+\gamma}{2(\alpha+\gamma)}e^{-(\alpha+\gamma)t}, \quad \rho_{01}(t) = \tfrac{1}{2}e^{i\Omega t-(\frac{1}{2}\alpha+\frac{1}{2}\gamma+2\delta)t},\\
&\rho_{10}(t) = \tfrac{1}{2}e^{-i\Omega t-(\frac{1}{2}\alpha+\frac{1}{2}\gamma+2\delta)t}, \text{ and } \rho_{11}(t) = \frac{\gamma}{\alpha+\gamma} + \frac{\alpha-\gamma}{2(\alpha+\gamma)}e^{-(\alpha+\gamma)t}.
\end{aligned} \tag{22.78}$$

Now, we need to rotate further to the z-axis with a second rotation in the same direction by another $\frac{\pi}{2}$, which requires us to multiply from the left by $\frac{1}{\sqrt{2}}(\mathbb{I}_2 - i\sigma_y)$ and on the right by $\frac{1}{\sqrt{2}}(\mathbb{I}_2 + i\sigma_y)$. This gives us

$$\begin{aligned}
\rho_{00}(t) &= \tfrac{1}{2}\left(1 - \cos(\Omega t)e^{-(\frac{1}{2}\alpha+\frac{1}{2}\gamma+2\delta)t}\right),\\
\rho_{01}(t) &= \frac{\alpha-\gamma}{2(\alpha+\gamma)}\left(1 - e^{-(\alpha+\gamma)t}\right) + \tfrac{i}{2}\sin(\Omega t)e^{-(\frac{1}{2}\alpha+\frac{1}{2}\gamma+2\delta)t}\\
\rho_{10}(t) &= \frac{\alpha-\gamma}{2(\alpha+\gamma)}\left(1 - e^{-(\alpha+\gamma)t}\right) - \tfrac{i}{2}\sin(\Omega t)e^{-(\frac{1}{2}\alpha+\frac{1}{2}\gamma+2\delta)t}\\
\rho_{11}(t) &= \tfrac{1}{2}\left(1 + \cos(\Omega t)e^{-(\frac{1}{2}\alpha+\frac{1}{2}\gamma+2\delta)t}\right).
\end{aligned} \tag{22.79}$$

The populations have a component that oscillates and decays with the decay rate given by T_2^*, so it provides a means to measure the decay by fitting the functional form to the measured population data as a function of time.

The final experiment we will discuss is the Hahn spin-echo experiment. This experiment is just like the Ramsey experiment, except, halfway through the time evolution, we apply a π rotation about the x-axis, which reverses the spins in the plane and allows them to accumulate to create an echo pulse at time t. The reversal allows us to average out the errors due to fluctuating magnetic fields in the local environment of the system. The analysis is as before; except, we take our time evolution of the dissipating system out to $\frac{t}{2}$ and then rotate with the rotation matrix $i\sigma_x$ before letting it evolve another $\frac{t}{2}$ in time before the final y-rotation and

measurement. The final value of the density matrix after this experimental sequence is

$$\rho_{00}(t) = \tfrac{1}{2}\left(1 - e^{-(\frac{1}{2}\alpha+\frac{1}{2}\gamma+2\delta)t}\right) = 1 - \rho_{11}(t)$$
$$\rho_{01} = \frac{\alpha-\gamma}{2(\alpha+\gamma)}\left(1 - 2e^{-(\alpha+\gamma)\frac{t}{2}} + e^{-(\alpha+\gamma)t}\right) = \rho_{10}(t). \tag{22.80}$$

You can see these results are cleaner than the Ramsey interference experiment—there are no oscillations in the populations here, and we get a clear measurement of T_2. The field of nuclear magnetic resonance has many more complicated pulse sequences that they consider. This just provides an introduction to some of the more common and simpler ones.

Section 22.5 Problems:

22.5.1 Ramsey Experiment Derivation

Perform the matrix multiplications to verify Eq. (22.79).

22.5.2 Hahn Spin-Echo Sequence

Derive the final result for the Hahn spin echo in Eq. (22.80).

22.6 Spin Squeezing and Spin-Phonon Entanglement in Ion Traps

Both quantum computing and quantum metrology have goals to generate highly entangled states—in quantum computing, these states are viewed as a resource for further computation, while in quantum metrology, they often improve the accuracy of measurements. Since qubits can be thought of as a collection of spins, this leads us to the notion of considering how to create squeezed spin states. One of the best platforms to do so is the Penning trap based quantum simulator. We have discussed this system briefly before, but because the analysis does not depend on the specific architecture, we won't discuss the details of the device further. We have the question in front of us, how do we create a squeezed spin state? We know we can create spin-dependent forces, and in the ion traps, those forces lead to spin-spin interactions when the phonons can be adiabatically eliminated. The basic idea is that if we take a spin state that is oriented in the z direction and flip it to the $x-y$-plane, then we

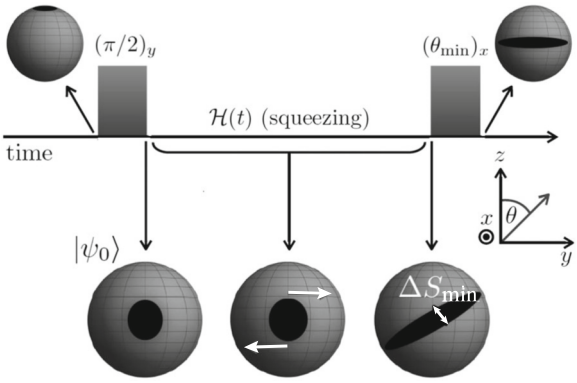

Fig. 22.1 Spin squeezing protocol used to generate squeezed spin states. Starting with the spins in the z-direction, they are rotated to the x-axis, then the spin-spin coupling is applied, followed by a rotation by θ to align the squeezed state with the $x-y$-plane (Figure is adapted from Dylewsky et al. (2015) © American Physical Society)

have a region of spin uncertainty that lives partly above and partly below the equator. By applying the spin-spin interaction, we can think of it, in a mean-field sense, as a magnetic field in the z direction, which precesses the spins above the equator in one direction and below the equator in another direction. This then shears the spin uncertainty circle into an ellipse, which then will have squeezing in a specific direction. The schematics for how this works can be seen in Fig. 22.1. Our spin squeezing discussion here closely follows Dylewsky et al. (2015).

It might not appear so obvious that applying the spin-spin interaction to the spin state rotated into the $x-y$-plane will necessarily lead to spin squeezing. To verify that it does, we need to calculate the uncertainty and show that we actually have a spin-squeezed state. What is remarkable is that we can exactly determine the expectation values needed to compute the spin squeezing for a system of effective spin qubits subject to the squeezing protocol as applied by a spin-dependent force. We now show how this is done.

We begin by recalling the final form we obtained for the time evolution operator when we apply a spin-dependent force and we have no extra magnetic field. The calculation is exact and is given in Eq. (21.57), with the coefficients in Eqs. (21.58) and (21.59). It includes a factor for the phonon time evolution, for the spin-phonon entanglement, and for the spin-spin time evolution. We wish to work in a basis where the Ising axis is in the z-direction, and we also want to separate the terms according to the factors that mutually commute with each other, so we rewrite the evolution operator as

$$\hat{U}(t) = \prod_{\alpha=1}^{N} e^{-i\omega_{\alpha x}t\left(\hat{a}_{\alpha x}\hat{a}_{\alpha x}+\frac{1}{2}\right)} \times \prod_{\alpha=1}^{N} e^{-i\sum_{j=1}^{N}\sigma_j^z\left(\Gamma_j^{(\alpha)}(t)\hat{a}_{\alpha x}+\Gamma_j^{(\alpha)*}(t)\hat{a}_{\alpha x}^\dagger\right)} \times \prod_{j=1}^{N}\prod_{j'=1}^{N} e^{-i\Phi_{jj'}(t)\sigma_j^z\sigma_{j'}^z}, \tag{22.81}$$

where we start the time evolution at $t_0 = 0$.

We begin with a state that is given by a spin state where all spins are in a product state, with the spins pointing in the positive direction along the x-axis and the phonons are all in the ground state (see Problem 22.6.4 for a discussion of how to extend the analysis to thermal states of phonons). This means the initial state is given by

$$|\psi\rangle = \frac{1}{\sqrt{2^N}} \sum_{\sigma_1,\cdots\sigma_n} \underbrace{|\sigma_1\rangle_z \otimes |\sigma_2\rangle_z \otimes \cdots \otimes |\sigma_N\rangle_z}_{\text{spin states}} \otimes \underbrace{|0\rangle_1 \otimes |0\rangle_2 \otimes \cdots \otimes |0\rangle_N}_{\text{phonon states}}. \tag{22.82}$$

We first note that the expectation values $\langle\psi(t)|\sigma_k^z|\psi(t)\rangle$ and $\langle\psi(t)|\sigma_k^z\sigma_{k'}^z|\psi(t)\rangle$ are both independent of time, because the σ_j^z operator commutes with the Hamiltonian at all times; indeed, they both vanish for this initial state. Thus, we need to compute the following time-dependent expectation values:

$$\langle\psi(t)|\sigma_k^{\pm}|\psi(t)\rangle,\ \langle\psi(t)|\sigma_k^z\sigma_{k'}^{\pm}|\psi(t)\rangle,\ \langle\psi(t)|\sigma_k^{\pm}\sigma_{k'}^{\pm}|\psi(t)\rangle, \text{and} \langle\psi(t)|\sigma_k^{\pm}\sigma_{k'}^{\mp}|\psi(t)\rangle, \tag{22.83}$$

where $k \neq k'$. One of the key issues in computing these different terms is that every factor in $\hat{U}(t)$ has a corresponding factor in the $\hat{U}^\dagger(t)$ that is its inverse. So, any of those factors that commute with the operator in the expectation value can be cancelled from both sides of the expectation value. So, the phonon term always cancels, the spin-phonon entangling term is always present, and the spin-spin term only contributes when j or j' is equal to the spatial index of at least one of the operators in the expectation value.

To evaluate these expectation values, we use our operator identities to simplify the expressions as much as we can. We now describe the details of how this is done. Let us begin with the spin-spin terms, which are easier to work with. They commute with the σ_j^z terms but not the raising or lowering terms. Noting that $\Phi_{jj'}(t)$ is real valued, we have

$$\begin{aligned}
&e^{i\sum_j\sum_{j'}\Phi_{jj'}(t)\sigma_j^z\sigma_{j'}^z}\sigma_k^{\pm}e^{-i\sum_j\sum_{j'}\Phi_{jj'}(t)\sigma_j^z\sigma_{j'}^z} \\
&= e^{i\sum_{j'\neq k}\Phi_{kj'}(t)\sigma_k^z\sigma_{j'}^z + i\sum_{j\neq k}\Phi_{jk}(t)\sigma_j^z\sigma_k^z}\sigma_k^{\pm}e^{-i\sum_{j'\neq k}\Phi_{kj'}(t)\sigma_k^z\sigma_{j'}^z - i\sum_{j\neq k}\Phi_{jk}(t)\sigma_j^z\sigma_k^z}, \\
&= e^{\pm 2i\sum_{j\neq k}\left(\Phi_{kj}(t)+\Phi_{jk}(t)\right)\sigma_j^z}\sigma_k^{\pm},
\end{aligned} \tag{22.84}$$

after using the Hadamard lemma for each of the factors. The two Φ terms are actually the same, because this coefficient is symmetric with respect to the exchange of the two indices j and j'. Now we examine the spin-phonon entangling term. To begin, we need to use the Weyl form of the Baker-Campbell-Hausdorff formula to separate the terms with $j = k$ and those with $j \neq k$. While the spin matrices always commute, the phonon creation and annihilation operators do not. Using

$$e^{-i\sum_{j=1}^{N}\sigma_j^z\left(\Gamma_j^{(\alpha)}(t)\hat{a}_{\alpha x}+\Gamma_j^{(\alpha)*}(t)\hat{a}_{\alpha x}^\dagger\right)}=e^{-i\sum_{j\neq k}\sigma_j^z\left(\Gamma_j^{(\alpha)}(t)\hat{a}_{\alpha x}+\Gamma_j^{(\alpha)*}(t)\hat{a}_{\alpha x}^\dagger\right)}$$
$$\times\, e^{-i\sigma_k^z\left(\Gamma_k^{(\alpha)}(t)\hat{a}_{\alpha x}+\Gamma_k^{(\alpha)*}(t)\hat{a}_{\alpha x}^\dagger\right)}e^{i\sum_{j\neq k}\sigma_k^z\sigma_j^z\mathrm{Im}\left(\Gamma_k^{(\alpha)}(t)\Gamma_j^{(\alpha)*}(t)\right)}, \tag{22.85}$$

we find that

$$e^{i\sum_{j=1}^{N}\sigma_j^z\left(\Gamma_j^{(\alpha)}(t)\hat{a}_{\alpha x}+\Gamma_j^{(\alpha)*}(t)\hat{a}_{\alpha x}^\dagger\right)}\sigma_k^{\pm}e^{-i\sum_{j=1}^{N}\sigma_j^z\left(\Gamma_j^{(\alpha)}(t)\hat{a}_{\alpha x}+\Gamma_j^{(\alpha)*}(t)\hat{a}_{\alpha x}^\dagger\right)}$$
$$=e^{\pm 2i\left(\Gamma_k^{(\alpha)}(t)\hat{a}_{\alpha x}+\Gamma_k^{(\alpha)*}(t)\hat{a}_{\alpha x}^\dagger\right)}e^{\pm 2i\sum_{j\neq k}\sigma_j^z\mathrm{Im}\left(\Gamma_k^{(\alpha)}(t)\Gamma_j^{(\alpha)*}(t)\right)}\sigma_k^{\pm}, \tag{22.86}$$

after evaluating two Hadamard lemmas. Now, if we look at the value for $\Gamma_j^{(\alpha)}(t)$ in Eq. (21.58), we see that it is actually real valued, because the eigenvectors b can always be chosen to be real for the Paul trap (in the Penning trap, some of the eigenvectors are complex valued and there this term needs to be kept). Hence, we drop the last exponential term as we will focus on the Paul trap here.

We now have all we need to compute the expectation values. We just have to assemble the results. We find

$$\langle\psi(t)|\sigma_k^{\pm}|\psi(t)\rangle=\langle\psi(t)|e^{\pm 4i\sum_{j\neq k}\Phi_{kj}(t)\sigma_j^z}\prod_{\alpha=1}^{N}e^{\pm 2i\left(\Gamma_k^{(\alpha)}(t)\hat{a}_{\alpha x}+\Gamma_k^{(\alpha)*}(t)\hat{a}_{\alpha x}^\dagger\right)}\sigma_k^{\pm}|\psi(t)\rangle. \tag{22.87}$$

The first exponential factor acts only on the spin part of the quantum state, while the second exponential factor acts only on the phonon part of the quantum state. The phonon part can be summarized in terms of coherent states using the displacement operator from Sect. 7.11. This yields

$$\langle\psi(t)|\sigma_k^{\pm}|\psi(t)\rangle=\frac{1}{2}\prod_{\substack{j=1\\j\neq k}}^{N}\cos\left(4\Phi_{kj}(t)\right)\prod_{\alpha=1}^{N}{}_\alpha\langle 0|\hat{D}\left(\pm 2i\Gamma_k^{(\alpha)}(t)\right)|0\rangle_\alpha. \tag{22.88}$$

This is easy to generalize for our next one, which gives

$$\langle\psi(t)|\sigma_k^{\pm}\sigma_{k'}^z|\psi(t)\rangle=\pm\frac{i}{2}\sin\left(4\Phi_{kk'}(t)\right)\prod_{\substack{j=1\\j\neq k,k'}}^{N}\cos\left(4\Phi_{kj}(t)\right)$$
$$\times\prod_{\alpha=1}^{N}{}_\alpha\langle 0|\hat{D}\left(\pm 2i\Gamma_k^{(\alpha)}(t)\right)|0\rangle_\alpha. \tag{22.89}$$

The plus or minus sign comes from the fact that the exponential from the spin-spin interaction has a plus or minus sign in the exponent, and so the sine function changes sign for the lowering operator versus the raising operator.

We have two more expectation values to work out. We start by computing

$$e^{i\sum_j\sum_{j'}\Phi_{jj'}(t)\sigma_j^z\sigma_{j'}^z}\sigma_k^{\pm}e^{-i\sum_j\sum_{j'}\Phi_{jj'}(t)\sigma_j^z\sigma_{j'}^z}e^{i\sum_j\sum_{j'}\Phi_{jj'}(t)\sigma_j^z\sigma_{j'}^z}$$
$$\sigma_{k'}^{\mp}e^{-i\sum_j\sum_{j'}\Phi_{jj'}(t)\sigma_j^z\sigma_{j'}^z}$$
$$= e^{\pm 4i\sum_{j\neq k}\Phi_{kj}(t)\sigma_j^z}\sigma_k^{\pm}e^{\mp 4i\sum_{j\neq k'}\Phi_{k'j}\sigma_j^z}\sigma_{k'}^{\mp}. \tag{22.90}$$

Now, we move the exponential operator in the middle to the left using a Hadamard lemma, and the operator with $j = k'$ in the leftmost exponential all the way to the right. This gives

$$e^{\pm 4i\sum_{j\neq kk'}\left(\Phi_{kj}(t)-\Phi_{k'j}(t)\right)\sigma_j^z}\sigma_k^{\pm}\sigma_{k'}^{\mp}e^{-8i\Phi_{kk'}(t)+8i\Phi_{kk'}(t)}$$
$$= e^{\pm 4i\sum_{j\neq kk'}\left(\Phi_{kj}(t)-\Phi_{k'j}(t)\right)\sigma_j^z}\sigma_k^{\pm}\sigma_{k'}^{\mp}. \tag{22.91}$$

The correction terms from rearranging the operators with the prefactor $\Phi_{kk'}(t)$ cancel. Next, we work on the spin-phonon entangling terms. Because the imaginary part of $\Gamma_j\Gamma_k^*$ vanishes, the individual terms of each spin-phonon entangling operator can be factored without requiring any corrections. This then allows us to immediately conclude that

$$e^{i\sum_{j=1}^N\sigma_j^z\left(\Gamma_j^{(\alpha)}(t)\hat{a}_{\alpha x}+\Gamma_j^{(\alpha)*}(t)\hat{a}_{\alpha x}^\dagger\right)}\sigma_k^{\pm}\sigma_{k'}^{\mp}e^{-i\sum_{j=1}^N\sigma_j^z\left(\Gamma_j^{(\alpha)}(t)\hat{a}_{\alpha x}+\Gamma_j^{(\alpha)*}(t)\hat{a}_{\alpha x}^\dagger\right)}$$
$$= e^{\pm 2i\left(\Gamma_k^{(\alpha)}(t)\hat{a}_{\alpha x}+\Gamma_k^{(\alpha)*}(t)\hat{a}_{\alpha x}^\dagger-\Gamma_{k'}^{(\alpha)}(t)\hat{a}_{\alpha x}-\Gamma_{k'}^{(\alpha)*}(t)\hat{a}_{\alpha x}^\dagger\right)}\sigma_k^{\pm}\sigma_{k'}^{\mp}. \tag{22.92}$$

We can now assemble these results to determine the expectation value. We find

$$\langle\psi(t)|\sigma_k^{\pm}\sigma_{k'}^{\mp}|\psi(t)\rangle = \frac{1}{4}\prod_{\substack{j=1\\ j\neq k,k'}}^{N}\cos\left(4\left(\Phi_{kj}(t)-\Phi_{k'j}(t)\right)\right)$$
$$\times\prod_{\alpha=1}^{N}{}_\alpha\langle 0|\hat{D}\left(\pm 2i\left(\Gamma_k^{(\alpha)}(t)-\Gamma_{k'}^{(\alpha)}(t)\right)\right)|0\rangle_\alpha. \tag{22.93}$$

By carefully following the signs, we can also show that

$$\langle\psi(t)|\sigma_k^{\pm}\sigma_{k'}^{\pm}|\psi(t)\rangle = \frac{1}{4}\prod_{\substack{j=1\\ j\neq k,k'}}^{N}\cos\left(4\left(\Phi_{kj}(t)+\Phi_{k'j}(t)\right)\right)$$
$$\times\prod_{\alpha=1}^{N}{}_\alpha\langle 0|\hat{D}\left(\pm 2i\left(\Gamma_k^{(\alpha)}(t)+\Gamma_{k'}^{(\alpha)}(t)\right)\right)|0\rangle_\alpha, \tag{22.94}$$

which you should be able to verify yourself. With all of the expectation values determined, we are now ready to compute the spin squeezing. Other expectation values can also be determined exactly in this approach, and we leave a discussion of those different cases to the problems.

The spin-squeezing protocol has already been illustrated in Fig. 22.1. After rotating the spin to the $x-y$-plane, we apply the spin-dependent force, which has the spin-spin interaction shear the spin distribution, producing a squeezed state with the squeezed axis rotated at an angle θ. To determine the amount of squeezing, and the angle of the orientation, we need to use our expectation values. We define a spin degree of freedom in the θ direction via $\hat{S}_\theta = \frac{\hbar}{2}(\cos\theta \sum_j \sigma_j^x + \sin\theta \sum_j \sigma_j^x)$ (note that this has a different sign than the σ_ϕ we used in the previous chapter). Then the uncertainty is given by

$$\left(\Delta \hat{S}_\theta\right)_{|\psi(t)\rangle} = \sqrt{\langle\psi(t)|\hat{S}_\theta^2|\psi(t)\rangle - \left(\langle\psi(t)|\hat{S}_\theta|\psi(t)\rangle\right)^2}. \tag{22.95}$$

The goal is to minimize $\xi_\theta = \left(\Delta \hat{S}_\theta\right)_{|\psi(t)\rangle} / \left(\sqrt{N}|\langle\psi(t)|\hat{S}_x|\psi(t)\rangle|\right)$ over all possible θ. This is because ξ_θ quantifies the level of squeezing of the system. We call the minimum of this ξ. A simple calculus exercise on minimizing the uncertainty gives us that it is minimized when

$$\theta = \frac{1}{2}\tan^{-1}\left(\frac{\langle\psi(t)|\hat{S}_x\hat{S}_y|\psi(t)\rangle + \langle\psi(t)|\hat{S}_y\hat{S}_z|\psi(t)\rangle}{\langle\psi(t)|\hat{S}_z\hat{S}_z|\psi(t)\rangle - \langle\psi(t)|\hat{S}_y\hat{S}_y|\psi(t)\rangle}\right). \tag{22.96}$$

The derivation is simple but requires calculus, so we do not show the details here.

We show an example of what the spin squeezing looks like when the system is detuned to the blue of the transverse center-of-mass mode of a linear Paul trap with 20 ions in Fig. 22.2. In units of the center-of-mass frequency $\omega_{\text{c.m.}}$, the detuning is 0.001, and the Rabi frequency is 0.00025. One can see that the time-averaged spin coefficients approximation and the case where we ignore spin-phonon entanglement both approximate the exact solution well here, but neither shows the oscillations in the exact data. As we raise the temperature of the phonons, the overall scale of the squeezing is reduced, and the approximations become less accurate. It is remarkable that this very complex quantum system can be solved *exactly* using the methods we have developed!

We now move on to discuss our final example, which describes how we can use spin-phonon entanglement to measure the number of phonons in specific phonon modes by performing a spin-echo-like experiment on the spins and fitting the response to our exact theoretical calculations. We have already seen this phenomena earlier, when we examined the interaction of a spin with photons at a temperature T—we saw that the relaxation rates depended on the average number of photons present. The analysis here is a bit more complicated than that, but the idea is similar, except we swap out the photons with the phonons in the ion trap system. Our discussion follows closely the work in Sawyer et al. (2015), where this protocol was used in a Penning trap to measure the average number of phonon excitations in

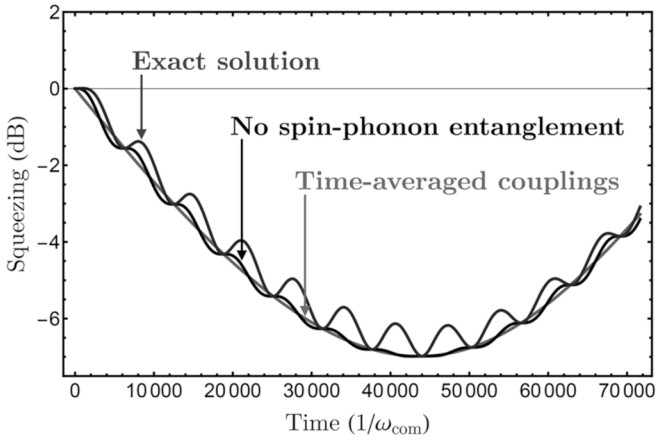

Fig. 22.2 Spin squeezing reported in decibels for a 20-spin system with driving near the center-of-mass mode. The exact calculation is the curve with the most oscillations, while the time-averaged curve has no oscillations. The squeezing parameter is normalized, so that a coherent state has 0 decibels of screening (Figure is adapted from Dylewsky et al. (2015) © American Physical Society)

each phonon mode of the system. In principle, this approach could also be used to measure very small displacements or forces acting on the ion crystal.

In the pulse sequence, we start from up in the z direction. Rotate to up in the x direction, evolve for a time τ, rotate from y to $-y$, evolve for a time τ, and finally rotate from x to $-z$. It is illustrated at the top of Fig. 22.3. If there was no time evolution in the system, the up spins would all be rotated to down spins in this process. But, with spin-phonon entanglement, some spins will remain in the up state. Our job is to determine how many.

We imagine the phonons are initiated in some arbitrary state. It could be a mixed state, such as a thermal state, or something else. But for now, we just generically treat it as a pure phonon state $|\psi_{\text{ph}}\rangle$ and we will sort out any issues with possible mixed states later. Our spins are all initialized in the $|\uparrow\rangle_z$ state, and we neglect the spin-spin (and phonon) parts of the time evolution. Using $|\uparrow\rangle_z \otimes \cdots |\uparrow\rangle_z = |\psi_{\text{sp}}\rangle$, the state preparation for the full spin-echo experiment is given by

$$|\psi_f\rangle = e^{-\frac{i\pi}{4}\sum_j \sigma_j^y} \hat{U}_{\text{sp-ph}}(2\tau,\tau) e^{-\frac{i\pi}{2}\sum_j \sigma_j^x} \hat{U}_{\text{sp-ph}}(\tau,0) e^{-\frac{i\pi}{4}\sum_j \sigma_j^y} |\psi_{\text{sp}}\rangle \otimes |\psi_{\text{ph}}\rangle, \tag{22.97}$$

where we now indicate the beginning and ending times for the time evolution operators. Now, the first step is to use the exponential reordering identity to move the x rotation to the left through the second time evolution operator. We can think of this as a multiply by one on the left of the evolution operator, followed by a Hadamard lemma evaluated with the braiding identity to move everything into the exponent. Since $\exp(-\frac{i\pi}{2}\sigma_j^x) = -i\sigma_j^x$, we see the exponent simply shifts $\sigma_j^z \to -\sigma_j^z$. Then, using the fact that $\exp(-\frac{i\pi}{4}\sigma_y) = \frac{1}{\sqrt{2}}(\mathbb{I}_2 - i\sigma_y)$ yields

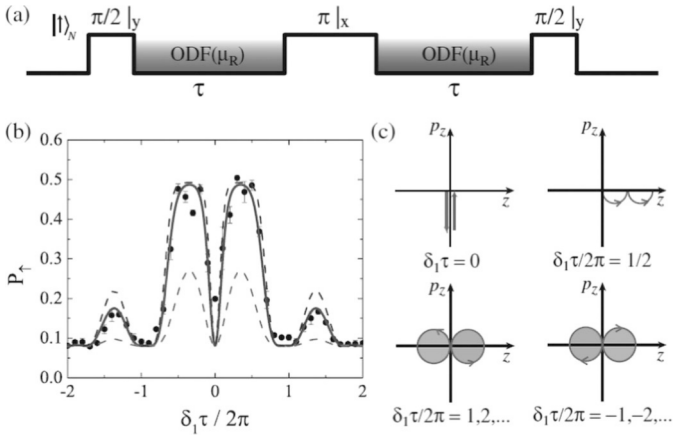

Fig. 22.3 (**a**) Spin echo pulse sequence involving a rotation from the z to x-axis, followed by time evolution, then a rotation from y to $-y$, followed by additional time evolution, finally we make a rotation from x to $-z$ and then measure. (**b**) The probability to find an up spin at one site as a function of $\delta\tau/2\pi$. (**c**) The evolution of the phonon in phase space for the pulse sequence (Figure is adapted from Sawyer et al. (2015) © American Physical Society)

$$
\begin{aligned}
|\psi_f\rangle = &\prod_{j=1}^{N}\left(\frac{i}{\sqrt{2}}\left(\sigma_j^z-\sigma_j^x\right)\right)\\
&\prod_{\alpha=1}^{N} e^{i\sum_{j=1}^{N}\left(\left(\Gamma_j^{(\alpha)}(2\tau)-2\Gamma_j^{(\alpha)}(\tau)\right)\hat{a}_{\alpha x}+\left(\Gamma_j^{(\alpha)*}(2\tau)-2\Gamma_j^{(\alpha)*}(\tau)\right)\hat{a}_{\alpha x}^{\dagger}\right)\sigma_j^z}\\
&\times\prod_{j=1}^{N}\left(\frac{1}{\sqrt{2}}\left(\mathbb{I}-i\sigma_j^y\right)\right)|\psi_{\rm sp}\rangle\otimes|\psi_{\rm ph}\rangle. \qquad (22.98)
\end{aligned}
$$

The rightmost spin matrices change $|\uparrow\rangle_z \rightarrow \frac{1}{\sqrt{2}}(|\uparrow\rangle_z + |\downarrow\rangle_z)$, for the spin degree of freedom for each ion. Then, if we schematically write each exponent as $e^{i\hat{\gamma}\sigma^z}$ for each ion, with $\hat{\gamma}$ a shorthand for the phonon operator we find that when it acts on the spin state, it takes us to $\frac{1}{\sqrt{2}}\left(e^{i\hat{\gamma}}|\uparrow\rangle_z + e^{-i\hat{\gamma}}|\downarrow\rangle_z\right)$. Now, to measure the probability to be in the up-spin state, we need to calculate the expectation value of $\left(\frac{1}{2}(\mathbb{I}+\sigma^z)\right)$, because this operator equals 1 for the $\uparrow$ state and 0 for the $\downarrow$ state; note that we are measuring the probability for the spin to be up at one site only. Multiplying out the remaining spin operators, we find the state at the jth ion becomes $-\sin\gamma|\uparrow\rangle_{jz}$. When we compute the matrix element of the up-spin projection operator, all spin terms on other sites cancel, because they appear as Hermitian conjugates on each side of the expectation value and they commute with the projection operator. Hence, the expectation value $\langle\psi_f|\frac{1}{2}(\mathbb{I}+\sigma_j^z)|\psi_f\rangle$ becomes

$$\begin{aligned}
&\langle\psi_f|\tfrac{1}{2}(\mathbb{I}+\sigma_j^z)|\psi_f\rangle \\
&= \langle\psi_{\rm ph}|\prod_{\alpha=1}^{N}\sin\Big(\big(\Gamma_j^{(\alpha)}(2\tau)-2\Gamma_j^{(\alpha)}(\tau)\big)\hat{a}_{\alpha x}+\big(\Gamma_j^{(\alpha)*}(2\tau)-2\Gamma_j^{(\alpha)*}(\tau)\big)\hat{a}_{\alpha x}^\dagger\Big) \\
&\quad\times\prod_{\alpha=1}^{N}\sin\Big(\big(\Gamma_j^{(\alpha)}(2\tau)-2\Gamma_j^{(\alpha)}(\tau)\big)\hat{a}_{\alpha x}+\big(\Gamma_j^{(\alpha)*}(2\tau)-2\Gamma_j^{(\alpha)*}(\tau)\big)\hat{a}_{\alpha x}^\dagger\Big)|\psi_{\rm ph}\rangle.
\end{aligned} \tag{22.99}$$

At this stage, the operator only acts on the phonon part of the quantum state.

Our next step is to evaluate this in a thermal state for the phonons. We first work this out for a single phonon mode, and then we can generalize to all of them. We have to compute

$$\begin{aligned}
&-\frac{1}{4}\frac{1}{Z}\sum_{n=0}^{\infty}e^{\beta\hbar\omega(n+\frac{1}{2})}\langle n|\left(e^{i\gamma\hat{a}+i\gamma^*\hat{a}^\dagger}-e^{-i\gamma\hat{a}-i\gamma^*\hat{a}^\dagger}\right) \\
&\quad\left(e^{i\gamma\hat{a}+i\gamma^*\hat{a}^\dagger}-e^{-i\gamma\hat{a}-i\gamma^*\hat{a}^\dagger}\right)|n\rangle \\
&=\frac{1}{2}-\frac{1}{4Z}\sum_{n=0}^{\infty}e^{\beta\hbar\omega(n+\frac{1}{2})}\langle n|\left(e^{2i\gamma\hat{a}+2i\gamma^*\hat{a}^\dagger}+e^{-2i\gamma\hat{a}-2i\gamma^*\hat{a}^\dagger}\right)|n\rangle,
\end{aligned} \tag{22.100}$$

with $Z = e^{-\frac{1}{2}\beta\hbar\omega}/(1-e^{-\beta\hbar\omega})$ and $\gamma = \Gamma(2\tau)-2\Gamma(\tau)$. We focus on the first summation. It gives

$$\sum_{n=0}^{\infty}e^{-\beta\hbar\omega n}\langle n|e^{2i\gamma\hat{a}+2i\gamma^*\hat{a}^\dagger}|n\rangle=\sum_{n=0}^{\infty}e^{-\beta\hbar\omega n}\langle n|e^{2i\gamma\hat{a}}e^{2i\gamma^*\hat{a}^\dagger}e^{2|\gamma|^2}|n\rangle, \tag{22.101}$$

after using the Weyl form of the Baker-Campbell-Hausdorff formula. Next, we expand the two exponentials as summations and act them on the harmonic oscillator states using the known action of the raising or lowering operator on an energy eigenstate. Then we recognize that the energy eigenstates are orthogonal to remove one of the summations. This gives us

$$\begin{aligned}
&\sum_{n=0}^{\infty}e^{-\beta\hbar\omega n}\langle n|e^{2i\gamma\hat{a}}e^{2i\gamma^*\hat{a}^\dagger}e^{2|\gamma|^2}|n\rangle \\
&\qquad=e^{2|\gamma|^2}\sum_{n=0}^{\infty}e^{-\beta\hbar\omega n}\sum_{m=0}^{\infty}\frac{(2i\gamma)^m}{m!}\sum_{m'=0}^{\infty}\frac{(2i\gamma^*)^{m'}}{m'!}\langle n|\hat{a}^m(\hat{a}^\dagger)^{m'}|n\rangle \\
&\qquad=e^{2|\gamma|^2}\sum_{n=0}^{\infty}e^{-\beta\hbar\omega n}\sum_{m=0}^{\infty}\frac{(-4|\gamma|^2)^m}{m!m!}\frac{(n+m)!}{n!}.
\end{aligned} \tag{22.102}$$

The next step requires us to change the order of the two summations (there is no difficulty to do this because the sums are both infinite and independent of each other). Then, we will use Newton's generalized binomial theorem (a great result to know, but not so simple to derive). We find

$$e^{2|\gamma|^2}\sum_{m=0}^{\infty}\frac{(-4|\gamma|^2)^m}{m!}\sum_{n=0}^{\infty}\frac{(n+m)!}{m!n!}e^{-\beta\hbar\omega n}=e^{2|\gamma|^2}\sum_{m=0}^{\infty}\frac{(-4|\gamma|^2)^m}{m!}\frac{1}{\left(1-e^{\beta\hbar\omega}\right)^{m+1}}$$
$$=\frac{e^{2|\gamma|^2}}{1-e^{\beta\hbar\omega}}e^{-\frac{4|\gamma|^2}{1-e^{-\beta\hbar\omega}}}=\frac{e^{-2|\gamma|^2(2\langle n\rangle_{\rm th}+1)}}{1-e^{\beta\hbar\omega}},\tag{22.103}$$

where

$$\langle n\rangle_{\rm th}=\frac{1}{e^{\beta\hbar\omega}-1},\tag{22.104}$$

is the average number of bosons of frequency ω and inverse temperature β when in a thermal state, as we found earlier.

Since the result of this summation is independent of the sign of γ, we have both results give the same thing, and then plugging into the final expression, by putting everything together, we find that

$$P_j^{\uparrow}=\frac{1}{2}\left(1-e^{-2\sum_{\alpha=1}^{N}|\Gamma_j^{(\alpha)}(2\tau)-2\Gamma_j^{(\alpha)}(\tau)|^2\left(2\langle n\rangle_{\alpha,\rm th}+1\right)}\right).\tag{22.105}$$

Note that you have to look carefully at the expectation values we are evaluating for the general case to obtain this result. Even though we have seen derivations before where thermal averages of bosons ended up in the exponent, the derivation here was a bit different from the previous case. There is an approximation that goes into evaluating a final result to compare with experiment. The system will have decoherence of the spins. We know that this corresponds to a decay of the signal be a factor of $e^{-\Gamma t}$, with $1/\Gamma$ being the relaxation time and having nothing to do with the spin-entangling time evolution. It is common to put such a factor in by hand, even if it has not been fully justified. This then yields the final formula of

$$P_j^{\uparrow}=\frac{1}{2}\left(1-e^{-2\Gamma\tau}e^{-2\sum_{\alpha=1}^{N}|\Gamma_j^{(\alpha)}(2\tau)-2\Gamma_j^{(\alpha)}(\tau)|^2\left(2\langle n\rangle_{\alpha,\rm th}+1\right)}\right).\tag{22.106}$$

It is the presence of the thermal occupancy of the phonons in the exponent that makes this experiment so sensitive to the average number of phonons in each phonon mode in the system.

Here is how we think of these experiments. If there is no spin-phonon entanglement, then all of the originally up spin ions end in the down spin state, and we measure 0 for the probability to see the ions in an up state. As the spin-phonon entangling term entangles the spins and the phonons, we have decoherence when

measuring the spin state at the end of the experiment, which gives us some up spins. The more phonons present in the initial state, the easier it is to entangle them with the spins, hence the enhancement of the signal at higher temperatures. As we detune near a resonance with the phonon, the entanglement hits its maximum, because we create many phonons. Interestingly though, whenever $\delta\tau = 2\pi n$ for integer n (including 0), the entanglement vanishes at the end of each leg of the echo sequence (illustrated in Fig. 22.3c), so we measure no up spin states (this is increased due to the spin decoherence in a real experiment). All of these results are shown in Fig. 22.3 (the dots in panel (b) are the measured results, while the lines are fits, found by adjusting the average number of phonons in the mode). Using this, one can measure the number of phonons in specific phonon modes by finding the best fit. What is interesting here is that the force is a uniform force applied to the spins, but it generates oscillations at small length scales due to the resonance condition. The displacements are quite small that can be detected, about 0.5 nm which is smaller than the extent of the ground-state wavefunction about each lattice site. I find it interesting how we can use the spin-phonon entanglement to selectively determine the average number of phonon excitations in individual phonon modes.

Section 22.6 Problems:

22.6.1 Verify the Expectation Value Computations

In the main text, we sketched how to perform the calculations of the different spin expectation values. Redo those calculations yourself, filling in all of the details. Be careful to follow signs that lead to cancellations very carefully, or you will not agree with the final results.

22.6.2 Describe How Higher-Order Spin Expectation Values Can Be Calculated

In the main text, we showed how the linear and quadratic spin-spin correlation functions could be calculated. Write an essay that describes how the approach can be generalized for arbitrary spin-spin correlation functions.

22.6.3 Show How to Calculate Phonon Expectation Values

In the expectation values, we need to determine the overlap of coherent states with the ground state, at zero temperature, and the matrix element needs to be generalized to excited states for a nonzero temperature calculation. Determine what $\langle 0|\hat{D}(\alpha)|0\rangle$

is for the ground state, and write a short essay describing how to generalize the calculation for higher-energy eigenstates.

22.6.4 How Do the Expectation-Value Calculations Change for Nonzero Phonon Temperatures?

Write an essay describing how one performs the calculation at a nonzero temperature. You do not need to actually work out the details but just describe how it works.

22.6.5 Details for Determining the Squeezing from the Expectation Values Worked Out in the Main Text

Convert the results for the optimal angle θ from expectation values with respect to the total spin operators to expectation values with respect to the linear and quadratic operators we worked with in the main text. You do not need to calculate the angle, just show how to reexpress the formula in terms of the expectation values we determined.

22.6.6 Verify the Equations Derived for the Metrology Application of the Average Phonon Numbers in the Main Text

Verify the derivation of the spin-echo sequence and what the probability is to be measured in an up spin at site j at the end of the sequence.

22.7 Damped Simple Harmonic Oscillator

One of the most iconic dissipative problems is the damped harmonic oscillator. This is often a problem solved in a junior-level classical mechanics class, but it usually is not discussed in quantum classes. We treat both here.

To start with the classical system, we need to understand how the position and momentum are updated at each time interval. Because the momentum is the mass times the velocity, and the velocity determines how the position changes, we have our first equation of motion which is

$$x(t+\Delta t) = x(t) + \Delta t \frac{p(t)}{m}, \tag{22.107}$$

where all objects are just numbers because we are describing the classical world. The second equation of motion is Newton's law, which says the force determines the change in the momentum, or

$$p(t+\Delta t) = p(t) + \Delta t\big(-\gamma p(t) - m\omega^2 x(t)\big), \tag{22.108}$$

which includes the harmonic oscillator restoring force for an oscillator (with mass m and frequency ω) and the damping force (proportional to the velocity or momentum) with a strength given by γ—it has dimensions of a frequency.

These equations are coupled equations that do not seem to be easy to solve. We can find their solution most easily by decoupling them. In analogy with the ladder operators from quantum mechanics, we consider the function

$$A(t) = \frac{1}{\sqrt{2m}}\big(p(t) + \alpha x(t)\big). \tag{22.109}$$

Our goal is to pick the constant α such that $A(t+\Delta t) = A(t) + \Delta t \lambda A(t)$, with λ a constant. So, we compute

$$\begin{aligned} A(t+\Delta t) &= \frac{1}{\sqrt{2m}}\big(p(t+\Delta t) + \alpha x(t+\Delta t)\big) \\ &= \frac{1}{\sqrt{2m}}\big(p(t) + \alpha x(t)\big) + \Delta t \frac{1}{\sqrt{2m}}\left(-\gamma p(t) - m\omega^2 x(t) + \alpha \frac{p(t)}{m}\right) \end{aligned} \tag{22.110}$$

$$= A(t) + \Delta t \frac{1}{\sqrt{2m}}\left(\frac{\alpha}{m} - \gamma\right)\left(p(t) - \frac{m^2\omega^2}{\alpha - m\gamma} x(t)\right), \tag{22.111}$$

which is of the form $A(t+\Delta t) = A(t) + \Delta t \lambda A(t)$ if we have

$$\alpha = -\frac{m^2\omega^2}{\alpha - m\gamma} \quad \text{or} \quad \alpha^2 - m\gamma\alpha + m^2\omega^2 = 0, \tag{22.112}$$

which can be immediately solved to yield

$$\alpha = m\left(\frac{\gamma}{2} \pm i\sqrt{\omega^2 - \frac{\gamma^2}{4}}\right), \tag{22.113}$$

where we assume that $\omega > \frac{\gamma}{2}$ for the underdamped regime. One can see we actually have two functions which are complex conjugates of each other $A(t)$ and $A^*(t)$. We

use the minus sign for A and the plus sign for A^*. We find the coefficient in front of the A term for its equation of motion becomes

$$\lambda = -\left(\frac{\gamma}{2} + i\sqrt{\omega^2 - \frac{\gamma^2}{4}}\right), \tag{22.114}$$

with the complex conjugate λ^* for the A^* function.

We have already seen how to determine the time-dependent solution to these problems for all times once we know how they behave for one time step. In particular, we find after N steps we have

$$\begin{aligned} A(t_0 + N\Delta t) &= \left(1 - \Delta t\left(\tfrac{\gamma}{2} + i\sqrt{\omega^2 - \tfrac{\gamma^2}{4}}\right)\right)^N A(t_0) \\ &\approx e^{-N\Delta t(\frac{\gamma}{2} + i\sqrt{\omega^2 - \frac{\gamma^2}{4}}) + O(N(\Delta t)^2)} A(t_0). \end{aligned} \tag{22.115}$$

Taking the limit where $\Delta t \to 0$, while $N\Delta t \to t$ then gives us

$$A(t) = e^{-\frac{\gamma t}{2} - i\sqrt{\omega^2 - \frac{\gamma^2}{4}}t} A(0), \tag{22.116}$$

where we set $t_0 = 0$. We have $A(0) = \left(p_0 + m(\frac{\gamma}{2} - i\sqrt{\omega^2 - \frac{\gamma^2}{4}})x_0\right)/\sqrt{2m}$ for the initial position x_0 and initial momentum p_0. You might say that this $A(t)$ is nice, but how did it solve the problem? After all, I wanted $x(t)$. We can obtain that as well. We have

$$\begin{aligned} x(t) = \frac{A^*(t) - A(t)}{i\sqrt{2m}\sqrt{\omega^2 - \frac{\gamma^2}{4}}} &= x_0 e^{-\frac{\gamma t}{2}} \cos\sqrt{\omega^2 - \frac{\gamma^2}{4}}t \\ &+ \frac{x_0\gamma}{2\sqrt{\omega^2 - \frac{\gamma^2}{4}}} e^{-\frac{\gamma t}{2}} \sin\sqrt{\omega^2 - \frac{\gamma^2}{4}}t \\ &+ \frac{p_0}{m\sqrt{\omega^2 - \frac{\gamma^2}{4}}} e^{-\frac{\gamma t}{2}} \sin\sqrt{\omega^2 - \frac{\gamma^2}{4}}t. \end{aligned} \tag{22.117}$$

This is plotted in Fig. 22.4 for some typical values of the initial condition and the damping. Note that the frequency of the oscillation changes with the damping rate γ, which is an important aspect of the motion.

If we look at the function $A(t)$, we can see that if we set $\gamma = 0$, then we find that it has the same functional form as the lowering operator for the quantum harmonic oscillator (using the Schrödinger form of the ladder operators). This motivates us to quantize our system and examine $\hat{A}^\dagger\hat{A}$. Using $\hat{A} = \left(\hat{p} + m(\frac{\gamma}{2} -\right.$

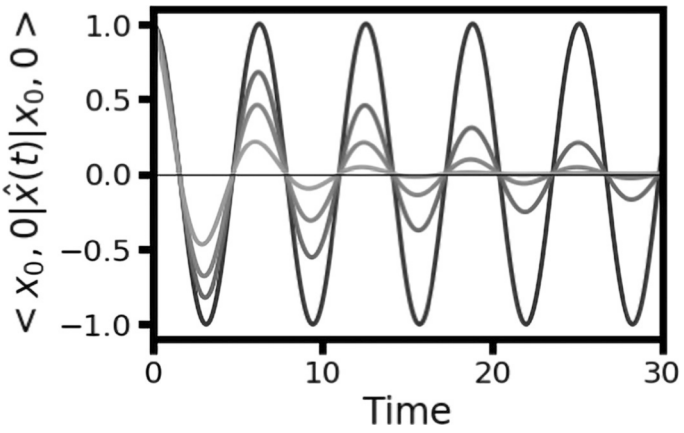

Fig. 22.4 Position versus time for a classical damped harmonic oscillator with $x_0 = 1$, $p_0 = 0$, $\omega = 1$, and $\gamma = 0$, 0.125, 0.25, and 0.5. You can see that as the damping increases, we also see the period grows as γ grows, although the effect is small in this example

$i\sqrt{\omega^2 - \frac{\gamma^2}{4}})\hat{x}\big)/\sqrt{2m}$, we find that

$$\hat{A}^\dagger \hat{A} = \frac{\hat{p}^2}{2m} + \frac{\gamma}{4}(\hat{x}\hat{p} + \hat{p}\hat{x}) + \tfrac{1}{2}m\omega^2\hat{x}^2 - \tfrac{1}{2}\hbar\sqrt{\omega^2 - \tfrac{\gamma^2}{4}}. \tag{22.118}$$

This can be rearranged to the form

$$\hat{H} = \hat{A}^\dagger \hat{A} + \tfrac{1}{2}\hbar\sqrt{\omega^2 - \tfrac{\gamma^2}{4}} = \underbrace{\frac{\hat{p}^2}{2m} + \tfrac{1}{2}m\omega^2\hat{x}^2}_{\hat{H}_0} + \underbrace{\frac{\gamma}{4}(\hat{x}\hat{p} + \hat{p}\hat{x})}_{\hat{H}_{\text{LS}}}. \tag{22.119}$$

We have separated the Hamiltonian into the original harmonic oscillator Hamiltonian $\hat{H}_0$ and the so-called Lamb-shift Hamiltonian $\hat{H}_{\text{LS}}$, which is proportional to the damping rate. The superpotential is now complex valued, because the Hamiltonian contains terms linear in the momentum, just as we discussed back when we described the rules for the factorization method in Chap. 11. You can see from the constant term on the left that the ground-state energy of $\hat{H}$ is proportional to the damping-rate-modified frequency $\sqrt{\omega^2 - \frac{\gamma^2}{4}}$. So, the addition of the Lamb-shift term changes the frequency of the harmonic oscillator, just as what happened in the classical case. You will show in Problem 22.7.1 how to find all of the energy eigenstates of this Hamiltonian. Previously, when we derived the Lindblad equation, we did not discuss the Lamb-shift term of the Hamiltonian, but we see here that the energies of the system change due to the damping (or decoherence), so we must include an extra term in the Hamiltonian to take into account this change in energies. Fortunately for us, the precise form arose naturally by using an analogy with the

classical results. In the general case, one must carefully determine the Lamb-shift Hamiltonian by "integrating out" the reservoir that provides the damping.

Because we have a dissipative system, we must describe its time evolution not just with the modified Hamiltonian $\hat{H}$ but also with the Lindblad terms, which provide dissipation to the density matrix. The general form of the master equation remains as given in Eq. (22.58). The only issue to pay attention to is that the $\hat{H}$ that appears in that equation is $\hat{H} = \hat{H}_0 + \hat{H}_{\rm LS}$.

While it remains true that we can solve the problem by solving for the density matrix as a function of time and using it to evaluate all relevant expectation values, it is often convenient to reformulate the master equation to the equivalent of the Heisenberg picture and to work directly with the time-dependent operators in the Heisenberg representation. To derive this picture, we first look at expectation values of general operators and then extract from them a general formulation for how the operators evolve in time. This is what we do next.

The strategy to do this is straightforward—we write down the matrix elements of the product of the density matrix and the operator in question. We then evolve the time by a time step Δt by using the Lindblad equation. Next, we use the invariance of the trace $\mathrm{Tr}(\hat{A}\hat{B}) = \mathrm{Tr}(\hat{B}\hat{A})$ to move the density matrix to the leftmost position in every term. Finally, we remove the trace and the density matrix to construct the operator equation for the time evolution of the operator in the Heisenberg picture. Let's get started. We have

$$\langle \hat{O}(t+\Delta t)\rangle = \mathrm{Tr}\hat{\rho}(t+\Delta t)\hat{O} = \mathrm{Tr}\hat{\rho}(t)\hat{O} + \Delta t \mathrm{Tr}\left(-\tfrac{i}{\hbar}[\hat{H},\hat{\rho}(t)]\hat{O} + \hat{L}\hat{\rho}(t)\hat{L}^\dagger\hat{O} - \tfrac{1}{2}\hat{\rho}(t)\hat{L}^\dagger\hat{L}\hat{O} - \tfrac{1}{2}\hat{L}^\dagger\hat{L}\hat{\rho}(t)\hat{O}\right). \tag{22.120}$$

Now, we use the invariance of the trace to move the density matrix operator all the way to the left in each term. This gives

$$\mathrm{Tr}\hat{\rho}(t+\Delta t)\hat{O} = \mathrm{Tr}\hat{\rho}(t)\hat{O} + \Delta t \mathrm{Tr}\left\{\hat{\rho}(t)\left(\tfrac{i}{\hbar}\hat{H}\hat{O} - \tfrac{i}{\hbar}\hat{O}\hat{H} + \hat{L}^\dagger\hat{O}\hat{L} - \tfrac{1}{2}\hat{L}^\dagger\hat{L}\hat{O} - \tfrac{1}{2}\hat{O}\hat{L}^\dagger\hat{L}\right)\right\}. \tag{22.121}$$

Next, we remove the trace operation, and we remove the density matrix from the left to find the equation of motion for the now time-dependent operator $\hat{O}(t+\Delta t)$ which becomes

$$\hat{O}(t+\Delta t) = \hat{O}(t) + \Delta t\left(\tfrac{i}{\hbar}[\hat{H},\hat{O}(t)] + \tfrac{1}{2}\hat{L}^\dagger[\hat{O}(t),\hat{L}] + \tfrac{1}{2}[\hat{L}^\dagger,\hat{O}(t)]\hat{L}\right). \tag{22.122}$$

We now apply this to the lowering operator $\hat{A}$ and force the quantum equation of motion to be the same as the classical one to properly determine the jump operators.

In order to perform this calculation, we will need the commutator of the ladder operators, which is worked out in Problem 22.7.2 below. We choose the jump

operator to be $\hat{L} = \beta\hat{A}$ for a constant β yet to be determined. In the classical solution, we saw that $A(t) = f(t)A(0)$, and we expect the same to occur here, so we find that

$$\begin{aligned}\hat{A}(t+\Delta t) &= \hat{A}(t) + \Delta t\left(\tfrac{i}{\hbar} + \tfrac{|\beta|^2}{2}\right)[\hat{A}^\dagger, \hat{A}(t)]\hat{A}. \\ &= \hat{A}(t) - \Delta t\left(\tfrac{\hbar|\beta|^2}{2}\sqrt{\omega^2 - \tfrac{\gamma^2}{4}} + i\sqrt{\omega^2 - \tfrac{\gamma^2}{4}}\right)\hat{A}(t). \end{aligned} \tag{22.123}$$

In order to reproduce the classical equation, we must have

$$\beta = \sqrt{\frac{\gamma}{\hbar\sqrt{\omega^2 - \frac{\gamma^2}{4}}}}. \tag{22.124}$$

Because the coefficient of the Δt term is a number, we can immediately find the time-evolved lowering operator, which becomes

$$\hat{A}(t) = e^{-\frac{\gamma t}{2} - i\sqrt{\omega^2 - \frac{\gamma^2}{4}}\,t}\hat{A}(0). \tag{22.125}$$

Indeed, this is a function of time multiplying the operator at $t = 0$, just as we said it would be. If we now take the expectation value with respect to the coherent state $|x_0, p_0\rangle$, which satisfies $\langle x_0, p_0|\hat{x}|x_0, p_0\rangle = x_0$ and $\langle x_0, p_0|\hat{p}|x_0, p_0\rangle = p_0$, then $\langle x_0, p_0|\hat{x}(t)|x_0, p_0\rangle$ is exactly equal to the $x(t)$ given in Eq. (22.117). This shows that the classical damped harmonic oscillator and the quantum damped harmonic oscillator have the same time-dependent behavior for their expectation values. Of course, one can examine other time-dependent expectation values for the quantum system as well.

Note that there is an alternative way to formulate the damped harmonic oscillator that has two independent jump operators $\hat{L}_1 = \beta\sqrt{(N+1)}\,\hat{A}$ and $\hat{L}_2 = \beta\sqrt{N}\,\hat{A}^\dagger$. One can think of N as the average number of phonons in the bath and the previous calculation we discussed as examining the $N = 0$ case. We can work out the equations of motion in this case as well, but we leave that for Problem 22.7.8 for you. This case corresponds to how the system would be described if the harmonic oscillator was in contact with a thermal bath which caused the damping. In that case, a specific functional form of N on temperature would be used corresponding to a Bose distribution.

Section 22.7 Problems:

22.7.1 Energy Eigenstates of $\hat{H}_0 + \hat{H}_{LS}$

Using the form of the ladder operators and the expression of the Hamiltonian as $\hat{A}^\dagger\hat{A} + \frac{1}{2}\hbar\sqrt{\omega^2 - \frac{\gamma^2}{4}}$, determine the energy eigenstates and the energy eigenvalues for $\hat{H}_0 + \hat{H}_{\rm LS}$.

22.7.2 Commutation Relation of the Ladder Operators

Show that $[\hat{A}, \hat{A}^\dagger] = \hbar\sqrt{\omega^2 - \frac{\gamma^2}{4}}$. Explain what happens as $\gamma \to 0$.

22.7.3 Quantum Equations of Motion for Position and Momentum

Derive the quantum equations of motion, expressing $\hat{x}(t + \Delta t)$ in terms of objects evaluated at time t and similarly for $\hat{p}(t + \Delta t)$.

22.7.4 Time Dependence of Momentum for the Damped Harmonic Oscillator

Choose reasonable values for x_0, p_0, ω, and γ, and plot the expectation value of $\hat{p}(t)$ in the appropriate coherent state. Compare it to the plot of $x(t)$ in Fig. 22.4.

22.7.5 Density Matrix of the Damped Harmonic Oscillator

In the main text, we showed how to determine the expectation values of Heisenberg-picture operators instead of solving for the time-dependent density matrix. Starting with the system in a pure state, given by the coherent state $|x_0, p_0\rangle$, solve for the time-dependent density matrix operator. Can you solve for $\hat{\rho}(t)$ if you start in a thermal state at a nonzero temperature?

22.7.6 Uncertainty in Damped Motion

Compute the uncertainty for position and momentum as a function of time in the damped harmonic oscillator. What happens to the uncertainty principle as a function of time? Recall that in a damped classical system, the momentum goes to zero, implying there is no remaining motion. In a damped quantum system, the average value of the momentum goes to zero, but this does not mean all motion has stopped, similar to what happens in the ground state of the undamped harmonic oscillator. Note that the final results you obtain for the uncertainty here are *different* from that of the harmonic oscillator ground state of $\hat{H}_0$. See if you can figure out why.

22.7.7 Driven-Dissipative Quantum Harmonic Oscillator

Add a driving force to the initial equations of motion that are a function of time, given by $F_0 \cos \omega_0 t$. Determine $\hat{x}(t)$ now. If you know how to solve the problem in the classical case, compare the solutions.

22.7.8 Alternative Master Equation Approach

Set up the master equation with the two jump operators described at the end of the text. Determine the β value that produces the same results as the classical case when $N = 0$ for the time evolution of position and momentum. Here, we have additional N dependence that was not present before. If brave, compute the uncertainty as well.

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Chapter 23
Real Single Quanta Measurements

Quantum mechanical phenomena do not occur in a Hilbert space. They occur in a laboratory.

Asher Peres, Physicist

The theoretical description of measurement in quantum mechanics is one of the most controversial topics in physics. It is deeply entrenched in the interpretations of quantum mechanics and combines both physics and philosophy into the discussion. Often it is described as the *measurement problem* because of some issues associated with the theory of measurement introduced by John von Neumann in 1932, related to the fact that the projection postulate is not unitary and so is not described by the time-dependent Schrödinger equation. We discuss that theory later in this chapter, but we start from a simpler approach focused on what is actually being done in most single quantum experiments. Along the way, we will encounter a number of counterintuitive and subtle issues, before we explain why we refuse to tackle the measurement problem itself. See a summary of all of the juicy details at https://tinyurl.com/yc2syjb4.

In a classical world, we can, in principle, find measuring instruments smaller than whatever we seek to measure and have them hardly disturb the object. This no longer works in the quantum world, because the measurement of individual quanta requires us to measure the smallest things in the universe. So we cannot use detectors smaller than the quanta we seek to measure in order to discover their properties. This alone implies we need to rethink how we approach measurement in the first place. Every measurement must be amplified to make the signal visible to the experimenter. Our focus in this chapter will be on single or few quanta measurements as these are the most relevant for quantum phenomena. The approach we take is quite different from that of others. I hope it will provide a fresh new perspective on the problem of measurement.

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23.1 General Strategy to Measure Individual Quanta

In a quantum measurement, we usually have probabilities for a number of different measurement results to occur for a given quantum state. We have described the measurement process using the projection rule and the Born rule, which both comprise the so-called *Copenhagen interpretation* of quantum mechanics. But, when we think of what a measurement is trying to do mathematically, it is to transform the original quantum state (pure, mixed, or in between) into a fully mixed state with no off-diagonals in the density matrix, when expressed in the eigenbasis corresponding to the quantity being measured. Such a state can be described in words as having different probabilities to be in the different eigenstates, so if we have the precision to measure these different states, then our measurement task will be complete (as we can directly measure the different values and their probabilities to occur without ever worrying about collapse). Note that we usually do not need to worry about what the system goes into after the measurement to describe the measurement itself. Most quantum experiments are not repeated, so we need not know about what happens afterward in those cases. But in some cases, it is important, and we will also discuss that later in the chapter when we describe the quantum Zeno effect.

One of the simplest ways to do this for all quanta is via a counting experiment, since nearly all quantum states preserve the quantum number of the state, and single-quanta experiments involve states with particle numbers of zero or one only, the density matrix essentially starts off in the form required for measurement (as we will see below, it assumes a "block-diagonal" density matrix form). There are exceptions, such as coherent and squeezed states, superconducting states, and so forth.

The fact that a mixed state of this form can be thought of as a probabilistic collection of the different experimentally measured values, with the probability coming from the square of the amplitude of the wavefunction, allows us to interpret the Born rule directly, without ever needing any complicated collapse of the wavefunction. In this sense, one can view the Copenhagen interpretation as a mathematical shortcut to achieving this result. In general, it is not so simple to transform the system to this type of mixed state, and we discuss some ways to achieve this below—counting experiments almost always achieve this form with respect to the number states, which is why they are the most common way to measure individual quanta. They simply are preset to work all the time if we can achieve single-quanta sensitivity. This leads us to the three most critical criteria needed for quantum experiments—single-quantum sensitivity, an ability to amplify the signal so it can be measured with classical detectors, and the requirement that the measuring equipment is reproducible. No detectors work perfectly, so the efficiency of the measuring device always needs to be taken into account as well.

Let us elaborate in these principles a bit more. In order to detect single quanta, we need the sensitivity to respond to the presence of a single quanta. We have seen examples of this before with photodetectors and particle detectors, which have a response to the presence of the quanta being measured. As an example, in a

photodiode, the process involves the conversion of a single photon to an electron via the photoelectric effect, while in a photon avalanche diode, it involves the excitation of the electron from a lower energy band to a higher energy band (by absorbing the photon), where the electron can conduct. Then the signal must be amplified. In the photodiode, the amplification effect comes from the chain of dynodes that multiply the number of electrons at each step, while in an avalanche detector, a high voltage accelerates the electron, which then scatters and creates additional electrons from impact ionization processes. Finally, we detect the current of the electrons using classical ammeters with high sensitivity. The reproducibility condition is that the device will respond the same way to the same incident quanta after it is reset. In many cases, these detectors serve also as energy filters, as the quanta needs to have a large enough energy to trigger the device. As we have seen, these different detectors can operate efficiently (but not perfectly) in counting individual quanta. Another example we discussed is the cycling transition in atoms. Here, light of the right frequency is shone onto a collection of atoms (corresponding to a transition from the ground state to an excited state). The light is absorbed and then reemitted by the atom shortly thereafter (typically a few nanoseconds). By repeating this process many times, we generate enough photons that they can be picked up by a camera that records the position of the atom. This is an example where repeating the measurement is the process that amplifies the signal.

If our main tool is a detector that counts individual quanta, we need to choose our experimental design to be able to reveal what we want to measure via the setup and the geometry of the experiment. We have again seen many examples of this before. A good one to discuss is a time-of-flight measurement, which detects when a particle is measured in a detector via the time elapsed from the start of the event to the detection. Then we infer the momentum from the distance traveled divided by the elapsed time and multiplied by the particle mass. The detector just tells us when to stop the clock. This type of strategy, where clever design is critical to determining what is actually being measured, is the key to how experimental work is done in quantum mechanics. This is often the single most important step in designing a new quantum experiment and is the true "measurement problem." Yet, curiously, it usually is not part of most theories of measurement! Nor, is it often discussed in quantum textbooks.

Let us discuss counting experiments in a bit more detail. While I mentioned above that counting experiments automatically produce block-diagonal density matrices (in the Fock basis for the photons) for single-photon experiments, this might not be clear just how it is done, so we will get more concrete. An experiment is simply a state preparation followed by a projection. The easiest way to see the block-diagonal nature of the density matrix is to consider the state preparation as the creation of the photon state, its manipulation through the experiment, followed by its measurement. We illustrate with single-mode photons. The single-mode single-photon state is created from the vacuum via $\hat{a}_\omega^\dagger|0\rangle$, where we use the label ω to indicate the monochromatic source with frequency ω. We consider traveling through a Mach-Zehnder interferometer illustrated in Fig. 23.1. As the photon moves through the device to the detector, the beam splits into two, the photon is

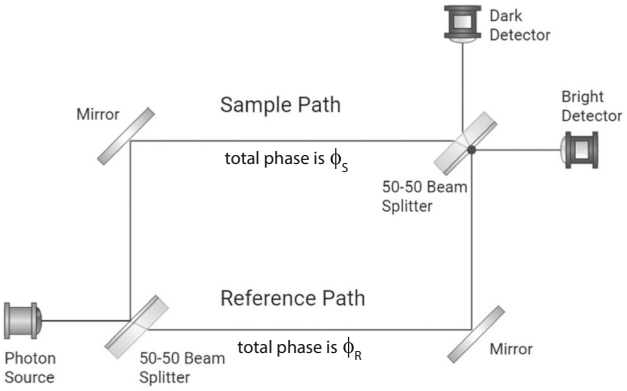

Fig. 23.1 Schematic of a Mach-Zehnder interferometer with the sample and reference paths and two ordinary beam splitters. The total phase along the sample and reference paths (not including the mirrors) is $\phi_{S,R}$

in an indeterminate state traveling on the sample and reference paths, and then it reaches the final beam splitter, traveling to the dark and bright detectors (also in an indeterminate state). If $\phi = \frac{1}{2}(\phi_S - \phi_R)$ is the difference in the phase for the sample and reference paths, then we have the state that approaches the detectors is in a pure-state superposition of the form

$$\begin{aligned} |\psi\rangle &= e^{\frac{i}{2}(\phi_R+\phi_S)}\big(\cos\phi|B\rangle - i\,\sin\phi|D\rangle\big) \\ &= e^{\frac{i}{2}(\phi_R+\phi_S)}\left(\cos(\phi)\hat{a}_B^\dagger|0\rangle - i\,\sin(\phi)\hat{a}_D^\dagger|0\rangle\right). \end{aligned} \tag{23.1}$$

Note that we colloquially call the state inside the Mach-Zehnder interferometer as a superposition between the two paths, but in quantum optics, they prefer the language to be more specific and refer to it as a superposition of the state with one photon on the sample path and no photon on the reference path and the state with no photon on the sample path and one photon on the reference path; this does not change any of the formal expressions we use to describe these experiments. We let $\hat{a}_{B,D}$ denote the destruction operator for a photon at the bright (B) or dark (D) detector, then the state the measurement projects onto will be $\langle 0|\hat{a}_{B,D}$; this also follows from using the electric field operator to measure the field at the detector. The density matrix becomes block diagonal in that it has nonzero matrix elements only between the prepared state and the single-photon states corresponding to the detectors. Note that it is not a diagonal matrix within this single-photon space, because we obtain a nonzero off-diagonal matrix element for the off-diagonal coherences (as happens with any pure-state superposition). So the diagonal nature of the density matrix is a block diagonal form according to the number states of the photons. At the moment of measurement, the photon is destroyed, which leaves no doubt about precisely when the experiment occurred. And the probability is given by the diagonal element of the density matrix. There is no ambiguity and no issues with interpretation for

such an experiment. The block-diagonal form of the density matrix provides the appropriate "block mixed state" such that the measurement process is unambiguous, because the off-diagonal elements cannot enter into the measurement probabilities, with the way that the single-photon detection occurs for these detector states. This is because the detectors are located at disjoint spatial positions and single-photon states can be detected once and only once. The only way to measure the coherences would be to bring both beams back to a beam splitter again, but that defeats the purpose of the original measurement. This type of analysis also works well for the Stern-Gerlach experiment and all destructive forms of single-photon detection of single-photon states. When we use the cycling transition of an atom to locate it in space, the experiment is nondestructive and can be repeated. We will discuss this example in more detail later when we discuss the quantum Zeno effect.

Is such a "block-diagonal mixed state" the same as a true mixed state? No, they are different, because all coherences have not been taken to zero. But, if we focus solely on the particle number in the analysis, and the experiment does not allow measurement of any operators that would be sensitive to off-diagonal elements, then we cannot tell the difference between them. This is the spirit under which these types of counting experiments require no additional theory to describe them. They occur at a definite moment in time, and they have clearly determined probabilities. The approach also works if the photon state is in a superposition, which we will discuss in more detail when we analyze the two-slit experiment.

One of the most important experiments is measuring the frequency of a transition (usually an atomic transition). The way such an experiment is set up is to correlate a position on a detector screen with the energy (or frequency) of the photon. We achieve this by first directing the light onto a diffraction grating. Then, by placing photon detectors on a screen away from the grating, we can correlate the measurement of position with a frequency. This is essentially a counting experiment, even if it is often performed with states of light that contain many photons. If, however, we perform it with a single-photon source, then we would see the signal buildup over time as the different photons are detected and counted. Hence, the amplification step for this experiment can be thought of as simply repeating it enough to obtain a clear signal of the results.

Often, experiments are described as entangling the quantum degrees of freedom with the classical degrees of freedom of the measuring apparatus. We will discuss this scenario in detail later in the chapter. As you might expect, it is not a realistic way of performing most experiments. But, there is a similar type of experiment that employs interference instead of entanglement to perform the measurement and that is balanced homodyne detection. In such an experiment, we use an intense local oscillator field that contains many photons, so we can think of it as being a classical state of light (indeed, it usually comes from a laser, so it really is a coherent classical state of light). Then we let the single-photon state (or more commonly a low-amplitude signal) to come into the other port of the beam splitter. We count the difference in the detections of the two detectors coupled to the output ports, and the oscillating signal tells us about the low-amplitude photon states. One usually uses avalanche detectors for this experiment, and they can reach close to 100% efficiency.

I will now take a moment again to reiterate the importance of high precision experiments for advancing science. These experiments are almost uniformly quantum in their nature, and require us to amplify the weak signals after we have devised the experimental geometry to yield the information we are seeking. They usually require us to measure rare events, which is why high precision is required, so the signal can rise above the noise. As an example of how this works, we revisit the Lamb-shift experiment to describe how remarkable it is. As a reminder, this experiment is designed to measure the difference in energy between the $2S_{\frac{1}{2}}$ and $2P_{\frac{1}{2}}$ states in atomic hydrogen. These states are degenerate in the relativistic Dirac theory but split their energies due to quantum field theory effects. One might have thought that one simply measures the frequencies of the transitions from these states to the $n = 1$ shell, but this is problematic, because the transition from 2S to 1S is forbidden and requires a two-photon transition. Furthermore, the precision needed is far beyond the precision we can obtain with most measurements of the frequency of light. Lamb's idea was different and ingenious. He used the fact that the 2P state immediately decays, while the 2S state is metastable. This allows him to easily create a beam that has 2S states in it, but not 2P states, and he has a detector that can detect 2S hydrogen as different from 1S hydrogen. Then he uses a radio frequency pulse to create transitions from the 2S to the 2P state. These hydrogen atoms immediately decay to the 1S state (within a nanosecond or so) and hence by measuring the reduction in the 2S hydrogen atoms at the detector, he is able to find the resonance frequency for the transition, which is the energy splitting. Even though the experiment was done in 1946, with equipment available then, it could find the energy splitting to two to three digits of accuracy. It is all about being creative in your experimental design to achieve remarkable results. Note how this experiment too is a counting experiment.

One of the key issues for the mainstream measurement problem is that our quantum theory does not tell us when the measurement occurs. This is not an issue for destructive measurements though. This is because the destructive nature of the experiment occurs at a precise time, and hence it signals precisely when the measurement process occurred. As we know, photon destruction (as with other quanta) is not a unitary process. But, this is OK, because measurement is inherently irreversible (in most cases), and hence this nonunitary nature leads to a clear identification for when the measurement occurs. It does require us to work with second quantization and quantum field theory to properly describe it. Furthermore, the theory does not predict when the experiment will occur. That is always uncertain to some degree depending upon the process involved. We will discuss this point in more detail later as well.

Nevertheless, the issue of what is collapse and what does it mean has a number of different connotations. If you are an *ontic*, then you believe that the quantum state, or even the wavefunction, is the real physical object and collapse is a physical effect that corresponds to a real change in the system. As we have discussed when thinking of photon detection coming from a distant star, such an interpretation has difficulty with relativity. A more common view is a statistical interpretation. Here, we view the

quantum formalism as just a mathematical tool used to predict the probabilities of measurements, which are realized only after repeating the experiment many times and then examining the statistical distribution of the measurement results, which should be close to the probabilities that the mathematical theory predicted. For the statistical viewpoint, collapse is just an update to the information about the quantum state, which is modified by the measurement in the experiment. It then is not a significant physical effect that changes the quanta themselves. It just changes the information we have about the quanta. Furthermore, the statistical interpretation allows us to be looser on determining precisely when a transition occurs, because the theory predicts the average, statistically, over many repeated measurements, rather than what happens with each individual shot.

One aspect of the whole measurement process that is often not discussed is how long does it take for a transition to occur? Is it instantaneous, or does it take time to occur? How do we picture the process? To further complicate the issues, in quantum mechanics, we know that very small objects can create objects that are much larger in size. Consider an atom, which is on the order of an Å in size that emits a photon. The lifetime of the excited state is typically on the order of a nanosecond, creating a photon whose size is on the order of a meter (the size of the photon is the overall spatial extent of the electromagnetic field it generates). How can this be? Can we instantaneously create a photon of such a size from the atomic transition? Our theory so far has been silent on this issue. So let us now discuss it in more detail.

To begin, we need to be clear that saying what happens to something when we do not measure it is a dangerous prospect in quantum mechanics. We have seen time and time again how hard it is to figure out how the quantum mysteries work. Nevertheless, in this case, we can make some observations about what might happen by bringing in classical ideas and incorporating them into our quantum language. Let us focus on photon absorption in an atom starting from the ground state. We have a photon incident on the atom, which has a nonzero electric and magnetic field associated with it. The electric field will accelerate the electron, which can be modeled by it creating a superposition between the ground state and the states it is coupled to. This time-dependent superposition oscillates, creating an electric field that starts to cancel the field of the photon. Then at some randomly chosen time, near the lifetime of the excited state, the transition occurs, with the system moving into the excited state. One can think of this as generating a "pre-transition state" prior to the transition taking place. We still view the transition itself as being an instantaneous transition, where the photon is destroyed and the system settles into the excited state. But we physically can understand the behavior a bit better by keeping such a picture in mind. The emission process is similar. One starts to have a superposition with the ground state, which generates an electric field that ultimately becomes the photon once the transition occurs at a randomly chosen time. The key to this description is that it gives a reasonable picture for how a photon many times larger than the atom is shed from it or absorbed by it. Note that this picture is very different from an instantaneous transition mediated by a particle picture. This discussion is not fully rigorous but does provide an interesting viewpoint about how quantum transitions may be thought to occur. It would be

intriguing to try to measure this directly, and it is possible the cavity QED work of Serge Haroche might shed light on just these types of processes. We can also obtain similar behavior if we assume the quantum state represents the results in a statistical fashion for a large number of atoms. In this case, the state that represents the statistical collection of atoms will have some atoms that have undergone the transition, and some that have not. These numbers change with time until all have transitioned for times much longer than the lifetime of the state. Finally, this picture helps us understand why so many people incorrectly say light "travels as a wave but is detected as a particle." This is because such words are used when one does not understand how a small quantum object can absorb a much larger sized photon in an instant. But, now that you know, please do not ever say quanta travel as waves but are detected like particles—the language is too imprecise for this complicated event.

No discussion of experiment would be complete without discussing how we measure momentum. I have written an article that discusses this in more detail (Freericks, 2023). Recall that this is most often done with a time-of-flight experiment. Let us recap how this works. We start a clock when an event occurs to release the particle (could be a photon pulse releasing an electron from a photoelectric effect, a particle-particle collision, dropping an atomic trap and so on) and then stop it when the particle is detected in a detector. If it moved in a force-free region, then we can determine the momentum from the velocity, found by dividing the distance from source to detector by the elapsed time. Note that this experiment always measures position as well, because the detector is physically located somewhere. Uncertainty plays no role in the measurement of each shot of the experiment and only enters when we look at the variance of the momentum after repeating the experiment many times; the uncertainty in the momentum is unchanged from what it was in the source. You can see that measuring momentum and position is not exclusionary—instead, we use the measurement of position to infer the momentum. Indeed, every detector is located at a position in space, so it will *de facto* always measure position along with whatever else is inferred from the geometry of the experimental setup.

As experiments become more sophisticated, it has become possible to perform *multimodal* experiments which measure more than one thing at the same time. Some examples are counting photons while also measuring their polarization, or measuring the momentum of an electron along with the projection of its spin. We describe the photon experiment carefully. There are two common ways that one can measure the polarization while counting a photon. The first one uses a calcite crystal, which is a birefringent crystal, having two paths the photon can go through in the crystal—one for each of the two perpendicular polarizations of light. We then place photodetectors where the two light beams emerge from the crystal and can then use it to determine the polarization *and* count the presence of the photon. One could also use a polarizing beam splitter for the same task. The second one involves using a time-of-flight detector to measure the momentum and then using a spin-sensitive detector to also measure the spin of the electron. These types of multimodal experiments allow for more information to be determined about the quanta than those that measure just one property at a time.

Quantum computers are one of the best platforms for examining single-quanta experiments. One can view a quantum code as the preparation of a quantum state, typically one that is in an entangled or product-state superposition, followed by a measurement of the probabilities to be in the different product states in the z-basis at the end. Mid-circuit measurements allow one to also create mixed states in the computation. The key to designing any specific quantum circuit lies in preparing the right state for the given computation. And doing it efficiently, because quantum computers often suffer from decoherence and other effects. While one can use tomography to measure all aspects of a quantum state, this is rarely done. Instead, we try to determine the answer we need by the probabilities which measure the diagonal elements of the density matrix of the final prepared state in the z-basis.

Section 23.1 Problems:

23.1.1 Fock State Single-Photon Measurements Are Block Diagonal Too

Show that if the initial state had exactly n photons in it, the density matrix still assumes a "block-diagonal" form. What happens to it if the initial photon state is a coherent state or a squeezed state of light?

23.1.2 Counting Experiments

Identify some of the counting experiments we have discussed in earlier chapters. Have we discussed any experiments that are not counting experiments? If so, identify them and describe what they measure.

23.1.3 How Long for the Transition?

The hyperfine transition has a lifetime of millions of years. Does this mean the transition takes such a long time to take place, similar to what we described in the text? What is the spatial extent of a photon that emerges from a hyperfine transition? Does it help to think of this in terms of the statistical modeling for a large number of atoms? Is there any way to verify "what is exactly going on with this transition?"

23.1.4 Quantum Beats

Quantum beats is a very intriguing phenomena. It shows oscillations in the decay of an atom from excited states if there are two excited states that decay to one (a

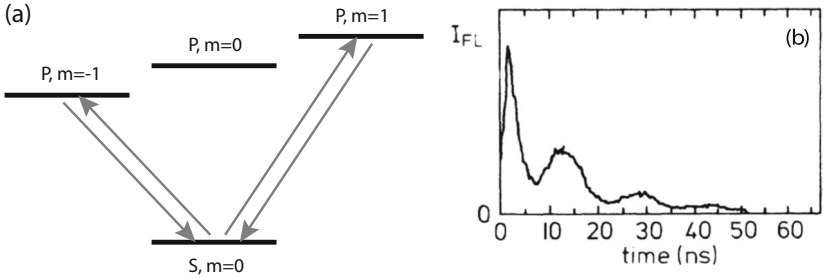

Fig. 23.2 (**a**) Energy–level diagram of a type I atom and (**b**) population of the ground state after excitation of the atom showing the oscillations, or quantum beats, in the population (Experimental data figure in panel (**b**) from Hellmuth et al. (1987) © American Physical Society)

so-called type I atom). But if we have an atom with two nearly degenerate ground states and one excited state (a so-called type II atom), then there are no beats—one only sees exponential decay in the population of the excited state. A semiclassical theory, however, always predicts that both cases will show quantum beats. So, this is a phenomenon that has quantum effects suppress the beats for the type II atoms. We describe this phenomena here using the full quantum theory. Our approach follows closely the development given in Greenstein and Zajonc (1997).

In Fig. 23.2, we show the energy-level diagram. This diagram arises when we have a singlet state as the ground state and a triplet state as an excited state. If we add a magnetic field in the z-direction, then the p-levels are split by the Zeeman effect. If the laser light is shone onto the light, with the polarization in the x-y-plane, only the $|p, m=\pm 1\rangle$ states can couple to the $|s, m{=}0\rangle$ state because of the selection rules for light polarized in the x-y-plane.

We start describing the system in the excited state, using our statistical approach, and the Heisenberg representation, where the time dependence for the dynamical phase is moved to the operators. Then at time $t = 0$, we have the state takes the form

$$|\psi(0)\rangle = A_+(0)|p, m{=}1\rangle \otimes |0\rangle + A_-(0)|p, m{=}-1\rangle \otimes |0\rangle + A_g(0)|s, m{=}0\rangle \otimes |0\rangle, \tag{23.2}$$

where $|0\rangle$ is the photon vacuum state. The fraction excited is given by $|A_\pm(0)|^2$, while the probability to still be in the ground state is $|A_g(0)|^2$. Now, because we can have transitions from the excited states to the ground state, at the time t, the state becomes

$$|\psi(t)\rangle = A_+(t)|p, m{=}1\rangle \otimes |0\rangle + A_-(t)|p, m{=}-1\rangle \otimes |0\rangle + A_g(t)|s, m{=}0\rangle \otimes |0\rangle + B_+(t)|s, m{=}0\rangle \otimes |1_\alpha\rangle + B_-(t)|s, m{=}0\rangle \otimes |1_\beta\rangle, \tag{23.3}$$

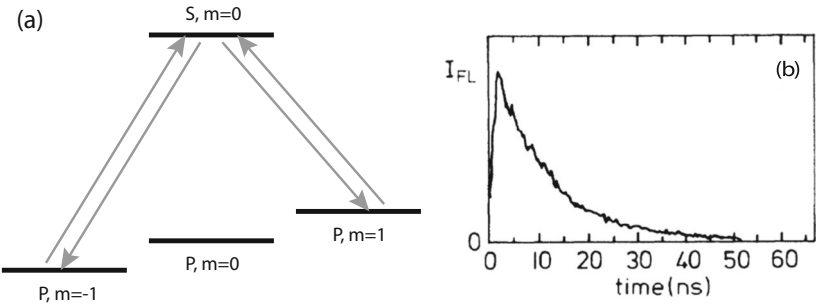

Fig. 23.3 (**a**) Energy-level diagram of a type II atom and (**b**) population of the ground state after excitation of the atom showing no oscillations, or quantum beats, in the population (Experimental data figure in panel (**b**) from Hellmuth et al. (1987) © American Physical Society)

where ω_α is the frequency of light for the transition from $|p, m{=}1\rangle$ to the ground state and ω_β is the frequency of light for the transition from $|p, m{=}-1\rangle$ to the ground state. The amplitudes change with time, even in the Heisenberg representation, as we deplete the excited states and add weight to the ground state.

Use the formalism we have developed to determine the probability to measure a photon in a detector by operating with $E^{(+)}$ and computing the norm-squared of the state. You should find a number of direct terms, and two terms that behave as $e^{\pm i(\omega_\alpha - \omega_\beta)t}$. Those terms determine the quantum beats, and their amplitude is nonzero, because $\langle s, m{=}0|s, m{=}0\rangle = 1$. Note that the photon counter acts as a high-pass filter, so it counts *all* photons with energy larger than some cut-off frequency, which is lower than ω_α and ω_β. In this case, the measurement cannot tell what the energy of the photon is, so it does not know which photon it has counted.

Repeat the calculation, now for a type II atom shown in Fig. 23.3. Show here that the beats disappear, because $\langle p, m{=}1|p, m{=}-1\rangle = 0$. Note that we must assume that the photon states that are created for the transitions between states are multimodal single photons with peaks located at ω_α and ω_β, respectively, and with spreads large enough that $\langle 0|\hat{a}_\alpha^\dagger \hat{a}_\beta|0\rangle$ is nonzero and is not exponentially small. This can be achieved by making the magnetic field small enough, so that the Zeeman splitting is smaller than the inverse lifetime of the states.

In a semiclassical calculation, there is no orthogonality condition that arises, so both cases are predicted to have quantum beats. For more details, you can consult (Greenstein and Zajonc, 1997).

23.1.5 Quantum Computation

Explain how quantum simulation/computing works by carefully preparing a specific quantum state and then measuring it. What kind of states might be interesting to

prepare and measure? What states might be easy for a quantum computer to prepare but difficult for a classical computer?

23.2 Entanglement, Product States, and Delayed Choice

Entanglement plays three prominent roles in quantum mechanics: (i) highly entangled systems display "spooky action at a distance" type correlations between its degrees of freedom; (ii) entanglement can tag a property of the system that is measured later via measuring the tag; and (iii) when a system interacts with an environment, the system-environment entanglement degrades the interference effects in the system. The first two of these effects are ones I want to have, and the last effect is one I want to minimize. In this section, we will explore all of these ideas about entanglement more closely, with an eye to the role it plays in measurement.

One of the key issues we have already seen in measurement is that single-state superpositions or product-state superpositions can display interference, while entangled superpositions do not. While the arguments are straightforward about how this works, these conceptual ideas do not seem to be well known by most practitioners unless they are discussed in the context of "which-way" information. As an example, consider a Mach-Zehnder interferometer with conventional beam splitters and incident light that is polarized along the diagonal direction as shown in Fig. 23.4. We will see the probabilities in the bright and dark detectors change as we vary the path length along the sample path. The system is in a product state, because we take the state of light and tensor it with the diagonal polarization, which is the same on both paths. Hence, we have $|\psi\rangle = \frac{1}{2}\left(e^{i\phi_S}|S\rangle - e^{i\phi_R}|R\rangle\right) \otimes (|V\rangle + |H\rangle)$ for the state incident on the second beam splitter. This state arises from the equal

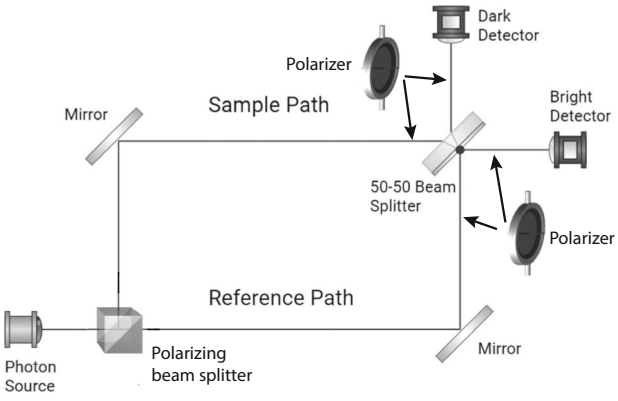

Fig. 23.4 Schematic of a Mach-Zehnder interferometer with the sample and reference paths and one polarizing beam splitter. We also show where polarizers can be placed inside the path or just before the detectors. The polarizer orientation can also be adjusted (not shown)

splitting into a superposition at the first beam splitter, followed by attaching the total phase evolved along the two different paths. We have no which-way information, because it is in a product state, so we see interference, with the probability changing as the phase changes. To complete the calculation, we simply pass through the second beam splitter, yielding

$$|\psi\rangle = \tfrac{1}{\sqrt{2}} e^{\frac{1}{2}(\phi_S+\phi_R)} \left(\cos\left(\tfrac{1}{2}(\phi_S-\phi_R)\right)|B\rangle + i\sin\left(\tfrac{1}{2}(\phi_S-\phi_R)\right)|D\rangle\right) \otimes (|V\rangle + |H\rangle). \tag{23.4}$$

Applying the projection onto the detector and polarization states and squaring the amplitude, we find a probability of $\cos^2\phi$ at the bright detector and $\sin^2\phi$ at the dark detector, with $\phi = \frac{1}{2}(\phi_S - \phi_R)$.

Contrast this to a Mach-Zehnder interferometer that has a polarizing beam splitter instead of an ordinary beam splitter as the first element (this reflects light with a horizontal polarization and transmits light with a vertical polarization). Now, the polarization tags the light on each path, resulting in the state $|\psi\rangle = \frac{1}{\sqrt{2}}\big(e^{i\phi_S}|S\rangle \otimes |H\rangle - e^{i\phi_R}|R\rangle \otimes |V\rangle\big)$ just before the second (ordinary) beam splitter, which is an entangled state that has tagged the path traveled by the polarization. Now, passing through the beam splitter, we obtain the state

$$|\psi\rangle = \tfrac{1}{2}\left(e^{i\phi_S}|B\rangle \otimes |H\rangle + e^{i\phi_R}|B\rangle \otimes |V\rangle + e^{i\phi_S}|D\rangle \otimes |H\rangle + e^{i\phi_R}|D\rangle \otimes |V\rangle\right) \tag{23.5}$$

Such an entangled state has no interference as a function of the phase on the sample path inside the interferometer, and so we always see half of the photons arriving in the bright detector and half in the dark detector. We have discussed this before in terms of having which-way information. Nevertheless, we do see that the entangled state is the sum of two pieces, one for each detector, with a complicated superposition of the polarization states in a product state with each detector ket. The key is that when we apply our projections and take the modulus squared, we find we always take the modulus squared of complex phases, which are equal to one, and do not interfere. Hence, the tagging of the system produced a final state that destroys the interference effects. We can bring the interference back by shining the light through a diagonal polarizer just before each detector; if we add the polarizer after the photon has gone through the interferometer, but before it goes through the detectors, this is a delayed-choice experiment. Half of the photons will be absorbed. For those not absorbed, the state just before measurement becomes

$$|\psi\rangle = \tfrac{1}{2\sqrt{2}}\left(\left(e^{i\phi_S} + e^{i\phi_R}\right)|B\rangle \otimes (|V\rangle + |H\rangle) + \left(e^{i\phi_S} - e^{i\phi_R}\right)|D\rangle \otimes (|V\rangle + |H\rangle)\right), \tag{23.6}$$

which supports interference effects, so the probabilities at the detectors now depend on the phase along the sample path once again. Note how the issue is more nuanced than just that the state is in an entangled state or a product state at each detector prior

to the measurement—it is the structure of the superposition that determines whether we have interference or not! Colloquially, you may hear of the light referred to as acting like a particle when there are no interference effects and as a wave when there are interference effects. Be careful of people who talk this way, because they often will say the character of the light is decided at the first beam splitter, which is a fallacy, because we create a superposition rather than perform a measurement there. Using that incorrect logic, they then will incorrectly discuss that the system reaches backward in time depending on whether there is a second beam splitter or not inserted into the system. At no point in time do we need retrocausality—the result of each experiment simply follows from the explicit state of the system just before the measurement and applying our projection hypothesis. Often, when a delayed choice experiment is discussed without Dirac notation and state vectors, it is hard to sort out exactly how the state evolves. This is especially so when one discusses the photon as being a particle or a wave, depending on the experiment. Instead, when the entanglement structure is properly formulated, it becomes quite easy to see how it works. You will find you will not make any mistakes in your analysis if you always write down the evolution of the quantum state through the experiment. It is surprising how many researchers neglect to do this. But you never will!

A detailed discussion of how a beam splitter works is critical for performing an analysis of interferometers. This goes under the name *measurement fallacy* in that one can confuse the proper nature of the beam splitter—that it creates a superposition, putting the system into an indeterminate state—with the incorrect nature of the beam splitter, that it performs a measurement, where the photon travels down one path or the other in a distinguishable fashion. We already know everything we need to know to analyze this situation. We start from the Grangier experiment (Grangier et al., 1986), where a photon from a single-photon source is measured after passing through a beam splitter. We find that it always is measured on one detector or another and never both. This can be used to establish that a photon can be measured once and only once. But it might make us think that the single photon is placed into a definite state of traveling on one path or another rather than an indeterminate superposition. We verify that it is a superposition state by adding mirrors and the second beam splitter to make a Mach-Zehnder interferometer. If it was definitely traveling on one path or another, it could not interfere, so that it will appear equally often in the bright and dark detectors. The fact that we do see the interference indicates that the photon must have been in a superposition state all along. This point is critical for properly analyzing interferometers.

John Wheeler introduced the idea of adding in the second beam splitter after the photon has passed through the first one (Wheeler, 1983) to show that the proper analysis of these systems requires us to follow what the quantum state is at each step through the interferometer. Our proper analysis of these experiments shows that one should never think of light in the colloquially popular fashion "travels as waves but is detected as particles." Wavefunction@Wave-particle duality is more complex than this. Quanta act simultaneously as waves and particles all the time. It is only via the choice of the experiment that we make that we see whether we

get a result that explores the particle nature (no interference) or the wave nature (interference). Obviously we can never see both in one experiment, because the two results are exclusionary. We never know precisely which character we will find until we analyze the full experiment. There are circumstances where we see partial interference, however, and these cases can be thought of as representing the intertwined wave-particle nature of quanta in more detail. Problem 23.2.2 discusses this in more detail.

Entanglement plays an important role in decoherence as well. Since we typically have that entangled states will not have interference, but product states will (although there are violations of this statement depending on the precise nature of the coefficients in the expansions of the states), the entanglement of a system with other degrees of freedom (the environment) will affect the system. A common way to reason about this is that even if the system starts off independent of the environment, because they interact, they will eventually form entangled states. Because we do not measure the environment, we trace out those degrees of freedom, creating a reduced density matrix for the state, which usually has reduced purity. When the entanglement with the environment is strong, we find the reduced density matrix of the system will have essentially no off-diagonal matrix elements, and hence the system becomes a fully mixed state. In this situation, we find that the reduced density matrix of a system has nonzero elements only on the diagonal (a fully mixed state with no coherences), and then this system can be thought of as a probabilistic mixture of the states used in the basis of the density matrix. This is precisely the way in which we must prepare a system to have a simple theory for measurement. Many view this decoherence from the environment as being the solution to the so-called measurement problem. But this approach does not apply to all measurements, so it can only partially solve the problem.

The prevailing view of decoherence is that the longer we have the system and environment interact, the less coherences we have in the density matrix. But, of course, if we think about the time evolution of the system plus environment, we can create situations where after some period of time the system and environment go to a product state again, so the reduced density matrix will have full coherences again. We can prevent that by making repeated measurements of the system at different times. Each measurement acts to lock in the decoherence of the system at that point in time, so that one cannot reverse the effects. One can also ask the question of how likely is it for the system and environment to become a product state again if we think of their interactions as being random in nature (as we often think of noise to be)? For random systems, the likelihood of creating a product state again is extremely unlikely. And this is the aspect usually invoked to say that decoherence will always reduce the coherences in the reduced density matrix. It is similar to taking a drop of colored paint and mixing it into a jar of white paint, the random interactions acting like stirring the paint. Once mixed, we cannot find any way to unmix the paint. In an analogous fashion, the quantum state loses its coherences due to entanglement from random interactions.

Section 23.2 Problems:

23.2.1 Measurement with an Entangled Superposition Versus a Product-State Superposition

We have seen a general rule that when we measure a system in an entangled superposition, it suppresses interference effects. Product state superpositions can support interference effects, but they can also provide situations where there is no interference, as we saw with the tagged photons in a Mach-Zehnder interferometer with one polarizing beam splitter. Write an essay carefully explaining the situations for when we can and when we cannot see interference effects. You can focus on the Mach-Zehnder interferometer only, if desired, but try to make a more general argument if you can.

23.2.2 Single Polarizing Beam Splitter Mach-Zehnder Interferometer

Consider a Mach-Zehnder interferometer with a polarizing beam splitter in the input port and the input light from the source having a diagonal polarization. We next put a polarizer at an angle of θ with respect to the vertical in front of each detector. Show that for nearly all θ, the system displays partial interference. Explain how this behavior relates to Wavefunction@Wave-particle duality.

23.2.3 Double Polarizing Beam Splitter Mach-Zehnder Interferometer

Analyze the Mach-Zehnder setup when we have two polarizing beam splitters, and we include diagonal polarizers on each arm of the interferometer and also before each detector (four polarizers in total). You should find that the probability to observe photons in each detector is now the same and depends on the phase difference in a simple way.

Recalculate the probabilities in the detectors when the polarizers placed in front of the detectors are at an angle θ to the vertical (you previously considered the case with $\theta = 45°$). You should find a more complex formula for the probabilities (still the same for each detector). Verify that when $\theta = 45°$, your general result agrees with the special result.

23.2.4 System-Environment Interaction

Consider a two qubit system with the first qubit the system and the second the environment. Start with the system qubit in the state $\alpha|\uparrow\rangle+\beta|\downarrow\rangle$, with $|\alpha|^2+|\beta|^2 = 1$, and the environment in the state $|\uparrow\rangle$. Compute the 4×4 density matrix. Take the partial trace with respect to the environment and show that the system density matrix remains in a pure state, because the initial state is a product state.

Let the system-environment interaction be given by a Heisenberg model, with a spin-exchange J. Expand the time evolution operator in a power series in time out to the second-order term and act it on your state. Show that it produces an entangled state, in general.

Form the density matrix of your system, and compute the partial trace with respect to the environment degrees of freedom. Show that the system is in a mixed state now for nonzero t, with a purity less than 1.

This is how system-environment interactions create entangled states and how those states decohere the system.

23.3 Theory of Measurement for the Two-Slit Experiment with a Single-Photon Light Source

In this section, we analyze the two-slit experiment in great detail, illustrating the concept of complementarity—when we have which-way information, there is no interference, and when we do not, we restore interference. The ideas for this section are nicely summarized in a *Scientific American* article illustrating how one can perform delayed choice two-slit experiments in an amateur laboratory (Hillmer & Kwiat, 2007). You may want to try to perform them yourself!

As with most single-quantum experiments, the two-slit experiment with a single photon source is a counting experiment. One sends photons through the device one-by-one and measures them in an array of detectors located on a "screen" far from the mask with the slits in them. We can describe the quantum state as a superposition of the quanta moving through the left and right slit $|\psi\rangle = e^{i\phi_L}|\psi_L\rangle + e^{i\phi_R}|\psi_R\rangle$, where the states corresponding to each slit have norms that represent the fraction of photons that make it through each slit (we chose the amplitude to be 0.1 in most of our discussions of this problem) and the phases are determined by the wavevector of the photon and the distance traveled to the screen. In this case, we explicitly choose just one path through each slit; this can be extended to include more paths if desired, but the analysis becomes more complex if we choose to do so—see the problems for more details. Using the simple approximation, we find the probability to measure the quanta at position x on the screen is given by

$$P(x) = |\langle x|\psi\rangle|^2 = (0.1)^2 4\cos^2\left(\frac{\phi_L(x)-\phi_R(x)}{2}\right) = 0.04\cos^2\left(\frac{\phi_L(x)-\phi_R(x)}{2}\right), \tag{23.7}$$

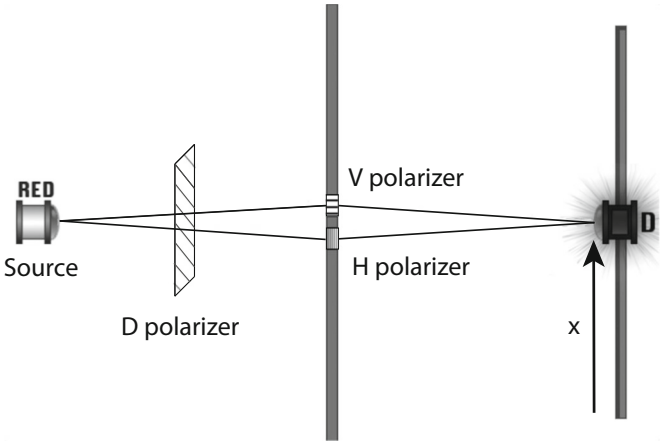

Fig. 23.5 Schematic of a two-slit experiment with crossed polarizers at the slits (viewed from the top)

where we assume that the amplitude of each term in the superposition is 0.1. The phases can be computed simply from the distance traveled from each slit to the position x on the screen, but we do not go into those details here. In this case, we cannot tell which slit the photon travels through. We say the state is an indeterminate state, cautioning that this does not mean it travels through one, or the other, or both, but instead is the quantum state where we do not know precisely how it moves through the slit except that it does so in superposition.

Our next step is to add crossed polarizers at the slits, say a V polarizer at the left slit and a H polarizer on the right slit (see Fig. 23.5). The source is set up to have diagonal polarization $\frac{1}{\sqrt{2}}(|V\rangle + |H\rangle)$ for the light incident on the slits. As the light passes through the slits, it is a measurement, because the polarizers irreversibly change the photon state. We normally assume that measurement collapses the quantum state into a specific result given by what we precisely measured. This case is a bit more nuanced. Here, half of the time the photon is absorbed. If it is not absorbed, then it travels through the slits in superposition. After passing through the slits, we have the state $|\psi\rangle = \frac{1}{\sqrt{2}}\big(|\psi_L\rangle \otimes |V\rangle + |\psi_R\rangle \otimes |H\rangle\big)$, which is an entangled superposition that has tagged the which-way information. This is an example of a measurement that does not collapse onto a single state but instead collapses the system onto an entangled superposition. This is achieved, because the measurement is measuring a parity of the system. If we map $\psi_L \to 0$, $\psi_R \to 1$, $V \to 0$, and $H \to 1$, then the polarizers enforce us to be in an even parity sector, given by the evenness of the sum of the two quantum numbers (here $0 + 0 = 0$ and $1 + 1 = 2$). The states with odd parity $|\psi_L\rangle \otimes |H\rangle$ and $|\psi_R\rangle \otimes |V\rangle$ ($0+1 = 1$ and $1+0 = 1$) are not present in the system after the measurement. They have been projected out. And this is exactly what the measurement has done–it has created a superposition from the measurement. These types of operations are important in quantum computing,

where we can work in a computation that always stays in a sector of definite parity. Then, if we measure the parity of the state, which can be thought of as filtering the state into a definite parity sector, then we do not disturb the state if it is of a definite parity. A generalization of these ideas is used in fault-tolerant quantum computation. The computational space is assigned to a specific symmetry sector, and parity measurements are used to see if the system remains in each parity sector. If it has moved out of the desired sector, a correction is applied to move it back into the right sector. The two-slit experiment is not a fault-tolerant quantum computer, but when we add polarizers, we are implementing ideas from fault tolerance into our system.

Now that we have tagged the quantum states, so we have which-way information, we find that if we measure at the screen, the interference is gone. We will go through the analysis of this experiment carefully. When we reach the screen, our state has become

$$|\psi\rangle = \tfrac{1}{\sqrt{2}}\big(e^{i\phi_L}|\psi_L\rangle \otimes |V\rangle + e^{i\phi_R}|\psi_R\rangle \otimes |H\rangle\big), \tag{23.8}$$

with the same phases as before when we had no polarizers. But now when we measure, we find that we must project onto the two different polarizations as well, even if we do not directly measure the polarization. This then creates a probability given by

$$P(x) = \tfrac{1}{2}(0.1)^2 \times (|e^{i\phi_L}|^2 + |e^{i\phi_R}|^2) = \tfrac{1}{2}(0.1)^2 \times 2 = 0.01. \tag{23.9}$$

The interference is now gone. We can restore it by adding a diagonal polarizer in front of the detector before the measurement. This projects us onto the polarization state $\frac{1}{\sqrt{2}}(|V\rangle + |H\rangle)$, absorbing another half of the photons. At the detector, we now have the state

$$|\psi\rangle = \tfrac{1}{2\sqrt{2}}\left(e^{i\phi_L}|\psi_L\rangle + e^{i\phi_R}|\psi_R\rangle\right) \otimes (|V\rangle + |H\rangle). \tag{23.10}$$

When we measure, we obtain the same interference pattern we had before, with a reduction in the intensity due to the absorbed photons. Indeed, if we look at the pattern with a diagonal polarizer before the detector, and with an antidiagonal polarizer before the detector, the sum of the two results yields the result with no polarizer, which has no interference (see Fig. 23.6). In some sense, we can view the pre-detector polarizers as filters that remove the photons in a nontrivial fashion, creating the interference pattern. This can also be thought of as a post-selection processing of the data. If we did a multimodal measurement at each pixel on our screen (by using a calcite detector or a polarizing beam splitter), then we can obtain the probability to arrive at each pixel as a function of the polarization of the light. By plotting the results of each polarization, we would then see the results shown in the figure.

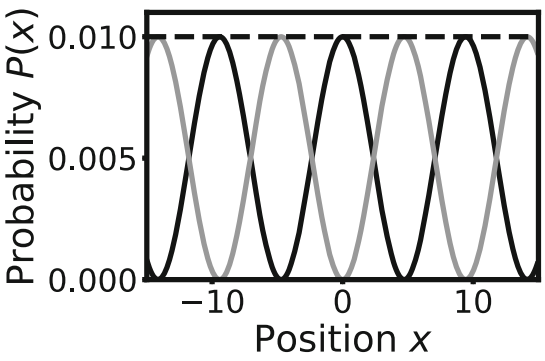

Fig. 23.6 Probability at the screen showing the full probability (straight dashed line), the diagonal polarization result (black line) and the antidiagonal polarization result (gray line). The sum of the two solid lines yields the dashed line, showing how the diagonal polarizers filter out the results nonuniformly, depending on the phase difference between the two slits

One of the benefits of the two-slit experiment is its simplicity and the ease with which we can describe it. The experiment has a lot of interesting aspects to it, but the key point is that whenever we have which-way information, we lose the interference effects and when we do not, they return. We can even have partial interference by rotating the angle of the polarizer before the detector from 0° to 90°. When we use polarizers to tag the path the photon went through, the disturbance itself of the measurement (going through the polarizer) does not directly give us information about which path it went through unless we also measure the polarization. So, interference is not destroyed, because we had to disturb the system in order to determine which slit the photon went through. This was the old-fashioned way of explaining the quantum mystery using the uncertainty principle to explain the loss of interference. But it is incorrect as a way to describe all experiments that measure at the slits. Instead, it is simply the question of whether you are in a product state superposition or an entangled superposition that determines the behavior here.

Section 23.3 Problems:

23.3.1 Changing the Polarization Basis in the Calculation of the Probability

We work on the two-slit experiment with crossed polarizers at the slits. Consider the state of the system just before entering the detector (with no polarizer in front of the detector). The procedure to measure the probability at x is to project onto $|x\rangle\langle x| \otimes |V\rangle\langle V|$, take the modulus squared and add that to the case where we project onto $|x\rangle\langle x| \otimes |H\rangle\langle H|$, and take the modulus squared.

The result should not depend on which basis we take for the polarization states. Redo the calculation using polarizer states at the angle θ from the vertical and $90° + \theta$. Show that your final result does not depend on the angle θ chosen.

23.3.2 Modifying the Analysis with Multiple Paths Through Each Slit

Consider the case with two paths through each slit. Now we will have phases ϕ_L and $\phi_L + \delta\phi_L$ and similar for the right slit. Here $\delta\phi$ is small in magnitude and essentially independent of x. Describe how the analysis changes in this case for the description of the two slit experiment.

23.3.3 Differences Between Conventional and Entangled Superposition and How Interference Arises

Write an essay that explains why we have different results when we have a simple superposition (two-slit experiment with no polarizers) and the entangled superposition (two-slit experiment with crossed polarizers). Carefully explain why interference is lost when we have polarizers at the slits even if we do not measure the polarization.

23.3.4 Pattern as a Function of the Angle of the Polarizer at the Detector

Consider the two-slit experiment with polarizers at the slits. We place a polarizer with an angle of θ to the vertical in front of the detector at position x. Determine the pattern you will see on the screen as a function of θ. Explain the cases $\theta = 0°, 45°$, and $90°$, as well as the general case.

23.3.5 Measurement as a Filtering Operation

Consider the two-slit experiment with polarizers at the slits, and imagine we have a multimodal detector that both counts photons and measures their polarization at the position x. Assume we measure the polarization in the V and H basis. Explain what the total pattern is, what the V polarized pattern is, and what the H polarized pattern is. Do your answers change if we measure diagonal and antidiagonal polarizations? Explain.

23.4 Interaction-Free Measurements and the Quantum Zeno Effect

In quantum mechanics, we have often discussed how quantum objects travel in superposition until they are measured. This invokes the notion that they do not move on single well-defined particle trajectories. But this has been known to be incorrect ever since the beginning of quantum mechanics, because we have had cloud chamber images of particle trajectories from high-energy physics experiments. We do not see random positions of the particles. We instead see that the particles have well-defined trajectories, which can be analyzed to determine their velocity, charge, and so on. So, what is going on? How come the measurement did not collapse the wavefunction and localize the particle?

The answer is that the measurement at each point on the trajectory is a *weak measurement*. It occurs by the high-energy particle ionizing molecules in the gas, which then act as seed points for the cloud formation that creates the trajectory imaged in the chamber. We now call such a process a *weak measurement*, because the high-energy particle has a large energy and it loses just a little bit to ionize the molecules, so it can ionize many of them without significantly changing its quantum state. The details for how this works are quite complicated, and we refer to Ballesteros et al. (2021) for further details. These weak measurements provide limited information about the quanta at each measurement and hence disturb it only slightly. Nevertheless, we always see nice trajectories of the particles when they are observed this way. This makes it appear that we should think of the quanta as traveling on trajectories. Such a perspective is used in the consistent quantum histories approach of Griffiths (2002). But we have not fully adopted that picture in this work. We also won't discuss weak measurements further.

Moving from weak measurements, where we gently interact with our quanta many times, we now consider *interaction-free measurements*, where there is no collapse at all of the wavefunction. The experiments work in a different fashion. The first experiment of this kind that I am aware of was done more than 100 years ago, and it is called the Millikan oil drop experiment. The history of this experiment is complicated, with a graduate student, Harvey Fletcher, having significant contributions to the Nobel-prize winning work, but not being a co-author on the original contribution (Perry, 2007).

The oil-drop experiment is an ingenious approach to determining the charge on a single electron. It again shows the importance of good experimental design to measure individual quanta. The way it works is that the amplification part of the experiment is the use of an oil drop, which is visible when viewed through a low-power microscope, to which a number of electrons are attached. The charged oil drop is then placed inside a parallel plate capacitor, which creates an electric field that can oppose the downward pull of gravity. By measuring the size of the drop (to determine its mass), and tuning the magnitude of the field to get the drop to float, we can determine the total electric charge on the drop (because the electric and gravitational forces are matched). Millikan and Fletcher found that the charges

were always integer multiples of a fundamental charge, which they identified as the charge of the electron. The experiment is wonderful, because it amplifies the small quantum signal to a measurable signal in an ingenious way and it involves no direct interaction with the quanta, at least in the sense that we never directly measure the electron—we do it indirectly. Since the quantum state of the electron remains undisturbed throughout the experiment, it is a true interaction-free experiment.

We have discussed a number of other interaction-free experiments already. The most prominent one was the quantum seeing-in-the-dark experiment, where we used a Michelson-Morley interferometer to measure whether or not we had an obstruction present in one of the arms by a photon that never interacted with the obstruction. The success probability of the experiment improved as we made it have more and more steps, which shows that it uses the ideas of the *quantum Zeno effect* in the experimental design. We will review a number of these ideas next to understand how to design these types of interaction-free experiments.

The key to designing such experiments is to have the system weakly coupled to the probe state (usually via a superposition) and then by repeating the experiment many times (in a weak-measurement or quantum Zeno-effect fashion), allowing us to obtain increasing sensitivity for the measurement. This design is not so easy to carry out without careful thought about how to achieve the goal. We now describe the quantum seeing-in-the-dark experiment that we covered in Sect. 4.4, which is a nice modern example of an interaction-free experiment. We use a small admixture of horizontal polarization into a vertically polarized photon, and then when we pass through a polarizing beam splitter, only a small weight of the quantum state reflects. If there is no blockage, then the horizontal and vertical components recombine in the beam splitter, and we continue to rotate the polarization until we have converted vertical to horizontal polarized light. If there is blockage, then the horizontal component is "absorbed," and the vertical component remains vertically polarized. Now we put absorbed in quotes, because we analyze the experiment as a beam splitter with detectors on the two paths—there we find that the photon really travels on one path or the other, as it does here. We detect the vertical polarization only if the photon is not absorbed. Of course, the experiment is set up to only detect horizontal polarized photons, so if it is absorbed, or is vertical polarized, we will not see it. The key is that by making the number of stages in the experiment as large as possible, we can increase the success rate (the photon is not absorbed) to be as high as we want. The key to this design is that we control the weak measurement by the amount of horizontal polarization admixture that we have at each pass through the device. Then, as the experiment goes through many repetitions, we finally complete the experiment. This experiment has again a very clever design that allows us to make the weak measurements and also allows us to have a high chance for success. The quantum Zeno effect enters via the measurement of the vertical polarization by the polarizing beam splitter during each pass through the device when one arm is blocked. The polarization of the photon in this case does not rotate due to the large number of measurements.

The second experiment that we will discuss is the measurement of the average number of phonons in a trapped ion system by measuring the spin dynamics, as

we discussed in Sect. 22.6. This experiment is not really an interaction-free measurement. Instead it is transducing the phonon properties onto the spin properties via entanglement. We discuss it in this section, because it has the nice property of measuring the property of interest (phonon occupancy) via a measurement of a heralded property (the spin dynamics), and this type of measurement shares some similarities with interaction-free experiments. We pump the spins into the all up state, and they are initially not entangled with the phonons. Then we run a spin-echo-like sequence: The spins are rotated into the x-y plane they evolve in time for some time τ; we flip them in the plane and then evolve again for the same time. If there was no spin-phonon entanglement during the evolution, the spins return to the state they were originally in. It is how the state changes that determines the number of phonons in the system. This arises from simple decoherence—we entangle the spins with the phonons and then measure the spins without detecting the phonons, and we see the spin properties degraded due to spin-phonon entanglement. It shares the design principle with the seeing-in-the-dark experiment in that it is the character of the final state that tells us about the presence of the phonons. It differs in that we are using entanglement rather than superposition to determine our measured result, and so it is not quite interaction-free.

The best example of the quantum Zeno effect that we have discussed is the time-of-flight experiment by Cindy Regal's laboratory that was used to measure the harmonic oscillator wavefunctions in Sect. 14.6. In that experiment, we use a cycling transition of the atom to have it absorb and reemit many photons over a short period of time. Each measurement fixes the location of the atom. Because the cycling transition occurs so rapidly, the atom is frozen in space and cannot move. This allows us to determine the position precisely, which then allows us to infer the momentum, as we always do in time-of-flight experiments. In general, this is how the quantum Zeno effect works—we have a state that is given by the initial state plus a small admixture of other states. Measuring it forces it to go back to the initial state with high probability. In some cases, such as the quantum seeing-in-the-dark experiment, the polarizing beam splitter is guaranteed to always restore the original state, while in other cases, there is only a high probability of success at each step.

Section 23.4 Problems:

23.4.1 Kasevich-Chu Atom Interferometry

Interferometry can be employed to measure the local gravitational field. This was first done using neutrons moving on two paths and falling as they moved. Kasevich-Chu atom interferometry is a more advanced and controllable way of doing the same experiment. Describing it takes a bit of time, but the analysis is simple if we use our operator methods. Our discussion follows closely (Kleinert et al., 2015). We provide a somewhat simplified analysis for how the experiment works.

Atom interferometry works by applying two pulses to a "lambda" structured atom that transition from the ground state to an excited state and back, without ever occupying the virtual state. Because the light involves two different frequencies, a net momentum $\hbar k$ is also imparted to the system. The picture is that we have ground state atoms that initially have no kinetic energy. We apply a pulse that acts like a beam splitter—it creates a 50-50 superposition of the ground and excited state with the excited state given a momentum kick upward. We wait a time T over which the atoms fall due to gravity and then we apply a mirror pulse, which transforms ground to excited and *vice versa*. We then allow them to fall due to gravity again for the same time interval, and finally we recombine them via second beam-splitter pulse. We measure the populations of ground and excited states at the end to determine how the states interfered. By applying these results on a cloud of ultracold atoms, we enhance the signal. By letting them fall for a long time, we can measure the phase due to the gravitational force, including small deviations due to nearby moving massive objects. In this problem, our focus is in using quantum mechanics to describe the different parts of the interferometer and understand how it can be used.

We start with describing the beam splitter and mirror pulses. We have seen similar things before when we discussed the superpositioner and the exciter–de-exciter for the Stern-Gerlach experiment. The beam splitter takes $|g\rangle \to \frac{1}{\sqrt{2}}(|g\rangle - ie^{i\phi(t)+ik\hat{z}}|e\rangle)$, with the excited state receiving a momentum kick of $\hbar k$ and both receiving opposite phase shifts of $\phi(t)$ from the lasers. The momentum shift changes depending on the position of the atom at the time of the shift. Similarly, the excited state satisfies $|e\rangle \to \frac{1}{\sqrt{2}}(-ie^{-i\phi(t)-ik\hat{z}}|g\rangle + |e\rangle)$. The mirrors act similarly. We have $|g\rangle \to -ie^{i\phi(t)+ik\hat{z}}|e\rangle$ and $|e\rangle \to -ie^{-i\phi(t)-ik\hat{z}}|g\rangle$.

We are now ready to move forward and calculate the change of the quantum states in the interferometer. We have a lower path and an upper path. We follow only the states that end up in the ground state when working out the evolution operators. We find that the upper path involves a transition from ground to excited, falling, a transition to ground, falling, and nothing at the last beam splitter. The lower path involves nothing, falling, a transition to upper, falling, and a transition to ground. In equations, we have the upper path state is given by $-\frac{1}{2}\hat{U}_{\text{upper}}|g\rangle$, with

$$\hat{U}_{\text{upper}} = \underbrace{e^{-\frac{i}{\hbar}\hat{H}T}}_{\text{fall}}\underbrace{e^{-i\phi(T)-ik\hat{x}}}_{\text{mirror}}\underbrace{e^{-\frac{i}{\hbar}\hat{H}T}}_{\text{fall}}\underbrace{e^{i\phi(0)+ik\hat{x}}}_{\text{beam splitter}} = e^{-i\phi(T)+i\phi(0)}e^{-\frac{i}{\hbar}\hat{H}T}e^{e^{-ik\hat{k}}\hat{H}e^{ik\hat{x}}T}. \tag{23.11}$$

The lower path state is given by $-\frac{1}{2}\hat{U}_{\text{lower}}$, with

$$\begin{aligned}\hat{U}_{\text{lower}} &= \underbrace{e^{-i\phi(2T)-ik\hat{k}}}_{\text{beam splitter}}\underbrace{e^{-\frac{i}{\hbar}\hat{H}T}}_{\text{fall}}\underbrace{e^{i\phi(T)+ik\hat{x}}}_{\text{mirror}}\underbrace{e^{-\frac{i}{\hbar}\hat{H}T}}_{\text{fall}}|g\rangle \\ &= e^{-i\phi(2T)+i\phi(T)}e^{e^{-ik\hat{z}}\hat{H}e^{ik\hat{z}}T}e^{-\frac{i}{\hbar}\hat{H}T}.\end{aligned} \tag{23.12}$$

Now, at the end of the interferometer, the ground state piece of the superposition is given by $-\frac{1}{2}(\hat{U}_{\text{upper}}+\hat{U}_{\text{lower}})|g\rangle$, so to find the probability to be in the ground state, we take the modulus squared of this state, or, because the $\hat{U}$ operators are unitary, we find

$$P_g(2T) = \frac{1}{2}\left(1+\frac{1}{2}\langle g|\hat{U}^\dagger_{\text{lower}}\hat{U}_{\text{upper}}|g\rangle + \frac{1}{2}\langle g|\hat{U}^\dagger_{\text{upper}}\hat{U}_{\text{lower}}|g\rangle\right). \tag{23.13}$$

Verify this equation. Then, use the Hadamard lemma and the Baker-Campbell-Hausdorff formula in the Weyl form to show that $\langle g|\hat{U}^\dagger_{\text{lower}}\hat{U}_{\text{upper}}|g\rangle = e^{i\alpha}$. Determine α. You may assume that $\hat{H} = \frac{\hat{p}^2}{2M} + Mg\hat{z}$.

You should find your final answer depends on the phases imparted to the state by the laser pulse at the different times and the value of the gravitational constant. So, by measuring the fraction of the population in the ground state, we can measure g. While this might seem like an odd way to do this, it can be exceedingly accurate. So much so, that one can detect small deviations to g in the local region of the interferometer if the time of the run is long enough. There is even talk of using this type of device as a gravitational wave detector. Of course for any of this to work, we need to know the phases from the laser quite accurately.

23.5 Uncertainty, Complementarity, and Collapse

We have discussed uncertainty extensively in this book. In this section, we will summarize these discussions as well as the discussions of complementarity, which are closely related to uncertainty. We also dispel a number of colloquial sayings about quantum mechanics that are plain wrong. While many others may say these things in public you will not. And if you want to, you can even correct them about what they are saying wrong.

We begin with the time-of-flight experiment to measure momentum. There are a few salient points we want to summarize about this. The experiment measures position to infer momentum, so there is no restriction in measuring position and momentum at the same time, as opposed to the common folklore. Uncertainty enters only when we consider many measurements, and we will find the momentum values yield a distribution of the momentum, which is the same as the momentum distribution of the original quantum state at the time of release for the time-of-flight experiment, because the momentum distribution is unchanged when the system is in a force-free region. The question of whether we actually know the position and momentum at the moment that we make the position measurement in the experiment is a matter of philosophy—for sure, there is no quantum state that has definite position and momentum. And equally certain, there is nothing in quantum mechanics that forbids us to use a position measurement to determine the momentum. Never say that "we cannot measure position and momentum at the same

time." The uncertainty principle does not have anything to say about that. Instead, it is safe to say "we can never have a quantum state with definite position and definite momentum."

The two-slit experiment with polarizers is the best example to discuss complementarity, although a similar discussion could be made with a Mach-Zehnder interferometer with a single polarizing beam splitter. When the polarizer is used to tag the paths, we have which-way information. Whenever we have such information, the interference effects disappear, and the system has a more particle nature to it. The which-way information can be erased with an appropriate polarizer inserted before the detector (typically a diagonal polarizer). This erases the which-way information allowing us to observe the interference, and the system has a more wave nature to it. While it is tempting to say that an entangled state always behaves like a particle with no interference and a product state superposition always behaves like a wave with interference, this is not always the case, as we saw when analyzing the Mach-Zehnder interferometer. It is really the question of whether or not we have which-way information that is critical, and this behavior is somewhat different from entangled or product state. Most of the time, we can differentiate the behavior using entangled and product state reasoning, but one needs to be careful, because it can fail.

Since passing through a polarizer is a measurement, we do not see the collapse of the system to a single state, because the measurement is a parity measurement. These types of measurements often have more than one state with the same eigenvalue. In this case, we can have arbitrary superpositions of the states with the same eigenvalue in the state that survives after the measurement. Be sure you are not confused by this behavior as it is an important concept to master.

It is very common for people to say that quanta travel as waves and are detected as particles. But this statement is patently incorrect. Wavefunction@Wave-particle duality is all about the quanta simultaneously having both properties, not their choosing to have one or the other depending upon what they are doing at a given instant of time. Wheeler's delayed choice experiment of the Mach-Zehnder interferometer, where we have a single beam splitter, we find the quanta travel on one path or the other only, versus the case where we add the second beam splitter and we have interference, indicating they are traveling on both paths, is the best example of this. To explain it properly, we know that the quantum state is an indeterminant state when passing through the first beam splitter, and then depending on what we do we either measure this state or modify it by going through a second beam splitter before measuring. But, whichever case we choose, we can determine what happens directly from what the quantum state is at the time of the measurement. We can ask, how does the photon know whether it travels on a definite path or in an indeterminate fashion, but this is not how we analyze the experiment—instead, we say that the state of the photon is always in a superposition after passing through a beam splitter. How it can be further described depends on how we further prepare the state prior to measurement. Of course, the details of how the quanta actually carry out this behavior remain to be the quantum mystery, even if our formalism always describes exactly what is measured.

The notion of "travels like waves but is detected like particles" arises from the challenge in reconciling how a large wave can be absorbed by a small atom or molecule (as in photography). We have discussed this point already with regard to photoproduction of atoms including an intuitive picture that helps us make sense of the process, which involves the dual nature of the quanta. So don't let someone else try to convince you that a small object cannot absorb or emit a large wave. It can. And now you know how to describe this behavior correctly.

Most single-quanta experiments involve a destructive measurement of the quanta, especially when measuring light. In this case, we do not need to worry at all about collapse of a wavefunction, because there is nothing after the measurement to describe. We also know, unambiguously, exactly when the measurement took place. Collapse becomes important when we are using phenomena such as the quantum Zeno effect. If we instead perform weak measurements, the collapse effect is also weakened, because the probe does not disturb the system strongly or change the quantum state significantly. We even can have measurements that do not affect a quantum superposition, as in the parity experiments we discussed for the two-slit experiment with polarizers. Again, we simply point out that every case is different and can have nuanced behavior that must be analyzed carefully.

Finally, we have multimodal experiments that can simultaneously measure two properties at once, such as position and polarization, or momentum and spin. These measurements open up new doors to experiments that can provide correlated information about the two quantities being measured rather than average values of each. This will lead in the future to much more advanced experiments that can do amazing things. I wish they had been available when I was an undergraduate. I worked on a project to analyze time-reversal invariance tests in muon capture. After spending nearly a year working on my project, I found the term that we wanted to measure exactly vanished! I spoke with a professor about this, and he convinced me that I had made a mistake—only later did he provide me with a simple explanation for why I was right. The only resolution was to measure the momentum and spin of the tritium nucleus that is produced after the muon capture. But this was deemed to be essentially impossible then. If only we had the multimodal experimental capabilities for that measurement!

Section 23.5 Problems:

23.5.1 Travels Like Waves and Is Detected Like Particles

Write a short essay that explains why the notion "travels like waves but is detected like particles" is too simplistic to describe what is really occurring in the quantum world.

23.6 Von Neumann Theory of Measurement and the Measurement Problem

John von Neumann (see Fig. 23.7) wrote his masterpiece on quantum mechanics in 1932 (von Neumann, 1932). The book has a significant treatment on the theory of measurement. The specific way in which his theory is constructed is that we have a classical device (typically described as a large machine with a dial that points to different values), and we somehow have a strategy to entangle the classical system with the quantum degrees of freedom in such a way that the different components of the quantum state, corresponding to the different eigenvalues that can be measured, are distinctly coupled to different classical states that point to the different eigenvalues corresponding to the possible results of the measurement. Then, at some point in time, the quantum state collapses onto one of these values, with the pointer pointing to the measured value, and we have performed the measurement. The measurement problem is all about trying to determine how this collapse occurs (or in some cases whether a collapse is needed at all).

Having seen a lot of different types of quantum measurement in this book, it should immediately be clear to you that while von Neumann's theory certainly would work, in principle, it is not the way any real quantum experiment actually works. We have no idea how to make such a classical device that can interact with individual quanta in the desired fashion! Most experiments are counting experiments, and they simply work in a different way, without the need for a classical device that the quantum systems become entangled with. The theory did not attract much attention when it was first released. Even two decades later in the 1950s, many people did not pay much attention to von Neumann's theory. Schrödinger was particularly critical, writing that "With great acuity he

Fig. 23.7 John von Neumann, the mathematical physicist who first put quantum mechanics on a rigorous mathematical footing (from the 1950s). He also described a theory for quantum measurements that underpins the so-called quantum measurement problem (Image courtesy of the AIP Emilio Segrè Visual Archives; modified to look like a sketch by James Freericks)

[von Neumann] constructs one analytical example. It does not refer to any actual experiment; it is purely analytical. He indicates in a simple case a supplementary operator which, when added to the internal wave operator, would with any desired approximation turn the wave function as time goes on into an eigenfunction of the observable that is measured. He found it necessary to show that such a mechanism is analytically possible. The idea has not been taken up and worked out since—in about 20 years or more. Indeed I do not think it would pay. *I do not believe any real measuring device is of this kind* (Schrödinger, 1995) (emphasis added is mine). This expresses a similar attitude to how we have been describing the von Neumann measurement problem—it is not formulated in a way that actually connects to experiments, even if it can be mathematically analyzed. Interestingly, in the past 40 years, the measurement problem has become a large field that many people work in. For this group of people, the measurement problem is *the fundamental problem* in quantum mechanics. Most solutions of the problem involve adopting different interpretations of quantum mechanics, and we will discuss these interpretations in a bit more detail below.

Within the von Neumann paradigm, the goal of the measurement problem is to find a way for the entangled quanta-classical device quantum state to become a mixed state, with no coherences. Then the measurement process will immediately follow as a simple probabilistic mixture of the measured states. If we stick with the unitary time evolution that we use in quantum mechanics, this cannot be done. If we think of the classical device as a sort of environment, with many degrees of freedom that we cannot formally describe, and we focus on the behavior of the system only, then after entanglement occurs, the natural decoherence will create such a mixed state for the system. But this does not really solve the problem, because we cannot create the initial entanglement in any clear way.

This shows that the measurement problem, when couched in the von Neumann paradigm, just is not the right way to look at how measurement is done. As we have discussed many times, measurement requires a clever design of the experiment to reveal the properties you are seeking. Key elements include single quanta sensitivity, amplification of the signal so it can be measured with "classical" equipment, and reproducibility of the measurement scheme. I believe this is where one needs to focus to design new experiments, rendering the measurement problem itself as a much less important question, certainly not *the* remaining unsolved question of quantum mechanics.

Nevertheless, this does not mean that these questions are unimportant. Just like other quantum mysteries that we have encountered, and not resolved, this problem is an important one to understand better. In particular, one would like to find a way to remove the collapse postulate from the description and have a better way to describe the way transitions occur. What is remarkable is that even without having such a detailed theory, the predictions we can make still agree well with the experimental results, and this is why many people adopt a "shut up and calculate" attitude.

Dag has a wonderful video https://tinyurl.com/47thwma5 on this theme (you need to watch until the very end).

All is not lost however. There are a number of recent developments that are attempting to solve this problem. In my opinion, there are two current formulations that show promise in better describing how measurement works. One is developed by Wiseman, and we will use that approach to describe some of the experiments of cavity QED below (Wiseman and Milburn, 2010). The other is the consistent history approach of Griffiths, which we have mentioned earlier, but will not discuss in any further detail here.

We discuss Wiseman's approach a bit more. He has developed a more refined version of the von Neumann approach, where the coupling of the classical system to the quantum system can be described in terms of operators acting on the system. The formulation is similar to the exact way that one computes Kraus operators when discussing decoherence—we let the combined apparatus and system evolve in time until they have coupled together forming the appropriate entangled states, then we compute the matrix elements of the time evolution with respect to the apparatus, leaving measurement operators that act on the system. Then, if the system is now decoupled from the apparatus, it encodes the experimental result, Wiseman says that we need not invoke any collapse of the system, because the classical apparatus can just be described classically and the results of measurements on the apparatus are given by classical random numbers. This approach shares some similarities with the many-worlds interpretation below but has some sophisticated twists to it. I encourage you to look further into this work if it interests you. Wiseman's approach is best suited to quantum non-demolition experiments, which we cover in Problem 23.6.1 below.

Philosophers and physicists have proposed a number of different resolutions to the measurement problem. One of them is the decoherence model that we have discussed above. Another is the many-worlds interpretation, which never has collapse, instead it says that all possible results occur in the different worlds that bifurcate every time a measurement is performed. One final solution we will discuss is a spontaneous collapse model, which says collapse occurs at a rate such that it is rare for isolated quanta, but common for large collections of quanta, so that once we go from a small to a mesoscopic system, collapse occurs essentially all the time. How can we actually decide on the right interpretation? It really requires a well-thought-out experimental design (similar in spirit to what Bell did for hidden variables) to find a way to determine which interpretation is correct. So this work has real merit, especially for researchers trying to find experiments that can differentiate between the different interpretations. Otherwise it is just argument, which is an exercise left better for the comedians of the world as expertly done by Monty Python's Flying Circus (Monty Python's Flying Circus, 1972) in their Argument Sketch.

Section 23.6 Problems:

23.6.1 Quantum Non-demolition Experiment

A quantum non-demolition experiment is an experiment that transfers the measurement from the object being measured to an ancillary system. The ancilla is decoupled from the system and then measured. In this fashion, we learn complete information about the system, without ever having any collapse occur. The experiments of cavity QED by Serge Haroche are one of the best examples of quantum non-demolition experiments. In this experiment, they measure the presence of a single photon nondestructively. These ideas have advanced significantly, and now cavity QED experimental protocols exist for measuring many photons simultaneously and nondestructively.

The Haroche experiment uses atom interferometry along with a clever way to transfer knowledge of the presence of a photon to the ancilla for measurement. There are two main steps for how this works. The experiment uses a "ground state" $|g\rangle$ which is an $n = 50$ Rydberg state of a Rb atom, an "excited state" $|e\rangle$, which is an $n = 51$ Rydberg state, and an ancilla state $|i\rangle$, which is an $n = 49$ Rydberg state and is used for the measurement and the atom interferometry (yes, it should be denoted as $|a\rangle$, but it was called $|i\rangle$ in the literature, so we stick with that nomenclature).

The cavity is set up in such a way that the two state system $|g\rangle$ and $|e\rangle$ evolves according to $\hat{U}(t) = e^{-\frac{i}{2}\Omega t\sigma_x}$ when a photon is in the cavity; the state $|i\rangle$ is too far detuned from the cavity to interact with it. Because of this resonant tuning of the cavity, the system is constantly emitting and absorbing a photon, transitioning from $|g\rangle$ to $|e\rangle$ and back in a Rabi oscillation.

(a) Starting from a state in $|g\rangle$ at $t = 0$, use the generalized Euler relation to find the state as a function of time t. You should find that if there is initially a photon in the cavity, so the initial state is $|g\rangle \otimes |1\rangle$, it then evolves to $|e\rangle \otimes |0\rangle$, and back. Take the time $t = 2\pi/\Omega$, and show that if there is a single photon in the cavity, the state returns to $-|g\rangle$ after the atom passes through the cavity. The cavity is set up for this to be precisely what happens. Note that after the passage through the cavity, the atom is decoupled from the photon. If the cavity has no photons in it, then the ground state simply stays as the ground state for the entire transit through the system, because there is no photon to absorb and make a superposition with the excited state. So, we go from $|g\rangle \otimes |0\rangle$ to $|g\rangle \otimes |0\rangle$ with no accumulation of any phase.

(b) We use an atom interferometry sequence with the ancilla state to encode the change in phase. Before entering the cavity, we pulse the atom to place it in the state $\frac{1}{\sqrt{2}}(|g\rangle + |i\rangle)$. The ancilla state cannot couple to the photon, because it is too far off resonance with the cavity, so the cavity does nothing to it, but its energy is different from that of the $|g\rangle$ state, so it picks up a different dynamical phase from the $|g\rangle$ state as a function of time. We define the phase ϕ, such that, after the transit through

the cavity, the state evolves to $\frac{1}{\sqrt{2}}(|g\rangle + e^{i\phi}|i\rangle)$ ignoring an overall global phase. Then we apply another atom interferometry pulse which takes $|g\rangle \rightarrow \frac{1}{\sqrt{2}}(|g\rangle + |i\rangle)$ and $|i\rangle \rightarrow \frac{1}{\sqrt{2}}(|g\rangle - |i\rangle)$. Show that the probability to measure the final state in the $|g\rangle$ state is $\frac{1}{2}(1 + \cos\phi)$ and in the $|i\rangle$ state is $\frac{1}{2}(1 - \cos\phi)$. The phase ϕ can be adjusted by changing the relative energies of the states by applying a static electric or magnetic field to the entire system.

(c) We put both pieces together to get the entire experiment. Write an essay explaining how we can pick a ϕ value such that when we measure the state after going through the cavity, it is either in the $|g\rangle$ state when there is no photon or in the $|i\rangle$ state when there is exactly one photon in the cavity. What value of ϕ does this? Can the experiment work for other values of ϕ? If so, which ones? Summarize how this experiment measures the presence of a photon without ever disturbing it. How does this experiment compare with quantum seeing in the dark and the delayed choice Stern-Gerlach experiment?

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Correction to: Quantum Mechanics Done Right

Correction to:
J. K. Freericks, *Quantum Mechanics Done Right*, https://doi.org/10.1007/978-3-031-87845-9

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Epilogue

We Did It

So long and thanks for all the fish.

Douglas Adams, Science Fiction Writer

Congratulations!

You have completed a long journey and can now say that you are a true *quantum mechanic*. In my opinion, this is a huge milestone for those who complete it. Quantum mechanics is known as the most accurate scientific theory, period. It is perhaps the most remarkable intellectual achievement of humankind. Yet, only a very small fraction of people actually know what it is. By making your way through this book, you are now one of those people. *Welcome to the club!* It is a very select one. Enjoy an overview of the journey at https://tinyurl.com/4wkbcj8s.

In working through the material in this book, you have improved your ability to solve problems and to think critically. These are skills that can be applied in many different areas. If you choose to go further in quantum mechanics, you will need to develop your math skills a bit more. Most more advanced topics (many-body physics, quantum field theory, and so on) do require use of more advanced mathematics, and there is no way to avoid that. You are ready to take that leap now and I encourage you to go forward if that is what you would like to do.

One question that might arise is do I need to go back and learn quantum mechanics using the differential equation and calculus-based approach? In my opinion definitely not! In many respects, the way you have learned quantum mechanics is more sophisticated and provides you with more powerful tools than the traditional approach. However, because many others only know quantum mechanics from the differential equation paradigm, it will be easier to talk with them if you share a common background. So having some knowledge of these ideas would be helpful. Once you have the requisite math background, you will be able to move through that material rather quickly given your extant quantum knowledge.

In particular, there are some topics we have not covered, because they really are inseparable from calculus. This includes perturbation theory (which is a Taylor expansion for the energy), potential scattering theory (which lays the groundwork

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for how high energy and other scattering-based experiments work and requires integration for constructing wavepackets), and semiclassical approaches to quantum mechanics (which also require calculus and help show the transition to classical mechanics). In addition, we did not fully cover the interaction of atoms with fields, or simple applications to atomic physics and solid state physics. You should be able to pick up those ideas as you need them, especially if you examine a more standard approach or go on to study these different fields in more detail.

Perhaps the most important question is what can you do with this new knowledge that you have? Believe it or not, you have a strong enough background to allow you to engage in research using quantum mechanics ideas (see how at https://tinyurl.com/y2kh8yjx). You would be most successful with mentored research, where you work closely with someone who will direct your work. I have done this with a large number of citizen scientists who have taken my course and who have read earlier versions of this book (about a dozen people so far; this research is summarized at https://tinyurl.com/56c643fj). They have accomplished a lot (many of my references involve work with people who have had a similar training to you), and you can too! I wish you the best of luck in how you apply your new knowledge.

Before ending, I want to let you watch videos about the history of the animations (at https://tinyurl.com/ya2fp3u3) and of the artwork (at https://tinyurl.com/bp9fx6j4). Creating these resources helped enhance much of the work in this project.

And finally, I have one last video which touches on an important, but often not discussed topic of how do we separate the outstanding science of many of the scientific leaders from their sometimes quite poor characters and behaviors as people during their lives. This is a difficult subject to discuss, but an important one, and my thoughts are summarized in this video https://tinyurl.com/4zjc73kb.

The playlist *Quantum Mechanics Done Right* https://tinyurl.com/yt2895u3 is likely to have additional videos placed there after the book is published, so please visit it and subscribe if you want automatic notices of updates.

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References

Abbott, B. P., et al., (LIGO Scientific Collaboration and Virgo Collaboration). (2016). Observation of gravitational waves from a binary black hole merger. *Physical Review Letters, 116*, 061102. http://doi.org/10.1103/PhysRevLett.116.061102.

Adams, B. G. (1994). *Algebraic approach to simple quantum systems: With applications to perturbation theory*. Springer-Verlag.

Adams, D. (1989). *The long dark tea-time of the soul*. Simon & Schuster.

Andrianov, A. A., Borisov, N. V., & Ioffe, M. V. (1984). The factorization method and quantum systems with equivalent energy spectra. *Physics Letters, 105A*, 19–22. http://doi.org/10.1016/0375-9601(84)90553-X.

Arecchi, F. T., Courtens, E. Gilmore, R., & Thomas, H. (1972). Atomic coherent states in quantum optics. *Physical Review A, 6*, 2211–2237. http://doi.org/10.1103/PhysRevA.6.2211.

Arnold, J. T., & Packard, M. E. (1951). Variations in absolute chemical shift of nuclear induction signals of hydroxyl groups of methyl and ethyl alcohol. *The Journal of Chemical Physics, 19*, 1608–1608. http://doi.org/10.1063/1.1748134.

Aspect, A. (2002). Bell's theorem: The naive view of an experimentalist. In R. A. Bertlmann, & A. Zeilinger, *Quantum (un)speakables: From Bell to quantum information*. Springer-Verlag.

Aspect, A., Grangier, P., & Roger, G. (1981). Experimental tests of realistic local theories via Bell's theorem. *Physical Review Letters, 47*, 460–463. http://doi.org/10.1103/PhysRevLett.47.460.

Avery, J. S. (2010). Harmonic polynomials, hyperspherical harmonics, and atomic spectra. *Journal of Computational and Applied Mathematics, 233*, 1366–1379. http://doi.org/10.1016/j.cam.2009.02.057.

Baker, G. A., Jr. (1956). Degeneracy of the n-dimensional, isotropic, harmonic oscillator. *Physical Review, 103*, 1119–1120. http://doi.org/10.1103/PhysRev.103.1119.

Balasubramanian, S., Han, S., Yoshimura, B. T., & Freericks, J. K. (2018). Bang-bang shortcut to adiabaticity in trapped-ion quantum simulators. *Physical Review A, 97*, 022313. http://doi.org/10.1103/PhysRevA.97.022313.

Ballentine, L. E. (2015). *Quantum mechanics: A modern development* (2nd ed.). World Scientific.

Ballesteros, M., Benoist, T., Fraas, M., & Fröhlich, J. (2021). The appearance of particle tracks in detectors. *Communications in Mathematical Physics, 385*, 429–463. http://doi.org/10.1007/s00220-021-03935-0.

Bell, J. S. (1964). On the Einstein Podolsky Rosen Paradox. *Physics, Physique, Fizika, 1*, 195–290. http://doi.org/10.1103/PhysicsPhysiqueFizika.1.195.

Bennett, C. H., & Brassard, G. (1984). Quantum cryptography: Public key distribution and coin tossing. In *International Conference on Computers, Systems & Signal Processing* (pp. 175–179), Bangalore, 10–12 December 1984.

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J. K. Freericks, *Quantum Mechanics Done Right*,
https://doi.org/10.1007/978-3-031-87845-9

Berry, M. (2023) Smooth and oscillatory geometric phase corrections for driven systems. *European Journal of Physics, 44*, 065402. http://doi.org/10.1088/1361-6404/acf81e.

Berry, M. V. (1984). Quantal phase factors accompanying adiabatic changes. *Proceedings of the Royal Society of London, Series A, 392*, 45–57. http://doi.org/10.1098/rspa.1984.0023.

Bertlmann, R., & Friis, N. (2023). *Modern quantum theory: From quantum mechanics to entanglement and quantum information*. Oxford University Press.

Bethe, H. (1931). Zur Theorie der Metalle. I. Eigenwerte und Eigenfunktionen der linearen Atomkette. *Zeitschrift für Physik, 71*, 205–226. http://doi.org/10.1007/BF01341708.

Bethe, H. E., & Salpeter, E. E. (1977). *One- and two-electron atoms*. Plenum Publishing.

Bigham, N., Denchy, M., & Kegerreis J. (2018). A simplified technique for the collection of an HCl/DCl gas mixture. *Journal of Chemical Education, 95*, 2298–2300. http://doi.org/10.1021/acs.jchemed.8b00702.

Blinder, S. M. (2011). Nuclear magnetic resonance spectrum of ethanol. http://demonstrations.wolfram.com/NuclearMagneticResonanceSpectrumOfEthanol/. Wolfram Demonstrations Project.

Bohm, A. (1994). *Quantum mechanics: Foundations and applications* (3rd ed.). Springer.

Bohm, D. (1951). *Quantum theory*. Prentice Hall, Inc.

Bohr, N. (1911). Studier over Metallernes Elektrontheori, Thesis, Københavns Universitet. (Republished in The Doctor's Dissertation (Text and Translation). In L. Rosenfeld & J. Rud Nielsen (Eds.) (1972). *Early Works (1905–1911). Niels Bohr Collected Works* (Vol. 1, pp. 163–393). Elsevier. http://doi.org/10.1016/S1876-0503(08)70015-X.

Bohr, N. (1935). Can quantum-mechanical description of physical reality be considered complete? *Physical Review, 48*, 696–702. http://doi.org/10.1103/PhysRev.48.696.

Boretsky, J. Z., Cohn, J. R., & Freericks, J. K. (2018). Spin and pseudospin towers of the Hubbard model on a bipartite lattice. *International Journal of Modern Physics B, 32*, 1840021. http://doi.org/10.1142/S0217979218400210.

Born, M., & Fock, V. A. (1928). Beweis des Adiabatensatzes. *Zeitschrift für Physik A, 51*, 165–180. http://doi.org/10.1007/BF01343193.

Born, M., Heisenberg, W., & Jordan, P. (1926). Zur Quantenmechanik. II. *Zeitschrift für Physik, 35*, 557–615. http://doi.org/10.1007/BF01379806.

Born, M., & Jordan, P. (1930). *Elementare Quantenmechanik: Zweiter Band der Vorlesungen über Atommechanik*. Springer.

Bransden, B. H., & Joachain, C. J. (2003). *Physics of atoms and molecules*. Prentice Hall.

Breitenbach, G., Schiller, S., & Mlynek, J. (1997). Measurement of the quantum states of squeezed light. *Nature, 387*, 471–485. http://doi.org/10.1038/387471a0.

Brickwedde, F. G. (1982). Harold Urey and the discovery of deuterium. *Physics Today, 35*, 34–39. http://doi.org/10.1063/1.2915259.

Brown, M. O., Muleady, S. R., Dworschack, W. J., Lewis-Swan, R. J., Rey, A. M., Romero-Isart, O., & Regal, C. A. (2023). Time-of-flight quantum tomography of an atom in an optical tweezer. *Nature Physics, 19*, 569–573. http://doi.org/10.1038/s41567-022-01890-8.

Chabris, C., & Simons, D. (2010). *The invisible gorilla: And other ways our intuitions deceive us*. Crown.

Chen, J., Cheng, H.-P., & Freericks, J. K. (2021). Quantum-inspired algorithm for the factorized form of unitary coupled cluster theory. *Journal of Chemical Theory and Computation, 17* 841–847. http://doi.org/10.1021/acs.jctc.0c01052.

Cirac, J. I., & Zoller, P. (1995). Quantum computations with cold trapped ions. *Physical Review Letters, 74*, 4091–4094. http://doi.org/10.1103/PhysRevLett.74.4091.

Čížek, J. (1966). On the correlation problem in atomic and molecular systems. Calculation of wavefunction components in Ursell-type expansion using quantum-field theoretical methods. *The Journal of Chemical Physics, 45*, 4256–4266. http://doi.org/10.1063/1.1727484.

Cohn, J., Safavi-Naini, A., Lewis-Swan, R. J., Bohnet, J. G., Garttner, M., Gilmore, K. A., Jordan, J. E., Rey, A. M., Bollinger, J. J., & Freericks, J. K. (2018). Bang-bang shortcut to adiabaticity in the Dicke model as realized in a Penning trap experiment. *New Journal of Physics, 20*, 055013. http://doi.org/10.1088/1367-2630/aac3fa.

Commins, E. D. (2014). *Quantum mechanics: An experimentalist's approach*. Cambridge University Press.
Cooper, F., Khare, A., & Sukhatme, U. (2001). *Supersymmetric quantum mechanics*. World Scientific.
Corry, S. (2017). *Symmetry and quantum mechanics*. Chapman & Hall/CRC Press.
Courtney, W. F., Vieira, L. B., Julienne, P. S., & Freericks, J. K. (2020). Incorporating the Stern-Gerlach delayed-choice quantum eraser into the undergraduate quantum mechanics curriculum. *American Journal of Physics, 88*, 298–307. http://doi.org/10.1119/10.0000519.
Curtwright, T. L., Fairlie, D. B., & Zachos, C. K. (2014). A compact formula for rotations as spin matrix polynomials. *Symmetry Integrability and Geometry: Methods and Applications, 10*, 084. http://doi.org/10.3842/SIGMA.2014.084.
De Lange, O. L., & Raab, T. E. (1991). *Operator methods in quantum mechanics*. Clarendon Press.
Dicke, R. H. (1954). Coherence in spontaneous radiation processes. *Physical Review, 93*, 99–110. http://doi.org/10.1103/PhysRev.93.99.
Dirac, P. A. M. (1926). On the theory of quantum mechanics. *Proceedings of the Royal Society A (London), 112*, 661–677. http://doi.org/10.1098/rspa.1926.0133.
Dirac, P. A. M. (1930). *The principles of quantum mechanics*. Clarendon Press.
Dirac, P. A. M. (1939). A new notation for quantum mechanics. *Mathematical Proceedings of the Cambridge Philosophical Society, 35*, 416–418. https://doi.org/10.1017/S0305004100021162.
Doyle, A. C. (1890). *The sign of four*. Spencer Blackett.
Dylewsky, D., Freericks, J. K., Wall, M. L., Rey, A. M., & Foss-Feig, M. (2015). Nonperturbative calculation of phonon effects on spin squeezing. *Physical Review A, 93*, 013415. http://doi.org/10.1103/PhysRevA.93.013415.
Einstein, A., Podolsky, B., & Rosen, N. (1935). Can quantum-mechanical description of reality be considered complete? *Physical Review, 47*, 777–780. http://doi.org/10.1103/PhysRev.47.777.
Elitzur, A. C., & Vaidman, L. (1993). Quantum mechanical interaction-free measurements. *Foundations of Physics, 23*, 987–997. http://doi.org/10.1007/BF00736012.
Epstein, J. H., & Epstein, S. T. (1962). Some applications of hypervirial theorems to the calculation of average values. *American Journal of Physics, 30*, 266–268. http://doi.org/10.1119/1.1941987.
Epstein, L. C. (1983). *Relativity visualized*. Insight Press.
Farhi, E., Goldstone, J., & Gutmann, S., (2014). A quantum approximate optimization algorithm. Preprint. https://arxiv.org/abs/1411.4028.
Feinberg, J. (1983). Interferometer with a self-pumped phase-conjugating mirror. *Optics Letters, 8*, 569–571. http://doi.org/10.1364/ol.8.000569.
Feynman, R. P. (1964). *Probability and uncertainty—The quantum mechanical view of nature*. The British Broadcasting Organization.
Feynman, R. P. (1965). *The character of the physical law*. The British Broadcasting Organization.
Feynman, R. P. (1985). *QED: The strange theory of light and matter*. Princeton University Press.
Feynman, R. P., Leighton, R. B., & Sands, R. (1965). *The Feynman lectures in physics, volume III: Quantum mechanics*. California Institute of Technology.
Fock, V. N. (1935). Zur Theorie des Wasserstoffatoms. *Zeitschrift für Physik, 98*, 145–154. http://doi.org/10.1007/BF01336904.
Foulis, D. L. (2017). The algebra of complex 2×2 matrices and a general closed Baker–Campbell–Hausdorff formula. *Journal of Physics A: Mathematical and Theoretical, 50*, 305204. http://doi.org/10.1088/1751-8121/aa7954.
Fowles, G. R. (1962). Solution of the Schrödinger equation for the hydrogen atom in rectangular coordinates. *American Journal of Physics, 30*, 308–309. http://doi.org/10.1119/1.1941997.
Franson, J. D. (1989). Bell inequality for position and time. *Physical Review Letters, 62*, 2205–2208. http://doi.org/10.1103/PhysRevLett.62.2205.
Freericks, J. K. (2022). Quantum number towers for the Hubbard and Holstein models. In R. L. Frank, A. Laptev, M. Lewin, & R. Seiringer (Eds.), *The physics and mathematics of Elliott Lieb: The 90th anniversary*. EMS Press.

Freericks, J. K. (2022). Operator relationship between conventional coupled cluster and unitary coupled cluster. *Symmetry, 14*, 494. http://doi.org/10.3390/sym14030494.

Freericks, J. K. (2023). How to measure the momentum of single quanta. *European Physical Journal: Special Topics, 232*, 3285–3294. http://doi.org/10.1140/epjs/s11734-023-00967-6.

Freericks, J. K., & Lieb, E. H. (1995). The ground state of a general electron-phonon Hamiltonian is a spin singlet. *Physical Review B, 51*, 2812–2821. http://doi.org/10.1103/PhysRevB.51.2812.

Friedrich, B., & Herschbach, D. (2003). Stern and Gerlach: How a bad cigar helped reorient atomic physics. *Physics Today, 56*, 53–59. http://doi.org/10.1063/1.1650229.

Frisch, R., & Segrè, E. (1933). Über die einstellungder richtungsquantelung. II. *Zeitschrift für Physik, 80*, 610–616. http://doi.org/10.1007/BF01335699.

Gabor, D. (1961). Light and information. In E. Wolf (Ed.), *Progress and optics* (Vol. 1, pp. 109–153). North-Holland. http://doi.org/10.1016/S0079-6638(08)70609-7.

Gaillard, M. K. (2015). *A singularly unfeminine profession: One woman's journey in physics*. World Scientific Publishing Co.

Glauber, R. J. (1965). *Quantum optics and electronics (Les Houches 1964)*, C. de Witt, A. Blandin, C. Cohen-Tannoudji (Eds.). Gordon and Breach Science Publishers, Inc.

Glauber, R. J., & Martin, P. C. (1958). Relativistic theory of radiative orbital electron capture. *Physical Review, 109*, 1307–1325. https://doi.org/10.1103/PhysRev.109.1307.

Grangier, P., Roger, G., & Aspect, A. (1986). Experimental evidence for a photon anticorrelation effect on a beam splitter—A new light on single-photon interferences. *Europhysics Letters, 1*, 173–179. http://doi.org/10.1209/0295-5075/1/4/004.

Green, H. (1965). *Matrix mechanics*. Groningen, P. Noordhoff, Ltd.

Greenstein, G., & Zajonc, A. G. (1997). *The quantum challenge: Modern research on the foundations of quantum mechanics*. Jones and Bartlett Publishers.

Griffiths, R. B. (2002). *Consistent quantum mechanics*. Cambridge University Press.

Grosswald, E. (1978). *Bessel polynomials*, A. Dold & B. Eckmann (Eds.). *Lecture Notes in Mathematics* (Vol. 698). Springer, Inc.

Guttieres, L. A. J., Petrović, M. D., & Freericks, J. K. (2023). Computational projects with the Landau-Zener problem in the quantum mechanics classroom. *American Journal of Physics, 91*, 885–892. http://doi.org/10.1119/5.0139717.

Hahn, E. L. (1950). Spin echoes. *Physical Review, 80*, 580–594. http://doi.org/10.1103/PhysRev.80.580.

Heilmann, O. J., & Lieb, E. H. (1971). Violation of the noncrossing rule: The Hubbard Hamiltonian for Benzene. *Annals of the New York Academy of Sciences, 172*, 584–617. http://doi.org/10.1111/j.1749-6632.1971.tb34956.x.

Heisenberg, W. (1925). Über quantentheoretische Umdeutung kinematischer und mechanischer Beziehungen. *Zeitschrift für Physik, 33*, 879–893. http://doi.org/10.1007/BF01328377.

Heisenberg, W. (1926). Mehrkörperproblem und Resonanz in der Quantenmechanik. *Zeitschrift für Physik, 38*, 411–426. http://doi.org/10.1007/BF01397160.

Hellmuth, T., Walther, H., Zajonc, A., & Schleich, W. (1987). Delayed-choice experiments in quantum interference. *Physical Review A, 35*, 2532–2541. http://doi.org/10.1103/PhysRevA.35.2532.

Hillmer, R., & Kwiat, P. (2007). A do-it-yourself quantum eraser. *Scientific American, 296*(May issue), 90–95.

Hong, C. K., Ou Z. Y., & Mandel, L. (1987). Measurement of subpicosecond time intervals between two photons by interference. *Physical Review Letters, 59*, 2044–2047. http://doi.org/10.1103/PhysRevLett.59.2044.

Hubbard, J. (1963). Electron correlations in narrow energy bands. *Proceedings of the Royal Society, 276*, 238–257. http://doi.org/10.1098/rspa.1963.0204.

Hulthén, L. (1938). Über das Austauschproblem eines Kristalles. *Arkiv för Matematik, Astronomi och Fysik, 26A*, 1–106.

Hylleraas, E. A. (1932). Die Wellengleichung des Keplerproblems im Impulsraume. *Zeitschrift für Physik, 74*, 216–224. http://doi.org/10.1007/BF01342375.

Infeld, L., & Hull, T. E. (1951). The factorization method. *Reviews of Modern Physics, 23*, 21–68. http://doi.org/10.1103/RevModPhys.23.21.

Ising, E. (1925). Beitrag zur Theorie des Ferromagnetismus. *Zeitschrift für Physik, 31*, 253–258. http://doi.org/10.1007/BF02980577.

Islam, R., Edwards, E. E., Kim, K., Korenblit, S., Noh, C.,Carmichael, H., Lin, G.-D., Duan, L.-M., Wang, C.-C. Joseph, Freericks, J. K., & Monroe, C. (2011). Onset of a quantum phase transition with a trapped ion quantum simulator. *Nature Communications, 2*, 377. http://doi.org/10.1038/ncomms1374.

James, D. F. V. (1998). Quantum dynamics of cold trapped ions with application to quantum computation. *Applied Physics B, 66*, 181–190. http://doi.org/10.1007/s003400050373.

Jaynes, E. T., & Cummings, F. W. (1963). Comparison of quantum and semiclassical radiation theories with application to the beam maser. *Proceedings of the IEEE, 51*, 89–109. http://doi.org/10.1109/PROC.1963.1664.

Jordan, P., & Wigner, E. (1928). Über das Paulische Aquivalenzverbot. *Zeitschrift für Physik, 47*, 631–651. http://doi.org/10.1007/BF01331938.

Judd, B. R. (1975). *Angular momentum theory for diatomic molecules*. Academic Press, Inc.

Karbach, M., & Müller, G. (1997). Introduction to the Bethe ansatz I. *Computational Physics, 11*, 36–43. http://doi.org/10.1063/1.4822511.

Kazlikowski, D. (2014). *Seeing without looking*, youtube video, https://www.youtube.com/watch?v=I7oZAo8hJnU.

Kennard, E. H. (1927). Zur Quantenmechanik einfacher Bewegungstypen. *Zeitschrift für Physik, 44*, 326–352. http://doi.org/10.1007/BF01391200.

Kiely, T. G., & Freericks, J. K. (2018). Relationship between the transverse-field Ising model and the XY model via the rotating-wave approximation. *Physical Review A, 97*, 023611. http://doi.org/10.1103/PhysRevA.97.023611.

Kim, K., Chang, M.-S., Korenblit, S., Islam, R., Edwards, E. E., Freericks, J. K., Lin, G.-D., Duan, L.-M., & Monroe, C. (2010). Quantum simulation of frustrated Ising spins with trapped ions. *Nature, 465*, 590–593. http://doi.org/10.1038/nature09071.

Kim, K., Korenblit, S., Islam, R., Edwards, E. E., Chang, M.-S., Noh, C., Carmichael, H., Lin, G.-D., Duan, L.-M., Wang, C.-C. Joseph, Freericks, J. K., & Monroe, C. (2011). Quantum simulation of the transverse Ising model with trapped ions. *New Journal of Physics, 3*, 105003. http://doi.org/10.1088/1367-2630/13/10/105003.

Kleinert, S., Kajari, E., Roura, A., & Schleich W. P. (2015). Representation-free description of light-pulse atom interferometry including non-inertial effects. *Physics Reports, 605*, 1–50. http://doi.org/10.1016/j.physrep.2015.09.004.

Kramers, H. A. (1957). *Quantum mechanics*. North-Holland.

Kramida, A. E. (2010). A critical compilation of experimental data on spectral lines and energy levels of hydrogen, deuterium, and tritium. *Atomic Data and Nuclear Data Tables, 96*, 586–644. http://doi.org/10.1016/j.adt.2010.05.001.

Kutzelnigg, W. (1977). Pair correlation theories. In H. F. Schaefer (Ed.), *Methods of electronic structure theory* (Chapter 5, pp. 129–188). Springer.

Landau, L. D. (1932). On the theory of transfer of energy at collisions II. *Physikalische Zeitschrift der Sowjetunion, 2*, 46–51.

Landau, L. D., & Lifshitz, E. M. (1977). *Quantum mechanics: Non-relativistic theory* (3rd ed.). Pergamon Press.

Lenz, W. (1920). Beitrag zum Verständnis der magnetischen Erscheinungen in festen Körpern. *Physikalische Zeitschrift, 21*, 613–615.

Li, J., Zgid, D. K., & Freericks, J. K. (2025). Operator-based derivation of the wavefunctions of the Morse potential. *European Journal of Physics, 46*, 055801. https://doi.org/10.1088/1361-6404/adeee3.

Lieb, E. H., Two theorems on the Hubbard model. *Physical Review Letters, 62*, 1201–1204. http://doi.org/10.1103/PhysRevLett.62.1201.

Lieb, E. H., & Mattis, D. (1962a). Ordering energy levels in interacting spin chains. *Journal of Mathematical Physics, 3*, 749–751. http://doi.org/10.1063/1.1724276.

Lieb, E., & Mattis, D. (1962b). Theory of ferromagnetism and the ordering of electronic energy levels. *Physical Review, 125*, 164–172. http://doi.org/10.1103/PhysRev.125.164.

Lieb, E., Schultz, T., & Mattis, D. (1961). Two soluble models of an antiferromagnetic chain. *Annals of Physics (New York), 16*, 407–466. http://doi.org/10.1016/0003-4916(61)90115-4.

Lipkin, H. J., Meshkov, N., & Glick. A. J. (1965). Validity of many-body approximation methods for a solvable model: (I). Exact solutions and perturbation theory. *Nuclear Physics, 62*, 188–198. http://doi.org/10.1016/0029-5582(65)90862-X.

Lohmann, B., & Weigold, E. (1981). Direct measurement of the electron momentum distribution in atomic hydrogen. *Physics Letters, 86A*, 139–141. http://doi.org/10.1016/0375-9601(81)90851-3.

Lyu, X., Daniel, C., & Freericks, J. K. (2022). Cartesian operator factorization method for hydrogen. *Atoms, 10*, 14. http://doi.org/10.3390/atoms10010014.

Machluf, S., Japha, Y., & Folman, R. (2013). Coherent Stern–Gerlach momentum splitting on an atom chip. *Nature Communications, 4*, 2424. http://doi.org/10.1038/ncomms3424.

Majorana, E. (1932). Atomi orientati in campo magnetico variabile. *Il Nuovo Cimento, 9*, 43–50. http://doi.org/10.1007/BF02960953.

Manenti, R., & Motta, M. (2023). *Quantum information science*. Oxford University Press.

Marquet, C., Schmidt-Kaler, F., & James, D. F. V. (2003). Phonon–phonon interactions due to non-linear effects in a linear ion trap. *Applied Physics B, 76*, 199–208. http://doi.org/10.1007/s00340-003-1097-7.

Marshall, W. (1955). Antiferromagnetism. *Proceedings of the Royal Society A (London), 232*, 48–68. http://doi.org/10.1098/rspa.1955.0200.

Mattis, D. C. (1981). *The theory of magnetism I: Statics and dynamics*. Springer.

Monty Python's Flying Circus (1972). *The argument clinic*, text available at https://python.mzonline.com/sketches/argument/ and video at https://youtu.be/ohDB5gbtaEQ.

Munguía-González, E., Rego, S., & Freericks, J. K. (2021). Making squeezed-coherent states concrete by determining their wavefunction. *American Journal of Physics, 89*, 885–896. https://doi.org/10.1119/10.0004872.

Nagaoka, Y. (1965). Ground state of correlated electrons in a narrow almost half-filled s band. *Solid State Communications, 3*, 409–412. http://doi.org/10.1016/0038-1098(65)90266-8.

Nagaoka, Y. (1966). Ferromagnetism in a narrow, almost half-filled *s* band. *Physical Review, 147*, 392–405. http://doi.org/10.1103/PhysRev.147.392.

Noonan, J. R., Shah, M. R., Xu, L., & Freericks, J. K. (2024). Employing an operator form of the Rodrigues formula to calculate wavefunctions without differential equations. *American Journal of Physics, 92*, 270–279. http://doi.org/10.1119/5.0177925.

Nord, P. (2011). *Quantum Eraser*, youtube video, https://www.youtube.com/watch?v=o4s28XVfQaU.

Ohanian, H. C. (1990). *Principles of quantum mechanics*. Prentice Hall.

Onsager, L. (1944). Crystal statistics. I. A two-dimensional model with an order-disorder transition. *Physical Review, 65*, 117–149. http://doi.org/10.1103/PhysRev.65.117.

Onsager, L. (1949). As a comment on p. 261 in Rushbrooke, G. S. (1949). On the theory of regular solutions. *Il Nuovo Cimento (Suppl.), 6*, 251–263. http://doi.org/10.1007/BF02780989.

Orjuela, A. M., & Freericks, J. K. (2023) Free expansion of a Gaussian wavepacket using operator manipulations. *American Journal of Physics, 91*, 463–469. http://doi.org/10.1119/5.0083964.

Pasternack, S. (1937). On the mean value of r^s for Keplerian systems. *Proceedings of the National Academy of Sciences, 23*, 91–94. http://doi.org/10.1073/pnas.23.2.91.

Pauli, W., Jr. (1926). Über das Wasserstoffspektrum vom Standpunkt der neuen Quantenmechanik. *Zeitschrift für Physik A, 36*, 336–363. http://doi.org/10.1007/BF01450175.

Peierls, R. E. (1929). Uber die Existenz stationärer Züstande (On the existence of stationary states). *Zeitschrift für Physik, 58*, 59–62. http://doi.org/10.1007/BF01347930.

Peierls, R. E. (1936). On Ising's model of ferromagnetism. *Mathematical Proceedings of the Cambridge Philosophical Society, 32*, 477–481. http://doi.org/10.1017/S0305004100019174.

Perry, M. F. (2007). Remembering the oil-drop experiment. *Physics Today, 60*, 56–60. http://doi.org/10.1063/1.2743125.

Peruzzo, A., McClean, J., Shadbolt, P., Yung, M.n-H., Zhou, X.-Q., Love, P. J., Aspuru-Guzik, A., & O'Brien, J. L. (2014). A variational eigenvalue solver on a photonic quantum processor. *Nature Communications, 5*, 4213. http://doi.org/10.1038/ncomms5213.

Pohl, R., Antognini, A., Nez, F., Amaro, F. D., Biraben, F., Cardoso, J. M. R., Covita, D. S., Dax, A., Dhawan, S., Fernandes, L. M. P., Giesen, A., Graf, T., Hänsch, T. W., Indelicato, P., Julien, L., Kao, C.-Y., Knowles, P., Le Bigot, E.-O., Liu, Y.-W., ... Kottmann, F. (2010). The size of the proton. *Nature, 466*, 213–216. http://doi.org/10.1038/nature09250.

Rayo, A. (2019). *On the brink of paradox: Highlights from the intersection of mathematics and philosophy*. Massachusetts Institute of Technology Press.

Rigden, J. S. (2003). *Hydrogen: The essential element*. Harvard University Press.

Robinett, R. W. (2006). *Quantum mechanics: Classical results, modern systems, and visualized examples*. Oxford University Press.

Robinson, T. R. (2021). Natural number dynamics: Reconstructing physics from generalized atomicity. *European Journal of Physics, 42*, 025402. http://doi.org/10.1088/1361-6404/abc432.

Rushka, M., & Freericks, J. K. (2020). A completely algebraic solution of the simple harmonic oscillator. *American Journal of Physics, 88*, 976–985. http://doi.org/10.1119/10.0001702.

Rushka, M., Esrick, M. A., Mathews, W. N., & Freericks, J. K. (2021). Converting translation operators into plane polar and spherical coordinates and their use in determining quantum-mechanical wavefunctions in a representation-independent fashion. *Journal of Mathematical Physics, 62*, 072102. http://doi.org/10.1063/5.0021013.

Rutherford, E. (1911). The scattering of α and β-particles by matter and the structure of the atom. *Philosophical Magazine, 21*, 669–688. http://doi.org/10.1080/14786440508637080.

Safavi-Naini, A., Lewis-Swan, R. J., Bohnet, J. G., Garttner, M., Gilmore, K. A., Jordan, J. E., Cohn, J., Freericks, J. K., Rey, A. M., & Bollinger, J. J. (2018). Verification of a many-ion simulator of the Dicke model through slow quenches across a phase transition. *Physical Review Letters, 121*, 040503. http://doi.org/10.1103/PhysRevLett.121.040503.

Saulson, P. R. (2017). *Fundamentals of interferometric gravitational wave detectors* (2nd ed.). World Scientific, Inc.

Sawyer, B. C., Britton, J. W., Keith, A., Wang, C.-C. Joseph, Freericks, J. K., Uys, H., Biercuk, M. J., & Bollinger, J. J. (2012). Spectroscopy and thermometry of drumhead modes in a mesoscopic trapped-ion crystal using entanglement. *Physical Review Letters, 108*, 213003. http://doi.org/10.1103/PhysRevLett.108.213003.

Schrödinger, E. (1926a). Quantisierung als Eigenwertproblem (Erste Miteilung). *Annalen der Physik, 384*, 361–376. http://doi.org/10.1002/andp.19263840404.

Schrödinger, E. (1926b). Der stetige Ubergang von der Mikro-zur Makromechanik. *Naturwissenschaften, 14*, 664–666. http://doi.org/10.1007/BF01507634.

Schrödinger, E. (1940). A method of determining quantum-mechanical eigenvalues and eigenfunctions. *Proceedings of the Royal Irish Academy, Section A, 46*, 9–16. https://www.jstor.org/stable/20490744.

Schrödinger, E. (1995). *The interpretation of quantum mechanics* (p. 83). Ox Bow Press.

Schwinger, J. (2015). *On angular momentum*. Dover Publications, Inc.

Shen, S.-Q. (1996). Applications of reflection positivity in strongly correlated electron systems. *Physical Review B, 54*, 4397–4400. http://doi.org/10.1103/PhysRevB.54.4397.

Shen, S.-Q. (1998). Strongly correlated electron systems: Spin-reflection positivity and some rigorous results. *International Journal of Modern Physics B, 12*, 709–779. http://doi.org/10.1142/S0217979298000442.

Sørensen, A., & Mølmer, K. (1999). Quantum computation with ions in thermal motion. *Physical Review Letters, 82*, 1971–974. http://doi.org/10.1103/PhysRevLett.82.1971.

Stodolna, A. S., Rouzée, A., Lépine, F., Cohen, S., Robicheaux, F., Gijsbertsen, A., Jungmann, J. H., Bordas, C., & Vrakking, M. J. J. (2013). Hydrogen atoms under magnification: Direct observation of the nodal structure of Stark states. *Physical Review Letters, 110*, 213001. http://doi.org/10.1103/PhysRevLett.110.213001.

Stoppard, T. (1994). *Hapgood: A play*. Faber and Faber, Inc.

Stueckelberg, E. C. G. (1932). Theorie der unelastischen Stösse zwischen Atomen. *Helvetica Physica Acta, 5*, 369–422. http://doi.org/10.5169/seals-110177.
Styer, D. F. (2000). *The strange world of quantum mechanics*. Cambridge University Press.
Szymanski, T., & Freericks, J. K. (2021). Algebraic derivation of Kramers-Pasternack relations based on the Schrodinger factorization method. *European Journal of Physics, 42*, 025409. http://doi.org/10.1088/1361-6404/abd228.
Tasaki, H. (2020). *Physics and mathematics of quantum many-body systems*. Springer Nature.
Tavis, M., & Cummings, F. W. (1968). Exact solution for an N-molecule radiation-field Hamiltonian. *Physical Review, 170*, 379–384. http://doi.org/10.1103/PhysRev.170.379.
Taylor, E. F., & Wheeler, J. A. (1992). *Spacetime physics* (2nd ed.). W. H. Freeman and Co.
Taylor, G. I. (1909). Interference fringes with feeble light. *Proceedings of the Cambridge Philosophical Society, 15*, 114–115. http://doi.org/10.1016/B978-0-08-029160-4.50015-4.
Technical Services Group, Massachusetts Institute of Technology, video on diffraction (2012). Hosted on youtube https://www.youtube.com/watch?v=9D8cPrEAGyc.
Toprak, E., Zgid, D., & Freericks, J. K. (2025). Quantum description of the asymmetric top using average and relative coordinates in an operator-based formalism. *American Journal of Physics, 93*, 897–904. https://doi.org/10.1119/5.0258534.
Transnational College of LEX Staff (2009). Japanese Language Services Staff and translated by J. Nambu (Ed.), *What is quantum mechanics?: A physics adventure* (2nd ed.). Language Research Foundation.
Tran, J., Doughty, L., & Freericks, J. K. (2025). The 1925 revolution of matrix mechanics and how to celebrate it in modern quantum mechanics classes. *American Journal of Physics*, *93*, 14–20. https://doi.org/10.1119/5.0195658
Van Leeuwen, H. J. (1921). Problèmes de la théorie électronique du magnétisme. *Journal of Physics Radium, 2*, 361–377. http://doi.org/10.1051/jphysrad:019210020012036100.
Vijayan, S. (2012). Awesome home made experiment of delayed-choice quantum eraser. Hosted on youtube https://www.youtube.com/watch?v=R-6St1rDbzo.
von Neumann, J. (1932). *Mathematische Grundlagen der Quantenmechanik*. Julius Springer. Translated into English in 1955 and published by Princeton University Press.
Wang, C.-C. Joseph, & Freericks, J. K. (2012). Intrinsic phonon effects on analog quantum simulators with ultracold trapped ions. *Physical Review A, 86*, 032329. http://doi.org/10.1103/PhysRevA.86.032329.
Weigold, E., & McCarthy, I. (1999). *Electron momentum spectroscopy*. Kluwer Academic/Plenum Publishing.
Weinberg, S. (2012). *Lectures on quantum mechanics*. Cambridge University Press.
Weitzman, M., & Freericks, J. K. (2018). Calculating spherical harmonics without derivatives. *Condensed Matter Physics, 21*, 33002. http://doi.org/10.5488/CMP.21.33002.
Wheeler, J. A. (1983). Law without law. In J. A. Wheeler & W. H. Zurek (Eds.), *Quantum theory and measurement* (pp. 182–214). Princeton University Press.
Wigner, E. P. (1959). *Group theory and its application to the quantum mechanics of atomic spectra*. Academic Press.
Wilcox, R. M. (2017). Exponential operators and parameter differentiation in quantum physics. *Journal of Mathematical Physics, 8*, 962–982. http://doi.org/10.1063/1.1705306.
Winter, R. G. (1979). *Quantum physics*. Wadsworth Publishing Company.
Wiseman, H. M., & Milburn, G. J. (2010). *Quantum measurement and control*. Cambridge University Press.
Witten, E. (1982). Constraints on supersymmetry breaking. *Nuclear Physics B, 202*, 253–316. http://doi.org/10.1016/0550-3213(82)90071-2.
Yamanouchi, K. (2012). *Quantum mechanics of molecular structures*. Springer Verlag.
Yang, C. N. (1952). The spontaneous magnetization of a two-dimensional Ising model. *Physical Review, 85*, 808–816. http://doi.org/10.1103/PhysRev.85.808.
Yun, M., Ho, P., & Lo, K. (1994). A high-resolution image of atomic hydrogen in the M81 group of galaxies. *Nature, 372*, 530–532. http://doi.org/10.1038/372530a0.

Zener, C. (1932). Non-adiabatic crossing of energy levels. *Proceedings of the Royal Society of London. Series A, 137*, 696–702. http://doi.org/10.1098/rspa.1932.0165.

Zwiebach, B. (2022). *Mastering quantum mechanics: Essentials, theory, and applications*. Massachusetts Institute of Technology Press.

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